The continuum limit of three-dimensional SU(2) lattice Yang–Mills theory on a torus

P. M. Aronow

Patrick Lopatto

AUTHORS' NOTE

With the exception of this note, the entire document is machine-written. We have taken care to supervise this writing to ensure that the proofs meet a minimum standard of readability and that previous literature is appropriately cited.

The manuscript began as an attempt to pursue the program of Magnen, Rivasseau, and Sénéor [132] using large language models. Our hope was that understanding the problem on the torus might offer some insight into the general case. The construction that resulted from this starting point is three-dimensional and lattice-based, and draws extensively on Bałaban's renormalization group methods.

As we were verifying these results, we learned that Sussman and Alpöge announced that YM4 has been done, following Bałaban's program (`https://github.com/ethanwsussman/Yang--Mills-LLM-experiment`). We are thrilled and look forward to learning more about these exciting developments. We hope that this manuscript nonetheless contributes to understanding constructive approaches to three-dimensional Yang–Mills theory. We did not inspect and, to the best of our knowledge, the LLMs we used did not access the ongoing work of Sussman and Alpöge.

Although we have done extensive machine auditing of these proofs, including adversarial passes with multiple models, the proof is too complex for either (i) Lean autoformalization given the authors' resources or (ii) our own understanding of the results. Thus we are less confident in correctness than we are in our other work. We welcome suggestions concerning mathematical corrections, omitted citations, and attribution of ideas.

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ABSTRACT

We consider the Wilson lattice gauge measure for SU(2) on a periodic three-dimensional torus of fixed size, with bare coupling $g^2 = g_3^2 a$ proportional to the lattice spacing a, as dictated by the superrenormalizability of three-dimensional Yang–Mills theory. Observables are Wilson loops of the lattice configuration evolved by the Wilson flow, the lattice gradient flow of the Wilson action, at positive flow time. We prove that, at every fixed value of the coupling, the joint laws of finitely many flowed Wilson loops converge as the lattice spacing tends to zero along the sequences of lattice sides nM^k, for fixed odd $M \geq 3$ and fixed positive n divisible by four. For each such refinement class, the limits are realized by a unique probability measure on the space of gauge orbits of Yang–Mills heat-flow trajectories at positive times, a version of the state space introduced by Cao and Chatterjee. Under this measure the energy at positive times is tight, and the measure is invariant under the lattice symmetries and the center symmetries. We also show that, at short distances, the rescaled deficits $(1 - W)/(g_3^2 \varepsilon)$ of the normalized fundamental Wilson loops W of loops $x + \varepsilon \ell$, flowed for time $\sigma \varepsilon^2$ with fixed $\sigma > 0$, converge to explicit quadratic functions of the heat-smoothed Gaussian free field. As the coupling tends to zero, the measure concentrates on flat connections, with Gaussian fluctuations of explicit covariance near regular commuting holonomies and fluctuations governed by the constant-mode quartic form near the central holonomies. The proof is a multiscale renormalization group analysis in the tradition of Bałaban, carried out on a small-field event whose complement we show to have small probability, uniformly in the lattice spacing, using reflection positivity and chessboard estimates. It is combined with stochastic estimates for the lattice heat flow and a deterministic analysis of the Wilson flow for rough initial data.

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PART I

Introduction

SECTION 1

Introduction

1.1 Yang–Mills theory and its lattice formulation

Yang and Mills introduced non-Abelian gauge theories in 1954 [156]. In the Euclidean formulation of the quantum theory, one would like to define a probability measure on connections A on a three- or four-dimensional manifold, formally given by

$$d\mu(A) = Z^{-1} \exp\Big(-\frac{1}{2g^2} \int |F_A|^2 \, dx\Big) \, \mathcal{D}A. \tag{1.1}$$

Here F_A is the curvature of A, g is the coupling constant, and $\mathcal{D}A$ stands for a Lebesgue measure on the infinite-dimensional space of connections, which does not exist. The measure should be invariant under gauge transformations, so it is really a measure on gauge orbits. The natural observables are gauge-invariant functions, such as traces of holonomies around closed loops, which are called Wilson loops.

Wilson's lattice regularization [153] replaces the connection by variables U_e in the gauge group, one for each link e of a lattice with mesh a. The action becomes the sum of $1 - \operatorname{tr} U_p/2$ over the plaquettes p of the lattice, where U_p is the product of the link variables around p. For every $a > 0$ this defines a probability measure that is exactly invariant under lattice gauge transformations. Osterwalder and Seiler established reflection positivity and other basic properties of these measures [134]; see also Seiler's monograph [141]. The central problem is to remove the regularization: one must show that suitable observables have limits as $a \to 0$, with the bare coupling tuned appropriately, and describe the limit. In four dimensions this is part of the problem described by Jaffe and Witten [114]; see [77] for a recent review by Douglas.

In this paper we consider the gauge group $\mathrm{SU}(2)$ in three dimensions, on a periodic torus $\mathbb{T}^3_L = (\mathbb{R}/L\mathbb{Z})^3$ of fixed side L. In three dimensions the coupling g_3^2 in (1.1) has the dimension of inverse length. The dimensionless strength of the interaction at length scale ε is $g_3^2\varepsilon$. In the sense of power counting, the theory becomes weakly coupled at short distances; it is superrenormalizable. The bare lattice coupling at mesh a is $g^2 = g_3^2 a$. With $N = L/a$ lattice sites along each side, this means that

$$g_N^2 = \frac{\gamma}{N}, \qquad \gamma = g_3^2 L, \tag{1.2}$$

where the dimensionless number γ is the coupling measured in units of the size of the torus.

Three-dimensional Yang–Mills theory is of independent physical interest. It describes the magnetic sector of four-dimensional gauge theory at high temperature and long distances; see Ginsparg [98] and Appelquist and Pisarski [5], and also Linde [126] and Jackiw and Templeton [113] on the infrared problems of its perturbative treatment. It is also expected to have a mass gap and to confine. Feynman gave qualitative arguments for both [88]. Karabali, Kim, and Nair developed a gauge-invariant Hamiltonian analysis of the gap and an approximate prediction of the string tension [115, 116], and Teper computed the spectrum and string tensions numerically, with continuum extrapolations [150]. Our results concern only the ultraviolet problem in a fixed finite volume.

Even in three dimensions, the ultraviolet problem is not only a question of bounds. In the Gaussian (Maxwell) approximation to (1.1), after gauge fixing and with the constant modes treated separately, a connection in three dimensions is a random distribution of Hölder–Besov regularity α for every $\alpha < -1/2$, and no better. Holonomies along general fixed loops then require an additional regularization [42, 43]. A convenient way to obtain gauge-invariant observables with limits is to smooth the field first. The gradient flow of the Yang–Mills action, $\partial_t A = -D_A^* F_A$, smooths connections at length scale $\sqrt{t}$ and commutes with gauge transformations. Its lattice version is the gradient flow of the Wilson action. Narayanan and Neuberger studied continuum limits of smeared Wilson loops numerically [133], and Lüscher developed this lattice flow systematically under the name Wilson flow [128, 129]. In [131], Lüscher and Weisz showed that in four dimensions the correlation functions of the flowed field are finite to all orders in renormalized perturbation theory. Wilson loops of the flowed field at positive flow time are bounded gauge-invariant functions of the lattice configuration, and we use them as our observables.

The geometric study of Yang–Mills connections and their flow has a long history, including work of Atiyah and Bott [6] and Donaldson [75, 76]. In three dimensions, global existence and regularity for smooth initial data are due to Råde [139]. Charalambous and Gross developed the flow for rough initial data and proposed it as a way to define gauge fields [47–49, 105]. In [43], Cao and Chatterjee made this proposal precise. They defined a state space for three-dimensional Yang–Mills theory, consisting of gauge orbits of trajectories of the Yang–Mills heat flow at positive times. They showed that it carries a natural topology in which Wilson loops of the flowed field are continuous, and they gave a tightness criterion for measures on it. Their arguments rest on Uhlenbeck's compactness theorem [151] and on Sengupta's result that holonomies separate gauge orbits [144]. In [42] they also proved local existence of the flow in the DeTurck gauge for a class of random distributional initial data that includes the Gaussian free field. Their decomposition into a rough linear part and a smoother nonlinear part follows the random-data method of Bourgain [32] and of Da Prato and Debussche [63]. The construction of a Yang–Mills measure on this state space from the lattice theory was left open. Our main result is such

a construction.

1.2 Previous rigorous results

We summarize the rigorous results most closely related to ours. The literature is large, and Chatterjee's survey [51] gives an overview from the point of view of probability.

Abelian theories, Higgs models, and two dimensions. In [104], Gross proved that three-dimensional pure $U(1)$ lattice gauge theory converges to the free electromagnetic field, for the Villain action and, in the electric sector, for the Wilson action. Brydges, Fröhlich, and Seiler constructed the Abelian Higgs model in two dimensions [36, 37, 40, 41]. King constructed the Abelian Higgs model in two and three dimensions by proving convergence of effective actions [117, 118]. Chatterjee constructed a Gaussian scaling limit of $\mathrm{SU}(2)$ lattice Yang–Mills theory coupled to a fixed-length Higgs field in the fundamental representation, in every dimension $d \geq 2$ [53]. The result uses unitary gauge, sends the Higgs length to infinity and the gauge coupling to zero rapidly, and yields independent massive Proca fields; it concerns a different scaling regime from the pure gauge theory considered here. In two dimensions, Yang–Mills theory is exactly solvable. Gross, King, and Sengupta [106] and Driver [78] gave probabilistic constructions in the plane, and Driver proved that the lattice Wilson loop expectations converge. Sengupta [142, 143, 145] and Lévy [124, 125] treated compact surfaces, and Fine gave a different geometric approach [89, 90]. In [154], Witten related the weak-coupling limit on surfaces to volumes of moduli spaces of flat connections. In [55], Chevyrev realized the two-dimensional measure on the torus as a random distributional connection.

The renormalization group for lattice gauge theories. Bałaban developed a renormalization group method for lattice gauge theories in a long series of papers. He proved estimates for lattice Green's functions and propagators [9–11, 13], introduced a gauge-covariant averaging operation for group-valued fields [12], and analyzed regular configurations and the variational problem that defines the background field [14, 16]. The culmination is ultraviolet stability on a fixed torus: bounds on renormalized effective densities and suitably normalized partition functions that are uniform in the lattice spacing. He proved it in three dimensions in [15], and the later papers on effective actions and large fields developed the four-dimensional program [17, 19–21]. His treatment of the complete effective densities also gave convergent renormalization expansions for superrenormalizable gauge models [18]. Bałaban, Imbrie, and Jaffe explained the exact renormalization group for gauge theories and treated Higgs models [22–25]. Federbush developed a different phase-cell approach to the stability problem [81–86]. Ultraviolet stability is a statement about bounds, and it does not by itself give convergence of observables or a limiting measure.

In [132], Magnen, Rivasseau, and Sénéor developed a constructive analysis of the Schwinger functions of four-dimensional pure $\mathrm{SU}(2)$ Yang–Mills theory with a fixed infrared cutoff, in the trivial topological sector. They work in a regularized axial gauge, with counterterms that restore gauge invariance, and they prove large-field estimates and nonperturbative Slavnov identities. Dimock gave a detailed exposition of Bałaban's method for the scalar ϕ^4_3 model [67–69], and applied related ideas to gauge fixing and to quantum electrodynamics in three dimensions [70–73]. More broadly, our methods descend from the block-spin renormalization group of Gawędzki and Kupiainen [94, 95, 97]. They also use the cluster expansion methods described by Glimm and Jaffe [99], Brydges [38, 39], and Rivasseau [140]. More specifically, we use the convergence criterion of Kotecký and Preiss [120] and the tree estimates going back to Penrose, as explained by Fernández and Procacci [87] and recorded in the graph estimate of Dimock [68].

Stochastic quantization. A different approach constructs the Yang–Mills measure as the invariant measure of a Langevin dynamics. This idea goes back to Parisi and Wu [135], with the gauge-fixing term introduced by Zwanziger [157]. Using regularity structures [107], Chandra, Chevyrev, Hairer, and Shen constructed a gauge-covariant Langevin dynamics and a state space of gauge orbits in two dimensions [45]. They also constructed local-in-time dynamics for the Yang–Mills–Higgs model in three dimensions [46]. Chevyrev and Shen proved that the Yang–Mills measure on the trivial bundle over the two-dimensional torus is invariant for this dynamics [57]. They also proved that the mass renormalization of the Yang–Mills field that yields gauge-covariant solutions of the stochastic three-dimensional Yang–Mills–Higgs equations is unique [58]. Bringmann and Cao gave a paracontrolled approach in two dimensions [35]. For the non-Abelian theory in three dimensions, these results give local dynamics, and they do not yet construct an invariant measure. For the Abelian theory, Chevyrev, Qu, and Shen proved convergence of gauge-fixed three-dimensional lattice Langevin dynamics to the one-form stochastic heat equation for a class of plaquette actions including Wilson, Manton, and Villain actions, under assumptions on the initial data [59]. We refer to Chevyrev's survey [56] for more on this approach.

Lattice estimates. Many recent results concern lattice gauge theories at a fixed lattice spacing, for instance at strong coupling or large rank [2, 26, 44, 52, 54, 146]. Closer to the present work are results on the free energy and on bounds that are uniform in the lattice spacing. In [50], Chatterjee identified the leading term of the $U(N)$ lattice Yang–Mills free energy, and in [34], Brennecke computed its Maxwell constant. Faria da Veiga and O'Carroll proved stability bounds and bounds for generating functions of scaled plaquette fields [64, 65]. These results control the normalization and local plaquette observables. They do not identify the continuum laws of macroscopic flowed Wilson loops, and asymptotics of the free energy that are extensive in the volume do not

by themselves control normalized observables.

1.3 Main results

We now describe our results informally; the precise statements are in Section 2. Let $\mu_{N,\gamma}$ denote the Wilson measure for $\mathrm{SU}(2)$ on the periodic lattice in $\mathbb{T}_L^3$ with N^3 sites and mesh $a_N = L/N$, with the coupling (1.2). For a configuration U and a physical time $t > 0$, we write $U(t)$ for the Wilson flow of U run for the lattice time $4t/a_N^2$; with this normalization, $U(t)$ approximates the continuum Yang–Mills heat flow at time t. Let ℓ be a piecewise C^1 closed loop in $\mathbb{T}_L^3$, and let ℓ_N be a lattice approximation of ℓ of bounded length, in the sense of Definition 2.1. The flowed Wilson loop is $\operatorname{tr} \mathrm{Hol}_{\ell_N} U(t)/2$, the normalized trace of the product of the flowed link variables along ℓ_N.

We fix an odd integer $M \geq 3$ and an integer n divisible by four, and consider the lattice sides $N_k = nM^k$ for $k \geq 0$. The restriction to such sequences comes from the renormalization group, which divides the lattice into blocks of side M.

- *Existence of the continuum limit* (Theorem 2.3). Fix $L > 0$ and $\gamma > 0$. Consider finitely many flowed Wilson loops, for arbitrary piecewise C^1 loops, characters of $\mathrm{SU}(2)$, and positive flow times. As $k \to \infty$, their joint laws under $\mu_{N_k,\gamma}$ converge, for every choice of admissible lattice approximations, and the limits do not depend on this choice. All these limits are realized by a single probability measure λ on the space $\mathcal{X}_L$ of gauge orbits of Yang–Mills heat-flow trajectories at positive times. That is, the limiting law of each finite family of flowed Wilson loops is the law of the corresponding continuum holonomies under λ. The energy at each positive flow time is tight. The measure λ depends only on L, γ, and the class of n under multiplication by powers of M, and it is invariant under translations, rotations, and reflections of the lattice, and under the center symmetries.
- *A bound on large fields that is uniform in the coupling* (Theorem 2.7). We carry out the renormalization group analysis on a small-field event, an explicit set of configurations that are close to a gauge transform of the identity at every scale of a multiscale decomposition. Let u be the side of the coarsest lattice in the decomposition. We prove that the complement of the event has probability at most $Cu^3 \exp(-cu/\gamma)$, plus a term of order $u^3 \exp(-c \log^2(u/\gamma))$. The bound is uniform in the number of renormalization group steps and in γ below any fixed ceiling.
- *Short distances* (Theorem 2.8). Fix a smooth loop ℓ in $\mathbb{R}^3$ and $\sigma > 0$, and let W be the flowed Wilson loop of the loop $x + \varepsilon\ell$ at flow time $\sigma\varepsilon^2$. As $\varepsilon \to 0$, the normalized deficit $(1 - W)/(g_3^2 \varepsilon)$ converges in law to $|C|^2/8$, where C is an explicit Gaussian vector given by the free field. This holds at every fixed coupling and uniformly in the position of the loop. Jointly for finitely many fixed loop shapes, the limit is a quadratic function of a Gaussian vector, and it is asymptotically independent of every fixed

bounded measurable function on $\mathcal{X}_L$. This makes precise, for flowed Wilson loops, the statement that the theory is weakly coupled at short distances. In three dimensions this is a consequence of superrenormalizability, rather than of the logarithmic running of the coupling found in four dimensions by Gross and Wilczek [102] and Politzer [137].

- *Weak coupling* (Theorems 2.9 to 2.11). As $\gamma \to 0$, every subsequential limit of the measures λ_γ is supported on stationary flat connections. In particular, the holonomies (H_1, H_2, H_3) around the three coordinate cycles concentrate on commuting triples. On compact regular patches near the identity, the four normal components of the logarithms of the flowed holonomies have asymptotically Gaussian fluctuations of size $\sqrt{\gamma}$, with an explicit covariance, in a sense made precise in Theorem 2.10. At the central triples the flat connections are singular, and the logarithms of the holonomies live on the scale $\gamma^{1/4}$. Ratios of expectations of compactly supported invariant functions of the rescaled logarithms converge to the corresponding ratios of integrals against $e^{-Q(Y)}\,dY$, where $Q(Y) = \sum_{i<j} |y_i \times y_j|^2/2$ is the constant-mode quartic form. The error is $O(\gamma^{1/4})$ for Lipschitz test functions. The measure $e^{-Q(Y)}\,dY$ has infinite total mass, so this is not a statement about a limiting probability measure. In both regimes we obtain explicit statistics that distinguish λ_γ from specific Gaussian and Abelian reference measures; the leading coefficient in the regular regime is itself the coefficient of the Gaussian approximation around a regular flat connection.

Section 2 also contains further results, including continuity of λ in γ, with convergence that is uniform on compact sets of couplings, and exact scaling in L. We also prove convergence of the energy and loop processes on compact intervals of positive times, and convergence in the Sobolev topology of Cao and Chatterjee. Finally, for $0 < \gamma \le 1$ and along the subsequences of Remark 2.4, the limit does not change when the Wilson flow is replaced by the gradient flow of $\sum_p [q(U_p) + c\, q(U_p)^2]$, uniformly for c in a fixed bounded interval of nonnegative numbers.

We emphasize that the main theorem holds at every fixed coupling $\gamma > 0$, not only at small coupling. This reflects superrenormalizability: at lattice side u the effective coupling is γ/u, so we can run the renormalization group from the finest lattice down to a coarse lattice whose side u is large compared to γ, where the field is still weakly coupled. By the large-field bound, the probability discarded by this procedure tends to zero as $u \to \infty$, uniformly in the lattice spacing. The results on weak coupling concern the opposite limit, $\gamma \to 0$ at fixed u, taken after the continuum limit.

1.4 Relation to previous work

Compared to Bałaban's three-dimensional theorem [15], which bounds renormalized effective densities and suitably normalized partition functions uniformly in the cutoff, our

main theorem identifies the limits of normalized observables and constructs a limiting measure. Our proof follows Bałaban's general strategy. We use several of his constructions: the averaging operation, the background field obtained by minimizing the action subject to block constraints, and estimates for the associated propagators. However, we organize the analysis differently. Bałaban decomposes the configurations into small-field and large-field regions at every scale. Instead, we condition on a single small-field event. We bound the probability of its complement directly under the Wilson measure, using the reflection positivity of Osterwalder and Seiler [134] and the chessboard estimates of Fröhlich, Israel, Lieb, and Simon [92, 93], in the form given by Biskup [29]. Large fields enter only in this probability estimate, whose proof uses a localized version of the renormalization group. Beyond stability, we also need convergence of the conditioned measures at every fixed number of renormalization group steps. Further, we need estimates for the reconstruction of the fine field from the coarse fields, which control the flowed observables. The localization and fluctuation estimates adapt Combes–Thomas localization [61], the Brascamp–Lieb inequality [33], the Bakry–Émery criterion [8], the Helffer–Sjöstrand representation of covariances [110], and the Herbst argument as described by Ledoux [123]. The quadratic fluctuation bounds are of Hanson–Wright type [108], in the form of Adamczak [1], and the maximal estimates use chaining [74, 79].

Compared to the work of Magnen, Rivasseau, and Sénéor [132], our setting is three-dimensional and superrenormalizable. They work in four dimensions, with gauge-fixed Schwinger functions, a fixed infrared cutoff, the trivial topological sector, and counterterms that restore gauge invariance. We work with the Wilson lattice measure without counterterms, and our observables are gauge-invariant flowed Wilson loops. Cao and Chatterjee [43] constructed the state space and proved conditional tightness results; we construct a measure on a version of their state space from the lattice theory. We use their deterministic results on the flow and on the topology of the state space, but not their results for random initial data. The stochastic quantization approach gives local dynamics in three dimensions for the Yang–Mills–Higgs model, including the pure gauge case [46, 58]. We construct an equilibrium measure for the pure gauge theory and prove nothing about dynamics. Gross's three-dimensional Abelian result [104] and King's construction of the Abelian Higgs model [117, 118] are earlier continuum limits of three-dimensional lattice gauge theories; our result concerns a non-Abelian theory.

Our weak-coupling theorems are related to the analysis of Yang–Mills theory on a small torus. The center symmetries fit into the framework of electric and magnetic flux of 't Hooft [149]; we divide out strictly periodic gauge transformations, treat the center operations as symmetries, and include no twist. In the Hamiltonian formulation of the four-dimensional theory on a small spatial torus, Lüscher [127], Lüscher and Münster [130], and Koller and van Baal [119] showed that the constant modes must be treated nonperturbatively. The leading classical interaction of the constant modes is quartic at

the trivial flat connection, and integrating out the other modes adds further terms to the effective Hamiltonian; see van Baal's review [152]. Near regular flat connections our analysis is a version of Laplace's method around a nondegenerate critical manifold, in the sense of Bott [31]. González-Arroyo, Jurkiewicz, and Korthals Altes [101], and Coste and these authors [62], studied the zero-momentum modes of the Euclidean lattice theory in a finite box. In this literature the periodic flat connections are often called torons [100]. Fodor and collaborators combined an exact treatment of the constant modes with perturbation theory for gradient-flow observables on the four-dimensional torus [91]. The integral of e^{-Q} over the constant modes is the bosonic Yang–Mills integral of three matrices, studied by Krauth, Nicolai, and Staudacher [122], by Krauth and Staudacher [121], and by Austing and Wheater [7]. For $\mathrm{SU}(2)$ and three matrices it diverges, and for this reason our theorem at the central triples is stated for compactly supported test functions. Our weak-coupling theorems give rigorous versions of these pictures for the measure we construct, at positive flow time and in the order of limits described above.

1.5 Scope

We record what we do not prove. The limit is taken along the sequences $N_k = nM^k$. We do not compare the limits for different classes of n or different M; we expect, but do not prove, that λ does not depend on these choices. The torus is fixed: we do not take the infinite-volume limit, and we prove nothing about a mass gap or confinement. The observables are flowed Wilson loops at positive flow time. We do not construct a random distributional connection at time zero, and we do not prove reflection positivity of the limit or the Osterwalder–Schrader axioms. We do not give a rate of convergence in the lattice spacing. The large-field bound and the weak-coupling results are uniform in γ below fixed ceilings. The convergence in the main theorem, however, is not shown to be uniform as $\gamma \to 0$ or as $\gamma \to \infty$.

1.6 Organization

Section 2 states our results precisely, and Section 3 gives a detailed outline of the proof. Section 4 collects notation and conventions. The proof occupies the remaining sections, grouped into parts. Part II contains finite-dimensional tools: integration over $\mathrm{SU}(2)$, the block averaging map, reflection positivity, gauge fixing, and the linearized theory. Part III constructs the background fields. Part IV constructs the renormalization group transformation on the small-field event, and proves convergence at each fixed resolution. Part V estimates the reconstruction of the fine field. Part VI proves the bound on large fields. Part VII studies the lattice and continuum gradient flows, and Part VIII proves the stochastic estimates for the flowed field. Part IX constructs the limiting measure and

proves the main theorem. Parts X and XI prove the results on short distances and on weak coupling.

SECTION 2

Statement of results

This section states our results precisely. Section 2.1 defines the lattice model, the Wilson flow, and the lattice approximations of loops, and Section 2.2 defines the space on which the limiting measure lives. The main theorem is stated in Section 2.3, and further properties of the limit in Section 2.4. Section 2.5 states the bound on large fields that underlies the proof. The results on short distances and on weak coupling are in Sections 2.6 and 2.7.

2.1 The lattice model

Fix a side length $L > 0$ and an integer $N \geq 4$, and let $a_N = L/N$. Let $\Lambda_N = a_N\mathbb{Z}^3/L\mathbb{Z}^3$ be the periodic cubic lattice of mesh a_N in the torus $\mathbb{T}^3_L = \mathbb{R}^3/L\mathbb{Z}^3$. A positive link is a pair $(x, x + a_N e_i)$ with $x \in \Lambda_N$ and $i \in \{1,2,3\}$, and a positive plaquette is the positively oriented boundary of the square spanned by $a_N e_i$ and $a_N e_j$ at x, with $i < j$. Let $\mathcal{E}_N$ and $\mathcal{P}_N$ denote the sets of positive links and positive plaquettes; each has $3N^3$ elements. A configuration is an element $U = (U_e)_{e\in\mathcal{E}_N}$ of $\mathcal{U}_N = \mathrm{SU}(2)^{\mathcal{E}_N}$. For the reversed link we set $U_{\bar{e}} = U_e^{-1}$, and for a path $e_1 \cdots e_m$ in the lattice we write $U_{e_1} \cdots U_{e_m}$ for the transport along it. For a plaquette p, U_p is the transport around its boundary.

Let H be the Haar probability measure on $\mathrm{SU}(2)$, and let H_N be the product of copies of H over $\mathcal{E}_N$. Set

$$q(U) = 1 - \tfrac{1}{2}\operatorname{tr} U, \qquad U \in \mathrm{SU}(2).$$

For $\gamma > 0$ we define the Wilson measure

$$d\mu_{N,\gamma}(U) = \frac{1}{Z_{N,\gamma}} \exp\Big(-\frac{4N}{\gamma} \sum_{p\in\mathcal{P}_N} q(U_p)\Big)\, dH_N(U), \tag{2.1}$$

where $Z_{N,\gamma}$ is the normalizing constant. In the usual notation the inverse coupling is $\beta = 4/g_N^2$ with $g_N^2 = \gamma/N$, as in (1.2). A lattice gauge transformation $k \in \mathrm{SU}(2)^{\Lambda_N}$ acts by $(U^k)_{(x,y)} = k_x U_{(x,y)} k_y^{-1}$, and $\mu_{N,\gamma}$ is invariant under this action.

We write $\mathfrak{su}(2)$ for the Lie algebra of $\mathrm{SU}(2)$, with the inner product $\langle X, Y\rangle = -2\operatorname{tr}(XY)$ and norm $|X| = \langle X, X\rangle^{1/2}$. With this normalization $q(e^X) = |X|^2/8 + O(|X|^4)$. On $\mathcal{U}_N$ we use the product of the corresponding bi-invariant Riemannian metrics.

The Wilson flow. Let Φ^{W}_s be the flow on $\mathcal{U}_N$ of the vector field $-\nabla \sum_{p\in\mathcal{P}_N} q(U_p)$, the negative gradient of the Wilson action. It is defined for all $s \geq 0$ and commutes with lattice gauge transformations. For a physical time $t \geq 0$ we write

$$\Psi^{(N)}_t = \Phi^{\mathrm{W}}_{4t/a_N^2},$$

and call $\Psi^{(N)}_t U$ the flowed configuration at time t. We choose the factor $4t/a_N^2$ so that $\Psi^{(N)}_t U$ approximates the Yang–Mills heat flow at time t: if $U_e = \exp(a_N A(x)\cdot e_i)$ for a smooth connection A, then $\Psi^{(N)}_t U$ approximates the heat flow of A; see Section 24. This is the flow introduced by Lüscher [128], with time measured in physical units. We also use the lattice energy

$$\mathsf{E}_{a_N}(U) = \frac{8}{a_N} \sum_{p\in\mathcal{P}_N} q(U_p), \tag{2.2}$$

which is nonincreasing along the flow and approximates the continuum energy (2.3) below.

Lattice approximations of loops. A loop is a piecewise C^1 map $\ell : [0,1] \to \mathbb{T}^3_L$ with $\ell(0) = \ell(1)$. We approximate loops by closed paths in the lattice in the following sense.

Definition 2.1. Let ℓ be a loop, and let $(N_k)_{k\geq 0}$ be an increasing sequence of lattice sides. A sequence $(\ell_k)_{k\geq 0}$ of closed lattice paths, with ℓ_k in Λ_{N_k}, is an *admissible approximation* of ℓ if the physical lengths of the ℓ_k are bounded, and if for all sufficiently large k there exists a piecewise C^1 based homotopy between ℓ_k and ℓ (together with a path joining their base points, if these differ) whose swept area, counted with absolute value, tends to zero as $k \to \infty$.

Every loop has an admissible approximation (Lemma 29.10), and a sequence of lattice paths of bounded length that converges uniformly to ℓ, after reparametrization, is an admissible approximation of ℓ (Theorem 29.23). For a closed lattice path ℓ_k with base point x we write $\mathrm{Hol}_{\ell_k} U \in \mathrm{SU}(2)$ for the transport along ℓ_k; it changes by conjugation under gauge transformations, so $\chi(\mathrm{Hol}_{\ell_k} U)$ is gauge invariant for every character χ of a finite-dimensional representation of $\mathrm{SU}(2)$. The basic observables of this paper are the flowed Wilson loops

$$\chi\big(\mathrm{Hol}_{\ell_k} \Psi^{(N_k)}_t U\big), \qquad t > 0.$$

2.2 The trajectory space

Let $\mathcal{A}_L = C^\infty(\mathbb{T}^3_L; \mathfrak{su}(2)^3)$ be the space of smooth connections on the trivial bundle over $\mathbb{T}^3_L$, and let $\mathcal{G}_L = C^\infty(\mathbb{T}^3_L; \mathrm{SU}(2))$ act by $A^k = kAk^{-1} - (dk)k^{-1}$. For $A \in \mathcal{A}_L$ we write

$F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j]$ and

$$E_L(A) = \sum_{i<j} \int_{\mathbb{T}^3_L} |F_{ij}(x)|^2 \, dx \tag{2.3}$$

for the Yang–Mills energy; with this normalization the exponent in (2.1) approaches $E_L/(2g_3^2)$ for smooth configurations. The Yang–Mills heat flow $\partial_t A_i = \sum_j D_j F_{ji}$, with $D_j = \partial_j + [A_j, \cdot]$, has a global smooth solution for every smooth initial datum on $\mathbb{T}^3_L$, and it induces a semigroup $(\Phi_t)_{t\geq 0}$ on the orbit space $\mathcal{A}_L/\mathcal{G}_L$ (Råde [139]; see Section 24). Following Charalambous and Gross [47] and Cao and Chatterjee [43], we consider trajectories of this flow at positive times.

Definition 2.2. The trajectory space is

$$\mathcal{X}_L = \big\{ X : (0,\infty) \to \mathcal{A}_L/\mathcal{G}_L : \ X(t) = \Phi_{t-s} X(s) \text{ for all } 0 < s < t \big\}.$$

For a loop ℓ, a character χ, and $t > 0$, let $W_{\ell,\chi,t}(X) = \chi(\mathrm{Hol}_\ell\, X(t))$, where Hol_ℓ is the holonomy of any representative of $X(t)$ along ℓ. The Wilson topology $\mathcal{T}_W$ on $\mathcal{X}_L$ is the weakest topology for which all the functions $W_{\ell,\chi,t}$ are continuous.

The space $\mathcal{X}_L$ is a version of the state space of Cao and Chatterjee [43]. In Theorem 29.3 we choose a countable family $(\ell_j, t_j)_{j\geq 1}$ of loops and positive times and set $w_j(X) = \mathrm{tr}\,\mathrm{Hol}_{\ell_j} X(t_j)/2$. We show that

$$d_c(X,Y) = \sum_{j\geq 1} 2^{-j} |w_j(X) - w_j(Y)|$$

is a metric on $\mathcal{X}_L$ whose topology is coarser than $\mathcal{T}_W$, that both topologies are separable and have the same Borel σ-algebra, and that a Borel probability measure on $\mathcal{X}_L$ is determined by the joint laws of finitely many of the w_j. For this reason we call the w_j determining coordinates. The energy $X \mapsto E_L(X(t))$ is a Borel function for every $t > 0$. For every sequence $(R_j)_{j\geq 1}$ of finite nonnegative numbers, the set $\{X : E_L(X(L^2 2^{-j})) \leq R_j \text{ for all } j \geq 1\}$ is compact for both topologies (Theorem 29.6).

2.3 The main theorem

Fix an odd integer $M \geq 3$. For integers $n, n' \geq 1$ we write $n \sim_M n'$ if $nM^a = n'M^b$ for some integers $a, b \geq 0$, and $[n]_M$ for the equivalence class of n.

Theorem 2.3. *Fix $L > 0$, $\gamma > 0$, an odd integer $M \geq 3$, and a positive integer n divisible by four. Set $N_k = nM^k$ and $\mu_k = \mu_{N_k,\gamma}$. Then the following statements hold.*

1. *Let $\ell_1, \dots, \ell_m$ be loops, $\chi_1, \dots, \chi_m$ characters of finite-dimensional representations of $\mathrm{SU}(2)$, and $t_1, \dots, t_m > 0$, and for each i let $(\ell_{i,k})_{k\geq 0}$ be an admissible approximation of ℓ_i. Then the law under μ_k of the random vector*

$$\Big(\chi_i\big(\mathrm{Hol}_{\ell_{i,k}} \Psi_{t_i}^{(N_k)} U\big)\Big)_{i=1}^{m} \tag{2.4}$$

   converges as $k \to \infty$, and the limit does not depend on the choice of admissible approximations. For every $R > 0$, the expectations of $\exp\big(\sum_i z_i \chi_i(\mathrm{Hol}_{\ell_{i,k}} \Psi_{t_i}^{(N_k)} U)\big)$ converge uniformly in $z \in \mathbb{C}^m$ with $\sum_i |z_i| \leq R$.
2. *There exists a unique Borel probability measure $\lambda = \lambda_{L,\gamma,[n]_M}$ on $\mathcal{X}_L$ such that, for every choice of data as in (1), the limit law of (2.4) is the law of $(W_{\ell_i,\chi_i,t_i})_{i=1}^m$ under λ.*
3. *There exist Borel maps $\pi_k : \mathcal{U}_{N_k} \to \mathcal{X}_L$, invariant under all lattice gauge transformations, such that the measures $(\pi_k)_\# \mu_k$ converge to λ against every bounded $\mathcal{T}_W$-continuous function and weakly with respect to d_c, and such that*

$$\lim_{k\to\infty} \mathbb{E}_{\mu_k} \big| w_j(\pi_k U) - \tfrac{1}{2} \operatorname{tr} \mathrm{Hol}_{\ell_{j,k}} \Psi_{t_j}^{(N_k)} U \big| = 0$$

   for every $j \geq 1$, where $(\ell_{j,k})_k$ is a fixed admissible approximation of ℓ_j.
4. *The energies are tight at positive times: for every $t > 0$,*

$$\lim_{R\to\infty} \sup_{k\geq 0} \mu_k\big(E_L((\pi_k U)(t)) > R\big) = 0.$$

5. *The measure λ depends only on L, γ, and the class $[n]_M$; in particular it does not depend on the auxiliary parameters used in its construction.*

The proof is completed in Theorem 29.28. The maps π_k do not preserve holonomies exactly: they are measurable selections of smooth trajectories whose Wilson loops approximate the flowed lattice Wilson loops, and we construct them in Theorem 29.19. Their role is to realize all limits on one probability space. Part (2) and the determining property of the coordinates w_j show that λ is characterized by the limits in part (1).

Remark 2.4. The proof first establishes the theorem along the subsequences $N_K = n_* B^K$ with $B = M^J$, for a fixed integer J chosen at the start, and for n_* larger than a threshold depending on γ; see Theorem 29.24. For every finite Γ, the parameters of the construction, including J and the threshold, can be chosen before $\gamma \in (0, \Gamma]$ is fixed. This uniformity is used in the results on weak coupling. The extension to the full sequence nM^k and to every n uses two consecutive values of J and the fact that the corresponding sequences of lattice sides share infinitely many terms.

2.4 Further properties of the limit

The following theorem collects further properties of the limit. For $0 < \tau < T$ we consider the lattice energy (2.2) and the flowed Wilson loops as continuous functions of the flow time.

Theorem 2.5. *Assume the hypotheses of Theorem 2.3, and let $\lambda = \lambda_{L,\gamma,[n]_M}$.*

1. *(Energy and loop processes.) Let $0 < \tau < T < \infty$, and let ℓ_i, χ_i, and $(\ell_{i,k})_k$, $1 \le i \le m$, be as in Theorem 2.3. Under μ_k, the processes*

$$t \mapsto \Big(\mathsf{E}_{a_{N_k}}(\Psi_t^{(N_k)} U),\ (\chi_i(\mathrm{Hol}_{\ell_{i,k}} \Psi_t^{(N_k)} U))_{i=1}^m \Big), \qquad t \in [\tau, T],$$

   converge in distribution in $C([\tau,T]; \mathbb{R} \times \mathbb{C}^m)$ to $t \mapsto (E_L(X(t)), (W_{\ell_i,\chi_i,t}(X))_i)$ with X distributed according to λ. For two admissible approximations of the same loops, the expected supremum over $[\tau, T]$ of the difference of the corresponding loop processes tends to zero.
2. *(Sobolev topology.) The image measures of $(\pi_k)_\# \mu_k$ and λ in the space of trajectories of Sobolev connections in H^1 modulo gauge transformations in $W^{2,2}$, with the topology of Cao and Chatterjee described in Definition 29.1, converge against every bounded continuous function.*
3. *(Symmetries.) The measure λ is invariant under the maps induced by the isometries $x \mapsto Qx + y$ of $\mathbb{T}_L^3$ with $Q \in O(3,\mathbb{Z})$ and $y \in \mathbb{T}_L^3$. It is also invariant under the three center symmetries: for $z \in \{\pm I\}^3$, the map $[A] \mapsto [A^h]$, where $h : \mathbb{R}^3 \to \mathrm{SU}(2)$ is smooth with $h(x + Le_i) = z_i h(x)$. In particular, if $\ell_1, \dots, \ell_m$ are loops with winding vectors $\nu(\ell_j) \in \mathbb{Z}^3$, $t_1, \dots, t_m > 0$, and $b_1, \dots, b_m \ge 0$ are integers with $\sum_j b_j \nu(\ell_j) \neq 0$ in $(\mathbb{Z}/2\mathbb{Z})^3$, then $\mathbb{E}_\lambda \prod_j \big(\mathrm{tr}\, \mathrm{Hol}_{\ell_j} X(t_j)/2\big)^{b_j} = 0$.*
4. *(Scaling.) Let D_L map a trajectory X on the unit torus to the trajectory $(D_L X)(t) = L^{-1} X(t/L^2)(\cdot/L)$ on $\mathbb{T}_L^3$. Then $\lambda_{L,\gamma,[n]_M} = (D_L)_\# \lambda_{1,\gamma,[n]_M}$.*
5. *(Continuity in the coupling.) On every compact interval $I \subset (0,\infty)$ of couplings, the convergence in Theorem 2.3(1) holds uniformly in $\gamma \in I$ in the bounded-Lipschitz distance, and the maps π_k can be chosen independently of $\gamma \in I$ so that the convergence in Theorem 2.3(3) and the tightness in Theorem 2.3(4) hold uniformly in $\gamma \in I$. The map $\gamma \mapsto \lambda_{L,\gamma,[n]_M}$ is weakly continuous on $(0,\infty)$ for both topologies on $\mathcal{X}_L$.*

Part (1) is Theorem 30.4, part (2) is Theorem 30.5, part (3) is Theorem 30.6, part (4) is Theorem 30.1, and part (5) is Theorem 30.10. By parts (4) and (5), at fixed physical coupling $g_3^2 = \gamma/L$ the measures on different tori, transported to the unit torus, depend continuously on L. Since $\gamma = g_3^2 L$, part (4) is not an infinite-volume statement.

Other smoothing flows. The limit does not depend on the details of the smoothing within the following family of flows. For $c \ge 0$, let $\Psi_t^{(N,c)}$ be the flow of the vector field

$-\nabla \sum_p [q(U_p) + c\, q(U_p)^2]$ run for the lattice time $4t/a_N^2$; for $c = 0$ this is the Wilson flow. The measure $\mu_{N,\gamma}$ is unchanged.

Theorem 2.6. *Fix $L > 0$ and $0 < \gamma \le 1$, and let $N_K = n_* B^K$ be as in Remark 2.4. Let $C_0 \ge 0$, let ℓ_i, χ_i, t_i, and $(\ell_{i,K})_K$ be as in Theorem 2.3, and write $W^c_{K,i} = \chi_i(\mathrm{Hol}_{\ell_{i,K}} \Psi^{(N_K,c)}_{t_i} U)$. Then*

$$\lim_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} \sup_{0\le c\le C_0} \max_i |W^c_{K,i} - W^0_{K,i}| = 0.$$

Further, for $0 < \tau < T$, the joint fields of lattice energies and Wilson loops along the modified flows converge in distribution in $C([0,C_0]\times[\tau,T];\mathbb{R}\times\mathbb{C}^m)$ to the field $(c,t) \mapsto (E_L(X(t)), (W_{\ell_i,\chi_i,t}(X))_i)$, $X \sim \lambda$, which does not depend on c.

This is Theorems 30.13 and 30.15.

2.5 The bound on large fields

The proof of Theorem 2.3 rests on a renormalization group analysis of $\mu_{N,\gamma}$ restricted to a small-field event. We describe the event briefly; the precise definition is Definition 4.4. Fix an odd integer $M \ge 3$ and an integer $J \ge 1$, and set $B = M^J$. For $N = uB^K$, let T_M be the block averaging map of Definition 4.3, which maps configurations on $\Lambda_{N'}$ to configurations on $\Lambda_{N'/M}$; it is a version of Bałaban's averaging operation [12]. Let $V^j = T_M^j U$ be the averaged configurations. A renormalization group step consists of J consecutive averaging steps. At the step from V^{kJ} to $V^{(k+1)J}$, the fine configuration is described by the coarse configuration and by normalized fluctuation variables w_i, one for each of the J intermediate scales. In free logarithmic coordinates z_i on the fiber of the averaging map, in a fixed gauge-fixed chart, they are defined by $z_i = z_i^* + g_i K_i^{-1/2} w_i$. Here z_i^* is the conditional constrained minimizer of the effective classical action, K_i is its constrained Hessian, and g_i is the coupling at that scale; see Section 3.2. The small-field event $S_{N,u,\gamma}$ is the set of configurations U such that U and each $V^{(k+1)J}$, $0 \le k < K$, are gauge equivalent to configurations all of whose link variables are close to the identity, such that each renormalization group step is defined at U, and such that every normalized fluctuation variable of the k-th step has size at most $A_{\mathrm{w}} \log(1/t_k)$, where $t_k^2 = \gamma/(M n_{(k+1)J})$ is the largest effective coupling of that step and $n_j = N/M^j$.

Theorem 2.7. *Fix $L > 0$, an odd integer $M \ge 3$, and $0 < \Gamma < \infty$. There exist an integer $J \ge 1$, a choice of the remaining parameters in the definition of $S_{N,u,\gamma}$, and an integer n_Γ such that the following holds with $B = M^J$. There exist constants $C, C_{\mathrm{w}}, c_*, a_{\mathrm{w}} > 0$ and $H_0, H_2 \ge 0$ such that, for every $\gamma \in (0,\Gamma]$, every $u \ge n_\Gamma$ divisible by four, and every $K \ge 0$,*

$$\mu_{uB^K,\gamma}(S^c_{uB^K,u,\gamma}) \le C u^3 \exp\Big(H_0 + H_2 \log\frac{\Gamma}{\gamma} - \frac{c_* u}{\gamma}\Big) + C_{\mathrm{w}} u^3 \exp\Big(-a_{\mathrm{w}} \log^2 \sqrt{\frac{Mu}{\gamma}}\Big). \quad (2.5)$$

The constants depend only on L, M, Γ, and the chosen parameters. In particular, for every fixed $u \geq n_\Gamma$ and every $q > 0$ there exists $C_{u,q}$ with $\sup_{K\geq 0} \mu_{uB^K,\gamma}(S^c_{uB^K,u,\gamma}) \leq C_{u,q}\gamma^q$ for all $\gamma \in (0,\Gamma]$, and the right side of (2.5) tends to zero as $u \to \infty$, uniformly in $\gamma \in (0,\Gamma]$. Further, for each such u there exist $c_u, C_u > 0$ and $\gamma_u > 0$ such that

$$e^{-C_u(1+\log(1/\gamma))^2} \leq \sup_{K\geq 0} \mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big) \leq e^{-c_u \log^2(1/\gamma)} \qquad (0 < \gamma \leq \gamma_u).$$

This is proved in Theorems 23.5 and 23.10 and Corollary 23.13. By the last display, at fixed u the supremum over K of the discarded probability lies between $\exp(-C_u(1+\log(1/\gamma))^2)$ and $\exp(-c_u \log^2(1/\gamma))$; in particular it is much larger than $\exp(-c/\gamma)$. The lower bound comes from configurations that violate the cutoff on the normalized fluctuation variables, and it already holds for $K = 1$. The first term in (2.5) bounds the probability that some averaged configuration has a plaquette with large curvature. We prove it using reflection positivity and chessboard estimates under the unconditioned Wilson measure. The second term bounds the probability that some normalized fluctuation variable exceeds its cutoff.

2.6 Short distances

For a loop ℓ in $\mathbb{R}^3$, a point $x \in \mathbb{T}^3_L$, and $\varepsilon > 0$ small, let $x + \varepsilon\ell$ be the scaled and translated loop in $\mathbb{T}^3_L$, and let

$$W_{x+\varepsilon\ell,s}(X) = \tfrac{1}{2} \operatorname{tr} \operatorname{Hol}_{x+\varepsilon\ell} X(s)$$

be the normalized fundamental Wilson loop under the measure λ of Theorem 2.3.

Theorem 2.8. *Fix $L > 0$, $\gamma > 0$, an odd integer $M \geq 3$, and n divisible by four, let $\lambda = \lambda_{L,\gamma,[n]_M}$, and set $g_3^2 = \gamma/L$. Let $\ell_1, \dots, \ell_m$ be smooth loops in $\mathbb{R}^3$ and $\sigma > 0$. For $x \in \mathbb{T}^3_L$ and $\varepsilon > 0$ set*

$$Z_\varepsilon(x) = \Big(\frac{1 - W_{x+\varepsilon\ell_i,\sigma\varepsilon^2}}{g_3^2\varepsilon}\Big)_{i=1}^m.$$

Let $C^a = (C^a_i)_{i=1}^m$, $a = 1,2,3$, be independent centered Gaussian vectors with covariance

$$\Sigma_{ij} = (2\pi)^{-3} \int_{\mathbb{R}^3} \frac{e^{-2\sigma|k|^2}}{|k|^2} \overline{\widehat{J}_i(k)} \cdot \widehat{J}_j(k)\, dk, \qquad \widehat{J}_i(k) = \int_0^1 e^{-ik\cdot\ell_i(s)} \dot{\ell}_i(s)\, ds, \tag{2.6}$$

and set $Z_G = (|C_i|^2/8)_{i=1}^m$, where $C_i = (C^1_i, C^2_i, C^3_i)$. Then the following statements hold.

1. *As $\varepsilon \to 0$, the law of $Z_\varepsilon(x)$ under λ converges to the law of Z_G in the bounded-Lipschitz distance, uniformly in $x \in \mathbb{T}^3_L$.*
2. *If $m = 1$ and $v = \Sigma_{11} > 0$, then*

$$\liminf_{\varepsilon\to 0} \inf_{x\in\mathbb{T}^3_L} \frac{\operatorname{Var}_\lambda W_{x+\varepsilon\ell_1,\sigma\varepsilon^2}}{(g_3^2\varepsilon)^2} \geq \frac{3v^2}{32}.$$

3. *For every bounded Borel function* $G : \mathcal{X}_L \to \mathbb{R}$,

$$\lim_{\varepsilon \to 0} \sup_{x \in \mathbb{T}_L^3} \sup_f \left| \mathbb{E}_\lambda[f(Z_\varepsilon(x))G] - \mathbb{E}[f(Z_G)]\, \mathbb{E}_\lambda[G] \right| = 0,$$

where the supremum is over functions f *with* $\|f\|_\infty \le 1$ *and Lipschitz constant at most one.*

This is proved in Theorems 32.13 and 32.17 and Corollary 32.14. In the notation of Section 31, $C_i = G_i$ and $(Z_G)_i = \mathsf{Z}_i$. The vector C is the heat-smoothed circulation of a Gaussian free field (Maxwell field) with three independent colors, so the theorem says that at short distances the flowed Wilson loops of λ behave as those of the free field, up to the quadratic map $V \mapsto |V|^2/8$ coming from $1 - \operatorname{tr} e^V/2 = |V|^2/8 + O(|V|^4)$. As an intermediate step, we prove a Gaussian limit for the normalized heat-smoothed circulations of the reconstructed field. This holds in a fixed gauge, under the conditioned Wilson measure, at every sufficiently large fixed coarse side (Theorem 32.8). These circulations are not gauge-invariant functions on $\mathcal{X}_L$; the gauge-invariant conclusion is the theorem above. Part (2) is a lower bound only, since the proof does not give uniform integrability.

2.7 Weak coupling

We now let $\gamma \to 0$ with L, M, and n fixed, and write $\lambda_\gamma = \lambda_{L,\gamma,[n]_M}$. Fix $\theta > 0$ and the flow time $t = \theta L^2$. For $X \in \mathcal{X}_L$, let $H = (H_1, H_2, H_3) \in \mathrm{SU}(2)^3$ be the holonomies of $X(t)$ around the three coordinate cycles of $\mathbb{T}_L^3$ based at the origin; the triple H is defined up to simultaneous conjugation. Every continuous function of H that is invariant under simultaneous conjugation is a continuous function of the seven traces of H_i, $H_i H_j$ $(i < j)$, and $H_1 H_2 H_3$ (Lemma 5.6). Such functions are bounded and $\mathcal{T}_W$-continuous on $\mathcal{X}_L$. In the proofs we fix, before γ varies, a coarse side in the class $[n]_M$ that is larger than certain thresholds; by Theorem 2.3(5) this choice does not change λ_γ. Each of the following theorems holds for all sufficiently small γ, with a threshold that depends on the theorem, and in each the continuum limit is taken first, at fixed γ.

A connection is flat if $F_A = 0$. Flat connections on $\mathbb{T}_L^3$ modulo gauge are classified by their holonomies, which form a commuting triple; the moduli space is

$$\mathrm{Hom}(\mathbb{Z}^3, \mathrm{SU}(2))/\mathrm{SU}(2) \simeq (\mathbb{R}/4\pi\mathbb{Z})^3/\{\boldsymbol{\vartheta} \sim -\boldsymbol{\vartheta}\},$$

where $\boldsymbol{\vartheta}$ parametrizes the triple $(e^{\vartheta_1 u}, e^{\vartheta_2 u}, e^{\vartheta_3 u})$ for a unit vector $u \in \mathfrak{su}(2)$. The eight central triples $H \in \{\pm I\}^3$ are the singular points.

Theorem 2.9. *Fix* $L > 0$, *an odd integer* $M \ge 3$, *and* n *divisible by four. Then the following statements hold.*

1. *For every* $t > 0$ *and* $\delta > 0$, $\lambda_\gamma(E_L(X(t)) > \delta) \to 0$ *as* $\gamma \to 0$.

2. *There exists* $\gamma_0 \in (0,1]$ *such that the family* $(\lambda_\gamma)_{0<\gamma\le\gamma_0}$ *is tight: for every* $\eta > 0$ *there exist* $R_j < \infty$ *such that* $\lambda_\gamma(E_L(X(L^2 2^{-j})) \le R_j \text{ for all } j \ge 1) \ge 1-\eta$ *for all* $0 < \gamma \le \gamma_0$. *The sets in this display are compact for* $\mathcal{T}_W$ *and for* d_c, *so every sequence* $\gamma_k \to 0$ *has a subsequence along which* λ_{γ_k} *converges against bounded* $\mathcal{T}_W$*-continuous functions and weakly for* d_c.
3. *Every subsequential limit of* λ_γ *as* $\gamma \to 0$, *against bounded* $\mathcal{T}_W$*-continuous functions or weakly for* d_c, *is supported on the stationary flat trajectories* $\{X : X(t) = [A] \text{ for all } t > 0,\ F_A = 0\}$.

This is Theorem 34.13. The theorem does not identify the limiting distribution on the moduli space of flat connections. The next two theorems describe the fluctuations near the regular and the singular points of this moduli space.

Regular commuting triples. In a fixed neighborhood of the identity, write $X = (x_1, x_2, x_3) = (\operatorname{Log} H_1, \operatorname{Log} H_2, \operatorname{Log} H_3) \in \mathfrak{su}(2)^3 \simeq \mathbb{R}^9$. The commuting triples correspond to the cone $\mathcal{M} = \{(\vartheta_1 u, \vartheta_2 u, \vartheta_3 u) : \boldsymbol{\vartheta} \in \mathbb{R}^3,\ u \in S^2\}$. On a regular patch $0 < r_0 \le |\boldsymbol{\vartheta}| \le r_1 < \pi$ the normal space to $\mathcal{M}$ at $y = (\vartheta_i u)_i$ is

$$\mathcal{N}_y = \Big\{ b = (b_1, b_2, b_3) :\ b_i \perp u,\ \sum_i \vartheta_i b_i = 0 \Big\},$$

which has dimension four, and near the patch we write $X = y + b$ with $y \in \mathcal{M}$ and $b \in \mathcal{N}_y$. Let $\Sigma_\theta(y)$ be the normal covariance, the positive definite covariance on $\mathcal{N}_y$ defined in (4.12). It is the covariance, divided by γ, of the linear response of the flowed logarithms X at the flat connection y, and it does not depend on γ. Let ψ be a smooth function of X that is invariant under simultaneous conjugation and supported in a small tube around a compact regular patch. We assume that ψ has the form (tangential bump)$\times$(normal cutoff equal to one near $b = 0$), and that its tangential factor takes values in $[0,1]$, vanishes on a nonempty open subset, and is positive on another. Set

$$q_*(y+b) = \tfrac{1}{2}\, b^{\mathsf{T}} \Sigma_\theta(y)^{-1} b, \qquad B_{\mathrm{obs}} = \psi, \qquad A_{\mathrm{obs}} = \psi\, q_*,$$

where the formula for q_* is used on a fixed normal tube that contains the supports of all the functions ψ and $\widetilde{w}$ below, and q_* is extended outside this tube by a bounded nonnegative invariant function. The functions A_{obs} and B_{obs} are extended by zero outside the chart. We define the normalized regression residual variance

$$\mathscr{C}_\gamma = \frac{\operatorname{Var}(A_{\mathrm{obs}}) - \operatorname{Cov}(A_{\mathrm{obs}}, B_{\mathrm{obs}})^2 / \operatorname{Var}(B_{\mathrm{obs}})}{\mathbb{E}(B_{\mathrm{obs}}^2)},$$

with all moments under λ_γ. This is the mean square error of the best affine predictor of A_{obs} from B_{obs}, normalized by $\mathbb{E} B_{\mathrm{obs}}^2$.

Theorem 2.10. *Fix $L > 0$, an odd integer $M \geq 3$, n divisible by four, $\theta > 0$, and ψ as above with a sufficiently small tube. Then the following statements hold.*

1. *There exists $c > 0$ such that $\mathbb{E}_{\lambda_\gamma} B_{\mathrm{obs}}^2 \geq c$ and $\mathrm{Var}_{\lambda_\gamma} B_{\mathrm{obs}} \geq c$ for all sufficiently small $\gamma > 0$, and*
$$\mathscr{C}_\gamma = 2\gamma^2 + o(\gamma^2) \qquad (\gamma \to 0).$$
2. *Let $w \geq 0$ be a nonzero smooth invariant function on $\mathcal{M}$ supported in the interior of the regular patch, let χ_w be a smooth normal cutoff invariant under simultaneous conjugation, with $0 \leq \chi_w \leq 1$, supported in the tube and equal to one near $b = 0$, and set $\widetilde{w}(y + b) = w(y)\chi_w(b)$. Then $\mathbb{E}_{\lambda_\gamma} \widetilde{w} \geq c_w > 0$ for small γ, and for every bounded continuous $f : [0, \infty) \to \mathbb{R}$,*
$$\frac{\mathbb{E}_{\lambda_\gamma}[\widetilde{w}\, f(q_*/\gamma)]}{\mathbb{E}_{\lambda_\gamma} \widetilde{w}} \longrightarrow \int_0^\infty f(s)\, s\, e^{-s}\, ds \qquad (\gamma \to 0).$$
3. *More generally, on $\{\widetilde{w} \neq 0\}$ write $X = y_\gamma + b_\gamma$ and $z_\gamma = \gamma^{-1/2} \Sigma_\theta(y_\gamma)^{-1/2} b_\gamma$, and define products with $\widetilde{w}$ to be zero where $\widetilde{w} = 0$. For every bounded continuous function $F(y, z)$ on the normal bundle that is invariant under simultaneous conjugation,*
$$\frac{\mathbb{E}_{\lambda_\gamma}\left[\widetilde{w}\left(F(y_\gamma, z_\gamma) - \overline{F}(y_\gamma)\right)\right]}{\mathbb{E}_{\lambda_\gamma} \widetilde{w}} \longrightarrow 0, \qquad \overline{F}(y) = \int F(y, z)\, \mathcal{N}(0, I_{\mathcal{N}_y})(dz).$$

This is Theorems 35.2 to 35.4. Part (3) is a Gaussian approximation for the normal component of the flowed holonomies near a regular commuting triple, with covariance $\gamma \Sigma_\theta(y)$, in the sense of the displayed ratio of weighted expectations. The tangential component keeps its own law under the weighting, and we do not claim that this law converges. Part (2) is the special case $F = f(|z|^2/2)$, since $|z|^2/2$ has density se^{-s} when z is a standard Gaussian vector in four dimensions. The coefficient 2 in part (1) comes from the first two moments 2 and 6 of this density. It is the coefficient of the Gaussian approximation around a regular commuting flat background, and the theorem does not identify corrections beyond it. For comparison, consider the reference measures of Definition 4.13. For the Abelian Maxwell measure with Haar cycle holonomies, the holonomy triples commute and the statistic vanishes identically. For the Gaussian holonomies with independent Haar factors, $H_i = U_i^0 \operatorname{Hol}_i(a)$, where a is a smoothed Gaussian field and U_1^0, U_2^0, U_3^0 are independent Haar distributed elements of $\mathrm{SU}(2)$, independent of a, the triple is exactly Haar distributed, and the statistic is a positive constant independent of γ (Lemma 35.17).

Central triples. Near the central triples the cone of commuting triples is singular, and the constant modes are no longer Gaussian. Let
$$Q(Y) = \tfrac{1}{2} \sum_{i<j} |y_i \times y_j|^2, \qquad Y = (y_1, y_2, y_3) \in \mathbb{R}^9, \qquad r_\gamma = \gamma^{1/4},$$

where $\times$ is the cross product, which corresponds to the Lie bracket under $\mathfrak{su}(2) \simeq \mathbb{R}^3$. We call Q the constant-mode quartic form. The measure $e^{-Q(Y)}dY$ on $\mathbb{R}^9$ is locally finite but has infinite total mass. For a function φ on $\mathbb{R}^9$ we write $\mathcal{I}_{\gamma,\varphi} = \varphi(X/r_\gamma)$, extended by zero outside the fixed neighborhood of the identity in which X is defined. A radial cutoff is a function $\chi(|Y|)$ with $\chi \geq 0$ smooth, compactly supported, not identically zero, and constant near zero.

Theorem 2.11. *Fix $L > 0$, an odd integer $M \geq 3$, n divisible by four, and $\theta > 0$. Then the following statements hold.*

1. *For every continuous compactly supported function φ on $\mathbb{R}^9$ invariant under simultaneous conjugation and every radial cutoff χ,*

$$\lim_{\gamma\to 0} \frac{\mathbb{E}_{\lambda_\gamma}\mathcal{I}_{\gamma,\varphi}}{\mathbb{E}_{\lambda_\gamma}\mathcal{I}_{\gamma,\chi(|\cdot|)}} = \frac{\int_{\mathbb{R}^9} \varphi(Y)e^{-Q(Y)}\,dY}{\int_{\mathbb{R}^9} \chi(|Y|)e^{-Q(Y)}\,dY}. \tag{2.7}$$

   If $\varphi \geq 0$ is not identically zero, then $c_\varphi r_\gamma \leq \mathbb{E}_{\lambda_\gamma}\mathcal{I}_{\gamma,\varphi} \leq C_\varphi r_\gamma$ for small γ.
2. *For every $R < \infty$ and radial cutoff χ there exist $C, \gamma_0 > 0$ such that for all $0 < \gamma \leq \gamma_0$ and all invariant Lipschitz functions φ supported in the ball of radius R, the difference of the two sides of* (2.7) *is at most $C(\|\varphi\|_\infty + \operatorname{Lip}\varphi)\gamma^{1/4}$.*
3. *Let $W_\chi(Y) = [4Q(Y) - 9]\chi(|Y|) - |Y|\chi'(|Y|)$. Then*

$$\frac{\mathbb{E}_{\lambda_\gamma}\mathcal{I}_{\gamma,W_\chi}}{\mathbb{E}_{\lambda_\gamma}\mathcal{I}_{\gamma,\chi(|\cdot|)}} = O(\gamma^{1/4}).$$

4. *The same statements hold, with the same constants, near each of the eight central triples $z \in \{\pm I\}^3$, with X replaced by $(\operatorname{Log}(z_i^{-1}H_i))_{i=1}^3$; at each fixed γ the corresponding expectations near different central triples are equal.*

This is proved in Theorems 36.7, 36.10, and 36.12. Part (3) is the dilation integration-by-parts identity, $\int \operatorname{div}(Y\chi(|Y|)e^{-Q(Y)})\,dY = 0$, which follows from the homogeneity $Y \cdot \nabla Q = 4Q$. For the reference measures of Definition 4.13, and for radial cutoffs with $\chi(0) > 0$, the same statistic has the following limits (Theorem 36.19). It tends to -6 for the Abelian Maxwell measure with Haar cycle holonomies, and to -9 for the Gaussian holonomies with Gaussian constant modes. For the mixture of normalized conditional normal Gaussian measures over commuting backgrounds, with a fixed prior density that is continuous and positive at the tip of the cone of commuting triples, its limit is strictly larger than 4. Hence the statistic distinguishes λ_γ from these comparison measures. In Section 36 we also prove joint versions of (2.7) at finitely many flow times, for test functions that are invariant under independent simultaneous conjugations at the different times (Theorem 36.15). Finally, let $0 < \tau < T$, and weight λ_γ by $\mathcal{I}_{\gamma,\chi(|\cdot|)}$ evaluated at time τ, normalized to a probability measure. Under this measure, the rescaled holonomy paths

on $[\tau, T]$, defined with a fixed smooth equivariant extension of the logarithm and taken modulo conjugation at each time, converge in the uniform topology to constant paths. The value of the limiting path has the law with density proportional to $\chi(|Y|)e^{-\mathrm{Q}(Y)}$, and the error against bounded Lipschitz path functionals is $O(\gamma^{1/4})$ (Theorem 36.18).

Constants in Theorems 2.10 and 2.11 may depend on L, M, n, θ, and the test functions, but not on γ.

SECTION 3

Outline of the proof

This section explains the main steps of the proof and the ideas behind them. It is meant to be read before the technical parts, and it fixes the vocabulary used there. We describe the proof of Theorem 2.3 in Sections 3.1 to 3.7. The proofs of the results on short distances and on weak coupling are outlined in Sections 3.8 and 3.9. Throughout, L and γ are fixed, $g_3^2 = \gamma/L$, and $M \geq 3$ is a fixed odd integer.

3.1 Strategy

The Wilson measure $\mu_{N,\gamma}$ at fine mesh a_N lives on a space whose dimension grows like N^3. The observables we care about, flowed Wilson loops at a fixed positive time t, depend on the configuration at length scales of order $\sqrt{t}$ and larger. The renormalization group organizes the integration over the configuration scale by scale, from the finest lattice to a coarse lattice of fixed side. At each scale, we only need to control a weakly coupled and approximately Gaussian integral. In three dimensions the effective coupling at lattice side m is $g_m^2 = \gamma/m$, which is small when m is large. For this reason the renormalization group is a convergent procedure in the ultraviolet, as Bałaban showed [15]. However, this is true only as long as the field is close to a pure gauge at every scale. Configurations for which this fails, the large fields, have to be treated separately.

We organize the proof as a limit in two stages.

1. We fix a coarse side u and a large integer K, and let $N = uB^K$, where $B = M^J$ is the blocking factor of one renormalization group step. We condition $\mu_{N,\gamma}$ on the small-field event $S_{N,u,\gamma}$ of Section 2.5, which gives the conditioned Wilson measure $\nu_{N,u,\gamma} = \mu_{N,\gamma}(\,\cdot \mid S_{N,u,\gamma})$. On this event we can run the renormalization group exactly from side N down to side u, without any analysis of large fields. We prove that, for each fixed u, the laws of flowed Wilson loops under $\nu_{uB^K,u,\gamma}$ converge as $K \to \infty$.
2. By Theorem 2.7, we have $\mu_{uB^K,\gamma}(S^c_{uB^K,u,\gamma}) \leq F(u)$ for all K, where $F(u) \to 0$ as $u \to \infty$. For a bounded observable G we have the elementary inequality

$$\left|\mathbb{E}_{\mu_{N,\gamma}} G - \mathbb{E}_{\nu_{N,u,\gamma}} G\right| \leq 2\|G\|_\infty \, \mu_{N,\gamma}(S^c_{N,u,\gamma}). \tag{3.1}$$

If $u = nB^r$, the sequence of sides uB^K is the sequence nB^K shifted by r. Then the first stage shows that, under $\mu_{nB^K,\gamma}$, the expectations of a fixed bounded continuous function of finitely many flowed Wilson loops form a Cauchy sequence up to an

error $4\|G\|_\infty F(u)$. Letting $u \to \infty$, we obtain convergence under the unconditioned Wilson measure.

This order of limits, first $K \to \infty$ at fixed u and then $u \to \infty$, appears throughout the paper. Configurations outside the small-field event enter only through the bound on large fields. We prove that bound directly under $\mu_{N,\gamma}$, without conditioning. In particular, we never need to show that a conditioned measure is reflection positive.

The rest of the proof of Theorem 2.3 consists of three tasks. The first is the renormalization group analysis on the small-field event, and its consequences at a fixed number of steps (Sections 3.2 and 3.3). The second is the bound on large fields (Section 3.4). The third is the passage from the renormalization group variables to flowed Wilson loops, and then to a measure on the trajectory space (Sections 3.5 to 3.7).

3.2 The renormalization group on the small-field event

Averaging. The renormalization group step is built from Bałaban's averaging operation for lattice gauge fields [12]. Fix a configuration on a lattice of side Mm. We divide the lattice into cubic blocks of M^3 sites, centered at the points of a sublattice of side m. For each link c of the coarse lattice, we consider the M^3 paths of length M parallel to c that start at the sites of the block at the initial point of c. They are joined to the endpoints of c by fixed connecting paths along the coordinate directions. We take the logarithms of their transports relative to the transport along c, average them, and multiply the exponential of the average by the transport along c; see (4.3). When some relative transport equals $-I$, the logarithm is not defined, and we use the reference transport instead. In this way the block averaging map T_M is a Borel map on the whole configuration space. It is gauge covariant, and it commutes with translations of the coarse lattice and with reflections of coordinates that preserve it. (Because the connecting paths change coordinates in a fixed order, we do not claim that it commutes with permutations of coordinates.) Since M is odd, the blocks are centered, and the blocks of successive averaging steps are nested. As a consequence, iterated averages of a configuration supported on one side of a reflection plane depend only on that side (Section 6). In Part II we prove the basic properties of the block averaging map, including an exact formula for the integral over a fiber $\{U : T_M U = b\}$ in local coordinates and an estimate that controls the curvature of $T_M U$ by that of U (Theorem 7.3). One renormalization group step consists of J consecutive averaging steps, $V^{(k+1)J} = T_M^J V^{kJ}$. The integer J is fixed at the start of the proof.

Background fields. Let c be a coarse configuration close to the identity. The integral over fine configurations U with $T_M^J U = c$ is dominated by a neighborhood of the minimizer of the Wilson action subject to this constraint. Following Bałaban [16], we call this minimizer the background field. We construct it in a gauge-fixed chart near the identity, by solving the constrained Euler–Lagrange equation with a contraction argument (Part III). It is a

specific branch of local constrained minimizers; we do not claim that it minimizes the action over the whole fiber of the averaging map. We prove that it depends analytically on c, with estimates that are uniform in the number of averaging steps between the fine and coarse lattices. The value of the constrained minimum defines the effective classical action $E_d(c)$ at depth d, the number of averaging steps. Minimizing in stages gives the nested minimization identity

$$E_{d+1}(c) = M \min_z E_d(b(c,z)).$$

Here $b(c,z)$ parametrizes, in the chart, the configurations on the intermediate lattice whose average is c, the minimum is over that chart, and the factor M is the change of units in three dimensions. The effective classical action E_0 is the Wilson action. Since $g_{d+1}^2 = Mg_d^2$, the ratio E_d/g_d^2 is invariant under this change of scale. The main technical point is that the analytic norms of E_d and of the background field do not grow with d. This rests on two facts. The first is the linear theory: the effective Maxwell quadratic form has an exponentially decaying kernel, and its restriction to the next constrained fluctuation space has a positive lower bound and an exponentially decaying inverse, uniformly in the depth and the lattice side (Theorem 8.3). Bałaban's propagator estimates [10, 11, 13] are used separately, in the constrained linear extension and the scalar gauge inverse of Theorem 9.3 and Lemma 9.8. The second is that the linearized averaging operator, acting on connections after a gauge correction, has row sums one and column sums M^{-3} ((10.12)). As a result the nonlinear errors are summable over scales.

Fluctuations and the effective interaction. We fix a renormalization group step from side $M^J m$ to side m. We parametrize the fine configuration by the coarse configuration c and by free logarithmic coordinates z_i at each of the J intermediate scales. We then center these coordinates at the background field and normalize them by the constrained Hessian:

$$z_i = z_i^*(c, z_{>i}) + g_i K_i(c, z_{>i})^{-1/2} w_i.$$

Here z_i^* is the conditional constrained minimizer, K_i is the constrained Hessian of the action, and g_i is the coupling at that scale. We call the w_i the normalized fluctuation variables. The change of variables is triangular, from the coarsest intermediate scale to the finest. We compute its Jacobian exactly, including the Haar density and the Jacobian of the constraint (Section 12). In the variables w the integrand is $\exp(-V(c,w))$. The interaction $U(c,w) = V(c,w) - |w|^2/2$ contains the higher-order terms of the action, the Jacobians, and the effective interaction inherited from finer scales. On the small-field event every w_i is bounded by $A_{\mathrm{w}} \log(1/t)$, where t is the largest coupling of the step. On this product of balls the interaction U is small, and we expand the integral over w in a convergent polymer expansion. We use the cluster expansion in the form of Kotecký and Preiss [120], with the tree estimates that go back to Penrose [136], as presented by Fernández

and Procacci [87] and Dimock [68]. After imposing the small-field tests through the k-th renormalization group step, we obtain an exact formula, relative to product Haar measure, for the unnormalized stopped density of the averaged configuration:

$$\mathbf{1}_{D_r}(c)\exp\Big(\Lambda_k - \frac{E_{kJ,\mathrm{sat}}(c)}{g_{kJ}^2} - \overline{R}_{kJ,\mathrm{sat}}(c)\Big). \tag{3.2}$$

Here $g_{kJ}^2 = \gamma/n_{kJ}$ is the coupling of the lattice of side n_{kJ}, and D_r, the gauge-invariant small-field domain, is the set of configurations that are gauge equivalent to configurations with all link variables close to the identity. The subscript sat denotes the gauge-invariant extension to D_r of a function defined near the identity. The function E_{kJ} is the effective classical action. The centered effective interaction $\overline{R}_{kJ}$ is generated by the integration of the fluctuations, and it is normalized so that $\overline{R}_{kJ}(I) = 0$. The normalization constant Λ_k collects all factors that do not depend on the field, including the vacuum energy (Sections 13 and 14). At $k = K$ the normalized stopped density is the law of the coarse field V^{JK} under the conditioned Wilson measure. At intermediate steps it does not include the remaining coarser restrictions of the small-field event.

Analytic norms and contraction. We control $\overline{R}_{kJ}$ by a norm that sums the absolute values of all Taylor coefficients at the identity (Definition 4.6). The coefficients of order k are weighted by $\rho^{k-2}/k!$ and by $\exp(a_{\mathrm{sp}}\mathrm{t})$, where t is the length of the shortest network that connects the lattice points on which the coefficient depends. The weighted coefficients measure spatial locality. Connected products and compositions are controlled by the coefficient calculus, using output row bounds and volume-normalized input column bounds, under the required image conditions for compositions (Lemma 13.1); derivative estimates hold on strictly smaller analytic domains. Products of interactions whose coefficient networks are not connected, such as products of two extensive interactions, are not controlled in this way. The key estimate is Theorem 13.5, which says that one renormalization group step maps a centered effective interaction R to

$$R_+ = Q + \mathcal{P}R + \mathcal{D}R, \qquad \|R_+\| \le C_G + q\|R\| + C_T t^2 \|R\|(p^4 + \|R\|p^2).$$

In the notation of that theorem, $Q = Q^{\mathrm{w}}_{d_0,J}$, $\mathcal{P} = \mathcal{P}_\delta$, and $\mathcal{D} = \mathcal{D}_J$ for the step under consideration; in this paragraph only, Q does not denote the constant-mode quartic form. The term Q is the newly generated effective interaction, produced by the step itself. The term $\mathcal{P}R$, the pullback at zero fluctuation, is R evaluated on the background field. It contracts by a factor $q = O(M^{-J})$, because R is gauge invariant, invariant under translations, global color rotations, and inversion, and has no linear term at the identity (Theorem 11.10). The term $\mathcal{D}R$ is the fluctuation correction to the pullback. It gains two powers of the coupling, because the linear term of the fluctuation integral vanishes on each connected support. We choose J so that $q < 1/4$, and then we choose the coupling

small. The ball $\|R\| \leq E_*$ is then invariant, uniformly in the number of steps. Gauge invariance also enters in another way. At each step we choose a gauge transformation that depends only on the coarse configuration, and that brings the whole reconstructed fine configuration, fluctuations included, into a fixed analytic neighborhood of the identity (Theorem 13.2).

The conditional law of the fluctuations. Given the coarse configuration c, the normalized fluctuation variables w of one step have a conditional law, the conditional fluctuation kernel, with density proportional to $\mathbf{1}_{\Omega_p}(w)e^{-V_c(w)}$. Here Ω_p is a product of three-dimensional balls, and $D_w^2 V_c \geq I/2$ (Theorem 13.8). We emphasize that this law is not Gaussian, and that the variables are not independent. It is, however, uniformly log-concave on a convex set. Using the Bakry–Émery criterion [8] with reflecting boundary conditions and the Helffer–Sjöstrand representation of covariances [110], we obtain a logarithmic Sobolev inequality and exponentially decaying covariance estimates, with constants that do not depend on the dimension (Theorem 5.4). These estimates, together with bounds on the conditional means, are the probabilistic input for everything that follows. The conditional means are small, of order $t\|\log c\|_\infty$; at $c = I$ they vanish by the global $\mathrm{SU}(2)$ symmetry (Theorem 13.10).

3.3 Convergence at fixed resolution and reconstruction

Convergence of the effective interactions. The effective interaction after one step depends on the effective interaction at finer scales only through the contraction above. The influence of the finest scales therefore decays geometrically with the number of steps. In Section 15 we use this to prove convergence at fixed resolution. We fix the coarse side and a number of steps counted from the coarse side. Then the effective interactions, the conditional laws of the fluctuations, and the joint laws of the coarse configuration and the fluctuations of these steps converge as the total number of steps $K \to \infty$. This is a statement about finitely many variables. A subtle point is that gauge invariance allows different choices of gauge representatives at different K. We fix, once and for all, a family of gauge transformations that depend only on the coarse configuration and are chosen by a compactness argument. For this family the reconstructed configurations at each fixed resolution converge (Theorem 15.12).

Reconstruction of the fine field. From the coarse configuration and the fluctuations of all steps, we reconstruct the fine configuration step by step, from coarse to fine. We write A for the logarithm of the reconstructed fine configuration in the chosen gauge, divided by the mesh, so that A approximates a connection. In Part V we estimate how A changes when we add the fluctuations of one more step. The change has three parts: a conditionally centered linear part, a nonlinear part, and the difference between the

background field and the interpolation of the coarser field. The key deterministic fact concerns the last part. For each input direction and color, the linearized background field has the same signed volume-normalized column integral as the piecewise constant embedding of the coarse field. Their difference therefore has mean zero in each input column, and after heat smoothing at time s it gains a factor $\min(1, a/\sqrt{s})$, where a is the coarse mesh (Section 17). The key probabilistic fact is concentration for Lipschitz functions of the strongly log-concave law of the fluctuations (Section 16). Together, these two facts show that the heat evolution of A is controlled in norms of negative regularity, with summable contributions from the fine scales.

3.4 The bound on large fields

Theorem 2.7 has two parts. The second term in (2.5) accounts for configurations whose curvature is small at every scale, but for which some normalized fluctuation variable exceeds its cutoff $A_{\mathrm{w}} \log(1/t)$. By concentration for the law of the fluctuations, this event has conditional probability at most $\exp(-a \log^2(1/t))$ at each step. The sum over steps converges, because the effective coupling t decreases geometrically toward the ultraviolet. The point that needs care is that we condition only on the preceding scales being good, and never on good behavior at later scales. The probability of the preceding event is multiplied, never divided (Sections 22 and 23).

The first term accounts for configurations with a plaquette of large curvature at some scale. Here the argument has three steps.

Step 1: exponential moments of blocked actions. Under the fixed-parameter hypotheses of Theorem 22.1, fix $0 < \Gamma < \infty$, take $u \in 4\mathbb{N}$ with $u \geq n_{\min}(\Gamma)$, and let $N = uB^K$ and $1 \leq j \leq JK$. Fix a tiling of the lattice of side n_j into cells. Let Q_j be the Wilson action, divided by the coupling, of the averaged configuration V^{j-1} on the plaquettes that determine the curvature of the faces inside the cells. We prove that

$$\log \mathbb{E}_{\mu_{N,\gamma}} e^{sQ_j} \leq C_v n_{j-1}^3 + C_s n_{j-1}^2 \log(1/g_{n_{j-1}}) + C_b, \qquad 0 \leq s \leq s_0,$$

with constants that do not depend on the number of averaging steps or on $\gamma \in (0, \Gamma]$ (Theorem 22.1). The surface term comes from the difference between the numbers of independent plaquette variables of the periodic and the open lattice, in a comparison of partition functions, and it has to be kept. This is a stability bound in the spirit of Bałaban, but for an exponential moment under the unconditioned measure. Its proof is the most involved part of the paper. We partition the configurations according to the regions where some averaged plaquette has large curvature, recorded scale by scale; we call this partition the large-field decomposition (Theorem 21.5). Outside these regions we run the renormalization group in a localized form. We construct background fields, fluctuation integrals, and effective interactions on arbitrary unions of blocks, with the link variables

in the bad regions held fixed as parameters (Sections 18 to 20). The estimates are uniform in the shape, the number of components, and the age of the bad regions. In Section 21 we bound the density of the fine configurations on each piece of this partition by the localized effective density with the same normalization. The bound holds pointwise on gauge sections that contain every point of the relevant fibers (Theorems 21.9 and 21.12). The difference between the localized effective interactions and those on the full torus decays exponentially away from the bad regions. The missing coordinates, the boundaries, and the frozen coordinates contribute to the normalization constant, and we keep track of these contributions separately. Inside the bad regions we use only a crude bound: the effective classical action of the coarse configuration dominates both the Wilson action of the omitted links and a positive cost for each recorded bad plaquette (Section 22). This cost grows linearly in n_j/γ, while the entropy of the bad regions grows only like the square of a logarithm.

Step 2: chessboard estimates. The Wilson measure is reflection positive with respect to reflections in lattice planes through sites and through links, as Osterwalder and Seiler showed [134]. We use this separately for each of the eight shifted tilings of the lattice of side n_j into cells of $2 \times 2 \times 2$ sites. For a cell C, let I_0 be the event that some internal plaquette of C at level j has large curvature. Let R_0 be the event that every plaquette at a finer level whose full dependence set on the fine lattice lies in C has small curvature; plaquettes whose dependence sets cross the boundary of the cell are not tested. Both indicators are bounded, gauge-invariant Borel functions of the fine links in the cell. Because the blocks of the averaging map are centered and nested, the reflections between cells map these dependence sets and tests to their reflected placements in the other cells. Then the chessboard estimate of Fröhlich, Israel, Lieb, and Simon [92, 93], in the form given by Biskup [29], applies to this tiling (Theorem 6.6 and Lemma 22.11). Let R and I_{all} be the intersections of the reflected placements of R_0 and of I_0 over all cells of the tiling. The chessboard estimate gives

$$\mu_{N,\gamma}(R_0 \cap I_0) \le \mu_{N,\gamma}(R \cap I_{\mathrm{all}})^{8/n_j^3}.$$

The regularity events stay inside the root, so no division by their probability occurs. On R, the estimate on the curvature of averaged configurations shows that each cell with a plaquette of large curvature contributes at least a fixed multiple of n_j/γ to Q_j. By Step 1 and Markov's inequality, the probability of $R \cap I_{\mathrm{all}}$ is then at most $\exp(-cn_j^4/\gamma + Cn_j^3 + Cn_j^2 \log(1/g_{n_j}))$. Taking the root, we find that $R_0 \cap I_0$ has probability at most $\exp(-c'n_j/\gamma)$, once n_j/γ is large.

Step 3: first failure. We decompose the event that some scale has a bad cell according to the first scale, counted from the finest lattice toward the coarse lattice, at which a bad cell occurs. On each piece of this decomposition all finer scales are good, which implies

the regularity events R_0 of Step 2. We sum the bounds of Step 2 over cells and scales (Theorem 23.6). Since the bound at side n_j decays like $\exp(-cn_j/\gamma)$ and the sides grow geometrically toward the ultraviolet, the sum is dominated by the coarsest side u, which gives the first term in (2.5). On the complementary event the curvature is small at every scale. By Theorem 7.7, at every scale there exists a gauge representative with all link variables close to the identity. The remaining conditions of the small-field event reduce to the cutoffs on the fluctuations treated above.

The parameters are the integer J, the analytic radii, the cutoff amplitude A_{w}, the widths of the boundary layers in the localized analysis, and the minimum coarse side. We choose them in a fixed order: first the quantities that are uniform in J, then J, then the constants that depend on J, and finally the minimum side (Theorem 23.2). One of the tasks of Section 23 is to check that this order is free of circularity.

3.5 The gradient flow

To pass from the renormalization group variables to flowed Wilson loops, we need a theory of the lattice gradient flow for rough initial data that is uniform in the mesh. We write a lattice configuration near the identity as $U_e = \exp(aA_e)$. We apply a time-dependent gauge transformation that adds a damping term, in the spirit of the method of Zwanziger [157], DeTurck [66], and Donaldson [76]. Then the Wilson flow becomes a parabolic equation whose linear part is the lattice heat equation,

$$\partial_t A = \Delta_a A + \mathcal{N}_a(A, A) + (\text{higher order}),$$

where $\mathcal{N}_a$ is a quadratic nonlinearity involving one lattice derivative. For the initial data of interest, A has regularity just below $-1/2$, and the quadratic term is not defined by naive multiplication. We follow the approach of Da Prato and Debussche [63], which Cao and Chatterjee used for the Yang–Mills heat flow with random data [42], and write

$$A(t) = x(t) + y(t) + r(t), \qquad x(t) = e^{t\Delta_a} A(0), \qquad y(t) = \int_0^t e^{(t-s)\Delta_a} \mathcal{N}_a(x(s), x(s))\, ds.$$

We then solve for the remainder r, which is more regular. We call the pair (x, y), consisting of the linear heat evolution and the quadratic Picard term, the flow data. For a fixed $\alpha \in (1/2, 2/3)$ and horizon T, we measure it in the norms

$$\sup_{0 \le t \le T} \big\{ (t + a^2)^{\alpha/2} \|x(t)\|_\infty + (t + a^2)^{(\alpha+1)/2} \|D_a x(t)\|_\infty \big\}$$

and

$$\sup_{0 \le t \le T} \big\{ (t + a^2)^{\alpha - 1/2} \|y(t)\|_\infty + (t + a^2)^{\alpha} \|D_a y(t)\|_\infty \big\},$$

where D_a includes the lattice differences and shifts that the equation requires (Definition 4.8). Part VII proves the deterministic theory. It consists of local existence of r on a short time interval with bounds uniform in the mesh (Theorem 24.5); smoothing after a positive time; continuation to any fixed time interval, by comparison with finitely many smooth solutions of the continuum flow (Theorems 24.10 and 25.7); and Lipschitz dependence of the flowed holonomies on the flow data (x, y) (Corollary 25.11). For smooth initial data the lattice flow converges to the continuum flow. Here we use the deterministic results of Cao and Chatterjee [42, 43], which rest on the global theory of Råde [139] and of Charalambous and Gross [47].

3.6 Stochastic estimates and convergence of flowed loops

In Part VIII we prove two facts about the flow data (x, y) of the reconstructed fine field under the conditioned Wilson measure. First, it is bounded in the norms of Section 3.5 with high probability, uniformly in the number of steps. Second, it is well approximated by the flow data of the field reconstructed from finitely many steps. At each step the new fluctuations contribute four terms: a conditionally centered linear term, a conditionally centered nonlinear term, an interpolation defect, and a predictable term (Lemma 26.2). We estimate all sixteen ordered products of these terms in the quadratic Picard term y (Sections 26 and 27). We keep their conditional means, which are nonzero, and we treat both orders of each product, since $\mathcal{N}_a$ is not symmetric. At mesh a the linear fluctuation contributes $g_3 a^{\alpha-1/2}\sqrt{\log(2 + L/a)}$, which is summable over the geometric sequence of meshes since $\alpha > 1/2$.

Given these estimates, we compare the flowed Wilson loops of the fine configuration with those of the configuration reconstructed from the coarse field and the fluctuations of a fixed number q of steps. They are close in probability, uniformly in the total number of steps. The latter configuration is a smooth interpolation of a finite-dimensional random variable whose law converges as $K \to \infty$, by Section 3.3. Further, the flow of a fixed smooth field converges to the continuum flow. Letting first $K \to \infty$ and then $q \to \infty$, we find that the laws of flowed Wilson loops under the conditioned Wilson measure converge (Theorem 28.1). Together with (3.1) and the bound on large fields, this gives convergence under the Wilson measure (Theorem 28.2). We first prove these results when the coupling of the coarse lattice is below a small ceiling, and then extend them to every fixed γ by taking the coarse side large (Lemma 28.11 and Theorem 29.21). This proves part (1) of Theorem 2.3 along the subsequences $N_K = n_* B^K$.

3.7 Construction of the limiting measure

Convergence of finite families of flowed Wilson loops does not by itself produce a measure on $\mathcal{X}_L$. We obtain one as follows (Section 29). For each fixed number q of steps, the field reconstructed from q steps is smooth after interpolation, and its continuum flow defines a random element of $\mathcal{X}_L$. These random trajectories have energies that are tight uniformly in q at each of the times $s_j = L^2 2^{-j}$. Indeed, the lattice energy along the flow is controlled by the stochastic estimates, and it does not increase in time. The sets of trajectories whose energies at the times s_j are bounded by given numbers R_j, for all j, are compact, by results of Cao and Chatterjee that rest on Uhlenbeck's compactness theorem. Hence these laws converge along subsequences, and the joint laws of flowed Wilson loops identify the limit. The joint convergence of energies and Wilson loops follows from a comparison with the smooth fields reconstructed from finitely many steps, together with the continuity of the energy after a positive time in the Sobolev setting of Cao and Chatterjee (Theorem 29.18).

We also want to realize the limit as a limit of the lattice measures themselves. For this purpose we construct measurable gauge-invariant maps π_K from lattice configurations to $\mathcal{X}_L$. We enumerate the trajectories of the continuum flow started from trigonometric polynomials with rational coefficients. Given a lattice configuration, π_K selects the first trajectory in this list with the following two properties: its first m determining coordinates are within 2^{-m} of the flowed lattice Wilson loops, and its energies at m fixed positive times are at most the lattice energies plus 2^{-m}. If there is no such trajectory, π_K selects the zero trajectory. With $m = m(K)$ increasing slowly, these maps have the properties in part (3) of Theorem 2.3, and the energy constraint gives the tightness in part (4) (Theorems 29.19 and 29.20).

Finally, the passage from the subsequences $n_* B^K$ to the full sequence nM^k is arithmetic. Since the construction works for every sufficiently large J, we may take two consecutive values J and $J+1$. Every residue class of k modulo J contains infinitely many k for which nM^k also belongs to a sequence of the form $n' M^{(J+1)K}$, and hence the limits along the different residue classes coincide (Theorem 29.28). We prove the remaining parts of Theorem 2.5 in Section 30.

3.8 Short distances

For the short-distance theorem we fix γ and a coarse side. We study the heat-smoothed circulation of the reconstructed field around a small loop $x + \varepsilon\ell$, normalized by $\sqrt{g_3^2 \varepsilon}$. Adding the fluctuations step by step, we write this circulation as the sum of three terms: a contribution from the coarse field, a martingale whose increments are linear in the new normalized fluctuations, and a small nonlinear error (Section 31). A useful observation is that the linearized successive background fields differ from a single background field

by a periodic gradient, which the closed circulation does not see. We compare each martingale increment to a Gaussian, by an integration by parts adapted to the cutoff on the fluctuations, in the spirit of Stein's method [147]. The sum of the conditional covariances converges to the covariance (2.6) of the free field, as $\varepsilon \to 0$ after the lattice spacing (Theorem 31.6 and Section 32). The contribution of the coarse field vanishes in the limit because the loop shrinks. We then compare the flowed Wilson loop with the circulation, using the stability of the flow and the expansion $1 - \operatorname{tr} e^V/2 = |V|^2/8 + O(|V|^4)$ (Lemma 32.11). We carry out this analysis under the conditioned Wilson measure and in the fixed gauge of the reconstruction. To obtain the gauge-invariant statement under λ, we take the limits in the following order: the lattice spacing to zero; then $\varepsilon \to 0$ at a fixed coarse side and with a fixed cutoff on the flow data; then the removal of that cutoff; and finally the coarse side to infinity. The asymptotic independence from macroscopic observables has a simple reason: the Gaussian approximation holds uniformly conditionally on any fixed number of coarse steps, while a macroscopic flowed observable is close to a function of finitely many coarse steps.

3.9 Weak coupling

For the weak-coupling results we fix, before γ varies, the parameters of the construction and a coarse side u in the class $[n]_M$, larger than certain thresholds; by Theorem 2.3(5) this does not change λ_γ. As $\gamma \to 0$ at this fixed coarse side, every scale of the hierarchy becomes weakly coupled, and the analysis takes place on the coarse lattice. Section 33 develops the geometry near flat connections and Laplace's method on the coarse lattice. The density of the coarse configuration at the end of the renormalization group is proportional to $\mathbf{1}_{D_r} \exp(-uE_d/\gamma - R + R(I))$, and its normalizing constant is comparable to γ^{3u^3-1}, uniformly in the number of finer steps (Theorem 33.10). Near a regular flat connection, the integral over the directions transverse to the flat connections is Gaussian. In Section 34 we compare the reconstructed field with the field in which all fluctuations beyond a fixed number of steps are set to zero. Under the conditioned Wilson measure, their flowed holonomies differ by $O(\sqrt{\gamma})$ in L^p, uniformly in the number of steps and in small γ (Theorems 34.9 and 34.10). This gives the concentration on flat connections in Theorem 2.9.

For Theorem 2.10, recall that the normal covariance Σ_θ is the covariance of the linear response of the flowed logarithms at a regular flat connection. We compute it explicitly in Theorem 33.11. The theorem then follows from Gaussian integration at finitely many scales and the transfer estimates of Section 34 (Section 35). We take the limits in the following order: the continuum limit, then $\gamma \to 0$ at a fixed number of scales, and then the number of scales to infinity.

For Theorem 2.11, the starting point is an exact Taylor expansion of the effective classical action of a constant configuration, $E_d(e^{X/u}) = Q(X)/u + O(|X|^5)$, where Q is the constant-mode quartic form and the remainder is uniform in the depth d (Theorem 36.2). We integrate out the nonconstant modes with Gaussian estimates. This leaves a nine-dimensional integral against $e^{-Q(Y)}$ on the scale $Y = X/\gamma^{1/4}$, multiplied by a positive factor that is bounded above and below (Theorem 36.6). This factor is the same for all test functions, so it cancels in the ratios (2.7). The expectation of a compactly supported test function at this scale is of order $\gamma^{1/4}$, which comes from the volume $\gamma^{9/4}$ of the constant modes, compared with the order γ^2 of the normalization. We then transfer the result from the reconstructed holonomies to the flowed holonomies of the field. For this we use an estimate that is conditional on the coarse configuration (Theorem 34.11). An estimate averaged over the coarse configuration would lose the factor $\gamma^{1/4}$ when divided by this small expectation.

3.10 Order of choices and dependence of constants

Many statements in the paper are uniform in some parameters and not in others, and the order in which parameters are chosen matters. We describe the convention used throughout in Definition 4.14. In brief, we choose the integer J first, using only estimates that are uniform in J. Then we compute all local constants at that J. They are uniform in the lattice side, the number of averaging steps, the coupling below the relevant ceiling, and, in the localized analysis, the shape of the regions. Finally we choose the minimum coarse side. Statements at fixed γ may have constants that depend on γ; these include the convergence of flowed loops, the construction of λ, and the short-distance theorem. The bound on large fields and the weak-coupling estimates are uniform in γ below fixed ceilings.

SECTION 4

Notation and conventions

This section fixes the notation and conventions used in the rest of the paper. Section 2 introduced the lattice model, the Wilson flow, admissible approximations of loops, and the trajectory space, and Section 3 described the objects of the renormalization group analysis informally. Here we give the precise definitions that the later sections cite. A definition here fixes the meaning of a symbol; the existence of the object, its analytic properties, and the probability estimates for it are proved where it is constructed, and we say where in each case. A choice of letter never implies a uniformity: the parameters on which a constant depends are listed in each statement, or by reference to an item of Definition 4.14. We write $\mathbb{N} = \{1, 2, \dots\}$; depths and the index K may also be zero.

Sections 4.1 to 4.3 fix the lattice model, the block averaging map, and the measures of the renormalization group analysis: the Wilson measure, the small-field event, the conditioned Wilson measure, the stopped measure, and the conditional fluctuation kernel. Sections 4.4 to 4.9 fix the analytic coefficient norms, the conventions for polymer expansions and exponential moments, the flow conventions, the heat-smoothed circulation, the continuum trajectory space, and the weak-coupling observables, and Section 4.10 describes the dependence of constants.

4.1 The periodic lattice and the Wilson measure

The following definition restates the model of Section 2.1 in the notation of the later sections, where n denotes a lattice side.

Definition 4.1. Let $L > 0$ and let $n \geq 4$ be an integer. Set

$$\Lambda_n = (L/n)\mathbb{Z}^3/(L\mathbb{Z})^3, \qquad a_n = L/n.$$

A positive link is a pair $(x, x + a_n e_i)$ with $x \in \Lambda_n$ and $i \in \{1, 2, 3\}$, and $\mathcal{E}_n$ is the set of these $3n^3$ links. A positive plaquette is the positively oriented boundary $p_{ij}(x)$ of the square spanned by $a_n e_i$ and $a_n e_j$ at x, with $i < j$, and $\mathcal{P}_n$ is the set of these $3n^3$ plaquettes. A configuration is an element U of $\mathcal{U}_n = \mathrm{SU}(2)^{\mathcal{E}_n}$. For the reversed link $\bar{e}$ of a positive link e we set $U_{\bar{e}} = U_e^{-1}$, and the transport along a path $e_1 \cdots e_m$ is the ordered product $U_{e_1} \cdots U_{e_m}$. Writing $U_i(x)$ for the variable on the link $(x, x + a_n e_i)$, we have

$$U_{p_{ij}(x)} = U_i(x) U_j(x + a_n e_i) U_i(x + a_n e_j)^{-1} U_j(x)^{-1}.$$

Let H be the Haar measure on $\mathrm{SU}(2)$ with $H(\mathrm{SU}(2)) = 1$, and let $H_n = \bigotimes_{e \in \mathcal{E}_n} H$. Let $\gamma > 0$, and set

$$q(U) = 1 - \frac{1}{2}\operatorname{tr} U, \qquad S_W(U) = 4 \sum_{p \in \mathcal{P}_n} q(U_p), \qquad h_n = g_n^2 = \frac{\gamma}{n}, \qquad \gamma = g_3^2 L.$$

The Wilson measure is the probability measure

$$d\mu_{n,\gamma}(U) = Z_{n,\gamma}^{-1} e^{-S_W(U)/h_n}\, dH_n(U)$$

on $\mathcal{U}_n$, where $Z_{n,\gamma}$ is the normalizing constant.

Since $S_W/h_n = (4n/\gamma)\sum_p q(U_p)$, this is the measure (2.1). The parameter γ is the dimensionless coupling, $g_3^2 = \gamma/L$ is the continuum coupling, and $h_n = g_n^2 = g_3^2 a_n$ is the lattice coupling at side n; in the usual notation the inverse coupling is $\beta_n = 4/h_n$. The letter n always denotes a lattice side, the number of sites in each coordinate direction, and never a depth. We write d and d_0 for classical depths (numbers of preceding averaging steps over which the classical action has been minimized), and ℓ for the index of an averaging step. The plaquette action is Wilson's [153]; the factor four and the identification $h_n = g_3^2 a_n$ are our normalization, explained in the following remark.

Remark 4.2. Let A be a smooth $\mathfrak{su}(2)^3$-valued function on $\mathbb{T}_L^3$, and set $U_e = \exp(a_n A_i(x))$ for $e = (x, x + a_n e_i)$. By the Baker–Campbell–Hausdorff formula, $\operatorname{Log} U_{p_{ij}(x)} = a_n^2 F_{ij}(x) + O(a_n^3)$ uniformly in x, where $F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j]$. By (4.2) below, $q(e^X) = |X|^2/8 + O(|X|^4)$. Since $h_n = g_3^2 a_n$ and $|\mathcal{P}_n| = 3n^3$, we obtain

$$\frac{S_W(U)}{h_n} = \frac{1}{2g_3^2} \sum_{x \in \Lambda_n} \sum_{i<j} a_n^3 |F_{ij}(x)|^2 + O(a_n) \longrightarrow \frac{E_L(A)}{2g_3^2} \qquad (n \to \infty),$$

where $E_L(A) = \sum_{i<j} \int_{\mathbb{T}_L^3} |F_{ij}|^2\, dx$ is the Yang–Mills energy (2.3). This computation concerns smooth configurations only; it says nothing about typical configurations under $\mu_{n,\gamma}$.

Gauge transformations. A lattice gauge transformation is an element k of $\mathcal{G}_n = \mathrm{SU}(2)^{\Lambda_n}$. It acts on configurations by

$$(U^k)_{xy} = k_x U_{xy} k_y^{-1}, \qquad (U^k)^l = U^{lk}. \tag{4.1}$$

A based gauge transformation is a lattice gauge transformation equal to the identity at specified vertices, its roots. Under a gauge transformation, the transport around a closed path is conjugated by the value at its basepoint. Hence, among the functions of a tuple of holonomies along closed paths with a common basepoint, only those invariant under simultaneous conjugation are gauge invariant, while a based gauge transformation whose roots include the basepoint leaves these holonomies unchanged. All gauge transformations in this paper are periodic; we do not consider twisted boundary conditions.

The Lie algebra. Let $\tau_a = -i\sigma_a/2$, where $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices. For $X = x^a\tau_a$ and $Y = y^a\tau_a$ in $\mathfrak{su}(2)$ we use

$$[\tau_a, \tau_b] = \epsilon_{abc}\tau_c, \qquad \langle X, Y\rangle = -2\,\mathrm{tr}(XY), \qquad |X| = |x|.$$

Then (τ_a) is an orthonormal basis, and the bracket corresponds to the cross product in $\mathbb{R}^3$. This is the inner product of Section 2.1, and on $\mathcal{U}_n$ we use the product of the corresponding bi-invariant Riemannian metrics. The group distance from the identity is

$$\rho_G(U) = \min\{|X| : e^X = U\} \in [0, 2\pi].$$

In analytic calculations, Log denotes the logarithm near the identity, continued holomorphically only inside a specified complex neighborhood of the identity. The principal real logarithm is defined on $\mathrm{SU}(2) \setminus \{-I\}$ and satisfies $|\mathrm{Log}\, U| < 2\pi$. There is no continuous logarithm on all of $\mathrm{SU}(2)$.

In these coordinates the Haar measure and the plaquette function are given by

$$\begin{aligned} dH(e^X) &= \frac{1}{16\pi^2}\,\mathrm{sinc}^2(|X|/2)\,d^3X \quad (|X| < 2\pi), \\ q(e^X) = 1 - \cos(|X|/2) &= \frac{|X|^2}{8} + O(|X|^4), \qquad \mathrm{sinc}\, z = \frac{\sin z}{z}, \quad \mathrm{sinc}\, 0 = 1. \end{aligned} \tag{4.2}$$

Integrating the first formula in polar coordinates, we find $H\{\rho_G < \varepsilon\} = (\varepsilon - \sin\varepsilon)/(2\pi)$ for $0 \le \varepsilon \le 2\pi$; for $\varepsilon \ge 2\pi$ this probability is one. These identities, and the estimates for products of finitely many group elements near the identity, are proved in Lemma 5.1. We write $c_{\mathrm{Haar}} = 1/(16\pi^2)$ for the constant in the coordinate density and $j_H(X) = \mathrm{sinc}^2(|X|/2)$ for its variable factor. The constant c_{Haar} is unrelated to the lower-bound constants in partition-function estimates with open boundary conditions.

4.2 Lattice refinements and the block averaging map

Fix an odd integer $M = 2R_M + 1 \ge 3$ and an integer $J \ge 1$, and set $B = M^J$. At total depth $K \ge 0$ the fine side is $N = n_* B^K$ with $n_* \in 4\mathbb{N}$, and for $0 \le \ell \le JK$ we set

$$n_\ell = N/M^\ell, \qquad a_{n_\ell} = L/n_\ell, \qquad g_\ell^2 = h_\ell = \gamma/n_\ell, \qquad V^\ell = T_M^\ell U,$$

where T_M is the block averaging map of Definition 4.3 below. Increasing ℓ moves toward the infrared. Since M is odd, all the sides n_ℓ are divisible by four. Schematically,

$$U \in \mathcal{U}_N \xrightarrow{T_M} V^1 \in \mathcal{U}_{n_1} \xrightarrow{T_M} \cdots \xrightarrow{T_M} V^{JK} \in \mathcal{U}_{n_*},$$

and the scales are as follows.

averaging step ℓ	side n_ℓ	mesh a_{n_ℓ}	coupling g_ℓ^2
0	N	L/N	γ/N
ℓ	$NM^{-\ell}$	LM^ℓ/N	$\gamma M^\ell/N$
JK	n_*	L/n_*	γ/n_*

As in Section 3.2, one application of T_M is an averaging step, and a renormalization group step (RG step) consists of J consecutive averaging steps, $V^{(k+1)J} = T_M^J V^{kJ}$, with blocking factor B.

We also use compositions of s averaging steps, $1 \le s \le J$, from an input side $M^s m$ to an output side $m \ge 4$, with an arbitrary preceding classical depth d_0. The fields on the s lattices of sides $Mm, \dots, M^s m$ are integrated. We call these lattices the integrated scales of the composition, and reserve the term intermediate scales for the J lattices of an RG step. The largest coupling of an integrated scale is attained at side Mm and is denoted by

$$t_{\max}^2 = \frac{\gamma}{Mm}.$$

If the integrated scales are numbered $i = 0, \dots, s-1$, from side $M^s m$ for $i = 0$ to side Mm for $i = s-1$, then the coupling g_i of scale i (not its square) satisfies $g_i / t_{\max} = M^{(i-s+1)/2}$. These ratios lie between $\lambda_* = M^{-(J-1)/2}$ and 1. The physical mesh is always denoted by a, never by h.

The equivalence relation $n \sim_M n'$ and the classes $[n]_M$ were defined in Section 2.3, and the index $[n_*]$ of the continuum measure $\lambda_{L,\gamma,[n_*]}$ refers to the class $[n_*]_M$. The measure is first constructed along the sides $n_* B^K$; by Theorem 29.28, it is the measure attached to the whole class. The small-field event and the bounds on its complement are always formulated with RG steps of J averaging steps.

Definition 4.3. We work in fine lattice units, so that the fine lattice is $\mathbb{Z}^3$ with its periodic identification and the coarse vertices are the points of $M\mathbb{Z}^3$. For a coarse vertex v, the centered block of v is

$$\mathsf{B}_M(v) = v + \big([-R_M, R_M]^3 \cap \mathbb{Z}^3\big).$$

For $x \in \mathsf{B}_M(v)$, let $\sigma_{v,x}$ be the lattice path from v to x that changes the coordinates in the order $1, 2, 3$, each monotonically. For the coarse link $c = (v, v + Me_i)$, let C_c be the straight path from v to $v + Me_i$, called the reference path, and for $x \in \mathsf{B}_M(v)$ let

$$F_{c,x} = \sigma_{v,x}[x, x + Me_i]\, \sigma^{-1}_{v+Me_i, x+Me_i}.$$

The paths $F_{c,x}$ and C_c both go from v to $v + Me_i$, so $U(F_{c,x})U(C_c)^{-1}$ is the holonomy of the closed path $F_{c,x}C_c^{-1}$ based at v; we call these matrices the based loop holonomies of c. If none of the based loop holonomies $U(F_{c,x})U(C_c)^{-1}$, $x \in \mathsf{B}_M(v)$, equals $-I$, define

$$(T_M U)_c = \exp\Big\{ M^{-3} \sum_{x \in \mathsf{B}_M(v)} \operatorname{Log}\big(U(F_{c,x})U(C_c)^{-1}\big) \Big\} U(C_c), \tag{4.3}$$

where Log is the principal logarithm. If one of these based loop holonomies equals $-I$, set $(T_M U)_c = U(C_c)$. The variable on a reversed coarse link is the inverse of the variable on the positive link. For $J \geq 1$, T_M^J denotes the J-fold iterate of T_M.

The second clause, the Borel extension of the block averaging map, is part of the definition: T_M is defined on the whole configuration space. The iterate T_M^J is never replaced by a separately defined average with blocking factor B.

This construction follows Bałaban's averaging operation [12, (15), p. 19, and (42), p. 23]: one transports the fine link variables to a reference path, averages the logarithms of the based loop holonomies, and exponentiates. The centered blocks, the particular paths, and the Borel extension are specific to this paper. Under a gauge transformation k, every based loop holonomy is conjugated by k_v. The principal logarithm commutes with conjugation, and the exceptional value $-I$ is invariant under conjugation. Hence $T_M(U^k) = (T_M U)^{k'}$, where k' is the restriction of k to the coarse vertices. The following properties are proved in Proposition 6.2: T_M is Borel and gauge covariant; it is covariant under translations by coarse vectors and under the reflections preserving the coarse lattice, including at the exceptional values; the identity $T_M U_0(b) = b$ below holds; and the blocks of iterates of T_M are nested. The constraint coordinates described next are constructed in Lemma 6.3.

Constraint coordinates. Let b be a coarse configuration. The coarse-field lift $U_0(b)$ is the fine configuration equal to I on every link inside a block and to b_c on every link from $\mathsf{B}_M(v)$ to $\mathsf{B}_M(v + Me_i)$, for $c = (v, v + Me_i)$. Each path $F_{c,x}$, and the path C_c, crosses exactly one such interface link, so all based loop holonomies of $U_0(b)$ equal I and $T_M U_0(b) = b$. In each block, the links of the paths $\sigma_{v,x}$ form a spanning tree rooted at the coarse vertex (Proposition 6.2). A based gauge transformation with roots at the coarse vertices sets the variables on these block trees equal to I; this is the block axial gauge. For the coarse link $c = (v, v + Me_i)$, the eliminated edge is the interface link

$$e_c^* = (v + R_M e_i,\, v + (R_M + 1)e_i)$$

on the reference path. We write $U = e^X U_0(b)$ on the links outside the block trees. The block averaging constraint for c reads

$$\mathsf{C}_{b,c}(X) = \mathrm{Log}\big((T_M U)_c b_c^{-1}\big) = 0, \qquad D_{X_{e_c^*}} \mathsf{C}_{b,c}(0) = M^{-2} I_3.$$

To compute the derivative, note that $X_{e_c^*}$ enters $U(C_c)$ and, through $U(C_c)^{-1}$, the $M^3 - M$ based loop holonomies with x off the line through v in direction e_i; those with x on this line do not depend on $X_{e_c^*}$. At $X = 0$ the derivative is therefore $(1 - (M^3 - M)M^{-3})I_3 = M^{-2} I_3$. Near $X = 0$ the constraints are solved by local constraint coordinates $X = \Gamma_b(z)$, where the free logarithmic coordinates z are the logarithms on the links outside the block

trees other than the eliminated edges. On this branch, the Haar disintegration density with respect to $dH_m(b)\,dz$ is

$$\mathscr{J}_b(z) = c_{\text{Haar}}^f \frac{\prod_{e \text{ outside the block trees}} j_H(\Gamma_{b,e}(z))}{\prod_c \det D_{X_{e_c^*}} \mathsf{C}_{b,c}(\Gamma_b(z))}, \tag{4.4}$$

where f is the number of independent free group variables; for one averaging step from side Mm to side m, $f = 2((Mm)^3 - m^3)$. The numerator includes the Haar factors j_H of the eliminated edges, and there is no additional Haar density for the coarse field, no Gram determinant, and no residual gauge-volume factor. The formula is used only on the local branch, on which the determinants are positive, and with all domain indicators retained; the precise statement, with its domain, is Lemma 6.3. This change of variables is the spanning-tree-gauge form of the disintegration of Haar measure for this averaging constraint. Spanning-tree gauges are standard in lattice gauge theory, and Bałaban used them in the renormalization group context [14].

Under product Haar measure, the integral over the gauge transformations at the vertices of the block trees other than their roots has mass one. We keep this integral explicitly until the gauge invariance of the integrand has been established. In the constructions on block regions of Sections 18 to 21, some group variables are frozen coordinates. The based transport of a frozen coordinate b_e, which carries it to its basepoint, is $\widehat{b}_e = U(\sigma_e)b_e$. If the variables of the block region are held fixed, the change $b_e \mapsto \widehat{b}_e$ is a triangular Haar translation, and it preserves product Haar measure. Frozen coordinates remain indexed by the lattice on which they were frozen.

4.3 Effective actions, the small-field event, and conditioned measures

The effective classical action and the effective interaction. For $d \geq 0$, E_d denotes the effective classical action at preceding classical depth d, and $E_0 = S_W$. It is the value of the constrained minimization problem on the local branch constructed in Sections 9 and 10, and it satisfies

$$E_{d+1}(c) = M \min_z E_d(b(c,z))$$

on the common domain of these local constructions (Theorem 10.10). Here $b(c,z)$ parametrizes the configurations on the intermediate lattice whose average is c, and the factor M is the change of units in three dimensions. This notation does not refer to an unrestricted minimum over the compact fiber of T_M^d. Following Bałaban [16], we call the constrained minimizer the background field. We write R_d for the effective interaction that accompanies E_d in the effective density, and Λ for the normalization constant in the log density, which collects the factors independent of the field. The analytic norm of R_d controls no additive constant.

Normalized fluctuation variables. In an RG step, the free logarithmic coordinates z_i of the intermediate scale i are written as

$$z_i = z_i^*(c, z_{>i}) + g_i K_i(c, z_{>i})^{-1/2} w_i, \tag{4.5}$$

where c is the coarse field, $z_{>i}$ are the coordinates of the coarser intermediate scales, and z_i^* is the conditional constrained minimizer. Further, K_i is the constrained Hessian (a positive definite matrix) in the free coordinates of scale i at z_i^*, $K_i^{-1/2}$ is the inverse of its positive square root, and g_i is the coupling of scale i. The w_i are the normalized fluctuation variables; they differ from z_i/g_i. The map $w \mapsto z$ is triangular, with diagonal blocks $g_i K_i^{-1/2}$. Its Jacobian is therefore exactly

$$\prod_i g_i^{3f_i} \det K_i(c, z_{>i})^{-1/2},$$

where f_i is the number of free group variables of scale i. Its value at $w = 0$ and the logarithm of its ratio to that value are two parts of this single factor; neither is counted twice. The change of variables is carried out in Lemma 12.3.

Reconstruction maps. We write $\widehat{\mathscr{R}}$ for the fixed family of reconstruction maps of Theorem 15.12. These maps differ from the reconstructions in the depth-dependent gauge representatives by gauge transformations that depend only on the coarse field and equal the identity at the coarse roots. On each fixed lattice, the depths at which the family switches representatives are finite and deterministic, and they do not depend on the coupling, on configurations, or on observables. The family preserves the conditional fluctuation kernels and the fluctuation cutoffs defined below. Its construction, and the preservation of the analytic coefficient bounds on the larger domains, are in Section 15. The hat does not assert equality with the depth-dependent representatives.

The small-field event. The following definition makes the description in Section 2.5 precise.

Definition 4.4. Let $0 < r < \rho$, where ρ is the analytic radius of Section 4.4, and set

$$B_r = \{\exp A : \max_e |A_e| < r\}, \qquad D_r = \mathcal{G}B_r,$$

where $\mathcal{G}B_r$ is the set of all U^k with $U \in B_r$ and k a lattice gauge transformation. We call B_r the small-link domain and D_r the gauge-invariant small-field domain. Fix the parameters of Definition 14.4; in particular J, r, ρ, and $A_{\mathrm{w}} > 0$ are fixed. Let $u \in 4\mathbb{N}$, $K \geq 0$, $N = uB^K$, $V^\ell = T_M^\ell U$, and let $\gamma > 0$ be such that $t_k < 1$ for $0 \leq k < K$, where

$$t_k^2 = \frac{\gamma}{M n_{(k+1)J}}, \qquad p_{\mathrm{w}}(t) = A_{\mathrm{w}} \log(1/t).$$

For $0 \le k < K$, let $\mathcal{A}_{k,J}(U)$ be the event that the k-th RG step, from V^{kJ} to $V^{(k+1)J}$ at preceding classical depth kJ, is defined at U. This means that each pair formed by an intermediate coarse field and the free coordinates of the next finer scale lies in the local branch of the constraint coordinates and in the domains of the background field and of the Haar disintegration density, with positive Jacobian determinants, as specified in Definition 14.4. On $\mathcal{A}_{k,J}(U)$, let $w_{k,i,e}(U)$ be the normalized fluctuation variables of this step, obtained by inverting the step. The small-field event is

$$S_{N,u,\gamma} = \{U \in D_r\} \cap \bigcap_{k=0}^{K-1} \Big[\mathcal{A}_{k,J}(U) \cap \Big\{ \max_{i,e} |w_{k,i,e}(U)| \le p_{\mathrm{w}}(t_k) \Big\} \cap \{V^{(k+1)J} \in D_r\} \Big]. \tag{4.6}$$

For $K = 0$, the event is $\{U \in D_r\}$.

The domain conditions in $\mathcal{A}_{k,J}$ are the inverse-branch, classical-domain, and positive-Jacobian conditions of the construction, with their fixed radii. There is no additional D_r condition at an intermediate scale inside an RG step. The coupling t_k is the largest coupling of an integrated scale in the k-th RG step, and it is largest for $k = K - 1$, where $t_{K-1}^2 = \gamma/(Mu)$. The condition $t_k < 1$ for all k therefore means $\gamma < Mu$ when $K \ge 1$, and it ensures that the thresholds $p_{\mathrm{w}}(t_k)$ are positive. The analysis in Section 14 restricts γ further, by requiring $t_k \le t_*$ for a small upper bound t_*.

Lemma 4.5. *Under the assumptions of Definition 4.4, $\mu_{N,\gamma}(S_{N,u,\gamma}) > 0$.*

Proof. We show that $S_{N,u,\gamma}$ contains a nonempty open set. Near the identity configuration I all based loop holonomies in (4.3) are close to I, so the first clause of Definition 4.3 applies and the iterates T_M^ℓ are real analytic on a neighborhood of I, with $T_M^\ell(I) = I$. At $U = I$, every intermediate coarse field equals I and every free coordinate vanishes. The background field of the coarse field I is I, since I minimizes the Wilson action, so the conditional constrained minimizers vanish as well. Each domain condition in Definition 14.4 is a strict inequality on the logarithms of the coarse fields, on the free coordinates, on $\Gamma_c(z)$, and on the classical data, or the positivity of a determinant. At $U = I$ these quantities equal zero, and the determinants equal $\det(M^{-2} I_3) = M^{-6} > 0$. By Lemma 6.3 and the constructions of Sections 10 and 12, the inverse coordinates of the RG steps are continuous functions of U near I. Hence there exists a neighborhood O_1 of I contained in $\bigcap_{k<K} \mathcal{A}_{k,J}$. On O_1, the variables $w_{k,i,e}$ are continuous and vanish at I. Since $p_{\mathrm{w}}(t_k) > 0$ for every k, there exists a smaller neighborhood $O_2 \subset O_1$ of I on which $\max_{i,e} |w_{k,i,e}| \le p_{\mathrm{w}}(t_k)$ for all $k < K$. Finally, B_r is open, $I \in B_r \subset D_r$, and the maps $T_M^{(k+1)J}$ are continuous at I with value I. Hence there exists a neighborhood $O \subset O_2$ of I on which U and all $V^{(k+1)J}$ lie in B_r, and $O \subset S_{N,u,\gamma}$.

Since $0 \le q \le 2$, we have $S_W \le 24N^3$, and $Z_{N,\gamma} \le 1$. The density of $\mu_{N,\gamma}$ with respect to H_N is therefore at least e^{-24N^3/h_N}. The Haar measure of a compact group gives positive mass to every nonempty open set, so

$$\mu_{N,\gamma}(S_{N,u,\gamma}) \ge e^{-24N^3/h_N} H_N(O) > 0. \qquad \square$$

The three measures. Let $m \ge 4$ be a lattice side and $d \ge 0$ a preceding classical depth for which E_d and R_d have been constructed. Let $E_{d,\mathrm{sat}}$ be the gauge-invariant extension of E_d to D_ρ, $E_{d,\mathrm{sat}}(U) = E_d(U^k)$ for $U^k \in B_\rho$ (Theorem 11.6), and let $\overline{R}_d = R_d - R_d(I)$ be the centered effective interaction, extended to D_r in the same way. The stopped measure at side m and depth d is

$$d\widehat{\nu} = \widehat{Z}^{-1} \mathbf{1}_{D_r} e^{-E_{d,\mathrm{sat}}/g_m^2 - \overline{R}_d}\, dH_m, \qquad \overline{R}_d = R_d - R_d(I).$$

Under the hypotheses of Theorem 14.8, we have $E_{d,\mathrm{sat}} \ge \kappa_* E_0 \ge 0$ and $|\overline{R}_d| \le C$ on D_r, for constants κ_*, C depending on the fixed lattice. The integrand is then a strictly positive bounded Borel function. Since D_r contains the nonempty open set B_r, we have $0 < \widehat{Z} < \infty$. The stopped measure is defined only in this situation.

Whenever $\mu_{N,\gamma}(S_{N,u,\gamma}) > 0$, in particular under the assumptions of Definition 4.4, we compare the three measures

$$\mu_{N,\gamma}, \qquad \nu_{N,u,\gamma} = \mu_{N,\gamma}(\,\cdot \mid S_{N,u,\gamma}), \qquad \widehat{\nu}, \tag{4.7}$$

namely the Wilson measure, the conditioned Wilson measure, and the stopped measure just defined. At an intermediate stopping side, the marginal of the conditioned Wilson measure generally differs from the stopped measure, because it also includes the restrictions of the remaining coarser RG steps. At the terminal side $m = u$ no such restriction remains, and the law of V^{KJ} under $\nu_{N,u,\gamma}$ is the stopped measure at depth KJ (Theorem 14.8(9)). An exponential-moment estimate over a range of scales that starts from the unconditioned Wilson measure uses $\mu_{N,\gamma}$ itself, without a D_r restriction.

The auxiliary integral. For coarse fields outside B_r, in the larger analytic domain, the effective interaction is defined recursively by an analytic, positive product integral, the auxiliary integral of Theorem 14.6. It is not the conditional integral of the preceding restricted measure, although the two agree for coarse fields in B_r. The reconstructions it uses for coarse fields in the larger domain need only take values in B_ρ. This distinction between the radii r and ρ is part of the construction in Section 14.

The conditional fluctuation kernel. Fix a coarse field c in the domain of an RG step. The conditional fluctuation kernel of the normalized fluctuation variables $w = (w_{i,e})$ of the step is

$$k(c, dw) = \frac{\mathbf{1}_{\Omega_p}(w) e^{-V(c,w)}\, dw}{\int_{\Omega_p} e^{-V(c,w)}\, dw}, \qquad \Omega_p = \prod_{i,e} \{w_{i,e} \in \mathbb{R}^3 : |w_{i,e}| \le p_i\}. \tag{4.8}$$

Here the conditional fluctuation potential $V(c, w)$ contains $|w|^2/2$, all higher-order terms of the classical action in the fluctuations, the terms from the Haar disintegration density and from the Jacobian of (4.5) that depend on w, and the input effective interaction, produced by the integrations at finer scales, evaluated at the reconstructed field. Terms that depend on c but not on w cancel in the normalized kernel, and only there. For the kernels arising from the small-field event, $p_i = p_{\mathrm{w}}(t_k)$ for all intermediate scales of the k-th RG step. Products of truncated Gaussian measures serve as reference measures in expansions; they are not the conditional law. All conditional means, covariances, and derivatives of the conditional log-density refer to $k(c, dw)$ on its convex domain Ω_p. In particular, for a differentiable parameter λ on which V depends, with the coupling, the radii p_i, and the coordinates w held fixed, the normalized log-density derivative on Ω_p is $\partial_\lambda \log k = -\partial_\lambda V + \mathbb{E}_k \partial_\lambda V$ (Lemma 13.6). The kernel and its potential are constructed in Theorem 13.8.

The comparison lattice. For the index $s \geq 0$ of the comparison lattice we write $n_s = n_* B^s$ for its side, $H_s = L/n_s$ for its mesh, and $\nu_{K,s} = \nu_{N_K, n_s, \gamma}$ for $K \geq s$, where $N_K = n_* B^K$. The corresponding backward reconstruction sequence has physical meshes $a_j = H_s B^{-j}$, $0 \leq j \leq K - s$, and preceding classical depths $J(K - s - j)$. Once the coarse field of a step is fixed, its conditional fluctuation kernel is not changed by the restrictions at coarser scales, since these are functions of the coarse field alone.

4.4 Analytic coefficient norms

Effective interactions, background fields, and reconstruction maps are controlled by weighted sums of absolute values of Taylor coefficients. A bound of this kind controls derivatives on smaller domains (Lemma 5.3), is stable under products and substitutions, and measures locality through an exponential weight in the length of a network connecting the locations of the coefficient. Norms of this type are used throughout constructive renormalization group methods [17, 38, 96]; the coefficient norm with boundary decay (4.11) is specific to this paper. All indices below are scalar indices, which include orientation and color.

Definition 4.6. *Indices and networks.* Let I_σ be finite index sets, each with a location map $x : I_\sigma \to \mathbb{T}_L^3$, a mesh $a_\sigma > 0$, and the volume $v_\sigma = a_\sigma^3$, and write $v_i = v_\sigma$ for $i \in I_\sigma$. The assigned location $x(i)$ of an index is its position on the lattice on which it was introduced. Different indices may have the same location, and index sets introduced at different scales remain different. Distances and network lengths in the coefficient weights are measured in units of the current output lattice spacing a_{ref}, that is, $d(x, y) = d_{\mathrm{phys}}(x, y)/a_{\mathrm{ref}}$, where d_{phys} is the periodic Euclidean distance on $\mathbb{T}_L^3$; all index sets are embedded in this common metric, and dist below denotes d. The boundary graph distance d_Ω and the

boundary widths below are also expressed in output-mesh units. For a finite set S of locations let

$$\mathtt{t}(S) = \inf\{\mathrm{length}(\mathcal{N}) : \mathcal{N} \text{ a finite connected metric network containing } S\}.$$

Lengths are measured with d. Internal vertices are allowed, networks may wrap around the torus, and $\mathtt{t}(\{x\}) = 0$.

Tensor norms. Let $a_{\mathrm{sp}} \geq 0$. For a tensor $T(o; i_1, \dots, i_k)$ with an output index o and input indices $i_1, \dots, i_k$, set

$$\begin{aligned} R_{a_{\mathrm{sp}}}(T) &= \sup_{o} \sum_{i_1,\dots,i_k} e^{a_{\mathrm{sp}}\mathtt{t}(x(o),x(i_1),\dots,x(i_k))} |T(o; i_1, \dots, i_k)|, \\ C_{a_{\mathrm{sp}},r}(T) &= \sup_{i_r} \sum_{o,(i_s)_{s\neq r}} \frac{v_o}{v_{i_r}} e^{a_{\mathrm{sp}}\mathtt{t}(x(o),x(i_1),\dots,x(i_k))} |T(o; i_1, \dots, i_k)|, \qquad 1 \leq r \leq k, \\ |T|_{a_{\mathrm{sp}}} &= \max\{R_{a_{\mathrm{sp}}}(T), C_{a_{\mathrm{sp}},1}(T), \dots, C_{a_{\mathrm{sp}},k}(T)\}. \end{aligned} \tag{4.9}$$

The row norm $R_{a_{\mathrm{sp}}}$ fixes the output and sums over all inputs; the volume-normalized column sum $C_{a_{\mathrm{sp}},r}$ fixes the r-th input and sums over all other indices, including the output. For a kernel $K(x, y)$ from mesh a_{in} to mesh a_{out}, this reads

$$\|K\|_{\mathrm{row},a_{\mathrm{sp}}} = \sup_x \sum_y e^{a_{\mathrm{sp}}\,\mathrm{dist}(x,y)} |K(x,y)|, \qquad \|K\|_{\mathrm{col},a_{\mathrm{sp}}} = \sup_y \frac{a_{\mathrm{out}}^3}{a_{\mathrm{in}}^3} \sum_x e^{a_{\mathrm{sp}}\,\mathrm{dist}(x,y)} |K(x,y)|.$$

For positive input radii r_i we write $T[r](o; \mathbf{i}) = T(o; \mathbf{i}) \prod_s r_{i_s}$, so that every input, including a fixed one, contributes its radius. Normalizations of fields by couplings are written separately.

Majorant series. For an analytic map

$$F_o(u) = \sum_{k\geq 0} \frac{1}{k!} \sum_{i_1,\dots,i_k} F_k(o; i_1, \dots, i_k) u_{i_1} \cdots u_{i_k}, \qquad F_k = D^k F(0),$$

an absolute majorant is a convergent series $\widehat{F}(s) = \sum_k f_k s^k$ with $f_k \geq 0$ and $|F_k|_{a_{\mathrm{sp}}}/k! \leq f_k$. There is exactly one Taylor factorial.

Scalar functions. For a symmetric tensor $A(i_1, \dots, i_k)$ without output, $k \geq 1$, set

$$|A|^{\mathrm{sc}}_{a_{\mathrm{sp}}} = \max_r \sup_{i_r} \sum_{(i_s)_{s\neq r}} e^{a_{\mathrm{sp}}\mathtt{t}(x(i_1),\dots,x(i_k))} |A(i_1, \dots, i_k)|, \tag{4.10}$$

and write $\|D^k R(0)\|_{a_{\mathrm{sp}},\mathrm{root}} = |D^k R(0)|^{\mathrm{sc}}_{a_{\mathrm{sp}}}$. For an analytic scalar function R of the link coordinates on the torus with $DR(0) = 0$, set

$$\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R) = \sum_{k\geq 2} \frac{\rho^{k-2}}{k!} \|D^k R(0)\|_{a_{\mathrm{sp}},\mathrm{root}}.$$

Functions on a block region. Let Ω be a set of scalar labels of the variables that are still varied or integrated in a block region, let W be its boundary as in Section 18.1, and let $0 < \phi_i \le \rho$ be a radius profile. Here $d_\Omega(i)$ is the graph distance, on the refined position graph of Section 18.2, from the position of i to the set of positions outside Ω, as in Definition 18.1; it is expressed in units of the current output mesh. It differs from the periodic Euclidean distance d used for the connecting networks. Sections 18–20 use the graph-metric variant of (4.11) fixed before Definition 18.1, in which network lengths are also measured in the ℓ^1 metric of that graph; a coefficient bound at a Euclidean decay rate is used there at that rate divided by $\sqrt{3}$. Let b_{rec} range over the admissible set of real frozen coordinates of Definition 18.1. For $f(v; b_{\mathrm{rec}}) = \sum_\nu f_\nu(b_{\mathrm{rec}}) v^\nu$, with $v = (v_i)_{i\in\Omega}$ and $\phi^\nu = \prod_i \phi_i^{\nu_i}$, set

$$\mathcal{X}^{\phi}_{\Omega}(f) = \sup_{b_{\mathrm{rec}}\ \mathrm{real}} \rho^{-2} \sum_{k\ge 1} \sup_{i\in\Omega} e^{\alpha_{\mathrm{bd}} d_\Omega(i)} \sum_{|\nu|=k} \frac{\nu_i}{k} |f_\nu(b_{\mathrm{rec}})| e^{a_{\mathrm{sp}}\mathfrak{t}(\nu)} \phi^\nu, \tag{4.11}$$

where $\mathfrak{t}(\nu)$ is the network length of the locations of the indices in the support of ν, and $a_{\mathrm{sp}}, \alpha_{\mathrm{bd}} > 0$. The order of the operations in (4.11), namely the supremum over frozen coordinates, the sum over degrees, the supremum over the fixed label, and the sum over coefficients, is part of the definition. Constants have norm zero, and degree one is included. If Ω is empty, the norm is zero. If Ω has no boundary, (4.11) is applied only to functions with vanishing nonconstant part, whose norm is then zero; effective interactions on the full torus are measured by $\mathcal{B}_{\rho,a_{\mathrm{sp}}}$. The classical version of these norms has its own radius ρ_{cl} and its own profile and rate parameters.

A bound on $|T|_{a_{\mathrm{sp}}}$ controls the row and every volume-normalized column; it is stronger than a bound on the output rows alone or on an operator norm of T contracted with given directions. In (4.10) the sums with one index fixed are counting sums. We write $|A|^{\mathrm{sc},*}_{a_{\mathrm{sp}}}$ for the convention that multiplies each summand with the index i_r fixed by v_*/v_{i_r}, for a reference volume v_*. Then

$$|A|^{\mathrm{sc}}_{a_{\mathrm{sp}}} \le \max_i \frac{v_i}{v_*} |A|^{\mathrm{sc},*}_{a_{\mathrm{sp}}}, \qquad |A|^{\mathrm{sc},*}_{a_{\mathrm{sp}}} \le \max_i \frac{v_*}{v_i} |A|^{\mathrm{sc}}_{a_{\mathrm{sp}}},$$

where the maxima are over the possible fixed indices. On one mesh we choose v_* equal to its volume, and the two conventions agree; on several meshes the conversion is kept explicitly. The assumption $DR(0) = 0$ in the definition of $\mathcal{B}_{\rho,a_{\mathrm{sp}}}$ concerns the effective interactions. A constant is stored separately in Λ and is never bounded by $\mathcal{B}_{\rho,a_{\mathrm{sp}}}$. The precise definitions on block regions, including the frozen coordinates and the labels, are in Definition 18.1.

Remark 4.7. The weight ν_i/k in (4.11) is the sum (4.10) with the index i fixed, for a single monomial. Indeed, let $|\nu| = k$ and $\nu_i > 0$. The derivative $D_i D_{i_2} \cdots D_{i_k} (f_\nu v^\nu)(0)$ is nonzero

for exactly $(k-1)!/[(\nu_i-1)!\prod_{j\neq i}\nu_j!]$ ordered tuples $(i_2,\dots,i_k)$, and each nonzero value has modulus $|f_\nu|\prod_j \nu_j!$. Hence

$$\frac{1}{k!}\sum_{i_2,\dots,i_k}|D_iD_{i_2}\cdots D_{i_k}(f_\nu v^\nu)(0)| = \frac{\nu_i}{k}|f_\nu|.$$

Multiplying by ϕ^ν shows that (4.11) is the profile version of the scalar norm (4.10), with the index i fixed. Replacing ν_i/k by ν_i, or adding a factor $1/k!$ to the multiindex coefficients, would change the norm.

Rescaling an input transfers its locations, its network lengths, and its volume factors explicitly. Replacing a shortest connecting network by a tree on the locations of the external indices reduces the exponent once, from a_{sp} to $a_{\mathrm{sp}}/2$ (Lemma 5.3); there is no loss at every derivative order. Derivatives in analytic series are divided by $k!$. A bound on the values of a holomorphic function, or a pointwise bound on its Hessian, does not replace a bound on its absolute coefficients.

Radius profiles. For a boundary width w_{bd}, the spatially varying analytic radius is

$$\phi(i) = \begin{cases} \rho_\partial, & d_\Omega(i) < w_{\mathrm{bd}}, \\ \rho, & d_\Omega(i) \geq w_{\mathrm{bd}}, \end{cases} \qquad \psi_{\mathrm{prof}} = \sqrt{K_*}\,\phi, \qquad \widehat{\phi} = K_*\phi,$$

where $0 < \rho_\partial \leq \rho$ and $K_* \geq 2$ is a fixed enlargement factor of the analytic radii. The classical profile is defined separately. The symbol ψ without a subscript is reserved for an observable cutoff (Definition 4.11). We keep the following radii distinct.

r	radius of the small-link domain B_r in the small-field event
ρ, ρ_{cl}	analytic radii for the effective interaction and the effective classical action
R_{raw}	larger radius of the initial construction
$R_{\mathrm{buf}} = C_{\mathrm{map},J}r$	radius of the analytic extension around coarse fields
$R_{\mathrm{cell}} = \lceil A_8/r \rceil$	side length of the enlarged cells
R_W	radius imposed on the variables $g_i w_i$
R_{en}	radius of the enclosing product domain
β_{rec}	bound on the frozen coordinates

Here A_8 is the sufficiently large constant of Theorem 21.3. None of these radii is the inverse coupling $\beta_n = 4/h_n$. The width s_{er} of a removed boundary layer, the separation widths, and the numbers of elapsed RG steps are distinct geometric variables. Frozen coordinates from earlier steps are fixed real parameters; they are neither differentiated nor continued to complex values. Newly frozen variables were analytic variables before the substitution.

Normalization constants. Effective densities have the form $\exp(\Lambda - E/g^2 - R)$, multiplied by domain indicators. The centered effective interaction is $R - R(I)$, and in such a density the constant $R(I)$ is moved into Λ. If some variables are fixed at the value c by substitution, we set

$$R_{\mathrm{new}}(v) = R(v,c) - R(0,c), \qquad \Lambda_{\mathrm{new}} = \Lambda - R(0,c),$$

which is the identity $e^{\Lambda - R(v,c)} = e^{\Lambda - R(0,c)} e^{-[R(v,c) - R(0,c)]}$. The classical action is kept in full. A substitution does not remove a first derivative by translating a conditional minimizer, so R_{new} has in general terms of degree one, which (4.11) includes. The full determinant and the normalization factor of every eliminated Gaussian coordinate are included exactly once in the density. Their field-independent factors are collected in Λ; their remaining field dependence is kept in the effective interaction and in the conditional fluctuation potential. This is proved in Lemma 18.9 and Theorem 20.13. A comparison of densities on block regions is first a statement at representatives satisfying the reconstruction identities of Theorem 21.12; only the resulting gauge-invariant bound in terms of the Wilson action is afterwards extended to all representatives (Theorem 22.10).

4.5 Polymers and exponential moments of blocked actions

Polymer expansions. A polymer $\mathfrak{X}$ is a nonempty connected set of the specified block labels. Polymer activities $\mathfrak{z}(\mathfrak{X}; c)$ arise from the finite product expansion displayed at each use. The hard-core polymer partition function is

$$\Xi(c) = \sum_{\{\mathfrak{X}_1, \dots, \mathfrak{X}_m\} \text{ pairwise compatible}} \prod_{i=1}^{m} \mathfrak{z}(\mathfrak{X}_i; c),$$

where the empty family contributes one. Compatibility is determined by the incompatibility relation specified at each expansion (disjointness of supports, or the exclusion relation). The norm of the polymer activities combines absolute coefficient majorants on the given radii, the spatial weight $e^{a_{\mathrm{sp}} \mathfrak{t}}$, and a size weight $e^{\kappa_{\mathrm{pol}} |\mathfrak{X}|}$, and it takes the supremum over sites v of the sum over the polymers containing v, $\sup_v \sum_{\mathfrak{X} \ni v}$. The precise incompatibility relation and size weight are part of the corresponding theorem (Lemma 12.9 and Theorem 12.6). Boundary visits and fixed inputs of a coefficient are kept in the networks through the expansion of the product into polymer activities (Lemma 12.9) and the cluster expansion of $\log \Xi$ (Theorem 12.6), and through their analogues on block regions in Section 20. The support of a polymer activity alone does not give a new decay rate. We use the cluster expansion in the form of Kotecký and Preiss [120], with the tree-graph bound as presented by Fernández and Procacci [87] and by Dimock [68, Appendix B, Theorem 27, p. 39, and (323), p. 42]; see also Brydges [38].

Reflected block tilings. These conventions are used in the bound on large fields in Sections 22 and 23. At averaging step j, a reflected block tiling $\mathcal{T}$ consists of $\mathfrak{m}_j = n_j^3/8$ disjoint coarse cells of $2 \times 2 \times 2$ sites. Each cell contains the sets of finer plaquettes on which the internal coarse plaquettes of the cell depend; the plaquettes crossing between cells are not tested in this tiling. The eight translates of $\mathcal{T}$ cover all plaquettes. We write

$$Q_j^{\mathcal{T}} = \frac{4}{h_{n_{j-1}}} \sum_{p \in \mathcal{P}_{j-1}^{\mathcal{T}}} q(V_p^{j-1})$$

for the blocked action of the finer plaquettes of the cells, and $R_{<j}^{\mathcal{T}}(d_{\mathrm{cut}})$ for the product over the cells of the indicators of the earlier small-field conditions. The threshold for a bad cell is $d_* = d_{\mathrm{cut}}/2$. The indicator stays inside the expectation that is raised to the power $1/\mathfrak{m}_j$ in the chessboard estimate. Reflection positivity of the unconditioned Wilson measure was proved by Osterwalder and Seiler [134]. By Theorem 6.6, it holds for the bounded Borel gauge-invariant functions of the fine links with both endpoints in the union of the blocks over the coarse vertices of one open half torus. For an observable supported in a reflection half, the pullback by the iterated block averaging map belongs to this class and intertwines the reflection, including at the exceptional values. This gives reflection positivity of the averaged measure. We never use reflection positivity for a fixed fiber or for a conditioned measure. The chessboard estimate is that of Fröhlich, Israel, Lieb, and Simon [92, 93], in the form of Biskup [29, Theorem 5.8, p. 44].

Exponential moments. The bound on the exponential moments of blocked actions has the form

$$\log \mathbb{E}_{\mu_{N,\gamma}}\big[e^{s Q_j^{\mathcal{T}}}; R_{<j}^{\mathcal{T}}\big] \le 3 C_v n_{j-1}^3 + C_s n_{j-1}^2 \log(1/g_{n_{j-1}}) + C_b', \qquad 0 \le s \le s_0,$$

with fixed constants C_v, C_s, C_b', s_0. It is proved, together with the induction over averaging steps under the Wilson measure, in Theorem 22.1 and Section 23. If a stopped side is $m = \chi_{\mathrm{stop}} n_{j-1}$, a coefficient 9 of m^2 becomes $9\chi_{\mathrm{stop}}^2$ in units of n_{j-1}^2. At the finest, unconditioned scale the bound contains the correction $36 \log \beta_N / N$, with $\beta_N = 4/h_N$, which is kept until the parameters are chosen uniformly in the coupling (Theorem 6.6).

4.6 Heat semigroup, Wilson flow, and flow data

For a physical mesh a, let $D_{a,i}^{\pm}$ be the componentwise forward and backward differences in direction i, in physical units, and let $\Delta_a = \sum_i D_{a,i}^- D_{a,i}^+$. The lattice heat semigroup is the componentwise scalar semigroup $P_t^a = e^{t\Delta_a}$, which retains the constant mode. The cardinal interpolation of directed-edge fields (Lemma 16.3) preserves constants and integrals of columns, and it is the identity when the input and output meshes coincide; its volume-normalized columns use its own input mesh.

The Wilson flow Φ^{W}_s is the flow of the vector field $-\nabla \sum_p q(U_p)$, where the gradient is taken with respect to the product metric on $\mathcal{U}_n$ given by $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$. Its version in physical time is

$$\Psi^{(N)}_t = \Phi^{\mathrm{W}}_{4t/a_N^2},$$

as in Section 2.1. This conversion of time contains no coupling factor; the factor $4/a_N^2$ is the normalization for which the linear part of the flow is the heat equation in physical units. The flow is Lüscher's [128]. In logarithmic coordinates $U = e^{aA}$, after we add the gauge-damping term of Lemma 24.2 (a lattice version of the gauge fixing of Zwanziger [157], DeTurck [66], and Donaldson [76]), the linear part of the equation is the scalar heat equation $\partial_t A = \Delta_a A$ in each component. We write $\mathcal{N}_a$ for the bilinear coefficient of this gauge-damped lattice equation, in the order in which its arguments appear and with all its lattice shifts (Lemma 24.2). Set

$$\mathcal{Q}_a(u,v)(t) = \int_0^t P^a_{t-s}\mathcal{N}_a(u(s),v(s))\,ds, \qquad \mathcal{B}_a(u,v) = \mathcal{Q}_a(u,v) + \mathcal{Q}_a(v,u).$$

The coefficient $\mathcal{N}_a$ and the bilinear Duhamel term $\mathcal{Q}_a$ need not be symmetric in their two arguments; their symmetrization $\mathcal{B}_a$ is symmetric. The letter Q alone is reserved for the constant-mode quartic form of Section 4.9.

Definition 4.8. Fix $1/2 < \alpha_{\mathrm{fl}} < 2/3$, and set $\beta_{\mathrm{fl}} = \alpha_{\mathrm{fl}} - 1/2$ and $\kappa_{\mathrm{fl}} = 1 - 3\alpha_{\mathrm{fl}}/2 > 0$. For a fixed time horizon $T > 0$ and a mesh a, let $v_a(t) = t + a^2$, and define, with $\alpha = \alpha_{\mathrm{fl}}$,

$$\mathsf{X}^a_\alpha(x) = \sup_{0\le t\le T}\big\{v_a(t)^{\alpha/2}\|x(t)\|_\infty + v_a(t)^{(\alpha+1)/2}\|D_a x(t)\|_\infty\big\},$$
$$\mathsf{Y}^a_\alpha(y) = \sup_{0\le t\le T}\big\{v_a(t)^{\beta_{\mathrm{fl}}}\|y(t)\|_\infty + v_a(t)^{\alpha}\|D_a y(t)\|_\infty\big\}.$$

Here $\|D_a x\|_\infty$ is the maximum over all componentwise differences $D^{\pm}_{a,i}x$ and their shifts occurring in the Wilson-flow equation. For $0 < \sigma_{\mathrm{sm}} < 1 - \alpha_{\mathrm{fl}}$ and $\sigma = \sigma_{\mathrm{sm}}$, set

$$\mathsf{H}^a_\sigma(f) = \sup_{0\le t\le T}\big\{v_a(t)^{\sigma/2}\|P^a_t f\|_\infty + v_a(t)^{(\sigma+1)/2}\|D_a P^a_t f\|_\infty\big\},$$
$$\mathsf{V}^a_\sigma(f) = \sup_{0\le t\le T} v_a(t)^{\sigma/2}\|P^a_t f\|_\infty.$$

For a physical initial potential A_0 on mesh a, the linear heat evolution is $x(t) = P^a_t A_0$, the quadratic Picard term is $y = \mathcal{Q}_a(x,x)$, and together they form the linear and quadratic flow data, measured in the norms X^a_α and Y^a_α.

The weights express heat scaling: $x(t)$ is of size $v_a(t)^{-\alpha/2}$ and $y(t)$ of size $v_a(t)^{1/2-\alpha}$, and each difference costs a further factor $v_a(t)^{-1/2}$. Flow data on different meshes are compared by the cross-mesh comparison (24.11) of Definition 24.3, whose four components are interpolated separately and weighted in time.

Further exponents. The constant C_{CP}, which controls finitely many derivatives of the conditional constrained minimizers and of their propagated contributions, is uniform in J. It is chosen before J, including every enlargement used later in an amplitude bound or a derivative bound, and J is then chosen with $C_{\mathrm{CP}}^2/B \le 1/4$. We write $a_{\mathrm{cap}} = C_{\mathrm{CP}}/B$ for the contraction factor in the amplitude bound (15.20) of the backward reconstruction (Lemma 15.10). The decay exponent at the coarse end of the reconstruction is a different number: $\kappa_{\mathrm{term}} = -\log a_{\mathrm{cap}}/\log B > 1/2$ when $a_{\mathrm{cap}}\sqrt{B} < 1$, and the contraction factor of the propagated contributions of coarser scales is $q_{\mathrm{anc}} = B a_{\mathrm{cap}}^2 < 1$. A localization exponent is denoted by ξ, not by γ. A tail exponent τ_{tail} is chosen with

$$0 < \tau_{\mathrm{tail}} < \min\{\sigma_{\mathrm{sm}}, 1/2, -\log q_{\mathrm{anc}}/\log B\}.$$

Logarithms along the flow. Let $A = x + y + r$ be the solution on $[0,T]$ of the gauge-damped flow equation on mesh a whose linear and quadratic flow data satisfy $\mathsf{X}_\alpha^a(x) + \mathsf{Y}_\alpha^a(y) \le R$, with the remainder r bounded as in Theorem 24.5. Then

$$a \sup_{t\le T}|A(t)| \le C_R a^{1-\alpha_{\mathrm{fl}}} + C_R a^{2-2\alpha_{\mathrm{fl}}} + C_R a (T + a^2)^{1-3\alpha_{\mathrm{fl}}/2} \longrightarrow 0 \qquad (a \to 0),$$

at fixed T. The first two terms come from $a\|x(t)\|_\infty \le R a\, v_a(t)^{-\alpha/2}$ and $a\|y(t)\|_\infty \le R a\, v_a(t)^{1/2-\alpha}$, evaluated at $t = 0$. The last term increases with time; its maximum is $O_R(a)$, attained at $t = T$, and does not keep the mesh power of its value at $t = 0$. The smooth solutions of the continuum flow are constructed from deterministic results for the flow with smooth data and from heat-kernel calculus [42, Theorem 1.2 and Lemma 1.3, p. 3; Lemmas C.2–C.3, pp. 55–56; Corollary C.6, p. 58; Lemmas C.12–C.13, p. 61]. We do not use the theorem of Cao and Chatterjee on random initial data as a hypothesis on Wilson fields.

4.7 The heat-smoothed circulation

The short-distance results are proved for the following linear functional of the reconstructed field.

Definition 4.9. Fix $L > 0$, $\gamma > 0$, $g_3^2 = \gamma/L$, $\sigma > 0$, smooth loops $\ell_1, \dots, \ell_m$ in $\mathbb{R}^3$, and the index s of the comparison lattice, and let $K \ge s$ and $N_K = n_* B^K$. Let $\widehat{\theta}_{K-s}$ be the link logarithm, in the gauge fixed by the reconstruction maps $\widehat{\mathscr{R}}$, of the configuration on Λ_{N_K} reconstructed from the lattice of side n_s by $K - s$ RG steps, and let $A^{K,s} = a_{N_K}^{-1}\widehat{\theta}_{K-s}$ be the corresponding physical initial potential. For $a_{N_K} \le \varepsilon$ and $x \in \mathbb{T}_L^3$, let $\ell_{i,K,\varepsilon}$ be the closed lattice path in Λ_{N_K} approximating $x + \varepsilon\ell_i$ that is specified in Definition 31.1. The heat-smoothed circulation is

$$I_i^{K,s}(\varepsilon, x) = \frac{a_{N_K}}{\sqrt{g_3^2\varepsilon}} \sum_{e \in \ell_{i,K,\varepsilon}} \mathrm{sgn}(e)\big(P_{\sigma\varepsilon^2}^{a_{N_K}} A^{K,s}\big)_{\mathrm{dir}(e)}\big(x_{\mathrm{positive}}(e)\big).$$

Here $\mathrm{dir}(e)$ is the direction of the link e, $x_{\mathrm{positive}}(e)$ is the starting point of its positive orientation, and $\mathrm{sgn}(e) = 1$ if e is traversed in the positive direction and -1 otherwise; a backward edge thus contributes the negative of the value on its positive edge.

In this range, the paths $\ell_{i,K,\varepsilon}$ have physical length and diameter at most $C_\ell \varepsilon$, and they have zero winding for $\varepsilon \le \varepsilon_{\mathrm{loop}}$ of Definition 31.1. The functional annihilates periodic lattice gradients. Indeed, if $A_j(y) = D^+_{a,j} f(y)$ with $a = a_{N_K}$, then $P^a_t A = D^+_a P^a_t f$. Each traversed edge contributes $\pm(P^a_t f(\mathrm{end}) - P^a_t f(\mathrm{start}))/a$, with the sign making it the increment along the path, and these increments sum to zero around a closed path. The zero winding also removes the pairing with constant one-forms. The scalar heat semigroup acts on the field at the finest mesh. It is not the nonlinear Wilson flow, and the comparison with flowed Wilson loops is a separate result (Lemma 32.11). The normalization $1/\sqrt{g_3^2 \varepsilon}$ is the scale of the circulation of the Gaussian free field after heat smoothing at time $\sigma\varepsilon^2$; the analysis is in Sections 31 and 32.

4.8 Gauge orbits and the trajectory space

Definition 4.10. A smooth connection is an element A of $\mathcal{A}_L = C^\infty(\mathbb{T}^3_L; \mathfrak{su}(2)^3)$. We write

$$F_{ij}(A) = \partial_i A_j - \partial_j A_i + [A_i, A_j], \qquad D_i = \partial_i + [A_i, \cdot\,].$$

A gauge transformation $k \in \mathcal{G}_L = C^\infty(\mathbb{T}^3_L; \mathrm{SU}(2))$ acts by $A^k = kAk^{-1} - (dk)k^{-1}$. The parallel transport along a piecewise C^1 path ℓ is the solution of $H'(s) = H(s)A(\ell(s))\dot{\ell}(s)$, $H(0) = I$; it transforms by $H^k(s) = k(\ell(0))H(s)k(\ell(s))^{-1}$. The Yang–Mills heat flow is $\partial_t A_i = \sum_j D_j F_{ji}$, and $(\Phi_t)_{t \ge 0}$ is the semigroup it induces on $\mathcal{A}_L/\mathcal{G}_L$. The trajectory space $\mathcal{X}_L$, the regularized Wilson loops $W_{\ell,\chi,t}$, and the Wilson topology $\mathcal{T}_W$ are as in Definition 2.2. Let $(\ell_j, t_j)_{j \ge 1}$ be the fixed enumeration used in Theorem 29.3, formed from its countable dense family of piecewise C^1 loops, which contains loops in every winding class, and the times in $L^2\mathbb{Q}_{>0}$. Set $w_j(X) = \mathrm{tr}\,\mathrm{Hol}_{\ell_j} X(t_j)/2$ and

$$d_c(X, Y) = \sum_{j \ge 1} 2^{-j} |w_j(X) - w_j(Y)|, \qquad X, Y \in \mathcal{X}_L.$$

Differentiating $k(\ell(0))H(s)k(\ell(s))^{-1}$ shows that it solves the transport equation for A^k, which confirms the sign of the gauge action and the order of the transport. Our gauge transformation k is the inverse of the gauge variable used by Cao and Chatterjee [43, (1.1), p. 2]. The trajectory space follows the proposal of Charalambous and Gross [47] and is a version of the state space of Cao and Chatterjee [43]. By definition d_c is a pseudometric. In Theorem 29.3 we prove that it separates points of $\mathcal{X}_L$, that both topologies are separable, and that they have the same Borel σ-algebra. We do not assert that $(\mathcal{X}_L, d_c)$ is complete.

The Yang–Mills energy is

$$E_L(A) = \sum_{i,j} \int_{\mathbb{T}^3_L} -\operatorname{tr}\big(F_{ij}(A)^2\big)\, dx = \sum_{i<j} \int_{\mathbb{T}^3_L} |F_{ij}(A)|^2\, dx,$$

since $-\operatorname{tr}(F_{ij}^2) = |F_{ij}|^2/2$. It carries no coupling factor; for smooth configurations the exponent of the Wilson measure tends to $E_L/(2g_3^2)$ (Remark 4.2). Under the rescaling to the unit torus, $\widetilde{A}(y) = LA(Ly)$, the flow time becomes t/L^2 and $E_1(\widetilde{A}) = LE_L(A)$.

The maps π_K of Theorem 29.19 are Borel maps on the lattice configuration space, invariant under all periodic lattice gauge transformations. They are chosen by a deterministic diagonal sequence of matching levels, fixed before any configuration is sampled. Their Wilson-loop coordinates approximate the flowed lattice Wilson loops in the sense of Theorem 2.3(3); the maps do not embed lattice holonomies exactly. We use the compactness of energy sublevel sets and the continuity of the flow semigroup under the hypotheses of the cited results, which concern the quotient of Sobolev connections in H^1 by gauge transformations in $W^{2,2}$ [43, Theorem 3.51, pp. 35–36; Lemma 3.68, p. 40].

4.9 Observables near commuting holonomies

Definition 4.11. Fix $\theta > 0$ and the flow time $t = \theta L^2$. Let $H_1, H_2, H_3 \in \mathrm{SU}(2)$ be the holonomies of $X(t)$ around the three coordinate cycles, based at a common point. Their seven trace coordinates are the normalized traces of H_i, of H_iH_j for $i < j$, and of $H_1H_2H_3$. In a fixed neighborhood of the identity write $X = (x_i)_{i=1}^3 = (\operatorname{Log} H_i)_{i=1}^3 \in \mathfrak{su}(2)^3 \simeq \mathbb{R}^9$. The cone of commuting triples is

$$\mathcal{M} = \{(y_i) = (\vartheta_i u)_{i=1}^3 : \boldsymbol{\vartheta} \in \mathbb{R}^3,\ u \in S^2\}, \qquad (\boldsymbol{\vartheta}, u) \sim (-\boldsymbol{\vartheta}, -u).$$

Fix $0 < r_0 < r_1 < \pi$, and let $\mathsf{P} = \{(\vartheta_i u)_i \in \mathcal{M} : r_0 \le |\boldsymbol{\vartheta}| \le r_1\}$, a compact subset of the regular part of $\mathcal{M}$ with nonempty relative interior. For $y = (\vartheta_i u)_i \in \mathsf{P}$, the normal space to $\mathcal{M}$ at y is

$$\mathcal{N}_y = \Big\{ b = (b_1, b_2, b_3) : b_i \perp u,\ \sum_i \vartheta_i b_i = 0 \Big\}, \qquad \dim \mathcal{N}_y = 4,$$

and on a fixed tubular neighborhood of P we use the unique coordinates $X = y + b$ with $y \in \mathcal{M}$ and $b \in \mathcal{N}_y$. Let $\Pi_{\boldsymbol{\vartheta}} = I - \boldsymbol{\vartheta}\boldsymbol{\vartheta}^{\mathsf{T}}/|\boldsymbol{\vartheta}|^2$. In unit-torus variables put, for $m \in \mathbb{Z}^3$,

$$k_m = 2\pi m + \boldsymbol{\vartheta}, \qquad \lambda_m = |k_m|^2, \qquad R_i(m, \boldsymbol{\vartheta}) = \begin{cases} 1, & m_i = 0, \\ \vartheta_i/(2\pi m_i + \vartheta_i), & m_i \neq 0, \end{cases}$$

$$d_i(\boldsymbol{\vartheta}, \theta) = \sum_{m \in \mathbb{Z}^3} \frac{e^{-2\theta\lambda_m}}{\lambda_m} R_i(m, \boldsymbol{\vartheta})^2, \qquad D = \operatorname{diag}(d_1, d_2, d_3),$$

and define the normal covariance

$$\Sigma_\theta(y) = \big(\Pi_{\boldsymbol{\vartheta}} D\Pi_{\boldsymbol{\vartheta}}\big)\big|_{\boldsymbol{\vartheta}^\perp} \otimes I_{u^\perp}. \tag{4.12}$$

Let $0 \le \psi_0 \le 1$ be a smooth function on P, invariant under simultaneous conjugation and compactly supported in the relative interior of P, that vanishes on a nonempty open set and is bounded below by a positive constant on another nonempty open set. Multiply it by a fixed smooth function χ of the normal variable, invariant under simultaneous conjugation, supported in the tube, with $0 \le \chi \le 1$ and $\chi = 1$ near $b = 0$ (for example a function of $|b|$), to obtain $\psi(y + b) = \psi_0(y)\chi(b)$. Fix a larger tube containing the support of ψ and the supports of all tangential insertions used later, and set

$$q_*(y + b) = \frac{1}{2}\, b^{\mathsf{T}} \Sigma_\theta(y)^{-1} b$$

throughout this larger tube, with a fixed bounded, continuous, nonnegative extension outside it that is invariant under simultaneous conjugation. All these functions are invariant under simultaneous conjugation, and they are extended by zero outside the identity chart, so that they define invariant functions of (H_1, H_2, H_3). The observables $B_{\mathrm{obs}} = \psi$ and $A_{\mathrm{obs}} = \psi q_*$ are fixed bounded functions, and the normalized regression residual variance $\mathscr{C}_\gamma$ is defined from them as in Section 2.7.

The tangent vectors to $\mathcal{M}$ at a regular point are $(\delta\vartheta_i u + \vartheta_i \delta u)_i$ with $\delta u \perp u$; their orthogonal complement is $\mathcal{N}_y$. Hence the regular part of the cone has dimension five and its normal space has dimension four. By the tubular neighborhood theorem, the coordinates $X = y + b$ near P are unique; the tip of the cone is excluded. The matrix $\Sigma_\theta(y)$ is the covariance, divided by γ, of the linearized response of the flowed logarithms at the flat connection y. It is derived and compared with the lattice in Theorem 33.11; its definition does not assert that the nonlinear law of b is Gaussian. The regression coefficient $\mathrm{Cov}(A_{\mathrm{obs}}, B_{\mathrm{obs}})/\mathrm{Var}(B_{\mathrm{obs}})$ in $\mathscr{C}_\gamma$ depends on the measure, and $\mathscr{C}_\gamma$ is defined for measures under which $\mathrm{Var}(B_{\mathrm{obs}}) > 0$ and $\mathbb{E}B_{\mathrm{obs}}^2 > 0$. We prescribe q_* on the larger tube, and not only on the support of ψ, so that q_* equals the prescribed quadratic form on the support of every insertion of Theorem 2.10(2).

Remark 4.12. The series defining d_i converges uniformly on P, and $\Sigma_\theta(y)$ is positive definite. Indeed, $|\vartheta_i| \le |\boldsymbol{\vartheta}| < \pi$, so $|k_m| \ge 2\pi|m| - \pi \ge \pi|m|$ for $m \ne 0$, while $k_0 = \boldsymbol{\vartheta} \ne 0$. For $m_i \ne 0$ we have $|2\pi m_i + \vartheta_i| > \pi > |\vartheta_i|$, so $|R_i| \le 1$. Since $\theta > 0$, the sum of $e^{-2\theta\lambda_m}/\lambda_m$ over $m \ne 0$ converges uniformly. The summand with $m = 0$ gives $d_i \ge e^{-2\theta|\boldsymbol{\vartheta}|^2}|\boldsymbol{\vartheta}|^{-2} > 0$ for each i, hence $\Pi_{\boldsymbol{\vartheta}} D\Pi_{\boldsymbol{\vartheta}} \ge e^{-2\theta|\boldsymbol{\vartheta}|^2}|\boldsymbol{\vartheta}|^{-2}$ on $\boldsymbol{\vartheta}^\perp$. The hypothesis $\theta > 0$ is needed: at $\theta = 0$, the summands with $m_i = 0$ have $R_i = 1$, and the sum of $1/|2\pi m + \boldsymbol{\vartheta}|^2$ over this plane diverges logarithmically.

Central triples. The central triples are the triples $(\pm I, \pm I, \pm I)$ of central elements; they are points of the moduli space of flat connections. Near them we set $r_\gamma = \gamma^{1/4}$ and define the constant-mode quartic form

$$Q(Y) = \frac{1}{2}\sum_{i<j}|y_i \times y_j|^2, \qquad Y = (y_1, y_2, y_3) \in \mathbb{R}^9.$$

The quartic reference measure $m_Q = e^{-Q(Y)}\,dY$ is locally finite and has infinite total mass; it is never normalized globally. The divergence of the corresponding reduced SU(2) integral is known from the matrix-integral literature [7, 122]; we give a proof in our coordinates in Sections 33 and 36. Test functions are compactly supported and invariant under the simultaneous adjoint rotation of the three vectors y_i. A radial cutoff is a function $\chi(|Y|)$ with χ smooth, nonnegative, not identically zero, compactly supported, and constant near zero; the comparison of the dilation statistic with the reference measures (Theorem 36.19) additionally requires $\chi(0) > 0$. Every observable expressed in logarithmic coordinates vanishes outside a fixed identity chart. We write $\rho_{\mathrm{rad}} = |X|/r_\gamma$ for the rescaled radial variable, to distinguish it from the analytic radius ρ.

Definition 4.13. The following comparison measures are distinct constructions. Each depends on the fixed choices stated in it, and the results about them hold for every such fixed choice.

1. *Abelian Maxwell measure with Haar cycle holonomies.* Embed a fixed $U(1)$ subgroup in SU(2). Take the transverse Maxwell Gaussian field of the nonconstant modes with values in this subgroup, with a fixed normalization of its covariance, and independent uniformly distributed constant phases in $(\mathbb{R}/4\pi\mathbb{Z})^3$. Smooth at the fixed observation time and take the Abelian holonomies around the three coordinate cycles. All these triples commute. A common random conjugation does not change invariant test functions.
2. *Gaussian holonomies with independent Haar factors.* Let a be a centered Gaussian field with three colors and covariance γC, for a fixed covariance C, and let S_t be a fixed smoothing. Let U_1^0, U_2^0, U_3^0 be independent Haar distributed elements of SU(2), independent of a, and set $H_i = U_i^0\,\mathrm{Hol}_i(\mathsf{S}_t a)$. The triple (H_1, H_2, H_3) has the product Haar law at every coupling.
3. *Gaussian holonomies with Gaussian constant modes.* Set $H_i = \mathrm{Hol}_i(\mathsf{S}_t a)$, without the Haar factors, where a includes Gaussian constant modes with covariance proportional to γ. Assume that the scalar color projections $(x, i, v) \mapsto \langle v, (\mathsf{S}_t a)_i(x)\rangle/\sqrt{\gamma}$, indexed by $\mathbb{T}_L^3 \times \{1,2,3\} \times S^2$, have a centered separable Gaussian realization with continuous paths and bounded variance, whose law does not depend on γ. Its supremum is then measurable and almost surely finite. The resulting Gaussian supremum estimate is Lemma 35.16, and it is used only for this measure.

4. *Conditional normal Gaussian measure.* On the patch P of Definition 4.11, choose a fixed smooth positive probability density for y that charges two nonempty open sets O_0 and O_+, invariant under simultaneous conjugation, with $\psi_0 = 0$ on O_0 and ψ_0 bounded below by a positive constant on O_+, as in Definition 35.1. Conditionally on y, let $b \sim N(0, \gamma\Sigma_\theta(y))$ on $\mathcal{N}_y$, and push the joint law forward by $X = y + b$. A fixed tube cutoff equal to one near the zero section, followed by normalization, changes bounded moments by exponentially small amounts. For these choices $\mathscr{C}_\gamma$ is defined for all sufficiently small γ and has leading term $2\gamma^2$ (Lemma 35.17).
5. *Mixture of normalized conditional normal Gaussian measures.* Let y have a fixed probability density $\alpha(y)\, d\mathrm{vol}_{\mathcal{M}}(y)$ on a small neighborhood of the tip of $\mathcal{M}$, with $\alpha \geq 0$ continuous and strictly positive at the tip. Given $y \neq 0$, let $b \sim N(0, \gamma\Sigma_\theta(y))$ be the normalized Gaussian measure on $\mathcal{N}_y$, and push the joint law forward by $X = y + b$. Assume that $|y|^2\Sigma_\theta(y) \to I_{\mathcal{N}_y}$ as $y \to 0$, uniformly in the direction of y, with the corresponding uniform upper and lower bounds on $|y|^2\Sigma_\theta(y)$; here $|y| = |\boldsymbol{\vartheta}|$. The associated weighted normal-bundle measure is

$$d\mu_{13} \;\propto\; |y|^4 e^{-|y|^2|b|^2/2}\, d\mathrm{vol}_{\mathcal{M}}(y)\, db.$$

Expectations of the form $\langle\cdot\rangle_{\chi,\mu_{13}}$ are normalized by $\int \chi(|y+b|)\, d\mu_{13}$.

The factor $|y|^4$ in μ_{13} comes from the normalization of the four-dimensional Gaussian measure: $\det(\gamma\Sigma_\theta(y))^{-1/2}$ is of order $\gamma^{-2}|y|^4$, since $|y|^2\Sigma_\theta(y) \to I$. Under the scaling $(y, b) = \gamma^{1/4}(Y, B)$, the tangential volume, the normal volume, and this normalization contribute the factor $\gamma^{5/4}$. After division by $\gamma^{5/4}$, the joint densities converge for $Y \neq 0$ to $(2\pi)^{-2}\alpha(0)|Y|^4 e^{-|Y|^2|B|^2/2}$ relative to $d\mathrm{vol}_{\mathcal{M}}(Y)\, dB$. Since $y \perp b$, we have $|y+b|^2 = |y|^2 + |b|^2$, and the uniform covariance bounds give dominated convergence on compact sets of outputs $Y + B$; this is the scaling limit of Theorem 36.19. The weighted volume element has dilation degree $5 + 4 + 4 = 13$, which explains the index of μ_{13}. The mixture differs from the quartic reference measure m_Q, and omitting the conditional normalization would change the comparison. The value of α at $y = 0$ is irrelevant, since $\{y = 0\}$ is a null set.

4.10 Dependence of constants

Each quantitative statement in this paper either lists the parameters on which its constants depend, or refers to one of the items of the following definition, with any additional fixed parameters listed in the statement. In quantifier form, all fixed data precede the choice of the constants, and every quantity said to vary follows that choice.

Definition 4.14. A constant has the dependence described in item (k) if the corresponding statement holds as follows.

(1) *Finite-dimensional estimates.* Algebraic identities hold in every finite dimension stated. Constants in analytic estimates for products of a bounded number of group elements depend only on the number of factors and on the distance to the boundary of the logarithmic chart. Estimates for measures on products of convex balls depend only on the stated bounds on the Hessian of the potential in the free coordinates and on the weighted interaction, and not on the dimension or on the radii of the individual balls.

(2) *Classical estimates on the torus.* Constants depend on M, on the fixed larger physical and analytic domains, on the spatial decay rates, and on fixed upper bounds for the norms involved. Bounds stated uniformly in depth are independent of the lattice side, of the preceding classical depth, and of the coarse field in the stated domain. The explicitly named constants controlling finitely many derivatives, and the Taylor coefficient majorant Z of the aligned zero-fluctuation minimizer in Lemma 11.9, together with its radius, its coefficient constant, and its spatial rate, are uniform in J: they are fixed before J is chosen. Joint coefficient bounds in the coarse field and the free coordinates, and coefficient bounds for the gauge transformations of the reconstruction maps, are bounds at fixed J; for the reconstruction maps, only the specified finitely many derivatives of the zero-fluctuation reconstruction are uniform in J.

(3) *Local estimates on block regions.* Each constant C_J, c_J, λ_J, local radius, or upper bound for an output norm depends only on M and J, on the fixed larger initial and analytic domains and their radius ratios, on the spatial and boundary decay rates and the stated margins, on a bound on the number of indices per lattice cell, and on fixed upper bounds for the norms on the torus and of the perturbations. It is independent of L, γ, m, N, K, d_0, of the number of elapsed RG steps, of the shape and topology of the block region, of the numbers of connected components of the block region and of the fixed complement, and of the values of the real frozen coordinates. Where a theorem says so, it is also independent of the later boundary widths, widths of removed boundary layers, separation widths, and enlarged-cell widths; such a constant is evaluated on the larger domain before these widths are chosen. This applies only to the theorems on block regions proved in this paper; it does not extend an estimate on the full torus to arbitrary regions.

(4) *Estimates for the small-field failure probability.* Constants may depend on L, on the coupling ceiling γ_0 and the parameter tuple Θ of Theorem 23.2, on M and J, and on the fixed allowed number of averaging steps and enlargement of the stopping scale, but not on the coupling γ, on the lattice side, on the number of RG steps on the fine side or the preceding depth, or on the age, geometry, or widths of a large-field decomposition. The constant $C_{u,q}$ may in addition depend on fixed u and q. In a statement uniform in $0 < \gamma \leq \Gamma$, these constants and the minimal admissible

side may in addition depend on the fixed ceiling $\Gamma < \infty$; they are independent of $\gamma \in (0, \Gamma]$. This does not change the dependence of the constants in items (3) and (7).

(5) *Short-distance estimates.* Constants may depend on fixed L, γ, M, J, on the family of loops, on σ, and on the fixed index s of the comparison lattice. They are uniform in the remaining number of RG steps on the fine side, in the point x, and in the allowed lattice offsets, in the range $a \le \varepsilon \le \varepsilon_0(s)$. No uniformity as $s \to \infty$ or as $\gamma \downarrow 0$ is asserted.

(6) *Weak-coupling estimates.* Constants may depend on fixed n, L, θ, M, J, on the compact subsets of the regular part of $\mathcal{M}$, on the test functions, and on the larger margins, all fixed before γ. They are independent of the number of RG steps on the fine side, of the preceding depth, and of the coupling below the ceiling. Constants C_m attached to a fixed number m of RG steps may grow with m; constants in tail bounds at the coupling-dependent scale may not. These are estimates on the full torus.

(7) *Flow estimates.* Deterministic estimates for the flow on a set where the linear and quadratic flow data are bounded by R may depend on R, on positive waiting times, on the time horizon, and on the margins within the chosen coordinate charts. Constants in stochastic estimates and in convergence statements at fixed coupling, and the deterministic diagonal sequence of matching levels, may also depend on that coupling, on the coarse side, and on the observables. No rate that is uniform as a restart time tends to zero, or as the coupling varies, is implied.

We claim no useful bound as $J \to \infty$ for the joint constants at fixed J or for the constants on block regions; this concerns only these fixed-J constants, and not the constants in item (2) that are stated to be uniform in J. The order in which the parameters are chosen is established in Theorem 23.2: first the constants uniform in J, then J, then the local constants at fixed J, and then the geometric widths. In particular, J is not enlarged after the constants on block regions at fixed J have been computed.

Probabilistic conventions. For a random vector in a finite-dimensional space, the bounded-Lipschitz distance uses the test functions bounded by one with Lipschitz constant at most one with respect to the Euclidean norm. For probability measures we use $\|\mu - \nu\|_{\mathrm{TV}} = \sup_A |\mu(A) - \nu(A)|$, so that $|\mathbb{E}_\mu f - \mathbb{E}_\nu f| \le 2\|f\|_\infty \|\mu - \nu\|_{\mathrm{TV}}$ for every bounded measurable f. The covariance $\mathrm{Cov}(F, G) = \mathbb{E}(FG) - \mathbb{E}F\,\mathbb{E}G$ is complex bilinear when complex sources are used. Whenever a real minimum or a positive density is asserted, the holomorphic continuations involved take real values at real arguments, that is, $f(\bar{z}) = \overline{f(z)}$ in the chosen coordinates.

PART II

Lattice gauge measures and block averaging

SECTION 5

Finite-dimensional integration and concentration inequalities

This section collects the finite-dimensional tools used throughout the renormalization group analysis. Each statement concerns a fixed finite lattice, a finite set of coordinates, or a fixed finite-dimensional measure; where we say so, the constants do not depend on the dimension. The results fall into three groups. The first group concerns gauge integration and Gaussian algebra. Lemma 5.1 records the normalization of Haar measure on $\mathrm{SU}(2)$, exact integration in a spanning-tree gauge, and analytic estimates for products of group elements near the identity, and Lemma 5.2 decomposes closed periodic one-cochains and eliminates positive definite quadratic blocks. The second group is the calculus of the analytic coefficient norms of Definition 4.6: compositions, localized inverses and square roots, logarithms of determinants, and implicit functions (Lemma 5.3). The third group is probabilistic. Theorem 5.4 proves logarithmic Sobolev, covariance, and quadratic concentration inequalities for uniformly log-concave measures on products of balls, Lemma 5.5 controls conditioning, parameter derivatives, and gauge-invariance identities, and Lemma 5.6 shows that seven traces determine a triple of $\mathrm{SU}(2)$ matrices up to simultaneous conjugation.

The only inputs are the definitions of Section 4: the Wilson measure and the normalization of the Lie algebra (Definition 4.1), the coefficient norms (Definition 4.6), and the convention on constants (Definition 4.14). Spanning-tree gauges and the estimates for products of group elements enter the construction of the block averaging map and the gauge fixing in Sections 6 and 7, and the local Wilson flow in Section 24. The Gaussian elimination is used for the linearized theory in Section 8. The coefficient calculus underlies all analytic estimates in Sections 6, 8 to 10, 12, and 18. The concentration theorem is the probabilistic input for the conditional fluctuation kernels in Sections 13, 16, 17, 20, 26 to 28, and 34. The conditioning lemma is used in Sections 15, 28, 29, and 32, and the seven traces in Sections 33, 35, and 36.

5.1 Haar measure, spanning-tree gauges, and products of group elements

The following facts are standard for the compact group $\mathrm{SU}(2)$ and for lattice gauge theory; we record them with the normalizations of Section 4.1, which fix the constants

used later. Gauge fixing along a spanning tree is the maximal-tree gauge, which we call the spanning-tree gauge. Bałaban used such gauges systematically in the renormalization group context [14].

Lemma 5.1. *Let $\mathcal{K}$ be a finite oriented cubical complex, with vertex set $\mathcal{V}$, set of positive links $\mathcal{E}$, and set of positive plaquettes $\mathcal{P}$. Then the following statements hold.*

1. *For every $\beta \geq 0$,*

$$Z_{\mathcal{K}}(\beta) = \int e^{-\beta \sum_{p \in \mathcal{P}} q(U_p)} \prod_{e \in \mathcal{E}} dH(U_e) \quad \text{satisfies} \quad e^{-2\beta|\mathcal{P}|} \leq Z_{\mathcal{K}}(\beta) \leq 1,$$

   and the normalized density defines a gauge-invariant Borel probability measure.

2. *The measure H is the Haar probability measure, and for $X = x^a \tau_a$ with $0 \leq r = |X| < 2\pi$,*

$$\frac{1}{2} \operatorname{tr} e^X = \cos(r/2), \qquad dH(e^X) = c_{\mathrm{Haar}} j_H(X)\, d^3x,$$

   where $c_{\mathrm{Haar}} = 1/(16\pi^2)$ and $j_H(X) = \operatorname{sinc}^2(r/2)$. For $0 \leq u \leq 2\pi$ and $U, V \in \mathrm{SU}(2)$,

$$H\{\rho_G < u\} = \frac{u - \sin u}{2\pi}, \qquad \frac{\rho_G(U)^2}{2\pi^2} \leq q(U) \leq \frac{\rho_G(U)^2}{8},$$

   and $\rho_G(UV) \leq \rho_G(U) + \rho_G(V)$. Further, $q(e^X) \geq (r^2/8)(1 - r^2/48)$ for $0 \leq r \leq 2\pi$, and on every ball $\{|X| \leq r_0\}$ with $r_0 < 2\pi$ we have $q(e^X) = r^2/8 + O(r^4)$ and $\log j_H(X) = -r^2/12 + O(r^4)$, with constants depending only on r_0. For $0 < u \leq 1$,

$$c_{\mathrm{ball}} u^3 \leq H\{\rho_G < u\} \leq \frac{u^3}{12\pi}, \qquad c_{\mathrm{ball}} = \frac{\operatorname{sinc}^2(1/2)}{12\pi} \geq \frac{1}{16\pi}.$$

3. *Left and right translations preserve Haar measure. For every $k \in \mathrm{SU}(2)$, $\mathrm{Ad}(k) \in \mathrm{SO}(3)$; it preserves the norm of $\mathfrak{su}(2)$, Lebesgue measure, and the balls centered at the origin. The map $\mathrm{Ad} : \mathrm{SU}(2) \to \mathrm{SO}(3)$ is surjective. These statements concern $\mathrm{SU}(2)$, not conjugation by elements of its complexification $\mathrm{SL}(2, \mathbb{C})$.*

4. *Choose a spanning tree in each connected component of $\mathcal{K}$, with one root per component. Let h_x be the transport from the root to x along the tree, and set $V_{xy} = h_x U_{xy} h_y^{-1}$. The map*

$$U \longmapsto \big((h_x)_{x \text{ not a root}}, (V_e)_{e \text{ not in the forest}}\big)$$

   is a smooth bijection, and it carries product Haar measure to product Haar measure; $V_e = I$ on the forest. Hence, for an integrand invariant under the gauge transformations equal to I at the tree roots, integration over the tree variables contributes a factor one and leaves the same integrand expressed in the remaining variables V. If $\mathcal{K}$ is connected, there remain $|\mathcal{E}| - |\mathcal{V}| + 1$ independent group variables, together with the simultaneous conjugation at the root, and no stabilizer volume is divided out. The same statements hold for a partition of $\mathcal{K}$ into blocks, each equipped with a rooted spanning tree, with gauge transformations equal to I at every tree root.

5. *Fix a length ℓ, a distance from the singularity of the logarithm (the eigenvalue -1), and a smaller complex coordinate polydisc inside the larger domain. Every map given by a product of at most ℓ factors of the form $e^{\pm X_j}$, and every logarithm of such a product relative to a reference product on the specified chart, has absolute Taylor derivatives of order s bounded by $s!C^s$ for all $s \geq 1$, where C depends only on ℓ, these domains, and the norm of $\mathfrak{su}(2)$. Factors in $\mathrm{SU}(2)$ may be included if the relative products keep the same distance from the singularity. For small complex $u, v, a \in \mathfrak{su}(2)_{\mathbb{C}}$ and $0 \leq \epsilon \leq 1$,*

$$\epsilon^{-1} \operatorname{Log}(e^u e^{\epsilon a} e^{-u-\epsilon v}) = \operatorname{Ad}(e^u)a - \int_0^1 \operatorname{Ad}(e^{tu})v \, dt + \epsilon \mathscr{R}_\epsilon(u, v, a),$$

where the left side at $\epsilon = 0$ is defined by analytic continuation, and the remainder $\mathscr{R}_\epsilon$ is holomorphic, has bounds of the same type, vanishes to order at least two in (a, v), and satisfies $\mathscr{R}_\epsilon(u, 0, a) = 0$.

All constants have the dependence described in item (1) of Definition 4.14; for a fixed length ℓ they do not depend on the number of cells, on L, on the coupling, or on the depth.

Proof. Quaternions and Haar measure. Let $t_a = 2\tau_a$. Multiplying the Pauli matrices, we find

$$t_a t_b = -\delta_{ab} I + \epsilon_{abc} t_c, \qquad (aI + v \cdot t)(bI + w \cdot t) = (ab - v \cdot w)I + (aw + bv + v \times w) \cdot t. \tag{5.1}$$

Then $\mathrm{SU}(2)$ is the unit sphere $\{a^2 + |v|^2 = 1\}$ in $\mathbb{R}^4$, and inversion is $(a, v) \mapsto (a, -v)$. Left and right multiplications by a unit quaternion are linear maps of $\mathbb{R}^4$ preserving the norm, so the normalized surface measure on the sphere is left and right invariant. It is the unique invariant probability measure. Indeed, if H_1, H_2 are two such measures, integrating a continuous function $f(uv)$ against $H_1(du)H_2(dv)$ in the two orders gives $H_1(f) = H_2(f)$. Equality on continuous functions identifies the Borel measures, since continuous approximations of the indicators of closed and open sets give equality on these sets, and hence on the σ-algebra they generate.

Since $(x \cdot \tau)^2 = -r^2 I/4$, we have $e^{x \cdot \tau} = \cos(r/2)I + \sin(r/2)(x/r) \cdot t$, with the value at $r = 0$ obtained by continuity; in particular $\operatorname{tr} e^X/2 = \cos(r/2)$. In the polar coordinates $a = \cos\vartheta$, $v = \sin\vartheta\,\omega$, $0 < \vartheta < \pi$, the area element of the sphere is $\sin^2\vartheta \, d\vartheta \, d\omega$, where ω ranges over the unit two-sphere of area 4π; the total area is $2\pi^2$. Substituting $r = 2\vartheta$ and $d^3x = r^2 dr \, d\omega$, we obtain

$$\frac{\sin^2(r/2)}{4\pi^2 r^2} \, d^3x = \frac{1}{16\pi^2} \operatorname{sinc}^2(r/2) \, d^3x.$$

The omitted point $-I$ has measure zero. The radial density is $\pi^{-1}\sin^2(r/2)\,dr$, and integrating it from 0 to u, we obtain the formula for $H\{\rho_G < u\}$, including $u = 0$ and $u = 2\pi$.

The distance. The quantity r is twice the round-sphere distance from I. By the exponential formula, every logarithm of $U = aI + v \cdot t$ has norm at least $2 \arccos a$, and

the logarithm displayed above attains this value. Great-circle arcs are shortest: for a curve $z(s)$ on the sphere, the derivative of its polar angle has absolute value at most $|z'(s)|$, as one sees by differentiating its scalar product with the initial point. Left multiplication preserves the round metric. Concatenating a shortest arc from I to U with the left translate by U of a shortest arc from I to V, we obtain $\rho_G(UV) \le \rho_G(U) + \rho_G(V)$.

The plaquette function. For $r \in [0, 2\pi]$ we have $q(e^X) = 2\sin^2(r/4)$. On $[0, \pi/2]$, the sine lies above its chord by concavity, $\sin x \ge 2x/\pi$, and $\sin x \le x$. These two inequalities imply the quadratic lower and upper bounds. Integrating $\sin x \ge x - x^3/6$, we obtain $1 - \cos x \ge x^2/2 - x^4/24$, and the inequality for the sine follows by integrating $x - \sin x \ge 0$ twice. Setting $x = r/2$ yields the additional lower bound. By Taylor's formula with integral remainder, $|q(e^X) - r^2/8| \le r^4/384$. The function $g(r) = \log \operatorname{sinc}^2(r/2) + r^2/12$ is even and satisfies $g(0) = g''(0) = 0$, as the expansion $\operatorname{sinc}(r/2) = 1 - r^2/24 + O(r^4)$ shows; its odd derivatives at 0 vanish as well. Its fourth derivative is bounded on every interval $[-r_0, r_0]$ with $r_0 < 2\pi$, and by the same remainder formula $|g(r)| \le C(r_0) r^4$. For $0 < u \le 1$, the function $\operatorname{sinc} s$ decreases on $[0, 1/2]$, since $(s\cos s - \sin s)' = -s \sin s \le 0$. Writing $\sin^2(r/2) = (r^2/4)\operatorname{sinc}^2(r/2)$ in the radial density, we obtain the two bounds on $H\{\rho_G < u\}$. Finally $\operatorname{sinc}(1/2) \ge 1 - 1/24 = 23/24$, whose square exceeds $3/4$; this implies $c_{\mathrm{ball}} \ge 1/(16\pi)$.

Partition functions. The function q is continuous on $\mathrm{SU}(2)$ with values in $[0, 2]$. Every plaquette transport is a product of link variables and their inverses, and its trace is invariant under gauge transformations, since the intermediate vertex factors cancel. Under a gauge transformation each link variable undergoes a left and a right translation, so product Haar measure is invariant. The integrand lies between $e^{-2\beta|\mathcal{P}|}$ and 1. Integrating, we obtain the bounds on $Z_{\mathcal{K}}(\beta)$, and dividing by $Z_{\mathcal{K}}(\beta)$ gives a gauge-invariant probability measure. Neither connectedness nor periodicity of $\mathcal{K}$ is needed.

The adjoint representation. Conjugation preserves the inner product $-2\operatorname{tr}(XY)$ and the bracket, so $\mathrm{Ad}(k)$ is orthogonal on $\mathfrak{su}(2)$. Its determinant is $+1$, since $\mathrm{SU}(2)$ is connected and the determinant equals 1 at I. In particular, $\mathrm{Ad}(k)$ preserves the norm, Lebesgue measure, and the balls centered at the origin. For surjectivity, note that $R(\vartheta) = \mathrm{Ad}(e^{\vartheta n \cdot \tau})$ solves $R'v = n \times Rv$; it is the rotation by ϑ about the unit vector n. Since every element of $\mathrm{SO}(3)$ has a fixed unit vector (its nonreal eigenvalues come in conjugate pairs, and the remaining real eigenvalue is 1) and acts on the orthogonal complement as a plane rotation, it is one of these rotations.

Spanning-tree gauges. We orient each forest link from the parent to the child, replacing U_e by U_e^{-1} when the given orientation is opposite; inversion preserves Haar measure. Set $h_o = I$ at every root and recursively $h_y = h_x U_{xy}$. Then $V_{xy} = h_x U_{xy} h_y^{-1} = I$ on the forest. Conversely, arbitrary h_x at the vertices that are not roots and arbitrary V_e off the forest determine U uniquely by $U_{xy} = h_x^{-1} V_{xy} h_y$, with $V_{xy} = I$ on forest links. All these maps are finite products and inverses, hence smooth. We list the vertices in increasing distance

from their roots along the forest. Conditionally on the earlier forest links, $h_y = h_x U_{xy}$ is Haar distributed and independent of all earlier h's. By induction, the joint law of the forest variables is product Haar measure. Conditionally on them, every link off the forest is translated on the left and on the right by fixed matrices. Its Haar factor is therefore unchanged, and these factors remain independent. This proves the product identity, first for products of functions and then by approximation for all integrable functions.

If the integrand is invariant under the gauge transformations equal to I at the roots, substituting the gauge transformation h makes it a function of V alone, and the product Haar integral over h contributes a factor one. A connected complex has $|\mathcal{V}| - 1$ tree links, which gives the stated number of remaining group variables. For a gauge transformation k we have $h_x(U^k) = k_o h_x(U) k_x^{-1}$ and $V_e(U^k) = k_o V_e(U) k_o^{-1}$, which leaves the conjugation at the root. We do not take a quotient by it, and no stabilizer factor appears. Applying the recursion in every block, with the links between blocks among the variables off the forest, proves the version for blocks with rooted spanning trees.

Products of group elements. Matrix exponentials are given by their absolutely convergent series, and inversion on $\mathrm{SL}(2,\mathbb{C})$ is polynomial in the matrix entries. In a complex neighborhood of a compact set of unitary matrices whose eigenvalues stay at a fixed distance from -1, the formula

$$\operatorname{Log} W = \int_0^1 (W - I)\big((1-s)I + sW\big)^{-1}\, ds \tag{5.2}$$

defines a holomorphic logarithm. Indeed, on the compact real set the matrices $(1-s)I + sW$ have smallest singular value bounded below uniformly in $s \in [0,1]$, and half of this bound persists on a fixed smaller complex neighborhood, by the identity for differences of inverses. For diagonal unitary matrices the formula reduces to the scalar principal logarithm, by direct integration; conjugation extends it to all unitary matrices of the set, and the identity $e^{\operatorname{Log} W} = W$ continues to the neighborhood by analytic continuation. On a connected neighborhood of I its trace vanishes, since $\det W = 1$ and the trace is continuous with value zero at I.

A product of at most ℓ factors involves at most 3ℓ scalar coordinates. On the larger polydisc its matrix norm and the norms of the relevant logarithms are bounded by a constant B depending on ℓ and the domains: each exponential is bounded by the exponential of the matrix norm of its exponent, each logarithm by (5.2), and factors in $\mathrm{SU}(2)$ have matrix norm one. If the smaller polydisc has coordinate margin $\delta > 0$, the Cauchy integral in each coordinate bounds a derivative of multi-index α by $B\alpha!\delta^{-|\alpha|}$. To see this, we insert the absolutely convergent power series into the circle integrals; the integral of z^k/z^{j+1} over a circle around zero vanishes for $k \neq j$ and equals $2\pi i$ for $k = j$. Summing all ordered derivatives of total order s introduces at most a factor $(3\ell)^s$, which gives the bound $s!C^s$ for the absolute derivative tensor. Where the branches are

defined, $\operatorname{Log}(kWk^{-1}) = k(\operatorname{Log} W)k^{-1}$ and $\operatorname{Log}(W^{-1}) = -\operatorname{Log} W$, by the same formula and analytic continuation.

The expansion with a removable singularity. Let $B(\epsilon) = \operatorname{Log}(e^{u}e^{\epsilon a}e^{-u-\epsilon v})$. For small fixed complex u, v, a it is holomorphic in $|\epsilon| < 1 + \delta_0$, with uniform bounds, and $B(0) = 0$. Differentiating the exponential series, we obtain

$$\frac{d}{d\epsilon} e^{-u-\epsilon v}\Big|_{\epsilon=0} = -\int_0^1 e^{-(1-t)u} v e^{-tu}\, dt.$$

Since the derivative of Log at I is the identity, $B'(0) = \operatorname{Ad}(e^{u})a - \int_0^1 \operatorname{Ad}(e^{tu})v\, dt$. The product depends on a, v only through $(\epsilon a, \epsilon v)$, and the coefficient of ϵ^k in B is therefore homogeneous of degree k in (a, v). Hence

$$\mathscr{R}_\epsilon = \int_0^1 (1-t) B''(t\epsilon)\, dt$$

is holomorphic, vanishes to order at least two in (a, v), and satisfies the bounds just proved. Using Taylor's formula along the segment $t(a, v)$ and the bound on second derivatives, we find on a fixed smaller complex ball

$$\|\mathscr{R}_\epsilon(u, v, a)\| \le C(|a| + |v|)^2, \qquad 0 \le \epsilon \le 1, \tag{5.3}$$

with C depending only on the product, the larger complex domain, and its margin. At $v = 0$ the product is $\exp(\epsilon \operatorname{Ad}(e^{u})a)$, its logarithm is $\epsilon \operatorname{Ad}(e^{u})a$, and the remainder vanishes. Every argument concerns a fixed product or an exact identity, and the stated dependence of the constants follows. □

5.2 Closed one-cochains and Gaussian elimination

Parts (1) and (2) of the next lemma are the discrete Poincaré lemma and the fact that the first real cohomology of the three-torus is $\mathbb{R}^3$; part (3) is Gaussian elimination by Schur complements [27]. The harmonic part c is kept as data throughout.

Lemma 5.2. *The following statements hold.*

1. *On every finite contractible rectangular box in $\mathbb{Z}^3$, a real closed one-cochain is the coboundary of a function on the vertices.*
2. *On the periodic cubic lattice of side $n \ge 4$, every real closed one-cochain α has a unique decomposition $\alpha = \nabla f + c$, where f is periodic with mean zero and c is constant in each link direction. Moreover $c_i = n^{-3} \sum_x \alpha_i(x)$. The decomposition holds in each color separately and after rescaling to physical units. Thus requiring the spatial mean of α to vanish removes exactly c, and not a gauge degree of freedom.*

3. *Let a real symmetric nonnegative quadratic form on a finite-dimensional space of fixed coordinates c and integration coordinates $z \in \mathbb{R}^f$ have matrix*

$$\mathsf{A} = \begin{pmatrix} A & B \\ B^{\mathsf{T}} & K \end{pmatrix}, \qquad K > 0.$$

For every c, including all specified harmonic coordinates,

$$\frac{1}{2}(c,z)^{\mathsf{T}}\mathsf{A}(c,z) = \frac{1}{2}c^{\mathsf{T}}(A - BK^{-1}B^{\mathsf{T}})c + \frac{1}{2}(z + K^{-1}B^{\mathsf{T}}c)^{\mathsf{T}}K(z + K^{-1}B^{\mathsf{T}}c),$$

and

$$\int_{\mathbb{R}^f} e^{-\frac{1}{2}(c,z)^{\mathsf{T}}\mathsf{A}(c,z)}\,dz = (2\pi)^{f/2}(\det K)^{-1/2}e^{-\frac{1}{2}c^{\mathsf{T}}(A-BK^{-1}B^{\mathsf{T}})c}.$$

Successive minimization, or successive Gaussian integration, over the blocks of any ordered decomposition of z gives the same Schur complement and the same product of determinants as simultaneous elimination. Conditionally on c, the Gaussian measure in z has mean $-K^{-1}B^{\mathsf{T}}c$ and covariance K^{-1}. A Gaussian measure defined on a specified orthogonal complement of a null space, by a quadratic form positive there, has covariance equal to the inverse on that complement and zero on the null space.

These are exact finite-dimensional statements; in particular, no lower bound on eigenvalues uniform in the side n is asserted.

Proof. Boxes. We regard a one-cochain α as an antisymmetric function on oriented links; closedness means that its signed sum around every unit square vanishes. On $\mathbb{Z}^3$, a finite path can be sorted by coordinate direction by adjacent interchanges of steps in different directions. Each interchange crosses exactly one unit square and preserves the integral of α, and adjacent opposite steps in one direction cancel. Then every closed path has integral zero, since after sorting its total displacement in each direction is zero and all steps cancel. Inside a rectangular box we integrate from a fixed corner, first in direction 1, then in direction 2, then in direction 3. Interchanging an additional final step in direction i with the later coordinate segments uses only the squares of the intervening rectangle, which lies in the box, and closedness makes the two path integrals equal. The resulting vertex function has coboundary α on every link. Boxes with a side of length zero are covered by omitting the absent direction.

The periodic lattice. We lift a periodic closed cochain to $\mathbb{Z}^3$. The argument just given produces a function f_0 on $\mathbb{Z}^3$ with $f_0(x + e_i) - f_0(x) = \alpha_i(x)$. Since α is periodic, $f_0(x + ne_i) - f_0(x) = P_i$ does not depend on x, because its difference across every link is zero; equivalently, P_i is the integral of α along any straight cycle winding once in direction i. Set $c_i = P_i/n$. The function $f(x) = f_0(x) - \sum_i c_i x_i$ is periodic and satisfies $\alpha = \nabla f + c$; subtracting its average makes its mean zero. If $\nabla f + c = 0$, then summing over a winding cycle gives $nc_i = 0$. Then all differences of f vanish, and the mean-zero condition implies

$f = 0$, which proves uniqueness. Because the sum of $\nabla_i f$ over all periodic vertices telescopes to zero, we have $c_i = n^{-3} \sum_x \alpha_i(x)$, and the mean condition removes exactly the three constants. For cochains with values in $\mathfrak{su}(2)$, the argument applies to each real component. For physical potentials we use $\nabla_a = a^{-1}\nabla$ and integrate $a\alpha$ along paths; the conclusion has the same form, with $c_i = P_i/(na)$.

Gaussian elimination. If we expand the right side of the completion of the square, its cross term is $c^{\mathsf{T}} B z$, and the added constant $c^{\mathsf{T}} B K^{-1} B^{\mathsf{T}} c/2$ cancels the subtracted one. After the translation $z \mapsto z - K^{-1} B^{\mathsf{T}} c$, an orthogonal diagonalization of the positive matrix K reduces the integral to products of $\int_{\mathbb{R}} e^{-kt^2/2} dt = \sqrt{2\pi/k}$. To diagonalize K, we maximize its Rayleigh quotient on the unit sphere to obtain an eigenvector, restrict to the orthogonal complement of this eigenvector, and induct on the dimension. The value of the one-dimensional integral follows by squaring it and using polar coordinates in the plane. We obtain the Gaussian formula, with the convention that the integral is one if $f = 0$. The same completion of the square with a linear term in z, differentiated at zero, gives the conditional mean and covariance.

For successive elimination, write $z = (z_1, z_2)$ and $K = \left(\begin{smallmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{smallmatrix}\right)$. Both K_{11} and $K_{22} - K_{21} K_{11}^{-1} K_{12}$ are positive definite, since their quadratic forms are the restriction of the form of K and the minimum of that form over z_1. The change of variables $z_1 \mapsto z_1 + K_{11}^{-1} K_{12} z_2$ has determinant one, and by block multiplication we have

$$\det K = \det K_{11} \, \det\bigl(K_{22} - K_{21} K_{11}^{-1} K_{12}\bigr).$$

The unique joint minimizer in z solves $Kz = -B^{\mathsf{T}} c$; solving the first block and substituting gives the successive minimizers. The minimum value is therefore the same Schur form. By the determinant identity and the Gaussian formula, successive integration gives the same normalization. Induction over the blocks proves the statement for every ordered decomposition; for nonnegative integrands the repeated integrals are justified directly. No harmonic coordinate is discarded, since the vector c was arbitrary throughout.

On the given orthogonal complement of a null space, we diagonalize the restricted positive form. In the resulting coordinates the Gaussian measure is a product of independent one-dimensional Gaussian measures with variances equal to the inverse eigenvalues; embedding this measure by setting the null coordinates equal to zero, we obtain the stated covariance. This construction lives on the complement; it is not a Lebesgue integral along null directions. □

5.3 Analytic coefficient bounds and implicit functions

The following lemma is the calculus of the coefficient norms of Definition 4.6. Its first parts are combinatorial. A Taylor coefficient of a composite map is a finite or absolutely convergent sum of products of Taylor coefficients of the constituents. We bound the

absolute value of the sum by the sum of absolute values. The spatial weight survives because the connecting networks of the factors can be joined at the contracted indices, and the volume factors in the column sums telescope. The simplest instance is a product of two kernels, T from mesh a_{mid} to mesh a_{out} and S from mesh a_{in} to mesh a_{mid}. In counting coordinates the kernel of TS is $\sum_j T_{ij}S_{jk}$, without any volume factor, and its column sum at a fixed input k carries the factor

$$\frac{a_{\mathrm{out}}^3}{a_{\mathrm{in}}^3} = \frac{a_{\mathrm{out}}^3}{a_{\mathrm{mid}}^3} \cdot \frac{a_{\mathrm{mid}}^3}{a_{\mathrm{in}}^3},$$

so that column norms are submultiplicative. For a composition with one fixed input, we use column sums along the chain of factors that joins this input to the output, and row sums for all other factors. The later parts concern inverses and square roots of matrices with a spectral gap. The exponential decay of an inverse follows from the conjugation argument of Combes and Thomas [61], and square roots are represented by the standard resolvent integral [27]; the proof keeps track of the spatial weights and of all fixed indices. Bounds by coefficientwise majorant series are a classical tool of constructive renormalization group methods; the volume-normalized columns, the arbitrary fixed inputs, and the traces with a fixed external index are specific to our use.

Throughout this section, all indices are scalar indices and all sums are counting sums. A contraction identifies two indices of the same kind (same lattice and scale of introduction); an identification of indices of different kinds must first be represented by an explicitly bounded transfer kernel. We fix a rate $a_{\mathrm{sp}} \geq 0$ and write $|T| = |T|_{a_{\mathrm{sp}}}$ for the norm (4.9), $R(T)$ for its row norm, and $C_r(T)$ for its volume-normalized column sums. For a scalar tensor we use the sums (4.10) with one index fixed; on several meshes, a scalar tensor may instead be normalized by a reference volume v_*, which multiplies each summand with the index i_r fixed by v_*/v_{i_r}. Distances and network lengths are those of Definition 4.6, measured in units of the current output lattice spacing. For a matrix $B = (B_{xy})$ of blocks indexed by a finite metric space (X, d), we write

$$\|B\|_c = \max\Big\{\sup_x \sum_y e^{cd(x,y)}\|B_{xy}\|,\ \sup_y \sum_x e^{cd(x,y)}\|B_{xy}\|\Big\}, \qquad c \geq 0,$$

where $\|B_{xy}\|$ is the operator norm of the block.

Lemma 5.3. *The following statements hold.*

1. Substitution. *For tensors $T(o; j_1, \dots, j_k)$ and $S_r(j_r; \mathbf{l}_r)$, $1 \leq r \leq k$, the tensor*

$$U(o; \mathbf{l}_1, \dots, \mathbf{l}_k) = \sum_{j_1,\dots,j_k} T(o; j_1, \dots, j_k) \prod_{r=1}^{k} S_r(j_r; \mathbf{l}_r)$$

has every output-row norm and every volume-normalized input-column norm bounded by $|T| \prod_r |S_r|$. In particular compatible matrix products are submultiplicative in the row norm

and in the volume-normalized column norm; for counting-coordinate kernels from mesh a_{in} to mesh a_{out} the column factor is exactly $a_{\mathrm{out}}^3/a_{\mathrm{in}}^3$, and a contraction introduces no further volume factor. The same bound holds for a scalar outer tensor normalized by a reference volume. For a scalar outer tensor with counting sums on mesh a_{old}, the sum with a new input l fixed acquires the factor v_l/v_{old}, maximized over the possible kinds of the replaced inputs. Pointwise products with a common output index satisfy the product bound, and a shared index may be represented by the equality tensor $\mathbf{1}_{o=i_1=\cdots=i_r}$, whose norm is one when all its ports are of the same kind.

2. Contraction diagrams. *Consider a finite connected diagram of tensors, in which shared indices are represented by equality tensors and every contracted index identifies two ports, and fix one external index. Choose a spanning tree of the graph of factors, rooted at the factor containing the fixed index, and attach every other factor V to its parent through a port $j(V)$. For a vector factor V with output $o(V)$, let $A_V = |V|$ if $j(V) = o(V)$ and $A_V = (v_{j(V)}/v_{o(V)})|V|$ if $j(V)$ is an input; for the factor containing the fixed index use the same rule at that index; for a scalar factor use its sum with the corresponding port fixed. Then the sum over all indices except the fixed one of the absolute value of the diagram, weighted by $e^{a_{\mathrm{sp}}\mathtt{t}}$ of the locations of the external indices, is at most $\prod_V A_V$. If the fixed index is an external input l, the volume-normalized column of the diagram is obtained by multiplying by v_o/v_l, where o is the output of the diagram; for the substitution trees of part (1) the volume ratios cancel exactly. A diagram without a fixed external index is not covered; for example* $\mathrm{tr}\, I$ *equals the number of indices. An externally differentiated trace is covered if the differentiated tensor has the bound with the input fixed and* all *of its matrix ports and other Taylor inputs summed; a row bound alone does not suffice.*

3. Majorant series. *Products, substitutions, and summable connected diagrams of analytic maps with absolute majorants (in the sense of Definition 4.6) are bounded coefficientwise by the corresponding products, compositions, and diagram sums of the nonnegative scalar majorants, with the bounds of parts (1) and (2). The domain of the conclusion is the common product of polydiscs on which these scalar series converge absolutely. For $F \circ G$ it suffices that the scalar majorant of G maps the domain strictly inside the coefficient domain of F; for formal substitution take $G(0) = 0$, and otherwise require this absolute image condition, so that the substitutions into the constant terms also converge. The coefficients with one fixed input are bounded by the derivative of the same scalar majorant; no second Taylor factorial occurs.*

4. Smaller domains. *If $A(s) = \sum_n A_n s^n$ with $A_n \geq 0$ converges at $R > 0$, then for $0 < r < R$ and every fixed $j \geq 0$,*

$$A^{(j)}(r) \leq R^{-j} \sup_{n \geq j} (n)_j (r/R)^{n-j} A(R),$$

where $(n)_j = n(n-1)\cdots(n-j+1)$. *If the majorant* $\sum_k b_k s^k$ *satisfies* $\sum_k b_k R^k \le B$, *the coefficients of degree* s *of its translate to a background of size* $r_1 < R$ *are at most* $B(R-r_1)^{-s}$. *Thus evaluation, translation to a background, and a fixed finite number of derivatives are controlled on a smaller coefficient domain, at the same spatial rate. These statements concern absolute coefficient series; they are not Cauchy estimates for directional values.*

5. Resolution into occurrences. *Every expression built by these operations has an exact coefficient resolution, indexed by its constituent coefficient occurrences, its contracted indices, and its partitions of derivatives. Assign to each constituent occurrence a connected network through the locations of its ports and Taylor indices, and to each contraction the common location; the network of a resolved occurrence is the union of these networks. Its external monomial has its exact multi-index, and its fluctuation index set is the set of blocks of fluctuation variables with a positive exponent; the locations of background variables need not lie in that set. The weighted total variation of the resolution satisfies the same scalar diagram bounds, provided the assigned networks satisfy the constituent bounds. Let* W *be a specified boundary. If a designated constituent occurrence is assigned, by hypothesis, a connected network meeting* W, *then every resolved occurrence containing this designated factor has a connected network meeting* W, *and its weighted total variation satisfies the corresponding diagram bound, provided the constituent bounds include the weights of these assigned networks. Marking an input for differentiation does not by itself provide such a visit.*

6. Networks and trees. *Let* $\mathrm{t}_{\mathrm{MST}}(S)$ *be the length of a minimal spanning tree on a finite set* S *of locations. Then* $\mathrm{t}(S) \le \mathrm{t}_{\mathrm{MST}}(S) \le 2\mathrm{t}(S)$. *A bound at rate* a_{sp} *with shortest networks therefore gives a bound at rate* $a_{\mathrm{sp}}/2$ *with spanning trees on the locations of the external indices; this reduction is made once, independently of the order of the tensors.*

7. Inverses. *Let* (X,d) *be a finite metric space, with blocks of a fixed finite dimension, such that* $\#\{y : d(x,y) \le R\} \le D(1+R)^v$ *for all* $x \in X$ *and* $R \ge 0$. *Let* $K = K^* \ge \lambda I > 0$ *satisfy, for some* $a > 0$,

$$\max\Big\{\sup_x \sum_{y\neq x} e^{ad(x,y)}\|K_{xy}\|,\ \sup_y \sum_{x\neq y} e^{ad(x,y)}\|K_{xy}\|\Big\} \le H.$$

Let $b = \min\{a/2, a\lambda/(4H)\}$ *if* $H > 0$ *and* $b = a/2$ *if* $H = 0$; *then the same maximum with weights* $e^{bd(x,y)} - 1$ *is at most* $\lambda/2$. *For every* $0 < b' < b$,

$$\|(K^{-1})_{xy}\| \le 2\lambda^{-1}e^{-bd(x,y)}, \qquad \|K^{-1}\|_{b'} \le 2\lambda^{-1}\mathcal{S}(b-b';D,v),$$

where $\mathcal{S}(c;D,v) = D\sum_{k\ge 0}(k+2)^v e^{-ck}$. *No upper bound on the diagonal of* K *is needed, and the bounds do not depend on* $|X|$.

8. Perturbations and square roots. *In the setting of part (7), put* $C_{\mathrm{inv}} = \|K^{-1}\|_{b'}$. *For every complex matrix* E *with* $\|E\|_{b'} \le (2C_{\mathrm{inv}})^{-1}$, *we have* $\|(K+E)^{-1}\|_{b'} \le 2C_{\mathrm{inv}}$, *and the*

inverse is a holomorphic function of E. If also $K \le \Lambda I$, there exists $\eta_0 > 0$, depending only on $a, H, \lambda, \Lambda, D, v, b'$, such that for $\|E\|_a < \eta_0$ the branches $(K+E)^{1/2}$ and $(K+E)^{-1/2}$ continuing the positive roots exist and are bounded in $\|\cdot\|_{b'/2}$ by constants depending only on these data. If $E = E(\theta)$ has an absolute majorant with all fixed indices on a fixed analytic ball, the coefficients of these roots in θ have absolute bounds with every fixed index, with constants depending in addition only on that majorant and on the ratios of the analytic radii.

9. Logarithms of determinants. *In the setting of part (8), $D \log\det K = \operatorname{tr}(K^{-1}DK)$, and the derivatives of positive order of $\log\det(K+E(\theta))$ in θ have absolute bounds with any fixed external index, with one reduction of the rate independent of the order. The undifferentiated $\log\det K$ is not bounded independently of $|X|$.*

10. Implicit functions. *Let $\Phi(\theta, u)$ be an analytic map of input variables $\theta = (\theta_j)$ and unknowns $u = (u_o)$, with values indexed like u, defined on a product of polydiscs, with $\Phi(0,0) = 0$. Assume that for all $k, l \ge 0$ the output-row norm and every input-column norm, at any fixed input θ_j or u_o, of the coefficient tensor $D_\theta^k D_u^l \Phi(0,0)/(k!\,l!)$ are at most the coefficient of $t^k Y^l$ in*

$$\Phi^+(t,Y) = At + \kappa Y + \frac{A(t+Y)^2}{1 - C(t+Y)}, \qquad A, C > 0, \quad 0 \le \kappa < 1.$$

In particular the linear map $D_u\Phi(0,0)$ has row and column norms at most κ. Let $\delta = 1-\kappa$, $r = \min\{1, (4C)^{-1}, \delta/(32A)\}$, and $t_0 = \min\{r, \delta r/(4A)\}$, and assume that the domain of Φ contains a neighborhood of the closed product $\{\max_j|\theta_j| \le t_0,\ \max_o|u_o| \le r\}$. Then the scalar equation $Y = \Phi^+(t,Y)$ has a unique solution $Y(t)$ with $|Y(t)| \le r$ for $|t| < t_0$; it is analytic, $Y(0) = 0$, and its Taylor coefficients are nonnegative. For $\max_j|\theta_j| < t_0$ the system $u = \Phi(\theta, u)$ has a unique solution $u(\theta)$ with $\max_o|u_o| \le r$, and for every k the output-row norm and every input-column norm of $D^k u(0)/k!$ are at most the coefficient of t^k in $Y(t)$. For a fixed input θ_j, the derivative $\partial_{\theta_j} u$ has coefficient tensors bounded by the coefficients of

$$Y'(t) = \frac{A + A\varphi'(t+Y(t))}{1 - \kappa - A\varphi'(t+Y(t))}, \qquad \varphi(s) = \frac{s^2}{1 - Cs},$$

whose denominator satisfies $|1 - \kappa - A\varphi'(t+Y(t))| \ge 5\delta/8$ for $|t| < t_0$. In particular, all derivatives of positive order satisfy bounds $s! C_1^s$ at the same spatial rate, also after translation to a background in a fixed smaller ball, where C_1 depends only on A, C, κ and the radius ratios, and not on the number of coordinates. If a parameter enters Φ with a variation of majorant $H(t,Y)$, the variation of the solution has majorant $Z(t)$ with $Z(t) \le H(t, Y(t))/(1 - \partial_Y \Phi^+(t, Y(t)))$ for $0 \le t < t_0$; this bounds the value of the majorant series at t, and it is not a coefficientwise inequality.

The constants in parts (1)–(6) are as displayed. In parts (7)–(10) they have the dependence described in item (1) of Definition 4.14, through the displayed data. When the counting bound for

the indices depends on J (with $D = D_J$), the constants depend on this fixed J, and no uniformity as $J \to \infty$ follows. Bounds obtained from part (10) for a class of expressions require one common scalar majorant for the whole class; expressions of arbitrary algebraic complexity are not covered without such a majorant.

Proof of Lemma 5.3. We first prove parts (1)–(6), which are bookkeeping for absolute sums, then the inverse estimates (7)–(9), and finally the implicit function statement (10).

Step 1: substitution. For matrices, the row norm of $\sum_j T_{ij} S_{jk}$ satisfies

$$\sup_i \sum_k e^{a_{\mathrm{sp}} d(i,k)} \Big\| \sum_j T_{ij} S_{jk} \Big\| \le R(T) R(S),$$

since $d(i,k) \le d(i,j) + d(j,k)$. For columns, the exact identity

$$\frac{v_i}{v_k} = \frac{v_i}{v_j} \frac{v_j}{v_k}, \qquad \sup_k \sum_i \frac{v_i}{v_k} e^{a_{\mathrm{sp}} d(i,k)} \Big\| \sum_j T_{ij} S_{jk} \Big\| \le C(T) C(S) \tag{5.4}$$

holds. If indices of several kinds occur, the sums include them with their volumes. This proves submultiplicativity and explains why the column factor of a counting-coordinate kernel must be exactly $a_{\mathrm{out}}^3 / a_{\mathrm{in}}^3$.

For the substitution tensor U, the union of the networks of T and of the S_r, joined at the contracted indices, connects the locations of all external indices, which implies

$$\mathsf{t}(\text{locations of the external indices}) \le \mathsf{t}(o, \mathbf{j}) + \sum_r \mathsf{t}(j_r, \mathbf{l}_r).$$

We take absolute values before any summation. For an output row, we sum each inner branch by its row norm and then the outer tensor by its row norm. If an external input l belonging to branch r is fixed, we sum every other branch by its row norm, use the column of branch r at l, and the j_r-column of the outer tensor. The volume factors telescope, since $v_o / v_l = (v_o / v_{j_r})(v_{j_r} / v_l)$ by (5.4). For a scalar outer tensor normalized by a reference volume the same computation applies, and for counting sums the only change is the factor v_l / v_{old}. At a common output index the networks meet at that output, and a bound with an input fixed uses the column of the factor carrying the input and rows of the other factors. This proves part (1).

Step 2: diagrams. We replace every edge of the diagram outside the chosen spanning tree by two independently summed copies of its index, deleting the equality constraint between them; for nonnegative sums this can only increase the result. When a split edge is present we may enlarge the external network weight to include the locations of both copies, since the connected tree of factors still spans the locations of the given external indices. We orient the tree from the factor containing the fixed index. Starting at its

leaves, we sum each factor over all its ports except the port of attachment to its parent, using the bounds A_V; we continue toward this factor and finally sum it over all ports except the fixed index. This proves the bound without any count of the internal indices. In a substitution tree, the factors other than rows lie on the path from the fixed input to the output, and the product of their volume ratios is v_l/v_o, which the external column normalization cancels. A diagram without a fixed external index has no factor at which to start, and it is excluded. This proves part (2).

Step 3: majorant series and smaller domains. We first consider Taylor polynomials and finite diagrams. For $G(0) = 0$, the ordered derivative formula reads

$$D^N(F \circ G)(0)[u_1, \dots, u_N] = \sum_{n=0}^{N} \frac{1}{n!} \sum_{(B_1,\dots,B_n)} F_n\big[G_{|B_1|}[u_{B_1}], \dots, G_{|B_n|}[u_{B_n}]\big],$$

where $(B_1, \dots, B_n)$ runs over the ordered partitions of $\{1, \dots, N\}$ into nonempty blocks, and the term $N = n = 0$ is understood separately. For block sizes $m_1 + \cdots + m_n = N$ the number of ordered partitions is $N!/\prod_r m_r!$. Dividing by $N!$ therefore gives exactly the coefficient of s^N in $\widehat{F}(\widehat{G}(s))$, and parts (1) and (2) bound each summand. Marking one input chooses its block and its position in the block, exactly as in $(\widehat{F} \circ \widehat{G})' = (\widehat{F}' \circ \widehat{G})\widehat{G}'$. This shows that the coefficient of order N with one marked input is bounded by Nh_N, where h_N is the Taylor-divided coefficient of the composite majorant. Inputs of different kinds may be marked by separate scalar variables. For infinite orders and diagrams, the nonnegative majorant series converge on the stated domain, and Tonelli's theorem and absolute convergence justify every grouping, contraction, substitution, and derivative; equality of the analytic functions follows from equality of their coefficients. The case $G(0) \neq 0$ is handled in the same way under the absolute image condition. This proves part (3).

For part (4), we write $r^{n-j} = R^{n-j}(r/R)^{n-j}$ termwise; then $A^{(j)}(r) = \sum_{n \geq j} (n)_j A_n R^{n-j} (r/R)^{n-j} \leq R^{-j} \sup_{n \geq j} (n)_j (r/R)^{n-j} \sum_n A_n R^n$. The coefficient of degree s of the translate of the majorant to a background of size $r_1 < R$ is the binomial sum $\sum_{k \geq s} \binom{k}{s} b_k r_1^{k-s}$. By the binomial theorem, $\binom{k}{s} r_1^{k-s} (R - r_1)^s \leq R^k$ for each $k \geq s$; multiplying by b_k, summing, and using $\sum_k b_k R^k \leq B$, we obtain $\sum_{k \geq s} \binom{k}{s} b_k r_1^{k-s} \leq B(R - r_1)^{-s}$. Evaluating extra inputs at a background can only remove locations of external indices from a network, and the same spatial rate therefore remains valid.

For part (5), we keep a separate signed occurrence for every summand of the derivative formula, every tuple of contracted indices, and every refinement of a constituent network. Its coefficient is the product of the constituent coefficients and the displayed combinatorial factor, and its monomial is the product of their monomials. Grouping occurrences with equal multi-indices or equal networks is optional and can only decrease the total variation. The network inequality of Step 1 implies the weighted bound, and the convergence

argument of this step justifies the infinite sums. If a designated constituent occurrence has an assigned network meeting W, the network of a resolved occurrence containing it is a union that contains this assigned network, and it retains the visit to W.

Step 4: networks and trees. The inequality $\mathtt{t}(S) \le \mathrm{t}_{\mathrm{MST}}(S)$ holds since a spanning tree on S is a connecting network. Conversely, we delete the cycles of a competing network without increasing its length, traverse the resulting tree with every edge twice, shortcut the tour between successive points of S by the triangle inequality, and delete its last edge. The result is a path through S of length at most twice the length of the network. Taking infima, we find $\mathrm{t}_{\mathrm{MST}}(S) \le 2\mathtt{t}(S)$, which implies $e^{(a_{\mathrm{sp}}/2)\mathrm{t}_{\mathrm{MST}}} \le e^{a_{\mathrm{sp}}\mathtt{t}}$. Joining the constituent trees once proves the assertion that the rate is halved only once. This proves part (6).

Step 5: inverses. For a block matrix B with absolute row sums at most r and column sums at most c, the Cauchy–Schwarz inequality gives

$$\sum_{x}\Big|\sum_{y} B_{xy}u_y\Big|^2 \le r\sum_{x,y}\|B_{xy}\|\,|u_y|^2 \le rc\sum_{y}|u_y|^2,$$

which implies $\|B\|_{2\to 2} \le \sqrt{rc}$. For $0 < b \le a/2$ and $s \ge 0$ we have $e^{bs}-1 \le bse^{bs} \le (2b/a)e^{as}$, and with the choice of b in part (7) the weighted off-diagonal sums are at most $(2b/a)H \le \lambda/2$; any smaller positive choice also works. Fix $y_0 \in X$ and let $W_x = e^{bd(x,y_0)}I$ on the block of x. The matrix

$$B = WKW^{-1} - K, \qquad B_{xz} = \big(e^{b(d(x,y_0)-d(z,y_0))}-1\big)K_{xz},$$

has zero diagonal, and by the triangle inequality each scalar factor is at most $e^{bd(x,z)}-1$ in absolute value. By the bound just proved, $\|B\|_{2\to 2} \le \lambda/2$. Since $K \ge \lambda I$, we have $\|K^{-1}\|_{2\to 2} \le 1/\lambda$, and the geometric series for $(I+K^{-1}B)^{-1}$ yields $\|(WKW^{-1})^{-1}\|_{2\to 2} \le 2/\lambda$. The (x,y_0) block of this inverse is $e^{bd(x,y_0)}(K^{-1})_{xy_0}$, which proves the pointwise bound. The largest value of the weight never enters, and the diagonal of K cancels from B.

For $c > 0$, decomposing into the shells $k \le d(x,y) < k+1$ and using the counting bound, we obtain

$$\sup_{x}\sum_{y} e^{-cd(x,y)} \le D\sum_{k=0}^{\infty}(k+2)^{v}e^{-ck} = \mathcal{S}(c;D,v) < \infty. \tag{5.5}$$

With $c = b - b'$ this bounds both weighted row and column sums of K^{-1} by $2\lambda^{-1}\mathcal{S}(b-b';D,v)$. The fixed block dimension does not change the argument, since all entries were bounded in block operator norm. This proves part (7).

Step 6: perturbations and square roots. By submultiplicativity of $\|\cdot\|_{b'}$ (Step 1, without volume factors), when $\|E\|_{b'} \le (2C_{\mathrm{inv}})^{-1}$ the series

$$(K+E)^{-1} = \sum_{r=0}^{\infty} (-K^{-1}E)^r K^{-1} \tag{5.6}$$

converges absolutely in this norm, with sum bounded by $2C_{\mathrm{inv}}$. Uniform convergence on every smaller ball implies holomorphy by termwise differentiation, and multiplication of partial sums shows that the sum is the inverse.

For the square roots, let $c_2 = b'/2$. We apply the pointwise bound of Step 5 to $K + tI \ge (\lambda + t)I$, whose off-diagonal part is that of K and whose commutator bound $\lambda/2 \le (\lambda + t)/2$ persists. This gives, for all $t \ge 0$,

$$\|(K+tI)^{-1}\|_{c_2} \le \frac{C_0}{\lambda + t}, \qquad C_0 = 2\mathcal{S}(b - c_2; D, v). \tag{5.7}$$

Fix $\eta_0 < \min\{\lambda/(2C_0), \lambda/2\}$. Since $c_2 < a$, we have $\|E\|_{c_2} \le \|E\|_a < \eta_0$, and expanding the shifted resolvents as in (5.6), we obtain $\|(K+E+tI)^{-1}\|_{c_2} \le 2C_0/(\lambda+t)$. Hence the absolutely convergent integral

$$S(E) = \frac{1}{\pi} \int_0^\infty t^{-1/2} (K + E + tI)^{-1}\, dt \tag{5.8}$$

has $\|S(E)\|_{c_2} \le 2C_0\lambda^{-1/2}$ and is holomorphic in E. The scalar identity $\pi^{-1}\int_0^\infty t^{-1/2}(k+t)^{-1}dt = k^{-1/2}$ (substitute $t = ks^2$ and integrate $2/(1+s^2)$) and diagonalization show that $S(E)$ is the positive inverse square root when $K + E$ is real symmetric and positive. For an arbitrary complex, possibly nonsymmetric E, let $M = K + E$. By the bound of Step 5, $\|E\|_{2\to 2} < \eta_0 < \lambda/2$, which implies $\mathrm{Re}\langle u, Mu\rangle \ge (\lambda/2)|u|^2$. For real $d > \|M\|_{2\to 2}^2/\lambda$,

$$\|(I - M/d)u\|^2 \le \big(1 - \lambda/d + \|M\|_{2\to 2}^2/d^2\big)|u|^2 < |u|^2.$$

With $T = I - M/d$, expanding $(M + tI)^{-1}$ in powers of T and integrating termwise in (5.8), which is justified by $\|T\|_{2\to 2} < 1$, identifies $S(E)$ with $d^{-1/2}\sum_{r\ge 0} c_r T^r$, where $c_r = 4^{-r}\binom{2r}{r}$ are the coefficients of $(1-z)^{-1/2}$; indeed, differentiating $(1-z)^{-1/2}$ at zero gives $c_0 = 1$ and $c_{r+1} = (r + 1/2)c_r/(r+1)$. The square of this scalar series is $(1-z)^{-1}$, which gives $S(E)^2 M = I$; all factors commute with M, being limits of its resolvents. This proves the root identity without a symmetry assumption on E. Multiplying by $K + E$ gives the square root; since $K \le \Lambda I$ implies $\|K\|_{c_2} \le \Lambda + H$, this multiplication is bounded uniformly. All choices depend only on $a, H, \lambda, \Lambda, D, v, b'$.

Suppose now that $E = E(\theta)$ has an absolute majorant with all fixed indices on a fixed analytic ball. We insert its Taylor series into the shifted Neumann series before taking any norm. Every coefficient is a contraction, with one external index fixed, of resolvents

and differentiated factors E, to which Steps 1 and 2 apply with any fixed external index. For r insertions, the resolvent integral is bounded by

$$\frac{1}{\pi}\int_0^\infty t^{-1/2}\Big(\frac{C_0}{\lambda+t}\Big)^{r+1}dt \le C_0^{r+1}\lambda^{-r-1/2}.$$

Replacing E by its majorant and using this bound, we obtain a nonnegative scalar series that bounds every absolute coefficient of $S(E(\theta))$ with any index fixed; its radius does not depend on the number of indices. Multiplication by $K+E$ gives the statement for the square root. The constant coefficient is controlled by (5.7), and every coefficient of positive degree contains a differentiated factor E. This proves part (8).

Step 7: logarithms of determinants. Expanding the determinant in columns, we have $D\det K = (\det K)\operatorname{tr}(K^{-1}DK)$, since the linear term of $\det(I+tB)$ is $t\operatorname{tr}B$. To localize the trace, we keep its two matrix indices x, y open. By Steps 1 and 2, a joint coefficient $T_s(x,y;u_1,\dots,u_s)$ of the differentiated product has a weighted output-row bound B_s at some fixed rate $c>0$. Fix an external input $u_j = u$ and close the trace by setting $y = x$. The network length before the trace is closed is at least both the network length of the locations of the remaining external indices and $d(x,u)$. Using half the rate for each, we find that the sum over the other external inputs, with their external weight at rate $c/2$, is at most $B_s e^{-cd(x,u)/2}$ for fixed x. If $u \notin X$, let $x_0 \in X$ be a nearest point; then $d(x,x_0) \le 2d(x,u)$, and (5.5) sums the last bound over x with a factor at most $\mathcal{S}(c/4; D, v)$; for $u \in X$ take $x_0 = u$. This gives an absolute bound with any external index fixed, with one reduction of the rate independent of s. For any volume normalization of the fixed external index, the column factors are propagated by (5.4); no total count of internal indices is inserted. Higher derivatives follow by differentiating the inverse, or from

$$\log\det(K+E) - \log\det K = \sum_{r\ge1}\frac{(-1)^{r+1}}{r}\operatorname{tr}\big((K^{-1}E)^r\big),$$

which holds in the neighborhood of part (8); to see this, differentiate in a scalar multiple of E, use the geometric series for the inverse, and integrate from zero. At positive order a derivative of E provides the fixed external index. Without such an index the bound fails: $K = 2I$ gives $\log\det K = |X|\log 2$ times the block dimension. This proves part (9).

Step 8: the scalar equation. We write $\varphi(s) = s^2/(1-Cs)$, so that $\Phi^+(t,Y) = At + \kappa Y + A\varphi(t+Y)$. For $|t| \le t_0$ and $|Y| \le r$ we have $|t+Y| \le 2r$, $C|t+Y| \le 1/2$, and

$$|At| + A|\varphi(t+Y)| \le \frac{\delta r}{4} + 8Ar^2 \le \frac{\delta r}{2}.$$

Moreover $|\varphi'(s)| \le 2|s|/(1-C|s|) + C|s|^2/(1-C|s|)^2 \le 12r$ for $|s| \le 2r$. The map $Y \mapsto \Phi^+(t,Y)$ therefore maps the closed disc of radius r into itself, since $\kappa r + \delta r/2 < r$,

and its derivative is at most $\kappa + 12Ar \le 1 - 5\delta/8 < 1$. The iteration started at $Y = 0$ converges uniformly, since successive differences decrease by this ratio, and the same estimate implies uniqueness in the disc. Since each iterate is holomorphic, the limit is holomorphic, and $Y(0) = 0$. Every iterate has nonnegative Taylor coefficients, and they increase along the iteration; the limit therefore has nonnegative coefficients. Equivalently, the coefficient of degree s of Y is $(1-\kappa)^{-1}$ times a nonnegative expression in the earlier coefficients, with the source term at degree one.

Step 9: the system. For $\max_j|\theta_j| \le t$ and $\max_o|u_o| \le Y$, the row bounds of the hypothesis imply $\max_o|\Phi_o(\theta,u)| \le \Phi^+(t,Y)$, since the weights $e^{a_{\mathrm{sp}}\mathfrak{t}}$ are at least one; similarly the Lipschitz constant of $u \mapsto \Phi(\theta,u)$ in the maximum norm is at most $\partial_Y\Phi^+(t,Y)$. By Step 8, the Picard iteration $u^{(0)} = 0$, $u^{(n+1)} = \Phi(\theta,u^{(n)})$ stays in the ball of radius r and converges, and the small solution is unique. By part (3), each iterate $u^{(n)}$ has coefficient tensors bounded, in every output-row norm and every input-column norm, by the coefficients of the scalar iterate $Y^{(n)}$, where $Y^{(n+1)} = \Phi^+(t,Y^{(n)})$; the factorials cause no loss, since after division by $s!$ the counting of ordered partitions in Step 3 reproduces ordinary composition of power series. Passing to the limit coefficientwise, we obtain the bounds for $u(\theta)$. For the derivative in a fixed input θ_j, we mark this input in the same expansion. The branch containing it uses column sums and all other branches use row sums. By Step 3, the resulting scalar recursion is the derivative of the scalar equation, which is solved by the displayed formula for $Y'(t)$. By the reverse triangle inequality and $A|\varphi'| \le 12Ar$, its denominator has modulus at least $1 - \kappa - 12Ar \ge 5\delta/8$, and no further loss of the spatial rate occurs. All comparisons are made coefficient by coefficient, before summing over degrees.

Translation to a background of size r_1 in a smaller ball is handled by part (4): the coefficients of order s are bounded by $B(R-r_1)^{-s}$, and multiplying back by $s!$ and absorbing the prefactor for $s \ge 1$ gives $s!C_1^s$, with C_1 depending only on the majorants and radius ratios. For a parameter entering Φ with variation majorant $H(t,Y)$, differentiating the fixed-point equation, we obtain the coefficientwise inequality $Z \preceq H + (\partial_Y\Phi^+)Z$ between majorant series, and repeated substitution bounds Z by the Neumann series in $\partial_Y\Phi^+(t,Y(t))$. Evaluating at $0 \le t < t_0$ gives the value bound $Z(t) \le H(t,Y(t))/(1 - \partial_Y\Phi^+(t,Y(t)))$; it does not give $Z_n \le H_n/(1-q)$ for the individual coefficients. Since the only counting estimate in the proof was (5.5), all constants depend on the fixed J when the counting constant is D_J. This proves part (10). □

5.4 Concentration on products of balls

The conditional fluctuation kernels of (4.8) are measures of the form $Z^{-1}e^{-V}dw$ on a product of Euclidean balls, with a potential whose Hessian is bounded below. The next theorem gives the concentration inequalities for such measures used in later sections,

with constants depending only on the lower Hessian bound. The variance, entropy, and gradient arguments are those of Brascamp and Lieb [33], Gross [103], and Bakry and Émery [8]. The representation of covariances by a one-form operator is the method of Helffer and Sjöstrand [110], whose relation to the Brascamp–Lieb inequality is discussed by Helffer [109]. The passage from entropy to exponential moments is the Herbst argument [123, Section 2.3, Proposition 2.3, p. 149, and Corollary 2.4, p. 151]. We must check that these arguments survive the boundary of the product of balls, including the intersections of faces. They do because the balls are convex: in the Bochner identity for the Neumann Laplacian, the second fundamental form of each sphere contributes the nonnegative term $p_x^{-1}|(\nabla_x h)^{\tan}|^2$, and the faces of different factors meet transversally. We prove this realization explicitly rather than importing a statement on the whole space.

Theorem 5.4. *Let X be a finite set, and for $x \in X$ let $q_x \geq 1$ be integers and $p_x > 0$. Let*

$$\Omega = \prod_{x \in X} \{w_x \in \mathbb{R}^{q_x} : |w_x| \leq p_x\}, \qquad d\mu = Z^{-1} e^{-V(w)}\, dw \quad \text{on } \Omega,$$

where V is real and of class C^2 on a neighborhood of Ω, and $D^2 V \geq mI$ for some $m > 0$. Let $q = \sum_x q_x$ and $\xi = w - \mathbb{E}_\mu w$. Then the following statements hold.

1. *For every real $f \in H^1(\mu)$ and every real function F that is Lipschitz for the Euclidean norm,*

$$\operatorname{Ent}_\mu(f^2) \leq \frac{2}{m} \mathbb{E}_\mu |\nabla f|^2, \qquad \operatorname{Var}_\mu(f) \leq \frac{1}{m} \mathbb{E}_\mu |\nabla f|^2,$$

$$\log \mathbb{E}_\mu e^{t(F - \mathbb{E}_\mu F)} \leq \frac{t^2 \operatorname{Lip}(F)^2}{2m} \qquad (t \in \mathbb{R}),$$

   where $\operatorname{Ent}_\mu(h) = \mathbb{E}_\mu[h \log h] - \mathbb{E}_\mu h \log \mathbb{E}_\mu h$. These constants do not depend on $|X|$, on the q_x and p_x, or on upper bounds for the diagonal blocks of the Hessian.

2. *Let d be a metric on X and $b > 0$, and write $V_{xy} = D_{w_x} D_{w_y} V$. If*

$$\max\Big\{ \sup_x \sum_{y \neq x} (e^{bd(x,y)} - 1) \sup_\Omega \|V_{xy}\|,\ \sup_y \sum_{x \neq y} (e^{bd(x,y)} - 1) \sup_\Omega \|V_{xy}\| \Big\} \leq \frac{m}{2},$$

   then for all real or complex $F, G \in H^1(\mu)$,

$$|\operatorname{Cov}_\mu(F, G)| \leq \frac{2}{m} \sum_{x,y} e^{-bd(x,y)} \|\nabla_x F\|_{L^2(\mu)} \|\nabla_y G\|_{L^2(\mu)},$$

   where the covariance is bilinear for complex functions. In particular, if $V_{xy} = 0$ for $d(x,y) > r_0$ and the mixed blocks have row and column sums at most H_0, the hypothesis holds whenever $H_0(e^{br_0} - 1) \leq m/2$; exponentially decaying mixed Hessians are covered by the displayed condition.

3. *For every real symmetric $q \times q$ matrix A, the quadratic form $Q_A = \xi^{\mathsf{T}} A \xi$ satisfies*

$$\log \mathbb{E}_\mu e^{t(Q_A - \mathbb{E}_\mu Q_A)} \le \frac{8t^2}{m^2} \|A\|_{\mathrm{HS}}^2, \qquad |t| \le \frac{m}{4\|A\|_{\mathrm{op}}},$$

and for every $\zeta \in \mathbb{R}^q$,

$$\log \mathbb{E}_\mu e^{\zeta\cdot\xi + t(Q_A - \mathbb{E}_\mu Q_A)} \le \frac{|\zeta|^2}{m} + \frac{16t^2}{m^2} \|A\|_{\mathrm{HS}}^2, \qquad |t| \le \frac{m}{8\|A\|_{\mathrm{op}}},$$

where the range of t is all of $\mathbb{R}$ if $A = 0$.

4. *Suppose $q_x = 3$ for all x. Let $\mathcal{F}$ be a family of distinct pairs (e, f) of blocks such that every block belongs to at most D_0 pairs of $\mathcal{F}$, counting both orientations if both occur, and let $j_{ef} \in \mathbb{R}^3$. Then $\sum_{(e,f)\in\mathcal{F}} j_{ef} \cdot (\xi_e \times \xi_f) = \xi^{\mathsf{T}} A_j \xi$ for a real symmetric matrix A_j with*

$$\|A_j\|_{\mathrm{op}} \le \frac{D_0}{2} \|j\|_\infty, \qquad \|A_j\|_{\mathrm{HS}}^2 \le D_0 \sum_{(e,f)\in\mathcal{F}} |j_{ef}|^2.$$

If μ is invariant under the simultaneous rotation $w \mapsto (Rw_x)_x$ for every $R \in \mathrm{SO}(3)$, then this quadratic form has mean zero.

All statements concern the measure μ on the fixed convex set Ω, with its full potential V, and they are centered at the mean under μ. They give no convexity statement for marginals of μ or for measures on nonconvex sets.

Steps 1–6 construct the Neumann realization $\mathcal{L}$ of $-\Delta + \nabla V \cdot \nabla$ and a realization $\mathcal{A}_1$ of the corresponding operator on vector fields, whose form contains the Hessian of V and a nonnegative curvature term from the spheres. They also prove the commutation $\nabla \mathcal{L}^{-1} = \mathcal{A}_1^{-1} \nabla$, and establish the semigroup bound $|\nabla P_t h| \le e^{-mt} P_t |\nabla h|$. Steps 7 and 8 derive the entropy bound and, from $\mathrm{Cov}(F, G) = \langle \nabla F, \mathcal{A}_1^{-1} \nabla G \rangle$ and exponential conjugation, the covariance bound. Step 9 derives the quadratic moments by comparison with an auxiliary Gaussian vector and absorption of the relative entropy, without independent coordinates or a Hessian bound for the tilted measure, and Step 10 treats cross products. The construction of the operators in Steps 1 and 2 uses constants that may depend on the dimension, the radii, and the bounds of e^{-V} on the fixed compact set Ω; these constants do not enter the inequalities.

Proof of Theorem 5.4. All Hilbert spaces in this proof use the measure μ unless indicated otherwise. We write $B_x = \{|w_x| < p_x\}$, and n_x for the outer normal of the face $\partial B_x \times \prod_{z \neq x} B_z$.

Step 1: the scalar operator. The Sobolev space $H^1(\Omega)$ consists of the L^2 functions with first distributional derivatives in L^2; the weighted and unweighted norms are equivalent, since e^{-V} is bounded above and below on Ω. Smooth functions on $\overline{\Omega}$ are dense in $H^1(\Omega)$, and the embedding $H^1(\Omega) \to L^2(\Omega)$ is compact. For both facts we extend functions to a fixed ambient rectangle. On a ball of radius p, we reflect the radial coordinate $r \in (p, 3p/2)$ to $2p - r \in (p/2, p)$, multiply the reflected function by a smooth cutoff equal to one near $r = p$ and zero near $r = 3p/2$, and set it zero beyond. The Jacobians on the annulus are bounded above and below and the values match at $r = p$; differentiation in the radial and tangential directions shows that this extension is bounded on H^1. For an interval we reflect at each endpoint. Applying these extensions successively in the finitely many factors gives a bounded extension operator into H^1 of a rectangle, and convolution with smooth approximate identities proves density. For compactness, we replace an extended function u by its averages over a grid of cubes of side ε. Integrating $u(t) - u(s) = \int_s^t u'$ in each coordinate and using Cauchy–Schwarz, we bound the squared L^2 error by $C\varepsilon^2 \|\nabla u\|_2^2$. The averages lie in a fixed finite-dimensional space. From a bounded sequence in H^1 we may therefore extract, by a diagonal argument over $\varepsilon = 2^{-k}$, a subsequence converging in L^2.

The form $q_0(h,k) = \mathbb{E}_\mu \nabla h \cdot \nabla k$ on $H^1(\mu)$ is closed. For $f \in L^2(\mu)$, minimizing $q_0(h,h) + \|h\|_2^2 - 2\langle f, h\rangle$ gives a unique weak solution of $(I + \mathcal{L})h = f$. Indeed, a minimizing sequence is bounded, and the parallelogram identity shows that it is Cauchy in the form norm. Its limit lies in H^1 since distributional derivatives pass to L^2 limits, the first variation gives the weak equation, and strict positivity of the form gives uniqueness. The solution operator $T = (I + \mathcal{L})^{-1}$ is positive, self-adjoint, and compact by the compact embedding. It has a complete orthonormal eigenbasis. Indeed, we maximize $\langle Tu, u\rangle$ on the unit sphere. A maximizing sequence has a weakly convergent subsequence, and the compactness of T lets us pass to the limit in $\langle Tu, u\rangle$; positivity forces the limit to have norm one, and the first variation shows that it is an eigenvector. We repeat this in the orthogonal complement of the eigenvectors already found. The eigenvalues tend to zero, since otherwise the images under T of these orthonormal vectors would have no convergent subsequence. On the orthogonal complement of all the eigenvectors the supremum of $\langle Tu, u\rangle$ is zero. By polarization T vanishes there, and since T is injective this complement is trivial. Hence $\mathcal{L}$ has an orthonormal basis of eigenfunctions with eigenvalues $0 \le \lambda_\nu \to \infty$. Where its derivatives are defined, $\mathcal{L} = -\Delta + \nabla V \cdot \nabla$, with the weak Neumann condition. The kernel of $\mathcal{L}$ consists of the constants. Indeed, a function with zero energy has zero distributional gradient. It is therefore constant on each ball contained in Ω, and these constants agree since Ω is connected. There also exists a positive gap on functions of mean zero, since otherwise normalized functions of mean zero with gradients tending to zero would converge, by compactness, to a constant of mean zero and norm one. This preliminary gap is used only for the existence of solutions

of Poisson equations and for convergence to equilibrium.

Step 2: a core for the unweighted Laplacian. Let A_0 be the unweighted Neumann Laplacian on Ω, with respect to $L^2(dw)$. On a single ball, the construction of Step 1 gives a Neumann eigenbasis. Its eigenfunctions are smooth up to the boundary, and weak Neumann solutions are in H^2. In the interior this is the standard difference-quotient argument. Testing the weak equation with $-D_{-h,j}(\chi^2 D_{h,j}u)$, where χ is a cutoff and $D_{h,j}$ a difference quotient, gives

$$c\|\chi D_{h,j}\nabla u\|_2^2 \le \frac{c}{2}\|\chi D_{h,j}\nabla u\|_2^2 + C\big(\|f\|_2^2 + \|u\|_{H^1}^2\big)$$

for uniformly elliptic Lipschitz coefficients a. Here the leading term on the left is $\int \chi^2 a(\cdot + he_j)D_{h,j}\nabla u \cdot D_{h,j}\nabla u$, and all remaining terms contain either $D_{h,j}a$, which is bounded by the Lipschitz constant of a, or a derivative of χ. The right side $\int f[-D_{-h,j}(\chi^2 D_{h,j}u)]$ is bounded by $C\|f\|_2(\|\chi D_{h,j}\nabla u\|_2 + \|\nabla u\|_2)$, by the line-integral bound for a difference quotient, and the inequality $rs \le \epsilon r^2 + s^2/(4\epsilon)$ absorbs these terms. Uniformly bounded difference quotients yield weak second derivatives, since a subsequence converges weakly in L^2, and testing against smooth compactly supported functions identifies the limit as the distributional derivative. Near the boundary we use the inward normal distance $s \ge 0$ and coordinates y on the sphere; the metric has no mixed normal-tangential term and normal coefficient one, and its volume density J is smooth and positive on a thin collar. The weak Neumann equation reads $-\partial_s(J\partial_s u) - \partial_{y_\alpha}(Jg^{\alpha\beta}\partial_{y_\beta}u) = Jf$. Reflecting $u, J, g^{\alpha\beta}, f$ evenly in s gives Lipschitz, uniformly elliptic coefficients and a weak equation across $s = 0$, since adding its tests on the two half-collars gives exactly the Neumann weak equation. The interior estimate then gives H^2 up to the boundary, and integration by parts identifies the zero normal derivative. For an eigenfunction, $f = \lambda u$, we repeat the tangential difference-quotient argument in the half-collar; at each order the commutators differentiate only the smooth metric coefficients and lower derivatives already controlled, so all tangential derivatives gain one more H^1 derivative. The equation then solves for $\partial_s^2 u$, and differentiating it recovers successively the derivatives with two, three, and more normal indices; by induction $u \in H^k$ for every k. To deduce smoothness, we extend u from a half-chart by a finite linear combination $\sum_j c_j u(y, -js)$ of reflections, whose coefficients solve the invertible Vandermonde system $\sum_j c_j(-j)^l = 1$, $0 \le l < k$; this gives an H^k extension. We multiply it by an interior cutoff in an ambient rectangle with q coordinates. Repeated integration in the q coordinate directions and the Cauchy–Schwarz inequality bound the supremum of a derivative of order r of a smooth function by $C\|u\|_{H^{r+q}}$ on the rectangle. For example, the function itself is the iterated integral of $\partial_1 \cdots \partial_q u$ from each coordinate to the far side of the support. Applying this bound to smooth approximations of the extension and to their differences, we see that their derivatives through order r converge uniformly. It follows that the H^{r+q} function has continuous derivatives of order

r. Since k is arbitrary, u is smooth. For an interval the same conclusions follow from $-u'' = f$ and the Neumann conditions at the endpoints.

Finite products of these eigenfunctions are smooth, satisfy the Neumann condition on every face, and form an orthonormal basis of $L^2(dw)$ on Ω, as one sees by expanding successively in each factor using Fubini's theorem. Their eigenvalues for A_0 are sums of eigenvalues of the factors. They form a core for A_0. Indeed, the weak equation shows that every $h \in D(A_0)$ has coefficients with $\sum_\nu (1+\lambda_\nu^2)|h_\nu|^2 < \infty$; conversely, if this sum is finite, the partial sums solve the weak equation and converge in the graph norm, which shows that $h \in D(A_0)$. The same argument with $1+\lambda_\nu$ describes the form domain; in particular, a function orthogonal to all product eigenfunctions in the form norm has $(1+\lambda_\nu)h_\nu = 0$ for all ν and hence vanishes, which proves density in the form norm. Truncating the expansion therefore gives convergence in the graph norm and in the form norm.

Step 3: the boundary identity. For a function h in this core, two integrations by parts, one factor at a time, give

$$\|D^2 h\|^2_{L^2(dw)} + \sum_x \int_{\partial B_x \times \prod_{z\neq x} B_z} p_x^{-1} |(\nabla_x h)^{\tan}|^2 \, dS = \|A_0 h\|^2_{L^2(dw)}. \tag{5.9}$$

Indeed, the difference of the two bulk terms is the boundary integral of $(\Delta h)\partial_n h - D^2 h(n, \nabla h)$. The first term vanishes. Differentiating $\partial_{n_x} h = 0$ tangentially along the face of x, we find $D^2 h(n_x, \nabla h) = -p_x^{-1}|(\nabla_x h)^{\tan}|^2$, because $D_v n_x = p_x^{-1} v_x^{\tan}$. For an interval the curvature term is zero. By Fubini's theorem on the individual balls, the identity holds on the product. Since intersections of distinct faces have zero surface measure, there is no integral over a corner. Applied to differences of core functions, the identity shows that convergence in the graph norm of A_0 implies convergence in H^2 and in the boundary term.

Step 4: the operator on vector fields. Let $\mathcal{H}_1$ be the space of H^1 vector fields $u = (u_x)_{x\in X}$ with $u_x \cdot n_x = 0$ on the face of x, in the sense of traces. Traces are continuous on H^1: in the collar, $|u(y,0)|^2 \le C\int_0^\varepsilon (|u(y,s)|^2 + |u(y,s)||\partial_s u(y,s)|)\, ds$, and Cauchy–Schwarz applies. Smooth vector fields satisfying the face conditions are dense in $\mathcal{H}_1$. To see this, we reflect the normal component of u_x oddly and its tangential components evenly in the local normal frame of the face of x. The odd component has zero trace, and no jump occurs. Near an intersection of faces we do this separately in each active ball factor; the reflections commute, since the normal frames of different factors depend on different variables. Convolving with a kernel that is even in every reflected normal coordinate and restricting to the product half-chart preserves the zero normal components. A partition of unity, constant in the normal coordinate near each face, and ordinary convolution in the interior complete the approximation, including near all corners.

On $\mathcal{H}_1$ we define the symmetric form

$$q_1(u,v) = \mathbb{E}_\mu \sum_i \partial_i u \cdot \partial_i v + \mathbb{E}_\mu u^\mathsf{T}(D^2V)v \\ + \sum_x \int_{\partial B_x \times \prod_{z\neq x} B_z} p_x^{-1} u_x^{\tan} \cdot v_x^{\tan}\, Z^{-1} e^{-V}\, dS. \tag{5.10}$$

By the trace estimate, the boundedness of D^2V, and $D^2V \geq mI$, its norm is equivalent to the H^1 norm, and the form is therefore closed. The associated self-adjoint operator $\mathcal{A}_1$, obtained as in Step 1, satisfies

$$q_1(u,u) \geq m\|u\|_2^2, \qquad \|\mathcal{A}_1^{-1}\|_{2\to 2} \leq m^{-1}. \tag{5.11}$$

Only this lower bound enters the final constants. Formally $\mathcal{A}_1 = \mathcal{L}I + D^2V$; on the face of x the tangential part of the block u_x satisfies the Robin condition $\partial_{n_x} u_x^{\tan} + p_x^{-1} u_x^{\tan} = 0$, while the other blocks satisfy the Neumann condition. At intersections of faces, the closed form (5.10) is the definition.

Step 5: gradient commutation. Let $\mathcal{L}h = g$ weakly. Testing with e^V times an unweighted H^1 test function, we find that $A_0 h = g - \nabla V \cdot \nabla h$ weakly in $L^2(dw)$. Since the right side is in L^2, we have $h \in D(A_0)$, and there exist core functions h_j converging to h in the graph norm of A_0. By (5.9), $h_j \to h$ in H^2 and in the boundary term, and $\mathcal{L}h_j \to g$ in $L^2(\mu)$. For a smooth vector field v satisfying the face conditions, differentiation and integration by parts give

$$q_1(\nabla h_j, v) = \langle \mathcal{L}h_j, d_V^* v\rangle, \qquad d_V^* v = -\operatorname{div} v + \nabla V \cdot v. \tag{5.12}$$

Indeed, $\nabla \mathcal{L}h_j = \mathcal{L}\nabla h_j + D^2V \nabla h_j$, and on the face of x the normal derivative of the gradient, paired with v, equals $-p_x^{-1}(\nabla_x h_j)^{\tan} \cdot v_x^{\tan}$, while the other blocks have zero normal derivative there; these are exactly the boundary terms in (5.10). The scalar integration by parts produces no further term, since the normal traces of v vanish. By density, (5.12) extends to $v \in \mathcal{H}_1$. Passing to the limit $j \to \infty$, and taking $g \in H^1$, we obtain

$$\nabla h = \mathcal{A}_1^{-1} \nabla g. \tag{5.13}$$

This uses only the C^2 regularity of V: ∇V is bounded, and the differentiation in (5.12) uses only continuous second derivatives of V.

Step 6: semigroups and the gradient bound. Using the eigenbasis of Step 1, we define $P_t = e^{-t\mathcal{L}}$ by multiplying the coefficients by $e^{-t\lambda_\nu}$. It preserves constants, positivity, and the interval $[0,1]$. For the last two properties, we solve $(I + s\mathcal{L})h = f$ and test with h^- and $(h-1)^+$. If $0 \leq f \leq 1$, the first test gives $h^- = 0$ and the second $(h-1)^+ = 0$; since $P_t = \lim_k (I + (t/k)\mathcal{L})^{-k}$ in L^2, these properties pass to P_t. Symmetry and $P_t 1 = 1$ imply the invariance of μ, and the preliminary gap implies $P_t h \to \mathbb{E}_\mu h$ in L^2.

The variational construction applied to q_1 on vector-valued L^2, with form domain $\mathcal{H}_1$ (dense, since it contains the compactly supported smooth vector fields, and compactly embedded by the scalar argument), gives the semigroup $e^{-t\mathcal{A}_1}$. For $h \in H^1$ and $t > 0$, the spectral expansion puts $P_t h$ and $\mathcal{L} P_t h$ in the required domains, and (5.12) shows that $u_t = \nabla P_t h$ solves $\partial_t u_t = -\mathcal{A}_1 u_t$ weakly, with initial value ∇h. Uniqueness follows by differentiating the squared norm of the difference of two solutions and using (5.11). We conclude that $\nabla P_t h = e^{-t\mathcal{A}_1} \nabla h$.

We claim that

$$|e^{-t\mathcal{A}_1} u| \le e^{-mt} P_t |u|. \tag{5.14}$$

Let $u_t = e^{-t\mathcal{A}_1} u$ and $h_\varepsilon = (|u_t|^2 + \varepsilon^2)^{1/2}$. In the weak equation for u_t we use the test field $\psi u_t / h_\varepsilon$, with $\psi \in H^1$ bounded and nonnegative; it belongs to $\mathcal{H}_1$, since multiplication by a scalar preserves the face conditions. The kinetic term equals

$$\nabla h_\varepsilon \cdot \nabla \psi + \psi \Big(\frac{|\nabla u_t|^2}{h_\varepsilon} - \frac{|\nabla h_\varepsilon|^2}{h_\varepsilon} \Big),$$

whose second summand is nonnegative by Cauchy–Schwarz. The Hessian term is at least $m\psi |u_t|^2 / h_\varepsilon \ge m\psi (h_\varepsilon - \varepsilon)$, and every boundary curvature term is nonnegative. Hence $(\partial_t + \mathcal{L} + m)(h_\varepsilon - \varepsilon) \le 0$ in the weak scalar sense. We compare with $e^{-mt} P_t |u|$ and test the difference with its positive part; the squared L^2 norm of this positive part does not increase from zero. This comparison is justified first on intervals $[\delta, T]$, where the spectral solutions have time derivatives in L^2 and lie in the form domains, and then by continuity in L^2 as $\delta \downarrow 0$; truncation of the tests and approximation in the form norm justify the chain rule, whose derivatives are bounded for fixed ε. Letting $\varepsilon \downarrow 0$ proves (5.14). Combined with the previous paragraph, this gives

$$|\nabla P_t h| \le e^{-mt} P_t |\nabla h|.$$

Step 7: entropy, variance, and exponential moments. Let h be smooth, positive, and bounded above and below. Applying the positive operator P_t to $(f - ch)^2 / h \ge 0$ for every rational c and minimizing over c, we obtain the pointwise inequality $(P_t f)^2 \le P_t(f^2/h)\, P_t h$. With $f = |\nabla h|$ and the gradient bound of Step 6, we obtain

$$\frac{|\nabla P_t h|^2}{P_t h} \le e^{-2mt} P_t \Big(\frac{|\nabla h|^2}{h} \Big). \tag{5.15}$$

By the weak equation and the chain rule for $\log P_t h$, we have

$$-\frac{d}{dt} \mathbb{E}_\mu [(P_t h) \log P_t h] = \mathbb{E}_\mu \frac{|\nabla P_t h|^2}{P_t h}.$$

Since the bounds on h persist under P_t, convergence of $P_t h$ to its mean in L^2 implies convergence of the entropy integral to $(\mathbb{E} h) \log \mathbb{E} h$. Integrating in time and using (5.15)

and the invariance of μ, we find $\mathrm{Ent}_\mu(h) \le (2m)^{-1}\mathbb{E}_\mu|\nabla h|^2/h$. For bounded smooth f, substituting $h = f^2 + \varepsilon$ and letting $\varepsilon \downarrow 0$, we obtain $\mathrm{Ent}_\mu(f^2) \le (2/m)\mathbb{E}_\mu|\nabla f|^2$. For general $f \in H^1$, we truncate, approximate by smooth functions, and use that the entropy is lower semicontinuous under convergence in L^2. Indeed, $s \log s$ is bounded below, and Fatou's lemma applies along an almost everywhere convergent subsequence, while the mean term converges in L^1. This proves the logarithmic Sobolev inequality.

Expanding at $f = 1 + \varepsilon g$ with g bounded, we find $\mathrm{Ent}((1+\varepsilon g)^2) = 2\varepsilon^2 \operatorname{Var}(g) + o(\varepsilon^2)$. Then the logarithmic Sobolev inequality implies the Poincaré inequality with constant $1/m$; truncation and density extend it to H^1. For a real Lipschitz F let $\varphi(t) = \log \mathbb{E}_\mu e^{tF}$. Applying the logarithmic Sobolev inequality to $e^{tF/2}$, we find $t\varphi'(t) - \varphi(t) \le t^2 \operatorname{Lip}(F)^2/(2m)$. Integrating $(\varphi(t)/t)'$ from zero proves the exponential moment bound for $t > 0$, and applying it to $-F$ gives $t < 0$. To justify the chain rule, we extend F to a neighborhood by $\inf_{y\in\Omega}(F(y) + \operatorname{Lip}(F)|x-y|)$ and smooth; this preserves the Lipschitz constant and converges uniformly on Ω; no differentiability of F at the boundary is needed. This proves part (1).

Step 8: covariances. For real $G \in H^1$, we solve $\mathcal{L}h = G - \mathbb{E}G$ with $\mathbb{E}h = 0$, using the preliminary gap. By weak integration by parts and (5.13), we have

$$\operatorname{Cov}_\mu(F, G) = \langle \nabla F, \mathcal{A}_1^{-1} \nabla G \rangle_{L^2(\mu)}, \tag{5.16}$$

first for smooth F and then for $F \in H^1$ by density. Complexifying gives the bilinear identity for complex F, G, with neither function conjugated; its absolute value is bounded by the Cauchy–Schwarz inequality in the complex Hilbert space.

Fix $y_0 \in X$ and set $W_x = e^{bd(x,y_0)} I_{q_x}$ on the components of the block x. This weight does not depend on w; it preserves the face conditions and commutes with the kinetic and boundary forms and with every diagonal block of the Hessian. In the closed-form realization, we obtain

$$W\mathcal{A}_1 W^{-1} = \mathcal{A}_1 + B, \qquad B_{xz} = \big(e^{b(d(x,y_0)-d(z,y_0))} - 1\big) V_{xz}(w).$$

Each block multiplier has norm at most $(e^{bd(x,z)} - 1)\sup_\Omega \|V_{xz}\|$. By the row-column bound of Step 5 in the proof of Lemma 5.3 and the hypothesis of part (2), we have $\|B\|_{2\to 2} \le m/2$ on the direct sum of the block spaces $L^2(\mu)$. By (5.11) and a Neumann series, $\|(W\mathcal{A}_1 W^{-1})^{-1}\| \le 2/m$, and the (x, y_0) block satisfies

$$\|(\mathcal{A}_1^{-1})_{xy_0}\|_{L^2\to L^2} \le \frac{2}{m} e^{-bd(x,y_0)}.$$

Inserting this bound into (5.16) and applying Cauchy–Schwarz to each block pairing, we obtain the covariance bound of part (2), for real and complex functions. For an interaction of range r_0, every off-diagonal weight is at most $e^{br_0} - 1$, which gives the stated criterion. Neither the individual p_x, q_x nor an upper bound on the diagonal Hessian blocks entered.

Step 9: quadratic moments. The classical Hanson–Wright inequality concerns independent coordinates [108], and extensions based on concentration allow dependent vectors [1, Theorem 2.5, p. 4]; here we derive the required bounds directly from the logarithmic Sobolev inequality for μ. Applying part (1) to the linear function $v \cdot w$, we obtain

$$\mathbb{E}_\mu e^{v\cdot\xi} \le e^{|v|^2/(2m)}. \tag{5.17}$$

If $A = 0$, all quadratic statements follow from this. Let $A \neq 0$, and let Z_0 be a standard Gaussian vector in $\mathbb{R}^q$, independent of w. Since $e^{\alpha|y|^2} = \mathbb{E}e^{\sqrt{2\alpha}Z_0\cdot y}$ (completion of the square, as in Lemma 5.2) and all integrands are positive, we have for $\alpha \ge 0$

$$\mathbb{E}_\mu e^{\alpha|A\xi|^2} = \mathbb{E}_{Z_0}\mathbb{E}_\mu e^{\sqrt{2\alpha}(AZ_0)\cdot\xi} \le \mathbb{E}_{Z_0} e^{(\alpha/m)|AZ_0|^2} = \det\bigl(I - (2\alpha/m)A^2\bigr)^{-1/2},$$

provided $2\alpha\|A\|_{\mathrm{op}}^2 < m$. At $\alpha = m/(4\|A\|_{\mathrm{op}}^2)$ all eigenvalues of $(2\alpha/m)A^2$ are at most $1/2$, and since $-\log(1-s) \le 2s$ on $[0,1/2]$,

$$\log \mathbb{E}_\mu e^{\alpha|A\xi|^2} \le \frac{2\alpha}{m}\|A\|_{\mathrm{HS}}^2. \tag{5.18}$$

Now let $Q = \xi^{\mathsf{T}} A\xi$, $\psi(t) = \log \mathbb{E}_\mu e^{tQ}$, and $d\mu_t = e^{tQ-\psi(t)}d\mu$. All these quantities are finite and smooth in t, since the domain is compact. The relative entropy of μ_t with respect to μ is $D_t = t\psi'(t) - \psi(t)$; with $r = d\mu_t/d\mu$, integrating $r\log r \ge r - 1$ against μ, we find $D_t \ge 0$. Applying the logarithmic Sobolev inequality to $e^{tQ/2}$ and using $\nabla Q = 2A\xi$, we find $D_t \le (2t^2/m)\mathbb{E}_{\mu_t}|A\xi|^2$. For any real random variable Y, the entropy inequality $\alpha\mathbb{E}_{\mu_t}Y \le D_t + \log \mathbb{E}_\mu e^{\alpha Y}$ holds (Jensen's inequality for the logarithm, applied to $e^{\alpha Y}d\mu/d\mu_t$ under μ_t). With $Y = |A\xi|^2$ and (5.18), we obtain

$$D_t \le \frac{8t^2\|A\|_{\mathrm{op}}^2}{m^2}D_t + \frac{4t^2}{m^2}\|A\|_{\mathrm{HS}}^2.$$

For $|t| \le m/(4\|A\|_{\mathrm{op}})$ the first coefficient is at most $1/2$, which implies $D_t \le 8t^2\|A\|_{\mathrm{HS}}^2/m^2$. For $t > 0$, integrating $(\psi(t)/t)' = D_t/t^2$ from zero, where $\psi(t)/t \to \mathbb{E}_\mu Q$, proves the first bound of part (3); replacing A by $-A$ gives $t < 0$. For the joint bound, by the Cauchy–Schwarz inequality and (5.17), we have

$$\log \mathbb{E}_\mu e^{\zeta\cdot\xi + t(Q-\mathbb{E}Q)} \le \frac{1}{2}\log \mathbb{E}_\mu e^{2\zeta\cdot\xi} + \frac{1}{2}\log \mathbb{E}_\mu e^{2t(Q-\mathbb{E}Q)} \le \frac{|\zeta|^2}{m} + \frac{16t^2}{m^2}\|A\|_{\mathrm{HS}}^2,$$

for $|t| \le m/(8\|A\|_{\mathrm{op}})$. This proves part (3).

Step 10: cross products. We write $J(j)v = j \times v$. A pair (e,f) contributes the symmetric matrix with blocks $A_{ef} = -J(j_{ef})/2$ and $A_{fe} = A_{ef}^{\mathsf{T}}$, because $\xi_e^{\mathsf{T}}[-J(j_{ef})]\xi_f = j_{ef}\cdot(\xi_e \times \xi_f)$. Since $\|J(j)\|_{\mathrm{op}} = |j|$ and $\|J(j)\|_{\mathrm{HS}}^2 = 2|j|^2$, a single pair has block row sums

$|j_{ef}|/2$ and squared Hilbert–Schmidt norm $|j_{ef}|^2$. Since at most D_0 pairs meet a block, the row-column bound implies $\|A_j\|_{\mathrm{op}} \le D_0\|j\|_\infty/2$; using $|\sum_{k=1}^r b_k|^2 \le r\sum_k |b_k|^2$ with $r \le D_0$ in each repeated matrix entry, we obtain $\|A_j\|_{\mathrm{HS}}^2 \le D_0 \sum |j_{ef}|^2$. Pairs with $e = f$ contribute zero. If μ is invariant under simultaneous rotations, then $\mathbb{E}_\mu w$ is invariant and ξ transforms as a vector. The law of ξ therefore has the same invariance. The vector $\mathbb{E}(\xi_e \times \xi_f)$ is then fixed by every rotation, hence zero (a rotation by π about an axis perpendicular to a nonzero vector reverses it). This proves part (4). □

5.5 Conditioning, exponential moments, and gauge identities

The next lemma gives conditioning bounds and identities for parameter derivatives and gauge transformations; the bound on large fields enters the proof through the comparison of a measure with its conditioning (see (3.1)). For a link e and $a \in \{1,2,3\}$ we write L_e^a and R_e^a for the derivatives of a function of U along $s \mapsto e^{s\tau_a}U_e$ and $s \mapsto U_e e^{s\tau_a}$ at $s = 0$, all other link variables being fixed.

Lemma 5.5. *Let μ be a probability measure on a measurable space, and let S be a measurable event with $e = \mu(S^c) < 1$. Put $\nu = \mu(\cdot \mid S)$. Then the following statements hold.*

1. *With $\|\mu - \nu\|_{\mathrm{TV}} = \sup_A |\mu(A) - \nu(A)|$, we have $\|\mu - \nu\|_{\mathrm{TV}} = e$, and*

$$|\mathbb{E}_\mu F - \mathbb{E}_\nu F| \le 2e\|F\|_\infty$$

   for every bounded complex measurable F; if $0 \le F \le A$, the bound is eA. The image measures of μ and ν under a common measurable map satisfy the same bound on the total variation distance. If $d\mu = Z^{-1} f\, d\omega$ with $f \ge 0$, then the restricted partition function $Z_S = \int_S f\, d\omega$ satisfies $Z_S/Z = 1 - e$, and $|\log(Z_S/Z)| \le 2e$ if $e \le 1/2$.

2. *For nested events $S_0 \supset S_1 \supset \cdots \supset S_k$ of positive probability and $0 \le i \le k$,*

$$\mu(S_k) = \mu(S_i) \prod_{j=i}^{k-1} \mu(S_{j+1} \mid S_j).$$

   With $r_j = \mu(S_j \setminus S_{j+1} \mid S_j)$, it follows that $1 - \mu(S_k) \le 1 - \mu(S_i) + \sum_{j=i}^{k-1} r_j$; the successive conditional failure probabilities are added without dividing by $\mu(S_i)$.

3. *For a real function J with $|J| \le R$, the tilted probability measure $\mu_J = e^J \mu / \mathbb{E}_\mu e^J$ satisfies $\mu_J(A) \le e^{2R}\mu(A)$ for every event A.*

4. *Let $O_1, \dots, O_r$ be real measurable functions with $|O_i| \le 1$, and set $M_\mu(z) = \mathbb{E}_\mu e^{\sum_i z_i O_i}$ and $M_\nu(z) = \mathbb{E}_\nu e^{\sum_i z_i O_i}$ for $z \in \mathbb{C}^r$. If $\sum_i |z_i| \le R$, then*

$$|M_\mu(z) - M_\nu(z)| \le 2e^R e.$$

If moreover $R < \log 2$, the branches of $\log M_\mu$ and $\log M_\nu$ vanishing at $z = 0$ satisfy

$$|\log M_\mu(z) - \log M_\nu(z)| \le \frac{2e^R}{2 - e^R}\, e,$$

and their derivatives satisfy the corresponding Cauchy bounds on every smaller polydisc, with constants depending only on the two polydiscs and the order of the derivative.

5. *Let ω be a measure on a fixed domain, let $V(\theta, w)$ be real with $0 < Z(\theta) = \int e^{-V(\theta,w)}\, d\omega(w) < \infty$, and let $F(\theta, w)$ be a function such that both e^{-V} and Fe^{-V} can be differentiated in θ under the integral. A sufficient condition is pointwise differentiability in θ together with an ω-integrable envelope, uniform for θ near the given parameter, for*

$$e^{-V}\big(1 + |\partial_\theta V| + |F| + |\partial_\theta F| + |F\partial_\theta V|\big);$$

joint C^1 regularity of V and F on a fixed compact domain and a neighborhood of the parameter suffices. Then, with $p_\theta = Z(\theta)^{-1} e^{-V(\theta,\cdot)}$,

$$\partial_\theta \log Z = -\mathbb{E}_\theta \partial_\theta V, \qquad \partial_\theta \log p_\theta = -\partial_\theta V + \mathbb{E}_\theta \partial_\theta V,$$

$$\partial_\theta \mathbb{E}_\theta F = \mathbb{E}_\theta \partial_\theta F - \mathrm{Cov}_\theta(F, \partial_\theta V).$$

These are identities for the full normalized density, and the covariance is bilinear for complex F.

6. *Let μ be the Wilson measure on a periodic lattice, or more generally a probability measure of the form in Lemma 5.1(1), and let $\mathscr{G}_v^a = \sum_{e \text{ starts at } v} L_e^a - \sum_{e \text{ ends at } v} R_e^a$ be the gauge generator at the vertex v. Then for all C^1 functions F and J, possibly complex,*

$$\mathbb{E}_\mu\big[e^J(\mathscr{G}_v^a F + F\mathscr{G}_v^a J)\big] = 0.$$

In a spanning-tree gauge, only the corresponding identity for the simultaneous conjugation at the root remains. If S is invariant under the gauge transformations, the same identity holds under $\mu(\cdot \mid S)$, without any regularity of the boundary of S. The same conclusion holds in any finite joint coordinate representation whose full normalized measure is invariant. In particular, a triangular change of variables $w_i' = O_i(\theta, w_{i+1}, \dots, w_r) w_i$ by orthogonal matrices O_i, independent of w_i, has Jacobian of absolute value one, preserves every product of balls centered at the origin, and its infinitesimal generator has zero flux through every face; the parameter-dependent normalizing factors remain in the joint derivative of the log-density.

7. *If F is integrable but possibly unbounded and $e > 0$, there is only the exact identity*

$$\mathbb{E}_\mu F - \mathbb{E}_\nu F = e\big(\mathbb{E}_{\mu(\cdot|S^c)} F - \mathbb{E}_\nu F\big);$$

an error tending to zero requires additional control of these integrals.

All constants are explicit as displayed.

Proof. Parts (1)–(3). If $e = 0$, then $\mu = \nu$ and there is nothing to prove. For $e > 0$ let $\nu^c = \mu(\cdot \mid S^c)$. Splitting every integral over S and S^c, we obtain

$$\mu = (1 - e)\nu + e\nu^c. \tag{5.19}$$

For an event A, we have $\mu(A) - \nu(A) = e(\nu^c(A) - \nu(A))$, where both $\nu(A)$ and $\nu^c(A)$ lie in $[0, 1]$. This bounds the total variation distance by e, and $A = S$ attains it. For complex F, the two conditional expectations have absolute value at most $\|F\|_\infty$, which gives $2e\|F\|_\infty$; if $0 \le F \le A$, their difference is at most A. A common measurable map replaces each test event by its preimage and therefore does not increase the total variation distance. In the density notation, $Z_S = \int_S f \, d\omega = Z\mu(S)$, and for $0 \le e \le 1/2$ we have $-\log(1 - e) = \int_0^e (1 - t)^{-1} dt \le 2e$.

For nested events, consecutive conditional probabilities cancel:

$$\prod_{j=i}^{k-1} \mu(S_{j+1} \mid S_j) = \prod_{j=i}^{k-1} \frac{\mu(S_{j+1})}{\mu(S_j)} = \frac{\mu(S_k)}{\mu(S_i)}.$$

Induction on the number of factors gives $1 - \prod_j (1 - r_j) \le \sum_j r_j$, the induction step being $1 - (1 - a)(1 - b) = a + b - ab \le a + b$. Since $\mu(S_k) = \mu(S_i) \prod_j (1 - r_j) \ge \mu(S_i) - (1 - \prod_j (1 - r_j))$, we obtain $1 - \mu(S_k) \le 1 - \mu(S_i) + \sum_j r_j$. For $|J| \le R$, the density of μ_J with respect to μ is at most $e^R / e^{-R} = e^{2R}$.

Part (4). Let $T(z) = \sum_i z_i O_i$; on the ball $\sum_i |z_i| \le R$ we have $|T(z)| \le R$ and $|e^{T(z)}| \le e^R$. Applying part (1) to $e^{T(z)}$, we obtain $|M_\mu - M_\nu| \le 2e^R e$. By the exponential series, we have $|M_\mu(z) - 1| \le e^R - 1$ and $|M_\nu(z) - 1| \le e^R - 1$. If $R < \log 2$, these values lie in the open disc of center 1 and radius $e^R - 1 < 1$, where the series of $\log(1 + w)$ defines the branch vanishing at the origin. Integrating its derivative $1/u$ along the segment between the two values, on which $|u| \ge 2 - e^R$, we find

$$|\log M_\mu(z) - \log M_\nu(z)| \le (2 - e^R)^{-1} |M_\mu(z) - M_\nu(z)|.$$

The functions are holomorphic, since bounded observables allow termwise integration of the exponential series. If a smaller polydisc has coordinate distances $\delta_i > 0$ to a larger one inside the ball, the Cauchy formula gives

$$|\partial^\alpha (\log M_\mu - \log M_\nu)| \le \alpha! \prod_i \delta_i^{-\alpha_i} \frac{2e^R}{2 - e^R} \, e$$

at every point of the smaller polydisc.

Part (5). Fix θ_0 and a neighborhood on which the integrable envelope $B(w)$ is valid. We write $g_\theta = e^{-V(\theta, \cdot)}$ and $h_\theta = F(\theta, \cdot) e^{-V(\theta, \cdot)}$. Their pointwise derivatives are $\partial_\theta g_\theta = -(\partial_\theta V) e^{-V}$ and $\partial_\theta h_\theta = (\partial_\theta F - F \partial_\theta V) e^{-V}$. The envelope bounds $|g_\theta|$, $|h_\theta|$,

and both derivatives by B, and the mean value theorem, applied to real and imaginary parts, bounds the difference quotients by $2B$. By dominated convergence, with $N(\theta) = \int F(\theta, w) e^{-V(\theta,w)}\, d\omega(w)$, we have

$$\partial_\theta Z = -\int (\partial_\theta V) e^{-V}\, d\omega, \qquad \partial_\theta N = \int (\partial_\theta F - F \partial_\theta V) e^{-V}\, d\omega. \tag{5.20}$$

For a vector parameter the argument applies to each coordinate derivative. Since $0 < Z < \infty$, dividing by Z gives the first identity, and differentiating $\log p_\theta = -V - \log Z$ gives the second. The quotient rule for $\mathbb{E}_\theta F = N/Z$ yields the third; the covariance is finite because the envelope also controls $|F \partial_\theta V| e^{-V}$. For the sufficient condition on a compact domain Ω, choose a compact neighborhood of the parameter inside the domain of C^1 regularity. Joint continuity bounds $|V|$, $|F|$, $|\partial_\theta V|$, and $|\partial_\theta F|$ by some $C \geq 1$ there. Moreover $e^{-V(\theta_0,w)} \geq e^{-C}$ and $Z(\theta_0) < \infty$ imply $\omega(\Omega) \leq e^C Z(\theta_0) < \infty$. Then the constant $e^C(1 + 3C + C^2)$ is an integrable envelope, uniformly on this neighborhood. No boundary term occurs, since the domain is fixed. In particular, a term of V that depends only on θ cancels in the derivative of $\log p_\theta$ only together with its contribution to $\log Z$; it cannot be deleted from a joint density.

Part (6). Let $k(s)$ equal $e^{s\tau_a}$ at v and I at all other vertices. By Lemma 5.1, μ is invariant under $U \mapsto U^{k(s)}$, which gives

$$\int (F e^J)(U^{k(s)})\, d\mu(U) = \int F e^J\, d\mu.$$

Differentiation at $s = 0$ is valid for C^1 functions on the compact manifold of configurations. On a link starting at v the derivative is L_e^a, and on a link ending at v it is $-R_e^a$; their sum is $\mathscr{G}_v^a$, and the product rule gives the identity, for real or complex J. In a spanning-tree gauge, the conjugation at the root described in Lemma 5.1 acts on every remaining link by $U_e \mapsto k U_e k^{-1}$; its derivative is $\sum_{e \notin T} (L_e^a - R_e^a)$, and the same change of variables gives the remaining identity. No generator at a vertex other than the root is used on the gauge-fixed configurations. If S is invariant, we insert $\mathbf{1}_S$ into the integral before the change of variables and divide by $\mu(S) > 0$. Neither the regularity of ∂S nor a derivative of $\mathbf{1}_S$ is needed, since the indicator is unchanged for every s.

For the triangular change of variables, given the transformed variables of higher index, we solve for w_i using O_i^{-1}, proceeding downward. This shows that the map is invertible, and it preserves the product of balls. Its Jacobian matrix is block triangular with orthogonal diagonal blocks O_i, and its determinant therefore has absolute value one. For a differentiable one-parameter family of such maps, the generator is $Y_i = B_i(\theta, w_{>i}) w_i$ with $B_i^{\mathsf{T}} = -B_i$. Hence

$$\operatorname{div}_{w_i} Y_i = \operatorname{tr} B_i = 0, \qquad w_i \cdot Y_i = 0 \quad \text{on } |w_i| = p_i,$$

and the flux through every face vanishes; intersections of faces carry no additional flux, since the map preserves each radius $|w_i|$. Under the assumed invariance of the full

joint probability measure, we obtain the gauge-invariance identity by differentiating the change-of-variables identity. If this measure is written with conditional factors $Z(\theta)^{-1}e^{-V(\theta,w)}$, their parameter derivatives are the derivatives of the normalized log-densities computed in part (5); a derivative of an unnormalized conditional density is never substituted into a joint identity.

Part (7). The identity (5.19) applies to an integrable F when $e > 0$, and both conditional integrals are finite since both events have positive probability. Without further control there is no small error: on a two-point space, give S^c probability e and take $F = e^{-2}\mathbf{1}_{S^c}$; then $\mathbb{E}_\nu F = 0$ and $\mathbb{E}_\mu F = e^{-1}$. □

5.6 Invariant functions of three holonomies

The weak-coupling results concern continuous functions of the three holonomies H_1, H_2, H_3 around the coordinate cycles of the torus, with a common basepoint, that are invariant under simultaneous conjugation. The next lemma shows that such functions are exactly the continuous functions of seven traces. If we write $H_i = a_i I + v_i \cdot t$, the traces of H_i and H_iH_j determine the scalar parts a_i and the Gram matrix of the vectors v_i. The trace of $H_1H_2H_3$ determines in addition the orientation $\det(v_1, v_2, v_3)$, and a triple of vectors is determined up to rotation by its Gram matrix and its orientation. These seven coordinates are the case of three generators in the classical theory of trace coordinates of Horowitz [111]; the argument below proves the separation and the continuous factorization for all triples in $\mathrm{SU}(2)$, including those of deficient rank.

Lemma 5.6. *Define* $\mathfrak{p} : \mathrm{SU}(2)^3 \to [-1,1]^7$ *by the normalized traces of* $H_1, H_2, H_3, H_1H_2, H_1H_3, H_2H_3, H_1H_2H_3$, *in this order. For all* $H, H' \in \mathrm{SU}(2)^3$,

$$\mathfrak{p}(H) = \mathfrak{p}(H') \quad \Longleftrightarrow \quad \text{there exists } k \in \mathrm{SU}(2) \text{ with } H_i' = kH_ik^{-1} \text{ for } i = 1,2,3.$$

This includes all triples of deficient rank and the central triples. Explicitly, if $H_i = a_i I + 2v_i^a \tau_a$ *with* $a_i^2 + |v_i|^2 = 1$, *the normalized trace of* H_iH_j *is* $a_ia_j - v_i \cdot v_j$, *and that of* $H_1H_2H_3$ *is*

$$a_1a_2a_3 - a_3 v_1 \cdot v_2 - a_1 v_2 \cdot v_3 - a_2 v_1 \cdot v_3 - \det(v_1, v_2, v_3).$$

The map induced by $\mathfrak{p}$ *from the quotient of* $\mathrm{SU}(2)^3$ *by simultaneous conjugation onto* $\mathfrak{p}(\mathrm{SU}(2)^3)$ *is a homeomorphism. Every continuous function* $F : \mathrm{SU}(2)^3 \to \mathbb{C}$ *invariant under simultaneous conjugation has the form* $F = f \circ \mathfrak{p}$, *where* f *is continuous on* $\mathfrak{p}(\mathrm{SU}(2)^3)$ *and extends to a continuous function on* $[-1,1]^7$ *with* $\|f\|_\infty \le \|F\|_\infty$. *In particular, if the trace vectors* $\mathfrak{p}(H^{(k)})$ *of random triples* $H^{(k)}$ *converge jointly in distribution, then for every such* F *the expectations* $\mathbb{E}F(H^{(k)})$ *converge, to the expectation of* f *under the limiting law, which is supported on* $\mathfrak{p}(\mathrm{SU}(2)^3)$.

The lemma concerns one triple of matrices and its seven traces; it does not by itself relate lattice holonomies to continuum observables.

Proof. By (5.1), $H_i = a_i I + v_i \cdot t$ with $t_a = 2\tau_a$; the scalar part a_i is the normalized trace, and $a_i^2 + |v_i|^2 = 1$. The scalar part of $H_i H_j$ is $a_i a_j - v_i \cdot v_j$. Multiplying once more, we find that the scalar part of $H_1 H_2 H_3$ is

$$(a_1 a_2 - v_1 \cdot v_2) a_3 - (a_1 v_2 + a_2 v_1 + v_1 \times v_2) \cdot v_3,$$

which expands to the stated formula, with the minus sign in front of $\det(v_1, v_2, v_3) = (v_1 \times v_2) \cdot v_3$ for our choice of τ_a.

Suppose $\mathfrak{p}(H) = \mathfrak{p}(H')$. Then $a_i = a_i'$, the Gram matrices $(v_i \cdot v_j)$ and $(v_i' \cdot v_j')$ agree (including the diagonal entries $1 - a_i^2$), and $\det(v_1, v_2, v_3) = \det(v_1', v_2', v_3')$. We define a map between the spans by $T(\sum_i c_i v_i) = \sum_i c_i v_i'$. Equality of the Gram matrices shows that T is well defined, since if $\sum_i c_i v_i = 0$ then $|\sum_i c_i v_i'|^2$ is the same quadratic form in c and vanishes, and that T is an isometry onto the span of the v_i'. Completing orthonormal bases of the two spans to bases of $\mathbb{R}^3$, we extend T to an orthogonal map R. If the common rank is three, equality of the nonzero determinants implies $\det R = 1$. If the rank is at most two and $\det R = -1$, we compose R with the reflection in a direction perpendicular to the span of the v_i'; this fixes every v_i' and changes the determinant to $+1$. Rank zero covers the central triples. In every case some $R \in \mathrm{SO}(3)$ maps each v_i to v_i'. By the surjectivity of Ad (Lemma 5.1) there exists $k \in \mathrm{SU}(2)$ with $\mathrm{Ad}(k) = R$. Since conjugation does not change the scalar parts, we have $H_i' = k H_i k^{-1}$. Conversely, simultaneous conjugation preserves the trace of every product, by cyclicity of the trace.

Let Q_0 be the quotient of $\mathrm{SU}(2)^3$ by simultaneous conjugation, with the quotient topology, and let $A = \mathfrak{p}(\mathrm{SU}(2)^3)$. Then Q_0 is compact, as a continuous image of a compact space, and so is A. Since the polynomial map $\mathfrak{p}$ is continuous and constant on equivalence classes, it induces a continuous map $Q_0 \to A$, which is bijective by what we just proved. A closed subset of Q_0 is compact, and its image is compact and hence closed in the metric space A. Thus the inverse map is continuous, and the induced map is a homeomorphism.

A continuous invariant function F descends to a continuous function on Q_0, which defines a continuous function f_A on A with $\|f_A\|_\infty = \|F\|_\infty$. We construct a continuous extension to the cube. For a real continuous function g on the compact set A with $|g| \le B$, set $g_T(x) = \inf_{y \in A}\{g(y) + T|x - y|\}$ for $x \in [-1, 1]^7$. By the triangle inequality g_T is T-Lipschitz, and $g_T \le g$ on A. Given $\varepsilon > 0$, choose $\delta > 0$ from the uniform continuity of g, and T with $T\delta \ge 2B$. For $x \in A$, the terms with $|x - y| \le \delta$ are at least $g(x) - \varepsilon$, and the remaining terms are at least $-B + T\delta \ge B \ge g(x)$. Thus $g - \varepsilon \le g_T \le g$ on A, and clipping g_T to $[-B, B]$ preserves its continuity and this approximation, and it bounds the result by B on the cube. Let $M = \|f_A\|_\infty$. If $M = 0$, extend by zero. Otherwise let $r_0 = f_A$. If $\|r_k\|_\infty \le B_k = 2^{-k} M$, apply this construction to the real and imaginary parts of r_k, with errors at most $B_k/(2\sqrt{2})$; this gives a continuous complex function h_k on the cube with $\|h_k\|_\infty \le \sqrt{2} B_k$ and $\|r_k - h_k|_A\|_\infty \le B_k/2$, and we set $r_{k+1} = r_k - h_k|_A$.

Since $\sum_k \sqrt{2}B_k < \infty$, the series $h = \sum_k h_k$ converges uniformly on the cube. Then h is continuous, and $h = f_A$ on A since the residuals tend to zero uniformly. Composing h with the radial retraction of $\mathbb{C}$ onto the closed disc of radius M,

$$z \longmapsto \begin{cases} z, & |z| \leq M, \\ Mz/|z|, & |z| > M, \end{cases}$$

which is continuous and fixes the values of f_A, gives a continuous extension f with $\|f\|_\infty \leq M$. Finally, $\mathbb{E}F(H^{(k)}) = \mathbb{E}f(\mathfrak{p}(H^{(k)}))$. Since f is bounded and continuous on the cube, convergence in distribution of the traces implies convergence of the expectations. The limiting law, a probability measure on the cube, is supported on the closed set A, since the nonnegative continuous function $\operatorname{dist}(\cdot, A)$ has expectation zero under every approximating law and hence under the limit. The limit of the expectations is therefore the integral of the intrinsically defined function f_A. □

SECTION 6

Block averaging and chessboard estimates

In this section, we prove the finite-dimensional properties of the block averaging map T_M of Definition 4.3 that are used throughout the renormalization group analysis, and the reflection positivity estimates for the Wilson measure on which the bound on large fields rests. Section 6.1 shows that T_M is Borel, gauge covariant, and covariant under the translations and reflections preserving the coarse lattice, also where its logarithm is not defined, and that its iterates have nested local dependence sets (Proposition 6.2). Section 6.2 constructs local constraint coordinates near the coarse-field lift $U_0(b)$ for every coarse configuration b, restores the block gauge variables, and computes Haar measure on the resulting coordinate domain exactly (Lemma 6.3); this formula is the starting point of every fluctuation integral in Parts IV and VI. Section 6.3 proves that the curl bounds the ℓ^2 norm on the linearized constrained fluctuation space, uniformly in the volume (Lemma 6.5); this is an input of Section 8. Section 6.4 contains reflection positivity and chessboard estimates for the Wilson measure and its images under T_M^ℓ, and exponential moment bounds for plaquette energies on the finest lattice (Theorem 6.6); these are used in Sections 7, 22, and 23. The inputs are the conventions of Definitions 4.1 and 4.3, the Haar measure, tree gauge, and finite-word estimates of Lemma 5.1, and the coefficient calculus of Lemma 5.3. No later result is used.

Throughout this section, we measure distances and dependence sets in fine lattice units; physical rescaling multiplies every spatial coordinate by the fine mesh. Unless a more specific dependence is stated, constants depend only on M, on the stated coordinate radii, and, for derivative estimates, on the prescribed finite derivative order. They are independent of L, the lattice side, the coupling, and the number of preceding averaging steps; this is the dependence described in item (1) of Definition 4.14.

6.1 The block averaging map

The block averaging map is an instance of Bałaban's averaging operation for lattice gauge fields [12, eq. (15)]: the fine link variables of a block are transported to a common reference path, their logarithms are averaged, and the average is exponentiated. It is defined in Definition 4.3; we fix the additional notation used in this section and the next.

Definition 6.1. Let $M = 2R_M + 1 \geq 3$ be odd, let the fine periodic lattice have side Mm with $m \geq 4$, work in fine lattice units, and let T_M be the block averaging map of Definition 4.3, with the centered blocks $\mathsf{B}_M(v)$, $v \in M\mathbb{Z}^3$, the root paths $\sigma_{v,x}$, the reference paths C_c, and the paths

$$F_{c,x} = \sigma_{v,x}\,[x, x + Me_i]\,\sigma^{-1}_{v+Me_i,x+Me_i}, \qquad c = (v, v + Me_i),\ x \in \mathsf{B}_M(v),$$

defined there; we call the $F_{c,x}$ translated paths with endpoint connectors. The logarithm in the first clause of (4.3) is the principal logarithm on $\mathrm{SU}(2) \setminus \{-I\}$, with $|\mathrm{Log}\, V| < 2\pi$, and we call the second clause, $(T_M U)_c = U(C_c)$ when some based loop holonomy $U(F_{c,x})U(C_c)^{-1}$ equals $-I$, the Borel extension of T_M. For odd $s \geq 1$ and a vertex v set

$$\mathsf{B}_s(v) = v + \{-(s-1)/2, \dots, (s-1)/2\}^3,$$

so that $\mathsf{B}_1(v) = \{v\}$. The links of the paths $\sigma_{v,x}$, $x \in \mathsf{B}_M(v)$, form a spanning tree of the block (Proposition 6.2), which we call the block tree, with root v. For $\ell \geq 1$, T_M^ℓ denotes the ℓ-fold composition of T_M; one renormalization group step is the composition T_M^J, not an average over blocks of side M^J. For a configuration b on the coarse lattice, the coarse-field lift $U_0(b)$ is the fine configuration that equals I on every link with both endpoints in one block, and equals b_c on every link from $\mathsf{B}_M(v)$ to $\mathsf{B}_M(v + Me_i)$, where $c = (v, v + Me_i)$.

Figure 6.1 shows these paths for two neighboring blocks.

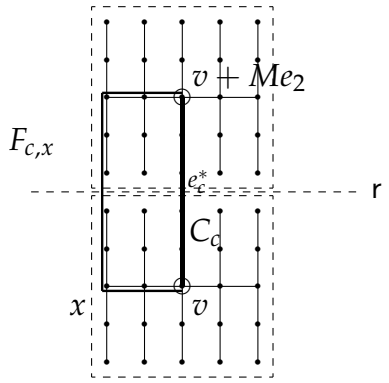


Figure 6.1. Two neighboring centered blocks for $M = 5$, in the plane spanned by e_1 and e_2, with $c = (v, v + Me_2)$. Dashed squares: the blocks $\mathsf{B}_M(v)$ and $\mathsf{B}_M(v + Me_2)$. Thin lines: the block trees. Thick line: the reference path C_c; it contains the eliminated edge e_c^* of Section 6.2. Medium line: the translated path with endpoint connectors $F_{c,x}$ for $x = v - 2e_1$. Dotted line: the plane of a reflection r bisecting the links between the two blocks; each block lies on one side of it.

Proposition 6.2. *Let M and m be as in Definition 6.1. Then the following statements hold.*

1. *The centered blocks partition the fine vertices. In each block the links of the paths $\sigma_{v,x}$ form a spanning tree with $M^3 - 1$ links, and the paths from the root v in this tree are the $\sigma_{v,x}$.*
2. *T_M is a Borel map on the whole configuration space $\mathcal{U}_{Mm}$.*
3. *For every periodic gauge transformation k, $T_M(U^k) = (T_M U)^{k|_{M\mathbb{Z}^3}}$.*
4. *T_M commutes with the translations by vectors of $M\mathbb{Z}^3$ and with every coordinate reflection $x_j \mapsto a_j - x_j$ with $a_j \in M\mathbb{Z}$, in particular with the spatial inversion $x \mapsto -x$. This includes the configurations at which the Borel extension is used. These reflections map block trees onto block trees; no invariance under permutations of the coordinates is needed or claimed.*
5. *For every coarse configuration b, without any restriction on its holonomies, $T_M \mathcal{U}_0(b) = b$.*
6. *Let $\ell \geq 0$, $m_\ell = M^\ell$, and let the fine side be $N = m_\ell n$ with $n \geq 4$. The value of $T_M^\ell U$ on the coarse link $(v, v + m_\ell e_i)$ depends only on the fine links with both endpoints in*

$$\mathsf{B}_{m_\ell}(v) \cup \mathsf{B}_{m_\ell}(v + m_\ell e_i).$$

For $\ell \geq 1$ the blocks are nested:

$$\bigcup_{\delta \in \{-R_M, \dots, R_M\}^3} \mathsf{B}_{M^{\ell-1}}\big(v + M^{\ell-1}\delta\big) = \mathsf{B}_{M^\ell}(v). \tag{6.1}$$

All statements are deterministic. The sets in (6) do not depend on L, the coupling, the number of preceding averaging steps, or the fields outside them.

Proof. Blocks and trees. In one coordinate, the intervals $Mk + \{-R_M, \dots, R_M\}$, $k \in \mathbb{Z}$, consist of M consecutive integers and have centers at distance M. They therefore partition $\mathbb{Z}$, and also $\mathbb{Z}/Mm\mathbb{Z}$. Taking products, we see that the blocks partition the vertices, including across the periodic identifications. For a vertex $x \neq v$ of $\mathsf{B}_M(v)$, we define the parent of x by decreasing the absolute value of the last nonzero component of $x - v$ by one. Since the ℓ^1 distance to v decreases strictly along parents and every vertex reaches v, the links from vertices to their parents form a tree with $M^3 - 1$ links. Read from v, the path to x in this tree changes first the first coordinate, then the second, then the third; it is $\sigma_{v,x}$. This proves (1).

Measurability and gauge covariance. Every transport along a translated path with endpoint connectors or a reference path is a finite product of link matrices and their inverses. The set of configurations at which none of the finitely many based loop holonomies equals $-I$ is open. Since the principal logarithm is continuous on $\mathrm{SU}(2) \setminus \{-I\}$, the first clause of (4.3) defines a continuous map on this open set. On its closed complement, $(T_M U)_c = U(C_c)$ is continuous. This proves (2); no value of $\mathrm{Log}(-I)$ is used. Let k be a periodic gauge transformation. The paths $F_{c,x}$ and C_c both start at v and end at $v + Me_i$, so each based loop holonomy $U(F_{c,x})U(C_c)^{-1}$ is conjugated by k_v. On $\mathrm{SU}(2) \setminus \{-I\}$ the principal logarithm satisfies $\mathrm{Log}(kVk^{-1}) = \mathrm{Ad}(k)\,\mathrm{Log}\,V$, since both sides exponentiate to kVk^{-1} and lie in the ball of radius 2π on which the exponential

map is injective. The condition that some based loop holonomy equals $-I$ is invariant under conjugation. Multiplying by the reference transport $k_v U(C_c) k_{v+Me_i}^{-1}$ gives (3), in both cases of the definition.

Translations and reflections. A translation by a vector of $M\mathbb{Z}^3$ maps blocks, root paths, translated paths with endpoint connectors, and reference paths to the corresponding objects. For reflections we use the following identity. Let F_x, $x \in \mathsf{B}_M(v)$, and C be elements of $\mathrm{SU}(2)$ with $F_x C^{-1} \neq -I$ for all x, and let $A = M^{-3} \sum_x \operatorname{Log}(F_x C^{-1})$. Then

$$\begin{aligned} \operatorname{Log}(F_x^{-1} C) &= -\operatorname{Ad}(C^{-1}) \operatorname{Log}(F_x C^{-1}), \\ \exp\Big(M^{-3} \sum_x \operatorname{Log}(F_x^{-1} C)\Big) C^{-1} &= C^{-1} e^{-A} = (e^A C)^{-1}. \end{aligned} \tag{6.2}$$

The first line follows from $F_x^{-1} C = C^{-1} (F_x C^{-1})^{-1} C$, the conjugation property of Log just used, and $\operatorname{Log}(V^{-1}) = -\operatorname{Log} V$; the second follows from the first. If one of the elements $F_x C^{-1}$ equals $-I$, then its conjugated inverse $F_x^{-1} C$ also equals $-I$, and the two values given by the Borel extension are C and C^{-1}, which are again inverse to each other.

We write a coordinate reflection preserving $M\mathbb{Z}^3$ as $x_j \mapsto a_j - x_j$ with $a_j \in M\mathbb{Z}$. It maps blocks to blocks and centers to centers, and it changes the signs of the steps of a path without changing the order in which the coordinates are changed. In particular, it maps each block tree and each root path $\sigma_{v,x}$ to the corresponding block tree and root path. If $j \neq i$, it maps $F_{c,x}$ and C_c to the corresponding paths of the reflected positive link. If $j = i$, let s denote the reflection, let $c' = (s(v + Me_i), s(v))$ be the positive link associated with the image of c, and let $y = s(x + Me_i)$, which lies in the initial block of c'. Substituting the reflected endpoint paths, we find

$$s F_{c,x} = F_{c',y}^{-1}, \qquad s C_c = C_{c'}^{-1},$$

and $x \mapsto y$ is a bijection between the two initial blocks. Then (6.2), including its exceptional case, implies reflection covariance. Products of the three coordinate reflections give the spatial inversion. This proves (4).

Right inverse. Each straight segment $[x, x + Me_i]$ passes from the initial block of c into the final block exactly once; its transverse coordinates stay in the common transverse range of the two blocks, and both root paths stay in their blocks. The same holds for the reference path. For $U_0(b)$, every link inside a block has transport I and the single interface link crossed has transport b_c. This gives $U_0(b)(F_{c,x}) = U_0(b)(C_c) = b_c$ for every x. Every based loop holonomy then equals I, and $(T_M U_0(b))_c = b_c$. This proves (5).

Nesting and dependence sets. Let $s = M^{\ell-1}$. In one coordinate, the intervals $s\delta + \{-(s-1)/2, \dots, (s-1)/2\}$, $-R_M \leq \delta \leq R_M$, are consecutive disjoint integer intervals, with extreme endpoints $\pm(sR_M + (s-1)/2) = \pm(Ms - 1)/2$. Taking products in three coordinates yields (6.1), including the case $s = 1$. For $\ell = 1$, the dependence statement in (6) follows from the description of the three pieces of each translated path with endpoint

connectors in the previous paragraph, and from the fact that the reference path lies in the two blocks. Suppose that it holds for $\ell - 1$. The last averaging step reads only intermediate links with both endpoints in the two centered blocks of intermediate vertices around the final endpoints. By the induction hypothesis, each of these intermediate links depends only on fine links in the blocks of side $M^{\ell-1}$ around its endpoints. By (6.1) the union of these blocks lies in the two claimed blocks of side M^ℓ. The decision whether the Borel extension is used reads the same based loop holonomies and does not enlarge the dependence set. By induction, this proves (6) for the composition T_M^ℓ. Since $n \geq 4$, the union of two blocks of side M^ℓ has extent $2M^\ell < M^\ell n$ in each coordinate, and all these sets lift to boxes in the covering lattice $\mathbb{Z}^3$, also across the periodic identifications. □

6.2 Local constraint coordinates and Haar measure

We now parametrize the configurations near a level set $\{T_M U = b\}$. In each block we fix the gauge by the block tree, with gauge transformations equal to I at the block centers; we call this the block axial gauge. For each positive coarse link $c = (v, v + Me_i)$ we then eliminate one link variable, on the central interface link

$$e_c^* = \big(v + R_M e_i,\, v + (R_M + 1)e_i\big),$$

by solving the equation $(T_M U)_c = b_c$. This edge, which we call the eliminated edge, lies on the reference path C_c (Figure 6.1), and C_c consists of e_c^* and of links of the two block trees.

The equation can be solved with radii independent of b. To see this, write $U = e^X U_0(b)$ on the links not in the block trees, and suppose that only $X_{e_c^*} = \omega$ is nonzero. Exactly M of the M^3 paths $F_{c,x}$ cross e_c^*, namely those with $x - v \in \mathbb{Z}e_i$, while the reference path always crosses it. Then $M^3 - M$ of the based loop holonomies equal $e^{-\omega}$, the others equal I, and

$$(T_M U)_c = e^{-(1-M^{-2})\omega} e^{\omega} b_c = e^{M^{-2}\omega} b_c.$$

The derivative of the constraint in the eliminated variable is therefore $M^{-2} I_3$ at every real coarse field b, including coarse fields with large holonomies. We construct the local inverse in a neighborhood of $U_0(b)$ rather than of the identity configuration.

The variables are counted as follows; here f denotes the number of remaining SU(2) variables.

fine links	$3(Mm)^3$
links of the block trees	$(Mm)^3 - m^3$
links not in the block trees	$2(Mm)^3 + m^3$
eliminated edges e_c^*, one for each positive coarse link	$3m^3$
remaining SU(2) variables	$f = 2((Mm)^3 - m^3)$
real dimension of the free logarithmic coordinates z	$3f$

We obtain the Haar formula in Lemma 6.3 by two successive changes of variables. First, with the free link matrices fixed, we replace the eliminated link matrices by the coarse output $T_M U$. Then we translate the free link matrices on the right by $U_0(b)^{-1}$ and pass to their logarithms. In the first change, the constant c_{Haar} of the output chart cancels that of the eliminated edges, but their Haar density j_H remains. For this reason the numerator of the Haar disintegration density $\mathscr{J}_b$ below contains j_H for every link not in the block trees. Recall from Lemma 5.1 that $dH(e^X) = c_{\mathrm{Haar}}\, j_H(X)\, d^3X$ for $|X| < 2\pi$, with $c_{\mathrm{Haar}} = (16\pi^2)^{-1}$ and $j_H(X) = \mathrm{sinc}^2(|X|/2)$.

Lemma 6.3. *Let M, m, and the block trees be as in Definition 6.1, and fix the block axial gauge. For a coarse configuration b write $U_e = e^{X_e} U_{0,e}(b)$ on the links e not in the block trees, and $U_e = I$ on the links of the block trees. For each positive coarse link c define*

$$\mathsf{C}_{b,c}(X) = \mathrm{Log}\big((T_M(e^X U_0(b)))_c\, b_c^{-1}\big).$$

Call the variables X_e with e not in the block trees and not an eliminated edge the free logarithmic coordinates, and denote their vector by z.

1. *Among the eliminated variables $X_{e^*_{c'}}$, the function $\mathsf{C}_{b,c}$ depends only on $X_{e^*_c}$. It depends on at most $6M^3$ links, all with both endpoints in the two blocks of c, and each link enters at most six of the functions $\mathsf{C}_{b,c}$. For every real b,*

$$\mathsf{C}_{b,c}(0) = 0, \qquad D_{X_{e^*_c}} \mathsf{C}_{b,c}(0) = M^{-2} I_3.$$

2. *There exist $r_M, \widetilde{r}_M > 0$, depending only on M, with the following property. For every real b and every vector z of free logarithmic coordinates with $\max_e |z_e| < r_M$, there exists a unique X whose free entries equal z, whose eliminated entries have norm less than $\widetilde{r}_M$, and which satisfies $\mathsf{C}_{b,c}(X) = 0$ for all c. Writing $X = \Gamma_b(z)$, we have*

$$\Gamma_b(0) = 0, \qquad (\Gamma_b(z))_{\mathrm{free}} = z, \qquad T_M\big(e^{\Gamma_b(z)} U_0(b)\big) = b.$$

   The map Γ_b is real analytic and has a holomorphic extension, jointly in the free logarithmic coordinates and in local complex coordinates of the target around every real b. The eliminated entry of Γ_b at e^*_c depends only on the free logarithmic coordinates in the two blocks of c. For each order $s \geq 1$, the derivatives of order s of Γ_b on a fixed smaller complex polydisc have absolute row and column sums bounded in terms of M, s, and the ratio of the radii of the two polydiscs. These bounds do not depend on m, on b, or on the number of preceding averaging steps. All determinants $\det D_{X_{e^*_c}} \mathsf{C}_{b,c}(\Gamma_b(z))$ are positive.

3. *Let $f = 2((Mm)^3 - m^3)$ be the number of free $\mathrm{SU}(2)$ variables, and set*

$$\mathscr{J}_b(z) = c_{\mathrm{Haar}}^f \, \frac{\prod_{e \notin \text{block trees}} j_H\big((\Gamma_b(z))_e\big)}{\prod_c \det D_{X_{e^*_c}} \mathsf{C}_{b,c}(\Gamma_b(z))}.$$

For a Borel set $\mathcal{Z} \subset \{\max_e |z_e| < r_M\}$, let $\mathcal{D}_{\mathcal{Z}}$ be the set of configurations $(e^{\Gamma_b(z)} U_0(b))^k$ with b a coarse configuration, $z \in \mathcal{Z}$, and k a gauge transformation equal to I at every block center. Then for every nonnegative Borel function F on $\mathcal{U}_{Mm}$,

$$\int_{\mathcal{D}_{\mathcal{Z}}} F(U)\, dH_{Mm}(U) = \int dH_m(b) \int_{\mathcal{Z}} \mathscr{J}_b(z)\, dz \int F\big((e^{\Gamma_b(z)} U_0(b))^k\big)\, dH_{\mathrm{for}}(k),$$

where dz is Lebesgue measure on $\mathbb{R}^{3f}$ and H_{for} is the product Haar probability measure on the values of k at the vertices other than block centers. For gauge-invariant F the inner integral equals $F(e^{\Gamma_b(z)} U_0(b))$. The numerator of $\mathscr{J}_b$ contains the Haar density of every eliminated edge, and no further factor (no density in a coarse logarithm, no Gram determinant, and no volume of a gauge orbit) occurs.

4. *Let $g > 0$. The substitution $z = g\eta$ turns $\mathscr{J}_b(z)\, dz$ into $g^{3f}\, \mathscr{J}_b(g\eta)\, d\eta$. If F is the Wilson weight $\exp[-4g^{-2} \sum_p q(U_p)]$ times the indicator of an event, the integrand in η is $g^{3f}\, \mathscr{J}_b(g\eta) \exp[-4g^{-2} \sum_p q((e^{\Gamma_b(g\eta)} U_0(b))_p)]$ times that indicator and the indicator of $\{g\eta \in \mathcal{Z}\}$. After integrating over η and normalizing, one obtains the image under T_M of the restricted measure; in general it differs from the image of the unrestricted Wilson measure, since the mass of the restriction depends on the target.*
5. *At fixed real b, a coarse gauge transformation h, extended to be constant on each block, acts on the free logarithms by $X_e \mapsto \mathrm{Ad}(h_v) X_e$, where v is the center of the block containing the initial vertex of e. The spatial inversion acts, together with the induced permutation of links, by*

$$X'_{e'} = -\,\mathrm{Ad}\big(U_{0,e}(b)^{-1}\big) X_e.$$

These orthogonal maps commute with Γ and preserve the Wilson action and $\mathscr{J}$ exactly. They map products of three-dimensional Euclidean balls to the products with permuted radii; products of balls with a common radius are invariant. Although the maps depend on b, the joint change of variables (b, z) has a block-triangular derivative whose diagonal block in z has absolute determinant one.

All constants depend only on M, the stated coordinate radii, and, where applicable, the prescribed derivative order. The only restriction involving g is that $g\eta$ lies in the coordinate domain of (2).

The uniqueness in (2) holds in the stated neighborhood of $U_0(b)$, not on the whole level set $\{T_M U = b\}$, and the formula in (3) concerns only the set $\mathcal{D}_{\mathcal{Z}}$. We construct Γ_b by a contraction argument in one coarse link at a time; this is why the radii do not depend on the number of blocks.

Proof. We proceed in six steps.

Step 1: locality of the constraints. We apply the block version of the tree identity of Lemma 5.1, with every block center fixed; the block trees have $(Mm)^3 - m^3$ links. In this gauge every reference path has exactly one link not in a block tree, namely its eliminated

edge. Since a straight segment crosses only the interface of c and a root path never crosses an interface, no constraint contains the eliminated variable of another constraint. The remaining variables of $\mathsf{C}_{b,c}$ are links in the two blocks of c and on their common interface, at most $6M^3$ positive links. A free link inside a block can affect only the constraints of the six coarse links incident to the center of that block, and a free interface link affects only its own constraint. In particular, every variable enters at most six constraints.

Step 2: the linear coefficient. If all free variables vanish and only $X_{e_c^*} = \omega$ is varied, then exactly M of the M^3 straight segments cross e_c^*: their two transverse offsets vanish and their longitudinal offset has M possible values. The translated paths with endpoint connectors that contain them, and the reference path, have transport $e^{\omega} b_c$; all other such paths have transport b_c. For ω small every based loop holonomy different from I equals $e^{-\omega}$. Hence

$$(T_M(e^X U_0(b)))_c = e^{-(1-M^{-2})\omega} e^{\omega} b_c = e^{M^{-2}\omega} b_c, \qquad \mathsf{C}_{b,c}(\omega, 0) = M^{-2}\omega. \tag{6.3}$$

This gives the value and the derivative in (1) at every real b, including coarse fields whose holonomies do not commute.

Step 3: analytic neighborhoods. In local target coordinates we write $b'_c = e^{\theta_c} b_c$ with b_c real. The word $F_{c,x} C_c^{-1}$ has at most $5M - 3$ fine factors, and at $X = 0$ it equals I, also for the complex target b', since the identity $U_0(b)(F_{c,x}) = U_0(b)(C_c)$ is algebraic. There exists $\epsilon_M > 0$, depending only on M, such that for $\max_e |X_e| < \epsilon_M$ and $\max_c |\theta_c| < \epsilon_M$ all these words and the outer transport $T_M(e^X U_0(b'))_c b_c'^{-1}$ have matrix distance less than $1/4$ from I. Indeed, we insert the factors e^{X_e} one at a time into the fixed word: real compact factors have operator norm one, the complex target factors and their inverses have norm at most $e^{|\theta_c|}$, and $\|e^Y - I\| \le e^{|Y|} - 1$. The resulting bound tends to zero uniformly in the real background. The outer transport tends to I by the same bound applied to the finite sum of inner logarithms. On this domain every logarithm is given by the convergent series $\sum_{k \ge 1} (-1)^{k+1} (V - I)^k / k$. By the finite-word estimates of Lemma 5.1, we have first and second derivative bounds there, uniform in the real background.

Step 4: construction of the inverse. We solve one constraint at a time. In one constraint let $\omega = X_{e_c^*}$ and let z denote its free logarithmic coordinates, with $\|z\|_\infty$ the maximum of their norms. After decreasing ϵ_M, we fix $A_M^{\mathrm{an}}, K_M^{\mathrm{an}} \ge 1$ such that, for all local targets with $\max|\theta| < \epsilon_M/2$ and $|\omega|, \|z\|_\infty < \epsilon_M/2$,

$$\begin{aligned} |\mathsf{C}_{b',c}(0, z)| &\le A_M^{\mathrm{an}} \|z\|_\infty, \\ \|D_\omega \mathsf{C}_{b',c}(\omega, z) - M^{-2} I\| &\le K_M^{\mathrm{an}} \big(|\omega| + \|z\|_\infty\big). \end{aligned} \tag{6.4}$$

The first line follows from the first derivative bound and $\mathsf{C}_{b',c}(0, 0) = 0$. The identity (6.3) continues to the complex target coordinates, so $D_\omega \mathsf{C}_{b',c}(0, 0) = M^{-2} I$, and the second line

follows from the second derivative bound. All these norms concern at most $6M^3$ links, independently of m. Fix

$$\delta_M \le \min\{\epsilon_M/4,\ (8M^2K_M^{\mathrm{an}})^{-1},\ 1/4\}, \qquad \widetilde{r}_M = \delta_M, \qquad r_M = \frac{\delta_M}{4M^2A_M^{\mathrm{an}}}.$$

For $\|z\|_\infty < r_M$, the second line of (6.4) implies that the map $\Psi(\omega) = \omega - M^2\mathsf{C}_{b',c}(\omega, z)$ has Lipschitz constant at most $1/4$ on $\{|\omega| \le \delta_M\}$, and by the first line we have $|\Psi(0)| \le \delta_M/4$. Then Ψ maps this closed ball into the ball of radius $\delta_M/2$. Starting from $\omega_0 = 0$, the iterates satisfy $|\omega_{j+1} - \omega_j| \le 4^{-j}\delta_M/4$ and converge uniformly to a solution of $\mathsf{C}_{b',c}(\omega, z) = 0$. Two solutions in the ball of radius δ_M differ by at most a quarter of their distance, so they coincide. We define Γ_b by solving these independent equations, one for each c, and keeping the free entries z.

Each iterate is holomorphic. On every strictly smaller polydisc in the free logarithmic coordinates and the target variables the uniform limit is holomorphic, since the one-variable Cauchy formula in each coordinate passes to the uniform limit. Iterated Cauchy formulas give the derivative bounds. If the margins between the two polydiscs are $d_z, d_\theta > 0$, a coefficient of order s in one constraint is bounded by $\delta_M s!\,(C_M/\min(d_z, d_\theta))^s$, after enlarging C_M to account for the at most $3(6M^3+1)$ scalar variables of the constraint. The free entries of Γ_b are coordinate functions. Each eliminated entry uses only the variables of its own constraint, and each variable belongs to at most six constraints, by Step 1. This proves the row and column bounds of every fixed order in (2); it is also the finite-range case of Lemma 5.3. Real coefficients and uniqueness imply real analyticity in real variables. The same construction applies in target coordinates centered at every real b, with the radii just chosen. On the solution, $\|M^2 D_\omega \mathsf{C} - I\| \le 1/4$, so the determinant does not vanish; by continuity from $X = 0$ it is positive for real data, and it lies between $M^{-6}(3/4)^3$ and $M^{-6}(5/4)^3$. This proves (2).

Step 5: the Haar formula. There are $d_{\mathrm{nf}} = 3(Mm)^3 - ((Mm)^3 - m^3) = 2(Mm)^3 + m^3$ links not in the block trees, of which $3m^3$ are eliminated edges, so $f = d_{\mathrm{nf}} - 3m^3 = 2((Mm)^3 - m^3)$ and dz is Lebesgue measure in dimension $3f$.

We first keep the untranslated free link matrices V_e fixed and replace the matrices on the eliminated edges by the output $T_M V$. At a configuration with output b we use the output chart $y_c = \mathrm{Log}((T_M V)_c b_c^{-1})$. Since $y_c = 0$ at that point, the coarse Haar measure has density $c_{\mathrm{Haar}}\, d^3 y_c$ there. In the input chart $V_{e_c^*} = e^{X_{e_c^*}} b_c$ the Haar measure of the eliminated edge has density $c_{\mathrm{Haar}}\, j_H(X_{e_c^*})\, d^3 X_{e_c^*}$. With the free matrices fixed, the derivative $D_{X_{e^*}} y$ is block diagonal with blocks $D_{X_{e_c^*}} \mathsf{C}_{b,c}$, by Step 1, and its determinant is positive. This gives the density factor

$$\prod_c \frac{j_H(X_{e_c^*})}{\det D_{X_{e_c^*}} \mathsf{C}_{b,c}(X)} \tag{6.5}$$

relative to the coarse Haar probability measure and the Haar measure of the untranslated free matrices. The constants c_{Haar} of the eliminated edges cancel against those of the output chart, not against the factors j_H.

At fixed b we next write $V_e U_{0,e}(b)^{-1} = e^{z_e}$ for the free links. Right translation preserves Haar measure. Made jointly with b, this change of variables is triangular, so the derivatives of z in b do not affect its determinant. Passing to logarithms gives the factor $c_{\mathrm{Haar}}^{f} \prod_{e \text{ free}} j_H(z_e)\, dz$. Combining this with (6.5), we obtain $\mathscr{J}_b$, with the factor j_H of every link not in the block trees in its numerator. Since no coarse logarithm has been integrated against Lebesgue measure, there is no additional coarse factor j_H. Since we used independent coordinates throughout, no surface measure or Gram determinant occurs.

The parametrization is injective: the target is recovered by applying T_M, and then the free logarithms are recovered from the free links. The image of a Borel set $\mathcal{Z}$ is Borel. Indeed, with $b = T_M V$, a configuration in the block axial gauge belongs to the image if and only if its logarithms relative to $U_0(b)$ on the links not in the block trees exist in the neighborhoods above, its free logarithms belong to $\mathcal{Z}$, and its eliminated logarithms have norm less than $\widetilde{r}_M$. The constraint then holds, and uniqueness identifies these logarithms with Γ_b; each condition is Borel. The pointwise Jacobian computation therefore gives the integral formula, first on coordinate neighborhoods and then on a disjoint measurable refinement of countably many of them. Countably many suffice, since a finite product of matrix groups has a countable base of coordinate neighborhoods. By monotone approximation, the formula holds for all nonnegative Borel F. Finally we restore the block-tree links by the tree identity of Lemma 5.1; they contribute the integral over k against H_{for}. For a gauge-invariant integrand this integral equals the value at $k = I$. Since the tree gauge and its inverse are finite products of transports, restoring it preserves the Borel description. This proves (3).

The substitution $z = g\eta$ multiplies Lebesgue measure in dimension $3f$ by g^{3f}, and multiplication by the Wilson weight gives the stated factor together with every indicator. Integrating out η and normalizing gives the image of the restricted measure. This proves (4).

Step 6: symmetries. We extend a coarse gauge transformation h to be constant on each block. The coarse-field lift transforms into the lift of b^h, the block-tree links remain equal to I, and the relative logarithm on a link whose initial vertex lies in the block of v becomes $\mathrm{Ad}(h_v) X_e$. This is an orthogonal map of $\mathbb{R}^3$ with determinant one. It acts in the same way on an eliminated input and on the output of its constraint, so $D_{X_{e_c^*}} \mathsf{C}_{b,c}$ is conjugated and its determinant is unchanged; uniqueness implies the equivariance of Γ.

Under the spatial inversion, let e' be the positive link representing the reversed image of e. The lift is covariant under inversion, and

$$\big(e^{X_e} U_{0,e}\big)^{-1} = e^{-\operatorname{Ad}(U_{0,e}^{-1})X_e}\, U_{0,e}^{-1},$$

which gives the formula in (5). At fixed real b this is an orthogonal signed permutation of the free and the eliminated variables. The reflected constraint for the reversed coarse link is

$$\operatorname{Log}\big((T_M V)_c^{-1} b_c\big) = -\operatorname{Ad}(b_c^{-1})\, \mathsf{C}_{b,c}(X),$$

and its eliminated input is transformed by the same orthogonal map, because $U_{0,e_c^*} = b_c$. Its derivative is therefore conjugated, and its determinant is unchanged. Norms and the Haar factors j_H are unchanged, and so is the Wilson action, since a reflection changes a plaquette transport only by conjugation and possibly inversion, both of which preserve q. Uniqueness implies the intertwining of Γ. A product of three-dimensional balls is mapped to the product with relabeled radii, and a product with a common radius is invariant. Although these orthogonal matrices depend on the target, the joint transformation of (b, z) has a triangular derivative, and its diagonal block in z has absolute determinant one. This proves (5). □

6.3 Coercivity on the constrained fluctuation space

At the identity configuration the constraint $T_M U = I$ linearizes to $DT_M(I)a = 0$, where a is a real one-cochain with values in $\mathfrak{su}(2) \simeq \mathbb{R}^3$. On this space, after the block axial gauge fixing, the curl controls the ℓ^2 norm with a constant depending only on M, for a local reason. The value of a gauge-fixed cochain on a link e not in a block tree is the circulation of a around the short loop formed by e and two root paths. On interface links, the loop has to be closed through the reference path C_c and therefore also involves the eliminated edge e_c^*. The constraint $DT_M(I)a = 0$ expresses $a_{e_c^*}$ as an average of circulations around the short loops $F_{c,x} C_c^{-1}$. Each circulation around a short loop is bounded by the curl in a small box, by an explicit filling (Lemma 6.4). The tree gauge and the constraint remove the directions in which the curl vanishes; the coarse field itself is not controlled.

Lemma 6.4. *Let P be a box in $\mathbb{Z}^3$ with side lengths d_1, d_2, d_3, measured in links. Every closed lattice path ξ of length h in P is the boundary of an integer combination $\sum_p n_p p$ of oriented elementary plaquettes of P with*

$$\sum_p |n_p| \le h(d_1 + d_2 + d_3).$$

In particular, every real $\mathfrak{su}(2)$-valued one-cochain a satisfies

$$|a(\xi)| \le h(d_1 + d_2 + d_3) \Big(\sum_{p \subset P} |(\mathrm{d}a)_p|^2 \Big)^{1/2},$$

where $a(\xi)$ is the signed sum of a along ξ and $(\mathrm{d}a)_p$ is the signed sum of a around p.

Proof. Let o be the lower corner of P and let τ_x be the path from o to x that changes the coordinates in the order $1, 2, 3$. For a positive link $e = (x, y)$ in direction i, we move the final step of $\tau_x e$ backwards through the steps of τ_x in the directions $j > i$. Each interchange crosses one elementary ij plaquette inside P; the result is τ_y, and the number of plaquettes crossed is $\sum_{j>i}(x_j - o_j) \le d_1 + d_2 + d_3$. This gives an integer filling of the closed chain $\tau_x + e - \tau_y$ with that mass bound. For a negatively oriented step we use the negative of the filling of the reversed positive link. When we sum over all steps of a closed path, the terms τ_x cancel and the oriented edge chain of ξ remains. Adding the mass bounds, we obtain the first assertion, also for paths that retrace links or repeat plaquettes. Pairing the chain identity with a gives $a(\xi) = \sum_p n_p (\mathrm{d}a)_p$, and the second assertion follows from the triangle inequality and $\max_p |(\mathrm{d}a)_p| \le (\sum_{p \subset P} |(\mathrm{d}a)_p|^2)^{1/2}$. The constant does not depend on the components of a. □

Lemma 6.5. *Fix an odd $M \ge 3$. There exist $0 < c_M \le C_M < \infty$, depending only on M, with the following properties. Let $m \ge 4$, and let a be a real $\mathfrak{su}(2)$-valued one-cochain on the periodic fine lattice of side Mm that vanishes on the block trees and satisfies $DT_M(I)a = 0$. Then*

$$\sum_e |a_e|^2 \le C_M \sum_p |(\mathrm{d}a)_p|^2,$$

where $\mathrm{d}a$ is the signed elementary curl in fine lattice units. Further, let Γ_I be the map of Lemma 6.3(2) at the coarse configuration all of whose links equal I, and let $E_0 = D_z \Gamma_I(0)$ be the induced linear embedding of the free logarithmic coordinates. Then

$$c_M I \le E_0^{\mathsf{T}} \mathrm{d}^{\mathsf{T}} \mathrm{d} E_0 \le C_M I.$$

The rows and columns of E_0 have finite range and absolute sums bounded in terms of M. These estimates include all periodic identifications and all three components of $\mathfrak{su}(2)$, and the constants do not depend on m or on the number of preceding averaging steps.

Proof. The linearized constraint. We differentiate (4.3) at I. The contribution of the reference path to the mean of the logarithms of the based loop holonomies cancels the derivative of the final reference factor, and we find

$$(DT_M(I)a)_c = M^{-3} \sum_{x \in \mathsf{B}_M(v)} a(F_{c,x}). \tag{6.6}$$

Since a vanishes on the block trees, $a(C_c) = a_{e_c^*}$, and the constraint gives

$$a_{e_c^*} = -M^{-3} \sum_x \big(a(F_{c,x}) - a(C_c) \big). \tag{6.7}$$

Local bounds. The closed path $F_{c,x}C_c^{-1}$ lies in the box P_c formed by the two blocks of c, and its length is at most $5M-3$: each root path has at most $3R_M$ links, the straight segment has M links, and C_c has M links. The side lengths of P_c, in links, sum to $(2M-1)+2(M-1)=4M-3$. By Lemma 6.4 and (6.7), with the common constant $A_M^{\mathrm{fill}}=20M^2\geq(5M-3)(4M-3)$, we have

$$|a_{e_c^*}|\leq A_M^{\mathrm{fill}}\Big(\sum_{p\subset P_c}|(\mathrm{d}a)_p|^2\Big)^{1/2}. \tag{6.8}$$

A link $e=(x,y)$ not in the block tree with both endpoints in the block of v has the closed path $\sigma_{v,x}e\sigma_{v,y}^{-1}$, of length at most $3M-2$. Its circulation is a_e, since a vanishes on the tree. The side lengths of the block sum to $3M-3$, and $(3M-2)(3M-3)\leq A_M^{\mathrm{fill}}$, so the same filling bound controls $|a_e|$ by the curl in this single block. For an interface link $e=(x,y)$ from the block of v to the block of $v+Me_i$, the closed path $\sigma_{v,x}e\sigma_{v+Me_i,y}^{-1}C_c^{-1}$ has length at most $4M-2$ and circulation $a_e-a_{e_c^*}$, and $(4M-2)(4M-3)\leq A_M^{\mathrm{fill}}$. Combining these bounds with (6.8), we find, for every fine link e,

$$|a_e|^2\leq 4\big(A_M^{\mathrm{fill}}\big)^2\sum_{p\subset P_e}|(\mathrm{d}a)_p|^2, \tag{6.9}$$

where P_e is the block containing e or, for an interface link, the box of the two blocks. Links of the block trees satisfy the same inequality, since a vanishes on them.

Summation. A plaquette has its vertices in at most four blocks. A box of one or two blocks containing the plaquette contains one of these blocks, and for each block there are at most seven such boxes: the block itself, or its union with one of its six neighbors. Each such box has at most $2M^3$ vertices and hence at most $6M^3$ positive links to which it is assigned. When (6.9) is summed over e, a fixed plaquette is counted at most $D_M^{\mathrm{ov}}=4\cdot 7\cdot 6M^3=168M^3$ times, and

$$\sum_e|a_e|^2\leq C_M^{\mathrm{gap}}\sum_p|(\mathrm{d}a)_p|^2,\qquad C_M^{\mathrm{gap}}=4\big(A_M^{\mathrm{fill}}\big)^2D_M^{\mathrm{ov}}. \tag{6.10}$$

Since a box of two blocks has longest side $2M-1<Mm/2$ for $m\geq 4$, it lifts to a box in the covering lattice even when it meets the periodic identifications. The fillings above are therefore valid with the same bounds. All arguments are identities for chains, applied to each of the three components of a, with the Euclidean norm on $\mathfrak{su}(2)$.

The embedding. In the block axial gauge we write (6.6) as

$$M^{-2}a_{e_c^*}+\sum_{e\ \mathrm{free}}q_{ce}a_e=0.$$

The coefficient of $a_{e_c^*}$ is M^{-2}, since e_c^* occurs in exactly M of the paths $F_{c,x}$. The sum of the absolute values of the other coefficients is at most $M^{-3}\sum_x\mathrm{length}(F_{c,x})\leq 4M$.

Differentiating the identity $\mathsf{C}_{I,c}(\Gamma_I(z)) = 0$ of Lemma 6.3 at $z = 0$, we find that the free rows of E_0 are the identity and that the eliminated rows are

$$(E_0 z)_{e_c^*} = -M^2 \sum_e q_{ce} z_e,$$

with absolute row sums at most $4M^3$; in particular each entry of an eliminated row is at most $4M^3$ in absolute value. A free link inside a block occurs only in the six constraints incident to its block, and a free interface link only in its own constraint. Hence the absolute row and column sums of E_0 are at most $1 + 4M^3$ and $1 + 24M^3$, respectively, and its rows and columns have finite range. For any matrix A, by the weighted Cauchy–Schwarz inequality, we have

$$\sum_x \Big| \sum_y A_{xy} z_y \Big|^2 \le \Big(\sup_x \sum_y |A_{xy}| \Big) \Big(\sup_y \sum_x |A_{xy}| \Big) \sum_y |z_y|^2,$$

and this also applies to scalar coefficients tensored with the identity on $\mathbb{R}^3$. In particular, $\|E_0\|_2^2 \le (1 + 4M^3)(1 + 24M^3)$. Since the curl matrix has four signed entries in each row and each link lies in four plaquettes, we have $\|\mathrm{d}\|_2 \le 4$. This proves the upper bound with the constant $16(1 + 4M^3)(1 + 24M^3)$.

For the lower bound, note that the free entries of $E_0 z$ are the entries of z, and that $E_0 z$ vanishes on the block trees and satisfies the linearized constraints. Applying (6.10) to $a = E_0 z$, we obtain

$$\|\mathrm{d} E_0 z\|_2^2 \ge \big(C_M^{\mathrm{gap}}\big)^{-1} \|E_0 z\|_2^2 \ge \big(C_M^{\mathrm{gap}}\big)^{-1} \|z\|_2^2.$$

We take $c_M = (C_M^{\mathrm{gap}})^{-1}$ and let C_M be the larger of C_M^{gap} and $16(1 + 4M^3)(1 + 24M^3)$. Both depend only on M. □

6.4 Reflection positivity and chessboard estimates

Osterwalder and Seiler proved in [134] that the Wilson measure is reflection positive. The chessboard estimate is the consequence of reflection positivity developed by Fröhlich, Israel, Lieb, and Simon in [92, 93]; we follow the formulation and proof of Biskup in [29, Theorem 5.8 and (5.34)–(5.35)]. We need these results for the images of the Wilson measure under T_M^ℓ, for gauge-invariant observables, for reflections through planes bisecting links, and for disjoint cells of two sites in each direction. The treatment in [29] concerns spin systems with blocks that share boundary sites. We therefore verify reflection positivity directly for the observables we need, show that it survives the block averaging map because the blocks are centered and nested, and give the chessboard argument in full. At the end of the section, we prove exponential moment bounds for plaquette energies under the Wilson measure by combining the chessboard estimate with elementary bounds on partition functions.

Reflections and observables. Let $n \in 4\mathbb{N}$ and consider the periodic lattice of side n, in its own lattice units. A bond reflection is a map r of the vertices of the form $x_j \mapsto 2b + 1 - x_j$ for one coordinate j and some $b \in \mathbb{Z}$, with the other coordinates unchanged. It fixes the two planes $x_j = b + 1/2$ and $x_j = b + 1/2 + n/2$, which bisect links, and these planes divide the vertices into two open halves of width $n/2$ that are exchanged by r. It acts on configurations by $(\mathsf{r}U)_{(x,y)} = U_{(\mathsf{r}x,\mathsf{r}y)}$, with the convention $U_{(y,x)} = U_{(x,y)}^{-1}$, and on functions by $(F \circ \mathsf{r})(U) = F(\mathsf{r}U)$. All observables below are bounded complex-valued Borel functions of the link variables. An observable supported in a reflection half of r is such a function that is invariant under all periodic gauge transformations and depends only on links with both endpoints in one of the two open halves of r. In particular, it does not depend on the links crossing the two planes.

We tile the vertices by the cells $Q_t = 2t + \{0,1\}^3$, $t \in (\mathbb{Z}/(n/2)\mathbb{Z})^3$. There are $\mathfrak{m} = n^3/8$ cells, and the cell indices form a torus of side $n/2$, which is even. An observable supported in Q_t is a bounded Borel function, invariant under all periodic gauge transformations, of the links with both endpoints in Q_t. For a cell index t, let g_t be the map of the vertices with coordinate maps

$$g_{t,j}(x_j) = x_j + 2t_j \quad \text{if } t_j \text{ is even}, \qquad g_{t,j}(x_j) = 2t_j + 1 - x_j \quad \text{if } t_j \text{ is odd},$$

modulo n. It maps Q_0 onto Q_t, and in each coordinate j it is the composition of the t_j reflections $x_j \mapsto 4k + 3 - x_j$, $0 \le k < t_j$, through the planes $x_j = 2k + 3/2$ between consecutive cells; since $n/2$ is even, g_t depends only on t modulo $n/2$. For an observable f supported in Q_0, the reflected translate $\vartheta_t f$ is the observable supported in Q_t defined by $\vartheta_t f(U) = f(U \circ g_t)$, where $(U \circ g_t)_{(x,y)} = U_{(g_t x, g_t y)}$ for the links (x,y) of Q_0. If r is the reflection through the planes between two consecutive cells and t' is the image of the cell index t under r, a direct computation gives $\mathsf{r} \circ g_t = g_{t'}$, and hence $(\vartheta_t f) \circ \mathsf{r} = \vartheta_{t'} f$. For a shift $\eta \in \{0,1\}^3$ we use in the same way the shifted tiling by the cells $\eta + 2t + \{0,1\}^3$, with $g_{t,j}(x_j) = x_j + 2t_j$ if t_j is even and $g_{t,j}(x_j) = 2\eta_j + 2t_j + 1 - x_j$ if t_j is odd.

For $\beta \ge 0$ and an even side $N \ge 4$, let

$$Z_N(\beta) = \int e^{-\beta \sum_{p \in \mathcal{P}_N} q(U_p)} \, dH_N(U), \qquad d\mu_N^\beta = Z_N(\beta)^{-1} e^{-\beta \sum_{p \in \mathcal{P}_N} q(U_p)} \, dH_N.$$

The Wilson measure of Definition 4.1 is $\mu_{N,\gamma} = \mu_N^\beta$ with $\beta = 4N/\gamma$.

Theorem 6.6. *Let $M \ge 3$ be odd, $\ell \ge 0$, $n \in 4\mathbb{N}$, and $N = M^\ell n$. For $\beta \ge 0$ let $\mu^{(\ell)} = (T_M^\ell)_* \mu_N^\beta$ be the image of μ_N^β on the coarse lattice of side n; for $\ell = 0$ this is μ_n^β.*

1. Reflection positivity. *For every bond reflection r of the coarse lattice and every observable F supported in a reflection half of r,*

$$\mathbb{E}_{\mu^{(\ell)}}\left[F \, \overline{F \circ \mathsf{r}}\right] \ge 0.$$

The same holds under μ_N^β for every bounded Borel function on $\mathcal{U}_N$ that is invariant under all periodic gauge transformations and depends only on fine links with both endpoints in the union of the blocks $\mathsf{B}_{M^\ell}(v)$ over the coarse vertices v of one open half; this class contains $F \circ T_M^\ell$ for every observable F supported in a reflection half of r. The configurations at which the Borel extension of T_M is used are included.

2. Chessboard estimates. *For nonnegative observables $f_1, \dots, f_k$ supported in Q_0 and distinct cell indices $t_1, \dots, t_k$,*

$$\mathbb{E}_{\mu^{(\ell)}} \prod_{j=1}^{k} \vartheta_{t_j} f_j \le \prod_{j=1}^{k} \Big[\mathbb{E}_{\mu^{(\ell)}} \prod_{t} \vartheta_t f_j \Big]^{1/\mathfrak{m}}.$$

The same holds for each of the eight shifted tilings, and every coarse plaquette lies in a cell of at least one of them. In particular, writing $V = T_M^\ell U$ and $D_t(u) = \{\max_{p \subset Q_t} \rho_G(V_p) > u\}$ for $u \ge 0$, where the maximum is over the plaquettes with all vertices in Q_t, we have for every set T of cell indices

$$\mu^{(\ell)}\Big(\bigcap_{t \in T} D_t(u) \Big) \le \mu^{(\ell)}\Big(\bigcap_{t} D_t(u) \Big)^{|T|/\mathfrak{m}},$$

and

$$\mu_N^\beta \Big\{ \max_p \rho_G\big((T_M^\ell U)_p\big) > u \Big\} \le n^3 \Big[\frac{1}{Z_N(\beta)} \int e^{-\beta \sum_p q(U_p)} \mathbf{1}_{\cap_t D_t(u)}(T_M^\ell U)\, dH_N(U) \Big]^{8/n^3}.$$

More generally, the chessboard inequality holds under μ_N^β for the cells $C_t^\eta = \bigcup_{\zeta \in \{0,1\}^3} \mathsf{B}_{M^\ell}(M^\ell(\eta + 2t + \zeta))$ of the fine lattice, for every shift η, and for nonnegative observables supported in C_0^η, with the reflected translates defined in Section 6.4. The inequality for $\mu^{(\ell)}$ is the special case of the functions $f \circ T_M^\ell$.

3. Plaquette energies under the Wilson measure. *Let $\ell = 0$ and $C_0 = 48 + 24 \log(1/c_{\mathrm{ball}})$, where $c_{\mathrm{ball}} = \operatorname{sinc}^2(1/2)/(12\pi)$ is the constant of Lemma 5.1. For $\beta \ge 2$ and $0 < s \le 1/2$ set*

$$A_s(n, \beta) = C_0 + 24 \log \frac{1}{1 - s} + \frac{36 \log \beta}{n}.$$

Then for every deterministic set S of plaquettes,

$$\mathbb{E}_{\mu_n^\beta} \exp\Big(\frac{s}{8} \sum_{p \in S} \beta q(U_p) \Big) \le e^{A_s(n,\beta)|S|/8}.$$

For $K \ge 0$ and a cell Q let $D_Q(K) = \{\max_{p \subset Q} \beta q(U_p) \ge K\}$. For distinct cells $Q_{t_1}, \dots, Q_{t_k}$ of one tiling,

$$\mu_n^\beta\Big(\bigcap_{j=1}^{k} D_{Q_{t_j}}(K) \Big) \le \big[6 e^{A_s(n,\beta) - sK} \big]^k$$

and

$$\mu_n^\beta\Big\{\max_p \beta q(U_p) \ge K\Big\} \le \min\big\{1, 6n^3 e^{A_s(n,\beta)-sK}\big\}.$$

If $\beta \le b_+ n$ for some $b_+ > 0$, then $36\log\beta/n \le 36b_+/\mathrm{e}$.

4. Conditioning. *If D is an event with $\mu_n^\beta(D) \ge 1 - \varepsilon_D > 0$, then $\mathbb{E}_{\mu_n^\beta}[X \mid D] \le (1-\varepsilon_D)^{-1}\mathbb{E}_{\mu_n^\beta} X$ for every nonnegative random variable X.*

The constant C_0 is absolute; no constant in (1)–(4) depends on L, β, n, N, or ℓ.

Remark 6.7. Reflection positivity is asserted here for the Wilson measure and its images. For $\ell > 0$, the integral in part (2) is estimated in Sections 22 and 23. The term $36\log\beta/n$ in A_s comes from the comparison of the periodic lattice with the open cube in Lemma 6.11. After tree gauge fixing, the periodic lattice has $r_P = 2n^3 + 1$ independent link variables and the open cube with n vertices per side has $r_O = 2n^3 - 3n^2 + 1$. Each variable contributes a factor of order $\beta^{-3/2}$, and the chessboard exponent is $1/\mathfrak{m} = 8/n^3$, so that

$$\frac{8}{n^3}\cdot\frac{3}{2}\,(r_P - r_O)\log\beta = \frac{36\log\beta}{n}.$$

This term cannot be absorbed into C_0 when n and β vary independently, so we keep it.

We prove part (1) in Section 6.4, part (2) in Section 6.4, and parts (3) and (4) in Section 6.4.

Reflection positivity

Lemma 6.8. *Let $N \ge 4$ be even, $\beta \ge 0$, and let r be a bond reflection of the periodic lattice of side N. Then μ_N^β is invariant under r, and $\mathbb{E}_{\mu_N^\beta}[F\,\overline{F\circ\mathsf{r}}] \ge 0$ for every observable F supported in a reflection half of r.*

Proof. Since the two planes of r divide the vertices into halves of width $N/2 \ge 2$, a vertex is an endpoint of at most one link crossing the planes, and the crossing links form a matching. We write the integral of the gauge-invariant function $G(U) = F(U)\overline{F(\mathsf{r}U)}e^{-\beta\sum_p q(U_p)}$ against H_N as an iterated integral, first over the links not crossing the planes and then over the crossing links. For fixed values of the crossing links, let k be the gauge transformation with $k_y = U_e$ at the endpoint y in the plus half of each crossing link e, oriented from the minus half to the plus half, and $k = I$ at all other vertices. It is well defined because the crossing links form a matching. Then $(U^k)_e = I$ on every crossing link, while every other link undergoes a left or right translation by a matrix depending only on the crossing links. These translations preserve the product Haar measure of the noncrossing links, and $G(U^k) = G(U)$. Hence the integral equals the integral of G with all crossing links set to I, against the Haar measure of the noncrossing links. The integrals over the crossing links give one. No cycle around the torus is removed by this gauge fixing.

A plaquette either has all its vertices in one half or crosses one of the planes. After the crossing links are set to I, the transport around a crossing plaquette is, up to conjugation and inversion, UV^{-1}, where U is a link of the plus half parallel to the plane and adjacent to it, and V is the reflected link of the minus half. We identify $\mathrm{SU}(2)$ with the unit quaternions, $U \leftrightarrow u \in S^3 \subset \mathbb{R}^4$ and $V \leftrightarrow v$. Then $\mathrm{tr}(UV^{-1})/2 = u \cdot v$, so

$$e^{-\beta q(UV^{-1})} = e^{-\beta} \sum_{r=0}^{\infty} \frac{\beta^r}{r!} \sum_{a_1,\dots,a_r=1}^{4} u_{a_1} \cdots u_{a_r} v_{a_1} \cdots v_{a_r}. \tag{6.11}$$

For each r the sum of the absolute values of the terms is at most $(\sum_a |u_a v_a|)^r \le 1$. The series and its finite products over the crossing plaquettes therefore converge absolutely and uniformly, and every coefficient is nonnegative for $\beta \ge 0$. The plaquettes inside the minus half and the Haar measure of its links are the reflected copies of those of the plus half. Integrating $F\overline{F \circ \mathsf{r}}$ against the expanded weight, we obtain

$$Z_N(\beta)\, \mathbb{E}_{\mu_N^\beta}\left[F\, \overline{F \circ \mathsf{r}}\right] = e^{-\beta |\mathcal{P}_{\mathrm{cross}}|} \sum_{\alpha} c_\alpha \left| \int F(U_+) e^{-\beta A_+(U_+)} \phi_\alpha(U_+)\, dH_+(U_+) \right|^2 \ge 0. \tag{6.12}$$

Here $\mathcal{P}_{\mathrm{cross}}$ is the set of crossing plaquettes, α runs over the degrees and component indices in (6.11) for all crossing plaquettes, c_α is the product of the corresponding coefficients $\beta^r/r!$, U_+ denotes the links with both endpoints in the plus half, H_+ their product Haar measure, A_+ the sum of $q(U_p)$ over the plaquettes in the plus half, and ϕ_α the product of the corresponding real quaternion components of the plus links adjacent to the planes. Absolute convergence justifies the interchange of sum and integral for bounded Borel F. The partition function $Z_N(\beta)$ is positive by Lemma 5.1, and the term $r = 0$ covers $\beta = 0$. Finally, μ_N^β is invariant under r, since Haar measure is preserved by permutations of links and by inversion, and q is preserved by conjugation and inversion. □

Proof of Theorem 6.6(1). For $\ell = 0$ this is Lemma 6.8. Let $\ell \ge 1$ and $a = M^\ell$, so that $N = an$. In fine units, a bond reflection of the coarse lattice acts in the reflected coordinate by $x \mapsto (2b+1)a - x$ for some $b \in \mathbb{Z}$. Since a is odd, both planes are bond planes of the fine lattice. For $0 \le k < \ell$, the intercept divided by the mesh M^k of the lattice after k averaging steps is $(2b+1)M^{\ell-k}$, an odd multiple of M. By Proposition 6.2(4), each averaging step therefore commutes with this reflection, including at the configurations where the Borel extension is used, and so does T_M^ℓ. By Proposition 6.2(6), a coarse link with both endpoints in one open half depends only on fine links in the union of the blocks of side a of that half. Since the block nearest a plane ends half a fine lattice spacing before it, these fine links do not cross the fine planes. Since T_M^ℓ is Borel, the composition with T_M^ℓ of an observable supported in a reflection half is bounded and Borel. By gauge covariance (Proposition 6.2(3)), it is invariant under all periodic gauge transformations of the fine lattice. Applying Lemma 6.8 to $F \circ T_M^\ell$ and to the other functions in the stated class

proves (1). By the same covariance, $\mu^{(\ell)}$ is invariant under the translations of the coarse lattice and under its bond reflections, since μ_N^β is invariant under the corresponding fine maps. □

The chessboard estimate

For an odd integer $a \geq 1$, $n \in 4\mathbb{N}$, and a shift $\eta \in \{0,1\}^3$, consider the periodic lattice of side an and the cells

$$C_t^\eta = \bigcup_{\zeta \in \{0,1\}^3} \mathsf{B}_a\big(a(\eta + 2t + \zeta)\big), \qquad t \in (\mathbb{Z}/(n/2)\mathbb{Z})^3.$$

They are cubes with $2a$ vertices in each direction, and they partition the vertices, since the blocks $\mathsf{B}_a(av)$, $v \in \mathbb{Z}^3$, do; for $a = 1$ they are the cells $\eta + 2t + \{0,1\}^3$ above. In the coordinate j, the plane between the cells with indices k and $k+1$ is $x_j = (2\eta_j + 4k + 3)a/2$. It bisects links, since a is odd, and the reflection through it, $x_j \mapsto (2\eta_j + 4k + 3)a - x_j$, maps each cell onto a cell. The second fixed plane of this reflection lies at distance $an/2 = 2a \cdot (n/4)$ from the first, which is a multiple of the side $2a$ of the cells. It is also a plane between consecutive cells, and every cell lies in one open half of the reflection. Observables supported in C_t^η are defined as for $a = 1$; they depend only on the links with both endpoints in C_t^η. The maps g_t now have the coordinate maps $g_{t,j}(x_j) = x_j + 2at_j$ if t_j is even and $g_{t,j}(x_j) = (2\eta_j + 2t_j + 1)a - x_j$ if t_j is odd, modulo an; they map C_0^η onto C_t^η and are compositions of t_j reflections through planes between consecutive cells in each coordinate j. Reflected translates are defined by $\vartheta_t f(U) = f(U \circ g_t)$ for observables f supported in C_0^η. If r is the reflection through the plane between the cells with indices k and $k+1$ in the coordinate j, it maps the index t to the index t' with $t'_j = 2k + 1 - t_j$ and the other components unchanged. Further, $\mathsf{r} \circ g_t = g_{t'}$, since for even t_j, $(2\eta_j + 4k + 3)a - (x_j + 2at_j) = (2\eta_j + 2t'_j + 1)a - x_j$, and for odd t_j, $(2\eta_j + 4k + 3)a - ((2\eta_j + 2t_j + 1)a - x_j) = x_j + 2at'_j$. This implies $(\vartheta_t f) \circ \mathsf{r} = \vartheta_{t'} f$.

Lemma 6.9. *Let $a \geq 1$ be odd, $n \in 4\mathbb{N}$, and $\eta \in \{0,1\}^3$. Let ν be a probability measure on $\mathcal{U}_{an}$ such that, for every bond reflection r through the planes between consecutive cells C_t^η, ν is invariant under r and $\mathbb{E}_\nu[F\,\overline{F \circ \mathsf{r}}] \geq 0$ for every observable F supported in a reflection half of r. Then for nonnegative observables $f_1, \dots, f_k$ supported in C_0^η and distinct cell indices $t_1, \dots, t_k$,*

$$\mathbb{E}_\nu \prod_{j=1}^k \vartheta_{t_j} f_j \leq \prod_{j=1}^k \Big[\mathbb{E}_\nu \prod_t \vartheta_t f_j\Big]^{1/\mathfrak{m}}, \qquad \mathfrak{m} = n^3/8.$$

The hypotheses concern only the reflections between the cells of the given tiling. In the applications below, all bond reflections are available.

Proof. The Schwarz inequality. Let r be a reflection through planes between consecutive cells, and call one of its open halves the plus half. For observables G, H supported in

the plus half, let $B_{\mathsf{r}}(G,H) = \mathbb{E}_\nu[G\,\overline{H\circ\mathsf{r}}]$. Since ν is invariant under the involution r, we have $B_{\mathsf{r}}(H,G) = \mathbb{E}_\nu[(H\circ\mathsf{r})\,\overline{G}] = \overline{B_{\mathsf{r}}(G,H)}$, so B_{r} is Hermitian, and by assumption it is positive semidefinite. Expanding $B_{\mathsf{r}}(G+zH, G+zH) \geq 0$ for complex z, we find $|B_{\mathsf{r}}(G,H)|^2 \leq B_{\mathsf{r}}(G,G)B_{\mathsf{r}}(H,H)$. If a diagonal entry vanishes, varying the real and imaginary parts of z shows that the mixed entry vanishes as well, so no nondegeneracy is needed. For an assignment $\boldsymbol{f} = (f_t)_t$ of nonnegative observables supported in C_0^η to the cell indices, we set

$$\mathcal{F}(\boldsymbol{f}) = \mathbb{E}_\nu \prod_t \vartheta_t f_t, \qquad \mathcal{F}(\boldsymbol{f})^2 \leq \mathcal{F}(R_+\boldsymbol{f})\,\mathcal{F}(R_-\boldsymbol{f}), \tag{6.13}$$

where $R_+\boldsymbol{f}$ assigns f_t both to each index t of the plus half and to its image t' under r, and $R_-\boldsymbol{f}$ is defined in the same way from the minus half. To prove the inequality, let $G = \prod_t \vartheta_t f_t$ and $H = \prod_t \vartheta_t f_{t'}$, both products over the indices t of the plus half. Since every cell lies in one open half of r, the functions G and H are observables supported in the plus half: they are bounded, Borel, gauge invariant, and depend only on links with both endpoints in that half. Since $(\vartheta_t f_{t'})\circ\mathsf{r} = \vartheta_{t'} f_{t'}$, the product $H\circ\mathsf{r}$ is the product over the minus half, so that $\mathcal{F}(\boldsymbol{f}) = B_{\mathsf{r}}(G,H)$, $\mathcal{F}(R_+\boldsymbol{f}) = B_{\mathsf{r}}(G,G)$, and $\mathcal{F}(R_-\boldsymbol{f}) = B_{\mathsf{r}}(H,H)$. These identities show that (6.13) is the Schwarz inequality.

One direction. Consider first a cyclic array of q positions, q even, in which the object at each position may be an entire product over a transverse slab of cells, and reflections through the planes between positions act as above. Fix a finite collection of strictly positive bounded functions, and for its assignments $(f_0,\dots,f_{q-1})$ set

$$\mathcal{G}(f_0,\dots,f_{q-1}) = \frac{\mathcal{F}(f_0,\dots,f_{q-1})}{\prod_{j=0}^{q-1}\mathcal{F}(f_j,\dots,f_j)^{1/q}},$$

where $\mathcal{F}(f_j,\dots,f_j)$ is the value at the assignment that places f_j at every position. The denominator is positive. Under each of the two copies in (6.13) every function of one half appears twice. The product of the two new denominators therefore equals the square of the old one, and $\mathcal{G}$ satisfies (6.13). Let $\mathcal{G}_*$ be the maximum of $\mathcal{G}$ over the finite set of assignments; $\mathcal{G}_* \geq 1$, since a constant assignment gives the value one. Among the maximizers we choose one with a longest cyclic run of one function, of length k. If $k < q$, we cut just before this run and at the opposite position; these two planes are the two fixed planes of one reflection through planes between cells. We copy the half that begins with the run to the other half. If $k < q/2$, the new array has a run of length at least $2k$; if $k \geq q/2$, that half is constant and the new array is constant. The two copied arrays have values at most $\mathcal{G}_*$, and (6.13) at the maximizer forces both values to equal $\mathcal{G}_* > 0$. This contradicts the maximality of the run unless $k = q$. Hence a constant assignment is a maximizer, and $\mathcal{G}_* = 1$. This proves the one-dimensional inequality $\mathcal{F}(f_0,\dots,f_{q-1}) \leq \prod_j \mathcal{F}(f_j,\dots,f_j)^{1/q}$ for every even q, without assuming that q is a power of two.

Three directions. Let $q = n/2$. We group the cell functions into the q products over the transverse slabs with fixed first cell index and apply the one-dimensional inequality. Each resulting expectation repeats one slab, by reflected translates, in the first direction, and carries the exponent $1/q$. We apply the inequality in the second direction to each of these expectations, grouping over the first and third indices. Now the first two directions are homogeneous, and each exponent is $1/q^2$. Applying it in the third direction, grouping over the first two, leaves the products over all reflected translates of single cell functions, with exponent $1/q^3 = 1/\mathfrak{m}$. Every slab product used is bounded, nonnegative, gauge invariant, and supported in a reflection half of the reflections in question. To allow functions with zeros, we replace each f by $f + \varepsilon$ and let $\varepsilon \downarrow 0$; the finite products converge boundedly. For partial assignments, we place the constant function one in the unused cells. This proves the inequality for the tiling with shift η. □

Proof of Theorem 6.6(2). We apply Lemma 6.9 in two ways.

Coarse cells. For every shift η, the reflections through the planes between consecutive cells $\eta + 2t + \{0,1\}^3$ are bond reflections of the coarse lattice. By part (1) and the invariance proved with it, $\nu = \mu^{(\ell)}$ is invariant under every bond reflection of the coarse lattice and reflection positive for observables supported in its reflection halves. Then the lemma with $a = 1$ implies the functional inequality for $\mu^{(\ell)}$ and every shifted tiling.

Fine cells. Let $a = M^\ell$. For every shift η, the planes between consecutive cells C_t^η are $x_j = (2\eta_j + 4k + 3)M^\ell/2$, and the reflections through them are bond reflections of the fine lattice, since $(2\eta_j + 4k + 3)M^\ell$ is odd. An observable supported in a reflection half of such a reflection is a bounded Borel gauge-invariant function of the fine links with both endpoints in one open half. Since the blocks partition the vertices and the planes run between blocks, these are the fine links with both endpoints in the union of the blocks $\mathsf{B}_{M^\ell}(v)$ over the coarse vertices v of that half. By Lemma 6.8, $\nu = \mu_N^\beta$ is invariant under every bond reflection of the fine lattice and reflection positive for these observables. By the lemma with $a = M^\ell$, the inequality holds for nonnegative observables supported in C_0^η, for every shift η.

The coarse inequality as a special case. Let f be an observable supported in the coarse cell $\eta + \{0,1\}^3$. By Proposition 6.2(2), (3), and (6), $f \circ T_M^\ell$ is bounded, Borel, invariant under all periodic gauge transformations of the fine lattice, and depends only on fine links with both endpoints in the union of the blocks $\mathsf{B}_{M^\ell}(M^\ell v)$ over the vertices v of the coarse cell, that is, in C_0^η. Written in fine units, the coarse map g_t is the fine map g_t with $a = M^\ell$: in each coordinate it is a translation by $2M^\ell t_j$ or a reflection $x_j \mapsto (2\eta_j + 2t_j + 1)M^\ell - x_j$. After $k < \ell$ averaging steps these are, in the units of that lattice, a translation by a multiple of M or a reflection with intercept $(2\eta_j + 2t_j + 1)M^{\ell-k} \in M\mathbb{Z}$. By Proposition 6.2(4), each averaging step commutes with them, and so does T_M^ℓ. We conclude that $(\vartheta_t f) \circ T_M^\ell =$

$\vartheta_t(f \circ T_M^\ell)$ and

$$\mathbb{E}_{\mu^{(\ell)}} \prod_j \vartheta_{t_j} f_j = \mathbb{E}_{\mu_N^\beta} \prod_j \vartheta_{t_j}(f_j \circ T_M^\ell),$$

so the inequality for $\mu^{(\ell)}$ is the special case of the functions $f \circ T_M^\ell$ of the fine inequality.

Bad cells. The reflections permute the six plaquettes of a cell and conjugate or invert their transports, which preserves ρ_G; hence the reflected translate of $\mathbf{1}_{D_0(u)}$ is $\mathbf{1}_{D_t(u)}$. As a function of V, this indicator is a nonnegative observable supported in Q_0, since ρ_G is continuous and invariant under conjugation. Its composition with T_M^ℓ is Borel and gauge invariant by Proposition 6.2(2) and (3). Taking $f_j = \mathbf{1}_{D_0(u)}$ for the cells of T gives the bound for $\bigcap_{t\in T} D_t(u)$. A coarse ij plaquette is contained in a cell of the tiling whose shift matches the parities of its starting vertex in the directions i and j, with either shift in the third direction. The eight tilings therefore cover all plaquettes, with $8\mathfrak{m} = n^3$ cells in total. The event that all cells of a shifted tiling are bad is a translate of the corresponding event for $\eta = 0$, and $\mu^{(\ell)}$ is invariant under translations of the coarse lattice. The bound for a single cell and a union bound over the n^3 cells give the maximum inequality, since

$$\mu^{(\ell)}\Big(\bigcap_t D_t(u)\Big) = \frac{1}{Z_N(\beta)} \int e^{-\beta \sum_p q(U_p)} \mathbf{1}_{\cap_t D_t(u)}(T_M^\ell U)\, dH_N(U).$$

□

Partition functions and plaquette energy moments

We now derive the moment bounds of part (3). We combine the chessboard estimate with bounds on partition functions obtained by tree gauge fixing, by integrating independent plaquette variables, and by dropping nonnegative plaquette terms. Elementary bounds of this kind are in the tradition of the stability estimates of Faria da Veiga and O'Carroll in [64, 65]; we derive the precise periodic correction needed here.

Lemma 6.10. *For every $\beta > 0$,*

$$z_1(\beta) = \int_{\mathrm{SU}(2)} e^{-\beta q(U)}\, dH(U) = \frac{2}{\pi} \int_0^\pi e^{-\beta(1-\cos t)} \sin^2 t\, dt$$
$$= \frac{2e^{-\beta} I_1(\beta)}{\beta} \le c_G \beta^{-3/2}, \qquad c_G = \sqrt{2/\pi},$$

where $I_1(\beta) = \sum_{k\ge 0} (\beta/2)^{2k+1}/(k!(k+1)!)$, and $z_1(0) = 1$ by continuous extension. If S is a random variable with density $s^{1/2}e^{-s}/\Gamma(3/2)$ on $s > 0$, then

$$\frac{z_1(\beta)}{c_G \beta^{-3/2}} = \mathbb{E}\sqrt{(1 - S/(2\beta))_+}.$$

For an integer $n \ge 1$ let O_n be the open cube with vertices $\{0, \dots, n\}^3$, all links and plaquettes between them, and no prescribed boundary links, and let $Z_{O_n}(\beta)$ be its partition function with

respect to the product Haar probability measure. Let $r_n = 2n^3 + 3n^2$ *be the number of links of* O_n *not in a spanning tree, and let* $P_n = 3n^2(n+1)$ *be the number of its plaquettes. Then for every* $\beta \geq 1$,

$$-\tfrac{3}{2} r_n \log \beta + r_n \log c_{\mathrm{ball}} - 2P_n \leq \log Z_{O_n}(\beta) \leq r_n \log z_1(\beta) \leq -\tfrac{3}{2} r_n \log \beta + r_n \log c_G,$$

where $c_{\mathrm{ball}} = \mathrm{sinc}^2(1/2)/(12\pi)$. *The constants are absolute and the inequalities hold uniformly in* n *and* β; *the bound* $Z_{O_n}(\beta) \leq z_1(\beta)^{r_n}$ *holds for every* $\beta > 0$. *For an open cube with* n *vertices in each direction the number of links not in a spanning tree is* $2n^3 - 3n^2 + 1$. *Finally, the partition function of the periodic lattice of side* $n+1$ *at any* $\beta > 0$ *is at most* $Z_{O_n}(\beta)$.

The last statement compares unnormalized integrals. It bounds a periodic partition function from above, but it does not replace that partition function in the denominator of a normalized expectation.

Proof. One plaquette. By the radial form of Haar measure in Lemma 5.1, with $t = |X|/2$, we have $z_1(\beta) = (2/\pi) \int_0^\pi e^{-\beta(1-\cos t)} \sin^2 t \, dt$. To compute the series, let $J_k = \int_0^\pi \sin^2 t \cos^{2k} t \, dt$, so $J_0 = \pi/2$. For $k \geq 1$, integrating the derivative of $\sin^3 t \cos^{2k-1} t$, whose boundary values vanish, gives $(2k+2)J_k = (2k-1)J_{k-1}$, and therefore $J_k = \pi(2k)!/(2 \cdot 4^k k!(k+1)!)$. Odd powers of $\cos t$ integrate to zero against $\sin^2 t$, by the substitution $t \mapsto \pi - t$. Since the exponential series converges uniformly on $[0, \pi]$, integrating termwise, we find

$$z_1(\beta) = e^{-\beta} \sum_{k \geq 0} \frac{(\beta/2)^{2k}}{k!(k+1)!} = \frac{2e^{-\beta} I_1(\beta)}{\beta}.$$

The integral is at most one and tends to one as $\beta \downarrow 0$, which gives the continuous extension. For $\beta > 0$, we substitute $s = \beta(1 - \cos t)$. Since $\sin t = (2s/\beta)^{1/2}(1 - s/(2\beta))^{1/2}$ and $ds = \beta \sin t \, dt$, we obtain

$$z_1(\beta) = \frac{2\sqrt{2}}{\pi} \beta^{-3/2} \int_0^{2\beta} s^{1/2} e^{-s} \sqrt{1 - \frac{s}{2\beta}} \, ds. \tag{6.14}$$

Dividing by $c_G \beta^{-3/2}$ and using $\Gamma(3/2) = \sqrt{\pi}/2$ gives the identity with the Gamma variable S. The integrand $\sqrt{(1 - S/(2\beta))_+}$ lies between zero and one, which proves $z_1(\beta) \leq c_G \beta^{-3/2}$ for every $\beta > 0$.

Counting. The open cube has $(n+1)^3$ vertices, $3n(n+1)^2$ links, and $P_n = 3n^2(n+1)$ plaquettes, and $3n(n+1)^2 - (n+1)^3 + 1 = r_n$. The following links form a spanning tree: all links in direction 1, the links in direction 2 in the plane $x_1 = 0$, and the links in direction 3 on the line $x_1 = x_2 = 0$. Indeed every vertex reaches $(0,0,0)$ by decreasing first x_1, then x_2, then x_3, and there are $n(n+1)^2 + n(n+1) + n = (n+1)^3 - 1$ such links. By Lemma 5.1, integrating out the gauge in this tree gives the factor one. In this gauge

we write $Y_{x,y,z}$ and $Z_{x,y,z}$ for the links in directions 2 and 3 at the vertex (x, y, z); then $Y_{0,y,z} = I$ and $Z_{0,0,z} = I$.

Upper bound. Consider the set of plaquettes consisting of all 12 plaquettes, all 13 plaquettes, and the 23 plaquettes in the plane $x_1 = 0$. With positive orientations, their transports are

$$A_{x,y,z} = Y_{x+1,y,z} Y_{x,y,z}^{-1}, \qquad B_{x,y,z} = Z_{x+1,y,z} Z_{x,y,z}^{-1}, \qquad D_{y,z} = Z_{0,y+1,z} Z_{0,y,z}^{-1}.$$

For A and B the coordinate x runs from 0 to $n-1$ and the other coordinates over their full ranges, so there are $n^2(n+1)$ plaquettes of each of these two kinds, and there are n^2 plaquettes D; in total r_n. From the D variables we reconstruct the links $Z_{0,y,z}$ successively in y, starting from $Z_{0,0,z} = I$. From the A variables we reconstruct every Y successively in x, starting from $Y_{0,y,z} = I$. From the B variables, with $Z_{0,y,z}$ already known, we reconstruct the remaining Z successively in x. In each step a new link equals a new plaquette variable times a link already fixed. By invariance of Haar measure and iterated integration, these plaquette variables are independent and Haar distributed. The plaquettes not in this set contribute factors at most one, since $q \geq 0$. Dropping them, we find $Z_{O_n}(\beta) \leq z_1(\beta)^{r_n}$ for every $\beta > 0$.

Lower bound. Let $\beta \geq 1$. We restrict each of the r_n links not in the tree to $\{\rho_G(U_e) < \beta^{-1/2}\}$, where $\beta^{-1/2} \leq 1$. By Lemma 5.1 each such ball has Haar measure at least $c_{\mathrm{ball}} \beta^{-3/2}$. A plaquette contains at most four links not in the tree, so by the triangle inequality for ρ_G we have $\rho_G(U_p) \leq 4\beta^{-1/2}$, and $q(U_p) \leq \rho_G(U_p)^2/8 \leq 2/\beta$; the first inequality is an inequality between group distances and holds even if its right side exceeds the radius of the logarithm chart. Hence the Boltzmann factor is at least e^{-2P_n} on this set, whose product Haar measure is at least $(c_{\mathrm{ball}} \beta^{-3/2})^{r_n}$. Taking logarithms gives the lower bound, and the bound of the previous paragraph together with $z_1(\beta) \leq c_G \beta^{-3/2}$ gives the remaining inequality. Since the boundary spheres of the balls have Haar measure zero, open and closed balls give the same estimate.

Replacing n by $n-1$, we find that the number of links not in a spanning tree of an open cube with n vertices in each direction is $2(n-1)^3 + 3(n-1)^2 = 2n^3 - 3n^2 + 1$. Finally, deleting from the periodic lattice of side $n+1$ the plaquettes that contain a link joining the vertex layers $x_j = n$ and $x_j = 0$, for some j, leaves the open cube O_n and the links joining those layers, which no remaining plaquette uses. The integrals of the latter against Haar probability measure equal one, and deleting plaquettes with nonnegative cost can only increase the integral. This proves the last statement. □

Lemma 6.11. *Let $n \in 4\mathbb{N}$, put $V = n^3$ and $\mathfrak{m} = V/8$, and let $Z_P(n, \beta) = Z_n(\beta)$ be the partition function of the periodic lattice of side n. Put $r_P = 2V + 1$ and $r_O = 2V - 3n^2 + 1$. Then*

$$\log Z_P(n, \beta) \geq -\tfrac{3}{2} r_P \log \beta + r_P \log c_{\mathrm{ball}} - 6V, \qquad \beta \geq 1, \tag{6.15}$$

$$Z_P(n, \beta') \leq Z_{O_{n-1}}(\beta') \leq \left(c_G (\beta')^{-3/2}\right)^{r_O}, \qquad \beta' > 0. \tag{6.16}$$

As a consequence, for $\beta \geq 2$ *and* $0 < s \leq 1/2$,

$$\begin{aligned} \frac{1}{\mathfrak{m}} \log \frac{Z_P(n,(1-s)\beta)}{Z_P(n,\beta)} &\leq \frac{12(r_P - r_O)}{V} \log \beta + \frac{12 r_O}{V} \log \frac{1}{1-s} \\ &\quad + \frac{8}{V}\left(r_O \log c_G - r_P \log c_{\mathrm{ball}} + 6V\right) \\ &\leq \frac{36 \log \beta}{n} + 24 \log \frac{1}{1-s} + C_0 = A_s(n,\beta), \end{aligned} \tag{6.17}$$

with

$$C_0 = 48 + 24 \log(1/c_{\mathrm{ball}}). \tag{6.18}$$

Proof. A spanning tree of the periodic lattice has $V - 1$ links, so $3V - (V-1) = r_P$ links are not in it; by Lemma 5.1 the gauge in the tree integrates to one. We restrict all links not in the tree, including those carrying the holonomies around the torus, to balls $\{\rho_G(U_e) < \beta^{-1/2}\}$ with $\beta^{-1/2} \leq 1$. As in the proof of Lemma 6.10, each ball has Haar measure at least $c_{\mathrm{ball}} \beta^{-3/2}$ and each plaquette costs at most $q(U_p) \leq 2/\beta$. There are $3V$ plaquettes, which gives (6.15). By the last statement of Lemma 6.10, applied with side $n = (n-1)+1$, the periodic partition function is at most that of the open cube with n vertices in each direction, which has r_O links not in a spanning tree. The upper bound of Lemma 6.10 then implies (6.16). This upper bound needs only $\beta' > 0$; below we use it with $\beta' = (1-s)\beta \geq 1$, where every inequality of the lemma applies.

Subtracting (6.15) from the logarithm of (6.16) with $\beta' = (1-s)\beta$ and multiplying by $1/\mathfrak{m} = 8/V$, we obtain the first inequality in (6.17). For the second we use $r_P - r_O = 3n^2$, $r_O/V \leq 2$, $r_P/V \leq 3$, $\log \beta \geq 0$, and $\log c_G < 0$: the first term equals $36 \log \beta / n$, the second is at most $24 \log(1/(1-s))$, and the third is at most $(8/V)(3V \log(1/c_{\mathrm{ball}}) + 6V) = C_0$. We keep the surface term $36 \log \beta / n$ in this form, before any relation between n and β is imposed. □

Proof of Theorem 6.6(3) and (4). We write $Y_p = \beta q(U_p) \geq 0$ and $\mathfrak{m} = n^3/8$. For a cell Q_0 let $f = \exp(s \sum_{p \subset Q_0} Y_p)$, the sum running over the six plaquettes of the cell. This is a nonnegative bounded observable supported in Q_0. The product of its reflected translates over all cells uses each plaquette inside a cell exactly once, so its exponent is at most $s \sum_p Y_p$, the sum over all plaquettes. By Lemma 6.11, we have

$$\mathbb{E}_{\mu_n^\beta} \prod_t \vartheta_t f \leq \frac{Z_P(n,(1-s)\beta)}{Z_P(n,\beta)} \leq e^{A_s(n,\beta)\mathfrak{m}}.$$

By part (2), the product of f over h distinct cells of one tiling has expectation at most $e^{A_s(n,\beta)h}$. Now let S be a deterministic set of plaquettes. We assign each $p \in S$, by a fixed rule, to a cell containing it in one of the eight shifted tilings. Let S_a be the plaquettes assigned to the tiling a, and let h_a be the number of cells of that tiling to which some

plaquette is assigned. Enlarging each assigned sum to all six plaquettes of the cell, we find $\mathbb{E}\exp(s\sum_{p\in S_a} Y_p) \le e^{A_s(n,\beta)h_a}$. By Hölder's inequality with eight exponents equal to eight, we have

$$\mathbb{E}e^{(s/8)\sum_{p\in S} Y_p} \le \prod_{a=1}^{8}\Big[\mathbb{E}e^{s\sum_{p\in S_a} Y_p}\Big]^{1/8} \le e^{A_s(n,\beta)\sum_a h_a/8} \le e^{A_s(n,\beta)|S|/8}.$$

In the last inequality we used $\sum_a h_a \le |S|$ and $A_s(n,\beta) \ge 0$.

For the bad cells, note the pointwise bound $\mathbf{1}_{D_Q(K)} \le \sum_{p\subset Q} e^{s(Y_p-K)}$. Multiplying over the $\mathfrak{m}$ cells of a tiling and expanding, we obtain $6^{\mathfrak{m}}$ terms, one for each choice of a plaquette in every cell. The chosen plaquettes are distinct, and the sum of their energies is at most the sum over all plaquettes. Therefore

$$\mu_n^\beta\Big(\bigcap_t D_{Q_t}(K)\Big) \le 6^{\mathfrak{m}} e^{-sK\mathfrak{m}}\,\frac{Z_P(n,(1-s)\beta)}{Z_P(n,\beta)} \le \big[6e^{A_s(n,\beta)-sK}\big]^{\mathfrak{m}}.$$

Applying part (2) to the indicator of $D_{Q_0}(K)$ in k distinct cells, we obtain the joint bound. A union bound over the n^3 cells of the eight tilings, together with the trivial bound by one, gives the bound for the maximum. All constants in these bounds are absolute.

If $\beta \le b_+ n$, then

$$\frac{\log\beta}{n} = \frac{\beta}{n}\cdot\frac{\log\beta}{\beta} \le \frac{b_+}{\mathrm{e}},$$

since $(\log x)/x$ attains its maximum at $x = \mathrm{e}$; this simplification uses the additional relation between β and n. Finally, for a nonnegative X and an event D with $\mu_n^\beta(D) \ge 1-\varepsilon_D > 0$,

$$\mathbb{E}[X \mid D] = \frac{\mathbb{E}[X\mathbf{1}_D]}{\mu_n^\beta(D)} \le (1-\varepsilon_D)^{-1}\mathbb{E}X.$$

This bound does not depend on the number of factors in X, and it does not use reflection positivity of the conditioned measure. This completes the proof of Theorem 6.6. □

SECTION 7

Periodic gauge fixing and energy estimates

In this section, we prove deterministic estimates for the block averaging map and for gauge fixing, and one probability bound. Together they show that configurations with small curvature at every scale lie in the domains where the renormalization group analysis applies. The two main estimates are the following. We write $\delta(U) = \max_p \rho_G(U_p)$ for the maximal curvature and $\mathscr{E}(U) = \sum_p q(U_p)$ for the Wilson energy of a configuration U. If $\delta(U)$ is small, then

$$\mathscr{E}(T_M U) \le M\big(1 + C_M \delta(U)\big)\mathscr{E}(U), \qquad \delta(T_M U) \le M^2 \delta(U) + c_M \delta(U)^2$$

(Theorem 7.3). Since the squared coupling grows by the factor M in each averaging step, the first inequality bounds the output Wilson action divided by the squared coupling by $1 + C_M\delta(U)$ times its input value. The product of these upper-bound factors is bounded under the squared-mesh curvature hypothesis of part (5) of that theorem. Second, on a periodic cubic lattice of side $n \ge 4$, every configuration with $\delta(U) \le \delta \le \delta_{\mathrm{SG}}$ has a periodic gauge transform U^k with

$$\max_e \rho_G(U^k_e) \le C_{\mathrm{SG}}\big(\sqrt{\delta} + n^{-1}\big)$$

(Theorem 7.7). In particular, $\delta(U) \le \delta_r$ and $n \ge 4C_{\mathrm{SG}}/r$ imply that U lies in the gauge-invariant small-field domain D_r, with δ_r defined in that theorem, and the plaquette moments of Theorem 6.6 bound the probability of the complement of D_r under the Wilson measure.

Section 7.1 gives sufficient conditions, small links or small plaquettes, under which the logarithms of a configuration relative to the coarse-field lift are the local constraint coordinates of Lemma 6.3 (Lemma 7.2), and Section 7.2 proves the energy estimate; these results are used in the classical analysis of Sections 8, 9, and 11 and in Sections 14, 22, and 23. Section 7.3 describes the dependence sets of iterated averages and the eight shifted tilings by which the chessboard estimates are applied to blocked plaquettes (Lemma 7.4). Section 7.4 constructs periodic gauges by gluing local axial gauges (Lemmas 7.5 and 7.6), and Section 7.5 proves the small-curvature gauge estimate and the probability bound; these results are used in the large-field analysis of Sections 18 and 21 to 23. The inputs are the block averaging map, its local constraint coordinates, and the chessboard estimates of Section 6, together with the metric, word, and conditioning estimates of Lemmas 5.1 and 5.5. No result on conditional constrained minimizers or on normalized fluctuation variables is used.

Conventions. The distance between group elements is $\operatorname{dist}(g,h) = \rho_G(gh^{-1})$. We work in lattice units until a physical mesh is displayed. A rectangular patch contains all its vertices, all links with both endpoints among those vertices, and all plaquettes with their four vertices among them; in particular its boundary links are included. A plaquette is counted once in every sum of energies, and a maximum over an empty set of plaquettes is zero. When an estimate concerns only the norm of a logarithm of a group element V, it refers to a shortest logarithm, whose norm is $\rho_G(V)$; every logarithm that is later used as an analytic coordinate lies in a fixed ball strictly inside the domain of the principal logarithm. Constants depend only on M and on such fixed radii, with additional dependencies stated explicitly; this is the dependence described in item (1) of Definition 4.14.

7.1 Local inverse coordinates for block averaging

We begin with the elementary discrete form of the non-Abelian Stokes theorem in a box: in an axial gauge, every link is a product of transported plaquette variables along strips of width one.

Lemma 7.1. *Let $Q = \prod_{i=1}^{3}\{0,\dots,b_i\}$ with integers $b_i \geq 0$, assign arbitrary values in $\mathrm{SU}(2)$ to its links, and put $b = b_1 + b_2 + b_3$. Let $q_Q(x)$ be the transport from the lower corner of Q to x along the path that changes the coordinates in the order $1,2,3$. After the gauge transformation q_Q, every link of Q is a product of at most b conjugates of plaquette variables of Q or their inverses. In particular, writing $d_p = \rho_G(U_p)$,*

$$\max_{e\subset Q} \rho_G\big(U_e^{q_Q}\big) \leq b \max_{p\subset Q} d_p, \qquad \Big(\sum_{e\subset Q} \rho_G\big(U_e^{q_Q}\big)^2\Big)^{1/2} \leq b\sqrt{3|Q|}\Big(\sum_{p\subset Q} d_p^2\Big)^{1/2}.$$

If the first bound is smaller than a fixed radius below 2π, these are bounds for the principal logarithms of the links. Every closed path of s links in Q has a transport that is a product of at most sb conjugated plaquette variables. These statements also hold in a periodic lattice for a box that lifts injectively to the covering lattice.

Proof. For two successive orthogonal positive steps we have the identity

$$U_i(x)U_k(x+e_i) = U_{p_{ik}(x)}U_k(x)U_i(x+e_k), \qquad i<k.$$

Changing an orientation replaces the plaquette factor by a conjugate of its inverse, and in a longer path the transport of the preceding subpath conjugates the plaquette factor. Each interchange of two adjacent orthogonal steps therefore introduces exactly one transported plaquette and leaves the other factors unchanged. Repeating it gives the corresponding identity for a strip of width one, including its boundary links.

The path from the corner to x first runs in direction 1, then 2, then 3. For a link in direction 3, the path to its endpoint is the path to its initial point followed by the link,

and its transformed value is I. For a link in direction 2, we move it backwards through the direction-3 part of the path, of length x_3. This uses the x_3 plaquettes of a 23 strip of width one, and the remaining path is the path to the endpoint, which cancels. For a link in direction 1, we move it first through the direction-3 part, using x_3 plaquettes of a 13 strip, and then through the direction-2 part in the lower plane, using x_2 plaquettes of a 12 strip. The direction-1 parts then cancel. These moves stay in the closed box and use at most b plaquettes. Reversing a link inverts its word. Using the bi-invariance of ρ_G, the triangle inequality $\rho_G(UV) \le \rho_G(U) + \rho_G(V)$ of Lemma 5.1, and invariance under inversion, we obtain the bound on the maximum.

Since each d_p is at most $(\sum_{p \subset Q} d_p^2)^{1/2}$, each transformed link is bounded by b times this quantity, and there are at most $3|Q|$ links. This gives the second bound. In a closed path the gauge factors cancel except for a conjugation at the starting vertex, and substituting the words just obtained for its links gives at most sb factors. In an injective periodic lift every interchange uses a lifted plaquette, and the same proof applies. If Q has no plaquettes, all words are empty. □

For a coarse link $c = (v, v + Me_i)$, its two blocks form a rectangular patch with $2M$ vertices in direction i and M vertices in the other directions. For a coarse ik plaquette, the four blocks at its corners form a rectangular patch with $2M$, $2M$, and M vertices in the directions i, k, and the third direction. Their diameters in the lattice path metric are at most $4M$ and $5M$. Since on a coarse lattice of side $m \ge 4$ every coordinate extent of these patches is smaller than the period Mm, each patch lifts injectively to a box of the covering lattice, also when it meets the periodic identifications. Every translated path with endpoint connectors, every reference path, and every strip used below lies in such a lift. Figure 7.1 shows these patches.

We now recall the setting of Lemma 6.3. Let h be the block axial gauge transformation, $h_x = U(\sigma_{v,x})$ for $x \in \mathsf{B}_M(v)$, so that $h_v = I$ at every block center and $U^h = I$ on the block trees. By gauge covariance, $b = T_M U^h = T_M U$. On the links not in the block trees we write

$$U_e^h = e^{X_e} U_{0,e}(b), \qquad z = X_{\text{free}}.$$

Lemma 6.3 provides radii r_M and $\widetilde{r}_M$ for the free and the eliminated coordinates, independent of the coarse configuration b, within which the constraint has the unique solution $X = \Gamma_b(z)$. The next lemma identifies the logarithms X of the given configuration with $\Gamma_b(z)$. The existence of a solution of the constraint near $U_0(b)$ would not suffice, since later sections integrate functions of the given configuration in these coordinates. We distinguish three normalizations: the relative logarithms X and their free part z, which are controlled here; the coupling-rescaled coordinates z/g; and the normalized fluctuation variables w of Lemma 12.3, defined by $z_i = z_i^* + g_i K_i^{-1/2} w_i$ after the conditional constrained minimizer has been subtracted, which do not appear in this section.

Lemma 7.2. *Fix an odd $M \geq 3$ and a fine side Mm with $m \geq 4$, and let T_M, the block trees, and Γ_b be as in Definition 6.1 and Lemma 6.3. There exist $u_M, \delta_M > 0$ and $C_M < \infty$, depending only on M and on fixed radii strictly inside the coordinate domain of Lemma 6.3 and the domain of the principal logarithm, with the following properties.*

1. *If $\max_e \rho_G(U_e) \leq u$ and $(5M-3)u < 2\pi$, then no based loop holonomy $U(F_{c,x})U(C_c)^{-1}$ in one application of T_M equals $-I$, and*

$$\max_c \rho_G\big((T_M U)_c\big) \leq (6M-3)u \leq 10Mu.$$

   In the block axial gauge, the logarithms relative to $U_0(b)$, $b = T_M U$, of the links not in the block trees satisfy $\max_e |X_e| \leq C_M u$, where shortest logarithms are meant. If moreover $u \leq u_M$, then all these logarithms, including those on the eliminated edges, are principal logarithms in the coordinate domain of Lemma 6.3, and $X = \Gamma_b(X_{\text{free}})$.
2. *Let $J \geq 1$ be fixed, and let the fine side be $M^J m_{\text{out}}$ with an integer $m_{\text{out}} \geq 4$. Suppose that $\max_e \rho_G(U_e) \leq u$ and $(10M)^J u \leq u_M$. Then the conclusions of (1) with $u \leq u_M$ hold for each of the J averaging steps, with u replaced by $(10M)^s u$ at the step from $T_M^s U$ to $T_M^{s+1} U$, and*

$$\max_e \rho_G\big((T_M^s U)_e\big) \leq (10M)^s u, \qquad 0 \leq s \leq J.$$

3. *Suppose only that $\max_p \rho_G(U_p) \leq \delta \leq \delta_M$; no bound on the links is assumed. Then all based loop holonomies in one application of T_M lie in a fixed ball strictly inside the domain of the principal logarithm, and in the block axial gauge*

$$\max_e |X_e| \leq C_M \delta, \qquad X = \Gamma_{T_M U}(X_{\text{free}}), \qquad \max_P \rho_G\big((T_M U)_P\big) \leq C_M \delta,$$

   where P runs over the coarse plaquettes. For every $g > 0$ the coupling-rescaled coordinate $\eta = z/g$ satisfies $\|\eta\|_\infty \leq C_M \delta / g$.
4. *Each statement is local: it holds for a given coarse link or plaquette if its hypothesis is imposed only on the patch formed by the two blocks of the link, or by the four blocks at the corners of the plaquette, with all their links and plaquettes. Links outside these patches and the holonomies around the noncontractible cycles of the torus do not affect the estimates.*

All constants are independent of m, of the physical scale, of the coupling, and of the number of preceding averaging steps.

Proof. Small links. Recall that $R_M = (M-1)/2$. Since each root path has at most $3R_M$ links, the path $F_{c,x}$ has at most $M + 6R_M = 4M - 3$ links and $F_{c,x} C_c^{-1}$ at most $5M - 3$. By the triangle inequality, a product of s links of distance at most u from I has distance at most su. If $(5M-3)u < 2\pi$, every based loop holonomy therefore differs from $-I$. With

$$A_c = M^{-3} \sum_{x \in \mathsf{B}_M(v)} \operatorname{Log}\big(U(F_{c,x})U(C_c)^{-1}\big), \qquad b_c = e^{A_c} U(C_c),$$

we have $|A_c| \le (5M-3)u$ and hence $\rho_G(b_c) \le (6M-3)u \le 10Mu$.

In the block axial gauge every transformed link not in a block tree is a product of at most $1 + 6R_M = 3M - 2$ links, and its distance from I is at most $(3M-2)u$. Since its value under the coarse-field lift is I or one of the b_c, every relative element $U_e^h U_{0,e}(b)^{-1}$ has distance at most $(9M-5)u$ from I. Taking shortest logarithms gives the first bound in (1) with any $C_M \ge 9M$.

Let $r_M, \widetilde{r}_M$ be the radii of Lemma 6.3 for the free and the eliminated coordinates. Fix smaller positive radii inside them and inside a fixed ball strictly within the domain of the principal logarithm, and choose $u_M > 0$ with

$$(5M-3)u_M < 2\pi, \qquad C_M u_M < \tfrac{1}{2}\min\{r_M, \widetilde{r}_M, 1\},$$

decreasing u_M further if the fixed ball requires it. For $u \le u_M$ all relative logarithms are unique principal logarithms. By gauge covariance, and since $h = I$ at the block centers, $T_M U^h = b$. Hence the vector X solves the constraint at the target b, and both its free and its eliminated entries lie in the uniqueness neighborhood of Lemma 6.3. Uniqueness implies $X = \Gamma_b(X_{\text{free}})$; this identifies every eliminated entry, not only the free ones. This proves (1).

Several averaging steps. Let the initial side be $M^J m_{\text{out}}$, and define

$$n_s = M^{J-s} m_{\text{out}} \quad (0 \le s \le J), \qquad n_s = M n_{s+1}, \quad n_{s+1} \ge m_{\text{out}} \ge 4 \quad (0 \le s < J). \tag{7.1}$$

At each step, the input side is a multiple of M and the output side is at least four, so the hypotheses of Definition 6.1 and Lemma 6.3 hold and every patch used above lifts injectively. Let $u_s = (10M)^s u$. The assumed bound $\max_e \rho_G(U_e) \le u$ gives the link bound for $s = 0$, and the hypothesis $(10M)^J u \le u_M$ implies $u_s \le u_M$, and hence $(5M-3)u_s < 2\pi$, for $0 \le s \le J$. We apply (1) to $T_M^s U$ on the lattice of side n_s, for $s = 0, 1, \dots, J-1$ in turn. This identifies the relative coordinates of each step with the solution Γ at its target and gives the link bound $10Mu_s = u_{s+1}$ for the next step. The constants are the same at every step, since they do not depend on n_s. This proves (2).

Small plaquettes. We now assume only a bound on the plaquettes, and let $K_M^{(0)} = 100M^2$. Every based loop holonomy is the transport around a closed path of at most $5M-3$ links in a patch of two blocks, whose side lengths sum to at most $4M$. By Lemma 7.1, it is a product of at most $20M^2 \le K_M^{(0)}$ conjugated plaquette variables, so

$$\rho_G\big(U(F_{c,x})U(C_c)^{-1}\big) \le K_M^{(0)}\delta, \qquad |A_c| \le K_M^{(0)}\delta. \tag{7.2}$$

Here and below, δ need only bound the plaquettes of the indicated patch. It is chosen so small that the first bound lies in a fixed ball strictly inside the domain of the principal logarithm.

Consider a link not in the block tree with both endpoints in one block. Its gauge-transformed value is the transport around the closed path formed by the link and two

root paths, of length at most $6R_M + 1$ in one block. The same argument bounds its distance from I by $K_M^{(0)}\delta$. For an interface link $e = (x, y)$ between the blocks of c, we write

$$W_e = U(\sigma_{v,x})\, U_e\, U(\sigma_{v+Me_i,y})^{-1} = U_e^h.$$

The closed path for $W_e U(C_c)^{-1}$ lies in the patch of the two blocks and has length at most $6R_M + 1 + M$. As before, we have $\rho_G(W_e U(C_c)^{-1}) \le K_M^{(0)}\delta$. Since $b_c = e^{A_c} U(C_c)$,

$$\rho_G\big(W_e b_c^{-1}\big) \le \rho_G\big(W_e U(C_c)^{-1}\big) + |A_c| \le 2K_M^{(0)}\delta. \tag{7.3}$$

This includes the eliminated edge. The coarse plaquette formed by the four reference paths has distance at most $M^2\delta$ from I, since its M by M square can be reduced by successive interchanges of steps that use each of its M^2 plaquettes once. Replacing its four reference transports by the values b_c changes this distance by at most $4K_M^{(0)}\delta$. Here we used that the bi-invariant metric satisfies $\operatorname{dist}(\prod_i a_i, \prod_i b_i) \le \sum_i \operatorname{dist}(a_i, b_i)$, as one sees by replacing one factor at a time, and that inverting both elements preserves each term. We obtain

$$\rho_G\big((T_M U)_P\big) \le \big(M^2 + 4K_M^{(0)}\big)\delta. \tag{7.4}$$

We increase C_M so that it dominates the three constants in (7.2), (7.3), and (7.4), and choose $\delta_M > 0$ with $C_M\delta_M < \min\{r_M, \widetilde{r}_M, 1\}/2$, also respecting the fixed logarithm radius. Then the relative vector lies in the uniqueness neighborhood of Lemma 6.3, and, as in the proof of (1), we obtain $X = \Gamma_{T_M U}(X_{\mathrm{free}})$. Its free part satisfies $\|z\|_\infty \le C_M\delta$, and dividing by $g > 0$ gives the bound for η. This proves (3).

Locality. All paths and all fillings used above lie in the patches of two or four blocks. A gauge transformation on such a patch can be extended arbitrarily to the other vertices of the torus, and by covariance of T_M no link outside the patch enters these computations; the block axial gauge uses only the paths in its own block. The local statements therefore do not constrain any holonomy around a noncontractible cycle. For a local statement on the inverse coordinates, the equality with Γ_b refers to the coordinates in question: we extend the local free data by zero to a free vector in the coordinate domain. Each eliminated coordinate is determined by its own constraint, which involves only its two blocks. By the locality and uniqueness statements of Lemma 6.3, the corresponding entry of Γ_b does not depend on the extension. The local eliminated value solves the same constraint in the same uniqueness neighborhood; by uniqueness, the two agree. This proves (4). All radii and constants were chosen from fixed finite patches and from the radii of Lemma 6.3; they depend only on M and the fixed radii. □

7.2 Energy estimates and physical curvature

The leading factor M in the energy estimate has a simple origin. At the linear level, the curvature of a coarse plaquette is an average over M^3 translates of the sum of the M^2

fine curvatures in an M by M square ((7.5) below). A coarse curvature is therefore about M^2 times a typical fine curvature, while the number of plaquettes decreases by the factor M^3. The sum of the squared curvatures, and with it the Wilson energy, is multiplied by $M^4/M^3 = M$. Since the squared coupling satisfies $g_{j+1}^2 = Mg_j^2$, this factor cancels in the action $4\mathscr{E}/g^2$. Part (3) bounds the nonlinear error by the maximal curvature times the local ℓ^2 norm of the curvature, which keeps the estimate uniform in the volume. The bound on the maximal curvature in part (1) and its iteration in parts (5) and (6) adapt the estimates of Bałaban for his averaging operation in [12, Section B, Propositions 1 and 2], which also use the four blocks at the corners of a coarse plaquette. We give the proofs for the paths and the metric used here. The estimate for the Wilson energy with leading factor M requires in addition the exact column sums of the linearized kernel and the local form of the remainder.

Theorem 7.3. *For each odd $M \geq 3$ there exist $\delta_*(M) > 0$ and $C_M, c_M < \infty$ with the following properties, for every $m \geq 4$ and every configuration U on the fine lattice of side Mm. Write $\delta(U) = \max_p \rho_G(U_p)$ and $\mathscr{E}(U) = \sum_p q(U_p)$.*

1. *If $\delta(U) \leq \delta_*$, then*

$$\mathscr{E}(T_M U) \leq M\big(1 + C_M\delta(U)\big)\mathscr{E}(U), \qquad \delta(T_M U) \leq M^2\delta(U) + c_M\delta(U)^2.$$

2. *The derivative of T_M at the identity acts on curvatures by the nonnegative kernel*

$$(\mathsf{L}_M f)_{ik}(v) = M^{-3} \sum_{r \in \{-R_M,\dots,R_M\}^3} \sum_{s,t=0}^{M-1} f_{ik}(v + r + se_i + te_k),$$

that is, $\mathrm{d}\, DT_M(I) = \mathsf{L}_M \mathrm{d}$ on one-cochains. Its row sums equal M^2 and its column sums equal M^{-1}, and $\|\mathsf{L}_M f\|_2^2 \leq M\|f\|_2^2$ for every array f indexed by plaquettes with values in a Euclidean space.

3. *For a set $\mathcal{A}$ of positive coarse plaquettes, let $\mathrm{Pa}(\mathcal{A})$ be the union, counted once, of the fine plaquettes in the patches of four blocks of the plaquettes in $\mathcal{A}$. If $\delta = \max_{p \in \mathrm{Pa}(\mathcal{A})} \rho_G(U_p) \leq \delta_*$, then*

$$\sum_{P \in \mathcal{A}} q\big((T_M U)_P\big) \leq M(1 + C_M\delta) \sum_{p \in \mathrm{Pa}(\mathcal{A})} q(U_p).$$

For each $P \in \mathcal{A}$ there is a rooted axial gauge on its patch $Q(P)$ of four blocks in which

$$\mathrm{Log}\big((T_M U)_P\big)^{(P)} = \sum_p \mathsf{L}_M(P,p)\,\mathrm{Log}(U_p)^{(P)} + e_P,$$

with

$$|e_P| \leq C_M\delta\Big(\sum_{p \subset Q(P)} \rho_G(U_p)^2\Big)^{1/2},$$

where the superscript (P) refers to this gauge; the gauges for different P need not agree. Each fine plaquette lies in at most 48 of the patches $Q(P)$, and the restriction of L_M to output plaquettes in $\mathcal{A}$ has row sums at most M^2 and column sums at most M^{-1}.

4. *Let $U^{j+1} = T_M U^j$, $0 \le j < d$, be configurations on lattices all of whose coarse sides are at least four, and let $g_j > 0$ satisfy $g_{j+1}^2 = M g_j^2$. If $\delta_j = \delta(U^j) \le \delta_*$ for $j < d$, then*

$$\frac{4\mathscr{E}(U^d)}{g_d^2} \le \prod_{j<d}(1 + C_M\delta_j)\,\frac{4\mathscr{E}(U^0)}{g_0^2} \le e^{C_M \sum_{j<d}\delta_j}\,\frac{4\mathscr{E}(U^0)}{g_0^2}.$$

5. *Measure lengths in units of the mesh of the coarsest lattice, so that U^j lives on a lattice of mesh $\epsilon_j = M^{j-d}$. Let*

$$0 < \varepsilon \le \varepsilon_M, \qquad \varepsilon_M \le \min\bigl\{\delta_*/2,\ M^2(M^2-1)/(4c_M)\bigr\}.$$

If $\delta(U^0) \le \varepsilon\epsilon_0^2$, then each of $U^0, \dots, U^{d-1}$ satisfies the hypothesis of Lemma 7.2(3), and

$$\delta(U^j) \le 2\varepsilon\epsilon_j^2 \quad (0 \le j \le d), \qquad \sum_{j<d}\delta(U^j) \le \frac{2\varepsilon}{M^2-1},$$

and

$$\frac{4\mathscr{E}(U^d)}{g_d^2} \le e^{2C_M\varepsilon/(M^2-1)}\,\frac{4\mathscr{E}(U^0)}{g_0^2}.$$

6. *Let U^0 be obtained by sampling a fixed smooth periodic connection, with $\delta(U^0) \le C_A a_0^2$, where a_j denotes the physical mesh of U^j and $a_* = a_d$, so that $a_j = a_*\epsilon_j$. If $\varepsilon = C_A a_*^2 \le \varepsilon_M$, the conclusions of (5) hold, and the free logarithmic coordinates z_j of U^j in Lemma 7.2(3) satisfy*

$$\|z_j/g_j\|_\infty \le C_{A,M}\,a_j^{3/2}/g_3, \qquad g_j = g_3\sqrt{a_j}, \qquad 0 \le j < d,$$

with $C_{A,M} = 2C_M C_A$.

All constants are independent of d, of the volume, of the physical side, and of the coupling, except for the constant C_A of the smooth connection. The iterated bound in (4) is uniform under a bound on the sum of the curvatures, not merely under a bound on each curvature separately.

In the proof, we first compute the derivative of T_M at the identity and the row and column sums of its kernel on curvatures. Then, in an axial gauge on the patch of each coarse plaquette, we compare $\operatorname{Log}((T_M U)_P)$ with its linearization by Taylor's formula. The remainder is quadratic in the logarithms of the links, and the axial gauge bounds these by the curvatures in the patch. Summing over P with the bounded overlap of the patches gives the energy estimate, and a single row gives the curvature recursion. The iteration in (4)–(6) is then elementary.

Proof. The linearized kernel. Let a be a real $\mathfrak{su}(2)$-valued one-cochain and let $U_e(t) = e^{ta_e}$. For a coarse vertex v define

$$r(v) = M^{-3} \sum_{x \in \mathsf{B}_M(v)} \sum_{e \in \sigma_{v,x}} a_e, \qquad k_i(v) = M^{-3} \sum_{r \in \{-R_M, \dots, R_M\}^3} \sum_{s=0}^{M-1} a_i(v + r + se_i),$$

where the sum along a path takes orientations into account. We differentiate the formula for T_M. The derivative of the reference transport cancels its negative occurrence in the logarithm of every based loop holonomy, and we find

$$(DT_M(I)a)_i(v) = r(v) + k_i(v) - r(v + Me_i).$$

Around a coarse plaquette the four terms involving r cancel. For each r, the four segment sums in k form the boundary of the translated M by M square, and summing the signed curls of its plaquettes cancels every interior link. With d the signed curl in the respective lattice units, this gives

$$\big(\mathrm{d}\, DT_M(I)a\big)_{ik}(v) = M^{-3} \sum_{r \in \{-R_M, \dots, R_M\}^3} \sum_{s,t=0}^{M-1} (\mathrm{d}a)_{ik}(v + r + se_i + te_k). \tag{7.5}$$

This proves the kernel identity in (2); repeated occurrences of a fine plaquette are added in its entry. Since each row has $M^3 M^2$ terms of weight M^{-3}, the row sums equal M^2. Fix a fine plaquette at x with orientation ik. For each $(s,t) \in \{0, \dots, M-1\}^2$ the vertex $x - se_i - te_k$ has exactly one representation $v + r$ with v a coarse vertex and $r \in \{-R_M, \dots, R_M\}^3$. This gives exactly M^2 contributions in the column, and its sum is M^{-1}; this count includes contributions across the periodic identifications. For an array f with values in a Euclidean space, by the weighted Cauchy–Schwarz inequality, we have

$$\sum_P \Big| \sum_p \mathsf{L}_M(P,p) f_p \Big|^2 \le M^2 \sum_{P,p} \mathsf{L}_M(P,p) |f_p|^2 = M \sum_p |f_p|^2. \tag{7.6}$$

Restricting to output plaquettes in $\mathcal{A}$ keeps the row sums equal to M^2 and decreases the column sums, so the restricted kernel satisfies the stated bounds and the same inequality.

The local Taylor estimate. Fix an output plaquette P. We use the rooted axial gauge of Lemma 7.1 on its patch $Q(P)$ of four blocks and set $A_e = \operatorname{Log}(U_e^{q_{Q(P)}})$. The patch has at most $12M^3$ links and at most $12M^3$ plaquettes, and the sum of its side lengths is at most $5M$. By Lemma 7.1, with constants depending only on M,

$$\|A\|_\infty \le C_M \delta, \qquad \|A\|_2 \le C_M \Big(\sum_{p \subset Q(P)} \rho_G(U_p)^2 \Big)^{1/2}. \tag{7.7}$$

The local gauge can be extended to the torus, and by Proposition 6.2(6) its values outside the patch do not affect $(T_M U)_P$. In this gauge, we consider the functions

$$\Phi_P(A) = \operatorname{Log}\big((T_M e^A)_P\big), \qquad f_p(A) = \operatorname{Log}\big((e^A)_p\big),$$

and

$$G_P(A) = \Phi_P(A) - \sum_p \mathsf{L}_M(P,p) f_p(A).$$

Let $\delta_* \le \delta_M$ be so small that all link, path, and plaquette transports in the patch stay in a fixed ball strictly inside the domain of the principal logarithm whenever $\|A\|_\infty \le C_M \delta_*$. Since the distance of a product from I is at most its length times $\|A\|_\infty$, this choice is made on a fixed finite patch and does not depend on the side of the torus. On a slightly larger fixed ball, the finite-word bounds of Lemma 5.1 bound the Hessian of G_P in the Euclidean operator norm by a constant H_M. Indeed, each averaged link is a composition of finitely many word maps, logarithms, an arithmetic mean, and an exponential, whose first two derivatives are bounded on these balls, and the chain rule applies. The value and the first derivative of G_P vanish at zero, the latter by (7.5). By Taylor's formula along the segment tA, we have

$$|G_P(A)| \le \int_0^1 (1-t) H_M \|A\|_2^2 \, dt \le \tfrac{1}{2} H_M \sqrt{12M^3}\, \|A\|_\infty \|A\|_2,$$

where we used $\|A\|_2 \le \sqrt{12M^3}\|A\|_\infty$. Inserting (7.7), we obtain the estimate for $e_P = G_P(A)$ in (3). Here $\operatorname{Log}(U_p)^{(P)} = f_p(A)$ is the logarithm of the plaquette transport in the gauge of this patch, at the usual base point of p. Its norm is $\rho_G(U_p)$, although its direction in $\mathfrak{su}(2)$ may depend on P. By covariance of T_M, the coarse plaquette is conjugated by the axial gauge at its starting vertex, and we also have $|\Phi_P(A)| = \rho_G((T_M U)_P)$.

Overlap. A fine vertex belongs to exactly one block, and a block center is a corner of twelve positive coarse plaquettes: three orientations and four choices of the corner. If a fine plaquette lies in $Q(P)$, its vertices lie in the four corner blocks of P, so one of the at most four block centers of its vertices is a corner of P. Hence there are at most $4 \cdot 12 = 48$ such P; this count may be redundant, and it holds on every periodic lattice considered. Combining this count with the error estimate in (3), we find

$$\sum_{P \in \mathcal{A}} |e_P|^2 \le C_M \delta^2 \sum_{p \in \operatorname{Pa}(\mathcal{A})} \rho_G(U_p)^2, \tag{7.8}$$

where the sum on the right is over the union, without multiplicity.

Energy and curvature bounds. Let $F^2 = \sum_{p \in \operatorname{Pa}(\mathcal{A})} \rho_G(U_p)^2$. Although the gauges differ from one P to another, the Cauchy–Schwarz inequality in each row uses only the norms $|f_p| = \rho_G(U_p)$, so (7.6) bounds the squared norm of the linear term by MF^2. By (7.8), the error term has norm at most $C_M \delta F$. By the triangle inequality in the finite-dimensional Euclidean space of outputs, we have

$$\sum_{P \in \mathcal{A}} \rho_G\big((T_M U)_P\big)^2 \le \big(\sqrt{M} + C_M \delta\big)^2 F^2 \le M(1 + C_M \delta) F^2, \tag{7.9}$$

after enlarging C_M, for $0 \le \delta \le \delta_* \le 1$. On a single row, the error estimate and the row sum give

$$\rho_G\big((T_M U)_P\big) \le M^2\delta + C_M\delta\sqrt{12M^3}\,\delta \le M^2\delta + c_M\delta^2, \tag{7.10}$$

with $c_M \ge 1$. The linear coefficient M^2 is exact.

To pass to the Wilson energy, recall from Lemma 5.1 that, for a fine plaquette with $\rho_G(U_p) \le \delta$ and for every coarse plaquette,

$$\frac{\rho_G(U_p)^2}{8}\Big(1 - \frac{\delta^2}{48}\Big) \le q(U_p), \qquad q\big((T_M U)_P\big) \le \frac{\rho_G((T_M U)_P)^2}{8}.$$

The energy of the output plaquettes in $\mathcal{A}$ is therefore at most $M(1 + C_M\delta)(1 - \delta^2/48)^{-1}$ times the energy of $\mathrm{Pa}(\mathcal{A})$. Decreasing $\delta_* \le 1$ and enlarging C_M, we absorb the last factor into $1 + C_M\delta$. This proves the restricted inequality in (3); with all output plaquettes it proves the energy inequality in (1), and (7.10) proves the curvature inequality in (1). All estimates used only the plaquettes in the patches.

Iteration. Dividing the one-step energy inequality by $g_{j+1}^2 = Mg_j^2$, multiplying by 4, and iterating, we obtain the product in (4); since $1 + x \le e^x$ for $x \ge 0$, the exponential bound follows. For $d = 0$ the product is empty and (4) is an equality.

For (5), let $y_j = \delta(U^j)/\epsilon_j^2$. As long as the preceding inputs satisfy $\delta(U^i) \le \delta_*$, (7.10) and $\epsilon_{j+1} = M\epsilon_j$ imply

$$y_{j+1} \le y_j + \frac{c_M}{M^2}\epsilon_j^2 y_j^2, \qquad \sum_{j=0}^{d-1} \epsilon_j^2 = \frac{1 - M^{-2d}}{M^2 - 1} \le \frac{1}{M^2 - 1}. \tag{7.11}$$

The initial value is at most ε. Suppose that $y_i \le 2\varepsilon$ for all $i \le j$. Summing the recurrence up to j bounds y_{j+1} by

$$\varepsilon + \frac{4c_M\varepsilon^2}{M^2(M^2 - 1)} \le 2\varepsilon,$$

by the choice of ε_M. Each preceding curvature is then at most $2\varepsilon\epsilon_i^2 \le 2\varepsilon \le \delta_* \le \delta_M$, which verifies the hypothesis needed for the next application of the recurrence and of Lemma 7.2(3). This induction reaches $j = d$. Summing $\delta(U^j) \le 2\varepsilon\epsilon_j^2$ over $j < d$, we find the bound on the sum of the curvatures, and inserting it into (4) gives the last bound in (5).

For (6), note that $a_0 = a_* M^{-d} = a_*\epsilon_0$, so the hypothesis $\delta(U^0) \le C_A a_0^2$ is the hypothesis of (5) with $\varepsilon = C_A a_*^2$. Part (5) then implies $\delta(U^j) \le 2C_A a_j^2$. Applying Lemma 7.2(3) to each U^j, $j < d$, we find

$$\|z_j/g_j\|_\infty \le \frac{C_M\delta(U^j)}{g_j} \le \frac{2C_M C_A}{g_3}a_j^{3/2}, \qquad 0 \le j < d,$$

where $g_j = g_3\sqrt{a_j}$, in agreement with $g_n^2 = \gamma/n = g_3^2 a_n$ from Definition 4.1. If $C_A = 0$, the curvature recursion gives $\delta(U^j) = 0$ at every step and Lemma 7.2(3) gives $z_j = 0$, and the formulas hold directly. Every constant in the proof came from fixed patches of blocks, from the radii of the logarithm and the inverse coordinates, or from the constant C_A. □

7.3 Dependence sets and reflected cell partitions

To apply the chessboard estimates of Theorem 6.6 to plaquettes of the blocked configurations $V^j = T_M^j U$, we need to know on which fine links such a plaquette depends. We also need a partition of the lattice into cells, invariant under reflections, such that each cell contains these dependence sets. The relevant sets are shown in Figure 7.1. Because the blocks are centered and nested, the iterated dependence set of a coarse plaquette stays inside the four blocks at its corners, at every scale. Since a single tiling by cells of two coarse sites in each direction does not contain every plaquette in a cell, we use eight tilings, shifted by the parities of the starting vertex.

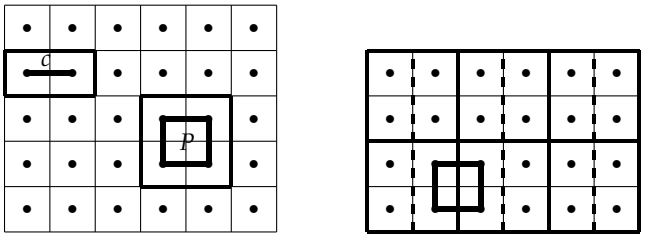


Figure 7.1. Left: the blocks of side m_j (thin squares) around the vertices of the level-j lattice, in a plane spanned by two coordinate directions. The patch of the coarse link c consists of its two blocks, and the patch of the coarse plaquette P consists of the four blocks at its corners (thick outlines). Right: cell boundaries of two of the eight shifted tilings, with shifts $(0,0)$ (solid) and $(1,0)$ (dashed) in the two directions shown; each cell is the union of the blocks of $2 \times 2 \times 2$ level-j vertices. The thick plaquette crosses a cell boundary of the first tiling, but lies in a cell of the second.

Lemma 7.4. *Fix an odd $M \geq 3$, integers $D \geq 0$ and $0 \leq j \leq D$, and the fine side $N = M^D n$ with $n \in 4\mathbb{N}$. Work in fine lattice units, put $m_j = M^j$ and $n_j = N/m_j$, and write $V^\ell = T_M^\ell U$.*

1. *For a plaquette P of the lattice at level j, define recursively $\mathcal{P}_j(P) = \{P\}$ and $\mathcal{P}_i(P) = \mathrm{Pa}(\mathcal{P}_{i+1}(P))$ for $i < j$, with* Pa *defined in Theorem 7.3(3); we call $\mathcal{P}_0(P)$ the energy-estimate predecessor set of P. Every fine plaquette in $\mathcal{P}_0(P)$ has its vertices in the union of the four blocks B_{m_j} at the corners of P. This union also contains all fine links on which V_P^j depends,*

including through the Borel extension of T_M, and all paths and plaquettes used in the patches of the intermediate steps. Further,

$$|\mathcal{P}_0(P)| \le C_M m_j^3, \qquad \operatorname{diam} \mathcal{P}_0(P) \le C_M m_j, \qquad \sup_p \#\{P : p \in \mathcal{P}_0(P)\} \le 48.$$

The same statements hold between any two intermediate levels, with the ratio of their meshes in place of m_j.

2. *For each shift $\eta \in \{0,1\}^3$, tile the vertices of the level-j lattice by the sets $\eta + 2t + \{0,1\}^3$, and define the cells of the fine lattice*

$$C_{j,t}^{\eta} = \bigcup_{\zeta \in \{0,1\}^3} \mathsf{B}_{m_j}\big(m_j(\eta + 2t + \zeta)\big).$$

These are disjoint cubes with $2m_j$ fine vertices in each direction, and there are $\mathfrak{m}_j = n_j^3/8$ of them in each tiling. If a level-j plaquette has all its vertices in the level-j vertices of a cell, then the plaquette and its energy-estimate predecessor set depend only on fine links with both endpoints in the cell. The reflections between adjacent cells are $x_i \mapsto (2\eta_i + 4t_i + 3)m_j - x_i$; their planes and the opposite planes bisect fine links, and these reflections map the blocks, the patches, and the sets $\mathcal{P}_\ell(P)$ at every level $\ell \le j$ onto the corresponding sets. Every level-j plaquette lies in a cell of at least one of the eight tilings.

3. *Let $j \ge 1$. For a cell C of one tiling $\mathcal{T}$, let $\mathcal{P}_{j-1}(C)$ be the union of the sets $\mathrm{Pa}(\{P\})$ of the six level-j plaquettes P in C. Then*

$$|\mathcal{P}_{j-1}(C)| \le 24M^3.$$

The sets obtained from $\mathcal{P}_{j-1}(C)$ by successive applications of Pa*, for distinct cells C of $\mathcal{T}$, are disjoint, and for individual level-j plaquettes the overlap bound 48 of (1) applies. For $0 \le \ell < j$ let*

$$\mathcal{F}_\ell(C) = \Big\{ p \in \mathcal{P}_{n_\ell} : \bigcup_{v \text{ a vertex of } p} \mathsf{B}_{m_\ell}(m_\ell v) \subset C \Big\},$$

where the vertices v are written in level-ℓ lattice units, and for $d_{\mathrm{cut}} > 0$ let

$$R_{<j}(C; d_{\mathrm{cut}}) = \bigcap_{\ell<j} \bigcap_{p \in \mathcal{F}_\ell(C)} \big\{ q(V_p^\ell) < d_{\mathrm{cut}} \big\}, \qquad R_{<j}^{\mathcal{T}}(d_{\mathrm{cut}}) = \bigcap_{C \in \mathcal{T}} R_{<j}(C; d_{\mathrm{cut}}).$$

These events impose the conditions on every $\mathcal{P}_\ell(P)$, $\ell < j$, for the six level-j plaquettes P in C. Further, put

$$\mathcal{P}_{j-1}^{\mathcal{T}} = \bigcup_{C \in \mathcal{T}} \mathcal{P}_{j-1}(C), \qquad Q_j^{\mathcal{T}} = \frac{4}{h_{n_{j-1}}} \sum_{p \in \mathcal{P}_{j-1}^{\mathcal{T}}} q(V_p^{j-1}),$$

where $h_n = g_n^2 = \gamma/n$.

4. *Put* $Q_{j,C} = (4/h_{n_{j-1}}) \sum_{p \in \mathcal{P}_{j-1}(C)} q(V_p^{j-1})$. *The following functions are nonnegative observables supported in the cell* C*, in the sense of Theorem 6.6(2), that is, bounded Borel functions of the fine links with both endpoints in* C *that are invariant under all periodic gauge transformations: the indicator of* $R_{<j}(C; d_{\mathrm{cut}})$*, the plaquette-energy sum* $Q_{j,C}$*, the functions* $\exp(\lambda Q_{j,C})$ *for* $\lambda \geq 0$*, and their products. Plaquettes crossing the boundaries of the cells of a tiling are not tested by the events of that tiling.*

All bounds are deterministic and do not depend on the coupling or on D*; the constant* C_M *depends only on* M*, not on any region of the lattice or on widths chosen later.*

Proof. Nesting. For odd $s \geq 1$ and $-R_M \leq \alpha \leq R_M$, the intervals of s consecutive integers centered at $v + s\alpha$ are disjoint and adjacent, with extreme endpoints $v - (Ms-1)/2$ and $v + (Ms-1)/2$. Taking products gives

$$\bigcup_{\alpha \in \{-R_M, \dots, R_M\}^3} \mathsf{B}_s(v + s\alpha) = \mathsf{B}_{Ms}(v). \tag{7.12}$$

This is the nesting identity of Proposition 6.2(6); it holds on the periodic lattice, since the blocks partition its vertices.

Containment. At a single step, a plaquette of $\mathrm{Pa}(\{P\})$ has its four vertices in the four corner blocks of P. Replacing each of these vertices by its own finer block keeps the union inside the same four larger blocks, by (7.12). By induction backwards from level j, we obtain the containment for $\mathcal{P}_0(P)$ and, in the same way, at every intermediate level. The same induction shows that the dependence sets are contained: at each step the dependence set of a link given by Proposition 6.2(6), its reference path, and the patch of four blocks used in Theorem 7.3 are included. The decision whether the Borel extension is used reads only the same based loop holonomies, and the value it assigns reads only the reference path. No continuity of T_M at such configurations is used.

Counting and overlap. The four corner blocks of side m_j contain $4m_j^3$ vertices and fit into a box with $2m_j$, $2m_j$, and m_j vertices in the three directions. Since at most three positive plaquettes start at each vertex, we have $|\mathcal{P}_0(P)| \leq 12m_j^3$, and the diameter in the lattice path metric is at most $5m_j$. Periodic identifications can only decrease these bounds. For the overlap, fix a fine plaquette p; each of its four vertices lies in one block of side m_j. If $p \in \mathcal{P}_0(P)$, all its vertices lie in corner blocks of P, so one of the at most four centers of these blocks is a corner of P. A center is a corner of twelve positive plaquettes, so there are at most $4 \cdot 12 = 48$ such P, also when some centers coincide. Starting the argument at an intermediate level replaces m_j by the ratio of the meshes and changes nothing else.

Cells and reflections. For the shift η, the coordinate interval of $C_{j,t}^{\eta}$ in direction i starts at $l_i = m_j(\eta_i + 2t_i) - (m_j - 1)/2$ and ends at $l_i + 2m_j - 1$; it contains $2m_j$ vertices, and the next interval starts at $l_i + 2m_j$. Since n_j is even, these intervals partition the coordinate circle, and their products give $(n_j/2)^3 = n_j^3/8$ disjoint cells. A level-j plaquette with all its vertices in a cell has its four corner blocks in the cell. By the containment just proved,

its energy-estimate predecessor set and all its dependencies are in the cell. Every link involved has both endpoints in the cell, including links in the boundary planes of the cell. The plane between the interval above and the next one is $x_i = l_i + 2m_j - 1/2$, and the reflection in it is $x_i \mapsto c_i - x_i$ with

$$c_i = 2l_i + 4m_j - 1 = (2\eta_i + 4t_i + 3)m_j.$$

Since m_j is odd, this number is odd, and the plane bisects fine links. The opposite plane has intercept larger by N, which is even, and it also bisects fine links. Since, for every level $\ell \le j$, the intercept divided by m_ℓ is still an odd integer, the reflection preserves the partition into blocks of side m_ℓ. By Proposition 6.2(4) it preserves the order of the coordinates in the root paths, and it maps a closed rectangular patch, with all its links and plaquettes, onto the reflected patch. Repeating this at every level proves the statement for all the sets $\mathcal{P}_\ell(P)$.

Coverage. Given a level-j *ik* plaquette with starting vertex v, we choose $\eta_i = v_i \bmod 2$ and $\eta_k = v_k \bmod 2$. Its two vertices in each of these directions then lie in one interval of the tiling. In the third direction either shift works, since the interval containing v contains the plaquette. Hence every plaquette lies in a cell of at least one of the eight tilings.

The sets of one cell. Let $j \ge 1$. Every plaquette of $\mathcal{P}_{j-1}(C)$ has its vertices among the $8M^3$ level-$(j-1)$ vertices whose blocks of side m_{j-1} form C, and there are at most $3 \cdot 8M^3 = 24M^3$ positive plaquettes with vertices there; the bound is for the union of the six patches, whose overlaps are not counted. Every backward expansion of this set stays in C, by (7.12). Since distinct cells have disjoint sets of vertices, the sets obtained from them by successive applications of Pa, and the links on which these depend, are disjoint. For individual target plaquettes the bound 48 of (1) still applies. When a plaquette occurs in one of the sets $\mathcal{P}_\ell(P)$, $\ell < j$, of a plaquette P in C, the union of the blocks of side m_ℓ of its four vertices lies in C, by the same nesting argument; this is the defining condition of $\mathcal{F}_\ell(C)$. The events $R_{<j}(C; d_{\mathrm{cut}})$ therefore impose the conditions on all these sets, and $Q_j^{\mathcal{T}}$ is the sum over the union of the sets $\mathcal{P}_{j-1}(C)$, each plaquette counted once, with the normalization $4/h_{n_{j-1}}$ of the Wilson action.

Observables. Each function $q(V_p^\ell)$ is Borel, since T_M is Borel (Proposition 6.2(2)), and gauge invariant, by gauge covariance of T_M and conjugation invariance of q. For $p \in \mathcal{F}_\ell(C)$ it depends only on links with both endpoints in C. The indicator of $R_{<j}(C; d_{\mathrm{cut}})$ is a finite product of Borel indicators of such events; it is Borel with values in $\{0, 1\}$. For $j \ge 1$, the sum $Q_{j,C}$ is a Borel gauge-invariant function of the links in C, and $0 \le q \le 2$ gives

$$0 \le Q_{j,C} \le \frac{8|\mathcal{P}_{j-1}(C)|}{h_{n_{j-1}}} \le \frac{8 \cdot 24M^3}{h_{n_{j-1}}}. \tag{7.13}$$

Hence, for every fixed $\lambda \ge 0$, $\exp(\lambda Q_{j,C})$ is bounded by the exponential of λ times this bound, and the indicator, the exponential, and their product are nonnegative bounded

Borel observables supported in C. The reflections between cells map each set of tested plaquettes onto the reflected set; they may reverse a plaquette or move its base point, and $q(HWH^{-1}) = q(W^{-1}) = q(W)$. Finally, the parity condition of Theorem 6.6 holds, since $n_j = M^{D-j}n \in 4\mathbb{N}$. The constants do not depend on the coupling, on D, or on any region or width. □

7.4 Lipschitz extensions and periodic gauge fixing

Regular lattice configurations and gauge fixing conditions are part of the renormalization group program of Bałaban for lattice gauge theories [14]. Here we need a particular quantitative statement: a configuration with small curvature on a periodic lattice has a periodic gauge transform with uniformly small links, with no restriction on its holonomies around the noncontractible cycles. The following example shows the form such a bound must take. Let $\theta \in \mathfrak{su}(2)$ with $0 < |\theta| < 2\pi$, and let U be the configuration on the lattice of side n with $U_e = e^{\theta/n}$ on the links in direction 1 and $U_e = I$ on all other links. Its curvature vanishes, and the transport around a cycle in direction 1 is e^{θ}. For every gauge transform U^k this transport is conjugate to e^{θ}. By the triangle inequality, some link satisfies $\rho_G(U_e^k) \geq |\theta|/n$. A bound of the form $C\sqrt{\delta}$ in terms of the maximal curvature δ alone is therefore impossible, and a term of order $1/n$ is needed. Spreading the holonomy evenly over the links of the cycle, as in the example, attains this order. At positive curvature δ the relevant length scale is $\delta^{-1/2}$. An axial gauge on a box of side R produces links of size about δR, while gluing such boxes requires gauge transformations varying by about $1/R$ per link; both are of order $\sqrt{\delta}$ when R is of order $\delta^{-1/2}$.

The construction glues local axial gauges on a partition of the torus into cells. In the gluing we extend group-valued maps successively over the edges, faces, and interiors of the cells. This is possible because $\mathrm{SU}(2) \simeq S^3$ has vanishing first and second homotopy groups, and Lemma 7.5 below gives Lipschitz bounds for the extensions. We prove the bounds on the extension, interpolation, and gluing directly, without an external extension theorem.

Lemma 7.5. *Use the bi-invariant metric of* $\mathrm{SU}(2) \simeq S^3$*, which is twice the round metric. There is an absolute constant C with the following properties.*

1. *Every map from the unit circle S^1 to* $\mathrm{SU}(2)$ *with Lipschitz constant B_0 extends to the unit disk with Lipschitz constant at most $C(1 + B_0)^2$, and every such map from the unit sphere S^2 extends to the unit ball of $\mathbb{R}^3$ with Lipschitz constant at most $C(1 + B_0)^5$; the extensions agree with the given maps on the boundary. For $A_0 \geq 1$ and a rectangle or a rectangular box with all side lengths between R/A_0 and A_0R, the corresponding boundary-preserving extensions of a map with Lipschitz constant B_0 on the boundary have Lipschitz constants at most*

$$C(A_0)R^{-1}(1 + B_0R)^2 \qquad \text{and} \qquad C(A_0)R^{-1}(1 + B_0R)^5.$$

2. *Fix the triangulation of every lattice cube into the six simplices determined by the orderings of the coordinates; it is consistent on common faces and invariant under translations, so it descends to periodic lattices, and it induces triangulations of the faces of lattice boxes. There is an absolute $c_0 > 0$ such that, if unit quaternions are prescribed at the vertices of a simplex with pairwise distances at most c_0, their normalized affine interpolation $\sum_i \lambda_i u_i / |\sum_i \lambda_i u_i|$ is defined, with denominator at least $1/2$. If vertex data on a lattice of mesh a satisfy $\operatorname{dist}(u_x, u_y) \le B_0 a$ for all nearest neighbors x, y, and $B_0 a \le c(A_0)$, then the interpolation has Lipschitz constant at most $C(A_0)B_0$, and the interpolations on two simplices agree on their common face.*
3. *For every $L > 0$, $\mathscr{K} \ge 0$, integer $n \ge 4$, and $a = L/n$, every periodic configuration on the lattice of side n and mesh a with $\max_p \rho_G(U_p) \le \mathscr{K} a^2$ admits a periodic gauge transformation k with*

$$\max_e \rho_G(U_e^k) \le \frac{Ca}{L}\bigl(1 + \mathscr{K} L^2\bigr)^{10}.$$

All holonomies around the cycles of the torus are allowed. If $a \le cL(1 + \mathscr{K} L^2)^{-2}$ for an absolute $c > 0$, the construction uses a partition of the torus into closed rectangular cells Q with side lengths between $L/8$ and $L/2$, each with a rooted axial gauge q_Q such that $\rho_G(U_e^{q_Q}) \le C\mathscr{K} La$ on every link of its closure, and the transition functions $t_{QR} = q_Q q_R^{-1}$ on common links (x, y) satisfy

$$t_{QR}(y) = \bigl(U_{xy}^{q_Q}\bigr)^{-1} t_{QR}(x)\, U_{xy}^{q_R}, \qquad \operatorname{dist}\bigl(t_{QR}(x), t_{QR}(y)\bigr) \le C\mathscr{K} La.$$

On coarser meshes the bound on $\max_e \rho_G(U_e^k)$ follows from the diameter of $\mathrm{SU}(2)$, after an absolute enlargement of C. When this bound is smaller than 2π, it bounds the principal logarithms of the links of U^k.

The constants in (1) and (2) depend only on A_0 and the triangulation, and the constant in (3) is absolute: it does not depend on n, L, $\mathscr{K}$, or the holonomies. No regularity of k as a function of U is claimed, and no estimate for differences of the link logarithms beyond the bound on each link is asserted.

We prove parts (1) and (2) first. Part (3) follows from the gluing estimate of Lemma 7.6 below.

Proof of Lemma 7.5(1) and (2). We identify $\mathrm{SU}(2)$ with the unit quaternions in $\mathbb{R}^4$. Its metric is twice the round metric of the unit sphere: the quaternion $\cos(t/2) + \omega \sin(t/2)$ is at distance $|t|$ from I for $0 \le |t| \le 2\pi$. Comparing arc and chord on a circle, we obtain, for $u, v \in S^3$,

$$2|u - v| \le \operatorname{dist}(u, v) \le \pi |u - v|. \tag{7.14}$$

All constants in the arguments on spheres below are absolute.

Covering. A unit sphere S^m, $m = 1, 2$, can be covered by at most $C_m r^{-m}$ balls of radius $r \le 1$. Indeed, consider a maximal set of points with mutual distances greater than r.

The balls of radius $r/2$ around them are disjoint, and each has volume at least $c_m r^m$, as one sees by integrating the polar density $(\sin t)^{m-1}$ and using $2t/\pi \le \sin t \le t$ for $0 \le t \le \pi/2$. A separated set is finite, since it has at most one point in each cube of a sufficiently fine subdivision of a cube containing the sphere, and maximality gives the cover. The same polar integration shows that a ball of radius 2ε in S^3 has volume at most $C\varepsilon^3$ for $\varepsilon \le c$.

Extension from a sphere. Let $f : S^m \to S^3$ have Lipschitz constant B_0. A cover of S^m by balls of radius $\varepsilon/(1+B_0)$ gives a cover of the image by at most $C((1+B_0)/\varepsilon)^m$ balls of radius ε, so the ε-neighborhood of the image has volume at most $C(1+B_0)^m \varepsilon^{3-m}$. Let

$$\varepsilon = c_m (1+B_0)^{-m/(3-m)}$$

with $c_m > 0$ so small that this volume is less than half the volume of S^3. Then some point p has distance at least ε from the image, and after a rotation we may take it to be the north pole. The stereographic projection from p is $\pi_p(u) = u_\perp/(1-u\cdot p)$, where $u_\perp$ is the component of u orthogonal to p. For points at round distance at least $\varepsilon/2$ from p it satisfies

$$|\pi_p(u)| \le C\varepsilon^{-1}, \qquad |\pi_p(u) - \pi_p(v)| = \frac{|u-v|}{\sqrt{(1-u\cdot p)(1-v\cdot p)}} \le C\varepsilon^{-2}|u-v|. \quad (7.15)$$

The equality follows by squaring both sides and using $|u| = |v| = 1$, and the inequalities follow from $1 - \cos t = 2\sin^2(t/2) \ge ct^2$ for $0 \le t \le \pi$; the factor two between the group metric and the round metric only changes constants. We conclude that $\psi = \pi_p \circ f$ has supremum at most $C\varepsilon^{-1}$ and Lipschitz constant at most $C\varepsilon^{-2}B_0$. On the source sphere the arc distance is at most $\pi/2$ times the chord distance, so this is also a Lipschitz bound with respect to the Euclidean metric, after an absolute change of the constant.

We extend ψ to the unit ball by $\Psi(r\omega) = r\psi(\omega)$, with $\Psi(0) = 0$. For $r \ge s$,

$$|r\psi(\omega) - s\psi(\eta)| \le (r-s)\|\psi\|_\infty + s\,\mathrm{Lip}(\psi)|\omega - \eta|,$$

and $r - s \le |r\omega - s\eta|$, $s|\omega - \eta| \le 2|r\omega - s\eta|$, so the Lipschitz constant of Ψ is at most $\|\psi\|_\infty + 2\,\mathrm{Lip}(\psi)$. The inverse stereographic projection is

$$\pi_p^{-1}(z) = \frac{(2z, |z|^2 - 1)}{1 + |z|^2}.$$

Its differential has Euclidean norm at most 2. Following the image of a line segment and using its round length, and doubling for the group metric, we find that π_p^{-1} has Lipschitz constant at most 4. Hence $\pi_p^{-1} \circ \Psi$ extends the given boundary map and has Lipschitz constant at most

$$C\big(\varepsilon^{-1} + B_0\varepsilon^{-2}\big) \le \begin{cases} C(1+B_0)^2, & m = 1, \\ C(1+B_0)^5, & m = 2. \end{cases}$$

This proves the extension bounds for spheres, including the case $B_0 = 0$.

Rectangles and boxes. In dimension $k = 2, 3$, the radial map $r\omega \mapsto r\omega/\|\omega\|_\infty$ takes the Euclidean unit ball onto $[-1,1]^k$ and the unit sphere onto its boundary. On the sphere, $1/\sqrt{k} \le \|\omega\|_\infty \le 1$ and $|\|\omega\|_\infty - \|\eta\|_\infty| \le |\omega - \eta|$, so the map is Lipschitz on the sphere, and so is its inverse, which has radial factor $\|\omega\|_\infty$ on each ray. The radial estimate used for Ψ shows that both maps are Lipschitz on the balls as well. A diagonal affine map onto a box with side lengths between R/A_0 and $A_0 R$ changes Lipschitz constants by factors at most $C(A_0)R$ and $C(A_0)/R$. Composing the extensions for spheres with these maps gives the bounds for rectangles and boxes. The same comparison bounds the intrinsic distance on the boundary of a box by $C(A_0)$ times the ambient distance: on a sphere, a shortest arc is at most $\pi/2$ times its chord, and the boundary maps just discussed transfer this inequality. This proves (1).

Interpolation. Let $c_0 > 0$ be so small that pairwise distances at most c_0 between the vertex values of a simplex imply $u_0 \cdot u_i \ge 3/4$; this follows from $u_0 \cdot u_i = 1 - |u_0 - u_i|^2/2$ and (7.14). For barycentric coordinates λ_i,

$$\Big|\sum_i \lambda_i u_i\Big| \ge u_0 \cdot \sum_i \lambda_i u_i \ge \frac{3}{4} \ge \frac{1}{2},$$

so the normalization is defined. The derivative of $z \mapsto z/|z|$ has operator norm at most 2 on $\{|z| \ge 1/2\}$. On a simplex of the triangulation, the affine interpolation has derivative at most $C(A_0)a^{-1}\max_{i,l}|u_i - u_l|$, as one sees by writing it in barycentric coordinates, whose gradients are bounded by $C(A_0)/a$ on the finitely many simplex shapes. Two vertices of a simplex are joined by at most three lattice links, so the maximum is at most CB_0a, and the derivative of the normalized interpolation, hence its Lipschitz constant for the group metric, is at most $C(A_0)B_0$. The condition $B_0 a \le c(A_0)$ ensures the smallness condition on the simplices. Since the restriction of the barycentric formula to a common face uses only the values at the common vertices, the interpolations agree there. This proves (2). □

Lemma 7.6. *Let $A_0 \ge 1$. There exist $c(A_0) > 0$ and $C(A_0) < \infty$ with the following property. Consider a periodic cubic lattice of mesh a and a face-to-face decomposition of the torus into closed coordinate boxes: each lifted cell has the form $\prod_{i=1}^3 [am_i, an_i]$, with integers $m_i < n_i$. Assume that every side length lies between R/A_0 and $A_0 R$ and that each cell lifts injectively to the covering lattice. Here face-to-face means that the intersection of two distinct cells is empty or a whole face, edge, or vertex of both cells. Suppose that each cell Q has a gauge transformation q_Q, defined on its vertices, such that*

$$\rho_G\big(U_{xy}^{q_Q}\big) \le B_0 a \qquad \text{for every link } (x,y) \text{ of the closure of } Q. \tag{7.16}$$

No bound on the gauge transformations q_Q themselves is assumed. Then there is a periodic gauge transformation k with

$$a \le c(A_0) R (1 + B_0 R)^{-2} \quad \Longrightarrow \quad \max_e \rho_G(U_e^k) \le \frac{C(A_0)\, a}{R} (1 + B_0 R)^{10}. \tag{7.17}$$

The constants do not depend on the number of cells.

Proof. On a link (x, y) common to two cells Q and R', let $t_{QR'} = q_Q q_{R'}^{-1}$. Direct multiplication gives

$$t_{QR'}(y) = \big(U_{xy}^{q_Q}\big)^{-1} t_{QR'}(x)\, U_{xy}^{q_{R'}}, \qquad \operatorname{dist}\big(t_{QR'}(x), t_{QR'}(y)\big) \le 2B_0 a. \tag{7.18}$$

This applies also to cells that meet only along an edge. We assign a single gauge transformation k at all vertices. Its coordinates in the frame of Q are $s_Q = k q_Q^{-1}$, so that $s_{R'} = s_Q t_{QR'}$. By bi-invariance, multiplying two group-valued maps adds their Lipschitz bounds, so passing from one incident frame to another increases the bound on neighbor distances by at most $2B_0 a$.

Vertices and edges of the cells. We set $k = I$ at every corner of a cell, once on the periodic lattice. For each edge of the cell partition, we choose one incident cell Q and join the two prescribed values of s_Q at its endpoints by a minimizing geodesic of $\mathrm{SU}(2)$, parametrized proportionally to arc length over the edge. Such a geodesic is an arc of a great circle in S^3; if its endpoints are antipodal, any great circle through them may be used. Since the diameter of $\mathrm{SU}(2)$ is 2π and the edge has length at least R/A_0, the speed is at most $C(A_0)/R$. At the lattice vertices of the edge, we set $k = s_Q q_Q$. These assignments reproduce the given values at the corners. In every incident frame the resulting vertex data satisfy $\operatorname{dist}(s(x), s(y)) \le B_1 a$ for neighbors on the edge, with

$$B_1 \le C(A_0)/R + 2B_0.$$

Faces of the cells. For each face of the cell partition, we choose one incident cell, express the values already assigned on the boundary of the face in its frame, and interpolate them along the lattice links of the boundary by Lemma 7.5(2). Provided $aB_1 \le c(A_0)$, this is a continuous loop with Lipschitz constant at most $C(A_0)B_1$. The comparison of intrinsic and ambient distances in the proof of Lemma 7.5(1) gives this bound also between points on different sides of the face. The extension over the disk, rescaled to the face, gives an extension with Lipschitz constant

$$B_2 \le \frac{C(A_0)}{R} (1 + B_1 R)^2.$$

We use it to assign $k = s_Q q_Q$ at the remaining lattice vertices of the face. Every vertex value assigned before is kept. Each face is treated once, including faces that meet the

periodic identifications. The vertices of an edge shared by several faces were assigned before and have a single value throughout.

Interiors of the cells. Consider a cell Q. On a face that was treated in the frame of a cell Q_F, the boundary values in the frame of Q are $s_Q = s_{Q_F} t_{Q_F Q}$, so by (7.18) neighboring values satisfy $\operatorname{dist}(s_Q(x), s_Q(y)) \le B_\partial a$ with

$$B_\partial \le B_2 + 2B_0 \le \frac{C(A_0)}{R}(1 + B_0 R)^2.$$

We interpolate the vertex values on the boundary of Q by the triangulation of Lemma 7.5(2). If $aB_\partial \le c(A_0)$, this is a continuous map on the boundary sphere with Lipschitz constant at most $C(A_0)B_\partial$. It is single valued at every edge and vertex, since the gauge transformation was assigned there only once. The extension over the ball gives an extension in the frame of Q with Lipschitz constant

$$B_3 \le \frac{C(A_0)}{R}(1 + B_\partial R)^5 \le \frac{C(A_0)}{R}(1 + B_0 R)^{10}.$$

At each interior vertex of Q, we assign $k = s_Q q_Q$ from this extension. Interior vertices belong to only one cell, and the boundary values are those assigned before. The result is a gauge transformation on the whole periodic lattice.

Links. Each lattice link lies in a closed cell, and the extension constructed for that cell represents both endpoint values, also when one or both endpoints lie on the boundary of the cell. Therefore

$$\rho_G\big(k_x U_{xy} k_y^{-1}\big) = \rho_G\big(s_Q(x) U_{xy}^{q_Q} s_Q(y)^{-1}\big) \le (B_0 + B_3)a.$$

Both smallness conditions $aB_1 \le c(A_0)$ and $aB_\partial \le c(A_0)$ follow from $a \le c(A_0)R(1 + B_0R)^{-2}$ after decreasing $c(A_0)$, and $(B_0 + B_3)a$ is at most the right side of (7.17). The constants do not depend on the number of cells: each change of frame compares two incident frames by a single transition (7.18), and no chain of comparisons through other cells is taken. Continuous extensions in different frames were required to agree only at the lattice vertices assigned earlier, and every later extension kept these values. This suffices for the bound on every link. □

Proof of Lemma 7.5(3). Suppose first that $n \ge 8$. We divide each coordinate circle at the vertices with indices 0, $\lfloor n/4 \rfloor$, $\lfloor n/2 \rfloor$, $\lfloor 3n/4 \rfloor$, and n; each interval has physical length between $L/8$ and $L/2$. The resulting 4^3 closed cells form a product, hence face-to-face, decomposition of the torus; they have bounded aspect ratios and lift injectively to the covering lattice, also when they meet the periodic identifications. On each cell, we use the rooted axial gauge of Lemma 7.1. Each link not in its axial tree is a product of at most CL/a conjugated plaquette variables along strips of width one, each at distance at most $\mathscr{K} a^2$ from I. Then (7.16) holds with $B_0 = C\mathscr{K} L$ and $R = L$. These gauges are rooted and

give the stated bound on every link of the closure of a cell; their transitions satisfy (7.18), which gives the stated transition bounds.

With the fixed aspect bound, the condition in (7.17) follows from $a \le cL(1+\mathscr{K}L^2)^{-2}$ after decreasing the absolute constant c; taking $c \le 1/8$ also guarantees $n \ge 8$. The conclusion of (7.17) is the asserted bound. If the condition on the mesh fails, we take $k = I$. Then $\rho_G(U_e) \le 2\pi$, while

$$\frac{a}{L}\big(1+\mathscr{K}L^2\big)^{10} > c\big(1+\mathscr{K}L^2\big)^{8} \ge c,$$

so an absolute enlargement $C \ge 2\pi/c$ gives the same bound on all such meshes, including $4 \le n < 8$. If the bound is smaller than 2π, the links of U^k differ from $-I$, and their principal logarithms have the same norms.

All assignments were made on the periodic lattice, not by identifying the values of independently chosen gauges afterwards, so arbitrary holonomies remain in the local gauges and in the assigned values throughout. □

7.5 Small-curvature gauge fixing and probability bounds

We apply the gluing lemma at cell size $R \approx \delta^{-1/2}$, and then estimate the probability of D_r^c using Theorem 6.6. Recall from Definition 4.4 the small-link domain and the gauge-invariant small-field domain; for $0 < r < 2\pi$ they are

$$B_r = \big\{V : \max_e \rho_G(V_e) < r\big\}, \qquad D_r = \big\{U : U^k \in B_r \text{ for some } k\big\},$$

where k runs over the periodic gauge transformations. The deterministic part of the following theorem, parts (1) and (2), is used at every scale in Sections 22 and 23; the probability bound in parts (3) and (4) concerns only the restriction of the Wilson measure on the finest lattice.

Theorem 7.7. *There exist absolute constants $C_{\mathrm{SG}} \ge 1$ and $\delta_{\mathrm{SG}} > 0$ with the following properties.*

1. *For every periodic cubic lattice of side $n \ge 4$, every $0 \le \delta \le \delta_{\mathrm{SG}}$, and every configuration U with $\max_p \rho_G(U_p) \le \delta$, there exists a periodic gauge transformation k with*

$$\max_e \rho_G(U_e^k) \le C_{\mathrm{SG}}\big(\sqrt{\delta} + n^{-1}\big).$$

   This includes $\delta = 0$ and arbitrary holonomies around the cycles of the torus.

2. *Fix $0 < r < 2\pi$ and set*

$$\delta_r = \min\big\{\delta_{\mathrm{SG}},\, (r/(4C_{\mathrm{SG}}))^2\big\}, \qquad d_r = \frac{\delta_r^2}{2\pi^2} = \min\Big\{\frac{\delta_{\mathrm{SG}}^2}{2\pi^2},\, \frac{r^4}{512\pi^2 C_{\mathrm{SG}}^4}\Big\}.$$

If $n \geq 4C_{SG}/r$, then

$$\{\max_p \rho_G(U_p) \leq \delta_r\} \subset D_r, \qquad \{\max_p q(U_p) \leq d_r\} \subset D_r.$$

Some lower bound on the side is necessary: for $n < 2\pi/r$, the flat configurations of the example at the beginning of Section 7.4 with $rn \leq |\theta| < 2\pi$ do not lie in D_r. The set D_r is open and gauge invariant, so these inclusions do not require a measurable choice of the gauge transformation.

3. *Let $n \in 4\mathbb{N}$ with $n \geq 4C_{SG}/r$, and let $\gamma > 0$ be such that $\beta_n = 4n/\gamma \geq 2$. Then the Wilson measure satisfies $\mu_{n,\gamma}(D_r^c) \leq \varepsilon_{n,r,\gamma}$, where*

$$\varepsilon_{n,r,\gamma} = \min\Big\{1,\ 3n^3 \exp\Big[\frac{1}{8}\Big(C_0 + 24\log 2 + \frac{36\log\beta_n}{n}\Big) - \frac{\beta_n \delta_r^2}{32\pi^2}\Big]\Big\}$$

and C_0 is the absolute constant of Theorem 6.6(3). The surface term $36\log\beta_n/n$ is kept for every (n,γ); no relation between n and γ is assumed.

4. *Under the assumptions of (3), $\mu_{n,\gamma}(D_r) > 0$, the conditioned measure $\mu_{n,\gamma}(\,\cdot \mid D_r)$ has total variation distance exactly $\mu_{n,\gamma}(D_r^c)$ from $\mu_{n,\gamma}$, and the partition function Z_{D_r} restricted to D_r satisfies*

$$\frac{Z_{D_r}}{Z} = \mu_{n,\gamma}(D_r), \qquad \Big|\log\frac{Z_{D_r}}{Z}\Big| \leq 2\varepsilon_{n,r,\gamma} \quad \text{if } \varepsilon_{n,r,\gamma} \leq 1/2.$$

The constants C_{SG} and δ_{SG} do not depend on L, γ, n, the number of averaging steps, or any widths of regions. The probability bound concerns the restriction to D_r of the Wilson measure on the full periodic lattice; it is not a conditional bound at a later scale.

Proof. Part (1). We use lattice spacing one. For $\delta = 0$, by Lemma 7.5(3) with $a = 1$, $L = n$, and $\mathscr{K} = 0$, we have $\max_e \rho_G(U_e^k) \leq C/n$ for every $n \geq 4$, including arbitrary flat holonomies. For $0 < \delta \leq 1$, let $R = \lceil \delta^{-1/2} \rceil$. Then

$$1 \leq \delta R^2 \leq 4. \tag{7.19}$$

If $n < 8R$, the same lemma with $\mathscr{K} = \delta$ gives

$$\max_e \rho_G(U_e^k) \leq \frac{C}{n}\big(1 + \delta n^2\big)^{10} \leq \frac{C'}{n},$$

with C' absolute, since $\delta n^2 < 64\delta R^2 \leq 256$.

Suppose now that $n \geq 8R$, and let $b = \lfloor n/R \rfloor \geq 8$. We divide each coordinate circle at the vertices $\lfloor in/b \rfloor$, $0 \leq i \leq b$. Since $R \leq n/b < R(1 + 1/b)$, the intervals have lengths between $R/2$ and $2R$ when $R \geq 2$. The closed cells form a product, hence face-to-face, decomposition; they have bounded aspect ratios, and they lift injectively since $2R \leq n/4$. By Lemma 7.1, the rooted axial gauges on these cells satisfy

$$\max_{e \subset Q} \rho_G\big(U_e^{q_Q}\big) \leq C\delta R.$$

Indeed, a link in direction 1 not in the axial tree is moved through one strip of width one in direction 3 and one in direction 2, of total length at most $4R$, and the counts for the other directions are smaller. Then Lemma 7.6 applies with mesh one, cell size R, and $B_0 = C\delta R$, and (7.19) implies $B_0 R \le 4C$. Fix an absolute $R_0 \ge 2$ so large that

$$1 \le c(A_0) R_0 (1 + 4C)^{-2}$$

for these fixed aspect bounds, and let $0 < \delta_{\mathrm{SG}} \le \min\{1, R_0^{-2}\}$. Then $R \ge R_0$, and the condition on the mesh in (7.17) holds. Its conclusion is

$$\max_e \rho_G(U_e^k) \le \frac{C}{R}(1 + B_0 R)^{10} \le \frac{C''}{R} \le C''\sqrt{\delta}.$$

As in the gluing lemma, these constants do not depend on the number b^3 of cells. Taking $C_{\mathrm{SG}} \ge \max\{1, C, C', C''\}$, we obtain (1) in all cases, with absolute constants and no condition on the holonomies.

Part (2). For δ_r and n as stated, the gauge transformation of part (1) satisfies

$$\max_e \rho_G(U_e^k) \le C_{\mathrm{SG}}\sqrt{\delta_r} + C_{\mathrm{SG}}/n \le r/4 + r/4 = r/2 < r < 2\pi.$$

These links have unique principal logarithms and lie in B_r, which proves the first inclusion, with a margin even at the closed threshold. By Lemma 5.1, $\rho_G(U_p)^2 \le 2\pi^2 q(U_p)$, so the event $\{\max_p q(U_p) \le d_r\}$ with $d_r = \delta_r^2/(2\pi^2)$ is contained in the first event. Squaring the two values in the minimum defining δ_r gives the formula for d_r. For fixed k the action of k is a homeomorphism of the configuration space, so D_r, a union of translates of the open set B_r, is open, and it is gauge invariant because gauge transformations compose. In particular, the existence of the constructed gauge transformation gives membership in a Borel set without a measurable choice. The example at the beginning of Section 7.4 shows that the term n^{-1} and the corresponding condition on the side are needed, also at zero curvature.

Part (3). Let $n \in 4\mathbb{N}$ and $\beta_n = 4n/\gamma \ge 2$, with $n \ge 4C_{\mathrm{SG}}/r$ as in part (2). In the notation of Definition 4.1 the exponent of the Wilson measure is $-\beta_n \sum_p q(U_p)$, so $\mu_{n,\gamma} = \mu_n^{\beta_n}$ in the notation of Theorem 6.6. We apply Theorem 6.6(3) with $\ell = 0$, the face set $S = \{p\}$, and $s = 1/2$; its hypotheses $n \in 4\mathbb{N}$ and $\beta_n \ge 2$ hold. The function $\exp(\beta_n q(U_p)/16)$ is gauge invariant, Borel, and between 1 and $e^{\beta_n/8}$, and p lies in a cell of one of the eight shifted tilings, by Lemma 7.4 with $j = 0$. We obtain

$$\mathbb{E}_{\mu_{n,\gamma}} \exp\big(\beta_n q(U_p)/16\big) \le \exp\Big[\frac{1}{8}\Big(C_0 + 24\log 2 + \frac{36\log\beta_n}{n}\Big)\Big].$$

Since $\{\rho_G(U_p) > \delta_r\} \subset \{q(U_p) > \delta_r^2/(2\pi^2)\}$, by Markov's inequality, we have

$$\mu_{n,\gamma}\{\rho_G(U_p) > \delta_r\} \le \exp\Big[\frac{1}{8}\Big(C_0 + 24\log 2 + \frac{36\log\beta_n}{n}\Big) - \frac{\beta_n \delta_r^2}{32\pi^2}\Big].$$

There are $3n^3$ positive plaquettes, and by part (2) the complement of D_r is contained in the union of these events. The union bound and the trivial bound by one give $\varepsilon_{n,r,\gamma}$. The calculation keeps the surface term for every n and γ and uses no bound on β_n/n.

Part (4). By Lemma 5.1, the open set B_r contains a product of balls of positive Haar measure. Since the density of the Wilson measure is strictly positive, we have $\mu_{n,\gamma}(D_r) > 0$. The identity for the total variation distance (in the convention $\|\mu - \nu\|_{\mathrm{TV}} = \sup_A |\mu(A) - \nu(A)|$), the identity $Z_{D_r}/Z = \mu_{n,\gamma}(D_r)$, and the bound $|\log(Z_{D_r}/Z)| \leq 2\mu_{n,\gamma}(D_r^c)$ when $\mu_{n,\gamma}(D_r^c) \leq 1/2$ are the conditioning identities of Lemma 5.5, applied with $S = D_r$. Since $\mu_{n,\gamma}(D_r^c) \leq \varepsilon_{n,r,\gamma}$, the last bound gives the stated one. The deterministic constants came only from the axial strips, the extension bounds, and the aspect bounds, and the probability constant is the absolute constant C_0. □

SECTION 8

Linearized averaging and effective quadratic forms

This section proves deterministic estimates for the block averaging map T_M of Definition 4.3. We compute the derivative of T_M^d at the identity, the transverse Gaussian covariance that it induces from an auxiliary Maxwell field, and the effective Maxwell quadratic form (Theorem 8.3). We also bound small translation-invariant kernels that annihilate gradients and commute with spatial inversion by the lattice Maxwell form (Lemma 8.6). We then construct the gauge-corrected iteration near a background with small curvature and prove locality and absolute derivative bounds at each fixed order, including volume-normalized column sums, uniformly in the number of averaging steps (Lemmas 8.10 and 8.11). The Gaussian fields in this section are auxiliary, and no statement concerns the Wilson measure. The estimates are used to construct the background fields in Part III.

We distinguish two statements with different proofs. The first is linear and exact: the derivative of T_M^d at the identity is the average K_{M^d} along straight segments of length M^d plus a lattice gradient,

$$D(T_M^d)(I) = K_{M^d} + \nabla \mathsf{R}_d$$

(Lemma 8.1). The second is nonlinear. Near a background configuration U_0 whose plaquettes are close to the identity, we multiply the output of each averaging step by gauge transformations at the two endpoints of each coarse link. In logarithmic coordinates divided by the lattice spacing, the resulting gauge-corrected step is a transported segment average plus errors that are summable over scales. The gauge-corrected iteration coincides with the iterated block averaging map after an explicit composed gauge transformation (Lemma 8.10). The gradient in the first statement and the gauge transformations in the second play analogous roles, but neither statement implies the other.

The inputs are the Hodge decomposition and Schur complement identities of Lemma 5.2, the weighted product and inverse estimates of Lemma 5.3, the coercivity of the curl on the constrained fluctuation space in Lemma 6.5, and the curvature iteration of Theorem 7.3. The lattice, the norms, and the conventions for constants are those of Definitions 4.1, 4.6, and 4.14. The results are used in Section 9 to construct coordinates on the constraint fibers (Theorems 9.3, 9.6, and 9.9), in Theorem 10.10 to identify the Hessian of the effective classical action, in Lemma 11.11 to factor perturbations of that Hessian, and in Section 31 for the normalization of the heat-smoothed circulations (Theorem 31.6).

The linear theory is part of Bałaban's analysis of Gaussian block integration for lattice gauge fields. The averages with and without endpoint paths, their action on gradients, their composition, and their Fourier representation appear in [10, (1.8)–(1.13), p. 19, Section B, and (1.29)–(1.32), p. 23]. We use centered blocks, a fixed order of coordinate paths, and a Borel extension at the singularity of the logarithm, and we prove the estimates below for these choices. Two points require care. First, the meromorphic formula for the inverse of the projected covariance has apparent singularities on the complex characteristic set $\Lambda_m(z) = 0$, which meets every neighborhood of the origin. We cancel these singularities before deducing exponential decay of the kernel. Second, the nonnegative scalar kernel of the transported segment average has row sums exactly one and counting column sums exactly M^{-3}. These masses must be kept exactly, since a fixed factor larger than one at each scale would destroy the uniformity in the number of averaging steps.

8.1 Linearized block averaging

Conventions. Unless physical units are displayed, a Fourier calculation in this section uses lattice coordinates with spacing one. For a finite matrix whose entries are color blocks, the row norm is the supremum over outputs of the sum over inputs of the operator norms of the blocks. The column norm is the supremum over inputs of the sum over outputs. Both use counting measure, and the subscript row + col denotes their sum, which is at least their maximum and at most twice it. A rate in the subscript, as in $\|\cdot\|_{a,\mathrm{row}+\mathrm{col}}$, means that the entry with indices x, y is multiplied by $e^{a\,\mathrm{dist}(x,y)}$ before summing. Distances between links are periodic ℓ^1 distances between their initial vertices. Moving the location of a link from its initial vertex to its midpoint changes these distances by at most one, and it changes any fixed exponentially weighted bound by at most a fixed factor. For the nonlinear maps in Sections 8.4 and 8.5, distances are measured in the physical units stated there.

Paths in the block averaging map. We recall the paths of Definition 4.3. We write $M = 2R_M + 1$. For a vertex $v \in M\mathbb{Z}^3$ of the coarse lattice, its block is $\mathsf{B}_M(v) = v + ([-R_M, R_M]^3 \cap \mathbb{Z}^3)$, and for $x \in \mathsf{B}_M(v)$ the coordinate path $\sigma_{v,x}$ from v to x changes the coordinates in the order $1, 2, 3$. For a coarse link $c = (v, v + Me_i)$, the reference path C_c is the straight path from v to $v + Me_i$, and for $x \in \mathsf{B}_M(v)$ the translated path with endpoint connectors is

$$F_{c,x} = \sigma_{v,x}\,[x, x + Me_i]\,\sigma^{-1}_{v+Me_i,x+Me_i};$$

see Figure 8.1. The block averaging map is

$$(T_M U)_c = \exp\Big\{M^{-3} \sum_{x \in \mathsf{B}_M(v)} \mathrm{Log}\big(U(F_{c,x})U(C_c)^{-1}\big)\Big\}U(C_c),$$

where $U(\eta)$ is the transport along a path η, and $(T_M U)_c = U(C_c)$ if one of the based loop holonomies $U(F_{c,x})U(C_c)^{-1}$ equals $-I$.

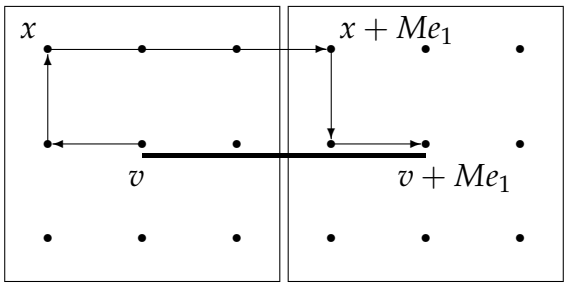


Figure 8.1. A two-dimensional slice of two neighboring blocks for $M = 3$. The thick line is the reference path C_c of the coarse link $c = (v, v + Me_1)$. The arrows show the translated path with endpoint connectors $F_{c,x}$: the coordinate path $\sigma_{v,x}$, the segment $[x, x + Me_1]$, and the reversed coordinate path $\sigma^{-1}_{v+Me_1,x+Me_1}$. At the identity the derivatives of the two connectors produce the gradient term in (8.1); in the gauge-corrected average of Section 8.4 they are canceled by gauge transformations at v and $v + Me_1$.

Lemma 8.1. *Fix an odd integer $M \geq 3$ and integers $d \geq 0$ and $n \geq 4$, and set $m = M^d$. On the fine torus of side mn, with coarse coordinates Y, let $\mathsf{I}_m = \{-(m-1)/2, \dots, (m-1)/2\}^3$ and define, for a one-cochain x and a scalar function f,*

$$(K_m x)_i(Y) = m^{-3} \sum_{r \in \mathsf{I}_m} \sum_{s=0}^{m-1} x_i(mY + r + se_i),$$
$$(\mathsf{A}_m f)(Y) = m^{-3} \sum_{r \in \mathsf{I}_m} f(mY + r).$$

For $d = 0$ both maps are the identity. Then the following statements hold.

1. *With the forward gradient $\nabla f(x, x + e_i) = f(x + e_i) - f(x)$,*

$$K_m \nabla_{\text{fine}} f = \nabla_{\text{coarse}} \mathsf{A}_m f, \qquad K_{m_2} K_{m_1} = K_{m_1 m_2}, \qquad \mathsf{A}_{m_2} \mathsf{A}_{m_1} = \mathsf{A}_{m_1 m_2},$$

   where in the last two identities m_1 and m_2 are odd, the first map averages by the factor m_1, and the second acts on the resulting lattice.

2. *There exists a linear map R_d from fine one-cochains to periodic scalar functions on the coarse torus such that*

$$D(T_M^d)(I)x = K_m x + \nabla_{\text{coarse}} \mathsf{R}_d x.$$

   In particular $\langle J, D(T_M^d)(I)x \rangle = \langle J, K_m x \rangle$ for every divergence-free coarse current J, that is, for every one-chain with zero signed incidence at each vertex.

3. *In units in which the coarse spacing is one, so that the fine spacing is $\epsilon = M^{-d}$, define the average of physical potentials by*
$$Q_d = \frac{1}{m} K_m.$$
Then Q_d maps every constant vector field to itself, $Q_d \nabla_{\text{phys,fine}} = \nabla_{\text{phys,coarse}} Q'_d$ with the scalar average $Q'_d = \mathsf{A}_m$, and if $d = d_1 + d_2$, then Q_d is the composition of the physical averages with factors M^{d_1} and M^{d_2} on the corresponding lattices.
4. *Translating the averaging box from $\{0, \dots, m-1\}^3$ to the centered box I_m multiplies the Fourier multiplier of K_m by the common phase $\exp[-i(m-1)\sum_j k_j/2]$, which is the same for all output directions and cancels in every product of the multiplier with its adjoint.*

These are identities for the derivative of T_M^d at the identity connection $U = I$; they do not identify a nonlinear average with factor m with T_M^d. They involve no coupling constant and no volume factor.

Proof. For a path σ, let $p_\sigma(x)$ be the signed sum of the one-cochain x along σ, so that a reversed link contributes the negative of the value on the positively oriented link. At the identity, the derivative of a path transport is the path sum. In the formula for T_M, the derivative of the final factor $U(C_c)$ cancels the reference-path sums inside the mean of the logarithms. The derivative on $c = (v, v + Me_i)$ is

$$M^{-3} \sum_{r \in \mathsf{I}_M} \Big\{ p_{\sigma_{v,v+r}}(x) + \sum_{s=0}^{M-1} x_i(v + r + se_i) - p_{\sigma_{v+Me_i, v+Me_i+r}}(x) \Big\}.$$

Let $p_v(x) = M^{-3} \sum_r p_{\sigma_{v,v+r}}(x)$. In the forward gradient convention of the statement we obtain

$$DT_M(I)x = K_M x - \nabla p(x). \tag{8.1}$$

The scalar p is periodic and linear in x. Its sign matters only for the definition of R_d and is fixed by (8.1).

For a scalar f, the inner sum over s in $K_m \nabla f$ telescopes to $f(mY + r + me_i) - f(mY + r)$, and averaging over r proves the gradient identity. For two successive odd factors, we expand both sums. The offsets in the box are $m_1 r_2 + r_1$, and the offsets along the segment are $m_1 s_2 + s_1$. The latter run once through $0, \dots, m_1 m_2 - 1$. In one coordinate, the former run once through the centered interval of length $m_1 m_2$: the intervals of length m_1 centered at the points $m_1 r_2$ are adjacent, and their extreme endpoints are $\pm(m_1 m_2 - 1)/2$. Taking products over the three coordinates and multiplying the weights gives both composition identities, including across the periodic seams.

Let $F_j = D(T_M^j)(I)$, and let $\mathsf{R}_0 = 0$. Suppose that $F_j = K_{M^j} + \nabla \mathsf{R}_j$. Since $T_M^j(I) = I$, using the chain rule, (8.1) at the next lattice, and the gradient identity, we obtain

$$F_{j+1} = K_{M^{j+1}} + \nabla\{\mathsf{A}_M \mathsf{R}_j - pF_j\}.$$

This defines the periodic linear map R_{j+1}. If J has zero signed incidence at every vertex, then $\nabla^{\mathsf{T}} J = 0$ and $\langle J, \nabla \mathsf{R}_d x \rangle = \langle \nabla^{\mathsf{T}} J, \mathsf{R}_d x \rangle = 0$.

A constant link-angle cochain is multiplied by m under K_m. If a physical potential on the fine lattice of spacing a/m is A, its link angles are $(a/m)A$, and dividing the coarse angles by a, we obtain $Q_d A = K_m A/m$. For a physical scalar gradient, the fine difference is multiplied by m/a and the coarse difference by $1/a$; the gradient identity becomes $Q_d \nabla_{\mathrm{phys,fine}} = \nabla_{\mathrm{phys,coarse}} \mathsf{A}_m$. The composition identity, with these divisions by the lattice spacings, gives the composition rule for Q_d.

Finally, moving the averaging box from the lower-corner position to the centered position translates each of its three coordinates by $-(m-1)/2$. Its Fourier multiplier is multiplied by $\exp[-i(m-1)\sum_j k_j/2]$, independently of the output direction, and multiplication by the adjoint cancels this phase at each frequency. All identities were obtained by differentiating at the identity or by finite sums, and no coupling or volume enters their coefficients. □

8.2 The effective Maxwell quadratic form

Fix $L > 0$, $\gamma > 0$, an odd integer $M \geq 3$, and integers $d \geq 0$ and $n \geq 4$. Set $m = M^d$, let $a = L/n$ be the coarse spacing, so that the fine spacing is a/m, and write $g_n^2 = g_3^2 a = \gamma/n$ and $g_{mn}^2 = \gamma/(mn) = g_n^2/m$ for the couplings at the coarse and fine lattices. For one color, let x be the centered Gaussian field of link angles on the fine torus of side mn with covariance $g_{mn}^2 (\mathrm{d}_{\mathrm{fine}}^{\mathsf{T}} \mathrm{d}_{\mathrm{fine}})^{\dagger}$, where $\mathrm{d}_{\mathrm{fine}}$ is the elementary curl from links to plaquettes and † denotes the Moore–Penrose inverse. By Lemma 5.2, the kernel of $\mathrm{d}_{\mathrm{fine}}$ consists of gradients and constant link fields, and this covariance vanishes on gradients and on constant fields. We push x forward by $D(T_M^d)(I)$ and project orthogonally onto the coarse transverse space, the orthogonal complement of the gradients and constant fields on the coarse torus. By Lemma 8.1, the projection removes the gradient term $\nabla \mathsf{R}_d x$, and the projected field is the projection of $K_m x$. By Lemma 5.2, the inverse covariance of a linear image of a Gaussian field is obtained by minimizing the quadratic form of the field under the corresponding linear constraint. This is why the inverse covariance computed below reappears as the Hessian at the identity of the effective classical action in Theorem 10.10.

Definition 8.2. Let $m = M^d$ with M and d as fixed above. For $z \in \mathbb{C}^3$ set

$$d_i(z) = 1 - e^{iz_i}, \qquad d_i^-(z) = 1 - e^{-iz_i}, \qquad \Lambda_m(z) = 4m^2 \sum_j \sin^2\big(z_j/(2m)\big),$$

and regard $d = (d_i)_i$ as a column vector and $d^- = (d_i^-)_i$ as a row vector; for real z we have $d^- = d^*$. For real $\theta \notin 2\pi\mathbb{Z}^3$ set

$$\lambda(\theta) = \sum_i |d_i(\theta)|^2, \qquad P(\theta) = I - \frac{d(\theta) d(\theta)^*}{\lambda(\theta)},$$

and for real k

$$b_{m,i}(k) = m^{-1} \sum_{s=0}^{m-1} e^{isk_i}, \qquad B_m(k) = \prod_j b_{m,j}(k).$$

For real $\theta \notin 2\pi\mathbb{Z}^3$ define

$$D_{m,i}(\theta) = \sum_r \frac{|B_m(k)|^2 |b_{m,i}(k)|^2}{\Lambda_m(\theta + 2\pi r)}, \qquad k = \frac{\theta + 2\pi r}{m},$$

where r runs over a complete set of representatives of $\mathbb{Z}^3/m\mathbb{Z}^3$; the points k are the m^3 fine frequencies with $mk \equiv \theta$ modulo 2π, called the aliases of θ, and each summand depends only on the class of r. Write $D = \operatorname{diag}(D_{m,i})$. Let

$$\beta_m(w) = \Big[\frac{\sin(w/2)}{m \sin(w/(2m))}\Big]^2,$$

extended by continuity at its removable singularities, let $\mathrm{I}_m^{(1)} = \{-(m-1)/2, \dots, (m-1)/2\}$, and define for $z \in \mathbb{C}^3$, wherever the denominators do not vanish,

$$D_{m,i}(z) = \sum_{r \in (\mathrm{I}_m^{(1)})^3} \frac{\prod_j \beta_m(z_j + 2\pi r_j)\, \beta_m(z_i + 2\pi r_i)}{\Lambda_m(z + 2\pi r)}, \tag{8.2}$$

and let $R_i(z)$ be the same sum restricted to $r \neq 0$. Finally set

$$\Lambda_0 = \Lambda_m(z), \qquad q_m(z) = \prod_j \beta_m(z_j), \qquad c_i(z) = q_m(z)\beta_m(z_i),$$

$$A_i(z) = c_i(z) + \Lambda_0 R_i(z), \qquad \omega_m(z) = q_m(z)^{-1} - \sum_i \frac{d_i^-(z) d_i(z) R_i(z)}{c_i(z) A_i(z)},$$

and $c_{\mathrm{av}} = (2/\pi)^{10}$, $C_{\mathrm{av}} = 1 + 3(1 + \pi^2/3)^3$.

For real $\theta \notin 2\pi\mathbb{Z}^3$ the two definitions of $D_{m,i}(\theta)$ agree; this is checked in Step 4 of the proof below. In the following theorem the coarse curl is denoted d, and $\mathrm{d}^{\mathsf{T}}\mathrm{d}$ is the lattice Maxwell form on the coarse torus.

Theorem 8.3. *Let $L > 0$, $\gamma > 0$, let $M \geq 3$ be odd, let $d \geq 0$ and $n \geq 4$ be integers, and let m, a, g_n, and the Gaussian field x be as above. Then the following statements hold.*

1. Covariance. *In unitary discrete Fourier coordinates on the coarse torus, at every nonzero frequency $\theta \in (2\pi/n)\mathbb{Z}^3/(2\pi\mathbb{Z})^3$, the covariance of the projected coarse field is*

   $$g_n^2 P(\theta) \operatorname{diag}_i\big(D_{m,i}(\theta)\big) P(\theta),$$

   and its zero-frequency component vanishes. For all values of the parameters, all i, and all $\theta \neq 0$,

   $$\frac{c_{\mathrm{av}}}{\lambda(\theta)} \leq D_{m,i}(\theta) \leq \frac{C_{\mathrm{av}}}{\lambda(\theta)}.$$

2. Inverse covariance. *Let $\mathsf{H}_m(\theta)$ be the inverse covariance of the projected coarse field after dividing the field by g_n, taken on the nonzero transverse space and extended by zero on gradients and constant fields. For $\theta \neq 0$,*

$$\mathsf{H}_m = D^{-1} - D^{-1} d \Big(\sum_i d_i^- d_i / D_{m,i} \Big)^{-1} d^- D^{-1}.$$

There exist absolute constants $\delta, a_0, C_0 > 0$ such that, for every m, this symbol has a 2π-periodic holomorphic continuation to the strip $\{z \in \mathbb{C}^3 : |\operatorname{Im} z_i| < \delta\}$, including a neighborhood of $z = 0$, bounded uniformly in m. On a neighborhood of the origin that does not depend on m, we have $A_i(z) = 1 + O(|z|^2)$ and $\omega_m(z) = 1 + O(|z|^2)$ uniformly in m, and

$$\mathsf{H}_m(z) = \Lambda_0 \operatorname{diag}(A_i^{-1}) - \frac{\operatorname{diag}(A_i^{-1})\, d\, d^- \operatorname{diag}(A_i^{-1})}{\omega_m(z)}.$$

The continuation satisfies $\mathsf{H}_m(z) d(z) = 0$ and $d^-(z) \mathsf{H}_m(z) = 0$, and $\mathsf{H}_m(0) = 0$. We call the corresponding periodic kernel $\mathsf{H}_{m,n}$ on the coarse torus the effective Maxwell quadratic form; it satisfies

$$\sup_x \sum_y e^{a_0 \operatorname{dist}_n(x,y)} \|\mathsf{H}_{m,n}(x,y)\| + \sup_y \sum_x e^{a_0 \operatorname{dist}_n(x,y)} \|\mathsf{H}_{m,n}(x,y)\| \le C_0.$$

As quadratic forms on coarse link fields,

$$C_{\mathrm{av}}^{-1} \mathrm{d}^{\mathsf{T}} \mathrm{d} \le \mathsf{H}_{m,n} \le c_{\mathrm{av}}^{-1} \mathrm{d}^{\mathsf{T}} \mathrm{d},$$

both sides vanishing on gradients and constant fields. With $\mathsf{H}_{\mathrm{Max}} = \lambda I - d d^-$ there exist absolute constants $\theta_0, C > 0$ such that $\|\mathsf{H}_m(\theta) - \mathsf{H}_{\mathrm{Max}}(\theta)\| \le C|\theta|^4$ for $|\theta| \le \theta_0$ and all m.

3. Next constrained fluctuation space. *Suppose that $n = M n_+$ with $n_+ \ge 4$, and let E_0 be the embedding of the next constrained fluctuation space of Lemma 6.5, on the torus of side n. Then*

$$\mathsf{K}_{m,n} = E_0^{\mathsf{T}} \mathsf{H}_{m,n} E_0$$

satisfies

$$c_M I \le \mathsf{K}_{m,n} \le C_M I, \qquad \|\mathsf{K}_{m,n}\|_{a_M,\mathrm{row+col}} + \|\mathsf{K}_{m,n}^{-1}\|_{a_M,\mathrm{row+col}} \le C_M$$

for constants $a_M, c_M, C_M > 0$ that depend only on M.

4. Normalization. *Let A be a smooth periodic connection on $\mathbb{T}_L^3$, and let $U^{(a)}$ be its link transports on the lattice of spacing a. Then*

$$\frac{S_W(U^{(a)})}{g_3^2 a} \longrightarrow \frac{1}{2g_3^2} \int_{\mathbb{T}_L^3} \sum_{i<j} |F_{ij}(A)|^2 \, dx \qquad (a \downarrow 0).$$

Let $A = g_3\mathcal{A}$ be the centered Gaussian transverse random distribution with zero constant mode, constructed by requiring that each finite symmetric Fourier truncation have density proportional to the exponential of minus the quadratic part of this action at $A = 0$, restricted to its real transverse coordinates. With the Fourier convention $\widehat{\mathcal{A}}(k) = \int_{\mathbb{T}^3_L} e^{-ik\cdot x}\mathcal{A}(x)\,dx$, the three colors of $\mathcal{A}$ are independent and

$$\mathbb{E}\widehat{\mathcal{A}}_i^{\alpha}(k)\widehat{\mathcal{A}}_j^{\beta}(k') = L^3\mathbf{1}_{\{k'=-k\}}\delta_{\alpha\beta}|k|^{-2}\Big(\delta_{ij} - \frac{k_ik_j}{|k|^2}\Big), \qquad k,k' \in (2\pi/L)\mathbb{Z}^3 \setminus \{0\}.$$

The constants $c_{\mathrm{av}}, C_{\mathrm{av}}, \delta, a_0, C_0, \theta_0$, and C are absolute, and a_M, c_M, C_M depend only on M; none of them depends on L, γ, n, or d.

Remark 8.4. The constant modes are set to zero only in the auxiliary Gaussian field. In the classical minimization of Section 10, coarse fields with nonzero constant modes are allowed, and the Schur complement identities of Lemma 5.2 apply to them. Accordingly, an inverse of $\mathsf{H}_{m,n}$ is asserted in part (3) only on the next constrained fluctuation space, where the averaging constraint removes the constant modes; no bound on an inverse of $\mathsf{H}_{m,n}$ on all transverse coarse fields, uniform in n, is asserted.

Remark 8.5. At each fixed nonzero θ, the alias sums have the limit

$$\lim_{d\to\infty} D_{M^d,i}(\theta) = D_{\infty,i}(\theta) = \sum_{r\in\mathbb{Z}^3} \frac{\prod_j \operatorname{sinc}^2\big((\theta_j + 2\pi r_j)/2\big)\ \operatorname{sinc}^2\big((\theta_i + 2\pi r_i)/2\big)}{|\theta + 2\pi r|^2}.$$

Indeed, choose $\theta_j \in [-\pi, \pi)$ and aliases with $\kappa_j = \theta_j + 2\pi r_j \in [-\pi m, \pi m)$. For a fixed integer vector r, as $m = M^d \to \infty$, the four squared averaging factors $|b_{m,j}(k)|^2 = \sin^2(\kappa_j/2)/(m^2\sin^2(\kappa_j/(2m)))$ converge to the four squared sinc factors, and $\Lambda_m(\kappa) \to |\kappa|^2$; each fixed alias is present for large m. By Step 2 of the proof below, after extension by zero outside the finite set of aliases, every nonzero-alias summand is bounded by $\prod_j w(r_j)/4$, where $w(0) = 1$ and $w(r) = r^{-2}$ for $r \neq 0$, and this bound is summable over $\mathbb{Z}^3$. Fix a finite cube of aliases outside which the sum of these bounds is less than a given $\eta > 0$, pass to the limit on that cube, and let $\eta \downarrow 0$. The zero alias is nonsingular at the fixed $\theta \neq 0$. This proves the limit.

Proof of Theorem 8.3. We begin with an outline. Part (1) is a Fourier computation followed by elementary bounds on the alias sums. For part (2), the formula for H_m is the inverse of the projected covariance on its range, and the comparison with the Maxwell form follows by inverting the bounds of part (1). The main point is the holomorphic continuation of H_m to a strip whose width does not depend on m; exponential decay of the kernel then follows by shifting contours. Near the origin, $D_{m,i} = A_i/\Lambda_0$, and the second term in the formula for H_m contains the inverse of $\sum_i d_i^- d_i/D_{m,i} = \Lambda_0 \sum_i d_i^- d_i/A_i$. The zero set of the holomorphic function $\Lambda_0 = \Lambda_m(z)$ near the origin is a complex hypersurface through 0,

and it contains nonreal points arbitrarily close to the origin. Although $\mathsf{H}_m(\theta) \to 0$ as real $\theta \to 0$, this does not show that H_m is holomorphic near 0. We rewrite H_m exactly so that its only denominators are A_i and ω_m, which equal $1 + O(|z|^2)$ uniformly in m. Part (3) combines the Maxwell comparison with Lemma 6.5 and the weighted inverse estimate of Lemma 5.3, and part (4) is a direct calculation.

Step 1: Fourier conventions and the covariance. On a torus of side N we use the unitary transform

$$\widehat{x}_i(k) = N^{-3/2} \sum_{z \in \mathbb{Z}^3_N} e^{-ik\cdot z} x_i(z), \qquad x_i(z) = N^{-3/2} \sum_k e^{ik\cdot z} \widehat{x}_i(k).$$

The gradient has symbol $-d(k)$, and the elementary curl has symbol

$$(\mathrm{d}x)^{\wedge}_{ij}(k) = -d_i(k)\widehat{x}_j(k) + d_j(k)\widehat{x}_i(k), \qquad i < j.$$

Expanding the three squares, we find for $k \neq 0$ that

$$\mathrm{d}(k)^*\mathrm{d}(k) = \lambda(k) I - d(k)d(k)^* = \lambda(k) P(k). \tag{8.3}$$

This identifies the null modes used in the Gaussian convention: the multiples of $d(k)$ at $k \neq 0$, which are the gradients, and all of $k = 0$, which are the constant fields, in agreement with Lemma 5.2.

For the lower-corner box, before restriction to the coarse lattice, the operator K_m is diagonal in Fourier space with multiplier $T_i(k) = mB_m(k)b_{m,i}(k)$. The centered box adds the common phase computed in Lemma 8.1. We substitute the inverse fine transform into $K_m x(Y)$ and take the coarse transform. Only the frequencies with $mk \equiv \theta$ modulo 2π survive the sum over coarse vertices, and the restriction to the coarse lattice contributes a factor $m^{-3/2}$ for each of these m^3 aliases. Summing the finite geometric series, we obtain

$$m b_{m,i}(k) d_i(k) = d_i(\theta).$$

Then $T(k)d(k) = B_m(k)d(\theta)$, with the same common phase in the centered convention, and $P(\theta)T(k)d(k) = 0$. The longitudinal part of the fine covariance vanishes after the projections on both sides, alias by alias.

The fine coupling is $g^2_{mn} = g^2_n/m$. The fine frequency $k = (\theta + 2\pi r)/m$ has $\lambda(k) = \Lambda_m(\theta + 2\pi r)/m^2$, and the inverse of the fine Laplacian at k is $m^2/\Lambda_m(\theta + 2\pi r)$. Combining the coupling, the factor from the restriction to the coarse lattice, the square of the segment multiplier m, and the inverse Laplacian, we obtain

$$\frac{g^2_n}{m} \cdot m^{-3} \cdot m^2 \cdot m^2 = g^2_n.$$

Since $P(\theta)$ removes the gradient term $\nabla \mathsf{R}_d x$ of Lemma 8.1, this is the covariance of the projection of $D(T^d_M)(I)x$, which proves the covariance formula. At $\theta = 0$, every alias

with $r \not\equiv 0$ modulo m has a vanishing factor in B_m, since $\sum_{s=0}^{m-1} e^{2\pi i r_j s/m} = 0$ when $r_j \not\equiv 0$ modulo m. The remaining alias is the fine constant mode, which has zero covariance. This shows that the zero-frequency component vanishes.

Step 2: Real bounds for the alias sums. We take $\theta_j \in [-\pi, \pi)$ and aliases with $\kappa_j = \theta_j + 2\pi r_j \in [-\pi m, \pi m)$. For the alias $r = 0$ we have

$$|b_{m,j}(\theta/m)| = \frac{|\sin(\theta_j/2)|}{m|\sin(\theta_j/(2m))|} \geq \frac{2}{\pi}, \qquad \lambda(\theta) \leq \Lambda_m(\theta) \leq |\theta|^2 \leq \frac{\pi^2}{4}\lambda(\theta),$$

where the value at a zero coordinate is understood by continuity. The first inequality for Λ_m follows from the concavity of the sine, $\sin(t/m) \geq \sin(t)/m$ for $0 \leq t \leq \pi/2$, and the last from $|\sin t| \geq 2|t|/\pi$ for $|t| \leq \pi/2$. These inequalities, together with the factor $(2/\pi)^8$ from the four squared averaging factors, prove the lower bound. The contribution of $r = 0$ to the upper bound is at most $1/\lambda(\theta)$.

For every other alias, the same chord inequality implies that

$$|b_{m,j}(k)| \leq \min\{1, \pi/|\kappa_j|\}, \qquad \Lambda_m(\kappa) \geq (4/\pi^2)|\kappa|^2.$$

If $r_j \neq 0$, then $|\kappa_j| \geq \pi|r_j|$. With $w(0) = 1$ and $w(r) = r^{-2}$ for $r \neq 0$, the numerator without its extra factor $|b_{m,i}(k)|^2$ is at most $\prod_j w(r_j)$, the extra factor is at most one, and the denominator is at least four. Since $\sum_{r \geq 1} r^{-2} = \pi^2/6$, we have $\sum_{r \in \mathbb{Z}} w(r) = 1 + \pi^2/3$, and the nonzero aliases contribute at most $(1 + \pi^2/3)^3/4$. Since $\lambda \leq 12$, this is at most $3(1 + \pi^2/3)^3/\lambda$, and the upper bound with the constant C_{av} follows.

Step 3: The transverse inverse and the Maxwell comparison. Fix real $\theta \neq 0$, write $D^{-1} = \mathrm{diag}(D_{m,i}^{-1})$, and set $s = d^- D^{-1} d = \sum_i d_i^- d_i / D_{m,i}$, which is positive. The matrix

$$H = D^{-1} - D^{-1} d\, s^{-1} d^- D^{-1}$$

is Hermitian, satisfies $Hd = 0$ and $d^- H = 0$, and $HD = I - D^{-1} d s^{-1} d^*$, so that $HDP = P$ since $d^* P = 0$. Since $HP = PH = H$, the matrix H is the inverse of PDP on $\operatorname{ran} P$. This proves the inverse formula. Note that we invert the projected covariance PDP, not the diagonal matrix D. Inverting the inequalities

$$\frac{c_{\mathrm{av}}}{\lambda} I_{\operatorname{ran} P} \leq (PDP)|_{\operatorname{ran} P} \leq \frac{C_{\mathrm{av}}}{\lambda} I_{\operatorname{ran} P}$$

from part (1) and using (8.3), we obtain the comparison with the Maxwell form at each nonzero frequency. Both sides vanish on gradients and at zero frequency, which proves the comparison of quadratic forms.

Step 4: Uniform bounds for the continued alias sums. The identity

$$\beta_m(w) = \Big(m^{-1}\sum_{s=0}^{m-1} e^{isw/m}\Big)\Big(m^{-1}\sum_{t=0}^{m-1} e^{-itw/m}\Big)$$

shows that the singularities of β_m are removable and that β_m is entire and even; no complex conjugation enters. For real θ each squared modulus in the real alias sum is the product of the multipliers at opposite frequencies, $|b_{m,j}(k)|^2 = \beta_m(\theta_j + 2\pi r_j)$, and (8.2), with the residue set $(\mathrm{I}_m^{(1)})^3$, agrees with the real formula. Increasing a coordinate of z by 2π permutes the aliases; the alias that leaves the residue set changes its coordinate by $2\pi m$, and both β_m and Λ_m are invariant under this change. This shows that (8.2) is 2π-periodic in each coordinate.

Let $z = x + iy$ with $|x_j| \le \pi$ and $|y_j| \le \delta_0$, for a sufficiently small absolute $\delta_0 > 0$. For our residue set, $|x_j + 2\pi r_j|/(2m) \le \pi/2$. Using $|\sin(u+iv)|^2 = \sin^2 u + \sinh^2 v$, we find for $r_j \neq 0$ that

$$|m\sin\big((z_j + 2\pi r_j)/(2m)\big)| \ge m|\sin\big((x_j + 2\pi r_j)/(2m)\big)| \ge |x_j + 2\pi r_j|/\pi \ge |r_j|.$$

The numerator $\sin((z_j + 2\pi r_j)/2)$ is bounded on this tube, and for $r_j = 0$ the finite-sum representation shows that β_m is uniformly bounded there. We obtain

$$|\beta_m(z_j + 2\pi r_j)| \le C w(r_j). \tag{8.4}$$

The extra factor $\beta_m(z_i + 2\pi r_i)$ is bounded by the same finite-sum formula. Since $\operatorname{Re}\sin^2(u+iv) = \sin^2 u - \cos(2u)\sinh^2 v$ and $\sinh|v| \le 2|v|$ for $|v| \le 1$, we have

$$\operatorname{Re}\Lambda_m(z + 2\pi r) \ge (4/\pi^2)|x + 2\pi r|^2 - 4|y|^2.$$

For $r \neq 0$ this is at least $2|r|^2$ after choosing δ_0 small. In particular no denominator of a nonzero alias vanishes on the tube, and its summand is bounded by $C\prod_j w(r_j)/|r|^2$. Summing and using (8.4), we obtain a uniform bound for every R_i on the whole tube over the real cube, with holomorphy in its interior. Bounds for derivatives on smaller neighborhoods follow from Cauchy's formula on fixed circles in each coordinate. For $m = 1$ these remainders vanish. For the zero alias the lower bound reads

$$\operatorname{Re}\Lambda_m(z) \ge (4/\pi^2)|x|^2 - 4|y|^2.$$

On any real region $|x| \ge \rho > 0$, the functions $D_{m,i}$ and their first derivatives are bounded and holomorphic, uniformly in m, in a tube whose width depends only on ρ.

Step 5: Cancellation on the complex characteristic set. On a small polydisc about zero that does not depend on m, the finite-sum formula and evenness imply that $\beta_m(z_j) = 1 + O(|z_j|^2)$ uniformly in m. We obtain

$$q_m = 1 + O(|z|^2), \qquad c_i = 1 + O(|z|^2), \qquad \Lambda_0 = O(|z|^2), \qquad A_i = 1 + O(|z|^2);$$

for the last assertion we used the bounds for R_i from Step 4. Since $d_i^-(z)d_i(z) = 4\sin^2(z_i/2)$ and $\beta_m(z_i) = \sin^2(z_i/2)/(m^2\sin^2(z_i/(2m)))$, we have the exact identity

$$\sum_i \frac{d_i^- d_i}{c_i} = q_m^{-1} \sum_i \frac{4\sin^2(z_i/2)}{\beta_m(z_i)} = \frac{\Lambda_0}{q_m},$$

valid also at the removable values. Using $A_i^{-1} = c_i^{-1} - \Lambda_0 R_i/(c_i A_i)$, we obtain

$$\sum_i \frac{d_i^- d_i}{A_i} = \Lambda_0 \omega_m, \qquad \omega_m = q_m^{-1} - \sum_i \frac{d_i^- d_i R_i}{c_i A_i} = 1 + O(|z|^2). \tag{8.5}$$

We fix an absolute radius $\rho > 0$ such that q_m, c_i, A_i, and ω_m have modulus at least $1/2$ on the polydisc $|z_j| < 2\rho$; this is possible uniformly in m by the preceding quadratic estimates. The zero alias of (8.2) is c_i/Λ_0. Where the alias sum is regular, we have

$$D_{m,i} = \frac{A_i}{\Lambda_0}, \qquad \sum_i \frac{d_i^- d_i}{D_{m,i}} = \Lambda_0^2 \omega_m.$$

Substituting these identities into the inverse formula, we obtain exactly the expression for $\mathsf{H}_m(z)$ in part (2). Its denominators A_i and ω_m do not vanish on the polydisc, including the points where $\Lambda_0 = 0$. This proves holomorphy there and the bound $\|\mathsf{H}_m(z)\| \le C|z|^2$, since Λ_0 and dd^- are $O(|z|^2)$. Multiplying by d on the right and by d^- on the left and using (8.5), we obtain $\mathsf{H}_m d = 0$ and $d^- \mathsf{H}_m = 0$; the value at zero is zero.

Step 6: The strip, the periodic kernel, and the comparison at low frequency. On the real complement of the smaller neighborhood $|z_j| < \rho$, part (1) implies that $D_{m,i} \ge c_{\mathrm{av}}/12$ and

$$\sum_i d_i^- d_i / D_{m,i} \ge \lambda^2 / C_{\mathrm{av}} \ge c_\rho > 0.$$

By the bounds of Step 4, the first derivatives there have a common bound C_ρ. We shrink the tube so that $D_{m,i}$ changes by less than $c_{\mathrm{av}}/24$ from its real value; its reciprocal then has bounded value and bounded first derivatives. After one more shrinking, we may assume that $\sum_i d_i^- d_i / D_{m,i}$ changes by less than $c_\rho/2$. The inverse formula of part (2) is then holomorphic and uniformly bounded on this tube. Across the faces of the real cube we use the periodicity of (8.2); the same real lower bounds hold there.

We fix an absolute width $\delta > 0$ below these widths and so small that points whose real parts lie in $|x_j| < \rho$ remain in the polydisc $|z_j| < 2\rho$. The two expressions for H_m agree on their common regular set; multiplying by their nonvanishing denominators shows that they agree throughout each overlap, including removable points. They define one bounded periodic holomorphic function on the strip of width δ.

Let its Fourier coefficients be

$$h_m(\zeta) = (2\pi)^{-3} \int_{[-\pi,\pi]^3} e^{i\theta\cdot\zeta} \mathsf{H}_m(\theta)\, d\theta, \qquad \zeta \in \mathbb{Z}^3.$$

For $0 < \delta_1 < \delta$, shift the three integration contours by $\pm i\delta_1$, with the signs chosen according to the signs of the components of ζ so as to decrease the exponential factor. The vertical sides cancel by periodicity, and from the uniform bound on the strip we obtain

$$\|h_m(\zeta)\| \le C e^{-\delta_1 |\zeta|_1}.$$

The Fourier series of H_m converges absolutely. Sampling it at the frequencies of the torus of side n and using the orthogonality of the finite characters, we obtain

$$\mathsf{H}_{m,n}(x,y) = \sum_{\zeta \equiv x-y \pmod{n}} h_m(\zeta).$$

For each representative, $\mathrm{dist}_n(x,y) \le |\zeta|_1$. For any $0 < a_0 < \delta_1$, each of the two weighted sums in part (2) is at most

$$C \sum_{\zeta \in \mathbb{Z}^3} e^{-(\delta_1 - a_0)|\zeta|_1} = C\Big(\frac{1 + e^{-(\delta_1 - a_0)}}{1 - e^{-(\delta_1 - a_0)}}\Big)^3,$$

which gives the kernel bound with a constant C_0 independent of n and d.

The comparison at low frequency follows from the same identities. On the real axis near zero, $\mathrm{diag}(A_i^{-1}) = I + O(|\theta|^2)$ and $\omega_m^{-1} = 1 + O(|\theta|^2)$. Since the Taylor remainder of the sine is uniform in $m \ge 1$, we have $\Lambda_m(\theta) - \lambda(\theta) = O(|\theta|^4)$. Since $dd^- = O(|\theta|^2)$, the two terms in the formula of part (2) differ from λI and dd^-, respectively, by $O(|\theta|^4)$. This proves $\|\mathsf{H}_m(\theta) - \mathsf{H}_{\mathrm{Max}}(\theta)\| \le C|\theta|^4$.

Step 7: Restriction to the next constrained fluctuation space. Let $a = E_0 z$. The embedding of Lemma 6.5 vanishes on the links of the centered block trees, which are the links of the coordinate paths $\sigma_{v,x}$, and every path sum $p_{\sigma_{v,x}}(a)$ in (8.1) vanishes. The linearized constraint $DT_M(I)a = 0$ says $K_M a = 0$. A fixed input link occurs exactly M times in the straight segments of the coarse sum, each time with weight M^{-3}. Summing over the coarse links, we find

$$\sum_Y (K_M a)_i(Y) = M^{-2} \sum_x a_i(x).$$

This shows that a has zero spatial mean in each component and has no constant mode. By the comparison with the Maxwell form in part (2) and the lower bound of Lemma 6.5, we have

$$\langle z, E_0^{\mathsf{T}} \mathsf{H}_{m,n} E_0 z \rangle \ge C_{\mathrm{av}}^{-1} \|\mathrm{d} E_0 z\|_2^2 \ge c_M \|z\|_2^2,$$

after renaming the constant. The upper comparison and the upper bound of the same lemma imply that $\mathsf{K}_{m,n} \le C_M I$. The rows and columns of E_0 have finite range and bounded sums depending only on M, by the same lemma. They transfer the weighted bounds for $\mathsf{H}_{m,n}$ to $\mathsf{K}_{m,n}$: each endpoint of a contributing stencil moves by at most a distance depending on M. The index set of the free coordinates has counting growth

$\#\{z : \operatorname{dist}(z, z_0) \le R\} \le D_M(1+R)^3$. We now apply the inverse estimate of Lemma 5.3. Its hypotheses are a self-adjoint matrix with a positive lower bound, bounded off-diagonal row and column sums with an exponential weight, and polynomial counting growth of the index set. We have verified them with lower bound c_M, rate a_0, a bound on the weighted sums depending only on M, and growth exponent three. The lemma gives exponentially weighted bounds for $\mathsf{K}_{m,n}^{-1}$ at a smaller rate. Taking a_M to be the smaller of the two rates and enlarging C_M proves part (3).

Step 8: Normalization. Let $U_i^{(a)}(x)$ be the transport of the smooth connection A along the link $(x, x + ae_i)$. Iterating the integral equation of the transport twice, we find uniformly in x that

$$U_i^{(a)}(x) = I + aA_i(x) + \tfrac{1}{2}a^2\{\partial_i A_i(x) + A_i(x)^2\} + O_A(a^3).$$

The remaining ordered integrals are bounded by $C_A a^3$, uniformly in x. We insert the four expansions, with the shifted arguments on the two shifted sides, into the plaquette product. The terms of degree one cancel, and the terms of degree two are $\partial_i A_j - \partial_j A_i + [A_i, A_j]$; hence $\operatorname{Log} U_{p_{ij}(x)} = a^2 F_{ij}(x) + O_A(a^3)$. For real $X = x^a \tau_a$, the Pauli identity implies that $X^2 = -|x|^2 I/4$. Then $\operatorname{tr}(e^X)/2 = \cos(|x|/2)$ and $4q(e^X) = |X|^2/2 + O(|X|^4)$. We obtain

$$\frac{S_W(U^{(a)})}{g_3^2 a} = \frac{1}{2g_3^2} a^3 \sum_x \sum_{i<j} |F_{ij}(x)|^2 + O_{A,L,g_3}(a),$$

and the Riemann sum converges because the integrand is smooth on the fixed torus. This proves the first assertion of part (4).

The quadratic part of the continuum action at $A = 0$ is $(2g_3^2)^{-1} \int \sum_{i<j} |\partial_i A_j - \partial_j A_i|^2 \, dx$. With the Fourier convention of the statement, we have $A(x) = L^{-3} \sum_k e^{ik\cdot x} \widehat{A}(k)$. Integrating the products of exponentials, we find that on any finite symmetric set of nonzero frequencies the quadratic part is

$$\frac{1}{2g_3^2 L^3} \sum_{k \neq 0} \widehat{A}(k)^* \big(|k|^2 I - kk^{\mathsf{T}}\big) \widehat{A}(k).$$

We restrict to $\widehat{A}(k) \perp k$ and impose the reality condition $\widehat{A}(-k) = \overline{\widehat{A}(k)}$. For one representative of each pair $\{k, -k\}$, the corresponding term is $|k|^2 |\widehat{A}(k)|^2 / (g_3^2 L^3)$. For a transverse real or imaginary coordinate u, the one-variable density is therefore proportional to $\exp[-|k|^2 u^2 / (g_3^2 L^3)]$, and differentiating the Gaussian integral gives the variance $g_3^2 L^3 / (2|k|^2)$; different coordinates, pairs, and colors are independent. We realize the field by independent centered real Gaussian variables with these variances, one for each transverse real and imaginary coordinate of one representative of each pair and each color. The remaining coordinates are defined by the reality condition. Then $A = g_3 \mathcal{A}$ has exactly the covariance displayed in part (4); in particular $\mathbb{E}\widehat{\mathcal{A}}(k)\widehat{\mathcal{A}}(k)^{\mathsf{T}} = 0$ for $k' = k \neq -k$, since the real and imaginary parts have equal variances.

These coordinates define a random distribution almost surely. Indeed,

$$\mathbb{E} \sum_{k \neq 0} (1 + |k|^2)^{-2} |\widehat{\mathcal{A}}(k)|^2 < \infty :$$

the summands are nonnegative with expectations at most $C_L(1 + |k|)^{-6}$, and the number of frequencies in the shell $R \leq |k| < R + 1$ is at most $C_L(1 + R)^2$, as one sees by covering their disjoint cubes of fixed volume by a shell of fixed extra thickness. Markov's inequality applied to the increasing partial sums shows that their supremum is finite almost surely, and on that event the Cauchy–Schwarz inequality pairs the Fourier sequence continuously with every smooth test function. Alternatively, each such pairing converges in mean square, since repeated integration by parts bounds the Fourier coefficients of a smooth test function f by $C_{f,s}(1 + |k|)^{-s}$ for every s. Limits of the characteristic functions of the finite truncations show that these pairings are jointly Gaussian with the stated covariance, and the covariance identity for arbitrary pairs of frequencies already holds at every finite truncation. The three colors are independent because their quadratic forms add.

The constants in Steps 1–6 are absolute; only the stencils, the lower bound, and the counting growth in Step 7 introduce dependence on M. No step used L, γ, n, or d in a constant. This completes the proof. □

8.3 Gradient annihilation and spatial inversion

The effective Maxwell quadratic form of Theorem 8.3 vanishes identically on the constant modes because of the convention for the auxiliary Gaussian field. The Hessians that occur later are perturbations of such forms by small translation-invariant kernels that annihilate gradients (by gauge invariance) and commute with spatial inversion, but need not vanish on the constant modes. The following lemma shows that such a kernel is bounded by the lattice Maxwell form, with an exponentially small error from its zero-frequency block. It is used in Lemma 11.11.

Spatial inversion and midpoint coordinates. Spatial inversion maps the vertex x of the periodic lattice of side n to $-x$. It maps the positive link $(x, x + e_i)$ to the reversed link of $(-x - e_i, -x)$, so it acts on link fields by

$$(\mathsf{I}v)_i(x) = -v_i(-x - e_i).$$

Let H be a translation-invariant kernel on link fields, written $H_{ij}(x, y) = h_{ij}(y - x)$, where $h_{ij}(t)$ is the block from a j-link with initial vertex y to an i-link with initial vertex x. The midpoint displacement of this entry is

$$t = y - x + (e_j - e_i)/2 \mod n\mathbb{Z}^3.$$

We lift it to $\mathbb{R}^3$ by choosing in each coordinate a representative of smallest absolute value. If a coordinate has two such representatives, which happens when it equals $n/2$ modulo n, we divide the coefficient equally between them. With several such coordinates, we take the product of these equal divisions. For example, in one dimension with $n = 4$ a coefficient at displacement 2 is split into two halves placed at 2 and -2. Denote the lifted coefficients by $\widetilde{h}_{ij}(t)$, $t \in \mathbb{R}^3$, and define the midpoint interpolation

$$\widetilde{H}_{ij}(k) = \sum_t \widetilde{h}_{ij}(t) e^{ik\cdot t}, \qquad k \in \mathbb{R}^3.$$

At a frequency $k \in (2\pi/n)\mathbb{Z}^3$ of the torus, lifts differing by elements of $n\mathbb{Z}^3$ have the same phase. Then $\widetilde{H}(k)$ is the Fourier block of H in midpoint coordinates, which differs from the Fourier block in initial-vertex coordinates by diagonal unitary phases on the links.

Lemma 8.6. *Fix a rate $a > 0$ and a finite color dimension. For every periodic side $n \geq 4$, let H be a real, self-adjoint, translation-invariant kernel on link fields of the torus of side n such that*

$$H\nabla = 0 = \nabla^{\mathsf{T}} H, \qquad H\mathsf{I} = \mathsf{I}H, \qquad \|H\|_{a,\mathrm{row+col}} \leq \varepsilon.$$

Then there exist $C(a), c(a) > 0$, depending only on a and the color dimension, such that with $w_n = (1+n)^3 e^{-c(a)n}$ the following hold.

1. *The zero-frequency block satisfies $\|\widehat{H}(0)\| \leq C(a)\varepsilon w_n$.*
2. *For every real or complex link field v, with the Hermitian pairing for complex v,*

$$|\langle v, Hv\rangle| \leq C(a)\varepsilon \|\mathrm{d}v\|_2^2 + C(a)\varepsilon w_n \|v\|_2^2.$$

3. *The midpoint interpolation is even, $\widetilde{H}(-k) = \widetilde{H}(k)$ for $k \in \mathbb{R}^3$, and at every frequency k of the torus, represented with $k_j \in [-\pi, \pi]$,*

$$\|\widetilde{H}(k)\| \leq C(a)\varepsilon\Big(\sum_i 4\sin^2(k_i/2) + w_n\Big).$$

The proof uses inversion in all three coordinates; a reflection in a single coordinate would not make the first derivative of $\widetilde{H}$ vanish at zero. The zero-frequency block is estimated, not assumed to vanish.

Proof. Let r be the color dimension; constants below may depend on r. The weighted hypothesis bounds each row sum and each column sum of H by ε.

We first estimate the zero-frequency block, without using inversion. Let $n \geq 32$, represent vertex coordinates in $[-n/2, n/2)^3$, and fix a Lipschitz function χ on $\mathbb{R}$ with $0 \leq \chi \leq 1$, equal to one on $[-1/8, 1/8]$, and vanishing outside $(-1/4, 1/4)$. The periodic function

$$f_j(x) = x_j \prod_{\ell=1}^{3} \chi(x_\ell/n)$$

has discrete first differences bounded by a fixed constant. The difference of x_j contributes at most one, and a difference of a cutoff contributes at most $\mathrm{Lip}(\chi)/n$ times a coordinate of size at most $n/2$. Its gradient coincides with the constant j-directed link field $\mathbf{1}_j$ on links whose initial vertices have ℓ^∞ norm at most $n/8-1$. For a color vector b with $|b|=1$, applying the identity $H\nabla=0$ to f_jb in the row of an i-link with initial vertex zero, we obtain

$$(H(\mathbf{1}_jb))_i(0)=\big(H(\mathbf{1}_jb-\nabla(f_jb))\big)_i(0).$$

Every nonzero input on the right is at periodic distance at least $n/16$ from this row and is bounded by a fixed constant. By the weighted row bound, the right side is at most $C\varepsilon e^{-an/16}$. The left side is the ij block of $\widehat{H}(0)b$. Summing over the finitely many blocks, we obtain $\|\widehat{H}(0)\|\le C\varepsilon e^{-an/16}$. For $4\le n<32$, the unweighted row bound implies that $\|\widehat{H}(0)\|\le\varepsilon$, and enlarging the constant covers these finitely many sides. With $c(a)=a/32$ both ranges give part (1).

We next use inversion. Both oriented links of an entry acquire a minus sign, and their initial vertices become $-x-e_i$ and $-y-e_j$. In terms of the entries, $H\mathsf{I}=\mathsf{I}H$ reads

$$h_{ij}(y-x)=h_{ij}(-y+x+e_i-e_j),$$

and the midpoint displacement of the right side is $-t$. Since ties were split symmetrically, we have $\widetilde{h}_{ij}(t)=\widetilde{h}_{ij}(-t)$. Then $\widetilde{H}$ is even and $D\widetilde{H}(0)=0$. The lifts of each coefficient sum to that coefficient, and the weighted absolute sums of the lifted coefficients are at most $C_a\varepsilon$, since the shortest midpoint distance exceeds the vertex distance by at most one. All representatives have $|t_\ell|\le n/2$. Since $\sup_{u\ge0}u^2e^{-au}<\infty$, the second derivatives satisfy

$$\sup_{k\in\mathbb{R}^3}\|D^2\widetilde{H}(k)\|\le C\sum_{i,j,t}|t|_1^2\|\widetilde{h}_{ij}(t)\|\le C(a)\varepsilon.$$

At $k=0$ all phases are one, and $\widetilde{H}(0)=\widehat{H}(0)$. By Taylor's formula along the segment from 0 to k and part (1), we have $\|\widetilde{H}(k)\|\le C(a)\varepsilon(w_n+|k|^2)$. For $k_j\in[-\pi,\pi]$ we have $|k|^2\le(\pi^2/4)\sum_j 4\sin^2(k_j/2)$, which proves part (3).

To pass to quadratic forms, set $q_i(k)=2\sin(k_i/2)$ and $\lambda_q=\sum_i q_i^2$. After we move the phases from the initial vertices to the midpoints of the links and plaquettes, the gradient becomes multiplication by iq, and the curl from links to plaquettes becomes

$$(D(k)u)_{ij}=i(q_iu_j-q_ju_i),\qquad D(k)^*D(k)=\lambda_qI-qq^{\mathsf{T}}.$$

At a frequency $k\neq0$, the identities $H\nabla=0=\nabla^{\mathsf{T}}H$ say that $\widetilde{H}(k)$ vanishes on q and maps into $q^\perp$. If u_T is the component of u orthogonal to q, then Hermitian symmetry and part (3) imply that

$$|\langle u,\widetilde{H}(k)u\rangle|\le C(a)\varepsilon\lambda_q|u_T|^2+C(a)\varepsilon w_n|u|^2=C(a)\varepsilon|D(k)u|^2+C(a)\varepsilon w_n|u|^2.$$

At $k=0$ we use part (1). The unitarity of the finite Fourier transform and the triangle inequality over the frequencies imply part (2) for real and complex link fields. □

Remark 8.7. On the transverse fields with nonzero frequency, the quadratic form of H factors through the curl with a bounded factor: there is an operator T on plaquette fields, diagonal in Fourier space, with $\|T\|_{\mathrm{op}} \le C(a)\varepsilon$ uniformly in n, such that $\langle v, Hv\rangle = \langle \mathrm{d}v, T\mathrm{d}v\rangle$ for every v orthogonal to the gradients and the constant fields. Indeed, at each frequency $k \neq 0$ set

$$T(k) = \lambda_q^{-2} D(k)\widetilde{H}(k)D(k)^*,$$

and set $T(0) = 0$. Since $D^*D = \lambda_q P_q$, where P_q is the orthogonal projection onto $q^\perp$, and $\widetilde{H}(k) = P_q \widetilde{H}(k) P_q$, we have $D(k)^* T(k) D(k) = \widetilde{H}(k)$, and $T(k)$ vanishes on the plaquette directions orthogonal to the range of $D(k)$. Since $\|D(k)\| = \lambda_q^{1/2}$, part (3) implies that $\|T(k)\| \le C(a)\varepsilon(1 + w_n/\lambda_q)$. Every nonzero frequency has $\lambda_q \ge 4\sin^2(\pi/n) \ge 16/n^2$ by the chord inequality, and $\sup_{n\ge 4} n^2(1+n)^3 e^{-c(a)n} < \infty$, so $\|T\|_{\mathrm{op}} \le C(a)\varepsilon$ uniformly in n. The zero-frequency block of H is not part of this factorization; it is estimated separately in part (1). This construction controls only the operator norm of T, and no exponential decay of the kernel of T follows from it. A factorization with exponentially local factors and an exponentially small remainder is proved separately in Lemma 11.11.

8.4 The gauge-corrected block average

In this section and the next we study the iterated block averaging map near a background configuration with small curvature. The background is fixed, and all derivatives are taken with respect to a fluctuation. Gauge-corrected averages, in which the output of each averaging step is multiplied by gauge transformations at the endpoints of the coarse links, occur in Bałaban's averaging construction [12, (61)–(63), p. 28], and their composition is treated in [12, (78)–(92), pp. 30–31; (97)–(100), p. 32]. We use the centered blocks of Definition 4.3, and we derive below the derivative of the corrected step, its endpoint terms, and the order of the composed gauge transformations.

We explain why the estimates are uniform in the number d of averaging steps. We use units in which the coarse spacing after d steps is one, so that the spacing at level j is $\epsilon_j = M^{j-d}$. If $\delta(U_0) \le \varepsilon\epsilon_0^2$, then by Theorem 7.3 the plaquette angles δ_j of $T_M^j U_0$ satisfy $\delta_j \le 2\varepsilon\epsilon_j^2$; they scale like the square of the spacing, as for a smooth field. One gauge-corrected step, written for the potential $A_j = \epsilon_j^{-1} \operatorname{Log} V_j$, has the form

$$A_{j+1} = Q_{U_j} A_j + O_M\big(\epsilon_{j+1}|A_j|^2\big),$$

where the row sums of Q_{U_j} are at most $1 + C_M\delta_j$. The relative errors at level j are of order $\delta_j + \epsilon_{j+1}|A_j|$, and

$$\sum_{j<d} \delta_j \le \frac{2\varepsilon}{M^2-1}, \qquad \sum_{j<d} \epsilon_{j+1} \le \frac{M}{M-1}$$

are bounded independently of d. This is why the constants below do not depend on d, and why the scalar weights of the transported segment average must sum to exactly one in each row, rather than to $1 + c$ for a fixed $c > 0$. All smallness assumptions concern plaquettes and holonomies of contractible closed paths; the background may have arbitrary holonomies around the cycles of the torus.

Definition 8.8. Let U be an $\mathrm{SU}(2)$-valued configuration on a periodic lattice of side Mn' with $n' \geq 4$, and let $V = (V_e)_e$ be a link field with values in $\mathrm{SL}(2,\mathbb{C})$. For a coarse vertex v set

$$\mathsf{h}_{v;U}(V) = \exp\Big[-M^{-3} \sum_{x \in \mathsf{B}_M(v)} \mathrm{Log}\big((VU)(\sigma_{v,x})U(\sigma_{v,x})^{-1}\big)\Big],$$

where $VU = (V_e U_e)_e$, and write $\mathsf{h}_U(V)$ for the gauge transformation $(\mathsf{h}_{v;U}(V))_v$ on the coarse vertices. For a coarse link $c = (v, s)$, $s = v + Me_i$, define the gauge-corrected block average

$$(\mathscr{M}_U(V))_c = \mathsf{h}_{v;U}(V)\,(T_M(VU))_c\,\mathsf{h}_{s;U}(V)^{-1}\,(T_M U)_c^{-1}.$$

For a fine spacing $\epsilon > 0$ and a link field A of potentials, set

$$\Psi_{U,\epsilon}(A) = (M\epsilon)^{-1}\,\mathrm{Log}\,\mathscr{M}_U(e^{\epsilon A}).$$

All logarithms are identity-chart logarithms; Lemma 8.9 shows that they are defined in the domains used below.

In other words, $\mathscr{M}_U(V)\,T_M U$ is the gauge transform of $T_M(VU)$ by $\mathsf{h}_U(V)$, and $\mathscr{M}_U(V)$ is its transport relative to the background output $T_M U$. The endpoint gauge transformation at v is chosen so that its derivative cancels the contribution of the connectors $\sigma_{v,x}$ in Figure 8.1. For a gauge transformation k on the vertices of a lattice, we write $k|_{\mathrm{roots}}$ for its restriction to the vertices of the next coarser lattice, which are the centers of the blocks.

Lemma 8.9. *Fix an odd $M \geq 3$, a radius $0 < \rho_{\log} < 2\pi$ inside the real identity chart of $\mathrm{SU}(2)$, and a radius $0 < r_{\log} < 1$ for the matrix logarithm series. There exist $b_M, \delta_M > 0$ and $q_M, C_M < \infty$, depending only on these data, such that the following holds for every application of T_M from a periodic lattice of side Mn' to the lattice of side n', $n' \geq 4$.*

1. *For each coarse link $c = (v, v + Me_i)$, the paths C_c, $F_{c,x}$, $\sigma_{v,x}$, and $\sigma_{v+Me_i,x+Me_i}$, $x \in \mathsf{B}_M(v)$, have at most q_M links and lie in a neighborhood of c of radius depending only on M, and each fine link belongs to at most a number depending only on M of these neighborhoods.*
2. *If U is $\mathrm{SU}(2)$-valued and $\delta(U) = \max_p \rho_G(U_p) \leq \delta$ with $0 \leq \delta < \delta_M$, then every based loop holonomy $U(F_{c,x})U(C_c)^{-1}$ has an identity-chart logarithm of norm at most $C_M\delta$.*
3. *For such U and complex link variables with $\max_e |X_e| < b_M$, the path products with links $e^{X_e}U_e$, their relative logarithms, and the logarithm $\mathrm{Log}\,\mathscr{M}_U(e^X)$ of the gauge-corrected average are holomorphic in X. On a fixed smaller ball, each partial derivative of fixed order*

in the finitely many relevant X_e is bounded by a constant depending only on M, the order, and the radii $\rho_{\log}$, $r_{\log}$. The logarithm $\operatorname{Log} \mathscr{M}_U(e^X)$ vanishes at $X = 0$.

These bounds are independent of the holonomies of U around the cycles of the torus and of n', and they include reversed links.

Proof. Each coordinate path has length at most $3(M-1)/2$, and the middle segment has length M. The union of the two adjacent centered blocks has extent $2M-1$ in the direction of the link and $M-1$ in the other two. Since the fine side is $Mn' \geq 4M$, this union lifts to a box in $\mathbb{Z}^3$ without winding. All occurrences of a fixed fine link lie in a bounded number of such neighborhoods: their coarse vertices are within a distance depending on M, and there are three orientations and boundedly many paths.

We next prove a filling estimate. In a lifted box of maximal extent D, we choose its lower corner and the coordinate paths, ordered $1, 2, 3$, from it to every vertex. For a positive link in direction i, we append the link to the ordered path of its initial vertex and move it to the left past the portions of the path in directions greater than i. Each interchange replaces two adjacent perpendicular links by the other two links of their elementary square. There are at most $3D$ interchanges, all within the box. Inserting a square changes the transport by a conjugate of the plaquette transport or its inverse. Hence, in the frame given by the ordered paths, the transport of this link is a product of at most $3D$ conjugated plaquette transports. For a reversed link we take the inverse. Along a closed path of length q the frames cancel at successive vertices, and the transport around the path is, up to one conjugation, a product of at most $3Dq$ conjugated plaquette transports. We apply this to $F_{c,x}C_c^{-1}$ in the box above, with $D \leq 2M-1$ and $q \leq q_M$, which gives at most $s_M = 3(2M-1)q_M$ plaquette factors. A product of s elements of $\mathrm{SU}(2)$, each within group distance δ of the identity, is within group distance $s\delta$ of the identity, as one sees by concatenating the shortest one-parameter subgroups and using the bi-invariance of the metric. Let $0 < \delta_M \leq \rho_{\log}/(2s_M)$. For $0 \leq \delta < \delta_M$, every based loop holonomy is within group distance $s_M\delta < \rho_{\log}$ of the identity, so its identity-chart logarithm exists and has norm at most $s_M\delta$. This proves part (2), and further decreases of δ_M below preserve it.

For the complex bounds, write a positive link as $e^{X_e}U_e$ and a reversed link as $U_e^{-1}e^{-X_e}$. For a path product W, moving the fixed $\mathrm{SU}(2)$-valued factors to the right writes $W(X)W(0)^{-1}$ as a product of at most q_M exponentials of $\pm X_e$, each conjugated by an $\mathrm{SU}(2)$-valued partial transport. On the complexified color space these conjugations are unitary, so in a fixed matrix norm

$$\|W(X)W(0)^{-1} - I\| \leq \exp\bigl(Cq_M \max_e |X_e|\bigr) - 1,$$

as one sees by expanding each exponential absolutely and multiplying the bounds. For a based loop holonomy, the value at $X = 0$ is within $C_M\delta$ of the identity by part (2).

Choosing b_M and δ_M small puts all these matrices in the ball $\|Z - I\| < r_{\log}$, on which $\operatorname{Log} Z = \sum_{\ell \geq 1} (-1)^{\ell+1} (Z-I)^{\ell}/\ell$ converges absolutely. Its derivatives on smaller balls are bounded by the differentiated series, and the exponential series has the same property everywhere. The endpoint gauge transformations equal the identity at $X = 0$, and there the output of T_M equals the background output, so $\mathscr{M}_U(e^0) = I$. After decreasing b_M, its logarithm lies in the same chart. At most q_M distinct link variables enter, even when a link occurs repeatedly. Bounds for every partial derivative of fixed order follow by differentiating these finite products and series, or by Cauchy's formula on fixed circles in these finitely many complex variables. Neither method introduces a dependence on the side or the depth. □

Lemma 8.10. *Fix an odd $M \geq 3$ and the radii $\rho_{\log}, r_{\log}$ of Lemma 8.9. There exist $\varepsilon_*, a_*, \delta_{\mathrm{av}}, C_M > 0$, depending only on these data, with $2\varepsilon_* \leq \delta_{\mathrm{av}} < \delta_M$, such that the following holds for all integers $d \geq 0$ and $n \geq 4$. Use units in which the spacing of the coarse lattice of side n is one, so that the lattice at level j has side $M^{d-j}n$ and spacing $\epsilon_j = M^{j-d}$, $0 \leq j \leq d$.*

1. One step. *Let U be an $\mathrm{SU}(2)$-valued configuration on a periodic lattice of side Mn', $n' \geq 4$, with $\delta(U) = \delta \leq \delta_{\mathrm{av}}$, and let $0 < \epsilon \leq 1/M$. On the complex ball $\|A\|_\infty < 2a_*$,*

$$\Psi_{U,\epsilon}(A) = Q_U A + \mathscr{E}_{U,\epsilon}(A), \qquad \|Q_U\|_{\mathrm{row}} \leq 1 + C_M\delta,$$
$$\|\mathscr{E}_{U,\epsilon}(A)\|_\infty \leq C_M M\epsilon \|A\|_\infty^2, \qquad \|D\mathscr{E}_{U,\epsilon}(A)\|_{\mathrm{row}} \leq C_M M\epsilon \|A\|_\infty,$$

   where $Q_U = D\Psi_{U,\epsilon}(0)$ does not depend on ϵ. Its principal part is the transported segment average

$$(Q_U^{\mathrm{main}} A)_c = M^{-4} \sum_{x \in \mathsf{B}_M(v)} \sum_{e \in [x, x+Me_i]} \operatorname{Ad}\big(U(\sigma_{v,x}[x, e^-])\big) A_e, \qquad c = (v, v + Me_i),$$

   where e^- is the initial vertex of e and $\sigma_{v,x}[x, e^-]$ is the path $\sigma_{v,x}$ followed by the segment from x to e^-. The scalar averaging weights in each row of Q_U^{main} sum to exactly one, the norms of its color blocks in each row sum to at most one, and $Q_U - Q_U^{\mathrm{main}}$ has row norm at most $C_M\delta$.

2. Iteration. *Let U_0 be an $\mathrm{SU}(2)$-valued configuration on the lattice of side $M^d n$ with $\delta(U_0) \leq \varepsilon\epsilon_0^2$ for some $0 \leq \varepsilon \leq \varepsilon_*$, and set $U_j = T_M^j U_0$. Then $\delta_j = \delta(U_j) \leq 2\varepsilon\epsilon_j^2$ for $0 \leq j \leq d$. For $\|A\|_\infty < a_*$ define*

$$V_0 = e^{\epsilon_0 A}, \qquad V_{j+1} = \mathscr{M}_{U_j}(V_j), \qquad A_j = \epsilon_j^{-1} \operatorname{Log} V_j.$$

   The recursion is defined and holomorphic on this ball, and for $0 \leq j \leq d$

$$\|A_j\|_\infty \leq 2\|A\|_\infty, \qquad \|D_A A_j\|_{\mathrm{row}} \leq \exp\Big[C_M \sum_{\ell < j} (\delta_\ell + \epsilon_{\ell+1}\|A_\ell\|_\infty)\Big] \leq C_M.$$

The coordinate $(A_j)_c$ depends only on the variables A_e with $\operatorname{dist}(c,e) \le C_M\epsilon_j$, where distances are measured in the units above, in which the coarse spacing is one. For every fixed $a_{\mathrm{sp}} \ge 0$ we have $\|D_A A_j\|_{a_{\mathrm{sp}},\mathrm{row}} \le C_{M,a_{\mathrm{sp}}}$ on the same ball. For every fixed $s \ge 1$, on the ball $\|A\|_\infty < a_*/2$, every coarse link c and all link fields $B_1,\dots,B_s$,

$$\left|\left(D_A^s A_j(A)[B_1,\dots,B_s]\right)_c\right| \le C_{M,s,a_{\mathrm{sp}}} \prod_{\ell=1}^{s} \sup_e e^{-a_{\mathrm{sp}}\operatorname{dist}(c,e)} |(B_\ell)_e|.$$

3. Composed gauge transformation. *Define gauge transformations on the vertices of the lattice at level j by*

$$\mathsf{G}_0 = I, \qquad \mathsf{G}_{j+1} = \mathsf{h}_{U_j}(V_j)\,(\mathsf{G}_j|_{\mathrm{roots}}).$$

Then

$$V_j U_j = \left(T_M^j(e^{\epsilon_0 A}U_0)\right)^{\mathsf{G}_j}, \qquad 0 \le j \le d.$$

For real A the gauge transformations G_j are $\mathrm{SU}(2)$-valued. On a fixed smaller complex ball they and their inverses have uniformly bounded holomorphic logarithms; the factors satisfy $\|\operatorname{Log}\mathsf{h}_{U_j}(V_j)\| \le C_M\epsilon_j\|A_j\|_\infty$, whose sum over j is bounded.

The constants in parts (1)–(3) do not depend on n, n', d, j, the holonomies of the background around the cycles of the torus, L, or γ; the constants $C_{M,a_{\mathrm{sp}}}$ and $C_{M,s,a_{\mathrm{sp}}}$ depend in addition only on a_{sp} and s.

Proof. We begin with the curvature along the background trajectory. Let $0 < \varepsilon_* \le \varepsilon_M$, with ε_M as in Theorem 7.3. At level j the background has side $M^{d-j}n$, and every step $j < d$ goes from side $M \cdot M^{d-j-1}n$ to side $M^{d-j-1}n \ge n \ge 4$, with the factor M and spacings $\epsilon_j = M^{j-d}$. For $0 < \varepsilon \le \varepsilon_*$, the hypothesis $\delta(U_0) \le \varepsilon\epsilon_0^2$ is the hypothesis of the curvature iteration in Theorem 7.3, which shows that T_M is given by its logarithmic formula at every step and yields the curvature bounds below. If $\varepsilon = 0$, then $\delta(U_0) = 0$, and for every $\eta \in (0,\varepsilon_M]$ the same background satisfies $\delta(U_0) \le \eta\epsilon_0^2$. We apply the same theorem with η in place of ε. The iterates $U_j = T_M^j U_0$ do not depend on η, and each application bounds the same fields by $\delta_j \le 2\eta\epsilon_j^2$; letting $\eta \downarrow 0$, we obtain $\delta_j = 0$ for every j, and any one application with positive η shows that the logarithmic formula applies. When $d = 0$ the bound is the hypothesis and the sums below are empty. In all cases, for $0 \le \varepsilon \le \varepsilon_*$,

$$\delta_j \le 2\varepsilon\epsilon_j^2, \qquad \sum_{j<d} \delta_j \le \frac{2\varepsilon}{M^2-1}. \tag{8.6}$$

We fix $0 < \delta_{\mathrm{av}} < \delta_M$, with δ_M from Lemma 8.9, and impose $2\varepsilon_* \le \delta_{\mathrm{av}}$. Every application below with $0 \le \delta \le \delta_{\mathrm{av}}$ then satisfies the curvature hypothesis $\delta < \delta_M$ of that lemma. We decrease ε_*, δ_{av}, and the fluctuation radius finitely many times below.

Step 1: The one-step differential and its endpoint terms. Fix an $\mathrm{SU}(2)$-valued U with $\delta(U) = \delta \le \delta_{\mathrm{av}}$, and write X for a link field of infinitesimal fluctuations. For a path η, the product rule and $(e^{tX_e}U_e)^{-1} = U_e^{-1}e^{-tX_e}$ give the left-trivialized derivative of the transport,

$$p_\eta(X) = \sum_{\substack{\text{positive}\\ \text{occurrences } e}} \mathrm{Ad}\big(U(\eta_{\text{before } e})\big)X_e - \sum_{\substack{\text{negative}\\ \text{occurrences } \bar e}} \mathrm{Ad}\big(U(\eta_{\text{before } \bar e})U_e^{-1}\big)X_e,$$

where $\eta_{\text{before } e}$ is the part of η preceding the occurrence. All transports in this formula are $\mathrm{SU}(2)$-valued, hence orthogonal on colors.

To differentiate T_M without discarding a term, let $F_x = U(F_{c,x})$, $C = U(C_c)$, $L_x = F_xC^{-1}$, $Z = M^{-3}\sum_x \operatorname{Log} L_x$, and $O = e^ZC = (T_MU)_c$. Let $f_x = \dot F_x F_x^{-1}$ and $c_0 = \dot C C^{-1}$, where the dot denotes the derivative in the direction X, and introduce the left-trivialized differentials

$$\mathcal{E}_Z(Y) = (D\exp_Z Y)e^{-Z}, \qquad \mathcal{L}_{L_x}(Y) = D\operatorname{Log}_{L_x}(YL_x).$$

By the product rule, we have

$$\dot O O^{-1} = \mathcal{E}_Z\Big[M^{-3}\sum_x \mathcal{L}_{L_x}\big(f_x - \mathrm{Ad}(L_x)c_0\big)\Big] + \mathrm{Ad}(e^Z)c_0. \tag{8.7}$$

By Lemma 8.9, $|\operatorname{Log} L_x| \le C_M\delta$ and $|Z| \le C_M\delta$. Using the convergent exponential and logarithm series, we obtain $\mathcal{E}_Z = I + O_M(\delta)$, $\mathcal{L}_{L_x} = I + O_M(\delta)$, $\mathrm{Ad}(L_x) = I + O_M(\delta)$, and $\mathrm{Ad}(e^Z) = I + O_M(\delta)$ as operators. Their absolute local coefficient sums have the same bounds, because at most q_M local labels occur. Replacing these four operators by I in (8.7) cancels the two terms containing c_0 and leaves $M^{-3}\sum_x f_x$. The replacement error, as a linear map of X, has absolute row norm at most $C_M\delta$.

We split the derivative along the path at its two endpoint connectors:

$$f_x = p_{\sigma_{v,x}}(X) + p^{\mathrm{seg}}_{v,x}(X) - \mathrm{Ad}(F_x)p_{\sigma_{s,x+Me_i}}(X), \qquad s = v + Me_i.$$

The middle term is the sum of the variations of the M links of the segment $[x, x+Me_i]$, transported first along $\sigma_{v,x}$ and then along the segment up to the link. Differentiating the endpoint gauge transformation, we find

$$\dot{\mathsf{h}}_{v;U}(I) = -M^{-3}\sum_x p_{\sigma_{v,x}}(X).$$

The left-trivialized derivative of the gauge-corrected output is the derivative of T_M minus this mean over the initial connectors, plus $\mathrm{Ad}(O)$ times the corresponding mean over the terminal connectors. The terms of the initial connectors cancel exactly. The remaining terminal terms are

$$M^{-3}\sum_x \{\mathrm{Ad}(O) - \mathrm{Ad}(F_x)\}p_{\sigma_{s,x+Me_i}}(X);$$

here the average over $x + Me_i$ is exactly the average over $\mathsf{B}_M(s)$ in the definition of $\mathsf{h}_{s;U}$, with the same paths. These terms have absolute row norm at most $C_M\delta$, since $OF_x^{-1} = e^Z L_x^{-1}$ is within $C_M\delta$ of the identity and $\mathrm{SU}(2)$-valued transports have norm one.

The remaining segment derivative is

$$M^{-3} \sum_{x \in \mathsf{B}_M(v)} \sum_{e \in [x, x+Me_i]} \mathrm{Ad}\big(U(\sigma_{v,x}[x, e^-])\big) X_e.$$

For $X = \epsilon A$, division by the coarse length $M\epsilon$ gives Q_U^{main}. There are M^4 occurrences of weight M^{-4}, each an orthogonal map on colors. The scalar weights in each row therefore sum to one, and the norms of the color blocks in each row sum to at most one. The endpoint and coefficient errors above, divided by M, are still bounded by $C_M\delta$. We conclude that $Q_U = D\Psi_{U,\epsilon}(0)$ does not depend on ϵ and satisfies the row estimates of part (1).

Step 2: The one-step remainder. Let $F_U(X) = \mathrm{Log}\, \mathscr{M}_U(e^X)$, which vanishes at $X = 0$. By Lemma 8.9, its second derivative has absolute row norm at most C_M on a fixed complex ball of X, uniformly in U. Here the row sums over both inputs, and the bounded size of the stencils justifies this absolute bound. Since $\Psi_{U,\epsilon}(A) = (M\epsilon)^{-1} F_U(\epsilon A)$, on a fixed complex ball of A with $\epsilon \le 1/M$,

$$\|D_A^2 \Psi_{U,\epsilon}\|_{\mathrm{row}} \le C_M \frac{\epsilon^2}{M\epsilon} = C_M \epsilon / M \le C_M M\epsilon.$$

The quadratic bound for $\mathscr{E}_{U,\epsilon}$ follows from Taylor's formula along the segment tA, $0 \le t \le 1$. Integrating the second derivative along the same segment gives the bound for its first derivative. This proves part (1) on $\|A\|_\infty < 2a_*$, after choosing $2a_*$ inside that ball.

Step 3: Uniform iteration. Let C dominate the constants of part (1). Since $A_{j+1} = \Psi_{U_j,\epsilon_j}(A_j)$, $M\epsilon_j = \epsilon_{j+1}$, and $\epsilon_j \le 1/M$ for $j < d$, the norms $b_j = \|A_j\|_\infty$ satisfy, whenever the iteration is defined,

$$b_{j+1} \le b_j\big(1 + C\delta_j + C\epsilon_{j+1} b_j\big), \qquad \sum_{j<d} \epsilon_{j+1} \le \frac{M}{M-1}.$$

We choose the radii so small that

$$C\Big(\frac{2\varepsilon_*}{M^2 - 1} + \frac{2a_* M}{M-1}\Big) < \log(3/2). \tag{8.8}$$

For an input with $\|A\|_\infty = t < a_*$, as long as $b_\ell \le 2t$ for $\ell < j$, multiplying the inequalities and using $\log(1+u) \le u$, we obtain

$$b_j \le t \exp\Big[C \sum_{\ell<j} \delta_\ell + 2Ct \sum_{\ell<j} \epsilon_{\ell+1}\Big] < \tfrac{3}{2} t.$$

In particular b_j never reaches $2t$, so the bound holds for all $j \le d$, and each step stays in the domain $b_j < 2a_*$ of part (1). Since a finite composition of holomorphic maps is holomorphic, the recursion is holomorphic on the input ball at every finite depth. For $t = 0$ it vanishes identically, and for $d = 0$ it is the identity.

The derivative of one step at A_j, including the nonlinear remainder, has row norm at most $1 + C_M(\delta_j + \epsilon_{j+1} b_j)$. By the submultiplicativity of absolute row sums, we obtain the bound for $D_A A_j$ in part (2). By (8.6) and the bootstrap, its exponent is at most

$$C_M\Big(\frac{2\varepsilon_*}{M^2-1} + \frac{2a_*M}{M-1}\Big),$$

which does not depend on d, n, or j.

Step 4: Locality and directional derivatives. An output of one step depends only on inputs within a neighborhood of radius $r_M^{\mathrm{loc}}\epsilon_j$, where $r_M^{\mathrm{loc}} < \infty$ is a fixed stencil radius. If r_j is the dependence radius at level j, measured in the units in which the coarse spacing is one, then $r_{j+1} \le r_j + r_M^{\mathrm{loc}}\epsilon_j$ and $r_0 = 0$. Summing the geometric series, we find $r_j \le C_M\epsilon_j$. The same recursion bounds the background links on which the coefficients depend, since each preceding application of T_M has a neighborhood of the same scale. The assertion holds on the torus, since each neighborhood can be lifted to a box before returning to the torus. Inserting the weight $e^{a_{\mathrm{sp}}\,\mathrm{dist}(c,e)}$ multiplies the bound for the first derivative by at most $e^{a_{\mathrm{sp}}C_M\epsilon_j} \le e^{a_{\mathrm{sp}}C_M}$.

For the higher derivatives, take the smaller ball $\|A\|_\infty < a_*/2$. For link fields $B_1, \dots, B_s$ with $\|B_\ell\|_\infty \le 1$, the map $(t_1, \dots, t_s) \mapsto A_j(A + \sum_\ell t_\ell B_\ell)$ is bounded by $2a_*$ for $|t_\ell| < a_*/(4s)$. Cauchy's formula on these circles bounds its mixed coefficient, and hence $\|D_A^s A_j(A)[B_1, \dots, B_s]\|_\infty$, by $2a_*(4s/a_*)^s$. This bounds a multilinear operator norm; absolute sums of coefficients over the inputs are treated in Lemma 8.11. For the weighted bound, restrict each B_ℓ to the dependence neighborhood of the fixed output c, which does not change the derivative by locality. On that neighborhood $|(B_\ell)_e| \le e^{a_{\mathrm{sp}}C_M} \sup_{e'} e^{-a_{\mathrm{sp}}\,\mathrm{dist}(c,e')}|(B_\ell)_{e'}|$, so the multilinear bound acquires at most the factor $e^{sa_{\mathrm{sp}}C_M}$.

Step 5: The composed gauge transformation. For a gauge transformation k, a transport along a path from v to s transforms as $k_v U(\eta) k_s^{-1}$. The based loop holonomies $U(F_{c,x})U(C_c)^{-1}$ are conjugated by k_v. Within the identity chart the logarithm series and the exponential are equivariant under conjugation, and substituting into the formula for T_M, we obtain

$$T_M(W^k) = (T_M W)^{k|_{\mathrm{roots}}}.$$

For SU(2)-valued gauge transformations this also holds at the configurations where the Borel extension is used, since the exceptional matrix $-I$ is central. For complex gauge transformations we use only the domain where all logarithms are given by their series.

Let $Z_j = T_M^j(e^{\epsilon_0 A}U_0)$. If $V_jU_j = Z_j^{\mathsf{G}_j}$, then by the definition of $\mathscr{M}$ and the gauge identity we have

$$V_{j+1}U_{j+1} = \big(T_M(V_jU_j)\big)^{\mathsf{h}_{U_j}(V_j)} = (T_MZ_j)^{\mathsf{h}_{U_j}(V_j)(\mathsf{G}_j|_{\mathrm{roots}})},$$

where we used $(W^k)^l = W^{lk}$. Starting at $j = 0$, induction proves the identity of part (3) with the stated order of the factors. For real A, every relative transport in an endpoint gauge transformation is $\mathrm{SU}(2)$-valued with small logarithm in $\mathfrak{su}(2)$. Then every factor and every G_j is $\mathrm{SU}(2)$-valued.

For complex A, Lemma 8.9 and the definition of h imply that

$$\|\mathrm{Log}\,\mathsf{h}_{U_j}(V_j)\| \le C_M\epsilon_j b_j, \qquad \sum_{j<d}\epsilon_j b_j \le \frac{2\|A\|_\infty}{M-1}.$$

It follows that all partial products and their inverses have norm at most $\exp(C_M\|A\|_\infty)$ and differ from the identity by at most $\exp(C_M\|A\|_\infty) - 1$. We decrease a_* once so that the last quantity is less than $1/2$. The logarithms then exist by the same convergent series and are holomorphic and uniformly bounded.

It remains to check that the gauge identities do not leave the logarithm chart. At level j, every based loop holonomy of V_jU_j differs from the identity by at most $C_M(\delta_j + \epsilon_j b_j)$, by Lemma 8.9. The corresponding based loop holonomy of Z_j is its conjugate by G_j^{-1}, and its distance to the identity in matrix norm is larger by at most the factor $\exp(C_M\|A\|_\infty)$. The bounds $\delta_j \le 2\varepsilon_*\epsilon_j^2$ and $b_j \le 2a_*$ allow one further decrease of ε_* and a_* after which both based loop holonomies lie strictly inside the chart. The conjugation identity for the logarithm series then holds term by term, and the induction above applies at every level on this ball. This shows that the complex gauge relation is an identity of holomorphic maps, not only of Taylor coefficients. All choices involved only M and the radii $\rho_{\log}$, $r_{\log}$. □

8.5 Volume-normalized column sums and derivative bounds

We now prove bounds on absolute sums of derivative coefficients of the iterates $A_j = \Phi_j(A)$ of Lemma 8.10, with either the output or any one input held fixed. Two normalizations of column sums occur. For a matrix P from link fields at spacing ϵ_0 to link fields at spacing ϵ_j, the counting column norm is $\sup_e \sum_c \|P(c,e)\|$. The volume-normalized column norm of Definition 4.6, with $a_{\mathrm{out}} = \epsilon_j$ and $a_{\mathrm{in}} = \epsilon_0$, is $(\epsilon_j/\epsilon_0)^3$ times the counting column norm. A fine link contributes to few coarse links, each time with a small weight. The counting column sums of $D\Phi_j$ are therefore of order $(\epsilon_0/\epsilon_j)^3 = M^{-3j}$, and the volume-normalized column sums are bounded. For derivatives of higher order with one input fixed, exactly one such volume factor appears. For example, with one fixed input e_1,

$$D^2(\Psi\circ\Phi)[e_1,e_2] = D^2\Psi\big[D\Phi\,e_1, D\Phi\,e_2\big] + D\Psi\,D^2\Phi[e_1,e_2];$$

in the first term, the branch that contains e_1 is summed with a column sum of $D\Phi$ and the other branch with a row sum, and only one factor $(\epsilon_0/\epsilon_j)^3$ is produced. The proof below organizes this observation by an induction over the order, using the set-partition form of the chain rule.

Lemma 8.11. *Assume the hypotheses of part (2) of Lemma 8.10, and let A range over a fixed smaller complex ball on which the bounds of Lemmas 8.9 and 8.10 hold with a fixed margin. Write $\Phi_j(A) = A_j$ and, for $j < d$, $e_j = \delta_j + \epsilon_{j+1}\|A_j\|_\infty$. The derivative of one step decomposes as*

$$L_j = D\Psi_{U_j,\epsilon_j}(A_j) = K_j + E_j, \qquad K_j = Q_{U_j}^{\mathrm{main}},$$

and the nonnegative scalar kernel of occurrences

$$k_M(c,e) = M^{-4}\#\big\{(x,s) : x \in \mathsf{B}_M(v),\ 0 \le s < M,\ e = (x + se_i, x + (s+1)e_i)\big\}$$

for $c = (v, v + Me_i)$, has row sums exactly 1 and column sums exactly M^{-3}; the sums of the norms of the color blocks of K_j are no larger. With counting column norms,

$$\begin{aligned} \|L_j\|_{\mathrm{row}} &\le 1 + C_M e_j, & \|L_j\|_{\mathrm{col}} &\le M^{-3}(1 + C_M e_j),\\ \|E_j\|_{\mathrm{row}} &\le C_M e_j, & \|E_j\|_{\mathrm{col}} &\le M^{-3} C_M e_j. \end{aligned}$$

Set $S_j = \sum_{\ell<j} e_\ell$, $P_j = L_{j-1}\cdots L_0 = D\Phi_j$, and $P_j^0 = K_{j-1}\cdots K_0$, empty products being the identity. Then

$$\begin{aligned} \|P_j\|_{\mathrm{row}} &\le e^{C_M S_j}, & \|P_j\|_{\mathrm{col}} &\le (\epsilon_0/\epsilon_j)^3 e^{C_M S_j},\\ \|P_j - P_j^0\|_{\mathrm{row}} &\le C_M S_j e^{C_M S_j}, & \|P_j - P_j^0\|_{\mathrm{col}} &\le C_M (\epsilon_0/\epsilon_j)^3 S_j e^{C_M S_j}, \end{aligned}$$

and these bounds follow from the identity

$$P_j - P_j^0 = \sum_{\ell<j} K_{j-1}\cdots K_{\ell+1} E_\ell L_{\ell-1}\cdots L_0.$$

For every fixed $a_{\mathrm{sp}} \ge 0$, the coefficients of the last iterate satisfy

$$\|D_{A_e}\Phi_d(c)\| \le C_{M,a_{\mathrm{sp}}} \epsilon_0^3 e^{-a_{\mathrm{sp}}\,\mathrm{dist}(c,e)},$$

and $D\Phi_j$ is bounded from ℓ^2 with weight ϵ_0^3 to ℓ^2 with weight ϵ_j^3, uniformly. For every fixed $s \ge 1$ and $0 \le j \le d$, with $\mathtt{t}(c, e_1, \dots, e_s)$ the length of a shortest connecting network of the locations of $c, e_1, \dots, e_s$, measured in the units in which the coarse spacing is one,

$$\begin{aligned} \sup_c \sum_{e_1,\dots,e_s} e^{a_{\mathrm{sp}}\mathtt{t}(c,e_1,\dots,e_s)} \|D_{A_{e_1}}\cdots D_{A_{e_s}}\Phi_j(c)\| &\le C_{M,s,a_{\mathrm{sp}}},\\ \sup_{e_1} \sum_{c,e_2,\dots,e_s} e^{a_{\mathrm{sp}}\mathtt{t}(c,e_1,\dots,e_s)} \|D_{A_{e_1}}\cdots D_{A_{e_s}}\Phi_j(c)\| &\le C_{M,s,a_{\mathrm{sp}}} (\epsilon_0/\epsilon_j)^3. \end{aligned}$$

In the second bound any one input may be fixed, and multiplying by $(\epsilon_j/\epsilon_0)^3$ gives a bounded volume-normalized column sum in the sense of Definition 4.6. All orientations, repeated inputs, and periodic identifications are included. The constants may depend on M, s, a_{sp}, the radii of Lemma 8.9, and the ratio of the nested balls, but not on j, d, n, L, γ, or the holonomies of the background. The bounds hold for each fixed order s; no bound uniform in s, such as a factorial bound, and no factor ϵ_0^{3s} for individual coefficients of order s, is asserted.

Proof. The background configurations $U_0, \dots, U_d$ are fixed and are not differentiated. We shrink the ball of fluctuations inside the ball of Lemma 8.10, so that the derivative bounds of Lemma 8.9 hold with a fixed margin.

Step 1: Exact incidence and the first derivative of one step. A coarse i-link has M^3 segment starts and M positively oriented links in each segment, and its scalar row sum is $M^4M^{-4} = 1$. For a fixed fine link $e = (z, z + e_i)$, the segment starts whose segments contain e are exactly $x = z - se_i$, $0 \le s < M$. Each start lies in exactly one centered block, and the vertex of that block determines exactly one coarse link in direction i. There are exactly M occurrences, and the scalar column sum is $MM^{-4} = M^{-3}$. If several occurrences share a coefficient block, combining their orthogonal transports cannot increase the sum of the norms, so the color-block sums of K_j obey the same upper bounds. The count also holds across the periodic seams: the fine side of a single averaging step is at least $4M$; the M starts are distinct, and the centered blocks partition the vertices. No reversed link occurs in K_j; reversed links and mixed orientations in the other paths are covered by the bound on overlaps in Lemma 8.9.

By the proof of Lemma 8.10, we have $\|E_j\|_{\mathrm{row}} \le C_M(\delta_j + \epsilon_{j+1}\|A_j\|_\infty) = C_M e_j$. Let J_M be the largest number of coarse output stencils that contain a given fine input link, counting all orientations and repetitions. Each coefficient norm is bounded by the row sum of its output, so $\|E_j\|_{\mathrm{col}} \le J_M C_M e_j$. Since M is fixed, we replace the constant by $M^3 J_M C_M$ and write this as $\|E_j\|_{\mathrm{col}} \le M^{-3} C_M e_j$, which keeps the factor M^{-3} of the column sum of K_j. Adding the bounds for K_j proves the four one-step estimates. This also applies for complex A_j, since orthogonality was used only for K_j, whose transports are those of the background.

Step 2: Products and the ordered telescope. For counting norms, $\|(AB)_{xz}\| \le \sum_y \|A_{xy}\|\|B_{yz}\|$ gives submultiplicativity of row sums and of column sums. As in Step 3 of the proof of Lemma 8.10, we obtain

$$\|P_j\|_{\mathrm{row}} \le \prod_{\ell<j}(1 + C_M e_\ell) \le e^{C_M S_j}, \qquad \|P_j\|_{\mathrm{col}} \le M^{-3j} e^{C_M S_j},$$

and $M^{-3j} = (\epsilon_0/\epsilon_j)^3$. For the telescope, we replace the leftmost factor L_{j-1} by $K_{j-1} + E_{j-1}$ and keep the term with E_{j-1}; then we make the same replacement in the next factor of the remaining product of factors K, and so on. After the last replacement the remaining

product is P_j^0, which gives the identity in the statement. Its summand with index ℓ has row norm at most $C_M e_\ell e^{C_M S_j}$. Its column norm contains exactly j factors M^{-3}, including the one in E_ℓ, and is at most $M^{-3j} C_M e_\ell e^{C_M S_j}$. Summing over ℓ proves the bounds for $P_j - P_j^0$; the error is proportional to S_j rather than to the number of levels. We also recall from the proof of Lemma 8.10 the uniform bound

$$S_j \le \frac{2\varepsilon_*}{M^2-1} + \frac{2a_* M}{M-1}. \tag{8.9}$$

Step 3: Pointwise and ℓ^2 bounds. Each coefficient $D_{A_e}\Phi_d(c)$ is bounded by the counting column sum containing it, hence by $C_M\epsilon_0^3$, since $\epsilon_d = 1$. It vanishes unless e is within distance C_M of c, by the locality in Lemma 8.10. Multiplying by $e^{a_{\mathrm{sp}} C_M}$ proves the pointwise exponential bound for every fixed $a_{\mathrm{sp}} \ge 0$. For a block matrix P with row bound R and column bound C, the weighted Cauchy–Schwarz inequality at each output implies that

$$\sum_c \Big| \sum_e P_{ce} u_e \Big|^2 \le R \sum_{c,e} \|P_{ce}\| |u_e|^2 \le RC \sum_e |u_e|^2.$$

For $P = P_j$ we have $RC \le C_M(\epsilon_0/\epsilon_j)^3$. Multiplying the left side by ϵ_j^3, we find that it is at most $C_M\epsilon_0^3 \sum_e |u_e|^2$, as required.

Step 4: Absolute bounds for higher derivatives of one step. For $r \ge 2$, we apply the derivative bounds of Lemma 8.9 to $\Psi_{U_j,\epsilon_j}(A) = \epsilon_{j+1}^{-1} F_{U_j}(\epsilon_j A)$. Every partial derivative of order r carries the factor $\epsilon_j^r/\epsilon_{j+1} = \epsilon_j^{r-1}/M$, which is at most ϵ_{j+1} since $\epsilon_j \le 1/M$. Summing over the at most q_M^r tuples of local inputs, we obtain, with $\Psi_j = \Psi_{U_j,\epsilon_j}$ evaluated at A_j,

$$\begin{aligned} \sup_c \sum_{b_1,\dots,b_r} \|D_{b_1}\cdots D_{b_r}\Psi_j(c)\| &\le C_{M,r}\epsilon_{j+1}, \\ \sup_{b_1} \sum_{c,b_2,\dots,b_r} \|D_{b_1}\cdots D_{b_r}\Psi_j(c)\| &\le M^{-3} C_{M,r}\epsilon_{j+1}. \end{aligned} \tag{8.10}$$

The second estimate uses the bound J_M on output stencils and absorbs $M^3 J_M$ into the constant, as in Step 1. These absolute sums use only that finitely many local variables enter, not directional Cauchy estimates. By symmetry of derivatives, any input may be the fixed one in the second line.

Step 5: Induction over the order. We first omit the spatial weights and set

$$R_s(j) = \sup_c \sum_{e_1,\dots,e_s} \|D_{e_1}\cdots D_{e_s}\Phi_j(c)\|, \qquad C_s(j) = \sup_{e_1} \sum_{c,e_2,\dots,e_s} \|D_{e_1}\cdots D_{e_s}\Phi_j(c)\|,$$

and $\widehat{C}_s(j) = (\epsilon_j/\epsilon_0)^3 C_s(j)$. Step 2 bounds R_1 and $\widehat{C}_1$. For $s \geq 2$ both vanish at $j = 0$, since $\Phi_0(A) = A$. By the set-partition form of the chain rule (Faà di Bruno's formula), the derivatives of $\Phi_{j+1} = \Psi_j \circ \Phi_j$ are

$$D^s\Phi_{j+1}[u_1, \dots, u_s] = \sum_{\pi} D^{|\pi|}\Psi_j(A_j)\Big[D^{|B|}\Phi_j(A)[u_\ell : \ell \in B] : B \in \pi\Big], \tag{8.11}$$

where π runs over the set partitions of $\{1, \dots, s\}$. The formula follows by induction on s: a further derivative either differentiates the outer tensor, creating a new block, or differentiates exactly one inner tensor, enlarging its block, and these alternatives produce each partition once.

For the sums with the output fixed, the partition with one block contributes at most $(1 + C_M e_j)R_s(j)$. Every other partition has at least two blocks, and all inner orders are smaller than s. Summing the inner coefficients at each fixed intermediate label uses $R_{|B|}(j)$, and summing the outer coefficients then uses the first line of (8.10). By induction on s we obtain, with a constant $B_{M,s}$ determined by the lower orders,

$$R_s(j+1) \leq (1 + C_M e_j)R_s(j) + B_{M,s}\epsilon_{j+1}.$$

For the sums with one input fixed, fix e_1 and, in each partition, let B_* be the block containing it. We sum the inner tensor of B_* using $C_{|B_*|}(j)$ and every other inner tensor using $R_{|B|}(j)$. With the intermediate input of B_* fixed, the remaining outer sum uses the second line of (8.10). This is the rule for contractions with a fixed input in Lemma 5.3, applied with counting volumes. The partition with one block contributes at most $M^{-3}(1 + C_M e_j)C_s(j)$, and, by the induction bounds for the lower orders, the other partitions contribute at most

$$M^{-3}B_{M,s}\epsilon_{j+1}(\epsilon_0/\epsilon_j)^3.$$

Multiplying the recurrence by $(\epsilon_{j+1}/\epsilon_0)^3 = M^3(\epsilon_j/\epsilon_0)^3$ cancels the factor M^{-3} exactly, and we find

$$\widehat{C}_s(j+1) \leq (1 + C_M e_j)\widehat{C}_s(j) + B_{M,s}\epsilon_{j+1}.$$

This is why exactly one volume factor appears.

Both recurrences start at zero. Expanding the products of their first-order factors and using (8.9), we find that their solutions are bounded by

$$B_{M,s}\sum_{\ell<j}\epsilon_{\ell+1}\exp\Big(C_M\sum_{\ell<j}e_\ell\Big) \leq B_{M,s}\frac{M}{M-1}\exp\Big[C_M\Big(\frac{2\varepsilon_*}{M^2-1} + \frac{2a_*M}{M-1}\Big)\Big].$$

There are finitely many partitions for each fixed s, so the induction closes at every fixed order and gives bounds for $R_s(j)$ and $\widehat{C}_s(j)$ that are uniform in j, d, and n. The same argument applies with any fixed input, either by selecting its block or by the symmetry of derivatives.

Step 6: Spatial weights. A nonzero derivative coefficient of $\Phi_j(c)$ has all its inputs within distance $C_M\epsilon_j$ of c, by the locality in Lemma 8.10. Joining each input to c by a shortest periodic path gives a connected network of length at most $sC_M\epsilon_j \le sC_M$, and the length of a shortest connecting network is no larger. Inserting the weight $e^{a_{\mathrm{sp}}\mathrm{t}}$ multiplies each of the sums above by at most $e^{a_{\mathrm{sp}}sC_M}$, which proves the two weighted bounds. Repeated inputs are ordinary terms of these ordered sums, and all orientations and periodic identifications were included in the one-step bounds. Multiplying the column bound by $\epsilon_j^3/\epsilon_0^3$ gives the volume-normalized column sum of Definition 4.6. This completes the proof. □

PART III

Classical background fields

SECTION 9

Constraint coordinates and local gauge fixing

Fix an odd integer $M \geq 3$, integers $n \geq 4$ and $d \geq 0$, and set $\epsilon = M^{-d}$. Let b be a configuration close to the identity on the coarse lattice of side n. In this section we describe the configurations near the identity on the fine lattice of side $M^d n$ whose image under d steps of the gauge-corrected averaging of Lemma 8.10 is b, choose a gauge compatible with this constraint, and prove that the Wilson action is uniformly strictly convex in the remaining coordinates. The statements are deterministic and finite-dimensional, and all constants are independent of d and n. The existence of the constrained minimizer (the classical background field) is proved in Section 10 (Theorem 10.6); here we show only that a critical point in the chart is unique and is a strict minimum there.

We explain the construction of the coordinates. With background connection $U_0 = I$, write $\mathsf{F}_d(A)$ for the logarithm of the output of d gauge-corrected averaging steps applied to the fine field $e^{\epsilon A}$. By Lemmas 8.1 and 8.10, $\mathsf{F}_d(A) = Q_d A + \mathscr{E}_d^{\mathrm{av}}(A)$, where Q_d is the linear average of physical potentials and $\mathscr{E}_d^{\mathrm{av}}$ is quadratically small. If H_d is a linear right inverse of Q_d, then for a fine field w and a coarse field η,

$$\mathsf{F}_d(w - H_d\eta) = Q_d w - \eta + \mathscr{E}_d^{\mathrm{av}}(w - H_d\eta).$$

The constraint $\mathsf{F}_d(w - H_d\eta) = Q_d w$ reduces to the coarse fixed-point equation $\eta = \mathscr{E}_d^{\mathrm{av}}(w - H_d\eta)$, which we solve by a contraction; the fine parameter $(I - H_d Q_d)w$ is unchanged. For the contraction to be uniform in d, the right inverse must have exponentially decaying kernels, with bounds on the potential and its first differences. Further, the counting column sums of the derivatives of $\mathscr{E}_d^{\mathrm{av}}$, which are of size ϵ^3, must cancel the counting column sums of H_d, which are of size ϵ^{-3}. We take for H_d the minimizing linear extension of Bałaban [10, 11], translated to our centered averages.

The gauge condition is $\mathsf{P}_d \operatorname{div}_\epsilon A = 0$, where P_d is the orthogonal projection onto $\mathscr{L}_\epsilon \ker Q'_d$, $\mathscr{L}_\epsilon$ is the positive scalar Laplacian, and Q'_d is the scalar block average. At $(A, \lambda) = (0, 0)$, the derivatives with respect to the gauge parameter λ of the divergence map and of the logarithmic coarse gauge map are $-\mathscr{L}_\epsilon \lambda$ and $Q'_d \lambda$, respectively. Thus $\ker Q'_d$ consists of the infinitesimal parameters preserving the coarse gauge at the identity, and their divergence variations span $\mathscr{L}_\epsilon \ker Q'_d$. The corresponding nonlinear equations, with prescribed coarse gauge, are solved in Theorem 9.9. This projected condition is the one used by Bałaban [10, (1.39)–(1.44), pp. 24–25]; it is not the ordinary condition $\operatorname{div}_\epsilon A = 0$.

After fixing the lattices and norms in Section 9.1, we transfer Bałaban's linear estimates through finite covers (Section 9.2) and construct H_d (Theorem 9.3). We then construct the constraint coordinates (Theorem 9.6), prove the constrained Hessian bound (Theorem 9.7), solve the scalar and nonlinear gauge equations (Lemma 9.8 and Theorem 9.9), and give the trial configuration (Corollary 9.10); the plaquette estimates used throughout are collected in Lemma 9.4.

The inputs are the weighted product and inverse estimates of Lemma 5.3, the inverse branch of the averaging constraint and its domain in Lemma 6.3, the identification of the coordinates of a configuration with small curvature with that branch in Lemma 7.2, the linear identities and nonlinear estimates of Lemmas 8.1, 8.10, and 8.11, and Bałaban's estimates for the linear theory [10, 11, 13]. The results are used in Section 10 to construct the background field and prove its symmetries and the nested minimization identity for the effective classical action (Theorems 10.6 and 10.10 and Lemma 10.8), in Section 11, and in the quartic expansion of Theorem 36.2.

Our choice of gauge representatives and our nonlinear solution of the averaging constraint follow the background-field program of Bałaban [14, 16], in particular his strategy of fixing the gauge and linearizing the constraint [14, Theorem 2, p. 83], [16, Theorem 1, p. 279]. The linear estimates are taken from [10, 11, 13] after the identifications checked below. The nonlinear right inverse, the coordinates on the fibers, and the gauge fixing with prescribed coarse gauge are constructed here for the centered averaging map. For comparison, the axial gauge of Dimock [70] concerns an Abelian theory and does not supply these non-Abelian statements.

9.1 Lattices, scalar products, and norms

Fix an odd integer $M \geq 3$, a depth $d \geq 0$, and a coarse side $n \geq 4$. We set $m = M^d$ and $\epsilon = M^{-d}$, and we use the fine torus $\epsilon\mathbb{Z}^3/(n\mathbb{Z})^3$ and the coarse torus $\mathbb{Z}^3/(n\mathbb{Z})^3$; lengths are measured in units in which the coarse spacing is one. The vector average $Q_d = K_m/m$ and the scalar average $Q'_d = \mathsf{A}_m$ are those of Lemma 8.1. These definitions are used for every torus side that occurs, including the covers of Section 9.2.

The coefficient of a positive fine link $e = (x, x + \epsilon e_i)$ is located at x, and that of a fine plaquette at its lower vertex. Changing these locations within their cells changes each spatial estimate by a constant independent of d and n. Scalar, link, and plaquette scalar products on the fine torus are ϵ^3 times the counting sums, each positive orientation counted once, and on the coarse torus they are counting sums. We write

$$(\nabla_{\epsilon,i} f)(x) = \frac{f(x + \epsilon e_i) - f(x)}{\epsilon}, \qquad \operatorname{div}_\epsilon A(x) = \sum_i \frac{A_i(x - \epsilon e_i) - A_i(x)}{\epsilon},$$

$$(\mathrm{d}_\epsilon A)_{ij}(x) = \nabla_{\epsilon,i} A_j(x) - \nabla_{\epsilon,j} A_i(x), \qquad i < j,$$

and $\mathscr{L}_\epsilon = \operatorname{div}_\epsilon \nabla_\epsilon = -\Delta_\epsilon \geq 0$. Then $\operatorname{div}_\epsilon = \nabla_\epsilon^\sharp$, where $^\sharp$ always denotes the adjoint with respect to the scalar products just described; adjoints between the fine and the coarse torus contain the ratio of the cell volumes. These formulas fix the sign of the scalar Laplacian and of the linearized gauge action. We use the color norm of Definition 4.1, and set

$$\|v\|_\epsilon^2 = \epsilon^3 \sum_e |v_e|^2, \qquad \|v\|_{H^1_\epsilon}^2 = \|v\|_\epsilon^2 + \sum_\mu \|\nabla_{\epsilon,\mu} v\|_\epsilon^2.$$

Let P_d be the orthogonal projection onto $\mathscr{L}_\epsilon \ker Q'_d$, and let

$$\mathcal{V}_d = \{v : Q_d v = 0,\ \mathsf{P}_d \operatorname{div}_\epsilon v = 0\}.$$

Unless stated otherwise, constants in this section depend only on M and on the following fixed choices. In an estimate on a ball of radius r inside a larger analytic ball of radius $R > r$, constants may depend on the chosen radii and on $R - r$. In a weighted composition estimate they may also depend on the input and output exponential decay rates and on their positive difference. This is the dependence described in item (2) of Definition 4.14. They do not depend on d, n, L, γ, or on the integer J that fixes the length of a renormalization group step. Additional parameters are listed where they occur.

Lemma 9.1. *Let K be a matrix from a field space on a lattice of spacing b_{in} to a field space on a lattice of spacing b_{out}, written in counting coordinates. If*

$$R = \sup_x \sum_y \|K(x,y)\|, \qquad C = \sup_y \frac{b_{\mathrm{out}}^3}{b_{\mathrm{in}}^3} \sum_x \|K(x,y)\|,$$

then the operator norm of K from ℓ^2 with weight b_{in}^3 to ℓ^2 with weight b_{out}^3 is at most $\sqrt{RC}$. The same holds for each component of a left gradient of K.

Proof. For each output, by the triangle inequality and the Cauchy–Schwarz inequality, we have

$$|(Ku)(x)|^2 \leq \Big(\sum_y \|K(x,y)\|\Big) \sum_y \|K(x,y)\| |u(y)|^2.$$

Multiplying by b_{out}^3, summing over x, and exchanging the finite sums, we find that the coefficient of $b_{\mathrm{in}}^3 |u(y)|^2$ is at most RC. The argument applies to color blocks, and a finite collection of gradient components changes only a dimensional constant. □

9.2 Transfer of periodic kernels through finite covers

Bałaban's linear estimates are proved for periodic lattices whose periods are compatible with an auxiliary system of blocks used in the proofs. A general side $n \geq 4$ need not have this property. We apply the estimates on a finite periodic cover whose side is a

suitable multiple of n, and transfer them to the torus of side n by the following lemma. The transfer uses uniqueness of the solutions and deck invariance, and it does not use any estimate for an inverse.

Lemma 9.2. *Fix an odd $M \geq 3$. Fix, as functions of M, integers R_0 and M_0, which are powers of M, and an integer K_{loc} such that $M_{\mathrm{loc}} = K_{\mathrm{loc}} R_0 M_0 \geq M_1(3, M)$, where the subdivisions and the lower bound $M_1(3, M)$ for the size of the localization cubes are those of [13, Section 3C, pp. 408–409]. Let $k = 6M_{\mathrm{loc}}$. Then the following statements hold for all integers $n \geq 4$ and $d \geq 0$.*

1. Geometry. *The periodic lattice of side kn at mesh $\epsilon = M^{-d}$, called the cover, satisfies $M_{\mathrm{loc}} \mid kn$ and $kn \geq 24M_{\mathrm{loc}}$. At every mesh $\epsilon_j = M^{j-d}$, $0 \leq j \leq d$, the whole cover is tiled by the auxiliary localization blocks of [13, Section 3C], and its localization cubes, of side at most $12M_{\mathrm{loc}}$, have injective lifts to coordinate cubes. With the domain at every scale equal to the whole torus, the separation conditions of [11, (2.1)–(2.4), p. 224] are vacuous.*
2. Projection and constrained space. *Let π^* map fields on the torus of side n to fields on the cover by periodic repetition, and let the deck group $\mathcal{D} = (\mathbb{Z}/k\mathbb{Z})^3$ act on the cover by translations by multiples of n. The averages Q_d and Q'_d, the physical differences, and their adjoints commute with π^*, and*

$$\mathsf{P}_d^{(kn)} \pi^* = \pi^* \mathsf{P}_d^{(n)}, \qquad \mathcal{V}_d^{(kn)} \cap \operatorname{Fix} \mathcal{D} = \pi^* \mathcal{V}_d^{(n)}.$$

3. Kernels. *Let T be the solution operator of one of the following linear problems at the identity connection, on the torus of side $r \in \{n, kn\}$: the scalar problem $(\mathscr{L}_\epsilon + \alpha (Q'_d)^\sharp Q'_d)\lambda = f$ with $\alpha > 0$; the scalar system with projected Laplacian and prescribed scalar average; the minimization of $\|\mathrm{d}_\epsilon A\|_\epsilon^2$ under the constraints $Q_d A = u$ and $\mathsf{P}_d \operatorname{div}_\epsilon A = 0$; or a problem $A \in \mathcal{V}_d$, $\langle \mathrm{d}_\epsilon A, \mathrm{d}_\epsilon v \rangle_\epsilon = \langle f, v \rangle_\epsilon$ for $v \in \mathcal{V}_d$, under the assumption that the curl form is positive definite on $\mathcal{V}_d$. Suppose that the problem has a unique solution on both tori. Then $T^{(kn)} \pi^* = \pi^* T^{(n)}$, and their matrices in the coordinate bases satisfy*

$$T^{(n)}(e, f) = \sum_{h \in \mathcal{D}} T^{(kn)}(\widetilde{e}, \widetilde{f} + nh),$$

where $\widetilde{e}, \widetilde{f}$ are fixed lifts; any physical source volume is already contained in these matrices. The same formula holds for the first physical differences of the output. In particular, for every $b > 0$,

$$\sum_f e^{b \operatorname{dist}_n(e,f)} \|T^{(n)}(e, f)\| \leq \sum_{f'} e^{b \operatorname{dist}_{kn}(\widetilde{e}, f')} \|T^{(kn)}(\widetilde{e}, f')\|,$$

and a bound $Ce^{-b \operatorname{dist}_{kn}}$ for the response to a source supported in one coarse block descends to the torus of side n with C replaced by $k^3 C$.

In the applications, estimates proved on the cover are transferred to the torus of side n by part (3), uniformly in d; each application checks the constraints and the uniqueness required there.

Proof. The auxiliary data are fixed before n and d. The integers R_0 and M_0 are powers of M chosen large enough for the auxiliary construction, and $M_{\mathrm{loc}} = K_{\mathrm{loc}} R_0 M_0$ is an integer multiple with $M_{\mathrm{loc}} \geq M_1(3, M)$. This is the subdivision of [13, Section 3C, pp. 408–409]: an auxiliary localization block is a union of $R_0 M_0$-blocks, and a localization cube is a union of auxiliary localization blocks. The required lower bounds depend only on M and the dimension three, and these choices can be made as functions of M; none uses n, d, a coupling, or J.

Let $n_{\mathrm{cov}} = kn = 6M_{\mathrm{loc}} n$. It is a multiple of M_{loc} and $n_{\mathrm{cov}} \geq 24 M_{\mathrm{loc}}$. At the mesh $\epsilon_j = M^{j-d}$ the number of sites in each coordinate direction is $n_{\mathrm{cov}}/\epsilon_j = knM^{d-j}$, which is divisible by M_{loc} for $0 \leq j \leq d$. The whole torus at every scale has a periodic partition into auxiliary localization blocks, and the subdivision into $R_0 M_0$-blocks is periodic as well. We take the domain sequence $\Omega_0 = \Omega_1 = \cdots = \Omega_d$ equal to the whole fine torus, with $A_j = \varnothing$ for $j < d$ and A_d the whole coarse torus. This whole-torus specialization is allowed in [11, (2.1)–(2.4), p. 224], and each complement in the separation condition (2.2) there is empty. The enlarged localization cubes of the background-field construction have side $(2 + 2r) M_{\mathrm{loc}} \epsilon_j$ with $r \leq 5$ [13, Section 3C, pp. 408–409]. This side is at most $12 M_{\mathrm{loc}} < n_{\mathrm{cov}}$, and each cube has an injective lift to a coordinate cube of the cover. The tori of [10, pp. 17–18] have general rectangular periods $2L_\mu$, and we take $L_\mu = n_{\mathrm{cov}}/2 = 3M_{\mathrm{loc}} n$; that definition does not require the period to be a power of M. The auxiliary localization blocks are used only in the proofs of the cited estimates and do not enter the definitions of the averages, the projection, or the solution operators. A deck translation need not preserve the auxiliary partition; the invariance of the operators is proved next from the averaging partitions and their equations.

For each scalar, link, or plaquette space, $\pi^* u$ is the field u repeated in the k^3 translates of a fundamental cell of side n, and the image of π^* is $\operatorname{Fix} \mathcal{D}$. Since the fine meshes agree on both tori, we have

$$\langle \pi^* u, \pi^* v \rangle_{\epsilon, kn} = k^3 \langle u, v \rangle_{\epsilon, n}, \tag{9.1}$$

and the coarse counting products satisfy the same identity. Forward differences, their adjoints, and the curl are given by the same periodic finite stencils on both tori, and they commute with π^*. The formulas for Q_d and Q'_d in Lemma 8.1 use the same centered block and segment at each coarse vertex. A deck translation moves the fine lattice by nm sites and preserves the averaging partitions at every scale. Repeating an input repeats the outputs. The adjoints for the scalar products of Section 9.1 also commute with π^*: if $Q_d(c, e)$ denotes the matrix of Q_d in the coordinate bases, then

$$(Q_d^\sharp b)(e) = \epsilon^{-3} \sum_c Q_d(c, e)^* b(c), \qquad ((Q'_d)^\sharp b)(x) = b(Y) \quad \text{for } x \text{ in the block of } Y,$$

and the occurrences and volume factors are the same on the cover. The translation between the centered and the lower-corner averages commutes with deck translations and with π^* for the same reason.

For the projection, define $S^{(r)} = \mathscr{L}_\epsilon^{(r)} \ker Q_d'^{(r)}$ for $r = n, kn$, and $\mathcal{A}_\mathcal{D} = k^{-3} \sum_{h \in \mathcal{D}} \tau_{nh}$, where τ_{nh} is the translation by nh. The deck translations are unitary, preserve $S^{(kn)}$, and commute with $\mathscr{L}_\epsilon$ and Q_d'. First, $\pi^* S^{(n)} \subset S^{(kn)} \cap \operatorname{Fix} \mathcal{D}$. Conversely, let $f = \mathscr{L}_\epsilon^{(kn)} \lambda$ with $Q_d'^{(kn)} \lambda = 0$ and $\tau_{nh} f = f$ for all h. Then

$$Q_d'^{(kn)} \mathcal{A}_\mathcal{D} \lambda = 0, \qquad \mathscr{L}_\epsilon^{(kn)} \mathcal{A}_\mathcal{D} \lambda = f.$$

The averaged preimage is invariant, hence equal to $\pi^* \lambda_0$ for some λ_0, and since π^* is injective, we have $Q_d'^{(n)} \lambda_0 = 0$. We conclude that

$$S^{(kn)} \cap \operatorname{Fix} \mathcal{D} = \pi^* S^{(n)}. \tag{9.2}$$

Since $S^{(kn)}$ and its orthogonal complement are invariant under the deck group, the projection $\mathsf{P}_d^{(kn)}$ commutes with every deck translation. For a fine scalar u, $\mathsf{P}_d^{(kn)} \pi^* u$ lies in the left side of (9.2), and $\pi^* u - \mathsf{P}_d^{(kn)} \pi^* u$ is orthogonal to $\pi^* S^{(n)}$. By (9.1), the unique vector with these two properties is $\pi^* \mathsf{P}_d^{(n)} u$. This proves

$$\mathsf{P}_d^{(kn)} \pi^* = \pi^* \mathsf{P}_d^{(n)}. \tag{9.3}$$

Applying it to the divergence and using the commutation for Q_d, we obtain the identity for the invariant part of $\mathcal{V}_d$.

We next consider one of the problems in part (3). For the scalar problem with penalty $\alpha > 0$, the weak equation is

$$\langle \nabla_\epsilon \lambda, \nabla_\epsilon \phi \rangle_\epsilon + \alpha \langle Q_d' \lambda, Q_d' \phi \rangle = \langle f, \phi \rangle_\epsilon \quad \text{for every fine scalar } \phi,$$

and all its operators commute with the deck group. For the scalar system with prescribed scalar average and projected Laplacian, the same commutation follows from (9.3). For the minimizing extension, the affine constraints are $Q_d A = u$ and $\mathsf{P}_d \operatorname{div}_\epsilon A = 0$, and the stationarity equation is $\langle \mathrm{d}_\epsilon A, \mathrm{d}_\epsilon v \rangle_\epsilon = 0$ for every $v \in \mathcal{V}_d$. For the constrained curl problem, if the curl form is positive definite on the finite-dimensional space $\mathcal{V}_d$, its Gram matrix in an orthonormal basis of $\mathcal{V}_d$ is invertible, and the problem is uniquely solvable.

On the cover with repeated data, a deck translate of a solution solves the same equation with the same data. By uniqueness, the solution is invariant and equals $\pi^* A_0$ (or $\pi^* \lambda_0$). The identities above show that the descended field satisfies the constraints and the coarse data on the torus of side n. For every test field on the torus of side n, its repetition is a test field on the cover, and by (9.1) every quadratic and source pairing is multiplied by the same factor k^3. Canceling this factor, we obtain the variational equation on the torus of side n, and uniqueness identifies the solution operator applied to the data. This proves $T^{(kn)} \pi^* = \pi^* T^{(n)}$; for the scalar system the same argument uses the

commutation in its two equations directly. An invertible coarse Schur matrix formed from commuting factors also commutes with π^*, and applying it to both sides gives the corresponding identity for its inverse. When a scalar source is naturally restricted to $\operatorname{Ran} \mathsf{P}_d$, we use the source $\mathsf{P}_d f$ for an arbitrary fine field f. This fixes its matrix in the coordinate basis on the whole coordinate space, and (9.3) preserves this convention.

Finally we write the operators in the coordinate bases. For an input u on the torus of side n, expanding $(T^{(kn)}\pi^* u)(\widetilde{e})$ first over a fundamental set of lifts of the sources and then over their k^3 deck images, we obtain

$$\sum_f \Big(\sum_{h \in \mathcal{D}} T^{(kn)}(\widetilde{e}, \widetilde{f} + nh) \Big) u(f).$$

Comparing coefficients proves the formula for $T^{(n)}$. A physical source volume is part of the matrix coefficient and is the same on both tori. Physical first differences of the output commute with repetition and with finite sums, which gives their formula. For every rate $b > 0$,

$$\begin{aligned} \sum_f e^{b \operatorname{dist}_n(e,f)} \| T^{(n)}(e,f) \| &\le \sum_{f,h} e^{b \operatorname{dist}_{kn}(\widetilde{e}, \widetilde{f}+nh)} \| T^{(kn)}(\widetilde{e}, \widetilde{f} + nh) \| \\ &= \sum_{f'} e^{b \operatorname{dist}_{kn}(\widetilde{e}, f')} \| T^{(kn)}(\widetilde{e}, f') \|, \end{aligned} \tag{9.4}$$

since the lifts of the sources partition the cover and periodic distances can only decrease on passing to the smaller torus. A weighted row bound on the cover descends with the same constant. A source in one coarse block has k^3 images on the cover, and by the triangle inequality for their responses, we obtain the bound $Ck^3 e^{-b \operatorname{dist}_n(e, \text{source block})}$. These arguments apply also to the first output differences. All these factors depend only on the auxiliary data, hence only on M. Neither uniqueness nor this transfer produces an estimate for first differences; such an estimate must be proved for the operator on the cover. □

The specialization of Bałaban's linear estimates. We use the estimates for the Gaussian theory at the identity connection collected in Table 9.1. They are applied on the cover of Lemma 9.2 in the following specialization, which we check once here and use in the proofs of Theorem 9.3 and Lemma 9.8. The geometry is periodic, with one region, equal to the whole torus at every scale, so that the separation conditions are vacuous by part (1) of Lemma 9.2. The coarse spacing is one, the fine scalar products carry the volume ϵ^3, and the coarse ones are counting sums. The background is the identity connection. The gradient of the Wilson action at the background vanishes, the regularity conditions [13, (3.35)–(3.36), p. 396] hold with zero left sides, and the smallness requirement of [13, Theorem 3.1] holds for every sufficiently small positive regularity parameter. The

averages in these papers are the lower-corner averages, which are related to ours by the translation T_d of Theorem 9.3. The auxiliary localization size M_{loc} is fixed as a function of M; it is distinct from the blocking factor M. In [11, p. 249], the bound following Corollary 2.8 is printed with the number (1.151); in the sequential numbering of that paper it is (2.151), and we cite it as "the bound printed as (1.151)".

Object here	Corresponding statements in Bałaban's papers	Relation
Q_d, Q'_d	lower-corner vector and scalar averages [10, (1.11), p. 19; (1.18), p. 20]	$Q_d = Q_d^{\mathrm{pub}}\mathsf{T}_d$
P_d	projection onto the Laplacian image of the kernel of the scalar average [10, (1.39)–(1.44), pp. 24–25]	conjugation by T_d
$\mathscr{D}_d^0$	vector operator with penalty one and its lower bound [10, Proposition 1.1, (1.89)–(1.90), p. 33]	conjugation by T_d
H_d	minimizing extension [10, (1.60)–(1.69), pp. 28–29; (1.103), p. 34]; kernel bounds [11, Proposition 2.7, (2.149); Corollary 2.8, (2.150) and the bound printed as (1.151), p. 249]	$H_d = \mathsf{T}_d^{-1} H_d^{\mathrm{pub}}$
G, T, P_d (scalar)	scalar algebra [10, (1.43)–(1.45), pp. 25–26], [11, (2.17), p. 225]; bounds [13, Theorem 3.1, (3.42), p. 397; (3.47), p. 398; Theorem 3.2, (3.48), p. 398; (3.49), p. 399]	conjugation by T_d; penalty converted

Table 9.1. Linear estimates of Bałaban used in this section, at the identity connection, and their relation to the operators defined here.

9.3 The constrained linear extension

Theorem 9.3. *Use the lattices, scalar products, and operators of Section 9.1. Let* $\mathsf{T}_d A(x) = A(x - \epsilon t_m)$, *where* $t_m = ((m-1)/2, (m-1)/2, (m-1)/2)$ *in fine lattice units, and let* Q_d^{pub} *and* $(Q'_d)^{\mathrm{pub}}$ *be the lower-corner physical averages of [10, (1.11), p. 19; (1.18), p. 20] at the identity connection. Then the following statements hold.*

1. $Q_d = Q_d^{\mathrm{pub}}\mathsf{T}_d$ *and* $Q'_d = (Q'_d)^{\mathrm{pub}}\mathsf{T}_d$.
2. *Set*

$$\mathscr{D}_d^0 = \mathrm{d}_\epsilon^\sharp \mathrm{d}_\epsilon + \nabla_\epsilon \mathsf{P}_d \,\mathrm{div}_\epsilon + Q_d^\sharp Q_d.$$

There exist $c_0, C, a_0 > 0$, *depending only on* M, *such that*

$$\mathscr{D}_d^0 \geq c_0(I + \nabla_\epsilon^\sharp \nabla_\epsilon), \qquad \|\mathrm{d}_\epsilon v\|_\epsilon^2 \geq c_0 \|v\|_{H_\epsilon^1}^2 \quad (v \in \mathcal{V}_d),$$

where $\nabla_\epsilon^\sharp \nabla_\epsilon$ *acts componentwise.*

3. *The map Q_d is onto, and the operator*

$$H_d = (\mathscr{D}_d^0)^{-1} Q_d^\sharp \big[Q_d (\mathscr{D}_d^0)^{-1} Q_d^\sharp \big]^{-1}$$

satisfies

$$H_d = \mathsf{T}_d^{-1} H_d^{\mathrm{pub}}, \qquad Q_d H_d = I, \qquad \mathsf{P}_d \operatorname{div}_\epsilon H_d = 0,$$

where H_d^{pub} is the extension of [10, (1.60)–(1.64), pp. 28–29; (1.103), p. 34]. For each coarse field u, $H_d u$ is the unique minimizer of $\|\mathrm{d}_\epsilon A\|_\epsilon^2$ among the fields A with $Q_d A = u$ and $\mathsf{P}_d \operatorname{div}_\epsilon A = 0$; in particular $\langle \mathrm{d}_\epsilon H_d u, \mathrm{d}_\epsilon v \rangle_\epsilon = 0$ for $v \in \mathcal{V}_d$. Constant fields are preserved: $H_d u = u$ for every constant coarse field u.

4. *The kernel of H_d satisfies*

$$|H_d(e,c)| + |\nabla_\epsilon H_d(e,c)| \le C e^{-a_0 \operatorname{dist}(e,c)},$$

and for every $0 < a < a_0$

$$\sup_e \sum_c e^{a \operatorname{dist}(e,c)} \big(|H_d(e,c)| + |\nabla_\epsilon H_d(e,c)| \big) \le C_a,$$
$$\sup_c \epsilon^3 \sum_e e^{a \operatorname{dist}(e,c)} \big(|H_d(e,c)| + |\nabla_\epsilon H_d(e,c)| \big) \le C_a.$$

It follows that $\|H_d u\|_\infty + \|\nabla_\epsilon H_d u\|_\infty \le C \|u\|_\infty$ and $\|H_d u\|_{H^1_\epsilon} \le C \|u\|_2$.

The constant C_a depends only on M and a. The translation T_d relates the linear averages at the identity; it does not relate nonlinear block averages.

The positivity in part (2) is a form estimate. The kernel bounds in part (4) are a separate regularity estimate, which we take from [11]; they do not follow from the form estimate.

Proof. We treat $d \ge 1$ first and verify the case $d = 0$ at the end without the cited kernel estimate.

Step 1: The averages. We write a fine point as $x = \epsilon r'$, and let $\mathsf{I}_m = \{-(m-1)/2, \dots, (m-1)/2\}^3$. By Lemma 8.1,

$$(Q_d A)_i(Y) = m^{-4} \sum_{r \in \mathsf{I}_m} \sum_{s=0}^{m-1} A_i\big(Y + \epsilon(r + s e_i)\big), \qquad (Q_d' f)(Y) = m^{-3} \sum_{r \in \mathsf{I}_m} f(Y + \epsilon r). \tag{9.5}$$

There are m^4 vector occurrences, and a constant physical potential is mapped to itself. The lower-corner formula in [10, (1.11), p. 19; (1.18), p. 20] has coefficient $m^{-4} = \epsilon^4$ and $r \in \{0, \dots, m-1\}^3$. Subtracting the integer vector t_m maps this box bijectively onto I_m, which proves part (1), including at the periodic seams.

Since fine translations are unitary for the physical scalar products and commute with the physical differences, we have

$$\mathsf{P}_d = \mathsf{T}_d^{-1}\mathsf{P}_d^{\mathrm{pub}}\mathsf{T}_d, \qquad \mathscr{D}_d^0 = \mathsf{T}_d^{-1}(\mathscr{D}_d^0)^{\mathrm{pub}}\mathsf{T}_d. \tag{9.6}$$

Indeed, the translation maps the Laplacian image of the kernel of the lower-corner scalar average onto $\mathscr{L}_\epsilon \ker Q_d'$, and orthogonality gives the first identity. Taking adjoints, we obtain $Q_d^\sharp = \mathsf{T}_d^{-1}(Q_d^{\mathrm{pub}})^\sharp$, and the second identity follows term by term.

Step 2: The cited estimates on the cover. We apply the estimates of Table 9.1 first on the cover of side kn of Lemma 9.2, in the specialization described after that lemma; the penalty in the vector operator is one. The projected gauge of [10, (1.39)–(1.44), pp. 24–25] projects onto the Laplacian image of the kernel of the scalar average. By [10, Proposition 1.1, (1.89)–(1.90), p. 33], the vector operator with penalty one is bounded below by $c_0(I + \nabla_\epsilon^\sharp \nabla_\epsilon)$. The extension of [10, (1.60)–(1.64), pp. 28–29; (1.103), p. 34] is a right inverse of the vector average, has zero projected divergence, minimizes the curl form under these two constraints, and equals

$$\big((\mathscr{D}_d^0)^{\mathrm{pub}}\big)^{-1}(Q_d^{\mathrm{pub}})^\sharp\big[Q_d^{\mathrm{pub}}\big((\mathscr{D}_d^0)^{\mathrm{pub}}\big)^{-1}(Q_d^{\mathrm{pub}})^\sharp\big]^{-1}.$$

These statements concern one real color, and we apply them to each of our three colors. No restriction to mean zero is made on coarse data; the constant fields are kept. Conjugation by T_d gives the same statements for the centered operators on the cover.

Step 3: Coercivity on the torus of side n. Before transferring any regularity estimate, we prove the finite-dimensional statements on the torus of side n. The commutation of P_d, of the differences, and of the averages and their adjoints with π^* (Lemma 9.2) implies that $\mathscr{D}_d^{0,(kn)}\pi^* = \pi^*\mathscr{D}_d^{0,(n)}$. For every fine vector field v on the torus of side n, we then have

$$\langle v, \mathscr{D}_d^{0,(n)} v\rangle_{\epsilon,n} = k^{-3}\langle \pi^* v, \mathscr{D}_d^{0,(kn)}\pi^* v\rangle_{\epsilon,kn} \geq c_0 \|v\|^2_{H^1_{\epsilon,n}}, \tag{9.7}$$

where we used (9.1) also for each first difference. This shows that $\mathscr{D}_d^{0,(n)}$ is positive definite. For $v \in \mathcal{V}_d^{(n)}$, the averaging and projected divergence terms vanish, which leaves the curl coercivity in part (2).

Step 4: Construction and identification of the extension. For a coarse field u on the torus of side n, let $A_{\mathrm{cov}} = H_d^{(kn)}\pi^* u$ be the centered cover extension. The cover minimizer is unique: the difference of two minimizers lies in $\mathcal{V}_d^{(kn)}$; subtracting their stationarity identities tested on this difference shows that its curl vanishes, and by the coercivity on the cover the difference vanishes. Deck translations preserve both constraints and the curl energy, and they fix the data. By uniqueness, A_{cov} is invariant and equals $\pi^* A_0$. By

the commutation relations, we have $Q_d^{(n)} A_0 = u$ and $\mathsf{P}_d^{(n)} \operatorname{div}_\epsilon A_0 = 0$. For every $v \in \mathcal{V}_d^{(n)}$, the repetition $\pi^* v$ lies in the constrained space on the cover, and by stationarity there we have

$$0 = \langle \mathrm{d}_\epsilon A_{\mathrm{cov}}, \mathrm{d}_\epsilon \pi^* v \rangle_{\epsilon,kn} = k^3 \langle \mathrm{d}_\epsilon A_0, \mathrm{d}_\epsilon v \rangle_{\epsilon,n}.$$

The energy at $A_0 + v$ exceeds the energy at A_0 by $\|\mathrm{d}_\epsilon v\|^2_{\epsilon,n}$, and the coercivity of Step 3 shows that A_0 is the unique constrained minimizer on the torus of side n. We set $H_d^{(n)} u = A_0$; this also shows that $Q_d^{(n)}$ is onto.

The formula for H_d also descends. For $r = n, kn$, let $G_0^{(r)} = (\mathscr{D}_d^{0,(r)})^{-1}$ and $C_0^{(r)} = Q_d^{(r)} G_0^{(r)} (Q_d^{(r)})^\sharp$. Both are positive definite, the latter because the average is onto. The commutation relations and uniqueness of these finite-dimensional inverses imply that

$$G_0^{(kn)} \pi^* = \pi^* G_0^{(n)}, \qquad (C_0^{(kn)})^{-1} \pi^* = \pi^* (C_0^{(n)})^{-1};$$

for example, the first identity follows by applying $\mathscr{D}_d^{0,(kn)}$ to both sides, and the second uses in addition the commutation of Q_d and $Q_d^\sharp$ with π^* and the positivity of C_0. The formula on the cover identifies the descended operator with $G_0^{(n)} Q_d^\sharp (C_0^{(n)})^{-1}$, and translation identifies it with $\mathsf{T}_d^{-1} H_d^{\mathrm{pub}}$ on the torus of side n. This proves all identities of part (3) without using any regularity estimate. If u is constant, the constant fine field equal to u satisfies both constraints and has zero curl, and by uniqueness $H_d u = u$.

Step 5: Kernel bounds. We apply [11, Proposition 2.7, (2.149), and Corollary 2.8, (2.150) and the bound printed as (1.151), p. 249] on the cover. In this whole-torus specialization the receiving and source regions both have coarse spacing one. The source integration factor in (2.150) and the scale factors for the potential and its first gradient in the bound printed as (1.151) are equal to one. We obtain

$$|H_d^{\mathrm{pub},(kn)}(e,c)| + |\nabla_\epsilon H_d^{\mathrm{pub},(kn)}(e,c)| \le C e^{-a_0 \operatorname{dist}_{kn}(e,c)},$$

including the constant fields. If the auxiliary penalty at the coarse scale is written as a_d in the multiscale convention of [10], it adds only the term $a_d \|u\|_2^2$ on the set of fields with fixed average u. The minimizing extension and the projected gauge subspace are therefore unchanged. Since $|\epsilon t_m| \le \sqrt{3}/2$, the centered cover kernel and its first gradient satisfy the same bound after multiplying C by at most $e^{a_0\sqrt{3}/2}$.

The minimization property, the uniqueness, and both constraints were verified in Step 4, and part (3) of Lemma 9.2 applies to the extension. For its matrix in the coordinate bases and the first physical differences of its output,

$$\begin{aligned} H_d^{(n)}(e,c) &= \sum_{h \in \mathcal{D}} H_d^{(kn)}(\widetilde{e}, \widetilde{c} + nh), \\ \nabla_\epsilon H_d^{(n)}(e,c) &= \sum_{h \in \mathcal{D}} \nabla_\epsilon H_d^{(kn)}(\widetilde{e}, \widetilde{c} + nh). \end{aligned} \tag{9.8}$$

Since every distance on the cover is at least the corresponding distance on the torus of side n, the pointwise estimate of part (4) follows with at most an additional factor k^3, which depends only on M. Weighted rows also descend without this factor, by (9.4).

To control the fine column sums, note that for a periodic mesh $b \le 1$ with period at least four and every $\eta > 0$,

$$\sup_x b^3 \sum_y e^{-\eta \operatorname{dist}(x,y)} \le C_\eta. \tag{9.9}$$

Indeed, choose representatives of the mesh points in a fundamental cube centered at x and assign to them disjoint mesh cubes. Distances vary by at most $\sqrt{3}b$ on such a cube, so the sum is bounded by $e^{\eta\sqrt{3}}$ times the integral of $e^{-\eta|z|}$ over $\mathbb{R}^3$. The same argument with $b = 1$ bounds the coarse counting sums, and the fixed number of orientations changes only a dimensional constant. Applying (9.9) with $\eta = a_0 - a$ to the pointwise estimate, we obtain the weighted rows and the weighted columns with the factor ϵ^3. The rows imply the supremum estimate, and Lemma 9.1, applied also to the gradient, gives the H^1_ϵ estimate.

Step 6: Depth zero. At $d = 0$ we have $Q_0 = H_0 = I$ and $\mathsf{P}_0 = 0$ directly. Also $\sum_\mu \|\nabla_{1,\mu} v\|_2^2 \le 12\|v\|_2^2$, so after reducing c_0 to at most $1/13$ the inequality of part (2) holds at unit mesh. □

9.4 Plaquette expansions

The following elementary estimates for the logarithm of a plaquette transport are used for the curvature and action bounds in Sections 9.6, 9.8, and 9.9. They involve only the four link variables of the plaquette and no differences of the field.

Lemma 9.4. *There exist $\rho_0, C > 0$ such that the following holds for every $\epsilon \in (0,1]$ and every fine plaquette p. For a complex link field A with $\|A\|_\infty \le \rho_0$, let $X_p(A) = \operatorname{Log}((e^{\epsilon A})_p)$ be the logarithm of the transport of the configuration $e^{\epsilon A}$ around p. For a complex link field z, write $Z_p = \sum_{e\in\partial p} |z_e|$. Then*

$$|X_p(A) - \epsilon^2 (\mathrm{d}_\epsilon A)_p| \le C\epsilon^2 \sum_{e\in\partial p} |A_e|^2, \tag{9.10}$$

$$|DX_p(A)[z] - \epsilon^2 (\mathrm{d}_\epsilon z)_p| \le C\epsilon^2 \|A\|_\infty Z_p, \tag{9.11}$$

$$|D^2 X_p(A)[z,z]| \le C\epsilon^2 Z_p^2. \tag{9.12}$$

Moreover, in the color normalization of Definition 4.1, the function $\varphi(X) = q(e^X)$ satisfies

$$D\varphi(X) = \tfrac{1}{4}X + O(|X|^3), \qquad D^2\varphi(X) = \tfrac{1}{4}I + O(|X|^2)$$

for small X, where $D\varphi$ is the gradient with respect to the color inner product, and $q(e^X) \le C|X|^2$ for real X in the identity chart.

Proof. Regard the four link variables of p as independent, with signs fixed by the orientation of the boundary, and write

$$X_p = \operatorname{Log}\bigl(e^{\epsilon A_1} e^{\epsilon A_2} e^{-\epsilon A_3} e^{-\epsilon A_4}\bigr).$$

This is a holomorphic function of four variables on a fixed product ball that does not depend on ϵ. Its Taylor term of degree one is $\epsilon(A_1 + A_2 - A_3 - A_4) = \epsilon^2 (\mathrm{d}_\epsilon A)_p$, and every monomial of degree at least two contains at least two factors ϵ. Since the series in these four variables converges absolutely on a larger ball, its tail is bounded by $C\epsilon^2 (\sum_e |A_e|)^2$, the first derivative of the tail by $C\epsilon^2 (\sum_e |A_e|) Z_p$, and the second derivative of X_p by $C\epsilon^2 Z_p^2$. There are four links, which gives (9.10), (9.11), and (9.12). The formulas for φ follow by differentiating the convergent series of $1 - \cos(|X|/2)$, which is an analytic function of $|X|^2$; at $X = 0$ they refer to its analytic quadratic extension. The last bound holds since $1 - \cos t \le t^2/2$. □

9.5 Local coordinates on constraint fibers

We use the following form of the contraction principle several times.

Lemma 9.5. *Let* B *be a closed ball in a finite-dimensional complex normed space, let* $\mathcal{O}$ *be an open subset of a finite-dimensional complex space, and let* $\Phi(t, x)$ *be holomorphic on* $\mathcal{O} \times \mathcal{U}$ *for an open set* $\mathcal{U} \supset \mathsf{B}$*. Suppose that* $\Phi(t, \cdot)$ *maps* B *into itself and is Lipschitz on* B *with constant* $\kappa < 1$*, uniformly in* $t \in \mathcal{O}$*. Then for every* $t \in \mathcal{O}$ *there is a unique fixed point* $x(t) \in \mathsf{B}$*, and* $t \mapsto x(t)$ *is holomorphic. If, in addition,* B *is invariant under complex conjugation and* $\Phi(\bar t, \bar x) = \overline{\Phi(t, x)}$ *in the real coordinates, then* $x(t)$ *is real for real* t*.*

Proof. We start the iteration $x_{k+1}(t) = \Phi(t, x_k(t))$ at a point $x_0 \in \mathsf{B}$ that does not depend on t. Then $\|x_{k+1} - x_k\| \le \kappa^k \|x_1 - x_0\|$, and the geometric sum shows that the iterates converge uniformly on $\mathcal{O}$ to a fixed point, which is unique in B since $\kappa < 1$. Since the iterates are holomorphic and Cauchy's formula on coordinate polydiscs passes to their uniform limit, the fixed point is holomorphic. At a real parameter, the complex conjugate of the fixed point is another fixed point in B when B is invariant under conjugation and $\Phi(\bar t, \bar x) = \overline{\Phi(t, x)}$. By uniqueness, the fixed point is real. No constant in this argument depends on the dimension of the space. □

In the following theorem, w denotes a fine link field; it is unrelated to the normalized fluctuation variables of Section 12. For the sums of absolute derivative coefficients we use the conventions of Definition 4.6. For each fixed order, the row is the sum over all inputs with the output fixed, and the column is the sum over the output and the other inputs with one input fixed. Both are weighted by $e^{a\mathsf{t}}$, where t is the length of a shortest connecting network of the locations, measured in units in which the coarse spacing is

one. A counting column sums with counting measure over fine outputs; a column with fine outputs summed with the weight ϵ^3 is volume-normalized.

Theorem 9.6. *Use the geometry and operators of Theorem 9.3, and let $V_d(A)$ be the gauge-corrected iteration of Lemma 8.10 with background connection $U_0 = I$, applied to the fine field $e^{\epsilon A}$. Write*

$$\mathsf{F}_d(A) = \operatorname{Log} V_d(A) = Q_d A + \mathscr{E}_d^{\mathrm{av}}(A), \qquad \mathscr{E}_d^{\mathrm{av}}(0) = D\mathscr{E}_d^{\mathrm{av}}(0) = 0.$$

There exist $r_*, \theta_*, C > 0$, independent of d and n, such that the following statements hold.

1. The remainder. *F_d is holomorphic on $\|A\|_\infty < r_*$, and*

$$\|\mathscr{E}_d^{\mathrm{av}}(A)\|_\infty \le C\|A\|_\infty^2, \qquad \|D\mathscr{E}_d^{\mathrm{av}}(A)\|_{\mathrm{row}} \le C\|A\|_\infty,$$
$$\|D\mathscr{E}_d^{\mathrm{av}}(A)\|_{\text{counting col}} \le C\epsilon^3\|A\|_\infty.$$

   These bounds, and the first-order bounds below, also hold with one fixed positive exponential weight in the distance, on fixed smaller domains.

2. An analytic right inverse. *The radii can be chosen so that $2\|H_d\|_{\infty\to\infty}\theta_* < r_*$ and so that, for every complex coarse field θ with $\|\theta\|_\infty < \theta_*$, the equation*

$$q_d(\theta) = \theta - \mathscr{E}_d^{\mathrm{av}}\big(H_d q_d(\theta)\big)$$

   has a unique solution in the coarse ball of radius $2\theta_*$, and $\|q_d(\theta)\|_\infty \le 2\|\theta\|_\infty$. The map $A_{\mathrm{sec},d}(\theta) = H_d q_d(\theta)$ is holomorphic, real for real θ, and satisfies

$$\mathsf{F}_d(A_{\mathrm{sec},d}) = \theta, \qquad \|A_{\mathrm{sec},d}\|_\infty + \|\nabla_\epsilon A_{\mathrm{sec},d}\|_\infty \le C\|\theta\|_\infty, \qquad \|q_d(\theta)\|_2 \le C\|\theta\|_2.$$

   For every fixed $s \ge 1$, the sums of the absolute derivative coefficients of order s of $A_{\mathrm{sec},d}$, and of $\nabla_\epsilon A_{\mathrm{sec},d}$, with the output fixed or with any one coarse input fixed, are bounded by C_s with the weight $e^{a_s \mathsf{t}}$ for some $a_s > 0$; in the sums with a coarse input fixed, the fine outputs are summed with the weight ϵ^3. The constants C_s and a_s depend only on s and on the radii and decay rates, as in the convention of Section 9.1.

3. Coordinates reducing the constraint to its linear part. *There exist $r_{\mathrm{st}}, r_D > 0$ with $r_{\mathrm{st}} < r_*/4$ such that, for every fine field w with $\|w\|_\infty < r_{\mathrm{st}}$, there is a unique coarse correction $\mathscr{D}_d(w)$ with $\|\mathscr{D}_d(w)\|_\infty < r_D$ satisfying*

$$\mathscr{D}_d(w) = \mathscr{E}_d^{\mathrm{av}}\big(w - H_d\mathscr{D}_d(w)\big).$$

   It is holomorphic in w, and $\Xi_d(w) = w - H_d\mathscr{D}_d(w)$ satisfies

$$\mathsf{F}_d(\Xi_d(w)) = Q_d w.$$

   Further, $\|\mathscr{D}_d(w)\|_\infty \le C\|w\|_\infty^2$, the row norm of $D\mathscr{D}_d(w)$ is at most $C\|w\|_\infty$, and its counting column norm is at most $C\epsilon^3\|w\|_\infty$. For every fixed $s \ge 2$, the derivatives of order

s of $\mathscr{D}_d$ have row sums at most C_s and counting column sums, with any one input fixed, at most $C_s\epsilon^3$, with a weight $e^{a_s \mathsf{t}}$, $a_s > 0$. Even if w has no control of its differences,

$$\|\Xi_d(w) - w\|_\infty + \|\nabla_\epsilon(\Xi_d(w) - w)\|_\infty \le C\|w\|_\infty^2.$$

4. Coordinates on the fibers. *There exists $0 < r_{\mathrm{fib}} < r_{\mathrm{st}}$, independent of d and n, such that every fine field A with $\|A\|_\infty < r_{\mathrm{fib}}$ and $\mathsf{F}_d(A) = \theta$, $\|\theta\|_\infty < \theta_*$, can be written uniquely as*

$$A = \Xi_d(a + H_d\theta), \qquad a \in \ker Q_d,$$

namely with $a = (I - H_dQ_d)A$, and then $\|a + H_d\theta\|_\infty < r_{\mathrm{st}}$. Conversely, $\mathsf{F}_d(\Xi_d(a + H_d\theta)) = \theta$ for every $a \in \ker Q_d$ with $\|a + H_d\theta\|_\infty < r_{\mathrm{st}}$. Moreover $\mathsf{P}_d \operatorname{div}_\epsilon \Xi_d(a + H_d\theta) = \mathsf{P}_d \operatorname{div}_\epsilon a$, so restricting a to $\mathcal{V}_d$ gives exactly the fields of this form that satisfy $\mathsf{P}_d \operatorname{div}_\epsilon A = 0$.

The configuration with prescribed image under T_M^d obtained from $A_{\mathrm{sec},d}$, and its action, are described in Corollary 9.10.

The following list summarizes the roles of the radii in Theorem 9.6; it does not replace the inequalities in the statement.

Radius	Domain or role
r_*	Fine input ball on which F_d is holomorphic and the remainder bounds of part (1) hold.
θ_*, $2\theta_*$	Coarse target ball, and coarse ball in which $q_d(\theta)$ is the unique fixed point.
r_{st}, r_D	Fine parameter ball on which Ξ_d is defined, and coarse ball in which $\mathscr{D}_d(w)$ is the unique correction.
r_{fib}	Fine ball of solutions of $\mathsf{F}_d(A) = \theta$ covered by the coordinates of part (4); their parameters lie in the ball of radius r_{st}, but the image of that whole parameter ball is not claimed to lie in the ball of radius r_{fib}.

Proof. We first extract the quadratic remainder of F_d from Lemma 8.11, then solve the two coarse fixed-point equations by Lemma 9.5, and finally prove the bounds for derivatives of higher order by induction over the order.

Step 1: The remainder. We apply Lemma 8.10 with background connection $U_0 = I$, and denote the intermediate potentials by A_j. Since $\epsilon_d = 1$, the last potential is $\mathsf{F}_d(A)$. At the identity every background curvature vanishes, and the transported segment averages K_j are the averages K_M/M of physical potentials, whose product is Q_d by Lemma 8.1. Also

$$\sum_{j<d} \epsilon_{j+1}\|A_j\|_\infty \le 2\|A\|_\infty \frac{M}{M-1}.$$

Using the telescope of Lemma 8.11 with $j = d$, we obtain

$$\|D\mathsf{F}_d(A) - Q_d\|_{\mathrm{row}} \le C\|A\|_\infty, \qquad \|D\mathsf{F}_d(A) - Q_d\|_{\text{counting col}} \le C\epsilon^3\|A\|_\infty. \tag{9.13}$$

All terms have a dependence radius at most C_M, and a fixed exponential weight multiplies their coefficients by at most a fixed factor. We shrink the complex ball once and call its radius r_*. Integrating $D\mathsf{F}_d(tA) - Q_d$ against A for $0 \le t \le 1$ gives the quadratic bound for $\mathscr{E}_d^{\mathrm{av}} = \mathsf{F}_d - Q_d$ and its two vanishing Taylor coefficients. By Lemma 8.11, every derivative of fixed order of this remainder has a bounded row sum and a counting column sum with a fixed input at most $C_s\epsilon^3$; subtracting the linear part affects only the first order.

Step 2: The coarse remainder. Fix a spatial rate smaller than the rate of the kernel of H_d. Let B_H be a common upper bound for $\|H_d\|_{\infty\to\infty}$ and for the weighted row and volume-normalized column norms of H_d given by Theorem 9.3, and set $r(q) = \mathscr{E}_d^{\mathrm{av}}(H_d q)$. Composing, with a further fixed reduction of the rate if necessary, we obtain

$$\|r(q)\|_\infty \le C_r\|q\|_\infty^2, \qquad \|Dr(q)\|_{\mathrm{row}} + \|Dr(q)\|_{\mathrm{col}} \le C_r\|q\|_\infty, \tag{9.14}$$

where here and below the norms in derivative estimates may carry the spatial weight. The column composition is explicit: with $E = D\mathscr{E}_d^{\mathrm{av}}(H_d q)$,

$$\sum_c \|(EH_d)(c,b)\| \le \sum_e \Big(\sum_c \|E(c,e)\|\Big)\|H_d(e,b)\| \le C\epsilon^3\|q\|_\infty \sum_e \|H_d(e,b)\| \le C\|q\|_\infty.$$

The row composition uses the two row bounds. In the weighted version, $\mathrm{dist}(c,b) \le \mathrm{dist}(c,e) + \mathrm{dist}(e,b)$, and the same calculation applies. No power of the fine volume remains.

Step 3: The analytic right inverse. We enlarge C_r so that it controls both weighted norms, and choose θ_* so small that

$$2B_H\theta_* < r_*, \qquad 2C_r\theta_* \le \tfrac{1}{4}.$$

The map $q \mapsto \theta - r(q)$ sends the closed ball $\|q\|_\infty \le 2\theta_*$ into itself, since $\|r(q)\|_\infty \le 4C_r\theta_*^2 \le \theta_*/2$, and it is a contraction with constant at most $1/4$. By Lemma 9.5, there exists a unique fixed point, which is holomorphic and real for real θ. Since $r(0) = 0$, the derivative bound implies that $\|q\|_\infty \le \|\theta\|_\infty + \|q\|_\infty/4$, which gives $\|q\|_\infty \le 2\|\theta\|_\infty$.

In both the weighted row norm and the weighted column norm on the coarse lattice,

$$U^{-1} = (I + Dr(q))^{-1} = \sum_{k\ge 0}(-Dr(q))^k, \qquad \|U^{-1}\| \le \tfrac{4}{3},$$

by submultiplicativity (Lemma 5.3). Differentiating the fixed-point equation, we find $D_\theta q = U^{-1}$. Lemma 9.1, applied with counting measure on both coarse lattices, bounds its ℓ^2 operator norm by $4/3$. Integrating along $t\theta$ with $q(0) = 0$, we obtain $\|q(\theta)\|_2 \le C\|\theta\|_2$. The identity $\mathsf{F}_d(A_{\mathrm{sec},d}) = Q_d H_d q + r(q) = \theta$ follows from $Q_d H_d = I$ and the fixed-point equation, and the bounds for $A_{\mathrm{sec},d}$ and its first differences follow from the row bounds of the extension.

Step 4: Derivatives of higher order of the right inverse. For each fixed order s, we compose a remainder tensor of order $p \le s$ with p copies of H_d. With the output fixed, all copies use rows. With a coarse input fixed, the copy on the branch of that input has counting column sum at most $C\epsilon^{-3}$, which is canceled by the counting column sum $C_p\epsilon^3$ of the remainder tensor with a fine input fixed. All other branches use rows. The union of the connecting networks of the components connects all external locations, and the weight of a shortest connecting network is bounded by the product of the weights of the components, as in Lemma 5.3. We obtain finite bounds for $D^p r$, for $1 \le p \le s$, with any index fixed, at a common positive rate. For $s \ge 2$, differentiating the equation for q, we obtain

$$U\, D^s q[h_1, \dots, h_s] = - \sum_{\substack{\pi \text{ a partition of } \{1,\dots,s\} \\ |\pi| \ge 2}} D^{|\pi|} r(q) \Big[D^{|B|} q[h_i : i \in B] : B \in \pi \Big], \tag{9.15}$$

and every block in the sum has size smaller than s. Starting from the bound for Dq, multiplication by U^{-1} and induction on s give the bounds with the output or any coarse input fixed. The number of partitions and the constants up to order s depend only on s and on the radii and decay rates fixed above. A final composition with H_d, or with $\nabla_\epsilon H_d$, proves the bounds for the section; with a coarse input fixed and fine outputs, this last column uses $\epsilon^3 \sum_e$.

Step 5: The coordinates Ξ_d. We now construct Ξ_d without any assumption on the differences of w. Let C_E dominate the quadratic and derivative constants of Step 1 on a smaller ball. We choose $r_{\mathrm{st}} < r_*/4$ and set $r_D = 8C_E r_{\mathrm{st}}^2$, decreasing r_{st} until

$$B_H r_D \le r_{\mathrm{st}}, \qquad 2C_E B_H r_{\mathrm{st}} \le \tfrac{1}{4}.$$

For $\|w\|_\infty < r_{\mathrm{st}}$ and $\|\mathscr{D}\|_\infty \le r_D$ we have $\|w - H_d\mathscr{D}\|_\infty \le 2r_{\mathrm{st}}$. Therefore the map $\mathscr{D} \mapsto \mathscr{E}_d^{\mathrm{av}}(w - H_d\mathscr{D})$ sends the closed ball of radius r_D into the concentric ball of radius $r_D/2$ and is a contraction with constant at most $1/4$. Lemma 9.5 constructs $\mathscr{D}_d(w)$, unique in that ball. Comparing its equation with the value at $\mathscr{D} = 0$, we find

$$\|\mathscr{D}_d(w)\|_\infty \le C_E \|w\|_\infty^2 + \tfrac{1}{4}\|\mathscr{D}_d(w)\|_\infty, \tag{9.16}$$

and absorbing the last term gives the quadratic bound. Using $Q_d H_d = I$ in the defining equation, we obtain

$$\mathsf{F}_d\big(w - H_d\mathscr{D}_d(w)\big) = Q_d w - \mathscr{D}_d(w) + \mathscr{E}_d^{\mathrm{av}}\big(w - H_d\mathscr{D}_d(w)\big) = Q_d w.$$

For the derivative bounds, write $X = \Xi_d(w)$, $E_1 = D\mathscr{E}_d^{\mathrm{av}}(X)$, and $W = I + E_1 H_d$. The coarse weighted row and column norms of $E_1 H_d$ are $O(\|w\|_\infty)$, and a Neumann series bounds W^{-1} uniformly. Differentiating, we obtain

$$D\mathscr{D}_d = W^{-1} E_1, \qquad D\Xi_d = I - H_d D\mathscr{D}_d. \tag{9.17}$$

The right factor E_1 has counting column sum at most $C\epsilon^3\|w\|_\infty$, which the first formula keeps, and its row sum is at most $C\|w\|_\infty$. The second formula has bounded rows and columns from fine fields to fine fields: in the column composition, the factor ϵ^{-3} from H_d cancels against the factor ϵ^3 from $D\mathscr{D}_d$. For $s \geq 2$,

$$W\, D^s \mathscr{D}_d[h_1, \dots, h_s] = \sum_{\substack{\pi \text{ a partition of } \{1,\dots,s\} \\ |\pi| \geq 2}} D^{|\pi|} \mathscr{E}_d^{\mathrm{av}}(X) \Big[D^{|B|} \Xi_d[h_i : i \in B] : B \in \pi \Big], \quad (9.18)$$

where $D^t \Xi_d = -H_d D^t \mathscr{D}_d$ for $t \geq 2$. By induction, these tensors from fine fields to fine fields have bounded rows and columns. With an external fine input fixed in a term of (9.18), we follow its branch: the tensor $D\Xi_d$ or $D^t\Xi_d$ on that branch uses its column, and the remainder tensor then uses its column with a fine input fixed, which supplies one factor ϵ^3. All other branches use rows. The coarse inverse W^{-1} preserves this factor. By the symmetry of derivatives any input may be the fixed one. This proves, by induction, the bounds C_s for rows and $C_s\epsilon^3$ for counting columns, with positive rates.

Step 6: Coordinates on the fibers. Let $r_{\mathrm{fib}} > 0$ be so small that

$$r_{\mathrm{fib}} < r_{\mathrm{st}}/2, \qquad r_{\mathrm{fib}} + B_H C_E r_{\mathrm{fib}}^2 < r_{\mathrm{st}}, \qquad C_E r_{\mathrm{fib}}^2 < r_D.$$

If $\mathsf{F}_d(A) = \theta$, set $a = (I - H_d Q_d)A$. Then $Q_d a = 0$ and

$$w = a + H_d\theta = A + H_d \mathscr{E}_d^{\mathrm{av}}(A).$$

By these inequalities, w lies in the ball of radius r_{st} and $\mathscr{E}_d^{\mathrm{av}}(A)$ in the ball of radius r_D in which the correction is unique. Since $w - H_d\mathscr{E}_d^{\mathrm{av}}(A) = A$, the field $\mathscr{E}_d^{\mathrm{av}}(A)$ solves the correction equation at w, and uniqueness implies that $\mathscr{D}_d(w) = \mathscr{E}_d^{\mathrm{av}}(A)$ and $\Xi_d(w) = A$. Conversely, $Q_d(a + H_d\theta) = \theta$, and the identity of part (3) yields the constraint for every such parameter. Applying $I - H_d Q_d$ to $\Xi_d(a + H_d\theta) = a + H_d\theta - H_d\mathscr{D}_d(a + H_d\theta)$ recovers a, which proves uniqueness of the parameter. Since $\mathsf{P}_d \operatorname{div}_\epsilon H_d = 0$,

$$\mathsf{P}_d \operatorname{div}_\epsilon \Xi_d(a + H_d\theta) = \mathsf{P}_d \operatorname{div}_\epsilon a,$$

which proves the statement about $\mathcal{V}_d$. Finally $\Xi_d(w) - w = -H_d\mathscr{D}_d(w)$. Using the row bounds of H_d and $\nabla_\epsilon H_d$ and (9.16), we obtain the bound for the correction even when w has no controlled differences. □

9.6 The constrained Wilson Hessian

We now show that the Wilson action, written in the coordinates $a \in \mathcal{V}_d$ of Theorem 9.6, has a Hessian comparable to the curl form. The domain is controlled by the supremum norms of the field and of its curl only, so that fields without control of all first differences

are allowed. The Hessian of the composition contains, besides the Hessian of the action in the direction of the tangent map, the first variation of the action applied to the second derivative of the coordinate map; this second term must be kept.

Theorem 9.7. *In the setting of Theorem 9.6, let*

$$\mathscr{S}_\epsilon(A) = 4\epsilon^{-1}\sum_p q\big((e^{\epsilon A})_p\big), \qquad \mathcal{S}_{d,\theta}(a) = \mathscr{S}_\epsilon\big(\Xi_d(a + H_d\theta)\big).$$

There exist $r_0, C_*, C > 0$ such that the following holds with the constant c_0 of Theorem 9.3. Let $0 < r_{\mathrm{ph}} \le r_0$, let θ be a real coarse field with $\|\theta\|_\infty < r_{\mathrm{ph}}/C_*$, and let $a \in \mathcal{V}_d$ be real with

$$\|a\|_\infty + \|\mathrm{d}_\epsilon a\|_\infty < r_{\mathrm{ph}}.$$

1. *The field $A = \Xi_d(a + H_d\theta)$ satisfies $\|A\|_\infty + \|\mathrm{d}_\epsilon A\|_\infty \le C r_{\mathrm{ph}}$.*
2. *For every $v \in \mathcal{V}_d$,*
$$\frac{c_0}{2}\|v\|^2_{H^1_\epsilon} \le D^2_a\mathcal{S}_{d,\theta}(a)[v,v] \le C\|v\|^2_{H^1_\epsilon}.$$
3. *With $w = a + H_d\theta$ and $T = D\Xi_d(w) = I - H_d D\mathscr{D}_d(w)$, the Hessian is*
$$D^2_a\mathcal{S}_{d,\theta}(a)[v,v] = D^2\mathscr{S}_\epsilon(A)[Tv,Tv] - D\mathscr{S}_\epsilon(A)\big[H_d D^2\mathscr{D}_d(w)[v,v]\big].$$
4. *The following bounds hold, where e_c denotes a unit coarse link-color vector:*
$$\big|D^2\mathscr{S}_\epsilon(A)[z,z] - \|\mathrm{d}_\epsilon z\|^2_\epsilon\big| \le C r_{\mathrm{ph}}\|z\|^2_{H^1_\epsilon},$$
$$\sup_c |D\mathscr{S}_\epsilon(A)[H_d e_c]| \le C r_{\mathrm{ph}}, \qquad \sum_c |D^2\mathscr{D}_{d,c}(w)[v,v]| \le C\|v\|^2_\epsilon.$$

The same conclusions hold, after a fixed adjustment of the constants, when the hypothesis on $\|\mathrm{d}_\epsilon a\|_\infty$ is replaced by the same hypothesis on $\|\nabla_\epsilon a\|_\infty$. The domain of a described above is convex. If it contains a critical point $a_*(\theta)$ of $\mathcal{S}_{d,\theta}$ on $\mathcal{V}_d$, then a_* is the only critical point in the domain, and for every a in the domain

$$\mathcal{S}_{d,\theta}(a) \ge \mathcal{S}_{d,\theta}(a_*) + \frac{c_0}{4}\|a - a_*\|^2_{H^1_\epsilon}.$$

For the Wilson action at coarse coupling g_n, every energy and Hessian in these statements is multiplied by g_n^{-2}. The constants do not depend on r_{ph}, d, n, L, γ, g_n, or J.

The bound on $D\mathscr{S}_\epsilon(A)[H_d e_c]$ concerns the first variation tested against the extensions of unit coarse fields; no bound on the first variation at individual fine links is used.

Proof. We first bound the Hessian and the tested first variation of $\mathscr{S}_\epsilon$ near the identity, then control the two terms of the chain rule separately, and finally deduce the convexity statements.

Step 1: The Hessian of the Wilson action. Suppose that $\|A\|_\infty + \|\mathrm{d}_\epsilon A\|_\infty \le \rho \le \rho_0$ with a fixed small ρ_0, and use the notation of Lemma 9.4. By (9.10) and the triangle inequality, $|X_p(A)| \le C\epsilon^2\rho$. Let $b_p = (\mathrm{d}_\epsilon z)_p$ and $E_p = DX_p(A)[z] - \epsilon^2 b_p$. The second derivative of the term of p in $\mathscr{S}_\epsilon$ is

$$\frac{4}{\epsilon}\Big(D^2\varphi(X_p)\big[\epsilon^2 b_p + E_p, \epsilon^2 b_p + E_p\big] + D\varphi(X_p)\big[D^2X_p(A)[z,z]\big]\Big).$$

Since $D^2\varphi = I/4 + O(|X|^2)$, its leading term is exactly $\epsilon^3|b_p|^2$. By (9.11), the cross term and the square of E_p are bounded by $C\epsilon^3\rho(|b_p|^2 + Z_p^2)$, since $\rho \le \rho_0$. The error in $D^2\varphi$ is bounded by the same expression, since $\epsilon \le 1$ and $|X_p| \le C\epsilon^2\rho$. By (9.12) and $|D\varphi(X_p)| \le C|X_p|$, the last term, the first variation applied to the second variation of X_p, is bounded by $C\epsilon^3\rho Z_p^2$. Summing over plaquettes, using the fixed overlap of links and plaquettes, $\|\mathrm{d}_\epsilon z\|_\epsilon \le C\|\nabla_\epsilon z\|_\epsilon$, and $\epsilon^3\sum_p Z_p^2 \le C\|z\|_\epsilon^2$, we obtain

$$\big|D^2\mathscr{S}_\epsilon(A)[z,z] - \|\mathrm{d}_\epsilon z\|_\epsilon^2\big| \le C\rho\|z\|_{H^1_\epsilon}^2. \tag{9.19}$$

In particular $D^2\mathscr{S}_\epsilon(0)[z,z] = \|\mathrm{d}_\epsilon z\|_\epsilon^2$.

Step 2: The first variation tested against the extension. Let $z = H_d e_c$. Using the same first-variation formula, $|D\varphi(X_p)| \le C\epsilon^2\rho$, and (9.11), we obtain

$$|D\mathscr{S}_\epsilon(A)[H_d e_c]| \le C\rho\,\epsilon^3 \sum_p \Big(|(\mathrm{d}_\epsilon H_d e_c)_p| + \rho \sum_{e\in\partial p} |H_d(e,c)|\Big) \le C\rho.$$

In the last step we used the column bounds with the factor ϵ^3 for the kernels of H_d and $\nabla_\epsilon H_d$ in Theorem 9.3, and the fixed overlap of plaquettes. The bound is uniform in the coarse link c.

Step 3: The size of the chart. Let B_0 and B_1 bound the rows of the kernels of H_d and $\nabla_\epsilon H_d$. For $w = a + H_d\theta$, the hypotheses imply that

$$\|w\|_\infty \le r_{\mathrm{ph}} + B_0 r_{\mathrm{ph}}/C_*, \qquad \|\mathrm{d}_\epsilon w\|_\infty \le r_{\mathrm{ph}} + CB_1 r_{\mathrm{ph}}/C_*.$$

We take $C_* > 4(1 + B_0 + CB_1)$ and r_0 so that these fields lie in the ball of radius r_{st} of Theorem 9.6 and their images lie in the potential ball of Step 1. By the bound for $\Xi_d(w) - w$ in that theorem, we have

$$\|A\|_\infty + \|\mathrm{d}_\epsilon A\|_\infty \le C_1 r_{\mathrm{ph}} = \rho, \tag{9.20}$$

with constants fixed before r_{ph}. If the hypothesis is instead the bound on $\|\nabla_\epsilon a\|_\infty$, the inequality $\|\mathrm{d}_\epsilon a\|_\infty \le C\|\nabla_\epsilon a\|_\infty$ gives the same estimate after enlarging C_1 by this dimensional factor. We choose r_0 below with the enlarged constant; the Hessian bounds then hold under both hypotheses.

Step 4: The term with the tangent map. Let $D_1 = D\mathscr{D}_d(w)$ and $T = I - H_d D_1$. By Theorem 9.6, the row sum of D_1 is at most Cr_{ph} and its counting column sum is at most $C\epsilon^3 r_{\mathrm{ph}}$. By Lemma 9.1 and the H^1_ϵ bound for the extension, we have

$$\|D_1 v\|_2 \le Cr_{\mathrm{ph}}\|v\|_\epsilon, \qquad \|Tv - v\|_{H^1_\epsilon} \le Cr_{\mathrm{ph}}\|v\|_\epsilon. \tag{9.21}$$

Applying (9.19) to Tv and expanding the difference of the two curl squares, we obtain

$$\left|D^2\mathscr{S}_\epsilon(A)[Tv, Tv] - \|\mathrm{d}_\epsilon v\|^2_\epsilon\right| \le C_2 r_{\mathrm{ph}}\|v\|^2_{H^1_\epsilon}. \tag{9.22}$$

Step 5: The term with the second derivative of the coordinates. This term requires an ℓ^1 bound on a tensor. Let $B_{c;e,f}$ denote the color block of the second derivative of $\mathscr{D}_{d,c}$ at w with respect to the fine inputs e, f. By part (3) of Theorem 9.6 with $s = 2$,

$$\sup_e \sum_{c,f} \|B_{c;e,f}\| \le C\epsilon^3.$$

Since the tensor is symmetric in its two fine inputs, we have

$$\begin{aligned}\sum_c |D^2\mathscr{D}_{d,c}(w)[v,v]| &\le \sum_{c,e,f} \|B_{c;e,f}\||v_e||v_f| \le \frac{1}{2}\sum_{c,e,f}\|B_{c;e,f}\|\left(|v_e|^2 + |v_f|^2\right)\\ &\le C\epsilon^3 \sum_e |v_e|^2 = C\|v\|^2_\epsilon.\end{aligned} \tag{9.23}$$

Combining this with Step 2, we obtain

$$\left|D\mathscr{S}_\epsilon(A)\left[H_d D^2\mathscr{D}_d(w)[v,v]\right]\right| \le C_3 r_{\mathrm{ph}}\|v\|^2_\epsilon. \tag{9.24}$$

No norm of the first variation summed over the coarse lattice enters this pairing.

Step 6: The chain rule and the conclusion. With θ fixed, $D_a A = T$ and $D^2_a A[v,v] = -H_d D^2\mathscr{D}_d(w)[v,v]$, and the second-order chain rule gives the formula of part (3). By (9.22) and (9.24), the Hessian differs from $\|\mathrm{d}_\epsilon v\|^2_\epsilon$ by at most $C_4 r_{\mathrm{ph}}\|v\|^2_{H^1_\epsilon}$. We shrink r_0 so that $C_4 r_0 \le c_0/2$. For $v \in \mathcal{V}_d$, the lower bound of part (2) follows from the coercivity of the curl in Theorem 9.3, and the componentwise bound $\|\mathrm{d}_\epsilon v\|_\epsilon \le C\|\nabla_\epsilon v\|_\epsilon$ implies the upper bound. This also proves the bounds of part (4), with $\rho = C_1 r_{\mathrm{ph}}$ absorbed into the constants.

The domain of a is the intersection of the linear space $\mathcal{V}_d$ with the open ball for the sum of a norm and a seminorm, which is convex. If a_* is a critical point in the domain and $h = a - a_*$, then the derivative at a_* vanishes on $h \in \mathcal{V}_d$, and by Taylor's formula along the segment we have

$$\mathcal{S}_{d,\theta}(a) - \mathcal{S}_{d,\theta}(a_*) = \int_0^1 (1-t) D^2_a \mathcal{S}_{d,\theta}(a_* + th)[h,h]\, dt \ge \frac{c_0}{4}\|h\|^2_{H^1_\epsilon}.$$

If there were two critical points, integrating the Hessian along the segment between them and pairing the difference of their derivatives with their difference would give a positive number and also zero, so they coincide. Finally, since the fine coupling satisfies $g_0^2 = g_n^2 \epsilon$, the Wilson action at coarse coupling g_n equals $g_n^{-2} \mathscr{S}_\epsilon$, and energies and derivatives are multiplied by g_n^{-2}. □

9.7 A scalar constrained inverse including constant modes

The linearization of the gauge-fixing problem of Section 9.8 is the scalar system $(\mathsf{P}_d \mathscr{L}_\epsilon \lambda, Q_d' \lambda) = (f, b)$. The following lemma inverts it, with estimates for λ, its first differences, and its Laplacian. Bounds on all second differences of λ are not needed and are not proved; the Laplacian is controlled through exact identities.

Lemma 9.8. *In the geometry of Theorem 9.3, let $Q = Q_d'$ and $\mathscr{L} = \mathscr{L}_\epsilon$, and define, with the adjoints of Section 9.1,*

$$G = (\mathscr{L} + Q^\sharp Q)^{-1}, \qquad T = QG^2Q^\sharp, \qquad \mathsf{P}_d = I - GQ^\sharp T^{-1} QG,$$
$$\mathscr{B} = G - G^2 Q^\sharp T^{-1} QG, \qquad \mathscr{E} = G^2 Q^\sharp T^{-1} + \mathscr{B} Q^\sharp.$$

Then the following statements hold.

1. *G is defined, T is positive definite, and the formula above for P_d is the orthogonal projection onto $\mathscr{L} \ker Q$.*
2. *We have*

$$Q\mathscr{B} = 0, \quad \mathscr{L}\mathscr{B} = \mathsf{P}_d, \quad Q\mathscr{E} = I, \quad \mathsf{P}_d \mathscr{L}\mathscr{E} = 0, \quad \mathscr{L}\mathscr{E} = GQ^\sharp T^{-1}(I - QGQ^\sharp).$$

3. *For every fine scalar field $f \in \operatorname{Ran} \mathsf{P}_d$ and every coarse scalar color field b, the unique solution of*

$$(\mathsf{P}_d \mathscr{L} \lambda, Q\lambda) = (f, b)$$

is $\lambda = \mathscr{B} f + \mathscr{E} b$, and in the norm

$$\|\lambda\|_{\mathsf{Y}} = \|\lambda\|_\infty + \|\nabla_\epsilon \lambda\|_\infty + \|\mathscr{L}\lambda\|_\infty$$

it satisfies $\|\lambda\|_{\mathsf{Y}} \le C(\|f\|_\infty + \|b\|_\infty)$.

4. *The maps $\mathscr{B}$ and $\mathscr{E}$, their left first differences, and $\mathscr{L}\mathscr{B}$ and $\mathscr{L}\mathscr{E}$ have absolute row sums bounded with a positive exponential weight in the distance between coarse blocks.*
5. *For a constant coarse field b, $\mathscr{E} b = b$; the constant scalar mode is fixed by the coarse datum, not removed.*

The constants depend only on M, the localization size M_{loc} of Lemma 9.2, and the decay rates and radii, as in the convention of Section 9.1; they do not depend on d, n, L, γ, or J.

The algebra in parts (1) and (2) is the scalar algebra of [10, (1.43)–(1.45), pp. 25–26] and the projection of [11, (2.17), p. 225], and the analytic input is [13, Theorems 3.1 and 3.2]; see Table 9.1. We give the finite-dimensional proof with our adjoints.

Proof. We first prove the finite-dimensional identities, then transfer the cited estimates for penalty α_d from the cover to the torus of side n, convert them to penalty one, and compose.

Step 1: Algebra. The centered scalar blocks partition the fine sites, and

$$(Qf)(Y) = \epsilon^3 \sum_{x \in \mathsf{B}_m(Y)} f(x), \qquad (Q^\sharp b)(x) = b(Y) \quad (x \in \mathsf{B}_m(Y)),$$

where $\mathsf{B}_m(Y)$ is the block of Y. Then $QQ^\sharp = I$, Q is onto, and $Q^\sharp$ is injective. If the nonnegative form of $\mathscr{L} + Q^\sharp Q$ vanishes at u, then $\nabla_\epsilon u = 0$ and $Qu = 0$; a periodic function with zero gradient is constant, and its block average is that constant, so $u = 0$. This shows that G exists for all finite d and n. Further, $\langle b, Tb\rangle = \|GQ^\sharp b\|_\epsilon^2 > 0$ for $b \neq 0$, and T is positive definite.

Let $Y = GQ^\sharp$, whose adjoint is $Y^\sharp = QG$. Then $Y^\sharp Y = T$, and $YT^{-1}Y^\sharp$ is a self-adjoint idempotent with range $\operatorname{Ran} Y$. We identify the orthogonal complement of this range. For $u \in \ker Q$,

$$\langle \mathscr{L}u, GQ^\sharp b\rangle_\epsilon = \langle u, (I - Q^\sharp QG)Q^\sharp b\rangle_\epsilon = 0,$$

where we used $\mathscr{L}G = I - Q^\sharp QG$; both terms of the second expression vanish since $Qu = 0$. The restriction of $\mathscr{L}$ to $\ker Q$ is injective, since its null vectors would be constants with zero block average. Its image has codimension equal to the number of coarse scalar variables, which is the dimension of the range of the injective map Y. Together with the orthogonality just proved, this implies that $(\operatorname{Ran} Y)^\perp = \mathscr{L} \ker Q$, which proves part (1).

Multiplying in the stated order, we find $Q\mathscr{B} = QG - (QG^2Q^\sharp)T^{-1}QG = 0$, and hence

$$\mathscr{L}\mathscr{B} = (G^{-1} - Q^\sharp Q)\mathscr{B} = I - GQ^\sharp T^{-1}QG = \mathsf{P}_d. \tag{9.25}$$

Let $Z = G^2Q^\sharp T^{-1}$. Then $QZ = I$ and $\mathscr{L}Z = GQ^\sharp T^{-1} - Q^\sharp$, which gives $Q\mathscr{E} = I$ and

$$\mathscr{L}\mathscr{E} = GQ^\sharp T^{-1} - Q^\sharp + \mathsf{P}_d Q^\sharp = GQ^\sharp T^{-1}(I - QGQ^\sharp).$$

Since $\mathsf{P}_d GQ^\sharp = 0$, this implies that $\mathsf{P}_d \mathscr{L}\mathscr{E} = 0$, which completes part (2). In particular, $\lambda = \mathscr{B}f + \mathscr{E}b$ solves the two equations for every $f \in \operatorname{Ran} \mathsf{P}_d$. For a solution of the homogeneous system, $Q\lambda = 0$ implies $\mathscr{L}\lambda \in \operatorname{Ran} \mathsf{P}_d$, the projected equation gives $\mathscr{L}\lambda = 0$, and λ is a constant with zero block average, hence zero. This proves uniqueness. If b is constant, the constant fine field equal to b solves the system with $f = 0$, and by uniqueness $\mathscr{E}b = b$.

At $d = 0$, substituting $Q = I$, we find $T = G^2$, $\mathsf{P}_0 = 0$, $\mathscr{B} = 0$, and $\mathscr{E} = I$. The unit-mesh first difference and Laplacian have finite stencils with bounded absolute rows, and all estimates hold directly, with $f = 0$. In the rest of the proof we assume $d \geq 1$.

Step 2: The cited estimates and the penalty. The uniform estimates are [13, Theorem 3.1, the potential and first-gradient entries of (3.42), p. 397, and the global bounds (3.47), p. 398]. Further, [13, Theorem 3.2, (3.48), p. 398] gives the exponential decay of the kernel of the corresponding coarse inverse, which is $\widehat{T}^{-1}$ in the notation introduced below, and (3.49), p. 399, gives the kernel of the complementary projection. For a source supported in one coarse block of unit size, the two entries of (3.42) bound the potential and its first physical gradient in another coarse block by $Ce^{-a\,\mathrm{dist(blocks)}}$ times the supremum of the source, and the coarse inverse in (3.48) has the same exponential bound. All scale factors are one in this whole-torus specialization with unit coarse spacing. We apply these estimates on the cover of side kn of Lemma 9.2, in the specialization described after that lemma. The local gauge transformations in [13, (3.35)–(3.36), p. 396] are the identity with potential zero, and the scalar average is the lower-corner average translated by T_d as in (9.5) and (9.6). Conjugation by T_d translates the scalar inverse and the kernels of the projection and preserves the coarse Schur operator. The fine measure is ϵ^3, and coarse kernels are summed with counting measure.

In the whole-torus convention of [13, (3.24), p. 394], the parameter $a = 1$ gives a coarse penalty coefficient α_d with

$$\alpha_0 = 1, \qquad \alpha_d = \frac{1 - M^{-2}}{1 - M^{-2d}} \quad (d \geq 1), \qquad 1 - M^{-2} \leq \alpha_d \leq 1.$$

Indeed, the recurrence $a_{j+1} = a_j/(M^{-2} + a_j)$ with $a_1 = 1$ is solved by taking reciprocals and summing the geometric progression. The operators to which the cited estimates apply directly are

$$\widehat{G} = (\mathscr{L} + \alpha_d Q^\sharp Q)^{-1}, \qquad \widehat{T} = Q\widehat{G}^2 Q^\sharp.$$

The estimates hold for $\widehat{G}^{(kn)}$ and $(\widehat{T}^{(kn)})^{-1}$ on the cover, uniformly in d and n. We next transfer them to the torus of side n and then convert to penalty one.

Step 3: Transfer to the torus of side n. For every $\alpha > 0$, on either torus, the form $\|\nabla_\epsilon \lambda\|_\epsilon^2 + \alpha\|Q\lambda\|_2^2$ is positive at $\lambda \neq 0$, since a field with zero gradient is constant and Q preserves constants. Hence $\mathscr{L} + \alpha Q^\sharp Q$ is invertible. The operators $\mathscr{L}$, Q, and $Q^\sharp$ commute with π^*, and α_d is the same on both tori. Applying the positive operator to both sides gives the first identity below; the second follows by composition, and the third by positivity and uniqueness of the coarse inverse:

$$\widehat{G}^{(kn)}\pi^* = \pi^*\widehat{G}^{(n)}, \qquad \widehat{T}^{(kn)}\pi^* = \pi^*\widehat{T}^{(n)}, \qquad (\widehat{T}^{(kn)})^{-1}\pi^* = \pi^*(\widehat{T}^{(n)})^{-1}. \tag{9.26}$$

Here $\langle b, \widehat{T}b\rangle = \|\widehat{G}Q^\sharp b\|_\epsilon^2 > 0$ for $b \neq 0$, since $Q^\sharp$ is injective. The nonlocal constraint is also the same on both tori: by Lemma 9.2, $\mathsf{P}_d^{(kn)}\pi^* = \pi^*\mathsf{P}_d^{(n)}$, where both are the

projections onto the corresponding $\mathscr{L}\ker Q$. These checks justify the scalar applications of that lemma. Their matrices in the coordinate bases and the first output differences satisfy its formulas, weighted rows descend without the factor k^3, and an estimate for a source in one coarse block descends with at most the factor $k^3 = (6M_{\mathrm{loc}})^3$. We conclude that all cited estimates hold on the torus of side n.

At the identity connection these scalar kernels act separately and identically on the colors. At a fixed output site and output component, choosing the signs of a real source in one source block identifies the supremum operator norm with the absolute row sum of the kernel over that block. Hence the estimates for sources in one block bound the absolute row sums, also for each component of $\nabla_\epsilon \widehat{G}$. Summing the exponential block estimates over the coarse unit blocks and using (9.9), we obtain bounds for the rows of $\widehat{G}$, $\nabla_\epsilon \widehat{G}$, and $\widehat{T}^{-1}$, and for their weighted block rows at a smaller fixed rate. Both Q and $Q^\sharp$ have rows of norm one and bounded dependence radius.

Step 4: Conversion to penalty one. We set $\delta = 1 - \alpha_d$, $S = Q\widehat{G}Q^\sharp$, and $K = I + \delta S$. The coarse matrix S is self-adjoint and nonnegative. Its exponentially weighted rows are bounded by the composition bounds above, and so are its columns, by self-adjointness. Since $0 \le \delta \le M^{-2}$, we have $K \ge I$ and the weighted off-diagonal row and column sums of K are bounded. The coarse lattice has unit mesh and polynomial counting growth independent of n. By the inverse estimate of Lemma 5.3 with lower bound one, we obtain bounded weighted rows and columns of K^{-1} at a smaller positive rate, independent of d and n. By direct multiplication, we have

$$G = \widehat{G} - \delta \widehat{G} Q^\sharp K^{-1} Q \widehat{G}, \qquad GQ^\sharp = \widehat{G} Q^\sharp K^{-1}, \qquad T = K^{-1}\widehat{T}K^{-1}, \qquad T^{-1} = K\widehat{T}^{-1}K. \tag{9.27}$$

For example, multiplying the first expression by $\mathscr{L} + Q^\sharp Q = \widehat{G}^{-1} + \delta Q^\sharp Q$ leaves the extra terms $\delta Q^\sharp Q\widehat{G} - \delta Q^\sharp K K^{-1} Q\widehat{G} = 0$. Its product with $Q^\sharp$ is the second expression, since $I - \delta K^{-1} S = K^{-1}$. Taking the adjoint and forming $(GQ^\sharp)^\sharp(GQ^\sharp)$ gives the formula for T, and inverting it, we obtain the last one. These identities transfer the bounds for the potential and first gradient from $\widehat{G}$ to G, and the bounds for the coarse inverse from $\widehat{T}^{-1}$ to T^{-1}, with one further fixed reduction of the rate. A left gradient in the first identity acts only on the leftmost factor $\widehat{G}$.

Step 5: The solution maps and the Laplacian. In each term of $\mathscr{B}$ and $\mathscr{E}$, these operators are composed in the displayed order. Row sums are submultiplicative, and for a left gradient only the leftmost factor G is differentiated. Therefore $\mathscr{B}$, $\mathscr{E}$, and their left first differences have bounded rows and bounded exponentially weighted block rows. For example, the left gradient of the second term of $\mathscr{B}$ is $(\nabla_\epsilon G)GQ^\sharp T^{-1}QG$, all of whose factors have the required rows; the triangle inequality for block distances gives the weighted version. For the Laplacian we use (9.25) and the formula for $\mathscr{L}\mathscr{E}$ in part (2). The projection $\mathsf{P}_d = I - GQ^\sharp T^{-1}QG$ has a bounded weighted block row, including its

identity part, and $\mathscr{L}\mathscr{E}$ is a composition of operators whose rows are already bounded. Adding the bounds for the potential, the first differences, and the Laplacian proves part (3), and the block rows prove part (4). No estimate for the first differences of the identity part of P_d applied to a general source is used. □

9.8 Local gauge fixing with prescribed coarse gauge

We now bring a small fine field into the chart of Theorem 9.7 by a small fine gauge transformation. The transformation is fixed by two conditions: the projected divergence of the transformed field vanishes, and the composed coarse gauge transformation of the gauge-corrected averaging, multiplied by the coarse values of the fine gauge transformation, is a prescribed coarse gauge transformation e^c. The divergence of a gauge-transformed field involves the Laplacian of the gauge parameter, and a direct estimate would require bounds on all second differences of the parameter. We avoid this by an exact formula on each link in which the inverse lattice spacing cancels. We measure the gauge parameter in the norm $\|\cdot\|_{\mathsf{Y}}$ of Lemma 9.8, which controls the Laplacian but not the individual second differences. The map $\mathscr{H}_d$ below is not related to the linear extension H_d.

Theorem 9.9. *Use the geometry of Theorem 9.3 at the identity connection, and set*

$$\|A\|_{\mathsf{X}} = \|A\|_\infty + \|\mathrm{div}_\epsilon A\|_\infty, \qquad \|\lambda\|_{\mathsf{Y}} = \|\lambda\|_\infty + \|\nabla_\epsilon \lambda\|_\infty + \|\mathscr{L}_\epsilon \lambda\|_\infty.$$

Let $\mathsf{G}_d(A)$ be the composed gauge transformation of Lemma 8.10 with background connection $U_0 = I$, applied to the fine field $e^{\epsilon A}$. There exist $r_1, C > 0$, independent of d and n, such that the following statements hold.

1. *On the complex balls $\|A\|_{\mathsf{X}} < 2r_1$ and $\|\lambda\|_{\mathsf{Y}} < 2r_1$, the maps*

$$A^{e^\lambda}_{xy} = \epsilon^{-1} \operatorname{Log}\bigl(e^{\lambda_x} e^{\epsilon A_{xy}} e^{-\lambda_y}\bigr), \qquad \mathscr{H}_d(A,\lambda) = \operatorname{Log}\bigl[\mathsf{G}_d(A^{e^\lambda})\, e^{\lambda|_{\mathrm{roots}}}\bigr]$$

are holomorphic, with bounds on their operator derivatives of each fixed order on smaller balls. They satisfy

$$\begin{aligned} \mathrm{div}_\epsilon A^{e^\lambda} &= \mathrm{div}_\epsilon A - \mathscr{L}_\epsilon \lambda + \mathscr{R}_D(A,\lambda), \\ \mathscr{H}_d(A,\lambda) &= \operatorname{Log} \mathsf{G}_d(A) + Q_d' \lambda + \mathscr{R}_H(A,\lambda), \end{aligned}$$

where both remainders vanish at $\lambda = 0$, and for $0 < r_{\mathrm{loc}} \le r_1$ and $\|A\|_{\mathsf{X}}, \|\lambda\|_{\mathsf{Y}}, \|\lambda'\|_{\mathsf{Y}} \le r_{\mathrm{loc}}$,

$$\|\mathscr{R}_D(A,\lambda) - \mathscr{R}_D(A,\lambda')\|_\infty + \|\mathscr{R}_H(A,\lambda) - \mathscr{R}_H(A,\lambda')\|_\infty \le C r_{\mathrm{loc}} \|\lambda - \lambda'\|_{\mathsf{Y}}.$$

In particular $D_\lambda \mathscr{H}_d(0,0) = Q_d'$, including the constants and across the periodic seams. The order of the factors in $\mathscr{H}_d$ follows from $(U^k)^l = U^{lk}$.

2. *There exist $0 < r_{\mathsf{Y}}, r_{\mathsf{X}}, r_c < r_1$, independent of d and n, such that for every fine field A with $\|A\|_{\mathsf{X}} < r_{\mathsf{X}}$ and every coarse field c with $\|c\|_\infty < r_c$ there is a unique λ with $\|\lambda\|_{\mathsf{Y}} < r_{\mathsf{Y}}$ satisfying*
$$\mathsf{P}_d \operatorname{div}_\epsilon A^{e^\lambda} = 0, \qquad \mathsf{G}_d(A^{e^\lambda})\, e^{\lambda|_{\mathrm{roots}}} = e^c.$$
This solution depends holomorphically on (A, c) on smaller balls, is real for real data, and satisfies $\|\lambda(A,c)\|_{\mathsf{Y}} \le C(\|A\|_{\mathsf{X}} + \|c\|_\infty)$. If A_* lies in these domains, $\mathsf{P}_d \operatorname{div}_\epsilon A_* = 0$, and $c = \operatorname{Log} \mathsf{G}_d(A_*)$, then $\lambda(A_*, c) = 0$.
3. *With $b = T_M^d(e^{\epsilon A})$, the gauge-corrected output of the transformed field is $V_d(A^{e^\lambda}) = b^{e^c}$.*
4. *For real data the transformation preserves the plaquette distances ρ_G, and*
$$\|A^{e^\lambda}\|_\infty + \|\mathrm{d}_\epsilon A^{e^\lambda}\|_\infty \le C\big(\|A\|_{\mathsf{X}} + \|c\|_\infty + \|\mathrm{d}_\epsilon A\|_\infty\big).$$
Further, for every fixed $r_{\mathrm{ph}} \le r_0$, if the supremum norms of A, of $\operatorname{div}_\epsilon A$, of $\mathrm{d}_\epsilon A$, and of c are smaller than r_{ph} divided by a sufficiently large constant, then $A' = A^{e^\lambda}$ lies in the domain of Theorem 9.7: $A' = \Xi_d(a + H_d\theta)$ with $a \in \mathcal{V}_d$, $\|a\|_\infty + \|\mathrm{d}_\epsilon a\|_\infty < r_{\mathrm{ph}}$, and $\|\theta\|_\infty < r_{\mathrm{ph}}/C_*$.

The constants depend only on M, the localization size M_{loc} of Lemma 9.8, and the radii of the balls in the statement and the larger balls containing them; they do not depend on d, n, L, γ, or J. A bound on all first differences of the output is not required.

We also summarize the remaining radii of Theorems 9.7 and 9.9.

Radius	Domain or role
r_0, C_*	Upper bound on the radius r_{ph} of the convex chart in Theorem 9.7, and the factor in the target condition $\|\theta\|_\infty < r_{\mathrm{ph}}/C_*$.
$2r_1$, r_{loc}	Complex input radii for the gauge maps $A \mapsto A^{e^\lambda}$ and $\mathscr{H}_d$ in the norms $\|\cdot\|_{\mathsf{X}}$ and $\|\cdot\|_{\mathsf{Y}}$, and radius $r_{\mathrm{loc}} \le r_1$ for the Lipschitz bounds of the remainders.
r_{X}, r_c, r_{Y}	Fine input radius in $\|\cdot\|_{\mathsf{X}}$, radius of the prescribed coarse gauge c, and radius of the ball in $\|\cdot\|_{\mathsf{Y}}$ in which the gauge parameter λ is unique.

Proof. We first derive an exact formula for the gauge transformation on one link, then expand the divergence and the coarse gauge to first order in λ, and finally solve the two equations by a contraction in the norm $\|\cdot\|_{\mathsf{Y}}$.

Step 1: The gauge transformation on one link. For a positive link $e = (x, x + \epsilon e_i)$ set
$$u = \lambda_x, \qquad v = \nabla_{\epsilon,i}\lambda(x), \qquad a = A_i(x), \qquad F_\epsilon(u,v,a) = \epsilon^{-1} \operatorname{Log}\big(e^u e^{\epsilon a} e^{-u-\epsilon v}\big).$$

The product inside the logarithm is the identity at $\epsilon = 0$. Its logarithm is a holomorphic function vanishing there, and the quotient by ϵ therefore has a removable analytic extension. Differentiating at $\epsilon = 0$, we obtain

$$F_\epsilon(u,v,a) = C(u)a - J(u)v + \epsilon R_\epsilon(u,v,a), \qquad C(u) = \operatorname{Ad}(e^u), \quad J(u)v = \int_0^1 \operatorname{Ad}(e^{su})v\,ds. \tag{9.28}$$

For example, the derivative of $e^{-u-\epsilon v}$ at $\epsilon = 0$, multiplied on the left by e^u, is $-J(u)v$, which fixes the sign and the order of the adjoints. On fixed small complex balls of (u,v,a) the products stay in the identity chart for $0 \le \epsilon \le 1$, and the analytic extension has bounded Taylor coefficients and derivatives, uniformly in ϵ. Its terms linear in (a,v) are exactly $C(u)a - J(u)v$, independently of ϵ, and R_ϵ starts at degree two in (a,v). Further, conjugation shows that $F_\epsilon(u,0,a) = C(u)a$ exactly, and $R_\epsilon(u,0,a) = 0$. Using its convergent Taylor series on a smaller ball, we obtain

$$|R_\epsilon(u,v,a)| \le C|v|\big(|a| + |v|\big), \tag{9.29}$$

with derivatives in (a,v) bounded by $C(|a| + |v|)$ and derivative in u bounded by $C|v|(|a| + |v|)$. In particular $\|A^{e^\lambda}\|_\infty \le C(\|A\|_\infty + \|\nabla_\epsilon \lambda\|_\infty)$.

Step 2: The divergence. We abbreviate $C_x = C(\lambda_x)$, $J_x = J(\lambda_x)$, $v_i(x) = \nabla_{\epsilon,i}\lambda(x)$, and $R_i(x) = R_\epsilon(\lambda_x, v_i(x), A_i(x))$. Substituting (9.28) into the backward divergence of Section 9.1 and freezing the coefficients at x, we obtain

$$\begin{aligned} \operatorname{div}_\epsilon A^{e^\lambda}(x) = {} & C_x \operatorname{div}_\epsilon A(x) - J_x \mathscr{L}_\epsilon \lambda(x) \\ & + \sum_i \frac{C_{x-\epsilon e_i} - C_x}{\epsilon} A_i(x - \epsilon e_i) - \sum_i \frac{J_{x-\epsilon e_i} - J_x}{\epsilon} v_i(x - \epsilon e_i) \\ & + \sum_i \big(R_i(x - \epsilon e_i) - R_i(x)\big). \end{aligned} \tag{9.30}$$

The last line has no factor ϵ^{-1}: the factor ϵ in (9.28) has canceled it. Its size is bounded by the two values in (9.29), and no difference of A is needed. For the differences of the coefficients, the integral formula along the segment from λ_x to $\lambda_{x-\epsilon e_i}$ gives a bound $C\|\nabla_\epsilon \lambda\|_\infty$; also $\|C_x - I\| + \|J_x - I\| \le C\|\lambda\|_\infty$. It follows that the difference between the first line of (9.30) and $\operatorname{div}_\epsilon A - \mathscr{L}_\epsilon \lambda$, together with the other two lines, defines a holomorphic remainder $\mathscr{R}_D$ that vanishes at $\lambda = 0$.

For the Lipschitz bound, take λ, λ' with $\|\lambda\|_Y, \|\lambda'\|_Y \le r_{\mathrm{loc}}$ and set $\delta\lambda = \lambda - \lambda'$. In $(C - I)\operatorname{div}_\epsilon A$, the change is at most $C\|\operatorname{div}_\epsilon A\|_\infty \|\delta\lambda\|_\infty$. In $(J - I)\mathscr{L}_\epsilon \lambda$, the product rule for differences bounds the change by

$$C\big(\|\lambda\|_\infty \|\mathscr{L}_\epsilon \delta\lambda\|_\infty + \|\mathscr{L}_\epsilon \lambda'\|_\infty \|\delta\lambda\|_\infty\big).$$

In the second line of (9.30), the same integral formula and the bounded second derivatives of C and J bound the changes of the difference quotients by $C(\|\nabla_\epsilon \delta\lambda\|_\infty + r_{\mathrm{loc}}\|\delta\lambda\|_\infty)$.

Their other factors have norm at most r_{loc}, and changing v_i in the second sum gives one more product of the same type. The derivative bounds following (9.29) bound the change of the last line by $Cr_{\mathrm{loc}}\|\delta\lambda\|_{\mathsf{Y}}$. These estimates prove the Lipschitz bound for $\mathscr{R}_D$, as well as holomorphy and bounds on the operator derivatives of fixed order of the divergence map from $\mathsf{X}\times\mathsf{Y}$ to the supremum norm on smaller balls. Only the scalar Laplacian, not the individual second differences, was used. Applying P_d preserves these estimates, since P_d has bounded rows by Lemma 9.8.

Step 3: The composed coarse gauge transformation. By Lemma 8.10, $\operatorname{Log}\mathsf{G}_d(A)$ is holomorphic and bounded on a fixed complex ball in the supremum norm, since the sizes of its factors are at most $C_M\epsilon_j\|A_j\|_\infty$, whose sum over j is a geometric sum. After a fixed decrease of r_1, the bound on A^{e^λ} in Step 1 puts A^{e^λ} inside that ball, and $\lambda|_{\mathrm{roots}}$ has supremum norm at most $\|\lambda\|_{\mathsf{Y}}$. The product defining $\mathscr{H}_d$ lies in a fixed identity chart and is a bounded holomorphic function. Its operator derivatives of fixed order are bounded independently of the dimension. To see this, we restrict the bounded holomorphic map to $x+\sum_{i=1}^s z_ih_i$ with $\|h_i\|\le 1$ and $|z_i|<\delta/s$ at distance at least δ from the boundary of the ball on which the map is bounded. Cauchy's formula then bounds the mixed derivative by $C(s/\delta)^s$. This bounds an operator norm; it does not sum absolute coefficients.

We compute the derivative in λ at $(0,0)$. At $A=0$, take the pure fine gauge transformation $\Omega_x=e^{t\lambda_x}$. At one level, each path transport is $\Omega_v\Omega_x^{-1}$, and the infinitesimal logarithm of the endpoint gauge transformation at the coarse vertex v is

$$-M^{-3}\sum_{x\in\mathsf{B}_M(v)}(\lambda_v-\lambda_x)=-\lambda_v+M^{-3}\sum_{x\in\mathsf{B}_M(v)}\lambda_x,$$

and the pure gauge transformation itself contributes $+\lambda_v$ at that vertex. Their sum is the centered scalar block average of λ. Iterating this calculation composes the scalar block averages, whose composition is Q'_d by Lemma 8.1. Every intermediate vertex of the paths cancels, also for paths across the periodic seams, and a constant is averaged to itself. For $d=0$ the composed gauge transformation is the identity and the restriction to the roots is $Q'_0=I$. In all cases

$$D_\lambda\mathscr{H}_d(0,0)=Q'_d. \tag{9.31}$$

The order of the factors follows from $(U^k)^l=U^{lk}$: the fine gauge transformation is applied first and $\mathsf{G}_d(A^{e^\lambda})$ second, and at the coarse vertices their product is $\mathsf{G}_d(A^{e^\lambda})e^{\lambda|_{\mathrm{roots}}}$.

We define $\mathscr{R}_H=\mathscr{H}_d(A,\lambda)-\mathscr{H}_d(A,0)-Q'_d\lambda$, which vanishes at $\lambda=0$; note $\mathscr{H}_d(A,0)=\operatorname{Log}\mathsf{G}_d(A)$. By the bounds on the second operator derivatives and (9.31), we have

$$\|D_\lambda\mathscr{H}_d(A,\lambda)-Q'_d\|\le C\big(\|A\|_{\mathsf{X}}+\|\lambda\|_{\mathsf{Y}}\big),$$

and integrating along the segment from λ' to λ gives the Lipschitz bound for $\mathscr{R}_H$. Together with Step 2 this proves part (1).

Step 4: Solving the two equations. On $\operatorname{Ran}\mathsf{P}_d \times \{\text{coarse scalar fields}\}$ we use the product supremum norm and define $\mathcal{L}\lambda = (-\mathsf{P}_d\mathscr{L}_\epsilon\lambda, Q_d'\lambda)$. By Lemma 9.8, its inverse is $\mathcal{L}^{-1}(f,b) = -\mathscr{B}f + \mathscr{E}b$, with norm at most C_I into Y. We set

$$\mathsf{d}(A,c) = \big(\mathsf{P}_d \operatorname{div}_\epsilon A,\ \operatorname{Log}\mathsf{G}_d(A) - c\big), \qquad R(A,\lambda) = \big(\mathsf{P}_d\mathscr{R}_D(A,\lambda),\ \mathscr{R}_H(A,\lambda)\big).$$

By part (1), the two equations of part (2) are equivalent to

$$\lambda = -\mathcal{L}^{-1}\big(\mathsf{d}(A,c) + R(A,\lambda)\big), \tag{9.32}$$

where the first component of the argument lies in $\operatorname{Ran}\mathsf{P}_d$, so that $\mathcal{L}^{-1}$ is applied within its domain. Let C_R dominate the Lipschitz constants of the remainders, including the row bound of P_d, and choose $0 < r_\mathsf{Y} < r_1/2$ with $C_I C_R r_\mathsf{Y} \le 1/4$. The composed gauge transformation is the identity at $A = 0$ and has a bounded first operator derivative, and $\|\mathsf{d}(A,c)\| \le C_d(\|A\|_\mathsf{X} + \|c\|_\infty)$. Let $r_\mathsf{X}, r_c < r_\mathsf{Y}$ satisfy $C_I C_d(r_\mathsf{X} + r_c) < r_\mathsf{Y}/4$. Since $R(A,0) = 0$, the right side of (9.32) maps the closed ball of radius r_Y in Y into the concentric ball of radius $r_\mathsf{Y}/2$ and has Lipschitz constant at most $1/4$. By Lemma 9.5, the solution exists, is unique in the ball, depends holomorphically on the data on smaller balls, and is real for real data. Absorbing the Lipschitz term, we obtain

$$\|\lambda(A,c)\|_\mathsf{Y} \le \tfrac{4}{3} C_I C_d\big(\|A\|_\mathsf{X} + \|c\|_\infty\big).$$

If $\mathsf{P}_d \operatorname{div}_\epsilon A_* = 0$ and $c = \operatorname{Log}\mathsf{G}_d(A_*)$, then $\lambda = 0$ solves (9.32), and by uniqueness $\lambda(A_*, c) = 0$. The logarithm and the exponential are inverse to each other in the chart, so the logarithmic equation for the coarse gauge is equivalent to the group equation in part (2).

Step 5: The gauge-corrected output. By part (3) of Lemma 8.10 with background connection $U_0 = I$, we have

$$V_d(A^{e^\lambda}) = \big[T_M^d\big((e^{\epsilon A})^{e^\lambda}\big)\big]^{\mathsf{G}_d(A^{e^\lambda})} = \big[b^{e^\lambda|_{\text{roots}}}\big]^{\mathsf{G}_d(A^{e^\lambda})} = b^{\mathsf{G}_d(A^{e^\lambda})e^\lambda|_{\text{roots}}} = b^{e^c}.$$

The second equality is the gauge covariance of T_M, which follows from the cancellation of the gauge transformations at the interior vertices of the paths in its definition. The small relative logarithms in the complex charts transform analytically under conjugation, so the calculation also holds on the holomorphic domains; for real data it is the gauge covariance for $\mathrm{SU}(2)$-valued gauge transformations.

Step 6: Real data and the chart. Let the data be real and set $A' = A^{e^\lambda}$. Plaquette transports transform by conjugation, so $\rho_G((e^{\epsilon A'})_p) = \rho_G((e^{\epsilon A})_p)$. Since both potentials are small, their plaquette transports lie in the same real identity chart, where the logarithms

are related by an adjoint at the base point of the plaquette. Hence $|X_p(A')| = |X_p(A)|$. Using (9.10) for A' with the reverse triangle inequality and for A with the triangle inequality, we obtain

$$\begin{aligned} \epsilon^2 |(\mathrm{d}_\epsilon A')_p| &\le |X_p(A')| + C\epsilon^2 \sum_{e\in\partial p} |A'_e|^2 = |X_p(A)| + C\epsilon^2 \sum_{e\in\partial p} |A'_e|^2 \\ &\le \epsilon^2 |(\mathrm{d}_\epsilon A)_p| + C\epsilon^2 \sum_{e\in\partial p} \big(|A_e|^2 + |A'_e|^2\big). \end{aligned} \tag{9.33}$$

Dividing by ϵ^2 and taking the supremum over plaquettes, we obtain

$$\|\mathrm{d}_\epsilon A'\|_\infty \le \|\mathrm{d}_\epsilon A\|_\infty + C\big(\|A\|_\infty^2 + \|A'\|_\infty^2\big).$$

The bound on A^{e^λ} from Step 1 and the bound on λ from Step 4 imply that $\|A'\|_\infty \le C(\|A\|_{\mathsf{X}} + \|c\|_\infty)$. Since these quantities lie in a fixed small ball, the squares can be absorbed into a linear bound, which proves the estimate of part (4).

To verify the statement about the chart, take the data so small that $\|A'\|_\infty < r_{\mathrm{fib}}$, and set

$$\theta = \mathsf{F}_d(A'), \qquad a = (I - H_d Q_d)A'.$$

Then $Q_d a = 0$, and $\mathsf{P}_d \operatorname{div}_\epsilon a = 0$ by the solved gauge equation and $\mathsf{P}_d \operatorname{div}_\epsilon H_d = 0$. By part (4) of Theorem 9.6, we have $A' = \Xi_d(a + H_d\theta)$. The row bound for the average and the quadratic bound for its remainder imply that $\|\theta\|_\infty \le C\|A'\|_\infty$, and by the row bounds of H_d and $\nabla_\epsilon H_d$, we have

$$\|a\|_\infty + \|\mathrm{d}_\epsilon a\|_\infty \le C\|A'\|_\infty + \|\mathrm{d}_\epsilon A'\|_\infty.$$

For any fixed $r_{\mathrm{ph}} \le r_0$, if the input potential, divergence, curl, and coarse gauge are smaller than r_{ph} divided by a sufficiently large constant, both conditions of Theorem 9.7 hold, including $\|\theta\|_\infty < r_{\mathrm{ph}}/C_*$. All radii depend only on M, the localization data, and the radii and decay rates fixed above. □

9.9 A trial configuration with prescribed coarse field

The analytic right inverse of Theorem 9.6 produces, after a gauge transformation, a fine configuration with prescribed image under T_M^d and Wilson action at most $C g_n^{-2}\|\theta\|_2^2$, where e^θ is the prescribed coarse field.

Corollary 9.10. *In the setting of Theorem 9.6, after a fixed decrease of θ_*, let θ be a real coarse field with $\|\theta\|_\infty < \theta_*$, let $A_{\mathrm{sec}} = A_{\mathrm{sec},d}(\theta)$, and let $U = e^{\epsilon A_{\mathrm{sec}}}$. Let k be any periodic* SU(2)*-valued gauge transformation on the fine lattice whose values at the coarse vertices are those of the composed gauge transformation $\mathsf{G}_d(A_{\mathrm{sec}})$ of Lemma 8.10 with background connection $U_0 = I$, and set $U_{\mathrm{sec}} = U^k$. Then the following statements hold.*

1. *$T_M^d U_{\mathrm{sec}} = e^\theta$, the plaquette distances of U_{sec} equal those of U, and for every coarse coupling $g_n > 0$*
$$4g_n^{-2}\epsilon^{-1}\sum_p q\big((U_{\mathrm{sec}})_p\big) \le C g_n^{-2}\|\theta\|_2^2.$$
2. *Let $\epsilon_j = M^{j-d}$ and $g_j = g_n\sqrt{\epsilon_j}$, so that the fine coupling is $g_0 = g_n\sqrt{\epsilon}$. For $0 \le j < d$, let z_j be the free logarithmic coordinates of $T_M^j U_{\mathrm{sec}}$ relative to $T_M^{j+1}U_{\mathrm{sec}}$ in the block axial gauge, given by the inverse branch of Lemma 6.3 for one averaging step. Then*
$$\|z_j/g_j\|_\infty \le C\frac{\|\theta\|_\infty}{g_n}\epsilon_j^{3/2} \qquad (0 \le j < d).$$
3. *Let $p_j > 0$ satisfy*
$$C\frac{\|\theta\|_\infty}{g_n}\epsilon_j^{3/2} \le p_j/2, \qquad g_j p_j < r_M \qquad (0 \le j < d),$$
where r_M is the radius of Lemma 6.3. Then the inverse branches are defined on the whole products $\prod_e\{|z_{j,e}| \le g_j p_j\}$, and the coordinates of U_{sec} lie in the interior of the product $\prod_{j,e}\{|z_{j,e}/g_j| \le p_j\}$, in its concentric half. For a common radius $p_j = p$, the condition $\|\theta\|_\infty \le c g_n p$ suffices, together with $\|\theta\|_\infty < \theta_*$ and $g_j p < r_M$.
4. *The infimum of the Wilson action $4g_n^{-2}\epsilon^{-1}\sum_p q(U_p)$ over any set of configurations that contains U_{sec} lies between 0 and $C g_n^{-2}\|\theta\|_2^2$.*

The gauge transformation k may be rough; the derivative bounds of Theorem 9.6 refer to A_{sec}, before the gauge transformation. The constants do not depend on d, n, L, γ, g_n, or J.

Proof. We first verify the image under T_M^d, then the bounds on the coordinates, and finally the bound on the action.

Step 1: The image under T_M^d. Since $\mathsf{F}_d(A_{\mathrm{sec}}) = \theta$, part (3) of Lemma 8.10 with background connection $U_0 = I$ reads
$$e^\theta = (T_M^d U)^{\mathsf{G}_d(A_{\mathrm{sec}})}.$$
One choice of k assigns to each fine vertex the value of this gauge transformation at the center of its block on the coarse lattice. This is a periodic fine gauge transformation with the required values. For any such k, the gauge covariance of T_M implies that $T_M^d(U^k) = (T_M^d U)^{k|_{\mathrm{roots}}} = e^\theta$, where $k|_{\mathrm{roots}}$ is the restriction to the coarse vertices.

Step 2: The choice of k does not affect the coordinates. In a block with center v, the block axial gauge uses the path transports $k_x^{\mathrm{ax}} = U(\sigma_{v,x})$. Under a real fine gauge transformation l, this transport becomes $l_v U(\sigma_{v,x}) l_x^{-1}$, and the two gauge-fixed fields differ by the gauge transformation equal to l_v on the whole block. Since the coarse-field

lift of the target transforms by the same blockwise constant gauge transformation, the relative logarithms transform as $X_e \mapsto \mathrm{Ad}(l_v)X_e$, and their norms are unchanged. The same calculation applies at each level, with the restriction of l to the vertices of that level. All the coordinates z_j of U and U^k have the same norms.

Step 3: The coordinates. By (9.10) and the bound $\|A_{\mathrm{sec}}\|_\infty + \|\nabla_\epsilon A_{\mathrm{sec}}\|_\infty \le C\|\theta\|_\infty$,

$$\delta(U) \le C\epsilon^2\|\theta\|_\infty. \tag{9.34}$$

We apply part (2) of Lemma 8.10, now regarding the real configuration U as the background. Its curvature hypothesis holds with $\varepsilon = C\|\theta\|_\infty$ after a fixed decrease of θ_*, and it yields for the iterates

$$\delta(T_M^j U) \le C\epsilon_j^2\|\theta\|_\infty \qquad (0 \le j \le d).$$

We decrease θ_* once more so that all these curvatures lie below the threshold of Lemma 7.2. We use its clause for one averaging step under a curvature hypothesis at each $j < d$, not its clause for J consecutive steps under a bound on the links. The lattice sides at this use are

$$n_j = M^{d-j}n = Mn_{j+1}, \qquad n_{j+1} = M^{d-j-1}n \ge n \ge 4, \tag{9.35}$$

so the local patches have the contractible lifts required there. That lemma identifies the relative coordinates in the block axial gauge with the inverse branch for one averaging step and bounds their free components by $C\epsilon_j^2\|\theta\|_\infty$. Dividing by $g_j = g_n\sqrt{\epsilon_j}$, we obtain part (2) for U, and Step 2 transfers it to U_{sec}.

The inequalities with $p_j/2$ imply strict membership in the product. For the statement about the whole product, the target $b = T_M^{j+1}U_{\mathrm{sec}}$ is $\mathrm{SU}(2)$-valued, and Lemma 6.3 defines Γ_b at every free input with $\max_e|z_e| < r_M$, uniformly in the target. The condition $g_j p_j < r_M$ puts the closed product $\prod_e\{|z_e| \le g_j p_j\}$ strictly inside this domain, and the coordinates of U_{sec} lie in the concentric half product. For a common radius we have $\epsilon_j^{3/2} \le 1$, and choosing the constant c small enough gives the sufficient condition in part (3). At depth zero there are no coordinates to check.

Step 4: The action. On the ball of the section, Lemma 9.4 implies that

$$q(U_p) \le C\epsilon^4\Big(|(\mathrm{d}_\epsilon A)_p|^2 + \|A\|_\infty^2 \sum_{e\in\partial p}|A_e|^2\Big), \qquad A = A_{\mathrm{sec}}. \tag{9.36}$$

Indeed, the linear term of $X_p(A)$ is exactly $\epsilon^2(\mathrm{d}_\epsilon A)_p$, $q(e^X) \le C|X|^2$, and the square of the remainder is bounded using $(\sum_{e\in\partial p}|A_e|^2)^2 \le 4\|A\|_\infty^2\sum_{e\in\partial p}|A_e|^2$. Each positive link belongs to a fixed number of plaquette boundaries, and $\|\mathrm{d}_\epsilon A\|_\epsilon \le C\|\nabla_\epsilon A\|_\epsilon$. We obtain

$$4\epsilon^{-1}\sum_p q(U_p) \le C\big(\|\nabla_\epsilon A\|_\epsilon^2 + \|A\|_\infty^2\|A\|_\epsilon^2\big) \le C\|q_d(\theta)\|_2^2 \le C\|\theta\|_2^2. \tag{9.37}$$

In the middle inequality we used the H^1_ϵ bound for the extension and the supremum bound on the ball of the section, and in the last one part (2) of Theorem 9.6. Since gauge transformations preserve the group distance of each plaquette, and hence its value of q, the same bound holds for U_{sec}. Since the fine coupling satisfies $g_0^2 = g_n^2 \epsilon$, multiplication by g_n^{-2} gives part (1). Since the Wilson action of an SU(2)-valued configuration is nonnegative, every infimum over a set containing U_{sec} lies between zero and this bound, which proves part (4). □

SECTION 10

Classical background fields and analytic estimates

In this section we construct the classical background field on the periodic torus and prove the analytic estimates for it that the renormalization group analysis uses. Fix a coarse lattice of side $n \geq 4$ and a depth $d \geq 0$, and consider the fine lattice of side nM^d, so that d averaging steps lead from the fine lattice to the coarse one. For a coarse configuration e^θ close to the identity, we minimize the Wilson action, rescaled to coarse lattice units, in a local gauge-fixed constraint chart. A gauge transform of the resulting field is a local minimum of the action on the fiber $\{U : T_M^d U = e^\theta\}$, modulo gauge transformations. Following Bałaban [16], we call the minimizer the classical background field. Its action is the effective classical action $E_d(e^\theta)$; divided by the square of the coupling, it is the classical part of the effective densities of Sections 13 and 14.

We prove four results. Theorem 10.6 constructs the background field in a gauge-fixed chart and shows that it depends analytically on θ and that its response to a change of θ decays exponentially. It also proves the local minimality on the fiber $\{U : T_M^d U = e^\theta\}$ described above. Lemma 10.8 shows that E_d is invariant under small coarse gauge transformations, lattice translations, the spatial inversion, and constant color rotations. Theorem 10.10 proves the nested minimization identity

$$E_{d+1}(c) = M \min_{\|z\|_\infty \leq r_z} E_d(b(c,z)),$$

where $b(c,z)$, for $\|z\|_\infty \leq r_z$, parametrizes a local branch of the fiber $\{b : T_M b = c\}$ on the intermediate lattice, in block axial gauge. The minimizer $z_d^*(c)$ is an interior point, it is analytic in c, and the Hessian in z has a positive lower bound that does not depend on d. The factor M is the change of units in three dimensions: with $g_+^2 = Mg^2$ the identity reads $E_{d+1}(c)/g_+^2 = g^{-2} \min_z E_d(b(c,z))$. Finally, Theorem 10.11 proves factorial bounds, in weighted row and column norms of the derivative tensors, for E_d, the background field, its block averages at every intermediate scale (in explicitly given gauges), and its plaquette logarithms divided by the square of the mesh. In all these statements the radii, decay rates, and constants are the same for every d and n.

The background field and the conditional constrained minimizer z^* center the fluctuation coordinates of Section 12, and the constrained Hessian normalizes them. The uniform estimates are the classical input to the contraction estimates of Section 11, and they are used throughout the later parts of the paper, from the analysis on block regions in Section 18 to the weak-coupling analysis in Section 36.

The inputs are the block averaging map of Definition 4.3 and the coefficient norms of Definition 4.6, the estimates for weighted absolute Taylor coefficients in Lemma 5.3, and the effective Maxwell quadratic form of Theorem 8.3. From Section 9 we use the linear constrained extension (Theorem 9.3), the nonlinear constraint parametrization (Theorem 9.6), the convexity of the constrained action (Theorem 9.7), the gauge slice with prescribed coarse gauge (Theorem 9.9), and the transfer of kernels from periodic covers (Lemma 9.2). The only other input from the literature is Bałaban's estimate for the constrained Green operator at the identity background [13, Theorem 3.13, p. 426].

There are two main ideas. The first is to invert the constrained linear curl form from the supremum norm to the C^1 norm, using Bałaban's estimate, and to solve the Euler equation in the constraint coordinates by a contraction argument. The strict convexity proved in Theorem 9.7 then identifies the solution as the minimum. The second concerns the derivative bounds. At the linear level, one averaging step has row sums one and column sums M^{-3}. Its nonlinear part at mesh $\epsilon_j = M^{j-d}$ carries a factor ϵ_j, and these factors are summable over j. Composing nonnegative majorant series with these linear masses, we obtain bounds in which no radius and no decay rate is lost at any scale.

The construction follows Bałaban's treatment of the variational problem in [16]. There the minimizing configuration with prescribed averages is constructed in a projected Landau gauge [16, Theorem 1, p. 279], and its analytic dependence on the averages and the exponential localization of its derivative are studied in [16, Section G, pp. 305–309, and Proposition 9, p. 309]. Our averaging map, coordinate domains, and coefficient norms differ from those of [16], and we give the proofs in full. Uniform regularity and decay of lattice Green functions go back to [9].

After the notation of Section 10.1, the proof proceeds through the constrained Green operator (Section 10.2), the block maps and constraint coordinates (Section 10.3), the minimization (Sections 10.4 to 10.7), and the estimates at intermediate scales (Section 10.8).

10.1 Rescaled fields, scalar products, and norms

Units and constants. Throughout this section $M \geq 3$ is a fixed odd integer, $n \geq 4$ is the side of the coarse lattice, and $d \geq 0$ is the depth. We measure lengths in coarse lattice units, in which the coarse lattice spacing is one. The fine lattice then has side nM^d and mesh $\varepsilon_d = M^{-d}$, and all distances are periodic distances in these units. On the torus of side L the coarse spacing is $a_n = L/n$, and the physical length of the fine mesh is $a_n \varepsilon_d$. The coupling plays no role in this section; in particular ε_d is unrelated to g_n^2.

All constants, radii, and decay rates in this section depend only on M, on fixed larger complex and real coordinate domains and the ratios of their radii, and on finitely many spatial decay rates chosen in the proofs, as in item (2) of Definition 4.14. In particular, they do not depend on d, n, L, or γ. The integer J of a renormalization group step does

not enter the constructions of this section, and the constants are also independent of J. A constant that carries a derivative order s as an index may depend on s. In proofs, K denotes a constant of this kind whose value may change from line to line. The auxiliary localization size used in Lemma 10.2 is a fixed function of M.

We identify $\mathfrak{su}(2)$ with $\mathbb{R}^3$ as in Section 4. A color block, that is, an element of $\mathfrak{su}(2)$ or a multilinear map between copies of $\mathfrak{su}(2)$, has its Euclidean norm. Complex fields take values in the complexification of $\mathfrak{su}(2)$, and the scalar products below are extended bilinearly to them. For a field and its first differences in the three coordinate directions we use the maximum of the norms of the four components. This maximum and the C^1_ε norm below are equivalent up to a fixed factor, which enters only at the beginning and at the end of an estimate.

Fields and scalar products. The operators Q_d, $\mathsf{P}_d^{\mathrm{div}}$, H_d, and D_d of the following definition are constructed in Section 9; here we only name them.

Definition 10.1. Fix $n \geq 4$ and $d \geq 0$, and put $\varepsilon = \varepsilon_d = M^{-d}$. A rescaled lattice connection is a function A on the positive links of the fine lattice of side nM^d, with values in $\mathfrak{su}(2)$ or its complexification; it determines the configuration $e^{\varepsilon A}$, $(e^{\varepsilon A})_e = \exp(\varepsilon A_e)$. Let ∇_ε be the collection of all componentwise forward differences divided by ε, and let d_ε be the lattice exterior derivative (the curl) divided by ε. Set

$$\mathscr{S}_\varepsilon(A) = 4\varepsilon^{-1} \sum_p q((e^{\varepsilon A})_p), \qquad \langle A, B\rangle_\varepsilon = \varepsilon^3 \sum_e \langle A_e, B_e\rangle, \qquad \|A\|_{C^1_\varepsilon} = \|A\|_\infty + \|\nabla_\varepsilon A\|_\infty,$$

where p runs over the positive fine plaquettes and e over the positive fine links. We also write $\mathscr{S}_\varepsilon(U)$ for the same sum $4\varepsilon^{-1}\sum_p q(U_p)$ at a fine configuration U. On coarse fields we use the counting scalar product $\sum_c \langle \theta_c, \eta_c\rangle$, where c runs over the positive coarse links.

Let Q_d be the centered linear block average, $\mathsf{P}_d^{\mathrm{div}}$ the orthogonal projection onto $\mathscr{L}_\varepsilon \ker Q'_d$, where Q'_d is the scalar centered box average and $\mathscr{L}_\varepsilon$ the positive scalar Laplacian, and H_d the linear constrained extension, all as in Theorem 9.3, where they are denoted Q_d, P_d, and H_d. Thus

$$\mathsf{Q}_d \mathsf{H}_d = I, \qquad \mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon \mathsf{H}_d = 0.$$

Write

$$\mathcal{V}_d = \ker \mathsf{Q}_d \cap \ker(\mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon).$$

The constrained Green operator C_d is defined by the conditions $\mathsf{C}_d f \in \mathcal{V}_d$ and

$$\langle d_\varepsilon v, d_\varepsilon \mathsf{C}_d f\rangle_\varepsilon = \langle v, f\rangle_\varepsilon \qquad (v \in \mathcal{V}_d);$$

its existence is proved in Lemma 10.2. Let D_d be the coarse correction and $\mathsf{P}_d^{\mathrm{cl}}$ the nonlinear constraint parametrization of Theorem 9.6, where they are denoted $\mathscr{D}_d$ and Ξ_d:

$$\mathsf{P}_d^{\mathrm{cl}}(w) = w - \mathsf{H}_d \mathsf{D}_d(w), \qquad w = v + \mathsf{H}_d\theta, \quad v \in \mathcal{V}_d.$$

Here $\mathsf{D}_d(0) = D\mathsf{D}_d(0) = 0$, and the field $\mathsf{P}_d^{\mathrm{cl}}(w)$ has gauge-corrected block average e^θ and projected divergence $\mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon \mathsf{P}_d^{\mathrm{cl}}(w) = 0$; the gauge-corrected block average is defined by the recursion (10.9) below. A superscript $\sharp$ denotes the adjoint with respect to the fine and coarse scalar products just defined, with the bilinear transpose on complexifications. In particular

$$((D\mathsf{D}_d(w))^\sharp u)_e = \varepsilon^{-3} \sum_c (D_{w_e} \mathsf{D}_{d,c}(w))^T u_c.$$

None of these objects removes the constant (harmonic) part of the coarse field θ.

The letter d denotes the depth, and d_ε the exterior derivative. The constrained Green operator is C_d, while in Section 10.7 the symbol $\mathsf{C}_{c'}$, indexed by a coarse link c', denotes the block averaging constraint of Definition 4.3. In Sections 10.3 and 10.8 we write $\epsilon_j = M^{j-d}$ for the mesh of the lattice reached after j averaging steps from the fine lattice; thus $\epsilon_0 = \varepsilon_d$ is the fine mesh, and the coarse mesh is $\epsilon_d = 1$. The variable w of Definition 10.1 is a fine field; it is not one of the normalized fluctuation variables of later sections, which do not occur here.

Coefficient norms. We use the weighted absolute coefficient norms of Definition 4.6. Let T be a tensor with one output index x and input indices $y_1, \dots, y_s$. Typically T is the derivative of order s, at a given point, of a map between spaces of fields, and an index consists of a link of some lattice, a color, and, where relevant, a component type. Each index has a spatial location. Let $\mathsf{t}(x; y_1, \dots, y_s)$ be the length, in coarse lattice units, of a shortest connected network containing the locations of all the indices; internal vertices are allowed. For $a \geq 0$ the output row and the counting input column of T are

$$\|T\|_{a,\mathrm{out}} = \sup_x \sum_{y_1,\dots,y_s} e^{a\mathsf{t}} |T(x; y_1, \dots, y_s)|,$$
$$\|T\|_{a,\mathrm{input,count}} = \sup_{y_1} \sum_x \sum_{y_2,\dots,y_s} e^{a\mathsf{t}} |T(x; y_1, \dots, y_s)|,$$

where $|\cdot|$ is the norm of the color block and $\mathsf{t} = \mathsf{t}(x; y_1, \dots, y_s)$. A derivative tensor is symmetric in its inputs, so fixing y_1 is the same as fixing any one input. If the output lives on a lattice of mesh a_{out} and the inputs on a lattice of mesh a_{in}, the volume-normalized input column of Definition 4.6 is $(a_{\mathrm{out}}/a_{\mathrm{in}})^3$ times the counting column. For a tensor with coarse inputs and outputs at mesh ε we write $\|T\|_{a,\mathrm{input};\varepsilon^3}$ for the volume-normalized column, which is ε^3 times the counting column. For a scalar function E of a coarse field, $\|D^s E\|_{a,\mathrm{root}}$ is the supremum over one argument of the weighted sum of $|D^s E|$ over the remaining arguments; by symmetry every argument may serve as the fixed one. For a matrix T, $\|T\|_{a,\mathrm{row+col}}$ is the sum of the output row and the counting input column. For matrices whose input and output indices have the same mesh, counting and volume-normalized columns coincide. For these matrices, and with the same spatial weights, the sum used here lies between the maximum used in Sections 4, 14, and 31 and

twice that maximum. The bounds imported from Theorem 8.3 use the same sum. The matrices measured in this sum in Lemma 10.9 and Theorem 10.10 have their input and output indices on one lattice, and their upper bounds therefore imply the corresponding maximum-norm bounds in Section 14 with unchanged constants. For matrices between different meshes, the column measures must first be related by the factor $(a_{\mathrm{out}}/a_{\mathrm{in}})^3$. We omit a from the notation when $a = 0$. These norms are sums of absolute Taylor coefficients. A bound on directional derivatives, or on the values of an analytic map, does not control them, which is why several proofs below work with majorant series for the coefficients.

For example, $D_\theta^2 A(\theta)$, for a map $\theta \mapsto A(\theta)$ from coarse fields to fine fields, has a fine output link e and coarse input links c_1, c_2. Its output row is $\sup_e \sum_{c_1,c_2} e^{a\mathsf{t}(e;c_1,c_2)} |D_\theta^2 A(\theta)(e;c_1,c_2)|$, and its volume-normalized input column is

$$\sup_{c_1} \varepsilon^3 \sum_e \sum_{c_2} e^{a\mathsf{t}(e;c_1,c_2)} |D_\theta^2 A(\theta)(e;c_1,c_2)|.$$

There are of order ε^{-3} fine links within distance one of c_1, and the factor ε^3 makes their contribution of order one. Since $\mathsf{t}(e;c_1,c_2)$ is at least each of the distances $\mathrm{dist}(e,c_1)$, $\mathrm{dist}(e,c_2)$, and $\mathrm{dist}(c_1,c_2)$, bounds on these norms give exponential decay in all three distances.

Normalizations. The following factors occur repeatedly.

1. The fine scalar product carries the factor ε^3 and the coarse one does not. For a linear map L from fine to coarse fields with kernel $L(c,e)$, the adjoint is $(L^\sharp u)_e = \varepsilon^{-3} \sum_c L(c,e)^T u_c$, and for a linear map L from coarse to fine fields, $(L^\sharp f)_c = \varepsilon^3 \sum_e L(e,c)^T f_e$.
2. By Theorem 9.3, the kernels of H_d and $\nabla_\varepsilon \mathsf{H}_d$ have bounded weighted rows and bounded volume-normalized columns

$$\sup_c \varepsilon^3 \sum_e e^{a\,\mathrm{dist}(e,c)} \big(|\mathsf{H}_d(e,c)| + |\nabla_\varepsilon \mathsf{H}_d(e,c)| \big) \le C_a;$$

   their counting columns are of order ε^{-3}.
3. The derivatives of D_d have bounded output rows, and their counting columns with a fine input fixed carry the factor ε^3 (Theorem 9.6 and Lemma 10.4 below). When H_d and a derivative of D_d are contracted over a fine index, the factors ε^{-3} and ε^3 cancel. In the adjoint $(D\mathsf{D}_d)^\sharp$, the factor ε^{-3} of item (1) cancels the factor ε^3 of the column.
4. The fine lattice at depth d over a coarse lattice of side Mm is also the fine lattice at depth $d+1$ over the coarse lattice of side m. In the new coarse units the fine mesh is ε/M. The rescaled connection of a fixed configuration is multiplied by M, its first differences by M^2, and $\mathscr{S}_{\varepsilon/M} = M\mathscr{S}_\varepsilon$. With the couplings $g_+^2 = Mg^2$, the action divided by the square of the coupling is unchanged.

10.2 The constrained Green operator

We invert the Euler equation of the variational problem with the operator C_d of Definition 10.1. A bound on C_d from the supremum norm to the supremum norm of the field and its first differences, uniform in d and n, does not follow from the coercivity of the curl form on $\mathcal{V}_d$. We obtain it from Bałaban's estimate for the constrained Green operator. A unit block below is a cube of side one in coarse lattice units, and $\|u\|_{C^1_\varepsilon(x)}$ is the supremum of $|u|$ and $|\nabla_\varepsilon u|$ over the fine links in the unit block x.

Lemma 10.2. *The operator C_d of Definition 10.1 exists, is real, and is self-adjoint with respect to $\langle\cdot,\cdot\rangle_\varepsilon$. There exist $b, K > 0$ such that the counting kernels of C_d and of $\nabla_{\varepsilon_d}\mathsf{C}_d$ satisfy*

$$\sup_e \sum_f e^{b\,\mathrm{dist}(e,f)} \big(|\mathsf{C}_d(e,f)| + |\nabla_{\varepsilon_d}\mathsf{C}_d(e,f)|\big) \le K. \tag{10.1}$$

Further, there exist $a, C > 0$ such that, for every fine field f,

$$\|\mathsf{C}_d f\|_{C^1_{\varepsilon_d}} \le C\|f\|_\infty,$$

and, if f is supported in one unit block y, then for every unit block x

$$\|\mathsf{C}_d f\|_{C^1_{\varepsilon_d}(x)} \le C e^{-a\,\mathrm{dist}(x,y)}\|f\|_\infty.$$

The counting kernel is defined by $(\mathsf{C}_d f)_e = \sum_f \mathsf{C}_d(e,f) f_f$; since the variational definition uses $\langle\cdot,\cdot\rangle_\varepsilon$, it already contains the factor ε^3 of the source. The first bound in the lemma follows from (10.1).

Proof of Lemma 10.2. Existence and self-adjointness are finite-dimensional linear algebra (Step 1). The estimates come from [13, Theorem 3.13, p. 426], which applies to periodic tori whose sides satisfy divisibility conditions relative to fixed auxiliary localization sizes. We first work on a finite periodic cover of side kn which satisfies these conditions (Step 2), identify C_d on the cover with Bałaban's constrained Green operator (Step 3), apply his estimate (Step 4), and convert it into absolute row bounds (Step 5). Finally we restrict the cover operator to fields invariant under the deck translations (Step 6) and transfer the estimates to side n (Step 7). Throughout, $\varepsilon = \varepsilon_d$.

Step 1: finite solvability. On a torus of any finite side $s \ge 4$, the curl coercivity of Theorem 9.3 makes the form $\langle d_\varepsilon u, d_\varepsilon v\rangle_\varepsilon$ positive definite on $\mathcal{V}_d^{(s)}$, the space $\mathcal{V}_d$ on that torus. Inverting this form after restricting the source pairing $v \mapsto \langle v, f\rangle_\varepsilon$ to $\mathcal{V}_d^{(s)}$, we obtain a unique operator $\mathsf{C}_d^{(s)}$ with the variational definition, and it is real and self-adjoint. If $d = 0$, then $\mathsf{Q}_0 = I$, $\mathcal{V}_0 = \{0\}$, and $\mathsf{C}_0 = 0$, and all assertions hold. We assume $d \ge 1$ from now on.

Step 2: a cover satisfying the conditions of the literature. Fix the admissible auxiliary integers $R_0, M_0, K_{\mathrm{loc}}$ of Lemma 9.2, and define

$$M_{\mathrm{loc}} = K_{\mathrm{loc}} R_0 M_0 \geq M_1(3, M), \qquad k = 6M_{\mathrm{loc}}, \qquad s = kn.$$

These integers depend only on M, not on n or d. At mesh ε the cover has fine side knM^d. The periodic torus of [10, Section A, pp. 17–18] has sides $2L_\mu$ with general side parameters; we take $2L_\mu = kn$ in coarse units, which is possible since k is even. For the scale domains of [11, (2.1)–(2.2), p. 224] we take every domain to be the whole cover. Since their complements are empty, the separation condition imposes no boundary restriction. At scale j the side of the cover, in the lattice units of that scale, is knM^{d-j}, which is divisible by M_{loc}. Both endpoints of the symmetric fundamental interval are boundaries of auxiliary blocks, since $knM^{d-j}/2 = 3M_{\mathrm{loc}} nM^{d-j}$. Hence the auxiliary blocks tile every scale domain.

The domain condition in [13, Section A, p. 393] is the same. The subdivisions there use the auxiliary sizes R_0, M_0, and $K_{\mathrm{loc}} R_0 M_0$. The local regularity step of [13, Section C, pp. 408–409] uses enlarged localization cubes $\widetilde{\square}^5$ of side $12M_{\mathrm{loc}}$ in the units of their own scale. Their side in coarse units is at most $12M_{\mathrm{loc}}$. Since $kn \geq 24M_{\mathrm{loc}}$, these cubes have embedded lifts in the cover. This verifies the subdivision, divisibility, and size requirements and the empty boundary condition, which are the geometric conclusions recorded in Lemma 9.2. A lower bound on n alone would not give these requirements on the torus of side n; this is why we pass to the cover.

Step 3: identification of the operator on the cover. We use the identity background $U = I$, zero background force, and zero local background potentials. Both regularity conditions (3.35)–(3.36) of [13, pp. 396–397] then have vanishing left sides. The dimension is 3, the blocking factor is our odd M, and the auxiliary localization size is M_{loc}. The constrained Gaussian covariance is defined in [13, (3.148), p. 425] and represented in [13, (3.153), p. 426]; at this background the additional background quadratic terms vanish.

The constraints match ours. By Theorem 9.3, the centered translation T_d conjugates the published vector average to Q_d, that is, $\mathsf{Q}_d = Q_d^{\mathrm{pub}} \mathsf{T}_d$. The scalar centered box average Q'_d is obtained from the published scalar average by the same fine translation. Since the translation is orthogonal and commutes with the scalar Laplacian, it also conjugates the orthogonal projection onto $\mathscr{L}_\varepsilon \ker Q_d'^{\mathrm{pub}}$ to $\mathsf{P}_d^{\mathrm{div}}$. The two constraints become $\mathsf{Q}_d v = 0$ and $\mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon v = 0$. The translation has bounded length in coarse units and preserves the fine scalar product. It changes localization distances and block assignments by at most a fixed collar, which is absorbed into the constants.

The penalty used in [13] does not change the operator on $\mathcal{V}_d$. For a penalty $\alpha > 0$, we define

$$\mathscr{D}^0_{d,\alpha} = d_\varepsilon^* d_\varepsilon + \nabla_\varepsilon \mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon + \alpha \mathsf{Q}_d^\sharp \mathsf{Q}_d.$$

For $u, v \in \mathcal{V}_d$ its pairing is $\langle d_\varepsilon v, d_\varepsilon u\rangle_\varepsilon$, since the last two terms vanish by the two constraints. In particular the covariance on $\mathcal{V}_d$ is the same whether the positive homogeneous penalty of [13], with its parameter equal to one, or the penalty $\alpha = 1$ is used. The Gaussian quadratic form is $\|d_\varepsilon v\|_\varepsilon^2/2$, with the source pairing $\langle v, f\rangle_\varepsilon$. No constant mode has to be removed separately: a constant field with zero block average is zero. By uniqueness of this finite variational problem, the translated published covariance on the cover equals $\widehat{\mathsf{C}}_d = \mathsf{C}_d^{(kn)}$.

Step 4: Bałaban's estimate on the cover. We use a special case. On a periodic torus satisfying the conditions verified in Step 2, at the identity background with zero background force and zero local background potentials, [13, Theorem 3.13, p. 426] transfers [13, Theorem 3.3, p. 399] to the constrained covariance (3.148). We use only the entries of [13, (3.42), p. 397] for the potential and its first differences, in their global form [13, (3.47), p. 398]. The scale in these entries is $M^d\varepsilon = 1$, so both scale factors equal one. With the hypotheses checked in Steps 2 and 3, the result on the cover is: there exist $K_0, b_0 > 0$ such that

$$\|\widehat{\mathsf{C}}_d f\|_{C^1_{\hat\varepsilon}(\hat x)} \le K_0 e^{-b_0 \operatorname{dist}_{kn}(\hat x,\hat y)} \|f\|_\infty \quad \text{if } \operatorname{supp} f \text{ is contained in the unit block } \hat y. \tag{10.2}$$

The entry for the covariant Laplacian, which is excepted in Theorem 3.13, is not used.

Step 5: absolute rows on the cover. Fix one scalar output component at a fine link $\hat e$ and one source unit block, and let each scalar source coordinate in that block be the sign of the real kernel coefficient of that output. The supremum norm of this source is bounded by a fixed color factor, and its output at $\hat e$ is the sum of the absolute values of the coefficients in the block. Applying (10.2) to this source, and repeating for the finitely many color components and first-difference outputs, we find that the sum of the absolute kernel coefficients over a unit block $\hat y$ is at most $K_0' e^{-b_0 \operatorname{dist}_{kn}(\hat x,\hat y)}$, where $\hat x$ is the unit block containing $\hat e$. For $0 < b < b_0$, summing over source blocks, we obtain

$$\sup_{\hat e} \sum_{\hat f} e^{b \operatorname{dist}_{kn}(\hat e,\hat f)} \left(|\widehat{\mathsf{C}}_d(\hat e,\hat f)| + |\nabla_\varepsilon \widehat{\mathsf{C}}_d(\hat e,\hat f)|\right) \le K_1. \tag{10.3}$$

In the summation we used that the number of unit blocks in a shell of radius R and thickness one is at most $K(1+R)^2$, uniformly on periodic tori, and that the remaining exponential is summable. The counting kernel already contains the factor ε^3 of the source, and the choice of signs introduces no factor that counts the fine source coordinates.

Step 6: restriction to deck-invariant fields. Let π^* repeat fields from side n to side kn, and let $\mathcal{D} = (\mathbb{Z}/k\mathbb{Z})^3$ act on the cover by the translations by $n\eta$, $\eta \in \mathcal{D}$, in coarse units. These translations are integer multiples of every lattice spacing M^{j-d}, $0 \le j \le d$, so they preserve the centered blocks, their patterns at all scales, and the fine translation T_d. The

vector and scalar averages, the differences, and the adjoints therefore intertwine π^*. Both the fine and the coarse scalar products satisfy

$$\langle \pi^* u, \pi^* v \rangle_{kn} = k^3 \langle u, v \rangle_n. \tag{10.4}$$

The scalar projection needs an additional argument, since it is not a local operator. We write $\mathcal{S}^{(s)} = \mathscr{L}_\varepsilon \ker Q_d'^{(s)}$; the cover space $\mathcal{S}^{(kn)}$ is invariant under $\mathcal{D}$. Let $g = \mathscr{L}_\varepsilon \lambda$ be an element of $\mathcal{S}^{(kn)}$ invariant under $\mathcal{D}$. The average $\overline{\lambda}$ of λ over $\mathcal{D}$ is of the form $\pi^* \lambda_0$, it still has zero scalar average, and $\mathscr{L}_\varepsilon \overline{\lambda} = g$. Since π^* is injective and $\pi^* Q_d'^{(n)} \lambda_0 = Q_d'^{(kn)} \pi^* \lambda_0 = 0$, we have $Q_d'^{(n)} \lambda_0 = 0$. Conversely, the repetition of every element of $\mathcal{S}^{(n)}$ is an invariant element of $\mathcal{S}^{(kn)}$. We conclude that

$$\mathcal{S}^{(kn)} \cap \operatorname{Fix} \mathcal{D} = \pi^* \mathcal{S}^{(n)}.$$

Let g_0 be an arbitrary scalar field on the torus of side n. Since the orthogonal projection onto $\mathcal{S}^{(kn)}$ commutes with the unitary action of $\mathcal{D}$, the projection of $\pi^* g_0$ is invariant. It is therefore $\pi^* h$ for some $h \in \mathcal{S}^{(n)}$, and $\pi^*(g_0 - h)$ is orthogonal to $\pi^* \mathcal{S}^{(n)}$. By (10.4), $g_0 - h$ is orthogonal to $\mathcal{S}^{(n)}$, that is, $h = \mathsf{P}_d^{\mathrm{div},(n)} g_0$. Combining this with the identities for the averages and the divergence, we obtain

$$\begin{aligned} \mathsf{P}_d^{\mathrm{div},(kn)} \pi^* &= \pi^* \mathsf{P}_d^{\mathrm{div},(n)}, \\ \mathcal{V}_d^{(kn)} \cap \operatorname{Fix} \mathcal{D} &= \pi^* \mathcal{V}_d^{(n)}. \end{aligned} \tag{10.5}$$

This verifies the nonlocal constraint in Lemma 9.2.

Let f be a field on the torus of side n, and let $\widehat{u} = \widehat{\mathsf{C}}_d \pi^* f$. A deck translation preserves $\mathcal{V}_d^{(kn)}$, the curl form, and the source $\pi^* f$, so it maps $\widehat{u}$ to a solution of the same variational problem on the cover. By uniqueness, $\widehat{u} = \pi^* u_0$ for some u_0, and (10.5) implies that $u_0 \in \mathcal{V}_d^{(n)}$. Testing the equation on the cover against $\pi^* v$, $v \in \mathcal{V}_d^{(n)}$, and using (10.4), we have

$$k^3 \langle d_\varepsilon v, d_\varepsilon u_0 \rangle_{\varepsilon,n} = k^3 \langle v, f \rangle_{\varepsilon,n}.$$

Canceling k^3, we see that u_0 solves the variational problem on the torus of side n. By uniqueness,

$$\widehat{\mathsf{C}}_d \pi^* = \pi^* \mathsf{C}_d^{(n)}. \tag{10.6}$$

This establishes the hypotheses of the constrained curl case of Lemma 9.2.

Evaluating (10.6) on a coordinate source, we obtain for any lifts $\widetilde{e}, \widetilde{f}$ the identity of counting kernels

$$\mathsf{C}_d^{(n)}(e, f) = \sum_{\eta \in \mathcal{D}} \widehat{\mathsf{C}}_d(\widetilde{e}, \widetilde{f} + n\eta). \tag{10.7}$$

It also holds after a first difference in the output: differences commute with repetition, also across periodic seams, where the neighboring output is taken at a neighboring lift. There is no factor k^{-3}, because repetition repeats a source rather than averaging it.

Step 7: the estimates on the torus of side n. For every image of the source, $\mathrm{dist}_n(e,f) \le \mathrm{dist}_{kn}(\widetilde{e},\widetilde{f}+n\eta)$. By the triangle inequality in (10.7) and summation over f and η, the weighted row in (10.1) is bounded by the left side of (10.3), since the images $\widetilde{f}+n\eta$ run over the positive source links of the cover exactly once. The identity for first differences gives the same bound for $\nabla_\varepsilon \mathsf{C}_d^{(n)}$. This proves (10.1), without a multiplicity factor, and the bound from the supremum norm to C^1_ε follows.

If f is supported in one unit block y, then $\pi^* f$ is the sum of the sources in the k^3 image blocks of y. At a lift $\widetilde{x}$ of x, using (10.2) and $\mathrm{dist}_{kn}(\widetilde{x},\widetilde{y}+n\eta) \ge \mathrm{dist}_n(x,y)$, we obtain

$$\|\mathsf{C}_d^{(n)} f\|_{C^1_\varepsilon(x)} \le K_0 \sum_{\eta\in\mathcal{D}} e^{-b_0\,\mathrm{dist}_{kn}(\widetilde{x},\widetilde{y}+n\eta)} \|f\|_\infty \le k^3 K_0 e^{-b_0\,\mathrm{dist}_n(x,y)} \|f\|_\infty .$$

Since the factor k^3 is a fixed function of M, this is the bound for sources in one unit block, for every $n \ge 4$. Taking the smaller of the decay rates and the larger of the constants proves all assertions for real sources. Complex sources follow by splitting into real and imaginary parts. □

10.3 Analytic block maps and constraint coordinates

In this section we prove coefficient bounds for the composition of d gauge-corrected averaging steps and for the coarse correction D_d, with one radius and one decay rate for all derivative orders and all depths. These bounds strengthen the fixed-order estimates of Lemma 8.10 and Theorem 9.6 to convergent majorant series. We first isolate the composition argument, since it is used again in Section 10.8.

Let $d \ge 1$ and $\epsilon_j = M^{j-d}$. By level j we mean the lattice of mesh ϵ_j, that is, of side nM^{d-j}. A label at level j is a pair consisting of a positive link of this lattice and a component type from a fixed finite set (which may consist of one element); a field at level j is a function on the labels at level j with values in the complexification of $\mathfrak{su}(2)$, and $\|A\|_\infty$ is the maximum of $|A|$ over the labels. The norms below are those of Section 10.1, with the component type as part of the index.

Lemma 10.3. *Let $d \ge 1$, and for $0 \le j < d$ let Φ_j be a holomorphic map from the complex fields A at level j with $\|A\|_\infty < \rho$ to the fields at level $j+1$. Let $K \ge 1$ and $0 < R < \rho/2$, and assume that for $0 \le j < d$, for every complex field A at level j with $\|A\|_\infty \le 2R$, and for every $s \ge 2$, the following hold:*

1. $\|\Phi_j(A)\|_\infty \le \|A\|_\infty + K\epsilon_j \|A\|_\infty^2$;
2. $\|D\Phi_j(A)\|_{\mathrm{out}} \le 1 + K\epsilon_j R$ *and* $\|D\Phi_j(A)\|_{\mathrm{input,count}} \le M^{-3}(1+K\epsilon_j R)$;
3. $\|D^s\Phi_j(A)\|_{\mathrm{out}} \le s! K^s \epsilon_j$ *and* $\|D^s\Phi_j(A)\|_{\mathrm{input,count}} \le M^{-3} s! K^s \epsilon_j$;
4. *each output label of Φ_j depends only on the input labels at distance at most $K\epsilon_j$ from it.*

Put $S = 1/(M-1)$ and assume that $8KSR < 1$. For a complex field A at level 0 with $\|A\|_\infty \le R$ let $A_0 = A$ and $A_{j+1} = \Phi_j(A_j)$. Then A_j is defined and

$$\|A_j\|_\infty \le \frac{\|A\|_\infty}{1 - KS\|A\|_\infty} \le 2\|A\|_\infty \qquad (0 \le j \le d).$$

Further, for every $b \ge 0$ there exists K_1, depending only on K, M, R, and b, such that for all such A, all $0 \le j \le d$, and all $s \ge 1$, the derivatives of the map $A \mapsto A_j$ at A satisfy

$$\|D^s A_j\|_{b,\mathrm{out}} \le K_1^{s+1} s!, \qquad \|D^s A_j\|_{b,\mathrm{input,count}} \le (\epsilon_0/\epsilon_j)^3 K_1^{s+1} s!.$$

Each output label of $A \mapsto A_j$ depends only on the labels at level 0 at distance at most $K\epsilon_j/(M-1)$ from it.

Proof of Lemma 10.3. We begin with the values. Let $a_j = \|A_j\|_\infty$. If some a_j vanishes, then by hypothesis (1) every later a_l vanishes, and their bounds are immediate. We may therefore assume that the iterates under consideration are nonzero when we take reciprocals. As long as $a_j \le 2R$, hypothesis (1) implies $a_{j+1} \le a_j + K\epsilon_j a_j^2$, and hence, if $a_{j+1} > 0$,

$$\frac{1}{a_{j+1}} \ge \frac{1}{a_j(1 + K\epsilon_j a_j)} \ge \frac{1}{a_j} - K\epsilon_j.$$

Since $\sum_{l<d} \epsilon_l = (1 - M^{-d})/(M-1) \le S$, by induction on j we have $1/a_j \ge 1/a_0 - KS$, that is, $a_j \le a_0/(1 - KSa_0)$. As $KSa_0 \le KSR < 1/8$, this is at most $2a_0 \le 2R$, which keeps the induction going.

Next we bound the rows. Fix A with $\|A\|_\infty \le R$, let $R_s(j)$ be the unweighted output row of $D^s A_j$ at A, and let $F_j(t) = \sum_{s \ge 1} R_s(j) t^s/s!$, a power series with nonnegative coefficients. Then $F_0(t) = t$, since $A_0 = A$. We write $F \preceq G$ if every coefficient of $G - F$ is nonnegative. By the chain rule, $D^s(\Phi_j \circ A_j)$ is the sum, over all partitions of $\{1, \dots, s\}$ into blocks $B_1, \dots, B_k$, of $D^k\Phi_j(A_j)[D^{|B_1|}A_j, \dots, D^{|B_k|}A_j]$. By Lemma 5.3, with output rows on all factors, the output row of each term is at most $\|D^k\Phi_j(A_j)\|_{\mathrm{out}} \prod_i R_{|B_i|}(j)$. The sum over partitions with k blocks has generating function $F_j^k/k!$. Since $\|A_j\|_\infty \le 2R$, hypotheses (2) and (3) give

$$F_{j+1} \preceq (1 + K\epsilon_j R)F_j + K^2\epsilon_j \frac{F_j^2}{1 - KF_j}, \tag{10.8}$$

where we used $\sum_{k \ge 2} K^k \epsilon_j F_j^k = K^2\epsilon_j F_j^2/(1 - KF_j)$. Let $P_j = \prod_{l<j}(1 + K\epsilon_l R)$, so that $P_j \le P = \exp(KRS)$, and let $f_j = F_j/P_j$. At a fixed $t > 0$ with $KF_j(t) \le 1/2$, (10.8) implies

$$f_{j+1}(t) \le f_j(t) + 2K^2 P \epsilon_j f_j(t)^2,$$

because $P_j^2/P_{j+1} \le P$. Fix $t_0 > 0$ with $4KPt_0 \le 1$ and $8K^2PSt_0 \le 1$. By the reciprocal inequality used for the values and induction on j, we have $1/f_j(t_0) \ge 1/t_0 - 2K^2PS \ge$

$3/(4t_0)$. In particular, $f_j(t_0) \le 2t_0$ and $F_j(t_0) \le 2Pt_0 \le 1/(2K)$, which keeps the induction going. Since the coefficients are nonnegative, $R_s(j) \le s!\, t_0^{-s} F_j(t_0) \le 2Pt_0\, s!\, t_0^{-s}$.

For the columns, let $C_s(j)$ be the unweighted counting column of $D^s A_j$ at A, with one input at level 0 fixed, and define $\widehat{C}_s(j) = (\epsilon_j/\epsilon_0)^3 C_s(j)$ and $H_j(t) = \sum_{s\ge 1} \widehat{C}_s(j) t^{s-1}/(s-1)!$. Then $H_0 = 1$. In the chain rule, the block containing the fixed input contributes the counting column of $D^{|B|}A_j$, the corresponding slot of $D^k\Phi_j$ contributes the counting column of $D^k\Phi_j$, and all other blocks contribute output rows; this is the contraction rule of Lemma 5.3 with the column on the branch containing the fixed index. The factor M^{-3} in hypotheses (2) and (3) cancels the change from $(\epsilon_j/\epsilon_0)^3$ to $(\epsilon_{j+1}/\epsilon_0)^3 = M^3(\epsilon_j/\epsilon_0)^3$. With $\psi(z) = Kz^2/(1-Kz)$ we obtain

$$H_{j+1} \preceq \big[1 + K\epsilon_j R + K\epsilon_j \psi'(F_j)\big] H_j.$$

At t_0 we have $KF_j(t_0) \le 1/2$ and $\psi'(F_j(t_0)) \le 4$, and therefore $H_j(t_0) \le \exp(KS(R+4))$. Since the coefficients are nonnegative, $C_s(j) \le (\epsilon_0/\epsilon_j)^3 (s-1)!\, t_0^{1-s} H_j(t_0)$.

Finally we add the spatial weights. By hypothesis (4) and induction, each output label of A_j depends only on labels at level 0 at distance at most $\sum_{l<j} K\epsilon_l \le K\epsilon_j/(M-1)$. A nonzero coefficient of $D^s A_j$ therefore has connecting length $\mathrm{t} \le sK\epsilon_j \le sK$, and the weight $e^{b\mathrm{t}}$ is at most e^{bKs}. Combining this with the unweighted bounds and enlarging the constant, we obtain the claim. □

We now apply this lemma to the gauge-corrected block averages. They are defined by the explicit path products of the following lemma, which also gives the gauge transformation relating them to the iterates of T_M.

Lemma 10.4. *Let $d \ge 0$ and $\epsilon_j = M^{j-d}$. For a field A on the lattice of mesh ϵ, with $V = e^{\epsilon A}$, and for a vertex r of the next coarser lattice, set*

$$\begin{aligned} k_\epsilon(A)_r &= \exp\Big\{-M^{-3} \sum_{x \in \mathsf{B}_M(r)} \operatorname{Log} V(\sigma_{r,x})\Big\}, \\ (\mathcal{M}_\epsilon(V))_{rs} &= k_\epsilon(A)_r (T_M V)_{rs} k_\epsilon(A)_s^{-1}, \\ \Psi_\epsilon(A) &= (M\epsilon)^{-1} \operatorname{Log} \mathcal{M}_\epsilon(e^{\epsilon A}), \qquad A_{j+1} = \Psi_{\epsilon_j}(A_j), \quad A_0 = A. \end{aligned} \tag{10.9}$$

The block $\mathsf{B}_M(r)$, the paths $\sigma_{r,x}$, and the coarse link (r,s) are those of Definition 4.3, at the current fine mesh. This is the gauge-corrected averaging step of Lemma 8.10 at the identity background, and e^{A_d} is the gauge-corrected block average of $e^{\epsilon_0 A}$; in the notation of Theorem 9.6, $A_d = \mathsf{F}_d(A)$. There exist $R, R', b, K > 0$ such that the following hold for every complex fine field A with $\|A\|_\infty \le R$.

1. *For $0 \le j \le d$ and $s \ge 1$, the map $A \mapsto A_j$ satisfies $\|D^s A_j\|_{b,\mathrm{out}} \le K^{s+1} s!$ and $\|D^s A_j\|_{b,\mathrm{input,count}} \le (\epsilon_0/\epsilon_j)^3 K^{s+1} s!$.*

2. *The nonlinear remainder* $\mathscr{E}_d^{\mathrm{av}}(A) = A_d - \mathsf{Q}_d A$ *of Theorem 9.6 has no constant or linear term. Its first derivative has output row at most* $K\|A\|_\infty$ *and counting input column at most* $\epsilon_0^3 K\|A\|_\infty$*, and its derivatives of order* $s \geq 2$ *have counting input columns at most* $\epsilon_0^3 K^{s+1} s!$*.*
3. *For the coarse correction* D_d*, which has fine inputs and coarse outputs, let* $\|\cdot\|_{b,\mathrm{input,count}}$ *be the counting sum over the coarse output and all remaining fine inputs with one fine input fixed. Since* $\epsilon_0 = \varepsilon_d$*, the volume-normalized column is* ϵ_0^{-3} *times this sum. For complex fine fields* w *with* $\|w\|_\infty \leq R'$*,*

$$\begin{aligned} \|D^s \mathsf{D}_d\|_{b,\mathrm{out}} &\leq K^s s!, \qquad \|D^s \mathsf{D}_d\|_{b,\mathrm{input,count}} \leq \epsilon_0^3 K^s s! \quad (s \geq 2), \\ \|D\mathsf{D}_d(w)\|_{b,\mathrm{out}} &\leq K\|w\|_\infty, \quad \|D\mathsf{D}_d(w)\|_{b,\mathrm{input,count}} \leq \epsilon_0^3 K\|w\|_\infty. \end{aligned} \tag{10.10}$$

4. *There is a gauge transformation* G_j *at the vertices of level* j*, holomorphic in* A*, such that*

$$e^{\epsilon_j A_j} = (T_M^j e^{\epsilon_0 A})^{G_j}, \qquad G_0 = I, \quad G_{j+1} = k_{\epsilon_j}(A_j)\, G_j|_{\text{next roots}}, \tag{10.11}$$

where $G_j|_{\text{next roots}}$ *is the restriction of* G_j *to the vertices of level* $j+1$*. Its final logarithm satisfies* $\|\mathrm{Log}\, G_d\|_\infty \leq K\|A\|_\infty$*.*

For $d = 0$ *the recursion is empty,* $G_0 = I$*, and* $\mathsf{D}_0 = 0$*.*

We write $G_d(A)$ for the gauge transformation G_d of (10.11) determined by the fine field A; it is the composed gauge transformation denoted $\mathsf{G}_d(A)$ in Theorem 9.9. The definition (10.9) fixes the representatives used later. They are not the iterates of the block averaging map, to which they are related through (10.11).

Proof of Lemma 10.4. For $d = 0$ there is nothing to prove, so let $d \geq 1$. We first prove bounds for one step (Steps 1–3), then apply Lemma 10.3 (Step 4), identify the linear part of the composition and the remainder (Step 5), solve the equation for D_d (Step 6), and finally prove the gauge identity (Steps 7 and 8).

Step 1: local analyticity. Every expression in (10.9) for one output involves at most q_M distinct link variables and paths of length at most q_M, for an integer q_M depending only on M. We shrink a complex polydisc of link angles until every exponential, relative logarithm, and final logarithm in (10.9) is holomorphic there. Cauchy estimates on smaller closed polydiscs and the geometric series for the matrix logarithm give a bound depending only on q_M and the ratio of the radii. Cauchy's formula in these $3q_M$ scalar variables bounds the sum of the absolute values of the derivatives of order s by $s!K_M^s$. The factor from counting the local variables is exponential in s and does not depend on the size of the lattice.

Step 2: the linear part at zero. Before composing the steps, we compute the linear masses exactly. At zero, the derivative of $\mathrm{Log}\, T_M V$ is the average of the sums of A along the paths $F_{c,x}$. The derivative of the reference path enters each relative logarithm with a

negative sign and cancels the derivative of the final reference factor. The derivative of k_ϵ cancels the averaged sums along the initial root paths, and the derivative of its inverse at the other endpoint cancels the averaged sums along the final root paths. After division by $M\epsilon$ there remains the positive operator

$$(QA)_{(r,r+Me_i)} = M^{-4} \sum_{x \in \mathsf{B}_M(r)} \sum_{l=0}^{M-1} A_i(x + le_i). \tag{10.12}$$

Its row mass is 1. A fixed positive fine link $(y, y + e_i)$ of orientation i occurs for the M starting points $x = y - le_i$, $0 \le l < M$, each of which lies in exactly one centered block. Its counting column mass is $M \cdot M^{-4} = M^{-3}$. The starting points are distinct on every input lattice of side at least $4M$, and the same count holds across periodic seams.

Step 3: bounds for one step. We write $\Psi_\epsilon(A) = (M\epsilon)^{-1} P(\epsilon A)$, where P is the fixed local analytic expression of Step 1. Its linear part is Q, and its derivatives of order $s \ge 2$ carry the factor ϵ^{s-1}/M. Together with Steps 1 and 2, this gives, on $\|A\|_\infty \le 2R$,

$$\begin{aligned} &\|D\Psi_\epsilon(A)\|_{\text{out}} \le 1 + K_M\epsilon R, \quad \|D\Psi_\epsilon(A)\|_{\text{input,count}} \le M^{-3}(1 + K_M\epsilon R), \\ &\|D^s\Psi_\epsilon(A)\|_{\text{out}} \le s!K_M^s\epsilon, \qquad \|D^s\Psi_\epsilon(A)\|_{\text{input,count}} \le M^{-3}s!K_M^s\epsilon \quad (s \ge 2). \end{aligned} \tag{10.13}$$

For the columns of the nonlinear part, we note that a fixed input lies in at most q'_M output stencils. Bounding each such coefficient by the row bound and absorbing $M^3 q'_M$ into K_M, we obtain the column bounds. This counting of incidences is applied only to the nonlinear part; the leading factor M^{-3} is the count of Step 2. Since Q has nonnegative scalar coefficients and row mass one, $\|QA\|_\infty \le \|A\|_\infty$. Using this, the Taylor series of $\Psi_\epsilon - Q$ at zero, and (10.13), we obtain, after enlarging K_M,

$$\|\Psi_\epsilon(A)\|_\infty \le \|A\|_\infty + K_M\epsilon\|A\|_\infty^2.$$

Each output of Ψ_ϵ depends only on the fine links in the two blocks of its coarse link, that is, on labels within distance $K_M\epsilon$.

Step 4: composition. By Step 3, the maps $\Phi_j = \Psi_{\epsilon_j}$ satisfy the hypotheses of Lemma 10.3 with $K = K_M$. We choose R so that $8K_MSR < 1$ and the polydisc of Step 1 contains the ball of radius $2R$. By the lemma, the recursion is defined on the common complex ball $\|A\|_\infty \le R$, with $\|A_j\|_\infty \le 2\|A\|_\infty$, and assertion (1) holds at a fixed exponent $b > 0$. Every composed expression depends only on initial inputs within distance $K_M\epsilon_j$ of its output. Its connecting length is therefore at most $K_M s\epsilon_j$ at order s, and the spatial weight does not reduce the rate at any scale or order.

Step 5: the linear map at level d and the remainder. At $j = d$ the linear part of $A \mapsto A_d$ is the product of the kernels (10.12), and it equals Q_d. The centered sums of the base-M

digits give the centered box, and the segment sums give its convolution in the direction of the link. This is also the linear map of Theorem 9.6. Subtracting it, we obtain $\mathscr{E}_d^{\mathrm{av}}$, which has no constant or linear term. Its derivatives of order $s \geq 2$ are those of $A \mapsto A_d$, and assertion (1) at $j = d$, where $\epsilon_d = 1$, gives the factor ϵ_0^3 in their counting columns. Integrating the second derivative from zero, we find that the first derivative of $\mathscr{E}_d^{\mathrm{av}}$ has row $O(\|A\|_\infty)$ and counting input column $O(\epsilon_0^3 \|A\|_\infty)$. This proves assertion (2).

Step 6: the coarse correction. By Theorem 9.6, D_d is the unique small solution of

$$\mathsf{D}_d(w) = \mathscr{E}_d^{\mathrm{av}}(w - \mathsf{H}_d \mathsf{D}_d(w)).$$

The rows of H_d and its volume-normalized columns are bounded by Theorem 9.3. When H_d is contracted with $\mathscr{E}_d^{\mathrm{av}}$ over a fine index, the counting column $O(\epsilon_0^{-3})$ of H_d cancels the counting input column $O(\epsilon_0^3)$ of $\mathscr{E}_d^{\mathrm{av}}$. In a tensor whose fixed input is a fine input w_e, the factor ϵ_0^3 of $\mathscr{E}_d^{\mathrm{av}}$ is kept instead. The weighted tensor composition inequality of Lemma 5.3 applies at a rate below the decay rate of H_d. It shows that all output rows and all columns with a fixed fine input are majorized by the solution of the scalar equation

$$x = K \frac{(t + Kx)^2}{1 - K(t + Kx)}.$$

This is the implicit-series assertion of Lemma 5.3, with zero linear coefficient in x, and it gives (10.10) at $w = 0$, including the factor ϵ_0^3 at a fixed fine input. To pass to a nonzero point w of a smaller ball of radius r, we translate the series: a series with coefficients bounded by K^{s+1} acquires the binomial sum $\sum_{l \geq 0} \binom{s+l}{l} (Kr)^l = (1 - Kr)^{-s-1}$. Contracting the additional slots with the fixed point w does not increase the connecting length of the external indices. One smaller ball and one rate therefore suffice for all the bounds in (10.10). By the uniqueness statement of Theorem 9.6, the solution of the majorized system is D_d. This proves assertion (3).

Step 7: gauge covariance of the block averaging map. For real configurations and compact gauge transformations, the gauge covariance below, including the Borel extension at the logarithm singularity, is stated in Proposition 6.2(3). We recall the argument, since we also need its holomorphic version. Let ω be a periodic vertex gauge transformation and $P = e_1 \cdots e_l$ a path from r to s. When we multiply the transformation laws of the links, the adjacent factors $\omega_x^{-1} \omega_x$ cancel, and we obtain

$$(U^\omega)(P) = \omega_r U(P) \omega_s^{-1}.$$

This also covers reversed links, whose transports are the inverses of the transports of the corresponding positive links. Since both $F_{c,x}$ and C_c run from r to s, their relative product transforms by conjugation:

$$(U^\omega)(F_{c,x})(U^\omega)(C_c)^{-1} = \omega_r \big[U(F_{c,x}) U(C_c)^{-1} \big] \omega_r^{-1}.$$

On the identity chart of the logarithm, conjugation commutes with Log, as one sees by conjugating each term of the power series of $\mathrm{Log}(I+X)$; it also commutes with the exponential. For compact real gauge transformations, the principal real logarithm away from $-I$ has the same property: the conjugated logarithm exponentiates to the conjugated group element and has the same norm, less than 2π, so it is the principal logarithm of that element. The mean logarithm in the definition of T_M is then conjugated at r, and the final reference factor transforms from r to s. This proves

$$T_M(U^\omega) = (T_M U)^{\omega|_{\mathrm{roots}}}, \tag{10.14}$$

where $\omega|_{\mathrm{roots}}$ is the restriction to the vertices of the coarser lattice. The exceptional case is preserved as well: a relative product equals $-I$ if and only if its conjugate does. In that case both sides use the reference path, for which covariance was just proved. The identity (10.14) therefore holds for the block averaging map with its Borel extension. For complex gauge transformations it holds on the domains where the logarithms before and after the transformation exist, and the power series argument gives its holomorphic version.

Step 8: the composed gauge transformation. Let $V_j = e^{\epsilon_j A_j}$, and suppose that $V_j = (T_M^j V_0)^{G_j}$, which holds for $j=0$ with $G_0 = I$. By (10.14), $T_M V_j = (T_M^{j+1} V_0)^{G_j|_{\text{next roots}}}$. The definition (10.9) applies the gauge transformation $k_{\epsilon_j}(A_j)$ to this field. Since $(U^a)^b = U^{ba}$, the new gauge transformation is $G_{j+1} = k_{\epsilon_j}(A_j) G_j|_{\text{next roots}}$. This proves (10.11), with the displayed order of the factors, and its holomorphic continuation.

Each $\mathrm{Log}\, k_{\epsilon_j}(A_j)$ has norm at most $K_M \epsilon_j \|A_j\|_\infty \le 2K_M \epsilon_j \|A\|_\infty$, and the sum of these bounds is at most $K\|A\|_\infty$. After reducing R, products of factors this close to the identity stay in a fixed ball of the logarithm. The power series of the product and of its logarithm give the holomorphy and the bound on $\mathrm{Log}\, G_d$, uniformly in j, d, and n. □

10.4 The Euler equation in constraint coordinates

We now write the condition for a critical point of $v \mapsto \mathscr{S}_\varepsilon(\mathsf{P}_d^{\mathrm{cl}}(v + \mathsf{H}_d\theta))$, $v \in \mathcal{V}_d$, as a fixed point equation that can be solved by contraction.

We begin with a heuristic. If A is the restriction to the lattice of a smooth connection, then the logarithm of the plaquette variable is $\varepsilon^2(\partial_i A_j - \partial_j A_i + [A_i, A_j]) + O(\varepsilon^3)$. Since $q(e^X) = |X|^2/8 + O(|X|^4)$, the action $\mathscr{S}_\varepsilon(A)$ is a Riemann sum for one half of $\sum_{i<j}\int |F_{ij}|^2$, and its gradient should be $d_\varepsilon^* d_\varepsilon A$ plus terms of the form $A\nabla A$ and A^3. However, a direct expansion of the lattice gradient produces a quadratic term with the prefactor ε^{-1}, which is of size $\varepsilon^{-1}|A|^2$ unless it cancels. The first part of the following lemma shows that this term vanishes on fields that are constant in each direction, and hence is of order $\|A\|_\infty \|\nabla_\varepsilon A\|_\infty$. The second part records the gradient of the action in the constraint coordinates. Differentiating these coordinates produces the term $(D\mathsf{D}_d)^\sharp \mathsf{J}_d$ below, which must be kept.

Lemma 10.5. *There exist $r_0, a_0, C > 0$ with the following properties for every $d \geq 0$ and $n \geq 4$, with $\varepsilon = \varepsilon_d$.*

1. *Define the gradient $G_\varepsilon(A)$ of the action by $D\mathscr{S}_\varepsilon(A)[v] = \langle G_\varepsilon(A), v\rangle_\varepsilon$. Then*

$$G_\varepsilon(A) = d_\varepsilon^* d_\varepsilon A + N_\varepsilon(A),$$

where d_ε^* is the adjoint of d_ε with respect to the fine scalar products. On the complex C^1_ε ball of radius r_0 the map N_ε is holomorphic, and for $0 < r \leq r_0$,

$$\|N_\varepsilon(A)\|_\infty \leq C\|A\|^2_{C^1_\varepsilon}, \qquad \|N_\varepsilon(A) - N_\varepsilon(B)\|_\infty \leq Cr\|A - B\|_{C^1_\varepsilon} \quad (\|A\|_{C^1_\varepsilon}, \|B\|_{C^1_\varepsilon} \leq r).$$

2. *There is a holomorphic function $\widetilde{N}_\varepsilon(A, G)$ of a fine field A and a triple $G = (G_\mu)_{\mu=1,2,3}$ of independent fine fields, defined on a fixed complex ball, with $\widetilde{N}_\varepsilon(A, \nabla_\varepsilon A) = N_\varepsilon(A)$, where $G_{i,\mu}$ takes the place of $\nabla_{\varepsilon,\mu} A_i$. It vanishes to second order at the origin, and its derivatives of order $s \geq 2$ have output rows and counting input columns at most $C^{s+1}s!$ at the exponent a_0. Each output of $\widetilde{N}_\varepsilon$ depends only on inputs at distance at most $C_M\varepsilon$.*
3. *For $w = v + \mathsf{H}_d\theta$ with $v \in \mathcal{V}_d$, and $A = \mathsf{P}^{\mathrm{cl}}_d(w)$, put*

$$\Delta_d = \mathsf{H}_d^\sharp d_\varepsilon^* d_\varepsilon \mathsf{H}_d, \qquad \mathsf{J}_d(w) = \Delta_d[\theta - \mathsf{D}_d(w)] + \mathsf{H}_d^\sharp N_\varepsilon(A).$$

If $\|v\|_{C^1_\varepsilon} + \|\theta\|_\infty \leq r \leq r_0$, then

$$\|\mathsf{J}_d(w)\|_\infty \leq C(\|\theta\|_\infty + r^2), \qquad \|(D\mathsf{D}_d(w))^\sharp\|_{\infty\to\infty} \leq Cr.$$

The second bound follows from $\sup_e \sum_c e^{a_0 \operatorname{dist}(e,c)} |D_{w_e}\mathsf{D}_{d,c}(w)| \leq C\varepsilon^3 r$. At fixed θ, the maps $v \mapsto N_\varepsilon(A)$ and $v \mapsto (D\mathsf{D}_d(w))^\sharp \mathsf{J}_d(w)$ are Lipschitz from the C^1_ε norm to the supremum norm, with constant Cr.
4. *For every $u \in \mathcal{V}_d$,*

$$D_v\mathscr{S}_\varepsilon(\mathsf{P}^{\mathrm{cl}}_d(w))[u] = \langle d_\varepsilon u, d_\varepsilon v\rangle_\varepsilon + \langle u, N_\varepsilon(A) - (D\mathsf{D}_d(w))^\sharp \mathsf{J}_d(w)\rangle_\varepsilon.$$

Consequently $D_v\mathscr{S}_\varepsilon(\mathsf{P}^{\mathrm{cl}}_d(w))[u] = 0$ for all $u \in \mathcal{V}_d$ if and only if

$$v = -\mathsf{C}_d\big\{N_\varepsilon(A) - (D\mathsf{D}_d(w))^\sharp \mathsf{J}_d(w)\big\}.$$

In the proof we show that $\mathsf{J}_d(w) = \mathsf{H}_d^\sharp G_\varepsilon(A)$: it is the gradient of the action pulled back to the coarse variables. The bounds of the lemma concern local rows and volume-normalized columns. The sum of $|\mathsf{J}_d(w)|$ over all coarse links is not bounded independently of the volume, and it is not used.

Proof of Lemma 10.5. Parts (1) and (2) are proved in Steps 1–4, part (3) in Steps 5 and 6, and part (4) in Step 7.

Step 1: expansion of the gradient. Fix a positive fine link. Differentiating the finitely many plaquette variables containing it, in the link angles $X_e = \varepsilon A_e$, we see that the gradient $G_\varepsilon(A)$ at that link is $4\varepsilon^{-3}$ times the angle derivative of $\sum_p q$. The common power series of these local expressions has no constant term. Its linear part is $d_\varepsilon^* d_\varepsilon A$, because the quadratic part of the action is $\varepsilon^3 \sum_p |d_\varepsilon A(p)|^2/2$, by $q(e^X) = |X|^2/8 + O(|X|^4)$. A term of degree k in A in the gradient carries the factor ε^{k-3}. After the linear part is subtracted, only the quadratic term can carry a negative power of ε; we write it as $\varepsilon^{-1} P_e(A)$, where P_e is a quadratic polynomial in the values of A near e.

Step 2: the quadratic term vanishes on constant fields. Let $A_i(x) = a_i$ on the whole lattice, for constants a_1, a_2, a_3. A plaquette variable is then the group commutator of $e^{\varepsilon a_i}$ and $e^{\varepsilon a_j}$. Its logarithm begins at degree two, so its value of q begins at degree four, and the cubic part of the action vanishes identically on such fields. At a constant field, translation invariance makes the gradient of the cubic part of the action the same at all links of a given orientation. Differentiating the vanishing cubic polynomial in a_i shows that the number of links of orientation i times this common coefficient is zero. We conclude that $P_e(a) = 0$ for every link e.

Step 3: the nonlinear bounds. Fix a vertex near e as base point, and replace each value $A_i(x)$ in the stencil of P_e by the value $a_i = A_i(\text{base point})$. The differences $A_i(x) - a_i$ are sums of at most K_M first differences times ε. Subtracting the two quadratic polynomials and using $P_e(a) = 0$, we obtain

$$|P_e(A)| \le K\varepsilon \|A\|_\infty \|\nabla_\varepsilon A\|_\infty,$$

and the differential of P_e at A in the direction v is bounded by

$$K\varepsilon\big(\|A\|_\infty \|\nabla_\varepsilon v\|_\infty + \|\nabla_\varepsilon A\|_\infty \|v\|_\infty\big).$$

These identities cancel the apparent inverse power of the mesh before any norm is taken. All terms of degree at least three in the gradient carry nonnegative powers of ε and are given by a uniformly convergent local series; their value and differential are bounded by $K\|A\|_\infty^3$ and $K\|A\|_\infty^2 \|v\|_\infty$. Integrating along the segment from B to A, we obtain the two inequalities of part (1).

Step 4: a version for independent value and difference variables. In the difference of the two quadratic polynomials used in Step 3, we change the arguments one at a time. Every resulting term contains a factor

$$A_i(x) - a_i = \varepsilon \sum_l \pm G_{i,\mu_l}(x_l),$$

a sum of first differences along a path from the base point to x. Treating the remaining local values of A and these variables G as independent, we obtain a polynomial extension

of $\varepsilon^{-1}P_e$ which is quadratic in (A, G). Adding the terms of degree at least three, which are local analytic functions of A, defines $\widetilde{N}_\varepsilon(A, G)$. It agrees with $N_\varepsilon(A)$ when $G = \nabla_\varepsilon A$. Cauchy estimates on a fixed smaller polydisc in finitely many variables bound each absolute derivative tensor of order s by $K^{s+1}s!$. Since a fixed input belongs to a bounded number of stencils, this bound holds with any input fixed as well as with the output fixed. Since the stencils have radius $K_M\varepsilon$, a spatial weight contributes a factor at most $e^{a_0 K_M s}$. Since these arguments use polynomial identities and convergent power series, they hold on the complex balls. This proves part (2).

Step 5: the coarse gradient. We use H_d and its minimizing property from Theorem 9.3: $\langle d_\varepsilon \mathsf{H}_d \eta, d_\varepsilon v\rangle_\varepsilon = 0$ for every coarse field η and every $v \in \mathcal{V}_d$. The kernel of Δ_d is $\Delta_d(c, c') = \varepsilon^3 \sum_p (d_\varepsilon \mathsf{H}_d)(p, c)^T (d_\varepsilon \mathsf{H}_d)(p, c')$, the pairing with the fine measure of two curl kernels of H_d. Since each curl kernel is controlled by the bounds of Theorem 9.3 on first differences, Δ_d has bounded weighted rows and columns. The same volume-normalized column bound implies $\|\mathsf{H}_d^\sharp f\|_\infty \le K\|f\|_\infty$. Since $A = v + \mathsf{H}_d[\theta - \mathsf{D}_d(w)]$, the minimizing property gives the identity

$$\mathsf{H}_d^\sharp G_\varepsilon(A) = \Delta_d[\theta - \mathsf{D}_d(w)] + \mathsf{H}_d^\sharp N_\varepsilon(A) = \mathsf{J}_d(w).$$

The bounds on H_d and on D_d (Theorems 9.3 and 9.6) place A in a C^1_ε ball of radius Kr and give $\|\mathsf{D}_d(w)\|_\infty \le Kr^2$. Together with part (1), this proves $\|\mathsf{J}_d(w)\|_\infty \le K(\|\theta\|_\infty + r^2)$.

Step 6: the adjoint of the derivative of the correction. By (10.10), the counting column of $D\mathsf{D}_d(w)$ with a fine input fixed is at most $K\varepsilon^3 r$, and the same holds with the exponential weight. Taking the adjoint multiplies this column by exactly ε^{-3} (item (1) of the normalizations in Section 10.1), which proves $\|(D\mathsf{D}_d(w))^\sharp\|_{\infty\to\infty} \le Kr$, together with its weighted version. The column bound for the second derivative of D_d bounds the difference of these adjoints at w and $w' = v' + \mathsf{H}_d\theta$ by $K\|v - v'\|_{C^1_\varepsilon}$. At fixed θ, the parametrized fields $\mathsf{P}_d^{\mathrm{cl}}(w)$ and $\mathsf{P}_d^{\mathrm{cl}}(w')$ differ by at most $K\|v - v'\|_{C^1_\varepsilon}$ in C^1_ε, the nonlinear terms N_ε by at most $Kr\|v - v'\|_{C^1_\varepsilon}$ by part (1), and J_d by the same amount. By the product rule and $\|\theta\|_\infty \le r$, the difference of $(D\mathsf{D}_d)^\sharp \mathsf{J}_d$ at w and w' is at most $Kr\|v - v'\|_{C^1_\varepsilon}$. This proves part (3).

Step 7: the tested equation. For $u \in \mathcal{V}_d$, the derivative of $v \mapsto \mathsf{P}_d^{\mathrm{cl}}(v + \mathsf{H}_d\theta)$ in the direction u is $u - \mathsf{H}_d D\mathsf{D}_d(w)u$. Testing the derivative of the action against it and using the adjoints, we have

$$\langle u, G_\varepsilon(A)\rangle_\varepsilon - \sum_c \langle (D\mathsf{D}_d(w)u)_c, \mathsf{J}_d(w)_c\rangle = \langle d_\varepsilon u, d_\varepsilon v\rangle_\varepsilon + \langle u, N_\varepsilon(A) - (D\mathsf{D}_d(w))^\sharp \mathsf{J}_d(w)\rangle_\varepsilon.$$

Here we used $\langle u, d_\varepsilon^* d_\varepsilon A\rangle_\varepsilon = \langle d_\varepsilon u, d_\varepsilon v\rangle_\varepsilon$, which follows from the minimizing property of H_d; only the linear contribution of the extension is removed by this property. By the

defining identity of C_d, the right side vanishes for every $u \in \mathcal{V}_d$ if and only if v satisfies the displayed fixed point equation; uniqueness follows from the coercivity of the curl form on $\mathcal{V}_d$. We decrease r_0 and a_0 to the minima of the finitely many radii and rates used above, and take C larger than the corresponding constants. □

10.5 The classical background field and local minimality

The two descriptions of the constraint. The renormalization group prescribes the block average $T_M^d U$ of a fine configuration. The coordinates of Section 9 prescribe its gauge-corrected block average instead. For a fine field A the gauge-corrected block average is $e^{\mathsf{F}_d(A)}$, where $\mathsf{F}_d(A)$ is the d-th iterate in (10.9). Taking $j = d$ in (10.11), we obtain

$$e^{\mathsf{F}_d(A)} = (T_M^d e^{\varepsilon A})^{G_d(A)}, \qquad \text{that is,} \qquad T_M^d e^{\varepsilon A} = \big(e^{\mathsf{F}_d(A)}\big)^{G_d(A)^{-1}}.$$

Thus a field A with gauge-corrected block average e^θ lies in the fiber of T_M^d over $(e^\theta)^{G_d(A)^{-1}}$. Let Ω be any fine gauge transformation whose values at the coarse vertices are $G_d(A)$. Applying (10.14) d times and using $(U^k)^l = U^{lk}$, we have

$$T_M^d\big((e^{\varepsilon A})^\Omega\big) = \big((e^\theta)^{G_d(A)^{-1}}\big)^{G_d(A)} = e^\theta, \qquad \mathscr{S}_\varepsilon\big((e^{\varepsilon A})^\Omega\big) = \mathscr{S}_\varepsilon(A).$$

We construct the background field in the chart $\{\mathsf{P}_d^{\mathrm{cl}}(v + \mathsf{H}_d\theta) : v \in \mathcal{V}_d\}$. All points of this chart have gauge-corrected block average e^θ and projected divergence zero, and we transfer the conclusions to the fiber $\{U : T_M^d U = e^\theta\}$ with such a gauge transformation. In the opposite direction, we use Theorem 9.9 to move a field near the fiber into the chart by a gauge transformation, in part (6) of the next theorem and in Lemma 10.7.

By Lemma 10.5, the critical points in the chart are the fixed points of the map

$$\mathcal{T}_\theta(v) = -\mathsf{C}_d\big\{N_\varepsilon(A) - (D\mathsf{D}_d(w))^\sharp \mathsf{J}_d(w)\big\}, \qquad w = v + \mathsf{H}_d\theta, \quad A = \mathsf{P}_d^{\mathrm{cl}}(w).$$

In the following theorem, r_{ph} and C_* denote the constants of Theorem 9.7, and $\mathcal{T}$ is called the Euler map.

Theorem 10.6. *There exist $r_v, r_\theta, a, C > 0$ such that the following hold for every $d \geq 0$ and $n \geq 4$, with $\varepsilon = \varepsilon_d$.*

1. *For every complex coarse field θ with $\|\theta\|_\infty < r_\theta$, the Euler map $\mathcal{T}_\theta$ maps the closed complex ball $\{v \in \mathcal{V}_d : \|v\|_{C_\varepsilon^1} \leq r_v\}$ strictly into itself and is a contraction with factor at most $1/2$. Its unique fixed point $v_d^*(\theta)$ is holomorphic in θ and real for real θ, and*

$$\|v_d^*(\theta)\|_{C_\varepsilon^1} \leq C\|\theta\|_\infty^2, \qquad A_d(\theta) = \mathsf{P}_d^{\mathrm{cl}}(v_d^*(\theta) + \mathsf{H}_d\theta),$$
$$\|A_d(\theta)\|_{C_\varepsilon^1} \leq C\|\theta\|_\infty, \qquad DA_d(0) = \mathsf{H}_d.$$

We call $A_d(\theta)$ the classical background field with coarse field e^θ.

2. *For real θ with $\|\theta\|_\infty < r_\theta$, the point $v_d^*(\theta)$ is the unique minimum of $v \mapsto \mathscr{S}_\varepsilon(\mathsf{P}_d^{\mathrm{cl}}(v + \mathsf{H}_d\theta))$ on the closed ball $\{v \in \mathcal{V}_d : \|v\|_{C_\varepsilon^1} \le r_v\}$. It is also the unique minimum of this function on the convex set $\{v \in \mathcal{V}_d : \|v\|_\infty + \|d_\varepsilon v\|_\infty < r_{\mathrm{ph}}\}$, for every r_{ph} allowed in Theorem 9.7 with $\|\theta\|_\infty < r_{\mathrm{ph}}/C_*$.*
3. *For a fine output link e and a coarse input link c, and $\|\theta\|_\infty < r_\theta$,*

$$|D_{\theta_c} A_d(\theta)_e| + |\nabla_\varepsilon D_{\theta_c} A_d(\theta)_e| \le C e^{-a\,\mathrm{dist}(e,c)}.$$

4. *For every integer $s \ge 1$ there exist $a_s, C_s > 0$, depending on s, such that on $\|\theta\|_\infty < r_\theta/2$,*

$$\|D_\theta^s(A_d, \nabla_\varepsilon A_d)\|_{a_s,\mathrm{out}} + \|D_\theta^s(A_d, \nabla_\varepsilon A_d)\|_{a_s,\mathrm{input};\varepsilon^3} \le C_s.$$

5. *Define the effective classical action by*

$$E_d(e^\theta) = \mathscr{S}_\varepsilon(A_d(\theta)), \qquad E_0 = S_W.$$

It is holomorphic on a common ball, nonnegative for real θ, and $E_d(I) = DE_d(I) = 0$. With $w = v_d^*(\theta) + \mathsf{H}_d\theta$, its gradient in θ is

$$\nabla_\theta E_d = (I - D\mathsf{D}_d(w)\mathsf{H}_d)^T \mathsf{J}_d(w).$$

For each $s \ge 2$ there exist $a_s, C_s > 0$ such that $\|D^s E_d\|_{a_s,\mathrm{root}} \le C_s$. For $g > 0$ let $W_{d,g}(B) = g^{-2} E_d(e^{gB})$. If $\|gB\|_\infty < r_\theta/2$, then

$$\|D_B^s W_{d,g}\|_{a_s,\mathrm{root}} \le C_s g^{s-2} \quad (s \ge 2), \qquad \|D_B W_{d,g}\|_\infty \le C\|B\|_\infty.$$

6. *Let θ be real with $\|\theta\|_\infty < r_\theta$, and let Ω_* be a fine gauge transformation whose values at the coarse vertices are $G_d(A_d(\theta))$. Then $U_* = (e^{\varepsilon A_d(\theta)})^{\Omega_*}$ satisfies $T_M^d U_* = e^\theta$, and U_* is a local minimum of $\mathscr{S}_\varepsilon$ on the fiber $\{U : T_M^d U = e^\theta\}$ modulo fine gauge transformations. More precisely, there is a neighborhood of U_* in this fiber on which $\mathscr{S}_\varepsilon(U) \ge \mathscr{S}_\varepsilon(U_*)$, with equality only if $U = U_*^h$ for a fine gauge transformation h whose values at the coarse vertices leave e^θ fixed. Every tangent vector to the fiber at U_* has zero first variation. The neighborhood may depend on the fine lattice; the radii and constants in (1)–(5) do not.*
7. *The constrained Green operator C_d satisfies the bounds of Lemma 10.2: $\|\mathsf{C}_d f\|_{C_\varepsilon^1} \le C\|f\|_\infty$, and $\|\mathsf{C}_d f\|_{C_\varepsilon^1(x)} \le C e^{-a\,\mathrm{dist}(x,y)}\|f\|_\infty$ if f is supported in the unit block y. They give weighted absolute rows for C_d and its first differences, and they are the specialization of [13, Theorem 3.13, p. 426] described in the proof of Lemma 10.2.*

The fixed-order bounds in (4) and (5) follow from the factorial bounds (10.17) and (10.20) proved below, which are also collected in Theorem 10.11. The minimality statements in (2) and (6) concern real coarse fields; for complex θ, the field $A_d(\theta)$ and the value $E_d(e^\theta)$ are holomorphic continuations of the critical point and the critical value.

Proof of Theorem 10.6. Step 1 constructs the fixed point by contraction. Steps 2 and 3 prove factorial coefficient bounds for A_d, which give (3) and (4). Step 4 proves (2). Steps 5 and 6 treat the effective classical action, and Step 7 proves the local minimality on the fiber. Part (7) is Lemma 10.2, applied on the torus of side n. Since the geometry of the cover, both constraints, and the variational identity were checked there, no divisibility condition or additional lower bound on n enters.

Step 1: contraction. Let K be larger than the constants of the extension H_d, of the correction D_d, of the nonlinear remainder of the action gradient in Lemma 10.5, and of the Green operator in Lemma 10.2, on fixed larger domains. By Lemmas 10.2 and 10.5, for $\|v\|_{C^1_\varepsilon}, \|v'\|_{C^1_\varepsilon} \le r_v$ and $\|\theta\|_\infty \le r_\theta$,

$$\begin{aligned} \|\mathcal{T}_\theta(v)\|_{C^1_\varepsilon} &\le K(\|v\|_{C^1_\varepsilon} + \|\theta\|_\infty)^2, \\ \|\mathcal{T}_\theta(v) - \mathcal{T}_\theta(v')\|_{C^1_\varepsilon} &\le K(r_v + r_\theta)\|v - v'\|_{C^1_\varepsilon}. \end{aligned} \tag{10.15}$$

Fix $r_v > 0$ with $16Kr_v \le 1$ and so small that the closed C^1_ε ball of radius r_v in $\mathcal{V}_d$ lies inside all the larger domains. We also require that $3r_v$ is an allowed value of r_{ph} in Theorem 9.7. Since $\|d_\varepsilon v\|_\infty \le 2\|\nabla_\varepsilon v\|_\infty$, the closed C^1_ε ball of radius r_v is then contained in the convex domain $\{v \in \mathcal{V}_d : \|v\|_\infty + \|d_\varepsilon v\|_\infty < 3r_v\}$ of that theorem. Next fix $0 < r_\theta \le r_v$ so that $\mathsf{H}_d\theta$ and the correction lie in their domains with fixed margins, uniformly in d and n, and so that $r_\theta < 3r_v/C_*$ and $6Kr_\theta \le C_*$. Then (10.15) implies that $\|\mathcal{T}_\theta(v)\|_{C^1_\varepsilon} \le 4Kr_v^2 \le r_v/4$, and it gives a contraction factor at most $2Kr_v \le 1/8$. Iteration from zero converges, since successive differences are bounded by a geometric series. The limit lies in the closed linear subspace $\mathcal{V}_d$, and it is the unique fixed point $v_d^*(\theta)$ in the closed ball. Comparing the fixed point with the first iterate $\mathcal{T}_\theta(0)$, we obtain

$$\|v_d^*(\theta)\|_{C^1_\varepsilon} \le \tfrac{1}{2}\|v_d^*(\theta)\|_{C^1_\varepsilon} + K\|\theta\|_\infty^2.$$

This proves the quadratic bound $\|v_d^*(\theta)\|_{C^1_\varepsilon} \le 2K\|\theta\|_\infty^2$. Since $A_d(\theta) = v_d^*(\theta) + \mathsf{H}_d\theta - \mathsf{H}_d\mathsf{D}_d(w)$ and $\|\mathsf{D}_d(w)\|_\infty \le K\|w\|_\infty^2$, it also implies the bound on A_d and $DA_d(0) = \mathsf{H}_d$. Since the equation is preserved by complex conjugation of all fields, uniqueness shows that $v_d^*(\theta)$ is real for real θ. Holomorphy follows from Step 2.

Step 2: factorial bounds for the background field. We prove analyticity and locality through the coefficients. Let $p = (v, V_1, V_2, V_3)$ be independent variables for a fine field and its first differences in the three directions. In the nonlinear expressions we replace A and its differences by

$$\Big(v + \mathsf{H}_d[\theta - \mathsf{D}_d(w)], \quad V_\mu + \nabla_{\varepsilon,\mu}\mathsf{H}_d[\theta - \mathsf{D}_d(w)]\Big), \qquad w = v + \mathsf{H}_d\theta, \tag{10.16}$$

and we use the extension $\widetilde{N}_\varepsilon$ of Lemma 10.5. If f denotes the negative of the resulting nonlinear constrained first variation, we define the output to be the pair $(\mathsf{C}_d f, \nabla_\varepsilon \mathsf{C}_d f)$.

This is an analytic map on a product ball. Its output is a field together with its own first differences, its field component lies in $\mathcal{V}_d$, and when $V_\mu = \nabla_{\varepsilon,\mu} v$ it coincides with the Euler map.

Every factor has a majorant series with nonnegative coefficients at one rate $b > 0$: (10.10) for the correction, part (2) of Lemma 10.5 for the nonlinear remainder of the action gradient, (10.1) for the pair $(\mathsf{C}_d, \nabla_\varepsilon \mathsf{C}_d)$, and the rows and volume-normalized columns of Theorem 9.3 for H_d and $\nabla_\varepsilon \mathsf{H}_d$. The kernel Δ_d is the contraction of two curl kernels of H_d with the fine measure, as in Step 5 of the proof of Lemma 10.5, and $\mathsf{H}_d^\sharp$ carries the factor ε^3 in its sum. The only apparent inverse power of the volume is in

$$((D\mathsf{D}_d(w))^\sharp u)_e = \varepsilon^{-3} \sum_c (D_{w_e} \mathsf{D}_{d,c}(w))^T u_c.$$

An output row of this expression fixes a fine input w_e of the tensor of D_d, and the column bound in (10.10) supplies the factor ε^3 that cancels ε^{-3}. Taking s further derivatives produces a factor at most $(s+1)! K^{s+1}$; after division by $s!$, the additional factor $s + 1 \le 2^{s+1}$ changes only the constant.

Since the pair map has no constant and no linear term in (θ, p), the part of Lemma 5.3 concerning output rows majorizes the implicit solution by the solution of $x = K(t + x)^2/(1 - K(t + x))$ on a fixed small disc. The comparison can also be read off from Picard iteration: the coefficient of degree s of the solution depends only on coefficients of lower degree, and the nonnegative scalar recursion bounds all of them. The convergent series defines a holomorphic solution on every finite lattice, and the contraction of Step 1 identifies it with v_d^*. Composing with the majorants of H_d and D_d, we obtain, at $\theta = 0$ and on a smaller common ball of coarse fields,

$$\|D_\theta^s (A_d, \nabla_\varepsilon A_d)\|_{b,\mathrm{out}} \le K^{s+1} s! \qquad (s \ge 1). \tag{10.17}$$

To pass from $\theta = 0$ to the smaller ball, we use the binomial sum of Step 6 of the proof of Lemma 10.4, at the same rate.

Step 3: from rows to columns. The rows also give the volume-normalized columns. Fix a coarse input c_1 and $0 < b' < b$. For a coefficient $T_s(e; c_1, \dots, c_s)$ of order s, the connecting length satisfies $\mathsf{t} \ge \mathrm{dist}(e, c_1)$. Therefore

$$\begin{aligned} &\varepsilon^3 \sum_e \sum_{c_2,\dots,c_s} e^{b'\mathsf{t}} |T_s(e; c_1, \dots, c_s)| \\ &\qquad \le K^{s+1} s!\, \varepsilon^3 \sum_e e^{-(b-b')\,\mathrm{dist}(e,c_1)} \le K_b K^{s+1} s!. \end{aligned} \tag{10.18}$$

In the first inequality we used $e^{b'\mathsf{t}} \le e^{b\mathsf{t}} e^{-(b-b')\,\mathrm{dist}(e,c_1)}$ and (10.17). The last inequality follows by summing over fine cubes in annuli around c_1: ε^3 times the number of fine links within distance R is at most $K(1+R)^3$, on every periodic torus and for every mesh

at most one. By symmetry any c_i may be the fixed input. Taking $s = 1$ in (10.17) gives part (3). Taking $a_s = b'$ and enlarging C_s, we obtain part (4), with no loss of exponent depending on the order.

Step 4: minimality in the chart. Let θ be real. The fixed point $v_d^*(\theta)$ is an interior point of the convex domain $\{v \in \mathcal{V}_d : \|v\|_\infty + \|d_\varepsilon v\|_\infty < 3r_v\}$, and its tested Euler equation vanishes by Lemma 10.5. The formula for the Hessian and the stability inequality of Theorem 9.7 then show that $v_d^*(\theta)$ is the unique critical point and the unique minimum throughout this domain, and in particular on the closed C^1_ε ball of radius r_v. Now let r_{ph} be any value allowed in Theorem 9.7 with $\|\theta\|_\infty < r_{\mathrm{ph}}/C_*$. By Step 1, $\|v_d^*\|_\infty + \|d_\varepsilon v_d^*\|_\infty \le 3\|v_d^*\|_{C^1_\varepsilon} \le 6K\|\theta\|_\infty^2$, and since $6K\|\theta\|_\infty \le C_*$ this is at most $C_*\|\theta\|_\infty < r_{\mathrm{ph}}$. Thus $v_d^*(\theta)$ is an interior point of the larger domain as well, all its constrained first variations vanish, and the same stability inequality gives uniqueness and minimality there. This proves part (2).

Step 5: the effective classical action. The finite sum defining $\mathscr{S}_\varepsilon$ is holomorphic on the common ball and nonnegative for real fields. At $\theta = 0$ we have $A_d(0) = 0$ and $\mathsf{J}_d = 0$; in particular $E_d(I) = 0$, and $DE_d(I) = 0$ once the gradient formula is proved. We differentiate at fixed v before using stationarity. The derivative in θ of the parametrized field $v + \mathsf{H}_d\theta - \mathsf{H}_d\mathsf{D}_d(v + \mathsf{H}_d\theta)$ is $\mathsf{H}_d(I - D\mathsf{D}_d(w)\mathsf{H}_d)$. The contribution of Dv_d^* is a variation in $\mathcal{V}_d$, and its derivative of the action vanishes because v_d^* is critical. Using $\mathsf{J}_d(w) = \mathsf{H}_d^\sharp G_\varepsilon(A)$, we obtain

$$\nabla_\theta E_d = (I - D\mathsf{D}_d(w)\mathsf{H}_d)^T \mathsf{J}_d(w), \tag{10.19}$$

where T is the transpose for the counting scalar product on coarse fields. Note that this formula keeps the contribution of the constraint parametrization, and that it gives the derivatives of E_d without estimating the size of the total action.

Step 6: coefficient bounds for the action. We now derive coefficient bounds at one rate from (10.19). We write $(\nabla_\theta E_d)_c = \mathsf{J}_d(w)_c - \mathsf{T}_c$, where

$$\mathsf{T}_c = \sum_{c'} \sum_{e} \mathsf{H}_d(e,c)^T \big(D_{w_e}\mathsf{D}_{d,c'}(w)\big)^T \mathsf{J}_d(w)_{c'}, \qquad w = v_d^*(\theta) + \mathsf{H}_d\theta,$$

with c, c' coarse links and e a fine link. In the term $\mathsf{H}_d^\sharp N_\varepsilon$ of J_d, a fixed coarse index c sees the volume-normalized column of H_d, which is bounded. In T_c the fine index e is summed between H_d and $D\mathsf{D}_d$, and the factors ε^{-3} and ε^3 cancel. We carry out the bookkeeping for the first derivative, which shows all the mechanisms. The derivative of T_c in θ_{c_1} is $\mathsf{T}_c^{(1)}(c_1) + \mathsf{T}_c^{(2)}(c_1)$, where

$$\mathsf{T}_c^{(1)}(c_1) = \sum_{c',e} \mathsf{H}_d(e,c)^T \big(D_{w_e}\mathsf{D}_{d,c'}(w)\big)^T D_{\theta_{c_1}}\big[\mathsf{J}_d(w)_{c'}\big],$$

$$\mathsf{T}_c^{(2)}(c_1) = \sum_{c',e,e'} \mathsf{H}_d(e,c)^T \big(D^2_{w_e w_{e'}}\mathsf{D}_{d,c'}(w)\, D_{\theta_{c_1}} w_{e'}\big)^T \mathsf{J}_d(w)_{c'}.$$

Let $b'' > 0$ be smaller than all rates used, and write $\omega(x,y) = e^{b''\,\mathrm{dist}(x,y)}$. By the triangle inequality, $e^{b''\mathsf{t}(c;c_1)} = \omega(c,c_1)$ is at most the product of the weights ω along the chain c, e, c', c_1, or c, e, e', c_1. We first consider the output row, with c fixed. For $\mathsf{T}^{(1)}$ we sum in the order c_1, c', e:

$$\sum_{c_1} \omega(c,c_1)|\mathsf{T}^{(1)}_c(c_1)| \\ \leq \sum_e \omega(c,e)|\mathsf{H}_d(e,c)| \sum_{c'} \omega(e,c')|D_{w_e}\mathsf{D}_{d,c'}(w)| \sup_{c'} \sum_{c_1} \omega(c',c_1)|D_{\theta_{c_1}}\mathsf{J}_d(w)_{c'}|.$$

The first sum is a counting column of H_d, of order ε^{-3}. For each fixed fine e, the second sum is a counting column of $D\mathsf{D}_d$ with the fine input fixed, of order $\varepsilon^3 r$ by (10.10). The last factor is a row of the coarse tensor $D_\theta \mathsf{J}_d$, which is bounded by Step 2. The product is of order r, with no power of ε. For $\mathsf{T}^{(2)}$ we bound $|\mathsf{J}_d(w)_{c'}| \leq \|\mathsf{J}_d(w)\|_\infty$ and sum in the order c_1, (c', e'), e: the counting column of H_d at c, of order ε^{-3}, multiplies the counting column of $D^2\mathsf{D}_d$ with the fine input e fixed, of order ε^3, and the row of $D_\theta w = Dv^*_d + \mathsf{H}_d$, which is bounded by Step 2. Next we consider the input column, with c_1 fixed. For $\mathsf{T}^{(1)}$ the fixed input lies in $D_\theta \mathsf{J}_d$. We use its column $\sum_{c'} \omega(c',c_1)|D_{\theta_{c_1}}\mathsf{J}_d(w)_{c'}|$, and then the rows $\sup_{c'} \sum_e \omega(e,c')|D_{w_e}\mathsf{D}_{d,c'}(w)| \leq Kr$ and $\sup_e \sum_c \omega(c,e)|\mathsf{H}_d(e,c)| \leq K$; no power of ε occurs. For $\mathsf{T}^{(2)}$ the fixed input lies in $D_\theta w$, whose counting column $\sum_{e'} \omega(e',c_1)|D_{\theta_{c_1}} w_{e'}|$ is of order ε^{-3}; it multiplies the counting column of $D^2\mathsf{D}_d$ with the fine input e' fixed, of order ε^3, the row of H_d, and $\|\mathsf{J}_d(w)\|_\infty$.

The derivatives of higher order are bounded in the same way. In each term of the chain rule, the branch containing the fixed input, whether it lies in w or in J_d, uses the input column of that factor, and every other branch uses its output row; these are the contraction rules of Lemma 5.3. Every contraction over a fine index pairs a counting column of order ε^{-3}, of H_d or of $D_\theta w$, with a counting column of order ε^3 of a derivative of D_d. Together with (10.17), this gives, on a fixed smaller ball and at the rate b'',

$$\|D^s_\theta \nabla_\theta E_d\|_{b'',\mathrm{root}} \leq K^{s+1} s! \qquad (s \geq 1), \tag{10.20}$$

where the norm is the maximum of the output row and the input columns. Since the derivatives of a scalar function are symmetric, every argument may be the fixed one. This proves the bounds $\|D^s E_d\|_{a_s,\mathrm{root}} \leq C_s$ of part (5). Integrating the row of the Hessian along $t\theta$, $0 \leq t \leq 1$, and using $\nabla_\theta E_d(0) = 0$, we obtain $\|\nabla_\theta E_d\|_\infty \leq K\|\theta\|_\infty$. The substitution $\theta = gB$ contributes exactly the factor g^{s-2} at order s of $g^{-2}E_d(e^{gB})$, and its first derivative is $g^{-1}\nabla_\theta E_d(gB)$, which gives $\|D_B W_{d,g}\|_\infty \leq K\|B\|_\infty$.

Step 7: local minimality on the fiber. Let θ be real, and write $A_* = A_d(\theta)$, $G_* = G_d(A_*)$, and $c_* = \mathrm{Log}\, G_*$. Let Ω_* be a periodic fine gauge transformation with values G_* at the coarse vertices. By the discussion at the beginning of this section, $U_* = (e^{\varepsilon A_*})^{\Omega_*}$ satisfies $T^d_M U_* = e^\theta$. On the fixed finite lattice, let U be a configuration in this fiber close to U_*,

and write $U^{\Omega_*^{-1}} = e^{\varepsilon A}$. Since $A = A_*$ for $U = U_*$, continuity in finitely many variables shows that A lies in the ball of inputs of Theorem 9.9 if U is close enough to U_*. Its block average is $T_M^d e^{\varepsilon A} = (e^\theta)^{G_*^{-1}}$.

We apply Theorem 9.9 to A with prescribed coarse datum $c = c_*$. For $A = A_*$ the gauge transformation of that theorem is the identity, since $\mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon A_* = 0$ and $c_* = \operatorname{Log} G_d(A_*)$. The transformed field A' has projected divergence zero and gauge-corrected block average

$$\big[(e^\theta)^{G_*^{-1}}\big]^{G_*} = e^\theta,$$

where the order of the factors comes from $(U^k)^l = U^{lk}$. By that theorem $\|A'\|_\infty + \|d_\varepsilon A'\|_\infty$ is small, and the parametrization of all nearby solutions of the constraint in Theorem 9.6 gives

$$v' = (I - \mathsf{H}_d \mathsf{Q}_d) A' \in \mathcal{V}_d, \qquad A' = \mathsf{P}_d^{\mathrm{cl}}(v' + \mathsf{H}_d \theta).$$

By continuity and the supremum and curl bounds for $\mathsf{H}_d \mathsf{Q}_d A'$, we see that v' lies in the larger convex domain of Step 4. By the stability inequality of Theorem 9.7, we have

$$\mathscr{S}_\varepsilon(U) - \mathscr{S}_\varepsilon(U_*) \geq \frac{c_0}{4} \|v' - v_d^*(\theta)\|_{H_\varepsilon^1}^2,$$

where we used that all the gauge transformations involved preserve the action $\mathscr{S}_\varepsilon$. Equality forces $v' = v_d^*(\theta)$, hence $A' = A_*$. Undoing the two fine gauge transformations shows that U and U_* lie on the same fine gauge orbit. Since both configurations lie in the fiber over e^θ, the values of the relating gauge transformation at the coarse vertices leave e^θ fixed. The construction applies to all points of a sufficiently small neighborhood of U_* in the fiber on the given finite lattice. Applying the inequality along a differentiable curve in the fiber through U_*, for both signs of the parameter, shows that every tangent vector to the fiber has zero first variation. This proves part (6) and completes the proof. □

10.6 Gauge extensions and symmetries of the effective action

In this section we prove that E_d is invariant under small coarse gauge transformations, the lattice symmetries, and constant color rotations. A symmetry applied to the background field moves it out of the gauge slice in which it was constructed. To compare, we bring the transformed field back into the slice with Theorem 9.9 and use the uniqueness of critical points in the convex domain of Theorem 9.7. The following lemma isolates this argument; it is used here and again in the proof of the nested minimization identity.

Lemma 10.7. *There exists $r_{\mathrm{tr}} > 0$ such that the following holds for every $d \geq 0$ and $n \geq 4$, with $\varepsilon = \varepsilon_d$. Let $U = e^{\varepsilon A}$ be a real fine configuration with $\|A\|_{\mathsf{C}_\varepsilon^1} < r_{\mathrm{tr}}$, and let $b = T_M^d U$ satisfy $\|\operatorname{Log} b\|_\infty < r_{\mathrm{tr}}$. Assume that U is a critical point of $\mathscr{S}_\varepsilon$ on the fiber $\{U' : T_M^d U' = b\}$, that is, $(d/dt)\mathscr{S}_\varepsilon(U(t))|_{t=0} = 0$ for every differentiable curve $t \mapsto U(t)$ in this fiber with $U(0) = U$.*

Then, with $\theta = \operatorname{Log} b$, there is a real fine gauge transformation h such that $U^h = e^{\varepsilon A_d(\theta)}$. In particular $\mathscr{S}_\varepsilon(U) = E_d(b)$.

Proof of Lemma 10.7. If $d = 0$, the fiber is the single point $U = b$, so $A = \theta$; since $\mathcal{V}_0 = \{0\}$, $\mathsf{H}_0 = I$, and $\mathsf{D}_0 = 0$, we have $A_0(\theta) = \theta$, and $h = I$ works. Let $d \geq 1$. We choose r_{tr} so small that the following three steps are possible.

Step 1: moving U into the gauge slice. We have $\|A\|_\infty + \|\operatorname{div}_\varepsilon A\|_\infty \leq 3\|A\|_{C^1_\varepsilon}$ and $\|d_\varepsilon A\|_\infty \leq 2\|\nabla_\varepsilon A\|_\infty$. For small r_{tr}, Theorem 9.9 therefore applies to A with prescribed coarse datum 0. It gives a real fine scalar field λ such that the field $A' = A^{e^\lambda}$, defined by $e^{\varepsilon A'} = U^{e^\lambda}$, satisfies

$$\mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon A' = 0, \qquad G_d(A')\, e^{\lambda|_{\mathrm{roots}}} = I,$$

and has gauge-corrected block average b and $\|A'\|_\infty + \|d_\varepsilon A'\|_\infty \leq K r_{\mathrm{tr}}$. By the parametrization of all nearby solutions of the constraint in Theorem 9.6, $A' = \mathsf{P}_d^{\mathrm{cl}}(a' + \mathsf{H}_d\theta)$ with $a' = (I - \mathsf{H}_d\mathsf{Q}_d)A' \in \ker \mathsf{Q}_d$. Since $\mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon A' = 0$ and $\mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon \mathsf{H}_d = 0$, also $\mathsf{P}_d^{\mathrm{div}} \operatorname{div}_\varepsilon a' = 0$, that is, $a' \in \mathcal{V}_d$. The bounds on H_d give $\|a'\|_\infty + \|d_\varepsilon a'\|_\infty \leq K r_{\mathrm{tr}}$.

Step 2: criticality in the chart. Let $u \in \mathcal{V}_d$, and for small real t let $a(t) = a' + tu$ and $A(t) = \mathsf{P}_d^{\mathrm{cl}}(a(t) + \mathsf{H}_d\theta)$. Since the gauge-corrected block average of $A(t)$ is $e^\theta = b$ for all t, the relation at the beginning of Section 10.5 gives $T_M^d e^{\varepsilon A(t)} = b^{G_d(A(t))^{-1}}$. For a fine vertex x let $r(x)$ be the coarse vertex whose centered block at depth d contains x, and let $\Omega(t)_x = G_d(A(t))_{r(x)}$. Then $\Omega(t)$ has the values $G_d(A(t))$ at the coarse vertices, and it depends differentiably on t because G_d is holomorphic. The curve $U(t) = (e^{\varepsilon A(t)})^{\Omega(t)}$ lies in the fiber over b, and $\mathscr{S}_\varepsilon(U(t)) = \mathscr{S}_\varepsilon(A(t))$. At $t = 0$ we have $e^{\varepsilon A'} = U^{e^\lambda}$, so $U(0) = U^{h_0}$ with $h_0 = \Omega(0)e^\lambda$, by $(U^k)^l = U^{lk}$. The values of h_0 at the coarse vertices are $G_d(A')e^{\lambda|_{\mathrm{roots}}} = I$. Since a fine gauge transformation equal to the identity at the coarse vertices maps the fiber over b onto itself and preserves $\mathscr{S}_\varepsilon$, the configuration $U(0)$ is critical on the fiber, as U is. It follows that

$$\frac{d}{dt}\mathscr{S}_\varepsilon(A(t))\Big|_{t=0} = \frac{d}{dt}\mathscr{S}_\varepsilon(U(t))\Big|_{t=0} = 0.$$

Since $u \in \mathcal{V}_d$ was arbitrary, a' is a critical point of $a \mapsto \mathscr{S}_\varepsilon(\mathsf{P}_d^{\mathrm{cl}}(a + \mathsf{H}_d\theta))$ on $\mathcal{V}_d$.

Step 3: identification. Fix an allowed value r_{ph} of Theorem 9.7, and choose r_{tr} so that $r_{\mathrm{tr}} < r_\theta$, $K r_{\mathrm{tr}} < r_{\mathrm{ph}}$, and $r_{\mathrm{tr}} < r_{\mathrm{ph}}/C_*$. Then a' is a critical point in the interior of the convex domain $\{a \in \mathcal{V}_d : \|a\|_\infty + \|d_\varepsilon a\|_\infty < r_{\mathrm{ph}}\}$. By Theorem 10.6 and Lemma 10.5, $v_d^*(\theta)$ is also an interior critical point of this domain. By the uniqueness of interior critical points in Theorem 9.7, $a' = v_d^*(\theta)$. Hence $A' = A_d(\theta)$, that is, $U^{e^\lambda} = e^{\varepsilon A_d(\theta)}$, and $\mathscr{S}_\varepsilon(U) = \mathscr{S}_\varepsilon(A_d(\theta)) = E_d(b)$, since gauge transformations preserve the action. □

The next lemma constructs a smooth extension of coarse gauge transformations to the fine lattice by cardinal interpolation, and proves the symmetries of E_d. The first

differences of the extension are of the size of the coarse gauge transformation, not of its size divided by the mesh. For a coarse vertex field κ and a point x of the torus we write $\sum_c \chi_c(x)\kappa_c$, where c runs over the coarse vertices.

Lemma 10.8. *There exist smooth periodic functions χ_c on the torus, indexed by the coarse vertices c, such that $\sum_c \chi_c = 1$, $\chi_c(c') = \delta_{cc'}$ for all coarse vertices c', the supports of the χ_c have diameter at most $2\sqrt{3}$, and*

$$\sup_x \sum_c |\partial^j \chi_c(x)| \le C_j \qquad (0 \le j \le 3);$$

their restrictions to the fine vertices satisfy $\varepsilon_d^3 \sum_x |\partial^j \chi_c(x)| \le C_j$. There exist $r_{\mathrm{lift}}, r_{\mathrm{sym}}, \kappa_{\mathrm{sym}}, C > 0$ with $2Cr_{\mathrm{sym}} < \min(r_{\mathrm{lift}}, \kappa_{\mathrm{sym}})$, where C also bounds the constants of Theorem 10.6 and the constant in the bound on $\operatorname{Log} G_d$ in Lemma 10.4, such that the following hold for every $d \ge 0$ and $n \ge 4$, with $\varepsilon = \varepsilon_d$.

1. *Let $k_c = e^{\kappa_c}$ be a coarse gauge transformation with $\|\kappa\|_\infty < \kappa_{\mathrm{sym}}$, and let $\Omega(x) = \exp(\sum_c \chi_c(x)\kappa_c)$ at the fine vertices x. Then Ω equals k at the coarse vertices, and for fine fields A with $\|A\|_{C^1_\varepsilon} < r_{\mathrm{lift}}$,*

$$\|\varepsilon^{-1} \operatorname{Log}[(e^{\varepsilon A})^\Omega]\|_{C^1_\varepsilon} \le C(\|A\|_{C^1_\varepsilon} + \|\kappa\|_\infty).$$

   In particular, for $\|\theta\|_\infty < r_{\mathrm{sym}}$ the gauge orbit of $e^{\varepsilon A_d(\theta)}$ contains a representative U, analytic in θ, with $T^d_M U = e^\theta$ and $\|\varepsilon^{-1} \operatorname{Log} U\|_{C^1_\varepsilon} \le C\|\theta\|_\infty$.
2. *On common smaller complex domains,*

$$E_d(b^k) = E_d(b), \qquad E_d(\sigma b) = E_d(b), \qquad E_d(b^u) = E_d(b) \quad (u \in \mathrm{SU}(2) \text{ constant}),$$

   where b and b^k lie in the chart $\{\|\operatorname{Log} b\|_\infty < r_{\mathrm{sym}}\}$, $\|\operatorname{Log} k\|_\infty < \kappa_{\mathrm{sym}}$, and σ is a translation of the coarse lattice or the spatial inversion $x \mapsto -x$, which maps links to reversed links and inverts their variables. Differentiating the first identity, we obtain the infinitesimal gauge-invariance identities $(d/dt)E_d(b^{e^{t\xi}})|_{t=0} = 0$ for every field ξ on the coarse vertices with values in $\mathfrak{su}(2)$.

The representative in part (1) is used in the proof of Theorem 10.10. Part (2) concerns small gauge transformations only. Gauge transformations far from the identity are treated in Section 11.

Proof of Lemma 10.8. We construct the cardinal functions (Step 1), prove the bound on the gauge transform (Steps 2 and 3), and then prove the three identities (Steps 4–6). Step 7 fixes the radii.

Step 1: cardinal functions. Let

$$\eta(t) = \begin{cases} \exp\bigl(1 - (1-t^2)^{-1}\bigr), & |t| < 1, \\ 0, & |t| \ge 1, \end{cases}$$

and

$$D(t) = \sum_{l \in \mathbb{Z}} \eta(t-l), \qquad \phi_l(t) = \eta(t-l)/D(t).$$

The function D is positive and periodic, so its minimum on $[0,1]$ is positive. The derivatives of each fixed order of η are bounded and vanish at the endpoints of its support. At most two translates of η are nonzero at a noninteger point and only one at an integer point, and the sums of their derivatives are bounded in terms of the order. It follows that $\sum_l \phi_l = 1$ and $\phi_l(k) = \delta_{lk}$ for integers k. Taking tensor products of the ϕ_l in the three coordinates and periodizing with period n, we obtain functions χ_c whose supports have diameter at most $2\sqrt{3}$ in the metric of the torus. Since $n \geq 4$, the periodization keeps the finite overlap and the values at the coarse vertices. The bounds on the sums of derivatives follow from the product rule. For fixed c, the support of χ_c contains at most $K\varepsilon^{-3}$ fine vertices and the derivatives of χ_c are bounded, which proves the bounds with the factor ε^3. Finite differences divided by ε satisfy the same bounds, since they are averages of the corresponding derivatives over intervals of length ε.

Step 2: the gauge transform has a bounded rescaled connection. Let $\lambda(x) = \sum_c \chi_c(x)\kappa_c$, so that $\Omega = e^{\lambda}$. By Step 1, the value of λ and its first two differences divided by ε are bounded by $K\|\kappa\|_\infty$. For a fine link xy we write $u = \lambda_x$, $v = (\lambda_y - \lambda_x)/\varepsilon$, and $a = A_{xy}$. The analytic function of finitely many variables

$$\varepsilon^{-1} \operatorname{Log}\bigl(e^{u} e^{\varepsilon a} e^{-u-\varepsilon v}\bigr) = \operatorname{Ad}(e^{u})a - \Bigl(\int_0^1 \operatorname{Ad}(e^{tu})\,dt\Bigr)v + \varepsilon R_\varepsilon(u,v,a)$$

has a removable singularity at $\varepsilon = 0$, uniformly on a fixed small complex ball of (u,v,a). Indeed, the product of the three exponentials is the identity at $\varepsilon = 0$, and its first derivative in ε at $\varepsilon = 0$ gives the two displayed principal terms. The remainder R_ε and its derivatives are bounded by the convergent series in finitely many variables on a smaller ball. The difference of the first principal term involves $\nabla_\varepsilon A$ and $\nabla_\varepsilon \lambda$, and the difference of the second involves the first two differences of λ. For the last term,

$$|\nabla_\varepsilon(\varepsilon R)_e| \leq |R_{e+\varepsilon e_\mu}| + |R_e|,$$

with no inverse power of the mesh. Since the remainder vanishes at $(a,v) = 0$, after shrinking the domain its value is at most $K(|a| + |v|)$. This proves the bound of part (1), on a radius r_{lift} independent of the fine mesh.

Step 3: a representative with block average e^{θ}. By Lemma 10.4 and part (1) of Theorem 10.6, $\|\operatorname{Log} G_d(A_d(\theta))\|_\infty \leq K\|\theta\|_\infty$. We apply the cardinal extension of Step 2 with these values at the coarse vertices. By the discussion at the beginning of Section 10.5, the resulting field U satisfies $T_M^d U = e^{\theta}$, and Step 2 implies that $\|\varepsilon^{-1}\operatorname{Log} U\|_{C^1_\varepsilon} \leq K\|\theta\|_\infty$. The formulas in finitely many variables and the holomorphic maps already constructed show that U is analytic in θ. The radii and the bound are the same for all d and n. This proves part (1).

Step 4: gauge invariance. We first let b and k be real with $\|\operatorname{Log} b\|_\infty$ and $\|\operatorname{Log} k\|_\infty$ small, and write $b = e^\theta$ and $G = G_d(A_d(\theta))$. Using the local logarithm of kG, we apply the cardinal extension of Step 2 to the coarse gauge transformation kG, and we apply the resulting fine gauge transformation to $e^{\varepsilon A_d(\theta)}$. Since $T_M^d e^{\varepsilon A_d(\theta)} = b^{G^{-1}}$, the transformed field U_k satisfies $T_M^d U_k = (b^{G^{-1}})^{kG} = b^k$. By Step 2 its rescaled connection has C^1_ε norm at most $K(\|\theta\|_\infty + \|\operatorname{Log} k\|_\infty)$. The field U_k is obtained from the configuration U_* of part (6) of Theorem 10.6 by a fine gauge transformation, which maps the fiber over b onto the fiber over b^k and preserves $\mathscr{S}_\varepsilon$. Since U_* is critical on its fiber, U_k is critical on the fiber over b^k. By Lemma 10.7, $\mathscr{S}_\varepsilon(U_k) = E_d(b^k)$. Since all the transformations used preserve the action, we conclude that $E_d(b^k) = \mathscr{S}_\varepsilon(e^{\varepsilon A_d(\theta)}) = E_d(b)$.

Step 5: translations, inversion, and constant color rotations. Translations of the coarse lattice preserve the centered blocks and their coordinate-ordered paths. They lift to translations of the fine lattice and commute with the block averaging map. For the inversion, let $c = (r, r + Me_i)$, let $c' = (-r - Me_i, -r)$, and let $y = -x - Me_i$. The inversion maps the paths as follows: $\sigma_{r,x} \mapsto \sigma_{-r,-x}$, $F_{c,x} \mapsto F^{-1}_{c',y}$, and $C_c \mapsto C^{-1}_{c'}$. For group elements F, C in the domain of the logarithm,

$$\operatorname{Log}(F^{-1}C) = -\operatorname{Ad}(C^{-1}) \operatorname{Log}(FC^{-1}).$$

Inserting this identity into the exponential average defining T_M, we find that the averaged reversed link is the inverse of the averaged link. The same holds for the Borel extension when a ratio equals $-I$, although that case does not occur in the present chart. The inversion therefore also commutes with the block averaging map. A translation preserves the C^1_ε norm of the rescaled connection; the inversion replaces each component by its negative at a translated link and also preserves that norm. Let σ be one of these maps and let U be the representative of part (1) for b, which is critical on its fiber by part (6) of Theorem 10.6. Then σU has block average σb and the same action, it is critical on the fiber over σb, and its rescaled connection has the same small norm. By Lemma 10.7, $E_d(\sigma b) = \mathscr{S}_\varepsilon(\sigma U) = \mathscr{S}_\varepsilon(U) = E_d(b)$. This proves the spatial identities on a common smaller domain.

A constant $u \in \mathrm{SU}(2)$ acts by the orthogonal color rotation $\operatorname{Ad}(u)$ on every path product, every linear operator, the constraints, and the action. It preserves all the domains above. By uniqueness of the fixed point, $v_d^*(\operatorname{Ad}(u)\theta) = \operatorname{Ad}(u) v_d^*(\theta)$ and $A_d(\operatorname{Ad}(u)\theta) = \operatorname{Ad}(u) A_d(\theta)$, which gives the identity $E_d(b^u) = E_d(b)$ for every such u.

Step 6: complex continuation. For translations, the inversion, and each fixed constant u, both sides of the identity are holomorphic functions of the coarse logarithm. Since they agree on a real open ball, every coefficient of the Taylor series of their difference vanishes. For the gauge identity the same argument applies jointly in $\operatorname{Log} b$ and $\operatorname{Log} k$.

Step 7: the radii and the infinitesimal identities. Let C be a constant larger than the constants for the minimum, for the gauge transformation G_d, and for the extension. First fix r_{lift} and κ_{sym} inside the domains used above, and then fix r_{sym} so small that $2Cr_{\mathrm{sym}} < \min(r_{\mathrm{lift}}, \kappa_{\mathrm{sym}})$ and all the requirements of Lemma 10.7 and of Steps 4 and 5 hold. This order gives the dependence of the constants stated in Section 10.1. Differentiating the gauge identity along $k_x(t) = e^{t\xi_x}$ gives the infinitesimal identity for every coarse vertex field ξ; for t in a small interval about zero, both b and $b^{k(t)}$ stay in the chart. □

10.7 One averaging step and nested minimization

We now consider one averaging step, from a lattice of side Mm to the lattice of side $m \geq 4$. We minimize E_d over the configurations on the lattice of side Mm whose block average is a given coarse field c. We first recall the local coordinates on the fiber $\{b : T_M b = c\}$ and prove the bounds we need for them; then we prove the nested minimization identity.

Coordinates for one averaging step. The coordinates are those of Definition 4.3 and Lemma 6.3; the eliminated edges are denoted $e^*_{c'}$ here. In this paragraph c denotes a coarse field and c' a positive coarse link. The coarse-field lift $U_0(c)$ is the identity on the internal links of the blocks and equals $c_{c'}$ on the interface links of the coarse link c'; thus $T_M U_0(c) = c$. In each block, the paths $\sigma_{r,x}$ from the block center r form a rooted spanning tree, and we let $\mathcal{F}$ be the set of positive fine links whose unoriented edges belong to these trees. For the coarse link $c' = (r, r + Me_i)$, the eliminated edge is $e^*_{c'} = (r + R_M e_i, r + (R_M + 1)e_i)$, where $M = 2R_M + 1$. We write a fine configuration as $e^X U_0(c)$, and we let

$$\mathsf{C}_{c'}(X) = \operatorname{Log}\big((T_M(e^X U_0(c)))_{c'} c_{c'}^{-1}\big)$$

be the block averaging constraint at c'. The free logarithmic coordinates z are the values X_e on the positive fine links $e \notin \mathcal{F}$ that are not eliminated edges, and $X_e = 0$ for $e \in \mathcal{F}$. The local constraint coordinates $X = \Gamma_c(z)$ solve the constraints $\mathsf{C}_{c'}(X) = 0$ for the eliminated-edge variables $X_{e^*_{c'}}$, with $\Gamma_c(z)_e = z_e$ on the free links. We set

$$b(c,z) = e^{\Gamma_c(z)} U_0(c), \qquad B(c,z) = \operatorname{Log} b(c,z), \qquad \mathsf{E}_{\mathrm{f}} = D_z B(I,0).$$

Let f be the number of free coordinates, each a group variable, let $c_{\mathrm{Haar}} = 1/(16\pi^2)$ and $j_H(X) = \operatorname{sinc}^2(|X|/2)$ be as in (4.2), and define the Haar disintegration density

$$\mathcal{J}_c(z) = c_{\mathrm{Haar}}^f \frac{\prod_{e \notin \mathcal{F}} j_H(\Gamma_c(z)_e)}{\prod_{c'} \det D_{X_{e^*_{c'}}} \mathsf{C}_{c'}(\Gamma_c(z))},$$

where e runs over the positive fine links and c' over the positive coarse links. In complex formulas $|X|^2$ denotes the bilinear sum of the squares of the three coordinates of X. This

is the density of Lemma 6.3; the numerator contains the Haar factors of the eliminated edges, which lie outside $\mathcal{F}$. We recall its derivation in the proof below, since the estimates use its explicit form.

Lemma 10.9. *There exist $R'_M, b, K_M > 0$ such that the following hold for every $m \geq 4$.*

1. *On the complex product $\|\theta\|_\infty < R'_M$, $\|z\|_\infty < R'_M$, with $c = e^\theta$, the maps $\Gamma_c(z)$ and $B(c,z)$ are holomorphic, their absolute derivative tensors have factorial bounds and finite stencils, and*

$$\|B(c,z)\|_\infty \leq K_M(\|\theta\|_\infty + \|z\|_\infty), \qquad \|D_z B(c,z) - \mathsf{E}_{\mathrm{f}}\|_{b,\mathrm{row+col}} \leq K_M(\|\theta\|_\infty + \|z\|_\infty).$$

2. *On a smaller real product, the constraint Jacobian determinants $\det D_{X_{e^*_{c'}}} \mathsf{C}_{c'}(\Gamma_c(z))$ are bounded below by a fixed positive constant, and*

$$\|D_z^2 \log \mathcal{J}_c(z)\|_{b,\mathrm{row+col}} \leq K_M.$$

3. *Let $t \mapsto b(t)$ be a differentiable curve in the fiber $\{T_M b = c\}$, and let $k(t)$, equal to the identity at the coarse vertices, be the block axial gauge transformation of $b(t)$, so that $b(t)^{k(t)} = b(c, z(t))$, with $z(t)$ in the domain of (1). Put $k_0 = k(0)$ and $z_0 = z(0)$, and let $\mathcal{G}_h(U) = U^h$. Then*

$$b'(0) = D\mathcal{G}_{k_0^{-1}}\big|_{b(c,z_0)} D_z b(c,z_0) z'(0) + \frac{d}{dt}\Big[b(c,z_0)^{k(t)^{-1}}\Big]\Big|_{t=0}.$$

The second term is the tangent vector at $b(0)$ of a curve of fine gauge transformations along the block trees which are the identity at the coarse vertices. If $b(0) = b(c,z_0)$, then $k_0 = I$, and the first term is $D_z b(c,z_0) z'(0)$.

Proof of Lemma 10.9. We construct the block axial gauge (Step 1), solve the constraint for the eliminated edges (Steps 2 and 3), derive the density and bound its Hessian (Steps 4 and 5), and prove the decomposition of tangent vectors (Step 6). The density formula is the tree-gauge specialization of the disintegration of Haar measure for this averaging constraint.

Step 1: the block axial gauge. We gauge the links of $\mathcal{F}$ to the identity with $k_r = I$ at each block center r. On a tree edge directed from a parent x to a child y we prescribe

$$k_y = k_x U_{xy}, \qquad k_x = U(\sigma_{r,x}). \tag{10.21}$$

If this direction reverses a positive link, U_{xy} is the inverse of the variable of that link, as in the path convention. Then $k_x U_{xy} k_y^{-1} = I$ on every tree edge. By recursion along the tree, we see that this gauge transformation is unique given the values at the roots and that each k_x is an analytic path product. By (10.14), a gauge transformation equal to the identity at the coarse vertices preserves the coarse field. In these coordinates each straight reference path $C_{c'}$ has only one link that is not gauged to the identity, the eliminated edge $e^*_{c'}$.

Step 2: the derivative in the eliminated edge. We note that in these coordinates the constraint at c' contains, among all the eliminated variables, only its own. Indeed, the root paths are identities, and a straight middle segment crosses only the interface from its starting block to the neighboring block in direction i. Exactly M of the M^3 segments cross $e^*_{c'}$: their transverse offset is zero, and the longitudinal start has M choices. At $z = 0$, let only $X = X_{e^*_{c'}}$ vary. Then the M contours through $e^*_{c'}$ have transport $e^X c_{c'}$, all other contours have transport $c_{c'}$, and the reference path has transport $e^X c_{c'}$. The relative logarithms are 0 for the first M contours and $-X$ for the others, so their mean is $-(1 - M^{-2})X$, and the output is exactly

$$\exp\bigl(-(1 - M^{-2})X\bigr) e^X c_{c'} = e^{M^{-2}X} c_{c'}.$$

Thus the derivative of the constraint in the eliminated edge is $M^{-2} I_3$ at $z = 0$, for every sufficiently small c, not only for $c = I$.

Step 3: solving the constraints. Each constraint involves a bounded number of variables, and a fixed free link meets a bounded number of constraints, both depending only on M. The derivatives of these path products on a fixed smaller polydisc are bounded by $s! K_M^s$. We consider the equation

$$X_{e^*_{c'}} = X_{e^*_{c'}} - M^2 \mathsf{C}_{c'}(X, z).$$

By Step 2, the derivative of its right side in $X_{e^*_{c'}}$ vanishes at $(X, z) = (0, 0)$. On a fixed small product this derivative has norm less than $1/4$, and the constant term is $O_M(\|z\|_\infty)$. Hence the right side maps a ball of radius $K_M \|z\|_\infty$ strictly into itself, and iteration gives the unique small holomorphic solution. Majorizing by the scalar quadratic implicit series of Lemma 5.3, we obtain factorial bounds on its derivatives. Since the equations are independent in their eliminated unknowns, the solution on the whole lattice has the same bounds and a fixed stencil, with no factor depending on the dimension. At $z = 0$ the solution vanishes identically. Multiplying by $U_0(c)$ and taking the local logarithm, we obtain the bounds on B and on $D_z B - \mathsf{E}_\mathrm{f}$ from Taylor's formula. Since a stencil has bounded diameter, a fixed spatial weight changes only the constant. A fixed input enters a bounded number of outputs, which gives the column bounds. This proves part (1).

Step 4: the density. To derive the density, we first eliminate the gauge transformations along the trees recursively, by left and right translations of the Haar probability measure; their integral is one. With the remaining links fixed, we change each eliminated link to the output of its constraint; the Jacobian of this change is the displayed 3×3 determinant. The remaining exponential coordinates lie outside $\mathcal{F}$ and contribute their Haar factors to the numerator of $\mathcal{J}_c$, including those of the eliminated edges. Since these changes of variables are triangular, a derivative in an earlier coordinate contributes no further determinant. More explicitly, we fix c as a parameter before the change of variables. A nearby output

is $e^{\varphi}c$, and the Haar probability measure in this coordinate is $c_{\mathrm{Haar}}^{q}\prod_{c'} j_H(\varphi_{c'})\,d\varphi$, where q is the number of positive coarse links. On the fiber $\varphi = 0$ these factors j_H equal one. There are $f + q$ input links outside $\mathcal{F}$, so dividing their Haar constants by those of the output leaves c_{Haar}^{f}. Right translation by the fixed configuration $U_0(c)$ preserves the Haar measure of the inputs. This gives the displayed density at each fixed coarse field. Letting this parameter vary afterwards introduces no further Jacobian and no additional Haar factor for the coarse field.

Step 5: the Hessian of the logarithm of the density. We shrink the product until the Haar factors are bounded away from zero and the derivatives of the constraints in the eliminated edges differ from $M^{-2}I_3$ by less than $M^{-2}/4$. Then the real determinants are positive, and their logarithms and the logarithms of the Haar factors have bounded local analytic derivatives. The function $\log \mathcal{J}_c$ is the sum of these local terms and of the constant $f \log c_{\mathrm{Haar}}$. Each differentiated row meets a bounded number of terms, and each term involves a bounded number of coordinates. This proves the bounds on the rows and columns of the Hessian in part (2); the undifferentiated density depends on the volume.

Step 6: tangent vectors to the fiber. For a curve $b(t)$ in the fiber as in part (3), we apply to each $b(t)$ its unique block axial gauge transformation $k(t)$, which is the identity at the coarse vertices and depends differentiably on t, since each $k_x(t)$ is a path product. The gauged curve has the same block average c, and its free coordinates are some $z(t)$. By uniqueness of the solution of the constraints, all its remaining coordinates are those of $\Gamma_c(z(t))$, that is, $b(t)^{k(t)} = b(c, z(t))$. Hence $b(t) = \mathcal{G}_{k(t)^{-1}}(b(c, z(t)))$. Differentiating at $t = 0$ by the product rule, we obtain the displayed formula: the first term comes from the derivative of $b(c, z(t))$ at the fixed gauge transformation k_0^{-1}, and the second from the derivative of $k(t)^{-1}$ at the fixed configuration $b(c, z_0)$. If $b(0) = b(c, z_0)$, the uniqueness of the block axial gauge transformation gives $k_0 = I$. The argument is local and requires no global choice on the compact fiber. □

In the following theorem, E_{d_0} is the effective classical action of depth d_0 over the lattice of side Mm, in the coarse lattice units of that lattice, and E_{d_0+1} is the effective classical action of depth $d_0 + 1$ over the lattice of side m. The coordinates z are the free logarithmic coordinates of Lemma 10.9; no coupling has been inserted.

Theorem 10.10. *There exist $r_c, r_z, m_0, C_0, a, C, g_* > 0$ such that the following hold for every $d_0 \geq 0$ and every $m \geq 4$. Let*

$$F_{d_0}(c, z) = E_{d_0}(b(c, z)), \qquad K_{d_0}^{0} = \mathsf{E}_{\mathrm{f}}^{T} D^2 E_{d_0}(I) \mathsf{E}_{\mathrm{f}}.$$

Here $D^2 E_{d_0}(I)$ is the effective Maxwell quadratic form of Theorem 8.3; its gauge and constant null directions are kept.

1. *On the real space of free coordinates,*

$$m_0 I \le K^0_{d_0} \le C_0 I, \qquad \|K^0_{d_0}\|_{a,\mathrm{row+col}} + \|(K^0_{d_0})^{-1}\|_{a,\mathrm{row+col}} \le C_0.$$

2. *On the real product $\|\mathrm{Log}\, c\|_\infty < r_c$, $\|z\|_\infty \le r_z$, the configuration $b(c,z)$ lies in the domain of E_{d_0} and in the domain of positivity of the Haar disintegration density with fixed margins. The Hessian*

$$D^2_z F_{d_0} = (D_z B)^T D^2 E_{d_0}(B) D_z B + D E_{d_0}(B) D^2_z B, \qquad B = B(c,z),$$

satisfies

$$\|D^2_z F_{d_0} - K^0_{d_0}\|_{a,\mathrm{row+col}} \le C(\|\mathrm{Log}\, c\|_\infty + \|z\|_\infty), \qquad D^2_z F_{d_0} \ge \tfrac{1}{2} m_0 I.$$

3. *For real c with $\|\mathrm{Log}\, c\|_\infty < r_c$ there is a unique minimizer $z^*_{d_0}(c)$ of $z \mapsto F_{d_0}(c,z)$ on the closed real ball $\|z\|_\infty \le r_z$. It is an interior point, it depends real analytically on c, and it has a holomorphic continuation to the complex ball $\|\mathrm{Log}\, c\|_\infty < r_c$. Further,*

$$D_z F_{d_0}(c, z^*_{d_0}(c)) = 0, \qquad \|z^*_{d_0}(c)\|_\infty \le C\|\mathrm{Log}\, c\|_\infty < r_z/8,$$

$$E_{d_0+1}(c) = M \min_{\|z\|_\infty \le r_z} F_{d_0}(c,z) = M F_{d_0}(c, z^*_{d_0}(c)).$$

We call $z^*_{d_0}(c)$ the conditional constrained minimizer.

4. *There exists $0 < r'_c \le r_c$, independent of d_0 and m, such that for every integer $s \ge 1$ there exist $a_s, C_s > 0$, depending on s but independent of d_0 and m, for which the following bound holds on the complex ball $\|\theta\|_\infty < r'_c$:*

$$\|D^s_\theta z^*_{d_0}(e^\theta)\|_{a_s,\mathrm{out}} + \|D^s_\theta z^*_{d_0}(e^\theta)\|_{a_s,\mathrm{input,count}} \le C_s,$$

where the outputs are the free coordinates, the inputs are the coarse links, and distances are measured in the coarse lattice units of the lattice of side m. The volume-normalized column differs from the counting column by the fixed factor M^{-3}.

5. *The identity $E_{d_0+1}(c) = M F_{d_0}(c, z^*_{d_0}(c))$ holds for complex c in the ball of (3), with the holomorphic continuation of $z^*_{d_0}$. For real c and couplings $g^2_+ = Mg^2$, the identity in (3) reads*

$$\frac{E_{d_0+1}(c)}{g^2_+} = \frac{1}{g^2} \min_{\|z\|_\infty \le r_z} F_{d_0}(c,z).$$

6. *Let c be real with $\|\mathrm{Log}\, c\|_\infty < r_c$, let $p \ge 1$, and let $0 < g \le g_*$ with $gp < r_z/4$. Then the function*

$$y \longmapsto \frac{1}{g^2}\big[F_{d_0}(c, z^*_{d_0} + gy) - F_{d_0}(c, z^*_{d_0})\big] - \log \frac{\mathcal{J}_c(z^*_{d_0} + gy)}{\mathcal{J}_I(0)}$$

of real y has Hessian at least $m_0 I/4$ on the product $\prod_e \{|y_e| \le p\}$.

Since the radii and the lower bounds in the theorem do not decrease with d_0, the theorem can be iterated over arbitrarily many averaging steps. The minimization statements and the lower bounds on Hessians concern real coarse fields and real free coordinates. The identity in (5) for complex c and the bounds in (4) concern the holomorphic continuations of the critical point and the critical value. In (6) the denominator $\mathcal{J}_I(0)$ is a constant, so the value of the numerator at the shifted point is kept.

Proof of Theorem 10.10. We identify the Hessian of E_{d_0} at the identity (Step 1), obtain the bounds on $K^0_{d_0}$ (Step 2), and bound the Hessian away from the identity (Step 3). We then choose the radii (Step 4), construct the minimizer (Step 5), identify the nested minimum with the background field of depth $d_0 + 1$ (Steps 6–8), and prove part (6) (Step 9).

Step 1: the Hessian of the action at the identity. Differentiating the action twice at $\theta = 0$ in Theorem 10.6, using $DA_{d_0}(0) = \mathsf{H}_{d_0}$ and the vanishing of the gradient of $\mathscr{S}_\varepsilon$ at 0, we obtain

$$D^2 E_{d_0}(I) = \mathsf{H}^\sharp_{d_0} d^*_\varepsilon d_\varepsilon \mathsf{H}_{d_0}. \tag{10.22}$$

We show that this is the effective Maxwell quadratic form of Theorem 8.3, which is defined with the linearization of $T^{d_0}_M$ rather than with the gauge-corrected average. Let $\ell(a)$ be the derivative at $a = 0$ of $\operatorname{Log} T^{d_0}_M e^{\varepsilon a}$, and let $\gamma(a)$ be the derivative at $a = 0$ of $\operatorname{Log} G_{d_0}(a)$, a field on the coarse vertices. For a coarse vertex field κ we write $(\nabla^{\mathrm{c}}\kappa)_{(r,s)} = \kappa_s - \kappa_r$. Linearizing (10.11) at $j = d_0$ and (10.14), we obtain, for fine fields a and fine scalar fields λ,

$$\mathsf{Q}_{d_0} a = \ell(a) - \nabla^{\mathrm{c}}\gamma(a), \qquad \ell(\nabla_\varepsilon \lambda) = \nabla^{\mathrm{c}}(\lambda|_{\mathrm{roots}}).$$

Thus the linearized block average and Q_{d_0} differ by a coarse gradient, and a fine gradient changes the linearized block average by a coarse gradient and has zero curl. Let a have $\ell(a) = \eta$. Linearizing the gauge slice of Theorem 9.9 with prescribed coarse datum zero, we obtain a fine scalar field λ such that $a' = a - \nabla_\varepsilon \lambda$ has $\mathsf{P}^{\mathrm{div}}_{d_0} \operatorname{div}_\varepsilon a' = 0$ and $\gamma(a') + \lambda|_{\mathrm{roots}} = 0$. Then $d_\varepsilon a' = d_\varepsilon a$ and $\mathsf{Q}_{d_0} a' = \eta - \nabla^{\mathrm{c}}(\lambda|_{\mathrm{roots}}) - \nabla^{\mathrm{c}}\gamma(a') = \eta$. Conversely, let a' have $\mathsf{Q}_{d_0} a' = \eta$, and let λ' be a fine scalar field with $\lambda'|_{\mathrm{roots}} = \gamma(a')$, for instance the linearization of the cardinal extension of Lemma 10.8. Then $a'' = a' - \nabla_\varepsilon \lambda'$ has $d_\varepsilon a'' = d_\varepsilon a'$ and $\ell(a'') = \eta$. We conclude that the minimum of $\|d_\varepsilon a\|^2_\varepsilon$ over the fields with $\ell(a) = \eta$ equals its minimum over the fields with $\mathsf{Q}_{d_0} a = \eta$ and $\mathsf{P}^{\mathrm{div}}_{d_0} \operatorname{div}_\varepsilon a = 0$. Every field of the latter kind is $\mathsf{H}_{d_0}\eta + v$ with $v \in \mathcal{V}_{d_0}$, and the minimizing property of H_{d_0} shows that the minimum equals the right side of (10.22) evaluated at η. This identifies $D^2 E_{d_0}(I)$ with the effective Maxwell quadratic form of Theorem 8.3, including its gauge and constant null directions. The function $W_{d_0,g}(B) = g^{-2} E_{d_0}(e^{gB})$ has the same Hessian at $B = 0$, since the two factors g from the derivatives cancel g^{-2}.

Step 2: the constrained Hessian at the identity. We apply the statement of Theorem 8.3 about the next constrained fluctuation space, with the lattice of side Mm as the coarse lat-

tice of that theorem, the next lattice of side m, depth d_0, and the embedding $E_0 = D_z\Gamma_I(0)$ of Lemma 6.5, which equals E_f because $U_0(I) = I$. That theorem gives $m_0, C_0, a_0 > 0$, independent of d_0 and m, for the lower bound, the upper bound, and the weighted bounds on $K^0_{d_0}$ and on its inverse. The inverse is asserted only on the next constrained fluctuation space, on which the vanishing of the linearized next average removes the constant fluctuation. If distances are measured in the coarse units of the lattice of side m rather than in those of the lattice of side Mm, they are divided by M, and we decrease the exponent once. This proves part (1).

Step 3: the Hessian away from the identity. We write $t = \operatorname{Log} c$, $H_0 = D^2E_{d_0}(I)$, and $B = B(c, z)$. The Hessian in the coordinates z is

$$D^2_z F_{d_0} = (D_z B)^T D^2 E_{d_0}(B) D_z B + DE_{d_0}(B) D^2_z B. \tag{10.23}$$

By the bounds on the third derivatives and on the gradient in Theorem 10.6, we have

$$\|D^2E_{d_0}(B) - H_0\|_{a,\mathrm{row+col}} \le K\|B\|_\infty, \qquad \|DE_{d_0}(B)\|_\infty \le K\|B\|_\infty.$$

For the second term in (10.23), we fix a free coordinate and use the column of $D^2_z B$ with this input fixed; the remaining input and the output are summed against the bound on the components of $DE_{d_0}(B)$. This term is $O(\|B\|_\infty)$ in both the row and the column norm. For the first term we expand around $\mathsf{E}_\mathrm{f}^T H_0 \mathsf{E}_\mathrm{f}$, using the bound on $D_z B - \mathsf{E}_\mathrm{f}$ in Lemma 10.9 and the displayed bound on $D^2E_{d_0}(B) - H_0$. Using the contraction rule of Lemma 5.3 and $\|B\|_\infty \le K(\|t\|_\infty + \|z\|_\infty)$, we obtain

$$\|D^2_z F_{d_0} - K^0_{d_0}\|_{a,\mathrm{row+col}} \le K(\|t\|_\infty + \|z\|_\infty). \tag{10.24}$$

Every sum here has one index fixed; no sum of $|DE_{d_0}|$ over all links, and no factor of the volume, occurs.

Step 4: the radii. We take a complex radius for the coordinates of Lemma 10.9 that leaves fixed margins for four requirements: the relative logarithms, the solution of the constraints, the positivity of the real density, and the larger domain of E_{d_0}. Let C_inv bound the weighted norm of the inverse of $K^0_{d_0}$. Fix $r_z > 0$ so small that the entire product lies inside these domains and

$$2Kr_z \le m_0/2, \qquad 2C_\mathrm{inv} K r_z \le 1/2.$$

Next we take $r_c \le r_z$ inside the domains for the coarse field. Since the weighted bound (10.24) also bounds the Euclidean operator norm, the real Hessian is at least $m_0 I/2$ on the entire product. This proves part (2). The radii use only the bounds on the fixed larger domains of Theorem 10.6, which do not depend on d_0, and not a recursively reduced domain of E_{d_0}; the same radii work for every d_0.

Step 5: the conditional constrained minimizer. At $t = 0$ and $z = 0$ the gradient in z vanishes. Integrating its mixed derivative in the coarse field, we obtain $\|D_z F_{d_0}(c, 0)\|_\infty \leq K_1 \|t\|_\infty$. The equation for the minimizer is the fixed point equation

$$z = -(K^0_{d_0})^{-1}\big[D_z F_{d_0}(c, z) - K^0_{d_0} z\big].$$

By (10.24), the right side has Lipschitz constant at most $1/2$ on the complex ball of radius r_z. If we reduce r_c so that $2C_{\mathrm{inv}} K_1 r_c < r_z/8$, the right side maps this ball strictly into itself, and its fixed point satisfies

$$\|z^*_{d_0}(c)\|_\infty \leq 2C_{\mathrm{inv}} K_1 \|t\|_\infty < r_z/8.$$

Using iteration and the convergent majorant series of Lemma 5.3, we obtain the holomorphy on the complex ball of coarse fields and the bounds on the derivative tensors of every fixed order. For the rows we use the rows of $(K^0_{d_0})^{-1}$, of the derivatives of E_{d_0} of each fixed order in Theorem 10.6, and of the derivatives of B. For the columns we follow the fixed input through the weighted inverse and the mixed derivatives of $D_z F_{d_0}$. Since the derivative tensors of the action and of B have both row and column bounds, the same implicit-series argument applies. The rates obtained depend on the order through the rates a_s of Theorem 10.6. We take $r'_c = r_c/2$, where r_c denotes its final value after the reductions in this step and in Step 7. The implicit-series clause of Lemma 5.3 includes the translation to backgrounds in such a fixed smaller ball, so the coefficient bounds of each fixed order hold on $\|\theta\|_\infty < r'_c$. This proves part (4). Strict convexity in the real variables shows that this interior critical point is the unique minimum on the closed real ball $\|z\|_\infty \leq r_z$.

Step 6: the nested minimizer is critical on the longer fiber. We now identify the minimum with the effective classical action of depth $d_0 + 1$. Let $b_* = b(c, z^*_{d_0}(c))$, so that $\|\mathrm{Log}\, b_*\|_\infty \leq K\|t\|_\infty$. For coarse fields b near b_*, let $s_{d_0}(b)$ be the representative of part (1) of Lemma 10.8: it lies in the gauge orbit of the background field of depth d_0, it depends analytically on b, its rescaled connection is small in C^1_ε, and $T^{d_0}_M s_{d_0}(b) = b$. Its action is $E_{d_0}(b)$. The fine configuration $U_* = s_{d_0}(b_*)$ satisfies

$$T^{d_0}_M U_* = b_*, \qquad T^{d_0+1}_M U_* = c.$$

Let $\dot U$ be a tangent vector at U_* to the fiber $\{T^{d_0+1}_M U = c\}$, and let $\dot b = D(T^{d_0}_M)\dot U$; then $DT_M\, \dot b = 0$. The vector $\dot U - Ds_{d_0}(b_*)\dot b$ is tangent to the fiber $\{T^{d_0}_M U = b_*\}$. By part (6) of Theorem 10.6, the first variation of the action in this direction vanishes. Indeed, U_* is of the form considered there: it is obtained from the background field with coarse field b_* by the cardinal extension of Step 3 of the proof of Lemma 10.8, whose values at the coarse vertices are $G_{d_0}(A_{d_0}(\mathrm{Log}\, b_*))$. The remaining first variation, in the direction $Ds_{d_0}(b_*)\dot b$, equals $DE_{d_0}(b_*)\dot b$, in the coarse units of the lattice of side Mm, since the action

on the analytic family s_{d_0} equals E_{d_0}. Since $b_* = b(c, z^*_{d_0})$ is already in block axial gauge, the fixed transport in part (3) of Lemma 10.9 is the identity. That part then shows that $\dot b$ is the sum of a tangent vector $D_z b(c, z^*_{d_0})\dot z$ of the free coordinates and of a tangent vector of gauge transformations along the block trees that are the identity at the coarse vertices. The first is annihilated by $D_z F_{d_0}(c, z^*_{d_0}) = 0$, and the second by the infinitesimal gauge-invariance identities of Lemma 10.8. This shows that U_* is critical on the fiber $\{T_M^{d_0+1} U = c\}$.

Step 7: identification with the background field of depth $d_0 + 1$. In the coarse units of the lattice of side m, the same fine lattice has mesh ε/M, where $\varepsilon = M^{-d_0}$. By item (4) of the normalizations in Section 10.1, the rescaled connection of U_* is multiplied by M and its first differences by M^2, so its $C^1_{\varepsilon/M}$ norm is at most $K_M \|t\|_\infty$. We decrease r_c once more so that $K_M r_c < r_{\mathrm{tr}}$ and $r_c < r_{\mathrm{tr}}$, the radius of Lemma 10.7. That lemma, at depth $d_0 + 1$, shows that U_* lies in the gauge orbit of the background field of depth $d_0 + 1$ with coarse field c, and that $\mathscr{S}_{\varepsilon/M}(U_*) = E_{d_0+1}(c)$.

Step 8: the change of units. By item (4) of the normalizations, we have $\mathscr{S}_{\varepsilon/M} = M \mathscr{S}_\varepsilon$ for the same fine configuration, and therefore

$$E_{d_0+1}(c) = \mathscr{S}_{\varepsilon/M}(U_*) = M \mathscr{S}_\varepsilon(U_*) = M E_{d_0}(b_*) = M \min_{\|z\|_\infty \le r_z} F_{d_0}(c, z).$$

This proves part (3) on the real ball of coarse fields. The value $M F_{d_0}(c, z^*_{d_0}(c))$ at the holomorphic minimizer and the effective classical action $E_{d_0+1}(c)$ are both holomorphic, and their difference has vanishing real Taylor coefficients on a real open ball. Its convergent complex Taylor series therefore vanishes as well, which proves the first statement of part (5). The second statement follows by dividing by $g_+^2 = M g^2$.

Step 9: the Hessian with the density. Let c and y be real. Under $gp < r_z/4$ we have $\|z^*_{d_0} + gy\|_\infty < 3r_z/8$. Two derivatives in y cancel the prefactor g^{-2} of the classical part, while two derivatives of the logarithm of the density give $g^2 D_z^2 \log \mathcal{J}_c$. By part (2) and Lemma 10.9, the Hessian of the function in part (6) is

$$D_z^2 F_{d_0}(c, z^*_{d_0} + gy) - g^2 D_z^2 \log \mathcal{J}_c(z^*_{d_0} + gy) \ge \tfrac{1}{2} m_0 I - K_M g^2 I.$$

Taking $g_* \le 1$ with $K_M g_*^2 \le m_0/4$, we obtain part (6) on the entire convex product. All the choices above have the dependence stated in Section 10.1. □

10.8 Analytic estimates at intermediate scales

The last theorem collects the factorial bounds proved so far and extends them to the block averages of the background field at every intermediate scale and to their curvature. At

the intermediate scales we use the gauge-corrected averages of (10.9) as representatives; they differ from the iterates of T_M by the explicit gauge transformations G_j of (10.11). As in Section 10.3, $\epsilon_j = \varepsilon_{d,j} = M^{j-d}$.

Theorem 10.11. *There exist $r_*, a_*, C_* > 0$ such that the following hold for every $d \geq 0$, $n \geq 4$, $0 \leq j \leq d$, and every real or complex coarse field θ with $\|\theta\|_\infty < r_*$. All connecting lengths are measured in coarse lattice units.*

1. *For $s \geq 2$,*
$$\|D_\theta^s E_d(e^\theta)\|_{a_*,\mathrm{root}} \leq C_*^{s+1} s!, \qquad \|DE_d(e^\theta)\|_\infty \leq C_* \|\theta\|_\infty.$$
2. *Put $\varepsilon_{d,j} = M^{j-d}$, and let $A_{d,j}(\theta)$ be the j-th iterate of* (10.9) *started at $A_{d,0}(\theta) = A_d(\theta)$. Then $e^{\varepsilon_{d,j} A_{d,j}(\theta)}$ is a representative of the gauge orbit of $T_M^j(e^{\varepsilon_d A_d(\theta)})$, related to it by the gauge transformation G_j of* (10.11)*, and for every $s \geq 1$,*
$$\|D_\theta^s(A_{d,j}, \nabla_{\varepsilon_{d,j}} A_{d,j})\|_{a_*,\mathrm{out}} + \|D_\theta^s(A_{d,j}, \nabla_{\varepsilon_{d,j}} A_{d,j})\|_{a_*,\mathrm{input};\varepsilon_{d,j}^3} \leq C_*^{s+1} s!.$$
In the second norm one coarse input is fixed, and the intermediate outputs and the remaining inputs are summed, with the output measure $\varepsilon_{d,j}^3$.
3. *For the coarse correction D_d we use the counting input columns $\|\cdot\|_{a_*,\mathrm{input,count}}$ of Lemma 10.4; multiplying them by ε_d^{-3} gives the volume-normalized columns. Then*
$$\|D_w^s \mathsf{D}_d\|_{a_*,\mathrm{out}} \leq C_*^s s!, \qquad \|D_w^s \mathsf{D}_d\|_{a_*,\mathrm{input,count}} \leq \varepsilon_d^3 C_*^s s! \quad (s \geq 2),$$
and, on a fixed smaller ball of fine fields w, the first derivative satisfies the corresponding bounds $C_*\|w\|_\infty$ and $\varepsilon_d^3 C_* \|w\|_\infty$.
4. *For a plaquette f of the lattice of mesh $\varepsilon_{d,j}$ put*
$$\mathcal{F}_{d,j,f}(\theta) = \varepsilon_{d,j}^{-2} \operatorname{Log}\big[(e^{\varepsilon_{d,j} A_{d,j}(\theta)})_f\big].$$
It vanishes at $\theta = 0$, and it satisfies the bound of (2) for every $s \geq 1$: the sum of its output row and its volume-normalized input column with output measure $\varepsilon_{d,j}^3$ is at most $C_*^{s+1} s!$. The underlying local fact is that the function
$$\varepsilon^{-2} \operatorname{Log}\big(e^{\varepsilon a} e^{\varepsilon b + \varepsilon^2 v} e^{-\varepsilon a - \varepsilon^2 u} e^{-\varepsilon b}\big)$$
of independent variables (a, b, u, v) in a small ball extends holomorphically to $\varepsilon = 0$, with value $v - u + [a, b]$ there, uniformly for $0 \leq \varepsilon \leq 1$.
5. *In the coupling-rescaled variables $\theta = g_+ B$, with $g_j = \sqrt{\varepsilon_{d,j}}\, g_+$, let*
$$X_1(B) = \varepsilon_{d,j} A_{d,j}(g_+ B)/g_j, \qquad X_2(B) = \operatorname{Log}\big[(e^{\varepsilon_{d,j} A_{d,j}(g_+ B)})_f\big]/g_j$$

be the normalized connection and curvature. Then, for every $s \geq 1$,

$$\|D_B^s X_1\|_{a_*,\mathrm{out}} + \|D_B^s X_1\|_{a_*,\mathrm{input};\varepsilon_{d,j}^3} \leq C_*^{s+1} s!\, \sqrt{\varepsilon_{d,j}}\, g_+^{s-1},$$
$$\|D_B^s X_2\|_{a_*,\mathrm{out}} + \|D_B^s X_2\|_{a_*,\mathrm{input};\varepsilon_{d,j}^3} \leq C_*^{s+1} s!\, \varepsilon_{d,j}^{3/2} g_+^{s-1}.$$

The columns use the output measure $\varepsilon_{d,j}^3$, as in (2).

If the connecting lengths are replaced by the lengths of spanning trees on the external indices, the bounds hold with a_* replaced by $a_*/2$. No reduction of the rate or of the radius depends on the order s.

The bounds in (2)–(5) concern the representatives $e^{\varepsilon_{d,j} A_{d,j}}$ on the stated domain of fields and first differences; in other gauges the iterates $T_M^j(e^{\varepsilon_d A_d(\theta)})$ need not satisfy them.

Proof of Theorem 10.11. Parts (1) and (3), and part (2) at $j = 0$, follow from the earlier coefficient bounds (10.20), (10.17), (10.18), and (10.10). It remains to prove the estimates at intermediate scales and for the curvature. We first construct a map on independent field and difference variables that describes one averaging step (Step 1). We check its mesh powers (Step 2) and its domain (Step 3), apply Lemma 10.3 (Step 4), compose with the background field (Step 5), treat the curvature (Steps 6 and 7), and collect the constants (Step 8). For $d = 0$ only $j = 0$ occurs, so let $d \geq 1$.

Step 1: one step on field and difference variables. We consider one step at mesh $\epsilon \leq 1/M$. A shift of the coarser lattice by one coarse spacing $M\epsilon$ is a shift of the fine lattice by M sites, and it preserves the centered partition and every ordered path in (10.9). Let $S_{M,\mu}$ be this shift of the fine lattice and S'_μ the shift by one spacing of the coarser lattice, in direction μ. Then

$$S'_\mu \Psi_\epsilon(A) = \Psi_\epsilon(S_{M,\mu} A),$$

also at periodic seams. For a triple $G = (G_\mu)$ of fine fields, we define

$$(T_\mu G_\mu)_i(x) = M^{-1} \sum_{l=0}^{M-1} G_{i,\mu}(x + l\epsilon e_\mu).$$

This average of shifts on the fine lattice has counting row and column mass one. If $G = \nabla_\epsilon A$, telescoping gives $S_{M,\mu} A - A = M\epsilon\, T_\mu G_\mu$, and by the fundamental theorem of calculus in finitely many variables we have

$$\nabla_{M\epsilon,\mu} \Psi_\epsilon(A) = \int_0^1 D\Psi_\epsilon\big(A + tM\epsilon\, T_\mu G_\mu\big)\,[T_\mu G_\mu]\, dt. \tag{10.25}$$

We define the right side for independent (A, G) and let $\mathcal{P}_\epsilon$ be the map whose outputs are $\Psi_\epsilon(A)$ and these three expressions. It is holomorphic on a common small ball of

(A, G): since $M\epsilon \leq 1$, the auxiliary arguments $A + tM\epsilon T_\mu G_\mu$ stay in the domain of the local expressions. By (10.25), on the fields with $G = \nabla_\epsilon A$ its outputs are the next pair $(\Psi_\epsilon(A), \nabla_{M\epsilon}\Psi_\epsilon(A))$.

The linear part of $\mathcal{P}_\epsilon$ at zero is

$$\operatorname{diag}(Q, QT_1, QT_2, QT_3),$$

with Q from (10.12). For each of the four component types, its row mass is 1 and its counting column mass is M^{-3}; for the latter, we sum first over the outputs of Q, whose column mass is M^{-3}, and then use the column mass one of T_μ. We keep the four component types as separate labels, so that no factor comparing norms of different components enters at any scale.

Step 2: the mesh powers. We record the powers of the mesh in the nonlinear part, rather than differentiating a bound on values. With the fixed local expression P of the proof of Lemma 10.4, the integrand in (10.25) is

$$M^{-1}DP\big(\epsilon A + tM\epsilon^2 T_\mu G_\mu\big)[T_\mu G_\mu]. \tag{10.26}$$

The constant coefficient $DP(0)$ contributes the linear term $QT_\mu G_\mu$, and every other term contains ϵA or $M\epsilon^2 T_\mu G_\mu$. On a ball $\|A\|_\infty \leq a$, $\|G\|_\infty \leq b$, the nonlinear part is bounded by $K_M(\epsilon ab + \epsilon^2 b^2)$, with first derivatives $O_M(\epsilon b)$ in A and $O_M(\epsilon a + \epsilon^2 b)$ in G. We consider a derivative of order s of the integrand, with p derivatives in A and q in G. If $q \geq 1$ and one of the derivatives in G falls on the external factor $T_\mu G_\mu$, the explicit factor is $\epsilon^p (M\epsilon^2)^{q-1}$. If none falls on it, the factor is $\epsilon^p (M\epsilon^2)^q$ times an undifferentiated G. When $p = 0$ and $q = 1$, the contribution from $DP(0)$ is the linear operator separated above; the remaining terms contain ϵA or $M\epsilon^2 T_\mu G_\mu$. When $q = 0$ and $p \geq 2$, the factor is ϵ^p times G. For the field output the factor at order s is ϵ^{s-1}/M. Using Cauchy estimates in finitely many variables on the fixed polydisc, we obtain, for $s \geq 2$,

$$\begin{aligned} &\|D\mathcal{P}_\epsilon\|_{\text{out}} \leq 1 + K_M\epsilon(a+b), \quad &&\|D\mathcal{P}_\epsilon\|_{\text{input,count}} \leq M^{-3}[1 + K_M\epsilon(a+b)], \\ &\|D^s\mathcal{P}_\epsilon\|_{\text{out}} \leq s!K_M^s\epsilon, &&\|D^s\mathcal{P}_\epsilon\|_{\text{input,count}} \leq M^{-3}s!K_M^s\epsilon. \end{aligned} \tag{10.27}$$

For the columns, the nonlinear terms have a fixed enlarged stencil of inputs, which includes the M shifts in T_μ. A fixed input lies in at most q''_M such stencils; this count enters as a factor only in the nonlinear part, while the linear masses are those computed in Step 1. Integration in t preserves all the bounds.

Step 3: a common domain under iteration. Let u_j be the maximum of the norms of the field and difference variables after j steps. By the bound on the field in Step 3 of the proof of Lemma 10.4 and the bound on the nonlinear part of the difference outputs in Step 2, we have $u_{j+1} \leq u_j + K_M\epsilon_j u_j^2$, after enlarging K_M. As in Lemma 10.3, with

$\sum_j \epsilon_j < 1/(M-1)$, the radius stays below twice its initial value if that value is small enough. On the fields with $G = \nabla_\epsilon A$, the argument $A + tM\epsilon T_\mu G_\mu$ in (10.25) is a convex combination of A and its shift. This shows directly that the difference variables after step $j+1$ have norm at most $(1 + K_M\epsilon_j \|A_j\|_\infty)$ times their norm after step j. This common domain justifies treating A and G as independent variables in (10.27).

Step 4: composition of the steps. By Steps 2 and 3, the maps $\mathcal{P}_{\epsilon_j}$ satisfy the hypotheses of Lemma 10.3, with the nonlinear coefficient ϵ_j and the linear column factor M^{-3} as before. The labels consist of a link and one of the four component types, and the joint radius takes the place of R. Each output depends only on inputs within distance $K_M\epsilon_j$. Applying the lemma, we obtain, for the composition of j steps,

$$R_s(j) \le K^{s+1}s!, \qquad C_s(j) \le (\epsilon_0/\epsilon_j)^3 K^{s+1}s!, \tag{10.28}$$

where $R_s(j)$ and $C_s(j)$ are the output row and the counting input column of the derivative of order s. This holds for all orders and all $0 \le j \le d$. Since the enlarged stencils have total radius at most $K_M\epsilon_j$, a nonzero tensor has connecting length at most $K_M s\epsilon_j$, and a fixed spatial weight only changes K^s.

Step 5: composition with the background field. We start the recursion from the pair $(A_d(\theta), \nabla_{\epsilon_0} A_d(\theta))$. Its value is $O(\|\theta\|_\infty)$, and it has the majorant (10.17). We take the radius of θ so small that this pair lies strictly inside the common domain of Step 3. Composing the two majorant series, we obtain factorial output rows at one fixed smaller radius and one rate. For a column with a coarse input fixed, we follow the branch of the composition containing that input. The initial tensor has counting column $O(\epsilon_0^{-3})$, by (10.18), and the composed map has counting column $O((\epsilon_0/\epsilon_j)^3)$ by (10.28); their product is $O(\epsilon_j^{-3})$, and all other branches use output rows. The output measure ϵ_j^3 cancels the factor ϵ_j^{-3}. By the contraction rule of Lemma 5.3, the same bound holds for every fixed coarse input. The field outputs of the composition are the fields $A_{d,j}(\theta)$ of part (2), and their gauge transformation is the G_j of (10.11), evaluated at the initial field $A_d(\theta)$. These fields represent the gauge orbits of $T_M^j(e^{\epsilon_d A_d(\theta)})$, including for $j = 0$ and $j = d$. For complex θ the gauge transformations are the analytic continuations of these path products, with values in $\mathrm{SL}(2,\mathbb{C})$; for real θ they are unitary. This proves part (2).

Step 6: the local curvature formula. It remains to treat the curvature and its factor ϵ^{-2}. At a plaquette in the directions μ, ν based at x, let

$$a = A_\mu(x), \qquad b = A_\nu(x), \qquad u = \nabla_{\epsilon,\nu} A_\mu(x), \qquad v = \nabla_{\epsilon,\mu} A_\nu(x).$$

The plaquette variable is

$$P_\epsilon = e^{\epsilon a} e^{\epsilon b + \epsilon^2 v} e^{-\epsilon a - \epsilon^2 u} e^{-\epsilon b}.$$

We treat a, b, u, v as independent and fix a small complex ball of these variables on which this product has a logarithm for all complex $|\epsilon| \le 2$. The constant and the linear terms in ϵ of $P_\epsilon - I$ vanish identically. Expanding the four factors to second order, we obtain

$$\operatorname{Log} P_\epsilon = \epsilon^2 (v - u + [a, b]) + O(\epsilon^3).$$

The quotient $\epsilon^{-2} \operatorname{Log} P_\epsilon$ therefore has a removable singularity at $\epsilon = 0$, with the value stated in part (4). Cauchy's formula on the disc $|\epsilon| \le 2$, or the Taylor series beginning at degree two, bounds the quotient uniformly for $|\epsilon| \le 1$. Cauchy's formula in the finitely many variables (a, b, u, v) on a strictly smaller ball then bounds its coefficients by $K^{s+1} s!$, uniformly for $0 \le \epsilon \le 1$. This division by ϵ^2 uses no second difference of the field.

Step 7: composition with the intermediate pair. We compose this local quotient with the majorant of Step 5 for the intermediate pair. Each plaquette involves a fixed number of field and difference entries, at distance $O(\epsilon_j)$, and each entry occurs in a fixed number of plaquettes. The composition rule gives the factorial row for the curvature. With a coarse input fixed, the branch containing it uses the volume-normalized column of the pair, with the factor ϵ_j^3, and the other branches use rows. Hence the plaquette output has the same volume-normalized column, with no inverse power of the mesh. Its value at $\theta = 0$ is zero, because all fields and differences vanish there. This proves part (4).

Step 8: constants and coupling-rescaled variables. We take a_* smaller than the finitely many rates used for the extension H_d, the Green operator, the correction, the background field, the envelope formula for the action, and the passage from rows to columns. We take r_* inside all the finitely many common balls, and we increase C_* to dominate their constants and the fixed counts of incidences and comparisons of norms. These choices are independent of d, n, and j. With them, parts (1) and (3) are (10.20), the bound $\|\nabla_\theta E_d\|_\infty \le K \|\theta\|_\infty$ of the proof of Theorem 10.6, and (10.10). Since $\epsilon_j = \varepsilon_{d,j}$ and $g_j = \sqrt{\epsilon_j}\, g_+$,

$$\frac{\epsilon_j A_{d,j}(g_+ B)}{g_j} = \sqrt{\epsilon_j}\, g_+^{-1} A_{d,j}(g_+ B), \qquad \frac{\operatorname{Log}[(e^{\epsilon_j A_{d,j}(g_+ B)})_f]}{g_j} = \epsilon_j^{3/2} g_+^{-1} \mathcal{F}_{d,j,f}(g_+ B).$$

Each of the s derivatives in B produces a factor g_+, which gives the powers g_+^{s-1} in part (5), with the same row and column conventions. Finally, by Lemma 5.3, the length of a spanning tree on the external indices is at most twice the connecting length, and replacing a_* by $a_*/2$ proves the last assertion. □

SECTION 11

Classical continuum limits and contraction estimates

This section proves four deterministic properties of the effective classical actions E_d and of the classical background fields constructed in Section 10. They are the classical inputs to the fluctuation integration in Part IV and to the large-field analysis in Part VI.

1. Section 11.2 fixes the coarse lattice and the coarse field and lets the number d of averaging steps tend to infinity, so that the fine lattice spacing M^{-d} tends to zero. We prove that a summably corrected version of E_d is nondecreasing in d, that E_d converges, that E_d is bounded below by a fixed multiple of the Wilson action, and that, at a fixed coarse lattice, E_d converges in the weighted analytic coefficient norm. We also prove that the reconstruction maps through finitely many averaging steps converge, together with all the matrices and densities entering them. This is Theorem 11.2; it is used in Sections 14, 15, and 22.
2. Section 11.3 proves that two small representatives of the same periodic gauge orbit have the same effective classical action, even when the gauge transformation relating them has no small logarithm. This implies that E_d extends to a real analytic function on the gauge-invariant small-field domain D_ρ (Theorem 11.6). This is used in Sections 13, 14, and 19.
3. Section 11.4 constructs, for one renormalization group step of J averaging steps, a representative of the gauge orbit of the classical background field that satisfies the block averaging constraint exactly and whose potential and first differences have coefficient bounds uniform in J. For each fixed J it identifies this representative, after an explicit gauge transformation, with the zero-fluctuation reconstruction (Lemma 11.9). This is used in Sections 13, 15, 23, and 31.
4. Section 11.5 proves that evaluating an effective interaction R at the classical background field of one renormalization group step contracts its weighted analytic norm by a factor $O(M^{-J})$, at the same analytic radius and the same spatial decay rate, provided that R is invariant under lattice translations, constant gauge transformations, spatial inversion, and infinitesimal gauge transformations (Theorem 11.10). This contraction allows us to choose J in Theorem 13.5.

The inputs are the results of Section 10 on the classical background field, its symmetries, the nested minimization identity for E_d, and its derivative estimates (Theorems 10.6, 10.10, and 10.11 and Lemma 10.8); the gauge slice with prescribed coarse gauge of Theorem 9.9;

the finite-torus estimate for gauge-invariant quadratic forms of Lemma 8.6; and the one-step energy estimate of Theorem 7.3 and Lemma 7.2.

The classical background fields are constrained minimizers in the sense of Bałaban's variational construction [16, Theorem 1, p. 279], which rests on his averaging operations and gauge-fixing conditions [12, 14]. The constraints, coordinate domains, and coefficient norms used here are those of Sections 9 and 10, and the results of this section are proved here.

11.1 Setting and conventions

Units and norms. Fix an odd integer $M \geq 3$. Throughout this section we measure lengths in units of the coarse lattice, which has integer side $n \geq 4$ (or $m \geq 4$ when we consider one renormalization group step) and spacing one. For $d \geq 0$ the fine lattice has side nM^d and spacing

$$\varepsilon_d = M^{-d}$$

in these units; its physical spacing on the torus $\mathbb{T}_L^3$ is $a_n \varepsilon_d$. For a configuration $U = e^{\varepsilon A}$ on a lattice of spacing ε we write

$$\mathscr{S}_\varepsilon(U) = 4\varepsilon^{-1} \sum_p q(U_p), \qquad \|A\|_{C^1_\varepsilon} = \|A\|_\infty + \|\nabla_\varepsilon A\|_\infty,$$

where ∇_ε is the collection of all componentwise forward differences divided by ε, as in Definition 10.1. We also write $\mathscr{S}_\varepsilon(A)$ for $\mathscr{S}_\varepsilon(e^{\varepsilon A})$. All spatial statements in this section concern complete periodic tori. In analytic coefficient norms, a and b denote spatial decay rates. We use ε_d, t, δ, and δ_l for dimensionless lattice spacings and a_n for the physical coarse spacing.

Tensor norms, their Taylor factorials, and their volume normalizations are those of Definition 4.6. For a derivative tensor of a scalar function, $\|\cdot\|_{a,\mathrm{root}}$ denotes the coefficient bound with each external index fixed in turn: it fixes any one index and sums the absolute values of the coefficients over the other indices, with weight $e^{a\mathrm{t}}$, where t is the length of a shortest network connecting the locations of all indices (internal vertices allowed). Let F be an analytic field of a coarse variable θ, with values in the link variables of a lattice of spacing ε, and let $v = \varepsilon^3$. The output-row norm $\|D_\theta^s F\|_{b,\mathrm{out}}$ fixes one output index and sums the absolute values of the coefficients over all coarse input indices, with weight $e^{b\mathrm{t}}$. The volume-normalized column norm $\|D_\theta^s F\|_{b,\mathrm{input};v}$ fixes one coarse input index, sums over the output index and the remaining inputs with the same weight, and multiplies the result by the output volume v.

Constants. Unless stated otherwise, a constant in this section depends only on M and on the fixed larger domains, spatial decay rates, and coefficient bounds of Sections 9 and 10, with the dependence described in item (2) of Definition 4.14. Such a constant does not

depend on the coarse side, on the depth d, on the coarse field, on L, on γ, or on J. We identify explicitly the constants that may depend on J, on a finite lattice, or on a fixed number of averaging steps.

Real and complex arguments. Variational comparisons and lower bounds on Hessians are applied only at real coarse fields and real free coordinates. At complex arguments the effective classical action and the conditional constrained minimizers denote the holomorphic continuations of the critical values and critical points from real data, as in Theorems 10.6 and 10.10.

The effective classical action and the gauge orbit of the classical background field. We recall the objects of Section 10. For a small real coarse logarithm θ, Theorem 10.6 constructs the classical background field $A_d(\theta)$ on the lattice of spacing ε_d. It is the unique minimizer of $\mathscr{S}_{\varepsilon_d}$ among the fields whose gauge-corrected block average (see below) equals e^θ, in a convex coordinate domain inside the gauge slice $\mathsf{P}_d \operatorname{div}_{\varepsilon_d} A = 0$, where P_d is the projection of Theorem 9.9. The effective classical action is its value,

$$E_d(e^\theta) = \mathscr{S}_{\varepsilon_d}(A_d(\theta)), \qquad E_0 = S_W.$$

We distinguish three objects.

1. The value $E_d(c)$, a function of the coarse field c. It is the minimum on the locally unique branch constructed in Sections 9 and 10; it is not asserted to be the infimum of $\mathscr{S}_{\varepsilon_d}$ over the whole fiber $\{U : T_M^d U = c\}$.
2. The gauge orbit of the classical background field, that is, the orbit of $e^{\varepsilon_d A_d(\theta)}$ under periodic gauge transformations of the fine lattice. By gauge invariance of the Wilson action, every point of this orbit has action $E_d(e^\theta)$.
3. Representatives at the fine and intermediate resolutions. On the fine lattice, the slice representative $e^{\varepsilon_d A_d(\theta)}$ and the representative with $T_M^d U = e^\theta$ from Lemma 10.8 belong to the orbit in item (2). For each j, the field $e^{\varepsilon_{d,j} A_{d,j}(\theta)}$, with $\varepsilon_{d,j} = M^{j-d}$, represents the gauge orbit of $T_M^j(e^{\varepsilon_d A_d(\theta)})$ on its intermediate lattice, as in Theorem 10.11. The representative $C_{d_0,J}(\theta)$ and its orbit on the fine lattice of one renormalization group step are defined in Section 11.4.

We prove convergence of the values and of specified finite reconstruction maps. We do not prove convergence of the representatives in a fixed gauge as $d \to \infty$, and we do not identify a continuum variational problem.

Block averages and gauge-corrected block averages. Two maps from fine to coarse configurations appear below. The first is the iterated block averaging map T_M^d. The second is the gauge-corrected block average of Lemma 8.10, which at each averaging step multiplies the output of T_M by explicit vertex gauge factors at the coarse vertices. Let

$V_d(U)$ denote the gauge-corrected block average of a configuration U after d steps, and let $\mathsf{G}_d(U)$ denote the accumulated gauge transformation at the coarse vertices. Then

$$V_d(U) = \big(T_M^d U\big)^{\mathsf{G}_d(U)}, \tag{11.1}$$

for configurations in the domain of Lemma 8.10. This follows from the gauge covariance of T_M (Section 6), induction over the d steps, and the composition rule $(U^k)^l = U^{lk}$ of (4.1); for the representatives of Theorem 10.11 we verify it explicitly in (11.12) below. The classical background field is defined through V_d, but the block averaging constraint in the renormalization group is $T_M^d U = c$.

The gauge slice of Theorem 9.9 connects the two. Given a small field A on the lattice of spacing ε_d and a small prescribed coarse gauge logarithm ζ, it produces a gauge transformation e^λ such that A^{e^λ} satisfies $\mathsf{P}_d \operatorname{div}_{\varepsilon_d} A^{e^\lambda} = 0$ and

$$V_d\big(e^{\varepsilon_d A^{e^\lambda}}\big) = b^{e^\zeta}, \qquad b = T_M^d\big(e^{\varepsilon_d A}\big).$$

With prescribed coarse gauge logarithm $\zeta = 0$, the gauge-corrected block average of the transformed field is exactly the block average b of the original field. Since gauge transformations preserve the Wilson action, this converts a configuration U with $T_M^d U = b$ into a field in the gauge slice with gauge-corrected block average b and the same action. We use this in the comparison of Section 11.2, the transfer of criticality in Section 11.3, and the identification in Section 11.4. We also use that the transformed field has small potential and curl (Theorem 9.9). It therefore lies in the larger domain of Theorem 9.7, on which the background field is still the unique minimum (Theorem 10.6). We do not need an estimate on all first differences of the output.

Majorant series. For power series with nonnegative coefficients we write $P \preccurlyeq Q$ if every coefficient of P is at most the corresponding coefficient of Q. For an analytic field F of the coarse variable θ, a decay rate b, and an output volume v, we define

$$\begin{aligned} \mathscr{R}_b(F;u) &= \sum_{s\ge 1} \frac{u^s}{s!} \|D_\theta^s F(0)\|_{b,\mathrm{out}}, \\ \mathscr{C}_{b,v}(F;u) &= \sum_{s\ge 1} \frac{u^{s-1}}{(s-1)!} \|D_\theta^s F(0)\|_{b,\mathrm{input};v}. \end{aligned} \tag{11.2}$$

For a group-valued F, only $F - F(0)$ enters these series, so the identity matrix does not contribute to a series vanishing at $u = 0$. All comparisons of series below are coefficientwise comparisons of absolute tensor sums, not comparisons of values. The following lemma gives the counting rule for compositions of such series.

Lemma 11.1. *Let $k \geq 1$, let T be a symmetric k-linear map between finite-dimensional spaces, let F be holomorphic near $0 \in \mathbb{C}^D$ with $F(0) = 0$, and put $G(\theta) = T[F(\theta), \dots, F(\theta)]/k!$. For $s \geq 1$ and $h_1, \dots, h_s \in \mathbb{C}^D$,*

$$\frac{1}{s!} D^s G(0)[h_1, \dots, h_s] = \frac{1}{k!} \cdot \frac{1}{s!} \sum_{(S_1, \dots, S_k)} T\big[D^{|S_1|} F(0)[h_{S_1}], \dots, D^{|S_k|} F(0)[h_{S_k}]\big], \tag{11.3}$$

where the sum runs over the ordered partitions of $\{1, \dots, s\}$ into k nonempty sets and h_S is the list of the h_i with $i \in S$. For given sizes $s_1 + \cdots + s_k = s$ there are $s!/\prod_i s_i!$ such partitions.

Proof. Since $F(0) = 0$, inserting the Taylor series of F in each argument of T, we find that the homogeneous part of degree s of G at h is

$$\frac{1}{k!} \sum_{\substack{s_1 + \cdots + s_k = s \\ s_i \geq 1}} T\Big[\frac{D^{s_1} F(0)[h^{\otimes s_1}]}{s_1!}, \dots, \frac{D^{s_k} F(0)[h^{\otimes s_k}]}{s_k!}\Big].$$

The right side of (11.3) is symmetric in $h_1, \dots, h_s$, and at $h_1 = \cdots = h_s = h$ it equals this expression, because the partitions with sizes $(s_1, \dots, s_k)$ number $s!/\prod_i s_i!$. A symmetric multilinear form is determined by its values on the diagonal, which proves (11.3). □

We use Lemma 11.1 as follows. In (11.3), the factor $1/k!$ cancels the reorderings of the k slots of T. After division by $s!$, the partitions with sizes $(s_1, \dots, s_k)$ carry exactly the weights $\prod_i 1/s_i!$ of the Taylor coefficients of the arguments. Taking absolute values before summing over labels, we bound the absolute order-s coefficients of G, divided by $s!$, by the products of the absolute order-s_i coefficients of F, divided by $s_i!$, summed over $s_1 + \cdots + s_k = s$, with the weight $1/k!$ and no further factorial. This also holds when some labels coincide. Consider an absolute weighted sum over labels in which the vertex T contributes a factor τ and each argument contributes the coefficients of a series $f(u)$ with nonnegative coefficients. Then the series of G is dominated by $\tau f(u)^k/k!$. If one label is fixed, it lies in exactly one of the k groups, and a coefficient of order j containing the fixed label has denominator $(j-1)!$ rather than $j!$. Summing over the k choices replaces one factor f by its termwise derivative f', which gives $\tau f(u)^{k-1} f'(u)/(k-1)!$. Any external index may serve as the fixed one. The spatial weights are handled separately in each application: the network connecting the external indices of a composition is contained in the union of the networks of T and of the arguments, so the weight $e^{b\mathrm{t}}$ of the composition is at most the product of the weights of the factors.

11.2 Classical actions at a fixed coarse lattice

Theorem 11.2. *Fix an odd integer $M \geq 3$. There exist $r_{\mathrm{CK}}, a_0, C > 0$, depending only on M and the fixed domains, and in particular independent of the coarse side $n \geq 4$ and of $d \geq 0$, with the following properties.*

1. *Let $0 < r \le r_{\mathrm{CK}}$ and let $c = e^{\theta}$ be a real coarse field with $\|\theta\|_\infty < r$. Then, with both actions evaluated at the same coarse field on the same coarse lattice,*

$$E_d(c) \le (1 + CrM^{-2(d+1)})E_{d+1}(c) \qquad (d \ge 0).$$

Set $P_0 = 1$, $P_d = \prod_{\ell=0}^{d-1}(1 + CrM^{-2(\ell+1)})$, and $\kappa_* = P_\infty^{-1} > 0$. Then $P_d E_d(c)$ is nondecreasing in d, $0 \le E_d(c) \le Cn^3\|\theta\|_\infty^2$, the limit $E_\infty(c) = \lim_{d\to\infty} E_d(c)$ exists, and

$$E_d(c) \ge P_d^{-1}E_0(c) \ge \kappa_* E_0(c), \qquad E_\infty(c) \ge (P_d/P_\infty)E_d(c),$$
$$E_d(c) - E_\infty(c) \le C_M r M^{-2d} E_d(c).$$

In particular these inequalities apply to $c = U^k$ whenever U is a coarse configuration and k is a fixed periodic gauge transformation with U^k in the stated domain; the identification of the values for different choices of k is Theorem 11.6.

2. *For each fixed n, the whole sequence E_d converges locally uniformly and holomorphically on a common complex polydisc of coarse logarithms to a holomorphic function E_∞. For $0 < a \le a_0$ and every sufficiently small fixed $\rho > 0$ inside that polydisc,*

$$\mathcal{B}_{\rho,a}(E_d - E_\infty) \longrightarrow 0 \qquad (d \to \infty).$$

Every derivative of fixed order converges on strictly smaller neighborhoods. The radii, the coefficient bounds, and κ_* are chosen uniformly in n; the convergence modulus is asserted only at fixed n.

3. *Fix two lattices of sides $M^s m$ and m, with $1 \le s < \infty$, and fix the couplings. Consider the reconstruction maps through s averaging steps in the free logarithmic coordinates, built from E_{d+i}, $0 \le i < s$, as in Theorem 10.10. As $d \to \infty$ they converge, and so do the conditional constrained minimizers, the constrained Hessians, their positive square roots and inverse square roots, every Haar disintegration density, and the whole triangular reconstruction map. The convergence holds with every derivative of fixed order in the coarse and free variables, uniformly on every compact set of real coarse fields and real free coordinates that lies in the domains of Theorem 10.10 with a fixed strict margin, and on a complex neighborhood of each such set. On these fixed finite lattices it also holds in an absolute coefficient norm with a smaller radius. The constants and convergence moduli in this item may depend on the two lattices, on s, on the couplings, and on the margins.*

The proof uses two lemmas. The first controls one application of T_M; the second is a form of Montel's theorem with a statement about coefficient norms.

Lemma 11.3. *There exist $b_M > 0$ and $C_M < \infty$, depending only on M and on fixed margins of the identity chart of the logarithm, with the following properties. Let U be a configuration on a periodic lattice of side Mn, with $n \ge 4$, such that $\eta = \max_p \rho_G(U_p) \le b_M$. Then every based*

loop holonomy entering the definition of the plaquettes of $T_M U$ lies in a fixed strict identity chart of the logarithm, and

$$\sum_P q((T_M U)_P) \le M(1 + C_M \eta) \sum_p q(U_p). \tag{11.4}$$

If, in addition, the spacing of this lattice is $t \le M^{-1}$ in coarse units, $U = e^{tA}$, and $\|A\|_{C^1_t} \le R \le b_M$, then

$$T_M U = e^{MtB}, \qquad \|B\|_{C^1_{Mt}} \le C_M R, \qquad \max_p \rho_G(U_p) \le C_M t^2 (R + R^2). \tag{11.5}$$

All constants are independent of the spacing and of the side.

Proof. Let $b_M \le \min(\delta_*(M), \delta_M)$, where $\delta_*(M)$ and δ_M are the thresholds of Theorem 7.3 and Lemma 7.2, reduced further for the fixed margins of the logarithm if needed. Theorem 7.3, applied to the averaging step from side Mn to side n with maximal plaquette distance η, implies (11.4). The part of Lemma 7.2 that assumes only a bound on plaquette distances places every based loop holonomy in a fixed strict identity chart, without a bound on the link variables themselves. Its constants depend only on M and the fixed margins, and not on t, n, or any coupling.

For the bound on the potential, consider the averaging word (4.3) with every fine link variable written as e^{ta_e}. Its identity logarithm vanishes at $t = 0$ for all small a. Division by Mt is removable, since after expanding the absolutely convergent product and logarithm series, every nonconstant monomial contains at least one factor t. We obtain a local analytic function of the finitely many a_e, with first derivative bounded uniformly for $0 \le t \le M^{-1}$ and value zero at $a = 0$. Its value is bounded by $C_M R$. To compare the words at two neighboring coarse vertices, we translate every input edge by Mt. By telescoping M fine differences, each input changes by at most $Mt\|\nabla_t A\|_\infty$. Integrating the derivative of the local function along the segment between the two lists of inputs gives the bound on $\nabla_{Mt} B$ in (11.5). Finally, in a fine plaquette the linear sum of the four exponents tA is bounded by $Ct^2\|\nabla_t A\|_\infty$, and the remainder of the product series is at most $Ct^2\|A\|_\infty^2$. The logarithm series preserves this bound, which gives the last assertion of (11.5). □

Lemma 11.4. *Fix $D \ge 1$, and let (f_j) be holomorphic functions on a fixed open polydisc in $\mathbb{C}^D$, uniformly bounded on every strictly smaller closed polydisc.*

1. *Every sequence of the f_j has a subsequence converging locally uniformly to a holomorphic function, together with all derivatives of fixed order.*
2. *If (f_j) converges pointwise on a real open subpolydisc, then the whole sequence converges locally uniformly on the polydisc.*
3. *Under the convergence hypothesis of item (2), let f be the locally uniform limit. If also $\|D^k f_j(0)\|_{a,\mathrm{root}} \le C^{k+1} k!$ for all $k \ge 2$ and all j, then for $C\rho < 1$ we have $\mathcal{B}_{\rho,a}(f_j - f) \to 0$.*

Here D and the finite spatial weights are fixed; only the coefficient bound C may be uniform over a family of dimensions.

Proof. Consider a polydisc of radius R on which $|f_j| \le B_R$. By the iterated one-variable Cauchy integral for the Taylor coefficient c_α, we have $|c_\alpha| \le B_R R^{-|\alpha|}$. (The Cauchy formula follows by integrating the power series on a smaller circle, where it converges uniformly, and then enlarging the circle inside the domain of holomorphy.) Fix an increasing sequence of radii approaching the outer radius. By successive selection over the countably many multi-indices, the bounded complex coefficients have a common convergent subsequence. On a smaller radius $r < R$, the tails of the series are bounded uniformly by $B_R \sum_{k>K} \binom{D+k-1}{k} (r/R)^k$, which tends to zero as $K \to \infty$. This implies that the subsequence converges locally uniformly to the corresponding power series. Multiplying the coefficients by a fixed power of k keeps these tails summable, which gives the convergence of all derivatives of fixed order. This proves (1).

For (2), two subsequential limits agree on the real open subpolydisc. Fix all but one coordinate at real values there. A holomorphic function of one variable that vanishes on a real interval vanishes on its disc: if it had a first nonzero Taylor coefficient, factoring out the corresponding power would make its zeros isolated. Repeating this for each coordinate, we find that the two limits agree on a complex subpolydisc. By continuation along line segments through overlapping Taylor discs, they agree on the whole polydisc. Suppose that the whole sequence fails to converge uniformly on some compact subset. Then a subsequence stays at a fixed positive distance from any limit there. By (1), it has a further locally convergent subsequence, whose limit is the common limit just identified. This contradiction proves (2).

For (3), at fixed D, convergence of the finitely many entries of order k implies convergence of every absolute sum at that order with a fixed index, and the limit inherits the assumed bound. The term of order k in $\mathcal{B}_{\rho,a}(f_j - f)$ is therefore at most $2C^{k+1}k!\rho^{k-2}/k! \le 2C^3(C\rho)^{k-2}$. Its tail is a geometric series, uniformly in j, while every finite initial sum converges. This proves (3). For derivatives at other centers on a strictly smaller neighborhood, we use the Cauchy integrals about those centers on a larger compact neighborhood; by the uniform convergence just proved, these integrals converge uniformly. □

Proof of Theorem 11.2. The proof has four steps: the comparison of consecutive depths, the monotone limit and its consequences, the analytic convergence at fixed n, and the convergence of the finite reconstruction maps.

Step 1: comparison of consecutive depths. Let θ be real and lie in a ball of coarse logarithms inside the common domains of Theorems 10.6, 10.10, and 10.11 and Lemma 10.8

and of the gauge slice of Theorem 9.9. By Lemma 10.8, the gauge orbit of the classical background field of depth $d+1$ has a real representative

$$U = e^{tA^{\mathrm{ord}}}, \qquad t = M^{-(d+1)}, \qquad T_M^{d+1}U = c, \qquad \|A^{\mathrm{ord}}\|_{C^1_t} \le C\|\theta\|_\infty,$$

with action $\mathscr{S}_t(U) = E_{d+1}(c)$. Let $b = T_M U$, a configuration on the lattice of spacing $Mt = \varepsilon_d$. Since the fine side is $nM^{d+1} = M(nM^d)$, the map $U \mapsto b$ is one averaging step with integer coarse side $nM^d \ge 4$, and $T_M^d b = T_M^{d+1}U = c$. If $C\|\theta\|_\infty \le b_M$, which holds after reducing r_{CK}, then Lemma 11.3 implies that $b = e^{\varepsilon_d B}$ with $\|B\|_{C^1_{\varepsilon_d}} \le C_M\|\theta\|_\infty$. The divergence and the curl of B are also bounded by this number. Let $r_{\mathrm{CK}} > 0$ be so small that these three bounds meet the hypotheses of Theorem 9.9 with prescribed coarse gauge logarithm zero, and that the output of the slice lies strictly inside the larger domain of Theorem 9.7 on which Theorem 10.6 applies. Each of these requirements divides a fixed positive radius by a fixed constant, so r_{CK} is independent of d and n. By the discussion in Section 11.1, the slice output has gauge-corrected block average exactly c and the same action as b. By the minimizing property of the background field in that larger domain, we have

$$E_d(c) \le \mathscr{S}_{\varepsilon_d}(b).$$

For $d = 0$ this is an equality, since $b = c$ and $E_0 = S_W$. No stationarity of b is needed.

Let $\|\theta\|_\infty < r \le r_{\mathrm{CK}}$. By the last bound of (11.5), we have $\eta = \max_p \rho_G(U_p) \le Crt^2$. Inserting this into (11.4), with the normalization of $\mathscr{S}$, we obtain

$$\mathscr{S}_{Mt}(T_M U) \le \frac{4}{Mt} M(1 + C\eta) \sum_p q(U_p) \le (1 + CrM^{-2(d+1)}) E_{d+1}(c).$$

Combining the last two displays, we obtain the comparison in item (1).

Step 2: the monotone limit. We integrate the bound $\|DE_d(e^\theta)\|_\infty \le C_*\|\theta\|_\infty$ of Theorem 10.11 along the segment $u\theta$, $0 \le u \le 1$, and use $E_d(I) = 0$ (Theorem 10.6). Since there are $3n^3$ link variables, each with a fixed number of color components, on both the real and the complex balls of coarse logarithms we have

$$|E_d(e^\theta)| \le Cn^3\|\theta\|_\infty^2. \tag{11.6}$$

Nonnegativity on real data is part of Theorem 10.6. Since $P_{d+1} = P_d(1 + CrM^{-2(d+1)})$, the comparison of Step 1 is equivalent to $P_d E_d \le P_{d+1}E_{d+1}$. Further, using $\log(1+x) \le x$ and $\sum_{\ell \ge d} M^{-2(\ell+1)} = M^{-2d}/(M^2-1)$, we have

$$1 \le P_d \le P_\infty \le \exp\Big(\frac{Cr}{M^2-1}\Big), \qquad 0 \le \log(P_\infty/P_d) \le \frac{CrM^{-2d}}{M^2-1}.$$

The sequence $P_d E_d(c)$ is nondecreasing and, by (11.6) and $P_d \le P_\infty$, bounded. It therefore has a limit, and dividing by $P_d \to P_\infty$ shows that $E_\infty(c) = \lim_d E_d(c)$ exists. The inequality

$P_d E_d \geq P_0 E_0$ implies $E_d \geq P_d^{-1} E_0 \geq P_\infty^{-1} E_0$, and $P_\infty E_\infty = \lim_j P_j E_j \geq P_d E_d$ implies $E_\infty \geq (P_d/P_\infty) E_d$. We conclude that

$$E_d - E_\infty \leq \big(1 - P_d/P_\infty\big) E_d \leq \log(P_\infty/P_d)\, E_d \leq \frac{Cr}{M^2-1} M^{-2d} E_d,$$

where we used $1 - e^{-u} \leq u$ for $u \geq 0$. This is the one-sided error bound in item (1). We obtain no bound of the same order for $E_\infty - E_d$.

Step 3: analytic convergence at fixed n. Fix n. The coarse field has finitely many coordinates, and (11.6) bounds the holomorphic functions E_d uniformly on the common complex polydisc of Theorem 10.11. By Step 2, they converge on real points. By part (2) of Lemma 11.4, the whole sequence converges locally uniformly and holomorphically. Let $a_0 \leq a_*$, and let the norm radius ρ satisfy $C_* \rho < 1/2$ and lie strictly inside that polydisc, where a_*, C_* are the constants of Theorem 10.11. Since $\|D^k E_d\|_{a_*,\mathrm{root}} \leq C_*^{k+1} k!$ for $k \geq 2$, part (3) of the same lemma implies that $\mathcal{B}_{\rho,a}(E_d - E_\infty) \to 0$ for $0 < a \leq a_0$. Cauchy integrals on larger compact neighborhoods give the convergence of derivatives of fixed order on strictly smaller neighborhoods. The radii, the coefficient bound, and κ_* were chosen before n was fixed; only the subsequence argument and the finite list of coefficients use the fixed value of n.

Step 4: finite reconstruction maps. Fix the two lattices, the number s of averaging steps, and the couplings. For the i-th averaging step write

$$F_{d+i}(c,z) = E_{d+i}(b(c,z)), \qquad D_z F_{d+i}(c, z^*_{d+i}(c)) = 0,$$
$$K_{d+i}(c) = D_z^2 F_{d+i}(c, z^*_{d+i}(c)),$$

where $b(c,z)$ is the configuration of Theorem 10.10 with coarse field c and free logarithmic coordinates z. At real c, the conditional constrained minimizers $z^*_{d+i}(c)$ are real minima; at complex c, they are the holomorphic continuations solving the displayed critical equation. The map $b(c,z)$ does not depend on d, and Theorem 10.10 provides its coordinate neighborhoods and the uniform lower bound on the real Hessian. Since E_{d+i} converges on each fixed lattice by Step 3, F_{d+i} and all its derivatives of fixed order converge on every compact complex product with a strict margin. On real products the Hessian is at least $m_0 I/2$. We subtract the critical equations at depths $d+i$ and $d'+i$, and integrate the Hessian of $F_{d'+i}$ along the segment joining the two minimizers, which lies in the convex ball of free coordinates. We obtain

$$|z^*_{d+i}(c) - z^*_{d'+i}(c)|_2 \leq \frac{2}{m_0} \sup_z |D_z F_{d+i}(c,z) - D_z F_{d'+i}(c,z)|_2.$$

This shows that the conditional constrained minimizers are uniformly Cauchy on compact sets of real coarse fields.

For complex coarse fields, we freeze the Hessian K at a real minimizer, so that $\|K^{-1}\|_2 \le 2/m_0$. On a sufficiently small complex neighborhood in the coarse and free variables, the uniform analytic bounds on third and mixed derivatives give $\|I - K^{-1}D_z^2 F_{d+i}\|_2 \le 1/2$. These bounds are uniform in d; their conversion between finite-dimensional norms may depend on the fixed lattice. The map $z \mapsto z - K^{-1}D_z F_{d+i}(c, z)$ is therefore a contraction there and maps a smaller closed ball into itself. Its iterates are holomorphic and converge geometrically, and they define a holomorphic critical point on a common neighborhood. For two late depths, we may use the matrix K of the first depth for both equations, since their Hessians converge. If Φ_d and $\Phi_{d'}$ denote the two contraction maps, then

$$|z_d - z_{d'}| \le |\Phi_d(z_d) - \Phi_d(z_{d'})| + |\Phi_d(z_{d'}) - \Phi_{d'}(z_{d'})| \le \frac{1}{2}|z_d - z_{d'}| + \sup|\Phi_d - \Phi_{d'}|,$$

which implies that the difference of the fixed points is at most twice the uniform difference of the maps. A finite cover of the compact set of real coarse fields gives common complex neighborhoods, and on overlaps the branches agree by uniqueness of the fixed point. This proves locally uniform convergence of the complex minimizers, and Cauchy integrals give the convergence of each derivative of fixed order. The constrained Hessians K_{d+i} then converge with their derivatives. Here the chain rule reads

$$D_z^2 F = (D_z B)^{\mathsf{T}} D^2 E(B) D_z B + DE(B) D_z^2 B, \qquad B = \operatorname{Log} b,$$

and the second term is kept throughout.

The positive square root and its inverse are holomorphic operations on the common matrix neighborhood in use. For a real symmetric $K \ge m_0 I/2$,

$$K^{-1/2} = \frac{2}{\pi}\int_0^\infty (K + t^2 I)^{-1}\, dt, \qquad K^{1/2} = K K^{-1/2}, \tag{11.7}$$

as one checks by diagonalizing K and using $\int_0^\infty (\lambda + t^2)^{-1} dt = \pi/(2\sqrt{\lambda})$. On a complex neighborhood on which the Hermitian part of K is at least $m_0 I/4$, the resolvent norm is at most $(m_0/4 + t^2)^{-1}$. Its Neumann series and the differentiated series converge on smaller neighborhoods, and every derivative is bounded by a finite product of these integrable resolvent bounds. Integrating over $[0, T]$ and using the uniform integrable tails, we obtain the convergence in (11.7) with every derivative of fixed order.

The reconstruction through the s averaging steps has the explicit triangular form

$$z_i = z_{d+i}^*(c_i) + g_i K_{d+i}(c_i)^{-1/2} w_i, \qquad c_{i-1} = b(c_i, z_i),$$

with the indexing adjusted to the input and output sides of each step. Starting at the coarsest field, we prove by finite induction over the s steps the convergence of each z_i and c_{i-1} and of every joint derivative of fixed order in the coarse and free variables. At

each step, the strict margin keeps the argument in a compact subset of the domain of the next map. The Haar disintegration density of one step is the product of the Haar factors $\operatorname{sinc}^2(|\Gamma_e|/2)$ over the non-tree edges, divided by the constraint determinant at the eliminated edges, times the constant Haar normalization ((4.2) and the formula following Definition 4.3). For complex arguments, the Haar factor is defined by its even power series in the bilinear expression $\Gamma_e \cdot \Gamma_e$, whose restriction to real arguments is the real formula; no Hermitian modulus is continued. All the arguments just shown to converge enter these fixed functions, and the fixed margin keeps the denominators uniformly away from zero. These factors and their derivatives therefore converge. The triangular factors $g_i^{3f_i} \det K_{d+i}^{-1/2}$ converge as well.

Finally, on these fixed finite lattices the finitely composed maps have a common holomorphic bound on a strictly larger product. The Cauchy estimate in the proof of Lemma 11.4 then gives a common absolute majorant on a smaller product. The spatial weight and the volume factor are finite here: the weight of a coefficient of degree k is at most $e^{aD_0 k}$, where D_0 is the diameter of the fixed finite set of indices. Let the radius of the coefficient norm be smaller than the larger radius times e^{-aD_0}. Then the tail of degree k is a polynomial in k times a fixed geometric sequence, and the tails are uniformly small, while every finite list of coefficients converges. This proves convergence in an absolute coefficient norm with a smaller radius. All constants in this last step may depend on the fixed lattices, on s, on the couplings, and on the margins, as asserted. □

Remark 11.5. Item (3) of Theorem 11.2 asserts complex convergence on a neighborhood of each compact set of real data; this is what the proof constructs, and it is what is used in Section 15. If, on a polydisc centered at a real point of such a set, all the maps in item (3) are holomorphic and bounded uniformly in d, then part (2) of Lemma 11.4 extends the locally uniform convergence to that polydisc.

11.3 Gauge invariance on overlapping charts

The classical background field exists only in a chart around the identity, but the renormalization group needs a well-defined value of E_d on the gauge-invariant small-field domain $D_\rho = \mathcal{G}B_\rho$. The representatives in B_ρ of a configuration in D_ρ are related by periodic gauge transformations that need not have a small global logarithm.

Theorem 11.6. *There exists $\rho_{\mathrm{ov}} > 0$, depending only on M and the fixed domains, with the following properties. Let $n \geq 4$, $d \geq 0$, and $0 < \rho \leq \rho_{\mathrm{ov}}$, and let k be a real periodic gauge transformation of the coarse lattice such that c and c^k both lie in B_ρ. Then*

$$E_d(c^k) = E_d(c).$$

No smallness of a global logarithm of k is assumed. Moreover the following statements hold.

1. *At adjacent coarse vertices x, y, $\rho_G(k_x k_y^{-1}) < 2\rho$.*
2. *There is a smooth periodic map $\mathscr{K}$ from the torus (in coarse units) to* $\mathrm{SU}(2)$*, obtained in each unit cell by normalizing a nonnegative cardinal average of the values of k at the corners of the cell, regarded as unit quaternions, such that*

$$\mathscr{K} = k \text{ at the coarse vertices}, \qquad \|D\mathscr{K}\|_\infty + \|D^2\mathscr{K}\|_\infty \le C\rho.$$

   The map $k \mapsto \mathscr{K}$ commutes with left and right multiplication by constant elements of $\mathrm{SU}(2)$*, with translations of the coarse lattice, and with spatial inversion. Near every real k satisfying the hypotheses with a fixed strict margin, $\mathscr{K}$ has a holomorphic continuation in the values of k whose absolute Taylor coefficients of order s are bounded by $C^{s+1}s!$, together with the corresponding volume-normalized column sums; these constants depend on the margin but not on d or n. On the fine lattice of spacing ε, the gauge transformation by $\mathscr{K}$ sends a small potential A to a potential whose C^1_ε norm is at most $C(\|A\|_{C^1_\varepsilon} + \rho)$, uniformly in ε.*
3. *Consequently the function $E_{d,\mathrm{sat}}(U) = E_d(U^k)$, where k is any periodic gauge transformation with $U^k \in B_\rho$, is well defined and real analytic on D_ρ, and its local holomorphic expressions agree on every strict real overlap. If, in addition, $\rho \le r_{\mathrm{CK}}$, then $E_{d,\mathrm{sat}} \ge \kappa_* E_0$ on D_ρ, with κ_* as in Theorem 11.2. The same identities and the same extension hold for E_∞ at each fixed coarse side n.*

Remark 11.7. The following picture describes the interpolation in item (2). In a unit cell of the coarse lattice, the values of k at the eight corners are unit quaternions that lie pairwise within angle 3ρ of each other, although they may be far from 1. Their weighted average $m(x)$ lies in the convex hull of these eight points, hence at distance more than $1/2$ from the origin, and $\mathscr{K}(x)$ is its radial projection back to the unit sphere $\mathrm{SU}(2) \cong S^3$. The holomorphic continuation in item (2) concerns the dependence on the corner values; the weights are smooth, but not analytic, functions of the position.

Proof of Theorem 11.6. The proof has seven steps. Steps 1–5 construct the interpolation $\mathscr{K}$ and prove the estimates in items (1) and (2). Step 6 proves $E_d(c^k) = E_d(c)$, and Step 7 proves item (3).

Step 1: neighboring values of k. Let $c' = c^k$. On a coarse edge (x, y) we have $c'_{xy} = k_x c_{xy} k_y^{-1}$, so

$$k_x k_y^{-1} = c'_{xy}\big(k_y c_{xy}^{-1} k_y^{-1}\big), \qquad \rho_G(k_x k_y^{-1}) \le \rho_G(c'_{xy}) + \rho_G(c_{xy}) < 2\rho,$$

where we used that ρ_G is invariant under conjugation and satisfies the triangle inequality. The eight values of k at the corners of any unit cell of the coarse lattice are therefore pairwise within 6ρ. This estimate concerns only differences and does not restrict a logarithm of k.

Step 2: the interpolation. Let

$$b(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \le 0, \end{cases} \qquad f(t) = \frac{b(t)}{b(t) + b(1-t)} \quad (0 \le t \le 1).$$

The function f is smooth, flat at both endpoints, takes values in $[0,1]$, and satisfies $f(1-t) = 1 - f(t)$. In a cell $x_0 + [0,1]^3$, we write $x = x_0 + t$ and, for $v \in \{0,1\}^3$, define

$$W_v(t) = \prod_{i=1}^{3} \big[v_i f(t_i) + (1 - v_i)(1 - f(t_i)) \big], \quad m(x) = \sum_v W_v(t) k_{x_0+v}, \quad \mathscr{K}(x) = \frac{m(x)}{|m(x)|}. \tag{11.8}$$

Here elements of $\mathrm{SU}(2)$ are represented by unit quaternions in $\mathbb{R}^4$. The weights W_v are nonnegative, sum to one, and are cardinal at the corners. With our normalization of $\mathfrak{su}(2)$, the angle between two unit quaternions is half the group distance between the corresponding elements. Fix one corner value k_0. By Step 1, every corner value lies within angle 3ρ of k_0, so

$$\langle m(x), k_0 \rangle \ge \cos(3\rho) > 1/2$$

after one fixed reduction of ρ. In particular, $|m(x)| > 1/2$. On a face of the cell, the weights of the corners off that face vanish, and the formulas of the two adjacent cells reduce to the same four-corner formula. Since f is flat at the endpoints, all normal derivatives of every order match across the face. We conclude that (11.8) defines a smooth periodic map, also across the periodic seams.

Step 3: derivative bounds and symmetries. For every spatial derivative of positive order j we have $\sum_v D^j W_v = 0$, so

$$D^j m = \sum_v D^j W_v \, (k_{x_0+v} - k_0).$$

The derivatives of the weights of each fixed order are bounded independently of the cell and of the side, and $|k_{x_0+v} - k_0| \le C\rho$ by Step 1. For $j = 1, 2$, the last display is therefore bounded by $C\rho$. The normalization $m \mapsto m/|m|$ has bounded derivatives on $|m| > 1/2$, and the second derivative of $\mathscr{K}$ contains, besides $D^2 m$, only products of two first derivatives of m. This proves

$$\|D\mathscr{K}\|_\infty + \|D^2\mathscr{K}\|_\infty \le C\rho. \tag{11.9}$$

Since left and right quaternion multiplications by fixed elements of $\mathrm{SU}(2)$ are orthogonal linear maps of $\mathbb{R}^4$, they commute with the average and its normalization. Integer translations commute with the cell formula. Spatial inversion reverses each coordinate of the cell and exchanges v with $1 - v$; the identity $f(1-t) = 1 - f(t)$ then gives the asserted equivariance. We do not assert a multiplication law for the interpolation of two variable gauge transformations.

Step 4: holomorphic continuation. We regard a quaternion as a 2×2 matrix. On real inputs,

$$\mathscr{K} = m(\det m)^{-1/2}, \qquad \det m = |m|^2 > 1/4.$$

We use exponential coordinates for the eight corner values about their fixed real values. By the equivariance under left multiplication, multiplying all eight on the left by the inverse of one real corner value puts them in a common fixed neighborhood of the identity. For sufficiently small complex increments of the coordinates, $\det m$ stays in a disc around its positive real value that avoids zero. The absolutely convergent binomial series then continues the branch of the square root that is positive on real data. On a smaller ball of increments, the exponential and binomial series give an absolute majorant $C/(1-Ct)$ for this function and for each of its first two spatial derivatives. Since there are only eight inputs, the absolute tensors of order s are bounded by $C^{s+1}s!$. The constants depend only on the fixed margins of the determinant and of the coordinates. Since an input vertex belongs to at most eight cells, each of bounded diameter, the tensors have finite range and satisfy the weighted bounds at every fixed decay rate. For a fine output lattice of spacing ε, the column sums with a fixed input index are bounded, because ε^3 times the number of fine vertices in these cells is at most a fixed constant. The same conclusions hold for fixed spatial differences, by integrating the corresponding spatial derivatives of the cell formula. Strict margins allow a common choice of all these neighborhoods. This proves the holomorphic and coefficient assertions of item (2), uniformly in d and n.

Step 5: the effect on a fine potential. For a fine edge in direction i, with $y = x + \varepsilon e_i$, we set

$$V_i(x) = \varepsilon^{-1} \operatorname{Log}\big(\mathscr{K}(x)^{-1}\mathscr{K}(y)\big).$$

By (11.9), the ratio $\mathscr{K}(x)^{-1}\mathscr{K}(y)$ has group distance at most $C\varepsilon\rho$ from the identity, so this logarithm is taken on the identity branch. To see that the factor ε^{-1} is harmless, let $R(x,s) = \mathscr{K}(x)^{-1}\mathscr{K}(x + s\varepsilon e_i)$; then

$$V_i(x) = \int_0^1 D\operatorname{Log}_{R(x,s)}\big[\mathscr{K}(x)^{-1}\partial_i\mathscr{K}(x + s\varepsilon e_i)\big]\, ds.$$

All derivatives of this local logarithm are bounded. Differentiating in x uses only the first and second derivatives in (11.9): differentiating R contributes a factor at most $C\rho$, and differentiating the other factor contributes at most $C(\rho + \rho^2)$. Using the mean value integral in x for the lattice differences, we have $\|V\|_\infty + \|\nabla_\varepsilon V\|_\infty \le C\rho$, uniformly in ε.

Since $\mathscr{K}(y) = \mathscr{K}(x)e^{\varepsilon V_i(x)}$, the transformed link variable $\mathscr{K}(x)e^{\varepsilon A_i(x)}\mathscr{K}(y)^{-1}$ has logarithmic potential

$$\varepsilon^{-1}\operatorname{Log}[(e^{\varepsilon A})^{\mathscr{K}}_{xy}] = \operatorname{Ad}(\mathscr{K}(x))\big(A_i(x) - V_i(x) + \varepsilon\mathcal{R}_\varepsilon(A_i(x), V_i(x))\big). \tag{11.10}$$

Here the remainder is defined by the product and logarithm series for $\varepsilon^{-1}\operatorname{Log}(e^{\varepsilon A}e^{-\varepsilon V})$ after subtracting $A - V$. Since all remaining monomials contain at least one factor ε, the

series and its first derivatives are bounded on a fixed small ball of (A, V), for $0 \le \varepsilon \le 1$. The real factor $\mathrm{Ad}(\mathscr{K}(x))$ is an isometry, and its first difference is bounded by $C\rho$. Taking a first difference in (11.10), by integrating the derivative of the local function between the neighboring arguments, we obtain

$$\big\|\varepsilon^{-1}\,\mathrm{Log}[(e^{\varepsilon A})^{\mathscr{K}}]\big\|_{C^1_\varepsilon} \le C(\|A\|_{C^1_\varepsilon} + \rho) \tag{11.11}$$

on a fixed small ball of potentials. The constants do not depend on the distance of $\mathscr{K}$ from the identity.

Step 6: identification of the minima. Let $U^{\mathrm{ord}}(c)$ be the real representative of the gauge orbit of the classical background field with $T^d_M U^{\mathrm{ord}}(c) = c$ from Lemma 10.8. Its potential has $C^1_{\varepsilon_d}$ norm at most $C\rho$, its action is $E_d(c)$, and by Theorem 10.6 every tangent vector to the fiber $\{U : T^d_M U = c\}$ at $U^{\mathrm{ord}}(c)$ has zero first variation. The restriction of $\mathscr{K}$ to the fine vertices takes the values k at the coarse vertices. The block averaging map is gauge covariant: the vertex factors cancel between adjacent edges of each defining path, and the logarithm of each based loop holonomy is conjugated at its basepoint. Hence

$$T^d_M\big[(U^{\mathrm{ord}}(c))^{\mathscr{K}}\big] = c^k = c'.$$

A fixed real gauge transformation preserves the Wilson action and maps the fiber over c diffeomorphically onto the fiber over c'. It therefore preserves constrained criticality. By (11.11), the potential, the divergence, and the curl of $(U^{\mathrm{ord}}(c))^{\mathscr{K}}$ are bounded by $C'\rho$. Let ρ_{ov} be smaller than the fixed radii of Theorems 9.7 and 9.9 divided by C', and smaller than the thresholds used in Steps 2–5. Then the gauge slice of Theorem 9.9, with prescribed coarse gauge logarithm zero, sends $(U^{\mathrm{ord}}(c))^{\mathscr{K}}$ into the larger domain of Theorem 9.7, with gauge-corrected block average exactly c' (Section 11.1).

We verify that the resulting field is critical in that domain, rather than identifying the two constraint maps. Consider a curve $\gamma(t)$ through it in the gauge slice, with fixed gauge-corrected block average c'. By (11.1), its block average is $T^d_M\gamma(t) = (c')^{\mathsf{G}(t)^{-1}}$, where $\mathsf{G}(t)$ is the accumulated gauge transformation of the gauge-corrected average along the curve. Since all defining words have strict margins in the identity chart of the logarithm, $\mathsf{G}(t)$ is differentiable. We interpolate $\mathsf{G}(t)$ smoothly to the fine vertices with the cardinal interpolation of Lemma 10.8, and apply the resulting gauge transformation to $\gamma(t)$. The new curve has block average fixed at c' and the same action as $\gamma(t)$, and its value at $t = 0$ lies on the critical gauge orbit of $(U^{\mathrm{ord}}(c))^{\mathscr{K}}$. The constrained criticality on the fiber over c' therefore kills its derivative at $t = 0$. The additional term coming from the field-dependent gauge transformation is a gauge tangent and has zero first variation as well. Hence the curve γ has zero first variation of the action. By Theorem 10.6, the unique critical point in the convex domain is the classical background field for c'. Since all transformations used preserve the action, we conclude that $E_d(c') = E_d(c)$. For $d = 0$, this identity follows directly by conjugating the plaquette variables.

Step 7: extension to D_ρ. If $U^k, U^l \in B_\rho$, then $U^l = (U^k)^{lk^{-1}}$, and Step 6 gives the same value in both charts. This proves that $E_{d,\mathrm{sat}}$ is well defined on D_ρ. Each fixed k defines an open set on which $U \mapsto E_d(U^k)$ is real analytic. On a strict real overlap, two such local holomorphic expressions agree on a real open set, and the coordinatewise uniqueness argument in the proof of part (2) of Lemma 11.4 identifies their germs. They therefore patch to a real analytic function. The lower bound follows from Theorem 11.2:

$$E_{d,\mathrm{sat}}(U) = E_d(U^k) \geq \kappa_* E_0(U^k) = \kappa_* E_0(U),$$

since the Wilson action E_0 is gauge invariant. For a fixed coarse side, passing to the limit $d \to \infty$ in the two chart values by Theorem 11.2, we find the same identities and the same extension for E_∞. □

11.4 Gauge alignment of the successive constrained minimizers

In this section, we consider one renormalization group step. Let $J \geq 1$, $d_0 \geq 0$, and $m \geq 4$, and let $\delta = M^{-J}$. The step goes from the fine lattice of side $m/\delta = mM^J$, of spacing δ in coarse units, to the coarse lattice of side m, through the intermediate lattices of spacings $\delta_l = M^{l-J}$, $0 \leq l \leq J$. The effective classical action on the fine lattice of the step is E_{d_0}.

At each step, the renormalization group needs a representative of the gauge orbit of the classical background field that satisfies the block averaging constraint $T_M^J(\cdot) = e^\theta$ exactly. Indeed, the small-field event and the reconstruction maps are defined through T_M, not through the gauge-corrected average. Its coefficient bounds must be uniform in J, since J is chosen only after these bounds are known. The identification with the reconstruction maps of Sections 12 and 13 is needed only for fixed J.

The classical background field of the step. Let θ be a small coarse logarithm. We apply Theorem 10.11 with coarse side $n = m$, depth $d = d_0 + J$, and intermediate index $j = d_0 + l$, $0 \leq l \leq J$. The intermediate spacing is then exactly $M^{j-d} = \delta_l$, and the fine side is $mM^J = m/\delta$. Since these are complete periodic tori of side at least four, Theorem 10.11 applies in its form for arbitrary sides. We denote the resulting representatives by

$$D_l(\theta) = e^{\delta_l A_l(\theta)}, \qquad 0 \leq l \leq J,$$

so that $D_J = e^\theta$ and $D_0 = e^{\delta A_0(\theta)}$ is a configuration on the fine lattice of the step. For real θ, these are gauge-corrected block averages of the classical background field of depth $d_0 + J$; on the common complex ball, they are holomorphic continuations. We refer to the gauge orbit of $D_0(\theta)$ as the gauge orbit of the classical background field of the step. By Theorem 10.11, the pairs $(A_l, \nabla_{\delta_l} A_l)$ have one common factorial majorant, for output-row norms and for column norms with output volume δ_l^3, independent of d_0, J, l, and m.

The next lemma identifies this orbit, for each fixed J, with the configuration obtained by successive conditional constrained minimization. For a configuration V on the fine

lattice of the step in the domain of Lemma 10.8 (applied with depth d_0 and coarse side m/δ), let $U^{\mathrm{ord}}_{d_0}(V)$ denote the representative of the gauge orbit of the classical background field of depth d_0 with $T_M^{d_0} U^{\mathrm{ord}}_{d_0}(V) = V$.

Lemma 11.8. *Let $J \geq 1$, $d_0 \geq 0$, $m \geq 4$, and $\delta = M^{-J}$.*

1. *For $0 \leq j \leq J$ define functions Φ_j of configurations c_j on the lattice of spacing δ_j by $\Phi_0 = E_{d_0}$ and $\Phi_{j+1}(c_{j+1}) = \Phi_j\big(b(c_{j+1}, z^*)\big)$, where $b(c,z)$ is the configuration of Theorem 10.10 for the averaging step from spacing δ_j to δ_{j+1} and z^* is the conditional constrained minimizer of $z \mapsto \Phi_j(b(c_{j+1}, z))$. Then $\Phi_j = M^{-j} E_{d_0+j}$, wherever the successive minimizers lie in the domains of Theorem 10.10. For real data this is an identity between minima on the branch of Theorem 10.10; it extends holomorphically.*
2. *Let $\widehat{c}_0(\theta)$ be the configuration on the fine lattice of the step obtained by successive conditional constrained minimization, starting from e^θ, that is, the zero-fluctuation reconstruction before gauge transformation, and let $\widehat{U}(\theta) = U^{\mathrm{ord}}_{d_0}(\widehat{c}_0(\theta))$. For each fixed J there is $r'_J > 0$, depending on J and the fixed domains but not on d_0, m, L, or γ, such that the following holds for real $\|\theta\|_\infty < r'_J$. The configuration $\widehat{c}_0(\theta)$ is a critical point of E_{d_0} on the fiber $\{V : T_M^J V = e^\theta\}$. The configuration $\widehat{U}(\theta)$ satisfies $T_M^{d_0+J}\widehat{U} = e^\theta$, every tangent vector to the fiber $\{U : T_M^{d_0+J} U = e^\theta\}$ at $\widehat{U}(\theta)$ has zero first variation of the Wilson action, its action in coarse units equals $E_{d_0+J}(e^\theta)$, and it lies in the gauge orbit of the classical background field of depth $d_0 + J$ at e^θ. Finally, $\widehat{c}_0(\theta)$ and $D_0(\theta)$ are gauge equivalent, and $E_{d_0}(\widehat{c}_0(\theta)) = E_{d_0}(D_0(\theta))$.*

Proof. For (1), by Theorem 10.10 we have $E_{d+1}(c) = M \min_z E_d(b(c,z))$ on the branch constructed there, for every depth d. If $\Phi_j = M^{-j} E_{d_0+j}$, then

$$\Phi_{j+1}(c_{j+1}) = M^{-j} \min_z E_{d_0+j}(b(c_{j+1}, z)) = M^{-j-1} E_{d_0+j+1}(c_{j+1}),$$

which proves (1) by induction; the factor M^{-j} comes from applying Theorem 10.10 j times. The minima are taken for real coarse fields and real free coordinates, and the critical-value identities extend holomorphically, as stated in Theorem 10.10.

For (2), we first show that $\widehat{c}_0$ is critical on its fiber. The tangent space at a configuration on a lattice of one averaging step decomposes according to the coordinates of Theorem 10.10. These coordinates gauge the configuration by the transports along the root paths of the block trees, take the remaining free logarithmic coordinates, and solve the eliminated edges by the inverse branch. Differentiating them, we split a tangent vector into a tangent to the gauge action of the block trees, a derivative of $b(c,z)$ in the free coordinates z, and a derivative through the next coarser field c. Consider a tangent vector to the fiber $\{T_M^J V = e^\theta\}$ at $\widehat{c}_0$. At the first averaging step, the gauge part has zero derivative of E_{d_0} by the local gauge invariance of Lemma 10.8. The free part is killed by the critical equation of the conditional constrained minimizer. The remaining derivative

is the derivative of the partially minimized value $\Phi_1 = M^{-1}E_{d_0+1}$ in the next coarser field. Repeating this at each of the J averaging steps, and using that the coarsest field e^θ is fixed, we find that the derivative vanishes. This proves the first assertion. The same induction, preceded by one more step, applies to $\widehat{U}$. A variation of $\widehat{U}$ at fixed coarse field splits into the variation of $U^{\mathrm{ord}}_{d_0}$ induced by the variation of its field on the fine lattice of the step, and a variation that keeps that field fixed. The latter has zero derivative of the action by Theorem 10.6. The former has derivative DE_{d_0} applied to the induced variation, up to the fixed scale factor, and this vanishes by the first assertion. We conclude that every tangent to the fiber $\{T_M^{d_0+J}U = e^\theta\}$ at $\widehat{U}$ has zero first variation.

Changing from units in which the fine lattice of the step has spacing one to coarse units multiplies potentials by δ^{-1} and first differences by δ^{-2}, which are fixed factors for fixed J. The potential of $\widehat{U}(\theta)$ therefore has C^1 norm at most $C_J\|\theta\|_\infty$ in coarse units. By (1), its action in coarse units is M^J times $\Phi_J = M^{-J}E_{d_0+J}$, that is, $E_{d_0+J}(e^\theta)$. Let r'_J be so small that the gauge slice of Theorem 9.9, with prescribed coarse gauge logarithm zero, places $\widehat{U}(\theta)$ in the larger domain of Theorem 9.7. The argument of Step 6 in the proof of Theorem 11.6 transfers the constrained criticality to that domain. The uniqueness in Theorem 10.6 then identifies the gauge orbit of $\widehat{U}(\theta)$ with that of the classical background field of depth $d_0 + J$.

Finally, we write $\widehat{U} = (e^{\varepsilon A_{d_0+J}})^g$ for a fine gauge transformation g, where $\varepsilon = M^{-d_0-J}$. By gauge covariance of T_M, $\widehat{c}_0 = T_M^{d_0}\widehat{U}$ is the gauge transform of $T_M^{d_0}e^{\varepsilon A_{d_0+J}}$ by the restriction of g, and by (11.1) the latter is gauge equivalent to D_0. Thus $\widehat{c}_0(\theta)$ and $D_0(\theta)$ are gauge equivalent, and after reducing r'_J both lie in B_ρ with $\rho \le \rho_{\mathrm{ov}}$. Theorem 11.6 then implies that $E_{d_0}(\widehat{c}_0) = E_{d_0}(D_0)$. □

We write $\mathscr{C}_{\mathrm{raw}\,d_0,J}(\theta, 0) = \widehat{c}_0(\theta)$ for the zero-fluctuation reconstruction of the step before gauge transformation. It is built from the successive conditional constrained minimizers of Theorem 10.10 and their normalization by the positive inverse square roots of the constrained Hessians, with every normalized fluctuation variable equal to zero.

Lemma 11.9. *Let $J \ge 1$, $d_0 \ge 0$, and $m \ge 4$, and put $\delta = M^{-J}$.*

1. *On a common complex ball $\|\theta\|_\infty < r_{\mathrm{cen}}$ there is an analytic representative*

$$C_{d_0,J}(\theta) = e^{\delta\overline{A}_{d_0,J}(\theta)}, \qquad T_M^J C_{d_0,J}(\theta) = e^\theta, \qquad \overline{A}_{d_0,J}(0) = 0,$$

   of the gauge orbit of the classical background field of the step.

2. *There exist $r_{\mathrm{cen}}, a_{\mathrm{cen}}, C_{\mathrm{cen}} > 0$, depending only on M and the fixed domains and independent of J, d_0, and m, such that for every integer $s \ge 1$ and every $\|\theta\|_\infty < r_{\mathrm{cen}}$,*

$$\|D^s_\theta(\overline{A}_{d_0,J}, \nabla_\delta\overline{A}_{d_0,J})\|_{a_{\mathrm{cen}},\mathrm{out}} + \|D^s_\theta(\overline{A}_{d_0,J}, \nabla_\delta\overline{A}_{d_0,J})\|_{a_{\mathrm{cen}},\mathrm{input};\delta^3} \le C_{\mathrm{cen}}^{s+1}s!.$$

   In particular $\|\overline{A}_{d_0,J}(\theta)\|_{C^1_\delta} \le C_{\mathrm{cen}}\|\theta\|_\infty$.

3. *The representative is $C_{d_0,J} = D_0^L$, where L is the smooth interpolation, by the cardinal weights of Lemma 10.8, of the accumulated gauge transformation G at the coarse vertices that relates the gauge-corrected and the ordinary block averages of D_0; at the coarse vertices $L = G$ (not G^{-1}), the multiplication order being forced by $(U^k)^l = U^{lk}$. The real gauge path connecting D_0 to $C_{d_0,J}$ has logarithms of the fine link variables bounded by $C\delta\|\theta\|_\infty$.*
4. *After one fixed reduction of r_{cen} with $Cr_{\mathrm{cen}} < 1/2$,*

$$\mathscr{R}_{a_{\mathrm{cen}}}\big((\overline{A}_{d_0,J}, \nabla_\delta \overline{A}_{d_0,J}); u\big) \preccurlyeq Z(u) = \frac{Cu}{1 - Cu}, \qquad \mathscr{C}_{a_{\mathrm{cen}},\delta^3}\big((\overline{A}_{d_0,J}, \nabla_\delta \overline{A}_{d_0,J}); u\big) \preccurlyeq Z'(u),$$

on $0 \le u < r_{\mathrm{cen}}$, with C and r_{cen} independent of J, d_0, and m. The accumulated gauge transformation G, its inverse, and its interpolation L have bounds of the same type, uniform in J. This item gives no bound uniform in J for the full reconstruction in coarse and free coordinates or for the gauge transformation F_J of item (6), and no column bound for arbitrary fine inputs to these gauge transformations.
5. *For each fixed J, the representative $C_{d_0,J}$, the gauge transformation L and its inverse, and the gauge transformation F_J of item (6) have weighted absolute coefficient bounds of factorial type, with every index fixed in turn, on a fixed larger domain of radius $r_J > 0$, with constants C_J. A positive spatial decay rate for these bounds can be chosen before J; r_J and C_J may depend on J and on the chosen analytic domains, but not on d_0, m, L, or γ. These constants are not used to choose J.*
6. *For the same fixed J, after reducing r_J only, let $F_J(\theta)$ be the gauge transformation obtained from $C_{d_0,J}(\theta)$ by imposing, in the order $j = J-1, J-2, \dots, 0$, the block axial gauge of the level-j blocks of the field after the preceding gauge transformations, each factor being extended constantly through the finer block partitions. With $(U^k)^l = U^{lk}$ its ordered product is $F_J = K_0 K_1 \cdots K_{J-1}$, and*

$$F_J = I \text{ at the coarse vertices}, \qquad F_J(0) = I, \qquad C_{d_0,J}(\theta)^{F_J(\theta)} = \mathscr{C}_{\mathrm{raw}\, d_0,J}(\theta, 0).$$

This identity holds on the complex ball of radius r_J, with strict margins in all intermediate charts. The gauge transformations F_J and F_J^{-1} depend only on θ and satisfy the bounds of item (5), uniformly in d_0, m, L, and γ; no bound on F_J uniform in J is asserted.

The bounds of item (2) hold at every order s; the reconstruction maps of Section 15 use only finitely many orders of such uniform bounds. Table 11.1 lists the order in which the constants of this section enter the choice of parameters.

Proof of Lemma 11.9. Steps 1–6 construct $C_{d_0,J}$ and prove items (1)–(4) with constants independent of J. Step 7 constructs F_J and proves item (5), and Step 8 proves the identity in item (6).

Constants	Dependence and role
a_*, C_*, r_*, m_0; $r_{\mathrm{CK}}, \rho_{\mathrm{ov}}$	Fixed geometry, decay rates, and classical coefficient bounds (Theorems 10.10, 10.11, 11.2, and 11.6); independent of J.
$a_{\mathrm{cen}}, r_{\mathrm{cen}}, C_{\mathrm{cen}}$ and Z	Items (2)–(4) of Lemma 11.9; independent of J, d_0, and m.
a, $\rho_0(a), \delta_0(a), C_P(a)$	Theorem 11.10; for the application to $C_{d_0,J}$, $0 < a < a_{\mathrm{cen}}/8$, $2\rho_0 < \min(r_{\mathrm{cen}}, r_*)$, and δ_0 chosen so that the image condition holds for both D_0 and $C_{d_0,J}$; independent of J, d_0, m, and of the later choice of $\rho \le \rho_0$.
J	Chosen after the preceding constants, for instance so that $C_P(a)M^{-J} < 1/4$ (Theorem 13.5; see also Theorem 23.2).
r_J, C_J	Items (5)–(6) of Lemma 11.9; depend on J but not on d_0, m, L, or γ; never used to choose J.
Later radii and coupling bounds	Chosen in Parts IV and VI, after J.

Table 11.1. Order of the constants in Section 11.

Step 1: conventions. We use the representatives D_l described before Lemma 11.8, with their common factorial majorant from Theorem 10.11. Fix $0 < a_{\mathrm{cen}} < a_*/32$. The larger multiples of this rate used below give spatial decay rates strictly larger than the working rate, which we use to sum columns. This choice does not depend on J, d_0, or m.

Step 2: the accumulated gauge transformation. We bound the gauge transformation relating the ordinary and the gauge-corrected block averages, rather than replacing the block averaging constraint by the gauge-corrected one. At level l, the vertex factor of the gauge-corrected average at a block center r is

$$H_l(r) = \exp\Big\{-M^{-3} \sum_{x \in \mathsf{B}_M(r)} \operatorname{Log} D_l(\sigma_{r,x})\Big\},$$

the identity-background factor of Lemma 8.10; by its definition, $D_{l+1} = (T_M D_l)^{H_l}$. By gauge covariance of T_M, the accumulated gauge transformations G_l satisfy

$$G_0 = I, \qquad G_{l+1} = H_l\,(G_l|_{\text{next coarser vertices}}), \qquad D_l = (T_M^l D_0)^{G_l};$$

the last identity follows by induction from $(U^k)^l = U^{lk}$. In particular, with $G = G_J$,

$$e^{\theta} = (T_M^J D_0)^G, \qquad T_M^J D_0 = (e^{\theta})^{G^{-1}}. \tag{11.12}$$

We now bound the coefficients of $\kappa = \operatorname{Log} G$ by a series that does not depend on J. Each path in H_l has at most C_M edges with variables $e^{\delta_l A_{l,e}}$. Expanding the product, the logarithm, and the exponential, we obtain the coefficientwise bound

$$H_l - I \preccurlyeq \delta_l \beta(Z(t)), \qquad \beta(z) \le \frac{Cz}{1 - Cz}, \quad \beta(0) = 0, \tag{11.13}$$

where $Z(t) = Ct/(1-Ct)$ is the majorant of the pairs $(A_l, \nabla_{\delta_l} A_l)$ and $\preccurlyeq$ refers to absolute output-row sums at a specified smaller decay rate. The factor δ_l appears because every nonconstant monomial of the local word contains at least one exponent $\delta_l A_l$, and $\delta_l^k \le \delta_l$. The finitely many path variables contribute a fixed factor per degree. The paths lie in a collar of radius $C_M \delta_l$, and joining it to the output vertex contributes another fixed factor per degree, absorbed in C. The absolute exponential, logarithm, and product series converge for one small fixed t. This proves (11.13) as an estimate on coefficients.

Restriction to coarser vertices cannot increase an output row. Using $\sum_{l=0}^{J-1} \delta_l < 1/(M-1)$, we obtain

$$G - I \preccurlyeq \prod_{l=0}^{J-1} (1 + \delta_l \beta(Z)) - 1 \preccurlyeq \exp\Big(\frac{\beta(Z)}{M-1}\Big) - 1,$$

and we denote the right side by $Q(t)$. Fix a radius of coarse logarithms such that $Q(t) < 1/2$. The logarithm series then bounds the absolute row series of κ by $\sum_{k\ge 1} Q(t)^k/k$. This series converges, vanishes at zero, and is dominated coefficientwise by $Ct/(1-Ct)$ after enlarging C. Indeed, the coefficient of degree s of a series with nonnegative coefficients that converges at $t_0 > 0$ is at most its value at t_0 times t_0^{-s}. The radius and the constants are independent of J, d_0, and m. The inverse needs no implicit differentiation: since $\det G = 1$, for 2×2 matrices

$$G^{-1} = \operatorname{adj} G, \qquad \operatorname{adj}(I + X) - I = (\operatorname{tr} X) I - X. \tag{11.14}$$

This is linear in X, with absolute rows at most a fixed multiple of those of its argument, and on the common branch $\operatorname{Log} G^{-1} = -\kappa$. Enlarging the constant of the rational majorant, we bound κ, $G - I$, and $G^{-1} - I$ simultaneously, at every order, before J is chosen.

Step 3: from rows to columns. We work with row bounds at twice the decay rate used in the conclusion. For an output vertex x and a fixed coarse input u, the length of a shortest connecting network is at least $\operatorname{dist}(x, u)$. We remove the sum over that input from an absolute output row and use half of the rate. Then the remaining weighted sum of order s is at most $C^{s+1} s! e^{-a \operatorname{dist}(x,u)}$. For the meshes that occur here,

$$\sup_u \sup_{\substack{\varepsilon = M^{-j} \\ j \in \mathbb{Z}_{\ge 0}}} \varepsilon^3 \sum_{x \in \varepsilon \mathbb{Z}^3/(m\mathbb{Z})^3} e^{-a \operatorname{dist}(x,u)} \le C_a, \tag{11.15}$$

where m is an integer, $m/\varepsilon = mM^j \in \mathbb{N}$, and u is a vertex of the fine lattice; this covers the coarse spacing 1 and every intermediate spacing δ_l. To prove (11.15), we take the shortest periodic representatives and bound the torus sum by the sum over the infinite lattice. By equivalence of the fixed lattice metrics, this reduces to a product of three one-dimensional sums. For $b > 0$, by monotonicity of e^{-bt} we have $\varepsilon \sum_{k \in \mathbb{Z}} e^{-b\varepsilon|k|} \le$

$1 + 2\int_0^\infty e^{-bt}dt = 1 + 2/b$, uniformly in the side. Hence κ has bounded counting columns with coarse outputs, and the same argument gives volume-normalized columns for outputs on a lattice of spacing ε. The input in this argument is the coarse logarithm θ, through the representatives D_l.

Step 4: the interpolated gauge transformation and the exact constraint. With the smooth periodic cardinal weights χ_r of Lemma 10.8, we define on the fine vertices

$$\lambda_x(\theta) = \sum_r \chi_r(x)\kappa_r(\theta), \qquad L_x(\theta) = e^{\lambda_x(\theta)}, \qquad C_{d_0,J}(\theta) = D_0(\theta)^{L(\theta)}. \tag{11.16}$$

Since $L = G$ at the coarse vertices, gauge covariance of T_M^J and (11.12) give $T_M^J C_{d_0,J} = ((e^\theta)^{G^{-1}})^G = e^\theta$ exactly. This is why we interpolate G, and not G^{-1}. The interpolation need not be the identity at the coarse vertices, since it corrects the block average; the gauge transformation F_J constructed in Step 7 will be the identity there.

The weights and their first two spatial derivatives have bounded absolute rows and volume-normalized columns, with a fixed collar of support. Integrating these derivatives, we obtain the same bounds for the first and second divided differences of their restrictions to the lattice. By Steps 2 and 3, λ, $\nabla_\delta\lambda$, and all second divided differences of λ have factorial rows and columns in θ, uniformly in J. If $Y(u)$ is their common row majorant, the exponential series bounds $L - I$ by $e^{Y(u)} - 1$. Since $\det L = 1$, (11.14) applies to L and bounds $L^{-1} - I$. Differentiating the exponential series spatially once or twice inserts a first difference of λ, or a second difference and products of first differences, which have convergent majorant series of the same type. Row bounds at a spatial decay rate strictly larger than the working rate, together with (11.15), give their columns with one fixed coarse input and output volume δ^3. For G, the output measure is counting measure on the coarse lattice. This proves the bounds of item (4) for G, its inverse, and its interpolation, all with the coarse logarithm θ as input.

Step 5: removal of the inverse mesh. We remove the inverse mesh in the gauge word before taking spatial or θ derivatives. For $u = \lambda_x$, $v = (\lambda_{x+\delta e_i} - \lambda_x)/\delta$, and $a = A_{0,i}(x)$, we define

$$F_\delta(u, v, a) = \delta^{-1}\operatorname{Log}(e^u e^{\delta a} e^{-u-\delta v}). \tag{11.17}$$

The product equals I at $\delta = 0$, for all small u, v, a. The absolutely convergent exponential and logarithm series are jointly analytic in (δ, u, v, a) on a small ball, also for $|\delta| \le 2$, and the zero at $\delta = 0$ makes the division removable. After division, the absolute series have a common majorant for $0 \le \delta \le 1$. Differentiating at $\delta = 0$, we find

$$F_0(u, v, a) = \operatorname{Ad}(e^u)a - \int_0^1 \operatorname{Ad}(e^{su})v\, ds, \tag{11.18}$$

as follows from $De^X[Y] = \int_0^1 e^{(1-s)X} Y e^{sX} ds$. Thus $F_\delta = F_0 + \delta R_\delta$ with a uniformly convergent absolute analytic remainder. We define

$$\overline{A}_{d_0,J,i}(x) = F_\delta(\lambda_x, \nabla_{\delta,i}\lambda_x, A_{0,i}(x)).$$

Since $L_x e^{\delta A_{0,i}(x)} L^{-1}_{x+\delta e_i} = e^{\lambda_x} e^{\delta A_{0,i}(x)} e^{-\lambda_x - \delta \nabla_{\delta,i}\lambda_x}$, this is the logarithmic potential of $C_{d_0,J}$ in (11.16). Its first divided difference is the integral of the derivative of F_δ along the segment between the neighboring lists of arguments, whose differences are $\nabla_\delta \lambda$, $\nabla^2_\delta \lambda$, and $\nabla_\delta A_0$. All of these have the majorants established above. Composing majorant series with the counting of Lemma 11.1, we obtain factorial rows for $(\overline{A}, \nabla_\delta \overline{A})$ at one fixed smaller decay rate. Here the union of the local collars and of the input networks connects every external index. Applying (11.15) to rows at a strictly larger rate gives the columns with output volume δ^3. All decay rates are chosen before J, and no rate is reduced from one averaging step to the next.

Step 6: one majorant for all bounds. We work first at twice the final rate a_{cen}. Let $Y(u) = C_0 u/(1 - C_0 u)$ dominate the absolute rows of $(A_0, \nabla_\delta A_0, \lambda, \nabla_\delta \lambda, \nabla^2_\delta \lambda)$, where C_0 is increased to account for the finitely many types of components. For a suitable fixed c, the series of F_δ and of its first derivative in its local arguments are bounded by $cz/(1-cz)$ and $c/(1-cz)^2$. Using the integral representation of the spatial difference, we obtain

$$\mathscr{R}_{2a_{\mathrm{cen}}}((\overline{A}, \nabla_\delta \overline{A}); u) \preccurlyeq P(u), \qquad P(u) = \frac{cY(u)}{1 - cY(u)} + \frac{cY(u)}{(1 - cY(u))^2}, \tag{11.19}$$

by multiplication of absolute Taylor series and Lemma 11.1. In the second term, the last factor Y accounts for the spatial difference of a local argument. The majorant is independent of δ and of J. At order s, since the rows are bounded at a larger rate than the columns, (11.15) bounds the column by $C_{a_{\mathrm{cen}}}$ times the row; in the notation (11.2),

$$\mathscr{C}_{a_{\mathrm{cen}},\delta^3}((\overline{A}, \nabla_\delta \overline{A}); u) \preccurlyeq C_{a_{\mathrm{cen}}} P'(u).$$

The same conversion applies to the series for G, G^{-1}, and L. We add all these row majorants, together with P and the fixed constants of the column conversions, into one series $S(u) = \sum_{s \geq 1} S_s u^s$ with nonnegative coefficients, convergent at some $u_* > 0$ independent of J, d_0, and m. Then $S_s \leq S(u_*) u_*^{-s}$. Choosing $C \geq \max\{1, u_*^{-1} \max\{1, S(u_*)\}\}$, we have $S_s \leq C^s$ for every $s \geq 1$, and hence

$$\begin{aligned} \mathscr{R}_{a_{\mathrm{cen}}}((\overline{A}, \nabla_\delta \overline{A}); u) &\preccurlyeq Z(u) = \frac{Cu}{1 - Cu}, \\ \mathscr{C}_{a_{\mathrm{cen}},\delta^3}((\overline{A}, \nabla_\delta \overline{A}); u) &\preccurlyeq Z'(u). \end{aligned} \tag{11.20}$$

The same Z and Z' dominate the nonconstant parts of the series of G, G^{-1}, and L, with output volume 1 for coarse outputs and δ^3 for fine outputs. The final Z may be larger than

the majorant used in (11.13); all earlier estimates remain valid on the smaller common ball.

With this rate, let $r_{\mathrm{cen}} > 0$ be such that the series just used converge on a ball strictly larger than $\|\theta\|_\infty < r_{\mathrm{cen}}$ and $Cr_{\mathrm{cen}} < 1/2$. These constants are fixed before J. The bounds also hold at every point of the ball, not only at its center. Indeed, if a tensor of order k at zero is bounded by $C^{k+1}k!$, then translating the argument by θ bounds the tensor of order s at θ by

$$\sum_{l\geq 0} \frac{\|\theta\|_\infty^l}{l!} C^{s+l+1}(s+l)! = \frac{C^{s+1}s!}{(1-C\|\theta\|_\infty)^{s+1}}.$$

Since contracting l inputs only shortens the external network, the same inequality holds for each coefficient bound with a fixed external index and for each volume-normalized column. A fixed reduction of the radius bounds the right side by $C_{\mathrm{cen}}^{s+1}s!$ for every $s \geq 1$, with no loss in the radius or the decay rate that depends on s. Enlarging C_{cen}, we also bound the first derivative itself. Since the pair $(\overline{A}, \nabla_\delta \overline{A})$ and the nonconstant parts of the gauge transformations vanish at $\theta = 0$, integrating the first derivative, we find $\|\overline{A}\|_{C^1_\delta} \leq C_{\mathrm{cen}}\|\theta\|_\infty$. Replacing λ by $t\lambda$ in (11.17), uniformly for $0 \leq t \leq 1$, we find that the real gauge path from D_0 to $C_{d_0,J}$ has fine logarithms at most $C\delta\|\theta\|_\infty$. This completes the proof of items (1)–(4); item (1) refers to this gauge orbit because $C_{d_0,J}$ is a gauge transform of D_0.

Step 7: the block axial gauge transformation for fixed J. For a configuration V on the fine lattice of one averaging step, we define on each block with center r

$$f(V)_x = V(\sigma_{r,x}), \qquad f(V)_r = I.$$

For each edge (x, y) of the block tree, oriented away from the center, the root path gives $f(V)_y = f(V)_x V_{xy}$, and hence $(V^{f(V)})_{xy} = I$. Starting from $V = C_{d_0,J}(\theta)$, we perform the following finite procedure for $j = J-1, J-2, \ldots, 0$. We compute $V_j = T_M^j V$ for the field V after the preceding gauge transformations, form $f(V_j)$, extend this gauge transformation constantly through the j finer block partitions to the fine vertices, and apply the resulting gauge transformation K_j to V. The factor K_j is the identity at the vertices of level $j+1$. By gauge covariance, it therefore leaves unchanged every coarser iterated block average and each block axial gauge condition already imposed there. The resulting gauge transformation is

$$F_J = K_0 K_1 \cdots K_{J-1}, \qquad F_J = I \text{ at the coarse vertices.} \tag{11.21}$$

Each factor is evaluated on the field transformed by the preceding factors, and the order of the product is forced by $(U^k)^l = U^{lk}$.

For fixed J, these operations are finite products and the local logarithms of the block averaging map. At $\theta = 0$, every intermediate field and every factor is the identity. Finite-word absolute series and the uniform series of $C_{d_0,J}$ therefore give a radius $r_J > 0$ with

strict margins in all intermediate charts. The lengths of the words depend on M and J, not on d_0 or m, and all factorial row constants may depend on J. The collars are bounded by a fixed multiple of $\sum_{j<J} M^{j-J} < 1/(M-1)$; extension through finer block partitions enlarges them only by the same geometric sum. The union of networks preserves a chosen positive decay rate at every composition, and the number of variables per coarse unit cell is finite for fixed J. Row bounds at a spatial decay rate strictly larger than the working rate and (11.15) give the volume-normalized columns. Since inversion of a matrix of determinant one is a fixed polynomial in its entries, the same statements hold for F_J^{-1}, for L, and for L^{-1}. This proves item (5).

Step 8: identification with the zero-fluctuation reconstruction. The stationarity and convexity arguments below use real coarse fields and real free coordinates; the final identity is then continued holomorphically. Consider the free logarithmic coordinates $z = (z_0, \dots, z_{J-1})$ of the J averaging steps in the nested coordinates of Theorem 10.10. Consider also the function of z obtained by composing E_{d_0} with the reconstruction of the fine field of the step from θ and z. By part (1) of Lemma 11.8, partial minimization over the first j steps gives $M^{-j}E_{d_0+j}$. At $\theta = 0$, let δz_j be a free tangent vector of step j, let δc_{j+1} be the induced tangent vector of the next coarser field, and set

$$\eta_j = \delta z_j - Dz^*_{d_0+j}(I)[\delta c_{j+1}].$$

We complete the square at each step. The critical equation of the conditional minimizer removes the mixed term, and differentiating the identity of part (1) gives the remaining term. We obtain the Hessian of this function at $\theta = 0$, $z = 0$:

$$\sum_{j=0}^{J-1} M^{-j} \langle \eta_j, K^0_{d_0+j} \eta_j \rangle, \qquad \delta c_J = 0, \tag{11.22}$$

where $K^0_{d_0+j}$ are the constrained Hessians of Theorem 10.10 at the identity. For a quadratic form $\langle z, Kz\rangle/2 + \langle z, Bc\rangle + \langle c, Dc\rangle/2$, the derivative of the minimizer is $-K^{-1}B$. Substituting $z = \eta - K^{-1}Bc$ leaves $\langle \eta, K\eta\rangle/2 + \langle c, (D - B^{\mathsf{T}}K^{-1}B)c\rangle/2$, where the last matrix, the Schur complement, is the Hessian of the partially minimized value. Iterating gives (11.22). In particular, the blocks are $M^{-j}K^0_{d_0+j}$, not $K^0_{d_0+j}$.

The map from (δz_j) to (η_j) is triangular with identity diagonal. Its inverse is given by backward substitution and involves only finitely many compositions of the mixed derivatives of the minimizers and of the geometric one-step maps. By their row and column bounds in Theorems 10.10 and 10.11, the rows and columns of the inverse are bounded by C_J. Its ℓ^2 norm is then at most C_J, since by the Cauchy–Schwarz inequality

$$\|Tv\|_2^2 \le \Big(\sup_x \sum_y |T_{xy}|\Big)\Big(\sup_y \sum_x |T_{xy}|\Big)\|v\|_2^2.$$

Since $K^0_{d_0+j} \geq m_0 I$, the Hessian (11.22) is bounded below by $m_0 M^{-(J-1)} C_J^{-2} I$. By the bounds on third derivatives for fixed J, with each external index fixed in turn, the Hessian at (θ, z) differs from this one by at most $C_J(\|\theta\|_\infty + \max_j \|z_j\|_\infty)$ in row and column norm. We reduce the radius for this fixed J so that this is less than half the lower bound. Then the function is strictly convex on a smaller real convex ball of coordinates, uniformly in d_0 and m.

The successive conditional constrained minimizers give a critical point of this function. Indeed, at the first minimizer the derivative in its free variables vanishes. The derivative in the remaining variables is the derivative of $M^{-1}E_{d_0+1}$, whose derivative in the next free variables vanishes at its minimizer, and so on through the J steps. This critical point corresponds to $\widehat{c}_0(\theta) = \mathscr{C}_{\mathrm{raw}\, d_0, J}(\theta, 0)$. On the other hand, by part (2) of Lemma 11.8, $C_{d_0,J}(\theta)$ is a gauge transform of $\widehat{c}_0(\theta)$, by a gauge transformation whose restriction to the coarse vertices fixes e^θ. Since $\widehat{c}_0(\theta)$ is critical for E_{d_0} on the fiber $\{T^J_M V = e^\theta\}$ and E_{d_0} is gauge invariant on D_ρ (Theorem 11.6), so is $C_{d_0,J}(\theta)$. The gauge transformation $F_J(\theta)$ is the identity at the coarse vertices and preserves this criticality. The field $C_{d_0,J}(\theta)^{F_J(\theta)}$ satisfies the block axial gauge condition at every level. It is therefore the reconstruction of the step from θ and some free coordinates z, and it is a critical point of the function above. It lies in the ball of strict convexity when r_J is small, because it equals the identity at $\theta = 0$ and its coordinates have rows bounded for fixed J. By strict convexity, it is the unique critical point, so

$$C_{d_0,J}(\theta)^{F_J(\theta)} = \widehat{c}_0(\theta) = \mathscr{C}_{\mathrm{raw}\, d_0, J}(\theta, 0)$$

for real θ. The identity extends to the complex ball by the coordinatewise uniqueness argument of Lemma 11.4. Thus F_J^{-1} is the gauge transformation, depending only on θ and equal to the identity at the coarse vertices, that takes the zero-fluctuation reconstruction to the representative $C_{d_0,J}$. The gauge transformations F_J, F_J^{-1} have the bounds of item (5). We shrink r_J inside a larger domain on which these bounds hold, so that they remain available later. Their constants depend only on M, J, and the chosen domains and rates, never on d_0, m, L, or γ. The constants r_{cen}, a_{cen}, and C_{cen} were fixed without them, so the constants of item (5) are not used to choose J. □

11.5 Contraction of the interaction under the classical background field

In the renormalization group step of Section 13, the effective interaction R of the fine lattice is evaluated at the classical background field of the step. We show that this evaluation contracts the weighted analytic norm by a factor $O(\delta)$, $\delta = M^{-J}$, at the same radius and the same decay rate. A term of degree k in the link logarithms ξ of the fine lattice, evaluated at $\xi = \delta A$ with A of order one, gains δ^k. The sum over the position of a fixed fine index, measured against the coarse lattice, contributes a factor δ^{-3}. Terms

of degree at least four therefore gain $\delta^{k-3} \leq \delta$. The quadratic and cubic terms need additional cancellations. Gauge invariance and spatial inversion make the quadratic term a quadratic form in first differences of ξ, up to an exponentially small finite-period remainder. Each first difference of δA gains an extra factor δ. Spatial inversion and translation invariance make the first derivative of the cubic term vanish at fields that are constant in each orientation, which yields one first difference. This use of gauge invariance is standard in renormalization group analyses of gauge theories; we prove the absolute kernel factorization, the cubic cancellation, and the composition estimate at unchanged radius below.

We use the following notation. On the fine lattice of side $q = m/\delta$, in units in which its spacing is one, ∇ denotes the collection of all componentwise first differences of link fields, $(\nabla v)_{\ell,i}(x) = v_i(x + e_\ell) - v_i(x)$ for each direction ℓ and each orientation and color component i. Its adjoint is ∇^*. We write grad for the lattice gradient from vertex fields to link fields, $(\operatorname{grad}\lambda)_{xy} = \lambda_y - \lambda_x$, and I for spatial inversion acting on link fields, $(\mathsf{I}v)_i(x) = -v_i(-x - e_i)$, as in Lemma 8.6. For a link logarithm ξ and a vertex field λ, let $\mathcal{D}_\xi\lambda$ be the infinitesimal gauge variation, that is, the derivative of $t \mapsto \operatorname{Log}\big((e^{\xi})^{e^{t\lambda}}\big)$ at $t = 0$; then $\mathcal{D}_0\lambda = -\operatorname{grad}\lambda$. In this section the letter q denotes the side of the fine lattice; the plaquette energy $q(U)$ does not occur. For a translation-invariant kernel K, $\|K\|_{b,\mathrm{row}+\mathrm{col}}$ denotes the sum of its weighted absolute row and column norms at rate b, using the periodic Euclidean distance between initial vertices, consistently with Definition 4.6. The sum convention agrees with Lemma 8.6; that lemma uses the periodic ℓ^1 distance instead, so its rate must be converted when we apply it. These kernels act on a single lattice, so their columns carry no volume factor. For the same distance, the sum lies between the maximum of the row and column norms, used in Sections 4, 14, and 31, and twice that maximum; in particular, the upper bounds on the sum proved below also hold for the maximum.

Theorem 11.10. *Let a_* be the decay rate of Theorem 10.11 and fix $0 < a < a_*/8$. There exist $\rho_0(a), \delta_0(a), C_P(a) > 0$, depending on a and on the fixed classical data (including M), and independent of the later choice of $0 < \rho \leq \rho_0$, of J, of d_0, and of the side of the lattice, with the following property. Let $\delta = M^{-J} \leq \delta_0$, let $m \geq 4$ be the side of the coarse lattice, so that the fine lattice of the renormalization group step has side m/δ. Let R be a scalar function, holomorphic in the logarithms ξ of the fine link variables on $\|\xi\|_\infty < \rho$, such that*

$$DR(I) = 0, \qquad \mathcal{B}_{\rho,a}(R) \leq \epsilon,$$

and such that R is invariant under translations of the fine lattice, under constant gauge transformations, and under spatial inversion, and satisfies the infinitesimal gauge-invariance identity $DR(\xi)[\mathcal{D}_\xi\lambda] = 0$ for all ξ in its domain and all vertex fields λ. These hypotheses concern the function R as a whole, not the individual terms of an expansion of R.

1. *Set $w_q = (1+q)^3 e^{-c_a q}$, where $c_a > 0$ is fixed using part of the spatial decay rate. Then, exactly on the finite torus,*

$$D^2 R(I) = \nabla^* W \nabla + \mathsf{E}_{\mathrm{per}},$$
$$\|W\|_{a/2,\mathrm{row+col}} \le C_a \epsilon, \qquad \|\mathsf{E}_{\mathrm{per}}\|_{a/2,\mathrm{row+col}} \le C_a \epsilon w_{m/\delta}.$$

2. *Let R_3 be the cubic term of the Taylor expansion of R at I. The first derivative of R_3 vanishes at every field that is constant in each orientation and color component, and it has the exact representation through first differences along coordinate paths given in* (11.26) *and* (11.27) *below.*
3. *Let $e^{\delta A_{d_0,J}(\theta)}$ be the representative $D_0(\theta)$ of the gauge orbit of the classical background field of the step from Section 11.4, with the bounds of Theorem 10.11 on the output rows and volume-normalized columns of $(A_{d_0,J}, \nabla_\delta A_{d_0,J})$. Assume that the image of $\|\theta\|_\infty \le 2\rho$, measured by absolute Taylor series, lies in the ball of radius $\rho/2$ of fine link logarithms; a sufficient condition is $\delta Z(2\rho) \le \rho/2$, where $Z(u) \le Cu/(1-Cu)$ is the majorant series of $(A_{d_0,J}, \nabla_\delta A_{d_0,J})$. Set $\mathcal{P}_\delta R(\theta) = R(e^{\delta A_{d_0,J}(\theta)})$. Then*

$$\mathcal{B}_{\rho,a}(\mathcal{P}_\delta R) \le C_a[\delta + \delta^{-1} w_{m/\delta}] \mathcal{B}_{\rho,a}(R) \le C_P(a)\delta\, \mathcal{B}_{\rho,a}(R).$$

   The radius ρ and the rate a are the same on both sides; in the norm of R distances are measured in units of the fine lattice, and in the norm of $\mathcal{P}_\delta R$ in units of the coarse lattice. The constant $C_P(a)$ includes the bound on the finite-period remainder, uniformly for $m \ge 4$.
4. *More generally, let $e^{\delta \widetilde{A}(\theta)}$ be another representative of the gauge orbit of the classical background field of the step (for instance one satisfying $T_M^J(\cdot) = e^\theta$ exactly), holomorphic on $\|\theta\|_\infty < \widetilde{r}$, and suppose that for some rate $b > 8a$ the output rows and the volume-normalized columns, with one fixed coarse input and output volume δ^3, of the Taylor coefficients at $\theta = 0$ of $(\widetilde{A}, \nabla_\delta \widetilde{A})$ are dominated at rate b by a series $\widetilde{Z}(u) \le \widetilde{C}u/(1-\widetilde{C}u)$ and by its derivative $\widetilde{Z}'$, where $\widetilde{r}$ and $\widetilde{C}$ do not depend on J, d_0, or m. Then the conclusion of item (3) holds with $e^{\delta \widetilde{A}}$ in place of $e^{\delta A_{d_0,J}}$, provided $2\rho_0 < \widetilde{r}$ and the image condition holds with $\widetilde{Z}$ in place of Z. The constants ρ_0, δ_0, and C_P then depend also on b, $\widetilde{C}$, and $\widetilde{r}$, and they remain independent of J, d_0, m, and of the later choice of $\rho \le \rho_0$. No conclusion is asserted at output rates $a \ge b/8$. The equality of $\mathcal{P}_\delta R$ with the evaluation of R at a specified reconstruction in block axial gauge coordinates requires, in addition, either a real gauge path inside the domain of R connecting the two representatives, or the invariance of R under the gauge transformations between overlapping charts.*

We record the application to the representative $C_{d_0,J}$ of Lemma 11.9. By items (2) and (4) of that lemma, $C_{d_0,J}$ satisfies the hypotheses of item (4) with $b = a_{\mathrm{cen}}$, $\widetilde{Z} = Z$, and $\widetilde{r} = r_{\mathrm{cen}}$. For $C_{d_0,J}$, the theorem therefore gives the estimate of item (3) at every output rate $0 < a < a_{\mathrm{cen}}/8$, and at no larger rate. Since $a_{\mathrm{cen}} < a_*/32$, such an a also satisfies $a < a_*/8$, and the estimate for D_0 holds at the same rate. To treat both representatives

with one set of constants, let $0 < a < a_{\mathrm{cen}}/8$ and $2\rho_0 < \min(r_{\mathrm{cen}}, r_*)$, where r_* is the radius of Theorem 10.11. We use one enlarged majorant $Cu/(1-Cu)$ at rate a_{cen} for both pairs; the bounds of Theorem 10.11 at rate a_* imply the same bounds at the smaller rate a_{cen}. Finally, δ_0 is chosen so that the image condition holds for both pairs for every $\rho \le \rho_0$. Steps 3–9 of the proof then give constants independent of J, d_0, and m for both representatives, before J is chosen. This is the form in which the theorem is used in Theorem 13.5. The proof of item (1) rests on the following kernel factorization, which strengthens the quadratic-form bound of Lemma 8.6 to a factorization with exponentially decaying kernels.

Lemma 11.11. *Fix $a > 0$ and a finite number of color components. Let $q \ge 4$, and let H be a real self-adjoint translation-invariant kernel on the link fields of the periodic lattice of side q such that*

$$H \operatorname{grad} = 0 = \operatorname{grad}^{\mathsf{T}} H, \qquad H\mathsf{I} = \mathsf{I}H, \qquad \|H\|_{a,\mathrm{row+col}} \le b.$$

Then there exist translation-invariant kernels W and E such that, exactly on the torus of side q,

$$H = \nabla^* W \nabla + \mathsf{E}, \qquad \|W\|_{a/2,\mathrm{row+col}} \le C_a b, \qquad \|\mathsf{E}\|_{a/2,\mathrm{row+col}} \le C_a b w_q.$$

The differences act on every orientation and color component, and the constants depend only on a, the fixed dimension, and the norm conventions, not on q, H, or b. The conclusion also holds for a complex symmetric kernel whose real and imaginary parts satisfy these hypotheses.

Proof. A bound on the quadratic form would not suffice here, since we need a factorization of the kernel. We write $h_{ij}(r)$ for the translation kernel, with color indices suppressed, and attach each positive link to its lower endpoint. The midpoint displacement of this entry is $p = r + (e_j - e_i)/2$. We lift it from the torus by its shortest representatives, splitting tied representatives equally, and use opposite lifts for the entry related by inversion. Since spatial inversion gives $h_{ij}(r) = h_{ij}(-r + e_i - e_j)$, this midpoint interpolation is even. It preserves every allowed Fourier sample: two lifts differ by qz with $z \in \mathbb{Z}^3$, and $e^{\mathrm{i}k\cdot qz} = 1$ at the allowed frequencies k. The lifted integer coefficients, attached to lower endpoints, have absolute exponentially weighted sum at most $C_a b$, since moving an endpoint to a midpoint changes distances by a fixed amount. We denote by $H(k)$ the trigonometric interpolation of the lower-endpoint coefficients and by $\widetilde{H}(k)$ that of the midpoint coefficients. Entrywise,

$$\widetilde{H}_{ij}(k) = e^{\mathrm{i}k\cdot(e_j-e_i)/2} H_{ij}(k), \qquad D_\ell \widetilde{H}(0) = 0,$$

so $D_\ell H_{ij}(0) = -\mathrm{i}(e_j - e_i)_\ell H_{ij}(0)/2$. Write d_2 and d_1 for the periodic Euclidean and ℓ^1 distances. Since $d_2 \le d_1 \le \sqrt{3} d_2$, the hypothesis at Euclidean rate a implies the same kernel bound at ℓ^1 rate $a/\sqrt{3}$. Applying the estimate on the Fourier block at zero frequency in part (1) of Lemma 8.6 at that rate, with $c_a = c(a/\sqrt{3}) > 0$ in w_q, we obtain

$$\|H(0)\| + \sum_{\ell} \|D_\ell H(0)\| \le C_a b w_q. \tag{11.23}$$

Since the symmetric choice of lifts preserves self-adjointness, $H(k)$ and its derivatives are Hermitian at real k. We set

$$E(k) = H(0) + \sum_{\ell=1}^{3} \sin(k_\ell) D_\ell H(0).$$

This is a Hermitian trigonometric polynomial of range one with the same value and first derivatives at zero as H, and its kernel norm is at most $C_a b w_q$. Let $h'(r)$ be the finitely supported lifted coefficients of $H - E$. They satisfy the two exact moment cancellations $\sum_r h'(r) = 0$ and $\sum_r r_\ell h'(r) = 0$.

Next, we divide by two differences, with an absolute bound. Let $t_i = e^{\mathrm{i}k_i}$ and $z_i = t_i - 1$. For each integer m, we write $t^m - 1 = z S_m(t)$ and $t^m - 1 - mz = z^2 B_m(t)$. For $m \geq 0$,

$$S_m(t) = \sum_{j=0}^{m-1} t^j, \qquad B_m(t) = \sum_{j=0}^{m-2} (m-1-j) t^j,$$

with empty sums equal to zero, and for $m = -p < 0$,

$$S_{-p}(t) = -\sum_{j=1}^{p} t^{-j}, \qquad B_{-p}(t) = \sum_{j=1}^{p} (p-j+1) t^{-j}.$$

Multiplying by $t - 1$, respectively $(t-1)^2$, we verify all four identities term by term. The sums of the absolute coefficients are at most $|m|$ and $C(1+|m|)^2$. We now telescope $t^r - 1 = \sum_i (t_i^{r_i} - 1) \prod_{j<i} t_j^{r_j}$. Substituting $t_i^{r_i} - 1 = r_i z_i + z_i^2 B_{r_i}(t_i)$ and telescoping each $\prod_{j<i} t_j^{r_j} - 1$ in the same way, we obtain

$$t^r - 1 - \sum_i r_i z_i = \sum_{i,j} z_i z_j P_{r,ij}(t). \tag{11.24}$$

The sum of the absolute coefficients of all $P_{r,ij}$ is at most $C(1+|r|)^2$: the pure terms contribute $\sum_i C(1+|r_i|)^2$ and the mixed terms $\sum_{j<i} |r_i||r_j|$. Their supports lie in the coordinate box from 0 to r, enlarged by a fixed distance, so their displacements are at most $|r| + C$ in the usual lattice metrics, with a fixed comparison of metrics absorbed in the constants.

By the moment cancellations, we may apply (11.24) termwise in $\sum_r h'(r) t^r$. The unused half of the exponential weight absorbs the factor $(1+|r|)^2$, since $\sup_{u \geq 0} (1+u)^2 e^{-au/2} < \infty$. Hence the coefficients in $H - E = \sum_{i,j} z_i L_{ij} z_j$ have weighted absolute sum at most $C_a b$ at rate $a/2$. Since $z_i = -t_i(t_i^{-1} - 1)$, setting $W_{ij} = -t_i L_{ij}$ gives $H - E = \nabla^* W \nabla$. Hermitian symmetrization of W preserves this identity and its bound. We periodize these kernels. Their Fourier samples, and hence their kernels, agree exactly with those of the original operator on the torus. Periodization does not increase the periodic weighted

absolute sums, because each distance on the torus is at most the distance of its lift. Taking E to be the periodized kernel of E proves the lemma. For a complex kernel, we apply the argument to the real and imaginary parts, each of norm at most b, and add the two factorizations. □

Proof of Theorem 11.10. We first prove the two cancellations, items (1) and (2), then the composition estimate of item (3), and finally item (4). Throughout, the norm of R is taken in units of the fine lattice, and $q = m/\delta$.

Step 1: the quadratic term. We subtract the constant $R(I)$, which does not enter $\mathcal{B}_{\rho,a}$, and write $T_k = D^k R(I)$ in the logarithmic coordinates of the fine lattice. By the definition of $\mathcal{B}_{\rho,a}$, we have

$$\frac{1}{k!}\|T_k\|_{a,\mathrm{root}} \le \epsilon\rho^{2-k} \qquad (k \ge 2). \tag{11.25}$$

We differentiate the identity $DR(\xi)[\mathcal{D}_{\tilde\xi}\lambda] = 0$ in $\tilde\xi$ at $\tilde\xi = 0$. The term containing the derivative of $\mathcal{D}_{\tilde\xi}\lambda$ is multiplied by $DR(0) = 0$, and we are left with $T_2\,\mathrm{grad} = 0 = \mathrm{grad}^{\mathsf{T}}\,T_2$. Translation invariance and invariance under spatial inversion give the remaining hypotheses of Lemma 11.11, with $b \le C\epsilon$. Since both norms use the periodic Euclidean distance, the constant accounts only for the color components and for adding the row and column bounds. For a complex-valued R, the real and imaginary parts of the coefficients satisfy the same linear identities. Lemma 11.11 then implies the factorization and the remainder bound in item (1). Its constant c_a is fixed in that proof from the rate $a/\sqrt{3}$ in Lemma 8.6, and is independent of all later radii.

Step 2: the cubic term. Let $R_3(\xi) = T_3[\xi,\xi,\xi]/6$. By linearity of I and of the translations, R_3 has the same invariances as R. On a field a that is constant in each orientation and color component, spatial inversion acts as $\mathsf{I}a = -a$. Hence $R_3(a) = R_3(-a) = -R_3(a)$, and $R_3(a) = 0$. By translation invariance, the first derivative of R_3 at such a field is the same at every link of a given orientation and color component. Differentiating the vanishing restriction with respect to one constant component gives q^3 times this common value, which therefore vanishes. We conclude that $DR_3(a) = 0$ at every link. For a link e with lower endpoint x_e, let $a_i = \xi_i(x_e)$ and let a be the corresponding constant field. Subtracting $DR_3(a) = 0$, we obtain componentwise the exact identity

$$(DR_3(\xi))_e = \frac{1}{2}\sum_{f,g}(T_3)_{efg}(\xi_f\xi_g - a_{i_f}a_{i_g}), \tag{11.26}$$

where i_f is the orientation of the link f and the products denote the bilinear color contractions with the tensor. We expand the bracket as $(\xi_f - a_{i_f})\xi_g + a_{i_f}(\xi_g - a_{i_g})$. For a coordinate path $x_0 = x_e, x_1, \dots, x_l = x_f$ on the fine lattice,

$$\xi_f - a_{i_f} = \sum_{j=0}^{l-1}(\xi_{i_f}(x_{j+1}) - \xi_{i_f}(x_j)). \tag{11.27}$$

We use shortest coordinate displacements on the torus, with one fixed choice in case of ties. The path and its reverse stay within a fixed multiple of the length of a shortest network connecting the three links. Together with Euler's identity $3R_3 = \sum_e \xi_e (DR_3)_e$, (11.26) and (11.27) are the representation asserted in item (2).

Step 3: majorants of the background field and power counting. We write $A = A_{d_0,J}$. For real θ, this is the potential of the representative of item (3); for complex θ, we use its holomorphic continuation and make no variational comparison. By Theorem 10.11, the Taylor tensors of $(A, \nabla_\delta A)$ have output rows, and columns with one fixed coarse input and output volume δ^3, that are dominated at a decay rate larger than the one used here by one series

$$Z(t) \le \frac{Ct}{1 - Ct}, \qquad A(0) = \nabla_\delta A(0) = 0. \tag{11.28}$$

We enlarge C to cover both members of the pair and the finitely many orientation and color types. The derivative $Z'(t)$ bounds the series with one fixed coarse input, because a Taylor coefficient of order s then has denominator $(s-1)!$ instead of $s!$.

We expand each vertex $T_k[\xi, \dots, \xi]/k!$, with $\xi = \delta A(\theta)$, and replace each argument by its Taylor series in θ. Lemma 11.1 implies that the resulting absolute coefficients are controlled by products of the argument series, with no factorials beyond those of the Taylor coefficients, for any fixed external index. For the spatial and volume bookkeeping, at a vertex we fix the relative positions of its fine indices and sum the remaining position with the fixed-index coefficient bound of T_k. We use output rows for every argument except the one containing the fixed coarse input; for that argument, its volume-normalized column bounds the sum over the translated fine index by $\delta^{-3} Z'(t)$. Since the union of the network of the coefficient and the networks of the arguments connects all external coarse indices, the external weight is at most the product of the weights. A fine displacement has length δ times its length in fine units. Let δ_0 be smaller than a fixed geometric constant, so that half of the fine weight remaining after a polynomial or path moment controls the same output rate a. The networks of the arguments need only the rate a, which is smaller than the rate a_* of their bounds, since $a < a_*/8$. Any enlargement caused by coordinate paths is absorbed by decreasing this fixed δ_0, independently of the order. In particular, no decay rate depends on the number of Taylor inputs. Table 11.2 summarizes the resulting powers of δ, which are verified in Steps 4–6.

Step 4: the quadratic contribution. A first difference of $\xi = \delta A$ over one fine step, in fine units, is exactly δ^2 times a component of $\nabla_\delta A$, translated and with a sign if the step is negative. The two difference factors in $\nabla^* W \nabla$ therefore supply δ^4, and the single sum over a fine index supplies δ^{-3}. We find that the quadratic contribution to the series with one fixed coarse input is at most $C_a \epsilon \delta Z(t) Z'(t)$. Since the finite-period remainder has two undifferentiated factors δA, it contributes $C_a \epsilon \delta^{-1} w_q Z(t) Z'(t)$.

Contribution	Field factors	Sum over a fine index	Result
Quadratic term, two differences	$\delta^2 \cdot \delta^2$	δ^{-3}	δ
Cubic term, one difference	$\delta^2 \cdot \delta \cdot \delta$	δ^{-3}	δ
Degree $k \geq 4$	δ^k	δ^{-3}	δ^{k-3}
Finite-period remainder	$\delta \cdot \delta \cdot w_{m/\delta}$	δ^{-3}	$\delta^{-1} w_{m/\delta} = O(\delta)$

Table 11.2. Powers of $\delta = M^{-J}$ in the proof of Theorem 11.10.

Step 5: the cubic contribution. In the representation of Step 2, the factor given by (11.27) supplies $\delta^2 \nabla_\delta A$, and the two other field factors supply δ^2. The counting column again supplies δ^{-3}. The length of the path and the number of its vertices are absorbed by a polynomial moment of the unused fine weight, since $l \leq C(1 + \mathrm{t})$. If the fixed coarse derivative falls on one of the undifferentiated factors, we use the column of that factor. If it falls on the difference factor, we first fix the relative fine configuration and the position along the path, and then translate the summed fine index to that vertex of the path. This translation is a bijection of the periodic lattice. Translation invariance of T_3 leaves the coefficient unchanged, and the same column with volume δ^3 controls the sum. By (11.25), we find the bound $C_a \epsilon \delta \rho^{-1} Z(t)^2 Z'(t)$, for every choice of the fixed external input.

Step 6: terms of degree at least four. No cancellation is needed. The k field factors and the one counting column give exactly δ^{k-3}. The vertex is bounded by (11.25), and the fixed derivative can fall on any of the k slots. Thus these terms are bounded by

$$C_a \epsilon \delta \rho^{-2} Z(t)^3 Z'(t) \sum_{k \geq 4} k \big(\delta Z(t)/\rho\big)^{k-4}.$$

Combining Steps 4–6, we find that the majorant $\mathcal{D}(t)$ of the series of the composed function with one fixed coarse input satisfies

$$\mathcal{D}(t) \leq C_a \epsilon \delta \Big\{ Z Z' + \rho^{-1} Z^2 Z' + \rho^{-2} Z^3 Z' \sum_{k \geq 4} k (\delta Z/\rho)^{k-4} \Big\} + C_a \epsilon \delta^{-1} w_q Z Z', \qquad (11.29)$$

where all series are evaluated at t. We used only symmetries of the tensors T_k, never symmetries of an individual term of an expansion of R.

Step 7: summation. Let $\rho_0(a) > 0$ lie inside the domain of Theorem 10.11 and satisfy $Z(2\rho) \leq C_1 \rho$ and $Z'(\rho) \leq C_1$ for $0 < \rho \leq \rho_0$. Let $\delta_0(a)$ be smaller than the geometric threshold of Step 3 and satisfy $\delta_0 C_1 \leq 1/2$. Through the sufficient condition $\delta Z(2\rho) \leq \rho/2$, these choices give the image condition of item (3) simultaneously for all later $\rho \leq \rho_0$. At $t = \rho$, the series in (11.29) converges, because $\delta Z(\rho)/\rho \leq 1/2$. Since $DR(I) = 0$ and $A(0) = 0$, we also have $D\mathcal{P}_\delta R(0) = 0$. Since a coefficient of order s with one fixed input

has denominator $(s-1)!$, the series $\rho^{-1}\mathcal{D}(\rho)$ dominates $\sum_{s\geq 2}\rho^{s-2}\|D^s\mathcal{P}_\delta R(0)\|_{a,\mathrm{root}}/s!$. Using $Z(\rho)\leq C_1\rho$ in each term of (11.29), we obtain

$$\mathcal{B}_{\rho,a}(\mathcal{P}_\delta R)\leq C_a(\delta+\delta^{-1}w_{m/\delta})\epsilon.$$

These constants do not depend on the later reduction of ρ. Replacing ϵ by $\mathcal{B}_{\rho,a}(R)$ gives the first inequality of item (3); the case $\mathcal{B}_{\rho,a}(R)=0$ follows by continuity.

Step 8: the finite-period remainder. Since $q=m/\delta\geq 4/\delta$,

$$\delta^{-2}w_q\leq\frac{1}{16}q^2(1+q)^3e^{-c_aq}\leq C_a.$$

The last supremum is finite, for example because $q^je^{-c_aq}\leq j!c_a^{-j}$ by the exponential series. Thus $\delta^{-1}w_q\leq C_a\delta$, uniformly for $m\geq 4$, and we enlarge $C_P(a)$ to include this constant. This proves the second inequality of item (3).

Step 9: other representatives. About the representative, Steps 3–8 used only its domain of holomorphy, the output rows and volume-normalized columns of $(A,\nabla_\delta A)$, their majorant Z, and the fact that these bounds hold at a rate larger than $8a$. Repeating Steps 3–8 with $\widetilde{A}$, $\widetilde{Z}$, $\widetilde{r}$, and the rate $b>8a$ in place of A, Z, r_*, and a_*, we obtain the estimate in item (4), with constants depending on these data and independent of J, d_0, and m. To compare with the evaluation of R at a specified reconstruction, we integrate the infinitesimal gauge-invariance identity along a real gauge path inside the domain of R. Even if the path is only continuous, at each of its points the strict margin of the domain allows all sufficiently small exponential gauge increments. Integrating the identity along each such increment, we find that R is constant on that neighborhood in the gauge variable. Continuity of the path and a finite subcover of its parameter interval identify the values at the two endpoints, and their holomorphic continuations agree by the uniqueness argument of Lemma 11.4. Alternatively, invariance of R under the gauge transformations between overlapping charts identifies the values at the two real endpoints directly. Neither conclusion follows from the bound on the image alone, which concerns only $e^{\delta A}$. This proves item (4). □

PART IV

Fluctuation integration and conditional measures

SECTION 12

Fluctuation integration and cluster expansions

In this section, we integrate out the fluctuations of a bounded number of averaging steps around the classical background field. We consider $s \leq J$ consecutive averaging steps, from a lattice of side $M^s m$ to a coarse lattice of side $m \geq 4$, starting from the effective classical action E_{d_0} of some depth $d_0 \geq 0$; for $s = J$ this is one renormalization group step. Given a coarse field c, the fine configurations whose average is c are described by free logarithmic coordinates at each intermediate scale. We center these coordinates at the conditional constrained minimizer, normalize them by the coupling and the constrained Hessian, and integrate the resulting normalized fluctuation variables w over a fixed product Ω_p of three-dimensional balls.

There are three main results. The first is an exact formula for this integral, for one averaging step in Lemma 12.3 and for $s \leq J$ averaging steps in Theorem 12.13. The second is a convergent polymer expansion of the logarithm of the fluctuation partition function, with volume-independent bounds on all its derivatives of positive order (Theorems 12.6 and 12.7 and Proposition 12.10). The third, Lemma 12.16, concerns an input effective interaction R, produced by the integrations at finer scales. It shows that integrating out the fluctuations replaces R by its value on the zero-fluctuation reconstruction plus a correction of second order in the coupling. We do not estimate the probability of the domain of integration under the Wilson measure here; this is done in Section 23.

We use the conventions of Section 4, the absolute coefficient calculus of Lemma 5.3, the constraint coordinates and the exact Haar disintegration of Lemma 6.3, and the conditional constrained minimizers, the nested minimization identity, and the coefficient bounds for E_d of Theorems 10.10 and 10.11. In Section 13, the results are used to construct the renormalization group recursion on the small-field event and to describe the conditional fluctuation kernel; they are also used in Sections 14, 15, and 23. Sections 18 and 20 reuse the triangular change of variables and the graph combinatorics, and Sections 31, 33, and 35 use the coordinates of Theorem 12.13 and its dimension counts.

The integral over one averaging step. We first describe the identity of Lemma 12.3 informally. Let $b(c, z)$ be the fine configuration with average c and free logarithmic coordinates z, and let $F(c, z) = E_{d_0}(b(c, z))$. Let $z^*(c)$ be the minimizer of $F(c, \cdot)$ and $K(c) = D_z^2 F(c, z^*(c))$ the constrained Hessian. The substitution

$$z = z^*(c) + gK(c)^{-1/2} w$$

has Jacobian $g^{3f} \det K(c)^{-1/2}$, where $3f$ is the number of real free logarithmic coordinates. Since z^* is stationary and $K^{-1/2} K K^{-1/2} = I$, we have

$$\frac{F(c,z)}{g^2} = \frac{F(c,z^*(c))}{g^2} + \frac{1}{2}|w|^2 + O(g|w|^3),$$

and $F(c,z^*(c))/g^2 = E_{d_0+1}(c)/(Mg^2)$ by the nested minimization identity. The Haar density of the free logarithmic coordinates is a reference constant times $e^{-\mathcal{L}_c(z)}$, and $\mathcal{L}_c(z) = \mathcal{L}_c(z^*(c)) + O(g|w|)$. Thus the integral over $w \in \Omega_p$ is the product of three factors that depend on c (the exponential of the effective classical action at the coarse scale, the determinant $\det K(c)^{-1/2}$, and the Haar density at the minimizer) and of the integral of $e^{-|w|^2/2 - U(c,w)}$, where U is small on Ω_p. The determinant and the Haar density at the minimizer become part of the effective interaction as functions of c, and $E_{d_0+1}/(Mg^2)$ remains a separate term. We control the integral of $e^{-|w|^2/2-U}$ by a polymer expansion about the product of truncated Gaussian measures on the balls. This product is only a reference measure. The conditional law of w, the conditional fluctuation kernel studied in Section 13, has density proportional to $e^{-|w|^2/2-U}$.

For several averaging steps the change of variables is triangular, from the coarsest intermediate scale to the finest. The minimizers, Hessians, and Haar densities at a finer scale depend on the fluctuations at all coarser scales, and the exact formula of Theorem 12.13 keeps these dependences.

Organization. Section 12.1 defines the coefficient norms and proves the exact formula for one averaging step. Section 12.2 proves the cluster expansion in an absolute coefficient algebra, in a form that applies to arbitrary interactions with connected supports. Section 12.3 constructs connected supports for the interaction of one averaging step and estimates the resulting effective interaction. Section 12.4 treats $s \le J$ averaging steps, and Section 12.5 treats an input effective interaction. The order in which the constants are chosen is summarized in Remark 12.17.

Conventions. All results of this section concern the periodic torus without a fixed complement. Distances are measured in units of the coarse lattice spacing. Unless stated otherwise, constants in the statements about one averaging step have the dependence described in item (2) of Definition 4.14. That is, they depend only on M, on the fixed larger domains of the classical and Haar disintegration charts, and on fixed radius ratios, spatial decay rates, and norm bounds. They are independent of the depth d_0, of the side m, and of the coupling and cutoff parameters satisfying the stated inequalities. For $s \le J$ averaging steps, the constants may also depend on J, on the finitely many changes of scale between the lattices involved, and on the fixed box geometry, and no uniformity as $J \to \infty$ is claimed. Every factor that is independent of the field, and every factor that depends only on the coarse field, is kept explicitly in the formulas.

We recall the coefficient norms of Definition 4.6. A tensor is bounded *at every fixed index* if, at the spatial decay rate b, its output-fixed absolute row sum and each of its input-fixed absolute column sums, the latter with the volume normalization of Definition 4.6, are bounded. Every sum carries the weight $e^{b\mathsf{t}}$, where t is the length of a shortest connected network joining the locations of all indices of the coefficient; internal vertices are allowed. For a scalar function, we fix any one input index and sum over the others with the same weight, and we write $\|\cdot\|_{b,\mathrm{root}}$ for the resulting norm of a derivative tensor. Scalar values are never included in these norms. A *majorant series* of a family of tensors is a power series with nonnegative coefficients that dominate the corresponding sums of absolute coefficients.

Attribution. The organization of the integral into an effective classical action and connected local fluctuation terms follows Bałaban's renormalization group analysis of lattice gauge theories [17, 19]; its three-dimensional antecedent is the stability analysis in [15]. We adapt this organization to the block averaging map, the constraint coordinates, and the truncated domains used here, and we prove the expansion below. In particular, we do not use the four-dimensional coupling renormalization of [17]. The cluster expansion combines the Mayer expansion with a convergence criterion of the type of Kotecký and Preiss [120]. The bound on connected graph sums by spanning trees comes from Penrose's partition [136], in the form described by Fernández and Procacci [87, Section 4.1]; see also Brydges' exposition [38]. Dimock records the finite graph estimate in the constructive renormalization group setting [68, Appendix B, Theorem 27, p. 39, and (323), p. 42]. His analysis concerns a scalar field, and no other result of [68] is used. The weighted resolvent bounds use the Combes–Thomas method [61], as in Lemma 5.3.

12.1 Coefficient norms and normalized fluctuation coordinates

Definition 12.1. Let $\mathscr{L} \geq 1$ be an integer, the box scale; it is unrelated to the side length L of the torus. Partition each periodic coordinate direction of the coarse lattice into integer intervals of lengths between $\mathscr{L}$ and $2\mathscr{L}$; a direction that is too short for such a partition forms a single interval. The products of these intervals are the *boxes*. A *support* is a nonempty connected set X of boxes; sets that wind around the torus are allowed, and $|X|$ denotes the number of boxes in X. A function is *supported in* X if it depends only on coordinates θ and w (or θ and q, when those variables are used) whose locations lie in X.

For a convergent series $f(\theta, w) = \sum_\nu f_\nu(w)\theta^\nu$ in the coarse-field coordinates θ, whose coefficients are functions of w on a fixed product Ω_p, and for a family $\{f_X\}$ indexed by supports, set

$$\|f\|_{u,\mathrm{coeff};p} = \sum_\nu u^{|\nu|} \sup_{w\in\Omega_p} |f_\nu(w)|, \qquad \|\{f_X\}\|_{u,\kappa;p} = \sup_x \sum_{X\ni x} e^{\kappa|X|}\|f_X\|_{u,\mathrm{coeff};p}, \tag{12.1}$$

where x ranges over the boxes. The coefficients are taken in the scalar link and color coordinates and include the Taylor factorials, that is, f_ν is the coefficient of the monomial θ^ν; replacing the scalar absolute value by a fixed norm on the three color components changes only fixed constants. When the functions do not depend on w we omit p from the notation.

For an extensive scalar holomorphic function $f(\theta, q)$ of the coarse-field coordinates θ and of variables q, with $q_i = g_i w_i$ in the applications below, set

$$\mathcal{J}_{u,v,b}(f) = \sum_{r+s\geq 1} \frac{u^r v^s}{r!\, s!} \| D_\theta^r D_q^s f(0,0) \|_{b,\mathrm{root}}, \tag{12.2}$$

where the tensor norm fixes any one index of θ or of q, sums over all other indices, and uses the weight $e^{b\mathfrak{t}}$ and the volume normalization of Definition 4.6. The constant coefficient of f is not restricted by $\mathcal{J}_{u,v,b}$. For a reconstructed field, which is vector valued, we use the analogous majorant series for the output-fixed rows and for all input-fixed volume-normalized columns. In a composition of s averaging steps the variables w_i and q_i of the i-th step have mesh M^{i-s}, $0 \leq i < s$, and the number of their indices, for all these steps, located in a unit cube is at most $C_M s M^{3s}$.

In (12.1), we take the supremum over $w \in \Omega_p$ of each coefficient in θ and then sum with the weights $u^{|\nu|}$. This norm dominates the supremum of the function. The symbol $\mathcal{J}_{u,v,b}$ is unrelated to the Haar disintegration densities $\mathcal{J}_c$ introduced below.

We now fix one averaging step, from side Mm to side $m \geq 4$, with input action E_{d_0} of depth $d_0 \geq 0$. By Lemma 6.3 and Theorem 10.10, the fine configuration with average c is $b(c,z) = e^{\Gamma_c(z)} U_0(c)$, where $U_0(c)$ is the coarse-field lift, Γ_c solves the block averaging constraint for the eliminated edges e_a^*, $a \in \mathcal{E}_m$, and z are the free logarithmic coordinates. We write

$$B(\theta,z) = \operatorname{Log} b(e^\theta, z), \qquad F(\theta,z) = E_{d_0}(b(e^\theta,z)), \qquad \mathcal{L}(\theta,z) = -\log \frac{\mathcal{J}_{e^\theta}(z)}{\mathcal{J}_I(0)},$$

where $\mathcal{J}_c(z)$ is the Haar disintegration density of Lemma 12.3 below; when c rather than θ is the variable we also write $F(c,z)$. Let $z^*(c)$ be the conditional constrained minimizer of Theorem 10.10 and $K(\theta) = D_z^2 F(\theta, z^*(e^\theta))$.

Lemma 12.2. *There exist $a, A, R > 0$, with the dependence described in item (2) of Definition 4.14 and independent of $d_0 \geq 0$ and $m \geq 4$, and fixed products of coarse-field coordinates and free logarithmic coordinates, contained with fixed positive margins in the domains of Lemma 6.3 and Theorems 10.10 and 10.11, on which the following bounds hold at the spatial decay rate a.*

1. *The map $B(\theta,z)$ has output-fixed absolute row sums and input-fixed volume-normalized column sums whose coefficients of order k are bounded by $A^{k+1}k!$.*

2. *The scalar functions $F(\theta, z)$ and $\mathcal{L}(\theta, z)$ have coefficients of every order $k \geq 1$ bounded by $A^{k+1}k!$ with any one input index fixed.*
3. *The minimizer $z^*(e^\theta)$ vanishes at $\theta = 0$ and has majorant series for its rows and its columns of all orders, with the same factorial bounds.*
4. *The coefficients of positive degree in θ of $K(\theta)$ may be bounded with either matrix index fixed, or with one coarse-field index fixed; in the latter case both matrix indices are summed. Their generating series is bounded by $Ax/(1 - Ax)$ for $0 \leq x < R$.*

Proof. The output side is $m \geq 4$, and the input action lives on the side $Mm \geq 4$ and has depth $d_0 \geq 0$, so Theorems 10.10 and 10.11 apply. We fix $R > 0$ such that the products $\|\theta\|_\infty \leq 2R$, $\|z\|_\infty \leq 2R$ lie inside the domain of the constraint coordinates and the domain of Theorem 10.10, and such that $\|B(\theta, z)\|_\infty < r_*/2$ there, where r_* is the radius of Theorem 10.11. Such a choice exists since $B(0, 0) = 0$ and by the first-order bounds of Lemma 6.3. The last condition places the argument of E_{d_0}, and not only the coarse field, in the domain where its coefficient bounds hold. The resulting margins depend only on the data of one averaging step, and all later reductions of R in this proof are made before J, g, and p are chosen.

Recall that if $\sum_\nu a_\nu z^\nu$ converges on a neighborhood of a product of circles of radii R_j, then by Cauchy's formula $|a_\mu| \prod_j R_j^{\mu_j}$ is at most the supremum of the function on these circles. The same holds for coefficients with values in a Banach space.

Each component of the solution Γ in Lemma 6.3 depends on a finite stencil of at most n_M scalar coordinates, of diameter at most d_M in the output lattice. At fixed M there are only finitely many local types of such stencils, and the periodic identifications only identify members of this finite list. On a closed complex product strictly inside the domain, the holomorphic solutions are bounded by a fixed constant A_0. If the circle radius in each scalar coordinate is R_0, the coefficient of a multiindex ν is then at most $A_0 R_0^{-|\nu|}$. There are $\binom{n_M+k-1}{k} \leq (2n_M)^k$ multiindices of degree $k \geq 1$. Summing absolute coefficients therefore introduces a factor exponential in k, and multiplying by $k!$ gives the bound $A^{k+1}k!$ for the ordered derivative tensors. A fixed input belongs to at most C_M stencils, which gives the column bounds. A tree on the $k + 1$ locations of a coefficient has length at most $d_M k$, so the exponential weight contributes another factor exponential in k. This proves the row and column bounds; no bound on directional derivatives is used. Finite products, exponentials, and the logarithm in the identity chart preserve such bounds, since their majorant series are dominated by products, by e^x, and by $\sum_{j\geq 1} x^j/j$ on a fixed small disc.

These are volume-normalized column bounds as well as counting bounds. In units of the coarse lattice spacing, the two meshes are 1 and M^{-1}. The column factor, the output volume divided by the input volume, is therefore 1, M^{-3}, or M^3, and we include these fixed factors in A. In compositions we use these factors as in Lemma 5.3, and the

intermediate volumes cancel at each contraction. Since lengths measured in the units of the input lattice are multiplied by M^{-1}, the exponential bounds of the input lattice still hold at one fixed positive rate.

The same reasoning applies to each term of $\mathcal{L}$. Each nonforest link contributes $-\log j_H(\Gamma_e)$, and each coarse link a contributes $\log\det(M^2 D_{X_{c_a^*}} \mathsf{C}_a)$. Here $j_H(0) = 1$, and the matrix in the second logarithm equals I at $z = 0$. On a smaller product, their differences from 1 and from I have norm less than $1/2$, which ensures that the scalar and matrix logarithm series converge with uniform local coefficient bounds. The constant $\mathcal{J}_I(0)$ cancels exactly these values at zero. Each index belongs to a bounded number of these local terms, which gives the asserted bounds for their sum with one index fixed. This argument also specifies the real logarithms and their holomorphic continuations.

For F, the product chosen above places every argument B inside the radius r_*, and we use the factorial bounds for E_{d_0} of Theorem 10.11. In a coefficient of the composition, we join the network of the outer tensor to the networks of the inserted B tensors. If an input index is fixed, we use the column of the insertion containing it and rows for the other insertions. The contraction rule of Lemma 5.3 then sums every internal index without a volume factor, and the exponential generating series is bounded by the composition of two convergent majorant series. Differentiating twice to form a Hessian introduces a factor $(k+1)(k+2)$ at coefficient degree k, which is absorbed on a fixed smaller radius.

To bound all orders of the minimizer, let $K^0 = D_z^2 F(0,0)$ and write the stationarity equation as the implicit system

$$z = \Phi(\theta, z), \qquad \Phi(\theta, z) = -(K^0)^{-1}\{D_z F(\theta, z) - K^0 z\}. \tag{12.3}$$

We have $\Phi(0,0) = 0$, and the linear part of Φ in z vanishes at the origin. Its part linear in θ is bounded by $A\|\theta\|$, and the remaining majorant series is bounded by $A(\|\theta\| + x)^2/[1 - A(\|\theta\| + x)]$ when the unknown has majorant x. The rows and columns of $(K^0)^{-1}$ are bounded uniformly by Theorem 10.10. We apply the implicit-series clause of Lemma 5.3 with the majorant

$$x = A\|\theta\| + \frac{A(\|\theta\| + x)^2}{1 - A(\|\theta\| + x)},$$

that is, with $\kappa = 0$ in its notation, and we obtain a convergent majorant at every fixed index. Its unique small solution is the minimizer of Theorem 10.10, by the uniqueness statement there.

Finally we compose the tensor $D_z^2 F$, which has two indices of z, with the minimizer. A coefficient of positive degree in θ contains a coarse-field index either in the outer tensor or in an inserted copy of z^*. Fixing that index and using the appropriate columns along its path, we sum over the two indices of z. The contractions and the majorant series just described prove the stated bound for $K - K^0$. Throughout the contractions we keep the

shortest connected networks, and we replace them by trees on the external indices only once, as permitted by Lemma 5.3. This fixes one positive rate for every order. □

In the next lemma, U denotes a scalar function of the coarse field and of w, not a configuration.

Lemma 12.3. *Fix an odd integer $M \geq 3$. There exist $u_0, v_0, b, m_0, C, \sigma_0 > 0$ and $p_0 \geq 1$, with the dependence described in item (2) of Definition 4.14 and independent of $d_0 \geq 0$ and $m \geq 4$, such that the following holds for the reconstruction $b(c,z)$ of Theorem 10.10. Put*

$$F(c,z) = E_{d_0}(b(c,z)), \quad z^*(c) = z^*_{d_0}(c), \quad K(c) = D_z^2 F(c, z^*(c)), \quad S(c) = K(c)^{-1/2}.$$

1. *There are $f = 2(M^3-1)m^3$ independent $\mathrm{SU}(2)$ variables, hence $3f$ real free logarithmic coordinates. Their density relative to the product Haar measure on the coarse configuration, including the Haar factors of the eliminated edges, is the Haar disintegration density*

$$\mathcal{J}_c(z) = c_{\mathrm{Haar}}^f \frac{\prod_{e \text{ nonforest}} j_H(\Gamma_{c,e}(z))}{\prod_{a \in \mathcal{E}_m} \det D_{X_{e_a^*}} \mathsf{C}_{c,a}(\Gamma_c(z))},$$

   where $j_H(X) = \mathrm{sinc}^2(|X|/2)$; its value at $c = I$, $z = 0$ is $\mathcal{J}_I(0) = c_{\mathrm{Haar}}^f M^{6|\mathcal{E}_m|}$. The integral over the forest variables has mass one. Set $\mathcal{L}_c(z) = -\log(\mathcal{J}_c(z)/\mathcal{J}_I(0))$ and $\mathcal{L}_0(c) = \mathcal{L}_c(z^*(c))$.
2. *For real c the matrix $S(c)$ is real, symmetric, and positive definite, and S extends holomorphically to $c = e^\theta$, $\|\theta\|_\infty < u_0$. With $K^0 = K(I)$, for every real $x \geq 0$,*

$$\|(K^0 + xI)^{-1}\|_{b,\mathrm{row+col}} \leq \frac{C}{m_0 + x}, \qquad S(c) = \frac{1}{\pi} \int_0^\infty x^{-1/2} (K(c) + xI)^{-1}\, dx.$$

   The coefficients of S satisfy factorial bounds at every fixed index, with exponentially weighted rows and columns, and each coefficient of positive degree in θ may also be bounded with one coarse-field index fixed and both matrix indices summed.
3. *There exist $a_1, A, R > 0$, chosen before J and independent of d_0, m, g, p, and fixed products of radius R contained with fixed positive margins in the larger classical and Haar disintegration domains, on which $B(\theta, z)$, $F(e^\theta, z)$, $\mathcal{L}_{e^\theta}(z)$, and $z^*(e^\theta)$ satisfy the bounds of Lemma 12.2 at the rate a_1, with coefficients of order $k \geq 1$ bounded by $A^{k+1}k!$; for B these are output-fixed rows and all input-fixed volume-normalized columns, and for the scalar functions any one input index is fixed. Scalar values are excluded, and no coupling or cutoff enters. The coefficients of positive degree of $K(e^\theta)$, $K^{1/2} = KS$, and $S = K^{-1/2}$ are bounded by $A^{k+1}k!$ with either matrix index fixed, or with a coarse-field index fixed and then both matrix indices and all other inputs summed; their series of positive degree are bounded by $Au/(1-Au)$. All these bounds also hold after translation to any background point of a fixed smaller product. The volume normalizations are those of Definition 4.6 for this pair of lattices.*

4. *Let $g > 0$, $p \geq p_0$, and $\|\theta\|_\infty < u_0$, and assume*

$$Cgp < \tfrac{1}{2}, \qquad \sigma = C\Big\{\frac{gp^3}{1 - Cgp} + gp\Big\} \leq \sigma_0. \tag{12.4}$$

Then, for every $w \in \Omega_p = \prod_{e=1}^f \{w_e \in \mathbb{R}^3 : |w_e| \leq p\}$, the point $z = z^*(c) + gS(c)w$ satisfies $\|z\|_\infty < v_0/2$, and $b(c, z)$ lies in the domain of the classical coefficient bounds. Let $z_p = \int_{|w| \leq p} e^{-|w|^2/2}\, d^3w$, let π_p be the product reference measure on Ω_p with density $z_p^{-f} \prod_e \mathbf{1}_{|w_e| \leq p} e^{-|w_e|^2/2}$, and set

$$U(c, w) = \frac{F(c, z^* + gSw) - F(c, z^*)}{g^2} - \frac{1}{2}|w|^2 + \mathcal{L}_c(z^* + gSw) - \mathcal{L}_0(c). \tag{12.5}$$

Then, with $g_+^2 = Mg^2$,

$$\int_{z^* + gS\Omega_p} e^{-F(c,z)/g^2}\, \mathcal{J}_c(z)\, dz = \mathcal{J}_I(0) g^{3f} e^{-E_{d_0+1}(c)/g_+^2} \det S(c)\, e^{-\mathcal{L}_0(c)} z_p^f\, \Xi(c), \tag{12.6}$$

where $\Xi(c) = \mathbb{E}_{\pi_p} e^{-U(c,w)}$. These identities are exact for real c. For complex c the left side is defined as the holomorphic continuation obtained by integrating over the same real product of w.

Proof. We take $p_0 = 1$; any fixed larger p_0 also works. We use the products of Lemma 12.2 inside the domain of Theorem 10.10, and we reduce their radii as follows.

Step 1: the weighted resolvent at the identity. Let $\lambda = m_0$ be the uniform lower bound of Theorem 10.10 for the Hessian K^0 on the real space of free logarithmic coordinates; it holds for every $m \geq 4$. The indices of z have bounded density and polynomial growth, with constants depending only on M. We apply the weighted inverse clause of Lemma 5.3 to $K^0 + xI$, $x \geq 0$. Its lower bound is $\lambda + x$, and its off-diagonal entries are those of K^0. We fix $0 < \beta_0 < a$ by the condition of that clause for the lower bound λ; this choice works for every $x \geq 0$. We obtain

$$|(K^0 + xI)^{-1}_{ef}| \leq \frac{2e^{-\beta_0 \operatorname{dist}(e,f)}}{\lambda + x}.$$

For $0 < b_0 < \beta_0$, summing with the weight $e^{b_0 \operatorname{dist}(e,f)}$ and using the polynomial growth just recalled, we obtain the row and column bound $C/(\lambda + x)$. The constant and the rate are independent of x, d_0, and m.

Step 2: the inverse square root. Let $\Delta K = K(c) - K^0$. On a smaller coarse-field product, Lemma 12.2(4) implies that $\|(K^0 + xI)^{-1}\Delta K\|_{b_0} < 1/2$ simultaneously for all $x \geq 0$. Hence

$$(K(c) + xI)^{-1} = \sum_{j \geq 0} \big[-(K^0 + xI)^{-1}\Delta K\big]^j (K^0 + xI)^{-1} \tag{12.7}$$

converges in the weighted matrix algebra. We may integrate its majorant against $\pi^{-1}x^{-1/2}\,dx$, since $x^{-1/2}/(\lambda+x)$ is integrable. For a real eigenvalue $k>0$, the substitution $x=ky^2$ and $2\int_0^\infty(1+y^2)^{-1}dy=\pi$ give $\pi^{-1}\int_0^\infty x^{-1/2}(k+x)^{-1}dx=k^{-1/2}$. Diagonalizing the real symmetric matrix $K(c)$, we find that the integral is its positive symmetric inverse square root. The same integral of the normally convergent series defines the holomorphic continuation. By continuation from real data we have $S^{\mathsf T}=S$ and $S^{\mathsf T}KS=I$, with the transpose rather than the Hermitian adjoint.

Step 3: coefficients of S. Each coefficient of positive degree in θ of (12.7) differentiates at least one factor ΔK. We fix the coarse-field index on that factor, sum both of its matrix indices by Lemma 12.2(4), and contract the remaining resolvents using their rows and columns. Integrating the resulting convergent majorant series proves the asserted bounds for S, including the sum over both matrix indices when a coarse-field index is fixed. The undifferentiated term needs only the rows and columns of $(K^0+xI)^{-1}$. All radius reductions and the final rate are chosen here, before g and p.

Step 4: joint bounds and the positive square root. The tensors of B, F, $\mathcal{L}$, and z^* in part (3) are those of Lemma 12.2; no fluctuation variable or cutoff enters their construction. For the matrix products, we fix $a_1>0$ such that $2a_1$ is at most both the rate of Lemma 12.2 and b_0; this leaves the factor two needed to replace a shortest connected network by a tree on the external indices. Since S is a function of K, it commutes with K. The holomorphic matrix

$$P(c)=K(c)S(c) \tag{12.8}$$

equals the positive symmetric square root $K(c)^{1/2}$ for real data, and $P^2=K$ on the complex product since $S^2K=I$. Then P is the continuation of the positive square root, and its row and column bounds follow from submultiplicativity.

We check the coefficient bounds for P. For a coarse-field multiindex ν we have $P_\nu=\sum_{\mu\le\nu}K_\mu S_{\nu-\mu}$, with the Taylor factorials included in the coefficients. If a matrix index is fixed, we use the row or column bound of the factor containing it and the compatible bound of the other factor. If a coarse-field index is fixed, we distinguish its position in the convolution. If it belongs to K_μ, we use the bound for K summing both matrix indices and then a row of S at the common index. If it belongs to $S_{\nu-\mu}$, we use the corresponding bound for S and the column of K at the common index. Each contraction joins two networks at the shared index of z, so the exponential weight and the volume normalization are those of Lemma 5.3.

To treat all orders at once, let $k(x),s(x)$ be majorant series of the coefficients of K and S with a matrix index fixed, including their constant terms, and let $k_{\mathrm t}(x),s_{\mathrm t}(x)$ be majorant series of their coefficients of positive degree with a coarse-field index fixed and both matrix indices summed. These four series converge, and the last two have no

constant term. The contractions just described bound the coefficients of positive degree of P by those of

$$k(x)s(x) - k(0)s(0), \qquad k_{\mathrm{t}}(x)s(x) + k(x)s_{\mathrm{t}}(x), \tag{12.9}$$

for a fixed matrix index and a fixed coarse-field index, respectively. If such a series is bounded by C_0 on a circle of radius r_0, its coefficient of degree k is at most $C_0 r_0^{-k}$. Enlarging A beyond $\max\{1, (1 + C_0)/r_0\}$ for the finitely many series involved, we bound these coefficients by A^k for $k \geq 1$. This bounds the series of positive degree by $Au/(1 - Au)$ and the ordered tensors of order k by $A^{k+1}k!$. The same enlargement covers B, F, $\mathcal{L}$, z^*, K, and S.

These bounds persist after translation to any background point of a fixed smaller product. Indeed, suppose that a tensor of order n, with one index fixed, is bounded by $C^{n+1}n!$. Expanding at a background point of radius r, we obtain for $k \geq 1$ the order-k bound

$$\sum_{j \geq 0} \frac{r^j}{j!} C^{k+j+1}(k + j)! = \frac{C^{k+1}k!}{(1 - Cr)^{k+1}} \leq (2C)^{k+1}k! \qquad (Cr \leq 1/2).$$

The fixed index remains a differentiated input or a matrix index, and the other legs at the background point are summed using the same rows or columns; removing these legs cannot increase the length of the remaining network. Thus one radius reduction and one enlargement of A cover all orders and all these background points, without changing a_1. We take the least common radius R and the least rate a_1, and we reduce u_0 to this radius if necessary. This proves part (3), with constants chosen before J, g, and p.

Step 5: the Haar disintegration. By Lemma 6.3 there are $f = 2((Mm)^3 - m^3) = 2(M^3 - 1)m^3$ independent SU(2) variables, and the density is exactly $\mathcal{J}_c$. At $c = I$ and $z = 0$ each of the $|\mathcal{E}_m|$ determinants $\det D_{X_{e_a^*}} \mathsf{C}_{I,a}(0) = \det(M^{-2} I_3)$ equals M^{-6}, and each factor j_H equals one, which gives $\mathcal{J}_I(0) = c_{\mathrm{Haar}}^f M^{6|\mathcal{E}_m|}$. The Haar probability measure on the gauge transformations of the block forests has mass one. The forest integral contributes no further factor, since the integral uses the action at the representative in the block axial gauge. This proves part (1).

Step 6: the domain. By the bound $\|z^*(c)\| \leq C\|\theta\|$ and the row bound for S, we have $\|z^* + gSw\| \leq C(u_0 + gp)$. We choose u_0 and the threshold for gp such that this is less than half of the least chart radius v_0 and the reconstruction stays inside the domain of the classical coefficient bounds. We achieve these requirements by enlarging the constant C in (12.4) before it is fixed; none of them depends on p, g, d_0, or m.

Step 7: the change of variables. For real c, the change of variables $z = z^* + gSw$ has Jacobian $g^{3f} \det S$. Stationarity removes the linear term of $F(c, \cdot)$ at z^*, and $S^{\mathsf{T}} K S = I$ turns its quadratic term into $|w|^2/2$. By the nested minimization identity $E_{d_0+1}(c) = MF(c, z^*(c))$ of Theorem 10.10 and $g_+^2 = Mg^2$, we have $F(c, z^*)/g^2 = E_{d_0+1}(c)/g_+^2$. Substituting

$\mathcal{J}_c(z) = \mathcal{J}_I(0)e^{-\mathcal{L}_c(z)}$ and extracting $e^{-\mathcal{L}_0(c)}$ leaves the integrand $e^{-|w|^2/2-U(c,w)}$, whose integral over Ω_p is $z_p^f \Xi(c)$ by the definition of π_p. This proves (12.6), with every scalar factor displayed. For complex c, we integrate over the same real product of w. Since the uniform chart margins and the convergent series justify integrating coefficient by coefficient, the result is the holomorphic continuation of the real identity. □

12.2 Polymer expansions in the coefficient algebra

In this subsection, we fix a finite set of boxes as in Definition 12.1. A *polymer* is a support, that is, a nonempty connected set of boxes. We prove the cluster expansion of a hard-core polymer gas whose activities take values in the coefficient algebra of (12.1). We then prove a version for arbitrary interactions with connected supports, which includes holomorphic dependence on the interaction and an estimate for the derivative with respect to an inserted interaction. These statements use only the norms and the independence of fluctuation variables in disjoint supports. They are applied to one averaging step in Section 12.3 and to several in Section 12.4. The weighted incidence condition below is a version of the convergence criterion of Kotecký and Preiss [120], and we prove the bound we need by majorizing with rooted trees. The successive support weights $e^{8|X|}$, $e^{5|P|}$, and $e^{2|Y|}$ below are chosen for convenience; only the gaps between them matter.

Lemma 12.4. *Let $\mathcal{S}$ be a finite family of nonempty sets of boxes, not necessarily connected, with nonnegative weights h_X, $X \in \mathcal{S}$. Let $\kappa \geq 0$ and suppose*

$$\eta = \sup_x \sum_{X \ni x} h_X e^{(\kappa+1)|X|} \leq 1.$$

Put $a_X = h_X e^{\kappa|X|}$. For $X \in \mathcal{S}$ let T_X be the sum over finite rooted trees whose vertices carry elements of $\mathcal{S}$, repetitions allowed, whose root carries X, and in which the sets carried by adjacent vertices intersect, of the product of the weights a_Y of all vertices, where a vertex with j children receives the factor $1/j!$. Then $T_X \leq a_X e^{|X|}$, and $\sum_{X \ni x} T_X \leq \eta$ for every box x. In particular, the sum of $e^{\kappa|\cup \mathcal{F}|} \prod_{X \in \mathcal{F}} h_X$ over all families $\mathcal{F}$ of distinct elements of $\mathcal{S}$ whose union contains x and whose intersection graph is connected is at most η.

Proof. For trees of height zero, we set $T_X^{(0)} = a_X$, and we define iteratively

$$T_X^{(n+1)} = a_X \exp\Big(\sum_{Y:\, Y \cap X \neq \varnothing} T_Y^{(n)} \Big). \tag{12.10}$$

Expanding the exponential assigns the factor $1/j!$ to the j children of each vertex, and induction on the height shows that $T_X^{(n)}$ is the sum over the trees of height at most n described in the statement. Equivalently, for trees with k vertices and a fixed root, one may label the other vertices $2, \dots, k$ and divide by $(k-1)!$; grouping the labels among

the subtrees of the root gives exactly the same factors $1/j!$. This verifies the count also when several vertices carry the same set.

If $T_Y^{(n)} \le a_Y e^{|Y|}$ for all Y, then

$$\sum_{Y:\, Y\cap X\neq\varnothing} T_Y^{(n)} \le \sum_{x\in X}\sum_{Y\ni x} a_Y e^{|Y|} \le \eta|X| \le |X|,$$

which implies $T_X^{(n+1)} \le a_X e^{|X|}$. Their limit T_X satisfies the same bound, since the iterates increase. Summing over $X \ni x$, we obtain $\sum_{X\ni x} T_X \le \sup_x \sum_{X\ni x} h_X e^{(\kappa+1)|X|} = \eta$. A family $\mathcal{F}$ of k distinct sets with connected intersection graph and union containing x has a member containing x and an intersection spanning tree rooted at it. Listing its other members in all orders cancels the factor $1/(k-1)!$, so the rooted tree sum counts $\prod_{X\in\mathcal{F}} a_X$ at least once. Finally, the inequality $|\cup\mathcal{F}| \le \sum_{X\in\mathcal{F}}|X|$ transfers the weight of the union to the vertex weights. □

Lemma 12.5. *For every finite simple graph G,*

$$\Big|\sum_{H}(-1)^{|E(H)|}\Big| \le \#\{\textit{spanning trees of } G\},$$

where the sum is over the connected spanning subgraphs H of G.

This inequality is used by Dimock in [68, Appendix B, Theorem 27, p. 39, and (323), p. 42]. We prove it with the partition of connected subgraphs introduced by Penrose [136]; see also [87, Section 4.1].

Proof. For a graph with one vertex both sides equal one, and for a disconnected graph both sides vanish. Otherwise we order the vertices and root the graph at vertex 1. For a connected spanning subgraph H, let $d_H(v)$ be the graph distance from v to 1 in H, and join each $v \neq 1$ to its least neighbor in H in the preceding distance layer. These edges form a spanning tree $T(H)$ with the same distances to 1. For a spanning tree T, let $R(T)$ consist of the edges of T, of every edge of G joining two vertices in the same distance layer of T, and of every edge of G joining successive layers whose endpoint in the lower layer is not smaller than the parent in T of its endpoint in the upper layer. No other edge belongs to $R(T)$. Then

$$T(H) = T \quad\Longleftrightarrow\quad T \subset H \subset R(T).$$

Indeed, an edge of H cannot skip a distance layer of H, and it cannot be joined to a smaller vertex of the preceding layer than the chosen parent. Conversely, adding edges of $R(T)\setminus T$ to T changes no distance, since each such edge changes the layer by at most

one, and it does not change the least parent. Thus the connected spanning subgraphs are partitioned into these intervals, and each interval contributes

$$\sum_{T\subset H\subset R(T)} (-1)^{|E(H)|} = (-1)^{k-1}(1-1)^{|E(R(T))\setminus E(T)|},$$

where k is the number of vertices. Its absolute value is at most one, and summing over spanning trees proves the inequality. □

Theorem 12.6. *Fix a finite set of boxes and $u > 0$. For each polymer P let $\mathfrak{z}(P)$ be a convergent power series in the coarse-field coordinates located in P, with $\|\mathfrak{z}(P)\|_{u,\mathrm{coeff}} < \infty$, and assume*

$$\sup_x \sum_{P\ni x} e^{5|P|}\|\mathfrak{z}(P)\|_{u,\mathrm{coeff}} \le 2\sigma, \tag{12.11}$$

where $\sigma \le \sigma_0$ for an absolute constant $\sigma_0 > 0$. Let

$$\Xi = \sum_{\{P_1,\dots,P_k\}\ \text{pairwise disjoint}} \prod_{i=1}^{k} \mathfrak{z}(P_i)$$

be the partition function of the hard-core polymer gas, in which two polymers are compatible if they are disjoint; the empty family contributes one. Then the following statements hold.

1. *We have $\log\Xi = \sum_Y \Psi_Y$, where*

$$\Psi_Y = \sum_{k\ge 1}\frac{1}{k!}\sum_{P_1\cup\cdots\cup P_k = Y} \varphi^T(P_1,\dots,P_k)\prod_{i=1}^{k}\mathfrak{z}(P_i).$$

   The inner sum is over ordered tuples of polymers, repetitions allowed, and $\varphi^T(P_1,\dots,P_k) = \sum_H (-1)^{|E(H)|}$, the sum being over the connected spanning subgraphs H of the incompatibility graph on $\{1,\dots,k\}$, in which $i \ne j$ are joined if $P_i \cap P_j \ne \varnothing$. A single polymer has coefficient one, and $|\varphi^T|$ is at most the number of spanning trees of the incompatibility graph by Lemma 12.5. All these series converge absolutely in the coefficient algebra.

2. *Each Ψ_Y depends only on the coarse-field coordinates located in Y, and*

$$\sup_x \sum_{Y\ni x} e^{2|Y|}\|\Psi_Y\|_{u,\mathrm{coeff}} \le C\sigma.$$

3. *$\Xi \ne 0$ on $\|\theta\|_\infty < u$, and $\log\Xi$ is the branch that vanishes at zero activity.*

4. *Suppose that the number of coordinate labels in each box is bounded and that, for every support Y, all labels located in Y can be joined by a network of length at most $C_{\mathrm{geom}}|Y|$. Then there exists $b > 0$ such that, on $\|\theta\|_\infty \le u/2$, for every $s \ge 1$,*

$$\|D_\theta^s \log\Xi\|_{b,\mathrm{root}} \le C^s s!\,\sigma.$$

The constants in (1)–(3) are absolute; in particular they do not depend on the number of polymers. In (4) the constants depend in addition on C_{geom}, on the bound on the number of labels per box, and on u, but not on the number of boxes or on s.

Proof. We take $\sigma_0 = 1/4$.

Step 1: the formal identity. We give each polymer P a formal variable z_P. The partition function is

$$\sum_{k\geq 0}\frac{1}{k!}\sum_{P_1,\dots,P_k}\prod_i z_{P_i}\prod_{i<j}(1-\mathbf{1}_{P_i\cap P_j\neq\varnothing}),$$

since the last product removes all tuples with an intersecting pair, including repeated polymers. Expanding this product gives a sum over all graphs on $\{1,\dots,k\}$, with the weight $-\mathbf{1}_{P_i\cap P_j\neq\varnothing}$ for each edge. Each graph has a unique partition of its vertices into connected components. For components of sizes $k_1,\dots,k_l$, distributing the labels gives the factor $k!/(l!\prod_j k_j!)$, where the factor $1/l!$ accounts for equal components. Hence the sum over all graphs is the exponential of the sum over connected graphs, and comparing coefficients, we obtain the formula for Ψ_Y.

Step 2: the bound. Let $h_P = \|\mathfrak{z}(P)\|_{u,\mathrm{coeff}}$ and $a_P = h_P e^{2|P|}$. For every tuple, the incompatibility graph is a finite simple graph on the distinct indices $1,\dots,k$, even if some of the listed polymers coincide, and its edges are the intersecting pairs. Thus Lemma 12.5 applies to every coefficient φ^T. If the union contains x, we choose a root polymer containing x, which replaces the factor $1/k!$ by the larger factor $1/(k-1)!$. By Lemma 12.4 with $\kappa = 2$, we have

$$\sup_x \sum_{Y\ni x} e^{2|Y|}\|\Psi_Y\|_{u,\mathrm{coeff}} \leq \sup_x \sum_{P\ni x} h_P e^{3|P|} \leq 2\sigma,$$

provided $2\sigma \leq 1$. This proves the absolute convergence in the coefficient algebra, the bound in (2), and the fact that Ψ_Y depends only on coordinates in Y.

Step 3: the analytic branch. We replace every activity by $\alpha\mathfrak{z}(P)$ with $\alpha \in \mathbb{C}$. Since $\sigma \leq 1/4$, the bound of Step 2 holds for $|\alpha| \leq 2$. Since the sum of all these bounds is finite at finite volume, the connected series and its exponential are convergent power series in α on this disc. By Step 1, their exponential and the polynomial $\Xi(\alpha)$ have the same Taylor coefficients at $\alpha = 0$, which implies that they agree on the whole disc. At $\alpha = 1$, this proves $\Xi = \exp(\sum_Y \Psi_Y)$, which gives $\Xi \neq 0$ and identifies the branch. This argument does not use a volume-independent bound on the scalar $\log \Xi$.

Step 4: derivatives. For a scalar series $f_Y(\theta) = \sum_\nu f_{Y,\nu}\theta^\nu$ and $\|\theta\|_\infty \leq u/2$, the sum over all ordered derivative indices of order s of the absolute values of the derivatives of f_Y is bounded by

$$\sum_{n\geq s}\sum_{|\nu|=n}|f_{Y,\nu}|(n)_s(u/2)^{n-s} \leq 2^{s+1}s!\,u^{-s}\|f_Y\|_{u,\mathrm{coeff}}, \tag{12.12}$$

where $(n)_s = n(n-1)\cdots(n-s+1)$. To check the constant, we write $n = s+j$ and bound each $\binom{s+j}{s}2^{-j}$ by $\sum_{j\geq 0}\binom{s+j}{s}2^{-j} = 2^{s+1}$. Fixing one derivative index only reduces the sum, and makes it zero unless that index is located in Y. By hypothesis, all distinct labels in Y can be joined by a network of length $C_{\mathrm{geom}}|Y|$, and repeated labels add no length. We choose b with $bC_{\mathrm{geom}} \leq 1$, multiply (12.12) by the weight $e^{bt} \leq e^{|Y|}$, and

use the bound in (2), in which the fixed index determines a box. This proves (4), with a single factorial. For the boxes of Definition 12.1, whose labels have bounded density, the geometric hypothesis holds. Indeed, we join the centers of adjacent boxes of Y and, inside each box, join its finitely many labels by a fixed tree; the total length is at most $C_{M,\mathscr{L}}|Y|$. □

In the next theorem, we apply two expansions to an arbitrary interaction with connected supports: first the connected-component expansion of $\prod_X (1 + (e^{-U_X} - 1))$, which produces a hard-core polymer gas, and then the cluster expansion of Theorem 12.6.

Theorem 12.7. *Fix a finite set of boxes, a fixed product domain Ω for the fluctuation variables, each fluctuation variable being located in one box, and a product probability measure π on Ω. Fix $u > 0$, and let $\mathscr{A}$ be the algebra of convergent series $f(\theta, w) = \sum_\nu f_\nu(w)\theta^\nu$ with bounded measurable coefficients and finite norm $\|f\|_{\mathscr{A}} = \sum_\nu u^{|\nu|} \sup_{w\in\Omega} |f_\nu(w)|$. For $h \geq 0$ let $\mathscr{S}_h$ be the Banach space of families $\boldsymbol{U} = (U_X)$, indexed by supports, such that each $U_X \in \mathscr{A}$ depends only on the fluctuation and coarse-field coordinates located in X, with the norm*

$$\|\boldsymbol{U}\|_{\mathscr{S}_h} = \sup_x \sum_{X \ni x} e^{h|X|} \|U_X\|_{\mathscr{A}}.$$

Constant coefficients are allowed. There exist absolute constants $\eta_0, C > 0$, independent of the number of boxes, of the dimension of the fluctuation variables, of u, and of π, such that the following holds for $\|\boldsymbol{U}\|_{\mathscr{S}_8} < \eta_0$. Define the polymer activities

$$\mathfrak{z}(P) = \sum_{\mathcal{F}} \mathbb{E}_\pi \prod_{X\in\mathcal{F}} \big(e^{-U_X} - 1\big), \tag{12.13}$$

the sum being over the families $\mathcal{F}$ of distinct supports whose intersection graph is connected and whose union is P, and set

$$\mathcal{L}_Y(\boldsymbol{U}) = -\sum_{k\geq 1} \frac{1}{k!} \sum_{P_1\cup\cdots\cup P_k = Y} \varphi^T(P_1, \dots, P_k) \prod_{i=1}^k \mathfrak{z}(P_i),$$

with φ^T as in Theorem 12.6. Then $\mathcal{L}_Y(\boldsymbol{U})$ depends only on the coarse-field coordinates located in Y, and

$$\sum_Y \mathcal{L}_Y(\boldsymbol{U}) = -\log \mathbb{E}_\pi e^{-\sum_X U_X}, \qquad \|\boldsymbol{\mathcal{L}}(\boldsymbol{U})\|_{\mathscr{S}_2} \leq C\|\boldsymbol{U}\|_{\mathscr{S}_8}, \tag{12.14}$$

where $\boldsymbol{\mathcal{L}}(\boldsymbol{U}) = (\mathcal{L}_Y(\boldsymbol{U}))_Y$. Here $\mathbb{E}_\pi e^{-\sum_X U_X} \neq 0$, and the logarithm is the branch that vanishes at $\boldsymbol{U} = 0$. Thus each connected cluster of interactions is assigned to the union of their supports. The map $\boldsymbol{U} \mapsto \boldsymbol{\mathcal{L}}(\boldsymbol{U})$, from the open ball of radius η_0 in $\mathscr{S}_8$ to $\mathscr{S}_2$, is Fréchet holomorphic. For $\|\boldsymbol{U}\|_{\mathscr{S}_8} \leq \eta_0/2$ and every $\boldsymbol{A} \in \mathscr{S}_8$,

$$\|D\boldsymbol{\mathcal{L}}(\boldsymbol{U})[\boldsymbol{A}]\|_{\mathscr{S}_2} \leq C\|\boldsymbol{A}\|_{\mathscr{S}_8}. \tag{12.15}$$

If the segment $\boldsymbol{U} + s\boldsymbol{A}$, $0 \le s \le 1$, lies in the closed ball of radius $\eta_0/2$, then $\boldsymbol{\mathcal{L}}(\boldsymbol{U}+\boldsymbol{A}) - \boldsymbol{\mathcal{L}}(\boldsymbol{U})$ is the integral of this derivative along the segment, and its $\mathscr{S}_2$ norm is at most $C\|\boldsymbol{A}\|_{\mathscr{S}_8}$. If $\boldsymbol{U}$ depends holomorphically on a parameter in a Banach space, so does $\boldsymbol{\mathcal{L}}(\boldsymbol{U})$. If θ is real and all U_X and A_X are real valued, then

$$\sum_Y D\mathcal{L}_Y(\boldsymbol{U})[\boldsymbol{A}] = \frac{\mathbb{E}_\pi\big[(\sum_X A_X)e^{-\sum_X U_X}\big]}{\mathbb{E}_\pi e^{-\sum_X U_X}}.$$

The supports of the A_X may lie anywhere relative to the box at which the norm $\mathscr{S}_2$ is evaluated. The constants are absolute, as in item (1) of Definition 4.14; when these norms are converted to the torus norms of Definition 4.6, the conversion constants have the dependence described in item (2).

Proof. By coefficient convolution, we have $\|fg\|_{\mathscr{A}} \le \|f\|_{\mathscr{A}}\|g\|_{\mathscr{A}}$. The coefficientwise expectation $\mathbb{E}_\pi$ has norm at most one, and functions of the fluctuation variables in disjoint sets of boxes are independent under π, so their product has the product expectation. These are the operations of Lemma 5.3. At a fixed finite number of boxes, the spaces $\mathscr{S}_h$ are complete, since they are finite products of closed subspaces of weighted ℓ^1 spaces of bounded measurable functions, with an equivalent finite maximum norm.

Step 1: the connected-component expansion. We write $F_X = e^{-U_X} - 1$, $h_X = \|F_X\|_{\mathscr{A}}$, and $\eta = \|\boldsymbol{U}\|_{\mathscr{S}_8}$. For η_0 small, the exponential series in $\mathscr{A}$ implies that $h_X \le e^{\|U_X\|_{\mathscr{A}}} - 1 \le 2\|U_X\|_{\mathscr{A}}$, and therefore $\sup_x \sum_{X\ni x} e^{8|X|}h_X \le 2\eta$. Since there are finitely many supports, we may expand $\mathbb{E}_\pi e^{-\sum_X U_X} = \mathbb{E}_\pi \prod_X (1+F_X)$ as the sum over families of distinct supports of $\mathbb{E}_\pi \prod_{X\in\mathcal{F}} F_X$. Each family has a unique decomposition into components whose intersection graphs are connected. Different components have disjoint unions, hence disjoint fluctuation variables, and the expectation factors. Grouping the families according to the unions of their components, we obtain exactly the hard-core gas with the activities (12.13); the empty family contributes one, and the coarse-field variables enter only as parameters. Since the expectation contracts the norm and $\sup_x \sum_{X\ni x} h_X e^{6|X|} \le 2\eta \le 1$, Lemma 12.4 with $\kappa = 5$ gives

$$\sup_x \sum_{P\ni x} e^{5|P|}\|\mathfrak{z}(P)\|_{\mathscr{A}} \le 2\eta.$$

Step 2: the cluster expansion. Taking $\eta_0 \le \sigma_0$, we apply Theorem 12.6 to these activities, with $\sigma = \eta$. We obtain the absolute convergence of the series defining $\mathcal{L}_Y$, the statement about their supports, the nonvanishing of $\mathbb{E}_\pi e^{-\sum_X U_X}$, the branch, and (12.14). The repetitions are those of Theorem 12.6. Since each connected union is fixed before the coefficients are summed, these formulas define a family, although its sum could also be distributed over the supports in other ways.

Step 3: holomorphy. At a fixed number of boxes the activities are finite sums of products of entire functions of the U_X in $\mathscr{A}$, hence holomorphic. We choose η_0 so small that the bounds of Step 2 still hold after every activity is multiplied by 2, uniformly for $\|\boldsymbol{U}\|_{\mathscr{S}_8} \le \eta_0$. Then the sum of all terms of the logarithm with $k \ge n$ polymers has $\mathscr{S}_2$ norm at most $C\eta_0 2^{-n}$. Each sum with fixed k is finite and holomorphic, and by this bound the series converges uniformly on the ball. To see that the limit is holomorphic, we fix around any point of the open ball a closed ball of radius r inside it. By Cauchy's formula, the homogeneous Taylor polynomial of degree j of each partial sum is bounded by $C\eta_0(\|v\|/r)^j$, uniformly in the increment v. The uniform convergence implies that these polynomials converge uniformly on smaller balls, and the limits are continuous homogeneous polynomials, since polarization of the partial sums passes to the limit. For $\|v\| < r$, their series converges geometrically, and by the one-variable Cauchy formula on the line through v, it equals the limit. This proves Fréchet holomorphy, and holomorphy after composition with holomorphic parameter families.

Step 4: the derivative. Let $a = \|\boldsymbol{A}\|_{\mathscr{S}_8} > 0$ and $r_v = \eta_0/(4a)$. For $|v| \le r_v$, the output has norm at most $C\eta_0$, since the input $\boldsymbol{U} + v\boldsymbol{A}$ has norm at most $3\eta_0/4$. By Cauchy's formula, we have

$$D\boldsymbol{\mathcal{L}}(\boldsymbol{U})[\boldsymbol{A}] = \frac{1}{2\pi i}\int_{|v|=r_v} \frac{\boldsymbol{\mathcal{L}}(\boldsymbol{U}+v\boldsymbol{A})}{v^2}\,dv, \qquad \|D\boldsymbol{\mathcal{L}}(\boldsymbol{U})[\boldsymbol{A}]\|_{\mathscr{S}_2} \le 4Ca.$$

For $a = 0$ there is nothing to prove. Renaming C, we obtain (12.15), and integrating along the segment proves the statement about differences. The norm sums all supports containing the chosen box after the circle integral. An inserted interaction need not contain that box, since the derivative acts at every position of the inserted interaction within a connected cluster.

Step 5: real data. At a fixed number of boxes the coefficient sums bound $\sum_X U_X$ and $\sum_X A_X$ uniformly on Ω. The integrable constant envelope $|\sum_X A_X| e^{|\sum_X U_X| + v_0|\sum_X A_X|}$ justifies differentiating $-\log \mathbb{E}_\pi e^{-\sum_X (U_X + vA_X)}$ under the integral at $v = 0$, and the sum of the components of $D\boldsymbol{\mathcal{L}}(\boldsymbol{U})[\boldsymbol{A}]$ is this derivative, which is the displayed formula. The envelope depends on the volume; we use it only to identify this sum, not in the estimates for the family. □

The derivative with respect to an inserted interaction gains a second power of a small parameter when the first-order term of the insertion has mean zero under π on every support. This is the mechanism behind Lemma 12.16.

Corollary 12.8. *Assume the setting of Theorem 12.7. For $s \in [0,1]$ let $\zeta \mapsto \boldsymbol{U}_s(\zeta)$ and $\zeta \mapsto \boldsymbol{A}(\zeta)$ be holomorphic maps into $\mathscr{S}_8$ on a neighborhood of the closed disc $|\zeta| \le R$ such that, uniformly in s,*

$$\|\boldsymbol{U}_s(\zeta)\|_{\mathscr{S}_8} \le \eta_0/2, \qquad \|\boldsymbol{A}(\zeta)\|_{\mathscr{S}_8} \le a|\zeta|, \qquad \boldsymbol{U}_s(0) = \boldsymbol{A}(0) = 0.$$

Assume that $\mathbb{E}_\pi A'_X(0) = 0$ *for every support* X*, as an identity of coefficients in* θ*, where* $A' = dA/d\zeta$*. Set* $\boldsymbol{F}_s(\zeta) = D\boldsymbol{\mathcal{L}}(\boldsymbol{U}_s(\zeta))[\boldsymbol{A}(\zeta)]$*. Then* $F_{s,Y}(0) = F'_{s,Y}(0) = 0$ *for every* s *and every support* Y*, coefficient by coefficient, and*

$$\|\boldsymbol{F}_s(t)\|_{\mathscr{S}_2} \le \frac{2Ca\,t^2}{R}, \qquad 0 < t \le R/2, \tag{12.16}$$

where C *is the constant of Theorem 12.7. The same conclusion holds if the uniform bounds are assumed only on an open disc.*

Proof. By (12.15), we have $\|\boldsymbol{F}_s(\zeta)\|_{\mathscr{S}_2} \le Ca|\zeta|$. At zero interaction the derivative of the family is $\mathbb{E}_\pi A_Y$, support by support: in the first expansion $e^{-U_X} - 1 = -U_X + O(U_X^2)$, and in the second only the terms with a single polymer are linear. Terms with two interaction factors are of degree at least two in ζ. Since $\boldsymbol{U}_s(0) = \boldsymbol{A}(0) = 0$, the coefficient of ζ in $\boldsymbol{F}_s(\zeta)$ is $(\mathbb{E}_\pi A'_Y(0))_Y = 0$. Thus $F_{s,Y}(0) = F'_{s,Y}(0) = 0$ for every support and every coefficient in θ; no cancellation between different supports is used.

The family $\boldsymbol{B}_s(\zeta) = \boldsymbol{F}_s(\zeta)/\zeta$ extends holomorphically to $\zeta = 0$, vanishes there, and has norm at most Ca. By Cauchy's formula for Banach-valued functions, we have $\|[\zeta^k]\boldsymbol{B}_s\|_{\mathscr{S}_2} \le CaR^{-k}$. For $0 < t \le R/2$, summing the series, we find

$$\|\boldsymbol{F}_s(t)\|_{\mathscr{S}_2} \le \frac{Ca\,t^2}{R(1 - t/R)} \le \frac{2Ca\,t^2}{R}.$$

If the bounds hold only on an open disc, we apply this on smaller circles and pass to the limit. No regularity in s is used. □

12.3 Connected supports and the effective interaction of one averaging step

We now return to the fluctuation interaction $U(c, w)$ of one averaging step, defined in (12.5). The coarse-field coordinates and the normalized fluctuation variables have locations on the lattice, and the coefficient bounds of Lemmas 12.2 and 12.3 decay exponentially in the length of a network joining these locations. In the next lemma, we convert this decay into a decomposition of U into pieces supported on connected sets of boxes, with the summability over supports required by the norm (12.1), as needed to apply Theorems 12.6 and 12.7. A large box scale $\mathscr{L}$ makes the factor $e^{8|X|}$, exponential in the number of boxes met by a network, compatible with the available spatial decay.

Lemma 12.9. *Assume the setting of Lemma 12.3. Choose the box scale* $\mathscr{L}$ *sufficiently large, using a fixed part of the spatial decay rate of the classical coefficient bounds, and then reduce the coarse-field radius* u *and the threshold for* gp*, once. Assign every joint Taylor monomial of* U*, in* θ *and* w*, to the set of boxes met by a rectilinear tree joining the locations of all its coarse-field coordinates and normalized fluctuation variables, chosen by a fixed deterministic rule described in*

the proof. This yields an exact decomposition $U = \sum_X U_X$ *into functions supported on connected sets of boxes, including sets winding around the torus, with the following properties.*

1. *For degree* r *in* θ *and degree* s *in* w*, the absolute coefficient sums with one index of* w *fixed and with the exponential weight of a connecting network at a fixed rate are bounded by*

$$C^{r+s} g^{s-2} \quad (s \geq 3) \text{ for the classical part}, \qquad C^{r+s} g^{s} \quad (s \geq 1) \text{ for the part from } \mathcal{L}.$$

   The classical part of U*, the first two terms of* (12.5)*, has no terms of degree zero, one, or two in* w*, and every term of* U *has degree at least one in* w*.*

2. *The support assigned to a monomial satisfies* $|X| \leq C(r + s + 1 + \mathtt{t}/\mathscr{L})$*, and*

$$\|\{U_X\}\|_{u,8;p} \leq \sigma = C\Big\{\frac{gp^3}{1 - Cgp} + gp\Big\}.$$

3. *If* $\sigma \leq \sigma_0$*, then, with* $F_X = e^{-U_X} - 1$*, the fluctuation partition function is the hard-core polymer gas*

$$\Xi = \sum_{\{P_1,\dots,P_k\} \text{ disjoint}} \prod_{i=1}^{k} \mathfrak{z}(P_i), \qquad \mathfrak{z}(P) = \sum_{\mathcal{F}} \mathbb{E}_{\pi_p} \prod_{X \in \mathcal{F}} F_X, \qquad \|\{\mathfrak{z}(P)\}\|_{u,5} \leq 2\sigma,$$

   where $\mathcal{F}$ *ranges over the families of distinct supports whose intersection graph is connected and whose union is* P*. The empty family contributes one, and each activity depends only on coarse-field coordinates located in* P*.*

4. *Statements (1)–(3) remain valid, with* g *replaced by* $|\zeta|$ *in all bounds and in* (12.4)*, for the interaction*

$$U(c, w; \zeta) = \frac{F(c, z^* + \zeta S w) - F(c, z^*)}{\zeta^2} - \frac{1}{2}|w|^2 + \mathcal{L}_c(z^* + \zeta S w) - \mathcal{L}_0(c)$$

   with a complex parameter ζ*, extended holomorphically across* $\zeta = 0$*; at* $\zeta = g > 0$ *this is* $U(c, w)$*.*

The constants have the dependence described in item (2) of Definition 4.14; they are independent of d_0*, of* m*, of* g, p *satisfying* (12.4)*, and of the number of supports. The only independence used is that of the fluctuation variables in distinct boxes under the product measure* π_p*. The exact formula* (12.6) *concerns real* g *only.*

Proof. We write $c = e^{\theta}$ and proceed in four steps.

Step 1: expansion in the fluctuation variables. At fixed θ, we expand in the increment gSw and obtain the normally convergent series

$$\frac{F(c, z^* + gSw) - F(c, z^*)}{g^2} - \frac{1}{2}|w|^2 = \sum_{s \geq 3} \frac{g^{s-2}}{s!} D_z^s F(c, z^*)[Sw, \dots, Sw], \tag{12.17}$$

$$\mathcal{L}_c(z^* + gSw) - \mathcal{L}_c(z^*) = \sum_{s \geq 1} \frac{g^s}{s!} D_z^s \mathcal{L}_c(z^*)[Sw, \dots, Sw]. \tag{12.18}$$

The classical terms of degree one and two vanish by stationarity and by $S^{\mathsf{T}} K S = I$. Both series vanish at $w = 0$ for every coarse field.

Step 2: coefficient bounds. We expand the coefficients further in θ and apply Lemmas 12.2 and 12.3 and the contraction rule of Lemma 5.3. If an index of w is fixed, the column of the factor S containing it carries the fixed index to a tensor of F or of $\mathcal{L}$ with one index fixed, and all other insertions use rows. The networks of the factors meet at the contracted indices, so their union joins all external locations. The joint coefficient of degree (r, s) has the generating factor $1/(r!\, s!)$, and passing from total derivatives gives only the factor $(r+s)!/(r!\, s!) \le 2^{r+s}$. Summing the composed majorant series proves the two coefficient bounds in (1), at one fixed rate $a > 0$. The choice of the fixed occurrence of w, including multiplicities and colors, contributes at most another factor exponential in $r + s$, which is absorbed into C.

Step 3: the supports. Let $\mathscr{L} \ge 1$ be an integer. If a periodic direction of the coarse lattice has length at least $2\mathscr{L}$, we divide it into integer intervals of lengths between $\mathscr{L}$ and $2\mathscr{L}$; this is possible by distributing the remainder among $\lfloor m/\mathscr{L} \rfloor$ intervals. A shorter direction forms one interval. The resulting boxes have bounded adjacency degree and contain at most $D_{M,\mathscr{L}}$ scalar labels.

Consider a monomial with $k = r + s \ge 1$ coordinate locations. We order all metric spanning trees of these locations and all shortest periodic coordinate paths by one fixed rule, choose a shortest metric spanning tree, and replace each of its edges by the chosen rectilinear path. The resulting network has length at most a fixed multiple of the length t of a shortest connected network joining the locations. Indeed, traversing every edge of such a network twice gives a closed walk of length $2\mathsf{t}$ through all locations. Shortcutting this walk yields a metric spanning tree of length at most $2\mathsf{t}$, and replacing an edge by a coordinate path changes its length by at most a factor depending on the dimension. We add the intersection points of the paths as vertices, delete cycles by the fixed rule, and prune branches that contain no location. The result is a rectilinear tree of no greater length. Let X be the set of boxes met by this tree, including the boxes of the locations. It is connected, since the fixed rule assigns boundary points to adjacent boxes. A coordinate segment of length l meets at most $C(1 + l/\mathscr{L})$ boxes. The unpruned paths consist of at most $3(k-1)$ segments, and the deletions do not add boxes. Applying the bound to these segments, we obtain

$$|X| \le C_0\big(k + 1 + \mathsf{t}/\mathscr{L}\big). \tag{12.19}$$

Supports winding around the torus occur, since the paths are periodic. If $x \in X$ and e is the location of the fixed index of w, then

$$\operatorname{dist}(x, e) \le C_1\big\{\mathsf{t} + \mathscr{L}(k+1)\big\},$$

where the distance to a box is measured to its center; another choice changes the constant by the diameter of a box.

We choose $\mathscr{L}$ so large that $8C_0/\mathscr{L} < a/4$, and then $b_1 > 0$ so small that $b_1 C_1 < a/4$. Combining the two geometric bounds, we obtain the support-weight estimate

$$\mathbf{1}_{x\in X} e^{8|X|} \le C D_{\mathscr{L}}^{k} e^{a\mathrm{t}/2} e^{-b_1 \operatorname{dist}(x,e)}, \tag{12.20}$$

with $D_{\mathscr{L}} = e^{8C_0 + b_1 C_1 \mathscr{L}}$. Since the indices of w have bounded density and polynomial growth, we have $\sup_x \sum_e e^{-b_1 \operatorname{dist}(x,e)} \le C_{M,\mathscr{L},b_1}$, uniformly in the side of the torus. This includes the case of a short direction, in which the number of labels in that direction is bounded by the box scale.

Step 4: summation and the polymer gas. We group the monomials, with their coefficients, according to their supports X; the sum U_X depends only on coordinates located in X. We apply (12.20) before summing over the location e of the fixed index of w, and then the coefficient bounds of Step 2. This bounds the norm with support weight $e^{8|X|}$ by

$$C \sum_{r\ge 0} (Cu)^r \Big\{ \sum_{s\ge 3} C^s g^{s-2} p^s + \sum_{s\ge 1} (Cgp)^s \Big\},$$

where the constants include the factors $D_{\mathscr{L}}^{r+s}$. We reduce u until $Cu \le 1/2$ and the threshold for gp until $Cgp < 1/2$. Summing the geometric series and enlarging the constant in (12.4), we obtain the bound $C\{gp^3/(1 - Cgp) + gp\}$. The absolute convergence proves both the identity $U = \sum_X U_X$ and the bound in (2). This estimate controls the sum of the separate coefficient suprema required by (12.1).

The bounds of Steps 1–4 use g only through the factors g^{s-2} and g^s of the series (12.17) and (12.18), and the domain condition $\|gSw\| \le C|g|p$. For a complex parameter ζ in place of g, the same series define $U(c, w; \zeta)$, which is holomorphic across $\zeta = 0$ since the classical series starts at ζ^1. The same estimates hold with $|\zeta|$ in place of g. This proves (4) for the bounds in (1) and (2), and the argument for (3) below applies without change.

For (3), we apply Step 1 of the proof of Theorem 12.7 with $\pi = \pi_p$ and with η replaced by σ. We obtain $\|F_X\|_{u,\mathrm{coeff};p} \le 2\|U_X\|_{u,\mathrm{coeff};p}$, the hard-core representation of Ξ with the displayed activities, and the bound $\|\{\mathfrak{z}(P)\}\|_{u,5} \le 2\sigma$. This requires only the independence of the fluctuation variables in disjoint sets of boxes under π_p; the coarse-field variables are parameters. All constants were chosen using M, the fixed domains, and the spatial decay rates, so they have the stated dependence. □

We now combine the exact formula (12.6) with the cluster expansion. We define the effective interaction newly generated by the averaging step as

$$Q_{d_0}^{\mathrm{w}}(c) = \mathcal{L}_0(c) + \tfrac{1}{2} \log\det K(c) - f \log z_p - \log \Xi(c). \tag{12.21}$$

Since $\det S(c) = (\det K(c))^{-1/2}$, Lemma 12.3 can be written as

$$\int_{z^*+gS\Omega_p} e^{-F(c,z)/g^2}\, \mathcal{J}_c(z)\, dz = \mathcal{J}_I(0) g^{3f} \exp\Big(-\frac{E_{d_0+1}(c)}{g_+^2} - Q^{\mathrm{w}}_{d_0}(c)\Big).$$

Proposition 12.10. *Assume the setting of Lemma 12.9, with $\sigma \le \sigma_0$. There exist $u_* \in (0, u_0]$ and $b > 0$, with the dependence described in item (2) of Definition 4.14, such that the following holds on $\|\theta\|_\infty \le u_*$, uniformly in d_0, m, and in g, p satisfying* (12.4).

1. *For every $s \ge 1$,*
$$\|D^s_\theta Q^{\mathrm{w}}_{d_0}\|_{b,\mathrm{root}} \le C^s s!.$$
The value of $Q^{\mathrm{w}}_{d_0}$ is not bounded uniformly in the volume.
2. *For real c and $w \in \Omega_p$, the Hessian of $w \mapsto |w|^2/2 + U(c,w)$ is $I + E$, where E is bounded by $C(gp + g^2)$ in a fixed exponentially weighted row-plus-column norm.*

Proof. By Lemma 12.9(3), the activities satisfy (12.11), and the labels have bounded density. Thus Theorem 12.6(4) implies that $\|D^s_\theta \log \Xi\|_{b,\mathrm{root}} \le C^s s! \sigma$ on $\|\theta\|_\infty \le u/2$.

We next treat the determinant. Let $A(\theta) = (K^0)^{-1}(K(c) - K^0)$, and reduce the coefficient radius such that the row and column series of A have norm less than $1/2$. On this radius, we have

$$\tfrac{1}{2} \log\det K(c) = \tfrac{1}{2} \log\det K^0 + \tfrac{1}{2} \sum_{j \ge 1} \frac{(-1)^{j+1}}{j} \operatorname{tr} A(\theta)^j, \tag{12.22}$$

where the logarithm is the branch that is real for real c. To verify this, we replace A by hA. Differentiating the determinant along its columns, we find $(d/dh) \log\det(I + hA) = \operatorname{tr}[(I + hA)^{-1} A]$, and integrating the Neumann series from $h = 0$ to $h = 1$ gives the formula. A coefficient of positive degree in θ differentiates at least one factor A. We fix the coarse-field index on such a factor and sum over its two matrix indices using Lemma 12.2(4) and the rows and columns of $(K^0)^{-1}$; every other factor is contracted by rows and columns. The choice of the distinguished factor cancels the factor $1/j$. If $a(x) = Cx/(1 - Cx)$ is a majorant series for A, the trace coefficients with one coarse-field index fixed are bounded by those of $Ca'(x)/(1 - a(x))$ on a fixed smaller disc, and these coefficients are at most C_1^{k+1}. Translating to a background point in a fixed smaller product, we obtain factorial bounds for all derivatives of positive order. The network of a trace may contain a cycle. Deleting an edge leaves a network joining the same locations, so the weight of a shortest connected network is still available. The constant $\log\det K^0$ is extensive and is not estimated. The term $\mathcal{L}_0$ is the composition of $\mathcal{L}$ with the minimizer, and by Lemma 12.2 it has the same bounds with one index fixed. The constant $f \log z_p$ has no derivatives of positive order. Combining these bounds on $\|\theta\|_\infty \le u_*$, where u_* is the smaller of $u/2$ and the radius chosen for A, we obtain (1). Here the determinant term enters (12.21) with the positive sign, and the constant $-f \log z_p$ is kept.

For (2), we fix a real coarse field. Up to a term independent of w, the potential is $g^{-2}F(c, z^* + gSw) + \mathcal{L}_c(z^* + gSw)$. Since S depends only on c, its Hessian minus the identity is

$$S^{\mathsf{T}}\{D_z^2 F(c, z^* + gSw) - K(c)\}S + g^2 S^{\mathsf{T}} D_z^2 \mathcal{L}_c(z^* + gSw)S.$$

The third derivative of F and the second derivative of $\mathcal{L}_c$ have fixed weighted bounds on the product, and $\|gSw\| \le Cgp$ by the row bound for S. We conclude that the displayed matrix is bounded by $C(gp + g^2)$ in a fixed weighted row-plus-column norm. □

The last result for one averaging step concerns symmetries.

Proposition 12.11. *In the setting of Lemma 12.3, there exist fixed products of coarse-field logarithms θ, free logarithmic coordinates z, and gauge parameters κ, independent of d_0 and m and contained in the domains of that lemma, on which both the original and the transformed reconstructed configurations lie in the domains of Lemma 6.3 and Theorems 10.10 and 10.11. We call them the common symmetry domains; they are constructed in the proof. Let $c = e^{\theta}$ be real, with θ in the common symmetry domain, and let τ be one of the following real operations on coarse fields: a translation by a vector of the coarse lattice; the spatial inversion, under which reversed links carry inverted matrices; a constant color conjugation; or a coarse gauge transformation $c \mapsto c^k$ with $\mathrm{SU}(2)$-valued $k = e^{\kappa}$ and κ real in the corresponding fixed parameter product. Then there exists an orthogonal map $O(c)$ of the free logarithmic coordinates, which permutes the three-component blocks, acts on each block by an orthogonal matrix, and does not depend on z, such that*

$$z^*(\tau c) = O(c)z^*(c), \qquad K(\tau c) = O(c)K(c)O(c)^{\mathsf{T}}, \qquad S(\tau c) = O(c)S(c)O(c)^{\mathsf{T}}.$$

The induced map $w \mapsto O(c)w$ of the normalized fluctuation variables preserves Ω_p, $|w|^2$, Lebesgue measure, and π_p, and the Haar disintegration density and $\det K$ *are preserved. Consequently the integral* (12.6) *and $Q^{\mathrm{w}}_{d_0}$ take the same values at c and at τc. The resulting identities between analytic functions of the coarse field extend to the corresponding common complex domains.*

Proof. At depth zero the Wilson action is invariant under periodic gauge transformations, translations, the spatial inversion, and constant color conjugations: the plaquette variables are conjugated, relabeled, or inverted, and $q(U^{-1}) = q(U)$. The transformations of the constraint coordinates in Lemma 6.3 intertwine the reconstruction, and on the common symmetry domains they send z to $O(c)z$, where $O(c)$ is as in the statement.

We first prove the invariance of E_d as an identity of analytic germs at the identity, by induction on the depth d. If the germ at depth d is invariant, then the intertwining of the constraint coordinates carries the constrained objective to the transformed objective near $(c, z) = (I, 0)$. The transformed stationary point is $O(c)z^*(c)$, which lies in the neighborhood where the minimizer of Theorem 10.10 is unique. The two stationary values agree, and multiplying by M gives the invariance of the germ at depth $d + 1$. Only

a nonempty neighborhood is used in this induction, and no uniform radius is inferred from it.

To obtain a uniform domain, we use the common holomorphic domains of Lemma 6.3 and Theorems 10.10 and 10.11. We choose fixed smaller products in (θ, z, κ) on which both the original and the transformed reconstructed fields lie in these domains; these are the common symmetry domains of the statement. The finite-word bounds for the reconstruction and the local gauge word $e^{\kappa_x} e^{\theta_{xy}} e^{-\kappa_y}$ allow this choice independently of the depth. The differences between the two action values, and between the two constrained objectives, are holomorphic there and have zero germ by the induction. We claim that they vanish on the whole smaller product. Indeed, we restrict such a difference to the complex line $h \mapsto h(\theta, z, \kappa)$ through a fixed point of the product. The fixed margins allow a disc $|h| < 1 + \eta$ for some $\eta > 0$. Its Taylor series vanishes near $h = 0$, and continuation through overlapping discs along $[0, 1]$ shows that it vanishes at $h = 1$. Translations and the spatial inversion are treated in the same way without κ, and a constant color conjugation preserves the radii and is kept fixed in the induction. Thus the identity of the objectives holds on the fixed products, and the uniqueness of the minimizer there implies that $z^*(\tau c) = O(c) z^*(c)$ on the common domain, without a loss of radius depending on the depth.

Differentiating the objective identity twice, we obtain $K(\tau c) = OKO^{\mathsf{T}}$, and the integral formula for the inverse square root implies that $S(\tau c) = OSO^{\mathsf{T}}$. Then $w = S^{-1}(z - z^*)/g$ transforms as $w \mapsto O(c) w$, which preserves the product of balls, $|w|^2$, Lebesgue measure, and the product reference measure π_p. By the exact Haar disintegration of Lemma 6.3 and $\det(OKO^{\mathsf{T}}) = \det K$, every remaining factor of the integral is preserved. This proves the identities for the integral and for $Q^{\mathrm{w}}_{d_0}$ at real coarse fields, without requiring the individual pieces U_X to be invariant. The statements about balls, Lebesgue measure, and π_p concern the real change of variables. The identities for z^*, K, S, and $Q^{\mathrm{w}}_{d_0}$ hold between functions that are holomorphic on the common complex symmetry domains, so they extend there by the identity theorem. □

The gauge parameter κ in Proposition 12.11 ranges over a fixed small product, and Theorem 12.13(6) imposes this restriction at every scale. Transitions between coarse fields in the comparison domains of Section 13 that are related by an arbitrary periodic gauge transformation are given by Theorem 13.4(3).

12.4 Several averaging steps

We now compose $s \le J$ averaging steps. At each scale, from the coarsest to the finest, we perform the change of variables of Lemma 12.3 with the coarse field of that scale, which depends on the fluctuations already introduced. The following example shows the structure of the resulting change of variables in the simplest case.

Example 12.12. Let $s = 2$, and let $c_2 = e^\theta$ be the coarse field. The coordinates of the coarser intermediate scale are $z_1 = z_1^*(c_2) + g_1 S_1(c_2) w_1$, and they determine the field $c_1 = b_1(c_2, z_1)$, so that $c_1 = c_1(w_1)$. The coordinates of the finer scale are

$$z_0 = z_0^*(c_1(w_1)) + g_0 S_0(c_1(w_1)) w_0.$$

In the order (w_1, w_0) the derivative of $w \mapsto (z_1, z_0)$ is block lower triangular,

$$D_w z = \begin{pmatrix} g_1 S_1(c_2) & 0 \\ D_{w_1} z_0 & g_0 S_0(c_1(w_1)) \end{pmatrix}, \qquad D_{w_1} z_0 = D z_0^*(c_1) D_{w_1} c_1 + g_0 D S_0(c_1)[D_{w_1} c_1] w_0.$$

We write D_i for the number of real coordinates of z_i, $i = 0, 1$. The off-diagonal block does not change the determinant, $\det D_w z = g_1^{D_1} \det S_1(c_2)\, g_0^{D_0} \det S_0(c_1(w_1))$, but it enters all later derivatives. The determinant of the finer scale depends on w_1. Let $\widehat{c}_1 = c_1(0) = b_1(c_2, z_1^*(c_2))$ be the zero-fluctuation reconstruction of c_1. We split

$$\log\det K_0(c_1(w_1)) = \log\det K_0(\widehat{c}_1) + \big[\log\det K_0(c_1(w_1)) - \log\det K_0(\widehat{c}_1)\big].$$

The first term depends only on c_2 and becomes part of the function $\mathsf{B}_{\mathrm{ref}}$ of Theorem 12.13. The second depends on w_1 and is part of the fluctuation interaction $H_{d_0,2}$. The negative logarithmic Haar density at the minimizer of the finer scale is split in the same way. Replacing $c_1(w_1)$ by $\widehat{c}_1$ in these terms would not give an exact formula. Under the product reference measure, w_1 and w_0 are independent, but the interacting density need not factor. Its classical and Haar-density remainders can involve both variables, while the determinant and minimizer-density corrections displayed here depend only on w_1 and change its marginal weight.

In the theorem below, the field c_i lives on the lattice of side mM^{s-i}, $0 \le i \le s$, so that c_0 is the finest field and $c_s = e^\theta$ is the coarse field. The i-th averaging step, from side mM^{s-i} to side mM^{s-i-1}, has input action E_{d_0+i}; we write $b_i(c, z)$ for its reconstruction, $\mathcal{J}_i(c, z)$ for its Haar disintegration density, and $z_i^*(c)$ for its conditional constrained minimizer, as in Lemma 12.3. The coupling of the coarsest integrated scale is t, and the couplings decrease by the factor $M^{-1/2}$ toward finer scales.

Theorem 12.13. *Fix an odd integer $M \ge 3$ and an integer $J \ge 1$. There exist $u_J, v_J, b_J, c_J, C_J, \sigma_0 > 0$, uniform in the depth $d_0 \ge 0$ and in the side $m \ge 4$, such that the following holds for every integer $1 \le s \le J$. For $0 \le i < s$ put*

$$f_i = 2(M^3 - 1)(mM^{s-i-1})^3, \qquad D_i = 3 f_i, \qquad g_i = t M^{(i-s+1)/2},$$

and $g_s^2 = M g_{s-1}^2$. Given $c_s = e^\theta$, define successively, for decreasing i,

$$F_i(c, z) = E_{d_0+i}(b_i(c, z)), \qquad K_i(c) = D_z^2 F_i(c, z_i^*(c)), \qquad S_i = K_i^{-1/2},$$
$$z_i = z_i^*(c_{i+1}) + g_i S_i(c_{i+1}) w_i, \qquad c_i = b_i(c_{i+1}, z_i).$$

Let $\Omega_p^{\mathrm{all}} = \prod_{i<s} \prod_e \{w_{i,e} \in \mathbb{R}^3 : |w_{i,e}| \le p\}$, *and let* π_p^{all} *be the product reference measure on it with density proportional to* $e^{-\sum_i |w_i|^2/2}$. *Assume* $p \ge p_0$, $\|\theta\|_\infty < u_J$, *and*

$$C_J t p < \tfrac{1}{2}, \qquad \sigma_J = C_J \frac{t p^3 + t p}{1 - C_J t p} \le \sigma_0. \tag{12.23}$$

Then the following statements hold.

1. *For every* $w \in \Omega_p^{\mathrm{all}}$ *the reconstruction lies, with fixed margins, in every classical and Haar disintegration chart, and the recursion is a jointly holomorphic map of* θ *and of the variables* $q_i = g_i w_i$ *on a common larger product.*
2. *In the order from coarser to finer scales,*

$$\det D_w z = \prod_{i=0}^{s-1} g_i^{D_i} \det S_i(c_{i+1}).$$

3. *Put*

$$U_i^{\mathrm{cl}} = \frac{F_i(c_{i+1}, z_i) - F_i(c_{i+1}, z_i^*(c_{i+1}))}{g_i^2} - \frac{1}{2}|w_i|^2.$$

Then, exactly,

$$\frac{E_{d_0}(c_0)}{g_0^2} = \frac{E_{d_0+s}(c_s)}{g_s^2} + \frac{1}{2}\sum_i |w_i|^2 + \sum_i U_i^{\mathrm{cl}}.$$

4. *Put* $\mathcal{L}_i(c, z) = -\log(\mathcal{J}_i(c, z)/\mathcal{J}_i(I, 0))$ *and*

$$\mathsf{B}_i(c) = \mathcal{L}_i(c, z_i^*(c)) + \tfrac{1}{2}\log\det K_i(c).$$

Let $\widehat{c}_s = c_s$ *and* $\widehat{c}_i = b_i(\widehat{c}_{i+1}, z_i^*(\widehat{c}_{i+1}))$ *be the zero-fluctuation reconstruction, and set* $\mathsf{B}_{\mathrm{ref}}(c_s) = \sum_i \mathsf{B}_i(\widehat{c}_{i+1})$. *Define the fluctuation interaction*

$$H_{d_0,s}(\theta, w) = \sum_i \Big[U_i^{\mathrm{cl}} + \mathcal{L}_i(c_{i+1}, z_i) - \mathcal{L}_i(c_{i+1}, z_i^*(c_{i+1})) + \mathsf{B}_i(c_{i+1}) - \mathsf{B}_i(\widehat{c}_{i+1}) \Big]. \tag{12.24}$$

If $\mathscr{D}_s(c_s)$ *is the image of* Ω_p^{all} *under* $w \mapsto (z_i)_{i<s}$, *then*

$$\begin{aligned} &\int_{\mathscr{D}_s(c_s)} e^{-E_{d_0}(c_0)/g_0^2} \prod_i \mathcal{J}_i(c_{i+1}, z_i)\, dz_i \\ &\qquad = \Big(\prod_i \mathcal{J}_i(I, 0) g_i^{D_i} \Big) e^{-E_{d_0+s}(c_s)/g_s^2 - \mathsf{B}_{\mathrm{ref}}(c_s)} z_p^{\sum_i f_i}\, \Xi_{d_0,s}(c_s), \end{aligned} \tag{12.25}$$

where $\Xi_{d_0,s} = \mathbb{E}_{\pi_p^{\mathrm{all}}} e^{-H_{d_0,s}}$.

5. *There is an exact decomposition* $H_{d_0,s} = \sum_X H_X$ *into functions supported on connected sets of boxes with* $\|\{H_X\}\|_{u_J', 8; p} \le \sigma_J$, *for a fixed coarse-field radius* $u_J' \in (0, u_J]$ *that also lies*

inside the domain of the coefficient series of $\mathsf{B}_{\mathrm{ref}}$ *described in the proof. Both expansions of Theorem 12.7 apply;* $\Xi_{d_0,s} \neq 0$ *on* $\|\theta\|_\infty < u'_J$*, and the effective interaction*

$$Q^{\mathrm{w}}_{d_0,s} = \mathsf{B}_{\mathrm{ref}} - \Big(\sum_i f_i\Big) \log z_p - \log \Xi_{d_0,s}$$

satisfies, on $\|\theta\|_\infty \le u'_J/2$, $\|D^k_\theta Q^{\mathrm{w}}_{d_0,s}\|_{b_J,\mathrm{root}} \le C_J^{k+1} k!$ *for every* $k \ge 1$*. The function* $\mathsf{B}_{\mathrm{ref}}$ *of the coarse field is kept in full.*

6. *Let* c_s *be real, and let* τ *be one of the operations of Proposition 12.11, applied to* c_s*. Assume that, for every* $w \in \Omega^{\mathrm{all}}_p$ *and every* i*, the field* c_{i+1} *and the operation induced by* τ *on it satisfy the hypotheses of Proposition 12.11 for the averaging step* i*, that is, they lie in the common symmetry domains of that step. Then the induced changes of variables* $w'_i = O_i(c_{i+1}) w_i$ *are triangular, orthogonal at each scale for fixed coarser variables, preserve every ball, and have absolute joint Jacobian one; their dependence on the coarser fluctuations is retained. They transport every factor of* (12.25)*.*

The constants depend only on M*,* J*, the larger chart margins, the box geometry, and the spatial decay rates; they do not depend on* d_0*,* m*, or on* t*,* p *satisfying* (12.23)*.*

In the proof, we first construct one product of coordinates on which all compositions are controlled, before any integration. We then compute the Jacobian and the integral exactly, and finally we estimate the fluctuation interaction and apply the expansions.

Proof. For each $0 \le i < s$ the averaging step i has output side $m_i = mM^{s-i-1} \ge 4$, input side Mm_i, and depth $d_0 + i \ge 0$. Thus Lemmas 12.2 and 12.3 apply to every step, and the construction below checks their domain conditions at every argument reached by the reconstruction.

Step 1: a common product. In logarithmic coordinates, we define the map of one step

$$T_i(x, q) = \mathrm{Log}\, b_i\big(e^x, z_i^*(e^x) + S_i(e^x) q\big).$$

Its value at $(0, 0)$ is zero. By Lemmas 12.2 and 12.3, its output rows and all its input columns are bounded by one convergent majorant series, which we enlarge to $P(h) = Ah/(1 - Ah)$; these bounds are uniform in $d_0 + i$ and in m_i. We choose a smaller domain for each step, with a fixed positive margin R_1 to every classical and Haar disintegration boundary and to every denominator in these series.

For a scalar variable h, we set $x_s(h) = h$ and successively $x_i(h) = P(x_{i+1}(h) + h)$. There are only $s \le J$ compositions, and their denominators equal one at $h = 0$. We choose $r_J > 0$ such that, for every $s \le J$ and $0 \le i < s$, all denominators are at least $1/2$ and $x_{i+1}(h) + h < R_1/4$ for $0 \le h \le 2r_J$. These are finitely many continuous inequalities that hold strictly at $h = 0$, so such an r_J exists; it depends only on the data listed in the statement. By coefficientwise composition and the contraction rule of Lemma 5.3, these

scalar series majorize the recursion with independent variables q_i. At a fixed input index, we use the column along its path to the output and the rows on the other branches, and the scalar compositions enumerate the partitions of the other inputs with their Taylor factorials. Thus all joint absolute coefficients are controlled, not only the values of the maps at real arguments.

Step 2: units and volumes. The output mesh of step i is M^{i+1-s} in units of the coarse lattice spacing. A length l in the units of step i therefore becomes $M^{i+1-s}l$, and the exponential bound of that step yields a bound at a fixed smaller rate in coarse units. The meshes of all inputs and outputs belong to the finite set $\{M^{-J}, \dots, 1\}$. The column factors are the ratios of the corresponding volumes, as in Lemma 5.3. Passing between counting sums and volume-normalized sums introduces at most a factor M^{3J} for each participating index, hence at most $M^{3J(k+1)}$ for a coefficient of order k, which is absorbed by a fixed smaller joint radius. The number of indices of the variables q_i of the finitely many scales per unit cube is at most $C_M s M^{3s}$. In particular, no factor M^{d_0} and no factor of the total volume occurs. We keep the composed networks until a single final replacement by trees on the external indices, which gives one fixed rate $a_J > 0$.

Step 3: the domain of the fluctuation variables. We choose u_J and v_J below r_J and below the last radius reduction. For $\|\theta\|_\infty < u_J$ and $\max_i \|q_i\|_\infty \le tp$, we enlarge C_J so that (12.23) implies $tp < v_J/2$. By Step 1, all charts then have fixed margins for every $w \in \Omega_p^{\mathrm{all}}$, and the recursion is jointly holomorphic in (θ, q) on the larger product. Taking maxima over $1 \le s \le J$ fixes all these choices at once. This proves (1).

Step 4: the Jacobian. By Lemma 6.3, the scale i has $2(M^3-1)(mM^{s-i-1})^3 = f_i$ independent $\mathrm{SU}(2)$ variables, hence $D_i = 3f_i$ real coordinates. We order the scales as $s-1, s-2, \dots, 0$. The field c_{i+1} depends only on the w_l with $l > i$, so

$$D_{w_i} z_i = g_i S_i(c_{i+1}), \qquad D_{w_l} z_i = 0 \quad (l < i).$$

For $l > i$, the derivative $D_{w_l} z_i$ contains both $Dz_i^*(c_{i+1}) D_{w_l} c_{i+1}$ and $g_i DS_i(c_{i+1})[D_{w_l} c_{i+1}] w_i$, and we keep these entries. The derivative matrix is block triangular, and its determinant is as stated in (2); for real data, each diagonal determinant is positive. The recursion is one-to-one onto its image, since one recovers w_{s-1} from z_{s-1} first and proceeds to finer scales, using the invertibility of each S_i. Hence it is a change of variables on its image.

Step 5: the exact integral. By the nested minimization identity of Theorem 10.10 at scale i, and since $g_{i+1}^2 = M g_i^2$ (including $g_s = \sqrt{M}t$), we have

$$\frac{E_{d_0+i}(c_i)}{g_i^2} - \frac{E_{d_0+i+1}(c_{i+1})}{g_{i+1}^2} = \frac{F_i(c_{i+1}, z_i) - F_i(c_{i+1}, z_i^*)}{g_i^2} = \frac{1}{2}|w_i|^2 + U_i^{\mathrm{cl}}.$$

Summing over i proves (3). The product of the Haar disintegration densities and of the Jacobian is

$$\prod_i \mathcal{J}_i(I,0) g_i^{D_i} \exp\Big\{ -\sum_i \Big[\mathcal{L}_i(c_{i+1}, z_i) + \tfrac{1}{2} \log\det K_i(c_{i+1}) \Big] \Big\}.$$

For each i the exponent splits exactly as

$$\begin{aligned} \mathcal{L}_i(c_{i+1}, z_i) + \tfrac{1}{2}\log\det K_i(c_{i+1}) = {} & \mathsf{B}_i(\widehat{c}_{i+1}) + \mathcal{L}_i(c_{i+1}, z_i) - \mathcal{L}_i(c_{i+1}, z_i^*(c_{i+1})) \\ & + \mathsf{B}_i(c_{i+1}) - \mathsf{B}_i(\widehat{c}_{i+1}). \end{aligned}$$

We add the terms U_i^{cl} from (3). The first terms on the right sum to $\mathsf{B}_{\mathrm{ref}}$, and all other terms together form $H_{d_0,s}$. Integrating the Gaussian factor over each ball contributes $z_p^{\sum_i f_i}$, and the remaining expectation is $\Xi_{d_0,s}$. This proves (12.25); the off-diagonal derivatives of the triangular change of variables contribute no additional determinant. The fluctuations of the determinants and of the Haar densities at the minimizers of the finer scales are exactly the terms $\mathsf{B}_i(c_{i+1}) - \mathsf{B}_i(\widehat{c}_{i+1})$. For complex coarse fields, the formula again means integration over the same real product of the w_i.

Step 6: bounds for the fluctuation interaction. All coefficients of positive degree of B_i have factorial bounds with one index fixed; for the determinant we use the series (12.22), and for the negative logarithmic Haar density Lemma 12.2. We compose these scalar series and the other tensors of one step with the maps bounded in Step 1. Fixing a scalar coefficient at any input and propagating the fixed index along columns, we sum all intermediate indices; the bounded density of the indices of the variables q_i changes only the constants.

By (12.17), a classical term from scale i has at least three w_i inputs and the factor g_i^{k-2} for its degree $k \geq 3$ in w_i. Expanding its coarse field c_{i+1} adds one factor $g_l w_l$ for each input from a coarser scale $l > i$. Since $0 < g_i \leq t$, the sum over these degrees and over all coarse-field degrees is bounded, on a smaller fixed coarse-field radius, by $C_J t p^3/(1 - C_J t p)$. By (12.18), a difference of negative logarithmic Haar densities contains at least one factor $g_i w_i$. A difference $\mathsf{B}_i(c_{i+1}) - \mathsf{B}_i(\widehat{c}_{i+1})$ has no term independent of the fluctuations, since at $q = 0$ both arguments are the zero-fluctuation reconstruction. Each of its terms contains a factor $g_l w_l$ with $l > i$. These two kinds of differences together have the majorant $C_J t p/(1 - C_J t p)$. For each of these terms, a fixed index of q either fixes the outer scalar coefficient or reaches it through the column of an insertion, so these are bounds with one index fixed. Summing over the at most J scales, we obtain the majorant

$$C_J \frac{t p^3 + t p}{1 - C_J t p}, \tag{12.26}$$

and every monomial of $H_{d_0,s}$ contains at least one fluctuation variable.

We apply the construction of supports of Lemma 12.9, with (12.19) and (12.20), in units of the coarse lattice spacing, with a_J in place of a, and with the density of the indices of q stated above in the sum over the location of the fixed index. We choose the box scale such that $8C_0/\mathscr{L} < a_J/4$, then the rate b_1, and then we reduce the joint radius to absorb the resulting factor $D_{\mathscr{L}}^{r+k}$. The geometric series of Step 4 of that proof now has the majorants (12.26) in place of those of one step. After enlarging C_J, we obtain an exact decomposition $H_{d_0,s} = \sum_X H_X$ with norm at most σ_J for the support weight $e^{8|X|}$. All the conditions in (12.23) use this final C_J.

Step 7: the expansions and the effective interaction. The connected-component expansion uses only the coefficient bound, the exact supports, and the independence of the fluctuation variables in disjoint supports, which holds for π_p^{all}. Thus Step 1 of the proof of Theorem 12.7 yields activities with $\|\{\mathfrak{z}(P)\}\|_{u'_J,5} \le 2\sigma_J$, and by Theorem 12.6 we have $\Xi_{d_0,s} \neq 0$ and the bound with support weight $e^{2|Y|}$ for the logarithm. The bounded density of the indices also provides a network of length at most $C_J|Y|$ joining all labels of a support. Then (12.12) implies factorial bounds for the derivatives of $\log \Xi_{d_0,s}$ on $\|\theta\|_\infty \le u'_J/2$, at one fixed rate $b_J > 0$, which we take no larger than a_J.

The zero-fluctuation reconstruction is the map of Step 1 at $q = 0$. We compose its majorant series with the series of the coefficients of positive degree of the B_i. This bounds the coefficients of positive degree of $\mathsf{B}_{\mathrm{ref}}$, with one coarse-field index fixed, at the rate a_J on a fixed product of coarse fields. Translating to background points of a smaller product, we bound every derivative of positive order of $\mathsf{B}_{\mathrm{ref}}$ by $C_J^{k+1}k!$ there; the value of the extensive trace is not bounded. We choose the radius u'_J of (5) such that $\|\theta\|_\infty < u'_J$ lies inside this smaller product, and we restrict the derivative bounds to $\|\theta\|_\infty \le u'_J/2$, where (12.12) applies. Restoring this function and the truncated Gaussian constants in (12.25) proves the formula for $Q^{\mathrm{w}}_{d_0,s}$ and its bounds.

Step 8: symmetries. For each operation τ, the hypothesis of (6) allows us to apply Proposition 12.11 at scale i, which shows that $z_i^*(\tau c_{i+1}) = O_i z_i^*(c_{i+1})$, $K_i(\tau c_{i+1}) = O_i K_i O_i^{\mathsf{T}}$, and $S_i(\tau c_{i+1}) = O_i S_i O_i^{\mathsf{T}}$, with $O_i = O_i(c_{i+1})$ independent of w_i. Proceeding from coarser to finer scales, we obtain $w'_i = O_i(c_{i+1})w_i$, which depends on every coarser fluctuation. This map is a bijection of Ω_p^{all}, since one inverts the block of the coarsest scale first and proceeds to finer scales. Its diagonal Jacobian blocks are the O_i, so its absolute Jacobian is one. Each block preserves its balls and $\sum_i |w_i|^2$, and each Haar density and determinant is transported by the identity of one step. Since at zero fluctuations the transformed zero-fluctuation reconstruction is the zero-fluctuation reconstruction of τc_s, the function $\mathsf{B}_{\mathrm{ref}}$ is transported as well. This proves (6). All choices used only the fixed data and finitely many scales, which proves the stated uniformity; nothing is claimed as $J \to \infty$. □

12.5 Perturbation by an input effective interaction

In the recursion of Section 13, the density at the finer scale contains, besides the effective classical action, an input effective interaction R, produced by the integrations at still finer scales and evaluated at the reconstructed fine field. In this subsection, we fix $s = J$, so that the J averaging steps form one renormalization group step, and we compute the effect of R on the integral of Theorem 12.13. We set $\lambda_i = M^{(i-J+1)/2}$, so that $g_i = t\lambda_i$, and let $\mathscr{C}_{\mathrm{raw}}(\theta, q)$ be the fine field c_0 produced by the recursion of Theorem 12.13 with q_i in place of $g_i w_i$. That is, $z_i = z_i^*(c_{i+1}) + S_i(c_{i+1})q_i$, and the matrices S_i remain in the map. This is the reconstruction before the gauge transformation used in Section 13.

The following heuristic explains why the effect of R, beyond its value R_{ref} on the zero-fluctuation reconstruction, is of second order in the coupling.

Remark 12.14. Suppose that $H = \zeta H_1 + \zeta^2 H_2 + \cdots$ and $A = \zeta A_1 + \zeta^2 A_2 + \cdots$ are interactions depending on a small parameter ζ, and that $\mathbb{E}_\pi A_1 = 0$. Expanding the logarithms to second order gives

$$-\log \mathbb{E}_\pi e^{-H-A} + \log \mathbb{E}_\pi e^{-H} = \zeta^2 \Big(\mathbb{E}_\pi A_2 - \mathrm{Cov}_\pi(H_1, A_1) - \tfrac{1}{2} \mathrm{Var}_\pi(A_1) \Big) + O(\zeta^3).$$

For the inserted interaction $A = R(c_0) - R_{\mathrm{ref}}$, the first-order term A_1 is linear in the normalized fluctuation variables, and its mean under the symmetric reference measure vanishes. The difference of the logarithms is therefore of order ζ^2, and ζ will be set equal to t. Lemma 12.16 proves a uniform version of this statement for the families of Theorem 12.7, using Corollary 12.8 instead of a finite-order expansion.

To apply Corollary 12.8 we vary the size of the fluctuations through a complex parameter ζ, which takes the value t for the integral of Theorem 12.13.

Definition 12.15. For a complex parameter ζ, substitute $q_i = \zeta \lambda_i w_i$ in the recursion, and write $c_i(\zeta)$ and $z_i(\zeta) = z_i^*(c_{i+1}(\zeta)) + \zeta \lambda_i S_i(c_{i+1}(\zeta)) w_i$ for the resulting fields and coordinates, so that $c_i(t) = c_i$ and $c_i(0) = \widehat{c}_i$. The auxiliary fluctuation interaction $H_{d_0,J}(\theta, w; \zeta)$ is obtained from (12.24) by replacing c_{i+1} and z_i by $c_{i+1}(\zeta)$ and $z_i(\zeta)$, and the classical term U_i^{cl} by

$$\frac{F_i\big(c_{i+1}(\zeta), z_i^* + \zeta \lambda_i S_i w_i\big) - F_i\big(c_{i+1}(\zeta), z_i^*\big)}{\zeta^2 \lambda_i^2} - \frac{1}{2}|w_i|^2, \tag{12.27}$$

where z_i^* and S_i are evaluated at $c_{i+1}(\zeta)$. Thus the explicit inverse square of the coupling varies together with the increments, while the classical functions, the balls, and the cutoff are held fixed. Put

$$\Xi_0(\theta; \zeta) = \mathbb{E}_{\pi_p^{\mathrm{all}}} e^{-H_{d_0,J}(\theta, w; \zeta)}, \tag{12.28}$$

where $\log \Xi_0$ is the branch of Theorem 12.7, with $\log \Xi_0(\theta; 0) = 0$.

By stationarity and $S_i^{\mathsf{T}} K_i S_i = I$, the expression (12.27) equals

$$\sum_{k\geq 3} \frac{\zeta^{k-2}\lambda_i^{k-2}}{k!} D_z^k F_i\big(c_{i+1}(\zeta), z_i^*\big)\big[(S_i w_i)^{\otimes k}\big],$$

so it extends holomorphically across $\zeta = 0$. The differences of negative logarithmic Haar densities and of the functions B_i in $H_{d_0,J}(\theta, w; \zeta)$ are evaluated at the same arguments $c_{i+1}(\zeta)$ and $z_i(\zeta)$. At $\zeta = t$, the auxiliary interaction is the interaction $H_{d_0,J}(\theta, w)$ of Theorem 12.13, and $\Xi_0(\theta; t) = \Xi_{d_0,J}(e^\theta)$. At $\zeta = 0$, every term vanishes, since the classical series starts at ζ^1, $z_i(0) = z_i^*(\widehat{c}_{i+1})$, and $c_{i+1}(0) = \widehat{c}_{i+1}$.

Lemma 12.16. *Fix J and the products, norms, and constants of Theorem 12.13 with $s = J$, and fix $u, v, b > 0$. Let R be a scalar function of the fine field such that its pullback*

$$\widehat{R}(\theta, q) = R\big(\mathscr{C}_{\mathrm{raw}}(\theta, q)\big)$$

is holomorphic on a product $\{\|\theta\|_\infty < u_2,\ \max_i \|q_i\|_\infty < v_2\}$ with $u_2 > u$ and $v_2 > v$, contained in the common product of Theorem 12.13(1), and satisfies $\mathcal{J}_{u,v,b}(\widehat{R}) \leq \epsilon$. The constant term of $\widehat{R}$ is unrestricted. Define

$$R_{\mathrm{ref}}(\theta) = \widehat{R}(\theta, 0), \qquad A_R(\theta, w; \zeta) = \widehat{R}\big(\theta, (\zeta\lambda_i w_i)_i\big) - R_{\mathrm{ref}}(\theta),$$
$$\Xi_{J,R}(\theta; \zeta) = \mathbb{E}_{\pi_p^{\mathrm{all}}} e^{-H_{d_0,J}(\theta, w; \zeta) - A_R(\theta, w; \zeta)}, \qquad \mathcal{D}_J R(\theta; \zeta) = -\log \Xi_{J,R}(\theta; \zeta) + \log \Xi_0(\theta; \zeta),$$

both logarithms being the branches of Theorem 12.7 that vanish at $\zeta = 0$. All parameters internal to R, the balls, and the classical functions are held fixed when ζ varies. There exist $u', b', C_J, c_J > 0$, depending only on M, J, u, v, b, the larger margins, and the spatial decay rates, and not on d_0, m, t, p, or ϵ, such that the following holds for every $p \geq p_0$, with $R_\zeta = c_J/(p^3 + \epsilon p)$.

1. *For every complex ζ with $|\zeta| \leq R_\zeta$ and every $w \in \Omega_p^{\mathrm{all}}$, the variables $q_i = \zeta\lambda_i w_i$ satisfy $\|q_i\|_\infty \leq c_J \leq \min\{v, v_J\}/2$. Consequently, for $\|\theta\|_\infty \leq u'$, the point (θ, q) lies in the product on which $\widehat{R}$ is holomorphic and in the common product of Theorem 12.13(1), and $H_{d_0,J}(\theta, w; \zeta)$ and $A_R(\theta, w; \zeta)$ are holomorphic in ζ on a neighborhood of the closed disc $|\zeta| \leq R_\zeta$.*
2. *Let $0 < t \leq c_J/[2(p^3 + \epsilon p)]$. Then the conditions* (12.23) *hold, and the integral* (12.25) *with the additional factor $e^{-R(c_0)}$ in the integrand equals the integral* (12.25) *times $\exp(-R_{\mathrm{ref}}(\theta) - \mathcal{D}_J R(\theta; t))$.*
3. *Let $\boldsymbol{\mathcal{D}}(\zeta) = \boldsymbol{\mathcal{L}}(\boldsymbol{H}(\zeta) + \boldsymbol{A}_R(\zeta)) - \boldsymbol{\mathcal{L}}(\boldsymbol{H}(\zeta))$ be the difference of the families of Theorem 12.7, at the coefficient radius u', for the supported decompositions $\boldsymbol{H}(\zeta)$ and $\boldsymbol{A}_R(\zeta)$ constructed in the proof; its sum over supports is $\mathcal{D}_J R(\theta; \zeta)$. Its constant and linear coefficients in ζ vanish, support by support. For $0 < t \leq c_J/[2(p^3 + \epsilon p)]$,*

$$\|\boldsymbol{\mathcal{D}}(t)\|_{\mathscr{S}_2} \leq C_J t^2 \epsilon (p^4 + \epsilon p^2),$$

and, on $\|\theta\|_\infty \le u'$*, for every* $k \ge 1$,

$$\|D_\theta^k \mathcal{D}_J R(\cdot;t)\|_{b',\mathrm{root}} \le C_J^{k+1} k!\, t^2 \epsilon (p^4 + \epsilon p^2).$$

The hypothesis concerns $\widehat{R}$ on a product containing all points described in (1). A bound for R on a smaller set of fine fields, which misses part of the range of the reconstruction over this product, does not suffice. The hypothesis is verified for the effective interactions of the recursion in Theorem 13.5.

Proof. We use Definition 12.15 throughout. Since $H_{d_0,J}(\theta, w; 0) = 0$ and $A_R(\theta, w; 0) = 0$ for every coarse field, both partition functions equal one at $\zeta = 0$.

Step 1: supported decompositions. We expand $\widehat{R}$ in the variables θ and q. Subtracting $\widehat{R}(\theta, 0)$ removes exactly the terms of degree zero in q. Every remaining monomial therefore contains a variable q_i, and its coefficient sum with that index fixed, including the volume normalization, is controlled by $\mathcal{J}_{u,v,b}(\widehat{R}) \le \epsilon$. The support-weight estimate (12.20), in the form used in Step 6 of the proof of Theorem 12.13, yields the bound $CD^{r+k} e^{b\mathrm{t}/2} e^{-b_1 \operatorname{dist}(x,e)}$ for the weight $e^{8|X|}$ of a monomial of degree r in θ, degree k in q, and fixed location e of an index of q. The sum over e contributes only the density of these indices. We choose $u_1 < u/(2D)$ and a radius $v_1 < v/(2D)$ for the variables q, and we reduce them further to absorb the fixed color factors and the choice of the fixed index of q, which contributes at most a factor $k \le 2^k$. A term of $\mathcal{J}_{u,v,b}(\widehat{R})$ is then multiplied by at most $(Du_1/u)^r (C_J|\zeta|p/v)^k$ with $k \ge 1$. Summing, we obtain an exact supported decomposition $\boldsymbol{A}_R(\zeta)$ of A_R with

$$\|\boldsymbol{A}_R(\zeta)\|_{\mathscr{S}_8} \le \frac{C_J \epsilon |\zeta| p}{1 - C_J|\zeta|p}, \tag{12.29}$$

an absolute coefficient bound at the coarse-field radius u_1. We use one box scale, chosen with the smaller of the two available rates, for H and A_R. Repeating the calculation of Step 6 in the proof of Theorem 12.13, with ζ in place of t, we obtain

$$\|\boldsymbol{H}(\zeta)\|_{\mathscr{S}_8} \le \frac{C_J|\zeta|(p^3 + p)}{1 - C_J|\zeta|p}. \tag{12.30}$$

Indeed, the classical terms have degree at least three in the fluctuation variables of their own scale, with the factor ζ^{k-2}. The differences of negative logarithmic Haar densities and of the functions B_i have degree at least one in the fluctuation variables. Joint composition and the support-weight estimate then give exactly these two geometric sums. The assignment of supports does not depend on ζ or on the interpolation parameter s below.

Step 2: the auxiliary disc. Let $R_\zeta = c_J/(p^3+\epsilon p)$. Since $p \geq 1$, we have $R_\zeta p \leq c_J$, $R_\zeta(p^3+p) \leq 2c_J$, and $R_\zeta \epsilon p \leq c_J$. We choose c_J so small that $C_J c_J \leq 1/2$, that $6C_J c_J \leq \eta_0/2$, that $c_J \leq \min\{v_1, v_J\}/2$, and that $4C'_J c_J \leq \sigma_0$, where η_0 is the constant of Theorem 12.7 and C'_J, σ_0 are the constants in (12.23). For $t \leq R_\zeta/2$, we have $tp \leq tp^3 \leq c_J/2$, hence $C'_J tp < 1/2$ and $C'_J(tp^3+tp)/(1-C'_J tp) \leq 2C'_J c_J \leq \sigma_0$, which are the conditions (12.23). Then (1) holds, the denominators in (12.29) and (12.30) are at least $1/2$ on a neighborhood of the closed disc $|\zeta| \leq R_\zeta$, and

$$\|\boldsymbol{H}(\zeta) + s\boldsymbol{A}_R(\zeta)\|_{\mathscr{S}_8} \leq 4C_J c_J + 2C_J c_J \leq \eta_0/2 \qquad (0 \leq s \leq 1).$$

Step 3: cancellation of the first order. By Theorem 12.7 and (12.29), the derivative family $D\boldsymbol{\mathcal{L}}(\boldsymbol{H}+s\boldsymbol{A}_R)[\boldsymbol{A}_R]$ has norm at most $C_J\epsilon p|\zeta|$ on the disc. The coefficient of ζ in A_R is $\sum_i \lambda_i D_{q_i}\widehat{R}(\theta,0)[w_i]$. Each monomial assigned to a support is linear in one fluctuation block, whose mean under the symmetric reference measure is zero. Thus its expectation vanishes before any sum over supports or coarse-field coefficients, and the hypothesis of Corollary 12.8 holds. Applying that corollary with $R = R_\zeta$, and integrating over $s \in [0,1]$, we obtain for the difference $\boldsymbol{\mathcal{D}}(\zeta)$ the bounds

$$\|\boldsymbol{\mathcal{D}}(\zeta)\|_{\mathscr{S}_2} \leq C_J\epsilon p|\zeta|, \qquad \|\boldsymbol{\mathcal{D}}(t)\|_{\mathscr{S}_2} \leq C_J\epsilon p\frac{t^2}{R_\zeta} = C_J t^2\epsilon(p^4+\epsilon p^2), \tag{12.31}$$

the second for $t \leq R_\zeta/2$. If $\epsilon = 0$, then $\widehat{R}$ is constant, $A_R = 0$, and the difference vanishes identically. Both terms $t^2\epsilon p^4$ and $t^2\epsilon^2 p^2$ in the bound are needed.

Step 4: derivatives. With $u' = u_1/2$, the bound (12.12) bounds the sum of the derivatives of order k of each support term by $2^{k+1}k!u_1^{-k}$ times its coefficient norm. All coarse-field labels in a support Y can be joined by a network of length at most $C_{\mathrm{geom},J}|Y|$, and we choose b' with $b'C_{\mathrm{geom},J} \leq 1$. Fixing a derivative index restricts the sum to the supports containing its box, and the factor $e^{2|Y|}$ dominates the network weight. This proves the bound for the derivatives, with a single factorial. Reducing the coefficient radius from u_1 to u' preserves the bound for $\boldsymbol{\mathcal{D}}(t)$.

Step 5: proof of (2). At $\zeta = t$, we have $q_i = g_i w_i$, so $R(c_0) = \widehat{R}(\theta,q) = R_{\mathrm{ref}}(\theta) + A_R(\theta,w;t)$. By Theorem 12.13, the integral with the factor $e^{-R(c_0)}$, divided by the integral without it, equals $e^{-R_{\mathrm{ref}}}\Xi_{J,R}(\theta;t)/\Xi_0(\theta;t)$, since the two integrals have the same partition function without insertion and the same classical, Haar density, determinant, and Gaussian prefactors. The negative logarithm of this ratio is $R_{\mathrm{ref}} + \mathcal{D}_J R(\theta;t)$, which proves (2). The radii and the constants C_J, c_J were chosen before p, ϵ, and t, which enter only through the conditions $p \geq p_0$ and $t \leq c_J/[2(p^3+\epsilon p)]$. □

Remark 12.17. The constants of this section are chosen in the following order. We first fix M and the larger domains of the classical and Haar disintegration charts. Next, we

choose the radii and rates of one averaging step, in Lemmas 12.2 and 12.3; these do not depend on J. Then we fix J. For this J, we choose the domain r_J of the compositions in Theorem 12.13, the box scale $\mathscr{L}$, and the rate b_1, we reduce the coefficient radius, and we fix C_J, σ_0, and the thresholds in (12.23). Finally, d_0, m, t, and p vary subject to the stated inequalities. In Lemma 12.16, the numbers u, v, b and the larger product of holomorphy are fixed before C_J and c_J, and ϵ, p, and t enter only through the final smallness conditions. This section does not close the recursion; Section 13 shows that the hypothesis of Lemma 12.16 is preserved from one renormalization group step to the next, at the same radii.

SECTION 13

Conditional measures and analytic estimates

In this section, we set up the renormalization group recursion for the Wilson measure restricted to the small-field event. We prove that the effective interactions stay in the same ball of an analytic coefficient norm at every scale, and we prove estimates for the conditional law of the normalized fluctuation variables given the coarse field. One renormalization group step consists of J consecutive applications of the block averaging map T_M, from a lattice of side mM^J to a coarse lattice of side $m \geq 4$. Its fluctuations are integrated with the results of Section 12. The probability of the restriction event under the Wilson measure and the continuum limit are treated in later sections.

The exact formula for one step. The central identity is Theorem 13.4. Let the density at the fine scale be $\mathbf{1}_{D_r} e^{-E_{d_0,\mathrm{sat}}/g_0^2 - R_{\mathrm{sat}}}$ with respect to the product Haar measure, where E_{d_0} is the effective classical action of depth d_0 and R is the input effective interaction, produced by the integrations at finer scales. Restricting to the event on which the step is defined and the fluctuations are cut off, and pushing forward by T_M^J, we obtain

$$\mathbf{1}_{D_r}(c)\, A_{\mathrm{coord}}\, e^{-E_{d_0+J,\mathrm{sat}}(c)/g_J^2 - \mathsf{B}_{\mathrm{ref,sat}}(c) - R_{\mathrm{ref,sat}}(c)} \Big(\int_{\Omega_p} e^{-V_c(w)}\, dw \Big)\, dH(c). \tag{13.1}$$

Here $\Omega_p = \prod_{i,e} \{ w_{i,e} \in \mathbb{R}^3 : |w_{i,e}| \leq p \}$, where e runs over the f_i normalized fluctuation variables of the intermediate scale i, each with three components, and dH is the product Haar measure on the coarse configuration. The factors are defined precisely in Theorem 13.4. Besides the effective classical action E_{d_0+J} at the coarse scale, the formula involves the reconstructed fine field, written through its logarithm $\varphi_\theta(w)$ in the gauge chosen in Theorem 13.2. The values at the identity of the Haar disintegration densities and Jacobians are collected in the constant A_{coord}. The terms $\mathsf{B}_{\mathrm{ref}}$ and R_{ref} are the determinant and Haar density terms and the input effective interaction, respectively, evaluated on the zero-fluctuation reconstruction. The conditional fluctuation potential V_c contains everything that depends on the fluctuations, and dividing by the integral gives the conditional fluctuation kernel $k(c, dw)$ of Theorem 13.8. The formula is first proved in local gauge charts and then glued across charts by exact, finite gauge invariance. Iterating it identifies the measures of the recursion with the conditioned Wilson measure, pushed forward to the coarse scales.

Main results. In Theorem 13.2, we construct a gauge representative of the reconstruction, depending only on the coarse field, on the whole domain of coarse fields and fluctuations. Theorem 13.4 proves (13.1) and identifies the iterated measures with the conditioned Wilson measure. Theorem 13.5 proves that one step maps the effective interaction into a fixed ball of the norm $\mathcal{B}_{\rho,a}$, at the same radius and rate. The new effective interaction is the sum of a newly generated effective interaction, a contracting pullback of the input effective interaction at zero fluctuation, and a fluctuation correction to the pullback, of second order in the coupling. Theorem 13.8 proves convexity and concentration for the conditional fluctuation kernel, and Theorem 13.10 estimates its means, its covariance, and the coefficients of the reconstruction at its mean.

The inputs are the concentration theorem on convex products of balls and the Haar measure identities of Theorem 5.4 and Lemma 5.1, the constrained minimizers and their coefficient bounds of Theorems 10.10 and 10.11, and the gauge invariance of the effective classical action, the contraction estimate, and the background field of Theorems 11.6 and 11.10 and Lemma 11.9. We also use the coordinates, exact integrals, and cluster expansions of Lemmas 12.3 and 12.16 and Theorems 12.6, 12.7, and 12.13. The results are used in Sections 14 and 15 to construct the small-field restriction and the backward reconstruction, and in Sections 16 and 17 for conditional estimates of the reconstruction. They are also used in Sections 19, 20, and 23, and, through the conditional law and its moments, in Sections 26 to 28, 31, and 35.

Domains. Several neighborhoods of the identity appear, with different roles.

Set	Role
B_r	coarse fields with link logarithms of norm less than r; here the integral is identified with the restricted Wilson measure
B_ρ	a larger set of coarse fields, $\rho \geq r$, on which the auxiliary integral and the analytic recursion are defined
$B_{\rho/2}$	contains the gauge-transformed reconstruction for all coarse fields and fluctuations in the enlarged product; the input effective interaction is evaluated here
$D_r = \mathcal{G}B_r$	the gauge-invariant small-field domain; the reconstructed fine field must lie in it

The gauge-transformed reconstruction lies in $B_{\rho/2}$ by the coefficient bounds of Theorem 13.2. That the reconstructed fine field lies in D_r, for every coarse field in B_r and every fluctuation in Ω_p, is a separate hypothesis, the support-inclusion hypothesis of Definition 13.3. A sufficient condition for it is $\mathfrak{M}_J(r, tp) < r/2$. Since the part of $\mathfrak{M}_J$ of positive degree in q does not vanish as $r \to 0$, this condition may require a further reduction of the coupling for each fixed r.

Order of choices. The parameters are chosen in the order

$$M,\ \text{initial domains and decay rates} \longrightarrow a, b, \mathscr{L}, K_*, \rho_0 \longrightarrow J \longrightarrow \rho, s_J \longrightarrow E_*, A_{\mathrm{w}}, t_*.$$

The rate a is taken below $a_{\mathrm{cen}}/8$, where a_{cen} is the rate of the bounds in Lemma 11.9, and $b > a$ is taken below the rate b_* of Theorem 13.2. No constant depending on J is used to choose J. The support radius $r \le \rho$ and its support-inclusion hypothesis are treated separately.

Conventions. All distances within a renormalization group step are measured in units of the coarse lattice spacing. The input mesh is $\delta = M^{-J}$, the normalized fluctuation variables w_i and the variables $q_i = g_i w_i$ of the intermediate scale $i \in \{0, \dots, J-1\}$ have mesh M^{i-J}, and the couplings are $g_i = t\lambda_i$ with $\lambda_i = M^{(i-J+1)/2} \in [M^{-(J-1)/2}, 1]$; also $g_J^2 = M g_{J-1}^2$. The number of scalar indices of θ and of w located in a unit cell is at most $D_J \le C_M(1+J)M^{3J}$; this is a density bound, not a bound on the number of variables. Unless a bound is stated to hold before J is chosen, a constant in this section may depend on M, J, the larger construction domains and their margins, the spatial decay rates, ρ, a, and the bound E_* on the effective interaction. Such a constant does not depend on the depth d_0, the side m, L, or γ. All results concern the whole periodic torus, not regions with fixed exterior data.

We use ordered scalar tensors with their Taylor factorials divided out; for example, a symmetric scalar tensor A of degree k contributes $A[\theta^{\otimes k}]/k!$. Fixing one input index and summing over the remaining indices defines the norm of a scalar tensor with one index fixed. For a vector tensor one may instead fix its output, or fix any input and sum over the output with its volume normalization, as in Definition 4.6. Every such sum carries the weight $e^{b\mathsf{t}}$, where t is the length of a shortest connected network joining the locations of the indices, with internal vertices allowed. At each degree we take the maximum over the positions of the fixed index. Additive constants are excluded from these norms. As in Section 12, a majorant series is a power series with nonnegative coefficients that dominates the corresponding sums of absolute coefficients.

Attribution. The construction follows Bałaban's renormalization group method for lattice gauge theories. The logarithmic averaging with reference transports adapts his averaging operations [12]; the use of small gauge representatives belongs to the program of [14]; and the constrained minimizer as a background field is the construction of [16]. The exponential decay of the propagators, used through Sections 10 and 11, adapts [10, 11, 13]. The iteration of effective densities with Gaussian-normalized fluctuations, separated normalization constants, and local effective interactions follows [15, 17, 19]; the four-dimensional coupling renormalization of [17] is not used. None of these works proves the identities for the centered block averaging map and the full periodic reconstruction used here, and we prove them below. The polymer expansions are those of Section 12, which

uses the criterion of Kotecký and Preiss [120] and the finite graph estimate recorded by Dimock [68, Appendix B, Theorem 27, p. 39, and (323), p. 42]. The variance, entropy, and covariance estimates for the conditional law come from Theorem 5.4, which belongs to the framework of Brascamp and Lieb [33], Gross [103], Bakry and Émery [8], and Helffer and Sjöstrand [109, 110]; here we verify its hypotheses for the conditional fluctuation potential.

13.1 Weighted coefficient bounds and majorant series

We first describe how the coefficient norms behave under the operations used in this section. The essential restriction is that the networks of the coefficients must be connected through shared indices.

Lemma 13.1. *Suppose that analytic tensor families have convergent majorant series, at one spatial decay rate b, for all their output-fixed rows and all their input-fixed volume-normalized columns. Consider finite products, contractions, and compositions in which the networks of the coefficients are connected through common outputs or contracted indices. Such expressions are bounded by the corresponding products and compositions of the majorant series. Products of extensive scalar interactions without such a connection are excluded: they require an additional connecting kernel with its own bounds, or separate bounds without a fixed index. To bound a composition with a specified input fixed, we use the derivative of the scalar composition of majorants with respect to the input variable, $(\widehat{F}\circ\widehat{G})' = (\widehat{F}'\circ\widehat{G})\,\widehat{G}'$, which accounts for the choice of the factor containing that input and of its position there. The volume normalization at the fixed input is that of its own lattice, and no second derivative factorial occurs. On a fixed smaller analytic product, evaluation and any fixed number of derivatives satisfy the same bounds.*

Let $T(o; i_1, \dots, i_k)$ be a tensor of fixed Taylor degree whose output-fixed rows are bounded by A at the rate $2b$, and let v_o and v_i denote the volumes of the meshes of the output and of an input. Then, for every fixed input i,

$$\sum_{o,\ \text{other inputs}} \frac{v_o}{v_i} e^{b\mathfrak{t}} |T(o; i, \dots)| \le \frac{A}{v_i} \sum_o v_o e^{-b\,\mathrm{dist}(o,i)}. \tag{13.2}$$

If i is an input of the coarse lattice, then $v_i = 1$, and if the outputs have spacing $\eta \le 1$ and bounded density, then $\eta^3 \sum_o e^{-b\,\mathrm{dist}(o,i)} \le C_b$; thus summing over the output converts the row bound into a column bound at the rate b, with a constant depending only on b and the density of the labels. An input of mesh $\eta_{\mathrm{in}} < 1$ keeps the factor η_{in}^{-3}, which within one renormalization group step is at most M^{3J}. Conversions between counting sums at the finitely many meshes of one renormalization group step cost at most a fixed factor depending on J.

Finally, for an analytic vector map $F_e(u) = \sum_\nu f_{e;\nu} u^\nu$ and a radius $r > 0$, define the entrywise envelope of its derivative

$$\mathcal{C}_{ex}(r) = \sum_{\nu:\, \nu_x > 0} \nu_x |f_{e;\nu}|\, r^{|\nu|-1}. \tag{13.3}$$

Then $\sup_{\|u\|_\infty \le r} |\partial_x F_e(u)| \le \mathcal{C}_{ex}(r)$*, and the sums of* $\mathcal{C}_{ex}(r)$ *over* x *at fixed* e*, and over* e *at fixed* x *with the volume factor* v_e / v_x*, both with the weight* $e^{b\mathsf{t}}$ *of the corresponding coefficients, are bounded by the derivatives at* r *of the majorant series of the output-fixed rows and of the input-fixed columns at* x*, respectively. This is an entrywise bound uniform on the product, not a bound on an evaluated operator norm.*

Proof. We first consider finite Taylor polynomials. For a pointwise product at a common output, we join the networks of the factors at that output; for a contraction we join them at the contracted index. Repeating these joins along the assumed connected structure, we obtain a connected union joining all uncontracted locations. Its length bounds the external t, which implies that the external exponential weight is at most the product of the weights of the factors. We fix the external index and sum over the contracted indices, starting with the factor containing the fixed index. At each attachment, we use an output row or the column at the attached input. The volume factor of that column is exactly the one needed for the contraction. This proves the bound for each connected product of coefficient tensors, with the product of their bounds. The argument gives no such bound for two scalar factors with disjoint networks and independent summed indices.

For a composition, we partition the ordered derivative indices among the inner factors. If the parts have sizes $k_1, \dots, k_l$, there are $k!/(k_1! \cdots k_l!)$ assignments, and division by $k!$ leaves the ordinary coefficients of the composition of exponential generating series. The derivative of the scalar composition bounds every coefficient series with a fixed input, including the columns, since it counts the choice of the inner factor and of the position of the fixed input in it. All bounds concern nonnegative sums, and passing to the limit of finite Taylor truncations gives the statements for the convergent series.

For the radius reduction, if $A_k \ge 0$ and $\sum_k A_k R^k < \infty$, then for $r < R$ the series obtained by l derivatives satisfies

$$\sum_{k \ge l} (k)_l A_k r^{k-l} \le R^{-l} \sup_{k \ge l} (k)_l (r/R)^{k-l} \sum_k A_k R^k < \infty,$$

where $(k)_l = k(k-1)\cdots(k-l+1)$. This applies only to series that are already absolute coefficient series; it does not infer absolute coefficients from a bound on directional values.

To prove (13.2), note that $\mathsf{t} \ge \operatorname{dist}(o, i)$ implies $e^{b\mathsf{t}} \le e^{2b\mathsf{t}} e^{-b \operatorname{dist}(o,i)}$. Applying the row bound separately at each output o, we obtain the inequality. For an input i of the coarse lattice, uniformly on periodic tori,

$$\eta^3 \sum_o e^{-b \operatorname{dist}(o,i)} \le C_b.$$

Indeed, the set of o with $k \le \operatorname{dist}(o, i) < k+1$ has counting volume, multiplied by η^3, at most $C(k+2)^3$, and $\sum_{k \ge 0} C(k+2)^3 e^{-bk}$ converges; a sum over the torus is bounded

by the corresponding periodized sum over the infinite lattice. This proves the column bound. An input of mesh $\eta_{\mathrm{in}} < 1$ keeps its factor η_{in}^{-3}, and replacing volume-normalized sums by counting sums at any of the fixed meshes changes a measure by at most M^{3J} for each conversion.

For the envelope, $|u^{\nu-\mathbf{e}_x}| \le r^{|\nu|-1}$ on the product, and the pointwise bound follows. Summing $\mathcal{C}_{ex}(r)$ over x at fixed e, we obtain $\sum_\nu |\nu||f_{e;\nu}| r^{|\nu|-1}$ with the weights, which is the derivative at r of the majorant for the output-fixed rows. At a fixed input x, for $|\nu| = k$ and $\nu_x > 0$, the remaining $k-1$ derivative indices can be ordered in $(k-1)!\,\nu_x / \prod_j \nu_j!$ ways, and each ordered derivative of $f_{e;\nu} u^\nu$ at zero has modulus $|f_{e;\nu}| \prod_j \nu_j!$; dividing by $k!$ gives $(\nu_x/k)|f_{e;\nu}|$ (see also (13.11) below). This shows that the column majorant at x has the coefficients $c_k = \sum_e \sum_{|\nu|=k} (\nu_x/k)|f_{e;\nu}|(v_e/v_x)e^{b\mathrm{t}}$, and the weighted sum of $\mathcal{C}_{ex}(r)$ over e equals $\sum_k k c_k r^{k-1}$, the derivative of this majorant at r. □

13.2 Gauge representatives on the reconstruction domain

The reconstruction $\mathscr{C}_{\mathrm{raw}\,d_0,J}(\theta, q)$ of Section 12.5 expresses the fine field in the block axial gauges of the intermediate scales. Even at zero fluctuation it need not lie in a small neighborhood of the identity in which the input effective interaction is defined. We now apply a gauge transformation, depending only on the coarse field, after which the reconstruction of the whole product of coarse fields and fluctuations lies in such a neighborhood. Here $\Omega(\theta)$ denotes a gauge transformation; it is unrelated to the product of balls Ω_p.

Theorem 13.2. *Fix an odd integer $M \ge 3$. There exist a spatial decay rate $b_* > 0$ and a majorant $Z(u) \le Cu/(1-Cu)$, both chosen before J, with the following property. Let $J \ge 1$, $d_0 \ge 0$, and $m \ge 4$, and put $\delta = M^{-J}$. There exist $r_J, C_J > 0$, independent of d_0, m, L, and γ, and a gauge transformation $\Omega_{d_0,J}(\theta)$ of the fine lattice, holomorphic in θ, with values in $\mathrm{SU}(2)$ for real θ, equal to the identity at every vertex of the coarse lattice, and with $\Omega_{d_0,J}(0) = I$, such that*

$$\Phi_{d_0,J}(\theta, q) = \mathrm{Log}\big[\mathscr{C}_{\mathrm{raw}\,d_0,J}(\theta, q)^{\Omega_{d_0,J}(\theta)}\big], \qquad T_M^J e^{\Phi_{d_0,J}(\theta,q)} = e^{\theta}, \qquad \Phi_{d_0,J}(\theta, 0) = \delta\overline{A}_{d_0,J}(\theta),$$

where $e^{\delta\overline{A}_{d_0,J}(\theta)}$ is the background field of Lemma 11.9. Here q_i replaces $g_i w_i$, and the matrices $K_i^{-1/2}$ remain inside the recursion. Moreover the following statements hold.

1. *On a fixed larger product with $u + v < r_J$, the map $\Phi_{d_0,J}$ has the majorant series*

$$\mathfrak{M}_J(u, v) = \delta Z(u) + \frac{C_J v}{1 - C_J(u+v)}$$

for its output-fixed rows. For the input-fixed columns, with the volume normalization of the fixed input, let $c_\theta(u, v)$ and $c_q(u, v)$ be the majorant series at a fixed coarse-field input and at a fixed input of q, in which every input, including the fixed one, is weighted by its

radius (u for coarse-field inputs and v for inputs of q). Then $c_\theta \le u\,\partial_u \widetilde{\mathfrak{M}}_J$ *and* $c_q \le v\,\partial_v \widetilde{\mathfrak{M}}_J$ *for a majorant* $\widetilde{\mathfrak{M}}_J(u,v) = \delta \widetilde{Z}(u) + C_J v/(1 - C_J(u+v))$ *of the same form, where* $\widetilde{Z}$ *has the properties of Z; in particular* $c_q(u,0) = 0$*. Thus the column bounds are the derivatives of majorants of the same form with respect to the variable of the fixed input, multiplied by its radius. The gauge transformation and its inverse have factorial coefficient bounds at every fixed index on the same larger product of coarse fields. After one fixed choice of radius,* Ω *has the logarithm* κ_Ω *in the identity chart, with* $\kappa_\Omega(0) = 0$*,* $\|\kappa_\Omega(\theta)\|_\infty \le C_J \|\theta\|_\infty$*, coefficient bounds of the same type, and* $\kappa_\Omega = 0$ *at every vertex of the coarse lattice. A positive spatial decay rate and the constants in the bounds for the first three derivatives in* θ *at zero fluctuation can be chosen before J; the joint constants and* r_J *may depend on J, the larger construction domains, and the decay rates.*

2. *For every constant* $u \in \mathrm{SU}(2)$*,* $\Omega(\mathrm{Ad}_u\,\theta) = u\Omega(\theta)u^{-1}$ *and* $\Phi(\mathrm{Ad}_u\,\theta, \mathrm{Ad}_u\,q) = \mathrm{Ad}_u\,\Phi(\theta,q)$*, on real data and for the holomorphic continuations on the domains above.*
3. *For every* $K_* \ge 2$*, one can choose J using only the constants chosen before J, and then* $\rho, s_J > 0$ *such that*

$$4K_*\rho < r_J, \qquad 4K_* C_J(\rho + s_J) < 1, \qquad \mathfrak{M}_J(4K_*\rho, 4K_* s_J) < \rho/2.$$

Then the image of the whole complex product with these radii consists of fields with link logarithms of norm less than $\rho/2$*, and all intermediate classical and Haar disintegration maps have fixed margins. For real coarse fields with* $\|\theta\|_\infty \le r \le \rho$ *and a radius Q for the variables q such that* $r + Q < r_J$*,* $1 - C_J(r+Q) > 0$*, all intermediate margins hold, and* $\mathfrak{M}_J(r,Q) < r/2$*, the image of the entire real product lies in* $B_{r/2}$*. These are statements about the whole product, not only about its center.*

4. *Suppose that two coarse fields related by a periodic gauge transformation have reconstructions* $C' = C^{\widetilde{k}}$*, related by the exact triangular transition of the fluctuation variables. Then*

$$(C')^{\Omega(\theta')} = \big(C^{\Omega(\theta)}\big)^{\Omega(\theta')\widetilde{k}\Omega(\theta)^{-1}}.$$

The gauge transformation changes only the representative at which the input effective interaction is evaluated; it does not change the fluctuation variables, the cutoff domain, the Haar disintegration densities, or the triangular determinant.

Proof. We first fix domains on which the maps of each averaging step have coefficient bounds, then construct the gauge transformation from the background field of Lemma 11.9, compose the steps, and finally choose the radii.

Step 1: the maps of one averaging step. At the intermediate scale i, $0 \le i < J$, the depth is $d_0 + i$ and the sides are $mM^{J-i} \to mM^{J-i-1}$, with output side at least four. Thus Theorem 10.10 applies at every scale with the same constants m_0, C_0, r_c, r_z. Statements about order, minima, and positive Hessians concern only real coarse fields and real free

logarithmic coordinates. For complex data, we use the holomorphic minimizer and the analytic continuation of the nested minimization identity, without any order statement. The free logarithmic coordinates are the logarithms in the block axial gauge, and

$$m_0 I \le K_i(I) \le C_0 I, \qquad \|K_i(I)\|_{\mathrm{row+col}} + \|K_i(I)^{-1}\|_{\mathrm{row+col}} \le C_0.$$

This concerns the free logarithmic coordinates only; no lower bound in the directions of coarse gauge transformations or of constant harmonic one-forms is used.

We choose smaller products of coarse fields and free logarithmic coordinates, with a common initial radius R_{prim} independent of d_0 and m, strictly inside the domains of Theorem 10.10 and the larger classical and Haar disintegration domains of Lemma 12.3. Part (3) of that lemma applies on these products without a cutoff or a coupling. It provides absolute coefficient bounds at every fixed index for the logarithmic reconstruction B_i, the objective F_i, the logarithmic Haar density, the minimizers z_i^*, and $K_i^{\pm 1/2}$. For every vector tensor these bounds include all its volume-normalized columns, and a coefficient of positive degree of $K_i^{\pm 1/2}$ may instead be bounded with a coarse-field index fixed and both matrix indices summed. The scalar tensors have one input fixed and exclude values. The actions in these objectives are the E_{d_0+i} of Theorem 10.11, which applies at depth $d_0 + i$ and side $mM^{J-i} \ge 4$; it provides factorial bounds for their derivatives at one rate, and $DE_{d_0+i}(I) = 0$. We take a positive rate below the rates of these results. This rate is valid at every scale, since each mesh is at most one in units of the coarse lattice spacing. All later reductions of these products are fixed reductions inside these domains.

Step 2: the zero-fluctuation reconstruction and the gauge. By Lemma 11.9, the background field $C(\theta) = C_{d_0,J}(\theta) = e^{\delta \overline{A}_{d_0,J}(\theta)}$ satisfies $T_M^J C(\theta) = e^{\theta}$, and the output rows of $(\overline{A}_{d_0,J}, \nabla_\delta \overline{A}_{d_0,J})$ have the majorant $Z(u) = Cu/(1 - Cu)$, uniformly before J. Each column at a coarse input, with output volume δ^3, is bounded by the corresponding derivative of Z. By the same lemma, for fixed J, there exists a block axial gauge $F(\theta) = F_J(\theta)$, fixed successively on the intermediate scales, such that

$$C(\theta)^{F(\theta)} = \mathscr{C}_{\mathrm{raw}\, d_0,J}(\theta, 0) \tag{13.4}$$

on a complex ball of fixed radius. The gauge transformation F takes values in $\mathrm{SU}(2)$ for real data, equals the identity at the vertices of the coarse lattice and at $\theta = 0$, and F and F^{-1} have factorial coefficient bounds at every fixed index. These bounds are uniform in d_0, m, L, and γ for fixed M and J, and they are not used to choose J.

We set $\Omega = F^{-1}$. Since Ω equals the identity at the vertices of the coarse lattice, it preserves the coarse field. By (13.4) and the rule $(U^k)^l = U^{lk}$, we have $\mathscr{C}_{\mathrm{raw}}(\theta, 0)^{\Omega} = C(\theta)$ and $\mathrm{Log}[\mathscr{C}_{\mathrm{raw}}(\theta, 0)^{\Omega}] = \delta \overline{A}_{d_0,J}(\theta)$. We reduce the radius so that the values of Ω lie in one identity chart. The absolutely convergent series $\mathrm{Log}(I + X) = \sum_{l \ge 1} (-1)^{l+1} X^l / l$ and Lemma 13.1 give the bounds for κ_Ω. Its constant coefficient vanishes, and its majorant is

$C_J u/(1 - C_J u)$ after increasing $C_J \geq 1$. At the vertices of the coarse lattice the matrix is I, and the logarithm vanishes there.

Step 3: composition of the steps. The recursion uses the independent variables $q_i = g_i w_i$ in

$$z_i = z_i^*(c_{i+1}) + K_i(c_{i+1})^{-1/2} q_i,$$

that is, the square root is not absorbed into q_i. We apply the bounds of Step 1 at every scale. If X_{i+1} is a majorant for the rows of $\operatorname{Log} c_{i+1}$, the majorants of the minimizer, of S_i, and of B_i give

$$X_i \preccurlyeq G(X_{i+1}, v), \qquad G(x, y) = \frac{A(x+y)}{1 - A(x+y)}, \qquad X_J = u,$$

after one enlargement of the constant A. All maps vanish at $(\theta, q) = (0, 0)$, except for the bounded constant matrix $S_i(I)$ multiplying q_i. There are only J compositions, and we choose $r_J > 0$ so that these scalar compositions converge and all intermediate values stay below the radii of Step 1. The networks are connected, since each composition joins the outputs of one step to the inputs of the next and the factor $S_i q_i$ contracts its index of z. The derivative of the same composition with respect to a fixed input controls each column at the mesh of that input. We obtain the joint absolute coefficient series of the reconstruction, independently of d_0 and m; the constants and the radius may depend on J.

To include the factors of the gauge transformation at the endpoints of each link, we transfer arrays indexed by vertices to the start or to the end of each link by the local incidence maps. These maps have one entry in each output row and a bounded number of entries in each input column. The corresponding segments have length at most δ, and these maps therefore have bounded series at the fixed rate. The three resulting factors indexed by links have a common output, and their connected pointwise product is the matrix

$$\Omega_x(\theta)\, \mathscr{C}_{\mathrm{raw}\, xy}(\theta, q)\, \Omega_y(\theta)^{-1}$$

on the link (x, y). Reducing r_J so that the logarithm of the identity chart applies, we obtain a joint absolute series for its logarithm at every fixed index. At $q = 0$ this series is exactly $\delta\overline{A}$. Subtracting this series coefficient by coefficient leaves only terms of positive degree in q. Any convergent nonnegative series in two variables with this property is bounded, on a smaller fixed product, by

$$\frac{C_J v}{1 - C_J(u + v)}.$$

Indeed, its coefficient of degree (r,s), $s \geq 1$, is at most AR^{-r-s} on a smaller disc of absolute convergence, and if $C_J R \geq \max(1, A)$, then

$$[u^r v^s] \frac{C_J v}{1 - C_J(u+v)} = C_J^{r+s} \binom{r+s-1}{r} \geq AR^{-r-s} \qquad (r \geq 0,\ s \geq 1),$$

including the coefficient of v itself. We reduce the final radius so that $C_J r_J < 1/2$. The part of degree zero in q has the separate majorant $\delta Z(u)$. This proves the majorant $\mathfrak{M}_J$ as a statement about coefficients on the larger product.

Step 4: columns and rates. Every input is kept as a fixed input in this composition, and its column has the volume normalization of its own lattice. Weighting each input, including the fixed one, by its radius turns the derivative of the composed majorant with respect to the variable of the fixed input into $u\,\partial_u$ or $v\,\partial_v$ of that majorant. This gives the column majorants c_θ and c_q of (1). Since every term of the part of positive degree in q contains a variable q_i and the part of degree zero does not depend on q, we have $c_q(u,0) = 0$. The sum over the output carries the factor δ^3; converting it to a sum with a coarse-field index fixed therefore introduces no factor of the total volume. Summing over the kinds of indices of q costs only D_J. The spatial decay rate in coarse units can be fixed before J. Indeed, a step of mesh $\eta \leq 1$ turns a rate b_0 in its own units into the rate $b_0/\eta \geq b_0$ in coarse units, and contractions use the union of networks at this one rate. The block forests of all scales together extend over a distance at most $C_M \sum_{l=1}^{J} M^{-l} < C_M/(M-1)$. A fixed part of the decay rate is used once for the conversion from rows to columns in Lemma 13.1, not at every scale. The constants for the first three derivatives at zero fluctuation, which are uniform before J, are those of Lemma 11.9.

Step 5: color equivariance. All operations in the construction of F commute with constant color conjugation: ordered transports, logarithms in the identity chart, the minimizers, and the positive square root. For the square root, conjugation carries the positive square root to the positive square root of the conjugated matrix, by uniqueness. Equivariance for real data extends to the holomorphic continuations by comparing Taylor coefficients. This proves (2).

Step 6: the choice of radii. We first take ρ_0 with $4K_* C\rho_0 < 1/2$ for the uniform majorant Z, so that $Z(4K_*\rho) \leq 8K_* C\rho$ for $\rho \leq \rho_0$. We choose J with $8K_* C\delta < 1/4$. After this choice we take $\rho < \min(\rho_0, r_J/(8K_*), (32K_* C_J)^{-1})$, and finally $0 < s_J < r_J/(8K_*)$ with $4K_* C_J s_J < \rho/16$, reduced further if needed for the fixed construction margins. Then the denominator in $\mathfrak{M}_J(4K_*\rho, 4K_* s_J)$ is larger than $1/2$, the first term is less than $\rho/4$, and the second term is less than $\rho/8$. Together, these bounds show that the majorant is less than $\rho/2$. The coefficient bound holds on the entire complex product, since also $4K_*(\rho + s_J) < r_J$. For real radii (r, Q) we require $r + Q < r_J$, $1 - C_J(r+Q) > 0$, and

the intermediate classical and Haar disintegration margins on this product; only then is $\mathfrak{M}_J(r, Q)$ a convergent majorant. If its value is less than $r/2$, domination of the coefficients places the image of the entire real product in $B_{r/2}$. This proves (3).

Step 7: overlaps. If $C' = C^{\widetilde{k}}$, then by the rule $(U^k)^l = U^{lk}$ we have

$$(C')^{\Omega(\theta')} = C^{\Omega(\theta')\widetilde{k}} = \bigl(C^{\Omega(\theta)}\bigr)^{\Omega(\theta')\widetilde{k}\Omega(\theta)^{-1}}.$$

No fluctuation variable changes in this calculation. The coordinate map, the cutoff domain, and the triangular determinant are therefore those of Theorem 12.13. This proves (4) and completes the proof. □

13.3 Cutoffs and gauge invariance on overlapping charts

We compute the integral over one renormalization group step in a chart, that is, for a chosen small representative of the coarse field. Different representatives are related by gauge transformations whose logarithms need not be small. The following definition states the invariance property used to glue the local formulas and the two hypotheses on the reconstruction.

Definition 13.3. Fix the larger construction domains and the coefficient bounds of Theorem 13.2.

1. A scalar function R on configurations of a periodic lattice is *gauge invariant on* B_ρ if it is real analytic on B_ρ and
$$R(U^k) = R(U)$$
for every periodic $\mathrm{SU}(2)$-valued gauge transformation k of that lattice with $U, U^k \in B_\rho$. Its gauge-invariant extension is $R_{\mathrm{sat}}(U) = R(U^k)$ whenever $U^k \in B_\rho$; it is defined on $D_\rho = \mathcal{G}B_\rho$.
2. The *enlarged-domain reconstruction hypothesis* consists of the coefficient inequality $\mathfrak{M}_J(4K_*\rho, 4K_* s_J) < \rho/2$, including the column bounds at every fixed input, together with fixed margins for every intermediate classical, constraint, and Haar disintegration map, and fixed margins for the classical comparison domains under the exact transitions between gauge-related charts. It does not depend on the support radius.
3. Let $0 < r \le \rho$. A fixed real product Ω_p satisfies the *support-inclusion hypothesis* if, for every $c = e^\theta \in B_r$ and every $w \in \Omega_p$, the variables $q_i = g_i w_i$ lie in the enlarged product of Theorem 13.2, all intermediate domain conditions hold, and $\mathscr{C}_{\mathrm{raw}\, d_{0,J}}(\theta, q) \in D_r$. The condition $\mathfrak{M}_J(r, tp) < r/2$, together with the margins of the enlarged product, is sufficient.

Both hypotheses are required at every depth that is used. Neither imposes a condition D_r at an intermediate scale of the renormalization group step.

The *restriction event* of a renormalization group step, at a given small representative of the coarse field, is the image of Ω_p under the sequential reconstruction of Theorem 12.13, together with the domain conditions of all intermediate classical, constraint, and Haar disintegration maps, the condition that the fine field lie in D_r, and the condition that the coarse field lie in D_r. We write $\mathbf{1}_{\mathrm{cut}}$ for its indicator.

Theorem 13.4. *Let $0 < r \le \rho$, and suppose that both hypotheses of Definition 13.3 hold for the renormalization group step under consideration. Let R be gauge invariant on B_ρ, with $DR(I) = 0$ and $\mathcal{B}_{\rho,a}(R) \le E_*$, and invariant under translations, the spatial inversion, and constant color conjugations. Then the following statements hold for every depth d_0.*

1. *The extension R_{sat} is well defined and real analytic on D_ρ, and*

$$|R_{\mathrm{sat}}(U) - R(I)| \le C|\mathcal{E}|E_*\rho^2,$$

   where $|\mathcal{E}|$ is the number of links. The input density

$$P_{\mathrm{in}}(U) = \mathbf{1}_{D_r}(U)\, e^{-E_{d_0,\mathrm{sat}}(U)/g_0^2 - R_{\mathrm{sat}}(U)}$$

   has a finite and strictly positive integral on every finite torus.
2. *At a chosen representative of the coarse field, the restriction event is exactly the image of the fluctuation product described above. Its union over the gauge transformations of the fine lattice that equal the identity at the vertices of the coarse lattice is a Borel event, and it does not depend on the representative.*
3. *Whenever two coarse fields related by a periodic gauge transformation lie, with margins, in the comparison domains, the exact transitions $w_i' = O_i(c_{i+1})w_i$ are triangular from coarser to finer scales, with orthogonal block permutations on the diagonal; they preserve every ball and the Gaussian term and have absolute Jacobian one. They identify the full integrals, including all Jacobians, Haar disintegration densities, and values of R. In particular the output density defined by this integral is a pointwise defined function, consistent on overlapping charts, and positive at every $c \in B_r$. Its effective interaction, obtained by subtracting E_{d_0+J}/g_J^2 from the negative logarithm of the density, is gauge invariant in the same sense, has the same symmetries, and has vanishing derivative at the identity. The auxiliary integral on B_ρ satisfies the same identities, but it is identified with the integral of the restricted input measure only on B_r.*
4. *Let $\mathsf{B}_{\mathrm{ref}}$ and $H_{d_0,J}$ be as in Theorem 12.13 with $s = J$, let $\varphi_\theta(w) = \Phi_{d_0,J}(\theta, (g_i w_i)_i)$ and $R_{\mathrm{ref}}(c) = R(e^{\varphi_\theta(0)})$, and set*

$$V_c(w) = \tfrac{1}{2}|w|^2 + H_{d_0,J}(\theta, w) + R(e^{\varphi_\theta(w)}) - R_{\mathrm{ref}}(c), \qquad A_{\mathrm{coord}} = \prod_i \mathcal{J}_i(I,0) g_i^{3f_i}.$$

With respect to the product Haar measure dH on the coarse configuration,

$$(T_M^J)_*(\mathbf{1}_{\mathrm{cut}} P_{\mathrm{in}}\, dH) = \mathbf{1}_{D_r}(c)\, A_{\mathrm{coord}}\, e^{-E_{d_0+J,\mathrm{sat}}(c)/g_J^2 - \mathsf{B}_{\mathrm{ref,sat}}(c) - R_{\mathrm{ref,sat}}(c)} \Big(\int_{\Omega_p} e^{-V_c(w)}\, dw \Big)\, dH(c). \tag{13.5}$$

The right side is defined by gluing the whole expression across charts. The terms $\mathsf{B}_{\mathrm{ref}}$ and R_{ref} need not be separately gauge invariant, and the subscript sat *on them is notation for the glued expression. Every indicator in $\mathbf{1}_{\mathrm{cut}}$ is kept in its definition; the support-inclusion hypothesis shows that the indicator of D_r for the fine field equals one on the domain of integration.*

5. *Start with $d_0 = 0$, $R_0 = 0$, and the measure $\mathbf{1}_{D_r} e^{-S_{\mathrm{W}}/g_N^2}\, dH_N$. Then any finite iteration of these positive operators gives the pushforward of the Wilson measure restricted to the initial condition D_r and to the intersection of the pulled-back restriction events of the steps. After normalization this is the Wilson measure conditioned on this event, pushed forward to the current scale.*

The constants depend only on M, J, ρ, a, E_*, and the fixed construction margins; they do not depend on the side, the depth, L, γ, or the number of steps.

Proof. We glue local formulas, using the gauge transformation and the coefficient bounds of Theorem 13.2 and the two hypotheses of Definition 13.3. The main point is that the transition between two charts is an exact, orthogonal, and triangular change of the fluctuation variables, so that the full integrals agree pointwise and not only almost everywhere.

Step 1: gluing and normalization. If $U^k, U^l \in B_\rho$, then $U^l = (U^k)^{lk^{-1}}$, and gauge invariance on B_ρ implies that $R(U^l) = R(U^k)$. Thus R_{sat} is well defined. The set D_ρ is open, as the union of the images of B_ρ under fixed gauge transformations. On the open set where $U^k \in B_\rho$ the function $U \mapsto R(U^k)$ is real analytic. These functions agree on intersections, and together they define a real analytic function on D_ρ. Translations and the spatial inversion preserve B_ρ and intertwine the gauge action, and the corresponding identities glue as well. For the effective classical action, the gluing is Theorem 11.6 on the classical domain.

We write $R(e^x) = R(I) + \sum_{k\geq 2} \sum_{|\alpha|=k} r_\alpha x^\alpha$ near zero, in scalar coordinates. The degree-k sum of $|r_\alpha|$ is at most $3|\mathcal{E}|$ times the corresponding sum with one index fixed, and the norm $\mathcal{B}_{\rho,a}(R)$ sums these at radius ρ. This implies that the Taylor series converges absolutely on the complex product $\|x\|_\infty < \rho$. It agrees with R on the real ball, since both are real analytic on this connected ball and agree near zero. This also specifies the complex continuation used below. By Euler's identity,

$$\sum_{|\alpha|=k} |r_\alpha| = \sum_i \sum_{|\alpha|=k} \frac{\alpha_i}{k} |r_\alpha|.$$

There are $3|\mathcal{E}|$ scalar indices i, and the inner sum is the sum of the ordered derivatives with i fixed, divided by $k!$, without its spatial weight, which is at least one. Summing over the degrees, we obtain

$$\sup_{B_\rho} |R - R(I)| \leq 3|\mathcal{E}|\rho^2 \mathcal{B}_{\rho,a}(R). \tag{13.6}$$

The same bound holds for R_{sat}, which takes the same values. The effective classical action is nonnegative, since it is a minimum of the Wilson action multiplied by a positive normalization. Hence $P_{\mathrm{in}} \leq \exp[-R(I) + 3|\mathcal{E}|E_*\rho^2]$, and since the Haar measure has mass one, the integral of P_{in} is finite. The classical action and the effective interaction are finite and continuous near the identity, and D_r contains an open neighborhood of the identity, of positive Haar measure. Integrating over a smaller closed neighborhood shows that the integral is strictly positive. This argument keeps the constant $R(I)$. This proves (1).

Step 2: the exact transition between charts. Fix real coarse fields c and $c' = c^k$ in the comparison domains, with margins. The overlap statement of Theorem 11.6 applies on every comparison domain required by the enlarged-domain reconstruction hypothesis. In a block forest, we transport each link variable to the root before taking its logarithm. Changing the frame at the root by k conjugates the transported variables by the value of k at the root, and the factors of the paths at intermediate vertices cancel in pairs. Thus a free logarithmic coordinate changes by the orthogonal adjoint action on its block. A translation or the spatial inversion in addition relabels the oriented blocks, and the reversal of an orientation acts by an orthogonal map on the color components. We denote by $O_i(c_{i+1})$ the resulting orthogonal map of the free logarithmic coordinates of scale i, which permutes the blocks. It depends on the coarser field and not on the free logarithmic coordinates of scale i. These are the transformations of Proposition 12.11 and Theorem 12.13.

By Theorem 11.6 the two classical objectives agree on these domains. Their minimizers are unique, and the transformation of the coordinates carries one to the other. Differentiating the objective in the orthogonal coordinates, we obtain

$$z_i^{*\prime} = O_i z_i^*, \qquad K_i' = O_i K_i O_i^{\mathsf{T}}, \qquad S_i' = O_i S_i O_i^{\mathsf{T}};$$

the last identity follows by taking the positive square root of real matrices, and it is continued analytically only on the fixed neighborhoods. Substituting in $z_i = z_i^* + g_i S_i w_i$ gives $w_i' = O_i(c_{i+1}) w_i$. We proceed from scale $J-1$ to finer scales; when O_i is evaluated, every coarser field has already been transformed. This proves $C' = C^{\widetilde{k}}$ for the reconstructions, with an $\mathrm{SU}(2)$-valued gauge transformation $\widetilde{k}$ of the fine lattice, as well as the triangular transition of the fluctuation variables.

In this order the derivative matrix of the transition is triangular. Its diagonal blocks are the O_i, and its off-diagonal blocks contain all derivatives of the O_i with respect to the coarser fluctuations. Therefore

$$|\det D_w w'| = \prod_i |\det O_i| = 1, \qquad \sum_i |w_i'|^2 = \sum_i |w_i|^2,$$

and each ball is carried onto the corresponding ball. The identity for block forests in Lemma 5.1 preserves the product Haar measure, gives mass one to the forest variables, and keeps every vertex of the coarse lattice fixed. The changes of the free logarithmic coordinates and of the determinants of the constraints are exactly the changes in the Haar disintegration formula of Lemma 6.3; since that formula is the disintegration of the Haar measure, the product of these factors is preserved. Equivalently, we may substitute $z_i' = O_i z_i$ in each integral over the free logarithmic coordinates and use the conjugation identity for S_i; the diagonal determinants and all Haar disintegration factors agree. No off-diagonal derivative is dropped in either computation.

Step 3: values of the input effective interaction. The two gauge-transformed reconstructions lie in $B_{\rho/2}$, and they are related by the gauge transformation $\Omega(\theta')\widetilde{k}\Omega(\theta)^{-1}$ of Theorem 13.2(4). The values of R agree by gauge invariance on B_ρ, not by smallness of the logarithm of this gauge transformation. In particular,

$$R_{\mathrm{sat}}\big(\mathscr{C}_{\mathrm{raw}}(\theta, (g_i w_i)_i)\big) = R\big(e^{\varphi_\theta(w)}\big). \tag{13.7}$$

For the auxiliary coarse fields in B_ρ this identity uses D_ρ only. For coarse fields in B_r, the support-inclusion hypothesis also places the reconstructed fine field in D_r. This removes the indicator of D_r for the fine field from the integral over the fluctuation product.

Step 4: the restriction event. On a chart of the coarse field, we take the image of the fluctuation product under the triangular reconstruction, with its classical, constraint, and Haar disintegration domain conditions. We keep the conditions D_r for the fine field and for the coarse field. We take its orbit under the gauge transformations of the fine lattice that equal the identity at the vertices of the coarse lattice. Changing the small representative of the coarse field changes this local set by the bijective triangular transition of Step 2. Extending also by the gauge transformations of the coarse lattice defines one event, independent of the chart; representatives are not counted.

This event is Borel. To see this, we exhaust the open ball of logarithms of the coarse field by smaller closed balls, and each construction margin by closed positive margins. On each such compact set the finite reconstruction and its domain data are continuous. The product of such a set with the closed balls of fluctuations and the compact group of gauge transformations therefore has a compact image in the finite-dimensional configuration manifold. By Step 2, the union of the local events over all charts coincides with the

local event on every chart; as the countable union of these compact images, it is Borel. Intersecting with the Borel conditions on the fine and the coarse field yields a Borel event. No measurable global choice of gauge is used. This proves (2).

Step 5: the density. We use the action identity and the triangular determinant of Theorem 12.13, and we insert (13.7). The normalization of the Haar disintegration is $A_{\mathrm{coord}} = \prod_i \mathcal{J}_i(I,0) g_i^{3f_i}$. The exponent depending on w is

$$\tfrac{1}{2}|w|^2 + H_{d_0,J}(\theta, w) + R(e^{\varphi_\theta(w)}) - R(e^{\varphi_\theta(0)}),$$

and the terms removed from it are exactly $E_{d_0+J}/g_J^2 + \mathsf{B}_{\mathrm{ref}} + R_{\mathrm{ref}}$. Including them, we obtain (13.5) with respect to the product Haar measure on the coarse configuration. There is no additional Haar density of the coarse field and no additional factor for the volume of gauge orbits.

The integral is finite and strictly positive at every coarse field satisfying the hypotheses, since the integrand is positive and continuous on the compact product of balls, which has positive Lebesgue measure. By Step 2, the entire integral, including the terms evaluated at the zero-fluctuation reconstruction and the constant A_{coord}, takes the same value in overlapping charts. It is therefore a pointwise defined version of the density, which is stronger than an identity almost everywhere. Taking the real logarithm and subtracting the gauge-invariant classical term proves the gauge invariance of the effective interaction on overlapping charts. The same argument for translations, the spatial inversion, and constant color conjugations proves its other symmetries. If ℓ is its derivative at the identity, color symmetry shows that ℓ is a linear functional on each link variable that is invariant under all rotations of the color components. Rotations by π about two coordinate axes force all three components to vanish. Hence $DR_+(I) = 0$, where R_+ denotes the output effective interaction.

On B_ρ the formula over the fixed product is the auxiliary integral. All comparisons of charts used above stay inside the enlarged construction domains, so the same identities hold for it. The condition D_r for the fine field was used only on B_r; outside B_r the auxiliary integral is not identified with the integral of the restricted input measure. The subscript sat on the terms evaluated at the zero-fluctuation reconstruction is only notation for the glued expression, and the argument does not require their separate invariance. This proves (3) and (4).

Step 6: iteration. Let A_j be the restriction event of the step applied to the configuration obtained after j completed steps, and let η_j be the unnormalized measure after j steps. For every nonnegative Borel function F, by the definition of the pushforward, we have

$$\int F \, d\eta_{j+1} = \int \mathbf{1}_{A_j}(U) F(T_M^J U) \, d\eta_j(U).$$

Starting from the restricted Wilson density and substituting this identity repeatedly, we obtain, since all integrands are nonnegative,

$$\int F\, d\eta_k = \int \mathbf{1}_{D_r}(U) \prod_{j=0}^{k-1} \mathbf{1}_{A_j}(T_M^{jJ}U)\, F(T_M^{kJ}U)\, e^{-S_W(U)/g_N^2}\, dH_N(U).$$

Every normalization is positive by Step 5. Dividing this identity by its value at $F = 1$, we obtain exactly the Wilson measure conditioned on the stated intersection of events. The constants used above depend only on the data stated in the theorem, and the extensive factor in (13.6) is explicit. This proves (5). □

13.4 Preservation of analytic bounds under iteration

We now show that one renormalization group step maps an effective interaction with $\mathcal{B}_{\rho,a}(R) \le E_*$ to one with the same bound, at the same radius ρ and rate a. The new effective interaction has three parts: the newly generated effective interaction $Q^{\mathrm{w}}_{d_0,J}$, which is bounded by Theorem 12.13; the pullback $\mathcal{P}_\delta R$ of R at zero fluctuation, which contracts by Theorem 11.10; and the fluctuation correction $\mathcal{D}_J R$ to the pullback, which is of second order in t by the argument of Lemma 12.16. Two points require care. First, the supported estimates must be converted back to the norm $\mathcal{B}_{\rho,a}$ without reducing ρ, since otherwise the radius would shrink at every step; this uses the translation invariance of the total effective interaction. Second, the composition of R with the reconstruction must be estimated at the level of coefficients, on the whole range of the reconstruction.

Theorem 13.5. *Fix an odd integer $M \ge 3$ and the bounds for the classical coefficients and for the reconstruction. Then the following statements hold.*

1. *There exist a spatial decay rate $a > 0$, a larger rate $b > a$, a box scale $\mathscr{L}$, a factor $K_* \ge 2$, and $\rho_0 > 0$, all chosen before J; the density of the coefficient indices affects only constants that depend on J. Let $C_{\mathrm{geom}} \ge 1$ be a constant, depending only on the metric conventions, that bounds the lengths of the rectilinear networks used to construct supports by the lengths of shortest connecting networks. The rate a is chosen below $a_{\mathrm{cen}}/8$, where a_{cen} is the rate of the bounds of Lemma 11.9. Choose J such that*

$$q_P = C_P(a) M^{-J} < \tfrac{1}{4}, \qquad M^{-J} Z(4K_*\rho) \le \rho/4 \quad (0 < \rho \le \rho_0), \qquad C_{\mathrm{geom}} b M^{-J} \le a,$$

where $C_P(a)$ is the constant of Theorem 11.10 and Z is the majorant of Theorem 13.2. Then choose $0 < \rho \le \min(\rho_0, r_J/(8K_*))$, with $\rho C_B \le 1/2$ for a constant C_B depending only on M, J, and the larger domains (defined in the proof), and $s_J > 0$ such that the enlarged-domain reconstruction hypothesis of Definition 13.3 holds strictly. These choices use only the fixed larger domains, not the support radius r.

2. *Let $\{\Psi_Y\}$ be a family of functions of the coarse field indexed by supports, with coefficient norms at radius ρ, whose sum $f = \sum_Y \Psi_Y$ is invariant under the translations of the coarse lattice. Suppose that $a\mathfrak{t} \le 2|Y|$ for the coarse-field labels of each coefficient of Ψ_Y. Then*

$$\mathcal{B}_{\rho,a}(f) \le 9\rho^{-2} \sup_x \sum_{Y \ni x} e^{2|Y|} \|\Psi_Y\|_{\rho,\mathrm{coeff}}$$

in scalar orientation and color coordinates. The individual Ψ_Y need not be translation invariant. Without translation invariance of f, a bound of this kind requires a strictly larger coefficient radius.

3. *There exist $C_G, C_T, c_* > 0$, depending on M, J, ρ, a, b, K_*, s_J, and the larger domains, and independent of the side and of d_0, such that the following holds. Let R be gauge invariant on B_ρ, invariant under translations, the spatial inversion, and constant color conjugations, with $DR(I) = 0$ and $\mathcal{B}_{\rho,a}(R) \le \epsilon$. Then the pullback $R(e^{\Phi_{d_0,J}(\theta,q)})$ has the joint coefficient bound $C_{J,\rho,a}\epsilon$ at every fixed index on the enlarged product of coarse fields and variables q. Define the auxiliary recursion by the integral over the fixed product of fluctuation variables and the formula* (13.5)*, using this pullback on the gauge-transformed image in $B_{\rho/2}$, and without any indicator of D_r at coarse fields outside B_r. Let $p \ge p_0$, and assume*

$$0 < t \le \frac{c_*}{2(p^3 + \epsilon p)}. \tag{13.8}$$

Then, with $R_{\mathrm{aux}} = c_*/(p^3 + \epsilon p)$, for every complex ζ with $|\zeta| \le R_{\mathrm{aux}}$ and every $w \in \Omega_p$ the variables $\zeta\lambda_i w_i$ lie in the enlarged product of Theorem 13.2, the conditions (12.23) *hold at (t, p), and the supported families of the auxiliary interaction and of the inserted interaction constructed in the proof lie in the ball of radius $\eta_0/2$ of Theorem 12.7. Under these conditions, apart from the field-independent factor A_{coord}, the output effective interaction is*

$$R_+ = Q^{\mathrm{w}}_{d_0,J} + \mathcal{P}_\delta R + \mathcal{D}_J R, \qquad \mathcal{B}_{\rho,a}(R_+) \le C_G + q_P\epsilon + C_T t^2 \epsilon(p^4 + \epsilon p^2). \tag{13.9}$$

Here $\mathcal{P}_\delta R(\theta) = R(e^{\varphi_\theta(0)}) = R(e^{\delta\overline{A}_{d_0,J}(\theta)})$ is the pullback at the zero-fluctuation reconstruction, $\mathcal{D}_J R$ is the difference of the logarithms of the fluctuation partition functions without and with the inserted interaction, as in Lemma 12.16 at the parameter t, and $Q^{\mathrm{w}}_{d_0,J}$ is the effective interaction of Theorem 12.13. The output R_+ is gauge invariant on B_ρ, invariant under translations, the spatial inversion, and constant color conjugations, and $DR_+(I) = 0$. Coupling parameters internal to R are held fixed when the auxiliary parameter varies.

4. *Fix $E_* \ge 2C_G/(1 - q_P)$ and $p(t) = A_{\mathrm{w}} \log(1/t)$ with $A_{\mathrm{w}} \ge 1$. There exists $t_* > 0$, depending only on the fixed data, such that for every $0 < t \le t_*$ we have $p(t) \ge p_0$, $tp(t) < s_J$, the enlarged-domain hypotheses, and*

$$t \le \frac{c_*}{2(p(t)^3 + E_* p(t))}, \qquad C_T t^2 \big(p(t)^4 + E_* p(t)^2\big) \le \frac{1 - q_P}{2}.$$

Consequently $\mathcal{B}_{\rho,a}(R) \le E_*$ implies $\mathcal{B}_{\rho,a}(R_+) \le E_*$.

5. *The auxiliary recursion starting at $R_0 = 0$ exists, with the same ρ, a, and E_*, for any finite number of renormalization group steps. For any $r \le \rho$ for which the support-inclusion hypothesis holds at every step, the restricted densities after k steps are exactly*

$$Z_k^{-1} \mathbf{1}_{D_r} e^{-E_{kJ,\mathrm{sat}}/g_{kJ}^2 - R_{k,\mathrm{sat}}}, \qquad \mathcal{B}_{\rho,a}(R_k) \le E_*,$$

for every k for which the output sides are at least four and the couplings of the integrated scales are at most t_*. Nothing is asserted about the integral of the restricted input measure at coarse fields outside B_r.

Proof. We use the coefficient bounds of Theorem 13.2. We first fix the rates and the supports, then prove (2), then estimate the three parts of R_+, and finally choose the coupling threshold.

Step 1: rates, supports, and the order of choices. At an intermediate scale of mesh $\eta_i \le 1$, a weight $e^{A\mathrm{t}}$ with t measured in the units of that scale equals $e^{(A/\eta_i)\mathrm{t}}$ with t measured in coarse units. In particular, it gives any fixed rate below A in coarse units. The union of networks in Lemma 13.1 keeps one rate under finite composition and does not divide it by the number of scales. Sums over the kinds of coordinates change constants by D_J.

We separate these two effects with the following geometric bounds. For a monomial with n coordinate locations, we choose a rectilinear network of length at most $C_{\mathrm{geom}}\mathrm{t}$. To that end, we first replace a shortest connected network by a tree on the locations, of at most twice its length, by traversing its edges and shortcutting the walk. Then we replace each edge of the tree by its three coordinate segments. This also shows that the number of segments is at most a fixed multiple of $n+1$. We partition the torus into boxes with integer side lengths between $\mathscr{L}$ and $2\mathscr{L}$, a shorter direction forming one interval. Since a coordinate segment of length l meets at most $2 + l/\mathscr{L}$ intervals in its direction, the connected set X of boxes met by the network satisfies

$$|X| \le c_0\big(n + 1 + C_{\mathrm{geom}}\mathrm{t}/\mathscr{L}\big). \tag{13.10}$$

All constructions are periodic and include sets winding around the torus, and c_0 depends only on the dimension, not on the density of labels.

We choose $\mathscr{L}$ so large that $8c_0C_{\mathrm{geom}}/\mathscr{L}$ is at most one quarter of the available rate. The factor $e^{8c_0(n+1)}$ in $e^{8|X|}$ is absorbed by enlarging $K_* \ge 2$ once, since the radii carry the factor K_*^n. A second quarter of the available rate is used for the sums over the location of the fixed index. If $x \in X$ and u is any location of the monomial, the network from u to x has length at most $C_{\mathrm{geom}}\mathrm{t} + C_{\mathscr{L}}$, which leaves a factor that decays exponentially in $\mathrm{dist}(x,u)$. Summing over u then costs only D_J. These choices give the support weights used in Theorem 12.6 without a factor depending on J in the degree.

Each box contains at most $C\mathscr{L}^3$ coarse-field labels. Joining them inside each box and then joining adjacent boxes of a connected set Y, we obtain

$$\mathsf{t}(\text{coarse-field labels in } Y) \le C_{\mathscr{L}}|Y|,$$

and repeated labels do not increase this length. The bounds of Lemma 11.9 for the background field $e^{\delta\overline{A}_{d_0,J}}$ hold at a rate a_{cen}, and for this representative Theorem 11.10 provides the contraction at every output rate $0 < a < a_{\mathrm{cen}}/8$, and at no larger rate. We therefore choose a with $0 < a < a_{\mathrm{cen}}/8$ and $aC_{\mathscr{L}} \le 2$; such an a also satisfies $a < a_*/8$, where a_* is the rate of Theorem 10.11. We choose $b > a$ below the rate b_* of Theorem 13.2, with enough room for the support and location estimates above. This is possible since b_* is a fixed positive rate chosen before J, and a may be taken as small as needed. The condition $C_{\mathrm{geom}}bM^{-J} \le a$ is then imposed when J is chosen, after a and b are fixed. All these choices precede J, and the metric conventions are included once in C_{geom}.

The coefficient bounds for the reconstruction $\Phi_{d_0,J}$, for the auxiliary interaction H, and for $\mathsf{B}_{\mathrm{ref}}$ hold at a common rate $b_* > 0$ in coarse units, chosen before J. Indeed, the bounds of one averaging step in Lemma 12.3 hold at a rate chosen before J. This rate does not decrease when lengths are measured in coarse units, since every mesh is at most one, and composition through connected networks keeps one rate, by Lemma 13.1. The rate b_* of Theorem 13.2 is chosen below these rates. The inserted interaction is treated separately in Step 3: the network of R has the rate a/δ in coarse units, the condition $C_{\mathrm{geom}}b\delta \le a$ supplies the rate b needed for the support estimates, and the attached networks of the reconstruction have rate at least $b_* > b$. Since

$$0 < a < b < b_*,$$

all four families of bounds hold at the rate a of the norm $\mathcal{B}_{\rho,a}$ and at the rate b used for the supports. The rate b_J of Theorem 12.13(5), which may depend on J, is not used; this is why the bound on $Q^{\mathrm{w}}_{d_0,J}$ in Step 5 is proved directly.

The constant $C_P(a)$ of Theorem 11.10 does not change under later reductions of the radius. We choose ρ_0 in the range allowed by that theorem and in the range of convergence of Z, so that $Z(4K_*\rho) \le C_1\rho$ for $\rho \le \rho_0$. We then take J so large that $q_P < 1/4$, that $M^{-J}Z(4K_*\rho) \le \rho/4$, that $C_{\mathrm{geom}}bM^{-J} \le a$, and that M^{-J} is below the threshold $\delta_0(a)$ of Theorem 11.10. This is the only choice of J. Next we choose ρ and s_J as in Theorem 13.2(3), with strict inequalities and with $\rho C_B \le 1/2$. Here C_B is the constant in the coefficient bound for $\mathsf{B}_{\mathrm{ref}}$ of Step 5, which depends only on M, J, and the larger domains. Reducing these radii does not change q_P, and none of these choices uses r. This proves (1).

Step 2: conversion at the same radius. For a monomial $c_\alpha \theta^\alpha$ of degree k, the ordered derivatives with a scalar input i fixed satisfy

$$\frac{1}{k!} \sum_{i_2,\dots,i_k} |D_i D_{i_2} \cdots D_{i_k} (c_\alpha \theta^\alpha)(0)| = \frac{\alpha_i}{k} |c_\alpha|. \tag{13.11}$$

Indeed, the remaining indices can be ordered in $(k-1)!/((\alpha_i - 1)! \prod_{j \neq i} \alpha_j!)$ ways, and each derivative equals $c_\alpha \prod_j \alpha_j!$. For sums of supported terms the triangle inequality gives the corresponding upper bound. A term differentiated at i has $i \in Y$, and $e^{a\mathfrak{t}} \leq e^{2|Y|}$ by hypothesis.

At each degree the translation-invariant function f has the same norm with a fixed index at all translates of a given scalar orientation and color type. We choose one reference index for each of the $3 \cdot 3 = 9$ types, replace the maximum of the nine nonnegative degree sums by their sum, and only then sum over the degrees. Applying (13.11) at these fixed indices, we obtain

$$\mathcal{B}_{\rho,a}(f) \leq 9\rho^{-2} \sup_x \sum_{Y \ni x} e^{2|Y|} \|\Psi_Y\|_{\rho,\mathrm{coeff}}.$$

The individual Ψ_Y need not be translation invariant. Without translation invariance of f, a coefficient norm at a radius $u > \rho$ bounds the degree-k norm with a fixed index by u^{-k} times the norm at u, and summing $\rho^{k-2}u^{-k}$ over $k \geq 2$ gives $u^{-2}(1-\rho/u)^{-1}$. This explains the need for a strictly larger radius in that case. A supremum over the fixed index cannot be interchanged with the infinite sum over degrees at the same radius. This proves (2).

Step 3: the coefficients of the composed input interaction. Let $\epsilon = \mathcal{B}_{\rho,a}(R)$. We write the logarithms of the fine field as $A = (A_j)$, where the labels j of the fine lattice have volume $v_{\mathrm{old}} = \delta^3$, and we expand

$$\begin{aligned} R(e^A) &= R(I) + \sum_{n \geq 2} \frac{1}{n!} \sum_{j_1,\dots,j_n} R_n(j_1,\dots,j_n) \prod_{r=1}^n A_{j_r}, \\ P_j(\theta, q) &= \Phi_{d_0,J}(\theta, q)_j = \sum_{|\xi|+|\eta| \geq 1} p_{j;\xi\eta} \theta^\xi q^\eta; \end{aligned} \tag{13.12}$$

there is no term with $n = 1$ since $DR(I) = 0$. For a multiindex (ν, η) of positive total degree $N = |\nu| + |\eta| \geq 1$, the coefficient of $\theta^\nu q^\eta$ in $R(e^P)$ is

$$[\theta^\nu q^\eta] R(e^P) = \sum_{n=2}^N \frac{1}{n!} \sum_{j_1,\dots,j_n} \sum_{\substack{\sum_r \xi_r = \nu,\ \sum_r \eta_r = \eta \\ |\xi_r|+|\eta_r| \geq 1}} R_n(j_1,\dots,j_n) \prod_{r=1}^n p_{j_r;\xi_r\eta_r}, \tag{13.13}$$

where the inner sum is over ordered allocations. No further multinomial factor occurs, since each $p_{j;\xi\eta}$ is already the coefficient of a monomial. For example, the allocation

$(\xi_1, \eta_1) = (\mathbf{e}_i, 0)$, $(\xi_2, \eta_2) = (0, \mathbf{e}_l)$ with $n = 2$ contributes $(1/2) \sum_{j_1, j_2} R_2(j_1, j_2) p_{j_1;\mathbf{e}_i,0}\, p_{j_2;0,\mathbf{e}_l}$, and the exchanged allocation contributes the other half. We keep each summand of (13.13) with the union of the network of R_n and the networks of the attached coefficients of the reconstruction. These are connected through the indices j_r, and Lemma 13.1 applies; no product of independent extensive scalars occurs.

Let $r_n = |R_n|_a / n!$, where $|R_n|_a$ is the norm with one index fixed, and $F(s) = \sum_{n \geq 2} r_n s^n$. Then $\sum_n r_n \rho^{n-2} = \epsilon$, so each $r_n \rho^{n-2} \leq \epsilon$, and for $0 \leq m \leq \rho/2$,

$$F'(m) \leq \epsilon\rho \sum_{n \geq 2} n 2^{-(n-1)} = 3\epsilon\rho. \tag{13.14}$$

At the radii $U_0 = 4K_*\rho$ and $V_0 = 4K_* s_J$ the majorant for the output-fixed rows satisfies $\mathfrak{M}_J(U_0, V_0) < \rho/2$; that is, every attached factor has majorant less than $\rho/2$. Fix an external input i. We choose the factor containing it and its position there, and we use the column of that factor and rows for all other factors. The column carries the factor v_{old}/v_i, while the norm of R_n with a fixed index has no volume factor, and the bound for the composed scalar with i fixed therefore contains $v_i/v_{\text{old}} \leq \Gamma_J$, where

$$\Gamma_J = \max_i \frac{v_i}{v_{\text{old}}} \leq M^{3J},$$

the maximum being over indices of θ and of q. Let $c_\theta(U, V)$ and $c_q(U, V)$ be the majorant series of Theorem 13.2(1) for the columns at a fixed coarse-field input and at a fixed input of q, in which the fixed input carries its radius U or V. In particular, $c_q(U, 0) = 0$. Then the series with the input i fixed is bounded by

$$\Gamma_J\, F'\big(\mathfrak{M}_J(U, V)\big)\big(c_\theta(U, V) + c_q(U, V)\big), \tag{13.15}$$

and by (13.11) the series without a distinguished input is no larger. The column majorants are finite at (U_0, V_0). Combining (13.14) and (13.15), we obtain the joint coefficient bound $C_{J,\rho,a}\epsilon$ on the enlarged product. It controls coefficients with a coarse-field index fixed, with an index of q fixed, and mixed ones, before any evaluation, and the constant $R(I)$ never enters.

A network of locations of the fine lattice has length δ^{-1} times its length in coarse units. Its weight at the rate a therefore gives the rate a/δ in coarse units, and the condition $C_{\text{geom}} b\delta \leq a$ provides the rate b on this part of the composed network. Each attached network of the reconstruction has its own rate, which is at least b. Their union provides the rates used for the supports and for the location sums in Step 1, while the output norm still has the rate a. Conversions between the fixed meshes cost factors depending only on J, not volume factors.

For real coarse fields and fluctuations, the value of this pullback is $R_{\text{sat}}(\mathscr{C}_{\text{raw}})$, by (13.7). Thus $R(e^{\Phi_{d_0,J}(\theta,q)})$ is the holomorphic pullback required in Lemma 12.16, and its hypothesis is verified on the whole range of the reconstruction, not only at its center. This proves the first assertion of (3).

Step 4: the inserted interaction at the radius ρ. We now show that the estimate of Lemma 12.16 holds at the radius ρ itself, rather than at a smaller radius. We fix p, every parameter internal to R, the balls, and the classical functions, and we substitute $q_i = \zeta\lambda_i w_i$. For each summand o of (13.13) with $|\eta| \geq 1$, where o specifies n, the indices j_r, the allocation, and the refinements of the networks, we set

$$a_o(\zeta) = \frac{\zeta^{|\eta|}}{n!} R_n(j_1, \dots, j_n) \prod_{r=1}^{n} p_{j_r;\xi_r\eta_r} \prod_l \lambda_{s(l)}^{\eta_l},$$

where $s(l)$ is the scale of the index l of q. Let Y_o be the set of boxes assigned to the external locations of o by the construction of Lemma 12.9, and set $(A_R)_Y(\theta, w; \zeta) = \sum_{o:\, Y_o = Y} a_o(\zeta)\theta^\nu w^\eta$. Then, exactly,

$$\sum_Y (A_R)_Y = R\big(e^{\Phi(\theta,(\zeta\lambda_i w_i)_i)}\big) - R\big(e^{\Phi(\theta,0)}\big),$$

supports winding around the torus being included; the construction is linear in R. We replace the supremum over w in the norm $\mathscr{S}_8$ of Theorem 12.7, at the coefficient radius ρ, by the sum of absolute coefficients at $|w_l| \leq p$. We then bound each monomial by the sum over its variables w_l of the corresponding terms with that variable fixed. The support-weight estimate (12.20) costs a fixed constant and a factor K^N for a monomial with N locations, which we absorb by multiplying all radii by K. This enlargement is included in K_*, so that $K\rho \leq U_0$ and $K|\zeta|p \leq V_0$ in the range used below. The sum over the location of the fixed index of w costs a constant depending on the density. By (13.14), (13.15), and the bound $c_q(U, V) \leq L_q V$ for $U \leq U_0$, with $L_q = \sup_{0<V\leq V_0} c_q(U_0, V)/V < \infty$, we obtain, after enlarging C_J,

$$\|\boldsymbol{A}_R(\zeta)\|_{\mathscr{S}_8} \leq \frac{C_J|\zeta|\epsilon p}{1 - C_J|\zeta|p} \qquad (C_J|\zeta|p < 1). \tag{13.16}$$

The bound is linear in ϵ, since it comes from the single tensor family (R_n). In the same way, repeating Step 6 of the proof of Theorem 12.13 at the coefficient radius ρ and with the boxes of Step 1, we obtain for the auxiliary interaction of Definition 12.15

$$\|\boldsymbol{H}(\zeta)\|_{\mathscr{S}_8} \leq \frac{C_J|\zeta|(p^3 + p)}{1 - C_J|\zeta|p},$$

since its classical terms contain at least three fluctuation variables of their own scale and its Haar density and determinant terms at least one.

Let $s_{\mathrm{aux}} > 0$ be such that $Ks_{\mathrm{aux}} \leq V_0$, and choose

$$0 < c_* < \min\Big\{s_{\mathrm{aux}}, \frac{1}{2C_J}, \frac{\eta_0}{12C_J}, \frac{\sigma_0}{2C_J}\Big\}, \qquad R_{\mathrm{aux}} = \frac{c_*}{p^3 + \epsilon p}, \tag{13.17}$$

where η_0 is the constant of Theorem 12.7, σ_0 is the constant of Theorem 12.13, and C_J is enlarged to dominate the constant of that theorem. Since $p \geq 1$, we have $R_{\mathrm{aux}} p \leq c_*$, so for $|\zeta| \leq R_{\mathrm{aux}}$ the substituted variables satisfy $|\zeta \lambda_i w_i| \leq c_* \leq s_{\mathrm{aux}}$, the denominators above are at least $1/2$, and, since $p^3 + p + \epsilon p \leq 2(p^3 + \epsilon p)$,

$$\|\boldsymbol{H}(\zeta) + s\boldsymbol{A}_R(\zeta)\|_{\mathscr{S}_8} \leq 2C_J R_{\mathrm{aux}}(p^3 + p + \epsilon p) \leq 4C_J c_* < \eta_0/2 \qquad (0 \leq s \leq 1).$$

Under (13.8) we have $tp \leq tp^3 \leq c_*/2$, which implies $C_J tp < 1/2$ and $C_J(tp^3 + tp)/(1 - C_J tp) \leq 2C_J c_* \leq \sigma_0$; these are the conditions (12.23). This proves the statements about the domain in (3).

By Theorem 12.7 and (13.16), the family $\boldsymbol{F}_s(\zeta) = D\boldsymbol{\mathcal{L}}(\boldsymbol{H}(\zeta) + s\boldsymbol{A}_R(\zeta))[\boldsymbol{A}_R(\zeta)]$ satisfies $\|\boldsymbol{F}_s(\zeta)\|_{\mathscr{S}_2} \leq C|\zeta|\epsilon p$ on the disc. Every term of H and of A_R vanishes at $\zeta = 0$. Their coefficients of order one in ζ are cubic polynomials in the fluctuation variables for the classical terms, and linear polynomials for the Haar density and determinant terms and for the terms from R. Since each of them changes sign under $w \mapsto -w$, while the reference product π_p^{all} is invariant under this reflection, each supported integral of first order vanishes. Terms with two interaction factors are of order at least two. Thus $\boldsymbol{F}_s(0) = \boldsymbol{F}_s'(0) = 0$ on every support. The family $\boldsymbol{F}_s(\zeta)/\zeta$ is holomorphic, vanishes at zero, and is bounded by $C\epsilon p$. Cauchy's formula then gives, for $0 < t \leq R_{\mathrm{aux}}/2$,

$$\|\boldsymbol{F}_s(t)\|_{\mathscr{S}_2} \leq C\epsilon p \, \frac{t^2}{R_{\mathrm{aux}}(1 - t/R_{\mathrm{aux}})} \leq \frac{2C}{c_*} t^2 \epsilon(p^4 + \epsilon p^2).$$

Integrating over $s \in [0,1]$, we find that the difference $\boldsymbol{\mathcal{D}}(t) = \boldsymbol{\mathcal{L}}(\boldsymbol{H}(t) + \boldsymbol{A}_R(t)) - \boldsymbol{\mathcal{L}}(\boldsymbol{H}(t))$ satisfies

$$\|\boldsymbol{\mathcal{D}}(t)\|_{\mathscr{S}_2} \leq C\epsilon p \frac{t^2}{R_{\mathrm{aux}}} \leq Ct^2 \epsilon(p^4 + \epsilon p^2), \tag{13.18}$$

at the coefficient radius ρ, without differentiation in the coarse field and without reduction of the radius. The dependence on ϵ^2 comes from the radius of the disc.

The auxiliary integral with the inserted interaction, the one without it, and the pullback at the zero-fluctuation reconstruction have the identities of Theorem 13.4 under gauge transformations, translations, the spatial inversion, and color conjugations. Subtracting, we find that the total difference of logarithms $\mathcal{D}_J R$ is translation invariant. Applying part (2) to (13.18), we obtain

$$\mathcal{B}_{\rho,a}(\mathcal{D}_J R) \leq C_T t^2 \epsilon(p^4 + \epsilon p^2).$$

The constant term of $\mathcal{D}_J R$ is not controlled by this norm; it is transferred to the normalization constant Λ, with densities written as $\exp(\Lambda - E/g^2 - R)$.

Step 5: the pullback at zero fluctuation and the newly generated effective interaction. Gauge invariance on B_ρ implies the local gauge-invariance identities required in

Theorem 11.10. Indeed, for a fixed real small field, we take a small gauge variation at one vertex whose endpoints remain in the chart and differentiate the exact invariance identity. The other symmetries and $DR(I) = 0$ are hypotheses. The background field $e^{\delta\overline{A}_{d_0,J}(\theta)}$ has the first-derivative bounds of Lemma 11.9, and by the choice of J its image under the reconstruction majorant on $\|\theta\|_\infty \le 2\rho$ lies in the ball of radius $\rho/2$, since $\delta Z(2\rho) \le \delta Z(4K_*\rho) \le \rho/4$. Theorem 11.10, applied to this representative at the rate $a < a_{\mathrm{cen}}/8$ chosen in Step 1, with $2\rho_0$ below the radius of Lemma 11.9, implies that

$$\mathcal{B}_{\rho,a}(\mathcal{P}_\delta R) \le q_P \epsilon.$$

Its value is the value of R at the zero-fluctuation reconstruction that appears in the integral. By gauge invariance on B_ρ, this identification holds even if the representative in the block axial gauge is not in B_ρ.

The newly generated effective interaction, from Theorem 12.13, is

$$Q^{\mathrm{w}}_{d_0,J} = \mathsf{B}_{\mathrm{ref}} - \Big(\sum_i f_i\Big) \log z_p - \log \Xi_{d_0,J}.$$

We bound its three terms in the norm $\mathcal{B}_{\rho,a}$ directly, using only rates at least b_*; the rate b_J of Theorem 12.13(5), which may depend on J, is not used.

The function $\mathsf{B}_{\mathrm{ref}}(c) = \sum_i \mathsf{B}_i(\widehat{c}_{i+1})$ depends on the coarse field; it is not a constant. Its coefficients of positive degree are obtained by composing the coefficient series of the terms B_i with the zero-fluctuation reconstruction $\theta \mapsto \widehat{c}_{i+1}$, that is, with the reconstruction of Step 3 of the proof of Theorem 13.2 at $q = 0$. The coefficients of positive degree of B_i, with one index fixed, are bounded by the logarithmic determinant series of the proof of Proposition 12.10 and by the bounds for the logarithmic Haar density and the minimizer of Lemma 12.3(3), at the rate of one averaging step. The reconstruction has coefficient bounds at the rate b_*, with its columns. The networks of the composed coefficients are connected through the indices of the fields $\widehat{c}_{i+1}$, and Lemma 13.1 applies. The conversion from the column normalization of these indices to the counting normalization of the scalar norm costs at most the factor $\Gamma_J \le M^{3J}$, as in Step 3. We conclude that there exists C_B, depending only on M, J, and the larger domains, such that

$$\frac{1}{k!}\|D^k_\theta \mathsf{B}_{\mathrm{ref}}(0)\|_{b_*,\mathrm{root}} \le C_B^{k+1} \qquad (k \ge 1),$$

uniformly in d_0 and m. Since $a < b_*$ and $\rho C_B \le 1/2$ by the choice of ρ in Step 1,

$$\mathcal{B}_{\rho,a}(\mathsf{B}_{\mathrm{ref}}) \le \sum_{k\ge 2} \rho^{k-2} C_B^{k+1} \le 2C_B^3.$$

For the logarithm of the fluctuation partition function, Step 4 gives an exact supported decomposition $\boldsymbol{H}(t)$ of $H_{d_0,J}$, at the coefficient radius ρ and with the boxes of Step 1,

with $\|\boldsymbol{H}(t)\|_{\mathscr{S}_8} \le \eta_0/2$ under (13.8). By Theorem 12.7, $-\log \Xi_{d_0,J} = \sum_Y \mathcal{L}_Y(\boldsymbol{H}(t))$ with $\|\boldsymbol{\mathcal{L}}(\boldsymbol{H}(t))\|_{\mathscr{S}_2} \le C\eta_0$, and each $\mathcal{L}_Y$ depends only on the coarse-field coordinates in Y. The integral (12.25), the action E_{d_0+J}, and the function $\mathsf{B}_{\mathrm{ref}}$ are invariant under the translations of the coarse lattice, by Theorem 12.13(6) and its proof, so $\log \Xi_{d_0,J}$ is translation invariant. Since $aC_{\mathscr{L}} \le 2$, the support condition of part (2) holds, and by part (2) we have

$$\mathcal{B}_{\rho,a}(\log \Xi_{d_0,J}) \le 9C\eta_0\rho^{-2}.$$

The Gaussian normalization $(\sum_i f_i)\log z_p$ is a constant; it has zero norm but remains in the density. Altogether $\mathcal{B}_{\rho,a}(Q^{\mathrm{w}}_{d_0,J}) \le C_G$ with $C_G = 2C_B^3 + 9C\eta_0\rho^{-2}$, uniformly in the side, in d_0, and in t, p satisfying (13.8). Taking the negative logarithm of the integral formula (13.5) at the parameter t, and separating $-\log A_{\mathrm{coord}}$, we obtain

$$R_+ = Q^{\mathrm{w}}_{d_0,J} + \mathcal{P}_\delta R + \mathcal{D}_J R.$$

Adding the three bounds proves (13.9). The symmetries of the integral give the stated symmetries of R_+, and the color argument of Step 5 of the proof of Theorem 13.4 shows that $DR_+(I) = 0$. This proves (3).

Step 6: one coupling threshold and the induction. Let E_* and $p(t) = A_{\mathrm{w}}\log(1/t)$ be as in (4). For $k > 0$ and $l \ge 0$ we have $t^k \log(1/t)^l \to 0$ as $t \downarrow 0$. There are finitely many strict requirements: the image conditions, the cluster conditions, the condition $t \le c_*/(2(p(t)^3 + E_* p(t)))$, the condition on $C_T t^2(p(t)^4 + E_* p(t)^2)$, the bound $tp(t) < s_J$, and $p(t) \ge p_0$. For $r \le \rho$, the bound $tp(t) < s_J$ implies $r + tp(t) < r_J$ and a positive denominator in $\mathfrak{M}_J(r, tp(t))$, by the inequalities for the enlarged product, and the intermediate maps have the margins already fixed. The additional inequality $\mathfrak{M}_J(r, tp(t)) < r/2$ is still required whenever the support-inclusion hypothesis is invoked; it is not implied by a coupling threshold independent of r. We choose $t_* > 0$ below the thresholds of all these requirements. For $\epsilon \le E_*$, condition (13.8) holds, and by the recursion we have

$$C_G + q_P\epsilon + C_T t^2 \epsilon(p^4 + \epsilon p^2) \le \frac{1-q_P}{2}E_* + q_P E_* + \frac{1-q_P}{2}E_* = E_*.$$

This proves (4). Starting with $R_0 = 0$, we construct the next effective interaction at the same ρ, a, and E_* from the auxiliary integral, the nonvanishing of its continuation, and this bound, and we repeat this for any finite number of steps. Where the support-inclusion hypothesis holds, Theorem 13.4 identifies each restricted density with the displayed expression, and the induction keeps every initial and pulled-back indicator. This proves (5). □

13.5 The conditional fluctuation potential

Dividing (13.5) by the integral over the fluctuations gives the conditional law of the normalized fluctuation variables given the coarse field. We show that its potential is

uniformly convex, which implies dimension-free concentration and covariance estimates. The heuristic reason is the following count of powers of the coupling. By stationarity and the normalization by the constrained Hessian,

$$\frac{F(z^* + gK^{-1/2}w) - F(z^*)}{g^2} = \frac{1}{2}|w|^2 + O(g|w|^3),$$

while the Haar density terms, the determinant terms, and the input effective interaction contribute $O(g|w|) + O(g^2|w|^2) + \cdots$. Substituting the fluctuations of coarser scales into the backgrounds of finer scales adds one factor $g_l w_l$ for each such variable and preserves these powers. Thus the Hessian of the interaction is small when tp is small, while the complex polymer expansion of Section 13.4 needs tp^3 small. On a larger product with radius $P = q_*/t$, in which only $tP = q_*$ is small, real convexity still holds although the polymer expansion is not available.

We first record some facts about probability measures on a fixed product of balls. Here the product does not move when parameters vary. Differentiation of a family of observables means differentiation of continuous functions with continuous derivatives on a neighborhood of the product, and the same formulas hold for families with dominated difference quotients and integrable derivatives.

Lemma 13.6. *Let Ω be a fixed product of closed balls, let $I \subset \mathbb{R}$ be an interval, and let $(V_u)_{u\in I}$ be real functions such that $V_u(w)$ and $\partial_u V_u(w)$ are continuous on $I \times N$ for a neighborhood N of Ω. Let $\mathbb{E}_u$, Cov_u refer to the probability measure on Ω with density proportional to e^{-V_u}, and let F_u be a family of observables as above.*

1. *We have*

$$\partial_u \mathbb{E}_u F_u = \mathbb{E}_u \partial_u F_u - \mathbb{E}_u(F_u \partial_u V_u) + \mathbb{E}_u F_u \, \mathbb{E}_u \partial_u V_u = \mathbb{E}_u \partial_u F_u - \mathrm{Cov}_u(F_u, \partial_u V_u), \tag{13.19}$$

   and $\partial_u \log \int_\Omega e^{-V_u}\, dw = -\mathbb{E}_u \partial_u V_u$.

2. *Let $m_u = \mathbb{E}_u w$, $\xi = w - m_u$, $\Sigma_u = \mathbb{E}_u \xi\xi^{\mathsf{T}}$, and $Q_u = \partial_u V_u - \mathbb{E}_u \partial_u V_u$. Then*

$$\partial_u \Sigma_u(i,j) = -\mathbb{E}_u \xi_i \xi_j Q_u, \tag{13.20}$$

   and

$$\mathbb{E}\xi_i\xi_j Q_u = \mathrm{Cov}(\xi_i\xi_j, Q_u) = \mathrm{Cov}(\xi_i, \xi_j Q_u) = \mathrm{Cov}(\xi_j, \xi_i Q_u), \tag{13.21}$$

   including the case $i = j$.

3. *For a matrix A, $\|A\|_{\mathrm{op}} \le (\|A\|_{\mathrm{row}}\|A\|_{\mathrm{col}})^{1/2}$, where $\|A\|_{\mathrm{row}} = \sup_i \sum_j |A_{ij}|$ and $\|A\|_{\mathrm{col}} = \sup_j \sum_i |A_{ij}|$. If an array (b_{eij}) satisfies $\sup_e \sum_{i,j} |b_{eij}| \le C$ and $\sup_{i,j} \sum_e |b_{eij}| \le C$, then $\|\sum_e a_e b_{e\cdot\cdot}\|_{\mathrm{HS}} \le C\|a\|_2$ for every array (a_e).*

Proof. On a compact interval of u, the functions e^{-V_u}, F_u, and their u-derivatives are bounded on Ω. The fundamental theorem of calculus applied to difference quotients,

followed by integration over the fixed set Ω, differentiates the integrals without a boundary term, and the quotient rule gives (13.19). Taking $F = 1$ before normalizing gives the formula for the logarithm. No integration by parts in w is used. For families with dominated difference quotients, the same argument applies with the integrable dominating function.

For (2), the explicit u-derivative of the observable $\xi_i\xi_j$ has expectation $-(\partial_u m_i)\mathbb{E}\xi_j - (\partial_u m_j)\mathbb{E}\xi_i = 0$. Then (13.19) implies (13.20). Since ξ_i, ξ_j, and Q_u are centered, expanding the definition of the covariance gives (13.21).

For (3), by the Cauchy–Schwarz inequality, we have

$$\sum_i \Big|\sum_j A_{ij}v_j\Big|^2 \le \Big(\sup_i \sum_j |A_{ij}|\Big) \sum_{i,j} |A_{ij}||v_j|^2 \le \|A\|_{\mathrm{row}}\|A\|_{\mathrm{col}} \sum_j |v_j|^2.$$

Regard (b_{eij}) as a matrix T with rows indexed by e and columns by pairs (i,j). Since its row sums and column sums are at most C, we have $\|T\|_{\mathrm{op}} \le C$, and the same holds for its transpose. The array $\sum_e a_e b_{e\cdot\cdot}$ is $T^{\mathsf{T}}a$, and its Hilbert–Schmidt norm is $\|T^{\mathsf{T}}a\|_2 \le C\|a\|_2$. This uses the column sums; a bound for each output e separately would not suffice. □

The next lemma bounds conditional means along an interpolation that starts from a measure with zero means. We use it twice below.

Lemma 13.7. *Let Ω be a fixed product of balls in the scalar coordinates w_i, and for $s \in [0,1]$ let π_s be the probability measure on Ω with density proportional to e^{-V_s}, where V_s satisfies the hypotheses of Lemma 13.6 in s, with continuous derivatives $\partial_{w_j}\partial_s V_s$. Let $m(s) = \max_i |\mathbb{E}_{\pi_s} w_i|$. Assume that*

1. *$m(0) = 0$;*
2. *there is a matrix $L = (L_{ij})$ with nonnegative entries such that, for every s and i, $|\mathrm{Cov}_{\pi_s}(w_i, \partial_s V_s)| \le \sum_j L_{ij}\|\partial_{w_j}\partial_s V_s\|_{L^2(\pi_s)}$, with $\sup_i \sum_j L_{ij} \le \Lambda_0$;*
3. *for every s, $\sup_j \|\partial_{w_j}\partial_s V_s\|_{L^2(\pi_s)} \le \beta(1 + m(s)^2)$.*

If $2\Lambda_0\beta < 1$, then $m(s) < 1$ and $m(s) \le 2\Lambda_0\beta s$ for every $s \in [0,1]$.

Proof. By (13.19) with $F = w_i$, we have $d\mathbb{E}_{\pi_s} w_i/ds = -\mathrm{Cov}_{\pi_s}(w_i, \partial_s V_s)$. Since the entries of L are nonnegative, (2) and (3) give $|d\mathbb{E}_{\pi_s} w_i/ds| \le \Lambda_0\beta(1 + m(s)^2)$. Integrating from 0 and using (1), we obtain $m(s) \le \Lambda_0\beta \int_0^s (1 + m(v)^2)\, dv$. The function m is continuous. If m reached the value one, then at the first such time $s_1 \le 1$ we would have $1 = m(s_1) \le 2\Lambda_0\beta s_1 < 1$, which is impossible. Thus $m < 1$ on $[0,1]$, and inserting this in the integral, we obtain $m(s) \le 2\Lambda_0\beta s$. □

Theorem 13.8. *Fix the parameters and the coupling threshold of Theorem 13.5, and an effective interaction R as in part (4) of that theorem, with $\mathcal{B}_{\rho,a}(R) \le E_*$. Consider a renormalization group step whose restriction event satisfies both hypotheses of Definition 13.3. Then the following statements hold for every $c = e^{\theta} \in B_r$, every depth, and every side $m \ge 4$.*

1. *The conditional law of the normalized fluctuation variables given the coarse field is the conditional fluctuation kernel*

$$k(c, dw) = \frac{\mathbf{1}_{\Omega_p}(w) e^{-V_c(w)}\, dw}{\int_{\Omega_p} e^{-V_c(w)}\, dw}, \qquad V_c = \tfrac{1}{2}|w|^2 + H_{d_0,J} + R(e^{\varphi_\theta(w)}) - R(e^{\varphi_\theta(0)}).$$

   The domain is the fixed product of three-dimensional balls, independent of c. The conditional fluctuation potential V_c contains all classical terms, the differences of Haar densities, the dependence of the determinants of finer scales on coarser fluctuations, and the full chain rule for the input effective interaction. If the whole hierarchy of restrictions is conditioned on this coarse field, the conditional law of the fluctuations of this step is the same kernel: the restrictions at coarser scales depend only on c and cancel in the normalization.

2. *There exist $b, C > 0$, depending only on M, J, ρ, a, E_*, and the larger domains, such that the entrywise envelope $\mathcal{H}_{ij} = \sup_{w\in\Omega_p} |\partial_{w_i}\partial_{w_j}(V_c - |w|^2/2)|$ satisfies*

$$\|\mathcal{H}\|_{b,\mathrm{row+col}} \le \frac{C[tp + t^2(1+E_*)]}{(1-Ctp)^3} \le \frac{1}{2}.$$

   In particular $D_w^2 V_c \ge I/2$ on the closed product, and V_c is smooth on a neighborhood of it. For the Neumann realization on the convex product Ω_p, every real $F \in H^1(k)$ satisfies

$$\mathrm{Var}_k F \le 2\mathbb{E}_k|\nabla F|^2, \qquad \mathrm{Ent}_k(F^2) \le 4\mathbb{E}_k|\nabla F|^2,$$

   and for real or complex F, G, with the covariance taken bilinear,

$$|\mathrm{Cov}_k(F,G)| \le C\sum_{i,j} e^{-b'\,\mathrm{dist}(i,j)} \|\partial_i F\|_2 \|\partial_j G\|_2$$

   for a fixed $b' > 0$. The constants do not depend on the dimension, on the radii of the balls, on the side, on the depth, or on the coupling satisfying the conditions of Theorem 13.5.

3. *Choose $q_* > 0$ inside the larger domain of the variables q and put $P(t) = q_*/t$. On $\Omega_{P(t)}$ define the same smooth potential, whether or not this larger product satisfies the support-inclusion hypothesis. If*

$$\frac{C[q_* + t^2(1+E_*)]}{(1-Cq_*)^3} \le \frac{1}{2},$$

   then for each $s \in [0,1]$ the potential $V^P_{s,c} = |w|^2/2 + s(V_c - |w|^2/2)$ has Hessian at least $I/2$ on $\Omega_{P(t)}$ and satisfies the inequalities of (2). If $\pi^P_{s,c}$ denotes the normalized measure, then, for a sufficiently small fixed t_*,

$$\sup_{s,c,i} |\mathbb{E}_{\pi^P_{s,c}} w_i| \le Ct(E_* + 2), \qquad \|w_i\|_{L^4(\pi^P_{s,c})} \le C\big(1 + \sup_j |\mathbb{E}_{\pi^P_{s,c}} w_j|\big).$$

Proof. We use the coefficient bounds of Theorem 13.2 and estimate the potential directly, before taking any logarithm of a partition function.

Step 1: the conditional kernel. In the density of Theorem 13.4, the factors A_{coord} and $\exp[-E_{d_0+J}/g_J^2 - \mathsf{B}_{\mathrm{ref}} - R_{\mathrm{ref}}]$ do not depend on w. Dividing the joint density by its positive integral over w cancels exactly these factors, and we obtain $k(c, dw)$. The domain is the product of balls of the normalized fluctuation variables, which does not depend on c. The terms $\mathsf{B}_i(c_{i+1}) - \mathsf{B}_i(\widehat{c}_{i+1})$ in $H_{d_0,J}$ keep the dependence of the determinants and of the Haar densities at the minimizers on the coarser fluctuations, and its other terms are the differences of the Haar densities. Thus $k(c, dw)$ is the full conditional law, not a perturbation of a Gaussian reference measure.

The same calculation can be expressed without choosing a version of the conditional law. To see this, we multiply the joint density by a bounded Borel function of c and a bounded Borel function of w and integrate first in w and then in c. The result is the joint integral of Theorem 13.4. This verifies the identity for the conditional kernel, and its pointwise version is the positive continuous integral. In a complete hierarchy, every restriction coarser than this step is a function of c and of its further averages. Such factors multiply both the numerator and the normalization of the conditional law by the same function of c and cancel. They do not cancel from the unnormalized law of the coarse field. This proves (1).

Step 2: powers of the coupling. Let $U = V_c - |w|^2/2$. We fix the coarse field in its larger ball but do not freeze the coarser fluctuations. At scale i the classical Taylor remainder has degree $s \geq 3$ in the variables of its own scale, and the normalization of the action gives the factor g_i^{s-2}. Substituting the coarser fluctuations into the background adds r_l fluctuation variables from each coarser scale $l > i$. With total degree $n = s + \sum_{l>i} r_l$ in w, the coupling factor is exactly

$$g_i^{s-2} \prod_{l>i} g_l^{r_l} = t^{n-2} \lambda_i^{s-2} \prod_{l>i} \lambda_l^{r_l}, \qquad n \geq 3. \tag{13.22}$$

No negative power of a ratio $\lambda_i \leq 1$ occurs, since $s \geq 3$. Centering at the minimizer removes the term of degree one in the variables of its own scale. The normalization by the positive square root of the Hessian removes the quadratic term, at every value of the coarser fluctuations. Thus no classical term of total degree one or two in w survives the substitution.

Each difference of Haar densities and each difference of determinant terms has positive total degree n in w and the coupling factor t^n. The composed difference of the input effective interaction obeys the same rule, with amplitude bounded by $C\epsilon \leq CE_*$. The joint coefficient bounds, evaluated on a slightly larger coarse-field radius and a slightly larger product of the variables q, therefore give the following bounds for the absolute

coefficients with one index fixed:

contribution	degree in w	coefficient bound
classical remainder	$n \geq 3$	$C^n t^{n-2}$
Haar density and determinant differences	$n \geq 1$	$C^n t^n$
difference of $R \circ e^{\varphi}$	$n \geq 1$	$C^n E_* t^n$.

(13.23)

These bounds include every substitution of a background and every derivative of the matrices S_i. The fixed ratios λ_i and the fixed radii are included in C.

Step 3: the Hessian. We take two derivatives in w and sum their weighted rows. The multiplicities of the two derivatives are at most $n(n-1)$, and the other coordinates have absolute value at most p. With $x = Ctp < 1$, the classical sum is

$$\sum_{n \geq 3} n(n-1) C^n t^{n-2} p^{n-2} = 2C^2 \big\{ (1-x)^{-3} - 1 \big\} \leq \frac{6C^3 tp}{(1-Ctp)^3},$$

where we used $1 - (1-x)^3 \leq 3x$. The sums for the Haar density and determinant terms and for the terms from R are bounded by

$$(1 + E_*) \sum_{n \geq 2} n(n-1) C^n t^n p^{n-2} = \frac{2C^2 t^2 (1 + E_*)}{(1-Ctp)^3},$$

and the terms of degree one contribute nothing. The network weights give the decay in the distance between the two remaining indices, and by symmetry the column sums have the same bound. These sums of absolute coefficients, evaluated at the radius p, are uniform in $w \in \Omega_p$, so they bound the entrywise envelope $\mathcal{H}_{ij}$; this is the bound (13.3) of Lemma 13.1 applied to the first derivatives of U. Increasing C, we obtain the stated bound for $\|\mathcal{H}\|_{b,\mathrm{row+col}}$. For the part coming from R, both terms of

$$D^2(R \circ e^{\varphi}) = D^2(R \circ \exp)(\varphi)[D\varphi, D\varphi] + D(R \circ \exp)(\varphi)[D^2\varphi]$$

are included in its series of degree n. In particular, the first derivative of R multiplying the second derivative of the reconstruction is kept.

We reduce the coupling threshold of Theorem 13.5 so that this bound is at most $1/2$. By Lemma 13.6(3), the operator norm is bounded by the row and column sums, and we obtain $D_w^2 V_c \geq I/2$. The margin in the coefficient bounds provides a smooth extension of the potential to an open neighborhood of the closed product.

Step 4: the inequalities of the Neumann realization. The balls are convex, have dimension three, and have radii independent of c, and the potential has the smoothness and the lower bound on the Hessian just proved. By Theorem 5.4 with $m = 1/2$, we have the variance constant 2 and the entropy constant 4, including the corners of the product. To

check its spatial hypothesis, recall that the weighted row-plus-column norm of $\mathcal{H}$ at the rate b is at most $1/2$. The hypothesis concerns the operator norms of the mixed blocks of the three-dimensional balls, so we group the scalar indices into their three-component blocks. The operator norm of a block is at most the sum of the absolute values of its scalar entries. Hence the weighted row and column sums of the operator norms of the mixed blocks are at most three times the scalar sums, that is, at most $3/2$. We set $b' = b/8$. For $z \geq 0$, convexity gives

$$e^{z/8} - 1 \leq \tfrac{1}{8}(e^z - 1) \leq \tfrac{1}{8} e^z.$$

Both block sums with the weight $e^{b' \operatorname{dist}(x,y)} - 1$ are therefore at most $3/16$, which is below $m/2 = 1/4$; the diagonal Gaussian blocks have no mixed contribution. Thus the spatial hypothesis of Theorem 5.4, which involves the suprema over Ω_p of the mixed blocks, holds at the rate b'. Expanding the block gradient norms into their scalar components, we obtain the stated scalar covariance inequality, with a fixed change of its constant. Distinct blocks with the same location can be treated by separating them by an arbitrarily small distance, applying the inequality at a slightly smaller rate, and passing to the limit; the strict inequality $3/16 < 1/4$ leaves room for this perturbation. All sums are finite at this stage, and the weighted bounds remain uniform in the limit. This proves (2).

Step 5: the larger product. The proof of the Hessian bound evaluated the majorant series at the radius tp of the variables q; it did not use smallness of tp^3. Let $0 < q_* < s_J$ lie in the same domain of the variables q, and set $P = q_*/t$. Then $\rho + q_* < r_J$, the denominator of the reconstruction majorant is positive, and all intermediate margins follow from the enlarged product. The gauge-transformed reconstruction lies in $B_{\rho/2}$, where the input effective interaction is defined, and no inclusion in D_r is needed. If desired, we reduce the coupling threshold so that $tp(t) < q_*$; then this product contains Ω_p. Repeating the two sums with p replaced by P, we obtain

$$\|\mathcal{H}^P\|_{b,\mathrm{row+col}} \leq \frac{C[q_* + t^2(1 + E_*)]}{(1 - Cq_*)^3},$$

for the entrywise envelope $\mathcal{H}^P$ of $D_w^2 U$ on Ω_P. The potential on this product is the smooth potential of the auxiliary integral, and no indicator of D_r is inserted. Under the displayed condition, $|w|^2/2 + sU$ has Hessian at least $I/2$ for every $s \in [0,1]$, its mixed blocks satisfy the same bound, and Step 4 applies on this product. No complex expansion at the radius P is used.

Step 6: the means on the larger product. It remains to bound the means without a loss of a factor P^2. By the concentration statement of Theorem 5.4, we have for each scalar coordinate $\mathbb{P}(|w_i - \mathbb{E}w_i| > u) \leq 2e^{-u^2/4}$, hence $\mathbb{E}|w_i - \mathbb{E}w_i|^4 \leq \int_0^\infty 8u^3 e^{-u^2/4}\,du \leq C$. Thus, with $m(s) = \max_i |\mathbb{E}_{\pi_{s,c}^P} w_i|$,

$$\|w_i\|_{L^4(\pi_{s,c}^P)} \leq C(1 + m(s)), \tag{13.24}$$

and the bound for three-component blocks changes only C.

If we differentiate a classical monomial of degree n in U with respect to one fluctuation variable, there remain $n-1 \geq 2$ fluctuation factors. We place two of them in L^4 and the others in L^∞, and Hölder's inequality bounds their product in L^2 by $C(1+m(s)^2)P^{n-3}$. The coupling factor becomes

$$t^{n-2}P^{n-3} = tq_*^{n-3}. \tag{13.25}$$

The absolute coefficient series converges at q_*, and the margin in the radius absorbs the multiplicities of the derivative and the choice of the two factors placed in L^4. The derivatives of the Haar density and determinant terms and of the terms from R of degree n give $t^n P^{n-1} = tq_*^{n-1}$, hence the bound $Ct(1+E_*)$. It follows that

$$\sup_j \|\partial_{w_j} U\|_{L^2(\pi^P_{s,c})} \leq Ct(1+E_*+m(s)^2) \leq Ct(1+E_*)(1+m(s)^2). \tag{13.26}$$

These bounds hold with one index of w fixed, with a profile that decays exponentially in the distance to that index. Summing the covariance kernel of Step 4 over its second index costs only a constant depending on the density. We apply Lemma 13.7 to $V_s = |w|^2/2 + sU$, for which $\partial_s V_s = U$. At $s=0$ the measure is the even product of truncated Gaussian measures, and $m(0)=0$. Hypothesis (2) of the lemma holds with the nonnegative kernel $L_{ij} = Ce^{-b'\operatorname{dist}(i,j)}$ of the covariance bound of Step 4, whose row sums are at most a constant Λ_0 depending only on the fixed data, and hypothesis (3) holds with $\beta = Ct(1+E_*)$ by (13.26). After one fixed reduction of the coupling threshold, $2\Lambda_0\beta < 1$, and the lemma implies that $m(s) \leq Ct(1+E_*) \leq Ct(E_*+2)$, uniformly in s, c, and the volume. Together with (13.24) this proves (3). All bounds are bounds at a fixed coordinate, and the total dimension does not appear. □

13.6 Conditional means and coarse-field derivatives

We now study how the conditional fluctuation kernel depends on the coarse field. The main point is that its means are small, of order $t\|\theta\|_\infty$, even though the kernel is interacting. At the identity coarse field they vanish by the global color symmetry. We also control the derivative of the conditional fluctuation covariance, by a three-point decay estimate, and the first two Taylor coefficients of the reconstruction at the conditional mean, which are used in the later reconstruction estimates. We first prove the three-point estimate under abstract hypotheses.

Lemma 13.9. *Let Ω be a fixed product of balls of radii at most p, $p \geq 1$, in scalar coordinates w_i, and let V_θ be potentials depending on parameters θ_u, with the probability measures of density proportional to e^{-V_θ} on Ω, along the segment $\{s\theta : 0 \leq s \leq 1\}$. Let Σ_θ be the covariance matrix*

of w. Suppose that $D_0 \ge 2$, $A_{\mathrm{sc}} \ge 0$, and that along the segment, for every location x and every $R \ge 0$,

$$\#\{i : \operatorname{dist}(i,x) \le R\} + \#\{u : \operatorname{dist}(u,x) \le R\} \le D_0(1+R)^3,$$
$$|\operatorname{Cov}(F,G)| \le \sum_{i,j} L_{ij}\|\partial_i F\|_2 \|\partial_j G\|_2, \qquad L_{ij} \le D_0 e^{-\operatorname{dist}(i,j)/D_0},$$
$$\sup_{\Omega} |\partial_{w_i}\partial_{\theta_u} V| \le A_{\mathrm{sc}} D_0 e^{-\operatorname{dist}(i,u)/D_0}.$$

Let $\tau_3(i,j,u)$ be the sum of the two smallest of the distances $\operatorname{dist}(i,j)$, $\operatorname{dist}(i,u)$, and $\operatorname{dist}(j,u)$. Then, for an absolute constant C,

$$|\partial_{\theta_u}\Sigma_\theta(i,j)| \le C p A_{\mathrm{sc}} D_0^6 e^{-\tau_3(i,j,u)/(4D_0)}, \qquad \|\Sigma_\theta - \Sigma_0\|_{\mathrm{op}} \le C p A_{\mathrm{sc}} D_0^{14} \|\theta\|_\infty.$$

Proof. We write $D = D_0$ and $A = A_{\mathrm{sc}}$. The counting hypothesis implies, for each fixed $c > 0$,

$$\sup_x \sum_i e^{-\operatorname{dist}(i,x)/(cD)} + \sup_x \sum_u e^{-\operatorname{dist}(u,x)/(cD)} \le C_c D^4. \tag{13.27}$$

To prove it, let $N_x(R) = \#\{i : \operatorname{dist}(i,x) \le R\}$ and use $e^{-r/(cD)} = (cD)^{-1}\int_r^\infty e^{-R/(cD)}\,dR$; summing this identity over i bounds the first sum by

$$\frac{1}{cD}\int_0^\infty e^{-R/(cD)} D(1+R)^3\,dR \le C_c D^4,$$

and the sum over u is bounded in the same way. The argument also works on small periodic tori. By the triangle inequality,

$$\sum_l e^{-\operatorname{dist}(i,l)/D - \operatorname{dist}(l,u)/D} \le C D^4 e^{-\operatorname{dist}(i,u)/(2D)}; \tag{13.28}$$

indeed, we extract the exponential on the right using half of the two distances and sum the remaining factor $e^{-\operatorname{dist}(i,l)/(2D)}$ by (13.27).

Since each coordinate and its mean have absolute value at most p, we have $|\xi_i| \le 2p$, where $\xi = w - \mathbb{E}w$. Moving between two points of the convex product along a segment and using the bound on the mixed derivatives, we find that the oscillation of $\partial_{\theta_u} V$ is at most $2pAD\sum_l e^{-\operatorname{dist}(l,u)/D}$. Subtracting the expectation does not increase this bound, and the centered potential derivative $Q_u = \partial_{\theta_u} V - \mathbb{E}\partial_{\theta_u} V$ satisfies

$$\|Q_u\|_\infty \le C p A D^5, \qquad |\partial_l Q_u| \le A D e^{-\operatorname{dist}(l,u)/D}. \tag{13.29}$$

By (13.20), $\partial_{\theta_u}\Sigma(i,j) = -\mathbb{E}\xi_i\xi_j Q_u$, and we bound this by each of the three covariances in (13.21). For the first, the gradient of $\xi_i\xi_j$ is $\partial_l(\xi_i\xi_j) = \delta_{li}\xi_j + \delta_{lj}\xi_i$. Using $\|\xi_i\|_2 \le 2p$, the amplitudes D and AD of the two kernels, and (13.28), we obtain

$$|\mathbb{E}\xi_i\xi_j Q_u| \le C p A D^6 \big\{ e^{-\operatorname{dist}(i,u)/(2D)} + e^{-\operatorname{dist}(j,u)/(2D)} \big\}.$$

For the second, the gradient of ξ_i is the coordinate vector at i, and $\partial_l(\xi_j Q_u) = \delta_{lj} Q_u + \xi_j \partial_l Q_u$. The first term costs $L_{ij}\|Q_u\|_2 \le CpAD^6 e^{-\operatorname{dist}(i,j)/D}$, and the second costs the convolution (13.28) times $2pAD^2$. We obtain

$$|\mathbb{E}\xi_i\xi_j Q_u| \le CpAD^6\big\{e^{-\operatorname{dist}(i,j)/(2D)} + e^{-\operatorname{dist}(i,u)/(2D)}\big\}.$$

For the third, the coordinate vector is at j, the gradient of the product is $\delta_{li} Q_u + \xi_i \partial_l Q_u$, and the two resulting terms are bounded by

$$CpAD^6\big\{e^{-\operatorname{dist}(i,j)/(2D)} + e^{-\operatorname{dist}(j,u)/(2D)}\big\}.$$

Using only the first of these three bounds would not control the separation of i from j.

We order the three distances as $x \le y \le z$. The smallest of the three braces is at most $e^{-y/(2D)} + e^{-z/(2D)} \le 2e^{-y/(2D)}$, and since $\tau_3 = x + y \le 2y$, this is at most $2e^{-\tau_3/(4D)}$. This proves the bound on $\partial_{\theta_u}\Sigma(i,j)$.

Since τ_3 is the smallest length of the three trees with two edges on the points i, j, u, the factor $e^{-\tau_3/(4D)}$ is at most the sum of the three corresponding products of exponentials. For fixed i, summing one such tree over (j,u) removes its two leaves successively, and (13.27) gives a factor at most CD^8. Hence

$$\sup_i \sum_{j,u} |\partial_{\theta_u}\Sigma(i,j)| \le CpAD^{14}.$$

Integrating along the segment $s\theta$, $0 \le s \le 1$, and using $|\theta_u| \le \|\theta\|_\infty$, we bound the row sums of $\Sigma_\theta - \Sigma_0$. By symmetry in i, j, the column sums have the same bound, and Lemma 13.6(3) implies the operator bound. □

In the following theorem the conditional law is the kernel $k(c, dw)$ of Theorem 13.8 on the product Ω_p, and the derivatives are taken with the couplings, and hence the radii of the balls, held fixed. We write $\mathbb{E}_\theta$, $\operatorname{Cov}_\theta$ for this kernel at $c = e^\theta$.

Theorem 13.10. *Use the conditional fluctuation kernel of Theorem 13.8 with $p = A_{\mathrm{w}} \log(1/t)$, and reduce the coupling threshold once so that tp^3 is small. Let $m_\theta = \mathbb{E}_\theta w$, $\xi = w - m_\theta$, $\Sigma_\theta = \mathbb{E}_\theta \xi\xi^{\mathsf{T}}$, and $\varphi_\theta(w) = \Phi_{d_0,J}(\theta, (g_i w_i)_i)$. There exist $C, b > 0$ such that the following statements hold uniformly for all $\|\theta\|_\infty < r \le \rho$ for which the support-inclusion hypothesis holds, all depths, and all sides.*

1. *The conditional means satisfy*

$$\|m_\theta\|_\infty \le Ct\|\theta\|_\infty, \qquad \sup_e |\mathbb{E}_\theta \varphi_{\theta,e}(w) - \varphi_{\theta,e}(0)| \le Ct^2\|\theta\|_\infty,$$

and both vanish at $\theta = 0$ by the global color symmetry.

2. *For $0 \le s \le 1$ let $\dot V_s = dV_{s\theta}/ds$ on the same product. Then, uniformly in s,*

$$\sup_l \|\partial_{w_l} \dot V_s\|_{L^2(k(e^{s\theta},\cdot))} \le Ct\|\theta\|_\infty.$$

This bound holds at each fixed index of w, with the exponentially summable profile of the joint coefficient bounds; it is not a bound on a norm of the extensive function $\dot V_s$. It includes the classical, Haar density, and determinant terms of the potential, and the terms from R.

3. *For every family F_θ of observables on this fixed product satisfying the differentiability conditions stated before Lemma 13.6 (continuity with continuous derivatives on a neighborhood of the product, or dominated difference quotients with integrable derivatives), and every coordinate θ_u, the identities* (13.19), (13.20), *and* (13.21) *hold; in particular*

$$\partial_u \mathbb{E}_\theta F_\theta = \mathbb{E}_\theta \partial_u F_\theta - \mathrm{Cov}_\theta(F_\theta, \partial_u V_\theta), \qquad \partial_u \log \int_{\Omega_p} e^{-V_\theta}\, dw = -\mathbb{E}_\theta \partial_u V_\theta,$$

and $\partial_u \Sigma_\theta(i,j) = -\mathbb{E}_\theta \xi_i \xi_j Q_u$ with $Q_u = \partial_u V_\theta - \mathbb{E}_\theta \partial_u V_\theta$. The domain does not move, so there is no boundary term.

4. *The hypotheses of Lemma 13.9 hold with a fixed D_0 and with $A_{\mathrm{sc}} = Ctp^2$; more precisely,*

$$\sup_{\Omega_p} |\partial_{w_i} \partial_{\theta_u} V_\theta| \le Ctp^2 e^{-b\,\mathrm{dist}(i,u)},$$
$$\|\Sigma_\theta - \Sigma_0\|_{b',\mathrm{row+col}} \le Ctp^3\|\theta\|_\infty, \qquad \|\Sigma_\theta\|_{\mathrm{op}} \le 2,$$

for a fixed $b' > 0$. Here $\Sigma_0 = \Sigma_{\mathrm{sp}} \otimes I_3$ is the covariance of the conditional fluctuation kernel at the identity coarse field; this kernel is interacting and truncated, and Σ_{sp} includes all its spatial covariances.

5. *Put $g = g_0$, $\mathsf{E}_\theta = g^{-1} D_w \varphi_\theta(m_\theta)$, and $\mathsf{B}_\theta = (2g^2)^{-1} D_w^2 \varphi_\theta(m_\theta)$. For every $w = m_\theta + \xi \in \Omega_p$,*

$$\varphi_\theta(m_\theta + \xi) = \varphi_\theta(m_\theta) + g\mathsf{E}_\theta \xi + g^2 \mathsf{B}_\theta(\xi,\xi) + \mathcal{R}_\theta(\xi), \qquad \|\mathcal{R}_\theta\|_\infty \le Cg^3 p^3.$$

Moreover $\|\mathsf{E}_\theta\|_{\mathrm{op}} \le C$, the scalar coefficients b_{eij} of B_θ satisfy

$$\sup_e \sum_{i,j} |b_{eij}| + \sup_i \sum_{e,j} |b_{eij}| \le C, \qquad \Big\| \sum_e a_e b_{e\cdot\cdot} \Big\|_{\mathrm{HS}} \le C\|a\|_2$$

for every array (a_e), and

$$\|\mathsf{E}_\theta - \mathsf{E}_0\|_{\mathrm{op}} + \|\mathsf{B}_\theta - \mathsf{B}_0\|_{\mathrm{abs,root}} \le C\|\theta\|_\infty,$$
$$\|\mathsf{E}_\theta \Sigma_\theta \mathsf{E}_\theta^{\mathsf{T}} - \mathsf{E}_0 \Sigma_0 \mathsf{E}_0^{\mathsf{T}}\|_{\mathrm{op}} \le C\|\theta\|_\infty (1 + tp^3), \qquad \sup_e |\mathbb{E}_\theta \mathsf{B}_{\theta,e}(\xi,\xi)| \le C\|\theta\|_\infty,$$

where $\|\cdot\|_{\mathrm{abs,root}}$ is the sum of absolute coefficients with one index fixed. At $\theta = 0$ the matrix E_0 is scalar in the color components, and the contraction of the color-equivariant quadratic tensor B_0 with Σ_0 vanishes.

6. *For the unnormalized integral of one step,*

$$\widetilde{F}(\theta) = A_{\mathrm{coord}} e^{-E_{d_0+J}(e^\theta)/g_J^2 - \mathsf{B}_{\mathrm{ref}}(\theta) - R_{\mathrm{ref}}(\theta)} \int_{\Omega_p} e^{-V_\theta(w)}\, dw,$$

we have

$$D \log \widetilde{F} = -D\big(E_{d_0+J}/g_J^2 + \mathsf{B}_{\mathrm{ref}} + R_{\mathrm{ref}}\big) - \mathbb{E}_\theta DV_\theta.$$

The derivatives of the terms evaluated at the zero-fluctuation reconstruction disappear only from normalized conditional expectations.

The constants depend only on M, J, ρ, a, E_*, A_{w}, the ratios of the couplings, and the fixed domains; they are independent of the side, L, γ, the depth, and the coarse field satisfying the hypotheses. Converting volume-normalized column bounds into bounds for operators in counting coordinates costs a fixed factor depending on J.

Proof. We use the coefficient bounds of Theorem 13.2 throughout; the conditional law is always the kernel on Ω_p of Theorem 13.8. We first prove (3) and (4), then the bounds on the means and on the derivative of the potential by Lemma 13.7, and finally the statements about the reconstruction.

Step 1: differentiation and the mixed derivatives. With the couplings, and hence the radii of the balls, fixed, the potential V_θ and its derivatives in θ are continuous on a neighborhood of the closed product, and Lemma 13.6 applies and proves (3).

We differentiate the series of (13.23) once in a coordinate θ_u of the coarse field and once in a fluctuation variable w_i. A derivative in θ on the larger product of coarse fields preserves the powers of the coupling and the coefficient bounds. A classical term of degree n leaves $n-1$ fluctuation factors after the derivative in w_i, so its sum is bounded by

$$Ctp^2 \sum_{n\geq 3} n(Ctp)^{n-3} \leq \frac{Ctp^2}{(1-Ctp)^2}.$$

The Haar density and determinant terms and the terms from R give at most $C(1+E_*)t/(1-Ctp)^2$, which is absorbed in the same bound since E_* is fixed and $p \geq 1$. Keeping both indices i and u in the network of each coefficient gives the factor $e^{-b\,\mathrm{dist}(i,u)}$. Reducing the coupling threshold so that Ctp is below a fixed margin, we obtain

$$\sup_{\Omega_p} |\partial_{w_i} \partial_{\theta_u} V_\theta| \leq Ctp^2 e^{-b\,\mathrm{dist}(i,u)}. \tag{13.30}$$

The Gaussian term contributes nothing to this derivative in θ, taken with all couplings fixed, since the normalization by the constrained Hessian made it equal to $|w|^2/2$ for every coarse field.

The density of the indices and the covariance bound of Theorem 13.8 satisfy the hypotheses of Lemma 13.9 with a fixed D_0 depending only on the fixed data, and by

(13.30) we may take $A_{\mathrm{sc}} = Ctp^2$. The proof of that lemma is linear in A_{sc}, and it yields the bound $Ctp^3 e^{-b_1\tau_3}$ for the derivative of Σ. We keep an exponential weight in the final sums. Since $\operatorname{dist}(i,j) \le \tau_3$ by the triangle inequality, we may choose $0 < b' < b_1/2$ and absorb $e^{b' \operatorname{dist}(i,j)}$ into half of the exponential of the tree. The remaining sums over two-edge trees converge by the fixed density. Integrating along the segment of coarse fields, we obtain

$$\|\Sigma_\theta - \Sigma_0\|_{b',\mathrm{row+col}} \le Ctp^3 \|\theta\|_\infty. \tag{13.31}$$

Applying the variance bound of Theorem 13.8 to $\sum_i v_i w_i$, we find $v^{\mathsf{T}}\Sigma_\theta v \le 2|v|^2$, hence $\|\Sigma_\theta\|_{\mathrm{op}} \le 2$. This proves (4), except for the structure of Σ_0, which is proved in Step 5.

Step 2: the conditional means. The concentration bound used in (13.24) implies, on Ω_p, that $\|w_i\|_{L^4} \le C(1 + m(s))$, where now $m(s) = \max_i |\mathbb{E}_{s\theta} w_i|$. We set $\dot V_s = \sum_u \theta_u \partial_{\theta_u} V_{s\theta}$ and take one derivative in w in its joint series. Every coarse field $s\theta$ lies in B_r, so the support-inclusion hypothesis holds along the whole segment. The larger product of coarse fields of Theorem 13.5 controls the derivative in θ, also when $\|\theta\|_\infty$ is close to r. The couplings, the radii of the balls, and the parameters internal to R are fixed while s varies. From a classical term of degree $n \ge 3$ we keep two of the remaining factors in L^4 and bound the other $n-3$ by p, which gives the coupling factor $t^{n-2}p^{n-3} = t(tp)^{n-3}$. The derivative in θ and its column sum give the factor $\|\theta\|_\infty$. Thus

$$\sup_j \|\partial_{w_j} \dot V_s\|_2 \le Ct\|\theta\|_\infty (1 + E_* + m(s)^2), \tag{13.32}$$

with all norms in $L^2(k(e^{s\theta}, \cdot))$. More precisely, we keep the index j of w and a scalar coarse-field index v in the network of each coefficient before contracting v with θ_v. The exponential weight then gives the factor $e^{b \operatorname{dist}(j,v)}$, while the two remaining fluctuation factors are estimated in L^4. The same summation shows that

$$\|\partial_{w_j} \dot V_s\|_{L^2(k(e^{s\theta},\cdot))} \le Ct(1 + E_* + m(s)^2) \sum_v e^{-b \operatorname{dist}(j,v)} |\theta_v|, \qquad \sup_j \sum_v e^{-b \operatorname{dist}(j,v)} \le C, \tag{13.33}$$

the last sum being over coarse-field indices of fixed density, bounded uniformly in the side by the count used in (13.27). The Haar density and determinant terms and the terms from R contribute $Ct(1 + E_*)\|\theta\|_\infty$, since one derivative in w of a term of degree $n \ge 1$ leaves $t^n p^{n-1} = t(tp)^{n-1}$. The margins in the coefficient bounds absorb all multiplicities of the derivatives. Summing the exponential covariance kernel against this profile costs only a constant depending on the density.

At the identity coarse field, a constant color conjugation rotates every three-component block of w and preserves the balls and the potential. A vector fixed by all such rotations vanishes, so $m(0) = 0$. We apply Lemma 13.7 to $V_s = V_{s\theta}$, with the nonnegative exponential kernel and the constant Λ_0 of the covariance bound of Theorem 13.8 and

with $\beta = Ct\|\theta\|_\infty(1+E_*)$, by (13.32) and $1+E_*+m^2 \le (1+E_*)(1+m^2)$. Since $\|\theta\|_\infty < \rho$, after one fixed reduction of the coupling threshold we have $2\Lambda_0\beta \le Ct\rho(E_*+2) < 1$, and the lemma implies that $m(s) < 1$ and $m(1) \le Ct\|\theta\|_\infty$. This proves the first bound in (1), for the interacting conditional means. Inserting $m(s) < 1$ in (13.32) and (13.33), and absorbing the factor E_*+2 into C, we obtain

$$\sup_j \|\partial_{w_j} \dot V_s\|_{L^2(k(e^{s\theta},\cdot))} \le Ct\|\theta\|_\infty \qquad (0 \le s \le 1),$$

which is (2). It uses the same coupling threshold as the bound on the means, and no norm of the extensive function $\dot V_s$.

Step 3: the mean of the reconstruction. Let $G_e(\theta, w) = \varphi_{\theta,e}(w) - \varphi_{\theta,e}(0)$. Its coefficients of degree n in w carry the factor t^n, and every term has positive degree in w. The same holds after a derivative in θ on the larger product. Its rows in the fluctuation variables are bounded by Ct on Ω_p. For $\dot G_e = dG_e(s\theta, w)/ds$, the expectation of the part of degree one is at most $Ct\|\theta\|_\infty m(s)$. For degrees at least two we keep two fluctuation factors in L^4 and bound the others by p. Summing the series with one index fixed, we obtain

$$|\mathbb{E}_{s\theta} \dot G_e| \le C\|\theta\|_\infty \big\{ tm(s) + t^2(1+m(s)^2) \big\}. \tag{13.34}$$

The covariance of G_e with $\dot V_s$ is at most $Ct^2\|\theta\|_\infty(1+E_*+m(s)^2)$, by the bound Ct on the rows of G_e, the profile (13.32), and the summable covariance kernel. By Step 2, $tm(s) \le Ct^2\|\theta\|_\infty$. Applying (13.19) to G_e, we find

$$\Big| \frac{d}{ds} \mathbb{E}_{s\theta} G_e(s\theta, w) \Big| \le Ct^2\|\theta\|_\infty.$$

At the identity coarse field $\Omega(0) = I$, and the reconstructed logarithm is color equivariant by Theorem 13.2(2). Its expectation is then a vector invariant under all rotations of the color components, which must vanish. Integrating from 0 to 1 proves the second bound in (1). The reference value $\varphi_\theta(0) = \delta\overline{A}_{d_0,J}(\theta)$ is the zero-fluctuation reconstruction.

Step 4: the expansion at the conditional mean. Since each block of the mean lies in its closed convex ball, the segment $m_\theta + v\xi$, $0 \le v \le 1$, lies in Ω_p whenever $m_\theta + \xi = w \in \Omega_p$. Three derivatives in w of the reconstruction have row sums at most Cg^3, since the ratios g_i/g are fixed. By Taylor's formula, we have

$$\mathcal{R}_\theta(\xi) = \frac{1}{2} \int_0^1 (1-v)^2 D_w^3 \varphi_\theta(m_\theta + v\xi)[\xi,\xi,\xi]\, dv,$$

and since $\|\xi\|_\infty \le 2p$, we obtain $\|\mathcal{R}_\theta\|_\infty \le Cg^3p^3$. The first two Taylor coefficients are exactly E_θ and B_θ, including the factor $1/2$ in the latter.

The row bounds and the volume-normalized column bounds of E_θ and B_θ follow from the absolute coefficient series of the reconstruction. Replacing the volume factors by

counting measures costs only a fixed factor depending on J. By Lemma 13.6(3), we have $\|\mathsf{E}_\theta\|_{\mathrm{op}} \le C$. The quadratic tensor satisfies $\sup_e \sum_{i,j} |b_{eij}| \le C$ and $\sup_i \sum_{e,j} |b_{eij}| \le C$; the second bound implies $\sup_{i,j} \sum_e |b_{eij}| \le C$, and Lemma 13.6(3) implies the Hilbert–Schmidt bound.

The normalized first and second derivatives of the reconstruction have bounded first derivatives, in their coefficient norms with one index fixed, on a still larger product. Comparing them at $(\theta, (g_i m_{\theta,i})_i)$ and at $(0,0)$ along the straight segment and integrating their derivatives, we obtain the bound

$$C\{\|\theta\|_\infty + g\|m_\theta\|_\infty\} \le C\|\theta\|_\infty.$$

The same bound holds for their rows and their columns, so the first derivative also satisfies the operator bound in (5). No derivative of the mean as a function of the coarse field is needed for this comparison.

Step 5: color structure and contractions. At the identity coarse field, the potential and the gauge-transformed reconstruction commute with every constant rotation of the color components, and $m_0 = 0$ by Step 2. Therefore each 3×3 block of Σ_0 and of E_0 commutes with every rotation. Such a matrix is a multiple of the identity: rotations by π about the coordinate axes force its off-diagonal entries to vanish, and rotations exchanging two axes equate its diagonal entries. This proves $\Sigma_0 = \Sigma_{\mathrm{sp}} \otimes I_3$ and that E_0 is scalar in the color components.

We expand

$$\mathsf{E}_\theta \Sigma_\theta \mathsf{E}_\theta^{\mathsf{T}} - \mathsf{E}_0 \Sigma_0 \mathsf{E}_0^{\mathsf{T}} = (\mathsf{E}_\theta - \mathsf{E}_0)\Sigma_\theta \mathsf{E}_\theta^{\mathsf{T}} + \mathsf{E}_0(\Sigma_\theta - \Sigma_0)\mathsf{E}_\theta^{\mathsf{T}} + \mathsf{E}_0 \Sigma_0 (\mathsf{E}_\theta - \mathsf{E}_0)^{\mathsf{T}}.$$

The bounds of Step 4, $\|\Sigma_\theta\|_{\mathrm{op}} \le 2$, and (13.31) give the bound $C\|\theta\|_\infty(1 + tp^3)$.

For one scalar output component e, let $B_{\theta,e}$ be the symmetric matrix of the quadratic form $\mathsf{B}_{\theta,e}$. Each scalar matrix unit has nuclear norm one, so the sum of the absolute coefficients bounds the nuclear norm, and

$$\|B_{\theta,e}\|_* \le C, \qquad \|B_{\theta,e} - B_{0,e}\|_* \le C\|\theta\|_\infty.$$

The contraction of the color-equivariant quadratic tensor at the identity coarse field with the color-scalar matrix Σ_0 is a vector invariant under all rotations of the color components, hence zero. This includes all spatial covariances of the kernel at the identity. Subtracting this zero contraction, we obtain

$$|\mathbb{E}_\theta \mathsf{B}_{\theta,e}(\xi,\xi)| \le \|B_{\theta,e} - B_{0,e}\|_* \|\Sigma_\theta\|_{\mathrm{op}} + \|B_{0,e}\|_* \|\Sigma_\theta - \Sigma_0\|_{\mathrm{op}} \le C\|\theta\|_\infty(1 + tp^3) \le C\|\theta\|_\infty,$$

where in the last inequality we used the reduced coupling threshold. This proves (5).

Step 6: the unnormalized integral. The constant A_{coord} does not depend on the coarse field. Differentiating $\log \widetilde{F}$ and using the formula for the derivative of the logarithm of the normalization in (3), we obtain (6). It keeps the derivatives of the classical, determinant, and Haar density terms and of R_{ref}, evaluated at the zero-fluctuation reconstruction, which cancel only in normalized conditional expectations. All estimates used sums with one index fixed, at a fixed density and fixed J. The constants therefore have the stated dependence and do not depend on the depth or the volume. □

SECTION 14

Small-field conditioning and partition-function bounds

This section connects the analytic recursion of Section 13 with the Wilson measure conditioned on the small-field event. We prove four results. First, consider one RG step and a fine configuration such that it and its coarse field both lie in the gauge-invariant small-field domain D_r. After a hierarchical tree gauge fixing, every such configuration has unique normalized fluctuation coordinates, whether or not it satisfies the fluctuation cutoff (Theorem 14.3). Second, with the parameters chosen in Definition 14.4, the recursion of Theorem 13.5, started at the Wilson action, produces effective interactions R_k whose densities are exactly those of the pushforwards of the Wilson measure restricted by the successive tests of the small-field event $S_{N,u,\gamma}$. Every normalization constant is accounted for (Theorem 14.6). Third, for coarse fields in D_r, if $d_t < 1$ and $\epsilon_t < 1$, then imposing the fluctuation cutoff changes the unnormalized density by a factor whose reciprocal lies between 1 and $(1-\epsilon_t)^{-1}$, with explicit errors d_t and ϵ_t (Theorem 14.7). Fourth, on the stopping lattice of side n, we bound the partition function of the stopped measure from below by a tree-gauge trial integral over the same domain D_r. We also bound exponential moments of the Wilson action under the stopped measure with a single boundary term $9n^2 \log(1/g)$ (Theorem 14.8). Finally, every flat configuration, including those whose holonomies form one of the eight central triples, belongs to the small-field event once the coarse side is large enough (Lemma 14.9).

Two points need care. First, the effective interactions are holomorphic on the ball B_ρ of configurations with link logarithms of norm less than ρ, while the small-field event uses a smaller radius $r < \rho/8$. On all of B_ρ, the recursion is defined by an auxiliary integral over the fluctuation cutoff domain. This integral is defined because the reconstruction maps the whole cutoff domain into $B_{\rho/2}$, where the incoming effective interaction is analytic. For coarse fields in B_r, the auxiliary integral coincides with the fiber integral of the restricted Wilson density. For coarse fields in $B_\rho \setminus B_r$, it is only an analytic continuation, with no probabilistic meaning. Second, the fiber of the incoming restricted measure over a coarse field need not be convex. However, it is contained in a product domain $\mathcal{K}_t$, independent of the coarse field, on which the conditional fluctuation potential is uniformly convex. Concentration for the normalized measure on $\mathcal{K}_t$ then passes to the fiber by conditioning, since the two densities have the same prefactor, which depends on

the coarse field but not on the fluctuation variables.

In the partition-function bounds, the denominator is bounded below on the same domain D_r as the stopped measure. Only the numerator is enlarged, to an integral over an open cube. After tree gauge fixing, the periodic lattice has $2n^3 + 1$ independent link variables and the open cube has $2n^3 - 3n^2 + 1$. Since each variable contributes three powers of g, the comparison leaves exactly the factor g^{-9n^2}.

We use the following inputs. Sections 5 to 7 supply the coefficient calculus, tree gauges, the Haar disintegration over the block averaging constraint, the partition-function bound on an open box, and the local inverse branch of the block averaging map. Sections 10 and 11 supply the classical background fields, the nested minimization identity for the effective classical action, its comparison with the Wilson action, and its gauge invariance on chart overlaps. Section 12 supplies the joint coordinate calculus and the sequential density formula. Section 13 supplies the reconstruction gauge and its coefficient majorant, the invariant analytic ball of the recursion, the cutoff identity for the restricted density, and the convexity of the conditional fluctuation potential. In turn, Section 15 uses the exact recursion and the conditional fluctuation kernels; Sections 18 and 21 to 23 use the fiber coordinates, the classical lower bound, the normalization, and the cutoff comparison; and Sections 33 to 36 use the estimates on the stopping lattice and the flat configurations. This section does not show that the small-field event has probability close to one; that is done in Section 23. In particular, we do not estimate here the ratio of auxiliary integrals that bounds the exit probability of the coarse field in Theorem 14.7.

Separating the field into regular gauge-field coordinates and then performing successive convergent integrations is the constructive lattice gauge program of Bałaban [14, 18]. Within this framework, we prove the coordinates for every point of the small-field fiber and the normalization identities below for the specific recursion and event used in this paper.

Section 14.1 fixes notation. Section 14.2 collects the classical estimates on a common domain and derives the exact density of one RG step in fluctuation coordinates. Section 14.3 proves the coordinate theorem and states the choice of parameters and the small-field event. Section 14.4 identifies the recursion with the restricted Wilson measure, and Section 14.5 proves the cutoff comparisons. Section 14.6 contains the partition-function bounds, and Section 14.7 treats flat configurations.

14.1 Conventions

One RG step. Fix an odd integer $M \geq 3$ and an integer $J \geq 1$, and set $B = M^J$. One RG step maps a configuration on the lattice of side $Bm = M^J m$ to a configuration on the lattice of side $m \geq 4$ by T_M^J. Its levels are indexed by $i = 0, \dots, J$; the integrated intermediate scales have $i < J$. The lattice at level i, which we also call a scale, has side

$M^{J-i}m$. We write c_i for the configuration at level i, so that c_0 is the fine configuration and $c_J = c$ is the coarse one, and we write $c = e^{\theta}$ when c lies in a chart of the identity. If the fine lattice is at classical depth d_0, the averaging step from scale i to scale $i+1$ is performed at depth $d_0 + i$. The couplings satisfy $g_{i+1}^2 = Mg_i^2$. The integrated couplings are $g_0, \dots, g_{J-1}$, and we write

$$t = \max_{0 \le i < J} g_i = g_{J-1}, \qquad t^2 = \frac{\gamma}{Mm}, \qquad \lambda_* = M^{-(J-1)/2} \le \frac{g_i}{t} \le 1, \qquad g_J^2 = Mt^2 = \frac{\gamma}{m}.$$

The normalized fluctuation variables w_i of the step from scale i to scale $i+1$ are defined by

$$z_i = z_i^*(c_{i+1}) + g_i K_i(c_{i+1})^{-1/2} w_i,$$

where z_i are the free logarithmic coordinates, z_i^* is the conditional constrained minimizer, and K_i is the constrained Hessian (Section 14.2). A block index (i, e) labels one vector $w_{i,e} \in \mathbb{R}^3$; a scalar index also specifies its color component. Distances are measured in units of the coarse lattice of side m. For fixed M and J, the number of coordinate labels in a ball of radius R is at most $C_{M,J}(1+R)^3$, uniformly on every periodic lattice, since the smallest label spacing is M^{-J} and there are finitely many scale and orientation types. This bound controls all sums of exponential kernels below.

Norms. For a matrix A indexed by labels and a rate $b > 0$, the weighted row-and-column norm is

$$\|A\|_{b,\mathrm{row+col}} = \max\Big\{ \sup_x \sum_y e^{b\,\mathrm{dist}(x,y)} |A_{xy}|,\ \sup_y \sum_x e^{b\,\mathrm{dist}(x,y)} |A_{xy}| \Big\}.$$

By Schur's test, it bounds the operator norm of A on ℓ^2. This is the maximum of the row bound and the counting column bound. In Section 10, the same two bounds are added instead. That sum lies between this norm and twice it, so the bounds imported from there hold for this norm with the same constants. No column-volume factor enters this comparison, since both conventions use counting columns. For a tensor T, $\|T\|_{b,\mathrm{root}}$ denotes the weighted absolute coefficient sum with one index fixed, maximized over the choice and position of the fixed index, with spatial weight $e^{b\mathsf{t}}$, where t is the length of a shortest connecting network of the coefficient locations (Definition 4.6). Input-fixed sums are volume-normalized unless we say that they are counting sums. For a function R of the link logarithms with $DR(I) = 0$, the weighted analytic coefficient norm is

$$\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R) = \sum_{k \ge 2} \frac{\rho^{k-2}}{k!} \|D^k R(I)\|_{a_{\mathrm{sp}},\mathrm{root}},$$

as in Definition 4.6; it does not control the constant $R(I)$.

Gauge-invariant extensions. Let $B_\rho = \{\exp A : \max_e |A_e| < \rho\}$ and $D_\rho = \mathcal{G}B_\rho$, the set of configurations gauge equivalent to a configuration in B_ρ. A real analytic function R on B_ρ is *gauge invariant on overlaps* if $R(U^k) = R(U)$ whenever U and U^k lie in B_ρ for a periodic lattice gauge transformation k (Definition 13.3). Then $R_{\mathrm{sat}}(U) = R(U^k)$, for any k with $U^k \in B_\rho$, is a well-defined function on D_ρ, which is gauge invariant and real analytic. The same subscript denotes the gauge-invariant extension $E_{d,\mathrm{sat}}$ of the effective classical action from its chart (Theorem 11.6). Global color rotations act by $U_e \mapsto uU_eu^{-1}$ with constant $u \in \mathrm{SU}(2)$, and spatial inversion is $x \mapsto -x$ with the induced action on oriented links.

Constants. Unless a parameter is displayed separately, the constants $C, c, b, \dots$ of this section depend only on M, the fixed J, the larger classical and analytic domains fixed below and the positive distances between them, the positive spatial rates, and the fixed bounds on coefficients and effective interactions. They do not depend on the lattice side, on L or γ, on the depth, or on the number of RG steps. All estimates concern the whole torus; we make no statement about block regions. The value $R(I)$ of an effective interaction at the identity is always kept separately. Only constants explicitly identified as uniform before the choice of J are used to choose J.

Radii. Several radii occur. The following table lists them, and Definition 14.4 fixes the inequalities between them.

Radius	Role
r_{C}	larger domain of the classical estimates (Section 14.2)
ρ	effective interactions are holomorphic on B_ρ
$r < \rho/8$	radius of the small-field domain D_r
$R_{\mathrm{buf}} = (10M)^J r$	$T^J_M(D_r) \subset D_{R_{\mathrm{buf}}}$
$P_{\mathrm{phys}} < Q_{\mathrm{phys}}$	physical radii of products of fluctuations contained in every fiber
$R_{\mathrm{en}} = \mathscr{L}_{\mathrm{en}} r$, $\mathscr{L}'_{\mathrm{en}} r$	physical radii of the enclosing products
$p_{\mathrm{w}}(t) = A_{\mathrm{w}} \log(1/t)$	radius of the fluctuation cutoff in the variables w

A product of fluctuation balls of radius a/t in the variables w is said to have physical radius a; on it $|g_i w_{i,e}| \le a$ for all i, e, since $g_i \le t$.

14.2 Classical coordinates and the conditional density

This subsection puts the classical estimates of Sections 10 to 12 on one common domain, proves the coordinate calculus needed below, and derives the density of one RG step in

the normalized fluctuation variables. We prove the density formula on the whole branch of the local constraint coordinates, before imposing any fluctuation cutoff.

Classical estimates on a larger domain

For an averaging step at depth d, with coarse field c and free logarithmic coordinates z, we write

$$b(c,z) = e^{\Gamma_c(z)} U_0(c), \qquad B(c,z) = \operatorname{Log} b(c,z), \qquad F_d(c,z) = E_d(b(c,z)).$$

Here $U_0(c)$ is the coarse-field lift, which is the identity on block interiors and has the prescribed coarse links on the interface links, and Γ_c are the local constraint coordinates solving the block averaging constraint for the eliminated edges (Lemma 6.3). We always write the lift with its argument, to distinguish it from a fine configuration U_0.

Fix a larger classical ball of radius r_{C} strictly inside the radii of Theorems 10.10, 10.11, 11.2, and 11.6; in particular $r_{\mathrm{C}} < \min(r_{\mathrm{CK}}, \rho_{\mathrm{ov}}, r_*)$, where r_{CK}, ρ_{ov}, and r_* are the radii of Theorem 11.2, Theorem 11.6, and Theorem 10.11. We take the products of coarse fields and free coordinates of one averaging step strictly inside the joint products of Lemma 12.3, and small enough that $B(c,z)$ and every representative related to it by a gauge transformation used below stay in $B_{r_{\mathrm{C}}}$. These reductions depend only on the fixed construction domains, not on the side or the depth. On these products, we use the following data:

$$m_0 I \le K_d^0 \le C_0 I, \qquad \|K_d^0\|_{a_0,\mathrm{row+col}} + \|(K_d^0)^{-1}\|_{a_0,\mathrm{row+col}} \le C_0, \tag{14.1}$$

$$\|D^s E_d\|_{a_0,\mathrm{root}} \le C_0^{s+1} s! \quad (s \ge 2), \qquad E_d(I) = DE_d(I) = 0, \tag{14.2}$$

$$E_{d+1}(c) = MF_d(c, z_d^*(c)), \qquad E_d(b^k) = E_d(b), \tag{14.3}$$

the last identity on the larger overlaps, for periodic gauge transformations k with $b, b^k \in B_{r_{\mathrm{C}}}$. Here

$$K_d^0 = (D_z B(I,0))^{\mathsf{T}} D^2 E_d(I)\, D_z B(I,0),$$

and $z_d^*(c)$ is the conditional constrained minimizer, an interior stationary point. The constants m_0, C_0 and the rate $a_0 > 0$ are independent of d and the side. This spatial rate a_0 is local to the present subsection and unrelated to the trial amplitude a_0 of Theorem 14.8. The lower bound in (14.1) bounds the Hessian in the free coordinates z of the constraint; we assert no lower bound for the quadratic form in the coarse variables.

These data come from the cited results as follows. Theorem 10.10 gives (14.1), the conditional constrained minimizer, and the first identity in (14.3), including the factor M, on the whole real product of free coordinates of that theorem. The minimizing property and the positivity of Hessians are used only for real coarse fields and free coordinates. The stationary equation and the identity for the stationary value then extend holomorphically to smaller complex products, where no minimum is asserted. The sides

in that theorem are Mm and m with $m \geq 4$, and we infer nothing from it about iterated steps. Theorem 10.11 gives the tensors in (14.2) at one rate, and its bound on the first derivative gives $DE_d(I) = 0$. We take only the derivatives of the scalar E_d from that theorem, not its counting input columns for the coordinate maps. The upper bound of Theorem 11.2 at the identity gives $E_d(I) = 0$. We take one common smaller rate a_0 and enlarge C_0 to cover both results. Since the two representatives of each classical overlap lie in $B_{r_C} \subset B_{\rho_{ov}}$, Theorem 11.6 implies the second identity in (14.3) for every periodic gauge transformation, without a global logarithm of the gauge transformation.

Finally, Lemma 12.3 supplies joint tensor bounds for B, F_d, the negative logarithmic Haar density $\mathcal{L}$ defined below, z_d^*, K_d, and $K_d^{\pm 1/2}$ on the chosen products. For a matrix-valued derivative with a fixed coarse-field index, it sums both matrix indices, and its input-fixed column sums are volume-normalized. Since the volume ratios for one averaging step, or for the J steps of one RG step, are fixed powers of M, passing to counting column sums costs only a constant depending on M and J. These bounds control no field-independent value. The exact determinant and the fluctuation interaction $H_{d_0,J}$ on cutoff products are given by Theorem 12.13; below we also derive the underlying pointwise identities on the larger coordinate domain just fixed.

Lemma 14.1. *On the larger domains just fixed, there are smaller radii $r_c, r_z > 0$ with the following properties, for every depth $d \geq 0$ and every averaging step from side Mm to side $m \geq 4$. The conditional constrained minimizer $z_d^*(c)$ is unique, $\|z_d^*(e^\theta)\|_\infty \leq C\|\theta\|_\infty$, and throughout the real product $\|\theta\|_\infty < r_c$, $\|z\|_\infty \leq r_z$ we have*

$$D_z^2 F_d \geq m_0 I/2, \qquad \|D_z^2 F_d\|_{b,\mathrm{row+col}} \leq C,$$
$$\|K_d^{1/2}\|_{b,\mathrm{row+col}} + \|K_d^{-1/2}\|_{b,\mathrm{row+col}} \leq C, \qquad K_d = D_z^2 F_d(c, z_d^*(c)).$$

The minimizer z_d^* and the two square roots extend holomorphically to a common smaller complex product, and their weighted absolute Taylor coefficients of degree s, for output rows and for volume-normalized input columns, are bounded by $C^{s+1}s!$. Compositions of these maps over the J averaging steps of one RG step have the same property, with constants depending on J.

Proof. For every $d \geq 0$ and every averaging step $Mm \to m$ with $m \geq 4$, we apply Theorem 10.10 on the intersection of its product with the larger products fixed above. This gives the unique real interior minimizer, its linear bound in the coarse field, and the Hessian estimate on the whole real product. Keeping the local labels, we have

$$D_z^2 F_d = (D_z B)^{\mathsf{T}} D^2 E_d(B)\, D_z B + DE_d(B)\, D_z^2 B, \tag{14.4}$$

and

$$\|D_z^2 F_d - K_d^0\|_{b,\mathrm{row+col}} \leq C(\|\theta\|_\infty + \|z\|_\infty), \tag{14.5}$$

with $D_z^2 F_d \geq m_0 I/2$ for real arguments. The second term in (14.4), which contains the first derivative of E_d, is kept. Here b is a positive rate smaller than the rates of Theorem 10.10

and of the joint calculus. Combining the upper bound in (14.1) with (14.5), we obtain the upper bound for the Hessian.

Next, we use the clause of Lemma 12.3 for coordinates of fixed physical radius, which does not involve the coupling. It supplies the holomorphic minimizer and the matrices $K_d^{1/2}$ and $K_d^{-1/2}$, with factorial bounds on their weighted absolute coefficients for rows, columns, and coarse-field indices, summing both matrix indices. Its radii and constants are chosen before the number J of averaging steps in an RG step, and they are uniform in d and m. We choose r_z inside those products and then reduce r_c so that $Cr_c < r_z/8$, keeping all earlier strict inclusions. For one averaging step, passing from volume-normalized to counting columns costs the factor M^3, which is included in C. The two square roots are the positive square roots for real coarse fields and their holomorphic continuations; no positivity is asserted for complex coarse fields.

For fixed J, we compose these tensors using Lemma 5.3, with a volume-normalized input column along the branch that contains the fixed input index and output rows elsewhere. Since each contraction attaches an output to an input, the coefficient networks remain connected. We pass to trees on the external indices only at the end. The finite list of meshes in one RG step has label density bounded in terms of M and J, and the volume ratios cancel at contracted indices. A fixed loss in the spatial rate and one fixed smaller radius, both depending on J, then give the factorial bounds for all the compositions. The resulting constant C_J is independent of the side, the depth, L, and γ. □

Haar disintegration and the fluctuation potential

Fix a coarse field c and the block forests rooted at its coarse vertices. By the block version of Lemma 5.1, the passage to forest-fixed coordinates is a unique based change of Haar variables; the tree variables have total mass one, and there is no variable at a coarse vertex. On the branch of the local constraint coordinates, Lemma 6.3, or equivalently the coarea clause of Lemma 12.3, gives the Haar disintegration density

$$\mathcal{J}_c(z) = c_{\mathrm{Haar}}^{f} \frac{\prod_{e \text{ not in the forest}} j_H(\Gamma_{c,e}(z))}{\prod_a \det D_{X_{e_a^*}} \mathsf{C}_{c,a}(\Gamma_c(z))}, \qquad j_H(X) = \operatorname{sinc}^2(|X|/2). \tag{14.6}$$

Here $f = 2(M^3-1)m^3$ is the number of free group variables, $c_{\mathrm{Haar}} = 1/(16\pi^2)$, a runs over the coarse links, e_a^* is the eliminated edge of the constraint $\mathsf{C}_{c,a} = 0$ at a, and $X_{e_a^*}$ is its logarithmic coordinate. The density is taken relative to the product Haar measure on coarse configurations and includes the Haar factor of every eliminated edge. It holds for nonnegative Borel integrands on every measurable subset of this branch, with all indicators of the branch conditions kept. It requires neither smallness of the interaction nor nonvanishing of any partition function.

For the sequential change of variables, we set $S_i = K_i^{-1/2}$ and order the variables from scale $J-1$ down to scale 0. Since c_{i+1} depends only on the fluctuation variables of the

coarser scales $i+1,\dots,J-1$, the Jacobian matrix is block triangular and

$$\det D_w z = \prod_{i=0}^{J-1} g_i^{3f_i} \det S_i(c_{i+1}). \tag{14.7}$$

Here f_i is the number of free group variables at scale i; the off-diagonal derivatives of S_i and z_i^* do not change the triangular determinant. We define

$$\begin{aligned}
\mathcal{L}_i(c,z) &= -\log[\mathcal{J}_i(c,z)/\mathcal{J}_i(I,0)],\\
\mathsf{B}_i(c) &= \mathcal{L}_i(c,z_i^*(c)) + \tfrac12 \log\det K_i(c),\\
\widehat{c}_J &= c, \qquad \widehat{c}_i = b_i(\widehat{c}_{i+1}, z_i^*(\widehat{c}_{i+1})), \qquad \mathsf{B}_{\mathrm{ref}}(c) = \sum_i \mathsf{B}_i(\widehat{c}_{i+1}),\\
U_i^{\mathrm{cl}} &= g_i^{-2}[F_i(c_{i+1},z_i) - F_i(c_{i+1},z_i^*(c_{i+1}))] - \tfrac12 |w_i|^2,\\
H_{d_0,J} &= \sum_i \big[U_i^{\mathrm{cl}} + \mathcal{L}_i(c_{i+1},z_i) - \mathcal{L}_i(c_{i+1},z_i^*(c_{i+1})) + \mathsf{B}_i(c_{i+1}) - \mathsf{B}_i(\widehat{c}_{i+1})\big].
\end{aligned}$$

The fields $\widehat{c}_i$ form the zero-fluctuation reconstruction of c. The determinant term B_i of a finer scale is evaluated at the coarse field c_{i+1} obtained from the given fluctuation variables of the coarser scales. Only its value along the zero-fluctuation reconstruction is subtracted. The nested minimization identity (14.3) and $g_{i+1}^2 = M g_i^2$ give the telescoping identity

$$\frac{E_{d_0}(c_0)}{g_0^2} = \frac{E_{d_0+J}(c_J)}{g_J^2} + \tfrac12 |w|^2 + \sum_i U_i^{\mathrm{cl}}. \tag{14.8}$$

Indeed $F_i(c_{i+1},z_i) = E_{d_0+i}(c_i)$ and $F_i(c_{i+1},z_i^*(c_{i+1})) = E_{d_0+i+1}(c_{i+1})/M$, so the ith term of the sum equals $E_{d_0+i}(c_i)/g_i^2 - E_{d_0+i+1}(c_{i+1})/g_{i+1}^2 - |w_i|^2/2$.

Let $\Phi(\theta,q) = \Phi_{d_0,J}(\theta,q)$ be the logarithm of the fine configuration reconstructed from $c = e^\theta$ and the variables $q_i = g_i w_i$, in the gauge given by the gauge transformation of Theorem 13.2. This gauge transformation depends only on c and is the identity at the coarse vertices, so $T_M^J e^{\Phi(\theta,q)} = e^\theta$. We write gw for $(g_i w_i)_i$ and $R_{\mathrm{ref}}(c) = R(e^{\Phi(\theta,0)})$ for an incoming effective interaction R. Multiplying (14.6) over the scales and by (14.7), and then inserting (14.8), we find the following. On every measurable set of fluctuation variables, the density of the measure $e^{-E_{d_0}/g_0^2 - R}\, dH$ in the coordinates (c,w), relative to $dH(c)\, dw$, has a common prefactor, independent of the fluctuation variables and of the integration subset, namely

$$\prod_i \mathcal{J}_i(I,0) g_i^{3f_i} \exp[-E_{d_0+J}(c)/g_J^2 - \mathsf{B}_{\mathrm{ref}}(c) - R_{\mathrm{ref}}(c)], \tag{14.9}$$

multiplied by $e^{-V_c(w)}$, where $V_c(w) = |w|^2/2 + H_{d_0,J}(\theta,w) + R(e^{\Phi(\theta,gw)}) - R_{\mathrm{ref}}(c)$. To check this, we write $\mathcal{J}_i(c_{i+1},z_i) = \mathcal{J}_i(I,0) e^{-\mathcal{L}_i(c_{i+1},z_i)}$ and $\det S_i = e^{-(\log\det K_i)/2}$, and regroup $\sum_i [\mathcal{L}_i(c_{i+1},z_i) + (\log\det K_i(c_{i+1}))/2]$ as $H_{d_0,J} - \sum_i U_i^{\mathrm{cl}} + \mathsf{B}_{\mathrm{ref}}$. The term of the

effective interaction remaining in the integrand is $R(e^{\Phi(\theta,gw)}) - R_{\mathrm{ref}}(c)$. By gauge invariance on overlaps, at a point of the fiber $R(e^{\Phi(\theta,gw)})$ is the value of the gauge-invariant extension R_{sat} at the fine configuration. Theorem 13.4 states exactly this formula on the cutoff product whenever its hypothesis on the image of the coarse fields in B_r holds, and there the identities agree term by term with Theorem 12.13. The derivation above uses only the branch of the local constraint coordinates. It does not extend the small-interaction or nonvanishing conclusions of Theorem 12.13 to a product of fixed physical radius.

The next lemma counts powers of the coupling in the Taylor coefficients of the potential. We use it for the convexity of the potential on the enclosing product and for the mean bound in Theorem 14.7, where one derivative in the coarse field is needed.

Lemma 14.2. *Fix the ball B_ρ and a function R holomorphic there, with $DR(I) = 0$ and $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R) \le E_*$, gauge invariant on overlaps, and invariant under global color rotations, lattice translations, and spatial inversion. Use fixed smaller products of coarse fields and of free coordinates of fixed physical radius on which the coefficient majorant of Theorem 13.2 holds with image in $B_{\rho/2}$, and on which the classical and coarea conditions hold with strict inequality. Define*

$$V_\theta(w) = \tfrac{1}{2}|w|^2 + H_{d_0,J}(\theta, w) + R(e^{\Phi(\theta,(g_i w_i)_i)}) - R(e^{\Phi(\theta,0)}), \qquad W_\theta = V_\theta - \tfrac{1}{2}|w|^2.$$

After the Taylor series in the coarse field has been summed on a strictly smaller radius, the following holds on these domains. The classical terms of W_θ of total degree k in w vanish for $k \le 2$, and for $k \ge 3$ they have the form t^{k-2} times coefficients whose weighted absolute sums, with any free or coarse-field index fixed and with a fixed positive rate, are at most C_J^{k+1}. The coarea, determinant, and effective-interaction terms of total degree $k \ge 1$ have the form t^k times coefficients with the corresponding bound $C_J^{k+1}(1 + E_*)$. The same statements hold for one derivative in the coarse field, with a further fixed constant. Consequently, on the product $\{|w_{i,e}| \le p\}$ with $tp \le q$, where q is at most the fixed physical radius of these domains and $Cq < 1$,

$$\|D_w^2 W_\theta\|_{b,\mathrm{row+col}} \le \frac{C[q + t^2(1 + E_*)]}{(1 - Cq)^3}. \tag{14.10}$$

Proof. For real coarse fields, the Taylor expansion of $F_i - F_i(z_i^*)$ at the conditional constrained minimizer has no zeroth or first order term, and normalization by the positive square root of K_i makes its second order term exactly $g_i^2|w_i|^2/2$. By the stationary equation and the square-root identity of Lemma 14.1, these coefficient identities extend holomorphically to the smaller complex products. After division by g_i^2, a term of order k in w_i therefore has $k \ge 3$ and the factor g_i^{k-2}. Each dependence on the fluctuation variables of a coarser scale j through the field c_{i+1} carries an additional factor $g_j w_j$. Since $\lambda_* \le g_i/t \le 1$, dividing by two such couplings changes constants by at most the fixed factor λ_*^{-2}. Every classical monomial of total degree k thus carries t^{k-2}. Each difference

of $\mathcal{L}_i$ or of B_i has positive degree in w and no inverse power of the coupling, so it carries t^k.

We control these compositions by the weighted coefficient bounds of Lemma 14.1: we fix the external index, sum the indices of the innermost differentiated map, and proceed outward, using the volume-normalized column at an input index. By the triangle inequality, the union of the intermediate connecting networks bounds the external network length. A fixed loss in the spatial rate absorbs the bounded label density of each of the J kinds of labels. For $\log\det K$, we write, on a smaller domain,

$$\log\det K = \log\det K^0 + \sum_{h\geq 1} \frac{(-1)^{h+1}}{h} \operatorname{tr}\big[((K^0)^{-1}(K-K^0))^h\big].$$

Here the perturbation $(K^0)^{-1}(K-K^0)$ has weighted norm below $1/2$. A derivative of positive order fixes one factor of this cycle, and its two matrix indices can then be summed by the weighted bounds. The constant trace belongs to $\mathsf{B}_{\mathrm{ref}}$ and is not bounded by this calculation. For the effective interaction, since the image of the coefficient majorant lies in $B_{\rho/2}$, the joint pullback bound of Theorem 13.5 applies. Its constant term is subtracted as R_{ref}, and each free input supplies its own factor g_i. The same weighted contractions give the factor t^k.

We divide the factorial Taylor bounds by the Taylor factorial and sum the series in the coarse field on a fixed, strictly smaller radius. This leaves geometric majorants C_J^{k+1}, and one derivative in the coarse field is controlled on the same smaller radius. Two derivatives in w cost at most $k(k-1)$. For the classical part we sum $C^k t^{k-2} p^{k-2}$ over $k \geq 3$, and for the other part we sum $C^k t^k p^{k-2}$ over $k \geq 2$. The identity $\sum_{h\geq 0}(h+1)(h+2)x^h = 2/(1-x)^3$, obtained by differentiating the geometric series twice, then gives (14.10). This is a convergent Taylor calculation at the small physical radius q; it does not assert a cluster expansion on a fluctuation product of radius q/t. □

14.3 Coordinates for the small-field fiber and the choice of parameters

We now show that the normalized fluctuation variables describe every point of the fiber of one RG step whose fine and coarse fields lie in D_u, and not only the points near the conditional constrained minimizers or inside the fluctuation cutoff domain. We then fix the parameters of the construction in a definite order and define the small-field event.

Theorem 14.3. *Fix an odd integer $M \geq 3$, an integer $J \geq 1$, and the larger classical domains and charts of local constraint coordinates of Section 14.2. By the* common branch *we mean the intersection of the branch of the local constraint coordinates (Lemma 7.2) with these classical domains. There exist $r_{\mathrm{geom}}, C_{\mathrm{TG}} > 0$, and a positive distance of all coordinates below from the boundary of the common branch, independent of the depth $d_0 \geq 0$, the coarse side $m \geq 4$, L, γ, and*

the couplings, such that the following holds for every $0 < u < r_{\mathrm{geom}}$. Let U_0 be a configuration on the lattice of side $M^J m$, let $U_i = T_M^i U_0$, a configuration on the lattice of side $M^{J-i} m$, and suppose that $U_0 \in D_u$ and $U_J \in D_u$. Let $c \in B_u$ be any configuration gauge equivalent to U_J.

1. *There is a periodic gauge transformation of the fine lattice, equal to the identity at every coarse vertex when $U_J = c$, after which the intermediate configurations have link logarithms of norm at most $C_{M,J} u$ and every averaging step lies in the common branch. One may take $10M$ as the growth factor of link logarithms in one averaging step, and $C_{M,J} = 2(10M)^J + 1$ before the constants of the forest coordinates are included.*
2. *After the unique hierarchical fixing of the block forests rooted at the coarse vertices, the free logarithmic coordinates and the normalized fluctuation variables*

$$z_i = z_i(U_0), \qquad w_i = g_i^{-1} K_i(c_{i+1})^{1/2} (z_i - z_i^*(c_{i+1})), \qquad g_i \|w_i\|_\infty \le C_{\mathrm{TG}} u,$$

   reconstruct every such fiber point, including points outside the fluctuation cutoff domain. They use one based Haar disintegration and the triangular determinant of Theorem 12.13; no sum over choices of the auxiliary gauge transformation and no multiplicity factor occurs.
3. *For any two configurations $c, c' \in B_u$ gauge equivalent to U_J, the transition between the two coordinate systems is triangular, its diagonal blocks are orthogonal on each three-dimensional color block, and it preserves every $|w_{i,e}|$.*
4. *If $K_i \le C_K I$ and $D_z^2 F_i \ge m_0 I/2$ are the fixed classical bounds on the chart, then*

$$\frac{E_{d_0,\mathrm{sat}}(U_0)}{g_0^2} - \frac{E_{d_0+J,\mathrm{sat}}(c)}{g_J^2} \ge \frac{m_0}{4 C_K} \sum_i \|w_i\|_2^2.$$

   These classical values are evaluated on the larger classical chart, irrespective of the radius ρ of the effective interactions.
5. *Let R be gauge invariant on overlaps, with $DR(I) = 0$ and $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R) \le E_*$. For $0 < \eta < 1$ and $\|\xi\|_\infty \le (1-\eta)\rho$,*

$$\|DR(e^\xi)\|_{a_{\mathrm{sp}},\mathrm{root}} \le C E_* \rho \eta^{-1}, \qquad \|D^2 R(e^\xi)\|_{a_{\mathrm{sp}},\mathrm{root}} \le C E_* \eta^{-2}.$$

   Suppose that U_0 is as above and also lies in $D_{(1-\eta)\rho}$, fix one gauge transformation taking it into $B_{(1-\eta)\rho}$, and hold this gauge transformation fixed while differentiating. Let $U_0(w)$ be the fine configuration reconstructed from c and w. Then, for one fixed smaller rate $b > 0$,

$$\|D_w^2 [R_{\mathrm{sat}}(U_0(w))]\|_{b,\mathrm{row+col}} \le C_J E_* t^2 (\eta^{-2} + \rho \eta^{-1}), \qquad t = \max_i g_i.$$

   For the domain D_r one may take $\eta = 1 - r/\rho$.

The bounds in (5) hold at each point. They do not assert that the fiber is convex, and they do not place the whole fiber in one chart of the effective interaction in the gauge of Theorem 13.2.

The point of (2) is that the coordinates cover the whole fiber over c of the measure restricted to D_u, so that the density formula of Section 14.2 applies at every point of this fiber. Part (4) is a quadratic lower bound for the excess classical action in terms of the normalized fluctuation variables.

Proof. Steps 1 and 2 construct the gauge transformation in (1), using only the local inverse-branch estimate of Lemma 7.2 and the nesting of centered blocks; no classical estimate is used there. Steps 3 to 6 fix the forests, bound the fluctuation variables, and prove (2) and (3). Steps 7 and 8 prove (4) and (5).

Step 1: a small representative and its averages. Let $L_M = 10M$, and fix a periodic gauge transformation k_0 such that $V_0 = U_0^{k_0} \in B_u$. The fine side is $M^J m$ with $m \geq 4$, and scale s has side $M^{J-s}m$. Each of the J averaging steps therefore has an integer coarse side at least $m \geq 4$, as required by the clause of Lemma 7.2 on iterated steps. We choose the threshold r_{geom} so that $(10M)^J u \leq u_M$, where u_M is the radius of that lemma, and so that the finitely many conditions on logarithms of based loop holonomies hold. Then, by that lemma, we have

$$V_s = T_M^s V_0, \qquad \max_e \rho_G((V_s)_e) \leq L_M^s u \qquad (0 \leq s \leq J).$$

Gauge covariance can also be read off from (4.3): every based loop holonomy is conjugated at its basepoint, the starting coarse vertex, and so is its logarithm, while the reference transport acquires the gauge factors at its two endpoints. In particular, the branch of based loop holonomies is preserved.

Step 2: alignment with the given representative. Let h be a gauge transformation of the coarse lattice with $V_J^h = c$; if $U_J = c$, we take $h = (k_0|_{\Lambda_m})^{-1}$. On a coarse bond xy, we have $h_x h_y^{-1} = c_{xy}\, h_y (V_J)_{xy}^{-1} h_y^{-1}$. Since ρ_G is conjugation invariant and satisfies the triangle inequality, we obtain

$$\rho_G(h_x h_y^{-1}) \leq \rho_G((V_J)_{xy}) + \rho_G(c_{xy}) < (L_M^J + 1)u.$$

We lift h to a gauge transformation $\widetilde{h}$ of the fine lattice that is constant on each centered block of M^J sites per direction around a coarse vertex. These blocks partition the fine torus. They are compatible with the blocks at every intermediate scale. Indeed, in one coordinate, the union of the M centered intervals of M^{s-1} sites centered at $M^{s-1}j$, $-R_M \leq j \leq R_M$, where $M = 2R_M + 1$, is exactly the interval with integer endpoints $\pm(M^s - 1)/2$; taking Cartesian products and inducting in s proves the nesting. The nesting implies that a link at scale s either has both endpoints in one block of $\widetilde{h}$ or crosses one boundary between neighboring blocks. In the second case, the increment of $\widetilde{h}$ between its endpoints obeys the preceding bound. Since $(U^k)^l = U^{lk}$, the composite gauge transformation is $\widetilde{h}k_0$, and we obtain

$$\max_e \rho_G((V_s^{\widetilde{h}})_e) \leq L_M^s u + (L_M^J + 1)u \leq (2L_M^J + 1)u.$$

At scale J, the sharper bound $c \in B_u$ holds. When $U_J = c$, the composite gauge transformation is the identity at every coarse vertex. No global logarithm of h is used.

Step 3: forest coordinates. We fix the based block forests successively from the finest scale. Each tree path has at most $3R_M$ links. On a field with link bound v, a link outside the forest has, after forest fixing, distance at most $(6R_M + 1)v$ from the identity. Removing its coarse-field lift increases this bound by at most the bound for the coarse links. Hence all relative logarithms, including those of the eliminated edges, are at most $C_{M,J}u$. We reduce r_{geom} so that these inequalities, and the finitely many inequalities for intermediate logarithms, hold strictly inside the uniqueness neighborhood of Lemma 7.2 and inside the larger classical charts. Since the field satisfies its block averaging constraint, each eliminated-edge coordinate is the value of $\Gamma_{c_{i+1}}(z_i)$ on that branch. This excludes a second local solution for any point of the fiber.

Step 4: bounds on the fluctuation variables. Since the pairs (c_{i+1}, z_i) just obtained lie in the larger products of Section 14.2, Lemma 14.1 implies $\|z_i^*(c_{i+1})\|_\infty \le C_{M,J}u$ and a uniform row bound on $K_i^{1/2}$. By the definition of w_i, we have

$$g_i\|w_i\|_\infty \le \|K_i^{1/2}\|_{\infty\to\infty}(\|z_i\|_\infty + \|z_i^*\|_\infty) \le C_{\mathrm{TG}}u.$$

Starting at $c_J = c$ and recovering z_i and then c_i in decreasing order of i, we recover the input configuration.

Step 5: the based Haar disintegration. The measure statement in (2) requires the based conclusion of Step 2, not just the existence of a small representative. The union of the nested forests has one connected component for each coarse vertex: a vertex reaches the root of its block at each stage, and reaches its coarse vertex after at most J stages. Since this union has no cycle, the based gauge transformation that fixes it is unique. The proof of (14.6) applies at every stage. The auxiliary gauge transformation k_0 serves only to verify membership in the local branch after this unique forest fixing; it is not a coordinate and is not summed over. The single change of free coordinates has the Jacobian (14.7), which is the determinant of Theorem 12.13. Finally, the pointwise identities of Section 14.2 hold on the whole branch, before any fluctuation radius is imposed, and the present coordinates lie in this branch. This proves the measure assertion.

Step 6: change of representative. Let $c, c' \in B_u$ be gauge equivalent to U_J. The connecting gauge transformation has increments less than $2u$ on coarse bonds, and its lift that is constant on blocks keeps both coordinate neighborhoods in the same larger classical domain, after a reduction of r_{geom}. A gauge transformation fixed at the coarse vertices acts on a relative logarithm by the adjoint action at its source block; reversing the orientation of a link permutes the coordinates and changes their sign, with an adjoint factor. These maps are orthogonal on each color block. By uniqueness of the local inverse, they

intertwine the solved coordinates. The classical overlap identity in (14.3) then sends the conditional constrained minimizer and the constrained Hessian to Oz_i^* and OK_iO^{T}. The positive square root is sent to $OK_i^{1/2}O^{\mathsf{T}}$, since the square of the latter is the transformed positive matrix. This gives $w_i' = O_i(c_{i+1})w_i$. Since O_i depends on the field of the coarser scales, the joint change is triangular, and its diagonal blocks have determinants of absolute value one. This proves the norm preservation and the absence of a multiplicity factor.

Step 7: the quadratic lower bound. At each scale, let $h_i = z_i - z_i^*$. By stationarity and the classical Hessian bound, we have

$$\begin{aligned} F_i(c_{i+1}, z_i) - F_i(c_{i+1}, z_i^*) &= \int_0^1 (1-s) D_z^2 F_i(c_{i+1}, z_i^* + sh_i)[h_i, h_i]\, ds \\ &\geq \frac{m_0}{4}\|h_i\|_2^2 \geq \frac{m_0 g_i^2}{4C_K}\|w_i\|_2^2. \end{aligned}$$

In the last inequality, we used $h_i = g_i K_i^{-1/2} w_i$ and $K_i^{-1} \geq C_K^{-1} I$. Both endpoints, and the whole segment between them, lie in the fixed product of free coordinates. Subtracting successive values of the nested minimization identity and using $g_{i+1}^2 = Mg_i^2$, we see that all intermediate values cancel, as in (14.8). This gives (4), with the coefficient $m_0/(4C_K)$.

Step 8: derivatives of the effective interaction. Let $A_s \geq 0$ be the contribution of order s to the weighted coefficient norm, so that $\sum_{s\geq 2} A_s \leq E_*$. On $\|\xi\|_\infty \leq v\rho$ with $v = 1 - \eta$, contracting the undifferentiated indices bounds the first two derivatives by

$$C\rho \sum_{s\geq 2} s v^{s-1} A_s, \qquad C \sum_{s\geq 2} s(s-1) v^{s-2} A_s.$$

The multipliers are bounded by C/η and C/η^2: the first is at most $\sum_{j\geq 0} v^j$, and applying the same comparison to ordered pairs bounds the second by $2(\sum_{j\geq 0} v^j)^2$. This proves the two bounds on DR and D^2R in (5), independently of $R(I)$.

At a point in the strict interior, we fix a gauge transformation k sending it into $B_{(1-\eta)\rho}$ and hold k fixed. Since the image lies at positive distance from the boundary of B_ρ, there exists a neighborhood on which the transformed field stays in B_ρ. On it, gauge invariance on overlaps identifies R_{sat} with this analytic representative. Fixed group multiplications on a link have bounded differentials, and the logarithm has bounded first two differentials on a strictly smaller chart; these operations create no sum over remote links. By the coefficient bounds for the reconstruction in Lemma 14.1, with one or two fluctuation indices fixed, we have

$$\|D_w \xi\|_{b,\mathrm{root}} \leq C_J t, \qquad \|D_w^2 \xi\|_{b,\mathrm{root}} \leq C_J t^2.$$

The two terms of the chain rule are $D^2R(\xi)[D_w\xi, D_w\xi]$ and $DR(\xi)D_w^2\xi$. Summing first at the fixed input indices and using a smaller positive rate, we bound their row and

column sums by $C_J E_* t^2 \eta^{-2}$ and $C_J E_* \rho t^2 \eta^{-1}$. The gauge transformation k has not been differentiated. Taking $\eta = 1 - r/\rho$ gives the stated choice for D_r. □

We now fix the parameters. The order matters: first J, using only constants that are uniform before J; then the fixed-J analytic domains; then the small-field radius r; then the physical radii of fluctuation products; and finally the coupling bound.

Definition 14.4. The parameters of the construction are chosen in the following order.

1. Fix M, the spatial rates, and the support-enlargement, polymer-summability, and coefficient constants of Theorem 13.5, all of which are uniform before J. Choose J once so that
$$C_P(a_{\mathrm{sp}})M^{-J} < 1/4, \qquad C_{\mathrm{cen}}M^{-J} < 1/8,$$
and so that the other conditions on J in Theorem 13.5 hold. Here $q_P = C_P(a_{\mathrm{sp}})M^{-J}$ is the contraction factor of the pullback of the effective interaction to the background field (Theorems 11.10 and 13.5), and C_{cen} is the coefficient constant of the zero-fluctuation background field in Lemma 11.9; both are uniform before J.
2. Fix ρ, ρ_{cl}, s_J, E_*, and the larger construction domains so that the image of the reconstruction majorant and all classical, constraint-coordinate, and coarea conditions of Theorem 13.5 hold. In particular, with the constants r_J, C_J, the analytic-radius enlargement factor K_*, and the majorant $\mathfrak{M}_J$ of Theorem 13.2, we require
$$4K_*(\rho + s_J) < r_J, \qquad C_J 4K_*(\rho + s_J) < \tfrac{1}{2}, \qquad \mathfrak{M}_J(4K_*\rho, 4K_* s_J) < \rho/2. \tag{14.11}$$
These domains contain one open branch for an averaging step. Here $r_{\mathrm{CO}} > 0$ is a fixed radius for the relative logarithms $X = \Gamma_c(z)$, including those of the eliminated edges, chosen so that the branch lies strictly inside the uniqueness domain of Lemma 7.2 and the domain of the Haar disintegration of Lemma 6.3. The branch is
$$\begin{gathered}\|\mathrm{Log}\, c\|_\infty < r_c, \quad \|z\|_\infty < r_z, \quad \|\Gamma_c(z)\|_\infty < r_{\mathrm{CO}}, \\ \det D_{X_{e_a^*}} \mathsf{C}_{c,a}(\Gamma_c(z)) > 0 \quad (a \text{ a coarse link}),\end{gathered}$$
strictly inside the larger construction domains, with r_c, r_z as in Lemma 14.1. The square root in the normalized fluctuation variables is the positive square root of the constrained Hessian at the conditional constrained minimizer. All geometric and Hessian constants on these domains are fixed, including r_{geom} and C_{TG} of Theorem 14.3, which we apply with this branch as its coordinate domain. We decrease r_{geom}, before r is chosen, so that the bounds on all intermediate coarse logarithms, free logarithms, and relative logarithms in the proof of that theorem lie strictly below r_c, r_z, and r_{CO}, respectively, and so that every constraint determinant $\det D_{X_{e_a^*}} \mathsf{C}_{c,a}(\Gamma_c(z))$ is positive. These bounds are linear in u, and the determinants are

continuous and equal to $\det(M^{-2}I_3) > 0$ at the identity, so this is one fixed reduction. Thus, whenever $u < r_{\mathrm{geom}}$ and both endpoint fields lie in D_u, the coordinates of Theorem 14.3 satisfy precisely the branch conditions used in $\mathcal{A}_{k,J}$ below.

3. Put $\lambda_* = M^{-(J-1)/2}$ and $C_{\mathrm{map},J} = (10M)^J$. Choose $\mathscr{L}_{\mathrm{en}} > 2C_{\mathrm{TG}}/\lambda_*$, and choose $\mathscr{L}'_{\mathrm{en}} \geq \mathscr{L}_{\mathrm{en}}$ larger than $2C_{\mathrm{TG}}C_{\mathrm{map},J}/\lambda_*$ and than the fixed geometric enlargement constants of the larger chart of coarse fields.
4. Let $r_{\max} > 0$, depending only on the data fixed so far, be such that every $0 < r \leq r_{\max}$ satisfies the following conditions, where $R_{\mathrm{buf}} = C_{\mathrm{map},J}r$ and $R_{\mathrm{en}} = \mathscr{L}_{\mathrm{en}}r$: $r < \rho/8$; $r < r_{\mathrm{geom}}$ and $R_{\mathrm{buf}} < r_{\mathrm{geom}}$; $C_{\mathrm{map},J}r < \rho/2$; the physical radii R_{en} and $\mathscr{L}'_{\mathrm{en}}r$ lie strictly inside the products of free coordinates fixed in (2);

$$\begin{aligned} R_{\mathrm{buf}} + \mathscr{L}'_{\mathrm{en}}r < r_J, \qquad & C_J(R_{\mathrm{buf}} + \mathscr{L}'_{\mathrm{en}}r) < \tfrac{1}{2}, \\ R_{\mathrm{buf}} < 4K_*\rho, \qquad & \mathscr{L}'_{\mathrm{en}}r < 4K_*s_J; \end{aligned} \tag{14.12}$$

   the inequality

$$\frac{C\mathscr{L}'_{\mathrm{en}}r}{(1 - C\mathscr{L}'_{\mathrm{en}}r)^3} < \frac{1}{4},$$

   with C the constant of (14.10); and $M^{-J}Z(r) < r/4$, where Z is the majorant of the zero-fluctuation reconstruction in Theorem 13.2. Choose $r \in (0, r_{\max}]$.
5. Choose $0 < P_{\mathrm{phys}} < Q_{\mathrm{phys}}$, fixed multiples of r, such that $\mathfrak{M}_J(r, Q_{\mathrm{phys}}) < r/2$ and every point of the product of physical radius Q_{phys} satisfies the intermediate coordinate conditions.
6. Choose $A_{\mathrm{w}} \geq 1$, put $p_{\mathrm{w}}(t) = A_{\mathrm{w}} \log(1/t)$, and choose a coupling bound $t_* \in (0,1)$ such that, for all $0 < t \leq t_*$, the smallness conditions on t in Theorems 13.5, 13.8, and 13.10 and in the proofs of Theorems 14.6 and 14.7 hold. These include $tp_{\mathrm{w}}(t) < P_{\mathrm{phys}}$ and $p_{\mathrm{w}}(t) \geq p_0$, where p_0 is the lower bound for the cutoff radius in Theorem 13.5.

The integer J is not changed after step (1), and r is not changed after step (4).

Let $u \in 4\mathbb{N}$, $K \geq 0$, and $N = uB^K$, put $n_\ell = N/M^\ell$, and let $\gamma > 0$ be such that $t_k^2 = \gamma/(Mn_{(k+1)J}) \leq t_*^2$ for $0 \leq k < K$. Put $V^\ell = T_M^\ell U$. For the kth RG step, from V^{kJ} to $V^{(k+1)J}$, let $\mathcal{A}_{k,J}$ be the event that, in the based inverse coordinates of this step, every pair (c_{i+1}, z_i) lies in the branch of (2) at depth $kJ + i$, with its conditional constrained minimizer and the normalization by the positive square root of its constrained Hessian, and with the classical and coarea domains fixed above. This event is transported between representatives of the coarse field by the coordinate transitions of Theorem 14.3(3), not by choosing a new fiber measure. Let $w_{k,i,e}$ be the normalized fluctuation variables of

this step, with couplings $g_{k,i} = t_k M^{(i-J+1)/2}$. The small-field event and the conditioned Wilson measure are

$$S_{N,u,\gamma} = \{U \in D_r\} \cap \bigcap_{k=0}^{K-1} \Big[\mathcal{A}_{k,J} \cap \{\max_{i,e} |w_{k,i,e}| \le A_{\mathrm{w}} \log(1/t_k)\} \cap \{V^{(k+1)J} \in D_r\} \Big],$$

$$\nu_{N,u,\gamma} = \mu_{N,\gamma}(\,\cdot \mid S_{N,u,\gamma}).$$

This definition makes precise the event of Definition 4.4. There is no D_r test at the intermediate scales of an RG step. For $K = 0$ the event is the initial test $\{U \in D_r\}$. The stopped measure after k RG steps is the normalized pushforward of the Wilson measure restricted by the tests of the first k steps. The conditioned Wilson measure also contains the tests of the remaining coarser steps, so its marginal at an intermediate scale is in general not the stopped measure.

Lemma 14.5. *The choices in Definition 14.4 can be made in the stated order. The integer J depends only on constants uniform before J; ρ and s_J as in step (2) exist; there is $r_{\max} > 0$ as in step (4); in step (5) one can take $Q_{\mathrm{phys}} = \zeta r$ and $P_{\mathrm{phys}} = Q_{\mathrm{phys}}/2$, with $\zeta > 0$ independent of $r \in (0, r_{\max}]$, and then the fluctuation cutoff domain satisfies the same conditions as the product of physical radius Q_{phys} whenever $t \le t_*$; and there is t_* as in step (6). The event $S_{N,u,\gamma}$ is a Borel set invariant under lattice gauge transformations, and for $K = 0$ it equals D_r.*

Proof. Only the constants that are uniform before J enter the choice of J. The additional conditions of Theorem 13.5 on the image of the majorant, on support enlargement, and on polymer summability are of this kind, so we can take one J satisfying all of them and the two displayed contractions. We then fix the larger domains, ρ, s_J, E_*, and every coefficient bound at this J. When choosing the larger domains, we take the strict inclusions of Theorem 13.2 small enough that (14.11) holds. The classical products of Section 14.2 and Theorem 14.3 fix C_{TG} before r is chosen. These choices are made on the larger domains, before r; the contraction of the pullback uses only the constants fixed before J.

Every radius in step (4) is a fixed multiple of r, and the rational expression in step (4) also tends to zero with r. Since there are finitely many strict inequalities, their set of solutions contains an interval $(0, r_{\max}]$ after a strict reduction of its endpoint. The constants used in this reduction have all been fixed already. By the contraction $C_{\mathrm{cen}} M^{-J} < 1/8$ and the form of the majorant Z, the zero-fluctuation term satisfies $M^{-J} Z(r) < r/4$ on a sufficiently small such interval.

We choose $\zeta > 0$ so small that $C_J \zeta / [1 - C_J(r + \zeta r)] < 1/4$ uniformly for $0 < r \le r_{\max}$, that all intermediate conditions hold on the products of the corresponding physical radii, and that $\zeta < \mathscr{L}_{\mathrm{en}}$. We set $Q_{\mathrm{phys}} = \zeta r$ and $P_{\mathrm{phys}} = Q_{\mathrm{phys}}/2$. By (14.12), we have $r + Q_{\mathrm{phys}} < r_J$, the denominator of the majorant is positive, and the corresponding product of coarse fields and free coordinates lies inside the products of step (2). Every

intermediate classical and coarea condition was included in the choice of ζ. The clause of Theorem 13.2 for coarse fields in B_r therefore applies, and with $\delta = M^{-J}$ it gives

$$\mathfrak{M}_J(r, Q_{\mathrm{phys}}) = \delta Z(r) + \frac{C_J Q_{\mathrm{phys}}}{1 - C_J(r + Q_{\mathrm{phys}})} < \frac{r}{4} + \frac{r}{4} = \frac{r}{2}.$$

On the fluctuation cutoff domain, we have $|g_i w_{i,e}| \le t p_{\mathrm{w}}(t) < P_{\mathrm{phys}}$, so this domain is contained in the product of physical radius Q_{phys} and has the same properties.

For any fixed $A_{\mathrm{w}} \ge 1$, all remaining conditions on t compare a fixed positive quantity with expressions $t^{\alpha} \log^h(1/t)$, where $\alpha > 0$ and h is a fixed nonnegative integer. These expressions tend to zero: with $x = \log(1/t)$, the expression is $e^{-\alpha x} x^h$, which the exponential series bounds by $C_{\alpha,h}/x$ for large x. Also, $p_{\mathrm{w}}(t) \to \infty$ as $t \to 0$. We conclude that one $t_* < 1$ enforces every condition, including those of Theorem 13.5. None of them requires increasing J or changing r.

The event keeps the indicators of the branch, determinant, and classical conditions, and then only the initial and coarse D_r tests. Every map in its finite definition is continuous on the open branch and is used together with the Borel indicator of the branch, and the event is false outside the branch. Hence the event is Borel. The orthogonal transitions of Theorem 13.4 on overlaps preserve its coordinate conditions and its fluctuation balls. It is therefore one event invariant under lattice gauge transformations, defined without a measurable choice among auxiliary gauge transformations. For $K = 0$, the intersection over k is empty and only the initial test remains. □

14.4 The restricted recursion and the enclosing product

This subsection identifies the recursion of Theorem 13.5 with the Wilson measure restricted by the small-field event. It also shows that the fiber of the incoming restricted density over a coarse field lies in a product domain on which the conditional fluctuation potential is uniformly convex.

Fix one RG step from depth d_0, with coarse side m and incoming effective interaction R. The incoming restricted density is

$$P_{\mathrm{in}}(U) = \mathbf{1}_{D_r}(U)\, e^{-E_{d_0,\mathrm{sat}}(U)/g_0^2 - R_{\mathrm{sat}}(U)}$$

relative to the product Haar measure, as in Theorem 13.4. For a coarse field $c \in B_{R_{\mathrm{buf}}}$, let S_c be the set of normalized fluctuation variables w such that the pairs (c_{i+1}, z_i) defined successively by $c_J = c$, $z_i = z_i^*(c_{i+1}) + g_i K_i(c_{i+1})^{-1/2} w_i$, and $c_i = b_i(c_{i+1}, z_i)$ lie in the common branch, and the reconstructed fine configuration c_0, with the forest variables in their gauge-fixed position, lies in D_r. For the fixed coarse representative c, Theorem 14.3 gives these coordinates for every configuration $U \in D_r$ with $T_M^J U = c$, modulo fine gauge transformations equal to the identity at every coarse vertex; the hierarchical forest

fixing chooses one representative. We apply the theorem with $u = r$ for $c \in B_r$ and with $u = R_{\mathrm{buf}}$ for $c \in B_{R_{\mathrm{buf}}}$. If the coarse image is only gauge equivalent to c, it is first aligned with c as in that theorem; we do not assert that this auxiliary alignment is unique. The set S_c is Borel, since it is the preimage of the open set D_r under a continuous map on an open domain. Since P_{in} is invariant under lattice gauge transformations, the forest variables integrate out with total mass one (Lemma 5.1), and all densities in w below refer to this reduced integral. They are statements about integrals of gauge-invariant functions, not a pointwise coarea formula for other functions. They do not define a global gauge fixing on the configuration space.

Theorem 14.6. *Use the parameters of Definition 14.4 and start with $R_0 = 0$.*

1. *For every finite number of RG steps, every coarse side $m \geq 4$, and all couplings with $t \leq t_*$ at each step, there are effective interactions R_k, holomorphic on B_ρ, with*

$$DR_k(I) = 0, \qquad \mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_k) \leq E_*,$$

gauge invariant on overlaps and invariant under global color rotations, lattice translations, and spatial inversion.

2. *Consider one RG step from depth d_0, with incoming effective interaction $R = R_k$ and coarse field $c = e^\theta$, and set*

$$\widetilde{F}_+(c) = A_{\mathrm{coord}}\, e^{-E_{d_0+J}(c)/g_J^2 - \mathsf{B}_{\mathrm{ref}}(c) - R_{\mathrm{ref}}(c)} \int_{\Omega_p^{\mathrm{w}}} e^{-V_c(w)}\, dw,$$

$$R_{\mathrm{ref}}(c) = R(e^{\Phi(\theta,0)}), \qquad V_c(w) = \tfrac{1}{2}|w|^2 + H_{d_0,J}(\theta, w) + R(e^{\Phi(\theta,(g_i w_i)_i)}) - R_{\mathrm{ref}}(c),$$

$$A_{\mathrm{coord}} = \prod_i \mathcal{J}_i(I,0) g_i^{3f_i}, \qquad \Omega_p^{\mathrm{w}} = \prod_{i,e} \{|w_{i,e}| \leq p_{\mathrm{w}}(t)\}.$$

For $c \in B_\rho$ this is an auxiliary integral, positive for real c, in which R is evaluated on the image of the reconstruction, which lies in $B_{\rho/2}$. With

$$z_p = \int_{|x| \leq p_{\mathrm{w}}(t)} e^{-|x|^2/2}\, d^3x, \qquad \pi_p^{\mathrm{all}}(dw) = z_p^{-\sum_i f_i} \mathbf{1}_{\Omega_p^{\mathrm{w}}}(w) e^{-|w|^2/2}\, dw,$$

and $\Xi_R = \mathbb{E}_{\pi_p^{\mathrm{all}}} e^{-H_{d_0,J} - R(e^{\Phi(\theta, gw)}) + R_{\mathrm{ref}}}$, we have

$$R_+ = \mathsf{B}_{\mathrm{ref}} + R_{\mathrm{ref}} - \Big(\sum_i f_i\Big) \log z_p - \log \Xi_R, \qquad \widetilde{F}_+ = A_{\mathrm{coord}} e^{-E_{d_0+J}/g_J^2 - R_+}.$$

For real c, $\log \Xi_R$ is the real logarithm; for complex c it is the holomorphic logarithm given by Theorem 13.5.

3. *For every $c \in B_r$, the reconstruction maps the whole fluctuation cutoff domain Ω_p^{w} into D_r, and all coordinate conditions of the step hold there. Consequently $\widetilde{F}_+(c)$ is the density at c of*

the pushforward under T_M^J of the incoming density restricted by the coordinate conditions and the fluctuation cutoff of this step, and the output density, including the coarse test, is $\mathbf{1}_{D_r}\widetilde{F}_{+,\mathrm{sat}}\,dH$. By induction, the stopped density after k RG steps is proportional to

$$\mathbf{1}_{D_r} e^{-E_{kJ,\mathrm{sat}}/g_{kJ}^2 - R_{k,\mathrm{sat}}}\,dH,$$

where $g_{kJ}^2 = \gamma/n_{kJ}$ is the coupling of the lattice of side n_{kJ}.

4. *The field-independent factor of this density is the product of the factors A_{coord} of the steps performed, together with the constants removed by centering, which are transferred to the normalization constant. Precisely, if the incoming effective interaction is centered and the incoming density carries the factor e^{Λ_k}, then centering the output of this step means*

$$R_{k+1} = R_+ - R_+(I), \qquad \Lambda_{k+1} = \Lambda_k + \log A_{\mathrm{coord}} - R_+(I).$$

If R_+ is kept uncentered, its constant stays in the exponent. The normalizing constant is finite and strictly positive. After k RG steps, normalizing the density in (3) gives the pushforward under T_M^{kJ} of the Wilson measure conditioned on the initial D_r test and the tests of the first k steps. After all K steps, this measure is $(T_M^{KJ})_*\nu_{N,u,\gamma}$, with $\nu_{N,u,\gamma}$ the conditioned Wilson measure of Definition 14.4.

5. *Let $b_{\mathrm{w}} = \sum_i f_i = 2(B^3-1)m^3$, so that $\dim_{\mathbb{R}} w = 3b_{\mathrm{w}}$, and define, for $a \in \{P_{\mathrm{phys}}, Q_{\mathrm{phys}}\}$,*

$$\Omega_a^{\mathrm{phys}} = \prod_{i,e}\{|w_{i,e}| \le a/t\}, \qquad \mathcal{K}_t = \prod_{i,e}\{|w_{i,e}| \le R_{\mathrm{en}}/t\}.$$

Then, for every $c \in B_r$,

$$\Omega_p^{\mathrm{w}} \subset \Omega_{P_{\mathrm{phys}}}^{\mathrm{phys}} \subset \Omega_{Q_{\mathrm{phys}}}^{\mathrm{phys}} \subset S_c \subset \operatorname{int}\mathcal{K}_t.$$

6. *The same potential V_c is smooth on a neighborhood of the product $\mathcal{K}_t$, which does not depend on c, and*

$$\|D_w^2(V_c - \tfrac{1}{2}|w|^2)\|_{b,\mathrm{row+col}} \le \frac{C[R_{\mathrm{en}} + t^2(1+E_*)]}{(1-CR_{\mathrm{en}})^3} \le \tfrac{1}{2}, \qquad D_w^2 V_c \ge I/2.$$

7. *For $c \in B_r$, the fiber of the incoming measure $P_{\mathrm{in}}\,dH$ over c, without fluctuation cutoff, has the density*

$$A_{\mathrm{coord}}\, e^{-E_{d_0+J}(c)/g_J^2 - \mathsf{B}_{\mathrm{ref}}(c) - R_{\mathrm{ref}}(c)}\, \mathbf{1}_{S_c}(w) e^{-V_c(w)}\,dw.$$

The prefactor is the same for every subset of S_c that is integrated. The set S_c need not be convex.

8. *On Ω_p^{w} the normalized density is the conditional fluctuation kernel of Theorem 13.8 with the effective interaction R, and the bounds of Theorem 13.10 on means, derivatives of the potential, and covariances apply to it. The tests of the coarser RG steps cancel when one*

conditions on the coarse field. A gauge transformation of the reconstructed fine field that depends only on c and equals the identity at every coarse vertex leaves c, the normalized fluctuation variables, the conditional fluctuation kernels, and the cutoffs unchanged, provided that the coefficient bounds of Theorem 13.2 continue to hold in the new representative.

All constants depend only on the parameters of Definition 14.4 and the larger domains; they do not depend on the side, the coupling, the depth, or the number of RG steps. The values of $\widetilde{F}_+$ at coarse fields outside B_r are not identified with pushforwards of the restricted Wilson measure.

Proof. Steps 1 to 6 concern the recursion and its identification with the restricted Wilson measure; they prove (1) to (4). Steps 7 to 10 concern the fiber without cutoff and prove (5) to (8).

Step 1: the auxiliary integral on B_ρ. The image of the coefficient majorant on the enlarged product was fixed in Theorem 13.2. By (14.11), the enlarged product of coarse fields and free coordinates lies before the pole of the majorant and within $u + v < r_J$, with positive denominator. On the whole auxiliary ball of coarse fields, every reconstructed argument of the effective interaction lies in $B_{\rho/2}$, and every intermediate classical and coarea argument satisfies its conditions with strict inequality. Then the formula in (2) defines the same auxiliary operator as Theorem 13.5, with the present parameters and incoming effective interaction, and that theorem supplies its holomorphic logarithm and the common bound on its analytic norm. Positivity of the integral refers to real coarse fields. For complex coarse fields, $\widetilde{F}_+$ is the holomorphic continuation, and the existence of its logarithm is part of Theorem 13.5, not a consequence of positivity.

Step 2: normalization constants. Since Ω_p^{w} is a product of $\sum_i f_i$ balls in $\mathbb{R}^3$, each of Gaussian mass z_p, substituting V_c gives $\int_{\Omega_p^{\mathrm{w}}} e^{-V_c}\, dw = z_p^{\sum_i f_i} \Xi_R$. Taking the logarithm, with the real branch on real coarse fields, gives the stated R_+ and A_{coord}; this is also the formula of Theorem 13.4. In particular, the normalization z_p is not absorbed into an unspecified term depending on the coarse field. Subtracting $R_+(I)$ changes no nonconstant coefficient, no overlap identity, and no symmetry, and the unnormalized densities agree:

$$e^{\Lambda_k} A_{\mathrm{coord}} e^{-E_{d_0+J}/g_J^2 - R_+} = e^{\Lambda_k + \log A_{\mathrm{coord}} - R_+(I)} e^{-E_{d_0+J}/g_J^2 - [R_+ - R_+(I)]}.$$

This proves the sign and every constant in (4).

Step 3: identification on B_r. Let $c \in B_r$. On the whole cutoff product, we have $\|gw\|_\infty \le t p_{\mathrm{w}}(t) < P_{\mathrm{phys}} < Q_{\mathrm{phys}}$. By Lemma 14.5, the radius Q_{phys} satisfies (14.12), $r + Q_{\mathrm{phys}} < r_J$, the positivity of the denominator of the majorant, and all intermediate conditions. The image bound of Theorem 13.2 for coarse fields in B_r therefore puts the fine field in $B_{r/2}$, in the gauge of that theorem, with all intermediate coordinate conditions. The indicator of D_r for the fine field is identically one there. All hypotheses of Theorem 13.4 now hold:

R is gauge invariant on overlaps, its first derivative at the identity vanishes, its norm is at most E_*, and it has the required symmetries. This implies that, at coarse fields in B_r, the auxiliary integral is the fiber integral of the restricted incoming density, including the branch, coarea, cutoff, and coarse tests; we do not use this conclusion outside B_r.

Step 4: overlaps, symmetries, and the first derivative. On an overlap, the triangular orthogonal change of Theorem 14.3(3) preserves the fluctuation product, Lebesgue measure, the Haar disintegration densities, and the determinants. The two reconstructed arguments of the effective interaction are related by a gauge transformation and both lie in $B_{\rho/2}$, so the values of R agree. A change of variables shows that the two auxiliary integrals are equal, including all reference factors. Translations and spatial inversion act by the corresponding orthogonal permutations, and constant color rotations also preserve the integral. At the identity, each component of the first derivative of R_+ at a link is a vector in $\mathfrak{su}(2)$ fixed by all adjoint rotations, and hence zero: rotations by a half-turn about the coordinate axes reverse each component. This verifies the overlap and first-derivative hypotheses at the next step.

Step 5: the analytic norm. The induction step for the norm is the inequality

$$\begin{aligned}\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_+) &\le C_G + q_P E_* + C_T t^2 E_*(p_{\mathrm{w}}^4 + E_* p_{\mathrm{w}}^2)\\ &\le C_G + \tfrac{1}{2}(1+q_P)E_* \le E_*.\end{aligned}$$

In the first inequality, we used the bound of Theorem 13.5; in the second, its condition $C_T t^2(p_{\mathrm{w}}^4 + E_* p_{\mathrm{w}}^2) \le (1-q_P)/2$; and in the last, $C_G \le (1-q_P)E_*/2$. Since $R_0 = 0$ satisfies all hypotheses, this proves (1) at every finite depth kJ.

Step 6: iteration and the normalizing constant. Iterating these positive operators, and using the last clause of Theorem 13.4, we obtain the pushforward of the Wilson density restricted by exactly the tests of Definition 14.4, and the stopped density after any number of steps. This proves (3). On a finite lattice, the identity configuration and a sufficiently small open neighborhood of it pass all of the finitely many tests strictly. Indeed, the inverse coordinates are continuous on the open branch, the image bound holds on the whole cutoff product, and every normalized fluctuation variable of the identity configuration is zero. That neighborhood has positive Haar measure. All densities are bounded on smaller chart neighborhoods, and the bound on the oscillation of R_{sat} in Theorem 13.4, together with the finiteness of Haar measure on the compact configuration space, gives finiteness. Hence the normalizing constant is finite and strictly positive. When all K tests are imposed, normalizing the restricted Wilson measure before pushforward gives $\nu_{N,u,\gamma}$, and normalizing its pushforward gives the corresponding stopped measure.

Step 7: the enclosure. We now consider the fiber without cutoff. The numbers of free variables at the intermediate scales are

$$f_i = 2(M^3-1)(mM^{J-i-1})^3, \qquad \sum_{i=0}^{J-1} f_i = 2m^3(M^{3J}-1),$$

where the equality follows by summing the finite geometric series. Since there are three real color components per block, we have $\dim_{\mathbb{R}} w = 3b_{\mathrm{w}}$. For a point of S_c, Theorem 14.3 with $u = r$ implies that

$$|w_{i,e}| \le \frac{C_{\mathrm{TG}} r}{g_i} \le \frac{C_{\mathrm{TG}} r}{\lambda_* t} < \frac{R_{\mathrm{en}}}{2t},$$

where in the last inequality we used $\mathscr{L}_{\mathrm{en}} > 2C_{\mathrm{TG}}/\lambda_*$. This proves that S_c lies in the interior of $\mathcal{K}_t$, before any cutoff. Conversely, if $|w_{i,e}| \le Q_{\mathrm{phys}}/t$ for all i, e, then $|g_i w_{i,e}| \le Q_{\mathrm{phys}}$. The image estimate of Theorem 13.2 puts the fine field in $B_{r/2}$, with all intermediate coordinate conditions, on this whole closed product, which is therefore a subset of S_c. The other two inclusions follow from $P_{\mathrm{phys}} < Q_{\mathrm{phys}}$ and $t p_{\mathrm{w}}(t) < P_{\mathrm{phys}}$. Since all inequalities on the coarse field and the branch hold strictly, the assertions also hold on the boundaries of these closed products.

Step 8: convexity on $\mathcal{K}_t$. Throughout $\mathcal{K}_t$, each input $g_i w_{i,e}$ has norm at most $R_{\mathrm{en}} \le \mathscr{L}'_{\mathrm{en}} r$, and together with the coarse-field radius $r \le R_{\mathrm{buf}}$ it satisfies (14.12). The image condition of step (2) of Definition 14.4 therefore puts the image of the reconstruction, and an open neighborhood of it, in the analytic domain of R. The same V_c is therefore smooth there, also at points not in S_c. The clause of Theorem 13.8 for products of fixed physical radius applies with $q_* = R_{\mathrm{en}}$ and gives the Hessian estimate in (6); equivalently, this is the convergent coefficient calculation of Lemma 14.2 with $q = R_{\mathrm{en}}$. The choice of $r_{\max}$ makes the term with R_{en} at most $1/4$, and the choice of t_* then makes the term with $t^2(1+E_*)$ at most $1/4$. By Schur's test, the weighted row-and-column bound controls the operator norm, which gives $D_w^2 V_c \ge I/2$ on the whole convex product, whether or not S_c is convex.

Step 9: the density on the whole fiber. By Theorem 14.3(2) and the discussion before the theorem, the uncut fiber has exactly one representation in the based coordinates. Multiplying (14.6), (14.7), and (14.8) gives the factor (14.9) times $\mathbf{1}_{S_c} e^{-V_c(w)}\, dw$. Gauge invariance on overlaps identifies the value of the effective interaction at every point of S_c. All other factors are contained in (14.9) or in $H_{d_0,J}$, and none of them depends on which subset of the fiber is integrated. This proves (7).

Step 10: conditional kernels and changes of representative. On the cutoff product, the normalized density is the kernel of Theorem 13.8 with the effective interaction just constructed; the hypotheses of that theorem, including the fixed domain Ω_p^{w} and the smallness of $t p_{\mathrm{w}}(t)^3$, were verified above. The effective interaction is holomorphic on B_ρ and gauge invariant on overlaps, and it has vanishing first derivative at the identity,

the required symmetries, and norm at most E_*. Further, the product of coarse fields and free coordinates satisfies the image condition of (14.11). The hypotheses of Lemma 14.2 then also hold, and Theorem 13.10 applies to this kernel, including its local L^2 bound along the interpolation. The segment $s\theta$, $0 \le s \le 1$, stays in B_r, and the product and the couplings are fixed along it. If the Wilson measure conditioned on the whole small-field event is conditioned further on the coarse field of this step, the tests of the coarser steps are functions of that coarse field and of its deterministic averages. Their common factor therefore cancels in the normalized conditional fluctuation kernel wherever these tests pass.

Finally, a gauge transformation depending only on the coarse field and equal to the identity at the coarse vertices changes the representative of the fine field after reconstruction. The free coordinates, all fluctuation balls, the Haar disintegration density, and the triangular Jacobian are unchanged, and by gauge invariance on overlaps the value of the effective interaction agrees. Hence the potential and the kernel are the same. The proviso on coefficient bounds allows the analytic estimates, including those on derivatives of the potential, to be used in the new representative. Every reduction and every bound used only the fixed parameters, and none involved the length of the preceding recursion. □

14.5 Conditional comparisons and exit probabilities

We now compare three normalized measures on the fluctuation variables over a coarse field $c \in B_r$: the conditional fiber measure of the incoming restricted density, its restriction to the fluctuation cutoff domain, and a comparison measure on the enclosing product $\mathcal{K}_t$. The last one has a uniformly convex potential on a convex domain, so it satisfies a logarithmic Sobolev inequality and a local covariance bound, and these estimates pass to the fiber by conditioning. We also bound the probability that the coarse field leaves D_r by a ratio of auxiliary integrals.

Theorem 14.7. *Use the parameters of Definition 14.4, consider one RG step from depth d_0 with incoming effective interaction R given by Theorem 14.6, and for every $c = e^\theta \in B_r$ define the normalized enclosing-product comparison measure ν_c and the conditional fiber measure κ_c^{old} by*

$$\nu_c(dw) = \frac{\mathbf{1}_{\mathcal{K}_t}(w) e^{-V_c(w)}\, dw}{\int_{\mathcal{K}_t} e^{-V_c(w)}\, dw}, \qquad \kappa_c^{\mathrm{old}}(dw) = \frac{\mathbf{1}_{S_c}(w) e^{-V_c(w)}\, dw}{\int_{S_c} e^{-V_c(w)}\, dw}.$$

1. *Uniformly in the depth, the side, and c,*

$$\sup_i |\mathbb{E}_{\nu_c} w_i| \le Ct \|\theta\|_\infty, \qquad \kappa_c^{\mathrm{old}} = \nu_c(\,\cdot \mid S_c).$$

2. *Put $m_{\mathrm{cv}} = 1/2$ and*

$$d_t = 6 b_{\mathrm{w}} e^{-m_{\mathrm{cv}} \mathrm{Q}_{\mathrm{phys}}^2/(24t^2)}, \qquad \delta_t = 6 b_{\mathrm{w}} e^{-m_{\mathrm{cv}} p_{\mathrm{w}}(t)^2/24}, \qquad \epsilon_t = \frac{\delta_t}{1 - d_t}.$$

If $d_t < 1$, then

$$\kappa_c^{\mathrm{old}}\big((\Omega^{\mathrm{phys}}_{P_{\mathrm{phys}}})^c\big) \le \frac{6b_{\mathrm{w}}e^{-m_{\mathrm{cv}}P_{\mathrm{phys}}^2/(24t^2)}}{1-d_t}, \qquad \kappa_c^{\mathrm{old}}\big((\Omega_p^{\mathrm{w}})^c\big) \le \epsilon_t.$$

3. *Let σ be the probability measure $Z_{\mathrm{in}}^{-1}P_{\mathrm{in}}\,dH$, where $Z_{\mathrm{in}} = \int P_{\mathrm{in}}\,dH$; this is the normalized incoming restricted density of this step. Let $G(c)$ be the density at c of the pushforward of the unnormalized measure $P_{\mathrm{in}}\,dH$ under T_M^J, given by the based coarea formula of Theorem 14.6(7), and let $\widetilde{F} = \widetilde{F}_+$ be the auxiliary integral of Theorem 14.6(2), both with the same convention for field-independent factors. Then, on D_r,*

$$\frac{\widetilde{F}(c)}{G(c)} = \kappa_c^{\mathrm{old}}(\Omega_p^{\mathrm{w}}),$$

without any smallness assumption, and $1 \le G(c)/\widetilde{F}(c) \le (1-\epsilon_t)^{-1}$ if $d_t < 1$ and $\epsilon_t < 1$.

4. *If $d_t < 1$, the normalized measures $\mathbf{1}_{D_r}G\,dH/\int_{D_r} G\,dH$ and $\mathbf{1}_{D_r}\widetilde{F}\,dH/\int_{D_r}\widetilde{F}\,dH$ differ in total variation $\sup_A|\mu(A)-\nu(A)|$ by at most ϵ_t, and*

$$\sigma(\text{the next RG step fails}) \le \sigma(T_M^J U \notin D_r) + \epsilon_t,$$

where the next RG step fails if U violates at least one of the coordinate conditions of the step, the fluctuation cutoff, and the condition $T_M^J U \in D_r$.

5. *We have $T_M^J(D_r) \subset D_{R_{\mathrm{buf}}}$ with $R_{\mathrm{buf}} = (10M)^J r$. Let $\mathcal{K}_t'$ be the product of physical radius $\mathscr{L}_{\mathrm{en}}' r$, and define*

$$H(c) = C(c)\int_{\mathcal{K}_t'} e^{-V_c(w)}\,dw, \qquad C(c) = A_{\mathrm{coord}}\,e^{-E_{d_0+J}(c)/g_J^2 - \mathsf{B}_{\mathrm{ref}}(c) - R_{\mathrm{ref}}(c)},$$

with the same convention for field-independent factors as G and $\widetilde{F}$. If $\delta_t < 1$, then

$$G(c) \le H(c) \le \frac{\widetilde{F}(c)}{1-\delta_t} \qquad (c \in D_{R_{\mathrm{buf}}}),$$

and, with $Z_{\mathrm{aux}}(A) = \int_A e^{-E_{d_0+J,\mathrm{sat}}/g_J^2 - R_{+,\mathrm{sat}}}\,dH$,

$$\sigma(T_M^J U \notin D_r) \le \frac{1}{1-\delta_t}\,\frac{Z_{\mathrm{aux}}(D_{R_{\mathrm{buf}}} \setminus D_r)}{Z_{\mathrm{aux}}(D_r)}.$$

In (5) no inclusion of the cutoff product in S_c is required for coarse fields in $D_{R_{\mathrm{buf}}} \setminus D_r$. The pointwise statements refer to the explicit density given by the based coarea formula; for an arbitrary disintegration they hold almost everywhere. The constants do not depend on the volume, but the explicit factor b_{w} and the conditions $d_t < 1$, $\epsilon_t < 1$, and $\delta_t < 1$ cannot be dropped. At fixed L and $\gamma > 0$, along refinement of the lattice, $b_{\mathrm{w}} = O_{M,J,\gamma}(t^{-6})$, so the errors d_t, δ_t, and ϵ_t are smaller than every power of the mesh. The theorem gives no bound on the ratio of auxiliary integrals in (5).

The mean bound in (1) concerns the comparison measure on the enclosing product, not the kernel on the cutoff product, whose means are bounded in Theorem 13.10. Its proof must avoid the large radius R_{en}/t of $\mathcal{K}_t$. Heuristically, a classical cubic term in the potential has size tw^3, so its mixed derivative in w and in the coarse field has size tw^2, whose L^2 norm is at most $t\|w\|_4^2$. The fourth moments are bounded independently of the radius of the product, by concentration. In a term of degree $k \geq 3$ the remaining $k-3$ factors are bounded by R_{en}/t, which leaves $t^{k-2}(R_{\mathrm{en}}/t)^{k-3} = tR_{\mathrm{en}}^{k-3}$, a summable series. Bounding all factors by the radius of the product, or using a global Cauchy–Schwarz bound for the extensive derivative of the potential, would lose this smallness.

Proof. Steps 1 to 4 prove the mean bound in (1): Step 1 recalls the convexity input, Step 2 derives concentration and fourth moments, Step 3 bounds the mixed derivative of the potential, and Step 4 integrates along the segment from 0 to θ. Steps 5 and 6 prove the conditioning identity and (2). Steps 7 and 8 prove (3) and (4). Steps 9 and 10 prove (5), and Step 11 proves the remaining statements.

Step 1: inputs. We use the coefficient bounds of Lemma 14.2. They apply because the effective interaction, constructed in Theorem 14.6, has the required holomorphy, vanishing first derivative, gauge invariance on overlaps, symmetries, and norm bound. The whole enclosing product and the segment of coarse fields satisfy (14.12). The convexity and the product inequalities are supplied by the clause of Theorem 13.8 for products of fixed physical radius, which rests on Theorem 5.4. Their constants depend on the lower Hessian bound $m_{\mathrm{cv}} = 1/2$ and on the weighted bounds on the interaction, but not on the radius R_{en}/t of the product or on the dimension. These inequalities go back to Gross [103], Bakry and Émery [8], and Helffer and Sjöstrand [110]; the version for reflecting boundary conditions on products of balls is proved in Section 5.

Step 2: concentration of one coordinate. Let X be a scalar coordinate. The logarithmic Sobolev inequality of Step 1 reads $\mathrm{Ent}(f^2) \leq (2/m_{\mathrm{cv}})\mathbb{E}|\nabla f|^2$. We apply it to $f = e^{\lambda X/2}$, for which $|\nabla f|^2 = \lambda^2 e^{\lambda X}/4$, and set $\psi(\lambda) = \log \mathbb{E}e^{\lambda X}$. Dividing by $\mathbb{E}e^{\lambda X}$, we obtain

$$\lambda\psi'(\lambda) - \psi(\lambda) \leq \frac{\lambda^2}{2m_{\mathrm{cv}}}.$$

The left side is λ^2 times the derivative of $\psi(\lambda)/\lambda$, whose limit at $\lambda = 0$ is $\mathbb{E}X$. Integrating from 0 to λ gives $\log \mathbb{E}e^{\lambda(X-\mathbb{E}X)} \leq \lambda^2/(2m_{\mathrm{cv}})$. This is the argument of Herbst; see [123, Section 2.3]. The exponential Markov inequality with $\lambda = m_{\mathrm{cv}}v$ gives

$$\mathbb{P}(|X - \mathbb{E}X| \geq v) \leq 2e^{-m_{\mathrm{cv}}v^2/2}. \tag{14.13}$$

The calculation is valid for either sign of λ on the bounded product. Integrating $4v^3$ times this tail bound gives

$$\|X\|_4 \leq C(1 + |\mathbb{E}X|). \tag{14.14}$$

In particular, the estimates below contain no factor equal to the radius of the product.

Step 3: the mixed derivative of the potential. Let $\theta_s = s\theta$ for $0 \le s \le 1$, let v_s be the corresponding comparison measure, and let $M(s) = \max_\alpha |\mathbb{E}_{v_s} w_\alpha|$, the maximum over scalar coordinates. In this proof, the running maximum $M(s)$ of the means should not be confused with the odd integer M. Since the product $\mathcal{K}_t$ does not depend on s, differentiating the finite positive integral by the quotient rule gives

$$\frac{d}{ds}\mathbb{E}_{v_s} w_\alpha = -\operatorname{Cov}_{v_s}(w_\alpha, \partial_s V_{\theta_s}), \tag{14.15}$$

and there is no boundary term.

In a classical monomial of total degree $k \ge 3$ in w, the derivative $D_w \partial_s V$ leaves $k-1$ fluctuation factors. We keep two of them in L^4 and bound the other $k-3$ by $p = R_{\mathrm{en}}/t$. By Hölder's inequality and (14.14), the L^2 norm of their product is at most $C(1+M(s))^2 p^{k-3}$, and the coupling factor of the monomial becomes

$$t^{k-2} p^{k-3} = t R_{\mathrm{en}}^{k-3}.$$

Since the derivative in the coarse field is taken in the fixed direction θ, the summation on a strictly smaller radius in Lemma 14.2 shows that the summed weighted majorant in the coarse field costs $C\|\theta\|_\infty$. A nonclassical term of degree one gives $Ct(1+E_*)\|\theta\|_\infty$ directly. A term of degree two gives $Ct^2(1+E_*)(1+M(s))\|\theta\|_\infty$, from its single L^2 factor. For degree $k \ge 3$ the preceding argument with two factors in L^4 gives the coupling factor $t^k p^{k-3} = t^3 R_{\mathrm{en}}^{k-3}$. Summing these convergent geometric series, and the weighted sums over the indices in the coarse field, we obtain the local weighted estimate

$$\sup_\alpha \|D_{w_\alpha} \partial_s V_{\theta_s}\|_{2,\mathrm{loc}} \le Ct\|\theta\|_\infty (1 + E_* + M(s)^2). \tag{14.16}$$

Here the local norm fixes the index α of the derivative in w and sums all coefficient labels and indices in the coarse field with their weights $e^{b\mathfrak{t}}$, and the L^2 norm is taken under v_s. This definition and the weighted coefficient bounds prevent a volume factor in this estimate.

Step 4: the mean bound. We apply the local covariance bound of Step 1 to (14.15). The gradient of w_α is the unit vector at α, and the exponentially weighted sum over the other index is finite by the bound on the number of labels in a ball. We obtain

$$\Big|\frac{d}{ds}\mathbb{E}_{v_s} w_\alpha\Big| \le Ct\|\theta\|_\infty (1 + E_* + M(s)^2).$$

At $s = 0$, every vector mean vanishes and $M(0) = 0$, since the potential and each color ball are invariant under simultaneous adjoint rotations. At fixed finite volume, the map

$s \mapsto M(s)$ is continuous. If $M(s)$ first reached the value one at some $s_0 \le 1$, integration up to s_0 would give $M(s_0) \le Ctr(2 + E_*) < 1/2$, after the corresponding reduction of t_* in Definition 14.4(6), a contradiction. Hence $M(s) < 1$ for all $s \in [0,1]$. Inserting this into the last differential inequality, we obtain $M(1) \le Ct\|\theta\|_\infty$. The constants may absorb the fixed E_*. This proves the mean bound in (1).

Step 5: the conditioning identity. By the description of the support and the density of the fiber in Theorem 14.6(5) and (7),

$$\kappa_c^{\mathrm{old}}(A) = \frac{v_c(A \cap S_c)}{v_c(S_c)}$$

for every Borel set A. The denominator is strictly positive, since $\Omega^{\mathrm{phys}}_{Q_{\mathrm{phys}}} \subset S_c$ has positive Lebesgue measure and the density of v_c is positive. This identity requires no convexity of S_c, and it proves the second statement of (1).

Step 6: tail bounds. We reduce t_* so that every scalar mean under v_c is at most $P_{\mathrm{phys}}/(2\sqrt{3}t)$ and at most $p_{\mathrm{w}}(t)/(2\sqrt{3})$. This is possible since the means are $O(tr)$ by Step 4, P_{phys} is a fixed positive multiple of r, and $p_{\mathrm{w}}(t) \ge p_0$. Let $a = P_{\mathrm{phys}}$ or $a = Q_{\mathrm{phys}}$. If $|w_{i,e}| \ge a/t$, then one of its three scalar coordinates has magnitude at least $a/(\sqrt{3}t)$, and hence deviates from its mean by at least $a/(2\sqrt{3}t)$. Applying (14.13) to both tails of each scalar coordinate and summing over the $3b_{\mathrm{w}}$ scalar coordinates, we obtain

$$v_c\big((\Omega_a^{\mathrm{phys}})^c\big) \le 6b_{\mathrm{w}} e^{-m_{\mathrm{cv}} a^2/(24t^2)}. \tag{14.17}$$

The same calculation with the threshold $p_{\mathrm{w}}(t)$ gives

$$v_c\big((\Omega_p^{\mathrm{w}})^c\big) \le 6b_{\mathrm{w}} e^{-m_{\mathrm{cv}} p_{\mathrm{w}}(t)^2/24} = \delta_t. \tag{14.18}$$

Now suppose $d_t < 1$. Since $\Omega^{\mathrm{phys}}_{Q_{\mathrm{phys}}} \subset S_c$, (14.17) with $a = Q_{\mathrm{phys}}$ implies $v_c(S_c) \ge 1 - d_t$. Dividing the bounds (14.17) with $a = P_{\mathrm{phys}}$ and (14.18) by this denominator and using Step 5, we obtain both bounds in (2), the second with exactly ϵ_t.

Step 7: density ratios. Let $C(c)$ denote the prefactor (14.9), as in (5), with the chosen convention for constants; it depends on c but not on w. By Theorem 14.6(2) and (7), on B_r we have

$$G(c) = C(c) \int_{S_c} e^{-V_c}\, dw, \qquad \widetilde{F}(c) = C(c) \int_{\Omega_p^{\mathrm{w}}} e^{-V_c}\, dw.$$

Since the cutoff product is a subset of S_c, dividing proves the identity in (3) without any smallness condition. If $d_t < 1$ and $\epsilon_t < 1$, then (2) implies the two-sided bound in (3). Both formulas are compatible with the orthogonal changes of representative of Theorem 14.3(3), and hence hold on D_r for this explicit density.

Step 8: total variation and failure of the next step. For the comparison of normalized measures and the failure bound, we assume only $d_t < 1$, so that ϵ_t is a nonnegative upper bound; the condition $\epsilon_t < 1$ is not needed. Let η be the joint law of the coarse field and the fluctuation variables under the uncut fiber densities, conditioned on $c \in D_r$, and let A be the event that $w \in \Omega_p^{\mathrm{w}}$. Integrating the second bound of (2) over c, we obtain $\eta(A^c) \le \epsilon_t$. For any probability measure η and any event A of positive probability, $\eta = \eta(A)\eta(\cdot \mid A) + \eta(A^c)\eta(\cdot \mid A^c)$, and the two conditional measures are carried by disjoint events. Hence

$$\sup_B |\eta(B) - \eta(B \mid A)| = \eta(A^c) \sup_B |\eta(B \mid A^c) - \eta(B \mid A)| = \eta(A^c).$$

Taking the image under a measurable map does not increase this distance, since an event for the coarse field pulls back to an event in the joint space. The images of η and $\eta(\cdot \mid A)$ under the projection to the coarse field are the two normalized measures in (4), so they differ in total variation by at most ϵ_t. For the failure bound, note that σ is carried by D_r. By Theorem 14.3 with $u = r$, and the reduction of r_{geom} in step (2) of Definition 14.4, every configuration $U \in D_r$ with $T_M^J U \in D_r$ satisfies the coordinate conditions of the step, so the next step can fail only if $T_M^J U \notin D_r$ or if $T_M^J U \in D_r$ and the fluctuation cutoff fails. The probability of the second event is $\sigma(T_M^J U \in D_r)\, \eta(A^c) \le \epsilon_t$. This proves (4).

Step 9: the larger coarse fields. We now prove (5), without using any inclusion of the cutoff product in S_c for coarse fields outside D_r. Since the fine side is $M^J m$ and the coarse side is $m \ge 4$, each averaging step has coarse side $M^{J-i-1} m \ge 4$. Moving an input configuration in D_r into B_r and applying Lemma 7.2 J times, we find, by gauge covariance, $T_M^J(D_r) \subset D_{R_{\mathrm{buf}}}$ with $R_{\mathrm{buf}} = (10M)^J r$. For $c \in B_{R_{\mathrm{buf}}}$, applying Theorem 14.3 with $u = R_{\mathrm{buf}}$ and using $D_r \subset D_{R_{\mathrm{buf}}}$, we obtain $|w_{i,e}| \le C_{\mathrm{TG}} R_{\mathrm{buf}}/(\lambda_* t)$ for every point of S_c. Since $\mathscr{L}'_{\mathrm{en}} > 2C_{\mathrm{TG}} C_{\mathrm{map},J}/\lambda_*$, the set S_c lies in the interior of $\mathcal{K}'_t$.

Both $\mathcal{K}'_t$ and the fluctuation cutoff domain satisfy the conditions fixed for the larger coarse fields in Definition 14.4. More precisely, by (14.12), we have $R_{\mathrm{buf}} + \mathscr{L}'_{\mathrm{en}} r < r_J$, the denominator of the majorant is at least $1/2$, and both radii lie in the products of step (2). Therefore the image condition of Theorem 13.2 also holds on this larger product. The Hessian argument of Theorem 14.6(6) applies with the physical radius $\mathscr{L}'_{\mathrm{en}} r$ in place of R_{en}, using the inequality for $C\mathscr{L}'_{\mathrm{en}} r$ in step (4) of Definition 14.4. Steps 2 to 4 apply with the coarse-field radius R_{buf} in place of r, with the fixed product and the weighted sums otherwise unchanged. These are the conditions imposed when choosing $r_{\max}$ and t_*. They give scalar means of size at most $Ct R_{\mathrm{buf}}$ and the tail bound (14.18) on the larger product, with the same exponent. With $H(c) = C(c) \int_{\mathcal{K}'_t} e^{-V_c}\, dw$ as in (5), the inclusion $S_c \subset \mathcal{K}'_t$ implies $G(c) \le H(c)$, also when S_c is empty. The ratio $\widetilde{F}(c)/H(c)$ is the probability of the cutoff product under the normalized measure on $\mathcal{K}'_t$, so the concentration bound gives

$\widetilde{F}(c)/H(c) \geq 1-\delta_t$, that is, $H(c) \leq \widetilde{F}(c)/(1-\delta_t)$ when $\delta_t < 1$. No inclusion of the cutoff product in S_c has been used outside D_r.

Step 10: the exit probability. Assume $\delta_t < 1$. The law of the coarse field under σ has density $G/\int G\,dH$, and by Step 9 it is carried by $D_{R_{\mathrm{buf}}}$. Its exit probability satisfies

$$\frac{\int_{D_{R_{\mathrm{buf}}}\setminus D_r} G\,dH}{\int_{D_{R_{\mathrm{buf}}}} G\,dH} \leq \frac{1}{1-\delta_t}\,\frac{\int_{D_{R_{\mathrm{buf}}}\setminus D_r} \widetilde{F}\,dH}{\int_{D_r} \widetilde{F}\,dH}.$$

In the numerator, we used the upper bound of Step 9, and in the denominator, we used $G \geq \widetilde{F}$ on D_r, from (3). Since $\widetilde{F} = A_{\mathrm{coord}} e^{-E_{d_0+J}/g_J^2 - R_+}$, the common constant A_{coord} cancels, and we obtain the ratio of the integrals Z_{aux} in (5). The same cancellation holds for any convention for centering.

Step 11: versions of the density and asymptotics. All pointwise statements used the explicit density given by the based coarea formula. Any other disintegration agrees with it almost everywhere on the set of coarse fields where it is positive, which gives the stated qualification. The explicit factor b_{w} cannot be absorbed into a constant independent of the volume. Along a refinement at fixed γ, we have $t^2 = \gamma/(Mm)$ and hence $m^3 = (\gamma/M)^3 t^{-6}$, so that

$$b_{\mathrm{w}} = 2(B^3-1)(\gamma/M)^3 t^{-6}.$$

With $x = \log(1/t)$, the logarithm of δ_t is at most $C + 6x - cA_{\mathrm{w}}^2 x^2$, which is below $-hx$ for every fixed $h > 0$ once x is large. The tail with a physical radius has the stronger exponent $-c/t^2$. Since the mesh is $L/m = (LM/\gamma)t^2$, both errors are smaller than every fixed power of the mesh along such a sequence, and the denominators $1-d_t$ tend to one. □

14.6 Partition-function bounds on the stopping lattice

After k RG steps, the stopped measure lives on the stopping lattice Λ_n, $n = n_{kJ}$, at depth $d = kJ$. This subsection bounds its partition function from below and the exponential moments of the Wilson action under it from above. It also compares effective classical actions with the Wilson action at every depth, records an exact identity for normalization constants, and bounds derivatives of the centered effective interactions in fixed charts. The lower bound is a trial integral over the same domain D_r as the stopped measure, in a periodic tree gauge with all non-tree link variables of size $a_0 g$. The upper bound enlarges only the numerator, to an integral over an open cube, for which Lemma 6.10 gives an explicit bound. The ranks of the two gauge-fixed integrals are as follows.

Integral	Group variables	Power of g
periodic tree trial inside D_r	$2n^3+1$	$g^{3(2n^3+1)}$
open cube, upper bound for the numerator	$2n^3-3n^2+1$	$g^{3(2n^3-3n^2+1)}$
ratio	difference $3n^2$	g^{-9n^2}

Since the numerator and the denominator are each estimated once, the boundary term $9n^2\log(1/g)$ appears once. These bounds, obtained by elementary gauge integration, are in the tradition of the stability estimates of Faria da Veiga and O'Carroll [64, 65]; here we need the precise periodic correction.

Theorem 14.8. *Use the parameters of Definition 14.4. There are $\kappa_*, a_0, g_* > 0$ and $C_A, C_R, C_\# > 0$, independent of the side $n \geq 4$, of the depths $d = kJ$ reached by the recursion, of the number of RG steps, and of $0 < g \leq g_*$, with the following properties. The integration domain is exactly D_r on the stopping lattice Λ_n. Let $R = R_k$ be the effective interaction of Theorem 14.6 at depth $d = kJ$, and put*

$$D = 3n^3, \qquad b_n = 2n^3+1, \qquad c = R(I), \qquad \overline{R} = R - c,$$

$$F_g = \frac{E_0}{g^2} = \frac{4}{g^2}\sum_p q(U_p), \qquad Z^{\mathrm{raw}} = \int_{D_r} e^{-E_{d,\mathrm{sat}}/g^2 - R_{\mathrm{sat}}}\, dH_n, \qquad \widehat{Z}_1 = e^c Z^{\mathrm{raw}}.$$

Here D is the number of links, and c is a constant, not a coarse field.

1. *For every integer $d \geq 0$, $E_{d,\mathrm{sat}} \geq \kappa_* E_0$ on the gauge-invariant extension of the larger classical domain, whether or not an effective interaction is attached to the depth d.*
2. *For the effective interactions of the stopped measures, $|\overline{R}| \leq C_R D E_* \rho^2$ on D_r. On the trial set of configurations which, in a periodic spanning tree gauge based at a root, have every non-tree link logarithm of norm at most $a_0 g < \min(r/2, 1)$,*

$$E_d \leq C_A D a_0^2 g^2, \qquad |\overline{R}| \leq C_R D E_* a_0^2 g^2,$$

$$Z^{\mathrm{raw}} \geq \Big(\frac{a_0 g - \sin(a_0 g)}{2\pi}\Big)^{b_n} e^{-c - C_A D a_0^2 - C_R D E_* a_0^2 g^2}.$$

3. *Let $\widehat{\nu} = \widehat{Z}_1^{-1} \mathbf{1}_{D_r} e^{-E_d/g^2 - \overline{R}}\, dH_n$ be the stopped measure, and put $\lambda_{\mathrm{en}} = \kappa_*/2$. Then*

$$\widehat{\nu}(F_g \geq u) \leq \exp(C_\# D + 9n^2\log(1/g) - \lambda_{\mathrm{en}} u), \qquad u \geq 0,$$

$$\mathbb{E}_{\widehat{\nu}} e^{sF_g} \leq \exp(C_\# D + 9n^2\log(1/g)), \qquad 0 \leq s \leq \lambda_{\mathrm{en}},$$

$$\mathbb{E}_{\widehat{\nu}} F_g \leq [C_\# D + 9n^2\log(1/g) + 1]/\lambda_{\mathrm{en}}.$$

The open cube with n vertices per side has rank $b_n^{\mathrm{op}} = 2n^3 - 3n^2 + 1$. Only an upper bound for a numerator is enlarged to the Haar integral over this open cube; the denominator remains the trial integral over D_r. The unmatched power of the coupling is exactly $g^{-3(b_n - b_n^{\mathrm{op}})} = g^{-9n^2}$.

4. *Fix $0 < \lambda_0 \le 1$, let $\lambda \in [\lambda_0, 1]$, and put*

$$\widehat{Z}_\lambda = \int_{D_r} e^{-\lambda E_d/g^2 - \lambda \overline{R}}\, dH_n, \qquad d\tau^{(\lambda)} = \widehat{Z}_\lambda^{-1} \mathbf{1}_{D_r} e^{-\lambda E_d/g^2 - \lambda \overline{R}}\, dH_n.$$

   After one reduction of g_* depending on λ_0, for $0 \le s \le \lambda_0 \kappa_*/2$,

$$\log \mathbb{E}_{\tau^{(\lambda)}} e^{sF_g} \le CD + 9n^2 \log(1/g), \qquad \log\Big\{ \frac{\widehat{Z}_\lambda}{\widehat{Z}_1} \mathbb{E}_{\tau^{(\lambda)}} e^{sF_g} \Big\} \le CD + 9n^2 \log(1/g).$$

   The second bound is obtained from one numerator and one denominator, not as the sum of bounds for the two factors.

5. *For every fixed $\alpha_0 > 0$ and $0 < \lambda_0 \le \lambda \le \lambda_1 < \infty$,*

$$\log \frac{\int_{\mathcal{U}_n} e^{-(\lambda\alpha_0 - s)F_g}\, dH_n}{\widehat{Z}_1} \le C_{\lambda_0,\lambda_1,\alpha_0} D + 9n^2 \log(1/g), \qquad 0 \le s \le \lambda_0 \alpha_0/2,$$

   where g_* is reduced so that the effective inverse coupling $4(\lambda\alpha_0 - s)/g^2$ is at least one. No condition on charts or small fields is imposed on this numerator.

6. *Let an initial probability density with unnormalized normalizing constant $\mathcal{Z}_{\mathrm{base}}$ be restricted by the small-field tests and pushed forward by finitely many RG steps of Theorem 14.6 to the lattice Λ_n, and suppose that the resulting subprobability measure is written in the centered convention as*

$$d\xi_T = \mathcal{Z}_{\mathrm{base}}^{-1} e^{\Lambda_T^{\mathbb{T}}} \mathbf{1}_{D_r} e^{-E_d/g^2 - \overline{R}}\, dH_n, \qquad p_T = \xi_T(\mathcal{U}_n) \le 1.$$

   Then, exactly,

$$\frac{e^{\Lambda_T^{\mathbb{T}}}}{\mathcal{Z}_{\mathrm{base}}} = \frac{p_T}{\widehat{Z}_1} = \frac{p_T e^{-c}}{Z^{\mathrm{raw}}} \le \widehat{Z}_1^{-1}.$$

   All normalization factors of integrated Gaussian coordinates, coarea constants, constant terms of cluster expansions, and constants removed from effective interactions belong to this same normalization constant $\Lambda_T^{\mathbb{T}}$.

7. *The constants depend only on the parameters of the recursion, the constants of Theorem 11.2 and Lemma 6.10, and the displayed parameters $\lambda_0, \lambda_1, \alpha_0$. For every fixed $\gamma_0 > 0$, one minimum side makes these estimates hold simultaneously for $g^2 = \gamma/n$, all $0 < \gamma \le \gamma_0$, and every larger side. The estimates involve no block regions.*

8. *The effective interactions $R_{d,\gamma} = R_k$ satisfy $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_{d,\gamma}) \le E_*$ and $DR_{d,\gamma}(I) = 0$, and they are gauge invariant on overlaps, as in Theorem 14.6. Fix a side n, a number $0 < \eta < 1$, and a finite family of smooth charts Ψ_α of configurations on Λ_n, defined on parameter domains $\widetilde{W}_\alpha$, with relatively compact subdomains $W_\alpha \Subset \widetilde{W}_\alpha$ and smooth based gauge transformations, such that the charts and the gauge transformations extend smoothly to neighborhoods of the*

closures $\overline{W}_\alpha$ and the gauge-transformed representatives of $\Psi_\alpha(y)$, $y \in W_\alpha$, lie in $B_{(1-\eta)\rho}$. Then every derivative of fixed positive order of $\overline{R}_{d,\gamma,\mathrm{sat}} \circ \Psi_\alpha$ on W_α is bounded uniformly in $d = kJ$ and $0 < \gamma \le \gamma_0$. In link logarithms, the bound of order $j \ge 1$ in the weighted norm is $C_j E_* \rho^{2-j} \eta^{-j}$, with $C_j = j!$. Passing to a fixed chart costs its fixed derivative bounds and a norm conversion in fixed finite dimension. The charts are composed with fixed smooth gauge transformations, never with a differentiated Borel selection of gauges. The constants may depend on n, j, η, the charts, and the larger domains, but not on the depth or the coupling.

9. *The law of the coarsest field $V^{KJ} = T_M^{KJ} U$, a configuration on Λ_u, under the conditioned Wilson measure $\nu_{N,u,\gamma}$ is the stopped measure after all K RG steps.*

Part (1) holds at every integer depth, and part (8) gives bounds on derivatives from the analytic coefficient norm itself; a bound on the oscillation of $\overline{R}$ alone would not control its derivatives. Any later comparison of normalization constants on block regions must be made with the normalization constant $\Lambda_T^{\mathbb{T}}$ of part (6) on the whole torus.

Proof. Step 1 relates the stopping lattice to the recursion. Steps 2 and 3 prove (1) and the first bound in (2). Step 4 proves the trial lower bound, and Step 5 the open-cube upper bound. Steps 6 to 8 combine them to prove (3) to (5), Step 9 proves (6), and Step 10 proves (7). Steps 11 and 12 prove (8), and Step 13 proves (9).

Step 1: depths and couplings. At a depth $d = kJ$, the effective interaction in this proof is the function $R_{d,\gamma} = R_k$ of Theorem 14.6, in its chosen convention for constants. By that theorem, it is holomorphic on B_ρ and gauge invariant on overlaps, with $DR_{d,\gamma}(I) = 0$ and $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_{d,\gamma}) \le E_*$, uniformly in k and in the coupling. Its real density is the stopped density on D_r. Subtracting the value at the identity changes no derivative of positive order and no analytic norm. We take g_* so small that $g_*/\sqrt{M} \le t_*$. Since $g_J^2 = Mt^2$, at the stopping coupling g the largest integrated coupling of the last RG step is $g/\sqrt{M}$, and all earlier integrated couplings are smaller. This implies that every run of the recursion in the theorem satisfies the conditions on the coupling in Definition 14.4; with zero RG steps there is no such condition.

Step 2: the classical lower bound. We apply the comparison of Theorem 11.2 at one stopping field and one mesh, for every integer depth, without reference to an effective interaction. With the larger radius r_{C} fixed in Section 14.2, the comparison product of that theorem satisfies

$$P_\infty \le \exp\Big(C r_{\mathrm{C}} \sum_{j \ge 0} M^{-2(j+1)} \Big) < \infty.$$

We conclude that $E_d \ge \kappa_* E_0$ with $\kappa_* = P_\infty^{-1}$. For an argument in the gauge-invariant extension of the classical domain, we fix a gauge transformation taking it into the corresponding classical ball inside $B_{r_{\mathrm{C}}}$ and use the clause of Theorem 11.2 on transport by a fixed gauge transformation. Theorem 11.6 identifies the transported values with $E_{d,\mathrm{sat}}$

for every $d \geq 0$, while E_0 is gauge invariant. This proves (1) on the whole gauge-invariant domain and at every integer depth. Only the assertions involving $R_{d,\gamma}$ restrict the depth to the values $d = kJ$ reached by the recursion. If the larger radius requires a smaller κ_*, it is fixed before the final reductions of the coupling bound.

Step 3: oscillation of the effective interaction. Let $A_s \geq 0$ denote the summand of order s in the weighted coefficient norm of R, so that $\sum_{s\geq 2} A_s \leq E_*$. For $\|\xi\|_\infty \leq \epsilon \leq \rho$ we expand $R(e^\xi) - R(I)$, fix one scalar link and color label in each coefficient, and sum the remaining labels. Since there are $3D$ choices of the fixed label and the spatial weights are at least one, absolute summation of the Taylor series gives

$$|R(e^\xi) - c| \leq C_R D\epsilon^2 \sum_{s\geq 2} A_s(\epsilon/\rho)^{s-2} \leq C_R D E_* \epsilon^2. \tag{14.19}$$

By gauge invariance on overlaps, this bound holds at every representative. With $\epsilon = r < \rho$, we obtain the bound $|\overline{R}| \leq C_R D E_* \rho^2$ on D_r. No estimate of c is involved.

Step 4: the trial lower bound. We choose a spanning tree of the periodic lattice graph and a root. By Lemma 5.1, the integral of a gauge-invariant integrand over a gauge-invariant set is an integral over exactly $b_n = D - (n^3 - 1) = 2n^3 + 1$ independent non-tree link variables, each with the Haar probability measure; the tree variables have total mass one, no root variable is integrated, and no volume factor of a stabilizer appears. We fix $a_0 > 0$ and reduce g_* so that $\epsilon = a_0 g < \min(r/2, 1)$ for all $0 < g \leq g_*$. Since all tree-fixed logarithms on the trial set are at most ϵ, the trial set lies strictly inside the same domain D_r. By the upper bound of Theorem 11.2 at radius ϵ, we have $E_d \leq C_A D\epsilon^2$ uniformly in d and n, and (14.19) implies $|\overline{R}| \leq C_R D E_* \epsilon^2$. By (4.2), the exact Haar mass of the ball for one non-tree link is

$$m(\epsilon) = \frac{1}{16\pi^2} \int_{|X|<\epsilon} \operatorname{sinc}^2(|X|/2)\, d^3X = \frac{1}{\pi} \int_0^\epsilon \sin^2(s/2)\, ds = \frac{\epsilon - \sin\epsilon}{2\pi}.$$

Integrating the lower bound for the density over the product of these b_n balls gives the lower bound for Z^{raw} in (2), with the factor e^{-c} kept.

We will use the weaker form

$$\widehat{Z}_1 \geq e^{-C_L D} g^{3b_n}. \tag{14.20}$$

To prove it, note that $\operatorname{sinc}(s/2) \geq \operatorname{sinc}(1/2) > 0$ for $0 \leq s \leq 1$, since the numerator $x\cos x - \sin x = -\int_0^x y \sin y\, dy$ of the derivative of $\sin x/x$ is nonpositive on $[0, 1/2]$. The radial integral therefore gives $m(\epsilon) \geq c_{\mathrm{ball}}\epsilon^3$, with $c_{\mathrm{ball}} = \operatorname{sinc}^2(1/2)/(12\pi)$. Since $b_n \leq D$ and $g_* \leq 1$, all fixed factors in the trial bound have logarithms bounded in absolute value by a fixed multiple of D. This proves (14.20).

Step 5: the open-cube upper bound. For $\alpha \geq \alpha_{\min} > 0$, let

$$I_\alpha(g) = \int_{\mathcal{U}_n} e^{-\alpha F_g}\, dH_n.$$

Since $q \geq 0$, deleting every plaquette that crosses a periodic seam can only increase the integrand. The remaining complex has n vertices per direction, $3(n-1)n^2$ links, and n^3 vertices, so its tree rank is

$$b_n^{\mathrm{op}} = 3(n-1)n^2 - n^3 + 1 = 2n^3 - 3n^2 + 1.$$

Each unused seam link contributes a Haar probability integral equal to one. Lemma 6.10, applied to this open cube at inverse coupling $\beta = 4\alpha/g^2$, which is legitimate when $4\alpha_{\min}/g_*^2 \geq 1$, gives

$$I_\alpha(g) \leq c_G^{b_n^{\mathrm{op}}} (4\alpha/g^2)^{-3b_n^{\mathrm{op}}/2} \leq e^{C_U D} g^{3b_n^{\mathrm{op}}}, \tag{14.21}$$

where $c_G = \sqrt{2/\pi}$ is the constant of that lemma. The constant C_U depends on $\alpha_{\min}$ and c_G, and is independent of n, d, and g.

Step 6: moments under the stopped measure. We use (1) and the oscillation bound of Step 3 in the numerator. For $0 \leq s \leq \lambda_{\mathrm{en}} = \kappa_*/2$ we have $-E_d/g^2 + sF_g \leq -(\kappa_* - s)F_g$, and hence

$$\int_{D_r} e^{-E_d/g^2 - \overline{R} + sF_g}\, dH_n \leq e^{C_R D E_* \rho^2} I_{\kappa_* - s}(g) \leq e^{C_U D} g^{3b_n^{\mathrm{op}}},$$

after including the oscillation constant in C_U. Here $\alpha_{\min} = \kappa_*/2$ works, including at the endpoint $s = \lambda_{\mathrm{en}}$. Dividing by (14.20) leaves exactly

$$3(b_n - b_n^{\mathrm{op}}) = 9n^2$$

unmatched powers of g^{-1}. This proves the bound on the exponential moment in (3), with $A = C_\# D + 9n^2 \log(1/g)$, after enlarging $C_\#$ if necessary. The exponential Markov inequality at $s = \lambda_{\mathrm{en}}$ gives the tail bound. Since $A \geq 0$, integrating the minimum of the tail bound and one gives

$$\mathbb{E}_{\widehat{\nu}} F_g \leq \int_0^\infty \min(1, e^{A - \lambda_{\mathrm{en}} u})\, du = \frac{A+1}{\lambda_{\mathrm{en}}}.$$

The exponential moment at the endpoint $s = \lambda_{\mathrm{en}}$ was proved through its defining integral, not by integrating a tail bound. In the uncentered convention, the same factor e^{-c} appears in the numerator and in the denominator and cancels.

Step 7: tempered measures. Fix $\lambda_0 \leq \lambda \leq 1$. On the same tree trial set, the exponent $\lambda E_d/g^2 + \lambda \overline{R}$ is bounded above by CD, uniformly in λ, and hence

$$\widehat{Z}_\lambda \geq e^{-CD} g^{3b_n}. \tag{14.22}$$

For $0 \le s \le \lambda_0\kappa_*/2$, the numerator is bounded above by

$$e^{CD} I_{\lambda\kappa_* - s}(g) \le e^{CD} g^{3b_n^{\mathrm{op}}}, \qquad \lambda\kappa_* - s \ge \lambda_0\kappa_*/2,$$

and one fixed reduction of g_* makes the inverse coupling of Lemma 6.10 at least one throughout this interval. Dividing by (14.22) gives the first bound in (4). For the second bound, we keep the exact identity

$$\frac{\widehat{Z}_\lambda}{\widehat{Z}_1} \mathbb{E}_{\tau^{(\lambda)}} e^{sF_g} = \frac{\int_{D_r} e^{-\lambda E_d/g^2 - \lambda\overline{R} + sF_g}\, dH_n}{\widehat{Z}_1},$$

and use the last upper bound for the numerator and (14.20), once each. The difference of ranks is still $3n^2$, so the exponent is again $9n^2 \log(1/g)$. Estimating the two factors on the left separately would count this term twice.

Step 8: the unrestricted numerator. Let $\alpha = \lambda\alpha_0 - s \ge \lambda_0\alpha_0/2$. Applying (14.21) to the unrestricted Wilson numerator and (14.20) to $\widehat{Z}_1$ proves (5), after the corresponding fixed reduction of g_*. The constants may depend on λ_0, λ_1, and α_0, but not on the depth or the side.

Step 9: the identity for normalization constants. Integrating the subprobability measure in (6) gives exactly

$$p_T = \mathcal{Z}_{\mathrm{base}}^{-1} e^{\Lambda_T^{\mathbb{T}}} \widehat{Z}_1.$$

Both $\mathcal{Z}_{\mathrm{base}}$ and $\widehat{Z}_1$ are positive. The total mass p_T is at most one, since restricting a probability measure and pushing it forward cannot increase its total mass. Solving this equality, and using $\widehat{Z}_1 = e^c Z^{\mathrm{raw}}$, proves the identities and the inequality in (6). Removing a Gaussian coordinate, a reference Haar disintegration factor, a constant term, or the constant of an effective interaction changes the representation of the density and the normalization constant together; it cannot change this identity for the total mass.

Step 10: uniformity in the coupling. All constants so far were chosen from the parameters of the recursion and the constants of Theorem 11.2 and Lemma 6.10. For a fixed $\gamma_0 > 0$, we take a minimum side at least γ_0/g_*^2, rounded up to an integer, and to a multiple of four when the RG steps require it. Then $g^2 = \gamma/n \le g_*^2$ for every larger side and every $0 < \gamma \le \gamma_0$. Since the earlier integrated couplings of the recursion are smaller, the same reductions hold at every depth. This proves (7).

Step 11: derivatives in link logarithms. We write $\mathcal{R}_{d,\gamma}(\xi) = R_{d,\gamma}(e^\xi) - R_{d,\gamma}(I)$ and

$$A_{k,d,\gamma} = \frac{\rho^{k-2}}{k!} \|D^k \mathcal{R}_{d,\gamma}(0)\|_{a_{\mathrm{sp}},\mathrm{root}}, \qquad \sum_{k\ge 2} A_{k,d,\gamma} \le E_*.$$

Let $j \geq 1$, $v = 1 - \eta$, and $\|\xi\|_\infty \leq v\rho$. We differentiate the absolutely convergent Taylor series on this strictly smaller ball. Contracting its $k - j$ undifferentiated indices does not increase the weight of the network on the remaining indices, so

$$\|D^j \mathcal{R}_{d,\gamma}(\xi)\|_{a_{\mathrm{sp}},\mathrm{root}} \leq \rho^{2-j} \sum_{k \geq \max(2,j)} A_{k,d,\gamma} \frac{k!}{(k-j)!} v^{k-j}.$$

Each multiplier satisfies

$$\frac{k!}{(k-j)!} v^{k-j} = j! \binom{k}{j} v^{k-j} \leq j! \eta^{-j},$$

because $\binom{k}{j}\eta^j v^{k-j}$ is one nonnegative term of $(\eta + v)^k = 1$. It follows that

$$\|D^j \mathcal{R}_{d,\gamma}(\xi)\|_{a_{\mathrm{sp}},\mathrm{root}} \leq j! E_* \rho^{2-j} \eta^{-j} \qquad (j \geq 1). \tag{14.23}$$

The uncontrolled constant $R_{d,\gamma}(I)$ disappears through differentiation; no derivative bound is deduced from the oscillation bound alone.

Step 12: derivatives in charts. For $y \in W_\alpha$, we write the gauge-transformed representative of $\Psi_\alpha(y)$ as $e^{\xi_\alpha(y)} \in B_{(1-\eta)\rho}$. The chart, the subdomain, and the gauge transformation are fixed independently of d and γ. By continuity, the extended representatives on $\overline{W}_\alpha$ lie in the closed ball of radius $(1 - \eta)\rho$, strictly inside the domain of the logarithm, and compactness gives a neighborhood of $\overline{W}_\alpha$ on which the same logarithm is smooth. For $1 \leq h \leq j$, let

$$C_{\alpha,h} = \sup_{y \in \overline{W}_\alpha} \|D^h \xi_\alpha(y)\|_{\mathrm{op},\infty} < \infty;$$

finiteness follows from continuity of these derivatives on the compact closure. The derivatives of the gauge transformation, of the group multiplication, and of the logarithm are included in these constants. By gauge invariance on overlaps, $\overline{R}_{d,\gamma,\mathrm{sat}} \circ \Psi_\alpha = \mathcal{R}_{d,\gamma} \circ \xi_\alpha$ on W_α. The chain rule for higher derivatives gives one term for each partition π of $\{1, \ldots, j\}$, with outer derivative $D^{|\pi|}\mathcal{R}_{d,\gamma}$ and inner factors $D^{|B|}\xi_\alpha$, $B \in \pi$; this follows by induction, since a new derivative either joins an existing block or forms a new singleton block. Since there are $9n^3$ scalar link and color labels, fixing the first outer index and summing the others bounds a full contraction by $9n^3$ times the weighted norm. Using (14.23), we obtain

$$\|D^j(\overline{R}_{d,\gamma,\mathrm{sat}} \circ \Psi_\alpha)\|_{\mathrm{op}} \leq 9n^3 E_* \sum_\pi |\pi|! \, \rho^{2-|\pi|} \eta^{-|\pi|} \prod_{B \in \pi} C_{\alpha,|B|}. \tag{14.24}$$

Every factor on the right is fixed before the depth and the coupling vary, and for a finite family of charts we take the maximum. To cover a compact subset of the gauge-invariant

domain with this interior property, we choose at each of its points a local chart and a fixed gauge transformation whose representatives stay in $B_{(1-\eta)\rho}$. Since these conditions are open, there exists a smaller parameter neighborhood with compact closure inside the chart on which they still hold. These neighborhoods cover the compact set, and a finite subcover gives subcharts of the required kind, with extensions of the charts and gauge transformations past their closures. We include the derivatives of their logarithmic charts in the same maximum. This proves (8).

Step 13: the coarsest marginal. At the coarsest lattice Λ_u, no tests of coarser RG steps remain. The small-field event is then exactly the event whose normalized pushforward defines the stopped density after K steps, so the law of V^{KJ} under $\nu_{N,u,\gamma}$ is this stopped measure. □

14.7 Flat configurations

A configuration U is flat if $U_p = I$ for every plaquette p. Flat configurations are the minima of the Wilson action, and the weak-coupling analysis of Sections 33 to 36 takes place near them. The lemma below first records the classification of flat configurations on a periodic lattice by their holonomies; this is the usual description of flat connections on $\mathbb{T}_L^3$ by representations of $\pi_1(\mathbb{T}_L^3) = \mathbb{Z}^3$, and we include an elementary proof on the lattice. The rest of the lemma is specific to our construction: all flat configurations, including those whose holonomies form one of the eight central triples, satisfy every test of the small-field event, with vanishing fluctuation variables, once the coarse side is large. The reason is that a flat configuration is gauge equivalent to a constant configuration with link variables $e^{H_i^{\log}/n}$, which are close to the identity when n is large, although the holonomies $e^{H_i^{\log}}$ around the cycles of the torus can be far from it.

Lemma 14.9. *1. On every periodic lattice Λ_n, the flat configurations modulo periodic lattice gauge transformations are classified by their commuting holonomies around the three coordinate cycles, up to simultaneous conjugation:*

$$\mathrm{Hom}(\mathbb{Z}^3, \mathrm{SU}(2))/\mathrm{SU}(2) \simeq (\mathbb{R}/4\pi\mathbb{Z})^3/\{\boldsymbol{\vartheta} \sim -\boldsymbol{\vartheta}\},$$

where $\boldsymbol{\vartheta}$ parametrizes the triple $(e^{\vartheta_1 u}, e^{\vartheta_2 u}, e^{\vartheta_3 u})$ for a unit vector $u \in \mathfrak{su}(2)$. There are exactly eight central points, the central triples, at which each coordinate is 0 or 2π modulo 4π. Every flat configuration is gauge equivalent to a configuration with constant commuting link variables $e^{H_i^{\log}/n}$ in direction i, where the $H_i^{\log} \in \mathfrak{su}(2)$ commute, $|H_i^{\log}| \le 2\pi$, and $e^{H_i^{\log}}$ are the cycle holonomies. The block averaging map T_M raises these link variables to their Mth powers and preserves the three cycle holonomies. Taking commuting Mth roots gives exact flat refinements.

2. *Fix $r_{\mathrm{flat}} > 0$ strictly smaller than r and than the radii of all real coordinate charts and gauge transformations of one RG step fixed in Definition 14.4. If the coarse side u satisfies $2\pi/u < r_{\mathrm{flat}}$, then, for every $K \geq 0$, $N = uB^K$, and every coupling allowed by Definition 14.4, every flat configuration on Λ_N belongs to the small-field event $S_{N,u,\gamma}$. In the successive based forests the coarse-field lift is flat, and*

$$z_i = 0, \qquad z_i^*(c_{i+1}) = 0, \qquad w_i = 0, \qquad E_{d,\mathrm{sat}}(c_i) = 0$$

at every intermediate scale and every depth d. The same holds for products of fluctuations of any positive physical radius that fit the fixed domains. All initial and coarse D_r tests and all intermediate classical and coarea conditions hold. Changes of representative based at the coarse vertices preserve the holonomies and the membership in the event. The gradient of the Wilson action vanishes at flat configurations, so they are stationary for the Wilson flow, and their three cycle holonomies are unchanged at every time.

3. *Under the side and parameter hypotheses of (2), at each fixed finite $N = uB^K$ and each allowed coupling, the compact set of flat configurations has an open neighborhood contained in $S_{N,u,\gamma}$; equivalently, there is $\delta_{N,u,\gamma} > 0$ with*

$$\{\max_p \rho_G(U_p) < \delta_{N,u,\gamma}\} \subset S_{N,u,\gamma}.$$

No bound on $\delta_{N,u,\gamma}$ uniform in N or in the coupling is asserted. Separately, on a fixed stopping lattice whose side n satisfies $2\pi/n < r_{\mathrm{flat}}$, the set of flat configurations lies in the open set D_r, with a neighborhood that depends only on n and r, not on the depth or the coupling.

No central triple is removed, and no projectively commuting or twisted flat data are introduced: all gauge transformations are periodic and all cycle holonomies commute.

Proof. Steps 1 to 4 prove (1), Steps 5 to 8 prove the membership statement in (2), Step 9 proves the statement on the Wilson flow, and Step 10 proves (3).

Step 1: path independence. We lift the periodic configuration to $\mathbb{Z}^3$. If all plaquette variables are the identity, the transport along a path is unchanged when an edge followed by its reverse is deleted, and when two adjacent coordinate steps are replaced by the other two sides of their elementary square. These moves connect any two paths with the same endpoints in $\mathbb{Z}^3$: one interchanges neighboring steps in different directions until the steps are grouped in coordinate order, and then cancels opposite steps in each group. This implies that a path and its reverse define inverse transports, and the transport $P_U(x)$ from the origin to a vertex x of the lift does not depend on the path.

Step 2: the holonomy triple. The cycle holonomies are $P_U(ne_i)$. By periodicity, the transport from ne_i to $ne_i + ne_j$ equals $P_U(ne_j)$, so path independence applied to the two orders of going around the i and j directions implies $P_U(ne_i)P_U(ne_j) = P_U(ne_j)P_U(ne_i)$.

A periodic loop whose lift ends at $n(k_1, k_2, k_3)$ has transport $P_U(ne_1)^{k_1} P_U(ne_2)^{k_2} P_U(ne_3)^{k_3}$. A periodic gauge transformation conjugates the triple by its value at the origin.

A commuting triple in SU(2) lies in one maximal torus. Indeed, suppose that one of the matrices is not central. Then it is unitarily diagonalizable with two distinct eigenvalues, and a matrix commuting with it preserves both eigenlines and is diagonal in this basis. The determinant condition then puts all three diagonal matrices in the same circle. If all three are central, any orthonormal basis works. Hence we can choose commuting logarithms $H_i^{\log} \in \mathfrak{su}(2)$ with $e^{H_i^{\log}} = P_U(ne_i)$ and $|H_i^{\log}| \le 2\pi$, and we set $a_i = e^{H_i^{\log}/n}$.

Step 3: an explicit gauge equivalence. Let $P_a(x)$ be the transport of the constant configuration with link variables a_i on $\mathbb{Z}^3$, and set $k_x = P_a(x)^{-1} P_U(x)$. By flatness and periodicity, we have $P_U(x + ne_i) = e^{H_i^{\log}} P_U(x)$, and the same identity holds for P_a, since $a_i^n = e^{H_i^{\log}}$. Since the two factors cancel in k_{x+ne_i}, the gauge transformation k is periodic. On an edge xy, we have $P_U(y) = P_U(x) U_{xy}$, so

$$k_x U_{xy} k_y^{-1} = P_a(x)^{-1} P_U(x) U_{xy} P_U(y)^{-1} P_a(y) = P_a(x)^{-1} P_a(y) = a_{xy}.$$

This gauge transformation is the identity at the origin. Conversely, a constant commuting configuration is flat and has the given cycle holonomies. Hence two flat configurations with the same based holonomy triple are equivalent under a gauge transformation based at the origin, and flat configurations modulo all periodic gauge transformations are classified by their triples modulo simultaneous conjugation.

Step 4: the angle coordinates. Writing the common element of the Cartan subalgebra as a multiple of a unit vector u gives the three angles, which have period 4π in our normalization. A change of the common eigenbasis either fixes the two eigenlines or interchanges them, and the interchange reverses all three angles. There is no independent change of sign for a single noncentral coordinate. At a central triple each angle is 0 or 2π modulo 4π, which gives exactly $2^3 = 8$ points. The sign involution fixes these points and removes none of them. This proves the stated quotient, including the central points.

For constant commuting link variables, every displaced contour in (4.3) and its straight reference path have the same net displacement Me_i, and hence the same transport a_i^M. The corresponding based loop holonomy is the identity, so the block averaging map involves no exceptional logarithm and returns a_i^M. The holonomy around a cycle of the coarse lattice is $(a_i^M)^{n/M} = a_i^n$, so the three cycle holonomies are preserved. Taking $a_i = e^{H_i^{\log}/(Mn')}$ refines a constant configuration on the lattice of side n' exactly to the lattice of side Mn'. This completes the proof of (1).

Step 5: flat lifts. We now check the minimizer and the membership statements on the larger classical products of Section 14.2. The coarse-field lift $U_0(c)$ of a flat coarse field

c is flat. Indeed, a fine plaquette inside one block has only identity factors, a plaquette crossing one interface has two equal and opposite interface factors, and a plaquette crossing two interfaces has exactly the four factors of a coarse plaquette. In all three cases, the product is the identity.

Step 6: zero classical action. For a constant commuting coarse field on a lattice of side n, there exists a zero-action configuration at every depth d, obtained by placing $e^{H_i^{\log}/(nM^d)}$ on the links of the finer lattice. Its logarithmic potential in units of the coarse lattice is the constant $H_i^{\log}/n$, with vanishing differences. Since the coarse link logarithms have norm at most $2\pi/n < r_{\text{flat}}$, this configuration, together with all the fixed changes of chart and of gauge, lies strictly inside the domains of the classical minimization. Its Wilson action is zero. Since the effective classical action is nonnegative and is defined as a constrained minimum, its value at this coarse field is zero at every depth. To transport this value to the coarse-field lift, we use Theorem 11.6: by the choice of r_{flat}, the constant representative and the forest-fixed representative both lie strictly inside B_{r_C}.

Step 7: vanishing fluctuation variables. Consider one averaging step. The block-tree gauge transformation of the constant configuration is based at every coarse vertex. If $x = v + \mathsf{r}$ lies in the centered block of the coarse vertex v, the value of this gauge transformation there is $\prod_j a_j^{\mathsf{r}_j}$. On a link inside a block, the commuting factors cancel to the identity. On an interface link in direction i, the offsets of the two endpoints differ by $1 - M$ in the ith coordinate, and the transformed value is $a_i^M = c_i$. Hence the forest-fixed configuration is exactly $U_0(c)$, and all its relative logarithms, including those of the eliminated edges, vanish. Its free coordinate is $z = 0$, which is a zero of the nonnegative function $F_d(c, \cdot)$ on the branch. The strict convexity and the uniqueness in Lemma 14.1 force $z_d^*(c) = 0$, and the normalization by the positive square root of K_d then gives $w = 0$. This proves the statement on minimizers.

Step 8: all tests. At every scale with side $n_s \geq u$, the constant link variables have logarithms of norm at most $2\pi/n_s \leq 2\pi/u < r_{\text{flat}} < r$. This proves the initial and coarse D_r tests, with strict inequality. The choice of r_{flat} below the radii of all fixed charts and gauge transformations of one RG step gives every intermediate classical, constraint-coordinate, and coarea condition. There are no further D_r tests at intermediate scales, and every product of fluctuation balls of positive radius contains the zero vector. These facts hold for every number of RG steps, since the smallest coarse side controls the same inequalities at each step. An arbitrary flat configuration is periodically gauge equivalent to such a constant configuration, and the coordinate transitions on overlaps and the gauge invariance of the event (Lemma 14.5) carry the membership over to it. A change of representative based at the coarse vertices leaves the based cycle holonomies fixed and preserves the norms of the fluctuation variables and the event. A change at the common

root only conjugates the triple simultaneously.

Step 9: the Wilson flow. We vary one link in a Lie algebra direction. At a flat configuration, each differentiated plaquette variable is the identity, and its derivative is tangent to $\mathrm{SU}(2)$ at the identity, hence traceless. Thus $Dq(I) = 0$, every summand of the gradient of the Wilson action vanishes, and the constant trajectory solves the Wilson flow equation. The Wilson flow is the flow of a smooth vector field on the compact manifold $\mathcal{U}_N$, so its solutions are unique. Indeed, locally, iterating the integral equation h times bounds the difference of two solutions with the same initial value by $C^h t^h / h!$ times a fixed supremum, which tends to zero, and the argument can be repeated on successive time intervals. Hence flat configurations are stationary, and their cycle holonomies are unchanged at every time.

Step 10: neighborhoods. Assume the hypotheses of (2), and fix a finite $N = uB^K$ and an allowed coupling. At each flat configuration, all classical and coarea inequalities hold strictly by Steps 5 to 8, the domains D_r for the fine and coarse fields are open, and every fluctuation vector is zero, inside a ball of positive radius. The maps defining the event and the inverse coordinates are continuous on their open branches. A finite intersection of their preimages therefore gives an open neighborhood of this configuration contained in the event, after replacing the fluctuation balls by smaller open balls around zero. The union of the gauge images of an open set is open, since each gauge transformation acts by a homeomorphism. Taking the union of these neighborhoods gives an open set $\mathcal{O}$ with $\mathcal{F}_N \subset \mathcal{O} \subset S_{N,u,\gamma}$, where $\mathcal{F}_N$ is the set of flat configurations.

The configuration space is compact, and $f(U) = \max_p \rho_G(U_p)$ is continuous with zero set exactly $\mathcal{F}_N$. If $\mathcal{O}^c$ is nonempty, f attains a strictly positive minimum on this compact set; otherwise any positive threshold works. Choosing $\delta_{N,u,\gamma}$ below this minimum proves the inclusion in (3). The minimum is taken separately at each N and each coupling, and the argument gives no uniform lower bound for it.

On a fixed stopping lattice whose side satisfies the same condition, Step 8 shows that the compact set of flat configurations lies in D_r. The set D_r is open, being a union of gauge images of the open set B_r. For any fixed metric on the finite product, the compact set of flat configurations therefore has positive distance from the closed complement of D_r, unless this complement is empty. The resulting neighborhood depends only on n and r; neither set involves the depth or the coupling. Throughout the argument all gauge transformations were periodic and all cycle holonomies commuted, so no projectively commuting or twisted data have been introduced. □

SECTION 15

Conditional kernels and reconstruction maps

In this section we prove that the renormalization group construction on the small-field event converges at every fixed resolution. That is, we fix the side n_0 of the coarse lattice and a number l of RG steps counted from the coarse side, and we let the total number k of RG steps tend to infinity. At a fixed resolution all objects are finite dimensional. In Theorem 15.6, we show that one RG step contracts effective interactions in the weighted analytic coefficient norm, uniformly in the lattice side and in the preceding depth, and we deduce the convergence of the effective interactions and of the stopped measures at each fixed level. Theorem 15.9 proves the convergence of the conditional fluctuation kernels, and Lemma 15.10 writes the conditioned Wilson measure as an iteration of these kernels. In Theorem 15.12, we construct the reconstruction maps $\widehat{\mathscr{R}}$, and in Theorem 15.13 we prove that the finite-scale reconstructions converge in law. Lemma 15.15 gives amplitude bounds for the reconstruction that are uniform in the mesh and in the coupling, and Theorem 15.16 makes the convergence uniform for the coupling in compact subintervals of $(0, \infty)$. The inputs are the classical convergence and the bounds for the zero-fluctuation reconstruction of Theorem 11.2 and Lemma 11.9, and the density and the cluster expansion of one RG step (Theorems 12.7 and 12.13 and Corollary 12.8). We also use the gauge transformations and the invariant ball of Theorems 13.2 and 13.5, and the identification of the auxiliary integral with the conditioned Wilson measure (Theorems 14.3 and 14.6). The later sections approximate the flowed field by the field reconstructed from finitely many RG steps (Sections 3.3 and 3.6). Sections 16, 17, 26 to 29, and 31 to 36 use the reconstruction maps, the conditional kernels, the convergence of finite-scale reconstructions, and the amplitude bounds, and Section 23 uses the deterministic constants and the order in which they are chosen.

The splitting of the fine field into a constrained minimizer and normalized fluctuations, and the integration of the fluctuations scale by scale, are Bałaban's [15, 16, 18]. We combine the standard renormalization group contraction with the cluster expansion estimates of Section 12 to obtain the limits of the conditional kernels and a compatible family of representatives for this construction. These estimates go back to Kotecký and Preiss [120] and use the finite tree-graph bound in Dimock's exposition [68, Appendix B]. The selection of a holomorphic subsequence is the normal-family argument in finitely many variables, and the limits follow from the contraction in the usual way.

15.1 Scales, measures, and conventions

Throughout this section $L > 0$ is fixed, $M \geq 3$ is odd, $J \geq 1$, and $B = M^J$. One averaging step is one application of the block averaging map T_M of Definition 4.3, and one RG step consists of J averaging steps. We fix the side $n_0 \in 4\mathbb{N}$ of the coarse lattice, let $H = L/n_0$, and for $j \geq 0$ set

$$n_j = n_0 B^j, \qquad a_j = HB^{-j}, \qquad g_j^2 = \frac{\gamma}{n_j} = g_3^2 a_j.$$

We call Λ_{n_j} the lattice of level j. If the finest lattice has level k, the lattices form the chain

$$\Lambda_{n_0} \xleftarrow{T_M^J} \Lambda_{n_1} \xleftarrow{T_M^J} \cdots \xleftarrow{T_M^J} \Lambda_{n_k}.$$

The fluctuations are integrated from right to left, toward coarser lattices, and the fine field is reconstructed from left to right. The number of averaging steps between the finest lattice and level j is

$$d_j^k = J(k - j),$$

and we call it the preceding depth at level j. The RG step from level j to level $j-1$ has J intermediate scales, indexed by $0 \leq i < J$, with couplings

$$g_{j,i} = t_j M^{(i-J+1)/2}, \qquad t_j = M^{(J-1)/2} g_j = g_{j-1}/\sqrt{M}.$$

Here t_j is the largest of these couplings, and the ratios $g_{j,i}/t_j$ are fixed powers of M. With $p(t) = A_{\mathrm{w}} \log(1/t)$ we write $p_j = p(t_j)$. The normalized fluctuation variables $w_j = (w_{j,i,e})$ of this RG step are restricted to the fluctuation cutoff domain Ω_{p_j}, the product of the three-dimensional balls $\{|w_{j,i,e}| \leq p_j\}$; it does not depend on k. We write $q_{j,i} = g_{j,i} w_{j,i}$ for the coupling-scaled fluctuation inputs of the reconstruction. Since the inverse square roots of the constrained Hessians remain inside the reconstruction maps, $q_{j,i}$ is not a free logarithmic coordinate of the fiber.

The three measures. Let

$$\nu^k = \nu_{n_0 B^k, n_0, \gamma}$$

be the conditioned Wilson measure of Definition 4.4, with the small-field event of (4.6) and Definition 14.4. In the comparison notation, the comparison index s gives $n_s = n_* B^s$, $H_s = L/n_s$, and $\nu_{K,s} = \nu_{N_K, n_s, \gamma}$ with $N_K = n_* B^K$. Then $n_0 = n_s$ and $k = K - s$, the meshes of the reconstruction are $a_j = H_s B^{-j}$, $0 \leq j \leq K - s$, and the preceding depths are $J(K - s - j)$. At level j three different measures occur.

1. The marginal of ν^k at level j, that is, the law of $V^{d_j^k}$ under ν^k.
2. The stopped measure $\widehat{\nu}_j^k$ of Theorem 15.6. It is obtained by integrating out the finer levels and imposing the small-field restrictions only up to level j.

3. The conditional fluctuation kernel $k_j^k(\theta, dw)$ of Theorem 15.9, the conditional law of the fluctuation variables of the RG step from level j to level $j-1$ given the field e^θ at level $j-1$.

The marginal in (1) also carries the restrictions at the coarser levels $j-1,\dots,0$, and in general it differs from the stopped measure. At the coarse level $j=0$ no coarser restriction remains, and (1) and (2) coincide (Lemma 15.11). The kernel (3) does not change when coarser restrictions are imposed, once the field at level $j-1$ has been fixed (Theorem 15.9).

Effective interactions. We write functions of a configuration near the identity in logarithmic coordinates, $R(e^\xi)$, and use the norm

$$\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R) = \sum_{m\geq 2} \frac{\rho^{m-2}}{m!} \|D^m R(0)\|_{a_{\mathrm{sp}},\mathrm{root}}$$

of Definition 4.6. The tensor norm fixes one index, sums the absolute values of the coefficients over the remaining indices with the weight $e^{a_{\mathrm{sp}}\mathsf{t}}$, where t is the length of a shortest connected network through the locations of the indices, and takes the supremum over the fixed index. For maps between lattices of different meshes the input-fixed sums carry the volume factor $a_{\mathrm{out}}^3/a_{\mathrm{in}}^3$ of Definition 4.6; we call them volume-normalized column sums. A function R on B_ρ is invariant under gauge transformations within B_ρ if $R(U^h) = R(U)$ whenever U and U^h both lie in B_ρ, as in Definition 13.3; then $R_{\mathrm{sat}}(U) = R(U^h)$ for $U^h \in B_\rho$ defines its gauge-invariant extension to D_ρ. For a lattice Λ and the parameters ρ, a_{sp}, E fixed in Theorem 15.6, let $\mathcal{A}_E = \mathcal{A}_E(\Lambda)$ be the set of functions R given by an absolutely convergent Taylor series on B_ρ that are real on real configurations, invariant under lattice translations, the spatial inversion, global color rotations, and gauge transformations within B_ρ, and satisfy

$$R(I) = 0, \qquad DR(I) = 0, \qquad \mathcal{B}_{\rho,a_{\mathrm{sp}}}(R) \leq E.$$

These are the symmetries of the recursion of Theorems 13.5 and 14.6. The norm $\mathcal{B}_{\rho,a_{\mathrm{sp}}}$ vanishes on constants.

Constants. A constant denoted C_J, c_J, or similar depends only on M, J, the fixed larger construction domains and their radius ratios, the spatial exponents, the factor K_*, and the norm ceilings. It is independent of the lattice side and of the preceding depth; this is the dependence described in item (2) of Definition 4.14. A subscript Λ allows, in addition, dependence on a fixed lattice pair. Such constants occur only in convergence moduli and never in the estimates that are used to choose J. We write $\|\mu-\nu\|_{\mathrm{TV}} = \sup_A |\mu(A)-\nu(A)|$ for the total variation distance.

15.2 Finite-dimensional analytic bounds and product domains

We begin with three elementary facts about total variation, which are used repeatedly below.

Lemma 15.1. *1. Let μ be a σ-finite measure on a measurable space X, and let V, V' be real measurable functions on X with $\sup_X |V - V'| \le \delta$ and $0 < \int e^{-V} d\mu < \infty$. Let π and π' be the probability measures proportional to $e^{-V}\mu$ and $e^{-V'}\mu$. Then the ratio of their normalizers lies in $[e^{-\delta}, e^{\delta}]$, the density $d\pi'/d\pi$ takes values in $[e^{-2\delta}, e^{2\delta}]$, and*

$$\|\pi - \pi'\|_{\mathrm{TV}} \le e^{2\delta} - 1.$$

2. Let η, η' be probability measures on a measurable space X_0, and for $1 \le j \le l$ let K_j, K_j' be probability kernels from $X_0 \times \cdots \times X_{j-1}$ to X_j. Then $P = \eta K_1 \cdots K_l$ and $P' = \eta' K_1' \cdots K_l'$ satisfy

$$\|P - P'\|_{\mathrm{TV}} \le \|\eta - \eta'\|_{\mathrm{TV}} + \sum_{j=1}^{l} \sup_h \|K_j(h,\cdot) - K_j'(h,\cdot)\|_{\mathrm{TV}}.$$

3. Let P, P' be probability measures on a measurable space X, and let F, F' be measurable maps from X into a set $\mathcal{K}$ of diameter at most D_* in a finite-dimensional normed space. Put $\|F - F'\|_\infty = \sup_X \|F - F'\|$. For every bounded real function f that is uniformly continuous on $\mathcal{K}$, with modulus of continuity ω_f,

$$\left| \int f(F)\, dP - \int f(F')\, dP' \right| \le 2\|f\|_\infty \|P - P'\|_{\mathrm{TV}} + \omega_f(\|F - F'\|_\infty),$$

and

$$W_1(F_*P, F_*'P') \le \|F - F'\|_\infty + D_* \|P - P'\|_{\mathrm{TV}}.$$

Proof. For (1), let Z, Z' be the two normalizers. Since $e^{-\delta}e^{-V} \le e^{-V'} \le e^{\delta}e^{-V}$, we have $e^{-\delta}Z \le Z' \le e^{\delta}Z$. The density $d\pi'/d\pi = e^{V-V'}Z/Z'$ therefore lies in $[e^{-2\delta}, e^{2\delta}]$. For an event A we have $|\pi(A) - \pi'(A)| \le \int |1 - d\pi'/d\pi|\, d\pi \le e^{2\delta} - 1$, where we used $1 - e^{-2\delta} \le e^{2\delta} - 1$.

For (2), first observe that a measurable function u with values in $[0,1]$ satisfies $|\int u\, d\mu - \int u\, d\mu'| \le \|\mu - \mu'\|_{\mathrm{TV}}$ for probability measures μ, μ'; indeed $u = \int_0^1 \mathbf{1}_{\{u > s\}}\, ds$. We replace first η by η', and then the kernels one at a time. Fix an event A in the product space. Since the function obtained by integrating $\mathbf{1}_A$ against $K_1 \cdots K_l$ takes values in $[0,1]$, the first replacement changes the probability of this event by at most $\|\eta - \eta'\|_{\mathrm{TV}}$. In the replacement of K_j by K_j', the integral of $\mathbf{1}_A$ against the later kernels is again a $[0,1]$-valued function of (h, x_j), and the change is at most $\sup_h \|K_j(h,\cdot) - K_j'(h,\cdot)\|_{\mathrm{TV}}$. Summing these changes proves (2).

For (3), we have

$$\left| \int f(F)\, dP - \int f(F')\, dP' \right| \le \int |f(F) - f(F')|\, dP + \left| \int f(F')\, dP - \int f(F')\, dP' \right|.$$

The first term is at most $\omega_f(\|F - F'\|_\infty)$. For the second, the function $(f(F') + \|f\|_\infty)/(2\|f\|_\infty)$ takes values in $[0,1]$, and the observation in the proof of (2) bounds it by $2\|f\|_\infty \|P - P'\|_{\mathrm{TV}}$. For the Wasserstein bound, let $\lambda = P + P'$, let u, u' be the densities of P, P' with respect to λ, and set $m = \min(u, u')$ and $\vartheta_0 = \int m \, d\lambda = 1 - \|P - P'\|_{\mathrm{TV}}$. If $\vartheta_0 = 1$ then $P = P'$ and we couple x with itself. Otherwise let Γ be the sum of the image of $m\lambda$ under $x \mapsto (x, x)$ and of the product measure $(1 - \vartheta_0)^{-1}(u - m)\lambda \otimes (u' - m)\lambda$. Its marginals are P and P', and it gives mass at least ϑ_0 to the diagonal. Then

$$\int \|F(x) - F'(x')\| \, d\Gamma(x, x') \le \|F - F'\|_\infty + (1 - \vartheta_0) D_*,$$

and $(F \times F')_* \Gamma$ couples $F_* P$ and $F'_* P'$. This completes the proof. □

In particular, for the bounded-Lipschitz distance, whose test functions are bounded by one and have Lipschitz constant at most one, part (3) gives the bound $2\|P - P'\|_{\mathrm{TV}} + \|F - F'\|_\infty$.

Box supports. The cluster expansion of Section 12 assigns interactions to connected sets of boxes. We partition each periodic coordinate direction of the coarse lattice into intervals of integer length between $\mathscr{L}$ and $2\mathscr{L}$, or use one interval when the direction is shorter than $\mathscr{L}$, as in Definition 12.1. The products of these intervals are the boxes. Each scalar variable has a fixed location in the torus, as in Definition 4.6; the coarse field has nine scalar components (three orientations and three colors) at each vertex. The first lemma bounds the length of a network through the locations of the coarse-field variables in a connected set of boxes.

Lemma 15.2. *Fix an integer $\mathscr{L} \ge 1$ in units of the coarse lattice. There exists a constant $C_{\mathrm{box}}(\mathscr{L})$, depending only on $\mathscr{L}$ and on the fixed metric and location conventions, such that every nonempty connected set Y of boxes satisfies*

$$\mathsf{t}(\textit{locations of the coarse-field variables in } Y) \le C_{\mathrm{box}}(\mathscr{L})|Y|. \tag{15.1}$$

The constant is independent of the side of the torus, the preceding depth, J, the couplings, the degree of a coefficient, and the density of the fluctuation variables. The bound also holds for every subset or repeated list of these locations, and for every union of connected box sets whose intersection graph is connected.

Proof. Set $q = 2\mathscr{L} + 2$, and let $c_{\mathrm{met}} \ge 1$ bound the length of a coordinate segment of unit length and the distance from the location of a link variable to an endpoint of the link. A lifted box, together with a collar of one cell for the locations of its link variables, lies in an integer cube of side at most q. The coordinate grid of this cube has at most $3q(q+1)^2$ edges. There are at most $9(q+1)^3$ locations of scalar variables in the cube, and we attach each of them to a grid point by a segment of length at most c_{met}. This gives a connected network of length at most $12 c_{\mathrm{met}}(q+1)^3$ containing every such location in the box.

Fix a grid vertex in each box. A traversal of the connected box graph selects $|Y| - 1$ adjacencies that connect all boxes. For each selected adjacency we join the two chosen grid vertices by a periodic coordinate path. Since the boxes are adjacent, this path has length at most $c_{\mathrm{met}}(6q + 3) \le 9c_{\mathrm{met}}q$. These paths and the grids form one connected network containing all these locations in Y, so we may take

$$C_{\mathrm{box}}(\mathscr{L}) = c_{\mathrm{met}}\{12(q+1)^3 + 9q\}.$$

In a periodic direction shorter than $\mathscr{L}$ we project the grid of the cube to the torus, which can only decrease its length; for an adjacency across the periodic boundary we use a shortest periodic coordinate path, which obeys the same bound. No lower bound on the side in terms of $\mathscr{L}$ was used. Repeated labels occupy the same location, and restricting the list of locations requires no new segment. Finally, a union of connected box sets whose intersection graph is connected is itself a connected box set, so the bound applies to it with its own cardinality. □

The next lemma makes the support assignment of Lemma 12.9 explicit. It assigns a connected box set to a coefficient and bounds the resulting support-counting factor by the length of the network of the coefficient.

Lemma 15.3. *Fix deterministic choices of shortest paths. Let $N \ge 1$ points be given, together with a connected network of length ℓ containing them. Join the points by a minimal spanning tree, replace each edge of the tree by its three coordinate segments along shortest periodic displacements, and let Y be the set of boxes met by the resulting network, including the boxes of the points. Then Y is a nonempty connected box set. There exist constants $c_0, C_{\mathrm{geom}} \ge 1$ depending only on the metric conventions, and a constant $C_{\mathscr{L}}$ depending only on $\mathscr{L}$ and the metric conventions, such that*

$$|Y| \le c_0\Big(N + 1 + \frac{C_{\mathrm{geom}}\ell}{\mathscr{L}}\Big), \qquad \operatorname{dist}(x,u) \le C_{\mathrm{geom}}\ell + C_{\mathscr{L}} \tag{15.2}$$

for every box $x \in Y$ and every given point u, where the distance is measured from a fixed point of the box. Consequently, for $h, \mu > 0$ and

$$K \ge e^{hc_0}, \qquad \beta \ge C_{\mathrm{geom}}\Big(\frac{hc_0}{\mathscr{L}} + \mu\Big), \qquad C_h = e^{hc_0 + \mu C_{\mathscr{L}}},$$

we have

$$\mathbf{1}_{\{x \in Y\}} e^{h|Y|} \le C_h K^N e^{\beta\ell} e^{-\mu \operatorname{dist}(x,u)}. \tag{15.3}$$

Proof. A minimal spanning tree on the N points has at most $N - 1$ edges, and its length is at most twice the length of any connected network containing the points; indeed, traversing such a network twice and shortcutting between successive points gives a path through all of them. The comparison between coordinate segments and the metric, together with this factor two, is included in C_{geom}. A coordinate segment of length l'

meets at most a fixed multiple of $2 + l'/\mathscr{L}$ boxes. Summing over the at most $3(N-1)$ segments gives the first bound in (15.2). The second follows by moving along the network from u to the box x and then inside the box. Inserting (15.2) into $e^{h|Y|}$ and using $\mu \operatorname{dist}(x,u) \le \mu C_{\mathrm{geom}} \ell + \mu C_{\mathscr{L}}$, we obtain (15.3). □

Order of the spatial choices. We now fix the spatial parameters, in an order that keeps them independent of J. Fix a decay rate $b > 0$ strictly below the common spatial rate of the classical and image bounds of Lemma 11.9 and Theorems 13.2 and 13.5. We choose $\mathscr{L}$ using this rate alone, so that $8c_0 C_{\mathrm{geom}}/\mathscr{L} \le b/4$. We fix $\mu > 0$ with $C_{\mathrm{geom}}\mu \le b/4$; this sets aside a part of b for summing over the distance between a box and the location of a variable. With $h = 8$ and $K = e^{8c_0}$ in Lemma 15.3, the factor K^N is absorbed by multiplying every input radius by K; we take the factor $K_* \ge 2$ of Theorem 13.5 so large that $K < 4K_*$, so that these enlarged radii stay inside the products of Lemma 15.4 below. Next we choose the spatial exponent a_{sp} of the effective interactions so that

$$\begin{aligned} 0 < a_{\mathrm{sp}} < \min\{b/4, a_{\mathrm{cen}}/8\}, \qquad & a_{\mathrm{sp}} C_{\mathrm{box}}(\mathscr{L}) \le 2, \\ C_P(a_{\mathrm{sp}})/B < 1/4, \qquad & C_{\mathrm{geom}} b/B \le a_{\mathrm{sp}}, \end{aligned} \tag{15.4}$$

where C_P is the constant of Theorem 11.10. The condition $a_{\mathrm{sp}} < a_{\mathrm{cen}}/8$, where a_{cen} is the rate of Lemma 11.9, is the restriction imposed in Theorem 13.5. Indeed, the zero-fluctuation reconstruction used below is the representative of Lemma 11.9, and Theorem 11.10 gives the contraction for this representative only at output rates below $a_{\mathrm{cen}}/8$, with the constant $C_P(a_{\mathrm{sp}})$ at the rate a_{sp}. The first two conditions determine a_{sp} after $\mathscr{L}$. Only then do we choose J, so that the last two conditions and the condition $M^{-J} Z(4K_*\rho) \le \rho/4$ of Theorem 13.5 hold. All constants in these conditions are fixed before J, and $M^{-J} \to 0$, so they hold for all large J. The scale $\mathscr{L}$ is not chosen again after a_{sp} or J, and no exponent is decreased during the iteration. In the common parameter choice of Theorem 23.2, the torus exponent denoted a_{sp} in this section is $a_{\mathbb{T}}$, distinct from the smaller regional boundary exponent chosen after J. By (15.1), these choices give

$$a_{\mathrm{sp}} \mathsf{t} \le 2|Y|$$

for the locations of the coarse-field variables in every connected box set Y, including subsets and repetitions of these locations. This inequality does not involve a radius.

Nested analytic product domains. The next lemma fixes the complex products on which the reconstruction is controlled.

Lemma 15.4. *Let $C_{\mathrm{im},J}$ and r_J be the constants of Theorem 13.2, and let*

$$\mathfrak{M}_J(u,v) = B^{-1} Z(u) + \frac{C_{\mathrm{im},J} v}{1 - C_{\mathrm{im},J}(u+v)}$$

be its row majorant. Before the radius r of the small-field domain and the coupling ceiling are chosen, the common product of coarse-field and fluctuation-input variables can be fixed with radii $u_\circ = 4K_*\rho$ *and* $v_\circ = 4K_* s_J$*, such that*

$$\begin{aligned} u_\circ + v_\circ < r_J, \qquad & C_{\mathrm{im},J}(u_\circ + v_\circ) \le 1 - \eta_{\mathrm{im}} < 1, \\ B^{-1}Z(u_\circ) \le \rho/4, \qquad & \mathfrak{M}_J(u_\circ, v_\circ) < \rho/2 \end{aligned} \tag{15.5}$$

for a fixed $\eta_{\mathrm{im}} > 0$*, and every intermediate classical, logarithm, and Haar disintegration domain condition holds with a margin. Every smaller product lies in this domain and has denominator at least* η_{im}*. For each* $0 < r \le \rho$ *there exists* $Q_{\mathrm{act}} > 0$*, depending on r, at most* s_J*, and at most any previously chosen positive bound on the inputs q, such that*

$$r + Q_{\mathrm{act}} < r_J, \qquad 1 - C_{\mathrm{im},J}(r + Q_{\mathrm{act}}) > 0, \qquad \mathfrak{M}_J(r, Q_{\mathrm{act}}) < r/2. \tag{15.6}$$

After the coupling ceiling is reduced so that $tp(t) \le Q_{\mathrm{act}}$*, these conditions hold on every product* $\{\|\theta\|_\infty < r\} \times \{\max_i \|g_i w_i\|_\infty \le tp(t)\}$ *used in the construction. All these choices are deterministic and independent of the coupling, the lattice side, and the preceding depth.*

Proof. The series Z of Lemma 11.9 is chosen before J, and the choice of J in Theorem 13.5 gives $B^{-1}Z(4K_*\rho) \le \rho/4$. After J has been fixed, we take ρ strictly below $r_J/(8K_*)$ and so small that $4K_* C_{\mathrm{im},J}\rho < 1/4$, within the common classical and Haar disintegration domains. Then we choose $s_J > 0$ so small that $u_\circ + v_\circ < r_J$, that $C_{\mathrm{im},J}(u_\circ + v_\circ) < 1/2$, and that the second term of $\mathfrak{M}_J(u_\circ, v_\circ)$ is strictly below $\rho/4$. This reduction keeps the intermediate margins of Theorem 13.2. No radius is changed during a recursion. We denote the positive margin of the denominator by η_{im}. Since the coefficients of all majorants are nonnegative, the domain and denominator conditions remain valid on smaller products.

The ratio $Z(u)/u$ is nondecreasing, because the series of Z has nonnegative coefficients and no constant term. Hence $B^{-1}Z(r) \le r/(16K_*)$ for $r \le \rho$. We take Q_{act} strictly below

$$\min\Big\{ s_J,\ \frac{r_J - r}{2},\ \frac{1 - C_{\mathrm{im},J} r}{2C_{\mathrm{im},J}},\ \frac{r(1 - C_{\mathrm{im},J} r)}{16 C_{\mathrm{im},J}} \Big\},$$

and below any previously chosen bound on the inputs; every entry of the minimum is positive. The denominator is then at least $(1 - C_{\mathrm{im},J} r)/2$, and the second term of $\mathfrak{M}_J(r, Q_{\mathrm{act}})$ is less than $r/8$. Thus $\mathfrak{M}_J(r, Q_{\mathrm{act}}) < r/(16K_*) + r/8 < r/2$. Since the product lies in the outer product, it has all intermediate margins. Finally, $tp(t) \to 0$ as $t \downarrow 0$, so one ceiling works on the whole interval of smaller couplings. □

A Taylor tail estimate. The last lemma of this section compares values, derivatives, and coefficient norms at a fixed lattice size.

Lemma 15.5. *On a fixed finite lattice Λ with edge set $\mathcal{E}_\Lambda$, let $D_\Lambda = 3|\mathcal{E}_\Lambda|$. Every function F given by a Taylor series in logarithmic coordinates with $F(I) = 0$ and $DF(I) = 0$ satisfies*

$$\sup_{\|\xi\|_\infty \le \rho} |F(e^\xi)| \le D_\Lambda \rho^2 \mathcal{B}_{\rho, a_{\mathrm{sp}}}(F).$$

The same bound holds on D_ρ if F is invariant under gauge transformations within B_ρ. Values on a closed coordinate ball mean the continuous extension of the absolutely convergent Taylor series. For $0 < \sigma < 1$ and $s \ge 1$, the sum over all indices of the absolute values of the coefficients of $D^s F$ at points of $B_{\sigma\rho}$ is at most

$$s!\, D_\Lambda \rho^{2-s} (1-\sigma)^{-s} \mathcal{B}_{\rho, a_{\mathrm{sp}}}(F).$$

Suppose a family F_α has a common bound on $\mathcal{B}_{\rho_1, a_{\mathrm{sp}}}$ for some $\rho_1 > \rho$, with the same spatial exponent. If each fixed Taylor coefficient tensor of F_α is Cauchy, uniformly in an additional parameter, then F_α is Cauchy in $\mathcal{B}_{\rho, a_{\mathrm{sp}}}$, uniformly in that parameter. This also holds for vector-valued maps and for joint products of coarse-field and fluctuation-input variables, including every volume-normalized column sum and every fixed spatial difference, provided the inner and outer radii of each group of variables have a common ratio less than one.

Proof. We write $F(e^\xi) = \sum_{s \ge 2} D^s F(0)[\xi^{\otimes s}]/s!$. Summing first over one scalar coordinate and then over the remaining ones gives at most D_Λ times the supremum of the sums with one fixed index. Dropping the spatial weights, which are at least one, and using $|\xi_e| \le \rho$, we obtain the value bound. Differentiation produces the falling factorial $m(m-1)\cdots(m-s+1)$ in degree m. Its product with σ^{m-s} is at most $s!(1-\sigma)^{-s}$, because $\binom{m}{s}(1-\sigma)^s \sigma^{m-s} \le 1$ is one term of the binomial expansion of $((1-\sigma)+\sigma)^m = 1$. Summing against the coefficient norms gives the derivative bound. Gauge invariance within B_ρ changes none of these values, since a single gauge transformation is used at the point of evaluation.

Let $u_m = \|D^m F(0)\|_{a_{\mathrm{sp}}, \mathrm{root}}/m!$. If $\mathcal{B}_{\rho_1, a_{\mathrm{sp}}}(F) \le C$, then

$$\sum_{m > h} \rho^{m-2} u_m \le C(\rho/\rho_1)^{h-1}. \tag{15.7}$$

For a difference of two members of the family we use twice the common bound. For fixed h there are finitely many coefficients, indices, and labels, and their weighted sums converge. We choose h first in (15.7) and then the indices of the family. This proves the Cauchy property without decreasing ρ or a_{sp}.

For a joint vector series we group the monomials by total degree. If the ratio of each inner radius to the corresponding outer radius is at most $\zeta < 1$, the part of degree greater than h has norm at most $C\zeta^{h+1}$. A fixed input contributes a degree factor, which is absorbed by using an intermediate ratio between ζ and one. At a fixed lattice size the volume factors, the spatial weights, and the difference matrices are fixed finite numbers, so the argument applies to all these sums and differences. It also shows that locally

uniform holomorphic convergence, together with a common absolute bound on a larger product, gives convergence in these norms. □

Here and below the Cauchy formula in several variables means the iterated one-variable formula on circles inside the polydisc. If a function is bounded by C on these circles, the formula bounds the coefficient of ξ^α by $Cr^{-\alpha}$. In coordinate dimension D there are $\binom{m+D-1}{D-1}$ multi-indices of degree m, a polynomial in m. On every strictly smaller polydisc, the sums of the absolute values of the coefficients are then finite. This justifies the finite-dimensional uses of locally uniform holomorphic convergence below.

15.3 Contraction and limits at fixed resolution

The auxiliary transformation. Fix a lattice pair of one RG step: a coarse lattice of side $m \geq 4$ and the fine lattice of side Bm, with $d \geq 0$ preceding averaging steps. Fix $0 < t \leq g_*$, and let $g_i = t\lambda_i$ with $\lambda_i = M^{(i-J+1)/2}$ for $0 \leq i < J$. We write $c = e^\theta$ for the coarse field and $q_i = g_i w_i$ for the inputs. Let $\mathscr{C}_{\mathrm{raw}\,d}(\theta, q)$ be the reconstruction of the fine field in the block axial gauge, built from the conditional constrained minimizers, the constrained Hessians, and their inverse square roots, as in Lemma 12.3 and Theorem 12.13. Let $\Omega_d(\theta)$ be the gauge transformation of Theorem 13.2; it depends only on θ and equals the identity at the vertices of the coarse lattice. Define

$$\phi_d(\theta, q) = \mathrm{Log}\big[\mathscr{C}_{\mathrm{raw}\,d}(\theta, q)^{\Omega_d(\theta)}\big], \qquad P_d R(\theta) = R(e^{\phi_d(\theta,0)}).$$

Thus $P_d R$ is the pullback of R by the zero-fluctuation reconstruction. Let $H_{d,J}(\theta, w)$ be the fluctuation exponent of Theorem 12.13. It contains the classical remainder, the nonconstant parts of the Haar disintegration densities, and the differences of the determinant terms; the determinant of a finer intermediate scale is evaluated at the argument determined by the coarser fluctuation variables. Let π_p be the normalized product of standard Gaussian measures on $\mathbb{R}^3$ restricted to balls of radius $p = p(t)$, one factor for each three-dimensional block of fluctuation variables. Let

$$z_p = \int_{\{v \in \mathbb{R}^3 : |v| \leq p\}} e^{-|v|^2/2}\, dv,$$

let f_i be the number of free group variables at the intermediate scale i, so that the cutoff domain is a product of $\sum_i f_i$ balls, and let $\mathsf{B}_{\mathrm{ref}}(\theta)$ be the zero-fluctuation Jacobian term of Theorem 12.13. For an input effective interaction R, produced by the integrations at finer

scales, we set

$$\begin{aligned} A_R(\theta, w) &= R(e^{\phi_d(\theta, (g_i w_i))}) - P_d R(\theta), \\ \Xi_R(\theta) &= \int e^{-H_{d,J}(\theta, w) - A_R(\theta, w)} \, d\pi_p(w), \\ D_d R &= -\log \Xi_R + \log \Xi_0, \\ Q_d &= \mathsf{B}_{\mathrm{ref}} - \Big(\sum_i f_i\Big) \log z_p - \log \Xi_0. \end{aligned} \tag{15.8}$$

By Theorems 12.13 and 14.6, $Q_d + P_d R + D_d R$ is the effective interaction produced by one RG step from R, before normalization at the identity. The three terms are the newly generated effective interaction, the pullback at zero fluctuation, and the fluctuation correction to the pullback. In the gauge of ϕ_d, they are the terms $Q^{\mathrm{w}}_{d,J}$, $\mathcal{P}_\delta R$, and $\mathcal{D}_J R$ of (13.9), with $d_0 = d$, up to field-independent constants. We define the auxiliary transformation

$$\widehat{\mathcal{M}}_{d,t} R = (Q_d + P_d R + D_d R) - (Q_d + P_d R + D_d R)(I).$$

The field-independent factors $\prod_i \mathcal{J}_i(I, 0) g_i^{3 f_i}$, the Haar disintegration density at the identity, and the powers of the couplings, are kept in the normalization constant of Theorem 14.6, together with the subtracted value at the identity; they are never estimated by $\mathcal{B}_{\rho, a_{\mathrm{sp}}}$. The integral Ξ_R carries no indicator of D_r on the fine lattice, and it is defined for complex θ in the larger analytic product domain of Lemma 15.4. At a coarse field in B_r, the image condition of the next theorem implies that this indicator would equal one on the whole cutoff domain. There Ξ_R is the normalized Gaussian fluctuation factor in the fiber formula of Theorem 14.6(2). Multiplying it by the classical, zero-fluctuation, Gaussian, and coordinate prefactors in that formula gives the density of the pushforward under T^J_M, before global normalization (Theorem 14.6).

Theorem 15.6. *Fix odd $M \geq 3$, $J \geq 1$, $B = M^J$, and the parameters of Theorems 13.5 and 14.6: the radii $0 < r < \rho$, the spatial exponent $a_{\mathrm{sp}} > 0$, the ceiling E for effective interactions, the coefficient $A_{\mathrm{w}} > 0$, and the coupling ceiling $g_* > 0$, with $p(t) = A_{\mathrm{w}} \log(1/t) > 0$ for $0 < t \leq g_*$. Fix a lattice pair of one RG step with coarse side $m \geq 4$, a coupling $0 < t \leq g_*$, and the couplings $g_i = t\lambda_i$ of its intermediate scales. Assume that the image under ϕ_d of the complex product of Lemma 15.4, with radii $4K_* \rho$ and $4K_* s_J$, lies strictly inside $B_{\rho/2}$; that for $\|\theta\|_\infty < r$ the real product $\{\max_i \|q_i\|_\infty \leq t p(t)\}$ is mapped into $B_{r/2}$; and that all intermediate classical, logarithm, and Haar disintegration domain conditions hold. Then, after one initial reduction of g_*, all $R_1, R_2 \in \mathcal{A}_E$ satisfy*

$$\begin{aligned} \mathcal{B}_{\rho, a_{\mathrm{sp}}}(\widehat{\mathcal{M}}_{d,t} R_1 - \widehat{\mathcal{M}}_{d,t} R_2) &\leq \{q_{\mathrm{pr}} + C_J t^2 [p(t)^4 + (\epsilon_1 + \epsilon_2) p(t)^2]\} \mathcal{B}_{\rho, a_{\mathrm{sp}}}(R_1 - R_2) \\ &\leq \vartheta \, \mathcal{B}_{\rho, a_{\mathrm{sp}}}(R_1 - R_2), \end{aligned}$$

where $\epsilon_i = \mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_i)$, $0 \le q_{\mathrm{pr}} < 1/4$, *and* $\vartheta = (1+q_{\mathrm{pr}})/2 < 1$. *The reduction of* g_* *imposes* $p(t) \ge 1$, *by taking* $g_* \le e^{-1/A_{\mathrm{w}}}$, *and*

$$t[p(t)^3 + 2Ep(t)] \le c_J/2, \qquad C_J t^2[p(t)^4 + 2Ep(t)^2] \le (1-q_{\mathrm{pr}})/2$$

for all $0 < t \le g_*$. *At this fixed lattice pair and fixed* t,

$$\sup_{R\in\mathcal{A}_E} \mathcal{B}_{\rho,a_{\mathrm{sp}}}\big(\widehat{\mathcal{M}}_{d,t}R - \widehat{\mathcal{M}}_{d',t}R\big) \longrightarrow 0 \qquad (d,d' \to \infty),$$

at the same radius and exponent. The conditional constrained minimizers, the constrained Hessians, the analytic branches of their square roots and inverse square roots that are positive at real arguments, the Haar disintegration densities, and the reconstructions $\mathscr{C}_{\mathrm{raw}\,d}$ *converge locally uniformly and holomorphically on smaller product domains, with every fixed derivative.*

Now fix $n_0 \in 4\mathbb{N}$ *and* $\gamma > 0$ *for which the construction of Theorem 14.6 applies, and use the hierarchy of Section 15.1. For* $0 \le j \le k$ *let* R_j^k *be the effective interaction at level* j *given by Theorem 14.6; thus* $R_k^k = 0$, $R_{h-1}^k = \widehat{\mathcal{M}}_{J(k-h),t_h} R_h^k$ *for* $1 \le h \le k$, *and* $R_h^k \in \mathcal{A}_E(\Lambda_{n_h})$. *The stopped measure at level* j *is*

$$d\widehat{\nu}_j^k = \widehat{Z}_{j,k}^{-1} \mathbf{1}_{D_r} \exp\{-E_{J(k-j),\mathrm{sat}}/g_j^2 - R_{j,\mathrm{sat}}^k\}\, dH_{n_j}, \qquad R_j^k(I) = 0.$$

For every fixed j *there exists* $R_j^\infty \in \mathcal{A}_E(\Lambda_{n_j})$ *such that*

$$\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_j^k - R_j^\infty) \longrightarrow 0, \qquad \|\widehat{\nu}_j^k - \widehat{\nu}_j^\infty\|_{\mathrm{TV}} \longrightarrow 0 \qquad (k \to \infty),$$

where $\widehat{\nu}_j^\infty$ *has the density of the same form with* $E_{\infty,\mathrm{sat}}$ *and* $R_{j,\mathrm{sat}}^\infty$, *the same indicator* $\mathbf{1}_{D_r}$, *and a positive finite normalizer. The contraction constants have the dependence described in item (2) of Definition 4.14; they are uniform in the side and in the preceding depth for fixed analytic parameters. The continuity moduli and the convergence thresholds may in addition depend on the fixed lattices and on* j, t, $p(t)$, L, *and* γ. *No rate that is uniform in the resolution, in the number of RG steps, or as* $\gamma \to 0$ *is asserted.*

Here E_∞ is the limit of the effective classical actions E_d at the fixed lattice (Theorem 11.2), and $E_{\infty,\mathrm{sat}}$ is its gauge-invariant extension.

Remark 15.7. The second-order gain in the coupling has the following heuristic origin. Suppose that a reconstruction has the expansion $F(\theta, tw) = F(\theta,0) + tL_\theta w + O(t^2)$. For a difference Δ of two effective interactions we then have

$$\Delta(F(\theta,tw)) - \Delta(F(\theta,0)) = tD\Delta(F(\theta,0))[L_\theta w] + O(t^2).$$

Since the first term is linear in w, it has mean zero under the reference measure π_p, which is invariant under $w \mapsto -w$. The derivative of the negative logarithm of the fluctuation

integral in the direction Δ is the expectation of this difference under the interacting conditional measure. Its leading term is the reference expectation, which vanishes, and the interaction contributes another factor of t. Lemma 15.8 and the cluster expansion of Theorem 12.7 make this argument uniform in the volume and in the degree of the coefficients, in the norm $\mathcal{B}_{\rho,a_{\mathrm{sp}}}$. The cancellation is a property of the reference measure on each support; it does not use any symmetry of the interacting conditional measure.

Local interaction decompositions. For a family $\boldsymbol{A} = (A_Y)$ indexed by nonempty connected box sets Y, where $A_Y(\theta, w) = \sum_\nu A_{Y,\nu}(w)\theta^\nu$ depends only on the coarse-field and fluctuation variables located in Y, we use the norm

$$\|\boldsymbol{A}\|_{\mathscr{S}_h} = \sup_x \sum_{Y \ni x} e^{h|Y|} \sum_\nu \rho^{|\nu|} \sup_{w \in \Omega_p} |A_{Y,\nu}(w)| \tag{15.9}$$

of Theorem 12.7, with coarse-field radius ρ and the cutoff radius $p \geq 1$ of the RG step; the supremum is over boxes x. The next lemma gives an explicit decomposition of the fluctuation interactions into such families, with a bound on the difference family that is linear in the norm of the difference.

Lemma 15.8. *Assume the hypotheses of Theorem 15.6 at a fixed lattice pair and preceding depth d, with $p = p(t) \geq 1$. Let $R_1, R_2 \in \mathcal{A}_E$, $\Delta = R_1 - R_2$, $\delta = \mathcal{B}_{\rho,a_{\mathrm{sp}}}(\Delta)$, $\epsilon_i = \mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_i)$, and $R_s = R_2 + s\Delta$ for $0 \leq s \leq 1$. For $z \in \mathbb{C}$ replace the couplings g_i by $z\lambda_i$, both in the inputs $q_i = z\lambda_i w_i$ of ϕ_d and in the classical remainder of $H_{d,J}$, as in Definition 12.15, and keep fixed the cutoff radius p and all parameters on which the input effective interaction depends. There exists C_J such that, for $C_J|z|p < 1$, there exist families $\boldsymbol{H}(z)$, $\boldsymbol{A}_s(z)$, and $\boldsymbol{A}_\Delta(z)$, holomorphic in z with values in $\mathscr{S}_8$, with the following properties.*

1. *Their sums over Y are, respectively, the exponent $H_{d,J}$ with $g_i = z\lambda_i$, the function $R_s(e^{\phi_d(\theta,(z\lambda_i w_i))}) - R_s(e^{\phi_d(\theta,0)})$, and the same function with R_s replaced by Δ.*
2. *$\boldsymbol{A}_s = \boldsymbol{A}_0 + s\boldsymbol{A}_\Delta$, and $\boldsymbol{H}(0) = \boldsymbol{A}_s(0) = \boldsymbol{A}_\Delta(0) = 0$.*
3. *On each Y the coefficient of z in $A_{\Delta,Y}(z)$ is a linear function of the fluctuation variables.*
4. *They satisfy*

$$\begin{aligned} \|\boldsymbol{H}(z)\|_{\mathscr{S}_8} &\leq \frac{C_J|z|(p^3+p)}{1 - C_J|z|p}, \\ \|\boldsymbol{A}_s(z)\|_{\mathscr{S}_8} &\leq \frac{C_J|z|(\epsilon_1+\epsilon_2)p}{1 - C_J|z|p}, \\ \|\boldsymbol{A}_\Delta(z)\|_{\mathscr{S}_8} &\leq \frac{C_J|z|\delta p}{1 - C_J|z|p}. \end{aligned} \tag{15.10}$$

Supports that wrap around the torus are allowed.

Proof. The family $\boldsymbol{H}$ is the local interaction decomposition, constructed in the proof of Lemma 12.16, of the auxiliary fluctuation interaction of Definition 12.15 at the parameter $\zeta = z$; it is built from the decompositions of Lemma 12.9, whose item (4) allows a complex parameter. By stationarity of the conditional constrained minimizers and the normalization of the fluctuation variables by the constrained Hessians, the classical remainder has no terms of degree zero, one, or two in the fluctuation variables, and its coefficients of degree $m \geq 3$ carry the factor z^{m-2}. The coefficients of the Haar disintegration and determinant terms of degree $m \geq 1$ carry the factor z^m. The coefficient bounds of Lemma 12.9(4) are therefore bounds for a power series in z, and the bound (12.30) in the proof of Lemma 12.16 is the first inequality in (15.10), after enlarging C_J. In particular $\boldsymbol{H}(0) = 0$.

We now construct $\boldsymbol{A}_\Delta$. The construction is linear in the interaction, and we define $\boldsymbol{A}_s$ by the same construction applied to R_s; this proves the first assertion of (2). Since $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_s) \leq (1-s)\epsilon_2 + s\epsilon_1 \leq \epsilon_1 + \epsilon_2$, the bound for $\boldsymbol{A}_s$ follows from the bound for $\boldsymbol{A}_\Delta$ with δ replaced by $\epsilon_1 + \epsilon_2$.

Coefficients. We write

$$\Delta(e^{\xi}) = \sum_{n\geq 2} \frac{1}{n!} \sum_{j_1,\dots,j_n} \Delta_n(j_1,\dots,j_n) \prod_{r=1}^{n} \xi_{j_r}, \qquad \phi_{d,j}(\theta,q) = \sum_{|\alpha|+|\eta|\geq 1} \phi_{j;\alpha\eta}\, \theta^\alpha q^\eta,$$

where j, j_r range over the scalar coordinates of the fine lattice, and α and η are multi-indices in the coarse-field variables and the inputs. Since $\phi_d(0,0) = 0$, substitution gives, for $|\alpha| + |\eta| = N$,

$$[\theta^\alpha q^\eta]\, \Delta(e^{\phi_d(\theta,q)}) = \sum_{n=2}^{N} \frac{1}{n!} \sum_{j_1,\dots,j_n} \sum_{\substack{\alpha_1+\cdots+\alpha_n=\alpha,\ \eta_1+\cdots+\eta_n=\eta \\ |\alpha_r|+|\eta_r|\geq 1}} \Delta_n(j_1,\dots,j_n) \prod_{r=1}^{n} \phi_{j_r;\alpha_r\eta_r}. \tag{15.11}$$

No further multinomial factor occurs, since each $\phi_{j;\alpha\eta}$ is already the coefficient of a monomial. An occurrence o consists of n, the indices $j_1,\dots,j_n$, and an ordered allocation $(\alpha_r,\eta_r)_{r\leq n}$ in (15.11). To $\Delta_n(j_1,\dots,j_n)$ we assign a shortest connected network through the locations of $j_1,\dots,j_n$, and to each $\phi_{j_r;\alpha_r\eta_r}$ a shortest connected network through the location of j_r and the locations of its inputs. The union $\mathcal{N}_o$ is a connected network through the locations of all external variables of o, and its length ℓ_o is at most the sum of the lengths of these networks. Let N_o be the number of external variables of o, counted with multiplicity.

The families. We keep the occurrences with $|\eta| \geq 1$. For such an occurrence, let

$$a_o(z) = \frac{z^{|\eta|}}{n!} \Delta_n(j_1,\dots,j_n) \prod_{r=1}^{n} \phi_{j_r;\alpha_r\eta_r} \prod_{l} \lambda_{i(l)}^{\eta_l},$$

where l runs over the inputs and $i(l)$ is the intermediate scale of the input l. Let Y_o be the box set assigned by Lemma 15.3 to the locations of the external variables of o and the network $\mathcal{N}_o$, and define

$$A_{\Delta,Y}(\theta, w; z) = \sum_{o:\, Y_o = Y} a_o(z)\, \theta^{\alpha} w^{\eta}.$$

The occurrences with $\eta = 0$ sum to $\Delta(e^{\phi_d(\theta,0)})$. Hence, once absolute convergence is established below,

$$\sum_{Y} A_{\Delta,Y}(z) = \Delta(e^{\phi_d(\theta,(z\lambda_i w_i))}) - \Delta(e^{\phi_d(\theta,0)}),$$

which is (1) for $\boldsymbol{A}_\Delta$. Since every coefficient carries a positive power of z, we have $\boldsymbol{A}_\Delta(0) = 0$, and the coefficient of z collects the occurrences with $|\eta| = 1$, which are linear in w. This proves (2) and (3).

Spatial weights. The networks of Δ_n have lengths measured in units of the fine lattice; in units of the coarse lattice these lengths are divided by B. By the last condition in (15.4) and $C_{\mathrm{geom}} \geq 1$, a weight $e^{b\ell}$ on this part, with ℓ in coarse units, is at most the weight $e^{a_{\mathrm{sp}}\mathfrak{t}}$ in fine units, which is included in δ. The networks attached to the coefficients of ϕ_d are weighted by the spatial rate of the bounds of Theorem 13.2, which exceeds b. With $h = 8$, $K = e^{8c_0}$, and the choices of $\mathscr{L}$ and μ made before (15.4), the exponent β in Lemma 15.3 is at most $b/2$, so the weight $e^{\beta\ell_o}$ is covered by these two weights.

The bound. In (15.9) we bound the supremum over $w \in \Omega_p$ by the sum of the absolute values of the coefficients of w^η times $p^{|\eta|}$; since $|\lambda_i| \leq 1$, the input l then has radius at most $|z|p$. For an occurrence o and a box $x \in Y_o$ we apply (15.3), with u the location of an input l of o, and we sum over all inputs l with their multiplicities $\eta_l \geq 1$; this overcounts each occurrence. The factor K^{N_o} is absorbed by multiplying the coarse-field radius ρ and the input radius $|z|p$ by K. For a fixed input l, the weighted sum over the occurrences containing l is bounded by the calculus of nonnegative coefficient bounds of Lemma 5.3. The outer coefficient Δ_n is summed with one index fixed, the branch r containing l is summed by its volume-normalized column sum at l, and the remaining branches by their row sums. Let

$$F_\Delta(y) = \sum_{n \geq 2} \frac{\|D^n \Delta(0)\|_{a_{\mathrm{sp}},\mathrm{root}}}{n!} y^n,$$

let $c_q(U, V)$ be the majorant of the input-fixed column sums of ϕ_d at a fixed free input, in which every input, including the fixed one, is weighted by its radius, as in Theorem 13.2(1), and let

$$\Gamma_J = \max_l \frac{v_l}{v_{\mathrm{fine}}} \leq M^{3J},$$

where v_l is the cell volume at the mesh of the input l and v_{fine} the cell volume of the fine lattice. The factor Γ_J converts the volume-normalized column sum of ϕ_d into the counting sum used in the norm of Δ. Choosing the branch that contains l produces the derivative

of F_Δ. With $S_J(\mu) = D_J \sum_{r\ge 0}(r+2)^3 e^{-\mu r}$, where $D_J(r+2)^3$ bounds the number of inputs at distance between r and $r+1$ from a point, we obtain

$$\|\boldsymbol{A}_\Delta(z)\|_{\mathscr{S}_8} \le C_8 S_J(\mu)\Gamma_J F'_\Delta\big(\mathfrak{M}_J(K\rho, K|z|p)\big)\, c_q(K\rho, K|z|p). \tag{15.12}$$

Let $U_0 = 4K_*\rho$ and $V_0 = 4K_* s_J$ be the radii of Lemma 15.4. Since $\sum_{n\ge 2}\rho^{n-2}\|D^n\Delta(0)\|_{a_{\mathrm{sp}},\mathrm{root}}/n! = \delta$, we have, for $0 \le y \le \rho/2$,

$$F'_\Delta(y) \le \delta\rho \sum_{n\ge 2} n 2^{1-n} = 3\delta\rho.$$

By Theorem 13.2(1), $c_q \le V\partial_V \widetilde{\mathfrak{M}}_J$ for a majorant $\widetilde{\mathfrak{M}}_J(U,V) = B^{-1}\widetilde{Z}(U) + C_{\mathrm{im},J}V/(1 - C_{\mathrm{im},J}(U+V))$ of the same form. Since $\partial_V[C_{\mathrm{im},J}V/(1 - C_{\mathrm{im},J}(U+V))] = C_{\mathrm{im},J}(1 - C_{\mathrm{im},J}U)/(1 - C_{\mathrm{im},J}(U+V))^2$, and the denominator is at least η_{im} on the product of Lemma 15.4, the quantity $L_J = \sup_{0<V\le V_0} c_q(U_0,V)/V$ is at most $C_{\mathrm{im},J}\eta_{\mathrm{im}}^{-2}$. Let

$$B_\Delta = 3\Gamma_J\rho\, C_8 S_J(\mu) L_J K,$$

and choose $C_J \ge \max\{1, B_\Delta, K/V_0\}$. If $C_J|z|p < 1$, then $K|z|p < V_0$, and $K\rho < U_0$ because $K < 4K_*$; hence $\mathfrak{M}_J(K\rho, K|z|p) \le \mathfrak{M}_J(U_0,V_0) < \rho/2$ by (15.5). Inserting these bounds into (15.12), we find

$$\|\boldsymbol{A}_\Delta(z)\|_{\mathscr{S}_8} \le B_\Delta\delta|z|p \le \frac{C_J|z|\delta p}{1 - C_J|z|p}.$$

The factor δ enters through the single tensor Δ_n, never through the sum of two norms. The bound is uniform in $|z| \le r_0$ for every $r_0 < (C_J p)^{-1}$. The series defining $\boldsymbol{A}_\Delta$ therefore converges absolutely in $\mathscr{S}_8$ and is holomorphic in z, and all regroupings used above are justified. This proves (4). □

Proof of Theorem 15.6. The proof has seven steps. Step 1 identifies the transformation. Step 2 extracts the contraction of the pullback P_d from the estimate of Theorem 13.5. Step 3 proves the Lipschitz bound, using Lemma 15.8 and the second-order cancellation. Steps 4 and 5 compare two preceding depths at a fixed lattice pair: Step 4 shows that each fixed coefficient tensor of the output converges, uniformly over $\mathcal{A}_E$, and Step 5 deduces convergence in $\mathcal{B}_{\rho,a_{\mathrm{sp}}}$ from a bound at a larger radius. Step 6 combines the contraction with this continuity to construct R_j^∞, and Step 7 passes to the stopped measures.

Step 1: the transformation. All products used in this proof are subproducts of the products of Lemma 15.4. In particular, on the cutoff domain $\max_i\|g_i w_i\|_\infty \le tp(t) \le Q_{\mathrm{act}}$, so that $r + Q_{\mathrm{act}} < r_J$ and the denominator of the majorant is positive. The complex product of the auxiliary integral satisfies (15.5), with its intermediate domain conditions. Thus the rational bound of Theorem 13.2 is used only inside its domain. As explained before the theorem, $Q_d + P_d R + D_d R$ is the effective interaction produced from R before normalization, $\widehat{\mathcal{M}}_{d,t}R$ subtracts its value at the identity, and at a coarse field in B_r the prefactors of Theorem 14.6(2) times Ξ_R give the conditional fiber density before global normalization.

Step 2: contraction of the pullback. Let F have the symmetries of $\mathcal{A}_E$, be invariant under gauge transformations within B_ρ, and satisfy $F(I) = 0$, $DF(I) = 0$, and $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(F) < \infty$; we do not require $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(F) \le E$. For each fixed $h > 0$, the estimate of Theorem 13.5 applies to the input hF on a sufficiently small interval of couplings, which may depend on h. Its constants C_G, C_T, and q_P are fixed before the ceiling of the invariant ball is chosen. The estimate holds at the rate a_{sp}. Indeed, $a_{\mathrm{sp}} < a_{\mathrm{cen}}/8$ by (15.4), and the pullback P_d is the evaluation at the representative $e^{B^{-1}\overline{A}_{d,J}}$ of Lemma 11.9 (Theorem 13.2), for which Theorem 11.10 applies at every rate below $a_{\mathrm{cen}}/8$. At the fixed lattice pair, as $t \downarrow 0$ with $p = A_{\mathrm{w}} \log(1/t)$, the fluctuation exponent and the term containing the input tend to zero uniformly on a smaller complex product domain: they are bounded by $C_{\Lambda,J}\, t(p^3 + p)$ and $C_{\Lambda,J}\, h\mathcal{B}_{\rho,a_{\mathrm{sp}}}(F)\, tp$, respectively. Here the factors g_i^{-2} in the classical remainder have already canceled against its first two orders in the inputs. Since the coarse-field radius of this product exceeds ρ, Lemma 15.5 implies that the same convergence holds in $\mathcal{B}_{\rho,a_{\mathrm{sp}}}$. This gives $D_d(hF) \to 0$ in this norm. The estimate with zero input implies $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(Q_d) \le C_G$. Using the estimate with input hF and the triangle inequality, we find

$$h\mathcal{B}_{\rho,a_{\mathrm{sp}}}(P_d F) \le 2C_G + q_P h\mathcal{B}_{\rho,a_{\mathrm{sp}}}(F) + C_T t^2 h\mathcal{B}_{\rho,a_{\mathrm{sp}}}(F)\big(p^4 + h\mathcal{B}_{\rho,a_{\mathrm{sp}}}(F)p^2\big) + \mathcal{B}_{\rho,a_{\mathrm{sp}}}(D_d(hF)).$$

The pullback P_d does not depend on t or p. Letting $t \downarrow 0$, we obtain $h\mathcal{B}_{\rho,a_{\mathrm{sp}}}(P_d F) \le 2C_G + q_P h\mathcal{B}_{\rho,a_{\mathrm{sp}}}(F)$. Dividing by h and letting $h \to \infty$, we conclude that

$$\mathcal{B}_{\rho,a_{\mathrm{sp}}}(P_d F) \le q_{\mathrm{pr}} \mathcal{B}_{\rho,a_{\mathrm{sp}}}(F), \qquad q_{\mathrm{pr}} = q_P < 1/4. \tag{15.13}$$

All auxiliary limits in this step are taken on one fixed lattice, and the resulting constant q_P is the constant of Theorem 13.5, which is the same for every lattice and depth. Inputs of arbitrary finite norm were used only in that estimate, before its restriction to $\epsilon \le E$. The order of the two limits matters: the small-coupling limit is taken at fixed h.

Step 3: the Lipschitz bound. Let $R_1, R_2, \Delta, \delta, \epsilon_i, R_s$ be as in Lemma 15.8. The segment R_s stays in the class of real, symmetric, gauge-invariant functions with $R_s(I) = 0$, $DR_s(I) = 0$, and $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_s) \le \epsilon_1 + \epsilon_2$. We hold fixed all parameters of the input effective interaction and the cutoff radius, and we vary only the current couplings, through $g_i = z\lambda_i$. Set

$$R_z = \frac{c_J}{p^3 + (\epsilon_1 + \epsilon_2)p}, \tag{15.14}$$

where $c_J \le \min\{s_J, (2C_J)^{-1}, \eta_0/(8C_J)\}$, with C_J from Lemma 15.8 and η_0 from Theorem 12.7. For $|z| \le R_z$ we have $|z|p \le c_J/p^2 \le c_J$, so $C_J|z|p \le 1/2$, and the inputs with radii enlarged by K stay inside (15.5), with all intermediate domain conditions and denominators keeping strict margins. By (15.10) and $p \le p^3$, the family $\boldsymbol{U}_s(z) = \boldsymbol{H}(z) + \boldsymbol{A}_s(z)$ satisfies

$$\|\boldsymbol{U}_s(z)\|_{\mathscr{S}_8} \le 2C_J|z|(p^3 + p + (\epsilon_1 + \epsilon_2)p) \le 4C_J|z|(p^3 + (\epsilon_1 + \epsilon_2)p) \le 4C_J c_J \le \eta_0/2.$$

This choice depends only on J. Since the sum of $\boldsymbol{U}_s$ at $z = t$ is $H_{d,J} + A_{R_s}$, Theorem 12.7 implies that $-\log \Xi_{R_s} = \sum_Y \mathcal{L}_Y(\boldsymbol{U}_s(t))$. Since $\partial_s \boldsymbol{A}_s = \boldsymbol{A}_\Delta$, the derivative of the fluctuation correction along the segment is given by the family

$$\boldsymbol{F}_s(z) = D\boldsymbol{\mathcal{L}}(\boldsymbol{U}_s(z))[\boldsymbol{A}_\Delta(z)],$$

and by (12.15) and (15.10), we have

$$\|\boldsymbol{F}_s(z)\|_{\mathscr{S}_2} \le C_J |z| \delta p \qquad (|z| \le R_z), \tag{15.15}$$

after enlarging C_J. At $z = t$ the sum of $\boldsymbol{F}_s(t)$ over Y is the expectation of A_Δ under the interacting conditional measure, not under the reference measure.

Both $\boldsymbol{U}_s$ and $\boldsymbol{A}_\Delta$ vanish at $z = 0$. By Lemma 15.8(3), the coefficient of z in $\boldsymbol{A}_\Delta$ is linear in the fluctuation variables on each support, so its expectation under the symmetric reference measure π_p vanishes before the sums over supports and over coefficients are taken. The terms containing a further factor of the interaction, including the differences of the Haar disintegration and determinant terms, are of degree at least two in z. Applying Corollary 12.8 with $a = 2C_J \delta p$, which bounds $\|\boldsymbol{A}_\Delta(z)\|_{\mathscr{S}_8}/|z|$ on the disc by (15.10), and with $R = R_z$, we obtain

$$\|\boldsymbol{F}_s(t)\|_{\mathscr{S}_2} \le C_J t^2 \delta [p^4 + (\epsilon_1 + \epsilon_2) p^2], \qquad 0 < t \le R_z/2, \tag{15.16}$$

after enlarging C_J once more.

The fluctuation correction, its value at zero fluctuation, and its derivative along the segment are invariant under translations and global color rotations, by Theorem 12.13. The coordinate changes map zero fluctuations to zero fluctuations, and the first derivatives with respect to the coarse field vanish at the identity. Every support of the logarithm is a union of supports of the decomposition whose intersection graph is connected. By Lemma 15.2 and (15.4), we have $a_{\mathrm{sp}}\mathrm{t} \le 2|Y|$ for the locations of the coarse-field variables in each such support, including subsets and repetitions. The conversion of local interaction decompositions into the norm $\mathcal{B}_{\rho, a_{\mathrm{sp}}}$ in Theorem 13.5, with nine types of scalar indices, therefore applies to (15.16) at the radius ρ, with the factor $9\rho^{-2}$. We integrate in s and add (15.13). The common term Q_d cancels in the difference, and the subtraction of the values at the identity does not change $\mathcal{B}_{\rho, a_{\mathrm{sp}}}$. This gives the factor

$$q_{\mathrm{pr}} + C_J t^2 [p^4 + (\epsilon_1 + \epsilon_2) p^2],$$

after enlarging C_J. One reduction of g_* imposes $p(t) \ge 1$, by taking $g_* \le e^{-1/A_{\mathrm{w}}}$, together with $t[p(t)^3 + 2Ep(t)] \le c_J/2$, which gives $t \le R_z/2$ when $\epsilon_1 + \epsilon_2 \le 2E$, and $C_J t^2 [p(t)^4 + 2Ep(t)^2] \le (1 - q_{\mathrm{pr}})/2$, for all $0 < t \le g_*$. The factor is then at most $\vartheta = (1 + q_{\mathrm{pr}})/2$.

Step 4: changing the preceding depth. By Theorem 11.2, all classical data of the RG step converge on strictly smaller product domains as the preceding depth tends to infinity. These data are the conditional constrained minimizers, the constrained Hessians, the analytic branches of their square roots and inverse square roots that are positive at real arguments, every Haar disintegration density, and the triangular reconstruction. The Hessians here are those of the intermediate scales, with their couplings g_i; no inverse of a Hessian in the coarse-field variables is needed. In particular $H_{d,J}$ and $\mathsf{B}_{\mathrm{ref}}$ converge with every fixed derivative, since their formulas in Theorem 12.13 are finite compositions of these data on products where all domain conditions hold.

It remains to control the term with the input effective interaction, uniformly in $R \in \mathcal{A}_E$. For two depths d, d' we use the same gauge transformation Ω_d on both reconstructions $\mathscr{C}_{\mathrm{raw}\,d}$ and $\mathscr{C}_{\mathrm{raw}\,d'}$. The arguments stay in the common product (15.5). The product with radii enlarged by K has radii strictly below $u_\circ$ and $v_\circ$, so their sum is less than r_J and the denominator of the majorant is at least η_{im}. Every intermediate map keeps its strict domain margin on this product. The bounds of Theorem 13.2 for the gauge transformation and its inverse, together with the convergence of the reconstructions, put the image $\mathrm{Log}[\mathscr{C}_{\mathrm{raw}\,d'}^{\Omega_d}]$ in $B_{3\rho/4}$ for all sufficiently large pairs of depths, while ϕ_d has image in $B_{\rho/2}$. All these bounds hold on a smaller complex product which still contains the output radius ρ and the cutoff domain. At depth d' we compare $R(\exp \mathrm{Log}[\mathscr{C}_{\mathrm{raw}\,d'}^{\Omega_d}])$ with $R(e^{\phi_{d'}})$. On real inputs these are evaluations of the gauge-invariant extension R_{sat} at two gauge-equivalent representatives of the same reconstruction, both in B_ρ, so they agree. Both functions are holomorphic on the comparison product, and their difference vanishes on an open real box. By the one-variable identity theorem applied in each coordinate, it vanishes identically. We may therefore compare the depths d and d' in the common gauge Ω_d. Since the derivative bound of Lemma 15.5 on $B_{3\rho/4}$ is uniform over $\mathcal{A}_E$, this term and each of its fixed derivatives change by $o(1)$ as $d, d' \to \infty$, uniformly over $\mathcal{A}_E$, without any assumption on the behavior of the sequence Ω_d.

Since the cutoff domain and its couplings do not change, the integrands on this fixed finite product, together with their fixed derivatives in the coarse field, are uniformly Cauchy. By Theorem 12.7, the logarithm of each partition function is the sum of a local interaction decomposition over the finitely many boxes of the lattice, with a bound in $\mathscr{S}_2$. This sum is bounded on the chosen complex product, and the modulus of each partition function has a positive lower bound there. Its branch agrees with the real logarithm at real coarse fields. Comparing the ratio of two close partition functions on a disc about one, we find that the logarithms, normalized at the identity, converge together with every fixed coefficient tensor of the output, uniformly in $R \in \mathcal{A}_E$.

Step 5: convergence in the norm. Fix $\rho < \rho_1 < 2\rho$ so close to ρ that the products with coarse-field radius ρ_1, enlarged by K, still lie in the larger analytic product domain. The argument of the input effective interaction stays below its input radius ρ, also at

the output radius ρ_1. Positive composition and the two expansions of Lemma 12.9 and Theorem 12.6 use the same connected box sets and their unions with connected intersection graph. Hence Lemma 15.2 still gives $a_{\mathrm{sp}}\mathrm{t} \le 2|Y|$ at this larger output radius. The output is translation invariant and has vanishing first derivative at the identity, as in Step 3. The conversion of Theorem 13.5, now with factor $9\rho_1^{-2}$, gives

$$\sup_{d,\, R\in\mathcal{A}_E} \mathcal{B}_{\rho_1,a_{\mathrm{sp}}}\big(\widehat{\mathcal{M}}_{d,t}R\big) \le C_{\Lambda,J,t,p,E}. \tag{15.17}$$

Only sufficiently large d are needed here; the finitely many smaller depths have finite bounds and are included after enlarging the constant. At the radius ρ_1 the bound need not be E. Applying Lemma 15.5 to (15.17) and to the convergent coefficient tensors of Step 4, we obtain the continuity statement at exactly the radius ρ and the exponent a_{sp}. Nothing in this argument identifies the auxiliary integral at coarse fields outside B_r with a conditional integral of the conditioned measure.

Step 6: contraction over finitely many scales. Fix j and a positive integer m. If $k, k' \ge j + m$, the two hierarchies have the same lattices and couplings at the levels $j, \dots, j+m$, and their effective interactions at level $j+m$ lie in $\mathcal{A}_E$ by Theorem 14.6. For $h = j+1, \dots, j+m$, define

$$\varepsilon_h(k,k') = \sup_{R\in\mathcal{A}_E} \mathcal{B}_{\rho,a_{\mathrm{sp}}}\big(\widehat{\mathcal{M}}_{J(k-h),t_h}R - \widehat{\mathcal{M}}_{J(k'-h),t_h}R\big).$$

For fixed m and h this tends to zero as $k, k' \to \infty$, by Step 5. Inserting and subtracting at each level the first operator applied to the second input, and using the contraction and the triangle inequality, we obtain

$$\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_j^k - R_j^{k'}) \le 2E\vartheta^m + \sum_{h=j+1}^{j+m} \vartheta^{h-j-1}\varepsilon_h(k,k'). \tag{15.18}$$

We let $k, k' \to \infty$ at fixed m, and then $m \to \infty$. The space of coefficient sequences with finite norm is complete: each coefficient tensor has a limit, Fatou's lemma bounds the weighted sums of the limit, and the Cauchy property makes the tails uniform. The limit series converges absolutely on the same ball and defines R_j^∞ with $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_j^k - R_j^\infty) \to 0$. Its value and first derivative at the identity vanish. Reality, the symmetries, and the identities expressing gauge invariance within B_ρ pass to the pointwise limit, which defines $R_{j,\mathrm{sat}}^\infty$ on D_ρ. Thus $R_j^\infty \in \mathcal{A}_E$.

Step 7: stopped measures. The stopped density in the statement is the density of Theorem 14.6 after $k - j$ RG steps, with preceding depth $J(k-j)$. Its classical term converges by Theorem 11.2 on this fixed lattice, and the gauge-invariant extension preserves the

uniform bound on B_r. Combining this with Step 6 and the value bound of Lemma 15.5, we obtain

$$\delta_{j,k} = \sup_{D_r} \Big| \frac{E_{J(k-j),\mathrm{sat}} - E_{\infty,\mathrm{sat}}}{g_j^2} + R^k_{j,\mathrm{sat}} - R^\infty_{j,\mathrm{sat}} \Big| \longrightarrow 0. \tag{15.19}$$

Here $g_j > 0$ is fixed. At this finite volume, the limiting weight is bounded above and below by positive constants on D_r. Since D_r contains a real open neighborhood of the identity of positive Haar measure, the limiting normalizer is positive and finite. By Lemma 15.1(1), the total variation distance between $\widehat{\nu}^k_j$ and $\widehat{\nu}^\infty_j$ is at most $e^{2\delta_{j,k}} - 1$, which proves the convergence of the stopped measures. The identification $n_0 = n_s$, $k = K - s$ follows from Definition 4.4. All the nonuniform constants introduced in (15.17)–(15.19) concern fixed finite resolutions or fixed positive couplings, as asserted. □

15.4 Conditional fluctuation kernels and the coarse-level measure

We now identify the conditional law of the fluctuation variables of one RG step, given the field at the coarser level, and prove its convergence. We then write the conditioned Wilson measure as an iteration of these kernels, starting from its marginal on the coarse lattice.

Theorem 15.9. *In the hierarchy of Theorem 15.6, fix $j \geq 1$ and $k \geq j$, and recall $d^k_j = J(k-j)$, $t_j = M^{(J-1)/2} g_j$, $p_j = p(t_j)$, and the cutoff domain Ω_{p_j} of the RG step from level j to level $j-1$, which does not depend on k. For a field $c = e^\theta \in B_r$ at level $j-1$, the conditional fluctuation kernel of level j is*

$$k^k_j(\theta, dw) = \frac{\mathbf{1}_{\Omega_{p_j}}(w)\, e^{-V^k_j(\theta,w)}\, dw}{\int_{\Omega_{p_j}} e^{-V^k_j(\theta,v)}\, dv},$$

with the conditional fluctuation potential

$$\begin{aligned} V^k_j(\theta, w) = {} & \tfrac{1}{2}|w|^2 + H_{d^k_j, J}(\theta, w) \\ & + R^k_{j,\mathrm{sat}}\big(\mathscr{C}_{\mathrm{raw}\, d^k_j}(\theta, (g_{j,i} w_i)_i)\big) - R^k_{j,\mathrm{sat}}\big(\mathscr{C}_{\mathrm{raw}\, d^k_j}(\theta, 0)\big). \end{aligned}$$

Here $H_{d,J}$ is the fluctuation exponent of Theorem 12.13, including every difference of determinant terms that depends on coarser fluctuation variables; only terms independent of w have been canceled. The reconstruction uses the inputs $g_{j,i} w_i$, with their fixed ratios, and contains every inverse square root of the constrained Hessians. There exists a limiting potential V^∞_j on the same product such that

$$\Delta_{j,k} = \sup_{e^\theta \in B_r,\, w \in \Omega_{p_j}} |V^k_j(\theta, w) - V^\infty_j(\theta, w)| \longrightarrow 0, \qquad \sup_{e^\theta \in B_r} \|k^k_j(\theta, \cdot) - k^\infty_j(\theta, \cdot)\|_{\mathrm{TV}} \leq e^{2\Delta_{j,k}} - 1.$$

The reconstructions $\mathscr{C}_{\mathrm{raw}\, d_j^k}$ converge uniformly on this product. The limiting density and the limiting reconstruction are Borel and continuous in θ on the smaller coordinate chart, and all normalizers are positive. The formula defines a version of the conditional law at every field in B_r, including fields of zero probability. Restrictions at coarser levels are functions of the field at level $j-1$, and they cancel from this conditional law wherever it is used. The constants have the fixed-resolution dependence of Theorem 15.6. No total variation statement is made for the images of these kernels under reconstruction maps that depend on k.

Proof. Here $t_j = g_{j-1}/\sqrt{M} = M^{(J-1)/2} g_j$ is the largest coupling of the intermediate scales, and $g_{j,i} = t_j M^{(i-J+1)/2}$, $0 \le i < J$. Thus the cutoff domain, its radii, and all ratios of couplings are fixed once j, L, γ, and the hierarchy are fixed.

We apply Theorem 12.13 with preceding depth d_j^k. Its telescoping identity for the classical action contributes $|w|^2/2$ and the classical remainder. Its triangular determinant contributes the factors $g_{j,i}^{3f_i}$, the zero-fluctuation terms $(1/2)\log\det K_i$, and the differences of these terms that depend on coarser fluctuation variables. Its Haar disintegration densities contribute their values at the identity and all nonconstant differences. The unnormalized fiber is therefore exactly

$$C_j^k(c)\, \mathbf{1}_{\Omega_{p_j}}(w)\, e^{-V_j^k(\theta,w)}\, dw,$$

where $C_j^k(c) > 0$ collects the factors that depend only on the field c, including the value of the input effective interaction at zero fluctuation, and V_j^k is the displayed potential. By Theorem 14.6, the indicator of D_r at level j and all intermediate domain indicators equal one on the whole product, at every field in B_r. Dividing this fiber by its own integral gives the formula for the kernel.

For the convergence, we first compare the same input effective interaction R at two preceding depths, using the common gauge transformation of Step 4 in the proof of Theorem 15.6. Both images lie in $B_{3\rho/4}$ on a common complex product. The difference of the reconstructions tends uniformly to zero, and the derivative bounds for R on $B_{3\rho/4}$ are uniform over $\mathcal{A}_E$. Next we replace R_j^k by R_j^∞ on this image. By the value bound of Lemma 15.5, this changes the potential by at most $D_{\Lambda_j}\rho^2 \mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_j^k - R_j^\infty)$, where D_{Λ_j} is the number of scalar link coordinates at level j. We apply these two comparisons at w and at $w = 0$. The classical and Haar disintegration terms converge by Theorem 11.2. Thus the potentials are uniformly Cauchy on the closed product of fields and fluctuation variables, which gives $\Delta_{j,k} \to 0$.

To define the limiting potential, we fix any sufficiently late depth d and use its gauge transformation Ω_d as a comparison gauge. It puts the image of the limiting reconstruction in $B_{3\rho/4}$, so R_j^∞ can be evaluated holomorphically there. Two such choices agree on real data, by gauge invariance within B_ρ, and hence as holomorphic functions, by the

coordinatewise identity argument of Step 4. The convergence of the reconstructions themselves follows from Theorem 11.2 and does not involve this interaction.

By Lemma 15.1(1), applied to Lebesgue measure on Ω_{p_j}, the total variation distance is at most $e^{2\Delta_{j,k}} - 1$, uniformly in the field. The integrals are positive, since the product has positive Lebesgue measure. On every closed smaller coordinate chart the limiting potential is continuous and bounded, so its exponential has a positive integral which depends continuously on θ. This proves that the normalized density is Borel and continuous in θ, and that the limiting reconstruction is continuous. In particular the density defines a version of the conditional law even at fields of zero probability.

Finally, once the field at level $j-1$ is fixed, a restriction at a coarser level has the form $a(c) \in \{0,1\}$, because all further averaging maps are deterministic. At a field used by the conditioned measure we have $a(c) = 1$, and it cancels between the fiber and its normalizer. For a general nonnegative factor depending only on c the same cancellation holds wherever the factor is positive. This proves the last assertion. □

Lemma 15.10. *Fix $k \geq 1$ and the conditioned Wilson measure ν^k, with meshes $a_j = HB^{-j}$ and preceding depths $d_j^k = J(k-j)$. Fix a Borel map S_0, independent of k, from D_r on the coarse lattice Λ_{n_0} to $\{\theta : \|\theta\|_\infty < r\}$ such that $e^{S_0(U)}$ is gauge equivalent to U; such a map exists and is constructed in the proof. Let $\theta_0 = S_0(V^{Jk})$, where V^{Jk} is the coarse field of $U \sim \nu^k$. Given $\theta_0, w_1, \dots, w_{j-1}$, let w_j have the law $k_j^k(\theta_{j-1}, \cdot)$, and put*

$$U_0 = e^{\theta_0}, \qquad U_j = e^{\theta_j}, \quad \theta_j = \Phi_{d_j^k, j}\big(\theta_{j-1}, (g_{j,i} w_{j,i})_i\big), \qquad T_M^J U_j = U_{j-1} \quad (1 \leq j \leq k).$$

Here $\Phi_{d,j}$ is either the map ϕ_d of Section 15.3, built with the gauge transformations of Theorem 13.2, or the logarithmic coordinate representation $\widehat{\Phi}_{d,\Lambda_j}$ of the reconstruction map of Theorem 15.12 below; in both cases the gauge transformation depends only on the field at level $j-1$ and equals the identity at the vertices of $\Lambda_{n_{j-1}}$. Then the following hold.

1. *The gauge orbits $([U_0], \dots, [U_k])$ have the law of the gauge orbits $([V^{Jk}], [V^{J(k-1)}], \dots, [V^0])$ under ν^k.*
2. *If the field at level $j-1$ is replaced by another gauge-equivalent representative in B_r, the kernels correspond under a triangular change of fluctuation coordinates with orthogonal diagonal blocks, which preserves the balls of the cutoff domain and the normalized density.*
3. *Restrictions at coarser levels that are functions of the field at level $j-1$ do not change the conditional law at level j.*

The map S_0 need not be constant on gauge orbits.

Suppose in addition that, on the products used in the construction,

$$\|\Phi_{d,j}(\theta, q)\|_\infty \leq a_{\mathrm{cap}} \|\theta\|_\infty + C_{\mathrm{rec},J} \max_i \|q_i\|_\infty, \qquad a_{\mathrm{cap}} = C_{\mathrm{CP}}/B, \qquad a_{\mathrm{cap}} \sqrt{B} < 1,$$

where $C_{\mathrm{CP}} \geq 1$ is fixed before J. Put $b_j = \|\theta_j\|_\infty$. Then on every sample, uniformly in k, j, and d_j^k,

$$b_j \leq a_{\mathrm{cap}} b_{j-1} + C_{\mathrm{rec},J} t_j p_j, \qquad b_j \leq r a_{\mathrm{cap}}^j + C_{\mathrm{cap},J} g_3 \sqrt{a_j}\, p_j,$$

for any $C_{\mathrm{cap},J} \geq M^{(J-1)/2} C_{\mathrm{rec},J}/(1 - a_{\mathrm{cap}}\sqrt{B})$. For $2 \leq l \leq k$,

$$\sum_{j=l}^{k} b_{j-1} \leq C_J \{ r a_{\mathrm{cap}}^{l-1} + g_3 \sqrt{a_{l-1}}\,(p_l + 1) \};$$

for $l = 1$ one adds the term $b_0 < r$. These constants have the dependence described in item (2) of Definition 4.14, and they may depend on $(1 - a_{\mathrm{cap}}\sqrt{B})^{-1}$; the physical parameters and the mesh factors are displayed explicitly.

Proof. We first construct S_0. The gauge group of the coarse lattice is a finite product of compact matrix groups and has a countable dense subset $(h_m)_{m \geq 1}$. For $U \in D_r$ the set of gauge transformations carrying U into the open set B_r is nonempty and open, so some h_m does so; we choose the least such m and set $S_0(U) = \mathrm{Log}(U^{h_m})$. The sets $\{U : U^{h_m} \in B_r\}$ are open, and removing from each of them the finitely many earlier ones gives Borel sets on which the choice is constant. Thus S_0 is Borel and independent of k, and it carries every field in D_r into B_r.

At fixed k, the density of ν^k disintegrates, by the fiber formula of Theorem 14.6, into its coarse marginal and the normalized kernels of Theorem 15.9. To verify this without choosing an abstract version of the conditional law, we multiply the unnormalized fiber by a bounded test function of the coarse field and a bounded test function of the fluctuation variables, and integrate. By Fubini's theorem and the exact fiber normalizer, we obtain the proposed conditional integral. We repeat this finite calculation from the coarse side to the fine side. At each repetition, all remaining restrictions at coarser levels are factors depending only on the coarser field, and they cancel. Integrating the resulting iterated product against the coarse marginal constructs the finite random hierarchy of the statement.

For changes between coarse representatives in B_r, we use part (3) of Theorem 13.4. The enlarged-domain reconstruction hypothesis inherited from Theorem 15.6 places both intermediate reconstructions in the required comparison domains, with margins. The transition is triangular in the intermediate scales and has orthogonal color rotations and block permutations on its diagonal, so it preserves every ball, the Gaussian term $|w|^2/2$, and Lebesgue measure, with absolute Jacobian one. The representative change after reconstruction is given by Theorem 13.2(4). For the initial coarse field, which lies in D_r but possibly outside B_r, the passage to the representative $e^{S_0(U)}$ uses the gauge invariance of the input density and of the restriction event in parts (1)–(3) of Theorem 13.4, not the chart theorem of Theorem 12.13. The Haar disintegration density and the gauge-invariant extension of the input effective interaction transform by the same identity, so the

normalized conditional density is preserved. This proves (2). Once the field at level $j-1$ is fixed, the gauge transformation in Φ does not depend on the fluctuation variables and equals the identity at every vertex of $\Lambda_{n_{j-1}}$. By gauge covariance of the block averaging map we have

$$T_M^J\big(\mathscr{C}_{\mathrm{raw}\,d}(c,q)^{\Omega(c)}\big) = \big(T_M^J \mathscr{C}_{\mathrm{raw}\,d}(c,q)\big)^{\Omega|_{\Lambda_{n_{j-1}}}} = c,$$

which proves the relation $T_M^J U_j = U_{j-1}$ at every step. The same computation applies to the reconstruction maps of Theorem 15.12, which have these properties by construction. We conclude that every function of the tuple that is invariant under gauge transformations at each level has the same expectation as under ν^k, which is (1). Assertion (3) is the last part of Theorem 15.9.

On the cutoff domain $\max_i \|g_{j,i} w_{j,i}\|_\infty \le t_j p_j$, and the assumed bound on Φ implies the first recurrence. We write $a = a_{\mathrm{cap}}$, $\xi = a\sqrt{B} < 1$, and $c_B = M^{(J-1)/2}$. Since $b_0 < r$ and $t_i = c_B g_3 \sqrt{a_i} = c_B g_3 \sqrt{a_j} B^{(j-i)/2}$, iteration gives

$$\begin{aligned} b_j &\le r a^j + C_{\mathrm{rec},J} \sum_{i=1}^{j} a^{j-i} t_i p_i = r a^j + c_B C_{\mathrm{rec},J} g_3 \sqrt{a_j} \sum_{i=1}^{j} \xi^{j-i} p_i \\ &\le r a^j + \frac{c_B C_{\mathrm{rec},J}}{1-\xi} g_3 \sqrt{a_j}\, p_j. \end{aligned} \tag{15.20}$$

In the last inequality we used $p_i \le p_j$, which holds since $p_j = p_1 + (j-1)c_p$ with $c_p = A_{\mathrm{w}} \log B/2 > 0$. This gives the second bound on b_j, on every sample. Only a constant depending on J is enlarged here; C_{CP} and a_{cap} stay fixed.

For $l \ge 2$, let $x = B^{-1/2}$. Using $\sqrt{a_h} = \sqrt{a_{l-1}}\, x^{h-l+1}$, $p_h = p_{l-1} + (h-l+1)c_p$, and the sums $\sum_{h\ge 0} x^h = (1-x)^{-1}$ and $\sum_{h\ge 0} h x^h = x(1-x)^{-2}$, we obtain

$$\begin{aligned} \sum_{j=l}^{k} b_{j-1} &\le \frac{r a^{l-1}}{1-a} + C_{\mathrm{cap},J} g_3 \sqrt{a_{l-1}} \Big(\frac{p_{l-1}}{1-x} + \frac{c_p x}{(1-x)^2} \Big) \\ &\le C_J \big\{ r a^{l-1} + g_3 \sqrt{a_{l-1}}\,(p_l + 1) \big\}, \end{aligned} \tag{15.21}$$

where $C_J \ge \max\{(1-a)^{-1}, C_{\mathrm{cap},J}[(1-x)^{-1} + c_p x (1-x)^{-2}]\}$. For $l = 1$ we separate $b_0 < r$ and apply this calculation from $j = 2$. □

Lemma 15.11. *In the hierarchy of Theorem 15.6, fix n_0, $L, \gamma > 0$, and the Borel map S_0 of Lemma 15.10, independent of k. Let $\nu^k_{0,\mathrm{full}}$ be the marginal of ν^k on the coarse lattice, that is, the law of V^{Jk}. Then $\nu^k_{0,\mathrm{full}} = \widehat{\nu}^k_0$, and the laws $\eta^k_0 = (S_0)_* \nu^k_{0,\mathrm{full}}$ and $\eta^\infty_0 = (S_0)_* \widehat{\nu}^\infty_0$ of the coarse logarithm satisfy*

$$\|\eta^k_0 - \eta^\infty_0\|_{\mathrm{TV}} \le \|\widehat{\nu}^k_0 - \widehat{\nu}^\infty_0\|_{\mathrm{TV}} \longrightarrow 0.$$

The limiting stopped measure has a positive normalizer. The map S_0 need only be Borel. The convergence has the dependence of Theorem 15.6 at the fixed coarse side.

Proof. We integrate the density of ν^k down to the coarse lattice, using Theorem 14.6. All k restrictions on the fluctuation variables and the indicator of D_r on the coarse lattice have been included, and no restriction at a coarser level remains. The normalized density is exactly that of $\widehat{\nu}_0^k$. Applying Theorem 15.6 with $j = 0$, we obtain its total variation limit and the positive limiting normalizer. For every Borel set A,

$$|\eta_0^k(A) - \eta_0^\infty(A)| = |\widehat{\nu}_0^k(S_0^{-1}A) - \widehat{\nu}_0^\infty(S_0^{-1}A)| \le \|\widehat{\nu}_0^k - \widehat{\nu}_0^\infty\|_{\mathrm{TV}}.$$

Taking the supremum over A, as in Lemma 5.5, completes the proof; no continuity of S_0 is needed. □

15.5 A fixed family of reconstruction maps

The gauge transformations Ω_d of Theorem 13.2 are auxiliary: they change only the representative in which the input effective interaction is evaluated, and the conditional measures do not depend on them. However, Theorem 13.2 asserts nothing about their convergence as the preceding depth tends to infinity, so the reconstructed fields need not converge. To fix this, we use that at a fixed lattice pair the gauge transformations form a bounded family of holomorphic functions of finitely many variables, so they have a convergent subsequence. We choose one subsequential limit and use it at every preceding depth beyond a deterministic threshold. The exact averaging identities and the conditional densities are preserved, because the new gauge transformation is the identity at the vertices of the coarse lattice and this interaction is gauge invariant. This is not a gauge fixing on the space of gauge orbits. The threshold must be large enough to preserve certain bounds, and we keep track of the order in which the constants are chosen:

$$\begin{aligned} C_{\mathrm{CP}} \text{ and the spatial parameters } &\longrightarrow J \longrightarrow \text{ joint coefficient bounds on larger domains} \\ &\longrightarrow \beta_J,\ r_{\mathrm{eff}},\ r \longrightarrow Q_{\mathrm{act}},\ \text{coupling ceiling, thresholds } d_0(\Lambda). \end{aligned} \tag{15.22}$$

The constants β_J and r_{eff} appear in Lemma 15.15. The thresholds can be increased at the last step without changing any earlier constant. The choice of the limiting gauge depends on the lattice pair and on the domains, and not on the coupling.

Theorem 15.12. *Fix nested complex domains $\mathscr{D}_{\mathrm{in}} \Subset \mathscr{D}_{\mathrm{mid}} \Subset \mathscr{D}_{\mathrm{out}}$ of coarse-field and input variables, with radius ratios less than one, such that all real products of coarse fields and inputs used in the construction lie in $\mathscr{D}_{\mathrm{in}}$ with a margin. The domains and the coefficient ceilings depend only on M, J, and the larger construction domains, and not on the lattice side or the preceding depth.*

1. *On each fixed lattice pair Λ of one RG step, choose once a holomorphic subsequential limit $\Omega_{\infty,\Lambda}$, on the middle domain of coarse fields, of the gauge transformations $\Omega_{d,\Lambda}$ of*

Theorem 13.2 as $d \to \infty$. It is periodic, takes values in $\mathrm{SU}(2)$ at real arguments, equals the identity at $\theta = 0$ and at the vertices of the coarse lattice, is equivariant under global color rotations, and has a bounded holomorphic inverse on that domain. Let $\mathscr{C}_{\mathrm{raw}\infty,\Lambda}$ be the limit of the reconstructions $\mathscr{C}_{\mathrm{raw}d,\Lambda}$ given by Theorem 15.6, and put

$$\Phi_{\infty,\Lambda}(c,q) = \mathrm{Log}\big[\mathscr{C}_{\mathrm{raw}\infty,\Lambda}(c,q)^{\Omega_{\infty,\Lambda}(c)}\big].$$

Then the whole sequence $\mathrm{Log}[\mathscr{C}_{\mathrm{raw}}{}_{d,\Lambda}^{\Omega_{\infty,\Lambda}}]$ converges to $\Phi_{\infty,\Lambda}$ on $\mathscr{D}_{\mathrm{in}}$ in every fixed derivative and every fixed spatial difference, and in the weighted analytic coefficient norm, including all output-fixed row sums and all volume-normalized column sums. The modulus of convergence may depend on Λ.

2. *Fix first tolerances, independent of J, for the finitely many quantities bounded in (3) below, and a tolerance depending on J for the joint coefficient norm on $\mathscr{D}_{\mathrm{in}}$. There exist finite deterministic thresholds $d_0(\Lambda)$, which also enforce the image conditions into B_r at real coarse fields in B_r and into B_ρ on the complex product, such that*

$$\widehat{\Omega}_{d,\Lambda}(c) = \begin{cases} \Omega_{d,\Lambda}(c), & d < d_0(\Lambda), \\ \Omega_{\infty,\Lambda}(c), & d \geq d_0(\Lambda), \end{cases}$$

and

$$\widehat{\mathscr{R}}_{d,\Lambda}(c,q) = \mathscr{C}_{\mathrm{raw}d,\Lambda}(c,q)^{\widehat{\Omega}_{d,\Lambda}(c)}, \qquad \widehat{\Phi}_{d,\Lambda} = \mathrm{Log}\,\widehat{\mathscr{R}}_{d,\Lambda},$$

define maps $\widehat{\Phi}_{d,\Lambda}$ that satisfy the same strict domain conditions on the whole products and have common joint coefficient bounds, with all row and column sums, uniformly in Λ and d. The thresholds depend only on the lattice pair, the deterministic domains, and the tolerances; they do not depend on the coupling, on samples, on sources, on an observation scale, on a position, or on the total number of RG steps. We call $\widehat{\mathscr{R}}_{d,\Lambda}$ the reconstruction map and $\widehat{\Phi}_{d,\Lambda}$ its logarithmic coordinate representation.

3. *Write $\widehat{\Phi}_{d,\Lambda}(c,0) = B^{-1}\overline{A}_{d,\Lambda}(c)$, and let $\nabla_{1/B}$ denote the first differences at spacing $1/B$, in units of the coarse lattice. The field $\overline{A}_{d,\Lambda}$ is a new representative of the gauge orbit of the classical background field; it is not claimed to equal the field of Lemma 11.9. Set*

$$G_{d,\Lambda}(\theta) = \big(\overline{A}_{d,\Lambda}(\theta), \nabla_{1/B}\overline{A}_{d,\Lambda}(\theta)\big),$$

and let $\mathscr{D}_\theta = \{\|\theta\|_\infty \leq r_\theta\}$ be a closed polydisc of coarse fields contained in the coarse-field factor of $\mathscr{D}_{\mathrm{in}}$ and in the ball $\|\theta\|_\infty < r_{\mathrm{cen}}$ of Lemma 11.9. There exists $C_{\mathrm{CP}} \geq 1$, independent of J, Λ, d, and r_θ, with the following properties. For every $\theta \in \mathscr{D}_\theta$ we have $\|G_{d,\Lambda}(\theta)\|_\infty \leq C_{\mathrm{CP}}\|\theta\|_\infty$. The output-fixed row sum and every input-fixed column sum of $D_\theta G_{d,\Lambda}(0)$ are at most C_{CP}. For every $\theta \in \mathscr{D}_\theta$ and $1 \leq s \leq 3$, the output-fixed row sum and every input-fixed column sum of $D_\theta^s G_{d,\Lambda}(\theta)$ are at most $s!\, C_{\mathrm{CP}}^{s+1}$. These sums are those of Definition 4.6, with the spatial weight at the rate a_{cen} of Lemma 11.9, which is fixed before J,

and with the column volume factor B^{-3}. In particular the linear coefficient of $c \mapsto \widehat{\Phi}_{d,\Lambda}(c,0)$ at $c = I$ has row and column sums at most C_{CP}/B. The full joint constants may depend on J. The constant C_{CP} is enlarged before J is chosen, and J is then chosen so that

$$a_{\mathrm{cap}} = C_{\mathrm{CP}}/B, \qquad q_{\mathrm{anc}} = C_{\mathrm{CP}}^2/B < 1,$$

with a margin; one may require $q_{\mathrm{anc}} \le 1/4$. The tolerances preserve these conditions.

4. *Let $\widehat{L}_{d,\Lambda}$ be the derivative at $c = I$ of $c \mapsto \widehat{\Phi}_{d,\Lambda}(c,0)$. In this item and in Step 4 of the proof, let H denote the mesh of the coarse lattice of the pair Λ (not the mesh L/n_0 of level zero), and let $a = H/B$ be the mesh of its fine lattice. Then for every coarse logarithm θ, every orientation i, and every color,*

$$a^2 \sum_x (\widehat{L}_{d,\Lambda}\theta)_i(x) = H^2 \sum_u \theta_i(u),$$

where x runs over the vertices of the fine lattice and u over those of the coarse lattice. In words, for each coarse link and color, the linearized zero-fluctuation reconstruction has the same componentwise integral as the piecewise constant embedding of the coarse potential.

5. *The reconstruction maps preserve the exact averaging identity $T_M^J e^{\widehat{\Phi}_{d,\Lambda}(c,q)} = c$, the gauge orbit of the zero-fluctuation reconstruction, the cutoff domains, the Haar disintegration densities, and the conditional fluctuation kernels.*

The constants have the dependence described in item (2) of Definition 4.14; among them, only the finitely many bounds in (3) are chosen before J. No convergence or uniqueness of the sequence $\Omega_{d,\Lambda}$ itself is asserted.

Proof. The proof has five steps: we choose the limiting gauge transformation, prove convergence of the whole sequence in the fixed limiting gauge, choose the thresholds, prove the identity (4), and verify the preservation statements.

Step 1: a limiting gauge transformation at a fixed lattice pair. On a fixed lattice pair Λ there are finitely many coarse-field coordinates and finitely many matrix entries at the vertices. We take the three domains inside the common product of Lemma 15.4, with strict nested margins. Their sums of radii are less than r_J, the denominator of the rational majorant is at least η_{im}, and all intermediate classical, logarithm, and Haar disintegration domain conditions hold. By Theorem 13.2, the gauge transformations have holomorphic logarithms κ_d in the identity chart, uniformly bounded on the outer domain. By Cauchy's formula, every Taylor coefficient is bounded there. We enumerate the coefficients of all matrix entries and choose nested subsequences along which the first coefficient, the second, and so on converge; along the diagonal subsequence, all coefficients converge. Such subsequences exist, since each coefficient lies in a fixed closed disc of $\mathbb{C}$.

On every strictly smaller polydisc the Cauchy bounds make the tail of the coefficient series uniformly summable: in coordinate dimension D there are $\binom{m+D-1}{D-1}$ monomials of

degree m, and the ratio of radii is bounded by a number less than one. Hence the diagonal series converges uniformly on compact subsets of the middle domain, and so do its fixed derivatives after a further reduction of the domain. Its sum κ_∞ is holomorphic there, and we set $\Omega_{\infty,\Lambda} = \exp \kappa_\infty$. The exponential series gives convergence of the corresponding gauge transformations. The bounded inverse entries have the same subsequential limit, and passing to the limit in $\Omega_d \Omega_d^{-1} = I$, we see that the limiting inverse is holomorphic and bounded. Periodicity, the identity at zero and at the coarse vertices, equivariance under global color rotations, and the values in $\mathrm{SU}(2)$ at real arguments are closed conditions and pass to the limit; for the last one, a limit of unitary matrices of determinant one is unitary with determinant one. We choose one such subsequence for each Λ.

By Theorem 11.2, the reconstructions $\mathscr{C}_{\mathrm{raw}\,d,\Lambda}$ converge along the whole sequence of depths. Along the chosen subsequence both the gauge transformations and the reconstructions converge, so their gauge-transformed logarithms converge to $\Phi_{\infty,\Lambda}$. The logarithm stays on its branch: the bound on the images has a strict gap to the boundary of the branch, and the same bound holds in the limit. Passing to the limit coefficient by coefficient in each nonnegative weighted sum and applying Fatou's lemma, we obtain for $\Phi_{\infty,\Lambda}$ the joint coefficient bounds of the unmodified maps, including its column sums; these bounds may depend on J. In the same way, we obtain separately the bounds of Lemma 11.9(2) for the pair $G_{\infty,\Lambda}$ formed by the field $B\Phi_{\infty,\Lambda}(\cdot, 0)$ and its first differences $\nabla_{1/B}$ in units of the coarse lattice. These are the value bound and the row and column sums of its first three derivatives in the coarse field at every point of $\mathscr{D}_\theta$. Indeed, Lemma 11.9(2) holds at every point of the ball $\|\theta\|_\infty < r_{\mathrm{cen}}$, and at a fixed lattice pair these derivatives are finite arrays whose entries converge along the chosen subsequence. Although Lemma 11.9 gives bounds of all orders that are independent of J, only the finitely many bounds listed in (3) are kept with a constant independent of J in the construction below.

Step 2: the whole sequence in the fixed limiting gauge. We define

$$\Psi_d(c, q) = \mathrm{Log}\big[\mathscr{C}_{\mathrm{raw}\,d,\Lambda}(c, q)^{\Omega_{\infty,\Lambda}(c)}\big]. \tag{15.23}$$

For all sufficiently large d this map is defined on a neighborhood of the inner product contained in the larger analytic domain, and it converges locally uniformly and holomorphically to $\Phi_{\infty,\Lambda}$, with every fixed derivative, because the gauge transformation in (15.23) does not depend on d. We now show that it also converges in the coefficient norm.

All operations needed below preserve nonnegative coefficient bounds with fixed indices; this is the calculus of nonnegative coefficient bounds of Lemma 5.3. For a product of coefficient tensors, the union of their networks connects the external indices, so the exponential weight of the product is at most the product of the weights. An output-fixed sum uses the row sums of all factors; an input-fixed sum uses the column sums of the factors on the path from that input to the output, and the row sums of the

others. The intermediate volume factors cancel, and only the ratio of the volumes of the external output and of the fixed input remains. Expanding derivatives of compositions, with factorials divided out, sums the partitions as in the composition of nonnegative power series. Matrix products contribute only a factor depending on the fixed matrix dimension. The series

$$e^X = \sum_{m\ge 0} X^m/m!, \qquad \log(I+Y) = \sum_{m\ge 1} (-1)^{m+1} Y^m/m$$

give the same conclusion for the exponential and the logarithm on products strictly inside their domains of convergence; the second series is used only where $\|Y\| < 1$ with a fixed margin.

To apply these facts uniformly before any threshold is imposed, we recover a link of the reconstruction from the gauge-transformed link by

$$(\mathscr{C}_{\mathrm{raw}\,d})_e = \Omega_d(x)^{-1} e^{\phi_{d,e}} \Omega_d(y), \qquad e = (x,y). \tag{15.24}$$

The gauge transformation Ω_d, its inverse, and the map ϕ_d have the common bounds of Theorem 13.2 on a larger product domain. The input product is strictly inside (15.5), so its denominator is positive and its intermediate margins hold. The limiting gauge transformation has the same bounds, by Fatou's lemma. Using (15.24), then multiplying by the limiting gauge transformation and applying the logarithm series, we obtain a joint bound, depending on J, for (15.23) on an intermediate product domain. The construction domains are chosen inside this product at the outset, with these margins. This preliminary bound need not carry the small factor B^{-1} of the zero-fluctuation reconstruction. Its constants and radii are independent of Λ, of d, and of the working radius chosen later. At fixed Λ each fixed coefficient tensor of (15.23) converges, and Lemma 15.5 controls the tail in the degree on $\mathscr{D}_{\mathrm{in}}$. Then the whole sequence converges in the coefficient norm, including every volume-normalized column sum. Each fixed spatial difference is a fixed finite difference matrix at this lattice size, so the same argument applies to it. The modulus may depend on Λ. This proves (1).

Step 3: finitely many derivatives, then thresholds. Let $C_0 \ge 1$ be a common constant for the quantities in (3) of the unmodified maps, that is, for the pair G_d formed by $B\phi_d(\cdot,0)$ and its first differences $\nabla_{1/B}$. These quantities are the value bound on $\mathscr{D}_\theta$, the row and column sums of $D_\theta G_d(0)$, and the row and column sums of $D_\theta^s G_d(\theta)$ for $1 \le s \le 3$ and every $\theta \in \mathscr{D}_\theta$. Since the unmodified zero-fluctuation map is $B^{-1}\overline{A}_{d,J}$ of Lemma 11.9 (Theorem 13.2), and Lemma 11.9(2) holds at every point of the ball $\|\theta\|_\infty < r_{\mathrm{cen}}$, the constant C_0 is independent of J, Λ, and d. We prescribe errors at most one in this finite list of scaled quantities, smaller if necessary to keep a chosen strict condition. We enlarge C_{CP} to cover C_0 and these errors. Then C_{CP} serves directly as the constant in the value bound and as the bound for the row and column sums of the linear coefficient tensor,

and it can be chosen so that the bounds of orders $1 \le s \le 3$ have the form $s!\, C_{\mathrm{CP}}^{s+1}$. This enlargement uses no analytic radius depending on J and no modulus of convergence. We choose J using this enlarged constant, with $C_{\mathrm{CP}}^2/B \le 1/4$ and the earlier conditions of Theorem 13.5 and (15.4). The construction of Steps 1 and 2 is then carried out for this J.

We also prescribe one error, depending on J, in the full joint coefficient norm on the inner product. By the convergence just proved, for each fixed Λ there exists a finite $d_0(\Lambda)$ such that, for every $d \ge d_0(\Lambda)$, the map (15.23) differs from $\Phi_{\infty,\Lambda}$ by at most this error, and the quantities in (3) for $B\Psi_d(\cdot, 0)$ and its first differences differ from those of $G_{\infty,\Lambda}$ by at most the prescribed errors, uniformly in $\theta \in \mathscr{D}_\theta$. At a fixed lattice pair these are finitely many entries of finitely many arrays, and they converge uniformly on the closed set $\mathscr{D}_\theta$ by Step 2, since $\mathscr{D}_\theta$ lies in the inner domain. We include in this finite list of requirements the image conditions on the real and complex products and every intermediate logarithm, classical, and Haar disintegration margin. The limiting images have strict margins, inherited from the common bounds of the unmodified maps, and the convergence on these product domains allows us to keep half of each margin. All image requirements concern the inputs q_i, on fixed products that contain every product $\{\max_i \|g_i w_i\|_\infty \le t p(t)\}$ used in the construction, so they do not involve the coupling. For the image into B_r we take the whole product $\{\|\theta\|_\infty \le r\} \times \{\max_i \|q_i\|_\infty \le Q_{\mathrm{act}}\}$ of (15.6), including $r + Q_{\mathrm{act}} < r_J$ and the positive denominator. On this product, the limiting image satisfies the strict bound inherited from the majorant of the unmodified maps, and the convergence in the fixed gauge keeps half of its margin above the threshold. The image into B_ρ is treated in the same way on a smaller product domain inside (15.5). The intermediate domain conditions belong to the reconstruction $\mathscr{C}_{\mathrm{raw}\,d,\Lambda}$, which is unchanged. These requirements use Q_{act} itself, not the varying radius $t p(t)$.

The piecewise family of (2) now has the asserted bounds: below the threshold we use the bounds of the unmodified maps, and above it the bounds of the limit plus the prescribed errors. The maximum of the two is independent of Λ and d, even if the thresholds diverge with the lattice size. Each threshold is a deterministic property of finitely many classical maps, domains, and tolerances.

In (3), $\overline{A}_{d,\Lambda} = B\widehat{\Phi}_{d,\Lambda}(\cdot, 0)$ is a representative of the gauge orbit of the classical background field. Its preserved bounds give $a_{\mathrm{cap}} = C_{\mathrm{CP}}/B$ and, under the choice of J in (3), $a_{\mathrm{cap}}\sqrt{B} \le 1/2$. The full joint constants may depend on J.

Step 4: preservation of componentwise integrals. We prove the identity in (4) directly, by differentiating one averaging step at I. For an array X of logarithms on the links of the fine lattice, we write $X(\sigma)$ for the signed sum of X along a path σ, and $X_i(x)$ for the value on the link from x to $x + e_i$. Differentiating the exponent and the reference path in

Definition 4.3, we obtain

$$\begin{aligned}(DT_M(I)X)_i(v) &= M^{-3}\sum_{x\in \mathsf{B}_M(v)}\sum_{s=0}^{M-1} X_i(x+se_i)+h_X(v)-h_X(v+Me_i),\\ h_X(v) &= M^{-3}\sum_{x\in \mathsf{B}_M(v)} X(\sigma_{v,x}).\end{aligned} \tag{15.25}$$

Here the derivative of the reference transport cancels the derivative of the reference transport subtracted inside the logarithm, and the paths $\sigma_{v+Me_i,x+Me_i}$ are translates of $\sigma_{v,x}$. Summing over the periodic coarse vertices v removes the last two terms. For each s the blocks $\mathsf{B}_M(v)$ partition the fine vertices, and translation by se_i permutes them. Thus every oriented fine link is counted exactly M times in the double sum, and

$$\sum_v (DT_M(I)X)_i(v) = M^{-2}\sum_x X_i(x). \tag{15.26}$$

Multiplying by the square of the coarse mesh converts this into the equality of the integrals of the two piecewise constant potentials, logarithm divided by mesh. Applying (15.26) J times gives the factor B^{-2} for T_M^J.

Every reconstruction map satisfies the exact averaging identity, since its gauge transformation is the identity at the coarse vertices. At zero fluctuation this reads $T_M^J e^{\widehat{\Phi}_{d,\Lambda}(\theta,0)} = e^{\theta}$, and its derivative at $\theta = 0$ gives $D(T_M^J)(I)\widehat{L}_{d,\Lambda}\theta = \theta$. With the coarse mesh H and the fine mesh $a = H/B$, (15.26) yields, for each orientation and color,

$$a^2\sum_x (\widehat{L}_{d,\Lambda}\theta)_i(x) = H^2\sum_u \theta_i(u). \tag{15.27}$$

If $\theta = Hv$ for a coarse potential v in physical units, the two sides are exactly the integrals of the fine linearized reconstruction $a^{-1}\widehat{L}_{d,\Lambda}\theta$ and of the piecewise constant embedding of v. Taking v supported on one oriented coarse link and one color proves (4).

Step 5: preservation of the conditional measures. At a fixed coarse field, the reconstruction map is a fixed fine gauge transformation applied after the unchanged reconstruction $\mathscr{C}_{\mathrm{raw}\,d,\Lambda}$. It fixes the coarse vertices, preserves the product Haar measure and the gauge orbit of the classical background field, and changes neither the fluctuation variables nor the cutoff domain. The Haar disintegration density, the triangular determinant, and the fluctuation exponent of Theorem 12.13 are those of $\mathscr{C}_{\mathrm{raw}\,d,\Lambda}$. The input effective interaction takes the same value at the two representatives, by gauge invariance within B_ρ, since both representatives lie in its domain. Thus the fiber and its normalizer are unchanged, and so is the conditional fluctuation kernel. This proves (5). □

15.6 Convergence of finite-scale reconstructions

We now use the reconstruction maps of Theorem 15.12 in every hierarchy. For the RG step from level j to level $j-1$ let Λ_j be its lattice pair, and use the preceding depth

$d_j^k = J(k-j)$. With the coarse logarithm θ_0 and the fluctuation variables $w_1, \dots, w_l$ of Lemma 15.10, the finite-scale reconstruction is defined by

$$F_0^k = \theta_0, \qquad F_j^k = \widehat{\Phi}_{d_j^k, \Lambda_j}(F_{j-1}^k, (g_{j,i} w_{j,i})_i) \qquad (1 \le j \le l).$$

Theorem 15.13. *Use the reconstruction maps of Theorem 15.12 at the preceding depths $d_j^k = J(k-j)$ in every hierarchy.*

1. *For every finite k, the replacement of Ω_d by $\widehat{\Omega}_d$ is a gauge transformation that depends only on the coarse field and equals the identity at the coarse vertices, applied after the same reconstruction $\mathscr{C}_{\mathrm{raw}\, d}$. At each coarse field in B_r it preserves the small-field event, every ball of the cutoff domain, the coordinates of the reconstruction, the effective classical action, the Haar disintegration density, the triangular determinant, the gauge-invariant extension of the input effective interaction, the normalizer, and the conditional fluctuation kernel. It preserves the exact averaging identities at every level, and the gauge invariance within B_ρ after the triangular change of fluctuation coordinates that accompanies a change of the coarse representative.*
2. *Fix $l < \infty$, $L, \gamma > 0$, and the Borel map S_0 of Lemma 15.10. As $k \to \infty$, the joint law P_l^k of $(\theta_0, w_1, \dots, w_l)$ converges in total variation. The law of the finite-scale reconstruction $(F_0^k, \dots, F_l^k)$ converges weakly and in bounded-Lipschitz distance. Since these reconstructions have a common compact range, it also converges in W_1 for any norm on the finite-dimensional space of logarithmic fields. The limiting laws are compatible: they are the finite iterations of one limiting coarse-level law, the kernels k_j^∞, and the maps Φ_{∞,Λ_j}, and the limit for l is a marginal of the limit for $l+1$. All these objects are Borel.*

No total variation statement is made for the laws of the reconstructed fields. The constants and moduli have the dependence of Theorem 15.6 at fixed l. The construction requires neither a limit of fields in a space of distributions nor a modulus that is uniform in l.

Remark 15.14. The restriction to weak convergence for the reconstructed fields is necessary. The reconstruction maps depend on k, and pushforwards by maps that move with k do not preserve total variation convergence. For example, if $x_k \to x$ with $x_k \neq x$, then $\delta_{x_k} \to \delta_x$ weakly, while $\|\delta_{x_k} - \delta_x\|_{\mathrm{TV}} = 1$. The coordinates $(\theta_0, w_1, \dots, w_l)$, on the other hand, live on fixed products, and there total variation convergence holds.

Proof. At a fixed coarse field, the replacement changes no coordinate of the reconstruction, no fluctuation variable, no ball, and no Jacobian. The indicators of D_r are gauge invariant, and the new representative satisfies the strict domain conditions of Theorem 15.12. Gauge covariance preserves the averaging identity, and gauge invariance within B_ρ preserves the effective classical action and the input effective interaction. By Theorem 14.6, every factor of the fiber, the normalizer, and the conditional fluctuation kernel are therefore preserved.

For a change of the coarse representative, we write $C' = C^{\widetilde{h}}$ for the reconstructions, after the triangular change of fluctuation coordinates. The convention (4.1) gives

$$(C')^{\widehat{\Omega}(c')} = \big(C^{\widehat{\Omega}(c)}\big)^{\widehat{\Omega}(c')\widetilde{h}\widehat{\Omega}(c)^{-1}}. \tag{15.28}$$

The change of coordinates preserves every ball and has absolute Jacobian one, by Theorem 13.4(3), on the comparison domains of the enlarged-domain reconstruction hypothesis. Both representatives lie in the domain of this interaction, so its values at them agree. This proves the gauge invariance within B_ρ without any equivariance of the limiting gauge transformation under local gauge transformations, and completes the proof of (1).

By Lemma 15.11, $\eta_0^k \to \eta_0^\infty$ in total variation. Fix $l < \infty$ and define

$$X_l = \{\|\theta_0\|_\infty \le r\} \times \prod_{j=1}^{l} \Omega_{p_j}.$$

The domain of each reconstruction map strictly contains its closed product. By Lemma 15.10 and Theorem 15.12, the reconstructions F_j^k are defined on X_l and have a common compact range. For k large, each of the finitely many depths d_j^k, $j \le l$, exceeds its threshold $d_0(\Lambda_j)$. The gauge transformation is then the fixed limiting one, and Theorem 15.12(1) implies uniform convergence of each map. Let $u_{j,k} = \|F_j^k - F_j^\infty\|_\infty$, let $v_{j,k}$ be the supremum distance between $\widehat{\Phi}_{d_j^k,\Lambda_j}$ and Φ_{∞,Λ_j} on the smaller product domain, and let ω_{Φ_j} be the modulus of continuity of Φ_{∞,Λ_j} in its first argument there. Adding and subtracting $\Phi_{\infty,\Lambda_j}(F_{j-1}^k, \cdot)$, we obtain

$$u_{j,k} \le v_{j,k} + \omega_{\Phi_j}(u_{j-1,k}), \qquad u_{0,k} = 0, \qquad v_{j,k} \longrightarrow 0. \tag{15.29}$$

By induction on j, $F_j^k \to F_j^\infty$ uniformly on X_l for every $j \le l$.

By Theorem 15.9, the limiting density is continuous on the compact product of fields and fluctuation variables, and its continuous positive normalizer has a positive minimum. Hence the limiting kernel is uniformly continuous in the field, in $L^1(dw)$ and therefore in total variation; let ω_{k_j} be its modulus. For points $h = (\theta_0, w_1, \dots, w_{j-1})$ of the coordinate space, define

$$K_j^k(h, dw) = k_j^k(F_{j-1}^k(h), dw), \qquad K_j^\infty(h, dw) = k_j^\infty(F_{j-1}^\infty(h), dw).$$

The two kernels are evaluated at different reconstructed fields, so the comparison contains a term for this discrepancy. By Theorem 15.9,

$$\sup_h \|K_j^k(h, \cdot) - K_j^\infty(h, \cdot)\|_{\mathrm{TV}} \le e^{2\Delta_{j,k}} - 1 + \omega_{k_j}(u_{j-1,k}) \longrightarrow 0. \tag{15.30}$$

Every term of the conditional fluctuation potential is kept here, including the input effective interaction, the Haar disintegration terms, and the differences of the determinant terms.

By Lemma 15.10, $P_l^k = \eta_0^k K_1^k \cdots K_l^k$ is the law of the coordinates. We define $P_l^\infty = \eta_0^\infty K_1^\infty \cdots K_l^\infty$. Lemma 15.1(2) implies that

$$\|P_l^k - P_l^\infty\|_{\mathrm{TV}} \le \|\eta_0^k - \eta_0^\infty\|_{\mathrm{TV}} + \sum_{j=1}^{l} \sup_h \|K_j^k(h,\cdot) - K_j^\infty(h,\cdot)\|_{\mathrm{TV}} \longrightarrow 0. \tag{15.31}$$

We write $F^k = (F_0^k, \dots, F_l^k)$. The ranges lie in a common compact subset of the finite-dimensional space of logarithmic fields, of diameter D_l. For every continuous f on this set, Lemma 15.1(3) gives

$$\left| \int f(F^k)\, dP_l^k - \int f(F^\infty)\, dP_l^\infty \right| \le 2\|f\|_\infty \|P_l^k - P_l^\infty\|_{\mathrm{TV}} + \omega_f(\|F^k - F^\infty\|_\infty), \tag{15.32}$$

which proves weak and bounded-Lipschitz convergence. For a fixed norm, Lemma 15.1(3) also gives

$$W_1\big((F^k)_* P_l^k, (F^\infty)_* P_l^\infty\big) \le \|F^k - F^\infty\|_\infty + D_l \|P_l^k - P_l^\infty\|_{\mathrm{TV}} \longrightarrow 0.$$

The limiting coarse-level law is a Borel pushforward, and the limiting maps and kernels are continuous on their product domains. Integrating the last probability kernel to one proves the compatibility. Only finitely many lattices, thresholds, and moduli were used. □

15.7 Deterministic amplitude bounds for the reconstruction

The bounds of Lemma 15.10 assume a linear bound on the reconstruction maps. We now derive such bounds, in a nonlinear form, from the joint coefficient bounds of Theorem 15.12. The constants are uniform in the mesh of the coarse lattice and in the coupling; this uniformity is used in Sections 17, 23, 31, and 34.

Lemma 15.15. *For the reconstruction maps of Theorem 15.12 write $\widehat{P}_{d,\Lambda}(c) = \widehat{\Phi}_{d,\Lambda}(c,0)$ and $\widehat{L}_{d,\Lambda} = D\widehat{P}_{d,\Lambda}(0)$. Fix the larger joint domains and one common joint coefficient bound, with all row sums and volume-normalized column sums of Definition 4.6, before the working radius and the radius r are chosen. Let a_m be the larger of the row and column norms of the homogeneous part of degree m of $\widehat{P}_{d,\Lambda}$. There exist $A_J, R_J > 0$, uniform in Λ and d, such that*

$$a_1 \le \alpha_* = C_{\mathrm{CP}}/B, \qquad a_m \le A_J R_J^{-m} \quad (m \ge 2).$$

After a fixed reduction of the radius, which absorbs the degree factors produced by fixing an input, choose

$$\beta_J \ge \max\{1, R_J^{-1}, A_J/(\alpha_* R_J^2)\}.$$

Then, uniformly in Λ and d,

$$\widehat{P} \prec \varphi(z) = \frac{\alpha_* z}{1-\beta_J z}, \qquad \widehat{P} - \widehat{L} \prec \frac{\alpha_* \beta_J z^2}{1-\beta_J z}.$$

Here $\prec$ means that each a_m (with $m \geq 2$ in the second relation) is at most the coefficient of z^m of the series on the right, and that a fixed input is bounded by the corresponding derivative of the same series. The full joint series in the coarse field and the inputs, not only these coefficients, has a common domain depending on J, with its column sums, and the bounds (3) and the identity (4) of Theorem 15.12 hold at the same time.

Choose J, using the constant C_{CP}, so that $B\alpha_*^2 = q_{\mathrm{anc}} \leq 1/4$, and only afterwards choose

$$0 < r_{\mathrm{eff}} < \rho, \qquad \frac{\beta_J r_{\mathrm{eff}}}{1-\alpha_*} < \frac{1}{8}, \qquad 0 < r < r_{\mathrm{eff}}/4.$$

On the working ball $\{\|\theta\|_\infty \leq r_{\mathrm{eff}}\}$,

$$\|D\widehat{P}\| \leq a_{\mathrm{map}} = \frac{\alpha_*}{(1-\beta_J r_{\mathrm{eff}})^2} \leq \frac{64}{49}\alpha_*, \qquad a_{\mathrm{map}}\sqrt{B} \leq \frac{32}{49} < 1.$$

The bounds on the inputs and the coupling ceiling can then be chosen so that every image of the reconstruction, and every repeated zero-fluctuation reconstruction, lies in the working domain. Increasing the thresholds $d_0(\Lambda)$ to obtain these smaller image margins changes none of C_{CP}, A_J, R_J, β_J, or the joint bound. These constants have the dependence described in item (2) of Definition 4.14; they are independent of r, of the thresholds, of the coupling, of the side, of the depth, and of the lattice offset, and among them only C_{CP} is independent of J.

The same joint bounds give the following amplitude bound. Let $H > 0$, $a_j = HB^{-j}$, $t_j = c_B g_3 \sqrt{a_j}$ with $c_B = M^{(J-1)/2}$, and $p_j = A_{\mathrm{w}} \log(1/t_j) \geq 1$. Consider any finite chain $\theta_j = \widehat{\Phi}_{d_j,\Lambda_j}(\theta_{j-1}, q_j)$, $1 \leq j \leq k$, of logarithmic representations of reconstruction maps in the working ball, with $b_0 = \|\theta_0\|_\infty \leq r$ and inputs $\max_i \|q_{j,i}\|_\infty \leq t_j p_j$, and put $b_j = \|\theta_j\|_\infty$. There exists a constant $C_{\mathrm{free},J}$, determined by the joint bounds, such that

$$b_j \leq \frac{\alpha_* b_{j-1}}{1-\beta_J b_{j-1}} + C_{\mathrm{free},J} t_j p_j.$$

Choose a coupling ceiling τ, after the working radii, so that these chains stay in the working ball. Then, whenever $t_1 \leq \tau$, uniformly in H, g_3, k, the preceding depths, the sides, and the lattice offset,

$$b_j \leq C_{\mathrm{cap},J} r \alpha_*^j + C_{\mathrm{cap},J} g_3 \sqrt{a_j}\, p_j, \qquad \sum_{j=1}^{k} b_{j-1} \leq C_{\mathrm{cap},J}\Big[\frac{r}{1-\alpha_*} + g_3\sqrt{H}\,(p_1+1)\Big].$$

The constant $C_{\mathrm{cap},J}$ may depend on the joint bounds, A_{w}, M, J, r_{eff}, β_J, and τ, but not on H or on the coupling. These bounds hold on every sample; they are not moment bounds.

Proof. We first build the majorant φ from the joint bounds and choose the radii. We then prove the amplitude bounds in two passes. The first pass uses the ratio a_{map} and bounds the sum of the amplitudes uniformly; the second uses this sum to recover the smaller ratio α_*, which is what makes the forcing summable.

Step 1: the majorant. We fix the common joint coefficient bound of Theorem 15.12 on a larger product domain, before any working radius is chosen, and restrict it to $q = 0$. If its coarse-field radius is R_0 and its bound is A_0, then the norm of degree m is at most $A_0 R_0^{-m}$. The row and every volume-normalized column use the same bound. Fixing an input produces at most a degree factor m, which is absorbed by replacing R_0 with $R_0/2$, since $m2^{-m}$ is bounded; a fixed finite list of such factors is absorbed by a further fixed reduction. This gives $A_J, R_J > 0$ as in the statement for every $m \geq 2$. The linear coefficient has the separate bound $a_1 \leq C_{\mathrm{CP}}/B = \alpha_*$ from Theorem 15.12(3). Only this last bound is needed uniformly before J is fixed.

We choose β_J at least the displayed maximum; it may be increased to cover the inverse radius of the fixed joint product as well. For $m \geq 2$,

$$\alpha_* \beta_J^{m-1} \geq \frac{A_J}{R_J^2} \beta_J^{m-2} = A_J R_J^{-m} (\beta_J R_J)^{m-2} \geq A_J R_J^{-m}. \tag{15.33}$$

These are the coefficients of $\varphi(z) = \alpha_* z/(1 - \beta_J z)$, whose first coefficient is treated separately. Subtracting the linear term gives the second majorant. In a nonnegative coefficient series, a derivative with respect to a fixed input marks that input and differentiates the same series. The same calculation therefore proves the statement for fixed inputs, with the volume normalization of the column of that input. The full joint series and the identities of Theorem 15.12 are those already proved there; restricting to $q = 0$ does not replace them.

Step 2: the working radii. We first choose J with $C_{\mathrm{CP}}^2/B \leq 1/4$. In particular $\alpha_* < 1$ and $\alpha_* \sqrt{B} \leq 1/2$. All constants defining β_J are now fixed. We choose r_{eff} as small as necessary inside the coarse-field radius of the common product domain, with the displayed inequality, and then $0 < r < r_{\mathrm{eff}}/4$. Such choices exist since β_J is finite. Differentiating the nonnegative series, we have on this working ball

$$\|D\widehat{P}\| \leq \varphi'(r_{\mathrm{eff}}) = \frac{\alpha_*}{(1 - \beta_J r_{\mathrm{eff}})^2}.$$

The working inequality implies $\beta_J r_{\mathrm{eff}} < (1 - \alpha_*)/8 < 1/8$. The last display is therefore at most $(64/49)\alpha_*$, and its product with $\sqrt{B}$ is at most $32/49$. Since $\widehat{P}(0) = 0$, the zero-fluctuation image of the working ball lies in the strictly smaller ball of radius $a_{\mathrm{map}} r_{\mathrm{eff}}$.

For the remaining choices, we note that the common joint series gives a continuous nonnegative majorant $G_J(u,v)$, with $G_J(u,0)=0$, for $\widehat{\Phi}(\theta,q)-\widehat{P}(\theta)$ on the working ball $\|\theta\|_\infty \le r_{\mathrm{eff}}$. We choose a fixed $v_*>0$ so small that

$$G_J(r_{\mathrm{eff}},v_*) < (1-a_{\mathrm{map}})r_{\mathrm{eff}}/4,$$

and, after a further reduction, the image condition into B_r and every intermediate domain margin hold. For example, at a coarse field in B_r the zero-fluctuation reconstruction is at most $a_{\mathrm{map}}r < r/2$, and we choose the bound on the inputs so that their contribution is less than $r/2 - a_{\mathrm{map}}r$. This number is positive, since $a_{\mathrm{map}} \le 32/(49\sqrt{B}) < 1/2$ for $B \ge 3$. The coupling ceiling is then reduced so that $tp(t)$ is below that bound for every smaller coupling. These choices of fixed domains and ceilings do not depend on the coupling.

If necessary, we increase each threshold $d_0(\Lambda)$ to impose these more restrictive image conditions, by the same convergence in the fixed gauge. For depths below the enlarged threshold we use the unmodified family, whose bounds satisfy the chosen requirements. For depths above it we keep the prescribed tolerances for the joint norm and for the quantities in Theorem 15.12(3), and add only these image requirements. Thus increasing a threshold changes none of the earlier coefficient constants. The order of the choices is the joint bound, then β_J, then r_{eff} and r, and then the bounds on the inputs and the coupling ceiling, as in (15.22). This proves the first part of the lemma.

Step 3: a uniform bound for the inputs and a first recursion. All products in the rest of the proof are closed subproducts of the common product of Lemma 15.4; in particular their sums of radii are less than r_J and the denominators of the majorants are positive. Fix $v_0>0$ such that the product with coarse-field radius r_{eff} and input radius $2v_0$ lies in the common joint coefficient domain, with a margin. Differentiating the nonnegative joint series in one input, and using the factor two in the radius, we find that every output-fixed absolute row sum of the derivative with respect to all fluctuation inputs is bounded by one constant $C_{\mathrm{free},J}<\infty$, uniformly in the lattice pair, the preceding depth, and the lattice offset. Integrating $D_q\widehat{\Phi}(\theta,sq)q$ over $0\le s\le 1$ and using its absolute row sum, we obtain

$$\|\widehat{\Phi}_{d,\Lambda}(\theta,q)-\widehat{\Phi}_{d,\Lambda}(\theta,0)\|_\infty \le C_{\mathrm{free},J}\max_i\|q_i\|_\infty, \qquad \|\theta\|_\infty\le r_{\mathrm{eff}},\quad \max_i\|q_i\|_\infty\le v_0. \tag{15.34}$$

The number of output links is never summed in this estimate. The inverse square roots of the constrained Hessians remain inside $\widehat{\Phi}$, whose inputs are exactly $q_i=g_iw_i$. Combining (15.34) with the majorant of $\widehat{P}$, which gives $\|\widehat{P}(\theta)\|_\infty\le\varphi(\|\theta\|_\infty)$, we obtain

$$b_j \le \varphi(b_{j-1})+f_j, \qquad f_j=C_{\mathrm{free},J}t_jp_j, \qquad \varphi(b)=\frac{\alpha_* b}{1-\beta_J b}. \tag{15.35}$$

Let $a=a_{\mathrm{map}}$, $x=B^{-1/2}$, and $\kappa_{\mathrm{w}}=A_{\mathrm{w}}\log B/2$. We choose the common ceiling $\tau>0$, after the working radii, so that

$$\tau\le e^{-1}, \qquad A_{\mathrm{w}}\log(1/\tau)\ge 1, \qquad \tau p(\tau)\le v_0, \qquad C_{\mathrm{free},J}\tau p(\tau)\le (1-a)r_{\mathrm{eff}}/4. \tag{15.36}$$

We also keep every previously imposed image and domain condition, by decreasing τ if necessary. For $0 < t \le \tau$ both $tp(t)$ and $t(p(t)+1)$ are increasing functions of t, since their derivatives are $A_{\mathrm{w}}(\log(1/t)-1)$ and $A_{\mathrm{w}}(\log(1/t)-1)+1$. Thus these conditions hold for every t_j whenever $t_1 \le \tau$. Since $\varphi(b) \le ab$ on the working ball, (15.35) implies

$$b_j \le a b_{j-1} + f_j.$$

It also keeps the chain in the working ball. Starting from $b_0 \le r < r_{\mathrm{eff}}/4$, if $b_{j-1} \le r_{\mathrm{eff}}/2$, then (15.36) gives $b_j \le a r_{\mathrm{eff}}/2 + (1-a) r_{\mathrm{eff}}/4 < r_{\mathrm{eff}}/2$. By induction, every evaluation in (15.35) takes place in the same working ball.

The identities $t_j = t_1 x^{j-1}$ and $p_j = p_1 + \kappa_{\mathrm{w}}(j-1)$ give

$$\sum_{j\ge 1} t_j p_j = t_1 \Big\{ \frac{p_1}{1-x} + \frac{\kappa_{\mathrm{w}} x}{(1-x)^2} \Big\} \le D_J t_1 (p_1+1), \qquad D_J = \frac{1}{1-x} + \frac{\kappa_{\mathrm{w}} x}{(1-x)^2}. \tag{15.37}$$

For a finite chain, the infinite sum is an upper bound for the forcing. Iterating the linear recursion and summing its nonnegative terms, we obtain

$$\sum_{h=0}^{k-1} b_h \le \frac{r + C_{\mathrm{free},J} D_J t_1 (p_1+1)}{1-a} \le S_* = \frac{r_{\mathrm{eff}}/4 + C_{\mathrm{free},J} D_J \tau (p(\tau)+1)}{1-a}. \tag{15.38}$$

This bound is independent of H, g_3, k, the lattice pairs, and their preceding depths. In particular S_* contains neither p_1 nor H as a varying parameter.

Step 4: the sharper ratio. Set

$$K_\beta = \frac{1}{1-\beta_J r_{\mathrm{eff}}} \le \frac{8}{7}, \qquad \zeta_h = \frac{\alpha_*}{1-\beta_J b_h}, \qquad C_{\mathrm{prod}} = \exp(K_\beta \beta_J S_*).$$

For $0 \le y \le \beta_J r_{\mathrm{eff}}$ we have $-\log(1-y) = \int_0^y (1-s)^{-1} ds \le K_\beta y$. Since $b_h \le r_{\mathrm{eff}}/2$, every finite product satisfies

$$\prod_{h=u}^{v} \zeta_h \le \alpha_*^{v-u+1} \exp\Big(K_\beta \beta_J \sum_{h=u}^{v} b_h \Big) \le C_{\mathrm{prod}} \alpha_*^{v-u+1}, \tag{15.39}$$

where we used (15.38); for an empty product we use $1 \le C_{\mathrm{prod}}$. Since $\varphi(b_h) = \zeta_h b_h$, successive substitution in $b_j \le \zeta_{j-1} b_{j-1} + f_j$ gives

$$\begin{aligned} b_j &\le b_0 \prod_{h=0}^{j-1} \zeta_h + \sum_{i=1}^{j} f_i \prod_{h=i}^{j-1} \zeta_h \le C_{\mathrm{prod}} r \alpha_*^j + C_{\mathrm{prod}} C_{\mathrm{free},J} \sum_{i=1}^{j} \alpha_*^{j-i} t_i p_i \\ &\le C_{\mathrm{prod}} r \alpha_*^j + \frac{C_{\mathrm{prod}} C_{\mathrm{free},J}}{1 - \alpha_* \sqrt{B}}\, t_j p_j. \end{aligned} \tag{15.40}$$

In the last inequality we used $t_i = t_j B^{(j-i)/2}$, $p_i \le p_j$, and $\alpha_* \sqrt{B} \le 1/2$. Substituting $t_j = c_B g_3 \sqrt{a_j}$, we obtain the pointwise bound of the statement, with the ratio α_* rather than a.

Summing (15.40) over $h = 1, \dots, k-1$, treating $b_0 \le r$ separately, and using (15.37), we obtain

$$\begin{aligned} \sum_{h=0}^{k-1} b_h &\le \frac{C_{\mathrm{prod}} r}{1-\alpha_*} + \frac{C_{\mathrm{prod}} C_{\mathrm{free},J} D_J}{1-\alpha_* \sqrt{B}}\, t_1 (p_1 + 1) \\ &= \frac{C_{\mathrm{prod}} r}{1-\alpha_*} + \frac{C_{\mathrm{prod}} C_{\mathrm{free},J} D_J c_B x}{1-\alpha_* \sqrt{B}}\, g_3 \sqrt{H}\, (p_1 + 1). \end{aligned} \tag{15.41}$$

For the last two displays, we may take any

$$C_{\mathrm{cap},J} \ge C_{\mathrm{prod}} \max\Big\{1, \frac{C_{\mathrm{free},J} c_B}{1-\alpha_* \sqrt{B}}, \frac{C_{\mathrm{free},J} D_J c_B x}{1-\alpha_* \sqrt{B}}\Big\}. \tag{15.42}$$

It may also be enlarged to cover the constant $C_{\mathrm{cap},J}$ of Lemma 15.10. This constant depends only on the joint bounds, the working radii, and the ceiling. Thus it is uniform in H and g_3 subject to the single condition $t_1 \le \tau$, and in the length of the chain, the sides, the lattice offset, and the preceding depths. The constant C_{CP} is not enlarged in this proof. □

15.8 Uniformity on compact intervals of couplings

In this section the coupling varies in a compact interval $I = [\gamma_-, \gamma_+]$ with $0 < \gamma_- \le \gamma_+ < \infty$. The lattice of level j has side $n_j = nB^j$ for a fixed coarse side $n \in 4\mathbb{N}$, and $a_j = (L/n) B^{-j}$, $g_j^2 = \gamma / n_j$, $t_j = M^{(J-1)/2} g_j$, and $d_j^k = J(k-j)$. On a fixed lattice pair Λ the largest coupling of the intermediate scales is $t = c_\Lambda \sqrt{\gamma}$ for a constant $c_\Lambda > 0$, with the fixed ratios λ_i. Constants with a subscript I may depend on both endpoints of I, on n, L, M, J, the fixed parameter collection Θ of Theorem 23.2, the analytic parameters and exponents, moment orders, and fixed lists of times and observables; constants for a fixed lattice pair may also depend on that pair. None of them depends on the value of $\gamma \in I$, on the preceding depth, or on a finer mesh. This is the dependence described in item (7) of Definition 4.14; these constants are neither the constants of item (3) nor constants chosen before J. Since the cutoff radius $p(t_j)$ depends on γ, we use the rescaled fluctuation variables $v_j = w_j / p(t_j)$, which live on the fixed product Ω_1 of unit balls.

Theorem 15.16. *Fix the parameters and products of Theorem 15.6, with its coupling ceiling g_*, a coarse side $n \in 4\mathbb{N}$, and $I = [\gamma_-, \gamma_+]$ with $0 < \gamma_- \le \gamma_+ < \infty$. Assume that the image, product, and coupling conditions of Theorem 15.6 hold for every $\gamma \in I$. Then the following hold.*

1. *On every fixed lattice pair Λ, at the same radius and exponent,*

$$\lim_{d,d'\to\infty} \sup_{\gamma\in I} \sup_{R\in\mathcal{A}_E} \mathcal{B}_{\rho,a_{\mathrm{sp}}}\big(\widehat{\mathcal{M}}_{d,t}R - \widehat{\mathcal{M}}_{d',t}R\big) = 0. \tag{15.43}$$

   The contraction factor $\vartheta < 1$ of Theorem 15.6 is the same for all $\gamma \in I$. At each finite d the transformation is jointly continuous in (γ, R) in this norm. After the substitution $w = p(t)v$, the classical data and the reconstruction maps converge uniformly in $\gamma \in I$ on every fixed smaller product domain, with every fixed derivative and every fixed spatial difference.
2. *For every fixed j, the effective interactions R_j^k and the stopped measures $\widehat{\nu}_j^k$ converge uniformly in $\gamma \in I$, in $\mathcal{B}_{\rho,a_{\mathrm{sp}}}$ and in total variation respectively, and the limits are continuous in γ in these norms.*
3. *For every fixed l, the joint law of the coarse logarithm and of the rescaled fluctuation variables $(v_1, \dots, v_l)$ converges in total variation, uniformly in $\gamma \in I$. The laws of the finite-scale reconstructions converge in bounded-Lipschitz distance and in W_1, for any norm on the finite-dimensional space of logarithmic fields, uniformly in $\gamma \in I$. The limiting laws are continuous in γ in these distances, and they are compatible.*

All moduli have the dependence described above, at fixed l. No rate that is uniform in l, and no statement as $\gamma \to 0$, is asserted.

The hypothesis holds in particular when $I \subset (0, \gamma_0]$ and the hierarchy satisfies the hypotheses of Theorem 15.6 for every $\gamma \le \gamma_0$. The theorem does not by itself guarantee that these hypotheses hold for a given coarse side and an arbitrary upper endpoint γ_+.

Proof. The proof follows the proofs of Theorems 15.6, 15.9, and 15.13. The new point is that every estimate is proved uniformly in $\gamma \in I$ on fixed products; uniform convergence is never deduced from pointwise convergence and compactness of I.

Step 1: rescaling. On a fixed lattice pair, let $t = c_\Lambda\sqrt{\gamma}$, $t_\pm = c_\Lambda\sqrt{\gamma_\pm}$, $p_- = p(t_+) > 0$, and $p_+ = p(t_-) < \infty$. Let D be the finite number of scalar fluctuation variables, and we change variables $w = p(t)v$, $v \in \Omega_1$. The Jacobian $p(t)^D$ cancels from normalized integrals and from logarithms normalized at the identity. For the Gaussian reference the remaining normalizer is

$$Z_p^{\mathrm{unit}} = \int_{\Omega_1} e^{-p^2|v|^2/2}\,dv, \qquad |\Omega_1| e^{-p_+^2 \sup_{\Omega_1}|v|^2/2} \le Z_p^{\mathrm{unit}} \le |\Omega_1|. \tag{15.44}$$

The scalar factor $z_p^{D/3}$ of (15.8) satisfies the exact relation $z_p^{D/3} = p^D Z_p^{\mathrm{unit}}$, so no normalization is discarded. Every input is $q_i = \lambda_i t p(t) v_i$, with fixed $\lambda_i > 0$ and $t \in [t_-, t_+]$. By Lemma 15.4 and the hypothesis, these inputs lie in one closed subproduct of the larger analytic product domain, and we also use its complex enlargement in the coarse field and a slightly larger input product. Their margins were chosen before the coupling.

Step 2: classical data. In Lemma 12.3, the conditional constrained minimizer minimizes $E_d \circ b$, the constrained Hessian is its second derivative in the free variables, and its square root is the analytic branch that is positive at real arguments. These objects do not depend on γ; the coupling enters only through $q_i = g_i w_i$ and the explicit divisors g_i^2 of the action. The finitely many ratios of couplings are fixed powers of M. By Theorem 11.2, the minimizers, the constrained Hessians, their square roots and inverses, the Haar disintegration densities, and the reconstructions converge with every fixed derivative on this one product of inputs. The formulas of Theorem 12.13 are finite compositions of these data, with factors $g_i^{-1} \le (\lambda_i t_-)^{-1}$ and bounded p and p^{-1}. Thus their substitutions converge uniformly in $\gamma \in I$, including every fixed derivative in the coarse field of the classical and Haar disintegration exponent, with all dependences of the determinant terms on coarser fluctuation variables. This proves the last assertion of (1).

Step 3: the input effective interaction. We use the same gauge transformation Ω_d of Theorem 13.2 on the two reconstructions being compared; its derivatives in the coarse field and its inverse are bounded on the common product domain. One image lies in $B_{\rho/2}$, and the uniform convergence of the reconstructions puts the other in $B_{3\rho/4}$ for all sufficiently large pairs of depths, uniformly over the product of Step 1 and over $\gamma \in I$. By Lemma 15.5, every $R \in \mathcal{A}_E$ satisfies

$$\sup_{B_{3\rho/4}} |D^s R| \le C_{s,\Lambda} \rho^{2-s} E. \tag{15.45}$$

Using the chain rule and the integral of DR along the segment between the two images, we find that every fixed derivative of the term with the input effective interaction converges, uniformly in R and in γ. Gauge invariance within B_ρ identifies these evaluations with the values of R_{sat}, and the coordinatewise identity argument of Step 4 in the proof of Theorem 15.6 identifies the holomorphic extensions. No convergence of Ω_d is needed.

Step 4: the partition functions. The integrands on Ω_1 are now uniformly Cauchy, together with each fixed derivative in the coarse field. On the finite set $\mathcal{Q}_\Lambda$ of boxes of the lattice, let $\boldsymbol{U}_{d,t,R}$ be the local interaction decomposition of Lemma 15.8, after the zero-fluctuation terms and the Gaussian factor have been removed. Its reference measure is the product probability

$$d\pi_p^{\mathrm{unit}}(v) = (Z_p^{\mathrm{unit}})^{-1} e^{-p^2|v|^2/2} \mathbf{1}_{\Omega_1}(v)\, dv.$$

The estimates of Lemma 15.8 for the supports, the support-counting factors, and the radii give $\|\boldsymbol{U}_{d,t,R}\|_{\mathscr{S}_8} \le \eta_0/2$ on the same enlarged product of coarse fields, uniformly in $\gamma \in I$, d, and R. These smallness conditions hold throughout I by hypothesis; the bounds involve only the fixed domains, their radius ratios, and bounded p, t, t^{-1}, and E. Theorem 12.7 applies to this family of coefficients in the coarse field, allows scalar

coefficients, and has constants independent of the reference probability and of the number of fluctuation variables. It gives a family $\boldsymbol{\mathcal{L}}$ with

$$-\log \Xi_{d,t,R} = \sum_{Y} \mathcal{L}_Y, \qquad \|\boldsymbol{\mathcal{L}}\|_{\mathscr{S}_2} \le C \|\boldsymbol{U}_{d,t,R}\|_{\mathscr{S}_8}.$$

Every support is nonempty, so on the complex product

$$\Big|\sum_{Y} \mathcal{L}_Y\Big| \le \sum_{Y} \sum_{\nu} \rho^{|\nu|} |\mathcal{L}_{Y,\nu}| \le |\mathcal{Q}_\Lambda| \|\boldsymbol{\mathcal{L}}\|_{\mathscr{S}_2} \le C |\mathcal{Q}_\Lambda| \eta_0/2.$$

The branch of the logarithm vanishes at zero interaction and agrees with the real logarithm on real data. This proves that the normalized complex partition functions do not vanish and satisfy

$$|\log \Xi_{d,t,R}| \le C_{\Lambda,I,E}, \qquad |\Xi_{d,t,R}| \ge e^{-C_{\Lambda,I,E}}. \tag{15.46}$$

Comparing the ratio of two close partition functions on a disc about one, we find that their logarithms normalized at the identity, and their fixed coefficient tensors, are uniformly Cauchy. The uniform lower bound (15.46) is not inferred from positivity of the normalizers at real arguments.

Step 5: the transformations. Fix $\rho < \rho_1 < 2\rho$ so close to ρ that the enlarged product with coarse-field radius ρ_1 still lies in the larger analytic product domain. Using the same composition bounds and bounds for local interaction decompositions, followed by the conversion used for (15.17), we obtain

$$\sup_{\gamma \in I,\, d,\, R \in \mathcal{A}_E} \mathcal{B}_{\rho_1, a_{\mathrm{sp}}}\big(\widehat{\mathcal{M}}_{d,t} R\big) \le C_{\Lambda,I,E}. \tag{15.47}$$

Indeed, the argument of the input effective interaction stays below its input radius ρ, the fixed connected box sets still satisfy $a_{\mathrm{sp}} t \le 2|Y|$, and the conversion factor is $9\rho_1^{-2}$; none of these facts depends on t. The finitely many small depths are included in the bound, on their own product domains. We split the difference of two outputs at a Taylor degree h. Its tail is at most $2C_{\Lambda,I,E}(\rho/\rho_1)^{h-1}$, by Lemma 15.5. We choose h first, and then d, d' so large that the finitely many remaining tensors are close, by Step 4. This proves (15.43) in the coefficient norm at the unchanged ρ and a_{sp}.

At a finite depth d the rescaled integrand and Z_p^{unit} depend continuously on $t > 0$. The same logarithm and tail argument proves continuity in the norm for fixed R. The contraction of Theorem 15.6, applied at a fixed parameter to two inputs, implies continuity in R. Its conditions on t, $p(t)$, and E hold for every $\gamma \in I$ by hypothesis, so the factor ϑ is unchanged. This proves (1).

Step 6: effective interactions and stopped measures. For fixed j and m we take the supremum over I in (15.18). By (1),

$$\sup_{\gamma\in I} \mathcal{B}_{\rho,a_{\mathrm{sp}}}(R_j^k - R_j^{k'}) \le 2E\vartheta^m + \sum_{h=j+1}^{j+m} \vartheta^{h-j-1}\varepsilon_{h,I}(k,k'), \qquad \varepsilon_{h,I}(k,k') \longrightarrow 0, \tag{15.48}$$

where $\varepsilon_{h,I}$ is the supremum over $\gamma \in I$ of the quantity ε_h of (15.18). We choose m first and then k, k'. By the completeness of the coefficient space, proved in Step 6 of the proof of Theorem 15.6, we obtain the uniform limit $R_j^\infty(\gamma) \in \mathcal{A}_E$. Each finite effective interaction is continuous in γ, since it is a finite composition of transformations that are continuous by (1), starting from the zero interaction on the finest lattice. The uniform limit is then continuous as well.

On the fixed domain D_r at level j, the difference of the classical terms in (15.19) is divided by $g_j^2 \ge \gamma_-/n_j$. Using the value bound of Lemma 15.5, we obtain

$$\delta_{j,k,I} = \sup_{\gamma\in I}\sup_{D_r}\Big|\frac{E_{J(k-j),\mathrm{sat}} - E_{\infty,\mathrm{sat}}}{g_j^2} + R_{j,\mathrm{sat}}^k - R_{j,\mathrm{sat}}^\infty\Big| \longrightarrow 0. \tag{15.49}$$

At this fixed volume all these exponents are bounded uniformly in $\gamma \in I$. This common domain contains an open neighborhood of the identity of positive Haar measure, so the normalizers have a positive lower bound uniform in γ. By Lemma 15.1(1), the total variation distance is at most $e^{2\delta_{j,k,I}} - 1$, which proves the uniform convergence of the stopped measures. The same bounds and the fixed domain prove continuity in γ in total variation, both at finite k and in the limit. In particular, with the fixed Borel map S_0 of Lemmas 15.10 and 15.11, the coarse-level laws η_0^k and η_0^∞ converge in total variation uniformly in γ and depend continuously on γ. No continuity of S_0 is needed. This proves (2).

Step 7: kernels and finite-scale reconstructions. For the potential of Theorem 15.9, with all its terms, we write

$$\widetilde{V}_j^k(\gamma,\theta,v) = V_j^k(\gamma,\theta,p_j(\gamma)v), \qquad v \in \Omega_1.$$

By Steps 2 and 3 and (15.49), we have

$$\Delta_{j,k,I} = \sup_{\gamma,\theta,v}|\widetilde{V}_j^k - \widetilde{V}_j^\infty| \longrightarrow 0, \tag{15.50}$$

where all suprema are over closed smaller product domains. Each finite potential is jointly continuous in (γ,θ,v), and so is the uniform limit, which is bounded there. The integral of $e^{-\widetilde{V}_j^\infty}$ over Ω_1 is at least $|\Omega_1|\exp(-\sup|\widetilde{V}_j^\infty|) > 0$. Thus the limiting normalized kernel on the unit product is jointly continuous in (γ,θ) in total variation, with one modulus

$\omega_{j,I}$ in the field. By Lemma 15.1(1), the finite kernel differs from it in total variation by at most $e^{2\Delta_{j,k,I}} - 1$, uniformly in the field and in γ.

The thresholds $d_0(\Lambda_j)$ of Theorem 15.12 do not depend on the coupling. For a fixed l, all gauge transformations are therefore the fixed limiting ones once k is large. After the substitution $q_i = g_{j,i}(\gamma)p_j(\gamma)v_i$, the reconstruction maps converge uniformly on the common compact product of γ and the coordinates, and their first derivatives in the coarse field are uniformly bounded. Let F_j^k be the finite-scale reconstruction, and let $u_{j,k,I} = \sup_{\gamma,h} |F_j^k(\gamma, h) - F_j^\infty(\gamma, h)|$. Then

$$u_{0,k,I} = 0, \qquad u_{j,k,I} \le v_{j,k,I} + C_{j,I} u_{j-1,k,I}, \qquad v_{j,k,I} \longrightarrow 0, \tag{15.51}$$

where $v_{j,k,I}$ is the uniform distance between the maps and $C_{j,I}$ bounds the first derivative of the limiting map. Since there are finitely many $j \le l$, all $u_{j,k,I}$ tend to zero. Evaluated along the reconstruction, the kernels differ by at most $e^{2\Delta_{j,k,I}} - 1 + \omega_{j,I}(u_{j-1,k,I})$. Using the disintegration of Lemma 15.10 and Lemma 15.1(2), we obtain

$$\sup_{\gamma \in I} \|P_l^k(\gamma) - P_l^\infty(\gamma)\|_{\mathrm{TV}} \le \sup_{\gamma \in I} \|\eta_0^k - \eta_0^\infty\|_{\mathrm{TV}} + \sum_{j=1}^{l} \left\{ e^{2\Delta_{j,k,I}} - 1 + \omega_{j,I}(u_{j-1,k,I}) \right\} \longrightarrow 0. \tag{15.52}$$

This bound accounts for the whole finite-scale reconstruction, including the dependence of later fields on earlier fluctuation variables. The logarithmic fields have a common compact range, by the amplitude bounds. Lemma 15.1(3), as in (15.32), implies uniform convergence in bounded-Lipschitz distance, and its coupling bound gives uniform convergence in W_1, with error at most $\|F^k - F^\infty\|_\infty + D_l \|P_l^k - P_l^\infty\|_{\mathrm{TV}}$. The continuity in γ of the coarse-level laws and of the kernels in total variation, and the continuity of the limiting reconstruction maps, prove the continuity of the limiting laws. Compatibility is the disintegration identity. This proves (3). □

PART V

Heat estimates and backward reconstruction

SECTION 16

Heat-kernel estimates, interpolation, and concentration

This section proves the analytic and probabilistic estimates through which the conditional fluctuation kernels of Sections 13 and 15 control the reconstructed field in norms of negative regularity. The main result is Theorem 16.6. It concerns one renormalization group step, from a coarse field c to the lattice of mesh a, and the lattice potential $A = a^{-1}\Phi$, where Φ is the logarithm of the reconstructed fine field in a gauge that depends only on c. Conditionally on c, the fluctuation $F = A - \mathbb{E}[A \mid c]$ of A about its conditional expectation satisfies, for $0 < \alpha < 3/2$ and every $p \geq 1$,

$$\Big(\mathbb{E}\Big[\Big(\sup_{0<s\leq T_0} s^{\alpha/2}\|P_s F\|_\infty\Big)^p \;\Big|\; c\Big]\Big)^{1/p} \leq C_p g_3 a^{\alpha-1/2}\sqrt{\log(2+L/a)},$$

where P_s is the continuum heat semigroup applied to F, viewed as a function that is constant on lattice cells. The bound is uniform in c and in the number of averaging steps that precede the step. For $\alpha > 1/2$ it tends to zero with the mesh and is summable over a geometric sequence of meshes. It controls only the fluctuation about the conditional expectation; the conditional expectation itself is estimated in Lemma 17.1.

The auxiliary results are deterministic heat and interpolation estimates and probabilistic maximal inequalities. We prove bounds for the heat kernel on the periodic lattice and on the torus (Lemma 16.1), and we construct a smooth cardinal interpolation between lattices whose meshes have an integer ratio, which preserves the integral of every column exactly (Lemma 16.3). We also prove an elementary chaining lemma (Lemma 16.4), estimates for time convolutions (Lemma 16.8), and the convergence of the discrete heat evolution of a fixed bounded finite-dimensional family of smooth fields to the continuum heat evolution (Lemma 16.9). All heat estimates are stated on a fixed finite time interval and keep the constant mode.

The scale in the main estimate. The factor $g_3 a^{\alpha-1/2}$ has a simple origin. Let $g^2 = g_3^2 a$ be the coupling at mesh a. The Jacobian of the map from the normalized fluctuation variables w to the lattice potential A has counting ℓ^2 operator norm at most $C_J g/a = C_J g_3 a^{-1/2}$. At a time $s \geq a^2$, the heat kernel averages over about $(\sqrt{s}/a)^3$ cells, and a linear function of w with this averaging has Lipschitz constant smaller by the factor $a^{3/2}s^{-3/4}$. When $\alpha < 3/2$, the product $s^{\alpha/2}a^{3/2}s^{-3/4}$ is largest at $s \asymp a^2$, where it equals a^α. By the concentration

inequality for the conditional fluctuation kernel, the fluctuations at each point have size $g_3 a^{\alpha-1/2}$, and the maximum over space and time costs the factor $\sqrt{\log(2+L/a)}$.

Three linear operations. Three linear operations on lattice arrays occur in this and the following sections.

Operation	Input and output	Use
Continuum heat semigroup P_s applied to the cell embedding	An array on the lattice of mesh a, extended as a constant on each half-open cell; a smooth function on $\mathbb{T}_L^3$	Negative-regularity norms of reconstructed fields (Theorem 16.6, Section 17)
Discrete heat semigroup P_t^a	An array on the lattice of mesh a; an array on the same lattice	Heat bounds for the lattice flow (Lemmas 16.1 and 16.9)
Cardinal interpolation $I_{a_0 \to a}$	An array on the lattice of mesh a_0; an array on the finer lattice of mesh $a = a_0/m$	Comparison of fields from different scales on a common lattice; values and difference arrays are interpolated separately (Lemmas 16.3 and 16.9)

Two points that need care. First, the conditional fluctuation kernel is supported on a product of balls, but it is not a product measure: the normalized fluctuation variables interact, and they are not Gaussian. We use only that its potential V satisfies $D_w^2 V \geq I/2$ on the convex support. This implies a logarithmic Sobolev inequality and Gaussian concentration for Lipschitz functions, with constants independent of the dimension. Second, the column sums of the reconstruction Jacobian must be normalized by the cell volume of the lattice on which each fluctuation variable lives. A uniform bound on the row sums alone does not imply the ℓ^2 operator bound used in the concentration argument, and the column bound with the correct input volume produces the factor $a^{3/2}$ above.

Inputs, uses, and attribution. The conventions for meshes, heat semigroups, analytic norms, and flow norms are those of Section 4. The probabilistic input is the concentration theorem on convex products of balls, Theorem 5.4. It rests on the variance inequality of Brascamp and Lieb [33], the logarithmic Sobolev inequality of Gross [103], and the curvature criterion of Bakry and Émery [8], with the boundary of the product treated in Section 5. The conditional fluctuation kernel and the bounds on the reconstruction and its derivatives come from Theorems 13.2 and 13.8; the reconstruction maps and the preservation of the kernels under them come from Theorems 15.12 and 15.13. The concentration theorem is used in Section 17; the heat bounds, the interpolation, and

the chaining lemma are used in Sections 17, 24, 26, 27, 29, 31, and 34; the convolution estimates in Section 24; and the consistency lemma in Section 25.

The heat-kernel bounds are standard; we prove them by Fourier analysis because the normalizations matter. The maximal inequalities use chaining in the sense of Dudley [79]. For general tail bounds by generic chaining, see Dirksen [74, Theorems 3.2 and 3.5]; the second of these theorems treats increments with mixed sub-Gaussian and exponential tails. Those results are stated for semimetrics satisfying the triangle inequality. Our version assumes only the covering and continuity properties that we use, and we include its short proof. Only the verification that the conditional fluctuation kernel and the reconstruction satisfy the hypotheses of these general estimates, uniformly in the coarse field and in the depth, depends on the construction.

Notation. Throughout this section, $L > 0$ is the side length of the torus $\mathbb{T}_L^3$. For a mesh $a = L/n$ with an integer $n \geq 4$, we write $\Gamma_a = (a\mathbb{Z}/L\mathbb{Z})^3$ for the vertex set of Λ_n in physical units, and we use the measure $a^3 \sum_{x \in \Gamma_a}$ in all norms L_a^r. Arrays on positive links are identified with arrays on vertices by attaching the link $(x, x + ae_i)$ to its starting vertex x; the direction i and the color form a fixed finite set of components, and all operators below act componentwise. For a direction i, we write $D_{a,i}^+ f(x) = (f(x + ae_i) - f(x))/a$ and $D_{a,i}^- f(x) = (f(x) - f(x - ae_i))/a$ for the physical forward and backward differences, $\Delta_a = \sum_i D_{a,i}^- D_{a,i}^+$, and $P_t^a = e^{t\Delta_a}$; this semigroup keeps the constant mode. For an integer $q \geq 0$, we write D_a^q for any ordered product of q physical forward or backward differences (the identity if $q = 0$). When the particular product matters, we write D_a^ϵ, and ∂^ϵ for the corresponding product of partial derivatives; bounds stated for D_a^q hold uniformly over the finitely many products of order q. The continuum heat semigroup is $P_t = e^{t\Delta}$ on $\mathbb{T}_L^3$, with kernel $p_t(x - y)$. The cell embedding of an array f on Γ_a is the function on $\mathbb{T}_L^3$ equal to f_y on the half-open cell $Q_y = y + [0, a)^3$. In this section, the letter q denotes an order of differentiation or an exponent; it is unrelated to the plaquette energy. Constants C may change from line to line, and depend only on the parameters listed in the statement being proved.

16.1 Heat-kernel bounds

The following lemma collects the bounds on the discrete and continuum heat kernels used in the rest of the paper. Its last assertion is a maximal inequality for sub-Gaussian processes. We state it here because it is applied to processes built from the heat kernel; it is a special case of Lemma 16.4. For a set I, a nonnegative symmetric function d on $I \times I$, and $v > 0$, we write $N(I, d, v)$ for the smallest cardinality of a set $\mathcal{N} \subset I$ such that every $i \in I$ has some $i' \in \mathcal{N}$ with $d(i, i') \leq v$.

Lemma 16.1. *Fix $L, T > 0$. Let $a = L/n$ with an integer $n \geq 4$, and let $p_t^a(x,y)$ be the density of P_t^a with respect to the measure $a^3 \sum_y$. Then $p_0^a(x,y) = a^{-3}\mathbf{1}_{x=y}$, $p_t^a \geq 0$, and $a^3 \sum_y p_t^a(x,y) = 1$. For every integer $q \geq 0$, every product D_a^ϵ of q physical forward or backward differences acting on the variable y, every $x \in \Gamma_a$, and every $0 \leq t \leq T$,*

$$\begin{aligned} \|D_a^\epsilon p_t^a(x,\cdot)\|_{L_a^1} &\leq C_q (t + a^2)^{-q/2}, \\ \|D_a^\epsilon p_t^a(x,\cdot)\|_{L_a^2} &\leq C_q (t + a^2)^{-3/4 - q/2}, \\ \|D_a^\epsilon p_t^a(x,\cdot)\|_{\infty} &\leq C_q (t + a^2)^{-(3+q)/2}. \end{aligned} \tag{16.1}$$

The continuum heat kernel p_t is smooth for $t > 0$ and satisfies the same three bounds with $a = 0$, for $0 < t \leq T$, with D_a^ϵ replaced by a product of q partial derivatives. The constants depend only on q, the fixed sets of directions and colors, and T/L^2.

For $y \in \Gamma_a$, $0 < t \leq T$, and $x \in \mathbb{T}_L^3$ let

$$\omega_y(t,x) = \int_{Q_y} p_t(x - z)\, dz.$$

Then

$$\|\omega(t,x)\|_{\ell^2} \leq C \min\{1, a^{3/2} t^{-3/4}\} \qquad (0 < t \leq T),$$

and

$$\|\omega(t,x) - \omega(t,x')\|_{\ell^2} \leq C a^{3/2} t^{-3/4} \min\{1, \operatorname{dist}(x,x')/\sqrt{t}\} \qquad (a^2 \leq t \leq T).$$

More generally, for every multi-index ν and every integer $r \geq 0$, the array of cell integrals of $\partial^\nu \partial_t^r p_t(x - \cdot)$ has ℓ^2 norm at most $C_{\nu,r} a^{3/2} t^{-3/4 - |\nu|/2 - r}$ for $0 < t \leq T$. Thus every additional spatial derivative costs a factor $t^{-1/2}$ and every time derivative a factor t^{-1}, and the corresponding increment bounds follow; for example, with $v = \min\{s,t\}$,

$$\|\omega(t,x) - \omega(s,x)\|_{\ell^2} \leq C a^{3/2} v^{-3/4} \min\{1, |t-s|/v\} \qquad (0 < s, t \leq T).$$

For $0 < t \leq a^2$ we use instead that $\|P_t f\|_\infty \leq \max_y |f_y|$ for the cell embedding of an array f.

Finally, let $(X_i)_{i \in I}$ be a centered real process and let d be a nonnegative symmetric function on $I \times I$ with $\sup d \leq \sigma$. Suppose that

$$\mathbb{E} e^{z X_i} \leq e^{z^2 \sigma^2/2}, \qquad \mathbb{E} e^{z(X_i - X_j)} \leq e^{z^2 d(i,j)^2/2} \qquad (z \in \mathbb{R},\ i, j \in I),$$

that $N(I, d, \eta\sigma) \leq A \eta^{-v}$ for every $0 < \eta < 1$, where $A \geq 1$ and $v \geq 0$, and that the process has a version which is continuous with respect to d and whose supremum equals its supremum over a countable d-dense subset of I (this holds automatically if I is finite). Then there is a universal constant C such that, for $u \geq 0$,

$$\mathbb{P}\Big\{ \sup_{i \in I} |X_i| > C\sigma \big[\sqrt{\log(2A)} + \sqrt{v} + \sqrt{u} \big] \Big\} \leq 2 e^{-u},$$

$$\mathbb{E} \sup_{i \in I} |X_i| \leq C\sigma \big[\sqrt{\log(2A)} + \sqrt{v} \big].$$

No Gaussian or independence assumption is made.

Remark 16.2. The finite horizon T cannot be removed, because the semigroups keep the constant mode. For instance, as $t \to \infty$ the density $p_t^q(x, \cdot)$ tends to the constant L^{-3}, whose L_a^2 norm is $L^{-3/2}$, while $(t + a^2)^{-3/4} \to 0$. In the same way, for $E_\theta = I$ and $a = a_0$ in Lemma 16.3 below, the undifferentiated row of W_t tends to the constant row n^{-3}, whose counting ℓ^2 norm is $n^{-3/2}$. The bound $Ca_0^{3/2}(t + a_0^2)^{-3/4}$ therefore cannot hold with a fixed constant on an unbounded time interval. For this reason, all row and covering estimates in this section are stated for times in $[0, T]$, and their constants depend on T/L^2.

Proof of Lemma 16.1. We use Euclidean norms on the fixed spaces of directions and colors; changing between equivalent norms costs a fixed factor. All heat times below lie in $[0, T]$, and they are positive for the continuum kernel and the cell integrals.

The discrete kernel and its L^1 bounds. We begin with unit spacing on $\mathbb{Z}$. Let S_+ and S_- be the two shifts. The kernel r_s of

$$e^{s(S_+ + S_- - 2I)} = e^{-2s} \sum_{k,l \ge 0} \frac{s^{k+l}}{k!\, l!} S_+^k S_-^l$$

is nonnegative and has sum one; the series converges in operator norm on bounded arrays. Its Fourier multiplier on $[-\pi, \pi]$ is $e^{-4s \sin^2(\theta/2)}$. Let ∇^ϵ be a product of r unit forward or backward differences in this coordinate, and set $h_s = \nabla^\epsilon r_s$. Up to a sign, the multiplier of h_s is

$$m_s(\theta) = e^{-il\theta}(e^{i\theta} - 1)^r e^{-4s \sin^2(\theta/2)}, \qquad 0 \le l \le r,$$

where l is the number of backward differences. For $s \le 1$, we have $\sum_j |h_s(j)| \le 2^r$ by the difference stencil. For $s \ge 1$, the inequalities $|e^{i\theta} - 1| \le |\theta|$ and $\sin^2(\theta/2) \ge \theta^2/\pi^2$ on $[-\pi, \pi]$ imply

$$\int_{-\pi}^{\pi} |m_s|^2 \, d\theta \le C_r s^{-r-1/2}, \qquad \int_{-\pi}^{\pi} |m_s'|^2 \, d\theta \le C_r s^{-r+1/2}. \tag{16.2}$$

Here and below, the finitely many sign patterns are included in C_r. To verify the second bound, note that m_s' is a sum of terms bounded by

$$C_r\big(|\theta|^r + \mathbf{1}_{r \ge 1}|\theta|^{r-1} + s|\theta|^{r+1}\big)e^{-cs\theta^2};$$

the first term is absent when $l = 0$, and only the last term is present when $r = 0$. Squaring, integrating, and substituting $v = \sqrt{s}\,\theta$, we obtain (16.2), since $\int_{\mathbb{R}} |v|^m e^{-cv^2} \, dv < \infty$ for every $m \ge 0$. For this finiteness, we split $[0, \infty)$ into the intervals $[k, k+1]$, $k \ge 0$; on each of them, $|v|^m e^{-cv^2} \le (k+1)^m e^{-ck^2}$, and these bounds form a summable sequence. The

negative half-line is treated by symmetry. The Fourier coefficients of m_s' are $ijh_s(j)$. By Parseval's identity, we have

$$\sum_j |h_s(j)|^2 \le C_r s^{-r-1/2}, \qquad \sum_j j^2 |h_s(j)|^2 \le C_r s^{-r+1/2}.$$

These identities may be applied first to the absolutely convergent kernel series, or obtained by approximation with trigonometric polynomials in L^2. For $s \ge 1$, we have $\sum_{j\in\mathbb{Z}}(1 + j^2/s)^{-1} \le C\sqrt{s}$; to see this, we separate $|j| \le \sqrt{s}$ from $|j| > \sqrt{s}$ and use $s\sum_{j>\sqrt{s}} j^{-2} \le C\sqrt{s}$. Then the weighted Cauchy–Schwarz inequality implies

$$\sum_j |h_s(j)| \le \Big(\sum_j (1 + j^2/s)^{-1}\Big)^{1/2} \Big(\sum_j (1 + j^2/s)|h_s(j)|^2\Big)^{1/2} \le C_r s^{-r/2}. \tag{16.3}$$

The kernel on $\mathbb{Z}^3$ is the product of three copies of r_s. A product of q differences puts r_i differences on coordinate i, with $r_1 + r_2 + r_3 = q$, and the absolute counting L^1 norm of the differentiated kernel is the product of the three one-dimensional norms. By (16.3), it is at most $C_q(1 + s)^{-q/2}$ for every $s \ge 0$. Periodizing with period n does not increase this norm, by the triangle inequality, and it preserves positivity and mass. At spacing a, the time is $s = t/a^2$, the density carries the factor a^{-3}, and a product of q physical differences carries the factor a^{-q}. We obtain

$$a^3 \sum_y |D_a^\epsilon p_t^a(x, y)| \le C_q a^{-q}(1 + t/a^2)^{-q/2} = C_q(t + a^2)^{-q/2}.$$

The periodized kernel solves the finite heat equation with initial density $a^{-3}\mathbf{1}_{x=y}$, and by uniqueness for this finite linear system it equals p_t^a. This proves the L^1 bound and the assertions on positivity and mass, including at $t = 0$.

The L^2 and supremum bounds. Fix one representative of each $k \in (\mathbb{Z}/n\mathbb{Z})^3$ with $|k_i| \le n/2$. Then

$$p_t^a(x, y) = L^{-3} \sum_k e^{-t\lambda_a(k)} e^{2\pi i k\cdot(x-y)/L}, \qquad \lambda_a(k) = 4a^{-2} \sum_i \sin^2(\pi k_i/n). \tag{16.4}$$

Indeed, these exponentials are eigenfunctions of Δ_a, and their orthogonality on Γ_a yields the required initial density at $t = 0$. For these representatives,

$$c|k|^2/L^2 \le \lambda_a(k) \le C|k|^2/L^2,$$

and the multiplier of D_a^ϵ has absolute value at most $C_q(|k|/L)^q$. If $0 \le t \le a^2$, we obtain the supremum bound $2^q a^{-3-q}$ and the L^2_a bound $2^q a^{-3/2-q}$ from the difference stencil and the mass identity; since $t + a^2 \le 2a^2$, these are the asserted bounds. If $t \ge a^2$, we extend the sum over nonzero frequencies to $\mathbb{Z}^3 \setminus \{0\}$. For every $r \ge 0$ and $c > 0$,

$$L^{-3} \sum_{k\ne 0} (|k|/L)^r e^{-ct|k|^2/L^2} \le C_{r,c} t^{-(3+r)/2}. \tag{16.5}$$

To check this estimate, we set $u = t/L^2$ and group the frequencies with $j \le |k| < j+1$. There are at most $C(j+1)^2$ of them, since the disjoint unit cubes centered at these points lie in an annulus of fixed extra width. For $u \le 1$, the sum of $(j+1)^{r+2}e^{-cuj^2}$ over $j \ge 1$ is at most $C_{r,c}u^{-(r+3)/2}$: on the interval $[j, j+1]$, we may replace j^2 by $x^2/4$ and $(j+1)^{r+2}$ by $C_r x^{r+2}$, and then integrate. For $u \ge 1$, we have $e^{-cuj^2} \le e^{-cu/2}e^{-cj^2/2}$ for $j \ge 1$, and $e^{-cu/2} \le C_{r,c}u^{-(r+3)/2}$. Rescaling gives (16.5).

The triangle inequality in (16.4), together with (16.5) for $r = q$, implies the supremum bound. By orthogonality on Γ_a, the square of the L^2_a norm equals L^{-3} times the sum over k of the squared multipliers times $e^{-2t\lambda_a(k)}$, and (16.5) with $r = 2q$ bounds it. For $q = 0$, the frequency $k = 0$ contributes L^{-3} both to the supremum and to the square of the L^2_a norm. On the horizon, we have $L^{-3} \le (T/L^2)^{3/2}t^{-3/2}$, and this contribution is at most $C(T/L^2)t^{-3/2}$ in both cases; every difference of positive order annihilates it. Since $t + a^2 \le 2t$ in this range, these are the asserted bounds, with constants depending only on q and T/L^2.

The continuum kernel and the cell integrals. The continuum kernel is the periodization of

$$G_t(z) = (4\pi t)^{-3/2}e^{-|z|^2/(4t)},$$

which has mass one. For each multi-index ν, the derivative $\partial^\nu G_t$ equals $t^{-(3+|\nu|)/2}$ times a fixed polynomial in $z/\sqrt{t}$ times $e^{-|z|^2/(4t)}$. A change of variables yields $\|\partial^\nu G_t\|_1 \le C_\nu t^{-|\nu|/2}$, and periodization does not increase the absolute L^1 norm. The periodic Fourier coefficients are $L^{-3}e^{-4\pi^2 t|k|^2/L^2}$, obtained by integrating the periodized Gaussian over one period. The other two bounds follow from (16.5), with the term $k = 0$ treated as before. For $t > 0$, all these series and their derivatives converge absolutely; multiplying Fourier coefficients then verifies the heat equation and the semigroup property.

By positivity and mass, we have $\sum_y \omega_y(t,x) = 1$, and hence $\|\omega(t,x)\|_2 \le 1$. Using the Cauchy–Schwarz inequality on each cell, we find

$$\sum_y \Big| \int_{Q_y} \partial^\nu \partial_t^r p_t(x-z)\, dz \Big|^2 \le a^3 \|\partial^\nu \Delta^r p_t\|_2^2 \le C_{\nu,r} a^3 t^{-3/2-|\nu|-2r}, \tag{16.6}$$

where we used the heat equation $\partial_t p_t = \Delta p_t$ and, in the last inequality, the continuum L^2 bound with $q = |\nu| + 2r$. The case $\nu = 0$, $r = 0$ is the second bound on $\|\omega(t,x)\|_2$. Integrating one spatial derivative along a shortest segment in $\mathbb{T}^3_L$ from x to x', we bound the difference of the two cell arrays by $Ca^{3/2}t^{-5/4}\operatorname{dist}(x,x')$. The value bound, used twice, bounds it by $Ca^{3/2}t^{-3/4}$, and the minimum of the two is the increment bound. In the same way, (16.6) controls every fixed spatial and time derivative. For the time increment, we integrate the time derivative over $[v, v + |t-s|]$ when $|t-s| \le v$, and we use the two value bounds otherwise. The same reasoning, applied to a differentiated array of cell integrals, yields its increment bounds, with the additional factors displayed

in (16.6). Finally, for $t \le a^2$, positivity and unit mass imply $\|P_t f\|_\infty \le \max_y |f_y|$ for every cell embedding f; this needs no regularity across cell boundaries.

The chaining assertion is a special case of Lemma 16.4; we deduce it at the end of Section 16.3. □

16.2 Cardinal interpolation and volume-normalized columns

We next define the cardinal interpolation used to compare arrays on different lattices.

Fix an even nonnegative function $b \in C_c^\infty([-1,1])$ that is positive on $(-1,1)$; all its derivatives vanish at ± 1. Set

$$\zeta(v) = \frac{b(v)}{\sum_{m\in\mathbb{Z}} b(v-m)} \quad (v \in \mathbb{R}), \qquad \zeta_3(u) = \prod_{i=1}^{3} \zeta(u_i) \quad (u \in \mathbb{R}^3).$$

Let $a_0 = L/n$ with $n \ge 4$, let $m \ge 1$ be an integer, and let $a = a_0/m$. For $x \in \Gamma_{a_0}$ let

$$\psi_x(z) = \sum_{l\in\mathbb{Z}^3} \zeta_3((z - x + Ll)/a_0) \qquad (z \in \mathbb{T}_L^3)$$

be the periodization of the translated function $\zeta_3((\cdot - x)/a_0)$. The cardinal interpolation of an array f on Γ_{a_0} is the array on Γ_a defined componentwise by

$$(I_{a_0\to a} f)_i(y) = \sum_{x\in\Gamma_{a_0}} \psi_x(y) f_i(x) \qquad (y \in \Gamma_a), \tag{16.7}$$

where, as above, each positive link is attached to its starting vertex. The continuous interpolant is $I_{a_0} f(z) = \sum_x \psi_x(z) f(x)$ for $z \in \mathbb{T}_L^3$. In (16.7) we also write $\zeta_3((y-x)/a_0)$ for $\psi_x(y)$.

For a matrix K from scalar variables ξ attached to the cells of the lattice of mesh a_0 to arrays on the lattice of mesh a, we use the row sums $\sup_y \sum_\xi |K(y,\xi)|$ and the volume-normalized column sums $\sup_\xi (a^3/a_0^3) \sum_y |K(y,\xi)|$, as in Definition 4.6. Throughout, a column is normalized by the cell volume a_0^3 of its input.

Lemma 16.3. *Fix $L, T > 0$ and the function b. Let $a_0 = L/n$ with $n \ge 4$, let $m \ge 1$ be an integer, and let $a = a_0/m$. All times below lie in $[0,T]$.*

1. *The map $I_{a_0\to a}$ is local within distance $\sqrt{3}a_0$, commutes with constant color rotations, preserves constants, satisfies $(I_{a_0\to a} f)(x) = f(x)$ for $x \in \Gamma_{a_0}$, and $I_{a\to a}$ is the identity. For every integer $q \ge 0$,*

$$\|D_a^q I_{a_0\to a} f\|_\infty \le C_q a_0^{-q} \|f\|_\infty, \qquad a^3 \sum_y \psi_x(y) = a_0^3,$$

$$a^3 \sum_y \operatorname{dist}(y,x) \psi_x(y) \le C a_0^4, \qquad \|D_a^q P_t^a I_{a_0\to a} f\|_\infty \le C_q (t + a_0^2)^{-q/2} \|f\|_\infty.$$

The continuous interpolant $I_{a_0} f$ satisfies the corresponding derivative bounds, and $\int_{\mathbb{T}^3_L} \psi_x(z)\, dz = a_0^3$.

2. *Let E_θ be a deterministic matrix, depending on a parameter θ, from scalar variables ξ attached to the cells of the lattice of mesh a_0, at most v_J of them per cell, to arrays on that lattice. Suppose that, uniformly in θ, its row sums and volume-normalized column sums are at most C_J, that is,*

$$\sup_x \sum_\xi |E_\theta(x,\xi)| \le C_J, \qquad \sup_\xi a_0^3 \sum_x |E_\theta(x,\xi)| \le C_J a_0^3, \tag{16.8}$$

and that its columns decay exponentially at scale a_0, with fixed rate and amplitude. Set $W_t = P_t^a I_{a_0 \to a} E_\theta$. Then for every integer $q \ge 0$,

$$\sup_{x,\theta} \|D_a^q W_t(x,\cdot)\|_{\ell^2} \le C_{J,q} a_0^{3/2} (t + a_0^2)^{-3/4 - q/2},$$
$$\|D_a^q W_t\|_{\ell^2 \to L^2_a} \le C_{J,q} a_0^{3/2} (t + a_0^2)^{-q/2},$$

where ℓ^2 refers to the variables ξ. For $q \in \{0,1\}$, put $A_q(t) = a_0^{3/2}(t+a_0^2)^{-3/4-q/2}$ and $v = \min\{s,t\}$. Then

$$\|D_a^q W_t(x,\cdot) - D_a^q W_s(x,\cdot)\|_{\ell^2} \le C_{J,q} A_q(v) \min\{1, |t-s|/(v+a_0^2)\},$$
$$\|D_a^q W_t(x,\cdot) - D_a^q W_t(y,\cdot)\|_{\ell^2} \le C_{J,q} A_q(t) \min\{1, \mathrm{dist}(x,y)/\sqrt{t+a_0^2}\}.$$

At fixed θ, for $q \in \{0,1\}$ and $0 < \eta < 1$, the set $\Gamma_a \times [0,T]$ with the distance $a_0^q \|D_a^q W_t(x,\cdot) - D_a^q W_s(y,\cdot)\|_{\ell^2}$ between (x,t) and (y,s) has an η-net with at most

$$C_{J,q} (1 + L/(\eta a_0))^3 (1 + T/(\eta a_0^2))$$

points. For the normalized rows $D_a^q W_t(x,\cdot)/A_q(t)$ the same holds, up to a factor which is a fixed power of $1 + T/a_0^2$.

3. *Let u be an array on the lattice of mesh a_1 with $a_1/a \in \mathbb{N}$. Then, componentwise,*

$$a^3 \sum_{y \in \Gamma_a} I_{a_1 \to a} u(y) = a_1^3 \sum_{x \in \Gamma_{a_1}} u(x).$$

Hence, if arrays originating on lattices of different meshes are interpolated separately from their own lattices to a common finer lattice before they are added or subtracted, every componentwise integral is preserved exactly; in particular every prescribed zero integral is preserved.

All the estimates are uniform in the ratio $m = a_0/a$, in n, and in θ, at fixed J and fixed coefficient constants. The constants depend only on b, q, T/L^2, v_J, C_J, and the decay constants of the columns. The hypotheses of the second part hold, with a constant C_J depending only on J, the fixed construction domains, and the order of differentiation, for the normalized Jacobian of the reconstruction and for each of its mixed derivatives of fixed order in the coarse field and the fluctuation variables, uniformly in the coarse field and in the depth; see Lemma 16.5.

Proof. All matrices in this proof are written in counting coordinates, and all times lie in $[0, T]$; we apply Lemma 16.1 with the same L and T. The parameter θ is fixed during the construction of each net, and the constants are uniform in θ; the covering assertion is not an entropy estimate for the family indexed by θ.

The basis functions and their masses. Let $d_b(u) = \sum_{m\in\mathbb{Z}} b(u-m)$. Since only finitely many terms are nonzero near any point, d_b is smooth and one-periodic. It is strictly positive, since some integer has distance at most $1/2$ from u, and its minimum over one period is positive. The function $\zeta = b/d_b$ is smooth, nonnegative, even, and supported in $[-1,1]$; all its derivatives vanish at ± 1, and

$$\zeta(k) = \mathbf{1}_{k=0} \quad (k \in \mathbb{Z}), \qquad \sum_{k\in\mathbb{Z}} \zeta(u-k) = 1. \tag{16.9}$$

The first identity holds because $b(k) = 0$ for integers $k \neq 0$ and $d_b(0) = b(0)$; the second follows by replacing every $d_b(u-k)$ by $d_b(u)$. The periodization in ψ_x accounts for every periodic seam. These functions sum to one as x ranges over Γ_{a_0}, they are supported within torus distance $\sqrt{3}a_0$ of x, and $\psi_x(x') = \mathbf{1}_{x=x'}$ for $x' \in \Gamma_{a_0}$. The map preserves constants and samples, and it is the identity when $a = a_0$. Since it acts by the same scalar coefficients on each direction and color, it commutes with every constant color rotation.

Since at any point of $\mathbb{T}^3_L$ at most a fixed number of the functions ψ_x are nonzero, we have

$$\|\partial^\epsilon I_{a_0} f\|_\infty \le C_q a_0^{-q} \|f\|_\infty$$

for every product ∂^ϵ of q partial derivatives. For a smooth periodic function h, each physical forward or backward difference is an average of the corresponding derivative,

$$D^{\pm}_{a,i} h(z) = \int_0^1 \partial_i h(z \pm r a e_i)\, dr. \tag{16.10}$$

Applying this identity repeatedly, we write a product of q differences as an average of the corresponding derivative of order q. The preceding bound therefore implies the bound on $D^q_a I_{a_0\to a} f$, uniformly in m.

For the mass, we write an integer l as $l = mk + r$ with $0 \le r < m$. By (16.9), we have

$$\sum_{l\in\mathbb{Z}} \zeta(l/m) = \sum_{r=0}^{m-1} \sum_{k\in\mathbb{Z}} \zeta(k + r/m) = m.$$

Taking the product over the three coordinates and regrouping the periodization, we obtain

$$a^3 \sum_{y\in\Gamma_a} \psi_x(y) = a^3 m^3 = a_0^3. \tag{16.11}$$

Since $\psi_x \ge 0$ is supported within distance $\sqrt{3}a_0$ of x, the first moment is at most $\sqrt{3}a_0^4$. Integrating the second identity in (16.9) over $[0,1]$, we find $\int_{\mathbb{R}} \zeta(u)\, du = 1$, and the

product over the coordinates gives $\int_{\mathbb{T}_L^3} \psi_x(z)\,dz = a_0^3$. These arguments also prove the assertions on the continuous interpolant.

Heat smoothing. If $t \le a_0^2$, we commute the differences with the discrete heat semigroup and use that P_t^a contracts the supremum norm, together with the bound on $D_a^q I_{a_0 \to a} f$; this gives $C_q a_0^{-q} \|f\|_\infty$. If $t \ge a_0^2$, we move the differences onto the heat kernel and use its L_a^1 bound from Lemma 16.1, together with $\|I_{a_0 \to a} f\|_\infty \le \|f\|_\infty$; since $a \le a_0$, we obtain $C_q t^{-q/2} \|f\|_\infty$. Both bounds are at most $C_q (t + a_0^2)^{-q/2} \|f\|_\infty$. The same argument, with continuous derivatives and the continuum heat kernel, proves the corresponding continuous estimate.

Row sums and volume-normalized column sums. We write ξ for the scalar input variables. The finitely many variables per cell and the color components are included in C_J. The estimates below use only (16.8), and they apply to every matrix obeying (16.8). The interpolation, whose range is $\sqrt{3} a_0$, preserves the exponential decay of the columns, but this decay is not needed for the numerical bounds.

Let $K = I_{a_0 \to a} E_\theta$. For every product of q differences,

$$\sup_y \sum_\xi |D_a^q K(y, \xi)| \le C_{J,q} a_0^{-q}, \qquad \sup_\xi a^3 \sum_y |D_a^q K(y, \xi)| \le C_{J,q} a_0^{3-q}. \tag{16.12}$$

For the first bound, we expand K over the coarse vertices. By (16.10) and the bounded overlap of the differentiated continuous basis functions, we have $\sum_x |D_a^q \psi_x(y)| \le C_q a_0^{-q}$, and the row bound in (16.8) implies the claim. For the second bound, each differentiated basis function has magnitude at most $C_q a_0^{-q}$ and support within distance $C_q a_0$ of its vertex, which contains at most $C_q (a_0/a)^3$ fine vertices. This also holds if the support wraps around the torus, by summing over its lifts. We conclude that $a^3 \sum_y |D_a^q \psi_x(y)| \le C_q a_0^{3-q}$, and summing over x with the column bound in (16.8) gives the second bound.

For $t \le a_0^2$, the heat semigroup is a contraction on absolute row sums and on volume-normalized column sums, and (16.12) holds for $D_a^q W_t$. In particular, the ℓ^2 norm of a row of $D_a^q W_t$ is at most its absolute row sum $C_{J,q} a_0^{-q} = C_{J,q} A_q(0)$. For $t \ge a_0^2$, we use the undifferentiated K in the heat convolution and move all differences to the heat kernel. The L_a^1 bound and the supremum bound of Lemma 16.1 yield, respectively,

$$\sup_x \sum_\xi |D_a^q W_t(x, \xi)| \le C_{J,q} t^{-q/2}, \qquad \sup_{x,\xi} |D_a^q W_t(x, \xi)| \le C_{J,q} a_0^3 t^{-(3+q)/2}. \tag{16.13}$$

The square of the ℓ^2 norm of a row is at most the product of these two quantities, that is, at most $C_{J,q} a_0^3 t^{-3/2-q}$. Together with the bound for $t \le a_0^2$, this proves the row bound with $t + a_0^2$ on the whole interval.

For the operator norm, we use Schur's test with the volume weights of the output lattice,

$$a^3 \sum_x \Big| \sum_\xi H(x,\xi) u_\xi \Big|^2 \le \Big(\sup_x \sum_\xi |H(x,\xi)| \Big) \Big(\sup_\xi a^3 \sum_x |H(x,\xi)| \Big) \sum_\xi |u_\xi|^2, \tag{16.14}$$

which follows from the Cauchy–Schwarz inequality in each row and an interchange of two finite sums. For $t \le a_0^2$, we use (16.12). For $t \ge a_0^2$, the first factor is bounded by (16.13), and the second is at most $C_{J,q} a_0^3 t^{-q/2}$, by the L^1_a bound on the heat kernel and the undifferentiated column bound for K. Taking the square root gives $C_{J,q} a_0^{3/2} (t + a_0^2)^{-q/2}$. The volume in this computation is the input cell volume a_0^3; refining the output lattice does not change it.

Increments and nets. Let $R_q(t,x)$ denote the row $D_a^q W_t(x,\cdot)$. The bounds already proved hold for every order of differentiation. Since $\partial_t W_t = \Delta_a W_t$, they imply

$$\|\partial_t R_q(t,x)\|_{\ell^2} \le C_{J,q} \frac{A_q(t)}{t + a_0^2}, \qquad \|D^{\pm}_{a,i} R_q(t,x)\|_{\ell^2} \le C_{J,q} \frac{A_q(t)}{\sqrt{t + a_0^2}}.$$

Since A_q is decreasing, integrating the first bound between s and t gives $C_{J,q} |t - s| A_q(v)/(v + a_0^2)$; the sum of the two row norms is at most $C_{J,q} A_q(v)$, and the minimum of the two bounds is the time increment. For the spatial increment, we join two fine vertices by a shortest periodic path along coordinate directions, of total physical length at most $\sqrt{3}\,\mathrm{dist}(x,y)$. Summing the second bound along this path gives the distance times the displayed derivative bound; the sum of the two value bounds is the alternative without the distance.

For the distance $d_q((x,t),(y,s)) = a_0^q \|R_q(t,x) - R_q(s,y)\|_{\ell^2}$, these estimates imply

$$d_q((x,t),(y,s)) \le C_{J,q} \Big(\frac{\mathrm{dist}(x,y)}{a_0} + \frac{|t-s|}{a_0^2} \Big).$$

We choose spatial points at physical separation at most $c_{J,q} \eta a_0$ and times at separation at most $c_{J,q} \eta a_0^2$. If a exceeds the proposed spatial separation, we use all vertices, whose number is then smaller than the bound below; otherwise, we round a continuum cubic covering to fine vertices, which at most doubles its radius. The resulting net has at most $C_{J,q}(1 + L/(\eta a_0))^3 (1 + T/(\eta a_0^2))$ points, independently of the number of fine vertices. For the normalized rows $R_q(t,x)/A_q(t)$, differentiation in time produces the additional term $A_q'(t)/A_q(t) = -(3/4 + q/2)/(t + a_0^2)$. The same estimates give a spatial Lipschitz bound $C_{J,q}/a_0$ and a time Lipschitz bound $C_{J,q}/a_0^2$ for the normalized rows, and the same net works. Alternatively, using only the global bounds costs powers of $a_0^{-q}/A_q(T) = (1 + T/a_0^2)^{3/4 + q/2}$.

Column integrals. Let u be an array on the lattice of mesh a_1 with $a_1/a \in \mathbb{N}$. Interchanging finite sums and using (16.11) with a_1 in place of a_0, we obtain

$$a^3 \sum_y I_{a_1 \to a} u(y) = \sum_x u(x)\, a^3 \sum_y \psi_x(y) = a_1^3 \sum_x u(x).$$

Applying this identity to each array on its own lattice before adding or subtracting arrays from different lattices preserves every signed componentwise integral exactly. This is an identity for arrays of components, not for lattice cochains or holonomies. All the constants above depend only on b, the order of differentiation, T/L^2, and the coefficient constants. □

16.3 Supremum bounds from covering numbers

The maximal inequalities of this paper are obtained by elementary chaining. We consider a process indexed by a set $\mathcal{T}$ with a nonnegative symmetric function d that controls the tails of the increments. We do not assume that d satisfies the triangle inequality; the proof uses only the covering numbers of d and the continuity of the process with respect to d. As in Section 16.1, for $v > 0$, a v-net is a set $\mathcal{N} \subset \mathcal{T}$ such that every $z \in \mathcal{T}$ has some $z' \in \mathcal{N}$ with $d(z, z') \le v$, and $N(\mathcal{T}, d, v)$ is the smallest cardinality of a v-net. A process $(F_z)_{z \in \mathcal{T}}$ is continuous with respect to d if $F_{z_k} \to F_z$ whenever $d(z_k, z) \to 0$.

Lemma 16.4. *Let $\mathcal{T}$ be a nonempty set and $d : \mathcal{T} \times \mathcal{T} \to [0, \infty)$ a symmetric function. Let $(F_z)_{z \in \mathcal{T}}$ be a real process which has a version that is continuous with respect to d and whose supremum equals its supremum over a countable d-dense subset of $\mathcal{T}$. Let $z_0 \in \mathcal{T}$, $s, b \ge 0$, $D < \infty$, $A \ge 1$, and $q_{\mathrm{ent}} \ge 0$, and assume that*

$$\sup_{z, z' \in \mathcal{T}} d(z, z') \le D, \qquad N(\mathcal{T}, d, v) \le A v^{-q_{\mathrm{ent}}} \quad (0 < v \le 1),$$

and that for every $u > 0$ and all $z, z' \in \mathcal{T}$,

$$\mathbb{P}\{|F_{z_0}| > s\sqrt{u} + bu\} \le 2e^{-u}, \qquad \mathbb{P}\{|F_z - F_{z'}| > d(z, z')(s\sqrt{u} + bu)\} \le 2e^{-u}.$$

Put $D_1 = \max\{1, D\}$ and $H = \log(2A) + q_{\mathrm{ent}}$. There is a universal constant C such that, for $u \ge 0$ and $1 \le p < \infty$,

$$\mathbb{P}\Big\{\sup_z |F_z| > C D_1 \big[s(\sqrt{H} + \sqrt{u}) + b(H + u)\big]\Big\} \le 2e^{-u},$$

$$\Big\|\sup_z |F_z|\Big\|_{L^p} \le C D_1 \big[s(\sqrt{H} + \sqrt{p}) + b(H + p)\big].$$

No Gaussian or independence assumption is made; in particular the lemma applies to centered processes.

Further, let $Y_1, \dots, Y_N$, $N \geq 1$, be real random variables with $\mathbb{P}\{|Y_i| > s\sqrt{u} + bu\} \leq 2e^{-u}$ for every $u > 0$, and let $M \geq 0$ be a random variable with $M \leq \max_{i \leq N} |Y_i| + e$ almost surely, for a deterministic $e \geq 0$. Then, for $u \geq 0$ and $1 \leq p < \infty$,

$$\mathbb{P}\big\{M > e + s\sqrt{\log(2N) + u} + b(\log(2N) + u)\big\} \leq e^{-u},$$
$$\|M\|_{L^p} \leq e + C\big\{s\big[\sqrt{\log(2N)} + \sqrt{p}\big] + b\big[\log(2N) + p\big]\big\}.$$

Every assertion holds under a fixed conditional probability kernel, with the same constants, if all the hypotheses hold for each fixed value of the conditioning variable. The constant C does not depend on the dimension, the meshes, the radii, the couplings, or the conditioning value. A bound which is uniform in the conditioning value requires that the parameters $s, b, D, A, q_{\mathrm{ent}}$ (and e) be bounded uniformly in that value.

Proof. We first normalize. Dividing d by D_1 gives a function of supremum at most one. At the radii 2^{-k}, $k \geq 1$, it has nets $\mathcal{T}_k$ with at most $A2^{kq_{\mathrm{ent}}}$ points: if $D_1 2^{-k} \leq 1$, we use the given net for d at radius $D_1 2^{-k}$, and otherwise we use the net at radius one. In the normalized function, the increment parameters become at most $D_1 s$ and $D_1 b$, while the parameters of the point z_0 are unchanged and are at most $D_1 s$ and $D_1 b$. If a net of minimal cardinality does not exist at some radius, we enlarge the radius by a factor two; the universal constant absorbs this factor. It therefore suffices to consider $D_1 = 1$ and nets $\mathcal{T}_k$ of radius at most 2^{1-k}.

We choose the nets and the connections deterministically. We connect every point of $\mathcal{T}_1$ to z_0, and each point of $\mathcal{T}_k$, $k \geq 2$, to one point of $\mathcal{T}_{k-1}$ within distance 2^{2-k}; such a point exists because $\mathcal{T}_{k-1}$ is a net of radius 2^{2-k} for all of $\mathcal{T}$. There are at most $A2^{kq_{\mathrm{ent}}}$ connections at level k. Set

$$h_0 = u + \log 8, \qquad h_k = u + \log(2A) + kq_{\mathrm{ent}} \log 2 + (k+3)\log 2 \quad (k \geq 1).$$

The point z_0 violates the bound $s\sqrt{h_0} + bh_0$ with probability at most $e^{-u}/4$. By a union bound over the connections at level k, the probability that one of them violates its bound is at most $e^{-u} 2^{-k-3}$. The sum of these probabilities is less than $2e^{-u}$.

On the complementary event, we follow the connections from a point of $\mathcal{T}_k$ back to z_0. The value of the process at every net point is then bounded by the bound at z_0, plus the bound for its connection at level one, plus

$$\sum_{k \geq 2} 2^{2-k}(s\sqrt{h_k} + bh_k).$$

Since the sums of 2^{-k}, $k2^{-k}$, and $\sqrt{k}2^{-k}$ converge, and $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$, this expression is at most

$$C\big\{s\big[\sqrt{\log(2A) + q_{\mathrm{ent}}} + \sqrt{u}\big] + b\big[\log(2A) + q_{\mathrm{ent}} + u\big]\big\}.$$

Every $z \in \mathcal{T}$ is the limit, with respect to d, of net points $z_k \in \mathcal{T}_k$, since $d(z, z_k) \le 2^{1-k}$. On the event where the given version is continuous, we have $F_{z_k} \to F_z$, and the bound extends to every point. The supremum equals a countable supremum and is therefore measurable. Restoring D_1 proves the tail bound. If $s = b = 0$, the same countable argument applies: the value at z_0 and every increment along a connection vanish almost surely. This also covers points at distance zero.

For the moments, we use that a nonnegative random variable X with $\mathbb{P}\{X > \alpha_0 + c\sqrt{u} + d_0 u\} \le 2e^{-u}$ for all $u \ge 0$ satisfies $\|X\|_{L^p} \le \alpha_0 + C(c\sqrt{p} + d_0 p) + C(c + d_0)$. To see this, let E be an exponential random variable with $\mathbb{P}(E > v) = e^{-v}$. Since $\min\{1, 2e^{-u}\} = \mathbb{P}(E + \log 2 > u)$, the variable X is stochastically dominated by $\alpha_0 + c\sqrt{E + \log 2} + d_0(E + \log 2)$. Minkowski's inequality and

$$\mathbb{E}E^r = \int_0^\infty t^r e^{-t}\, dt \le 2(2r/e)^r \qquad (r > 0)$$

imply the claim; the last inequality follows by maximizing $t^r e^{-t/2}$ and integrating the remaining factor $e^{-t/2}$, and we apply it with $r = p/2$ and $r = p$. In our case, $\alpha_0 = CD_1(s\sqrt{H} + bH)$, $c = CD_1 s$, and $d_0 = CD_1 b$, and $H \ge \log 2$ absorbs the term $C(c + d_0)$. This proves the L^p bound, including the bound on the expectation at $p = 1$.

For the finite family, we apply the tail bound for each Y_i with $u + \log(2N)$ in place of u and sum the N probabilities of failure; then we add the deterministic error e. The same comparison with an exponential random variable yields the moment bound.

Every step above is an inequality under one fixed probability measure, and we may therefore apply it separately under each conditional kernel. In the applications, the supremum is over rational times and a finite spatial grid, or over a set on which the process is continuous; its measurability is part of the hypotheses. The nets are constructed for the fixed value of the conditioning variable and require no random choice. Bounds on the parameters that are uniform in the conditioning value give bounds that are uniform in that value. □

Proof of the chaining assertion in Lemma 16.1. If $\sigma > 0$, we apply Lemma 16.4 to the function d/σ, with $s = \sqrt{2}\sigma$, $b = 0$, $D = 1$, $q_{\mathrm{ent}} = v$, the given A, and an arbitrary point $i_0 \in I$. The moment generating function bounds imply the required tails. Indeed, for a variable Y with $\mathbb{E}e^{zY} \le e^{z^2 r^2/2}$ for all real z, minimizing $e^{-zx}\mathbb{E}e^{zY}$ at $z = \pm x/r^2$ gives $\mathbb{P}\{|Y| > x\} \le 2e^{-x^2/(2r^2)}$, which is the required tail with $u = x^2/(2r^2)$. Since the one-point variance proxy is at most σ at every point, the choice of i_0 is arbitrary. The hypothesis on nets concerns radii $\eta < 1$; at radius one, a single point is a net, since $d/\sigma \le 1$. The lemma then gives the displayed tail and expectation bounds, where we use $\sqrt{\log(2A) + v} \le \sqrt{\log(2A)} + \sqrt{v}$. If $\sigma = 0$, every value of the process on the countable dense set vanishes almost surely by its moment generating function bound, and the claim follows from the continuity of the version. □

16.4 Concentration of conditionally centered reconstructions

One renormalization group step. Fix an odd integer $M \geq 3$, an integer $J \geq 1$, $B = M^J$, and the parameters of Theorems 13.8 and 15.13: the analytic radii, the cutoff amplitude A_{w}, the bound E_* on the effective interaction, and the larger construction domains. Fix $L > 0$ and the continuum coupling $g_3 > 0$. Let $a = L/n$ be a mesh such that $n/B \geq 4$ is an integer. We consider the renormalization group step from the lattice of mesh Ba, which we call the coarse lattice, to the lattice of mesh a, preceded by $d \geq 0$ averaging steps; we call d the depth. Its intermediate scales are indexed by $0 \leq i \leq J-1$. The intermediate scale i has mesh $a_i = M^i a$ and coupling $g_i = g_3\sqrt{a_i} = M^{i/2}g$, where $g = g_3\sqrt{a}$, so that $g^2 = g_3^2 a$. The largest coupling of the step is $t = g_{J-1} = M^{(J-1)/2}g$. The normalized fluctuation variables $w = (w_{i,e})$ take values in the fluctuation cutoff domain $\Omega_p = \prod_{i,e}\{w_{i,e} \in \mathbb{R}^3 : |w_{i,e}| \leq p_i\}$, whose radii satisfy $p_i \leq p(t) = A_{\mathrm{w}} \log(1/t)$. We write $q_i = g_i w_i$ for the corresponding coupling-scaled fluctuation inputs, which range over a product of balls of radius at most $Q = \max_i g_i p_i \leq t p(t)$.

The conditional fluctuation kernel. Let R_{supp} be a support radius, either $R_{\mathrm{supp}} = r$ or $R_{\mathrm{supp}} = \rho$ as described below, and let $c \in B_{R_{\mathrm{supp}}}$ be a coarse field, with $\theta = \operatorname{Log} c$. Its conditional fluctuation kernel is

$$\begin{aligned} V_{d,c}^{[R_{\mathrm{supp}}]}(w) &= \tfrac{1}{2}|w|^2 + H_{d,J}(\theta, w) + R_{\mathrm{in}}^{[R_{\mathrm{supp}}]}(e^{\varphi_\theta(w)}) - R_{\mathrm{in}}^{[R_{\mathrm{supp}}]}(e^{\varphi_\theta(0)}), \\ Z_{d,c}^{[R_{\mathrm{supp}}]} &= \int_{\Omega_p^{[R_{\mathrm{supp}}]}} e^{-V_{d,c}^{[R_{\mathrm{supp}}]}(w)}\, dw, \\ d\mu_{d,c}^{[R_{\mathrm{supp}}]}(w) &= \frac{\mathbf{1}_{\Omega_p^{[R_{\mathrm{supp}}]}}(w) e^{-V_{d,c}^{[R_{\mathrm{supp}}]}(w)}}{Z_{d,c}^{[R_{\mathrm{supp}}]}}\, dw. \end{aligned} \tag{16.15}$$

Here $\varphi_\theta(w) = \Phi_{d,J}(\theta, (g_i w_i)_i)$ is the reconstruction of Theorem 13.2, at which the effective interaction is evaluated, and $R_{\mathrm{in}}^{[R_{\mathrm{supp}}]}$ is the effective interaction produced by the d preceding averaging steps. The function $H_{d,J}(\theta, w)$ collects the remaining terms of the conditional fluctuation potential: the part of the effective classical action beyond its quadratic approximation, the Haar disintegration densities, and the logarithms of the Jacobian determinants of the conditional Hessian normalizations. The density in (16.15) is exactly the kernel $k(c, dw)$ of Theorem 13.8 (compare (4.8)). The subtraction of $R_{\mathrm{in}}^{[R_{\mathrm{supp}}]}(e^{\varphi_\theta(0)})$ depends only on c and cancels in the normalized kernel. The domain $\Omega_p^{[R_{\mathrm{supp}}]}$ does not depend on c. Since the potential is smooth on the compact closed product and the exponential is positive, $0 < Z_{d,c}^{[R_{\mathrm{supp}}]} < \infty$. When no confusion can arise, we write $\mu_{d,c}$, V_c, and Ω_p.

For $R_{\mathrm{supp}} = r$, the coarse field ranges over the small-link domain B_r, and $\mu_{d,c}$ is the conditional law, given the coarse field, of the fluctuation variables of the step under the

conditioned Wilson measure. Indeed, once the coarse field is fixed, the restrictions at coarser scales are fixed factors and cancel in the normalized kernel (Theorem 13.8). For $R_{\mathrm{supp}} = \rho$, we assume that a separate instance of Theorem 13.8 with support radius ρ is given. It has its own incoming effective interaction $R_{\mathrm{in}}^{[\rho]}$, its own fluctuation cutoff domain, normalized kernel, and change-of-gauge identity, and the image conditions of Theorem 13.2 hold for all coarse fields in B_ρ, including the margins of Lemma 16.5 below. This instance is a different conditional measure; it is not obtained by evaluating the auxiliary integral of Section 14 at coarse fields in $B_\rho \setminus B_r$, and we make no statement about that integral there. In Section 17, only the case $R_{\mathrm{supp}} = r$ is used.

The reconstruction. Let $\Phi(c, q)$ be the logarithm of a reconstruction of the fine field, in a gauge transformation that depends only on the coarse field c. We require the following properties on the whole product of coarse fields $c \in B_{R_{\mathrm{supp}}}$ and coupling-scaled fluctuation inputs of radius at most Q. The image of the reconstruction lies in the domain prescribed by Theorem 13.2. Its derivatives in q satisfy the absolute row bounds and the volume-normalized column bounds of that theorem, with each variable of the intermediate scale i normalized by its own cell volume a_i^3. The map $\Phi_{d,J}$ of Theorem 13.2 and the logarithmic form $\widehat{\Phi}_d$ of the reconstruction maps $\widehat{\mathscr{R}}$ of Theorem 15.12 have these properties; see Lemma 16.5. We set

$$A(w) = a^{-1}\Phi(c, (g_i w_i)_i),$$

the lattice potential of the reconstructed field, and we view it as a cell function: the component of A on a positive link $(y, y + ae_i)$ is extended as a constant on the cell Q_y.

The domain conditions. The next lemma collects the conditions under which the whole product of fluctuation variables lies in the domain where the coefficient bounds of Theorem 13.2 and the Hessian bound of Theorem 13.8 hold. It fixes the order of choices: the larger construction domains and coefficient constants first, then the support radius, and the coupling bound g_* last; J is fixed throughout.

Lemma 16.5. *Let $K_* \geq 2$, ρ, s_J, r_J, C_J, and the coefficientwise majorant series*

$$\mathfrak{M}_J(u, v) = B^{-1}Z(u) + \frac{C_J v}{1 - C_J(u + v)}$$

for the absolute row sums of the reconstruction, be as in Theorem 13.2. Thus Z has nonnegative coefficients and no constant term, and

$$4K_*\rho < r_J, \qquad C_J 4K_*(\rho + s_J) < 1, \qquad \mathfrak{M}_J(4K_*\rho, 4K_* s_J) < \rho/2.$$

For the reconstruction and conditional-potential assertions, fix a support radius $0 < R_{\mathrm{supp}} \leq \rho$ and a conditional-kernel instance on $B_{R_{\mathrm{supp}}}$. Assume that this ball of coarse fields satisfies the

coordinate-domain conditions of that instance, and that its product with the coupling-scaled fluctuation inputs lies in the domain of the chosen reconstruction, with the fixed radius ratios; for the reconstruction maps of Theorem 15.12, assume that this product lies in their common inner domain. In the application below $R_{\mathrm{supp}} = r$, or $R_{\mathrm{supp}} = \rho$ with the separately hypothesized instance and its own domains. For this fixed ball and instance, choose $Q_\circ$ with $0 < Q_\circ \le s_J$ and $Q_\circ < (r_J - \rho)/2$, inside every further domain of coarse fields and coupling-scaled fluctuation inputs used by the reconstruction, and set $\eta_\circ = 1 - C_J(\rho + s_J)$. Then the following statements hold.

1. *We have $\eta_\circ > 0$ and*

$$0 < R \le \rho, \quad 0 \le Q \le Q_\circ \quad \Longrightarrow \quad R + Q < r_J, \quad 1 - C_J(R+Q) \ge \eta_\circ. \tag{16.16}$$

2. *For $0 \le u \le \rho$ and $0 \le v \le Q_\circ$,*

$$\partial_v \mathfrak{M}_J(u,v) = \frac{C_J(1 - C_J u)}{[1 - C_J(u+v)]^2} \le C_J \eta_\circ^{-2}.$$

   On the fixed product of coarse fields $c \in B_{R_{\mathrm{supp}}}$ and coupling-scaled fluctuation inputs of radius at most $Q_\circ$, the normalized Jacobian $g^{-1} D_w \Phi(c, (g_i w_i)_i)$, with each fluctuation variable attached to a cell of the lattice of mesh a, and each of its mixed derivatives of fixed order in the coarse field and the fluctuation variables, satisfy (16.8) *with $a_0 = a$ and a constant depending only on J, the fixed construction domains, the order of differentiation, and $\eta_\circ$.*

3. *For $0 < R \le \rho$ put $Q_{\mathrm{gate}}(R) = \min\{Q_\circ, R\eta_\circ/(8C_J)\}$. If $0 \le Q \le Q_{\mathrm{gate}}(R)$, then $\mathfrak{M}_J(R,Q) < 3R/16 < R/2$.*

4. *Fix $0 < R \le \rho$. There is $g_* > 0$, depending only on the fixed construction data and on R, such that for all $0 < t \le g_*$,*

$$\begin{aligned} tp(t) &\le Q_{\mathrm{gate}}(R), \qquad 1 - Ctp(t) > 0, \\ \frac{C[tp(t) + t^2(1 + E_*)]}{[1 - Ctp(t)]^3} &\le \frac{1}{2}, \end{aligned} \tag{16.17}$$

   where $p(t) = A_{\mathrm{w}} \log(1/t)$ and C is the constant in the Hessian bound of Theorem 13.8 (not the row constant C_J).

In particular, for the fixed support radius and instance specified above, let g_* be as in the fourth part with $R = R_{\mathrm{supp}}$, let $0 < t \le g_*$, and let the radii of the fluctuation cutoff domain satisfy $p_i \le p(t)$. Then for every coarse field in $B_{R_{\mathrm{supp}}}$ the whole product of fluctuation variables, and not only the point $w = 0$, is mapped into the product of the first part; the image condition $\mathfrak{M}_J(R,Q) < R/2$ of Theorem 13.2 holds with $Q = \max_i g_i p_i$; and the conditional fluctuation potential satisfies $D_w^2 V_c \ge I/2$ on the closed product.

Proof. Since $K_* \geq 2$, we have $C_J(\rho + s_J) < 1/(4K_*) < 1$, and hence $\eta_\circ > 0$. If $0 < R \leq \rho$ and $0 \leq Q \leq Q_\circ$, then $R + Q \leq \rho + Q_\circ < \rho + (r_J - \rho)/2 < r_J$ and $1 - C_J(R + Q) \geq 1 - C_J(\rho + s_J) = \eta_\circ$. This proves (16.16). The choice of $Q_\circ$ is possible because $r_J - \rho > 0$, and it can be made strictly inside every further domain used by the map, since these domains are open and have fixed radius ratios. For the reconstruction maps of Theorem 15.12, the required room comes from the hypothesis that the fixed product of coarse fields and coupling-scaled fluctuation inputs lies in their common inner domain with fixed radius ratios. We use their coefficient bound for mixed derivatives at fixed J and their uniform interior-domain conditions, rather than a majorant uniform in J.

Every product in (16.16) is contained in the product with radii $4K_*\rho$ and $4K_*s_J$. By Theorem 13.2, the image of this larger product lies in $B_{\rho/2}$, with interior margins for the intermediate classical, logarithm, and Haar disintegration conditions; subsets inherit these margins. On this product, the formula for $\partial_v \mathfrak{M}_J$ follows by differentiation, and the bound follows from $1 - C_J u \leq 1$ and $1 - C_J(u + v) \geq \eta_\circ$. The derivatives of the column majorants with respect to one fixed input give the volume-normalized column sums, with the same lower bound $\eta_\circ$ on the denominator. Each further derivative of fixed order costs only a factorial, the order, and inverse powers of $\eta_\circ$; it does not depend on the number of lattice points. These are derivatives of the absolute coefficient series, not consequences of a bound on the values of a holomorphic function. It remains to convert the normalizations. A fluctuation variable of the intermediate scale i lives on the lattice of mesh $a_i = M^i a$, and its column is normalized by a_i^3. Attaching it to a cell of the lattice of mesh a changes the normalization by the factor $M^{3i} \leq B^3$, and replacing q_i by $g_i w_i$ and dividing by g multiplies the derivative by $g_i/g = M^{i/2}$. At most a number of variables depending only on J is attached to each cell. These conversions only enlarge C_J by a factor depending on J, and they yield (16.8) for the normalized Jacobian and its mixed derivatives of fixed order. This proves the second part.

For the third part, the coefficients of Z are nonnegative and $Z(0) = 0$. Comparing term by term and using $(R/(4K_*\rho))^k \leq R/(4K_*\rho)$ for every degree $k \geq 1$, we find $Z(R) \leq (R/(4K_*\rho))Z(4K_*\rho)$ for $0 < R \leq \rho$. Since $B^{-1}Z(4K_*\rho) \leq \mathfrak{M}_J(4K_*\rho, 4K_*s_J) < \rho/2$, we obtain

$$B^{-1}Z(R) \leq \frac{R}{4K_*\rho} B^{-1}Z(4K_*\rho) < \frac{R}{8K_*} \leq \frac{R}{16}.$$

If $Q \leq Q_{\mathrm{gate}}(R)$, then by (16.16),

$$\mathfrak{M}_J(R, Q) \leq B^{-1}Z(R) + C_J Q/\eta_\circ < \frac{R}{16} + \frac{R}{8} = \frac{3R}{16} < \frac{R}{2}.$$

For the fourth part, let $x = \log(1/t)$. Then $tp(t) = A_{\mathrm{w}} x e^{-x} \to 0$ as $t \downarrow 0$, since $e^x \geq x^3/6$ for $x > 0$, and $t^2(1 + E_*) \to 0$. By the definition of a limit, the three conditions in (16.17) hold for all t in an interval $(0, g_*]$, not only for one value of t. This is a single

reduction of the upper bound on the coupling, which uses only the fixed construction data and R; it introduces no dependence on the lattice side, the depth, or the coarse field.

Finally, let $R = R_{\mathrm{supp}}$, let $t \le g_*$, and let $p_i \le p(t)$. Since $g_i \le t$, we have $Q = \max_i g_i p_i \le t p(t) \le Q_{\mathrm{gate}}(R) \le Q_\circ$. Hence, for every coarse field in $B_{R_{\mathrm{supp}}}$, the whole product of fluctuation variables lies in the fixed product on which the reconstruction and the conditional potential of the chosen instance are defined, and in the product of the first part. The third part gives $\mathfrak{M}_J(R, Q) < R/2$. Shrinking Q does not enlarge the domain of coarse fields; the ball $B_{R_{\mathrm{supp}}}$ and its conditional kernel are given by the hypothesis. In particular, the majorant of Theorem 13.2 is never evaluated beyond this product or across its pole, and both the image condition and the domain conditions under which it is stated hold. The last condition in (16.17) is the hypothesis of the Hessian bound in Theorem 13.8, which implies $D_w^2 V_c \ge I/2$ on the closed product. □

Theorem 16.6. *Fix M, J, and the parameters of Theorems 13.8 and 15.13, and fix $L, T_0 > 0$. Let $R_{\mathrm{supp}} = r$, or $R_{\mathrm{supp}} = \rho$ under the additional hypothesis on the separate instance described above, and let g_* be as in Lemma 16.5 with $R = R_{\mathrm{supp}}$. Let $a = L/n$ be a mesh with $n/B \ge 4$, let $d \ge 0$, let $g^2 = g_3^2 a$, and assume that $t = M^{(J-1)/2} g \le g_*$. Let $c \in B_{R_{\mathrm{supp}}}$, let $\mu_{d,c}$ be the conditional fluctuation kernel* (16.15)*, and let Φ be a reconstruction with the properties described above; the reconstruction maps of Theorem 15.12 are an example. Put $A(w) = a^{-1}\Phi(c, (g_i w_i)_i)$, viewed as a cell function, and*

$$F = A - \mathbb{E}_{\mu_{d,c}} A, \qquad \|F\|_{-\alpha;T_0} = \sup_{0<s\le T_0} s^{\alpha/2} \|P_s F\|_\infty,$$

where P_s is the continuum heat semigroup acting componentwise. Then for every $0 < \alpha < 3/2$ and $u \ge 0$,

$$\mu_{d,c}\Big\{ \|F\|_{-\alpha;T_0} > C g_3 a^{\alpha-1/2} \sqrt{\log(2 + L/a) + u} \Big\} \le C e^{-u}.$$

For every $p \ge 1$ and $\lambda \ge 0$, with $\tau_a = C g_3 a^{\alpha-1/2}$,

$$\big(\mathbb{E}_{\mu_{d,c}} \|F\|_{-\alpha;T_0}^p\big)^{1/p} \le C_p g_3 a^{\alpha-1/2} \sqrt{\log(2 + L/a)},$$
$$\mathbb{E}_{\mu_{d,c}} e^{\lambda \|F\|_{-\alpha;T_0}} \le \exp\Big\{ C\lambda\tau_a \sqrt{\log(2 + L/a)} + C\lambda^2 \tau_a^2 \Big\}.$$

In particular, if $1/2 < \alpha < 3/2$ and $a \downarrow 0$ with g_3 fixed (equivalently, with L and $\gamma = g_3^2 L$ fixed), then for each fixed p the conditional L^p norms of $\|F\|_{-\alpha;T_0}$ tend to zero, for each fixed λ the exponential moments $\mathbb{E}_{\mu_{d,c}} e^{\lambda\|F\|_{-\alpha;T_0}}$ tend to one, and the bounds on $\mathbb{E}_{\mu_{d,c}} \|F\|_{-\alpha;T_0}$ are summable along every geometric sequence of meshes; these statements are uniform in c and d.

For every deterministic real $\varphi \in L^2(\mathbb{T}_L^3)$, paired componentwise with A,

$$\mathrm{Var}_{\mu_{d,c}} \langle A, \varphi \rangle \le C_J g_3^2 a^2 \|\varphi\|_2^2, \qquad \log \mathbb{E}_{\mu_{d,c}} e^{z\langle F, \varphi\rangle} \le C_J z^2 g_3^2 a^2 \|\varphi\|_2^2 \qquad (z \in \mathbb{R}).$$

For finitely many real $\varphi_1, \dots, \varphi_m \in L^2(\mathbb{T}_L^3)$ and every compact set $\mathcal{K} \subset \mathbb{C}^m$,

$$\sup_{z \in \mathcal{K}} \Big| \mathbb{E}_{\mu_{d,c}} \exp\Big(\sum_{j=1}^{m} z_j \langle F, \varphi_j \rangle \Big) - 1 \Big| \longrightarrow 0$$

as $a \downarrow 0$ with g_3 fixed.

If $c \in D_{R_{\mathrm{supp}}}$, choose a gauge transformation k on the vertices of the coarse lattice with $c^k \in B_{R_{\mathrm{supp}}}$, and transport the conditional fluctuation kernel by the corresponding change of coordinates. Then all the above estimates hold for the reconstruction evaluated at c^k, uniformly in c and in the choice of k.

The constants depend only on M, J, the fixed construction domains and their margins, the Hessian bound, the row and column constants of the derivatives of Φ, α, L, T_0, and, where displayed, p. They do not depend on n, d, or c, and they do not depend on the coupling below g_*, except through the displayed factors of g_3.

Remark 16.7. Theorem 16.6 bounds the fluctuation F about the conditional expectation under the conditional fluctuation kernel. It gives no bound on $\mathbb{E}_{\mu_{d,c}} A$, which is estimated in Lemma 17.1. The norm $\|\cdot\|_{-\alpha;T_0}$ is not invariant under gauge transformations that vary in space, and the estimates refer to the chosen gauge representative of the reconstruction.

Proof of Theorem 16.6. We fix c and the kernel $\mu = \mu_{d,c}$ for the whole proof; all expectations and probabilities are taken with respect to μ, and nothing in the proof differentiates in c. The constants have the dependence stated in the theorem, and C_J may increase by factors depending on M, J, and the fixed construction domains. The proof has seven steps. We verify the hypotheses of the concentration theorem and bound the Jacobian of $w \mapsto A(w)$. We then bound $P_s F$ at a fixed time by chaining over space, using the cell integrals of the heat kernel, take the supremum over times by a dyadic decomposition, and derive the moments, the exponential moments, and the limits. Finally, we estimate deterministic source pairings and their generating functions, and we transport the conclusions to coarse fields in $D_{R_{\mathrm{supp}}}$.

Step 1: the kernel and its convexity. Let R_{supp} and g_* be as in the statement. We apply Lemma 16.5 to this support radius and to the corresponding instance of the conditional kernel: for $R_{\mathrm{supp}} = r$ the kernel of Theorem 13.8 on B_r, and for $R_{\mathrm{supp}} = \rho$ the separately hypothesized instance on B_ρ, each with its own domains. By that lemma, for every $c \in B_{R_{\mathrm{supp}}}$, the whole product Ω_p is mapped into the fixed product on which the reconstruction and the potential are defined, and the image condition of Theorem 13.2 holds. For the reconstruction of Theorem 13.2, each intermediate classical, logarithm, and Haar disintegration input lies in the enlarged product of that theorem, and its interior margin remains valid. For the reconstruction maps of Theorem 15.12, the same inputs lie in their inner domain, where that theorem preserves the uniform interior-domain

conditions and the normalized conditional kernels. We use only the joint coefficient bound of that family at fixed J. In the case $R_{\mathrm{supp}} = \rho$, the image condition is required on all of B_ρ, including the margins of Lemma 16.5, and not only on B_r; the analytic radius and the coefficient constants of that instance cover B_ρ by hypothesis.

By the last condition in (16.17), Theorem 13.8 gives $D_w^2 V_c \geq I/2$ on the closed convex product, with a C^2 extension to an open neighborhood. Then μ satisfies the hypotheses of Theorem 5.4 with $m = 1/2$; that theorem includes the faces and corners of the product of balls. We use only the following two of its conclusions: for every function H of w in $H^1(\mu)$ and every real Euclidean-Lipschitz function H,

$$\mathrm{Var}_\mu H \leq 2\mathbb{E}_\mu |\nabla_w H|^2, \qquad \log \mathbb{E}_\mu e^{z(H-\mathbb{E}_\mu H)} \leq z^2 \,\mathrm{Lip}_w(H)^2 \quad (z \in \mathbb{R}). \tag{16.18}$$

These inequalities require no independence under μ.

The kernel does not depend on the gauge in which the observable A is computed: another gauge transformation depending only on c changes A but not V_c. The reconstruction maps of Theorem 15.12 are obtained from the same reconstruction before gauge transformation by a gauge transformation depending only on the coarse field. By Theorems 15.12 and 15.13, they preserve the fluctuation cutoff domain, the free logarithmic coordinates, the effective classical action, the Haar disintegration density, the Jacobian determinants of the conditional Hessian normalizations, the incoming effective interaction, the coarse field, and the normalization constant. Hence (16.15) defines the same conditional measure for the reconstruction maps, and its Hessian is that of the same potential.

Step 2: the Jacobian. We write ξ for a scalar fluctuation coordinate, e for a scalar component of the output on a positive link, and $\varphi(w) = \Phi(c, (g_i w_i)_i)$. By the derivative bounds of Theorem 13.2 on the whole product, converted as in the second part of Lemma 16.5, we have

$$\sup_{w,e} \sum_{\xi} |\partial_{w_\xi} \varphi_e(w)| \leq C_J g, \qquad \sup_{w,\xi} \sum_{e} |\partial_{w_\xi} \varphi_e(w)| \leq C_J g. \tag{16.19}$$

We convert the normalizations as in the proof of Lemma 16.5. The column of a variable of the intermediate scale i is normalized by a_i^3, and the column bound of Theorem 13.2 reads $a^3 \sum_e |\partial_{q_\xi} \Phi_e| \leq C_J a_i^3$. The substitution $q_\xi = g_i w_\xi$ contributes the factor $g_i = M^{i/2} g$, and the passage to counting coordinates costs the factor M^{3i}; the row bound is converted with the same bounded ratios of couplings. These factors, and the number of variables per cell, depend only on J, and (16.19) holds with no factor depending on the number of cells or on d. The gauge transformation depends only on c and has no derivative in w, and the derivatives of the conditional Hessian normalizations and of the reconstruction before gauge transformation are included in (16.19). For the reconstruction

maps of Theorem 15.12, that theorem provides the joint coefficient bound, every volume-normalized column, and the image conditions on their common domains, uniformly in the depth and with deterministic thresholds that do not depend on the coupling or on the sample. The same computation applies, with the constant of that family at fixed J.

By Schur's test (16.14) in counting coordinates, we have

$$\sup_{w} \|D_w A(w)\|_{\ell^2 \to \ell^2} \le K_a, \qquad K_a = C_J g / a = C_J g_3 a^{-1/2}, \tag{16.20}$$

where the norm is from the fluctuation variables to the components of A on the links. Since the product is convex, integrating the derivative along the segment from w to w' gives

$$\|A(w) - A(w')\|_{\ell^2} \le K_a \|w - w'\|_2. \tag{16.21}$$

Step 3: a fixed time. Fix one scalar component, given by a direction and a color, and let $X_s(x) = P_s F(x)$. Since F is constant on the half-open cells, we have

$$X_s(x) = \sum_y \omega_y(s,x)\big(A_y(w) - \mathbb{E}_\mu A_y\big), \tag{16.22}$$

by integration of the cell function against the continuum heat kernel. As a function of w, the right side is a linear combination of the components of A. By (16.20), its Lipschitz constant is at most $K_a \|\omega(s,x)\|_2$, and the Lipschitz constant of $X_s(x) - X_s(y)$ is at most $K_a \|\omega(s,x) - \omega(s,y)\|_2$. Both have mean zero, because the subtracted mean is the conditional expectation under μ. By (16.18) and Lemma 16.1 with horizon T_0, their sub-Gaussian parameters, in the normalization $e^{z^2\sigma^2/2}$, can be taken as

$$\sigma_s = C_J K_a \min\{1, a^{3/2} s^{-3/4}\} \tag{16.23}$$

and, when $s \ge a^2$,

$$d_s(x,y) = C_J K_a a^{3/2} s^{-3/4} \min\{1, \operatorname{dist}(x,y)/\sqrt{s}\}. \tag{16.24}$$

The constants include the factor $\sqrt{2}$ that converts (16.18) to this normalization. The function d_s is nonnegative and symmetric, and after enlarging the constants we have $\sup d_s \le \sigma_s$ for $s \ge a^2$.

For $s \ge a^2$, a cubic grid on $\mathbb{T}_L^3$ of spacing at most $c\eta\sqrt{s}$ is an $\eta\sigma_s$-net for d_s, which gives

$$N(\mathbb{T}_L^3, d_s, \eta\sigma_s) \le C(1 + L/\sqrt{s})^3 \eta^{-3} \qquad (0 < \eta < 1).$$

For each w, the function $x \mapsto P_s F(x)$ is smooth for $s > 0$. Its supremum is determined by a countable dense set, and the process is continuous with respect to d_s. We set $\ell_a = \log(2 + L/a)$. Applying the chaining assertion of Lemma 16.1 with $A = C(1 + L/\sqrt{s})^3$ and $v = 3$, we obtain

$$\mathbb{P}_\mu\Big\{ \|P_s F\|_\infty > C K_a a^{3/2} s^{-3/4} \sqrt{\ell_a + u} \Big\} \le C e^{-u}, \qquad \mathbb{E}_\mu \|P_s F\|_\infty \le C K_a a^{3/2} s^{-3/4} \sqrt{\ell_a}. \tag{16.25}$$

Here we used $\log(2 + C(1 + L/\sqrt{s})^3) \le C\ell_a$, which holds because $s \ge a^2$. There are nine scalar components; a union bound over them and the equivalence of norms on the components change only C.

Step 4: the supremum over times. Each scalar cell variable $A_y - \mathbb{E}_\mu A_y$ is centered and K_a-Lipschitz. The exponential bound in (16.18), applied to it and to its negative, yields

$$\mathbb{P}_\mu\{|A_y - \mathbb{E}_\mu A_y| > 2K_a\sqrt{h}\} \le 2e^{-h} \qquad (h \ge 0).$$

There are $9n^3$ scalar cell variables. Choosing $h = u + \log(18n^3)$ and summing, we obtain

$$\mathbb{P}_\mu\Big\{\|F\|_\infty > CK_a\sqrt{\ell_a + u}\Big\} \le Ce^{-u}. \tag{16.26}$$

Since the heat semigroup contracts the supremum norm, we have, for $0 < s \le \min\{a^2, T_0\}$,

$$s^{\alpha/2}\|P_sF\|_\infty \le a^\alpha\|F\|_\infty.$$

Together with (16.26), this implies the required bound at all these times, including the case $T_0 < a^2$.

Suppose now that $T_0 \ge a^2$. Let $s_j = 2^j a^2$ for $0 \le j \le r$, where r is the largest integer with $s_r \le T_0$. By the semigroup property and the contraction of the supremum norm, we have, for $s_j \le s \le \min\{2s_j, T_0\}$,

$$s^{\alpha/2}\|P_sF\|_\infty \le 2^{\alpha/2}s_j^{\alpha/2}\|P_{s_j}F\|_\infty. \tag{16.27}$$

The weighted scale at the time s_j satisfies

$$K_a a^{3/2}s_j^{\alpha/2-3/4} = K_a a^\alpha 2^{-j(3/4-\alpha/2)} \le K_a a^\alpha, \tag{16.28}$$

where we used $\alpha < 3/2$. Further, $r + 1 \le 1 + \log_2(T_0/a^2) \le C_{T_0/L^2}\ell_a$, since $n = L/a \ge 4$. We use (16.25) at every s_j with $u + \log(r + 2)$ in place of u, and (16.26) with the same parameter. The sum of the probabilities of failure is at most Ce^{-u}. Since $\log(r + 2) \le C_{T_0/L^2}\ell_a$, all the thresholds in (16.27) and (16.28) are bounded by

$$CK_a a^\alpha\sqrt{\ell_a + u} = Cg_3 a^{\alpha-1/2}\sqrt{\ell_a + u}.$$

This proves the tail bound for all meshes in the statement. The time supremum is measurable, because the heat field is continuous in (s, x) for $s > 0$ and a countable dense subset of $(0, T_0] \times \mathbb{T}_L^3$ determines it. For each finite lattice, the field is bounded, and the norm is finite.

Step 5: moments, exponential moments, and limits. Let $\tau_a = C g_3 a^{\alpha-1/2}$, with C large enough for the preceding bounds, and let $Z = \|F\|_{-\alpha;T_0}$. Since $\sqrt{\ell_a + u} \le \sqrt{\ell_a} + \sqrt{u}$, the positive part of $Z - C\tau_a\sqrt{\ell_a}$ has tail at most $C\exp[-x^2/(C\tau_a^2)]$ at height x. Multiplying this tail by px^{p-1}, integrating over $x > 0$, and substituting $x = \tau_a v$, we obtain $C_p\tau_a^p$. By the triangle inequality in L^p and $\ell_a \ge \log 2$, we have

$$\mathbb{E}_\mu Z \le C\tau_a\sqrt{\ell_a}, \qquad (\mathbb{E}_\mu Z^p)^{1/p} \le C_p\tau_a\sqrt{\ell_a}.$$

The norm Z is itself a Lipschitz function of w, with constant at most $C\tau_a$. Indeed, by (16.21) and the bound on the cell integrals of the heat kernel, the difference of its values at w and w' is at most

$$CK_a\|w - w'\|_2 \sup_{0<s\le T_0} s^{\alpha/2}\min\{1, a^{3/2}s^{-3/4}\} \le CK_a a^\alpha \|w - w'\|_2,$$

where in the last inequality we used $s^{\alpha/2} \le a^\alpha$ for $s \le a^2$, and that $s^{\alpha/2-3/4}$ is decreasing for $s \ge a^2$. The subtracted conditional mean does not depend on w. Applying (16.18) to Z, we obtain

$$\mathbb{E}_\mu e^{\lambda Z} \le \exp\{\lambda\mathbb{E}_\mu Z + C\lambda^2\tau_a^2\} \le \exp\Big\{C\lambda\tau_a\sqrt{\ell_a} + C\lambda^2\tau_a^2\Big\}.$$

Since the concentration theorem applies to every real Lipschitz function, no smoothness of Z is needed.

All limits and summability statements are at fixed L and g_3, equivalently at fixed L and $\gamma = g_3^2 L$. If $\beta = \alpha - 1/2 > 0$, then $a^\beta\sqrt{\log(2 + L/a)} \to 0$ as $a \downarrow 0$, and every fixed conditional L^p norm therefore tends to zero, uniformly in c and d. The exponential moment is at least one because $Z \ge 0$, and its upper bound tends to one for fixed λ. For a geometric sequence of meshes $a_k = a_*\mathfrak{b}^{-k}$ with a fixed $\mathfrak{b} > 1$, we have

$$a_k^\beta\sqrt{\log(2 + L/a_k)} \le a_*^\beta\mathfrak{b}^{-k\beta}\Big\{\sqrt{\log(2 + L/a_*)} + \sqrt{k\log\mathfrak{b}}\Big\},$$

and both resulting series converge; the ratio of consecutive terms in the second tends to $\mathfrak{b}^{-\beta} < 1$. This proves the summability of the bounds on the expected norms, without any independence between different meshes.

Step 6: deterministic sources and generating functions. For a real test field φ, let $q_y = \int_{Q_y}\varphi(x)\,dx$, with the pairing of directions and colors included. By the Cauchy–Schwarz inequality on each cell, we have

$$\sum_y |q_y|^2 \le a^3\|\varphi\|_2^2. \tag{16.29}$$

Combining this with (16.20), we find that the pairing $H_\varphi = \langle A, \varphi\rangle$ has Lipschitz constant at most

$$K_a a^{3/2}\|\varphi\|_2 = C_J g_3 a\|\varphi\|_2,$$

and its gradient satisfies the same bound pointwise. The two inequalities in (16.18) then imply

$$\operatorname{Var}_\mu H_\varphi \le C_J g_3^2 a^2 \|\varphi\|_2^2, \qquad \log \mathbb{E}_\mu e^{z\langle F,\varphi\rangle} \le C_J z^2 g_3^2 a^2 \|\varphi\|_2^2 \quad (z \in \mathbb{R}).$$

Next, fix finitely many real sources $\varphi_1, \dots, \varphi_m$ and a compact set $\mathcal{K} \subset \mathbb{C}^m$, and let $Y_z = \sum_j z_j \langle F, \varphi_j\rangle$. By the variance bound and the triangle inequality in L^2, we have

$$(\mathbb{E}_\mu |Y_z|^2)^{1/2} \le C_J g_3 a \sum_j |z_j| \|\varphi_j\|_2.$$

Since the real part of Y_z is the pairing of F with the real source $\sum_j \Re(z_j)\varphi_j$, the exponential bound gives

$$\mathbb{E}_\mu e^{2|\Re Y_z|} \le \mathbb{E}_\mu e^{2\Re Y_z} + \mathbb{E}_\mu e^{-2\Re Y_z} \le 2 \exp\Big\{ C_J g_3^2 a^2 \Big(\sum_j |z_j| \|\varphi_j\|_2 \Big)^2 \Big\}.$$

By the identity $e^y - 1 = y \int_0^1 e^{\varrho y}\, d\varrho$ and the Cauchy–Schwarz inequality, uniformly for $z \in \mathcal{K}$,

$$|\mathbb{E}_\mu e^{Y_z} - 1| \le (\mathbb{E}_\mu |Y_z|^2)^{1/2} (\mathbb{E}_\mu e^{2|\Re Y_z|})^{1/2} \longrightarrow 0$$

as $a \downarrow 0$ with g_3 fixed. For each finite lattice, these functions of z are entire. Indeed, the domain of the fluctuation variables is compact and the reconstruction is continuous on a neighborhood of it; hence the exponential of the source pairing has a uniformly convergent power series on every compact set of sources. Complex test fields are handled by including their real and imaginary parts among the real sources.

Step 7: coarse fields in $D_{R_{\mathrm{supp}}}$. Let R_{supp} be as in (16.15). For $c \in D_{R_{\mathrm{supp}}}$, choose a gauge transformation k on the vertices of the coarse lattice with $c' = c^k \in B_{R_{\mathrm{supp}}}$. The change-of-gauge identity of Theorem 15.13, together with the corresponding identity for the reconstruction in Theorem 13.2, uses a triangular change $\mathcal{T}_k$ of the fluctuation variables whose diagonal blocks are orthogonal. It preserves each ball of the fluctuation cutoff domain and has Jacobian of absolute value one, and

$$(\mathcal{T}_k)_* \mu_{d,c}^{[R_{\mathrm{supp}}]} = \mu_{d,c'}^{[R_{\mathrm{supp}}]}.$$

For $R_{\mathrm{supp}} = \rho$, this is the change-of-gauge identity of the separate instance, not an extension of the auxiliary integral. Since $\mathcal{T}_k$ preserves the radii of all the balls, the radius Q and all the domain conditions verified above are the same at c'. The transported representative is therefore exactly the case $c' \in B_{R_{\mathrm{supp}}}$ already treated, and the bounds are uniform in c and in the choice of k. Evaluating the transported reconstruction at c' gives the last assertion of the theorem. □

16.5 Weighted heat convolutions

The Duhamel integrals for the Wilson flow in Section 24 lead to time convolutions of two powers, each regularized by the same positive parameter ϵ; in the applications, $\epsilon = a_0^2$ for a mesh a_0. The following lemma estimates these scalar integrals.

Lemma 16.8. *Let $\epsilon > 0$, $t \ge 0$, $0 \le p < 1$, and $q \ge 0$, and define*

$$I^{\epsilon}_{p,q}(t) = \int_0^t (s+\epsilon)^{-p}(t-s+\epsilon)^{-q}\,ds.$$

For fixed p and q, uniformly in ϵ and t,

$$I^{\epsilon}_{p,q}(t) \le C_{p,q} \begin{cases} (t+\epsilon)^{1-p-q}, & 0 \le q < 1, \\ (t+\epsilon)^{-p}[1+\log(1+t/\epsilon)], & q = 1, \\ \epsilon^{1-q}(t+\epsilon)^{-p}, & q > 1. \end{cases}$$

For $0 \le t \le \epsilon$ we also have $I^{\epsilon}_{p,q}(t) \le t\epsilon^{-p-q}$, and for $p = q = 1/2$ we have $I^{\epsilon}_{1/2,1/2}(t) \le \pi$.

In particular, let $1/2 < \alpha_{\mathrm{fl}} < 2/3$, $p = (\alpha_{\mathrm{fl}}+1)/2$, and $\epsilon = a_0^2$ with $a_0 > 0$. Then

$$I^{a_0^2}_{p,3/4}(t) \le C(t+a_0^2)^{-\alpha_{\mathrm{fl}}/2-1/4}, \qquad I^{a_0^2}_{p,5/4}(t) \le Ca_0^{-1/2}(t+a_0^2)^{-p},$$

$$\sup_{t\ge 0}\Big\{(t+a_0^2)^{\alpha_{\mathrm{fl}}-1/2} I^{a_0^2}_{p,3/4}(t) + (t+a_0^2)^{\alpha_{\mathrm{fl}}} I^{a_0^2}_{p,5/4}(t)\Big\} \le C_{\alpha_{\mathrm{fl}}} a_0^{\alpha_{\mathrm{fl}}-3/2}.$$

The same bound holds if either of the two weights $(t+a_0^2)^{\alpha_{\mathrm{fl}}-1/2}$ and $(t+a_0^2)^{\alpha_{\mathrm{fl}}}$ is replaced by the same power of $t+a^2$, for some $0 < a \le a_0$. For $p = 3\alpha_{\mathrm{fl}}/2$, the integrals with $q = 0$ and $q = 1/2$, which correspond to the value and to one spatial difference of a heat convolution, are bounded by $C(t+\epsilon)^{1-3\alpha_{\mathrm{fl}}/2}$ and $C(t+\epsilon)^{1/2-3\alpha_{\mathrm{fl}}/2}$, respectively. The constants depend only on the displayed exponents.

Proof. All integrands are nonnegative. If $t = 0$, the integral is zero. If $0 < t \le \epsilon$, each factor of the integrand is at most its value with s or $t-s$ replaced by zero, which gives

$$I^{\epsilon}_{p,q}(t) \le t\epsilon^{-p-q}. \tag{16.30}$$

Since $\epsilon \le t+\epsilon \le 2\epsilon$ in this range, this implies the three general bounds, with a constant depending only on p and q; in the case $q = 1$, we use $1+\log(1+t/\epsilon) \ge 1$.

For $t \ge \epsilon$, we split the integral at $t/2$. On the first half, $t-s+\epsilon \ge (t+\epsilon)/2$, which gives

$$\int_0^{t/2} (s+\epsilon)^{-p}(t-s+\epsilon)^{-q}\,ds \le 2^q(t+\epsilon)^{-q}\int_0^{t/2}(s+\epsilon)^{-p}\,ds \le \frac{2^q}{1-p}(t+\epsilon)^{1-p-q}.$$

On the second half, we set $r = t - s$ and use $s + \epsilon \geq (t + \epsilon)/2$; this half is at most

$$2^p (t+\epsilon)^{-p} \int_0^{t/2} (r+\epsilon)^{-q}\, dr.$$

The last integral is at most $(t+\epsilon)^{1-q}/(1-q)$ if $q < 1$, it equals $\log(1 + t/(2\epsilon))$ if $q = 1$, and it is at most $\epsilon^{1-q}/(q-1)$ if $q > 1$. The first half is absorbed in the bounds for the second half: for $q = 1$, it is of order $(t+\epsilon)^{-p}$, and for $q > 1$ we use $(t+\epsilon)^{1-q} \leq \epsilon^{1-q}$. This proves the general estimate for all t and ϵ.

For $p = q = 1/2$ and $t > 0$, we decrease the two positive shifts to zero and obtain

$$I^{\epsilon}_{1/2,1/2}(t) \leq \int_0^t s^{-1/2}(t-s)^{-1/2}\, ds = \int_0^{\pi/2} 2\, d\vartheta = \pi,$$

where we substituted $s = t \sin^2 \vartheta$. At $t = 0$, the bound is trivial.

Now let $\alpha = \alpha_{\mathrm{fl}}$ and $p = (\alpha + 1)/2$. Then $p < 1$, and the cases $q = 3/4$ and $q = 5/4$ of the general estimate yield

$$I^{\epsilon}_{p,3/4}(t) \leq C_\alpha (t+\epsilon)^{-\alpha/2-1/4}, \qquad I^{\epsilon}_{p,5/4}(t) \leq C_\alpha \epsilon^{-1/4} (t+\epsilon)^{-(\alpha+1)/2}.$$

Multiplying by the two weights, we obtain

$$\begin{aligned} (t+\epsilon)^{\alpha-1/2} I^{\epsilon}_{p,3/4}(t) &\leq C_\alpha (t+\epsilon)^{\alpha/2-3/4}, \\ (t+\epsilon)^{\alpha} I^{\epsilon}_{p,5/4}(t) &\leq C_\alpha \epsilon^{-1/4} (t+\epsilon)^{(\alpha-1)/2}. \end{aligned} \tag{16.31}$$

Since both powers of $t + \epsilon$ on the right are negative for α in the stated range, both right sides are at most $C_\alpha \epsilon^{\alpha/2-3/4}$, which equals $C_\alpha a_0^{\alpha-3/2}$ for $\epsilon = a_0^2$. Replacing either weight by the corresponding power of $t + a^2$ decreases it, since $a \leq a_0$ and both powers are positive. The short-time bound (16.30) remains available; in particular, none of these bounds replaces an integral that vanishes at $t = 0$ by a singular expression.

Finally, $p = 3\alpha/2 < 1$, and the cases $q = 0$ and $q = 1/2$ of the general estimate give $(t+\epsilon)^{1-3\alpha/2}$ and $(t+\epsilon)^{1/2-3\alpha/2}$, respectively. All constants in this proof are built from 2^p, 2^q, $(1-p)^{-1}$, $|1-q|^{-1}$ (when $q \neq 1$), and fixed powers of two; they do not depend on any mesh ratio, coupling, lattice side, or depth. □

16.6 Heat consistency for finite-scale reconstructions

The last lemma of this section compares the discrete heat evolution of a sampled smooth field with the continuum heat evolution. It is applied in Section 25 to the cardinal interpolation of a finite-scale reconstruction, that is, of a field reconstructed from a fixed finite number of renormalization group steps. The family of fields is fixed before the mesh tends to zero, and no uniformity is claimed as the number of scales grows. The

comparison concerns the linear heat evolution only; the corresponding statement for the quadratic Picard term is proved in Section 25.

For a smooth periodic function f we write $\|f\|_{C^r}$ for the maximum of the supremum norms of its derivatives of order at most r. For a mesh $a = L/n$ we write $S_a f = f|_{\Gamma_a}$ for the restriction of f to the vertices.

Lemma 16.9. *Fix $L, T, H > 0$ with $L/H \in \mathbb{N}$, a finite collection of meshes $a_1, \dots, a_l$ commensurate with H, and a deterministic bound M_H. Let $\mathscr{F}_H$ be the set of functions $f = \sum_{j=1}^{l} I_{a_j} u^{(j)}$, where each $u^{(j)}$ is an array of link variables on the lattice of mesh a_j with $\|u^{(j)}\|_\infty \le M_H$; it is a bounded finite-dimensional family of smooth periodic functions. Let $a \downarrow 0$ run through meshes $a = L/n$ such that $a_j/a \in \mathbb{N}$ for every j. Then, uniformly in $f \in \mathscr{F}_H$ and $0 \le t \le T$,*

$$\|P_t^a S_a f - S_a P_t f\|_\infty \le C_H a^2,$$

and for every ordered product D_a^ϵ of $q \ge 1$ forward or backward differences, with corresponding product of partial derivatives ∂^ϵ,

$$\|D_a^\epsilon P_t^a S_a f - S_a \partial^\epsilon P_t f\|_\infty \le C_{H,q} a.$$

For a general smooth periodic function f the two right sides may be replaced by $CTa^2\|f\|_{C^4}$ and $C_q\big[a\|f\|_{C^{q+1}} + Ta^2\|f\|_{C^{q+4}}\big]$, respectively. These bounds include $t = 0$, the periodic seams, and the constant mode.

Let $u_a(t) = P_t^a S_a f$ and $u_a^\epsilon(t) = D_a^\epsilon P_t^a S_a f$, and let $I_a u_a(t)$ and $I_a u_a^\epsilon(t)$ be their cardinal interpolations on $\mathbb{T}_L^3$, each array being interpolated separately. Then $I_a u_a \to P_t f$ and $I_a u_a^\epsilon \to \partial^\epsilon P_t f$ uniformly on $[0, T] \times \mathbb{T}_L^3$, and the same convergence holds after multiplying the discrete and the continuum quantities by the weights $(t + a^2)^\delta$ and t^δ, respectively, for any $\delta > 0$. In particular, the separately interpolated value and difference arrays converge in the positive-time weighted heat norms of Definition 4.8.

The constants depend on L, T, q, the meshes $a_1, \dots, a_l$, the function b, and M_H, but not on the finer mesh a, the coupling, the depth, or the number of steps of the whole reconstruction.

Remark 16.10. The convergence in Lemma 16.9 concerns the value array and each difference array, interpolated separately. It is not a statement about the derivatives of the interpolated value field, and such a statement would be false. Indeed, $\zeta'(k) = 0$ for every integer k: at $k = 0$ because ζ is even, at $k = \pm 1$ because all derivatives of ζ vanish there, and elsewhere because ζ vanishes near k. It follows that $\partial_i(I_a u)$ vanishes at every vertex of Γ_a for every array u, and it cannot converge uniformly to $\partial_i f$ even when $u = S_a f$. The proof below uses no identity between D_a^ϵ and derivatives of I_a.

Proof of Lemma 16.9. The fields are deterministic. By Lemma 16.3, for every integer $r \ge 0$, the functions in $\mathscr{F}_H$ satisfy

$$\sup_{f \in \mathscr{F}_H} \|f\|_{C^r} \le C_{b,r} M_H \sum_{j=1}^{l} (1 + a_j^{-r}), \tag{16.32}$$

and we denote the right side by $K_{H,r}$. Since there are only finitely many link variables on the meshes a_j, the family $\mathscr{F}_H$ is bounded and finite-dimensional; no compactness argument is needed for the uniform estimates below.

Taylor's formula with integral remainder in the coordinate i gives, for a smooth periodic function f,

$$\|D^-_{a,i}D^+_{a,i}S_af - S_a\partial_i^2 f\|_\infty \le Ca^2\|\partial_i^4 f\|_\infty.$$

To see the cancellation, we expand $f(x+ae_i)$ and $f(x-ae_i)$ to order three at x; the odd terms cancel, and the two remainders of order four, divided by a^2, are at most $Ca^2\|\partial_i^4 f\|_\infty$. Summing over i, we obtain

$$\|\Delta_a S_a f - S_a\Delta f\|_\infty \le Ca^2\|f\|_{C^4}. \tag{16.33}$$

The Taylor expansions are carried out on the periodic lift of f to $\mathbb{R}^3$, so they apply at every seam with the same constants.

The discrete and the continuum semigroups both commute with their own constant-coefficient derivatives, and both contract the supremum norm; for the continuum semigroup, this follows from its convolution kernel in Lemma 16.1. Differentiating $P^a_{t-s}S_aP_sf$ in s and integrating, we obtain the identity

$$P^a_tS_af - S_aP_tf = \int_0^t P^a_{t-s}(\Delta_aS_a - S_a\Delta)P_sf\,ds. \tag{16.34}$$

The differentiation is legitimate because f is smooth and the discrete space is finite. Since $\|P_sf\|_{C^4} \le \|f\|_{C^4}$, inserting (16.33) into (16.34), we obtain the bound $CTa^2\|f\|_{C^4}$, uniformly on $[0,T]$.

Fix an ordered product of q differences D^ϵ_a and the corresponding product of derivatives ∂^ϵ. Using (16.10) repeatedly, we write $D^\epsilon_aS_af(x)$ as an average of $\partial^\epsilon f$ over shifts of x of total length at most qa. The fundamental theorem of calculus along these shifts then gives

$$\|D^\epsilon_aS_af - S_a\partial^\epsilon f\|_\infty \le C_qa\|f\|_{C^{q+1}}. \tag{16.35}$$

We commute the differences with P^a_t and compare the differentiated initial data:

$$D^\epsilon_aP^a_tS_af - S_a\partial^\epsilon P_tf = P^a_t\big(D^\epsilon_aS_af - S_a\partial^\epsilon f\big) + \big(P^a_tS_a\partial^\epsilon f - S_aP_t\partial^\epsilon f\big).$$

The first term is bounded by (16.35), and the second by the value estimate already proved, applied to $\partial^\epsilon f$, and we obtain $C_q\{a\|f\|_{C^{q+1}} + Ta^2\|f\|_{C^{q+4}}\}$. This argument does not divide an error in the values by a. It includes $t=0$, and both errors vanish identically for constant f. Together with (16.32), this proves the two estimates for the family $\mathscr{F}_H$; since $a \le L$, the term of order a^2 is absorbed into $C_{H,q}a$.

For the interpolation statement, we use the nonnegative partition of unity and its support radius: for every smooth periodic g and every array u,

$$\|I_aS_ag - g\|_\infty \le Ca\|\nabla g\|_\infty, \qquad \|I_au\|_\infty \le \|u\|_\infty. \tag{16.36}$$

We apply the first bound to $g = P_t f$ and $g = \partial^{\epsilon} P_t f$, and the second to the differences between the discrete arrays and the sampled continuum functions. The smooth norms of these continuum functions are bounded uniformly by (16.32) and heat contraction. Then the interpolated value array and each separately interpolated difference array converge uniformly on $[0, T] \times \mathbb{T}_L^3$ to the corresponding continuum functions.

For a positive power δ, the weights $(t + a^2)^{\delta}$ are bounded by $(T + L^2)^{\delta}$ and converge to t^{δ} uniformly on $[0, T]$ as $a \downarrow 0$, by the uniform continuity of $v \mapsto v^{\delta}$ on $[0, T + L^2]$. Multiplying the uniform errors in the values and the differences by the weights of Definition 4.8 therefore preserves the convergence, also when one compares the weighted discrete and the weighted continuum functions themselves. All constants depend only on the meshes a_j, M_H, b, L, T, and q. □

SECTION 17

Backward reconstruction and conditional quadratic moments

This section proves three estimates on the coarse-to-fine reconstruction of the fine field under the conditioned Wilson measure. They are used in Sections 26 and 27 to control the heat evolution of the reconstructed field and its quadratic Picard term, and also in Sections 28 and 31. Throughout, the torus, the coarsest mesh H, the continuum coupling $g_3 > 0$, and a time horizon T_0 are fixed, and all estimates are uniform in the number k of renormalization group steps between the coarsest and the finest lattice.

The purpose of the three estimates. Recall from Section 3.3 that the fine field is reconstructed from the coarsest field one renormalization group step at a time, from coarse to fine. At the jth step, the logarithm θ_j of the reconstructed field on the lattice of mesh a_j is obtained from the logarithm θ_{j-1} on the lattice of mesh $a_{j-1} = Ba_j$ and from the normalized fluctuation variables w_j of that step. Let $A_j = a_j^{-1}\theta_j$ be the lattice potential, and let C_j be the zero-fluctuation reconstruction in the same units, that is, the reconstruction with $w_j = 0$. We write $\overline{P}$, $\overline{Q}$, and $\overline{R}$ for the linear, quadratic, and cubic parts of the zero-fluctuation reconstruction defined in Lemma 17.2, so that $C_j = a_{j-1}^{-1}[\overline{P}\theta_{j-1} + \overline{Q}(\theta_{j-1}, \theta_{j-1}) + \overline{R}(\theta_{j-1})]$. Viewing all terms as functions that are constant on lattice cells, we obtain the exact decomposition

$$A_j - E_{a_{j-1}} A_{j-1} = \big(A_j - \mathbb{E}[A_j \mid \mathcal{F}_{j-1}]\big) + m_j + D_d A_{j-1} + a_{j-1}^{-1}\overline{Q}(\theta_{j-1}, \theta_{j-1}) + a_{j-1}^{-1}\overline{R}(\theta_{j-1}). \tag{17.1}$$

Here $E_{a_{j-1}}$ is the cell embedding on the coarser lattice, and $\mathcal{F}_{j-1}$ is generated by the coarsest field and the fluctuation variables of the first $j-1$ steps. The term $m_j = \mathbb{E}[A_j \mid \mathcal{F}_{j-1}] - C_j$ is the conditional mean correction (in the notation of Sections 26 and 34, $m_j = m_j^{\mathrm{ph}}$), and $D_d A_{j-1}$ is the difference between the linearized reconstruction of A_{j-1} and its cell embedding; the precise definitions are given below. The first term is controlled by Theorem 16.6. Lemma 17.1 gives a deterministic bound on θ_j, the reconstruction amplitude bound, which tends to zero with the mesh. It also bounds m_j and shows that the sum over j of $A_j - C_j$, which is the sum of the first two terms, is small in a negative-regularity norm. Lemma 17.2 shows that each column of the linear operator D_d has zero integral and is localized at the scale a_{j-1}, so that heat smoothing at time t gains the factor $a_{j-1}/\sqrt{t}$. Theorem 17.4 controls the quadratic term. The cubic term is

controlled by the reconstruction amplitude bound on θ_{j-1}.

Inputs and attribution. The deterministic inputs are the reconstruction maps of Theorem 15.12, with the representation of the conditioned Wilson measure in Lemma 15.10, the preservation of the conditional kernels in Theorem 15.13, and the derivative bounds of Lemma 15.15. The zero-fluctuation reconstruction is built from the classical background fields of Bałaban [16], and the averaging is a version of Bałaban's averaging operation [12]. The probabilistic inputs are the bounds on conditional means and covariances of Theorem 13.10, the concentration theorem Theorem 5.4, and the results of Section 16. The concentration theorem rests on the convexity methods of Brascamp and Lieb [33], Gross [103], and Bakry and Émery [8]. The gain from a column of zero integral is the standard cancellation of a localized kernel with vanishing mass, and the maximal inequalities are elementary chaining in the sense of Dudley [79] (see also [74]), in the form of Lemma 16.4.

17.1 Setting

Order of the parameters. We use the constructions of Section 15 with the following order of choices. The constant $C_{\mathrm{CP}} \geq 1$ of Theorem 15.12, which bounds the first three derivatives in the coarse field and the first spatial difference of the zero-fluctuation reconstruction, is fixed before J. Before J is chosen, it may be enlarged to include finitely many further bounds of the same kind, as in the proofs below. Then J is chosen so that $q_{\mathrm{anc}} = C_{\mathrm{CP}}^2/B < 1$, where $B = M^J$; Theorem 15.12 allows $q_{\mathrm{anc}} \leq 1/4$. The larger analytic domains and their coefficient bounds, which may depend on J, are fixed next. Then the radii are chosen as in Lemma 15.15,

$$0 < r < r_{\mathrm{eff}}/4, \qquad \frac{\beta_J r_{\mathrm{eff}}}{1-\alpha_*} < \frac{1}{8}, \qquad \alpha_* = C_{\mathrm{CP}}/B,$$

and the bounds on the coupling-scaled fluctuation inputs and on the coupling are chosen so that every reconstructed field lies in the working domain of that lemma. Finally, the upper bound g_* on the couplings is fixed. It includes the bound of Theorem 16.6 with $R_{\mathrm{supp}} = r$, it ensures that $p_j \geq 1$, and it makes $t_j p_j^3$ as small as required in Theorem 13.10, in the notation introduced next.

The coarse-to-fine reconstruction. Fix $L > 0$, $T_0 > 0$, a coarsest mesh $H = L/n_0$ with $n_0 \in 4\mathbb{N}$, and $g_3 > 0$ such that $t_1 \leq g_*$, in the notation introduced below; here $\gamma = g_3^2 L$. We also fix the Borel choice of gauge of Lemma 15.10, which carries every stopping field in D_r into B_r and does not depend on k, and the reconstruction maps $\widehat{\mathscr{R}}$ of Theorem 15.12. We write $\widehat{\Phi}_{d,j}$ for their logarithmic form at depth d, at the jth step. For $k \geq 1$, let $a_j = HB^{-j}$ and $d_j^k = J(k-j)$ for $0 \leq j \leq k$. For $1 \leq j \leq k$, we define

$$g_j = g_3\sqrt{a_j}, \qquad g_{j,s} = M^{s/2} g_j \quad (0 \leq s < J), \qquad t_j = M^{(J-1)/2} g_j, \qquad p_j = A_{\mathrm{w}} \log(1/t_j).$$

Since $t_j = t_1 B^{-(j-1)/2}$, we have $t_j \le t_1 \le g_*$ at every fluctuation step. Here g_j^2 is the lattice coupling at mesh a_j, $g_{j,s}$ is the coupling at the intermediate scale s of the jth step, and t_j is the largest of these couplings. Further, p_j is the radius of the fluctuation cutoff domain of the jth step, and d_j^k is the number of averaging steps that precede it, counted from the finest lattice. The reconstruction is

$$\theta_j = \widehat{\Phi}_{d_j^k,j}\big(\theta_{j-1}, (g_{j,s} w_{j,s})_{s=0}^{J-1}\big), \qquad \mathcal{F}_j = \sigma(\theta_0, w_1, \dots, w_j) \qquad (1 \le j \le k), \tag{17.2}$$

where θ_0 is the logarithm of the stopping field in the chosen gauge, and $w_j = (w_{j,s})_s$ are the normalized fluctuation variables of the jth step. The meshes and depths are

$$(\theta_0;\, H;\, Jk) \longrightarrow (\theta_1;\, H/B;\, J(k-1)) \longrightarrow \cdots \longrightarrow (\theta_k;\, H/B^k;\, 0).$$

The measure. Let $N = n_0 B^k$, and let $\nu = \nu_{N,n_0,\gamma}$ be the conditioned Wilson measure. By Lemma 15.10, the gauge orbits of the fields $(e^{\theta_0}, \dots, e^{\theta_k})$ have exactly the law of the averaged configurations under ν. The conditional law of w_j given $\mathcal{F}_{j-1}$ is the conditional fluctuation kernel $k(e^{\theta_{j-1}}, dw)$ of Theorem 13.8 at the coarse field $e^{\theta_{j-1}}$, and the reconstruction maps preserve it (Theorem 15.13). This kernel has a density $Z^{-1}\mathbf{1}_\Omega e^{-V}$ on a fixed product Ω of three-dimensional balls of radius p_j, where V is smooth on a neighborhood of the closed product and $D_w^2 V \ge I/2$ there. It is not a product of independent Gaussian measures. All conditional expectations below are taken under ν. The marginal law of θ_{j-1} under ν includes the restrictions at all coarser steps, and it is in general not a stopped measure. Once $\mathcal{F}_{j-1}$ is fixed, these restrictions are fixed factors, and they cancel in the conditional law of w_j.

At each step, the coarse field $e^{\theta_{j-1}}$ lies in B_r, after the change of gauge of Lemma 15.10 if it is given in D_r. By the strict domain inclusions established in Lemma 15.15 and Theorem 15.13, the reconstruction of Theorem 13.2 and the reconstruction maps act on the same product. All segments of coarse fields and of coupling-scaled fluctuation inputs used below lie in the larger domains on which the derivative bounds hold. We use Theorem 16.6 with $R_{\mathrm{supp}} = r$, together with its change-of-gauge assertion.

Norms. Every array is embedded as a componentwise constant function on the half-open cells of its own lattice, each positive link being attached to its starting vertex. We use the heat norm

$$\|f\|_{-\alpha;T_0} = \sup_{0<s\le T_0} s^{\alpha/2} \|P_s f\|_\infty, \tag{17.3}$$

where P_s is the continuum heat semigroup acting componentwise; it keeps the constant mode. Since the heat kernel is positive and has unit mass, we have

$$\|f\|_{-\alpha;T_0} \le T_0^{\alpha/2} \|f\|_\infty \qquad (\alpha > 0). \tag{17.4}$$

Discrete heat semigroups carry their mesh as a superscript, as in Section 16. For linear and quadratic coefficient arrays, we use the row sums and the volume-normalized column

sums of Definition 4.6: a column of an input on a lattice of mesh a_{in} is multiplied by $a_{\mathrm{out}}^3/a_{\mathrm{in}}^3$.

Constants. We vary the depth at fixed L, T_0, H, and g_3, and all limits are taken at these fixed values. Constants have the dependence specified in each result: the lemma on conditional means allows dependence on the upper bound on the coupling, the lemma on column cancellation has the independence stated there, and the theorem on quadratic moments allows dependence on the fixed positive g_3 and the fixed H. No stronger statement, uniform as H varies or as $g_3 \downarrow 0$, is claimed; the bounds of Lemma 15.15 that are uniform in H are not used to improve this dependence. A subscript J indicates a constant that may depend on J; such constants are chosen after J. The applications of Lemmas 16.1 and 16.3 use the horizon $T = T_0$, and their constants therefore depend on T_0/L^2.

17.2 Conditional means along the reconstruction

The first lemma bounds the reconstructed fields and their conditional mean corrections. The stopping-field contribution to the reconstruction amplitude bound is $r(a_j/H)^{\kappa_{\mathrm{term}}}$, where κ_{term} is determined by the contraction factor $a_{\mathrm{cap}} = C_{\mathrm{CP}}/B$ of the linearized reconstruction; the fluctuation contribution is bounded by $C_J g_3 \sqrt{a_j}\, p_j$. The conditional mean correction is at most $C_J g_3^2$ times the size of the coarser field. Indeed, by Theorem 13.10, the conditional mean of the reconstructed link variables differs from the zero-fluctuation reconstruction by $O(t_j^2 \|\theta_{j-1}\|_\infty)$, and t_j^2/a_j is a fixed multiple of g_3^2.

Lemma 17.1. *Assume the setting of Section 17.1, and let the constant $C_{\mathrm{CP}} \geq 1$, fixed before J, satisfy*

$$a_{\mathrm{cap}} = C_{\mathrm{CP}}/B, \qquad a_{\mathrm{cap}}\sqrt{B} < 1, \qquad \kappa_{\mathrm{term}} = -\frac{\log a_{\mathrm{cap}}}{\log B} > \frac{1}{2}.$$

For $1 \leq j \leq k$ set

$$A_j = a_j^{-1}\theta_j, \qquad C_j = a_j^{-1}\widehat{\Phi}_{d_j^k, j}(\theta_{j-1}, 0), \qquad m_j = \mathbb{E}[A_j \mid \mathcal{F}_{j-1}] - C_j,$$

each embedded as a componentwise constant function on the cells of its own lattice. Then, on every sample of the conditioned Wilson measure,

$$b_j = \|\theta_j\|_\infty \leq \overline{b}_j, \qquad \overline{b}_j = r(a_j/H)^{\kappa_{\mathrm{term}}} + C_J g_3 \sqrt{a_j}\, p_j, \qquad \|m_j\|_\infty \leq C_J g_3^2 b_{j-1}.$$

For $2 \leq l \leq k$ let

$$M_l = C_J g_3^2 \bigl\{ r a_{\mathrm{cap}}^{l-1} + g_3 \sqrt{a_{l-1}}\,(p_l + 1) \bigr\}.$$

Then $\sum_{j=l}^{k} \|m_j\|_\infty \le M_l$, *and* $M_l \to 0$ *as* $l \to \infty$, *uniformly in* $k \ge l$. *For every* $1/2 < \alpha < 3/2$ *and* $u \ge 0$,

$$\sum_{j=l}^{k} \mathbb{E}\|A_j - C_j\|_{-\alpha;T_0} \le C_{J,\alpha} g_3 a_l^{\alpha-1/2} \sqrt{\log(2 + L/a_l)} + C_{\alpha,T_0} M_l,$$

and, with probability at least $1 - Ce^{-u}$, *simultaneously for all terms of this finite sum,*

$$\sum_{j=l}^{k} \|A_j - C_j\|_{-\alpha;T_0} \le C_{J,\alpha} g_3 a_l^{\alpha-1/2} \big[\sqrt{\log(2 + L/a_l)} + \sqrt{u}\big] + C_{\alpha,T_0} M_l.$$

These estimates also hold after integration over the law of the stopping field under the conditioned Wilson measure, uniformly in k. *The constants depend on the fixed construction parameters, on* L, T_0, α, H, *and on the upper bound on the coupling, but not on* j, k, *the depth, or the choice of the representative of the stopping field.*

The factors g_3 and the powers of the mesh are displayed explicitly. The exponent κ_{term} is defined by the contraction of the linearized reconstruction and is unrelated to the exponent $\kappa_{\mathrm{fl}} = 1 - 3\alpha_{\mathrm{fl}}/2$ of the flow norms. The lemma controls $A_j - C_j$, the deviation of the reconstructed field from its zero-fluctuation reconstruction; the adjacent difference $A_j - A_{j-1}$ contains in addition the last three terms of (17.1), which are treated in Sections 17.3 and 17.4.

Proof. All constants have the dependence allowed in the statement. A constant with subscript J may include the finitely many ratios of couplings within a step and the factor $(1 - a_{\mathrm{cap}}\sqrt{B})^{-1}$. The proof has four steps: the reconstruction amplitude bound, its summation, the bound on the conditional mean in the gauge of the reconstruction maps, and the bounds on the centered part.

Step 1: the reconstruction amplitude bound. The recurrence and its summation in Steps 1 and 2 are also stated in Lemma 15.10. We repeat the short computation because the distinction between the constant $C_{\mathrm{rec},J}$ of the recurrence and the constant $C_{\mathrm{cap},J}$ of its solution is used again in the proof of Theorem 17.4. We first explain the choice of C_{CP}. Let K_1, K_2, and K_3 be the bounds, independent of J, from Lemma 15.15 for the linear term, the quadratic term, and the cubic remainder of the normalized zero-fluctuation reconstruction $\overline{A}_d$ (see Lemma 17.2) on its fixed inner ball. For a coarse field of norm $R \le 1$, the value of $\overline{A}_d$ is at most $K_1 R + K_2 R^2 + K_3 R^3 \le (K_1 + K_2 + K_3) R$. We include this sum in C_{CP}; the working ball is chosen inside the inner ball, with $R \le 1$, after J is fixed. No constant that depends on all orders of differentiation at fixed J is used to choose J. With this choice, the bound

$$\|\widehat{\Phi}_{d,j}(\theta, q)\|_\infty \le a_{\mathrm{cap}} \|\theta\|_\infty + C_{\mathrm{rec},J} \max_s \|q_s\|_\infty$$

of Lemma 15.10 holds with the same a_{cap}; here $C_{\mathrm{rec},J}$ is the constant of that lemma for the coupling-scaled fluctuation inputs.

Set $\varrho = a_{\mathrm{cap}}\sqrt{B} < 1$. Since $|g_{j,s}w_{j,s}| \le t_j p_j$ on the fluctuation cutoff domain, the last bound implies that, at every point of the support of the conditioned measure,

$$b_j \le a_{\mathrm{cap}} b_{j-1} + C_{\mathrm{rec},J} t_j p_j, \qquad b_0 \le r.$$

Let $c_B = M^{(J-1)/2}$, so that $t_j = c_B g_3 \sqrt{a_j}$, and fix

$$C_{\mathrm{cap},J} \ge \frac{c_B C_{\mathrm{rec},J}}{1-\varrho}. \tag{17.5}$$

Both $C_{\mathrm{rec},J}$ and $C_{\mathrm{cap},J}$ are chosen after J, and neither changes C_{CP}. Since $t_j = t_1 B^{-(j-1)/2}$, we have

$$p_j = p_1 + c_{\mathrm{w}}(j-1), \qquad c_{\mathrm{w}} = \tfrac{1}{2} A_{\mathrm{w}} \log B > 0; \tag{17.6}$$

in particular p_j is increasing in j. Iterating the recurrence, with no exceptional set, and using $a_{\mathrm{cap}}^{j-i}\sqrt{a_i} = \varrho^{j-i}\sqrt{a_j}$ and $p_i \le p_j$ for $i \le j$, we obtain

$$b_j \le r a_{\mathrm{cap}}^{j} + c_B C_{\mathrm{rec},J} g_3 \sqrt{a_j} \sum_{i=1}^{j} \varrho^{j-i} p_i \le r a_{\mathrm{cap}}^{j} + C_{\mathrm{cap},J} g_3 \sqrt{a_j}\, p_j.$$

Since $a_{\mathrm{cap}}^{j} = B^{-\kappa_{\mathrm{term}} j} = (a_j/H)^{\kappa_{\mathrm{term}}}$, this is the asserted bound, with a constant $C_J \ge C_{\mathrm{cap},J}$ that may also dominate the constants of the summation and of the mean below. The condition $\kappa_{\mathrm{term}} > 1/2$ is equivalent to $\varrho < 1$.

Step 2: summation of the amplitude bound. Let $q_B = B^{-1/2}$ and $j = l + h$. By the amplitude bound, (17.6), and $a_{j-1} = a_{l-1} B^{-h}$, we have

$$\sum_{j=l}^{k} b_{j-1} \le \frac{r a_{\mathrm{cap}}^{l-1}}{1-a_{\mathrm{cap}}} + C_{\mathrm{cap},J} g_3 \sqrt{a_{l-1}} \sum_{h \ge 0} q_B^h (p_l + c_{\mathrm{w}} h) \le C_J \{ r a_{\mathrm{cap}}^{l-1} + g_3 \sqrt{a_{l-1}} (p_l + 1) \}.$$

The two series equal $(1-q_B)^{-1}$ and $q_B(1-q_B)^{-2}$. The constant does not depend on k or on the sample.

Step 3: the conditional mean in the gauge of the reconstruction maps. Conditionally on $\mathcal{F}_{j-1}$, the coarse field is fixed, the gauge transformation of the reconstruction maps depends only on it, and the conditional fluctuation kernel is the same as for the reconstruction of Theorem 13.2. Equality of the kernels does not by itself identify the two reconstructed means, because the observable changes; we show that the bound on the mean survives the change of gauge. We write $\phi = \theta_{j-1}$ and $R = \|\phi\|_\infty$. Let $X_e(w)$ be the

logarithm of the reconstruction of Theorem 13.2 on one output link e, and let $x_e = X_e(0)$. By the bound on the reconstructed mean in Theorem 13.10, we have

$$|\mathbb{E}X_e - x_e| \le C_J g_j^2 R,$$

where we used $t_j^2 = c_{B\mathcal{G}}^2 g_j^2$. The variance of X_e is at most $C_J g_j^2$, by Theorem 16.6, applied under the conditional fluctuation kernel at the coarse field $e^\phi \in B_r$, with fine mesh a_j and horizon T_0. Indeed, we apply its variance bound to the test function $a_j^{-3}\mathbf{1}_{Q_e}$ in one color component of X/a_j, whose squared L^2 norm is a_j^{-3}, and multiply by a_j^2. Summing over the three colors, we obtain

$$\mathbb{E}|X_e - x_e|^2 \le C_J g_j^2 + C_J g_j^4 R^2 \le C'_J g_j^2. \tag{17.7}$$

The reconstruction of Theorem 13.2 satisfies the hypotheses of both theorems used here.

At the fixed coarse field, the reconstruction maps differ from this reconstruction by a gauge transformation given by matrices h_x at the vertices. On the link $e = (x, y)$, the logarithm of the reconstruction map is $H_e(X_e)$, where $H_e(X) = \mathrm{Log}(h_x e^X h_y^{-1})$. By the bounds of Theorems 13.2 and 15.12, the two zero-fluctuation reconstructions x_e and $\widehat{x}_e = H_e(x_e)$ have norm at most CR. The identity $e^{\widehat{x}_e} = h_x e^{x_e} h_y^{-1}$ therefore implies $|\mathrm{Log}(h_y^{-1}h_x)| \le CR$. Let $z_e = \mathrm{Log}(h_y^{-1}h_x)$. By conjugation, we have

$$H_e(X) = \mathrm{Ad}(h_x)\,\mathrm{Log}(e^X e^{z_e}).$$

On the fixed small ball of logarithms, the analytic series of the exponential and the logarithm give bounded derivatives of $(X, z) \mapsto \mathrm{Log}(e^X e^z)$ through order three. More precisely, we enlarge this ball within the fixed chart so that $\|e^X e^z - I\| \le q < 1$ in matrix norm. Termwise differentiation of $\mathrm{Log}(I + Z) = \sum_{m\ge1}(-1)^{m+1}Z^m/m$ is justified by the convergent majorants $\sum_{m\ge r} m^{r-1}q^{m-r}$, $r = 1, 2, 3$, and the derivatives of the exponential are bounded by its absolutely convergent series on this ball. These bounds depend only on the chart. At $z = 0$, the function equals X, and its second derivative in X vanishes there. Integrating the mixed derivative $D_z D_X^2$ along the segment sz_e, $0 \le s \le 1$, we therefore find $\|D_X^2 H_e\| \le CR$, while $\|D_X H_e\| \le C$. All segments used here lie in the fixed chart. Since the matrices $\mathrm{Ad}(h_x)$ are orthogonal, these constants do not depend on the matrices h_x or on the number of vertices. By Taylor's formula at x_e, the bound on the mean, and (17.7), we obtain

$$|\mathbb{E}H_e(X_e) - H_e(x_e)| \le C|\mathbb{E}(X_e - x_e)| + CR\,\mathbb{E}|X_e - x_e|^2 \le C_J g_j^2 R.$$

This is the bound on the mean for the reconstruction maps, on the same product of balls. Dividing by the fine mesh, we obtain

$$\|m_j\|_\infty \le C_J \frac{t_j^2}{a_j} b_{j-1} \le C_J g_3^2 b_{j-1}.$$

Together with Step 2, this implies $\sum_{j=l}^{k} \|m_j\|_\infty \le M_l$, after one enlargement of C_J. Both terms of M_l tend to zero as $l \to \infty$, because $a_{\mathrm{cap}} < 1$ and p_l grows linearly in l while $\sqrt{a_{l-1}}$ decays geometrically.

Step 4: the centered part. We define

$$F_j = A_j - \mathbb{E}[A_j \mid \mathcal{F}_{j-1}].$$

Theorem 16.6 applies to F_j conditionally on $\mathcal{F}_{j-1}$. Indeed, the fine mesh is a_j, the side of the coarse lattice of the step is $n_0 B^{j-1} \ge 4$, the depth is d_j^k, the largest coupling is $t_j \le g_*$, the coarse field is in B_r, the horizon is T_0, and the reconstruction maps satisfy the hypotheses of that theorem on the whole product. The same holds for the reconstruction of Theorem 13.2, which we used in Step 3. Let $\beta = \alpha - 1/2 > 0$ and $\ell_j = \log(2 + L/a_j)$. Taking the conditional expectation and then the expectation, we obtain

$$\mathbb{E}\|F_j\|_{-\alpha;T_0} \le C_{J,\alpha} g_3 a_j^\beta \sqrt{\ell_j}.$$

For $h \ge 0$, we have $\ell_{l+h} \le \ell_l + h \log B$ and $a_{l+h} = a_l B^{-h}$. It follows that

$$\sum_{h\ge 0} a_{l+h}^\beta \sqrt{\ell_{l+h}} \le a_l^\beta \Big\{ \frac{\sqrt{\ell_l}}{1 - B^{-\beta}} + \sqrt{\log B} \sum_{h\ge 0} B^{-\beta h} \sqrt{h} \Big\} \le C_{B,\beta} a_l^\beta \sqrt{\ell_l}. \tag{17.8}$$

The last series converges, for instance because $\sqrt{h} \le h + 1$, and $\ell_l \ge \log 2$ absorbs its contribution. Since $A_j - C_j = F_j + m_j$, the bound on the expectation follows from (17.4) applied to m_j and from Step 3.

For the bound in probability, we use the tail bound of Theorem 16.6 at level $j = l + h$ with $u + 2h \log 2$ in place of u. Integrating the conditional probability does not change its upper bound. By a union bound over the finitely many levels, the probability of failure is at most

$$C e^{-u} \sum_{h\ge 0} 4^{-h} \le C' e^{-u}.$$

On the complementary event, we sum the thresholds, using $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$, (17.8), and the convergent series $\sum_h B^{-\beta h}$ and $\sum_h B^{-\beta h}\sqrt{h}$, and we add the deterministic sum of the means by (17.4). This proves the bound in probability. No independence between levels has been used.

Since all conditional estimates are uniform in the coarse fields that occur and in the stopping field, the same inequalities hold after integration over the law of the stopping field under the conditioned Wilson measure. The argument concerns $A_j - C_j$ throughout, and C_j is never replaced by A_{j-1}. □

17.3 Column cancellation and heat-kernel estimates

The normalization of an input column is essential here. For an operator K from arrays on a lattice of mesh A to componentwise functions on the torus, the volume-normalized absolute column of an input x is $A^{-3}\int |K_x(y)|\,dy$. A zero integral of a column is a separate identity, which no absolute bound implies. On a lattice of mesh δ, the integral is replaced by $\delta^3\sum_y$, still divided by the input volume A^3.

The lemma below concerns the difference D_d between the linearized zero-fluctuation reconstruction and the cell embedding on the coarse lattice. Each column of D_d is concentrated within distance of order A of its input, and it has zero integral. At times $t \gg A^2$, the heat kernel is nearly constant on the scale A, and the zero integral allows us to replace $p_t(z-y)$ by $p_t(z-y)-p_t(z-x)$, which costs the distance from y to x times the gradient of the kernel. This yields the factor $A/\sqrt{t}$.

Lemma 17.2. *Assume the setting of Section 17.1. Let $d \geq 0$, let $A > 0$ be a mesh with $L/A \in \mathbb{N}$ and $L/A \geq 4$, let $a = A/B$, and let $\widehat{\Phi}_d$ be the logarithmic form of the reconstruction map of one renormalization group step at depth d, from the lattice of mesh A to the lattice of mesh a. In units of the coarse lattice, its zero-fluctuation reconstruction has the expansion*

$$\widehat{\Phi}_d(\theta,0) = B^{-1}\overline{A}_d(\theta), \qquad \overline{A}_d(\theta) = \overline{P}_d\theta + \overline{Q}_d(\theta,\theta) + \overline{R}_d(\theta),$$

where $\overline{P}_d = D\overline{A}_d(0)$, $\overline{Q}_d = D^2\overline{A}_d(0)/2$, and $\|\overline{R}_d(\theta)\|_\infty \leq C_{\mathrm{CP}}\|\theta\|_\infty^3$ on a fixed inner ball of coarse fields. The first three derivatives in θ, together with their first spatial differences $\nabla_{1/B}$ in units of the coarse lattice, have weighted row sums and volume-normalized column sums bounded by one constant, uniformly in d, the lattice side, and J, as in Lemma 15.15. Here $\nabla_{1/B} = A\nabla_a$, where ∇_a denotes the first differences at the physical mesh a; thus a physical first difference of $\overline{A}_d$ carries the factor A^{-1}, and a physical first difference of the potential $A^{-1}\overline{A}_d$ carries the factor A^{-2}. The cubic remainder and these finitely many bounds are included in C_{CP} before J is fixed. In physical units the zero-fluctuation reconstruction is $A^{-1}\overline{A}_d(\theta)$.

Let $E_A v$ be the cell embedding on the lattice of mesh A of a coarse lattice potential v, let $P_d v$ be the cell embedding on the lattice of mesh a of the array $A^{-1}\overline{P}_d(Av)$, and let $D_d = P_d - E_A$. For each input link x and input color b, and each output direction and color (i,c), the corresponding column of D_d satisfies

$$\int_{\mathbb{T}_L^3} (D_d)_{(i,c),(x,b)}(y)\,dy = 0, \qquad A^{-3}\int_{\mathbb{T}_L^3} \mathrm{dist}(y,x)\,\big|(D_d)_{(i,c),(x,b)}(y)\big|\,dy \leq C_{\mathrm{CP}}A.$$

Its volume-normalized column sums and its row sums are bounded by C_{CP}, with exponential localization at the scale A. In particular, for $0 < t \leq T_0$,

$$\|P_t D_d v\|_\infty \leq C\min\{1, A/\sqrt{t}\}\,\|v\|_\infty, \qquad \|D_d v\|_{-\alpha;T_0} \leq C_\alpha A^\alpha \|v\|_\infty \quad (0<\alpha<1).$$

Let $m \in \mathbb{N}$ and $\delta = a/m$. Let $P_d^{\mathrm{arr}} v$ be the array on the lattice of mesh a whose cell embedding is $P_d v$, and let

$$D_d^\delta = I_{a\to\delta} P_d^{\mathrm{arr}} - I_{A\to\delta},$$

where the two cardinal interpolations of Lemma 16.3 start from the lattices of meshes a and A, respectively. Then every componentwise column of D_d^δ has zero integral $\delta^3 \sum_y$, its absolute column sums and first moments satisfy the same bounds, and for every integer $q \geq 0$, every product D_δ^q of q physical forward or backward differences, and every $0 \leq t \leq T_0$,

$$\|D_\delta^q P_t^\delta D_d^\delta v\|_\infty \leq C_{J,q}(t + A^2)^{-q/2} \min\{1, A/\sqrt{t + A^2}\} \|v\|_\infty.$$

In the continuum estimates, C depends only on the column and moment bounds, on α where it appears, and on T_0/L^2. In the estimates on the lattice of mesh δ the constant may also depend on J, q, and the function b of the cardinal interpolation. No constant depends on δ/a, the lattice side, the depth, the coupling, or the coarse field. The zero integral is a linear identity for each input column; no zero integral of $\overline{Q}_d$ is asserted.

Proof. The proof has five steps: the expansion, the zero integrals, the localization, the continuum heat estimates, and the estimates on the finer lattice.

Step 1: the expansion and its units. By Lemma 15.15, the reconstruction maps keep the bounds on the first three derivatives of the zero-fluctuation reconstruction in the coarse field and on its first spatial difference in units of the coarse lattice, the volume-normalized columns, and the zero-integral identity of Theorem 15.12. Their common domain contains a fixed inner ball of coarse fields. Taylor's formula on this ball gives

$$\overline{A}_d(\theta) = D\overline{A}_d(0)\theta + \tfrac{1}{2} D^2\overline{A}_d(0)[\theta,\theta] + \overline{R}_d(\theta),$$
$$\overline{R}_d(\theta) = \tfrac{1}{2} \int_0^1 (1-s)^2 D^3\overline{A}_d(s\theta)[\theta,\theta,\theta]\, ds.$$

The value at $\theta = 0$ is zero by global color equivariance. The bound on the third derivative implies $\|\overline{R}_d(\theta)\|_\infty \leq (1/6)\sup\|D^3\overline{A}_d\|\, \|\theta\|_\infty^3$. We enlarge C_{CP} to include this bound, the finitely many derivative bounds, and their first spatial differences in units of the coarse lattice, before J is fixed. This uses the finitely many constants that are uniform in J, not the bounds on all derivatives at fixed J. The conversion to physical units is

$$a^{-1}\widehat{\Phi}_d(\theta, 0) = (A/B)^{-1} B^{-1} \overline{A}_d(\theta) = A^{-1}\overline{A}_d(\theta). \tag{17.9}$$

Step 2: zero integrals. We first restate the zero-integral identity of Theorem 15.12 in these units. The reconstruction satisfies the averaging constraint exactly. Differentiating this identity at $\theta = 0$, we find that the linearized reconstruction preserves the componentwise physical integral of each input column. For each physical input basis vector, at an input link x, with a direction and a color, this gives

$$\int_{\mathbb{T}_L^3} (P_d)_{(i,c),(x,b)}(y)\, dy = \int_{\mathbb{T}_L^3} (E_A)_{(i,c),(x,b)}(y)\, dy.$$

The direction of x is suppressed in the notation, as in the statement. The matching component on the right has integral A^3, and the other components have integral zero. Subtracting, we obtain the zero integral of each column of D_d. This is the exact identity of Theorem 15.12, not a consequence of its absolute bounds. In that identity, the periodic gradient part of the differentiated reconstruction has zero sum, including across the seams of the torus; hence no constant mode has been discarded.

Step 3: localization and the first moment. After a fixed reduction of the exponent of the spatial weight, there exist constants $C, c > 0$, independent of J, such that for every column and every integer $h \geq 0$,

$$A^{-3} \int_{hA \leq \operatorname{dist}(y,x) < (h+1)A} |(D_d)_x(y)|\, dy \leq Ce^{-ch}, \tag{17.10}$$

and the row sums are bounded. These statements follow by subtracting the coarse cell column from the volume-normalized column of P_d, weighted exponentially; moving a point of a cell to the vertex of the cell costs at most $\sqrt{3}A$. The same argument, applied to each weighted row, yields the localization of the rows. Multiplying (17.10) by $(h+1)A$ and summing over h, we obtain

$$A^{-3} \int \operatorname{dist}(y,x) |(D_d)_x(y)|\, dy \leq CA \sum_{h \geq 0} (h+1)e^{-ch} \leq C'A.$$

All sums over directions and colors are finite and included in C.

Step 4: heat-kernel cancellation in the continuum. In this step, all heat estimates use the horizon $T = T_0$ in Lemma 16.1. To pass from the moments of individual columns to a bound on the supremum over outputs, we use the following elementary inequality. For a smooth periodic function f and a cube Q of side A,

$$A^3 \sup_Q |f| \leq C \sum_{S \subset \{1,2,3\}} A^{|S|} \int_Q |\partial_S f(y)|\, dy. \tag{17.11}$$

Indeed, on an interval of length A, we have $|f(x)| \leq A^{-1} \int |f| + \int |f'|$, and applying this successively in the three variables yields (17.11); the endpoints are handled by continuity. If Q is enlarged to the set of points within distance $(h+2)A$, the enlargement is covered by at most $C(h+3)^3$ cubes of side A. Summing the resulting bounds on the supremum over the lattice of inputs costs the same overlap factor $C(h+3)^3$.

Let $A^2 \leq t \leq T_0$ and $f_z(y) = p_t(z-y)$. Since every column has zero integral, we have

$$(P_t D_d v)(z) = \sum_x v_x \int \big[p_t(z-y) - p_t(z-x)\big] (D_d)_x(y)\, dy.$$

On the hth annulus around x, the bracket is at most $(h+1)A$ times the supremum of $|\nabla p_t(z - \cdot)|$ on the enlarged cube around x; a shortest periodic path proves this also

when the annulus crosses a seam. Using (17.10), then (17.11) with the overlap bound, and finally the L^1 bounds on the derivatives of the heat kernel in Lemma 16.1, we obtain

$$\|P_t D_d v\|_\infty \le CA\|v\|_\infty \sum_{h\ge 0} e^{-ch}(h+3)^4 \sum_{r=0}^{3} A^r \|\nabla^{r+1} p_t\|_1 \le C\frac{A}{\sqrt{t}}\|v\|_\infty \qquad (A^2 \le t \le T_0),$$

where in the last inequality we used that each factor $(A/\sqrt{t})^r$ is at most one. For $0 < t \le \min\{A^2, T_0\}$, the contraction of the heat semigroup and the bound on the row sums imply $\|P_t D_d v\|_\infty \le C\|v\|_\infty$. This proves the value estimate, with constants depending only on the column bounds and on T_0/L^2. When $0 < \alpha < 1$, multiplying by $t^{\alpha/2}$ gives at most CA^α in both ranges of time; if $A^2 > T_0$, the bound for small times suffices. This proves the bound on the heat norm.

Step 5: the finer lattice. We now keep the lattices of meshes a and A distinct. Let $m \in \mathbb{N}$ and $\delta = a/m$, so that

$$\frac{a}{\delta} = m \in \mathbb{N}, \qquad \frac{A}{\delta} = Bm \in \mathbb{N}. \tag{17.12}$$

Since $L/A \in \mathbb{N}$ and $L/A \ge 4$, the numbers $L/a = B(L/A)$ and $L/\delta = Bm(L/A)$ are integers at least four. We therefore apply Lemma 16.3 to the pairs of meshes (a,δ) and (A,δ), with integer ratios m and Bm, the same function b, and the horizon $T = T_0$. For a fixed integer $q \ge 0$, D_δ^q denotes any ordered product of q physical forward or backward differences, and the identity if $q = 0$; these are exactly the products allowed in both lemmas of Section 16. Let $p_{z,x}$ be the coefficients of the array P_d^{arr}. By the mass identity of Lemma 16.3, we have, for each componentwise column,

$$\begin{aligned}\delta^3 \sum_y (D_d^\delta)_{y,x} &= \delta^3 \sum_y \sum_{z\in\Lambda_{L/a}} \zeta_3((y-z)/a)\,p_{z,x} - \delta^3 \sum_y \zeta_3((y-x)/A)\,I \\ &= a^3 \sum_z p_{z,x} - A^3 I = 0,\end{aligned} \tag{17.13}$$

where the last equality is the discrete form of the zero integral in Step 2. The identity matrices in this formula involve the matching direction and color only. Since the function b is nonnegative, the mass and moment bounds of the interpolation imply

$$\delta^3 \sum_y \Big|\sum_z \zeta_3((y-z)/a)\,p_{z,x}\Big| \le a^3 \sum_z |p_{z,x}|,$$

$$\delta^3 \sum_y \mathrm{dist}(y,x)\Big|\sum_z \zeta_3((y-z)/a)\,p_{z,x}\Big| \le a^3 \sum_z \big(\mathrm{dist}(z,x) + Ca\big)\,|p_{z,x}|.$$

The interpolated coarse column has mass A^3 and first moment at most CA^4. Both terms of D_d^δ therefore satisfy the bounds of the columns before interpolation, with a fixed enlargement of the constants. Since their supports are displaced by at most $Ca \le CA$,

(17.10) holds with the integral replaced by $\delta^3 \sum_y$. The row sums are bounded as well, by the positive partition of unity of each interpolation.

The argument of Step 4 applies on the lattice of mesh δ, and we only describe its new ingredient. On an interval of $m_A = A/\delta = Bm$ lattice points, averaging the inequality

$$|f(r)| \le |f(s)| + \delta \sum_{u \text{ between } r,s} |D_\delta f(u)|$$

over s yields the one-dimensional analog of (17.11). Taking the product over the coordinates, with an enlargement by one lattice step to include the differences at the boundary, we obtain (17.11) with sums in place of integrals and products of forward differences in place of derivatives. The overlap bound for the enlarged cubes is unchanged. To subtract the kernel, we use (17.13) and a shortest lattice path, whose length is at most a fixed multiple of the torus distance. Then the L^1 bounds on discrete differences of the heat kernel in Lemma 16.1 imply

$$\|P_t^\delta D_d^\delta v\|_\infty \le C \min\{1, A/\sqrt{t}\} \|v\|_\infty \qquad (0 < t \le T_0). \tag{17.14}$$

No constant here depends on a/δ.

For $0 \le t \le \min\{2A^2, T_0\}$, we commute the product of q differences with the heat semigroup and use the derivative bounds of the two interpolations:

$$\|D_\delta^q P_t^\delta D_d^\delta v\|_\infty \le \|D_\delta^q I_{a\to\delta} P_d^{\mathrm{arr}} v\|_\infty + \|D_\delta^q I_{A\to\delta} v\|_\infty \le C_q (a^{-q} + A^{-q}) \|v\|_\infty \le C_{J,q} A^{-q} \|v\|_\infty.$$

For $2A^2 \le t \le T_0$, we split the heat semigroup into two equal halves, use the L^1 bound on the differentiated heat kernel for the first half and (17.14) for the second:

$$\|D_\delta^q P_t^\delta D_d^\delta v\|_\infty \le C_q (t/2 + \delta^2)^{-q/2} \frac{A}{\sqrt{t/2}} \|v\|_\infty \le C_q A t^{-(q+1)/2} \|v\|_\infty.$$

Up to fixed constants, the two ranges of time give the asserted bound with $t + A^2$; they include $t = 0$ and every fixed order of differences. Only the estimate for small times introduces the factor B^q, through $a^{-q} = B^q A^{-q}$, and this factor is allowed in $C_{J,q}$. All the cancellations used in this proof are identities for the columns of linear operators; neither Taylor's formula nor the heat-kernel cancellation yields a zero integral of $\overline{Q}_d$. □

17.4 Conditional quadratic moments

Quadratic maps and their norms. Let Q be a quadratic map from arrays on a lattice of mesh a_{in} to arrays on a lattice of mesh $a_{\mathrm{out}} \le a_{\mathrm{in}}$. We write a scalar output component as

$$Q_y(v, v) = \sum_{x,z} T_{y;x,z} v_x v_z,$$

where the indices include directions and colors. We use the symmetric polarization: replacing $T_{y;x,z}$ by $(T_{y;x,z} + T_{y;z,x})/2$ leaves $Q(v,v)$ unchanged and does not increase any of the coefficient sums below. We write $Q(u,v)$ for the corresponding symmetric bilinear map, and $Q : \mathcal{M}$ for the array with components $\sum_{x,z} T_{y;x,z}\mathcal{M}_{xz}$, for a matrix $\mathcal{M}$ indexed by the inputs. Let $\mathrm{t}(y,x,z)$ be the length of a shortest connected network in $\mathbb{T}_L^3$ containing the locations of y, x, and z, with internal vertices allowed, as in Definition 4.6. We fix a spatial exponent $a'_{\mathrm{sp}} > 0$, smaller than the exponents of the coefficient bounds of Section 15 by a fixed amount, and set

$$\|T\|_{a_{\mathrm{in}},a_{\mathrm{out}}} = \max\Big\{\sup_y \sum_{x,z} e^{a'_{\mathrm{sp}}\mathrm{t}(y,x,z)/a_{\mathrm{in}}}|T_{y;x,z}|,\ \sup_x \frac{a_{\mathrm{out}}^3}{a_{\mathrm{in}}^3}\sum_{y,z} e^{a'_{\mathrm{sp}}\mathrm{t}(y,x,z)/a_{\mathrm{in}}}|T_{y;x,z}|\Big\}. \quad (17.15)$$

By symmetry, the column with the second input fixed has the same bound. In particular, if $a_{\mathrm{in}} = a_i$, $a_{\mathrm{out}} = \delta$, and $\|T\|_{a_i,\delta} \le K_Q$, then

$$\sup_y \sum_{x,z}|T_{y;x,z}| \le K_Q, \qquad \sup_x \frac{\delta^3}{a_i^3}\sum_{y,z}|T_{y;x,z}| \le K_Q. \quad (17.16)$$

The same exponent a'_{sp} is used at every level below; it is not reduced after each composition. At the coarsest level, we set $\overline{b}_0 = r$. For $i \ge 1$, $\overline{b}_i$ always denotes the reconstruction amplitude bound of Lemma 17.1, never a random bound left outside a conditional expectation.

Why a conditional decomposition is needed. For $j \ge 2$, by the reconstruction amplitude bound, the quadratic term $a_{j-1}^{-1}\overline{Q}(\theta_{j-1},\theta_{j-1})$ in (17.1) is at most $Ca_{j-1}^{-1}\overline{b}_{j-1}^2$, which is in general of order $g_3^2 p_{j-1}^2$ and is not summable in j. We therefore regard θ_{j-1} as reconstructed from θ_{j-2}, condition on $\mathcal{F}_{j-2}$, and separate a conditionally centered part, which concentrates, from a conditional expectation error, which is smaller. The conditional expectation is close to the quadratic term evaluated on the linearized reconstruction $L_{j-1}\theta_{j-2}$, and repeating the argument pulls the quadratic coefficient back through the linearized reconstructions. Each pullback contracts the coefficient by the factor a_{cap}^2, one factor for each argument, while dividing by the coarser mesh costs a factor B; the resulting ratio $q_{\mathrm{anc}} = Ba_{\mathrm{cap}}^2 < 1$ makes the final convolution summable. Volume-normalized columns are essential here: with counting column sums, each pullback would lose an additional factor B^3.

Three errors remain in the conditional expectation, and they explain the three terms in the first part of Theorem 17.4 below. Replacing the nonlinear zero-fluctuation reconstruction by its linearization costs $a_{\mathrm{cap}}^2\|\phi\|_\infty^3$. The conditional means of the fluctuation variables and the change of their covariance with the coarse field cost $g_i^2\|\phi\|_\infty$. Here we use that the conditional kernel at the zero coarse field is invariant under constant

color rotations, although it is spatially correlated; an expected adjoint vector built from it therefore vanishes. The terms of higher order in the fluctuations cost $g_i^3 p_i^3$.

The deterministic part of the pullback is the following lemma.

Lemma 17.3. *Let $1 \le i \le j$, let T be the symmetric coefficient array of a quadratic map from arrays on the lattice of mesh a_i to arrays on the lattice of mesh a_j, and let L be the coefficient array of a linear map from arrays on the lattice of mesh $a_{i-1} = Ba_i$ to arrays on the lattice of mesh a_i. Suppose that $\|T\|_{a_i,a_j} \le K$ and that*

$$\sup_x \sum_u e^{a'_{\mathrm{sp}} \operatorname{dist}(x,u)/a_{i-1}} |L_{x,u}| \le a_{\mathrm{cap}}, \qquad \sup_u \frac{a_i^3}{a_{i-1}^3} \sum_x e^{a'_{\mathrm{sp}} \operatorname{dist}(x,u)/a_{i-1}} |L_{x,u}| \le a_{\mathrm{cap}}. \quad (17.17)$$

Then the coefficient array

$$T'_{y;u,v} = \sum_{x,z} T_{y;x,z} L_{x,u} L_{z,v}$$

of the map $v \mapsto Q(Lv, Lv)$ is symmetric and satisfies $\|T'\|_{a_{i-1},a_j} \le K a_{\mathrm{cap}}^2$. If the quadratic map is equivariant under the adjoint action of constant color rotations and L commutes with them, then the map $v \mapsto Q(Lv, Lv)$ is equivariant.

Proof. We first prove the unweighted bounds. The output row of T' is at most Ka_{cap}^2, by the row bounds of T and of L. For the column with the first input u fixed, we use the volume-normalized column of L in the first argument and its row in the second:

$$\frac{a_j^3}{a_{i-1}^3} \sum_{y,v} |T'_{y;u,v}| \le \frac{a_i^3}{a_{i-1}^3} \sum_x |L_{x,u}| \, a_{\mathrm{cap}} \Big(\frac{a_j^3}{a_i^3} \sum_{y,z} |T_{y;x,z}| \Big) \le K a_{\mathrm{cap}}^2.$$

Fixing the second input instead repeats this calculation with x, z and u, v interchanged. The factor $a_i^3 / a_{i-1}^3 = B^{-3}$ cancels the factor B^3 in the counting column sum of L.

The weights cause no additional loss. A shortest network for (y, x, z), joined with shortest paths from x to u and from z to v, is a connected network containing (y, u, v), which implies $\mathtt{t}(y, u, v) \le \mathtt{t}(y, x, z) + \operatorname{dist}(x, u) + \operatorname{dist}(z, v)$; periodic shortest networks satisfy the same inequality. In the units $a_{i-1} > a_i$ of the new input mesh, the weight of the old network is at most its weight in the units a_i, and the weights in (17.17) are already in the units a_{i-1}. The new weight is therefore at most the product of the three old weights, and the preceding calculation applies with these weights inserted and the same exponent. Symmetry of T' follows from that of T. Equivariance is preserved because L commutes with the color rotations. □

Theorem 17.4. *Assume the setting of Section 17.1 and fix $k \ge 1$. Let $\bar{b}_i$, $1 \le i \le k$, be the reconstruction amplitude bounds of Lemma 17.1, and $\bar{b}_0 = r$. Let $a_{\mathrm{cap}} = C_{\mathrm{CP}}/B$ and $q_{\mathrm{anc}} = Ba_{\mathrm{cap}}^2 = C_{\mathrm{CP}}^2/B < 1$, and for $1 \le i \le k$ let $L_i = D_\theta \widehat{\Phi}_{d_i^k, i}(0, 0)$ be the linearized zero-fluctuation reconstruction of the ith step.*

Let $1 \le i \le k$, and let Q be a deterministic quadratic map from arrays on the lattice of mesh a_i to $\mathfrak{su}(2)$-valued arrays on a lattice of mesh $\delta \le a_i$, which is equivariant under the adjoint action of constant color rotations and satisfies $\|T\|_{a_i,\delta} \le K_Q$ for its coefficient array T, with all input and output directions and colors included. Then the following statements hold.

1. *For every coarse field $e^\phi \in B_r$, under the conditional fluctuation kernel of the ith step at the coarse field e^ϕ,*

$$\big\|\mathbb{E}[Q(\theta_i,\theta_i) \mid \theta_{i-1} = \phi] - Q(L_i\phi, L_i\phi)\big\|_\infty \le C_J K_Q \big\{a_{\mathrm{cap}}^2 \|\phi\|_\infty^3 + g_i^2\|\phi\|_\infty + g_i^3 p_i^3\big\}.$$

   Let $\mu_\phi = \mathbb{E}_\phi w$ and $\Sigma_\phi = \mathrm{Cov}_\phi(w,w)$ be the conditional mean and covariance of the fluctuation variables under this kernel (in the notation of Section 26, $\mu_\phi = m_\phi$), and let E_ϕ be the derivative at $w = \mu_\phi$ of the map $w \mapsto g_i^{-1}\widehat{\Phi}_{d_i^k,i}(\phi, (g_{i,s}w_s)_s)$, which includes the ratios $g_{i,s}/g_i$. Then the expectation of $Q(g_iE_\phi(w-\mu_\phi), g_iE_\phi(w-\mu_\phi))$ equals $g_i^2 Q : (E_\phi\Sigma_\phi E_\phi^{\mathsf{T}})$. This contraction vanishes at $\phi = 0$, and

$$\|E_\phi\Sigma_\phi E_\phi^{\mathsf{T}} - E_0\Sigma_0 E_0^{\mathsf{T}}\|_{\mathrm{op}} \le C_J\|\phi\|_\infty.$$

   The first estimate accounts for the value of the reconstruction at μ_ϕ, its displacement from the zero-fluctuation reconstruction, and all remainders of the means and of the covariance; it does not assume that the conditional expectation of a quadratic term vanishes when $\phi \ne 0$.

2. *For every $0 < \alpha < 3/2$, on every sample,*

$$\mathbb{E}\Big[\big\|Q(\theta_i,\theta_i) - \mathbb{E}[Q(\theta_i,\theta_i) \mid \mathcal{F}_{i-1}]\big\|_{-\alpha;T_0} \;\Big|\; \mathcal{F}_{i-1}\Big] \le C_{J,\alpha} K_Q g_i \bar{b}_i a_i^\alpha \sqrt{\log(2 + L/a_i)}.$$

3. *For $2 \le j \le k$ define quadratic maps $Q_{j,i}$ from arrays on the lattice of mesh a_i to arrays on the lattice of mesh a_j, for $0 \le i \le j-1$, by*

$$Q_{j,j-1} = \overline{Q}_{d_j^k,j}, \qquad Q_{j,i-1}(v,v) = Q_{j,i}(L_iv, L_iv) \quad (1 \le i < j),$$

   where $\overline{Q}_{d_j^k,j}$ is the quadratic part of the zero-fluctuation reconstruction of the jth step, as in Lemma 17.2. These maps are equivariant. Let $T_{j,i}$ be the symmetric coefficient array of $Q_{j,i}$, and set $K_{j,i} = \|T_{j,i}\|_{a_i,a_j}$. Then

$$K_{j,i} \le C a_{\mathrm{cap}}^{2(j-1-i)}, \qquad \frac{K_{j,i}}{a_{j-1}} \le C\frac{q_{\mathrm{anc}}^{j-1-i}}{a_i}.$$

4. *For $2 \le j \le k$ and $1 \le i \le j-1$ put*

$$Z_{j,i} = Q_{j,i}(\theta_i,\theta_i) - \mathbb{E}[Q_{j,i}(\theta_i,\theta_i) \mid \mathcal{F}_{i-1}],$$
$$e_{j,i} = \mathbb{E}[Q_{j,i}(\theta_i,\theta_i) \mid \mathcal{F}_{i-1}] - Q_{j,i-1}(\theta_{i-1},\theta_{i-1}).$$

Then, on every sample,

$$\overline{Q}_{d_j^k,j}(\theta_{j-1},\theta_{j-1}) = Q_{j,0}(\theta_0,\theta_0) + \sum_{i=1}^{j-1}(Z_{j,i}+e_{j,i}).$$

Further, with

$$v_i = g_i\overline{b}_i a_i^{\alpha-1}, \qquad u_i = \frac{q_{\mathrm{anc}}\overline{b}_{i-1}^3}{a_{i-1}} + g_3^2\overline{b}_{i-1} + g_3^3\sqrt{a_i}\,p_i^3,$$

we have, for $0<\alpha<3/2$,

$$\mathbb{E}\big\|a_{j-1}^{-1}\overline{Q}_{d_j^k,j}(\theta_{j-1},\theta_{j-1})\big\|_{-\alpha;T_0} \le \frac{Cr^2}{H}q_{\mathrm{anc}}^{j-1} + C\sum_{i=1}^{j-1} q_{\mathrm{anc}}^{j-1-i}\Big\{v_i\sqrt{\log(2+L/a_i)}+u_i\Big\}.$$

5. *For* $1/2<\alpha<3/2$ *and every* $0<\tau<\min\{1/2,-\log q_{\mathrm{anc}}/\log B\}$,

$$\sum_{j=l}^{k}\mathbb{E}\big\|a_{j-1}^{-1}\overline{Q}_{d_j^k,j}(\theta_{j-1},\theta_{j-1})\big\|_{-\alpha;T_0} \le C_\tau a_l^\tau,$$

uniformly in $k\ge l\ge 2$.

The constants depend on the fixed construction parameters and coefficient bounds, on L, H, T_0, α, *and* τ, *and on the fixed coupling* g_3, *but not on* k, *the depth, the ratio* a_i/δ, *or the sample. The identity in the fourth part does not identify compositions of the nonlinear zero-fluctuation reconstructions with products of their linearizations; their difference is contained in* $e_{j,i}$.

Proof. We fix k for the whole proof, so that all coefficients, zero-fluctuation reconstructions, and conditional kernels have the depths $d_i^k = J(k-i)$ of one hierarchy. Intermediate constants in the estimates for a single kernel may depend on J; all estimates are uniform in the depth. The proof has five steps. Steps 1 and 2 prove the first two parts, Step 3 proves the third, Step 4 the fourth, and Step 5 the last.

Step 1: the conditional expectation of a quadratic term. Fix a level i and a coarse field $e^\phi\in B_r$, and write $R=\|\phi\|_\infty$, $g=g_i$, $t=t_i$, and $p=p_i$. In this step, all expectations are under the conditional fluctuation kernel at e^ϕ. Let

$$\mu_\phi = \mathbb{E}_\phi w, \qquad \xi = w-\mu_\phi, \qquad \Sigma_\phi = \mathbb{E}_\phi \xi\xi^{\mathsf{T}}, \qquad \psi_\phi(w) = \widehat{\Phi}_{d_i^k,i}(\phi,(g_{i,s}w_s)_s).$$

The segment of coarse fields $u\phi$, $0\le u\le 1$, stays in B_r. The fluctuation cutoff domain, the free logarithmic coordinates, and the conditional fluctuation potential are those preserved by Theorem 15.13. The domains and bounds on the coupling-scaled fluctuation inputs are those fixed in Section 17.1; hence the whole segment satisfies the domain conditions.

The bounds on means and on the dependence on the coarse field in Theorem 13.10 apply to this kernel, and give

$$\|\mu_\phi\|_\infty \le C_J t R, \qquad \|\Sigma_\phi\|_{\mathrm{op}} \le 2, \qquad \|\Sigma_\phi - \Sigma_0\|_{\mathrm{row+col}} \le C_J t p^3 R, \tag{17.18}$$

where the last norm is the larger of the maximal row and column sums, possibly with a fixed smaller exponential weight; since it is at most their sum, the cited bound holds with either convention. The product of balls does not depend on ϕ. To make precise which normalization enters, let V_ϕ be the conditional fluctuation potential and $S_u = \partial_u V_{u\phi} - \mathbb{E}_{u\phi}\partial_u V_{u\phi}$ its centered potential derivative along the segment. In the notation of Lemma 13.6, $S_u = Q_u$ for this path, and $-S_u$ is the normalized log-density derivative. By differentiation, we have

$$\partial_u \mathbb{E}_{u\phi} F = \mathbb{E}_{u\phi}\partial_u F - \mathbb{E}_{u\phi}(F S_u), \qquad \partial_u \Sigma_{u\phi}(r,s) = -\mathbb{E}_{u\phi}(\xi_r \xi_s S_u). \tag{17.19}$$

Indeed, we differentiate the numerator and the denominator; this is legitimate because the support is compact and fixed and the integrands are smooth. In the second identity, the terms containing the derivative of $\mu_{u\phi}$ multiply $\mathbb{E}\xi = 0$ and cancel. Thus (17.18) refers to the covariance of the interacting truncated kernel and to its centered derivative; there is no boundary term and no reference Gaussian covariance.

By the joint coefficient bounds kept by Lemma 15.15, we may expand the reconstruction map by Taylor's formula at the conditional mean μ_ϕ. We define

$$v_\phi = \psi_\phi(\mu_\phi), \qquad E_\phi = g^{-1} D_w \psi_\phi(\mu_\phi), \qquad B_\phi = (2g^2)^{-1} D_w^2 \psi_\phi(\mu_\phi);$$

the ratios $g_{i,s}/g$ are included in these definitions. The mean lies in the convex product, and so does the segment from it to each w. Since $\|\xi\|_\infty \le 2p$, Taylor's formula with integral remainder and the bound on the third derivative in w imply

$$\theta_i = v_\phi + g E_\phi \xi + g^2 B_\phi(\xi,\xi) + \mathcal{R}_\phi(\xi), \qquad \|\mathcal{R}_\phi(\xi)\|_\infty \le C_J g^3 p^3. \tag{17.20}$$

Each derivative in w contributes its coupling factor, and the joint absolute bound sums the remaining coefficients. These bounds are at fixed J and impose no further condition on J.

By the bounds on the row sums and volume-normalized column sums kept by the reconstruction maps, converted between the finitely many types of fluctuation variables of a step, we have

$$\|E_\phi\|_{\mathrm{op}} + \|B_\phi\|_{\mathrm{abs}} \le C_J,$$

where $\|B_\phi\|_{\mathrm{abs}}$ is the larger of its absolute row and column sums. The joint bounds in the coarse field and the coupling-scaled fluctuation inputs also give

$$\|E_\phi - E_0\|_{\mathrm{op}} + \|B_\phi - B_0\|_{\mathrm{abs}} \le C_J(R + g\|\mu_\phi\|_\infty) \le C_J' R. \tag{17.21}$$

For this bound, we integrate the derivative in the coarse field along $s\phi$ with the coupling-scaled fluctuation inputs equal to zero, and then the derivative in these inputs from zero to $(g_{i,s}\mu_{\phi,s})_s$. Both segments lie in the fixed inner domain, where the normalized coefficients have bounded row and column sums. At the zero coarse field, we have $\mu_0 = 0$ by the global color symmetry, and the endpoint is exactly E_0, B_0. The row and column bounds imply the displayed operator bounds by Schur's test,

$$\sum_x \Big| \sum_y K_{xy} u_y \Big|^2 \le \Big(\sup_x \sum_y |K_{xy}| \Big) \Big(\sup_y \sum_x |K_{xy}| \Big) \sum_y |u_y|^2,$$

and the conversion from volume-normalized to counting column sums costs only a factor depending on J.

Let $\mathcal{Z}_\phi = E_\phi \Sigma_\phi E_\phi^{\mathsf{T}}$. Adding and subtracting $E_0 \Sigma_\phi E_\phi^{\mathsf{T}}$ and $E_0 \Sigma_0 E_\phi^{\mathsf{T}}$, and using (17.18) and (17.21), we obtain

$$\| \mathcal{Z}_\phi - \mathcal{Z}_0 \|_{\mathrm{op}} \le C_J R (1 + t p^3) \le C_J' R. \tag{17.22}$$

This is the covariance comparison of the first part, for the coefficients of the reconstruction maps themselves.

At the zero coarse field, the product of balls and the potential are invariant under every constant adjoint color rotation. This implies $\Sigma_0 = \Sigma_{\mathrm{sp}} \otimes I_3$ for a spatial matrix Σ_{sp}, and the linear coefficient E_0 acts as a scalar on the adjoint color representation. To check this, note that rotations by π about the three coordinate axes force each 3×3 color block commuting with them to be diagonal, and rotations exchanging the axes force its three diagonal entries to agree. The same invariance for the equivariant map Q implies

$$Q : \mathcal{Z}_0 = 0. \tag{17.23}$$

Indeed, each output component of the contraction is an adjoint vector fixed by every rotation, and rotations by π about two different axes force it to vanish. This argument is componentwise in the output and keeps every spatial entry of the covariance; it needs no translation invariance of Q.

The same reasoning shows that $B_0 : \Sigma_0 = 0$. Together with (17.21) and (17.18), this implies

$$\| \mathbb{E}_\phi B_\phi(\xi, \xi) \|_\infty \le C_J R (1 + t p^3) \le C_J' R. \tag{17.24}$$

Here we used that each scalar coefficient matrix of B_ϕ has absolute entry sum at most C_J, and that a contraction against a matrix of operator norm D has absolute value at most $C_J D$, since every entry of that matrix is bounded by D. The same remark justifies contractions against (17.22) without any dependence on the dimension.

The displacement of the reconstruction at the mean satisfies

$$\| v_\phi - \psi_\phi(0) \|_\infty \le C_J g \| \mu_\phi \|_\infty \le C_J g^2 R, \tag{17.25}$$

where we integrated the derivative in w, whose absolute row sums are at most $C_J g$, along the segment from 0 to μ_ϕ, and used that t/g depends only on J. By the bounds on the first three derivatives of the zero-fluctuation reconstruction and the majorant of Lemma 15.15, we have

$$\|\psi_\phi(0) - L_i\phi\|_\infty \le C_J a_{\mathrm{cap}} R^2, \qquad \|L_i\phi\|_\infty \le a_{\mathrm{cap}} R. \tag{17.26}$$

Only the constant in the linear bound, which is uniform in J, enters the choice of a_{cap}; the quadratic bound may have a constant depending on J. By (17.25), (17.26), and $\|Q(u,v)\|_\infty \le K_Q \|u\|_\infty \|v\|_\infty$, we have

$$\|Q(v_\phi, v_\phi) - Q(L_i\phi, L_i\phi)\|_\infty \le C_J K_Q (a_{\mathrm{cap}}^2 R^3 + g^2 R). \tag{17.27}$$

To see every power, let $\epsilon = v_\phi - L_i\phi$; then the left side is at most $2K_Q a_{\mathrm{cap}} R\|\epsilon\|_\infty + K_Q\|\epsilon\|_\infty^2$, with $\|\epsilon\|_\infty \le C_J(a_{\mathrm{cap}}R^2 + g^2 R)$. Since R and g are bounded by fixed constants, the term R^4 is absorbed into R^3 and the remaining terms into $g^2 R$.

We now expand $Q(\theta_i, \theta_i)$ using (17.20). The term linear in $gE_\phi\xi$ has mean zero. The term quadratic in it has mean

$$\mathbb{E}_\phi Q(gE_\phi\xi, gE_\phi\xi) = g^2 Q : \mathcal{Z}_\phi,$$

which is the identity of the first part; by (17.23) and (17.22), it is at most $C_J K_Q g^2 R$. The cross term of v_ϕ with $g^2 B_\phi(\xi,\xi)$ has mean at most $C_J K_Q g^2 R$, by (17.24) and the boundedness of v_ϕ. Every remaining product contains either one linear and one quadratic term in ξ, or a remainder, or two quadratic terms. Their pointwise bounds are $C_J K_Q g^3 p^3$, $C_J K_Q g^3 p^3$, and $C_J K_Q g^4 p^4$, respectively. Products of a remainder with a term of positive degree in ξ satisfy the same bound $C_J K_Q g^3 p^3$, since gp is bounded by a fixed constant, and so does the square of the remainder. In the term $Q(v_\phi, \mathcal{R}_\phi)$, only the boundedness of v_ϕ is needed. Further, $g^4 p^4 \le C_J g^3 p^3$. Combining these bounds with (17.27), we obtain the first estimate.

Step 2: the centered heat norm of a quadratic term. We prove the second part without assuming smoothness of the output. Fix a sample, so that $\phi = \theta_{i-1}$ is fixed and $\|\phi\|_\infty \le \overline{b}_{i-1}$. Let $Y(w) = Q(\theta_i(w), \theta_i(w))$ and $F = Y - \mathbb{E}_\phi Y$. On the whole product of balls, differentiating the polarized map, we find

$$D_w Y = Q(D_w\theta_i, \theta_i) + Q(\theta_i, D_w\theta_i).$$

The absolute row sums of the derivative of θ_i in w are at most $C_J g_i$, and its volume-normalized columns satisfy the same bound. At most a number of fluctuation variables depending only on J is attached to each cell of the lattice of mesh a_i. Further, $\|\theta_i(w)\|_\infty \le \overline{b}_i$ at every point w of the product. Indeed, the recurrence of Step 1 in the proof of

Lemma 17.1 gives $\|\theta_i(w)\|_\infty \le a_{\mathrm{cap}}\bar{b}_{i-1} + C_{\mathrm{rec},J}t_i p_i$, and this is at most $\bar{b}_i$ by (17.5), since $C_J \ge C_{\mathrm{cap},J}$. Let $\mathcal{A} = C_J K_Q g_i \bar{b}_i$. Then the derivative matrix $G = D_w Y$ satisfies

$$\sup_y \sum_r |G_{y,r}| \le \mathcal{A}, \qquad \sup_r \frac{\delta^3}{a_i^3} \sum_y |G_{y,r}| \le \mathcal{A}, \tag{17.28}$$

where r runs over the scalar fluctuation variables. For the column bound, we fix the differentiated fluctuation variable, use the column of Q for its differentiated argument, bound its other argument by $\bar{b}_i$, and then use the volume-normalized column of $D_w\theta_i$; the volume factors multiply to δ^3/a_i^3. Since both differentiated arguments satisfy the same bound, their sum is included in C_J. This holds for all directions and colors.

By Schur's test, the counting ℓ^2 operator norm of G is at most $\mathcal{A}(a_i/\delta)^{3/2}$. For $0 < s \le T_0$, let $\omega_y(s,x) = \int_{Q_y} p_s(x-z)\,dz$ for the output cells Q_y of side δ. For each scalar component of P_sY at a point x, the Lipschitz constant in w is at most

$$C\mathcal{A}\min\{1, a_i^{3/2}s^{-3/4}\}. \tag{17.29}$$

The first bound follows from the absolute row sums and $\sum_y \omega_y = 1$. For the second, we compose the operator bound with $\|\omega(s,x)\|_2 \le C\delta^{3/2}s^{-3/4}$, from Lemma 16.1; the factor $(a_i/\delta)^{3/2}$ cancels the factor $\delta^{3/2}$ of the output cells. By the derivative and time-increment bounds of the same lemma, one spatial derivative of the cell integrals costs $s^{-1/2}$ and one time derivative costs s^{-1} when $a_i^2 \le s \le T_0$, in which case $s \ge \delta^2$. These bounds also hold for differences of two evaluations, by integration along a shortest spatial path or a time segment.

We now take the supremum over times, beginning with times below a_i^2. Let $M_i = \mathbb{E}_\phi\theta_i$. By the amplitude bound, we have $\|M_i\|_\infty \le \bar{b}_i$ as well as $\|\theta_i\|_\infty \le \bar{b}_i$. The output row bound in (17.16) then gives

$$\|Q(\theta_i,\theta_i) - Q(M_i,M_i)\|_\infty \le 2K_Q\bar{b}_i\|\theta_i - M_i\|_\infty,$$

and subtracting the conditional expectation, we obtain

$$\|F\|_\infty \le 2K_Q\bar{b}_i\big(\|\theta_i - M_i\|_\infty + \mathbb{E}_\phi\|\theta_i - M_i\|_\infty\big). \tag{17.30}$$

This uses the output row; a bound on the input columns alone would not imply it. There are $N_i = 9(L/a_i)^3$ scalar coordinates of θ_i on its own lattice. By the bounds on the derivative rows of the reconstruction maps in Lemma 15.15, each of them is a Lipschitz function of w with Euclidean constant at most $C_J g_i$, since the ℓ^2 norm of a row of the derivative is at most its absolute sum. The conditional kernel is the one on a convex product described in Section 17.1 (Lemma 15.10 and Theorem 15.13), with Hessian at least $I/2$. By Theorem 5.4 with $m = 1/2$ and a union bound, we have

$$\mathbb{P}_\phi\{\|\theta_i - M_i\|_\infty > x\} \le \min\{1, 2N_i\exp[-x^2/(C_J g_i^2)]\}. \tag{17.31}$$

To integrate this bound, we take $x_0 = g_i\sqrt{C_J \log(2N_i)}$, and for this integral we let $A^2 = C_J g_i^2$. We use $\int_{x_0}^{\infty} e^{-x^2/A^2}\,dx \le A^2 e^{-x_0^2/A^2}/(2x_0)$, which follows by inserting the factor $x/x_0 \ge 1$ in the integrand. After enlarging the constant, we obtain

$$\mathbb{E}_\phi \|\theta_i - M_i\|_\infty \le C_J g_i \sqrt{\log(2 + L/a_i)}, \qquad \mathbb{E}_\phi \|F\|_\infty \le C_J K_Q g_i \bar{b}_i \sqrt{\log(2 + L/a_i)},$$

where we used $\log(2N_i) \le C \log(2 + L/a_i)$, which holds because $L/a_i \ge 4$. Since the heat semigroup contracts the supremum norm, the expected supremum over $0 < s \le \min\{a_i^2, T_0\}$ of $s^{\alpha/2}\|P_s F\|_\infty$ is at most $C\mathcal{A} a_i^\alpha \sqrt{\log(2 + L/a_i)}$. Only the coordinates on the lattice of mesh a_i have been counted; there is no union bound over the arbitrarily fine output lattice. The mean M_i is the conditional mean of the reconstruction, which need not vanish.

For the remaining times, let $S_h = 2^h a_i^2$, and consider each nonempty interval $[S_h, \min\{2S_h, T_0\}]$, $h \ge 0$. The centered process $s^{\alpha/2}(P_s F)(x)$, with its nine scalar components, has one-point sub-Gaussian parameter at most

$$\sigma_h = C_{J,\alpha} \mathcal{A} a_i^{3/2} S_h^{\alpha/2 - 3/4},$$

by (17.29) and (16.18). Its increments have parameter at most $C\sigma_h$ times

$$\min\{1, \operatorname{dist}(x, x')/\sqrt{S_h}\} + \min\{1, |s - s'|/S_h\};$$

this includes the derivative of $s^{\alpha/2}$, whose relative size on the interval is at most C_α/S_h. Between distinct components, we take the distance to be $C\sigma_h$; adding the Lipschitz constants of the two components justifies this choice. Using spatial nets of spacing $c\eta\sqrt{S_h}$ and temporal nets of spacing $c\eta S_h$, together with the nine components, we obtain covering numbers at most

$$C(1 + L/\sqrt{S_h})^3 \eta^{-4} \qquad (0 < \eta < 1).$$

Since the process is continuous for $s > 0$ and smooth in x on these intervals, a countable dense set of parameters determines its supremum. We apply Lemma 16.4 to the increment function divided by σ_h, with $s = C\sigma_h$, $b = 0$, $D = C$, $A = C(1 + L/\sqrt{S_h})^3$, and $q_{\text{ent}} = 4$; the point z_0 of that lemma has the same one-point bound. At $p = 1$, the lemma bounds the expected supremum over the interval by

$$C\sigma_h \sqrt{\log(2 + L/a_i)},$$

because $\sqrt{S_h} \ge a_i$, so that $A \le C(1 + L/a_i)^3$; no factor depending on the output mesh enters. Since $\sigma_h = C_{J,\alpha} \mathcal{A} a_i^\alpha 2^{-h(3/4 - \alpha/2)}$, the sum over the intervals converges with ratio $2^{\alpha/2 - 3/4} < 1$ when $\alpha < 3/2$. Adding the bound for small times, we obtain

$$\mathbb{E}_\phi \|F\|_{-\alpha; T_0} \le C_{J,\alpha} K_Q g_i \bar{b}_i a_i^\alpha \sqrt{\log(2 + L/a_i)}.$$

Conditionally on $\mathcal{F}_{i-1}$, all constants here are deterministic and uniform over the coarse fields that occur. This is the second part.

Step 3: the pulled-back coefficients. The linearized reconstruction L_i has output mesh a_i and input mesh $a_{i-1} = Ba_i$, and it satisfies (17.17). These are the weighted bounds on the first derivative of the zero-fluctuation reconstruction, with the constant C_{CP}, uniform in J, enlarged as in Section 17.1. By the bounds on the second derivative of the zero-fluctuation reconstruction, the coefficient $Q_{j,j-1} = \overline{Q}_{d_j^k,j}$ has norm at most a fixed constant C. By induction down to the input level i, using Lemma 17.3 at each step, we obtain

$$K_{j,i} \le C a_{\mathrm{cap}}^{2(j-1-i)}.$$

The global color equivariance of the zero-fluctuation reconstruction implies the equivariance of its quadratic part, and each L_i commutes with constant color rotations; hence every $Q_{j,i}$ is equivariant. With $m = j-1-i$, the identity $a_i = B^m a_{j-1}$ gives

$$\frac{K_{j,i}}{a_{j-1}} \le \frac{C B^m a_{\mathrm{cap}}^{2m}}{a_i} = \frac{C q_{\mathrm{anc}}^m}{a_i}. \tag{17.32}$$

Step 4: the telescoping identity and the convolution. Let $Y_i = Q_{j,i}(\theta_i, \theta_i)$ for $0 \le i \le j-1$. By the definitions in the fourth part,

$$Y_i - Y_{i-1} = \big\{ Y_i - \mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] \big\} + \big\{ \mathbb{E}[Y_i \mid \mathcal{F}_{i-1}] - Y_{i-1} \big\} = Z_{j,i} + e_{j,i}.$$

Summing this finite identity over $1 \le i \le j-1$, we obtain the telescoping identity on every sample. Since every conditional expectation is taken under the kernel of the same hierarchy that generated θ_i, Steps 1 and 2 apply with $\phi = \theta_{i-1}$ and the amplitude bound $\overline{b}_{i-1}$ on $\|\theta_{i-1}\|_\infty$.

The term from the coarsest level is bounded deterministically by

$$\Big\| \frac{Q_{j,0}(\theta_0, \theta_0)}{a_{j-1}} \Big\|_{-\alpha;T_0} \le \frac{C r^2}{H} q_{\mathrm{anc}}^{j-1},$$

by (17.4), (17.32) with $i = 0$, $a_0 = H$, and $\|\theta_0\|_\infty \le r$. For the centered terms, the second part and (17.32) give

$$\frac{\mathbb{E}\|Z_{j,i}\|_{-\alpha;T_0}}{a_{j-1}} \le C q_{\mathrm{anc}}^{j-1-i} g_i \overline{b}_i a_i^{\alpha-1} \sqrt{\log(2 + L/a_i)}.$$

For the conditional expectation errors, we use the first part, (17.4), and the three identities

$$\frac{a_{\mathrm{cap}}^2}{a_i} = \frac{q_{\mathrm{anc}}}{a_{i-1}}, \qquad \frac{g_i^2}{a_i} = g_3^2, \qquad \frac{g_i^3}{a_i} = g_3^3 \sqrt{a_i},$$

and obtain

$$\frac{\mathbb{E}\|e_{j,i}\|_{-\alpha;T_0}}{a_{j-1}} \le C q_{\mathrm{anc}}^{j-1-i} u_i.$$

By the triangle inequality, we obtain the convolution estimate of the fourth part, with v_i as defined there. No composition of the nonlinear zero-fluctuation reconstructions has been identified with a product of linearizations; the difference remains in $e_{j,i}$.

Step 5: summation. Let $1/2 < \alpha < 3/2$. By the reconstruction amplitude bound of Lemma 17.1 and (17.6), we have, with constants depending on the fixed H, g_3, and r,

$$\overline{b}_i \le C(1+i)B^{-i/2} \qquad (i \ge 0), \tag{17.33}$$

since $a_{\mathrm{cap}}^i = B^{-\kappa_{\mathrm{term}} i}$ with $\kappa_{\mathrm{term}} > 1/2$; the case $i = 0$ is included by enlarging C. More precisely, inserting the two terms of $\overline{b}_i$, we obtain

$$v_i \le C\big\{B^{-(\kappa_{\mathrm{term}}+\alpha-1/2)i} + (1+i)B^{-\alpha i}\big\},$$

and both exponents exceed $1/2$. In u_i, the part of $\overline{b}_{i-1}^3/a_{i-1}$ coming from the coarsest field decays with the exponent $3\kappa_{\mathrm{term}} - 1 > 1/2$, and the part coming from the fluctuations is at most $C(1+i)^3 B^{-i/2}$; the inequality $(x+y)^3 \le 4(x^3+y^3)$ handles the mixed terms. The other two terms of u_i are at most $C(1+i)^3B^{-i/2}$ directly. It follows that

$$v_i + u_i \le C(1+i)^3 B^{-i/2}. \tag{17.34}$$

For $i < j$, we use the cruder bound $\sqrt{\log(2+L/a_i)} \le \sqrt{\log(2+L/a_j)} \le C\sqrt{1+j}$.

Let $0 < \tau < \min\{1/2, -\log q_{\mathrm{anc}}/\log B\}$, and set $r_1 = q_{\mathrm{anc}} B^\tau < 1$ and $r_2 = B^{\tau-1/2} < 1$. Let $\mathcal{X}_j$ be the deterministic upper bound obtained from the convolution estimate by inserting (17.34) and $\sqrt{\log(2+L/a_i)} \le C\sqrt{1+j}$; it is defined for every $j \ge 2$ and does not depend on k. Writing $j = i + m + 1$, we have

$$\sum_{j\ge 2} B^{\tau j}\mathcal{X}_j \le C\sum_{j\ge 2} B^{\tau j} q_{\mathrm{anc}}^{j-1} + CB^\tau \sum_{i\ge 1}\sum_{m\ge 0} (2+i+m)^{1/2}(1+i)^3 r_1^m r_2^i < \infty.$$

All terms are nonnegative, and $(2+i+m)^{1/2} \le (2+i)(1+m)$ reduces the double series to the product of $\sum_i (2+i)^4 r_2^i$ and $\sum_m (1+m) r_1^m$, which converge by comparison with geometric series. The bound depends only on the fixed data and on the gaps in the strict inequalities for τ. The finite sequence for a given k is bounded by this deterministic majorant; no coefficients from different values of k are identified. We conclude that

$$\sum_{j=l}^{k} \mathbb{E}\big\| a_{j-1}^{-1} \overline{Q}_{d_j^k, j}(\theta_{j-1}, \theta_{j-1})\big\|_{-\alpha; T_0} \le B^{-\tau l} \sum_{j\ge 2} B^{\tau j}\mathcal{X}_j \le C_\tau H^{-\tau} a_l^\tau,$$

and the factor $H^{-\tau}$ is absorbed into C_τ. This is the last part. The proof used only $q_{\mathrm{anc}} < 1$ and the stated choice of τ; no condition of the form $q_{\mathrm{anc}} B^\alpha < 1$ is needed. □

PART VI

Large-field control and the small-field event

SECTION 18

Local coordinates and constrained minimizers on block regions

This section begins the large-field analysis. Its subject is a block region: a union of final blocks of $s \leq J$ consecutive averaging steps, possibly disconnected and with holes, on which the configuration is integrated while the configuration on the fixed complement is held fixed. On a block region, we construct coordinates adapted to the block averaging constraints, compute Haar measure in these coordinates, and construct the constrained minimizers of the effective classical action, with weighted bounds on all their Taylor coefficients. All estimates are uniform in the shape, the topology, and the number of components of the block region.

We need these objects for the following reason. As explained in Section 3.4, the exponential-moment bound for blocked Wilson actions (Theorem 22.1) runs the renormalization group in a localized form. The configurations are partitioned according to the regions where some averaged plaquette has large curvature (Theorem 21.5). Outside these regions, background fields, fluctuation integrals, and effective interactions are constructed on arbitrary unions of blocks, and the link variables in the excluded regions are held fixed as frozen coordinates. This section supplies the finite-dimensional part of that construction. Section 19 compares the constructions on a block region with those on the whole torus, and Section 20 integrates the fluctuations on block regions and tracks the normalization constants. Further, Section 21 uses the change of variables and the coercivity estimate at the first block integration to write exact densities, and Section 23 uses the substitution estimates in its choice of parameters.

Definition 18.1 introduces coefficient norms with exponential weights in the distance to the boundary of the block region, and Lemma 18.3 shows that these norms are stable under contractions, inverses, inverse square roots, and Schur complements. Lemma 18.4 proves that the iterated block averaging map is local for every output side $m \geq 4$. Lemma 18.7 computes the pushforward of Haar measure in the local constraint coordinates of a block region, under the hypothesis that the fiber is covered by the chosen chart. Lemma 18.8 records the derivatives of constrained minimum values, and Lemma 18.9 states how coefficient norms, normalization constants, and determinants change when variables are frozen. The main result is Theorem 18.12: the constrained Hessian of the effective classical action on a block region is bounded below by a constant that does

not depend on the region, and the constrained minimizer is analytic, with weighted coefficient bounds. Its coercivity estimate rests on the patch inequality of Lemma 18.11. Lemma 18.14 identifies the objective of the first block integration with the Wilson action on the block region and gives the exact form and range of its Hessian, as needed in Section 21. Finally, Lemma 18.15 shows that connected expressions containing a perturbation in the fixed complement decay exponentially into the block region, without a factor proportional to the volume of the complement.

The inputs are the coefficient calculus of Lemma 5.3, the local constraint coordinates and Haar disintegration of Lemma 6.3, the nested minimization identity and the weighted coefficient bounds for the effective classical actions in Theorems 10.10 and 10.11, the triangular change of variables of Theorem 12.13, the domains of Definition 14.4, and, from Theorem 14.8, only the inequality $E_{d,\mathrm{sat}} \geq \kappa_* E_0$ for the effective classical action at every depth $d \geq 0$. The geometry of Lemma 7.4 is used only for a comparison.

Two observations organize the argument. The first concerns coercivity. Let u be a linearized variation that satisfies the tree gauge conditions and the linearized block averaging constraints of the block region. If u has zero curl on a single final block, then u is a gradient there, the tree conditions make its potential constant on the block, and hence u vanishes on the block. On two adjacent final blocks, the constraint on the coarse bond between them equates the two constants. Coercivity on these two kinds of patches, a covering with bounded overlap, and an explicit extraction of the free coordinates from u give a lower bound on the constrained Hessian that does not see the topology of the region. The second observation concerns locality. A connected contraction of coefficient tensors that contains a perturbation in the fixed complement must connect each of its external indices in the block region to the complement. Its length therefore bounds the distance of these indices to the boundary, and summing over the positions in the complement with exponential weights gives decay without a volume factor.

The constructions are analytic near the identity. Whether every point of a fiber of the block averaging map lies in the chart is a hypothesis of Lemma 18.7; it is verified in Section 21. Frozen coordinates from earlier block integrations are fixed real parameters: we never differentiate in them or continue them to complex values. Constant terms are tracked separately from the nonconstant Taylor series, and the coefficient norms never bound them.

Charts adapted to the averaging constraints and constrained minimizers of the action (background fields) are central ingredients of Bałaban's renormalization group. Configurations with prescribed averages and the axial gauge conditions on them are introduced in [14, §§A–B]; the minimizers, their analytic dependence on the coarse field, and the decay of their derivatives are constructed in [16, Theorem 1, p. 279, and §G, Proposition 9, p. 309]. We follow that method, but the block regions, the centered averaging map, the constraint coordinates, and the coefficient norms are different, and we prove the estimates

needed here directly. The weighted inverse bounds adapt the exponential conjugation of Combes and Thomas [61]. Gauge fixing along trees by successive Haar translations, the analytic implicit function theorem, the formulas for derivatives of minimum values, Schur complements, the resolvent identity, and the integral formula for inverse square roots are standard.

18.1 Block regions, coordinates, and constants

Fix an odd integer $M = 2R_M + 1 \geq 3$ and an integer $J \geq 1$. Throughout this section $1 \leq s \leq J$ and $m \geq 4$ are integers, and we consider s consecutive averaging steps T_M^s, from the lattice of side $M^s m$ (the input lattice) to the lattice of side m (the output lattice). The lattices of sides $M^{s-i}m$, $0 \leq i \leq s$, are the levels, indexed by i; the integrated intermediate scales have $i < s$. The level $i = 0$ is the input lattice and $i = s$ the output lattice. Distances are measured in units of the output mesh. The final blocks are the centered blocks of M^s input sites per direction at the vertices of the output lattice; they are described precisely in Section 18.3. A block region is a union Ω of final blocks. The fixed complement consists of the input vertices and links not in Ω, and the boundary W of Ω is the set of positions of the input links with one endpoint in Ω and the other in the fixed complement. A block region may be empty, may be disconnected, may have holes, and may contain isolated final blocks.

Labels and positions. A scalar coordinate is one component of the logarithm of a link variable. It is specified by its link, its color, and the level on which the link lives; a frozen coordinate also records the scale at which it was introduced. Coordinates that differ in direction, color, scale, or scale of introduction are different labels, even if their positions coincide. Each label is assigned the position of its link, embedded in output units. Inner products on lists of coordinates are counting inner products. Volume-normalized column sums carry the factors $a_{\mathrm{out}}^3/a_{\mathrm{in}}^3$ of Definition 4.6. Changing the counting unit across at most J scales changes a counting sum by a factor depending only on M and J, and it introduces no volume factor. In analytic formulas we write $E_d(A)$ for $E_d(e^A)$ on the identity chart.

Coordinates. We use the coordinates listed in Table 18.1. The coarse field, the free coordinates, and the tree coordinates are defined precisely in Definition 18.6 and Lemma 18.7, and the normalized fluctuation variables in Lemma 18.9. The logarithms c of the coarse links, the free coordinates z, and the logarithms ξ of the frozen coordinates introduced at the current block integration are analytic variables, and they may be complexified on the construction chart. The frozen coordinates b_{rec} introduced at earlier block integrations are fixed real parameters. Figure 18.1 shows the different kinds of links in a two-dimensional sketch.

Symbol	Meaning	Reference measure	Analytic role
c	logarithms of the coarse links of Ω, that is, of the bonds joining two final blocks of Ω	Haar measure $dH(e^c)$ of the group values	variable; complexified
z_i	free logarithmic coordinates at intermediate scale i, on the free edges $\mathcal{F}_i$	Lebesgue measure, with the density J_i of Lemma 18.7	variable; complexified
w_i	normalized fluctuation variables, $z_i = a_i + g_i K_i^{-1/2} w_i$	Lebesgue measure	variable
$\mathfrak{f}$	tree coordinates (gauge transformations at the nonroot vertices of the trees)	product of Haar probability measures	integrated; never differentiated
ξ	logarithms of the frozen coordinates introduced at the current block integration	Haar measure on the group values	small; complexified
b_{rec}	frozen coordinates introduced at earlier block integrations, at their scales of introduction	product Haar measure	fixed and real; never differentiated

Table 18.1. The coordinates used on a block region.

Uniform constants. In this section a constant is called uniform if it depends only on the following data: M and J; the radius r_E of the identity chart on which the effective classical actions E_d are controlled, the bound C_E on their weighted Taylor coefficients with each index fixed, and their spatial rate $a_E > 0$, all from Theorem 10.11; the constant $\kappa_* > 0$ in the lower bound (18.20) below; the radii of the local constraint coordinates of Lemma 6.3, strictly smaller than their radii of definition, and the coefficient bounds of the finite group products that define them; the label density v_J of (18.1); and fixed ratios of radii and fixed differences of spatial rates, stated where they are used. Uniform constants are independent of L, γ, m, N, K, the depth d_0, the number of preceding block integrations, the number of components and the shape of the block region, the values of the frozen coordinates b_{rec}, and every width chosen later (boundary strips, interior restriction distances, protected neighborhoods, and cells). A bound for a general analytic input may depend in addition on the coefficient bound and the inverse bound assumed for that input. No uniformity as $J \to \infty$ is claimed. This is the dependence described in item (3) of Definition 4.14. Frozen coordinates are fixed before Taylor coefficients are taken or degrees are summed.

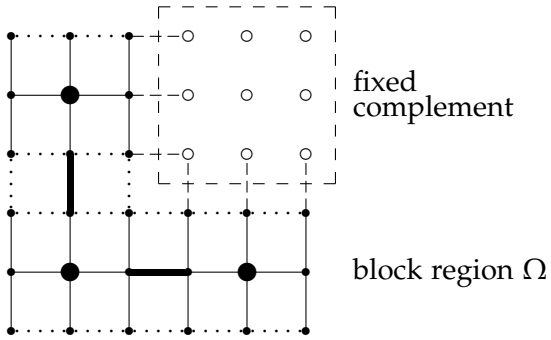


Figure 18.1. A two-dimensional sketch of a block region with $M = 3$ and $s = 1$: three final blocks (filled sites, with enlarged roots) and one final block of the fixed complement (open sites, dashed square). Solid lines are tree edges. The two thick edges are the eliminated edges of the two coarse bonds of Ω. Dotted lines are free edges. Dashed edges are links across the boundary; they are not coordinates of the block region, and they carry frozen coordinates.

18.2 Coefficient norms with boundary decay

In the regional estimates, distances between label positions are periodic shortest-path distances on the common refined position graph: the periodic cubic lattice of mesh $(2M^s)^{-1} \le 1$ in output units, which contains the endpoints and the midpoints of the links of all levels. Its vertices need not carry a coordinate. In this section dist denotes this graph distance, and the boundary distance uses it. The connecting-network length t uses the associated periodic ℓ^1 metric, with internal vertices allowed. If d_2 and t_2 denote the Euclidean distance and network length of Definition 4.6, then $d_2 \le \mathrm{dist} \le \sqrt{3} d_2$ and $\mathsf{t}_2 \le \mathsf{t} \le \sqrt{3}\mathsf{t}_2$. A coefficient bound imported at Euclidean rate b is therefore used at graph rate $b/\sqrt{3}$. The regional norms below are the graph-metric variants of the norms of Definition 4.6 and (4.11), written out for a block region.

Definition 18.1. Fix the data of Section 18.1 and a block region Ω. Assume that the positions of the labels satisfy the density bound

$$\#\{y : R \le \mathrm{dist}(x,y) < R+1\} \le v_J (R+2)^3 \qquad (R \ge 0) \tag{18.1}$$

for every label position x, where the count is over labels y. For a label i, let $d_\Omega(i)$ be the graph distance, on the refined position graph, from its position to the set of positions outside Ω. When a position is removed from a block region, all labels at that position are removed together. A different convention for the positions of link endpoints changes these distances by at most a fixed amount, which is absorbed in the widths below.

Let $\rho > 0$, $a_{\mathrm{sp}} > 0$, and $\alpha_{\mathrm{bd}} > 0$, and let ϕ be a spatially varying analytic radius on Ω, with $0 < \phi(i) \le \rho$. Let $f(v; b_{\mathrm{rec}}) = \sum_\nu f_\nu(b_{\mathrm{rec}}) v^\nu$ be a function of the variables

$v = (v_i)_{i\in\Omega}$ of the block region, given by its Taylor series in v, which depends on the frozen coordinates b_{rec}. Define

$$\mathcal{X}^{\phi}_{\Omega}(f) = \sup_{b_{\mathrm{rec}}} \rho^{-2} \sum_{k\geq 1} \sup_{i\in\Omega} e^{\alpha_{\mathrm{bd}} d_{\Omega}(i)} \sum_{|\nu|=k} \frac{\nu_i}{k} |f_{\nu}(b_{\mathrm{rec}})| e^{a_{\mathrm{sp}}\mathtt{t}(\nu)} \phi^{\nu}.$$

Here $\mathtt{t}(\nu)$ is the length of a shortest connecting network for the positions of the labels in $\operatorname{supp}\nu$, with internal vertices allowed, measured in the periodic ℓ^1 metric of the refined position graph (the graph-metric variant of the length in Definition 4.6), and $\phi^{\nu} = \prod_i \phi(i)^{\nu_i}$. The supremum over i is taken inside the sum over degrees, and the supremum over b_{rec} outside it; b_{rec} ranges over the given compact set of real values of the frozen coordinates, each at the scale at which it was introduced. The norm involves no derivative in b_{rec} and no complex value of b_{rec}. The constant term $f(0; b_{\mathrm{rec}})$ is not part of the norm: it is kept in the effective classical action or in the normalization constant Λ. All nonconstant degrees, including degree one, enter the norm. The norm of a function on the empty block region is zero. If the fixed complement is empty, a boundary perturbation is required to have zero nonconstant part, and its norm is zero. For the effective classical action we use the same definition with its own radius ρ_{cl}, its own radius function, and its own rates.

For $K_* \geq 2$, $w_{\mathrm{bd}} \geq 1$, and $0 < \rho_{\partial} \leq \rho$ we use the radius

$$\phi(i) = \begin{cases} \rho_{\partial}, & d_{\Omega}(i) < w_{\mathrm{bd}}, \\ \rho, & d_{\Omega}(i) \geq w_{\mathrm{bd}}, \end{cases} \qquad \psi_{\mathrm{prof}} = \sqrt{K_*}\,\phi, \qquad \widehat{\phi} = K_*\phi,$$

which is smaller in the boundary strip of width w_{bd}. The enlarged radii ψ_{prof} and $\widehat{\phi}$ are used in Sections 19 and 20, on correspondingly larger analytic domains.

For a coefficient tensor with a set A of external indices and Taylor multi-index ν, put $S = A \cup \operatorname{supp}\nu$ and

$$\mathtt{t}_W(S) = \inf_{w\in W} \mathtt{t}(S \cup \{w\}).$$

The bulk and boundary coefficient norms of such a tensor at rate a are the sums of the absolute values of its coefficients at the given radii, with weights $e^{a\mathtt{t}(S)}$ and $e^{a\mathtt{t}_W(S)}$ respectively, in which one external index is fixed and all other indices are summed; the norm is the maximum over the choice of the fixed index. Differentiated analytic inputs count as external indices. Column sums at input indices are volume-normalized as in Definition 4.6. A tensor has a bulk bound or a boundary bound if the corresponding norm is bounded by a stated constant. The algebraic properties of these norms are collected in Lemma 18.3.

Remark 18.2. The weight $e^{a\mathtt{t}_W(S)}$ of the boundary coefficient norm and the weight $e^{a_{\mathrm{sp}}\mathtt{t}(\nu)+\alpha_{\mathrm{bd}} d_{\Omega}(i)}$ in $\mathcal{X}^{\phi}_{\Omega}$ are different, and a boundary bound at rate a does not by itself

bound $\mathcal{X}_\Omega^\phi$ with $a_{\mathrm{sp}} = \alpha_{\mathrm{bd}} = a$. The two are related by the length inequalities (18.51) and (18.52) in the proof of Lemma 18.15, at the cost of a fixed fraction of the rate.

For example, freezing the variable y at the value c in the function $vy + y^2$ produces the linear term cv and the constant c^2. This is why the norm $\mathcal{X}_\Omega^\phi$ includes degree one, and why constants are tracked separately (see Lemma 18.9).

Lemma 18.3. *Use the norms of Definition 18.1. Fix radii and rates, together with strictly smaller radii and rates; the bounds below hold at the smaller radii and rates, with constants depending only on these choices, on v_J, and on the stated bounds of the factors.*

1. *For a scalar function f, a label i, and $k \geq 1$,*

$$\sum_{|\nu|=k} \frac{\nu_i}{k} |f_\nu| e^{a\mathfrak{t}(\nu)} = \frac{1}{k!} \sum_{i_2,\dots,i_k} |D^k f(0)_{i,i_2,\dots,i_k}| e^{a\mathfrak{t}(i,i_2,\dots,i_k)}. \tag{18.2}$$

   In particular, a bound on the Taylor tensors of f with each index fixed in turn is the same as a bound on the absolute values of its coefficients.

2. *A connected product, contraction, or analytic composition of tensors, in which one factor has a boundary bound and the other factors have bulk bounds, has a boundary bound at every remaining external index. This includes at most J Schur complements of constrained Hessians in the free coordinates.*

3. *For $u > 0$ put $S_J(u) = 8v_J \sum_{r\geq 0} (r+1)^3 e^{-ur}$. For a matrix K on a list of labels let $\|K\|_b$ be the maximum of its absolute row and column sums with weight $e^{b\,\mathrm{dist}(x,y)}$. Let $K_0 = K_0^{\mathsf{T}} \geq \lambda I$ be real with $\|K_0\|_b \leq Q$, and put*

$$\mu = \frac{b\lambda}{8(Q+\lambda)}, \qquad C_R = 2S_J(\mu/2).$$

   Let L be a complex symmetric matrix, analytic in further variables, with bulk coefficient bounds on a larger radius such that $C_R \|L\|_{\mu/2} \leq \lambda/4$, where the norm includes the absolute values of all coefficients at that radius. Then $(K_0 + L)^{-1}$ and the inverse square root $(K_0 + L)^{-1/2}$, obtained by holomorphic continuation from the positive inverse square root of K_0, have bulk coefficient bounds at rate $\mu/2$, depending only on b, λ, Q, v_J, and the ratio of the radii.

4. *For two matrices A and B as in (3) on a common list, a boundary bound on $A - B$ gives boundary bounds on $A^{-1} - B^{-1}$ and $A^{-1/2} - B^{-1/2}$, through the ordered identities*

$$A^{-1} - B^{-1} = -A^{-1}(A - B)B^{-1}$$

   and

$$A^{-1/2} - B^{-1/2} = -\frac{1}{\pi} \int_0^\infty t^{-1/2} (A + t)^{-1} (A - B)(B + t)^{-1}\, dt.$$

The lower bound λ is a hypothesis on the free coordinates only, not on the coarse field. All constants are uniform.

Proof. We begin with the identity (18.2). Fix ν with $|\nu| = k$ and $\nu_i > 0$. An ordered list $(i_2, \dots, i_k)$ whose multiplicities are $\nu - e_i$ occurs $(k-1)!/\prod_j (\nu - e_i)_j!$ times in the sum on the right, and for each such list the derivative is $\nu! f_\nu$. Multiplying these two numbers and dividing by $k!$, we obtain ν_i/k, which proves (18.2). In particular, a bound on Taylor tensors with each index fixed is an absolute coefficient bound, not a bound on a derivative in a single direction.

Next we prove (2). When two factors are contracted at a label, we join their connecting networks at that label. The union spans every remaining external label, and if the first factor carries a boundary bound, it also reaches the point of W of that factor. Hence

$$\mathrm{t}_W(S_{\mathrm{ext}}) \le \mathrm{t}_W(S_1) + \sum_{r \ge 2} \mathrm{t}(S_r), \tag{18.3}$$

where S_{ext} is the set of external labels of the contraction and S_r the label set of the r-th factor. We fix any remaining external index. On the branch leading to that label we use the corresponding input-column bound, and on the other branches we use output-row bounds; these are the contractions of Lemma 5.3, with the same counting measures. A connected matrix cycle with an external insertion can be cut at that insertion, and successive absolute row and column sums give the same bound. (This does not apply to a disconnected scalar trace.) Multiplication of majorant series with nonnegative coefficients sums over the partitions of the Taylor labels, and division by the Taylor factorials turns the multinomial factors into ordinary products of power series. The same reasoning, applied to a composition with its majorant on the given radius, proves (2) for compositions. Fixing a specified analytic input only changes the fixed index. An external index within distance R_0 of W may instead be joined to W at the price of a factor e^{aR_0}. No step involves a sum over the positions of W.

For (3) we first record a density sum. By the density bound (18.1), we have, for every $u > 0$ and every subset of a periodic list of labels,

$$\sup_x \sum_y e^{-u\,\mathrm{dist}(x,y)} \le 8 v_J \sum_{r \ge 0} (r+1)^3 e^{-ur} = S_J(u). \tag{18.4}$$

Here we divided the displacements according to $r \le \mathrm{dist}(x,y) < r+1$ and used $(r+2)^3 \le 8(r+1)^3$. The series is finite, since for large r the ratio of successive summands is at most $e^{-u/2} < 1$.

We now adapt the exponential conjugation of Combes and Thomas [61] to the matrix K_0. Let q be a real function that is one-Lipschitz for the distance on the labels, and

let $D_q = \operatorname{diag}(e^{\mu q(x)})$. The entries of $D_q K_0 D_q^{-1} - K_0$ are $(K_0)_{xy}(e^{\mu(q(x)-q(y))} - 1)$, so the maximum of its unweighted absolute row and column sums is at most

$$Q \sup_{r \geq 0} e^{-br}(e^{\mu r} - 1) \leq \frac{Q\mu}{b - \mu} \leq \frac{\lambda}{8}.$$

For the first inequality we used $e^{\mu r} - 1 \leq \mu r e^{\mu r}$ and $r e^{-(b-\mu)r} \leq (b - \mu)^{-1}$; the second follows from the definition of μ. The Euclidean operator norm is bounded by the row and column sums: by the Cauchy–Schwarz inequality,

$$\|Av\|_2^2 \leq \Big(\sup_x \sum_y |A_{xy}|\Big)\Big(\sup_y \sum_x |A_{xy}|\Big)\|v\|_2^2.$$

For $t \geq 0$ the matrix $(K_0 + t)^{-1}$ has operator norm at most $(\lambda + t)^{-1}$. The Neumann series for the conjugated resolvent therefore converges, and we obtain $\|D_q(K_0 + t)^{-1}D_q^{-1}\|_2 \leq 2/(\lambda + t)$. Taking $q(x) = \operatorname{dist}(x, y)$ for a fixed y and reading off the (x, y) entry, we find

$$|(K_0 + t)^{-1}_{xy}| \leq \frac{2e^{-\mu \operatorname{dist}(x,y)}}{\lambda + t}, \qquad \|(K_0 + t)^{-1}\|_{\mu/2} \leq \frac{C_R}{\lambda + t},$$

where the second bound follows from the first and (18.4).

Next let L be the perturbation in (3). Since $C_R\|L\|_{\mu/2}/\lambda \leq 1/4$, we have

$$(K_0 + L + t)^{-1} = \sum_{n \geq 0} [-(K_0 + t)^{-1}L]^n (K_0 + t)^{-1}, \qquad \|(K_0 + L + t)^{-1}\|_{\mu/2} \leq \frac{4C_R}{3(\lambda + t)}. \tag{18.5}$$

The series converges absolutely in the coefficient norms, uniformly for $t \geq 0$. Bounds with every index fixed, and on the derivatives in the further variables, follow by inserting the differentiated perturbation and its connecting network into this series, or by differentiating its scalar geometric majorant on a strictly smaller radius. Here we use that L has bulk coefficient bounds with every index fixed; a pointwise bound on a matrix-valued function would not suffice.

For the inverse square root we use the scalar identity $\pi^{-1} \int_0^\infty t^{-1/2}(x + t)^{-1}\, dt = x^{-1/2}$ for $x > 0$, which follows from the substitution $t = xu^2$. Diagonalization gives the same formula for a real positive matrix. Expanding the resolvents as in (18.5) proves the formula for the complex perturbations of (3), and shows that it defines the branch obtained by holomorphic continuation from the positive inverse square root. Multiplying out the two resolvents, we have

$$(A + t)^{-1} - (B + t)^{-1} = -(A + t)^{-1}(A - B)(B + t)^{-1}.$$

At $t = 0$ this is the first identity in (4), and integrating it against $\pi^{-1}t^{-1/2}\, dt$ gives the second. The order of the factors matters, since they need not commute. By (2) and

(18.5), the integrand of the second identity has a boundary bound by a constant times $t^{-1/2}(\lambda+t)^{-2}$ times the boundary norm of $A-B$, and $\int_0^\infty t^{-1/2}(\lambda+t)^{-2}\,dt = \pi/(2\lambda^{3/2})$. The integral therefore preserves absolute coefficient bounds and the boundary weight. This proves (3) and (4).

Finally we treat Schur complements, completing the proof of (2). If $K = \left(\begin{smallmatrix} A & B \\ B^{\mathsf{T}} & C \end{smallmatrix}\right) \geq \lambda I$, then $A \geq \lambda I$ and

$$v^{\mathsf{T}}(C - B^{\mathsf{T}}A^{-1}B)v = \min_u (u,v)^{\mathsf{T}}K(u,v) \geq \lambda|v|^2, \tag{18.6}$$

where the equality follows by substituting the minimizer $u = -A^{-1}Bv$. In the coefficient norms the Schur complement is bounded by $\|C\| + \|B^{\mathsf{T}}\|\|A^{-1}\|\|B\|$. To compare two Schur complements, we replace the three factors of the product one at a time; each of the resulting terms contains one difference, possibly the inverse difference of (4), and therefore has a boundary bound by (2). For at most J successive eliminations we obtain a finite recursion: starting from a bound Q_0, the next bound can be taken to be $Q_{k+1} = Q_k + Q_k^2 I_k$, where I_k is the preceding inverse bound from (18.5). We choose the finitely many intermediate rates and radii in advance and take their minimum; there is no loss depending on the Taylor order or on the number of sites. The lower bounds used here are lower bounds for matrices in the free coordinates, and all constants depend only on the quantities listed in the statement. □

18.3 Local coordinates and Haar disintegration on block regions

The coordinates of a block region are obtained by solving, scale by scale, only the block averaging constraints of the coarse bonds that join two blocks of the region, with the local constraint coordinates of Lemma 6.3.

We set $R^{(t)} = (M^t - 1)/2$ for $t \geq 0$, so that $R^{(0)} = 0$ and $R^{(1)} = R_M$. In input lattice units, the final block with anchor v has the vertices $v + [-R^{(s)}, R^{(s)}]^3 \cap \mathbb{Z}^3$. Balanced base-$M$ digits give the disjoint decomposition

$$[-R^{(s)}, R^{(s)}]^3 \cap \mathbb{Z}^3 = \bigsqcup_{x \in [-R^{(s-1)}, R^{(s-1)}]^3 \cap \mathbb{Z}^3} (Mx + [-R_M, R_M]^3 \cap \mathbb{Z}^3). \tag{18.7}$$

To see this in one coordinate, we write an integer uniquely as $Mq + r$ with $-R_M \leq r \leq R_M$; since $MR^{(s-1)} + R_M = R^{(s)}$, the integers in $[-R^{(s)}, R^{(s)}]$ are exactly those with $|q| \leq R^{(s-1)}$. We apply this in each of the three coordinates and iterate. The internal links of a block region have both endpoints in Ω, and the links across the boundary of the block region have exactly one endpoint in Ω. At every intermediate scale we use exactly the blocks of this nested partition that lie in Ω.

Lemma 18.4. *Let $M \geq 3$ be odd, and let $s \geq 1$ and $m \geq 4$ be integers. On the input lattice of side $M^s m$, the value of T_M^s at a final link with endpoints v and $v + M^s e_i$ depends only on the input links with both endpoints in the union of the two centered M^s-blocks at v and $v + M^s e_i$. The tree paths of all intermediate scales inside a final block stay in that block. The same holds for the Borel extension of T_M by the reference-path transport of Definition 4.3. No divisibility of m by four is needed.*

Proof. We begin with one averaging step and write $x = v + \delta$ with $\delta \in [-R_M, R_M]^3 \cap \mathbb{Z}^3$. The coordinate-ordered path $\sigma_{v,x}$ stays in the block at v, since each coordinate of each intermediate vertex lies between its initial and final value. The path $\sigma_{v+Me_i,x+Me_i}$ stays in the translated block for the same reason. Along the straight segment from x to $x + Me_i$, the transverse coordinates stay equal to δ_j, and the longitudinal coordinate lies in $[-R_M, M + R_M]$. Since $M = 2R_M + 1$, we have

$$[-R_M, M + R_M] \cap \mathbb{Z} = ([-R_M, R_M] \cap \mathbb{Z}) \sqcup ([M - R_M, M + R_M] \cap \mathbb{Z}).$$

Every vertex of the segment lies in one of the two endpoint blocks, and its only link between the blocks joins the two adjacent boundary vertices. The reference path from v to $v + Me_i$ satisfies the same containment. Therefore every contour, based loop holonomy, and tree path in the definition (4.3) of the output uses only links with both endpoints in the union of the two blocks. The test whether some based loop holonomy equals $-I$ reads these same holonomies, and the exceptional output is the reference transport, so the Borel extension has no additional dependence.

For the induction, suppose that the assertion holds for r averaging steps. A one-step formula for a link at the next scale reads links whose endpoints lie in its two endpoint M-blocks on the current lattice. Each such link, pulled back through r steps, reads only the two centered M^r-blocks at its endpoints, by the induction hypothesis. By (18.7), these blocks partition the two centered M^{r+1}-blocks at the endpoints. All links that are read, including links between the smaller blocks, therefore have their endpoints in the union of the two large blocks. A tree path inside one block of the current lattice pulls back to tree paths in its smaller blocks and to the links joining these, all inside the same final block. This proves both assertions for $r + 1$ steps, including the exceptional outputs. When $m \geq 4$ the two final blocks lift injectively to a rectangular box in $\mathbb{Z}^3$; alternatively, the same coordinate containment can be read periodically. No step used a tiling, a reflection, or a congruence condition on the side. □

Remark 18.5. When $m \in 4\mathbb{N}$, the dependence in Lemma 18.4 agrees with the geometry of Lemma 7.4, applied with $D = s$ and $n = m$. The reflection and tiling conclusions of that lemma are not needed here. For the other sides $m \geq 4$, and for all shapes of block regions, Lemma 18.4 is the dependence statement used below; it requires no filling across a hole.

When we solve only the block averaging constraints of the internal coarse bonds, their eliminated-edge equations keep their independence and their derivatives after restriction to Ω: by Lemma 18.4, an internal equation reads only links in its two endpoint blocks, and it contains no eliminated edge other than its own. The following definition makes the construction explicit and fixes the linear maps used in the coercivity estimates of Sections 18.6 and 18.7.

Definition 18.6. Fix $1 \le s \le J$ and $m \ge 4$, put $n = M^s m$, and work in input lattice units on $(\mathbb{Z}/n\mathbb{Z})^3$. For an anchor $a \in (M^s\mathbb{Z}/n\mathbb{Z})^3$, let $\mathsf{B}_s(a) = a + \{-R^{(s)}, \dots, R^{(s)}\}^3$ be the final block at a. Let $\mathcal{K}$ be the set of anchors of the final blocks of Ω, let $\Omega_0 = \bigcup_{a\in\mathcal{K}} \mathsf{B}_s(a)$ be the vertex set of Ω, and let $\Omega_i = \Omega_0 \cap (M^i\mathbb{Z}/n\mathbb{Z})^3$ for $0 \le i \le s$, so that $\Omega_s = \mathcal{K}$.

1. *Edges.* For $0 \le i \le s$, let $\mathcal{L}_i$ be the set of positive edges of length M^i with both endpoints in Ω_i. Thus $\mathcal{L}_0 = L_\Omega$ is the set of internal links of Ω, including the links that join two final blocks of Ω, and $\mathcal{L}_s$ is the set of coarse bonds. An input link with exactly one endpoint in Ω_0 is a link across the boundary of the block region; such links are not coordinates of the block region. A reversed edge carries the opposite cochain value and the inverse group value.
2. *Blocks and trees.* For $i < s$ the blocks $\mathsf{B}_i(a) = a + M^i\{-R_M, \dots, R_M\}^3$, $a \in \Omega_{i+1}$, partition Ω_i, by (18.7). Write $\pi_i(x) = a$ for $x \in \mathsf{B}_i(a)$. For $x \in \mathsf{B}_i(a)$, let $\sigma_{a,x}$ be the path from a to x with steps of length M^i that changes the coordinates in the order $1, 2, 3$. In each block these paths form a tree rooted at a, and $\mathcal{T}_i \subset \mathcal{L}_i$ is the union of these trees.
3. *Eliminated and free edges.* For a bond $c' = (a, a + M^{i+1}e_k) \in \mathcal{L}_{i+1}$, its eliminated edge is
$$e_i^*(c') = (a + R_M M^i e_k,\ a + (R_M + 1)M^i e_k) \in \mathcal{L}_i.$$
Let $\mathcal{D}_i = \{e_i^*(c') : c' \in \mathcal{L}_{i+1}\}$, and let $\mathcal{F}_i = \mathcal{L}_i \setminus (\mathcal{T}_i \cup \mathcal{D}_i)$ be the set of free edges at scale i. No constraint and no free coordinate is attached to a link across the boundary.
4. *Linearized maps.* For $c' = (a, a + M^{i+1}e_k) \in \mathcal{L}_{i+1}$ and $x \in \mathsf{B}_i(a)$, let
$$F_{i,c',x} = \sigma_{a,x}[x, x + M^{i+1}e_k]\,\sigma^{-1}_{a+M^{i+1}e_k, x+M^{i+1}e_k},$$
a path with steps of length M^i. With $[\omega : e]$ the signed number of traversals of the edge e by the path ω, define the scalar matrices
$$(Q_i)_{c'e} = M^{-3} \sum_{x\in\mathsf{B}_i(a)} [F_{i,c',x} : e], \qquad \Pi_i^{\mathcal{T}} u = u|_{\mathcal{T}_i},$$
which act on three colors by tensoring with I_3. Put $Q_{[0,0)} = I$ and $Q_{[0,i)} = Q_{i-1}\cdots Q_0$, and define
$$G_\Omega u = \big((\Pi_i^{\mathcal{T}} Q_{[0,i)} u)_{0\le i<s},\ Q_{[0,s)} u\big) \tag{18.8}$$

for edge cochains u on $\mathcal{L}_0$. The last component prescribes zero variation of every coarse bond of Ω; no bond to a final block outside Ω is constrained. The kernel $\ker G_\Omega$ is the space of linearized variations of the block region.

5. *Reconstruction.* For a group field b on $\mathcal{L}_{i+1}$, its coarse-field lift is

$$(\mathcal{H}_i b)_e = \begin{cases} I, & \pi_i(x) = \pi_i(y), \\ b_{(\pi_i(x),\pi_i(y))}, & \pi_i(x) \neq \pi_i(y), \end{cases} \qquad e = (x,y) \in \mathcal{L}_i, \tag{18.9}$$

with the orientation convention in the second line. Start with $U^{(s)} = \exp(c)$ on $\mathcal{L}_s$ and descend by

$$U_e^{(i)} = \exp\big(\Gamma_{U^{(i+1)},e}(z_i)\big)\,(\mathcal{H}_i U^{(i+1)})_e, \qquad e \in \mathcal{L}_i. \tag{18.10}$$

Here Γ_b is the map of local constraint coordinates of Lemma 6.3 at the target $b = U^{(i+1)}$, solving only the constraints indexed by $\mathcal{L}_{i+1}$ for the logarithms on the eliminated edges $\mathcal{D}_i$, with free entries $z_i \in (\mathbb{R}^3)^{\mathcal{F}_i}$ and zero entries on $\mathcal{T}_i$. All products are taken on a common identity chart, smaller than the chart of definition, on which these finitely many inverses exist. Write $\Phi_\Omega(c,z) = U^{(0)}$ for the resulting field on $\mathcal{L}_0$. Thus Φ_Ω receives the logarithm c of the coarse field on $\mathcal{L}_s$, not its group value. The logarithms c of the coarse links, the free coordinates $z = (z_0, \dots, z_{s-1})$, and the logarithms ξ of the current frozen coordinates are independent parameters of this chart. The reference measure of the coarse links is Haar measure of their group values, written $dH(c)$ in Lemma 18.7; in these coordinates it is $dH(e^c) = \prod_e c_{\mathrm{Haar}} j_H(c_e)\, d^3c_e$, with the product over the coarse links, not Lebesgue measure.

6. *Tree-path transports to the basepoints.* For $x \in \Omega_0$ let $x_0 = x$ and $x_{i+1} = \pi_i(x_i)$. For $1 \le k \le s$ and a configuration U on $\mathcal{L}_0$, let

$$\Theta_{x,k}(U) = U^{(k-1)}(\sigma_{x_k,x_{k-1}}) \cdots U^{(0)}(\sigma_{x_1,x_0}), \tag{18.11}$$

where $U^{(i)} = T_M^i U$ on $\mathcal{L}_i$; this is a transport with endpoints x_k and x. A frozen coordinate on a link across the boundary at x that is based at the level-k ancestor x_k is represented by its based transport, obtained by multiplying the link value from the left by $\Theta_{x,k}$; a link across the boundary frozen at the first block integration is based at the root of its final block, $k = s$. We write $\Theta_x = \Theta_{x,s}$.

On the reconstruction we have $T_M^i \Phi_\Omega(c,z) = U^{(i)}$ on $\mathcal{L}_i$, since each Γ solves the constraints of the internal bonds, and these read only internal links by Lemma 18.4. Each factor in (18.11) is a transport along a tree path, and the tree links of (18.10) are equal to I. Hence $\Theta_{x,k}(\Phi_\Omega(c,z)) = I$ identically, also at nonzero (c,z) in the chart. In particular, the based tree gauge transformations fix both endpoints of a frozen coordinate represented by its based transport. The intermediate factors with $k < s$ only describe

the based transport at an earlier scale; they are not additional frozen coordinates. The coarse output on a bond across the boundary of the block region is evaluated from the reconstructed field after this construction; it is not a prescribed target. The frozen coordinates supply the based transports of frozen links across the boundary and the links of the fixed complement, each at the scale at which it was introduced. Arguments of an analytic torus action outside the block region that were outside earlier block regions are set to zero (logarithm zero), as described in Theorem 18.12. Arbitrary data deep in the fixed complement are never arguments of an analytic function.

Lemma 18.7. *Fix $1 \le s \le J$, a block region Ω, and Borel sets $\mathcal{S}_{c,b_{\mathrm{rec}}}$ of free coordinates z in the chart of Definition 18.6, depending measurably on the coarse links c and on the frozen coordinates b_{rec}. For tree coordinates $\mathfrak{f}$, that is, a gauge transformation at the nonroot vertices of the trees of all scales, let $U(c, z, \mathfrak{f}; b_{\mathrm{rec}})$ be the configuration obtained from $\Phi_\Omega(c, z)$ by the based gauge transformation $\mathfrak{f}$ and completed outside Ω by the frozen coordinates. Let the incoming finite positive measure have a Borel density $p(U, b_{\mathrm{rec}})$ with respect to the product of Haar probability measures on the links of Ω and Haar measures on the independently introduced frozen coordinates, each at the scale at which it was introduced; any normalizing constant of the incoming measure is included in p. Let $\chi(U, b_{\mathrm{rec}})$ be the product of all indicators carried by the incoming measure: those of the large-field decomposition, of the coordinate domains, of the small-field conditions of Definition 14.4, and of the fluctuation cutoffs. Assume coverage of the fiber by the chart: every configuration U with $p(U, b_{\mathrm{rec}})\chi(U, b_{\mathrm{rec}}) > 0$ has the form $U(c, z, \mathfrak{f}; b_{\mathrm{rec}})$ with $z \in \mathcal{S}_{c,b_{\mathrm{rec}}}$, after the based tree gauge transformation that fixes its coarse field and the based transports of its frozen coordinates.*

Then the pushforward of the measure with density $p\chi$ to (c, b_{rec}) has the density

$$p^{\mathrm{out}}(c, b_{\mathrm{rec}}) = \int dH(\mathfrak{f}) \int_{\mathcal{S}_{c,b_{\mathrm{rec}}}} \chi(U(c, z, \mathfrak{f}; b_{\mathrm{rec}}), b_{\mathrm{rec}})\, p(U(c, z, \mathfrak{f}; b_{\mathrm{rec}}), b_{\mathrm{rec}}) \prod_{i=0}^{s-1} J_i \prod_{i=0}^{s-1} dz_i$$

with respect to $dH(c) \prod_e dH(b_{\mathrm{rec},e})$, where the product is over the independently introduced frozen coordinates, $dH(\mathfrak{f})$ is the product of Haar probability measures, and we use the letter c both for the coarse links and for their logarithms. If f_i is the number of free group variables at scale i, the Haar disintegration density at scale i is

$$J_i = c_{\mathrm{Haar}}^{f_i} \frac{\prod_{e \in \mathcal{L}_i \setminus \mathcal{T}_i} j_H(\Gamma_{i,e})}{\prod_{c' \in \mathcal{L}_{i+1}} \det D_{X_{e_i^*(c')}} \mathsf{C}_{b,c'}(\Gamma_i)}, \qquad j_H(X) = \operatorname{sinc}^2(|X|/2), \qquad c_{\mathrm{Haar}} = \frac{1}{16\pi^2},$$

where $\Gamma_i = \Gamma_{U^{(i+1)}}(z_i)$ and $\mathsf{C}_{b,c'}$ is the block averaging constraint of c' at the target $b = U^{(i+1)}$. All Haar factors of the eliminated edges appear in the numerator. The determinants are positive on the real chart and continue holomorphically, without zeros, to its complex neighborhood. There is no further Haar density of the coarse field, no Gram determinant, no gauge-orbit volume, and no

Jacobian of a field-dependent choice of gauge. The formula is an identity tested against nonnegative Borel functions, and the displayed integral defines the density at every covered target.

When a frozen coordinate is introduced on a link e across the boundary at a fixed field v of the block region, the change from the link value a_e to the based transport $\widehat{b}_e = U_v(\sigma_e)a_e$, where $U_v(\sigma_e)$ is the tree-path transport of e to its basepoint in the field v, is a Haar translation that is triangular in the coordinates. Frozen coordinates introduced earlier are unchanged, and a blocked image of an earlier frozen coordinate is a function of the existing coordinates, not an additional Haar coordinate. If $p\chi$ is invariant under the based tree gauge transformations that fix the coarse field and the frozen coordinates, the integral over $\mathfrak{f}$ equals one and may be omitted; otherwise it remains in the formula. The identities are finite-dimensional, and the analytic bounds of the chart are uniform. Coverage is a hypothesis of this lemma.

Proof. We first fix one scale and the frozen coordinates. On each block we can remove a leaf of its rooted tree: we replace the variable on that edge by the transport from the root to the leaf, and translate the variables on the other edges at the leaf accordingly. At fixed values of the remaining variables this is a left or right Haar translation. We remove the leaves successively. The inverse map reconstructs each edge variable from the transports to its endpoints and its based value, so this is a bijective triangular change of group coordinates. It gives Haar probability measure in each tree variable at a nonroot vertex and in each based link outside the trees. No division by a stabilizer or by an orbit volume occurs, also for a single isolated block.

For an internal bond its eliminated-edge equation is the equation of Lemma 6.3 on its two endpoint blocks. Its determinant equals $\det(M^{-2}I_3) > 0$ at zero and stays positive on the real chart. We compute the normalization when only internal equations are solved. Let n_i be the number of internal group links outside $\mathcal{T}_i$ and q_i the number of internal constraints, so that $f_i = n_i - q_i$. At a fixed target c_{i+1} we write $U_e = e^{X_e}U_{0,e}(c_{i+1})$, where U_0 is the coarse-field lift. Right translation preserves Haar measure, and these n_i factors have the density $c_{\mathrm{Haar}}^{n_i}\prod_e j_H(X_e)$ with respect to dX, by the coordinate formula for Haar measure in (4.2).

The tested pushforward can be computed by inserting a unit point mass at $T_\Omega U = c_{i+1}$ with respect to Haar measure on the targets, where T_Ω denotes the internal outputs of T_M. More explicitly, if δ_H is the density of the unit point mass at the identity, then

$$\int dH(c_{i+1})\,\delta_H\big((T_\Omega U)c_{i+1}^{-1}\big)\,G(c_{i+1},U) = G(T_\Omega U, U)$$

for nonnegative tests G. This notation only expresses a change of variables. In logarithmic coordinates Y at the identity it reads $\delta_H(e^Y) = c_{\mathrm{Haar}}^{-1}\delta_0(Y)$, since $j_H(0) = 1$. The logarithms of the constraints are $Y_{c'} = \mathsf{C}_{b,c'}(X)$. The local substitution $(X_{e_i^*(c')})_{c'} \mapsto (Y_{c'})_{c'}$, with the free logarithms fixed, has Jacobian equal to the product of the eliminated-edge

determinants, since each equation contains no other eliminated variable. We obtain

$$c_{\mathrm{Haar}}^{n_i-q_i}\frac{\prod_{e\in\mathcal{L}_i\setminus\mathcal{T}_i} j_H(\Gamma_{i,e})}{\prod_{c'}\det D_{X_{e_i^*(c')}}\mathsf{C}_{b,c'}(\Gamma_i)}\,dz_i = J_i\,dz_i,$$

with the Haar density of every solved eliminated edge still in the numerator. The point-mass calculation can equivalently be carried out by first making this invertible local substitution and then integrating the unit mass in Y; no global disintegration theorem is needed. This proves the identity for continuous tests on compact subsets of the chart. Exhaustion of the open chart and monotone approximation extend it to nonnegative Borel tests. The reference measure $dH(c_{i+1})$ is already present in this calculation; expressing it once more in logarithms and keeping its density as another factor would count it twice.

We apply this one-scale identity to a nonnegative Borel test of the coarse links and frozen coordinates, multiplied by the Borel function $p\chi$, keeping the tree variables in the integrand. The internal output Haar variables of scale i are the internal input Haar variables of scale $i+1$. Repeated nonnegative integration therefore gives the product of the J_i, Lebesgue measure in all free logarithms, Haar probability measure in each tree variable of each scale, and Haar measure on the coarse links. This proves the displayed formula against every nonnegative Borel test. Restricting each free integral to $\mathcal{S}_{c,b_{\mathrm{rec}}}$ and keeping χ records the domain of the chart. The coverage hypothesis says that the image of this domain under the based tree gauge transformations contains every point that is pushed forward; without it the argument proves only the identity restricted to the chart.

The local coordinate formulas are Borel, their domains are Borel, and their Jacobians are continuous on the chart. After extension of the integrand by zero outside these domains, the integral is a Borel function of (c, b_{rec}): this holds first for indicators of rectangles, then for nonnegative simple functions, and then by increasing approximation. At an exceptional target its value may be $+\infty$; since the incoming measure is finite, it is finite almost everywhere. No unspecified almost-everywhere density enters the pointwise formula.

When a frozen coordinate is introduced, we hold the field v of the block region fixed. For every nonnegative Borel function G,

$$\int G(v, U_v(\sigma_e)a_e)\,dH(a_e) = \int G(v, \widehat{b}_e)\,dH(\widehat{b}_e),$$

by left invariance of Haar measure. Iterating over the newly frozen links and then integrating over v proves the assertion on Haar translations, although the translations depend on v. If the orientation gives a right translation instead, right invariance gives the same identity. An earlier frozen coordinate is not changed by this operation, and a later blocked value computed from it is a function on the existing coordinate space, not a new independent coordinate. This proves the statement on the reference measure of the frozen coordinates; it makes no claim of probabilistic independence under the density.

If $p\chi$ is invariant under the based gauge transformations in the statement, its value in these coordinates does not depend on $\mathfrak{f}$, since these gauge transformations fix both the target and the endpoints of the based transports of the frozen coordinates. Each Haar probability factor of the tree variables then integrates to one. Finally, the determinants do not vanish on the complex neighborhood of the chart, by the local inverse bounds of Lemma 6.3. Since there are at most J scales, each with bounded stencils and fixed radii, the analytic constants of the chart are uniform. The measure identity itself needs no analyticity of p. □

18.4 Derivatives of constrained minima

The constrained minimizers below are stationary points of an objective in the free coordinates, with the coarse field and the current frozen coordinates as parameters. The identities below hold at complex arguments; positivity of the Hessian, and hence minimization, enters only at real data.

Lemma 18.8. *Let $F(u, y; b_{\mathrm{rec}})$ be holomorphic on a finite-dimensional complex product and real at real arguments, where u denotes free coordinates and y parameters. Let $a(y)$ be a holomorphic branch of solutions of $F_u(a(y), y; b_{\mathrm{rec}}) = 0$ along which $A = F_{uu}$ is invertible. All derivatives below are evaluated at $(a(y), y; b_{\mathrm{rec}})$, with the frozen coordinates b_{rec} fixed. For a direction η of the parameters (the coarse field or the current frozen coordinates), set $V_\eta = (-A^{-1}F_{uy}\eta, \eta)$, and let $F^{\min}(y) = F(a(y), y)$. Then*

$$Da\,\eta = -A^{-1}F_{uy}\eta, \qquad DF^{\min} = F_y, \qquad D^2F^{\min} = F_{yy} - F_{yu}A^{-1}F_{uy},$$

and

$$D^3F^{\min}[\eta_1, \eta_2, \eta_3] = D^3F[V_{\eta_1}, V_{\eta_2}, V_{\eta_3}].$$

At intermediate scale i the effective classical action is $M^iF^{\min}$, and each of its derivatives carries the same factor M^i. Successive and joint constrained minimizers agree wherever all the conditional branches and the joint branch are defined in their stated domains and unique there.

If $F = E \circ \Psi$, then for any directions u, v of the free coordinates or of the parameters,

$$F_{uv} = \Psi_u^{\mathsf{T}} E''(\Psi)\Psi_v + E'(\Psi)\Psi_{uv}.$$

The second term is present also at a constrained minimizer. In particular, differentiation in a current frozen coordinate ξ gives $D_\xi a = -F_{uu}^{-1}F_{u\xi}$, where $F_{u\xi}$ contains both terms. At complex arguments, transposes are bilinear transposes. Higher derivatives of a require only the inverse of F_{uu}. Weighted coefficient bounds on these derivatives hold in the coefficient norms on smaller fixed products, with constants depending on the coefficient bounds of F, on the bound for F_{uu}^{-1}, and on uniform data. No inverse of a Hessian in the coarse field or in the frozen coordinates is used, and no derivative in a frozen coordinate b_{rec} is taken.

Proof. We begin with the existence and uniqueness of the branch near a given point of it. At that point the matrix $A_* = F_{uu}$ is invertible, so the map $u \mapsto u - A_*^{-1} F_u(u, y)$ has derivative zero there. By continuity, there exist a ball of free coordinates on which this derivative has norm at most $1/2$ and a smaller neighborhood of parameters on which the map moves the base point by less than a quarter of the radius of that ball. The iteration is then a contraction of the ball into itself, and the analytic implicit-series statement of Lemma 5.3 gives its holomorphic solution. Local uniqueness identifies this solution with the given branch, and these local constructions agree on their common neighborhoods.

Let $L(y) = (a(y), y)$, with b_{rec} fixed. Differentiating $F_u \circ L = 0$, we obtain

$$A\, Da\, \eta + F_{uy}\eta = 0.$$

This proves the formula for Da and shows that $DL\,\eta = V_\eta$. The first derivative of $F \circ L$ is F_y, since $F_u = 0$. Differentiating this equality once more and inserting the formula for Da, we find the stated Schur complement formula for $D^2F^{\min}$.

For the third derivative it is convenient to keep the tangent vectors. For every vector h of free coordinates and every parameter direction η, we have

$$D^2F[(h,0), V_\eta] = h^{\mathsf{T}}(A\, Da\, \eta + F_{uy}\eta) = 0. \tag{18.12}$$

The second derivative can also be written $D^2(F \circ L)[\eta_1, \eta_2] = D^2F[V_{\eta_1}, V_{\eta_2}]$. Its derivative in the direction η_3 consists of $D^3F[V_{\eta_1}, V_{\eta_2}, V_{\eta_3}]$ and two further terms, in each of which one of the vectors V is differentiated. A differentiated V has the form $(D^2a[\eta_j, \eta_3], 0)$, so both further terms vanish by (18.12). This proves the polarized identity for the third derivative.

By the chain rule, before any stationarity is imposed, we have

$$D_u D_v (E \circ \Psi) = (D_u\Psi)^{\mathsf{T}} E''(\Psi) D_v\Psi + E'(\Psi) D_u D_v \Psi.$$

Stationarity of the constrained objective says only that $E'(\Psi)\Psi_u = 0$ for the free directions u; it does not remove $E'(\Psi)\Psi_{uv}$. Taking v to be the direction of a current frozen coordinate, we obtain the formula for $D_{\check{\zeta}} a$. Since these are polynomial identities between derivatives, their holomorphic versions use the same bilinear transpose, without complex conjugation of the parameters.

We now turn to higher derivatives. Differentiating $F_u \circ L = 0$ in r given directions and using the chain rule indexed by the partitions π of these directions, we obtain

$$0 = \sum_{\pi} D^{|\pi|} F_u(L)\big[D^{|B|} L[\eta_j : j \in B] : B \in \pi\big].$$

For $r \geq 2$ the partition with a single block contributes $A\, D^r a$, since the higher derivatives of the second component of L vanish. Every other partition contains only derivatives

of a of order less than r. The new derivative is therefore obtained by solving one linear equation with the matrix A. For $r = 1$ this is the formula for Da proved above.

A quantitative bound uses given coefficient majorants for F and for the inverse A^{-1}. On a product where these bounds hold, we divide the partition formula by $r!$, use the contraction bounds of Lemma 5.3, and pass to a strictly smaller product of parameters. After the derivative condition for the preconditioned nonlinear map has been imposed, the solution is bounded by the majorant series Y with nonnegative coefficients determined by

$$Y = Ct + \kappa Y + \frac{C(t+Y)^2}{1 - C(t+Y)}, \qquad \kappa < 1.$$

By the implicit-series statement of Lemma 5.3, we obtain the factorial bounds and their versions with a specified input index; the boundary-weighted version is Lemma 18.3(2). The constants depend only on these coefficient bounds, the bound for A^{-1}, and uniform data. The hypotheses of this quantitative application are verified for the objective of a block region in Theorem 18.12.

To compare successive and joint minimizers, we split the free coordinates as (u, v). At a stationary point of the partial minimum value as a function of v, the formula $DF^{\min} = F_y$ implies $F_v = 0$, while the conditional minimizer already satisfies $F_u = 0$. It is therefore the joint stationary point in the given domain. Conversely, the joint stationary point satisfies the conditional stationarity equations, and uniqueness of their branches identifies it with the successive construction. This argument applies only where the conditional minimizers and the joint point lie in their stated domains, and it asserts nothing about other branches. Finally, one application of the nested minimization identity of Theorem 10.10 multiplies the minimum value by M, so after i steps the factor is M^i. Multiplication by a positive constant does not change the minimizer, and every derivative of the value carries the same factor. □

18.5 Freezing variables

When a block region shrinks, the variables at the removed labels are frozen: they are replaced by fixed values, which become frozen coordinates. The next lemma describes how the coefficient norm, a torus interaction, the effective density, and the triangular determinant behave under this substitution.

Lemma 18.9. *Use the norms and the position convention of Definition 18.1. Let D be a set of labels of a block region, with the radius ϕ of Definition 18.1 for D, let $s_{\mathrm{er}} \geq \max\{2, w_{\mathrm{bd}}\}$, let $I \subset \{i \in D : d_D(i) \geq s_{\mathrm{er}}\}$, and let $F = D \setminus I$. The labels in F are frozen. Since $s_{\mathrm{er}} \geq w_{\mathrm{bd}}$, every label of D at which $\phi < \rho$ lies in F; the set I may omit further labels of D, which are additional holes. Let $c = (c_j)_{j \in F}$ satisfy $|c_j| \leq \eta\phi(j)$ with $0 \leq \eta \leq 1/4$. For a function v on I, write*

$v \oplus c$ for the variables on D equal to v on I and to c on F, and for an analytic function f set $T_c f(v) = f(v \oplus c) - f(0 \oplus c)$. The following statements hold.

1. *For $i \in I$ we have $d_D(i) \geq d_I(i) + s_{\mathrm{er}} - 1$, and*

$$\mathcal{X}_I^{\rho}(T_c f) \leq (1-\eta)^{-2} e^{-\alpha_{\mathrm{bd}}(s_{\mathrm{er}}-1)} \mathcal{X}_D^{\phi}(f) \leq 2 e^{-\alpha_{\mathrm{bd}}(s_{\mathrm{er}}-1)} \mathcal{X}_D^{\phi}(f).$$

   The norm on the left includes every remaining nonconstant degree, including degree one. The same statement holds with the separate classical radius and rates.

2. *Let $r_{\mathbb{T}}$ be a function on the torus with no linear term and $\mathcal{B}_{\rho,\bar{a}}(r_{\mathbb{T}}) \leq E$, where $\mathcal{B}_{\rho,\bar{a}}$ is the torus norm of Definition 4.6, with the Euclidean network length, and $\bar{a} \geq \sqrt{3}(a_{\mathrm{sp}} + \alpha_{\mathrm{bd}})$, and assume $|c_j| \leq \beta = \eta\rho$ for $j \in F$. The boundary-completion difference*

$$G_c r_{\mathbb{T}}(v) = r_{\mathbb{T}}(v,c) - r_{\mathbb{T}}(0,c) - r_{\mathbb{T}}(v,0) + r_{\mathbb{T}}(0),$$

   that is, the difference between the substitution of the frozen values c and the substitution of zero, satisfies $\mathcal{X}_I^{\rho}(G_c r_{\mathbb{T}}) \leq 8\eta E$. All torus labels outside D are set to zero in these evaluations. The analogous difference for the effective classical action has its own radius and bound.

3. *Consider a joint density, relative to its given reference measure, of the form*

$$Z^{-1} e^{\Lambda(b_{\mathrm{rec}})}\, W(b_{\mathrm{rec}})\, e^{-E(v,c;b_{\mathrm{rec}})/g^2 - R(v,c;b_{\mathrm{rec}})} \chi(v,c,b_{\mathrm{rec}}),$$

   where $W(b_{\mathrm{rec}}) \geq 0$ is a weight depending only on the frozen coordinates (not to be confused with the boundary W). Freezing c gives the same joint measure, with the same Z, the same indicators, and the same effective classical action E, with the enlarged frozen coordinates (b_{rec}, c), and with

$$R_{\mathrm{new}}(v) = R(v,c) - R(0,c), \qquad \Lambda_{\mathrm{new}} = \Lambda - R(0,c).$$

   If the effective classical action is centered as well, its factor $e^{-E(0,c;b_{\mathrm{rec}})/g^2}$ remains as an external factor. No partition function at fixed frozen coordinates is introduced. At uniform radius, write $R = r_{\mathbb{T}}|_D + f$ with $r_{\mathbb{T}}$ as in (2), and set

$$\epsilon_D(f) = \sum_{k \geq 2} \rho^{k-2} \sup_{i \in D} e^{\alpha_{\mathrm{bd}} d_D(i)} \sum_{|\nu|=k} \frac{\nu_i}{k} |f_\nu| e^{a_{\mathrm{sp}} \mathfrak{t}(\nu)}, \qquad \ell_D(f) = \sup_{i \in D} e^{\alpha_{\mathrm{bd}} d_D(i)} |f_{e_i}|.$$

   If $\|c\|_\infty \leq \beta \leq \rho/4$, then

$$|R(0,c;b_{\mathrm{rec}}) - R(0;b_{\mathrm{rec}})| \leq |F| \big\{ \beta \ell_D(f) + \beta^2 [\epsilon_D(f) + E] \big\}.$$

   This bounds the change of the constant term; it does not bound $R(0;b_{\mathrm{rec}})$ itself.

4. *Consider a sequential normalization by the constrained Hessians,*

$$z_i = a_i(c_{i+1}; b_{\mathrm{rec}}) + g_i K_i(c_{i+1}; b_{\mathrm{rec}})^{-1/2} w_i, \qquad c_{i+1} = c_{i+1}(w_{>i}),$$

where c_{i+1} denotes the coarser fields as functions of the coarser normalized fluctuation variables, and let $d_i = 3f_i$ be the number of scalar free coordinates at scale i. The Jacobian determinant of $w \mapsto z$ is

$$\prod_i g_i^{d_i} \det K_i(c_{i+1}(w_{>i}); b_{\mathrm{rec}})^{-1/2}.$$

Adding and subtracting half of $\log\det K_i(\widehat{c}_{i+1}; b_{\mathrm{rec}})$, where $\widehat{c}_{i+1} = c_{i+1}(0)$, splits this single determinant into its value at zero fluctuation and the logarithmic difference containing all the dependence on the coarser normalized fluctuation variables. Missing Gaussian coordinates and all factors at zero fluctuation remain explicit.

At frozen coordinates represented by their based transports, we have the identity $f(v \oplus c) = T_c f(v) + f(0 \oplus c)$. Every algebraic identity above is finite-dimensional. All bounds are uniform and independent of additional holes, of the number of preceding block integrations, of the widths chosen later, and of the number of frozen labels except through the displayed factor $|F|$.

Proof. We begin with the distance inequality in (1). If I is empty there is no nonconstant coefficient to estimate. If D has no boundary, the boundary perturbation is zero by the convention of Definition 18.1. Otherwise let $i \in I$, and let $h = (2M^s)^{-1} \le 1$ be the mesh of the refined position graph. We choose a shortest path $i = x_0, \dots, x_k$ of vertices of this graph with x_k outside D, so that $kh = d_D(i)$, and we set $s' = \lceil s_{\mathrm{er}}/h \rceil$. Since $kh = d_D(i) \ge s_{\mathrm{er}}$ and k is an integer, we have $k \ge s'$. The vertex $x_{k-s'+1}$ is at distance at most $(s'-1)h < s_{\mathrm{er}}$ from the complement of D, so it is not a position of I. We conclude that

$$d_I(i) \le (k - s' + 1)h \le d_D(i) - s_{\mathrm{er}} + h \le d_D(i) - s_{\mathrm{er}} + 1.$$

Additional holes can only decrease d_I. This proves the distance inequality, with the exact term -1. Removing all labels at a position together is what makes this a statement on the graph of positions. The fixed buffer of Definition 18.1 transfers the same argument to another convention for link endpoints.

Next we prove the coefficient bound in (1). We fix the earlier frozen coordinates and an allowed c, and for each input degree m we set

$$A_m = \rho^{-2} \sup_{i \in D} e^{\alpha_{\mathrm{bd}} d_D(i)} \sum_{|\nu| = m} \frac{\nu_i}{m} |f_\nu| e^{a_{\mathrm{sp}} \mathfrak{t}(\nu)} \phi^\nu.$$

A monomial of degree m splits uniquely as (ν_I, ν_F), of degrees $k \ge 1$ and $j \ge 0$ with $m = k + j$. Every remaining radius is ρ, since all labels with smaller radius are frozen. For a fixed index $i \in I$, the new factor ν_i/k is at most m/k times the old factor ν_i/m, and

$m/k \le 1 + j$. Substituting the frozen values costs at most a factor η^j, and $\mathrm{t}(\nu_I) \le \mathrm{t}(\nu)$. The distance inequality contributes the factor $q = \exp[-\alpha_{\mathrm{bd}}(s_{\mathrm{er}} - 1)]$. Combining these factors, we find that the outgoing summand of degree k is at most $q \sum_{j \ge 0} (1 + j)\eta^j A_{k+j}$; coinciding outgoing monomials are bounded by the triangle inequality. We sum over k after taking the suprema over the fixed index. Since all terms are nonnegative, we obtain

$$\sum_{k \ge 1} q \sum_{j \ge 0} (1 + j)\eta^j A_{k+j} \le q\Big(\sum_{j \ge 0} (1 + j)\eta^j\Big) \sum_{m \ge 1} A_m = q(1 - \eta)^{-2} \sum_{m \ge 1} A_m.$$

This includes $k = 1$, even when the linear term comes from a monomial of high degree. No binomial factor for the choice of frozen factors appears, since the two label sets are fixed. The suprema over the earlier frozen coordinates and over the allowed real values c are taken after this inequality and the degree sums. Since $(1 - \eta)^{-2} \le 16/9 < 2$, this proves the two inequalities in (1). For the classical statement we replace the three parameters by the classical radius and rates in the same calculation; the partition of labels and the ratios of the factors ν_i/k are unchanged.

We now explain why the outgoing norm has uniform radius. We write ϕ_{cl} for the classical radius function. When all labels with smaller radius are frozen, we have

$$\phi|_I \equiv \rho, \qquad \phi_{\mathrm{cl}}|_I \equiv \rho_{\mathrm{cl}}. \tag{18.13}$$

Indeed, $d_D(i) \ge s_{\mathrm{er}} \ge w_{\mathrm{bd}}$ excludes every remaining position from the boundary strip, and additional holes remove labels without changing the radius at the remaining ones. This means that $\mathcal{X}_I^{\rho}(T_c f)$ is the norm with the uniform radius ρ at every remaining variable, and its classical counterpart has uniform radius ρ_{cl}. These are the incoming norms at uniform radius for the next calculation on the same list of labels, with their existing scales of introduction. The new values c are fixed real frozen coordinates, not additional variables with the radius ρ, and no extension beyond an earlier analytic domain is asserted, since each remaining variable already had the radius ρ.

For (2), only monomials that meet both I and F contribute to $G_c r_{\mathbb{T}}$. For any fixed index $i \in I$, the network of such a monomial reaches the complement of I, so its ℓ^1 length is at least $d_I(i)$, as well as at least $\mathrm{t}(\nu_I)$. Since the torus norm uses the Euclidean network length t_2 and $\mathrm{t} \le \sqrt{3}\mathrm{t}_2$, we obtain

$$a_{\mathrm{sp}}\mathrm{t}(\nu_I) + \alpha_{\mathrm{bd}} d_I(i) \le (a_{\mathrm{sp}} + \alpha_{\mathrm{bd}})\mathrm{t}(\nu) \le \sqrt{3}(a_{\mathrm{sp}} + \alpha_{\mathrm{bd}})\mathrm{t}_2(\nu) \le \bar{a}\mathrm{t}_2(\nu).$$

We use (18.2) for the torus norm. The degree calculation above now has $j \ge 1$, and its incoming sum is at most E. We obtain the stronger estimate

$$\mathcal{X}_I^{\rho}(G_c r_{\mathbb{T}}) \le [(1 - \eta)^{-2} - 1]E \le 4\eta E \le 8\eta E.$$

For the middle inequality we wrote the bracket as $\eta(2 - \eta)/(1 - \eta)^2$ and used $\eta \le 1/4$. This calculation counts the linear term produced by the substitution. Setting the other

torus labels to zero only deletes coefficients. For the classical difference we replace the radius, the rates, and the bound by their classical counterparts in the same calculation, with the same factor $\sqrt{3}$ in the rate hypothesis.

For (3), the statement on the measure is an algebraic identity with the given reference measure. Indeed,

$$e^{\Lambda - R(0,c)} e^{-[R(v,c)-R(0,c)]} = e^{\Lambda - R(v,c)}.$$

The weight $W(b_{\mathrm{rec}})$, every indicator, and the term $E(v,c;b_{\mathrm{rec}})/g^2$ are unchanged. Splitting E into $E(v,c) - E(0,c)$ and $E(0,c)$ gives the external classical factor in the statement. Fubini's theorem with the given reference measure proves the identity of joint measures, with the same Z; no normalization at fixed frozen coordinates has taken place. If the frozen variables are based transports, the identity is evaluated at those same values; it is a substitution in existing variables.

Only monomials supported in F contribute to $R(0,c;b_{\mathrm{rec}}) - R(0;b_{\mathrm{rec}})$. At each degree m, we assign such a monomial to its labels with weights ν_i/m, whose sum is one, and drop the network and distance weights, which are at least one. In degree one the sum is at most $|F|\beta\ell_D(f)$. For $m \geq 2$ we use $\beta^m \leq \beta^2\rho^{m-2}$. Summing over $i \in F$ and then over m bounds the contribution of f by $|F|\beta^2\epsilon_D(f)$ and that of the torus part by $|F|\beta^2 E$. The constant terms cancel in this difference, which is why the bound says nothing about an arbitrary incoming $R(0;b_{\mathrm{rec}})$. This proves the bound on the change of the constant term.

Finally we prove (4). We order the blocks of free coordinates from the coarser scales to the finer ones. The sequential map has the blocks

$$z_i = a_i(c_{i+1};b_{\mathrm{rec}}) + g_i K_i(c_{i+1};b_{\mathrm{rec}})^{-1/2} w_i, \qquad c_{i+1} = c_{i+1}(w_{>i}).$$

Its derivative is block triangular, with diagonal blocks $g_i K_i(c_{i+1}(w_{>i});b_{\mathrm{rec}})^{-1/2}$, so the determinant is the product in the statement. This is the triangular argument of Theorem 12.13, applied to the list of free coordinates of the block region; the dimensions $d_i = 3f_i$ are those of this list, not of the torus. With $L_i(w) = \log\det K_i(c_{i+1}(w_{>i});b_{\mathrm{rec}})$ and $\widehat{L}_i = \log\det K_i(\widehat{c}_{i+1};b_{\mathrm{rec}})$, the splitting reads

$$\det D_w z = \Big(\prod_i g_i^{d_i}\Big) e^{-\frac{1}{2}\sum_i \widehat{L}_i} e^{-\frac{1}{2}\sum_i [L_i(w)-\widehat{L}_i]}.$$

The second exponential contains all the dependence on the coarser normalized fluctuation variables. This is one determinant, with its real positive branch and its analytic continuation. If some free labels are absent, their Gaussian coordinates are absent from this product and remain explicit in any comparison, and the factors at zero fluctuation remain constant factors. The only estimates were sums of coefficients with a fixed index and graph distances, and neither introduces a dependence on the number of preceding block integrations, the number of holes, or the later widths, beyond the displayed factor $|F|$. □

18.6 Coercivity and analytic constrained minimizers

We first prove the patch inequality, which bounds a linearized variation of a block region by its curl on single final blocks and on pairs of adjacent final blocks. The patches and the restriction maps are defined explicitly, so that the inequality applies both to the effective classical action (Theorem 18.12) and to the Wilson action at the first block integration (Lemma 18.14).

Definition 18.10. Use the notation of Definition 18.6. The set $\mathfrak{P}_\Omega$ of patches of Ω consists of every final block $\mathsf{B}_s(a)$ with $a \in \mathcal{K}$ and of every union $\mathsf{B}_s(a) \cup \mathsf{B}_s(a + M^s e_k)$ of two adjacent final blocks with both anchors in $\mathcal{K}$. For $P \in \mathfrak{P}_\Omega$, let $\mathcal{E}(P)$ be the set of positive input edges with both endpoints in P, and $\mathcal{P}(P)$ the set of elementary positive plaquettes with all four vertices in P; thus P is a complete cuboid, with its boundary edges and faces. Construct $\mathcal{L}_i(P)$, Q_i^P, and $\Pi_i^{\mathcal{T},P}$ by the rules of Definition 18.6, using only the final block or blocks of P, and let

$$G_P h = \big((\Pi_i^{\mathcal{T},P} Q^P_{[0,i)} h)_{0 \le i < s},\ Q^P_{[0,s)} h \big), \qquad \mathcal{V}_P = \ker G_P. \tag{18.14}$$

For a single block $\mathcal{L}_s(P)$ is empty, and for a pair it consists of its one coarse bond. The tree conditions are imposed at every scale, not only at the finest one. Let d_P be the signed plaquette incidence matrix (the curl) of the cuboid P. For an edge cochain u on $\mathcal{L}_0$, the restriction to the patch is $(R_P u)_e = u_e$ for $e \in \mathcal{E}(P)$. For a patch that crosses a periodic seam we use the orientations induced by its rectangular lift.

Lemma 18.11. *Use Definitions 18.6 and 18.10. Let $\dot{\mathcal{H}}_i$ be the derivative of the coarse-field lift $\mathcal{H}_i$ at the identity, let $I_{\mathcal{F}_i}$ and $I_{\mathcal{D}_i}$ be the inclusions of coordinates on $\mathcal{F}_i$ and $\mathcal{D}_i$ into cochains on $\mathcal{L}_i$, and let $\Pi_{\mathcal{F}_i}$ be the restriction to $\mathcal{F}_i$. Define*

$$\Gamma'_i = I_{\mathcal{F}_i} - M^2 I_{\mathcal{D}_i} Q_i I_{\mathcal{F}_i}, \qquad A_\Omega z = \sum_{i=0}^{s-1} \dot{\mathcal{H}}_0 \cdots \dot{\mathcal{H}}_{i-1} \Gamma'_i z_i, \tag{18.15}$$

where the product preceding Γ'_0 is the identity, and

$$(B_{\mathrm{ext}} v)_i = \Pi_{\mathcal{F}_i} (I - \dot{\mathcal{H}}_i Q_i) Q_{[0,i)} v. \tag{18.16}$$

For a matrix B write $\|B\|_\infty = \sup_x \sum_y |B_{xy}|$ and $\|B\|_1 = \sup_y \sum_x |B_{xy}|$. The following statements hold.

1. *The map A_Ω is the derivative of $\operatorname{Log} \Phi_\Omega(0, \cdot)$ at $z = 0$. It maps the free coordinates bijectively onto $\ker G_\Omega$, and B_{ext} is its inverse on $\ker G_\Omega$.*
2. *We have*
$$\|A_\Omega\|_\infty \le J(1 + 4M^3), \qquad \|A_\Omega\|_1 \le M^{2(J-1)}(1 + 24M^3),$$

$$\|B_{\mathrm{ext}}\|_\infty \le b_\infty = (1+4M)(4M)^{J-1}, \qquad \|B_{\mathrm{ext}}\|_1 \le b_1 = (1+24M^3)\sum_{i=0}^{J-1}(24M)^i.$$

Consequently $\|z\|_2 \le C_{\mathrm{ext},J}\|A_\Omega z\|_{\mathcal{L}_0}$ *with* $C_{\mathrm{ext},J} = \max(1, \sqrt{b_\infty b_1})$. *Every nonzero entry of* A_Ω *or* B_{ext} *joins two positions at periodic maximum distance at most* $R_{M,J} = 2\sum_{i=0}^{J-1} M^{i+1}$ *in input units, hence also in output units.*

3. *Let* $u \in \ker G_\Omega$, *and let* du *be the curl of any edge cochain on all input edges that agrees with* u *on* $\mathcal{L}_0$. *For every patch* P, *we have* $R_P u \in \mathcal{V}_P$ *and* $(d_P R_P u)_p = (du)_p$ *for* $p \in \mathcal{P}(P)$.
4. *There is* $\sigma_J > 0$, *depending only on* M *and* J, *such that* $\|h\|^2 \le \sigma_J^{-1}\|d_P h\|^2$ *for every patch* P *and every* $h \in \mathcal{V}_P$. *For every* $u \in \ker G_\Omega$,

$$\|u\|^2_{\mathcal{L}_0} \le \sum_{P\in\mathfrak{P}_\Omega} \|R_P u\|^2 \le \sigma_J^{-1} \sum_{P\in\mathfrak{P}_\Omega} \|d_P R_P u\|^2 \le \frac{7}{\sigma_J} \sum_{p\in\mathcal{P}_\Omega} |(du)_p|^2, \tag{18.17}$$

where $\mathcal{P}_\Omega = \bigcup_{P\in\mathfrak{P}_\Omega} \mathcal{P}(P)$.

All constants are independent of m, *of the block region, of its number of components and holes, and of its topology. For an empty list of free coordinates all statements hold with the conventions for zero-dimensional spaces.*

Proof. We begin with the derivatives of the block averaging map at the identity. Differentiating the averaging formula (4.3) at the identity gives Q_i: the derivative of the reference transport in the averaged logarithm cancels the derivative of the final reference factor. The exceptional rule plays no role, since all based loop holonomies are close to I in a neighborhood of the identity. For a scalar vertex function φ and a bond $c' = (a, a + M^{i+1}e_k)$, each contour $F_{i,c',x}$ telescopes, and hence

$$(Q_i \nabla \varphi)_{c'} = \varphi(a + M^{i+1}e_k) - \varphi(a). \tag{18.18}$$

This holds at every scale, and it uses only the contours of internal bonds; no gradient is extended across a hole.

The derivative $\dot{\mathcal{H}}_i$ copies a coarse value to each edge of the interface of its bond and vanishes on the interiors of the blocks. We have

$$Q_i \dot{\mathcal{H}}_i = I, \qquad \Pi_i^{\mathcal{T}} \dot{\mathcal{H}}_i = 0, \qquad Q_i I_{\mathcal{D}_i} = M^{-2} I. \tag{18.19}$$

For the first identity, each contour of c' crosses the interface of c' exactly once, with positive orientation, and its tree parts lie in block interiors. The second holds since tree edges lie in block interiors. For the third, a contour of a different bond never uses the eliminated edge of c', and among the M^3 contours of c' exactly M use $e_i^*(c')$, each once with positive orientation. Since tree paths stay in their blocks, these identities hold also for an internal bond next to a final block outside Ω.

The linearized constraint at the identity is $Q_i X = 0$ for a cochain X on $\mathcal{L}_i$ with zero tree entries, and (18.19) gives its unique solution with free entries z_i:

$$\Gamma_i' z_i = I_{\mathcal{F}_i} z_i - M^2 I_{\mathcal{D}_i} Q_i I_{\mathcal{F}_i} z_i, \qquad \Pi_i^{\mathcal{T}} \Gamma_i' = 0, \quad Q_i \Gamma_i' = 0, \quad \Pi_{\mathcal{F}_i} \Gamma_i' z_i = z_i.$$

No inversion of a matrix of unbounded range occurs, since the coefficient matrix of the eliminated edges is $M^{-2} I$. Thus Γ_i' is the derivative at zero of the local constraint coordinates at the identity target. Since $\Gamma_b(0) = 0$ for every target b, differentiating (18.10) at $(c, z) = (0, 0)$, we obtain

$$u_s = 0, \qquad u_i = \dot{\mathcal{H}}_i u_{i+1} + \Gamma_i' z_i, \qquad u_0 = A_\Omega z,$$

where u_i is the derivative of $\operatorname{Log} U^{(i)}$ in the direction z. By (18.19), we have $u_i = Q_{[0,i)} u_0$, $\Pi_i^{\mathcal{T}} u_i = 0$, and $u_s = 0$, so $u_0 \in \ker G_\Omega$. Conversely, let $v \in \ker G_\Omega$ and set

$$v_i = Q_{[0,i)} v, \qquad z_i = \Pi_{\mathcal{F}_i} (v_i - \dot{\mathcal{H}}_i v_{i+1}).$$

The cochain $v_i - \dot{\mathcal{H}}_i v_{i+1}$ has zero tree entries and satisfies $Q_i (v_i - \dot{\mathcal{H}}_i v_{i+1}) = v_{i+1} - v_{i+1} = 0$. By (18.19) its eliminated entries are determined by its free entries, so it equals $\Gamma_i' z_i$. Since $v_s = 0$, descending induction shows that $v = A_\Omega z$. We conclude that A_Ω maps the free coordinates bijectively onto $\ker G_\Omega$, with inverse (18.16). This is the successive extraction of the averaged fields and of their free coordinates, written as a matrix, and it proves (1).

Next we prove (2). A contour has at most $4M - 3$ steps, so

$$\|Q_i\|_\infty \le 4M, \qquad \|Q_i\|_1 \le 24M, \qquad \|\dot{\mathcal{H}}_i\|_\infty \le 1, \qquad \|\dot{\mathcal{H}}_i\|_1 \le M^2.$$

For the column bound for Q_i, an interior edge of a block occurs in the contours of at most the six bonds incident to its block, and an interface edge only in the contours of its own bond; each such row has absolute sum at most $4M$. A bond has M^2 interface edges, which gives the column bound for $\dot{\mathcal{H}}_i$. We obtain $\|\Gamma_i'\|_\infty \le 1 + 4M^3$ and $\|\Gamma_i'\|_1 \le 1 + 24M^3$, and the bounds for A_Ω follow from (18.15), since a column of A_Ω belongs to a single scale $i \le J - 1$. For B_{ext} we multiply the row and column bounds of the factors in (18.16); stacking over i takes the maximum of the row bounds and the sum of the column bounds. By the Cauchy–Schwarz inequality, we have $\|Bv\|_2^2 \le \|B\|_\infty \|B\|_1 \|v\|_2^2$, and therefore $\|z\|_2 = \|B_{\mathrm{ext}} A_\Omega z\|_2 \le C_{\mathrm{ext},J} \|A_\Omega z\|_{\mathcal{L}_0}$. These estimates apply to each color and to the Euclidean norms over the colors.

We now prove the range bound. We assign to each edge its midpoint. Each nonzero entry of Q_i, $\dot{\mathcal{H}}_i$, or Γ_i' joins positions at periodic maximum distance at most $2M^{i+1}$, so every product in (18.15) or (18.16) has range at most $R_{M,J}$. Dividing coordinates by $M^s \ge 1$ only decreases this range. All these estimates hold after restriction to a block region: only equations of internal bonds were used, and deleting other equations does not increase a row sum, a column sum, or a range.

We now prove (3). For a patch P and each i, let r_i^P select the edges of $\mathcal{L}_i(P)$ among those of $\mathcal{L}_i$. Since all contours of a bond stay in its two complete endpoint blocks (Lemma 18.4), we have the matrix identity

$$Q_i^P r_i^P = r_{i+1}^P Q_i,$$

where the right side has zero coefficients on the input edges outside P; these zeros come from the supports of the contours, not from an extension of u by zero. The same identity holds for the rows of the trees. Iterating it and using $G_\Omega u = 0$, we obtain $G_P R_P u = 0$, that is, $R_P u \in \mathcal{V}_P$. This holds for every patch, including the patches next to the boundary of Ω. Every plaquette in $\mathcal{P}(P)$ has its four edges in $\mathcal{E}(P) \subset \mathcal{L}_0$, so $(d_P R_P u)_p = (du)_p$ for $p \in \mathcal{P}(P)$.

Next we show that the curl is injective on each space $\mathcal{V}_P$. A single block or an adjacent pair lifts injectively to a rectangular cuboid in $\mathbb{Z}^3$: its longest side has at most $2M^s - 1$ links, which is less than $n/2$ since $m \geq 4$. Let $h \in \mathcal{V}_P$ with $d_P h = 0$. A closed cochain on a rectangular cuboid is a gradient, as we now recall. Fix a corner o, and for a vertex x let τ_x be the coordinate-ordered path from o to x. For an edge (x, y), the closed chain $\tau_x + (x, y) - \tau_y$ is an integer combination of elementary squares in the cuboid: we move the last coordinate step through the portions of τ_x in later coordinates, and all squares used lie between the relevant coordinate values, hence in the cuboid. Pairing with h shows that $\varphi(x) = h(\tau_x)$ satisfies $h_{xy} = \varphi(y) - \varphi(x)$. By (18.18), $Q^P_{[0,i)} h$ is the gradient of the values of φ at the anchors of scale i. The tree equations at the finest scale make φ constant on each block of the first scale. If φ is constant on the blocks of scale i, the tree equations at scale i equate these constants along a connected tree in each block of scale $i + 1$. After s steps φ is constant on each final block. On a single block this implies $h = 0$. On a pair, the remaining equation of G_P for its coarse bond says, by (18.18), that the two final constants agree, so $h = 0$ on the pair as well. No equation with an endpoint in a final block outside Ω was used.

For $1 \leq s \leq J$ there are finitely many cuboids of these types, including the directions of the pairs. For each one with $\mathcal{V}_P \neq \{0\}$, we take an orthonormal basis matrix B_P of $\mathcal{V}_P$ and let σ_P be the least eigenvalue of $B_P^\mathsf{T} d_P^\mathsf{T} d_P B_P$, which is positive by the injectivity just proved; if $\mathcal{V}_P = \{0\}$ we set $\sigma_P = 1$. Let σ_J be the minimum of these finitely many numbers. It depends only on M and J, and we have $\|h\|^2 \leq \sigma_J^{-1} \|d_P h\|^2$ for $h \in \mathcal{V}_P$.

It remains to prove the covering and overlap bounds in (18.17). Every edge of $\mathcal{L}_0$ has its endpoints either in one final block or in two adjacent final blocks of Ω; the first kind is covered by the single block and the second by the pair. This gives the first inequality, including for isolated blocks. The second inequality follows from (3) and the bound on each $\mathcal{V}_P$. For the third, fix a plaquette p and one of its vertices x. Every patch containing p contains x, hence the final block of x. There are at most seven such patches: that block and the pairs formed with its six possible neighbors. So each plaquette occurs in at most

seven of the sets $\mathcal{P}(P)$; a plaquette that meets three or four final blocks may occur in none. Using (3), we obtain (18.17). □

We now state the main result. Its objective is the effective classical action at depth d_0, evaluated on the reconstruction of a block region and completed outside Ω as described in the statement.

Theorem 18.12. *Fix M and J, and let $\kappa_* > 0$ be the constant of Theorem 14.8. Then*

$$E''_{d_0}(0) \geq \kappa_* \operatorname{curl}^{\mathsf{T}} \operatorname{curl} \qquad \text{for every depth } d_0 \geq 0, \tag{18.20}$$

as quadratic forms on real link variations. Let $1 \leq s \leq J$, $m \geq 4$, let Ω be a block region and $d_0 \geq 0$ a depth, and set

$$F_R(c, z; \xi) = E_{d_0}(\Psi_R(c, z; \xi)).$$

Here $\Psi_R(c, z; \xi)$ is the logarithm, on all links of the torus, of the reconstruction $\Phi_\Omega(c, z)$ of Definition 18.6, completed outside Ω by the small values with logarithms ξ prescribed by the current frozen coordinates (at the first block integration, by the completion of Definition 18.13), and with logarithm zero at the torus labels that lay outside earlier block regions. The parameters ξ may be complexified on the construction chart. Frozen coordinates from earlier block integrations and arbitrary data in the fixed complement are not arguments of this analytic action. The following statements hold with uniform constants and radii; in particular they are independent of the widths chosen later, of the shape of Ω (which may contain isolated final blocks), of d_0, and of the values of the frozen coordinates.

1. *On a common list of coordinates, the linearizations at zero E_1 and E_0 of the reconstruction on Ω and of the reconstruction on the whole torus with the complementary coordinates set to zero, defined in* (18.22)*, satisfy $E_1 = E_0 + D_W$; here E_0 and E_1 denote linear embeddings, not effective classical actions. Both indices of every nonzero entry of D_W lie in a band of fixed width L_J around the boundary W, the weighted rows and columns of D_W are at most C_J, and its rank is at most $C_J|W|$. On matched coordinates in the common chart, the nonlinear formulas for the outputs in Ω of the two reconstructions agree away from this band. Prescribed values in the fixed complement may differ; their effect is estimated as a bounded source in Lemma 18.15. Coarse outputs on bonds across the boundary of the block region that are not prescribed are holomorphic functions of $(c, z; \xi)$, with all their derivatives. This compares two embeddings; it does not identify their fibers.*
2. *There are $\nu_J, C_{\mathrm{ext},J}, \lambda_J, Q_J > 0$ with the following properties. Let $u = D_z\Psi_R(0)z$; its restriction to $L_\Omega = \mathcal{L}_0$ lies in* $\ker G_\Omega$*. Then*

$$\|\operatorname{curl} u\|_2^2 \geq \nu_J \|u\|_{L_\Omega}^2, \qquad \|z\|_2 \leq C_{\mathrm{ext},J} \|u\|_{L_\Omega},$$

$$K_R^0 = F_{R,zz}(0) \geq \lambda_J I, \qquad \lambda_J = \kappa_* \nu_J / C_{\mathrm{ext},J}^2, \qquad \|K_R^0\|_{a_J} \leq Q_J.$$

One may take $\nu_J = \sigma_J/7$ and $C_{\mathrm{ext},J}$ as in Lemma 18.11. Every principal block of K_R^0 and every real Schur complement of K_R^0 is bounded below by $\lambda_J I$. Let z_H be the free coordinates of the torus outside Ω. On the common list (z_L, z_H), the Hessian at $(c, z, \xi) = 0$ of the interpolation between the joint objective on the torus and $F_R + \|z_H\|_2^2/2$ is bounded below by $\lambda_{U,J} I$, where $\lambda_{U,J} = \min(\lambda_J, \lambda_{T,J}, 1) > 0$ and $\lambda_{T,J}$ is the corresponding flat bound on the torus.

3. *There are radii $r_{a,J}, r_{z,J}, \rho_J, \beta_{\mathrm{rec},J} > 0$, a rate $a_J > 0$, and an inverse bound I_J such that, on an open product containing*

$$\|c\|_\infty \le 2\rho_J, \qquad \|z\|_\infty \le r_{z,J}, \qquad \|\xi\|_\infty \le 2\beta_{\mathrm{rec},J},$$

the reconstruction lies in $B_{r_{a,J}}$, inside the identity chart on which E_{d_0} is controlled, and

$$\|F_{R,zz} - K_R^0\|_{a_J} \le Q_J(\|c\|_\infty + \|z\|_\infty + \|\xi\|_\infty) \le \lambda_J/8.$$

Mixed derivative matrices, and tensors of order $l \ge 2$ with each index fixed in turn, are bounded by $l! C_J^{l+1}$. The inverse of the complex free Hessian has weighted row-column norm at most I_J at a fixed smaller rate.

4. *Choose the parameter radii below the solution radius so that*

$$I_J Q_J(2\rho_J + 2\beta_{\mathrm{rec},J}) < r_{z,J}/16, \qquad 8G_J(2\rho_J + r_{z,J} + 2\beta_{\mathrm{rec},J}) < \min(r_{\Gamma,J}, r_E),$$

where G_J and $r_{\Gamma,J}$ are the coefficient bound and the radius of the reconstruction in (18.21)*. Then there exists a unique holomorphic stationary point $z_R = z_R(c;\xi)$ in the free box,*

$$F_{R,z}(c, z_R; \xi) = 0, \qquad \|z_R\|_\infty \le C_J(\|c\|_\infty + \|\xi\|_\infty) < r_{z,J}/8.$$

It has factorial coefficient bounds on output rows and on volume-normalized input columns on a smaller fixed product. For real data it is the unique minimizer of $F_R(c, \cdot\,; \xi)$ in the convex free box.

5. *Conditional minimizers over initial segments of the scales exist on smaller products of the coarser free coordinates around this joint minimizer, chosen so that the bounds* (18.31) *on the Hessian and on the inverse times the gradient hold. No claim is made on other slices of the joint box.*

Proof of Theorem 18.12. The proof has six steps. In the first, we construct the reconstruction with coefficient bounds that do not depend on the size of the lattice, and compare the embeddings on Ω and on the torus. In the second, the patch inequality gives the coercivity of the curl. In the third, we derive (18.20) from Theorem 14.8 and obtain the lower bound on K_R^0. In the fourth, we bound the Hessian on a complex product and choose the radii. In the fifth, we construct the minimizer by a contraction argument, and in the sixth, the conditional minimizers. All constants are constructed on larger products before any of the later widths is chosen. If the list of free coordinates is empty, the minimizer is empty and the matrices are zero-dimensional, so we assume that it is nonempty.

Step 1: local charts and comparison of embeddings. For one averaging step the reconstruction is the eliminated-edge solution of Lemma 6.3, restricted to internal bonds as in Definition 18.6. Let L_M bound its stencil radius in output units of one step. Its local scalar majorants can be chosen of the form $G_M t/(1 - G_M t)$ on a fixed radius. This absolute coefficient bound does not depend on the size of the lattice, for the following reason. Each defining group product reads a bounded number n_M of variables. Matrix exponentials are bounded coefficientwise by e^t, and on $\|V - I\| < 1$ the logarithm has the convergent series $\sum_{k\geq 1}(-1)^{k+1}(V-I)^k/k$. The inverse of the linear coefficient of an eliminated edge is $M^2 I_3$, and the nonlinear equation for that edge has zero constant term and a quadratic remainder after its linear part is removed. On a sufficiently small fixed radius, the implicit-series statement of Lemma 5.3 gives G_M and factorial coefficients for the local inverse. Since only n_M variables occur in each local equation, summing over its finite stencil, also with any input fixed, we obtain absolute row and column bounds.

By composing at most J such maps, with the embeddings of positions, we obtain numbers $G_J, L_J, r_{\Gamma,J} > 0$ such that the logarithm Ψ_R of the reconstruction is holomorphic, vanishes at zero, and has a majorant series with nonnegative coefficients on that radius, with the bounds

$$\|\Psi_R\|_{\text{coeff}} \leq \frac{G_J T}{1 - G_J T}, \qquad \|D^l \Psi_R\|_{\text{all indices}} \leq l! G_J^{l+1} \quad (l \geq 1) \tag{18.21}$$

on a fixed smaller product. Here $T = r_c + r_z + r_\xi$, where r_c, r_z, and r_ξ are the maximal radii of the coarse logarithms c, of the free coordinates z, and of the current frozen coordinates ξ, respectively; the second bound absorbs the fixed ratios of radii. The first norm is a row majorant for the image; the derivative bounds include every input column. One may take $L_J \geq \sum_{i=0}^{J-1} M^i L_M$, and enlarge G_J for the finitely many compositions. There are at most a fixed multiple of JM^{3J} labels per unit cube of the output lattice, and v_J includes this multiplicity. Volume-normalized input columns acquire their mesh-volume ratios under composition, by Lemma 5.3, and these factors are bounded by fixed powers of M^J.

We now compare the embeddings, before any minimization. Let $\Psi_T(c_T, z_T)$ be the logarithm of the reconstruction on the whole torus, and let P_L insert the free coordinates of Ω into the list of free coordinates of the torus, with zero in every other slot. At matched data we use $c_T = c \oplus 0$ and $z_T = P_L z$. The two linearized embeddings on this common list are

$$E_1 z = D_z \Psi_R(0, 0; 0) z, \qquad E_0 z = \frac{d}{dt} \Psi_T(0, t P_L z)\Big|_{t=0}. \tag{18.22}$$

The symbols E_0 and E_1 in this formula and in (18.23) denote linear embeddings, not effective classical actions. All current frozen coordinates and logarithms of the coarse links vanish in these derivatives, and both reconstructions vanish there. At one scale, the equations on Ω and on the torus for an internal bond have the same two endpoint blocks,

the same internal target, and the same free values, so their eliminated-edge solutions agree by local uniqueness. The additional equations of the torus concern bonds meeting the complement. Descending through the scales, a change in such an equation can propagate only through the accumulated stencil of width L_J of Lemma 18.4. Therefore, with matching free values in Ω and zero complementary torus values, the nonlinear formulas for the outputs in Ω agree outside a band of fixed width around W. Prescribed values in the fixed complement are separate formulas and may differ there. An input in Ω farther than one further stencil from the band cannot affect a changed formula, and we enlarge L_J once to include that stencil. Differentiating with the conventions of (18.22), we obtain

$$E_1 = E_0 + D_W, \qquad (D_W)_{ex} = 0 \text{ unless } e, x \text{ lie in the } L_J\text{-band around } W. \tag{18.23}$$

Here the L_J-band is the set of positions within distance L_J of W. The weighted row and column bounds follow from the finite range, (18.21), and the density bound. We cover the band by balls of radius $L_J + 1$ around the positions of W. Each ball contains at most $Cv_J(L_J + 2)^3$ labels, so the number of nonzero output rows of D_W, and hence its rank, is at most $C_J|W|$.

Every coarse output on a bond across the boundary of the block region that is not prescribed is obtained by applying the finite group products of T^s_M to the reconstructed field. In the chart it is a holomorphic function of (c, z, ξ), with all its derivatives given by these products and (18.21). No equation sets it to a prescribed value, so (18.23) compares two embeddings of a common list and asserts no equality of their fibers.

Step 2: coercivity of the curl. Let $u = D_z\Psi_R(0)z$. On $\mathcal{L}_0$ it equals $A_\Omega z$, by Lemma 18.11(1), so its restriction lies in $\ker G_\Omega$. Then (18.17) implies

$$\|u\|^2_{\mathcal{L}_0} \le \frac{7}{\sigma_J} \sum_{p \in \mathcal{P}_\Omega} |(du)_p|^2 \le \frac{7}{\sigma_J} \|\operatorname{curl} u\|_2^2,$$

where $\operatorname{curl} u$ is the curl of the completed variation on all plaquettes of the torus. In the last inequality we used that restricting a sum of squares to the plaquettes of the patches can only decrease it; since these plaquettes have all their edges in $\mathcal{L}_0$, variations of the completed boundary values induced by the tree-path transports to the basepoints do not enter. This proves the first inequality of (2) with $\nu_J = \sigma_J/7$. The second inequality is Lemma 18.11(2); it shows in particular that a nonzero vector of free coordinates is visible on the internal links, also for an isolated final block.

Step 3: the lower bound on the Hessian at the identity. The quadratic form $E''_{d_0}(0)$ is the Hessian at the identity, that is, the effective Maxwell quadratic form of Theorem 8.3 (see Theorem 10.10). We derive (18.20) from Theorem 14.8, which gives $E_{d,\mathrm{sat}} \ge \kappa_* E_0$ for every integer $d \ge 0$ on the domain of the gauge-invariant extension $E_{d,\mathrm{sat}}$ of E_d, whether or

not an effective interaction is attached to that depth. We choose a ball of link logarithms at the identity that is contained both in the construction chart of that extension and in the domain used here; on it, $E_{d,\mathrm{sat}}$ equals the analytic function E_d. For a fixed real variation u, we choose $|t|$ so small that tu lies in this ball. Then for every $d_0 \geq 0$ we have $E_{d_0}(tu) \geq \kappa_* E_0(tu)$, with the same κ_*. Both functions vanish at zero and are stationary there: for E_{d_0} this follows from Theorems 10.10 and 10.11 by induction from the Wilson action. Further, $E_0(tu) = t^2 \| \operatorname{curl} u\|_2^2/2 + O(t^3)$, by expanding the plaquette product and using $4q(e^X) = |X|^2/2 + O(|X|^4)$. Dividing the inequality by $t^2/2$ and letting $t \to 0$ proves (18.20) for every integer depth. Only this inequality from Theorem 14.8 has been used, and this step gives no positivity at a nonflat coarse field.

By Lemma 18.8, the Hessian of $E_{d_0} \circ \Psi_R$ at zero has no term involving the gradient of the action, since $DE_{d_0}(0) = 0$. Using (18.20) and Step 2, we obtain

$$z^{\mathsf{T}} K_R^0 z \geq \kappa_* \| \operatorname{curl} D_z \Psi_R(0) z\|_2^2 \geq \frac{\kappa_* \nu_J}{C_{\mathrm{ext},J}^2} \|z\|_2^2, \tag{18.24}$$

which is the claimed value of λ_J. The same patch construction with every final block of the torus gives a lower bound $\lambda_{T,J} > 0$ for the joint free Hessian on the torus. Every principal block of either joint matrix has the same lower bound, and (18.6) implies the same lower bound for its real Schur complements. On the common list (z_L, z_H), where z_L are the free coordinates of Ω, the interpolating Hessian at the identity is

$$H_t^0 = (1-t) H_T^0 + t(K_R^0 \oplus I_H), \qquad 0 \leq t \leq 1,$$

where H_T^0 is the joint Hessian at the identity on the torus and I_H the identity on the z_H coordinates. It is bounded below by $\lambda_{U,J} = \min(\lambda_J, \lambda_{T,J}, 1) > 0$. This is a convex combination of two objectives on a common list of coordinates, not an identification of fibers. All the flat and nonflat estimates below can be chosen uniformly in t, using this common lower bound and the maximum of the two coefficient bounds.

Step 4: analytic products and the inverse of the free Hessian. Let $b_J > 0$ be smaller than the rate a_E of the torus tensors after the transfer of positions to output units and the conversion to the graph metric; this conversion divides a Euclidean rate by $\sqrt{3}$. We include the finite-stencil weights of (18.21). We compose the torus tensors of Theorem 10.11 with the majorant series of Ψ_R, using Lemma 5.3. The scalar composition converges on a fixed product, and after enlarging C_J and Q_J we have

$$\|D^l F_R\|_{b_J,\text{all indices}} \leq l! C_J^{l+1} \quad (l \geq 2), \qquad \|K_R^0\|_{b_J} \leq Q_J. \tag{18.25}$$

The mixed matrices in z, c, and ξ satisfy the same bound after enlargement. The labels of ξ have their positions on the input lattice and their own density, not the density of the output lattice. The volume-normalized column bounds follow from the contraction rule, with the volume factors kept in each factor.

At a general point of the product, the Hessian is

$$F_{R,zz} = \Psi_{R,z}^{\mathsf{T}} E''_{d_0}(\Psi_R)\Psi_{R,z} + E'_{d_0}(\Psi_R)\Psi_{R,zz}, \tag{18.26}$$

and the last term, which contains the gradient of the action, is kept. The gradient bound $\|E'_{d_0}(A)\|_\infty \le C_E\|A\|_\infty$ is part of Theorem 10.11. We integrate the third-derivative bound of the composed objective along the segment from zero to (c, z, ξ), which lies in its product domain. In the weighted row-column norm, we obtain

$$\|F_{R,zz}(c, z; \xi) - K_R^0\|_{b_J} \le Q_J(\|c\|_\infty + \|z\|_\infty + \|\xi\|_\infty), \tag{18.27}$$

$$\|F_{R,z}(c, 0; \xi)\|_\infty \le Q_J(\|c\|_\infty + \|\xi\|_\infty). \tag{18.28}$$

These estimates also follow by expanding all the terms of (18.26); integrating the complete third derivative ensures that none of the terms involving the gradient of the action or the second derivative of the coordinates is omitted.

We apply Lemma 18.3(3) with the lower bound $\lambda_{U,J}$, the bound Q_J, and the rate b_J to every flat principal block of free coordinates and to every interpolating Hessian. We obtain a common smaller rate $a_J > 0$ and a flat inverse bound; we take I_J so large that it bounds these inverses and their perturbations. For example, the constants μ and C_R of that lemma give the flat bound $C_R/\lambda_{U,J}$ and the perturbed bound $4C_R/(3\lambda_{U,J})$ on a sufficiently small coefficient product. We enlarge I_J and Q_J if necessary; these enlargements precede every choice of radius. Bounds at the rate b_J imply the same bounds at the rate a_J.

We now choose the radii, on a product larger than the one in the statement. Let $r_0 = \min(r_{\Gamma,J}, r_E)$, where r_E is replaced by a strictly smaller radius of the torus chart if necessary. We first choose $r_{z,J} > 0$, and then $\rho_J, \beta_{\mathrm{rec},J} > 0$, so that, with $T_+ = 2r_{z,J} + 4\rho_J + 4\beta_{\mathrm{rec},J}$,

$$\begin{gathered} Q_J T_+ < \min\{\lambda_{U,J}/8, (8I_J)^{-1}\}, \qquad 8G_J T_+ < r_0, \\ I_J Q_J(4\rho_J + 4\beta_{\mathrm{rec},J}) < r_{z,J}/16. \end{gathered} \tag{18.29}$$

We also require that the nonconstant part of the Hessian on this product satisfies the condition of Lemma 18.3(3) with the factor $1/4$ in (18.5). By (18.25) this perturbation tends to zero with T_+, so this is one more upper bound on the radii. All these inequalities can be satisfied: we first choose $r_{z,J}$ to satisfy the two inequalities that involve it alone, with strict inequality, and then make the two parameter radii sufficiently small. We choose $r_{a,J} < r_E$ larger than the bound on the image in (18.21) at T_+, which is possible by the inequality $8G_J T_+ < r_0$.

These choices hold on the open product defined by $\|z\|_\infty < 2r_{z,J}$, $\|c\|_\infty < 4\rho_J$, and $\|\xi\|_\infty < 4\beta_{\mathrm{rec},J}$, which contains the closed product of the statement. In particular the right side of (18.27) is at most $\lambda_{U,J}/8 \le \lambda_J/8$ there, the image lies in $B_{r_{a,J}}$, and (18.29) implies both inequalities on the parameter radii in (4). The inverse of the complex free Hessian is bounded by I_J at its smaller rate by the resolvent series (18.5), not by a positivity argument at complex data. This proves (3).

Step 5: the constrained minimizer and its coefficients. On the free box we use the map

$$\mathcal{T}_{c,\xi}(z) = z - (K_R^0)^{-1} F_{R,z}(c, z; \xi).$$

By (18.27) and (18.29), its derivative has maximal row norm at most $1/8$. By (18.28), its value at zero is at most $I_J Q_J(\|c\|_\infty + \|\xi\|_\infty) < r_{z,J}/16$ on the larger parameter product. Thus it maps the closed box of radius $r_{z,J}$ into itself, strictly. Starting at zero, its iterates satisfy $\|z^{(n+1)} - z^{(n)}\|_\infty \le 8^{-n} r_{z,J}/16$. Their geometric sum converges to a fixed point in the box, and it is unique, since the distance between two fixed points would be at most one eighth of itself. Since $\|z_R\|_\infty \le \|\mathcal{T}_{c,\xi}(0)\|_\infty + \|z_R\|_\infty/8$, we have

$$\|z_R\|_\infty \le \frac{8}{7} I_J Q_J(\|c\|_\infty + \|\xi\|_\infty) < r_{z,J}/8,$$

which gives the stated C_J. The iterates are holomorphic, and their convergence is geometric and uniform on compact subsets of the product. By the Cauchy integral formula in each coordinate, the limit is holomorphic: the integral of the uniform tail tends to zero, also on each smaller contour used for derivatives.

For the absolute coefficient bounds with each index fixed, we use the majorant version of this iteration. We subtract the flat linear part. By (18.25), the nonlinear remainder is quadratic and has the majorant $C_J(t + Y)^2/[1 - C_J(t + Y)]$. The coefficients of the solution, and their versions with a specified input index, are therefore bounded by the majorant series Y with nonnegative coefficients determined by

$$Y = C_J t + \tfrac{1}{2} Y + \frac{C_J(t + Y)^2}{1 - C_J(t + Y)}. \tag{18.30}$$

All hypotheses of the implicit-series statement of Lemma 5.3 have now been checked: zero constant term, a bounded inverse of the linear part, tensor bounds with each index fixed, the bound $1/8$ on the derivative, and a strictly larger product. The lemma therefore gives factorial coefficient bounds on a smaller fixed product. Fixing a chosen current input index in the same majorant series, we obtain the bound on volume-normalized columns. One spatial rate is used throughout, with at most the single initial reduction for external trees described in Lemma 5.3.

At real data the objective and the chart are real. Complex conjugation maps a stationary point in the box to a stationary point in the same box, and uniqueness shows that z_R is real. Throughout the box the real Hessian is at least $7\lambda_{U,J} I/8$, by (18.27) and (18.29). For a real z in the convex box, by Taylor's formula at z_R with $h = z - z_R$, we have

$$F_R(c, z; \xi) - F_R(c, z_R; \xi) = \int_0^1 (1 - t) h^{\mathsf{T}} F_{R,zz}(c, z_R + th; \xi) h \, dt \ge \frac{7\lambda_{U,J}}{16} \|h\|_2^2.$$

We conclude that z_R is the unique minimizer in the box. This proves (4).

Step 6: domains for the conditional minimizers. It remains to specify the domains of the conditional minimizers. Let I be an initial segment of the scales, write $z = (u, y)$ with $u = z_I$ and $y = z_{I^c}$, and let $z^0 = z_R(c;\xi)$. We use the parameters $v = y - z^0_{I^c}$ and solve for $h = u - z^0_I$. The matrix $A_I = F_{R,uu}(c, z^0; \xi)$ has a common inverse bound I'_J, by the estimates for principal blocks proved above. We choose a solution radius $r_I > 0$ inside the larger domain, and then a smaller radius $\eta_I > 0$ for the coarser parameters v, such that throughout the corresponding product

$$\begin{aligned} \|A_I^{-1}(F_{R,uu} - A_I)\|_{\mathrm{coeff}} &\le 1/4, \\ \|A_I^{-1} F_{R,u}(c, z^0_I, z^0_{I^c} + v; \xi)\|_{\mathrm{coeff}} &\le r_I/8. \end{aligned} \tag{18.31}$$

The first condition holds after shrinking the product, by the convergent majorant of the third derivative. In the second, the gradient vanishes at $v = 0$, and its change is

$$\int_0^1 F_{R,uy}(c, z^0_I, z^0_{I^c} + tv; \xi) v \, dt.$$

Its coefficient norm is at most $Q'_J \|v\|$, by the bounds on mixed derivatives, so $I'_J Q'_J \eta_I \le r_I/8$ suffices. The solution radius is chosen first and the parameter radius second; they are not equal.

The map $h \mapsto h - A_I^{-1} F_{R,u}$ now has derivative at most $1/4$ and value at most $r_I/8$ at $h = 0$. It maps the box of radius $r_I/2$ into itself, and the convergence and majorant arguments of Step 5 apply. The conditional minimizer a_I is holomorphic, real at real data, and satisfies

$$a_I(c, z^0_{I^c}; \xi) = z^0_I, \qquad D_y a_I = -F_{R,uu}^{-1} F_{R,uy}.$$

Since there are at most J initial segments, their solution radii and parameter radii have a common positive minimum. For a sequential application we also choose a smaller radius for the fluctuation inputs. The finite composition of these conditional minimizers and of their inverse square roots equals the joint minimizer when all the inputs $g_i w_i$ vanish, and its deviation has a majorant $C_{\mathrm{seq},J} t / (1 - C_{\mathrm{seq},J} t)$. We choose $r_q > 0$ so that this expression is less than a quarter of the least of the preceding solution and parameter radii. All sequential images then lie in the domains of the conditional minimizers. This is a choice for finitely many scales, independent of the number of sites. Lemma 18.8 identifies successive and joint minimizers there and gives the factors M^i. The real lower bounds for principal blocks and Schur complements persist there by (18.6), and their complex coefficient bounds follow from the Schur calculus of Lemma 18.3.

The conditions (18.31) are part of the smaller products just constructed. Every constant used to obtain them comes from the finite local formulas, the tensor bounds on the torus, the constants of the patch inequality, and the density bound. Taking maxima and minima over $s \le J$ and over initial segments therefore gives uniform constants, independent in particular of isolated final blocks, of the later widths, and of the values of the frozen coordinates. This completes the proof. □

18.7 The first block integration

At depth zero, we identify the completed objective with the part of the Wilson action that depends on (c, z), and we give the exact form and range of its Hessian, uniformly in the arbitrary frozen group values deep in the fixed complement, as used in Section 21. The restriction and extraction maps are those of Lemma 18.11.

Definition 18.13. Use Definition 18.6 at the first block integration, and let $\mathcal{E}(\Lambda)$ be the set of all positive input links.

1. *Dependencies.* Collect in a finite list all contours $F_{i,c',x}$ with $c' \in \mathcal{L}_{i+1}$, their reference paths and tree paths, and the tree-path transports (18.11) to the basepoints of frozen coordinates on links across the boundary, with the edges of the coarser scales expanded recursively through the definition of T_M. The formulas of the local constraint coordinates and of the Haar disintegration densities are included in this list. Let $\mathcal{E}_\Omega^{\mathrm{dep}}$ be the union of the input links on which these formulas depend. If a further formula at the boundary is to be used analytically, its input links are added to this list before the near links are fixed.
2. *Near and deep links.* Let $\mathcal{P}_{\mathrm{inc}}$ be the set of plaquettes p with at least one edge in $\mathcal{L}_0$, and set
$$\mathcal{E}_\Omega^{\mathrm{near}} = \Big(\mathcal{E}_\Omega^{\mathrm{dep}} \cup \bigcup_{p \in \mathcal{P}_{\mathrm{inc}}} \partial p\Big) \setminus \mathcal{L}_0, \qquad \mathcal{E}_\Omega^{\mathrm{deep}} = \mathcal{E}(\Lambda) \setminus (\mathcal{L}_0 \cup \mathcal{E}_\Omega^{\mathrm{near}}). \tag{18.32}$$
Further links may be added to $\mathcal{E}_\Omega^{\mathrm{near}}$; none is needed below.
3. *Near values.* Assume that every near link carries a value in the identity chart. The near values b^{near} are independent small group values on $\mathcal{E}_\Omega^{\mathrm{near}}$: on a link with both endpoints outside Ω_0 the value itself, and on a link across the boundary its based transport. If a link (x, y) across the boundary is first given by its value a_{xy}, we set $b_e = \Theta_x(U) a_{xy}$ at fixed field of the block region; this is the Haar translation of Lemma 18.7, and all indicators are kept under this substitution. The logarithms of the near values are denoted ξ.
4. *Completion.* For a link $e = (x, y)$ across the boundary, with $x \in \Omega_0$, $y \notin \Omega_0$, and based transport b_e, set
$$U_{xy}^{\mathrm{act}} = \Theta_x(\Phi_\Omega)^{-1} b_e; \tag{18.33}$$
for a positive link whose endpoint in Ω_0 is its terminal endpoint, we apply this rule to the reversed link and invert. On the remaining links we set
$$U_e^{\mathrm{act}}(c, z; \xi, \eta) = \begin{cases} \Phi_{\Omega,e}(c, z), & e \in \mathcal{L}_0, \\ b_e, & e \in \mathcal{E}_\Omega^{\mathrm{near}} \text{ with both endpoints outside } \Omega_0, \\ \eta_e, & e \in \mathcal{E}_\Omega^{\mathrm{deep}}, \end{cases} \tag{18.34}$$

where $\eta \in \mathrm{SU}(2)^{\mathcal{E}_\Omega^{\mathrm{deep}}}$ are the frozen group variables in the fixed complement on the deep links (the deep frozen group variables); they are arbitrary real group values, not small analytic logarithms. The analytic completion $U^{\mathrm{an}}(c, z; \xi)$ is defined by the same rules with $\eta_e = I$ on the deep links, and we write $\Psi_R = \operatorname{Log} U^{\mathrm{an}}$, componentwise on the chart.

5. *Varying plaquettes.* Let $f_p(c, z; \xi) = 4q(U_p^{\mathrm{an}}(c, z; \xi))$, and

$$\mathcal{P}_{\min} = \{p \in \mathcal{P}_{\mathrm{inc}} : f_p(\cdot, \cdot\,; \xi) \text{ is nonconstant in } (c, z) \text{ for some small real } \xi\}. \quad (18.35)$$

Fix a list $\mathcal{P}_{\mathrm{var}}$ with $\mathcal{P}_{\min} \subseteq \mathcal{P}_{\mathrm{var}} \subseteq \mathcal{P}_{\mathrm{inc}}$, and define the objective of the first block integration

$$F_{\mathrm{bare}}(c, z; \xi) = \sum_{p \in \mathcal{P}_{\mathrm{var}}} f_p(c, z; \xi). \quad (18.36)$$

Every link across the boundary is a near link: if $x \in \Omega_0$ is the endpoint of such a link in direction k, a step from x in a transverse direction toward the interior of the final block of x gives a plaquette that contains the link and an edge of $\mathcal{L}_0$, since a final block has at least three vertices per side. This holds also at corners and at periodic seams. The requirement that the near values lie in the identity chart is the condition imposed on the protected neighborhoods in Section 21; the conversion between the value a_{xy} and the based transport b_e is a finite group product, whose smallness is also controlled there. Since $\Theta_x(\Phi_\Omega(c, z)) = I$, the value (18.33) of such a link equals b_e on the reconstruction. Hence, at fixed frozen coordinates, the derivatives in (c, z) of all links outside Ω vanish. Completion by zero means logarithm zero, that is, the group value I; it does not mean solving the block averaging constraints of the torus with the missing targets set to zero. Coarse outputs on bonds across the boundary of the block region are evaluated by T_M on the completion, and they are not imposed as additional equations. Exceptional values of the Borel extension of T_M at the deep frozen group variables are kept in the measure, and no logarithm of such a value enters U^{an}.

The list $\mathcal{P}_{\min}$ is fixed on the chart, not chosen anew at each value of the frozen coordinates. A plaquette function outside $\mathcal{P}_{\min}$ is also independent of the complex variables: its derivatives in (c, z) vanish on the real product, hence, by the identity theorem applied successively in each scalar coordinate, on the complex product. If the split is made before some internal constraint coordinates are eliminated, the list of varying plaquettes may be larger than $\mathcal{P}_{\min}$. All conclusions below hold for every fixed list between $\mathcal{P}_{\min}$ and $\mathcal{P}_{\mathrm{inc}}$.

For the statement of the next lemma, let $d_{\mathrm{var}} u = ((du)_p)_{p \in \mathcal{P}_{\mathrm{var}}}$, let ι_Ω be the extension by zero from edge cochains on $\mathcal{L}_0$ to all input links, and let $\widehat{A}_\Omega = \iota_\Omega A_\Omega$, with A_Ω as in (18.15). For a patch P define a linear map from cochains on the varying plaquettes to

cochains on the plaquettes of P by

$$(R_P^{(2)} v)_p = \begin{cases} v_p, & p \in \mathcal{P}(P) \cap \mathcal{P}_{\mathrm{var}}, \\ 0, & p \in \mathcal{P}(P) \setminus \mathcal{P}_{\mathrm{var}}. \end{cases} \tag{18.37}$$

Lemma 18.14. *Use Definition 18.13. The following statements hold.*

1. *For $p \in \mathcal{P}_{\mathrm{inc}}$, the plaquette variables of U^{act} and U^{an} agree for every η. The sum of the Wilson terms of the plaquettes not in $\mathcal{P}_{\mathrm{var}}$, evaluated at U^{act}, is independent of (c, z); denote it by $E_{\mathrm{hole}}(\xi, \eta)$. Then*

$$S_W(U^{\mathrm{act}}) = F_{\mathrm{bare}}(c, z; \xi) + E_{\mathrm{hole}}(\xi, \eta) \tag{18.38}$$

and

$$E_0(\Psi_R(c, z; \xi)) = F_{\mathrm{bare}}(c, z; \xi) + C_\Omega(\xi), \qquad C_\Omega(\xi) = \sum_{p \notin \mathcal{P}_{\mathrm{var}}} f_p(0, 0; \xi), \tag{18.39}$$

where $E_0 = S_W$ is the Wilson action. The function C_Ω is not the energy of the deep frozen group variables. Every derivative that involves (c, z) is the same on the two sides of (18.39).

2. *Let $u = D_z \operatorname{Log} U^{\mathrm{an}}(0, 0; 0) z$ and $K_\Omega = \partial_z^2 F_{\mathrm{bare}}(0, 0; 0)$. Only the variables of the block region and the near values vanish here; the deep frozen group variables η are arbitrary, and the same u on the varying plaquettes and the same K_Ω are obtained from U^{act}, independently of η. Then $u = \widehat{A}_\Omega z$, and for every patch P,*

$$R_P u \in \mathcal{V}_P, \qquad d_P R_P u = R_P^{(2)} d_{\mathrm{var}} u. \tag{18.40}$$

Further,

$$\|u\|_{\mathcal{L}_0}^2 \le \sum_{P \in \mathfrak{P}_\Omega} \|R_P u\|^2, \qquad \sum_{P \in \mathfrak{P}_\Omega} \|R_P^{(2)} d_{\mathrm{var}} u\|^2 \le 7 \|d_{\mathrm{var}} u\|^2, \tag{18.41}$$

$$\|d_{\mathrm{var}} u\|^2 \ge \frac{\sigma_J}{7} \|u\|_{\mathcal{L}_0}^2, \qquad z = B_{\mathrm{ext}} u|_{\mathcal{L}_0}, \qquad \|z\|_2 \le C_{\mathrm{ext},J} \|u\|_{\mathcal{L}_0}, \tag{18.42}$$

with σ_J, B_{ext}, and $C_{\mathrm{ext},J}$ as in Lemma 18.11.

3. *We have*

$$K_\Omega = \widehat{A}_\Omega^{\mathsf{T}} d_{\mathrm{var}}^{\mathsf{T}} d_{\mathrm{var}} \widehat{A}_\Omega, \qquad \lambda_{M,J} I \le K_\Omega \le Q_{M,J} I, \qquad \lambda_{M,J} = \frac{\sigma_J}{7 C_{\mathrm{ext},J}^2}, \tag{18.43}$$

where $Q_{M,J} = 16 a_1 a_\infty$ with $a_\infty = J(1 + 4M^3)$ and $a_1 = M^{2(J-1)}(1 + 24M^3)$. In positions measured in output units,

$$(K_\Omega)_{ab} = 0 \qquad \text{if } \operatorname{dist}_\infty(a, b) > D_{M,J} = 2R_{M,J} + 2. \tag{18.44}$$

Principal submatrices of K_Ω satisfy the same lower bound, the same upper bound, and the same range condition. Real Schur complements of K_Ω are bounded below by $\lambda_{M,J}$ and have uniform exponentially weighted row and column bounds; no finite range is asserted for them.

4. *There are a chart radius $\epsilon_{M,J} > 0$ and a constant C_{lip}, depending in addition only on a fixed choice of the local analytic chart, such that for real (c, z, ξ) with $\max(\|c\|_\infty, \|z\|_\infty, \|\xi\|_\infty) \le \epsilon_{M,J}$,*

$$\partial_z^2 F_{\mathrm{bare}}(c, z; \xi) \ge \tfrac{1}{2}\lambda_{M,J} I. \tag{18.45}$$

All constants depend only on M, J, and the coordinate conventions, and are independent of m, of the block region, of its number of components and holes, of its topology, of the widths of the protected neighborhoods, and of the deep frozen group variables. No uniformity as $J \to \infty$ is claimed. For an empty list of free coordinates the statements on matrices and on the extraction hold with the conventions for zero-dimensional spaces. By (18.39)*, K_Ω is the matrix K_R^0 of Theorem 18.12 at depth zero, where* (18.20) *holds with $\kappa_* = 1$, and* (18.43) *is the corresponding case of that theorem, with $\nu_J = \sigma_J/7$.*

Proof. We begin with (1). Let $p \in \mathcal{P}_{\mathrm{inc}}$. By (18.32), each edge of p lies in $\mathcal{L}_0$ or in $\mathcal{E}_\Omega^{\mathrm{near}}$, so the deep values do not enter U_p^{act}, and $U_p^{\mathrm{act}} = U_p^{\mathrm{an}}$ for every η. A plaquette outside $\mathcal{P}_{\mathrm{inc}}$ has no edge in $\mathcal{L}_0$. On the reconstruction its edges are constant in (c, z): links across the boundary are equal to b_e, other near links to b_e, and deep links to η_e. Its Wilson term is therefore independent of (c, z). A plaquette in $\mathcal{P}_{\mathrm{inc}} \setminus \mathcal{P}_{\mathrm{var}}$ is not in $\mathcal{P}_{\mathrm{min}}$, so its term is independent of (c, z) as well. This proves that E_{hole} is independent of (c, z), and (18.38) follows by splitting the sum over all plaquettes. The same argument for U^{an} shows that $E_0(\Psi_R) = F_{\mathrm{bare}} + \sum_{p \notin \mathcal{P}_{\mathrm{var}}} f_p(c, z; \xi)$ with each summand independent of (c, z) and hence equal to $f_p(0, 0; \xi)$. This proves (18.39). If the list $\mathcal{P}_{\mathrm{var}}$ contains additional plaquettes whose terms are constant in (c, z), only the complementary constants change.

Next we prove (2). On $\mathcal{L}_0$ the derivative u equals $A_\Omega z$, by Lemma 18.11(1). The links outside Ω are constant in z at zero near-link logarithms and fixed deep frozen group variables, as noted after Definition 18.13, so $u = \widehat{A}_\Omega z$. The first assertion of (18.40) is Lemma 18.11(3). Let $p \in \mathcal{P}(P)$. All four edges of p belong to $\mathcal{L}_0$, so $p \in \mathcal{P}_{\mathrm{inc}}$, and the link variables of U^{act} and U^{an} on p are equal, independently of the deep frozen group variables. At $(c, z, \xi) = 0$ all these links are equal to I. The Wilson expansion of a single term is

$$f_p(0, tz; 0) = \tfrac{1}{2} t^2 |(du)_p|^2 + O(t^3). \tag{18.46}$$

Indeed, the first derivative of the plaquette product is the signed sum of the derivatives of its four edge logarithms, and $4q(e^X) = |X|^2/2 + O(|X|^4)$. If $p \notin \mathcal{P}_{\mathrm{var}}$, then $f_p(0, tz; 0)$ is constant, equal to its value zero at $t = 0$, and (18.46) implies $(du)_p = 0$. This holds for every $u \in \ker G_\Omega$, since A_Ω maps onto $\ker G_\Omega$ by Lemma 18.11(1); it does not assert that every plaquette of a patch is a varying plaquette. On a varying plaquette of P the entries of $d_P R_P u$ are those of du, and on the other plaquettes of P both sides vanish. This proves the second assertion of (18.40).

The first inequality of (18.41) is the first inequality of (18.17). The second follows since every plaquette lies in at most seven of the sets $\mathcal{P}(P)$, as shown in the proof of Lemma 18.11. Using the bound $\|h\|^2 \le \sigma_J^{-1}\|d_P h\|^2$ on $\mathcal{V}_P$ and (18.40), we obtain

$$\|u\|_{\mathcal{L}_0}^2 \le \sum_P \|R_P u\|^2 \le \sigma_J^{-1} \sum_P \|R_P^{(2)} d_{\mathrm{var}} u\|^2 \le \frac{7}{\sigma_J}\|d_{\mathrm{var}} u\|^2.$$

Together with Lemma 18.11(1) and (2), this proves (18.42).

For (3), all varying plaquettes have only internal or near links, by (18.32). At their common zero they are identity plaquettes, even when the deep links are arbitrary. Summing (18.46) over the varying plaquettes proves the formula for K_Ω in (18.43). In particular, the term in the Hessian of a composition that involves the gradient of the action vanishes here, since each plaquette term has zero first derivative at the identity; it is not being dropped at a nonflat field. The lower bound follows from (18.42): we have $z^{\mathsf{T}} K_\Omega z = \|d_{\mathrm{var}} u\|^2 \ge (\sigma_J/7)\|u\|_{\mathcal{L}_0}^2 \ge \lambda_{M,J}\|z\|_2^2$.

For the upper bound and the range, note that each row of the curl matrix has four entries, and each positive edge lies in four plaquettes, so its row and column norms are at most 4. Using Lemma 18.11(2) for A_Ω, we find that $16 a_1 a_\infty$ bounds both the operator norm and the unweighted absolute row and column norms of K_Ω. If $(K_\Omega)_{ab} \ne 0$, then two edges in the supports of the columns a and b of A_Ω belong to a common plaquette, so their midpoints are within maximum distance 2 in input units. Since each column of A_Ω is supported within maximum distance $R_{M,J}$ of its label, this proves (18.44). This is a finite range after the elimination of the trees and of the eliminated edges, before any minimization over a whole scale; no orthonormal basis of $\ker G_\Omega$ was used as coordinates.

We record the cases at the boundary. If a final block of Ω is next to the fixed complement, all trees of that block and all equations used for its restriction still lie in the complete block, and a pair is used only when both of its blocks lie in Ω. No missing target of a link across the boundary is set to zero: such a link is fixed through (18.33), and its tree-path transport to the basepoint is identically I on the reconstruction. Plaquettes that contain an internal link and a near link belong to $\mathcal{P}_{\mathrm{inc}}$, have only small links, and contribute nonnegative squares of curls at zero; the estimate above is already obtained from the plaquettes of the patches. If a plaquette of a patch has become constant in (c, z) after the elimination of the constraints, (18.46) shows that its curl vanishes, so no missing energy term is needed, and no boundary condition is imposed on a face of a hole.

At the boundary between near and deep links, a plaquette containing a deep link cannot contain a link of $\mathcal{L}_0$, by (18.32). All its links are constant on the reconstruction, and it belongs to E_{hole} even when some of its other links are near links. Likewise, a deep link occurs in none of the formulas for the coordinates and the Haar disintegration densities. Changing η therefore changes neither A_Ω, B_{ext}, F_{bare}, nor any of their derivatives in (c, z). There is no smallness or branch condition on η, and no derivative in η is taken.

In particular, an exceptional value of the Borel averaging map at the deep frozen group variables cannot affect an internal formula or the matrices above.

Next we treat principal submatrices and Schur complements. Principal submatrices keep the lower bound, the upper bound, and the range. For a decomposition $K = \left(\begin{smallmatrix} K_{II} & K_{IR} \\ K_{RI} & K_{RR} \end{smallmatrix}\right)$ of a matrix with $\lambda I \le K \le Q I$, the real Schur complement satisfies

$$\lambda \|y\|^2 \le y^{\mathsf{T}} (K_{RR} - K_{RI} K_{II}^{-1} K_{IR}) y = \min_{x} (x,y)^{\mathsf{T}} K (x,y) \le Q \|y\|^2. \tag{18.47}$$

The lower inequality uses $\lambda(\|x\|^2 + \|y\|^2) \ge \lambda \|y\|^2$, and the upper one the choice $x = 0$. This proves the lower bound for the Schur complement of every initial segment of the scales. The factors M^i of the effective classical actions multiply whole quadratic forms and lie between 1 and M^J.

We now prove locality without asserting a finite range for Schur complements. In output units, the number of scalar labels in a ball of radius r is at most $v_{M,J}(r+2)^3$ for a fixed $v_{M,J}$. Indeed, there are at most J scales and at most $9M^{3J}$ scalar edge labels per scale and per unit cube of the output lattice, with a fixed enlargement for the boundaries of the cubes. For a principal matrix K of K_Ω let $a = 2Q_{M,J}$ and $T = I - K/a$. Its operator norm is at most $q = 1 - \lambda_{M,J}/a < 1$. The matrix T^k has range $k D_{M,J}$ in the maximum distance, hence range at most $3kD_{M,J}$ in the graph distance dist, its entries are bounded by q^k in absolute value, and its rows and columns are supported on at most $C_{M,J}(k+1)^3$ labels. For $b > 0$ so small that $q e^{3bD_{M,J}} < 1$, the series

$$K^{-1} = a^{-1} \sum_{k \ge 0} T^k, \qquad K^{-1/2} = a^{-1/2} \sum_{k \ge 0} \binom{2k}{k} 4^{-k} T^k \tag{18.48}$$

converge in the norm $\|\cdot\|_b$ of Lemma 18.3(3), uniformly, since their scalar majorants are constant multiples of $\sum_{k \ge 0} (k+1)^3 (q e^{3bD_{M,J}})^k$; the second series is the binomial series of $(1-x)^{-1/2}$, whose coefficients are at most one. The norm $\|\cdot\|_b$ is submultiplicative by the triangle inequality for the distance, so a Schur complement S satisfies

$$\|S\|_b \le \|K\|_b + \|K\|_b^2 \|K_{II}^{-1}\|_b, \qquad S \ge \lambda_{M,J} I,$$

and these bounds can be repeated through the at most J eliminations of initial segments.

Finally we bound the inverse and the inverse square root of such an exponentially localized matrix directly. Suppose $S \ge \lambda I$ and $\|S\|_b \le H$. For a one-Lipschitz real function f and $W_f = \operatorname{diag}(e^{\mu f})$ with $0 < \mu < b$, we have

$$\|W_f S W_f^{-1} - S\|_2 \le \frac{H\mu}{b - \mu},$$

by $|e^t - 1| \le |t| e^{|t|}$, the inequality $r e^{-(b-\mu)r} \le 1/(b-\mu)$, the row and column bounds, and the Cauchy–Schwarz inequality. We choose μ so that the right side is at most $\lambda/2$.

The Neumann series about $S + tI$ then bounds the conjugated resolvent by $2/(\lambda + t)$ for every $t \geq 0$. Taking $f = \operatorname{dist}(\cdot, y)$, we obtain

$$|(S + tI)^{-1}_{xy}| \leq \frac{2e^{-\mu \operatorname{dist}(x,y)}}{\lambda + t}.$$

The density of labels sums this bound in the weighted norm at rate $\mu/2$. At $t = 0$ it bounds S^{-1}, and the formula

$$S^{-1/2} = \frac{1}{\pi} \int_0^\infty t^{-1/2} (S + tI)^{-1} \, dt$$

bounds $S^{-1/2}$, since $\int_0^\infty t^{-1/2}(\lambda + t)^{-1} \, dt = \pi/\sqrt{\lambda}$. The integral formula follows by orthogonal diagonalization of the positive matrix and the substitution $t = \lambda x^2$ in the scalar integral. This proves the assertions on Schur complements in (3), without a finite range for them.

It remains to prove (4). We fix local constraint coordinates of Lemma 6.3 on strictly smaller charts. Their finite-dimensional analytic bounds and (18.10) give a common complex product $\max(\|c\|_\infty, \|z\|_\infty, \|\xi\|_\infty) < 2\delta$, for some $\delta > 0$ depending only on M, J, and the radii of these charts. To check that δ does not depend on the region, we start at scale s, reduce its target and free radii so that its image lies strictly inside the target chart of the next scale, and repeat at most J times. Since the inverse at one scale depends on at most $6M^3$ links, this finite sequence of choices does not depend on the number of blocks. The completion outside Ω is by constants on the reconstruction.

On this product each f_p depends on at most N_* scalar labels, and each label occurs in the dependency lists of at most N_{inc} plaquettes; both numbers depend only on the finite stencils and on M and J. There are finitely many patterns of final blocks of Ω in one stencil. Let C_3 be the maximum, over these patterns and all scalar indices, of the absolute values of the third derivatives of a single f_p on the closed product of radius δ. It is finite, since each local function is holomorphic on the larger product and the list of functions and of their arguments is finite; this also describes exactly how C_3 depends on the chart. Integrating the third derivative from zero to $x = (c, z, \xi)$ and summing over the second free index, we obtain

$$\|\partial_z^2 F_{\mathrm{bare}}(x) - K_\Omega\|_\infty \leq C_{\mathrm{lip}} \|x\|_\infty, \qquad C_{\mathrm{lip}} = N_{\mathrm{inc}} N_*^2 C_3.$$

The column bound is the same by symmetry, so this is also a bound on the operator norm. The calculation differentiates the whole composition, including the term with the gradient of the action at nonzero x. We take $\epsilon_{M,J} \leq \min(\delta, \lambda_{M,J}/(2\max(1, C_{\mathrm{lip}})))$. For real data the Hessian is real symmetric, and (18.45) follows. All plaquette functions in F_{bare} are independent of η before differentiation, so the uniformity in the deep frozen group variables persists. □

18.8 Decay of connected boundary perturbations

We finally consider expressions, built from the objects of Theorem 18.12, that contain one perturbation located in the fixed complement or near the boundary: for example the difference between the reconstructions on Ω and on the torus, or a bounded source in the fixed complement. A contraction network is a finite contraction, or a convergent sum of contractions, of the following tensors: the tensors of the chart and of the effective classical action, with coefficient bounds with each index fixed; inverses of constrained Hessians; at most J Schur complements; and the convergent Neumann and implicit-function series of these operations. Its coefficients are summed with the convergent coefficients of these operations, so a network may contain arbitrarily many inverse or implicit vertices. One constant cannot, however, bound expressions of unbounded algebraic complexity; multiplying an allowed expression by a large scalar already destroys any common bound. We therefore consider families of networks with one common scalar majorant. A perturbation factor has a stated bound on its amplitude. If it depends analytically on current variables, these variables occur among its external indices and satisfy the coefficient bounds of the family; a fixed real source needs only its amplitude bound. The perturbation factor must lie in the same connected component of the network as the external index in the block region.

Lemma 18.15. *Use the products of Theorem 18.12. Let H be the fixed complement, let H_{L_J} be its enlargement by the fixed width L_J, and for $b > 0$ put*

$$S_J(b) = Cv_J \sum_{r \geq 0} (r+1)^3 e^{-br} < \infty.$$

1. *On every periodic subset of labels we have $\sup_x \sum_y e^{-b\,\mathrm{dist}(x,y)} \leq S_J(b)$. For $0 < a < b$ and every position u in Ω,*

$$e^{a\,\mathrm{dist}(u,W)} \sum_{h \in H_{L_J}} e^{-b\,\mathrm{dist}(u,h)} \leq e^{aC_J} S_J(b-a).$$

2. *Let $\mathfrak{N}$ be a family of connected contraction networks, each with at least one external index in Ω and with a perturbation factor at a position of H_{L_J} in the same connected component. The perturbation may be a bounded source s_h at any position h of H_{L_J}, with $|s_h| \leq s_*$. Assume that the diagram sums of all members of $\mathfrak{N}$ are bounded, on one common coefficient product and including all mesh-volume factors, by one common scalar majorant $\mathcal{P}$ with nonnegative coefficients. Then there are $c_J, C_J > 0$, depending only on uniform data, on $\mathcal{P}$, and on s_*, such that every tensor T_l of order $l \geq 1$ in $\mathfrak{N}$ with a specified external index i_1 in Ω satisfies*

$$\sup_{i_1 \in \Omega} e^{c_J\,\mathrm{dist}(i_1,W)} \sum_{i_2,\dots,i_l} e^{c_J \mathfrak{t}(i_1,\dots,i_l)} |T_l(i_1,\dots,i_l)| \leq l! C_J^{l+1}.$$

A single fixed reduction of the spatial rate suffices, independently of the Taylor order and of the number of contractions.

3. *For a member of $\mathfrak{N}$ with output index e and Taylor series $T_e(v) = \sum_\nu f_{e;\nu} v^\nu$, and an input index x, let*

$$\mathcal{C}_{ex} = \sum_{\nu : \nu_x > 0} \nu_x |f_{e;\nu}| \, r^{\nu - e_x}$$

be the entrywise absolute coefficient envelope of $\partial_x T_e$ at the radii r of a strictly smaller coefficient product. Let v_e and v_x be the mesh volumes of the lattices on which e and x live. Then

$$\sup_{x \in \Omega} \sum_{e \in \Omega} \frac{v_e}{v_x} \, e^{c_J[\operatorname{dist}(e,W) + \operatorname{dist}(e,x)]} \, r_x \, \mathcal{C}_{ex} \le C_J. \tag{18.49}$$

In particular, if x and e lie in Ω, then $\mathcal{C}_{ex} \le C_J e^{-c_J[\operatorname{dist}(e,W) + \operatorname{dist}(e,x)]}$, where the fixed conversions of radii and mesh volumes are absorbed in C_J. For two Hessian indices e and f in Ω,

$$|T_{ef}| \le C_J e^{-c_J[\operatorname{dist}(e,f) + \operatorname{dist}(e,W) + \operatorname{dist}(f,W)]},$$

with factorial joint versions that include every differentiated background and free index in the external network. These statements also apply to derivatives in the current frozen coordinates on their construction chart, at the density of their labels.

Traces confined to H, which have no external index in Ω, remain sums over H; no bound on them independent of the size of H is asserted. Differences of coordinates deep in H need only be bounded on their construction domain; they need not be exponentially small. The constants do not depend on the volume of H, on the area of W, on the shape or number of components of Ω, on the number of preceding block integrations, or on the widths chosen later.

Proof. We begin with (1). The first bound is (18.4), with $C = 8$, possibly enlarged to account for a fixed change between positions of links and of their endpoints. It holds on every periodic subset, since restriction deletes terms from a sum of nonnegative terms. A shortest path from a position u in Ω to a position in H crosses W. If $h \in H_{L_J}$ lies in Ω, we add a path of length at most L_J to H and then its crossing of W. In both cases

$$\operatorname{dist}(u, W) \le \operatorname{dist}(u, h) + C_{\mathrm{band},J}, \tag{18.50}$$

where the fixed buffer for link endpoints is included in $C_{\mathrm{band},J}$. Multiplying termwise by $e^{a \operatorname{dist}(u,W)}$ and using (18.4), we obtain

$$e^{a \operatorname{dist}(u,W)} \sum_{h \in H_{L_J}} e^{-b \operatorname{dist}(u,h)} \le e^{aC_{\mathrm{band},J}} \sum_h e^{-(b-a) \operatorname{dist}(u,h)} \le e^{aC_{\mathrm{band},J}} S_J(b-a).$$

This proves (1), after enlarging C_J. The bound sums over all of H, with no separate constant for each face of the boundary.

Next we record how a network acquires its distance to the boundary. For the coefficient of each factor we choose a connected network spanning its external indices and Taylor labels, with length as close as desired to its shortest connecting length, and for each

matrix propagator a shortest path between its endpoints. Joining these networks at the contracted labels gives a connected network $\mathcal{N}$ of total length ℓ. If S is the set of its external positions, then

$$\mathrm{t}(S) \le \ell, \qquad \operatorname{dist}(i, W) \le \ell + C_{\mathrm{band},J} \quad (i \in S \cap \Omega). \tag{18.51}$$

For the second inequality, we follow a route in $\mathcal{N}$ from i to its perturbation vertex $h \in H_{L_J}$ and use (18.50). The same route reaches a point of W, after a segment of length at most $C_{\mathrm{band},J}$ if necessary, so also $\mathrm{t}_W(S) \le \ell + C_{\mathrm{band},J}$. Adding these inequalities, we obtain for all rates $a, \alpha \ge 0$ and every external index i in Ω

$$e^{a\mathrm{t}(S)+\alpha \operatorname{dist}(i,W)} \le e^{\alpha C_{\mathrm{band},J}} e^{(a+\alpha)\ell}, \qquad e^{a\mathrm{t}_W(S)} \le e^{aC_{\mathrm{band},J}} e^{a\ell}. \tag{18.52}$$

This is how the two weights of Remark 18.2 are related: both are bounded by the weight of the total length of the joined network, at a fixed rate. For two external indices e and f in Ω we have the three inequalities

$$\operatorname{dist}(e, f) \le \ell, \qquad \operatorname{dist}(e, W) \le \ell + C_{\mathrm{band},J}, \qquad \operatorname{dist}(f, W) \le \ell + C_{\mathrm{band},J} \tag{18.53}$$

simultaneously; they do not require a separate perturbation for each index. Summing them, we find, for every $c \ge 0$,

$$e^{c[\operatorname{dist}(e,f)+\operatorname{dist}(e,W)+\operatorname{dist}(f,W)]} \le e^{2cC_{\mathrm{band},J}} e^{3c\ell}. \tag{18.54}$$

We now divide the spatial rate. Let $b_0 > 0$ be a common rate for the coefficients of the factors, including the smaller rate of the inverse of the free Hessian. We take all contraction and implicit-series estimates first at the rate $b_0/2$, and we choose $0 < c_J \le b_0/32$. A factor $e^{b_0\ell}$ is available in the absolute product bounds at the rate b_0. We use at most $e^{b_0\ell/4}$ of it for the desired external network and the distances in (18.51) and (18.53). For example, the network length plus one distance to the boundary is at most $2\ell + C_{\mathrm{band},J}$, and the network length plus the three distances in (18.53) is at most $4\ell + 2C_{\mathrm{band},J}$. The choice of c_J covers either sum and leaves a positive fixed fraction of $b_0\ell$ for the internal summations. The fixed added lengths contribute at most the factor $e^{2c_J C_{\mathrm{band},J}}$.

Next we give the summation. We orient a finite tree of contractions toward the specified external index and sum the labels on the outermost branches first, using the corresponding row bounds. On the unique branch containing a specified input index we use the input-column bound instead. The network weights multiply by (18.3), and Lemma 5.3 gives the product of the bounds for the factors, with the normalization of the columns unchanged. A redundant edge or a matrix cycle with a fixed external index is bounded by a further absolute row sum after cutting it at an attached vertex; alternatively, its propagators can be expanded into their convergent chains. All internal sums then use

the same row and column bounds. In particular, a final sum over a bounded source s_h, $h \in H_{L_J}$, uses

$$\sum_{h \in H_{L_J}} |s_h| e^{-u \operatorname{dist}(i,h)} \leq \|s\|_\infty S_J(u),$$

or the weighted version proved in (1), rather than the bound $|H|\|s\|_\infty$. Only one source insertion is needed; further perturbation vertices already present in a coefficient do not require independent sums of this kind.

A fixed product or composition has a finite scalar majorant, obtained by multiplying or composing the majorants of the factors. Each inverse or resolvent series is an infinite sum of chains with geometric ratio at most $1/4$ on the larger product, as in (18.5). The perturbation in an inverse difference stays in its single difference factor. A difference of Schur complements is the finite telescoping sum of Lemma 18.3, and at most J such operations have a fixed complexity. The common majorant $\mathcal{P}$ bounds the number of summed chains and contractions. The division of the spatial rate is applied once to each complete joined network, not anew at each vertex of an infinite expansion.

For an implicit series we make this precise. Let $Y(t)$ be the solution with nonnegative coefficients of (18.30). A derivative in a parameter that inserts the perturbation once gives an equation whose majorant satisfies

$$Z \preceq H(t, Y(t)) + A(t) Z, \qquad A(t) = \tfrac{1}{2} + \partial_Y \frac{C_J(t+Y)^2}{1 - C_J(t+Y)}\Big|_{Y=Y(t)}. \tag{18.55}$$

Here $\preceq$ denotes domination of coefficients, and $H(t, Y)$ is the series with the source inserted (not to be confused with the fixed complement H); its constant and higher coefficients have the bounds with a fixed index just proved. The implicit series is obtained by repeated substitution, so all its coefficients are bounded by those of

$$\sum_{n \geq 0} A(t)^n H(t, Y(t)) = \frac{H(t, Y(t))}{1 - A(t)}.$$

We choose a radius $t_0 > 0$ strictly inside the radius of the solution such that $A(t_0) \leq 3/4$; this is possible since $Y(0) = 0$ and $A(0) = 1/2$. The series converges absolutely at t_0, and its value is at most $4H(t_0, Y(t_0))$. This is a bound on the value of a series with nonnegative coefficients; it is not the false coefficientwise inequality $Z_l \leq 4H_l$. A derivative in another analytic input differentiates this convergent series on a strictly smaller radius, where the additional geometric denominators are bounded; this gives the estimate with a specified input index. All indices are allowed, since all factor tensors and all inserted inverses were controlled with every index fixed.

For example, consider the interpolation from Step 3 of the proof of Theorem 18.12, and let $d_a = \Psi_R - \Psi_T$ on the common list of link variables. Its changed formulas in Ω are

confined to the band of width L_J, and its values in the fixed complement are bounded on the construction domain. The difference of the two objectives is

$$\Delta F = \int_0^1 DE_{d_0}(\Psi_T + t d_a)[d_a]\, dt + \tfrac{1}{2}\|z_H\|_2^2. \tag{18.56}$$

The segment of integration lies in the torus chart, by the choice of radii in that proof. If a derivative in a current or free variable acts on d_a, the changed formula keeps its perturbation. If it acts on the action or on another argument of the chart, the tensor of the action connects that derivative to the remaining position of d_a. The auxiliary quadratic form has only indices in the complement, and it reaches an index in Ω only through a connected contraction. A derivative at a remote background position therefore cannot disconnect the perturbation. Differentiating the stationarity equation in the interpolation parameter, we obtain $\dot z = -F_{zz}^{-1}(\Delta F)_z$, which is an instance of (18.55). This verifies that the implicit networks used here have a perturbation connected to every external index, rather than an unrelated scalar factor.

To pass from convergence to the factorial bounds in (2), we fix a common radius t_1 with $0 < t_1 < t_0$ and let D_J bound the preceding majorants with a fixed index there, after the division of the spatial rate. If a_l denotes their Taylor coefficients of order l, normalized as in (18.2), nonnegativity implies $a_l t_1^l \le D_J$. Hence the derivative tensors of order l are bounded by $l! D_J t_1^{-l}$. Taking $C_J \ge \max(D_J, t_1^{-1}, 1)$ proves the bound in (2). Every degree sum and every supremum over the fixed index in this argument has been taken at fixed frozen coordinates, and the uniform bound permits the final supremum over their real values.

For (3), we keep both the input index x and the output index e in the complete network before summing the other leaves. The length then covers $\operatorname{dist}(e, W)$ and $\operatorname{dist}(e, x)$ at the rate c_J. For each differentiated occurrence we have $\operatorname{dist}(e, W) + \operatorname{dist}(e, x) \le 2\ell + \mathrm{C}_{\mathrm{band},J}$, by (18.51). Summing the differentiated majorant series on a smaller product over e, with the input column of x and its volume normalization, we obtain (18.49). Each entry is then at most $(v_x / v_e) r_x^{-1} C_J e^{-c_J[\operatorname{dist}(e,W)+\operatorname{dist}(e,x)]}$; since at most J scales are involved and r_x is a fixed radius of the smaller product, this gives the entrywise bound after enlarging C_J. This bound includes the multiplicity of the input in the differentiated series; it is not only a bound on a first derivative at zero. For two Hessian indices we use (18.53). A background or free coefficient of order k is bounded by

$$\frac{(k+2)!}{k!} C_J^{k+3} R^k = C_J^3 (k+1)(k+2)(C_J R)^k,$$

whose sum over k is $2C_J^3/(1 - C_J R)^3$ for $C_J R < 1$. The external network at this order includes both Hessian indices and all k further leaves. Reducing R by a fixed factor gives the bound on T_{ef} and its joint factorial versions. Derivatives in the current frozen

coordinates use the same tensors and the same counting, at the density of the labels on the input lattice.

No step of the proof estimates a trace confined to the complement by fixing an index in Ω, and no step requires decay of a coordinate difference at a position deep in the complement. For example, a prescribed small link $\exp(\varepsilon\tau_a)$ in the complement can differ from the corresponding link I of the torus completion by the logarithm $\varepsilon\tau_a$, whatever its distance from W; its effect at an index in Ω is bounded instead by (18.50) and the convergent sum over sources. All choices used only fixed coefficient bounds, the density of labels, and fixed rates, so no volume, area, number of components, number of preceding block integrations, or later width enters c_J and C_J. The empty block region and the convention for a block region without boundary give the remaining degenerate cases. □

SECTION 19

Comparison of reconstructions on block regions

This section compares one renormalization group step on a block region with the same step on the whole torus. Fix a composition T_M^s of $s \leq J$ averaging steps and a union Ω of blocks of its output lattice. In Section 18 we reconstructed the fine configuration on Ω from the coarse field on Ω and free logarithmic coordinates, with the coordinates outside Ω held fixed. We also constructed the stationary point of the effective classical action in these coordinates and proved a gap for its free Hessian. Here we prove that the regional construction differs from the torus construction with the coarse field extended by zero outside Ω only by terms which decay exponentially with the distance to the boundary of Ω. We also prove that the regional construction is stable under small analytic perturbations of the incoming action. All estimates are bounds on absolute Taylor coefficients. Their constants do not depend on the shape of Ω, the number of its components, the depth, the number of preceding renormalization group steps, or the values of previously frozen coordinates.

We use these estimates in the localized renormalization group analysis behind the exponential-moment bound for blocked actions (Step 1 in Section 3.4). There the configurations are decomposed according to the regions where some averaged plaquette has large curvature, and the renormalization group is run outside these regions with the link variables inside them held fixed. Section 20 integrates the fluctuations on block regions using the reconstructions and perturbation estimates proved here. Section 21 uses the gauge transformation of Lemma 19.5 and the perturbation theorem Theorem 19.9 to construct large-field decompositions, and Section 23 uses the domain conditions of Lemmas 19.3 and 19.7 and Theorem 19.9 to choose the parameters.

The results, in order, are as follows: Lemma 19.3 treats successive minimization; Theorem 19.4 compares the regional and torus stationary points, together with the minimized actions, the Hessians, and the Haar disintegration densities; Lemma 19.5 constructs a gauge transformation which is the identity at prescribed vertices; Theorem 19.6 compares the gauge-transformed reconstructions and bounds their input columns; Lemma 19.7 collects the conditions on the radii; and Theorem 19.9 proves stability under perturbations of the incoming action.

The inputs are Theorem 18.12 (regional coordinates, the gap of the free Hessian, and the regional stationary point), Lemma 18.15 (coefficient bounds for connected networks which reach the fixed complement), Lemma 18.8 (derivatives of constrained stationary

values), Lemma 18.9 (freezing of coordinates), the coefficient algebra of Definition 18.1, the target-dependent gauge transformation of Theorem 13.2, the classical coefficient bounds of Theorem 10.11, and the equality of effective classical actions on compact gauge overlaps, Theorem 11.6.

We compare constrained minimizers through the implicit equation for their response, and we control the response by an exponentially decaying inverse of the constrained Hessian. This approach follows Bałaban's analysis of the variational problem for background fields [16, (182)–(190), pp. 307–308, and Proposition 9, p. 309], although our constraints, coordinate domains, and coefficient norms differ from his. The comparison between the regional and the torus problems and the gauge transformation fixing the prescribed endpoints are constructed here. The exponentially weighted resolvent bound, proved in Lemma 18.3, is the finite-dimensional version of the exponential-conjugation method of Combes and Thomas [61].

The mechanism. The regional problem and the torus problem with coarse field extended by zero have the same constraints away from a fixed neighborhood of the boundary of Ω. If we restrict the torus stationary point to the regional coordinates, the gradient of the regional objective at this point is therefore concentrated near the boundary, up to exponentially decaying tails. This residual gradient has two sources, which are the two terms of the formula for r_R in Theorem 19.4. First, the reconstructed fine field changes. Second, the tangent map of the constraints changes, since the regional problem omits the coarse constraints on links leaving Ω. Writing p for the restricted torus stationary point and z_R for the regional stationary point, we have $z_R - p = -\mathcal{A}_R^{-1} r_R$, where r_R is the residual gradient and $\mathcal{A}_R$, the regional Hessian averaged along the segment from p to z_R, has a positive gap. Since this inverse decays exponentially in the distance, the two stationary points differ by an amount which decays exponentially away from the boundary. The remaining work is to make this mechanism stable under differentiation, composition, and successive minimization. This requires bounds on all Taylor coefficients, with any one external index fixed. A bound on the values of a holomorphic map does not give the spatial weights and the fixed-index sums which are needed when these functions are composed later.

Figure 19.1 shows the sets which appear below. Two of them play different roles: the prescribed vertices, where the gauge transformation must be the identity, and the complement of the incoming list, which controls the spatial decay of the perturbation in Theorem 19.9.

19.1 Setting and conventions

Geometry. Throughout this section $M \geq 3$ is odd, $J \geq 1$ is fixed, $1 \leq s \leq J$, and $B_s = M^s$. The composition T_M^s of s averaging steps maps configurations of side $M^s m$ to

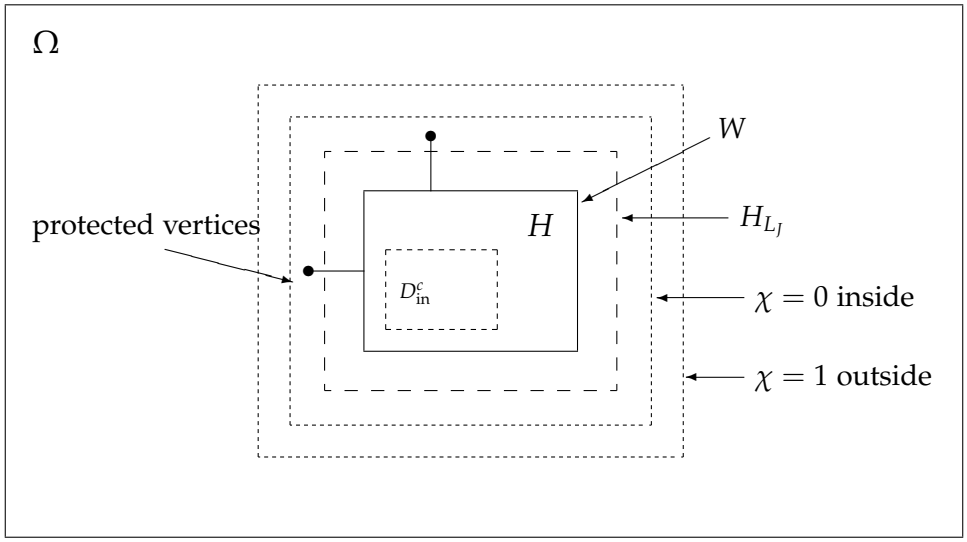


Figure 19.1. The geometry in the output lattice (schematic). The solid inner rectangle is the fixed complement $H = \Omega^c$, and its boundary is the interface W. The dashed rectangle bounds the enlargement H_{L_J} of H by the reconstruction stencil. The cutoff χ of Lemma 19.5 vanishes inside the inner dotted rectangle and equals one outside the outer dotted rectangle. The black dots are protected vertices, here the endpoints of the paths of two based transports of frozen coordinates. The small dotted rectangle inside H is the complement of the incoming list D_{in} of Theorem 19.9, which lies within a fixed distance of H.

configurations of side $m \geq 4$, and the prior depth $d_0 \geq 0$ is an arbitrary integer. When $s = J$ the composition is one renormalization group step, with blocking factor $B = M^J$. Positions are measured in units of the output lattice, of side m. Let Ω be a union of blocks of the output lattice, let $H = \Omega^c$ be its fixed complement, let W be the interface between Ω and H, and let H_{L_J} be the enlargement of H by the fixed reconstruction stencil L_J of Theorem 18.12. A coordinate is a label carrying a position, a direction, a color, and a scale index; its position is taken at the lattice where the coordinate was introduced and embedded in output units. In particular the newly frozen coordinates are counted with the density of the lattice on which they were introduced, not as coordinates of the output lattice. The density constant v_J of Definition 18.1 includes the colors, the directions, and the at most J scale indices. For a label i in Ω we write $\delta(i) = \operatorname{dist}(i, W)$. Here dist and t are the graph distance and the ℓ^1 network length of the refined position graph fixed before Definition 18.1; coefficient bounds imported at a Euclidean rate b are used at rate $b/\sqrt{3}$. If Ω is empty, every statement below with a fixed index in Ω is void. If H is empty, every boundary difference considered below is zero; we never use an expression of the form $0 \cdot e^{\infty}$.

Coordinates. We use the following coordinates.

- The coarse field c is the logarithm of the output links internal to Ω, and $c \oplus 0$ is its extension by zero to all output links of the torus.
- The free logarithmic coordinates $z = (z_i)_{0 \le i < s}$ of Theorem 18.12 parametrize the solutions of the block averaging constraints in Ω, with one block z_i for each intermediate scale i; here $i = 0$ is the finest scale. On the torus we write $z = (z_L, z_H)$, where z_L are the coordinates of the regional list and z_H are the remaining torus coordinates.
- $\Psi_R(c, z; \xi)$ is the logarithm of the regional reconstruction of the fine field before gauge transformation, including its prescribed small values outside Ω, and $\Psi_T(c_T, z_T)$ is the logarithm of the torus reconstruction before gauge transformation. We call the links of the input lattice of T_M^s fine links.
- In the successive reconstruction the free coordinates of scale i are $a_i + K_i^{-1/2} q_i$, where a_i is the conditional stationary point and K_i the constrained Hessian of that scale, and q_i is the fluctuation increment. The increments q_i replace $g_i w_i$, where w_i are the normalized fluctuation variables; no coupling is absorbed into K_i.
- In Section 19.7 we also use the jointly normalized coordinates u, defined by $z = z_0 + K_0^{-1/2} u$, where z_0 is the joint stationary point and K_0 the Hessian there.
- The newly frozen coordinates ξ are the small logarithmic coordinates frozen at the present step; they enter Ψ_R through its prescribed values outside Ω. The previously frozen coordinates b_{rec} are fixed real parameters, kept at the lattices on which they were introduced.

The points $z = 0$, $q = 0$, and $u = 0$ are different. Both $q = 0$ and $u = 0$ correspond to the joint stationary point $z = z_0$ (for q this is part of Lemma 19.3). The point $z = 0$ is the origin of the constraint coordinates.

Frozen coordinates and stationary points. No derivative in the previously frozen coordinates b_{rec} is taken or asserted in this section. In Theorems 19.4 and 19.6 the newly frozen coordinates ξ are analytic variables, which may be complexified and differentiated. In Theorem 19.9 the perturbed objects are differentiated only in the coarse field, the free coordinates, and the interpolation parameter, at fixed real (ξ, b_{rec}). All stationary points below are the holomorphic stationary branches in the coordinate domains of Theorem 18.12. At real data the positive free Hessian makes them local minima, and the conditional stationary points are the conditional constrained minimizers of the successive minimization.

Dependence of constants. All constants and radii in this section have the dependence described in item (3) of Definition 4.14. They may depend on M and J, on the larger classical and reconstruction domains and the ratios of their radii, on the coefficient bounds on these domains, on the gaps of the free Hessians and the bounds on their weighted inverses, on the fixed spatial decay exponents, on v_J, and on any fixed bound

on the size of the perturbation. In Lemma 19.7 and Theorem 19.9 they may also depend on the fixed analytic radii, on K_*, and on the ratios of these radii. They do not depend on L, γ, the lattice side, the depth d_0, the shape or topology of Ω, the number of components of Ω or H, the values of the previously frozen coordinates, or the number of preceding renormalization group steps. They are evaluated before the later choices of the boundary-layer width w_{bd}, the erosion width, the widths of the protected neighborhoods, the bound β_{rec} on newly frozen coordinates, and the enclosing product domains of Section 20. Each constant includes a finite maximum over $1 \le s \le J$ and over the finitely many initial segments. We do not assert uniformity as $J \to \infty$. Theorem 18.12 provides the gap of the free Hessian at every integer depth d_0, and when only this classical estimate is used, no condition on an effective interaction is imposed.

The torus construction domain. Every use of Theorem 13.2 below takes place on its construction product. For a composition of s averaging steps, we denote the radius and coefficient constant of that theorem by r_s and $C_s^{\mathbb{T}}$. Before the later profiles and widths are chosen, we fix positive U_T, Q_T in the common larger product of coarse fields and free coordinates such that

$$U_T + Q_T < \frac{1}{2} \min_{1 \le s \le J} r_s, \qquad C_T(U_T + Q_T) < \frac{1}{2}, \qquad C_T = \max_{1 \le s \le J} C_s^{\mathbb{T}}. \tag{19.1}$$

Decreasing U_T, Q_T if necessary, we may assume that the intermediate classical, inverse, and Haar-density margins of Theorem 13.2 hold. Later radii for coarse fields and free coordinates are chosen below U_T and Q_T, so each use of Theorem 13.2 has $u + v < r_s$, a positive denominator, and its intermediate margins. The rational row bound of that theorem alone does not show that a larger product is allowed.

Coefficient bounds. We use the coefficient algebra of Definition 18.1, Theorem 18.12, , and Lemma 18.15 and its coefficient bounds with each external index fixed in turn. A rooted coefficient sum of a tensor fixes one external index, its root, which is either an output or an input, and sums the absolute values of the Taylor coefficients over the remaining indices, with the weight $e^{a\mathsf{t}}$, where t is the ℓ^1 length of a shortest connected network spanning the positions involved. For a fixed input index the sum is the volume-normalized column sum of Definition 4.6, with the mesh of that input. A coefficient bound is a bound on all these sums, with any external index fixed. Taylor tensors at complex arguments use bilinear transposes. A majorant series is the series of absolute coefficients after the prescribed rooted sums and weights; the original coefficients need not be positive. In a scalar majorant the variable t is a radius after these rooted sums, never a sum of radii over all lattice sites. We say that a tensor, or a convergent series of tensors, has boundary-localized coefficient bounds if it satisfies the bounds of Lemma 18.15: with any index i in Ω fixed, its rooted coefficient sums carry the additional weight $e^{c\delta(i)}$ for a fixed $c > 0$.

A product or contraction joins the connecting networks of its factors at the contracted indices. We root the joined network at any external index and sum over the internal indices, using the corresponding rooted bounds of the factors. The boundary-localized estimate applies if one factor is boundary-localized, or if the joined network visits H_{L_J}. An index deep in H is bounded there, and the decay comes from its connection to the fixed index in Ω. In particular, for decay exponents $b > a > 0$, the sum over such indices is bounded by

$$e^{a\delta(e)} \sum_{h \in H_{L_J}} e^{-b\,\mathrm{dist}(e,h)} \le e^{aL'_J} S_J(b-a), \qquad S_J(u) = C v_J \sum_{r \ge 0} (r+1)^3 e^{-ur}, \tag{19.2}$$

where L'_J also includes the fixed convention for link endpoints. This is the case of a fixed index in Ω in Lemma 18.15; there is no factor of the volume of H.

Implicit series. The implicit equations below are solved in the absolute coefficient algebra. After the construction constant C_J is increased, the scalar majorant of the solution without boundary weight can be taken to satisfy

$$Y = C_J t + \frac{1}{2} Y + \frac{C_J (t+Y)^2}{1 - C_J (t+Y)}. \tag{19.3}$$

Fix $t_0 > 0$ such that, on $0 \le t \le t_0$ and $0 \le Y \le 4C_J t_0$, the rational term has Y-derivative at most $1/4$ and denominator at least $1/2$, and its value at $(t, 4C_J t)$ is at most $C_J t$. Such a t_0 exists and depends only on C_J, since these are finitely many strict inequalities which hold at $t = 0$. Iteration from zero produces series with nonnegative coefficients which stay below $4C_J t$. The iteration converges uniformly on every smaller disc, since the derivative in Y of the right side of (19.3) is at most $3/4$ on its majorant disc. The degree-l coefficients of the solution are bounded by $4C_J t_0^{1-l}$, and dividing Taylor tensors by $l!$ before this comparison gives factorial tensor bounds. The argument works in the absolute multivariate coefficient algebra because each product satisfies the rooted contraction bound just described. Uniform convergence of the power series implies holomorphy.

A derivative which inserts one difference factor, localized near H, has the majorant

$$Z \preceq H(t, Y(t)) + A(t) Z, \qquad A(t) = \frac{1}{2} + \partial_Y \frac{C_J (t+Y)^2}{1 - C_J (t+Y)}\bigg|_{Y = Y(t)}, \tag{19.4}$$

where $\preceq$ denotes coefficientwise domination and $H(t, Y)$ is the series with the difference factor inserted. On a smaller fixed radius $A(t) \le 3/4$, and the solution is majorized coefficientwise by $H(t, Y(t))/(1 - A(t))$; this is a convergent majorant series with one inserted difference factor (compare (18.55)). Fixing an analytic input differentiates this same series and keeps the index of that input. By Lemma 18.15, the connecting network and the route to the boundary are bounded once for the whole network, and their exponent is not divided by the Taylor order.

Matrix functions. We use matrix functions only on free coordinate spaces where a gap has been proved. The next lemma collects the facts we need.

Lemma 19.1. *Let $A^b \geq \lambda I$ be a real symmetric free matrix whose two-sided weighted Schur norm at exponent b is at most Q, and set*

$$\mu = \frac{b\lambda}{8(Q+\lambda)}, \qquad C_R = 2S_J(\mu/2).$$

1. *For every $t \geq 0$, the entries of $(A^b + t)^{-1}$ are bounded by $2e^{-\mu \operatorname{dist}(x,y)}/(\lambda + t)$, and its weighted Schur norm at exponent $\mu/2$ is at most $C_R/(\lambda + t)$; the constant C_R is the same for every subset of the label list.*
2. *If L is a complex coefficient perturbation with $C_R\|L\|_{\mu/2}/\lambda \leq 1/4$, then the series*

$$(A^b + L + t)^{-1} = \sum_{k\geq 0} [-(A^b + t)^{-1}L]^k (A^b + t)^{-1} \tag{19.5}$$

   converges normally, and its coefficient Schur norm is at most $4C_R/[3(\lambda + t)]$.
3. *For positive definite real matrices A, B,*

$$A^{-1/2} = \frac{1}{\pi}\int_0^\infty t^{-1/2}(A + t)^{-1}\,dt, \tag{19.6}$$

$$A^{-1/2} - B^{-1/2} = -\frac{1}{\pi}\int_0^\infty t^{-1/2}(A + t)^{-1}(A - B)(B + t)^{-1}\,dt, \tag{19.7}$$

$$\log A - \log B = \int_0^\infty (A + t)^{-1}(A - B)(B + t)^{-1}\,dt, \tag{19.8}$$

   and these identities extend holomorphically to the complex perturbations in (2).
4. *If $T = \begin{pmatrix} A & B \\ C & D_0 \end{pmatrix}$ is invertible with invertible blocks A and $D_0 - CA^{-1}B$, then*

$$(T^{-1})_{LL} - A^{-1} = A^{-1}B(D_0 - CA^{-1}B)^{-1}CA^{-1}. \tag{19.9}$$

Proof. Parts (1) and (2) are the Combes–Thomas bounds proved in Lemma 18.3(3); see (18.5). We recall the choice of constants. For a real one-Lipschitz function f on the label metric, conjugation by $\operatorname{diag}(e^{\mu f})$ changes A^b by a matrix of Schur norm at most $Q\mu/(b - \mu) \leq \lambda/8$; here we used $e^{\mu r} - 1 \leq \mu r e^{\mu r}$ and $\sup_{r\geq 0} r e^{-(b-\mu)r} \leq (b - \mu)^{-1}$. By the geometric resolvent series, the conjugated resolvent has operator norm at most $2/(\lambda + t)$ for $t \geq 0$. Taking $f(x) = \operatorname{dist}(x, y)$, we obtain the entry bound, and summing at exponent $\mu/2$ gives $C_R/(\lambda + t)$, uniformly over every subset of the label list. Under the condition in (2) each term of (19.5) after the first is smaller than the preceding one by the factor $1/4$, which gives the bound $4C_R/[3(\lambda + t)]$.

For (3), we diagonalize the positive matrices and substitute $t = \lambda u^2$ in each eigenvalue integral, which gives (19.6). The resolvent identity $(A + t)^{-1} - (B + t)^{-1} =$

$-(A+t)^{-1}(A-B)(B+t)^{-1}$ fixes the order of the three factors and gives (19.7) after integration against $\pi^{-1}t^{-1/2}\,dt$. For (19.8), we integrate $(1+t)^{-1}I-(A+t)^{-1}$ to obtain $\log A$ and subtract the corresponding formula for B. The series (19.5) extends these identities holomorphically to the complex perturbations in (2). Their majorants are integrable: they are bounded by $C_J t^{-1/2}(\lambda+t)^{-2}$ and $C_J(\lambda+t)^{-2}$ for the two differences, whose integrals are $\pi/(2\lambda^{3/2})$ and $1/\lambda$. Thus a boundary-localized middle factor $A-B$ gives boundary-localized differences of these functions, rooted at every index, and differentiation under the integral is justified by the same normally convergent majorants. Restricted diagonal sums are taken after this calculation; only a complete scalar trace permits a cyclic permutation of the factors. Part (4) follows by block multiplication. □

Additive constants. The coefficient seminorm with boundary decay $\mathcal{X}$ of Definition 18.1 vanishes on constants, so it never estimates additive constants. The constant term of a difference of minimized actions remains in the full action. Scalar determinants, Haar disintegration densities, and traces over H, which have no index in Ω, are kept as their exact finite values. The difference between two exterior extensions of the classical action is kept with all its constants (Theorem 19.9). After freezing at c, replacing an effective interaction $R(v,c)$ by $R(v,c)-R(0,c)$ changes the normalization constant from Λ to $\Lambda-R(0,c)$ and preserves the density, as in Lemma 18.9.

19.2 Conditional minimization and successive substitution

The reconstruction of the fine field proceeds from the coarsest intermediate scale to the finest. At scale i the free coordinates are $a_i(c,z_{>i})+K_i^{-1/2}q_i$, where a_i is a conditional stationary point which depends on the coarser coordinates $z_{>i}$ already reconstructed. We must check two things. First, for every fixed value of the coarser coordinates in some domain, the conditional stationary point exists. Second, the coarser coordinates produced by the earlier substitutions lie in that domain. The first does not imply the second, as the following example shows.

Example 19.2. Let $F_0(x,y)=(x^2+y^2)/2$ and $F_1(x,y)=F_0(x,y)-\delta y$ with $\delta>0$, and eliminate x first, so that y is the coarser coordinate. For every fixed y the conditional stationary point of F_1 in x is $x=0$, and the joint Hessian is the identity. The joint stationary point of F_1 is $(0,\delta)$. If the conditional stationary points are constructed only for $|y|<\eta$ with $\eta<\delta$, the successive substitution never reaches the joint stationary point, however small the fluctuation radius in x is chosen. The quantity δ_V of Lemma 19.3 detects this: for the initial segment consisting of both coordinates it equals δ, so the condition in part (3) of the lemma fails.

In the next lemma, I is an initial segment of the finer scale coordinates eliminated in the successive minimization, and $y = z_{I^c}$ denotes the coarser coordinates held as parameters. The segment containing all free coordinates is included.

Lemma 19.3. *Fix the larger construction domains of Theorem 18.12, within* (19.1)*, and the margins of their coefficient, inverse, and image bounds and of the lower bound for the real constrained Hessian. Let $F_0(c, z; \xi)$ be the unperturbed regional objective and $z^0(c; \xi)$ its joint stationary point. For each initial segment I write $y = z_{I^c}$, and let $a_I(c, y; \xi)$ be the conditional stationary point of F_0 in the variables z_I.*

1. *One can first choose a solution radius $r_I > 0$, and then a strictly smaller radius η_I for the deviation $y - z^0_{I^c}(c; \xi)$, uniformly over the coarse fields in the domain, such that*

$$a_I(c, z^0_{I^c}; \xi) = z^0_I, \qquad D_y a_I = -F^{-1}_{0,II} F_{0,II^c}, \qquad \|a_I - z^0_I\|_{\mathrm{coeff}} \le C_J \|y - z^0_{I^c}\|_{\mathrm{coeff}}.$$

   The derivative identity is evaluated at the conditional stationary point. We write $\mathcal{T}_I$ for the resulting domain of y.

2. *Let $F_\lambda = F_0 + \lambda V$, $0 \le \lambda \le 1$, be holomorphic and real at real arguments, with the same coefficient bounds on the larger domains. Set $A_I = F_{0,II}(c, a_I, y; \xi)$ and assume that, throughout the domains of part (1), in the operator norm of the coefficientwise majorant,*

$$\|A_I^{-1} F_{\lambda,I}(c, a_I, y)\| \le r_I/4, \qquad \|A_I^{-1}(F_{\lambda,II} - A_I)\| \le 1/2.$$

   Assume also that the real Hessian is perturbed by less than half of the lower bound for the constrained Hessian, and that the complex resolvent condition of Lemma 19.1(2) holds. Then for each I and each $y \in \mathcal{T}_I$ there is a unique holomorphic stationary point $a_{\lambda,I}(c, y; \xi)$ of F_λ in z_I within distance $r_I/2$ of a_I. It is real at real data and a local real minimum there, and it is unique in the convex domain on which the second inequality holds.

3. *Put $V = \partial_\lambda F_\lambda$ and*

$$\delta_V = \max_I \sup_{y \in \mathcal{T}_I} \|A_I^{-1} V_I(c, a_I(c, y; \xi), y)\|_{\mathrm{coeff}},$$

   where the coefficient norm includes the common product of coarse fields, uniformly at fixed real frozen coordinates, and let $\varepsilon_{\mathrm{tube}}$ be the minimum over I of the solution and parameter margins at I. Then, for every I and every $y \in \mathcal{T}_I$,

$$\|a_{\lambda,I}(c, y; \xi) - a_I(c, y; \xi)\|_{\mathrm{coeff}} \le 2\lambda \delta_V.$$

   Assume further that $\delta_V < \infty$, that the mixed coefficient bounds hold uniformly on the larger domains, and that $r_q > 0$ satisfies

$$C_{\mathrm{seq},J}(2\delta_V + r_q) < \frac{1}{2}\varepsilon_{\mathrm{tube}},$$

where $C_{\mathrm{seq},J}$ is the constant given in the proof. Let $a_{\lambda,i}$ be the conditional stationary point of F_λ at scale i, and let $K_{\lambda,i}$ be M^i times the Schur block of F_λ at scale i in input units. Then, whenever $\max_i \|q_i\|_\infty \le r_q$, the successive substitutions in the original free coordinates

$$z_i = a_{\lambda,i}(c, z_{>i}; \xi) + K_{\lambda,i}^{-1/2}(c, z_{>i}; \xi) q_i$$

stay in every parameter domain and every solution domain, with half of each margin unused. At $q = 0$ they give the joint stationary point.

The radii are chosen in the following order: first the unperturbed product, for which $\delta_V = 0$; then a bound on the size of the perturbation on the fixed larger unperturbed products, and a smaller perturbed product; only afterwards the smaller enclosing and fluctuation radii. Reducing r_q does not reduce the radius of the coarse field, since the chain at $q = 0$ is the joint stationary point. All radii and constants have the dependence of Section 19.1, independently of the later widths, the depth, and the geometry of Ω.

Proof. Part (1) is a quantitative implicit function argument about the joint stationary point, and part (2) is the same argument about a_I. The new point is part (3), which first bounds the displacement of the conditional stationary points at every fixed y and then propagates these bounds through the successive substitution by a finite recurrence.

Step 1: the unperturbed conditional stationary points. Fix I and write $y = z^0_{I^c}(c; \xi) + v$. This shift is an analytic change of coordinates with the coefficient bounds of Theorem 18.12. In these coordinates we set $u = z_I - z^0_I(c; \xi)$ and

$$B_I(c; \xi) = F_{0,II}(c, z^0(c; \xi); \xi).$$

By the bounds on the inverse and on the joint third derivative, there exist constants I_0, C_0 on the fixed larger products such that

$$\|B_I^{-1} F_{0,I}(c, z^0_I, z^0_{I^c} + v; \xi)\|_{\mathrm{coeff}} \le I_0 C_0 \|v\|_{\mathrm{coeff}}, \tag{19.10}$$

$$\|B_I^{-1}(F_{0,II}(c, z^0_I + u, z^0_{I^c} + v; \xi) - B_I)\|_{\mathrm{coeff}} \le I_0 C_0 (\|u\|_{\mathrm{coeff}} + \|v\|_{\mathrm{coeff}}). \tag{19.11}$$

For the first inequality we integrate F_{0,II^c} in v, starting from the joint stationarity $F_{0,I}(c, z^0; \xi) = 0$; for the second we integrate the third derivative tensor in (u, v). These are coefficient inequalities, since integration along a segment from the origin multiplies each term of positive degree by a number at most one, and the rooted bounds on the larger domains control the products.

Fix $r_I > 0$ within a fixed fraction of the solution margin of the chart, with $I_0 C_0 r_I < 1/8$. Then fix $0 < \eta_I < r_I$ within the parameter margin, with $I_0 C_0 \eta_I < r_I/8$ and $I_0 C_0 (r_I + \eta_I) < 1/4$. For $\|v\| \le \eta_I$, the map

$$u \longmapsto u - B_I^{-1} F_{0,I}(c, z^0_I + u, z^0_{I^c} + v; \xi)$$

has derivative $-B_I^{-1}(F_{0,II} - B_I)$, whose norm is less than $1/4$ on $\|u\| \le r_I/2$ by (19.11), and its value at $u = 0$ has norm less than $r_I/8$ by (19.10). It is therefore a coefficient contraction of the ball $\|u\| \le r_I/2$ into itself. Its Picard iterates converge in the coefficient norm, and their power series converge locally uniformly on the parameter product, which implies that the limit $a_I - z_I^0$ is holomorphic. At $v = 0$ the stationary point is the joint one, which proves the first identity of part (1). Differentiating the gradient equation at the conditional stationary point and using Lemma 18.8, we obtain

$$D_y a_I = -F_{0,II}^{-1} F_{0,II^c}.$$

The inverse and the mixed matrices have convergent coefficient bounds on a product larger than the chosen one. Integrating this identity from $y = z_{I^c}^0$ to y, we obtain the Lipschitz bound of part (1), with C_J a common bound for the product of these matrices. The choices are uniform in the coarse field, because the estimates were made on its fixed larger domain before v was introduced.

Step 2: the perturbed conditional stationary points. We write $z_I = a_I + Y$ and consider on the solution domain the map

$$\mathcal{T}_\lambda(Y) = Y - A_I^{-1} F_{\lambda,I}(c, a_I + Y, y). \tag{19.12}$$

Since $D_Y \mathcal{T}_\lambda = -A_I^{-1}(F_{\lambda,II} - A_I)$, the two inequalities of part (2) imply

$$\|\mathcal{T}_\lambda(0)\| \le r_I/4, \qquad \|D_Y \mathcal{T}_\lambda\| \le 1/2.$$

Then $\mathcal{T}_\lambda$ maps the coefficient ball of radius $r_I/2$ into itself and contracts differences by the factor $1/2$. Iterating from zero proves existence and convergence. The sum of the successive differences is a geometric series in the coefficient norm, so the limit is holomorphic. The differentiated series converge on a fixed smaller domain: they satisfy an equation $Z = H + TZ$ in which the majorant of T is at most $1/2$, and the solution is $\sum_{j \ge 0} T^j H$. This gives the coefficient bounds with any parameter index fixed.

If two stationary points lie in the convex domain on which the derivative bound holds, their difference is unchanged by (19.12) and has norm at most half of itself, so the two points coincide. Since F_λ is real at real arguments, every Picard iterate is real at real data, and so is the limit. The margin on the real Hessian keeps at least half of the lower bound for the constrained Hessian on a neighborhood of the stationary point. Integrating the real Hessian along segments shows a positive quadratic increase away from it, so it is a local real minimum. The complex resolvent condition is a hypothesis, which provides the inverse square roots used below.

Step 3: the displacement at fixed coarser coordinates. For part (3), note that $F_{0,I}(c, a_I, y) = 0$, so the map (19.12) satisfies $\mathcal{T}_\lambda(0) = -\lambda A_I^{-1} V_I(c, a_I, y)$. Writing $Y_{\lambda,I} = a_{\lambda,I} - a_I$ in the fixed point equation, we obtain for every $y \in \mathcal{T}_I$

$$\|Y_{\lambda,I}\|_{\text{coeff}} \le \lambda \delta_V + \frac{1}{2} \|Y_{\lambda,I}\|_{\text{coeff}}, \qquad \text{hence} \qquad \|Y_{\lambda,I}\|_{\text{coeff}} \le 2\lambda \delta_V. \tag{19.13}$$

This estimate is uniform on the whole domain $\mathcal{T}_I$, with the common product of coarse fields. In particular it can be used after moving coarser coordinates have been substituted; only their range enters, not a derivative in the frozen coordinates.

Step 4: the Lipschitz bound in the coarser coordinates. By the coefficient bounds on the larger domains, the mixed matrices F_{λ,II^c} are bounded by a common constant Q_{mix} and the reference inverse by I_0. The second inequality of part (2) and the Neumann series imply that $\|F_{\lambda,II}^{-1}\| \le 2I_0$. Implicit differentiation at the perturbed conditional stationary point yields

$$D_y a_{\lambda,I} = -F_{\lambda,II}^{-1} F_{\lambda,II^c}, \qquad \|D_y a_{\lambda,I}\|_{\mathrm{coeff}} \le 2I_0 Q_{\mathrm{mix}}. \tag{19.14}$$

The rooted versions follow from the same ordered products on a fixed smaller domain. From now on let C_0 denote a bound for these maps and their unperturbed versions, and let C_S bound the rows of the inverse square roots. These constants depend on the common bound for the mixed tensors, not on δ_V or on the later free radius.

Step 5: successive substitution. Let $\varepsilon_{\mathrm{tube}}$ be as in part (3). We use the successive maps in the original free coordinates, with $K_{\lambda,i}$ equal to M^i times the Schur blocks in input units. Let A_i bound all deviations of the coarser coordinates and all displacements of intermediate solutions at scale i, relative to the unperturbed joint stationary point. For the initial segment of all free coordinates, (19.13) bounds the perturbed joint displacement by $2\delta_V$. At each finer scale we first compare the perturbed and the unperturbed conditional stationary points at the same coarser argument, using (19.13), and then change the coarser argument, using (19.14). Inserting the free increment adds at most $C_S r_q$. We obtain the recurrence

$$A_{i-1} \le (1 + C_0) A_i + C_S r_q + 2\delta_V, \qquad A_s \le 2\delta_V, \tag{19.15}$$

which also holds for the images under the majorants, since it is obtained by composing majorants. After the finite conversion between the original free coordinates and the coordinates centered at the joint stationary point is included in C_0 and C_S, we may take

$$C_{\mathrm{seq},J} = 2(1 + C_S)(1 + C_0)^{J+1}\Big(1 + \sum_{j=0}^{J-1} (1 + C_0)^j\Big).$$

Indeed, iterating (19.15), we find

$$A_i \le 2\delta_V (1 + C_0)^J + (C_S r_q + 2\delta_V) \sum_{j=0}^{J-1} (1 + C_0)^j,$$

which is at most $C_{\mathrm{seq},J}(2\delta_V + r_q)$. The condition of part (3) therefore leaves half of each margin unused. The same bound applies to all Picard reference points and solution balls used in the construction from the coarsest scale downward. In particular, no gradient is evaluated outside the domains already reached.

Step 6: zero increments. At $q = 0$, the stationarity at the coarsest scale and the successive stationarities at the finer scales give the joint gradient equation. By uniqueness of its branch, this chain is the joint stationary point; Step 5 shows that the chain stays in the required domains.

Step 7: the order of choices. For $\delta_V = 0$ we first choose r_q by the condition of part (3). On the fixed larger unperturbed products, we then choose the perturbed product and the bound on the perturbation so that the same condition holds, and all later radii are chosen inside that product. Decreasing r_q changes only the terms of positive degree in q, and the image at $q = 0$ is the joint stationary point for every coarse field in the domain; hence no later reduction of the coarse-field radius follows. Finite minima and maxima over the initial segments preserve the dependence of the constants. □

19.3 Comparison of the regional and torus stationary points

We now compare the regional problem with the torus problem at a nonzero coarse field. Recall that $F_R(c, z; \xi) = E_{d_0}(\Psi_R(c, z; \xi))$ is the regional objective of Theorem 18.12, and that the torus objective is the effective classical action of the torus reconstruction with the coarse field extended by zero. For a target v of the torus we write $z_T(v)$ for the torus stationary point at v, and P for the restriction from the torus free list to the regional list.

Theorem 19.4. *Fix the larger domains of Theorem 18.12, an odd $M \geq 3$, $J \geq 1$, $1 \leq s \leq J$, $m \geq 4$, a depth $d_0 \geq 0$, and a union Ω of output blocks; positions are in output units. Use a smaller fixed product of $(c, z; \xi)$ on which the gap, inverse, and image margins of that theorem hold, with the previously frozen coordinates fixed. Write*

$$F_R = E_{d_0} \circ \Psi_R(c, z; \xi), \qquad F_T = E_{d_0} \circ \Psi_T(c \oplus 0, z_L, z_H), \qquad (p, h) = z_T(c \oplus 0),$$

put

$$a_T^0 = \Psi_T(c \oplus 0, p, h), \qquad \widehat{a}_R = \Psi_R(c, p; \xi), \qquad d_a = \widehat{a}_R - a_T^0,$$
$$J_R = \Psi_{R,z}(c, p; \xi), \qquad J_T = \Psi_{T,z_L}(c \oplus 0, p, h),$$

and let $r_R = F_{R,z}(c, p; \xi)$ be the regional gradient at the restricted torus stationary point. Then the following statements hold.

1. *(Gradient and response.) Let z_R be the regional stationary point. Then*

$$r_R = (J_R - J_T)^{\mathsf{T}} E'_{d_0}(a_T^0) + J_R^{\mathsf{T}} \int_0^1 E''_{d_0}(a_T^0 + u d_a) d_a \, du,$$
$$\mathcal{A}_R(z_R - p) = -r_R, \qquad \mathcal{A}_R = \int_0^1 F_{R,zz}(c, p + u(z_R - p); \xi) \, du.$$

The first term of r_R is the chain-rule contribution of the coarse constraints omitted in the regional problem. With $a_R = \Psi_R(c, z_R; \xi)$, for every free label or fine link i in Ω,

$$|(r_R)_i| + |(z_R - p)_i| + |(a_R - a_T^0)_i| \leq C_J(\|c\|_\infty + \|\xi\|_\infty) e^{-c_J \operatorname{dist}(i, W)},$$

each summand being read on its own list. These three differences have factorial boundary-localized coefficient bounds, as in Lemma 18.15. In particular, for a fixed input x and a fixed output e in Ω, their differentiated coefficient series are bounded by $C_J e^{-c_J[\operatorname{dist}(e,W)+\operatorname{dist}(e,x)]}$.

2. *(Full output.) Let $u_*(c,\xi) = T_M^s \exp a_R$ be the full output of the s averaging steps, including the values of the coarse links which are not constrained in Ω. It lies in the common torus target chart on this product. The bounds of part (1) on Ω, including those with a fixed input, remain true if $c \oplus 0$ is replaced by $\operatorname{Log} u_*$ and a_T^0 by the torus stationary field a_T^* at the target $\operatorname{Log} u_*$. All derivatives through u_*, including those through z_R, are included.*
3. *(Minimized action.) Let $V_R(c;\xi) = B_s F_R(c, z_R; \xi)$. There are fixed $\rho_0(J), a_W, \alpha_W, A_J^{\mathrm{cl}} > 0$ such that*
$$\mathcal{X}^{\rho_0,\mathrm{cl}}_{\Omega,a_W,\alpha_W}\big(V_R - E_{d_0+s}(c \oplus 0)\big) \le A_J^{\mathrm{cl}}.$$
This is the seminorm of Definition 18.1 with the displayed constant classical radius and rates; it includes all degrees $l \ge 1$. The constant term of the difference is kept in the full action.
4. *(Newly frozen coordinates.) For the newly frozen coordinates ξ, at their positions and with their density,*
$$D_\xi z_R = -F_{R,zz}^{-1} F_{R,z\xi}, \qquad \sum_f |(D_{\xi_f} z_R)_e| \le C_J e^{-a_W \operatorname{dist}(e,W)},$$
$$\|F_{R,z\xi}\|_{a_W} + \|F_{R,zc}\|_{a_W} \le C_J.$$
5. *(Hessians.) The constrained Hessians at the two stationary points satisfy*
$$\big|\big[F_{R,zz}(c, z_R; \xi) - P F_{T,zz}(c \oplus 0, z_T) P^{\mathsf{T}}\big]_{ef}\big| \le C_J e^{-a_W[\operatorname{dist}(e,f)+\operatorname{dist}(e,W)+\operatorname{dist}(f,W)]}.$$
The order-l coefficient tensors of this difference, with every coarse-field and free argument in the connecting network and any root in Ω fixed, are bounded by $(l+2)! C_J^{l+2}$ at fixed exponents.
6. *(Successive minima, inverse square roots, and Haar disintegration densities.) The statements of part (5) hold for each conditional partial minimum, on a smaller product of the coarser coordinates about the joint stationary point, with the exact normalization M^i at scale i. The inverse square roots of these Hessians and the nonconstant coefficients of the logarithms of the Haar disintegration densities satisfy the corresponding boundary-localized comparisons, with any index fixed. Scalar determinants and Haar disintegration densities which are not rooted at a label in Ω are kept as exact quantities.*

All constants have the dependence of Section 19.1 and do not depend on the later widths.

The theorem gives no inverse on fixed boundary coordinates and no rank estimate for the boundary terms at a nonzero coarse field.

Proof. We pose both problems on the torus free list by interpolating between the torus objective and the augmented regional objective. The two objectives then differ by one

inserted factor localized near H, and all estimates follow from the response of the interpolated stationary point to this factor and from the exponential decay of the inverse Hessian.

Step 1: one coordinate list. We write $z = (z_L, z_H)$ for the torus free list and set

$$\widetilde{F}_R(c, z; \xi) = F_R(c, z_L; \xi) + \frac{1}{2}|z_H|^2, \qquad F_t = (1-t)F_T + t\widetilde{F}_R, \quad 0 \le t \le 1, \qquad D = \widetilde{F}_R - F_T. \tag{19.16}$$

Since its minimum is zero, the added auxiliary quadratic does not change the regional value. Theorem 18.12 gives a positive gap at the identity for the Hessian of this interpolation, for all its principal blocks, and for its Schur complements. We reduce the construction product by a fixed amount so that the coefficient variation of each of these Hessians satisfies the condition of Lemma 19.1(2), uniformly in t. This is possible since the variation is at most $Q_J(\|c\| + \|z\| + \|\xi\|)$, and the gap at the identity of the interpolation is at least the minimum of the regional gap, the torus gap, and 1. All straight segments of fine fields used below then lie strictly inside the larger classical chart.

We retain a split resolvent margin for the later perturbation argument. Choose a common intrinsic exponent $b_{\mathrm{ref}} > 0$ and a constant $C_{0,J}$ such that every flat principal and scale Schur reference K_i^b has resolvent norm at most $C_{0,J}/(\lambda_J + t)$ at this exponent, including the factors M^i at the scale stages. When the larger unperturbed products are fixed, reduce them once so that their conditional matrices $K_{0,i}$ satisfy

$$C_{0,J}\|K_{0,i} - K_i^b\|_{b_{\mathrm{ref}}} \le \lambda_J/8$$

on those products, uniformly over the finitely many stages. We decrease the unperturbed reserve c_J if necessary so that $c_J \le b_{\mathrm{ref}}$. These choices use only the unperturbed constructions at fixed J and precede the incoming rates, perturbation threshold, and later widths.

Step 2: localization of the difference of the reconstructions. We use the comparison of the nonlinear words in Theorem 18.12. At common internal coarse values and common regional free values, we set

$$d(c, z; \xi) = \Psi_R(c, z_L; \xi) - \Psi_T(c \oplus 0, z_L, z_H).$$

On fine links in Ω, its defining words vanish outside the fixed interface band. The prescribed values on H can differ, and their difference is a bounded source on H with the separate coefficient bounds of the reconstruction charts. The difference is supported in H together with that band, hence in H_{L_J}; the two extensions need not agree on H. Its coefficient tensors have the corresponding finite-range bounds, rooted at any index. To check the statement about fine links in Ω at nonzero arguments, consider an internal eliminated-edge equation at the coarsest intermediate scale. On the two sides

it has the same endpoint blocks, the same coarse value, and the same free values, so by uniqueness on the common local branch its solution is the same. We repeat this down the s intermediate scales. Only equations meeting a block of H can differ, and the union of their at most J stencils is contained in H_{L_J}. This argument keeps the values of the coarse links across the boundary of the block region as functions of the common free values; it imposes no value on an omitted coarse constraint.

Step 3: the inserted difference. The difference in (19.16) is

$$D = \int_0^1 E'_{d_0}(\Psi_T + vd)[d]\, dv + \frac{1}{2}|z_H|^2. \tag{19.17}$$

Every derivative of the integral either differentiates d, whose fine position stays in H_{L_J}, or leaves such a factor undifferentiated. In the latter case the classical tensor of Theorem 10.11, which is bounded at every root, connects that position to every differentiated chart argument. A derivative in a coarse field or in a free variable in Ω therefore cannot disconnect the difference factor from the network. The auxiliary quadratic has only indices in H, and any connected contribution from it to an external index in Ω passes through an index in H and is bounded by (19.2). This verifies the hypotheses of Lemma 18.15 for the whole inserted difference, at every order and with any external root.

Step 4: the gradient formula and the bounds on values. Torus stationarity says that $J_T^{\mathsf{T}} E'_{d_0}(a_T^0) = 0$. Adding and subtracting $J_R^{\mathsf{T}} E'_{d_0}(a_T^0)$ in $F_{R,z}(c, p; \xi) = J_R^{\mathsf{T}} E'_{d_0}(\widehat{a}_R)$, and applying the fundamental theorem of calculus on the complex segment from a_T^0 to $\widehat{a}_R$, we obtain the formula for r_R. The difference $J_R - J_T$ is supported in the construction band, in its regional input indices and its fine output indices. Its product with the gradient of the action on the fine links is kept: torus stationarity makes only the projection of this gradient onto the torus free directions vanish. By the classical bounds, we have

$$\|E'_{d_0}(a_T^0)\|_\infty + \|d_a\|_\infty \le C_J(\|c\|_\infty + \|\xi\|_\infty);$$

for the first summand we used the bound on the torus stationary point, and for the second the word comparison of Step 2 and the same bound for (p, h). Contracting the two terms of r_R with their localized sources and using (19.2), we obtain

$$|(r_R)_i| \le C_J(\|c\|_\infty + \|\xi\|_\infty) e^{-c_J \delta(i)}. \tag{19.18}$$

Subtracting the regional gradient equations at z_R and at p and integrating along the segment between them, we obtain $\mathcal{A}_R(z_R - p) = -r_R$. The segment lies in the convex larger free box, and its averaged Hessian is the positive regional Hessian at the identity plus a coefficient perturbation satisfying the condition of Lemma 19.1(2). By that lemma, the inverse has entries bounded by $C_J e^{-b_J \operatorname{dist}(e,i)}$. Convolving this bound with (19.18)

and using $\delta(e) \le \delta(i) + \operatorname{dist}(e,i) + L'_J$, we obtain the bound on $z_R - p$, at a fixed smaller exponent. For the fine field we have the identity

$$a_R - a_T^0 = d_a + \int_0^1 \Psi_{R,z}(c, p + v(z_R - p); \xi)(z_R - p)\, dv. \tag{19.19}$$

The first term is supported in H_{L_J}, and the second is the product of a finite-range chart kernel, without boundary weight, with $z_R - p$. This proves the bound on fine links in Ω. No decay is asserted at a fine link deep in H.

Step 5: the coefficient bounds. We now prove the coefficient bounds directly, using the interpolation (19.16). On a common smaller product of parameters, the stationary point z_t of F_t is obtained by the implicit iteration with the inverse of the free Hessian at the identity, which converges coefficientwise. The radius condition and the bound on the inverse times the gradient follow from Theorem 18.12, and the scalar comparison equation is (19.3). Differentiating in t, we obtain

$$\dot z_t = -F_{t,zz}^{-1} D_z(c, z_t; \xi). \tag{19.20}$$

This expression contains exactly one inserted difference from (19.17), and its convergent majorant is (19.4). A further derivative in an input may hit the inverse, the inserted difference, or its argument z_t; in each case the joined network still contains the inserted difference and the index of the new input. We apply Lemma 18.15 and integrate from $t = 0$ to $t = 1$. Since the endpoints are z_T and $(z_R, 0)$, their difference on the regional list has factorial boundary-localized coefficient bounds, rooted at every index. Reading (19.19) as an identity of convergent majorant series, we obtain the corresponding bounds for the fine field, and the same argument applied to the formula for r_R gives its bounds. Keeping one input x fixed and summing over the remaining indices, we obtain the decay with a fixed input stated in part (1).

Step 6: the full output. The composition $T_M^s \exp \Psi_R$ is holomorphic on the common larger chart, by the construction in Theorem 18.12. At the stationary point it satisfies

$$\| \operatorname{Log} u_* \|_\infty \le C_J(\|c\|_\infty + \|\xi\|_\infty).$$

The internal coordinates of $\operatorname{Log} u_*$ are c, and its other coordinates are the values of the coarse links which are not constrained in Ω. These are analytic functions of $(c, z; \xi)$, whose differentiated series compose the coefficient series of the chart words with the series of the stationary point from Step 5. For instance, for any analytic direction v the first derivative is

$$D_v \operatorname{Log} u_* = D_c \mathcal{T}\, D_v c + D_z \mathcal{T}\, D_v z_R + D_\xi \mathcal{T}\, D_v \xi, \qquad \mathcal{T} = \operatorname{Log}(T_M^s \exp \Psi_R).$$

Since higher derivatives are compositions of these factors, all derivatives through the stationary point are included. We reduce the parameter radius so that the segment from $\operatorname{Log} u_*$ to $c \oplus 0$ lies in the common torus target chart. If $\mathcal{A}_T(v)$ denotes the torus stationary fine field at the target v, then

$$a_T^* - a_T^0 = \int_0^1 D\mathcal{A}_T(c \oplus 0 + t(\operatorname{Log} u_* - (c \oplus 0)))[\operatorname{Log} u_* - (c \oplus 0)]\, dt. \tag{19.21}$$

The difference of targets is supported on coarse links which are not internal to Ω. By the same implicit equation on the torus list, the lower bound for the torus constrained Hessian, the chart bounds, and the classical tensors give coefficient bounds without boundary weight for the response of $\mathcal{A}_T$. Each term of (19.21) with a root in Ω therefore passes through a coarse link outside Ω, and its composed network, including the derivatives of u_*, visits H_{L_J}. Lemma 18.15 and the finite word bounds then give the same amplitude and fixed-input bounds as before. The same integral with $\mathcal{A}_T$ replaced by the torus free stationary point z_T bounds $p^* - p$ on the regional free labels, where $(p^*, h^*) = z_T(\operatorname{Log} u_*)$. Thus $z_R - p^* = (z_R - p) - (p^* - p)$ satisfies the same bounds. For the new residual gradient, we subtract its value at p and integrate $F_{R,zz}$ along the segment from p to p^*. A Hessian without boundary weight multiplies this boundary-localized displacement, so the residual satisfies the same bounds. Its formula follows again from torus stationarity, now at the fixed target $\operatorname{Log} u_*$. These arguments are identities of convergent series in all variables, so they also give the fixed-input bounds at the full output. A representative in complementary forest coordinates, if needed, can use forest words of fixed depth based at the regional vertices and the coarse roots. These words change only the construction stencil outside Ω, their coefficient series are among the chart bounds, and the comparison in Ω is unchanged.

Step 7: the minimized action. Let $f_t(c;\xi) = F_t(c, z_t; \xi)$. By stationarity and Lemma 18.8, we have

$$\partial_t f_t = D(c, z_t; \xi). \tag{19.22}$$

At the endpoints, $B_s f_1 = V_R$ and $B_s f_0 = E_{d_0+s}(c \oplus 0)$. The second identity follows by iterating the recurrence $E_{d+1}(c) = M \min_z E_d(b(c,z))$ which defines the effective classical action: by Lemma 18.8 the successive minima equal the joint minimum, and the product of the s factors M is B_s. A nonconstant derivative of the right side of (19.22) has the network structure checked in Step 3. Therefore, for $k \geq 1$ and a root i in Ω,

$$\sum_{i_2,\dots,i_k} e^{c_J \mathfrak{t}(i,i_2,\dots,i_k)} |D^k[B_s(f_1 - f_0)](0;\xi)_{i,i_2,\dots,i_k}| \leq k! C_J^{k+1} e^{-c_J \delta(i)}, \tag{19.23}$$

where the factor $B_s \leq M^J$ is included in C_J. For a scalar series $f(c) = \sum_\nu f_\nu c^\nu$ we have, as in (18.2),

$$\sum_{|\nu|=k} \frac{\nu_i}{k} |f_\nu| e^{a\mathfrak{t}(\nu)} = \frac{1}{k!} \sum_{i_2,\dots,i_k} |D^k f(0)_{i,i_2,\dots,i_k}| e^{a\mathfrak{t}(i,i_2,\dots,i_k)}. \tag{19.24}$$

Indeed, the multiset obtained from ν by removing one occurrence of i has $(k-1)!/[(\nu_i - 1)!\prod_{j\neq i}\nu_j!]$ ordered listings, and $D^k f(0)$ evaluated on any of them is $\nu! f_\nu$. Fix $a_W, \alpha_W < c_J/4$, and then $\rho_0 > 0$ in the smaller construction domain with $C_J\rho_0 < 1/4$. Inserting (19.23) into (19.24), we bound the regional seminorm by

$$\sum_{k\geq 1} \rho_0^{k-2} C_J^{k+1} < \infty.$$

This proves part (3), including degree one. The value $B_s(f_1 - f_0)(0;\xi)$ is never subtracted in (19.22); it remains in the full action.

Step 8: derivatives in the newly frozen coordinates. Differentiating the regional gradient equation in ξ and using invertibility of the Hessian, we obtain $D_\xi z_R = -F_{R,zz}^{-1} F_{R,z\xi}$. The mixed derivative has exactly two terms,

$$F_{R,uv} = \Psi_{R,u}^{\mathsf{T}} E''_{d_0}(\Psi_R)\Psi_{R,v} + E'_{d_0}(\Psi_R)\Psi_{R,uv}, \tag{19.25}$$

and the bounds of Theorem 18.12 on the larger products control both terms for $(u,v) = (z,\xi)$ and $(u,v) = (z,c)$. An index of ξ is at a fine position outside Ω or in its fixed interface stencil. Contracting it with the free inverse and summing over the labels of ξ, we obtain the row sum $C_J e^{-a_W\delta(e)}$ by (19.2). This proves both bounds on the mixed matrices and the bound on $D_\xi z_R$, with no change of the density of the inputs.

Step 9: the Hessians. We compare the endpoints of (19.16) at their stationary points. Before any elimination, the total derivative is

$$\frac{d}{dt} F_{t,zz}(c, z_t; \xi) = D_{zz}(c, z_t; \xi) + F_t^{(3)}(c, z_t; \xi)[\dot z_t]. \tag{19.26}$$

Both terms contain the inserted difference, by (19.17) and (19.20), and all terms of (19.25), including the one with the gradient of the action, are part of these tensors. For two matrix roots in Ω, Lemma 18.15 gives the three distances in the Hessian bound of part (5), and at degree l in additional coarse-field and free arguments it gives the bound $(l+2)!C_J^{l+2}$ after C_J is increased. For example, the corresponding coefficient series of the matrix is dominated on radius R by

$$\sum_{l\geq 0} \frac{(l+2)!}{l!} C_J^{l+2} R^l = \frac{2C_J^2}{(1-C_J R)^3}, \qquad C_J R < 1.$$

The same majorant with a fixed input gives every version rooted at an input.

Step 10: successive minima. We restrict to smaller products for the initial segments before this comparison is used, following the construction of Lemma 19.3(1) uniformly in t. For an initial segment I, we take a solution radius r_I about the joint stationary point

and a smaller radius η_I for the deviation of the coarser coordinates. The principal inverse has bound I_J and the mixed tensors have bound C_J, so at the joint stationary point the inverse times the conditional gradient is at most $I_J C_J \eta_I$. We require $I_J C_J r_I < 1/8$ and $I_J C_J \eta_I < r_I/8$ within the margin of the chart, uniformly in t. Then the conditional Newton map is a contraction on the solution domain. We reduce the radii of the coarse field and of ξ so that $\sup_t \|z_t(c;\xi)\|_{\mathrm{coeff}} < \min_I \eta_I/8$. Then a common product $\|y\|_{\mathrm{coeff}} < \eta_I/2$ of coarser coordinates lies in the allowed domain for every t. On this product the conditional response is $\dot z_I = -F_{t,II}^{-1} D_I$ at fixed y, so the argument of Step 5 applies before a partial minimum is taken. Since there are finitely many initial segments, a common choice is possible, and this is the smaller product in part (6). At such a partial minimum, Lemma 18.8 gives

$$f_{yy} = F_{yy} - F_{yI} F_{II}^{-1} F_{Iy}, \qquad D^3 f[\eta_1, \eta_2, \eta_3] = D^3 F[V_{\eta_1}, V_{\eta_2}, V_{\eta_3}].$$

Differentiating the Schur formula in t replaces one factor by its derivative, which contains the inserted difference; an inverse contributes $-F_{II}^{-1} \dot F_{II} F_{II}^{-1}$, and evaluation at a moving stationary point contributes its response, which also contains the inserted difference. Each resulting network keeps all external roots and a visit to H_{L_J}. At scale i the formula is multiplied by the normalization M^i of the action, and a finite composition over $i < J$ changes only C_J.

Step 11: inverse square roots, and fixing versus eliminating. We apply (19.7) to the Hessians of Step 10. By the common gap and the resolvent condition, the differences of the inverse square roots have integrable boundary-localized coefficient bounds, rooted at every index. To compare $(K_{LL}^T)^{-1/2}$ with $((K^T)^{-1/2})_{LL}$, we set $C = K_{LL}^T \oplus K_{HH}^T$ and apply the resolvent identity to $K^T - C$ before integrating the resolvents. Each off-diagonal factor connects the two sides of the boundary, and hence gives the same boundary-localized estimate rooted in Ω. This also treats the difference between eliminating and fixing the torus free coordinates outside the regional list. For a free matrix T written in blocks with respect to the regional and the remaining torus coordinates, (19.9) expresses $(T^{-1})_{LL} - A^{-1}$ through the Schur complement $D_0 - CA^{-1}B$ on the remaining coordinates, which has the gap of Theorem 18.12. The factors B and C pass through the remaining coordinates and keep the visit to H_{L_J}, also after T is replaced by $T + t$. The same identity inside a larger Schur formula for one scale treats elimination instead of fixing.

Step 12: logarithms of the Haar disintegration densities. In the fixed coordinates, the Haar disintegration density of one intermediate scale is that of Lemma 18.7:

$$c_{\mathrm{Haar}}^f \frac{\prod_{e\ \mathrm{nonforest}} j_H(\Gamma_e)}{\prod_c \det D_{X_{e_c^*}} \mathsf{C}_{b,c}(\Gamma)}, \qquad j_H(X) = \mathrm{sinc}^2(|X|/2), \tag{19.27}$$

where e_c^* is the eliminated edge of the constraint at the coarse link c; the Haar factors of the solved eliminated edges are included in the numerator. At the common zero

the local factors of the numerator equal 1 and the local constraint determinants equal M^{-6}. By the word coefficients of fixed depth and the bounds on the implicit chart, there exists a uniform smaller radius on which each factor, divided by its nonzero value at zero, has nonconstant part with absolute coefficient row sum less than $1/2$. The series $\log(1+x) = \sum_{k\geq 1}(-1)^{k+1}x^k/k$ therefore converges absolutely, with the attached local networks. We compose these logarithms with the conditional stationary points and the inverse square roots. The words for fine links in Ω agree off the interface band, and the prescribed values on H remain bounded sources. Thus every difference has an output in H_{L_J}, and the derivatives of the other factors stay joined to it. Logarithmic matrix factors are treated by (19.8). A nonconstant scalar logarithmic coefficient has a differentiated index in Ω at which its trace cycle can be rooted, so the rooted sums above bound it. Pure scalar determinants, Haar disintegration densities, and traces over H have no such index and are kept as their exact finite values. All choices involved only the fixed construction data of Section 19.1, which proves the stated dependence of the constants. □

19.4 A gauge transformation fixing prescribed vertices

On the torus, Theorem 13.2 brings the reconstructed field into a small neighborhood of the identity by a gauge transformation which depends only on the coarse field. On a block region this gauge transformation must not change the field at the vertices to which frozen coordinates are attached. A frozen coordinate is stored as its based transport $U(\sigma_e)a_e$: the transport along a path σ_e from its basepoint to a link e, times the frozen value a_e of that link (Lemma 18.7). Under a gauge transformation G the frozen value transforms as $a_e^G = G_v a_e G_w^{-1}$, where v, w are the endpoints of e. We therefore multiply the logarithm of the torus gauge transformation by a cutoff which vanishes near H. To avoid a clash with the block region Ω, we write $\Omega^{\mathbb{T}}_{d_0,s}$ for the gauge transformation of Theorem 13.2, applied to a composition of s averaging steps at depth d_0.

Lemma 19.5. *Fix M, J, and the geometry of a composition of $s \leq J$ averaging steps as above. Let the prescribed protected set contain all vertices of H, every fine endpoint and every basepoint of each based transport of a newly or previously frozen coordinate, and all endpoints required in finer-lattice representations of these transports. Assume that its protected vertices outside H lie in the fixed interface band $\mathcal{B}$, in output units. Let $0 \leq \chi \leq 1$ be fixed by the geometry, zero on H and its neighborhood of width two blocks, one outside its neighborhood of width three blocks, and zero at every prescribed protected vertex. Define*

$$k(c) = \operatorname{Log} \Omega^{\mathbb{T}}_{d_0,s}(c \oplus 0), \qquad G^{\Omega}(c) = \exp(\chi k(c)).$$

There are $u_J, A_J, a_G > 0$ such that the following statements hold on $\|c\|_\infty < u_J$.

1. *The maps k and G^{Ω} are holomorphic, periodic, and equivariant under constant global color rotations. At real data they take values in $\mathfrak{su}(2)$ and $\mathrm{SU}(2)$, respectively, and $\|k(c)\|_\infty \leq$*

$A_J\|c\|_\infty$. Their order-l absolute coefficients are bounded by $l!C_J^{l+1}$, with the connecting-network weight at exponent a_G and with the volume-normalized column of every fixed input. Replacing the volume normalization of a column for a fine output by counting measure costs at most the factor $M^{3s} \le M^{3J}$.

2. *At fixed geometry G^Ω depends only on c, not on the free coordinates or the frozen coordinates. It is exactly I at $c = 0$, at the coarse roots, and at all protected vertices, also for complex c.*
3. *Consequently, for every reconstruction U in the chart before gauge transformation,*

$$(T_M^s U^{G^\Omega(c)})_{\text{internal}} = (T_M^s U)_{\text{internal}}, \qquad (U^{G^\Omega(c)})(\sigma_e) a_e^{G^\Omega(c)} = U(\sigma_e) a_e$$

for every based transport of a frozen coordinate. A frozen link across the boundary of the block region with both fine endpoints protected is also unchanged.

4. *The gauge transformation is applied after the reconstruction in the rooted spanning-forest coordinates. It does not change the free coordinates, the support restrictions, the Haar disintegration density, or the determinant of the normalization of the fluctuation variables.*

The constants have the dependence of Section 19.1 and do not depend on the later widths or on the size of the protected set. The existence of a large-field decomposition whose protected vertices satisfy these hypotheses is a geometric hypothesis, not a conclusion of the lemma.

Proof. We take the torus gauge transformation of Theorem 13.2 for each $s \le J$, with a common smaller domain of targets. Extending the target by zero on the coarse links outside Ω deletes coefficients from its majorant. After a construction constant A is increased, the gauge transformation minus the identity has absolute row series at most

$$U(t) = \frac{At}{1 - At};$$

here and below this includes the connecting network and the versions with a fixed input. Theorem 13.2 also provides the logarithm of the gauge transformation. To check the domain condition, we fix $u_J \le U_T$ inside the larger target domain with $Au_J < 1/4$. This is the case $Q_T = 0$ of (19.1), and it gives $u_J < r_s$, a positive denominator, and the fixed intermediate chart margins for every $s \le J$. The convergent matrix logarithm is bounded by

$$\sum_{r \ge 1} \frac{U(t)^r}{r} = \log(1 - At) - \log(1 - 2At) \le \frac{At}{1 - 2At} \le 2At \qquad (0 \le t \le u_J). \tag{19.28}$$

In the first inequality, we wrote the difference of logarithms as $\log(1 + At/(1 - 2At))$ and used $\log(1 + x) \le x$ for $x \ge 0$. Products join coefficient networks. Fixing one input differentiates this majorant series, and on a further fixed smaller radius we obtain order-l tensors bounded by $l!C_J^{l+1}$, for an output row or for the volume-normalized column of

any input. The same bounds hold for $k(c)$, and in particular $\|k(c)\|_\infty \le A_J \|c\|_\infty$. The fine mesh in output units is M^{-s}, so the volume-normalized column of a fine output carries the factor M^{-3s}. Replacing it by counting measure multiplies the bound by the reciprocal, at most M^{3J}; no factor depending on the depth or the volume occurs.

At fixed geometry, multiplication by $\chi(v) \in [0,1]$ can only decrease an absolute coefficient row. With $L(t) = \sum_{r\ge1} U(t)^r/r$, exponentiating χk gives the coefficientwise majorant $\exp L(t) = (1-At)/(1-2At)$. This series and its fixed-input derivative series converge on a fixed smaller radius and give the asserted factorial coefficient bounds for G^Ω. The numerical bound $L(t) \le 2At$ in (19.28) bounds values only and is not used in this coefficient comparison. Periodicity and global color equivariance hold for the torus gauge transformation and its logarithm by Theorem 13.2, and multiplication by the fixed scalar cutoff and exponentiation preserve them. For real targets the logarithm of the compact gauge transformation lies in $\mathfrak{su}(2)$, and its exponential lies in $\mathrm{SU}(2)$; the complex functions are their holomorphic continuations with values in $\mathfrak{sl}_2(\mathbb{C})$ and $\mathrm{SL}(2,\mathbb{C})$.

The torus gauge transformation is I at $c = 0$ and at each coarse root, so its logarithm vanishes at these points. At every other protected vertex $\chi = 0$ by hypothesis. Hence $G^\Omega = I$ at all these points, identically as a holomorphic function and not only at real arguments.

The transport along a path σ from v to w transforms as

$$U^G(\sigma) = G_v U(\sigma) G_w^{-1},$$

since all intermediate vertex factors cancel. A based transport $U(\sigma_e)a_e$ therefore transforms only at its basepoint and at the far fine endpoint of e. Both values of the gauge transformation there are I, which implies the second identity in part (3). If both fine endpoints of a frozen link across the boundary of the block region are protected, the gauge transformation also leaves its value unchanged.

For the coarse output, every based loop holonomy in the centered average (4.3) transforms by conjugation at its initial coarse root, and its reference path transforms at its two coarse endpoints. On the branch of the logarithm used here, the logarithm commutes with conjugation, term by term in its power series. Thus one averaging step is covariant, with the gauge transformation acting only at the coarse endpoints. Iterating s steps cancels the intermediate gauge factors and leaves only the gauge transformation at the two coarse roots of the output link, where it is I. This proves the first identity in part (3), also at complex data. No exceptional value of the Borel extension of T_M occurs inside these charts.

Finally, the gauge transformation acts on a field already reconstructed from its free coordinates in the rooted spanning-forest coordinates, and it changes neither these coordinates nor the integration variables. Hence the support restrictions, the Haar disintegration density, and the determinant of the normalization of the fluctuation variables remain the

same functions of the original variables. The cutoff enters only as a bounded fixed multiplier; neither the size of its support nor a derivative in its geometric parameters enters any bound. This proves that the constants do not depend on the size of the protected set or on the later widths, under the stated hypothesis on the protected vertices. □

19.5 The gauge-transformed reconstruction and its input columns

We now apply $G^{\Omega}(c)$ to the regional reconstruction and compare the result with the gauge-transformed torus reconstruction of Theorem 13.2. Let $a_R^{\mathrm{raw}}(c,q;\xi)$ be the logarithm of the regional reconstruction before gauge transformation, built from the successive regional conditional stationary points and constrained inverse square roots, and let a_T^{raw} be the corresponding torus reconstruction. For a fine link $e=(v,w)$ and a logarithmic gauge field k, we write

$$H_e^{\chi}(a,k) = \mathrm{Log}\big(e^{\chi(v)k(v)}e^{a_e}e^{-\chi(w)k(w)}\big).$$

Then $\Phi_R = H^{\chi}(a_R^{\mathrm{raw}},k(c))$ is the logarithm of the regional reconstruction transformed by $G^{\Omega}(c)$, and $\Phi_{\mathbb{T}} = H^{1}(a_T^{\mathrm{raw}},k(c))$ is the gauge-transformed torus reconstruction of Theorem 13.2.

Theorem 19.6. *On the fixed products of Theorem 19.4 and Lemma 19.5, build the reconstruction from the free coordinates with the successive regional conditional stationary points and constrained inverse square roots, with small newly frozen coordinates ξ. There are fixed larger construction radii and positive rates such that, on every smaller product of coarse fields and fluctuation increments satisfying the domain conditions,*

$$\Phi_R(c,q;\xi) = \Phi_{\mathbb{T}}(c\oplus 0, q\oplus 0)|_{\Omega} + \Delta(c,q;\xi),$$

where $q\oplus 0$ uses the same scale indices on the regional list and sets the remaining torus increments to zero. The following statements also hold.

1. *All terms have convergent majorant series in all variables. The correction Δ has factorial coefficient bounds with a root in Ω, a connecting network through every input, and a visit to H_{L_J}. In particular, let $\boldsymbol{R}$ be a radius profile on a smaller construction product, let $a>0$ be a fixed network exponent, and let $\Delta_{e,\nu}$ be the Taylor coefficients of Δ_e in its analytic variables, namely the coarse field, the increments q, and the newly frozen coordinates ξ. For a fixed input x, the differentiated series*

$$\mathcal{D}_{e,x}(\boldsymbol{R}) = \sum_{\nu:\nu_x\geq 1} \nu_x |\Delta_{e,\nu}| \boldsymbol{R}^{\nu-\mathbf{1}_x} e^{a\mathfrak{t}(\{e\}\cup\operatorname{supp}\nu)},$$

   with the measure conventions of Section 19.1, satisfies

$$\mathcal{D}_{e,x}(\boldsymbol{R}) \leq C_J e^{-a_W[\operatorname{dist}(e,W)+\operatorname{dist}(e,x)]}.$$

   Unlike $C_{e,x}$ in part (4), this series does not contain the factor R_x.

2. *At $q = 0$, the dependence of Δ on the newly frozen coordinates satisfies, in the coefficient norm in c,*

$$\|\Delta_e(\cdot, 0; \xi) - \Delta_e(\cdot, 0; 0)\|_{\text{coeff}} \le C_F \|\xi\|_\infty e^{-a_W \operatorname{dist}(e,W)}, \qquad \Delta(0,0;0) = 0.$$

3. *At radius s_q for the fluctuation increments, the absolute coefficient row sum of the terms of Δ of positive degree in q is at most $C_{\Delta,J} s_q$. The rows of the coarse-field terms near the boundary and all rows in the free variables are only bounded by finite constants depending on J; no contraction factor M^{-J} is asserted for them.*
4. *(Input columns.) Let $R_y \le K_* \rho$ be a construction profile, and choose $a_1, b_1 > 0$ with $b_1 > \alpha_{\text{bd}}$ and with $a_1 + b_1$ smaller than a decay exponent for which the full reconstruction has coefficient bounds without boundary weight. Define*

$$C_{e,x}(\boldsymbol{R}) = \sum_\nu \nu_x |\Phi_{R,e,\nu}(\xi)| \boldsymbol{R}^\nu e^{a_1 \mathfrak{t}(\{e\} \cup \operatorname{supp} \nu)},$$

where $\Phi_{R,e,\nu}(\xi)$ are the Taylor coefficients of Φ_R at the fine output e. Then $C_{e,x}(\boldsymbol{R}) \le D_J R_x e^{-b_1 \operatorname{dist}(e,x)}$, and

$$\sum_e e^{-\alpha_{\text{bd}} d_{\text{old}}(e)} C_{e,x}(\boldsymbol{R}) \le K_J \rho \, e^{-\alpha_{\text{bd}} d_{\text{out}}(x)},$$
$$K_J \le C_{M,J} D_J K_* e^{\alpha_{\text{bd}} C_J} \sum_{l \ge 0} (l+1)^3 e^{-(b_1 - \alpha_{\text{bd}}) l}.$$

Here d_{old} and d_{out} are the distances to the complements of the specified old and output position lists, respectively, both expressed in output units, and we assume that, for every summation label e and every input x,

$$d_{\text{out}}(x) \le d_{\text{old}}(e) + \operatorname{dist}(e,x) + C_J,$$

with a buffer C_J depending only on the fixed construction geometry. The sum runs over the full list of old output labels (the fine labels), at their creation-scale density. For an input among the free coordinates the same statement holds with its own radius R_x in place of $K_* \rho$ in the last bound. For a kernel from inputs to outputs, the column of a fixed input carries the volume of the output cells divided by the volume of the cell of that input. The same calculation applies to the profile $R_y = 2\rho$ within the auxiliary construction ball and to the smaller spatially varying radius profiles of Definition 18.1. These column bounds are different from the bounds on the image under the reconstruction majorant in Lemma 19.7.

The constants have the dependence of Section 19.1, are uniform in the depth and in the shape of Ω, and do not depend on the later widths. No equality with a perturbed incoming constrained problem is asserted.

Proof. We compare the successive regional and torus reconstructions before applying the gauge transformation with cutoff, and then estimate the differentiated coefficient series.

Step 1: the domain of the successive substitutions. We take a common larger construction product as in Theorem 19.4, with radii for coarse fields and free coordinates at most U_T and Q_T from (19.1). Then the torus reconstruction, its gauge transformation, and all column series below are evaluated with $u + v < r_s$ and denominator at least $1/2$, and the smaller product keeps the intermediate classical and Haar-density margins. For the finitely many conditional problems at the intermediate scales, we use the smaller parameter domains of Step 10 of the proof of Theorem 19.4. We choose the radius of the fluctuation increments so that every successive substitution lies strictly inside these domains. Such a radius exists: by Lemma 18.8, at $q = 0$ the successive conditional stationary points are the joint stationary point. Each conditional stationary point and each inverse square root has a convergent majorant series without boundary weight on a larger product. If A_i bounds the deviation of the coarser coordinates from the joint stationary point after the scales above i have been reconstructed, then by these series we have

$$A_{i-1} \le (1 + C_0)A_i + C_S s_q, \qquad A_s = 0, \tag{19.29}$$

on a fixed smaller free ball, where the constants are maxima of the rows of the conditional derivatives and of the inverse square roots on the larger products. The same recurrence holds in the coefficient norms. Iterating it at most J times, we obtain $A_i \le C_S s_q \sum_{j=0}^{J-1}(1 + C_0)^j$. Taking s_q smaller than a fixed fraction of every domain margin divided by this finite sum places the successive product inside the domains. This is the case $\delta_V = 0$ of Lemma 19.3(3), and this choice does not reduce the radius of the coarse field.

Step 2: one scale of the reconstruction. On the original list of free coordinates, the reconstruction at scale i is

$$\Gamma_i\big(c_{i+1}, a_i(c_{i+1}; \xi) + K_i(c_{i+1}; \xi)^{-1/2} q_i; \xi\big). \tag{19.30}$$

Here a_i is the conditional stationary point of the partial minimum after the finer scales have been eliminated, K_i is its constrained Hessian at scale i, and Γ_i is the local inverse chart of the block averaging constraint. The action at this scale is M^i times the partial minimum of the objective in input units, so K_i includes exactly the factor M^i. The variable q_i is the fluctuation increment, and no coupling is absorbed into K_i.

Step 3: the reconstructions before gauge transformation. On each common list the comparison of the stationary points is given by Theorem 19.4. To compare inverse square roots, we extend the regional matrix to the full list by its direct sum with a principal free block of the torus matrix on the complementary coordinates. Both summands have the gap. The internal difference is the difference of the regional and torus Hessians at nonzero coarse field, estimated in Theorem 19.4(5). The removed off-diagonal blocks of the torus matrix have their two roots on opposite sides of W, so they are boundary-localized. The identity (19.7) then gives the comparison on the common list, with every coarse-field and

free argument. When a list of finer coordinates is eliminated on one side and fixed on the other, the identity (19.9) inserts a chain through that list, as in Step 11 of the proof of Theorem 19.4, and that chain visits H_{L_J}. Thus the statement concerns the regional matrix itself, not the torus problem with coordinates deleted.

We subtract the regional and the torus versions of (19.30), with the torus free inputs outside the regional list set to zero. First we change the chart words at the same arguments; their difference lies in the fixed interface stencil. Then we change the conditional stationary point and the inverse square root, using the boundary-localized differences just proved. Finally we change their coarser arguments. The difference of an analytic function at two arguments is its derivative integrated along the common segment, that is, a tensor without boundary weight applied to a boundary-localized difference of arguments. At the coarsest scale these are all the terms. Each descent by one scale adds the boundary-localized difference from the scale above as one more difference of arguments. By induction over at most J scales, the difference $a_R^{\mathrm{raw}} - a_T^{\mathrm{raw}}$ is a convergent boundary-localized series. The values of the torus reconstruction on the complementary outputs, which can be nonzero at zero complementary free input, are joined by networks without boundary weight from their output in H to their inputs in Ω. Such networks cross W, and they provide the visit to H_{L_J} at the next substitution. No sum of sources over the volume of H without decay occurs.

Step 4: the gauge transformation. We apply the same torus gauge transformation $k(c)$ to both fields and cut it off only on the regional side. After the construction radius is chosen, all arguments of H_e^{χ} lie in one fixed larger identity chart, and the difference is

$$\begin{aligned} \Delta_e = {} & H_e^{\chi}(a_R^{\mathrm{raw}}, k) - H_e^{\chi}(a_T^{\mathrm{raw}}, k) \\ & + H_e^{\chi}(a_T^{\mathrm{raw}}, k) - H_e^{1}(a_T^{\mathrm{raw}}, k). \end{aligned} \tag{19.31}$$

The first line contains $a_R^{\mathrm{raw}} - a_T^{\mathrm{raw}}$ as a factor, after the derivative in a is integrated. The second line vanishes when both endpoint values of the cutoff equal 1, so its output root lies in a fixed interface band of width three blocks, including the buffer for the endpoints of fine links. Its other factors have the torus and gauge coefficient bounds of Theorem 13.2 and Lemma 19.5. These facts imply the boundary-localized estimate for both lines of (19.31). A coarse-field derivative which hits a gauge factor appends the network of that factor to the existing network and does not remove the visit to H_{L_J}. The torus term without the cutoff is $\Phi_{\mathbb{T}}$. This proves the decomposition and its factorial coefficient bounds.

Step 5: the cutoff at first order. The contribution of the cutoff can be read off at first order. Let $k_1 = Dk(0)c$. The cutoff part of the linearized correction on the link (v, w) is the difference of $(1 - \chi)k_1$ between its endpoints, which equals

$$(1 - \chi(v))(k_1(w) - k_1(v)) + (\chi(v) - \chi(w))k_1(w).$$

Both terms are contained in (19.31). In particular the second term, which contains the gradient of the cutoff, is not removed by treating the gauge transformation as constant in the target.

Step 6: the local logarithms. The local logarithms in this argument have uniform complex coefficient margins. Indeed, in a submultiplicative matrix norm the exponential series are dominated by e^t, and the logarithm of $I + X$ is dominated by $\sum_{l\ge 1} t^l/l$ on $\|X\| \le t < 1$. We take the rows of the reconstruction before gauge transformation and of the gauge transformation small enough that the product lies in such a disc with a fixed margin. Composition of majorants and the attached networks then bound every derivative of H_e^{χ}, with constants independent of $\chi \in [0,1]$.

Step 7: fixed inputs and the newly frozen coordinates. Fix an input x in the boundary-localized series of the reconstruction and keep its position while summing over the other indices. The connected network contains e, x, and a visit to H_{L_J}, so by the larger decay exponent and Lemma 18.15, we have $\mathcal{D}_{e,x}(\boldsymbol{R}) \le C_J e^{-a_W[\delta(e)+\operatorname{dist}(e,x)]}$. At $q = 0$ the dependence on ξ satisfies the coefficient identity

$$\Delta_e(c,0;\xi) - \Delta_e(c,0;0) = \int_0^1 \sum_f \xi_f D_{\xi_f} \Delta_e(c,0;t\xi)\, dt. \tag{19.32}$$

The derivatives in ξ come from its analytic input slots. The bound on their sum follows from part (4) of Theorem 19.4, the finite compositions of charts, and the fact that the gauge transformation does not depend on ξ. Integrating their coefficient series in c in (19.32), we obtain the bound $C_F\|\xi\|_\infty e^{-a_W\delta(e)}$ of part (2). At $c = q = \xi = 0$ the unperturbed stationary points vanish and both gauge transformations are the identity, so $\Delta(0,0;0) = 0$.

Step 8: terms of positive degree in q. If a convergent majorant series in all variables is bounded on a larger fixed free radius r_Q, then, for $s_q \le r_Q/2$, its terms of positive total degree in q are at most s_q times a fixed bound for its derivative majorant on the larger product. Applying this to (19.31), we obtain the bound $C_{\Delta,J} s_q$ of part (3). The rows near the boundary and the rows in the free variables are bounded by the corresponding convergent majorants without boundary weight, with no factor M^{-J}.

Step 9: the input columns. Fix the input x in the differentiated coefficient series. Removing one power of its radius R_x and using the column bound with any input fixed, we see that the connecting network contains a path from e to x. Assigning b_1 of the larger decay exponent to this path gives

$$C_{e,x}(\boldsymbol{R}) \le D_J R_x e^{-b_1 \operatorname{dist}(e,x)} \tag{19.33}$$

on every smaller construction profile in the statement; since the majorant has nonnegative coefficients, we may restrict to that profile before summing. A free input is included,

since it is an input of (19.30) and of every later composition. By hypothesis the distances satisfy

$$d_{\mathrm{out}}(x) \le d_{\mathrm{old}}(e) + \mathrm{dist}(e,x) + C_J,$$

where the additive constant is a fixed buffer. For an incoming list D_{in} whose complement lies in H up to the fixed stencil, with $d_{\mathrm{old}}(e) = d_{D_{\mathrm{in}}}(e)/B_s$, this condition follows from Lemma 19.8 after the incoming distance and the buffer are divided by B_s; we do not identify the incoming and the current fixed complements. Since $B_s \ge 1$, the condition also holds when d_{old} is measured in input units. Hence

$$\begin{aligned}\sum_e e^{-\alpha_{\mathrm{bd}} d_{\mathrm{old}}(e)} C_{e,x}(\boldsymbol{R}) &\le D_J R_x e^{\alpha_{\mathrm{bd}} C_J} e^{-\alpha_{\mathrm{bd}} d_{\mathrm{out}}(x)} \sum_e e^{-(b_1-\alpha_{\mathrm{bd}})\,\mathrm{dist}(e,x)} \\ &\le C_{M,J} D_J R_x e^{\alpha_{\mathrm{bd}} C_J} e^{-\alpha_{\mathrm{bd}} d_{\mathrm{out}}(x)} \sum_{l\ge 0} (l+1)^3 e^{-(b_1-\alpha_{\mathrm{bd}})l}.\end{aligned}$$

Using $R_x \le K_* \rho$, we obtain the bound on K_J; keeping R_x gives the version for a free input. The sum includes every old output label at its density; restricting it to a sublist only decreases it. For a kernel from inputs to outputs it is multiplied by the volume of the fine output cells divided by the volume of the cell of the fixed input, as in Definition 4.6. The colors and the finitely many scale indices enter $C_{M,J}$, not a factor of the lattice volume. Setting $R_y = 2\rho$ is allowed within the auxiliary ball since $K_* \ge 2$, and the smaller spatially varying radius profile is another restriction of the same nonnegative coefficient series. This completes the proof. □

19.6 Construction radii and bounds on the reconstruction majorant

The next lemma collects the conditions on the radii under which the reconstruction maps a product domain strictly inside the analytic domain used at the next step, in the following sense. For a map with a convergent Taylor series, the absolute coefficient row sum of the Taylor series, evaluated at the radii of the product domain, bounds the image of the domain under the majorant of the map. For the reconstruction we call this bound the image under the reconstruction majorant. For $c \ge 0$ we write S_c for this row sum with the weight $e^{c\ell}$, where ℓ is the length of the connecting network of a monomial including its output; thus S_0 is the unweighted row sum. A bound $S_c < \rho/2$ is stronger than a bound on the values of the map, and it is the hypothesis needed to substitute the reconstruction into a function whose coefficients are controlled at radius ρ. The lemma is applied twice, with the quantum radius ρ and with the classical radius ρ_{cl}.

Lemma 19.7. *Fix the constants of the unperturbed constructions before the later choices of the boundary-layer width, the erosion width, the widths of the protected neighborhoods, and the bound β_{rec} on newly frozen coordinates.*

1. *(Construction radius.) Let $v_{\mathrm{free},J} > 0$ be a free construction radius, distinct from the density constant v_J, with*
$$Q_J v_{\mathrm{free},J} < \lambda_J/64, \qquad I_J Q_J v_{\mathrm{free},J} < r_{z,J}/64.$$
There is a radius $R_{\mathrm{raw}} > 0$, inside both the chart of the block averaging constraint and the chart of Theorem 11.6, which may be chosen below
$$\min\Big\{\frac{\min(r_{\Gamma,J}, r_E)}{64G_J}, \frac{\lambda_J}{64Q_J}, \frac{r_{z,J}}{64I_J Q_J}, \frac{v_{\mathrm{free},J}}{64C_{\mathrm{lift},J}}\Big\}.$$
Here G_J and $r_{\Gamma,J}$ are the bound and the radius of the finite constraint words, r_E is the larger radius of the torus action, and $C_{\mathrm{lift},J}$ bounds the block-constant lift and the coarse-field lift, including all embeddings for the initial segments. If
$$C_{\mathrm{lift},J}[K_* \max(\rho, \rho_{\mathrm{cl}}) + r + \beta_{\mathrm{rec}} + q_{\mathrm{raw}}] < R_{\mathrm{raw}}/8,$$
where r is the radius of the small-link domain B_r and q_{raw} is the radius allowed for the fluctuation increments before gauge transformation, then all images of the reconstruction before gauge transformation lie in these fixed larger charts, and their real free Hessians have gap at least $\lambda_J/2$, with the complex inverse margins. No factor M^{-J} is required of the lift.
2. *(Zero fluctuations.) For each $\rho_\bullet \in \{\rho, \rho_{\mathrm{cl}}\}$, use its own spatially varying radius profile $\phi_\bullet$, equal to the boundary radius $\rho_{\partial,\bullet}$ at labels within distance w_{bd} of H and to $\rho_\bullet$ elsewhere, the outer profile $K_*\phi_\bullet$, and the outer boundary radius $R_{\partial,\bullet} = K_*\rho_{\partial,\bullet}$. For a full renormalization group step, $s = J$ and $B = M^J$, assume the torus contraction inequality (the first inequality below) and impose the four inequalities*
$$B^{-1}Z_\bullet(K_*\rho_\bullet) \le \rho_\bullet/16, \quad C_{\Delta,\bullet}C_{a,\bullet}R_{\partial,\bullet} \le \rho_\bullet/16,$$
$$C_{\Delta,\bullet}C_{a,\bullet}e^{-a_W w_{\mathrm{bd}}/2}K_*\rho_\bullet \le \rho_\bullet/16, \qquad C_{F,\bullet}\beta_{\mathrm{rec}} \le \rho_\bullet/16,$$
where $Z_\bullet$ is the classical row majorant of Theorem 13.2, $C_{\Delta,\bullet}$ and $C_{F,\bullet}$ are the constants of Theorem 19.6, and $C_{a,\bullet}$ is a summation constant fixed in the proof. These constants are evaluated on the larger auxiliary construction ball, before the radii and widths are chosen. The sum of the four left sides bounds the row sum S_0 at $q = 0$ by $\rho_\bullet/4$.
3. *(Fluctuation increments.) For a radius s_q of the fluctuation increments, the increment of S_0 is bounded by*
$$W_\bullet(s_q) = \frac{C_{T,\bullet}s_q}{1 - C_{T,\bullet}(K_*\rho_\bullet + s_q)} + C_{\Delta,\bullet}s_q.$$
Require the denominator to be at least $1/2$ and $W_\bullet(s_q) \le \rho_\bullet/8$ for both radii. The same rule applies to the product in the jointly normalized coordinates u, with its finite-composition free radius and its own denominator, which includes β_{rec} if it occurs in that majorant. The products for the successive substitution must lie in this joint product.

4. *(Spatial weight.) The resulting unweighted absolute row sum satisfies* $S_0 \le 3\rho_\bullet/8$. *If* $S_c \le C_J\rho_\bullet$ *for a fixed* $c > 0$, *then there is a fixed* $0 < \vartheta < 1$, *common to both radii and to the finitely many initial segments, such that*

$$S_{\vartheta c} \le S_0^{1-\vartheta} S_c^{\vartheta} \le \rho_\bullet (3/8)^{1-\vartheta} C_J^{\vartheta} < \rho_\bullet/2.$$

Thus, for both radii, the image under the reconstruction majorant, with connecting-network weight at one positive exponent, is less than half of the radius.

All constants and thresholds have the dependence of Section 19.1 and do not depend on the later widths. For a composition of $s < J$ *averaging steps, the conclusion of part (4) requires the torus contraction inequality with* $B_s = M^s$ *in place of* B, *which must be checked separately for that composition; the finiteness of the constants at fixed* J *does not give it.*

Proof. We choose the construction radius, bound the unweighted row sum at zero fluctuation and its increment, and then add the spatial weight by interpolation.

Step 1: the construction radius. We choose the finite word and lift constants on the larger products of Theorem 18.12 and Lemma 19.3. We increase $C_{\mathrm{lift},J}$ so that it bounds the block-constant lift of the target, the coarse-field lift, the extensions by zero and by the small newly frozen coordinates, and each of the finitely many embeddings for initial segments. Since a local lift uses only its block stencil of fixed depth, its row and input columns have one finite coefficient bound, independent of the number of components and of the later interface widths. Let $v_{\mathrm{free},J}$ satisfy the two strict inequalities in part (1), and let R_{raw} lie below the listed bounds and below the common radii of the local Haar-density chart and of the compact-overlap chart. The lift inequality then places all images strictly inside the larger inverse and torus action charts, their free arguments inside the radius $v_{\mathrm{free},J}$, and their stationary points inside the solution radius $r_{z,J}$. Indeed, the variation of the free Hessian is bounded by the sum of its contributions from the free radius and from the lifted arguments, which is at most

$$Q_J v_{\mathrm{free},J} + Q_J R_{\mathrm{raw}} < \lambda_J/32.$$

Enlarging the fixed lift constant before the radii are chosen accounts for any finite number of such types of arguments. The displacement of the stationary point is bounded by the corresponding inverse times gradient, which is less than $I_J Q_J (v_{\mathrm{free},J} + R_{\mathrm{raw}}) < r_{z,J}/32$. The real gap λ_J at the identity therefore implies a real gap at least $\lambda_J/2$ on all these products, with room for the complex margins. The term $E'_{d_0}\Psi_{zz}$, which contains the gradient of the action, is included in the bound on the variation of the Hessian in Theorem 18.12. The conditional stationary points are those of Lemma 19.3.

Step 2: the row sum at zero fluctuation. We carry out the following calculation for each $\rho_\bullet$, with its own profile and constants. The construction radii satisfy $K_*\rho_\bullet \le U_T$ and

$s_q \le Q_T$, so $K_*\rho_\bullet + s_q < r_s$ and all intermediate margins hold by (19.1); these domain checks precede each rational majorant. At $q = 0$ we use Theorem 19.6. By Theorem 13.2, the torus term has absolute row at most $B^{-1}Z_\bullet(K_*\rho_\bullet)$. We split the regional correction into its value at $\xi = 0$ and its increment in ξ. By Theorem 19.6(2), the latter is bounded by $C_{F,\bullet}\beta_{\mathrm{rec}}$ in the coefficient norm in the target. The former vanishes at zero target, and integrating its differentiated series with a fixed target input bounds it by

$$C_{\Delta,\bullet}\sum_x R_x e^{-a_W[\delta(e)+\mathrm{dist}(e,x)]}, \qquad R_x = (K_*\phi_\bullet)(x),$$

where $C_{\Delta,\bullet}$ is increased to include the fixed normalization of the derivative. Terms with $\delta(x) < w_{\mathrm{bd}}$ have $R_x = R_{\partial,\bullet}$, and their sum is at most $C_{a,\bullet}R_{\partial,\bullet}$, where $C_{a,\bullet}$ bounds the sum of the exponentials. For the other terms, by the conventions for endpoints, we have

$$\delta(e) + \mathrm{dist}(e,x) \ge \delta(x) - L'_J \ge w_{\mathrm{bd}} - L'_J.$$

We use half of the exponent for this inequality and sum the other half using the density bound. After $e^{a_W L'_J/2}$ is included in the fixed constant $C_{a,\bullet}$, these terms are at most $C_{a,\bullet}e^{-a_W w_{\mathrm{bd}}/2}K_*\rho_\bullet$. This bounds the row at $q = 0$ by the sum of the four terms of part (2). Since each of them is at most $\rho_\bullet/16$, the row is at most $\rho_\bullet/4$. All constants in this calculation were evaluated on the larger unperturbed auxiliary ball, before w_{bd} or the smaller boundary radius was chosen.

Step 3: terms of positive degree in the increments. For positive degree in the free coordinates, the torus row is bounded by

$$\frac{C_{T,\bullet}s_q}{1 - C_{T,\bullet}(K_*\rho_\bullet + s_q)},$$

which is the majorant of Theorem 13.2. The row of the correction of positive degree in q is at most $C_{\Delta,\bullet}s_q$, by Theorem 19.6(3). Adding the two, we obtain $W_\bullet(s_q)$, and under the two conditions of part (3) the increment is at most $\rho_\bullet/8$. Combining this with Step 2, we obtain the absolute coefficient row bound

$$S_0 \le 3\rho_\bullet/8. \tag{19.34}$$

The denominator cannot be discarded: its geometric series contains all higher degrees in the free coordinates and in the target.

Step 4: the jointly normalized coordinates. For the joint coordinate u, we substitute $z = z_0 + K_0^{-1/2}u$ into the maps in the original coordinates before taking the majorant. The successive increments are recovered from z by

$$q_i = K_i(c, z_{>i}; \xi)^{1/2}\big(z_i - a_i(c, z_{>i}; \xi)\big),$$

which inverts the unperturbed insertion at scale i. The factor $K_i^{1/2} = K_i K_i^{-1/2}$ has the same larger coefficient bounds. At $u = 0$ all these increments vanish, since the chain at zero fluctuation is the joint stationary point. By finite composition of majorants, there exists a joint radius for which the resulting majorant of the increments is at most Q_T. We take the product of u inside this radius and inside the solution domains of the initial segments. Its target argument is at most U_T, so (19.1) also verifies the torus domain and the denominator for this composition. The finite composition has a rational majorant of the same kind for its increment, with its own constants and argument; in particular, a contribution of the newly frozen coordinates to that argument remains in its denominator. Requiring this majorant to be at most $\rho_\bullet/8$ gives (19.34) for the joint product as well. The products for the successive substitution are restricted to the preimage of this product and to the domains of Lemma 19.3. This restricts a coefficient series which is already bounded.

Step 5: the spatial weight. For one output row, we write its terms as $b_\nu \geq 0$ and let ℓ_ν be the length of the attached network, including the output root. For finite partial sums, by Hölder's inequality with exponents $1/(1-\vartheta)$ and $1/\vartheta$, we have

$$\sum_\nu b_\nu e^{\vartheta c \ell_\nu} = \sum_\nu b_\nu^{1-\vartheta} (b_\nu e^{c\ell_\nu})^\vartheta \leq \Big(\sum_\nu b_\nu\Big)^{1-\vartheta} \Big(\sum_\nu b_\nu e^{c\ell_\nu}\Big)^\vartheta ,$$

and passing to increasing partial sums gives the inequality for the convergent series. Using $S_c \leq C_J \rho_\bullet$ and (19.34), we obtain the bound in part (4). The function $\vartheta \mapsto (3/8)^{1-\vartheta} C_J^\vartheta$ is continuous at $\vartheta = 0$, with value $3/8 < 1/2$. We fix $\vartheta \in (0,1)$ at which its value is below $1/2$ for both radii and for the finitely many initial segments. This proves both weighted bounds on the image under the reconstruction majorant at the same positive exponent. Because the weighted row bound is a bound of the unperturbed construction, this choice of exponent uses no later perturbative estimate.

Step 6: shorter compositions. For $s < J$, Step 1 and all estimates for finite compositions still apply, with $B_s = M^s$. The first inequality of part (2), however, becomes $B_s^{-1} Z_\bullet(K_* \rho_\bullet) \leq \rho_\bullet/16$, which follows only from a separate contraction inequality for that composition. This is the restriction for $s < J$ in the statement. □

19.7 Perturbations of the incoming action

The incoming action of a renormalization group step on a block region is the effective classical action plus a perturbation, defined on the list of variable labels of the previous step. This list and the current block region play different roles; in particular, the protected vertices of the gauge transformation alone do not determine the boundary of the incoming list. We first compare the distance to the boundary of the incoming list with the distance to the current boundary.

Lemma 19.8. *Fix $1 \le s \le J$, $B_s = M^s$, a list D_{in} of incoming positions, and a block region Ω. Use the periodic graph metric, rescaled linearly on positions of coordinates introduced at finer lattices, and freeze all components of a label at one position together. Write $\pi_s x = x/B_s$. Suppose that every position $x \notin D_{\mathrm{in}}$ lies in H up to the fixed chart stencil:*

$$\operatorname{dist}(\pi_s x, \Omega^c) \le L_{\mathrm{buf},J}.$$

Here $L_{\mathrm{buf},J}$ and the fixed buffer $c_{\mathrm{end},J}$ for the convention on link endpoints depend only on the construction geometry, and are fixed before the later widths. Let $d_{D_{\mathrm{in}}}(i)$ be the distance of i to the complement of D_{in}, in input units, and $d_{\mathrm{current}}(e)$ the distance of e to H, in output units. Then for every $i \in D_{\mathrm{in}}$ and every position e of a label in Ω,

$$d_{D_{\mathrm{in}}}(i) \ge B_s d_{\mathrm{current}}(e) - B_s \operatorname{dist}(e, \pi_s i) - C_{\mathrm{buf},J}, \qquad C_{\mathrm{buf},J} = \max_{1 \le s \le J} B_s (L_{\mathrm{buf},J} + c_{\mathrm{end},J}). \tag{19.35}$$

The same conclusion holds with $\operatorname{dist}(e, W)$ in place of $d_{\mathrm{current}}(e)$ after the fixed endpoint buffer is enlarged. It also holds if H is enlarged by additional blocks.

Proof. Suppose that the complement of D_{in} is nonempty, and choose a position $x \notin D_{\mathrm{in}}$ reached by a shortest path from i to this complement; its length is $d_{D_{\mathrm{in}}}(i)$. The scaled path has length $d_{D_{\mathrm{in}}}(i)/B_s$, including the position of x at its own lattice; scale indices and colors give no extra distance. By the triangle inequality and the hypothesis, we have

$$\begin{aligned} d_{\mathrm{current}}(e) &\le \operatorname{dist}(e, \pi_s x) + L_{\mathrm{buf},J} + c_{\mathrm{end},J} \\ &\le \operatorname{dist}(e, \pi_s i) + d_{D_{\mathrm{in}}}(i)/B_s + L_{\mathrm{buf},J} + c_{\mathrm{end},J}. \end{aligned}$$

Multiplying by B_s, we obtain (19.35). The endpoint buffer covers the difference between the distance to the positions in H and the distance to the interface W. Enlarging H can only decrease its distance from a fixed position in Ω, so additional blocks of H preserve both the hypothesis and the bound. If D_{in} has no complement, the finiteness of the boundary seminorm forces the nonconstant boundary part of the incoming perturbation to vanish, by the convention of Definition 18.1, and there is no incoming boundary dependence to transport. If D_{in} is empty there is no incoming root. □

The constant incoming radius in the next theorem is the radius produced by freezing, in the following form of Lemma 18.9. We start with a classical output profile ϕ of interior radius r_c on a set D of variable labels, keep $I \subset \{i \in D : d_D(i) \ge s_{\mathrm{er}}\}$ with $s_{\mathrm{er}} \ge \max(2, w_{\mathrm{bd}})$, and freeze $F = D \setminus I$, which contains the zone of reduced radius. For an analytic function f and frozen values $c^\sharp$, we write $T_{c^\sharp} f(v) = f(v \oplus c^\sharp) - f(0 \oplus c^\sharp)$. For $|c_j^\sharp| \le \eta \phi(j)$ with $\eta \le 1/4$, Lemma 18.9 implies

$$\phi|_I \equiv r_c, \qquad \mathcal{X}_I^{r_c}(T_{c^\sharp} f) \le 2 e^{-\alpha_{\mathrm{cl}}(s_{\mathrm{er}}-1)} \mathcal{X}_D^{\phi}(f). \tag{19.36}$$

The seminorms here have the classical normalization and the stated network and boundary rates. The identity $\phi|_I \equiv r_c$ holds because every remaining position lies outside the zone of reduced radius, and it is needed for the seminorm with constant profile on the left. The frozen values become fixed real parameters, not new variables of radius r_c. Applying (19.36) to the boundary correction produced at the previous step, with $D_{\mathrm{in}} = I$, gives the incoming seminorm at the next step. A new classical boundary-completion difference of the torus action must in addition satisfy its own bound in the seminorm with constant profile, by the separate statement on such differences in Lemma 18.9; erosion alone does not bound such a difference at an additional component of H. The theorem below assumes the bound on the incoming seminorm. That the regions and frozen values of a large-field decomposition satisfy its hypotheses is verified in the construction of that decomposition in Section 21.

Theorem 19.9. *Fix the domains, the protected geometry, the radius R_{raw}, the conditions on the image under the reconstruction majorant for both radii, and the products for the initial segments of Lemmas 19.3, 19.5, and 19.7, with $s = J$; compositions of $s < J$ averaging steps are treated at the end of the statement. Fix K_*, classical input rates $a_{\mathrm{cl}}, \alpha_{\mathrm{cl}} > 0$, and $r_c = \rho_{\mathrm{cl}}$. Let $b = (\xi, b_{\mathrm{rec}})$, where the newly frozen coordinates ξ are small and the previously frozen coordinates b_{rec} are arbitrary in their allowed real set.*

The incoming action. *Let $E_{\mathrm{in}}(a; b) = E_{d_0}(\iota_0 a) + P(a; b)$ be holomorphic and real at real arguments in its variable input a, with b_{rec} fixed, where ι_0 extends a by zero. Fix its incoming list D_{in} of variable labels, the positions of these labels at the lattices where they were introduced, and their cell volumes. The incoming seminorm is the seminorm $\mathcal{X}^{r_c}_{D_{\mathrm{in}}, a_{\mathrm{cl}}, \alpha_{\mathrm{cl}}}$ of Definition 18.1, with the constant profile $\phi_{\mathrm{in}} \equiv r_c$ on every label of D_{in} at which Φ_0 below is substituted; the labels of ξ are fixed parameters of this seminorm. The spatially varying radius profiles are used only for the output. Assume that the complement of D_{in} lies in H up to the fixed stencil, as in Lemma 19.8 (H may contain additional blocks), and that, in input units, for every position e of a label in Ω and every incoming position i,*

$$d_{D_{\mathrm{in}}}(i) \geq B_s d_{\mathrm{current}}(e) - B_s \operatorname{dist}(e, i/B_s) - C_{\mathrm{buf},J}, \qquad B_s = M^s,$$

where the positions on the right are in output units and the fixed buffer includes the chart stencils. If D_{in} has no complement, the nonconstant boundary part of the incoming perturbation is zero. All images of Φ_0 and all connecting segments lie strictly inside the ball of the incoming seminorm. A difference of two perturbations uses the same incoming list, weights, and profile.

The exterior extension. *If the unperturbed regional objective uses the extension ι_ξ, which inserts the newly frozen coordinates into their slots while all previously frozen torus slots remain zero, define the classical boundary-completion difference and the adjusted perturbation*

$$S^{\mathrm{cl}}_\xi(a) = E_{d_0}(\iota_\xi a) - E_{d_0}(\iota_0 a), \qquad \overline{P} = P - S^{\mathrm{cl}}_\xi,$$

and assume $\mathcal{X}^{r_c}_{D_{\mathrm{in}},a_{\mathrm{cl}},\alpha_{\mathrm{cl}}}(\overline{P}) \le 2\sigma_*$. *If the reference objective already uses the extension by zero, take* $S^{\mathrm{cl}}_{\xi} = 0$. *All constants of* S^{cl}_{ξ} *remain in the full classical action. The corresponding difference for the effective interaction, if the reference extension is changed there, is kept with the exact sign of the constant term in Lemma 18.9.*

The objective. *Let* z_0 *be the unperturbed joint stationary point,* $K_0 = F_{0,zz}(c, z_0; \xi)$, *and* $S_0 = K_0^{-1/2}$, *and put*

$$\begin{aligned} F_0(c,z;\xi) &= E_{d_0}(\iota_{\xi}\Psi_R(c,z;\xi)),\\ A_0(c,u;\xi) &= \Psi_R(c, z_0 + S_0 u;\xi),\\ \Phi_0(c,u;\xi) &= \mathrm{Log}[(\exp A_0(c,u;\xi))^{G^{\Omega}(c)}],\\ \mathcal{F}_{\lambda}(c,u;b) &= F_0(c, z_0 + S_0 u;\xi) + \lambda\overline{P}(\Phi_0(c,u;\xi);b), \qquad 0 \le \lambda \le 1, \end{aligned}$$

and $\mathcal{Q}_{\lambda}(c,z;b) = \mathcal{F}_{\lambda}(c, S_0^{-1}(z - z_0); b)$. *The perturbation is evaluated on the gauge-transformed reconstruction* Φ_0, *and the unperturbed Hessian of* $\mathcal{F}_{\lambda}$ *in* u *at* $u = 0$ *is* I. *Initial segments, the conditional stationary points* $a_{\lambda,I}$, *and the Hessians* $K_{\lambda,i}$ *refer to* $\mathcal{Q}_{\lambda}$, *in the original free coordinates.*

Then the following statements hold.

1. *(Incoming derivatives.) If the inputs at which* $\overline{P}$ *is evaluated have absolute coefficient row sums at most* $\kappa\rho_{\mathrm{cl}}$, *with* $0 < \kappa < 1$, *then for every* $j \ge 1$,

$$\|D^j\overline{P}\|_{\mathrm{root}} \le \frac{j!}{(1-\kappa)^{j+1}}(2\sigma_*)\rho_{\mathrm{cl}}^{2-j},$$

   with the incoming network and boundary weights, and with any differentiated input as root.

2. *(Stationary points.) There exist* $\sigma_{0,J}, Q_{0,J} > 0$ *and a fixed smaller domain of* u *such that the following holds for every* $0 \le \sigma_* \le \sigma_{0,J}$, *every* $0 \le \lambda \le 1$, *and every product of increments with* $\max_i \|q_i\|_{\infty} \le Q_{0,J}$ *whose substitutions stay in that domain. Every initial segment of the original free coordinates has a unique holomorphic stationary point of* $\mathcal{Q}_{\lambda}$ *in the domains of Lemma 19.3, and the inequalities of Lemma 19.3(2) on the preconditioned gradient and Hessian hold, not only a spectral bound. The stationary points are real at real data, the real free Hessians satisfy*

$$\frac{1}{2}\lambda_J I \le K_{\lambda,i} \le \Lambda_J I, \qquad \|u_{\lambda}\|_{\mathrm{coeff}} \le C_J\sigma_*\rho_{\mathrm{cl}},$$

   with the exact factors M^i *included in* $K_{\lambda,i}$, *and for every initial segment* I *and every* $y \in \mathcal{T}_I$,

$$\|a_{\lambda,I}(c,y;b) - a_I(c,y;\xi)\|_{\mathrm{coeff}} \le C_J\sigma_*\rho_{\mathrm{cl}}.$$

   The quantity δ_V *of Lemma 19.3 satisfies* $\delta_V \le I_0 C_P\sigma_*\rho_{\mathrm{cl}}$, *with constants* I_0, C_P *depending only on the fixed data, and* $\sigma_{0,J}, Q_{0,J}$ *are chosen so that the condition of Lemma 19.3(3) holds for all initial segments. The weighted complex resolvents are bounded by* $C_J/(\lambda_J + t)$ *for* $t \ge 0$ *at the allocated working rates of part (4).*

3. *(Differences.) The differences from $\lambda = 0$ of the stationary points, the Hessians, their inverses and inverse square roots, and the successive reconstructions have convergent boundary-localized coefficient bounds in all variables, including every coarse-field input in Ω and every fluctuation increment; their networks carry the incoming boundary dependence. These differences are $O_J(\sigma_*)$, and the displacements of stationary points are $O_J(\sigma_*\rho_{\mathrm{cl}})$ on the fixed product. For example, at fixed coarser coordinates y,*

$$\partial_\lambda z_{I,\lambda} = -\mathcal{Q}^{-1}_{\lambda,II} V_I,$$

where V_I is the I-component of the gradient of $z \mapsto \overline{P}(\Phi_0(c, S_0^{-1}(z - z_0); \xi); b)$. Along the successive chain this derivative contains the additional term $-\mathcal{Q}^{-1}_{\lambda,II}\mathcal{Q}_{\lambda,II^c}\partial_\lambda z_{I^c}$ and the full derivative $(\partial_\lambda K^{-1/2}_{\lambda,i})q_i$. The function $V = \overline{P} \circ \Phi_0$ satisfies $\|V_u\| \le C_J\sigma_*\rho_{\mathrm{cl}}$ and $\|V_{uu}\| \le C_J\sigma_*$; the term $\overline{P}'\Phi_{0,uu}$ is kept in the second derivative.

4. *(Output rates.) The estimates of part (3) hold at any fixed positive output rates $a_{\mathrm{out}}, \alpha_{\mathrm{out}}$ with*

$$a_{\mathrm{out}} + \alpha_{\mathrm{out}} < c_J / C_{\mathrm{geom}}, \qquad a_{\mathrm{out}} + \alpha_{\mathrm{out}} < B a_{\mathrm{cl}} / C_{\mathrm{geom}}, \qquad \alpha_{\mathrm{out}} < B\alpha_{\mathrm{cl}} / C_{\mathrm{geom}},$$

where C_{geom} is fixed before J, the rates, and the later widths: it depends only on the dimension, the block geometry, and the allocation of exponents, not on J or on the construction radii. Fixed stencil buffers depending on J affect the amplitudes C_J, not the coefficients of the distances. In particular, the perturbation and every external argument stay in one connected coefficient network.

5. *(Two perturbations.) For two perturbations $\overline{P}_0, \overline{P}_1$ as above, the bounds of parts (2) and (3) on the differences hold with σ_* replaced by $\mathcal{X}^{r_c}_{D_{\mathrm{in}}, a_{\mathrm{cl}}, \alpha_{\mathrm{cl}}}(\overline{P}_1 - \overline{P}_0)$. The corresponding minimized full actions $V_0^{\mathrm{out}}, V_1^{\mathrm{out}}$ satisfy a Lipschitz bound in the output seminorm with constant K_J^{cl}, namely* (19.51).

6. *(The full action.) The outgoing full action is $M^s\mathcal{F}_1(c, u_1; b)$, and on the connected chart*

$$\mathcal{F}_1(c, u; b) = E_{\mathrm{in}}(\Phi_0(c, u; \xi); b).$$

This identity uses the equality of classical actions on compact gauge overlaps only for the torus term, together with the cancellation of the exact classical boundary-completion difference. It does not assert that $\overline{P}(\Phi_0)$ equals $\overline{P}(A_0)$, and it uses no gauge identity for an incoming branch.

7. *(Frozen coordinates.) All perturbed estimates differentiate the coarse-field and free variables at fixed real $b = (\xi, b_{\mathrm{rec}})$; no derivative of a perturbed quantity in ξ or b_{rec} is asserted, and the unperturbed estimates in ξ of Theorem 19.4 are unchanged. If the incoming coefficients and the charts are Borel (respectively continuous) functions of the real parameter b, then the stationary points, the matrix functions, and the minimized values are Borel (respectively continuous) in b.*

All constants have the dependence of Section 19.1; they do not depend on the later widths, the number of preceding steps, the shape of Ω, the depth, or the smaller enclosing and fluctuation radii chosen later. For a composition of $s < J$ averaging steps, the same statements hold, with B_s in place of B, whenever the separate conditions of Lemma 19.7 for that composition hold; no repeated reduction of radii is used. The theorem is a local analytic statement and does not compare probability densities.

Proof. We extract the derivatives of the incoming perturbation from the incoming seminorm and transport its boundary dependence to the output. We then construct the stationary points with Lemma 19.3, first jointly and then for the initial segments. The remaining steps treat the differences, two perturbations, the full action, and the dependence on the frozen coordinates.

Step 1: conventions. All functions in this proof are considered at a fixed real $b = (\xi, b_{\mathrm{rec}})$ in the allowed set, including the newly frozen coordinates ξ. Every estimate is uniform over this set, and the supremum over it is taken after the sums over degrees. Only the coarse field in Ω, the free coordinates, and the interpolation parameter are differentiated, and the labels of ξ are not roots of the incoming seminorm.

Step 2: derivatives of the incoming perturbation. We write $r_c = \rho_{\mathrm{cl}}$ and $\overline{P}(a; b) = \sum_\nu p_\nu(b) a^\nu$, where every multi-index runs over D_{in}, with the positions of its labels at the lattices where they were introduced. Since $\phi_{\mathrm{in}}(i) = r_c$ for every such label, we define, for each degree k,

$$A_k(b) = r_c^{-2} \sup_{i \in D_{\mathrm{in}}} e^{\alpha_{\mathrm{cl}} d_{D_{\mathrm{in}}}(i)} \sum_{|\nu|=k} \frac{\nu_i}{k} |p_\nu(b)| r_c^k e^{a_{\mathrm{cl}} \mathfrak{t}_{\mathrm{in}}(\nu)}.$$

By hypothesis, $\sup_{b \text{ real}} \sum_{k \geq 1} A_k(b) \leq 2\sigma_*$. This uses the full input radius also at the incoming boundary; a reduced radius near the boundary appears only in the output product. By (19.36), the incoming action produced by freezing has this profile when the hypotheses of Lemma 18.9 on freezing the zone of reduced radius and on the frozen values hold. We consider inputs whose absolute coefficient row sums are at most κr_c, and fix the first differentiated input label i. In degree $k \geq j$, summing over the other $j-1$ ordered differentiated labels gives the factor

$$\frac{\nu_i}{k} (k)_j, \qquad (k)_j = k(k-1) \cdots (k-j+1),$$

and the remaining input powers are bounded by $\kappa^{k-j} r_c^{k-j}$. Keeping the original incoming network keeps all differentiated roots and all contracted input labels in it. Therefore the rooted derivative of order j is at most

$$r_c^{2-j} \sum_{k \geq j} A_k (k)_j \kappa^{k-j} \leq (2\sigma_*) r_c^{2-j} \frac{j!}{(1-\kappa)^{j+1}},$$

where we used $\sum_{k\geq j}(k)_j \kappa^{k-j} = j!/(1-\kappa)^{j+1}$, which follows by differentiating the geometric series j times. Any differentiated root can be chosen first, so this proves part (1), with the incoming network and the boundary weight of its root. Degree one is included. The constant coefficient p_0 does not appear in these derivatives and is not restricted.

Step 3: the normalized objective and the perturbing gradient. Let A_0, Φ_0, and $\mathcal{F}_\lambda$ be as in the statement, on the joint product of Lemma 19.7. The image of this product under the reconstruction majorant, with weight at one fixed positive exponent of the connecting network, is less than $r_c/2$, so part (1) applies with a fixed $\kappa < 1$, for example $\kappa = 1/2$. At $u = 0$ we have

$$D_u F_0(c, z_0 + S_0 u; \xi)\big|_{u=0} = 0, \qquad D_u^2 F_0(c, z_0 + S_0 u; \xi)\big|_{u=0} = S_0^{\mathsf{T}} K_0 S_0 = I. \tag{19.37}$$

The first identity is the joint stationarity, and the second uses the symmetric inverse square root on the branch connected to the identity. The coefficients of this function, and those of z_0, Ψ_R, and G^Ω, are bounded on a larger product by the unperturbed results above.

Let $V = \overline{P} \circ \Phi_0$. By the chain rule, we have

$$V_u = \sum_i \overline{P}_i(\Phi_0)(\Phi_0)_{i,u}, \tag{19.38}$$

$$V_{uv} = \sum_{i,j} \overline{P}_{ij}(\Phi_0)(\Phi_0)_{i,u}(\Phi_0)_{j,v} + \sum_i \overline{P}_i(\Phi_0)(\Phi_0)_{i,uv}. \tag{19.39}$$

By part (1) with $j = 1, 2$, together with the fixed rows and volume-normalized columns of the chart, we obtain

$$\|V_u\| \leq C_P \sigma_* r_c, \qquad \|V_{uu}\| \leq C_P \sigma_*. \tag{19.40}$$

The term $\overline{P}'\Phi_{0,uu}$ in (19.39) contributes $C_J \sigma_* r_c$, which is included in $C_P \sigma_*$ since r_c is fixed. We note that the perturbation need not have zero gradient at $u = 0$.

These bounds hold at all orders, as coefficient bounds with spatial weights. Let $B_J(t) = C_J t/(1 - C_J t)$ majorize the increment series of the chart in a fixed displacement direction, on a fixed larger domain. After division by the factorials of the derivatives, part (1) and the chain rule over partitions are bounded by the coefficients of

$$2\sigma_* r_c^2 \Big\{ \frac{1}{1 - \kappa - B_J(t)/r_c} - \frac{1}{1-\kappa} \Big\}. \tag{19.41}$$

Fix $t_* > 0$, a fixed multiple of $\min(r_c, 1)$, with $B_J(t_*) \leq (1-\kappa) r_c/2$. By nonnegativity of its coefficients, the coefficient of degree j of (19.41) is at most $C_J \sigma_* r_c^2 t_*^{-j}$, and restoring $j!$ gives the factorial bounds $j! C_J^{j+1} \sigma_* r_c^{2-j}$, with constants depending on the fixed radius. Fixing a coarse-field or free input uses the column version of the same composition, with the cell volume of that input. Thus all derivatives used here satisfy coefficient bounds on the product.

Step 4: transport of the incoming boundary dependence. A rooted input monomial carries its incoming network and a path from its chosen input root to the complement of D_{in}. The incoming list and its complement are those used in $A_k(b)$. By hypothesis the complement of D_{in} lies in H up to the fixed stencil, so Lemma 19.8 implies (19.35) with its fixed buffer, and hence the comparison assumed in the statement. We scale these paths by B^{-1} to output units and join every network of the chart at its fine input label. The union is connected and contains every external coarse-field and free argument. If the input root lies in Ω, its path to the complement of D_{in} reaches H after at most one additional leg of fixed length. If it lies in a newly removed region, the chart branch joining it to a root in Ω already reaches the boundary. Additional components of H can only shorten the required distance to the boundary.

To make this precise, let ℓ_0 be the length of the incoming network in input units, d_0^{in} the distance of the chosen root to the complement of D_{in}, and ℓ_β the lengths of the branches of the chart and of the free inverse in output units. Let A be the set of external positions and e any root in Ω. There exists a constant C_{dim}, depending only on the dimension and on the comparison of elementary paths, and fixed before J, such that

$$\begin{aligned} a_{\mathrm{out}}\mathsf{t}(A) + \alpha_{\mathrm{out}}\delta(e) \le C_{\mathrm{dim}}(a_{\mathrm{out}}+\alpha_{\mathrm{out}})\Big(\ell_0/B + \sum_\beta \ell_\beta\Big) \\ + C_{\mathrm{dim}}\alpha_{\mathrm{out}} d_0^{\mathrm{in}}/B + (a_{\mathrm{out}}+\alpha_{\mathrm{out}})L'_J(k+1). \end{aligned} \tag{19.42}$$

Indeed, the scaled incoming network and the chart branches alone join all external positions, and adding the scaled path to the complement of D_{in} also reaches the boundary. The first network bounds $\mathsf{t}(A)$, and the second bounds the distance of e to the boundary; this gives the separate coefficients of ℓ_0 and d_0^{in} in (19.42). Here k is the number of attached input factors, and the last term accounts for their fixed displacements by the stencil. To see why C_{dim} depends only on the dimension, we take shortest paths in the graph metric and their connected union. Using a tree on the external vertices increases the length by at most a factor two, by traversing each edge of a Steiner tree twice. Replacing segments by the fixed paths in coordinate order increases it only by the fixed comparison constant of the three coordinate directions. Scaling is exact, and neither operation depends on the number of scales. An offset depending on the scale changes only the additive term with L'_J, whose exponential $e^{CL'_J(k+1)}$ is absorbed in the factorial amplitude C_J^{k+1} on the fixed construction domain. More explicitly, we choose the working radii so that the larger unperturbed radii are at least ζ times the corresponding working radii, for a fixed $\zeta > 1$. Then a monomial of degree k gains the factor ζ^{-k}. We reduce c_J so that $e^{c_J L'_J} \le \zeta^{1/4}$, and the accumulated offset factors $e^{c_J L'_J(k+1)}$ are absorbed by these factors. This choice is made before the later profiles and widths, and it does not change C_{geom}.

We take $C_{\mathrm{geom}} \ge 8C_{\mathrm{dim}}$, increased according to the fixed allocation of exponents among the external network, the route to the boundary, and the internal sums, and we

keep part of the remaining exponent in each chart branch for its density sum $S_J(\varepsilon)$. Under the three strict inequalities of part (4), the incoming factors $e^{a_{\mathrm{cl}}\ell_0}$ and $e^{\alpha_{\mathrm{cl}}d_0^{\mathrm{in}}}$, together with the exponent c_J of the unperturbed branches, dominate the output network and boundary weights in (19.42), and a positive part of the exponent is left for the summations. The exponent in the bound on the image under the reconstruction majorant from Lemma 19.7 is included in the choice of c_J before these rates are specified. The allocation is applied to the whole network, so the exponent is not divided by the Taylor order or by the number of scales. This proves the boundary-localized versions of (19.40) and (19.41) at every allowed output rate.

Step 5: the joint stationary point. We now verify the domain conditions, working first in u near zero. By the unperturbed bound on the third derivative and (19.37), we have

$$\|D_u^2 F_0(c, z_0 + S_0 u; \xi) - I\| \le C_1 \|u\|_{\mathrm{coeff}}.$$

Fix $u_0 > 0$ inside the joint product with $C_1 u_0 < 1/8$, and fix $\sigma_{0,J} > 0$ with $C_P \sigma_{0,J} < 1/8$ and $C_P \sigma_{0,J} r_c < u_0/8$. Then $u \mapsto u - D_u \mathcal{F}_\lambda$ is a coefficient contraction with factor at most $1/4$ on the domain of radius u_0, and it maps the ball of radius $u_0/2$ into itself. Picard iteration gives a unique holomorphic joint stationary point, and from the fixed point inequality we obtain

$$\|u_\lambda\|_{\mathrm{coeff}} \le 2 C_P \sigma_* r_c. \tag{19.43}$$

The stationary point is real at real data, since $\mathcal{F}_\lambda$ is real at real arguments and the stationary point is unique.

Step 6: initial segments in the original coordinates. To treat initial segments, we return to the original free coordinates by the invertible affine map $u = S_0^{-1}(z - z_0)$ at fixed c and ξ. The pullback is

$$\mathcal{Q}_\lambda(c, z; b) = F_0(c, z; \xi) + \lambda \overline{P}\left(\mathrm{Log}[(\exp \Psi_R(c, z; \xi))^{G^{\Omega}(c)}]; b\right). \tag{19.44}$$

All subscripts referring to initial segments refer to this expression, with the original lists of scales. Because the affine map and its inverse have fixed coefficient bounds, (19.40) holds in the original coordinates with larger fixed constants. We take the solution domain of every initial segment from Lemma 19.3 small enough that its image under this affine map stays in the domain of radius u_0.

At an unperturbed conditional stationary point $a_I(c, y; \xi)$, the preconditioned perturbed gradient is

$$A_I^{-1} \mathcal{Q}_{\lambda,I}(c, a_I, y; b) = \lambda A_I^{-1} V_I(c, a_I, y; b),$$

whose coefficient norm is at most $I_0 C_P \sigma_* r_c$. This holds uniformly for every $y \in \mathcal{T}_I$, not only at the joint stationary point, since all these conditional reference points lie in the

image of the same product under the reconstruction majorant. If δ_V is defined for the pullback of V as in Lemma 19.3, the same calculation yields

$$\delta_V \le I_0 C_P \sigma_* r_c. \tag{19.45}$$

Across a conditional displacement of size r_I, the derivative bound is at most

$$\|A_I^{-1}(\mathcal{Q}_{\lambda,II} - A_I)\| \le I_0(C_1 r_I + C_P \sigma_*). \tag{19.46}$$

We first take r_I with $I_0 C_1 r_I < 1/8$, and then reduce $\sigma_{0,J}$ so that, for every initial segment,

$$I_0 C_P \sigma_{0,J} < 1/8, \qquad I_0 C_P \sigma_{0,J} r_c < r_I/8.$$

These inequalities are stronger than the two inequalities required in Lemma 19.3(2). We also require that the error of the real Hessian is less than $\lambda_J/8$. The coefficient resolvent condition is imposed at the allocated working rates of Step 4, using the unperturbed resolvents as references, as detailed in Step 8 below. The larger construction products, their radii, and the unperturbed half of the resolvent margin are fixed first; after J and these spatial rates are fixed, all perturbative constants in this condition are fixed before $\sigma_{0,J}$ is decreased and before the later widths and enclosing radii are chosen. Then Lemma 19.3 applies on each domain $\mathcal{T}_I$ of coarser coordinates, and its estimate (19.13) implies that, for every such y,

$$\|a_{\lambda,I}(c,y;b) - a_I(c,y;\xi)\|_{\mathrm{coeff}} \le 2\lambda I_0 C_P \sigma_* r_c \le C_J \sigma_* r_c. \tag{19.47}$$

This is the conditional bound of part (2), with one common $C_J \ge 2I_0 C_P$, and it holds before any moving coarser coordinates are substituted. The mixed coefficient bounds in the original coordinates follow from (19.41) and the affine inverse; together with the free inverse they verify (19.14) on the same products.

Step 7: successive substitution. The recurrence (19.15), with the conditional and matrix-function bounds just obtained, gives a constant $C_{\mathrm{seq},J}$ for a fixed rough bound on the size of the perturbation. Fix $Q_{0,J} > 0$ and, if necessary, reduce $\sigma_{0,J}$ further, so that

$$C_{\mathrm{seq},J}(2I_0 C_P \sigma_{0,J} r_c + Q_{0,J}) < \frac{1}{2}\varepsilon_{\mathrm{tube}}, \qquad Q_{0,J} \le r_c,$$

where the conversion between the original and the normalized coordinates is included in the minimal margin. The bound (19.45) verifies the condition of Lemma 19.3(3) for every y. The recurrence bounds every conditional reference point and every Newton iterate, not only the final solution, so the hypothesis on the image under the reconstruction majorant holds throughout the construction. This proves the statements on stationary points in part (2), for all initial segments simultaneously, on the product of increments.

Step 8: the Hessians. The unperturbed real Hessian in the original coordinates is at least $7\lambda_J/8$ on these products, since its variation from the Hessian at the identity is at most $\lambda_J/8$ by Theorem 18.12, and the perturbation changes it by less than $\lambda_J/8$. Its principal blocks and Schur complements have the same lower bound, since for a symmetric matrix $H \geq \lambda I$

$$v^{\mathsf{T}}(H_{yy} - H_{yI}H_{II}^{-1}H_{Iy})v = \min_{x}(x,v)^{\mathsf{T}}H(x,v) \geq \lambda|v|^2,$$

as in (18.6). After the normalization by $M^i \geq 1$, we obtain $K_{\lambda,i} \geq \lambda_J I/2$. The upper bounds on the tensors and on the inverses give a common upper bound Λ_J on the same product.

The complex resolvent condition is used at an allocated working exponent b_{w}, with strict reserve for the network and boundary weights of part (4). This exponent is smaller than b_{ref} and is within the transported incoming-rate allocation of Step 4. For each flat reference K_i^b, including the factor M^i when present, the unperturbed bounds give

$$\|(K_i^b + t)^{-1}\|_{b_{\mathrm{w}}} \leq \frac{C_{0,J}}{\lambda_J + t} \qquad (t \geq 0).$$

Put $L_i = K_{\lambda,i} - K_i^b$. At each fixed coarser argument on the conditional products, the fixed-reference Picard bounds of Steps 5 and 6 give the coefficient displacement of the conditional center. Together with Step 4 and the unperturbed mixed tensor bounds, they bound the perturbative matrix difference at b_{w} by $C_{P,b_{\mathrm{w}},J}\sigma_*$. For a Schur block, telescope its defining product and use the principal inverses already constructed. This estimate is obtained before successive substitution. Since $b_{\mathrm{w}} < b_{\mathrm{ref}}$, monotonicity preserves the unperturbed half-margin on the fixed products. Fix the perturbative constants at the working rates and then reduce $\sigma_{0,J}$ so that

$$C_{0,J}C_{P,b_{\mathrm{w}},J}\sigma_{0,J} \leq \lambda_J/8.$$

The two halves give $C_{0,J}\|L_i\|_{b_{\mathrm{w}}} \leq \lambda_J/4$ at every stage. With $R_i^b(t) = (K_i^b + t)^{-1}$, the ordered Neumann expansion

$$(K_{\lambda,i} + t)^{-1} = \sum_{n\geq 0}[-R_i^b(t)L_i]^n R_i^b(t)$$

converges in the coefficient norm at this same working exponent and gives $4C_{0,J}/[3(\lambda_J + t)]$. The bounds with an analytic input fixed follow by inserting the differentiated tensor in this normally convergent series on a fixed smaller product. Thus the inverse uses the intrinsic decay of the unperturbed resolvent and the allocated working decay of the incoming perturbation.

Step 9: the differences carry the perturbation. It remains to prove that the differences carry the incoming boundary dependence, including through the moving coarser arguments. At fixed y, differentiating the gradient equation of an initial segment, we obtain

$$\partial_\lambda z_{I,\lambda} = -\mathcal{Q}_{\lambda,II}^{-1} V_I. \tag{19.48}$$

The free inverse just constructed is a coefficient series at the allocated working rates without boundary weight, and V_I is the boundary-localized series of Steps 3 and 4. The majorant of the product is (19.4), and it is $O_J(\sigma_* r_c)$; integrating in λ proves the bound on the difference of stationary points. Let Z_λ be the stationary point at which a Hessian is evaluated, and let H denote here this Hessian of $\mathcal{Q}_\lambda$. Its total derivative is

$$\dot{H} = V_{zz} + \mathcal{Q}_\lambda^{(3)}[\dot{Z}_\lambda],$$

which is $O_J(\sigma_*)$ and boundary-localized. For a set L of finer eliminated coordinates, the Schur block of scale i is $S = H_{ii} - H_{iL} R H_{Li}$ with $R = H_{LL}^{-1}$, and its derivative is

$$\dot{S} = \dot{H}_{ii} - \dot{H}_{iL} R H_{Li} - H_{iL} R \dot{H}_{Li} + H_{iL} R \dot{H}_{LL} R H_{Li}. \tag{19.49}$$

Every term has exactly one factor which carries the perturbation, including the term coming from the derivative of the inverse. Multiplication by M^i gives the derivative of $K_{\lambda,i}$. The derivative of the inverse is $-K^{-1}\dot{K}K^{-1}$, and (19.7), or its differentiated integral, gives the derivative of the inverse square root, with integrable majorant $C_J t^{-1/2}(\lambda_J + t)^{-2}\|\dot{K}\|$. These identities prove the stated bounds on all differences of matrix functions.

Along the successive chain, the derivative in (19.48) becomes

$$\dot{z}_I = -\mathcal{Q}_{\lambda,II}^{-1}(V_I + \mathcal{Q}_{\lambda,II^c}\dot{z}_{I^c}).$$

An inserted increment has total derivative

$$\frac{d}{d\lambda}[a_{\lambda,i}(c, z_{>i}) + K_{\lambda,i}(c, z_{>i})^{-1/2} q_i] = \partial_\lambda a_{\lambda,i} + D_y a_{\lambda,i}\dot{z}_{>i} + \frac{d}{d\lambda} K_{\lambda,i}^{-1/2} q_i,$$

where the last derivative includes both the explicit change of the objective and the change of the coarser argument. We start at the coarsest conditional stationary point and descend. Each direct derivative carries the perturbation, and each new contribution from the coarser arguments contains a factor which is already known to carry it. By (19.49) and (19.7) this remains true for every inverse square root used at the next scale. Since $\|q_i\|_\infty \le Q_{0,J} \le r_c$, the finite recurrence gives $O_J(\sigma_* r_c)$ for the stationary points and the reconstructed displacements, and $O_J(\sigma_*)$ for their matrix coefficients. Differentiating in a coarse-field or free input fixes the index of that input in these convergent series, and the allocation of Step 4 keeps the perturbation and every external argument in one connected network. This proves part (3) on the product, and together with Step 4 it proves part (4).

Step 10: two perturbations. For two perturbations we interpolate between them. Since the coefficient bounds are convex and the fixed bound $\sigma_{0,J}$ holds for both, every interpolated objective stays in the same domains. Its derivative in the interpolation parameter is given by (19.48) with $V = (\overline{P}_1 - \overline{P}_0) \circ \Phi_0$. We set

$$\Delta_{\mathrm{in}} = \mathcal{X}^{r_c}_{D_{\mathrm{in}},a_{\mathrm{cl}},\alpha_{\mathrm{cl}}}(\overline{P}_1 - \overline{P}_0), \qquad \phi_{\mathrm{in}} \equiv r_c. \tag{19.50}$$

Since both perturbations use the same D_{in}, the same positions and cell volumes, and the same incoming boundary, all the preceding difference estimates are linear in Δ_{in}. We write $\mathcal{Q}_\lambda$ also for the interpolated objective. The derivative of the outgoing values in the interpolation parameter is

$$\partial_\lambda[M^s \mathcal{Q}_\lambda(c, z_\lambda; b)] = M^s(\overline{P}_1 - \overline{P}_0)(\Phi_0(c, u_\lambda; \xi); b).$$

Its nonconstant coefficient networks contain the same incoming difference, also when a derivative hits the stationary point. The fixed ratios of the radii, the composition bound, and (19.24) convert them into

$$\mathcal{X}^{\phi_{\mathrm{cl}},\mathrm{cl}}_{\Omega,a_{\mathrm{out}},\alpha_{\mathrm{out}}}(V_1^{\mathrm{out}} - V_0^{\mathrm{out}}) \le K_J^{\mathrm{cl}} \Delta_{\mathrm{in}}. \tag{19.51}$$

Here the outgoing classical seminorm uses the output profile ϕ_{cl} with interior radius r_c. The seminorm in (19.50) is the incoming seminorm with constant profile. The output seminorm includes degree one. A difference of incoming constants stays in the full value and has zero seminorm. This proves part (5).

Step 11: the full action. We prove the identity of part (6), with its constants. When Ψ_R is written on its regional list, ι_ξ inserts the newly frozen coordinates once; when it is written as a complete fine field, the same insertion is already included. In either notation, the exact identities of Lemma 19.5 at the endpoints imply the equality of complete group fields

$$\exp(\iota_\xi \Phi_0) = (\exp(\iota_\xi A_0))^{G^\Omega(c)},$$

since the basepoint and the fine endpoints of every based transport of a newly frozen coordinate are protected, and the gauge transformation is the identity on the previously frozen extension. For real data both complete fields lie in the common chart of Theorem 11.6: the field before gauge transformation by the choice of R_{raw}, and the gauge-transformed field by the bound on the image under the reconstruction majorant and the bound on the newly frozen coordinates. The gauge transformation is periodic and compact there. Thus Theorem 11.6 applies to the torus classical action, and only to it, and we obtain

$$\begin{aligned} \mathcal{F}_1 &= E_{d_0}(\iota_\xi \Phi_0) + P(\Phi_0; b) - S^{\mathrm{cl}}_\xi(\Phi_0) \\ &= E_{d_0}(\iota_0 \Phi_0) + P(\Phi_0; b) = E_{\mathrm{in}}(\Phi_0; b). \end{aligned}$$

These equalities keep $S^{\mathrm{cl}}_{\tilde{\zeta}}(0)$ as well as all nonconstant coefficients of $S^{\mathrm{cl}}_{\tilde{\zeta}}$; no constant Taylor coefficient has been subtracted. Both sides are holomorphic on the connected product, and they agree on its real open part. Expanding at a real interior point and differentiating the real identity in each coordinate, we find that all Taylor coefficients of the difference vanish. Continuation through overlapping polydiscs in the connected product gives the equality everywhere on it.

If the reference extension of the effective interaction is changed, the corresponding difference is treated by Lemma 18.9. In particular, replacing a resulting interaction $R(v, c)$ by $R(v, c) - R(0, c)$ changes the normalization constant from Λ to $\Lambda - R(0, c)$ and preserves the density; the constant of the extension is not deleted. The perturbation has been evaluated on Φ_0 throughout.

Step 12: dependence on the frozen coordinates. Fix the geometry, the finite lists of labels, and a strictly smaller analytic product, and suppose that the incoming coefficients and the charts are Borel functions of the real parameter b. A finite Taylor partial sum of any chart or incoming function is then jointly Borel in its analytic arguments and b. Since the common coefficient bounds on the larger radius give a geometric tail on the smaller product, uniformly in b, the limits of the series are jointly Borel. For the scalar incoming function this argument is applied to its part of positive degree, and the unrestricted coefficient $p_0(b)$ is kept as a separate Borel summand.

Every finite-dimensional matrix inverse used in a Picard iterate is a quotient of polynomials in its entries with nonzero determinant on the domain, hence Borel there, and each Picard iterate is a composition of such operations and of the jointly Borel functions just obtained. The uniform contraction ratio bounds the distance of the iterates to the stationary point by a geometric sequence, uniformly in b, which implies that the stationary point is jointly Borel. This applies first to the joint problem and then to every conditional problem, at arbitrary coarser arguments in the domains, and the finite successive substitution of these maps is Borel.

For matrix functions we use (19.5) and (19.6). Finite Neumann sums are Borel, and their uniform geometric limit gives the resolvent. On a compact interval of integration inside $(0, \infty)$ the Riemann sums of the integral for the inverse square root are Borel, and the uniform bounds on the gap and the resolvent make the tails at zero and at infinity small, uniformly in the parameters; hence the integral is Borel. The same argument applies to the differences and to the logarithmic factors on their fixed nonvanishing branches. Evaluating the full objective at the stationary point gives Borel minimized values, with the incoming constant restored exactly.

If the coefficients and charts are continuous in b, every finite partial sum, inverse, and Picard iterate in this argument is continuous. The common geometric bounds and contraction ratios make all limits uniform on the smaller products, and the tails of the integrals satisfy the same uniform bounds. The limits and the finite successive

compositions are therefore continuous in b. This proves part (7) without any derivative in ξ or b_{rec}.

Step 13: shorter compositions. For $s < J$ we replace B by B_s in the scaling and use the separate conditions on the radius and on the image under the reconstruction majorant for that composition; the proof for the initial segments is unchanged, with s in place of J. All constants were computed on fixed larger products, before the eventual free, enclosing, and geometric choices. □

SECTION 20

Fluctuation integration and normalization on block regions

In Sections 18 and 19 we set up the averaging steps of one renormalization group step on a block region. There, the coordinates attached to a union Ω of coarse blocks are analytic variables, and the coordinates on the fixed complement of Ω are frozen at real values chosen earlier. Those sections constructed the conditional constrained minimizers of the classical problem on Ω, proved a lower bound on the constrained Hessian independent of the shape of Ω, and controlled the effect of a small perturbation of the classical objective. The goal of this section is to integrate out the normalized fluctuation variables of such a step. We prove that the resulting effective interaction differs from the corresponding torus quantity only by terms that decay exponentially with the distance to the fixed complement, and we keep track of the normalization constant Λ by a separate argument. All estimates are uniform in the shape and topology of Ω, the number of its components, the preceding depth, the number of renormalization group steps already performed, and the values of the frozen coordinates.

Section 20.1 fixes the notation and the coefficient calculus. We then compare the conditional Hessians and Haar densities with their torus counterparts (Section 20.2) and derive the conditional fluctuation potential and its concentration bounds (Section 20.3). We obtain the effective interaction by a cluster expansion (Section 20.4) and control Λ by a separate comparison of real integrals (Section 20.5). Next, we choose the parameters and prove the finite iteration (Theorem 20.14). Finally, we sum the truncation errors for couplings in $(0, \Gamma]$ with any finite Γ (Section 20.7).

We use the norms and the dependence of constants of Section 4, the concentration estimates of Theorem 5.4, and the triangular change of variables and the finite graph inequality of Theorems 12.6 and 12.13. From Section 18 we use the norm with boundary decay, the network estimates, and the substitution identities, and from Section 19 the conditional domains, image bounds, perturbation theorem, and column bounds. The results are used in Section 21, which compares the integrals over the regions of the large-field decomposition with the analytic integrals constructed here, in Section 22, which proves the exponential-moment estimates for blocked actions, and in Section 23, which chooses the parameters.

All statements of this section concern analytic functions on stated domains and real integrals over stated products. We prove in Section 21 that the configurations of the large-field decomposition lie in these domains.

Heuristics. We first explain the orders of magnitude. We write the free logarithmic coordinates of one intermediate scale as $z = z^* + gK^{-1/2}w$, where z^* is the conditional constrained minimizer, K the constrained Hessian, g the coupling, and w the normalized fluctuation variable. Since the classical action enters the density divided by g^2, and its expansion around the minimizer has no linear term and a quadratic term equal to $|w|^2/2$, the remaining classical terms are

$$g^{-2}\,O\big((z-z^*)^3\big) = O(g|w|^3).$$

The Haar disintegration density, the logarithmic determinants of the higher scales, and the incoming effective interaction are not divided by g^2, and their dependence on w starts with a term of order $g|w|$. On fluctuation balls of radius $p \geq 1$, the interaction is therefore small when $g[p^3 + (1+H)p]$ is small, where H bounds the incoming effective interaction; this is the smallness condition of Theorem 20.9. The terms of first order in g are odd under $w \mapsto -w$. Their expectations under a symmetric reference measure vanish, and the fluctuation correction to the effective interaction is of order g^2. The conditional measure with the frozen coordinates held fixed is interacting and need not be symmetric. For this reason, we bound the means of the fluctuation variables separately in Lemma 20.5.

We control the normalization constant by two identities. First, a missing scalar Gaussian coordinate at an intermediate scale with coupling g_i removes one factor g_i from the density. This contributes $\log(1/g_i)$ to the difference between the regional and the torus normalization. Second, fixing coordinates at values c uses

$$e^{\Lambda - R(v,c)} = e^{\Lambda - R(0,c)} e^{-[R(v,c) - R(0,c)]},$$

which moves the value at $v = 0$ into Λ and leaves a centered function. Since the norms of the effective interaction vanish on constants, they cannot control Λ, and we estimate Λ by comparing real integrals.

Finally, fixing the coordinates of a boundary layer affects the boundary part of the effective interaction in two ways. First, the norm of the old boundary discrepancy is bounded by $2e^{-\alpha_{\mathrm{bd}}(s_{\mathrm{er}}-1)}$ times its previous norm, where s_{er} is the width of the frozen layer. Second, fixing coordinates at nonzero values creates a new mixed term from the torus interaction, of size at most a constant times $E\beta_{\mathrm{rec}}/\rho$, where E bounds the torus interaction and β_{rec} bounds the frozen values. The first estimate needs a wide layer, the second needs small frozen values, and neither replaces the other.

Attribution. The architecture used here, namely constrained minimization in a gauge-fixed chart, separation of a background field from normalized fluctuations, and integra-

tion of the fluctuations by a cluster expansion, is Bałaban's renormalization group for lattice gauge theories [18, 19]. We attribute the individual tools where they are used.

Order of the proofs. No conclusion is used before it is proved. In particular, Lemma 20.5 uses only the coefficient, product, and curvature bounds of Lemma 20.3, the density sum (20.1), and Theorem 5.4. It determines the constant C_1 in a hypothesis of Lemma 20.3, and the logarithmic radii of Theorem 20.14 enter it only as a sufficient condition for an inclusion.

20.1 Setting and coefficient calculus

Geometry and labels. Fix an odd integer $M \geq 3$, an integer $J \geq 1$, and $B = M^J$. We consider a composition of s averaging steps, $1 \leq s \leq J$, from a lattice of side $M^s m$ to a lattice of side $m \geq 4$, together with an arbitrary union Ω of blocks of the output lattice. When $s = J$, we speak of a full step; it is one renormalization group step. The intermediate scales are numbered $i = 0, \dots, s-1$. A label is a scalar coordinate: it specifies a direction, a color, an intermediate scale (or the output lattice), and the lattice on which the coordinate was introduced, and it has an assigned position. A fluctuation block is the three-component vector $w_{i,e} \in \mathbb{R}^3$ of normalized fluctuation variables attached to one free link e at one intermediate scale i; it is a single variable of the reference measures below, not three independent ones. Scalar labels and fluctuation blocks are counted separately.

The labels attached to Ω are unfixed: their coordinates are analytic variables. These are the coarse background variable $v = \operatorname{Log} c$ on the output lattice and the free logarithmic coordinates z_i at the intermediate scales. We write Ω^c for the fixed complement of the unfixed positions, W for the boundary (the interface between the unfixed positions and Ω^c), and $\Omega^c_{L_J}$ for the fixed enlargement of Ω^c determined by the construction stencils, as in Lemma 18.15. The fixed complement is denoted H in Section 18 and Section 19: in their notation, $\Omega^c = H$ and $\Omega^c_{L_J} = H_{L_J}$. In this section the letter H denotes instead the bound $E + \mathcal{X}^\rho(f)$ on the incoming effective interaction introduced in Lemma 20.7. All distances are graph distances on the refined position graph of Section 18.2, in units of the output lattice, and network lengths use its periodic ℓ^1 metric, with internal vertices allowed; positions of labels introduced on finer lattices are embedded in these units. We write d_g and t_g for this distance and network length, and d_E, t_E for the periodic Euclidean distance and network length of Definition 4.6. Then

$$d_E \leq d_g \leq \sqrt{3}\, d_E, \qquad \mathsf{t}_E \leq \mathsf{t}_g \leq \sqrt{3}\, \mathsf{t}_E,$$

since $|y|_2 \leq |y|_1 \leq \sqrt{3}|y|_2$ for $y \in \mathbb{R}^3$, applied to each periodic image and to each segment $[p, q]$ of a network, with $y = p - q$. Consequently, a coefficient bound stated

with Euclidean distances and network lengths, at spatial exponent c and boundary exponent c', holds with the graph quantities at exponents $c/\sqrt{3}$ and $c'/\sqrt{3}$. Conversely, a bound stated with the graph quantities implies the Euclidean bound at the same exponents. Every imported Euclidean coefficient bound is converted in this way before the rates of this section are chosen, and the fixed factor $\sqrt{3}$ is included in the geometric constants. Throughout this section, dist and t denote d_g and t_g, as in Section 18.2. The torus seminorm $\mathcal{B}_{\rho,a_{\mathbb{T}}}$ keeps the Euclidean convention of Definition 4.6. For an unfixed label i we write $d_\Omega(i)$ for its distance to Ω^c. We write Ω' for the set of unfixed scalar output labels. Separately, let $\partial_{\mathrm{blk}}\Omega$ be the set of anchors of retained final blocks that share a face with a final block of the fixed complement. This is a geometric set on the output lattice, counted once per anchor, regardless of directions, colors, or the presence of an output link. In particular it includes an isolated retained block, even though that block has no internal coarse bond and contributes no label to Ω'. It is empty when Ω or its fixed complement is empty. We assume the label-density bound

$$\#\{y : R \le \mathrm{dist}(x,y) < R+1\} \le v_J (R+2)^3,$$

where v_J accounts for directions, colors, intermediate scales, and the lattices on which labels are introduced.

Frozen coordinates. The frozen coordinates are written $b = (\xi, b_{\mathrm{rec}})$. Here ξ consists of the coordinates frozen at the end of the preceding step, written in the lattice on which they were frozen, and b_{rec} consists of the older frozen coordinates, each on the lattice on which it was frozen. Throughout this section the frozen coordinates are real and fixed, and only the unfixed background and free variables are differentiated or continued to complex values. The supremum over the frozen values is taken after every coefficient sum and every degree sum. We claim no derivative of a perturbed quantity in ξ or b_{rec}.

Norms. For a function $f(v; b_{\mathrm{rec}}) = \sum_\nu f_\nu(b_{\mathrm{rec}}) v^\nu$ of the unfixed labels, a radius $\rho > 0$, a positive profile $\phi \le \rho$, a spatial exponent $a_{\mathrm{sp}} > 0$, and a boundary exponent $\alpha_{\mathrm{bd}} > 0$, we use the coefficient norm with boundary decay of Definition 18.1, which is the graph-metric variant of (4.11):

$$\mathcal{X}^{\phi}_{\Omega}(f) = \sup_{b_{\mathrm{rec}}\ \mathrm{real}} \rho^{-2} \sum_{k\ge 1} \sup_{i\in\Omega} e^{\alpha_{\mathrm{bd}} d_\Omega(i)} \sum_{|\nu|=k} \frac{\nu_i}{k} |f_\nu(b_{\mathrm{rec}})| e^{a_{\mathrm{sp}} \mathrm{t}(\mathrm{supp}\,\nu)} \phi^\nu.$$

Here $\mathrm{t}(S) = \mathrm{t}_g(S)$ is the ℓ^1 length of a shortest connected network containing the positions in S, internal vertices allowed, and d_Ω is the graph distance, as in Definition 18.1. The supremum over the root i precedes the degree sum, and the supremum over the frozen values follows it. Constants and functions on an empty set of unfixed labels have norm zero, and the degree-one term is included. When Ω^c is empty, a boundary difference has zero nonconstant part and norm zero. We write $\mathcal{X}^{\rho}$ for the norm with the uniform

profile $\phi \equiv \rho$, and $\mathcal{X}^{\mathrm{cl}}$ for the classical version, which has its own radius ρ_{cl}, profiles, and exponents. For a torus function r with $Dr(0) = 0$ we use

$$\mathcal{B}_{\rho,a}(r) = \sum_{k \geq 2} \frac{\rho^{k-2}}{k!} \| D^k r(0) \|_{a,\mathrm{root}}$$

from Definition 4.6. This seminorm does not control the value $r(0)$, which is unrestricted. The three profiles are

$$\phi(i) = \begin{cases} \rho_\partial, & d_\Omega(i) < w_{\mathrm{bd}}, \\ \rho, & d_\Omega(i) \geq w_{\mathrm{bd}}, \end{cases} \qquad \psi_{\mathrm{prof}} = \sqrt{K_*}\phi, \qquad \widehat{\phi} = K_*\phi,$$

where $K_* \geq 2$ is a fixed analytic-radius enlargement factor, $0 < \rho_\partial \leq \rho$, and w_{bd} is the width of the boundary layer with the smaller radius. We call $\widehat{\phi}$ the outer profile, ψ_{prof} the intermediate profile, and ϕ the working profile. A bound on a coefficient tensor is rooted at an external index if it holds with that index fixed and all other indices summed, where for an input index the volume-normalized column factor $a_{\mathrm{out}}^3 / a_{\mathrm{in}}^3$ of Definition 4.6 is included. A coefficient bound with each external index fixed in turn is a bound rooted at every external index. All coefficient bounds concern absolute Taylor coefficients; we never substitute for them a bound on values of a holomorphic function or on an evaluated Hessian.

Admissible data. We work on the products of Lemmas 19.3 and 19.7: the conditional solution neighborhoods of the finitely many stages of the conditional minimization, the smaller neighborhoods of the higher parameters, and the product of sequential free coordinates contained in all of them. On these products, the images under the absolute reconstruction majorant satisfy strict bounds for the classical and quantum radii. A coarse background, a frozen value, or a product of fluctuation balls is called admissible if it lies in these products and satisfies these strict image bounds and the inequality on the conditional forces in Lemma 19.3.

Dependence of constants. Unless stated otherwise, every constant in this section (C_J, c_J, λ_J, radii, and bounds) has the dependence described in item (3) of Definition 4.14. It depends only on M, J, the fixed larger analytic domains and their radius ratios, the spatial and boundary exponents, the label density v_J, and the fixed torus and perturbation bounds. It does not depend on L, γ, m, N, K, the preceding depth d_0, the number of renormalization group steps already performed, the shape or topology of Ω, the number of components of Ω or Ω^c, or the values of the frozen coordinates. It is also independent of the widths chosen later (the width of the boundary layer, the width s_{er} of the frozen layer, the widths of protected neighborhoods, and the size of the cells of the large-field decomposition): every constant is evaluated on the fixed larger domains before these widths are chosen. We assert no bound uniform as $J \to \infty$. Constants in statements about summed errors have the dependence described in item (4) of Definition 4.14.

Density sums and boundary transport. Spatial sums use the label density in units of the output lattice. For $b > 0$ we use

$$S_J(b) = C v_J \sum_{l \geq 0} (l+1)^3 e^{-bl}, \qquad \sup_x \sum_y e^{-b\,\mathrm{dist}(x,y)} \leq S_J(b). \tag{20.1}$$

The constant C accounts for directions, colors, and the finitely many intermediate scales. This is the periodic subset estimate of Lemma 18.15; restricting any of the sets of labels only decreases the sum. Kernels between different meshes keep the volume-normalized column factors of Definition 4.6. For an unfixed root x and a position u in $\Omega^c_{L_J}$ we have

$$d_\Omega(x) \leq d_\Omega(u) + \mathrm{dist}(x,u) + C_J, \tag{20.2}$$

where $d_\Omega(u) = 0$ if $u \in \Omega^c$ and C_J accounts for the fixed enlargement. The weights for network length and for distance to Ω^c are both bounded by one fixed part of the available spatial exponent. The fixed stencil offset in (20.2) contributes a factor e^{bC_J} to the coefficient bound and no factor depending on a later width. We always use the version of Lemma 18.15 with every external index rooted. It performs all contractions with this single allocation of the exponent, independently of the number of vertices and of the Taylor order, instead of reducing an exponent at each contraction.

For a proper nonempty block region, we also use the boundary sum

$$\sum_{u \text{ unfixed}} e^{-c\, d_\Omega(u)} \leq C_J(c) |\partial_{\mathrm{blk}} \Omega|, \qquad c > 0. \tag{20.3}$$

Suppose first that Ω and its fixed complement are nonempty. A shortest route from an unfixed position to the fixed complement crosses an exposed face of a retained final block. The anchor of that block belongs to $\partial_{\mathrm{blk}}\Omega$ and is within a fixed distance C_{face} of the crossing point, in output units. The constant includes the fixed convention for link midpoints and endpoints. Thus each unfixed label u can be assigned to an anchor $a(u) \in \partial_{\mathrm{blk}}\Omega$ with

$$\mathrm{dist}(u, a(u)) \leq d_\Omega(u) + C_{\mathrm{face}}.$$

The density sum (20.1), applied also at these anchor positions, gives

$$\sum_u e^{-c d_\Omega(u)} \leq e^{cC_{\mathrm{face}}} \sum_{a \in \partial_{\mathrm{blk}}\Omega} \sum_u e^{-c\,\mathrm{dist}(u,a)} \leq e^{cC_{\mathrm{face}}} S_J(c) |\partial_{\mathrm{blk}}\Omega|.$$

An anchor need not itself carry a variable; the same density bound applies at every position of the refined graph, after a fixed enlargement of v_J if needed. This proves (20.3), without a regularity or component assumption on the boundary. If Ω is empty there is no label to sum. If its fixed complement is empty, boundary differences vanish and the bound is used only for those zero terms.

We also record the comparison with the interface labels used in the normalization count. For a fixed construction width $w_{\mathrm{band}} \geq 1$, let N_{band} count the scalar labels of the regional step whose positions are within that width of W, and set it to zero when W is empty. Then

$$|\partial_{\mathrm{blk}}\Omega| \leq N_{\mathrm{band}} \leq C_J|\partial_{\mathrm{blk}}\Omega|. \tag{20.4}$$

For the upper bound, a position within w_{band} of W is within $w_{\mathrm{band}} + C_{\mathrm{face}}$ of a boundary-block anchor. The label-density bound on this fixed-radius ball proves the claim. For the lower bound, choose one exposed face of each boundary block and a finest-scale M-block touching that face. There is a finest-scale free edge on the exposed face of this smaller block. Indeed, for the coordinate-ordered tree of Definition 18.6, edges in direction 1 are tree edges only on the line where coordinates 2 and 3 equal those of the root, and edges in direction 2 are tree edges only in the plane where coordinate 3 equals that of the root. On a face normal to direction 1, take an edge in direction 2 with coordinate 3 different from that of the root; on a face normal to direction 2 or 3, take an edge in direction 1 off the root line. Such edges exist since $M \geq 3$. Each lies inside a finest-scale block, so it is not an eliminated edge, which joins two such blocks. Its midpoint is within one input mesh unit, hence within $1/B_s \leq 1$, of a link of W. One fixed color component of this edge is therefore counted by N_{band}. The choices for distinct final blocks are distinct scalar labels. This gives the lower bound, including isolated final blocks. All constants depend only on M, J, v_J and the fixed construction width, not on any later width of an excluded or frozen neighborhood.

Two series. For $0 \leq q < 1$ and integers $j \geq 0$ we have

$$\sum_{k\geq 0}(k+1)(k+2)q^k = \frac{2}{(1-q)^3}, \qquad \sum_{k\geq j}(k)_j q^{k-j} = \frac{j!}{(1-q)^{j+1}}, \tag{20.5}$$

where $(k)_j = k(k-1)\cdots(k-j+1)$. They follow by differentiating $\sum_{k\geq 0} q^k = (1-q)^{-1}$ twice and j times, respectively, on compact subintervals of $[0,1)$. In particular a tensor of order $k+2$ with factorial bound $C^{k+2}(k+2)!$ becomes, after division by $k!$, a summable series of degree-k Hessian coefficients. Additional fixed external indices contribute fixed powers of $k+1$, which are absorbed by passing to one strictly smaller auxiliary product. No smaller radius is chosen at a later renormalization group step.

Coefficient resolution. The following lemma makes precise how the connected networks of coefficients behave under the operations used below. It combines the calculus of coefficientwise majorants of Lemma 5.3 with Definition 18.1. For a tensor V with output index $o(V)$ we write v_i for the cube of the mesh of the lattice of an index i, and $|V|_b$ for the maximum of its output-row norm and its volume-normalized input-column norms at spatial exponent b, as in Definition 4.6.

Lemma 20.1. *Consider a finite family of typed analytic coefficient tensors. Each tensor is supplied with a coefficient resolution whose total variation, weighted by the exponentials of the lengths of its assigned connected networks, satisfies the bounds of Definition 4.6 with each external index fixed in turn on one common larger product. Consider a sum of externally rooted connected contraction diagrams built from these tensors by substitution, contraction at shared indices (represented by equality tensors), finitely many Schur complements of positive constrained free matrices, and positive implicit or Neumann series. Products of factors with disjoint fluctuation supports and no shared index are not connected; they form separate connected components, each with its own network. Assume that every substitution $F \circ G$ is made on a product whose absolute image majorant for G lies strictly inside the coefficient domain of F, that all diagram majorants, including the mesh conversions of part (4), converge absolutely there, and that the inverses and Schur operations satisfy the imported uniform gap and coefficient Neumann conditions on that product. Then the following hold.*

1. *The sum has an exact coefficient resolution indexed by the occurrences of its constituent tensors, its contracted indices, and its derivative partitions. In each connected diagram component, the network of a resolved occurrence is the union of the assigned networks of its constituent occurrences, joined at the shared indices. This union is connected, contains every external position of that component, and*

$$\mathtt{t}(\textit{external positions of the component}) \leq \sum_{\textit{constituent occurrences}} \ell(\textit{assigned network}).$$

2. *The monomial of a resolved occurrence has an exact multi-index. Its fluctuation support is the set of whole fluctuation blocks on which its multi-index is positive. Background positions need not belong to this support, and no enlargement of the support to a connected geometric set is made.*
3. *If one constituent occurrence of a connected diagram component has a network that visits $\Omega^c_{L_J}$, then the union visits $\Omega^c_{L_J}$, and the visit lies in the same connected network as every external position of that component.*
4. *Fix an external root of a connected diagram component. For a vector factor V attached to its parent at an input index j use the counting bound $A_V(j) = (v_j / v_{o(V)})|V|_b$, and if it is attached at its output use $A_V(o(V)) = |V|_b$; for a scalar factor use its counting-root bound. Then the sum over the resolution of the absolute coefficients, weighted by the exponentials of the network lengths, is bounded by the product of these bounds, multiplied at the end by the external normalization v_o / v_{root} when the component has an output o. For a substitution tree the volume ratios cancel exactly and the bound is the product of the norms $|\cdot|_b$, in the row norm and in every volume-normalized column norm. For other diagrams the remaining ratios are kept in the scalar majorant. In particular, a counting scalar outer tensor on an old mesh, rooted at a new input l, carries the factor v_l / v_{old}.*

5. *If a connected network of length ℓ contains the external positions S and the unfixed index i, and visits $\Omega^c_{L_J}$, then $\mathsf{t}(S) \le \ell$ and $d_\Omega(i) \le \ell + L_0$, where L_0 is the fixed constant of the enlargement, and consequently*

$$e^{a\mathsf{t}(S)+\alpha d_\Omega(i)} \le e^{\alpha L_0} e^{(a+\alpha)\ell}.$$

Proof. We first work with finite Taylor polynomials and finite diagrams. For a substitution $U(o;\mathbf{l}_1,\dots,\mathbf{l}_k) = \sum_{j_1,\dots,j_k} T(o;j_1,\dots,j_k)\prod_r S_r(j_r;\mathbf{l}_r)$, the union of the network of T with the networks of the S_r, joined at the indices j_r, connects every external position, so

$$\mathsf{t}(o,\mathbf{l}_1,\dots,\mathbf{l}_k) \le \mathsf{t}(o,\mathbf{j}) + \sum_r \mathsf{t}(j_r,\mathbf{l}_r).$$

We take absolute values before summing. For an output row we sum the inner factors by their rows and then the outer tensor by its row. If a fixed external input belongs to the factor S_r, we sum the other inner factors by their rows, use the input column of S_r, and use the j_r-column of T. Along this path the volume factors multiply to $(v_o/v_{j_r})(v_{j_r}/v_l) = v_o/v_l$, which is exactly the normalization of the column of the composite at l. This proves the cancellation for substitution trees. Products with a common output are treated in the same way: the networks meet at the output, and an input-rooted estimate uses the column of the factor carrying the input and the rows of the others.

For a general connected diagram, we replace every edge outside a spanning tree by two independently summed copies of its index, which only increases the nonnegative sums. We orient the tree from the factor containing the fixed external root and, starting at the leaves, sum each factor over all its indices except the one attached to its parent. A factor attached at an input index j contributes at most its volume-normalized column bound times $v_j/v_{o(V)}$, since the column norm carries the factor $v_{o(V)}/v_j$; this is the counting bound $A_V(j)$. A factor attached at its output contributes its row bound. Finally summing the root factor over all its indices except the fixed root, we obtain the product of the counting bounds, with no count of the internal indices. The external normalization is applied only at the end. In a substitution tree the ratios on the path from the root input to the output multiply to v_l/v_o and cancel against this normalization; in a general diagram they are kept. Connectedness is essential: factors with disjoint supports and no shared index have no tree, and their networks remain separate. A cycle without an external index, such as an undifferentiated trace, is not bounded in this way.

For compositions we use the ordered derivative formula

$$D^N(F\circ G)(0)[u_1,\dots,u_N] = \sum_{n=0}^{N} \frac{1}{n!} \sum_{(B_1,\dots,B_n)} F_n\big[G_{|B_1|}[u_{B_1}],\dots,G_{|B_n|}[u_{B_n}]\big],$$

where $G(0) = 0$, the inner sum is over ordered partitions of $\{1,\dots,N\}$ into nonempty blocks, and $F_n = D^nF(0)$, $G_k = D^kG(0)$. If $G(0) \neq 0$, we write $G = G(0) + \widetilde{G}$ and

apply this formula to $\widetilde{G}$, with F_n replaced by $D^n F(G(0))$; the strict image condition justifies this translation and the infinite sums over the constant terms. Dividing by $N!$, we obtain the coefficient of s^N in $\widehat{F}(\widehat{G}(s))$, where $\widehat{F}, \widehat{G}$ are the absolute majorants. Fixing an external index chooses its branch and then its position in that branch, as in $(\widehat{F} \circ \widehat{G})' = (\widehat{F}' \circ \widehat{G})\widehat{G}'$; no second Taylor factorial occurs. We define the resolution in part (1) by retaining a separate signed occurrence for every summand of this formula, every tuple of contracted indices, and every network refinement of the constituent tensors; grouping occurrences can only decrease the total variation. Its monomial is the product of the constituent monomials, which proves part (2). In a connected component the union of networks contains the visit of a constituent occurrence and is connected, which proves part (3). The diagram bound above, applied to every resolved occurrence, proves part (4) for finite truncations; the gap and Neumann conditions cover the Schur complements, Neumann series, and implicit series. Finally, by the image condition the nonnegative majorant sums converge on the common product. Tonelli's theorem then justifies every grouping, contraction, substitution, and derivative, and the resulting analytic functions agree because their coefficients agree.

For part (5), the network contains every external position, so $\mathrm{t}(S) \le \ell$. A route in the network from i to the visited position $h \in \Omega^c_{L_J}$ has length at most ℓ, and at most L_0 more length reaches Ω^c. We conclude that $d_\Omega(i) \le \ell + L_0$, and adding the two inequalities with weights a and α completes the proof. □

When we apply Lemma 20.1, we sum an internal index by keeping a positive part b of the connecting exponent and using (20.1). We convert a visit to $\Omega^c_{L_J}$ by (20.2) into decay in the distance of every unfixed root to Ω^c. In the applications below, the mixed-mesh contractions are substitutions, chains of matrices on a single lattice, and the composition with the incoming interaction, where the remaining ratio is the fixed factor $\Gamma_J \le M^{3J}$ of Step 4 of the proof of Lemma 20.7; these factors are included in the common majorants.

20.2 Conditional minimizers, constrained Hessians, and Haar densities

We order the intermediate scales from the finest, $i = 0$, to the coarsest, $i = s - 1$. A stage of the conditional minimization is given by an initial segment I of the conditional integrations in this order. At that stage the free coordinates of the scales in I are minimized, with the coordinates $y = z_{I^c}$ of the remaining scales held fixed. The fluctuation increments in the following theorem are the coupling-scaled fluctuation inputs, $\zeta_i = q_i = g_i w_i$ in the notation of Section 19.

Theorem 20.2. *Fix an odd integer $M \ge 3$, an integer $J \ge 1$, $B = M^J$, the larger analytic domains and their strict margins, the label-density bound v_J, and the bounds for the classical and quantum torus norms. Use the conditional solution neighborhoods, the smaller neighborhoods*

of the higher parameters, and the product of sequential free coordinates of Lemma 19.3, and the perturbed objective of Theorem 19.9. In joint coordinates this objective is

$$\mathcal{F}(v,u;b_{\mathrm{rec}}) = F_0\big(v, z_0(v;\xi) + S_0(v;\xi)u;\xi\big) + \bar{P}\big(\Phi_0(v,u;\xi);\xi,b_{\mathrm{rec}}\big), \qquad S_0 = K_0^{-1/2}. \tag{20.6}$$

Here $v = \operatorname{Log} c$ is the coarse background variable, z_0 is the unperturbed joint constrained minimizer, K_0 is its constrained Hessian, u is the joint normalized free variable, and Φ_0 is the unperturbed reconstruction followed by the gauge transformation $G^{\Omega}(c)$ of Lemma 19.5, which depends only on c and is the identity at all protected endpoints. The function $\bar{P} = P - S^{\mathrm{cl}}_{\xi}$ is the incoming perturbation of the classical action with the classical boundary-completion difference S^{cl}_{ξ} subtracted, including its classical constant, as in Theorem 19.9. Assume

$$\mathcal{X}^{\mathrm{cl}}(\bar{P}) \le 2\sigma_*, \qquad 0 < \sigma_* \le \sigma_0,$$

where the preliminary perturbation bound $\sigma_0 > 0$ satisfies the margins of Lemma 19.3 for the preconditioned conditional forces, the Hessians, and the images. Here $\mathcal{X}^{\mathrm{cl}}$ is the incoming classical norm on the incoming set D_{in} of unfixed labels, with the uniform profile ρ_{cl}, as in Theorem 19.9. Assume the inequality on the conditional forces of Lemma 19.3, and assume that the incoming fixed complement lies on the current fixed side, in the sense that for every current unfixed position e and every incoming label i,

$$d_{D_{\mathrm{in}}}(i) \ge B_s d_{\mathrm{current}}(e) - B_s \operatorname{dist}(e, i/B_s) - C_{\mathrm{buf},J}. \tag{20.7}$$

Here $B_s = M^s$, $d_{D_{\mathrm{in}}}$ is measured in input units, $d_{\mathrm{current}} = d_{\Omega}$, i/B_s is the position of i in output units, and $C_{\mathrm{buf},J}$ is a fixed buffer that includes the chart stencils. Assume finally that both images of the incoming functions under the absolute reconstruction majorant, for the classical radius and for the quantum radius, are strictly smaller than half of that radius, as in Lemma 19.7. Derivatives are taken at fixed real (ξ, b_{rec}).

Then the following statements hold for every preceding depth $d_0 \ge 0$, every $1 \le s \le J$, every output side $m \ge 4$, every union Ω of output blocks, and every admissible real value of the frozen coordinates.

1. *Every stage of the conditional minimization has a holomorphic conditional constrained minimizer, and every scale has a positive constrained Hessian $K_i^{\Omega,P}$, $0 \le i < s$, which includes the exact factor M^i of its scale. Writing the unnormalized fluctuation increments as $\zeta_i = g_i w_i$, the free logarithmic coordinates at scale i are*

$$z_i = z_i^*(c_{i+1};b_{\mathrm{rec}}) + K_i(c_{i+1};b_{\mathrm{rec}})^{-1/2}\zeta_i,$$

where c_{i+1} is the reconstruction from the coarser scales, with their fluctuation increments inserted. There are $Q, a_1, \alpha_1, \lambda_J > 0$ and $C_J, \mathcal{L}_{K,J} < \infty$ such that, on the outer profile $\widehat{\phi}$ and for $|\zeta_{i,e}| \le Q$, the real matrices satisfy

$$\tfrac{1}{2}\lambda_J I \le K_i^{\Omega,P} \le \mathcal{L}_{K,J} I, \qquad \|(K_i^{\Omega,P} + t)^{-1}\|_{\mathrm{row,col},a_1} \le \frac{C_J}{\lambda_J + t} \quad (t \ge 0). \tag{20.8}$$

The complex continuation satisfies the same resolvent bound with a fixed larger C_J.

2. *Adjoin to the regional problem, as identical independent auxiliary variables with unit Hessian, the free labels of the matched torus problem that are absent from it, so that both problems have the same sets of free labels. Let $K_i^{\mathbb{T},\mathrm{cut}}$ be the restriction of the matched torus Hessian to the unfixed free labels, extended by the identity on the auxiliary labels and by zero on the mixed blocks. If*

$$\Delta_i(v,\zeta;b_{\mathrm{rec}}) = K_i^{\Omega,P} - K_i^{\mathbb{T},\mathrm{cut}} = \sum_{\nu,\kappa} \Delta_{i;\nu\kappa}(b_{\mathrm{rec}}) v^\nu \zeta^\kappa,$$

then

$$\sup_{u \text{ unfixed}} e^{\alpha_1 d_\Omega(u)} \sum_{v',\nu,\kappa} |\Delta_{i;uv',\nu\kappa}(b_{\mathrm{rec}})|\, \widehat{\phi}^\nu Q^{|\kappa|} e^{a_1 \mathfrak{t}(u,v',\mathrm{supp}\,\nu,\mathrm{supp}\,\kappa)} \le C_J. \tag{20.9}$$

The transposed bound holds, and so do the bounds rooted at any unfixed background or free external index, with the volume-normalized column factors of Definition 4.6. The underlying mixed tensors with k additional background or free derivatives have bounds $C_J^{k+2}(k+2)!$ at every external index, on a fixed larger product. The auxiliary blocks have zero defect. If Ω^c is empty, the defect has zero nonconstant part.

3. *The same rooted comparisons with boundary decay hold for the conditional minimizers of every stage, the inverse square roots, the logarithms of the Haar disintegration densities, and the diagonal densities $(\log K_i)_{xx}/2$, after composition with the reconstruction from every value of the coarser fluctuation variables. In particular, let L_i be the negative logarithm of the Haar disintegration density at scale i after its field-independent normalization has been moved into Λ, and put*

$$L_{0,i}(c_{i+1}) = L_i\big(c_{i+1}, z_i^*(c_{i+1})\big), \qquad B_i = L_{0,i} + \tfrac{1}{2}\log\det K_i.$$

The fluctuation-dependent part of the density contains

$$\sum_{i=0}^{s-1} \big[L_i(c_{i+1},z_i) - L_{0,i}(c_{i+1}) + B_i(c_{i+1}) - B_i(\widehat{c}_{i+1})\big], \tag{20.10}$$

where $\widehat{c}_{i+1}$ is the reconstruction with every fluctuation increment equal to zero. These differences start at degree one in ζ, whereas the classical remainders start at degree three. The comparison includes the correction $\log(T_{II}) - (\log T)_{II}$ that arises when the logarithm of a torus matrix T is restricted to the list I of unfixed labels. A torus diagonal density that is absent from the regional problem and has an unfixed external index has a network visiting Ω^c and is summable from that index; its constant part remains an explicit sum over omitted labels.

4. *All constants have the dependence described in item (3) of Definition 4.14, including independence of the later widths. They are evaluated on the fixed larger products before the final radius for the free coordinates is decreased. In particular they do not change at the renormalization group steps following the creation of a new component of* Ω^c.

Proof. Steps 1 and 2 verify that the incoming perturbation satisfies the hypotheses of Theorem 19.9 on all parameter domains. Steps 3 to 5 construct the conditional minimizers by a contraction argument that is uniform in the higher parameters, and check that every sequential substitution stays in the admissible domains. Steps 6 to 9 compare the regional and torus quantities, first without and then with the perturbation. Step 10 treats the matrix functions and the Haar densities, Step 11 the triangular determinant, and Step 12 the dependence of the constants on the frozen values.

We fix the real frozen value $b = (\xi, b_{\mathrm{rec}})$ and write $\mathcal{Q}_\theta = F_0 + \theta V$, where $V = \bar{P} \circ \Phi_0$ and $0 \le \theta \le 1$; here θ is an interpolation parameter. Returning from u to $z = z_0 + S_0 u$ is an invertible affine change of variables at fixed v. In the free logarithmic coordinates, the objective is

$$\mathcal{Q}_\theta(v,z) = F_0(v,z;\xi) + \theta \bar{P}\big(\mathrm{Log}[(\exp \Psi_R(v,z;\xi))^{G^{\Omega}(v)}];\xi, b_{\mathrm{rec}}\big),$$

where Ψ_R is the reconstruction before the gauge transformation. We use only this change of variables; we do not need the incoming perturbation to take the same value at the reconstruction before and after the gauge transformation.

Step 1: the incoming hypotheses. The incoming analytic labels form the set D_{in}, with their positions on the lattices where they were introduced and their volume factors. On this set the incoming profile is uniformly ρ_{cl}; the coordinates ξ are fixed parameters, and the spatially varying profiles are output profiles. The strict bounds on the image under the reconstruction majorant place Φ_0, on every admissible stage, inside this uniform input ball with a margin of half the radius. The same holds for every segment between two compared images, since coefficientwise a convex combination is bounded by the larger of the two absolute image bounds. In particular, the input jet bounds of Theorem 19.9 apply before any conditional minimization, on all its parameter neighborhoods.

Step 2: the incoming boundary comparison. For example, the inclusion of fixed complements used in Theorem 19.9 implies (20.7) as follows. Let h be an incoming fixed position nearest to i. A path from e to the current fixed complement can first go to i/B_s, then follow the scaled image of a shortest path from i to h, of length $d_{D_{\mathrm{in}}}(i)/B_s$, and finally cross the fixed buffer between the incoming and the current fixed complements, whose length is included in $C_{\mathrm{buf},J}/B_s$. Multiplying by B_s, we obtain (20.7). This argument uses no regularity or component assumption on the boundary. We keep (20.7) as a hypothesis on the geometry; choosing small radii does not establish it.

In particular, if $\alpha_{\mathrm{out}} \le B_s \alpha_{\mathrm{cl}}$ and a chart branch has spatial rate $b > \alpha_{\mathrm{out}}$, then

$$e^{\alpha_{\mathrm{out}} d_{\mathrm{current}}(e)} e^{-\alpha_{\mathrm{cl}} d_{D_{\mathrm{in}}}(i)} e^{-b\,\mathrm{dist}(e,i/B_s)} \le e^{\alpha_{\mathrm{out}} C_{\mathrm{buf},J}/B_s} e^{-(\alpha_{\mathrm{cl}}-\alpha_{\mathrm{out}}/B_s) d_{D_{\mathrm{in}}}(i)} e^{-(b-\alpha_{\mathrm{out}})\,\mathrm{dist}(e,i/B_s)}.$$

This is the transport of the input boundary weight used in the perturbation bound. Input network lengths are divided by B_s when expressed in output units. Column normalizations remain the ratios of the cubed mesh sizes of their output and input labels. Local rates may be chosen below the finite minimum of these bounds over $1 \le s \le J$; the rates returned after a full step use $s = J$ and $B_s = B$. If there is no incoming boundary, the incoming nonconstant boundary function is zero.

Step 3: conditional minimizers. For each stage I (an initial segment of the conditional integrations, with $y = z_{I^c}$ fixed), let $a_I(v, y)$ be the unperturbed conditional minimizer. By Lemma 19.3, there exist a fixed solution neighborhood, a smaller neighborhood of the higher parameters, and an invertible matrix $A_I = F_{0,II}(v, a_I, y)$. Let I_0 be a common bound for the coefficient norms of the inverses A_I^{-1} on the fixed larger neighborhoods. Using the uniform incoming jet bound with $j = 1$ and the fixed rows of the chart derivatives, we obtain, at every admissible higher parameter y and not only at the joint minimizer,

$$\|V_I(v, a_I, y; b)\|_{\mathrm{coeff}} \le C_P \sigma_* \rho_{\mathrm{cl}}, \qquad \delta_V \le I_0 C_P \sigma_* \rho_{\mathrm{cl}}, \tag{20.11}$$

where δ_V is the bound on the preconditioned conditional forces of Lemma 19.3. The constant C_P is evaluated on the larger products at the preliminary bound σ_0 and includes the conversion from z to u. The jet bound with $j = 2$ and the chain rule, including the term $\bar{P}' \Phi_{0,zz}$, give a uniform bound $C_P \sigma_*$ on the mixed second derivatives. The real gap, the complex coefficient margins, the inequality $\delta_V \le r_I/4$, where r_I is the radius of the solution neighborhood, and the derivative inequality below are then the hypotheses of Lemma 19.3. On that neighborhood the map

$$h \longmapsto h - A_I^{-1} \mathcal{Q}_{\theta,I}(v, a_I + h, y)$$

has derivative norm at most $1/2$ and value at zero of norm at most $r_I/4$. Hence it maps the closed ball of radius $r_I/2$ into itself, and its successive iterates converge: the difference of the iterates $n+1$ and n is at most 2^{-n} times the first difference. The same bounds on absolute coefficients hold for the analytic iterates, which gives normal convergence on every smaller product and a holomorphic limit. Uniqueness follows by subtracting two fixed point equations. Since the objective is holomorphic and real at real arguments, the minimizer is real at real data. The real Hessian and complex resolvent margins are those of Theorem 19.9; its weighted resolvent bounds rest on the exponential conjugation method of Combes and Thomas [61]. Taking the finite minimum over $s \le J$ and over the stages fixes λ_J and Q, and taking the finite maximum fixes $\mathcal{L}_{K,J}$ and the bound on the inverses. The factors M^i belong to these matrices before the extrema are taken.

Step 4: sequential substitution. For the sequential construction we use the inequality

$$C_{\mathrm{seq},J}(2\delta_V + Q) < \tfrac{1}{2}\varepsilon_{\mathrm{tube}}, \tag{20.12}$$

where $\varepsilon_{\mathrm{tube}}$ is the minimal margin of the parameter and solution neighborhoods of Lemma 19.3. By the contraction bound, the displacement from $a_I(v, y)$ is at most $2\theta\delta_V$, for every admissible y. The uniform mixed coefficient bounds and the bound on the free inverses give a finite common Lipschitz constant D_J of the conditional minimizers in y, and $\|K_i^{-1/2}\| \le C_{S,J}$. Let A_i be a bound for the deviations of the already reconstructed coarser coordinates from the unperturbed joint chain. The next insertion satisfies

$$A_{i-1} \le (1 + D_J)A_i + 2\delta_V + C_{S,J}Q, \qquad A_s = 0. \tag{20.13}$$

Iterating, we find that the total deviation is at most $\max\{1, C_{S,J}\}\sum_{k=0}^{J-1}(1 + D_J)^k(2\delta_V + Q)$. We choose $C_{\mathrm{seq},J}$ larger than the coefficient of this finite expression and than every fixed embedding constant of the stages. Then (20.12) keeps every substituted coarser argument in the admissible domains, even after the deviation is doubled. The mixed bounds and the constant $C_{\mathrm{seq},J}$ are fixed before the later widths. In particular the preliminary bound σ_0 must satisfy the same inequality with $\delta_V \le I_0 C_P \sigma_0 \rho_{\mathrm{cl}}$.

Step 5: dependence on the coarser coordinates. We explain why this construction controls coefficients when the coarser coordinates move. Differentiating the stationary equation at fixed y, we obtain

$$\partial_\theta z_I = -\mathcal{Q}_{\theta,II}^{-1} V_I. \tag{20.14}$$

If y moves, the right side has the additional term $-\mathcal{Q}_{\theta,II}^{-1}\mathcal{Q}_{\theta,II^c}\partial_\theta y$. A sequential insertion also has the term $(\partial_\theta K_i^{-1/2})\zeta_i$. All these terms are included in Theorem 19.9, and (20.13) allows their evaluation at every value of the coarser coordinates. At zero free input, conditional uniqueness identifies the sequential chain with the joint minimizer. Decreasing Q therefore does not change the admissible profile of the coarse background.

Step 6: the unperturbed boundary comparison. We use matched reconstruction formulas on the unfixed labels. Away from the fixed construction enlargement of Ω^c, the formulas defining the common unfixed outputs, the block axial gauge equations, and the choices of eliminated edges coincide. Their unique admissible solutions on the unfixed outputs therefore coincide there. This does not assert that the extended fields agree inside a component of Ω^c. The prescribed values in Ω^c are separate bounded sources; a source deep inside a component of Ω^c need not be small at its own position. For the torus action composed with two reconstructions $\Psi_R = \Psi_{\mathbb{T}} + d$, the exact difference is

$$F_R - F_{\mathbb{T}} = \int_0^1 DE_{d_0}(\Psi_{\mathbb{T}} + \tau d)[d]\, d\tau \tag{20.15}$$

on the common unfixed variables. If auxiliary missing variables are kept, their separate auxiliary quadratic form is kept as a term rooted in Ω^c. Every differentiated coefficient of the integral contains either a derivative of a changed formula or the factor d attached to an action tensor. The action and chart tensors on the larger domain are the tensors of Theorem 19.9, and their contractions with a changed formula are networks covered by Lemma 18.15. Rooting at any unfixed external index connects that index to the change. For example, the second derivative is

$$D_z^2(E_{d_0} \circ \Psi) = \Psi_z^{\mathsf{T}} E''_{d_0}(\Psi)\Psi_z + E'_{d_0}(\Psi)\Psi_{zz}.$$

Both terms are present: stationarity in z does not make $E'_{d_0}(\Psi)$ vanish. Differentiating (20.15) produces derivatives falling on E_{d_0} as well as derivatives falling on d; the former remain connected to the inserted change. Summing a bounded source in Ω^c from an unfixed root is justified by (20.1) and (20.2). This proves factorial bounds with a visit to Ω^c for the difference before minimization, rooted at every external index.

Step 7: minimization, Schur complements, and restriction. We pass to the conditional minimizers through their stationary equations. A force carrying a visit to Ω^c is multiplied by the free inverse, as in (20.14). The coefficientwise majorant series of the implicit equation converges on the admissible products: if Y denotes the majorant of the solution without the visit, the variation carrying the visit has the form $H(x, Y)/(1 - A(x, Y))$, where $A \le 3/4$ after the auxiliary radius has been fixed. This follows by substituting the majorant of the variation into the differentiated fixed point equation and summing $\sum_{n\ge 0} A^n$. The visit and all fixed external indices are kept. For a partial minimum the exact second derivative is the Schur complement

$$\mathcal{Q}_{yy} - \mathcal{Q}_{yI}\mathcal{Q}_{II}^{-1}\mathcal{Q}_{Iy}. \tag{20.16}$$

We subtract two such expressions by replacing the three factors of the product one at a time. An inverse difference is $A^{-1} - B^{-1} = -A^{-1}(A - B)B^{-1}$, so each summand keeps a factor with a visit to Ω^c. Only constrained free blocks are inverted. Positivity passes to the Schur complement by minimizing the positive quadratic form over the coordinates in I.

A separate correction compares restriction with elimination. Let A be a single torus free matrix, and let R and H be the lists of unfixed and of missing labels (in this paragraph R and H denote lists of labels). Block elimination gives

$$(A^{-1})_{RR} - A_{RR}^{-1} = A_{RR}^{-1}A_{RH}\big(A_{HH} - A_{HR}A_{RR}^{-1}A_{RH}\big)^{-1}A_{HR}A_{RR}^{-1}. \tag{20.17}$$

To check this identity, we solve $Ax = (f_R, 0)$, first for x_R in terms of x_H and then for x_H. Each term on the right of (20.17) passes through a missing index. Its coefficient network from an unfixed root therefore reaches Ω^c, and the part of the exponential decay not

used for the network weight sums over that index. Adding this correction to the regional comparison, we obtain the unperturbed defect at each stage. This operation compares restrictions of one torus matrix; it does not identify a regional problem with a torus problem whose complement is set to zero.

Step 8: from factorial bounds to coefficient bounds. The bounds just obtained for the tensors of order $k+2$, at every external index, are $C_J^{k+2}(k+2)!$. Dividing by the Taylor factorials and using (20.5), we find that on a fixed smaller auxiliary radius h the summed matrix coefficients are at most

$$C_J^2 \sum_{k\ge 0} (k+1)(k+2)(C_J h)^k = 2C_J^2 (1 - C_J h)^{-3}.$$

We choose the outer background and free products inside this radius and do not change them later. Since any index may be fixed before the sums are taken, we may fix a background or free input, which proves the column versions.

Step 9: adding the perturbation. We insert the unperturbed regional matrix at its own chain of coarser arguments. The defect in (20.9) is the sum of three terms: the change of the objective and of the minimizer at a common coarser argument, the change of that argument, and the unperturbed regional–torus defect. The first term has the rooted perturbation bound of Theorem 19.9, with its visit to Ω^c. The second term is the integral of an unperturbed matrix derivative against the difference of the coarser reconstructions, which carries a visit to Ω^c; both are defined on the same products by the statement on the sequential product in Lemma 19.3, and the network joins the new index to the visit. The third term was treated in Steps 6 to 8. The sum satisfies (20.9) for fixed smaller $a_1, \alpha_1 > 0$. The real matrices are symmetric, which gives the transposed bound with the same constants. Rooting each contraction at the specified background or free index gives the other bounds, with its volume factor. The auxiliary blocks are identical throughout and have zero difference; no matrix element rooted purely at an auxiliary label is assigned a boundary decay.

Step 10: matrix functions and Haar densities. For positive matrices, integrating the scalar resolvent and applying the resolvent identity, we obtain the ordered identities of holomorphic functional calculus (see [27])

$$A^{-1/2} - B^{-1/2} = -\frac{1}{\pi} \int_0^\infty t^{-1/2} (A + t)^{-1} (A - B)(B + t)^{-1} \, dt, \tag{20.18}$$

$$\log A - \log B = \int_0^\infty (A + t)^{-1} (A - B)(B + t)^{-1} \, dt. \tag{20.19}$$

Indeed $A^{-1/2} = \pi^{-1} \int_0^\infty t^{-1/2} (A + t)^{-1} \, dt$ and $\log A = \int_0^\infty [(1 + t)^{-1} I - (A + t)^{-1}] \, dt$ hold for positive numbers, hence for real symmetric positive matrices by orthogonal diag-

onalization. The common complex resolvent margins extend these identities holomorphically. The coefficient majorants of the integrands are $C_J t^{-1/2}(\lambda_J + t)^{-2}$ and $C_J(\lambda_J + t)^{-2}$, which are integrable at both endpoints. We may therefore integrate all contractions and derivatives termwise. Each difference keeps its factor $A - B$.

The logarithm of the Haar disintegration density is a sum of logarithms of the nonvanishing Haar factors of finitely many local formulas and of the determinants of the constraints at the eliminated edges, as in Definition 4.3. On the fixed chart we write each factor as $a(1 + h)$, with its nonzero value a at the base point and a fixed strict coefficient margin $|h| < 1$; the series $\log(1 + h) = \sum_{k\geq 1}(-1)^{k+1}h^k/k$ gives its absolute derivatives on every smaller chart. The base factors are moved into Λ. Away from the boundary the local formulas agree. In a difference of compositions, a changed local formula, or a changed minimizer, inverse square root, or coarser argument, is a factor carrying a visit to Ω^c. The preceding calculus therefore yields the comparison of the Haar densities at all orders.

A diagonal matrix-logarithm density keeps its index until an external unfixed index is chosen. Its nonconstant derivatives are chains of resolvents with a differentiated matrix inserted. Rooting at that insertion makes the trace sum summable by (20.1). If a torus diagonal density is missing from the regional problem, its network with any unfixed external index must reach the missing index, and the same argument sums over the missing index from the unfixed index. A pure constant has no such index and remains its exact sum over omitted labels.

To compare with the restriction of the torus logarithm, we set $T = K_i^{\mathbb{T}}$ and $C = T_{II} \oplus T_{HH}$, where $I = I_i$ and $H = H_i$ are the lists of unfixed and missing free labels at scale i. Using the integral representation of the logarithm, we obtain

$$\log T_{II} - (\log T)_{II} = \int_0^\infty \big[(T + t)^{-1} - (C + t)^{-1}\big]_{II}\, dt. \tag{20.20}$$

The integrand equals $-[(T + t)^{-1}(T - C)(C + t)^{-1}]_{II}$. Its middle factor connects unfixed and missing labels, so it has a visit to Ω^c, and it has the same integrable bound as (20.19). Thus this correction, composed with every value of the coarser fluctuation variables, is included in the comparison.

Step 11: the triangular determinant. The determinant of the triangular change of variables of Theorem 12.13, or of the identical finite triangular calculation for the present minimizers, is

$$\prod_i g_i^{3f_i} \det K_i(c_{i+1})^{-1/2},$$

where f_i is the number of free blocks at scale i. Adding and subtracting the logarithm of its value at zero fluctuation, we combine the Haar terms into (20.10). Only one

determinant occurs. For an analytic function G on the common product,

$$G(v,\zeta) - G(v,0) = \int_0^1 D_\zeta G(v,\tau\zeta)[\zeta]\, d\tau. \tag{20.21}$$

This keeps every positive degree in ζ. Stationarity and the exact normalization by $K_i^{-1/2}$ cancel the terms of degrees one and two of each classical remainder, so its degree is at least three. Composition with the coarser scales does not lower that degree.

Step 12: constants and measurability. The proof uses only margins on fixed domains, finitely many operations on the stages, (20.1), normally convergent coefficientwise majorant series, and the resolvent integrals above. These give the dependence stated in part (4). A new component of Ω^c changes the set of indices carrying a visit and the frozen values, but no constant. If Ω^c is empty, no formula changes and there is no input carrying a visit. This proves the stated special cases and uniformities.

We also record the regularity in the frozen values needed for finite iterations. If the incoming coefficients and the charts depend on the real value b in a Borel way, then so does every iterate of the contraction in Step 3: finite sums, products, and admissible inverses preserve this property, and a convergent coefficient composition is a limit of its polynomial truncations. The geometric contraction bound is uniform in b, so the coefficients of the limit are Borel. The resolvent integrals for square roots and logarithms are limits of finite Riemann sums with uniformly bounded tails, and preserve this property. With continuous incoming data on a fixed branch, the same uniform convergence gives continuity. None of these arguments differentiates in b. In the finite analytic recursion, the initial torus functions and the local formulas in the identity chart have this regularity, and substitution of a frozen value and the integrals over fixed products below preserve Borel dependence. Thus every coefficient family constructed by the recursion depends on the frozen values in a Borel way. □

20.3 The conditional fluctuation potential

Lemma 20.3. *Use the data and products of Theorem 20.2. Let the incoming effective interaction be*

$$R = r_{d_0}(\,\cdot \oplus 0) + f, \qquad Dr_{d_0}(0) = 0, \quad \mathcal{B}_{\rho,a_{\mathbb{T}}}(r_{d_0}) \le E, \quad \mathcal{X}^{\rho}(f) \le \sigma_* \le \sigma_0,$$

where r_{d_0} is the torus effective interaction at depth d_0, with its torus symmetries, evaluated with the fixed slots set to zero, and f is its boundary part. The value of r_{d_0} at zero is unrestricted. For $0 \le i < s \le J$ set

$$t_{\max} = \max_i g_i = \sqrt{\gamma/(Mm)}, \qquad g_i = t_{\max}\lambda_i, \qquad \lambda_i = M^{(i-s+1)/2} \in [\lambda_*, 1],$$

where $\lambda_* = M^{-(J-1)/2}$. Let F_i be the partially minimized objective at scale i, without the coupling factor and with its exact normalization M^i, and define

$$T_i(c_{i+1},\zeta_i) = F_i\big(c_{i+1}, z_i^* + K_i^{-1/2}\zeta_i\big) - F_i(c_{i+1}, z_i^*) - \tfrac{1}{2}|\zeta_i|^2.$$

For a fixed admissible real coarse background and fixed real frozen values, the conditional fluctuation potential, with its values at zero fluctuation moved into the prefactor, is exactly

$$\begin{aligned} V(w) = {} & \tfrac{1}{2}\sum_i |w_i|^2 + \sum_i g_i^{-2} T_i(c_{i+1}, g_i w_i) \\ & + \sum_i \big[L_i(c_{i+1}, z_i) - L_{0,i}(c_{i+1}) + B_i(c_{i+1}) - B_i(\widehat{c}_{i+1})\big] \\ & + R\big(\operatorname{Log} C(v, (g_i w_i)_i; b_{\mathrm{rec}})\big) - R\big(\operatorname{Log} C(v, 0; b_{\mathrm{rec}})\big), \end{aligned} \tag{20.22}$$

where $C(v,\zeta;b_{\mathrm{rec}})$ denotes the reconstructed configuration on the unfixed labels. Only logarithms of unfixed labels in the identity chart are taken. The Haar disintegration density, the degree-one terms of the incoming interaction, and all determinant terms of the coarser scales occur in (20.22). *The separate prefactor contains all constants of the block averaging constraint and of the Haar measure, the Gaussian powers of the couplings, the values at the minimizers, and the external compact factors.*

There are $A \geq 1$, $a' > 0$, and $C_{V,J}, C_0 < \infty$ with $C_0 \geq 2A^3$, and a fixed radius $s_W > 0$ for the increments ζ_i, such that on the product $|g_i w_{i,e}| \leq s_W$

$$V(w) = \tfrac{1}{2}|w|^2 + \sum_{k\geq 3} t_{\max}^{k-2} P_k(w) + \sum_{k\geq 1} t_{\max}^k Q_k(w), \qquad \|P_k\|_{a',1} \leq A^k, \quad \|Q_k\|_{a',1} \leq \mathcal{L}_V A^k, \tag{20.23}$$

where P_k, Q_k are homogeneous polynomials of degree k in the scalar coordinates of w and $\mathcal{L}_V = C_{V,J}(1 + E + \sigma_0) \geq 1$. For $T_k(w) = \sum_{|\eta|=k} t_\eta w^\eta$ the norm is

$$\|T_k\|_{a',1} = \sup_j \sum_{\eta_j > 0} \eta_j |t_\eta| e^{a' \mathrm{t}(\operatorname{supp}\eta)}.$$

The bounds (20.23) *also hold before the background is evaluated, in the coefficient algebra at the outer profile, with the background positions included in the network length.*

The following inequalities suffice for a common enclosing product. All constants below are computed at the preliminary bound σ_0, before σ_*, β_{rec}, r, and the later widths are fixed. Let $r_{q,U}, r_{q,P}, u_0$ be the radii of the unperturbed and perturbed sequential free products and of the joint product of Lemma 19.3, and $c_{\mathrm{tube},J}$ its fixed conversion constant. For $\rho_\bullet \in \{\rho, \rho_{\mathrm{cl}}\}$ let R_{raw}, $C_{\mathrm{lift},J}$, $s_{\mathrm{aux},\bullet}$, $C_{T,\bullet}$, $C_{\Delta,\bullet}$, and $A_{0,\bullet}$ be the radius of the larger construction domain, the bound on the lifts, the auxiliary radii, the constants of the image increments, and the bound on the image at zero fluctuation in Lemma 19.7. Choose $R_W > 0$ strictly below

$$\begin{aligned} \min\big\{ & s_W/2,\ r_{q,U}/2,\ r_{q,P}/2,\ c_{\mathrm{tube},J} u_0,\ R_{\mathrm{raw}}/(64 C_{\mathrm{lift},J}),\ 1/(4A),\ 1/(16 C_0), \\ & \min_{\rho_\bullet \in \{\rho,\rho_{\mathrm{cl}}\}} \{ s_{\mathrm{aux},\bullet}/2,\ 1/(4C_{T,\bullet}),\ \rho_\bullet/[8(2C_{T,\bullet} + C_{\Delta,\bullet})] \} \big\}. \end{aligned}$$

Impose the margins of the larger construction domain and of the stages in Lemma 19.7 and, for both radii, $A_{0,\bullet} \le \rho_\bullet/4$ *and* $C_{T,\bullet}K_*\rho_\bullet \le 1/4$. *Let* $C_{\mathrm{en},J} \ge 1$ *be a fixed constant (in the application it is the constant of the enclosure estimate Theorem 21.8; nothing else from that section is used), and require*

$$r,\ \beta_{\mathrm{rec}},\ \sigma_*\rho_{\mathrm{cl}} \le \frac{\lambda_* R_W}{6C_{\mathrm{en},J}}, \qquad \delta_{\mathrm{en}} = C_{\mathrm{en},J}(r + \beta_{\mathrm{rec}} + \sigma_*\rho_{\mathrm{cl}}), \qquad R_{\mathrm{en}} = 2\delta_{\mathrm{en}}/\lambda_*, \tag{20.24}$$

where r *is the radius of the small-link domain* B_r *and* β_{rec} *is the bound on the frozen values. Then the whole product*

$$\mathcal{K}_{\mathrm{en}} = \prod_{i,e} \{|w_{i,e}| \le R_{\mathrm{en}}/t_{\max}\}$$

is admissible. On it the absolute coefficient increment of either incoming image is bounded by

$$\frac{C_{T,\bullet}R_{\mathrm{en}}}{1 - C_{T,\bullet}(K_*\rho_\bullet + R_{\mathrm{en}})} + C_{\Delta,\bullet}R_{\mathrm{en}} \le \rho_\bullet/8,$$

the denominator is at least $1/2$, *and the image under the absolute reconstruction majorant at spatial exponent zero is at most* $3\rho_\bullet/8$. *If these bounds are imposed in a weighted coefficient row, the same bound holds in that row. In the alternative provided by Lemma 19.7, the unperturbed image row* $S^0_c \le C_J\rho_\bullet$ *at an exponent* $c > 0$, *larger than the working exponent, gives one fixed positive exponent* ϑc *with*

$$S^0_{\vartheta c} \le (3\rho_\bullet/8)^{1-\vartheta}(C_J\rho_\bullet)^{\vartheta} < (\sqrt{3}/4)\rho_\bullet.$$

For the unperturbed construction, this exponent is fixed before the regional rates. For the perturbed reconstruction, impose the output-rate conditions of Theorem 19.9 at the allocated working rates below this exponent and bound its coefficient difference from the unperturbed reconstruction there. Fix the larger-product difference-row constant $C_{P,\bullet}$ *before the final perturbation bound and require* $C_{P,\bullet}\sigma_*\rho_{\mathrm{cl}} \le \rho_\bullet/64$ *for both radii. The total image row is then less than* $(\sqrt{3}/4 + 1/64)\rho_\bullet < \rho_\bullet/2$ *at those rates; no bound for the perturbed image at the original higher exponent is required. Thus the margin of half the incoming radius is an absolute weighted coefficient bound, not only a bound on values.*

Let C_1 *be the constant of Lemma 20.5, which depends only on* A, a', *and* v_J. *For* $0 < t_{\max} \le t_* < e^{-1}$ *impose separately*

$$C_0 t_*^2 \mathcal{L}_V \le 1/16, \qquad C_1 t_*(\mathcal{L}_V + 2) < \frac{1}{4\sqrt{3}}. \tag{20.25}$$

Then on a neighborhood of the whole real product $\mathcal{K}_{\mathrm{en}}$,

$$\|D^2_w(V - |w|^2/2)\|_{\mathrm{row,col},a'} \le \frac{C_0(R_{\mathrm{en}} + t^2_{\max}\mathcal{L}_V)}{(1 - AR_{\mathrm{en}})^3} \le \frac{8}{27} < \frac{1}{2}. \tag{20.26}$$

If $p(t) = A_{\mathrm{w}} \log(1/t)$*, the products of balls with radii* $p_i \in [p(t_{\max}), p(t_{\max}) + A_{\mathrm{w}}(J-1)(\log M)/2]$ *lie in* $\mathcal{K}_{\mathrm{en}}$ *whenever*

$$t_{\max}\big[p(t_{\max}) + A_{\mathrm{w}}(J-1)(\log M)/2\big] < R_{\mathrm{en}}.$$

The cutoff of the small-field event (4.6) *is the case* $p_i = p(t_{\max})$ *for all* i*; the cutoffs* $p_i = p(g_i)$ *at the individual scales are a separate case. All constants have the dependence described in item (3) of Definition 4.14, and are also independent of the later widths, of the frozen values, and of the individual radii of the fluctuation balls.*

Proof. We keep $b = (\xi, b_{\mathrm{rec}})$ fixed and write $t = t_{\max}$. The data of Theorem 20.2 include the uniform incoming classical profile, (20.7), and (20.11) on all parameter neighborhoods. We choose the radius for the increments ζ below Q; then (20.12) holds at every stage. These are the hypotheses for the perturbed tensors used here, and no derivative in the frozen values is needed.

Step 1: the exact formula. At scale i, the minimum at fixed coarser arguments is $F_i(c_{i+1}, z_i^*)$. With $S_i = K_i^{-1/2}$, Taylor's formula with integral remainder gives

$$g_i^{-2} T_i(c_{i+1}, g_i w_i) = \frac{g_i}{2} \int_0^1 (1-\tau)^2 D_z^3 F_i(c_{i+1}, z_i^* + \tau g_i S_i w_i)[S_i w_i, S_i w_i, S_i w_i]\, d\tau. \tag{20.27}$$

The terms of orders zero, one, and two have been, respectively, subtracted, annihilated by stationarity, and normalized to $|g_i w_i|^2/2$ by $S_i^{\mathsf{T}} K_i S_i = I$. The successive partial minima satisfy $F_{i+1} = M \min F_i$, and the couplings satisfy $g_{i+1}^2 = M g_i^2$, so the values at the minimizers telescope in the exponent. In decreasing order of the scales, the map from w to z is triangular with diagonal blocks $g_i S_i$. This proves the classical part of (20.22) and its Gaussian term, by the same finite algebra as in Theorem 12.13. The Haar disintegration density and the single triangular determinant split as in part (3) of Theorem 20.2. Splitting $R(\operatorname{Log} C)$ at its value at zero fluctuation, we obtain the last two terms. All values removed at zero fluctuation, the Gaussian powers, and the constant Haar densities belong to the prefactor; none of them is set to one.

Step 2: the two series. We fix an auxiliary radius $h_0 > 0$ for the increments ζ, strictly inside every admissible stage product and its joint neighborhood. Theorem 20.2 gives tensors with bounds at every external index on a larger product. Dividing by the Taylor factorials, composing coefficientwise majorants (Lemma 20.1), and using (20.5), we bound the degree-k scalar coefficients, rooted at any free index, by $D_J H_J^k$ for fixed D_J, H_J. An additional multiplicity $\eta_j \le k$ is bounded by 2^k. For an incoming function, the strict image bound gives the jet bound

$$\|D^j R\|_{\mathrm{root}} \le (E + \sigma_0)\rho^{2-j} \frac{j!}{(1-\kappa)^{j+1}}, \qquad \kappa \le \tfrac{1}{2}, \quad j \ge 1, \tag{20.28}$$

at a point whose image under the absolute reconstruction majorant is at most $\kappa\rho$. To verify it, we fix the first differentiated input index. At original degree l, the fixed-index weight in the input norm and the sum over the other $j-1$ differentiated positions give the factor $(l)_j$, with the remaining factor $\kappa^{l-j}\rho^{2-j}$, and the second identity in (20.5) sums the result. The torus seminorm starts at degree two and the boundary norm at degree one; the same estimate bounds both. Constants of R disappear only in the fluctuation difference. The input networks are joined to the reconstruction networks, and a prescribed differentiated index stays fixed. This proves (20.28) in the coefficient algebra, before the background is evaluated.

Consider a monomial of T_i, and let η_j be its degree in the free coordinates of scale j. It has at least three indices at scale i, so $\eta_i \geq 3$ and $\sum_{j\geq i}\eta_j = k \geq 3$. After division by g_i^2 its coupling factor is

$$t^{k-2}\lambda_i^{\eta_i-2}\prod_{j>i}\lambda_j^{\eta_j}.$$

The product of ratios is at most one; the weaker bound λ_*^{-2} would also be an admissible fixed constant. By (20.21), every Haar, determinant, and incoming difference has at least one free index. A term of degree k there has the factor $t^k\prod_j\lambda_j^{\eta_j}$, whose product of ratios is again at most one. The values B_i in (20.22) are evaluated at the coarser reconstructions, and their coefficients in the coarser free variables belong to the same series. Summing over at most J scales, increasing $A \geq \max\{1, 2H_J, 2D_JH_J\}$, and decreasing h_0 if necessary, we obtain

$$\|P_k\|_{a',1} \leq A^k, \qquad \|Q_k\|_{a',1} \leq C_{V,J}(1+E+\sigma_0)A^k.$$

The first quantum coefficient Q_1 includes every degree-one term of the incoming interaction. Only powers of the fixed radii were absorbed in these constants. We choose $s_W < h_0$ with a strict margin in the larger domain. The series are then normally convergent on that radius, at the outer profile as well as after evaluation.

Step 3: the enclosing product is admissible. The three inequalities in (20.24) imply that

$$\delta_{\mathrm{en}} \leq \lambda_* R_W/2, \qquad R_{\mathrm{en}} \leq R_W.$$

For $w \in \mathcal{K}_{\mathrm{en}}$ we have $|\zeta_i| = t\lambda_i|w_i| \leq R_{\mathrm{en}}$. The minimum defining R_W places this whole product strictly inside the unperturbed and perturbed sequential products and the joint neighborhood; the constant $c_{\mathrm{tube},J}$ includes every finite sequential embedding. Indeed, the incoming displacement term is still $2\delta_V$ at the preliminary bound σ_0, and the new free radius is at most $R_{\mathrm{en}} \leq Q$. Then (20.13) implies that $C_{\mathrm{seq},J}(2\delta_V + R_{\mathrm{en}}) < \varepsilon_{\mathrm{tube}}/2$ at every coarser argument. The separate margin of the larger construction domain in Lemma 19.7 includes the background, the frozen values, and the free radius; it places all reconstructions before the gauge transformation in the larger charts of the Haar densities and of the action.

Step 4: the image increments. For either radius, we abbreviate $C_T = C_{T,\bullet}$ and $C_\Delta = C_{\Delta,\bullet}$. By hypothesis, we have $C_T(K_*\rho_\bullet + R_{\mathrm{en}}) \le 1/2$. The rational increment in the statement is therefore at most $(2C_T + C_\Delta)R_{\mathrm{en}} \le \rho_\bullet/8$. Adding the bound $A_{0,\bullet} \le \rho_\bullet/4$ at zero fluctuation, we obtain the bound $3\rho_\bullet/8$ at exponent zero. If these inequalities hold in a weighted coefficient row, each calculation holds in that row. Otherwise apply the following interpolation to the unperturbed image row at such an exponent c, with $S_c^0 \le C_J\rho_\bullet$. For nonnegative coefficient weights b_η and network lengths l_η, Hölder's inequality for finite sums gives

$$\sum_\eta b_\eta e^{\vartheta c l_\eta} \le \Big(\sum_\eta b_\eta\Big)^{1-\vartheta}\Big(\sum_\eta b_\eta e^{c l_\eta}\Big)^{\vartheta}.$$

Passing to increasing finite sums proves the stated interpolation. Its right side, divided by $\rho_\bullet$, tends to $3/8$ as $\vartheta \downarrow 0$. Hence one positive ϑ works for both radii and the finitely many stages of the unperturbed construction. Its exponent is fixed before the regional rates are chosen. At allocated working rates below this exponent and satisfying Theorem 19.9, that theorem bounds the coefficient row of the perturbed-minus-unperturbed reconstruction by $C_{P,\bullet}\sigma_*\rho_{\mathrm{cl}}$ on the same enclosing products. Fixing this constant before the final choice of σ_* and making that bound smaller than the remaining half-radius margin gives the asserted perturbed image bound. No interpolation at the unperturbed exponent is applied to inherited perturbative networks.

Step 5: curvature. For the curvature bound, we fix one derivative at the scalar coordinate j and sum over the second differentiated index. Differentiating a homogeneous monomial of degree k twice gives the factor $k-1$ relative to the norm $\|\cdot\|_{a',1}$. The matrix weight is bounded by the weight of the original network, since the distance between the two differentiated indices is at most the network length. At the radius $\varrho = R_{\mathrm{en}}$ in the variables ζ, the two contributions are bounded by

$$\sum_{k\ge 3}(k-1)A^k\varrho^{k-2} = A^2\big[(1-A\varrho)^{-2}-1\big] \le \frac{2A^3\varrho}{(1-A\varrho)^2},$$

$$t^2\mathcal{L}_V\sum_{k\ge 2}(k-1)A^k\varrho^{k-2} = \frac{A^2t^2\mathcal{L}_V}{(1-A\varrho)^2}.$$

Here we used that on $\mathcal{K}_{\mathrm{en}}$ the coordinates satisfy $|w_j| \le R_{\mathrm{en}}/t$, so that $t^{k-2}(R_{\mathrm{en}}/t)^{k-2} = R_{\mathrm{en}}^{k-2}$ for the classical terms and $t^k(R_{\mathrm{en}}/t)^{k-2} = t^2R_{\mathrm{en}}^{k-2}$ for the others. We enlarge $C_0 \ge 2A^3$ to include the fixed conversion between scalar and three-component block norms. The column bound follows from the symmetry of the Hessian. The weaker denominator $(1-AR_{\mathrm{en}})^{-3}$ in (20.26) applies to both contributions. Since $AR_{\mathrm{en}} \le 1/4$, $C_0R_{\mathrm{en}} \le 1/16$, and $C_0t^2\mathcal{L}_V \le 1/16$, the result is

$$\frac{1/16+1/16}{(1-1/4)^3} = \frac{8}{27} < \frac{1}{2}.$$

All inequalities have strict margins in the charts, so they hold on a neighborhood of the real product, including its faces and corners. The constant C_1 of Lemma 20.5 is determined from A, a', the label density, and the curvature bound just proved, so the second inequality in (20.25) is a separate requirement compatible with the present choices.

Step 6: logarithmic cutoffs. A ball of radius p_i lies in $\mathcal{K}_{\mathrm{en}}$ whenever $tp_i < R_{\mathrm{en}}$. The finite logarithmic offset in the statement bounds every such p_i, since $p(g_i) = p(t) + A_{\mathrm{w}} \log(1/\lambda_i)$ and $\lambda_i \geq \lambda_*$. This proves the inclusion both for the common cutoff and for the cutoffs at the individual scales. All constants were computed at σ_0 on fixed larger products, before any final enclosure quantity or width. The proof involves no derivative in the frozen values, no sum over earlier renormalization group steps, and no dependence on the individual radii of the fluctuation balls. □

Remark 20.4. The incoming boundary function f in Lemma 20.3 is measured in the norm $\mathcal{X}^{\rho}$ with the uniform profile ρ, and (20.28) is stated for this case. For a function measured with a nonuniform profile ϕ, the analogous estimate holds for the rescaled function $\widetilde{R}(x) = R((\phi_i x_i)_i)$: if its degree-wise rooted sum at unit profile is at most $H\rho^2$, then at a point whose image under the absolute reconstruction majorant is bounded by $\kappa\phi$, $0 \leq \kappa < 1$,

$$|D^j \widetilde{R}|_{\kappa,\mathrm{root}} \leq H\rho^2 \sum_{l \geq j} (l)_j \kappa^{l-j} = H\rho^2 \frac{j!}{(1-\kappa)^{j+1}}, \qquad j \geq 1.$$

Thus each differentiated slot carries its own radius ϕ_i. The unscaled form (20.28) does not follow for a spatially varying radius profile. Indeed, fix an unfixed index i at distance d_i from Ω^c. The function $R(v) = H\rho^2 e^{-\alpha_{\mathrm{bd}} d_i} v_i/\phi_i$ has norm H and derivative $H\rho^2 e^{-\alpha_{\mathrm{bd}} d_i}/\phi_i$, which is unbounded as the boundary radius $\phi_i = \rho_\partial$ decreases with d_i fixed. In the coefficient sums of Section 20.4 every differentiated background index is multiplied by its own radius, so no inverse power of the small boundary radius ρ_∂ enters.

Lemma 20.5. *The following statements hold.*

1. *Let $\mathcal{K}$ be a finite product of three-dimensional balls, let $S, \mathcal{W} \subset \mathcal{K}$ be Borel sets, let $\chi : \mathcal{K} \to [0,1]$ be Borel, let e^{-V} be a positive Borel function with $0 < Z_{\mathcal{K}} = \int_{\mathcal{K}} e^{-V}\, dw < \infty$, and let $P > 0$ be a constant prefactor (unrelated to the perturbation P of Theorem 20.2). Set*

$$G = P \int_S \chi e^{-V}\, dw, \qquad \widetilde{F} = P \int_{\mathcal{W}} e^{-V}\, dw, \qquad d\nu_{\mathcal{K}} = Z_{\mathcal{K}}^{-1} \mathbf{1}_{\mathcal{K}} e^{-V}\, dw.$$

   If $\nu_{\mathcal{K}}(\mathcal{W}^c) \leq \delta < 1$, then $G \leq (1-\delta)^{-1}\widetilde{F}$. Nonnegative functions of the coarse background and external factors may multiply both sides. No inclusion between $\mathcal{W}$ and S is needed. A product with no factors has mass one.

2. *Let $\mathcal{K} = \mathcal{K}_{\mathrm{en}}$ be the enclosing product of Lemma 20.3, under its hypotheses with $t_{\max} \le t_*$, at an admissible real coarse background and fixed real frozen values, and let V be the potential* (20.22). *There is a constant C_1, depending only on A, a', and v_J, such that*

$$D_w^2 V \ge I/2, \qquad \sup_j |\mathbb{E}_{\nu_{\mathcal{K}}} w_j| \le C_1 t_{\max}(\mathcal{L}_V + 2),$$

where j runs over the scalar coordinates. For every scalar function F that is 1-Lipschitz for the Euclidean norm and every $v \ge 0$,

$$\nu_{\mathcal{K}}(|F - \mathbb{E}F| \ge v) \le 2e^{-v^2/4}.$$

Let $p(t) = A_{\mathrm{w}} \log(1/t)$, and choose radii $p_i \ge p(t_{\max})$ with $t_{\max} \max_i p_i < R_{\mathrm{en}}$, so that the product $\mathcal{W} = \prod_{i,e}\{|w_{i,e}| \le p_i\}$ is contained in $\mathcal{K}$. If $C_1 t_{\max}(\mathcal{L}_V + 2) \le p(t_{\max})/(2\sqrt{3})$, which holds for all sufficiently small $t_{\max}$ and is one of the requirements on the coupling bound g_* in Theorem 20.14, then

$$\nu_{\mathcal{K}}(\mathcal{W}^c) \le 6\mathsf{b} \exp\{-p(t_{\max})^2/48\}, \qquad \mathsf{b} \le C_J m^3,$$

where b counts all fluctuation blocks of the step, jointly over all components of Ω and all scales. These are means under $\nu_{\mathcal{K}}$ at fixed frozen values; no symmetry of the frozen values is assumed. The constants have the dependence described in item (3) of Definition 4.14 and are independent of the later widths.

3. *Consider the stopping sides $m_t = mB^t$, $t \ge 0$, together with at most one additional side $\eta m B^j$, where η ranges over a fixed finite set of positive ratios (arising from a final composition of fewer than J averaging steps). Suppose that nonnegative errors at these sides satisfy*

$$\delta_t \le C_J m_t^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{M m_t/\gamma}}, \qquad z_t \le C_J m_t^3 e^{-b m_t/\gamma}, \qquad a_{\mathrm{w}}, b > 0,$$

and similarly at the additional side. By part (2), the first bound holds with $a_{\mathrm{w}} = A_{\mathrm{w}}^2/48$ for the comparison errors of this lemma. Then for every fixed $\Gamma \in (0, \infty)$, in particular for every $\Gamma = \gamma_0 \le 1$, there is a minimum side m_0, depending only on Γ, the fixed data, and the finite set of ratios, such that, uniformly for $m \ge m_0$ and $0 < \gamma \le \Gamma$, all errors are at most $1/2$ and

$$\sum_t (\delta_t + z_t) < \infty, \qquad \log \prod_t (1 - \delta_t)^{-1} \le 2 \sum_t \delta_t \le C_\delta.$$

The sums and the logarithm tend to zero as $m \to \infty$, uniformly in $0 < \gamma \le \Gamma$ and in the finite or infinite number of sides. The same conclusion holds for $\epsilon_t = \delta_t/(1 - z_t)$ after increasing m_0. There is one comparison for the whole product at each stopping side, not a separate factor for each component of Ω^c. The minimum side and C_δ have the dependence described in item (4) of Definition 4.14; none of them depends on the number of steps or on the large-field decomposition.

Proof. Step 1 proves part (1). Steps 2 to 6 prove part (2): we first check that the measure is well defined, then interpolate between the Gaussian product and the potential, bound the force of the interaction using fourth moments, integrate the derivative of the means along the interpolation, and finally bound the probability outside the chosen product. Step 7 reduces part (3) to Lemma 20.15.

Step 1: the abstract comparison. The Borel hypotheses make both integrands measurable, and since $0 \le \chi \le 1$ and $0 < Z_{\mathcal{K}} < \infty$, we have $0 \le G, \widetilde{F} \le PZ_{\mathcal{K}} < \infty$. Since $S \subset \mathcal{K}$, we have $G \le PZ_{\mathcal{K}}$. On the other hand

$$\widetilde{F} = PZ_{\mathcal{K}}\nu_{\mathcal{K}}(\mathcal{W}) \ge PZ_{\mathcal{K}}(1-\delta).$$

Dividing by the positive finite number $PZ_{\mathcal{K}}$ gives part (1); the common prefactor cancels before any estimate. Multiplying by nonnegative functions of the coarse background preserves the inequality pointwise. For a product with no factors we use the unit point mass, and the same proof applies.

Step 2: normalizability on the enclosing product. The potential of Lemma 20.3 is real and smooth on a neighborhood of the fixed compact product $\mathcal{K}_{\mathrm{en}}$. Its ball radii $R_{\mathrm{en}}/t_{\max}$ are positive and finite, so the product has positive finite Lebesgue volume. If $V_{\min}$ and $V_{\max}$ are the minimum and maximum of V on $\mathcal{K}_{\mathrm{en}}$, then

$$0 < e^{-V_{\max}} \operatorname{vol}(\mathcal{K}_{\mathrm{en}}) \le Z_{\mathcal{K}_{\mathrm{en}}} \le e^{-V_{\min}} \operatorname{vol}(\mathcal{K}_{\mathrm{en}}) < \infty.$$

Thus the integrand is positive, Borel, and normalizable. The chosen product $\mathcal{W}$ is closed, hence Borel; the comparison in part (1) keeps its Borel hypotheses on S and χ.

Step 3: convexity and concentration. We write $t = t_{\max}$ and $H(w) = V(w) - |w|^2/2$, and interpolate on the same product

$$V_\theta(w) = |w|^2/2 + \theta H(w), \qquad 0 \le \theta \le 1.$$

Each V_θ is real and smooth on the compact product, so the normalization bounds of Step 2 hold throughout the interpolation. By (20.26) the Hessian perturbation is at most $\kappa = 8/27$, so every V_θ has Hessian at least $(1-\kappa)I \ge I/2$; for $\theta = 1$ this is the first assertion. The absolute coefficient bound in (20.26) bounds the exponentially weighted sums over y of the suprema over the product of $|(V_\theta)_{xy}|$, in rows and in columns, by the same fixed constant; the three-component block form follows by including the fixed block conversion in C_0. Fix $0 < b \le a'$ so small that

$$\max\Big\{\sup_x \sum_{y\neq x} (e^{b\,\mathrm{dist}(x,y)} - 1) \sup\|(V_\theta)_{xy}\|,\ \sup_y \sum_{x\neq y} (e^{b\,\mathrm{dist}(x,y)} - 1) \sup\|(V_\theta)_{xy}\|\Big\} \le \tfrac{1}{4}.$$

Such a choice depends only on the fixed weighted bound, since the convexity of $u \mapsto e^u$ gives $e^{bd} - 1 \le (b/a')(e^{a'd} - 1)$ for $b \le a'$. The off-diagonal blocks of the Gaussian

part vanish. Thus all hypotheses of Theorem 5.4, which adapts the Bakry–Émery and Helffer–Sjöstrand methods [8, 110] to products of balls with reflecting boundary, hold with convexity constant $1/2$, on a neighborhood of the finite product; its ball faces and product corners are included in that theorem. It yields the scalar tail $2e^{-v^2/4}$ and the covariance bound

$$|\mathrm{Cov}_\theta(F,G)| \le 4\sum_{x,y} e^{-b\,\mathrm{dist}(x,y)}\|\nabla_x F\|_2\|\nabla_y G\|_2, \tag{20.29}$$

where Cov_θ and $\|\cdot\|_2$ refer to the normalized measure with density proportional to e^{-V_θ} on $\mathcal{K}_{\mathrm{en}}$. These constants do not depend on the number or the radii of the balls.

Step 4: the force. Let $M(\theta) = \max_j |\mathbb{E}_\theta w_j|$, where j is scalar (in this proof $M(\theta)$ is unrelated to the blocking factor M). The tail just obtained implies a universal bound on the fourth moment of $w_j - \mathbb{E}_\theta w_j$, by integrating $\int_0^\infty 4u^3\mathbb{P}(|w_j - \mathbb{E}_\theta w_j| > u)\,du$ against the Gaussian majorant. Hence

$$\|w_j\|_4 \le C_4 + M(\theta).$$

Consider a differentiated classical monomial of degree $k \ge 3$. After fixing its differentiated coordinate, we place two of its remaining $k-1$ factors in L^4 and the other $k-3$ factors in the supremum norm R_{en}/t of the product. Hölder's inequality, which applies also when the two factors are the same scalar coordinate, gives the factor

$$t^{k-2}(R_{\mathrm{en}}/t)^{k-3}(C_4 + M(\theta))^2 = tR_{\mathrm{en}}^{k-3}(C_4 + M(\theta))^2.$$

The coefficient sum with the differentiated index fixed is bounded by A^k, and the geometric sum is at most $C(A)t(1 + M(\theta)^2)$ because $AR_{\mathrm{en}} \le 1/4$. For the quantum terms, including degree one, a direct supremum bound gives

$$\sum_{k\ge 1} \mathcal{L}_V A^k t^k (R_{\mathrm{en}}/t)^{k-1} = \frac{\mathcal{L}_V A t}{1 - AR_{\mathrm{en}}}.$$

Combining the two bounds, we obtain

$$\sup_j \|\partial_j H\|_2 \le C(A)t[\mathcal{L}_V + 1 + M(\theta)^2]. \tag{20.30}$$

Step 5: the means. Differentiating the finite integrals on the fixed compact product, we obtain

$$\frac{d}{d\theta}\mathbb{E}_\theta w_j = -\,\mathrm{Cov}_\theta(w_j, H).$$

Indeed, the derivative of the numerator is $-\int w_j H e^{-V_\theta}$ and that of the denominator is $-\int He^{-V_\theta}$, and their quotient is the displayed centered expression. The integrands and their derivatives are bounded on this finite product, so the differentiation is justified

directly. This is the calculation of Theorem 13.10 on a fixed product, carried out here for the present measure; we do not use the conclusions of that theorem, whose hypotheses concern the torus products. Combining (20.29), (20.30), and (20.1), we obtain a fixed C_1 with

$$\Big|\frac{d}{d\theta}\mathbb{E}_\theta w_j\Big| \le C_1 t[\mathcal{L}_V + 1 + M(\theta)^2]. \tag{20.31}$$

It depends only on A, a', and the label density; the margin $\kappa < 1/2$ has already been imposed. This is the constant in the hypothesis (20.25) of Lemma 20.3.

At $\theta = 0$ the measure is a product of centered truncated Gaussian measures on balls, so every coordinate has mean zero, for every frozen value. All means are continuous in θ. Suppose that M reached the value one, and let θ_1 be the first time at which it does. Integrating (20.31) for each coordinate on $[0, \theta_1]$, where $M \le 1$, would give $M(\theta_1) \le C_1 t(\mathcal{L}_V + 2) < 1/(4\sqrt{3}) < 1$, a contradiction. Inserting $M < 1$ into (20.31) and integrating once more proves $M(1) \le C_1 t(\mathcal{L}_V + 2)$. This argument integrates the inequalities for the individual coordinates before taking the maximum; it does not use any differentiability of the maximum.

Step 6: the probability outside the chosen product. The condition $t \max_i p_i < R_{\mathrm{en}}$ puts the chosen product strictly inside $\mathcal{K}_{\mathrm{en}}$. For the logarithmic radii of Theorem 20.14 this condition follows from (20.47); for other radii it is the explicit inclusion hypothesis. If a fluctuation block lies outside its ball of radius $p_i \ge p(t)$, then some scalar coordinate has absolute value larger than $p(t)/\sqrt{3}$. Since the mean of that coordinate is at most $p(t)/(2\sqrt{3})$ in absolute value, the coordinate deviates from its mean by at least $p(t)/(2\sqrt{3})$. By the scalar tail and a union bound over the three coordinates, the probability that a given block lies outside its ball is at most $6e^{-p(t)^2/48}$. The step has at most $\mathsf{b} \le C_J m^3$ blocks: each of the at most J scales has a fixed finite number of free labels per output block, and removing blocks from the region only decreases this number. This proves the bound on $\nu_{\mathcal{K}}(\mathcal{W}^c)$, with no factor depending on the number of components. The enclosure and curvature hypotheses are used on the whole product $\mathcal{K}_{\mathrm{en}}$, which does not depend on the configuration.

Step 7: summation. Part (3) is the case $C = C_J$, $c = b$, $a = a_{\mathrm{w}}$ of Lemma 20.15, with the ratio B and the finite set of additional ratios, since $\log^2 \sqrt{Mm/\gamma} = z^2$ with $z = (1/2)\log(Mm/\gamma)$. With $p(t_{\max}) = A_{\mathrm{w}} \log \sqrt{Mm/\gamma}$, the bound of part (2) is of the stated form with $a_{\mathrm{w}} = A_{\mathrm{w}}^2/48$, since $6\mathsf{b} \le C_J m^3$. □

20.4 Cluster expansion with boundary-localized coefficients

We now expand the logarithm of the fluctuation integral, keeping two structures apart. Probabilistic independence is determined by the set of fluctuation blocks on which a

monomial depends: two monomials depending on disjoint sets of blocks are independent under a product reference measure, even if they depend on the same background coordinates. Spatial decay is carried instead by a connected network through all positions of the coefficient, background positions included, which may lie far from the fluctuation blocks. Overlapping monomials share a fluctuation block, so their networks meet there, and the union of their networks is connected without any additional connecting path.

The organization into connected polymer activities and a convergent logarithm follows the constructive cluster expansion; see Bałaban [18, 19] for its implementation in lattice gauge theories, Brydges [38] for a general exposition, and Kotecký and Preiss [120] for the abstract polymer criterion. The cancellation of connected graphs against rooted trees used in Theorem 12.6 is Penrose's partition [136]; its construction by generations and parents, and the positive tree recursions used below, are described in [87, Sections 2 and 4.1]. The finite graph inequality is imported in the form of [68, Appendix B, (323), p. 42], through Theorem 12.6; no other result of that paper, which concerns a scalar field theory, is used. We prove directly the features needed here: the two kinds of roots, the exact fluctuation supports, and the visits to the fixed complement.

Complex coupling parameter. In this section we replace the coupling powers in (20.23) by powers of a complex parameter z: the classical powers $t_{\max}^{q-2}$ become z^{q-2} and the other powers $t_{\max}^{q}$ become z^{q}. The incoming functions, the couplings of earlier steps, the ratios λ_i, and the normalized symmetric reference measures on balls are kept fixed. The incoming insertion becomes

$$A_R(z) = R\big(\operatorname{Log} C(v, (z\lambda_i w_i)_i; b_{\mathrm{rec}})\big) - R\big(\operatorname{Log} C(v, 0; b_{\mathrm{rec}})\big).$$

We call the interaction without insertion the sum of all terms of $V - |w|^2/2$ in (20.22) except the incoming insertion, after this substitution. For $p_x > 0$ attached to the fluctuation blocks, with $p_x = p_i$ for a block at scale i, we write $p = \max_i p_i$, and we always assume $p \geq 1$.

Definition 20.6. Let F be an interaction, given with a resolution into scalar monomials

$$U_\lambda = u_\lambda v^{\nu_\lambda} w^{\eta_\lambda}, \qquad |\eta_\lambda| \geq 1,$$

as in Lemma 20.1. A fluctuation index is a whole fluctuation block at its scale and position; a background index is an unfixed scalar background coordinate on the lattice where it was introduced. Write S_λ for the set of fluctuation blocks on which η_λ is positive (the fluctuation support of the monomial) and $k_\lambda = |\nu_\lambda|$ for its background degree. Each monomial carries a connected network through all of its fluctuation and background positions; for a monomial whose coefficient has a visit to $\Omega_{L_J}^c$, the network contains this visit. We write ℓ_λ for the total length of the network, including the visit. No enlargement of S_λ is made. Internal indices are summed, using a fixed part of the exponential decay

rate, before the resolution into monomials, and repeated occurrences keep their own network refinements. For the exponent $s_0 > 0$ set

$$b_\lambda(F) = |u_\lambda(F)|\psi_{\mathrm{prof}}^{\nu_\lambda}\prod_x p_x^{|\eta_{\lambda,x}|}e^{s_0\ell_\lambda},$$

and for $h \geq 0$

$$W_h(F) = \sup_x \sum_{\lambda: x\in S_\lambda} e^{h|S_\lambda|}b_\lambda(F),$$
$$D_h(F) = \sum_{k\geq 1}\sup_i \sum_{\lambda: k_\lambda = k} \nu_{\lambda,i}e^{h|S_\lambda|}b_\lambda(F).$$

In W_h the root x is a fluctuation block, and in D_h the root i is a background index. In D_h the supremum over the root is taken before the sum over the background degree, as in the coefficient norm with boundary decay.

Lemma 20.7. *Fix the admissible larger products and the preliminary perturbation ball of Theorem 20.2 and Lemma 20.3, including both strict margins for the images under the reconstruction majorant and the inequalities on the conditional forces at every stage. Fix the ratio K_* and the profiles $\widehat{\phi} = K_*\phi$ and $\psi_{\mathrm{prof}} = \sqrt{K_*}\phi$. The only restrictions on the rates needed here are a fixed allocation of the exponential decay rate available in the coefficient bounds of the potential and the restrictions on the transport from input to output of Theorem 19.9 and Lemma 18.15; let $s_0 = a + \alpha$ lie strictly below these bounds. Consider the following families of interactions, resolved into monomials as in Definition 20.6: the interaction without insertion, the incoming insertion, the differences between the regional families and the restricted torus families, and the mixed difference of the boundary change and the incoming insertion. Let $H = E + \mathcal{X}^\rho(f)$ and set*

$$\mathfrak{s}(z) = \frac{C_J|z|(p^3 + p + Hp)}{1 - C_J|z|p},$$

for z in the fixed larger radius with $C_J|z|p < 1$, inside both incoming image margins. Then, uniformly in the fixed real frozen values,

$$W_{16}(F) + D_{16}(F) \leq \epsilon_F$$

for each of these families F. One may take $\epsilon_F \leq C_J\mathfrak{s}(z)$ for the family without insertion and for the boundary differences, and

$$\epsilon_A + \epsilon_{\partial A} \leq \frac{C_J|z|Hp}{1 - C_J|z|p} \tag{20.32}$$

for the incoming insertion A and its boundary difference ∂A. In particular the latter bound holds with a visit to Ω^c and with an arbitrary background root. The constants have the dependence described in item (3) of Definition 4.14 and are independent of the later widths; the profiles and the input radii are not decreased at later steps.

Proof. Step 1 records the coefficient resolution of the terms of the potential and its basic bounds. Steps 2 and 3 write out, with every contraction, the resolution in the two typical cases of a classical monomial and of a monomial of the Haar density. Step 4 treats the incoming insertion and the boundary differences. Step 5 converts the coefficient bounds into the bounds on W_{16} and D_{16}, keeping the order of the supremum and the degree sum.

Step 1: the resolution. The tensors of Theorem 20.2, with bounds at every external index, and the coefficient version of Lemma 20.3 are available on the outer profile, before the background is evaluated. By Lemma 20.1, every term of the potential has an exact resolution into monomials carrying connected networks. Rooted at an arbitrary free index, this resolution has convergent majorants in the free degree q, namely

$$C_J^q |z|^{q-2} p^q \quad (q \geq 3) \qquad \text{and} \qquad C_J^q |z|^q p^q \quad (q \geq 1),$$

for the classical terms and for the other terms, respectively. For the incoming insertion the second majorant is multiplied by H. The ratios λ_i have modulus at most one, and J is fixed. For the last assertion the input is (20.28) with $E + \sigma_0$ replaced by H: fixing its first differentiated input and summing over the other inputs gives

$$\|D^j R\|_{\mathrm{root}} \leq H \rho^{2-j} \frac{j!}{(1-\kappa)^{j+1}}, \qquad \kappa \leq \tfrac{1}{2},$$

by applying the factor $(l)_j \kappa^{l-j}$ to the input norm at degree l with that input fixed and summing the geometric series. The constants of R disappear in the fluctuation difference; they are not bounded by H. The composition with every index rooted in Theorem 19.9, with its volume-normalized input columns, keeps a specified external background index as well as a specified free index. At a background root i, the sum of the monomials containing that index is bounded by $\widehat{\phi}(i)$ times the absolute coefficient majorant of $\partial_i F$. This produces the factor $\nu_{\lambda,i}$ in D_h. No division by the smaller boundary radius occurs.

Step 2: a classical monomial. Fix a scale i. By the definition of the partial minimization, $F_i(v, c_{i+1}, y)$ is the perturbed objective $\mathcal{Q}_1$ of the proof of Theorem 20.2, in the free logarithmic coordinates and multiplied by the factor M^i, evaluated with the coordinates of the finer scales at their conditional minimizers, at fixed frozen values. We write this as $F_i(v, c_{i+1}, y) = \mathcal{Q}^{(i)}(\Psi_i(v, c_{i+1}, y))$, where $\mathcal{Q}^{(i)} = M^i \mathcal{Q}_1$, y denotes the free logarithmic coordinates of scale i, $c_{i+1} = C_{i+1}(v, \zeta_{>i})$ is the coarser reconstruction, and Ψ_i maps (v, c_{i+1}, y) to the full list of coordinates $(v, z^*_{<i}, y, z_{>i})$, with the finer coordinates $z^*_{<i}$ at their conditional minimizers. The coefficient bounds for $\mathcal{Q}^{(i)}$ on the common product are those of Theorem 19.9, and those for Ψ_i, which contains the conditional minimizers of the finer stages, are those of Theorem 20.2. The factorization only displays the chain rule; the general term can equally be bounded through the tensors $D^r_y F_i$ of that theorem.

At fixed (v, c_{i+1}) the minimizer z_i^* satisfies $F_{i,y}(v, c_{i+1}, z_i^*) = 0$, and $K_i = F_{i,yy}(v, c_{i+1}, z_i^*)$, $S_i = K_i^{-1/2}$, $z_i = z_i^* + g_i S_i w_i$. Since $S_i^{\mathsf{T}} K_i S_i = I$, Taylor's formula in y on the common coefficient product gives

$$\frac{F_i(v, c_{i+1}, z_i) - F_i(v, c_{i+1}, z_i^*)}{g_i^2} - \frac{|w_i|^2}{2} = \sum_{r \geq 3} \frac{g_i^{r-2}}{r!} \sum_{a_1, \dots, a_r} \sum_{\beta_1, \dots, \beta_r} F^*_{i, a_1 \cdots a_r} \prod_{l=1}^{r} S_i(a_l, \beta_l) w_{i, \beta_l}.$$

The star denotes evaluation at $y = z_i^*$ (not at zero coarser input), and the β_l are scalar labels at scale i. This expression is analytic at $g_i = 0$: stationarity and the exact subtraction of the quadratic term remove the apparent singularity.

We display the term $r = 3$. For a multi-index $\tau = (\nu, \eta_{>i})$ in the background variables and the coarser free variables, we write $[\,\cdot\,]_\tau$ for the coefficient of $v^\nu \zeta_{>i}^{\eta_{>i}}$ after the substitution $c_{i+1} = C_{i+1}(v, \zeta_{>i})$, and set

$$e^{(m)}_{x_1 \cdots x_m; \tau} = [(D^m \mathcal{Q}^{(i)})_{x_1 \cdots x_m}(\Psi_i^*)]_\tau, \qquad P_{x; A; \tau} = [(\partial_{y_A} \Psi_{i,x})^*]_\tau, \qquad s_{a\beta; \tau} = [S_i(a, \beta)]_\tau,$$

where $\partial_{y_A} = \partial_{y_a} \partial_{y_b}$ for $A = (a, b)$, and similarly for longer A. The chain rule gives, for $T_{abc;\tau} = [F^*_{i,abc}]_\tau$,

$$\begin{aligned} T_{abc;\tau} = {} & \sum_{x_1, x_2, x_3} \sum_{\tau_0 + \tau_1 + \tau_2 + \tau_3 = \tau} e^{(3)}_{x_1 x_2 x_3; \tau_0} P_{x_1; a; \tau_1} P_{x_2; b; \tau_2} P_{x_3; c; \tau_3} \\ & + \sum_{x_1, x_2} \sum_{\tau_0 + \tau_1 + \tau_2 = \tau} e^{(2)}_{x_1 x_2; \tau_0} \big(P_{x_1; ab; \tau_1} P_{x_2; c; \tau_2} + P_{x_1; ac; \tau_1} P_{x_2; b; \tau_2} + P_{x_1; bc; \tau_1} P_{x_2; a; \tau_2} \big) \\ & + \sum_{x} \sum_{\tau_0 + \tau_1 = \tau} e^{(1)}_{x; \tau_0} P_{x; abc; \tau_1}. \end{aligned}$$

In particular, the Hessian is

$$F^*_{i,ab} = \sum_{x, x'} \mathcal{Q}^{(i)}_{xx'}(\Psi_i^*) \Psi^*_{i,x,a} \Psi^*_{i,x',b} + \sum_x \mathcal{Q}^{(i)}_x(\Psi_i^*) \Psi^*_{i,x,ab}.$$

Stationarity in y does not imply $(\mathcal{Q}^{(i)})'(\Psi_i^*) = 0$, so the term with $(\mathcal{Q}^{(i)})'$ and the last line of the formula for $T_{abc;\tau}$ cannot be deleted. Let η_i be a multi-index of degree three in the fluctuation variables of scale i. After the substitutions $g_l = z\lambda_l$ and $\zeta_{>i} = z\lambda_{>i} w_{>i}$, the coefficient of $v^\nu w_i^{\eta_i} w_{>i}^{\eta_{>i}}$ in this cubic term is

$$\frac{z^{1 + |\eta_{>i}|} \lambda_i}{6} \prod_{j=i+1}^{s-1} \lambda_j^{|\eta_j|} \sum_{a,b,c} \sum_{\substack{\beta_1, \beta_2, \beta_3 \\ \mathbf{e}_{\beta_1} + \mathbf{e}_{\beta_2} + \mathbf{e}_{\beta_3} = \eta_i}} \sum_{\tau_0 + \tau_1 + \tau_2 + \tau_3 = (\nu, \eta_{>i})} T_{abc; \tau_0} s_{a\beta_1; \tau_1} s_{b\beta_2; \tau_2} s_{c\beta_3; \tau_3}, \tag{20.33}$$

where η_j is the part of the multi-index $\eta_{>i}$ at scale j. The factor $1/6$ also accounts for repeated β's; no further factorial occurs. For each summand we take the networks of its

coefficients and join them at their common indices x and a, and at the coarser arguments. The network of the resulting monomial is this connected union, and its fluctuation support is

$$S_\lambda = \{\text{blocks of } \beta_1, \beta_2, \beta_3\} \cup \{\text{coarser blocks used by } \eta_{>i}\},$$

with repeated blocks counted once. The background positions in ν belong to the network even if they lie outside S_λ. Let $\mathcal{Q}_m^+$, P_m^+, and S^+ be the absolute scalar majorants of the jets and maps above on their common product, including the volume ratios of part (4) of Lemma 20.1. The formula for $T_{abc;\tau}$ and Lemma 20.1 give the cubic majorant

$$\tfrac{1}{6}\big(\mathcal{Q}_3^+(P_1^+)^3 + 3\mathcal{Q}_2^+P_2^+P_1^+ + \mathcal{Q}_1^+P_3^+\big)(S^+)^3,$$

and the same expression with a derivative falling on one factor bounds the sum rooted at a fluctuation or background index. Each product here corresponds to a connected contraction in the formula for $T_{abc;\tau}$, so no unrooted product over an extensive set appears. For general r the rule is the partition formula

$$\partial_{a_1}\cdots\partial_{a_r}(\mathcal{Q}^{(i)}\circ\Psi_i) = \sum_{\pi}\sum_{(x_B)_{B\in\pi}} (D^{|\pi|}\mathcal{Q}^{(i)})_{(x_B)}(\Psi_i)\prod_{B\in\pi}\partial_{a_B}\Psi_{i,x_B},$$

where π runs over the partitions of $\{1,\dots,r\}$. Together with Lemma 20.1, it gives a convergent coefficientwise majorant on a common smaller free product. Since $r \geq 3$, a term of total free degree $q = r + |\eta_{>i}|$ has the coupling power z^{q-2}, and the ratios of the scales have modulus at most one. This proves the cubic starting order before any expectation over the fluctuation variables is taken.

Step 3: a monomial of the Haar density. Let $\Gamma(v, c_{i+1}, y)$ be the chart of solved links at scale i, let $\mathcal{E}_i^\circ$ be its links outside the block axial gauge trees, and let $\mathfrak{c}$ run over the set $\mathfrak{C}_i$ of constraints solved at scale i, with eliminated edges $e_\mathfrak{c}^*$. We write

$$A_\mathfrak{c} = D_{X_{e_\mathfrak{c}^*}}\mathsf{C}_\mathfrak{c}(\Gamma), \qquad L_i = -\sum_{e\in\mathcal{E}_i^\circ}\log j_H(\Gamma_e) + \sum_{\mathfrak{c}\in\mathfrak{C}_i}\log\det A_\mathfrak{c},$$

with $j_H(X) = \operatorname{sinc}^2(|X|/2)$ as in Definition 4.3; only field-independent factors have been removed. The fluctuation-dependent part of L_i at scale i is

$$L_i(v, c_{i+1}, z_i) - L_i(v, c_{i+1}, z_i^*) = \sum_{r\geq 1}\frac{g_i^r}{r!}\sum_{a_1,\dots,a_r}\sum_{\beta_1,\dots,\beta_r} L^*_{i,a_1\cdots a_r}\prod_{l=1}^{r} S_i(a_l,\beta_l)w_{i,\beta_l}.$$

Its linear coefficient is

$$L^*_{i,a} = -\sum_{e\in\mathcal{E}_i^\circ}\sum_{\varsigma=1}^{3}(\partial_\varsigma\log j_H)(\Gamma_e^*)\Gamma^*_{e\varsigma,a} + \sum_{\mathfrak{c}\in\mathfrak{C}_i}\sum_{A,B=1}^{3}(A_\mathfrak{c}^{-1})^*_{BA}(\partial_{y_a}A_\mathfrak{c})^*_{AB},$$

where $(\partial_{y_a} A_{\mathfrak{c}})_{AB} = \sum_{e,\varsigma} (\partial_{X_{e\varsigma}} A_{\mathfrak{c}})_{AB} \Gamma_{e\varsigma,a}$, and all factors are evaluated at the minimizer and at the coarser reconstruction $C_{i+1}(v, \zeta_{>i})$. Taking the coefficient $[\,\cdot\,]_\tau$ of each factor and summing over all allocations of τ gives the coefficients $l_{a;\tau} = [L^*_{i,a}]_\tau$, with every link, constraint, color, integrated coordinate, and multi-index sum displayed. For $|\eta_i| = 1$ the coefficient after the coupling substitution is

$$z^{1+|\eta_{>i}|} \lambda_i \prod_{j=i+1}^{s-1} \lambda_j^{|\eta_j|} \sum_a \sum_{\beta : \mathbf{e}_\beta = \eta_i} \sum_{\tau_0 + \tau_1 = (\nu, \eta_{>i})} l_{a;\tau_0} s_{a\beta;\tau_1}.$$

The trace in $L^*_{i,a}$ is rooted at the index a through the derivative $\partial_{y_a} A_{\mathfrak{c}}$; its coefficient bound sums over $\mathfrak{c}, e, A, B, \varsigma$. It is not the unrooted norm of a sum of logarithmic determinants. The network is the connected union of the networks of the Haar factors, the inverse matrices, and the chart, joined to the network of S_i at a, and the support is the block of β together with the coarser blocks used by $\eta_{>i}$. Lemma 20.1 bounds these coefficients by the products of the majorants of the factors and their derivatives. The series starts at total free degree one, with coupling power equal to the total free degree.

The remaining part of the measure fluctuation in (20.10) is the difference $B_i(c_{i+1}) - B_i(\widehat{c}_{i+1})$, whose fluctuation support may lie entirely in coarser scales; it is not part of the field-independent prefactor. For a coordinate h_γ of the coarser argument, the total derivative is

$$\partial_{h_\gamma} B_i = L^*_{i,h_\gamma} + \sum_a L^*_{i,a} \partial_{h_\gamma} z^*_{i,a} + \frac{1}{2} \sum_{a,b} (K_i^{-1})_{ba} \partial_{h_\gamma} K_{i,ab},$$

where $\partial_{h_\gamma} z^*_i = -K_i^{-1} F^*_{i,y h_\gamma}$ and $\partial_{h_\gamma} K_{i,ab} = F^*_{i,ab h_\gamma} + \sum_{a'} F^*_{i,aba'} \partial_{h_\gamma} z^*_{i,a'}$. The difference equals the convergent coefficient expansion of

$$\int_0^1 \sum_{j=i+1}^{s-1} \sum_{\beta \in I_j} \zeta_{j,\beta} \sum_\gamma (\partial_{h_\gamma} B_i)\big(v, C_{i+1}(v, \tau\zeta_{>i})\big)\, (\partial_{\zeta_{j,\beta}} C_{i+1,\gamma})(v, \tau\zeta_{>i})\, d\tau,$$

since the integrand is the derivative in τ of $B_i(v, C_{i+1}(v, \tau\zeta_{>i}))$; here I_j is the set of unfixed scalar free labels at scale j, and $\partial_{\zeta_{j,\beta}} C_{i+1}$ is evaluated at $\tau\zeta_{>i}$ without an extra factor τ. Each Taylor coefficient of this integral carries the factor $1/(n+1)$ when the remaining integrand has degree n in τ. Likewise, at fixed couplings,

$$dz_i = (\partial_v z^*_i) dv + (\partial_c z^*_i) dc_{i+1} + g_i \{(\partial_v S_i) dv + (\partial_c S_i) dc_{i+1}\} w_i + g_i S_i\, dw_i,$$

and none of these terms may be dropped. The inverse square root S_i is treated as a constituent tensor with the bounds of Theorem 20.2; if it is expanded, (20.18) and the Neumann series of the resolvent around $K_i(0,0)$ give a chain of ordinary matrix products whose coefficients are its network refinements. The comparison of determinant matrices with the same indices also includes the correction (20.20), which is a further constituent tensor with a visit to Ω^c.

Step 4: the incoming insertion and the boundary differences. We write $\mathcal{A}(v,\zeta) = \operatorname{Log} C(v,\zeta;b_{\mathrm{rec}})$. Then

$$R(\mathcal{A}(v,\zeta)) - R(\mathcal{A}(v,0)) = \sum_{n\geq 1} \frac{1}{n!} D^n R(\mathcal{A}(v,0))\big[\mathcal{A}(v,\zeta)-\mathcal{A}(v,0),\dots,\mathcal{A}(v,\zeta)-\mathcal{A}(v,0)\big],$$

and each factor $\mathcal{A}(v,\zeta)-\mathcal{A}(v,0)$ has positive degree in ζ. The coefficient of $v^{\nu}\zeta^{\eta}$ is the sum over n, over the contracted indices $j_1,\dots,j_n$ of the old lattice, and over the allocations of the multi-indices to the n factors with positive free degree in each, of the products of the jet coefficients of R and the coefficients of the factors. No additional multinomial factor occurs, since each coefficient is already a multi-index coefficient. Each such occurrence keeps the union of the old network of R, scaled by B^{-1} in output units, and the attached reconstruction networks, and its fluctuation support is the set of blocks used by η. By the jet bound of Step 1 with $j=1$ at an image bounded by $\rho/2$, the derivative series of R at the image is at most $4H\rho$. The column majorant of the reconstruction with a free index fixed vanishes at zero free radius and is linear in the free radius on the fixed product, so it is at most $C_J|z|p$. Fixing an external input in the old counting convention contributes the fixed volume ratio $\Gamma_J = \max_i v_i/v_{\mathrm{old}} \leq M^{3J}$ between the input mesh and the old mesh. We obtain the incoming bound, linear in H and starting at $|z|p$.

For a difference between the regional and the restricted torus constructions we compare factor by factor, before any expansion of logarithms. In each product of coefficients we use the ordered identity

$$\prod_{r=1}^{m} A_r^1 - \prod_{r=1}^{m} A_r^0 = \sum_{r=1}^{m} \Big(\prod_{l<r} A_l^1\Big)(A_r^1 - A_r^0)\Big(\prod_{l>r} A_l^0\Big),$$

and for the incoming function at two reconstructed arguments C_0, C_1 in the identity chart (here C_0, C_1 denote arguments, not the constants of Lemma 20.3),

$$R(C_1) - R(C_0) = \int_0^1 DR\big(C_0 + s(C_1-C_0)\big)[C_1-C_0]\,ds.$$

Both images under the absolute reconstruction majorant lie in the common product with half the incoming radius, so the whole segment does. Terms involving the incoming boundary function use its incoming boundary weight; torus terms acquire the visit carried by the difference of the reconstructions. Fixed input slots are visits to Ω^c. Changed formulas, minimizers, inverse square roots, and Haar densities are treated by the same identity, with $-A^{-1}(A-B)B^{-1}$ for inverses and (20.18) for inverse square roots. Each term contains one changed factor with its visit to Ω^c. The versions with every index rooted in Theorem 19.9 keep every differentiated background or free index, even when the changed factor is not differentiated; the degree-one terms are therefore included.

We join the incoming networks, scaled by B^{-1}, to the chart networks at their contracted indices. The transport of the route to the fixed complement is exactly (19.42), and

Lemma 18.15 sums the remaining internal indices. We append the resulting route to the network of the monomial. One fixed allocation of the available exponents bounds $e^{s_0\ell_\lambda}$, including this route and the fixed stencil offsets; the offsets contribute a fixed factor in the degree, absorbed on the margin already provided by the construction. The allocation is the same for every order and every contraction, and no exponent is lost at each vertex. The scaled incoming networks and routes are bounded by the inequalities against Ba and $B\alpha$. Every term still contains an incoming factor, so its bound is linear in H and starts at $|z|p$. The arguments of Step 5 apply to this majorant with a visit to Ω^c and give (20.32). Finally, a torus monomial deleted because it uses a missing fluctuation block already has a visit at that block; if it has an unfixed root, its network provides the route to Ω^c.

Step 5: conversion to W_{16} and D_{16}. By Steps 1 to 4, there exist a free radius $V_1 > 0$, strictly inside all image and stage margins, and a common bound A_0 for the coefficientwise majorants of the diagrams on the outer profile $\widehat{\phi}$ and at free radius V_1, including each permitted fixed coefficient index and the volume ratios of part (4) of Lemma 20.1. For the incoming family the bound is A_0H, and for a difference it is linear in the bound of the changed factor. We write the resolved coefficients of free degree q and background degree k as $u_\lambda^0 v^{\nu_\lambda}\zeta^{\eta_\lambda}$, after removing the factor z^{q-2} for the classical terms and z^q for the others; the ratios λ_i have modulus at most one. The mixed factorial identity is

$$\frac{1}{k!\,q!}\sum_{\text{other ordered indices}}\left|D_v^k D_\zeta^q (u^0 v^\nu \zeta^\eta)\right| = \begin{cases} (\eta_l/q)|u^0|, & \text{one free input } l \text{ fixed},\\ (\nu_i/k)|u^0|, & \text{one background input } i \text{ fixed}, \end{cases}$$

for $q \geq 1$ and $k \geq 1$ respectively, and the radii of all inputs multiply these expressions. With $\widetilde{C} = \max\{1, V_1^{-1}\}$, Lemma 20.1 and the finite majorant at V_1 give

$$\sup_l \sum_{\lambda:(k_\lambda,q_\lambda)=(k,q)} \frac{\eta_{\lambda,l}}{q}|u_\lambda^0|\widehat{\phi}^{\nu_\lambda}e^{s_0\ell_\lambda} \leq A_0\widetilde{C}^q, \qquad \sup_i \sum_{\lambda:(k_\lambda,q_\lambda)=(k,q)} \frac{\nu_{\lambda,i}}{k}|u_\lambda^0|\widehat{\phi}^{\nu_\lambda}e^{s_0\ell_\lambda} \leq A_0\widetilde{C}^q.$$

These bounds concern the total variation of the resolution produced by composition of coefficientwise majorants.

Let $\vartheta = K_*^{-1/2} < 1$. Passing from $\widehat{\phi}$ to ψ_{prof} multiplies a term of background degree k by exactly ϑ^k, independently of the positions of its background indices in the spatially varying radius profile. Each term has $|S_\lambda| \leq q$, and selecting a fluctuation block is bounded by the sum of its three scalar multiplicities. At fixed k, q, the preceding bounds therefore contribute at most $3qA_0\widetilde{C}^q\vartheta^k$ for a fluctuation root and $kA_0\widetilde{C}^q\vartheta^k$ for a background root; the latter bound is uniform in the root at each degree. Taking the supremum over the root at each degree and only then summing over k, and using $q \leq 2^q$, we find that the sums in the definitions of W_h and D_h, at fixed q, are at most

$$A_0\widetilde{C}^q e^{hq}\Big(\frac{3q}{1-\vartheta} + \frac{\vartheta}{(1-\vartheta)^2}\Big) \leq \mathsf{A}\mathsf{B}^q, \qquad \mathsf{A} = A_0\Big(\frac{3}{1-\vartheta} + \frac{\vartheta}{(1-\vartheta)^2}\Big), \quad \mathsf{B} = 2e^h\widetilde{C}.$$

Multiplying by $|z|^{q-2}p^q$, $q \geq 3$, or by $|z|^q p^q$, $q \geq 1$, and summing the geometric series, we obtain on $\mathsf{B}|z|p < 1$

$$W_h^{\mathrm{cl}} + D_h^{\mathrm{cl}} \leq \frac{\mathsf{A}\mathsf{B}^3|z|p^3}{1-\mathsf{B}|z|p}, \qquad W_h^{\mathrm{Haar}} + D_h^{\mathrm{Haar}} \leq \frac{\mathsf{A}\mathsf{B}|z|p}{1-\mathsf{B}|z|p}, \qquad W_h^{\mathrm{inc}} + D_h^{\mathrm{inc}} \leq \frac{\mathsf{A}\mathsf{B}H|z|p}{1-\mathsf{B}|z|p}$$

for the classical, Haar-density, and incoming families. The same bounds hold for the differences with a visit to Ω^c, with their majorants. Note that the classical bound starts with $|z|p^3$: the two coupling powers removed by the division by g_i^2 are not restored. For $h = 16$, enlarging the constant to a common C_J and restricting the disc accordingly gives the assertions of the lemma. Choosing the auxiliary disc smaller by this fixed factor keeps it inside the same larger free radius, and neither an input radius nor the ratio K_* changes. Finitely many additional background marks contribute a polynomial factor in k, which is summable since $\sum_k (k+1)^m \vartheta^k < \infty$. The argument never interchanges $\sup_i \sum_k$ with $\sum_k \sup_i$. The bounds at fixed real frozen values are proved first and are uniform, so taking the supremum over the frozen values afterwards preserves them. All coefficientwise majorant series converge on the same fixed larger products, and finite truncations and the absolute majorants justify the displayed sums. □

Lemma 20.8. *Let $\pi = \bigotimes_x \pi_x$ be a finite product of probability measures on fluctuation blocks, each π_x supported in the ball $\{|w_x| \leq p_x\}$ with $p_x < \infty$; these are the radii used in b_λ, W_{16}, and D_{16}. Use the same working rates $a, \alpha > 0$ as in $\mathcal{X}^\phi$, and assume $s_0 \geq a + \alpha$. Consider monomial families as in Definition 20.6, with nonempty supports S_λ and networks through all of their positions, and put $\epsilon = W_{16}(U) + D_{16}(U)$. There is an absolute constant $\epsilon_0 > 0$ such that $\epsilon < \epsilon_0$ implies the following.*

1. *The product expansion followed by the hard-core polymer expansion, with incompatibility $S_\lambda \cap S_\mu \neq \varnothing$, gives a normally convergent logarithm of $\int e^{-\sum_\lambda U_\lambda}\, d\pi$, equal to zero at zero interaction, and this integral does not vanish in the coefficient ball. Neither spatial connectedness of S_λ nor membership of the background positions in S_λ is needed.*
2. *Every resolved term of the logarithm carries the connected union of the networks of its monomials, with no additional length. Its absolute sum rooted at a fluctuation block, with the weight $e^{4|S|}$ where S is the union of the supports of its monomials, is at most $C\epsilon$. After passage from ψ_{prof} to ϕ, the sum over the background degrees of its rooted coefficient suprema is at most $C\epsilon$.*
3. *For one marked family A the same bounds hold with $C\epsilon_A$. Let $U(s,r) = U + sB + rA + srC$, $0 \leq s, r \leq 1$, where B carries the boundary change, A the incoming insertion, and C both (here s, r are interpolation parameters and B, C are families of monomials). Assume that every occurrence in B and C has an assigned network visiting $\Omega^c_{L_J}$, with that visit included in ℓ_λ and in its weighted bound, and that every interpolated family lies in the same fixed small ball.*

Then the mixed derivative $\partial_s \partial_r$ of the logarithm and its four-corner difference are bounded, in both rooted norms after passage of profile, by

$$C(\epsilon_{\partial A} + \epsilon_{\partial} \epsilon_A), \tag{20.34}$$

where $\epsilon_{\partial}, \epsilon_A, \epsilon_{\partial A}$ are the values of $W_{16} + D_{16}$ for B, A, C. A term containing the boundary change keeps a visit to Ω^c, and a term containing both changes keeps an incoming factor. These bounds keep the weight of the network length. In particular, for a difference F of logarithms containing the boundary change, on the working profile,

$$\mathcal{X}^{\phi}(F) \le C_J(\epsilon_{\partial A} + \epsilon_{\partial} \epsilon_A)$$

in the case of two changes, and $\mathcal{X}^{\phi}(F) \le C_J \epsilon_{\partial}$ in the case of one change.

4. *The logarithm and its derivatives in the changes are holomorphic on this ball.*

The combinatorial constants are absolute, with the dependence described in item (1) of Definition 4.14; the constants of the passage of profile, of the density, and of the spatial conversion have the dependence described in item (3).

Proof. The proof follows the standard two-stage cluster expansion; we give the estimates so that the root and the support weights stay visible. We first truncate to finitely many monomials.

Step 1: the product expansion. Let $f_\lambda = e^{-U_\lambda} - 1$. Its absolute coefficient majorant is at most $e^{b_\lambda} - 1 \le 2b_\lambda$ for small ϵ. Indeed $|w_x| \le p_x$ holds π_x-almost surely, so the polynomial bounds defining b_λ bound the integrands on the support of the product measure. The normalized truncated Gaussian measures on balls used later satisfy this hypothesis. We expand the exact identity

$$e^{-\sum_\lambda U_\lambda} = \prod_\lambda (1 + f_\lambda).$$

We group each selected family of distinct indices by the connected components of the graph in which λ, μ are joined when $S_\lambda \cap S_\mu \neq \varnothing$. Different components depend on disjoint sets of fluctuation blocks, so their expectations factor, even if their background parameters coincide. This is an exact hard-core gas whose polymers are unions of fluctuation supports. Spatially disconnected sets may serve as polymers, because compatibility uses only set intersection and the independence of the reference variables.

Step 2: the tree majorant at the first stage. Let $n_\lambda = |S_\lambda|$ and $a_\lambda = 2b_\lambda e^{10 n_\lambda}$. The majorant by rooted trees is the increasing iteration from zero of

$$T_\lambda = a_\lambda \exp\Big(\sum_{\mu : S_\mu \cap S_\lambda \neq \varnothing} T_\mu \Big). \tag{20.35}$$

Every connected selected family has a spanning tree; expanding the exponential supplies the factorials for unordered children, and dropping the distinctness of the indices only increases this sum of nonnegative terms. If $T_\mu \le a_\mu e^{n_\mu}$ for all μ, then the exponent in (20.35) is at most

$$n_\lambda \sup_x \sum_{\mu : x \in S_\mu} a_\mu e^{n_\mu} \le 2\epsilon n_\lambda .$$

For $2\epsilon \le 1$ the candidate $a_\lambda e^{n_\lambda}$ therefore bounds every iterate. Its sum rooted at a fluctuation block is at most 2ϵ. Thus the polymer activities are absolutely summable with the weight $e^{10|S|}$.

Step 3: the second stage. We apply the finite graph inequality of Theorem 12.6 to this hard-core gas; only its bound on alternating sums over connected graphs is imported. At the level of polymers we repeat (20.35), using the weight $e^{4|S|}$ for the output and a factor $e^{|S|}$ of the support weight for the children. The estimate with weight $e^{10|S|}$ bounds the sum over the children by $C\epsilon$ times the size of the parent support. Decreasing ϵ_0 if necessary, we obtain the bound of part (2) with weight $e^{4|S|}$. Repeated polymer occurrences are allowed, as in the finite graph theorem. Thus the input weight is $e^{16|S|}$, the polymer activities have weight $e^{10|S|}$, and the logarithm has weight $e^{4|S|}$.

Step 4: the networks. We keep the resolution into monomials inside each exponential and each polymer. Within a connected family of the first stage, the networks of adjacent factors share a fluctuation position. At the second stage, adjacent polymers share a fluctuation block used by a factor in each of them, so their networks also meet at that position. By induction along a spanning tree at each stage, the union is connected and

$$\ell(\mathcal{N}_{\text{union}}) \le \sum_{\lambda \text{ occurrences}} \ell_\lambda . \tag{20.36}$$

Coincident positions of different color coordinates have distance zero. Copies of the same monomial in an exponential have the same support and network. A visit to Ω^c remains a visit in the union. Thus the network weights multiply, with no extra connecting length at either stage.

Step 5: the background root. Fix a background coordinate i in a coefficient of a product. The identity $\nu_i = \sum_\lambda \nu_{\lambda,i}$ assigns the derivative in i to one monomial occurrence, with its multiplicity. We root the spanning tree of the first stage at this factor and, at the second stage, at the polymer containing it. The estimates for the children just proved bound the remaining part of the expansion by $Ce^{cn_\lambda} b_\lambda$ with $c < 16$, so that the total support weight used is less than $e^{16 n_\lambda}$. At the distinguished factor we use the degree-k term in D_{16} instead of its norm rooted at a fluctuation block. This works even if the background position is not a fluctuation position. The other factors are summed through their intersections of fluctuation supports.

Fixing i and summing over all output degrees at the intermediate profile, we obtain the bound $C \sum_k d_k \le C\epsilon$, where d_k is the degree-k supremum in D_{16} for the distinguished monomial. Indeed, the remaining part of the expansion is summed at its fixed fluctuation root, and its total coefficient majorant there is bounded by the estimates for the children. We thus obtain a bound on $\sup_i \sum_K$ at the intermediate profile, which is not yet the coefficient norm with boundary decay. Let $q_* = K_*^{-1/2} < 1$. At output degree K, passing to the working profile multiplies every coefficient by q_*^K. Thus, taking the supremum separately at each degree,

$$\sum_{K \ge 1} \sup_i \{\text{output coefficients of degree } K \text{ rooted at } i, \text{ at } \phi\} \le C\epsilon \sum_{K \ge 1} q_*^K.$$

This is the required order of operations; a bound on $\sup_i \sum_K$ at the same radius would not imply a bound on $\sum_K \sup_i$. The second half of the ratio between the profiles is used here, once for each local map on the fixed profiles. The factor ν_i / K in the coefficient norm with boundary decay only decreases the multiplicity bound. Repeated copies in an exponential obey the same identity for ν_i, and differentiating the exponential accounts for all possible distinguished occurrences.

Step 6: one and two changes. The tree estimates also hold in the rooted norms with changes. Differentiating (20.35) once gives

$$T'_\lambda = e^{\sum_{\mu \sim \lambda} T_\mu} a'_\lambda + T_\lambda \sum_{\mu \sim \lambda} T'_\mu,$$

where $\mu \sim \lambda$ means $S_\mu \cap S_\lambda \ne \varnothing$. In the norm rooted at a fluctuation block, the last operator has norm at most $C\epsilon < 1/2$, since its incidence factor $n_\lambda \le e^{n_\lambda}$ is bounded by a factor e^{n_λ} of the support weight that is not used in the output bound. Its inverse is a Neumann series with nonnegative terms and norm at most two. Hence a first variation is bounded by the norm of the changed source. Writing $h_\lambda = \sum_{\mu \sim \lambda} T_\mu$, the mixed variation satisfies

$$T_{\lambda,sr} = e^{h_\lambda} a_{\lambda,sr} + e^{h_\lambda} (a_{\lambda,s} h_{\lambda,r} + a_{\lambda,r} h_{\lambda,s}) + T_\lambda h_{\lambda,s} h_{\lambda,r} + T_\lambda \sum_{\mu \sim \lambda} T_{\mu,sr}.$$

Its inhomogeneous terms are a source with both changes, products of the two first variations, and a source with one change times a first variation. Their bounds are $C(\epsilon_{\partial A} + \epsilon_\partial \epsilon_A)$, and the same inverse bounds the mixed variation. The polynomial incidence factors have degree at most two, and $\sup_{n \ge 1} n^2 e^{-n} < \infty$. In the bilinear interpolation, terms containing the family C are bounded by a constant times $\epsilon_{\partial A}$, since the other monomial norms are bounded on the fixed small ball. Integrating the mixed derivative over the unit square, we obtain the four-corner difference, with the same bound. These calculations apply at both stages of the expansion. For a background root, we

first sum the resulting rooted expressions at the distinguished factor at the intermediate profile, with the sequence d_k described above, and then apply the factor q_*^K at each output degree. If the background root lies on a factor different from both changes, the remaining expansion has at most one additional distinguished branch. Its cubic incidence factor is bounded by a further factor e^{n_λ} of the support weight, since $\sup_{n\ge1} n^3 e^{-n} < \infty$; the passages from $e^{16|S|}$ to $e^{10|S|}$ and from $e^{10|S|}$ to $e^{4|S|}$ leave these factors available. The source with both changes keeps its incoming factor, and the product source contains the first variation in the incoming insertion. This proves (20.34) in both rooted norms, including the case of a background root on an unchanged factor.

Step 7: spatial conversion. We keep the joined networks. By the hypothesis on B and C, every term containing the boundary change carries a visit to $\Omega^c_{L_J}$. If a coefficient with unfixed background root i carries such a visit, then by part (5) of Lemma 20.1

$$a\mathrm{t}(\nu) + \alpha d_\Omega(i) \le (a+\alpha)\ell(\mathcal{N}_{\mathrm{union}}) + C_J\alpha.$$

Since $s_0 \ge a+\alpha$, the product of the weights $e^{s_0\ell_\lambda}$ in b_λ dominates $e^{(a+\alpha)\ell(\mathcal{N}_{\mathrm{union}})}$, by (20.36). The fixed buffer is used only at the final visit to Ω^c. The factor ν_i/K in (4.11) is at most the multiplicity of the distinguished index. The bound at the intermediate profile and the degree sum $\sum_{K\ge1} q_*^K$ just proved, multiplied by the fixed factor ρ^{-2} and the buffer constant, give the stated bounds on $\mathcal{X}^\phi$.

Step 8: the logarithm. The bounds on the coefficientwise majorants prove the normal convergence of every resolved series and of the displayed derivatives. For a small multiple of the interaction, the two series are the Taylor expansion of the logarithm of the finite product integral. Their exponential agrees with the integral throughout the strict ball, by the identity theorem in that multiplier. Hence the integral does not vanish there. The uniform absolute bounds allow us to remove the truncations to finitely many monomials and to differentiate termwise in the parameters, and they prove holomorphy in the absolute coefficient norms. We have not used the hypothesis on background supports in the holomorphy theorem of Section 12. □

Theorem 20.9. *Fix the torus rate $a_{\mathbb{T}} > 0$ and the ratio K_* between the profiles $\widehat{\phi} = K_*\phi$ and $\psi_{\mathrm{prof}} = \sqrt{K_*}\phi$. There are geometric constants C_{alloc} and C_g, independent of J, with the following property. Choose $C_*^{\mathrm{geom}} \ge C_g$ and choose J once with*

$$B = M^J \ge B_{\mathrm{geom}}, \qquad \textit{where } B_{\mathrm{geom}} = 256 C_{\mathrm{alloc}}(C_*^{\mathrm{geom}} + 1).$$

After J has been fixed, choose positive rates

$$a = a_{\mathrm{cl}} = a_{\mathrm{sp}}, \qquad \alpha = \alpha_{\mathrm{cl}} = a/16, \qquad m_{\mathrm{rate}} = C_*^{\mathrm{geom}}(a+\alpha), \qquad b_1 = 8C_*^{\mathrm{geom}}\alpha$$

so that

$$m_{\mathrm{rate}} + b_1 < \min(c_{\mathrm{prim},J}, Ba)/C_{\mathrm{alloc}}, \qquad b_1 < B\alpha/C_{\mathrm{alloc}}, \qquad \sqrt{3}\,(a+\alpha) < a_{\mathbb{T}},$$

where $c_{\mathrm{prim},J}$ is the fixed positive reserve for the intrinsic tensors and the unperturbed weighted reconstruction images, in the graph metric of Section 20.1, chosen no larger than the intrinsic exponent c_J of Theorem 19.9. The inherited perturbative networks use the separate inequalities against Ba and Bα from Theorem 19.9; their images satisfy the half-radius bound at the allocated working rates.

Consider a full renormalization group step ($s = J$) on admissible data. Sum the internal indices first, keep networks through all positions of the monomials, and use the exact nonempty fluctuation supports of Definition 20.6 in both expansions. All factors and polymer occurrences that are joined share a fluctuation position; the union of their networks keeps every background index and every visit to Ω^c, and adds no length. Set $p = \max_i p_i \geq 1$, $x = \mathcal{X}^\rho(f)$, and $H = E + x$, and substitute the complex parameter z for the coupling powers as described before Definition 20.6, with the insertion $A_R(z)$. Let Ξ_0 and Ξ_R be the integrals of e^{-U} against the product of normalized symmetric reference measures on balls, for the interaction U without insertion and for U plus the insertion, respectively. For the matched torus potential, with the fixed background slots set to zero, define its restriction by deleting the monomials that use a missing fluctuation block, and adjoin the missing blocks with the identical normalized reference measures.

There are $\epsilon_J, c_{z,J} > 0$ and $C_J < \infty$ such that the following holds if

$$\mathfrak{s}(z) = \frac{C_J|z|(p^3 + p + Hp)}{1 - C_J|z|p} \leq \epsilon_J, \qquad C_J|z|p < 1.$$

The integrals without and with insertion, the restricted torus integrals, and the linearly interpolated integrals do not vanish, and their logarithms are the normally convergent expansions of Lemma 20.8. The regional and restricted families satisfy $W_{16} + D_{16} \leq C_J\mathfrak{s}(z)$; their boundary-difference families satisfy the same bound with networks visiting Ω^c. The comparisons of the nonconstant parts of the logarithms, including the deleted torus terms with an unfixed background index, satisfy

$$\mathcal{X}^\phi\big(\log \Xi_\bullet^\Omega - (\log \Xi_\bullet^{\mathbb{T}})|\big) \leq C_J\mathfrak{s}(z), \qquad \bullet \in \{0, R\},$$

where the bar denotes zero extension of the fixed background slots; constants supported in Ω^c remain separate. The mixed difference of the boundary change and the insertion is bounded by $C_J(\epsilon_{\partial A} + \epsilon_\partial \epsilon_A)$ in the norms of Definition 20.6, and every term keeps an incoming factor. The family extends holomorphically to $z = 0$, and its compared insertion logarithm F satisfies $F(0) = F'(0) = 0$. In particular, if

$$r_z = \frac{c_{z,J}}{p^3 + p + Hp}, \qquad 0 < t_{\max} \leq r_z/2,$$

then

$$\mathcal{X}^\phi\big(\mathsf{D}^\Omega R - (\mathsf{D}^{\mathbb{T}} r_{d_0})|\big) \leq C_J t_{\max}^2 H(p^4 + Hp^2), \qquad \mathsf{D}R = -\log \Xi_R + \log \Xi_0.$$

After the whole boundary layer with reduced radius has been frozen as in Lemma 18.9, the classical and quantum comparisons return the same rates a, α and the same input radii; the torus rate

remains $a_{\mathbb{T}}$. The constants have the dependence described in item (3) of Definition 4.14 and are independent of the later widths. The return of the same profile is asserted only for full steps with $B \geq B_{\mathrm{geom}}$, not for compositions of fewer than J averaging steps.

Proof. The constants C_{alloc} and C_g bound only the fixed contraction with every index rooted and the transport of the input boundary weight in Theorem 19.9 and Lemma 18.15; they include the factor $\sqrt{3}$ of the conversion between Euclidean and graph exponents in Section 20.1, and hence do not depend on J. The torus coefficients, bounded in $\mathcal{B}_{\rho,a_{\mathbb{T}}}$ with Euclidean network lengths, have graph exponent $a_{\mathbb{T}}/\sqrt{3} > a + \alpha$ by the last inequality in the statement. With $\alpha = a/16$ we have $m_{\mathrm{rate}} + b_1 = (25/16)C_*^{\mathrm{geom}} a$, and both restrictions against Ba and $B\alpha$ follow from the lower bound on B. At this fixed J, we decrease $a > 0$ to satisfy the bounds on the exponents of the coefficients of the potential, of the weighted images, and of the torus, including $\sqrt{3}\,(a + \alpha) < a_{\mathbb{T}}$. No constant depending on J is used to choose J. The comparison of the classical boundary networks uses the same allocation of the decay rate and no expansion with respect to a reference product. At zero fluctuation it works as follows. Consider a classical comparison, or the comparison of the pullback terms, with the boundary tensors rooted at every index. At background degree $k \geq 1$, its rooted absolute sum on $\widehat{\phi}$ is bounded by its tensor majorant for fixed J, and that allocation already accounts for the weights of its output network and of its distance to Ω^c. Passing to ϕ gives the factor K_*^{-k}. Summing over the degree, including any fixed polynomial factor for the choice of a distinguished index, we obtain the bound in the coefficient norm with boundary decay. For a pulled-back incoming boundary term, we use its jet majorant at half the radius and its distinguished input columns before this calculation. Thus the classical coefficients are converted without assigning a fluctuation support to a term at zero fluctuation, and these terms never enter either expansion.

We apply Lemma 20.7 at these rates and then Lemma 20.8. The hypotheses of the latter on the rates and on the boundary-change families hold, since Lemma 20.7 uses $s_0 = a + \alpha$ and gives the boundary differences with their networks visiting Ω^c. Restricting a torus family decreases its norms, which are sums of absolute values. For its difference from the full family, every deleted coefficient with an unfixed background root visits a missing fluctuation position and hence Ω^c, and the same lemma applies. Pure background constants without an unfixed index have zero norm with boundary decay and stay outside this comparison. The estimate with two changes implies the comparison of the logarithms with and without insertion. The logarithms and their derivatives in the parameters are determined by the finite identities and normal convergence proved above, not by a choice of a complex logarithm after integration.

We choose $c_{z,J}$ so that the disc $|z| \leq r_z$ lies strictly inside the margins of Lemma 20.7, of the free product, and of the logarithm. On this disc, (20.32) and the bounds with one and two changes give

$$\|F(z)\|_{\mathcal{X}^\phi} \leq C_J |z| Hp.$$

At $z = 0$ every fluctuating monomial vanishes. The coefficient of z is cubic in the fluctuation variables for the classical terms and linear for the quantum, Haar density, determinant, and incoming terms. Each of these is odd under the simultaneous reversal $w \mapsto -w$, so its expectation under the symmetric reference measures on balls vanishes. At first order the logarithm contains a single monomial occurrence, hence $F(0) = F'(0) = 0$ coefficientwise, at every real frozen value.

The normally convergent series $F(z)/z$ is holomorphic, vanishes at zero, and has norm at most $C_J H p$. Cauchy's coefficient bound on $|z| = r < r_z$, followed by $r \uparrow r_z$, gives for $t \le r_z/2$

$$\|F(t)\|_{\mathcal{X}^\phi} \le 2C_J t^2 H p / r_z \le C'_J t^2 H(p^4 + Hp^2),$$

where in the last step we used $p \ge 1$. The integral proof of the coefficient bound applies to the supremum over the fixed index at every background degree, and then to their sum, by the triangle inequality.

Finally, Lemma 18.9 restores the incoming radii when the whole zone with reduced radius is frozen with the stated bounds on the frozen values. All conversions of coefficients were made on the fixed outer and intermediate profiles, and none is iterated at later steps. This proves the return of the same profile and the stated independence of the constants. □

20.5 Effective interactions and normalization constants

Theorem 20.10. *Consider a full renormalization group step, $s = J$, on admissible data as in Theorem 20.9. Set*

$$\mathsf{P}^\Omega R(v) = R\big(\operatorname{Log} C^\Omega(v, 0; b_{\mathrm{rec}})\big), \qquad \mathsf{Q}^\Omega = B^\Omega_{\mathrm{ref}} - \sum_i f_i \log z_{p_i} - \log \Xi^\Omega_0,$$

$$B_{\mathrm{ref}} = \sum_i B_i(\widehat{c}_{i+1}),$$

where f_i is the number of free blocks at scale i and $z_p = \int_{|w|\le p} e^{-|w|^2/2}\, d^3w$, in the convention of Theorem 12.13. The field-independent factors of the block averaging constraint and of the Haar measure extracted from L_i are kept in Λ. Up to the exact constant kept in Λ, the new effective interaction $\mathsf{M}^\Omega R$ produced by the step satisfies

$$\mathsf{M}^\Omega R = \mathsf{Q}^\Omega + \mathsf{P}^\Omega R + \mathsf{D}^\Omega R.$$

Here Q^Ω is the newly generated term, $\mathsf{P}^\Omega R$ is the pullback of the incoming interaction by the zero-fluctuation reconstruction, and $\mathsf{D}^\Omega R$ is the fluctuation correction of Theorem 20.9. For the

matched torus reference with zero extension of the fixed slots, there are constants A_J, K_J, and C_J, independent of the later widths, such that

$$\mathcal{X}^{\phi}\big(\mathsf{Q}^{\Omega} - \mathsf{Q}^{\mathbb{T}}|_{\Omega'} + \mathsf{P}^{\Omega}(r_{d_0}|_{\Omega}) - (\mathsf{P}^{\mathbb{T}} r_{d_0})|_{\Omega'}\big) \le A_J(1+E),$$
$$\mathcal{X}^{\phi}(\mathsf{P}^{\Omega} f) \le K_J \mathcal{X}^{\rho}(f).$$

Here Ω' is the set of unfixed scalar output labels. Assume that the column bounds of Theorem 19.6 and their perturbed versions hold on the product with doubled output analytic radii, and that the image under the reconstruction majorant on this product is at most half the incoming radius, as in Lemma 19.7. Then one may take $K_J = 4(\log 2)K_{\mathrm{col},J}$. Here $K_{\mathrm{col},J}$ bounds, for one fixed output, the sum over all old input labels, weighted by their boundary decay, of the distinguished derivative series of the perturbed zero-fluctuation reconstruction; it is chosen uniformly on the preliminary perturbation ball by Theorem 19.9. The degree-one terms and every nonconstant dependence of an omitted torus density on the unfixed labels are included.

At every admissible complex value of the coarse background and every fixed real admissible frozen value, let I_i and H_i be the sets of unfixed and of missing scalar free labels at scale i. With matched torus arguments extended by zero,

$$\log\det K_i^{\Omega} = \sum_{x\in I_i} (\log K_i^{\mathbb{T}})_{xx} + \mathcal{B}_i, \qquad |(\log K_i^{\mathbb{T}})_{xx}| \le C_J, \qquad |\mathcal{B}_i| \le C_J |\partial_{\mathrm{blk}}\Omega|.$$

These determinant statements also hold in the coefficient norm before evaluation. Scalar determinant labels and fluctuation blocks are counted separately. All constants have the dependence described in item (3) of Definition 4.14 and are independent of the later widths.

Proof. Every step here is a full step, $s = J$; a final composition of fewer than J averaging steps that is evaluated deterministically adds no fluctuation integral.

Step 1: the exact factorization. The triangular calculation in the proof of Lemma 20.3 gives, at a fixed coarse background and fixed frozen values, the value of the integral

$$\Big(\prod_i \mathcal{J}_{i,\mathrm{ref}} g_i^{3f_i}\Big) e^{-F_{\mathrm{out}}/g_s^2 - B_{\mathrm{ref}} - R(\mathrm{Log}\, C(v,0;b_{\mathrm{rec}}))} \prod_i z_{p_i}^{f_i}\, \Xi_R(v), \tag{20.37}$$

where F_{out}/g_s^2 is the classical value of the step with the output coupling, and the $\mathcal{J}_{i,\mathrm{ref}}$ are the extracted constant Haar disintegration factors. Inserting the definitions of Q, P, and D into the exponent of (20.37), and writing $\log \Xi_R = \log \Xi_0 - \mathsf{D}R$, we obtain $\mathsf{M}R = \mathsf{Q} + \mathsf{P}R + \mathsf{D}R$. This proves the recursion, including its signs and its Gaussian convention. Centering moves the value at zero into Λ and changes no factor of this identity.

Step 2: the generated term and the torus pullback. The deterministic part of $\mathsf{Q}^{\Omega} - \mathsf{Q}^{\mathbb{T}}|$ is the difference of the Haar densities at the minimizers and of the diagonal matrix-logarithm densities along the two zero-fluctuation chains. Their rooted comparisons with

boundary decay are part (3) of Theorem 20.2. A density based at a missing torus label is not discarded: a nonconstant coefficient with an unfixed index has a network to that missing label, and its sum over the missing label is bounded from the unfixed index by (20.1). Its constant is left in Λ. The terms $\log z_{p_i}$ and the constant Haar factors have no nonconstant background coefficient. The difference of the logarithms without insertion has the same fixed rates by Theorem 20.9, with a bounded norm in the cluster expansion. We obtain a finite bound C_J for the comparison of the nonconstant generated terms.

For the torus pullback, denote by a_Ω the zero extension of $\operatorname{Log} C^\Omega(v, 0; b_{\mathrm{rec}})$ and by $a_{\mathbb{T}}$ the matched torus argument; in this paragraph $a_{\mathbb{T}}$ is an argument, not the torus rate. We have

$$r_{d_0}(a_\Omega) - r_{d_0}(a_{\mathbb{T}}) = \int_0^1 Dr_{d_0}\big(a_{\mathbb{T}} + \tau(a_\Omega - a_{\mathbb{T}})\big)\,[a_\Omega - a_{\mathbb{T}}]\,d\tau. \tag{20.38}$$

Both absolute images are below half the incoming radius, so the whole segment admits the coefficient expansion of the integrand. On an unfixed old input, the difference of the arguments is the unperturbed defect of the reconstruction followed by the gauge transformation (Theorem 19.6) plus the defect of the perturbation (Theorem 19.9). On a fixed old input, the difference is a visit to Ω^c. The action and interaction tensors join every output index to that visit, even if the source does not decay inside its own component of Ω^c. The jet bound (20.28) with $\sigma_0 = 0$ bounds the integrand by $C_J E$, with its visit. The conversion of networks and degrees in the proof of Theorem 20.9 therefore bounds its $\mathcal{X}^\phi$ norm by $C_J E$. Together these bounds prove the first inequality, with A_J fixed on the larger products. No first derivative of the output was canceled.

Step 3: the pullback of the boundary function. For the zero-fluctuation reconstruction $\Phi_e(v) = \operatorname{Log} C_e(v, 0; b_{\mathrm{rec}})$, we define

$$C_{e,x}(2\phi) = \sum_\nu \nu_x |\Phi_{e,\nu}|(2\phi)^\nu e^{a_1 \mathfrak{t}(\{e\}\cup \operatorname{supp}\nu)}.$$

All constants in this step refer to the perturbed map. Let D_J^0 bound the unperturbed distinguished series on the larger product with doubled output analytic radius. The difference of the reconstructions in Theorem 19.9, rooted at every index, with the uniform incoming norm and the inequalities on the conditional forces checked above, bounds the additional distinguished series by $\sigma_* D_J^1$ there. We fix a spatial decay rate $b > \alpha$ and define

$$D_J^{\mathrm{act}} = D_J^0 + \sigma_0 D_J^1, \qquad C_{e,x}(2\phi) \le D_J^{\mathrm{act}}(2\phi_x) e^{-b\,\mathrm{dist}(e,x)}. \tag{20.39}$$

The factor $2\phi_x$ appears because the input index is distinguished before the sum of absolute coefficients is taken. The rates are below the common exponent available in the coefficient bounds of the potential, and both D_J^0 and D_J^1 are computed at the preliminary bound on fixed larger products, before the later widths. Here and below, for an old input label e of R (a label of the incoming set D_{in}, on the input lattice of the step), we write

$d_{\mathrm{old}}(e) = d_{D_{\mathrm{in}}}(e)$ for its distance to the incoming fixed complement, measured in units of the input lattice as in the incoming norm. For a current unfixed background input x, we write $d_{\mathrm{out}}(x) = d_{\Omega}(x)$ for its distance to Ω^c, in units of the output lattice. Finally, $\mathrm{dist}(e, x)$ is the distance between x and the position e/B_s of e, in units of the output lattice. Applying (20.7) on the old input set of R, which is the same set of unfixed labels, with quantum radius ρ, and dividing by B_s, we obtain

$$d_{\mathrm{out}}(x) \le d_{\mathrm{old}}(e)/B_s + \mathrm{dist}(e, x) + C_{\mathrm{buf},J}/B_s.$$

In Theorem 19.6 the distance d_{old} is measured in units of the output lattice, and the inequality $d_{\mathrm{out}}(x) \le d_{\mathrm{old}}(e) + \mathrm{dist}(e, x) + C_J$ is an explicit hypothesis of that theorem. With d_{old} there equal to our $d_{\mathrm{old}}(e)/B_s$, the inequality just displayed verifies this hypothesis, with buffer $C_{\mathrm{buf},J}/B_s$; it holds because the incoming fixed complement lies on the current fixed side. Since $B_s \ge 1$, this also gives the weaker comparison with $d_{\mathrm{old}}(e)$ in place of $d_{\mathrm{old}}(e)/B_s$. The density of the old labels then implies

$$\sum_{e \text{ old unfixed}} e^{-\alpha d_{\mathrm{old}}(e)} C_{e,x}(2\phi) \le K_{\mathrm{col},J}\, \rho\, e^{-\alpha d_{\mathrm{out}}(x)}. \tag{20.40}$$

We define $K_{\mathrm{col},J}$ as a common bound for this inequality, uniformly on the preliminary perturbation ball. One admissible choice is any number at least $2D_J^{\mathrm{act}} e^{\alpha C_{\mathrm{buf},J}/B_s} S_J(b - \alpha)$, with the fixed volume factors included in S_J, and at least the unperturbed column bound; we take the finite maximum over the label types and the stages. The absolute image on the product with doubled output analytic radius is at most $\rho/2$. The construction used a ratio larger than two between the profiles, so no repeated reduction of radii is involved. This is the column calculation of Theorem 19.6, with its constant enlarged by (20.39). The hypotheses on the doubled product and on the incoming image at half the radius are used here.

Let $x_f = \mathcal{X}^{\rho}(f)$. At an old input root e, the distinguished derivative sum evaluated at $\rho/2$ satisfies

$$\sum_{\eta : \eta_e > 0} \eta_e |f_\eta| (\rho/2)^{|\eta|-1} e^{a\mathrm{t}(\mathrm{supp}\,\eta)} \le 4\rho x_f e^{-\alpha d_{\mathrm{old}}(e)}. \tag{20.41}$$

For degree l this follows by multiplying the normalized coefficient with this input fixed by $l 2^{1-l}\rho$, which is at most 4ρ; the case $l = 1$ is included. The chain rule for coefficientwise majorants, the union of networks, and (20.40) bound the distinguished coefficient sum of the output at 2ϕ by $4K_{\mathrm{col},J}\rho^2 x_f e^{-\alpha d_{\mathrm{out}}(x)}$. At degree k the fixed-index weight of the coefficient norm with boundary decay inserts the factor $1/k$, and evaluation at ϕ instead of 2ϕ inserts 2^{-k}. We take the supremum over the fixed index at each degree and then sum. Since integrating $\sum_{j \ge 0} u^j = (1-u)^{-1}$ over $[0, 1/2]$ gives $\sum_{k \ge 1} 2^{-k}/k = \log 2$, we obtain

$$\mathcal{X}^{\phi}(\mathsf{P}^{\Omega} f) \le 4(\log 2) K_{\mathrm{col},J} x_f.$$

This also shows that a finite K_J exists when no particular column constant is specified.

Step 4: the logarithmic determinants. The diagonal coefficient series of $\log K_i^{\mathbb{T}}$ is bounded uniformly for each scalar label. To bound the constant as well as the derivatives, we write

$$\log K = \int_0^\infty \left[(1+t)^{-1} I - (K+t)^{-1}\right] dt.$$

Near $t = 0$ the gap bounds the integrand. For large t, the resolvent identity with I bounds it by $C_J(1+t)^{-2}$, since $K - I$ has a bounded coefficient row on the larger product. This integrable bound and the derivative bounds with every index rooted allow us to integrate coefficient by coefficient. The exact normalization of the scales adds only the same term $i \log M$ for each scalar label, with $i < J$. Thus $|(\log K_i^{\mathbb{T}})_{xx}| \le C_J$ on the stated complex profile, not only at zero background.

The comparison of a regional logarithm with the restriction of the torus logarithm is the sum of (20.19) and (20.20). Its diagonal density at an unfixed root u is bounded in coefficient norm by $C_J e^{-c_J d_\Omega(u)}$. Summing these diagonal defects and using (20.3), we obtain the bound on $\mathcal{B}_i$. If Ω^c is empty, the defect is zero. □

To state the recursion for Λ we fix the counts that enter it.

Definition 20.11. Consider a full step on the block region Ω and the matched torus step. Every scalar label is counted on the lattice on which it is introduced.

1. For each scale i, $|H_i|$ is the number of missing scalar free labels at scale i, that is, free labels of the torus step that are not labels of the regional step.
2. N_{missing} is the number of scalar labels of the torus step (background labels on the output lattice and free labels at the intermediate scales) that are not labels of the regional step. We write $\mathcal{M}_{\text{block}}$ for the set of missing torus fluctuation blocks; since each such block consists of three missing scalar labels, $|\mathcal{M}_{\text{block}}| \le N_{\text{missing}}$.
3. N_{bd} is the number of scalar labels of the regional step whose positions lie within the fixed distance $w_{\text{band}} \ge 1$ of W, where w_{band} is the width of the interface band determined by the construction stencils; it does not depend on the later widths. This is N_{band} of (20.4), so $|\partial_{\text{blk}}\Omega| \le N_{\text{bd}} \le C_J|\partial_{\text{blk}}\Omega|$, including for isolated final blocks. Both counts are zero when W is empty.
4. N_{freeze} is the number of analytic scalar labels that are newly frozen at the end of the step, including, at the first step, the labels frozen at initialization.

When the steps are indexed by t, these counts are written $|H_i|$ for the scales i of step t, and $N_{\text{missing},t}$, $N_{\text{bd},t}$, $N_{\text{freeze},t}$.

Lemma 20.12. *Assume the hypotheses on the products and the small-interaction hypothesis of Theorem 20.9, in a full step, and fix a real admissible coarse background, a real admissible frozen value, and a real coupling $t = t_{\max}$. Use the potential without or with the insertion, from Lemma 20.3, and remove its entire zero-fluctuation part before forming the interaction $U_\bullet$,*

$\bullet \in \{0, R\}$. Let π be the product of the identical normalized three-dimensional Gaussian reference measures on balls, including the reference measures of all missing torus blocks, and write

$$\Xi_\bullet^{\mathbb{T}} = \int e^{-U_\bullet^{\mathbb{T}}}\, d\pi, \qquad \Xi_\bullet^{\Omega} = \int e^{-U_\bullet^{\Omega}}\, d\pi, \qquad \bullet \in \{0, R\}.$$

The regional interaction does not depend on the adjoined variables. With N_{missing} and N_{bd} as in Definition 20.11, there is a restricted interaction $U_\bullet^c$ on the same product such that

$$\|U_\bullet^{\mathbb{T}} - U_\bullet^c\|_\infty \le C_J \mathfrak{s}(t)(N_{\mathrm{missing}} + N_{\mathrm{bd}}),$$
$$\|U_\bullet^{\Omega} - U_\bullet^c\|_\infty \le C_J \mathfrak{s}(t) N_{\mathrm{bd}}.$$

Consequently, for the positive real logarithms,

$$|\log \Xi_\bullet^{\mathbb{T}} - \log \Xi_\bullet^{\Omega}| \le C_J \mathfrak{s}(t)(N_{\mathrm{missing}} + N_{\mathrm{bd}}). \tag{20.42}$$

This includes evaluation at coarse background zero, with the frozen values unchanged. Empty products have integral one.

Proof. The reference measure on a block (i, e) is $z_{p_i}^{-1} \mathbf{1}_{|w| \le p_i} e^{-|w|^2/2}\, d^3 w$. We take the product of these measures over the union of the two sets of blocks. Adjoining blocks multiplies the regional integral by one, even for blocks at different scales with different radii.

For this comparison of constants only, we resolve the torus monomials as in Definition 20.6, and let $\mathcal{D}_{\mathrm{mon}} = \{\lambda : S_\lambda \cap \mathcal{M}_{\mathrm{block}} \ne \varnothing\}$ be the set of deleted monomials. We call the remaining interaction U^c. The bound on W_{16} gives

$$\|U^{\mathbb{T}} - U^c\|_\infty \le \sum_{x \in \mathcal{M}_{\mathrm{block}}} \sum_{\lambda : x \in S_\lambda} b_\lambda \le C_J \mathfrak{s}\, |\mathcal{M}_{\mathrm{block}}|.$$

The set of monomials may be infinite; no bound on its cardinality is used. By Definition 20.11, $|\mathcal{M}_{\mathrm{block}}| \le N_{\mathrm{missing}} \le N_{\mathrm{missing}} + N_{\mathrm{bd}}$. This proves the first inequality.

For the regional and the restricted monomials, we compare coefficients on the common set of unfixed fluctuation labels, at the same background argument extended by zero. We use the comparison before the logarithm in Theorem 20.2 and Lemma 20.3: a reconstruction formula, a minimizer, an inverse square root, or a Haar density is replaced one factor at a time. A change of the coarser arguments is the integral of their derivative against the difference of the reconstructions, which carries a visit to Ω^c. For an incoming torus term we integrate Dr_{d_0} on the segment between the two absolute images, which lie below half the radius; fixed old input slots are visits to Ω^c, and unfixed slots use the defect of the reconstruction in Theorem 19.9. The boundary function f enters with its own incoming boundary weight. By the versions with every index rooted, these comparisons keep a visit to Ω^c and every specified unfixed fluctuation index, including all degree-one

terms. They are estimates on monomials before either logarithm is expanded, so the present restriction need not coincide with the one used in the earlier expansions.

Every remaining fluctuation monomial has an unfixed fluctuation index. We assign each compared monomial to its first such index y. The comparison of coefficientwise majorants with every index rooted bounds its absolute sum there by $C_J \mathfrak{s}(t) e^{-\alpha d_\Omega(y)}$. Summing over these indices, we obtain

$$\|U^\Omega - U^c\|_\infty \le C_J \mathfrak{s}(t) \sum_{y \text{ unfixed}} e^{-\alpha d_\Omega(y)} \le C'_J \mathfrak{s}(t) |\partial_{\mathrm{blk}} \Omega| \le C'_J \mathfrak{s}(t) N_{\mathrm{bd}},$$

by (20.3) and Definition 20.11. This proves the second inequality, at every fixed real frozen value.

The zero-fluctuation values of F_i, L_i, B_i, and R in Lemma 20.3 were subtracted before this sum and stay in the prefactor. In particular, a constant depending on the frozen values does not enter the boundary estimate. The convergent coefficientwise majorants justify all assignments and sums, by finite truncation and monotone convergence of absolute sums.

Finally, if real bounded U, V satisfy $|U - V| \le D$ pointwise, then $e^{-D} e^{-V} \le e^{-U} \le e^{D} e^{-V}$. Integrating against a probability measure and taking logarithms gives

$$\left| \log \int e^{-U} \, d\pi - \log \int e^{-V} \, d\pi \right| \le D.$$

All integrals here are finite and strictly positive because the potentials have bounded majorants. We apply this twice, through U^c, which proves (20.42). For a real interaction, the integral with the interaction multiplied by $0 \le \theta \le 1$ is positive and equals one at $\theta = 0$. The logarithm given by the cluster expansion vanishes at $\theta = 0$ and therefore agrees with the positive real logarithm along this path. The comparison therefore also bounds the constant used in the analytic recursion. □

Theorem 20.13. *Use full steps on admissible data, the exact extraction of field-independent factors, and the centered torus recursion of Theorem 20.10. Restrict the coarse backgrounds, the couplings, and the frozen values to real data on the chosen products of fluctuation balls. Index the full steps by $t = 0, 1, \ldots$, keep the exact incoming value of Λ, and write $\Delta\Lambda_t = \Lambda_t - \Lambda_t^{\mathbb{T}}$ for the difference between the regional and the torus normalization constants before step t. At the end of step t, the newly frozen analytic scalar labels form a set F, frozen simultaneously at the values $c^\sharp$. Assume the following.*

(a) The outgoing centered effective interaction of step t has the form $R^{\mathrm{out}} = r_{\mathbb{T}} + f$ with $R^{\mathrm{out}}(0) = 0$, $D r_{\mathbb{T}}(0) = 0$, $\mathcal{B}_{\rho,a}(r_{\mathbb{T}}) \le E$, and $\mathcal{X}^{\phi}_{\Omega}(f) \le Y$, where Y is a fixed bound.

(b) Every newly frozen value satisfies $|c^\sharp_j| \le \phi(j)/4$ for $j \in F$.

Then at each step, including the simultaneous restriction to fixed values at its end,

$$\Delta\Lambda_{t+1} \le \Delta\Lambda_t + \sum_{i \text{ in step } t} |H_i| \log(1/g_i) + C_J (N_{\mathrm{missing},t} + N_{\mathrm{bd},t} + N_{\mathrm{freeze},t}), \tag{20.43}$$

with the counts of Definition 20.11, where C_J may also depend on E and Y. Each Gaussian dimension occurs once, and the normalized auxiliary variables and the Haar measure of the block axial gauge trees have mass one. The restriction to fixed values is exactly

$$R_{\mathrm{new}}(v) = R^{\mathrm{out}}(v, c^\sharp) - R^{\mathrm{out}}(0, c^\sharp), \qquad \Lambda_{\mathrm{new}} = \Lambda^{\mathrm{out}} - R^{\mathrm{out}}(0, c^\sharp),$$

and the frozen constant satisfies

$$|R^{\mathrm{out}}(0, c^\sharp)| \le \rho^2 |F| (E/16 + Y/4). \tag{20.44}$$

The classical constants remain in the full classical action. A constant of the incoming interaction is first moved into Λ exactly, using $e^{\Lambda - R} = e^{\Lambda - R(0)} e^{-(R - R(0))}$; no nonconstant norm bounds this constant. No additional term proportional to $|\Omega| \log(1/g)$ appears. The restriction at initialization is included, with R^{out} the torus reference and $f = 0$. The constant C_J has the dependence described in item (3) of Definition 4.14 and is independent of the later widths, of the preceding depth, and of the number of steps. No inequality for complex values and no recursion for compositions of fewer than J averaging steps is asserted.

Proof. We compare the logarithms of the regional and the torus factorizations term by term: first the deterministic factors (Step 1), then the fluctuation integrals (Step 2), then the pullback at zero fluctuation (Step 3), and finally the constant produced by fixing coordinates (Step 4).

Step 1: coupling powers, determinants, and Haar factors. At fixed real frozen values, the factorization (20.37) contains exactly one factor g_i for each scalar free coordinate and one triangular determinant. Subtracting the logarithms of the regional and the torus factorizations, we find for the coupling powers

$$\sum_i \Big[\sum_{x \in I_i} \log g_i - \sum_{x \in I_i \cup H_i} \log g_i \Big] = \sum_i |H_i| \log(1/g_i).$$

The determinant statement of Theorem 20.10 bounds the difference of the logarithmic determinants by C_J for each missing scalar label plus $C_J |\partial_{\mathrm{blk}} \Omega| \le C_J N_{\mathrm{bd}}$, by (20.4). The same argument applies to the Haar densities at zero fluctuation, by the comparison in part (3) of Theorem 20.2. Their constant normalizations depend only on M and J and contribute C_J for each omitted or interface coordinate. Since $p_i \ge 1$ we have $0 < z_1 \le z_{p_i} \le (2\pi)^{3/2}$, so the terms $\log z_{p_i}$ contribute a bounded amount for each missing block. The Haar measure of the block axial gauge trees and the adjoined reference measures contribute no dimension.

Step 2: the fluctuation integrals. At coarse background zero, Lemma 20.12 bounds the difference of the fluctuation logarithms without insertion, and of those with insertion whenever the factorization uses them, by $C_J \mathfrak{s}(N_{\mathrm{missing}} + N_{\mathrm{bd}})$. In that proof, each missing

fluctuation block is counted among the missing labels and each comparison at unfixed roots in the interface count; no geometric crossing argument is needed. Using $\mathfrak{s} \le \epsilon_J$, we find that this step contributes $C_J(N_{\mathrm{missing}} + N_{\mathrm{bd}})$.

Step 3: the pullback at zero fluctuation. The zero-fluctuation value of the pullback has not entered Step 2. For its centered torus and boundary parts we assign each monomial of degree $l \ge 1$ to its input roots with the shares η_e/l, whose sum is one. The input norms of Definition 4.6 give $C_J E$ or $C_J \sigma_*$ for each input root, after the common image at half the radius has summed the degree series. Missing roots are counted as missing labels. Differences rooted at unfixed labels carry the visit in (20.38) and are summed by (20.3). The incoming constant was moved into Λ exactly before this calculation.

Step 4: fixing coordinates. Finally we apply Lemma 18.9. To prove (20.44), note that when the unfixed variables are set to zero, only monomials supported in F survive. On such a monomial of degree l we have $\sum_{j\in F} \nu_j/l = 1$, and by hypothesis (b) its frozen value has modulus at most $4^{-l}|f_\nu|\phi^\nu$. Summing over $j \in F$ and using the definition of $\mathcal{X}_\Omega^\phi$ with the weights larger than one dropped, we find that the boundary part contributes at most $|F|\rho^2 Y/4$, since $l \ge 1$. The torus part has $l \ge 2$ and $|c_j^\sharp| \le \rho/4$, so the seminorm bounds its contribution by $|F|\rho^2 E/16$. This proves (20.44); the bound contributes C_J for each newly frozen analytic label, degree one included. The value is subtracted from Λ with the sign in the statement, and its absolute value gives the upper bound. The same identity applies at initialization. Data deep inside the fixed complement stay in the external factors, and the classical constants stay in the classical action. Adding the finitely many contributions proves (20.43). Every estimate used a coefficient bound for each label or the boundary sum (20.3), which does not depend on the shape of the region; none of them estimates the incoming constant or introduces a factor depending on the number of components or of steps. □

20.6 Preservation of the analytic bounds under iteration

The following theorem is the main result of this section.

Theorem 20.14. *Fix the torus analytic radii and rates, the torus support weights and spatial margins, and M, and fix $A_{\mathrm{w}} \ge 1$. Choose J once so that the conditions on J of the torus construction used in Lemma 19.7 hold and $B \ge B_{\mathrm{geom}}$, as in Theorem 20.9. There exist positive fixed radii ρ, ρ_{cl}, R_{raw}, R_W, and R_{en}, rates a, α as in Theorem 20.9, a perturbation bound σ_*, the classical and quantum radius profiles defined in Section 20.1, a width s_{er} of the frozen layer, a bound β_{rec} on the frozen values, a support radius r, and a coupling bound $g_* < e^{-1}$, with the following property. One may take $\rho = \rho_{\mathrm{cl}}$; other fixed ratios are allowed if both image conditions are imposed. The bound g_* may depend on A_{w}.*

Consider a full step on an admissible block region satisfying the protected-neighborhood conditions of Lemma 19.5, at any preceding depth and after any number of earlier steps, with a real admissible frozen value and an input satisfying

$$\mathcal{X}^{\mathrm{cl}}(\bar{P}) \le 2\sigma_*, \qquad \mathcal{B}_{\rho,a_{\mathbb{T}}}(r_{d_0}) \le E, \qquad \mathcal{X}^{\rho}(f) \le \sigma_*, \qquad \max_i g_i \le g_*.$$

Assume the uniform incoming classical norm on D_{in}, the boundary comparison (20.7)*, and the inequality on the conditional forces of Theorem 19.9 and Lemma 19.3. Set $p(t) = A_{\mathrm{w}} \log(1/t)$ and require at the t-th step, with $t_t = \max_i g_{t,i}$,*

$$p(t_t) \le p_{t,i} \le p(t_t) + \tfrac{1}{2} A_{\mathrm{w}} (J-1) \log M.$$

Assume the inclusion of the products, the positivity of the rational image denominators, the small-interaction condition, and the conditions on the auxiliary half-disc of Lemma 20.3 and Theorem 20.9, with the bound chosen for this whole family of radii. The small-field event uses $p_{t,i} = p(t_t)$. Under these hypotheses the perturbed objective has all of the following properties: the coefficient bounds of the potential with each index fixed in turn, the gap of the constrained Hessians at every stage, and the strict bounds on the image under the reconstruction majorant, on the outer profile times the whole enclosing product; the comparisons with boundary decay of Theorem 20.2; the potential of Lemma 20.3; the expansions without and with insertion of Theorem 20.9, with input support weight $e^{16|S|}$ and output support weight $e^{4|S|}$ as in Lemma 20.8, and with the same returned rates and radii; the bounds on the generated term, the pullback, and the determinants of Theorem 20.10; and the recursion for Λ of Theorem 20.13. These properties are conclusions of the earlier results.

More explicitly, let $x_t = \mathcal{X}(f_t)$ and $x_t^{\mathrm{cl}} = \mathcal{X}^{\mathrm{cl}}(P_t)$ be the incoming boundary norms of the quantum and the classical parts at step t, the boundary-completion adjustment of the classical part being understood. Before coordinates are fixed, the outgoing boundary norms y_t and y_t^{cl} satisfy

$$\begin{aligned} y_t &\le A_J(1+E) + K_J x_t + C_J t_t^2 (E + x_t)\big[p_t^4 + (E+x_t)p_t^2\big], \\ y_t^{\mathrm{cl}} &\le A_J^{\mathrm{cl}} + K_J^{\mathrm{cl}} x_t^{\mathrm{cl}}, \qquad t_t = \max_i g_{t,i}, \quad p_t = \max_i p_{t,i}. \end{aligned}$$

Under the hypotheses on the doubled output analytic radius and on the incoming image at half the radius, $K_J = 4(\log 2) K_{\mathrm{col},J}$, where $K_{\mathrm{col},J}$ is the column bound of Theorem 20.10 for the perturbed zero-fluctuation reconstruction, uniform on the preliminary perturbation ball. The recursion for Λ is Theorem 20.13.

Whenever coordinates are fixed, fix every label with reduced radius and keep the remaining unfixed labels at distance at least s_{er} from the previous boundary. For $\bullet \in \{\mathrm{cl}, \mathrm{q}\}$, with $(\rho_{\mathrm{q}}, E_{\mathrm{q}}) = (\rho, E)$ and $(\rho_{\mathrm{cl}}, E_{\mathrm{cl}})$ the classical radius and the classical torus bound, require

$$C_{\mathrm{bc},\bullet} E_\bullet \beta_{\mathrm{rec}} / \rho_\bullet \le \sigma_*/4, \qquad |c_j^\sharp| \le \min\{\beta_{\mathrm{rec}}, \phi_\bullet(j)/4\}.$$

Here $C_{\mathrm{bc},\bullet} \geq 8$ is fixed before β_{rec} and the later widths, and includes any finite sum over types of boundary-completion differences and over conversions between coordinates; a single simultaneous restriction of the union of all newly frozen labels to their fixed values uses $C_{\mathrm{bc},\bullet} = 8$ in the coordinates in which the labels were introduced. Every frozen label, whether in the boundary layer, in a protected neighborhood, or among the boundary-crossing coordinates, and including the labels frozen at initialization, is covered by this bound. Here $c^\sharp$ is the value in the analytic coordinates in which the label was introduced; any conversion from group values or transported coordinates is included in the bound by β_{rec} beforehand. Assume $x_0, x_0^{\mathrm{cl}} \leq \sigma_*$. Then, including the new boundary-completion differences of the torus parts,

$$x_{t+1} \leq z_t + 2e^{-\alpha(s_{\mathrm{er}}-1)} y_t, \qquad x_{t+1}^{\mathrm{cl}} \leq z_t^{\mathrm{cl}} + 2e^{-\alpha(s_{\mathrm{er}}-1)} y_t^{\mathrm{cl}},$$

where z_t and z_t^{cl} are the norms of the boundary-completion differences of the torus parts created when coordinates are fixed at the end of step t. The choices can be made with

$$x_0, x_0^{\mathrm{cl}} \leq \sigma_*, \qquad z_t, z_t^{\mathrm{cl}} \leq \sigma_*/4,$$

and with the last terms of both recursions at most $\sigma_*/2$, uniformly on the input ball. In particular

$$x_t, x_t^{\mathrm{cl}} \leq 3\sigma_*/4 \qquad (t \geq 1)$$

after any finite number of steps, with no further loss of radius or rate. For $|v_i| \leq \phi(i)$ the oscillation of the centered effective interaction satisfies

$$|R_t(v; b_{\mathrm{rec}}) - R_t(0; b_{\mathrm{rec}})| \leq \rho^2 \Big[E|\Omega_t| + \sigma_* \sum_{j \in \Omega_t} e^{-\alpha d_{\Omega_t}(j)} \Big],$$

where, for this count and this sum, Ω_t denotes the set of unfixed scalar labels, directions and colors included. At the output of a step that is not followed by fixing coordinates, σ_* in this formula is replaced by the fixed output bound for y_t.

All constants have the dependence described in item (3) of Definition 4.14 and exclude the later widths. The output bounds are computed first and s_{er} afterwards; J is not enlarged by means of these constants for fixed J. The theorem concerns admissible data; that the configurations of the large-field decomposition are covered by them is proved separately in Section 21. For a composition of $s < J$ averaging steps the estimates at each stage hold on their admissible products, but a conclusion about the two images or about the enclosing product also requires the separate image condition for that shorter composition; the preservation of the profile is asserted only for full steps. An empty set of unfixed labels has nonconstant norm zero, determinant one, logarithmic determinant zero, and fluctuation integral one.

Proof. The proof has three parts. Steps 1 to 3 prove the bounds on y_t and y_t^{cl} before coordinates are fixed. Steps 4 to 10 make the parameter choices in the order summarized in Table 20.1 and check that each choice uses only quantities fixed before it. Steps 11 and 12 prove the iteration and the bound on the oscillation.

Quantity	Chosen after
K_*, torus support weights, and margins	fixed data
J	K_* and the torus data
$\rho, \rho_{\mathrm{cl}}, R_{\mathrm{raw}}$	J
a, α	J and the exponents of the coefficients of the potential
σ_0, r_q	radii and rates
$A, C_0, C_1, C_{\mathrm{en},J}, \mathcal{L}_V$, and all output, tensor, and image constants	σ_0
R_W, then σ_*	these constants
$\rho_{\partial,\bullet}$, w_{bd}	these constants
Y, Y^{cl}, then s_{er}	σ_*
$C_{\mathrm{bc},\bullet}$, $C_{\mathrm{word},J}$, $C_{\mathrm{conv},J}$	the larger charts
β_{rec}, then r, then R_{en}	σ_* and the conversion constants
g_*	all of the above, and A_{w}

Table 20.1. The order of the parameter choices in the proof of Theorem 20.14. The requirements are stated in Steps 4 to 10.

Step 1: the hypotheses at each step. We fix $A_{\mathrm{w}} \geq 1$ before the coupling bound. The coefficient constants of the potential and the torus and geometric conditions used to choose J do not depend on A_{w}; only the final coupling bound may depend on it. At each step, the incoming classical norm is the uniform norm on D_{in}, the boundary comparison (20.7) holds by assumption, with all label positions and volume factors on the lattices where the labels were introduced, and the conditional forces satisfy (20.11) and (20.12). The strict bounds on the two images place all segments of evaluation inside the uniform incoming products. Thus, at the fixed frozen value $b = (\xi, b_{\mathrm{rec}})$, both the perturbation and its comparison with zero satisfy the hypotheses of Theorem 19.9.

Step 2: the classical comparison before coordinates are fixed. Let a_R and a_T be the two admissible zero-fluctuation chains before the gauge transformation, the regional one with its missing slots set to zero. The comparison of values, before the common factor M^J, is

$$F_R(v, a_R) - F_T(v, a_T) = F_R(v, a_R) - F_T(v, a_R) + \int_0^1 D_z F_T\big(v, a_T + \tau(a_R - a_T)\big)\big[a_R - a_T\big]\, d\tau.$$

The fixed larger charts contain this segment, since the smaller solution neighborhoods have strict room in them. The first difference has the representation (20.15) through changed formulas. On unfixed free slots, the difference of the chains in the integral is the comparison of minimizers in Theorem 20.2, here with perturbation zero; on missing

free slots it is a bounded source in Ω^c. Every nonconstant coefficient with an unfixed output index therefore has a network to Ω^c, by the action tensors rooted at every index and Lemma 18.15. This includes derivatives falling on the action tensor while the source stays undifferentiated. We sum over the indices in Ω^c from the unfixed index using (20.1). Pure constants supported in Ω^c stay in the classical value. The conversion of coefficients and profiles in the proof of Theorem 20.9, without its expansions, now yields a finite bound A_J^{cl} for the centered comparison. The factor M^J depends only on J and is included in this constant. No interpolation between different fibers is used in this calculation of values.

Step 3: the perturbation and the quantum comparison. Next we interpolate in the adjusted input perturbation. By the envelope formula, the derivative of the minimum value is the evaluation of the perturbation at the minimizer. The strict bound on the image under the reconstruction majorant bounds its input jets, and joining its input boundary network to the reconstruction networks proves the output bound $K_J^{\mathrm{cl}} x^{\mathrm{cl}}$. Equivalently, this is the conclusion of Theorem 19.9 on the minimum values for two perturbations, with its exact boundary-completion adjustment. The fixed small boundary-completion difference is included with the unperturbed term in A_J^{cl}. This proves $y^{\mathrm{cl}} \le A_J^{\mathrm{cl}} + K_J^{\mathrm{cl}} x^{\mathrm{cl}}$. The coefficient networks in this argument are the classical ones, with the common rates chosen in Theorem 20.9.

We subtract the matched torus recursion from the recursion of Theorem 20.10. The difference is the sum of the comparison of the generated terms and torus pullbacks, the term $\mathsf{P}^{\Omega} f$, and the compared fluctuation correction. The first two are bounded in Theorem 20.10 by $A_J(1+E)$ and $K_J x$, and the third by Theorem 20.9. By the triangle inequality, we obtain the stated bound on y.

Step 4: radii, rates, and the perturbation bound. We fix K_* before J and choose J for the torus conditions and the geometric condition. We choose the base radii inside the fixed larger construction domains; one may choose $\rho = \rho_{\mathrm{cl}}$, and different fixed ratios simply keep both image conditions. We choose the common rates below the intrinsic coefficient and unperturbed image exponents at fixed J, and impose the separate output-rate conditions against Ba and $B\alpha$ for the inherited perturbation. Its controlled coefficient difference preserves the half-radius image at these working rates after the final perturbation bound is decreased. We decrease R_{raw} and the base radii if necessary, for the fixed margin of the larger construction domain and for $C_{T,\bullet} K_* \rho_\bullet \le 1/4$. The torus conditions are used with their stronger torus rate, not through an estimate involving an uncontrolled regional constant.

We choose a preliminary bound σ_0 below the admissible inverse, conditional force, and real gap margins, and a free radius r_q with

$$C_{\mathrm{seq},J}(2I_0 C_P \sigma_0 \rho_{\mathrm{cl}} + r_q) < \varepsilon_{\mathrm{tube}}/2.$$

The constants are those of the larger construction neighborhoods in (20.11), whose uniformity in all higher parameters was checked there. We compute all output, tensor, image, sequential, and enclosure constants uniformly for this bound on the fixed larger products. In particular we fix A, C_0, C_1, $C_{\mathrm{en},J}$, and $\mathcal{L}_V$, and the difference-row constants $C_{P,\bullet}$ at the allocated working rates, and choose R_W from the positive minimum in Lemma 20.3. Then we fix $\sigma_* \le \sigma_0$ so that, for both radii,

$$\sigma_* \rho_{\mathrm{cl}} \le \lambda_* R_W / (6 C_{\mathrm{en},J}), \qquad C_{P,\bullet} \sigma_* \rho_{\mathrm{cl}} \le \rho_\bullet / 64.$$

This order is necessary because this summand of the enclosure remains when r and β_{rec} are decreased.

Step 5: the boundary radii. We choose the two reduced boundary radii so that the bounds on the image at zero fluctuation near the boundary in Lemma 19.7 are at most $\rho_\bullet/16$. Then we choose a fixed width w_{bd} of the boundary layer so that the bounds on the tails far from the boundary are at most $\rho_\bullet/16$. The remaining bounds at zero fluctuation are the torus bulk bound and the bound for the frozen values. The first is a torus condition, and the second will be imposed on β_{rec}. All constants in these choices were computed independently of these widths and reduced radii.

Step 6: output bounds and the width of the frozen layer. We set

$$c_J^{\log} = \tfrac{1}{2} A_{\mathrm{w}} (J-1) \log M, \qquad p^+(t) = A_{\mathrm{w}} \log(1/t) + c_J^{\log}, \qquad H_* = E + \sigma_*. \tag{20.45}$$

By the hypothesis on the radii, we have $1 < p(t) \le p_{t,i} \le p^+(t)$ when $t < e^{-1}$. On the input ball we impose provisionally the requirement

$$C_J t^2 H_* \big[(p^+(t))^4 + H_* (p^+(t))^2 \big] \le 1,$$

which holds for small couplings. We then take the fixed output bounds

$$Y = A_J (1 + E) + K_J \sigma_* + 1, \qquad Y^{\mathrm{cl}} = A_J^{\mathrm{cl}} + K_J^{\mathrm{cl}} \sigma_*.$$

We choose the integer width of the frozen layer with

$$s_{\mathrm{er}} \ge \max\Big\{ 2,\ w_{\mathrm{bd}},\ 1 + \alpha^{-1} \log\big(4 \max\{Y, Y^{\mathrm{cl}}, 1\} / \sigma_* \big) \Big\}.$$

The bound may be enlarged to account for the fixed buffer between label positions in Lemma 18.9. Each output restricted to the interior of the frozen layer is then at most $\sigma_*/2$, since $2 e^{-\alpha(s_{\mathrm{er}}-1)} \max\{Y, Y^{\mathrm{cl}}\} \le \sigma_*/2$.

Step 7: the frozen values and the new boundary-completion differences. Whenever coordinates are fixed, let F be the union of all newly frozen analytic labels: the whole zone with reduced radius, the protected neighborhoods, and the boundary-crossing coordinates. The same convention covers the restriction at initialization. In the coordinates in which the labels were introduced, a single application of Lemma 18.9 to this union bounds the torus boundary-completion difference by $8E_\bullet\beta_{\mathrm{rec}}/\rho_\bullet$. The rate hypothesis of part (2) of that lemma holds with the torus rate $a_{\mathbb{T}}$ and the regional rates a, α, since $\sqrt{3}\,(a+\alpha) < a_{\mathbb{T}}$ by Theorem 20.9. No separate factor arises for each component of Ω^c. If a fixed finite list of types of boundary-completion differences is kept separately, with conversion factors $L_{k,\bullet}$, the triangle inequality yields instead the bound with $C_{\mathrm{bc},\bullet} = 8\sum_k L_{k,\bullet}$. We include the classical boundary-completion adjustment among these types whenever it is recorded separately. These constants are fixed on the larger charts, before β_{rec} and all later widths.

The bound concerns the analytic values $c_j^\sharp$ in the coordinates in which the labels were introduced. For a frozen group element in the identity chart, $|\operatorname{Log} U| = \rho_G(U)$ in the Euclidean norm of the Lie algebra, so this conversion has constant one. A longer fixed chart formula, in the coordinates before the gauge transformation or in transported coordinates, has a finite bound $C_{\mathrm{word},J}$ on its derivative rows on the larger construction domain. Integrating along the input segment, we bound its logarithm in the identity chart by $C_{\mathrm{word},J}$ times the maximal input norm. These conversion constants are fixed before the bounds on the frozen values are chosen. If a separate estimate for the initial frozen values is provided with a constant $C_{\mathrm{ad},J}$, we choose $C_{\mathrm{conv},J} \geq \max\{1, C_{\mathrm{ad},J}, C_{\mathrm{word},J}\}$ before using a condition relating the support radius to the frozen values. The theorem assumes the bound on every $c_j^\sharp$. It does not deduce this bound from a small support radius in an arbitrary gauge representative, and it transports a norm bound between gauge frames for the frozen coordinates only when the change of frame is the norm-preserving adjoint action of a block.

We choose $\beta_{\mathrm{rec}} > 0$ below the bound for the frozen values in the image, below the enclosure bound $\lambda_* R_W/(6C_{\mathrm{en},J})$, and, for both $\bullet \in \{\mathrm{cl}, \mathrm{q}\}$, below

$$\min\Big\{\rho_{\partial,\bullet}/4,\ \rho_\bullet/4,\ \frac{\sigma_*\rho_\bullet}{4C_{\mathrm{bc},\bullet}(E_\bullet+1)}\Big\}. \tag{20.46}$$

These are upper bounds on a quantity chosen after their constants, and they remain positive when a bound $E_\bullet$ is zero. They imply the two inequalities $C_{\mathrm{bc},\bullet}E_\bullet\beta_{\mathrm{rec}}/\rho_\bullet \leq \sigma_*/4$ in the statement. The hypothesis $|c_j^\sharp| \leq \min\{\beta_{\mathrm{rec}}, \phi_\bullet(j)/4\}$ is imposed on all of F, not only on the old boundary zone.

Step 8: initialization. The assumption $x_0, x_0^{\mathrm{cl}} \leq \sigma_*$ of the iteration can be met: starting from the torus reference, both boundary parts are zero before initialization, and the same restriction of the union to fixed values, together with (20.46), gives initial boundary parts

of norm at most $\sigma_*/4$. For other prescribed inputs the bounds on the initial norms are kept as hypotheses. The coverage of the configurations and the estimate for the initial frozen values are separate inputs of the later applications.

Step 9: the support radius. We choose $r > 0$ inside all margins of the small-link domain and of the larger construction domain, below $\lambda_* R_W/(6C_{\mathrm{en},J})$, and, when an estimate for the frozen values in terms of the support radius is provided, with $C_{\mathrm{conv},J} r \le \beta_{\mathrm{rec}}$, imposed after the conversion constant has been fixed. Then $R_{\mathrm{en}} = 2C_{\mathrm{en},J}(r + \beta_{\mathrm{rec}} + \sigma_* \rho_{\mathrm{cl}})/\lambda_*$ is positive and at most R_W. Both image denominators and the curvature bound on the whole product now hold by Lemma 20.3.

Step 10: the coupling bound. Finally, we choose $g_* < e^{-1}$ so that the following inequalities hold for every $0 < t \le g_*$, with the largest constant C_J of Lemma 20.7 and the fixed $\epsilon_J, c_{z,J}$ of Theorem 20.9:

$$\begin{gathered} tp^+(t) < R_{\mathrm{en}}, \qquad C_J t p^+(t) \le \tfrac{1}{2}, \\ 2C_J t D^+(t) \le \epsilon_J, \qquad tD^+(t) \le c_{z,J}/2, \qquad D^+(t) = (p^+(t))^3 + (1 + H_*)p^+(t), \\ C_0 t^2 \mathcal{L}_V \le 1/16, \qquad C_1 t(\mathcal{L}_V + 2) < 1/(4\sqrt{3}), \\ C_J t^2 H_* \big[(p^+(t))^4 + H_*(p^+(t))^2\big] \le 1. \end{gathered} \tag{20.47}$$

The image denominators are at least $1/2$ on the whole enclosing product by the earlier image bounds, which do not involve the coupling. For any radii in the stated logarithmic range we have $p_t \le p^+(t)$ and $E + x_t \le H_*$. The first line implies the inclusion of the products. The next line implies $\mathfrak{s}(t) \le \epsilon_J$ and $t \le c_{z,J}/(2[p_t^3 + p_t + (E + x_t)p_t]) = r_z/2$. The third line implies (20.25); since $p(t) > 1$, it also implies the condition $C_1 t(\mathcal{L}_V + 2) \le p(t)/(2\sqrt{3})$ on the means in Lemma 20.5. The last line is the provisional requirement of Step 6. The choice is simultaneous, because every left side that has to be small is a fixed combination of $t \log^q(1/t)$ and $t^2 \log^q(1/t)$ for finitely many integers q. Writing $t = e^{-u}$, we have $e^{-u} u^q \to 0$; for example the power series of $e^{u/2}$ bounds $u^q e^{-u}$ by a constant times $e^{-u/2}$. These upper bounds hold for all sufficiently small positive t, so a single bound works for all scales of every full step in the prescribed family of radii. The common cutoff has $p_{t,i} = p(t)$, and the cutoffs at the individual scales, $p(g_i) = p(t) + A_{\mathrm{w}} \log(1/\lambda_i)$, lie in the same range. The expansions are used on these chosen products, not on the enclosing product $\mathcal{K}_{\mathrm{en}}$. This final choice does not change the output bounds or the width s_{er}, which were computed before.

Step 11: the iteration. By Lemma 18.9 with $\eta = 1/4$, fixing the coordinates of the union F bounds the old boundary part by $2e^{-\alpha(s_{\mathrm{er}}-1)} y_t$, since $|c_j^\sharp| \le \phi_\bullet(j)/4$ and all labels with reduced radius are frozen. Independently, the boundary-completion differences of the torus parts satisfy, by the hypothesis on the frozen values,

$$z_t^\bullet \le C_{\mathrm{bc},\bullet} E_\bullet \beta_{\mathrm{rec}}/\rho_\bullet \le \sigma_*/4, \tag{20.48}$$

where $z_t^{\mathrm{q}} = z_t$. Fixing coordinates at an additional interior component of Ω^c satisfies the same estimate, because its values are also bounded by β_{rec}. Combining the two bounds, we obtain

$$x_{t+1} \le z_t + 2e^{-\alpha(s_{\mathrm{er}}-1)} y_t, \qquad x_{t+1}^{\mathrm{cl}} \le z_t^{\mathrm{cl}} + 2e^{-\alpha(s_{\mathrm{er}}-1)} y_t^{\mathrm{cl}}.$$

These estimates hold for arbitrary additional components of Ω^c, by Lemma 18.9. If both input norms are at most σ_*, then $y_t \le Y$ and $y_t^{\mathrm{cl}} \le Y^{\mathrm{cl}}$, and by Step 6 the right sides are at most $3\sigma_*/4$. Starting from the initial bounds, finite induction proves the bounds on x_t, x_t^{cl} after every number of steps. The older frozen values are simply fixed in the next application. The remaining labels keep their input radii because the whole zone with reduced radius was frozen: the conclusion of Lemma 18.9 is the norm with the uniform radius on the remaining labels. This verifies the hypothesis of a uniform incoming profile at the next step. The classical boundary-completion adjustment satisfies

$$\mathcal{X}^{\mathrm{cl}}(\bar{P}) \le \mathcal{X}^{\mathrm{cl}}(P) + \mathcal{X}^{\mathrm{cl}}(S_{\bar{\xi}}^{\mathrm{cl}}) \le \sigma_* + \sigma_*/4 < 2\sigma_*,$$

by the included bound on the boundary-completion differences. The inclusion of the old fixed complement and the positions of the labels in the next geometry are hypotheses of that step, not conclusions of this norm inequality. The inequality on the conditional forces then holds at every coarser parameter of the next step, by (20.11). Every operation preserves the Borel dependence on the frozen values, as proved in Theorem 20.2. No analytic loss is summed over earlier steps.

Step 12: the oscillation. We write the centered effective interaction as its torus part plus its boundary part. At each degree we assign each monomial to its roots with the shares $\nu_i/|\nu|$, whose sum is one. Evaluating at $|v_i| \le \phi(i)$ and dropping the network weight, we obtain for the boundary part

$$|f_t(v) - f_t(0)| \le \rho^2 \mathcal{X}(f_t) \sum_{i \in \Omega_t} e^{-\alpha d_{\Omega_t}(i)}.$$

The degree-one term is included in this assignment. For the torus part the same assignment and its seminorm give $\rho^2 E|\Omega_t|$. These are the stated bounds on the oscillation; before coordinates are fixed we use Y instead of σ_*.

The preceding results, with the choices just made, imply every assertion of the theorem. All local constants were evaluated before the later widths, and none of them was used to choose J. A composition of fewer than J averaging steps keeps its estimates at each stage on its admissible products, but its conclusions about the full radius require the separate image condition. The iteration for Λ and for the norms uses only full steps. A final composition of fewer than J averaging steps may remain as a deterministic function; no recursion for Λ is asserted for it here. The assertions for an empty set of unfixed labels are the usual conventions for empty sums, products, and determinants. This completes the proof. □

20.7 Summation of truncation errors

Lemma 20.15. *Fix $0 < \Gamma < \infty$, $M > 1$, a finite set $\mathcal{R}$ of ratios $R > 1$, a finite set of positive numbers η, and finitely many positive constants a, c, C, and ε.*

1. *For every finite list of nonnegative integers k, there is a minimum side m_0, independent of $0 < \gamma \le \Gamma$, such that*

$$C[1 + \log(m/\gamma)]^k \le \varepsilon m/\gamma \qquad (m \ge m_0).$$

2. *With $z = (1/2)\log(Mm/\gamma)$, the functions*

$$X_c(m,\gamma) = m^3 e^{-cm/\gamma}, \qquad W_a(m,\gamma) = m^3 e^{-az^2}$$

are decreasing in m and increasing in γ for $m \ge m_0$ and $0 < \gamma \le \Gamma$, after increasing m_0, and they satisfy $X_c(Rm,\gamma) \le X_c(m,\gamma)/2$ and $W_a(Rm,\gamma) \le W_a(m,\gamma)/2$ for $R \in \mathcal{R}$. Their sums over the sides mR^k, $k \ge 0$, are at most twice their first terms, and tend to zero as $m \to \infty$, uniformly on $(0,\Gamma]$.

3. *Let $m_k = mR^k$, $k \ge 0$, for some $R \in \mathcal{R}$, together with at most one additional side $\eta m R^j$, and let nonnegative errors δ and z_{ext} at these sides satisfy $\delta \le CW_a$ and $z_{\mathrm{ext}} \le CX_c$. After increasing m_0, depending only on Γ, M, $\mathcal{R}$, the set of η, and the constants, the following holds uniformly for $m \ge m_0$, $0 < \gamma \le \Gamma$, and the number of sides, finite or infinite. Every δ is at most $1/2$, every z_{ext} is at most $1/4$, and $\epsilon = \delta/(1 - z_{\mathrm{ext}})$ satisfies $\epsilon \le (4/3)\delta \le 1/2$. The sums of the δ, of the z_{ext}, and of the ϵ are finite, and*

$$\log \prod (1-\delta)^{-1} \le 2\sum \delta, \qquad \log \prod (1-\epsilon)^{-1} \le 2 \sum \epsilon,$$

also for every finite subproduct. These sums are bounded by any prescribed positive number after increasing m_0, and they tend to zero as $m \to \infty$, uniformly on $(0,\Gamma]$.

In particular, part (3) of Lemma 20.5 holds. All constants in this lemma have the dependence described in item (4) of Definition 4.14; the local constants of Lemma 20.5 keep their own dependence.

Proof. Let $x = m/\gamma \ge m_0/\Gamma$. For every fixed k we have $(1+\log x)^k/x \to 0$ as $x \to \infty$: with $y = \log x$, the exponential series bounds $y^k e^{-y/2}$ on a half-line $y \ge y_0$. This proves part (1), for finitely many k simultaneously, and allows us to assume $x \ge e$ throughout.

For part (2), let $\Delta_R = (1/2)\log R > 0$. Since $z(Rm) = z(m) + \Delta_R$, the ratios are

$$\frac{W_a(Rm,\gamma)}{W_a(m,\gamma)} = R^3 e^{-a(2\Delta_R z + \Delta_R^2)}, \tag{20.49}$$

$$\frac{X_c(Rm,\gamma)}{X_c(m,\gamma)} = R^3 e^{-c(R-1)m/\gamma}. \tag{20.50}$$

Their maxima over $m \geq m_0$ and $0 < \gamma \leq \Gamma$ occur at the smallest m and the largest γ, and they tend to zero as m_0 increases, since $z \geq (1/2)\log(Mm_0/\Gamma) \to \infty$ and $m/\gamma \geq m_0/\Gamma \to \infty$. We choose m_0 so that both ratios are at most $1/2$ for every $R \in \mathcal{R}$. The logarithmic derivatives in $\log m$ are $3 - cm/\gamma$ for X_c and $3 - az$ for W_a, and those in $\log \gamma$ are cm/γ and az. Hence X_c and W_a are increasing in γ, and they are decreasing in m once $cm_0/\Gamma > 3$ and $(a/2)\log(Mm_0/\Gamma) > 3$. The logarithms of the first terms at $\gamma = \Gamma$ are $3\log m - cm/\Gamma$ and $3\log m - a\log^2(Mm/\Gamma)/4$, and both tend to $-\infty$. By monotonicity in γ, the first terms tend to zero uniformly on $(0, \Gamma]$. Summing the series with ratio at most $1/2$ shows that the sums over mR^k are at most twice their first terms, which proves part (2).

For part (3), the additional side $\eta m R^j$ is handled in the same way. Let $m_{\mathrm{low}} = \min\{1, \min \eta\} m$. For sufficiently large m, monotonicity in m bounds the term at the additional side by the corresponding function at m_{low}, whether or not $\eta < 1$. Since m_{low} is a fixed positive multiple of m, the sums over all sides are at most three times the first terms evaluated at m_{low}, and these tend to zero uniformly on $(0, \Gamma]$ as $m \to \infty$, for finitely or infinitely many sides. We take the maximum of the finitely many minimum sides so obtained. Increasing m_0 once more, we make every δ at most $1/2$, every z_{ext} at most $1/4$, and the sums smaller than any prescribed positive number. Then $\epsilon = \delta/(1 - z_{\mathrm{ext}}) \leq (4/3)\delta$, so the sum and the maximum of the ϵ can be made as small as desired as well, in particular every ϵ at most $1/2$. For $0 \leq u \leq 1/2$,

$$-\log(1-u) = \int_0^u (1-v)^{-1}\, dv \leq 2u.$$

Summing this inequality bounds the logarithms of all reciprocal products, including every finite subproduct; the infinite products are defined by the increasing finite sums of logarithms. This proves part (3).

Part (3) of Lemma 20.5 is the case $R = B$, with $C = C_J$, $a = a_{\mathrm{w}}$, $c = b$, and the finite set of additional ratios, since $\log^2 \sqrt{Mm/\gamma} = z^2$; its bound $\epsilon_t \leq 2\delta_t$ follows from $\epsilon_t \leq (4/3)\delta_t$. The local concentration statements of Lemma 20.5 use only its hypotheses on the products and on the coupling at the given step; the restriction to a coupling bound enters only in the summation just proved, so those local statements and their prefactors are unchanged. □

SECTION 21

Large-field decompositions and conditional densities

This section connects the renormalization group on unions of blocks, constructed in Sections 18 to 20, with the two measures to which it is applied in Sections 22 and 23: the Wilson measure and the stopped measure. All statements concern a fixed lattice and are finite-dimensional.

The analytic theorems of Section 20 integrate the normalized fluctuation variables of one RG step over a fixed product of balls. They work in local coordinates near the identity, on a block region from which a fixed complement has been removed, with the link variables of that complement held fixed as frozen coordinates. To apply them to the Wilson measure, we must solve three problems. The first is coverage. A fiber of the block averaging map consists of all fine configurations with a given average. It is a Borel subset of the compact configuration space, can contain many gauge-equivalent representatives, and need not lie in a coordinate chart. We show that every point of every fiber carrying mass becomes, after a gauge transformation that fixes all prescribed vertices, a point of the domain of the local inverse coordinates, and that its normalized fluctuation coordinates lie in an enclosing product domain on which the conditional fluctuation potential is uniformly convex. The second problem is the choice of a density. The later estimates evaluate densities at particular gauge representatives. These form sets of Haar measure zero, so a Radon–Nikodym derivative defined only almost everywhere does not suffice. We construct an explicit Borel density from the Haar disintegration formulas and prove that it is invariant at every point under the relevant gauge transformations. The third problem is the preservation of frozen coordinates. When the integration stops near a region of large curvature, the link variables there are kept on the lattice where they were frozen. The gauge transformations used afterward must not change them, and the analytic estimates require their logarithms to be small.

Section 21.1 proves locality of the iterated block averaging map (Lemma 21.1) and a gauge-fixing theorem on arbitrary unions of large closed cells (Theorem 21.3). The last part of that theorem is a rule that protects the endpoints of all frozen coordinates. Section 21.2 defines a deterministic sequence of curvature tests, partitions the configuration space according to their outcomes into large-field decompositions (Theorem 21.5), and verifies the hypotheses of the analytic theorems of Sections 19 and 20 on each decomposition (Lemma 21.7). Section 21.3 proves the coverage of every supported fiber point and constructs, at all stages, the representatives satisfying the reconstruction identities

(Theorem 21.8): they are obtained from the coarsest scale downward by the reconstruction maps, so that each reproduces the output field above it and contains the frozen values. At the first RG step under the Wilson measure, this section uses the lower bound for the constrained Hessian of Lemma 18.14, which is uniform in the region and in the frozen variables. Section 21.4 constructs the invariant densities (Theorem 21.9), and Section 21.5 proves the identity relating the densities before and after freezing (Lemma 21.10). Section 21.6 first compares an integral over a fiber with an integral over the fluctuation cutoff domain (Lemma 21.11), and then combines these results in the induction of Theorem 21.12.

The inputs are the quantitative extension of maps into $\mathrm{SU}(2)$ in Lemma 7.5; the stopped measure and its coordinates on the torus (Theorems 14.3 and 14.6); the Haar disintegration on block regions (Lemma 18.7); the gauge transformation fixed at prescribed endpoints and the perturbation theorem of Lemma 19.5 and Theorem 19.9; and the analytic integration, the comparison of normalization constants, and the concentration estimate of Theorems 20.13 and 20.14 and Lemma 20.5. Theorems 21.5, 21.9, and 21.12 are used in Section 22 for the bounds on Wilson densities and the exponential moment bounds, and Theorems 21.3, 21.5, and 21.8 are used in the choice of parameters in Section 23.

The organization by nested regions, fields frozen on their complements, and effective densities on the remaining regions follows Bałaban's large-field renormalization [18, 20, 21]. Dimock's account of Bałaban's method for the scalar ϕ_3^4 model describes the partition into small-field and large-field regions, the unintegrated complement, and localized block averaging [67, §§1–2]; the summation of the resulting expansion is in [69]. These works concern a different model and different coordinates. The deterministic tests for the block averaging map T_M, the gauge construction on arbitrary unions of cells, the protection of frozen endpoints, the coverage of every fiber point, and the invariant density versions below are proved here.

Conventions. Throughout, $M \geq 3$ is odd, $J \geq 1$, and $B = M^J$. One application of T_M (Definition 4.3), including its Borel extension, is an averaging step, and an RG step consists of J consecutive averaging steps. A group of $1 \leq s \leq J$ consecutive averaging steps with output side m has intermediate scales $0 \leq i < s$ and couplings

$$g_i = t_{\max} M^{(i-s+1)/2}, \qquad t_{\max} = \sqrt{\gamma/(Mm)}, \qquad \lambda_* = M^{-(J-1)/2} \leq g_i/t_{\max} \leq 1.$$

The angle of $U \in \mathrm{SU}(2)$ is $\rho_G(U)$, its distance from I in the bi-invariant metric of Section 4; we have $\rho_G(U_1 \cdots U_l) \leq \sum_j \rho_G(U_j)$ and $|\operatorname{Log} U| = \rho_G(U)$ when $\rho_G(U) < \pi$. We write c for the output field. In analytic functions and in the coordinates of Definition 18.6, the coarse argument is its identity-chart logarithm $\operatorname{Log} c$, which we also denote by c when no confusion can arise; since $|\operatorname{Log} c_e| = \rho_G(c_e)$ for every output link e, bounds on the two agree. A block region Ω is a union of blocks of the output lattice of the current group

of averaging steps; its complement is the fixed complement. The frozen coordinates are written b_{rec}. They consist of frozen links and of based transports of frozen links across the boundary of the block region (defined in Section 21.2), each kept on the lattice on which it was frozen. A link is across the boundary of the block region if exactly one of its endpoints lies in it.

Fix the cutoff amplitude $A_{\mathrm{w}} \geq 1$ at the outset. Every application of Theorem 20.14 below uses the coupling ceiling g_* chosen after this amplitude, and g_* may depend on A_{w}. The analytic constants and the conditions used to choose J do not depend on A_{w}.

Unless stated otherwise, constants written C_J, c_J, u_J, ϵ_J, and similar depend only on M, J, the fixed larger construction and analytic domains and their radius ratios, the spatial and boundary decay rates and their margins, the bounded density of labels, and the fixed ceilings on actions and perturbations. They are independent of L, γ, all lattice sides, the total and the preceding depth, the number of RG steps since a coordinate was frozen, the shape, topology, and number of components of the regions, and the values of the frozen coordinates. This is the dependence described in item (3) of Definition 4.14. Where stated, these constants are also independent of the widths of boundary bands, removed bands, protected neighborhoods, and cells chosen later; such widths enter only through explicit geometric counts. No uniformity as $J \to \infty$ is asserted. The constant C_δ in Theorem 21.12 has instead the dependence described in item (4) of Definition 4.14.

21.1 Local path products and closed cell complexes

The full dependence set of a link of an averaged configuration is the set of links read by the block averaging map, including its Borel extension at exceptional values, by its reference path, and by every path product used to check that the logarithms involved lie in the correct branch. For an iterate we replace every link recursively by its full dependence set, and for a plaquette we take the union of the sets of its four links. These lists are treated as sets when we define excluded label sets; repetitions are kept only to estimate the lengths of path products. Geometric neighborhoods use the periodic maximum coordinate distance. Replacing it by the lattice distance used in the analytic norms changes only dimensional comparison constants.

Lemma 21.1. *Fix M and J. There exist finite constants D_J, L_J, v_J, depending only on M and J, and constants $u_J > 0$ and $C_J < \infty$, with the following properties for every group of $s \leq J$ averaging steps.*

1. *The full dependence set of a link, of a plaquette, or of a prescribed based transport of an averaged configuration is contained in the neighborhood of radius D_J of its output label, in output lattice units, and contains at most v_J input labels per output unit cube. The expanded hierarchical tree paths have length at most L_J.*

2. *Suppose that every input link of such a full dependence set has angle at most $u \le u_J$. Then the links of the i-th iterate have angle at most $(10M)^i u$ for $i \le s$, and the relative logarithmic coordinates of the block averaging constraint, including the eliminated coordinates, have size at most $C_J u$. After one fixed reduction of u_J, they belong to the domain of the local inverse of the block averaging constraint, with a fixed positive margin.*

These conclusions do not depend on the region or on its components.

Proof. An ordered path from the center of an M-block to a vertex of the block has length at most $3(M-1)/2$. The path $F_{c,x}$ of Definition 4.3 has length at most $4M-3$, and $F_{c,x}C_c^{-1}$ has length at most $5M-3$. If $(5M-3)u < 2\pi$, every relative transport $U(F_{c,x})U(C_c)^{-1}$ is in the identity chart, and the exponential of the average of their logarithms has angle at most $(5M-3)u$. Multiplying by $U(C_c)$, we obtain the bound $(6M-3)u \le 10Mu$. Iteration is permitted when $(10M)^J u$ is smaller than the fixed positive margin of one averaging step. The Borel extension reads the same finite list of links, so the locality of the list does not require this smallness.

The based tree gauge on a block is obtained by multiplying transports along its ordered tree paths. It sets each tree link to I, equals I at the root, and is unique with these properties, since the equation on an oriented tree edge determines the gauge at the child from the gauge at the parent. At the next intermediate scale we make the same construction and lift the new gauge as a constant on each of the previous blocks. Such a lift conjugates each previously fixed tree link and hence preserves the value I. The conditions at coarser scales are fixed before the finer variables are expressed. The resulting hierarchical transports are products of these tree factors at the various scales; no eliminated constraint edge is inserted into a tree. Each factor equals I on the hierarchical tree gauge section. There are at most J scales, each with finitely many paths in a fixed block, and by substitution we obtain a finite maximal path length L_J and a finite maximal radius D_J of the dependence neighborhoods. Equivalently, we may take these constants to be the maxima over the finitely many oriented dependence neighborhoods for $1 \le s \le J$ on the infinite cubic lattice: restriction to a region only deletes path products, and passage to a periodic quotient only identifies labels. The maximal number of input labels in the same dependence neighborhoods gives v_J.

After the tree gauge transformation, every link outside the trees is a product of a number, bounded in terms of M and J, of the small links just bounded, and the coarse-field lift of the target is another such product. The relative vector is therefore bounded by $C_J u$. The constraints have the eliminated-edge inverse and the uniqueness neighborhood, with its positive margin, used in Lemma 18.7 and Theorem 20.14. We choose u_J so that the preceding bound is less than half the radius of that neighborhood, taking the minimum over $s \le J$. Then the relative solution, including its eliminated entries, is the unique solution given by that inverse. This identifies the branch of a given solution; it

does not solve the constraints anew. Every path used above stays in a block of at most J scales, and none follows a tree spanning a whole component. □

The next corollary gives the initial bounds for a stopped measure; it is used in the proofs of Theorems 21.5 and 21.8.

Corollary 21.2. *Before the bound β_{rec} for frozen coordinates and the support radius r are fixed, one can fix $C_{\mathrm{ad},J} < \infty$, depending only on M, J, and the fixed tree conventions, with the following property. Let $U \in D_r$ be an initial configuration of a stopped measure on a lattice of side $M^s m$, with $s \le J$, $m \ge 4$, and no previously frozen coordinates, and assume that r is below the fixed margin of the finite path-product charts. Then, for every union of blocks, there exists a gauge representative in which the output field and all frozen links and based transports at the input level are bounded by $C_{\mathrm{ad},J} r$. This is a geometric existence statement, uniform in the preceding depth, the side, and the region.*

Proof. A gauge transformation taking U into B_r gives the bound r, not $r/2$, at the input. By Lemma 21.1, all intermediate internal links have angle at most $(10M)^J r$. The expanded hierarchical tree paths have at most L_J factors of these links. A link after the tree gauge transformation is a product of a path, a link, and an inverse path, and has angle at most $(2L_J + 1)(10M)^J r$. A based transport is a path times its link across the boundary of the block region and has the smaller bound $(L_J + 1)(10M)^J r$. Links lying entirely in the fixed complement remain below r, since the regional tree gauge is the identity there. The output roots are fixed by the tree gauge, and the output bound is $(10M)^J r$. We conclude that $C_{\mathrm{ad},J} = 2(2L_J + 3)(10M)^J$ exceeds twice each of these bounds. We fix it now, and only afterward require $C_{\mathrm{ad},J} r$ to be below the fixed identity and inverse chart radii. □

We now turn to gauge fixing on large closed cells. Small curvature does not force small links: a flat configuration with nontrivial holonomy around a noncontractible cycle cannot be gauge transformed to the identity. The following theorem shows that on a union of closed cells of side about R, plaquette angles of order R^{-2} allow a gauge in which all links have angle of order R^{-1}. The term R^{-1} comes from distributing compact gauge values over the edges of the cell grid, which have length at least $R/2$; the curvature contributes $O(\delta R)$ through an axial gauge in each cell. Transitions between the frames of neighboring cells are compared only on common edges and are never multiplied along chains of cells, so the constants do not depend on the topology of the union. We combine standard axial frames and the extension of maps into $\mathrm{SU}(2) \simeq S^3$ with the quantitative patching estimate of Lemma 7.5, preserving all prescribed vertices. The gauge fixing of lattice configurations with small plaquettes goes back to Bałaban [14]; the version below, on arbitrary unions of cells and with prescribed vertex values, is proved here.

Theorem 21.3. *There exist universal constants $C, C', R_0 < \infty$ and $c > 0$ with the following properties. Work in lattice units, let $R \ge R_0$ be an integer and $n \ge 8R$, and partition each*

coordinate circle of the periodic lattice of side n into intervals of lengths in $[R/2, 2R]$. Let P be any union of the resulting closed three-dimensional cells, and let U be a configuration on the links of P such that

$$\rho_G(U_p) \le \delta \quad \text{for every plaquette of every cell of } P, \qquad \delta R^2 \le 1,$$

including the plaquettes in the boundaries of the cells. These hypotheses are in force in all parts below.

1. *There exists a gauge transformation k on the vertices of P with*

$$\max_{e \subset P} \rho_G(U_e^k) \le C(\delta R + R^{-1}) \le C'/R.$$

This holds for disconnected P, for cells meeting only at vertices or edges, and for arbitrary holes, handles, periodic wrapping, and holonomy; the constants depend on none of these features, nor on n, L, the coupling, the depth, or J. If $q(U_p) < d_*$ for every such plaquette, one may take $\delta = \pi\sqrt{2d_*}$. For a fixed sufficiently large A_8, the choice $R = \lceil A_8/r \rceil$ together with $\pi\sqrt{2d_*}R^2 \le 1$ gives the bound $\rho_G(U_e^k) < r/2$ for every link.

2. *Let $\Gamma \subset P$ be a union of closed cells of the grid, and let k_Γ be a gauge transformation on the vertices of Γ with*

$$\max_{e \subset \Gamma} \rho_G(U_e^{k_\Gamma}) \le \eta, \qquad K_R = 1 + \eta R + \delta R^2, \qquad K_R^2/R \le c.$$

Then there exists an extension k equal to k_Γ at every vertex of Γ with

$$\max_{e \subset P} \rho_G(U_e^k) \le C K_R^{10}/R.$$

Every link contained in a cell with no vertex in Γ satisfies the stronger bound C'/R. In particular, this bound holds after removing two layers of cells around Γ.

3. *For fixed M and J, let $C_B \ge (10M)^J$ also bound the lengths of the finite tree path products. Assume $\eta \le C_B r$, $r \le 1$, and $R = \lceil A_8/r \rceil$, and set*

$$K_B = 2 + C_B(A_8 + 1), \qquad C_{\mathrm{col}} = C K_B^{10}/A_8.$$

If $K_B^2 r/A_8 \le c$ and $C_{\mathrm{col}} r < \beta_{\mathrm{rec}}/2$, then all links of P are smaller than $\beta_{\mathrm{rec}}/2$, and, after one fixed increase of A_8, the links in cells with no vertex in Γ are smaller than $r/2$. The constants in this part have the dependence described in item (3) of Definition 4.14. These statements assert the existence of gauge transformations; no configuration-dependent gauge transformation is used as a change of variables in an integral.

4. *At later scales, use the following endpoint protection rule. On the lattice on which each frozen coordinate was frozen, mark both endpoints of every frozen link, both endpoints of every based transport, and every prescribed basepoint or root. A block of the current output*

lattice is excluded whenever its full centered preimage on the lattice of some mark contains that mark. Then every gauge transformation supported on the remaining blocks has a lift, constant on blocks, which equals I at every mark, and this lift preserves every frozen link and every based transport exactly. This holds at every finite depth, for coordinates frozen at any earlier step and at periodic seams, with no sum over the steps at which the coordinates were frozen. A newly chosen root of the Haar disintegration is fixed separately by the current based gauge. For the gauge transformation of Lemma 19.5, the current endpoints of the links across the boundary of the block region are included among the fixed vertices as well.

Proof. We give the absolute and the relative constructions on the same skeleton.

Step 1: axial frames. In each cell Q of P, we fix a corner o_Q, let τ_x be the coordinate-ordered path in Q from o_Q to x, and define $q_Q(x) = U(\tau_x)$. Then $U_{xy}^{q_Q}$ is the transport around the loop $\tau_x(x,y)\tau_y^{-1}$. To bound it, we compare the ordered paths to the adjacent vertices x and y. Moving the last elementary step through the ordered path inserts at most two rectangular strips, each of width one and length at most $2R$. After canceling consecutive inverse edges, the resulting loop is a product of conjugated elementary plaquettes. There are at most CR of them, all in Q, including its boundary. By the product inequality for ρ_G, we have

$$\rho_G(U_{xy}^{q_Q}) \le C_0 \delta R. \tag{21.1}$$

Here C_0 is absolute, since the aspect ratio of a cell is at most four. This local estimate is the closed-cell clause of Lemma 7.5; the strip argument shows that it applies to a cell cut out of a torus. A cell has an injective periodic lift because $2R \le n/4$.

Step 2: transitions. For two cells Q, Q' containing a common lattice edge (x,y), we define $t_{QQ'}(x) = q_Q(x) q_{Q'}(x)^{-1}$. Direct multiplication gives

$$t_{QQ'}(y) = (U_{xy}^{q_Q})^{-1} t_{QQ'}(x) U_{xy}^{q_{Q'}}, \qquad d(t_{QQ'}(x), t_{QQ'}(y)) \le 2C_0 \delta R, \tag{21.2}$$

where we used (21.1) and bi-invariance. This estimate holds whether the cells share a face or only an edge. When two cells share only a vertex, only the equality of the assigned value at that vertex is required.

Step 3: the edges of the grid. We write the desired gauge in the frame of Q as $s_Q(x) = k(x) q_Q(x)^{-1}$ and set $k = I$ at every vertex of the cell grid. On each edge of the grid, we fix one incident cell frame, interpolate the two endpoint values of s_Q by a shortest geodesic in $\mathrm{SU}(2)$, and sample it at the lattice vertices. Since the group has diameter 2π and the edge has length at least $R/2$, the speed is at most $4\pi/R$. Each edge of the grid is assigned once, regardless of the number of incident cells. By (21.2), changing to another incident frame adds at most $C\delta R$ per lattice edge. Since $\delta R^2 \le 1$, the assigned frame values on every edge of the grid have lattice Lipschitz constant at most C_1/R.

Step 4: the faces of the grid. On a face of the grid, we convert the assigned boundary values into the frame of one incident cell, and we use a single consistent fine triangulation on all faces of the grid. Since the vertices of each fine simplex can be joined by a bounded number of lattice edges in the boundary, their distances are at most C_2/R. We take R_0 so large that the normalized affine interpolation clause of Lemma 7.5 applies, with denominator at least $1/2$. The continuous boundary map then has Lipschitz constant C_3/R. The disk extension clause, rescaled to this rectangle, bounds the Lipschitz constant of its extension by $CR^{-1}(1+C_3)^2$. Sampling the extension at all lattice vertices of the face, we recover $k = s_Q q_Q$. This is done once for each face, and previously assigned vertices are preserved exactly.

Step 5: the cells. For a cell Q of P, we convert the assigned boundary values into the frame of Q. On every lattice edge in the boundary their difference is at most C_4/R, by the face bound and (21.2). The same consistent triangulation and normalized interpolation give a boundary map from the sphere with Lipschitz constant C_5/R; we increase R_0 once to satisfy its simplex margin. By the ball extension clause, we obtain a map with Lipschitz constant $CR^{-1}(1+C_5)^5$, and sampling it gives k at the remaining vertices. For an edge of Q, the final comparison is the sum

$$\begin{aligned} \rho_G(k_x U_{xy} k_y^{-1}) &= \rho_G\big(s_Q(x) U_{xy}^{q_Q} s_Q(y)^{-1}\big) \\ &\le \rho_G(U_{xy}^{q_Q}) + d(s_Q(x), s_Q(y)) \le C(\delta R + R^{-1}). \end{aligned} \tag{21.3}$$

In the last inequality, we used (21.1) and the Lipschitz bound of the extension. Since $\delta R \le R^{-1}$, this proves the absolute estimate with $C' = 2C$, after enlarging C to include the finitely many aspect and simplex constants. Since the assignments on the skeleton agree at every common vertex, they can be made directly on the periodic quotient and on any union of cells. Handles, the number of components, and the holonomy have not entered any constant.

Step 6: the relative construction. We keep every lattice vertex of Γ at its prescribed value. In the frame of a cell Q, the prescribed values satisfy, on each prescribed lattice edge,

$$d(k_\Gamma(x) q_Q(x)^{-1}, k_\Gamma(y) q_Q(y)^{-1}) \le \eta + C_0 \delta R \le C K_R / R. \tag{21.4}$$

This follows by applying the first inequality in (21.3) in reverse to the two transformed link matrices, whose angles are both known; it imposes no bound on the increments of k_Γ in the original frame. At the vertices of the grid not in Γ, we set $k = I$. An edge of the grid in Γ is kept, and every other edge of the grid is a geodesic in one incident frame between its assigned endpoint values. Since Γ is a union of closed cells of the grid, its intersection with an edge of the grid is either the whole edge or a subset of its endpoints, so this rule respects all prescribed values. Its lattice increments in any cell frame are bounded by CK_R/R.

We keep each face of the grid lying in Γ. On every other face, the boundary interpolation and the disk filling give the bound CK_R^2/R; the interpolation is legitimate when CK_R/R is below its fixed simplex margin. On the boundary of a cell not in Γ, all assigned frame increments, including those on a face kept from Γ, are now at most CK_R^2/R. The universal c in the hypothesis is taken smaller than both required interpolation margins divided by these fixed constants. Since $K_R \geq 1$, the condition $K_R^2/R \leq c$ implies the first margin as well. The boundary interpolation on the sphere and the ball filling then give

$$\operatorname{Lip}(s_Q) \leq CR^{-1}(1 + CK_R^2)^5 \leq CK_R^{10}/R.$$

All cells of Γ are kept unchanged. Using (21.3) and $\delta R \leq R^{-1} \leq K_R^{10}/R$, we obtain the relative bound.

Step 7: cells away from Γ. If Q has no vertex in Γ, no face or edge of Q is prescribed. Each edge of the grid in the boundary of Q is then a shortest geodesic of speed at most C/R, even if its endpoint values were assigned in the frame of another incident cell, and frame changes add at most C/R. The disk extensions on the faces of Q then have Lipschitz constant C/R, and so does the ball extension in Q. By (21.3), we obtain the stronger conclusion for every edge of Q. Removing two layers of cells around Γ removes every cell with a vertex in Γ.

Step 8: the choice of δ and of A_8. If $U = e^X$ and $x = |X| \leq 2\pi$, then $q(U) = 2\sin^2(x/4)$ and $\sin(x/4) \geq x/(2\pi)$. Hence $x^2 \leq 2\pi^2 q(U)$, and the stated value of δ is valid. Let A_8 be larger than $4C'$ and R_0, increased by the fixed geometric factors used later in the excluded sets. Then $R \geq A_8/r \geq R_0$ and $C'/R \leq (C'/A_8)r < r/2$. For part (3) we have $A_8/r \leq R \leq (A_8+1)/r$, so

$$K_R \leq 2 + C_B(A_8+1) = K_B, \qquad CK_R^{10}/R \leq C_{\mathrm{col}} r, \qquad K_R^2/R \leq K_B^2 r/A_8.$$

The two smallness conditions in part (3) then imply the stated bounds. Since the constant C_B is fixed before r, these conditions are upper bounds on r and a lower bound on the ratio β_{rec}/r, and they are not circular.

Step 9: endpoint protection. In one coordinate, the balanced-digit identity

$$\Big\{ \sum_{j=0}^{s-1} M^j a_j : a_j \in \{-(M-1)/2, \dots, (M-1)/2\} \Big\} = \{-(M^s-1)/2, \dots, (M^s-1)/2\}$$

holds: existence follows by successively taking the balanced residue modulo M, and uniqueness by reducing two such expressions modulo M and dividing their difference by M. Taking the product over the three coordinates and then the periodic quotient, we find that each vertex of the lattice of a mark has exactly one ancestor in any current centered partition, also at a seam. A mark in the preimage of a block would exclude that block. A

gauge transformation supported on the remaining blocks is I at every such ancestor, and its lift, constant on blocks, is I at the mark. For any path, the gauge factors at the interior vertices cancel in order, and only the two endpoint factors remain. This proves that every marked frozen link and every based transport is preserved exactly, for all marks at once and with no sum over earlier steps. Finally, the cutoff in Lemma 19.5 vanishes at the prescribed endpoints of the links across the boundary of the block region as well, which proves the last statement. □

21.2 Deterministic large-field decompositions

In this section, we partition the support of the initial measure according to the outcomes of a deterministic sequence of curvature tests, and we show that on each piece every block region admits a representative with small links and small newly frozen coordinates.

Based transports and frozen coordinates. Let $e = (u_e, v_e)$ be a link across the boundary of the block region, oriented from its endpoint u_e in the block region to its endpoint v_e in the fixed complement, and let σ_e be the hierarchical tree path of Lemma 21.1 from the root α_e of the block containing u_e to u_e. The based transport of e is $\widehat{b}_e = U(\sigma_e)U_e$, the transport from α_e to v_e; its endpoints are α_e and v_e, while the endpoints of the link itself are u_e and v_e. When integration stops at a boundary, such links are frozen through their based transports, and the other links leaving the integration are frozen as links. We write $c^\sharp$ for the analytic coordinates of newly frozen coordinates: in the coordinates we use, the logarithm $\operatorname{Log} b$ of the frozen group value b, or a fixed analytic function of finitely many such logarithms (see (21.5)). Group values of frozen coordinates are never compared with analytic radii directly.

Parameters. We fix M, J, the construction domains of Theorem 20.14, and A_8 sufficiently large as in Theorem 21.3. Then we fix the frozen-coordinate conversion constant $C_{\mathrm{conv},J}$ of (21.7) below. It is computed from the relative extension of Theorem 21.3, the finite path products of the constraints and trees, the constant $C_{\mathrm{ad},J}$ of Corollary 21.2, the constant converting the near values at the first step under the Wilson measure, and the constants converting frozen group values into analytic coordinates. It is fixed before the two radii below. Then $r > 0$ and the bound $\beta_{\mathrm{rec}} > C_{\mathrm{conv},J} r$ for frozen coordinates are taken so small that every newly frozen analytic coordinate satisfies

$$|c_j^\sharp| \le \min\{\beta_{\mathrm{rec}}, \phi_\bullet(j)/4\}, \qquad C_{\mathrm{bc},\bullet} E_\bullet \beta_{\mathrm{rec}}/\rho_\bullet \le \sigma_*/4, \qquad \bullet \in \{\mathrm{cl}, \mathrm{q}\},$$

for the classical and the quantum spatially varying analytic radii $\phi_{\mathrm{cl}}, \phi_{\mathrm{q}}$, with the radii $\rho_\bullet$, ceilings $E_\bullet$, and boundary-completion constants $C_{\mathrm{bc},\bullet}$ of Theorem 20.14. These are conditions on deterministic parameters, imposed before the coupling. The deep frozen group variables at the first step under the Wilson measure are arbitrary elements of

SU(2), not small analytic logarithms; they are kept as fixed parameters and are never inserted into this analytic freezing. We set

$$R_{\mathrm{cell}} = \lceil A_8/r \rceil, \qquad W = 8BR_{\mathrm{cell}}, \qquad \pi\sqrt{2d_*}R_{\mathrm{cell}}^2 \le 1.$$

The width W is measured in output units of the current RG step; in output units of the next RG step it is $W/B = 8R_{\mathrm{cell}}$. The excluded sets below also contain the band of width s_{er} removed at each boundary (Theorem 20.14) and the protected neighborhoods of the finite path products. We may decrease r initially so that $W \ge s_{\mathrm{er}}$.

At each boundary between RG steps, the previous fixed complement and the earlier frozen basepoints lie outside the next block region, up to the fixed buffer of the dependence neighborhoods of the path products. Equivalently, their incoming and current boundary distances satisfy the comparison required in Theorem 19.9; this is proved as (21.10). The endpoint protection rule of Theorem 21.3 contains these basepoints and the newly frozen labels, and additional excluded regions preserve the inclusion. At the first step, there is no previous fixed complement, and the condition is vacuous.

Cell grids. At each lattice side $n \ge 8R_{\mathrm{cell}}$, we fix the cell grid deterministically. For example, we divide each coordinate circle into $\lceil n/R_{\mathrm{cell}} \rceil$ intervals, distributing the remainder among intervals of the two nearest integer lengths. For $R_{\mathrm{cell}} \ge 2$ these lengths lie in $[R_{\mathrm{cell}}/2, 2R_{\mathrm{cell}}]$. When an input grid must be aligned with the blocks, we perform the same construction in units of the blocking grid and round its scale up to a multiple of the block side. Since B is fixed and R_{cell} may be taken larger than B initially, this changes the scale by at most a fixed factor.

Definition 21.4. Fix a finite range of scales consisting of complete RG steps and an initial configuration U. The ordered tests are the following finite procedure.

1. Before any gauge transformation, compute the iterates $V^\ell = T_M^\ell U$ at all levels of the current RG step, with the Borel extension of T_M. Go through these levels in increasing order and, at each level, through the plaquettes in a fixed order.
2. A plaquette f fails if $q_f = q(V_f^\ell) \ge d_*$ and its full dependence set does not meet the current excluded label set. When f fails, all variables of its full dependence set that are not yet frozen are frozen at the input of its RG step, together with those of a fixed neighborhood of the dependence set. This neighborhood is aligned with the blocks of the output lattice and has output radius $2W$. It includes the fixed margins of the dependence neighborhoods and the protected neighborhood of the endpoint protection rule. Previously frozen variables stay where they are. The neighborhood is added to the excluded label set. A plaquette whose dependence set meets the excluded label set is not tested; it is excluded, not declared good.
3. After the tests of an RG step, retain only the closed cells of the output cell grid whose full dependence sets and protected neighborhoods avoid the excluded label set, and

add every touching excluded cell to the fixed complement. Freeze the full output boundary band: the layer of retained cells next to the excluded set, its incident cells, all labels of reduced analytic radius, the removed band of width s_{er}, and the margins of the dependence neighborhoods. Enlarge the previous fixed complement by an output neighborhood of width W, and then by the protected neighborhood of the endpoint protection rule of Theorem 21.3, using every endpoint and basepoint of every frozen coordinate on the lattice where it was frozen. A cell is retained only if its closure avoids all these exclusions, so a plaquette in the boundary of a cell is tested on the same terms as an interior plaquette.

4. For the next RG step, keep only the blocks that lie, together with their full dependence neighborhoods, in the retained interior. The discarded band is frozen. Excluded regions are combined by taking unions. If the block region becomes empty, its frozen coordinates are kept and the procedure stops.

When the initial measure is the Wilson measure, the first RG step is modified as in Theorem 21.5(4). The outcome of the procedure, consisting of the failing plaquettes, the excluded regions, and the label lists of the coordinates frozen at all steps, is a large-field decomposition, denoted $\mathcal{X}$. The values of these coordinates are variables of the stage measures and are not part of the finite index $\mathcal{X}$. We write $\mathcal{A}_{\mathcal{X}}$ for the set of initial configurations for which the procedure produces $\mathcal{X}$.

Theorem 21.5. *Make the parameter choices above.*

1. *For every finite range of scales, every initial side, every preceding depth $d_0 = aJ$ with $a \geq 0$, and every normalized stopped measure*

$$d\widehat{\nu} = \widehat{Z}^{-1}\mathbf{1}_{D_r} e^{-E_{d_0,\mathrm{sat}}/g^2_{m_{\mathrm{base}}} - [R_{d_0,\mathrm{sat}} - R_{d_0}(I)]}\, dH_{m_{\mathrm{base}}},$$

   the sets $\mathcal{A}_{\mathcal{X}}$ form a finite deterministic Borel partition of the support of $\widehat{\nu}$, provided that the output side of every complete RG step is at least $8R_{\mathrm{cell}}$. The same holds when the initial measure is the Wilson measure, with the first step described in part (4). Every tested plaquette that is not excluded satisfies $q_f < d_*$. The gauge transformations used below to exhibit small representatives do not enter the tests.
2. *At every boundary between RG steps there exists a gauge representative in which the links in the interior of the block region are smaller than $r/2$, and the newly frozen links and based transports are smaller than β_{rec} on the lattice where they are frozen. All previously frozen coordinates are preserved together with their endpoints; no smallness of their later averaged images is asserted. A block region that becomes empty keeps its frozen coordinates, and the procedure stops there.*
3. *Every added cell and every frozen label can be assigned to a failing plaquette by a local rule. For a failing plaquette f, the radius $R_t(f)$ of the neighborhood assigned to f, measured at*

the outputs of successive RG steps, satisfies

$$R_{t+1}(f) \le B^{-1} R_t(f) + C_J(W + R_{\mathrm{cell}} + 1), \qquad R_{\mathrm{first}}(f) \le C_J(W + R_{\mathrm{cell}} + 1),$$

where $R_{\mathrm{first}}(f)$ refers to the RG step in which f fails. The number of scalar labels assigned to f is at most $C_J(BW + R_{\mathrm{cell}} + 1)^3$ in that RG step and at most $C_J(W + R_{\mathrm{cell}} + 1)^3$ in each later RG step. The final plaquettes whose full dependence sets meet a neighborhood assigned to f are at most

$$F_J = C_J(W + R_{\mathrm{cell}} + 1)^3$$

in number, with a fixed enlargement for each of the finitely many remaining numbers of averaging steps after the last RG step (Theorem 21.12). These estimates are subadditive under unions. A set of failing plaquettes that can be produced by the ordered tests determines at most one large-field decomposition. The constants C_J have the dependence stated at the beginning of this section and are independent of the later widths; the displayed polynomials exhibit the dependence of the counts on the widths.

4. *For the first RG step under the Wilson measure, the ordered tests also test the plaquettes of the input configuration, before any representative is chosen. On a fixed input cell grid of scale S, aligned with the blocks, with $C_J^{-1} R_{\mathrm{cell}} \le S \le C_J R_{\mathrm{cell}}$, retain only the closed cells whose full dependence sets avoid the final excluded set. Move the block region inward by one full layer of cells and by the full dependence neighborhood of the constraint, tree, and basing path products, and require*

$$n_{\mathrm{input}} \ge 8S, \qquad \pi\sqrt{2d_*}\,S^2 \le 1.$$

Then every plaquette of the resulting closed cell complex, including its boundary plaquettes, is a tested plaquette with $q < d_*$. The complex contains every plaquette whose Wilson term depends on an integration variable and every constraint and tree path product of the block region. There are no earlier frozen vertices at this step. The gauge transformation of Theorem 21.3(1) on this complex makes the integration variables and the near values small; it extends to the remaining vertices, whose links are kept as arbitrary elements of $\mathrm{SU}(2)$ (the deep frozen group variables). The relative extension at the output gives the bounds of part (2) for the newly frozen coordinates. All additional excluded sets are assigned to the same failing plaquettes and satisfy the bounds of part (3) after one enlargement of C_J.

5. *For either initial measure ν define the stage measures*

$$\eta_t^{\mathcal{X}} = (\mathcal{O}_t^{\mathcal{X}})_{\#}(\mathbf{1}_{\mathcal{A}_{\mathcal{X}}}\nu),$$

where $\mathcal{O}_t^{\mathcal{X}}$ maps a configuration to the field on the current block region together with all frozen links and based transports, each on the lattice where it was frozen. All the tests remain in these measures. For every Borel set S (an event, not the grid scale of part (4)), in particular for every small-field event of the form (4.6) *on the same finite hierarchy,*

$$\mathbf{1}_S = \sum_{\mathcal{X}} \mathbf{1}_S \mathbf{1}_{\mathcal{A}_{\mathcal{X}}} \quad \nu\text{-almost surely.}$$

The partition does not change S and imposes no additional restriction on the Wilson measure. All statements are uniform in the holonomy, the number of RG steps since coordinates were frozen, the topology of the regions, and the deep frozen group variables. A remaining group of fewer than J averaging steps at the end of the range stops before the geometric conditions on the minimal side fail.

The following lemma records which changes of representative preserve the bounds on frozen coordinates; it is used in the proof of Theorem 21.5 and afterward.

Lemma 21.6. *Consider a representative as in Theorem 21.5(2) whose analytic frozen coordinates satisfy* (21.8).

1. *Every gauge transformation supported on the current block region and lifted under the endpoint protection rule equals I at the endpoints of all earlier frozen links and based transports. It fixes their group values and their logarithms exactly.*
2. *A gauge transformation equal to I at the roots of the block region and at the outer endpoints v_e of the links across the boundary of the block region fixes every based transport $\widehat{b}_e$, although it may move the inner endpoint u_e. The gauge transformation of Lemma 19.5 fixes u_e as well.*
3. *A simultaneous conjugation $b \mapsto kbk^{-1}$ of the frozen coordinates preserves* (21.8).
4. *A change $b_{xy} \mapsto k_x b_{xy} k_y^{-1}$ with independent k_x, k_y need not preserve* (21.8).

In the following, analytic estimates are transported between representatives only by the changes in parts (1)–(3). Changes as in part (4) are used only for the invariance of densities in Theorem 21.9. None of these conventions changes the tests or restricts the initial measure.

Proof. Part (1) is Theorem 21.3(4). For part (2), the gauge factors at the interior vertices of the path σ_e and of the link e cancel, so $\widehat{b}_e$ changes only by the factors at α_e and v_e, which are I; the factor at u_e cancels. The gauge transformation of Lemma 19.5 is I at the protected endpoints of the links across the boundary of the block region, including u_e. For part (3), we have $\operatorname{Log}(kbk^{-1}) = \operatorname{Ad}_k \operatorname{Log} b$ on the identity chart, and Ad_k is an isometry of $\mathfrak{su}(2)$. For part (4), take $b = I$ and $k_y = I$; then $\rho_G(k_x b k_y^{-1}) = \rho_G(k_x)$ is arbitrary. □

Proof of Theorem 21.5. We first fix all margins and constants that do not depend on a configuration (Steps 1 and 2). We then show that the tested plaquettes that are not excluded are good (Step 3), construct the small representatives by induction over the RG steps (Step 4), and compare boundary distances (Step 5). Steps 6 and 7 prove the counts and the measurability, Step 8 treats the first step under the Wilson measure, and Step 9 identifies the stage measures.

Step 1: fixed margins. All excluded sets in this proof are sets of labels, chosen by their locations and dependence sets; their construction uses no link representative. We fix

D_J, L_J, v_J from Lemma 21.1 and $C_{\mathrm{ad},J}$ from Corollary 21.2. The protected neighborhoods include one full copy of each dependence neighborhood, its incident cells, and the removed band of width s_{er}, and A_8 includes the factors required in Theorem 21.3. The support radius will be required to give $R_{\mathrm{cell}} \geq R_0$ and W larger than s_{er} and than the radii of these finitely many dependence neighborhoods. The number of copies and their enlargement factors depend only on M, J, and the fixed construction domains; in particular, they are fixed before W or R_{cell} is evaluated.

Step 2: constants for frozen coordinates. We specify the margins for frozen coordinates before using any representative. On the lattice where it is frozen, the analytic coordinate of a frozen link b_e or based transport $\widehat{b}_e$ is its identity-chart logarithm. If its angle is below π, then

$$|\mathrm{Log}\, b_e| = \rho_G(b_e), \qquad |\mathrm{Log}\, \widehat{b}_e| = \rho_G(\widehat{b}_e). \tag{21.5}$$

Here $|\cdot|$ is the adjoint-invariant Euclidean norm on $\mathfrak{su}(2)$, and the conversion constant is one. If the fixed coordinate convention also uses a longer path product or a relative eliminated coordinate, let $C_{\mathrm{cr},J} \geq 1$ bound its path length and the row sums of the first derivative of its fixed inverse chart on the larger construction domain. Such maps vanish at the configuration with all links equal to I. By the product inequality for angles, or by integration of the first derivative along the straight segment, their size is at most $C_{\mathrm{cr},J}$ times the bound on their inputs. This constant depends only on the fixed charts and on M, J, not on any width or number of steps.

We fix $C_{\mathrm{conv},J}$ by (21.7) below, after A_8 and before either radius; it includes $C_{\mathrm{ad},J}$ and $C_{\mathrm{cr},J}$. Let $\rho_{\partial,\mathrm{cl}}$ and $\rho_{\partial,\mathrm{q}}$ be the boundary values of the two spatially varying radii, and define

$$b_{\min} = \tfrac{1}{4} \min\{\rho_{\mathrm{cl}}, \rho, \rho_{\partial,\mathrm{cl}}, \rho_{\partial,\mathrm{q}}\}.$$

We take $\beta_{\mathrm{rec}} < \min\{\pi, \rho/4, \rho_{\mathrm{cl}}/4\}$ satisfying the two conditions $C_{\mathrm{bc},\bullet} E_\bullet \beta_{\mathrm{rec}}/\rho_\bullet \leq \sigma_*/4$, and then require

$$C_{\mathrm{conv},J} r < \min\{\beta_{\mathrm{rec}}, b_{\min}\}. \tag{21.6}$$

These are the conditions on frozen coordinates in the choice of parameters. All quantities on the right side are fixed positive parameters, so this upper bound on r is compatible with its other upper bounds.

Step 3: the tests and the goodness of retained plaquettes. We first consider a stopped measure. At the input of the first RG step, a gauge transformation into B_r gives all links smaller than r. By Corollary 21.2, which precedes this construction, the initial frozen coordinates at the input are bounded by $C_{\mathrm{ad},J} r$ in a common representative. Since $C_{\mathrm{conv},J} \geq C_{\mathrm{ad},J}$, their analytic coordinates satisfy (21.6) as well. At later inputs, by the induction in Step 4, the links in the interior of the block region are small, with all earlier frozen coordinates fixed. Before any gauge transformation, we compute the iterates at

every level of the current RG step and perform the ordered tests of Definition 21.4. The tests always use the values of the iterates themselves, also when a path product that is not used has an exceptional value of the Borel extension.

After the tests, the retained cells are those of Definition 21.4(3). A dependence set avoiding the final excluded set also avoided the smaller excluded set at the time of its test. If that plaquette had value at least d_*, its failure would have put the dependence set into the excluded set. This contradiction shows that every tested plaquette that is not excluded is good, that is, $q_f < d_*$. Adding excluded sets after the tests can only reduce this set of plaquettes.

Step 4: small representatives. We prove part (2) by induction over the boundaries. Let P be the retained closed cell complex. Every link read for an output link of P comes from links of the current interior, before the new freezing at the input. By Lemma 21.1, its angle in the incoming representative is at most $C_B r$, where C_B includes the finite tree path lengths. By Step 3 and Step 8 of the proof of Theorem 21.3, all plaquettes of P have angle at most $\delta = \pi\sqrt{2d_*}$. Let Γ consist of one layer of retained cells next to the excluded set, enlarged to contain all prescribed boundary vertices. We prescribe the identity gauge on Γ in this representative; its link bound is $\eta \le C_B r$. After the initial reduction $K_B^2 r/A_8 \le c$, parts (2) and (3) of Theorem 21.3 apply. The resulting gauge transformation is exactly I at every prescribed lattice vertex, has link bound $C_{\mathrm{col}} r$ on P, and has the bound $< r/2$ in cells without prescribed vertices. It extends by I outside P, because every boundary vertex of P is prescribed. We lift it as a constant through the current RG step. By the endpoint protection rule, the lift is I at all earlier marks and at the variables already frozen at the input in this RG step. Hence every earlier frozen coordinate, its endpoints, and every newly frozen input variable keep their values.

We freeze the full output boundary band, including Γ, its incident cells, all labels of reduced analytic radius, the removed band, and the margins of the dependence neighborhoods. Since the remaining interior is separated from Γ by two layers of cells, its links are smaller than $r/2$. The newly frozen links have angle at most $\max(C_B, C_{\mathrm{col}})r$. A based transport multiplies such a link by a hierarchical transport with at most L_J factors, and its angle is at most the sum of their angles. As announced before the radii were chosen, we may therefore fix

$$C_{\mathrm{conv},J} > 2C_{\mathrm{cr},J}(1+L_J)\max\{C_B, C_{\mathrm{col}}, C'/A_8, C_{\mathrm{ad},J}\}, \tag{21.7}$$

with any additional fixed factors of dependence neighborhoods absorbed once. Every analytic coordinate of the union of the newly frozen boundary band, boundary layer, and based transports then has size at most $C_{\mathrm{conv},J} r$. By (21.5) and (21.6), we have simultaneously

$$|c_j^\sharp| < \beta_{\mathrm{rec}}, \qquad |c_j^\sharp| \le \phi_\bullet(j)/4 \quad (\bullet \in \{\mathrm{cl}, \mathrm{q}\}). \tag{21.8}$$

This includes every label of reduced radius, not only the boundary links of a layer. The constant is computed from A_8, C_B, and the universal relative extension before r and β_{rec} are chosen.

For the next RG step, we retain only the blocks that lie, together with their full dependence neighborhoods, in the remaining interior. The discarded band has thickness at most C_J in that input lattice and consists of links that were small before discarding. The fixed factor coming from its dependence neighborhood was included in $C_{\mathrm{conv},J}$ before the radii were fixed, so the band may be frozen with the same bound. The previous excluded region was enlarged by W in the current output lattice. After the next RG step, this is $W/B = 8R_{\mathrm{cell}}$ units, which suffices for the required layer of cells and the fixed incidences; the full dependence neighborhood is included in addition. None of these operations requires a later averaged image of a frozen coordinate to be small, since the coordinate stays on its own lattice. This completes the induction.

Step 5: incoming and current boundary distances. For the next RG step, let D_{in} be its list of remaining analytic input labels after the whole freezing, and let Ω' be the set of its current retained blocks. Each incoming label is kept at its location on the lattice where it was created, with its original volume factor. A current block is retained only when its full input dependence neighborhood is contained in D_{in}. An omitted incoming label lies outside the block region, apart from the fixed displacement between the location of a chart label and its path product. A newly frozen boundary label, or a label of a link across the boundary of the block region, is omitted from D_{in} and satisfies the same statement. By the endpoint protection rule, an earlier frozen endpoint or basepoint has its current centered ancestor in an excluded block. By the balanced-digit identity, the difference between its rescaled position and that ancestor is less than one current unit, uniformly in the number of steps. Including the finite dependence neighborhoods of the charts, we may take $L_{\mathrm{buf},J} = 3D_J + 6$, so that every omitted input label x satisfies

$$\operatorname{dist}(x/B_s, (\Omega')^c) \le L_{\mathrm{buf},J}, \qquad B_s = M^s. \tag{21.9}$$

In the induction $s = J$. The same inclusion holds for a single group of $s \le J$ averaging steps when its own domain conditions are imposed. Enlarging the excluded regions can only decrease the distance to $(\Omega')^c$, so it preserves the inclusion.

If the incoming fixed complement is nonempty, let i be an incoming retained label and x a nearest omitted input label. For a current retained label e we use the same distance as in the two boundary-weighted norms; the factor 3 in $L_{\mathrm{buf},J}$ covers the comparison with the maximum-distance convention for dependence neighborhoods. By the triangle inequality and (21.9), we have

$$d_{\mathrm{current}}(e) \le \operatorname{dist}(e, i/B_s) + B_s^{-1} d_{D_{\mathrm{in}}}(i) + L_{\mathrm{buf},J} + 1,$$

where the last term 1 accounts for the choice of a cell rather than a vertex as location. Rearranging, we obtain

$$d_{D_{\mathrm{in}}}(i) \geq B_s d_{\mathrm{current}}(e) - B_s \operatorname{dist}(e, i/B_s) - C_{\mathrm{buf},J}, \qquad C_{\mathrm{buf},J} = B_s(L_{\mathrm{buf},J} + 1). \tag{21.10}$$

Rescaling coordinates in this calculation does not rescale the counting measures of the labels or discard their volume factors in a column norm; these are the same as in the incoming analytic norm. If there is no incoming fixed complement, the boundary-weighted part of the input is zero, and no nearest omitted label is needed. A new excluded region satisfies the inclusion just proved once the initial split has been made. The buffer in (21.10) depends only on M, J, and the fixed charts, not on W, s_{er}, the number of steps, or the number of excluded components.

Step 6: locality and counts. We verify locality and the counts without counting connected components. We label each initially added neighborhood by its failing plaquette; when neighborhoods overlap, a label in their intersection is assigned to the first incident failing plaquette in the fixed order. The enlargement of a union by a fixed neighborhood is contained in the union of the corresponding enlargements of its members. A cell meeting such an enlargement lies within a further $4R_{\mathrm{cell}}$ of it, and a full dependence neighborhood costs at most $D_J + 1$ more. Projection of a mark to its current ancestor contracts distances by the blocking factor, with a rounding error of at most one current unit for the whole projection, not one for each earlier step. This follows from the balanced-digit identity in the proof of Theorem 21.3, since the whole preimage has radius less than half a block side.

If $R_t(f)$ bounds all current neighborhoods and projected marks assigned to f, the next RG step gives, with a fixed $a_J < \infty$,

$$R_{t+1}(f) \leq B^{-1} R_t(f) + a_J(W + R_{\mathrm{cell}} + 1). \tag{21.11}$$

The neighborhood created when f fails satisfies the same bound in the output units of its RG step. We take $C_J \geq a_J/(1 - B^{-1})$ and at least the corresponding constant for the step in which f fails. Iterating (21.11), we obtain $R_t(f) \leq C_J(W + R_{\mathrm{cell}} + 1)$ at every later boundary between RG steps. A remaining group of fewer than J averaging steps at the end is read as a fixed finite dependence neighborhood at the last boundary; it creates no additional integrated step and no update of the normalization constant. All endpoint marks added when boundary layers are frozen lie in the same neighborhoods, so the protected neighborhoods of the endpoint protection rule obey (21.11) too, even if the frozen coordinates contain marks on many lattices.

A ball of radius a in the maximum distance contains at most $(2a+3)^3$ vertices. The link direction, the color component, the intermediate scale, and the type of frozen coordinate multiply this by a fixed C_J. In the RG step in which f fails we convert the radius into

input units, which costs a factor B; at later RG steps we use the bound in current units just proved. We obtain the counts

$$C_J(BW + R_{\mathrm{cell}} + 1)^3, \qquad C_J(W + R_{\mathrm{cell}} + 1)^3$$

of scalar labels. The lists of missing labels, boundary-band labels, and frozen labels used in Theorem 20.13 are subsets of these lists of bounded density; deterministic images of earlier frozen coordinates are not new independent labels. A final plaquette whose full dependence set meets an excluded region lies in a further fixed dependence neighborhood of one of the assigned sets. Counting the three plaquette directions there, we obtain the bound F_J. The finitely many remaining numbers of averaging steps after the last RG step only enlarge the dependence neighborhood. On a periodic quotient these counts can only decrease. Since we use unions rather than hulls of components, all the bounds are subadditive, including for excluded regions that touch or wrap around the torus.

Step 7: Borel measurability. The outcome of the ordered tests is determined by its set of failing plaquettes: replaying the ordered list and adding the prescribed excluded sets at exactly those failures reproduces it. Hence there is at most one decomposition for each set of failing plaquettes that the procedure can produce. Each event $\mathcal{A}_{\mathcal{X}}$ is Borel, since it is a finite Boolean combination of tests of Borel iterates and of membership in deterministic dependence lists. These events are disjoint and exhaust the support. The small representatives were used only after these events had been specified.

Step 8: the first step under the Wilson measure. For the Wilson measure, we perform the same tests, including the plaquettes of the input configuration, without assuming a small input representative. Let F be the final excluded dependence set. On the fixed input cell grid of scale S, aligned with the blocks, we form the initial good-cell complex P_{in}, consisting of all closed cells whose full dependence sets avoid F. The blocks of the block region are chosen only after moving inward by one full layer of cells and by the full dependence neighborhood of the constraint, tree, basing, and plaquette path products. This choice implies that every input plaquette whose Wilson term depends on an integration variable, and every path product of the block region, is contained in P_{in}. For example, a point of a retained block together with its dependence neighborhood still lies in a retained cell or in its incident layer, and that layer was retained before moving inward. The near links are the links outside the block region read by these path products and plaquettes, as in (18.32), and their values are the near values. All remaining input links are deep links; their values are the deep frozen group variables, that is, the frozen group variables in the fixed complement that are not near values.

An input plaquette of P_{in}, including a boundary plaquette, avoids the final set F; by the argument of Step 3, it is good. The absolute part of Theorem 21.3 applies with $n_{\mathrm{input}} \geq 8S$ and $\delta S^2 \leq 1$, and we obtain the bound $C(\delta S + S^{-1}) \leq C'/S$. Fixing the

mesh factors when A_8 is chosen makes this smaller than the required fixed multiple of r. At this first input there are no earlier fixed vertices, and we extend the resulting gauge transformation arbitrarily to the other vertices. All transformed deep links are kept as arbitrary elements of $\mathrm{SU}(2)$ in the frozen data. The finite path-product estimate and the relative construction at the output of Step 4 give the claimed bounds for the new boundary layers and based transports. The near values are small in this representative, not in the original configuration. Every additional excluded set lies within a neighborhood of radius $C_J S$ of an excluded neighborhood, and we assign it to the same failing plaquette. Since $S \le C_J R_{\mathrm{cell}}$, the radius and label counts keep their displayed forms after one fixed enlargement. After this first step, the induction of Step 4 applies.

All analytically frozen adjacent variables of this first step lie in the complex P_{in}, and hence obey the same conversion and (21.8). A dependence set frozen because its averaged plaquette fails can contain arbitrary elements of $\mathrm{SU}(2)$. Such deep frozen group variables are fixed parameters, not variables of the analytic freezing, and their plaquettes involving only frozen variables remain in the external Wilson factor. For a stopped measure, the corresponding initial variables lie in its gauge-invariant small-field domain, by Corollary 21.2.

Step 9: stage measures. By Lemma 21.6, every gauge transformation used in this construction fixes the earlier frozen coordinates and their logarithms; this is the form in which the representatives of part (2) are used later. All stage maps are Borel, since they are compositions of restrictions, the block averaging map, and finite transport products; their pushforwards are the measures in part (5). Each frozen coordinate is recorded once, as an independent link or based transport, and carries all its dependence through the density; no statement about independence under the probability measure is needed. Since the events $\mathcal{A}_{\mathcal{X}}$ partition the support of ν, multiplying their pointwise sum by $\mathbf{1}_S$ gives the identity in part (5), and no restriction is added to the Wilson measure. All geometric arguments were finite local constructions; no step involves L, γ, d_0, the depth, the number of steps since freezing, or the number of components. A final group of fewer than J averaging steps is stopped before its minimal-side condition fails, as stated. □

The next lemma verifies, on every large-field decomposition, the hypotheses of Theorems 19.9 and 20.14 that concern the incoming radius, the boundary distances, the conditional forces, and the cutoff domains. We use the notation of Lemma 19.3. For each initial segment I of the integrated intermediate scales, $y = z_{I^c}$ denotes the remaining free coordinates, $\mathcal{T}_I$ their domain, $a_I(c, y; \xi)$ the unperturbed conditional minimizer, and A_I the unperturbed conditional Hessian at a_I. Further, r_I is the radius of the conditional coordinate neighborhood, r_q the bound on the scaled fluctuations $g_i w_i$, $\varepsilon_{\mathrm{tube}}$ the fixed margin, and $C_{\mathrm{seq},J}$ the constant of the sequential construction. As in Theorem 19.9, Φ_0 is the reconstruction in the target-dependent gauge and $\overline{P}$ is the classical perturbation ad-

justed by the boundary-completion difference. Here $\Phi_0(c,u;\xi)$ has the joint normalized argument u used in that theorem: if z_0 is the unperturbed joint stationary point, K_0 the unperturbed Hessian there, and $S_0 = K_0^{-1/2}$, then $z = z_0 + S_0 u$. In the lemma below, the perturbing function in the original free coordinates is

$$V(c,z;b) = \overline{P}\big(\Phi_0(c, S_0^{-1}(z-z_0);\xi);b\big). \tag{21.12}$$

Thus V_I and V_{II} below are derivatives in the original free coordinates z, at fixed c and real b, and the constants C_P and C_P' include the fixed coefficient bounds of this affine change of coordinates. For a function f of the labels D and frozen values $c^\sharp$ on $F \subset D$, we write $T_{c^\sharp} f$ for the function of the labels $D \setminus F$ obtained by inserting these values, and $\mathcal{X}_D^\phi$ for the coefficient norm with boundary decay of Definition 18.1 with spatially varying radius ϕ.

Lemma 21.7. *Assume the conditions on frozen coordinates and the construction conditions of Theorem 21.5.*

1. *Freezing at the boundary of a complete RG step removes every label of reduced analytic radius and leaves the incoming perturbation on the remaining labels with a uniform analytic input radius, separately for the classical and the quantum radius. Its incoming and current boundary distances satisfy* (21.10).
2. *Fix real $b = (\xi, b_{\mathrm{rec}})$, and suppose that the adjusted classical perturbation $\overline{P}$ has uniform norm at most $2\sigma_*$. Then the preconditioned conditional force δ_V, taken over every initial segment I and every $y \in \mathcal{T}_I$ and defined by the first expression below with V as in* (21.12)*, satisfies*
$$\delta_V = \max_I \sup_{y\in\mathcal{T}_I} \|A_I^{-1} V_I(c, a_I(c,y;\xi), y; b)\|_{\mathrm{coeff}} \le I_0 C_P \sigma_* \rho_{\mathrm{cl}}, \tag{21.13}$$
where I_0 and C_P are fixed bounds on the larger domains, independent of the later widths.
3. *The enclosing product domain and every conditional coordinate neighborhood lie in the domains of Lemma 19.3 when the construction conditions include*
$$\delta_V < \tfrac{1}{4}\min_I r_I, \qquad R_{\mathrm{en}} < r_q, \qquad C_{\mathrm{seq},J}(2I_0 C_P \sigma_* \rho_{\mathrm{cl}} + r_q) < \tfrac{1}{2}\varepsilon_{\mathrm{tube}}, \tag{21.14}$$
together with the fixed Hessian conditions and the two reconstruction-image conditions (21.17).
4. *The fluctuation cutoff domain with common radius $p(t) = A_{\mathrm{w}} \log(1/t)$ is one of the cutoff domains allowed in Theorem 20.14. Its containment and expansion conditions hold below a single coupling ceiling, chosen after these fixed positive radii.*

None of these statements requires a derivative in ξ or in b_{rec}.

Proof. Steps 1 and 2 prove part (1) and the bound on the new boundary-completion differences, Step 3 proves part (2), Steps 4 and 5 verify the image and convexity conditions behind part (3), and Step 6 proves part (4).

Step 1: uniform radius on the remaining labels. The uniform radii are used in the freezing part of Theorem 20.14. We write D for the old output list, I for the remaining labels, and $F = D \setminus I$. All labels of reduced radius belong to F, and $d_D(i) \geq s_{\mathrm{er}}$ for $i \in I$. If there is no old boundary, the boundary-weighted part of the incoming norm is zero, and the estimate below is immediate. Otherwise, since the full neighborhood of width $s_{\mathrm{er}} - 1$ of the old boundary was removed, a shortest route from i to the old boundary meets F at least $s_{\mathrm{er}} - 1$ steps earlier, and additional excluded regions can only shorten the distance to F. Hence

$$d_D(i) \geq d_I(i) + s_{\mathrm{er}} - 1.$$

For a monomial of degree $k \geq 1$ in the remaining labels and degree $j \geq 0$ in the frozen labels, inserting the frozen values multiplies its absolute coefficient contribution by at most η^j with $\eta \leq 1/4$, by (21.8). The new fraction ν_i/k contributes at most a factor $(k+j)/k \leq 1 + j$, and the remaining connecting network is no longer than the original one. Since every remaining label had the full radius $\rho_\bullet$, the resulting series is estimated at the uniform radius $\rho_\bullet$ on I, not at a reduced boundary radius. Summing $\sum_{j\geq 0}(1+j)\eta^j = (1-\eta)^{-2} < 2$ and using the distance inequality, we obtain, including degree one,

$$\mathcal{X}_I^{\rho_\bullet}(T_{c^\sharp} f) \leq 2e^{-\alpha_\bullet(s_{\mathrm{er}}-1)} \mathcal{X}_D^{\phi_\bullet}(f). \tag{21.15}$$

Replacing the radius and rates by the classical radius and rates in this monomial calculation, we obtain the classical statement, with the same factor $(1-\eta)^{-2}$.

Step 2: new boundary-completion differences. The bound $|c_j^\sharp| \leq \phi_\bullet(j)/4$ controls the inherited boundary term. Freezing also inserts the frozen values into the torus functions, and this produces a new boundary-completion difference. For it we use the bound $|c_j^\sharp| \leq \beta_{\mathrm{rec}}$ from (21.8). Every contributing monomial has frozen degree $j \geq 1$ and degree $k \geq 1$ in the remaining labels. For a fixed remaining index i, the graph connecting network joins i to a frozen label, so $d_I(i) \leq \mathtt{t}(\nu)$ and $\mathtt{t}(\nu_I) \leq \mathtt{t}(\nu)$. Let $\mathtt{t}_2$ denote the Euclidean network length in the torus coefficient norm, and let $\bar{a}_\bullet$ be its Euclidean rate. The rate choices of Theorem 20.9 give $\bar{a}_\bullet \geq \sqrt{3}(a_\bullet + \alpha_\bullet)$. Therefore

$$a_\bullet \mathtt{t}(\nu_I) + \alpha_\bullet d_I(i) \leq (a_\bullet + \alpha_\bullet)\mathtt{t}(\nu) \leq \sqrt{3}(a_\bullet + \alpha_\bullet)\mathtt{t}_2(\nu) \leq \bar{a}_\bullet \mathtt{t}_2(\nu).$$

With $\eta_\bullet = \beta_{\mathrm{rec}}/\rho_\bullet \leq 1/4$, the ratio of the new and old fractions ν_i/k is at most $1 + j$. For each fixed original degree we split its absolute coefficient sum according to the frozen degree j, which is uniquely determined. Taking the supremum over the remaining index before the sum over degrees, we lose at most the factor

$$\sum_{j\geq 1}(1+j)\eta_\bullet^j = \frac{\eta_\bullet(2-\eta_\bullet)}{(1-\eta_\bullet)^2} \leq 4\eta_\bullet.$$

The normalization is $\rho_\bullet^{-2}$ in both norms, and the powers of $\rho_\bullet$ cancel against the factors of the frozen labels. This includes the new degree-one terms, and gives the bound $8E_\bullet\beta_{\mathrm{rec}}/\rho_\bullet$ for one simultaneous freezing of the union of the frozen labels. The fixed constant $C_{\mathrm{bc},\bullet}$ includes every required conversion and any finite sum of such terms. The total contribution therefore satisfies

$$z_t^\bullet \le C_{\mathrm{bc},\bullet}E_\bullet\beta_{\mathrm{rec}}/\rho_\bullet \le \sigma_*/4. \tag{21.16}$$

The boundary band, the new boundary layers, the based transports, and the initialization are all included in this single bound. For a new excluded region in the interior, we do not claim that the removed band alone makes this difference small. Equations (21.15) and (21.16) imply the uniform incoming norms in the recursion of Theorem 20.14. The frozen coordinates ξ are parameters of those norms and need not have the full incoming analytic radius.

Step 3: conditional forces. We use the absolute coefficient bounds for the derivatives of $\overline{P}$ in Theorem 19.9, on its uniform input image, which lies strictly inside the input ball. By the fixed construction hypotheses, every conditional coordinate neighborhood has the same image, together with its connecting segments. Composing with the row-sum bounds of the first derivative of Φ_0 and of the affine change of coordinates in (21.12), we find $\|V_I\|_{\mathrm{coeff}} \le C_P\sigma_*\rho_{\mathrm{cl}}$, uniformly in $y \in \mathcal{T}_I$. Multiplying by the conditional inverse bound I_0, we obtain (21.13). The second derivative also contains the term $\overline{P}'\Phi_{0,II}$, where $\Phi_{0,II}$ is the second z_I-derivative of $z \mapsto \Phi_0(c, S_0^{-1}(z - z_0); \xi)$; keeping it, we obtain $\|V_{II}\|_{\mathrm{coeff}} \le C_P'\sigma_*$. The unperturbed neighborhoods are fixed first, so that their own Hessian variation uses at most half of the preconditioned margin. Reducing the fixed ceiling for σ_* makes $I_0C_P'\sigma_*$ fit the remaining margin and preserves the real half-gap and the complex inverse margin. At an unperturbed conditional minimizer, we have $F_{0,I} = 0$, so the condition on inverse times force is precisely $\lambda\|A_I^{-1}V_I\| \le \delta_V < r_I/4$ for $0 \le \lambda \le 1$. This verifies the hypothesis of the fixed-parameter contraction at every y, and its solution moves by at most $2\lambda\delta_V$. The uniform mixed coefficient bounds of the same theorem control its dependence on the higher parameters. The sequential conclusion of Lemma 19.3 then applies under (21.14) to all intermediate scales. In particular, a product with $|g_iw_{i,e}| \le R_{\mathrm{en}} < r_q$ stays inside every parameter and solution neighborhood with margin. This is a check of the conditional force at moving parameters, not only at the joint minimizer.

Step 4: the image conditions. Both inherited image conditions are checked with the bound $s_q = R_{\mathrm{en}}$ for the scaled fluctuations. For each radius, the four zero-fluctuation terms of Lemma 19.7 satisfy

$$B^{-1}Z_\bullet(K_*\rho_\bullet) \le \rho_\bullet/16, \qquad C_{\Delta,\bullet}C_{a,\bullet}R_{\partial,\bullet} \le \rho_\bullet/16,$$
$$C_{\Delta,\bullet}C_{a,\bullet}e^{-a_Ww_{\mathrm{bd}}/2}K_*\rho_\bullet \le \rho_\bullet/16, \qquad C_{F,\bullet}\beta_{\mathrm{rec}} \le \rho_\bullet/16.$$

Their sum is $A_{0,\bullet}$. The initial good-cell complex at the first step under the Wilson measure and the endpoint protection rule change neither these analytic constants nor the bound on the current data. In the notation of the fixed construction constants of Lemma 19.7, the image conditions are

$$A_{0,\bullet} \le \rho_\bullet/4, \quad 1 - C_{T,\bullet}(K_*\rho_\bullet + R_{\mathrm{en}}) \ge \tfrac{1}{2}, \quad \frac{C_{T,\bullet}R_{\mathrm{en}}}{1 - C_{T,\bullet}(K_*\rho_\bullet + R_{\mathrm{en}})} + C_{\Delta,\bullet}R_{\mathrm{en}} \le \rho_\bullet/8. \tag{21.17}$$

Here $R_{\partial,\bullet} = K_*\rho_{\partial,\bullet}$. The margin in the larger construction domain of radius R_{raw} is, explicitly,

$$C_{\mathrm{lift},J}[K_* \max(\rho, \rho_{\mathrm{cl}}) + r + \beta_{\mathrm{rec}} + q_{\mathrm{raw}}] < R_{\mathrm{raw}}/8,$$

where q_{raw} is the fixed upper bound given by the row sums of the first derivative for the sequential product of scaled fluctuations and for all its initial segments; this bound is fixed before the eventual enclosing radius. Together with this margin, the image conditions give the image $3\rho_\bullet/8$ without spatial weights. The fixed interpolation of spatial rates in Theorem 20.14 then gives an absolute weighted image strictly below $\rho_\bullet/2$, so all evaluations and segments used for the force estimates above lie strictly in the uniform input ball, for both the classical and the quantum radius. These conditions on the parameters are already required by Theorem 21.5 and the analytic theorem, and they involve no test on the support of the measure.

Step 5: convexity on the enclosing product domain. The radius R_W of the scaled fluctuations is fixed independently of the logarithmic cutoff. We first compute its constants on the fixed larger perturbation domain. Then R_W is taken below the joint and sequential bounds, the bound of the larger construction domain, and the two inherited image bounds of the analytic theorem, as well as below $1/(4A)$ and $1/(16C_0)$, where A and C_0 are the constants of Lemma 20.3. The remaining deterministic choices satisfy

$$r,\ \beta_{\mathrm{rec}},\ \sigma_*\rho_{\mathrm{cl}} \le \frac{\lambda_* R_W}{6C_{\mathrm{en},J}}, \qquad R_{\mathrm{en}} = \frac{2C_{\mathrm{en},J}(r + \beta_{\mathrm{rec}} + \sigma_*\rho_{\mathrm{cl}})}{\lambda_*} \le R_W. \tag{21.18}$$

Here $C_{\mathrm{en},J}$ is evaluated on the fixed larger products, as in the proof of Theorem 21.8, before these radii are reduced. Then $AR_{\mathrm{en}} \le 1/4$ and $C_0R_{\mathrm{en}} \le 1/16$. When $C_0t^2\mathcal{L}_V \le 1/16$, the potential estimate of Theorem 20.14 yields

$$\|D_w^2(V - |w|^2/2)\|_{\mathrm{row,col},a'} \le \frac{C_0(R_{\mathrm{en}} + t^2\mathcal{L}_V)}{(1 - AR_{\mathrm{en}})^3} \le \frac{1/8}{(3/4)^3} = \frac{8}{27} < \frac{1}{2}.$$

This verifies the Hessian margin on the whole enclosing product domain, separately from any expansion on the smaller logarithmic cutoff domain.

Step 6: the logarithmic cutoff domain. Let $p(t) = A_{\mathrm{w}} \log(1/t)$, $c_J^{\log} = A_{\mathrm{w}}(J-1)(\log M)/2$, and $p^+(t) = p(t) + c_J^{\log}$. The common choice $p_i = p(t)$ lies in $[p(t), p^+(t)]$. Since $g_i/t = M^{(i-s+1)/2}$, the choice $p(g_i)$ at each intermediate scale lies in that interval as well; only the common choice is used here. With the fixed ceiling $H = 1 + E + \sigma_*$ and the construction constants of the analytic theorem, the relevant conditions on the coupling can be checked in the following sufficient form:

$$\begin{gathered} tp^+(t) < R_{\mathrm{en}}, \qquad C_J t p^+(t) < \tfrac{1}{2}, \\ \frac{C_J t[(p^+(t))^3 + p^+(t) + Hp^+(t)]}{1 - C_J t p^+(t)} \le \epsilon_J, \qquad 2t[(p^+(t))^3 + p^+(t) + Hp^+(t)] \le c_{z,J}, \\ C_0 t^2 \mathcal{L}_V \le \tfrac{1}{16}, \qquad C_1 t(\mathcal{L}_V + 2) < \min\{1/(4\sqrt{3}), p(t)/(2\sqrt{3})\}. \end{gathered} \tag{21.19}$$

The first line ensures that the chosen radii are enclosed, and the second checks the conditions on the interaction and on the auxiliary half-disc of Theorem 20.14. The last line checks the curvature and mean conditions of Lemma 20.5, in addition to the already fixed small radius R_W. Any needed provisional output remainder has the form $C_J t^2 (E + \sigma_*)[(p^+)^4 + (E + \sigma_*)(p^+)^2]$ and fits its fixed bound by the same reduction. Indeed, with $u = \log(1/t)$, each left side required to be small is bounded by a finite sum of terms $e^{-ku}(A_{\mathrm{w}} u + c_J^{\log})^l$ with $k > 0$, or by such a sum divided by a denominator at least $1/2$, and these tend to zero as $u \to \infty$. With the amplitude A_{w} fixed, we choose a finite U beyond all these thresholds and $g_* \le e^{-U} < e^{-1}$. Every condition then holds uniformly for $0 < t \le g_*$, and $p(t) \ge 1$. The ceiling may depend on A_{w}; it is independent of the side, the preceding depth, the number of steps, and the frozen coordinates. No expansion is applied on the larger radii R_{en}/t: that product is used only for the comparison of Lemma 21.11. □

21.3 Coordinate coverage and representatives satisfying the reconstruction identities

We show that every supported point of every relevant fiber has small relative coordinates in a suitable gauge, and that its normalized fluctuation coordinates lie in the enclosing product domain. We also construct, from the coarsest scale downward, the representatives satisfying the reconstruction identities, at which the densities will be evaluated. The comparison of the corresponding integrals with integrals over the fluctuation cutoff domain is Lemma 21.11 in Section 21.6. That comparison requires an incoming density bound, which is proved there by induction. The coverage statement proved here uses no density bound.

For a group of averaging steps, the conditional minimizers, Hessians K_i, and reconstruction maps C_t are those of Theorems 19.9 and 20.14, with the gauge transformation of

Lemma 19.5, which depends only on the target. For a supported fiber point, z^{act} denotes its free logarithmic coordinates and w^{act} its normalized fluctuation coordinates, obtained by inverting the reconstruction. The enclosing product domain is

$$\mathcal{K}_t = \prod_{i,e} \{|w_{i,e}| \le R_{\mathrm{en}}/t_{\max}\},$$

with R_{en} as in (21.18).

Theorem 21.8. *There exist $\epsilon_{\mathrm{en},J} > 0$ and $C_{\mathrm{en},J} < \infty$, with the dependence stated at the beginning of this section and independent of all later widths of boundary bands, removed bands, protected neighborhoods, and cells, such that the following holds for every large-field decomposition of Theorem 21.5. Assume its construction conditions, the two reconstruction-image conditions* (21.17)*, and $r + \beta_{\mathrm{rec}} < \epsilon_{\mathrm{en},J}$.*

1. *Let $1 \le s \le J$. Let (c, b_{rec}) be the output field and frozen coordinates of a configuration in the support of the stage measure, given either by a small representative of Theorem 21.5(2) (at the first step for a stopped measure, of Corollary 21.2), or by a representative obtained from such a representative by an output gauge change acting on the frozen coordinates as in Lemma 21.6(1)–(3). Assume that this representative and the associated fields before the target-dependent gauge transformation belong to the common larger construction domain of Theorem 20.14, and that its analytic frozen coordinates satisfy* (21.8)*. Let U be a configuration in the support of the fiber over (c, b_{rec}). Then there exists a correcting gauge transformation, equal to I at the roots of the block region, at the vertices of the frozen boundary layer on the side of the fixed complement (in particular at the endpoints v_e of the links across the boundary of the block region), and at all protected earlier endpoints and basepoints, with the following property. After the hierarchical tree gauge fixing, every relative constraint vector X_i of U, including its eliminated entries, and its free part z_i satisfy*

$$\max_{0 \le i < s} (\|X_i\|_\infty + \|z_i\|_\infty) \le C_J (r + \beta_{\mathrm{rec}}).$$

   They lie in the common domain of the local inverse of the block averaging constraint, with a fixed positive margin. The correcting gauge transformation is the composition of a smallness gauge and the lift, constant on blocks, of the inverse of its root values; its factors are not separately required to fix the roots. The inner endpoint u_e of a current link across the boundary of the block region may move before the tree gauge fixing, while the based transport is preserved, since its endpoints are fixed.
2. *Choose any small representative at the top of the decomposition, in the coordinate domains of part (1), whose frozen coordinates satisfy* (21.8)*. Define the reconstruction maps on their full spatially varying radii first, and descend through them, aligning the target at each step by the accumulated output gauge change. The unperturbed coordinates, conditional minimizers,*

and inverse square roots of the Hessians transform by blockwise adjoint orthogonal transformations, and the perturbing force is estimated separately. This constructs representatives satisfying the reconstruction identities at all stages, with

$$(T_M^s C_t(c, \zeta; b_{\mathrm{rec}}))|_{\Omega_t'} = c,$$

and for every point of every fiber over every resulting output datum, its normalized fluctuation coordinates satisfy

$$\max_{i,e} g_i |w_{i,e}^{\mathrm{act}}| \le \delta_{\mathrm{en}} = C_{\mathrm{en},J}(r + \beta_{\mathrm{rec}} + \sigma_* \rho_{\mathrm{cl}}).$$

With $R_{\mathrm{en}} = 2\delta_{\mathrm{en}}/\lambda_*$, each such point lies in the interior $|w_{i,e}^{\mathrm{act}}| \le R_{\mathrm{en}}/(2t_{\max})$ of the enclosing product domain $\mathcal{K}_t$. Each gauge transformation of Lemma 19.5 used here is I at all protected vertices, including the current endpoints of the links across the boundary of the block region. No estimate on the logarithm of the accumulated gauge transformation is used. The statement concerns every supported fiber point, including points near the boundary of the coordinate domains, not one selected point or almost every point under an auxiliary measure.

3. *At the first RG step for a stopped measure, every nonempty fiber over an output field and all input frozen coordinates that meets D_r has a representative in which all output links and input frozen coordinates are at most $C_{\mathrm{ad},J} r$. The scaled fluctuations of the unperturbed objective, $K_0^{1/2}(z^{\mathrm{act}} - z_0)$, are bounded by $C_J(r + C_{\mathrm{ad},J} r)$.*
4. *At the first RG step under the Wilson measure, let b^{near} be the near values, that is, the values on the near links of* (18.32)*, which are the links outside the block region read by a plaquette depending on an integration variable or by a path product of the block region in the initial good-cell complex. Let η_{deep} be the deep frozen group variables, the values on all remaining links. Then we have the exact identity*

$$S_W(U) = F_{\mathrm{bare}}(U_{\mathrm{int}}; b^{\mathrm{near}}) + E_{\mathrm{hole}}(b^{\mathrm{near}}, \eta_{\mathrm{deep}}),$$

where U_{int} denotes the integration variables and F_{bare} is the constrained Wilson-action objective of the first block integration, defined in Definition 18.13 and written in these variables; in the coordinates of Definition 18.6 this identity is (18.38)*. No smallness is imposed on η_{deep}; its factor $e^{-E_{\mathrm{hole}}/g_{\mathrm{base}}^2}$ remains with its original Haar integration and the original partition function. Every conditional minimizer, derivative, square root, and Haar disintegration factor of the integration variables is independent of η_{deep}. A plaquette containing only frozen links, near or deep, belongs to E_{hole}. At this step there is no inherited effective interaction, and the scaled fluctuations are bounded by $C_J(r + \beta_{\mathrm{rec}})$. The deep frozen group variables and exceptional values of the Borel extension in them are never logarithmized or inserted into a torus analytic action.*

The constants have the dependence stated at the beginning of this section, and they are uniform in the deep frozen group variables.

Proof. We first work at a small output representative and an arbitrary supported fiber point over it (Steps 1–3). Step 4 transports the estimate to the representatives produced from above, using only the unperturbed objective, and Step 5 estimates the perturbation. Step 6 constructs the representatives satisfying the reconstruction identities, Step 7 converts the bound to the enclosing product domain, and Steps 8 and 9 treat the first step for the two initial measures. For $s < J$, the image hypothesis includes the separately imposed interior image condition of that group of steps. Only its conditional local chart and force estimates are used here; for fewer than J steps there is no restoration of norms and no recursion of normalization constants.

Step 1: a smallness gauge for an arbitrary fiber point. We work at a small output representative (c, b_{rec}) supplied by Theorem 21.5, and keep this notation until Step 4. Its output links and the newly used near values and based data are bounded by $r/2$ and β_{rec}, respectively, and its analytic frozen coordinates obey (21.8). An arbitrary analytic background on a larger spatially varying radius need not satisfy the absolute bound on c used in this step. Fix any configuration U in the support of the fiber over this representative. By Theorem 21.5, there exists a smallness gauge S for the input links of the block region. We restrict S to the input vertices of the block region and set it equal to I elsewhere; internal links are unchanged by this restriction. By the endpoint protection rule, its lifts fix every earlier endpoint and basepoint. A current based transport, which is not a frozen coordinate, can move under this intermediate operation. We write $V = U^S$ on the block region and let u_{in} be its bound for interior links. At the initial input for a stopped measure we use $u_{\mathrm{in}} = r$; at later inputs $u_{\mathrm{in}} \le r/2$; at the first input under the Wilson measure $u_{\mathrm{in}} \le r$, after the fixed choice of A_8. By Lemma 21.1, for internal dependence neighborhoods reading only interior links, we have

$$\max_{e \text{ internal}} \rho_G((T_M^i V)_e) \le (10M)^i u_{\mathrm{in}}, \qquad 0 \le i \le s. \tag{21.20}$$

These dependence sets lie inside full blocks of the region. The display is not used for a path product reading a prescribed near value; such path products are bounded after the alignment below, using their frozen data as well as the interior bound.

Step 2: the root values. For each final root a of the region, let $h_a = S(a)^{-1}$. On an internal final link (a, a'), by the gauge covariance of the block averaging constraint, we have

$$c_{aa'} = h_a (T_M^s V)_{aa'} h_{a'}^{-1}.$$

Multiplying this identity by the appropriate inverse factor and using bi-invariance, we obtain

$$\rho_G(h_a h_{a'}^{-1}) \le \rho_G(c_{aa'}) + \rho_G((T_M^s V)_{aa'}) \le r + \beta_{\mathrm{rec}} + (10M)^s u_{\mathrm{in}}. \tag{21.21}$$

We lift h as a constant on each final block of the region, with value I off these blocks, and denote this lift by L (a gauge transformation, not the side of the torus). The total

correction is $H = LS$, since $(U^S)^L = U^{LS}$, and it equals I at every final root. At the vertices of the frozen boundary layer on the side of the fixed complement both factors are I, and the endpoint protection rule fixes the earlier frozen endpoints after lifting to their own lattices. An intermediate internal link either stays in one final block or crosses a single boundary between adjacent final blocks. In the first case L conjugates its value; in the second case its angle increases by at most the root increment in (21.21). Together with (21.20), this bounds all internal iterates by $C_J(r + \beta_{\mathrm{rec}})$. A component consisting of a single block has only the first case, and its root value causes no additional estimate.

The bound $\rho_G(c_{aa'}) \le r + \beta_{\mathrm{rec}}$ used in (21.21) belongs to this small representative. At the first step for a stopped measure, the representative of Corollary 21.2 has output bound $C_{\mathrm{ad},J} r < \beta_{\mathrm{rec}}$, by (21.6), so the displayed bound holds with $u_{\mathrm{in}} = r$.

Step 3: based transports and the branch. The endpoints of a current based transport are a root of the region and the endpoint of its link in the fixed complement, and the factors of H at both are I. The transport is therefore unchanged, although the endpoint u_e in the region can have moved. We now apply the hierarchical based tree gauge fixing of Lemma 21.1. Since its paths have length at most L_J, all internal links after the tree gauge fixing satisfy the same bound with a larger C_J. A link across the boundary of the block region that is read by a path product of the region is the inverse of such a bounded path times its prescribed based transport. It is therefore bounded by $C_J(r + \beta_{\mathrm{rec}})$ as well. Every path product of the region mixing integration variables and near values has at most L_J factors of these links and of its prescribed frozen data; its bound is $C_J(r + \beta_{\mathrm{rec}})$, including the contribution of the frozen data. We include the initial $C_{\mathrm{ad},J}$ and the constants for these mixed path products in this fixed C_J before choosing $\epsilon_{\mathrm{en},J}$ or the final radii. At the small representative, the relative vector X_i and its free part z_i then satisfy the estimate in part (1). We choose $\epsilon_{\mathrm{en},J}$ so that this bound is less than half the common uniqueness radius and all the fixed construction margins. Since the configuration solves the block averaging constraint, its eliminated entries equal the unique eliminated entries on this branch. This reasoning applies to the arbitrary point U itself, not to another point of its fiber, and neither S nor L has to fix the roots separately.

Step 4: transport to representatives produced from above. We carry the estimate to an output datum produced from above, using only the unperturbed objective with torus input. Let k be the accumulated output gauge change. Its lift, constant on blocks, preserves the tree identities, and the coarse-field lift of the target satisfies $U_0(c^k) = U_0(c)^{\widetilde{k}}$. Each relative factor therefore transforms as

$$e^{X_e} \longmapsto e^{\mathrm{Ad}_{k_a} X_e}, \qquad z_i \longmapsto A_{k,i} z_i, \tag{21.22}$$

where a is the root of its block and $A_{k,i}$ is block diagonal and orthogonal on the real color vectors. The equation for the eliminated edges transforms in the same way, as we see by

conjugating the constraint and using the uniqueness of its small inverse. We repeat this over the $s \le J$ intermediate scales. Since the changes at earlier marks are the identity, the prescribed frozen data are preserved. For a comparison of two representatives that are not related by a based change, the same formula uses their correspondingly transformed output data.

We apply this change separately to each fiber point and to the small representative of its gauge orbit. The given output datum and that small datum are outputs of the same configuration up to a gauge transformation, so the output change exists without choosing a different point of the fiber. Formula (21.22) transports the relative bound and the eliminated-edge inverse to every output datum as in part (1), even if its output links are bounded only by a larger analytic radius; no absolute bound $r/2$ on its iterates is asserted. All frozen coordinates used in this comparison satisfy (21.8), and only the changes of Lemma 21.6(1)–(3) are used.

All fields in this calculation, taken before the target-dependent gauge transformation, lie in the common larger construction domain of Theorem 20.14. In particular, the backgrounds at block boundaries, which need not be of order $M^{-s}\rho_{\mathrm{cl}}$, lie in that domain. The torus classical values used there are the gauge-invariant values on their common classical chart, as in Theorems 14.3 and 14.6, so their restrictions agree at such fields related by a compact gauge transformation in this chart. In a regional completion, the prescribed small frozen data transform at the endpoints of their paths. The earlier frozen coordinates that are completed by zero have both endpoints fixed by the endpoint protection rule, so appending them commutes with this change. For the unperturbed objective F_0 and the product color map A_k,

$$F_0(c^k, A_k z; b^k_{\mathrm{rec}}) = F_0(c, z; b_{\mathrm{rec}}). \tag{21.23}$$

This is the orthogonal transition of Theorem 14.3, restricted to the regional constraints and the fixed frozen data. Only the torus classical restriction is transported in this way.

By stationarity in z and the uniqueness of the unperturbed conditional minimizer, we have

$$z_0(c^k; b^k_{\mathrm{rec}}) = A_k z_0(c; b_{\mathrm{rec}}), \qquad K_0(c^k; b^k_{\mathrm{rec}}) = A_k K_0(c; b_{\mathrm{rec}}) A_k^{-1}. \tag{21.24}$$

The minimizer stays in the common uniqueness box; if boxes in scalar coordinates are used, the fixed comparison of color norms is included in the margin of the larger construction domain. An orthogonal conjugate of a positive square root is positive and squares to the conjugated matrix, so uniqueness of positive square roots gives the same identity for $K_0^{1/2}$ and $K_0^{-1/2}$. Applying this argument after each conditional minimization, we obtain it for every initial segment. At a small output representative, the unperturbed minimizers vanish when all current small parameters vanish, and the row sums of their

first derivatives are bounded by the constants of Theorem 20.14. Hence their size is $C_J(r + \beta_{\mathrm{rec}})$. Equations (21.22) and (21.24) carry the bound

$$\|K_0^{1/2}(z^{\mathrm{act}} - z_0)\|_\infty \le C_J(r + \beta_{\mathrm{rec}}) \tag{21.25}$$

to every output datum described in part (1). No bound on $\operatorname{Log} k$ is needed, since blockwise adjoint transformations have norm one for every compact value at the roots.

Step 5: the perturbation. We treat the perturbation separately, at fixed real $b = (\xi, b_{\mathrm{rec}})$. The uniform incoming radius, the boundary-distance comparison, and the conditional force estimates are Lemma 21.7; the analytic norm bounds are those of the recursion of Theorem 20.14, constructed without any density inequality. In the notation of Theorem 19.9, the preconditioned force is at most $C_J\sigma_*\rho_{\mathrm{cl}}$, the change of the constrained Hessian is at most $C_J\sigma_*$, and the rows of all square roots for initial segments are bounded by C_J. That theorem estimates the perturbation on the reconstructed field in the target-dependent gauge, including the full boundary-completion difference; it asserts no covariance of the perturbation, and none is needed. At each intermediate scale the inverse relation for the normalized fluctuations is

$$g_i w_i^{\mathrm{act}} = K_{P,i}(y_i; b_{\mathrm{rec}})^{1/2}\big(z_i^{\mathrm{act}} - z_{P,i}^*(y_i; b_{\mathrm{rec}})\big),$$

where y_i denotes the higher parameters of the fiber point, $z_{P,i}^*$ the perturbed conditional minimizer, and $K_{P,i}$ the perturbed conditional Hessian. On the smaller products for initial segments, the displacement of the minimizer from the unperturbed one is at most $2\delta_V \le C_J\sigma_*\rho_{\mathrm{cl}}$ at every y_i, by (21.13) and (21.14). The higher parameters of the fiber point belong to the required domains $\mathcal{T}_I$. Indeed, their unperturbed deviations satisfy the geometric bound just proved, and the conditional force condition leaves the factor-two margin of the neighborhoods for all scaled fluctuations up to R_{en}. We subtract and add the unperturbed minimizer, use (21.25), and apply the bounded rows of the square roots and inverse square roots. The substitutions of higher parameters have at most J stages. Their finite composition, including the derivatives of the higher minimizers in the integration and free higher parameters specified in Theorem 19.9, gives

$$\max_{i,e} g_i |w_{i,e}^{\mathrm{act}}| \le C_{\mathrm{en},J}(r + \beta_{\mathrm{rec}} + \sigma_*\rho_{\mathrm{cl}}). \tag{21.26}$$

The construction conditions ensure that these substitutions stay inside the conditional neighborhoods of the joint branch; we do not replace these neighborhoods by arbitrary slices of a joint box of free coordinates. The constants are evaluated on the fixed larger products and on a fixed perturbation ceiling, before the eventual small radii and widths. No perturbed derivative in a frozen coordinate ξ or in an earlier frozen coordinate is taken; these coordinates enter only as fixed real parameters of the uniform coefficient estimates.

Step 6: representatives satisfying the reconstruction identities. The construction downward is a finite induction. We start with any small top representative given by Theorem 21.5, including the bounds (21.8). At a boundary where coordinates are frozen, the hierarchical tree transports are I by Lemma 21.1, and the gauge transformation of Lemma 19.5 is I at their endpoints. Undoing the basing therefore inserts the already chosen new frozen values directly. At the next lower group of averaging steps, we lift the accumulated output change, take the free logarithmic coordinates of any configuration in its fiber, and invert the reconstruction defined beforehand. By the branch argument of Steps 1–4 and (21.26), these inverse coordinates lie in its domain. The reconstruction has the same target and the same free coordinates, and uniqueness of the eliminated-edge inverse identifies the reconstructed configuration, before the gauge transformation, with the configuration aligned by the accumulated output gauge change. We then apply the gauge transformation of Lemma 19.5, which depends only on the target; by that lemma, the target is unchanged and every protected vertex is fixed. This produces the present representative and permits the next step of the descent. By Lemma 21.6(1)–(2), the subsequent lifts and the based correcting and tree gauges fix every frozen coordinate chosen at this step, so (21.8) holds throughout the family, and all supported points over each fixed output datum are included. Other top representatives satisfying the same bounds on frozen coordinates are treated in the same way. Each step uses only the representative above and the analytic maps defined beforehand on their full radii, so the induction assumes no completed family of representatives. Keeping all choices gives all fibers over all resulting output data and the identity for the target in part (2).

Step 7: the enclosing product domain. At output side m, we have $g_i = t_{\max} M^{(i-s+1)/2} \geq \lambda_* t_{\max}$. Dividing (21.26) by g_i, we obtain

$$|w_{i,e}^{\mathrm{act}}| \leq \frac{\delta_{\mathrm{en}}}{\lambda_* t_{\max}} = \frac{R_{\mathrm{en}}}{2t_{\max}}.$$

Every point of $\mathcal{K}_t$ has $g_i |w_{i,e}| \leq R_{\mathrm{en}}$. Both the classical and the quantum image conditions are imposed on the whole product $\mathcal{K}_t$, not only on the supported subset or on the logarithmic cutoff domain.

Step 8: the first step for a stopped measure. We fix one point of a nonempty fiber over an output field and all input frozen coordinates, and apply Corollary 21.2 to it. In the resulting representative, all output links and all based input transports are at most $C_{\mathrm{ad},J} r$, and the remaining input frozen links are smaller than r. For any other point of the same fiber, we take its own gauge transformation into B_r, cancel its root values by the lift of the inverse root values, and cancel its gauge transformation at the vertices of the fixed complement. Equations (21.20) and (21.21) hold with $u_{\mathrm{in}} = r$ and the output increment $C_{\mathrm{ad},J} r + (10M)^s r$. All input variables used analytically, including all input frozen coordinates, are small after the tree gauge fixing. The unperturbed estimate is

$C_J(r + C_{\mathrm{ad},J} r)$. This is the regional form of the coordinates of Theorem 14.3 at the first step; it is uniform in the preceding depth and is not applied to frozen coordinates created at later steps.

Step 9: the first step under the Wilson measure. We use the initial good-cell complex of Theorem 21.5(4), together with the internal coordinates of Definition 18.6 and the completion of Definition 18.13: the links across the boundary of the block region are completed by (18.33), the other links by (18.34), and $\mathcal{P}_{\mathrm{var}}$ is a fixed list as in (18.35)–(18.36), namely the list of the split made in this step. The near links of (18.32) carry the near values b^{near}, and the deep links carry the deep frozen group variables η_{deep}. The hypothesis of Definition 18.13(3) that the near values lie in the identity chart is the conclusion of Theorem 21.5(4), since every near link lies in the initial good-cell complex. Indeed, the links of the path products in $\mathcal{E}_{\Omega}^{\mathrm{dep}}$ lie there because of the inset by the full dependence neighborhood. The plaquettes incident to $\mathcal{L}_0$ lie within one input step of a block of the region, while Theorem 21.5(4) moves inward by a full layer of cells and the full dependence neighborhood. The conversion between the value on a link across the boundary and its based value is included in $C_{\mathrm{conv},J}$.

In the hierarchical tree coordinates, we partition the plaquettes according to whether their Wilson term depends on an integration variable. By the definition of the near links, every term in the first class uses only integration variables and near values, and the sum of these terms is F_{bare}. Every remaining term is a function of frozen variables only and is assigned to E_{hole}, even if its plaquette contains both near and deep links. This partitions the sum $S_W = 4\sum_p q(U_p)$, and Lemma 18.14(1) implies the exact split (18.38), which is the identity in part (4). In particular, every objective of the integration variables and all its derivatives are independent of η_{deep}, without any estimate of a small remainder. The Haar disintegration path products also avoid the deep links, because of the inset by the full dependence neighborhood. By (18.39), the objective agrees with the objective of Theorem 18.12 at preceding depth zero, up to a function constant in the integration variables. The external factor $e^{-E_{\mathrm{hole}}/g_{\mathrm{base}}^2}$ remains the one in (18.38), with its original Haar variables and partition function.

We apply Lemma 18.14(2)–(3) to the block region of the first step, with the prescribed near completion. Its embedding of the free coordinates is A_Ω in (18.15), and its extraction matrix is (18.16). Equations (18.40)–(18.42) apply to the list $\mathcal{P}_{\mathrm{var}}$, including every case near the boundary. By (18.43), the flat joint Hessian in the free coordinates then satisfies

$$K_\Omega = \widehat{A}_\Omega^{\mathsf{T}} d_{\mathrm{var}}^{\mathsf{T}} d_{\mathrm{var}} \widehat{A}_\Omega \geq \lambda_{M,J} I,$$

It has uniformly bounded absolute rows and columns and the range (18.44), independently of the region and of the deep frozen group variables. By (18.39), this is also the case $d_0 = 0$, $\kappa_* = 1$ of Theorem 18.12. Every principal free matrix and every real Schur

complement of an initial segment keeps a fixed-J gap, by (18.47). No Hessian of a field outside the region is inverted.

All Wilson and constraint derivatives at this step are derivatives of finite local path products. Their absolute row sums are bounded on a fixed smaller identity chart, independently of the deep frozen group variables. We obtain the weighted inverses without a volume estimate. First let K be the flat joint matrix K_Ω, or one of its principal free submatrices, before the minimization over a lower scale. By Lemma 18.14(3), $\lambda_{M,J} I \le K \le Q_{M,J} I$, and K has range $D_{M,J}$. We set $a = 2Q_{M,J}$ and $T = I - K/a$. Then T has norm at most $q = 1 - \lambda_{M,J}/a < 1$, and the entries of T^k vanish beyond maximum distance $kD_{M,J}$, hence beyond graph distance $3kD_{M,J}$. By the bound on the density of labels in the proof of Lemma 18.14, each row of T^k has at most $C_J(k+1)^3$ nonzero entries, each of absolute value at most q^k. For $b > 0$ with $qe^{3bD_{M,J}} < 1$, the series (18.48) for K^{-1} and $K^{-1/2}$ have uniformly summable rows and columns, weighted in the graph distance; the second is the scalar binomial series applied to T, whose coefficients are at most one. Small complex perturbations are inverted by the Neumann series about these matrices. For the conditional matrices obtained after minimization over lower scales, we use (18.47) and the bounds for Schur complements in the proof of Lemma 18.14(3), which rest on the argument of Combes and Thomas [61]. These matrices and their inverse square roots have uniform exponential row and column bounds, although the Schur complements need not have finite range. The factors M^i are included in the corresponding matrices.

The equation for the conditional minimizers is a contraction on a fixed smaller box. We first prove this at the small representative used in Steps 1–3. There, its force at zero fluctuation is $C_J(r + \beta_{\mathrm{rec}})$, and by the third-derivative bound in the proof of Lemma 18.14(4) its derivative changes by at most C_J times the radius of the box. At this representative, the coarse argument of (18.45) is $\operatorname{Log} c$, and $\|\operatorname{Log} c\|_\infty$ and $\|\xi\|_\infty$ are at most $C_J(r + \beta_{\mathrm{rec}})$. In the choice of $\epsilon_{\mathrm{en},J}$, we require in addition $C_J(r + \beta_{\mathrm{rec}}) \le \epsilon_{M,J}$, and we require the radius of the box to be at most $\epsilon_{M,J}$, so that (18.45) applies on the whole box, including the coarse field. We reduce the radius of the box so that the inverse times the change of the derivative is at most $1/2$; the solution then has size $C_J(r + \beta_{\mathrm{rec}})$. The finitely many conditional products and their image margins are those required in Theorems 19.9 and 20.14, and their specialization at this step has no inherited effective interaction. Applying the inverse formula for the normalized fluctuations, we obtain the bound on the scaled fluctuations in part (4) at the small representative, without the perturbative term.

For an output representative transported from above, we use the output gauge change of Step 4, with the frozen values transformed consistently. The Wilson terms are traces of plaquette variables, so the exact split in part (4) gives for F_{bare} the same covariance identity as (21.23). Its conditional minimizers and Hessians therefore transform as in (21.24), and their square roots transform by the same orthogonal conjugations. Hence the minimizers at the transported representative are the images of those at the small

representative, and each normalized fluctuation $w_{i,e}$ is rotated by an orthogonal map, so the bound on the scaled fluctuations carries over. No bound of order $r + \beta_{\mathrm{rec}}$ on the transported coarse links is needed.

All constants arose from local path products, fixed construction products, and their row-sum bounds. □

21.4 Pointwise gauge-invariant densities

The densities of the stage measures will be evaluated at representatives satisfying the reconstruction identities, which form a set of Haar measure zero. We therefore construct an explicit Borel density and prove that it is gauge invariant at every point. The coverage hypothesis of Lemma 18.7 is supplied by Theorem 21.8(1), which uses no density bound.

Theorem 21.9. *Fix a finite lattice, a preceding depth that is a multiple of J, an initial measure (a stopped measure or the Wilson measure), and a large-field decomposition $\mathcal{X}$ of Theorem 21.5. At stage t, let X_t be the product of the link variables of the current block region and of all independent frozen links and based transports, each on the lattice where it was frozen. Let H_t be the product of the Haar probability measures on these factors, and let $\eta_t^{\mathcal{X}}$ be the stage measure, a subprobability measure. Let Γ_t be the group of consistent gauge assignments on the finite multiscale transport graph formed by these endpoints: repeated physical vertices are identified, and each recorded object transforms at the endpoints of its path. In particular, Γ_t contains arbitrary values at the output roots and their lifts, constant on the blocks of the region, together with the induced changes of earlier frozen coordinates used in Theorem 21.5. For a frozen link on the lattice of level ℓ and for a based transport, the action is*

$$b_{xy} \longmapsto k_x^{[\ell]} b_{xy} (k_y^{[\ell]})^{-1}, \qquad \widehat{b}_e \longmapsto k_{\alpha_e}^{[\ell]} \widehat{b}_e (k_{v_e}^{[\ell]})^{-1}, \qquad \widehat{b}_e = U(\sigma_e) b_e.$$

For each fixed assignment, this is a left and right multiplication in the independent group variables, and it preserves H_t.

There exists an explicitly defined nonnegative Borel density p_t such that

$$\eta_t^{\mathcal{X}} = p_t H_t, \qquad p_t(x^k) = p_t(x) \quad \text{for every } x \in X_t \text{ and } k \in \Gamma_t.$$

On the gauge-invariant chart domain containing the measure, p_t is given by the parameter integrals of the Haar disintegration formulas, with every indicator of the decomposition and every protected external factor retained, patched by their exact equalities on overlaps of fixed gauge translates. On the invariant complement of this domain, which has zero mass, its value is zero. In particular, the values of p_t at representatives satisfying the reconstruction identities are given by these formulas pointwise.

More explicitly, if $Z_{c,b_{\mathrm{rec}}}$ is the domain of the free coordinates of the fiber over an output datum (c, b_{rec}) satisfying the coordinate conditions of Theorem 21.8, then the outgoing density is

$$p_t^{\mathrm{out}}(c, b_{\mathrm{rec}}) = \int_{Z_{c,b_{\mathrm{rec}}}} \chi_t(c, z, b_{\mathrm{rec}})\, p_t(U_{c,b_{\mathrm{rec}}}(z), b_{\mathrm{rec}})\, J_{\mathrm{CO}}(c, z; b_{\mathrm{rec}})\, dz,$$

where χ_t is the product of the indicators of the decomposition and of the support, and the normalized integrals over the tree variables are kept until the covariance under based gauges has been applied. The factor at scale i of the Haar disintegration density J_{CO} is exactly

$$c_{\mathrm{Haar}}^{f_i} \frac{\prod_{e \text{ outside the trees}} j_H(\Gamma_{i,e})}{\prod_a \det D_{X_{e_a^*}} \mathsf{C}_a(\Gamma_i)}, \qquad j_H(X) = \mathrm{sinc}^2(|X|/2),$$

where the product in the denominator runs over the internal constraints, f_i is the number of free group variables at scale i, and the numerator includes the Haar factors of the eliminated edges. The reference measure already contains the Haar measure of the output field; no further coarse Haar density, Gram determinant, or orbit-volume factor occurs. For the stage densities and the tests of the decomposition, the covariance under based tree gauges makes each tree Haar integral equal to one; this elimination is not valid for arbitrary noncovariant Borel densities. Basing at a fixed field on the block region is the triangular Haar translation $b_e \mapsto U(\sigma_e) b_e$, with Jacobian one.

These statements concern the stage measures and the block averaging map with its Borel extension; they assert no gauge invariance or overlap identity for an auxiliary classical action or effective interaction. Deep frozen group variables, including exceptional values of the Borel extension, are allowed. All identities hold in each finite dimension and introduce no factor depending on the size, the number of steps, the shape, or the depth. The analytic chart bounds used in the construction have the dependence stated at the beginning of this section.

Proof. We construct the density and prove its pointwise invariance together. This avoids evaluating an unspecified Radon–Nikodym derivative on a set of representatives.

Step 1: the endpoint group. We identify repeated physical vertices in the finite list of endpoints. An endpoint assignment is an element of the resulting finite product of copies of $\mathrm{SU}(2)$, and any additional consistency identifications give the corresponding closed subgroup. Its action on an independent group variable is a left and a right multiplication by fixed matrices. Hence H_t is preserved by each fixed assignment, with no assumption of probabilistic independence of the frozen coordinates under $\eta_t^{\mathcal{X}}$.

Step 2: the initial density. At the initial scale we use the pointwise density of the Wilson measure, or the gauge-invariant density of the stopped measure, multiplied by the indicator of the decomposition. The initial density is invariant: the Wilson plaquette variables transform by conjugation, and the density of the stopped measure is the gauge-invariant density of Theorem 14.6. The indicator of the decomposition is invariant as well, since the decomposition is determined by traces of plaquette variables and by

deterministic dependence lists. The block averaging map is equivariant also on the exceptional set of its Borel extension: conjugation fixes $-I$, and the replacement reference transport transforms at the same two endpoints. These are pointwise statements. The first freezing is a splitting into independent variables followed by the triangular basing, treated in Step 5.

Step 3: the outgoing density on a chart. Assume that the incoming density has been constructed. At an output datum satisfying the coordinate conditions, we use Lemma 18.7 with all indicators; its coverage hypothesis is part (1) of Theorem 21.8. We keep the normalized tree variables $\mathfrak{f}$ at first. If J_i denotes the factor at scale i, we define the local outgoing value to be

$$\int dH(\mathfrak{f}) \int \mathbf{1}_{Z_{c,b_{\mathrm{rec}}}}(z,\mathfrak{f})\chi_t(c,z,\mathfrak{f},b_{\mathrm{rec}})p_t(U_{c,b_{\mathrm{rec}}}(z,\mathfrak{f}),b_{\mathrm{rec}})\prod_{i=0}^{s-1} J_i\,dz. \tag{21.27}$$

An indicator already contained in p_t is not counted again; it may equally be written again, since indicators are idempotent. This convention keeps all tests of the decomposition, including those whose information has already been integrated. By the disintegration identity, testing (21.27) against any nonnegative Borel function of the output gives the pushforward measure on this chart.

Step 4: pointwise covariance. We prove that these integrals are equal under the full group of consistent endpoint assignments. A fixed output assignment has a lift, constant on blocks, through the centered partitions of the block region; at earlier marks we use its prescribed values on the lattices of the marks. There is no conflict, since a mark in the preimage of a block of the region would have excluded that block, and repeated physical values are identified by consistency. Additional changes of the interior frames can be absorbed in the tree variables. The block averaging map and the map recording the frozen coordinates are equivariant, so the lift is a bijection of the fiber over (c, b_{rec}) onto the fiber over $(c^k, b_{\mathrm{rec}}^k)$. It preserves every test, including an equality in a threshold and every condition at earlier scales.

On the hierarchical tree section, the part of the lift that is constant on blocks acts as in (21.22). The adjoint action is orthogonal because it preserves $-2\,\mathrm{tr}(XY)$. Its determinant is 1, since it is a continuous function on the connected group $\mathrm{SU}(2) \simeq S^3$ with values in $\{-1,1\}$ and equals 1 at I. Hence the change of free coordinates preserves dz, and uniqueness of the eliminated-edge inverse transports every eliminated coordinate as well. The factors j_H are radial and unchanged. The derivative matrix of the eliminated edges is multiplied on its row and column frames by these adjoints of determinant one, so its determinant is unchanged, including its positive sign on the real branch. The constants $c_{\mathrm{Haar}}^{f_i}$ are the same. If the lift changes interior tree frames, the tree variables change by

left and right Haar translations, before any tree integral is removed. Using the inductive pointwise identity for p_t, we obtain

$$\begin{aligned} p_t^{\mathrm{out}}(c^k, b_{\mathrm{rec}}^k) &= \int_{Z_{c^k,b_{\mathrm{rec}}^k}} \chi_t(c^k, z', b_{\mathrm{rec}}^k)\, p_t(U_{c^k,b_{\mathrm{rec}}^k}(z'), b_{\mathrm{rec}}^k) J_{c^k,b_{\mathrm{rec}}^k}(z')\, dz' \\ &= \int_{Z_{c,b_{\mathrm{rec}}}} \chi_t(c, z, b_{\mathrm{rec}})\, p_t(U_{c,b_{\mathrm{rec}}}(z), b_{\mathrm{rec}}) J_{c,b_{\mathrm{rec}}}(z)\, dz = p_t^{\mathrm{out}}(c, b_{\mathrm{rec}}). \end{aligned} \tag{21.28}$$

The normalized tree integrals are understood in both lines when they have not yet been removed. This is an identity of explicit parameter integrals at each output datum; no uniqueness almost everywhere is used.

Step 5: freezing. Freezing has the required covariance too. We orient a link across the boundary of the block region from its endpoint u_e in the region to its frozen endpoint v_e, and let α_e be the root of its hierarchical transport. Cancellation at the interior vertices gives

$$\begin{aligned} U_{v^k}(\sigma_e) a_e^k &= k_{\alpha_e} U_v(\sigma_e) k_{u_e}^{-1} k_{u_e} a_e k_{v_e}^{-1} \\ &= k_{\alpha_e} \widehat{b}_e k_{v_e}^{-1}. \end{aligned} \tag{21.29}$$

At fixed v, the map from the independent variables a_e to their based transports is a product of left Haar translations. Together with the unchanged v, this is a triangular measure-preserving bijection; its inverse multiplies by $U_v(\sigma_e)^{-1}$. Recording any other link as a frozen link is Fubini's splitting of product Haar measure. The endpoint protection rule makes these endpoint changes consistent at every later stage. Equations (21.28) and (21.29) prove the inductive covariance for arbitrary values at the output roots and for all induced changes of earlier frozen coordinates.

Step 6: Borel patching. Before inserting the tests, we take the open chart domains of small block-region variables and small near values, with the positive margins of their inverses; the deep frozen group variables need no logarithm. Their gauge saturation $\mathcal{O}_t$ is open and invariant, and the proofs of Theorems 21.5 and 21.8 put all the mass in $\mathcal{O}_t$. A finite product of spheres is second countable and has a countable dense subset, for instance obtained by approximating quaternion coordinates by rational ones and normalizing; the finite endpoint subgroup has the same property after the consistency identifications. If a point is moved into an open chart by some fixed gauge assignment, the set of assignments doing so is open and meets this countable dense set. Hence countably many fixed gauge translates of the small charts cover $\mathcal{O}_t$, and their coordinate charts can be chosen countably as well.

On one such chart, the integrand of (21.27), extended by zero outside its domain in a fixed box of free and tree coordinates, is a nonnegative Borel function. Indeed, the Haar disintegration denominators and the reconstruction maps are continuous on their

branches, and the tests, including values of the Borel extension at the deep frozen group variables, are Borel. By Tonelli's theorem, its integral with respect to the finite product of Lebesgue and Haar measures on the box is a Borel function of (c, b_{rec}).

On an overlap, the fixed gauge assignments relating two charts give the change of variables in (21.28). Since both formulas include the same domain, they agree at every point of the overlap. We use the first chart in the countable cover containing a point to define p_t there, and set $p_t = 0$ on $X_t \setminus \mathcal{O}_t$. This function is Borel, it is a density by the disintegration identities, and it does not depend on the chart used. To check invariance under an arbitrary $k \in \Gamma_t$, we take charts containing x and x^k and perform the same change of variables; this proves equality of the two values also when k is not in the countable dense set. The complement of $\mathcal{O}_t$ is invariant and has zero mass, so the zero extension is invariant there as well. Countability was used only to define a Borel function, not to sum densities or to introduce a multiplicity of charts.

Step 7: removal of the tree integrals. Finally, at fixed target and frozen coordinates, we apply a based tree gauge fixing. The pointwise covariance just proved makes the density and its tests independent of the tree variables in (21.27), so each Haar probability integral over tree variables equals one. Removing them, we obtain the formula in the statement. Its factors at the scales i are exactly those of Lemma 18.7, in particular the numerator with the eliminated edges and the single inverse determinant of the eliminated-edge derivative. The Haar measure of the output is already the reference measure at the next scale, so this change of variables has no room for a further coarse Haar density, a Gram determinant, or an orbit-volume factor. For an arbitrary noncovariant Borel input density, the tree average would remain in (21.27), and its removal uses the covariance of the stage density. The argument uses the Haar disintegration before the change to normalized fluctuation variables. The deep frozen group variables, including exceptional values of the Borel extension, remain compact Borel data throughout. □

21.5 Freezing and substitution of frozen coordinates

When integration stops on part of the region, the corresponding variables become frozen coordinates. The next lemma shows that, at the representatives of the next stage satisfying the reconstruction identities, the inverse of the freezing map inserts the frozen values directly into the old variables. This identifies the density after freezing with the density before freezing at the same configuration, also on sets of Haar measure zero, and gives the exact substitution rules for the analytic functions.

Lemma 21.10. *At every boundary of a large-field decomposition where coordinates are frozen, use the hierarchical tree transports σ_e of Lemma 21.1, with the tree conditions at coarser scales fixed first. Split the old output into the remaining link variables v, the variables a_e on the links*

across the boundary of the block region, and the other links to be frozen. The freezing map $\mathcal{A}_t$ replaces each such variable by its based transport

$$\widehat{b}_e = U_v(\sigma_e)a_e$$

and records the other links as frozen links on that lattice. At fixed v this map preserves the product Haar measure and has Jacobian one.

Let $b^\sharp$ denote the new frozen group values and $\xi^\sharp = \operatorname{Log} b^\sharp$ their logarithms. For every reconstructed configuration of the next stage in the family of representatives satisfying the reconstruction identities,

$$v = C_{t+1}(c', \zeta'; b_{\mathrm{rec}}, b^\sharp),$$

the hierarchical tree transport has value I: each of its tree factors is I, and the gauge transformation of Lemma 19.5 is I at its two endpoints. All protected earlier endpoints are fixed as well. Hence

$$a_e = U_v(\sigma_e)^{-1}\widehat{b}_e = \widehat{b}_e, \qquad \mathcal{A}_t^{-1}(v, b_{\mathrm{rec}}, b^\sharp) = (v \oplus b^\sharp, b_{\mathrm{rec}}),$$

and the density versions of Theorem 21.9 satisfy the pointwise identity

$$p_{t+1}(v, b_{\mathrm{rec}}, b^\sharp) = p_t^{\mathrm{out}}(v \oplus b^\sharp, b_{\mathrm{rec}}).$$

At this configuration, all omitted old variables are exactly their new frozen values, based transports or links, with the same weights and indicators. In analytic functions, $\xi^\sharp$ is inserted in the corresponding analytic arguments. In particular, the freezing of the effective interaction subtracts only the value $R^{\mathrm{out}}(0 \oplus \xi^\sharp)$ and transfers it with a minus sign to Λ, and the freezing of the classical action is $E_{\mathrm{new}}(v) = E^{\mathrm{out}}(v \oplus \xi^\sharp)$. These are exact finite identities, uniform in the number of steps, the topology, the depth, and the coupling. They compare the same configuration and require no overlap identity between auxiliary functions at different gauges.

Proof. We fix the deterministic geometry of the freezing. The remaining variables and the independent variables on the links across the boundary are different coordinates of the old output product. At fixed v, left multiplication by each fixed matrix $U_v(\sigma_e)$ preserves the Haar probability measure of such a variable, and the inverse multiplication is defined for every value. Integrating first in these variables, we obtain the triangular statement with Jacobian one for every nonnegative Borel integrand; earlier frozen coordinates and the other frozen links do not change.

We verify the value of the transport at a reconstructed configuration. On the based section of one scale, the tree equations set each ordered tree factor to I, and the fixing gauges of the coarser scales, lifted as constants on the finer blocks, conjugate these factors and leave them equal to I. By induction over the scales, with the coarser conditions fixed first, every factor of the prescribed hierarchical transport is I on the hierarchical tree section. This is the finite algorithm of Lemma 21.1, and it uses no path through an

eliminated edge. Hence the product has value I for every target and every free coordinate in the domain, not only at zero fluctuation.

The reconstructed configuration is obtained from that section by the gauge transformation of Lemma 19.5. This gauge transformation multiplies a transport from α_e to u_e by its values at exactly these two vertices, and both values are I by that lemma, with the full set of protected vertices supplied by Theorem 21.5. The transport remains I. On the lattices of earlier frozen coordinates, the endpoint protection rule fixes all their endpoints. This proves, at this reconstructed configuration,

$$U_v(\sigma_e)^{-1}\widehat{b}_e = \widehat{b}_e.$$

Inverting the freezing map therefore inserts the new based transports directly into the old variables on the links across the boundary, and every other omitted variable is its new frozen link value. At this configuration, the inverse map is $(v, b_{\mathrm{rec}}, b^\sharp) \mapsto (v \oplus b^\sharp, b_{\mathrm{rec}})$.

The general formula for the density after freezing is the composition of the old outgoing density with the inverse triangular map, since its Haar Jacobian is one. We use the versions of Theorem 21.9 on both sides; that construction uses the same formulas and their pointwise overlap identities. Composing with the inverse, we obtain the pointwise identity, even when the reconstructed configuration lies in a set of representatives of Haar measure zero. All weights and indicators of the decomposition are values of the same functions at the same old configuration.

For a variable that was analytic before this freezing, its new group value lies in the small chart of frozen coordinates, and we insert $\xi^\sharp = \operatorname{Log} b^\sharp$ there. The deep frozen group variables at the first step under the Wilson measure are fixed parameters instead and are not part of this logarithmic substitution. For every old analytic function F we have the exact identity

$$F(v \oplus \xi^\sharp) = \big[F(v \oplus \xi^\sharp) - F(0 \oplus \xi^\sharp)\big] + F(0 \oplus \xi^\sharp). \tag{21.30}$$

Applying it to R^{out}, the factor $e^{\Lambda^{\mathrm{out}} - R^{\mathrm{out}}}$ becomes

$$e^{\Lambda^{\mathrm{out}} - R^{\mathrm{out}}(0\oplus\xi^\sharp)} e^{-[R^{\mathrm{out}}(v\oplus\xi^\sharp) - R^{\mathrm{out}}(0\oplus\xi^\sharp)]},$$

which proves the minus sign of the transfer to Λ. For the classical function we keep the whole value $E^{\mathrm{out}}(v \oplus \xi^\sharp)$, including its constant. Centering the effective interaction changes no first derivative: the new first derivative in the remaining variables is the old one evaluated at $(0, \xi^\sharp)$, and it is kept. □

21.6 Conditional integration and normalization on large-field decompositions

This section proves the density bounds used in Section 22. The induction has three stages. First, the analytic effective actions, effective interactions, normalization constants,

and reconstruction maps are defined from fine to coarse scales, by the recursion of Theorem 20.14, for every value of the frozen coordinates and without reference to any measure. Second, representatives satisfying the reconstruction identities are chosen from coarse to fine scales, using these maps. Third, the stage densities are bounded from fine to coarse scales at exactly these representatives. The construction is not circular, since each stage uses only objects defined in the preceding ones. One step of the third stage is the following comparison, which needs an incoming density bound; the bound is supplied by the induction itself.

Lemma 21.11. *Assume the hypotheses of Theorem 21.8, together with the conditions* (21.18) *and* (21.19)*, which give the hypotheses of Lemma 20.5 on the whole enclosing product domain. Fix a stage t and an output datum (c, b_{rec}) of the family of representatives satisfying the reconstruction identities in Theorem 21.8(2), with real frozen coordinates, and define the fluctuation cutoff domain*

$$\mathcal{W}_t = \prod_{i,e} \{|w_{i,e}| \le p(t_t)\}, \qquad t_t = \max_i g_{t,i}, \qquad p(t) = A_{\mathrm{w}} \log(1/t). \tag{21.31}$$

Let $V = V_{c,b_{\mathrm{rec}}}$ be the conditional fluctuation potential of Theorem 20.14 at this datum, and let $S_{c,b_{\mathrm{rec}}} \subset \mathcal{K}_t$ be the set of normalized fluctuation coordinates of the supported fiber points over (c, b_{rec}). For an invariant Borel function $0 \le \chi \le 1$, let $G_{t,\chi}(c, b_{\mathrm{rec}})$ be the outgoing density of Theorem 21.9 at (c, b_{rec}) with χ inserted in the integrand. Suppose that there exist a finite prefactor $P_t(c, b_{\mathrm{rec}}) \ge 0$ and a Borel function χ_t on $S_{c,b_{\mathrm{rec}}}$ with values in $[0,1]$ such that

$$G_{t,\chi}(c, b_{\mathrm{rec}}) \le P_t(c, b_{\mathrm{rec}}) \int_{S_{c,b_{\mathrm{rec}}}} \chi_t(w) e^{-V(w)}\, dw. \tag{21.32}$$

Then $S_{c,b_{\mathrm{rec}}}$ is a Borel set, $0 < \int_{\mathcal{K}_t} e^{-V}\, dw < \infty$, and

$$G_{t,\chi}(c, b_{\mathrm{rec}}) \le (1-\delta_t)^{-1} \widetilde{F}_t(c, b_{\mathrm{rec}}), \qquad \widetilde{F}_t = P_t(c, b_{\mathrm{rec}}) \int_{\mathcal{W}_t} e^{-V(w)}\, dw,$$

with $0 \le \delta_t < 1$, $\delta_t \le 6\mathsf{b}_t e^{-p(t_t)^2/48}$, and $\mathsf{b}_t \le C_J m^3$, where b_t is the number of three-dimensional free blocks of fluctuation variables of the step. Factors depending only on the output datum multiply both sides. The indicators are removed only in this upper bound. The constants have the dependence stated at the beginning of this section, and they are uniform in the deep frozen group variables.

At the first RG step, for either initial measure, (21.32) *holds with equality. In that case P_t consists of the inverse partition function of the initial measure, the external weights, the Haar disintegration constants, the powers of the couplings, the output classical factor, and the zero-fluctuation factors, and χ_t is the product of χ and the indicators composed with the inverse coordinates. At a later stage,* (21.32) *is obtained by inserting an incoming bound on the density at the representatives satisfying the reconstruction identities, with the incoming multiplicative loss included in the same prefactor P_t; this incoming bound is proved in Theorem 21.12.*

Proof. At an output datum, the set $S_{c,b_{\mathrm{rec}}}$ is defined with all indicators still inserted, and before any upper comparison the Haar disintegration is the exact formula of Lemma 18.7. The radii of $\mathcal{W}_t$ belong to the logarithmic family of Theorem 20.14 and satisfy $t_t p(t_t) < R_{\mathrm{en}}$ by (21.19); for a complete RG step $t_t = t_{\max}$. The product $\mathcal{K}_t$ is used only in this positive comparison, never as the product for the expansion defining the analytic output. At the first step, the Haar disintegration and the change to normalized fluctuation variables give the density integral in the form

$$G_{t,\chi}(c, b_{\mathrm{rec}}) = P_t(c, b_{\mathrm{rec}}) \int_{S_{c,b_{\mathrm{rec}}}} \chi_t(w) e^{-V_{c,b_{\mathrm{rec}}}(w)}\, dw, \qquad \widetilde{F}_t = P_t(c, b_{\mathrm{rec}}) \int_{\mathcal{W}_t} e^{-V_{c,b_{\mathrm{rec}}}(w)}\, dw, \tag{21.33}$$

with the prefactor described in the statement. At a later step the equality in the first formula is replaced by the inequality (21.32).

By Theorem 21.8(2), $S_{c,b_{\mathrm{rec}}} \subset \mathcal{K}_t$ for every such datum. We check the measure-theoretic hypotheses of Lemma 20.5 at this fixed real datum. The finite products $\mathcal{K}_t$ and $\mathcal{W}_t$ are compact Borel sets. The map from free logarithmic coordinates to normalized fluctuation coordinates is a homeomorphism onto its open image, with the continuous inverse already constructed, so the image of the Borel set of free coordinates of supported points is Borel. Composing the Borel tests with that inverse gives a Borel function χ_t, extended by zero outside $S_{c,b_{\mathrm{rec}}}$, with $0 \le \chi_t \le 1$. The conditional fluctuation potential is analytic on a neighborhood of $\mathcal{K}_t$ and real at its real points, by Theorem 20.14. Its finite minimum $\underline{V}$ and maximum $\overline{V}$ on this compact product give

$$0 < e^{-\overline{V}} \operatorname{vol}(\mathcal{K}_t) \le Z_{\mathcal{K}_t} = \int_{\mathcal{K}_t} e^{-V}\, dw \le e^{-\underline{V}} \operatorname{vol}(\mathcal{K}_t) < \infty. \tag{21.34}$$

The volume is finite and positive, since every ball has a finite positive radius. For an empty free product its zero-dimensional measure is one and $V = 0$, so $Z_{\mathcal{K}_t} = 1$. No uniform bound on $\underline{V}$ or $\overline{V}$ is needed here. The prefactor is finite at each real datum; if it vanishes, both sides vanish, and otherwise it is positive.

The hypotheses of Lemma 20.5 on the whole product give its potential estimate and, for the normalized enclosing-product comparison measure $\nu_{\mathcal{K}_t}$, the bound

$$\delta_t := \nu_{\mathcal{K}_t}(\mathcal{K}_t \setminus \mathcal{W}_t) \le 6\mathsf{b}_t e^{-p(t_t)^2/48}, \qquad \mathsf{b}_t \le C_J m^3.$$

Every ball of $\mathcal{W}_t$ has positive radius, so its volume is positive. Since V is finite on $\mathcal{K}_t$, we have $\int_{\mathcal{W}_t} e^{-V}\, dw > 0$, and hence $\nu_{\mathcal{K}_t}(\mathcal{W}_t) > 0$ and $0 \le \delta_t < 1$. This strict inequality concerns the actual missing mass; the displayed union-bound majorant need not be below one under the local hypotheses. For an empty free product, $\delta_t = 0$.

Since $0 \le \chi_t \le 1$, the right side of (21.32) is at most $P_t \int_{\mathcal{K}_t} e^{-V}$, and part (1) of Lemma 20.5 bounds the latter by $(1 - \delta_t)^{-1} \widetilde{F}_t$. No inclusion of $\mathcal{W}_t$ in $S_{c,b_{\mathrm{rec}}}$ is needed. The same reasoning applies if $P_t = 0$ or if the free list is empty. All constants arose from local path products, the fixed construction products, and their row-sum bounds. □

In the theorem below, E_t, R_t, and Λ_t are the effective classical action, the effective interaction, and the normalization constant at stage t; the last is the field-independent additive term in the log-density. Further, W_t is the product of the external weights (the factors of frozen variables that are not integrated), C_t is the reconstruction map, and ℓ_t is the level of the output of stage t. For the normalization comparison, $\Lambda_t^{\mathbb{T}}$ denotes the normalization constant of the corresponding recursion on the whole torus. Here H_t denotes the set of scalar Gaussian coordinates of stage t that are integrated in the corresponding renormalization group recursion on the whole torus but absent from the regional free list (it is not the Haar measure of Theorem 21.9), and $d_i(H_t)$ is the number of its elements at scale i. The numbers $N_{\mathrm{missing},t}$, $N_{\mathrm{bd},t}$, and $N_{\mathrm{freeze},t}$ count, on the lattices where they were created, the missing labels of field-independent generated terms, the labels of the fixed interface band, and the newly frozen analytic labels, as in Theorem 20.13.

Theorem 21.12. *Fix a large-field decomposition of Theorem 21.5, an initial measure ν (a stopped measure or the Wilson measure), an initial scale and a stopping scale differing by a whole number T of RG steps, and the cutoff domains and norm ball of Theorem 20.14. Every integrated group of averaging steps is a complete RG step, $s_t = J$. A remaining number $0 \le u_{\mathrm{ph}} < J$ of averaging steps within the given hierarchy is kept through the pullback $C \mapsto F(T_M^{u_{\mathrm{ph}}} C)$ of an observable F at the stopping scale; this requires $m_{\mathrm{stop}} = M^{u_{\mathrm{ph}}} m_{\mathrm{rem}}$ with $m_{\mathrm{rem}} \in 4\mathbb{N}$. For a fixed additional number $r_{\mathrm{stop}} \in \mathbb{N}_0$ of RG steps, the same condition is imposed with the total exponent $r_{\mathrm{stop}} J + u_{\mathrm{ph}}$, and all iterates must lie within the finite hierarchy. No recursion for normalization constants is asserted for an integrated group of fewer than J averaging steps. Assume the uniform incoming radii obtained by freezing every label of reduced radius, the boundary-distance comparison* (21.10)*, the conditional force conditions* (21.13) *and* (21.14)*, the cutoff radii of* (21.19)*, the initial norm bounds of Theorem 20.14, and the bounds* (21.8) *on frozen coordinates. Let $\mathcal{W}_t$ be the fluctuation cutoff domain* (21.31)*, which is the instance of Theorem 20.14 with common radius $p(t_t)$; the enclosing product domain $\mathcal{K}_t$ is used only in Lemma 21.11. Write Z_ν for the partition function of the initial measure, without centering: for a stopped measure*

$$Z_\nu = \int_{D_r} e^{-E_{d_0,\mathrm{sat}}/g_{\mathrm{base}}^2 - R_{d_0,\mathrm{sat}}}\, dH,$$

and $Z_\nu = Z_{m_{\mathrm{base}},\gamma}$ for the Wilson measure. Use the stage measures and density versions of Theorem 21.9. Then the following constructions and bounds hold at every finite depth and number of steps.

1. *(Analytic functions, from fine to coarse.) For every real value of the frozen coordinates satisfying* (21.8)*, the functions E_t, R_t, Λ_t, W_t, the classical objectives, the reconstruction maps C_t, and the auxiliary fluctuation integrals are defined on the full spatially varying radii by the recursion of Theorem 20.14, with the substitution of frozen coordinates of Lemma 21.10. This definition uses no density inequality and no choice of representatives.*

The effective interaction is centered only in its constant term, all its first derivatives are kept, and the full classical values are kept. For a stopped measure the normalization constant on the torus is $\Lambda_0^{\mathbb{T}} = -R_{d_0}(I)$*, and an initial freezing subtracts in addition the frozen value of the centered effective interaction. For the Wilson measure we use the exact split of Theorem 21.8(4), with its original external factor.*

2. *(Representatives satisfying the reconstruction identities, from coarse to fine.) Take all small top representatives in the coordinate domains of Theorem 21.8(1) whose frozen coordinates satisfy* (21.8)*, and construct such representatives downward with the maps of part (1). At a boundary with freezing, let* $\mathscr{D}_{t+1}$ *be the set of these representatives at the next stage, and define the set of output data satisfying the freezing and reconstruction identities*

$$\mathscr{P}_{t+1} = \mathcal{A}_t^{-1}(\mathscr{D}_{t+1}).$$

For each such datum, include every supported fiber point:

$$\mathscr{D}_t = \{(C_t(c,\zeta;b_{\mathrm{rec}}),b_{\mathrm{rec}}) : (c,b_{\mathrm{rec}}) \in \mathscr{P}_{t+1},\ \zeta \in \mathcal{Z}_{c,b_{\mathrm{rec}}}\},$$

where $\mathcal{Z}_{c,b_{\mathrm{rec}}}$ *is the set of supported fiber coordinates over* (c,b_{rec})*. At the last RG step, the set of output data consists of these small top representatives whose frozen coordinates satisfy* (21.8)*; every configuration in the support of the stage measure has such a representative by Theorem 21.5. Other representatives are compared only by the changes of Lemma 21.6(1)–(3) or by the pointwise invariance of Theorem 21.9. These sets contain all choices, and no probability law is assigned to a representative map.*

3. *(Density bounds, from fine to coarse.) For every real value of the frozen coordinates, every* $(c,b_{\mathrm{rec}}) \in \mathscr{P}_{t+1}$*, and every supported fiber coordinate* ζ *over it,*

$$\begin{aligned} p_t(C_t(c,\zeta;b_{\mathrm{rec}}),b_{\mathrm{rec}}) \le\ & A_t Z_v^{-1} e^{\Lambda_t(b_{\mathrm{rec}})} W_t(b_{\mathrm{rec}}) \\ & \times \exp\{-E_t(\mathrm{Log}\, C_t(c,\zeta;b_{\mathrm{rec}});b_{\mathrm{rec}})/g_{\ell_t}^2\} \\ & \times \exp\{-R_t(\mathrm{Log}\, C_t(c,\zeta;b_{\mathrm{rec}});b_{\mathrm{rec}})\}. \end{aligned}$$

Here $A_0 = 1$ *and*

$$A_{t+1} = A_t(1-\epsilon_t)^{-1}, \qquad \epsilon_t \le 6\mathsf{b}_t e^{-p(t_t)^2/48}, \quad \mathsf{b}_t \le C_J m_t^3, \qquad \log A_T \le C_\delta,$$

after the uniform choice of the minimal side in Lemma 20.5. Every step uses the exact Haar disintegration and then a single triangular change to normalized fluctuation variables, with determinant

$$\prod_{i=0}^{s_t-1} g_i^{3f_i} \det K_i(c_{i+1}(w_{>i});b_{\mathrm{rec}})^{-1/2}.$$

The resulting Borel set of fluctuation coordinates of supported points is contained in the enclosing product domain, and Lemma 21.11, with identical prefactors and the smaller cutoff

domain, gives the stated loss. The next stage evaluates the same configuration. The indicators are kept until this comparison, and no domination at arbitrary points of the enclosing product domain is required.

4. *(Normalization.) With* $\Delta\Lambda_t = \Lambda_t - \Lambda_t^{\mathbb{T}}$ *in the same centered renormalization group recursion on the whole torus, Theorem 20.13 gives*

$$\Delta\Lambda_{t+1} \le \Delta\Lambda_t + \sum_{i \text{ in } t} d_i(H_t)\log(1/g_i) + C_J(N_{\text{missing},t} + N_{\text{bd},t} + N_{\text{freeze},t}),$$

including the initial freezings. This comparison uses the same centered recursion on the whole torus and the same partition function Z_ν *throughout.*

5. *(External classical terms and the measure of frozen coordinates.) For the comparison of the output classical actions, the protected external classical exponents are recombined exactly once. At fixed frozen coordinates, with their original couplings included in* $S_{\text{ext}}(b_{\text{rec}})$,

$$e^{-S_{\text{ext}}(b_{\text{rec}})}e^{-E^{\text{int}}/g_t^2} = e^{-E^{\text{full}}/g_t^2}, \qquad E^{\text{full}} = E^{\text{int}} + g_t^2 S_{\text{ext}}(b_{\text{rec}}),$$

where E^{int} *is the effective classical action of the block region, and this factor is removed from the external weight. Under an RG step both the action and* g_t^2 *are multiplied by* M^{s_t}*, so each such exponent is preserved and counted once. At the top we keep the measure*

$$\rho_{\mathcal{X}}(db_{\text{rec}}) = w_{\mathcal{X}}(b_{\text{rec}})H_{\mathcal{X}}(db_{\text{rec}}) \le H_{\mathcal{X}}(db_{\text{rec}}), \qquad w_{\mathcal{X}} = I_{\mathcal{X}}^{\text{act}} I_{\mathcal{X}}^{\text{inv,end}},$$

on the frozen coordinates, where $H_{\mathcal{X}}$ *is their product Haar measure. The first indicator keeps every failure* $q_f(b_{\text{rec}}) \ge d_*$*, which is a function of the frozen coordinates since the whole dependence set of* f *was frozen. The second keeps any chosen invariant tests depending only on the frozen coordinates at the output. There is no second copy of the frozen-coordinate terms included in the full classical action. Optional noninvariant chart restrictions may be bounded by one.*

All local constants have the dependence stated at the beginning of this section, including independence of the later widths, while C_δ *has the dependence described in item (4) of Definition 4.14. No comparison of regional integrals from below, gauge covariance of auxiliary functions, or overlap identity for the classical action is part of this induction.*

Proof. Every integrated step has $s_t = J$, so $\ell_{t+1} = \ell_t + J$ and $g^2_{\ell_{t+1}} = Bg^2_{\ell_t}$. We reserve z_i for the free logarithmic coordinates and write $\zeta_i = g_i w_i$ for the scaled fluctuation arguments of C_t. All analytic functions below are functions of the identity-chart variables of the block region at fixed real (ξ, b_{rec}). We prove the five parts in order, followed by the statement on the remaining averaging steps.

Part (1): initialization. For a stopped measure, we set

$$r_{d_0} = R_{d_0} - R_{d_0}(I), \qquad \Lambda_0^{\mathbb{T}} = -R_{d_0}(I).$$

The partition function without centering then gives exactly $Z_\nu^{-1} e^{\Lambda_0^{\mathbb{T}}} e^{-E_{d_0}/g_{\mathrm{base}}^2 - r_{d_0}}$ at a small representative in D_r. In particular $\widehat{Z} = e^{R_{d_0}(I)} Z_\nu$ for the centered partition function in Theorem 21.5; we keep Z_ν throughout this proof. For an initial freezing at $\xi^\sharp$, we set

$$\begin{aligned} E_0(v; b_{\mathrm{rec}}) &= E_{d_0}(v \oplus \xi^\sharp), \\ R_0(v; b_{\mathrm{rec}}) &= r_{d_0}(v \oplus \xi^\sharp) - r_{d_0}(0 \oplus \xi^\sharp), \\ \Lambda_0(b_{\mathrm{rec}}) &= -R_{d_0}(I) - r_{d_0}(0 \oplus \xi^\sharp). \end{aligned} \tag{21.35}$$

Additional frozen coordinates are held as real parameters. For the Wilson measure, we set $E_0 = F_{\mathrm{bare}}$, $R_0 = 0$, $\Lambda_0 = 0$, and $W_0 = e^{-E_{\mathrm{hole}}/g_{\mathrm{base}}^2}$, together with any kept indicators at the output. The exact split in Theorem 21.8(4) implies this initialization with the partition function of the Wilson measure.

We verify the initial norm bounds. The initial analytic values obey (21.8), because $C_{\mathrm{conv},J} \geq C_{\mathrm{ad},J}$, which was fixed before Theorem 21.5, and the near values at the first step under the Wilson measure satisfy its conversion bound. For a stopped measure, the decomposition with zero completion has classical perturbation $P_0(v) = E_{d_0}(v \oplus \xi^\sharp) - E_{d_0}(v \oplus 0)$, including its constant. With the matched completion $\iota_{\xi^\sharp}$ of Theorem 19.9, this is exactly its boundary-completion difference, and the adjusted perturbation is $\overline{P}_0 = 0$. In the norm with zero completion, its nonconstant part is bounded by $C_{\mathrm{bc,cl}} E_{\mathrm{cl}} \beta_{\mathrm{rec}} / \rho_{\mathrm{cl}} \leq \sigma_*/4$, by (21.16). For the quantum input, splitting the centered function $r_{d_0}(v \oplus \xi^\sharp)$ into its restriction to the torus and its boundary-completion difference gives the bound $C_{\mathrm{bc,q}} E \beta_{\mathrm{rec}} / \rho \leq \sigma_*/4$. The constant in (21.35) is kept exactly and is not bounded by this norm.

At the first step under the Wilson measure, the plaquettes depending on integration variables use only the small near values of the initial good-cell complex. The zero-completed Wilson function evaluated at those data differs from the sum of these plaquette terms by plaquettes involving only frozen variables, which are constant in the integration variables. Hence the adjusted classical perturbation is constant at this step, and its nonconstant boundary-completion difference with zero completion obeys the same classical bound. These classical constants remain in F_{bare}, and the energy of the deep frozen group variables stays in W_0. The quantum input is zero. This proves $x_0, x_0^{\mathrm{cl}} \leq \sigma_*$ for either initial measure, on uniform radii of the remaining inputs. No deep frozen group variable has entered an analytic logarithm.

Part (1): the analytic recursion. At a later input, let D_{in} be the list of remaining analytic variables of the block region, with their original locations and volume factors; it does not include the frozen parameters. Since the preceding freezing of the union removed every label of reduced radius, the classical and quantum input radii are uniform by (21.15). The previous fixed complement and all earlier frozen basepoints obey (21.10). The initialization satisfies the same statement, with zero boundary-weighted input when

there is no incoming fixed complement, and additional excluded regions created by the tests preserve this inclusion. All completion adjustments use only newly frozen small analytic variables; previously nonintegrated torus variables remain zero, and the deep frozen group variables remain fixed external parameters.

We apply Theorems 19.9 and 20.14 inductively, with the fixed amplitude A_{w} and the coupling ceiling g_* chosen for it, to define the classical objective, the conditional minimizers for initial segments, their positive square roots, and the reconstruction map on the full radii. The hypotheses of each application are checked before any set of representatives is constructed. The input norm just restored bounds the adjusted perturbation by $2\sigma_*$. The images and every connecting segment lie strictly in the uniform input ball, by (21.17) and the fixed margin of the larger construction domain. The force and Hessian estimates, uniformly for $y \in \mathcal{T}_I$ for each initial segment I, followed by (21.14), give all conditional minimizers and sequential neighborhoods; these checks are Lemma 21.7. All perturbation derivatives are taken in the integration and free variables at fixed real b. The fixed rates and spatial margins are those of the common tuple of Theorem 20.14 and are not changed by freezing. The torus input is the member at accumulated classical depth $d_t = d_0 + tJ$ of its uniform analytic class, with ceiling E; a new nonconstant boundary term is accounted for by the two input norms rather than added to this ceiling. We include the exact boundary-completion difference when the unperturbed objective uses newly frozen input variables. By Theorem 19.9, this objective agrees on the connected chart with E_t evaluated at its reconstructed field; only the torus term uses compact overlap there, and the perturbation is evaluated at that field from the outset. The full outgoing action is defined to be the value at the conditional minimizer multiplied by M^{s_t}.

We next define the update of the effective interaction by an integral; this also fixes all normalization conventions. Let $J_t^0(c, b_{\mathrm{rec}})$ be the product of the Haar disintegration factors of the intermediate scales evaluated along the zero-fluctuation chain, and define

$$D_t^0(c, b_{\mathrm{rec}}) = J_t^0(c, b_{\mathrm{rec}}) \prod_i g_i^{3f_i} \det K_i(\widehat{c}_{i+1}; b_{\mathrm{rec}})^{-1/2}, \tag{21.36}$$

where $\widehat{c}_{i+1}$ denotes the corresponding higher argument at zero fluctuation. All field-independent constants of the Haar disintegration are included. Sequential minimization and normalization give the exact decomposition of the action: the output value divided by $g_{\ell_{t+1}}^2$, the quadratic term $\sum_i |w_i|^2/2$, and the higher classical remainders add up to the input classical exponent. Indeed, at each intermediate scale we subtract the conditional minimum and its quadratic Taylor term; the latter is $|w_i|^2/2$ by the definition of K_i, and the factor M in the minimum cancels the factor M in the next coupling. Repeating this over the s_t scales, we obtain the decomposition, including its constants. Adding the inherited difference of effective interactions, all logarithmic differences of the Haar disintegration factors, and all logarithmic differences of the determinants that depend on the

higher fluctuation variables, we obtain the conditional fluctuation potential $V_t(c, b_{\mathrm{rec}}; w)$, including its quadratic part; it is the potential of Theorem 20.14.

In this integral, we use the fluctuation cutoff domain (21.31): every radius is $p(t_t)$ with $t_t = \max_i g_{t,i}$. This is in the allowed logarithmic family; it is not the radius R_{en}/t_t of the enclosing product. The containment, positive denominator, interaction, and auxiliary half-disc conditions hold by (21.17) and (21.19); they are deterministic conditions on the parameters and on g_*, not restrictions on the support of the measure. On the fixed product $\mathcal{W}_t$, we define

$$\begin{aligned} \mathcal{I}_t(c; b_{\mathrm{rec}}) &= D_t^0(c, b_{\mathrm{rec}}) e^{-R_t(\mathrm{Log}\, C_t(c,0;b_{\mathrm{rec}});b_{\mathrm{rec}})} \int_{\mathcal{W}_t} e^{-V_t(c,b_{\mathrm{rec}};w)}\, dw, \\ Q_t(c; b_{\mathrm{rec}}) &= -\log \mathcal{I}_t(c; b_{\mathrm{rec}}), \\ R_t^{\mathrm{out}}(c; b_{\mathrm{rec}}) &= Q_t(c; b_{\mathrm{rec}}) - Q_t(0; b_{\mathrm{rec}}), \qquad \Lambda_t^{\mathrm{out}} = \Lambda_t - Q_t(0; b_{\mathrm{rec}}). \end{aligned} \tag{21.37}$$

The logarithm is real on real data, where it agrees with the real logarithm of the positive integral; its zero-free analytic continuation is supplied by Theorem 20.14, and its sign is unrestricted. The integrand and its domain are defined on the full radii before any support set is considered. We have the exact identity

$$Z_v^{-1} e^{\Lambda_t} W_t e^{-E_t^{\mathrm{out}}(c)/g_{\ell_{t+1}}^2} \mathcal{I}_t(c; b_{\mathrm{rec}}) = Z_v^{-1} e^{\Lambda_t^{\mathrm{out}}} W_t e^{-E_t^{\mathrm{out}}(c)/g_{\ell_{t+1}}^2 - R_t^{\mathrm{out}}(c;b_{\mathrm{rec}})}. \tag{21.38}$$

Changing the distribution of constant factors between D_t^0 and the definition of Q_t leaves this identity unchanged; we use the convention of Theorem 20.13, which assigns each field-independent factor once.

At a boundary we use the substitution of Lemma 21.10. This defines all functions at the next level for every real value of the frozen coordinates, independently of any density bound. The simultaneously frozen list is the union of the full boundary band of reduced radius, the removed band, the new boundary layers, and the based transports; the corresponding initial variables were included at the initial scale. Their analytic values satisfy (21.8) for both radii. At a representative of the next stage satisfying the reconstruction identities, the freezing identity inserts these values, not small values from a different frame, and the total boundary-completion difference obeys (21.16). The resulting norm bounds are exactly those of Theorem 20.14: the output ceilings before freezing are y_t and y_t^{cl}, the boundary-completion differences are at most $\sigma_*/4$, and the removed band makes each remaining term at most $\sigma_*/2$. With $\alpha_{\mathrm{q}} = \alpha$ and the corresponding classical rate, we have for both types

$$x_{t+1}^{\bullet} \le C_{\mathrm{bc},\bullet} E_{\bullet} \beta_{\mathrm{rec}}/\rho_{\bullet} + 2e^{-\alpha_{\bullet}(s_{\mathrm{er}}-1)} y_t^{\bullet} \le \sigma_*/4 + \sigma_*/2,$$

so $x_{t+1}, x_{t+1}^{\mathrm{cl}} \le 3\sigma_*/4$ whenever the input is in its ball. The output ceilings $y_t^{\bullet}$ are computed on the input ball before the width s_{er} is chosen. By induction from the initial

ball, we obtain the analytic recursion at every finite number of steps. It keeps the terms of degree one; no first derivative has been removed by recentering, and every restoration of the bounds happens at a complete RG step.

It remains to verify the Borel regularity in the real parameters required at each perturbation step. The initial coefficients and chart path products are Borel functions of the protected real frozen coordinates; the deep frozen group variables at the first step under the Wilson measure enter only the Borel external factor. Suppose this holds at an input. The fixed-point iterations for the conditional minimizers use Borel coefficients and converge uniformly on their fixed neighborhoods, so their limits are Borel in b. The positive matrix square roots and inverses are Borel as uniform limits of the matrix-function constructions. The resulting values, Haar disintegration determinants, and coefficients are Borel. By Tonelli's theorem, integration over the fixed finite product preserves Borel dependence; we apply it to the positive and negative parts and to the real and imaginary parts, which have finite integrals. The logarithm can be chosen jointly Borel without a branch choice depending on the frozen coordinates: on the star-shaped output domain,

$$\log \mathcal{I}_t(c;b) = \log \mathcal{I}_t(0;b) + \int_0^1 \frac{D_c \mathcal{I}_t(sc;b)[c]}{\mathcal{I}_t(sc;b)}\, ds,$$

where the first term is the real logarithm. Zero-freeness and holomorphy make this the required analytic branch, and the parameter integral is Borel. Finally, the substitution of frozen coordinates evaluates these Borel families at the Borel coordinates of the frozen values. This proves the Borel regularity inductively, without differentiating in a real frozen parameter.

Part (2): representatives satisfying the reconstruction identities. We start from all small representatives of top configurations, in the coordinate domains of Theorem 21.8(1), whose analytic frozen coordinates satisfy (21.8); they exist by Theorem 21.5. At a boundary with freezing, we pull the next set of representatives satisfying the reconstruction identities back by the freezing bijection $\mathcal{A}_t$; this defines $\mathscr{P}_{t+1}$, which is not enlarged to an arbitrary comparison set. For each such datum, we include every supported fiber coordinate, as in the definition of $\mathscr{D}_t$. Theorem 21.8, applied at this step of the descent, shows that each of them has its inverse normalized fluctuation coordinate in the interior half of the enclosing product, and that its representative has the correct target and protected frozen coordinates. This defines the whole finite family downward using only the maps defined by (21.37). The sets integrated below are the Borel sets of free coordinates of supported fiber points at fixed output data, so no measurable representative map is needed.

The bounds (21.8) persist along this descent, by Lemma 21.6(1)–(3) and, at a new boundary, the direct insertion of Lemma 21.10. Changes as in Lemma 21.6(4) are not

used to enlarge these sets. Over each output datum in $\mathscr{P}_{t+1}$, the set still contains every supported fiber point.

Part (3): density bounds. We use the densities of Theorem 21.9. For a stopped measure, the gauge-invariant density agrees with its analytic expression at each representative satisfying the reconstruction identities, and the indicators of the decomposition and of the support are between zero and one. Then (21.35) implies the claimed bound at the initial scale with $A_0 = 1$. For the Wilson measure, the exact split implies the same bound with the kept factor W_0. At a point outside the support, the density is zero, and no analytic identity is needed there.

Suppose that the inequality holds on $\mathscr{D}_t$, and fix an output datum $(c, b_{\mathrm{rec}}) \in \mathscr{P}_{t+1}$. We apply the exact Haar disintegration to the incoming density, keeping its tree variables at first. By Theorem 21.9, the covariance under based gauges makes their normalized Haar integrals equal to one. For this fixed target, the gauge transformation of Lemma 19.5 is a fixed gauge transformation, independent of the free variables and equal to I at the roots of the target and at every protected vertex. The invariance of the density therefore allows us to evaluate the integrand at $C_t(c, \zeta; b_{\mathrm{rec}})$ instead of at its representative before the gauge transformation, and the tests take the same values there. This changes neither the Haar disintegration nor any determinant, and it is not a covariance statement for the auxiliary functions on the right side of the inequality.

We now perform the single sequential change to normalized fluctuation variables for the classical objective. At scale i its derivative with respect to w_i is $g_i K_i^{-1/2}$, and its other dependence is on higher variables. Ordering the higher variables first makes the derivative block triangular, so its positive real determinant is exactly

$$\prod_{i=0}^{s_t-1} g_i^{3f_i} \det K_i(c_{i+1}(w_{>i}); b_{\mathrm{rec}})^{-1/2}. \tag{21.39}$$

In particular, the determinant at a varying higher argument is not replaced by its value at zero. Equation (21.36) and the corresponding logarithmic differences split this one determinant; they do not add a second copy. Injectivity and the inverse map on the supported domain follow from the branches for initial segments, with their positive margins, established before. As in the proof of Lemma 21.11, the image is a Borel set $S_{c,b_{\mathrm{rec}}}$ in the enclosing product, and the remaining indicators composed with the inverse form a Borel function χ_t on $\mathcal{K}_t$ with values in $[0,1]$. The potential has a finite nonzero integral over $\mathcal{K}_t$ at each fixed real datum, also for an empty product.

Every supported point of that domain belongs to $\mathscr{D}_t$. We insert the inductive bound there, and let $P_t^0(c, b_{\mathrm{rec}})$ be

$$Z_\nu^{-1} e^{\Lambda_t(b_{\mathrm{rec}})} W_t(b_{\mathrm{rec}}) D_t^0(c, b_{\mathrm{rec}}) e^{-E_t^{\mathrm{out}}(c;b_{\mathrm{rec}})/g^2_{\ell_{t+1}} - R_t(\mathrm{Log}\, C_t(c,0;b_{\mathrm{rec}});b_{\mathrm{rec}})}.$$

Tests depending only on the output datum are kept in this common factor. If this factor is zero, the outgoing upper bound is zero; otherwise it is finite and positive. With χ_t denoting the remaining indicators, the exact disintegration followed by the inductive bound gives

$$\begin{aligned} p_t^{\mathrm{out}}(c, b_{\mathrm{rec}}) &\le A_t P_t^0(c, b_{\mathrm{rec}}) \int_{S_{c,b_{\mathrm{rec}}}} \chi_t(w) e^{-V_t(w)}\, dw \\ &\le A_t P_t^0(c, b_{\mathrm{rec}}) \int_{\mathcal{K}_t} e^{-V_t(w)}\, dw \\ &\le \frac{A_t}{1-\epsilon_t} P_t^0(c, b_{\mathrm{rec}}) \int_{\mathcal{W}_t} e^{-V_t(w)}\, dw. \end{aligned} \tag{21.40}$$

Here ϵ_t is a common upper bound for the datum-dependent δ_t of Lemma 21.11 at stage t, over all admitted output data, after the uniform minimum-side choice in part (3) of Lemma 20.5. One may take $\epsilon_t = 6\mathsf{b}_t e^{-p(t_t)^2/48} \le 1/2$. The first inequality is the hypothesis (21.32) of that lemma with $\chi = 1$ and $P_t = A_t P_t^0$, and the remaining two inequalities are its conclusion. The middle step uses the enclosure of Theorem 21.8, positivity on the whole larger product, and only $0 \le \chi_t \le 1$. The last step is Lemma 20.5 with identical prefactors; its radii are the common logarithmic radii, enclosed by $t_t p(t_t) < R_{\mathrm{en}}$, and its Hessian and mean conditions hold by (21.19), in particular with mean at most $p(t_t)/(2\sqrt{3})$. The larger product $\mathcal{K}_t$ is used only in this comparison. This verifies the hypothesis of Lemma 21.11 at every later stage, with the incoming loss A_t in the common prefactor.

By (21.38), the last line of (21.40) is the analytic outgoing bound, defined independently, with the factor $A_{t+1} = A_t/(1-\epsilon_t)$. By Lemma 21.10, the inverse under freezing of a representative of the next stage is this output configuration. That lemma substitutes the based transports and frozen links directly, transfers $-R_t^{\mathrm{out}}(0 \oplus \xi^\sharp)$ to the normalization constant, and keeps the full classical value. This proves the incoming inequality at stage $t+1$ on exactly $\mathscr{D}_{t+1}$ and completes the induction.

There is one comparison per output datum, jointly over all components and scales of the step. The probability estimate used here is $\epsilon_t \le 6\mathsf{b}_t e^{-p(t_t)^2/48}$ with $\mathsf{b}_t \le C_J m_t^3$. Ordering the finite list of output data along the sequence of sides and taking the uniform minimal side of Lemma 20.5, we obtain

$$\log A_T = \sum_{t<T} -\log(1-\epsilon_t) \le 2 \sum_{t<T} \epsilon_t \le C_\delta.$$

Empty free products have determinant one, logarithmic determinant zero, and fluctuation integral one, and an empty block region has zero nonconstant norm. Such an empty step contributes no loss.

Part (4): normalization. Since $s_t = J$ and the initial and stopping scales differ by complete RG steps, every application below is within the scope of Theorem 20.13. The recursion just

constructed is the same centered recursion, with the same extraction of field-independent factors, as in that theorem. All its norm and domain hypotheses were verified in part (1), including the common logarithmic cutoffs and the freezing of the whole union. The column bound used for the analytic norm propagation in Theorem 20.10 is that of the perturbed zero-fluctuation map. Indeed, the doubled output radius and the inherited half radius are part of the two image conditions, and the differences with each index fixed in Theorem 19.9 give, in the same boundary-weighted sum over distinguished outputs,

$$K_{\mathrm{col},J}^{\mathrm{actual}} \le K_{\mathrm{col},J}^{0} + C_J \sigma_{0,J},$$

where $K_{\mathrm{col},J}^{\mathrm{actual}}$ and $K_{\mathrm{col},J}^{0}$ are the column ceilings of the perturbed and unperturbed maps. The fixed allocations of dependence neighborhoods and exponents are already included in C_J. More explicitly, the margin in the coefficient exponent, after the transfer of boundary distances in (21.10), leaves a positive exponent c' for summing over the incoming label at a fixed output, and this sum is bounded by $v_J \sum_{q\ge 0}(q+2)^3 e^{-c'q} < \infty$. The volume factors of the lattices where the labels were created are kept in this column sum. Deleting labels or adding excluded regions cannot increase this bound of bounded density. This proves that the enlargement has the dependence stated at the beginning of this section. We fix this enlarged ceiling on the whole larger perturbation domain before choosing s_{er} or later widths. In Theorem 20.10, its coefficient is $K_J = 4(\log 2) K_{\mathrm{col},J}$ for this enlarged ceiling; no column bound for the unperturbed map alone is substituted.

Applying the generated-increment comparison of Theorem 20.13 with the incoming $\Delta\Lambda_t = \Lambda_t - \Lambda_t^{\mathbb{T}}$, we obtain the increment in part (4), including the initial freezings. To check the sign of the dimension term, note that one scalar Gaussian coordinate integrated in the corresponding renormalization group recursion on the whole torus contributes a factor g_i. When that coordinate is absent from the regional free list, the logarithmic difference between the regional and torus terms is $-\log g_i = \log(1/g_i)$. Summing over the missing scalar coordinates, we obtain $d_i(H_t)\log(1/g_i)$, once for each scale. By that theorem, the remaining generated differences of Haar disintegration factors, minimizers, and cluster expansions, and the freezings, are bounded by C_J times the three label counts, taken on the lattices where the labels were created. This comparison is additional analytic information: neither the exact disintegration, nor (21.37), nor normalization would by itself bound these generated differences. The factor A_t remains separate and is not added again to Λ_t.

The remaining averaging steps. Let $a = \ell_T - \ell_0 = JT$ be the number of integrated averaging steps (here a is not a mesh), and let $\ell_{\max}$ be the last level of the given finite hierarchy. A remaining number $0 \le u_{\mathrm{ph}} < J$ of averaging steps is chosen only when it is available: at the stopping scale we require

$$m_{\mathrm{stop}} = M^{u_{\mathrm{ph}}} m_{\mathrm{rem}}, \qquad m_{\mathrm{rem}} \in 4\mathbb{N}, \qquad \ell_T + u_{\mathrm{ph}} \le \ell_{\max}.$$

For a fixed additional number $r_{\mathrm{stop}} \in \mathbb{N}_0$ of RG steps, we set $v = r_{\mathrm{stop}} J + u_{\mathrm{ph}}$ and require

$$m_{\mathrm{stop}} = M^v m_{\mathrm{rem}}, \qquad m_{\mathrm{rem}} \in 4\mathbb{N}, \qquad \ell_T + v \le \ell_{\max};$$

the case of u_{ph} alone is $r_{\mathrm{stop}} = 0$. For each $0 \le k \le v$, the intermediate side is $M^{v-k} m_{\mathrm{rem}} \in 4\mathbb{N}$, which is at least four. The initial side is $M^a m_{\mathrm{stop}}$, and the bound on the index keeps every intermediate level in the given hierarchy. Hence both T_M^v at the stopping scale and T_M^{a+v} at the initial scale exist; a lower bound on m_{stop} alone would not give these integer conditions. We work on whole configurations, before regional coordinates are introduced. The block averaging map with its Borel extension is Borel, and its iterates are compositions of that map, so for every initial configuration V, we have

$$F(T_M^{a+v} V) = (F \circ T_M^v)(T_M^a V),$$

and the function at the stopping scale is exactly $C \mapsto F(T_M^v C)$, in particular $C \mapsto F(T_M^{u_{\mathrm{ph}}} C)$ for u_{ph} alone. We define the measure $\mu_{a,\mathcal{X}} = (T_M^a)_{\#}(\mathbf{1}_{\mathcal{A}_{\mathcal{X}}} \nu)$ on the whole configurations at the stopping scale. For every nonnegative Borel F we have

$$\int \mathbf{1}_{\mathcal{A}_{\mathcal{X}}}(V) F(T_M^{a+v} V)\, d\nu(V) = \int (F \circ T_M^v)(C)\, d\mu_{a,\mathcal{X}}(C), \tag{21.41}$$

by the definition of the pushforward; the identity extends to integrable F through positive and negative parts. Each fixed additional number of steps uses the total exponent v whose existence was just checked, together with its finite enlarged dependence neighborhood. For the local link and plaquette observables of Theorem 21.5, these iterates use its already enlarged final dependence neighborhood. The decomposition keeps this observable on each piece. There is no Haar disintegration, change of variables, restoration of bounds, or update of normalization for v. When the integrated part of the decomposition is empty, the identity is the initial pushforward itself.

Part (5): external classical factors and failure indicators. At fixed frozen coordinates, let $S_{\mathrm{ext}}(b_{\mathrm{rec}})$ be the sum of the protected external classical exponents with their original couplings; for the Wilson measure it includes $E_{\mathrm{hole}}/g_{\mathrm{base}}^2$. The term $g_t^2 S_{\mathrm{ext}}$ is constant in all integration variables, so adding it to the full action of the block region changes no minimizer, Hessian, map, or nonconstant norm. We have the exact identity

$$e^{-S_{\mathrm{ext}}} e^{-E^{\mathrm{int}}/g_t^2} = e^{-(E^{\mathrm{int}} + g_t^2 S_{\mathrm{ext}})/g_t^2}.$$

We delete this external factor from W_t when the full classical action including the frozen-coordinate terms is used. After an RG step, the corresponding constant becomes $M^{s_t} g_t^2 S_{\mathrm{ext}} = g_{t+1}^2 S_{\mathrm{ext}}$, so its exponent is preserved exactly. At each classical freezing, the full value was kept. Induction over the finite decomposition shows that each frozen-coordinate term of the full classical action has one copy in the output action and none in

the remaining weight of the frozen coordinates. The constants from the freezing of the effective interaction remain separate exact terms of Λ.

For each failing plaquette f, the whole dependence set was frozen at the input of its RG step. Its value q_f is therefore a Borel function of the frozen coordinates on that lattice, computed with the same path products of the block averaging map, and later operations fix its marked data exactly. Its trace is invariant under the consistent endpoint action. We keep the product of all indicators $\{q_f \geq d_*\}$ as $I_{\mathcal{X}}^{\mathrm{act}}$, and any chosen invariant tests depending only on the frozen coordinates at the output as $I_{\mathcal{X}}^{\mathrm{inv,end}}$. Every other optional coordinate restriction is between zero and one and may be removed in an upper bound. After the classical recombination, this leaves exactly the measure on frozen coordinates in part (5), which is bounded by its product Haar measure. It contains no second copy of an energy that could be used for a lower bound on the output action or for an energy bound on failing plaquettes.

Every local bound has the dependence stated at the beginning of this section on the fixed larger domains, and only C_δ has the dependence of item (4) of Definition 4.14. □

SECTION 22

Exponential-moment bounds between renormalization scales

In this section, we prove exponential-moment bounds for sums of plaquette energies of block-averaged configurations. The bounds are uniform in the number of averaging steps between the lattice on which the measure is given and the lattice on which the energies are evaluated. The main result is Theorem 22.1. Under its hypotheses, fix a coupling ceiling $\Gamma > 0$ and take the smallest side $u \geq n_{\min}(\Gamma)$. Let $b \in J\mathbb{N}_0$ with $0 \leq b \leq j-1$, and let ν be either the Wilson measure on the lattice of side n_b or a normalized stopped measure there at a preceding depth in $J\mathbb{N}_0$. The theorem states that the exponential moments under ν of the sum $Q_j^{\mathcal{T}}$ of the energies of the predecessor plaquettes of a chessboard tiling, (22.2), are bounded by $\exp\{3C_v n_{j-1}^3 + C_s n_{j-1}^2 \log(1/g_{n_{j-1}}) + C_b\}$. The constants are independent of $\gamma \in (0, \Gamma]$, of the lattice sides, and of $j-1-b$, and the expectation contains no small-field or regularity indicator. We also prove a chessboard estimate in which the regularity indicators stay inside the root (Lemma 22.11). Finally, we compare the Wilson measure, restricted to the first a conditions of the small-field event, with the stopped measures; this bounds the probability that the next condition fails, without conditioning on later scales (Lemmas 22.12 and 22.13). The inputs are the large-field decomposition and the density bounds of Section 21 (Theorems 21.5, 21.9, and 21.12); the regional integration estimates of Section 20 (Theorems 20.13 and 20.14 and Lemma 20.5); the classical lower bound of Theorems 11.2 and 11.6; the energy transport and predecessor geometry of Theorem 7.3 and Lemma 7.4; the reference normalization of Theorem 14.8; the chessboard estimate of Theorem 6.6; and, for the comparison of restrictions, Theorems 7.7 and 14.7. The results are used in Section 23 to prove the bound on large fields, Theorem 2.7.

Section 22.1 states the main theorem and outlines its proof, Sections 22.2 to 22.6 prove the estimates for one large-field decomposition, Section 22.7 completes the proof, and Sections 22.8 and 22.9 contain the chessboard estimate and the comparison of restrictions.

22.1 Setting and the main estimate

Scales and couplings. Fix $L > 0$, an odd integer $M \geq 3$, an integer $J \geq 1$, and $B = M^J$. For $u \in 4\mathbb{N}$ and $K \in \mathbb{N}_0$ let $N = uB^K$, $n_i = N/M^i$, and $V^i = T_M^i U$ for $0 \leq i \leq JK$, where T_M is the block averaging map of Definition 4.3, including its Borel extension at

the exceptional values. Increasing i moves toward the infrared, and all the sides n_i are divisible by four. One application of T_M is an averaging step, and J consecutive averaging steps form an RG step. The levels $i \in J\mathbb{N}_0$ are the boundaries of RG steps. At side n the coupling is $h_n = g_n^2 = \gamma/n$, as in Definition 4.1, so that $h_{n_{i+1}} = M h_{n_i}$. All Haar measures in this section are probability measures.

Plaquette energies. For $A \in \mathrm{SU}(2)$ let $\rho_G(A) \in [0, 2\pi]$ be the distance from A to I in the bi-invariant metric of Section 4, and recall that $q(A) = 1 - (\operatorname{tr} A)/2$. On the whole group,

$$\frac{\rho_G(A)^2}{2\pi^2} \le q(A) \le \frac{\rho_G(A)^2}{8}, \qquad 0 \le q(A) \le 2. \tag{22.1}$$

Indeed, if $t = \rho_G(A)/4 \in [0, \pi/2]$, then $q(A) = 1 - \cos(2t) = 2\sin^2 t$ by (4.2), and $2t/\pi \le \sin t \le t$. For a set $\mathcal{P}$ of positive plaquettes at level i and a configuration V on the lattice of side n_i we write

$$\mathcal{S}_i^{\mathcal{P}}(V) = \frac{4}{h_{n_i}} \sum_{p \in \mathcal{P}} q(V_p).$$

If $\mathcal{P}$ is the set of all positive plaquettes, then $\mathcal{S}_i^{\mathcal{P}}(V) = S_W(V)/h_{n_i}$ is the Wilson exponent.

The plaquette-energy sum. For a set $\mathcal{A}$ of positive plaquettes at level $i+1$, let $\mathrm{Pa}(\mathcal{A})$ be the union, counted once, of the plaquettes at level i in the complete predecessor rectangles of the elements of $\mathcal{A}$ (Theorem 7.3). Let $1 \le j \le JK$. For each of the eight tilings $\mathcal{T}$ of the level-j lattice into cells of $2 \times 2 \times 2$ sites in Lemma 7.4, and each cell $C \in \mathcal{T}$, let $\mathcal{P}_{j-1}(C)$ be the union of the complete predecessor rectangles of the six internal level-j plaquettes of C, and let $\mathcal{P}_{j-1}^{\mathcal{T}} = \bigcup_{C \in \mathcal{T}} \mathcal{P}_{j-1}(C)$, counted once. By Lemma 7.4, $|\mathcal{P}_{j-1}(C)| \le 24M^3$. For a configuration V on the lattice of side n_{j-1} we set

$$Q_j^{\mathcal{T}}(V) = \frac{4}{h_{n_{j-1}}} \sum_{p \in \mathcal{P}_{j-1}^{\mathcal{T}}} q(V_p) = \mathcal{S}_{j-1}^{\mathcal{P}_{j-1}^{\mathcal{T}}}(V). \tag{22.2}$$

This is the function that enters the chessboard estimate of Section 22.8.

Initial measures. Every estimate of this section starts from a probability measure ν on the configurations of some side n_b of the hierarchy, which we call the initial measure. It is one of the following.

1. The Wilson measure $\nu = \mu_{n_b,\gamma}$, with partition function $Z_\nu = Z_{n_b,\gamma}$. It is not restricted to D_r.
2. A normalized stopped measure at a depth $d_0 \in J\mathbb{N}_0$,

$$d\nu = Z_\nu^{-1} \mathbf{1}_{D_r} e^{-E_{d_0,\mathrm{sat}}/h_{n_b} - R_{d_0,\mathrm{sat}}}\, dH_{n_b}, \qquad Z_\nu = \int_{D_r} e^{-E_{d_0,\mathrm{sat}}/h_{n_b} - R_{d_0,\mathrm{sat}}}\, dH_{n_b},$$

as in Definition 4.4 and Theorem 14.6. Here $E_{d,\mathrm{sat}}$ is the extension of the effective classical action E_d to D_ρ by gauge invariance (Theorem 11.6), and $R_{d,\mathrm{sat}}$ is the corresponding effective interaction. The partition function Z_ν is not centered: the value $R_{d_0}(I)$ is kept in it.

Large-field decompositions. We use the partition of Theorem 21.5. Given an initial measure ν and a stopping level, the support of ν is partitioned into finitely many Borel sets $\mathcal{A}_\mathcal{X}$, indexed by large-field decompositions $\mathcal{X}$. A decomposition is produced by testing the plaquettes of the iterates $T_M^\ell V$ in increasing order of the level ℓ, and in a fixed order within one level. A plaquette f at level ℓ_f is a recorded bad plaquette of $\mathcal{X}$ if $q_f \geq d_*$ and its full dependence set avoids the set excluded before it. The variables of that dependence set that are not yet frozen are then frozen at the beginning of the RG step containing the level ℓ_f, together with a neighborhood, the excluded neighborhood of f, which is enlarged at each later RG step. We write $n_f = n_{\ell_f}$ for the side on which f is recorded and $x_f = n_f/\gamma$. Away from the excluded regions, the renormalization group is run on block regions, in RG steps of J averaging steps each, from the initial level to the stopping level; both levels are boundaries of RG steps. At the stopping level, of side m, the decomposition leaves a stopping field $c \in \mathrm{SU}(2)^F$ on the set F of links of the block region at the stopping level. We write $H = \mathcal{E}_m \setminus F$ for the other links of the stopping lattice, including the links across the boundary of the block region. For $z \in \mathrm{SU}(2)^H$, we write $c \oplus z$ for the configuration on the stopping lattice that equals c on F and z on H, and we call it a completion of c. The frozen coordinates b_{rec} consist of all frozen group variables, each kept on the lattice of the scale at which it was frozen. The map $V \mapsto \mathscr{O}_T^\mathcal{X}(V) = (c, b_{\mathrm{rec}})$ records the stopping field and the frozen coordinates. For a nonnegative function Q and $s \geq 0$, we write

$$\eta_\mathcal{X} = (\mathscr{O}_T^\mathcal{X})_\#(\mathbf{1}_{\mathcal{A}_\mathcal{X}}\nu), \qquad \eta_\mathcal{X}^s = (\mathscr{O}_T^\mathcal{X})_\#(e^{sQ}\mathbf{1}_{\mathcal{A}_\mathcal{X}}\nu).$$

For a Wilson initial measure, the first RG step of Theorem 21.5 separates the frozen group variables adjacent to the block region from those deep inside the excluded regions. The latter are arbitrary elements of $\mathrm{SU}(2)$; they are never inserted into an analytic function and enter only through an exact Wilson factor.

The analytic estimates are applied at gauge representatives of the stopping field and the frozen coordinates that satisfy the domain conditions and the reconstruction identities of Theorem 21.12. They are obtained from the representatives with small frozen coordinates at the stopping level by applying the reconstruction maps of the successive RG steps, and they also satisfy the identities for the insertion of frozen coordinates stated there. Below, we call them representatives satisfying the reconstruction identities.

Table 22.1 lists the measures that occur in the proof of Theorem 22.1, with their total masses and normalizations. The measures $\eta_\mathcal{X}$, $\eta_\mathcal{X}^s$, and ξ_T are not renormalized; they keep the partition function Z_ν of the initial measure.

Table 22.1. Measures used in the proof of Theorem 22.1.

Measure	Space	Total mass	Normalization
Wilson initial measure $\mu_{n_b,\gamma}$	$\mathcal{U}_{n_b}$	1	$Z_{n_b,\gamma}$
Stopped initial measure $\eta_{\mathcal{X}}$	$D_r \subset \mathcal{U}_{n_b}$ stopping fields and frozen coordinates	1 $\nu(\mathcal{A}_{\mathcal{X}}) \le 1$	Z_ν, not centered that of ν
$\eta^s_{\mathcal{X}}$	same as $\eta_{\mathcal{X}}$	$\mathbb{E}_\nu[e^{sQ}; \mathcal{A}_{\mathcal{X}}]$	that of ν
Reference measure ξ_T (Theorem 22.4)	$D_r \subset \mathcal{U}_m$	$m_T^{\mathbb{T}} \le 1$	prefactor $Z_\nu^{-1} e^{\Lambda_T^{\mathbb{T}}}$
Comparison measure $\widetilde{\sigma}_{\lambda,d,g}$ (Theorem 22.10)	$\mathcal{U}_m$	finite	$\widehat{Z}_d(g)$, an integral over D_r
Frozen-coordinate measure $\rho_{\mathcal{X}} = w_{\mathcal{X}} H_{\mathcal{X}}$	frozen coordinates	≤ 1	Haar

Dependence of constants. The constants of the local analytic estimates have the dependence described in item (3) of Definition 4.14. They depend only on M, J, the larger construction domains and their radius ratios, the spatial and boundary decay rates, the density of variables, and the norm bounds. They are independent of L, γ, the lattice sides, the depth, the number of RG steps since a frozen coordinate was created, the shape and topology of the block regions, and the real values of the frozen coordinates in their domain. Where stated, they are also independent of the widths of the boundary layers and excluded neighborhoods chosen later; these widths enter only through explicit counts. The constants in the probability estimates have the dependence described in item (4) of Definition 4.14. They may in addition depend on L, a fixed coupling ceiling, the fixed parameters, and the finite family of stopping offsets, but not on the coupling below the ceiling, the sides, the depth, or the geometry, age, and frozen values of a decomposition. We claim no useful bound as $J \to \infty$.

Thresholds. Fix $d_{\mathrm{cut}} > 0$ and set

$$d_* = d_{\mathrm{cut}}/2, \qquad c_E = 1 + C_M \pi \sqrt{2 d_{\mathrm{cut}}},$$

where δ_* and C_M are the constants of Theorem 22.4(a). For the chessboard estimate of Lemma 22.11 we assume $\pi\sqrt{2d_{\mathrm{cut}}} \le \delta_*$. In Theorem 22.1, these numbers enter only through the range of s and the definition of the recorded bad plaquettes.

The main estimate. The constants $K_M, K_1, \kappa_*, \kappa_0, c_A^{\mathrm{act}}, \alpha_W$ in the following theorem are defined in Theorem 22.4; they depend only on M and on the fixed classical data.

Theorem 22.1. *Fix $L > 0$, an odd integer $M \geq 3$, an integer $J \geq 1$, and $B = M^J$, and fix parameters satisfying the hypotheses of Theorems 20.14, 21.5, and 21.12. These include the coefficient-norm bounds at initialization, both substitution bounds* (22.6)*, the boundary-distance comparison, and the product domains verified in Lemma 22.3. The cutoff amplitude A_{w} is fixed before the coupling ceiling $g_* < 1$. Fix a finite nonempty set $\mathscr{R} \subset \mathbb{N}_0$ of stopping offsets, all remainders $0 \leq u_{\mathrm{ph}} < J$, the buffers containing their full dependence sets, and the bound F_J of Theorem 21.5 for the number of affected source plaquettes. Fix $\chi_* \geq 1$ such that the ratio of every stopping side of this family to the requested side n_{j-1} is at most χ_*, and such that $n_b/n_{j-1} \leq \chi_*$ whenever the stopping level lies at or below the initial level. Every iterate of T_M that is used must belong to the given finite hierarchy, and only full RG steps are integrated.*

Let $K_M, K_1, \kappa_*, \kappa_0, c_A^{\mathrm{act}}, \alpha_W$ be as in Theorem 22.4, and put $K_{\mathrm{W}} = 48K_1/\alpha_{\mathrm{W}}$ and $\theta = 1/2$. Choose $s_0 > 0$, after F_J, such that

$$s_0 \leq \min\Big\{\frac{1}{2},\ \frac{\kappa_*}{2}c_E^{-(J-1)},\ \frac{1}{4K_W},\ \frac{\theta c_A^{\mathrm{act}} d_*}{16F_J},\ \frac{\kappa_0}{2K_M}\Big\}.$$

Any further fixed decrease of s_0 is allowed. Then for every $0 < \Gamma < \infty$ there exist $n_{\min} \in 4\mathbb{N}$ and finite nonnegative constants C_v, C_s, C_b with $C_s \geq 9\chi_*^2$, chosen before the coupling, the side, and the depth, such that the following holds.

Let $0 < \gamma \leq \Gamma$, $u \in 4\mathbb{N}$ with $u \geq n_{\min}$, $K \in \mathbb{N}_0$, $N = uB^K$, $n_i = N/M^i$, $1 \leq j \leq JK$, and $b \in J\mathbb{N}_0$ with $0 \leq b \leq j-1$. Let ν be either the Wilson measure $\mu_{n_b,\gamma}$ or a normalized stopped measure on $\mathcal{U}_{n_b}$ at a depth aJ, $a \in \mathbb{N}_0$. Then, for each of the eight tilings $\mathcal{T}$ and every $0 \leq s \leq s_0$,

$$\log \mathbb{E}_\nu \exp\{sQ_j^{\mathcal{T}}(T_M^{j-1-b}V)\} \leq 3C_v n_{j-1}^3 + C_s n_{j-1}^2 \log(1/g_{n_{j-1}}) + C_b.$$

This covers every remainder u_{ph}, stopping levels at or below the initial level, an empty block region at the stopping level, and decompositions whose block region becomes empty before the stopping level and which are continued by empty RG steps.

More precisely, a recorded bad plaquette f created at side n_f contributes at least $P_0 x_f$ to the classical part of the exponent, where $x_f = n_f/\gamma$ and $P_0 = 4\theta c_A^{\mathrm{act}} d_*$. The plaquette-energy sum costs at most $8sF_J x_f \leq P_0 x_f/4$ per recorded bad plaquette, and all the corrections assigned to f, from the normalization constants, the missing coordinates, the boundary bands, the substitutions, and the integration over the omitted links, are at most

$$C_{\mathrm{ch},J}^{\Gamma}(1 + \log x_f)^2 \leq P_0 x_f/4.$$

Here one may take $C_{\mathrm{ch},J}^{\Gamma} = (1 + \log_+ \Gamma) C_{\mathrm{ch},J}^{\mathrm{base}}$, where $C_{\mathrm{ch},J}^{\mathrm{base}}$ depends only on the fixed parameters and the family of stopping offsets. Each decomposition retains the factor $\prod_f e^{-a_{\mathrm{act}} n_f/\gamma}$, with $a_{\mathrm{act}} = P_0/2$, and the sum of these factors over all decompositions is at most

$$\exp\Big\{\sum_\ell 3n_\ell^3 e^{-a_{\mathrm{act}} n_\ell/\gamma}\Big\},$$

which is bounded uniformly in the depth once $u \geq n_{\min}$. At a stopping side $m = \chi n_{j-1}$ one has $g_m = g_{n_{j-1}}/\sqrt{\chi}$, and the only surface term is $9\chi^2 n_{j-1}^2 \log(1/g_{n_{j-1}})$.

The constants $n_{\min}, C_v, C_s, C_b$ have the dependence described in item (4) of Definition 4.14, with the fixed ceiling Γ, the fixed parameters, and the finite family of stopping offsets and remainders. They are independent of the coupling $\gamma \in (0, \Gamma]$, the sides, the depth, the position and the length $j - 1 - b$ of the range of scales, and the age, shape, topology, and frozen values of the decompositions. The constants of the local analytic estimates keep the dependence described in item (3) of Definition 4.14; the widths enter only through the displayed counts.

Remark 22.2. There is no restriction on $j - b$, and the expectation contains no indicator of a cell condition or of a condition of the small-field event. Taking $b = 0$ and $\nu = \mu_{N,\gamma}$ gives the estimate under the Wilson measure with its own partition function. The proof does not use the bounds on the Wilson measure at successive scales of Section 23, where the theorem is used in Theorems 23.2 and 23.6.

Outline of the proof. Fix j, b, and $\mathcal{T}$, and write $n = n_{j-1}$. If the planned stopping level lies at or below the initial level, then only a bounded number of averaging steps separates the initial lattice from level $j - 1$, and the estimate follows from the partition-function bounds on the initial lattice. Otherwise we integrate full RG steps from level b to a stopping level $t_{\text{stop}} \leq j - 1$ and keep the remaining $v = j - 1 - t_{\text{stop}}$ averaging steps inside the function $Q_j^{\mathcal{T}} \circ T_M^v$.

Step 1: classical lower bounds. Along a decomposition $\mathcal{X}$, the density of $\eta_{\mathcal{X}}$ at a representative satisfying the reconstruction identities is bounded by e^{-E_T/g_m^2} times normalization factors, where E_T is the effective classical action of the stopping field with all external classical exponents included (Theorem 21.12). We reconstruct one configuration U_* on the initial lattice with zero fluctuations. Using an energy transport estimate that is uniform in the number of averaging steps, we show that E_T/g_m^2 dominates both $\Psi_F(c)/g_m^2$, where $\Psi_F(c)$ is the minimum of $\alpha_W S_W(c \oplus z)$ over all values z of the links in H, and $A_{\mathcal{X}} = c_A^{\text{act}} \sum_f 4q_f/h_{n_f} \geq 4c_A^{\text{act}} d_* \sum_f x_f$ (Theorem 22.4). Since both lower bounds are gauge invariant, the density bound extends to all points (Theorem 22.10).

Step 2: the plaquette-energy sum. At every point of $\mathcal{A}_{\mathcal{X}}$, the plaquette-energy sum is at most $K_W \Psi_F(c)/g_m^2 + 8F_J \sum_f x_f$ (Lemma 22.8). Since $sK_W \leq 1/4$, the factor e^{sQ} is absorbed by the factor $e^{-\Psi_F(c)/(2g_m^2)}$, which leaves at least $e^{-\Psi_F(c)/(4g_m^2)}$. The minimum is then replaced by a Haar integral over the links in H, which adds $3|H| \log(1/(c_* g_m))$ to the exponent (Lemma 22.7).

Step 3: summation over decompositions. Each recorded bad plaquette contributes $-P_0 x_f$ to the exponent. All normalization corrections, boundary terms, substitution terms, and the correction from the integration over the omitted links can be assigned to recorded bad plaquettes, and the corrections assigned to f are at most $C(1 + \log x_f)^2$. For a large minimum side, they are absorbed by $P_0 x_f/4$. This leaves a factor $e^{-P_0 x_f/2}$ per recorded

bad plaquette, and the sum over decompositions converges uniformly in the depth. The remaining ratio of a Wilson integral to the reference partition function on the stopping lattice is at most $e^{Cm^3+9m^2\log(1/g_m)}$ (see Remark 22.5).

The number of ultraviolet scales causes no accumulated loss, for the following reason. A recorded bad plaquette created at side n_f has $q_f \geq d_*$, and therefore contributes at least a fixed multiple of $x_f = n_f/\gamma$ to the classical action. At each later RG step, its excluded neighborhood contains a number of variables bounded independently of the scale, and there are at most $O(1+\log n_f)$ later RG steps. Each missing Gaussian coordinate changes the normalization by $\log(1/g_i) = \log(n_i/\gamma)/2 \leq (\log x_f)/2 + O(1)$, and each other correction is bounded per variable. The total correction is therefore $O_\Gamma((1+\log x_f)^2)$, which is small compared to x_f once x_f is large.

We distinguish two sets attached to a recorded bad plaquette f; see Figure 22.1. The first is its energy-estimate predecessor set $\mathcal{B}_f$ on the initial lattice, which carries the classical lower bound for q_f. These sets are disjoint for recorded bad plaquettes at different levels and overlap at most 48 times at one level. The second is the full excluded neighborhood assigned to f, which is much larger and may overlap with the neighborhoods of other recorded bad plaquettes. Source plaquettes whose full dependence sets meet such a neighborhood are discarded, and their number per recorded bad plaquette is bounded by F_J.

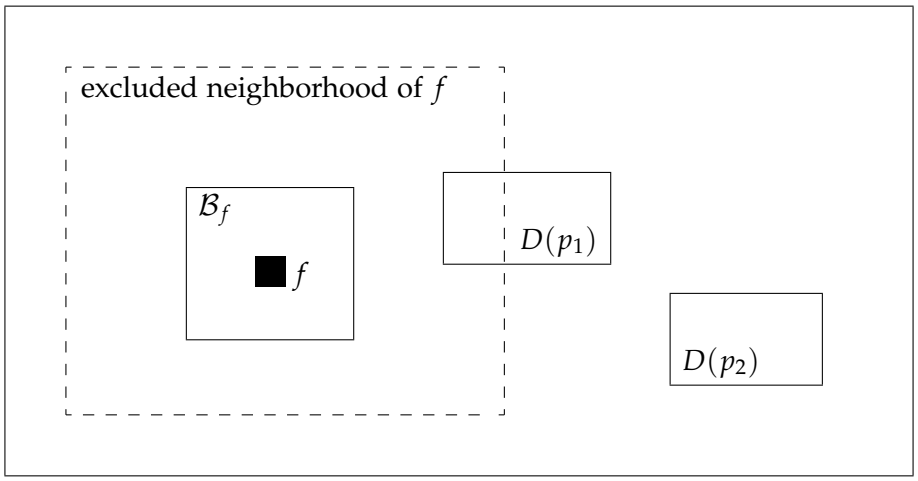


Figure 22.1. Schematic picture of the sets attached to a recorded bad plaquette f, drawn at one scale. The energy-estimate predecessor set $\mathcal{B}_f$ carries the lower bound for the action. The source plaquette p_1 is discarded because its full dependence set $D(p_1)$ meets the excluded neighborhood of f; the source plaquette p_2 is kept because $D(p_2)$ consists only of links of the block region at the stopping level.

22.2 Product domains along a large-field decomposition

The following lemma checks that the regional integration estimates of Section 20, on which the density bounds of Theorem 21.12 rest, apply along an arbitrary large-field decomposition. Only the small frozen coordinates are substituted into analytic functions (Theorems 21.5 and 21.12).

Lemma 22.3. *Fix $A_{\mathrm{w}} \geq 1$ first, and then fix parameters satisfying the hypotheses of Theorems 20.14 and 21.5, including the product domains and the coefficient-norm bounds at initialization in Theorem 21.12. Let $\mathcal{X}$ be a large-field decomposition whose initial and stopping levels are boundaries of RG steps, with T integrated RG steps. Then every integrated step consists of J averaging steps, the total number of integrated averaging steps is JT, and Theorems 20.13 and 20.14 and Lemma 20.5 apply at the gauge representatives of $\mathcal{X}$ satisfying the domain conditions and reconstruction identities of Theorem 21.12, with the frozen coordinates satisfying the bounds of Theorem 21.5. In each such application the fluctuation cutoff domain is*

$$\mathcal{W}_t = \prod_{i=0}^{J-1} \prod_{e} \{|w_{i,e}| \leq p(t_t)\}, \qquad t_t = \max_i g_{t,i}, \qquad p(t) = A_{\mathrm{w}} \log(1/t), \tag{22.3}$$

where $g_{t,i}$ is the coupling of the i-th averaging step of the t-th RG step. The coupling ceiling may depend on the fixed amplitude A_{w}; this does not change the constants of the local analytic estimates or the bounds that are uniform before J is chosen. The conclusion is uniform in the number of RG steps since the creation of the frozen coordinates and in the real values of the frozen coordinates in their stated domain, including the first step for a Wilson initial measure. No derivative with respect to a frozen coordinate is involved.

Proof. The cutoff amplitude is fixed before the coupling ceiling of Theorem 20.14 is chosen. By Theorem 21.12, every integrated step has J averaging steps, and (22.3) holds. If m_t is the output side of an RG step, then

$$t_t^2 = \gamma/(Mm_t), \qquad g_{t,i} = t_t M^{(i-J+1)/2} \quad (0 \leq i < J).$$

In particular, the radii $p_{t,i} = p(t_t)$ used in the small-field event belong to the family of radii allowed in Theorem 20.14,

$$p(t_t) \leq p_{t,i} \leq p(t_t) + c_J^{\log}, \qquad c_J^{\log} = \tfrac{1}{2} A_{\mathrm{w}} (J-1) \log M.$$

The larger enclosing product domain is used only in the comparison of Lemma 20.5; its radii do not replace the radii $p(t_t)$.

We now check the conditions on the coupling for this logarithmic family, which contains the common radius as a special case. Let $p^+(t) = p(t) + c_J^{\log}$. The condition that the cutoff domain lies in the enclosing product domain is $tp^+(t) < R_{\mathrm{en}}$. With the local

constants of Theorem 20.14, and with $E + \sigma_*$ for the norm bound of the input effective interaction, the small-interaction condition and the condition on the auxiliary complex parameter domain required there read

$$C_J t p^+(t) \le \tfrac{1}{2}, \qquad \frac{C_J t[(p^+(t))^3 + p^+(t) + (E+\sigma_*)p^+(t)]}{1 - C_J t p^+(t)} \le \epsilon_J, \qquad 2t[(p^+(t))^3 + p^+(t) + (E+\sigma_*)p^+(t)] \le c_{z,J}. \tag{22.4}$$

These are the expansion conditions of Theorem 20.14, and by the same choice of parameters the denominators of its rational image bounds are positive on the outer product domains. With $v = \log(1/t)$, every term of the form $t^k(p^+(t))^q$ with $k > 0$ equals $e^{-kv}(A_{\mathrm{w}} v + c_J^{\log})^q$, which tends to zero as $v \to \infty$. Hence one fixed ceiling on t gives the containment, all the inequalities (22.4), and the required smallness of the perturbative term

$$C_J t^2 (E+\sigma_*)\big[(p^+(t))^4 + (E+\sigma_*)(p^+(t))^2\big].$$

This ceiling is one of the fixed parameters. Since $g_* < e^{-1}$ and $A_{\mathrm{w}} \ge 1$, the radii $p(t_t)$ are at least one. At each fixed value of the frozen coordinates, the bound for the conditional mean in Lemma 20.5 stays uniformly below the cutoff radius: the ratio of the mean bound to $p(t)$ is at most $C_1 t(\mathcal{L}_V + 2)/p(t)$, which tends to zero. Together with the containment, this verifies the hypotheses of the tail estimate of Lemma 20.5 on the chosen cutoff domain.

We also verify the measure-theoretic hypotheses of that comparison, at every fixed real value of the frozen coordinates and every fixed coarse background. The enclosing product domain $\mathcal{K}_t$ is a finite product of closed balls of positive radius in $\mathbb{R}^3$, so it is Borel, compact, and of finite positive Lebesgue volume. The conditional fluctuation potential V given by Theorem 20.14 is real and of class C^2 on a neighborhood of $\mathcal{K}_t$, because the image bounds keep its inverse maps, Jacobians, and positive Hessian determinants within their domains. Thus there exist finite real numbers $v_- \le v_+$, which may depend on the fixed background and frozen coordinates, such that $v_- \le V \le v_+$ on $\mathcal{K}_t$. In particular,

$$0 < e^{-v_+}|\mathcal{K}_t| \le Z_{\mathcal{K}_t} = \int_{\mathcal{K}_t} e^{-V}\, dw \le e^{-v_-}|\mathcal{K}_t| < \infty. \tag{22.5}$$

The set S of supported fluctuation coordinates is the inverse image under a Borel coordinate map of the finitely many tests of the decomposition and of the coordinate domains, and the cutoff domain $\mathcal{W}_t$ is defined by finitely many inequalities; both are Borel. Every indicator or optional test $0 \le \chi \le 1$ that is kept is Borel, and e^{-V} is positive and Borel. The Haar disintegration, Gaussian, and constant prefactors are identical on both sides of the comparison. Since these checks are made at each fixed preceding output point and fixed frozen coordinates, no uniform bound on $v_\pm$ is needed. If there are no fluctuation

blocks, the product domain is a point of mass one; its partition function is positive and finite, its determinant is one, and its logarithmic determinant is zero.

For the substitution of frozen coordinates into analytic functions, Theorem 21.5 includes the initial bound for finite products of links and all conversion constants in $C_{\mathrm{conv},J} \geq C_{\mathrm{ad},J}$. At the representatives of the lemma, we have, at each scale at which variables are frozen and for both spatially varying radii $\phi_{\mathrm{cl}}, \phi_{\mathrm{q}}$,

$$|c_j^\sharp| \leq \min\{\beta_{\mathrm{rec}}, \phi_\bullet(j)/4\}, \qquad C_{\mathrm{bc},\bullet} E_\bullet \beta_{\mathrm{rec}}/\rho_\bullet \leq \sigma_*/4, \qquad \bullet \in \{\mathrm{cl}, \mathrm{q}\}. \tag{22.6}$$

For a frozen group variable b_j kept on the lattice where it was frozen, we have $c_j^\sharp = \operatorname{Log} b_j$ in the identity chart, and hence $|c_j^\sharp| = \rho_G(b_j)$. A longer fixed product of links uses the conversion included in $C_{\mathrm{conv},J}$, which is fixed before β_{rec} and r. At initialization, for a stopped initial measure, we use the bound by r coming from D_r, together with $C_{\mathrm{ad},J} r < \beta_{\mathrm{rec}}$; for a Wilson initial measure, the bound for the frozen variables adjacent to the block region is the other initialization clause of Theorem 21.5. The bound (22.6) covers all frozen variables, including the boundary band, the boundary neighborhoods, the links across the boundary of the block region, and the initialization, and its constant $C_{\mathrm{bc},\bullet}$ includes the sum over all these types. The arbitrary frozen group variables deep inside excluded regions are fixed external parameters and are not inserted into analytic functions.

All variables in the band of reduced analytic radius are frozen. Thus every variable that remains in the incoming block region $D_{\mathrm{in},t}$ has the classical radius ρ_{cl}, and the incoming norm of the classical perturbation in the coefficient-norm bounds is evaluated at that radius; the spatially varying radius is used only for the output. By Theorem 21.5, the old fixed complement and the points of creation of frozen variables remain in the fixed complement of the next block region, up to the fixed buffer of the finite link products. In the current units, the incoming and current distances to the fixed complement satisfy

$$d_{D_{\mathrm{in},t}}(i) \geq B d_{\mathrm{current}}(e) - B \operatorname{dist}(e, i/B) - C_{\mathrm{buf},J},$$

where i is the point of creation of an incoming variable, e is a current point, and $C_{\mathrm{buf},J}$ is the fixed buffer constant. This is the boundary-distance comparison required in Theorem 21.12 (compare Theorem 19.9); enlarging the fixed complement preserves the inclusion, and at initialization there is no boundary. The uniform bound on the conditional force kept in that induction is the following condition on the same product domains:

$$\begin{gathered} \delta_{V,t} = \max_I \sup_{y \in \mathcal{T}_I} \left\| A_I^{-1} V_I(c, a_I(c, y; \xi), y) \right\|_{\mathrm{coeff}}, \\ C_{\mathrm{seq},J}(2\delta_{V,t} + r_q) < \tfrac{1}{2}\varepsilon_{\mathrm{tube}}. \end{gathered} \tag{22.7}$$

Here I runs over the finitely many initial segments of the conditional integrations, A_I is the positive reference block of the segment I, a_I is its conditional constrained minimizer

without perturbation, V_I is the z_I-derivative, in the original free coordinates, of the added force V of (21.12), $\mathcal{T}_I$ is the domain of the remaining parameters, and r_q is the fixed radius of the free coordinates. The coefficient norm includes the common product domain of the coarse fields, uniformly at fixed real frozen coordinates. The condition (22.7) is a hypothesis on the product domains in Theorem 21.12 (see Lemma 19.3), uniform on the prescribed ball of classical perturbations; it is part of the fixed parameters and is verified in Section 23, see (23.29). The supremum is over every remaining parameter used in the sequential construction, not only over the point with zero fluctuations, and all real frozen coordinates are held fixed. Hence the image bounds and the sequential containment used in the induction hold on their stated domains.

Finally, $x_0, x_0^{\mathrm{cl}} \le \sigma_*$ are the coefficient-norm hypotheses of the induction at initialization, including the substitution at initialization. Its constants are those of the perturbed maps: $K_{\mathrm{col},J}$ in Theorem 20.14 bounds the columns of the perturbed zero-fluctuation map uniformly on the larger perturbation domain, and we use the resulting K_J in the output bound of that theorem and in its choice of the width s_{er} of the boundary layer removed at each step. By the substitution bounds (22.6), we have $z_t, z_t^{\mathrm{cl}} \le \sigma_*/4$, and the choice of s_{er} gives $2e^{-\alpha(s_{\mathrm{er}}-1)} y_t \le \sigma_*/2$, with the same inequality for the classical output bound. Thus both incoming norms at the next boundary are at most $3\sigma_*/4 < \sigma_*$, and by induction the coefficient-norm hypothesis holds after any number of RG steps. Since every integrated step is a full RG step, Theorem 20.13 applies at every step, including its incoming value and the initialization.

The sets of such representatives in Theorem 21.12 start from the representatives at which the bounds on the frozen coordinates just checked hold, and they contain every supported fiber point over each preceding output point. A gauge transformation acting independently at the endpoints of the frozen variables need not preserve the bound on their logarithms. We therefore use analytic estimates only on these sets, and we use the pointwise gauge covariance of Theorem 21.9 only after a gauge-invariant upper bound has been obtained. This verifies the hypotheses for both initial measures. □

22.3 Energy transport, feasible configurations, and the reference normalization

The next theorem collects the classical estimates: (a) is used in Sections 22.5 and 22.8, (b) in Section 22.6, and (c) in Section 22.7.

Theorem 22.4. *Fix an odd integer $M \ge 3$.*

(a) There are $\delta_*(M) > 0$ and $C_M < \infty$, depending only on M, such that for every level i, every set $\mathcal{P}'$ of positive plaquettes at level $i+1$, every configuration V at level i, and every δ,

$$\max_{p \in \mathrm{Pa}(\mathcal{P}')} \rho_G(V_p) \le \delta \le \delta_* \quad \Longrightarrow \quad \mathcal{S}_{i+1}^{\mathcal{P}'}(T_M V) \le (1 + C_M \delta)\, \mathcal{S}_i^{\mathrm{Pa}(\mathcal{P}')}(V).$$

Under this hypothesis no based loop holonomy in these predecessor rectangles equals $-I$, so all the logarithms in the definition of T_M there are the analytic ones. Define

$$B_* = \min\Big\{ \frac{\delta_*^2}{\pi^2}, \Big[\frac{(\sqrt{M}-1)\log 2}{\pi C_M} \Big]^2 \Big\}, \qquad \sigma(b) = \frac{\pi C_M \sqrt{b}}{\sqrt{M}-1}.$$

Let $\ell \ge 0$ be such that the side at level ℓ is at least four, let $\mathcal{P}_\ell$ be a set of positive plaquettes at level ℓ, and let $\mathcal{P}_i = \mathrm{Pa}(\mathcal{P}_{i+1})$ for $0 \le i < \ell$; we call the $\mathcal{P}_i$ the energy-estimate predecessor sets of $\mathcal{P}_\ell$. Then for every configuration U at level 0 and every $0 < b \le B_*$,

$$\mathcal{S}_\ell^{\mathcal{P}_\ell}(T_M^\ell U) \le \max\{ e^{\sigma(b)}, 8|\mathcal{P}_\ell|/b \}\, \mathcal{S}_0^{\mathcal{P}_0}(U). \tag{22.8}$$

This holds for every holonomy and with the Borel extension of T_M. If $\mathcal{S}_0^{\mathcal{P}_0}(U) \le b/h_{n_\ell}$, then for $0 \le i \le \ell$ every plaquette in $\mathcal{P}_i$ has angle at most $\pi\sqrt{b}\, M^{-(\ell-i)/2}$, and the factor in (22.8) *can be replaced by $e^{\sigma(b)} \le 2$. If $q < d_{\mathrm{cut}}$ for every plaquette in $\mathcal{P}_\ell$, then $8|\mathcal{P}_\ell|/b$ may be replaced by $4|\mathcal{P}_\ell| d_{\mathrm{cut}}/b$. Let $p_M \le 24M^3$ be the largest number of plaquettes in a set $\mathcal{P}_{j-1}(C)$, and put*

$$K_M = \max(2, 8p_M/B_*), \qquad K_1 = \max(2, 8/B_*).$$

Then for $0 \le k \le j-1$ and every configuration V at level k, and for every depth ℓ,

$$Q_j^{\mathcal{T}}(T_M^{j-1-k} V) \le K_M \frac{S_W(V)}{h_{n_k}}, \qquad \frac{S_W(T_M^\ell U)}{h_{n_\ell}} \le 48 K_1 \frac{S_W(U)}{h_{n_0}}.$$

These constants are independent of the depth, the side, J, L, γ, and the geometry. The factor K_M may be replaced by 2 only if $q < d_{\mathrm{cut}}$ for all the relevant plaquettes at level $j-1$ and $4p_M d_{\mathrm{cut}} \le B_*$.

(b) Let $\mathcal{X}$ be a large-field decomposition as in Theorem 21.12, with ℓ integrated averaging steps and stopping side m, and let $g_{\mathrm{base}}^2 = \gamma/n_b$ be the coupling of the initial lattice. At every representative $(\widehat{c}, \widehat{b_{\mathrm{rec}}})$ satisfying the domain conditions and reconstruction identities of Theorem 21.12, the zero-fluctuation reconstructions of the successive RG steps, with the frozen coordinates inserted at the scales at which they were frozen, give one configuration U_* on the initial lattice such that

$$(T_M^\ell U_*)|_F = \widehat{c}, \qquad \frac{E_T(\widehat{c}; \widehat{b_{\mathrm{rec}}})}{g_m^2} = \frac{\mathcal{E}_{\mathrm{base}}(U_*)}{g_{\mathrm{base}}^2}, \qquad g_m^2 = M^\ell g_{\mathrm{base}}^2.$$

Here E_T is the effective classical action of the stopping field after all external classical exponents have been incorporated once into the classical exponent. For a stopped initial measure, $U_* \in D_\rho$ and $\mathcal{E}_{\text{base}} = E_{d_0,\text{sat}}$ on this larger gauge-invariant domain; for a Wilson initial measure, U_* is a configuration in $\mathcal{U}_{n_b}$ and $\mathcal{E}_{\text{base}} = S_W$. Consequently

$$E_T(\widehat{c};\widehat{b_{\text{rec}}}) \geq M^\ell \inf\{\mathcal{E}_{\text{base}}(U) : U \in \mathcal{S},\ (T_M^\ell U)|_F = \widehat{c}\},$$

where $\mathcal{S} = D_\rho$ or $\mathcal{S} = \mathcal{U}_{n_b}$, respectively. No minimizer in D_r is asserted.
Let $\kappa_* > 0$ be such that $E_{d,\text{sat}} \geq \kappa_* S_W$ on D_ρ for every depth $d \geq 0$ (Theorems 11.2 and 11.6), and set

$$\kappa_0 = \min(1, \kappa_*), \qquad c_A^{\text{act}} = \alpha_W = \frac{\kappa_0}{48K_1}, \qquad \mathcal{V}_m = \alpha_W S_W.$$

The two names c_A^{act} and α_W denote the same number, in its two roles below. For $c \in \mathrm{SU}(2)^F$ and for frozen coordinates b_{rec} put

$$\Psi_F(c) = \mathcal{V}_{m;F}(c) = \min_{z \in \mathrm{SU}(2)^H} \mathcal{V}_m(c \oplus z), \qquad A_{\mathcal{X}}(b_{\text{rec}}) = c_A^{\text{act}} \sum_{f \in \mathcal{X}} \frac{4q_f(b_{\text{rec}})}{h_{n_f}},$$

where $q_f(b_{\text{rec}})$ is the energy of the recorded bad plaquette f, a function of the frozen coordinates because the full dependence set of f is frozen. Then, at the same representative,

$$\frac{E_T}{g_m^2} \geq \frac{\Psi_F(\widehat{c})}{g_m^2}, \qquad \frac{E_T}{g_m^2} \geq A_{\mathcal{X}}(\widehat{b_{\text{rec}}}), \tag{22.9}$$

and

$$A_{\mathcal{X}}(b_{\text{rec}}) \geq 4c_A^{\text{act}} d_* \sum_{f \in \mathcal{X}} \frac{n_f}{\gamma} \qquad \text{on } I_{\mathcal{X}}^{\text{act}} = \{q_f(b_{\text{rec}}) \geq d_* \text{ for all } f \in \mathcal{X}\}.$$

(c) Use the full-torus recursion of the stopped measures (Theorem 14.6) with the same RG steps as the decomposition; we call it the reference recursion. At a depth $d \geq 0$ of the reference recursion on the stopping lattice put

$$\widehat{Z}_d(g) = \int_{D_r} e^{-E_{d,\text{sat}}/g^2 - [R_{d,\text{sat}} - R_d(I)]}\, dH_m, \qquad g^2 = \gamma/m.$$

The restriction of the initial measure to the small-field conditions of the reference recursion and its pushforward to the stopping lattice has the form

$$d\xi_T = Z_\nu^{-1} e^{\Lambda_T^{\mathbb{T}}} \mathbf{1}_{D_r} e^{-E_{d,\text{sat}}/g^2 - [R_{d,\text{sat}} - R_d(I)]}\, dH_m,$$

where $\Lambda_T^{\mathbb{T}}$ is the normalization constant of the reference recursion; for a Wilson initial measure, only this reference measure is restricted to D_r at the start. With $m_T^{\mathbb{T}} = \xi_T(1) \leq 1$, one has exactly

$$\frac{e^{\Lambda_T^{\mathbb{T}}}}{Z_\nu} = \frac{m_T^{\mathbb{T}}}{\widehat{Z}_d(g)} \leq \widehat{Z}_d(g)^{-1}. \tag{22.10}$$

All Haar disintegration, Gaussian, cluster expansion, and substitution constants are included in $\Lambda_T^{\mathbb{T}}$, and for a stopped initial measure $\Lambda_0^{\mathbb{T}} = -R_{d_0}(I)$.
For every fixed $0 < \lambda_0 \le \lambda_1 < \infty$ there are $C, C', C'' < \infty$, uniform in the depth d of the reference recursion, in m, and in $0 < g \le g_* < 1$, such that

$$\widehat{Z}_d(g) \ge e^{-C3m^3} g^{3(2m^3+1)}, \qquad \int e^{-\lambda \mathcal{V}_m/g^2}\, dH_m \le e^{C'3m^3} g^{3(2m^3-3m^2+1)}$$

for $\lambda \in [\lambda_0, \lambda_1]$ with $4\lambda\alpha_W/g^2 \ge 1$. Consequently the logarithm of the ratio of these two integrals is at most $C''3m^3 + 9m^2\log(1/g)$. The normalized stopped measure $\widehat{\nu}$ on the stopping lattice, with the same support D_r, satisfies

$$\log \mathbb{E}_{\widehat{\nu}} e^{vS_W/g^2} \le C''3m^3 + 9m^2\log(1/g), \qquad 0 \le v \le \kappa_*/2.$$

For fixed $0 < \lambda_0 \le 1$, the quantity

$$\widehat{Z}_d(g)^{-1} \int_{D_r} e^{-\lambda(E_{d,\mathrm{sat}}/g^2 + R_{d,\mathrm{sat}} - R_d(I)) + vS_W/g^2}\, dH_m, \qquad \lambda \in [\lambda_0, 1], \quad 0 \le v \le \lambda_0\kappa_*/2,$$

satisfies the same bound with a larger volume constant. In each of these ratios the boundary correction $9m^2\log(1/g)$ occurs once. The denominator is always the integral over exactly D_r; the periodic boundary plaquettes are deleted only in the upper bound for the numerator.

The constants of the reconstruction and of the normalization constants have the dependence described in item (3) of Definition 4.14, excluding the widths chosen later. The constants in (c) depend on the fixed domains on the torus, r, the norm bounds, and the interval $[\lambda_0, \lambda_1]$ of action coefficients, not on the depth, the side, or the coupling. Applications to probability estimates have the dependence described in item (4) of Definition 4.14.

Remark 22.5. The factor g^{-9m^2} in (c) comes from a count of independent coordinates. The periodic lattice of side m has $3m^3$ links and m^3 sites, so the complement of a spanning tree has $2m^3 + 1$ links; restricting each of them to a ball of radius proportional to g in SU(2), a group of dimension three, gives the lower bound $g^{3(2m^3+1)}$ for $\widehat{Z}_d(g)$. The open cube with m sites per side, used for the numerator, has $3m^2(m-1)$ links and m^3 sites, hence $2m^3 - 3m^2 + 1$ links outside a spanning tree, which give the upper bound $g^{3(2m^3-3m^2+1)}$. The difference $3[(2m^3+1) - (2m^3 - 3m^2 + 1)] = 9m^2$ is a boundary correction in an inequality; it is not an identification of a physical quantity. At a side $m = \chi n$ it becomes $9\chi^2 n^2 \log(1/g_n) + (9/2)\chi^2 n^2 \log\chi$. Gauge fixing along a tree, deleting nonnegative plaquette terms, and factorizing one-plaquette integrals are elementary stability arguments in the tradition of Faria da Veiga and O'Carroll [64, 65]; the precise periodic and open counts used here are proved in Theorem 14.8, with the open-cube estimate of Lemma 6.10.

Proof of Theorem 22.4(a). We write $h_i = h_{n_i}$ within this proof.

Transport over one averaging step. Multiplying the restricted conclusion of Theorem 7.3 by four and dividing by $h_{i+1} = Mh_i$, we obtain the displayed implication. Its hypotheses are local: the maximum is over the union of the complete predecessor rectangles, and the side at level $i+1$ is at least four. We may decrease $\delta_*(M)$ so that the exceptional based loop holonomies are excluded as well. To check this, note that a contour $F_{c,x}C_c^{-1}$ in (4.3) is filled by moving its straight segment of length M across the coordinate segments of the path $\sigma_{v,x}$. At most $M\sum_{k=1}^{3}|x_k - v_k| \le 3M(M-1)/2$ elementary moves across one plaquette are needed, and each move inserts a conjugate of a plaquette transport or of its inverse. All these plaquettes belong to the complete predecessor rectangle, so by the bi-invariant triangle inequality the distance of this based loop holonomy from I is at most $3M^2\delta$. If we also require $3M^2\delta_* < 2\pi$, then this holonomy is not $-I$, and its logarithm is the analytic one. This decrease of δ_* changes only a constant depending on M.

Transport over many averaging steps. Let $V^i = T_M^i U$ and write

$$e_i = \mathcal{S}_i^{\mathcal{P}_i}(V^i), \qquad \delta_i = \max_{p\in\mathcal{P}_i} \rho_G(V_p^i), \qquad \Pi_i = \prod_{k<i}(1 + C_M\delta_k).$$

For an empty set of plaquettes we set the maximum to zero; the corresponding energy is also zero. Suppose first that $e_0 \le b/h_\ell$. We prove by induction on $i \le \ell$ that the first i steps satisfy

$$e_i \le \Pi_i e_0, \qquad \Pi_i \le \exp\Big\{\pi C_M\sqrt{b}\sum_{k<i} M^{-(\ell-k)/2}\Big\} \le 2. \tag{22.11}$$

The case $i = 0$ is immediate. Suppose that (22.11) holds for some i. Since all plaquette energies are nonnegative, (22.1) implies that, for every $p \in \mathcal{P}_i$,

$$\rho_G(V_p^i)^2 \le 2\pi^2 q(V_p^i) \le \frac{\pi^2 h_i}{2} e_i \le \frac{\pi^2}{2}\Pi_i b M^{-(\ell-i)} \le \pi^2 b M^{-(\ell-i)},$$

where we used $h_i/h_\ell = M^{-(\ell-i)}$ and $\Pi_i \le 2$. For $i < \ell$ this is at most $\pi^2 b \le \delta_*^2$. Then the transport over one averaging step applies with $\mathrm{Pa}(\mathcal{P}_{i+1}) = \mathcal{P}_i$; it gives $e_{i+1} \le (1 + C_M\delta_i)e_i$ and excludes all exceptional values used in that step. Further,

$$\log\Pi_{i+1} \le C_M\sum_{k\le i}\delta_k \le \pi C_M\sqrt{b}\sum_{r\ge 1} M^{-r/2} = \sigma(b) \le \log 2.$$

This closes the induction (22.11). The bound on individual plaquettes also holds at $i = \ell$, and no further transport step is needed there. This proves all the stated conclusions under the hypothesis $e_0 \le b/h_\ell$, including the case $\ell = 0$.

If instead $e_0 > b/h_\ell$, then since $q \le 2$, we have $e_\ell \le 8|\mathcal{P}_\ell|/h_\ell < (8|\mathcal{P}_\ell|/b)e_0$. If every plaquette of $\mathcal{P}_\ell$ has $q < d_{\mathrm{cut}}$, the upper bound is $4|\mathcal{P}_\ell|d_{\mathrm{cut}}/h_\ell$. These two cases prove (22.8) for arbitrary configurations. Since no logarithm is taken in the second case, the exceptional values of the Borel extension cause no difficulty. If $q < d_{\mathrm{cut}}$ for all the relevant

plaquettes and $4p_M d_{\mathrm{cut}} \le B_*$, we may choose $b = 4p_M d_{\mathrm{cut}}$; then for every cell the second factor in (22.8) is at most one and the first is at most two.

Cells and all plaquettes. We apply (22.8) with $b = B_*$ separately to the set $\mathcal{P}_{j-1}(C)$ of each cell, with level k in place of level 0. Each such set has at most p_M plaquettes, so the factor is at most K_M. By Lemma 7.4, the energy-estimate predecessor sets at level k of distinct cells of one tiling are disjoint, and the sum of their energies is therefore at most $S_W(V)/h_{n_k}$. This proves the bound with K_M. For an arbitrary set of plaquettes at level ℓ, we apply (22.8) to each plaquette separately, with $|\mathcal{P}_\ell| = 1$ and factor at most K_1. By the same lemma, a plaquette of the initial lattice lies in at most 48 of these predecessor sets, and summation gives the factor $48K_1$. Both arguments apply with any intermediate level as initial level. □

Proof of Theorem 22.4(b). The effective classical action along the decomposition. By Theorem 21.12, the initial and stopping levels are boundaries of RG steps, and there are T integrated RG steps of J averaging steps each; thus $\ell = JT$. The logarithmic product domains, the bounds on the frozen coordinates that are inserted into analytic functions, the coefficient-norm bounds, and the representatives are those checked in Lemma 22.3. For the t-th RG step, let C_t^0 be its reconstruction map with every normalized fluctuation variable set to zero, that is, its zero-fluctuation reconstruction. By the definition of the classical part of its objective, the outgoing effective classical action is

$$E_{t+1}^{\mathrm{out}}(c; b_{\mathrm{rec}}^{(t)}) = B\, E_t\big(C_t^0(c; b_{\mathrm{rec}}^{(t)}); b_{\mathrm{rec}}^{(t)}\big), \qquad B = M^J. \tag{22.12}$$

If new coordinates $c^\sharp$ are frozen at the end of the step, the next incoming action is

$$E_{t+1}(v; b_{\mathrm{rec}}^{(t)}, c^\sharp) = E_{t+1}^{\mathrm{out}}(v \oplus c^\sharp; b_{\mathrm{rec}}^{(t)}).$$

Thus freezing is a direct substitution in this identity; in particular, the value at $v = 0$ is kept together with the nonconstant part. For an external classical exponent S_{ext}, the convention of Theorem 21.12 is the exact identity

$$e^{-S_{\mathrm{ext}}} e^{-E_t^{\mathrm{int}}/g_t^2} = e^{-[E_t^{\mathrm{int}} + g_t^2 S_{\mathrm{ext}}]/g_t^2}.$$

It adds the term once to the effective classical action and removes it from the weight of the frozen coordinates. Under (22.12) the added action and g_t^2 are both multiplied by B, which leaves the exponent unchanged.

One feasible configuration. We start from such a representative at which the bounds on the frozen coordinates hold. We insert the frozen coordinates at each preceding scale at which they were frozen, and apply the maps C_t^0 downward. The reconstruction identities and the protection of all stored endpoints in Theorems 21.5 and 21.12 ensure that each insertion gives back the coordinate of the preceding output point, that each reconstruction

solves its block averaging constraint exactly, and that every older frozen dependence is preserved. The image bounds of Theorem 20.14, applied with the hypotheses verified in Lemma 22.3, keep this downward sequence of reconstructions in its product domains. For a stopped initial measure, the resulting representative on the initial lattice lies in the larger common chart, so its configuration U_* lies in D_ρ; this uses the larger image domain, not membership in D_r. For a Wilson initial measure, the frozen variables deep inside excluded regions remain arbitrary group elements, and the initial classical exponent is exactly $S_W(U_*)$. Composing the constraint identities, we obtain $(T_M^\ell U_*)|_F = \widehat{c}$. Dividing (22.12) by $g_{t+1}^2 = B g_t^2$ and iterating over the T RG steps gives the identity for E_T/g_m^2, and $JT = \ell$ implies $g_m^2 = M^\ell g_{\text{base}}^2$. Since U_* belongs to the constrained set, the lower bound by the infimum follows; no interchange of minima is involved.

The first lower bound. By the lower bound of Theorem 11.2 on the common radius ρ, transported by the fixed gauge transformation of a representative on the initial lattice (Theorem 11.6), we have

$$\frac{\mathcal{E}_{\text{base}}(U_*)}{g_{\text{base}}^2} \geq \kappa_0 \frac{S_W(U_*)}{g_{\text{base}}^2}. \tag{22.13}$$

The radius is within the domain of Theorem 11.2, and the constant κ_* there is valid for every integer depth, whether or not an effective interaction is attached to that depth, as also stated in Theorem 14.8; for a Wilson initial measure (22.13) holds with constant one. Combining this with the bound for all plaquettes in (a), we obtain

$$\frac{E_T}{g_m^2} \geq \frac{\kappa_0}{48K_1} \frac{S_W(T_M^\ell U_*)}{g_m^2} \geq \frac{\Psi_F(\widehat{c})}{g_m^2}.$$

In the last inequality we used that $T_M^\ell U_* = \widehat{c} \oplus z$ for some $z \in \mathrm{SU}(2)^H$, and that Ψ_F minimizes over all such z.

The second lower bound. For a recorded bad plaquette f, let $\mathcal{B}_f$ be the energy-estimate predecessor set of $\{f\}$ on the initial lattice. We first justify disjointness at different levels. On the input lattice of the RG step in which f is recorded, the endpoints of its full dependence list cover every vertex of the four centered blocks obtained by pulling back the corners of f to that input lattice. Indeed, each one-step formula reads the root paths $\sigma_{a,x}$ for every vertex x of each endpoint block; their trees span those blocks, and the same assertion follows recursively from the centered nesting identity. For a plaquette at the input level, its four boundary links already cover its four corner vertices. By Lemma 7.4, every plaquette of $\mathcal{B}_f$ lies in the centered preimages of these blocks.

All these dependence variables are frozen together with their excluded neighborhood by Definition 21.4(2). The neighborhood includes the protected endpoint margins, so it excludes the incident dependence links as well as the marked vertices; the endpoint protection rule of Theorem 21.3(4) preserves this exclusion at later scales. Suppose that f and f' are recorded at levels $\ell_f < \ell_{f'}$ and that their initial energy-estimate predecessor

sets share a plaquette. The centered partitions are nested. A shared vertex therefore places a marked vertex from a corner block of f in a corner block of f', when both are read on the input lattice where f was recorded. If the two tests belong to the same RG step, the full dependence list of f' meets the excluded neighborhood just added for f. If they belong to different RG steps, the centered ancestor of that mark is excluded by the endpoint protection rule, and again the full dependence list of f' meets the earlier excluded set. Both cases contradict the rule that a recorded test avoid that set. Thus the sets $\mathcal{B}_f$ at different levels are disjoint. This argument uses endpoint coverage and the protected neighborhoods; it does not require the minimal list of read links to contain every link of an energy-estimate patch.

At one level, a plaquette lies in at most 48 of the sets $\mathcal{B}_f$ by Lemma 7.4. Since plaquette energies are nonnegative, we have

$$\frac{S_W(U_*)}{g_{\mathrm{base}}^2} \geq \frac{1}{48} \sum_f \mathcal{S}_0^{\mathcal{B}_f}(U_*).$$

The comparison (22.8) for one plaquette and arbitrary configurations, proved in (a), applies at the level ℓ_f of each f with constant K_1. Since the feasible configuration U_* preserves every frozen dependence, the energy of f in it is $q_f(\widehat{b_{\mathrm{rec}}})$. Thus

$$\frac{S_W(U_*)}{g_{\mathrm{base}}^2} \geq \frac{1}{48K_1} \sum_f \frac{4q_f(\widehat{b_{\mathrm{rec}}})}{h_{n_f}}.$$

Multiplying by κ_0 and using (22.13), we find $E_T/g_m^2 \geq A_{\mathcal{X}}(\widehat{b_{\mathrm{rec}}})$. On $I_{\mathcal{X}}^{\mathrm{act}}$ we have $q_f \geq d_*$ and $h_{n_f} = \gamma/n_f$ for every f, which gives the lower bound for $A_{\mathcal{X}}$. □

Proof of Theorem 22.4(c). The reference identity. We apply the identity at the end of Theorem 14.8 to the same reference operators and the same partition function Z_ν as in Theorem 21.12. Every reference operator is a full RG step on the same levels. The depth d_0 of a stopped initial measure is a multiple of J, and the depth at the stopping level is $d = d_0 + JT$, with $d_0 = 0$ for a Wilson initial measure. Thus the reference recursion lies within the scope of Theorem 14.8. The stopping side is m, the preceding depth is d, and the centered effective interaction is $R_{d,\mathrm{sat}} - R_d(I)$. Restriction and pushforward give a subprobability measure, also for a Wilson initial measure. Its density is the displayed ξ_T, with mass at most one. Integrating it, we obtain (22.10). The initial value $\Lambda_0^{\mathbb{T}} = -R_{d_0}(I)$ for a stopped initial measure holds because Z_ν is not centered. All later constants are those of this one recursion.

The partition-function bounds. These are the bounds of Theorem 14.8 on the lattice of side m, with $D = 3m^3$. The trial configurations of that theorem have $2m^3 + 1$ links outside a spanning tree, each with logarithm of norm at most $a_0 g < \min(r/2, 1)$, and the tree

variables have Haar integral one. The bounds on the action and the effective interaction on this set, together with the volume of a ball in SU(2), give

$$\widehat{Z}_d(g) \geq \Big(\frac{a_0 g - \sin(a_0 g)}{2\pi}\Big)^{2m^3+1} e^{-C3m^3} \geq e^{-C_L 3m^3} g^{3(2m^3+1)},$$

where we used $x - \sin x \geq x^3/8$ for $0 \leq x \leq 1$ and included the powers of a_0 in C_L. This lower bound is an integral over exactly D_r. The upper bound for $\int e^{-\lambda \mathcal{V}_m/g^2}\, dH_m = \int e^{-\lambda \alpha_W S_W/g^2}\, dH_m$ is the bound for Wilson numerators in Theorem 14.8, with $\alpha_0 = \alpha_W$, $s = 0$, and inverse coupling $4\lambda\alpha_W/g^2 \geq 1$. In its proof, the plaquettes crossing the periodic boundary are deleted from the numerator, the unused links integrate to one, and on the remaining open cube with m sites per side the $2m^3 - 3m^2 + 1$ plaquettes selected along an axial tree have independent Haar-distributed transports; the one-plaquette bound $\int e^{-\beta q}\, dH \leq \sqrt{2/\pi}\, \beta^{-3/2}$ of Lemma 6.10 then gives the factor $g^{3(2m^3-3m^2+1)}$. Dividing by the lower bound for $\widehat{Z}_d(g)$ leaves $3[(2m^3+1) - (2m^3 - 3m^2 + 1)] = 9m^2$ powers of $1/g$.

The bound for the stopped measure is the exponential-moment bound of Theorem 14.8, with $\lambda_{\mathrm{en}} = \kappa_*/2$. It can also be read off from the same calculation: on D_r its integrand satisfies

$$e^{-E_{d,\mathrm{sat}}/g^2 - [R_{d,\mathrm{sat}} - R_d(I)] + vS_W/g^2} \leq e^{C3m^3} e^{-(\kappa_* - v)S_W/g^2}, \qquad \kappa_* - v \geq \kappa_*/2,$$

where we used $E_{d,\mathrm{sat}} \geq \kappa_* S_W$ and the bound on the oscillation of the centered effective interaction. For the quantity with the coefficient λ we replace the coefficient by $\lambda\kappa_* - v \geq \lambda_0\kappa_*/2$, and we bound the centered effective interaction by its uniform oscillation bound from Theorem 14.8. Both estimates then follow from the upper bound for the Wilson numerator and the single lower bound for the denominator. If the ceiling g_* has not yet been reduced so that the inverse coupling of these two numerators is at least one, then on the remaining range of g, below g_* and above that threshold, $\log(1/g)$ is bounded by a fixed constant; enlarging the volume constant gives the same bounds there. Alternatively, one may make the common reduction of the ceiling in Theorem 14.8. □

22.4 The minimum of the Wilson action over omitted links

The density bounds of Section 22.6 contain the factor $e^{-\lambda \Psi_F(c)/g^2}$, a minimum over all values of the links in H, which we compare with a Haar integral over these links. The comparison is a lower bound of Laplace type that uses only stationarity at a minimizer, an upper bound on the second derivative, and the Haar volume of a small ball; no nondegenerate Hessian is needed. We first record the continuity and gauge invariance of the minimum and of the corresponding integral.

Throughout this section $m \geq 4$, $H \subset \mathcal{E}_m$ is a set of positive links, $F = \mathcal{E}_m \setminus H$, and for $c \in \mathrm{SU}(2)^F$ and $z \in \mathrm{SU}(2)^H$ we write $c \oplus z$ for the configuration on Λ_m that equals c on

F and z on H. We write H_H for the product Haar measure on $\mathrm{SU}(2)^H$, and $\mathcal{V}_m = \alpha_W S_W$ and $\Psi_F(c) = \min_z \mathcal{V}_m(c \oplus z)$ for a fixed $\alpha_W > 0$, as in Theorem 22.4(b).

Lemma 22.6. *For $\lambda > 0$ and $g > 0$ let*

$$I_{\lambda,g}(c) = \int_{\mathrm{SU}(2)^H} e^{-\lambda \mathcal{V}_m(c \oplus z)/g^2}\, dH_H(z).$$

The functions Ψ_F and $I_{\lambda,g}$ are continuous on $\mathrm{SU}(2)^F$. Let $k \in \mathrm{SU}(2)^{\Lambda_m}$ be a lattice gauge transformation and let c^k be the transformed field on F, which depends only on the values of k at the endpoints of the links in F. Then $\Psi_F(c^k) = \Psi_F(c)$ and $I_{\lambda,g}(c^k) = I_{\lambda,g}(c)$. In particular, Ψ_F and $I_{\lambda,g}$ are invariant under every gauge transformation of the endpoints of the links in F, extended arbitrarily to the other sites, and the result does not depend on the extension.

Proof. For $c, c' \in \mathrm{SU}(2)^F$ we have

$$|\Psi_F(c) - \Psi_F(c')| \le \alpha_W \sup_{z \in \mathrm{SU}(2)^H} |S_W(c \oplus z) - S_W(c' \oplus z)|.$$

Since S_W is uniformly continuous on the compact group $\mathrm{SU}(2)^{\mathcal{E}_m}$, the right side tends to zero as $c' \to c$. The same argument, applied to the integrand, shows that $I_{\lambda,g}$ is continuous. Let k be a gauge transformation, and write $z \mapsto z^k$, $(z^k)_{xy} = k_x z_{xy} k_y^{-1}$, for its action on the links in H. This map is a bijection of $\mathrm{SU}(2)^H$ that acts on each factor by a fixed left and a fixed right translation, so it preserves H_H. Since $(c \oplus z)^k = c^k \oplus z^k$ and S_W is gauge invariant, we have

$$\mathcal{V}_m(c^k \oplus z^k) = \mathcal{V}_m(c \oplus z), \qquad dH_H(z^k) = dH_H(z).$$

Taking the minimum over z and changing variables in the integral, we conclude that Ψ_F and $I_{\lambda,g}$ are invariant. If a gauge transformation is given only at the endpoints of the links in F, any two extensions to the other sites give bijections of the whole product $\mathrm{SU}(2)^H$ as above, so the result does not depend on the extension. □

Lemma 22.7. *For every $m \ge 4$, every set $H \subset \mathcal{E}_m$ of positive links, $F = \mathcal{E}_m \setminus H$, every $c \in \mathrm{SU}(2)^F$, and every fixed $\alpha_W > 0$, the function $z \mapsto \mathcal{V}_m(c \oplus z) = \alpha_W S_W(c \oplus z)$ has a minimizer z_* on $\mathrm{SU}(2)^H$, and it is stationary at z_* in all directions of the links in H. For $X = (X_e)_{e \in H} \in \mathfrak{su}(2)^H$ and the curve $z_e(t) = z_{*,e} e^{tX_e}$,*

$$\Big| \frac{d^2}{dt^2} S_W(c \oplus z(t)) \Big| \le 16 \sum_{e \in H} |X_e|^2.$$

Hence, for every $0 < g \le 1$, the product neighborhood $\{z_{*,e} e^{X_e} : |X_e| < g \text{ for all } e \in H\}$ has Haar mass at least

$$(c_* g)^{d_H}, \qquad d_H = 3|H|, \qquad c_* = (16\pi)^{-1/3},$$

and on it $\mathcal{V}_m(c \oplus z) - \Psi_F(c) \le 8\alpha_W |H| g^2$. *For* $0 < \lambda_0 \le \lambda \le \lambda_1 < \infty$,

$$e^{-\lambda \Psi_F(c)/g^2} \le (c_* g)^{-d_H} e^{8\lambda\alpha_W |H|} \int_{\mathrm{SU}(2)^H} e^{-\lambda \mathcal{V}_m(c \oplus z)/g^2} \, dH_H(z),$$

and the exponential factor may be enlarged to $e^{8\lambda_1 \alpha_W |H|}$. *These bounds hold for arbitrary links across the boundary of the block region, arbitrary values of* c, *and arbitrary shape, topology, age, and depth; the displayed constants do not depend on these parameters or on the coupling. If* H *is empty, equality holds. No regularity, uniqueness, or measurable choice of a minimizer is required.*

Proof. The function $z \mapsto \mathcal{V}_m(c \oplus z)$ is continuous on the compact product $\mathrm{SU}(2)^H$, so it attains its minimum. Every curve $t \mapsto z_{*,e} e^{tX_e}$ through the minimizer stays in this product for both signs of t, and differentiating at $t = 0$ proves stationarity in all directions of the links in H.

We extend X by zero on F. Along the curve, the transport around a positive plaquette is a product of four unitary factors. A factor $z_{*,e} e^{tX_e}$ has first and second derivatives of operator norm at most $|X_e|/2$ and $|X_e|^2/4$, since $\|X_e\|_{\mathrm{op}} = |X_e|/2$ with our normalization of the inner product; an inverse factor $e^{-tX_e} z_{*,e}^{-1}$ has the same bounds. By the product rule, $|\mathrm{tr}\, A|/2 \le \|A\|_{\mathrm{op}}$, and the Cauchy–Schwarz inequality for four terms,

$$\Big| \frac{d^2}{dt^2} q\big((c \oplus z(t))_p\big) \Big| \le \frac{1}{4} \Big(\sum_{e \in \partial p} |X_e| \Big)^2 \le \sum_{e \in \partial p} |X_e|^2.$$

Here the sum is over the positive links underlying the oriented boundary of p; for $m \ge 4$ they are distinct. Since each positive link lies in exactly four positive plaquettes and $S_W = 4 \sum_p q(U_p)$, the second derivative of S_W is bounded by $16 \sum_{e \in H} |X_e|^2$ everywhere along the curve. By Taylor's formula with integral remainder, whose weight $\int_0^1 (1 - t)\, dt$ equals $1/2$, and stationarity at $t = 0$, we have

$$\mathcal{V}_m(c \oplus z_* e^X) - \Psi_F(c) \le 8\alpha_W \sum_{e \in H} |X_e|^2.$$

Thus on the product of balls of radius g the increase is at most $8\alpha_W |H| g^2$.

The Haar probability measure on $\mathrm{SU}(2)$ has radial density $(2/\pi) \sin^2 \vartheta \, d\vartheta$ in the angle $\vartheta = |X|/2$. Thus the ball $\{e^X : |X| < g\}$ has mass $(g - \sin g)/(2\pi)$. Taylor's formula for the sine gives, for $0 \le g \le 1$,

$$g - \sin g \ge g^3/6 - g^5/120 \ge g^3/8.$$

The exponential map is injective on this ball since $g \le 1 < 2\pi$. By translation invariance of Haar measure and the product structure, the product neighborhood has mass at least $(g^3/(16\pi))^{|H|} = (c_* g)^{d_H}$. On this neighborhood the integrand $e^{-\lambda \mathcal{V}_m(c \oplus z)/g^2}$ is at least $e^{-\lambda \Psi_F(c)/g^2 - 8\lambda\alpha_W |H|}$. Integrating over the neighborhood, enlarging the domain of

integration to the whole product, and rearranging, we obtain the displayed inequality. Replacing λ by λ_1 in the positive exponential only enlarges the right side. If H is empty, all products have mass one and the additional exponents vanish. Each bound is uniform on the whole compact group, and a minimizer was used separately for each c, not as a measurable function of c. □

22.5 Domination of the plaquette-energy sum by the Wilson minimum

Along a large-field decomposition, the function $Q_j^{\mathcal{T}} \circ T_M^{j-1-b}$ need not be a function of the stopping field and the frozen coordinates alone: some requested plaquettes may have dependence sets that meet excluded neighborhoods. The next lemma shows that this does not matter for an upper bound: the bound holds for every completion $c \oplus z$, and we may then take the infimum over completions.

Lemma 22.8. *Fix an odd integer $M \geq 3$, an integer $J \geq 1$, $B = M^J$, and the fixed analytic, geometric, and buffer data of Theorems 21.5 and 21.12. Fix a finite family of stopping offsets and remainders, with total exponents $v = r_{\mathrm{stop}} J + u_{\mathrm{ph}}$ and $0 \leq u_{\mathrm{ph}} < J$, and fix the corresponding enlargement F_J of the bound for the number of affected source plaquettes in Theorem 21.5, before the interval of s or any minimum side is chosen. Let $K_1, \alpha_W > 0$ be as in Theorem 22.4 and put $K_W = 48K_1/\alpha_W$.*

Let $L > 0$ and $\gamma > 0$, and let a finite hierarchy of sides in $4\mathbb{N}$, an initial measure (stopped, at a depth $d_0 \in J\mathbb{N}_0$, or Wilson), a large-field decomposition $\mathcal{X}$ from its initial level to its stopping level with $\ell \in J\mathbb{N}_0$ averaging steps, and an exponent v of the family be given. Write $m = M^v n_{\mathrm{req}}$ for the stopping side, where $n_{\mathrm{req}} \in 4\mathbb{N}$ is the requested side; every iterate that is used must belong to the finite hierarchy. Let F be the set of links of the block region at the stopping level, $H = \mathcal{E}_m \setminus F$, and let $\mathcal{P}_{\mathrm{orig}}$ be the requested list of plaquettes at side n_{req}, a union of the sets $\mathcal{P}_{j-1}(C)$ of cells, counted once. Set

$$Q_{\mathrm{orig}}(V) = \frac{4}{h_{n_{\mathrm{req}}}} \sum_{p \in \mathcal{P}_{\mathrm{orig}}} q\big((T_M^{\ell+v} V)_p\big), \qquad g_m^2 = \gamma/m, \qquad D_{\mathcal{X}} = 8F_J \sum_{f \in \mathcal{X}} \frac{n_f}{\gamma}.$$

Then every point $V \in \mathcal{A}_{\mathcal{X}}$, with $(c, b_{\mathrm{rec}}) = \mathcal{O}_T^{\mathcal{X}}(V)$, and every $z \in \mathrm{SU}(2)^H$ satisfy

$$Q_{\mathrm{orig}}(V) \leq 48K_1 \frac{S_W(c \oplus z)}{g_m^2} + D_{\mathcal{X}}. \tag{22.14}$$

Consequently, at every point of $\mathcal{A}_{\mathcal{X}}$,

$$Q_{\mathrm{orig}}(V) \leq \frac{K_W}{g_m^2} \min_{z \in \mathrm{SU}(2)^H} \{\alpha_W S_W(c \oplus z)\} + 8F_J \sum_{f \in \mathcal{X}} \frac{n_f}{\gamma} = \frac{K_W \Psi_F(c)}{g_m^2} + D_{\mathcal{X}}. \tag{22.15}$$

The minimum Ψ_F is continuous and invariant under gauge transformations of the endpoints of the links in F (Lemma 22.6). For either initial measure ν, with its partition function Z_ν, which is not centered, let $\eta_{\mathcal{X}}$ and $\eta_{\mathcal{X}}^s$ be as in Section 22.1, with $Q = Q_{\mathrm{orig}}$. For every $s \geq 0$,

$$\eta_{\mathcal{X}}^s(dc\, db_{\mathrm{rec}}) \leq e^{sK_W \Psi_F(c)/g_m^2 + sD_{\mathcal{X}}}\, \eta_{\mathcal{X}}(dc\, db_{\mathrm{rec}}). \tag{22.16}$$

This includes decompositions without recorded bad plaquettes, an empty set F, and decompositions whose block region becomes empty before the planned stopping level, continued by empty RG steps. No small-field event, support, or partition function is changed, and no smallness of the coupling is needed beyond the existence of the decomposition. The constant K_W depends only on M and on the fixed classical lower bound and its domain; it does not depend on the side, L, the coupling, the depth, the geometry, the age, or the frozen values. The constant F_J keeps the explicit polynomial dependence on the widths in Theorem 21.5, and its coefficient has the dependence described in item (3) of Definition 4.14, independently of the widths chosen later. The constants of later probability estimates have the dependence described in item (4) of Definition 4.14.

Proof. We fix the decomposition and the requested list first.

Step 1: dependence sets. For each requested plaquette p, let $D_m(p)$ be the full set of links of the stopping lattice on which $C \mapsto q((T_M^v C)_p)$ depends. It contains the links of the four boundary transports of p and, recursively, all reference paths, relative contours, filling plaquettes, and branch tests of every entry read in the v averaging steps; the clause on complete predecessor sets in Lemma 7.4 supplies these finite dependence sets. We also trace them to every scale of the integrated decomposition at which variables were frozen. We keep p only if $D_m(p) \subset F$ and the traced dependence sets avoid all the full excluded neighborhoods assigned in Theorem 21.5: the neighborhoods of the recorded bad plaquettes with their buffers, the adjacent cells, the frozen boundary bands, the links across the boundary of the block region, the enlarged earlier excluded regions, the buffers protecting earlier stored endpoints, and the protective sets of the first step for a Wilson initial measure. Their levels of creation and their assignment to recorded bad plaquettes are kept. The smaller energy-estimate predecessor sets $\mathcal{B}_f$ would not suffice for this definition.

Step 2: kept plaquettes are determined by the stopping field. If two configurations on the stopping lattice agree on $D_m(p)$, then for every entry needed in the first averaging step they read identical path products and identical branch tests. A regular entry has the same analytic logarithm in both configurations, and an exceptional entry has the same reference-path value. By induction over the v averaging steps, the final plaquette transports agree, and hence so do their traces. This proves locality for all group values, including the exceptional branches; for $v = 0$ it is the equality of the four boundary links.

By the identity for the observation map in Theorems 21.5 and 21.9, we have $(T_M^{\ell} V)|_F = c$, and therefore, for every completion $C = c \oplus z$,

$$\frac{4}{h_{n_{\mathrm{req}}}} \sum_{p \text{ kept}} q\big((T_M^{\ell+v} V)_p\big) = \frac{4}{h_{n_{\mathrm{req}}}} \sum_{p \text{ kept}} q\big((T_M^{v} C)_p\big).$$

This equality concerns the point V itself; no auxiliary feasible configuration or representative has been substituted.

Step 3: discarded plaquettes. We assign each discarded requested plaquette to the first full excluded neighborhood that its dependence set meets, ordering first by the level of creation and then by the fixed order of the tests. If $D_m(p) \not\subset F$, an omitted link lies in an excluded neighborhood at the stopping level; otherwise the exclusion comes from an earlier traced intersection. The counting clause of Theorem 21.5, enlarged for the finite family of stopping offsets and remainders, bounds the number of plaquettes assigned to each recorded bad plaquette f by F_J. The deterministic assignment to a single f is needed only because the neighborhoods may overlap; we never use disjointness of the full neighborhoods. Since $n_{\mathrm{req}} \le n_f$ and $0 \le q \le 2$, each discarded plaquette contributes at most $8n_{\mathrm{req}}/\gamma \le 8n_f/\gamma$ to Q_{orig}. Summing over the discarded plaquettes, we find, for every $C = c \oplus z$,

$$Q_{\mathrm{orig}}(V) \le \frac{4}{h_{n_{\mathrm{req}}}} \sum_{p \text{ kept}} q\big((T_M^{v} C)_p\big) + D_{\mathcal{X}} \le \frac{4}{h_{n_{\mathrm{req}}}} \sum_{p} q\big((T_M^{v} C)_p\big) + D_{\mathcal{X}},$$

where the last sum is over all positive plaquettes at side n_{req}.

Step 4: the Wilson action of a completion. We apply the bound for all plaquettes and arbitrary configurations in Theorem 22.4(a) with initial configuration C at side m and depth v. Its initial exponent is $S_W(C)/g_m^2$, and $h_{n_{\mathrm{req}}} = M^v g_m^2$. Together with Step 3, this gives (22.14), with no additional factor m/n_{req} and no regularity test at the requested level. The bound applies to every completion, whatever its holonomies or exceptional values. Taking the infimum over the whole compact product $\mathrm{SU}(2)^H$, we obtain (22.15). By continuity of S_W and compactness, the infimum is attained, but no measurable choice of a minimizer is needed. Continuity and invariance of Ψ_F are proved in Lemma 22.6.

Step 5: empty cases. If $\mathcal{X}$ has no recorded bad plaquettes, there are no excluded neighborhoods and $D_{\mathcal{X}} = 0$. If F is empty, then no plaquette is kept, since the dependence set of every plaquette is nonempty on sides at least four, and the assignment of Step 3 gives $Q_{\mathrm{orig}} \le D_{\mathcal{X}}$ directly; in this case $\Psi_{\varnothing} = 0$, attained by the identity configuration. If the block region becomes empty before the planned stopping level, we carry the frozen coordinates unchanged through the remaining RG steps. For these steps the fluctuation integral and the determinant equal one, the nonconstant norm is zero, and the simultaneous replacement $E \mapsto BE$, $g^2 \mapsto Bg^2$ leaves the exponent unchanged. The recurrence

$R_{t+1}(f) \le B^{-1}R_t(f) + C_J(W + R_{\mathrm{cell}} + 1)$ for the radii of the excluded neighborhoods in Theorem 21.5 still assigns every later omitted link to the same neighborhoods, even when their union is the whole torus. The finite dependence range of the requested plaquettes at the stopping level is included in F_J. Thus the same assignment and the same bound hold at the planned stopping level. These empty steps use only the clauses on empty products and on normalization constants in Theorems 20.13 and 20.14 and the partition of Theorem 21.5.

Step 6: integration. For a nonnegative Borel function $\varphi(c, b_{\mathrm{rec}})$, we integrate $\varphi(\mathscr{O}_T^{\mathcal{X}}(V))e^{sQ_{\mathrm{orig}}(V)}$ against $\mathbf{1}_{\mathcal{A}_{\mathcal{X}}}\, d\nu(V)$ and insert (22.15). The right side becomes the integral of $\varphi e^{sK_W\Psi_F/g_m^2 + sD_{\mathcal{X}}}$ against $\eta_{\mathcal{X}}$, which proves (22.16). Both sides carry the same factor Z_ν^{-1}. All integrands are nonnegative, the spaces are finite-dimensional, and Q_{orig} is bounded on them. The dependence of the constants follows from the bound for all plaquettes and from the explicit count F_J. □

The same argument applies to the Wilson exponent of all plaquettes at the requested side, with a larger bound for the number of affected plaquettes. This version is used in Section 23 for exponential moments of the full blocked action.

Corollary 22.9. *In the setting of Lemma 22.8, let the requested function at side n instead be $\mathcal{Q}_n(V) = S_W(T_M^{\ell+v}V)/g_n^2$, where $m = M^v n$, v belongs to a fixed finite family of exponents, and all iterates are available. Enlarge the bound for the number of affected plaquettes to a bound F_J^{A} for the list of all positive plaquettes, before choosing any interval of s. Then at every point of $\mathcal{A}_{\mathcal{X}}$ and for every completion z,*

$$\mathcal{Q}_n(V) \le 48K_1\frac{S_W(c\oplus z)}{g_m^2} + 8F_J^{\mathrm{A}}\sum_f \frac{n_f}{\gamma}, \qquad \text{hence} \qquad \mathcal{Q}_n(V) \le \frac{K_W\Psi_F(c)}{g_m^2} + 8F_J^{\mathrm{A}}\sum_f \frac{n_f}{\gamma}.$$

The corresponding inequality for the pushforward measures, analogous to (22.16)*, holds with the same partition function. This includes Wilson initial measures, empty sets F, and decompositions continued by empty RG steps. One may take $F_J^{\mathrm{A}} = C_J^{\mathrm{A}}(W + R_{\mathrm{cell}} + 1)^3$; this bound has the same polynomial dependence on the widths, and its coefficient the same dependence, as F_J, and K_W is unchanged.*

Proof. Steps 1, 2, and 4–6 of the proof of Lemma 22.8 apply verbatim with the list of all positive plaquettes at side n in place of $\mathcal{P}_{\mathrm{orig}}$, with n in place of n_{req}, and with F_J^{A} in place of F_J. In Step 3, each discarded plaquette still contributes at most $8n/\gamma \le 8n_f/\gamma$, since the recorded bad plaquette to which it is assigned lies at a level finer than the requested side. In Step 4, the Wilson exponent of $T_M^v(c\oplus z)$ at side n is bounded by Theorem 22.4(a) with no factor depending on the ratio of the sides.

It remains to justify the count in Step 3. We check explicitly that the counting clause of Theorem 21.5 extends to the list of all plaquettes. The family of exponents v

is finite, and each exponent has a finite fixed dependence range. A neighborhood of radius R is met by the dependence sets of at most a fixed multiple of $(R+1)^3$ positive plaquettes. The recurrence for the radii in Theorem 21.5 bounds R by a fixed multiple of $W + R_{\mathrm{cell}} + 1$ at later outputs, and the enlargement at the input of the RG step of creation is already included in the count at creation, which carries the factor B. Thus passing to all plaquettes changes only this incidence constant, and unions and overlaps are handled by subadditivity and by the assignment to a single recorded bad plaquette. This proves the bound $F_J^{\mathrm{A}} = C_J^{\mathrm{A}}(W + R_{\mathrm{cell}} + 1)^3$, with the enlargement for the finite family of exponents; the number of RG steps of the decomposition does not occur. □

22.6 Gauge-invariant domination of the densities

We now combine the density bound of Theorem 21.12, which holds at representatives satisfying the reconstruction identities, with the classical lower bounds of Theorem 22.4(b). Since the resulting bound has only gauge-invariant factors, the pointwise gauge covariance of the density in Theorem 21.9 extends it to every point.

Theorem 22.10. *Let $\mathcal{X}$ be a large-field decomposition for a stopped or a Wilson initial measure as in Theorem 21.12, with stopping side m, $g^2 = \gamma/m$, and $0 < g \le 1$. Whenever the uniform bounds of Theorem 22.4(c) on the stopping lattice are used, impose also their fixed ceiling on g. Let F be the set of links of the block region at the stopping level, H the set of all other links of the stopping lattice, including the links across the boundary of the block region, and $d_H = 3|H|$. Let $p_{\mathcal{X}}(c, b_{\mathrm{rec}})$ be the density of $\eta_{\mathcal{X}}$ with respect to $H_F \otimes H_{\mathcal{X}}$ given by Theorem 21.9, where H_F and $H_{\mathcal{X}}$ are the product Haar measures on the stopping field and on the frozen coordinates. Let $w_{\mathcal{X}}$ be the gauge-invariant weight of the frozen coordinates and $\rho_{\mathcal{X}} = w_{\mathcal{X}} H_{\mathcal{X}}$, as in Theorem 21.12; they keep the indicator $I_{\mathcal{X}}^{\mathrm{act}}$ of the tests of all recorded bad plaquettes. Fix $0 < \theta < 1$, put $\lambda = 1 - \theta$, and use the constants of Theorem 22.4.*

At every representative $(\widehat{c}, \widehat{b_{\mathrm{rec}}})$ in the support satisfying the domain conditions and reconstruction identities of Theorem 21.12, the density bound of Theorem 21.12 and its estimate for the normalization constant give

$$p_{\mathcal{X}}(\widehat{c}, \widehat{b_{\mathrm{rec}}}) \le \widehat{Z}_d(g)^{-1} e^{L_{\mathcal{X}}(\widehat{b_{\mathrm{rec}}})} w_{\mathcal{X}}(\widehat{b_{\mathrm{rec}}}) e^{-E_T(\widehat{c};\widehat{b_{\mathrm{rec}}})/g^2}, \qquad L_{\mathcal{X}} \le \overline{L}_{\mathcal{X}}.$$

One may take

$$L_{\mathcal{X}} = \log A_T + \Lambda_T - \Lambda_T^{\mathbb{T}} + C_{\mathrm{osc},J} m^3, \qquad \overline{L}_{\mathcal{X}} = C_\delta + 3C_J m^3 + C_\partial \,\mathrm{Surf}_{\mathcal{X}} + \sum_{f \in \mathcal{X}} \overline{\mathrm{ch}}_f.$$

Here Λ_T is the normalization constant of the regional recursion along $\mathcal{X}$, $\Lambda_T^{\mathbb{T}}$ is that of the reference recursion, and A_T is the accumulated comparison constant of Theorem 21.12. Further, $\mathrm{Surf}_{\mathcal{X}}$ is the number of variables in the boundary bands that enter the normalization constant, and

$\overline{\mathrm{ch}}_f$ *is the sum of the nonnegative upper bounds, assigned to* f *by Theorem 21.5, for the corrections from missing Gaussian coordinates, from missing constant terms of fluctuation integrals, and from substitutions of frozen coordinates. Each missing Gaussian coordinate contributes its own* $\log(1/g_i)$*, and each other such variable contributes a fixed bound* C_J*, including the variables at initialization. These are deterministic bounds in terms of the geometry and the couplings, independent of the chosen representative. The identity* (22.10) *supplies only the prefactor* $\widehat{Z}_d(g)^{-1}$*; the bound for* $\Lambda_T - \Lambda_T^{\mathbb{T}}$ *comes from the increment estimate of Theorem 20.13.*

Applying both lower bounds (22.9) *at the same representative gives a gauge-invariant upper bound which is defined everywhere. At every point* (c, b_{rec})*, with the density of Theorem 21.9, which vanishes off the gauge saturation of these representatives,*

$$p_{\mathcal{X}}(c, b_{\mathrm{rec}}) \le \widehat{Z}_d(g)^{-1} e^{\overline{L}_{\mathcal{X}}} w_{\mathcal{X}}(b_{\mathrm{rec}}) \exp\bigl\{-\lambda \Psi_F(c)/g^2 - \theta A_{\mathcal{X}}(b_{\mathrm{rec}})\bigr\}.$$

The minimum Ψ_F *and the corresponding integral over* H *are invariant under every gauge transformation of the endpoints of the links in* F*, extended to the other sites. The tests of the recorded bad plaquettes and the other tests in* $w_{\mathcal{X}}$ *are invariant under the action of consistent endpoint gauge transformations at all scales. Hence the pointwise covariance of Theorem 21.9 applies to this bound. Define the finite measure, which is in general not normalized,*

$$d\widetilde{\sigma}_{\lambda,d,g}(C) = \widehat{Z}_d(g)^{-1} e^{-\lambda \mathcal{V}_m(C)/g^2}\, dH_m(C),$$

and let $\widetilde{\sigma}_{\lambda,d,g;F}$ *be its marginal on the links of* F*. Integrating the Wilson factor over the links in* H *gives*

$$\begin{aligned}\eta_{\mathcal{X}}(dc\, db_{\mathrm{rec}}) \le\ & \exp\bigl\{\overline{L}_{\mathcal{X}} + d_H \log(1/(c_* g)) + 8\lambda \alpha_W |H| - \theta A_{\mathcal{X}}(b_{\mathrm{rec}})\bigr\} \\ & \times [\widetilde{\sigma}_{\lambda,d,g;F} \otimes \rho_{\mathcal{X}}](dc\, db_{\mathrm{rec}}).\end{aligned}$$

For $\theta = \lambda = 1/2$*, let* Q_{orig} *be a requested plaquette-energy sum as in Lemma 22.8, and let* $\eta_{\mathcal{X}}^s = (\mathscr{O}_T^{\mathcal{X}})_{\#}(e^{sQ_{\mathrm{orig}}} \mathbf{1}_{\mathcal{A}_{\mathcal{X}}} \nu)$*. For* $0 \le s \le (4K_W)^{-1}$ *and* $\lambda_s = 1/2 - sK_W \in [1/4, 1/2]$*,*

$$\eta_{\mathcal{X}}^s \le \exp\bigl\{\overline{L}_{\mathcal{X}} + sD_{\mathcal{X}} + d_H \log(1/(c_* g)) + 8\lambda_s \alpha_W |H| - A_{\mathcal{X}}(b_{\mathrm{rec}})/2\bigr\}\, [\widetilde{\sigma}_{\lambda_s,d,g;F} \otimes \rho_{\mathcal{X}}]. \tag{22.17}$$

The function Q_{orig} *need not be a function of the observation* (c, b_{rec})*, and the same partition function* Z_ν *and the same centered denominator* $\widehat{Z}_d(g)$ *are used for all* s*.*

All external classical exponents occur once in E_T *and are absent from* $\rho_{\mathcal{X}}$*. The intermediate auxiliary actions and effective interactions need only be defined at the analytic arguments given by these representatives; no global auxiliary density is asserted or integrated. The local constants in the bounds for the normalization constant and for the oscillation have the dependence described in item (3) of Definition 4.14, independently of the widths chosen later. The explicit counts contain the dependence on the geometry, and uniform probability constants have the dependence described in item (4) of Definition 4.14.*

Proof. We first check the hypotheses of the imported estimates, then bound the density at such a representative, extend the bound to every point by gauge invariance, and finally integrate over the omitted links.

Step 1: hypotheses. The decomposition in Theorem 21.12 has its initial and stopping levels at boundaries of RG steps and integrates only full RG steps. By Lemma 22.3, its logarithmic product domains satisfy the containment and the expansion conditions, its frozen coordinates satisfy the one-quarter radius bounds and the substitution bounds, and its coefficient norms at initialization remain within the same bounds. Here A_{w} is fixed before the ceiling, and (22.5) gives measurability and a positive finite partition function for every comparison with the enclosing product domain. This verifies the hypotheses of the integration estimate and of the comparison at every step, with all frozen coordinates real.

Step 2: the bound at a representative. We choose such a representative at which the bounds on the frozen coordinates hold, and let Z_ν be the partition function of Theorem 21.12. At this representative, after the external classical exponents have been incorporated once, the induction of that theorem gives

$$p_{\mathcal{X}}(\widehat{c}, \widehat{b_{\mathrm{rec}}}) \le A_T Z_\nu^{-1} e^{\Lambda_T(\widehat{b_{\mathrm{rec}}})} w_{\mathcal{X}}(\widehat{b_{\mathrm{rec}}}) e^{-E_T/g^2 - R_T(\widehat{c};\widehat{b_{\mathrm{rec}}})}.$$

The regional effective interaction R_T is centered at the origin of the block region at the stopping level. By Theorem 20.14, its part coming from the full torus and its boundary correction satisfy the oscillation bound

$$|R_T(v; b_{\mathrm{rec}})| \le \rho^2 \Big[E|\Omega_T| + y_* \sum_{i \in \Omega_T} e^{-\alpha d_{\Omega_T}(i)} \Big] \le C_{\mathrm{osc},J} m^3,$$

where Ω_T is the list of scalar variables of the block region at the stopping level, $d_{\Omega_T}(i)$ is the distance of i to its complement, and y_* is the fixed output bound of Theorem 20.14, which is used because no boundary layer is removed after the last RG step. There are at most $9m^3$ scalar variables. In the last inequality, we used $e^{-\alpha d} \le 1$, which avoids any assumption on the shape of the boundary. The linear terms are included in this oscillation bound. The constants are evaluated on the fixed analytic domains, before the widths are chosen.

Step 3: the normalization constants. Inserting (22.10), that is $e^{\Lambda_T^{\mathbb{T}}}/Z_\nu \le \widehat{Z}_d(g)^{-1}$, into this density bound, we obtain the stated expression for $L_{\mathcal{X}}$. To bound it, we sum over the RG steps the increment inequality of Theorem 20.13, as recorded in Theorem 21.12. Every summand comes from a full RG step, and the column bounds of the perturbed maps and all the product conditions have been checked in Lemma 22.3. At initialization the normalization constant equals the constant of the reference recursion plus the

increment from the substitution at initialization. Each missing scalar Gaussian coordinate contributes $\log(1/g_i)$, and each missing constant term of a fluctuation integral and each substituted variable contributes its fixed bound C_J. Using Theorem 21.5, we assign these nonnegative bounds to the recorded bad plaquette whose assigned neighborhood contains the variable. Variables in the boundary bands contribute $C_\partial \operatorname{Surf}_{\mathcal{X}}$. A variable that lies in several neighborhoods is assigned to the first recorded bad plaquette in the fixed order of the tests, and is therefore counted once, or overcounted by a bounded positive amount. Variables at initialization are assigned in the same way. Thus

$$\Lambda_T - \Lambda_T^{\mathbb{T}} \le C_\partial \operatorname{Surf}_{\mathcal{X}} + \sum_f \overline{\mathrm{ch}}_f,$$

after a fixed bound for the bulk at the stopping level has been absorbed into $3C_J m^3$; the same C_J may include $C_{\mathrm{osc},J}/3$. Finally, $\log A_T \le C_\delta$ by Theorem 21.12 and Lemma 20.5. These estimates prove the bound by $\overline{L}_{\mathcal{X}}$, in which only counts of geometric objects and the given couplings g_i occur, and no value at the representative.

Step 4: the classical lower bounds. At the same representative, by Theorem 22.4(b), we have $E_T/g^2 \ge \Psi_F(\widehat{c})/g^2$ and $E_T/g^2 \ge A_{\mathcal{X}}(\widehat{b_{\mathrm{rec}}})$. Multiplying these inequalities by λ and θ, respectively, adding them, and using $\lambda + \theta = 1$, we obtain

$$\frac{E_T}{g^2} \ge \frac{\lambda \Psi_F(\widehat{c})}{g^2} + \theta A_{\mathcal{X}}(\widehat{b_{\mathrm{rec}}}).$$

Inserting this into the bound of Step 2, we find the pointwise upper bound at every such representative.

Step 5: extension to every point. By Lemma 22.6, Ψ_F and the integral $I_{\lambda,g}$ are continuous and invariant under gauge transformations of the endpoints of the links in F, extended to the other sites. Each value $q_f(b_{\mathrm{rec}}) = 1 - \operatorname{tr}(U_f(b_{\mathrm{rec}}))/2$ of a recorded bad plaquette is computed from the transport $U_f(b_{\mathrm{rec}})$ around f, using its frozen full dependence set and the Borel extension of T_M; under an endpoint gauge transformation the intermediate endpoint factors cancel and the final transport is conjugated. Thus $A_{\mathcal{X}}$ and all tests in $w_{\mathcal{X}}$ are invariant under the action of consistent endpoint gauge transformations in Theorem 21.9. The bound $\overline{L}_{\mathcal{X}}$ is gauge invariant, since it depends only on the geometry and the couplings. It follows that the right side of the bound is a gauge-invariant Borel function G defined everywhere. For a point $x = (c, b_{\mathrm{rec}})$ in the gauge saturation of these representatives, we choose such a representative $\widehat{x} = x^k$ at which the bounds on the frozen coordinates hold; it exists by Theorem 21.5, and the bounds of Steps 2–4 hold at $\widehat{x}$. The endpoint action on arbitrary frozen variables need not preserve the bound on their logarithms, but G is defined by invariant factors only. By the pointwise covariance of the density in Theorem 21.9, we have

$$p_{\mathcal{X}}(x) = p_{\mathcal{X}}(\widehat{x}) \le G(\widehat{x}) = G(x).$$

This is a pointwise statement for each x, and the gauge transformation k may depend on x; neither a random choice according to Haar measure nor an integration over the set of representatives is needed. Off the gauge saturation the density vanishes, and the inequality holds there as well.

Step 6: integration over the omitted links. It remains to integrate the Wilson factor, which is now defined globally. The hypothesis of Lemma 22.7 is $0 < g \le 1$, which we have assumed; hence that lemma applies with the fixed λ. This condition at the stopping side is separate from the ceiling on the largest coupling of an integrated RG step: at the output of a full RG step, $g = \sqrt{M}\, t_{\max}$. The reference identity used in Step 3 needs no uniform bound on the ratio of partition functions; whenever that bound is used, its fixed ceiling is imposed as well, as required in the statement. Multiplying by $\widehat{Z}_d(g)^{-1}\, dH_F(c)$ gives exactly

$$\widehat{Z}_d(g)^{-1}\Big[\int_{\mathrm{SU}(2)^H} e^{-\lambda \mathcal{V}_m(c\oplus z)/g^2}\, dH_H(z)\Big]\, dH_F(c) = d\widetilde{\sigma}_{\lambda,d,g;F}(c).$$

All these are finite nonnegative integrals over compact products. Multiplying the pointwise inequality by $dH_F(c)\, dH_{\mathcal{X}}(b_{\mathrm{rec}})$ proves the inequality for $\eta_{\mathcal{X}}$, with $w_{\mathcal{X}}\, dH_{\mathcal{X}} = d\rho_{\mathcal{X}}$. Every external classical exponent has entered E_T exactly once. The invariant indicators of the recorded bad plaquettes remain in $\rho_{\mathcal{X}}$; hence $q_f \ge d_*$ holds at every point of integration used later. No global value or covariance property of E_T or R_T was used. The dependence of the constants follows from the local bounds for the oscillation and the normalization constant and from the displayed counts.

Step 7: the weighted measures. For the last assertion, we take $\theta = \lambda = 1/2$ and apply (22.16) to the bound for $\eta_{\mathcal{X}}$ obtained in Step 5, before the integration over H. This changes $-\Psi_F/(2g^2)$ into $-\lambda_s \Psi_F/g^2$ and adds $sD_{\mathcal{X}}$ to the exponent. We then apply Lemma 22.7 to this Wilson factor with $\lambda = \lambda_s$. Its constants are uniform for $\lambda_s \in [1/4, 1/2]$, and the marginal is exactly $\widetilde{\sigma}_{\lambda_s,d,g;F}$. This proves (22.17). Since the multiplication by $e^{sQ_{\mathrm{orig}}}$ was made before the integration over H, no two partition functions are compared. □

22.7 Summation over large-field decompositions

We now prove Theorem 22.1, following the outline in Section 22.1. The method of absorbing the normalization losses of excluded regions into the action of the large fields goes back to Bałaban's treatment of large fields [20, 21]; the decomposition, the frozen coordinates, and the constants used here are those of Section 21.

Proof of Theorem 22.1. Steps 1 and 2 fix the ceilings and the stopping level, Steps 3–6 treat a stopping level above the initial level, and Step 7 treats the remaining case.

Step 1: ceilings and the minimum side. We write $n = n_{j-1}$. The parameters, the buffers, and the count F_J are fixed first. We choose the coupling bound $g_{\mathrm{anc}} > 0$ for the bounds in Theorem 22.4(c), uniformly for $\lambda \in [1/4, 1/2]$, and impose

$$g_{\mathrm{end}} \le \min\{1, g_*, g_{\mathrm{anc}}, \sqrt{\alpha_W}, \sqrt{2\kappa_*}\}, \qquad n_{\min} \ge \max\{e\Gamma, \Gamma/g_{\mathrm{end}}^2, \Gamma/(Mg_*^2)\}. \tag{22.18}$$

We also include every fixed minimum side required by the geometry and by the first step for a Wilson initial measure in Theorem 21.5. Then, for every stopping side $m \ge n_{\min}$, we have $g_m \le 1$, as required in Lemma 22.7. Each ratio of partition functions satisfies its own coupling bound, the inverse-coupling conditions $4\lambda\alpha_W/g_m^2 \ge 1$ for $\lambda \ge 1/4$ and $2\kappa_*/g_m^2 \ge 1$ hold, and a full RG step with output side m has $t_{\max}^2 = \gamma/(Mm) \le g_*^2$. The identity $g_m = \sqrt{M}\, t_{\max}$ shows why the condition at the stopping side is checked separately. No restriction on γ enters a local estimate.

Step 2: the stopping level. Let $t_0 = J\lfloor (j-1)/J \rfloor$ and $u_{\mathrm{ph}} = j - 1 - t_0$. For a given $r_{\mathrm{stop}} \in \mathscr{R}$, we set $t_{\mathrm{stop}} = t_0 - r_{\mathrm{stop}} J$. If $t_{\mathrm{stop}} > b$, we integrate the $(t_{\mathrm{stop}} - b)/J$ full RG steps of the decomposition. The function at their endpoint is exactly

$$C \longmapsto Q_j^{\mathcal{T}}(T_M^v C), \qquad v = r_{\mathrm{stop}} J + u_{\mathrm{ph}} = j - 1 - t_{\mathrm{stop}}, \qquad m = n_{t_{\mathrm{stop}}} = M^v n. \tag{22.19}$$

Here $0 \le b < t_{\mathrm{stop}} \le j - 1 < JK$, and therefore all v iterates are available and all their sides are multiples of four. The depth $aJ + t_{\mathrm{stop}} - b$ of the reference recursion is a multiple of J. If $t_{\mathrm{stop}} \le b$, we use the initial measure directly: $j - 1 - b$ is an available nonnegative exponent, bounded by $J \max \mathscr{R} + J - 1$, and $n_b/n \le \chi_*$. This treats all remainders without integrating a partial RG step. Steps 3–6 concern the case $t_{\mathrm{stop}} > b$.

Step 3: one decomposition. We use the Borel partition of Theorem 21.5 and the density induction of Theorem 21.12. For a Wilson initial measure, the first step covers all of $\mathcal{U}_{n_b}$ and does not first restrict the Wilson measure to D_r. By Lemma 22.3, the product domains, the coefficient norms, the bounds on the frozen coordinates, and the boundary comparisons hold for every finite decomposition, without a bound on its number of RG steps.

Fix one decomposition $\mathcal{X}$. Let $\Psi_F = \min_z \alpha_W S_W(c \oplus z)$, and let $H_{\mathcal{X}} = \sum_f x_f$. In Steps 3–6, $H_{\mathcal{X}}$ denotes this scalar sum, not the product Haar measure on the frozen coordinates. At the observation, by Lemma 22.8, we have $Q_{\mathrm{orig}} \le K_W \Psi_F/g_m^2 + 8F_J H_{\mathcal{X}}$. We apply the bound of Theorem 22.10 with $\theta = 1/2$, before the integration over H. Multiplication by $e^{sQ_{\mathrm{orig}}}$ is justified by the inequality (22.16) between pushforward measures, without assuming that Q_{orig} is a function of the observation. With $\lambda_s = 1/2 - sK_W \in [1/4, 1/2]$, the exponent before the integration over H satisfies

$$sQ_{\mathrm{orig}} - \frac{\Psi_F}{2g_m^2} - \frac{A_{\mathcal{X}}}{2} \le -\frac{\lambda_s \Psi_F}{g_m^2} - (P_0 - 8sF_J)H_{\mathcal{X}}.$$

Indeed, by Theorem 22.4(b) and the invariant indicators of the recorded bad plaquettes, we have $A_{\mathcal{X}}/2 \geq 2c_A^{\mathrm{act}} d_* H_{\mathcal{X}} = P_0 H_{\mathcal{X}}$, and the condition $s \leq \theta c_A^{\mathrm{act}} d_*/(16F_J)$ gives $8sF_J \leq P_0/8 \leq P_0/4$. We integrate the Wilson minimum once over H by Lemma 22.7, which adds $3|H|\log(1/(c_* g_m)) + 8\lambda_s \alpha_W |H|$ to the exponent. The centered reference denominator and the partition function Z_ν of the initial measure are the same for every s.

Step 4: assignment of the corrections. We enlarge the local counting constant to account for the directions, the colors, the links across the boundary of the block region, and the finite family of remainders, and set

$$D_0 = C_J(BW + R_{\mathrm{cell}} + 1)^3, \qquad D_1 = C_J(W + R_{\mathrm{cell}} + 1)^3.$$

By Theorem 21.5, every variable outside the bulk that enters the normalization constant is assigned to a recorded bad plaquette at whose creation it was produced, and the assignment is subadditive when neighborhoods merge. At most D_0 variables are assigned to f at its creation, and at most D_1 at each later RG step. At creation their integration side is at most Bn_f, and the sides at later steps decrease by the factor B. We absorb the finite factors coming from the stopping offsets and the finite link products into a constant $\chi_{\mathrm{lay}} \geq 1$. For every missing Gaussian coordinate assigned to f,

$$\log(1/g_i) = \tfrac{1}{2}\log(n_i/\gamma) \leq \tfrac{1}{2}\log x_f + \tfrac{1}{2}\log \chi_{\mathrm{lay}}. \tag{22.20}$$

By Theorems 20.13 and 21.12, all other corrections per variable have fixed bounds. Thus a constant C_{lab}, with the dependence described in item (3) of Definition 4.14, bounds the correction of each variable by $C_{\mathrm{lab}}(1 + \log x_f)$. The dependence on the widths is entirely in D_0, D_1, and the fixed counts of affected plaquettes.

Let H_f be the number of RG steps after the creation of f, including empty steps. Counting the decreasing sides down to four, we find

$$H_f + 1 \leq C_{\mathrm{lay},J} + \#\{q \in \mathbb{N}_0 : 4B^q \leq \chi_{\mathrm{lay}} n_f\} \leq C'_{\mathrm{lay},J}(1 + \log n_f).$$

The finite additive term includes the conventions at creation and at the stopping level; the number of RG steps of the decomposition does not occur. Since $x_f \geq e$ and $\gamma \leq \Gamma$, we have $1 + \log n_f \leq (1 + \log_+ \Gamma)(1 + \log x_f)$. We write H_f^{term} for the set of omitted links at the stopping level assigned to f. The deterministic assignment of Theorem 21.5, to the first recorded bad plaquette in the fixed order of the tests, partitions H into these sets, and $|H_f^{\mathrm{term}}| \leq D_1$. The corrections assigned to f are listed in Table 22.2. For the integration over H we also used that $m \leq n_f$, so that $\log(1/(c_* g_m)) \leq \log(1/c_*) + (\log x_f)/2$.

The numbers of nonempty and of empty later RG steps in the table add up to at most H_f, so no step contributes twice. At the first RG step at which the block region is empty, all omitted variables are covered by assigned neighborhoods, and the recurrence $R_{t+1}(f) \leq B^{-1} R_t(f) + C_J(W + R_{\mathrm{cell}} + 1)$ of Theorem 21.5 continues to cover

Table 22.2. All corrections assigned to one recorded bad plaquette f created at side n_f. Here $L_f = 1 + \log x_f$, and the later RG steps include empty steps.

Stage	Corrections included, and where they are bounded	Bound
Creation	Missing Gaussian coordinates, missing constant terms of the Haar disintegration and fluctuation integrals, substitutions at initialization and at creation; Theorems 20.13 and 21.12	$D_0 C_{\mathrm{lab}} L_f$
Later nonempty RG steps	Missing coordinates and constant terms, boundary bands, and every later substitution, at the couplings of their integration; the same increment estimate	$D_1 C_{\mathrm{lab}} L_f$ per step
Integration over H	Omitted links, including links across the boundary of the block region: $3\lvert H_f^{\mathrm{term}}\rvert \log(1/(c_* g_m)) + 8\lambda_s \alpha_W \lvert H_f^{\mathrm{term}}\rvert$; Lemma 22.7	$D_1 C_{\mathrm{com}} L_f$, once
Empty RG steps	Determinant and fluctuation integral equal one; logarithmic determinant and centered effective interaction vanish. Missing reference variables use the same increment estimate	at most $D_1 C_{\mathrm{lab}} L_f$ per empty step
External exponents	Incorporated once into the classical exponent; the weight of the frozen coordinates is a subprobability; Theorem 21.12	no further positive correction
Plaquette-energy sum	Discarded plaquettes keep the coefficient $4/g_n^2$; Lemma 22.8	$8 s F_J x_f \le P_0 x_f / 4$

their coarse images, including the full finite dependence range, even when the union of the neighborhoods is the whole torus. We carry the frozen coordinates by the identity through the remaining full RG steps; both the effective classical action and g^2 acquire the factor B, which leaves each external exponent unchanged. The clauses on empty products in Theorems 20.13 and 20.14 therefore justify the row for empty steps.

Combining the first four rows of the table, we find

$$C_\partial \operatorname{Surf}_{\mathcal{X}} + \sum_f \overline{\mathrm{ch}}_f + 3|H| \log(1/(c_* g_m)) + 8\lambda_s \alpha_W |H| \le \sum_f C^{\Gamma}_{\mathrm{ch},J} (1 + \log x_f)^2. \tag{22.21}$$

For example, one may take for $C^{\mathrm{base}}_{\mathrm{ch},J}$ a fixed multiple of $(D_0 + D_1)(1 + C'_{\mathrm{lay},J})(C_{\mathrm{lab}} + C_{\mathrm{com}})$. The bulk term $3C_J m^3$ and the single comparison constant C_δ of Theorem 22.10 remain as explicit prefactors at the stopping level; they are not generated again at each step.

Step 5: absorption and the bound for one decomposition. For $x \ge e$, the function $(1 + \log x)^2/x$ decreases to zero. We increase the minimum side so that $C^{\Gamma}_{\mathrm{ch},J}(1 + \log x)^2 \le P_0 x/4$ for every $x \ge n_{\min}/\Gamma$. Since $n_f \ge m \ge u \ge n_{\min}$ and $\gamma \le \Gamma$, every x_f is in this range, and by Steps 3 and 4 at least $P_0 x_f/2$ of the classical lower bound remains for each recorded bad plaquette. By Theorem 21.12, the weight of the frozen coordinates, after the removal of the incorporated external exponents, has mass at most one. The marginal $\widetilde{\sigma}_{\lambda_s, d, g_m; F}$ has the mass of the whole Wilson numerator, $\widehat{Z}_d(g_m)^{-1} \int e^{-\lambda_s \mathcal{V}_m / g_m^2}\, dH_m$. Using the bound for the single ratio in Theorem 22.4(c), uniform for $\lambda_s \in [1/4, 1/2]$, we conclude that

$$\mathbb{E}_\nu \big[e^{sQ_{\mathrm{orig}}}; \mathcal{A}_{\mathcal{X}} \big] \le e^{3C_T m^3 + 9m^2 \log(1/g_m) + C_\delta} \prod_f e^{-a_{\mathrm{act}} n_f/\gamma}, \tag{22.22}$$

where C_T collects C'' of Theorem 22.4(c) and C_J. Only the reference integral in the denominator is restricted to D_r.

Step 6: summation over decompositions. Since the sets $\mathcal{A}_{\mathcal{X}}$ partition the support of ν, $\mathbb{E}_\nu e^{sQ_{\mathrm{orig}}}$ is the sum of the left sides of (22.22) over all decompositions. By Theorem 21.5, there is at most one decomposition for each attainable set of recorded bad plaquettes, and there are at most $3n_\ell^3$ positive plaquettes at level ℓ. Hence

$$\sum_{\mathcal{X}} \prod_f e^{-a_{\mathrm{act}} n_f/\gamma} \le \prod_\ell \big(1 + e^{-a_{\mathrm{act}} n_\ell/\gamma}\big)^{3n_\ell^3} \le \exp\Big\{ \sum_\ell 3 n_\ell^3 e^{-a_{\mathrm{act}} n_\ell/\gamma} \Big\},$$

which is the bound in the theorem. To bound its logarithm, we include all levels, also those inside RG steps, so that consecutive sides differ by the factor M, and note that

$$\frac{(Mm')^3 e^{-a_{\mathrm{act}} M m'/\gamma}}{m'^3 e^{-a_{\mathrm{act}} m'/\gamma}} = M^3 e^{-a_{\mathrm{act}}(M-1)m'/\gamma} \le \frac{1}{2}$$

for all sides m' above a fixed further minimum side, uniformly for $\gamma \le \Gamma$. We also require $n_{\min} \ge 3\Gamma/a_{\mathrm{act}}$, so that $x \mapsto x^3 e^{-a_{\mathrm{act}} x/\gamma}$ decreases for $x \ge n_{\min}$, and $6n_{\min}^3 e^{-a_{\mathrm{act}} n_{\min}/\Gamma} \le 1$.

Since all sides n_ℓ that occur are at least $u \geq n_{\min}$, the sum in the exponent is at most twice its first term, hence at most one, for every finite depth. Since $m = \chi n$, the remaining exponent is

$$3C_T\chi^3 n^3 + 9\chi^2 n^2 \log(1/g_n) + \tfrac{9}{2}\chi^2 n^2 \log\chi + C_\delta + 1.$$

The fixed term containing $\log\chi$ is absorbed into the volume constant, and the condition $C_s \geq 9\chi_*^2$ accommodates the single surface term.

Step 7: stopping level at or below the initial level. In this case $Q_{\mathrm{orig}} = Q_j^{\mathcal{T}} \circ T_M^{j-1-b}$, and by the bound for cells in Theorem 22.4(a), we have $Q_{\mathrm{orig}} \leq K_M S_W(V)/h_{n_b}$. For a stopped initial measure, the bound for stopped measures in Theorem 22.4(c) applies, because $sK_M \leq \kappa_0/2 \leq \kappa_*/2$. For a Wilson initial measure, we have $sK_M \leq 1/2$, and the moment is at most

$$\frac{\int e^{-(1-sK_M)S_W/h_{n_b}}\, dH_{n_b}}{\int e^{-S_W/h_{n_b}}\, dH_{n_b}}.$$

We use the integral over D_r of the reference recursion at depth zero only to bound the denominator from below, and the Wilson integral over the whole configuration space, with coefficient $1 - sK_M \in [1/2, 1]$, in the numerator. The same calculation of Theorem 22.4(c), that is, of Theorem 14.8, bounds this ratio by $e^{3C_T n_b^3 + 9n_b^2 \log(1/g_{n_b})}$. We convert units using $n_b/n \leq \chi_*$, as in Step 6. Finally, we enlarge the constants C_v, C_s, C_b to cover the finite family of stopping offsets and remainders, which proves the theorem. All increases of the minimum side are made after Γ is fixed and before the coupling is chosen. No step used a bound on $j - b$ or a bound on the probability, under the Wilson measure, of the complement of a small-field event. □

22.8 Chessboard estimates with regularity indicators

The chessboard estimate converts the exponential-moment bound of Theorem 22.1 into a bound for the probability that a given cell contains a bad plaquette at level j, given regularity at finer levels inside the cells. Reflection positivity of the Wilson measure with respect to reflections in lattice planes is due to Osterwalder and Seiler [134], and systematic treatments of reflection positivity and chessboard estimates were given by Fröhlich, Israel, Lieb, and Simon [92, 93]. We use the functional form of the chessboard estimate in [29, Theorem 5.8, (5.34)–(5.35)]. It applies to the algebra of bounded gauge-invariant functions of the links on one side of a lattice plane through links, and to the pullbacks of such functions by the block averaging maps with their Borel extension, as verified in Theorem 6.6. In the form proved there, it states the following. Let $\mu^{(\ell)}$ be the pushforward of the Wilson measure to level ℓ, tile the lattice at level ℓ into cells Q_t of $2 \times 2 \times 2$ sites, $\mathfrak{m}$ in number, and let ϑ_t denote the reflected placements in the cells. Then

for nonnegative bounded gauge-invariant functions $f_1, \dots, f_k$ of the links of one cell and distinct cells $t_1, \dots, t_k$,

$$\mathbb{E}_{\mu^{(\ell)}} \prod_{i=1}^{k} \vartheta_{t_i} f_i \le \prod_{i=1}^{k} \Big[\mathbb{E}_{\mu^{(\ell)}} \prod_{t} \vartheta_t f_i \Big]^{1/\mathfrak{m}},$$

and the same holds for each of the eight shifted tilings. Reflection positivity is used only for the Wilson measure and its pushforwards under the block averaging maps; a measure restricted to a fiber of the block averaging map or to a small-field event need not be reflection positive.

Lemma 22.11. *Let $N = uB^K$ with $u \in 4\mathbb{N}$, and let $1 \le j \le JK$. For each of the eight tilings at level j, let $\mathfrak{m}_j = n_j^3/8$ be the number of its disjoint cells of $2 \times 2 \times 2$ sites. All events below are gauge-invariant functions of the links of one cell, at the respective levels, pulled back to the fine lattice by the block averaging maps with their Borel extension, and they are considered under the Wilson measure $\mu = \mu_{N,\gamma}$. For two such events R_0 and I_0 attached to one cell, let $\vartheta_t R_0$ and $\vartheta_t I_0$ be their reflected placements in the cells t of the tiling, and set*

$$R = \bigcap_t \vartheta_t R_0, \qquad I_{\mathrm{all}} = \bigcap_t \vartheta_t I_0.$$

Then

$$\mu(R \cap I_0) \le \mu(R \cap I_{\mathrm{all}})^{1/\mathfrak{m}_j} \mu(R)^{1-1/\mathfrak{m}_j}, \qquad \mu(R_0 \cap I_0) \le \mu(R \cap I_{\mathrm{all}})^{1/\mathfrak{m}_j}.$$

If $\mu(R) > 0$, this gives $\mu(I_0 \mid R) \le \mu(I_{\mathrm{all}} \mid R)^{1/\mathfrak{m}_j}$.

Now let $R_0 = R_{<j}(C; d_{\mathrm{cut}})$ be the event of Lemma 7.4 that $q < d_{\mathrm{cut}}$ for exactly those plaquettes at finer levels whose full dependence sets lie in the cell C, and let $I_0 = I_j(C; d_*)$ be the event that at least one internal level-j plaquette of C has $q \ge d_*$, where $d_* = d_{\mathrm{cut}}/2$. The event R imposes no condition on plaquettes whose dependence sets cross the boundaries of the cells. Assume $\pi\sqrt{2d_{\mathrm{cut}}} \le \delta_*$, and let $c_E = 1 + C_M \pi \sqrt{2d_{\mathrm{cut}}}$. On $R \cap I_{\mathrm{all}}$, the function (22.2) *of the configuration at level $j-1$ satisfies*

$$Q_j^{\mathcal{T}} \ge \frac{4 d_* \mathfrak{m}_j}{c_E h_{n_j}}.$$

Consequently, for every $s > 0$,

$$\log \mu_{N,\gamma}(R_0 \cap I_0) \le -\frac{4 s d_* n_j}{c_E \gamma} + \mathfrak{m}_j^{-1} \log \mathbb{E}_{\mu_{N,\gamma}} \big[e^{s Q_j^{\mathcal{T}}}; R \big].$$

The expectation on the right is restricted to the event R and taken with the partition function of the Wilson measure. If its logarithm is at most $3C_v n_{j-1}^3 + C_s n_{j-1}^2 \log(1/g_{n_{j-1}}) + C_b^\circ$, then dividing this bound by $\mathfrak{m}_j$ gives, at $m = n_j$, exactly

$$\mathcal{A}_\gamma(m) = 24 C_v M^3 + \frac{4 C_s M^2}{m} \log(Mm/\gamma) + \frac{8 C_b^\circ}{m^3}.$$

No division by the probability of a regularity event occurs. Regularity of all plaquettes at finer levels implies the event R, but R implies neither regularity of the plaquettes whose dependence sets cross cell boundaries nor any condition of the small-field event. The inequalities contain no constant depending on the depth, the geometry, or the coupling, apart from c_E.

Proof. Each indicator of an event in a cell used below is bounded by one and Borel: its list of plaquettes is finite, q is continuous, and every iterate of the block averaging map with its Borel extension is Borel. Since it is gauge invariant and depends only on links internal to its cell, it belongs to the algebra of Theorem 6.6; the support condition, the reflection geometry for odd M, and the exact number of cells needed there are given by Lemma 7.4. In the functional inequality we place $\mathbf{1}_{R_0\cap I_0}$ in the distinguished cell and $\mathbf{1}_{R_0}$ in every other cell. Placing the first function at all reflected translates gives $\mathbf{1}_{R\cap I_{\mathrm{all}}}$, and placing each of the other functions at all reflected translates gives $\mathbf{1}_R$. The product on the left is $\mathbf{1}_{R\cap I_0}$. This proves the first inequality, including the case of zero probabilities. Applying the same inequality to the single function $\mathbf{1}_{R_0\cap I_0}$, with the constant function one in the other cells, proves the second inequality. If $\mu(R) > 0$, dividing the first inequality by $\mu(R)$ gives the conditional form, with the normalization still inside the root.

For the event R, every plaquette in a complete predecessor rectangle of an internal level-j plaquette is tested, since these lists are included in the tests of Lemma 7.4. By (22.1), the angles of these plaquettes at level $j-1$ are at most $\pi\sqrt{2d_{\mathrm{cut}}} \le \delta_*$. Applying the transport estimate over one averaging step in Theorem 22.4(a) to the union of the internal level-j plaquettes, we obtain

$$\frac{4}{h_{n_j}} \sum_{\text{internal } P} q(V_P^j) \le c_E Q_j^{\mathcal{T}}.$$

On I_{all} every cell has at least one internal plaquette with $q \ge d_*$. The internal plaquettes of distinct cells are distinct, so the sum on the left is at least $4d_*\mathfrak{m}_j/h_{n_j}$. This proves the lower bound for $Q_j^{\mathcal{T}}$. For $s > 0$, the exponential Markov inequality under the Wilson measure implies that

$$\mu(R\cap I_{\mathrm{all}}) \le \exp\bigl\{-4sd_*\mathfrak{m}_j/(c_E h_{n_j})\bigr\}\mathbb{E}_{\mu_{N,\gamma}}\bigl[e^{sQ_j^{\mathcal{T}}}; R\bigr].$$

Taking the $\mathfrak{m}_j$-th root, using the second chessboard inequality, and taking logarithms, with $\log 0 = -\infty$, we obtain the stated bound, since $h_{n_j} = \gamma/n_j$. Only one factor $d_* = d_{\mathrm{cut}}/2$ occurs.

Finally, we have $n_{j-1} = Mm$, $\mathfrak{m}_j = m^3/8$, and $\log(1/g_{n_{j-1}}) = \log(Mm/\gamma)/2$. Dividing the three terms of the assumed bound by $m^3/8$ gives $24C_vM^3$, $4C_sM^2\log(Mm/\gamma)/m$, and $8C_b^\circ/m^3$, respectively, which is $\mathcal{A}_\gamma(m)$. Regularity of all plaquettes at finer levels implies the event R by inclusion of the lists of plaquettes, while R omits every finer plaquette whose full dependence set crosses the boundary of its cell. □

22.9 Domination of successive small-field restrictions

The next lemma is the input for the second term in the bound on large fields, Theorem 2.7, which concerns configurations whose curvature is small at every scale but for which some fluctuation cutoff or coordinate-domain condition fails. The probability of the preceding conditions is multiplied, never divided, and no condition at later scales enters. The restricted measures and their dominations are indexed by $0 \le a \le K$, and quantities involving the next RG step by $0 \le a < K$.

Lemma 22.12. *Fix the small-field event of Definitions 4.4 and 14.4, with its prefactors and its fixed local parameters, together with $L > 0$, M, J, and $0 < \gamma_0 \le 1$. For $N = uB^K$ and $0 \le a \le K$, let S^a be the event consisting of the initial condition $U \in D_r$ and precisely the conditions of the first a RG steps in* (4.6)*, and put*

$$\eta_a = (T_M^{aJ})_{\#}(\mathbf{1}_{S^a}\mu_{N,\gamma}), \qquad \zeta_a = (T_M^{aJ})_{\#}\big(\mathbf{1}\{V^{bJ} \in D_r \text{ for } 0 \le b \le a\}\,\mu_{N,\gamma}\big).$$

For $0 \le a < K$, at the output side $m = n_{(a+1)J}$ of the next RG step, set

$$t_a^2 = \gamma/(Mm), \qquad \delta_a = 6\mathsf{b}_a e^{-m_{\mathrm{w}} p(t_a)^2/24}, \qquad z_a = 6\mathsf{b}_a e^{-m_{\mathrm{w}} Q_{\mathrm{DB}}^2/(24t_a^2)},$$
$$\epsilon_a = \delta_a/(1-z_a), \qquad \mathsf{b}_a \le C_J m^3,$$

where $p(t) = A_{\mathrm{w}} \log(1/t)$ and $a_{\mathrm{w}} = m_{\mathrm{w}} A_{\mathrm{w}}^2/24$. Here b_a is the number of three-dimensional fluctuation blocks of the RG step, $Q_{\mathrm{DB}} > 0$ is the radius denoted Q_{phys} in Definition 14.4, whose product of balls of radius Q_{DB}/t_a lies inside the supported fiber coordinates, and $m_{\mathrm{w}} > 0$ is the lower bound for the Hessian of the conditional fluctuation potential; neither Q_{DB} nor m_{w} is related to the plaquette-energy sum or to the logarithmic cutoff. Above the common minimum side described below, the comparison of Theorem 14.7, with identical unnormalized prefactors, gives

$$\mathbf{1}_{D_r}(T_M^J)_{\#}\eta_a \le (1-\epsilon_a)^{-1}\eta_{a+1}, \qquad \zeta_a \le \prod_{b<a}(1-\epsilon_b)^{-1}\eta_a.$$

It also gives the following bound for the failure of the next condition:

$$\mu_{N,\gamma}(S^a \cap A_m \cap W_m^c) \le \epsilon_a \mu_{N,\gamma}(S^a) \le C_J m^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mm/\gamma}},$$

where $A_m = \{V^{(a+1)J} \in D_r\}$ and $W_m = \mathcal{A}_{a,J} \cap \{\max_{i,e}|w_{a,i,e}| \le p(t_a)\}$ is the condition of the next RG step on the coordinate domains and the fluctuation cutoff, so that $S^{a+1} = S^a \cap A_m \cap W_m$. This bound keeps only the preceding conditions and the condition on the target; there is no conditioning on regularity at later scales.

There is one minimum side, valid simultaneously for all $0 < \gamma \le \gamma_0$ and every finite depth, above which $z_a, \epsilon_a \le 1/2$ and the product is at most e^{C_δ}; the logarithm of the product can be made at most one. Every event S^a has positive probability. Let $\widehat{\nu}_a = \eta_a/\mu_{N,\gamma}(S^a)$ be the normalized stopped measure. If $G_{\le aJ}$ is an event of regularity of the curvature which, by Theorem 7.7, implies

the conditions $V^{bJ} \in D_r$ for $0 \le b \le a$, then for every nonnegative Borel function F of the configuration at level aJ,

$$\mathbb{E}_{\mu_{N,\gamma}}[F(V^{aJ}); G_{\le aJ}] \le e^{C_\delta} \mu_{N,\gamma}(S^a) \mathbb{E}_{\widehat{\nu}_a} F \le e^{C_\delta} \mathbb{E}_{\widehat{\nu}_a} F.$$

This is a multiplicative domination, valid also for test functions with exponentially large values on the finite lattice; the probability $\mu_{N,\gamma}(S^a)$ is not inverted. It does not identify a stopped measure with a marginal of the conditioned Wilson measure.

The same minimum side can be chosen so that, in addition, any fixed finite collection of the following inequalities holds: for $k \in \{1, 2\}$, fixed $C < \infty$, and fixed $\varepsilon > 0$,

$$C[1 + \log(m/\gamma)]^k \le \varepsilon m/\gamma \qquad (m \ge n_0,\ 0 < \gamma \le \gamma_0).$$

For the event $\mathrm{FF}_0 = \{\max_p q(U_p) \ge d_*\}$ that some plaquette of the fine lattice has $q(U_p) \ge d_*$, the following bound holds at every $N \in 4\mathbb{N}$ with $\beta_N = 4N/\gamma \ge 2$:

$$\mu_{N,\gamma}(\mathrm{FF}_0) \le 3N^3 \exp\Big\{ \frac{\mathrm{C_{LT}} + 36 \log \beta_N / N}{8} - \frac{\beta_N d_*}{16} \Big\},$$

where $\mathrm{C_{LT}} = \max\{C_0 + 24 \log 2, 0\}$, with C_0 the constant of Theorem 6.6, is independent of N, β_N, and γ. Since

$$\sup_{\beta \ge 4N/\gamma_0} \frac{\mathrm{C_{LT}} + 36 \log \beta / N}{\beta} \le \frac{\mathrm{C_{LT}} \gamma_0}{4N} + \frac{9 \gamma_0 \log(4N/\gamma_0)}{N^2} \longrightarrow 0,$$

one obtains, above the common minimum side,

$$\mu_{N,\gamma}(\mathrm{FF}_0) \le 3N^3 e^{-d_* N/(8\gamma)}.$$

The bound for the supremum is used for $4N/\gamma_0 \ge e$. The numbers of free blocks can always be kept in the form $\mathsf{b} \le C_J m^3$; in terms of $t^2 = \gamma/(Mm)$ this also gives $\mathsf{b} \le C_J \gamma_0^3 t^{-6}/M^3$. All minimum sides and all constants of absorption and of the products have the dependence described in item (4) of Definition 4.14; they are independent of the coupling, the side, the depth, and the geometry and age of decompositions. The correction $36 \log \beta_N / N$ is kept exactly until the minimum side, uniform in the coupling, has been chosen.

Proof. Restricted measures are indexed by $0 \le a \le K$, and quantities of a transition by $0 \le a < K$. We first choose the common minimum side, so that all positivity conditions on denominators hold before a quantitative comparison is applied.

Step 1: measure-theoretic hypotheses. The amplitude $A_{\mathrm{w}} \ge 1$ is fixed before its coupling ceiling and before n_0. The product domains on the torus here are the instances of Lemma 22.3 without boundary. In particular, their sets of supported fluctuation coordinates, their cutoff domains, and their tests are Borel, the conditional fluctuation potential

is smooth on a neighborhood of the compact enclosing product domain, and (22.5) gives a positive finite partition function at each fixed target satisfying the coordinate-domain conditions. The initial Wilson density is positive, and every identical prefactor carries the normalization of the Wilson measure. These facts verify the measure-theoretic hypotheses of the comparison in Lemma 20.5 whenever its conclusion for products over endpoints is used below.

Step 2: the common minimum side. We choose all minimum sides uniformly in the coupling. Set

$$z(m,\gamma) = \tfrac{1}{2}\log(Mm/\gamma), \qquad a_{\mathrm{w}} = m_{\mathrm{w}}A_{\mathrm{w}}^2/24 > 0.$$

Since $p(t) = A_{\mathrm{w}}\log(1/t)$ and $\log(1/t_a) = z(m,\gamma)$, the errors satisfy

$$\delta_a \le C_J m^3 e^{-a_{\mathrm{w}} z(m,\gamma)^2}, \qquad z_a \le C_J m^3 e^{-b_{\mathrm{DB}} m/\gamma}, \qquad b_{\mathrm{DB}} = m_{\mathrm{w}} M Q_{\mathrm{DB}}^2/24 > 0.$$

For any fixed $R > 1$, the exponential envelope has ratio

$$R^3 e^{-b_{\mathrm{DB}}(R-1)m/\gamma} \le R^3 e^{-b_{\mathrm{DB}}(R-1)n_0/\gamma_0}$$

between the sides Rm and m, for $m \ge n_0$ and $\gamma \le \gamma_0$. For the envelope $W(m,\gamma) = C_J m^3 e^{-a_{\mathrm{w}} z(m,\gamma)^2}$, let $\Delta_R = (\log R)/2$. Since $z(Rm,\gamma) = z(m,\gamma) + \Delta_R$, its ratio is exactly

$$\frac{W(Rm,\gamma)}{W(m,\gamma)} = R^3 e^{-a_{\mathrm{w}}(2\Delta_R z + \Delta_R^2)},$$

and the right side decreases in z. Also

$$\frac{\partial \log W}{\partial \log m} = 3 - a_{\mathrm{w}} z, \qquad \frac{\partial \log W}{\partial \log \gamma} = a_{\mathrm{w}} z > 0 \quad \text{when } z > 0.$$

Thus, for $m \ge n_0$ and $0 < \gamma \le \gamma_0$, one sufficiently large n_0 makes both ratios at $R = B$ at most $1/2$. Both envelopes then decrease in m, they are maximized in the coupling at γ_0, and their first terms tend to zero as $n_0 \to \infty$. The same holds for each ratio in a fixed finite family of remainders, after taking the maximum of the required minimum sides.

We increase this minimum side until $z_a \le 1/2$ for every a and $W(n_0,\gamma_0) \le 1/4$. Then

$$z_a \le \tfrac{1}{2} < 1, \qquad \delta_a \le \tfrac{1}{4} < 1, \qquad \epsilon_a = \frac{\delta_a}{1 - z_a} \le \tfrac{1}{2} < 1 \qquad (0 \le a < K). \tag{22.23}$$

Thus the hypotheses $d_t < 1$ and $\epsilon_t < 1$ of Theorem 14.7 both hold before any error bound or reciprocal bound is used. We set $D_a = \prod_{0 \le b < a}(1 - \epsilon_b)^{-1}$ for $0 \le a \le K$, with $D_0 = 1$. The output sides of the RG steps of a finite sequence, ordered increasingly, form a segment of the sequence $m, mB, mB^2, \dots$, where $m \ge n_0$. Using $\epsilon_a \le 2\delta_a$ and $-\log(1-x) \le 2x$ for $0 \le x \le 1/2$, we obtain

$$\sum_a \epsilon_a \le 4W(n_0,\gamma_0), \qquad \log D_a \le 2\sum_b \epsilon_b \le 8W(n_0,\gamma_0).$$

Increasing n_0 makes the last quantity at most one, or arbitrarily small. This proves the assertion on C_δ and the bound by the exponential envelope, with an enlarged fixed C_J. The argument is an explicit instance of the conclusion for products over endpoints in Lemma 20.5.

Step 3: positivity. The events S^a have positive probability. Initially, D_r contains a product neighborhood of the identity of positive Haar measure, on which the Wilson density is positive. At each further RG step on the torus, the cutoff domain has positive volume and the integrand is real and positive. The recursion for the stopped measures used in Theorems 14.8 and 21.12 therefore multiplies a positive constant by a positive integral at the stopping level, which is bounded below by its strictly positive trial integral over D_r. By induction, $\mu_{N,\gamma}(S^a) > 0$ for every $0 \le a \le K$. The normalized pushforward is exactly the stopped measure $\widehat{\nu}_a$, at the preceding depth aJ.

Step 4: one RG step. At the initial side, $\eta_0 = \zeta_0 = \mathbf{1}_{D_r}\mu_{N,\gamma}$. The density of η_a is the unnormalized density of the stopped measure, including the probability of the preceding conditions and all its constant factors. For $0 \le a < K$, at a target of side $m = n_{(a+1)J}$ satisfying the coordinate-domain conditions, we apply Theorem 14.7 to this density. Its functions G and $\widetilde{F}$ have identical Haar disintegration, Gaussian, and constant prefactors, and multiplying both by the probability of the preceding conditions preserves their ratio. The quantities Q_{phys} and m_{cv} of that theorem are the present Q_{DB} and m_{w}. Its number of blocks is the number $\mathsf{b}_a \le C_J m^3$ of blocks of the RG step, its largest integrated coupling is $t_a^2 = \gamma/(Mm)$, and its logarithmic cutoff is the common $p(t_a)$ at every averaging step of this RG step. The condition $t_a p(t_a) < R_{\mathrm{en}}$ follows from the fixed ceiling for the logarithmic family, and the enclosing product domain is that of the fixed parameters of the stopped measures. The exact identity $\widetilde{F}_a/G_a = \kappa_c^{\mathrm{old}}(\Omega_p^{\mathrm{w}})$ needs no smallness of the errors. Its consequence for the tail uses $d_{t_a} = z_a < 1$ from (22.23) and gives, on D_r,

$$\widetilde{F}_a/G_a \ge 1 - \epsilon_a, \qquad \epsilon_a = \delta_a/(1 - z_a), \tag{22.24}$$

with the displayed δ_a and z_a. The reciprocal bound also uses $\epsilon_a < 1$ from (22.23). The ratio bound holds at every target satisfying the coordinate-domain conditions, and integrating it, we obtain $\mathbf{1}_{D_r}(T_M^J)_{\#}\eta_a \le (1-\epsilon_a)^{-1}\eta_{a+1}$. This comparison includes the coordinate-domain conditions of the next RG step, as required in $\widetilde{F}_a$.

Step 5: induction and the failure bound. For $0 \le a < K$, if $\zeta_a \le D_a\eta_a$, then, since pushforward and restriction preserve inequalities between measures,

$$\zeta_{a+1} = \mathbf{1}_{D_r}(T_M^J)_{\#}\zeta_a \le D_a \mathbf{1}_{D_r}(T_M^J)_{\#}\eta_a \le D_a(1-\epsilon_a)^{-1}\eta_{a+1}.$$

By induction, the stated domination holds for every $0 \le a \le K$. For $0 \le a < K$, the failure bound follows from the other consequence of (22.24), namely $G_a - \widetilde{F}_a \le \epsilon_a G_a$ on

D_r. Integrating over the targets in D_r, we obtain

$$\mu_{N,\gamma}(S^a \cap A_m \cap W_m^c) \le \epsilon_a \mu_{N,\gamma}(S^a \cap A_m) \le \epsilon_a \mu_{N,\gamma}(S^a).$$

Step 6: regular events and test functions. For $0 \le a \le K$, by the hypothesis on $G_{\le aJ}$, the deterministic inclusion of Theorem 7.7 places every configuration V^{bJ}, $0 \le b \le a$, on this event in D_r. Its hypotheses include the fixed threshold for the plaquette energies and the minimum side of that theorem, and arbitrary holonomies of flat connections are allowed. Thus the pushforward of the restriction of $\mu_{N,\gamma}$ to $G_{\le aJ}$ is bounded by ζ_a. For every nonnegative bounded Borel function F, the domination implies that

$$\mathbb{E}_{\mu_{N,\gamma}}[F(V^{aJ}); G_{\le aJ}] \le D_a \int F \, d\eta_a = D_a \mu_{N,\gamma}(S^a) \mathbb{E}_{\widehat{\nu}_a} F.$$

For an arbitrary nonnegative Borel function F, we apply this to $\min(F, k)$ and let $k \to \infty$. This proves the stated inequality also when the test function has exponentially large values on the finite lattice. The probability $\mu_{N,\gamma}(S^a)$ is at most one and may be dropped, without being inverted. When $K = 0$ there is only $\eta_0 = \zeta_0 = \mathbf{1}_{D_r} \mu_{N,\gamma}$ and $D_0 = 1$, and no transition error, next target, or failure event occurs.

Step 7: absorption of logarithms. For the remaining minimum sides, we set $x = m/\gamma \ge n_0/\gamma_0$. For $k = 1, 2$, we have

$$\frac{d}{dx} \frac{(1 + \log x)^k}{x} = \frac{(1 + \log x)^{k-1}(k - 1 - \log x)}{x^2}.$$

Thus this ratio decreases to zero once $x \ge e^{k-1}$. For a given finite collection of C, k, and ε, we choose n_0 so that the ratio at n_0/γ_0 is at most ε/C for each entry. This proves all the displayed absorptions simultaneously.

Step 8: the fine lattice. Finally, we use the bound of Theorem 6.6 for exponential moments of arbitrary sets of plaquettes at the fine scale, with a single plaquette and its parameter $s = 1/2$. At $\beta_N = 4N/\gamma \ge 2$ it reads

$$\mathbb{E}_{\mu_{N,\gamma}} e^{\beta_N q(U_p)/16} \le \exp\{[C_0 + 24 \log 2 + 36 \log \beta_N / N]/8\}.$$

By the exponential Markov inequality and a union bound over the $3N^3$ positive plaquettes, we obtain the displayed bound with $C_{\mathrm{LT}} = C_0 + 24 \log 2$, enlarged to be nonnegative if necessary. For $4N/\gamma_0 \ge e$, both $1/\beta$ and $(\log \beta)/\beta$ decrease on the ray $\beta \ge 4N/\gamma_0$; evaluating each at the left endpoint gives the displayed bound for the supremum. Its right side tends to zero as $N \to \infty$, independently of the coupling. We choose the common minimum side so that it is at most $d_*/4$ for every larger N. Then the positive part of the exponent is at most $\beta_N d_*/32$, which leaves

$$-\frac{\beta_N d_*}{16} + \frac{\beta_N d_*}{32} = -\frac{\beta_N d_*}{32} = -\frac{d_* N}{8\gamma}.$$

This is the asserted bound, in which the exact correction $36 \log \beta_N / N$ was kept until this absorption, which is uniform in the coupling.

Solving $t^2 = \gamma/(Mm)$ gives $m^3 = \gamma^3 t^{-6}/M^3 \le \gamma_0^3 t^{-6}/M^3$, which proves the last counting formula. Taking the maximum of the finitely many minimum sides constructed above, and rounding up to an element of $4\mathbb{N}$, proves the claim on the common minimum side, with the dependence stated in the lemma. □

The next lemma extends these conclusions to an arbitrary finite coupling ceiling Γ, which is the form used in Theorem 23.5.

Lemma 22.13. *Fix $0 < \Gamma < \infty$ and the parameters, the buffers, the count of affected plaquettes, the finite family of stopping offsets and remainders, and the interval of s of Theorem 22.1. The exponential-moment bounds and the conclusions on decompositions of that theorem hold for $0 < \gamma \le \Gamma$. Above one further minimum side, all conclusions of Lemma 22.12 on successive restrictions, on nonnegative test functions, and on the fine lattice also hold for $0 < \gamma \le \Gamma$, with the same small-field event and identical unnormalized prefactors. One may require*

$$m \ge \max\{e\Gamma, \Gamma/g_{\mathrm{end}}^2, \Gamma/(Mg_*^2)\},$$

with g_{end} as in (22.18)*, and $C_{\mathrm{ch},J}^{\Gamma} = (1 + \log_+ \Gamma) C_{\mathrm{ch},J}^{\mathrm{base}}$. In particular, for $0 \le a < K$, with $m = n_{(a+1)J}$ and with A_m and W_m as in Lemma 22.12,*

$$\mu_{N,\gamma}(S^a \cap A_m \cap W_m^c) \le \epsilon_a \mu_{N,\gamma}(S^a) \le C_J m^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mm/\gamma}}.$$

There is no conditioning on the curvature at later scales. All constants have the dependence described in item (4) of Definition 4.14, with the fixed ceiling Γ; the constants of the local estimates and the intervals of s and of the action coefficients do not change when the ceiling is increased.

Proof. The assertion on exponential moments was proved in Theorem 22.1 for the whole interval $0 < \gamma \le \Gamma$. For the assertions on successive restrictions, we use Theorem 14.7 with the same unnormalized prefactor and the count $\mathsf{b}_a \le C_J m^3$. With $t_a^2 = \gamma/(Mm)$, its cutoff and exterior errors are exactly $\delta_a = 6\mathsf{b}_a e^{-m_{\mathrm{w}} p(t_a)^2/24}$ and $z_a = 6\mathsf{b}_a e^{-m_{\mathrm{w}} Q_{\mathrm{DB}}^2/(24 t_a^2)}$. The bounds for a finite coupling ceiling in Lemma 20.15 make $z_a \le 1/4$, $\epsilon_a = \delta_a/(1 - z_a) \le 1/2$, and the logarithm of every reciprocal product at most one, above a minimum side independent of $\gamma \in (0, \Gamma]$. All transitions have $a < K$. Every finite sequence of satisfied conditions has positive probability. Indeed, at the identity configuration its coordinate-domain conditions hold strictly and its normalized fluctuation variables vanish; thus a finite intersection of neighborhoods of the identity on which the conditions hold has positive Haar measure, and the Wilson density is positive. Iterating the comparison of one RG step as in Steps 4–6 of the proof of Lemma 22.12, we obtain the domination and every consequence for nonnegative test functions. Multiplying the failure bound for one

RG step by the probability of the preceding conditions proves the displayed bound; the event keeps only the preceding conditions and the target.

The estimate on the fine lattice from Theorem 6.6, in the form used in Step 8 of the proof of Lemma 22.12, is

$$\mu(\mathrm{FF}_0) \le 3N^3 \exp\bigl\{(C_{\mathrm{LT}} + 36\log\beta/N)/8 - \beta d_*/16\bigr\}, \qquad \beta = 4N/\gamma.$$

For $4N/\Gamma \ge e$, the monotonicity of $1/\beta$ and $\log\beta/\beta$ implies that

$$\frac{C_{\mathrm{LT}} + 36\log\beta/N}{\beta} \le \frac{C_{\mathrm{LT}}\Gamma}{4N} + \frac{9\Gamma\log(4N/\Gamma)}{N^2} \longrightarrow 0.$$

We choose the minimum side so that this is at most $d_*/4$. The exponent is then at most $-\beta d_*/32 = -d_* N/(8\gamma)$, which proves the same bound on the fine lattice uniformly for $0 < \gamma \le \Gamma$. □

SECTION 23

Uniform bounds for the small-field event

This section proves that the small-field event $S_{N,u,\gamma}$ of Definition 4.4 has probability close to one under the Wilson measure, uniformly in the number of renormalization group (RG) steps and in the coupling below a fixed ceiling. The main result is Theorem 23.10. For every $0 < \Gamma < \infty$ it provides an integer J, with $B = M^J$, and all remaining parameters of the small-field event, chosen before the coupling, such that

$$\mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big) \le C' u^3 e^{-\bar{c}u/\gamma} + C_{\mathrm{w}} u^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mu/\gamma}} \tag{23.1}$$

for all $0 < \gamma \le \Gamma$, all coarse sides $u \in 4\mathbb{N}$ above a threshold n_Γ, and all $K \ge 0$. The first term bounds the probability that some averaged configuration $V^j = T_M^j U$ has a plaquette whose energy exceeds a fixed threshold d_*. The second term bounds the probability that the curvature is small at every scale, but the coordinate-domain conditions or the cutoffs $|w_{k,i,e}| \le A_{\mathrm{w}} \log(1/t_k)$ on the normalized fluctuation variables fail in some RG step. This is the bound on large fields, Theorem 2.7, which lets us pass from the conditioned Wilson measure $\nu_{N,u,\gamma}$ to the Wilson measure by (3.1).

Two regimes of (23.1) are used later. For a fixed coarse side u, the probability of S^c is smaller than every power of γ as $\gamma \to 0$, uniformly in K (Corollary 23.11); this is used in the weak-coupling analysis. For a fixed ceiling Γ, it tends to zero as $u \to \infty$ through the sides nB^a, uniformly in $\gamma \in (0, \Gamma]$ and K; this is used to remove the conditioning in the construction of the continuum measure. The logarithmic cutoff limits the rate: by Theorem 23.12, at a fixed coarse side, $\sup_K \mu(S^c)$ is of order $\exp(-c \log^2(1/\gamma))$, and a matching lower bound holds already for one RG step.

The principal antecedent of the present construction is Bałaban's proof of ultraviolet stability for three-dimensional lattice gauge theories [15]. The estimate proved here concerns the probability of the specific event $S_{N,u,\gamma}$. Its proof has two independent parts: a bound on large curvature under the Wilson measure itself, and a bound on the first failure of a coordinate-domain condition or of a cutoff. Both are described at the beginning of Section 23.4.

The other task of this section is the choice of parameters. The integer J, the analytic radii, the support radius r of B_r and D_r, the widths of the boundary layers of the large-field decompositions, the cutoff amplitude A_{w}, the coupling bound g_*, and the minimum coarse side must satisfy all hypotheses of the results of Sections 13 to 15 and 18 to 22, and these hypotheses refer to one another. Theorem 23.2 lists the choices in an order in

which each choice uses only quantities fixed before it, and Table 23.1 summarizes this order. The main point is that J is chosen using only constants that are uniform in J, and is never enlarged afterward.

Sections 23.1 and 23.2 contain the elementary inequalities and the statements on the choice of parameters, Sections 23.3 to 23.5 the probability estimates, Sections 23.6 and 23.7 the proofs of the parameter theorems, and Section 23.8 the extension to every sufficiently large J, which is needed in Section 29 to pass from the subsequences $n_* B^K$ to the full sequence nM^k.

The main inputs are the moment bounds for coarse actions and the comparison for initial segments of Section 22; the large-field decompositions and their pointwise support bounds for the normalized fluctuation variables in Section 21; the regional estimates of Sections 18 to 20; the recursion for the effective interaction on the torus and the reconstruction maps of Sections 13 to 15; and the chessboard and curvature estimates of Sections 6 and 7. The results of this section are used in Sections 28 and 29 to remove the conditioning and to construct the continuum measure, in Sections 31 and 32 at fixed coupling, and in Sections 34 to 36 at weak coupling.

Throughout, $L > 0$ and an odd integer $M \geq 3$ are fixed. We write $h_n = g_n^2 = \gamma/n$ for the lattice coupling at side n. The letter p with an argument denotes the cutoff radius $p(t) = A_{\mathrm{w}} \log(1/t)$ (written $p_{\mathrm{w}}(t)$ in Definition 4.4); a subscript p on a configuration denotes a plaquette.

23.1 Elementary inequalities

We use the following lemma repeatedly to choose a coupling bound and a minimum side. It states that powers of logarithms are dominated by positive powers, and that sums over geometrically growing sides are dominated by their first term.

Lemma 23.1. *Fix $0 < \gamma_0 \leq 1$, a finite set of ratios $R > 1$, finitely many positive constants a, c, C, ε, and finitely many nonnegative integers k. Then there exists $0 < g_* < e^{-1}$ such that each of the finitely many inequalities $Ct^{\varepsilon}[1 + \log(1/t)]^k \leq c$ holds for all $0 < t \leq g_*$. There exists m_0, independent of $\gamma \in (0, \gamma_0]$, such that each of the finitely many inequalities*

$$C[1 + \log(m/\gamma)]^k \leq \varepsilon m/\gamma$$

holds for all $m \geq m_0$ and $0 < \gamma \leq \gamma_0$. For a fixed $M > 1$ define

$$X_c(m,\gamma) = m^3 e^{-cm/\gamma}, \qquad W_a(m,\gamma) = m^3 e^{-az(m,\gamma)^2}, \qquad z(m,\gamma) = \frac{1}{2}\log(Mm/\gamma).$$

After increasing m_0, each of these functions is decreasing in $m \geq m_0$ and increasing in $\gamma \in (0, \gamma_0]$, and its value at Rm is at most half of its value at m. Consequently

$$\sum_{h \geq 0} X_c(mR^h, \gamma) \leq 2X_c(m,\gamma), \qquad \sum_{h \geq 0} W_a(mR^h, \gamma) \leq 2W_a(m,\gamma),$$

and the right sides tend to zero as $m \to \infty$, uniformly in $\gamma \in (0, \gamma_0]$. Finally, let $\delta > 0$, and let numbers $\epsilon_h \geq 0$, attached to the sides mR^h, be bounded by fixed finite sums of these functions. Then, after increasing m_0, we have $\log \prod_h (1-\epsilon_h)^{-1} \leq \delta$ for all $m \geq m_0$. The thresholds g_* and m_0 depend only on the data fixed in the statement.

Proof. Set $y = \log(1/t)$. For $\varepsilon > 0$ and an integer $l > k$ we have $e^{\varepsilon y/2} \geq (\varepsilon y/2)^l / l!$ for $y > 0$, which implies $e^{-\varepsilon y}(1+y)^k \to 0$ as $y \to \infty$. Then each inequality holds on a half-line $y \geq y_*$, that is, for $0 < t \leq g_*$, and we take the smallest of the finitely many thresholds. The same calculation with $y = \log x$ shows that $(1+\log x)^k / x \to 0$ as $x \to \infty$. Since $x = m/\gamma \geq m_0/\gamma_0$, the supremum of this ratio over the relevant half-line tends to zero as $m_0 \to \infty$, which gives the second assertion.

With $\Delta_R = (1/2)\log R > 0$, the two ratios are

$$\frac{X_c(Rm,\gamma)}{X_c(m,\gamma)} = R^3 e^{-c(R-1)m/\gamma}, \qquad \frac{W_a(Rm,\gamma)}{W_a(m,\gamma)} = R^3 e^{-a(2\Delta_R z + \Delta_R^2)}. \tag{23.2}$$

Their suprema over $m \geq m_0$ and $\gamma \leq \gamma_0$ are attained at $m = m_0$ and $\gamma = \gamma_0$, and tend to zero as $m_0 \to \infty$. The derivatives of $\log X_c$ and $\log W_a$ with respect to $\log m$ are $3 - cm/\gamma$ and $3 - az$, which are negative once m_0 is large; their derivatives with respect to $\log \gamma$ are cm/γ and az, which are positive in the same range. Taking logarithms shows that $m^3 e^{-cm/\gamma_0} \to 0$ and $m^3 e^{-a\log^2(Mm/\gamma_0)/4} \to 0$. The sums are bounded by geometric series with ratio $1/2$. Finally, for $0 \leq \epsilon \leq 1/2$ we have

$$-\log(1-\epsilon) = \int_0^\epsilon \frac{dv}{1-v} \leq 2\epsilon.$$

We increase m_0 so that every ϵ_h is at most $1/2$ and $2\sum_h \epsilon_h \leq \delta$. This proves the last assertion, also for finite sublists of the ϵ_h taken in either order. All thresholds depend only on the fixed data of the lemma. □

23.2 Choice of the parameters

The constants in the next theorem are those of the cited earlier results, evaluated on their larger construction domains before any of the later restrictions. In particular, q_{CA} is the contraction factor for the zero-fluctuation pullback of the effective interaction on the torus in Theorem 13.5, C_G is the constant bounding the part of the effective interaction generated by one RG step, and $Z_\bullet$ are the majorants for the absolute coefficient sums of the background-field map at zero fluctuation from Theorem 13.2. In Step 8 of Section 23.6, we use Theorem 14.6 to identify the recursion controlled by these constants with the recursion defining the small-field event. The number $q_{\mathrm{raw}} > 0$ is a radius reserved for the free logarithmic coordinates. The symbols A_J, R_J denote the coefficient bounds for the reconstruction maps $\widehat{\mathscr{R}}$ in Lemma 15.15; output constants denoted by the same letters

elsewhere can be enlarged to a common bound at fixed J. The normalization constant Λ is kept separate from the bound $\mathcal{L}_V$ for the conditional fluctuation potential. A subscript $\bullet$ stands for either of the two analytic radii, ρ for the effective interaction and ρ_{cl} for the effective classical action, and every condition carrying $\bullet$ is imposed for both.

The radii play different roles. The support radius r defines B_r and D_r; $R_{\mathrm{buf}} = C_{\mathrm{map},J} r$ is the larger radius of coarse fields on which the analytic recursion is defined; ρ and ρ_{cl} are analytic radii of coefficient norms; R_{raw} is the radius of the larger domain of free logarithmic coordinates; r_{eff} is the working radius for the zero-fluctuation reconstruction; $0 < P < Q_{\mathrm{DB}}$ are radii, proportional to r, for the unscaled fluctuation coordinates $\zeta_i = g_i w_i$ in the small-field construction; R_W bounds the unscaled fluctuation coordinates allowed in the regional estimates, and $R_{\mathrm{en}} \le R_W$ is the radius of the enclosing product domain; β_{rec} bounds the frozen coordinates of the large-field decompositions.

In applying Section 15.3, its generic torus interaction exponent a_{sp} is instantiated by $a_{\mathbb{T}}$. Its contraction and the construction and convergence of the torus effective interactions are used at that fixed exponent. The smaller regional exponent $a = a_{\mathrm{sp}}$ chosen below is used for boundary estimates; bounds and convergence of the already constructed torus interactions pass to it by monotonicity of the coefficient norms. No contraction on an entire ball at the weaker exponent is inferred from this monotonicity.

Theorem 23.2. *Fix $L > 0$, an odd integer $M \ge 3$, and $0 < \gamma_0 \le 1$. There exist an integer $J \ge 1$, with $B = M^J$, a tuple Θ of the parameters listed in items* (1)–(10) *below, a number $0 < g_* < e^{-1}$, and an integer $n_0 \in 4\mathbb{N}$, all chosen before the coupling, with the following property. For every*

$$0 < \gamma \le \gamma_0, \qquad u \in 4\mathbb{N},\ u \ge n_0, \qquad K \in \mathbb{N}_0, \qquad N = uB^K,$$

the same J, Θ, and reconstruction maps satisfy the local, moment, and scalar conditions of the results cited below; only the running couplings and the cutoff radii $p(t)$ are evaluated at γ. In particular, the parameters satisfy the hypotheses of Theorem 19.9 on uniform incoming radii, on boundary compatibility, and on conditional displacements, and the bounds of Theorem 20.14 on logarithmic fluctuation radii and on frozen coordinates. In the large-field decompositions only complete renormalization group steps are integrated; remainders modulo J and short scale intervals are treated by the composed insertions and the bounded side-ratio clauses of Theorem 22.1. The coefficient majorants on the torus and the recursion for the effective interaction are those of Theorems 13.2 and 13.5, and Theorem 14.6 identifies this recursion with the effective interactions of the stopped measures used in Theorem 22.1. The geometric hypotheses of Theorem 21.5 hold, including those of Theorem 21.3 for the neighborhoods in which gauge transformations are extended, the closed complex of tested cells at the initial step, and the protection of all older endpoints and tree roots attached to frozen records.

The parameters can be chosen in the order of items (1)–(10). *Each item uses only constants provided by the cited results and quantities fixed in earlier items.*

(1) The number J. *Fix the spatial decay rate $a_{\mathbb{T}}$ of the torus estimates with $0 < a_{\mathbb{T}} < a_{\mathrm{cen}}/8$, where a_{cen} is the J-uniform rate of the bounds of Lemma 11.9 (the contraction of Theorem 11.10 for the aligned representative holds only at such output rates). Fix also all larger spatial, analytic, and differentiation domains and the analytic-radius enlargement factor K_* needed for the connected-network estimates of Theorem 20.9. Enlarge the J-uniform constant C_{CP} of part (3) of Theorem 15.12 to include the reconstruction tolerances. In the notation of that theorem, it bounds the value and the first three coarse-field derivatives of the pair formed by $B\widehat{\Phi}(\theta,0)$ and its first spatial differences $\nabla_{1/B}B\widehat{\Phi}(\theta,0)$, with the row sums, volume-normalized column sums, and polydisc of coarse fields specified there. Bounds for derivatives with respect to the fluctuation inputs are chosen after J is fixed. Let $Z_\bullet(t) \le C_\bullet t/(1-C_\bullet t)$ be the J-uniform majorants for the absolute coefficient sums of the background-field map at zero fluctuation on the torus. Choose J so that*

$$B \ge B_{\mathrm{geom}}, \qquad q_{\mathrm{CA}} < 1/4, \qquad q_{\mathrm{CP}} \le 1/4, \qquad M^{-J} \le \delta_0(a_{\mathbb{T}}),$$

where $q_{\mathrm{CA}} = C_P(a_{\mathbb{T}})/B$ (denoted q_P in Theorem 13.5), $q_{\mathrm{CP}} = C_{\mathrm{CP}}^2/B$, and $C_P(a_{\mathbb{T}})$ and $\delta_0(a_{\mathbb{T}})$ are the constants of Theorem 11.10, and so that the finitely many linear image conditions on the torus hold with strict inequality. These include $C_\bullet K_*/B < 1/32$, which permits the later bound $B^{-1}Z_\bullet(K_*\rho_\bullet) \le \rho_\bullet/16$, and the two conditions (23.25) *on spatial rates. Here B_{geom} is the constant of Theorem 20.9. None of these lower bounds on B uses a constant of the regional estimates or of the effective-interaction estimates computed at fixed J.*

(2) Coordinate domains at fixed J. *Choose the product domains of Lemmas 19.3 and 19.7. In their notation choose $v_{\mathrm{free},J} > 0$ with*

$$Q_J v_{\mathrm{free},J} < \lambda_J/64, \qquad I_J Q_J v_{\mathrm{free},J} < r_{z,J}/64,$$

and choose R_{raw} strictly below the radii in those results for the identities on overlapping gauge charts and for the local constraint coordinates, and strictly below

$$\min\{\min(r_{\Gamma,J}, r_E)/(64G_J),\ \lambda_J/(64Q_J),\ r_{z,J}/(64I_JQ_J),\ v_{\mathrm{free},J}/(64C_{\mathrm{lift},J})\}.$$

For each initial segment of conditional integrations, fix first a neighborhood of its conditional constrained minimizer, then a smaller domain for the parameters of the later integrations, and then a common smaller product domain for the free logarithmic coordinates of the successive substitutions. These domains preserve real values on real arguments, the conditional displacement bound $r_I/4$, and the bound $1/2$ for the error of the preconditioned Hessian. All inverses are inverses of constrained Hessians in the free coordinates.

(3) Analytic radii and coefficient bounds. *Fix $\rho, \rho_{\mathrm{cl}} > 0$, all inner, middle, and outer domains of the reconstruction maps inside the larger construction domains, a bound E^{cl} for the effective classical action on the torus, and a bound*

$$E \ge 2C_G/(1-q_{\mathrm{CA}})$$

for the effective interaction. Set aside the inequality

$$C_{\mathrm{lift},J}[K_* \max(\rho, \rho_{\mathrm{cl}}) + r + \beta_{\mathrm{rec}} + q_{\mathrm{raw}}] < R_{\mathrm{raw}}/8,$$

in which the first summand, fixed now, is required to be below $R_{\mathrm{raw}}/64$; the other summands are made small by the later choices of q_{raw}, of β_{rec} in item (5)*, and of r in item* (6)*. Compute the coefficient constants A_J, R_J of the reconstruction maps on these larger domains, after one fixed enlargement, before the working and support radii are chosen; these constants and the bounds for the limiting maps do not depend on those radii. Then choose the regional rates as in Theorem 20.9: $a = a_{\mathrm{sp}} > 0$, $\alpha = \alpha_{\mathrm{bd}} = a/16$, equal classical rates, $m_{\mathrm{rate}} = C_*^{\mathrm{geom}}(a + \alpha)$, and $b_1 = 8C_*^{\mathrm{geom}}\alpha$, with*

$$m_{\mathrm{rate}} + b_1 < \min(c_{\mathrm{prim},J}, Ba)/C_{\mathrm{alloc}}, \qquad b_1 < B\alpha/C_{\mathrm{alloc}}, \qquad \sqrt{3}\,(a + \alpha) < a_{\mathbb{T}}.$$

Here $c_{\mathrm{prim},J}$ is the reserve for the intrinsic tensors and the unperturbed weighted reconstruction images in the graph metric of Section 20.1, obtained from the Euclidean exponents by dividing by $\sqrt{3}$. It includes the positive exponent obtained once from interpolation of the unperturbed coefficient bounds and is chosen no larger than the intrinsic exponent c_J of Theorem 19.9. Inherited perturbative networks use the separate inequalities against Ba and $B\alpha$. Decreasing a changes neither $a_{\mathbb{T}}$ nor J.

(4) Perturbation bounds and the fluctuation radius. *Choose a preliminary bound $\sigma_0 > 0$ for perturbations that are real on real arguments, on the larger product domains. Compute all local constants of the output, sequential, image, curvature, and mean estimates, and of the estimates of the support of the normalized fluctuation variables, uniformly for $\mathcal{X}^{\mathrm{cl}}(\overline{P}) \le 2\sigma_0$ and $\mathcal{X}(f) \le \sigma_0$. In particular fix*

$$A \ge 1, \quad C_0 \ge 2A^3, \quad C_1, \quad C_{\mathrm{en},J}, \quad \mathcal{L}_V = C_{V,J}(1 + E + \sigma_0) \ge 1,$$

the image constants $C_{T,\bullet}, C_{\Delta,\bullet}, C_{F,\bullet}, C_{P,\bullet}$, and all radii for free logarithmic coordinates that occur in Lemma 20.3. Fix also the geometric constants A_8, C_B, K_B, C_{col}, $C_{\mathrm{conv},J}$, and the mesh and interior-restriction factors of the initial step. None of these constants depends on the widths chosen later (of boundary layers, removed layers, protected neighborhoods, and large cells). Choose R_W below the full minimum in Lemma 20.3, including $1/(16C_0)$, both image bounds, both product domains, and the conversion to the joint solution neighborhood. Then fix

$$0 < \sigma_* \le \sigma_0, \qquad \sigma_*\rho_{\mathrm{cl}} < \lambda_* R_W/(6C_{\mathrm{en},J}), \qquad \lambda_* = M^{-(J-1)/2},$$

all margins for the preconditioned forces and displacements, and $C_{P,\bullet}\sigma_*\rho_{\mathrm{cl}} < \rho_\bullet/64$. This choice precedes item (5)*, and it is not replaced by taking r and β_{rec} small.*

(5) Boundary radii, removed layers, and frozen coordinates. *Fix the outer boundary radius $R_{\partial,\bullet}$, the inner radius $\rho_{\partial,\bullet} = R_{\partial,\bullet}/K_*$, and the width w_{bd} of the boundary layer so that the zero-fluctuation image bound satisfies $A_{0,\bullet} \le \rho_\bullet/4$. Specifically, impose*

$$B^{-1}Z_\bullet(K_*\rho_\bullet) \le \rho_\bullet/16, \qquad C_{\Delta,\bullet}C_{\rho,\bullet}R_{\partial,\bullet} \le \rho_\bullet/16,$$

$$C_{\Delta,\bullet}C_{\rho,\bullet}e^{-a_{W,\bullet}w_{\mathrm{bd}}/2}K_*\rho_\bullet \le \rho_\bullet/16, \qquad C_{T,\bullet}K_*\rho_\bullet \le 1/4,$$

together with the full condition

$$A_{0,\bullet} \le B^{-1}Z_\bullet(K_*\rho_\bullet) + C_{\Delta,\bullet}C_{\rho,\bullet}\big[R_{\partial,\bullet} + e^{-a_{W,\bullet}w_{\mathrm{bd}}/2}K_*\rho_\bullet\big] + C_{F,\bullet}\beta_{\mathrm{rec}} + C_{P,\bullet}\sigma_*\rho_{\mathrm{cl}} \le \rho_\bullet/4.$$

Here $C_{F,\bullet}$ and $C_{P,\bullet}$ are the image constants for frozen coordinates and for perturbed conditional constrained minimizers on the larger domains. Choose the width $s_{\mathrm{er}} \ge w_{\mathrm{bd}}$ of the layer removed after each renormalization group step so that

$$2e^{-\alpha(s_{\mathrm{er}}-1)}[A_J(1+E) + K_J\sigma_* + 1] \le \sigma_*/2, \qquad 2e^{-\alpha(s_{\mathrm{er}}-1)}[A_J^{\mathrm{cl}} + K_J^{\mathrm{cl}}\sigma_*] \le \sigma_*/2.$$

Choose the bound $\beta_{\mathrm{rec}} > 0$ for frozen coordinates below all coordinate-domain, product, and boundary radii, below one quarter of $\min\{\rho_\bullet, \rho_{\partial,\bullet}\}$, and below $\lambda_* R_W/(6C_{\mathrm{en},J})$. Impose the bounds of Lemma 18.9 for the substitution of frozen coordinates and for the boundary-completion differences, in particular

$$C\beta_{\mathrm{rec}}E_\bullet/\rho_\bullet \le \sigma_*/8, \qquad |\xi_j^\sharp| \le \phi_\bullet(j)/4, \qquad z_t, z_t^{\mathrm{cl}} \le \sigma_*/4,$$

including the substitution at the initial step, where C denotes the larger-domain constants of Lemma 18.9. Require also $C_{F,\bullet}\beta_{\mathrm{rec}} < \rho_\bullet/64$.

(6) Working radius and support radius. *Set $\alpha_* = C_{\mathrm{CP}}/B$ and choose*

$$\beta_J \ge \max\{1, R_J^{-1}, A_J/(\alpha_* R_J^2)\}.$$

Choose a working radius $r_{\mathrm{eff}} > 0$ inside the domains of item (3) *with $\beta_J r_{\mathrm{eff}}/(1-\alpha_*) < 1/8$. Then choose the support radius $r > 0$ so that*

$$r < r_{\mathrm{eff}}/4, \quad r < \beta_{\mathrm{rec}}/C_{\mathrm{conv},J}, \quad r < \lambda_* R_W/(6C_{\mathrm{en},J}), \quad (5M-3)(10M)^{J-1}r < \pi,$$

together with the fixed coordinate-domain and chart margins of the earlier constructions, in particular those of Definition 14.4. Set

$$R_{\mathrm{buf}} = C_{\mathrm{map},J}r, \qquad C_{\mathrm{map},J} \ge (10M)^J, \qquad r < R_{\mathrm{buf}} < \rho.$$

There exist radii $0 < P < Q_{\mathrm{DB}}$, proportional to r, such that every reconstruction from a coarse field e^v with $|v| \le r$ and unscaled fluctuation coordinates with $\max_i|\zeta_i| \le Q_{\mathrm{DB}}$ has

all link logarithms smaller than $r/2$. With $\mathscr{L}_{\mathrm{en}}$ the enclosing constant of Definition 14.4 and C the constant of its curvature bound, require $C\mathscr{L}_{\mathrm{en}} r/(1-C\mathscr{L}_{\mathrm{en}} r)^3 < 1/2$; the condition including the coupling is completed in item (8)*. Define*

$$\delta_{\mathrm{en}} = C_{\mathrm{en},J}(r + \beta_{\mathrm{rec}} + \sigma_* \rho_{\mathrm{cl}}), \qquad R_{\mathrm{en}} = 2\delta_{\mathrm{en}}/\lambda_* \le R_W.$$

On the entire enclosing product domain,

$$\frac{C_{T,\bullet} R_{\mathrm{en}}}{1 - C_{T,\bullet}(K_* \rho_\bullet + R_{\mathrm{en}})} + C_{\Delta,\bullet} R_{\mathrm{en}} \le \rho_\bullet/8, \qquad 1 - C_{T,\bullet}(K_* \rho_\bullet + R_{\mathrm{en}}) \ge 1/2.$$

Together with item (5)*, the image under the reconstruction majorant at spatial exponent zero is at most $3\rho_\bullet/8$. The margin of half the radius in the weighted norm is kept, either by the same bounds for weighted sums or by the interpolation of coefficient bounds in Lemma 20.3, whose positive exponent was included before a and α were chosen.*

(7) Geometric widths and the curvature threshold. *Set*

$$R_{\mathrm{cell}} = \lceil A_8/r \rceil, \quad W = 8BR_{\mathrm{cell}}, \quad \varrho = \max(s_{\mathrm{er}}, W), \quad r_F = \lceil \varrho/B \rceil.$$

If desired, r may be taken small enough in item (6) *that $W \ge s_{\mathrm{er}}$. The protected sets contain all lifted endpoints of older frozen links and transports and all tree roots, in the lattices where they were introduced. At the initial renormalization group step, which starts from the Wilson measure, use the block-aligned mesh $S \asymp_J R_{\mathrm{cell}}$, one full layer of large cells (the closed cells of side R_{cell} of Theorem 21.3), and the interior restriction by the full dependence radius of the constraint, tree, and transport paths. These choices fix the constant F_J counting affected plaquettes and the constant $C_{\mathrm{ch},J}$ for the normalization factors attached to threshold failures. Here a threshold failure is a bad plaquette, that is, a plaquette selected by the tests of a large-field decomposition because its energy is at least d_*, together with the neighborhood excluded because of it. Fix $0 < \delta_* \le 1$ for the curvature threshold of Theorem 7.3, as in its proof, and then choose*

$$0 < d_{\mathrm{cut}} \le \min\{d_{\mathrm{SG}}, B_*/(96M^3), \delta_*^2/(2\pi^2)\}, \qquad d_* = d_{\mathrm{cut}}/2, \qquad \pi\sqrt{2d_*}\, S^2 \le 1.$$

The threshold d_{SG}, with its side condition, gives membership in D_r with link bound $r/2$ by Theorem 7.7. Use the constants $K_M, K_1, \kappa_0, c_A^{\mathrm{act}}, \alpha_W$ of Theorem 22.4 for arbitrary fields, $K_W = 48K_1/\alpha_W$, and $\theta = \lambda = 1/2$.

(8) Cutoff amplitude and coupling bound. *Fix $A_{\mathrm{w}} \ge 1$ in the range allowed by Theorem 20.14 (no earlier choice depends on A_{w}, so it may be fixed at any earlier stage), and set $p(t) = A_{\mathrm{w}} \log(1/t)$ and $p^+ = p(t) + A_{\mathrm{w}}(J-1)\log M/2$. Choose $g_* \le t_* < e^{-1}$ such that, for every $0 < t \le g_*$,*

$$C_0 t_*^2 \mathcal{L}_V \le 1/16, \qquad C_1 t_* (\mathcal{L}_V + 2) < 1/(4\sqrt{3}), \qquad tp(t) < P, \qquad tp^+ < R_{\mathrm{en}},$$

$$C_T t^2[p(t)^4 + Ep(t)^2] \le (1 - q_{\mathrm{CA}})/2, \qquad C_J t^2 (E + \sigma_*)[p(t)^4 + (E + \sigma_*)p(t)^2] \le 1,$$

the same two inequalities with p^+ in place of $p(t)$,

$$t \le \frac{c_{z,J}}{2[(p^+)^3 + p^+ + (E + \sigma_*)p^+]}, \qquad \mathfrak{s}(t) \le \epsilon_J, \qquad \frac{C[\mathscr{L}_{\mathrm{en}} r + t^2(1 + E)]}{(1 - C\mathscr{L}_{\mathrm{en}} r)^3} \le \frac{1}{2},$$

and the strict interaction and Jacobian margins of the earlier product theorems with their finitely many bounds for free coordinates and for parameter derivatives of the normalized log density. Here the parameter derivatives are coarse-field derivatives on a fixed fluctuation product: the running couplings, the cutoff radii, and the real frozen data are held fixed. For such a derivative ∂ and $k = Z^{-1}e^{-V}$ on that product, $\partial \log k = -\partial V + \mathbb{E}_k[\partial V]$; the applicable bounds are those of Theorem 13.10 and their regional versions. Each remaining bound is a fixed estimate of the form $Ct^{\varepsilon} \log^k(1/t) \le c$ with $\varepsilon, c > 0$. Require in addition

$$g_*^2 \le \alpha_W, \qquad 2\kappa_*/g_*^2 \ge 1, \qquad a_0 g_* < \min(r/2, 1)$$

for a fixed positive constant a_0 of the tree trial configuration. The total error in the Hessian of the conditional fluctuation potential on the enclosing product domain is then at most $8/27$.

(9) Stopping levels, the insertion interval, and the moment constants. *Fix the finite family of stopping levels, the treatment of each of the J possible remainders modulo J, their side-ratio bound χ_*, and the conversion constant K_{src}. Include both functions whose exponential moments are bounded in Theorem 22.1 and Corollary 22.9, namely the predecessor action $Q_j^{\mathcal{T}}$ of Section 23.3 and the total Wilson action S_W/g^2; we call them the predecessor insertion and the total-action insertion. Enlarge F_J once so that it counts all plaquettes affected by both. Only then choose $s_0 > 0$ in the interval of Theorem 22.1, imposing also*

$$s_0 \le \frac{\kappa_0}{2\max\{K_M, 48K_1\}}.$$

The coefficient of the Wilson action ranges over $[1/4, 1/2]$, and the bounds for the stopping partition-function ratio and its numerator are fixed for that interval. Set

$$s_b = s_0/2, \qquad a_b = 4s_b d_*/c_E, \qquad 0 < \bar{c} \le \min\{d_*/8, a_b/2\}.$$

Compute C_v, C_s, C_b from Theorem 22.1, which places no restriction on the number of averaging steps, keeping $C_s \ge 9\chi_*^2$; these constants may be enlarged for the total-action insertion by Corollary 22.9 and the same proof. Fix the value $C_\delta = 1$ for the bound on the logarithm of the reciprocal products in Lemma 22.12; it is achieved by the minimum side in item (10)*. Set $C_b' = C_b + 1 + \log 2$; a larger fixed choice is allowed.*

(10) The minimum side. *Finally choose $n_0 \in 4\mathbb{N}$ above all fixed geometric, curvature-threshold, mesh, flat, and scalar thresholds, with*

$$n_0 \ge \max\{4, e\gamma_0, \gamma_0/g_{\mathrm{end}}^2, 8R_{\mathrm{cell}}, 8S\}, \qquad 0 < g_{\mathrm{end}} \le \min\{1, g_*, g_{\mathrm{anc}}, \sqrt{\alpha_W}, \sqrt{2\kappa_*}\},$$

where g_{anc} is the coupling bound for the stopping partition-function ratio of Theorem 22.4, uniform for coefficients in $[1/4, 1/2]$. Keep every fixed margin near flat configurations, in particular $2\pi/n_0 < \min\{r/2, r_{\mathrm{flat}}\}$ with strict inequality, where r_{flat} is fixed by Lemma 14.9. Choose n_0 so that the ratios of successive terms in the error sums at the endpoints and in the sums over large-field decompositions are at most $1/2$, the reciprocal products have logarithm at most $C_\delta = 1$, and

$$\mathcal{A}_\gamma(m) = 24C_v M^3 + \frac{4C_s M^2}{m}\log(Mm/\gamma) + \frac{8C_b'}{m^3} \le \frac{a_b m}{2\gamma} \qquad (m \ge n_0,\ 0 < \gamma \le \gamma_0). \tag{23.3}$$

Impose also $M^3 e^{-\bar{c}(M-1)n_0/\gamma_0} \le 1/2$ and the threshold at the initial scale in Lemma 22.12. All these are lower bounds on n_0 in terms of previously fixed constants. No bound on the number of averaging steps is imposed.

The reconstruction maps of Theorem 15.12 are fixed once on the countable family of pairs of finite lattices. Their joint coefficient bound and β_J are obtained before r_{eff} and r; a later refinement of thresholds only increases the finite deterministic thresholds of these maps and does not enlarge that bound. The prescription for the entire indexed family is fixed once and does not depend on γ, on the field, on observations, or on a later comparison index. Its members and their deterministic thresholds are indexed by the pair of finite lattices and the preceding classical depth; no choice is repeated when the coarse side changes. Write $\widehat{C}$ for the group-valued reconstruction and $\widehat{P}(v) = \operatorname{Log}\widehat{C}(e^v, 0)$ for the logarithm of its value at zero fluctuation on the active variables. At every preceding classical depth these maps are majorized by

$$\varphi(z) = \frac{\alpha_* z}{1-\beta_J z}, \qquad \sup_{|v|\le r_{\mathrm{eff}}} \|D\widehat{P}(v)\|_{\mathrm{root}} \le a_{\mathrm{map}} = \frac{\alpha_*}{(1-\beta_J r_{\mathrm{eff}})^2}, \qquad a_{\mathrm{map}}\sqrt{B} \le \frac{32}{49} < 1. \tag{23.4}$$

Here the row sums and volume-normalized column sums follow the conventions of Lemma 15.15. We use its common majorant as a_{map}, which gives $q_{\mathrm{anc}} = B\alpha_*^2 \le B a_{\mathrm{map}}^2 < 1$ as required in the later geometric summations. All constants of the full joint estimates and of the estimates for the gauge transformations of the reconstruction maps are constants at fixed J. The choice of J uses only the J-uniform geometric constants and torus rates, the J-uniform series Z of Lemma 11.9, and the finitely many coarse-field derivative bounds of the reconstruction maps at zero fluctuation.

For every $0 \le a < K$, on the event that the initial condition and the first a renormalization group steps pass the tests of the small-field event, the next J averaging steps, from side uB^{K-a} to side uB^{K-a-1}, avoid the exceptional set of T_M: no based loop holonomy in (4.3) equals $-I$, even if a later test fails. No assertion is made about a further averaging step at $a = K$. Indeed, every closed path of such a holonomy has length at most $5M - 3$, and one averaging step increases a bound on link distances by at most the factor $10M$. Consequently every coarse field produced by the coordinate construction of the small-field event equals the iterate T_M^J of the finer field. The gauge transformations of the reconstruction maps depend only on the coarse field and are

the identity at the tree roots; they preserve this identity and all conditional kernels, cutoffs, and Jacobians.

The local constants have the dependence described in item (3) of Definition 4.14, and do not depend on the widths chosen later. The probability constants and n_0 may depend on L, γ_0, M, J, Θ, and the fixed family of stopping levels, as in item (4) of Definition 4.14; they do not depend on γ, u, N, K, the preceding classical depth, or the regions, frozen coordinates, and numbers of elapsed renormalization group steps of a large-field decomposition. Neither J nor r is changed at a different coarse side. The analytic radii are restored to their full values, after removal of a boundary layer, only at the end of complete renormalization group steps; remainders modulo J are treated by the fixed insertions and side-ratio bounds.

Item	Quantities chosen	Determined by
(1)	J, $B = M^J$	J-uniform constants $a_{\mathbb{T}} < a_{\mathrm{cen}}/8$, $\delta_0(a_{\mathbb{T}})$, K_*, C_{CP}, $C_\bullet$, $C_P(a_{\mathbb{T}})$, B_{geom}, $C^{\mathbb{T}}_{\mathrm{geom}}$, $b_{\mathbb{T}}$, C_{cen}
(2)	$v_{\mathrm{free},J}$, R_{raw}, solution neighborhoods	J; constants of Section 19 at fixed J
(3)	ρ, ρ_{cl}, E^{cl}, E; A_J, R_J; a, α, m_{rate}, b_1	items (1)–(2); C_G, q_{CA}, $c_{\mathrm{prim},J}$, C_{alloc}, C^{geom}_*
(4)	σ_0; local and geometric constants; R_W; σ_*	items (1)–(3); Sections 20 and 21
(5)	$R_{\partial,\bullet}$, $\rho_{\partial,\bullet}$, w_{bd}; s_{er}; β_{rec}	items (1)–(4); Lemma 18.9
(6)	β_J, r_{eff}; r, R_{buf}; P, Q_{DB}; R_{en}	items (1)–(5)
(7)	R_{cell}, W, ϱ, r_F, S; F_J, $C_{\mathrm{ch},J}$; δ_*, d_{cut}, d_*	items (1)–(6); Sections 7, 21, and 22
(8)	A_{w}, t_*, g_*	items (1)–(7); Lemma 23.1
(9)	stopping levels, χ_*, enlarged F_J; s_0; s_b, a_b, $\bar{c}$; C_v, C_s, C_b, C'_b	items (1)–(8); Section 22
(10)	g_{anc}, g_{end}, n_0	items (1)–(9); Lemmas 23.1 and 23.4

Table 23.1. The order of the choices in Theorem 23.2. Each row uses only the quantities of the earlier rows and the constants of the cited results, evaluated on their larger construction domains. The amplitude A_{w} may be fixed at any stage before item (8).

We prove Theorem 23.2 in Section 23.6. The parts of the inequalities in items (3) and (6) reserved for r, β_{rec}, q_{raw}, and the coupling are filled by later upper bounds on these small parameters, so no cycle arises.

Remark 23.3. The parameters of Theorem 23.2 also allow one to fix, before the coupling, the exponents $\alpha_{\mathrm{fl}} = 7/12$ and $\sigma_{\mathrm{sm}} = 5/24$ of Definition 4.8 and

$$\tau_{\mathrm{tail}} = \frac{1}{2}\min\Big\{\frac{5}{24}, \frac{1}{2}, -\frac{\log q_{\mathrm{CP}}}{\log B}\Big\} > 0,$$

together with smaller positive losses in the heat-kernel estimates. These numbers are positive since $q_{\mathrm{CP}} \le 1/4$ and $B > 1$. They satisfy the conditions of Definition 4.8 and the tail condition $0 < \tau_{\mathrm{tail}} < \min\{\sigma_{\mathrm{sm}}, 1/2, -\log q_{\mathrm{anc}}/\log B\}$ of Section 4: indeed $1/2 < 7/12 < 2/3$ and $5/24 < 1 - 7/12$, and the contraction factor of the reconstruction amplitude is $a_{\mathrm{cap}} = C_{\mathrm{CP}}/B$ in the amplitude bound of Lemma 15.10, so that $q_{\mathrm{anc}} = Ba_{\mathrm{cap}}^2 = q_{\mathrm{CP}}$ and the factor $1/2$ makes the inequality strict. This is a numerical choice only; it asserts no stochastic estimate for the flow. The deterministic sequence of matching levels in Theorem 29.19 is chosen only after the convergence at fixed coupling.

Lemma 23.4. *Fix the parameters of items* (1)–(8) *of Theorem 23.2. As in item* (9)*, fix the family of stopping levels with its remainders, the full sets of plaquettes affected by both insertions with the enlarged count F_J, and s_0 in the interval of Theorem 22.1. Fix $0 < \Gamma < \infty$, and set*

$$s_b = s_0/2, \qquad a_b = 4s_b d_*/c_E, \qquad 0 < \bar{c} \le \min\{d_*/8, a_b/2\}.$$

Let C_v, C_s, C_b be the constants of Theorem 22.1 for the ceiling Γ, and let $C_b' \ge C_b$ (for example $C_b' = C_b + 1 + \log 2$). Then there exists $n_\Gamma \in 4\mathbb{N}$, chosen before γ, the coarse side, and the number of renormalization group steps, and larger than every previous geometric, flat, and local threshold, with the following properties. The conclusions of Theorem 22.1 and Lemma 22.13 hold for all coarse sides $u \ge n_\Gamma$ and $0 < \gamma \le \Gamma$, and

$$\mathcal{A}_\gamma(m) \le a_b m/(2\gamma) \qquad (m \ge n_\Gamma,\ 0 < \gamma \le \Gamma),$$

with $\mathcal{A}_\gamma(m)$ as in (23.3)*. For $m \ge n_\Gamma$, the ratios of successive terms in the exponential sums with exponents a_{act}, $\bar{c}$, and $c_* = \min\{a_b, d_*/8\}$ over sides with ratio M or B, and in the sums for the fluctuation cutoffs over sides with ratio B, are at most $1/2$; and every product $\prod(1-\epsilon)^{-1}$ required in Lemma 22.12 has logarithm at most one. No restriction on the number of averaging steps is imposed. The new constants depend only on L, Γ, M, J, Θ, and the family of stopping levels, as described in item* (4) *of Definition 4.14.*

Proof. Theorem 22.1 and Lemma 22.13 provide the thresholds for the endpoints, for the factors attached to threshold failures, and for the initial segments at the fixed parameters. None of their hypotheses is a bound on the probability of a failure at some scale. For $x = m/\gamma \ge e$ and $m \ge 4$ we have

$$\mathcal{A}_\gamma(m) \le D(1 + \log x), \qquad D = 1 + 24C_vM^3 + C_sM^2(1 + \log M) + C_b'/8.$$

Since $(1 + \log x)/x \to 0$, we may choose a half-line $x \ge n_\Gamma/\Gamma$ on which $D(1 + \log x) \le a_b x/2$. The ratio of successive terms with exponent c is $R^3 e^{-c(R-1)m/\gamma}$, with $R = M$ or $R = B$ as appropriate. For the sums of the fluctuation cutoffs we write $z = (1/2)\log(Mm/\gamma)$ and $\Delta = (1/2)\log R$; the ratio is $R^3 e^{-a(2\Delta z + \Delta^2)}$. Both ratios are at most $1/2$ after a further increase of the minimum side, uniformly for $\gamma \le \Gamma$, and the corresponding sums tend

to zero. If $\epsilon \le 1/2$, then $-\log(1-\epsilon) \le 2\epsilon$, and the products of reciprocal errors have logarithm as small as prescribed; this is also the content of Lemma 20.15. We take the maximum of these thresholds, of the threshold at the initial scale, and of all earlier geometric and flat thresholds, rounded up to an element of $4\mathbb{N}$; all of them are monotone conditions on the minimum side.

The structural parameters fixed in the hypotheses involve only the local running couplings and fixed larger products, and none of them is recomputed from C_v, C_s, C_b or from Γ. The same argument works for any prescribed J for which the parameters of items (1)–(8) have been chosen, and it keeps that J. This proves the lemma, with the stated quantifier order and with the margin $a_{\mathrm{map}}\sqrt{B} \le 32/49$ of the reconstruction maps unchanged. □

Theorem 23.5. *Fix $L > 0$, an odd integer $M \ge 3$, and $0 < \Gamma < \infty$. There exist an integer $J \ge 1$, with $B = M^J$, a tuple Θ, reconstruction maps, a number $0 < g_* < e^{-1}$, and an integer $n_\Gamma \in 4\mathbb{N}$, all chosen before γ, the coarse side, and the number of renormalization group steps, such that every structural, local, moment, scalar, flat, and reconstruction conclusion of Theorem 23.2 holds for all*

$$0 < \gamma \le \Gamma, \qquad u \in 4\mathbb{N},\ u \ge n_\Gamma, \qquad K \in \mathbb{N}_0, \qquad N = uB^K,$$

with γ_0 replaced by Γ in its conclusions and side conditions, and with the moment constants of Theorem 22.1 and Lemma 22.13 for the ceiling Γ. In particular $a_{\mathrm{map}}\sqrt{B} \le 32/49$, every coarse field produced by the coordinate construction is the corresponding iterate of T_M, and the reconstruction maps, the support radius, and the local constants are fixed before the coupling. The minimum side satisfies

$$n_\Gamma \ge \max\{4, e\Gamma, \Gamma/g_{\mathrm{end}}^2, 8R_{\mathrm{cell}}, 8S\}$$

and the coupling condition of Lemma 22.13; no restriction on the number of averaging steps is imposed. The parameters satisfy $0 < \delta_* \le 1$, fixed before d_{cut}, $g_*^2 \le \alpha_W$, and

$$0 < s_0 \le \min\Big\{\frac{1}{2},\ \frac{\kappa_*}{2}c_E^{-(J-1)},\ \frac{1}{4K_W},\ \frac{c_A^{\mathrm{act}}d_*}{32F_J},\ \frac{\kappa_0}{2\max\{K_M, 48K_1\}}\Big\}, \qquad K_W = 48K_1/\alpha_W,$$

and the stopping partition-function ratio is controlled uniformly for coefficients of the Wilson action in $[1/4, 1/2]$. The construction can be carried out with any prescribed J for which parameters as in Theorem 23.2 with $\gamma_0 = 1$ have been chosen, without changing J. The local constants have the dependence described in item (3) *of Definition 4.14, and the probability constants the dependence described in item* (4)*, with the fixed ceiling Γ. No uniformity as $\Gamma \to \infty$ is asserted.*

We give the proof in Section 23.7. When $\Gamma = \gamma_0 \le 1$, the parameters of Theorem 23.2 satisfy the conclusions of Theorem 23.5, and in that case we write n_0 for n_Γ.

23.3 Bounds under the Wilson measure

In this subsection, we bound the probability that some averaged configuration has a plaquette of large energy, and we bound exponential moments of coarse actions under the Wilson measure. We recall the notation of Theorem 22.1 and Lemma 22.11. For $N = uB^K$ and $n_j = N/M^j$, fix one of the eight tilings $\mathcal{T}$ of the lattice of side n_j into $r_j = n_j^3/8$ disjoint cells with two sites in each direction (this number is denoted $\mathfrak{m}_j$ in Lemma 22.11). Let $\mathcal{P}_{j-1}^{\mathcal{T}}$ be the union, counted once, of the complete predecessor patches at side n_{j-1} of the internal plaquettes of the cells, and for a configuration V at side n_{j-1} set

$$Q_j^{\mathcal{T}}(V) = \frac{4}{h_{n_{j-1}}} \sum_{p \in \mathcal{P}_{j-1}^{\mathcal{T}}} q(V_p).$$

For a cell C, let $R_{<j}(C; d_{\mathrm{cut}})$ be the local regularity event of Lemma 22.11, which tests at the threshold d_{cut} exactly the plaquettes of the finer scales whose full dependence sets lie in C; let $I_j(C; d_*)$ be the event that some internal plaquette of C at scale j has energy at least d_*; and let $R_{<j}^{\mathcal{T}}(d_{\mathrm{cut}})$ be the intersection of the events $R_{<j}(C; d_{\mathrm{cut}})$ over the cells of $\mathcal{T}$. These local events do not test the plaquettes on the boundaries of the cells. By Lemma 7.4, every plaquette at scale j is internal to a cell of one of the eight tilings.

The probability bounds below rest on the chessboard estimate of Fröhlich, Israel, Lieb, and Simon [92, 93], which we use in the form of Biskup [29, Theorem 5.8]. The Wilson measure is reflection positive [134]. Part (2) of Theorem 6.6 gives the chessboard inequality, for each of the eight shifted tilings, under the image measure $(T_M^j)_* \mu_{N,\gamma}$ for nonnegative observables supported in the coarse cells, and more generally under $\mu_{N,\gamma}$ for nonnegative observables supported in the fine cells formed by the blocks B_{M^j} of the vertices of a coarse cell. The side n_j is divisible by four, as that theorem requires. The event $I_j(C; d_*)$ is a function of $T_M^j U$ on the coarse cell C, and the event $R_{<j}(C; d_{\mathrm{cut}}) \cap I_j(C; d_*)$ is a gauge-invariant function of the fine links in the fine cell of C, since it tests only plaquettes whose full dependence sets lie in C; this is the form in which Lemma 22.11 applies the theorem. All estimates in this subsection are under the Wilson measure $\mu_{N,\gamma}$, normalized by its own partition function; no conditioned measure occurs.

Theorem 23.6. *Use the parameters and the minimum side n_Γ of Lemma 23.4; for $\Gamma \le 1$ these are the parameters of Theorem 23.2 with $\gamma_0 = \Gamma$, and for general finite Γ those of Theorem 23.5. Let $0 < \gamma \le \Gamma$, $u \in 4\mathbb{N}$ with $u \ge n_\Gamma$, and $K \in \mathbb{N}_0$, and let $N = uB^K$, $D = JK$, $n_j = N/M^j$, and $V^j = T_M^j U$. Define*

$$\mathrm{FF}_0 = \{\max_p q(U_p) \ge d_*\}, \qquad \mathrm{FF}_j = \{q(V_p^\ell) < d_* \text{ for all } \ell < j \text{ and all } p,\ \max_p q(V_p^j) \ge d_*\}$$

for $1 \le j \le D$. Then, simultaneously for all $0 \le j \le D$,

$$\mu_{N,\gamma}(\mathrm{FF}_j) \le 3 n_j^3 e^{-\bar{c} n_j/\gamma}. \tag{23.5}$$

For $1 \le j \le D$, each of the eight tilings $\mathcal{T}$, and $0 \le s \le s_0$,

$$\log \mathbb{E}_{\mu_{N,\gamma}}\big[e^{sQ_j^{\mathcal{T}}(V^{j-1})}; R_{<j}^{\mathcal{T}}(d_{\mathrm{cut}})\big] \le 3C_v n_{j-1}^3 + C_s n_{j-1}^2 \log(1/g_{n_{j-1}}) + C_b'.$$

The same bound holds without the indicator of $R_{<j}^{\mathcal{T}}(d_{\mathrm{cut}})$, with C_b in place of C_b'. The constants depend only on L, Γ, M, J, Θ, and the family of stopping levels, as described in item (4) *of Definition 4.14, and are uniform in γ, the side, the number of renormalization group steps, the scale j, and the tiling.*

Proof. At $j = 0$, by the threshold at the initial scale in item (10) of Theorem 23.2 (for general Γ, in Lemma 23.4 and Lemma 22.13), we have

$$\mu_{N,\gamma}(\mathrm{FF}_0) \le 3N^3 e^{-d_* N/(8\gamma)} \le 3N^3 e^{-\bar{c}N/\gamma}, \tag{23.6}$$

where we used $\bar{c} \le d_*/8$. For a scale $j \ge 1$, we apply Theorem 22.1 with the Wilson measure at side N as the starting measure (that is, with $b = 0$) and the same j. This gives the bound without the indicator directly at that scale, whatever its distance from the finest lattice. Since the integrand is nonnegative and $C_b \le C_b'$, the bound with the indicator follows. This proves both moment bounds, without any assumption on earlier failures.

Let $m = n_j$. We divide the logarithm of the moment bound with the indicator by the number $r_j = m^3/8$ of cells. Since $n_{j-1} = Mm$ and $\log(1/g_{Mm}) = (1/2)\log(Mm/\gamma)$, we have

$$\begin{aligned}\frac{3C_v(Mm)^3 + C_s(Mm)^2 \frac{1}{2}\log(Mm/\gamma) + C_b'}{m^3/8} &= 24C_v M^3 + \frac{4C_s M^2}{m}\log(Mm/\gamma) + \frac{8C_b'}{m^3} \\ &= \mathcal{A}_\gamma(m).\end{aligned}$$

This identity explains why a moment bound with a volume term and a surface term carrying a logarithm suffices. After the reciprocal-cell-count exponent in the chessboard estimate is applied, the entropy per cell is of order $1 + \log(m/\gamma)/m$, which is dominated by the action $a_b m/\gamma$ of a cell containing a bad plaquette. Then, by the estimate of Lemma 22.11, which combines Markov's inequality at $s = s_b$ with the chessboard estimate, we have

$$\log \mu_{N,\gamma}\big(R_{<j}(C; d_{\mathrm{cut}}) \cap I_j(C; d_*)\big) \le -a_b m/\gamma + \mathcal{A}_\gamma(m) \le -\bar{c}m/\gamma, \tag{23.7}$$

where in the last inequality we used $\mathcal{A}_\gamma(m) \le a_b m/(2\gamma)$ and $\bar{c} \le a_b/2$.

Since $d_* < d_{\mathrm{cut}}$, regularity at the threshold d_* at all earlier scales implies every local event $R_{<j}(C; d_{\mathrm{cut}})$. Every plaquette of scale j with energy at least d_* is internal to a cell of one of the eight tilings, by Lemma 7.4. Then FF_j is contained in the union of the $8r_j = m^3$ events in (23.7), and by a union bound we have $\mu_{N,\gamma}(\mathrm{FF}_j) \le m^3 e^{-\bar{c}m/\gamma}$. This

implies (23.5), where the constant three is kept only for a common formula with (23.6). The argument applies to each j separately, without induction over scales; if $K = 0$, only the assertion at $j = 0$ remains. □

The next theorem gives exponential moments of coarse actions without any indicator. It is not needed for the bound on S^c, but it is a quantitative consequence of the same estimates.

Theorem 23.7. *Fix $L > 0$, an odd integer $M \geq 3$, and $0 < \Gamma < \infty$. Use parameters as in Theorem 23.5, in which, as in item* (9) *of Theorem 23.2, the count F_J includes the plaquettes affected by both the predecessor insertion and the total-action insertion before s_0 is chosen. There exist $s_* = s_0 > 0$, $n_\Gamma \in 4\mathbb{N}$, and finite nonnegative constants C_v, C_s, C_b, chosen before γ, the side, and the number of renormalization group steps, such that for every $0 < \gamma \leq \Gamma$, $u \in 4\mathbb{N}$ with $u \geq n_\Gamma$, $K \in \mathbb{N}_0$, $N = uB^K$, $n_j = N/M^j$, and $0 \leq s \leq s_*$,*

$$\log \mathbb{E}_{\mu_{N,\gamma}} e^{sQ_j^{\mathcal{T}}(T_M^{j-1}U)} \leq 3C_v n_{j-1}^3 + C_s n_{j-1}^2 \log(1/g_{n_{j-1}}) + C_b, \qquad 1 \leq j \leq JK, \tag{23.8}$$

$$\log \mathbb{E}_{\mu_{N,\gamma}} e^{sS_W(T_M^j U)/g_{n_j}^2} \leq 3C_v n_j^3 + C_s n_j^2 \log(1/g_{n_j}) + C_b, \qquad 0 \leq j \leq JK. \tag{23.9}$$

The first inequality holds for each of the eight tilings $\mathcal{T}$. All expectations are under the Wilson measure, normalized by its partition function, and all iterates of T_M that occur are defined on the given finite sequence of lattices. The constants depend only on L, Γ, M, J, Θ, and the family of stopping levels, as described in item (4) *of Definition 4.14; they are uniform in γ, the side, the scale, and the number of renormalization group steps, and they do not depend on the large-field decompositions used in the proof. Moreover, with $\mathcal{B}_\gamma(n) = 3C_v n^3 + C_s n^2 \log(1/g_n) + C_b$, for every integer $p \geq 1$,*

$$\mathbb{E}_{\mu_{N,\gamma}}\Big[\big(S_W(T_M^j U)/g_{n_j}^2\big)^p\Big] \leq p!\, s_*^{-p} e^{\mathcal{B}_\gamma(n_j)},$$

and the analogous polynomial moment bound holds for $Q_j^{\mathcal{T}}(T_M^{j-1}U)$, with $\mathcal{B}_\gamma(n_{j-1})$.

Proof. Let $F_J' = \max(F_J, F_J^{\mathrm{A}})$ be the larger of the counts for the two insertions, enlarged also over the union of the two finite families of remainders. We take a common interval satisfying the conditions of item (9) of Theorem 23.2 with F_J' and the further restriction

$$0 < s_* = s_0 \leq \frac{\kappa_0}{2\max\{K_M, 48K_1\}}.$$

All entries are positive constants at this stage. The bound (23.8) is Theorem 22.1 with the Wilson measure as the starting measure ($b = 0$). For (23.9), we choose the stopping level $t = J\lfloor j/J \rfloor$, with $v = j - t \in \{0, \dots, J-1\}$, so that $m = n_t = M^v n_j$. All iterates from level t to level j are defined, including for $j = JK$, where $v = 0$ and $t = JK$.

Suppose first that $t > 0$. We integrate exactly t/J complete RG steps, starting from the Wilson measure, and use the bound of Corollary 22.9 for the total-action insertion

inside the invariant density bound without weight of Theorem 22.10. The coefficient of the Wilson action remains $\lambda_s = 1/2 - sK_W \in [1/4, 1/2]$, and the plaquettes affected by the insertion contribute at most $8sF_J' \sum_f n_f/\gamma \le P_0 \sum_f n_f/(4\gamma)$. The factors listed in Table 22.2 are unchanged except for their fixed incidence constants: the normalization terms generated by the integration and the completion terms concern the same large-field decompositions, and the finite dependence set of the insertion was enlarged before the interval was chosen. The quantitative conclusion of Theorem 22.1 on these factors applies, with the scale n_j and enlarged fixed constants. We complete the density bound by Lemma 22.7 and use the single stopping partition-function ratio of Theorem 22.4. Then each large-field decomposition contributes at most

$$e^{3C_T m^3 + 9m^2 \log(1/g_m) + C_\delta} \prod_f e^{-(P_0/2) n_f/\gamma}.$$

The bound of Theorem 22.1 for the sum over decompositions controls the sum of these contributions uniformly at the common minimum side. Converting from $m = \chi n_j$, with $\chi \le M^{J-1}$, leaves exactly a volume term, the surface term $9\chi^2 n_j^2 \log(1/g_{n_j})$, and a fixed volume contribution from $\log \chi$. We take the maximum of the constants for the two insertions.

Suppose now that $t = 0$. Then $0 \le j < J$ and $N/n_j = M^j$ is bounded. By the comparison of Theorem 22.4 for arbitrary fields, valid at all depths, we have $S_W(T_M^j U)/g_{n_j}^2 \le 48K_1 S_W(U)/g_N^2$. Since $48sK_1 \le \kappa_0/2 \le 1/2$, the ratio of the full compact numerator to the periodic trial denominator at depth zero, used for short scale intervals in Theorem 22.1, applies. This ratio has the required surface-order correction in the partition-function comparison at side N, and the bounded ratio $N/n_j \le M^{J-1}$ implies the stated bound at n_j. At $j = 0$ the deterministic inequality holds with the smaller constant one; this case includes $K = 0$. In both cases we bounded the integrals without indicators directly; no indicator of a regularity event was dropped from an upper bound.

Finally, for $x \ge 0$ we have $x^p \le p!\, s_*^{-p} e^{s_* x}$, by the exponential series. Inserting the exponential moment bounds, we obtain the last assertion. □

Corollary 23.8. *Use the parameters and constants of Theorem 23.7, increasing only the minimum side if necessary. There exists $c_{\mathrm{bad}} > 0$ with the following property. Let $0 < \gamma \le \Gamma$, $u \in 4\mathbb{N}$ with $u \ge n_\Gamma$, $K \in \mathbb{N}_0$, $N = uB^K$, and $0 \le j \le JK$, and let $\mathcal{T}$ be one of the eight tilings at scale j. We set*

$$r_j = n_j^3/8, \qquad q_j = e^{-c_{\mathrm{bad}} n_j/\gamma},$$

and for a cell C of $\mathcal{T}$ let I_C be the event that some internal plaquette P of C satisfies $q((T_M^j U)_P) \ge d_*$. Then for every deterministic set A of distinct cells of $\mathcal{T}$,

$$\mu_{N,\gamma}\Big(\bigcap_{C \in A} I_C\Big) \le q_j^{|A|}. \tag{23.10}$$

For $Y_j = \sum_{C\in\mathcal{T}} \mathbf{1}_{I_C}$, every integer $0 \le k \le r_j$, and every $\vartheta \ge 0$,

$$\mathbb{E}_{\mu_{N,\gamma}} \binom{Y_j}{k} \le \binom{r_j}{k} q_j^k, \tag{23.11}$$

$$\mathbb{E}_{\mu_{N,\gamma}} e^{\vartheta Y_j} \le [1 + (e^{\vartheta} - 1)q_j]^{r_j} \le \exp\{r_j q_j (e^{\vartheta} - 1)\}. \tag{23.12}$$

Equivalently, $\mathbb{E}(Y_j)_k \le (r_j)_k q_j^k$ for the falling factorials. The constants depend only on L, Γ, M, J, Θ, and the family of stopping levels (item (4) of Definition 4.14), and are uniform in γ, the side, the number of renormalization group steps, the scale, the tiling, and A.

The bounds (23.11) and (23.12) are those of a sum of independent Bernoulli variables with parameter q_j, but they are proved from the joint bounds (23.10) alone; no independence or stochastic domination is asserted.

Proof. Fix j and $\mathcal{T}$, and let $m = n_j$. The indicator of I_C is a nonnegative, bounded, gauge-invariant function of the configuration $T_M^j U$ on the internal plaquettes of the cell C, that is, an observable supported in the coarse cell C. Part (2) of Theorem 6.6, applied to the image measure $(T_M^j)_* \mu_{N,\gamma}$ and the tiling $\mathcal{T}$, applies to it; equivalently, its pullback is supported in the fine cell formed by the centered blocks of the vertices of C. The reflection positivity in part (1) of that theorem includes the configurations at which the Borel extension of T_M is used. Each I_C is the reflected translate of the corresponding event for the base cell of $\mathcal{T}$, since the energy of a plaquette is invariant under reversal of orientation. For nonempty A, the chessboard inequality with all observables equal to the indicator of this event gives

$$\mu_{N,\gamma}\Big(\bigcap_{C\in A} I_C\Big) \le \mu_{N,\gamma}(I_{\mathrm{all}})^{|A|/r_j}, \qquad I_{\mathrm{all}} = \bigcap_{C\in\mathcal{T}} I_C.$$

On I_{all}, we choose one internal plaquette with energy at least d_* in each cell, in a fixed deterministic order. The chosen plaquettes are distinct, since the cells have disjoint vertex sets and only internal plaquettes are used. This gives, pointwise on I_{all},

$$S_W(T_M^j U)/g_m^2 \ge 4 d_* r_j m/\gamma.$$

By Markov's inequality and (23.9) at $s = s_*$, we have

$$\frac{1}{r_j} \log \mu_{N,\gamma}(I_{\mathrm{all}}) \le -4 s_* d_* m/\gamma + 24 C_v + \frac{4C_s}{m} \log(m/\gamma) + \frac{8C_b}{m^3}.$$

Here the entropy terms are in the units of scale j itself, so no factor M^3 appears. With $x = m/\gamma \ge e$ and $m \ge 4$, their sum is at most $D_{\mathrm{A}}(1 + \log x)$ for a fixed D_{A}. We increase the minimum side so that $D_{\mathrm{A}}(1 + \log x) \le 2 s_* d_* x$ on the half-line $x \ge n_\Gamma/\Gamma$. Then

$c_{\mathrm{bad}} = 2s_* d_* > 0$ works. If $\mu_{N,\gamma}(I_{\mathrm{all}}) = 0$, the chessboard estimate already implies the result; otherwise we exponentiate the bound on its logarithm. For $A = \varnothing$ both sides of (23.10) equal one.

Pointwise we have $\binom{Y_j}{k} = \sum_{|A|=k} \prod_{C \in A} \mathbf{1}_{I_C}$, including for $k = 0$, and taking expectations proves (23.11). For $\vartheta \geq 0$ we expand the finite product

$$e^{\vartheta Y_j} = \prod_{C \in \mathcal{T}} \bigl[1 + (e^{\vartheta} - 1)\mathbf{1}_{I_C}\bigr].$$

Since every coefficient in the expansion is nonnegative, the joint bounds give (23.12); its last inequality uses $1 + y \leq e^y$ for $y \geq 0$. The case $j = 0$, including $K = 0$, is covered by the instance $j = 0$ of (23.9). □

23.4 Probability of the small-field event

We now bound the probability of the complement of the small-field event. Recall from Definitions 4.4 and 14.4 that, for $N = uB^K$ and $V^\ell = T_M^\ell U$,

$$S_{N,u,\gamma} = \{U \in D_r\} \cap \bigcap_{k=0}^{K-1} \Bigl[\mathcal{A}_{k,J}(U) \cap \{\max_{i,e} |w_{k,i,e}(U)| \leq p(t_k)\} \cap \{V^{(k+1)J} \in D_r\}\Bigr],$$

where $t_k^2 = \gamma/(Mn_{(k+1)J})$ and $p(t) = A_{\mathrm{w}} \log(1/t)$. Here $\mathcal{A}_{k,J}(U)$ is the event that, in the inverse coordinates of the k-th RG step at preceding classical depth kJ, each pair (c_{i+1}, z_i) lies in the coordinate domain of Definition 14.4, with its conditional constrained minimizer, its positive constrained Hessian, and a positive constraint Jacobian; and $w_{k,i,e}(U)$ are the normalized fluctuation variables of that step, defined by these inverse coordinates. There is no condition of the form $V^\ell \in D_r$ at levels ℓ that are not multiples of J. For $K = 0$ the event is $\{U \in D_r\}$.

The proof splits S^c according to the event G that every plaquette at every scale has energy below d_*. The probability of G^c is the sum of the probabilities of the first failures FF_j, which Theorem 23.6 bounds under the Wilson measure. On G, Theorem 7.7 implies all conditions $V^{kJ} \in D_r$, and S can fail only through the coordinate-domain conditions or the cutoffs of some RG step. We bound the first such failure using the comparison of Lemma 22.12, which keeps only the passing initial segment and the condition on the next coarse field. In this comparison, the probability of the initial segment is multiplied, never divided, and we never condition on regularity at later scales, which would change the conditional law.

Theorem 23.9. *Use the parameters, the renormalization group steps, and the reconstruction maps of Theorem 23.2. There exist constants $C', C, C_{\mathrm{w}}, a_{\mathrm{w}} > 0$ such that for every $u \in 4\mathbb{N}$ with $u \geq n_0$, every $K \in \mathbb{N}_0$, and every $0 < \gamma \leq \gamma_0$,*

$$\mu_{uB^K,\gamma}\bigl(S^c_{uB^K,u,\gamma}\bigr) \leq C' u^3 e^{-\bar{c}u/\gamma} + C_{\mathrm{w}} u^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mu/\gamma}}$$

and

$$\mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big) \le Cu^3 e^{H_0+H_2\log(1/\gamma)-c_*u/\gamma} + C_{\mathrm{w}}u^3 e^{-a_{\mathrm{w}}\log^2\sqrt{Mu/\gamma}},$$

where

$$H_0 = 24C_vM^3 + \frac{4C_sM^2}{n_0}\log(Mn_0) + \frac{8C_b'}{n_0^3}, \qquad H_2 = \frac{4C_sM^2}{n_0}, \qquad c_* = \min\{a_b, d_*/8\} > 0.$$

One may take a_{w} to be the smallest of the finitely many fixed positive exponents in the fluctuation bounds for the auxiliary measures and the stopped measures. The constants may depend on $L, \gamma_0, M, J, \Theta$, and the family of stopping levels, but not on γ, u, N, K, the preceding classical depth, or the large-field decompositions. The event, and hence these bounds, are the same when the inverse coordinates are computed with the reconstruction maps of Theorem 15.12.

The two theorems have a common proof.

Theorem 23.10. *Use the parameters of Theorem 23.5 for a fixed $0 < \Gamma < \infty$, and the small-field event $S_{N,u,\gamma}$ defined with them, for $0 < \gamma \le \Gamma$, $u \in 4\mathbb{N}$ with $u \ge n_\Gamma$, $K \in \mathbb{N}_0$, and $N = uB^K$. There exist constants $C', C, C_{\mathrm{w}}, a_{\mathrm{w}} > 0$ such that*

$$\mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big) \le C'u^3 e^{-\bar{c}u/\gamma} + C_{\mathrm{w}}u^3 e^{-a_{\mathrm{w}}\log^2\sqrt{Mu/\gamma}}, \tag{23.13}$$

$$\mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big) \le Cu^3 e^{H_0^\Gamma+H_2\log(\Gamma/\gamma)-c_*u/\gamma} + C_{\mathrm{w}}u^3 e^{-a_{\mathrm{w}}\log^2\sqrt{Mu/\gamma}}, \tag{23.14}$$

where

$$H_0^\Gamma = 24C_vM^3 + \frac{4C_sM^2}{n_\Gamma}\log(Mn_\Gamma/\Gamma) + \frac{8C_b'}{n_\Gamma^3} \ge 0,$$

$$H_2 = \frac{4C_sM^2}{n_\Gamma} \ge 0, \qquad c_* = \min\{a_b, d_*/8\} \ge \bar{c} > 0.$$

The constants may depend on the fixed Γ (item (4) of Definition 4.14), and do not depend on γ, the side, the number of renormalization group steps, the preceding classical depth, or the large-field decompositions. Moreover, for every fixed $u \in 4\mathbb{N}$ with $u \ge n_\Gamma$ and every $q > 0$,

$$\sup_{K\ge 0}\mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big) \le C_{u,q,\Gamma}\gamma^q \qquad (0 < \gamma \le \Gamma),$$

with the admissible constant

$$C_{u,q,\Gamma} = Cu^3 e^{H_0^\Gamma}\Gamma^{H_2}\Big(\frac{H_2+q}{c_*ue}\Big)^{H_2+q} + C_{\mathrm{w}}u^3(Mu)^{-q}e^{q^2/a_{\mathrm{w}}}.$$

Finally, for every fixed $n \in 4\mathbb{N}$ with $n \ge n_\Gamma$ and $u_a = nB^a$,

$$\lim_{a\to\infty}\sup_{0<\gamma\le\Gamma}\sup_{K\ge a}\mu_{nB^K,\gamma}\big(S^c_{nB^K,u_a,\gamma}\big) = 0.$$

Proof of Theorems 23.9 and 23.10. For $0 \le a \le K$, let S^a be the event formed by the initial condition $U \in D_r$ and the conditions of the first a RG steps in the definition of $S_{N,u,\gamma}$, and let

$$G = \{q(V_p^j) < d_* \text{ for every } 0 \le j \le JK \text{ and every plaquette } p\}.$$

The disjoint events FF_j, $0 \le j \le JK$, exhaust G^c. By Theorem 23.6 and the condition $M^3 e^{-\bar{c}(M-1)n_\Gamma/\Gamma} \le 1/2$, summing over the sides uM^h in increasing order, we have

$$\mu(G^c) \le 3 \sum_{h \ge 0} (uM^h)^3 e^{-\bar{c}uM^h/\gamma} \le 6u^3 e^{-\bar{c}u/\gamma}. \tag{23.15}$$

The sum over sides starts at the coarse side u, so no factor depending on K appears.

To keep the dependence on γ before absorption, we use (23.7) in the form $\log \mu(R_{<j}(C; d_{\mathrm{cut}}) \cap I_j(C; d_*)) \le -a_b m/\gamma + \mathcal{A}_\gamma(m)$. Since $\log(Mm/\gamma) = \log(Mm/\Gamma) + \log(\Gamma/\gamma)$ and $\log(Mm/\Gamma)/m$ is decreasing for $m \ge n_\Gamma \ge e\Gamma$, we have

$$\begin{gathered} \mathcal{A}_\gamma(m) \le H_0^\Gamma + H_2 \log(\Gamma/\gamma), \\ H_0^\Gamma = 24 C_v M^3 + \frac{4C_s M^2}{n_\Gamma} \log(M n_\Gamma/\Gamma) + \frac{8C_b'}{n_\Gamma^3}, \qquad H_2 = \frac{4C_s M^2}{n_\Gamma}, \end{gathered} \tag{23.16}$$

and this bound is nonnegative. Combining the bound at the initial scale with the bounds at positive scales, and summing the same geometric list of sides, we obtain

$$\mu(G^c) \le 6u^3 e^{H_0^\Gamma + H_2 \log(\Gamma/\gamma) - c_* u/\gamma}, \qquad c_* = \min\{a_b, d_*/8\} \ge \bar{c}. \tag{23.17}$$

For Theorem 23.9 we use instead $H_0 + H_2 \log(1/\gamma)$, which follows from the same calculation with Γ replaced by one in the logarithm; the function $\log(Mm)/m$ is decreasing for $m \ge 4$.

On G, the condition $U \in D_r$ and every condition $V^{(k+1)J} \in D_r$ hold by Theorem 7.7. By the identity of coarse fields in Theorems 23.2 and 23.5, on a passing initial segment each coarse field produced by the coordinate construction is the corresponding iterate of T_M. We conclude that

$$G \cap S^c \subset \bigcup_{0 \le a < K} \big(S^a \cap A_{n_{(a+1)J}} \cap W^c_{n_{(a+1)J}}\big), \tag{23.18}$$

where, for $m = n_{(a+1)J}$, A_m is the event that the coarse field $V^{(a+1)J}$ lies in D_r, and W_m is the event that all coordinate-domain conditions and fluctuation cutoffs of the $(a+1)$-th RG step hold. After this inclusion we discard G before estimating each term. By the comparison with identical unnormalized prefactors in Lemma 22.12, in its form for a fixed upper coupling bound Γ in Lemma 22.13, we have

$$\mu(S^a \cap A_m \cap W_m^c) \le \epsilon_a \mu(S^a) \le C_{\mathrm{rej},J} m^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mm/\gamma}}. \tag{23.19}$$

This is the comparison for coarse fields in D_r at preceding depth aJ; only $a < K$ is used, and the probability $\mu(S^a) \le 1$ of the initial segment is multiplied, never inverted. Summing the bound over the sides $u, uB, uB^2, \dots$ and using the ratio condition for the fluctuation cutoffs, we obtain

$$\mu(G \cap S^c) \le 2C_{\mathrm{rej},J} u^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mu/\gamma}}. \tag{23.20}$$

Adding (23.15) or (23.17) proves both bounds of each theorem, for example with $C = C' = 6$ and $C_{\mathrm{w}} = 2C_{\mathrm{rej},J}$. For $K = 0$ the sum (23.20) is empty, and the initial condition holds on the event that there is no failure at the initial scale. By Theorem 15.12, the reconstruction maps preserve the event, the transitions, the cutoffs, the Jacobians, and the kernels, so the same bounds hold in the coordinates of the reconstruction maps.

For the bound by powers of γ, we divide the curvature term of (23.14) by γ^q. For $\kappa = H_2 + q > 0$ and $A = c_* u > 0$ we have $\sup_{\gamma>0} \gamma^{-\kappa} e^{-A/\gamma} = (\kappa/(Ae))^\kappa$. For the fluctuation term we write $z = (1/2)\log(Mu/\gamma)$, and then

$$\gamma^{-q} e^{-a_{\mathrm{w}} z^2} = (Mu)^{-q} e^{2qz - a_{\mathrm{w}} z^2} \le (Mu)^{-q} e^{q^2/a_{\mathrm{w}}}.$$

These give the constant $C_{u,q,\Gamma}$. For $u_a = nB^a$, both terms of (23.13) decrease in u and increase in γ for sides above the minimum side, by Lemma 23.1. Their values at $\gamma = \Gamma$ tend to zero as $a \to \infty$, uniformly in $K \ge a$, since $nB^K = u_a B^{K-a}$. This proves the last assertion of Theorem 23.10. □

Corollary 23.11. *Use the parameters of Theorem 23.2, and fix $u \in 4\mathbb{N}$ with $u \ge n_0$ before varying the coupling. For every real $q > 0$ there exists a finite $C_{u,q}$ such that*

$$\sup_{K \ge 0} \mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big) \le C_{u,q}\gamma^q \qquad (0 < \gamma \le \gamma_0).$$

With the constants of the second bound of Theorem 23.9, an admissible constant is

$$C_{u,q} = Cu^3 e^{H_0}\Big(\frac{H_2 + q}{c_* ue}\Big)^{H_2+q} + C_{\mathrm{w}} u^3 (Mu)^{-q} e^{q^2/a_{\mathrm{w}}},$$

or any larger finite constant. In particular, if

$$F_u(\gamma) = \limsup_{K \to \infty} \mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big),$$

then $F_u(\gamma) = o(\gamma^\kappa)$ as $\gamma \to 0$ for every fixed $\kappa > 0$. Besides the probability constants, $C_{u,q}$ depends only on u and q; it does not depend on K.

Now fix $n \in 4\mathbb{N}$ with $n \ge n_0$, and set $u_a = nB^a$ and $N_K = nB^K$. For all integers $a \ge 0$ and $K \ge a$,

$$e_{K,a}(\gamma) = \mu_{N_K,\gamma}\big(S^c_{N_K,u_a,\gamma}\big) \le \mathcal{F}(u_a,\gamma), \qquad \mathcal{F}(u,\gamma) = C'u^3 e^{-\bar{c}u/\gamma} + C_{\mathrm{w}} u^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mu/\gamma}},$$

and

$$\lim_{a\to\infty} \sup_{0<\gamma\le\gamma_0} \mathcal{F}(u_a,\gamma) = 0.$$

These bounds use the same small-field events and reconstruction maps for all coarse sides; no radius, cutoff, map, or threshold is chosen again at a larger coarse side.

At each fixed positive coupling, the probability of the complement tends to zero along the sides u_a, while at each fixed side it is smaller than every power of the coupling. The corollary gives no rate, uniform in the coupling, for the convergence of flowed observables, and it does not interchange a limit in the cutoff or in an observable with the limit $a \to \infty$.

Proof. We keep u fixed and let $\tau = \log(1/\gamma)$ and $d_u = \log(Mu)$. Dividing the second bound of Theorem 23.9 by γ^q, its curvature term becomes

$$Cu^3 e^{H_0} \exp\{(H_2+q)\tau - c_* u e^{\tau}\}.$$

For $\kappa = H_2 + q > 0$, the derivative of $v^\kappa e^{-c_* u v}$ on $v > 0$ has the sign of $\kappa - c_* u v$. This function has the global maximum $[\kappa/(c_* u e)]^\kappa$, attained at $v = \kappa/(c_* u)$. Bounding the value at $v = e^\tau = 1/\gamma$ by this maximum, we obtain the first summand of $C_{u,q}$. For the fluctuation term, completing the square gives the identity

$$q\tau - \frac{a_{\mathrm{w}}}{4}(\tau + d_u)^2 = -\frac{a_{\mathrm{w}}}{4}\Big(\tau + d_u - \frac{2q}{a_{\mathrm{w}}}\Big)^2 + \frac{q^2}{a_{\mathrm{w}}} - q d_u,$$

and dropping the negative square gives the second summand. Both constants are finite for fixed u and q, and the bound of Theorem 23.9 is uniform in K. This proves the bound on the supremum. Given $\kappa > 0$, we choose $q > \kappa$; then

$$0 \le \gamma^{-\kappa} F_u(\gamma) \le C_{u,q}\gamma^{q-\kappa} \longrightarrow 0,$$

and no limit in the cutoff has been interchanged with the limit in the coupling.

Now fix n and set $u_a = nB^a$. For $K \ge a$ we have $N_K = u_a B^{K-a}$, and Theorem 23.9 applies directly with coarse side u_a and $K - a$ RG steps. No parameter of the event, of a local reconstruction, or of its finite thresholds is chosen again. We obtain $e_{K,a}(\gamma) \le \mathcal{F}(u_a,\gamma)$.

Since both terms of $\mathcal{F}(u,\gamma)$ increase with γ for $Mu/\gamma \ge Mu/\gamma_0 > 1$, we have

$$\sup_{0<\gamma\le\gamma_0} \mathcal{F}(u_a,\gamma) \le C' u_a^3 e^{-\bar{c}u_a/\gamma_0} + C_{\mathrm{w}} u_a^3 e^{-a_{\mathrm{w}}\log^2(Mu_a/\gamma_0)/4}.$$

Up to fixed constants, the logarithms of these two terms are $3\log u_a - \bar{c}u_a/\gamma_0$ and $3\log u_a - a_{\mathrm{w}}\log^2(Mu_a/\gamma_0)/4$, and both tend to $-\infty$ as $a \to \infty$, since $u_a \to \infty$. □

23.5 Two-sided bounds of logarithm-squared order

The fluctuation term in Theorem 23.9 has a simple heuristic origin. The cutoff radius is $p(t) \asymp \log(1/t)$, and a fluctuation variable with a Gaussian-type tail exceeds it with probability of order $\exp(-c\log^2(1/t))$. At a fixed coarse side u this gives an upper bound of order $\exp(-c\log^2(1/\gamma))$. The next theorem shows that this order is correct for the supremum over the number of RG steps. For the lower bound we fix one RG step, so that the problem is finite dimensional, and we exhibit a set of configurations of controlled Haar measure and controlled action on which one normalized fluctuation variable lies just beyond its cutoff while all other conditions hold.

Theorem 23.12. *Use the small-field event and the parameters of Theorems 23.2 and 23.9. For each fixed $u \in 4\mathbb{N}$ with $u \geq n_0$ set*

$$E_u(\gamma) = \sup_{K \in \mathbb{N}_0} \mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big).$$

There exist $c_u, C_u > 0$ and $0 < \gamma_u \leq \min\{\gamma_0, e^{-1}\}$, depending on u and the probability constants but not on γ or on the number of renormalization group steps, such that

$$e^{-C_u[1+\log(1/\gamma)]^2} \leq \mu_{uB,\gamma}\big(S^c_{uB,u,\gamma}\big) \leq E_u(\gamma) \leq e^{-c_u\log^2(1/\gamma)} \qquad (0 < \gamma \leq \gamma_u). \tag{23.21}$$

In particular, for fixed positive c, ε, and D, the bound $E_u(\gamma) \leq D\exp(-c\gamma^{-\varepsilon})$ does not hold for all sufficiently small $\gamma > 0$. No lower bound on $\limsup_{K\to\infty} \mu_{uB^K,\gamma}(S^c_{uB^K,u,\gamma})$ is asserted.

Proof. Upper bound. Let $x = \log(1/\gamma)$ and $\ell_u = (1/2)\log(Mu) > 0$. The first bound of Theorem 23.9, which is uniform in K, reads

$$E_u(\gamma) \leq C' u^3 e^{-\bar{c}ue^x} + C_{\mathrm{w}} u^3 e^{-a_{\mathrm{w}}(x/2+\ell_u)^2}.$$

For all sufficiently large x, the first term is at most $e^{-(a_{\mathrm{w}}/8)x^2}/2$, since $e^x/x^2 \to \infty$; the second term is also at most this quantity, since its exponent is a negative quadratic in x with leading coefficient $a_{\mathrm{w}}/4$. This proves the upper bound with, for example, $c_u = a_{\mathrm{w}}/8$, after decreasing γ_u.

A box in the coordinates of one RG step. Fix $K = 1$, $N = uB$, and preceding classical depth zero. For the rest of the proof this is a fixed finite-dimensional problem. We set

$$t = \sqrt{\gamma/(Mu)}, \quad p = p(t) = A_{\mathrm{w}}\log(1/t), \quad g_i = t\lambda_i, \quad \lambda_i = M^{(i-J+1)/2} \quad (0 \leq i < J).$$

The recursion of Theorem 12.13 is

$$z_i = z_i^*(c_{i+1}) + g_i S_i(c_{i+1}) w_i, \qquad c_i = b_i(c_{i+1}, z_i), \qquad c_J = c = e^\theta, \qquad S_i = K_i^{-1/2}.$$

We use it as a classical change of coordinates, before any fluctuation cutoff is imposed. This is valid beyond the region $\max|w| \leq p$: Lemma 12.3 gives product domains of fixed

positive radius for the coarse fields and the free coordinates, independent of g and p, and Theorems 12.13 and 13.2 give the composition of the J averaging steps on a common such domain, with the unscaled inputs $q_i = g_i w_i$. The conditional constrained minimizer at the identity is zero by Theorem 10.10, and the gauge transformation of Theorem 13.2 is the identity there. All intermediate Jacobian determinants are positive on a sufficiently small closed product domain.

Fix $0 < \epsilon < 1/4$ and one free link (i_*, e_*) of the step, with its three-dimensional fluctuation variable; such a link exists since $M \geq 3$ and $u \geq 4$. We identify the corresponding copy of $\mathfrak{su}(2)$ with $\mathbb{R}^3$ through the orthonormal basis of Section 4, and let e_1 be its first unit vector. We define the box $\mathcal{B}_\gamma$ of parameters by

$$\begin{gathered} |\theta_e| < \epsilon\sqrt{\gamma} \quad (e \in \mathcal{E}_u), \\ |w_{i,e}| < \epsilon \quad ((i,e) \neq (i_*, e_*)), \qquad |w_{i_*,e_*} - (p+1)e_1| < \epsilon. \end{gathered} \tag{23.22}$$

Each ball for a fluctuation variable has the same positive Lebesgue volume, independent of its center and of γ. The largest unscaled fluctuation input is at most $t(p+1+\epsilon) \to 0$, and $\|\theta\|_\infty \to 0$. Then the closure of the box lies inside the fixed larger coordinate domain, where no cutoff is imposed, for all sufficiently small γ. We do not assign a Gaussian law to fluctuation variables beyond the cutoff. Explicitly, one may take $\widetilde{p} = p + 2$ in Theorem 12.13: the box is contained in $\Omega^{\mathrm{all}}_{\widetilde{p}}$, while $C_J t\widetilde{p} < 1/2$ and $C_J(t\widetilde{p}^3 + t\widetilde{p})/(1 - C_J t\widetilde{p}) \to 0$. All hypotheses of that theorem on the coordinate domains and on the triangular Jacobian hold for this larger domain at small coupling. The radius $\widetilde{p}$ is used only to check the domain; the small-field event still uses exactly $p(t)$.

For the coordinate-domain conditions, we note that at small coupling the unscaled fluctuation inputs lie in the product domain of radius Q_{phys} of Definition 14.4 and Theorem 13.2, and the coarse field is closer to the identity than r. The reconstruction then has all input logarithms smaller than $r/2$, with the intermediate margins for the classical action, the constraint coordinates, and the Jacobians. Then $c_0 \in D_r$, $c \in D_r$, and $\mathcal{A}_{0,J}$ holds, and the averaged field is $T^J_M c_0 = c$. By the uniqueness and the identities on overlapping charts in Theorem 14.3 and Definition 14.4, the constructed w_i are the normalized fluctuation variables defined by the inverse coordinates, up to their norm-preserving triangular orthogonal changes. Let $\mathcal{D}_\gamma \subset \mathcal{U}_N$ be the set of configurations obtained from the configurations with coordinates in $\mathcal{B}_\gamma$ by all gauge transformations of the block axial gauge that equal the identity at the prescribed tree roots, that is, the union over the tree gauge variables of the images of $\mathcal{B}_\gamma$. Every configuration in $\mathcal{D}_\gamma$ fails the small-field event, since

$$|w_{i_*,e_*}| > p + 1 - \epsilon > p,$$

while it satisfies all other conditions of the step. In particular, $\mathcal{D}_\gamma$ is a subset of the complement of the small-field event itself.

Haar measure of the box. By repeated use of Lemma 6.3, followed by the triangular substitution from free coordinates to normalized fluctuation variables, we obtain the exact formula

$$H_N(\mathcal{D}_\gamma) = \int_{\mathcal{B}_\gamma} dH_u(c)\, dw \prod_{i=0}^{J-1} \big[\mathcal{J}_i(c_{i+1}, z_i)\, g_i^{D_i} \det S_i(c_{i+1})\big], \qquad D_i = 6(M^3-1)(uM^{J-i-1})^3. \tag{23.23}$$

The integral is over the Haar box of coarse fields and the balls of fluctuation variables; the tree gauge variables of $\mathcal{D}_\gamma$ are integrated out with Haar probability mass one. The determinant identity is that of Theorem 12.13, and its off-diagonal dependence on the coarser fluctuation variables is kept. The formula is a change of variables on the open coordinate domain, and its restriction to the box needs no indicator of the cutoff product; equivalently, it follows from successive Haar disintegration formulas for single averaging steps and Fubini's theorem for the nonnegative integrand. There is no additional density for the logarithm of the coarse field, no Gram determinant, and no gauge multiplicity. Here D_i is the number of real free coordinates of the i-th averaging step: besides the eliminated edges there are $3(M^3-1)(uM^{J-i-1})^3$ fine links, with three real coordinates each, and the gauge transformations equal to the identity at the tree roots remove $3(M^3-1)(uM^{J-i-1})^3$ of these coordinates.

On the fixed small closed product domain, all $\mathcal{J}_i$ and $\det S_i$ are continuous and strictly positive, by Lemmas 6.3 and 12.3. Their finite product has a positive lower bound depending on u and the fixed chart, but not on γ. Since $g_i = \sqrt{\gamma}\,(Mu)^{-1/2}\lambda_i$, and the box of coarse fields is a product over the $|\mathcal{E}_u| = 3u^3$ links, Lemma 5.1 and (23.23) imply

$$H_N(\mathcal{D}_\gamma) \geq c(u)\gamma^{\kappa_u}, \qquad \kappa_u = \frac{9u^3 + \sum_{i=0}^{J-1} D_i}{2} > 0, \tag{23.24}$$

where $\sum_i D_i = 6[(uB)^3 - u^3]$. This lower bound keeps the Haar densities and all powers of the couplings from the Jacobians.

The Wilson action on the box. By the majorant of Theorem 13.2, or its finite-dimensional derivative bound on the same smaller product, we have, in the gauge representative of that theorem,

$$\max_e |\operatorname{Log} c_{0,e}| \leq C_u\{\epsilon\sqrt{\gamma} + t(p+1+\epsilon)\} \leq C_u'\sqrt{\gamma}\,(1+p).$$

For each plaquette, the triangle inequality for the group distance in Lemma 5.1 bounds its distance from the identity by four times this quantity, and by the same lemma we have $q \leq \rho_G^2/8$. There are $3N^3$ plaquettes, so for a fixed finite A_u,

$$\frac{4N}{\gamma} \sum_P q((c_0)_P) \leq A_u(1+p)^2 \qquad (c_0 \in \mathcal{D}_\gamma).$$

By gauge invariance, the same bound holds on all of $\mathcal{D}_\gamma$. The partition function of the Wilson measure satisfies $Z_{N,\gamma} \le 1$, by Lemma 5.1. Combining these bounds, we obtain

$$\mu_{uB,\gamma}\big(S^c_{uB,u,\gamma}\big) \ge \mu_{uB,\gamma}(\mathcal{D}_\gamma) \ge c(u)\gamma^{\kappa_u}\exp\{-A_u(1+p)^2\}.$$

Finally, $p = (A_{\mathrm{w}}/2)[x + \log(Mu)]$ for $x = \log(1/\gamma) \ge 1$. Then $1 + p \le C''_u(1+x)$ and $\kappa_u x + |\log c(u)| \le C'''_u(1+x)^2$. After increasing C_u and decreasing γ_u to include every threshold of the finite chart and of the upper bound, this is the lower bound in (23.21). Since $e^{\varepsilon x}/(1+x)^2 \to \infty$, comparing logarithms excludes the bound $D\exp(-c\gamma^{-\varepsilon})$. □

Corollary 23.13. *Use the parameters of Theorem 23.5. For each fixed $u \in 4\mathbb{N}$ with $u \ge n_\Gamma$ there exist $c_{u,\Gamma}, C_{u,\Gamma} > 0$ and $0 < \gamma_{u,\Gamma} \le \min\{\Gamma, e^{-1}\}$ such that*

$$\begin{aligned} e^{-C_{u,\Gamma}[1+\log(1/\gamma)]^2} &\le \mu_{uB,\gamma}\big(S^c_{uB,u,\gamma}\big) \le \sup_{K\ge 0}\mu_{uB^K,\gamma}\big(S^c_{uB^K,u,\gamma}\big) \\ &\le e^{-c_{u,\Gamma}\log^2(1/\gamma)} \qquad (0 < \gamma \le \gamma_{u,\Gamma}). \end{aligned}$$

No lower bound is asserted for $\limsup_{K\to\infty}\mu_{uB^K,\gamma}(S^c_{uB^K,u,\gamma})$.

Proof. The upper bound follows from (23.13) exactly as in the proof of Theorem 23.12: with $x = \log(1/\gamma)$, the curvature term decays like e^{-ce^x}, while the fluctuation term has an exponent that is a negative quadratic in x with leading coefficient $a_{\mathrm{w}}/4$. After decreasing $\gamma_{u,\Gamma}$, the sum is at most $e^{-(a_{\mathrm{w}}/8)x^2}$.

For the lower bound we keep u and J fixed and use the coordinates of one RG step as in the proof of Theorem 23.12. Their coordinate domains, the classical change of coordinates, the Jacobians, and the cutoff of the small-field event are the same for the parameters of Theorem 23.5, since that theorem enlarges only the probability constants and the minimum side. On the box $|\theta_e| < \epsilon\sqrt{\gamma}$ of coarse fields, we place all fluctuation variables in balls of radius $\epsilon < 1/4$, except one, whose ball is centered at $(p(t)+1)e_1$, where $t^2 = \gamma/(Mu)$. Since $t(p(t)+2) \to 0$, all unscaled inputs and all Jacobian determinants lie in the same fixed domain without cutoff for sufficiently small γ. Every such point satisfies the other conditions and fails the cutoff. The exact formula (23.23) bounds the Haar measure of the corresponding set of configurations below by $c(u,\Gamma)\gamma^{\kappa_u}$, with $\kappa_u = (9u^3 + \sum_i D_i)/2$ and $D_i = 6(M^3-1)(uM^{J-i-1})^3$. The Wilson exponent on this set is at most $A(u,\Gamma)(1+p(t))^2$, and the partition function is at most one. The probability of the complement is then at least $c\gamma^{\kappa_u}e^{-A(1+p(t))^2}$. Since $p(t) = A_{\mathrm{w}}[x+\log(Mu)]/2$, we obtain the claimed lower bound. All domains and constants are fixed before γ; the restriction $\gamma \le \Gamma$ is obtained by decreasing $\gamma_{u,\Gamma}$ only. □

23.6 Proof of the choice of parameters

In this subsection we prove Theorem 23.2, verifying the choices in the order of the items and using each earlier estimate on its domains and with the quantifiers of its statement.

No probability estimate is used. Steps 1–7 make the choices of items (1)–(8), Steps 8 and 9 verify the hypotheses of the cited results without new choices, Steps 10 and 11 make the choices of items (9) and (10), and Step 12 fixes the reconstruction maps and proves the identity of the coarse fields with the iterates of T_M.

We fix $A_{\mathrm{w}} \geq 1$ in the range allowed by Theorem 20.14 at the start. This precedes every use of the existence of a coupling bound in that theorem. Its constants and the J-uniform conditions do not depend on A_{w}; the coupling bound g_* may depend on it.

Step 1: the choice of J. We apply Theorem 13.2 at the fixed odd M, and denote its J-uniform spatial exponent by b_* and its majorant for the conditional constrained minimizers at zero fluctuation by $Z(v) \leq Cv/(1-Cv)$. First we fix a rate $0 < b_{\mathbb{T}} < b_*$ below the remaining margin of the classical and image estimates. Next we fix a finite scale $\mathscr{L}_{\mathbb{T}}$ for boxes covering supports, and its support-counting factor, as in Theorem 13.5, so that the entropy of supports is compensated by the rate $b_{\mathbb{T}}$. At this stage we enlarge the entropy and support constants to cover the margin for the free degrees of freedom in Theorem 20.9. Only after fixing these boxes do we choose the rate $a_{\mathbb{T}} \in (0, b_{\mathbb{T}})$, with $a_{\mathbb{T}} < a_{\mathrm{cen}}/8$ and small enough for the network bound below. Here a_{cen} is the J-uniform rate of the bounds of Lemma 11.9; for the aligned representative, Theorem 11.10 gives the contraction at every output rate below $a_{\mathrm{cen}}/8$ and at no larger rate, so this restriction is necessary. It also gives $a_{\mathbb{T}} < a_*/8$ for the rate a_* of Theorem 10.11, as observed in Theorem 13.5. These choices precede J, and decreasing $a_{\mathbb{T}}$ does not change the boxes.

We check the support condition of the conversion from supported coefficient bounds to coefficient norms at the same analytic radius in Theorem 13.5. A connected set Y of boxes joins the coefficient locations of the background in its boxes by a network of length at most $C_{\mathscr{L}_{\mathbb{T}}}|Y|$, with $C_{\mathscr{L}_{\mathbb{T}}}$ independent of J. Indeed, we connect the centers of adjacent boxes along a spanning tree of Y, and connect the finitely many locations in each box by the coordinate grid inside that box. The same construction covers a connected union of intersecting boxes, and shortest periodic connections give the same bound in directions that wrap around the torus. The bound also covers the supports of the logarithms in the cluster expansion and the larger output radius. Repeated locations add no length. There are only nine types of orientation and color at a site, and the density of free variables does not change this count. With this $C_{\mathscr{L}_{\mathbb{T}}}$ we take $a_{\mathbb{T}} C_{\mathscr{L}_{\mathbb{T}}} \leq 2$, so that $a_{\mathbb{T}} \mathrm{t} \leq 2|Y|$ for every support used in the conversion. The effective interaction and its output as a whole, rather than the individual polymer activities, have the translation symmetry required by the conversion, and their color and inversion identities are those kept by Theorem 13.5.

Let $C^{\mathbb{T}}_{\mathrm{geom}}$ be the network comparison constant of that theorem. We include the conditions

$$C^{\mathbb{T}}_{\mathrm{geom}} b_{\mathbb{T}}/B \leq a_{\mathbb{T}}, \qquad C_{\mathrm{cen}}/B < 1/8 \tag{23.25}$$

in the choice of J. The first compensates the conversion from the trees on the finer lattice to the trees on the coarse lattice, while keeping the rate $a_{\mathbb{T}}$ on the torus; it is separate from the smaller regional rates of item (3). By part (3) of Theorem 15.12, one enlarged constant $C_{\mathrm{CP}} \geq 1$ bounds the value, the first spatial differences, and the first three coarse-field derivatives of the zero-fluctuation reconstruction, with all row sums and all volume-normalized column sums, uniformly before J is chosen; the enlargement includes the prescribed reconstruction tolerances. Derivatives with respect to the fluctuation inputs use the fixed-J joint bounds of Lemma 15.15, not this uniformity. The same results provide the signed conservation identities for the volume-normalized derivative kernels and the full bound at fixed J. The series Z of Lemma 11.9 and its spatial rate are J-uniform, as are these finite derivative bounds and the geometric constants. Neither the full joint norm nor the full norm of the gauge transformations of the reconstruction maps is J-uniform. We enlarge the J-uniform constant C_{alloc} of Theorem 20.9, which controls the distribution of the spatial decay rates, so that it also covers the corresponding geometric constant in Theorem 19.9, choose $C_*^{\mathrm{geom}} \geq \max(1, C_g)$, and keep $B_{\mathrm{geom}} = 256 C_{\mathrm{alloc}}(C_*^{\mathrm{geom}} + 1)$ from Theorem 20.9. All lower bounds on M^J in item (1) now have fixed finite right sides, and they hold simultaneously for large J, since $M^J \to \infty$.

It remains to separate the linear conditions that choose J from the later choice of radii. For existence we may take $\rho_{\mathrm{cl}} = \rho$, but we keep the subscript $\bullet$ to check both functions. The same classical map evaluates both the effective interaction and the effective classical action, and the series Z provides both Z_{q} and Z_{cl}; fixed different ratios of radii would only require the two restrictions of the same series. We choose a J-uniform bound ρ_0 with $4K_* C\rho_0 \leq 1/2$. With $Z_\bullet(v) \leq C_\bullet v/(1 - C_\bullet v)$, the conditions

$$C_\bullet K_*/B < 1/32, \qquad C_\bullet K_* \rho_\bullet \leq 1/2$$

give $B^{-1} Z_\bullet(K_* \rho_\bullet) < \rho_\bullet/16$, and at the enlarged radii, uniformly for $0 < \rho_\bullet \leq \rho_0$,

$$B^{-1} Z_\bullet(4K_* \rho_\bullet) \leq (8 C_\bullet K_*/B)\rho_\bullet < \rho_\bullet/4. \tag{23.26}$$

Only the first condition chooses J; the second is an upper bound on a later radius. Together with $q_{\mathrm{CA}} < 1/4$, the threshold $M^{-J} \leq \delta_0(a_{\mathbb{T}})$ of Theorem 11.10, and (23.25), these are the J-uniform conditions of Theorem 13.5, whose contraction constant $C_P(a_{\mathbb{T}})$ is uniform over the later radii $0 < \rho \leq \rho_0$. Since $a_{\mathbb{T}}$ is fixed before J and depends only on J-uniform data, these conditions are lower bounds on B with fixed right sides. The finitely many additional conditions on classical first derivatives use the same J-uniform series and C_{CP}, and we impose them with strict inequality in the same choice. No constant at fixed J, no radius for free coordinates, and no constant of the effective-interaction estimates has entered; no later step changes J.

Step 2: larger domains, solution neighborhoods, and rates. At this J, all entries in the minima defining $v_{\mathrm{free},J}$ and R_{raw} in Lemma 19.7 are positive and depend only on

the larger construction data, and we choose each radius strictly below its minimum. For each of the finitely many initial segments of conditional integrations, we choose by Lemma 19.3 a neighborhood of the conditional constrained minimizer and then a smaller domain for the later parameters. At zero perturbation we choose, with a positive margin, a smaller product domain for the free coordinates whose successive substitutions stay inside all the previously chosen domains; the condition with a perturbation is checked at every value of the later parameters in Step 3. Taking the minimum of the finitely many margins at zero perturbation preserves them. In particular, the conditional displacement bound and the bound on the preconditioned Hessian hold on these products, and a slice of a larger joint solution domain is never used in their place. The real lower bound for the constrained Hessian and the complex domain of its inverse in the two cited lemmas apply on the resulting product domains.

We choose the inner, middle, and outer domains of the reconstruction maps strictly inside these construction domains; Theorem 15.12 provides its bounds on these domains independently of the later support and working radii. We choose $\rho, \rho_{\mathrm{cl}} > 0$ inside them, small enough that the first summand in the inequality of item (3) is below $R_{\mathrm{raw}}/64$ and that both denominators of the torus images are positive. All coefficient sums used here can be bounded on the larger domains before the radii are restricted, since a restriction can only decrease a sum of absolute coefficients; this is the convention of Lemma 19.7. We also impose the hypotheses of Theorems 13.2 and 13.5 on the larger product. With r_J and C_J fixed on that product, we choose

$$\begin{aligned} 0 < \rho \le \min\{\rho_0, r_J/(16K_*)\}, \qquad & 4K_*C_J\rho < 1/4, \\ 0 < s_J < \min\{r_J/(16K_*), 1/(16K_*C_J), \rho/(64K_*C_J)\}. & \end{aligned} \tag{23.27}$$

These restrictions are compatible with the bounds on the coordinate and reconstruction domains just chosen. We have $4K_*(\rho + s_J) < r_J$ and $1 - 4K_*C_J(\rho + s_J) > 1/2$. Then the majorant of the free part is at most $8K_*C_J s_J < \rho/8$. Together with (23.26), the enlarged image is strictly below $3\rho/8 < \rho/2$. All intermediate margins for the classical action and for the constraint Jacobians are the strict margins of Theorem 13.2, since we work on a restriction of its product. We keep this larger product after r is fixed.

The two majorants for the input functions used here concern the torus map of Theorem 13.2. We do not assert a J-uniform bound of all orders for the gauge transformations of the reconstruction maps. The full bound at fixed J and the strict image margins of Theorem 15.12 apply on nested subproducts of the same domains, and their limiting maps and later refinements of thresholds preserve these margins. The only constants of the reconstruction maps used before J were the finitely many coarse-field derivative bounds at zero fluctuation.

Theorem 13.5 now provides finite constants $C_G, C_T > 0$, and the constant $c_* > 0$ for its auxiliary discs (not to be confused with the exponent $c_* = \min\{a_b, d_*/8\}$ of the

probability bounds), on the fixed data $(M, J, \rho, a_{\mathbb{T}}, b_{\mathbb{T}}, K_*, s_J)$. They do not depend on the side, the preceding depth, or the later support radius. We fix E^{cl} and take a finite $E \geq 2C_G/(1 - q_{\mathrm{CA}})$; the denominator is positive by item (1). We verify the invariance of this bound under the recursion in Step 8, after the coupling bound has been chosen; it is not assumed as a hypothesis of a regional estimate. For example, the radius $q_{\mathrm{raw}} > 0$ and the later r and β_{rec} can each be taken at most $R_{\mathrm{raw}}/(64C_{\mathrm{lift},J})$; then the sum in item (3) is strictly below $R_{\mathrm{raw}}/8$.

The interpolation of Lemma 19.7 provides a positive bound on the spatial rate of the unperturbed coefficients on these larger products, and we include it in $c_{\mathrm{prim},J}$ after the conversion of Section 20.1 to the graph metric. Its finite bound for sums with the remaining exponent can be written as $C_J\rho_\bullet$: we divide a common bound on the larger product by the smaller of the two positive radii and enlarge C_J to at least one. The bound is uniform over the later restrictions, which stay inside the larger product. For its remaining exponent $c > 0$ we choose $0 < \vartheta < 1$ with

$$\vartheta \max\{1, \log(8C_J/3)\} < \frac{1}{2}\log(4/3), \tag{23.28}$$

and take the minimum of the resulting positive exponents over the two radii and the finitely many initial segments. By Lemma 19.7, the bounds on the larger product are uniform in the later smaller radii and widths, so this choice precedes the choice of r.

We now set $\alpha = a/16$, $m_{\mathrm{rate}} = C_*^{\mathrm{geom}}(a + \alpha)$, and $b_1 = 8C_*^{\mathrm{geom}}\alpha$. Then $m_{\mathrm{rate}} + b_1 = (25/16)C_*^{\mathrm{geom}}a$, and the inequalities of item (3) containing Ba and $B\alpha$ reduce to

$$\frac{25}{16}C_*^{\mathrm{geom}} < B/C_{\mathrm{alloc}}, \qquad 8C_*^{\mathrm{geom}} < B/C_{\mathrm{alloc}},$$

which follow from $B \geq B_{\mathrm{geom}}$. We decrease $a > 0$ further so that $\sqrt{3}\,(a + \alpha) < a_{\mathbb{T}}$ and the conditions involving $c_{\mathrm{prim},J}$ and the unperturbed image exponent hold, including $m_{\mathrm{rate}} < \vartheta c$ for every radius and initial segment. The cluster expansion with exact supports imposes no further restriction on lengths. The rate $a_{\mathbb{T}}$ and J are unchanged.

Step 3: perturbation bounds, the fluctuation radius, and the final perturbation bound. We choose a positive preliminary bound σ_0 for perturbations that are real on real arguments, inside the margins for the force, for the lower bound of the constrained Hessian, and for the domain of its complex inverse in Lemma 19.3 and Theorem 19.9. Here and in every later application, the incoming classical norm is taken on the incoming list D_{in} of active labels, with the lattices where its coefficient locations were introduced and the volumes of its variables, at the uniform radius ρ_{cl}. The spatially varying radii are output radii. In Step 9 we verify, using the coverage by large-field decompositions and the substitution of frozen coordinates, that these are the incoming radii and compatible complements after any number of RG steps.

The perturbation must be controlled at every value of the later parameters, not only at the joint constrained minimizer. Let $\varepsilon_{\mathrm{tube}} > 0$ be the minimum of the margins of the unperturbed solution and parameter domains, and let I_0 and $C^{\mathrm{force}}_{P,J}$ be the constants on the larger product for the inverse and for the conditional force in Theorem 19.9; here V_I is the z_I-derivative, in the original free coordinates, of the function V in (21.12). The estimate of that theorem, uniform over all $y \in \mathcal{T}_I$, gives

$$\begin{gathered}\delta_V = \max_I \sup_{y\in\mathcal{T}_I} \|A_I^{-1} V_I(c, a_I(c,y;\xi), y)\|_{\mathrm{coeff}} \le I_0 C^{\mathrm{force}}_{P,J} \sigma_0 \rho_{\mathrm{cl}}, \\ C_{\mathrm{seq},J}(2\delta_V + r_{q,P}) < \varepsilon_{\mathrm{tube}}/2.\end{gathered} \tag{23.29}$$

To obtain the last strict inequality, we first decrease σ_0 so that $2C_{\mathrm{seq},J} I_0 C^{\mathrm{force}}_{P,J} \sigma_0 \rho_{\mathrm{cl}} < \varepsilon_{\mathrm{tube}}/8$ and $I_0 C^{\mathrm{force}}_{P,J} \sigma_0 \rho_{\mathrm{cl}} < \min_I r_I/4$. Then we choose $0 < r_{q,P} < \varepsilon_{\mathrm{tube}}/(8C_{\mathrm{seq},J})$ inside the fixed margin of the joint solution domain; the sum is then even less than $\varepsilon_{\mathrm{tube}}/4$. We decrease σ_0 also for the bound $1/2$ on the error of the preconditioned Hessian, for half of the real lower bound of the constrained Hessian, and for the domain of its complex inverse; these are the coefficient bounds of Theorem 19.9 on the same larger product. Its contraction at fixed parameters bounds every conditional displacement by $2\delta_V$, and the sequential conclusion of Lemma 19.3 applies at all such y with a margin of a factor two. In particular, the chain of conditional constrained minimizers at zero fluctuation satisfies the coordinate-domain conditions before the fluctuation radius is chosen. A smaller final σ_* preserves all these conditions.

All derivatives here are taken in the active background variables and the free coordinates, at fixed real frozen data $b = (\xi, b_{\mathrm{rec}})$. The incoming coefficients and charts are Borel functions of b. At the initial step, the input consists of finite products of link variables and the analytic charts of the classical action; at a step starting from a stopped measure, it is the effective interaction, which is invariant on overlapping charts; and each later input is obtained by substitution, finite products, and convergent operations on coefficients. Integration over a fixed compact product of fluctuation balls preserves Borel dependence on b. Indeed, after rescaling the balls to unit balls the measure is a fixed finite measure, and the integral of a nonnegative jointly Borel function against a fixed finite measure is a Borel function of the parameters; this is the monotone class argument given in the proof of Theorem 21.9. Then the uniformly convergent Picard iterations in Theorem 19.9 preserve Borel dependence on b. No derivative in a frozen real parameter is needed.

On these larger fixed products, Lemma 20.3 and Theorems 20.14 and 21.8 provide finite constants for the output, sequential, image, potential, and mean estimates, and for the estimates of the support of the normalized fluctuation variables, uniformly for the doubled classical bound $2\sigma_0$. Their dependence, described in item (3) of Definition 4.14, excludes all later widths and all values of frozen coordinates. We use these bounds, rather than bounds computed on the eventual small cutoff domain. The threshold $\epsilon_{\mathrm{en},J}$ and

the constant $C_{\mathrm{en},J}$ of Theorem 21.8 are fixed at this stage as well; the threshold already contains the smallness $C_J(r+\beta_{\mathrm{rec}}) \le \epsilon_{M,J}$ that its proof needs for the Hessian bound (18.45) of Lemma 18.14, at the small representative, before the orthogonal transport to the representatives used in the descent. The radius of the auxiliary box in that proof is chosen there. The constant $C_{\mathrm{en},J}$ is a bound for every covered large-field decomposition; its conclusion for every point is applied only after the coverage is established in Step 9.

We fix all geometric constants at this stage. Let C, C', R_0, c be the constants of Theorem 21.3. We take $A_8 \ge R_0 + 1$ so large that $C'/A_8 < 1/4$, and enlarge it for the fixed mesh at the initial step and the fixed margins for path lengths. This is a lower bound independent of r and β_{rec}. Let $C_B \ge (10M)^J$ cover the lengths of all tree and transport paths within one RG step, and set

$$K_B = 2 + C_B(A_8 + 1), \qquad C_{\mathrm{col}} = CK_B^{10}/A_8. \tag{23.30}$$

Every small frozen coordinate created at the first step after a stopped measure is a product of a fixed number of current small links and of their at most J iterates under T_M. By repeated use of the triangle inequality for the group distance and the factor $10M$ per averaging step, it is bounded by $C_{\mathrm{ad},J} r$, where $C_{\mathrm{ad},J}$ depends only on these fixed products. The frozen coordinates at the protected initial step satisfy the same bound. Older long transports are frozen group variables, not new small analytic coordinates. Let $C_{\mathrm{an},J} \ge 1$ bound all fixed conversions from group or transported coordinates to the analytic coordinates at the lattice where a frozen coordinate was introduced. For a link variable in the identity chart this constant is one, since $|\mathrm{Log}\, U| = \rho_G(U)$; a fixed longer product is bounded by its fixed length and its analytic derivative bound. We fix $C_{\mathrm{conv},J}$ large enough for these conversions, including

$$C_{\mathrm{conv},J} \ge C_{\mathrm{ad},J}, \qquad C_{\mathrm{conv},J} > 2C_{\mathrm{an},J}C_{\mathrm{col}}, \qquad C_{\mathrm{conv},J} > C_{\mathrm{an},J}C_{\mathrm{ad},J},$$

and enlarge it for the fixed transport paths and the protected frozen coordinates of Theorem 21.5. We fix the mesh multiplier $C_{\mathrm{mesh},J}$ of the initial step and the interior restriction by the full dependence radius at the same stage. None of these constants depends on a later width.

Every entry in the minimum defining R_W in Lemma 20.3 is positive at this J. We choose R_W strictly below this minimum, including the conversion to the joint solution neighborhood and both product domains. Include in $C_{P,\bullet}$ the larger-product absolute coefficient row constant for the difference of the perturbed and unperturbed reconstructions at output rates (m_{rate}, b_1) in Theorem 19.9. These rates satisfy its conditions by item (3), since C_{alloc} includes its geometric constant. Then we fix $\sigma_* > 0$ below σ_0, below all preconditioned displacement margins, and below

$$\frac{\lambda_* R_W}{6C_{\mathrm{en},J}\rho_{\mathrm{cl}}}, \qquad \min_{\bullet} \frac{\rho_\bullet}{64C_{P,\bullet}\rho_{\mathrm{cl}}}.$$

If a constant vanishes, the corresponding upper bound is not needed. Only fixed positive upper bounds occur. In particular, the perturbation term in the radius R_{en} of the enclosing product domain is accounted for before the removed layer is chosen.

Step 4: boundary radii, removed layers, and frozen coordinates. We choose each outer boundary radius so that its contribution is at most $\rho_\bullet/16$, and set $\rho_{\partial,\bullet} = R_{\partial,\bullet}/K_*$. Since $a_{W,\bullet} > 0$, a finite common width w_{bd} makes each contribution from distant parts of the boundary at most $\rho_\bullet/16$. The torus contribution was made smaller than this in Step 1. Once β_{rec} is chosen below with $C_{F,\bullet}\beta_{\mathrm{rec}} < \rho_\bullet/64$, and since $C_{P,\bullet}\sigma_*\rho_{\mathrm{cl}} < \rho_\bullet/64$ by Step 3, the sum at zero fluctuation satisfies

$$A_{0,\bullet} < \Big(\frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{64} + \frac{1}{64}\Big)\rho_\bullet = \frac{7}{32}\rho_\bullet < \rho_\bullet/4. \tag{23.31}$$

The individual bounds give the full sum condition of item (5), with every summand counted.

The brackets in the two inequalities for s_{er} are fixed finite numbers. For example, if $Y = A_J(1+E) + K_J\sigma_* + 1$, then $s_{\mathrm{er}} \geq 1 + \alpha^{-1}\log(4Y/\sigma_*)$ suffices for the first inequality, and the same calculation with $Y^{\mathrm{cl}} = A_J^{\mathrm{cl}} + K_J^{\mathrm{cl}}\sigma_*$ yields the second. We take an integer above these bounds and above $\max(2, w_{\mathrm{bd}})$. If a bracket vanishes, its inequality holds already.

We now choose $\beta_{\mathrm{rec}} > 0$ below the minimum of the coordinate-domain, product, and boundary radii and of the bound $\lambda_* R_W/(6C_{\mathrm{en},J})$ used for the radius of the enclosing product domain. On every label with reduced radius, β_{rec} below one quarter of the corresponding inner boundary radius makes the frozen fraction at most $1/4$. The bound for the boundary-completion difference in Lemma 18.9 is $8\beta_{\mathrm{rec}}E_\bullet/\rho_\bullet$ at uniform radius; the fixed conversion to the spatially varying radius changes only its constant on the larger domain. This is part (2) of that lemma. Its hypothesis on the torus rate holds for both functions, since their torus norms have the rate $a_{\mathbb{T}}$ and $\sqrt{3}\,(a+\alpha) < a_{\mathbb{T}}$ by item (3). We fix a constant $C_{\mathrm{bc},\bullet} \geq 8$ for the simultaneous substitution of all frozen coordinates and every required conversion, including the boundary layer, the neighborhoods where gauge transformations are extended at the lattices of introduction, the based transports of frozen links across the boundary, and the initial step. If these substitutions are made in several fixed stages, we take the sum of their converted bounds in this constant. The number of frozen labels, of components, and of earlier RG steps does not enter, since the estimate of Lemma 18.9 already bounds the union. We require

$$C_{\mathrm{bc},\bullet}E_\bullet\beta_{\mathrm{rec}}/\rho_\bullet \leq \sigma_*/8, \qquad \beta_{\mathrm{rec}} < \frac{1}{4}\min\{\rho_\bullet, \rho_{\partial,\bullet}\} \quad (\bullet \in \{\mathrm{q}, \mathrm{cl}\}). \tag{23.32}$$

Then the sum of these differences is below the bound $\sigma_*/4$ required in item (5). We also require $\beta_{\mathrm{rec}} < \min(\pi/4, \epsilon_{\mathrm{en},J}/4)$, where $\epsilon_{\mathrm{en},J}$ is the fixed margin of Theorem 21.8. The geometric argument of Step 6 provides these bounds in the analytic coordinates at the

lattice of introduction. We include the bound for frozen coordinates used in (23.31). All these choices are compatible, since each is an upper bound with a fixed positive right side.

For a large-field decomposition with the hypotheses on incoming radii and boundaries verified in Step 9, these conditions preserve the initial norm bound. If the input norm is at most σ_*, the last nonlinear term in Theorem 20.14 is at most one after the coupling bound is chosen in Step 7. Then its output is bounded by Y, and its output after removal of the layer of width s_{er} is bounded by $\sigma_*/2$. Adding the boundary-completion difference, we obtain at most $3\sigma_*/4$. The classical calculation uses Y^{cl} and gives the same bound $3\sigma_*/4$. The initial norm is at most σ_* once the initial substitution has been checked. Step 9 provides this initialization and the input hypotheses at every step, so no errors are summed over earlier RG steps.

Step 5: the majorant of the reconstruction maps, the working radius, and the support radius. The coefficient constants of Lemma 15.15 were computed on the fixed larger domains. Recall that $\alpha_* = C_{\mathrm{CP}}/B$. Since $B \geq 3$ and $C_{\mathrm{CP}}^2/B \leq 1/4$, we have $0 < \alpha_* \leq 1/(2\sqrt{B}) < 1$. For the coefficients in that lemma, the choice of β_J gives, for every $m \geq 2$,

$$\alpha_* \beta_J^{m-1} \geq \alpha_* \frac{A_J}{\alpha_* R_J^2} R_J^{-(m-2)} = A_J R_J^{-m},$$

where we used $\beta_J \geq A_J/(\alpha_* R_J^2)$ and $\beta_J \geq R_J^{-1}$. The linear term is also bounded by α_*. This proves the majorant φ in (23.4) and its derivative bound at every fixed input, in the volume normalization of that lemma. We choose $r_{\mathrm{eff}} > 0$ inside the domains of item (3) and below $(1-\alpha_*)/(8\beta_J)$. Then $\beta_J r_{\mathrm{eff}} < 1/8$, and we have

$$\sup_{|v| \leq r_{\mathrm{eff}}} \|D\widehat{P}(v)\|_{\mathrm{root}} \leq \varphi'(r_{\mathrm{eff}}) = \frac{\alpha_*}{(1-\beta_J r_{\mathrm{eff}})^2} \leq \frac{64}{49}\alpha_*, \qquad a_{\mathrm{map}}\sqrt{B} \leq \frac{32}{49}. \tag{23.33}$$

We choose r below all positive bounds displayed in item (6), including the bound from path lengths, the bound from frozen coordinates, and $r_{\mathrm{eff}}/4$. Further upper bounds used for the protected neighborhoods below are imposed in the same choice. We include $r < \min(1, \epsilon_{\mathrm{en},J}/4)$ and

$$K_B^2 r/A_8 < c, \qquad C_{\mathrm{an},J} C_{\mathrm{col}} r < \beta_{\mathrm{rec}}/2, \qquad C_{\mathrm{ad},J} r < \beta_{\mathrm{rec}}, \tag{23.34}$$

which hold after decreasing r, since all constants in (23.30) are fixed. In particular, $\beta_{\mathrm{rec}} > C_{\mathrm{conv},J} r$. With $R_{\mathrm{buf}} = C_{\mathrm{map},J} r$ and a fixed $C_{\mathrm{map},J} \geq (10M)^J$, the requirement $R_{\mathrm{buf}} < \rho$ and the margins of the larger charts are again upper bounds on r. No lower bound on r occurs.

We also choose the radii P and Q_{DB} explicitly. On a fixed smaller joint product, the bound of Lemma 15.15 provides a finite $C_{\mathrm{free},J} \geq 1$ such that the free increment of the reconstruction is at most $C_{\mathrm{free},J} \max_i |\zeta_i|$. For $|v| \leq r$ we have

$$|\widehat{P}(v)| \leq \frac{\alpha_* r}{1 - \beta_J r} \leq \frac{32}{31} \cdot \frac{r}{2\sqrt{B}} < \frac{r}{3},$$

where we used $\beta_J r < 1/32$ and, in the last inequality, $B \geq 3$. For the torus map of Theorem 13.2 we also take $Cr < 1/2$ and $r < \min(r_J/4, 1/(4C_J))$; by the J-uniform condition we have $B^{-1}Z(r) \leq 2Cr/B < r/3$. We choose

$$0 < Q_{\mathrm{DB}} = q_{\mathrm{DB},J} r, \qquad q_{\mathrm{DB},J} \leq \min\{1/2, 1/(12C_{\mathrm{free},J}), 1/(24C_J)\}. \tag{23.35}$$

Then $r + Q_{\mathrm{DB}} < r_J$ and $1 - C_J(r + Q_{\mathrm{DB}}) > 1/2$. The majorant of the free part of the torus map is then below $2C_J Q_{\mathrm{DB}} \leq r/12$, and the hypotheses of Theorem 13.2 for coarse fields in B_r hold, including the domain before the pole of its majorant and $\mathfrak{M}_J(r, Q_{\mathrm{DB}}) < r/2$. The same Q_{DB} works for the free part of the reconstruction maps, since $C_{\mathrm{free},J} Q_{\mathrm{DB}} \leq r/12$. We take $P = Q_{\mathrm{DB}}/2$, decreasing the fixed proportionality factor if necessary so that every intermediate coordinate-domain condition holds. Then every input logarithm is smaller than $r/3 + r/12 < r/2$. At zero coarse field and zero fluctuation all these intermediate maps vanish, and their finite coefficient bounds at fixed J and the positive margins of the coordinate and product domains justify this simultaneous decrease. It is made when r is chosen, before the coupling. The enclosing constant $\mathscr{L}_{\mathrm{en}}$ of Definition 14.4 can be bounded on the larger fixed domains, and taking r smaller first makes the part of its curvature bound without the coupling strictly less than $1/2$, as required in item (6). The remaining term $t^2(1 + E)$ is controlled in Step 7.

The three contributions to the radius of the enclosing product domain give

$$r + \beta_{\mathrm{rec}} + \sigma_* \rho_{\mathrm{cl}} \leq \frac{\lambda_* R_W}{2C_{\mathrm{en},J}}, \qquad R_{\mathrm{en}} = \frac{2C_{\mathrm{en},J}}{\lambda_*}(r + \beta_{\mathrm{rec}} + \sigma_* \rho_{\mathrm{cl}}) \leq R_W. \tag{23.36}$$

For either radius, each of the two terms subtracted in the denominator of item (6) is at most $1/4$, since $C_{T,\bullet} K_* \rho_\bullet \leq 1/4$ and, by the choice of R_W, $C_{T,\bullet} R_W \leq 1/4$. The denominator is then at least $1/2$, and the increment from the fluctuations is at most

$$(2C_{T,\bullet} + C_{\Delta,\bullet}) R_{\mathrm{en}} \leq (2C_{T,\bullet} + C_{\Delta,\bullet}) R_W \leq \rho_\bullet/8.$$

Together with (23.31), this gives the bound $3\rho_\bullet/8$ at spatial exponent zero. Write S^0_b for the absolute image row of the unperturbed reconstruction at exponent b, and ΔS_b for the absolute coefficient row of the difference of the perturbed and unperturbed reconstructions on the same products. The interpolation of Lemma 19.7 and (23.28), applied to the unperturbed coefficients, gives

$$S^0_{m_{\mathrm{rate}}} \leq S^0_{\vartheta c} < (\sqrt{3}/4)\rho_\bullet,$$

where the choice in Step 2 includes $m_{\mathrm{rate}} < \vartheta c$. Applying Theorem 19.9 at output rates (m_{rate}, b_1) and the bound on σ_* in Step 3 now gives

$$\Delta S_{m_{\mathrm{rate}}} \le C_{P,\bullet}\sigma_*\rho_{\mathrm{cl}} \le \rho_\bullet/64, \qquad S_{m_{\mathrm{rate}}} \le S^0_{m_{\mathrm{rate}}} + \Delta S_{m_{\mathrm{rate}}} < \left(\frac{\sqrt{3}}{4} + \frac{1}{64}\right)\rho_\bullet < \rho_\bullet/2.$$

This is the required absolute weighted image bound for both input functions on the entire enclosing product. The higher exponent included in $c_{\mathrm{prim},J}$ belongs to the unperturbed construction; inherited perturbative networks use the separate inequalities against Ba and $B\alpha$.

Step 6: widths, the curvature threshold, and the large-field decompositions. Let R_{cell}, W, ϱ, and r_F be as in item (7). We may also require $8BA_8/r \ge s_{\mathrm{er}}$ at the initial choice of r; replacing the width of the removed layer by ϱ only decreases the left sides of its exponential inequalities. We now verify the geometric hypotheses of Theorem 21.5, using Theorem 21.3 for the neighborhoods in which gauge transformations are extended. Here $R_{\mathrm{cell}} \ge R_0$, and the output side after each RG step must be at least $8R_{\mathrm{cell}}$; both follow from the choice of A_8 and from the minimum side in Step 11. Whenever $\delta R^2_{\mathrm{cell}} \le 1$ and $\eta \le C_B r$, the constants of the estimate on large cells satisfy

$$\begin{aligned} K_R &= 1 + \eta R_{\mathrm{cell}} + \delta R^2_{\mathrm{cell}} \le K_B, \\ K_R^2/R_{\mathrm{cell}} \le K_B^2 r/A_8 < c, &\qquad CK_R^{10}/R_{\mathrm{cell}} \le C_{\mathrm{col}} r < \beta_{\mathrm{rec}}/(2C_{\mathrm{an},J}), \end{aligned} \tag{23.37}$$

where we used $rR_{\mathrm{cell}} \le A_8 + r \le A_8 + 1$. Away from the two prescribed boundary layers of cells, the bound is $C'/R_{\mathrm{cell}} < r/4$. Then the fixed conversion for products of links and (23.34) give the margin $r/2$ in the bulk and the bound for frozen analytic coordinates required by Theorem 21.5. These small representatives are used only for their existence, not for a change of variables in a probability.

To protect older data, we mark, in each lattice where data were introduced, both endpoints of every older frozen link, both endpoints of every transported frozen coordinate, and all prescribed tree roots and coefficient locations attached to frozen records. We exclude every current block of the coarse lattice whose centered lift to any of these lattices meets a mark. A gauge transformation supported in the remaining active blocks has a lift that is constant on blocks and equals the identity at every mark, by the corresponding clause of Theorem 21.3; hence every older frozen link and transport is preserved exactly. The newly chosen roots of the current constraint coordinates are fixed separately by the current based gauge, as in that theorem; they are not added to the persistent exclusion marks unless attached to a frozen record. The endpoints of the current links across the boundary are fixed by the target-dependent gauge of Lemma 19.5, and their records are protected at subsequent steps if those links are frozen. The union of the persistent exclusions protects the data of all earlier RG steps at once, without a product of losses over these steps.

We fix the block-aligned input mesh $S = B\lceil C_{\mathrm{mesh},J} R_{\mathrm{cell}}/B\rceil$, where $C_{\mathrm{mesh},J} \geq 1$ was fixed in Step 3, large enough for the layer of large cells around the active blocks and for the full dependence radius of the constraint, tree, and transport paths. Then $R_{\mathrm{cell}} \leq S \leq (C_{\mathrm{mesh},J} + B)R_{\mathrm{cell}}$, so $S \asymp_J R_{\mathrm{cell}}$. At an initial RG step from the Wilson measure, the input side n_{input} is divisible by B. When $n_{\mathrm{input}} \geq 8S$, this mesh partitions each coordinate circle into block-aligned intervals with lengths in $[S/2, 2S]$: we distribute the n_{input}/B blocks among $\lfloor n_{\mathrm{input}}/S \rfloor$ intervals as evenly as possible.

At the initial step, the procedure of Theorem 21.5 tests the input plaquettes before a representative is chosen, and keeps only those closed large cells whose full dependence sets avoid the excluded dependencies. Every plaquette of the resulting complex, including each boundary plaquette, is tested good. We restrict the active blocks to the interior by one full layer of large cells and by the fixed dependence radius. This places every plaquette that depends on an active variable, and every relevant constraint, tree, and transport path, inside this closed complex. There are no older protected data at the initial step. The curvature condition below implies the estimate on large cells there, and the remaining complement consists of arbitrary compact data. This is the initial covering construction of Theorem 21.5; it is not a restriction of the Wilson measure to D_r.

The three upper bounds d_{SG}, $B_*/(96M^3)$, and $\delta_*^2/(2\pi^2)$ for d_{cut} are fixed and positive; the first is given explicitly below. Since $d_* = d_{\mathrm{cut}}/2$, the mesh condition $\pi\sqrt{2d_*}S^2 \leq 1$ is the additional bound $d_{\mathrm{cut}} \leq 1/(\pi^2 S^4)$. Since $S \geq R_{\mathrm{cell}}$, it also implies $\pi\sqrt{2d_*}R_{\mathrm{cell}}^2 \leq 1$ for the estimate on large cells. For the link bound $r/2$ we use Theorem 7.7 with support radius $r/2$; one may take

$$d_{\mathrm{SG}} = \min\{\delta_{\mathrm{SG}}^2/(2\pi^2),\ r^4/(8192\pi^2 C_{\mathrm{SG}}^4)\}, \qquad m \geq 8C_{\mathrm{SG}}/r,$$

where the second condition is a condition on the side m. Strict inequalities in the threshold and in the side give the link bound $r/2$. The constants $K_M, K_1, \kappa_0, c_A^{\mathrm{act}}, \alpha_W, K_W$ are those of Theorem 22.4 for arbitrary fields. Its hypotheses require the Borel extension of T_M and complete predecessor patches, both of which we use. We use no improvement of K_M based on tests at an untested stopping level; such an improvement would require an energy hypothesis at the base and tests at the top plaquettes.

With these conditions, Theorem 21.5 gives, for each finite sequence of RG steps starting from a stopped measure or from the protected initial step, a finite Borel partition of the support of the starting measure into large-field decompositions. Its tests use the iterates of T_M, including its Borel extension, and every test is kept. Its geometric constants $C_{\mathrm{cov},J}$ do not depend on the widths, and the counts keep the widths explicitly:

$$\begin{gathered} R_{t+1}(f) \leq B^{-1}R_t(f) + C_{\mathrm{cov},J}(W + R_{\mathrm{cell}} + 1), \qquad R_{\mathrm{first}}(f) \leq C_{\mathrm{cov},J}(W + R_{\mathrm{cell}} + 1), \\ N_{\mathrm{first}}(f) \leq C_{\mathrm{cov},J}(BW + R_{\mathrm{cell}} + 1)^3, \qquad N_{\mathrm{later}}(f) \leq C_{\mathrm{cov},J}(W + R_{\mathrm{cell}} + 1)^3, \\ F_J = C_{\mathrm{cov},J}(W + R_{\mathrm{cell}} + 1)^3. \end{gathered} \tag{23.38}$$

Here, for a threshold failure f, $R_t(f)$ is the radius of the neighborhood attributed to f at the output of the t-th following RG step, and $R_{\mathrm{first}}(f)$ is its radius at creation. The counts $N_{\mathrm{first}}(f)$ and $N_{\mathrm{later}}(f)$ bound the numbers of labels of normalization factors attributed to f at its creation and at each later step, and F_J bounds the number of plaquettes of the insertion at the stopping level affected by f. The affected sets include the full excluded neighborhoods: the protected neighborhoods, the touching cells, the boundary layers of frozen coordinates, the links across the boundary, the cores enlarged by the fixed layer, and the attachments at the lattices of introduction, together with the closed excluded sets at the initial step. They are not just the smaller predecessor sets used in the energy estimate. For the fixed finite family of stopping levels in Step 10, we enlarge $C_{\mathrm{cov},J}$ to count the discarded plaquettes of the insertion whose full dependence sets meet these excluded sets. The family of remainders is bounded independently of the number of averaging steps. This enlargement keeps the local dependence at fixed J, and all dependence on widths remains in the polynomials of (23.38). These counts are subadditive under unions of fixed complements, and they, rather than local norm constants that do not depend on widths, determine F_J and $C_{\mathrm{ch},J}$. The only side conditions used for this cover are $8R_{\mathrm{cell}}$ for outputs and $8S$ for the input at the initial step; both are imposed in Step 11. If the active region is empty, the construction stops, and all its frozen data are kept.

Step 7: the coupling bound. With A_{w} fixed, we set $c_J^{\log} = A_{\mathrm{w}}(J-1)\log M/2$, so that $p^+ = A_{\mathrm{w}}\log(1/t) + c_J^{\log}$. All fluctuation products used with Theorem 20.14 have radii $p(t) \le p_i \le p^+$. In the choice of the coupling bound we impose the two remainder conditions also with p^+:

$$\begin{aligned} C_T t^2[(p^+)^4 + E(p^+)^2] &\le (1-q_{\mathrm{CA}})/2, \\ C_J t^2(E+\sigma_*)[(p^+)^4 + (E+\sigma_*)(p^+)^2] &\le 1, \qquad tp^+ < R_{\mathrm{en}}. \end{aligned} \tag{23.39}$$

These imply the conditions of item (8) with the common radius $p(t)$ and the conditions for every product in the family of logarithmic radii. No radius of size R_{en}/t enters a cluster expansion: the enclosing product domain is used only for the Hessian, for the mean, and for the comparison with the auxiliary measure. Every inequality of item (8) involving t, after multiplication by a fixed positive denominator if necessary, is bounded by finitely many terms $Ct^{\varepsilon}[1+\log(1/t)]^k$. For the interaction radius, for example, we first impose $C_J t p^+ \le 1/2$; then its denominator is at least $1/2$, and

$$\mathfrak{s}(t) \le 2C_J t[(p^+)^3 + p^+ + (E+\sigma_*)p^+] \longrightarrow 0.$$

The other bound for discs has the same limit. We apply Lemma 23.1 to all these inequalities, to $tp(t) < P$ and $tp^+ < R_{\mathrm{en}}$, and to every fixed bound for free coordinates, for these coarse-field derivatives of the normalized log density on the fixed fluctuation product, and for Jacobians. We choose t_* for the two conditions on the potential and the mean,

and then $g_* \le t_*$ for the remaining inequalities. These include the curvature condition of item (8) involving $t^2(1+E)$, the conditions $g_*^2 \le \alpha_W$, $g_*^2 \le 2\kappa_*$, and $a_0 g_* < \min(r/2, 1)$, and the bound $t[p(t)^3 + Ep(t)] \le c_*/2$ for the auxiliary discs, with the constant c_* of Theorem 13.5. For the bound on the reconstruction amplitude in Lemma 15.15 we require in addition

$$C_{\mathrm{free},J}\, t p(t) < (1 - a_{\mathrm{map}}) r_{\mathrm{eff}}/2 \qquad (0 < t \le g_*), \tag{23.40}$$

which is possible by the same argument. Together with $b_0 \le r < r_{\mathrm{eff}}/4$ and $a_{\mathrm{map}} < 1$, this keeps the recursion for the amplitude bound inside the working ball, since $a_{\mathrm{map}} r_{\mathrm{eff}}$ plus this forcing term is strictly below r_{eff}. All these conditions hold on the interval $0 < t \le g_*$.

Step 8: the effective interaction on the torus. We now attach the constants to the recursion that defines the small-field event. We start with $R_0 = 0$ and use the common fluctuation product $p_i = p(t)$ and the full formula for the auxiliary integral in Theorem 13.5. At every step the input effective interaction is invariant on overlapping charts, has vanishing first derivatives at the identity, and has the total color, translation, and inversion symmetries; these hold at $R_0 = 0$ and are preserved by that theorem. Its hypothesis on the enlarged image is (23.27), and the unscaled fluctuation inputs lie in the product domain by the coupling conditions of Step 7. The conversion of coefficients at the same radius is valid by the checks of supports and symmetries in Step 1. If $\mathcal{B}_{\rho,a_{\mathbb{T}}}(R_k) \le E$, then we have

$$\begin{aligned} \mathcal{B}_{\rho,a_{\mathbb{T}}}(R_{k+1}) &\le C_G + q_{\mathrm{CA}} E + C_T t^2 E[p(t)^4 + Ep(t)^2] \\ &\le \frac{1}{2}(1 - q_{\mathrm{CA}})E + q_{\mathrm{CA}} E + \frac{1}{2}(1 - q_{\mathrm{CA}})E = E, \end{aligned} \tag{23.41}$$

where we used $C_G \le (1 - q_{\mathrm{CA}})E/2$ and item (8). The norm does not change when a constant is subtracted. This proves the bound, at the fixed radius and with the torus rate, at every finite depth of the form kJ. The constants C_G and C_T were chosen after J and the outer radii, and neither was used to choose J.

We apply Theorem 14.6 to the same parameters. The condition on the product for coarse fields in B_r was checked in (23.35), and $tp(t) < P$ and the intermediate margins for coordinates, the classical action, and the Jacobians hold. The support of the starting measure is D_r, the fluctuation cutoff is the same common radius, the constrained Hessians are the same positive Hessians of the classical action, and the triangular determinant and the constant A_{coord} are unchanged. At every coarse field in D_r, the integral of the Wilson density over the fiber of the block averaging map, restricted by the conditions of the small-field event, equals the auxiliary integral just used. At preceding classical depth aJ we write $r_{aJ} = R_a - R_a(I)$. The stopped measure used as a starting measure in Theorem 22.1 is then

$$\widehat{Z}^{-1} \mathbf{1}_{D_r} e^{-E_{aJ,\mathrm{sat}}/g^2 - r_{aJ,\mathrm{sat}}}\, dH, \qquad \mathcal{B}_{\rho,a_{\mathbb{T}}}(r_{aJ}) \le E, \tag{23.42}$$

where the subscript sat denotes the gauge-invariant extension to D_r. All removed values $R_a(I)$ and all constants from coordinates and Gaussian integrals stay in the normalization constant. With the convention that unnormalized densities are written $\exp(\Lambda - E/g^2 - R)$, the update is $\Lambda_{a+1} = \Lambda_a + \log A_{\mathrm{coord}} - R_+(I)$. The same holds at every allowed starting side and preceding depth, since the two cited theorems are uniform in these parameters. In particular, this is the recursion on the torus used in Theorem 22.1; no new analytic class is assumed.

The reconstruction maps give the same effective interaction. At coarse fields in B_r, invariance on overlapping charts and the gauge transformations depending only on the coarse field preserve the full fiber integral, including its constant, by Theorem 15.12. On the common outer domain both auxiliary formulas are holomorphic. Equality on a real open ball of coarse fields implies equality of all real Taylor derivatives there, and hence of the holomorphic series. Their nonconstant parts, and the bound (23.41), coincide; no J-uniform bound of all orders for the reconstruction maps is used in this identification.

Step 9: the hypotheses of the regional estimates. We take the large-field decompositions of Theorem 21.5, with the geometric conditions verified above. The controlled analytic functions are evaluated in the simultaneous gauge representatives, provided by the cover, in which all frozen coordinates are small. The protection of Step 6 preserves all older frozen data and their attachments exactly. The new analytic coordinates in the neighborhoods where gauge transformations are extended have the bound checked in Step 6, and $C_{\mathrm{conv},J} \geq C_{\mathrm{ad},J}$ covers the initial representative of size r (not $r/2$) after a stopped measure. Frozen logarithms of single links have Euclidean norm equal to their group distance, and the larger constant $C_{\mathrm{an},J}$ covers the other fixed products. We conclude that every substitution of analytic frozen coordinates, including the initial one, satisfies

$$|c_j^\sharp| \leq \min\{\beta_{\mathrm{rec}}, \phi_\bullet(j)/4\} \qquad \text{for both spatially varying radii.} \tag{23.43}$$

Plaquettes with energy of order one and arbitrary distant data are frozen group variables, and they are never substituted into an analytic coordinate. We do not claim that an arbitrary independent gauge transformation at the endpoints preserves the norm of a frozen logarithm. The bounds for small frozen coordinates are used in the simultaneous representatives provided by the cover, with older marks fixed; the later comparison of measures is the conclusion of Theorem 22.1 for representatives satisfying the reconstruction identities. Color rotations at the roots, without perturbation, preserve the relevant free norms, since they act by isometries of the adjoint representation.

At a step starting from a stopped measure, the incoming action on the torus and (23.42) have no boundary perturbation before the substitution. At a step starting from the Wilson measure, $R_0 = 0$ and the active Wilson action is the exact restriction; the action of the fixed complement stays with its compact variables and its normalizing constant. The first substitution of analytic coordinates creates only the stated boundary-completion

differences. The initial boundary norms are at most σ_*, by (23.32), with their constants kept in the full action and in the normalization constant. This proves the initialization $x_0, x_0^{\mathrm{cl}} \le \sigma_*$ for both kinds of starting measure.

After each substitution at the output, the zone of reduced radius is removed, and all remaining labels have distance at least $s_{\mathrm{er}} \ge w_{\mathrm{bd}}$ from the fixed complement. Lemma 18.9 then gives the uniform incoming radii ρ and ρ_{cl} on the next list D_{in} of active labels, with the lattices of introduction of its coefficient locations and the volumes of its variables. This is the uniform input norm required in Theorem 19.9; no derivative is estimated with a reduced radius at the boundary of the output. Its adjustment for the completion adds only the boundary-completion difference already accounted for, and we have $\mathcal{X}^{\mathrm{cl}}(\overline{P}) \le 2\sigma_*$. The input for the effective interaction has $\mathcal{X}(f) \le \sigma_*$, and its part on the torus is the effective interaction in (23.41).

Theorem 21.5 also places every older incoming complement and its older coefficient locations in the next current fixed complement, up to its fixed buffer for the dependence of paths. In input units this gives, with $B_s = B$ for each integrated step,

$$d_{D_{\mathrm{in}}}(i) \ge B d_{\mathrm{current}}(e) - B \operatorname{dist}(e, i/B) - C_{\mathrm{buf},J}. \tag{23.44}$$

Indeed, a point of the older complement nearest to i, divided by B, lies in the current fixed complement up to $C_{\mathrm{buf},J}/B$, so $d_{\mathrm{current}}(i/B) \le (d_{D_{\mathrm{in}}}(i) + C_{\mathrm{buf},J})/B$; the triangle inequality from e to i/B and a rearrangement yield (23.44). Additional fixed complements only enlarge the current complement and preserve this implication. At an initialization without boundary, the input marked by the boundary vanishes. The input norm and the norm of differences use this same list and uniform radius, and every input image and connecting segment lies strictly inside it, by the weighted image bound of half the radius and the product domains of Step 2.

We conclude that the hypotheses of Theorem 19.9 on conditional displacements, frozen real data, uniform radii, and boundary compatibility hold after any number of RG steps. The inequalities for the rates required there are included in the choice of rates in Step 2 and in $m_{\mathrm{rate}} + b_1 < c_{\mathrm{prim},J}/C_{\mathrm{alloc}}$. For each complete RG step we use the common fluctuation product, or one of the products with logarithmic radii in an auxiliary comparison. By the bounds (23.29), (23.32), (23.39), and (23.43), all hypotheses of Theorem 20.14 hold. Combining its output estimate, the removal of the boundary layer, and the new boundary-completion difference, we obtain the recursion with bound $3\sigma_*/4$ of Step 4. By induction, these hypotheses hold again at the next complete RG step.

We now apply Theorem 21.8 to each covered large-field decomposition. Its construction hypotheses and its hypotheses on the two images have been checked, and $r + \beta_{\mathrm{rec}} < \epsilon_{\mathrm{en},J}$ by our choices. For every point of every fiber in the support, over compatible fixed data, we have $g_i|w_i| \le \delta_{\mathrm{en}}$ by that theorem. Since $g_i \ge \lambda_* t$, this implies

$$|w_i| \le \delta_{\mathrm{en}}/(\lambda_* t) = R_{\mathrm{en}}/(2t). \tag{23.45}$$

Conversely, every point of the enclosing product domain satisfies $|g_i w_i| \le R_{\mathrm{en}}$. The full coordinate, sequential, and image margins apply on the enclosing product domain, while the points of the support keep a margin of half the radius.

By Lemma 20.3, on the enclosing product domain we have

$$\|D_w^2(V - |w|^2/2)\|_{\mathrm{row,col}} \le \frac{C_0(R_{\mathrm{en}} + t^2\mathcal{L}_V)}{(1 - AR_{\mathrm{en}})^3} \le \frac{1/16 + 1/16}{(1 - 1/4)^3} = \frac{8}{27} < \frac{1}{2}, \tag{23.46}$$

where we used $R_{\mathrm{en}} \le R_W \le 1/(16C_0)$, $C_0 t_*^2 \mathcal{L}_V \le 1/16$, and $AR_{\mathrm{en}} \le A/(16C_0) \le 1/4$, the last since $C_0 \ge 2A^3$ and $A \ge 1$. The separate condition for the mean also holds. Convexity is asserted only in the fluctuation variables; all inverses above are inverses of constrained Hessians in the free coordinates.

We also check the hypotheses of the comparison on the enclosing product domain in Lemma 21.11, which compares the outgoing density with the integral over the fluctuation cutoff domain. Its hypotheses are those of Theorem 21.8, verified above, the conditions on the enclosing product domain verified in Steps 5 and 7, and an incoming density bound (21.32). At the first RG step after either kind of starting measure, this bound holds with equality. At a later step, it is the incoming bound at the representatives satisfying the reconstruction identities provided by the induction from fine to coarse scales in Theorem 21.12, whose hypotheses are the conditions verified in this step. We next check the measurability required by the comparison. For fixed finite geometry and fixed real frozen data, the enclosing product domain $\mathcal{K} = \prod_i \overline{B}_{\mathbb{R}^3}(0, R_{\mathrm{en}}/t)$ is compact and has finite positive Lebesgue measure when an active fluctuation coordinate is present. The support and the fluctuation product are Borel sets: the tests of the decomposition use finitely many Borel iterates of T_M and finitely many inequalities, and the inverse coordinates and their dependence on frozen data are Borel by Step 3. Every indicator of a support or of an event of the decomposition is a Borel function χ with values in $[0, 1]$. The conditional fluctuation potential V is real and smooth on a neighborhood of $\mathcal{K}$, by the margins on the full product and the positive real constrained Hessians. In particular, for fixed data there exist finite numbers $V_{\min}$ and $V_{\max}$ bounding V on $\mathcal{K}$, and

$$0 < e^{-V_{\max}}|\mathcal{K}| \le Z_{\mathcal{K}} = \int_{\mathcal{K}} e^{-V(w)}\, dw \le e^{-V_{\min}}|\mathcal{K}| < \infty. \tag{23.47}$$

No uniform bound on these extrema is needed for the normalization. The integrand is jointly Borel, and integration over each fixed compact product, after rescaling its balls to unit balls, preserves Borel dependence on the parameters, by the fact on parameter integrals recalled in Step 3. The comparison then applies to a measurable finite normalizing constant. For an empty active region the product has mass one, the nonconstant norm vanishes, the determinant is one, and the fluctuation integral is one. All factors independent of the field keep their places in the normalization constant.

Step 10: stopping levels and the insertion interval. We fix the finite family of stopping levels and remainders of Theorem 22.1, with finite constants for side ratios and conversions. Explicitly, for a scale j and a starting level b divisible by J, let $t_{\mathrm{align}} = J\lfloor (j-1)/J \rfloor$ and $u_{\mathrm{ph}} = j - 1 - t_{\mathrm{align}} \in \{0, \dots, J-1\}$. For each fixed offset $r_{\mathrm{stop}} \in \mathbb{N}_0$ of the family, the integrated sequence of RG steps ends at $t_{\mathrm{stop}} = t_{\mathrm{align}} - r_{\mathrm{stop}} J \geq b$. Every integrated step consists of J averaging steps, and the remaining insertion at the stopping level is

$$C \longmapsto Q_j^{\mathcal{T}}\big(T_M^{r_{\mathrm{stop}}J+u_{\mathrm{ph}}} C\big), \qquad n_{t_{\mathrm{stop}}} = M^{r_{\mathrm{stop}}J+u_{\mathrm{ph}}} n_{j-1}, \qquad n_{j-1} \in 4\mathbb{N}. \tag{23.48}$$

Indeed $j - 1 - t_{\mathrm{stop}} = r_{\mathrm{stop}} J + u_{\mathrm{ph}}$, and every level between t_{stop} and $j-1$ belongs to the given finite sequence of lattices, whose last side is $n_{j-1} \geq u \geq 4$. These iterates of T_M are defined, with the required pairs of integer lattices; a large side alone would not justify this. We use the conversion of predecessor sets for arbitrary fields, with its fixed constant K_{src}, without adding a regularity test at the stopping level. The side ratio to n_{j-1} is $M^{r_{\mathrm{stop}}J+u_{\mathrm{ph}}} \leq \chi_*$, after taking the maximum over the fixed finite family. If no complete RG step fits, we apply the clause of the same theorem for a bounded side ratio at the starting measure, with the exponent $j-1-b$ and the side identity $n_b = M^{j-1-b} n_{j-1}$; in this case the exponent is bounded by the same fixed bound. Every remainder is kept as part of the insertion, and none requires an integrated step of fewer than J averaging steps, a short recursion for the normalization constant, or a new output radius.

We fix a covered large-field decomposition of length $\ell = t_{\mathrm{stop}} - b$, with the set F of coarse links kept at the stopping level. We use the list $\mathcal{P}_{\mathrm{orig}} = \mathcal{P}_{j-1}^{\mathcal{T}}$ at scale $j-1$ and $Q_{\mathrm{orig}}(V) = Q_j^{\mathcal{T}}(T_M^{j-1-b} V)$, with the coefficient $4/h_{n_{j-1}}$. The exponent and the side identity in (23.48) are the hypotheses of Lemma 22.8. Using the full excluded sets and the enlarged count in (23.38), we obtain, at every value (c, b_{rec}) of the coarse field and the frozen coordinates,

$$Q_{\mathrm{orig}}(V) \leq K_W \mathcal{V}_{m;F}(c)/g_m^2 + 8F_J \sum_{f \in \mathcal{X}} n_{\ell_f}/\gamma, \qquad m = n_{t_{\mathrm{stop}}}, \quad K_W = 48K_1/\alpha_W, \tag{23.49}$$

where $\mathcal{X}$ is the set of threshold failures of the decomposition, n_{ℓ_f} is the side at which f was created, and $\mathcal{V}_{m;F}$ is the compact minimal action of Lemma 22.8. We include both the predecessor insertion and the total-action insertion of Theorem 22.1 and Corollary 22.9, and enlarge F_J to count all their affected plaquettes. Only then do we choose $s_0 > 0$ in the full interval of Theorem 22.1, in particular with $s_0 K_W \leq 1/4$ and $s_0 \leq \theta c_A^{\mathrm{act}} d_*/(16F_J)$, where $\theta = 1/2$, and we impose $s_0 \leq \kappa_0/(2\max\{K_M, 48K_1\})$ for the total-action insertion. Each bound is fixed and positive before the coupling. We write $\Psi_F = \mathcal{V}_{m;F}$, $\lambda_s = 1/2 - sK_W \in [1/4, 1/2]$, $P_0 = 4\theta c_A^{\mathrm{act}} d_*$, and $H_{\mathcal{X}} = \sum_f n_{\ell_f}/\gamma$. At every value of the coarse field and the frozen coordinates, the action bound and the gauge-invariant threshold tests imply

$$sQ_{\mathrm{orig}}(V) - \frac{\Psi_F(c)}{2g_m^2} - \theta A_{\mathcal{X}}(b_{\mathrm{rec}}) \leq -\lambda_s \frac{\Psi_F(c)}{g_m^2} - \frac{3P_0}{4} H_{\mathcal{X}}. \tag{23.50}$$

Indeed $\theta A_{\mathcal{X}} \geq P_0 H_{\mathcal{X}}$ by the threshold tests, and $8sF_J \leq P_0/8 \leq P_0/4$. We use this inequality in the sense of image measures, as in (22.16); no factorization of the full insertion through the coarse variables is asserted. The invariant density bound without weight has been extended to the whole configuration space before this multiplication. We complete its Wilson factor once, at the coefficient λ_s, using (22.17). The scalar and completion factors attributable to the threshold failures are at most $P_0 H_{\mathcal{X}}/4$ after the same choice of minimum side, which leaves $a_{\mathrm{act}} H_{\mathcal{X}}$ with $a_{\mathrm{act}} = P_0/2$. The constants for the stopping partition-function ratio are chosen uniformly for $\lambda_s \in [1/4, 1/2]$, with $g_m^2 \leq \alpha_W$ and the fixed coupling bound g_{anc}. The denominator is the same centered trial integral on the torus, and the partition function Z_ν of the starting measure is unchanged. This argument uses only the positive action bound, and it adds no constant and no further surface logarithm. It includes the decompositions of Lemma 22.8 that become empty early.

We define s_b, a_b, and $\bar{c}$ as in item (9). The full affected sets were fixed before this sequence of choices: an enlargement for the finite family affects only these later choices and C_v, C_s, C_b, n_0, never J or an earlier analytic constant. We apply Lemma 23.4 with $\Gamma = \gamma_0$ to obtain the moment constants and their common minimum side; its hypotheses on the local product domains, the protection of older data, the norm bounds, and the insertions for the remainders have been checked above. Its surface coefficient is kept with $C_s \geq 9\chi_*^2$.

Step 11: the minimum side. All geometric thresholds, including $8R_{\mathrm{cell}}$, $8S$, $8C_{\mathrm{SG}}/r$, and every fixed margin near flat configurations, are finite lower bounds on the side. We also fix a coupling bound $0 < g_{\mathrm{anc}} < 1$ for the stopping partition-function ratios of Theorem 22.4, on the fixed intervals of coefficients of the Wilson action used in Theorem 22.1, whose lower endpoint is $1/4$. It depends on the fixed torus domains, r, the norm bounds, and that interval, and it is uniform in the depth, the side, and γ. We take $g_{\mathrm{end}} = \min\{1, g_*, g_{\mathrm{anc}}, \sqrt{\alpha_W}\} > 0$ and impose

$$n_0 \geq \gamma_0/g_{\mathrm{end}}^2, \qquad g_m = \sqrt{\gamma/m} \leq g_{\mathrm{end}} \qquad (m \geq n_0,\ 0 < \gamma \leq \gamma_0). \tag{23.51}$$

This verifies the completion at the stopping level, the uniform bound on the stopping partition-function ratio, and every coupling bound of an integrated averaging step. At the output of a complete RG step we have $g_m = \sqrt{M}\, t_{\max}$, and the coupling restriction at the stopping level is checked directly from the bound on the side; it is not inferred from $t_{\max} \leq g_*$ alone. The two side conditions of the cover hold separately: an output after an RG step has side at least $u \geq n_0 \geq 8R_{\mathrm{cell}}$, and an input at the initial step has side at least $u \geq n_0 \geq 8S$. They need no congruence condition on u beyond $u \in 4\mathbb{N}$; an integrated step has input side Bm with $m \in 4\mathbb{N}$, and every intermediate side is at least four.

We fix $r_{\mathrm{flat}} > 0$ from Lemma 14.9, below r and below all fixed margins of one RG step for the real classical action, the constraint, and the gauge transformations; these are

uniform in the preceding depth and are fixed before γ. We impose, with strict inequality,

$$2\pi/n_0 < \min\{r/2, r_{\mathrm{flat}}\}. \tag{23.52}$$

Lemma 14.9 then applies at every $u \geq n_0$ and every finite K for the same parameters: constant commuting links have distance at most $2\pi/u$ from the identity, the averaging steps preserve their holonomies, all free normalized fluctuation variables vanish, and all coordinate-domain conditions hold. This uses the uniform margin of that lemma near all flat configurations, not an open neighborhood of the flat set that depends on the cutoff.

The factors attached to threshold failures and the ratios of the sums over decompositions are controlled by Lemma 23.1, with $x = m/\gamma$ and $k \leq 2$. The errors at the endpoints in Lemma 22.12 have the forms $C_J X_c$ and $C_J W_a$ with fixed positive exponents; the finite family of remainders adds only finitely many constants and ratios of scales. Explicitly, for every transition $0 \leq a < K$, we take the minimum side so large that the exterior error z_a and the cutoff error δ_a of that lemma are at most $1/4$; the exterior error is the positivity condition on d_t in the comparison of Theorem 14.7. Then $z_a < 1$ and $\epsilon_a = \delta_a/(1 - z_a) \leq 1/3 < 1$, and every rejection estimate and every domination by reciprocals applies. The error ϵ_t of the regional comparison on the enclosing product domain in Lemma 21.11 is likewise made at most $1/4$; in particular, every reduction through the auxiliary measure and its annular region has error strictly below one. We make the ratios of successive envelopes at most $1/2$, and the sum of the ϵ_a so small that every reciprocal product has logarithm at most one; this achieves $C_\delta = 1$ before the final enlargement of n_0. The measures for the initial segments are defined up to $a = K$, but no quantity of a next transition is used at $a = K$; when $K = 0$ these products and lists are empty.

For the entropy, at $m \geq 4$ and $x = m/\gamma \geq e$, the quantity $\mathcal{A}_\gamma(m)$ of (23.3) satisfies

$$\mathcal{A}_\gamma(m) \leq D_0(1 + \log x), \qquad D_0 = 1 + 24C_v M^3 + C_s M^2(1 + \log M) + C_b'/8,$$

since $4/m \leq 1$, $8/m^3 \leq 1/8$, and $\log(Mm/\gamma) = \log M + \log x$. Increasing n_0 makes this at most $a_b x/2$, uniformly in γ. The ratio with exponent $\bar{c}$ is made at most $1/2$ by (23.2). These conditions become stronger as the minimum side increases.

Finally we keep the exact correction at the initial scale from Lemma 22.12. With $\beta = 4N/\gamma \geq 4N/\gamma_0 \geq e$, its positive term divided by β is at most

$$\frac{C_{\mathrm{LT}}\gamma_0}{4N} + \frac{9\gamma_0 \log(4N/\gamma_0)}{N^2} \longrightarrow 0,$$

where we used that $1/\beta$ and $\log\beta/\beta$ are decreasing for $\beta \geq e$. We require this to be at most $d_*/4$. The exponent at the initial scale is then at most $-\beta d_*/32 = -d_* N/(8\gamma)$, at all larger sides and all couplings in the interval. We choose $n_0 \in 4\mathbb{N}$ above this threshold and all preceding ones. The logarithmic correction at the initial scale is not replaced by a bound along a fixed sequence of sides.

Step 12: the reconstruction maps and the identity of coarse fields. The nested domains and the tolerances for coefficients and derivatives used in Theorem 15.12 are fixed. Its thresholds at fixed lattices can now be increased for the smaller inner image margins, by Lemma 15.15: below the threshold the earlier bound holds, and above it the bound for the limit plus the same fixed tolerance holds. This increase therefore cannot enlarge C_{CP}, A_J, R_J, or β_J. There are countably many pairs of finite lattices and preceding depths, and we fix the maps on all of them at once, rather than choosing a new limit or threshold at some coarse side. Each choice concerns classical maps on domains of free coordinates and has no argument depending on the coupling, the field, or an observation. The coarse fields, kernels, cutoffs, and Jacobians are preserved, by Theorem 15.12.

The bound on the reconstruction amplitude in Lemma 15.15 applies to these maps. To check its parameters, we fix $u \geq n_0$ and let $H = L/u$, $a_j = HB^{-j}$, and $g_3^2 = \gamma/L$. Then

$$t_j = M^{(J-1)/2} g_3 \sqrt{a_j}, \qquad t_1^2 = \gamma/(Mu) \leq g_*^2, \qquad t_{j+1} = t_j/\sqrt{B},$$

where $t_1^2 \leq \gamma_0/n_0 \leq g_{\mathrm{end}}^2 \leq g_*^2$ by (23.51). Here $p(t_j) \geq 1$. The free input is at most $t_j p(t_j)$, all products remain in the common working ball by (23.40), and $b_0 \leq r$. Then the amplitude bound and its sum give

$$\begin{aligned} b_j &\leq C_{\mathrm{cap},J}\big[r\alpha_*^j + g_3\sqrt{a_j}\, p(t_j)\big], \\ \sum_{j=1}^{k} b_{j-1} &\leq C_{\mathrm{cap},J}\big[r/(1-\alpha_*) + g_3\sqrt{H}\,(p(t_1)+1)\big]. \end{aligned} \tag{23.53}$$

The constant does not depend on H, u, γ, or the length of the finite chain. We used the deterministic bound of Lemma 15.15, not a stochastic bound at fixed H, and made no new choice at a coarse side.

Finally, the path-length condition gives the identity of coarse fields, even before a test that may fail. In a representative whose link distances are at most s, a product of l links has distance at most ls, by repeated use of the triangle inequality for the bi-invariant distance. The two tree paths have total length at most $3(M-1)$; adjoining the straight segment of M links and the inverse of the reference path of M links gives the closed paths of the based loop holonomies, of length at most $5M-3$. If $(5M-3)s < \pi$, all these based loop holonomies have principal logarithms. In (4.3), the exponential of the average logarithm then has distance at most $(5M-3)s$, and multiplication by the reference transport gives the bound $(6M-3)s \leq 10Ms$. Starting from $s = r$, induction gives the link bound $(10M)^i r$ before averaging step $i < J$, so the condition $(5M-3)(10M)^{J-1}r < \pi$ covers each of the next J averaging steps. For a next RG step with $0 \leq a < K$, the input side is $uB^{K-a} = M^J m$ with $m = uB^{K-a-1} \geq n_0 \geq 4$, so each of these J averaging steps has a valid pair of input and output lattices. There is no next step at $a = K$.

For an arbitrary input in D_r, we apply this argument to a small periodic representative. Under a gauge transformation, the based loop holonomies are conjugated at their

basepoint, and their distances and their membership in the exceptional set are unchanged. Conjugation commutes with the principal logarithm and with the exponential, and the reference transport transforms in the usual way at its endpoints. The same covariance holds for the Borel extension, since $-I$ is central. Then T_M and all its iterates are gauge covariant, and the conclusion transfers back without taking a logarithm of the gauge transformation. On an initial segment that passes the tests, the current coarse field lies in D_r, and the same argument applies to the next coarse field. The inverse coordinates in Definition 14.4 satisfy the identity for one averaging step on this coordinate domain; composing the J identities shows that the coarse field produced by the construction equals T_M^J of the finer field, even when a later cutoff fails. A gauge transformation depending only on the coarse field and equal to the identity at the tree roots preserves this identity, as well as the kernels and Jacobians, by Theorem 15.12.

All local estimates were evaluated before the later widths, and all side conditions used fixed constants and the bound γ_0. This proves the stated dependences, and the absence of any dependence on the number of elapsed RG steps, on shapes, or on depths. The single choice of J used only the J-uniform geometric constants and torus rates, the J-uniform series Z, and the finitely many coarse-field derivative bounds of the reconstruction maps at zero fluctuation; the full joint norms and the norms of the gauge transformations of the reconstruction maps were fixed after J. The full radii were restored only after complete RG steps, and the remainders were included in the fixed family of insertions. This completes the proof of Theorem 23.2.

23.7 Parameters below an arbitrary coupling ceiling

In this subsection we prove Theorem 23.5.

Proof of Theorem 23.5. We start with parameters given by Theorem 23.2 with $\gamma_0 = 1$. The argument works for every J obtained from that construction; it does not select a new J. We keep its J, the analytic domains and rates on the torus and on block regions, the margins of the product domains for free coordinates, the cutoff amplitude, the widths of boundary layers, removed layers, and protected neighborhoods, the margins near flat configurations, g_*, the interval for s, the family of remainders, and the numbers s_0, d_*, a_b, and $\bar{c}$. Before the curvature threshold is chosen, δ_* is normalized to $(0, 1]$, which keeps the range of Theorem 7.3. The parameters include K_W and $g_*^2 \le \alpha_W$, and the stopping partition-function ratio is controlled for coefficients in $[1/4, 1/2]$ before the side is enlarged. The deterministic column bounds come from Lemma 15.15; no stochastic estimate is used in this choice. All these quantities are constructed from the local results cited in Theorem 23.2, whose hypotheses concern g_i, t, and fixed products and contain no upper bound on γ. In particular, the classical constants, the coefficient constants of the reconstruction maps, and the J-uniform constants are not chosen again using a later

probability constant. The reconstruction maps of Lemma 15.15 and Theorem 15.12 are fixed on the same countable family of pairs of finite lattices; they depend on deterministic domains, not on the coupling. Their exact identities and the margin $32/49$ are unchanged.

At these parameters we apply Theorem 22.1 and Lemma 22.13 for the ceiling Γ, without a restriction on the number of averaging steps. We obtain C_v, C_s, C_b, and we set $C_b' = C_b + 1 + \log 2$. Only the later probability constants and thresholds use these constants, so their possible dependence on Γ creates no cycle with an earlier choice. We choose the final side at least $\Gamma/g_{\mathrm{end}}^2$ and above every earlier geometric and flat threshold. For every side $m \geq n_\Gamma$ of a stopping lattice we have $g_m^2 = \gamma/m \leq \Gamma/n_\Gamma \leq g_{\mathrm{end}}^2$, and also $t_k^2 = \gamma/(Mn_{(k+1)J}) \leq \Gamma/(Mn_\Gamma) \leq g_{\mathrm{end}}^2 \leq g_*^2$. These inequalities verify the coupling bounds at the stopping level and for the integrated steps, respectively. The separately imposed flat thresholds give $2\pi/n_\Gamma < \min(r/2, r_{\mathrm{flat}})$. All inequalities involving local small parameters hold on $0 < t \leq g_*$, including every fixed power of t multiplied by a logarithmic cutoff, and they hold at these sides on the new coupling interval.

We apply Lemma 23.4. At $x = m/\gamma \geq e$ we have

$$\mathcal{A}_\gamma(m) = 24C_vM^3 + \frac{4C_sM^2}{m}\log(Mm/\gamma) + \frac{8C_b'}{m^3} \leq D_0(1+\log x),$$

with the same $D_0 = 1 + 24C_vM^3 + C_sM^2(1+\log M) + C_b'/8$ as in Step 11 of Section 23.6, since $m \geq 4$. Lemma 20.15 makes this at most $a_b x/2$ uniformly on $x \geq n_\Gamma/\Gamma$. The same lemma makes all ratios at the endpoints at most $1/2$ and the exterior errors at most $1/4$, and it makes the cutoff errors and the logarithms of the reciprocal products as small as prescribed; in particular $C_\delta \leq 1$. The ratio over scales satisfies $M^3 e^{-\bar{c}(M-1)n_\Gamma/\Gamma} \leq 1/2$. The exact positive term at the initial scale, divided by β, is bounded by the vanishing expression in Lemma 22.13. Finally, the constant $C_{\mathrm{ch},J}^\Gamma$ for the factors attached to threshold failures at the ceiling Γ is absorbed there, before the moment constants and the final side are used. All these requirements become stronger as the minimum side increases, and we take their maximum, rounded up to an element of $4\mathbb{N}$.

The amplitude bounds of Lemma 15.15 require only $t_1^2 = \gamma/(Mu) \leq g_*^2$ and the common working ball. The coupling condition just verified provides this, and under this hypothesis the bound is uniform in $H = L/u$ and g_3. The conditions from path lengths and the coordinate-domain conditions are deterministic in r, M, and J, and they are unchanged. The conditions listed in this proof are the only places where the proof of Theorem 23.2 uses the bound $\gamma \leq \gamma_0$. Hence every conclusion of that theorem holds, with the same local dependences and with the fixed Γ entering only the probability constants. No support radius or reconstruction map changes at a larger coarse side. □

23.8 Every sufficiently large number of averaging steps

Section 29 compares the limits obtained along the sequences $n_* M^{JK}$ for two consecutive values of J, so it needs the construction of Theorem 23.2 for every sufficiently large J. No uniformity of the constants in J is asserted.

Lemma 23.14. *Fix $L > 0$, an odd integer $M \geq 3$, a ceiling $0 < \Gamma \leq 1$, and the J-uniform constants in item* (1) *of Theorem 23.2. There exists a finite J_0 such that, for every integer $J \geq J_0$, there exist a tuple Θ_J, a positive local coupling bound $g_{*,J}$, and $n_{\mathrm{sf},J} \in 4\mathbb{N}$, chosen before the coupling and the side, for which all conclusions of Theorems 23.2 and 23.9 hold with $B = M^J$, simultaneously for*

$$0 < \gamma \leq \Gamma, \qquad u \in 4\mathbb{N},\ u \geq n_{\mathrm{sf},J}, \qquad N = uB^K, \quad K \geq 0.$$

In particular, two consecutive values of J can be fixed before γ is varied. Neither the tuples nor any later constants or thresholds are asserted to be uniform in J.

Proof. We verify the additional quantifier "for all $J \geq J_0$" in the proof of Theorem 23.2; it is not inferred from the existence statement. We fix the outer rate on the torus, the boxes, and the analytic-radius enlargement factor K_*. The series for the conditional constrained minimizers at zero fluctuation of Lemma 11.9 and Theorem 13.2 has a J-uniform majorant $Z(v) \leq Cv/(1 - Cv)$ and a J-uniform spatial rate. The finitely many coarse-field derivative bounds of the reconstruction maps at zero fluctuation, as opposed to the full analytic norm of their gauge transformations, are also J-uniform, by Theorem 15.12 and Lemma 15.15. The additional identity at fixed J for hierarchical trees in Lemma 11.9 does not change its J-uniform series, and only that series is used here. Thus all constants in the following list are fixed before J:

$$\begin{gathered} B \geq B_{\mathrm{geom}}, \quad C_P(a_{\mathbb{T}})/B < 1/4, \quad C_{\mathrm{CP}}^2/B \leq 1/4, \\ C_\bullet K_*/B < 1/32, \quad C_{\mathrm{geom}}^{\mathbb{T}} b_{\mathbb{T}}/B \leq a_{\mathbb{T}}, \quad C_{\mathrm{cen}}/B < 1/8, \quad B^{-1} \leq \delta_0(a_{\mathbb{T}}). \end{gathered} \tag{23.54}$$

These are the conditions of item (1) of Theorem 23.2 and (23.25); the rate $a_{\mathbb{T}} < a_{\mathrm{cen}}/8$ and the threshold $\delta_0(a_{\mathbb{T}})$ are fixed before J. The finitely many additional image conditions on first derivatives use the same series and derivative bounds; they are also strict lower bounds on B with fixed finite right sides. The maximum of their thresholds is exceeded by M^J for every $J \geq J_0$. This uses the exception for J-uniform constants in item (2) of Definition 4.14, not a bound on a joint free map.

The constants C_{alloc} and C_g in Theorem 20.9 do not depend on J. We choose $C_*^{\mathrm{geom}} \geq \max(1, C_g)$ and keep $B_{\mathrm{geom}} = 256 C_{\mathrm{alloc}}(C_*^{\mathrm{geom}} + 1)$. For each fixed J above the threshold we set $\alpha = a/16$, $m_{\mathrm{rate}} = C_*^{\mathrm{geom}}(a + \alpha)$, and $b_1 = 8C_*^{\mathrm{geom}}\alpha$. The inequalities against Ba and $B\alpha$ reduce to

$$\frac{25}{16} C_*^{\mathrm{geom}} < B/C_{\mathrm{alloc}}, \qquad 8C_*^{\mathrm{geom}} < B/C_{\mathrm{alloc}}.$$

We decrease $a > 0$ below the positive bounds at fixed J from the initial spatial decay rate and the image exponent, and so that $\sqrt{3}\,(a + \alpha) < a_{\mathbb{T}}$. No further inequality on supports or connecting lengths remains.

We also check the later choices. At each fixed J there are finitely many product domains for initial segments. The positive lower bounds for the constrained Hessians and the radii of the inverses in Lemmas 19.3 and 19.7 have a positive finite minimum, and the coordinate, analytic, and reconstruction domains are chosen within them. The contraction on the torus satisfies $q_{\mathrm{CA}} < 1/4$, and the bound $E \geq 2C_G/(1 - q_{\mathrm{CA}})$ is finite by Theorem 13.5. The preliminary perturbation bound, the radius of the enclosing product domain, and the image bounds of Lemma 20.3 and Theorem 20.14 can be chosen positive on the fixed domains at this J. The widths of boundary layers and removed layers can be increased after their output constants are computed, and the bound for frozen coordinates, the working radius, and the support radius can then be decreased. In particular, $(5M - 3)(10M)^{J-1} r < \pi$ is an upper bound on r, not on J.

The bound on derivatives follows the same order of choices. Its linear coefficient is $\alpha_* = C_{\mathrm{CP}}/B$, and by (23.54) we have $\alpha_* \sqrt{B} \leq 1/2$. At this fixed J, β_J is finite by Lemma 15.15. The restriction $\beta_J r_{\mathrm{eff}}/(1 - \alpha_*) < 1/8$ implies $\beta_J r_{\mathrm{eff}} < 1/8$, and hence

$$a_{\mathrm{map}} \sqrt{B} \leq \frac{1/2}{(1 - \beta_J r_{\mathrm{eff}})^2} \leq \frac{32}{49}.$$

This requires no new choice of J. The remaining conditions for the fluctuations and the interaction are finitely many inequalities $C_J t^{\varepsilon} \log^b(1/t) \leq c_J$ with $\varepsilon, c_J > 0$, including the fixed additive term $A_{\mathrm{w}}(J - 1) \log M/2$ in a logarithmic radius. They hold after decreasing $g_{*,J} > 0$, by Lemma 23.1.

We next check items (7)–(10). We normalize $0 < \delta_* \leq 1$ before fixing d_{cut}. The product domains, the block-aligned initial mesh S, the classical lower bound $\alpha_W > 0$, the count F_J, and the constant for the normalization factors are all fixed at this J. By Lemma 22.8, $K_W = 48K_1/\alpha_W < \infty$ and the insertion satisfies

$$Q_{\mathrm{orig}} \leq K_W \mathcal{V}_{m;F}/g_m^2 + 8F_J \sum_f n_{\ell_f}/\gamma.$$

We enlarge F_J for the finitely many remainders. The interval of Theorem 22.1 allows

$$0 < s_0 \leq \min\Big\{\frac{1}{2},\ \frac{\kappa_*}{2} c_E^{-(J-1)},\ \frac{1}{4K_W},\ \frac{\theta c_A^{\mathrm{act}} d_*}{16F_J},\ \frac{\kappa_0}{2\max\{K_M, 48K_1\}},\ s_{\mathrm{short},J}\Big\},$$

where $s_{\mathrm{short},J} > 0$ is the minimum of its finitely many conditions for short scale intervals and $\theta = 1/2$. Every entry is positive once J is fixed, and $\lambda_s = 1/2 - sK_W \in [1/4, 1/2]$. The action bound keeps at least $3P_0/4$ of the coefficient of the threshold failures, with $P_0 = 4\theta c_A^{\mathrm{act}} d_*$, and the later scalar threshold uses at most $P_0/4$, so that $P_0/2$ remains.

The compact minimum in the action bound uses the partition function of the starting measure once; no factorization of the insertion through the coarse variables is needed. We decrease $g_{*,J} > 0$ to impose $g_{*,J}^2 \le \alpha_W$ and every local condition already listed. We choose the bounds for the stopping partition-function ratio and its numerator on the fixed interval $[1/4, 1/2]$, and let $g_{\mathrm{end},J} > 0$ be their minimum with 1 and $g_{*,J}$. Finally, we increase the minimum side to include $n_{\mathrm{sf},J} \ge \Gamma / g_{\mathrm{end},J}^2$ and all fixed geometric and scalar thresholds and the thresholds for initial segments. The latter are conditions on the minimum side, by Lemma 22.12; for $\Gamma \le 1$ their uniform scalar thresholds also follow from Lemma 23.1.

These are exactly the later choices in Theorem 23.2, and none of them requires enlarging J using a constant already computed at that J. By Theorems 23.6 and 23.9, we then obtain the bound on the small-field failure probability on $0 < \gamma \le \Gamma$. The positive exponent $\min(5/24, 1/2, -\log q_{\mathrm{CP}} / \log B)/2$ of Remark 23.3 is also available at each fixed J. □

Corollary 23.15. *Fix $L > 0$, an odd integer $M \ge 3$, and $0 < \Gamma < \infty$. There exists a finite J_0 such that, for every $J \ge J_0$, there are a tuple Θ_J, $g_{*,J} > 0$, and $n_{\Gamma,J} \in 4\mathbb{N}$, fixed before γ, the side, and the number of renormalization group steps, for which every conclusion of Theorems 23.5 and 23.10 holds on $0 < \gamma \le \Gamma$ with this prescribed J. In particular, consecutive values of J can be fixed before the coupling. No constant or threshold is asserted to be uniform in J.*

Proof. We apply Lemma 23.14 with the ceiling one. Its check of the J-uniform conditions provides J_0 and parameters for $\gamma \le 1$ for every $J \ge J_0$. The constant $K_W = 48K_1/\alpha_W$ is fixed after the classical lower bound α_W and before s_0, and the normalization $\delta_* \le 1$ is fixed before the curvature threshold; neither changes a J-uniform lower bound on $B = M^J$ or imposes an upper bound on J. The interval for s is a finite minimum of positive numbers at fixed J, and the coupling bound at the endpoints is positive. For each prescribed $J \ge J_0$, we apply the construction in the proof of Theorem 23.5 to these parameters. That construction changes only the later probability constants and the final minimum side, and it keeps this J and the deterministic bounds of Lemma 15.15. By Theorem 23.10, we then obtain the bounds for all finite numbers of RG steps on the interval $0 < \gamma \le \Gamma$. For two values J and $J+1$ one may take the larger of the two minimum sides, with no uniformity as J varies. □

PART VII

Wilson flow and continuum comparison

SECTION 24

Local Wilson flow and continuum solutions

This section proves the deterministic Wilson-flow estimates used to pass from the renormalization group variables to flowed Wilson loops. We also record an elementary comparison for exponential transforms of bounded random observables. The section has three parts.

The first part concerns the lattice. Let $a = L/n$ be the mesh and write a configuration close to the identity as $U_e = e^{aA_e}$. After a time-dependent gauge transformation, which adds a damping term, the Wilson flow $\Psi_t^{(n)} = \Phi^{\mathrm{W}}_{4t/a^2}$ becomes the parabolic equation

$$\partial_t A = \Delta_a A + \mathcal{N}_a(A, A) + F_a(A), \tag{24.1}$$

where Δ_a is the lattice Laplacian, $\mathcal{N}_a$ is an explicit bilinear map containing one lattice difference, and F_a collects the terms of degree at least three (Lemma 24.2). We write

$$A(t) = x_a(t) + y_a(t) + r_a(t), \qquad x_a(t) = P_t^a A(0), \qquad y_a(t) = \int_0^t P_{t-s}^a \mathcal{N}_a\big(x_a(s), x_a(s)\big)\, ds, \tag{24.2}$$

where $P_t^a = e^{t\Delta_a}$. As in Section 3.5, x_a is the linear heat evolution and y_a the quadratic Picard term, and (x_a, y_a) is the flow data of $A(0)$. Theorem 24.5 constructs a local solution on a time interval independent of the mesh, with remainder of order $(t + a^2)^{1-3\alpha/2}$, and proves Lipschitz dependence on these data. The probabilistic bounds on the flow data are proved in Sections 26 to 28, using the bilinear estimate of Lemma 24.6.

The second part, Section 24.6, concerns the Yang–Mills heat flow on $\mathbb{T}_L^3$ and is independent of the lattice. For smooth initial data near a compact family, it represents the flow on a finite time interval by finitely many local solutions of the DeTurck equation, joined by fixed gauge transformations.

The third part (Section 24.7) treats the flows of the actions $\sum_p (q(U_p) + c\, q(U_p)^2)$ with $0 \le c \le C_0$, which appear in Theorem 2.6. The additional force begins in degree three and is $O(a^3)$ on every fixed bounded C_a^1 ball. We show that these flows have the same flow data as the Wilson flow, and that for the same flow data their solutions differ from the Wilson-flow solution by $O(a^\varepsilon)$ on the local time interval.

The inputs are the conventions of Section 4, in particular the weighted norms of Definition 4.8 and the continuum conventions of Definition 4.10; the finite-word analyticity and the plaquette bounds of Lemma 5.1; and the heat kernel bounds, convolution estimates, and cardinal interpolation of Lemmas 16.1, 16.3, and 16.8. The lattice results are used in

Section 25 (continuation to fixed time intervals and stability), in Sections 26 to 28 (estimates for the flow data and convergence of flowed loops), in Sections 29 and 30 (lattice energy, trajectories, and alternative flows), and in Sections 32 and 34. The continuum theorem is used in Sections 25, 28 to 30, and 34.

The lattice flow is the Wilson flow of Lüscher [128, 129]; the factor $4/a^2$ in its time variable comes from our normalization of q, of the plaquette sum, and of the metric on $\mathrm{SU}(2)$. Zwanziger [157], DeTurck [66] (for the Ricci flow), and Donaldson [76] made gauge-invariant flows parabolic by adding a gauge velocity. The decomposition (24.2) into a rough linear part and smoother corrections is the lattice analogue of the decomposition used by Cao and Chatterjee [42, §§1.1 and 2.1] for the Yang–Mills heat flow with random initial data, an idea that goes back to Bourgain [32] and to Da Prato and Debussche [63]. Here the subtracted term is the quadratic Duhamel term of the lattice equation, with the bilinear map fixed in Lemma 24.2. The global smooth theory of the continuum flow is due to Råde [139] and Charalambous and Gross [47]; we use it in the form stated by Cao and Chatterjee [42, 43]. The explicit lattice coefficient, the weighted remainder estimates, and the finite continuation construction are proved in this section.

24.1 The local equation for the Wilson flow

Notation. Throughout this section $L > 0$ is fixed, $n \geq 4$ is an integer, and $a = L/n$. A field A on Λ_n assigns to each positive link $(x, x + ae_i)$ an element $A_i(x) \in \mathfrak{su}(2)$; we use the norm $|X| = (-2 \operatorname{tr} X^2)^{1/2}$ on $\mathfrak{su}(2)$, for which $|[X, Y]| \leq |X||Y|$, since the bracket corresponds to the cross product in $\mathbb{R}^3$. Let T_i be the translation $(T_i f)(x) = f(x + ae_i)$, and let

$$D^+_{a,i} = \frac{T_i - I}{a}, \qquad D^-_{a,i} = \frac{I - T_i^{-1}}{a}, \qquad \Delta_a = \sum_{i=1}^{3} D^-_{a,i} D^+_{a,i}$$

act on each link component separately, a link being identified with its starting vertex. The heat semigroup $P^a_t = e^{t\Delta_a}$ acts componentwise and includes the constant mode. We write $\|A\|_\infty = \max_e |A_e|$, and $\|D_a A\|_\infty$ for the maximum of $\|D^\pm_{a,j} A\|_\infty$ over $j \in \{1, 2, 3\}$, both signs, and all link components. We set

$$\|A\|_{C^1_a} = \|A\|_\infty + \max_{j \in \{1,2,3\},\, \epsilon \in \{+,-\}} \|D^\epsilon_{a,j} A\|_\infty. \tag{24.3}$$

Since translations permute the sites of the periodic lattice, they are isometries for these norms and commute with P^a_t and with all differences. More generally, D^m_a denotes the maximum over all ordered products of m differences $D^\pm_{a,k}$.

We identify tangent vectors at $U \in \mathrm{SU}(2)$ with $\mathfrak{su}(2)$ through $X \mapsto (d/d\varepsilon)\, e^{\varepsilon X} U|_{\varepsilon=0}$. A curve $U(s)$ then has velocity $\dot U U^{-1}$, and the gradient $\nabla f(U) \in \mathfrak{su}(2)$ of a function f on $\mathrm{SU}(2)$ is defined by $\langle \nabla f(U), X \rangle = (d/d\varepsilon)\, f(e^{\varepsilon X} U)|_{\varepsilon=0}$; since the metric is bi-invariant,

this is the Riemannian gradient. On $\mathcal{U}_n$ we use the product metric. As in Section 2.1, Φ^{W}_s is the flow of $-\nabla \sum_{p\in\mathcal{P}_n} q(U_p)$ and $\Psi^{(n)}_t = \Phi^{\mathrm{W}}_{4t/a^2}$; no coupling enters this time change.

All constants in Section 24.1 depend only on the fixed finite stencils, the orders of the derivatives involved, and the margins of the logarithm charts, and not on L, n, the coupling, or any probability law. These are the dependences of item (1) of Definition 4.14. Later constants may depend in addition on the quantities displayed in each statement.

Local analytic maps of a finite stencil. The terms of degree at least three in the flow equations of this section are functions of the link variables in a fixed finite neighborhood, with the same functional form at every link. We first record the estimates that such functions satisfy after the rescaling $\theta = aA$. Let S_0 be a finite set of pairs (η, j) with $\eta \in \mathbb{Z}^3$ and $j \in \{1,2,3\}$. For a field θ and a site x, we write $Z_x(\theta) = (\theta_j(x + a\eta))_{(\eta,j)\in S_0}$ for the stencil array of θ at x. A map G from fields to fields is a local analytic map with stencil S_0 if there exist functions g_1, g_2, g_3, real analytic in a neighborhood of 0 in $\mathfrak{su}(2)^{S_0}$, such that $G(\theta)_i(x) = g_i(Z_x(\theta))$ whenever θ is small. For a homogeneous polynomial P of degree m on $\mathfrak{su}(2)^{S_0}$ with values in $\mathfrak{su}(2)$, let $\|P\|_{\mathrm{coeff}}$ denote the sum of the norms of its coefficients in the monomial basis of the real coordinates. If g_i extends to a bounded analytic function on a complex polydisc around 0, then by the Cauchy estimates there exist constants K_0, K_1 such that the homogeneous parts $P_{m,i}$ of g_i satisfy $\|P_{m,i}\|_{\mathrm{coeff}} \le K_0 K_1^m$.

Lemma 24.1. *Let G be a local analytic map with stencil S_0 whose Taylor coefficients at zero vanish through degree two, and suppose that the homogeneous parts of the g_i satisfy $\|P_{m,i}\|_{\mathrm{coeff}} \le K_0 K_1^m$ for $m \ge 3$; on $\{Z : K_1\|Z\| < 1\}$, where $\|Z\|$ is the maximum of the norms of the entries, we use the convergent Taylor series $g_i = \sum_{m\ge3} P_{m,i}$ to define g_i and hence G. Fix $0 < \eta < 1$. For $0 < a \le 1$ and a field A with $K_1 a\|A\|_\infty \le \eta$, set $G_a(A) = a^{-3}G(aA)$ and $M_A = \|A\|_\infty$, and let*

$$b_j(M,a) = \begin{cases} M^{3-j}, & 0 \le j \le 3, \\ a^{j-3}, & j > 3. \end{cases}$$

Then the following statements hold, with constants depending only on S_0, K_0, K_1, η, and the indicated orders.

1. *The series $G_a(A)_i(x) = \sum_{m\ge3} a^{m-3} P_{m,i}(Z_x(A))$ converges absolutely.*
2. *For every $j \ge 0$ and all fields $V_1, \dots, V_j$,*

$$\|D^j G_a(A)[V_1,\dots,V_j]\|_\infty \le K_j\, b_j(M_A,a) \prod_{r=1}^{j} \|V_r\|_\infty.$$

3. *If B is another field with $K_1 a\|B\|_\infty \le \eta$, then $\|G_a(A) - G_a(B)\|_\infty \le K(\|A\|_\infty^2 + \|B\|_\infty^2)\|A - B\|_\infty$.*

4. For every $m \geq 1$,

$$\|D_a^m G_a(A)\|_\infty \leq K_m \sum_{j=1}^{m} b_j(M_A, a) \sum_{\substack{\nu_1+\cdots+\nu_j=m \\ \nu_r \geq 1}} \prod_{r=1}^{j} \|D_a^{\nu_r} A\|_\infty.$$

5. If B is as in (3) and $M = \max(\|A\|_\infty, \|B\|_\infty)$, then

$$\|D_a(G_a(A) - G_a(B))\|_\infty \leq K\big\{M^2 \|D_a(A-B)\|_\infty + M(\|D_a A\|_\infty + \|D_a B\|_\infty)\|A-B\|_\infty\big\}.$$

Proof. We write $g_i^a(Z) = a^{-3} g_i(aZ) = \sum_{m\geq 3} a^{m-3} P_{m,i}(Z)$, so that $G_a(A)_i(x) = g_i^a(Z_x(A))$; every entry of $Z_x(A)$ has norm at most M_A. For a homogeneous polynomial P of degree m, differentiating each monomial, we find

$$|D^j P(Z)[W_1, \ldots, W_j]| \leq m(m-1)\cdots(m-j+1)\, \|P\|_{\text{coeff}}\, \|Z\|^{m-j} \prod_{r=1}^{j} \|W_r\|,$$

where $\|Z\|$ is as in the statement; the right side is zero when $j > m$. For $j \leq 3$ we have $a^{m-3} K_1^m M_A^{m-j} = K_1^3 M_A^{3-j} (K_1 a M_A)^{m-3}$, and for $j > 3$ and $m \geq j$ we have $a^{m-3} K_1^m M_A^{m-j} = a^{j-3} K_1^j (K_1 a M_A)^{m-j}$. Since $K_1 a M_A \leq \eta < 1$, summing over m, we obtain (1) and (2), with

$$K_j = K_0 K_1^{\max(3,j)} \sum_{m \geq \max(3,j)} m(m-1)\cdots(m-j+1)\, \eta^{m-\max(3,j)} < \infty.$$

For (3), we write $G_a(A) - G_a(B) = \int_0^1 DG_a(B + u(A-B))[A-B]\, du$. Since the field $B + u(A-B)$ has norm at most $\max(\|A\|_\infty, \|B\|_\infty)$, (2) with $j = 1$ implies the claim.

For (4), we use that G_a has the same form at every site. Since $Z_{x+ae_k}(A) = Z_x(T_k A)$, by the fundamental theorem of calculus we have

$$D_{a,k}^+\big(G_a(A)\big) = \int_0^1 DG_a\big((1-u)A + uT_kA\big)\big[D_{a,k}^+ A\big]\, du,$$

and similarly $D_{a,k}^-(G_a(A)) = \int_0^1 DG_a((1-u)T_k^{-1}A + uA)[D_{a,k}^- A]\, du$. The fields $(1-u)A + uT_kA$ are convex combinations of translates of A, so they satisfy the same bound $K_1 a\|\cdot\|_\infty \leq \eta$. The map $(W, V) \mapsto DG_a(W)[V]$ is again of the same form at every site, so the same identity applies to it; differences of the two arguments are differences of W and of V, and differences of products obey $D_{a,k}^+(fg) = (D_{a,k}^+ f)\, T_k g + f\, D_{a,k}^+ g$ and $D_{a,k}^-(fg) = (D_{a,k}^- f)\, T_k^{-1} g + f\, D_{a,k}^- g$. By induction on m, $D_a^m G_a(A)$ is a finite sum, with a number of terms depending only on m, of integrals over $[0,1]^m$ of expressions $D^j G_a(W)[\widetilde{V}_1, \ldots, \widetilde{V}_j]$, where $1 \leq j \leq m$, W is a convex combination of translates of A, and each $\widetilde{V}_r$ is a convex combination of translates of a difference $D_a^{\nu_r} A$ with $\nu_r \geq 1$ and $\nu_1 + \cdots + \nu_j = m$. No

factor a^{-1} appears outside a difference of A. Since translations preserve the supremum norm, (2) implies (4).

For (5), we apply one difference to $\int_0^1 DG_a(B + u(A - B))[A - B]\,du$ in the same way. The terms in which the difference falls on the second argument are bounded by $K\,b_1(M,a)\|D_a(A - B)\|_\infty = KM^2\|D_a(A - B)\|_\infty$. The other terms contain D^2G_a, evaluated at convex combinations of translates of A and B, applied to a difference of $B + u(A - B)$ and to a translate of $A - B$; by (2) with $j = 2$ they are bounded by $KM(\|D_aA\|_\infty + \|D_aB\|_\infty)\|A - B\|_\infty$. □

The equation. We derive (24.1) and fix the bilinear map $\mathcal{N}_a$. In the lemma and its proof we use dimensionless differences; the symbols d_0, C_0, and H_0 have the meaning given there only in this lemma and its proof.

Lemma 24.2. *Let $L > 0$, $n \geq 4$, and $a = L/n$. Write $\nabla_i = T_i - I$, $d_0 = -\nabla$, $(C_0\theta)_{ij} = \nabla_i\theta_j - \nabla_j\theta_i$ for $i < j$, and $H_0 = \sum_i \nabla_i^*\nabla_i$, where adjoints are taken with respect to counting measure on sites, links, and positive plaquettes.*

1. *We have*
$$C_0^*C_0 + d_0d_0^* = H_0I, \qquad d_2C_0 = 0, \qquad \Delta_a = -a^{-2}H_0,$$
where d_2 is the next forward exterior difference.

2. *For θ in a sufficiently small fixed ball of link logarithms and the plaquette $p = p_{ij}(x)$, $i < j$, put*
$$(v_{p,1}, v_{p,2}, v_{p,3}, v_{p,4}) = \big(\theta_i(x), \theta_j(x + ae_i), -\theta_i(x + ae_j), -\theta_j(x)\big),$$
$$\mathfrak{q}_p(\theta) = \frac{1}{2}\sum_{k<l}[v_{p,k}, v_{p,l}],$$
so that $U_p = e^{v_{p,1}}e^{v_{p,2}}e^{v_{p,3}}e^{v_{p,4}}$ for $U = e^\theta$. Let $\|\theta\|_{\infty,p}$ be the maximum of $|\theta_e|$ over the four links e of p. Then, with positive plaquettes counted once,
$$\operatorname{Log} U_p = (C_0\theta)_p + \mathfrak{q}_p(\theta) + O(\|\theta\|_{\infty,p}^3),$$
$$\sum_p q(U_p) = \frac{1}{8}\|C_0\theta\|_2^2 + \frac{1}{4}\langle C_0\theta, \mathfrak{q}(\theta)\rangle + R_E(\theta), \qquad |R_E(\theta)| \leq C\sum_p \|\theta\|_{\infty,p}^4.$$
Each local remainder and each of its derivatives of fixed order is bounded on a smaller fixed ball, independently of n.

3. *Let $s = t/a^2$, so that $\Psi_t^{(n)} = \Phi_{4s}^{\mathrm{W}}$ and the Wilson flow has velocity $-4\nabla\sum_p q(U_p)$ in the variable s. Add to this velocity the infinitesimal gauge transformation*
$$(G_U\phi)_i(x) = \phi(x) - \operatorname{Ad}(U_i(x))\phi(x + ae_i), \qquad \phi = -d_0^*\theta.$$

With $\mathcal{J}(\theta) = (e^{\operatorname{ad}\theta} - I)/\operatorname{ad}\theta$, *the resulting velocity in logarithmic coordinates is* $\mathcal{J}(\theta)^{-1}(-4\nabla \sum_p q + G_U\phi)$, *and it has the expansion*

$$\partial_s\theta = -H_0\theta + N_2(\theta) + R_3(\theta), \qquad N_2 = -C_0^*\mathfrak{q} - D\mathfrak{q}(\theta)^* C_0\theta + G_2(\theta),$$
$$(G_2)_i(x) = \frac{1}{2}\big[\theta_i(x), (d_0^*\theta)(x) + (d_0^*\theta)(x + ae_i)\big].$$

The adjoint is given by

$$z_{p,k} = \frac{1}{2}\Big[\sum_{l>k} v_{p,l} - \sum_{l<k} v_{p,l},\, y_p\Big], \qquad D\mathfrak{q}(\theta)^* y = \sum_{p,k} \epsilon_{p,k} T_{p,k}^* z_{p,k},$$

where $v_{p,k} = \epsilon_{p,k} T_{p,k}\theta_{\mathrm{dir}(p,k)}$ *with* $\epsilon_{p,k} \in \{\pm 1\}$ *and a translation* $T_{p,k}$, *and* $T_{p,k}^*$ *also inserts the result in the direction* $\mathrm{dir}(p,k)$.

4. *Expand* N_2 *by the rules (i)–(iii) in Step 5 of the proof below. This gives a fixed finite list of terms, each a numerical multiple of* $[T_\eta\theta_i, T_\zeta\nabla_k\theta_j]$ *or* $[T_\eta\theta_i, T_\zeta\nabla_k^*\theta_j]$ *for translations* T_η, T_ζ. *Replacing the first entry of each bracket by* au, *the second by* av, *and dividing by* a^3, *defines a bilinear map* $\mathcal{N}_a$, *which satisfies*

$$a^{-3}N_2(aA) = \mathcal{N}_a(A,A), \qquad \mathcal{N}_a(u,v) = \sum_{\lambda=1}^{m_0} c_\lambda\big[T_{\lambda,1}u_{i_\lambda}, T_{\lambda,2}D_{a,k_\lambda}^{\pm} v_{j_\lambda}\big], \tag{24.4}$$
$$\|\mathcal{N}_a(u,v)\|_\infty \le C\|u\|_\infty\|D_a v\|_\infty,$$

where each term also carries the insertion in the output direction prescribed by its rule. The number m_0, *the coefficients* c_λ, *the directions, and the translations do not depend on* a, n, *or* L. *Further differences of* $\mathcal{N}_a(u,v)$ *are distributed by the exact rules* $D_{a,k}^+[u,v] = [D_{a,k}^+u, T_k v] + [u, D_{a,k}^+ v]$ *and* $D_{a,k}^-[u,v] = [D_{a,k}^-u, T_k^{-1}v] + [u, D_{a,k}^- v]$. *Every later occurrence of* $\mathcal{N}_a$ *refers to this map; it is not symmetric in its two arguments.*

5. *There exist numerical constants* $r_* > 0$ *and* $C < \infty$ *such that, if* $\|\theta\|_\infty, \|\theta'\|_\infty \le r_*$, *then*

$$\|R_3(\theta)\|_\infty \le C\|\theta\|_\infty^3, \qquad \|R_3(\theta) - R_3(\theta')\|_\infty \le C\big(\|\theta\|_\infty^2 + \|\theta'\|_\infty^2\big)\|\theta - \theta'\|_\infty.$$

Moreover, R_3 *is a local analytic map whose Taylor coefficients vanish through degree two and satisfy the hypotheses of Lemma 24.1. In particular, for* $0 < a \le 1$ *the physical remainder* $a^{-3}R_3(aA)$ *is the sum of a convergent series whose term of degree* m *carries the factor* a^{m-3}, $m \ge 3$, *and on a fixed smaller ball it satisfies the bounds of Lemma 24.1, uniformly in the mesh.*

6. *Let* A *be a solution of*

$$\partial_t A = \Delta_a A + \mathcal{N}_a(A,A) + a^{-3}R_3(aA)$$

on an interval $[0,T]$ *with* $a\|A(t)\|_\infty < r_*$ *throughout. Let* $k(t) \in \mathrm{SU}(2)^{\Lambda_n}$ *solve*

$$\partial_t k_x\, k_x^{-1} = a^{-2}\phi_x\big(aA(t)\big), \qquad k_x(0) = I,$$

for every site x. Then k is a periodic lattice gauge transformation and, for every positive link (x, y),

$$e^{aA_{xy}(t)} = k_x(t)\,\Psi_t^{(n)}\big(e^{aA(0)}\big)_{xy}\,k_y(t)^{-1}, \qquad 0 \le t \le T.$$

Proof. We first compute the linear algebra, then expand the plaquette logarithms and the energy, convert the group gradient into logarithmic coordinates, add the gauge velocity, fix the bilinear map, bound the remainder, and finally identify the flow.

Step 1: Hodge identities. With counting measure, $\nabla_i^* = T_i^{-1} - I$, and differences in different directions commute. We extend a function on positive plaquettes antisymmetrically by $z_{ji} = -z_{ij}$ and $z_{ii} = 0$. By summation by parts, we have

$$(C_0^* z)_i = -\sum_j \nabla_j^* z_{ij}, \qquad d_0^*\theta = -\sum_j \nabla_j^* \theta_j. \tag{24.5}$$

We obtain

$$(C_0^* C_0\theta)_i = H_0\theta_i - \nabla_i \sum_j \nabla_j^*\theta_j, \qquad (d_0 d_0^*\theta)_i = \nabla_i \sum_j \nabla_j^*\theta_j,$$

which proves the first identity. The component of $d_2 C_0\theta$ in the directions $i < j < k$ is

$$\nabla_i(\nabla_j\theta_k - \nabla_k\theta_j) - \nabla_j(\nabla_i\theta_k - \nabla_k\theta_i) + \nabla_k(\nabla_i\theta_j - \nabla_j\theta_i) = 0.$$

Finally $D_{a,i}^+ = a^{-1}\nabla_i$ and $D_{a,i}^- = -a^{-1}\nabla_i^*$, so $D_{a,i}^- D_{a,i}^+ = -a^{-2}\nabla_i^*\nabla_i$. This checks the signs of all three identities.

Step 2: plaquette logarithms and the energy. All matrix norms below are equivalent to the norm on $\mathfrak{su}(2)$, with numerical constants. We fix a complex neighborhood of zero in four copies of $\mathfrak{su}(2)_{\mathbb{C}}$ on which the product of the four exponentials lies in the domain of the identity logarithm chart. Expanding the exponentials through degree two, and then $\log(I + Z) = Z - Z^2/2 + \cdots$, we obtain

$$\operatorname{Log}\big(e^{v_1}e^{v_2}e^{v_3}e^{v_4}\big) = \sum_k v_k + \frac{1}{2}\sum_{k<l}(v_k v_l - v_l v_k) + \text{terms of degree at least three.}$$

The linear term is $\sum_k v_{p,k} = \theta_i(x) + \theta_j(x + ae_i) - \theta_i(x + ae_j) - \theta_j(x) = (C_0\theta)_p$. The finite-word analyticity of Lemma 5.1 applies, since the word has four letters and all its arguments lie in the same fixed identity chart. Alternatively, the two convergent series give the coefficient bounds directly on a smaller polydisc. In either description the radius and the constants are numerical. If $w_p = \|\theta\|_{\infty,p}$, the remainder has size at most Cw_p^3, and its derivatives of order $j \le 3$ are at most $C_j w_p^{3-j}$; derivatives of higher fixed order are bounded on each smaller chart.

By Lemma 5.1, $q(e^X) = |X|^2/8 + O(|X|^4)$. Substituting the expansion of the logarithm, we obtain on each plaquette

$$q(U_p) = \frac{1}{8}|(C_0\theta)_p|^2 + \frac{1}{4}\langle (C_0\theta)_p, \mathfrak{q}_p(\theta)\rangle + O(w_p^4).$$

Summing over positive plaquettes, each counted once, we obtain the expansion of the energy. These estimates are local: a link lies in four positive plaquettes, so differentiating the energy with respect to one link introduces at most four of these plaquette remainders, and no volume factor.

Step 3: the group gradient in logarithmic coordinates. For a differentiable curve $\theta(s)$,

$$\Big(\frac{d}{ds}e^{\theta}\Big)e^{-\theta} = \int_0^1 e^{u\theta}\dot\theta e^{-u\theta}\,du = \mathcal{J}(\theta)\dot\theta;$$

the integral formula follows by differentiating the exponential series term by term. The pullback of the bi-invariant metric to logarithmic coordinates is $g_\theta = \mathcal{J}(\theta)^*\mathcal{J}(\theta)$. Since the inner product is invariant, $\operatorname{ad}(\theta)^* = -\operatorname{ad}(\theta)$ for real θ, and hence

$$\mathcal{J}(\theta) = I + \tfrac{1}{2}\operatorname{ad}(\theta) + O(|\theta|^2), \qquad g_\theta = I + O(|\theta|^2).$$

We write $E_W = \sum_p q(U_p)$ and let ∇_θ denote its gradient in the Euclidean coordinates θ (only in the next formula). The velocity of $-4\nabla E_W$ in logarithmic coordinates is $-4g_\theta^{-1}\nabla_\theta E_W$. The metric correction first contributes in degree three, and by the energy expansion of Step 2 we have

$$\dot\theta_W = -C_0^*C_0\theta - C_0^*\mathfrak{q}(\theta) - D\mathfrak{q}(\theta)^*C_0\theta + O_{\mathrm{loc}}(\theta^3), \tag{24.6}$$

where $O_{\mathrm{loc}}(\theta^3)$ denotes a local analytic map vanishing to third order. The linear term $-\operatorname{ad}(\theta)/2$ of $\mathcal{J}^{-1}$ cannot be dropped unless the group gradient is also converted: the two corrections cancel only in the combination $g_\theta^{-1} = (\mathcal{J}^*\mathcal{J})^{-1}$.

We now verify the formula for $D\mathfrak{q}(\theta)^*$. Varying the kth entry of a plaquette by δv_k changes $\mathfrak{q}_p$ by

$$\frac{1}{2}\sum_{l>k}[\delta v_k, v_l] + \frac{1}{2}\sum_{l<k}[v_l, \delta v_k].$$

The invariant inner product satisfies $\langle y,[h,v]\rangle = \langle h,[v,y]\rangle$ and $\langle y,[v,h]\rangle = \langle h,[y,v]\rangle$. Pairing with y_p, we obtain exactly $\langle \delta v_k, z_{p,k}\rangle$. Since $\delta v_k = \epsilon_{p,k}T_{p,k}\delta\theta_{\mathrm{dir}(p,k)}$, transposing gives $\epsilon_{p,k}T_{p,k}^*z_{p,k}$, including the insertion in the direction $\mathrm{dir}(p,k)$. Adding the four entries and all plaquettes containing a given link proves the formula.

Step 4: the gauge velocity. If a lattice gauge transformation has velocity $\dot{k}_x k_x^{-1} = \phi_x$, then differentiating $k_x U_{xy} k_y^{-1}$ and multiplying on the right by its inverse gives the link velocity $\phi_x - \mathrm{Ad}(U_{xy})\phi_y$. For fixed ϕ, the corresponding logarithmic velocity on the link $(x, x + ae_i)$ is

$$\begin{aligned}&\big(I - \tfrac{1}{2}\,\mathrm{ad}(\theta_i) + O(|\theta_i|^2)\big)\big(\phi_x - \phi_{x+ae_i} - [\theta_i, \phi_{x+ae_i}] + O(|\theta_i|^2|\phi|)\big)\\ &\quad = (d_0\phi)_i - \tfrac{1}{2}[\theta_i, \phi_x + \phi_{x+ae_i}] + O_{\mathrm{loc}}(|\theta|^2|\phi|).\end{aligned}$$

We now set $\phi = -d_0^*\theta$. The linear part is $-d_0 d_0^*\theta$ and the quadratic part is G_2. Combining this with (24.6) and the identity $C_0^* C_0 + d_0 d_0^* = H_0$ proves the linear part $-H_0\theta$ and the formula for N_2. We define R_3 as the logarithmic velocity of part (3) minus $-H_0\theta + N_2(\theta)$.

Step 5: the bilinear map. We specify the expansion of N_2 by explicit rules, so that no choice of polarization is left to later applications. We write $T_\eta = \prod_{j=1}^3 T_j^{\eta_j}$ for $\eta \in \mathbb{Z}^3$, and apply the rules in the following order.

(i) Expand $-C_0^*\mathfrak{q}$ using (24.5), the antisymmetric extension, and the six terms $[v_k, v_l]/2$, $k < l$. For each resulting outer ∇_j^* use, in the order displayed, the two terms

$$\nabla_j^*[u, v] = [\nabla_j^* u, T_j^{-1} v] + [u, \nabla_j^* v].$$

In the first of these terms, move the difference into the second entry of the bracket by antisymmetry, keeping the minus sign.

(ii) Expand $-D\mathfrak{q}(\theta)^* C_0\theta$ using the formula for the adjoint in part (3), writing each $v_{p,l}$ as its signed translated link variable and each component of $C_0\theta$ as $\nabla_i\theta_j - \nabla_j\theta_i$. Distribute the translations over the brackets.

(iii) Expand G_2 using $d_0^*\theta = -\sum_j \nabla_j^*\theta_j$, and distribute its two translations.

Within each rule, order the directions and the entries increasingly, and keep repeated terms. This produces one fixed finite list. Every entry is a numerical multiple of $[T_\eta\theta_i, T_\zeta\nabla_k\theta_j]$ or $[T_\eta\theta_i, T_\zeta\nabla_k^*\theta_j]$; these rules, and not a symmetrization, define the two entries. The product identity in rule (i) is exact, since both sides equal $[T_j^{-1}u, T_j^{-1}v] - [u, v]$. The number of entries and their integer translations do not depend on n or L. The finite sum is read componentwise, and each term carries the insertion in the output direction prescribed by its rule.

In this list, we replace the first entry of each bracket by au and the second by av, and divide by a^3. Since $\nabla_k = aD_{a,k}^+$ and $\nabla_k^* = -aD_{a,k}^-$, the result has the form (24.4), and $a^{-3}N_2(aA) = \mathcal{N}_a(A, A)$ because the list is an exact expansion of the quadratic map N_2. The inequality $|[X, Y]| \le |X||Y|$ and the fact that translations are isometries give the bound in (24.4). A further forward difference of a bracket is distributed by

$$D_{a,k}^+[u, v] = \frac{[T_k u, T_k v] - [u, v]}{a} = [D_{a,k}^+ u, T_k v] + [u, D_{a,k}^+ v],$$

and a backward difference by $D^-_{a,k}[u,v] = [D^-_{a,k}u, T_k^{-1}v] + [u, D^-_{a,k}v]$. Iterating, we obtain the exact rules at each fixed order.

Step 6: the remainder. The logarithmic velocity $\mathcal{J}(\theta)^{-1}(-4\nabla E_W + G_U\phi)$ at a link is an analytic function of the link variables in a fixed finite neighborhood of that link, with the same form at every link. Indeed, the gradient of E_W at a link involves the four plaquettes containing it, and ϕ involves the links at the two endpoints. By Step 2, by the analyticity of $\mathcal{J}^{-1}$ and of Ad near the identity, and by the Cauchy estimates on a fixed complex polydisc, R_3 is a local analytic map whose homogeneous parts satisfy $\|P_{m,e}\|_{\text{coeff}} \le K_0 K_1^m$. Its Taylor coefficients vanish through degree two by Steps 3 and 4. For $0 < a \le 1$ we have

$$a^{-3}R_{3,e}(aA) = \sum_{m\ge 3} a^{m-3}P_{m,e}(A), \qquad \|P_{m,e}\|_{\text{coeff}} \le K_0 K_1^m, \tag{24.7}$$

where $P_{m,e}$ is homogeneous of degree m in the stencil of e. We choose r_* so that $K_1 r_* \le \eta < 1$ for a fixed η. Lemma 24.1 with $a = 1$ implies the two bounds on R_3 in part (5). Explicitly, by Taylor's formula with integral remainder, we have

$$|R_{3,e}(\theta)| \le C \max_f |\theta_f|^3, \qquad \|DR_{3,e}(\theta)\| \le C \max_f |\theta_f|^2,$$

where the maxima are over the stencil of e. Integrating the second bound along the segment from θ' to θ, which stays in the chart, and using $((1-u)b + uc)^2 \le b^2 + c^2$ for $b, c \ge 0$, we obtain the difference bound. Bounded incidence makes the constants independent of the number of links. Lemma 24.1 with $0 < a \le 1$ implies the bounds on the physical remainder. On $K_1 a\|A\|_\infty \le \eta$, its jth field derivative is at most $C_{j,\eta}\|A\|_\infty^{3-j}$ for $j \le 3$ and $C_{j,\eta}a^{j-3}$ for $j > 3$, and its spatial differences are finite sums of products of differences of A with these field derivatives, with no inverse power of a left over. The factors a^{m-3} in (24.7) are nonnegative powers of a for $m \ge 3$ and $a \le 1$.

Step 7: identification with the Wilson flow. On a finite lattice the Wilson vector field is smooth and gauge equivariant: the energy is gauge invariant, and the product metric is invariant under the action of each fixed gauge transformation. Let A be as in part (6). We solve $\dot{k}_x k_x^{-1} = \lambda_x$ with $\lambda = a^{-2}\phi(aA)$ and $k_x(0) = I$. The matrix integral equation has a convergent Picard series on every bounded time interval, since its mth iterated integral is at most $(\int\|\lambda\|)^m/m!$. Since λ_x is skew-Hermitian and traceless, $k_x^* k_x = I$ and $\det k_x = 1$ are preserved, and periodic coefficients give a periodic gauge transformation. In the variable $s = t/a^2$, the substitution $\theta = aA$ turns the equation of part (6) into $\partial_s\theta = -H_0\theta + N_2(\theta) + R_3(\theta)$, which by part (3) says that the links $U(t) = e^{aA(t)}$ move with velocity equal to the Wilson velocity (with factor $4/a^2$ in the variable t) plus the gauge velocity generated by λ. Differentiating $k_x^{-1}U_{xy}k_y$ cancels the gauge velocity, and the remaining velocity is the Wilson vector field with factor $4/a^2$,

by gauge equivariance; the initial links are $e^{aA(0)}$. By uniqueness for this smooth finite-dimensional ordinary differential equation (for example by the local Lipschitz integral inequality), $k^{-1}Uk = \Psi_t^{(n)}(e^{aA(0)})$, which is the claimed formula. The chart is needed only to define the damping generator, and no coupling enters. □

Substituting $\theta = aA$ and $s = t/a^2$ into part (3) shows that the equation of part (6) is (24.1) with $F_a(A) = a^{-3}R_3(aA)$, without approximation. We call the solution A in part (6) the *damped representative* of the Wilson flow started from $e^{aA(0)}$, and k its *damping gauge transformation*. The gauge velocity added in part (3) is the lattice analogue of the gauge damping of Zwanziger [157] and DeTurck [66]. By (24.5), $\phi = -d_0^*\theta = \sum_j \nabla_j^*\theta_j$, so the generator of the damping gauge transformation, which we call the *physical damping generator*, is

$$\lambda_a = a^{-2}\phi(aA) = -\sum_j D^-_{a,j}A_j, \tag{24.8}$$

a linear function of A.

24.2 Linear and quadratic flow data

We recall the weighted norms of Definition 4.8. Throughout the rest of this section $1/2 < \alpha < 2/3$, and

$$\beta = \alpha - \tfrac{1}{2}, \qquad \kappa = 1 - \tfrac{3}{2}\alpha;$$

in the applications $\alpha = \alpha_{\mathrm{fl}}$. Fix a horizon $T > 0$ and write $v_a(t) = t + a^2$. For paths x, y on $[0,T]$ with values in fields on Λ_n, and for a field f, set

$$\begin{aligned}
\mathsf{X}^a_\alpha(x) &= \sup_{0\le t\le T}\big\{v_a(t)^{\alpha/2}\|x(t)\|_\infty + v_a(t)^{(\alpha+1)/2}\|D_a x(t)\|_\infty\big\},\\
\mathsf{Y}^a_\alpha(y) &= \sup_{0\le t\le T}\big\{v_a(t)^{\beta}\|y(t)\|_\infty + v_a(t)^{\alpha}\|D_a y(t)\|_\infty\big\},\\
\mathsf{H}^a_\sigma(f) &= \sup_{0\le t\le T}\big\{v_a(t)^{\sigma/2}\|P^a_t f\|_\infty + v_a(t)^{(\sigma+1)/2}\|D_a P^a_t f\|_\infty\big\}, \qquad 0<\sigma<1-\alpha.
\end{aligned} \tag{24.9}$$

The horizon is part of the norms; when it matters we say that a bound holds on $[0,T]$. Each norm controls a value and every first difference, and translations do not change it. For paths u, v we write

$$\mathcal{Q}_a(u,v)(t) = \int_0^t P^a_{t-s}\mathcal{N}_a\big(u(s),v(s)\big)\,ds, \qquad \mathcal{B}_a(u,v) = \mathcal{Q}_a(u,v) + \mathcal{Q}_a(v,u),$$

with the bilinear map $\mathcal{N}_a$ of Lemma 24.2; in general $\mathcal{Q}_a(u,v) \neq \mathcal{Q}_a(v,u)$.

Definition 24.3. Fix $L, T > 0$ and α as above. Let $a = L/n$ and let A_0 be a field on Λ_n. The *flow data* of A_0 on $[0,T]$ are the linear heat evolution and the quadratic Picard term

$$x_a(t) = P^a_t A_0, \qquad y_a(t) = \mathcal{Q}_a(x_a,x_a)(t), \qquad 0 \le t \le T,$$

and we write $\|(x_a, y_a)\|_{\mathrm{fl},a} = \mathsf{X}^a_\alpha(x_a) + \mathsf{Y}^a_\alpha(y_a)$. The equation associated with A_0 is

$$\partial_t A = \Delta_a A + \mathcal{N}_a(A, A) + a^{-3} R_3(aA), \qquad A(0) = A_0, \tag{24.10}$$

considered as long as $a\|A(t)\|_\infty < r_*$. Differences of flow data on the same lattice are measured by the same two norms applied to the differences of the paths.

For comparisons between different lattices, let I_a be the continuous cardinal interpolation of Lemma 16.3 on Λ_n, applied to each component of a field on positive links, each link being anchored at its starting vertex. It is applied to the value array of a path and, separately, to each of its difference arrays $D^\pm_{a,j}$. For flow data (x, y) on the mesh a and $(\widetilde{x}, \widetilde{y})$ on the mesh $\widetilde{a}$, set

$$\begin{aligned} d_T\big((x,y),(\widetilde{x},\widetilde{y})\big) = \sup_{0<t\le T} \Big\{ & t^{\alpha/2}\|I_a x(t) - I_{\widetilde{a}}\widetilde{x}(t)\|_\infty \\ & + t^{(\alpha+1)/2} \max_{j,\pm} \|I_a(D^\pm_{a,j}x(t)) - I_{\widetilde{a}}(D^\pm_{\widetilde{a},j}\widetilde{x}(t))\|_\infty \\ & + t^{\beta}\|I_a y(t) - I_{\widetilde{a}}\widetilde{y}(t)\|_\infty \\ & + t^{\alpha} \max_{j,\pm} \|I_a(D^\pm_{a,j}y(t)) - I_{\widetilde{a}}(D^\pm_{\widetilde{a},j}\widetilde{y}(t))\|_\infty \Big\}. \end{aligned} \tag{24.11}$$

The interpolated difference array $I_a(D^\pm_{a,j}x)$ is a separate component of this comparison; it is not the derivative of $I_a x$. The definition involves neither a choice of gauge nor a comparison of the maps $\mathcal{N}_a$ on different lattices.

We record four elementary properties.

Interpolation. The one-dimensional cardinal weight ζ of Lemma 16.3 is nonnegative and satisfies $\sum_{m\in\mathbb{Z}} \zeta(u-m) = 1$. Multiplying three such identities and periodizing shows that the weights in each row of I_a are nonnegative and sum to one, so $\|I_a f\|_\infty \le \|f\|_\infty$. This applies separately to each difference component, and it uses no derivative estimate for an interpolant. The maps I_a do not depend on time.

Behavior at time zero. On a fixed lattice all arrays are finite dimensional and the heat matrices are continuous in t, so $x(t)$, $D^\pm_{a,j}x(t)$, $y(t)$, and $D^\pm_{a,j}y(t)$ are continuous and bounded on $[0, T]$. Since the four weights in (24.11) have positive exponents, each weighted component of d_T extends continuously by zero to $t = 0$.

Comparison of weights. Since $t^c \le (t + a^2)^c$ for $c > 0$, the interpolation bound implies

$$\begin{aligned} \sup_{0<t\le T} \Big\{ & t^{\alpha/2}\|I_a x(t)\|_\infty + t^{(\alpha+1)/2} \max_{j,\pm} \|I_a(D^\pm_{a,j}x(t))\|_\infty \\ & + t^{\beta}\|I_a y(t)\|_\infty + t^{\alpha} \max_{j,\pm} \|I_a(D^\pm_{a,j}y(t))\|_\infty \Big\} \le \|(x,y)\|_{\mathrm{fl},a}. \end{aligned}$$

In particular, bounds in the norms with weights $t + a^2$ imply bounds in the weights of d_T; the reverse does not hold uniformly near $t = 0$. The local existence arguments below use the norms with weights $t + a^2$.

Exactness. On a fixed lattice, the substitution $\theta = aA$, $s = t/a^2$ in Lemma 24.2(3) gives (24.10) without approximation, and the quadratic Duhamel term generated by A_0 in this equation is y_a, with the map $\mathcal{N}_a$ fixed there.

24.3 Local existence for the Wilson flow

We first explain the choice of exponents. Suppose that a field satisfies $\|A(t)\|_\infty \lesssim t^{-\alpha/2}$ and $\|D_a A(t)\|_\infty \lesssim t^{-(\alpha+1)/2}$, as the linear heat evolution of initial data of regularity $-\alpha$ does. Then the sources in (24.10) have the following sizes:

source	size	integrable at $t = 0$
$\mathcal{N}_a(A, A)$	$t^{-\alpha-1/2}$	no, since $\alpha > 1/2$
$a^{-3}R_3(aA)$	$t^{-3\alpha/2}$	yes, since $\alpha < 2/3$

The Duhamel integral of the cubic term has size $t^{1-3\alpha/2} = t^\kappa$. The quadratic term is too singular to be integrated with these bounds, and its Duhamel integral cannot be estimated from bounds on x_a alone. This is why the quadratic Picard term y_a is part of the data: it is the Duhamel integral of the most singular source, and in the applications its bound is proved in Sections 26 to 28 using the probabilistic structure of x_a. After x_a and y_a are subtracted, every remaining source is bounded by a multiple of $(t + a^2)^{-3\alpha/2}$ (see the table in the proof below), and the remainder is of order $(t + a^2)^\kappa$. The restriction $\alpha > 1/2$ corresponds to the regularity of the initial data in the applications (see Section 3.5), and it makes $\beta = \alpha - 1/2$ positive.

Lemma 24.4. *Let $0 < p < 1$, $a^2 \le T \le 1/2$, and $K \ge 0$, and let f be a measurable path on $[0, T]$ with values in fields on Λ_n and $\|f(t)\|_\infty \le K(t + a^2)^{-p}$ for $0 \le t \le T$. Then $z(t) = \int_0^t P^a_{t-s} f(s)\, ds$ satisfies, for $0 \le t \le T$,*

$$\|z(t)\|_\infty \le C_{p,L} K (t + a^2)^{1-p}, \qquad \|D_a z(t)\|_\infty \le C_{p,L} K (t + a^2)^{1/2-p}. \tag{24.12}$$

One may take $C_{p,L} = 6(1 + C_L)/(1 - p)$, where C_L depends only on L; in particular $C_{p,L}$ is bounded uniformly for p in each interval $(0, 1 - \delta]$, $\delta > 0$.

Proof. Since the heat kernel is nonnegative with mass one, P^a_s contracts the supremum norm, and

$$\|z(t)\|_\infty \le K \int_0^t (s + a^2)^{-p}\, ds \le \frac{K}{1 - p} (t + a^2)^{1-p}.$$

By Lemma 16.1 with one difference and horizon one, $\|D_a P^a_s h\|_\infty \le C_L (s + a^2)^{-1/2} \|h\|_\infty$ for $0 \le s \le 1$, where C_L depends only on L; this applies since $0 \le t - s \le T \le 1/2$. Hence

$$\|D_a z(t)\|_\infty \le C_L K \int_0^t (t - s + a^2)^{-1/2} (s + a^2)^{-p}\, ds.$$

The integral is bounded by a multiple of $(t+a^2)^{1/2-p}$ by Lemma 16.8, whose hypotheses are $0 \le p < 1$ and exponent $1/2 < 1$. To see the dependence on p, we split the integral at $s = t/2$. For $s \le t/2$ we have $t - s + a^2 \ge (t+a^2)/2$, and the integral of $(s+a^2)^{-p}$ over $[0, t/2]$ is at most $(t+a^2)^{1-p}/(1-p)$. For $s \ge t/2$ we have $(s+a^2)^{-p} \le 2^p (t+a^2)^{-p}$, and the integral of $(t-s+a^2)^{-1/2}$ over $[t/2, t]$ is at most $2(t+a^2)^{1/2}$. Combining the two parts, we obtain

$$\int_0^t (t-s+a^2)^{-1/2}(s+a^2)^{-p}\,ds \le \Big(\frac{\sqrt{2}}{1-p} + 2^{p+1}\Big)(t+a^2)^{1/2-p} \le \frac{6}{1-p}(t+a^2)^{1/2-p}.$$

Measurable sources with the stated bound are integrable since $a > 0$, and no other property of f is used. □

Theorem 24.5. *Fix $L > 0$, $1/2 < \alpha < 2/3$, and $R < \infty$, and let $\kappa = 1 - 3\alpha/2$. There exist $T_R \in (0,1]$, $a_R > 0$, and $C_R < \infty$, depending only on L, α, R, and the stencil and chart of Lemma 24.2, such that the following holds for every mesh $a = L/n \le a_R$. Let A_0 be a field on Λ_n whose flow data on $[0, T_R]$ satisfy $\|(x_a, y_a)\|_{\mathrm{fl},a} \le R$. Then* (24.10) *has a unique solution on $[0, T_R]$ of the form $A_a = x_a + y_a + r_a$ with $r_a(0) = 0$ and finite remainder norm*

$$\|r\|_{\mathsf{R}_a} = \sup_{0 \le t \le T_R} \big\{ (t+a^2)^{-\kappa} \|r(t)\|_\infty + (t+a^2)^{1/2-\kappa} \|D_a r(t)\|_\infty \big\}, \tag{24.13}$$

and $\|r_a\|_{\mathsf{R}_a} \le C_R$. Uniqueness holds among mild solutions with these flow data and finite remainder norm.

Let A_0' be a second field on Λ_n whose flow data (x_a', y_a') satisfy the same bound, the two fields being compared as arrays, without a gauge transformation between them, and let $d_a = \mathsf{X}_\alpha^a (x_a - x_a') + \mathsf{Y}_\alpha^a (y_a - y_a')$. Then

$$\|r_a - r_a'\|_{\mathsf{R}_a} \le C_R d_a, \qquad \sup_{s \le t \le T_R} \|A_a(t) - A_a'(t)\|_{C_a^1} \le C_{R,s} d_a \qquad (0 < s < T_R).$$

After reducing a_R, the whole interval lies in the logarithm chart:

$$a \sup_{t \le T_R} \|A_a(t)\|_\infty \le R a^{1-\alpha} + R a^{2-2\alpha} + C_R a (T_R + a^2)^\kappa < r_*.$$

Consequently, by Lemma 24.2(6), $e^{aA_a(t)}$ is a periodic time-dependent gauge transform, equal to the identity at $t = 0$, of the Wilson flow $\Psi_t^{(n)}(e^{aA_0})$ at physical time t. The constants do not depend on the coupling, on n beyond the restriction $a \le a_R$, or on any probability law. They depend on R and s, and they are not uniform as $R \to \infty$ or $s \downarrow 0$.

Here a mild solution is a continuous path A on $[0, T_R]$ with $a\|A(t)\|_\infty < r_*$ and $A(t) = P_t^a A_0 + \int_0^t P_{t-s}^a (\mathcal{N}_a(A,A) + a^{-3} R_3(aA))(s)\,ds$. Uniqueness in Theorem 24.5 holds among the mild solutions A for which $A - x_a - y_a$ has finite remainder norm.

Proof. We construct r_a as the fixed point of a contraction on a ball for the norm (24.13). The equation for r is obtained by subtracting the equations satisfied by x_a and y_a; this removes the most singular source $\mathcal{N}_a(x_a, x_a)$. We then check, term by term, that the remaining sources are bounded by $(t + a^2)^{-3\alpha/2}$, choose the radius, the time, and the mesh threshold in this order, and prove uniqueness, Lipschitz dependence, and the chart bound. In Steps 2–6 the cubic term enters only through the bounds (24.15) on the set $\{B : a\|B\|_\infty \le r_*/2\}$; we use this in the proof of Theorem 24.12.

Step 1: bounds on the flow data. We write

$$\beta = \alpha - \tfrac{1}{2}, \qquad p = \tfrac{3}{2}\alpha < 1, \qquad \kappa = 1 - p, \qquad \delta = \frac{1-\alpha}{2} > 0.$$

We may assume $R \ge 0$, since otherwise no field satisfies the hypothesis. For a tentative $T \le 1/2$ and $a^2 \le T$, we set $v = t + a^2$. If the flow data satisfy $\|(x, y)\|_{\mathrm{fl},a} \le R$ on $[0, T]$, then

$$\begin{aligned} &\|x(t)\|_\infty \le R v^{-\alpha/2}, \quad \|D_a x(t)\|_\infty \le R v^{-(\alpha+1)/2}, \\ &\|y(t)\|_\infty \le R v^{-\beta}, \qquad \|D_a y(t)\|_\infty \le R v^{-\alpha}, \end{aligned} \tag{24.14}$$

and these bounds include all translations. Since $v \le 2T \le 1$ and $\alpha/2 - \beta = \delta > 0$, we have in particular $\|x(t) + y(t)\|_\infty \le 2R v^{-\alpha/2}$.

Step 2: the equation for the remainder. Recall that $F_a(A) = a^{-3} R_3(aA)$. We take $a \le 1$. By Lemma 24.2(5) and Lemma 24.1(1) and (3), on the set $\{B : a\|B\|_\infty \le r_*/2\}$,

$$\|F_a(B)\|_\infty \le C\|B\|_\infty^3, \qquad \|F_a(B) - F_a(B')\|_\infty \le C\big(\|B\|_\infty^2 + \|B'\|_\infty^2\big)\|B - B'\|_\infty, \tag{24.15}$$

with C independent of a. Since $\partial_t x = \Delta_a x$ and $\partial_t y = \Delta_a y + \mathcal{N}_a(x, x)$ with $x(0) = A_0$, $y(0) = 0$, the path $A = x + y + r$ solves (24.10) if and only if $r(0) = 0$ and $\partial_t r = \Delta_a r + \mathcal{N}_a(A, A) - \mathcal{N}_a(x, x) + F_a(A)$. Expanding $\mathcal{N}_a(A, A)$ by bilinearity, we obtain the source

$$\begin{aligned} \mathscr{F}_{x,y}(r) = {} & \mathcal{N}_a(x, y) + \mathcal{N}_a(y, x) + \mathcal{N}_a(y, y) \\ & + \mathcal{N}_a(x, r) + \mathcal{N}_a(r, x) + \mathcal{N}_a(y, r) + \mathcal{N}_a(r, y) + \mathcal{N}_a(r, r) + F_a(x + y + r), \end{aligned} \tag{24.16}$$

where every occurrence of $\mathcal{N}_a$ uses the bilinear map fixed in Lemma 24.2. The fixed-point equation is

$$\mathscr{M}_{x,y}(r)(t) = \int_0^t P^a_{t-s} \mathscr{F}_{x,y}(r)(s)\, ds, \qquad r = \mathscr{M}_{x,y}(r). \tag{24.17}$$

Step 3: bounds on the sources. If $\|r\|_{\mathsf{R}_a} \le S$, then $\|r(t)\|_\infty \le S v^\kappa$ and $\|D_a r(t)\|_\infty \le S v^{\kappa - 1/2}$. By (24.4), we have $\|\mathcal{N}_a(u, w)\|_\infty \le C\|u\|_\infty \|D_a w\|_\infty$. Together with (24.14), this

gives the following bounds for the quadratic sources, up to a constant depending only on the stencil:

source	amplitude	power of v
$\mathcal{N}_a(x,y),\ \mathcal{N}_a(y,x)$	R^2	$-p$
$\mathcal{N}_a(y,y)$	R^2	$-p+\delta$
$\mathcal{N}_a(x,r),\ \mathcal{N}_a(r,x)$	RS	$-p+\delta$
$\mathcal{N}_a(y,r),\ \mathcal{N}_a(r,y)$	RS	$-p+2\delta$
$\mathcal{N}_a(r,r)$	S^2	$-p+3\delta$

For example, in the third row $\mathcal{N}_a(x,r)$, with the difference on r, has exponent $-\alpha/2+\kappa-1/2 = 1/2-2\alpha = -p+\delta$, and $\mathcal{N}_a(r,x)$, with the difference on x, has exponent $\kappa-(\alpha+1)/2$, which is the same number. In the fourth row the exponents of $\mathcal{N}_a(y,r)$ and $\mathcal{N}_a(r,y)$ are $-\beta+\kappa-1/2$ and $\kappa-\alpha$, both equal to $-p+2\delta$. In the first row the exponents of $\mathcal{N}_a(x,y)$ and $\mathcal{N}_a(y,x)$ are $-\alpha/2-\alpha$ and $-\beta-(\alpha+1)/2$, both equal to $-p$. In this way each quadratic term is estimated with its arguments in the order displayed in (24.16).

By (24.14) and (24.15), the cubic source at $r=0$ is at most CR^3v^{-p}. For r, r' in the ball of radius S, by (24.15) we have

$$\begin{aligned}\|F_a(x+y+r)-F_a(x+y+r')\|_\infty &\le C\{R^2v^{-\alpha}+S^2v^{2\kappa}\}v^{\kappa}\|r-r'\|_{\mathsf{R}_a}\\ &\le C(R^2+S^2)v^{-p+2\delta}\|r-r'\|_{\mathsf{R}_a},\end{aligned}$$

where we used $-\alpha+\kappa=-p+2\delta$ and $v^{3\kappa}\le v^{-p+2\delta}$, the exponents differing by $2-2\alpha = 4\delta>0$, with $v\le 1$. Subtracting each quadratic row gives the same table with one factor S replaced by $\|r-r'\|_{\mathsf{R}_a}$. Bounding every positive extra power of v by its maximum on $[0,T]$, we obtain

$$\begin{aligned}\|\mathscr{F}_{x,y}(0)(t)\|_\infty &\le K_R v^{-p},\\ \|\mathscr{F}_{x,y}(r)(t)-\mathscr{F}_{x,y}(r')(t)\|_\infty &\le K_{R,S}(2T)^{\delta}v^{-p}\|r-r'\|_{\mathsf{R}_a},\end{aligned}\tag{24.18}$$

where K_R depends only on R and the stencil and chart, and $K_{R,S}$ depends in addition on S.

Step 4: choice of the parameters. By Lemma 24.4 with $p=3\alpha/2$,

$$\|\mathscr{M}_{x,y}(0)\|_{\mathsf{R}_a}\le C_{\alpha,L}K_R,\qquad \|\mathscr{M}_{x,y}(r)-\mathscr{M}_{x,y}(r')\|_{\mathsf{R}_a}\le C_{\alpha,L}K_{R,S}(2T)^{\delta}\|r-r'\|_{\mathsf{R}_a}.$$

We choose $S=1+2C_{\alpha,L}K_R$, and then $T_R\in(0,1/2]$ so that $C_{\alpha,L}K_{R,S}(2T_R)^{\delta}\le 1/2$. Neither choice involves the mesh. We then reduce a_R so that $a_R\le\min(1,\sqrt{T_R})$ and

$$Ra_R^{1-\alpha}+Ra_R^{2-2\alpha}+Sa_R(2T_R)^{\kappa}<r_*/2.\tag{24.19}$$

All three exponents of a_R are positive. Since $v\ge a^2$, we have $av^{-\alpha/2}\le a^{1-\alpha}$, $av^{-\beta}\le a^{2-2\alpha}$, and $av^{\kappa}\le a(2T_R)^{\kappa}$, so every candidate $x+y+r$ with $\|r\|_{\mathsf{R}_a}\le S$ satisfies $a\|x+y+r\|_\infty < r_*/2$ for $a\le a_R$. In particular, (24.15) is never used outside its domain.

Step 5: existence and uniqueness. On a fixed lattice, the continuous paths r on $[0, T_R]$ with $r(0) = 0$ and finite R_a norm form a complete space: v is bounded below by $a^2 > 0$, there are finitely many coordinates, and the differences are continuous linear maps. By Step 4 and the choice of S, $\mathscr{M}_{x,y}$ maps the closed ball of radius S into itself and is a contraction with constant $1/2$ there. Starting from zero, successive iterates differ by at most 2^{-m} times the first difference, so they converge in this space to a fixed point. The same estimate gives uniqueness inside the ball. The heat convolution vanishes at $t = 0$, so $r(0) = 0$. This constructs the solution, and we take $C_R \geq S$.

Uniqueness in the larger class of the theorem does not require an a priori bound on the radius of a second remainder. On a fixed lattice, a continuous mild solution in the chart is differentiable, since its heat integral is finite dimensional, and it satisfies the ordinary differential equation (24.10). On a compact subinterval of the time interval on which both solutions stay in the chart, the analytic vector field has a finite Lipschitz constant K_a. For two solutions with the same initial value, $\sup_{u \leq t} |A(u) - A'(u)| \leq K_a t \sup_{u \leq t} |A(u) - A'(u)|$. Taking $K_a t < 1$ and restarting at time t proves equality on that interval, and finitely many such intervals give equality on the whole common interval. The flow data are determined by the initial value, so this proves uniqueness among mild solutions with the given flow data and finite remainder norm. The argument needs no bound on K_a that is uniform in a.

Step 6: Lipschitz dependence. For two flow data in the ball, (24.14) applies to their differences with R replaced by d_a. In each quadratic term of (24.16), we write the difference by changing first the first argument and then the second. The first three terms are then bounded by $C_R d_a v^{-p}$, and the terms involving a fixed remainder of norm at most S by $C_{R,S} d_a v^{-p}$, since $v \leq 1$. For the cubic term, the difference of $x + y$ is at most $2 d_a v^{-\alpha/2}$ and both arguments are bounded by $(2R + S) v^{-\alpha/2}$, so by (24.15) its difference is also at most $C_{R,S} d_a v^{-p}$. Writing $r - r' = (\mathscr{M}_{x,y}(r) - \mathscr{M}_{x',y'}(r)) + (\mathscr{M}_{x',y'}(r) - \mathscr{M}_{x',y'}(r'))$ and applying Lemma 24.4 to the first difference and the contraction estimate to the second, we obtain

$$\|r - r'\|_{\mathsf{R}_a} \leq C_{R,S,\alpha,L} d_a + \tfrac{1}{2} \|r - r'\|_{\mathsf{R}_a}.$$

We enlarge C_R to absorb the factor two. On $[s, T_R]$ all weights in (24.14) and in (24.13) are bounded by constants depending only on s, R, α, and L. Adding the bounds for $x - x'$, $y - y'$, and $r - r'$, we obtain the estimate for $\|A(t) - A'(t)\|_{C^1_a}$.

Step 7: the chart and the identification. Since the value bounds for x and y decrease in time and the value bound for the remainder increases, we have

$$a \sup_{t \leq T_R} \|x(t) + y(t) + r(t)\|_\infty \leq R a^{1-\alpha} + R a^{2-2\alpha} + C_R a (T_R + a^2)^\kappa.$$

If necessary we reduce a_R again after the last enlargement of C_R, so that this bound is less than r_* on the whole interval. The last term is of order a; it is not replaced by its smaller

value at $t = 0$. Part (6) of Lemma 24.2 now applies on $[0, T_R]$ and identifies $e^{aA_a(t)}$ with the gauge transform of the Wilson flow by the periodic damping gauge transformation, which is the identity at $t = 0$. All choices above have the stated dependence. The proof gives $T_R \le 1/2$ and $a_R \le \min(1, \sqrt{T_R})$. □

24.4 Bilinear estimates for the Duhamel integral

The probabilistic estimates of Sections 26 to 28 decompose the flow data into many pieces and estimate the Duhamel terms of their products. The following lemma bounds such a term when one argument is a heat path with the regularity of x_a and the other a heat path with the better exponent $\sigma < 1 - \alpha$. Since $\mathcal{N}_a$ is not symmetric, we estimate $\mathcal{Q}_a(u, v)$ and $\mathcal{Q}_a(v, u)$ separately.

Lemma 24.6. *Fix $L, T > 0$, $1/2 < \alpha = \alpha_{\mathrm{fl}} < 2/3$, and $0 < \sigma < 1 - \alpha$. For every mesh $a = L/n$, $n \ge 4$, and all fields f, g on Λ_n, let $u(t) = P_t^a f$ and $v(t) = P_t^a g$. Then, with the norms* (24.9) *on $[0, T]$,*

$$\mathsf{Y}_\alpha^a\big(\mathcal{Q}_a(u, v)\big) + \mathsf{Y}_\alpha^a\big(\mathcal{Q}_a(v, u)\big) \le C_{L,T,\alpha,\sigma}(T + a^2)^{(\alpha-\sigma)/2}\, \mathsf{X}_\alpha^a(u)\, \mathsf{H}_\sigma^a(g).$$

In particular the same right side bounds $\mathsf{Y}_\alpha^a(\mathcal{B}_a(u, v))$. The constant depends only on L, T, α, σ, and the stencil of $\mathcal{N}_a$; it does not depend on the mesh, the fields, the coupling, or any probability law. The estimate includes $t = 0$, all translations in the stencil, and every first difference. For $a \le 1$ it follows from heat kernel estimates uniformly in a; the finitely many lattices with $1 < a \le L/4$ are included by a finite maximum.

Proof. We first assume $a \le 1$. The horizon T is the same in the norms and in Lemma 16.1; every heat time below is $t - s \in [0, T]$, so the heat estimates are used only on this fixed horizon, with the constant mode included. Let $w(t) = t + a^2$, $K_u = \mathsf{X}_\alpha^a(u)$, and $K_g = \mathsf{H}_\sigma^a(g)$. By the definition of the norms, for $0 \le s \le T$,

$$\begin{aligned} \|u(s)\|_\infty &\le K_u w(s)^{-\alpha/2}, \quad \|D_a u(s)\|_\infty \le K_u w(s)^{-(\alpha+1)/2}, \\ \|v(s)\|_\infty &\le K_g w(s)^{-\sigma/2}, \quad \|D_a v(s)\|_\infty \le K_g w(s)^{-(\sigma+1)/2}. \end{aligned}$$

By (24.4), both $\mathcal{N}_a(u(s), v(s))$ and $\mathcal{N}_a(v(s), u(s))$ are bounded by $C K_u K_g w(s)^{-p}$, where

$$p = \frac{\alpha + \sigma + 1}{2} < 1.$$

The bound for $\mathcal{N}_a(u(s), v(s))$ uses the displayed bound on $D_a v$, and the bound for $\mathcal{N}_a(v(s), u(s))$ uses the displayed bounds on v and $D_a u$.

By Lemma 16.1 with zero and one difference, for either Duhamel term z,

$$\|z(t)\|_\infty \le C K_u K_g \int_0^t w(s)^{-p}\, ds \le C K_u K_g w(t)^{1-p},$$

$$\|D_a z(t)\|_\infty \le C_{L,T} K_u K_g \int_0^t (t - s + a^2)^{-1/2} w(s)^{-p}\, ds \le C_{L,T,p} K_u K_g w(t)^{1/2-p}.$$

In the last inequality we used Lemma 16.8, whose hypotheses are $0 < p < 1$ and exponent $1/2 < 1$. The weight of the value in Y_α^a adds $\beta = \alpha - 1/2$, and the weight of the difference adds α; both resulting powers of $w(t)$ equal $(\alpha - \sigma)/2 > 0$. Their suprema on $[0, T]$ are bounded by $(T + a^2)^{(\alpha-\sigma)/2}$. At $t = 0$ both integrals vanish. Every translation in the stencil is an isometry and commutes with the heat semigroup and with differences. Adding the bounds for $\mathcal{Q}_a(u, v)$ and $\mathcal{Q}_a(v, u)$ proves the inequality, and the triangle inequality implies the bound for $\mathcal{B}_a$. For $a \le 1$ the constant consists of the constant of the stencil, the two heat kernel constants on the horizon T, and the constant of the scalar convolution estimate.

It remains to include $a > 1$ at the fixed side L. Let

$$\mathcal{I}_L = \{n \in \mathbb{N} : 4 \le n < L\}, \qquad a_n = L/n.$$

This finite set consists of the possible coarse lattices, and it is empty when $L \le 4$. The formula (24.4) and its bound hold for every $a > 0$, since they use no smallness of a; let C_{st} be the constant in that bound. For every field h, each difference satisfies $\|D^{\pm}_{a,j} h\|_\infty \le 2a^{-1}\|h\|_\infty$, since translations preserve the supremum, and the heat semigroup contracts the supremum. Hence, for the two argument orders displayed below,

$$\|\mathcal{N}_a(P_s^a f, P_s^a g)\|_\infty,\ \|\mathcal{N}_a(P_s^a g, P_s^a f)\|_\infty \le 2C_{\mathrm{st}} a^{-1}\|f\|_\infty \|g\|_\infty.$$

For either Duhamel term z, we obtain

$$\|z(t)\|_\infty \le 2C_{\mathrm{st}} T a^{-1}\|f\|_\infty\|g\|_\infty, \qquad \|D_a z(t)\|_\infty \le 2a^{-1}\|z(t)\|_\infty,$$

and the second inequality holds for all translated difference components. With $\beta = \alpha - 1/2 > 0$ and $\eta = (\alpha - \sigma)/2 > 0$, the two weights of Y_α^a are maximal at $t = T$. The terms at $t = 0$ of the input norms give $K_u \ge a^\alpha \|f\|_\infty$ and $K_g \ge a^\sigma \|g\|_\infty$. The sum of the two output norms is then at most

$$4C_{\mathrm{st}} T a^{-1-\alpha-\sigma}\big\{(T + a^2)^\beta + 2a^{-1}(T + a^2)^\alpha\big\} K_u K_g. \tag{24.20}$$

We define

$$C_{\mathrm{coarse}} = \max_{n \in \mathcal{I}_L} 4C_{\mathrm{st}} T a_n^{-1-\alpha-\sigma}\big\{(T + a_n^2)^{\beta-\eta} + 2a_n^{-1}(T + a_n^2)^{\alpha-\eta}\big\},$$

with value zero if $\mathcal{I}_L$ is empty. It is finite and depends only on L, T, α, σ, and the stencil, and (24.20) is at most $C_{\mathrm{coarse}}(T + a^2)^\eta K_u K_g$ on every coarse lattice. Taking the maximum of C_{coarse} and the constant for $a \le 1$ proves the lemma. □

24.5 Gauge transformations, energy, and loop observables

We next control changes of gauge, holonomies, the lattice energy, and exponential transforms of bounded observables.

For a closed lattice path based at z, the holonomy of a configuration changes by conjugation at z under a lattice gauge transformation, so only conjugation-invariant functions of it, such as characters, are gauge invariant. For a smooth connection b and a smooth gauge transformation k we use $b^k = kbk^{-1} - (dk)k^{-1}$, as in Definition 4.10, and we write $b|_a$ for the field $(b|_a)_i(x) = b_i(x)$ obtained by restriction to the sites. The parallel transport along a path ℓ solves $H'(s) = H(s)b(\ell(s))\dot{\ell}(s)$, $H(0) = I$.

Lemma 24.7. *The following deterministic statements hold on the periodic lattice.*

1. *Let k be a smooth periodic gauge transformation and b a smooth connection. For a field A define*
$$\mathscr{G}_{a,k}(A)_i(x) = a^{-1}\operatorname{Log}\big(k(x)e^{aA_i(x)}k(x+ae_i)^{-1}\big),$$
the logarithm of the lattice gauge transform of e^{aA} by the restriction of k to the sites. On every fixed ball $\{\|A\|_{C^1_a} \le \rho\}$, for sufficiently small a,
$$\|\mathscr{G}_{a,k}(A) - \mathscr{G}_{a,k}(A')\|_{C^1_a} \le C_k\|A - A'\|_{C^1_a}, \qquad \|\mathscr{G}_{a,k}(b|_a) - (b^k)|_a\|_{C^1_a} \le C_{k,b}a.$$
The constants and the threshold for a depend on ρ, $\|k\|_{C^3}$, and, for the second bound, $\|b\|_{C^2}$; they do not depend on the mesh. For a coordinate edge from x to $x + ae_i$, the parallel transport of b along the edge and $e^{ab_i(x)}$ differ by at most $C\|b\|_{C^1}a^2$ in operator norm. The products of these matrices along a lattice path of physical length at most ℓ differ by at most $C_{\ell,b}a$, and so do their normalized traces.
2. *Let A_a and A'_a be the solutions of Theorem 24.5 for two fields A_0, A'_0 on the same lattice, compared as arrays, and let d_a be as there. For every closed lattice path of physical length at most ℓ and $s \le t \le T_R$, the normalized traces of the holonomies of $\Psi^{(n)}_t(e^{aA_0})$ and $\Psi^{(n)}_t(e^{aA'_0})$ along the path differ by at most $C_{R,s,\ell}d_a$. Moreover,*
$$\|D_a(A_a - A'_a)(t)\|_\infty \le C_R d_a\big\{(t+a^2)^{-(\alpha+1)/2} + (t+a^2)^{-\alpha} + (t+a^2)^{\kappa-1/2}\big\}.$$
Let $\lambda_a = a^{-2}\phi(aA_a)$ and λ'_a be the physical damping generators (24.8) *of the two solutions. Then $\|\lambda_a - \lambda'_a\|_\infty \le C\|D_a(A_a - A'_a)\|_\infty$ and $\int_0^{T_R}\|\lambda_a(t) - \lambda'_a(t)\|_\infty\,dt \le C_R d_a$, uniformly in a. The two damping gauge transformations $k, \widetilde{k}$ of Lemma 24.2(6), both equal to the identity at $t = 0$, satisfy $\|k_x(t) - \widetilde{k}_x(t)\|_{\mathrm{op}} \le C_R d_a$ for every site x and $0 \le t \le T_R$. For a closed path based at z, the holonomies $H_{\mathrm{d}}(t)$ of the damped links $e^{aA_a(t)}$ and $H_{\mathrm{W}}(t)$ of $\Psi^{(n)}_t(e^{aA_0})$ satisfy $H_{\mathrm{d}}(t) = k_z(t)H_{\mathrm{W}}(t)k_z(t)^{-1}$. Consequently the based holonomy matrices of the two Wilson flows along such a path differ by at most $C_{R,s,\ell}d_a$ for $s \le t \le T_R$.*

The constants in part (2) have the dependence of Theorem 24.5, and depend in addition on s and ℓ as indicated.

Proof. We use the operator norm for matrix products; it is equivalent to the norm on $\mathfrak{su}(2)$ with numerical constants.

Part (1): smooth gauge transformations. We define

$$h_{a,i}(x) = k(x)k(x + ae_i)^{-1}, \qquad q_{a,i}(x) = a^{-1} \operatorname{Log} h_{a,i}(x).$$

These are defined for all sufficiently small a, depending on $\|k\|_{C^1}$ and a fixed margin of the logarithm chart. We claim that $\|q_{a,i}\|_{C^2} \le C_k$ and

$$\|q_{a,i} + (\partial_i k)k^{-1}\|_{C^1} \le C_k a, \tag{24.21}$$

where here and below C^j norms of functions depending on the parameter a are norms of continuous functions on $\mathbb{T}^3_L$, taken before restriction to the sites. To verify this without dividing a remainder by a, we set $h_i(u,x) = k(x)k(x+ue_i)^{-1}$. Then $h_i(0,x) = I$ and $\partial_u h_i(0,x) = -(\partial_i k)k^{-1}$. Taylor's formula with integral remainder for $\operatorname{Log} h_i(u,x)$ in u gives (24.21), using two derivatives in u and one in x. For the bound in C^2, we use instead $q_{a,i} = \int_0^1 \partial_u \operatorname{Log} h_i(ua,x)\,du$ with two spatial derivatives. These derivatives involve at most three derivatives of k; since $k^{-1} = k^*$, the inverse has the same derivative bounds.

Let $B_i(x) = \operatorname{Ad}(k(x))A_i(x)$. The transformed link is $k(x)e^{aA_i(x)}k(x+ae_i)^{-1} = e^{aB_i(x)}e^{aq_{a,i}(x)}$. The function

$$F_a(B,q) = a^{-1}\operatorname{Log}\bigl(e^{aB}e^{aq}\bigr) = B + q + aK_a(B,q)$$

has a removable singularity at $a = 0$, and K_a and its derivatives of fixed order are bounded on each fixed bounded set of (B,q), for sufficiently small a. Indeed, the convergent product and logarithm series have linear term $a(B+q)$, all other terms carry at least a^2, and their geometric majorants can be differentiated on a smaller ball. We conclude that $\mathscr{G}_{a,k}(A)_i(x) = F_a(B_i(x), q_{a,i}(x))$, and the transformed links lie in the logarithm chart.

For the first bound of part (1), let A, A' lie in the ball. Then B, D_aB, and the corresponding quantities for A' are bounded in terms of ρ and k. We write

$$F_a(B,q) - F_a(B',q) = \int_0^1 \partial_B F_a\bigl(B' + u(B-B'),q\bigr)(B-B')\,du.$$

By the shifted product rule, a difference D_a of this expression falls either on $B - B'$ or on the coefficient under the integral. In the second case, the integral form of the chain rule used in the proof of Lemma 24.1 bounds it by $C(\|D_aB\|_\infty + \|D_aB'\|_\infty + \|D_aq\|_\infty)\|B - B'\|_\infty$. It follows that the C^1_a norm of the difference is at most $C_k\|A - A'\|_{C^1_a}$. Forward and backward differences obey the same calculation, with a backward translation in place of a forward one. For a sample $A = b|_a$ of a smooth connection, (24.21) in $F_a = B + q + aK_a$ shows that $\mathscr{G}_{a,k}(b|_a)_i$ is the restriction of $\operatorname{Ad}(k)b_i - (\partial_i k)k^{-1} + O_{k,b}(a) = (b^k)_i + O_{k,b}(a)$, the error being of order a in continuous C^1. Restriction to the sites preserves this bound for both differences, since $D^+_{a,j}f(x) = \int_0^1 \partial_j f(x + uae_j)\,du$ and the backward difference is the average of $\partial_j f$ over the preceding segment. This proves both bounds.

For the transport, let $V(s)$ solve $V' = Vb_i(x + se_i)$, $V(0) = I$, and let $E(s) = e^{sb_i(x)}$. The derivative of $V(s)E(a-s)$ is $V(s)(b_i(x+se_i) - b_i(x))E(a-s)$, whose outer factors are unitary. Integrating from 0 to a, we obtain

$$\|V(a) - E(a)\|_{\mathrm{op}} \le \int_0^a \|b_i(x+se_i) - b_i(x)\|_{\mathrm{op}}\, ds \le Ca^2\|b\|_{C^1}.$$

Reversed edges use inverses, and $\|V^{-1} - E^{-1}\| = \|V - E\|$. For a product of m such factors, we replace the factors one at a time; the other factors are unitary, so the errors add up to at most $mCa^2\|b\|_{C^1}$, and $ma \le \ell$ gives $C_{\ell,b}a$. A normalized trace has absolute value at most the operator norm, so the same bound holds for traces.

Part (2): holonomies and damping gauge transformations. For skew-Hermitian X, Y, differentiating $e^{(1-u)X}e^{uY}$ in u, we find

$$e^X - e^Y = \int_0^1 e^{(1-u)X}(X - Y)e^{uY}\, du,$$

and hence $\|e^{aA_e} - e^{aA'_e}\|_{\mathrm{op}} \le Ca|A_e - A'_e|$; inverses satisfy the same bound. Telescoping along a closed path with at most ℓ/a links, we obtain the bound $C\ell\|A(t) - A'(t)\|_\infty$ for the difference of the holonomies of the damped links. By Theorem 24.5 on $[s, T_R]$, this is at most $C_{R,s,\ell}d_a$. The holonomy of the Wilson flow is conjugate to the damped holonomy at the basepoint, so the two have the same trace, and the trace bound follows.

Before the time weights are suppressed, Step 6 of the proof of Theorem 24.5 gives exactly

$$\|D_a(A - A')(t)\|_\infty \le C_R d_a\big\{(t+a^2)^{-(\alpha+1)/2} + (t+a^2)^{-\alpha} + (t+a^2)^{\kappa-1/2}\big\},$$

the three terms coming from $x - x'$, $y - y'$, and $r - r'$. By (24.8), the damping generator is a linear combination of differences of A, so $\|\lambda_a - \lambda'_a\|_\infty$ is at most a numerical constant times the last display. The three negative powers have exponents $(\alpha+1)/2$, α, and $1/2 - \kappa$, all strictly less than one. For $0 \le q < 1$,

$$\int_0^{T_R} (t+a^2)^{-q}\, dt \le \int_0^{T_R} t^{-q}\, dt = \frac{T_R^{1-q}}{1-q},$$

which yields the bound on the integral of $\|\lambda_a - \lambda'_a\|_\infty$, uniformly in a.

Let k and $\widetilde{k}$ solve $\dot{k} = \lambda k$ and $\dot{\widetilde{k}} = \lambda'\widetilde{k}$ with identity initial values. Direct differentiation gives

$$\frac{d}{dt}\big(k^{-1}\widetilde{k}\big) = k^{-1}(\lambda' - \lambda)\widetilde{k},$$

and the outer factors are unitary. Integrating, we obtain $\sup_x\|k_x(t) - \widetilde{k}_x(t)\|_{\mathrm{op}} = \sup_x\|I - k_x(t)^{-1}\widetilde{k}_x(t)\|_{\mathrm{op}} \le \int_0^t\|\lambda - \lambda'\|_\infty \le C_R d_a$. No exponential factor depending

on the individual generators appears. For a closed path based at z, the damped links are $k_x U_{xy} k_y^{-1}$ by Lemma 24.2(6), so $H_{\mathrm{d}} = k_z H_{\mathrm{W}} k_z^{-1}$. Undoing the two dampings adds at most $2\|k_z - \widetilde{k}_z\|_{\mathrm{op}} \le 2C_R d_a$ to the difference of the damped holonomies, which implies the bound for the based holonomy matrices of the Wilson flows. Here we used that both damping gauge transformations start at the identity. □

Lemma 24.8. *Write*

$$\mathcal{E}_a(U) = \frac{4}{a}\sum_p q(U_p), \qquad F_{a,p} = a^{-2}\operatorname{Log} U_p, \qquad \|F_a\|_{2,a}^2 = a^3\sum_p |F_{a,p}|^2.$$

There exist numerical constants $n_{\mathrm{cov}}, r_{\mathrm{cov}}, c_0, c, C > 0$ such that the following holds for $n \ge n_{\mathrm{cov}}$. If $|\operatorname{Log} U_p| \le r_{\mathrm{cov}}$ for every plaquette, then $c\|F_a\|_{2,a}^2 \le \mathcal{E}_a(U) \le C\|F_a\|_{2,a}^2$. The Wilson flow $U(t) = \Psi_t^{(n)}(U(0))$ exists for all $t \ge 0$, and $\mathcal{E}_a(U(t))$ is nonincreasing. If the plaquette bound holds at a time t_0, $\mathcal{E}_a(U(t_0)) \le E_0$, and $aE_0 \le c_0$, then $|\operatorname{Log} U_p(t)| \le r_{\mathrm{cov}}/2$ for every plaquette and every $t \ge t_0$. No smallness of E_0 itself is assumed.

The normalization of $\mathcal{E}_a$ differs by a factor two from the lattice energy (2.2): $\mathcal{E}_a = \mathsf{E}_a/2$. For $U = e^{aA}$ with $A = b|_a$ the sample of a smooth connection, $\operatorname{Log} U_{p_{ij}(x)} = a^2 F_{ij}(x) + O(a^3)$ and $q(e^X) = |X|^2/8 + O(|X|^4)$, so $\mathcal{E}_a(U)$ approximates $(1/2)\sum_{i<j}\int_{\mathbb{T}_L^3}|F_{ij}|^2\,dx = E_L(b)/2$ and $\mathsf{E}_a(U)$ approximates $E_L(b)$; see Lemma 29.14.

Proof of Lemma 24.8. By Lemma 5.1, on the principal chart of the plaquette variables, $|\operatorname{Log} U_p|^2/(2\pi^2) \le q(U_p) \le |\operatorname{Log} U_p|^2/8$. Since $\|F_a\|_{2,a}^2 = a^{-1}\sum_p |\operatorname{Log} U_p|^2$, we may take $c = 2/\pi^2$ and $C = 1/2$. We choose any numerical $0 < r_{\mathrm{cov}} < 1$ and take $n_{\mathrm{cov}} = 4$.

On a fixed lattice the configuration space is a compact manifold and the Wilson vector field is smooth, so its flow exists for all times by the standard theorem on smooth vector fields on compact manifolds: a solution on a maximal interval with a finite endpoint would, by the bounded speed, have a limit at that endpoint, from which it could be continued. Along the flow,

$$\frac{d}{dt}\mathcal{E}_a(U(t)) = -\frac{16}{a^3}\|\nabla E_W(U(t))\|_2^2 \le 0, \qquad E_W = \sum_p q(U_p), \tag{24.22}$$

since $\mathcal{E}_a = 4E_W/a$ and $U(t)$ moves with velocity $-(4/a^2)\nabla E_W$. An added gauge velocity contributes zero to this derivative, since E_W is invariant under lattice gauge transformations at the link endpoints. Under the assumptions at t_0, each nonnegative plaquette term satisfies $q(U_p(t)) \le aE_0/4$ for $t \ge t_0$. We fix $0 < c_0 \le 2(1 - \cos(r_{\mathrm{cov}}/4))$. Then $q(U_p(t)) < 1 - \cos(r_{\mathrm{cov}}/4)$. Since $q(U) = 1 - \cos(\rho_G(U)/2)$ and this function is strictly increasing for $0 \le \rho_G(U) \le 2\pi$, we obtain $\rho_G(U_p(t)) < r_{\mathrm{cov}}/2$. These plaquette variables avoid $-I$, and their principal logarithms have norm $\rho_G(U_p(t)) < r_{\mathrm{cov}}/2$. This argument uses only the energy of the compact configuration; it requires no representative with small link variables and no curvature estimate at later times, and E_0 may be large. □

Lemma 24.9. *Let $W = (W_i)_{i=1}^m$ and $W' = (W_i')_{i=1}^m$ be random vectors in $\mathbb{R}^m$ on a common probability space with $|W_i|, |W_i'| \le 1$. For a number $U \ge 0$ and $u \in \mathbb{C}^m$ with $\sum_i |u_i| \le U$,*

$$\left|\mathbb{E}e^{u\cdot W} - \mathbb{E}e^{u\cdot W'}\right| \le e^U U\, \mathbb{E}\|W - W'\|_\infty.$$

More generally, for every event $\mathcal{O}$,

$$\left|\mathbb{E}e^{u\cdot W} - \mathbb{E}e^{u\cdot W'}\right| \le e^U U\, \mathbb{E}\left[\|W - W'\|_\infty \mathbf{1}_{\mathcal{O}^c}\right] + 2e^U \mathbb{P}(\mathcal{O}).$$

For real u with $\sum_i |u_i| \le U$, $\mathbb{E}e^{u\cdot W} \ge e^{-U}$. Let $U_* > 0$ satisfy $e^{U_*} - 1 < 1/2$. On the complex neighborhood $\mathcal{V} = \{u : \sum_i |u_i| < U_*\}$, the transform $\mathbb{E}e^{u\cdot W}$ has an analytic logarithm vanishing at $u = 0$, and if a sequence of such transforms converges uniformly on compact subsets of $\mathcal{V}$, then every partial derivative of fixed order of their logarithms converges uniformly on compact subsets of $\mathcal{V}$. For characters of representations of dimensions d_i, the same statements hold with the bounds $|W_i|, |W_i'| \le d_i$ and with $\sum_i |u_i|$ replaced by $\sum_i d_i |u_i|$.

Proof. The segment from W to W' stays in the unit cube. For complex u with $\sum_i |u_i| \le U$, by the fundamental theorem of calculus we have

$$\left|e^{u\cdot W} - e^{u\cdot W'}\right| \le \int_0^1 e^{\mathrm{Re}(u\cdot((1-v)W + vW'))} \sum_i |u_i||W_i - W_i'|\, dv \le e^U U \|W - W'\|_\infty.$$

Taking expectations, we obtain the first bound. On $\mathcal{O}$ each of the two integrands has modulus at most e^U, which gives the second bound. For real u the integrand is at least e^{-U}.

Since the observables are bounded, the exponential series and each of its derivatives converge uniformly on bounded sets of u, so $M(u) = \mathbb{E}e^{u\cdot W}$ is entire. Moreover $|M(u) - 1| \le e^{\sum_i |u_i|} - 1$, which is less than $1/2$ on $\mathcal{V}$. Hence

$$\log M(u) = \sum_{m \ge 1} \frac{(-1)^{m+1}}{m} \big(M(u) - 1\big)^m$$

converges uniformly on $\mathcal{V}$ and vanishes at the origin. Integrating $1/w$ along the segment between two points of the disc $\{|w - 1| < 1/2\}$ shows that the logarithms of two such transforms differ by at most twice the difference of the transforms. In particular, uniform convergence of transforms on a compact subset of $\mathcal{V}$ implies uniform convergence of their logarithms there. For a multi-index ν and a compact set in $\mathcal{V}$, we choose closed polydiscs of radii $\rho_j > 0$ around its points lying in a compact subset of $\mathcal{V}$. The Cauchy integral formula bounds the difference of the derivatives ∂^ν at the center by $\nu! \prod_j \rho_j^{-\nu_j}$ times the supremum of the difference of the logarithms on the distinguished boundary, and the convergence of derivatives follows. For characters of dimensions d_i, the same calculation applies with $\sum_i |u_i|$ replaced by $\sum_i d_i |u_i|$ and the bound 1 replaced by d_i. □

24.6 Smooth solutions of the Yang–Mills heat flow

In this subsection we work on the trivial periodic $\mathrm{SU}(2)$ bundle over $\mathbb{T}^3_L$. For an integer $r \geq 1$ let $\mathcal{A}^r = C^r(\mathbb{T}^3_L; \mathfrak{su}(2)^3)$, with the norm $\|b\|_{C^r}$ given by the maximum of all component derivatives of order at most r; replacing it by their finite sum changes only constants depending on r. We write $\mathcal{A}^\infty = \mathcal{A}_L$ for the smooth connections. The Yang–Mills heat flow $\partial_t A_i = \sum_j D_j F_{ji} = -(d_A^* F)_i$ and its semigroup Φ on smooth periodic gauge orbits are as in Definition 4.10. The DeTurck equation is

$$\partial_t b_i = \Delta b_i + \sum_j \big\{ 2[b_j, \partial_j b_i] - [b_j, \partial_i b_j] + [b_j, [b_j, b_i]] \big\}, \tag{24.23}$$

and we write its right side as $\Delta b + \mathcal{N}(b)$, with $\mathcal{N} = \mathcal{N}_2 + \mathcal{N}_3$ the sum of its quadratic and cubic parts. A mild solution on $[0, d]$ is a path $b \in C([0,d]; \mathcal{A}^r)$ with $b(t) = P_t b_0 + \int_0^t P_{t-s} \mathcal{N}(b(s))\, ds$, where P_t is the continuum heat semigroup acting on each component.

Inputs from the literature. We use the following deterministic statements; page numbers refer to the arXiv versions cited. These papers work on the unit torus with a compact Lie group and the trivial bundle, and $\mathrm{SU}(2)$ is allowed.

(C1) [43, Definition 3.13, pp. 24–25, and Theorem 3.14, p. 25]. For every initial connection in H^1 there exists a unique solution of the Yang–Mills heat flow in the following class: it is continuous with values in H^1, its curvature $F(t)$ lies in H^1 for $t > 0$, it has a strong L^2 time derivative equal to $-d_A^* F$, its curvature is bounded in L^∞ on compact intervals of positive times, and $t^{3/4}\|F(t)\|_\infty$ is bounded near $t = 0$. Smooth initial data give a solution that is smooth up to $t = 0$. The proof of this theorem [43, Appendix B, p. 58] obtains existence from Charalambous and Gross [47, Theorem 2.5] or Råde [139, Theorem 1], uniqueness from [47, Theorem 9.15], and smoothness for smooth data from [139, Theorem 1]; on the torus the boundary conditions of Charalambous and Gross are vacuous.

(C2) [42, Theorem 1.2 and Lemma 1.3, p. 3]: local existence and uniqueness of smooth solutions of the DeTurck equation for smooth periodic initial data.

(C3) [42, Lemma C.3, p. 56]: for $f \in C^q$, with $q \geq 0$ an integer, $\|P_t f\|_{C^q} \leq \|f\|_{C^q}$ and $\|P_t f\|_{C^{q+u}} \leq C_u t^{-u/2} \|f\|_{C^q}$ for $u > 0$ and $0 < t \leq 1$.

(C4) [42, Lemma C.2, p. 55]: $\|\mathcal{N}_2(b)\|_{C^q} \leq C_q \|b\|_{C^q} \|b\|_{C^{q+1}}$ and $\|\mathcal{N}_3(b)\|_{C^q} \leq C_q \|b\|^3_{C^q}$, and, with $M_q = \max(\|b\|_{C^q}, \|c\|_{C^q})$,

$$\|\mathcal{N}_2(b) - \mathcal{N}_2(c)\|_{C^q} \leq C_q \big\{ M_q \|b - c\|_{C^{q+1}} + M_{q+1} \|b - c\|_{C^q} \big\},$$
$$\|\mathcal{N}_3(b) - \mathcal{N}_3(c)\|_{C^q} \leq C_q M_q^2 \|b - c\|_{C^q}.$$

(C5) [42, (C.9) and Corollary C.6, p. 58; Lemmas C.12 and C.13, p. 61]: for a smooth initial datum and $0 < T \leq 1$, a fixed point in $C([0,T]; C^1)$ of the Duhamel map

(C.9) belongs to $C([0,T];C^q)$ for every q and is a smooth solution of the DeTurck equation on $[0,T)$; conversely, a smooth solution satisfies the Duhamel equation on each closed subinterval $[0,T']$ with $T'<T$.

All connections to which we apply these statements are smooth periodic connections on the trivial SU(2) bundle, and hence belong to H^1. The statements are transferred from the unit torus to $\mathbb{T}^3_L$ by

$$\widetilde{b}(y)=Lb(Ly),\qquad \widetilde{t}=t/L^2,\qquad F_{\widetilde{b}}(y)=L^2F_b(Ly),\qquad E_1(\widetilde{b})=LE_L(b).$$

Differentiating the first formula shows that both evolution equations are preserved under this change of variables, and C^j norms change by the factors L^{1+m}, $0\le m\le j$. Constants may depend on L; no coupling is rescaled. With the convention of Definition 4.10, our gauge transformation k corresponds to k^{-1} in [43, (1.1), p. 2]; the statements (C1)–(C5) involve no gauge transformation, and all gauge calculations below use our convention. The local estimates in C^r, the removal of the gauge fixing, and the construction of the neighborhoods in the following theorem are proved here.

Theorem 24.10. *The following statements hold.*

1. *For every integer $r\ge 1$ and $R<\infty$ there exists $d=d(r,R,L)>0$ such that, for every $b_0\in\mathcal{A}^r$ with $\|b_0\|_{C^r}\le R$, the equation* (24.23) *has a unique mild solution $D.b_0\in C([0,d];\mathcal{A}^r)$ (here D_t denotes the solution map, not a covariant derivative), and, for two such initial data,*

$$\sup_{t\le d}\|D_tb_0\|_{C^r}\le 2(R+1),\qquad \sup_{t\le d}\|D_tb_0-D_tb_0'\|_{C^r}\le 2\|b_0-b_0'\|_{C^r}.$$

   If b_0 is smooth, then, after reducing d once if necessary (depending only on r, R, and L), the solution is the restriction to $[0,d]$ of a smooth solution on $[0,2d)$. If b is such a smooth solution, then the solution k of

$$\partial_tk=-k\,d^*b=k\sum_i\partial_ib_i,\qquad k(0)=I,$$

   is a smooth periodic gauge transformation, and $b(t)^{k(t)}=\Phi_t(b_0)$. Finally, for every integer $q\ge 0$ there exists a polynomial $P_{q,L}$ with nonnegative coefficients such that every smooth periodic gauge transformation k with $B=A^k$, for smooth connections A and B, satisfies

$$\|k^{\pm1}\|_{C^{q+1}}\le P_{q,L}\bigl(1+\|A\|_{C^q}+\|B\|_{C^q}\bigr).$$

2. *For every integer $r\ge 1$, every finite horizon $T>0$, and every set $\mathcal{K}\subset\mathcal{A}^\infty$ that is compact in the C^r topology, there exist a C^r-open neighborhood $\mathcal{U}$ of $\mathcal{K}$, a finite open cover $(\mathcal{U}_\lambda)$ of $\mathcal{U}$, and a partition $0=t_0<\cdots<t_Q=T$ with the following property. For every smooth $b_0\in\mathcal{U}_\lambda$ there exist smooth solutions $b_{\lambda,i}(s;b_0)$, $0\le s\le t_{i+1}-t_i$, of* (24.23), *with*

$b_{\lambda,0}(0;b_0) = b_0$, *and smooth periodic gauge transformations* $k_{\lambda,i}$ *that do not depend on* $b_0 \in \mathscr{U}_\lambda$, *such that*

$$b_{\lambda,i+1}(0;b_0) = b_{\lambda,i}(t_{i+1}-t_i;b_0)^{k_{\lambda,i}}, \qquad [b_{\lambda,i}(s;b_0)] = [\Phi_{t_i+s}(b_0)].$$

The numbers of cover members and of intervals, the C^r *bounds of the* $b_{\lambda,i}$, *and their Lipschitz constants as maps from* $\mathscr{U}_\lambda$, *with the* C^r *norm, to* $C([0,t_{i+1}-t_i];\mathcal{A}^r)$ *are finite and uniform on* $\mathscr{U}$. *For every* q, *the* C^q *norms of the finitely many* $k_{\lambda,i}^{\pm 1}$ *are bounded. These constants depend on* L, T, r, $\mathscr{K}$, *and the construction, and on* q *where indicated; they do not depend on any coupling, lattice mesh, lattice side,* M, *or* J.

3. *Fix an integer* $r \geq 3$ *and a finite horizon* $T > 0$, *and let* b_0, b_0' *be smooth. As* $b_0' \to b_0$ *in* C^r, *or* $b_0' \to b_0^h$ *in* C^r *for a fixed smooth time-independent gauge transformation* h, *every finite vector of values* $W_{\ell_i,\chi_i,t}$ *of loops and characters along the trajectory of* b_0', *and the energies* $E_L(\Phi_t b_0')$, *converge to the corresponding quantities for* b_0 *uniformly for* $t \in [0,T]$, *and*

$$\sup_{0\leq t\leq T} \inf_{k\in C^\infty(\mathbb{T}^3_L;\mathrm{SU}(2))} \|\Phi_t b_0 - (\Phi_t b_0')^k\|_\infty \longrightarrow 0.$$

The map $b_0 \mapsto ([\Phi_t b_0])_{t>0}$ *from the smooth connections, with the* C^r *topology, to* $(\mathcal{X}_L, \mathcal{T}_W)$ *is continuous, and hence Borel. The energy is nonincreasing along the flow, and*

$$E_L(\Phi_t b_0) \leq E_L(b_0) \leq CL^3(K_1 + K_1^2)^2, \qquad K_1 = \|b_0\|_{C^1}.$$

4. *If* $\mathscr{K}$ *is a set of smooth connections that is compact in the* C^3 *topology, then there exist a neighborhood* $\mathscr{U}$ *as in part (2) with* $r = 3$ *and* $\eta > 0$ *such that every smooth connection at* C^3 *distance less than* η *from* $\mathscr{K}$ *lies in* $\mathscr{U}$. *In particular, smooth connections approaching* $\mathscr{K}$ *in* C^3 *eventually lie in* $\mathscr{U}$; *no assumption on the closure of the approximating family is needed.*

In part (2), the members $\mathscr{U}_\lambda$ of the cover are neighborhoods of initial data, while the index i labels time intervals; the construction is not a chart on the space of gauge orbits. On one member of the cover, the representative of the trajectory is obtained by alternating local solutions of the DeTurck equation and fixed gauge transformations:

$$b_0 = c_0 \xrightarrow{D_{h_0}} D_{h_0}c_0 \xrightarrow{(\cdot)^{k_0}} c_1 \xrightarrow{D_{h_1}} D_{h_1}c_1 \xrightarrow{(\cdot)^{k_1}} c_2 \longrightarrow \cdots \longrightarrow c_{Q-1} \xrightarrow{D_{h_{Q-1}}} D_{h_{Q-1}}c_{Q-1}, \tag{24.24}$$

where $h_i = t_{i+1} - t_i$ and c_i represents the orbit at time t_i.

Proof. We first prove the local estimates in C^r and identify smooth solutions with the solutions of (C2) and (C5). Next we remove the gauge fixing and prove the bounds on gauge transformations. The neighborhoods of part (2) are then constructed around one reference trajectory, by propagating errors through (24.24), and finally around a compact family. The remaining steps prove parts (3) and (4).

Step 1: the local solution map in C^r. By the product rule and $|[X,Y]| \le |X||Y|$, for $r \ge 1$ and $\|b\|_{C^r}, \|c\|_{C^r} \le S$,

$$\|\mathscr{N}(b)\|_{C^{r-1}} \le C_r(S^2+S^3), \qquad \|\mathscr{N}(b)-\mathscr{N}(c)\|_{C^{r-1}} \le C_r(S+S^2)\|b-c\|_{C^r}. \tag{24.25}$$

Indeed, for $|\nu| \le r-1$, $\partial^\nu[b_j, \partial_k b_i] = \sum_{\mu\le\nu} \binom{\nu}{\mu}[\partial^\mu b_j, \partial^{\nu-\mu}\partial_k b_i]$, and both factors involve at most r derivatives. For the cubic brackets, we distribute ν over the three factors. For differences, we replace one factor at a time by the corresponding factor of $b-c$ and keep the other factors at b or c. The constants depend only on r and the conventions for directions and $\mathfrak{su}(2)$; these formulas also give the estimates (C4).

By (C3), after the rescaling to $\mathbb{T}^3_L$, for $0 < h \le \min(1, L^2)$,

$$\|P_h f\|_{C^r} \le \|f\|_{C^r}, \qquad \|P_h f\|_{C^r} \le C_{r,L} h^{-1/2}\|f\|_{C^{r-1}}.$$

The heat semigroup is strongly continuous at zero in C^r: each derivative of order at most r is continuous and periodic, and convolution with the positive approximate identity converges uniformly to it. For a continuous path f with values in C^{r-1} and bounded by K, the heat convolution $\int_0^t P_{t-s} f(s)\, ds$ has C^r norm at most $2C_{r,L}\sqrt{t}\,K$. It is also continuous with values in C^r, including at $t = 0$. To see this, we split off the upper part of the integral, of length η, whose norm is at most $2C_{r,L}\sqrt{\eta}\,K$. On the remaining interval the heat times are bounded below, so the kernel and all derivatives needed are continuous and bounded, and dominated convergence applies. We then let $\eta \to 0$.

For $R \ge 0$ we set $S = 2(R+1)$ and use the Banach space $C([0,2d]; \mathcal{A}^r)$ with the supremum norm. On its closed ball of radius S, we consider

$$\mathscr{M}_{b_0}(b)(t) = P_t b_0 + \int_0^t P_{t-s}\mathscr{N}(b(s))\, ds.$$

We choose $d > 0$ such that $2d \le \min(1, L^2)$ and

$$2C_{r,L}C_r\sqrt{2d}\,(S^2+S^3) \le S - R, \qquad 2C_{r,L}C_r\sqrt{2d}\,(S+S^2) \le \tfrac{1}{2}. \tag{24.26}$$

These choices depend only on r, R, and L. By (24.25), the map preserves the ball and is a contraction with constant $1/2$. Its iterates converge, since successive differences form a geometric series. Subtracting the fixed-point equations for two initial data, we find

$$\sup_{t\le 2d}\|D_t b_0 - D_t b_0'\|_{C^r} \le \|b_0 - b_0'\|_{C^r} + \tfrac{1}{2}\sup_{t\le 2d}\|D_t b_0 - D_t b_0'\|_{C^r},$$

which implies the Lipschitz bound with constant two; the ball gives the size bound. Any two mild solutions in C^r are bounded on a common compact interval. On an interval of length h, by (24.25), their difference is at most $C\sqrt{h}$ times its supremum when the initial values agree. Choosing h so that this coefficient is less than one and restarting the mild equation by the semigroup property, finitely many intervals give uniqueness. (If $R < 0$ there are no initial data, and any choice made for $R = 0$ will do.)

Step 2: smooth initial data. Let b_0 be smooth. The solution just constructed is a fixed point in $C([0,2d];C^1)$ of the map (C.9) of [42, p. 58], and the time $2d$ corresponds to a unit-torus time at most one. By (C5) it belongs to $C([0,2d];C^q)$ for every q and is a smooth solution on $[0,2d)$; conversely, a smooth solution satisfies the mild equation on every closed subinterval of its interval of definition. The local smooth existence and uniqueness of (C2) describe the same solution. Restriction to $[0,d]$ leaves a smooth solution beyond the right endpoint, which is used at the endpoints of the intervals below.

Step 3: removal of the gauge fixing. Let $v = \sum_i \partial_i b_i = -d^* b$. Expanding the curvature and the covariant derivatives gives

$$\Delta b + \mathscr{N}(b) = -d_b^* F_b + d_b v. \tag{24.27}$$

Indeed, in components $\sum_j D_j F_{ji}$ has linear part $\Delta b_i - \partial_i v$, quadratic part $[v,b_i] + 2\sum_j [b_j,\partial_j b_i] - \sum_j [b_j,\partial_i b_j]$, and cubic part $\sum_j [b_j,[b_j,b_i]]$, and adding $(d_b v)_i = \partial_i v + [b_i,v]$ cancels the two terms $-\partial_i v$ and $[v,b_i]$.

At every point, we solve $\dot{k} = kv$, $k(0) = I$. The ordered integral series converges on each finite interval on which b is smooth. Its spatial derivatives also converge: after any fixed number of derivatives, their absolute majorant is a polynomial in the index of the iterated integral times the same factorially convergent series, with constants from finitely many derivatives of v. It follows that k is smooth in space and time. Its coefficient is traceless and skew-Hermitian, and differentiating kk^* and $\det k$ shows that $k \in \mathrm{SU}(2)$. Uniqueness for the matrix integral equation and periodicity of v give periodicity of k.

For a time-dependent gauge transformation, we set $\lambda = \dot{k}k^{-1}$. Differentiating the gauge action, we find

$$\partial_t (b^k) = \mathrm{Ad}(k)\partial_t b - d_{b^k}\lambda. \tag{24.28}$$

For example, the derivative of $-\partial_i k\, k^{-1}$ is $-\partial_i \lambda - [\lambda, \partial_i k\, k^{-1}]$, and combining it with the derivative $[\lambda, k b_i k^{-1}] + \mathrm{Ad}(k)\partial_t b_i$ of $k b_i k^{-1}$ gives the covariant term. Since $\dot{k} = kv$, we have $\lambda = \mathrm{Ad}(k)v$, and the covariance $\mathrm{Ad}(k) d_b v = d_{b^k} \mathrm{Ad}(k) v$ cancels the last term of (24.27). Hence b^k solves the Yang–Mills heat flow with initial value b_0.

The initial datum is a smooth periodic connection on the trivial $\mathrm{SU}(2)$ bundle, so (C1) applies, and the global solution is smooth up to $t = 0$. The connection b^k belongs to the same solution class on the interval where it is defined: it is smooth on a compact initial time interval, its strong time derivative is given by the equation, and its curvature is bounded there, which also bounds $t^{3/4}\|F(t)\|_\infty$ near zero. Uniqueness in (C1) implies that $b(t)^{k(t)} = \Phi_t(b_0)$.

Step 4: fixed gauge transformations. For a smooth time-independent gauge transformation h, differentiating the gauge formula, we find $F(A^h) = \mathrm{Ad}(h)F(A)$ and $d_{A^h}\mathrm{Ad}(h)z = \mathrm{Ad}(h)d_A z$. In particular, a fixed gauge transformation maps smooth solutions of the Yang–Mills heat flow to smooth solutions, and by the uniqueness in (C1)

we have $\Phi_t(A^h) = \Phi_t(A)^h$. Time translation and uniqueness give the semigroup property. These two consequences identify the orbits on successive intervals below.

If $B = A^k$, rearranging the formula for the gauge action in our convention, we obtain

$$\partial_i k = kA_i - B_i k, \qquad \partial_i k^{-1} = k^{-1}B_i - A_i k^{-1}. \tag{24.29}$$

The norms of order zero of k and k^{-1} are fixed. Applying ∂^ν to the first identity, we obtain

$$\partial^\nu \partial_i k = \sum_{\mu \le \nu} \binom{\nu}{\mu} \big((\partial^\mu k)(\partial^{\nu-\mu} A_i) - (\partial^\mu B_i)(\partial^{\nu-\mu} k) \big).$$

By induction on $|\nu| \le q$, all derivatives of k of order at most $q+1$ are bounded by a polynomial with nonnegative coefficients in $1 + \|A\|_{C^q} + \|B\|_{C^q}$. The second identity gives the same bound for k^{-1}, after enlarging the polynomial. This defines $P_{q,L}$; no normalization at a point is needed. For a fixed k, the identity $A^k - B^k = k(A-B)k^{-1}$ and the Leibniz rule give a Lipschitz bound in C^r for the action of k, involving at most r derivatives of k. The inhomogeneous term of the action involves at most $r+1$ derivatives of k.

Step 5: a neighborhood of one reference trajectory. Fix a smooth reference $b_* \in \mathscr{K}$. By (C1), $A_*(t) = \Phi_t b_*$ is smooth on $[0,T]$; we set

$$R_* = 1 + \sup_{t \le T} \|A_*(t)\|_{C^r} < \infty.$$

We choose a partition whose lengths $h_i = t_{i+1} - t_i$ are at most $d(r, R_*, L)$. On the ith interval, let $b_i^*(s)$ be the solution of the DeTurck equation started from $A_*(t_i)$. By Step 3 and the semigroup property, there exists a smooth periodic gauge transformation $k_i^*(s)$ with $(b_i^*(s))^{k_i^*(s)} = A_*(t_i + s)$. For each $i < Q-1$ set $k_i = k_i^*(h_i)$. These fixed gauge transformations depend only on the reference and the partition, and they map the right endpoints exactly: $(b_i^*(h_i))^{k_i} = A_*(t_{i+1})$. They and their inverses have bounded derivatives of every fixed order, by smoothness and compactness, or quantitatively by (24.29).

Let $L_i \ge 1$ be a Lipschitz constant in C^r for the action of k_i, and set $K_0 = 1$ and $K_{i+1} = 2L_iK_i$. We fix $\rho_* > 0$ so small that $K_i\rho_* < 1/2$ for every i. For b_0 in the open ball $\|b_0 - b_*\|_{C^r} < \rho_*$ define successively

$$c_0(b_0) = b_0, \qquad b_i(s; b_0) = D_s c_i(b_0), \qquad c_{i+1}(b_0) = \big(D_{h_i} c_i(b_0)\big)^{k_i}, \tag{24.30}$$

as in (24.24). By induction, $\|c_i(b_0) - A_*(t_i)\|_{C^r} \le K_i \|b_0 - b_*\|_{C^r} < 1/2$. Indeed, $c_i(b_0)$ lies in the ball of radius R_*, the local solution map has Lipschitz constant two by Step 1 and maps $A_*(t_i)$ to b_i^*, and the fixed gauge transformation k_i then contributes the factor L_i.

This proves the induction step, and it shows that every map in (24.30) is defined on the whole ball. Arguing in the same way for two data in the ball, we obtain

$$\|c_i(b_0) - c_i(b_0')\|_{C^r} \le K_i \|b_0 - b_0'\|_{C^r}, \qquad \sup_{s \le h_i} \|b_i(s; b_0) - b_i(s; b_0')\|_{C^r} \le 2K_i \|b_0 - b_0'\|_{C^r}.$$

The solutions on the intervals have C^r norm at most $2(R_* + 1)$. For smooth initial data all the solutions b_i and all the transformed endpoints are smooth.

On the first interval, Step 3 shows that $b_0(s; b_0)$ represents the orbit of $\Phi_s(b_0)$. If the right endpoint of the ith interval represents $[\Phi_{t_{i+1}} b_0]$, then the fixed gauge transformation k_i preserves this orbit, and by Step 3, Step 4, and the semigroup property the next solution represents $[\Phi_{t_{i+1}+s} b_0]$. By induction, the orbit identity holds on every interval. This argument provides, for each interval, a smooth periodic gauge transformation relating the representative to the Yang–Mills heat flow; no continuous choice of such gauge transformations in b_0 is needed.

Step 6: compact families. The open balls of Step 5 around the points of $\mathscr{K}$ cover $\mathscr{K}$. By compactness in C^r finitely many of them suffice; let $\mathscr{U}$ be their union and let these balls be the $\mathscr{U}_\lambda$. We take maxima of the finitely many bounds and Lipschitz constants; for each q, the finitely many fixed gauge transformations and their inverses have a finite maximal C^q norm. To obtain one partition for all members of the cover, we take the union of the finitely many partitions. When a new time subdivides an interval, we restrict the solution and translate its time origin; it still solves the autonomous DeTurck equation. We use the identity gauge transformation at each new endpoint and keep the original fixed gauge transformation at each old endpoint. The recursion, the orbit identities, and all bounds persist, and the number of intervals remains finite. This proves part (2). Only the finitely many references, their smooth trajectories, the fixed gauge transformations, and the local constants in C^r enter the dependence of the constants. Compactness in C^r is not used to bound higher derivatives of the whole family.

Step 7: observables and continuity. Let $r \ge 3$ and let smooth $b_0' \to b_0$ in C^r. We use one member of the cover containing b_0. All sufficiently close data lie in it, and by the bounds of Step 5 the representatives on each interval converge in C^r uniformly in time. For two representatives b, c and a piecewise C^1 loop ℓ, we subtract their transport equations in a fixed unitary representation ϱ. By the variation-of-constants argument of Lemma 24.7(1), we have

$$\|\mathrm{Hol}_\ell(b) - \mathrm{Hol}_\ell(c)\|_{\mathrm{op}} \le C_\varrho \,\mathrm{length}(\ell) \|b - c\|_\infty,$$

where C_ϱ is the norm of the differential of ϱ, a linear map on the finite-dimensional Lie algebra. Taking the trace costs at most the dimension of ϱ. By gauge invariance, these characters coincide at the endpoints of the intervals with the characters of the Yang–Mills heat flow. We conclude that every finite vector of loop and character values converges uniformly on $[0, T]$.

On a bounded set in C^1, by the formula for the curvature, we have

$$\|F_b - F_c\|_\infty \le C(1 + \|b\|_{C^1} + \|c\|_{C^1})\|b - c\|_{C^1}, \qquad \|F_b\|_\infty \le C(\|b\|_{C^1} + \|b\|_{C^1}^2).$$

Using $||X|^2 - |Y|^2| \le (|X| + |Y|)|X - Y|$ and integrating over the volume L^3, the energies of the representatives converge uniformly. By gauge invariance they are the energies of the Yang–Mills heat flow, also at the endpoints of the intervals.

To check the convergence of the orbit infimum without using any later result on metrics, we write the two representatives at time t as $b = (\Phi_t b_0)^h$ and $c = (\Phi_t b_0')^{h'}$ for smooth periodic gauge transformations h, h'. We apply h^{-1} to both. The inhomogeneous terms cancel in the difference, and the linear part is a pointwise isometry, so

$$\|\Phi_t b_0 - (\Phi_t b_0')^{h^{-1}h'}\|_\infty = \|b - c\|_\infty.$$

The infimum in part (3) is at most this quantity, and the bounds are uniform in t, which implies the uniform limit. If instead $b_0' \to b_0^h$ for a fixed smooth time-independent h, first use $\Phi_t(b_0^h) = \Phi_t(b_0)^h$ from Step 4; characters, energies, and the infimum take the same values.

By the global smooth semigroup of (C1), $t \mapsto [\Phi_t b_0]$ is a compatible trajectory at positive times, that is, a point of $\mathcal{X}_L$. Each function $W_{\ell,\chi,t}$ that generates $\mathcal{T}_W$ is a continuous function of b_0 by what we have just proved. Since $\mathcal{T}_W$ is the weakest topology for which these functions are continuous, the trajectory map is continuous; preimages of open sets are open, so it is Borel. No completeness or separability of a metric on orbits enters this argument.

Step 8: the energy. For a smooth solution of the Yang–Mills heat flow, differentiation and periodic integration by parts give

$$\begin{aligned}\frac{d}{dt}E_L(A) &= 2\sum_{i<j}\int \langle F_{ij}, D_i\dot{A}_j - D_j\dot{A}_i\rangle\, dx \\ &= -2\sum_i \int \Big\langle \sum_j D_j F_{ji}, \dot{A}_i \Big\rangle\, dx = -2\sum_i \int |\dot{A}_i|^2\, dx \le 0.\end{aligned}$$

The integration by parts for the brackets uses the invariance of the inner product, and periodicity eliminates boundary terms. The bound on $\|F_b\|_\infty$ in Step 7 at $t = 0$ then implies $E_L(b_0) \le CL^3(K_1 + K_1^2)^2$.

Step 9: approximation of a compact family. Let $\mathscr{K}$ be compact in C^3 and consist of smooth connections, and apply part (2) with $r = 3$. For each point of $\mathscr{K}$, we choose a ball around it contained in $\mathscr{U}$, cover $\mathscr{K}$ by finitely many of the balls with half these radii, and let η be smaller than half the least chosen radius. Every point within η of $\mathscr{K}$ then lies in one of the original balls, hence in $\mathscr{U}$. Smooth connections approaching $\mathscr{K}$ in C^3 eventually lie in $\mathscr{U}$. □

24.7 Modified plaquette actions

In this subsection $c \geq 0$ is a parameter of the action. For a fixed upper bound $C_0 \geq 0$ and $0 \leq c \leq C_0$ set

$$\mathcal{A}_c(U) = \sum_{p \in \mathcal{P}_n} \{q(U_p) + c\, q(U_p)^2\}, \qquad \Psi_t^{(n,c)} = \Phi^{\mathcal{A}_c}_{4t/a^2}, \qquad a = L/n,$$

where $\Phi^{\mathcal{A}_c}$ is the flow of $-\nabla \mathcal{A}_c$ for the product metric of Section 24.1. In particular $\Psi^{(n,0)} = \Psi^{(n)}$, and these are the flows of Theorem 2.6. The vector field is smooth on the compact configuration space, so its flow exists for all times, as for the Wilson flow. Since the action is gauge invariant and the metric is invariant under gauge transformations, the flow commutes with lattice gauge transformations.

The Taylor expansion of $q(U_p)^2$ in link logarithms starts in degree four, so its force begins in degree three. The linear operator and the bilinear map of the damped equation are unchanged, and the modified flows have the same flow data. On every fixed bounded C^1_a ball, the plaquette logarithms are $O(a^2)$ and the added physical force is $O(a^3)$, by (24.34) below. In Theorem 24.12 we use this factor near rough initial data by measuring the difference of the solutions in a norm with a slightly weaker time weight, which leaves a positive power of the mesh.

Lemma 24.11. *Let $C_0 \geq 0$, and use the stencil, $\mathcal{J}$, the gauge velocity, $\mathcal{N}_a$, and R_3 of Lemma 24.2. Decreasing r_* if necessary, depending only on the stencil, the following holds. The damped equation for $\mathcal{A}_c$, obtained as in Lemma 24.2(3) with $\sum_p q(U_p)$ replaced by $\mathcal{A}_c$ and the same gauge velocity, is, for $a\|A\|_\infty \leq r_*$,*

$$\partial_t A = \Delta_a A + \mathcal{N}_a(A, A) + F_a(A) + E_{a,c}(A), \qquad F_a(A) = a^{-3} R_3(aA). \tag{24.31}$$

For $0 \leq c \leq C_0$ and $0 < a \leq 1$, the following statements hold uniformly in the mesh and the volume.

1. *$E_{a,c}(A) = a^{-3}\mathscr{E}_c(aA)$, where $\mathscr{E}_c$ is a local analytic map whose Taylor coefficients vanish through degree two; after rescaling, its term of degree m carries the factor a^{m-3}, $m \geq 3$. If $M = \|A\|_\infty$ and aM lies in a fixed smaller ball, then for every $j \geq 0$*

$$\|D^j E_{a,c}(A)[V_1, \dots, V_j]\|_\infty \leq K_{j,C_0} \begin{cases} M^{3-j}, & 0 \leq j \leq 3, \\ a^{j-3}, & j > 3, \end{cases} \cdot \prod_{r=1}^{j} \|V_r\|_\infty, \tag{24.32}$$

   with the usual conventions for $j = 0$ and M^0. In particular,

$$\|E_{a,c}(A)\|_\infty \leq K_{C_0} M^3, \qquad \|E_{a,c}(A) - E_{a,c}(B)\|_\infty \leq K_{C_0}\big(\|A\|_\infty^2 + \|B\|_\infty^2\big)\|A - B\|_\infty.$$

2. *All forward and backward differences obey the composition rule of Lemma 24.1, with the exact lattice translations: for $m \geq 1$,*

$$\|D_a^m E_{a,c}(A)\|_\infty \leq K_{m,C_0} \sum_{j=1}^{m} b_j(M,a) \sum_{\substack{\nu_1+\cdots+\nu_j=m \\ \nu_r \geq 1}} \prod_{r=1}^{j} \|D_a^{\nu_r} A\|_\infty, \tag{24.33}$$

where $b_j(M,a) = M^{3-j}$ for $j \leq 3$ and $b_j(M,a) = a^{j-3}$ for $j > 3$. For two fields in the same smaller ball, with $M = \max(\|A\|_\infty, \|B\|_\infty)$,

$$\|D_a(E_{a,c}(A) - E_{a,c}(B))\|_\infty \leq K_{C_0}\big\{M^2 \|D_a(A-B)\|_\infty + M(\|D_a A\|_\infty + \|D_a B\|_\infty)\|A-B\|_\infty\big\}.$$

3. *For A in the smaller ball,*

$$\|E_{a,c}(A)\|_\infty \leq K_{C_0} a^3 \big(\|D_a A\|_\infty + \|A\|_\infty^2\big)^3. \tag{24.34}$$

Consequently $E_{a,c}(A) = O_{C_0,\rho}(a^3)$ on $\{\|A\|_{C_a^1} \leq \rho\}$.

The constants depend only on the stencil, the margin of the smaller ball, the orders, and C_0, and not on L, n, the coupling, or any probability law. Part (6) of Lemma 24.2 holds for (24.31) *with $\Psi^{(n,c)}$ in place of $\Psi^{(n)}$, with the same damping equation, started at the identity.*

Proof. We first compute the added force exactly on the group, then bound it with Lemma 24.1, prove the bound (24.34) from the size of the plaquette logarithms, and identify the flow.

Let e be a positive link and p a plaquette containing it. We vary U_e to $e^{\varepsilon X} U_e$. If P is the product preceding the entry of e in the oriented plaquette word, then

$$\frac{d}{d\varepsilon} U_p(\varepsilon)\Big|_{\varepsilon=0} U_p^{-1} = B_{p,e} X, \qquad B_{p,e} = \begin{cases} \mathrm{Ad}(P), & \text{if } e \text{ enters positively,} \\ -\mathrm{Ad}(P U_e^{-1}), & \text{if } e \text{ enters negatively.} \end{cases}$$

Each $B_{p,e}$ is an isometry. Direct differentiation, with $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$, gives

$$h(V) = \nabla q(V) = \frac{V - V^{-1}}{8}, \qquad h(e^X) = \frac{\sin(|X|/2)}{2|X|} X, \qquad \nabla_e q(U_p) = B_{p,e}^* h(U_p),$$

the quotient being continuous at $X = 0$. Since $\nabla(c\, q_p^2) = 2c\, q_p \nabla q_p$, the added velocity in logarithmic coordinates, in the variable $s = t/a^2$ and with the factor four, is exactly

$$\mathscr{E}_{c,e}(\theta) = -8c\, \mathcal{J}(\theta_e)^{-1} \sum_{p \ni e} q(U_p(\theta))\, B_{p,e}(\theta)^* h(U_p(\theta)). \tag{24.35}$$

This is the additional negative-gradient velocity expressed in logarithmic coordinates: the factor $\mathcal{J}(\theta_e)^{-1}$ converts the velocity on the group into the logarithmic velocity, and

the Euclidean gradient in the coordinates θ is not used. The gauge velocity is unchanged. Substituting $\theta = aA$ and $s = t/a^2$ and multiplying by a^{-3}, we obtain (24.31).

On a fixed small ball all functions of the finite plaquette words in (24.35) are analytic, and $\operatorname{Log} U_p = O(\|\theta\|_{\infty,p})$, $q(U_p) = O(\|\theta\|_{\infty,p}^2)$, $h(U_p) = O(\|\theta\|_{\infty,p})$. The inverse $\mathcal{J}^{-1}$ is analytic on a smaller fixed ball. We conclude that the Taylor coefficients of (24.35) vanish through degree two. Its Taylor series at an output link is a series in a fixed finite stencil, and, extending the finite exponential, inverse, and trace formulas to a smaller complex polydisc and using the Cauchy estimates there, its homogeneous part of degree m has coefficient sum at most $K_1 K_2^m C_0$. Lemma 24.1, with $K_0 = K_1 C_0$ and the smaller ball chosen so that $K_2 aM \le \eta < 1$, implies (24.32), the bound on the values and their differences, (24.33), and the estimate for the difference of two fields. In these estimates every intermediate stencil is a convex combination of translates of the fields, hence lies in the same ball, and no factor a^{-1} remains outside a difference of A.

For (24.34), note that the four entries of the plaquette $p_{ij}(x)$ for $\theta = aA$ have sum $a^2(D^+_{a,i}A_j - D^+_{a,j}A_i)(x)$. The Baker–Campbell–Hausdorff formula on the fixed ball, with quadratic error bounded by a constant times the square of the largest entry, gives

$$|\operatorname{Log} U_p| \le K a^2 \big(\|D_a A\|_\infty + \|A\|_\infty^2\big).$$

On the plaquette chart, $|q(e^X)| \le K|X|^2$ and $|h(e^X)| \le K|X|$. Inserting these bounds into (24.35), and using bounded incidence and the boundedness of $\mathcal{J}^{-1}$, we obtain $|\mathscr{O}_{c,e}(aA)| \le K C_0 a^6 (\|D_a A\|_\infty + \|A\|_\infty^2)^3$; multiplication by a^{-3} gives (24.34).

Finally, we remove the damping by solving the same equation for the damping gauge transformation, started at the identity. Gauge equivariance of $\nabla \mathcal{A}_c$ cancels the gauge velocity exactly, as in Step 7 of the proof of Lemma 24.2. The remaining ordinary differential equation is the one that defines $\Psi^{(n,c)}$, and uniqueness identifies the flows. This argument uses the force (24.35), not a truncation of it. All formulas involve boundedly many links, which gives the stated dependence of the constants. □

Theorem 24.12. *Fix $L > 0$, $1/2 < \alpha < 2/3$, $0 \le R < \infty$, and $0 \le C_0 < \infty$, and let $\kappa = 1 - 3\alpha/2$. There exist $T_{R,C_0} \in (0, 1/2]$, $a_{R,C_0} > 0$, and $K_{R,C_0} < \infty$ such that the following holds for every mesh $a \le a_{R,C_0}$, every $c \in [0, C_0]$, and every field A_0 whose flow data, as in Definition 24.3, satisfy $\|(x_a, y_a)\|_{\mathrm{fl},a} \le R$ on $[0, T_{R,C_0}]$. The equation* (24.31) *with $A(0) = A_0$ has a unique solution of the form*

$$A_a^c = x_a + y_a + r_a^c, \qquad r_a^c(0) = 0, \qquad \|r_a^c\|_{\mathsf{R}_a} \le K_{R,C_0},$$

where the remainder norm (24.13) *is taken on $[0, T_{R,C_0}]$; uniqueness holds in the class of Theorem 24.5. The Lipschitz bounds of Theorem 24.5 for two fields, at the same c, hold uniformly in $c \in [0, C_0]$. These solutions represent the flows $\Psi^{(n,c)}$ in their damping gauges, started at the identity.*

For every fixed $0 < \varepsilon < 2 - 3\alpha$ *put* $\kappa_\varepsilon = \kappa - \varepsilon/2 > 0$ *and*

$$\|w\|_{\mathsf{R}_a^{\kappa_\varepsilon}} = \sup_{0 \le t \le T_{R,C_0}} \big\{ (t+a^2)^{-\kappa_\varepsilon} \|w(t)\|_\infty + (t+a^2)^{1/2-\kappa_\varepsilon} \|D_a w(t)\|_\infty \big\}.$$

The time T_{R,C_0} *can be chosen independently of* ε *such that, for the same field* A_0 *and uniformly in* $c \in [0, C_0]$,

$$\|A_a^c - A_a^0\|_{\mathsf{R}_a^{\kappa_\varepsilon}} \le K_{R,C_0,\varepsilon} a^\varepsilon. \tag{24.36}$$

In particular, for every fixed $t_0 \in (0, T_{R,C_0}]$, $\|A_a^c(t_0) - A_a^0(t_0)\|_{C_a^1} \to 0$ *as* $a \to 0$, *uniformly in* c. *The damping gauge transformations of* A_a^c *and* A_a^0, *both equal to the identity at* $t = 0$, *differ by* $O_{R,C_0,\varepsilon}(a^\varepsilon)$ *at every site, uniformly on* $[0, T_{R,C_0}]$ *and in* c. *The constants depend only on* L, α, R, C_0, ε, *and the stencil and chart, and not on any coupling or probability law.*

Proof. The existence part repeats the contraction of Theorem 24.5 with the cubic term F_a replaced by $F_a + E_{a,c}$. The new information is the comparison: the difference of the two solutions satisfies a difference equation with the inhomogeneous source $E_{a,c}(A_a^c)$, which (24.34) bounds by a^ε times a power of $t + a^2$ that is still integrable.

Step 1: construction. We write $p = 3\alpha/2 < 1$, $\delta = (1-\alpha)/2$, and $v = t + a^2$, and let r_* be as in Lemma 24.11. By Lemma 24.2(5), Lemma 24.1, and Lemma 24.11(1), the map $F_a + E_{a,c}$ satisfies the bounds (24.15) on $\{B : a\|B\|_\infty \le r_*/2\}$, with a constant depending also on C_0 and uniform in $c \in [0, C_0]$. Subtracting the equations for x_a and y_a from (24.31) leaves the source

$$\begin{aligned} &\mathcal{N}_a(x,y) + \mathcal{N}_a(y,x) + \mathcal{N}_a(y,y) \\ &\quad + \mathcal{N}_a(x,r) + \mathcal{N}_a(r,x) + \mathcal{N}_a(y,r) + \mathcal{N}_a(r,y) + \mathcal{N}_a(r,r) + (F_a + E_{a,c})(x+y+r). \end{aligned}$$

The linear operator and the map $\mathcal{N}_a$ are the same as in Theorem 24.5. The five types of quadratic sources therefore have the exponents $-p$, $-p+\delta$, $-p+\delta$, $-p+2\delta$, and $-p+3\delta$, with the amplitudes KR^2, KR^2, KRS, KRS, and KS^2, respectively, as in the table in Step 3 of that proof, with the arguments in the displayed order. The cubic source at $r = 0$ is at most $K_{R,C_0} v^{-p}$, and its Lipschitz constant in r is at most $K_{R,S,C_0} v^{-p+2\delta} \|r - r'\|_{\mathsf{R}_a}$. For the latter, the possible term $S^2 v^{3\kappa}$ is bounded by this power, since $3\kappa - (-p+2\delta) = 4\delta > 0$ and $v \le 1$. Steps 3–7 of the proof of Theorem 24.5 now apply with constants depending also on C_0: the heat integral of the source at $r = 0$ has norm at most K_{R,C_0}, and on a ball of fixed radius S the contraction constant is at most $K_{R,S,C_0}(2T)^\delta$. We choose S larger than twice the first bound, then $T \le 1/2$ such that the second is at most $1/2$, and then $a_{R,C_0} \le \min(1, \sqrt{T})$ such that

$$R a^{1-\alpha} + R a^{2-2\alpha} + S a (2T)^\kappa < r_*/2.$$

Every candidate then stays in the ball where the estimates hold. The complete space of remainders on a fixed lattice and the Picard iteration give the fixed point. Subtracting the

sources for two fields with distance d_a gives an additional source bounded by $K_{R,S,C_0} d_a v^{-p}$, and the contraction yields the Lipschitz bound for the same c. Uniqueness in the class of Theorem 24.5, outside the ball, follows from uniqueness for the finite-dimensional ordinary differential equation on successive compact subintervals of the chart, as in Step 5 of that proof. After reducing a_{R,C_0} as in Step 7, the solution stays in the chart, and Lemma 24.11 identifies it with $\Psi^{(n,c)}$ in its damping gauge. The construction applies on every shorter time interval, so T_{R,C_0} may be decreased further below.

Step 2: bounds on the solutions and on the added force. For $v \le 1$ the constructed solutions satisfy

$$\|A_a^c(t)\|_\infty \le K v^{-\alpha/2}, \qquad \|D_a A_a^c(t)\|_\infty \le K v^{-(\alpha+1)/2}.$$

Indeed, the value bounds of x, y, and r^c are $Rv^{-\alpha/2}$, $Rv^{-\beta}$, and Sv^κ, and $-\beta > -\alpha/2$, $\kappa > 0$. The difference bounds are $Rv^{-(\alpha+1)/2}$, $Rv^{-\alpha}$, and $Sv^{\kappa-1/2}$; here $\alpha < (\alpha+1)/2$ since $\alpha < 1$, and $\kappa - 1/2 + (\alpha+1)/2 = 1 - \alpha > 0$. By (24.34) and $v^{-\alpha} \le v^{-(\alpha+1)/2}$, we obtain

$$\|E_{a,c}(A_a^c(t))\|_\infty \le K a^3 v^{-p-3/2} \le K a^\varepsilon v^{-p-\varepsilon/2},$$

where in the last inequality we used $a/\sqrt{v} \le 1$ and $\varepsilon < 2 - 3\alpha < 3$. Let $p_\varepsilon = p + \varepsilon/2$. Then $p_\varepsilon < 1$ exactly because $\varepsilon < 2 - 3\alpha$, and $\kappa_\varepsilon = 1 - p_\varepsilon$.

Step 3: the comparison. For $0 < \kappa' < 1$ define $\|\cdot\|_{\mathsf{R}_a^{\kappa'}}$ as in the statement with κ' in place of κ_ε; thus $\mathsf{R}_a^\kappa = \mathsf{R}_a$. By Lemma 24.4, for $0 < q < 1$,

$$\|f(t)\|_\infty \le H(t + a^2)^{-q} \quad \Longrightarrow \quad \Big\| \int_0^\cdot P^a_{\cdot - u} f(u)\, du \Big\|_{\mathsf{R}_a^{1-q}} \le K_{L,q} H, \tag{24.37}$$

including $t = 0$, where $K_{L,q} = 2C_{q,L}$ is bounded uniformly for q in every interval $(0, 1 - \delta']$, $\delta' > 0$.

The flow data cancel in $w = A_a^c - A_a^0 = r_a^c - r_a^0$, and $w(0) = 0$. Subtracting the two equations and treating $E_{a,c}(A_a^c)$ as an inhomogeneous source, we write the remaining source for this difference w as $\mathcal{L}(w)$, so that

$$\partial_t w = \Delta_a w + \mathcal{L}(w) + E_{a,c}(A_a^c), \qquad \mathcal{L}(w) = \mathcal{N}_a(w, A_a^c) + \mathcal{N}_a(A_a^0, w) + F_a(A_a^c) - F_a(A_a^0).$$

Expanding $A_a^c = x + y + r^c$ and $A_a^0 = x + y + r^0$, the terms of $\mathcal{L}(w)$ are $\mathcal{N}_a(w, x)$, $\mathcal{N}_a(x, w)$, $\mathcal{N}_a(w, y)$, $\mathcal{N}_a(y, w)$, $\mathcal{N}_a(w, r^c)$, $\mathcal{N}_a(r^0, w)$, and the difference of F_a.

Let $W = \|w\|_{\mathsf{R}_a^{\kappa_\varepsilon}}$, so that $\|w(t)\|_\infty \le W v^{\kappa_\varepsilon}$ and $\|D_a w(t)\|_\infty \le W v^{\kappa_\varepsilon - 1/2}$. The quadratic terms containing x and w have exponent $-p_\varepsilon + \delta$: for instance $-\alpha/2 + \kappa_\varepsilon - 1/2 = -p_\varepsilon + \delta$ and $\kappa_\varepsilon - (\alpha+1)/2 = -p_\varepsilon + \delta$. Those containing y and w have exponent $-p_\varepsilon + 2\delta$, and those containing r^c and w, or w and r^0, have exponent $-p_\varepsilon + 3\delta$. By (24.15) and the value bound of Step 2 for both solutions, the difference of F_a is at most $K v^{-\alpha} W v^{\kappa_\varepsilon}$, with exponent $-p_\varepsilon + 2\delta$. Since $v \le 1$, we obtain $\|\mathcal{L}(w)(t)\|_\infty \le K_{R,S,C_0} W v^{-(p_\varepsilon - \delta)}$.

We apply (24.37) with $q = p_\varepsilon - \delta$; the output is bounded in $\mathsf{R}_a^{\kappa_\varepsilon+\delta}$, and multiplying by $(2T)^\delta$ bounds it in $\mathsf{R}_a^{\kappa_\varepsilon}$, since $v^{-\kappa_\varepsilon} = v^\delta v^{-\kappa_\varepsilon-\delta}$. The constants $K_{L,q}$ are bounded uniformly, because $q = p_\varepsilon - \delta \in [p-\delta, 1-\delta)$ and $p - \delta = 2\alpha - 1/2 > 1/2$. Hence, decreasing T_{R,C_0} once, independently of ε, we may assume that the heat integral of $\mathcal{L}(w)$ has $\mathsf{R}_a^{\kappa_\varepsilon}$ norm at most $W/2$.

For the inhomogeneous term, (24.37) with $q = p_\varepsilon < 1$ and Step 2 give the bound $K_\varepsilon a^\varepsilon$ in $\mathsf{R}_a^{\kappa_\varepsilon}$. On a fixed lattice W is finite, since w is continuous, $w(0) = 0$, and $v \geq a^2$; so the inequality $W \leq W/2 + K_\varepsilon a^\varepsilon$ implies $W \leq 2K_\varepsilon a^\varepsilon$, which is (24.36). At a fixed $t_0 > 0$, this implies $\|w(t_0)\|_{C_a^1} \leq 2K_\varepsilon a^\varepsilon (1 + t_0^{\min(\kappa_\varepsilon - 1/2, 0)})$, which tends to zero uniformly in c.

Step 4: the damping gauge transformations. The gauge velocity is the same for all c, so by (24.8) both damping generators are the same linear combination of differences of the respective solutions. Their difference is bounded by $3\|D_a w(t)\|_\infty \leq K a^\varepsilon v^{\kappa_\varepsilon - 1/2}$. Since $\kappa_\varepsilon - 1/2 > -1/2$ and $T_{R,C_0} \leq 1/2$, the integral of $v^{\kappa_\varepsilon - 1/2}$ over $[0, T_{R,C_0}]$ is bounded uniformly in a, so the integral of the difference of the generators is $O(a^\varepsilon)$. For skew-Hermitian generators, the variation-of-constants argument in the proof of Lemma 24.7(2) bounds the difference of the two gauge transformations, started at the identity, by this integral. Together with Step 1, this proves the statements on the gauges and the identification of the flows. The choices of S, T_{R,C_0}, and a_{R,C_0} were made in this order and have the stated dependence. □

SECTION 25

Consistency and continuation of the Wilson flow

This section completes the deterministic theory of the Wilson flow used in the rest of the paper. It turns bounds on the flow data of a lattice initial potential, which the stochastic estimates of Part VIII provide, into statements about flowed Wilson loops at fixed positive times. Section 24 solved the damped Wilson equation on a short time interval for initial potentials with bounded flow data (Theorem 24.5). For a compact set of smooth initial data, it also constructed a continuation of the Yang–Mills heat flow using a finite reference family: finitely many smooth DeTurck solutions on consecutive time intervals, joined by fixed gauge transformations (Theorem 24.10). At a fixed mesh the Wilson flow exists for all times, since it is the gradient flow of a smooth function on a compact manifold. The problem is to obtain estimates in physical units that are uniform as $a \to 0$, together with logarithmic representatives of the flowed configuration in which these estimates hold.

We prove that every initial potential with bounded flow data can be continued to any fixed time interval $[\tau, T]$ with $\tau > 0$, uniformly in the mesh. We also prove that the based holonomy matrices and the characters of flowed Wilson loops are Lipschitz functions of the flow data (Lemma 25.9, Theorem 25.10, , and Corollary 25.11). For initial potentials depending on finitely many real parameters, we bound two parameter derivatives of the based holonomy matrices of the coordinate cycles uniformly in the mesh. For fixed smooth families of initial data, we identify the limits of these matrices and their derivatives (Theorem 25.13). Finally, we show that the flows of the plaquette potentials $q + cq^2$, $0 \le c \le C_0$, give asymptotically the same flowed loops as the Wilson flow (Lemma 25.15). The main tools are the consistency of the lattice vector field on smooth fields (Lemma 25.3 and Theorem 25.4), stability in C^1_a near smooth solutions (Lemma 25.5), and smoothing after a positive time together with compactness (Theorem 25.7 and Lemma 25.6).

The section contains three kinds of comparison, with different kinds of error.

Comparison	Hypothesis	Conclusion
Wilson flow from the exponentials of a sampled smooth solution, versus that solution	initial link-logarithm error $O(a^2)$	link-logarithm error $O(a^2)$; error $O(a)$ for loops of bounded length
Lattice solution versus a selected smooth DeTurck solution	initial error η in C_a^1	error $C(\eta + a)$ in C_a^1
Two lattice solutions on the same lattice	difference d_a of the flow data	difference Cd_a in C_a^1 after a waiting time, and in the based holonomy matrices

Only the last comparison has no additive error in the mesh; this makes the flowed loops Lipschitz functions of the flow data. For matrices, as opposed to characters, we must track every gauge transformation applied at the basepoint.

Two observations organize the proofs. First, we expand the lattice vector field completely before taking the continuum limit. The contributions of the two plaquettes on either side of a link cancel at order a^2, and their common part also cancels at order a^3; what remains at order a^3 is the DeTurck vector field. Second, after smoothing for a positive time, all solutions whose flow data obey a fixed bound are uniformly close to a compact set of smooth connections. Finitely many selected smooth DeTurck solutions then give continuation and stability uniformly over these solutions, without assuming that the initial potentials converge as the mesh varies. We work both in dimensionless units, with link logarithms $\theta = aA$ and time $s = t/a^2$, where the lattice structure is simplest, and in physical units. Since $\partial_s \theta = a^3 \partial_t A$, a dimensionless error $O(a^4)$ is a physical error $O(a)$, which explains the factors a^{-3} below.

We use the local formulas for the Wilson vector field, the local solution, the sampled gauge transformations, and the continuation of the smooth flow from Lemmas 24.2 and 24.7, Definition 24.3, , and Theorems 24.5 and 24.10; the estimates for the modified plaquette potential in Lemma 24.11 and Theorem 24.12; and the heat estimates Lemmas 16.3 and 16.9. All existence statements for the continuum flow used here are contained in Theorem 24.10, which rests on the deterministic smooth-data results of Cao and Chatterjee [42, 43]; no result about random initial data is used. The Wilson flow was introduced by Lüscher [128]. The damping gauge transformation that turns it into a parabolic equation is in the spirit of the method of Zwanziger [157], DeTurck [66], and Donaldson [76], and for the Wilson flow it is described in [128, Section 2]. The arguments are standard parabolic ones (weakly singular Volterra estimates, a bootstrap for higher differences, and compactness by the Arzelà–Ascoli theorem), carried out on the exact lattice equation with constants independent of the mesh. The results are used in Sec-

tions 28 and 29 for the convergence of flowed loops and energies and for the construction of the limiting measure, in Section 32 for macroscopic loops in the short-distance analysis, and in Sections 34 to 36 for holonomy matrices and their parameter derivatives at weak coupling.

25.1 Conventions and heat estimates

Throughout this section $a = L/n$, with $n \geq 4$, is a lattice mesh, and $\Lambda_{L/a}$ is the corresponding periodic lattice. For a continuous field f on $\mathbb{T}^3_L$ we write $S_a f = f|_{\Lambda_{L/a}}$ for its sample on this lattice. A link field A has components $A_i(x) \in \mathfrak{su}(2)$ on the positively oriented links $(x, x + ae_i)$. We locate each component at the starting vertex x of its link, and we write T_i for translation by ae_i. We write D_a for any physical forward or backward difference $D^{\pm}_{a,j}$, and D^{ν}_a for an ordered composition of such differences. In a norm, D^m_a stands for the maximum over all ordered compositions of order m and over all link and color components. For a composition ν we write ∂^ν for the corresponding composition of partial derivatives. As in Theorem 24.5, we use the norm

$$\|A\|_{C^1_a} = \|A\|_\infty + \max_{j,\epsilon\in\{+,-\}} \|D^\epsilon_{a,j} A\|_\infty.$$

Translations do not change these norms, since they permute the sites of the periodic lattice. In dimensionless coordinates we write $\nabla_i = T_i - I$ and $H_0 = \sum_i \nabla_i^* \nabla_i$, so that $\Delta_a = -a^{-2} H_0$, and $S^a_s = e^{-sH_0} = P^a_{a^2 s}$ for the heat semigroup in dimensionless time. This semigroup differs from the sampling S_a.

Flow data. As in Definition 24.3, the flow data of a lattice initial potential A_0 on mesh a are the linear heat evolution and the quadratic Picard term

$$x_a(t) = P^a_t A_0, \qquad y_a(t) = \mathcal{Q}_a(x_a, x_a)(t) = \int_0^t P^a_{t-r} \mathcal{N}_a(x_a(r), x_a(r))\, dr.$$

Here $\mathcal{N}_a$ is the ordered bilinear map fixed in Lemma 24.2; it is not symmetric, and we always take its arguments in the displayed order. On a time interval $[0, T_0]$ we write

$$\|(x_a, y_a)\|_{\mathrm{fl},a} = \mathsf{X}^a_\alpha(x_a) + \mathsf{Y}^a_\alpha(y_a),$$

with the norms of Definition 4.8 taken on $[0, T_0]$. These norms contain the weighted first differences, with all translated components. Differences of flow data on the same lattice are measured by the same norms applied to the difference. Throughout, $1/2 < \alpha < 2/3$, and we write

$$\beta = \alpha - \tfrac{1}{2}, \qquad \kappa = 1 - \tfrac{3}{2}\alpha, \qquad v_a(t) = t + a^2.$$

The remainder norm of Theorem 24.5 is

$$\|r\|_{\mathsf{R}_a} = \sup_{0\leq t\leq T_R} \{ v_a(t)^{-\kappa} \|r(t)\|_\infty + v_a(t)^{1/2-\kappa} \|D_a r(t)\|_\infty \}.$$

The chart and damped representatives. Let r_* be the radius of the link-logarithm chart in Lemma 24.2. A potential A on mesh a lies in the chart if $a\|A\|_\infty < r_*$. All local analytic bounds below are used on a fixed smaller chart $\{a\|A\|_\infty \le r'\}$ with $r' < r_*$, and the corresponding constants may depend on $r_* - r'$. In the chart, the damped Wilson equation in physical units reads

$$\partial_t A = \Delta_a A + \mathcal{N}_a(A, A) + a^{-3} R_3(aA),$$

with the analytic remainder R_3 of Lemma 24.2. By that lemma, a solution in the chart is related to the Wilson flow by

$$e^{aA_i(x,t)} = k(x,t)\,\big(\Psi_t^{(n)} U\big)_i(x)\,k(x + ae_i, t)^{-1}, \qquad U = e^{aA(0)},$$

where the damping gauge transformation k solves $\partial_t k\, k^{-1} = a^{-2}\phi(aA(t))$ with $k(0) = I$, and ϕ is the damping parameter of Lemma 24.2. As in Section 24, we call A the damped representative of the Wilson flow. The Wilson flow is used at physical time t, that is, at time $4t/a^2$ for the vector field $-\nabla \sum_p q(U_p)$; there is no coupling factor in this conversion. When the flow is restarted at a time t_0 from a gauge-transformed configuration, the new damping gauge transformation has value I at t_0.

Constants. A constant that comes only from the finite local formulas (the finitely many terms of $\mathcal{N}_a$, the analytic remainder R_3, and their derivatives) depends only on the chart radius r' and on the order of differentiation, as in item (1) of Definition 4.14. Other constants may in addition depend on L, on the displayed bounds, waiting times, and horizons, and on finitely many smooth DeTurck solutions selected in the proofs; this is the dependence described in item (7) of Definition 4.14. No constant in this section depends on the coupling, on a probability measure, on the number of averaging steps, or on M and J. When applying the local results of Section 24, we may decrease the chart radius and the local existence time, and increase the minimal lattice side; we do this once, before choosing the constants below. Every restriction $a \le a_*$ on the mesh is chosen after the displayed data and before the particular initial potentials to which it is applied.

Lemma 25.1. *Fix $L, H_1 > 0$ and an integer $q \ge 0$. The lattice heat semigroup P_u^a is positivity preserving, preserves constants, is a contraction for the supremum norm, and commutes with every translation and every difference. Further,*

$$\|D_a^q P_u^a f\|_\infty \le C_{q,H_1/L^2}(u + a^2)^{-q/2}\|f\|_\infty, \qquad 0 \le u \le H_1.$$

In particular, for $a \le 1$ and $0 \le s \le H_1/a^2$,

$$\|S_s^a\|_{\infty\to\infty} \le 1, \qquad \|\nabla S_s^a\|_{\infty\to\infty} \le C_{H_1/L^2}(1+s)^{-1/2}.$$

For $p < 1$, $0 \le q < 1$, and $\epsilon > 0$,

$$\int_0^t (r+\epsilon)^{-p}(t-r+\epsilon)^{-q}\,dr \le C_{p,q}(t+\epsilon)^{1-p-q}, \tag{25.1}$$

and the same bound holds for $\epsilon = 0$ and $t > 0$.

Proof. We have $\Delta_a = a^{-2}(\sum_i (T_i + T_i^{-1}) - 6I)$. Factoring out e^{-6u/a^2} and expanding the exponential of the remaining sum of translations shows that P_u^a is positivity preserving and preserves constants; in particular it is a contraction for the supremum norm. It commutes with translations and differences, since all translations commute. For the difference bound we apply Lemma 16.3 with equal input and output meshes, physical horizon $T = H_1$, and refinement ratio one. The cardinal interpolation is then the identity, and the heat estimate of that lemma is the displayed bound for every lattice array. (Where the lemma is used below for fixed smooth interpolants, the refinement ratio is the integer H/a, and the horizon is the displayed finite T.) Multiplying by a and setting $u = a^2 s$, we obtain the dimensionless bounds.

We now prove (25.1). Suppose first that $p < 0$. Then $(r+\epsilon)^{-p} \le (t+\epsilon)^{-p}$, and the integral of the remaining factor is at most $(t+\epsilon)^{1-q}/(1-q)$, which gives the bound. Suppose now that $p \ge 0$. If $t \ge \epsilon$, we bound the integrand by $r^{-p}(t-r)^{-q}$ and substitute $r = tv$. The integral of $v^{-p}(1-v)^{-q}$ over $[0,1]$ is finite, as we see by bounding $(1-v)^{-q}$ by 2^q on $[0,1/2]$ and v^{-p} by 2^p on $[1/2,1]$. Since $t \le t+\epsilon \le 2t$, we obtain the bound, and the same argument proves the case $\epsilon = 0$. If $t < \epsilon$, the integral is at most $t\epsilon^{-p-q} \le \epsilon^{1-p-q}$, which is at most a constant times $(t+\epsilon)^{1-p-q}$ because $\epsilon \le t+\epsilon \le 2\epsilon$. These arguments apply for either sign of $1-p-q$. □

We use the ordered expansion of $\mathcal{N}_a$ fixed in Lemma 24.2: a fixed finite sum of brackets of translated components, with one physical difference in the second argument of each bracket. Differences of brackets obey the exact product rules

$$D^+_{a,k}[u,v] = [D^+_{a,k}u, T_k v] + [u, D^+_{a,k}v], \qquad D^-_{a,k}[u,v] = [D^-_{a,k}u, T_k^{-1}v] + [u, D^-_{a,k}v]. \tag{25.2}$$

Repeated application distributes a fixed number of differences among the factors and produces finitely many terms with bounded translations. Whenever we take differences of a discrete nonlinearity, we use these identities and never differentiate an interpolated product.

25.2 Consistency for fixed smooth interpolants

The first consistency statement concerns the flow data of a fixed finite family of smooth cardinal interpolants. Such families arise in Section 28 from finite-scale reconstructions, the fields reconstructed from finitely many RG steps. The point is that the constants do not depend on the finer mesh on which the flow is computed.

Lemma 25.2. *Fix $L, T, H > 0$ with $L/H \in \mathbb{N}$, and $M_H < \infty$. Let $f = I_H v$ and $g = I_H w$ be the smooth cardinal interpolants, as in Lemma 16.3, of link fields v, w on the lattice of mesh H with $\|v\|_\infty + \|w\|_\infty \le M_H$. For every mesh $a = H/m$ with $m \in \mathbb{N}$, set*

$$x_a = P_\cdot^a(S_a f), \qquad z_a = P_\cdot^a(S_a g), \qquad x = P_\cdot f, \qquad z = P_\cdot g.$$

Let $\mathcal{N}_0$ be the continuum bilinear map obtained from the fixed expansion of $\mathcal{N}_a$ by replacing every translation by the identity and every physical difference by the corresponding partial derivative. Then

$$(\mathcal{N}_0(b,b))_i = \sum_j \{2[b_j, \partial_j b_i] - [b_j, \partial_i b_j]\}.$$

Uniformly over this family of inputs and over $0 \le t \le T$, the values and all first differences of $\mathcal{N}_a(x_a, z_a)$ differ from the samples of $\mathcal{N}_0(x, z)$ and of its corresponding first derivatives by at most $C_{H,M_H,L,T}\, a$, and the same holds for $\mathcal{N}_a(z_a, x_a)$ and $\mathcal{N}_0(z, x)$. The same estimate holds for

$$\mathcal{Q}_a(x_a, z_a)(t) - S_a \int_0^t P_{t-s} \mathcal{N}_0(x(s), z(s))\, ds$$

and its first differences, and for the reversed order of the arguments. Consequently, for every bounded finite sum f of such interpolants, the flow data of $S_a f$ converge, after separate interpolation of value and difference arrays, to the continuum pair $(x, \int_0^t P_{t-s}\mathcal{N}_0(x(s), x(s))\, ds)$ in the four weighted uniform norms of Definition 24.3, whose weights are powers of t, and $\|(x_a, y_a)\|_{\mathrm{fl},a}$ is bounded uniformly in a. The constants are deterministic, do not depend on a, and may grow arbitrarily as $H \downarrow 0$. The lemma compares two discretizations of the same smooth input on the mesh a; it does not compare the quadratic term $\mathcal{Q}_H$ on the mesh H with $\mathcal{Q}_a$.

The diagonal formula for $\mathcal{N}_0$ is the quadratic part of the DeTurck form of the Yang–Mills heat flow, written in Theorem 24.10; compare the gauge-fixed flow of Lüscher [128, Section 2.2]. We verify it from the lattice formula, since its sign and the factor two must match the lattice map exactly.

Proof. Step 1: bounds on the inputs. For each integer $q \ge 0$, by the bounds for cardinal interpolants in Lemma 16.3, we have

$$\|f\|_{C^q} + \|g\|_{C^q} \le C_q(1 + H^{-q}) M_H.$$

By heat contraction on differentiated smooth data, and its discrete counterpart, every fixed spatial difference of the input fields is bounded uniformly in a and $t \le T$. Applying Lemma 16.9 through order two, and not only through order one, we obtain

$$\max_{0 \le q \le 2} \sup_{t \le T} \big(\|D_a^q x_a - S_a \partial^q x\|_\infty + \|D_a^q z_a - S_a \partial^q z\|_\infty \big) \le C_{H,M_H,L,T}\, a. \tag{25.3}$$

The maximum includes every ordered choice of differences; for $q = 0$ the stronger error $O(a^2)$ is available. A translation by any vector of the fixed stencil changes a sampled smooth qth derivative by at most $Ca\|\partial^{q+1}x\|_\infty$, and the same holds for z.

Step 2: the diagonal of $\mathcal{N}_0$. Recall that $\mathcal{N}_0$ is defined by the termwise replacement in the statement. To identify its diagonal, let $(Cb)_{ij} = \partial_i b_j - \partial_j b_i$ and $Q(b)_{ij} = [b_i, b_j]$, extended antisymmetrically in i, j. At zero translation the logarithms of the four factors of the plaquette in Lemma 24.2 are $b_i, b_j, -b_i, -b_j$, and half the sum of their pairwise brackets is $Q(b)_{ij}$. By the adjoint formula in Lemma 24.2, we have

$$(C^*Q(b))_i = \sum_j \partial_j [b_i, b_j], \qquad (DQ(b)^*Cb)_i = \sum_j [b_j, \partial_i b_j - \partial_j b_i].$$

For the second identity, note that a variation h_i of b_i changes $[b_i, b_j]$ by $[h_i, b_j]$, and invariance of the inner product moves this bracket to $[b_j, (Cb)_{ij}]$. Combining the two orientations of each plaquette, we obtain the displayed formula. Further, since $d_0^*(ab)/a^2$ tends to $\sum_j \partial_j b_j$, the quadratic gauge term G_2 of Lemma 24.2, divided by a^3, tends to $[b_i, \sum_j \partial_j b_j]$. The diagonal coefficient is therefore

$$-C^*Q(b) - DQ(b)^*Cb + [b, \operatorname{div} b] = \Big(\sum_j \big\{ 2[b_j, \partial_j b_i] - [b_j, \partial_i b_j] \big\} \Big)_i,$$

where we used $[\partial_j b_i, b_j] = -[b_j, \partial_j b_i]$, and the terms $[b_i, \partial_j b_j]$ cancel. This verifies the sign and the factor two without symmetrizing the ordered map.

Step 3: the nonlinearity. Every summand of $\mathcal{N}_a(x_a, z_a)$ is a bracket of a factor of order zero and a factor of order one. Applying (25.2) once, we obtain factors of orders $(1,1)$ or $(0,2)$, with the corresponding translations. By (25.3) and Step 1, each factor differs from its continuum counterpart by $O(a)$, and every other factor is bounded by a constant depending on the family. Using the inequality $|[u, v]| \le |u||v|$ and subtracting the two products, we obtain the asserted $O(a)$ errors for the values and the first differences. Reversing the inputs replaces the factors by z_a, x_a in the same list; their bounds through order two and their errors are again those in (25.3), which gives the estimate for the reversed order.

Step 4: the Duhamel integrals. Let $h(t) = \mathcal{N}_0(x(t), z(t))$, a uniformly bounded smooth family at every fixed spatial order. In the value of the Duhamel integral we first replace $\mathcal{N}_a(x_a, z_a)$ by $S_a h$; by contraction this costs at most CTa. Then we replace $P^a_{t-r} S_a h(r)$ by $S_a P_{t-r} h(r)$. By the general smooth-input bound of Lemma 16.9, applied separately at each r, this costs at most $CT^2a^2 \sup_r \|h(r)\|_{C^4}$. For a first difference of the integral, we commute the difference through P^a_{t-r} onto the source. The replacement of the source difference by $S_a \partial h$ costs CTa, and the heat comparison on $\partial h(r)$ together with the comparison of $D_a S_a$ with $S_a \partial$ cost at most $C_{H,M_H,L,T}\, a$ after integration. This order of comparisons avoids dividing a value error by a. Replacing h by $\mathcal{N}_0(z, x)$ gives the assertion for the reversed order.

Step 5: weighted norms and convergence. We first bound the flow data near time zero, without using any convergence. Let $B = \|f\|_\infty + \|g\|_\infty$. By Lemma 25.1 and the bound $\|\mathcal{N}_a(u,v)\|_\infty \le C\|u\|_\infty \|D_a v\|_\infty$, we have

$$\|x_a(t)\|_\infty + \|z_a(t)\|_\infty \le B, \qquad \|D_a x_a(t)\|_\infty + \|D_a z_a(t)\|_\infty \le CB(t+a^2)^{-1/2},$$
$$\|\mathcal{Q}_a(x_a,z_a)(t)\|_\infty \le CB^2\sqrt{t}, \qquad \|D_a \mathcal{Q}_a(x_a,z_a)(t)\|_\infty \le CB^2.$$

In the last inequality we used $\int_0^t (t-r+a^2)^{-1/2}(r+a^2)^{-1/2}\,dr \le \pi$. For a fixed family one can instead bound all input and source differences by its smooth bounds, which also covers $t = 0$. Since $a \le H$ and all four weights $v_a^{\alpha/2}$, $v_a^{(\alpha+1)/2}$, v_a^{β}, v_a^{α} have positive exponents, multiplying by them we obtain a bound on $\|(x_a, y_a)\|_{\mathrm{fl},a}$ depending only on the data of the family. Interpolation of each array separately is a contraction, since the cardinal weights are nonnegative and sum to one. Because of its support radius, it has error at most $Ca\|\partial F\|_\infty$ for a smooth F. Then the uniform errors already proved imply convergence in all four weighted uniform norms. For the quadratic Picard term we take $g = f$, and for a finite sum of interpolants we use the corresponding bounded smooth family. All constants have the asserted dependence, and $\mathcal{Q}_H$ is never compared with $\mathcal{Q}_a$. □

25.3 Consistency and stability for smooth initial data

We now expand the exact lattice vector field on sampled smooth potentials. This vector field consists of the Riemannian gradient of the plaquette action for the bi-invariant product metric and the damping term, written in logarithmic coordinates by means of the inverse differential of the exponential. All three parts enter the term of order a^3.

Lemma 25.3. *Let $\mathcal{F}$ be the dimensionless damped logarithmic vector field of Lemma 24.2, so that the damped Wilson equation reads*

$$\partial_s \theta = \mathcal{F}(\theta) = -H_0\theta + N_2(\theta) + R_3(\theta)$$

in dimensionless time $s = t/a^2$. Let $K_0 > 0$, and let b be a smooth periodic connection with $\|b\|_{C^3} \le K_0$. If a is so small that $K_0 a$ lies in a fixed smaller chart, then

$$a^{-3}\mathcal{F}(aS_a b) = S_a\{\Delta b + \mathcal{N}_0(b,b) + \mathcal{N}_3(b)\} + O_{K_0}(a), \qquad (\mathcal{N}_3(b))_i = \sum_j [b_j,[b_j,b_i]],$$

and

$$a^{-3}R_3(aS_a b) = S_a \mathcal{N}_3(b) + O(a).$$

More generally, for every integer $q \ge 0$ the errors in both identities remain $O(a)$ after any q physical differences, compared with the samples of the corresponding derivatives, uniformly over families of b bounded in C^{q+4}. The constants depend only on q, on the smooth bounds, and on the chart.

Proof. Step 1: the exact vector field. For $i \neq j$, let

$$P_{ij}(x) = U_i(x)U_j(x+ae_i)U_i(x+ae_j)^{-1}U_j(x)^{-1}, \qquad p(V)^c = -2\,\mathrm{tr}(\tau_c V),$$

and identify the color vector $p(V)$ with its Lie-algebra element. For $V = e^f$ in the plaquette chart, $p(e^f) = 2\sin(|f|/2)f/|f| = f + O(|f|^3)$. When $j < i$ we have $P_{ij}(x) = P_{ji}(x)^{-1}$ and $q(V^{-1}) = q(V)$, so each unoriented plaquette is counted once. Varying $U_i(x)$ by $e^{\varepsilon w}U_i(x)$ in $4\sum_p q(U_p)$ gives $\langle w, p(P_{ij}(x))\rangle$ from the plaquette $P_{ij}(x)$ and

$$-\big\langle w, \mathrm{Ad}(U_j(x-ae_j)^{-1})\, p(P_{ij}(x-ae_j))\big\rangle$$

from the plaquette $P_{ij}(x-ae_j)$; the second formula follows by moving the varied inverse factor cyclically inside the trace. The negative gradient for the bi-invariant metric, in left trivialization, is therefore

$$V_{\mathrm{W},i}(x) = \sum_{j\neq i}\big\{\mathrm{Ad}(U_j(x-ae_j)^{-1})\, p(P_{ij}(x-ae_j)) - p(P_{ij}(x))\big\}. \tag{25.4}$$

This is an identity at every configuration. By Lemma 24.2, with $U_i(x) = e^{\theta_i(x)}$, the damping parameter, the damping velocity, and the logarithmic vector field are

$$\begin{aligned} \phi(x) &= \sum_j \big(\theta_j(x-ae_j) - \theta_j(x)\big), \\ V_{\mathrm{g},i} = \phi(x) - \mathrm{Ad}(e^{\theta_i(x)})\phi(x+ae_i), &\qquad \mathcal{F}_i(\theta) = \mathcal{J}(\theta_i)^{-1}\big(V_{\mathrm{W},i} + V_{\mathrm{g},i}\big), \end{aligned} \tag{25.5}$$

where $\mathcal{J}(\theta) = (e^{\mathrm{ad}\,\theta} - I)/\,\mathrm{ad}\,\theta$ is the differential of the exponential in left trivialization.

Step 2: expansion of one plaquette. Let $\theta = aS_a b$, that is, $U_i(x) = e^{ab_i(x)}$, and write

$$F_{ij} = \partial_i b_j - \partial_j b_i + [b_i, b_j], \qquad v = \sum_j \partial_j b_j.$$

By Taylor expansion at x through order a^3, the logarithms of the four factors of $P_{ij}(x)$ are

$$\begin{aligned} ab_i, \quad & ab_j + a^2\partial_i b_j + \tfrac{1}{2}a^3\partial_i^2 b_j + O_{K_0}(a^4), \\ & -ab_i - a^2\partial_j b_i - \tfrac{1}{2}a^3\partial_j^2 b_i + O_{K_0}(a^4), \quad -ab_j. \end{aligned}$$

Taking the finite analytic product and the logarithm, we obtain

$$p(P_{ij}(x)) = a^2 F_{ij}(x) + a^3 B_{ij}(x) + O_{K_0}(a^4), \qquad \|B_{ij}\|_{C^1} \leq C_{K_0}. \tag{25.6}$$

Here B_{ij} is the homogeneous coefficient of a^3 in the product of the four factors, with their logarithms replaced by the displayed jets. It is a fixed polynomial in the two-jets of b, and its first derivative involves only three-jets. Taylor's integral remainder for each translated b and the analytic remainder of fourth order of the finite product are bounded by $C_{K_0}a^4$.

The coefficient of a^2 is F_{ij}: these logarithms add up to $a^2(\partial_i b_j - \partial_j b_i)$ modulo $O(a^3)$, and half the sum of their pairwise brackets is $a^2[b_i, b_j]$ modulo $O(a^3)$, as in Step 2 of the proof of Lemma 25.2. Replacing the logarithm by p changes the result only by $O_{K_0}(a^6)$. This proves (25.6) without computing the polynomial B_{ij}.

Step 3: cancellation between the two plaquettes. At the neighboring basepoint, $F_{ij}(x - ae_j) = F_{ij}(x) - a\partial_j F_{ij}(x) + O_{K_0}(a^2)$ and $B_{ij}(x - ae_j) = B_{ij}(x) + O_{K_0}(a)$, and the adjoint in (25.4) is $I - a\,\mathrm{ad}(b_j(x)) + O_{K_0}(a^2)$. Subtracting the contribution of $P_{ij}(x)$, we obtain

$$V_{\mathrm{W},i}(aS_a b) = -a^3 \sum_j \big(\partial_j F_{ij} + [b_j, F_{ij}]\big) + O_{K_0}(a^4).$$

The terms of order a^2 and the common terms $a^3 B_{ij}$ cancel before the inverse differential $\mathcal{J}^{-1}$ is applied. For the damping parameter we have

$$\phi = -a^2 v + a^3 w + O_{K_0}(a^4), \qquad w = \tfrac{1}{2}\sum_j \partial_j^2 b_j, \qquad \|w\|_{C^1} \le C_{K_0}.$$

The translated terms $a^3 w$ cancel in V_{g}, and expanding the adjoint gives $V_{\mathrm{g},i} = a^3(\partial_i v + [b_i, v]) + O_{K_0}(a^4)$. Then the total group velocity is $O_{K_0}(a^3)$. Since $\mathcal{J}(ab_i)^{-1} = I + O_{K_0}(a)$, applying this inverse changes it only by $O_{K_0}(a^4)$. Finally,

$$-\sum_j \big(\partial_j F_{ij} + [b_j, F_{ij}]\big) + \partial_i v + [b_i, v] = \Delta b_i + \sum_j \big\{2[b_j, \partial_j b_i] - [b_j, \partial_i b_j] + [b_j, [b_j, b_i]]\big\},$$

and by Lemma 25.2 the right side is $(\Delta b + \mathcal{N}_0(b,b) + \mathscr{N}_3(b))_i$. This proves the first identity. The only condition on a is that $K_0 a$ lie in a fixed smaller chart, and all spatial remainders used at most three derivatives of b.

Step 4: the analytic remainder. By Lemma 24.2,

$$a^{-3}\mathcal{F}(aS_a b) = \Delta_a S_a b + \mathcal{N}_a(S_a b, S_a b) + a^{-3} R_3(aS_a b).$$

Central-difference consistency gives $\Delta_a S_a b = S_a \Delta b + O(a)$, and the calculation of Step 3 of the proof of Lemma 25.2, applied to the smooth field b, implies that $\mathcal{N}_a(S_a b, S_a b) = S_a \mathcal{N}_0(b,b) + O(a)$. Subtracting these from the first identity, we obtain the second.

Step 5: differences. For a first difference, we regard the finite sampled formula for $a^{-3}\mathcal{F}(aS_a b)$ as a smooth function $B_a(u)$ of a continuous basepoint $u \in \mathbb{T}^3_L$, evaluating every stencil entry at $u + a\xi$ instead of $x + a\xi$ for the corresponding stencil vector ξ. Differentiating the Taylor calculation of Steps 2 and 3 in u, we find that $\partial_k B_a$ differs from $\partial_k\{\Delta b + \mathcal{N}_0(b,b) + \mathscr{N}_3(b)\}$ by $O(a)$, since this calculation uses one additional bounded spatial derivative of b. Using the exact identity

$$D^+_{a,k} B_a(x) = \int_0^1 \partial_k B_a(x + rae_k)\, dr$$

we compare the difference with the sampled continuum derivative, at error $O(a)$; for a backward difference we integrate over $x - rae_k$ instead. Repeating this argument at a fixed order q, we obtain the assertion through order q on families bounded in C^{q+4}. The same argument applies to $a^{-3}R_3(aS_ab)$. There is no negative power of a in the remainder: its term of degree d in the analytic series has coefficient a^{d-3}, $d \geq 3$, by Lemma 24.2. On a smaller chart, the derivatives of this series converge absolutely after multiplication by polynomial factors in d of fixed order. Hence the product-rule argument applies to the whole analytic remainder, not only to its cubic truncation. □

Theorem 25.4. *Fix $L, T, K_0, C_{\mathrm{init}} > 0$. Let b be a smooth periodic solution on $[0, T]$ of the DeTurck equation of Theorem 24.10, continuously differentiable in time, with $\sup_{t \leq T} \|b(t)\|_{C^3} \leq K_0$. For $a = L/n$ write $s = t/a^2$, $S = T/a^2$, and $\theta^{\mathrm{ref}}(s, x) = ab(a^2 s, x)$. There is $a_* > 0$ such that the following statements hold for all $a \leq a_*$.*

1. *The residual of θ^{ref} for the exact vector field $\mathcal{F}$ of Lemma 24.2 satisfies*

$$\sup_{s \leq S} \|\partial_s \theta^{\mathrm{ref}}(s) - \mathcal{F}(\theta^{\mathrm{ref}}(s))\|_\infty \leq C_{K_0} a^4.$$

   Here $\mathcal{F}$ contains the Riemannian gradient of the plaquette action, the damping term, and the inverse differential of the exponential.

2. *Let $\vartheta(s)$ be a path on $[0, S]$, not necessarily a solution, with $\|\vartheta\|_\infty \leq K_0 a$ and $\|\nabla\vartheta\|_\infty \leq K_0 a^2$. Its linearized lower-order operator $\mathsf{A}_\vartheta = D\mathcal{F}(\vartheta) + H_0$ satisfies*

$$\|\mathsf{A}_\vartheta z\|_\infty \leq C_{K_0}\big(a\|\nabla z\|_\infty + a^2\|z\|_\infty\big).$$

   Define

$$\|z\|_{\mathsf{R}_{\mathrm{sm}}} = \sup_{s \leq S} \|z(s)\|_\infty + \sup_{s \leq S} \sqrt{1+s}\, \|\nabla z(s)\|_\infty$$

   and

$$\mathsf{L}_\vartheta z(s) = \int_0^s e^{-(s-r)H_0} \mathsf{A}_\vartheta(r) z(r)\, dr.$$

   Then $\|(I - \mathsf{L}_\vartheta)^{-1}\|_{\mathsf{R}_{\mathrm{sm}} \to \mathsf{R}_{\mathrm{sm}}} \leq C_{L,T,K_0}$, uniformly in a. A forcing F enters through its whole Volterra integral, and if $\sup_s \|F(s)\|_\infty \leq c$, then

$$\Big\| \int_0^{\cdot} e^{-(\cdot - r)H_0} F(r)\, dr \Big\|_{\mathsf{R}_{\mathrm{sm}}} \leq C_{L,T}\, c\, (1 + S).$$

3. *If $\|\theta(0) - \theta^{\mathrm{ref}}(0)\|_\infty \leq C_{\mathrm{init}} a^2$, then the damped lattice solution θ with this initial value exists in the chart on $[0, S]$, is a damped representative of the Wilson flow, and satisfies*

$$\|\theta - \theta^{\mathrm{ref}}\|_{\mathsf{R}_{\mathrm{sm}}} \leq C_{L,T,K_0,C_{\mathrm{init}}}\, a^2.$$

4. *Let b_0 be smooth, and let the partition $0 = t_0 < \cdots < t_Q = T$, the DeTurck solutions b_i on $[0, t_{i+1} - t_i]$, and the fixed smooth gauge transformations k_i be those given by Theorem 24.10 for the initial value b_0. Start the Wilson flow from $\exp(aS_a b_0)$, and at each time t_{i+1} apply the gauge transformation k_i, sampled at the lattice sites, to the links before restarting the damped equation. Then at each time $t \in [t_i, t_{i+1}]$ the link logarithms differ from $aS_a b_i(t - t_i)$ by $O(a^2)$, and hence the links differ from $\exp(aS_a b_i(t - t_i))$ by $O(a^2)$.*

The constants and a_* depend only on $L, T, K_0, C_{\mathrm{init}}$, and in (4) also on the smooth bounds and margins of the continuation. The hypothesis of (3) is an initial error of order a^2; an initial link error that is only $o(a)$ is not covered.

The proof follows the usual pattern for semilinear parabolic equations: a consistency estimate for the residual, a bound on the linearized Volterra operator in a weighted norm, and a contraction argument for the error. The only delicate point is the scaling. In dimensionless units the coefficient of the first difference in A_ϑ is $O(a)$, the time interval has length T/a^2, and the weight $e^{-\lambda s}$ must be taken with λ of order a^2.

Proof. Step 1: the residual. By Lemma 25.3,

$$\mathcal{F}(\theta^{\mathrm{ref}}(s)) = a^3 S_a \big\{ \Delta b + \mathcal{N}_0(b, b) + \mathscr{N}_3(b) \big\}(a^2 s) + O_{K_0}(a^4).$$

By the DeTurck equation the bracket is $\partial_t b$, and $\partial_s \theta^{\mathrm{ref}} = a^3 S_a \partial_t b$. This proves (1). The threshold only requires that $K_0 a$ lie in a fixed smaller chart.

Step 2: the linearized operator. Let ϑ satisfy the two bounds in (2). We have $\mathsf{A}_\vartheta = DN_2(\vartheta) + DR_3(\vartheta)$. By Lemma 24.2, the quadratic part N_2 is a fixed sum of translated products of the form $[u, \nabla v]$. In $DN_2(\vartheta)z$, the terms with the difference on z are bounded by $CK_0 a \|\nabla z\|_\infty$, and the terms with the difference on ϑ by $CK_0 a^2 \|z\|_\infty$. The derivative of the remainder satisfies $\|DR_3(\vartheta)z\|_\infty \le C\|\vartheta\|_\infty^2 \|z\|_\infty$. This proves the bound for A_ϑ.

Step 3: the Volterra inverse. Let $k(u) = (1 + u)^{-1/2}$, and for $\lambda > 0$ consider the equivalent weighted norm

$$\|z\|_\lambda = \sup_{s \le S} e^{-\lambda s} \|z(s)\|_\infty + \sup_{s \le S} e^{-\lambda s} \sqrt{1 + s}\, \|\nabla z(s)\|_\infty.$$

For every $\lambda > 0$,

$$\int_0^\infty e^{-\lambda u} k(u)\, du \le C\lambda^{-1/2}, \qquad \sup_s \int_0^s e^{-\lambda(s-r)} k(r)\, dr \le C\lambda^{-1/2}. \tag{25.7}$$

The first bound follows from $k(u) \le u^{-1/2}$ and rescaling. For the second, if $s \le \lambda^{-1}$ we discard the exponential and integrate $r^{-1/2}$. If $s > \lambda^{-1}$, the intervals $r \le s/2$ and $r \ge s/2$ contribute at most $C\sqrt{s} e^{-\lambda s/2}$ and $C/(\lambda\sqrt{s})$ respectively, and each of these is at most $C\lambda^{-1/2}$. We also use the elementary inequality

$$\sqrt{1 + s}\, k(s - r) k(r) \le k(s - r) + k(r), \qquad 0 \le r \le s,$$

which follows from $\sqrt{1+s} \le \sqrt{1+r} + \sqrt{1+s-r}$. We insert the bound for A_ϑ into the Volterra integral. For the value we use that $e^{-(s-r)H_0}$ is a contraction, and for the first difference the bound $\|\nabla e^{-(s-r)H_0}\|_{\infty\to\infty} \le Ck(s-r)$ of Lemma 25.1 with $H_1 = T$. Using (25.7) and the elementary inequality for the term with $a\|\nabla z\|_\infty$ in the differentiated estimate, we obtain

$$\|\mathsf{L}_\vartheta\|_{\lambda\to\lambda} \le C_{L,T,K_0}\Big\{\frac{a}{\sqrt{\lambda}} + \frac{a^2}{\lambda} + \frac{a^2\sqrt{1+S}}{\sqrt{\lambda}}\Big\}. \tag{25.8}$$

The last term comes from the coefficient of order zero in the differentiated estimate. We take $\lambda = \kappa_0 a^2$ with $\kappa_0 = \kappa_0(L,T,K_0)$ so large that the right side is at most $1/2$; here we used $a\sqrt{1+S} = \sqrt{a^2+T} \le \sqrt{1+T}$ for $a \le 1$. The series $\sum_{j\ge0} \mathsf{L}_\vartheta^j$ then converges, with weighted norm at most two, and removing the weight costs at most $e^{\lambda S} = e^{\kappa_0 T}$. This proves the bound on $(I - \mathsf{L}_\vartheta)^{-1}$ in R_{sm}.

Step 4: the forcing. An initial heat term $e^{-sH_0}z_0$ has R_{sm} norm at most $C_{L,T}\|z_0\|_\infty$. A forcing of uniform norm c has a value integral at most cs, and its weighted first difference is at most

$$Cc\sqrt{1+s}\int_0^s (1+s-r)^{-1/2}\,dr \le 2Cc(1+s).$$

This proves the forcing bound in (2), including the constant Fourier mode, on which the heat semigroup does not decay.

Step 5: the error. Let θ be the damped solution with the initial value in (3), set $z = \theta - \theta^{\mathrm{ref}}$, and let $\mathcal{R} = \partial_s\theta^{\mathrm{ref}} - \mathcal{F}(\theta^{\mathrm{ref}})$ be the residual of (1). Since N_2 is quadratic, the difference satisfies the exact equation

$$z' = -H_0 z + \mathsf{A}_{\theta^{\mathrm{ref}}} z + N_2(z) + Q_{\theta^{\mathrm{ref}}}(z) - \mathcal{R}, \qquad Q_\vartheta(z) = R_3(\vartheta+z) - R_3(\vartheta) - DR_3(\vartheta)z. \tag{25.9}$$

By analytic Taylor expansion on a smaller chart, we have

$$\|Q_\vartheta(z)\|_\infty \le C\big(K_0 a\|z\|_\infty^2 + \|z\|_\infty^3\big),$$

and the difference of Q_ϑ at two points of a ball of radius δ has the corresponding factor $K_0 a\delta + \delta^2$. If $\|z\|_{\mathsf{R}_{\mathrm{sm}}} \le \delta$, the quadratic source $N_2(z)$ has size at most $C\delta^2 k(s)$. Its Volterra integral has R_{sm} norm at most $C\delta^2\sqrt{1+S}$; for the differentiated component we use $\int_0^s k(s-r)k(r)\,dr \le \pi$ and multiply by $\sqrt{1+s}$. By Step 4, the integral of $Q_{\theta^{\mathrm{ref}}}(z)$ costs at most $C(K_0 a\delta^2 + \delta^3)(1+S)$. After the Volterra inverse, the Lipschitz constant of these nonlinear terms on the ball of radius δ is at most

$$C_{L,T,K_0}\big\{\delta\sqrt{1+S} + K_0 a\delta(1+S) + \delta^2(1+S)\big\}. \tag{25.10}$$

The initial heat term and the residual in (25.9) have R_{sm} norm at most $C_{L,T,K_0,C_{\mathrm{init}}}(a^2 + a^4(1+S)) \le Ca^2$, by Step 4 and (1). Let D be twice this constant multiplied by the bound

for the Volterra inverse, and set $\delta = Da^2$. Every term in (25.10) then tends to zero: the first two are $O_{L,T,K_0,D}(a)$ and the last is $O_{T,D}(a^2)$. For a small enough, the mild form of (25.9) defines a map of the ball of radius δ into itself which is a contraction with factor at most $1/2$. Iteration from zero converges in the complete space defined by the norm R_{sm}; this proves existence of the mild solution and the error bound in (3). The mild solution solves the finite-dimensional differential equation. Since its amplitude is at most $K_0 a + Da^2 < r_*$, it does not leave the chart. Local uniqueness for the differential equation and the gauge identification of Lemma 24.2 show that it is the damped representative of the Wilson flow.

Step 6: continuation by restarts. We use the continuation of Theorem 24.10 with $r = 3$ and its fixed margins. On one interval, Step 5 gives an error $O(a^2)$ for the link logarithms, and thus $O(a^2)$ for the links. At a restart time we apply the fixed sampled gauge transformation to the links. Left and right multiplication are isometries, and the logarithm is Lipschitz on the next smaller chart. By Lemma 24.7, applying the gauge transformation to the exponentials of a sample differs from sampling the transformed smooth connection by $O(a^2)$ per link. Then the initial error on the next interval is again of the form $C_{\mathrm{init},i}a^2$, which is the hypothesis of (3). Iterating over the finite partition preserves this order, with the maximum of finitely many constants and the minimum of finitely many restrictions on the mesh. This proves (4). □

25.4 Continuation near smooth solutions

We now pass to physical units and to the norm C^1_a, and we write $S_a b = b|_a$. In physical time, the damped equation of Lemma 24.2 reads

$$\partial_t B = \Delta_a B + N_a(B), \qquad N_a(B) = \mathcal{N}_a(B,B) + \mathscr{C}_a(B), \qquad \mathscr{C}_a(B) = a^{-3} R_3(aB).$$

We write

$$\mathscr{B}_a(B)V = \mathcal{N}_a(B,V) + \mathcal{N}_a(V,B) + D\mathscr{C}_a(B)V$$

for the linearization of N_a at B. Here D on the cubic-and-higher remainder $\mathscr{C}_a$ differentiates in the field, and D_a is a physical spatial difference. On the smaller chart,

$$\|D^j \mathscr{C}_a(B)[V_1,\dots,V_j]\|_\infty \le C\|B\|_\infty^{3-j} \prod_{i=1}^{j} \|V_i\|_\infty, \qquad 0 \le j \le 3, \tag{25.11}$$

where for $j = 0$ the empty product is one. Indeed, the derivative is $a^{j-3} D^j R_3(aB)$, and since the Taylor coefficients of R_3 vanish through order two, $D^j R_3(aB)$ has a factor $(a\|B\|_\infty)^{3-j}$; for $j = 3$ we use the bounded analytic third derivative on the smaller chart. All translated components are included. We use these bounds to show that, on a bounded set of the norm C^1_a, say $\|B\|_{C^1_a}, \|B'\|_{C^1_a} \le K$ in the chart,

$$\|N_a(B) - N_a(B')\|_\infty \le C_K \|B - B'\|_{C^1_a}. \tag{25.12}$$

For the quadratic term, we write the difference as $\mathcal{N}_a(B, B - B') + \mathcal{N}_a(B - B', B')$ and use $\|\mathcal{N}_a(u, v)\|_\infty \le C\|u\|_\infty \|D_a v\|_\infty$ from Lemma 24.2. For the remainder we use the Lipschitz bound for R_3 in that lemma, rescaled to physical units. No interpolation enters (25.12).

Lemma 25.5. *The following statements hold.*

1. *Fix $L, h, K > 0$. Let $B_a(t)$ be a damped lattice solution on $[0, h]$ in the smaller chart with $\sup_{t \le h} \|B_a(t)\|_{C_a^1} \le K$. Then the solution of*

$$V(t) = P_t^a V_0 + \int_0^t P_{t-r}^a \big\{ \mathscr{B}_a(B_a(r)) V(r) + f(r) \big\}\, dr$$

satisfies

$$\sup_{t \le h} \|V(t)\|_{C_a^1} \le C_{L,h,K} \big(\|V_0\|_{C_a^1} + \sup_{r \le h} \|f(r)\|_\infty \big). \tag{25.13}$$

The same bound holds if the right side is replaced by $C_{L,h,K}$ times the norm in $C([0, h]; C_a^1)$ of the whole inhomogeneous term $P_t^a V_0 + \int_0^t P_{t-r}^a f(r)\, dr$. The constant is uniform in small meshes, and no derivative of f is needed.

2. *Let b be a fixed smooth DeTurck solution on $[0, h]$ whose spatial derivatives through order five are bounded by K_b. There are $\eta_b, a_b > 0$ and $C_b < \infty$, depending only on L, h, K_b, and the chart, such that every lattice initial field with $\|B_a(0) - S_a b(0)\|_{C_a^1} \le \eta \le \eta_b$, on a mesh $a \le a_b$, has a damped solution through time h with*

$$\sup_{t \le h} \|B_a(t) - S_a b(t)\|_{C_a^1} \le C_b(\eta + a), \qquad \sup_{t \le h} \|B_a(t)\|_{C_a^1} \le C_b. \tag{25.14}$$

The same holds on each time interval of a fixed continuation as in Theorem 24.10, when the flow is restarted with its fixed sampled gauge transformations. In particular, this comparison supplies the C_a^1 bound that is the hypothesis of the smoothing on later time intervals in Theorem 25.7.

3. *If two initial fields satisfy the hypothesis of (2) with respect to the same solution b, and their solutions are restarted with the same fixed sampled gauge transformations, then*

$$\max_i \sup_{t \in I_i} \|B_a^{(i)}(t) - B_a'^{(i)}(t)\|_{C_a^1} \le C_b \|B_a(0) - B_a'(0)\|_{C_a^1}. \tag{25.15}$$

Here I_i are the finitely many time intervals of the continuation (or the single interval $[0, h]$), $B_a^{(i)}$ and $B_a'^{(i)}$ are the representatives on I_i, and the constant may also depend on the continuation.

4. *Fix $L, h > 0$ and a set $\mathscr{K} \subset C^\infty(\mathbb{T}_L^3; \mathfrak{su}(2)^3)$ that is compact in the C^3 topology. There are $\eta_{\mathscr{K}}, a_{\mathscr{K}} > 0$, $C_{\mathscr{K}} < \infty$, and finitely many continuations as in Theorem 24.10 with the following property. For every $b_0 \in \mathscr{K}$, every $a \le a_{\mathscr{K}}$, and any two lattice initial fields with*

$$\max\big(\|B_a(0) - S_a b_0\|_{C_a^1}, \|B_a'(0) - S_a b_0\|_{C_a^1} \big) \le \eta_{\mathscr{K}},$$

one of these continuations gives the bound $\sup \|B_a(t)\|_{C^1_a} \le C_{\mathscr{K}}$ *through time* h *for both solutions, and* (25.15) *with constant* $C_{\mathscr{K}}$*. The constants depend on* $L, h, \mathscr{K}$*, on the bounds of the finitely many selected smooth DeTurck solutions, and on the margins of the chart and of the continuations. Higher derivatives are only bounded for these finitely many selected solutions, not on a* C^3 *neighborhood of* $\mathscr{K}$*.*

In (2)–(4), every representative is gauge equivalent to the Wilson flow from its initial links.

Proof. Part (1). By the bound on $\mathcal{N}_a$ and (25.11), we have $\|\mathscr{B}_a(B)V\|_\infty \le C_K\|V\|_{C^1_a}$. We use Lemma 25.1 with physical horizon h. The heat semigroup is a contraction for C^1_a, since it commutes with differences, and for $t > 0$

$$\|P^a_t f\|_{C^1_a} \le C_{L,h}\big(1 + (t + a^2)^{-1/2}\big)\|f\|_\infty \le C_{L,h}(1 + t^{-1/2})\|f\|_\infty.$$

In the weighted norm $\|V\|_\lambda = \sup_{t\le h} e^{-\lambda t}\|V(t)\|_{C^1_a}$, the integral operator $V \mapsto \int_0^t P^a_{t-r}\mathscr{B}_a(B_a(r))V(r)\,dr$ has norm at most

$$C_{L,h,K}\int_0^\infty e^{-\lambda v}(1 + v^{-1/2})\,dv = C_{L,h,K}\big(\lambda^{-1} + \sqrt{\pi}\lambda^{-1/2}\big).$$

We choose λ, independently of the mesh, so that this is at most $1/2$. The Neumann series then gives an inverse of weighted norm at most two, and removing the weight costs $e^{\lambda h}$. The initial heat term has norm at most $\|V_0\|_{C^1_a}$, and the integral of the forcing has unweighted norm at most $C_{L,h}(h + 2\sqrt{h})\sup\|f\|_\infty$. This proves both linear assertions.

Part (2). Set $b_a = S_a b$ and $e_a = \partial_t b_a - \Delta_a b_a - N_a(b_a)$. The residual in Theorem 25.4(1) is $O(a^4)$ in dimensionless link-logarithm units. Since $\theta = ab$ and physical time is a^2 times dimensionless time, we divide by a^3 and obtain $\sup_t \|e_a(t)\|_\infty \le C_{K_b}a$. For $Z = B_a - b_a$ the mild difference equation is

$$Z(t) = P^a_t Z(0) + \int_0^t P^a_{t-u}\big\{N_a(b_a(u) + Z(u)) - N_a(b_a(u)) - e_a(u)\big\}\,du.$$

Up to the first time at which $\|Z\|_{C^1_a} = 1$, the difference of the nonlinear terms is at most $C_{K_b}\|Z\|_{C^1_a}$ by (25.12). Using the weighted estimate of Part (1) on every such interval, we obtain

$$\sup_t \|Z(t)\|_{C^1_a} \le C_b(\eta + a).$$

We choose the constants in the following order: first the reference solution and the horizon; then λ and C_b; then η_b and a_b so small that $C_b(\eta_b + a_b) \le 1/2$ and $a_b(\sup\|b\|_\infty + 1)$ is strictly below the radius of the smaller chart. No first time with $\|Z\|_{C^1_a} = 1$ exists, since it would contradict the bound $1/2$. At fixed mesh, the finite-dimensional analytic differential equation can be continued as long as the solution stays in a compact part of the chart, which gives existence through time h. The gauge identification of Lemma 24.2 applies through time h. This proves (25.14).

Part (3) and continuations. For two solutions we subtract their exact equations. The source is $N_a(B_a) - N_a(B'_a)$, and there is no residual. Using the common bound in C^1_a already proved and the estimate of Part (1), we obtain (25.15) on one interval. By Lemma 24.7, at a restart with the fixed smooth gauge transformation k of the continuation, we have

$$\|\mathscr{G}_{a,k}(B_a) - S_a(b^k)\|_{C^1_a} \le C_k \|B_a - S_a b\|_{C^1_a} + C_{k,b}\, a,$$
$$\|\mathscr{G}_{a,k}(B_a) - \mathscr{G}_{a,k}(B'_a)\|_{C^1_a} \le C_k \|B_a - B'_a\|_{C^1_a},$$

where $\mathscr{G}_{a,k}(A)_i(x) = a^{-1}\operatorname{Log}(k(x)e^{aA_i(x)}k(x+ae_i)^{-1})$. Let L_i be the constant of (25.14) or (25.15) on the ith interval, let C_i, C'_i be the Lipschitz constants of the restart, and let $d_i = C_{k_i,b_i}$ be the coefficient of a in the first restart estimate. For the errors e_i relative to the selected DeTurck solution at the restart times, we have the finite affine recurrence $e_{i+1} \le L_i C_i (e_i + a) + d_i a$. For the differences δ_i of a pair at the restart times, we have the homogeneous recurrence $\delta_{i+1} \le L_i C'_i \delta_i$. Their finite products and sums give the asserted constants. We choose the initial error and the mesh only afterwards, below the threshold of every interval divided by the resulting finite multiplier. This proves the assertions for continuations. It also provides the C^1_a bound needed for the later smoothing, without using that smoothing.

Part (4). For each $c \in \mathscr{K}$ we take one continuation through time h from Theorem 24.10. Its finitely many DeTurck solutions are smooth on their closed time intervals and have finite C^5 bounds, and its gauge transformations are fixed and smooth. Parts (2) and (3) give an initial radius $\varepsilon_c > 0$, a threshold $a_c > 0$, and finite constants for this continuation. The sets $\{b_0 : \|b_0 - c\|_{C^1} < \varepsilon_c/4\}$ are open in the C^3 topology and cover $\mathscr{K}$; we select a finite subcover. By the fundamental theorem of calculus along coordinate segments, sampling is a contraction from the norm of values and first derivatives into C^1_a. Thus, if $\eta_{\mathscr{K}} \le \min_c \varepsilon_c/4$ and $\|b_0 - c\|_{C^1} < \varepsilon_c/4$ for a selected c, both lattice initial fields are within $\varepsilon_c/2$ of $S_a c$. We apply Parts (2) and (3) with this c and take the maximum of the finitely many constants and the minimum of the finitely many thresholds. This proves Part (4). □

25.5 Smoothing after a positive waiting time

In the statements about loops below, a closed lattice path ℓ_a approximating a continuum loop ℓ is required to have bounded physical length. It must also be joined to ℓ by a based piecewise smooth homotopy whose swept area, counted with multiplicity, tends to zero, as in Definition 2.1. An area bound $O(a)$ suffices, and coordinate polygons whose vertices lie on the lattice can be taken exactly. The following estimate for a smooth connection b explains this condition.

Let $\Gamma(h,r)$, $h, r \in [0,1]$, be a smooth homotopy with $\Gamma(h,0)$ and $\Gamma(h,1)$ equal to a fixed basepoint, and let $V(h,r)$ be the parallel transport along $\Gamma(h,\cdot)$ up to r, so

that $\partial_r V = V\, b(\partial_r \Gamma)$ and $V(h,0) = I$, where $b(\xi) = \sum_i b_i \xi^i$. We define $X = \partial_h V - V\, b(\partial_h \Gamma)$. Differentiating the transport equation in h and subtracting the r-derivative of $V\, b(\partial_h \Gamma)$, we find $\partial_r X = X\, b(\partial_r \Gamma) + V\, F_b(\partial_h \Gamma, \partial_r \Gamma)$, where $F_b(\xi,\zeta) = \sum_{i,j} F_{ij} \xi^i \zeta^j$ and $F_{ij} = \partial_i b_j - \partial_j b_i + [b_i, b_j]$. Since the basepoint is fixed, $X(h,0) = 0$ and $\partial_h V(h,1) = X(h,1)$. Solving for XV^{-1}, we obtain that $\partial_h V(h,1) V(h,1)^{-1}$ is the integral over $r \in [0,1]$ of the conjugate $V F_b(\partial_h \Gamma, \partial_r \Gamma) V^{-1}$. Since conjugation by unitary matrices has norm one and $\|X\|_{\mathrm{op}} = |X|/2$ for $X \in \mathfrak{su}(2)$, this proves

$$\| \operatorname{Hol}_b(\Gamma(1,\cdot)) - \operatorname{Hol}_b(\Gamma(0,\cdot)) \|_{\mathrm{op}} \le C \|F_b\|_\infty \iint |\partial_h \Gamma \wedge \partial_r \Gamma|\, dr\, dh, \tag{25.16}$$

where C only converts the componentwise curvature norm into the norm of a two-form. Subdivision gives the same estimate for piecewise smooth homotopies. A moving basepoint only adds a conjugation, which does not change traces. The approximation error therefore tends to zero uniformly on each bounded set of smooth connections.

The next lemma is a compactness statement for lattice arrays with bounded differences. In its proof, limits of difference arrays are identified with derivatives by discrete summation by parts, not by differentiating an interpolation of the arrays.

Lemma 25.6. *Let $a_k \to 0$ be meshes, and let A_k be $\mathfrak{su}(2)$-valued link fields on Λ_{L/a_k}. Suppose that for every integer $m \ge 0$ there are $K_m < \infty$ and k_m such that $\max_{|\nu| \le m} \|D^\nu_{a_k} A_k\|_\infty \le K_m$ for all $k \ge k_m$, where ν ranges over ordered compositions of forward and backward differences.*

1. *There are a subsequence (k_j) and a smooth $b \in C^\infty(\mathbb{T}^3_L; \mathfrak{su}(2)^3)$ with $\|\partial^\nu b\|_\infty \le K_{|\nu|}$ for all ν such that, for every ν,*
$$\|D^\nu_{a_{k_j}} A_{k_j} - S_{a_{k_j}} \partial^\nu b\|_\infty \longrightarrow 0.$$
Equivalently, the cardinal interpolants of Lemma 16.3, applied separately to each difference array $D^\nu_{a_{k_j}} A_{k_j}$, converge to $\partial^\nu b$ uniformly on $\mathbb{T}^3_L$.
2. *Let $\mathscr{K}$ be the set of all $b \in C^\infty(\mathbb{T}^3_L; \mathfrak{su}(2)^3)$ with $\|\partial^\nu b\|_\infty \le K_{|\nu|}$ for every ν. Then $\mathscr{K}$ is compact in the C^3 topology, and it contains 0.*
3. *Suppose that for each mesh a a set $\mathcal{F}_a$ of link fields on $\Lambda_{L/a}$ is given such that, for every m, the bound $\max_{|\nu| \le m} \|D^\nu_a A\|_\infty \le K_m$ holds for all $A \in \mathcal{F}_a$ whenever $a \le a_m$. Then*
$$\sup_{A \in \mathcal{F}_a} \inf_{b \in \mathscr{K}} \|A - S_a b\|_{C^1_a} \longrightarrow 0 \qquad (a \to 0).$$

Proof. Part (1). An array g on mesh a whose first differences are bounded by K has oscillation at most $CK(|x-y| + a)$ between two lattice sites x, y, as we see by following a shortest coordinate path on the torus. Since cardinal interpolation is a convex combination of samples within distance Ca, the interpolated array has the same modulus of continuity, up to a fixed constant and an additional CKa. It also takes the given values at the lattice sites. Applying this to $D^\nu A_k$ with the bound $K_{|\nu|+1}$, we find that the interpolants

of the arrays of order $|\nu|$ are bounded by $K_{|\nu|}$ and have modulus $CK_{|\nu|+1}(|x-y|+a_k)$. We choose finite nets of $\mathbb{T}_L^3$ with spacing tending to zero. The bounded values at the points of one net have a convergent subsequence, and the modulus of continuity extends the convergence uniformly over the torus up to an error that vanishes as the spacing and a_k tend to zero. By successive extractions and a diagonal subsequence, we obtain, for every one of the countably many ν, a continuous limit b_ν of the interpolants of $D^\nu A_{k_j}$, with $\|b_\nu\|_\infty \le K_{|\nu|}$. Since the interpolants take the given values at the lattice sites, $\|D^\nu A_{k_j} - S_{a_{k_j}} b_\nu\|_\infty \to 0$.

Let $b = b_\varnothing$ be the limit of the arrays themselves. For a smooth periodic test function φ, discrete summation by parts gives

$$a^3 \sum_x D_a^\nu A(x)\varphi(x) = (-1)^{|\nu|} a^3 \sum_x A(x) \widetilde{D}_a^\nu \varphi(x),$$

where $\widetilde{D}_a^\nu$ reverses the order of the differences and exchanges forward and backward differences. By Taylor expansion of φ and Riemann sums, along the subsequence the left side tends to $\int b_\nu \varphi$ and the right side to $(-1)^{|\nu|} \int b\, \partial^\nu \varphi$. We conclude that $b_\nu = \partial^\nu b$ in the sense of distributions, for every forward or backward choice in ν. Further, the left side is bounded by $K_{|\nu|} a^3 \sum_x |\varphi(x)|$, which gives

$$|\langle \partial^\nu b, \varphi \rangle| \le K_{|\nu|} \|\varphi\|_{L^1}, \tag{25.17}$$

with all direction and color components included. To see that these derivatives are classical, we convolve b with a nonnegative smooth periodic approximate identity χ_ϵ of mass one. Then $\partial^\nu (b * \chi_\epsilon) = b_\nu * \chi_\epsilon$ is bounded by $K_{|\nu|}$ by (25.17), and it converges uniformly to b_ν as $\epsilon \to 0$, because b_ν is continuous; also $b * \chi_\epsilon \to b$ uniformly. By the fundamental theorem of calculus along coordinate segments, when a sequence of functions and their first derivatives converge uniformly, the limit is continuously differentiable with the limiting derivatives. Applying this order by order, we conclude that b is smooth, with $\partial^\nu b = b_\nu$ and $\|\partial^\nu b\|_\infty \le K_{|\nu|}$. The equivalence with uniform convergence of the interpolants follows from the modulus of continuity and the fact that the interpolants take the sampled values at the lattice sites.

Part (2). The bounds on the derivatives of order four imply that the derivatives of order at most three of the elements of $\mathscr{K}$ are bounded and equicontinuous. By the Arzelà–Ascoli theorem, $\mathscr{K}$ is relatively compact in C^3. Let $b_n \in \mathscr{K}$ converge in C^3 to b. The uniform bounds at every higher order allow a diagonal subsequence along which every derivative of b_n converges uniformly. The fundamental theorem of calculus along coordinate segments identifies the limits with the derivatives of b. Hence b is smooth and satisfies the same bounds, and $\mathscr{K}$ is closed in C^3. This uses the bounds at every order; a family controlled at only one finite order need not have a closure consisting of smooth fields.

Part (3). If the assertion fails, there are $\epsilon > 0$, meshes $a_k \to 0$, and $A_k \in \mathcal{F}_{a_k}$ with $\inf_{b \in \mathscr{K}} \|A_k - S_{a_k} b\|_{C^1_{a_k}} \geq \epsilon$. Part (1) implies that a subsequence satisfies $\|A_k - S_{a_k} b\|_\infty \to 0$ and $\|D_{a_k} A_k - S_{a_k} \partial b\|_\infty \to 0$ for some $b \in \mathscr{K}$. Since $\|D_a S_a b - S_a \partial b\|_\infty \leq Ca \|b\|_{C^2} \leq CaK_2$, it follows that $\|A_k - S_{a_k} b\|_{C^1_{a_k}} \to 0$, a contradiction. □

Theorem 25.7. *Fix L, $1/2 < \alpha < 2/3$, and R, and let T_R be the local existence time of Theorem 24.5. For every integer $m \geq 0$ and every $0 < s < T_R$ there are $a_{m,R,s} > 0$ and $C_{m,L,R,s,T_R} < \infty$ such that*

$$\sup_{a \leq a_{m,R,s}} \sup_{s \leq t \leq T_R} \max_{|\nu| \leq m} \|D_a^\nu A_a(t)\|_\infty \leq C_{m,L,R,s,T_R}$$

for all solutions A_a given by Theorem 24.5 whose flow data satisfy $\|(x_a, y_a)\|_{\mathrm{fl},a} \leq R$ on $[0, T_R]$. All ordered compositions of forward and backward differences and all translated components are included. More generally, the same conclusion holds on any smaller interval that begins after a positive waiting time inside a time interval on which a restarted damped representative is used, provided that this representative lies in the smaller chart and a uniform bound in C^1_a on the preceding interval is given. The constant then depends on that bound, on the waiting time, on the length of the interval, and on m.

Suppose that b is a continuous periodic field and that such solutions, along meshes $a_k \to 0$, satisfy $\|A_{a_k}(s) - S_{a_k} b\|_{C^1_{a_k}} \to 0$ at a fixed time $s \in (0, T_R)$. Then b is smooth, with bounds on its derivatives of every order depending only on the order and on L, R, s, T_R. The closure in C^3 of the set of all such limits is compact and consists of smooth connections, so Theorem 24.10 applies to it for every finite horizon. For each such limit b, the Wilson flow started from $\exp(aS_a b)$ has link error $O(a^2)$ on every time interval of a continuation as in Theorem 24.10, including the restarts with the sampled fixed gauge transformations. The normalized traces of its holonomies around closed lattice paths of bounded length that approximate continuum loops, in the sense described before Lemma 25.6, converge to the corresponding continuum traces.

On each later time interval of a continuation by restarts with fixed smooth gauge transformations, the first paragraph applies as soon as a uniform C^1_a bound for the restarted representative is known, for example from Lemma 25.5. Then every fixed difference order is bounded at every fixed positive later time. The C^1_a bound is a hypothesis of this statement; it is not deduced from the smoothing. The constants are deterministic and do not depend on the mesh below the stated thresholds. No bound is claimed as the waiting time tends to zero, and no bound on high norms at time zero follows from the compactness in C^3.

The proof of the first assertion is the standard parabolic bootstrap, carried out on the lattice equation. In a difference of order $m - 1$ of the nonlinearity, the only difference of order m of the solution appears linearly, with a bounded factor in front. The heat semigroup gains one difference at the integrable cost $(t - r)^{-1/2}$, and the resulting term can be absorbed on a short interval.

Proof. Step 1: the base case. By Theorem 24.5, on $[s/2, T_R]$ we have

$$\|A_a(t)\|_\infty + \|D_a A_a(t)\|_\infty \le K_1(L, R, s, T_R). \tag{25.18}$$

This proves the cases $m = 0, 1$. In the smaller chart the equation is

$$\partial_t A_a = \Delta_a A_a + \mathcal{N}_a(A_a, A_a) + a^{-3} R_3(a A_a). \tag{25.19}$$

Step 2: the highest difference. Let $m \ge 2$, and suppose that all differences through order $m-1$ are bounded on a preceding interval, by $K_1, \dots, K_{m-1}$. We show that

$$\big\| D_a^{m-1} \{ \mathcal{N}_a(A_a, A_a) + a^{-3} R_3(a A_a) \} \big\|_\infty \le B_m(K_1, \dots, K_{m-1}) + C_m K_1 \|D_a^m A_a\|_\infty. \tag{25.20}$$

Each summand of $\mathcal{N}_a$ contains exactly one physical difference. When we apply (25.2) $m-1$ times, a factor of order m occurs only when all these differences fall on the factor that already carries a difference; its companion is then a value of A_a, bounded by K_1. If at least one difference falls on the companion, both factors have orders at most $m-1$ and are bounded by the K_j. The number of terms and all translations depend only on m and on the fixed stencil. In the analytic remainder, the term of degree d in physical units has the factor a^{d-3}, $d \ge 3$, and a difference of order $m-1$ of a product involves differences of A_a of order at most $m-1$ only. To see that the series converges uniformly, we shrink the threshold so that aK_1 lies in a fixed smaller chart. A differentiated term of degree d is bounded by a fixed polynomial in d times its coefficient in the series times products of the finitely many K_j. The remaining undifferentiated factors keep a geometric factor from the smaller chart. The resulting series, even multiplied by the polynomial, is summable uniformly in a. This proves that the remainder contributes only to B_m, with no negative power of the mesh. These observations prove (25.20) for every ordered choice of forward and backward differences and all translated components.

Step 3: absorption. We start the mild equation at a time u_0 in the preceding interval, and define

$$Z_m(d) = \sup_{0 < v \le d} \sqrt{v}\, \|D_a^m A_a(u_0 + v)\|_\infty.$$

At each fixed mesh this number is finite, since the arrays are finite dimensional. We commute $m-1$ differences onto the initial value and onto the source, and place one difference on the heat semigroup, whose bound is $C(v + a^2)^{-1/2} \le C v^{-1/2}$. Multiplying the resulting mild estimate by $\sqrt{v}$ and using (25.20), we obtain

$$Z_m(d) \le C_m K_{m-1} + C_m B_m d + C_m K_1 \sqrt{d}\, Z_m(d). \tag{25.21}$$

For the last term we used $\int_0^v (v-r)^{-1/2} r^{-1/2}\, dr = \pi$, together with the additional factor $\sqrt{v} \le \sqrt{d}$; for the bounded source the corresponding factor is $2v \le 2d$. We choose $d \le 1$ such that $C_m K_1 \sqrt{d} \le 1/2$ and such that $[u_0, u_0 + d]$ lies in the preceding interval. Then

(25.21) implies that $Z_m(d) \le 2C_m(K_{m-1} + B_m d)$, and hence a uniform bound on $D_a^m A_a$ at time $u_0 + d$.

Step 4: the whole interval. For a given finite $m \ge 2$ we choose times $s/2 = s_1 < s_2 < \cdots < s_m = s$. At stage $j \ge 2$ we take $d_j > 0$ less than $s_j - s_{j-1}$ and satisfying the absorption condition of Step 3 for order j. For any $t \in [s_j, T_R]$ we apply Step 3 with $u_0 = t - d_j \ge s_{j-1}$ and evaluate at $v = d_j$. This gives one bound K_j for all such t. Since there are finitely many orders, the restrictions $a \le a_{j,R,s}$ have a positive minimum, and the constants are finite. This proves the first assertion. On a restarted time interval, the same proof starts from the given chart and the given C_a^1 bound on the preceding interval instead of (25.18). Inspection of the choices shows the stated dependence of the constant.

Step 5: limits at a positive time. Let $A_{a_k}(s)$ satisfy $\|A_{a_k}(s) - S_{a_k} b\|_{C^1_{a_k}} \to 0$. The first assertion implies that the arrays $A_{a_k}(s)$ satisfy the hypotheses of Lemma 25.6, with bounds $K_m = C_{m,L,R,s,T_R}$. By part (1) of that lemma, a subsequence converges, together with all its difference arrays, to a smooth field with these bounds. This field is b, since the arrays converge to $S_{a_k} b$. Hence every limit belongs to the set $\mathscr{K}$ of part (2) of that lemma, which is compact in C^3 and consists of smooth connections, and so does the C^3 closure of the limits.

Step 6: continuation and loops. We apply Theorem 24.10 with $r = 3$ to this compact set and a fixed horizon. Its open neighborhood contains every smooth connection sufficiently close to the compact set in C^3. Indeed, for each point of the compact set we choose a C^3 ball whose doubled radius lies in the neighborhood, take a finite subcover, and use the minimum of the finitely many radii. Theorem 25.4(4), applied on the finitely many time intervals, yields link error $O(a^2)$ for the Wilson flow started from $\exp(aS_a b)$, including the restarts with the sampled gauge transformations. The Wilson flow and its damped representatives have the same traces of holonomies around closed paths. Since a closed lattice path of length at most ℓ has at most ℓ/a links, unitary telescoping turns the link error $O(a^2)$ into an error $O(\ell a)$ for the holonomy. By Lemma 24.7, the product of the exponentials e^{ab_i} along the path differs from the smooth parallel transport along the same path by an error of the same order. Finally, (25.16) bounds the difference between the smooth transport along the lattice path and along the continuum loop. This proves the assertion on loops.

Step 7: later restarts. Suppose that lattice differences of A of order up to m are bounded and that k is a fixed smooth periodic gauge transformation. Then

$$Y_i(x) = k(x) e^{aA_i(x)} k(x + ae_i)^{-1} = I + O(a),$$

and every difference of fixed order of $Y - I$ keeps this factor a. To verify this, we split $Y - I$ into $k(x)(e^{aA_i(x)} - I)k(x + ae_i)^{-1}$ and $k(x)k(x + ae_i)^{-1} - I$. We handle the first part with the convergent exponential series and the exact product rule, and the second with $k(x + ae_i) - k(x) = \int_0^a \partial_i k(x + re_i)\, dr$. Both give a times a polynomial in the

given bounds and in smooth norms of k. In the analytic composition $a^{-1}\operatorname{Log} Y$, a term with $j \geq 1$ differentiated factors carries a^{j-1}, and the value is $a^{-1}O(a)$. The bounds for the composition are therefore uniform in the mesh, and such a gauge transformation preserves the bounds on lower differences. On each restarted interval, a given uniform C^1_a bound in the common gauge restarts the induction of Step 2, and waiting intervals before each given later time give every given finite order. At a restart time we use a preceding interval and the bound for the composition just proved to transfer the bounds on the differences. This proves the last assertion. □

The next lemma is the space-time version of Lemma 25.6, together with the passage to the limit in the Wilson equation.

Lemma 25.8. *Let $a_k \downarrow 0$, and let $J \subset (0,\infty)$ be a closed interval. Let B_{a_k} be time-dependent link fields on J such that, for every m, all differences of order at most m of B_{a_k} and their time derivatives are bounded on J uniformly in k. We interpolate each difference array separately by the cardinal interpolation of Lemma 16.3. Then every subsequence has a further subsequence along which these interpolated arrays converge, uniformly on $J \times \mathbb{T}^3_L$ at every fixed order, to the corresponding derivatives of one field B that is smooth in space for each time; by a diagonal argument the same holds for countably many intervals simultaneously. If, in addition, each B_{a_k} is a damped representative of the Wilson flow on J, that is, solves $\partial_t B = \Delta_a B + \mathcal{N}_a(B,B) + \mathscr{C}_a(B)$ in the smaller chart, then the limit B solves the DeTurck equation $\partial_t B = \Delta B + \mathcal{N}_0(B,B) + \mathscr{N}_3(B)$ on J. The same holds for the mild form of the equation with a fixed lower integration endpoint in J.*

The passage to the limit in the equation uses three properties of the Wilson equation. Its quadratic part is a fixed finite sum of translated brackets containing one difference (Lemma 24.2). The remainder $\mathscr{C}_a$ has local Lipschitz bounds that are uniform on bounded sets ((25.11)). Finally, $\mathscr{C}_a$ is consistent, uniformly on bounded families of smooth fields (Lemma 25.3). The same proof applies to any equation with these three properties.

Proof. We obtain the spatial modulus of continuity of each interpolated difference array from the bound on the next difference, as in the proof of Lemma 25.6(1). The bounds on the time derivatives give a uniform Lipschitz modulus in time. Using finite nets in space and time, successive subsequences of the bounded finite-dimensional values at their points, and a diagonal argument, we obtain uniform convergence on $J \times \mathbb{T}^3_L$, first for one array and then for the countable list of orders and positive intervals. As in the proof of Lemma 25.6(1), summation by parts against smooth test functions, mollification, and the fundamental theorem of calculus identify the limits with the corresponding classical derivatives.

For the nonlinear terms, every summand of $\mathcal{N}_a$ involves finitely many translations and one difference, and these converge to the corresponding untranslated value and

derivative. For the analytic remainder, the uniform bound on B_a on J implies that the analytic composition converges uniformly on the smaller chart. Its local Lipschitz constants are bounded by (25.11), and its values on the sample of the smooth limit converge to $S_a \mathscr{N}_3(B)$ by Lemma 25.3. Subtracting the value at the sample, we obtain the convergence at B_a itself. The same reasoning applies after any fixed difference for which bounds are given. The lattice Laplacian is a sum of second differences and converges to ΔB. Integrating the differential equations in time, and using uniform convergence to pass every term through the integral, we obtain the limiting equation.

For the mild form, the heat semigroup applied to a sampled smooth family converges by the explicit bounds for general smooth inputs in Lemma 16.9. For a convergent array, we first replace it by the sample of its smooth limit; the replacement error is controlled by contraction. For a first difference of the output, we either commute it onto a convergent difference of the source, or use the integrable heat bound of Lemma 25.1. These arguments pass each term of the mild equation with a positive lower endpoint to the limit. □

25.6 Continuation and stability for bounded flow data

We now apply smoothing and compactness to bounded flow data. After a short positive time u, Theorem 25.7 bounds all lattice differences of the solution, and Lemma 25.6 shows that the solutions at time u are uniformly close to samples of a compact set $\mathscr{K}$ of smooth connections. This compact set depends only on the bound R on the flow data, not on any convergence of the initial potentials. Part (4) of Lemma 25.5 then provides finitely many continuations, chosen before the initial potential. Along them we compare every solution with a selected smooth DeTurck solution, and two solutions with each other.

Lemma 25.9. *Fix $L, T_0, R > 0$, $1/2 < \alpha < 2/3$, and $0 < \tau \le T < \infty$. Let T_R be the local existence time of Theorem 24.5 for flow data of norm at most R, replaced by $\min(T_R, T_0)$, and choose $0 < u < \min(T_R, T_0, \tau)/4$. There are $a_*, K, C, \delta > 0$ and finitely many continuations as in Theorem 24.10, all chosen after these data, with the following properties. For every mesh $a = L/n \le a_*$ and every lattice initial potential whose flow data satisfy $\|(x_a, y_a)\|_{\mathrm{fl},a} \le R$ on $[0, T_0]$, the Wilson flow on $[u, T]$ has damped representatives, restarted at the times of one of these continuations with its sampled gauge transformations, whose C^1_a norms are at most K. If two such initial potentials on the same lattice, compared without any gauge transformation, satisfy*

$$d_a = \mathsf{X}^a_\alpha(x_a - x'_a) + \mathsf{Y}^a_\alpha(y_a - y'_a) \le \delta,$$

with both norms taken on $[0, T_0]$ and including all first differences and translated components, then the two flows can use the same continuation with the same sampled gauge transformations,

and on each of its time intervals

$$\sup_t \|B_a(t) - B'_a(t)\|_{C^1_a} \le C\, d_a.$$

Each representative is gauge equivalent to the Wilson flow $\Psi_t^{(n)}$. The constants and a_* depend only on $L, T_0, R, \alpha, \tau, T, u$, and the chart. In particular they do not depend on higher derivatives of the initial potentials, nor on any finite approximation from which these potentials are constructed. Only the finitely many selected smooth DeTurck solutions need bounds on higher derivatives.

Proof. Step 1: bounds at time u. By the individual bounds and the bound on differences in Theorem 24.5, we have

$$\|A_a(u) - A'_a(u)\|_{C^1_a} \le L_u d_a, \qquad \sup_{u/2\le t\le 2u} \|A_a(t)\|_{C^1_a} \le K_1,$$

and the same individual bound holds for the second initial potential. Theorem 25.7, applied with a preceding interval after a positive waiting time, yields for every fixed m a bound K_m on all differences of order m at time u, for all sufficiently small meshes. The restriction on the mesh may depend on m; along every sequence $a_j \to 0$ it is eventually satisfied at each fixed order.

Step 2: compactness. Let $\mathscr{K}$ be the set of smooth connections b with $\|\partial^\nu b\|_\infty \le K_{|\nu|}$ for all ν. By Lemma 25.6(2), $\mathscr{K}$ is compact in C^3 and contains 0; by part (1) of that lemma, it contains every possible limit of the endpoints $A_a(u)$. Applying part (3) of that lemma to the set $\mathcal{F}_a$ of all endpoints $A_a(u)$ of initial potentials with flow data bounded by R, we obtain

$$\sup \inf_{b\in\mathscr{K}} \|A_a(u) - S_a b\|_{C^1_a} \longrightarrow 0 \qquad (a \to 0),$$

where the supremum is over all such initial potentials.

Step 3: choice of the continuations. We apply part (4) of Lemma 25.5 to $\mathscr{K}$ with horizon $T - u$. Its neighborhood radius $\eta_{\mathscr{K}}$, its restriction $a \le a_{\mathscr{K}}$, its uniform bound, and its finitely many continuations are fixed before we choose either initial potential. We decrease a_* until every endpoint $A_a(u)$ lies within $\eta_{\mathscr{K}}/4$ of $S_a b_0$ for some $b_0 \in \mathscr{K}$, and we choose $\delta \le \eta_{\mathscr{K}}/(4\max(1, L_u))$. If $d_a \le \delta$, then $A'_a(u)$ lies within $\eta_{\mathscr{K}}/4 + L_u\delta \le \eta_{\mathscr{K}}/2$ of the same $S_a b_0$. Thus both endpoints lie in the neighborhood of one common element of $\mathscr{K}$.

Step 4: conclusion. The paired part of Lemma 25.5(4) propagates the initial difference $L_u d_a$ at time u through the finite continuation, using the same sampled gauge transformations for both flows. Its individual part bounds the representative of every initial potential, whether or not it is paired with another. All representatives are gauge equivalent to the Wilson flow by Lemma 24.2. Since the two compared equations are exact lattice equations, their difference has no consistency residual. This proves the asserted bounds and their dependence. □

For matrices, we must also compare the gauge transformations used to define the representatives. On the early interval the damping gauge transformation solves $\partial_t k = \lambda_a k$ with an anti-Hermitian generator. Since such equations have unitary propagators, the difference of two solutions is bounded by the integral of the difference of the generators, without any exponential of the size of the generators. By the estimates of Lemma 24.7, the difference of the generators has an integrable singularity at time zero.

Theorem 25.10. *Fix $L, T_0, R > 0$, $1/2 < \alpha < 2/3$, $0 < t_- \le t_+ \le T < \infty$, and $\mathfrak{l}_1, \dots, \mathfrak{l}_b > 0$. For each mesh $a = L/N$, let $\ell_{1,a}, \dots, \ell_{b,a}$ be closed paths in Λ_N with given basepoints and with physical lengths $aN_{\ell_{i,a}} \le \mathfrak{l}_i$, where N_ℓ is the number of links of ℓ. Let A_0, A_0' be lattice initial potentials on Λ_N whose flow data (x, y), (x', y') satisfy $\|(x,y)\|_{\mathrm{fl},a}, \|(x',y')\|_{\mathrm{fl},a} \le R$ on $[0, T_0]$, put $U = e^{aA_0}$ and $U' = e^{aA_0'}$, and let $d_a = \mathsf{X}^a_\alpha(x - x') + \mathsf{Y}^a_\alpha(y - y')$, with both norms on $[0, T_0]$. Then*

$$\sup_{t_- \le t \le t_+} \max_i \left\| \mathrm{Hol}_{\ell_{i,a}} \Psi^{(N)}_t U - \mathrm{Hol}_{\ell_{i,a}} \Psi^{(N)}_t U' \right\|_{\mathrm{op}} \le C\, d_a.$$

The holonomies are compared as matrices: no gauge transformation is applied to either configuration, and no conjugation at the basepoints is optimized. The constant depends only on $L, T_0, R, \alpha, t_\pm, T$, the lengths $\mathfrak{l}_i$, the chart, and the finitely many smooth DeTurck solutions selected in the proof. It holds for every mesh $a = L/N$ with $N \ge 4$, and it does not depend on the position or shape of the paths beyond their lengths, nor on higher derivatives of the initial potentials.

Let $h_R = \min(T_R, T_0)$, where T_R is the local existence time of Theorem 24.5. After decreasing the mesh bound a_R of that theorem if necessary, the following holds for $a \le a_R$ and $0 \le s \le h_R$. The damped representatives A, A' and their damping gauge transformations k, k' with initial value I satisfy, at every vertex,

$$\|D_a(A - A')(s)\|_\infty \le C_R d_a \big\{ v_a(s)^{-(\alpha+1)/2} + v_a(s)^{-\alpha} + v_a(s)^{\kappa - 1/2} \big\},$$

$$\|k(s) - k'(s)\|_\infty \le C \int_0^s \|D_a(A - A')(v)\|_\infty \, dv \le C_R d_a.$$

For $a \le a_*$ and $d_a \le \delta$, where a_* and δ are the constants of Lemma 25.9 with $\tau = t_-$, both flows can use the same continuation and the same sampled gauge transformations on the later intervals. The matrices are always expressed in the gauges of the initial configurations. For other pairs and meshes, the matrix bound follows from the diameter bound for unitary matrices or from the finite-dimensional Wilson flow, as in the proof.

Proof. Step 1: the early interval. Fix $u < \min(T_R, T_0, t_-)/4$. Since the flow data are bounded by R only on $[0, T_0]$, we apply Theorem 24.5 on $[0, h_R]$. Adding the bound on the difference of the remainders in Theorem 24.5 to the differences of the linear heat evolutions and of the quadratic Picard terms, we obtain exactly the three powers in the bound for $D_a(A - A')$. This bound is also stated in Lemma 24.7. By the formula for the generator in that lemma, the damping generator in physical units is the signed lattice

divergence of the potential, which gives $\|\lambda_a - \lambda'_a\|_\infty \le C\|D_a(A - A')\|_\infty$. The damping gauge transformations solve $\dot k = \lambda_a k$ and $\dot k' = \lambda'_a k'$ with $k(0) = k'(0) = I$. Since the generators are anti-Hermitian, the propagators are unitary, and variation of constants gives, at every vertex,

$$\|k(s) - k'(s)\|_\infty \le \int_0^s \|\lambda_a(v) - \lambda'_a(v)\|_\infty \, dv. \tag{25.22}$$

Indeed, $\partial_s(k^{-1}k') = k^{-1}(\lambda'_a - \lambda_a)k'$, and $\|k' - k\| = \|k^{-1}k' - I\|$. All three powers in the bound for $D_a(A - A')$ are integrable uniformly in the mesh, since $(\alpha + 1)/2 < 1$, $\alpha < 1$, and $1/2 - \kappa < 1$; for $0 < p < 1$ we have $\int_0^{h_R} (v + a^2)^{-p}\, dv \le h_R^{1-p}/(1-p)$. This proves the estimate for the gauge transformations on the early interval $[0, h_R]$. The representatives remain in the chart, since Theorem 24.5, applied on $[0, h_R]$, bounds the contribution of the remainder to the link angles by $C_R a(h_R + a^2)^\kappa$.

Step 2: later intervals and the composed gauge transformations. We apply Lemma 25.9 at time u. If $d_a \le \delta$, the paired estimate implies that $\sup \|B_a - B'_a\|_{C^1_a} \le C d_a$ on every later time interval. Let g, g' be the composed gauge transformations, from the initial configurations to the current damped representatives. At time u their difference is $O(d_a)$ by Step 1. On a time interval I_i of the continuation, $g = k_i g_i$, where g_i is the value at the start of I_i and k_i solves the damping equation with initial value I. The generators of k_i and k'_i differ by at most a constant times the C^1_a difference of the representatives. The inhomogeneous version of (25.22) therefore carries the incoming difference $\|g_i - g'_i\|$ with coefficient one and adds at most $C|I_i|d_a$. At a restart both composed gauge transformations are multiplied on the left by the same fixed unitary gauge transformation, which does not change the norm of their difference. By induction over the finitely many intervals, we obtain $\sup \|g - g'\|_\infty \le C d_a$ through time T, including the possibly different damping gauge transformations of the two flows.

Step 3: the matrices. On a common time interval, the exponential $e^{aB_{a,e}}$ of each link e changes by at most $Ca\|B_a - B'_a\|_\infty$, and unitary telescoping along a path with at most $\mathfrak{l}_i/a$ links gives

$$\|H_i(B_a) - H_i(B'_a)\|_{\mathrm{op}} \le C\mathfrak{l}_i \|B_a - B'_a\|_\infty,$$

where $H_i(B_a)$ is the holonomy of e^{aB_a} around $\ell_{i,a}$. The holonomy of the Wilson flow is $g(x_0)^{-1} H_i(B_a) g(x_0)$, where x_0 is the basepoint, and undoing the composed gauge transformation at x_0 adds at most $2\|g(x_0) - g'(x_0)\|_{\mathrm{op}}$. This proves the estimate for $d_a \le \delta$, in the gauge of the initial configurations. For $d_a > \delta$, the distance between the two matrices is at most two, and it suffices to take $C \ge 2/\delta$.

Step 4: coarse meshes. Only finitely many lattice sides have $a > a_*$. On each of them the Wilson vector field is smooth on the compact manifold $\mathrm{SU}(2)^{\mathcal{E}_N}$, and its solution map over a finite time is Lipschitz. To see this, we subtract the two integral equations, apply the integral form of Gronwall's inequality, and concatenate up to time $4T/a^2$. At $t = 0$

the first weighted norm implies that $\|A_0 - A_0'\|_\infty \le a^{-\alpha} d_a$. Then the initial links differ by at most $C a^{1-\alpha} d_a$. Telescoping after the finite-dimensional flow, with $N_{\ell_{i,a}} \le \mathfrak{l}_i / a$, gives a finite constant on each such side, and we take the maximum over these finitely many sides. This proves the estimate for all meshes. The assertions about the early damped representatives concern the interval $[0, h_R]$ and the mesh threshold of Theorem 24.5, as stated, and the common continuation is used only when $a \le a_*$ and $d_a \le \delta$. □

Corollary 25.11. *Fix $L, T_0, R > 0$, $1/2 < \alpha < 2/3$, $0 < \tau \le T < \infty$, lengths $\mathfrak{l}_1, \dots, \mathfrak{l}_b > 0$, and characters $\chi_1, \dots, \chi_b$ of finite-dimensional representations of $\mathrm{SU}(2)$. Let T_R be the local existence time of Theorem 24.5, replaced by $\min(T_R, T_0)$, and choose*

$$0 < u < \tfrac{1}{4} \min\{T_R, T_0, \tau\}.$$

There are $a_* > 0$ and $C < \infty$, chosen after these data, such that the following holds for every mesh $a = L/n \le a_*$. Let (x_a, y_a) and (x_a', y_a') be the flow data of two lattice initial potentials $A_a(0), A_a'(0)$ on the same lattice, with $\|(x_a, y_a)\|_{\mathrm{fl},a}, \|(x_a', y_a')\|_{\mathrm{fl},a} \le R$ on $[0, T_0]$. Write

$$d_a = \mathsf{X}_\alpha^a (x_a - x_a') + \mathsf{Y}_\alpha^a (y_a - y_a'),$$

with both norms on $[0, T_0]$ and including all first differences, and put $U_a = e^{a A_a(0)}$, $U_a' = e^{a A_a'(0)}$. Then for every choice of closed lattice paths $\ell_{r,a}$ with $a N_{\ell_{r,a}} \le \mathfrak{l}_r$ and every choice of times $t_r \in [\tau, T]$,

$$\Big(\sum_{r=1}^{b} \big| \chi_r\big(\mathrm{Hol}_{\ell_{r,a}} \Psi_{t_r}^{(n)} U_a\big) - \chi_r\big(\mathrm{Hol}_{\ell_{r,a}} \Psi_{t_r}^{(n)} U_a'\big) \big|^2 \Big)^{1/2} \le C\, d_a. \tag{25.23}$$

The threshold and the constant depend only on the displayed data, the chart, and the finitely many smooth DeTurck solutions selected in the proof. They do not depend on the position or shape of the paths beyond their lengths, nor on higher derivatives of the initial potentials or on any finite approximation from which they are constructed. No convergence of the flow data as the mesh varies is assumed, and there is no additive error in the mesh. For normalized fundamental traces, take $\chi_r(V) = (1/2) \operatorname{tr} V$.

Proof. We apply Theorem 25.10 with the given lengths and with $t_- = \tau$, $t_+ = T$. Every character χ of a finite-dimensional representation of $\mathrm{SU}(2)$ is a polynomial p_χ in $(1/2) \operatorname{tr} V$. Indeed, on the diagonal matrices $\operatorname{diag}(e^{i\theta}, e^{-i\theta})$ the character is $\sum_k m_k e^{ik\theta}$ with integer weights k and multiplicities $m_k = m_{-k}$, which is a combination of the functions $\cos(k\theta)$. The recurrence for $\cos(k\theta)$ expresses these as polynomials in $\cos\theta = (1/2) \operatorname{tr} V$. Then $\sup_{[-1,1]} |p_\chi'|$ bounds the Lipschitz constant of χ as a function of the normalized trace, and the normalized trace is Lipschitz in the matrix norm. Summing the finitely many bounds in Euclidean norm, we obtain (25.23). □

25.7 Parameter derivatives in a fixed initial gauge

In the weak-coupling analysis of Sections 34 to 36, the initial potential depends on finitely many real parameters, and we need two parameter derivatives of the flowed holonomy matrices, uniformly in the mesh. We first treat a single time interval on which the representative is bounded in C^1_a, and then the singular initial interval and the continuation.

Lemma 25.12. *Assume the hypothesis of Lemma 25.5(1) for every value of a finite-dimensional real parameter, and suppose that the initial field depends twice continuously differentiably on the parameter. For unit directions v, w write $V_v = \partial_v B_a$ and $W_{vw} = \partial_v \partial_w B_a$. If the initial C^1_a norms of V_v and W_{vw} are bounded by b_1 and b_2, then*

$$\sup_{t \le h} \|V_v(t)\|_{C^1_a} \le C_{L,h,K}\, b_1, \qquad \sup_{t \le h} \|W_{vw}(t)\|_{C^1_a} \le C_{L,h,K}(b_2 + b_1^2). \tag{25.24}$$

These estimates persist through finitely many restarts with fixed sampled smooth gauge transformations. After positive waiting times they also give bounds on every given fixed order of spatial differences of B_a, V_v, and W_{vw}. The corresponding assertions hold for smooth continuum solutions on the time intervals of a continuation as in Theorem 24.10.

Proof. Step 1: differentiability. At each fixed mesh, the initial fields and the finite-dimensional analytic vector field have two continuous derivatives. Difference quotients in the integral equation converge to the solutions of the differentiated linear integral equations. Indeed, Taylor's integral formula for the vector field leaves a remainder that tends to zero uniformly on compact sets of parameters and times, and the linear Volterra inverse of Lemma 25.5(1) bounds the effect of this remainder. Applying this first to the fields and then to their first derivatives, we obtain the differentiability. The uniform bounds below do not depend on the fixed mesh.

Step 2: the equations. The derivatives satisfy

$$\begin{aligned} \partial_t V_v &= \Delta_a V_v + \mathscr{B}_a(B_a) V_v, \\ \partial_t W_{vw} &= \Delta_a W_{vw} + \mathscr{B}_a(B_a) W_{vw} + f_{vw}, \\ f_{vw} &= \mathcal{N}_a(V_v, V_w) + \mathcal{N}_a(V_w, V_v) + D^2 \mathscr{C}_a(B_a)[V_v, V_w]. \end{aligned} \tag{25.25}$$

The Volterra estimate (25.13) implies the first bound in (25.24). By the bound on $\mathcal{N}_a$ and (25.11), we have $\|f_{vw}\|_\infty \le C_K \|V_v\|_{C^1_a} \|V_w\|_{C^1_a}$, and applying the Volterra estimate to the inhomogeneous second equation gives the second bound. Note that the second variation satisfies an inhomogeneous equation, and the forcing f_{vw} must be kept.

Step 3: restarts. For a fixed smooth gauge transformation k, the map $\mathscr{G}_{a,k}(B)_i(x) = a^{-1} \operatorname{Log}(k(x) e^{aB_i(x)} k(x + a e_i)^{-1})$ satisfies, on a bounded set of the norm C^1_a,

$$\|D^j \mathscr{G}_{a,k}(B)[Z_1, \dots, Z_j]\|_{C^1_a} \le C_{k,K} \prod_{i=1}^{j} \|Z_i\|_{C^1_a}, \qquad j = 1, 2, \tag{25.26}$$

where D differentiates in the field. The case $j = 1$ is the Lipschitz bound of Lemma 24.7; we check the case $j = 2$ directly. For $Y = k(x)e^{aB_i(x)}k(x + ae_i)^{-1}$ we have $Y - I = O_K(a)$, and the jth field derivative of Y has a factor a^j for $j = 1, 2$. By the exact product rule, the given C^1_a norms, and the bounded smooth differences of k, a further spatial difference keeps these factors. Since the derivatives of Log are bounded near I, a differentiated term in $a^{-1}\operatorname{Log} Y$ keeps at least one factor a to compensate the factor a^{-1}, also after one spatial difference. Terms with more differentiated factors carry further nonnegative powers of a. This proves (25.26), using only fixed smooth norms of k and a margin of the chart. Counting factors in the same way, with three field arguments, we bound $D^3\mathscr{G}_{a,k}$ on this set: it has the factor a^3 before the division by a, and products of derivatives of the logarithm have at least the same total number of field arguments; one spatial difference again keeps these factors. Hence the first two field derivatives in (25.26) are locally Lipschitz on this set, with uniform constants, which we use below to subtract second derivatives at restarts. The chain rule at a restart gives

$$V_v^+ = D\mathscr{G}_{a,k}(B)V_v, \qquad W_{vw}^+ = D\mathscr{G}_{a,k}(B)W_{vw} + D^2\mathscr{G}_{a,k}(B)[V_v, V_w].$$

These bounds and (25.24) can be composed finitely many times. Since the gauge transformations k are fixed, no parameter dependence of a gauge choice is differentiated in these formulas.

Step 4: higher differences. We apply Theorem 25.7 to B_a, whose C^1_a bound is given. After $m - 1$ differences of the first equation in (25.25), computed with the product rules (25.2), the only unknown term of highest order is bounded by $C_K\|D^m_a V_v\|_\infty$. The term containing $D^m_a B_a$ has the companion V_v, whose value is already bounded, and it belongs to the known forcing. Every other term contains only lower differences of V_v and bounded differences of B_a. The field derivative of the remainder involves no difference of order larger than $m - 1$ and no negative power of the mesh. This estimate has the form of (25.20), with a known bounded forcing. As in (25.21), the weighted quantity $\sup_v \sqrt{v}\|D^m_a V_v(u_0 + v)\|_\infty$ and the integral $\int_0^v (v - r)^{-1/2} r^{-1/2}\, dr = \pi$ then bound $D^m_a V_v$ after a positive waiting time. In the second equation the unknown term of highest order is bounded by $C_K\|D^m_a W_{vw}\|_\infty$. Its additional forcing involves differences of V_v and V_w of order at most m, which are already bounded by the argument for the first derivatives, and the same absorption gives order m for W_{vw}. Finitely many successive waiting intervals give any given finite list of orders, and the argument for restarts extends to these orders by the exact product rule.

Step 5: continuum solutions. For smooth continuum solutions on the time intervals of a continuation, the same proof applies, with the ordinary product rule and the heat estimates stated in Theorem 24.10, on a fixed physical horizon subdivided if necessary. Starting from a smooth finite-dimensional family of initial data, we take difference quotients of the continuum mild equation. The same Volterra inverse and Taylor's

integral remainder give their convergence, first at order one and then at order two. If bounds through a fixed order are needed up to the initial time, they hold for the smooth family at the initial time. We then apply the product rule one order below the highest desired derivative and the same Volterra estimate; the highest term is linear, with a bounded undifferentiated companion. Induction gives the required mixed bounds in space and in the parameters, down to the initial time of each such smooth solution. □

Theorem 25.13. *Fix $L > 0$, $1/2 < \alpha < 2/3$, $T_0 > 0$, a finite horizon $T > 0$, and $t_* \in (0, T]$, and consider the holonomies around the three coordinate cycles of $\mathbb{T}_L^3$, based at a fixed vertex x_0, at times in $[t_*, T]$. Let $U_0 \Subset U_1 \subset \mathbb{R}^d$ be bounded parameter domains, covered by finitely many smooth input charts with fixed positive margins; changes between these charts act on the initial potentials by gauge transformations equal to the identity at the basepoint. For each mesh $a = L/n$, let $u \mapsto \mathcal{E}_a(u) = (x_a(u), y_a(u))$ be the flow data of a lattice initial potential $A_{0,a}(u)$, twice continuously differentiable on U_1, with*

$$\sup_{a,\, u \in U_1} \sum_{r=0}^{2} \|D_u^r \mathcal{E}_a(u)\|_{\mathrm{fl},a} \le R_2 < \infty,$$

where the norms are taken on $[0, T_0]$ and contain both components and all their first physical differences, and the parameter derivatives are measured in a fixed Euclidean operator norm. For every u, start the damping gauge transformation at I, in the gauge of the initial configuration $e^{a A_{0,a}(u)}$. Let T_{R_2} be the local existence time of Theorem 24.5 for the bound R_2, replaced by $\min(T_{R_2}, T_0)$.

1. *There is $a_* > 0$, chosen after the displayed data and below the restrictions on the mesh in Theorem 24.5, such that the following holds for every mesh $a \le a_*$. On the early interval $[0, T_{R_2}]$, the remainder $r_a(u)$ and the damped representative $A_a(u, t)$ satisfy, for $r = 1, 2$,*

$$\sup_{a \le a_*,\, u \in U_0} \|D_u^r r_a(u)\|_{\mathsf{R}_a} \le C_{R_2},$$

$$\|D_u^r D_a A_a(u, t)\|_\infty \le C_{R_2} \big\{ (t + a^2)^{-(\alpha+1)/2} + (t + a^2)^{-\alpha} + (t + a^2)^{\kappa - 1/2} \big\}.$$

   All three powers of $t + a^2$ are integrable at zero, uniformly in the mesh. The first and second parameter derivatives of the damping gauge transformation are bounded uniformly at every vertex on this interval. The representatives remain in the chart, by the bound

$$a \sup_{t \le T_{R_2}} \|A_a(t)\|_\infty \le C_{R_2} \big\{ a^{1-\alpha} + a^{2-2\alpha} + a (T_{R_2} + a^2)^{\kappa} \big\} \longrightarrow 0.$$

2. *At a positive restart time, the solutions are smoothed by Theorem 25.7, and the continuations of Theorem 24.10 are used on a neighborhood of the compact closure of their possible limits; on each restart, the same fixed sampled gauge transformation is applied to all parameters in a neighborhood. The first and second parameter derivatives of the resulting based holonomy*

matrices of the three coordinate cycles on $[t_*, T]$, expressed in the gauge of the initial configuration, are bounded by a constant $C_{R_2,U_0,U_1,t_*,T}$ uniformly in the mesh. These matrices therefore have a uniform $C^{1,1}$ bound on smaller convex parameter charts. The conjugation at the basepoint by all damping and restart gauge transformations is included; this is a statement about matrices, not only about their traces.

3. *Let $f(u)$, $u \in U_1$, be a fixed compact family of finite sums of smooth cardinal interpolants, twice continuously differentiable in u, and let $A_{0,a}(u) = S_a f(u)$. Then the matrices of (2) converge, uniformly in $u \in U_0$ and $t \in [t_*, T]$, to the holonomy matrices, based at x_0, of the solution of $\partial_t A_i = \sum_j D_j F_{ji}$ with initial value $f(u)$; the proof shows that their first two parameter derivatives converge as well. The divergence generators of the damping gauge transformations converge on positive time intervals and in L^1 down to time zero, and the damping gauge transformations with initial value I converge at the basepoint. The same conclusions hold when the finitely many coefficients of the family depend on a and converge together with their first two parameter derivatives. No rate that is uniform over growing families is asserted.*

The constants in (1) and (2) depend only on d, U_0, U_1, and the margins of the input charts, on R_2, L, α, T_0, t_*, T, and on the chart and the margins of the continuations. The bounds for the based holonomy matrices in (2) hold for every mesh $a = L/n$, $n \geq 4$. The assertions about logarithmic representatives and their damping gauge transformations hold for $a \leq a_*$; finitely many additional meshes can be included when the corresponding representatives remain in the fixed smaller chart. The constants in (3) may also depend on the fixed family. The derivatives are real derivatives of the Wilson flow; neither a complex continuation of the flow nor a derivative of a Borel selection is involved.

We outline the proof. On the early interval, where the flow data are singular, we differentiate the fixed-point equation for the remainder $r = A - x - y$ of Theorem 24.5 twice in the parameters. The second derivative of the quadratic term contains products of two differentiated linear heat evolutions, whose singularity $v_a^{-\alpha-1/2}$ is not integrable. These products cancel against the corresponding products in the definition of r, and the remaining terms have the integrable singularity $v_a^{-3\alpha/2}$. We control the parameter derivatives of the damping gauge transformation by unitary variation of constants. After a positive time, the solutions are smooth uniformly in the mesh, and we continue them by comparison with finitely many selected smooth DeTurck solutions, as in Lemma 25.9. Along this continuation, Lemma 25.12 controls the parameter derivatives. Finally we return to the gauge of the initial configuration by conjugating with the composed gauge transformation at the basepoint, and differentiate this conjugation as well. For a fixed smooth family, the consistency results of Sections 25.2 and 25.3, differentiated twice in the parameters, identify the limit.

Proof. Step 0: setup. We work first in one of the finitely many input charts. Since changes between these charts are gauge transformations equal to the identity at the basepoint, they do not change the gauge in which the matrices are expressed. We choose an intermediate parameter domain V with $\overline{U}_0 \subset V \Subset U_1$. We take all difference quotients inside U_1 and first establish all bounds on V; the minimal parameter margin and the constants of the charts and of the conversion of Euclidean norms enter the final constant. In formulas involving directions, v, w have Euclidean norm at most one. Set

$$h_0 = T_{R_2}, \qquad p = \tfrac{3}{2}\alpha < 1,$$

and recall $\beta = \alpha - 1/2$, $\kappa = 1 - 3\alpha/2 = 1 - p$, and $v_a(t) = t + a^2$. Using the contraction proof of Theorem 24.5, we may decrease the local existence time while restricting the horizon of the flow data. All weighted norms and all the gains of the contraction then only decrease, and none of the bounds on the early remainder becomes larger. We choose h_0 small enough for the contractions below, and take $a^2 \le h_0$ and $a \le 1$. All heat integrals in the early argument have physical horizon h_0.

Step 1: the differentiated equation on the early interval. We write $x = x_a(u)$, $y = y_a(u)$, and $A = x + y + r$, and let

$$(\mathcal{T}_a f)(t) = \int_0^t P^a_{t-s} f(s)\, ds, \qquad F_a((x,y),r) = \mathcal{T}_a\big\{\mathcal{N}_a(A,A) - \mathcal{N}_a(x,x) + \mathscr{C}_a(A)\big\}.$$

The remainder satisfies the fixed-point equation $r = F_a((x,y),r)$ exactly, with no term of the vector field truncated. By Theorem 24.5, r lies in a ball of fixed radius $D = D(R_2)$ in the norm R_a. Using the estimates of the local contraction on a slightly larger fixed ball, we obtain

$$\|A(t)\|_\infty \le C_{R_2,D} v_a(t)^{-\alpha/2}, \qquad \|D_a A(t)\|_\infty \le C_{R_2,D} v_a(t)^{-(\alpha+1)/2}. \tag{25.27}$$

Indeed, the exponents β, α of y are smaller than the corresponding exponents $\alpha/2, (\alpha+1)/2$ of x, the remainder has exponents $\kappa, \kappa - 1/2$, and $v_a \le 2$. The derivative in the remainder variable is

$$\mathsf{L}_a \rho = D_r F_a((x,y),r)\rho = \mathcal{T}_a\{\mathscr{B}_a(A)\rho\}.$$

If $\|\rho\|_{\mathsf{R}_a} \le 1$, its source is bounded by

$$C_{R_2,D}\big\{v_a^{\kappa-(\alpha+1)/2} + v_a^{\kappa-\alpha}\big\};$$

the first term comes from the two ordered quadratic products and the second from $D\mathscr{C}_a(A)\rho$. Both exponents exceed -1. Bounding the value of the convolution and one difference of the heat semigroup, and then applying (25.1), we obtain

$$\|\mathsf{L}_a\|_{\mathsf{R}_a \to \mathsf{R}_a} \le C_{R_2,D}\big\{(h_0 + a^2)^{(1-\alpha)/2} + (h_0 + a^2)^{1-\alpha}\big\} \le \tfrac{1}{2}, \qquad \|(I - \mathsf{L}_a)^{-1}\| \le 2. \tag{25.28}$$

We choose h_0, and then the restriction $a \le a_*$ on the mesh, so that this holds on the slightly larger ball. The inverse is given by the norm-convergent Neumann series, and it acts on the whole Volterra integral.

For a direction (ξ, η) in the flow data, the derivative of the source with r fixed is

$$\begin{aligned} J_a(\xi,\eta) &= \mathcal{N}_a(x,\eta) + \mathcal{N}_a(\eta,x) \\ &\quad + \mathcal{N}_a(y+r,\xi+\eta) + \mathcal{N}_a(\xi+\eta,y+r) + D\mathscr{C}_a(A)(\xi+\eta); \end{aligned} \tag{25.29}$$

the products $\mathcal{N}_a(x,\xi)$ and $\mathcal{N}_a(\xi,x)$ cancel against the derivative of $\mathcal{N}_a(x,x)$. If $b = \|\xi\|_{\mathsf{X}^a_\alpha} + \|\eta\|_{\mathsf{Y}^a_\alpha}$, then

$$\|J_a(\xi,\eta)(t)\|_\infty \le C_{R_2,D}\, b\, v_a(t)^{-p}, \qquad \|\mathcal{T}_a J_a(\xi,\eta)\|_{\mathsf{R}_a} \le C_{R_2,D}\, b. \tag{25.30}$$

To verify the time exponents, recall that $\|\mathcal{N}_a(u,v)\|_\infty \le C\|u\|_\infty \|D_a v\|_\infty$. The products of one factor x or ξ with one factor y or η have worst exponents $-\alpha/2 - \alpha = -p$ and $-\beta - (\alpha+1)/2 = -p$. Two factors of the type y, η give the exponent $1/2 - 2\alpha$. A remainder factor against a factor of type x, ξ gives the same exponent $1/2 - 2\alpha$, and against a factor of type y, η the exponent $\kappa - \alpha$. All of these are at least $-p$ for $\alpha < 2/3$. Finally, (25.11) bounds the last term by $C\|A\|_\infty^2 \|\xi+\eta\|_\infty$, again with exponent $-p$. Since $\kappa = 1 - p$, the convolutions have exactly the powers v_a^κ and $v_a^{\kappa-1/2}$ of the norm R_a.

For fixed a, the initial value $A_0(u) = x_a(u,0)$ is twice continuously differentiable in u. The differential equation in the chart, or difference quotients of its contracting integral equation, allow us to differentiate $r = F_a((x,y),r)$ twice. To justify the second route, we subtract two fixed-point identities and integrate the derivative in the field along the segment joining them. The difference of the remainders is then obtained by applying an inverse of norm at most two to the difference in the flow data. By Taylor's integral remainder and (25.30), the first difference quotient converges. Repeating this with the first differentiated equation, and using the second field derivatives of the analytic vector field, we obtain the convergence of the second difference quotient. This justifies the differentiation before we derive the uniform bounds below.

We write $\xi_v = \partial_v x$, $\eta_v = \partial_v y$, $\rho_v = \partial_v r$, and $Z_v = \partial_v A$. The first derivative satisfies

$$(I - \mathsf{L}_a)\rho_v = \mathcal{T}_a J_a(\xi_v, \eta_v), \qquad Z_v = \xi_v + \eta_v + \rho_v. \tag{25.31}$$

By hypothesis, $b \le R_2$ here, and (25.28) and (25.30) imply that $\|\rho_v\|_{\mathsf{R}_a} \le C_{R_2}$. The second derivative satisfies, with all inhomogeneous terms,

$$\begin{aligned} (I - \mathsf{L}_a)\rho_{vw} &= \mathcal{T}_a\{J_a(\xi_{vw},\eta_{vw}) + K_{vw}\}, \\ K_{vw} &= \mathcal{N}_a(Z_v,Z_w) + \mathcal{N}_a(Z_w,Z_v) - \mathcal{N}_a(\xi_v,\xi_w) - \mathcal{N}_a(\xi_w,\xi_v) \\ &\quad + D^2\mathscr{C}_a(A)[Z_v,Z_w]. \end{aligned} \tag{25.32}$$

The two subtracted products of differentiated linear heat evolutions must be kept: bounding the two unsubtracted products separately would produce the exponent $-\alpha - 1/2$,

which is not integrable for $\alpha > 1/2$. We write $Z_v = \xi_v + \delta_v$ with $\delta_v = \eta_v + \rho_v$. Each ordered quadratic difference in K_{vw} then contains at least one factor δ; for instance,

$$\mathcal{N}_a(Z_v, Z_w) - \mathcal{N}_a(\xi_v, \xi_w) = \mathcal{N}_a(\xi_v, \delta_w) + \mathcal{N}_a(\delta_v, \xi_w) + \mathcal{N}_a(\delta_v, \delta_w).$$

The part η of δ has the exponents of the quadratic Picard term, and the part ρ has the exponents of the remainder, which were bounded above. By the exponent calculation for (25.30), the sum of these terms is bounded by $C_{R_2} v_a^{-p}$. For the last term we use $\|D^2\mathscr{C}_a(A)[Z_v, Z_w]\|_\infty \le C\|A\|_\infty \|Z_v\|_\infty \|Z_w\|_\infty$; each factor has the value bound $C v_a^{-\alpha/2}$, which again gives the exponent $-p$. In total, the second forcing has, before inversion, R_a norm at most C_{R_2}, and by (25.28) we have $\|\rho_{vw}\|_{\mathsf{R}_a} \le C_{R_2}$. These directional estimates give the asserted bounds on both parameter derivatives in Euclidean operator norm, with all directions and all first physical differences included.

Adding the differentiated flow data to these two estimates for the remainder, we find, for $r = 1, 2$,

$$\|D_u^r D_a A(u,t)\|_\infty \le C_{R_2}\big\{ v_a(t)^{-(\alpha+1)/2} + v_a(t)^{-\alpha} + v_a(t)^{\kappa-1/2} \big\}. \tag{25.33}$$

The same bound, with a possibly larger C_{R_2}, holds for $r = 0$. All three negative exponents are smaller than one in absolute value. In particular, the integrals of the right side over $[0, \epsilon]$ are at most

$$C_{R_2}\, \omega(\epsilon), \qquad \omega(\epsilon) = \epsilon^{(1-\alpha)/2} + \epsilon^{1-\alpha} + \epsilon^{\kappa+1/2}, \tag{25.34}$$

uniformly in a, for $0 < \epsilon \le h_0$. Finally, the representatives remain in the chart because

$$a \sup_{t \le h_0} \|A(t)\|_\infty \le C_{R_2}\big\{ a^{1-\alpha} + a^{2-2\alpha} + a(h_0 + a^2)^\kappa \big\} < r_*;$$

in the last term we used the maximum of the remainder over the interval, which is attained at the end and not at time zero. The differentiated mild solution is therefore, for every parameter, the damped representative of the Wilson flow.

Step 2: the damping gauge transformation on the early interval. The generator in physical units is linear in the potential:

$$\lambda_a = a^{-2}\phi(aA) = -\sum_i D_{a,i}^- A_i, \qquad \partial_t k_a = \lambda_a k_a, \qquad k_a(0) = I. \tag{25.35}$$

It is anti-Hermitian. Its parameter derivatives of order zero, one, and two are bounded in $L^1([0,h_0]; \ell^\infty)$ by (25.33), and their integrals over $[0,\epsilon]$ are bounded by (25.34). At any vertex write $l_v = \partial_v k_a$ and $m_{vw} = \partial_v \partial_w k_a$. Direct differentiation gives

$$\begin{aligned} \partial_t l_v &= \lambda_a l_v + (\partial_v \lambda_a) k_a, \qquad l_v(0) = 0, \\ \partial_t m_{vw} &= \lambda_a m_{vw} + (\partial_v \lambda_a) l_w + (\partial_w \lambda_a) l_v + (\partial_{vw} \lambda_a) k_a, \qquad m_{vw}(0) = 0. \end{aligned} \tag{25.36}$$

The homogeneous propagator is unitary. If $B_j = \int_0^{h_0} \|D_u^j \lambda_a\|_\infty \, dt$ for $j = 1, 2$, variation of constants gives

$$\sup_t \|l_v(t)\|_\infty \le B_1, \qquad \sup_t \|m_{vw}(t)\|_\infty \le B_2 + 2B_1^2.$$

These bounds hold at every vertex, uniformly in the mesh, with no exponential of the size of λ_a, and they include the two products of first derivatives in the second equation. Differentiating $k_a^{-1} k_a = I$, we also bound both derivatives of k_a^{-1}. This completes the proof of (1).

Step 3: bounds on later intervals. Fix $0 < s < \min(h_0/2, t_*/2)$, which leaves room for a positive waiting time before s. By the early estimates we have uniform C_a^1 bounds on A, Z_v, and Z_{vw} on $[s/2, h_0]$. We apply Theorem 25.7 to A, and then Lemma 25.12 on intervals after positive waiting times inside the early interval. At time s these give bounds on every given fixed order of differences of the three fields, with constants depending on that order, on L, R_2, s, and on the parameter data.

Next, we justify the use of finitely many selected smooth DeTurck solutions, although we have not assumed that the flow data converge as the mesh varies. Let $\mathscr{K}$ be the set of smooth connections whose derivatives of every order satisfy the bounds just obtained for $A_a(s, u)$; it depends only on L, R_2, s. By Lemma 25.6, it is compact in C^3, it contains every limit of a sequence $A_{a_j}(s, u_j)$ with $a_j \to 0$, and

$$\sup_u \inf_{b \in \mathscr{K}} \|A_a(s, u) - S_a b\|_{C_a^1} \longrightarrow 0 \qquad (a \to 0). \tag{25.37}$$

Only the spatial bounds at time s are needed here.

We apply part (4) of Lemma 25.5 to $\mathscr{K}$ with horizon $T - s$. It provides $\eta_{\mathscr{K}} > 0$ and finitely many continuations, built from the DeTurck solutions of Theorem 24.10 with $r = 3$, whose needed C^5 norms are finite. It also provides one restriction $a \le a_{\mathscr{K}}$ and one C_a^1 bound for all comparisons with lattice fields within $\eta_{\mathscr{K}}$ of the sample of an element of $\mathscr{K}$. Then (25.37) implies that for every sufficiently small mesh and every u there exists $b \in \mathscr{K}$ with $\|A_a(s, u) - S_a b\|_{C_a^1} \le \eta_{\mathscr{K}}/4$. By the early bound on the first parameter derivative, we have

$$\|A_a(s, u') - A_a(s, u)\|_{C_a^1} \le C_{R_2, s} |u' - u|$$

on every segment in V. Hence, for $|u' - u|$ smaller than the input margin and smaller than $\eta_{\mathscr{K}} / (4 C_{R_2, s})$, the field $A_a(s, u')$ lies within $\eta_{\mathscr{K}}/2$ of the same $S_a b$. Then all parameters in this neighborhood of u can use the same continuation, with the same fixed sampled gauge transformations. Then the individual estimate (25.14) gives the C_a^1 bound for the representatives through time T. This estimate applies to an incoming small discrepancy in C_a^1; it does not require a link error of order a^2.

The two initial parameter derivatives at time s already have uniform C_a^1 bounds. Applying (25.24) and (25.26) along this finite continuation, we obtain one uniform C_a^1

bound for the first and second parameter derivatives on every time interval, including its endpoints. Only after the C^1_a bound for the representative has been obtained in this way do we apply the last part of Theorem 25.7. Together with the last part of Lemma 25.12, it gives further spatial orders at every given positive waiting time.

Step 4: composed gauge transformations and matrices. On a later time interval I_j, let $g_j(t)$ be the damping gauge transformation with initial value I at the start of I_j. By (25.35) and the C^1_a bounds just proved, its generator and the first two parameter derivatives of the generator are bounded. Then the equations (25.36) bound the first two parameter derivatives of g_j on I_j. We keep track of the whole gauge transformation at every vertex, and not only of its value modulo a conjugation at the basepoint. If Q_j is the composed gauge transformation at the start of I_j, then

$$Q(t) = g_j(t)Q_j, \qquad Q_{j+1} = k_j\, g_j(h_j)Q_j, \qquad Q_0 = k_a(s), \tag{25.38}$$

where h_j is the length of I_j and k_j is the fixed sampled gauge transformation applied at its end. The order of these products follows from $(U^k)^l = U^{lk}$. Since all factors are unitary, the k_j do not depend on the parameter, and there are finitely many of them, the first and second product rules bound $D_u Q$ and $D_u^2 Q$ at every vertex. This includes the early damping gauge transformation with its initial value I.

Let $H^{\mathrm{rep}}_a(t,u)$ be the holonomy of one of the coordinate cycles in the current damped representative. If the cycle has N links and $aN \le \ell$, differentiating a single exponential gives

$$\|\partial_v e^{aA_e}\|_{\mathrm{op}} \le Ca\|Z_v\|_\infty, \qquad \|\partial_{vw} e^{aA_e}\|_{\mathrm{op}} \le Ca\|Z_{vw}\|_\infty + Ca^2\|Z_v\|_\infty\|Z_w\|_\infty;$$

these follow by integrating the derivative of the exponential once and twice along its unitary factors. An inverse factor satisfies bounds of the same type, by differentiating the identity for the inverse. The product rule over the links of the cycle gives

$$\begin{aligned} \|\partial_v H^{\mathrm{rep}}_a\|_{\mathrm{op}} &\le C\ell\|Z_v\|_\infty, \\ \|\partial_{vw} H^{\mathrm{rep}}_a\|_{\mathrm{op}} &\le C\ell\|Z_{vw}\|_\infty + C(\ell a + \ell^2)\|Z_v\|_\infty\|Z_w\|_\infty. \end{aligned} \tag{25.39}$$

The term ℓa counts the N second derivatives of a single factor, and the term ℓ^2 counts the at most $N(N-1)$ products of two first derivatives. Since every observation time is later than s, the fields in these estimates are uniformly bounded by Step 3. At the basepoint x_0, the holonomy matrix of the Wilson flow in the gauge of the initial configuration is

$$H_a(t,u) = Q(t,x_0)^{-1} H^{\mathrm{rep}}_a(t,u)\, Q(t,x_0). \tag{25.40}$$

Its two derivatives include the derivatives of both factors $Q(t,x_0)^{\pm 1}$. By the bounds for Q and Q^{-1} and (25.39), all these terms are bounded uniformly. At a restart time both representations describe the same holonomy matrix of the Wilson flow, so their parameter

derivatives agree there. The choice of a reference neighborhood was used only to prove a bound for the map $u \mapsto H_a(t,u)$, which is defined independently of this choice. The choice is constant on the parameter neighborhood where the derivatives were computed, and we take no derivative of it. Taking the maximum over the finitely many possible continuations, we obtain the uniform bounds on the first and second derivatives on U_0. On a smaller convex chart, the fundamental theorem of calculus along segments and the bound on the second derivative imply that $\|DH_a(u) - DH_a(u')\| \le C|u-u'|$. Together with $\|H_a\|_{\mathrm{op}} = 1$, this is the uniform $C^{1,1}$ bound in (2).

Step 5: smooth families, consistency of the fields. Let $f(u)$ be a fixed compact family as in (3), with its two parameter derivatives on the larger domain. Its spatial derivatives of every fixed order, and its mixed derivatives with up to two parameter derivatives, have uniform finite bounds. The same holds if its finitely many coefficients depend on a and converge together with their first two parameter derivatives. Lemma 25.2 and Lemma 16.9 apply to f, to its parameter derivatives, and to their ordered products. On meshes that do not refine the fixed input lattices, we use the general smooth-input bounds of Lemma 16.9 in place of its bounds for interpolants. The proof of Lemma 25.2 uses only these bounds and the fixed smooth norms of the inputs, so its conclusions hold on those meshes as well. In particular, the quadratic Picard term satisfies the exact identities

$$\begin{aligned}\partial_v y_a &= \mathcal{Q}_a(\partial_v x_a, x_a) + \mathcal{Q}_a(x_a, \partial_v x_a),\\ \partial_{vw} y_a &= \mathcal{Q}_a(\partial_{vw} x_a, x_a) + \mathcal{Q}_a(x_a, \partial_{vw} x_a) + \mathcal{Q}_a(\partial_v x_a, \partial_w x_a) + \mathcal{Q}_a(\partial_w x_a, \partial_v x_a).\end{aligned}$$

Hence the bounds on the flow data through two parameter derivatives, and the consistency of both components of the flow data, hold for this family, with a constant R_2 depending on the family. All heat comparisons are on the same fixed physical horizon. This also covers coefficients that converge as described.

Theorem 24.10 with $r = 3$ applies to the compact family of smooth initial data, and on each of its continuations the gauge transformations do not depend on u. By the continuum part of Lemma 25.12, the DeTurck solutions $b_j(u)$ on the time intervals have two parameter derivatives, with all the spatial bounds needed below. The bounds of the smooth initial family and the finitely many fixed gauge transformations give these bounds also at the left endpoints of the intervals. This verifies the regularity needed to differentiate the residual of the sampled smooth solution, without deducing it from the Lipschitz statement of Theorem 24.10 alone.

Let B_a be the lattice representative and $b_a = S_a b$ the sample of the selected DeTurck solution on a common time interval. Expanding the vector field, we obtain for $r = 0, 1, 2$

$$\sup_{u,t} \big\| D_u^r \{ \partial_t b_a - \Delta_a b_a - \mathcal{N}_a(b_a, b_a) - \mathscr{C}_a(b_a) \} \big\|_\infty \le C a. \tag{25.41}$$

This follows by differentiating the finite analytic expression and its Taylor remainder with respect to u. The integral remainder is bounded by the corresponding mixed spatial and

parameter derivatives of the fixed smooth family. The linear part is central-difference consistency, the quadratic part is the calculation for $\mathcal{N}_a$ in Lemma 25.2, and the remaining terms form the analytic remainder treated in Lemma 25.3. The first physical difference of the residual is consistent as well, since the smooth spatial derivatives of one more order are bounded.

We now prove stability of the derivatives. For the value we use (25.14). Recall that $\mathscr{B}_a(B)V$ is the linearization, and let $f_a(B;V,W)$ be defined by the last line of (25.25), with V, W as the two first-derivative arguments. On a common bounded set of the norm C^1_a, the displayed products and (25.11) imply

$$\|(\mathscr{B}_a(B) - \mathscr{B}_a(C))V\|_\infty \le C_K \|B - C\|_{C^1_a} \|V\|_{C^1_a},$$

and the difference of f_a at two triples is at most

$$C_K\big(\|B - C\|_{C^1_a}\|V\|_{C^1_a}\|W\|_{C^1_a} + \|V - V'\|_{C^1_a}\|W\|_{C^1_a} + \|V'\|_{C^1_a}\|W - W'\|_{C^1_a}\big)$$

when the remaining fields are bounded by K. For the remainder, this uses the bound on the third derivative in (25.11). We subtract the equations for the first derivatives, keep the linear operator at B_a, and place the difference of the coefficients and the differentiated residual in the forcing. By the Volterra inverse of Lemma 25.5(1), the C^1_a difference of the first derivatives is at most the initial difference plus $C(a + \sup \|B_a - b_a\|_{C^1_a})$. We subtract the equations for the second derivatives in the same way. By the bound for f_a, the additional forcing is at most a constant times the errors of the fields and of the first derivatives already controlled, plus the second differentiated residual. Applying the Volterra inverse again, we obtain the convergence of this difference as well. At a fixed sampled gauge transformation, the consistency error and its first two parameter derivatives are $O(a)$ in C^1_a. To see this, we expand $k(x + ae_i)$ to first order in the formula for $\mathscr{G}_{a,k}$ and differentiate twice in the bounded smooth family. The remainder of second order, before division by a, is $O(a^2)$, and one spatial difference uses one more smooth derivative. Together with (25.26), this propagates the three error bounds across the finite continuation.

For initial values that are exact samples, the three initial errors vanish. For converging finite coefficients they are $O(\delta_a)$ with $\delta_a \to 0$, in all needed fixed spatial norms. The preceding argument then gives, on every common time interval,

$$\sup_{u,t} \|D^r_u(B_a - S_a b)\|_{C^1_a} \le C(a + \delta_a), \qquad r = 0, 1, 2. \tag{25.42}$$

For $r = 0$, the sharper link consistency of Theorem 25.4 is also available, since its hypothesis of an initial link error $O(a^2)$ holds at every restart from an exact sample. The estimate in C^1_a above is used for the parameter derivatives and for the generators.

Step 6: smooth families, generators, gauge transformations, and matrices. We verify the convergence of the generators in L^1 on the early interval directly from the weighted

estimates, without a uniform unweighted estimate at time zero. Let $\lambda = -\sum_i \partial_i b_i$, with the sign fixed by (25.35). On every interval $[\epsilon, h_0]$, by the consistency of the values and first differences, we have

$$\sup_u \|D_u^r \lambda_a - S_a D_u^r \lambda\|_\infty \longrightarrow 0, \qquad r = 0, 1, 2.$$

By (25.34), the integral of each lattice term over $[0, \epsilon]$ is at most $C_{R_2}\omega(\epsilon)$. The limiting continuum bound has the same property, by the limit of the same estimates at positive times or by the stronger smooth bounds of the fixed family. Taking first $a \to 0$ at fixed ϵ and then $\epsilon \downarrow 0$, we obtain

$$\sup_u \int_0^{h_0} \|D_u^r \lambda_a - S_a D_u^r \lambda\|_\infty \, dt \longrightarrow 0, \qquad r = 0, 1, 2. \tag{25.43}$$

On later time intervals the generators and their two parameter derivatives are bounded by the C_a^1 estimates of Step 3, and (25.42) implies the analogous convergence in L^1 on each whole interval.

Let k solve $\partial_t k = \lambda k$ with $k(0) = I$ in the continuum. Then $\partial_t(k^{-1}) = k^{-1} \sum_i \partial_i b_i$, and k^{-1} is the gauge transformation that undoes the damping in Theorem 24.10, with the convention used there. At every vertex, unitary variation of constants gives

$$\sup_{t,u} \|k_a(t,u) - S_a k(t,u)\|_{\mathrm{op}} \le \sup_u \int \|\lambda_a - S_a \lambda\|_\infty \, dt \longrightarrow 0. \tag{25.44}$$

For the first derivative we use its integral formula from (25.36). Its homogeneous propagators $k_a(t)k_a(r)^{-1}$ converge uniformly by (25.44), its source generators converge in L^1 by (25.43), and k_a converges uniformly. Subtracting the two integrals, we bound the difference by the L^1 error of the generators plus a uniform error of the factors times a bounded L^1 norm, which proves the convergence of the first derivative. For the second derivative, the two additional sources are $(\partial_v \lambda_a) l_w$ and $(\partial_w \lambda_a) l_v$. Since the first derivatives just treated converge uniformly and the generators converge in L^1, the same subtraction proves the convergence of the second derivative; the term $(\partial_{vw} \lambda_a) k_a$ is handled by its own L^1 convergence. Since the fixed sampled gauge transformation at a restart has the same value at the basepoint as its continuum counterpart, the finite products (25.38) converge at the basepoint, together with their first two parameter derivatives.

On each of the three coordinate cycles, the physical path is the same coordinate circle, of length L and with the same basepoint, for every mesh. Using the estimate for smooth edge transport in Lemma 24.7 and unitary telescoping, we compare the product of the exponentials of the sampled selected DeTurck solution with its smooth holonomy. To include derivatives, we regard this product as the parallel transport, along the same coordinate circle, for a piecewise constant coefficient. This coefficient and its two parameter derivatives differ from the corresponding smooth coefficients by $O(a)$, using

one smooth spatial derivative. Differentiating the one-dimensional transport equation once and twice, we find the same linear and quadratic forcing structure as in (25.36), now over the fixed interval of arclength L. By unitary variation of constants and subtraction, the products and their two parameter derivatives converge at rate $O(a)$. The errors between the lattice fields and the samples in (25.42) add $O(a+\delta_a)$, by the same transport argument or by (25.39) and its version for differences. Finally we undo the gauge transformations at the basepoint by (25.40). The convergence at the basepoint proved above identifies the holonomy matrices of the Wilson flow with the holonomy matrices of the covariant continuum flow, in the gauge of the initial configuration, uniformly over the compact smooth family. This proves (3).

Step 7: dependence of the constants and coarse meshes. All restrictions on the mesh and all constants in Steps 5 and 6 may depend on the fixed smooth family and on the displayed bounds. All bounds in Steps 1–4 used only R_2, the parameter domains, the waiting times, the horizon, the torus, and the margins of the finitely many continuations. For the finitely many meshes with $a > a_*$, the initial fields and their two parameter derivatives are bounded by the weighted norm at time zero, since these meshes are bounded below. Since the Wilson vector field and its first two derivatives are bounded on the corresponding compact products of copies of $\mathrm{SU}(2)$, the real integral equations for the derivatives give finite uniform bounds on the fixed horizon. This gives the bounds for the based holonomy matrices on these meshes, and we take a maximum over this finite set. For any additional meshes on which the damped representatives remain in the fixed smaller chart, the positive chart margin also gives bounds for their logarithms and for the equations of their gauge transformations. □

25.8 Flows with the modified plaquette potential

We now treat the flows of the plaquette potential $q + cq^2$ with $0 \le c \le C_0$, which are used in Section 29 to show that the limit does not depend on the choice of smoothing within this family. As in Theorem 24.12, let

$$\mathcal{A}_c(U) = \sum_{p \in \mathcal{P}_N} \{ q(U_p) + c\, q(U_p)^2 \}, \qquad \Psi_t^{(N,c)} = \Phi^{\mathcal{A}_c}_{4t/a^2}, \qquad a = L/N,$$

where $\Phi^{\mathcal{A}_c}$ is the flow of the vector field $-\nabla \mathcal{A}_c$ for the product metric; thus $\Psi^{(N,0)} = \Psi^{(N)}$. By Lemma 24.11, the damped representatives of this flow solve

$$\partial_t A = \Delta_a A + \mathcal{N}_a(A, A) + F_a(A) + E_{a,c}(A)$$

in a smaller chart, where $F_a(A) = a^{-3} R_3(aA) = \mathscr{C}_a(A)$ is the cubic-and-higher remainder and $E_{a,c}$ is the additional analytic term coming from cq^2. Since the linear and quadratic parts are the same as for the Wilson flow, the flow data are unchanged. The additional

term satisfies the bounds on field derivatives (24.32), the bounds on differences (24.33), and the stronger bound (24.34), which makes it $O(a^3)$ on bounded sets of the norm C_a^1. Only bounded $c \geq 0$ is considered.

Lemma 25.14. *Fix $L > 0$, $0 \leq C_0 < \infty$, and a smaller chart.*

1. *For flow data bounded by R, the solutions of Theorem 24.12 have bounds on every fixed order of physical differences on every interval after a positive waiting time, uniformly in a and $c \in [0, C_0]$. The same holds on any restarted time interval for which a uniform C_a^1 bound is given, with the dependence stated in Theorem 25.7 and an additional dependence on C_0.*
2. *Let b be a fixed smooth periodic DeTurck solution on $[0, h]$ whose spatial derivatives through order five are bounded by K_b. There are $\eta_b, a_b > 0$ and $K_b' < \infty$, depending only on L, h, K_b, C_0, and the chart, such that, whenever*

$$\|B_a(0) - S_a b(0)\|_{C_a^1} \leq \eta \leq \eta_b, \qquad a \leq a_b,$$

the damped flow for $\mathcal{A}_c$ exists in the chart through time h and

$$\sup_{t \leq h} \|B_a(t) - S_a b(t)\|_{C_a^1} \leq K_b'(\eta + a), \qquad \sup_{t \leq h} \|B_a(t)\|_{C_a^1} \leq K_b'. \tag{25.45}$$

This holds uniformly for $c \in [0, C_0]$, and also on each time interval of a fixed continuation as in Theorem 24.10, with its fixed sampled gauge transformations at the restarts. In particular, every sufficiently small initial discrepancy in C_a^1 is allowed.

The constants are deterministic and do not depend on the mesh below the thresholds.

Proof. Part (1). The local solution of Theorem 24.12 gives the initial C_a^1 bound after a positive waiting time. Suppose that differences through order $m-1$ have been bounded by $K_1, \dots, K_{m-1}$ on a preceding interval. By the product rules (25.2), we have

$$\|D_a^{m-1} \mathcal{N}_a(B, B)\|_\infty \leq P_m(K_1, \dots, K_{m-1}) + C_m K_1 \|D_a^m B\|_\infty,$$

since the only factor of order m occurs when all additional differences fall on the factor that already carries a difference, and its companion is a bounded value of B. By the composition rule of Lemma 24.2 and by (24.33), both analytic terms F_a and $E_{a,c}$ contribute only a bounded function of $K_1, \dots, K_{m-1}$. Indeed, their arguments carry no difference to begin with, and all their powers of the mesh are nonnegative. We start the mild equation at the left endpoint u_0 of the preceding interval, commute $m-1$ differences onto the initial value and onto the source, and place one difference on the heat semigroup. For $Z_m(d) = \sup_{0<v\leq d} \sqrt{v} \|D_a^m B(u_0 + v)\|_\infty$, using the bound for one difference of the heat semigroup, we obtain

$$Z_m(d) \leq C_m K_{m-1} + C_m P_m \, d + C_m K_1 \sqrt{d} \, Z_m(d),$$

where for the highest term we used $\int_0^v (v-r)^{-1/2} r^{-1/2}\, dr = \pi$. At fixed mesh, $Z_m(d)$ is finite before the absorption. We choose d so that the last coefficient is at most $1/2$, and cover the smaller interval by intervals of this length. By induction, every fixed order is bounded, with uniform constants of the stated dependence.

Part (2). Let $b_a = S_a b$, and define

$$e_{a,c} = \partial_t b_a - \Delta_a b_a - \mathcal{N}_a(b_a, b_a) - F_a(b_a) - E_{a,c}(b_a).$$

The residual of Theorem 25.4(1) bounds the residual for the Wilson flow in physical units by $K_{K_b} a$, after dividing the dimensionless residual $O(a^4)$ of the link logarithms by a^3. By (24.34), the additional term is at most $K_{K_b,C_0} a^3$. Hence $\sup_t \|e_{a,c}\|_\infty \le Ka$, and no difference of this residual will be needed. Let $Z = B_a - b_a$, and stop at the first time at which $\|Z\|_{C^1_a} = 1$. Apart from $-e_{a,c}$, the source of order zero in the equation for Z is

$$\mathcal{N}_a(b_a, Z) + \mathcal{N}_a(Z, b_a) + \mathcal{N}_a(Z, Z) + (F_a + E_{a,c})(b_a + Z) - (F_a + E_{a,c})(b_a).$$

By the bound on $\mathcal{N}_a$ and the bounds on the cubic field derivatives, its supremum norm is at most $K_{K_b,C_0} \|Z\|_{C^1_a}$ on this interval. The stopped comparison with the smooth DeTurck solution b and the induction over restarts in the proof of Lemma 25.5 apply with this larger local Lipschitz constant and the residual just bounded. They prove (25.45), remove the stopping condition, and give the assertion for continuations, with the additional dependence on C_0. □

Lemma 25.15. *Fix $L > 0$, $1/2 < \alpha < 2/3$, $0 \le R, C_0 < \infty$, $0 < t_* \le T < \infty$, and $0 \le \ell < \infty$. There is a deterministic function $\omega(a)$ with $\omega(a) \to 0$ as $a \downarrow 0$ with the following property. Let $A_{0,a}$ be a lattice initial potential whose flow data on the common early interval $[0, T_{R,C_0}]$ of Theorem 24.12 have norm at most R, and let $U^c_a(t) = \Psi^{(N,c)}_t(e^{aA_{0,a}})$. Then, for every sufficiently small mesh,*

$$\sup_{0\le c\le C_0} \sup_{t_*\le t\le T} \sup_{\substack{\ell_a \text{ closed path} \\ aN_{\ell_a}\le \ell}} \left| \tfrac{1}{2} \operatorname{tr} \operatorname{Hol}_{\ell_a} U^c_a(t) - \tfrac{1}{2} \operatorname{tr} \operatorname{Hol}_{\ell_a} U^0_a(t) \right| \le \omega(a). \tag{25.46}$$

The same holds for any fixed finite collection of characters of finite-dimensional representations of SU(2). *No approximation or area hypothesis on the closed paths is needed, since paths in the same lattice are compared. The function ω may depend on the displayed data, on the representations, and on the chart; no rate is asserted.*

Proof. Fix $0 < s < \min(t_*, T_{R,C_0})$ such that there is a positive waiting time before s. The early comparison (24.36), evaluated at time s, gives

$$\sup_c \|A^c_a(s) - A^0_a(s)\|_{C^1_a} \longrightarrow 0$$

uniformly over the initial potentials considered. We prove the remaining uniform assertion by contradiction, which avoids the need for a single smooth continuation that is uniform over arbitrary families with bounds of finite order. Suppose that the assertion fails. Then there exist meshes $a_j \downarrow 0$, initial potentials with flow data bounded by R, parameters c_j, closed paths, and times that exhibit a fixed positive discrepancy. On a fixed interval before s, all fixed differences of the Wilson fields $A^0_{a_j}$ are bounded by Theorem 25.7.

By Lemma 25.6, applied at time s, there exist a subsequence and a smooth periodic b_s such that

$$\| A^0_{a_j}(s) - S_{a_j} b_s \|_{C^1_{a_j}} \longrightarrow 0.$$

Only the spatial bounds at time s are needed here. (Since the time derivatives of all differences are bounded by the equation and the bounds on the next two differences, Lemma 25.8 would give convergence on the whole interval before s as well.)

We use the continuation of Theorem 24.10 for the single initial value b_s, with $r \geq 5$, through the horizon $T - s$. Its finitely many DeTurck solutions have finite smooth bounds, and its gauge transformations are fixed and smooth. The endpoints at time s of the Wilson flow and of the flows for $\mathcal{A}_c$ approach the sample of b_s in C^1_a, the latter uniformly in c. We apply Lemma 25.14 on each time interval of the continuation (for $c = 0$ this is also Lemma 25.5). In the gauges of the continuation, both flows approach the sample of the same selected DeTurck solution uniformly in C^1_a on the finite continuation, with error $K(\eta_j + a_j)$, where $\eta_j \to 0$, uniformly over $c \in [0, C_0]$.

For anti-Hermitian X, Y, the integral formula for the derivative of the exponential implies that $\| e^{aX} - e^{aY} \| \leq Ka |X - Y|$. We telescope over the $N_{\ell_a} \leq \ell / a$ unitary factors of a closed path, including the inverse factors. The matrix error is then at most $K\ell \| B_a - S_a b \|_\infty$, where b is the selected DeTurck solution at the given time. Since a conjugation at the basepoint does not change the character, each of the two character values approaches the character of the product of the exponentials $e^{aS_a b}$ along the same path, uniformly in the path and in the time. For a fixed representation, the same calculation in its unitary matrices involves in addition the norm of the corresponding representation of $\mathfrak{su}(2)$ and its dimension. This contradicts the fixed positive discrepancy. The supremum of the bounded left side of (25.46) over the initial potentials considered defines a function of a that tends to zero, which proves the lemma. □

PART VIII

Fluctuation estimates and Wilson-loop convergence

SECTION 26

Concentration of reconstruction fluctuations

This section and the next control the heat evolution of the reconstructed field and its quadratic Picard term under the conditioned Wilson measure. Recall from Section 3.6 that the field on the finest lattice is reconstructed from the coarse field one renormalization group step at a time, from coarse to fine, by adding the normalized fluctuation variables of each step. At the step from mesh a_{j-1} to mesh a_j, the heat evolution x_j of the reconstructed field changes by $d_j = x_j - x_{j-1}$. In Lemma 26.2, we write d_j exactly as a sum of four terms:

- the linear fluctuation s_j, a linear function of the newly integrated fluctuation variables after subtraction of their conditional mean;
- the heat evolution z_j of the conditionally centered nonlinear reconstruction residual;
- an interpolation defect r_j, which compares the linearized reconstruction of the coarser field with the direct interpolation of that field;
- a predictable term u_j, which contains the nonlinear part of the zero-fluctuation reconstruction and the conditional mean correction, that is, the conditional mean of the difference between the reconstructed field and its zero-fluctuation reconstruction.

The quadratic Picard term $y_j = \mathcal{Q}_\delta(x_j, x_j)$ then changes by the two mixed terms $\mathcal{Q}_\delta(x_{j-1}, d_j)$, $\mathcal{Q}_\delta(d_j, x_{j-1})$, and by the sixteen ordered products of the four terms.

This section proves the estimates that involve the newly integrated fluctuation variables. Theorem 26.10 bounds the linear fluctuation s_j and the mixed Duhamel terms of s_j with a predictable background field, in either order; the backgrounds of interest are x_{j-1} and r_j. Theorem 26.11 bounds z_j and its mixed terms with x_{j-1}. Theorem 26.12 bounds the quadratic Duhamel term $\mathcal{Q}_\delta(s_j, s_j)$ and, separately, its conditional mean, which is not zero. The terms built only from r_j and u_j are treated in Section 27. Section 28 assembles the estimates of both sections into bounds on the linear and quadratic flow data of the reconstructed field, and the coefficient bounds proved here are also used in Sections 29 and 34.

At mesh $a = a_j$, the linear fluctuation has size $g_3 a^{\alpha - 1/2}\sqrt{\log(2 + L/a)}$ in the norm X^δ_α, and the heat evolution of the nonlinear reconstruction residual has size $g_3^2 a^\alpha \log(2 + L/a) + g_3^3 a^{1/2} p_j^3$, where p_j is the largest fluctuation cutoff radius of the step. The centered part of $\mathcal{Q}_\delta(s_j, s_j)$ has size $g_3^2 a^{2\alpha - 1}\log(2 + L/a)$ in the norm Y^δ_α, and its conditional mean is at most $C g_3^2 \|\theta_{j-1}\|_\infty$, where θ_{j-1} is the logarithm of the coarser reconstructed field. Since

$1/2 < \alpha < 2/3$ and p_j grows only logarithmically, the first three rates are summable over the geometric sequence of meshes. The mean is summable because the coarser fields are bounded by a summable deterministic sequence; see (26.8).

Heuristics. At times t of order a^2, a linear reconstruction fluctuation is a lattice potential of size $g_j/a = g_3 a^{-1/2}$, where $g_j^2 = g_3^2 a$ is the coupling at mesh a; multiplying by the weight a^α of X_α^δ gives $g_3 a^{\alpha-1/2}$. A quadratic reconstruction fluctuation has the coefficient $g_j^2/a = g_3^2$, and the weight a^η of the corresponding heat norm gives $g_3^2 a^\eta$. The source of the quadratic Duhamel term contains one lattice derivative, and for the linear fluctuation it has size $g_3^2 a^{-2}$. Integration over a time interval of length a^2 and the weight $a^{2\alpha-1}$ of Y_α^δ give $g_3^2 a^{2\alpha-1}$. The maximum over space and time contributes a factor $\sqrt{\log(2+L/a)}$ for linear forms in the fluctuation variables and $\log(2+L/a)$ for quadratic forms. Scaling alone does not give the long-time bounds. They come from the spatial averaging of the heat kernel, and their proof keeps track of the physical volume of each lattice cell.

Inputs and two points that need care. The probabilistic input is the conditional fluctuation kernel of Theorem 13.8: given the coarser field, the normalized fluctuation variables of one renormalization group step have the density $Z^{-1}\mathbf{1}_\Omega e^{-V}$ on a product Ω of three-dimensional balls, with $D_w^2 V \geq I/2$. By the logarithmic Sobolev inequality of Theorem 5.4, linear and quadratic functions of these variables concentrate, with constants independent of the dimension. We also use the bounds on conditional means and covariances of Theorem 13.10, and the reconstruction maps $\widehat{\mathscr{R}}$ of Theorem 15.12, with their disintegration (Lemma 15.10) and uniform bounds on their absolute Taylor coefficients. The deterministic inputs are the cardinal interpolation and the chaining lemma of Lemmas 16.3 and 16.4, the cancellation estimate of Lemma 17.2, the covariance estimate of Theorem 17.4, and the quadratic coefficient of the damped Wilson flow and its Duhamel estimates from Lemmas 24.2 and 24.6 and Definition 24.3.

Two points need care. First, the normalized fluctuation variables are neither Gaussian nor independent, and their conditional mean is not zero when the coarser field is not the identity. We subtract the conditional mean under the conditional fluctuation kernel itself, and we keep the nonzero mean of every quadratic term. Second, all fields are interpolated to the finest lattice, of mesh δ, while their randomness lives at the much coarser mesh a. To obtain the factor $a^{3/2}$ in the operator bounds, and logarithms of L/a rather than of L/δ in the maximal inequalities, we keep the cell volume of each coefficient on its own lattice and cover space and time at the scale a.

We separate the heat evolution of rough initial data from its quadratic Picard term as Cao and Chatterjee do for the Yang–Mills heat flow with random initial data in [42, Definition 2.8 and Theorem 2.10]; their treatment follows the approach of Bourgain [32] and Da Prato and Debussche [63]. The first nonlinear correction of Cao and Chatterjee also contains a cubic term, and their random-data theorem treats a class of initial data

that includes Gaussian free fields; here the estimates are proved from the conditional fluctuation kernels. The concentration inequalities use the convexity methods of Gross [103] and Bakry and Émery [8], in the form proved in Theorem 5.4. The quadratic tail bound is of Hanson–Wright type [108]; for dependent coordinates satisfying a concentration inequality, compare Adamczak [1, Theorem 2.5]. The maximal inequalities use elementary chaining in the sense of Dudley [79] (see also [74]), in the form of Lemma 16.4.

26.1 Setting and the exact decomposition of an increment

The hierarchy. Throughout this section we fix $L > 0$, $\gamma = g_3^2 L > 0$, an odd integer $M \geq 3$, and an integer $J \geq 1$ for which the constructions of Theorems 13.8 and 15.12 apply, and we set $B = M^J$. We fix a coarse side $n \in 4\mathbb{N}$ at least the threshold of Theorem 15.12. For $K \geq 0$ we set $N = nB^K$ and

$$a_j = (L/n)B^{-j}, \qquad g_j = g_3\sqrt{a_j} \qquad (0 \leq j \leq K), \qquad \delta = a_K,$$

so that $g_j^2 = \gamma/(nB^j)$ is the lattice coupling at mesh a_j. The jth renormalization group step from the coarse side maps the lattice of mesh a_j to the lattice of mesh $a_{j-1} = Ba_j$; it consists of J averaging steps, and its intermediate scales have couplings $g_{j,i}$, $0 \leq i \leq J-1$, whose ratios to g_j lie in an interval depending only on M and J. The largest of them is

$$t_j = M^{(J-1)/2} g_j, \qquad t_j^2 = \frac{\gamma}{M m_j}, \qquad m_j = nB^{j-1}.$$

Here $m_j \geq 4$ is the side of the coarser lattice of the step. The zero-fluctuation reconstruction at this step is built from a classical background field whose effective classical action has been obtained by $\mathfrak{d}_j = J(K-j)$ preceding averaging steps, counted from the finest lattice.

The measure and the filtration. We use the conditioned Wilson measure

$$\nu_K = \mu_{N,\gamma}(\,\cdot \mid S_{N,n,\gamma})$$

and the reconstruction maps $\widehat{\mathscr{R}}$ of Theorem 15.12. Let θ_0 be the logarithm of the coarsest configuration in the gauge chosen there, and let w_j be the normalized fluctuation variables of the jth renormalization group step from the coarse side. We write $\theta_j = a_j A_j$ for the link logarithm of the field reconstructed at mesh a_j, so that A_j is the corresponding lattice potential. Let $\mathscr{R}_j(\theta, w)$ be the logarithmic form of the reconstruction map at step j; thus $\theta_j = \mathscr{R}_j(\theta_{j-1}, w_j)$, and $\mathscr{R}_j(\theta, w) = \widehat{\Phi}_{\mathfrak{d}_j,j}(\theta, (g_{j,i} w_i)_i)$ in the notation of Section 17. Let $\mathcal{F}_j$ be the σ-algebra generated by the coarsest configuration and by $w_1, \dots, w_j$. All conditional statements below hold for every value of θ_{j-1} that occurs on the small-field event, and *uniformly in* θ_{j-1} refers to these values.

Each scalar fluctuation coordinate λ of the jth step is attached to a lattice location on one of its intermediate scales. We write a_λ for the mesh of that lattice, $v_\lambda = a_\lambda^3$ for its cell volume, and $g_\lambda = g_3\sqrt{a_\lambda}$ for its coupling. Then $C_J^{-1}a_j^3 \le v_\lambda \le C_J a_j^3$, and the ratios g_λ/g_j are bounded above and below by constants depending only on M and J. The fluctuation cutoff domain of the step is

$$\Omega_j = \prod_{i,e}\{w_{i,e} \in \mathbb{R}^3 : |w_{i,e}| \le p_{j,i}\}, \qquad p_j = \max_i p_{j,i} \ge 1,$$

where the radii follow the logarithmic prescription of Definition 4.4, up to an additive constant depending on M and J that accounts for the intermediate scales: $p_j \le A_{\mathrm{w}} \log(1/t_j) + c_J$.

Common domains. We assume that all integrated couplings lie below the common upper bound g_* of Theorems 13.8, 13.10, and 17.4 and Lemma 15.10; in particular, $t_j \le g_*$ for every j. We also assume that the coarser fields lie in the small-link domain B_r, the target set of Theorem 13.8, and that the reconstruction is evaluated in the larger analytic domain of Theorem 15.12, with the fixed positive distances between its nested domains. These choices are made before K, j, or any sample, and they do not change with K or j.

For a coarser field θ and a number $\mathfrak{d} \ge 0$ of preceding averaging steps, we write $\mu_{\mathfrak{d},\theta}$ for the conditional fluctuation kernel of Theorem 13.8 on Ω_j. For coarser fields in the small-link domain, $\mu_{\mathfrak{d},\theta}(dw) = k(e^\theta, dw)$ in the notation of (4.8), for a step preceded by $\mathfrak{d}$ averaging steps. Let $\mathbb{E}_\theta$, $\mathbb{P}_\theta$, and Cov_θ denote expectation, probability, and covariance under this kernel. We write

$$m_\theta = \mathbb{E}_\theta w_j, \qquad \zeta_j = w_j - m_\theta,$$

and Σ_θ for the conditional fluctuation covariance of w_j. (This Σ_θ is unrelated to the normal covariance of (4.12).)

Norms and adjoints. We fix a finite horizon $T > 0$ for all heat estimates. Coefficient arrays indexed by fluctuation coordinates carry the counting norm, which sums over every scalar coordinate of every intermediate scale; we write $\|l\|_2$ or $\|l\|_{\ell^2(w)}$ for it, and the transpose ^{T} is taken with respect to the counting inner product. Fields on a lattice of mesh b carry the physical norms

$$\|f\|_{L^2_b} = \Big(b^3 \sum_x |f(x)|^2\Big)^{1/2}, \qquad \|f\|_{L^1_b} = b^3 \sum_x |f(x)|,$$

and the adjoint W^* of a map $W : \ell^2(w) \to L^2_\delta$ is taken with respect to L^2_δ, so that matrix products with W^* contain the factor δ^3. The covariance matrix of the linear reconstruction coefficient is the counting matrix

$$\mathsf{Z}_\theta = E_\theta \Sigma_\theta E_\theta^{\mathsf{T}},$$

where E_θ is defined in Lemma 26.2; the corresponding physical operator is $a^3\mathsf{Z}_\theta$. We always use the counting matrix Z_θ between counting rows. The heat semigroup $P_t^\delta = e^{t\Delta_\delta}$, the cardinal interpolation $I_b = I_{b\to\delta}$ from mesh b to mesh δ, the differences D_δ, and the norms X_α^δ, Y_α^δ, H_σ^δ, and V_σ^δ are those of Definition 4.8, with $v_\delta(t) = t + \delta^2$; here D_δ includes every componentwise difference and every translation required by the quadratic coefficient of Lemma 24.2. For $0 < \eta < 3/2$ we also write

$$\mathsf{H}_\eta^\delta(f) = \mathsf{X}_\eta^\delta(P_\cdot^\delta f) = \sup_{0\le t\le T} \big\{ (t+\delta^2)^{\eta/2} \|P_t^\delta f\|_\infty + (t+\delta^2)^{(\eta+1)/2} \|D_\delta P_t^\delta f\|_\infty \big\};$$

for $0 < \eta < 1 - \alpha$ this is the norm of Definition 4.8. Finally,

$$\mathcal{Q}_\delta(u,v)(t) = \int_0^t P_{t-s}^\delta \mathcal{N}_\delta(u(s), v(s))\, ds, \qquad \mathcal{B}_\delta(u,v) = \mathcal{Q}_\delta(u,v) + \mathcal{Q}_\delta(v,u),$$

where $\mathcal{N}_\delta$ is the ordered quadratic coefficient of the damped Wilson flow on the lattice of mesh δ; it is not symmetric, and we always keep the order of its arguments.

Constants. Unless stated otherwise, constants C, C_J, and $C_{J,q}$ depend only on L, T, M, J, the fixed analytic domains and the distances between them, the bounds on the absolute Taylor coefficients of the reconstruction maps, the interpolation function of the cardinal interpolation, the exponents, and the indicated derivative orders. They do not depend on the lattice side, on K, on the number of preceding averaging steps, on the coarser field, on the coupling below g_*, or on the refinement ratio a/δ. The constants of maximal inequalities proved with the fluctuation cutoff (Lemma 26.9) may in addition depend on the fixed γ, n, and the prescription of the cutoff radii; the covering and chaining bounds of Lemmas 26.7 and 26.8 have no such dependence. Among the constants of this section, only the bound C_{CP} on finitely many derivatives of the zero-fluctuation reconstruction (see the proof of Lemma 26.2) is chosen before J; all other bounds are fixed-J bounds. All estimates concern the whole torus.

The conditional law and the Taylor coefficients

The first lemma identifies the conditional law of the fluctuation variables of one step and records the coefficient bounds used in all later estimates. Its proof mostly verifies the hypotheses of earlier results; the new ingredient is the passage from absolute coefficient bounds on a larger domain to derivative bounds, without loss in the volume-normalized column sums.

Lemma 26.1. *Assume the setting of Section 26.1. For every $K \ge 0$ and $1 \le j \le K$, the conditional law of w_j given $\mathcal{F}_{j-1}$ is the normalized density $Z_{j,\theta}^{-1} e^{-V_{j,\theta}(w)}\, dw$ on Ω_j, where $\theta = \theta_{j-1}$, the potential $V_{j,\theta}$ is smooth on a neighborhood of the closed product Ω_j, and*

$$D_w^2 V_{j,\theta} \ge \tfrac{1}{2} I \qquad \text{on } \Omega_j.$$

This density and the normalized fluctuation coordinates are the same before and after the change to the reconstruction maps $\widehat{\mathscr{R}}$ at a fixed coarser field. For each base point $w_* = 0$ or $w_* = m_\theta$, the normalized tensors $g_j^{-1} D_w \mathscr{R}_j(\theta, w_*)$ and $(2g_j^2)^{-1} D_w^2 \mathscr{R}_j(\theta, w_*)$ have uniformly bounded absolute output row sums and volume-normalized input column sums for every choice of fixed input, on common nested domains in the unscaled coarser field and the coupling-scaled fluctuation inputs $q_{j,i} = g_{j,i} w_i$. On a smaller domain these bounds extend to every fixed-order mixed derivative in the unscaled coarser field and the q coordinates, with a constant depending on the order. These assertions are uniform in the lattice side and in the number of preceding averaging steps, at fixed M, J, and domains.

Proof. At step j, the coarser lattice has side $m_j = nB^{j-1} \geq 4$, the number of preceding averaging steps is $\mathfrak{d}_j = J(K - j)$, and $t_j^2 = \gamma/(M m_j)$. By Lemma 15.10, the coarser field lies in the small-link domain B_r. The cutoff radii follow the logarithmic prescription, and the intermediate scales change their maximum by at most an additive constant depending on M and J. The common upper bound g_* includes the smallness conditions of Theorems 13.8 and 13.10. With $p = p_j$ and $t = t_j$, we have in particular

$$Ctp < 1, \qquad \frac{C[tp + t^2(1 + E_*)]}{(1 - Ctp)^3} \leq \frac{1}{2}, \qquad tp^3 \leq c_*. \tag{26.1}$$

Here E_* is the upper bound on the weighted analytic coefficient norm of the effective interaction in Theorem 13.8, and C and c_* are constants of that theorem and of Theorem 13.10. Since $p_{\max}(t) \leq A_{\mathrm{w}} \log(1/t) + c_J$ and $t[\log(1/t) + c_J]^k \to 0$ as $t \to 0$ for each fixed k, we may choose g_* to satisfy (26.1) before K, j, or any sample is chosen. The domain conditions on the product Ω_j, and the positive distances from the analytic images to the boundaries of their domains, are the common-domain hypotheses preserved in Theorem 15.12.

All hypotheses of Theorem 13.8 hold at this coarser side, number of averaging steps, coarser field, and product Ω_j. That theorem gives the potential, its smooth extension to a neighborhood, and the lower bound $I/2$ on its Hessian; the integration domain is Ω_j itself, not a larger enclosing product domain. By its last clause and Lemma 15.10, this law is the conditional law of w_j under ν_K, since the factors that depend only on the coarser field cancel. By Theorem 15.13, the reconstruction maps preserve, at this coarser field, the normalized fluctuation coordinates, the balls, the action, the Haar disintegration density, the triangular determinant, the input effective interaction produced by the finer-scale integrations, the normalizer, and the conditional fluctuation kernel. Two equal positive normalized densities have potentials that differ by a constant depending only on the coarser field. Fixing that constant, we obtain the same $V_{j,\theta}$, and hence the same Hessian, after the change of representatives. In particular, we never transport a Hessian through a nonlinear change of the fluctuation variables. The auxiliary integral of Theorem 14.6 over larger domains does not enter, because the coarser field lies in B_r.

For the analytic assertion, we use the norm of Theorem 15.12 on the absolute joint Taylor coefficients, including its volume-normalized column sums, rather than a bound on the values of a holomorphic map. The products of coarser fields and coupling-scaled inputs occurring here lie in the inner domain $\mathscr{D}_{\mathrm{in}}$ of that theorem, at a positive distance from its boundary, and the absolute coefficient norm is bounded uniformly on $\mathscr{D}_{\mathrm{in}}$. We choose an evaluation polydisc and a larger polydisc,

$$\mathscr{P}_- \Subset \mathscr{P}_+ \Subset \mathscr{D}_{\mathrm{in}}, \qquad \eta_\lambda = \mathrm{rad}_\lambda(\mathscr{P}_+) - \mathrm{rad}_\lambda(\mathscr{P}_-) > 0.$$

They may be centered at any point of evaluation, with gaps bounded below by fixed fractions of the distance to the boundary of $\mathscr{D}_{\mathrm{in}}$; inscribing coordinate polydiscs in the product balls needs only a fixed change of color coordinates. The nested domains of Theorem 15.12 give these gaps uniformly, and nothing depends on a modulus of convergence or on the size of the lattice. The coupling-scaled images $(g_{j,i}w_i)_i$ of every point of Ω_j, of the point $w = 0$, and of the conditional mean m_θ lie in such an evaluation polydisc, because the mean of a probability measure on a ball lies in that ball.

Differentiating the absolute joint series term by term and evaluating it on $\mathscr{P}_-$ introduces the factorial of the derivative order and inverse powers of the gaps. These operations preserve the column sum with a fixed input, because all other inputs, including the output, are already summed in that column sum. Equivalently, one may apply the Cauchy integral formula on $\mathscr{P}_+$ to the coefficients, in these finite-dimensional row and column norms. We obtain uniform bounds for the first, second, and every fixed higher derivative. A derivative in a fluctuation variable of intermediate scale i introduces the factor $g_{j,i}$, and the ratios $g_{j,i}/g_j$ and the ratios of cell volumes are bounded by constants depending only on M and J. Dividing the first and second derivatives in w by g_j and $2g_j^2$, respectively, we obtain the two normalized tensors of either Taylor expansion in Lemma 26.2, with their volume-normalized column sums. All constants in this argument are fixed-J constants; the constant C_{CP} chosen before J is not enlarged. □

The exact decomposition

Table 26.1 lists the four terms of the increment of the heat evolution; their formulas are in the next lemma.

The classical part uses the zero-fluctuation reconstruction $C_j = a_j^{-1}\mathscr{R}_j(\theta_{j-1}, 0)$, which is a minimizing reconstruction and not a conditional mean. Its linearization P_j maps a lattice potential on the coarser lattice, of mesh $H = a_{j-1}$, to a lattice potential on the lattice of mesh a_j. The interpolation defect compares this linearization, interpolated from mesh a_j, with the coarser field interpolated directly from its own lattice; the two interpolations are applied on their own lattices before they are subtracted. In the notation of Lemma 28.5, $C_j = a_j^{-1}\widehat{P}_j(\theta_{j-1})$ and $\mathsf{P}_j = B\widehat{L}_j$, so the linearized reconstruction is $H^{-1}\mathsf{P}_j\theta_{j-1} = a_j^{-1}\widehat{L}_j\theta_{j-1}$.

Lemma 26.2. *Assume the setting of Section 26.1, and fix $K \geq 0$.*

Term	Field before heat evolution $P_\cdot^\delta$	Conditional mean	Treated in
s_j	$I_{a_j}(g_j/a_j)E_\theta\zeta_j$, linear in ζ_j	zero	Theorem 26.10
z_j	$I_{a_j}Z_j$, nonlinear in ζ_j	zero	Theorem 26.11
r_j	$H^{-1}(I_{a_j}\mathsf{P}_j - I_H\mathsf{E}_H)\theta_{j-1}$	predictable	Lemma 27.1
u_j	$I_{a_j}U_j^0$, nonlinear classical part plus mean	predictable	Theorem 27.7

Table 26.1. The four terms of the increment $d_j = x_j - x_{j-1}$ in Lemma 26.2.

1. *For every $1 \le j \le K$, conditional on $\mathcal{F}_{j-1}$, the law of w_j is the conditional fluctuation kernel $\mu_{\mathfrak{d}_j,\theta_{j-1}}$ on the product Ω_j, with $\mathfrak{d}_j = J(K-j)$, including all Haar disintegration densities, determinants, input effective interactions, and indicators. For this fixed coupling and coarse side, $p_j \le C_\gamma(1+j)$. Every estimate below for a single kernel holds uniformly for any number $\mathfrak{d} \ge 0$ of preceding averaging steps, on the same domains; the conditioned Wilson measure uses $\mathfrak{d} = \mathfrak{d}_j$.*
2. *Write the Taylor expansion of the reconstruction at the conditional mean as*

$$\mathscr{R}_j(\theta, m_\theta + \zeta) = \mathscr{R}_j(\theta, m_\theta) + g_j E_\theta\zeta + g_j^2 \mathsf{B}_\theta(\zeta,\zeta) + R_\theta(\zeta), \tag{26.2}$$

where the ratios of the intermediate couplings to g_j are included in E_θ and B_θ. Let $C_j = a_j^{-1}\mathscr{R}_j(\theta_{j-1}, 0)$ and $m_j^{\mathrm{ph}} = \mathbb{E}(A_j - C_j \mid \mathcal{F}_{j-1})$. With $H = a_{j-1} = Ba_j$, the zero-fluctuation reconstruction has the exact expansion

$$C_j(\theta) = H^{-1}\big\{\mathsf{P}_j\theta + \overline{\mathsf{Q}}_j(\theta,\theta) + \overline{\mathsf{R}}_j(\theta)\big\}, \tag{26.3}$$

where P_j is linear, $\overline{\mathsf{Q}}_j$ is quadratic, and $\overline{\mathsf{R}}_j$ is the remainder. Let E_H be the identity map of arrays on the lattice of mesh H, and set

$$\begin{aligned}
x_j &= P_\cdot^\delta I_{a_j} A_j, \qquad d_j = x_j - x_{j-1},\\
s_j &= (g_j/a_j) P_\cdot^\delta I_{a_j} E_\theta\zeta_j,\\
Z_j &= A_j - \mathbb{E}(A_j \mid \mathcal{F}_{j-1}) - (g_j/a_j)E_\theta\zeta_j, \qquad z_j = P_\cdot^\delta I_{a_j} Z_j,\\
r_j &= H^{-1}P_\cdot^\delta(I_{a_j}\mathsf{P}_j - I_H\mathsf{E}_H)\theta_{j-1},\\
U_j^0 &= H^{-1}\big\{\overline{\mathsf{Q}}_j(\theta_{j-1},\theta_{j-1}) + \overline{\mathsf{R}}_j(\theta_{j-1})\big\} + m_j^{\mathrm{ph}}, \qquad u_j = P_\cdot^\delta I_{a_j} U_j^0,
\end{aligned} \tag{26.4}$$

with $\theta = \theta_{j-1}$ in E_θ. Then

$$d_j = s_j + z_j + r_j + u_j. \tag{26.5}$$

The terms s_j and z_j have conditional mean zero given $\mathcal{F}_{j-1}$, while r_j and u_j are $\mathcal{F}_{j-1}$-measurable; neither of the latter is recentered. With $y_j = \mathcal{Q}_\delta(x_j, x_j)$, for every $1 \le k \le l \le K$,

$$y_l - y_{k-1} = \sum_{j=k}^{l}\big\{\mathcal{B}_\delta(x_{j-1}, d_j) + \mathcal{Q}_\delta(d_j, d_j)\big\}. \tag{26.6}$$

Inserting (26.5) *expresses the right side through the four mixed terms* $\mathcal{B}_\delta(x_{j-1}, v)$, $v \in \{s_j, z_j, r_j, u_j\}$, *and the sixteen ordered products* $\mathcal{Q}_\delta(v, v')$, $v, v' \in \{s_j, z_j, r_j, u_j\}$; *equivalently, through the nine ordered products of* s_j, z_j, *and* $r_j + u_j$. *All these identities are finite and exact, uniformly in* K, $\mathfrak{d}_j$, *and* θ_{j-1}.

3. *There is also an expansion at* $w = 0$. *Set*

$$E_\theta^{(0)} = g_j^{-1} D_w \mathscr{R}_j(\theta, 0), \qquad \mathsf{B}_\theta^{(0)} = (2g_j^2)^{-1} D_w^2 \mathscr{R}_j(\theta, 0),$$

define $s_j^{(0)}$ *as* s_j *with* E_θ *replaced by* $E_\theta^{(0)}$, *and define* $Z_j^{(0)}$ *and* $z_j^{(0)}$ *by the same centered subtraction. Then* (26.5) *and* (26.6) *hold with* $s_j^{(0)}$, $z_j^{(0)}$ *in place of* s_j, z_j, *and with* r_j, u_j *unchanged. Define* $\widetilde{R}_\theta$ *by*

$$\mathscr{R}_j(\theta, m_\theta + \zeta) - \mathscr{R}_j(\theta, m_\theta) = g_j E_\theta^{(0)} \zeta + g_j^2 \mathsf{B}_\theta^{(0)}(\zeta, \zeta) + \widetilde{R}_\theta(\zeta). \tag{26.7}$$

Then $\|\widetilde{R}_\theta\|_\infty \le C_J g_j^3 p_j^3$ *on* Ω_j. *Theorems 26.10 to 26.12 hold for either expansion, with* E_θ, B_θ, R_θ *replaced consistently by* $E_\theta^{(0)}$, $\mathsf{B}_\theta^{(0)}$, $\widetilde{R}_\theta$ *in the second. This is a second decomposition of the same sample, under the same law.*

Proof. The proof consists of exact computations, apart from two estimates: the deterministic bound on the coarser fields in Step 2, which is used throughout the section, and the bound on $\widetilde{R}_\theta$ in Step 7.

Step 1: the conditional law. The disintegration of Lemma 15.10 applies with coarse mesh L/n, total number K of renormalization group steps, and the reconstruction maps $\widehat{\mathscr{R}}$, whose gauge transformations depend only on the coarser field. Every restriction imposed by the small-field event at coarser scales is measurable with respect to the coarser field and therefore cancels in the normalized conditional law of the finer variables. This proves the assertion about the law of w_j given $\mathcal{F}_{j-1}$, which is a statement about the joint law of all variables under ν_K. Each cutoff radius is a fixed multiple of $\log(1/t_j)$, up to the ratios between intermediate scales. Since $t_j = M^{(J-1)/2} g_3 \sqrt{L/n}\, B^{-j/2}$, their maximum satisfies $p_j \le C_{\gamma,n,M,J}(1 + j)$.

Step 2: a deterministic bound on the coarser fields. Let $C_{\mathrm{rec},J}$ be the fixed-J bound on the absolute row sums of the derivative of one reconstruction step in the coupling-scaled inputs. By the bound on the reconstruction in Lemma 15.10, we have

$$\|\theta_j\|_\infty \le a_{\mathrm{cap}} \|\theta_{j-1}\|_\infty + C_{\mathrm{rec},J} t_j p_j, \qquad a_{\mathrm{cap}} = C_{\mathrm{CP}}/B, \qquad a_{\mathrm{cap}} \sqrt{B} < 1.$$

Here C_{CP} bounds finitely many derivatives of the zero-fluctuation reconstruction and is chosen before J. We iterate this recursion, using $p_i \le p_j$ for $i \le j$ and $t_i = M^{(J-1)/2} g_3 \sqrt{a_j}\, B^{(j-i)/2}$, and obtain

$$\|\theta_j\|_\infty \le r a_{\mathrm{cap}}^j + M^{(J-1)/2} C_{\mathrm{rec},J} g_3 \sqrt{a_j}\, p_j \sum_{h=0}^{j-1} (a_{\mathrm{cap}} \sqrt{B})^h \le r a_{\mathrm{cap}}^j + C_{\mathrm{cap},J} g_3 \sqrt{a_j}\, p_j, \quad (26.8)$$

where r is the radius of B_r and $C_{\mathrm{cap},J} \ge M^{(J-1)/2} C_{\mathrm{rec},J} / (1 - a_{\mathrm{cap}} \sqrt{B})$ is a separate constant. A constant C_J in a deterministic bound on the coarser fields below may include $C_{\mathrm{cap},J}$; it is never used to enlarge C_{CP} or to change J. The conditional fluctuation kernel and the coefficient bounds used in the Taylor expansions below are verified in Lemma 26.1.

Step 3: the expansion at the conditional mean. By convexity, the product of balls contains the coordinatewise mean m_θ and the segment $m_\theta + v\zeta$, $0 \le v \le 1$. We apply Theorem 13.10 to the reconstruction on the domain of Lemma 26.1, including the ratios between intermediate scales. With

$$E_\theta = g_j^{-1} D_w \mathscr{R}_j(\theta, m_\theta), \qquad \mathsf{B}_\theta = (2 g_j^2)^{-1} D_w^2 \mathscr{R}_j(\theta, m_\theta),$$

Taylor's formula gives (26.2) exactly, with

$$R_\theta(\zeta) = \frac{1}{2} \int_0^1 (1-v)^2 D_w^3 \mathscr{R}_j(\theta, m_\theta + v\zeta)[\zeta, \zeta, \zeta]\, dv, \qquad \|R_\theta\|_\infty \le C_J g_j^3 p_j^3. \quad (26.9)$$

The derivatives here are taken in the normalized fluctuation variables. Writing an increment of the coupling-scaled input $q_{j,i}$ as $g_{j,i} w_i$ contributes its own factor $g_{j,i}$ at each input, and all ratios $g_{j,i}/g_j$ lie in an interval depending only on M and J. This accounts for the normalization of both tensors.

Step 4: the expansion of the zero-fluctuation reconstruction. We define $\overline{\mathsf{A}}_j(\theta) = B \mathscr{R}_j(\theta, 0)$, $\mathsf{P}_j = D\overline{\mathsf{A}}_j(0)$, and $\overline{\mathsf{Q}}_j = D^2 \overline{\mathsf{A}}_j(0)/2$, and let $\overline{\mathsf{R}}_j$ be what remains after subtracting these two terms from $\overline{\mathsf{A}}_j$. We have $\overline{\mathsf{A}}_j(0) = 0$, because the zero-fluctuation reconstruction of the identity is the identity. Since $B a_j = H$, we have $a_j^{-1} \mathscr{R}_j(\theta, 0) = H^{-1} \overline{\mathsf{A}}_j(\theta)$, which is (26.3).

Step 5: the four terms. By the definitions of C_j, m_j^{ph}, and Z_j,

$$A_j = C_j + m_j^{\mathrm{ph}} + (g_j/a_j) E_\theta \zeta_j + Z_j.$$

We apply I_{a_j} to this identity, subtract $I_H A_{j-1} = H^{-1} I_H \mathsf{E}_H \theta_{j-1}$, and apply P_t^δ. By (26.3), the linear part of C_j and the coarser field combine to $H^{-1} P_t^\delta (I_{a_j} \mathsf{P}_j - I_H \mathsf{E}_H) \theta_{j-1} = r_j(t)$. The remaining predictable terms are u_j, which proves (26.5). Since $\mathbb{E}_\theta \zeta_j = 0$, the definitions imply that $\mathbb{E}_\theta s_j = \mathbb{E}_\theta z_j = 0$, while r_j and u_j depend only on θ_{j-1}. The conditional mean correction m_j^{ph}, which is in general not zero, has not been removed.

Step 6: the quadratic telescope. For any bilinear map $\mathcal{Q}$ and any x, d,

$$\mathcal{Q}(x+d,x+d)-\mathcal{Q}(x,x)=\mathcal{Q}(x,d)+\mathcal{Q}(d,x)+\mathcal{Q}(d,d).$$

We apply this identity to $\mathcal{Q}_\delta$ on the lattice of mesh δ, with $x=x_{j-1}$ and $d=d_j$, and sum from $j=k$ to $j=l$. This is (26.6). Inserting $d_j=\sum_{v\in\{s_j,z_j,r_j,u_j\}}v$, we obtain $\mathcal{B}_\delta(x_{j-1},d_j)=\sum_v\mathcal{B}_\delta(x_{j-1},v)$ and $\mathcal{Q}_\delta(d_j,d_j)=\sum_{v,v'}\mathcal{Q}_\delta(v,v')$, which are the four mixed terms and the sixteen ordered products. Combining r_j+u_j gives the nine ordered products.

Step 7: the expansion at zero. We derive the second expansion from an exact Taylor computation; we do not assume that $m_\theta=0$. At $w=0$ we have

$$\mathscr{R}_j(\theta,v)=\mathscr{R}_j(\theta,0)+g_jE_\theta^{(0)}v+g_j^2\mathsf{B}_\theta^{(0)}(v,v)+R_\theta^{(0)}(v).$$

By Lemma 26.1, the third directional derivative is bounded, and we obtain $\|R_\theta^{(0)}(v)\|_\infty\le C_Jg_j^3\|v\|_\infty^3$. One may also obtain this bound directly from the one-variable Cauchy formula for the bounded reconstruction along the line through 0 and v, written in the coupling-scaled inputs $(g_{j,i}v_i)_i$; the radius available on that line is the distance to the boundary of the domain of these inputs divided by the supremum norm of the direction. By convexity, the segments from 0 to the points used here lie in the product. We subtract the expansions at $v=m_\theta+\zeta$ and at $v=m_\theta$ and obtain

$$\widetilde{R}_\theta(\zeta)=g_j^2\{\mathsf{B}_\theta^{(0)}(m_\theta,\zeta)+\mathsf{B}_\theta^{(0)}(\zeta,m_\theta)\}+R_\theta^{(0)}(m_\theta+\zeta)-R_\theta^{(0)}(m_\theta).$$

The bilinear form $\mathsf{B}_\theta^{(0)}$ is bounded in the supremum norm, by polarization of the same directional Taylor formula. Together with the bound $\|m_\theta\|_\infty\le C_Jt_j\|\theta\|_\infty$ of Theorem 13.10, this bounds the first term by $C_Jg_j^2t_j\|\theta\|_\infty p_j$. Since θ lies in a fixed ball, t_j/g_j is fixed, and $p_j\ge1$, this is at most $C_Jg_j^3p_j^3$. The two remainders $R_\theta^{(0)}$ obey the same bound. We define $z_j^{(0)}$ by the centered subtraction; the algebra of Steps 5 and 6 then applies with r_j and u_j unchanged. The row and column bounds and the Cauchy argument below are made on the common coefficient domain of Lemma 26.1, so they apply at either base point $w=m_\theta$ or $w=0$. This completes the proof. □

Remark 26.3. In this section s_j and z_j always denote the terms of the expansion (26.2) at the conditional mean, whose linear coefficient is $E_\theta=g_j^{-1}D_w\mathscr{R}_j(\theta,m_\theta)$; the terms of the expansion at $w=0$, with linear coefficient $E_\theta^{(0)}=g_j^{-1}D_w\mathscr{R}_j(\theta,0)$, are written $s_j^{(0)}$ and $z_j^{(0)}$. Both pairs use the same centered vector $\zeta_j=w_j-m_\theta$, and

$$s_j+z_j=s_j^{(0)}+z_j^{(0)}=P_\cdot^\delta I_{a_j}\big(A_j-\mathbb{E}(A_j\mid\mathcal{F}_{j-1})\big),$$

so the decomposition (26.5) and the terms r_j and u_j are the same in both conventions. Section 28 uses the expansion at $w = 0$: its terms s_j and z_j are the terms $s_j^{(0)}$ and $z_j^{(0)}$ of this section. By part (3) of Lemma 26.2, Theorems 26.10 to 26.12 hold for these terms as stated.

26.2 Deterministic estimates for heat kernels and coefficients

Standing conventions. In this section and the next, $a = a_j$ for some $1 \le j \le K$, $H = Ba = a_{j-1}$, and $\delta = a_K$. Then $0 < \delta \le a \le L/4$, and $a/\delta = B^{K-j}$ and $H/\delta = B^{K-j+1}$ are integers. We fix $1/2 < \alpha < 2/3$ and set

$$\beta = \alpha - 1/2, \qquad p_\alpha = (\alpha+1)/2, \qquad \Lambda_a = \log(2 + L/a),$$

where Λ_a is the logarithmic lattice-size factor, written $\Lambda_{a_j} = \ell_j$ in Section 17; it is unrelated to the normalization constant Λ. For $q > 0$ and $0 \le t \le T$, we also set

$$I_q^a(t) = \int_0^t (s + a^2)^{-p_\alpha} (t - s + a^2)^{-q}\, ds. \tag{26.10}$$

For a heat path $B(t) = P_t^\delta b$ on the lattice of mesh δ we write

$$\mathsf{R}_a(B) = \sup_{0 \le t \le T} \big\{ (t+a^2)^{\alpha/2} \|B(t)\|_\infty + (t+a^2)^{(\alpha+1)/2} \|D_\delta B(t)\|_\infty \big\} \tag{26.11}$$

for its radius at scale a; it is the norm $\mathsf{X}_\alpha^\delta(B)$ with δ^2 replaced by a^2 in the weights.

Let $f(t,x)$ be a field on $[0,T] \times \Lambda_N$ whose components are linear functions of a vector of counting inputs, with deterministic coefficients. Its *value row* at (t,x) is the coefficient vector of a scalar component of $f(t,x)$, and its *first-difference rows* are those of the components of $D_\delta f(t,x)$. For a field whose components are quadratic forms, rows are replaced by the symmetric coefficient matrices, measured in Hilbert–Schmidt and operator norm. Given exponents (κ_0, κ_1), the *weighted coefficient bound* of f is

$$\sup_{0 \le t \le T} \sup_x \max\big\{ (t+\delta^2)^{\kappa_0} \|\text{value row}\|,\ (t+\delta^2)^{\kappa_1} \|\text{first-difference row}\| \big\}.$$

We use $(\kappa_0, \kappa_1) = (\eta/2, (\eta+1)/2)$ for the norms X_η^δ and H_η^δ, and $(\kappa_0, \kappa_1) = (\beta, \alpha)$ for the norm Y_α^δ. A bound by S for the weighted coefficient bound means that every scalar component of the weighted value and first difference, evaluated at a fixed point (t,x), is a linear (or quadratic) form in the inputs with coefficient norm at most S.

The periodic heat kernel

We write $P_t^\delta f(x) = \delta^3 \sum_y p_t^\delta(x,y) f(y)$, so that $p_t^\delta(x,\cdot)$ is the heat kernel relative to the physical measure $\delta^3 \sum_y$, and we let $\|\cdot\|_1$, $\|\cdot\|_2$ be the norms of L_δ^1 and L_δ^2.

Lemma 26.4. *For every fixed integer $q \geq 0$ and $0 \leq t \leq T$,*

$$\|D_\delta^q p_t^\delta(x,\cdot)\|_1 \leq C_q (t+\delta^2)^{-q/2},$$
$$\|D_\delta^q p_t^\delta(x,\cdot)\|_2 \leq C_q (t+\delta^2)^{-3/4-q/2},$$
$$\|D_\delta^q p_t^\delta(x,\cdot)\|_\infty \leq C_q (t+\delta^2)^{-(3+q)/2}.$$

The kernel is nonnegative and has physical mass one. Fixed translations and adjoint differences obey the same estimates, with the same dependence of the constants.

Proof. We write $\Delta_\delta = \delta^{-2}\sum_i (T_i + T_i^{-1} - 2)$, where T_i is the translation by δe_i. The exponential series of the positive part $\delta^{-2}\sum_i (T_i + T_i^{-1})$, multiplied by e^{-6t/δ^2}, is a convolution with nonnegative coefficients whose sum is one. At $t = 0$ its density is $\delta^{-3}\mathbf{1}_{x=y}$. For $t \leq \delta^2$, contraction and the finite difference stencil give the three bounds, since $(t+\delta^2) \asymp \delta^2$ there.

We give the details for the L^1 bound on differences at longer times. On $\mathbb{Z}$, at dimensionless time $v = t/\delta^2 \geq 1$, the Fourier multiplier of the qth unscaled forward difference of the random walk kernel is

$$f_{v,q}(\xi) = (e^{i\xi} - 1)^q e^{-4v\sin^2(\xi/2)}, \qquad -\pi \leq \xi \leq \pi.$$

Backward differences change only a factor of modulus one. Since $\sin^2(\xi/2) \geq c\xi^2$ on this interval, bounding each derivative of a factor and substituting $z = \sqrt{v}\,\xi$, we find

$$\int |f_{v,q}|^2 \, d\xi \leq C_q v^{-q-1/2}, \qquad \int |\partial_\xi f_{v,q}|^2 \, d\xi \leq C_q v^{-q+1/2}.$$

For $q = 0$ only the derivative of the exponential occurs in the second integral, and its size is $Cv|\xi|e^{-cv\xi^2}$; for $q \geq 1$ the other term is $C_q|\xi|^{q-1}e^{-cv\xi^2}$, and both integrals have the displayed powers. By Parseval's identity, the difference kernel $h_{v,q}(m)$ satisfies

$$\sum_m (1 + m^2/v)|h_{v,q}(m)|^2 \leq C_q v^{-q-1/2}.$$

Using the Cauchy–Schwarz inequality with weights $1 + m^2/v$, together with $\sum_m (1 + m^2/v)^{-1} \leq C\sqrt{v}$, we obtain $\sum_m |h_{v,q}(m)| \leq C_q v^{-q/2}$. Taking products over the three coordinates, we obtain this bound for every difference in several directions on $\mathbb{Z}^3$. Periodization does not increase the absolute sum, and rescaling each difference by δ^{-1} gives the physical L^1 estimate.

The remaining estimates on the finite torus follow from the finite Fourier sum. Frequencies k in the principal cube satisfy $\omega_\delta(k) = 4\delta^{-2}\sum_i \sin^2(\pi\delta k_i/L) \geq c|k|^2/L^2$, and the multiplier of a difference of order q is bounded by $C_q(|k|/L)^q$. Grouping the integer frequencies in unit shells $m \leq |k| < m+1$, each of which contains at most $C(m+1)^2$ points,

and comparing the decreasing Gaussian tails with integrals after splitting at $m = L/\sqrt{t}$, we obtain

$$L^{-3}\sum_{k}(|k|/L)^{q}e^{-ct|k|^{2}/L^{2}} \le C_{q}t^{-(3+q)/2} + \mathbf{1}_{q=0}L^{-3}.$$

This bounds the absolute Fourier sum, and hence the L^{∞} norm. Parseval's identity with order $2q$ and time $2t$ gives the square of the L^{2} bound. When $q = 0$, the constant Fourier mode contributes L^{-3} to that square; since $t \le T$, it is absorbed into the stated estimate with a constant depending on T/L^{2}. Differences annihilate the constant mode when $q > 0$. □

Coefficient tensors

Lemma 26.5. *Suppose a tensor $b_{y;\lambda\mu}$, with y on the lattice of mesh a and λ, μ fluctuation coordinates of cell volumes v_{λ}, v_{μ}, satisfies*

$$\sup_{y}\sum_{\lambda,\mu}|b_{y;\lambda\mu}| \le C_{J}, \qquad \sup_{\lambda}\frac{a^{3}}{v_{\lambda}}\sum_{y,\mu}|b_{y;\lambda\mu}| \le C_{J}, \qquad v_{\lambda}/a^{3} \le C_{J},$$

and the same column estimate with the roles of the two inputs interchanged. Set $F_{t} = P_{t}^{\delta}I_{a}b$, regarding the ordered pair (λ, μ) as one counting input. Then, for every fixed integer $q \ge 0$,

$$\|D_{\delta}^{q}F_{t}\|_{\ell^{2}(\lambda,\mu)\to L^{2}_{\delta}} \le C_{J,q}a^{3/2}(t + a^{2})^{-q/2},$$
$$\sup_{x}\|D_{\delta}^{q}F_{t}(x,\cdot,\cdot)\|_{2} \le C_{J,q}a^{3/2}(t + a^{2})^{-3/4-q/2}.$$

Symmetrizing a scalar coefficient matrix increases neither its Hilbert–Schmidt norm nor the displayed bound for its operator norm. The same heat argument, for one input, applies to E_{θ}.

Proof. By the column hypothesis and $v_{\lambda}/a^{3} \le C_{J}$, we have

$$\sup_{\lambda,\mu}\sum_{y}|b_{y;\lambda\mu}| \le \sup_{\lambda}\sum_{y,\mu'}|b_{y;\lambda\mu'}| \le C_{J}.$$

In other words, the column of a pair is bounded, because the column sum with one input fixed already sums over the other input. For any array h indexed by pairs, the Cauchy–Schwarz inequality with weights $|b_{y;\lambda\mu}|$ implies that

$$\sum_{y}\Big|\sum_{\lambda,\mu}b_{y;\lambda\mu}h_{\lambda\mu}\Big|^{2} \le \Big(\sup_{y}\sum_{\lambda,\mu}|b_{y;\lambda\mu}|\Big)\Big(\sup_{\lambda,\mu}\sum_{y}|b_{y;\lambda\mu}|\Big)\sum_{\lambda,\mu}|h_{\lambda\mu}|^{2}.$$

Multiplying by a^{3}, we obtain the physical operator factor $a^{3/2}$. By Lemma 16.3, the cardinal weights are nonnegative, their row sums are one, and their column sums, weighted by the output cell volume, are a^{3}. By Jensen's inequality, $I_{a} : L^{2}_{a} \to L^{2}_{\delta}$ has norm

at most one. The absolute rows and columns of its differences are bounded by $C_q a^{-q}$ times those of I_a, because each column is a smooth bump at scale a.

For $t \le a^2$, we commute the differences through the heat semigroup. The absolute row is then $C_{J,q} a^{-q}$, the absolute column, weighted by the output cell volume, is $C_{J,q} a^{3-q}$, and the heat semigroup preserves these bounds. By the Schur test, we have the operator bound. The counting ℓ^2 norm of a row is at most its absolute row sum, which gives the pointwise bound in this range. For $t \ge a^2$, we put all differences on the heat kernel. Its L^1 estimate in Lemma 26.4 and the Schur test imply the operator bound, and its physical L^2 row, composed with the undifferentiated operator, gives the pointwise bound. The bounds of the two ranges agree at $t = a^2$ up to fixed constants. This proves the estimates for all orders, with no factor counting output sites.

For scalar coefficients, the counting norm of the vector indexed by ordered pairs is the Hilbert–Schmidt norm of the corresponding matrix. The orthogonal projection $A \mapsto (A + A^{\mathsf{T}})/2$ does not increase that norm, and the operator norm is at most the Hilbert–Schmidt norm. This proves the last assertion. □

Mixed Duhamel terms with a background field

The next lemma bounds the coefficients of the mixed Duhamel terms $\mathcal{Q}_\delta(B, F)$ and $\mathcal{Q}_\delta(F, B)$, where B is a background heat path, fixed after conditioning, and F is linear in the counting inputs. Both orders are treated, and for each the proof must decide where to put the difference contained in $\mathcal{N}_\delta$. Table 26.2 summarizes the choice, with $u = t - s$. Two differences of the background suffice for these row bounds; the four differences assumed in the lemma are used only for the covering bounds of Lemma 26.7.

Range	Source difference on	Placement
$u \ge a^2$	B	keep it on B; use the L^2_δ row of the outer heat kernel
$u \ge a^2$, $s \le u$	F_s	move it to the outer heat kernel by summation by parts
$s \ge u \ge a^2$	F_s	keep it on F_s
$u < a^2$	either	put all output differences on the source; contract by the heat semigroup

Table 26.2. Placement of differences in the proof of Lemma 26.6.

Lemma 26.6. *Assume the standing conventions of Section 26.2, and put $S = s + a^2$ and $U = t - s + a^2$. Suppose a coefficient map F_s, from a space of counting inputs to fields on the lattice of mesh δ, satisfies*

$$\|D^q_\delta F_s\|_{2 \to L^2_\delta} \le C_q h S^{-q/2}, \qquad \sup_x \|D^q_\delta F_s(x, \cdot)\|_2 \le C_q h S^{-3/4 - q/2}$$

for all fixed orders q required below, and that $B(s) = P_s^\delta b^\delta$ satisfies

$$\|D_\delta^q B(s)\|_\infty \le C_q R (s + a^2)^{-(\alpha+q)/2}, \qquad 0 \le q \le 4. \tag{26.12}$$

The constants C_q are fixed, and B is fixed after conditioning. The hypothesis (26.12) *holds for $B(s) = P_s^\delta I_H b$ with $R = \mathsf{R}_a(B)$ and C_q depending only on q, M, J, α, T/L^2, and the interpolation function. For either order $\mathcal{Q}_\delta(B, F)$ or $\mathcal{Q}_\delta(F, B)$, the coefficient rows of the mixed Duhamel term after $r = 0, 1$ output differences are bounded by*

$$C_J R h \, I^a_{3/4+r/2}(t),$$

and its weighted coefficient bound for Y_α^δ is at most $C_J R h \, a^{\alpha-3/2}$. These assertions hold with an ordered pair as the counting input, as well as with a single input.

Proof. Steps 3 and 4 are summarized in Table 26.2.

Step 1: backgrounds interpolated from mesh H. We first verify (26.12) for $B(s) = P_s^\delta I_H b$ and $R = \mathsf{R}_a(B)$; the radius then controls all higher differences. At $s = 0$, since I_H reproduces the values of b at the sites of the lattice of mesh H, we have $\|b\|_\infty \le \|I_H b\|_\infty \le R a^{-\alpha}$. For $s \le a^2$, by the bound on differences of the cardinal interpolation and heat contraction, we have $\|D_\delta^q B(s)\|_\infty \le C_q R H^{-q} a^{-\alpha} \le C_{q,B} R (s + a^2)^{-(\alpha+q)/2}$. For $s \ge a^2$, we write $B(s) = P_{s/2}^\delta B(s/2)$ and apply the L^1 bound of Lemma 26.4 to the differences of $P_{s/2}^\delta$, using $\|B(s/2)\|_\infty \le R(s/2 + a^2)^{-\alpha/2}$. This proves the same bound for every fixed q. The constant depends on $B = H/a$ and q, but not on b or R. For a general background, the bounds (26.12) are hypotheses. The row estimates below use only $0 \le q \le 2$.

Step 2: the quadratic coefficient. We use the expression of the quadratic coefficient in Lemma 24.2, on the periodic lattice of side $L/\delta = nB^K \ge 4$ and with physical difference scale δ. That lemma writes the quadratic coefficient of the damped Wilson flow as a fixed finite sum of terms $c[T_1 u_i, T_2 D_\delta^\pm v_j]$, with fixed translations T_1, T_2. We keep this order, these coefficients, and these translations in every Duhamel term of this section, including those in the estimates for the residual, for the square, and for the mean. Since this coefficient is a quadratic polynomial, no bound on the amplitude of the fields is needed to evaluate it. Subtracting $[f, g]$ from $[Tf, Tg]$ and dividing by δ gives the product rule

$$D^+[f, g] = [D^+ f, Tg] + [f, D^+ g],$$

and the backward rule follows by replacing T by T^{-1}. By this rule, a difference on F can be moved onto the whole bracket and onto B, with the translated arguments kept. Summation by parts moves a difference of the whole bracket to the outer heat kernel. All translations are isometries of L^2_δ and of the supremum norm.

Step 3: the range $u = t - s \geq a^2$. A source term whose difference falls on B has coefficient row at most

$$C_J R h \, S^{-p_\alpha} u^{-3/4-r/2},$$

by the L^2_δ bound of Lemma 26.4 on the differentiated outer heat kernel and the operator bound on F_s. If the difference of the source falls on F_s and $s \leq u$, summation by parts gives a term of this form and a term bounded by

$$C_J R h \, S^{-\alpha/2} u^{-5/4-r/2} \leq C_J R h \, S^{-p_\alpha} U^{-3/4-r/2},$$

since $S \leq U$ and $u \asymp U$. If $s \geq u \geq a^2$, we leave the difference on F_s; its operator bound gives the factor $S^{-1/2}$, and the same bound follows. For the order $\mathcal{Q}_\delta(F, B)$ the difference of the source falls on B, and the first case applies; for the order $\mathcal{Q}_\delta(B, F)$ the second and third cases apply.

Step 4: the range $u < a^2$. We put every output difference on the source and contract by the heat semigroup. At output order one, the products have the distributions of differences BD^2F, $(DB)DF$, and $(D^2B)F$, with their fixed translations; at output order zero the total order is one. Each coefficient row is bounded by

$$C_J R h \, S^{-p_\alpha - 3/4 - r/2} \leq C_J R h \, S^{-p_\alpha} U^{-3/4-r/2},$$

because $U \leq 2a^2 \leq 2S$. The color brackets contribute a constant factor. Integrating the rows of Steps 3 and 4 in s, we obtain the bounds by $C_J R h \, I^a_{3/4+r/2}(t)$.

Step 5: the convolution integrals. Set $e = a^2$ and $p = p_\alpha < 1$. For $t \leq e$, direct integration bounds $I^a_q(t)$ by te^{-p-q}. For $t \geq e$ we split at $t/2$. On the early half, $t - s + e \asymp t + e$ and $\int_0^{t/2} (s+e)^{-p} \, ds \leq C_p (t+e)^{1-p}$. On the late half, $s + e \asymp t + e$, and integrating the other factor yields $C(t+e)^{1/4}$ for $q = 3/4$ and $Ca^{-1/2}$ for $q = 5/4$. In both regimes we obtain

$$I^a_{3/4}(t) \leq C_\alpha (t + a^2)^{1/4 - p_\alpha}, \qquad I^a_{5/4}(t) \leq C_\alpha a^{-1/2} (t + a^2)^{-p_\alpha}. \tag{26.13}$$

Multiplying by $(t + \delta^2)^\beta$ and $(t + \delta^2)^\alpha$, respectively, gives at most the same expressions with δ^2 replaced by a^2. The remaining powers of $t + a^2$ are $\alpha/2 - 3/4$ and $(\alpha - 1)/2$, which are both negative. Evaluating at $t + a^2 = a^2$ bounds both weighted rows by $C_\alpha a^{\alpha - 3/2}$. This proves the final assertion, with all input factors contained in h. □

Covering numbers of coefficient rows

Lemma 26.7. *Assume the standing conventions of Section 26.2. Let $B(t) = P^\delta_t b^\delta$ be $\mathcal{F}_{j-1}$-measurable and satisfy*

$$\| D^q_\delta B(t) \|_\infty \leq C_q R (t + a^2)^{-(\alpha+q)/2}, \qquad 0 \leq q \leq 4,$$

where R is $\mathcal{F}_{j-1}$-measurable and the C_q are fixed constants. Put $F_t = (g_j/a)P_t^\delta I_a E_\theta$, for either expansion of Lemma 26.2. For either order of the mixed Duhamel term of B and F, let $l_r(t,x)$ be a scalar coefficient row after r output differences, $r = 0, 1$. Set

$$S = g_3 R a^\beta, \qquad A_r(t,x) = S^{-1}(t+\delta^2)^{\kappa_r} l_r(t,x), \qquad \kappa_0 = \beta, \quad \kappa_1 = \alpha.$$

If $R = 0$, all these rows vanish, and we define their normalized versions to be zero. Otherwise the distance

$$d((r,t,x),(r',s,y)) = \|A_r(t,x) - A_{r'}(s,y)\|_2,$$

on the index set that includes the finite list of scalar components and differences, has diameter bounded by a fixed constant, and its covering numbers satisfy

$$N(v) \le C(1 + L/a)^P v^{-Q}, \qquad 0 < v \le 1. \tag{26.14}$$

The same holds for the rows normalized pointwise, $l_r(t,x)/(g_3 R a\, I^a_{3/4+r/2}(t))$, extended continuously to $t = 0$. The constants depend only on the deterministic data and on $C_0, \dots, C_4$; they do not depend on the fluctuation cutoff, the coupling, the coarser field, the number of preceding averaging steps, or a/δ.

Proof. After conditioning, all rows are deterministic. By Lemmas 26.1 and 16.3, for every fixed order needed,

$$\sup_x \|D_\delta^q F_t(x,\cdot)\|_2 \le C_{J,q} g_3 a (t + a^2)^{-3/4 - q/2} \le C_{J,q} g_3 a^{-1/2-q}.$$

Each of the two ordered sources, denoted G_t, is the quadratic coefficient applied to $B(t)$ and F_t in that order. By the product rule of Lemma 24.2 and the four assumed differences of the background, we have

$$\sup_x \|D_\delta^q G_t(x,\cdot)\|_2 \le C_J R g_3 a^{-\alpha - 3/2 - q}, \qquad 0 \le q \le 3. \tag{26.15}$$

Order four suffices, because at most $q+1$ differences fall on either argument. The heat equations for B and F imply that, for $r = 0, 1$,

$$\sup_x \|D_\delta^r \partial_t G_t(x,\cdot)\|_2 \le C_J R g_3 a^{-\alpha - 7/2 - r}.$$

Indeed, a time derivative replaces one factor by a sum of its second differences; together with the difference of the source and the r output differences, the largest order on the background is $r + 3 \le 4$.

For $l_r(t) = \int_0^t P_{t-s}^\delta D_\delta^r G_s \, ds$, positivity and contraction of the heat semigroup also hold for rows in counting ℓ^2, by the triangle inequality. Thus

$$\sup_{t,x} \|D_\delta^h l_r(t,x)\|_2 \le C_J T R g_3 a^{-\alpha - 3/2 - r - h} \quad (h = 0, 1), \qquad \partial_t l_r(t) = D_\delta^r G_t + \Delta_\delta l_r(t).$$

Commuting the Laplacian onto G_s in the last integral bounds the time derivative by $C_J R g_3 [a^{-\alpha-3/2-r} + T a^{-\alpha-7/2-r}]$; the largest order of the source here is $r+2 \le 3$, which is covered by (26.15).

Let $\ell_* = 1 + L + \sqrt{T}$ and $\epsilon = a/\ell_* < 1$. Since $a \le L/4$, every power of a in these polynomial estimates may be replaced by the same power of ϵ, with a constant factor depending only on L and T. After division by $S = R g_3 a^{\alpha-1/2}$, the unweighted row, one spatial difference, and one time derivative are bounded by $C\epsilon^{-4}$, $C\epsilon^{-5}$, and $C\epsilon^{-6}$, respectively. For example, the spatial exponent is $2\alpha + 2 + r < 5$, and the time exponent is $2\alpha + 3 + r < 6$. By Lemma 26.6, the weighted row itself is bounded uniformly.

The weights have powers less than one. For $0 < k < 1$ and $h \ge 0$ we have $(z+h)^k - z^k \le h^k$ for $z \ge 0$, since the left side decreases in z. This gives $|(t+\delta^2)^k - (s+\delta^2)^k| \le |t-s|^k$, with no constant depending on δ. Applying this with $k = \beta$ and $k = \alpha$, and using a shortest lattice path on the torus for the spatial difference, we obtain

$$\|A_r(t,x) - A_r(s,y)\|_2 \le C\epsilon^{-8}\{\operatorname{dist}(x,y) + |t-s|^\beta\}. \tag{26.16}$$

Here $|t-s|$ and $|t-s|^\alpha$ are bounded by fixed multiples, depending on T, of $|t-s|^\beta$, and the exponent eight is a common bound for the preceding exponents. The triangle inequality and $\sup \|A_r\|_2 \le C$ bound the diameter, also between different members of the finite list of components.

We also check the second normalization, which avoids a division by t at the initial time. We write

$$l_r(t) = t U_r(t), \qquad U_r(t) = \int_0^1 P^\delta_{(1-v)t} D^r_\delta G_{vt}\, dv,$$

and $I^a_q(t) = t J_q(t)$ with

$$J_q(t) = \int_0^1 (vt + a^2)^{-p_\alpha} ((1-v)t + a^2)^{-q}\, dv.$$

This positive function has a lower bound depending only on L, T, α, and q, because $t + a^2 \le T + (L/4)^2$. Differentiating its integrand shows that $|J_q'(t)| \le C_\alpha a^{-2} J_q(t)$. By the formula for U_r and the heat equations, $\partial_t U_r$ is the integral over $v \in [0,1]$ of

$$(1-v) P^\delta_{(1-v)t} \Delta_\delta D^r_\delta G_{vt} + v P^\delta_{(1-v)t} D^r_\delta \partial_s G_s|_{s=vt}.$$

The bounds on the source control it by $CRg_3 a^{-\alpha-7/2-r}$, and one spatial difference of U_r is bounded by $CRg_3 a^{-\alpha-5/2-r}$. By the row bounds of Lemma 26.6, $U_r/(g_3 R a J_{3/4+r/2})$ is bounded uniformly. By the quotient rule, its moduli in time and space are bounded by fixed multiples of ϵ^{-8}. For example, the largest factor in the time derivative of the numerator is $a^{-\alpha-9/2-r}$, with exponent below seven, and the derivative of J_q contributes only a^{-2} times the already bounded quotient. The formula defines the continuous value at $t = 0$, and the pointwise normalized rows obey the modulus (26.16) as well.

For $0 < v \le 1$ we choose the spatial step $cv\epsilon^8$ and the time step $(cv\epsilon^8)^{1/\beta}$, where $c > 0$ depends only on the constant in the modulus. We choose nearest lattice sites when the spatial step is at least δ, and all lattice sites otherwise. Their number is at most $C(1 + L/(v\epsilon^8))^3$, and the time net has at most $C(1 + T/(v\epsilon^8)^{1/\beta})$ points. Both nets are deterministic, independent of the fluctuation variables and of the amplitude of the background. Multiplying these counts by each other and by the number of components and orders proves (26.14), for example with $P = \lceil 24 + 8/\beta \rceil$ and $Q = \lceil 3 + 1/\beta \rceil$. All constants have the stated independence, because only coefficient norms have been used and neither a sample nor the cutoff radius entered. □

26.3 Concentration and maximal inequalities

We now combine the deterministic estimates with concentration under the conditional fluctuation kernel. We control a supremum over space and time in two ways. The first uses the covering bound of Lemma 26.7 and chaining; its constants do not depend on the fluctuation cutoff or on the coupling, and it applies to mixed Duhamel terms with general backgrounds (Lemma 26.8). The second uses the fluctuation cutoff to bound moduli of continuity of the realized fields, followed by a union bound over a finite net at scale a (Lemma 26.9); its constants may depend on the fixed coupling and coarse side.

Lemma 26.8. *Assume the standing conventions of Section 26.2, and fix $\theta = \theta_{j-1}$ and $\mathfrak{d} \ge 0$. Let $(X_i)_{i \in \mathcal{T}}$ be a family of linear evaluations $X_i = l_i\zeta$, where $\zeta = w - \mathbb{E}_\theta w$ under $\mu_{\mathfrak{d},\theta}$, indexed by a finite list of components, the output sites, and the times in $[0, T]$, with coefficients continuous in time at each site. Suppose that $\sup_i \|l_i\|_2 \le C_0 S$, and that the distance $\|l_i - l_{i'}\|_2/S$ on $\mathcal{T}$ satisfies the covering bound* (26.14)*. Then, for every $u \ge 0$ and $1 \le p < \infty$,*

$$\mathbb{P}_\theta\Big\{\sup_i |X_i| > CS\sqrt{\Lambda_a + u}\Big\} \le Ce^{-u}, \qquad \Big\|\sup_i |X_i|\Big\|_{L^p(\mu_{\mathfrak{d},\theta})} \le CS(\sqrt{\Lambda_a} + \sqrt{p}).$$

The constants depend only on the lower bound $1/2$ for the Hessian and on the constants C_0, C, P, Q in the covering bound; in particular, they do not depend on the fluctuation cutoff.

Proof. By Lemma 26.1 and Theorem 5.4, for each coefficient row l, we have $\log \mathbb{E}_\theta e^{zl\zeta} \le z^2\|l\|_2^2$ for $z \in \mathbb{R}$, and the exponential Markov inequality shows that $l\zeta$ satisfies $\mathbb{P}_\theta\{|l\zeta| > 2\|l\|_2\sqrt{u}\} \le 2e^{-u}$; the same holds for every difference of two rows. We apply Lemma 16.4 with the distance $\|l_i - l_{i'}\|_2/S$, $s = CS$, $b = 0$, diameter at most $2C_0$, $A = C(1 + L/a)^P$, and entropy exponent Q. At a fixed reference index, the evaluation has coefficient norm at most $C_0 S$. By the Cauchy–Schwarz inequality in the finite-dimensional space of ζ, limits of rows in this distance correspond to limits of the evaluations. Because the coefficients are continuous in time and the lattice is finite, the supremum is determined by a countable dense set. Since $\log(2A) + Q \le C\Lambda_a$, the tail and moment bounds of Lemma 16.4 imply the assertions, with the explicit factor $\sqrt{p}$. If $S = 0$, all rows vanish. □

Lemma 26.9. *Assume the standing conventions of Section 26.2. Fix $\theta = \theta_{j-1}$ and the conditional fluctuation kernel of Lemma 26.1, and let $\zeta = w_j - m_\theta$.*

1. *A linear evaluation $l\zeta$ with $\|l\|_2 \le S$ satisfies $\mathbb{P}_\theta\{|l\zeta| > CS\sqrt{v}\} \le 2e^{-v}$ for $v > 0$.*
2. *A quadratic evaluation $\zeta^{\mathsf{T}} A\zeta - \mathbb{E}_\theta \zeta^{\mathsf{T}} A\zeta$, with A symmetric and $\|A\|_{\mathrm{HS}} \vee \|A\|_{\mathrm{op}} \le S$, satisfies*

$$\mathbb{P}_\theta\big\{|\zeta^{\mathsf{T}} A\zeta - \mathbb{E}_\theta \zeta^{\mathsf{T}} A\zeta| > C(\|A\|_{\mathrm{HS}}\sqrt{v} + \|A\|_{\mathrm{op}} v)\big\} \le 2e^{-v}, \qquad v > 0;$$

   in particular, the threshold $CS(1+v)$ suffices.
3. *Consider the following fields, all centered under $\mu_{\mathfrak{d}_j,\theta}$: the heat fields $P_\cdot^\delta I_a f$ of arrays f on the lattice of mesh a whose entries are linear or quadratic in ζ, measured in H_η^δ for some $0 < \eta < 3/2$; the mixed Duhamel terms of s_j, or of the quadratic part of z_j, with a background $P_\cdot^\delta I_H b$ fixed after conditioning, measured in Y_α^δ; and the quadratic Duhamel term $\mathcal{Q}_\delta(s_j, s_j) - \mathbb{E}_\theta \mathcal{Q}_\delta(s_j, s_j)$, measured in Y_α^δ. Suppose S bounds the weighted coefficient bound of the field and, for heat fields, also a^η times the coefficient norms of the entries of f. For a mixed Duhamel term let $R \ge \mathsf{R}_a(P_\cdot^\delta I_H b)$. In the mixed linear, mixed quadratic, and self-quadratic Duhamel cases, respectively, also require $S \ge Rg_3 a^{\alpha-1/2}$, $S \ge Rg_3^2 a^\alpha$, and $S \ge g_3^2 a^{2\alpha-1}$. These additional lower bounds are not imposed on heat fields. Then the norm of the field exceeds $CS\sqrt{\Lambda_a + u}$ in the linear case, and $CS(\Lambda_a + u)$ in the quadratic case, with probability at most Ce^{-u}, for every $u \ge 0$; the corresponding moment bounds hold for every $1 \le p < \infty$.*

The constants in (1) and (2) depend only on the lower bound $1/2$ for the Hessian. The constants in (3) may in addition depend on the fixed γ, n, and the prescription of the cutoff radii, as stated in Section 26.1.

Proof. The proof of (3) uses the fluctuation cutoff to bound the moduli of continuity of the realized fields. The short times $t \le a^2$, for heat fields, and $t \le \min(a^2, a^8)$, for Duhamel terms, are treated without a time net. On the remaining times, crude bounds by a fixed power of a^{-1} suffice, because the net then has polynomially many points in $1/a$ and only the logarithm of its size enters.

Step 1: pointwise tails. By Lemma 26.1, the potential is smooth on a neighborhood of the closed product Ω_j and $D_w^2 V \ge m_* I$ with $m_* = 1/2$. The domain is a finite product of Euclidean balls, with the reflecting boundary condition. These are the hypotheses of Theorem 5.4, applied to the potential itself, including its determinant and Haar disintegration terms. For a linear row l, that theorem implies that $\log \mathbb{E}_\theta e^{zl\zeta} \le z^2\|l\|_2^2/(2m_*)$, and optimizing Markov's inequality in z, we obtain the two-sided sub-Gaussian tail (1). For a symmetric matrix A, we write $a_2 = \|A\|_{\mathrm{HS}}$ and $a_\infty = \|A\|_{\mathrm{op}}$. By the same theorem, we have $\log \mathbb{E}_\theta e^{z(\zeta^{\mathsf{T}} A\zeta - \mathbb{E}_\theta \zeta^{\mathsf{T}} A\zeta)} \le 8z^2 a_2^2/m_*^2$ for $|z| \le m_*/(4a_\infty)$. Chernoff's inequality, applied with both signs and $z = \min(c_1\sqrt{v}/a_2, c_2/a_\infty)$, gives (2). Vanishing norms are

treated by continuity. Throughout, ζ is centered at the conditional mean m_θ under the same kernel.

Step 2: consequences of the cutoff. We now prove (3) by a finite net, including the endpoints of the time interval; this avoids any entropy at the scale δ. On Ω_j we have $\|\zeta\|_\infty \le C_J p_j$, and, at fixed γ and n, $p_j \le C_{\gamma,n}[1+\log(2+L/a)]$. In particular, $p_j^k \le C_{\gamma,n,k} a^{-1}$ for every fixed k when $a \le \min(1, \sqrt{T})$. The bounded remaining range of a is absorbed by enlarging the constants, which then depend on L, T, γ, and n.

Step 3: continuous extensions. We estimate the deterministic moduli through continuous extensions in space, starting from the smooth periodic cardinal interpolant on the torus. The exponential of the positive shift part defining P_t^δ acts on its translates for every real x, not only at lattice sites; this defines a smooth function whose restriction to the lattice is the discrete heat evolution. Continuous spatial derivatives commute with this convolution. Further,

$$\|\Delta_\delta f\|_{C^q} \le C \|f\|_{C^{q+2}},$$

by the second-order Taylor formula with integral remainder for each symmetric difference. Thus a time derivative costs two derivatives of the smooth data, never a factor δ^{-2}. The same extension applies to translated brackets and to their Duhamel integrals.

Step 4: unweighted bounds on the cutoff. Let the background $P_\cdot^\delta I_H b$ have radius at most R, and fix an order q of continuous spatial derivatives and an order k of time derivatives. On Ω_j, the supremum of $D^q \partial_t^k$ of each field or source satisfies the following bounds:

field or source	bound
$(g_j/a) P_t^\delta I_a E_\theta \zeta$	$C_{J,q,k} g_3 p_j a^{-1/2-q-2k}$
$P_t^\delta I_a \mathsf{B}_\theta(\zeta,\zeta)$	$C_{J,q,k} p_j^2 a^{-q-2k}$
$\mathcal{N}_\delta(B, s_j)$, $\mathcal{N}_\delta(s_j, B)$	$C_{J,q,k} R g_3 p_j a^{-\alpha-3/2-q-2k}$
$\mathcal{N}_\delta(B, g_3^2 P^\delta I_a \mathsf{B}_\theta(\zeta,\zeta))$, and the reversed order	$C_{J,q,k} R g_3^2 p_j^2 a^{-\alpha-1-q-2k}$
$\mathcal{N}_\delta(s_j, s_j)$	$C_{J,q,k} g_3^2 p_j^2 a^{-2-q-2k}$

To verify these entries, note that the absolute row bounds give $C_J p_j$ for $E_\theta \zeta$ and $C_J p_j^2$ for $\mathsf{B}_\theta(\zeta,\zeta)$. A spatial derivative of their cardinal interpolants costs $C_q a^{-q}$, and the heat semigroup contracts every continuous derivative. By its initial value and the heat equation, the background is bounded by $C_{J,q,k} R a^{-\alpha-q-2k}$. Using the quadratic coefficient of Lemma 24.2, we multiply the two factors in each bracket and distribute its difference; this gives the last three entries. By the product rule for differences, the same bounds hold for differences of these smooth extensions. A time derivative is the sum of the terms with one factor differentiated, and it satisfies the listed bound. Centering at the conditional mean doubles these bounds. For a Duhamel field, spatial derivatives are bounded by T

times the corresponding bound on the source, and its first time derivative is the source plus Δ_δ of the Duhamel field.

Step 5: heat fields. For a heat field measured with exponent $0 < \eta < 3/2$, we treat the times $0 \le t \le a^2$ without a time net. By heat contraction and the bound on differences of the cardinal interpolation, we have

$$\sup_{t \le a^2} \left\{ (t+\delta^2)^{\eta/2} \|P_t^\delta I_a f\|_\infty + (t+\delta^2)^{(\eta+1)/2} \|D_\delta P_t^\delta I_a f\|_\infty \right\} \le C_\eta a^\eta \max_y |f(y)|.$$

The maximum on the right is over at most $C_J (L/a)^3$ scalar entries, and we take a union bound over the tails of (1) or (2) at these entries. For $t \ge a^2$, the weights and their first time derivatives are bounded by fixed powers of a^{-1}. The following coefficient argument treats arbitrary admissible S, including for the special heat fields listed in the table. Let d_w be the number of scalar fluctuation coordinates of the step. Since there are J intermediate scales with fixed ratios of meshes, $d_w \le C_J (L/a)^3$. On the cutoff domain, we have $\|\zeta\|_2 \le C_J p_j \sqrt{d_w}$, and the assumed bounds on the coefficients of the entries of f give

$$\max_y |f_y| \le C_J S a^{-\eta} \times \begin{cases} p_j \sqrt{d_w} & \text{for centered linear forms,} \\ p_j^2 d_w & \text{for centered quadratic forms;} \end{cases}$$

in the quadratic case we apply the bound on the operator norm both to the value of the form and to its expectation. By the bounds on the cardinal interpolation, heat contraction, and the continuous extension of Step 3, we have

$$\|D^q \partial_t^k P_t^\delta I_a f\|_\infty \le C_{J,q,k} a^{-q-2k} \max_y |f_y|.$$

For $t \ge a^2$, the weighted value and first-difference components and their first spatial and time derivatives are bounded, after division by S, by a fixed power of a^{-1}, independently of a/δ. By the bounds of Step 2 on p_j, and since $d_w \le C a^{-3}$, the exponent 30 suffices. If $S = 0$, the coefficients of the entries of f vanish, and so does the field.

Step 6: Duhamel fields. For a Duhamel term, we set $t_0 = \min(T, a^2, a^8)$. If $t \le t_0 \le a^2$, its weighted value and first difference, divided by S, are bounded by $C p_j^k t/a^2$, with $k = 1$ for a mixed linear term and $k = 2$ for a quadratic term. Indeed, the last three source bounds of the table, multiplied by $t a^{2\alpha-1}$ and $t a^{2\alpha}$ at output orders zero and one, respectively, give this ratio for the natural scales $R g_3 a^{\alpha-1/2}$, $R g_3^2 a^\alpha$, and $g_3^2 a^{2\alpha-1}$, respectively. The additional lower bounds on S make the same bounds valid after division by S. If $R = 0$, each mixed field vanishes. Since $p_j^k t_0/a^2 \le p_j^k \min(1, a^6)$ is bounded at fixed coupling, using Step 2 also on the bounded coarse-mesh range, this is a deterministic contribution CS. If $t_0 < T$, then $t_0^{-1} = \max(a^{-2}, a^{-8}) \le \max\{1, (L/4)^6\} a^{-8}$. Thus on $t \ge t_0$ a differentiated weight costs at most $C a^{-8}$, with the allowed dependence on L. The table and the formula for

the time derivative of a Duhamel term again bound the first moduli in space and time by CSa^{-30}. The largest unweighted exponent of a source, after division by its scale, is less than 6; together with the factor a^{-8} from the weight and the growth of the cutoff, the total exponent stays below 30. If $t_0 = T$, the deterministic bound covers the whole interval.

Step 7: the net. We choose spatial and time steps ca^{30}, with c small depending on the constant in the modulus. A net of the torus and of the time interval then has at most

$$C(1 + La^{-30})^3(1 + Ta^{-30})$$

points, and the error in each normalized weighted component is at most one. When a step of the spatial net exceeds δ, we choose nearest lattice sites, and otherwise all lattice sites; the displayed count bounds either choice. The continuous extension is used only to bound the modulus between these sites, and all probability estimates are evaluated on the lattice. The logarithm of this count, and of the count of the short-time entries in Step 5, is at most $C\Lambda_a$, independently of a/δ. We apply the finite-family clause of Lemma 16.4 with $s = CS$ and $b = 0$ for a linear field, and with $s = CS$ and $b = CS$ for a quadratic field, using the tails of Step 1. The deterministic error from interpolation and from short times is $e \le CS$, and the logarithmic count is $C\Lambda_a$, with the dependence on the fixed γ and n stated above. This proves the maximal tails and all their moments. □

26.4 The linear fluctuation and the mixed Duhamel terms

We now prove the estimates for the linear fluctuation s_j and its mixed Duhamel terms with a predictable background. The background x_{j-1}, the heat evolution of the coarser field interpolated from mesh H, is covered by part (3) of the next theorem. The interpolation defect r_j, a predictable heat path on the finest lattice, is covered by parts (4) and (5). For the general class of part (4), we assume bounds on four differences of the background at scale a. These are needed for the covering bound of Lemma 26.7, and a bound on the value and the first difference alone does not give them uniformly at small times.

Theorem 26.10. *Assume the setting of Section 26.1 and the standing conventions of Section 26.2, and put $W_\theta(t) = P_t^\delta I_a E_\theta$, with $\theta = \theta_{j-1}$.*

1. *For every fixed integer $r \ge 0$, the coefficient rows and the operator of W_θ satisfy*

$$\sup_x \|D_\delta^r W_\theta(t,x,\cdot)\|_{\ell^2(w)} \le C_{J,r} a^{3/2}(t + a^2)^{-3/4-r/2},$$
$$\|D_\delta^r W_\theta(t)\|_{\ell^2(w)\to L^2_\delta} \le C_{J,r} a^{3/2}(t + a^2)^{-r/2}.$$

Each fluctuation coordinate enters with its own cell volume v_λ, which is comparable to a^3 up to a factor depending only on M and J. For the same scalar output component, times

$s, t \in [0, T]$ with $\tau = \min(s, t)$, and output sites x, y, the difference of the coefficient rows at (t, x) and at (s, y) is bounded in $\ell^2(w)$ by

$$C_{J,r} a^{3/2} (\tau + a^2)^{-3/4 - r/2} \min\Big\{1, \frac{\operatorname{dist}(x, y)}{\sqrt{\tau + a^2}} + \frac{|t - s|}{\tau + a^2}\Big\}.$$

2. *For every $1 \le p < \infty$, uniformly in θ_{j-1},*

$$\big\| \mathsf{X}^{\delta}_{\alpha}(s_j) \big\|_{L^p(\mu_{\mathfrak{d}_j, \theta})} \le C_J g_3 a^{\beta} (\sqrt{\Lambda_a} + \sqrt{p}),$$

with C_J independent of p.

3. *Let $B_j(t) = P^{\delta}_t I_H b$ be an $\mathcal{F}_{j-1}$-measurable heat evolution of an array b interpolated from the lattice of mesh H, and let $\mathsf{R}_j(B_j) = \mathsf{R}_a(B_j)$ be its radius* (26.11)*. For each fixed $r \ge 2$,*

$$\| D^r_{\delta} B_j(t) \|_{\infty} \le C_{r,B}\, \mathsf{R}_j(B_j) (t + a^2)^{-(\alpha + r)/2},$$

where $C_{r,B}$ depends only on r, M, J, α, T/L^2, and the interpolation function, and not on b or its radius. For each of the two orders $\mathcal{M}_j = \mathcal{Q}_{\delta}(B_j, s_j)$ and $\mathcal{M}_j = \mathcal{Q}_{\delta}(s_j, B_j)$, the value rows and first-difference rows of the coefficients are bounded by

$$C_J g_3 \mathsf{R}_j(B_j) a\, I^a_{3/4}(t) \qquad \text{and} \qquad C_J g_3 \mathsf{R}_j(B_j) a\, I^a_{5/4}(t),$$

respectively. Each order has conditional mean zero given $\mathcal{F}_{j-1}$. For every $u \ge 0$, on the event $\{\mathsf{R}_j(B_j) \le R\}$,

$$\mathbb{P}\Big\{ \mathsf{Y}^{\delta}_{\alpha}(\mathcal{M}_j) > C_J g_3 R a^{\beta} \sqrt{\Lambda_a + u} \;\Big|\; \mathcal{F}_{j-1} \Big\} \le C e^{-u}.$$

The sum $\mathcal{B}_{\delta}(B_j, s_j)$ of the two orders satisfies the same estimate with a larger C_J.

4. *The two mixed-term coefficient row bounds, conditional mean-zero assertions, and conditional maximal inequalities of (3) also hold for an $\mathcal{F}_{j-1}$-measurable heat path $B_j(t) = P^{\delta}_t b^{\delta}_j$ on the lattice of mesh δ, provided an $\mathcal{F}_{j-1}$-measurable radius $R^{\sharp}_j$ satisfies*

$$\| D^q_{\delta} B_j(t) \|_{\infty} \le C_q R^{\sharp}_j (t + a^2)^{-(\alpha + q)/2} \qquad (0 \le t \le T,\ 0 \le q \le 4),$$

with fixed constants $C_0, \dots, C_4$; the heat equation then supplies the increments in time. For this class we replace $\mathsf{R}_j(B_j)$ by $R^{\sharp}_j$ in both row bounds and in the conditional tail. The normalized coefficient distances of Lemma 26.7 have diameter bounded by a fixed constant and covering numbers $N(v) \le C(1 + L/a)^P v^{-Q}$, $0 < v \le 1$, with P, Q, C independent of a/δ, the lattice side, the number of averaging steps, θ_{j-1}, and the coupling. The covering is at scale a; no maximum over all sites of the finest lattice is used.

5. *The interpolation defect*

$$B_j(t) = H^{-1} P_t^{\delta}(I_a \mathsf{P}_j - I_H \mathsf{E}_H)\theta_{j-1} = r_j(t)$$

belongs to the class of (4), with $R_j^{\sharp} = C_J \|\theta_{j-1}\|_{\infty} H^{\alpha-1}$. *If* $\|\theta_{j-1}\|_{\infty} \le \bar{b}_{j-1}$ *on the small-field event, for a deterministic* $\bar{b}_{j-1}$, *then for every* $1 \le p < \infty$,

$$\big\|\mathsf{Y}_{\alpha}^{\delta}(\mathcal{Q}_{\delta}(s_j, r_j))\big\|_{L^p(\nu_K)} + \big\|\mathsf{Y}_{\alpha}^{\delta}(\mathcal{Q}_{\delta}(r_j, s_j))\big\|_{L^p(\nu_K)} \le C_{J,p} g_3 a^{\beta} (\sqrt{\Lambda_a} + \sqrt{p})\, \bar{b}_{j-1} H^{\alpha-1}.$$

All assertions hold for either expansion of Lemma 26.2, with the conditionally centered vector ζ_j *in both. The constants in the coefficient bounds depend only on* L, T, M, J, *the analytic domains, the column bounds,* α, *and* r; *they do not depend on* K, j, $\mathfrak{d}_j$, a/δ, *the dimension, the coupling, or* θ_{j-1}. *The constants in the conditional maximal inequalities may in addition depend on the fixed coupling, the coarse side, and the prescription of the cutoff radii. No bound on the expectation of a background radius is assumed or proved.*

Proof. We first prove the deterministic assertions, then the probability estimates, and then the application to r_j.

Step 1: coefficient rows. By Theorem 13.10, the first reconstruction tensor has bounded absolute rows and volume-normalized column sums. Each fluctuation coordinate has the volume $v_\lambda = a_\lambda^3$ of the lattice of its intermediate scale, and the finite ratios between intermediate scales imply that $C_J^{-1} a^3 \le v_\lambda \le C_J a^3$; there is no comparison with δ^3. This verifies the hypotheses of Lemma 16.3 with input mesh a and output mesh δ, and that lemma yields the two bounds of (1) for every fixed order r.

Step 2: increments of the rows. Fix r. A shortest coordinate path on the torus from x to y and the bound on rows of order $r+1$ bound the spatial increment by $C_{J,r}\,\mathrm{dist}(x,y) a^{3/2}(t + a^2)^{-5/4-r/2}$; the triangle inequality gives twice the bound of order r instead. For the time increment, we integrate $\partial_t D_\delta^r W_\theta = \Delta_\delta D_\delta^r W_\theta$ from $\min(s,t)$ to $\max(s,t)$ and use the bound of order $r+2$. Combining these linear increment bounds with twice the bound on values, we obtain the stated minimum. The coordinate path has length at most $\sqrt{3}$ times the distance on the torus, which is absorbed into $C_{J,r}$.

Step 3: the linear fluctuation. Since $g_j/a = g_3 a^{-1/2}$ and $\delta \le a$, the weighted coefficient bound of s_j for X_α^δ is at most the supremum over t of

$$(t+\delta^2)^{\alpha/2} g_3 a^{-1/2} a^{3/2} (t+a^2)^{-3/4} \quad \text{and} \quad (t+\delta^2)^{(\alpha+1)/2} g_3 a^{-1/2} a^{3/2} (t+a^2)^{-5/4}.$$

Both are bounded by $C_J g_3 a (t+a^2)^{\alpha/2-3/4} \le C_J g_3 a^{\alpha-1/2}$, because $\alpha < 3/2$. At a fixed coarser field, these evaluations are linear in ζ_j, and their conditional mean is zero. The entries of the initial array $(g_j/a) E_\theta \zeta_j$ have coefficient norms at most $C_J g_3 a^{-1/2}$, and a^α

times these norms is at most $C_J g_3 a^\beta$. The conditional fluctuation kernel and its Hessian bound $1/2$ are verified in Lemma 26.1. By Lemma 26.9, with $S = C_J g_3 a^\beta$, we have

$$\mathbb{P}_\theta\{\mathsf{X}_\alpha^\delta(s_j) > C_J g_3 a^\beta \sqrt{\Lambda_a + u}\} \le Ce^{-u}.$$

Integrating this tail, we obtain the moment bound of (2). In detail, $\sqrt{\Lambda_a + u} \le \sqrt{\Lambda_a} + \sqrt{u}$, and the positive part of the normalized norm beyond $\sqrt{\Lambda_a}$ has a tail bounded by Ce^{-cv^2} at level v. Its pth moment is at most $(C\sqrt{p})^p$: in the integral $p\int_0^\infty v^{p-1}e^{-cv^2}\,dv$, we bound $v^{p-1}e^{-cv^2/2}$ by its maximum and integrate $e^{-cv^2/2}$. The constants of Lemma 26.9 do not depend on p, so C_J does not depend on p.

Step 4: rows of the mixed terms. The bound on higher differences of $B_j = P_\cdot^\delta I_H b$ was proved in Step 1 of the proof of Lemma 26.6, from the values of the interpolation at the sites of the lattice of mesh H at time zero and from the splitting of the heat semigroup at later times. Its constant $C_{r,B}$ depends only on r, M, J, α, T/L^2, and the interpolation function, and not on the array b or its radius. We apply Lemma 26.6 with $F_s = (g_j/a)W_\theta(s)$ and $h = (g_j/a)a^{3/2} = g_3 a$; by (1), F_s satisfies its hypotheses. For each order separately, we obtain the value rows $C_J g_3 \mathsf{R}_j(B_j) a\, I^a_{3/4}(t)$ and the first-difference rows $C_J g_3 \mathsf{R}_j(B_j) a\, I^a_{5/4}(t)$. The two orders are linear functions of ζ_j with $\mathcal{F}_{j-1}$-measurable coefficients. Each therefore has conditional mean zero as a field in space and time, and no covariance needs to be computed.

Step 5: conditional maxima of the mixed terms. By Lemma 26.6, the weighted coefficient bound of each order is $C_J g_3 \mathsf{R}_j(B_j) a^\beta$. We fix a coarser field in the measurable event $\{\mathsf{R}_j(B_j) \le R\}$. The four differences of the background were just proved, and Lemmas 26.7 and 26.8 yield the conditional tail of (3) for each order. A union bound for the two tails and the triangle inequality for the sum give the assertion for $\mathcal{B}_\delta(B_j, s_j)$; the factor two is absorbed into C_J and C. If $R = 0$, then on this event $\mathsf{R}_j(B_j) = 0$ and $I_H b = B_j(0) = 0$; since the interpolation reproduces b at the sites of the lattice of mesh H, we have $b = 0$, and both terms vanish identically.

Step 6: general backgrounds. For the class of (4), we condition on $\mathcal{F}_{j-1}$ and use the radius $R_j^\sharp$ and the four assumed difference bounds. These are the hypotheses of Lemmas 26.6 and 26.7. The former implies both row bounds and the common weighted bound $C_J g_3 R_j^\sharp a^\beta$; its proof treats the difference in either argument. The latter implies the bounded diameter and the covering bound (26.14), independently of the cutoff and of the coupling; the heat equation gives the increments in time used there, with at most four differences of the background. Each term is linear in the conditionally centered ζ_j and has conditional mean zero. On $\{R_j^\sharp \le R\}$, Lemma 26.8 implies the conditional maximum and its moments. If the radius vanishes, the bound on the background of order zero shows that both terms vanish. Every conditional event is measurable: the reconstruction

and its coefficients are Borel, the lattice is finite, and suprema over continuous time are determined by rational times.

Step 7: the interpolation defect. We verify that r_j belongs to the class of (4), keeping the two lattices on which it is defined. Here $a = a_j$, $H = Ba$, and $\delta = a_K$, so

$$\delta = a/m, \qquad m = B^{K-j} \in \mathbb{N}, \qquad H/\delta = B^{K-j+1} \in \mathbb{N}.$$

We apply Lemma 17.2 with coarse mesh H, fine mesh a, $\mathfrak{d}_j$ preceding averaging steps, horizon T, and the integer refinements just displayed. By the expansion (26.3), its fine reconstruction of a lattice potential v on the coarser lattice is $\mathsf{P}_j v$, and its coarse embedding is $\mathsf{E}_H v$. It follows that its difference operator is $I_a \mathsf{P}_j - I_H \mathsf{E}_H$, and with $v = \theta_{j-1}/H$ its heat evolution is r_j. By that lemma, for every fixed integer q, we have

$$\|D_\delta^q r_j(t)\|_\infty \le C_{J,q} \frac{\|\theta_{j-1}\|_\infty}{H} (t + H^2)^{-q/2} \frac{H}{\sqrt{t + H^2}}.$$

For $z = t/H^2$, $0 < \alpha < 1$, and $B \ge 1$ we have

$$(z + B^{-2})^{(\alpha+q)/2} (1 + z)^{-(q+1)/2} \le (1 + z)^{(\alpha-1)/2} \le 1.$$

Multiplying the bound on $\|D_\delta^q r_j(t)\|_\infty$ by $(t + a^2)^{(\alpha+q)/2} = H^{\alpha+q}(z + B^{-2})^{(\alpha+q)/2}$, we obtain

$$(t + a^2)^{(\alpha+q)/2} \|D_\delta^q r_j(t)\|_\infty \le C_{J,q} \|\theta_{j-1}\|_\infty H^{\alpha-1}.$$

Choosing C_J large enough for $q = 0, 1, 2, 3, 4$, we obtain the radius $R_j^\sharp = C_J \|\theta_{j-1}\|_\infty H^{\alpha-1}$.

Step 8: moments under the conditioned Wilson measure. For either order separately, the conditional moment bound of (4) implies that

$$\Big(\mathbb{E}\big[\mathsf{Y}_\alpha^\delta(\mathcal{Q}_\delta(s_j, r_j))^p \mid \mathcal{F}_{j-1} \big] \Big)^{1/p} \le C_J g_3 a^\beta (\sqrt{\Lambda_a} + \sqrt{p}) \|\theta_{j-1}\|_\infty H^{\alpha-1},$$

and the same inequality holds with the two arguments interchanged, by the separate row computation for that order. We replace $\|\theta_{j-1}\|_\infty$ by $\bar{b}_{j-1}$, raise to the power p, and integrate over the law of θ_{j-1} under ν_K. By the tower property, we obtain both $L^p(\nu_K)$ bounds, and adding them gives (5) after changing $C_{J,p}$.

Step 9: the expansion at zero. By Lemma 26.1, we have the same rows and columns for $E_\theta^{(0)}$. Replacing E_θ by $E_\theta^{(0)}$ in the Schur, mixed-term, covering, and concentration arguments proves all assertions for the second expansion. The centered vector ζ_j, its domain, and its law are the same in both cases. □

26.5 The conditionally centered nonlinear residual

The conditionally centered nonlinear reconstruction residual Z_j is the reconstructed field after subtraction of its conditional mean and of the linear term; it is not a quadratic approximation. Its quadratic part is a centered quadratic form in ζ_j, to which the quadratic concentration of Lemma 26.9 applies, and the rest is bounded deterministically on the cutoff domain.

Theorem 26.11. *Assume the setting of Section 26.1 and the standing conventions of Section 26.2, and put $a = a_j$, $\theta = \theta_{j-1}$, and $\zeta = w_j - m_\theta$. Let $p_j \geq 1$ bound the cutoff radii of this step, so that $\|\zeta\|_\infty \leq 2C_J p_j$.*

1. *The initial residual is*

$$Z_j = g_3^2\big\{\mathsf{B}_\theta(\zeta,\zeta) - \mathbb{E}_\theta \mathsf{B}_\theta(\zeta,\zeta)\big\} + a^{-1}\big\{R_\theta(\zeta) - \mathbb{E}_\theta R_\theta(\zeta)\big\},$$

   and $\|a^{-1}(R_\theta - \mathbb{E}_\theta R_\theta)\|_\infty \leq C_J g_3^3 a^{1/2} p_j^3$.

2. *The coefficients $b_{y;\lambda\mu}$ of B_θ satisfy*

$$\sup_y \sum_{\lambda,\mu} |b_{y;\lambda\mu}| \leq C_J, \qquad \sup_\lambda \frac{a^3}{v_\lambda} \sum_{y,\mu} |b_{y;\lambda\mu}| \leq C_J$$

   for either fixed input, where v_λ is the cell volume of the coordinate λ and $v_\lambda / a^3 \leq C_J$. Consequently

$$\|\mathsf{B}_\theta\|_{\ell^2(\text{ordered pairs}) \to L^2_a} \leq C_J a^{3/2}.$$

   For each scalar component and every fixed $r \geq 0$, the symmetric coefficient matrix $\mathsf{A}^{(r)}_{t,x}$ of $D^r_\delta P^\delta_t I_a \mathsf{B}_\theta(\zeta,\zeta)$ at x satisfies

$$\|\mathsf{A}^{(r)}_{t,x}\|_{\mathrm{HS}} \vee \|\mathsf{A}^{(r)}_{t,x}\|_{\mathrm{op}} \leq C_{J,r} a^{3/2} (t + a^2)^{-3/4 - r/2}.$$

   Centering changes only the constant term of the quadratic form, not this matrix.

3. *For $0 < \sigma < 1 - \alpha$ and every $u \geq 0$,*

$$\mathbb{P}_\theta\Big\{\mathsf{H}^\delta_\sigma\big(I_a[\mathsf{B}_\theta(\zeta,\zeta) - \mathbb{E}_\theta \mathsf{B}_\theta(\zeta,\zeta)]\big) > C_J a^\sigma (\Lambda_a + u)\Big\} \leq C e^{-u},$$

   and the same bound holds for $\mathsf{H}^\delta_\alpha = \mathsf{X}^\delta_\alpha(P^\delta_\cdot\, \cdot\,)$ with a^α in place of a^σ. In particular, for every fixed $1 \leq p < \infty$, both conditionally at every θ_{j-1} and under ν_K,

$$\|\mathsf{H}^\delta_\sigma(I_a Z_j)\|_p \leq C_{p,J}\big\{g_3^2 a^\sigma (1+j) + g_3^3 a^{1/2} p_j^3\big\},$$
$$\|\mathsf{X}^\delta_\alpha(z_j)\|_p \leq C_{p,J}\big\{g_3^2 a^\alpha (1+j) + g_3^3 a^{1/2} p_j^3\big\}.$$

4. *Let $B_j = x_{j-1}$, with radius $\mathsf{R}_j = \mathsf{R}_a(x_{j-1})$ as in Theorem 26.10. Each of $\mathcal{Q}_\delta(B_j, z_j)$ and $\mathcal{Q}_\delta(z_j, B_j)$ has conditional mean zero given $\mathcal{F}_{j-1}$. Its part that is quadratic in ζ has weighted coefficient bound, in Hilbert–Schmidt and operator norm for Y^δ_α, at most $C_J \mathsf{R}_j g_3^2 a^\alpha$, and its remaining centered part has Y^δ_α norm at most $C_J \mathsf{R}_j g_3^3 a^{1/2} p_j^3$. In particular, on $\{\mathsf{R}_j \le R\}$, each order and their sum satisfy*

$$\mathbb{P}\Big\{ \mathsf{Y}^\delta_\alpha(\mathcal{B}_\delta(B_j, z_j)) > C_J R\big[g_3^2 a^\alpha (\Lambda_a + u) + g_3^3 a^{1/2} p_j^3 \big] \;\Big|\; \mathcal{F}_{j-1} \Big\} \le C e^{-u}.$$

The constants in the coefficient bounds have the dependence stated in Theorem 26.10; the constants in the maximal and L^p bounds may in addition depend on the fixed γ, n, p, and σ. All constants are uniform in $K \ge j$, in the number of preceding averaging steps, in θ_{j-1}, in the dimension, and in a/δ. Here $p_j \le C_\gamma(1+j)$, and the factor $1+j$ absorbs Λ_a at the fixed coarse side n; this does not give uniformity as $n \to \infty$. All norms include the first-difference component at short times. Both expansions of Lemma 26.2 are allowed.

Proof. The proof has five steps: the exact form of Z_j, the coefficient bounds, the maximal inequality for the quadratic part, the remainder and the moments, and the two mixed terms.

Step 1: the exact residual. We take the conditional expectation of the Taylor expansion (26.2); its linear term has expectation zero. Subtracting this conditional expectation, then the linear term, and dividing by a, we obtain

$$Z_j = \frac{g_j^2}{a}\big(\mathsf{B}_\theta(\zeta,\zeta) - \mathbb{E}_\theta \mathsf{B}_\theta(\zeta,\zeta)\big) + \frac{1}{a}\big(R_\theta(\zeta) - \mathbb{E}_\theta R_\theta(\zeta)\big).$$

Since $g_j^2/a = g_3^2$, this is the identity of (1). By (26.9) and $\|\zeta\|_\infty \le 2C_J p_j$, the centered remainder is at most $2C_J g_j^3 p_j^3/a = C_J g_3^3 a^{1/2} p_j^3$.

Step 2: coefficient bounds. We apply the row and column bounds of Theorem 13.10 to the scalar coefficients of B_θ. A fluctuation coordinate on its intermediate lattice contributes its own volume v_λ, and converting the output volume to a^3, we obtain the two bounds of (2); the same holds with μ as the fixed input. The normalization by g_j^2 introduces only the fixed ratios of intermediate couplings, and the coefficients do not depend on the realized value of w_j. Lemma 26.5 now implies the operator bound for pairs and the bounds on the matrices $\mathsf{A}^{(r)}_{t,x}$. The symmetric coefficient matrix represents the same quadratic polynomial, and subtracting the conditional mean changes only its constant term.

Step 3: the maximal inequality for the quadratic part. Let $f_y = \mathsf{B}_\theta(\zeta,\zeta)_y - \mathbb{E}_\theta \mathsf{B}_\theta(\zeta,\zeta)_y$. We first treat short times explicitly, including the first difference. For every $0 < \eta < 3/2$,

by heat contraction and the bound on differences of the cardinal interpolation, we have

$$\sup_{0\le t\le a^2}(t+\delta^2)^{\eta/2}\|P_t^\delta I_a f\|_\infty \le C_\eta a^\eta \max_y |f_y|,$$
$$\sup_{0\le t\le a^2}(t+\delta^2)^{(\eta+1)/2}\|D_\delta P_t^\delta I_a f\|_\infty \le C_\eta a^\eta \max_y |f_y|.$$

In the second line, we commute D_δ with the heat semigroup and use $\|D_\delta I_a f\|_\infty \le Ca^{-1}\max_y|f_y|$. There are only $C(L/a)^3$ scalar entries on the right, and the matrix of each entry has Hilbert–Schmidt and operator norm at most C_J, by its absolute row. For $t \ge a^2$, by (2), the weighted coefficient matrices have norms at most

$$C_{J,q}a^{3/2}(t+a^2)^{(\eta-3/2)/2} \le C_{J,q}a^\eta, \qquad q=0,1,$$

where the weight is $(t+\delta^2)^{\eta/2}$ for the value and $(t+\delta^2)^{(\eta+1)/2}$ for the difference. This proves the bound $C_J a^\eta$ for the weighted coefficient bound, including its first-difference component. By the Hessian bound verified in Lemma 26.1, the quadratic part of Lemma 26.9 applies; taking its union over short-time entries and its net at scale a, we obtain

$$\mathbb{P}_\theta\{\mathsf{H}_\eta^\delta(I_a f) > C_J a^\eta(\Lambda_a + u)\} \le Ce^{-u}.$$

We take $\eta = \sigma$ for the first bound of (3), and $\eta = \alpha$ for the second; in the latter case $\mathsf{H}_\alpha^\delta(I_a f)$ is the same heat expression $\mathsf{X}_\alpha^\delta(P_\cdot^\delta I_a f)$. No restriction on η coming from products is needed for this heat estimate; only the later products require $\alpha+\sigma<1$.

Step 4: the remainder and the moments. Let $e = a^{-1}(R_\theta - \mathbb{E}_\theta R_\theta)$ and $\epsilon_j = C_J g_3^3 a^{1/2} p_j^3$. By the bounds on the cardinal interpolation and on the heat kernel, we have

$$\|P_t^\delta I_a e\|_\infty \le \epsilon_j, \qquad \|D_\delta P_t^\delta I_a e\|_\infty \le C_J \epsilon_j (t+a^2)^{-1/2}.$$

In particular, both $\mathsf{H}_\sigma^\delta(I_a e)$ and $\mathsf{H}_\alpha^\delta(I_a e)$ are at most $C_{J,T,L,\alpha,\sigma}\epsilon_j$. For example, the difference component is bounded by $C\epsilon_j(t+a^2)^{\eta/2}$ after multiplication by its weight, for either $\eta=\sigma$ or $\eta=\alpha$, and the horizon and $a \le L/4$ bound the last factor. Integrating the quadratic tail of Step 3, we obtain the conditional moment bound $C_{p,J}a^\eta(\Lambda_a+p+1)$. Since $\Lambda_{a_j} \le C_{n,B}(1+j)$, combining this with the deterministic bound on the remainder, we find the two conditional L^p bounds of (3). For fixed p, we raise these conditional bounds to the power p and integrate over the law of $\mathcal{F}_{j-1}$ under ν_K; the tower property gives the same bounds under ν_K. This changes neither the law of the coarsest configuration nor the conditional kernels of the finer variables.

Step 5: the mixed terms. We first treat the quadratic part. Let $V_s = P_s^\delta I_a \mathsf{B}_\theta(\zeta,\zeta)$, regarded as a coefficient map from ordered pairs of fluctuation coordinates; its rows and operator norm are bounded by Lemma 26.5. We apply Lemma 26.6 with $F_s = g_3^2 V_s$ and $h = g_3^2 a^{3/2}$;

the background $x_{j-1} = P_\cdot^\delta I_H A_{j-1}$ satisfies its hypotheses with $R = \mathsf{R}_j$. The unweighted value rows and first-difference rows of either order are

$$C_J \mathsf{R}_j g_3^2 a^{3/2} I^a_{3/4}(t) \qquad \text{and} \qquad C_J \mathsf{R}_j g_3^2 a^{3/2} I^a_{5/4}(t),$$

respectively, and after weighting they are at most $C_J \mathsf{R}_j g_3^2 a^\alpha$. Rows indexed by pairs have the Hilbert–Schmidt norms of the corresponding matrices, and symmetrization gives the same bound for the operator norm. Centering the quadratic polynomial at its conditional expectation centers the Duhamel term, because, with the background fixed, the term is a fixed linear function of its initial input.

We prove the bound for the remaining part directly. Let $V_s^e = P_s^\delta I_a e$. By the bounds of Step 4 and the radius of the background, we have, in either order,

$$\|\mathcal{N}_\delta(B_j(s), V_s^e)\|_\infty + \|\mathcal{N}_\delta(V_s^e, B_j(s))\|_\infty \le C_J \mathsf{R}_j \epsilon_j (s + a^2)^{-p_\alpha}.$$

For the value of a Duhamel term, contraction and integration give $C\mathsf{R}_j\epsilon_j(t+\delta^2)^{1-p_\alpha}$. For its first difference, we put the difference on the outer heat kernel and use Lemma 26.4; this gives the bound

$$C_J \mathsf{R}_j \epsilon_j \int_0^t (t - s + \delta^2)^{-1/2} (s + a^2)^{-p_\alpha}\, ds \le C_J \mathsf{R}_j \epsilon_j (t + \delta^2)^{1/2 - p_\alpha}.$$

To check the last inequality, we replace $s + a^2$ by $s + \delta^2$ and split the integral at $t/2$; both exponents are below one, and for $t \le \delta^2$ the length of the interval suffices. The weights of Y_α^δ for the value and the difference leave the same nonnegative power $(t+\delta^2)^{\alpha/2}$. This proves the bound $C_J \mathsf{R}_j \epsilon_j$ for the Y_α^δ norm. The same bound follows from Lemma 24.6 applied on the lattice of mesh $\delta = L/(nB^K)$, with first argument the heat path B_j, bounded in X_α^δ, and second argument the heat evolution of $I_a e$, bounded in H_σ^δ: there $\mathsf{X}_\alpha^\delta(B_j) \le \mathsf{R}_j$, $\mathsf{H}_\sigma^\delta(I_a e) \le C\epsilon_j$ by Step 4, $\alpha + \sigma < 1$, and T is fixed. No hypothesis $\delta \le 1$ is needed there: when $\delta > 1$, the clause of that lemma for coarse meshes applies to the finitely many integers $4 \le nB^K < L$ at this fixed L, and the maximum of its constants over these meshes is an allowed constant.

Conditional on $\mathcal{F}_{j-1}$, the map from Z_j to either Duhamel term is linear, and $\mathbb{E}(Z_j \mid \mathcal{F}_{j-1}) = 0$. Hence the term, including the remaining part, has conditional mean zero. On $\{\mathsf{R}_j \le R\}$, the quadratic part of Lemma 26.9 implies the threshold $R g_3^2 a^\alpha (\Lambda_a + u)$, and the deterministic bound just proved yields the threshold $R g_3^3 a^{1/2} p_j^3$. A union bound over the two orders gives the estimate for each order and for their sum.

Finally, for the expansion at zero, Lemma 26.1 provides the second tensor $\mathsf{B}_\theta^{(0)}$ with the same column bounds, and the remainder $\widetilde{R}_\theta$ of Lemma 26.2 has, after centering, the same deterministic bound. The Schur and mixed-term arguments then apply to this second decomposition. □

26.6 The quadratic Duhamel term of the linear fluctuation

The last estimate of this section concerns $\mathcal{Q}_\delta(s_j, s_j)$. Its centered part is a quadratic form in ζ_j, and its conditional mean is in general not zero. The mean vanishes when the coarser field is the identity, by the invariance of the conditional law under global color rotations; at a nonzero coarser field we bound it by comparing the covariance of the linear reconstruction term with its value at the identity.

Theorem 26.12. *Assume the setting of Section 26.1 and the standing conventions of Section 26.2, and put $a = a_j$ and $r_\theta = \|\theta_{j-1}\|_\infty$. Either expansion of Lemma 26.2 may be used.*

1. *For each translated bracket in the quadratic coefficient $\mathcal{N}_\delta$, and for their sum, the matrix $\mathsf{C}^{(r)}_{s,x}$ of the quadratic form $D^r_\delta \mathcal{N}_\delta(s_j(s), s_j(s))(x)$ in ζ_j satisfies*

$$\|\mathsf{C}^{(r)}_{s,x}\|_{\mathrm{HS}} \vee \|\mathsf{C}^{(r)}_{s,x}\|_{\mathrm{op}} \le C_J g_3^2 a^2 (s + a^2)^{-2-r/2}, \qquad r = 0, 1.$$

   The corresponding matrices $\mathsf{A}^{(r)}_{t,x}$ of $D^r_\delta \mathcal{Q}_\delta(s_j, s_j)(t, x)$ satisfy

$$\|\mathsf{A}^{(r)}_{t,x}\|_{\mathrm{HS}} \vee \|\mathsf{A}^{(r)}_{t,x}\|_{\mathrm{op}} \le C_J g_3^2 \frac{t}{t + a^2}\, a^{3/2} (t + a^2)^{-3/4-r/2}, \qquad r = 0, 1,$$

   and hence the weighted coefficient bound of $\mathcal{Q}_\delta(s_j, s_j)$ for Y^δ_α is $C_J g_3^2 a^{2\alpha-1}$. These coefficients keep the physical factor δ^3 of the heat semigroup and the meshes of the input coordinates; no factor $(a/\delta)^{3/2}$ is lost. Every further fixed difference in space or time needed for the covering at scale a is bounded by a fixed power of a^{-1}, with a constant depending on its order, and with the same factor g_3^2.
2. *Regard the normalized first derivative of the reconstruction as a jointly analytic function of the unscaled coarser field and the coupling-scaled fluctuation inputs $q_{j,i} = g_{j,i} w_i$, before its base point in these inputs is evaluated at the conditional mean. Choose an inner polydisc whose gaps from the fixed larger polydisc are $\eta_\lambda > 0$ in each coordinate, and evaluate the bounds of (1) on the larger polydisc. For every finite multi-index $\mathbf{k}$, the bounds of (1) for the source and Duhamel coefficients hold for the derivatives $\partial^{\mathbf{k}}$, with the additional factor $\mathbf{k}! \prod_\lambda \eta_\lambda^{-k_\lambda}$. If an input coordinate on mesh a_λ is written instead as a lattice potential, or as a normalized fluctuation variable of coupling g_λ, it contributes the factor $a_\lambda^{k_\lambda}$, or $g_\lambda^{k_\lambda}$, respectively, and each difference at the location of that input may cost $2/a_\lambda$. Coordinates of the coarser field use the mesh H, the q coordinates use the mesh of their intermediate scale, and none uses δ. These derivatives are taken with the conditional centering held fixed; they are not derivatives of m_θ or of the conditional law.*
3. *Let Σ_θ be the conditional fluctuation covariance of w_j and $\mathsf{Z}_\theta = E_\theta \Sigma_\theta E_\theta^{\mathsf{T}}$. Then $\|\mathsf{Z}_\theta - \mathsf{Z}_0\|_{\mathrm{op}} \le C_J r_\theta$, and Z_0 is proportional to the identity in the color indices. The conditional*

mean of the Duhamel term satisfies

$$\begin{aligned}\|\mathbb{E}_\theta \mathcal{Q}_\delta(s_j,s_j)(t)\|_\infty &\le C_J g_3^2 r_\theta \frac{t}{t+a^2},\\ \|D_\delta \mathbb{E}_\theta \mathcal{Q}_\delta(s_j,s_j)(t)\|_\infty &\le C_J g_3^2 r_\theta (t+a^2)^{-1/2}\min(1,t/a^2),\\ \mathsf{Y}_\alpha^\delta(\mathbb{E}_\theta \mathcal{Q}_\delta(s_j,s_j)) &\le C_{J,T,\alpha} g_3^2 r_\theta.\end{aligned}$$

These means vanish when the coarser field is the identity, and they are kept when it is not.

4. *For every $u \ge 0$,*

$$\mathbb{P}_\theta\Big\{\mathsf{Y}_\alpha^\delta\big(\mathcal{Q}_\delta(s_j,s_j)-\mathbb{E}_\theta\mathcal{Q}_\delta(s_j,s_j)\big) > C_J g_3^2 a^{2\alpha-1}[\log(2+L/a)+u]\Big\} \le Ce^{-u}.$$

5. *Let $b_0=\rho_0$ and $b_i=\rho_0 a_{\mathrm{cap}}^i + C_J g_3\sqrt{a_i}\,p_i$ for $i\ge 1$ be the deterministic bounds on $\|\theta_i\|_\infty$ on the small-field event, as in Theorem 17.4, with $a_{\mathrm{cap}}\sqrt{B}<1$. Then*

$$\sup_{K\ge j}\mathbb{E}_{\nu_K}\mathsf{Y}_\alpha^\delta(\mathcal{Q}_\delta(s_j,s_j)) \le C_J g_3^2\big\{a_j^{2\alpha-1}(1+j)+b_{j-1}\big\}.$$

The right sides are summable over j, with tails that vanish uniformly in K, and each fixed higher moment satisfies the analogous bound.

The constants in the coefficient and mean bounds depend only on L, T, M, J, α, and the fixed analytic, covariance, and column bounds; they do not depend on the coupling, θ_{j-1}, K, j, the number of preceding averaging steps, or a/δ. The constants in the maximal inequalities may in addition depend on the fixed coupling, the coarse side, and the prescription of the cutoff radii.

The long-time bound in (1) is the main analytic point. Integrating the bound on the source alone would lose the averaging of the outer heat kernel over the cells of the input lattice. Instead, for $u = t-s \ge a^2$, we factor the coefficient matrix through physical space,

$$\ell^2(w) \xrightarrow{D_\delta W_s} L^2_\delta \xrightarrow{M_{D_\delta^r p_u^\delta(x,\cdot)}} L^2_\delta \xrightarrow{W_s^*} \ell^2(w),$$

where M_k denotes multiplication by k and W_s^* is the adjoint with respect to L^2_δ. In the computation of the mean, on the other hand, the covariance enters as the counting matrix Z_θ between counting rows.

Proof. The proof has eight steps: the source matrices, the Duhamel matrices for long and short times, their weighted bounds, higher differences, derivatives in unscaled coordinates, the conditional mean, and the maximal and summability claims.

Step 1: the source matrices. We use the quadratic coefficient of Lemma 24.2 on the same periodic lattice of side $nB^K \geq 4$ and mesh δ as in Lemma 26.6. Set $\lambda = g_j/a = g_3 a^{-1/2}$ and $W_s = P_s^\delta I_a E_\theta$. For one scalar color component of one translated bracket of the ordered source, the coefficient matrix is a fixed finite sum of rank-one matrices

$$\lambda^2 \, l_s \otimes m_s,$$

where l_s is a translated value row of W_s and m_s is a translated first-difference row of W_s. The color coefficients are the structure constants of $\mathfrak{su}(2)$, which correspond to the cross product on $\mathbb{R}^3$. The counting Hilbert–Schmidt norm of $l \otimes m$ is $\|l\|_2 \|m\|_2$. At output order $r = 0, 1$, the product rule distributes a total of $r+1$ differences over the two rows. By Theorem 26.10(1), at each (s, x),

$$\|\mathsf{C}_{s,x}^{(r)}\|_{\mathrm{HS}} \leq C_J \lambda^2 a^3 (s+a^2)^{-3/2-(r+1)/2} = C_J g_3^2 a^2 (s+a^2)^{-2-r/2}.$$

Symmetrization gives the same quadratic polynomial and does not increase this bound, and the operator norm is at most the Hilbert–Schmidt norm. This proves the first bound of (1) for each translated bracket and, by the triangle inequality, for the ordered source, which is their finite sum.

Step 2: long times. Let $u = t - s \geq a^2$, and fix one translated scalar summand. After r output differences on the heat kernel, its coefficient for the outer heat semigroup has the form

$$\lambda^2 W_s^* M_{D_\delta^r p_u^\delta(x,\cdot)} D_\delta W_s,$$

with the fixed translations and color projections inserted in their places. For a multiplier k, expanding in an orthonormal basis of $\ell^2(w)$, we find

$$\|M_k D_\delta W_s\|_{\mathrm{HS}}^2 = \delta^3 \sum_y |k(y)|^2 \|D_\delta W_s(y,\cdot)\|_2^2.$$

Translations and color projections change only the constant. The ideal inequality $\|AB\|_{\mathrm{HS}} \leq \|A\|_{\mathrm{op}} \|B\|_{\mathrm{HS}}$, obtained by summing $\|ABe_i\|^2 \leq \|A\|_{\mathrm{op}}^2 \|Be_i\|^2$ over the basis, implies that

$$\begin{aligned} \|\lambda^2 W_s^* M_{D_\delta^r p_u^\delta} D_\delta W_s\|_{\mathrm{HS}} &\leq C_J \lambda^2 a^{3/2} \, a^{3/2} (s+a^2)^{-5/4} u^{-3/4-r/2} \\ &= C_J g_3^2 a^2 (s+a^2)^{-5/4} u^{-3/4-r/2}. \end{aligned}$$

Here we used the operator norm of W_s from Theorem 26.10(1) and the L_δ^2 bound on the heat kernel from Lemma 26.4. The factor δ^3 is part of the identity, and therefore neither a counting norm over the output lattice nor a factor $(a/\delta)^{3/2}$ occurs.

Step 3: short times. For $0 \le u < a^2$, we commute the output differences onto the source and use heat contraction on the matrix-valued field. By the bound of Step 1, we have

$$\|\mathsf{A}^{(r)}_{u,s,x}\|_{\mathrm{HS}} \vee \|\mathsf{A}^{(r)}_{u,s,x}\|_{\mathrm{op}} \le C_J g_3^2 a^2 (s+a^2)^{-2-r/2},$$

where $\mathsf{A}^{(r)}_{u,s,x}$ is the matrix of the integrand at time s. The differences of the smooth cardinal interpolation up to order two justify this bound also at $s = 0$.

Step 4: integration in time and weights. For $t \le 2a^2$, the source bound and contraction give $C_J g_3^2 t a^{-2-r}$, which is at most a fixed multiple of the claimed bound and retains its factor $t/(t+a^2)$. For $t \ge 2a^2$, we split the integral at $s = t/2$. On the early half, $u \ge t/2 \ge a^2$, and

$$a^2 \int_0^{t/2} (s+a^2)^{-5/4} u^{-3/4-r/2}\, ds \le C a^{3/2} t^{-3/4-r/2},$$

because $\int_0^\infty (s+a^2)^{-5/4}\, ds = 4a^{-1/2}$. On the part of the late half with $u \ge a^2$, the bound is

$$C a^2 t^{-5/4} \int_{a^2}^{t/2} u^{-3/4-r/2}\, du.$$

For $r = 0$ this is at most $C a^2 t^{-1}$, which is at most $C a^{3/2} t^{-3/4}$ since $a^2 \le t$; for $r = 1$ the integral is at most $C a^{-1/2}$, and the bound is $C a^{3/2} t^{-5/4}$. The final interval $u < a^2$ contributes at most $C a^4 t^{-2-r/2}$, which is at most $C a^{3/2} t^{-3/4-r/2}$ for the same reason. Multiplying by g_3^2, and noting that $t/(t+a^2)$ lies between $1/2$ and 1 in this range, we obtain the bound of (1) at all times.

For either component of Y^δ_α, replacing the weight $t + \delta^2$ by the larger $t + a^2$ leaves the power $(t+a^2)^{\alpha-5/4}$ in that bound, and this exponent is negative. The supremum is therefore at most $C_J g_3^2 a^{3/2} (a^2)^{\alpha-5/4} = C_J g_3^2 a^{2\alpha-1}$. The factor $t/(t+a^2)$ remains in the unweighted bound; in particular the Duhamel term vanishes at $t = 0$.

Step 5: higher differences. Every further fixed spatial difference follows by distributing differences in the finite translated products and on the rows of the heat-smoothed interpolation kernel. For a polynomial bound, it suffices to put all outer differences on the smooth source. If the total spatial order is q and the time order is k, the source coefficients are bounded in Hilbert–Schmidt norm by $C_{J,q,k} g_3^2 a^{-2-q-2k}$ on $[0, T]$. The Duhamel coefficients are bounded by T times this bound at time order zero, and their time derivative is the source plus the discrete Laplacian of the Duhamel term. This proves the bounds by fixed powers of a^{-1}, with the factor g_3^2.

Step 6: derivatives in the unscaled coordinates. We use the bounds on the absolute joint Taylor coefficients verified in Lemma 26.1, and we choose the evaluation and larger polydiscs there, with the displayed gaps. Before the conditional mean is substituted,

we regard E as a function of the unscaled coarser field and of the base point q in the coupling-scaled fluctuation inputs. The scalar quadratic coefficient matrices above are holomorphic functions of these variables: they use bilinear transposes, and neither complex conjugation nor a conditional covariance. Repeating the absolute row, Schur, and heat arguments with the coefficient bounds valid on the larger polydisc, we find that the bounds of (1) hold there. For a finite multi-index $\mathbf{k}$, we iterate the Cauchy integral formula on circles of radii η_λ about the evaluation point. For a matrix $A(q)$ in finite dimensions, this gives

$$\partial^{\mathbf{k}} A(q) = \frac{\mathbf{k}!}{(2\pi i)^{|\operatorname{supp}\mathbf{k}|}} \int \cdots \int \frac{A(q+z)\prod_{\lambda\in\operatorname{supp}\mathbf{k}} dz_\lambda}{\prod_{\lambda\in\operatorname{supp}\mathbf{k}} z_\lambda^{k_\lambda+1}}.$$

This identity follows by expanding the power series on slightly smaller circles, integrating term by term, and noting that a nonconstant Fourier monomial integrates to zero over a circle; one then increases the circles to the chosen radii. By the triangle inequality in Hilbert–Schmidt or operator norm, the integral is at most the bound on the larger polydisc times $\mathbf{k}!\prod_\lambda \eta_\lambda^{-k_\lambda}$, and the constants of the heat estimates are unaffected. If the input coordinate is $q_\lambda = a_\lambda A_\lambda$ or $q_\lambda = g_\lambda w_\lambda$, the chain rule gives the factor $a_\lambda^{k_\lambda}$ or $g_\lambda^{k_\lambda}$. For an array at an input location on mesh a_λ, the bound $\|D_{a_\lambda} f\| \le 2a_\lambda^{-1}\|f\|$ gives the possible cost of a difference. Coordinates of the coarser field and the q coordinates keep their own meshes. Only after the differentiation do we evaluate at the real base point given by the conditional mean; this base point and the conditional law are held fixed throughout.

Step 7: the conditional mean. We first treat the expansion at the conditional mean. The covariance comparison is applied to the same conditional fluctuation kernel and reconstruction as in Lemma 26.1. By Theorem 13.10,

$$\|\mathsf{Z}_\theta - \mathsf{Z}_0\|_{\mathrm{op}} = \|E_\theta \Sigma_\theta E_\theta^{\mathsf{T}} - E_0 \Sigma_0 E_0^{\mathsf{T}}\|_{\mathrm{op}} \le C_J r_\theta (1 + t_j p_j^3) \le C_J r_\theta,$$

because the common upper bound on the couplings bounds $t_j p_j^3$, by (26.1). The same conclusion is part of Theorem 17.4. The reference Z_0 is built from the conditional law at the zero coarser field. The blocks of its covariance in the color indices are scalar, and so is the linear coefficient at the zero coarser field. Hence Z_0 is proportional to the identity in the color indices, with all its spatial correlations kept.

Here Z_θ is the counting matrix of Section 26.1, and its operator norm is the counting operator norm. The physical operator $a^3\mathsf{Z}_\theta$ is not inserted between counting rows, and the adjoint in the factorization of Step 2 is a different object.

Let $V_s = P_s^\delta I_a$ be the heat evolution of the cardinal interpolation with identity input. For one bracket and one color component, the expected source is the contraction of two rows of V_s against $\lambda^2\mathsf{Z}_\theta$. The contraction against Z_0 vanishes: its two color indices enter

through δ_{bc}, while the bracket coefficient ϵ_{bcd} is antisymmetric in b and c. This holds for every translation and difference in the quadratic coefficient. Equivalently, the expected vector is fixed by every adjoint color rotation, and such a vector is zero. We subtract this vanishing contraction and use the operator norm of $\mathsf{Z}_\theta - \mathsf{Z}_0$ between the rows of V_s. By Lemma 16.3 with identity input, or by the Schur argument of Lemma 26.5,

$$\|\mathbb{E}_\theta D_\delta^r \mathcal{N}_\delta(s_j(s), s_j(s))\|_\infty \le C_J g_3^2 r_\theta a^2 (s + a^2)^{-2-r/2}, \qquad r = 0, 1.$$

For the value, heat contraction and the exact integral

$$a^2 \int_0^t (s + a^2)^{-2}\, ds = \frac{t}{t + a^2}$$

give the first bound of (3). For the first difference at $t \le a^2$, the differentiated source integrates to $C_J g_3^2 r_\theta t a^{-3}$, which is at most $C_J g_3^2 r_\theta (t + a^2)^{-1/2} (t/a^2)$. For $t \ge a^2$, we split at $s = t/2$. On the early half, we put the difference on the outer heat kernel and use its L^1 bound; the contribution is at most

$$C_J g_3^2 r_\theta t^{-1/2} a^2 \int_0^{t/2} (s + a^2)^{-2}\, ds \le C_J g_3^2 r_\theta t^{-1/2}.$$

On the late half, we put the difference on the source and contract by the heat semigroup, which gives $C_J g_3^2 r_\theta a^2 t (t + a^2)^{-5/2} \le C_J g_3^2 r_\theta t^{-1/2}$. This proves the second bound of (3) in both regimes. The weight of the value has the positive exponent β, and the weight of the difference leaves the bounded power $(t + a^2)^{\alpha - 1/2}$; the fixed horizon and $a \le L/4$ then give the bound on Y_α^δ of the mean.

For the expansion at zero, the same covariance comparison follows from Lemma 26.1: we integrate the derivative of $E_\theta^{(0)}$ in the coarser field along the segment to the zero coarser field, and use the bound on $\Sigma_\theta - \Sigma_0$ of Theorem 13.10. At the zero coarser field, $m_0 = 0$, which implies that the reference covariance and the reference linear tensor are the same for both expansions. The contraction above then implies the same bounds on the mean.

Step 8: maxima and summability. At a fixed coarser field, the Duhamel term is a quadratic polynomial in the centered vector ζ_j. Its matrices are bounded in (1), and its conditional mean in (3). By the Hessian bound verified in Lemma 26.1, Lemma 26.9 applies with $S = C_J g_3^2 a^{2\alpha - 1}$; its proof uses the Duhamel factor at very short times and the polynomial bounds of Step 5 afterwards, and it covers both the value and the first-difference components. We obtain the centered tail (4), with $\Lambda_a = \log(2 + L/a)$.

For each fixed $p \ge 1$, integrating this conditional tail and adding the bound on the mean, we obtain

$$\Big(\mathbb{E}_\theta \mathsf{Y}_\alpha^\delta(\mathcal{Q}_\delta(s_j, s_j))^p\Big)^{1/p} \le C_{p,J} g_3^2 \big\{ a_j^{2\alpha - 1} (\Lambda_{a_j} + p + 1) + r_\theta \big\}.$$

The deterministic bound (26.8), from Lemma 15.10, with $C_J \geq C_{\mathrm{cap},J}$ and with the bound on the coarsest field denoted by ρ_0, implies that $r_\theta \leq b_{j-1}$ on the small-field event; this is the bound used in Theorem 17.4. We integrate the conditional bound on the pth moment over the law of θ_{j-1} under ν_K and use $\Lambda_{a_j} \leq C_{n,B}(1+j)$. For $p = 1$, we obtain the bound of (5), and the same computation gives its versions for fixed higher moments.

Finally, if $c > 0$ and $q \geq 0$, then, since $a_{m+h} = a_m B^{-h}$ and $1 + m + h \leq (1+m)(1+h)$,

$$\sum_{j \geq m} a_j^c (1+j)^q \leq a_m^c (1+m)^q \sum_{h \geq 0} B^{-ch} (1+h)^q < \infty.$$

The last series converges because geometric decay eventually dominates the polynomial factor, and the right side tends to zero as $m \to \infty$. We apply this with $c = 2\alpha - 1 > 0$ and $c = 1/2$, and we sum the geometric sequence $\rho_0 a_{\mathrm{cap}}^{j-1}$ separately; the condition $a_{\mathrm{cap}}\sqrt{B} < 1$ implies $a_{\mathrm{cap}} < 1$. Since $p_j \leq C_\gamma(1+j)$, every tail is summable, uniformly in K. The deterministic majorants do not depend on K or on the coarser field, and therefore the convergence over short scales is uniform in the number of steps, at fixed coupling and coarse side. □

Remark 26.13. The mixed terms with one conditionally centered input have conditional mean zero. The centered part of the contribution $\mathcal{Q}_\delta(s_j, s_j)$ to the quadratic Picard term is summable because $2\alpha - 1 > 0$; its mean is summable because $\|\theta_{j-1}\|_\infty \leq b_{j-1}$.

SECTION 27

Conditional means and propagated fluctuations

This section completes the estimates of Section 26 on the heat evolution of the reconstructed field and its quadratic Picard term under the conditioned Wilson measure. Recall the decomposition of Lemma 26.2,

$$x_j - x_{j-1} = s_j + z_j + r_j + u_j,$$

where s_j is the centered linear fluctuation term, z_j the centered nonlinear heat term, r_j the interpolation defect, and u_j the predictable term. The latter is $u_j = P_\cdot^\delta I_{a_j}[U_j^{\mathrm{cl}} + m_j^{\mathrm{ph}}]$, the heat evolution of the sum of the nonlinear part U_j^{cl} of the zero-fluctuation reconstruction and the conditional mean correction m_j^{ph}. Section 26 proved the fluctuation estimates for s_j and z_j; Section 28 combines them with the estimates proved here to bound all ordered products. Here we treat the terms built only from r_j and u_j. For the quadratic Picard term, these are $\mathcal{B}_\delta(x_{j-1}, r_j)$ and $\mathcal{Q}_\delta(r_j, r_j)$. The products involving u_j are handled in Section 28 by the bilinear estimate of Lemma 24.6, once u_j is bounded in the heat norm H_σ^δ of its initial field; we prove that bound here.

Three ordered terms. Let $H = a_{j-1}$ and $a = a_j$. The coarser field enters both x_{j-1} and r_j linearly: with

$$\mathsf{U} = P_\cdot^\delta I_a \mathsf{P}_j / H, \qquad \mathsf{V} = P_\cdot^\delta (I_a \mathsf{P}_j - I_H \mathsf{E}_H)/H,$$

we have $x_{j-1} = (\mathsf{U} - \mathsf{V})\theta_{j-1}$ and $r_j = \mathsf{V}\theta_{j-1}$. For any bilinear map $\mathcal{N}$,

$$\begin{aligned}
\mathcal{N}(U - V, V) + \mathcal{N}(V, U - V) &= \mathcal{N}(U, V) + \mathcal{N}(V, U) - 2\mathcal{N}(V, V),\\
\mathcal{N}(U, U) - \mathcal{N}(U - V, U - V) &= \mathcal{N}(U, V) + \mathcal{N}(V, U) - \mathcal{N}(V, V).
\end{aligned}$$

Applying these identities with $U = \mathsf{U}\theta_{j-1}$ and $V = \mathsf{V}\theta_{j-1}$, we obtain $\mathcal{B}_\delta(x_{j-1}, r_j) = \mathscr{T}^{UV} + \mathscr{T}^{VU} - 2\mathscr{T}^{VV}$ and $\mathcal{Q}_\delta(r_j, r_j) = \mathscr{T}^{VV}$, where $\mathscr{T}^{UV}$, $\mathscr{T}^{VU}$, and $\mathscr{T}^{VV}$ are the quadratic Duhamel terms of the ordered pairs $(\mathsf{U}\theta_{j-1}, \mathsf{V}\theta_{j-1})$, $(\mathsf{V}\theta_{j-1}, \mathsf{U}\theta_{j-1})$, and $(\mathsf{V}\theta_{j-1}, \mathsf{V}\theta_{j-1})$. Since $\mathcal{N}_\delta$ is not symmetric, the three terms are estimated separately; a bound on their sum would not bound the summands.

Propagation through coarser scales. The coarser field θ_{j-1} depends on the fluctuation variables of all coarser steps. We propagate both inputs back through the linearized reconstructions $\mathsf{L}_{i+1}, \dots, \mathsf{L}_{j-1}$ and telescope over the coarser scales i; each step of the

telescope contributes a conditionally centered term $Z_{j,i}^{\omega}$ and a conditional drift $e_{j,i}^{\omega}$ (see (27.30)). Theorem 27.5 bounds the coefficients and the drifts, Theorem 27.6 bounds and sums the centered terms, and Theorem 27.7 controls the predictable term.

The scaling. The maps between lattices are

$$G \xrightarrow{\mathsf{L}_{j-1}\cdots\mathsf{L}_{i+1}} H \xrightarrow{\mathsf{P}_j} a \xrightarrow{I_a} \delta \qquad \text{and} \qquad H \xrightarrow{I_H} \delta,$$

where $G = a_i$ and each arrow is labeled by its map and points to the mesh of its target. The interpolation defect is the difference of the two branches out of H. Each column of this difference with a fixed input on the lattice of mesh H has total mass zero (Lemma 27.1); after heat evolution for time s this gives the factor $d_H(s) = H(s + H^2)^{-1/2}$. The cancellation must be used at the scale H, before the earlier maps are composed: a first moment measured at the scale G would give the wrong length. Each linearization L_i has absolute row sums and volume-normalized column sums at most $a_{\mathrm{cap}} = C_{\mathrm{CP}}/B$. Dividing by the mesh H of the coarser lattice, we obtain the contraction

$$\frac{a_{\mathrm{cap}}^{2(j-1-i)}}{H} = \frac{q_{\mathrm{anc}}^{j-1-i}}{G}, \qquad q_{\mathrm{anc}} = B a_{\mathrm{cap}}^2 < 1,$$

for a quadratic term propagated from scale i. Finally, the Schur bound in counting norms gives a factor $(G/\delta)^{3/2}$, which cancels against the factor $\delta^{3/2}$ of the ℓ^2 row of the heat kernel on the finest lattice.

Inputs and constants. We use the setting and notation of Sections 26.1 and 26.2, including the conditioned Wilson measure ν_K, the filtration $\mathcal{F}_j$, the normalized fluctuation variables w_j, the norms X_α^δ, Y_α^δ, H_σ^δ, and V_σ^δ of Definition 4.8, and the four terms of Lemma 26.2. Throughout, $1/2 < \alpha < 2/3$, $\beta = \alpha - 1/2$, $0 < \sigma < 1 - \alpha$, and the horizon T is fixed. All heat kernels keep their constant mode, and all estimates concern the whole torus. The inputs are the conditional fluctuation kernel and the bounds on conditional means and covariances of Theorems 13.8 and 13.10; the reconstruction maps and the bounds on their absolute Taylor coefficients of Theorems 15.12 and 15.13; the derivatives of the zero-fluctuation reconstruction and the bounds on reconstructions of Lemmas 17.1 and 17.2 and Theorem 17.4; the interpolation and heat estimates of Lemmas 16.1, 16.3, and 16.4; the quadratic coefficient of Lemma 24.2; and the concentration inequalities of Theorem 5.4. The results are used in Section 28, which assembles the bounds on the linear and quadratic flow data, and in Sections 29 and 34.

We summarize the dependence of the constants in Table 27.1. The bounds on coefficients and on conditional means hold uniformly in the coupling below the common upper bound g_*. The maximal inequalities and the summations use the bound $b_i/g_i \le C_\gamma(1+i)$ on the coarser fields, which holds only at fixed positive coupling.

Estimate	Constants may depend on
Unweighted coefficients and derivatives in w (Lemma 27.1; Theorems 27.5 and 27.7, time-dependent rows)	L, T, M, J, the interpolation function, the analytic domains and column bounds, and the orders of differences
Pointwise conditional means and drifts, unweighted (Lemmas 27.3 and 27.4)	the analytic domains and coefficient bounds, M, J, and the common upper bound g_*
Weighted coefficients and drifts in $\mathsf{Y}^{\delta}_{\alpha}$ (Theorem 27.5)	in addition α and the coarse mesh $a_0 = L/n$
Drifts in $\mathsf{V}^{\delta}_{\sigma}$ (Theorem 27.7)	in addition σ; not on a_0
Maximal inequalities, moments, and summations (Theorems 27.6 and 27.7)	in addition the fixed γ, n, ρ_0, the moment, the exponents, and the prescription of the cutoff radii

Table 27.1. Dependence of constants in Section 27. No constant depends on K, on the scale indices i and j, on the number of preceding averaging steps, on a/δ, or on the values of the coarser fields.

The tools are standard: the weighted Schur test, cancellation for convolution with a signed kernel of mass zero, the logarithmic Sobolev inequality (Gross [103]; Bakry and Émery [8]) and the covariance representation of Helffer and Sjöstrand [110] in the forms proved in Theorem 5.4, the derivative formula for normalized expectations, and union bounds over finite nets. As in Section 26, we control a heat evolution together with its quadratic Picard term following Cao and Chatterjee [42, Section 2.1]; here the estimates are proved from the conditional fluctuation kernels.

27.1 Volume-normalized column sums and the interpolation defect

We use the following norms for matrices between arrays on lattices. Each index u of such an array has a location and a cell volume v_u: the cube of the mesh of its lattice. For a matrix A with entries A_{yu} we set

$$\mathsf{R}(\mathsf{A}) = \sup_{y} \sum_{u} |\mathsf{A}_{yu}|, \qquad \mathsf{C}(\mathsf{A}) = \sup_{u} \frac{v_{\mathrm{out}}}{v_{\mathrm{in}}} \sum_{y} |\mathsf{A}_{yu}|,$$

the absolute row sum and the volume-normalized column sum, where v_{in} and v_{out} are the cell volumes of the input and output lattices. All sums include the finitely many direction and color indices.

Lemma 27.1. *Assume the setting of Section 26.1.*

1. *The norms* R *and* C *are submultiplicative under composition. For a block with fixed input and output volumes,*

$$\|\mathsf{A}\|_{\ell^2_{\mathrm{count}} \to \ell^2_{\mathrm{count}}} \le \sqrt{\mathsf{R}(\mathsf{A})\mathsf{C}(\mathsf{A}) v_{\mathrm{in}}/v_{\mathrm{out}}}.$$

For several types of input or output indices, the column ratio in each summand is replaced by $v_{\mathrm{out}}(y)/v_{\mathrm{in}}(u)$. The operator norm from $\ell^2(v_{\mathrm{in}})$ to $\ell^2(v_{\mathrm{out}})$, the ℓ^2 spaces weighted by the cell volumes, is then at most $\sqrt{\mathsf{RC}}$, and counting norms use the corresponding conversions of volumes for each block. The ratios of volumes within one renormalization group step cost only C_J. For a pointwise product $c_{y;u,v} = \mathsf{A}_{yu}\mathsf{B}_{yv}$ of two matrices, the absolute row sum is at most $\mathsf{R}(\mathsf{A})\mathsf{R}(\mathsf{B})$, and the two column sums with a fixed input are at most $\mathsf{C}(\mathsf{A})\mathsf{R}(\mathsf{B})$ and $\mathsf{R}(\mathsf{A})\mathsf{C}(\mathsf{B})$.

2. *Let $a = a_j$, $H = a_{j-1} = Ba$, $\delta = a_K$, and put*

$$\mathsf{D}_j = I_a\mathsf{P}_j - I_H\mathsf{E}_H, \qquad d_H(s) = H(s+H^2)^{-1/2}.$$

Each column of D_j with a fixed input u on the lattice of mesh H has total physical mass zero, separately in every output direction and color, and first absolute moment at most CH^4 about the location of u:

$$\delta^3\sum_y(\mathsf{D}_j)_{yu} = 0, \qquad \delta^3\sum_y \mathrm{dist}(y,u)|(\mathsf{D}_j)_{yu}| \le CH^4.$$

The matrix D_j and the matrix $I_a|\mathsf{P}_j| + I_H|\mathsf{E}_H|$ of absolute values have bounded absolute row sums and volume-normalized column sums. For every $s \in [0,T]$ and $r = 0,1,2$,

$$(\mathsf{R}+\mathsf{C})(D_\delta^r P_s^\delta \mathsf{D}_j) \le C_r(s+H^2)^{-r/2}d_H(s),$$
$$(\mathsf{R}+\mathsf{C})(D_\delta^r P_s^\delta I_a\mathsf{P}_j) \le C_r(s+H^2)^{-r/2}.$$

The cancellation is used for each column with input on the lattice of mesh H, before any earlier linearization is composed.

3. *Let $\mathsf{L}_i = D_\theta\mathscr{R}_i(0,0)$ be the linearization at the zero field of the zero-fluctuation reconstruction of step i, acting on link logarithms, with $\mathsf{R}(\mathsf{L}_i), \mathsf{C}(\mathsf{L}_i) \le a_{\mathrm{cap}} = \mathsf{C}_{\mathrm{CP}}/B$ and $q_{\mathrm{anc}} = Ba_{\mathrm{cap}}^2 < 1$. For $0 \le i \le j-1$ set $m = j-1-i$, $G = a_i = B^m H$, $\mathsf{T}_{ji} = \mathsf{L}_{j-1}\cdots\mathsf{L}_{i+1}$ (the identity if $i = j-1$), and*

$$\mathsf{U}_s = P_s^\delta I_a\mathsf{P}_j\mathsf{T}_{ji}/H, \qquad \mathsf{V}_s = P_s^\delta\mathsf{D}_j\mathsf{T}_{ji}/H, \qquad \mathfrak{A}_{ji} = C_{\mathrm{CP}}^m/G.$$

If $\|\theta_i\|_\infty \le b_i$, then for $r = 0,1,2$,

$$\|D_\delta^r\mathsf{U}_s\theta_i\|_\infty \le C_r\mathfrak{A}_{ji}b_i(s+H^2)^{-r/2}, \qquad \|D_\delta^r\mathsf{V}_s\theta_i\|_\infty \le C_r\mathfrak{A}_{ji}b_i(s+H^2)^{-r/2}d_H(s).$$

For $i \ge 1$, the derivatives in w_i of $\mathsf{U}_s\theta_i$ and $\mathsf{V}_s\theta_i$, where $\theta_i = \mathscr{R}_i(\theta_{i-1}, w_i)$ is differentiated at the given point w_i of Ω_i, have absolute row sums and volume-normalized column sums bounded by the same expressions with b_i replaced by g_i. In bounds on derivatives, b_i may be any deterministic bound on $\|\theta_i(w)\|_\infty$ valid for every $w \in \Omega_i$ at the fixed coarser field θ_{i-1}; a bound sharper than the global one may be used when it holds on that whole set.

A derivative in the dimensionless input θ_i removes a factor b_i; for an input written as a lattice potential, $\theta_i = Gv_i$, the derivative in v_i acquires the factor G; and a derivative in a normalized fluctuation coordinate $w_{i,\lambda}$ acquires its own coupling $g_{i,\lambda}$. These conversions use the mesh of the input, never δ.

4. *Let $\mathscr{S}_s$ be any one of the three ordered sources*

$$\mathcal{N}_\delta(\mathsf{U}_s\theta_i, \mathsf{V}_s\theta_i), \qquad \mathcal{N}_\delta(\mathsf{V}_s\theta_i, \mathsf{U}_s\theta_i), \qquad \mathcal{N}_\delta(\mathsf{V}_s\theta_i, \mathsf{V}_s\theta_i),$$

or their linear combinations with coefficients $1, 1, -1$ or $1, 1, -2$. Then

$$(\mathsf{R}+\mathsf{C})(D_{w_i}D_\delta\mathscr{S}_s) \le C_J C_{\mathrm{CP}}^{2m} g_i b_i G^{-2}(s+H^2)^{-1} d_H(s).$$

The constants depend on L, T, M, J, the interpolation function, the analytic domains and column bounds, and the order of differences; they do not depend on K, j, i, the number of preceding averaging steps, a/δ, the coupling, or the coarser field. These are estimates on the whole torus.

Proof. Constants C_J may absorb the number of coordinate types and their volume ratios within one renormalization group step, but never a ratio between the mesh G of a coarser scale and the finest mesh. The horizon in every heat estimate of this section is the fixed T. The refinement hypotheses of the cardinal interpolation and of Lemma 17.2 hold because

$$\frac{a_j}{\delta} = B^{K-j} \in \mathbb{N}, \qquad \frac{a_{j-1}}{\delta} = B^{K-j+1} \in \mathbb{N}. \tag{27.1}$$

Every order of differences used below is a nonnegative integer. The coarse side $n \ge 4$ and all finer sides nB^l satisfy the hypotheses on lattice sides of the heat and interpolation estimates, including meshes larger than one.

Step 1: the matrix norms. We write v_z for the cell volume of an intermediate index z. By finite summation, we have

$$\sum_u |(\mathsf{AB})_{yu}| \le \sum_z |\mathsf{A}_{yz}| \sum_u |\mathsf{B}_{zu}|,$$

and

$$\frac{1}{v_u}\sum_y v_y |(\mathsf{AB})_{yu}| \le \sum_z \frac{v_z}{v_u}|\mathsf{B}_{zu}| \Big(\frac{1}{v_z}\sum_y v_y |\mathsf{A}_{yz}|\Big).$$

Taking suprema gives both composition inequalities. For an array f, the Cauchy–Schwarz inequality with weights $|\mathsf{A}_{yu}|$ implies that

$$\sum_y v_y |(\mathsf{A}f)_y|^2 \le \mathsf{R}(\mathsf{A}) \sum_u |f_u|^2 \sum_y v_y |\mathsf{A}_{yu}| \le \mathsf{R}(\mathsf{A})\mathsf{C}(\mathsf{A}) \sum_u v_u |f_u|^2.$$

For fixed input and output volumes, replacing both weighted norms by counting norms introduces exactly the factor $(v_{\mathrm{in}}/v_{\mathrm{out}})^{1/2}$. The same computation gives the assertion for

several types of indices. For $c_{y;u,v} = \mathsf{A}_{yu}\mathsf{B}_{yv}$, summing first over u and v gives the row bound. Fixing u and summing first over v, we obtain

$$\sup_u \frac{v_{\mathrm{out}}}{v_{\mathrm{in}}} \sum_{y,v} |\mathsf{A}_{yu}\mathsf{B}_{yv}| \le \mathsf{C}(\mathsf{A})\mathsf{R}(\mathsf{B}),$$

and fixing v gives the other column bound. The same computation applies to two different types of input indices with their volumes. Fixed translations of either output index are permutations of the output indices, so the proof applies to them with the same norms. A color bracket costs only the norm of a fixed bilinear map on $\mathbb{R}^3$.

Step 2: zero mass of the defect. We apply Lemma 17.2 with coarse mesh H and fine mesh $a = H/B$; its linear reconstruction is P_j of Lemma 26.2. The two interpolations are applied on their own lattices. By Lemma 16.3, after interpolation a column with input on the lattice of mesh a has total physical mass a^3, and a column with input on the lattice of mesh H has mass H^3. The identity for the total mass of the columns of P_j in Lemma 17.2 implies that

$$\delta^3 \sum_y (I_a \mathsf{P}_j - I_H \mathsf{E}_H)_{yu} = 0$$

in every output component, without summing over colors or directions. For an index z on the lattice of mesh a, we use the triangle inequality $\operatorname{dist}(y,u) \le \operatorname{dist}(y,z) + \operatorname{dist}(z,u)$ and the first moment of the column of the cardinal interpolation matrix at z to obtain

$$\delta^3 \sum_y \operatorname{dist}(y,u) |(I_a \mathsf{P}_j)_{yu}| \le C \sum_z a^3 \{a + \operatorname{dist}(z,u)\} |(\mathsf{P}_j)_{zu}| \le C H^4.$$

The term $I_H \mathsf{E}_H$ has the same bound, and periodic distances may be used in each triangle inequality. The rows and volume-normalized columns of Lemma 17.2, together with Step 1, also bound $I_a|\mathsf{P}_j| + I_H|\mathsf{E}_H|$.

Step 3: heat evolution of the defect. The estimate on the finest lattice in Lemma 17.2 gives the bound on the absolute rows of $D_\delta^r P_s^\delta \mathsf{D}_j$. We give a separate argument for the columns. For $s \le H^2$ we commute every difference through the heat semigroup and apply it to the smooth columns of the cardinal interpolation matrix. Their rth differences have volume-normalized column sums at most $C_{J,r} H^{-r}$, since $a = H/B$ with B fixed, and the heat semigroup does not increase absolute column sums. For $s \ge H^2$ we subtract the heat kernel at the location of the input u, which is allowed by the zero mass of the column:

$$(D_\delta^r P_s^\delta \mathsf{D}_j)_{yu} = \delta^3 \sum_z \big\{ D_\delta^r p_s^\delta(y,z) - D_\delta^r p_s^\delta(y,u) \big\} (\mathsf{D}_j)_{zu}.$$

Using a shortest coordinate path from z to u, the telescoping identity along it, and Lemma 16.1, we obtain

$$\delta^3 \sum_y |D_\delta^r p_s^\delta(y,z) - D_\delta^r p_s^\delta(y,u)| \le C_r \operatorname{dist}(z,u) (s + \delta^2)^{-(r+1)/2};$$

the coordinate path has length at most $\sqrt{3}\,\mathrm{dist}(z,u)$ on the torus. Dividing the resulting column bound by the input volume H^3 and using the first moment of Step 2, we obtain $C_r H s^{-(r+1)/2}$, which is comparable to $(s+H^2)^{-r/2} d_H(s)$ in this range. For $r=2$ the proof uses at most three differences of the heat kernel.

For $I_a \mathsf{P}_j$ we use differences of the cardinal interpolation for $s \le H^2$, and differences of the heat kernel for $s \ge H^2$, without subtraction; this proves both of its bounds. Thus the cancellation is used at the scale H; a first moment about a location at the scale G would not suffice.

Step 4: derivatives of the reconstruction at an arbitrary point. The coefficient bounds come from Theorem 15.12, not from the heat estimate of Lemma 16.3. We write $\mathscr{D}_{\mathrm{in}} \Subset \mathscr{D}_{\mathrm{mid}} \Subset \mathscr{D}_{\mathrm{out}}$ for its nested analytic domains, with fixed positive gaps between their radii. The products of coarser fields and free coordinates in Lemma 26.2 lie in $\mathscr{D}_{\mathrm{in}}$ at a fixed positive distance from its boundary, uniformly in the side of the coarser lattice and in the number of preceding averaging steps. On that domain the reconstruction has a bounded absolute coefficient norm, including every volume-normalized column sum with a fixed input. Differentiating its majorant series on a smaller domain, we obtain bounds on the first and second derivatives in the free coordinates and on the fixed mixed derivatives used below. Indeed, a derivative of order k of a monomial of degree l multiplies its majorant by at most l^k times a fixed inverse power of the radius. Since the ratio η of the radii is less than one, $\sup_{l \ge k} l^k \eta^{l-k}$ is finite. The same argument with a fixed input gives the column bounds, without summing over a new output volume. These are absolute coefficient bounds, not bounds on values of a holomorphic map.

The coupling-scaled fluctuation inputs are $q_{i,\lambda} = g_{i,\lambda} w_{i,\lambda}$, and the ratios $g_{i,\lambda}/g_i$ lie in a fixed interval determined by M and J. The chain rule supplies one such coupling for every derivative in w. The resulting coefficients satisfy the hypotheses of Lemma 16.3: a fixed number of types of coordinates, bounded rows and volume-normalized columns, and exponential decay in their own units. In particular, for the normalized derivative of the reconstruction,

$$W_i(w) = D_{w_i} \mathscr{R}_i(\theta_{i-1}, w), \qquad \mathsf{R}(W_i) + \mathsf{C}(W_i) \le C_J g_i. \tag{27.2}$$

Here the coarser field is fixed, and the coefficient is evaluated on the whole cutoff domain Ω_i, within the domain of Lemma 26.2, not only at $w=0$. Equivalently, Lemma 16.3 applies to $g_i^{-1} W_i$, with the ratios of intermediate couplings included. Its fixed mixed derivatives satisfy the same bounds, with the coupling factors of their inputs. The same conclusion can be reached from the mean. The segment from the conditional mean $\mathbb{E}w$ to a point w of the product lies in the product by convexity. Along it, Taylor's formula for the first derivative involves the second derivative in w, whose normalized absolute rows and columns are bounded by the same coefficient norm. Since $\|w - \mathbb{E}w\|_\infty \le 2p_i$, its

contribution to either norm is at most $C_J g_i^2 p_i$; the first derivative at the mean has either norm at most $C_J g_i$; and $g_i p_i$ is bounded by the smallness conditions of Lemma 26.1.

Step 5: the zero-fluctuation reconstruction and composition. By the bounds on the derivatives of the zero-fluctuation reconstruction in Lemma 17.2, with the constant C_{CP} chosen before J, we have

$$\begin{aligned} \|\mathscr{R}_i(\phi,0)\|_\infty &\le a_{\mathrm{cap}}\|\phi\|_\infty, \\ \|\mathscr{R}_i(\phi,0) - \mathsf{L}_i\phi\|_\infty &\le C a_{\mathrm{cap}}\|\phi\|_\infty^2. \end{aligned} \tag{27.3}$$

The first line follows by integrating the first derivative along $s\phi$, $0 \le s \le 1$, and the second from Taylor's formula with integral remainder and the second derivative in the coarser field. Both use the constant chosen before J; it is not enlarged after J is fixed.

By composition, we have $\mathsf{R}(\mathsf{T}_{ji}), \mathsf{C}(\mathsf{T}_{ji}) \le a_{\mathrm{cap}}^m$. The division is by the mesh H of the coarser lattice of step j. Thus

$$\frac{a_{\mathrm{cap}}^m}{H} = \frac{C_{\mathrm{CP}}^m}{G} = \mathfrak{A}_{ji}, \qquad \mathfrak{A}_{ji}^2 H = \frac{q_{\mathrm{anc}}^m}{G} = K_{ji}. \tag{27.4}$$

The bounds of (3) follow from those of Step 3. Replacing the rightmost input by W_i, we obtain the bounds on rows and columns of derivatives in w_i. At a fixed coarser field, a smaller bound b_i is allowed only if it bounds $\theta_i(w)$ at every point w of Ω_i. Differentiating in a dimensionless input removes its bound b_i, the substitution $\theta_i = G v_i$ multiplies that derivative by G, and for a fluctuation coordinate the chain rule supplies its own $g_{i,\lambda}$.

Step 6: one more difference of the source. By Lemma 24.2, it suffices to consider one translated summand $[T_\xi f, T_\zeta D_k^\pm h]$, with (f,h) equal to $(\mathsf{U}_s\theta_i, \mathsf{V}_s\theta_i)$, $(\mathsf{V}_s\theta_i, \mathsf{U}_s\theta_i)$, or $(\mathsf{V}_s\theta_i, \mathsf{V}_s\theta_i)$. For a forward difference D_l, the product rule is

$$D_l[T_\xi f, T_\zeta D_k^\pm h] = [T_\xi D_l f, T_{\zeta+\delta e_l} D_k^\pm h] + [T_\xi f, T_\zeta D_l D_k^\pm h].$$

Its derivative in w_i has four terms, obtained by differentiating either argument of either bracket; their orders of differences are $(1,1)$, $(1,1)$, $(0,2)$, and $(0,2)$. A backward difference is a translated forward difference up to sign, so the same four bounds apply to it. In every term one factor is the deterministic defect or the derivative in w_i of the defect. Multiplying the corresponding bounds of (3), we find

$$C_J \mathfrak{A}_{ji}^2 g_i b_i (s + H^2)^{-1} d_H(s).$$

For the row we use the row of the differentiated factor and the supremum of the other factor; for the column we use the volume-normalized column of that factor and the same supremum. For $\mathscr{S}_s = \mathcal{N}_\delta(\mathsf{V}_s\theta_i, \mathsf{V}_s\theta_i)$ there may be two factors $d_H(s)$; since $d_H(s) \le 1$, we may discard one. The two signed combinations cost at most the sum of the absolute values of their coefficients. Summing over the fixed list of summands proves (4), with the stated dependence of the constants. □

27.2 Time-dependent coefficients and conditional drifts

This section bounds the coefficients of the ordered quadratic terms propagated from scale i, and the conditional drifts in the telescope over scales. We first record an elementary summation, then the properties of the conditional fluctuation kernel that we need, with constants uniform in the coupling, and a bound on conditional means of quadratic functions of the reconstructed field. The main estimate is Theorem 27.5.

An elementary summation

We write $\lambda_q = -\log q_{\mathrm{anc}} / \log B > 0$, so that $q_{\mathrm{anc}} = B^{-\lambda_q}$, and recall that $a_i = a_0 B^{-i}$ with $a_0 = L/n$.

Lemma 27.2. *Let $r > 0$ and $d \geq 0$.*

1. *For every $0 < \zeta < \min(r, \lambda_q)$ there exists C, depending only on r, ζ, λ_q, B, and a_0, such that for all $j \geq 1$,*
$$\sum_{i=0}^{j-1} q_{\mathrm{anc}}^{j-1-i} a_i^r \leq C a_{j-1}^{\zeta}.$$
2. *For every $0 < \eta < \min(r, \lambda_q)$ there exists C, depending only on r, d, η, λ_q, B, and a_0, such that for all $k \geq 1$,*
$$\sum_{j \geq k} (1+j)^d \sum_{i=0}^{j-1} q_{\mathrm{anc}}^{j-1-i} a_i^r \leq C a_k^{\eta}.$$
In particular, the left side is finite and tends to zero as $k \to \infty$.

Proof. Let $0 < \zeta < \min(r, \lambda_q)$ and $m = j - 1 - i$. Since $j - 1 = m + i$,

$$q_{\mathrm{anc}}^m a_i^r = a_0^r B^{-\lambda_q m - r i} = a_0^r B^{-\zeta(j-1)} B^{-(\lambda_q - \zeta)m - (r-\zeta)i}.$$

The two remaining exponents are positive, so the sum over $0 \leq i \leq j-1$ of the last factor is at most $\sum_{m,i \geq 0} B^{-(\lambda_q-\zeta)m-(r-\zeta)i} < \infty$. This gives (1), since $B^{-\zeta(j-1)} = (a_{j-1}/a_0)^{\zeta}$. For (2) we choose ζ strictly between η and $\min(r, \lambda_q)$ and apply (1). Since $\zeta - \eta > 0$, the polynomial factor is absorbed: $(1+j)^d B^{-\zeta(j-1)} \leq C_{d,\zeta-\eta} B^{-\eta(j-1)}$, and summing over $j \geq k$ gives at most $C B^{-\eta k} = C(a_k/a_0)^{\eta}$. Polynomial factors in i or m are bounded by $(1+j)^d$, since $i, m \leq j$. □

The conditional fluctuation kernel

The next lemma collects the properties of the conditional law of w_i given $\mathcal{F}_{i-1}$ that we use. In the notation of the lemma, $\mu_{\mathfrak{d}_i,\phi} = k_{\mathfrak{d}_i}(e^{\phi}, dw)$ is the conditional fluctuation kernel of (4.8) after $\mathfrak{d}_i$ preceding averaging steps, and its mean $\mu_\phi = m_\phi$ is the conditional mean of Section 26.1. For the small-field event of Definition 4.4, all cutoff radii of one

renormalization group step are equal, $p_i = A_{\mathrm{w}} \log(1/t_i) \geq 1$. This is the case treated here; it is a special case of the prescription allowed in Section 26.1. The point of the lemma is that its constants do not depend on the coupling below the common upper bound g_*.

Lemma 27.3. *Assume the setting of Section 26.1. Fix $K \geq 1$ and $1 \leq i \leq K$, and let $\phi = \theta_{i-1}$.*

1. *The conditional law of w_i given $\mathcal{F}_{i-1}$ is the conditional fluctuation kernel $\mu_{\mathfrak{d}_i,\phi}$, $\mathfrak{d}_i = J(K-i)$, with density $Z_\phi^{-1} \mathbf{1}_{\Omega_i} e^{-V_{\mathfrak{d}_i,\phi}}$, and, uniformly in K, i, $\mathfrak{d}_i$, ϕ, and the coupling below g_*,*

$$1 - C t_i p_i > 0, \qquad \frac{C[t_i p_i + t_i^2(1+E_*)]}{(1 - C t_i p_i)^3} \leq \frac{1}{2}, \qquad t_i p_i^3 \leq \varepsilon_{\mathrm{sc}}, \tag{27.5}$$

   where E_* is the fixed upper bound on the weighted analytic coefficient norm of the effective interaction in Theorem 13.8, and $\varepsilon_{\mathrm{sc}}$ is fixed by the construction. The same holds at every point $u\phi$, $0 \leq u \leq 1$, of the segment from the zero field to ϕ, with the same product Ω_i.

2. *The potential is C^2 on a neighborhood of the closed product Ω_i, and*

$$D_w^2 V_{\mathfrak{d}_i,\phi} \geq \tfrac{1}{2} I \quad \text{on } \Omega_i. \tag{27.6}$$

   In particular, the conditional fluctuation covariance Σ_ϕ of w_i satisfies $\|\Sigma_\phi\|_{\mathrm{op}} \leq 2$, and every real function F on Ω_i that is Lipschitz for the Euclidean distance satisfies

$$\log \mathbb{E}_\phi e^{z(F - \mathbb{E}_\phi F)} \leq z^2 \operatorname{Lip}(F)^2 \qquad (z \in \mathbb{R}). \tag{27.7}$$

3. *Let $\mu_\phi = \mathbb{E}_\phi w_i$. Then*

$$\|\mu_\phi\|_\infty \leq C_J t_i \|\phi\|_\infty, \qquad \mu_0 = 0, \qquad \|\Sigma_\phi - \Sigma_0\|_{\mathrm{op}} \leq C_J t_i p_i^3 \|\phi\|_\infty, \qquad \|\Sigma_\phi\|_{\mathrm{op}} \leq 2. \tag{27.8}$$

4. *Let $E_\phi = g_i^{-1} D_w \mathscr{R}_i(\phi, \mu_\phi)$ and $\mathsf{Z}_\phi = E_\phi \Sigma_\phi E_\phi^{\mathsf{T}}$. Then*

$$\|E_\phi\|_{\mathrm{op}} + \|E_0\|_{\mathrm{op}} \leq C_J, \qquad \|E_\phi - E_0\|_{\mathrm{op}} \leq C_J \|\phi\|_\infty, \tag{27.9}$$

   and

$$\|\mathsf{Z}_\phi - \mathsf{Z}_0\|_{\mathrm{op}} \leq C_J (1 + t_i p_i^3) \|\phi\|_\infty \leq C_J \|\phi\|_\infty. \tag{27.10}$$

The constants depend only on the fixed analytic domains and coefficient bounds, M, J, and the common upper bound g_*; they do not depend on the coupling below g_*, on the lattice side, on K, i, $\mathfrak{d}_i$, or on ϕ.

Proof. For fixed K and i, Lemma 26.2 identifies the conditional law of w_i given $\mathcal{F}_{i-1}$ with the conditional fluctuation kernel at the coarser field ϕ, with $\mathfrak{d}_i = J(K-i)$ preceding averaging steps. The side of the coarser lattice of the step is $nB^{i-1} \geq 4$, the coarser field lies in B_r, and the coupling-scaled fluctuation inputs $q_{i,\lambda} = g_{i,\lambda} w_{i,\lambda}$ of every point of Ω_i

lie in the common analytic domain, as checked in the proof of Lemma 26.1. The domain conditions there include those on every intermediate classical background field, on the block averaging constraint, and on the Haar disintegration density, and the condition that the reconstruction of every point of the product lies in D_r. These are the common domains of Lemma 26.2, and Theorems 15.12 and 15.13 preserve them. In particular, the change to the reconstruction maps preserves each ball of fluctuation coordinates, the free logarithmic coordinates, the Haar disintegration density, the triangular determinant, the inherited normalization constant, and the normalizer, and hence the normalized kernel.

The hypotheses of Theorem 13.8 hold on this product. Let $t_i = M^{(J-1)/2} g_i$; the radii are $p_i = A_{\mathrm{w}} \log(1/t_i) \geq 1$, and the ratios of intermediate couplings are included in the reconstruction map. The common upper bound g_* of Lemma 26.2 is also the reduced bound of Theorem 13.10. Hence (27.5) holds uniformly in K, i, $\mathfrak{d}_i$, ϕ, and the coupling below g_*. The last condition is uniform in the coupling: since $t[A_{\mathrm{w}} \log(1/t)]^3 \to 0$ as $t \downarrow 0$, the reduced bound controls it on the whole interval of couplings. Every point $u\phi$ of the segment lies in B_r and satisfies the same product and domain conditions, so the theorem on means and covariances also applies along the segment.

We write the density as $Z_\phi^{-1} \mathbf{1}_{\Omega_i} e^{-V_{\mathfrak{d}_i,\phi}}\, dw$. By Theorem 13.8, the potential is smooth on an open neighborhood of the closed product, and (27.6) holds. The restrictions imposed at coarser scales depend only on the coarser field and cancel in this normalized conditional law, by the same theorem. The product is a finite product of convex balls in $\mathbb{R}^3$, the potential is C^2 on its neighborhood, and the Hessian bound holds on the whole product. These are the hypotheses of Theorem 5.4, with $m = 1/2$, and that theorem implies that $\|\Sigma_\phi\|_{\mathrm{op}} \leq 2$ and (27.7). All later uses of conditional concentration in this section are for this kernel, so their constants do not depend on the dimension, the number of averaging steps, or the coupling.

By Theorem 13.10, which applies by the preceding checks and the preservation of the kernel, we have (27.8). To bound the linear coefficient of the reconstruction map, we write the map as a function $\widehat{F}_i(\phi, q)$ of the coupling-scaled fluctuation inputs. Let $S_i = \mathrm{diag}(g_{i,\lambda}/g_i)$ and $q_\phi = (g_{i,\lambda}\mu_{\phi,\lambda})_\lambda$. Then

$$E_\phi = g_i^{-1} D_w \mathscr{R}_i(\phi, \mu_\phi) = D_q \widehat{F}_i(\phi, q_\phi) S_i, \qquad E_0 = D_q \widehat{F}_i(0,0) S_i,$$

and the norm of S_i is bounded by the fixed ratio of intermediate couplings. The inner domain contains these points and the segments joining them: the conditional mean lies in the convex product, and $u\phi$ lies in the ball of coarser fields. The second derivatives in the free coordinates, and the mixed second derivatives in the coarser field and the free coordinates, have bounded absolute row sums and volume-normalized column sums. This follows from Theorem 15.12 and the computation with majorant series in Step 4 of the proof of Lemma 27.1. Integrating these derivatives along the two segments, we obtain

$$(\mathsf{R} + \mathsf{C})(E_\phi - E_0) \leq C_J(\|\phi\|_\infty + \|q_\phi\|_\infty) \leq C_J(1 + t_i^2)\|\phi\|_\infty,$$

where we used $\|q_\phi\|_\infty \le t_i \|\mu_\phi\|_\infty$. Converting the volumes of the fluctuation coordinates to the output volume uses only fixed ratios within one renormalization group step, so the weighted Schur bound of Lemma 27.1(1) implies (27.9). For $\mathsf{Z}_\phi = E_\phi \Sigma_\phi E_\phi^{\mathsf{T}}$ we expand

$$\mathsf{Z}_\phi - \mathsf{Z}_0 = (E_\phi - E_0)\Sigma_\phi E_\phi^{\mathsf{T}} + E_0(\Sigma_\phi - \Sigma_0)E_\phi^{\mathsf{T}} + E_0\Sigma_0(E_\phi^{\mathsf{T}} - E_0^{\mathsf{T}}). \tag{27.11}$$

By (27.8) and (27.9), the three terms are bounded by $C_J\|\phi\|_\infty$, $C_J t_i p_i^3 \|\phi\|_\infty$, and $C_J\|\phi\|_\infty$, respectively, which gives (27.10). Every constant here comes from the common upper bound on the couplings and from the coefficient bounds of the reconstruction maps; none depends on the coupling. The covariance bound of Theorem 17.4, whose constants may depend on the fixed coupling, is not used. □

Conditional means of quadratic functions

The next lemma bounds the conditional mean of a quadratic function of the reconstructed field without using any spatial decay of its coefficients. This allows us to apply it after heat evolution, which spreads the coefficients over the torus. We say that a quadratic form Q on link logarithms, with values in arrays of $\mathfrak{su}(2)$-valued components, is equivariant if $Q(\mathrm{Ad}_h v, \mathrm{Ad}_h v) = \mathrm{Ad}_h Q(v, v)$ for every constant $h \in \mathrm{SU}(2)$ acting on all color indices.

Lemma 27.4. *Assume the setting of Lemma 27.3. Let Q be an equivariant quadratic form on link logarithms on the lattice of mesh a_i, fix one scalar output component of $Q(v, v)$, and let k_Q be the absolute sum of the coefficients of that component over both inputs. Put $r = \|\phi\|_\infty$ and $c = \mathscr{R}_i(\phi, 0)$. Then*

$$|\mathbb{E}_\phi Q(\theta_i, \theta_i) - Q(c, c)| \le C_J k_Q (g_i^2 r + g_i^3 p_i^3), \tag{27.12}$$

$$|Q(c, c) - Q(\mathsf{L}_i\phi, \mathsf{L}_i\phi)| \le C k_Q a_{\mathrm{cap}}^2 r^3. \tag{27.13}$$

The constants depend only on the fixed coefficient bounds, M, J, and the common upper bound g_*; they do not depend on the coupling below g_*, the lattice side, K, i, or the number of preceding averaging steps.

Proof. The matrix C_Q of the fixed scalar component has nuclear norm at most k_Q, since it is a sum of coordinate matrix units, each of nuclear norm one. Thus, for every matrix C,

$$|Q : C| = |\mathrm{tr}(C_Q C)| \le k_Q \|C\|_{\mathrm{op}}. \tag{27.14}$$

This needs no spatial weight on Q.

Let $\mu = \mathbb{E}_\phi w_i$, $\xi = w_i - \mu$, and $v = \mathscr{R}_i(\phi, \mu)$. Using the bound on the mean in (27.8), the expansion (26.2) of Lemma 26.2, and the bounds (27.2) on the derivatives of the reconstruction, we have

$$\begin{aligned} \theta_i &= v + g_i E_\phi \xi + g_i^2 \mathsf{B}_\phi(\xi, \xi) + R_\phi(\xi), \qquad \|R_\phi\|_\infty \le C_J g_i^3 p_i^3, \\ \|v - c\|_\infty &\le C_J g_i \|\mu\|_\infty \le C_J g_i t_i r \le C_J g_i^2 r, \qquad \|v\|_\infty \le C_J r. \end{aligned} \tag{27.15}$$

The normalized coefficients of E_ϕ and of each output component of B_ϕ have bounded absolute rows. By (27.14) and the bound $\|\Sigma_\phi\|_{\mathrm{op}} \le 2$, which follows from the Poincaré inequality, we obtain

$$\|\mathbb{E}_\phi \mathsf{B}_\phi(\xi,\xi)\|_\infty \le 2C_J. \tag{27.16}$$

We do not need the quadratic mean to vanish at a nonzero coarser field.

We insert (27.15) into both arguments of Q. The terms linear in ξ have mean zero, since ξ is centered under the same law. The term with $g_i E_\phi \xi$ in both arguments has mean exactly $g_i^2 Q : (E_\phi \Sigma_\phi E_\phi^{\mathsf{T}})$. By (27.14) and (27.10), it differs from $g_i^2 Q : \mathsf{Z}_0$ by at most $C_J k_Q g_i^2 r$, uniformly in the coupling. The vector-valued contraction $Q : \mathsf{Z}_0$ vanishes. Indeed, Z_0 is scalar in the color indices, so by equivariance the contraction is a vector fixed by all adjoint rotations; such a vector is zero, since a rotation through π about an axis perpendicular to a nonzero vector changes its sign.

The two terms that pair v with $g_i^2 \mathbb{E}_\phi \mathsf{B}_\phi(\xi,\xi)$ are bounded by $C_J k_Q g_i^2 r$, by (27.16). Replacing $Q(v,v)$ by $Q(c,c)$ costs at most $C_J k_Q g_i^2 r$, since the extra factor r is bounded on the fixed chart. Every remaining term contains R_ϕ, or one argument linear and one quadratic in ξ, or two arguments quadratic in ξ. On the cutoff domain $|\xi| \le 2p_i$ coordinatewise, so these terms are bounded by $C_J k_Q g_i^3 p_i^3$, $C_J k_Q g_i^3 p_i^3$, and $C_J k_Q g_i^4 p_i^4$, respectively; the last is absorbed into the first because $g_i p_i$ is bounded by (27.5). The factors v and $g_i p_i$ in products with R_ϕ are bounded on the same domain. This proves (27.12). For the classical error, by (27.3), we have

$$|Q(c,c) - Q(\mathsf{L}_i\phi, \mathsf{L}_i\phi)| \le k_Q(\|c\|_\infty + \|\mathsf{L}_i\phi\|_\infty)\|c - \mathsf{L}_i\phi\|_\infty \le C k_Q a_{\mathrm{cap}}^2 r^3,$$

which is (27.13). The only quantities involving the cutoff that were absorbed into constants are $g_i p_i$, $g_i t_i / g_i^2 = M^{(J-1)/2}$, and $t_i p_i^3$, all bounded by fixed data as checked in Lemma 27.3. □

Coefficients and drifts of the propagated terms

For $\omega \in \{UV, VU, VV\}$ and $0 \le i \le j-1$ we define the bilinear sources

$$S_s^{UV}(v,z) = \mathcal{N}_\delta(\mathsf{U}_s v, \mathsf{V}_s z), \qquad S_s^{VU}(v,z) = \mathcal{N}_\delta(\mathsf{V}_s v, \mathsf{U}_s z), \qquad S_s^{VV}(v,z) = \mathcal{N}_\delta(\mathsf{V}_s v, \mathsf{V}_s z),$$

with U_s, V_s as in Lemma 27.1(3), and the bilinear Duhamel terms

$$\mathscr{T}_{j,i}^{\omega}(v,z)(t) = \int_0^t P_{t-s}^{\delta} S_s^{\omega}(v,z)\, ds.$$

Taking $z = v$ gives the quadratic terms of the introduction; each argument has a fixed place, and no symmetry of $\mathcal{N}_\delta$ is used. Both arguments are propagated through the same linearizations, so $\mathscr{T}_{j,i-1}^{\omega}(v,v) = \mathscr{T}_{j,i}^{\omega}(\mathsf{L}_i v, \mathsf{L}_i v)$. We write $\mathscr{S}_s$ for any of the sources $S_s^{\omega}(\theta_i, \theta_i)$.

Theorem 27.5. *Assume the setting of Section 26.1 and the notation of Lemma 27.1, with $m = j - 1 - i$ and $G = a_i$. Set*

$$K_{ji} = C_{\mathrm{CP}}^{2m} H/G^2 = q_{\mathrm{anc}}^m/G, \qquad \ell_H(t) = 1 + \log(1 + t/H^2),$$
$$J_H(t) = \int_0^t (t-s)^{-1/2}(s + H^2)^{-1}\, ds.$$

1. *For $1 \le i \le j-1$, the derivative $\mathsf{F}_s = D_{w_i}\mathscr{S}_s$ in w_i, at any point of Ω_i, satisfies*

$$(\mathsf{R} + \mathsf{C})(\mathsf{F}_s) \le C_J K_{ji} g_i b_i (s + H^2)^{-1},$$
$$\|\mathsf{F}_s\|_{\ell^2(w_i)\to\ell^2(\delta)} \le C_J K_{ji} g_i b_i (s + H^2)^{-1}(G/\delta)^{3/2},$$

   where both ℓ^2 norms are counting norms. For $s \ge G^2$, the Euclidean norm of its value row improves to $C_J K_{ji} g_i b_i G^{3/2} s^{-7/4}$; no such improvement is asserted for its volume-normalized column sum.

2. *The rows of the derivatives in w_i of the Duhamel terms satisfy, for $0 \le t \le G^2$,*

$$\sup_x \|D_{w_i}\mathscr{T}^{\omega}_{j,i}(t,x)\|_2 \le C_J K_{ji} g_i b_i \log(1 + t/H^2),$$
$$\sup_x \|D_{w_i} D_\delta \mathscr{T}^{\omega}_{j,i}(t,x)\|_2 \le C_J K_{ji} g_i b_i J_H(t),$$

   and, for $t \ge G^2$ and $r = 0, 1$,

$$\sup_x \|D_{w_i} D_\delta^r \mathscr{T}^{\omega}_{j,i}(t,x)\|_2 \le C_J K_{ji} g_i b_i G^{3/2} t^{-3/4 - r/2} \ell_H(T).$$

   At small times the displayed integrals, which vanish at $t = 0$, are kept; in particular $J_H(t) \le C \min\{\sqrt{t}/H^2, (t + H^2)^{-1/2}\ell_H(t)\}$. The bounds on first differences use the outer heat kernel.

3. *For a quadratic coefficient $c_{y;u,v}$, let R_2 be the supremum over outputs of the absolute sum over both inputs, and, for a coefficient with input mesh G and output mesh h,*

$$\mathsf{C}_2(c) = \max\Big\{\sup_u \frac{h^3}{G^3}\sum_{y,v}|c_{y;u,v}|,\ \sup_v \frac{h^3}{G^3}\sum_{y,u}|c_{y;u,v}|\Big\}$$

   the larger of its two volume-normalized column sums with a fixed input. Here $h = \delta$; for $\mathsf{Q}_{j,i}$ in Theorem 27.7, before interpolation, $h = a_j$. For each ω,

$$(\mathsf{R}_2 + \mathsf{C}_2)(\mathscr{T}^{\omega}_{j,i}(t)) \le C K_{ji} \log(1 + t/H^2),$$
$$(\mathsf{R}_2 + \mathsf{C}_2)(D_\delta \mathscr{T}^{\omega}_{j,i}(t)) \le C K_{ji} J_H(t).$$

   With the weights of $\mathsf{Y}^{\delta}_{\alpha}$, that is, after multiplying the first bound by $(t + \delta^2)^{\beta}$ and the second by $(t + \delta^2)^{\alpha}$, the supremum over $t \le T$ of these coefficient norms is therefore at most

$C_{L,T,\alpha,a_0}(1+j)K_{ji}$; *we call this the weighted coefficient bound. On the whole cutoff domain* Ω_i *we also have, for* $r = 0, 1, 2, 3$,

$$\sup_{w_i,x} \|D_{w_i} D_\delta^r \mathscr{T}_{j,i}^\omega(t,x)\|_2 \le C_{J,r} C_{\mathrm{CP}}^{2m} g_i b_i G^{-2}\, t H^{-1-r},$$

and, for $r = 0, 1$,

$$\sup_{w_i,x} \|D_{w_i} \partial_t D_\delta^r \mathscr{T}_{j,i}^\omega(t,x)\|_2 \le C_{J,r} C_{\mathrm{CP}}^{2m} g_i b_i G^{-2}(H^{-1-r} + tH^{-3-r}).$$

These bounds keep the factor $g_i b_i$, *for later use in coverings.*

4. *Define the conditional drift*

$$e_{j,i}^\omega = \mathbb{E}\big[\mathscr{T}_{j,i}^\omega(\theta_i,\theta_i) \;\big|\; \mathcal{F}_{i-1}\big] - \mathscr{T}_{j,i-1}^\omega(\theta_{i-1},\theta_{i-1}).$$

With the deterministic bounds $b_0 = \rho_0$ *and* $b_i = \rho_0 a_{\mathrm{cap}}^i + C_J g_3 \sqrt{a_i}\, p_i$, $i \ge 1$, *on* $\|\theta_i\|_\infty$,

$$\mathsf{Y}_\alpha^\delta(e_{j,i}^\omega) \le C_J(1+j)\frac{q_{\mathrm{anc}}^{j-1-i}}{a_i}\big\{a_{\mathrm{cap}}^2 b_{i-1}^3 + g_i^2 b_{i-1} + g_i^3 p_i^3\big\},$$

and

$$\mathsf{Y}_\alpha^\delta(\mathscr{T}_{j,0}^\omega(\theta_0,\theta_0)) \le C\rho_0^2(1+j)q_{\mathrm{anc}}^{j-1}/a_0.$$

More precisely, for $\phi = \theta_{i-1}$ *and* $\theta_i^{\mathrm{cl}} = \mathscr{R}_i(\phi, 0)$, *write* $e_{j,i}^\omega = e_{j,i}^{\omega,\mathrm{cl}} + e_{j,i}^{\omega,\mathrm{q}}$, *where the linearization error and the fluctuation correction are*

$$\begin{aligned} e_{j,i}^{\omega,\mathrm{cl}} &= \mathscr{T}_{j,i}^\omega(\theta_i^{\mathrm{cl}},\theta_i^{\mathrm{cl}}) - \mathscr{T}_{j,i-1}^\omega(\phi,\phi),\\ e_{j,i}^{\omega,\mathrm{q}} &= \mathbb{E}\big[\mathscr{T}_{j,i}^\omega(\theta_i,\theta_i) \;\big|\; \mathcal{F}_{i-1}\big] - \mathscr{T}_{j,i}^\omega(\theta_i^{\mathrm{cl}},\theta_i^{\mathrm{cl}}). \end{aligned}$$

Then

$$\begin{aligned} \mathsf{Y}_\alpha^\delta(e_{j,i}^{\omega,\mathrm{cl}}) &\le C_J(1+j)K_{ji}a_{\mathrm{cap}}^2\|\phi\|_\infty^3,\\ \mathsf{Y}_\alpha^\delta(e_{j,i}^{\omega,\mathrm{q}}) &\le C_J(1+j)K_{ji}(g_i^2\|\phi\|_\infty + g_i^3 p_i^3). \end{aligned}$$

Both deterministic majorants have summable tails in j, *uniformly in* K. *When the coarser field is the identity, the vector-valued quadratic mean vanishes by the invariance of the conditional law under global color rotations; at a nonzero coarser field, the variation of the covariance, the nonlinear mean, and the classical Taylor error are all kept.*

The constants in the coefficient and mean bounds do not depend on the coupling, K, j, i, *the number of preceding averaging steps, or* a/δ; *they have the dependence of Lemma 27.1, with* α *added. The weighted coefficient bounds and the weighted bounds on drifts may in addition depend on the fixed coarse mesh* $a_0 = L/n$; *this does not apply to the unweighted time-dependent rows. The constants in the summability statements may depend on the fixed* γ, n, ρ_0, *and the prescription of the cutoff radii.*

Proof. We first prove the deterministic estimates on coefficients and on derivatives in w_i, and then the estimates on drifts, which use Lemmas 27.3 and 27.4. Throughout, K_{ji} is the exact value displayed in the statement.

Step 1: sources and late rows. Combining the bound on pointwise products in Lemma 27.1(1) with part (3) of that lemma without a derivative in w_i, and using (27.4), we find that the bilinear coefficient of the source satisfies

$$(\mathsf{R}_2 + \mathsf{C}_2)(S_s^{\omega}) \le C\mathfrak{A}_{ji}^2 (s + H^2)^{-1/2} d_H(s) = \frac{CK_{ji}}{s + H^2}, \tag{27.17}$$

One argument uses a volume-normalized column and the other a row; hence no factor of the output volume occurs. Differentiating the two arguments in w_i instead, we obtain

$$(\mathsf{R} + \mathsf{C})(\mathsf{F}_s) \le \frac{C_J K_{ji} g_i b_i}{s + H^2}. \tag{27.18}$$

The counting Schur bound of Lemma 27.1(1) then implies the operator bound of (1), with the factor $(G/\delta)^{3/2}$. For several types of fluctuation coordinates, the intermediate volumes are their own cell volumes, and replacing them by G^3 costs only C_J.

For $s \ge G^2$ we split the heat semigroup in either argument into two equal halves. The first half, applied to the derivative in w_i, has row and volume-normalized column at most $C_J \mathfrak{A}_{ji} g_i$; in the argument with the defect, both acquire the factor $d_H(s/2) \le \sqrt{2} d_H(s)$. The second half has counting row norm at most $C\delta^{3/2} s^{-3/4-r/2}$ after r differences, by Lemma 16.1. The Schur bound for the first half implies that

$$\begin{aligned} &\sup_x \|D_{w_i} D_\delta^r \mathsf{U}_s \theta_i(x)\|_2 \le C_{J,r} \mathfrak{A}_{ji} g_i G^{3/2} s^{-3/4-r/2}, \\ &\sup_x \|D_{w_i} D_\delta^r \mathsf{V}_s \theta_i(x)\|_2 \le C_{J,r} \mathfrak{A}_{ji} g_i G^{3/2} s^{-3/4-r/2} d_H(s) \qquad (r = 0, 1), \end{aligned} \tag{27.19}$$

where we used $\delta^{3/2}(G/\delta)^{3/2} = G^{3/2}$. In a translated bracket, the derivative in w_i falls either on the argument without a difference or on the argument with the difference. We use, respectively, the bound of (27.19) with $r = 0$ or $r = 1$, and the complementary deterministic bound of Lemma 27.1(3) for the other argument. Every term keeps one factor d_H. Since $d_H(s) \le Hs^{-1/2}$ and $\mathfrak{A}_{ji}^2 H = K_{ji}$, we find

$$\sup_x \|\mathsf{F}_s(x, \cdot)\|_2 \le C_J K_{ji} g_i b_i G^{3/2} s^{-7/4} \qquad (s \ge G^2). \tag{27.20}$$

For the order VV we use $d_H(s)^2 \le d_H(s)$. These bounds also hold on $G^2/2 \le s \le G^2$ after increasing the constant: on that interval the absolute row in (27.18) is bounded by the right side of (27.20). The heat semigroup conserves the mass of columns, so this argument gives no improvement at late times for the absolute column.

Step 2: the outer heat semigroup at small times. Let $\Lambda = C_J K_{ji} g_i b_i$. By heat contraction and (27.18), we have the value row $\Lambda \int_0^t (s + H^2)^{-1}\, ds = \Lambda \log(1 + t/H^2)$ at every t. For the first difference we use the bound on the outer heat kernel

$$\|D_\delta P_r^\delta\|_{\infty\to\infty} \le C(r + \delta^2)^{-1/2} \le C r^{-1/2} \qquad (r > 0),$$

which gives $C\Lambda J_H(t)$, without differentiating the source; the singularity at $s = t$ is integrable. The substitution $z = \sqrt{t - s}$ yields

$$J_H(t) = \frac{2}{\sqrt{t + H^2}} \operatorname{artanh} \sqrt{\frac{t}{t + H^2}}. \tag{27.21}$$

Bounding $(s + H^2)^{-1}$ by H^{-2}, we find $J_H(t) \le 2\sqrt{t}/H^2$ at all t. For $t \ge H^2$, the identity $2 \operatorname{artanh} z = \log((1 + z)/(1 - z))$ gives $J_H(t) \le C(t + H^2)^{-1/2} \ell_H(t)$; for $t \le H^2$ this bound also follows from $\sqrt{t}/H^2 \le C/H$. This proves the stated minimum, including its vanishing at $t = 0$.

Step 3: late times. For $2G^2 \le t \le T$ we split the Duhamel integral at $t/2$. On the early half we use the counting operator bound of (1) for F_s and the counting row of the heat kernel,

$$\|D_\delta^r P_{t-s}^\delta(x, \cdot)\|_2 \le C_r \delta^{3/2} t^{-3/4 - r/2}, \qquad r = 0, 1;$$

the integral is at most $C\Lambda G^{3/2} t^{-3/4 - r/2} \log(1 + t/H^2)$. On the late half we use (27.20) and the ℓ^1 row of $D_\delta^r P_{t-s}^\delta$; the integral is at most

$$C\Lambda G^{3/2} \int_{t/2}^t (t - s)^{-r/2} s^{-7/4}\, ds \le C\Lambda G^{3/2} t^{-3/4 - r/2}.$$

If $G^2 \le t \le 2G^2$, the bounds of Step 2 give the same result, since $t \asymp G^2$ and $J_H(t) \le C t^{-1/2} \ell_H(T)$. Thus the late estimates of (2) hold from $t = G^2$ on. The constants for the fixed horizon may depend on T/L^2.

Step 4: quadratic coefficients and weights. The outer heat kernel has absolute row and volume-normalized column equal to one, and its first difference has both norms at most $C(t - s)^{-1/2}$. Applying these facts to (27.17), we obtain the two coefficient bounds of (3), with the exact integral profiles. If $t \le H^2$, the weighted sum is at most

$$C K_{ji} \{H^{2\beta} t/H^2 + H^{2\alpha} \sqrt{t}/H^2\} \le C K_{ji} H^{2\alpha - 1}.$$

If $t \ge H^2$, it is at most $C_{T,\alpha} K_{ji} \ell_H(T)$, because both powers of the weights are positive and $\delta \le H$. In the first regime we use $H \le a_0$. Since $H = a_0 B^{-(j-1)}$,

$$\begin{aligned} \ell_H(T) &= 1 + \log\bigl(1 + (T/a_0^2) B^{2(j-1)}\bigr) \\ &\le 1 + \log(1 + T/a_0^2) + 2(j - 1) \log B \\ &\le C_{T, a_0, B}(1 + j), \end{aligned} \tag{27.22}$$

where we used $1 + xB^{2(j-1)} \le (1+x)B^{2(j-1)}$ for $x = T/a_0^2$, since $B^{2(j-1)} \ge 1$. The constant in the weighted coefficient bound may therefore depend on the fixed $a_0 = L/n$. The unweighted bounds on time-dependent coefficients and the bounds on rows of derivatives in w_i keep their stated dependence and their logarithmic profiles.

Step 5: crude bounds for coverings. For any fixed integer $r \ge 0$ we commute the differences through the heat semigroup and apply them to the cardinal interpolations on their own lattices before smoothing. The arguments then satisfy

$$\|D_\delta^r \mathsf{U}_s \theta_i\|_\infty + \|D_\delta^r \mathsf{V}_s \theta_i\|_\infty \le C_{J,r} \mathfrak{A}_{ji} b_i H^{-r},$$

and the rows of their derivatives in w_i have the same bound with b_i replaced by g_i. By the finite product rule, we have, for each fixed r,

$$\begin{aligned} \sup_{w_i,s,x} |D_\delta^r \mathscr{S}_s(x)| &\le C_{J,r} \mathfrak{A}_{ji}^2 b_i^2 H^{-1-r}, \\ \sup_{w_i,s,x} \|D_{w_i} D_\delta^r \mathscr{S}_s(x)\|_2 &\le C_{J,r} \mathfrak{A}_{ji}^2 g_i b_i H^{-1-r}. \end{aligned} \tag{27.23}$$

Heat contraction and integration in time give the factor t for the Duhamel term. For the time derivative we use the exact identity $\partial_t \mathscr{T}(t) = \mathscr{S}_t + \Delta_\delta \mathscr{T}(t)$ in finite dimensions. The Laplacian is a sum of three second differences, which are bounded by (27.23) with two more differences. This proves both crude bounds of (3), uniformly on Ω_i, and avoids a bound of size δ^{-2} for the generator.

Step 6: the drifts. We apply Lemma 27.4 to each output component of $\mathscr{T}_{j,i}^\omega(t)$ and of its first difference. Their deterministic coefficients are fixed before conditioning, they are equivariant, and their absolute sums k_Q are bounded by the unweighted bounds of (3). Conditional integration commutes with finite differences of outputs. With $\phi = \theta_{i-1}$ and $\mathscr{T}_{j,i-1}^\omega(\phi,\phi) = \mathscr{T}_{j,i}^\omega(\mathsf{L}_i\phi, \mathsf{L}_i\phi)$, the weighted coefficient bound of Step 4 implies that

$$\begin{aligned} \mathsf{Y}_\alpha^\delta(e_{j,i}^{\omega,\mathrm{cl}}) &\le C_{J,a_0}(1+j) K_{ji} a_{\mathrm{cap}}^2 \|\phi\|_\infty^3, \\ \mathsf{Y}_\alpha^\delta(e_{j,i}^{\omega,\mathrm{q}}) &\le C_{J,a_0}(1+j) K_{ji} (g_i^2 \|\phi\|_\infty + g_i^3 p_i^3). \end{aligned} \tag{27.24}$$

Here C_{J,a_0} has the deterministic dependence already listed, including a_0 from (27.22); it does not depend on the coupling, K, j, i, the number of averaging steps, or a/δ. In the weighted bounds on drifts in the statement, C_J denotes this constant, while the constants of (27.12) and (27.13) are unchanged. Adding the two bounds and using $\|\phi\|_\infty \le b_{i-1}$, we obtain the bound on $e_{j,i}^\omega$.

The identity $\mathscr{T}_{j,i-1}^\omega(v,v) = \mathscr{T}_{j,i}^\omega(\mathsf{L}_i v, \mathsf{L}_i v)$ holds because $\mathsf{T}_{j,i-1} = \mathsf{T}_{ji}\mathsf{L}_i$ in both arguments. Substituting θ_0, whose norm is at most ρ_0, into the weighted coefficient bound, we obtain the bound on the coarsest term, with the same dependence on a_0. When the

coarser field is the identity, the conditional law at the zero field and the reconstruction are equivariant under global color rotations, so the vector-valued quadratic mean itself vanishes, and not only its contribution from the linear covariance.

Step 7: bounds on the coarser fields and summability. Integrating (27.2) from $w = 0$ to w_i along the segment in the convex product, and using (27.3), we obtain

$$\|\theta_i\|_\infty \le a_{\mathrm{cap}}\|\theta_{i-1}\|_\infty + C_J g_i p_i.$$

Iterating, and using $g_{i-l} = B^{l/2} g_i$ and that the cutoff radii do not decrease with i, we obtain

$$\|\theta_i\|_\infty \le \rho_0 a_{\mathrm{cap}}^i + C_J g_i p_i \sum_{l=0}^{i-1} (a_{\mathrm{cap}}\sqrt{B})^l \le \rho_0 a_{\mathrm{cap}}^i + C_J g_3 \sqrt{a_i}\, p_i,$$

where the geometric series is included in the fixed C_J. For fixed γ and n, we obtain

$$b_i \le C_\gamma (1+i)\sqrt{a_i}, \qquad b_i/g_i \le C_\gamma(1+i), \tag{27.25}$$

because $a_{\mathrm{cap}}^i = (a_i/a_0)^{\kappa_{\mathrm{term}}}$ with $\kappa_{\mathrm{term}} = -\log a_{\mathrm{cap}}/\log B > 1/2$. In particular, the three contributions to the drifts, divided by a_i, satisfy

$$\frac{a_{\mathrm{cap}}^2 b_{i-1}^3 + g_i^2 b_{i-1} + g_i^3 p_i^3}{a_i} \le C_\gamma (1+i)^3 a_i^{1/2}. \tag{27.26}$$

For the first term we use $a_{i-1} = B a_i$; the other two are exactly $g_3^2 b_{i-1}$ and $g_3^3\sqrt{a_i}\, p_i^3$. The double series of majorants of the drifts is bounded by

$$C_\gamma \sum_{i\ge 1}\sum_{m\ge 0} (2+i+m)(1+i)^3 q_{\mathrm{anc}}^m a_i^{1/2},$$

and by Lemma 27.2(2), with $r = 1/2$, its tail over $j = i+1+m \ge k$ is at most $C a_k^\eta$ for some $\eta > 0$, and tends to zero; truncation at any K only decreases it. The majorant of the coarsest terms is summable because $\sum_j (1+j) q_{\mathrm{anc}}^{j-1} < \infty$. This proves the summability statement. Only this summation uses the bound (27.25), which holds at fixed coupling; the constants in the coefficient bounds and in the pointwise bounds on means are uniform in the coupling below g_*. □

27.3 Maximal estimates and ultraviolet tails of the ordered terms

We now bound the conditionally centered terms $Z_{j,i}^\omega$ of the telescope, sum the telescope, and obtain the bounds on $\mathcal{Q}_\delta(r_j, r_j)$ and $\mathcal{B}_\delta(x_{j-1}, r_j)$. The maximal inequality uses a finite net in space and time at the scale H, whose size is a fixed power of H^{-1}, independent of δ. It also uses the bound $b_i/g_i \le C_\gamma(1+i)$ of (27.25), so it holds at fixed positive coupling.

Theorem 27.6. *Assume the setting of Section 26.1 and the notation of Theorem 27.5, and fix $\gamma > 0$, L, n, T, M, J, and the domains before any ultraviolet limit is taken. Put*

$$Z^{\omega}_{j,i} = \mathscr{T}^{\omega}_{j,i}(\theta_i, \theta_i) - \mathbb{E}\big[\mathscr{T}^{\omega}_{j,i}(\theta_i, \theta_i) \mid \mathcal{F}_{i-1}\big], \qquad A_{ji} = K_{ji} g_i b_i G^{2\alpha-1} \ell_H(T).$$

1. *There exist $C_\gamma < \infty$ and a fixed finite exponent c such that, for every $u \geq 0$, every $1 \leq i \leq j-1 \leq K-1$, and each ω separately,*

$$\mathbb{P}\Big\{ \mathsf{Y}^{\delta}_{\alpha}(Z^{\omega}_{j,i}) > C_\gamma A_{ji} \sqrt{\log(2 + L/H) + u} \;\Big|\; \mathcal{F}_{i-1} \Big\} \leq C e^{-u}.$$

   It follows that, for every fixed $0 < p < \infty$,

$$\big\| \mathsf{Y}^{\delta}_{\alpha}(Z^{\omega}_{j,i}) \big\|_{L^p(\cdot \mid \mathcal{F}_{i-1})} \leq C_{\gamma,p} (1+j)^c q_{\mathrm{anc}}^{j-1-i} g_i b_i a_i^{2\alpha-2} \big[\sqrt{\log(2 + L/a_i)} + \sqrt{p} \big].$$

   The proof uses a net in physical space and time at the scale H, with at most a fixed power of H^{-1} points, independently of δ/H; a short initial interval of times is controlled by the factor t of the Duhamel term. The proof uses $b_i/g_i \leq C_\gamma(1+i)$, and no constant uniform in the coupling is asserted.

2. *At $i = j-1$ we have $(\mathsf{U} - \mathsf{V})\theta_{j-1} = x_{j-1}$ and $\mathsf{V}\theta_{j-1} = r_j$, and*

$$\mathcal{B}_\delta(x_{j-1}, r_j) = \mathscr{T}^{UV}_{j,j-1} + \mathscr{T}^{VU}_{j,j-1} - 2\mathscr{T}^{VV}_{j,j-1}, \qquad \mathcal{Q}_\delta(r_j, r_j) = \mathscr{T}^{VV}_{j,j-1},$$

   with θ_{j-1} in both arguments of each term on the right. For the full quadratic difference, $\mathcal{Q}_\delta(x_{j-1} + r_j, x_{j-1} + r_j) - \mathcal{Q}_\delta(x_{j-1}, x_{j-1}) = \mathscr{T}^{UV}_{j,j-1} + \mathscr{T}^{VU}_{j,j-1} - \mathscr{T}^{VV}_{j,j-1}$, so the coefficient of $\mathscr{T}^{VV}_{j,j-1}$ is -1.

3. *For sufficiently large fixed C_γ and c, let*

$$W_j = C_\gamma (1+j)^c \Big\{ \rho_0^2 q_{\mathrm{anc}}^{j-1}/a_0 + \sum_{i=1}^{j-1} q_{\mathrm{anc}}^{j-1-i} \Big[g_i b_i a_i^{2\alpha-2} \sqrt{\log(2 + L/a_i)} + \frac{a_{\mathrm{cap}}^2 b_{i-1}^3 + g_i^2 b_{i-1} + g_i^3 p_i^3}{a_i} \Big] \Big\}.$$

   For each ω separately, $\mathbb{E}_{\nu_K} \mathsf{Y}^{\delta}_{\alpha}(\mathscr{T}^{\omega}_{j,j-1}(\theta_{j-1}, \theta_{j-1})) \leq W_j$, including the coarsest term and all drifts. In particular,

$$\mathbb{E}_{\nu_K} \mathsf{Y}^{\delta}_{\alpha}(\mathcal{Q}_\delta(r_j, r_j)) \leq W_j, \qquad \mathbb{E}_{\nu_K} \mathsf{Y}^{\delta}_{\alpha}(\mathcal{B}_\delta(x_{j-1}, r_j)) \leq 4 W_j.$$

   For every $0 < \eta < \min\{2\alpha - 1, -\log q_{\mathrm{anc}} / \log B\}$ there exists $C_{\gamma,\eta}$ such that $\sum_{j \geq k} W_j \leq C_{\gamma,\eta} a_k^\eta$. By the triangle inequality, the same sum bounds the expected norms of all finite partial sums over $j \geq k$, uniformly in K.

All constants may depend on the fixed coupling, the coarse side, the moment, the exponents, the horizon, and the fixed analytic and column bounds; they do not depend on K, on the number of preceding averaging steps, on i and j, on a/δ, or on the values of the coarser fields. These are estimates on the whole torus.

Proof. Throughout, we use the conditional law and the properties verified in Lemma 27.3; see (27.5), (27.6), and (27.7). The proof has five steps: a pointwise Lipschitz bound, a deterministic net, concentration, the telescope, and the summation.

Step 1: the Lipschitz constant of one evaluation. Let $\Lambda = K_{ji} g_i b_i$. For $t \le H^2$, by the bounds of Theorem 27.5(2) for small times and $\delta \le H$, we have

$$(t+\delta^2)^{\beta} \Lambda \log(1 + t/H^2) + (t+\delta^2)^{\alpha} \Lambda J_H(t) \le C \Lambda H^{2\beta}.$$

For $H^2 \le t \le G^2$, the two terms are at most $C\Lambda t^{\beta} \ell_H(T)$. For $t \ge G^2$, the late bounds, multiplied by their weights, are at most

$$C \Lambda G^{3/2} t^{\alpha - 5/4} \ell_H(T) \le C \Lambda G^{2\alpha - 1} \ell_H(T),$$

since $\alpha < 2/3 < 5/4$ and $t \ge G^2$. These three ranges show that every scalar weighted evaluation, of the value or of a first difference, is Lipschitz in w_i with constant at most $C_J A_{ji}$ on the whole product Ω_i. The sum of the two maxima bounds the norm $\mathsf{Y}_{\alpha}^{\delta}$, at the cost of a factor two. Translations and the finite number of colors and directions cost a fixed factor.

Step 2: a deterministic net. This step uses the bound b_i on the coarser fields rather than a distance between coefficient rows, so its constants depend on the fixed coupling. It suffices to consider $H \le \min(1, \sqrt{T})$; the other levels form a finite set, independent of the number of remaining steps. By (27.23) and heat contraction, the uncentered Duhamel term and its first three differences satisfy

$$\| D_{\delta}^{r} \mathscr{T}(t) \|_{\infty} \le C_{J,r} \mathfrak{A}_{ji}^2 b_i^2 t H^{-1-r}, \qquad 0 \le r \le 3, \tag{27.27}$$

and the corresponding time derivatives for $r = 0, 1$ are bounded by $C_{J,r} \mathfrak{A}_{ji}^2 b_i^2 (H^{-1-r} + tH^{-3-r})$. These bounds are deterministic on the whole product Ω_i. Conditional expectation preserves them, and centering costs a factor two.

For $0 \le t \le H^6$, both weights are bounded by their values at $2H^2$, up to fixed powers of two. The centered weighted term on this interval is at most $C \mathfrak{A}_{ji}^2 b_i^2 H^{6+2\alpha-2}$, and its ratio to A_{ji} is at most

$$C \frac{b_i}{g_i} H^4 (H/G)^{2\alpha - 1} \ell_H(T)^{-1} \le C_{\gamma} (1 + j) H^4, \tag{27.28}$$

by (27.25). The right side tends to zero geometrically in j, uniformly in i and in $j - i$. For all sufficiently fine levels it is less than $1/8$, and enlarging the constant handles the finitely many other levels. This uses the factor t of the Duhamel term, not a bound on a source at time zero.

On $[H^6, T]$ we consider the two scalar processes

$$(t+\delta^2)^{\beta}\big(\mathscr{T} - \mathbb{E}[\mathscr{T} \mid \mathcal{F}_{i-1}]\big)(t,x), \qquad (t+\delta^2)^{\alpha} D_{\delta}\big(\mathscr{T} - \mathbb{E}[\mathscr{T} \mid \mathcal{F}_{i-1}]\big)(t,x).$$

Their Lipschitz constants in physical space follow from one more output difference in (27.27). For their time derivatives we use the bounds following that display and the product rule for the weights. Differentiating a weight costs no power of the mesh: for $\zeta = \beta$ or $\zeta = \alpha$,

$$t(t+\delta^2)^{\zeta-1} \le (t+\delta^2)^{\zeta} \le C_{T,\zeta}.$$

Thus all these Lipschitz constants are bounded by $C_{T,J}\mathfrak{A}_{ji}^2 b_i^2 H^{-4}$. Dividing by A_{ji}, we obtain the bound

$$C_{T,J}\frac{b_i}{g_i} H^{-5} G^{-(2\alpha-1)} \le C_{\gamma}(1+j) H^{-5-(2\alpha-1)} \le C_{\gamma} H^{-7},$$

after enlarging C_{γ}, since $2\alpha - 1 < 1/3$ and $j = O(1 + |\log H|)$ for the fixed sequence of lattices. After another fixed adjustment, these inequalities also hold on the finite set of coarse levels.

We choose a spacing $\varepsilon_H = c_{\gamma} H^{12}$ in physical space and time, with $c_{\gamma} > 0$ small. If $\delta \ge \varepsilon_H$, we take every lattice site; there are at most $C_L \varepsilon_H^{-3}$ of them. If $\delta < \varepsilon_H$, we partition the torus into cubes of side between $\varepsilon_H/2$ and ε_H, and choose one lattice site in each occupied cube. Every lattice site can then be joined to its chosen site by a coordinate lattice path of length at most $C\varepsilon_H$. The bound on physical differences applies along these paths, although the net points need not form a lattice. We add a time net of spacing at most ε_H, including both endpoints. The total number of points is at most

$$C_{\gamma,L,T}(1 + L/\varepsilon_H)^3 (1 + T/\varepsilon_H) \le C_{\gamma,L,T,a_0} H^{-48}. \tag{27.29}$$

After the choice of c_{γ}, the error of interpolation for the centered weighted processes is at most $A_{ji}/8$, and so is the contribution of short times. The finite numbers of components are included in the constant.

Step 3: concentration and conditional moments. Consider an evaluation F at a point of the net. The coarser field is fixed, the domain is the same finite product of balls, and F is a smooth real function that is Lipschitz for the Euclidean distance on a neighborhood of the product. We may apply (27.7) with Lipschitz constant at most $C_J A_{ji}$; Chernoff's inequality with positive and negative z then gives

$$\mathbb{P}\big\{|F - \mathbb{E}[F \mid \mathcal{F}_{i-1}]| > z A_{ji} \;\big|\; \mathcal{F}_{i-1}\big\} \le 2e^{-c_J z^2}.$$

We apply the finite-family clause of Lemma 16.4 with $s = C_J A_{ji}$, $b = 0$, and deterministic error $e \le C A_{ji}$. The logarithm of the number of points is at most $C_{\gamma} \log(2 + L/H)$,

which implies the tail of (1). The moment clause yields all $p \geq 1$, and for $0 < p < 1$ Jensen's inequality gives $\|M\|_p \leq \mathbb{E}M$. The moments in (1) then follow from deterministic comparisons between H, G, and j. By (27.22), $\ell_H(T) \leq C_{T,a_0,B}(1+j)$, and $\log(2+L/H) \leq C(1+j)\log(2+L/G)$. Inserting $K_{ji} = q_{\mathrm{anc}}^{j-1-i}/G$ and absorbing these polynomial factors, we obtain the moment bound of (1) with a fixed finite c independent of K, j, and i. The constant $C_{\gamma,p}$ may absorb $C_{T,a_0,B}$, because L and n are fixed and $a_0 = L/n$.

Step 4: the three orders and the telescope. At $i = j-1$ the product of linearizations is empty, and

$$(\mathsf{U} - \mathsf{V})\theta_{j-1} = H^{-1}P_\cdot^\delta I_H \theta_{j-1} = x_{j-1}, \qquad \mathsf{V}\theta_{j-1} = r_j.$$

For any bilinear map $\mathcal{N}$,

$$\begin{aligned}\mathcal{N}(U-V,V) + \mathcal{N}(V,U-V) &= \mathcal{N}(U,V) + \mathcal{N}(V,U) - 2\mathcal{N}(V,V),\\ \mathcal{N}(U,U) - \mathcal{N}(U-V,U-V) &= \mathcal{N}(U,V) + \mathcal{N}(V,U) - \mathcal{N}(V,V).\end{aligned}$$

Integrating against the same outer heat semigroup gives the identities of (2), including the coefficient -1 for the full difference.

For each order separately, we have the telescoping identity

$$\mathscr{T}^\omega_{j,j-1}(\theta_{j-1},\theta_{j-1}) = \mathscr{T}^\omega_{j,0}(\theta_0,\theta_0) + \sum_{i=1}^{j-1}\big(Z^\omega_{j,i} + e^\omega_{j,i}\big), \tag{27.30}$$

since $Z^\omega_{j,i} + e^\omega_{j,i} = \mathscr{T}^\omega_{j,i}(\theta_i,\theta_i) - \mathscr{T}^\omega_{j,i-1}(\theta_{i-1},\theta_{i-1})$. Each summand uses the conditional expectation $\mathbb{E}_{\nu_K}[\,\cdot \mid \mathcal{F}_{i-1}]$. We take the norm and the expectation, and use the moment bound of (1) with $p = 1$ and the bounds on drifts and on the coarsest term of Theorem 27.5(4). For sufficiently large fixed C_γ and c, we obtain W_j for each order. Here C_γ includes the dependence on a_0 of the weighted bounds on drifts and on the coarsest term. By the identities of (2) and the triangle inequality, we then obtain the bounds W_j and $4W_j$ for $\mathcal{Q}_\delta(r_j,r_j)$ and $\mathcal{B}_\delta(x_{j-1},r_j)$.

Step 5: the tail of the convolution. Let $r_* = 2\alpha - 1 \in (0,1/3)$ and $\lambda_q = -\log q_{\mathrm{anc}}/\log B > 0$. By (27.25), the centered summand of W_j without the factor q_{anc}^m is at most $C_\gamma(1+i)^d a_i^{r_*}$ for a fixed finite d; here $\sqrt{\log(2+L/a_i)} \leq C\sqrt{1+i}$. By (27.26), the summand of the drifts is at most $C_\gamma(1+i)^3 a_i^{1/2}$, and hence also satisfies this weaker bound, after adjusting a constant depending on a_0. The coarsest term is the term $i = 0$, up to a constant depending on a_0. Lemma 27.2(2), with $r = r_*$ and the polynomial factor $(1+j)^{c+d}$, implies that

$$\sum_{j\geq k} W_j \leq C_{\gamma,\eta} a_k^\eta$$

for every $0 < \eta < \min(r_*, \lambda_q)$. For any finite partial sum over $j \geq k$, the triangle inequality bounds its expectation by the corresponding part of this series, uniformly in K. At each

finite K the fields and their differences are continuous in t and have finitely many spatial components, so their suprema are determined by a countable dense set of times and are measurable. This completes the proof, under the conditioned Wilson measure ν_K. □

27.4 The predictable term on the finest lattice

The predictable term of Lemma 26.2 is $u_j = P_\cdot^\delta I_a U_j^0$, with

$$U_j^0 = F_j + H^{-1}\overline{\mathsf{R}}_j(\theta_{j-1}) + m_j^{\mathrm{ph}}, \qquad F_j = H^{-1}\overline{\mathsf{Q}}_j(\theta_{j-1},\theta_{j-1}).$$

In the notation of the introduction, $U_j^{\mathrm{cl}} = F_j + H^{-1}\overline{\mathsf{R}}_j(\theta_{j-1})$ and $U_j^0 = U_j^{\mathrm{cl}} + m_j^{\mathrm{ph}}$. We bound u_j in the norm H_σ^δ of its initial field, which controls both the value and the first difference of the heat evolution. The quadratic part F_j is treated by the same propagation and telescope as in Sections 27.2 and 27.3, now on the lattice of mesh a and without the outer Duhamel integral. The cubic part is bounded deterministically, and the conditional mean correction m_j^{ph} is bounded directly, with an extra factor of the coupling, from the derivative formula for normalized expectations.

Theorem 27.7. *Assume the setting of Section 26.1 and the notation of Theorem 27.5, and let $0 < \sigma < 1 - \alpha$. For $0 \le i \le j-1$ let*

$$\mathsf{Q}_{j,i}(v,z) = H^{-1}\overline{\mathsf{Q}}_j(\mathsf{T}_{ji}v, \mathsf{T}_{ji}z),$$

the quadratic coefficient of F_j propagated from scale i, including the factor $1/H$, on the lattice of mesh $a = a_j$.

1. *The absolute row sum and both volume-normalized column sums with a fixed input of $\mathsf{Q}_{j,i}$ satisfy $(\mathsf{R}_2 + \mathsf{C}_2)(\mathsf{Q}_{j,i}) \le CK_{ji} = Cq_{\mathrm{anc}}^{j-1-i}/a_i$. For $1 \le i \le j-1$ and $G = a_i$, the derivative in w_i after interpolation satisfies*

$$\sup_x \|D_{w_i} P_t^\delta I_a \mathsf{Q}_{j,i}(\theta_i,\theta_i)(x)\|_2 \le C_J K_{ji} g_i b_i \min\{1, G^{3/2}(t+G^2)^{-3/4}\}.$$

   After multiplication by $(t+\delta^2)^{\sigma/2}$ its supremum is at most $C_{T,J}K_{ji}g_i b_i G^\sigma$, for every $0 < \sigma < 3/2$. For every fixed pair of nonnegative integers r, v, the crude bounds

$$\sup_{w_i,t,x} \|D_{w_i}\partial_t^v D_\delta^r P_t^\delta I_a \mathsf{Q}_{j,i}(\theta_i,\theta_i)(x)\|_2 \le C_{J,r,v} K_{ji} g_i b_i a^{-r-2v}$$

   hold. The constant mode of the heat kernel is kept.

2. *For the conditionally centered term $Z_{j,i}^{\mathrm{st}} = \mathsf{Q}_{j,i}(\theta_i,\theta_i) - \mathbb{E}[\mathsf{Q}_{j,i}(\theta_i,\theta_i) \mid \mathcal{F}_{i-1}]$, the conditional logarithmic Sobolev inequality and a net at scale a give, for every $1 \le p < \infty$,*

$$\big\|\mathsf{V}_\sigma^\delta(I_a Z_{j,i}^{\mathrm{st}})\big\|_{L^p(\cdot|\mathcal{F}_{i-1})} \le C_{\gamma,p,J} K_{ji} g_i b_i G^\sigma \big[\sqrt{\log(2 + L/a)} + \sqrt{p}\big].$$

3. *For the conditional drift* $e^{\mathrm{st}}_{j,i} = \mathbb{E}[\mathsf{Q}_{j,i}(\theta_i,\theta_i) \mid \mathcal{F}_{i-1}] - \mathsf{Q}_{j,i-1}(\theta_{i-1},\theta_{i-1})$ *and for the coarsest term,*

$$\mathsf{V}^{\delta}_{\sigma}(I_a e^{\mathrm{st}}_{j,i}) \le C_{T,J}\frac{q_{\mathrm{anc}}^{j-1-i}}{a_i}\big(a_{\mathrm{cap}}^2 b_{i-1}^3 + g_i^2 b_{i-1} + g_i^3 p_i^3\big),$$
$$\mathsf{V}^{\delta}_{\sigma}(I_a \mathsf{Q}_{j,0}(\theta_0,\theta_0)) \le C_{T,J}\rho_0^2 q_{\mathrm{anc}}^{j-1}/a_0.$$

All three contributions to the drift are present. The drift has the same exact splitting into linearization error and fluctuation correction as in Theorem 27.5(4), with the factor $(1+j)K_{ji}$ *there replaced by* $C_{T,J}K_{ji}$ *in the bounds in* $\mathsf{V}^{\delta}_{\sigma}$*. In the bounds on derivatives in* w_i*,* b_i *may be replaced by any bound valid on the whole set* Ω_i *at the fixed coarser field.*

4. *For every array* f *on the lattice of mesh* δ*, including the times* $0 \le t \le \delta^2$*,*

$$\mathsf{H}^{\delta}_{\sigma}(f) \le C_{L,T,\sigma}\mathsf{V}^{\delta}_{\sigma}(f), \qquad \mathsf{X}^{\delta}_{\alpha}(P^{\delta}_{\cdot} f) \le C_{L,T,\alpha,\sigma}\mathsf{H}^{\delta}_{\sigma}(f).$$

5. *The cubic and mean terms of* U_j^0 *satisfy* $\|\overline{\mathsf{R}}_j/H\|_\infty \le C_J b_{j-1}^3/H$ *and* $\|m_j^{\mathrm{ph}}\|_\infty \le C_J g_3^2 b_{j-1}$*. For every* $p \in [1,\infty)$ *and every*

$$0 < \tau < \min\{\sigma, \tfrac{1}{2}, -\log q_{\mathrm{anc}}/\log B\},$$

there exist a finite exponent c *and a constant* $C_{\gamma,p,J,\tau}$ *such that*

$$\sup_{K\ge j}\Big\{\big\|\mathsf{H}^{\delta}_{\sigma}(I_a U_j^0)\big\|_{L^p(\nu_K)} + \big\|\mathsf{X}^{\delta}_{\alpha}(u_j)\big\|_{L^p(\nu_K)}\Big\} \le C_{\gamma,p,J,\tau}(1+j)^c a_{j-1}^{\tau}.$$

In particular,

$$\sup_K \sum_{j=k}^{K} \mathbb{E}_{\nu_K}\mathsf{H}^{\delta}_{\sigma}(I_{a_j}U_j^0) \le C_{\gamma,J,\tau}(1+k)^c a_k^{\tau},$$

and $\mathsf{H}^{\delta}_{\sigma}(I_a U_j^0) \le C_{\gamma,J}(1+j)^c$ *on the small-field event.*

The constants in the bounds on rows and means depend on L*,* T*,* M*,* J*, and the fixed analytic and column bounds; they do not depend on the coupling,* K*,* i*,* j*, the number of preceding averaging steps, or* a/δ*. The constants in the maximal and summed bounds may in addition depend on the fixed* γ*,* n*,* p*,* σ*,* τ*, and the prescription of the cutoff radii. No maximal estimate on regions or uniform in the coupling is asserted.*

Proof. All computations take place on the common lattice of mesh δ, and all uses of conditional concentration are for the kernel of Lemma 27.3. The conditional mean correction is identified below with the nonlinear mean of the reconstruction maps.

Step 1: the propagated quadratic coefficient. The quadratic Taylor coefficient of the zero-fluctuation reconstruction, in the normalization of (26.3), is

$$\overline{\mathsf{Q}}_j(v,z) = \frac{B}{2} D_\theta^2 \mathscr{R}_j(0,0)[v,z],$$

and its coefficient bounds are given by Lemma 17.2. Propagating both arguments through each L_i, we obtain

$$\mathsf{Q}_{j,i}(v,z) = H^{-1}\overline{\mathsf{Q}}_j(\mathsf{T}_{ji}v, \mathsf{T}_{ji}z), \qquad \mathsf{Q}_{j,i-1}(v,z) = \mathsf{Q}_{j,i}(\mathsf{L}_i v, \mathsf{L}_i z).$$

Here $\overline{\mathsf{Q}}_j$ is the symmetric bilinear Taylor coefficient just defined, while the sources of Section 27.2 keep their order. A row acquires the factor a_{cap}^2 at each step. For a column with a fixed input, the argument with the fixed input uses the volume-normalized column of L_i, and the other argument uses its row. The cell volume of the intermediate index then cancels, as in Lemma 27.1(1). Thus

$$(\mathsf{R}_2 + \mathsf{C}_2)(\mathsf{Q}_{j,i}) \le C\frac{a_{\mathrm{cap}}^{2m}}{H} = C\frac{q_{\mathrm{anc}}^{m}}{G} = CK_{ji}. \tag{27.31}$$

The same bound on the propagated coefficient, including its weighted version before heat evolution, is part of Theorem 17.4. Interpolation from the lattice of mesh a by I_a preserves these column bounds, since each column of I_a has physical mass exactly a^3.

Step 2: rows of derivatives and crude bounds. By (27.2), at any point of Ω_i, the derivative in w_i of the two arguments has absolute row and volume-normalized column at most $C_J K_{ji} g_i b_i$. For $t \le G^2$, the absolute row controls the counting ℓ^2 row. For $t \ge G^2$ we split the heat semigroup into two equal halves. For the first half, the Schur bound gives counting operator norm at most $C_J K_{ji} g_i b_i (G/\delta)^{3/2}$, and the second half has counting row norm at most $C\delta^{3/2} t^{-3/4}$. Their product is $C_J K_{ji} g_i b_i G^{3/2} t^{-3/4}$. The bounds for small and for late times give the profile with $t + G^2$ in (1), since this quantity is comparable to G^2 at small times and to t at late times.

For $0 < \sigma < 3/2$, the weighted supremum follows by splitting at G^2. Below that time the weight is at most CG^σ, because $\delta \le G$. Above it the profile gives

$$CG^{3/2} t^{\sigma/2 - 3/4} \le CG^\sigma.$$

For fixed r and v we commute all r differences and the v Laplacians through the heat semigroup and apply them to I_a before smoothing. Using the bound on differences of order $r + 2v$ of the cardinal interpolation and heat contraction, we obtain $C_{J,r,v} K_{ji} g_i b_i a^{-r-2v}$, which proves the crude bounds uniformly on Ω_i.

Step 3: the centered term. We write $Z = Z^{\mathrm{st}}_{j,i}$ and $S_i = K_{ji} g_i b_i G^\sigma$. Since the interpolation is positive and preserves constants, heat contraction gives, for $0 \le t \le a^2$,

$$\sup_{t \le a^2} (t+\delta^2)^{\sigma/2} \| P^\delta_t I_a Z \|_\infty \le C a^\sigma \max_y |Z(y)|, \tag{27.32}$$

where the maximum is over the sites y of the lattice of mesh a. Each scalar output $Z(y)$ is Lipschitz in w_i with constant at most $C_J K_{ji} g_i b_i$. By the conditional logarithmic Sobolev inequality for these finitely many outputs and a union bound over at most $C(L/a)^3$ of them, we control (27.32) at the scale $C_J K_{ji} g_i b_i a^\sigma [\sqrt{\log(2 + L/a)} + \sqrt{u}]$, which is at most the claimed scale since $a \le G$.

On $[a^2, T]$ the uncentered term has deterministic amplitude at most $C K_{ji} b_i^2$, and its interpolation and heat evolution have fixed derivatives in space and time bounded by $C_{J,r,v} K_{ji} b_i^2 a^{-r-2v}$. For $a \le 1$, the weighted process of values has Lipschitz constants in physical space and time at most $C_{T,J} K_{ji} b_i^2 a^{-2}$. For the time derivative of the weight we use $(t+\delta^2)^{\sigma/2-1} \le C_{T,\sigma} a^{-2}$ on this interval. Centering costs a factor two. Dividing these constants by S_i, we find

$$C_{T,J} \frac{b_i}{g_i} a^{-2} G^{-\sigma} \le C_\gamma (1+i) a^{-2-\sigma} \le C_\gamma a^{-4},$$

where we used (27.25), $i \le j$, the fixed sequence of lattices, and $\sigma < 1 - \alpha < 1/2$. A net in space and time of spacing $c_\gamma a^8$ therefore approximates every centered weighted value within $S_i/8$, and it has at most $C_{\gamma,L,T} a^{-32}$ points. To choose lattice sites when this spacing is smaller or larger than δ, we use the two constructions of Step 2 of the proof of Theorem 27.6, with H replaced by a and the exponent 12 by 8. The lengths of coordinate paths and the count follow from the same partition into cubes. If $a^2 > T$, the interval of small times covers all times, and the finitely many levels with $a > 1$ are included in the constant.

By the bounds of Step 2, at each point of the net, the weighted evaluation is Lipschitz in w_i with constant $C_J S_i$. We use the counts of entries on the lattice of mesh a and of the net, and the deterministic errors of interpolation above. The finite-family clause of Lemma 16.4 then implies the tail and the moments of (2). This net uses b_i/g_i, so it holds at fixed coupling. All bounds on rows remain valid with a smaller bound b_i that holds on the whole set Ω_i.

Step 4: the drifts and the coarsest term. The telescoping identity is

$$F_j = \mathrm{Q}_{j,0}(\theta_0, \theta_0) + \sum_{i=1}^{j-1} \big(Z^{\mathrm{st}}_{j,i} + e^{\mathrm{st}}_{j,i} \big). \tag{27.33}$$

Before interpolation, each propagated coefficient belongs to the weighted class of Theorem 17.4. By (27.31), its coefficient norm, including $1/H$, is bounded by $C K_{ji}$; we use

that theorem for its coefficient bound and its telescope. To obtain a constant uniform in the coupling, we instead apply Lemma 27.4 to each scalar output of $\mathsf{Q}_{j,i}$ on the lattice of mesh a, with $k_Q \le CK_{ji}$ and $\phi = \theta_{i-1}$. The propagated coefficients are equivariant under global color rotations, since the zero-fluctuation reconstruction and its linearizations are. The conditional law, the coefficient bounds, and the covariance are those of Lemma 27.3, see (27.5) and (27.10). We obtain

$$\|e^{\mathrm{st}}_{j,i}\|_\infty \le C_J K_{ji}\big(a_{\mathrm{cap}}^2 b_{i-1}^3 + g_i^2 b_{i-1} + g_i^3 p_i^3\big). \tag{27.34}$$

This bounds the whole drift with a constant uniform in the coupling, as in the computation of Theorem 27.5. Applying (27.12) and (27.13) separately gives the splitting into linearization error and fluctuation correction, with all three contributions kept. The output is still on the lattice of mesh a at this step. Using the contraction of the interpolation and of the heat semigroup, together with $\sup_{t\le T}(t+\delta^2)^{\sigma/2} \le C_{L,T,\sigma}$, we obtain the bounds in V^δ_σ of (3). The coarsest term is bounded by substituting $\|\theta_0\|_\infty \le \rho_0$ into (27.31) and applying the same contractions. These bounds do not contain the factor $1+j$ of Theorem 27.5.

Step 5: the first difference, including the smallest times. For every array f on the lattice of mesh δ and $t \ge \delta^2$, we split the heat semigroup at $t/2$ and use Lemma 16.1:

$$\|D_\delta P^\delta_t f\|_\infty \le C(t/2+\delta^2)^{-1/2}\|P^\delta_{t/2} f\|_\infty \le C(t+\delta^2)^{-(\sigma+1)/2}\mathsf{V}^\delta_\sigma(f).$$

For $0 \le t \le \delta^2$, the value term at time zero gives $\|f\|_\infty \le \delta^{-\sigma}\mathsf{V}^\delta_\sigma(f)$, and writing a difference as the difference of two contractions gives

$$\|D_\delta P^\delta_t f\|_\infty \le 2\delta^{-1}\|f\|_\infty \le 2\delta^{-\sigma-1}\mathsf{V}^\delta_\sigma(f).$$

Multiplication by $(t+\delta^2)^{(\sigma+1)/2}$ makes this bound uniform also at $t=0$. Including the value term, we obtain $\mathsf{H}^\delta_\sigma \le C\mathsf{V}^\delta_\sigma$. Since $\alpha > \sigma$, the two components of $\mathsf{X}^\delta_\alpha(P^\delta_\cdot f)$ are those of $\mathsf{H}^\delta_\sigma(f)$ multiplied by $(t+\delta^2)^{(\alpha-\sigma)/2}$, which is at most $[T+(L/4)^2]^{(\alpha-\sigma)/2}$ on the lattices considered. This proves the second comparison of (4), with the stated dependence on L, T, α, and σ. In particular, for a bounded array,

$$\mathsf{H}^\delta_\sigma(f) \le C_{L,T,\sigma}\|f\|_\infty. \tag{27.35}$$

Step 6: the rate of the quadratic part. Applying the conditional and then the unconditional Minkowski inequality to (27.33) for fixed $p \ge 1$, we obtain, with

$$v_i = \frac{a_{\mathrm{cap}}^2 b_{i-1}^3 + g_i^2 b_{i-1} + g_i^3 p_i^3}{a_i},$$

the bound

$$\begin{aligned}\|\mathsf{V}_\sigma^\delta(I_a F_j)\|_{L^p(\nu_K)} \le C_{\gamma,p,J}\Big\{ &\rho_0^2 q_{\mathrm{anc}}^{j-1}/a_0 + \sum_{i=1}^{j-1} q_{\mathrm{anc}}^{j-1-i} v_i \\ &+ \big\{\sqrt{\log(2+L/a)} + \sqrt{p}\big\} \sum_{i=1}^{j-1} q_{\mathrm{anc}}^{j-1-i} g_i b_i a_i^{\sigma-1}\Big\}.\end{aligned} \tag{27.36}$$

The factor in the last sum satisfies

$$g_i b_i a_i^{\sigma-1} \le C_J\big\{g_3\rho_0 a_0^{-\kappa_{\mathrm{term}}} a_i^{\kappa_{\mathrm{term}}+\sigma-1/2} + g_3^2 a_i^\sigma p_i\big\} \le C_\gamma(1+i)a_i^\sigma,$$

since $\kappa_{\mathrm{term}} > 1/2$, and v_i is bounded by (27.26), with rate $a_i^{1/2}$. Also $\log(2+L/a) \le C(1+j)$. For fixed $\tau < \min(\sigma, 1/2, \lambda_q)$ we choose ζ strictly between τ and this minimum, and apply Lemma 27.2(1) with $r = \sigma$ and with $r = 1/2$. The polynomial factors in i are bounded by powers of $1+j$, and the coarsest term is the term $i = 0$. We obtain

$$\sup_{K\ge j} \|\mathsf{H}_\sigma^\delta(I_a F_j)\|_{L^p(\nu_K)} \le C_{\gamma,p,J,\tau}(1+j)^c a_{j-1}^\tau, \tag{27.37}$$

where the comparison of Step 5 has replaced V_σ^δ by H_σ^δ, so the first difference needs no further probability estimate.

Step 7: the cubic term and the conditional mean. The cubic bound follows from the Taylor remainder of the zero-fluctuation reconstruction in Lemma 17.2, normalized as in (26.3):

$$\|\overline{\mathsf{R}}_j/H\|_\infty \le C_J b_{j-1}^3/H \le C_\gamma(1+j)^3 H^{1/2},$$

where the last inequality is (27.25) on the coarser lattice. For the conditional mean correction, we take in Lemma 17.1 the coarse mesh $a_0 = L/n$ and total number K of renormalization group steps. Its numbers of preceding averaging steps are $J(K-j)$, its reconstruction maps are those used here, and its filtration is $\mathcal{F}_j$. Its lattice potentials and zero-fluctuation reconstructions are therefore

$$A_j = a_j^{-1}\theta_j, \qquad C_j = a_j^{-1}\mathscr{R}_j(\theta_{j-1}, 0),$$

and its conditional mean correction $\mathbb{E}[A_j \mid \mathcal{F}_{j-1}] - C_j$ is m_j^{ph} of Lemma 26.2. This identifies the conditional mean correction, including its conditional law.

We prove the uniform bound directly from the derivative formula for normalized expectations, so that no stochastic constant depending on the fixed coarse mesh enters this deterministic bound. Fix $\phi = \theta_{j-1}$, write $r_\phi = \|\phi\|_\infty$, and set

$$\mathscr{G}_\phi(w) = \mathscr{R}_j(\phi, w) - \mathscr{R}_j(\phi, 0).$$

For $0 \le u \le 1$ let $\mathbb{E}_u$, Cov_u denote expectation and covariance under the conditional fluctuation kernel at $u\phi$. The segment of coarser fields and every point of the free coordinates remain in the same domain, as checked in Lemma 27.3, and the product of balls does not change. The bound on the mixed derivative of the potential in Theorem 13.10 and the covariance inequality of Theorem 13.8 apply. The majorant series of the joint Taylor coefficients used in (27.2) gives deterministic envelopes $a_{e,l} \ge 0$ such that, on the whole product and the whole segment, $|\partial_{w_l} \mathscr{G}_{u\phi,e}| \le a_{e,l}$ and $\sum_l a_{e,l} \le C_J t_j$. In particular $\sum_l \|\partial_{w_l} \mathscr{G}_{u\phi,e}\|_{L^2} \le C_J t_j$; this uses the absolute coefficient majorant, rather than an exchange of a supremum with a sum of L^2 norms. The bound on the mixed derivative of the potential is $\sup_l \|\partial_{w_l} \dot{V}_u\|_{L^2} \le C_J t_j r_\phi$, where $\dot{V}_u = \partial_u V_{u\phi}$. The covariance kernel decays exponentially and has a uniformly bounded sum at each index, in the units of the renormalization group step, because the number of fluctuation coordinates in each cell is bounded in terms of J. The covariance inequality then implies that

$$|\mathrm{Cov}_u(\mathscr{G}_{u\phi,e}, \dot{V}_u)| \le C_J t_j^2 r_\phi. \tag{27.38}$$

To see that no volume factor appears, we sum the covariance kernel over its second index first, using the uniform bound on the mixed derivative there, and then sum the absolute row of derivatives at the fixed output e.

The expectation of $\partial_u \mathscr{G}_{u\phi,e}$ has the same bound. Indeed, its series of absolute joint Taylor coefficients has positive degree in the free coordinates, and one insertion of the coarser field costs $C_J r_\phi$. We fix a radius $R_{\mathrm{free}} > 0$ in the free coordinates from the nested domains, with the segment of coarser fields in its inner domain and a fixed positive margin above every cutoff domain. The differentiated coefficient sum at this radius is bounded by $C_J r_\phi$, and $t_j p_j / R_{\mathrm{free}} \le \eta_* < 1$, with a fixed margin. For the term of degree one we use the bound $\|\mathbb{E}_u w\|_\infty \le C_J t_j r_\phi$ of Theorem 13.10; its expectation costs $C_J t_j^2 r_\phi^2$. For degree $k \ge 2$ we put two factors in L^2 and bound the other $k-2$ by the cutoff p_j. The L^2 factors are bounded uniformly: the Poincaré inequality and the bound on the mean give $\mathbb{E}_u |w_l|^2 \le 2 + |\mathbb{E}_u w_l|^2 \le C_J$ for scalar coordinates. After division by the radius of its coefficients, the term of degree k costs at most $C_J t_j^2 R_{\mathrm{free}}^{-2} \eta_*^{k-2}$ times its absolute coefficient mass at R_{free}. Summing this mass, and using the fixed bound on r_ϕ, we obtain

$$|\mathbb{E}_u \partial_u \mathscr{G}_{u\phi,e}| \le C_J t_j^2 r_\phi. \tag{27.39}$$

Repeated coordinates are handled by the same Cauchy–Schwarz inequality for two factors; no independence is used.

The derivative formula for normalized expectations of Theorem 13.10 implies that

$$\frac{d}{du} \mathbb{E}_u \mathscr{G}_{u\phi,e} = \mathbb{E}_u \partial_u \mathscr{G}_{u\phi,e} - \mathrm{Cov}_u(\mathscr{G}_{u\phi,e}, \dot{V}_u).$$

Its hypotheses hold because these maps and their derivatives are continuous on a neighborhood of the fixed compact product, so all difference quotients in finite dimensions

are dominated there. At $u = 0$, the conditional law at the zero coarser field and the reconstruction map are equivariant under global color rotations, so $\mathbb{E}_0 \mathscr{G}_{0,e} = 0$. Integrating (27.38) and (27.39), we obtain the bound on the nonlinear mean of the reconstruction,

$$\|\mathbb{E}_\phi \mathscr{R}_j(\phi, w) - \mathscr{R}_j(\phi, 0)\|_\infty \le C_J t_j^2 \|\phi\|_\infty. \tag{27.40}$$

This is the bound of Theorem 13.10 on the mean of the reconstruction, proved here for the reconstruction maps of this section.

Dividing by the mesh a_j of the reconstructed field keeps the factor of the intermediate scales:

$$\|m_j^{\mathrm{ph}}\|_\infty \le C_J \frac{t_j^2}{a_j} \|\theta_{j-1}\|_\infty = C_J M^{J-1} g_3^2 \|\theta_{j-1}\|_\infty \le C_J g_3^2 b_{j-1}, \tag{27.41}$$

where the last C_J absorbs only the fixed factor M^{J-1}. All constants in this pointwise bound on the mean are uniform in the lattice side, in the number of preceding averaging steps, and in the coupling below g_*. By this bound and (27.25), we have $\|m_j^{\mathrm{ph}}\|_\infty \le C_\gamma (1+j) H^{1/2}$.

Step 8: combination and summation. Contraction of the interpolation and (27.35) bound the contributions of the cubic and mean terms to $\mathsf{H}_\sigma^\delta(I_a U_j^0)$ by $C_\gamma (1+j)^3 H^{1/2}$, pathwise and hence in every L^p. Combining this with (27.37), and using $\tau < 1/2$, we obtain the bound in L^p of $\mathsf{H}_\sigma^\delta(I_a U_j^0)$ in (5). The comparison $\mathsf{X}_\alpha^\delta(P_\cdot^\delta f) \le C \mathsf{H}_\sigma^\delta(f)$ of (4) then implies the bound for u_j under the same law. For the sum we use the bound with $p = 1$ and

$$\sum_{j \ge k} (1+j)^c a_{j-1}^\tau \le B^\tau a_k^\tau (1+k)^c \sum_{l \ge 0} (1+l)^c B^{-\tau l} \le C_{B,\tau,c} (1+k)^c a_k^\tau.$$

The finite sum under ν_K is bounded by this deterministic infinite series, uniformly in K. Finally, by the deterministic bounds on the small-field event, we have

$$\|U_j^0\|_\infty \le C_J \{ b_{j-1}^2 / H + b_{j-1}^3 / H + g_3^2 b_{j-1} \} \le C_\gamma (1+j)^3,$$

and (27.35) gives the pathwise bound of (5). Only the maximal inequalities and the summations use constants depending on the fixed γ and n. □

Remark 27.8. These are the norms needed for the bilinear estimate of Lemma 24.6. If $x = P_\cdot^\delta f$, then x and $u_j = P_\cdot^\delta I_a U_j^0$ are heat evolutions on the same lattice, $\alpha \in (1/2, 2/3)$, and $\sigma \in (0, 1-\alpha)$, so that lemma applies in both orders with the norm H_σ^δ bounded in Theorem 27.7. Its mesh is $\delta = L/(nB^K)$, with integer side $nB^K \ge 4$. If $\delta \le 1$ its clause for the heat semigroup applies, and if $\delta > 1$ its clause for finitely many coarse sides applies, with a constant depending on the fixed L and T.

SECTION 28

Linear and quadratic flow data and convergence of Wilson loops

In this section we prove that the joint law of finitely many flowed Wilson loops under the Wilson measure $\mu_{N_K,\gamma}$ converges along the lattice sides $N_K = n_* B^K$, $B = M^J$, at fixed coupling γ (Theorem 28.2). This is part (1) of Theorem 2.3 along the subsequences of Remark 2.4. The passage to all lattice sides nM^k and the construction of the continuum measure on the trajectory space are carried out in Section 29.

The proof has three steps.

1. *Truncation.* Fix a coarse side n and condition $\mu_{nB^K,\gamma}$ on the small-field event. Under the conditioned Wilson measure the fine field is obtained from the coarse field by a coarse-to-fine reconstruction, one RG step at a time. We compare the linear and quadratic flow data (x, y) of the fine field with those of the finite-scale reconstruction from the first q RG steps, interpolated to the fine lattice. The difference tends to zero in probability as $q \to \infty$, uniformly in K, and the norms of the flow data are tight (Theorem 28.8).
2. *Comparison with smooth flows.* For fixed q the finite-scale reconstruction is a random field on a fixed lattice, with compact support, and its law converges as $K \to \infty$ (Theorem 15.13). Its smooth cardinal interpolant is a smooth initial datum, and the lattice Wilson flow started from it converges to the Yang–Mills heat flow (Lemma 28.4). On bounded sets of flow data, flowed Wilson loops are Lipschitz functions of the flow data, with constants that do not depend on q (Corollary 25.11). We choose first a bound R on the norm of the flow data, then a tolerance η, then q, and finally let $K \to \infty$. This gives convergence of the laws of flowed Wilson loops under the conditioned measure (Theorem 28.1).
3. *Removal of the conditioning.* At a fixed lattice side, conditioning on an event changes the expectation of a bounded function by at most twice the probability of the complement (Lemma 5.5). By Theorem 23.9, the small-field failure probability is bounded uniformly in K by a quantity that tends to zero as the coarse side grows. We let first $K \to \infty$ at a fixed coarse side $n_r = n_* B^r$, and then $r \to \infty$.

The main estimate behind the first step is Lemma 28.5. Each increment d_j of the linear heat evolution is split into four terms (Lemma 26.2): the centered linear fluctuation term s_j, the centered nonlinear heat term z_j, the interpolation defect r_j, and the predictable

term u_j. The increment of the quadratic Picard term consists of a cross term with the previous linear heat evolution and of the sixteen ordered products of these four terms. We bound the expected norm of each ordered product separately, keeping the conditional means of these products, which do not vanish. Since the bilinear Duhamel term is not symmetric, the two orders of each product are treated separately. The stochastic inputs are the estimates of Sections 26 and 27 for the conditional fluctuation kernel. The deterministic inputs are the ordered product inequality Lemma 24.6 and the consistency and stability results of Sections 24 and 25.

The strategy of separating a rough linear evolution from a smoother nonlinear part goes back to Bourgain [32] and to Da Prato and Debussche [63]. Cao and Chatterjee used probabilistic smoothing in this form for the Yang–Mills heat flow with random initial data [42, Section 1.1]. Here the quadratic Picard term is the ordered bilinear Duhamel term of the damped Wilson flow of Section 24, and the stochastic estimates concern the conditioned Wilson measure; no Gaussian hypothesis on the initial data is used. The lattice flow is the Wilson flow of Lüscher [128], in the physical time normalization of Section 2.1.

Section 29 uses Theorems 28.1, 28.2, and 28.8 and the approximation Lemma 28.9 to construct the continuum measure, and Lemma 28.11 to work with the parameters chosen below an arbitrary finite bound on the coupling. Sections 29 and 32 use Lemma 28.4 and the estimates of Lemma 28.10, which are uniform for couplings in a compact subinterval of $(0,\infty)$. Section 34 uses the bounds on the ordered products in Lemma 28.5.

The stochastic truncation estimates are stated at a fixed coarse side and, except in Lemma 28.10, at a fixed coupling. The convergence results for observables concern finitely many Wilson loops. We do not construct a limit of the flow data themselves as random fields, nor a solution map of the flow for rough initial data, and we do not claim a rate of convergence.

Section 28.1 fixes the notation and states the two main theorems. Section 28.2 collects deterministic estimates, including the comparison with smooth flows. Section 28.3 contains the four-term decomposition and the bounds on the ordered products, and Section 28.4 the truncation estimates. The two main theorems are proved in Sections 28.5 and 28.6. Section 28.7 treats compact intervals of couplings and couplings below a finite bound.

28.1 Setting and main results

Standing assumptions. Fix $L > 0$, an odd integer $M \geq 3$, and $0 < \gamma_0 \leq 1$, and let J, $B = M^J$, the tuple Θ of remaining parameters, g_*, and n_0 be as in Theorem 23.2; these are the parameters used in Theorem 23.9. Fix $0 < \gamma \leq \gamma_0$, set $g_3^2 = \gamma / L$, and fix a coarse

side $n \in 4\mathbb{N}$ with $n \geq n_0$. For an integer $K \geq 0$ we set

$$H = L/n, \qquad N = nB^K, \qquad a_j = HB^{-j} \quad (0 \leq j \leq K), \qquad a = a_K, \qquad g_j = g_3\sqrt{a_j}.$$

Thus a_j is the mesh of the lattice of side nB^j, and $g_j^2 = \gamma/(nB^j)$ is its lattice coupling. We work under the conditioned Wilson measure $\nu_{N,n,\gamma} = \mu_{N,\gamma}(\,\cdot \mid S_{N,n,\gamma})$ of Definition 4.4. All fields are defined on the whole torus. Tail indices k, l, m are positive integers, and a sum whose upper limit is smaller than its lower limit is zero.

The coarse-to-fine reconstruction. Let $\widehat{\mathscr{R}}$ be the family of reconstruction maps of Theorem 15.12, and consider its coarse-to-fine disintegration of $\nu_{N,n,\gamma}$ (Lemma 15.10) at the classical depths $J(K-j)$:

$$\widehat{\theta}_j = \widehat{\Phi}_j(\widehat{\theta}_{j-1}, g_{j,\cdot} w_j), \qquad A_j = \widehat{\theta}_j/a_j, \qquad \mathcal{F}_j = \sigma(\widehat{\theta}_0, w_1, \dots, w_j), \qquad 1 \leq j \leq K.$$

Here $\widehat{\theta}_0$ is the link logarithm of the coarse field on Λ_n, in the Borel gauge of Lemma 15.10, so that $\|\widehat{\theta}_0\|_\infty < r$, where r is the radius of the small-link domain B_r. Further, w_j is the vector of normalized fluctuation variables of the jth RG step counted from the coarse side, $g_{j,\cdot}$ are the couplings of the J intermediate scales of this step, $\widehat{\Phi}_j$ is the logarithmic reconstruction map of the family $\widehat{\mathscr{R}}$ for this step, $\widehat{\theta}_j$ is the link logarithm of the reconstructed field on the lattice of side nB^j, and A_j is the corresponding lattice potential on the mesh a_j. In the notation of Section 26.1, $\widehat{\Phi}_j(\theta, g_{j,\cdot} w) = \mathscr{R}_j(\theta, w)$, at the depth $\mathfrak{d}_j = J(K-j)$. Conditionally on $\mathcal{F}_{j-1}$, the law of w_j is the conditional fluctuation kernel (4.8) of the jth RG step at the coarser field $\widehat{\theta}_{j-1}$, including all action, Jacobian, determinant, and effective-interaction factors and the indicator of the fluctuation cutoff domain. This kernel is not Gaussian. All conditional expectations below refer to it, and we write $\bar{w}_j = \mathbb{E}(w_j \mid \mathcal{F}_{j-1})$; this is the conditional mean $m_{\widehat{\theta}_{j-1}}$ of Section 26.1.

Let $t_j = M^{(J-1)/2} g_j$ be the largest coupling of the intermediate scales of the jth RG step, and $p_j = A_{\mathrm{w}} \log(1/t_j) \geq 1$ the corresponding fluctuation cutoff. Let a_{cap} be the contraction factor of the reconstruction amplitude bound of Lemma 15.10, which satisfies $a_{\mathrm{cap}}\sqrt{B} < 1$, and set

$$q_{\mathrm{anc}} = B a_{\mathrm{cap}}^2 < 1, \qquad \lambda_{\mathrm{anc}} = -\log(q_{\mathrm{anc}})/\log B > 0.$$

Interpolation, heat semigroup, and norms. For meshes $a \leq h$ with h/a an integer, $I_{h\to a}$ is the cardinal interpolation from the lattice of mesh h to the lattice of mesh a, recalled in Section 28.2; $I_{a\to a}$ is the identity. The lattice heat semigroup $P_t^a = e^{t\Delta_a}$ acts componentwise and includes the constant mode. For an array f on the lattice of mesh a we write $P_\cdot^a f$ for the heat path $t \mapsto P_t^a f$. Let $\mathcal{N}_a$ be the ordered quadratic coefficient of the damped Wilson flow (Lemma 24.2), and recall the bilinear Duhamel term and its polarization,

$$\mathcal{Q}_a(u,v)(t) = \int_0^t P_{t-s}^a \mathcal{N}_a(u(s), v(s))\, ds, \qquad \mathcal{B}_a(u,v) = \mathcal{Q}_a(u,v) + \mathcal{Q}_a(v,u).$$

The term $\mathcal{Q}_a$ is not symmetric. We fix a horizon $T < \infty$ and exponents

$$1/2 < \alpha = \alpha_{\mathrm{fl}} < 2/3, \quad \beta = \beta_{\mathrm{fl}} = \alpha - 1/2, \quad 0 < \sigma < 1 - \alpha, \quad 0 < \tau < \min\{\sigma, 1/2, \lambda_{\mathrm{anc}}\}.$$

We use the norms of Definition 4.8 on $[0, T]$: with $v_a(t) = t + a^2$,

$$\begin{aligned}
\mathsf{X}^a_\alpha(x) &= \sup_{0\le t\le T} \big\{ v_a(t)^{\alpha/2} \|x(t)\|_\infty + v_a(t)^{(\alpha+1)/2} \|D_a x(t)\|_\infty \big\}, \\
\mathsf{Y}^a_\alpha(y) &= \sup_{0\le t\le T} \big\{ v_a(t)^{\beta} \|y(t)\|_\infty + v_a(t)^{\alpha} \|D_a y(t)\|_\infty \big\}, \\
\mathsf{H}^a_\sigma(f) &= \sup_{0\le t\le T} \big\{ v_a(t)^{\sigma/2} \|P^a_t f\|_\infty + v_a(t)^{(\sigma+1)/2} \|D_a P^a_t f\|_\infty \big\},
\end{aligned}$$

where D_a runs over all physical forward and backward first differences, applied also to the translated components that occur in $\mathcal{N}_a$; $\mathsf{V}^a_\sigma(f)$ is the first term of $\mathsf{H}^a_\sigma(f)$. The linear and quadratic flow data of the reconstructed field at level j, on the fine mesh a, are

$$x_j = P^a_\cdot I_{a_j \to a} A_j, \qquad y_j = \mathcal{Q}_a(x_j, x_j), \qquad \text{and we set} \qquad d_j = x_j - x_{j-1}.$$

Thus x_K is the linear heat evolution of the fine field A_K, and y_K is its quadratic Picard term. For $j < K$, the pair (x_j, y_j) is the flow data of the finite-scale reconstruction A_j, interpolated to the fine mesh. The norm of a pair is $\|(x,y)\|_{\mathrm{fl},a} = \mathsf{X}^a_\alpha(x) + \mathsf{Y}^a_\alpha(y)$, as in Definition 24.3.

By Theorem 15.13, for each fixed q the law of the array A_q under $\nu_{nB^K,n,\gamma}$ converges weakly as $K \to \infty$, and these arrays take values in a fixed compact set. We denote the limit law by π_q.

Dependence of constants. Unless stated otherwise, the constants in the stochastic estimates of this section depend only on L, n, γ, M, J, Θ, T, the exponents, and the moment orders. They do not depend on K, on the level j, on the classical depth $J(K-j)$, or on the ratio a/a_j. The letter $b < \infty$ denotes an exponent with the same dependence, which may increase finitely many times; in a displayed estimate we may replace finitely many such exponents by their maximum. This is the dependence described in item (7) of Definition 4.14. None of the stochastic estimates of this section is claimed to be uniform as $\gamma \to 0$ or as $n \to \infty$.

Wilson loops. Let $\ell_1, \dots, \ell_m$ be piecewise C^1 loops in $\mathbb{T}^3_L$, let $\chi_1, \dots, \chi_m$ be characters of finite-dimensional representations of $\mathrm{SU}(2)$, and let $t_1, \dots, t_m > 0$. For each i let $(\ell_{i,K})_{K\ge 0}$ be an admissible approximation of ℓ_i along the lattices Λ_{nB^K} (Definition 2.1) whose based homotopies to ℓ_i can be chosen with swept area, counted with multiplicity, of order $O(a_K)$. Under $\nu_{nB^K,n,\gamma}$ we consider the vector $W_K = (W_{K,1}, \dots, W_{K,m})$ with

$$W_{K,i} = \chi_i\big(\mathrm{Hol}_{\ell_{i,K}} \Psi^{(N)}_{t_i} U\big), \qquad N = nB^K.$$

Each $W_{K,i}$ takes values in $\{w \in \mathbb{C} : |w| \le \chi_i(I)\}$, where $\chi_i(I)$ is the dimension of the representation.

Main results. The first theorem gives convergence under the conditioned measure at a fixed coarse side; it is proved in Section 28.5.

Theorem 28.1. *Assume the standing assumptions, and let W_K be as above. Then the law of W_K under $\nu_K = \nu_{nB^K,n,\gamma}$ converges weakly as $K \to \infty$. The limit laws are consistent under finite unions of lists of loops and under coordinate projections. For each q and each array ϑ on the lattice of side nB^q, let $G_q(\vartheta) = (\chi_i(\mathrm{Hol}_{\ell_i}\,\Phi_{t_i}[f]))_{i=1}^m$, where f is the smooth cardinal interpolant of ϑ, viewed as a smooth connection, and $\Phi_t[f]$ is the gauge orbit at time t of the Yang–Mills heat flow started from f. Then the limit law of W_K is the weak limit, as $q \to \infty$, of the laws $(G_q)_\# \pi_q$. The moment generating functions $\mathbb{E}_{\nu_K} \exp(\sum_i z_i W_{K,i})$ converge uniformly on every bounded subset of $\mathbb{C}^m$, and their cumulants of every finite order converge, the logarithms being taken on a fixed neighborhood of the origin on which none of these functions vanishes.*

The second theorem removes the conditioning; it is proved in Section 28.6. Here the coarse side varies along the sequence $n_r = n_* B^r$, while the fine lattice Λ_{N_K}, the configuration, and the observable stay the same.

Theorem 28.2. *Assume the standing assumptions, except that the coarse side is not fixed. Fix $n_* \in 4\mathbb{N}$, put $n_r = n_* B^r$, $N_K = n_* B^K$, and $\nu_{K|r} = \nu_{N_K,n_r,\gamma}$ for $K \geq r$, and let W_K be the vector of flowed Wilson loops on Λ_{N_K} defined above, with $D = \max_i \chi_i(I)$. Let C', $\bar{c}$, C_{w}, and a_{w} be the constants of Theorem 23.9, and set*

$$F_\gamma(u) = C' u^3 e^{-\bar{c}u/\gamma} + C_{\mathrm{w}} u^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mu/\gamma}}.$$

Then there exists r_0 such that, for all $r \geq r_0$ and all $K \geq r$,

$$e_{K|r} = \mu_{N_K,\gamma}(S^c_{N_K,n_r,\gamma}) \leq F_\gamma(n_r),$$

and $F_\gamma(n_r) \to 0$ as $r \to \infty$. For $r \geq r_0$ let $Z^{\mathrm{sel}}_r(z)$ be the limit as $K \to \infty$ of $\mathbb{E}_{\nu_{K|r}} \exp(\sum_i z_i W_{K,i})$, which exists by Theorem 28.1. For every $R < \infty$ and all $r, r' \geq r_0$,

$$\sup_{\sum_i |z_i| \leq R} |Z^{\mathrm{sel}}_r(z) - Z^{\mathrm{sel}}_{r'}(z)| \leq 2e^{DR}\big[F_\gamma(n_r) + F_\gamma(n_{r'})\big].$$

There exists a probability measure ρ on $\prod_i \{w \in \mathbb{C} : |w| \leq \chi_i(I)\}$ such that, as $K \to \infty$,

$$\mathrm{Law}_{\mu_{N_K,\gamma}}(W_{K,1}, \dots, W_{K,m}) \Longrightarrow \rho, \qquad \mathbb{E}_{\mu_{N_K,\gamma}} e^{\sum_i z_i W_{K,i}} \longrightarrow \int e^{\sum_i z_i w_i}\, d\rho(w),$$

and the second convergence is uniform on $\{\sum_i |z_i| \leq R\}$ for every $R < \infty$. These limits are consistent under taking subfamilies of the loops. The events $S_{N_K,n_r,\gamma}$ for different r need not be nested.

28.2 Deterministic estimates

This section collects the deterministic facts used below: elementary properties of the cardinal interpolation and of the lattice heat semigroup, a geometric summation, and the comparison of lattice and continuum flows for a fixed family of smooth initial data.

Cardinal interpolation. We recall the cardinal interpolation of Lemma 16.3, explicitly enough to justify the elementary facts needed here. Let b_* be the even nonnegative smooth function of that lemma, denoted b there, which is supported in $[-1,1]$, positive on $(-1,1)$, and flat at ± 1, and set

$$\varphi(u) = \frac{b_*(u)}{\sum_{q\in\mathbb{Z}} b_*(u-q)}, \qquad \varphi_3(u) = \prod_{i=1}^{3} \varphi(u_i).$$

The denominator is positive, smooth, and one-periodic. Hence $\varphi(q) = \mathbf{1}_{q=0}$ for $q \in \mathbb{Z}$ and $\sum_q \varphi(u-q) = 1$. In the notation of that lemma, $\varphi = \zeta$ and $\varphi_3 = \zeta_3$. For an array f on the lattice of mesh h, its periodized smooth interpolant is

$$f_h(x) = \sum_{v\in\Lambda_{L/h}} \varphi_3((x-v)/h) f(v),$$

where the periodic images of v are understood. The formula is applied separately to every directed-link and color component, at the location of that component. For $a \le h$ with h/a an integer, $I_{h\to a} f$ is the restriction of f_h to the lattice of mesh a. For an array f on the lattice of mesh a, we write $J_a f = f_a$ for its interpolant, which is the continuous interpolant $I_a f$ of that lemma. At every point only a bounded number of the translates $\varphi_3((x-v)/h)$ are nonzero.

Lemma 28.3. *Let $a \le h$ be meshes with h/a an integer, let f be an array on the lattice of the indicated mesh, and let F be a smooth periodic field. Take $t \ge 0$ and an integer $q \ge 0$. Let D_a^q be any product of q physical forward or backward differences, possibly applied to translated components. Then*

$$\|J_a f\|_\infty \le \|f\|_\infty, \qquad \|D_a^q I_{h\to a} f\|_\infty \le C_q h^{-q} \|f\|_\infty, \tag{28.1}$$

$$\|J_a(F|_a) - F\|_\infty \le Ca \|\partial F\|_\infty, \tag{28.2}$$

$$\|P_t^a I_{h\to a} f\|_\infty \le \|f\|_\infty, \qquad \|D_a P_t^a I_{h\to a} f\|_\infty \le Ch^{-1} \|f\|_\infty, \tag{28.3}$$

where C_q and C depend only on q and b_*. Further, if $a_j = HB^{-j}$, $\delta > \delta' > 0$, and $c \ge 0$, then for every integer $k \ge 0$,

$$\sum_{j\ge k} (1+j)^c a_j^\delta \le C_{B,c,\delta} (1+k)^c a_k^\delta \le C_{H,B,c,\delta,\delta'} a_k^{\delta'}. \tag{28.4}$$

Proof. The first bound in (28.1) follows from nonnegativity of φ_3 and the partition of unity. For the second, a product of q physical differences of $I_{h\to a}f$ is an average of the corresponding qth derivative of f_h over q segments. Each derivative of order q of $\varphi_3((x-v)/h)$ is bounded by $C_q h^{-q}$, and only a bounded number of terms contribute at each point. This proves the bound, also for backward differences and translated components, with the same constant. For (28.2), every lattice site v that contributes to $J_a(F|_a)(x)$ lies within distance Ca of x. Subtracting $F(x)$ inside the partition sum and integrating the derivative of F along the segment from x to v gives the bound.

For (28.3), we write $\Delta_a = a^{-2}\sum_i (T_i + T_i^{-1} - 2I)$, where T_i is the translation by ae_i. Then $e^{t\Delta_a} = e^{-6t/a^2}\exp(ta^{-2}\sum_i (T_i + T_i^{-1}))$ is a series in translations with nonnegative coefficients of total mass one, so P_t^a is a contraction in the supremum norm. Differences commute with P_t^a. The two bounds then follow from (28.1).

For the first inequality in (28.4), set $j = k + i$ and use $1 + k + i \le (1 + k)(1 + i)$ and $a_{k+i} = a_k B^{-i}$; the remaining series $\sum_{i\ge 0}(1+i)^c B^{-i\delta}$ converges by the ratio test. The second inequality holds because $a_k^\delta = H^{\delta-\delta'}B^{-k(\delta-\delta')}a_k^{\delta'}$ and $\sup_k (1+k)^c B^{-k(\delta-\delta')} < \infty$. □

We use (28.3) only for a fixed finite-scale reconstruction or at times $t < h^2$; it does not transfer between meshes uniformly. Under the standing assumptions, $t_j = t_1 B^{-(j-1)/2}$, so the cutoff $p_j = A_{\mathrm{w}} \log(1/t_j)$ is an affine function of j with slope $A_{\mathrm{w}}(\log B)/2$. In particular

$$p_j \le C_{L,n,\gamma,M,J,A_{\mathrm{w}}}(1 + j). \tag{28.5}$$

Smooth initial data. The next lemma compares the lattice objects with their continuum counterparts for a fixed compact family of smooth initial data. It is used for the finite-scale reconstructions, whose interpolants form such a family. The continuum counterpart of $\mathcal{Q}_a$ is the Duhamel term $\int_0^t P_{t-s}\mathcal{N}(u(s), v(s))\,ds$, where $P_t = e^{t\Delta}$ and $\mathcal{N}$ is obtained from the finite sum of shifted brackets defining $\mathcal{N}_a$ by replacing translations by the identity and physical differences by derivatives.

Lemma 28.4. *Let $h = L/m$ for an integer $m \ge 4$, let $M_h < \infty$ and $0 < T < \infty$, and let $\mathscr{F}$ be the family of cardinal interpolants $f = \vartheta_h$ of arrays ϑ on the lattice of mesh h with $\|\vartheta\|_\infty \le M_h$. For meshes $a \le h$ with h/a an integer, put*

$$x_a = P_\cdot^a(f|_a), \qquad y_a = \mathcal{Q}_a(x_a, x_a), \qquad f \in \mathscr{F},$$

where $f|_a = I_{h\to a}\vartheta$ is the restriction of f to the lattice of mesh a. Then, as $a \to 0$, the four components of (x_a, y_a), embedded as in Definition 24.3, converge uniformly over $\mathscr{F}$ to the continuum heat path $P_\cdot f$, the continuum Duhamel term $\int_0^t P_{t-s}\mathcal{N}(P_s f, P_s f)\,ds$, and their spatial derivatives in the difference components. The norms $\|(x_a, y_a)\|_{\mathrm{fl},a}$ are bounded by a constant depending only on h, M_h, L, T, α, and b_*, and not on a.

Further, let U_a be the configuration on the lattice of mesh a with link variables $\exp(af|_a)$. Fix finitely many characters χ_i of finite-dimensional representations of $\mathrm{SU}(2)$, loops ℓ_i, and closed

lattice paths $\ell_{i,a}$, independent of time, whose physical lengths are bounded uniformly in a and which admit based homotopies to the ℓ_i, including connecting paths between base points, with swept areas counted with multiplicity tending to zero as $a \to 0$. Then for every compact interval $[\tau, T'] \subset (0, \infty)$,

$$\chi_i\big(\mathrm{Hol}_{\ell_{i,a}} \Psi_t^{(L/a)} U_a\big) \longrightarrow \chi_i\big(\mathrm{Hol}_{\ell_i} \Phi_t[f]\big) \qquad (a \to 0),$$

uniformly in $t \in [\tau, T']$ and $f \in \mathscr{F}$, where $\Phi_t[f]$ is the Yang–Mills heat flow started from f. The map from ϑ to the path $t \mapsto \chi_i(\mathrm{Hol}_{\ell_i} \Phi_t[\vartheta_h])$ is continuous. The constants in these convergences may depend on h, M_h, and the smooth bounds on $\mathscr{F}$; no uniformity as $h \to 0$ is asserted.

Proof. Since the cardinal basis is fixed, $\|f\|_{C^k} \le C_{k,h} M_h$ for every k and every $f \in \mathscr{F}$. Let S_a denote sampling on the lattice of mesh a. By Taylor's formula, we have

$$\|\Delta_a S_a P_t f - S_a \Delta P_t f\|_\infty \le C a^2 \|f\|_{C^4}, \qquad \|D_a^k S_a P_t f - S_a \partial^k P_t f\|_\infty \le C_k a \|f\|_{C^{k+1}},$$

where we used that P_t commutes with derivatives and contracts every C^k norm. By variation of constants, we have

$$P_t^a S_a f - S_a P_t f = \int_0^t P_{t-s}^a (\Delta_a S_a - S_a \Delta) P_s f \, ds,$$

so $\|P_t^a S_a f - S_a P_t f\|_\infty \le C t a^2 \|f\|_{C^4}$ by heat contraction. Commuting finitely many differences with P_t^a and using the second Taylor estimate, we obtain the same convergence, with error $O_{h,M_h}(a)$, for every fixed number of differences, uniformly over $\mathscr{F}$ and $t \in [0, T]$.

The ordered coefficient $\mathcal{N}_a$ of Lemma 24.2 is a fixed finite sum of shifted brackets with one physical difference. Its first difference is a sum of terms with $(1,1)$ or $(0,2)$ differences on the two factors. By the heat convergence just proved, the value and the first difference of $\mathcal{N}_a(x_a, x_a)$ agree with those of the continuum source $S_a \mathcal{N}(P_\cdot f, P_\cdot f)$ up to an error $O_{h,M_h}(a)$, uniformly. The same holds in each order of the two arguments. The error $y_a - S_a y$, where y is the continuum Duhamel term, solves the lattice heat equation with zero initial value and with source equal to this source error plus the heat residual $(\Delta_a S_a - S_a \Delta) y$ of the smooth continuum output. By heat contraction, also for the equation after commuting one difference, the value and the first differences of y_a converge, uniformly down to time zero. The bounds (28.1) and (28.2) then imply the convergence of the four embedded components. The same bounds on sources and derivatives, multiplied by the weights in the norms, bound $\|(x_a, y_a)\|_{\mathrm{fl},a}$ uniformly in a.

For the Wilson flow we use the consistency of the flow, with all its nonlinear terms, for smooth initial data, namely Lemma 25.2 and Theorem 25.4; no quadratic truncation of the equation is used. The family $\mathscr{F}$ is compact in every fixed smooth norm. Theorem 24.10 gives a finite reference family of local smooth continuum solutions that covers the flows started from $\mathscr{F}$ on a fixed time horizon. On each of these solutions, Lemma 25.5 compares

the lattice flow and the sampled smooth flow in values and first physical differences, in the same gauge frame. The consistency of smooth samples gives a residual of order a for the potential, or a^2 for the link variables, including each of the finitely many sampled gauge transformations between the local solutions. We choose this finite family and the corresponding thresholds at the fixed h, before letting $a \to 0$. Thus the lattice flow is uniformly close, on all local solutions, to the sampled smooth flow. Since transports are unitary, the error of the transport along a closed lattice path telescopes into a sum of edge errors; a path of physical length at most ℓ has at most ℓ/a edges, so the error in its character is $O(\ell a)$.

It remains to compare the lattice paths with the continuum loops for the smooth continuum flow; for this only bounds on the smooth curvature are needed. Differentiating parallel transport along a smooth piece of a based homotopy $H(u, v)$ gives an integral of $F_A(H)(\partial_u H, \partial_v H)$, conjugated by unitary transports. Its norm is at most $C\|F_A\|_\infty$ times the swept area of that piece, counted with multiplicity. Subdivision and smooth approximation in C^1 give the same bound for piecewise C^1 homotopies. The terms coming from moving base points are removed by the transport along the connecting paths. The transport along each edge of a lattice path, with the frozen sampled link variable, differs from the continuum parallel transport along that edge by $O(a^2)$, by the transport equation and a C^1 bound. Writing A_a for the swept area of the homotopy at mesh a, counted with multiplicity, the total error in each character is therefore at most $C_{h,M_h}(a + A_a)$, uniformly in $t \in [\tau, T']$ and in $f \in \mathscr{F}$. Only characters are used, so conjugations at the base point cancel. Continuity with respect to ϑ follows, on the same finite family of local solutions, from the smooth dependence of the flow in Theorem 24.10 and the continuous dependence of the transport equation on the connection. □

28.3 The four-term decomposition and the ordered bilinear terms

Since $\mathcal{Q}_a$ is bilinear and $y_j = \mathcal{Q}_a(x_{j-1} + d_j, x_{j-1} + d_j)$, we have

$$y_j - y_{j-1} = \mathcal{B}_a(x_{j-1}, d_j) + \mathcal{Q}_a(d_j, d_j), \qquad 1 \le j \le K. \tag{28.6}$$

Summing over j expresses the truncation error of the quadratic Picard term through the cross terms $\mathcal{B}_a(x_{j-1}, d_j)$ and the quadratic increments $\mathcal{Q}_a(d_j, d_j)$; both sums are taken on the same fine mesh and with the same ordered operator $\mathcal{Q}_a$. The next lemma splits d_j into four terms and bounds the expected norm of each of the sixteen ordered products $\mathcal{Q}_a(v_j, w_j)$. The centered linear fluctuation term s_j is linear in the centered new fluctuations $w_j - \bar{w}_j$. The centered nonlinear heat term z_j is the heat evolution of the rest of the conditionally centered part of A_j, which is nonlinear in w_j. The interpolation defect r_j is the difference between the linearized reconstruction of the coarser field and the cardinal interpolation of that field; it is a deterministic function of $\widehat{\theta}_{j-1}$. The predictable term u_j

is the nonlinear part of the zero-fluctuation reconstruction plus the conditional mean correction. The cross terms are treated in Section 28.4.

Lemma 28.5. *Assume the standing assumptions of Section 28.1. For $1 \le j \le K$ define*

$$\widehat{P}_j(v) = \widehat{\Phi}_j(v,0), \qquad \widehat{L}_j = D\widehat{P}_j(0), \qquad \mathsf{E}_{j,v} = g_j^{-1} D_w \widehat{\Phi}_j(v, g_{j,\cdot} w)|_{w=0},$$

and the following four heat paths on the mesh a:

$$\begin{aligned}
s_j &= \frac{g_j}{a_j} P_\cdot^a I_{a_j \to a} \mathsf{E}_{j,\widehat{\theta}_{j-1}}(w_j - \bar{w}_j),\\
z_j &= P_\cdot^a I_{a_j \to a}\big(A_j - \mathbb{E}[A_j \mid \mathcal{F}_{j-1}]\big) - s_j,\\
r_j &= P_\cdot^a \Big(I_{a_j \to a} \frac{\widehat{L}_j \widehat{\theta}_{j-1}}{a_j} - I_{a_{j-1} \to a} \frac{\widehat{\theta}_{j-1}}{a_{j-1}} \Big),\\
u_j &= P_\cdot^a I_{a_j \to a} U_j^0, \qquad U_j^0 = \mathbb{E}[A_j \mid \mathcal{F}_{j-1}] - \widehat{L}_j \widehat{\theta}_{j-1}/a_j.
\end{aligned}$$

Then the following statements hold.

1. *We have $d_j = s_j + z_j + r_j + u_j$. The paths s_j and z_j have zero conditional mean given $\mathcal{F}_{j-1}$, and r_j and u_j are $\mathcal{F}_{j-1}$-measurable. The difference defining r_j involves the two input meshes a_j and a_{j-1}.*
2. *For any two heat paths $v = P_\cdot^a v_0$ and $w = P_\cdot^a w_0$ on the mesh a, the two orders satisfy separately, without cancellation,*

$$\mathsf{Y}_\alpha^a(\mathcal{Q}_a(v,w)) + \mathsf{Y}_\alpha^a(\mathcal{Q}_a(w,v)) \le C_{L,T,\alpha,\sigma,J,\Theta}\, \mathsf{X}_\alpha^a(v) \mathsf{H}_\sigma^a(w_0).$$

   The constant does not depend on γ, n, or K.
3. *The reconstructed fields satisfy deterministic bounds $\|\widehat{\theta}_j\|_\infty \le \bar{b}_j$, where $\bar{b}_0 = r$ and*

$$\bar{b}_j \le C\big[r a_{\mathrm{cap}}^j + g_3 \sqrt{a_j}(p_j + 1)\big], \qquad a_{\mathrm{cap}}\sqrt{B} < 1 \qquad (j \ge 1).$$

   For every fixed $p \ge 1$ define

$$\begin{aligned}
S_{p,j} &= C_p g_3 a_j^\beta \big[\sqrt{\log(2 + L/a_j)} + \sqrt{p}\big],\\
Z_{p,j}^- &= C_p\big[g_3^2 a_j^\sigma (1+j) + g_3^3 a_j^{1/2} p_j^3\big],\\
Z_{p,j}^+ &= C_p\big[g_3^2 a_j^\alpha (1+j) + g_3^3 a_j^{1/2} p_j^3\big],\\
R_j &= C\bar{b}_{j-1} a_{j-1}^{\alpha-1}, \qquad U_{p,j} = C_{p,\tau}(1+j)^b a_j^\tau.
\end{aligned}$$

   Then $\|\mathsf{X}_\alpha^a(s_j)\|_p \le S_{p,j}$, $\|\mathsf{H}_\sigma^a(z_j(0))\|_p \le Z_{p,j}^-$, $\|\mathsf{X}_\alpha^a(z_j)\|_p \le Z_{p,j}^+$, $\mathsf{X}_\alpha^a(r_j) \le R_j$ pathwise, and $\|\mathsf{H}_\sigma^a(I_{a_j \to a} U_j^0)\|_p \le U_{p,j}$ and $\|\mathsf{X}_\alpha^a(u_j)\|_p \le U_{p,j}$. The first three bounds also hold conditionally on $\mathcal{F}_{j-1}$, for every value of the coarser field. Further, $\mathcal{U}_j = \mathsf{H}_\sigma^a(I_{a_j \to a} U_j^0)$ satisfies $\mathcal{U}_j \le C(1+j)^b$ pathwise.

4. *There is a nonnegative deterministic sequence* (W_j) *with* $\mathbb{E}\mathrm{Y}^a_\alpha(\mathcal{Q}_a(r_j,r_j)) \le W_j$ *and such that* $W_j \le C_\zeta(1+j)^b a_j^\zeta$ *for every* $0<\zeta<\min\{2\alpha-1,\lambda_{\mathrm{anc}}\}$.
5. *Put* $F_j = g_3^2[a_j^{2\alpha-1}(1+j)+\bar b_{j-1}]$. *For* $v,w\in\{s,z,r,u\}$, *the expectation* $\mathbb{E}\mathrm{Y}^a_\alpha(\mathcal{Q}_a(v_j,w_j))$ *is bounded by a constant times the entry in row* v *and column* w *of the following table:*

	s	z	r	u
s	F_j	$S_{2,j}Z^-_{2,j}$	$S_{1,j}R_j$	$S_{1,j}U_{1,j}$
z	$S_{2,j}Z^-_{2,j}$	$Z^+_{2,j}Z^-_{2,j}$	$R_jZ^-_{1,j}$	$U_{1,j}Z^-_{1,j}$
r	$R_jS_{1,j}$	$R_jZ^-_{1,j}$	W_j	$R_jU_{1,j}$
u	$U_{1,j}S_{1,j}$	$U_{1,j}Z^-_{1,j}$	$R_jU_{1,j}$	$(1+j)^bU_{1,j}$

6. *For every* $0<\zeta_2<\min\{2\alpha-1,\tau\}$,

$$\sup_{K\ge j}\sum_{v,w\in\{s,z,r,u\}}\mathbb{E}_{\nu_{N,n,\gamma}}\mathrm{Y}^a_\alpha(\mathcal{Q}_a(v_j,w_j)) \le C_{\zeta_2}(1+j)^b a_j^{\zeta_2}.$$

The constants in (3)–(6) have the dependence stated in Section 28.1; the same holds in the remaining results of this section.

The lemma uses the expansion at $w=0$. By Remark 26.3, s_j and $z_j = P^a_\cdot I_{a_j\to a}Z^{(0)}_j$ are the terms $s^{(0)}_j$ and $z^{(0)}_j$ of Lemma 26.2, while s_j+z_j, r_j, and u_j are the same as in the expansion at the conditional mean. In the notation of that lemma, $\widehat P_j(\theta)/a_j = C_j(\theta)$ is the zero-fluctuation reconstruction in physical units; the linearization $\widehat L_j$ of $\widehat P_j$ acts on link logarithms, and $\mathsf{P}_j = B\widehat L_j$ acts on lattice potentials, so that the linear part of C_j is $a_j^{-1}\widehat L_j\theta = a_{j-1}^{-1}\mathsf{P}_j\theta$.

Three groups of entries in the table require different inputs. The entry F_j for $\mathcal{Q}_a(s_j,s_j)$, which includes the conditional mean of this product at a nonzero coarser field, and the two entries $S_{1,j}R_j$ for $\mathcal{Q}_a(s_j,r_j)$ and $\mathcal{Q}_a(r_j,s_j)$ rely on the conditional concentration estimates of Section 26. The entry W_j for $\mathcal{Q}_a(r_j,r_j)$ is a separate estimate from Theorem 27.6. The remaining twelve entries follow from the product inequality (2) and the bounds in (3), combined either with the Cauchy–Schwarz inequality or with conditioning on $\mathcal{F}_{j-1}$. All entries decay like positive powers of a_j; by (28.12) below, $\bar b_{j-1}\le C(1+j)^b a_j^{1/2}$, R_j and $S_{p,j}$ are $O((1+j)^b a_j^\beta)$, and $2\beta = 2\alpha-1>0$.

Proof of Lemma 28.5. We first prove the decomposition and the bounds on the four factors, then the product inequality, then the table, and finally the summed bound (6).

Step 1: the decomposition. The arrays A_j are bounded at every finite K by the amplitude bounds in (3), so conditional expectation commutes with the fixed linear heat and cardinal maps. By the definition of z_j, we have

$$s_j+z_j = P^a_\cdot I_{a_j\to a}\big(A_j-\mathbb{E}[A_j\mid\mathcal{F}_{j-1}]\big).$$

Adding u_j replaces the conditional expectation by $\widehat{L}_j\widehat{\theta}_{j-1}/a_j$, and adding r_j cancels this linearized reconstruction and subtracts $P_{\cdot}^{a} I_{a_{j-1}\to a} A_{j-1} = x_{j-1}$. This proves $d_j = s_j + z_j + r_j + u_j$. The vector $w_j - \bar{w}_j$ has conditional mean zero under the conditional fluctuation kernel, and the derivative $\mathsf{E}_{j,\widehat{\theta}_{j-1}}$ and the conditional mean of A_j are $\mathcal{F}_{j-1}$-measurable. This gives the four assertions on conditional means and measurability.

Step 2: the linear fluctuation term and the centered nonlinear term. We use the convention of Theorems 26.10 to 26.12 in which the reconstruction is Taylor expanded at $w = 0$. The input mesh, coarser mesh, and common fine mesh of those theorems are a_j, $a_{j-1} = Ba_j$, and a_K, and the refinement ratios are the integers B and B^{K-j}. The classical depth is $J(K-j)$, and the coarser dimensionless field is $\widehat{\theta}_{j-1}$. The parameter choice of Theorem 23.2 gives $t_1 \le g_*$, where g_* is the bound on the coupling under which the conditional fluctuation kernel is controlled (Theorem 13.8). Hence $t_j = t_1 B^{-(j-1)/2} \le g_*$ for every $j \ge 1$. The coarser fields that occur lie in the inner domain of the small-field event, and their reconstructions are evaluated on the fixed larger analytic domain. All conditional estimates below use the conditional fluctuation kernel on its fluctuation cutoff domain and its mean $\bar{w}_j$. The larger domain supplies bounds on coefficients only; no conditional law on a larger domain of coarser fields is used. Thus the terms of those theorems in the convention at $w = 0$ are precisely s_j and z_j. By the conditional estimates of the first two theorems, we obtain $S_{p,j}$, $Z^{-}_{p,j}$, and $Z^{+}_{p,j}$, and taking expectations of their conditional pth moments gives the same unconditional bounds. The amplitude bounds in Theorem 26.12 give the displayed $\bar{b}_j$, where r is the radius of B_r.

Step 3: the interpolation defect. Let $H_j = a_{j-1}$ and $d_{H_j}(t) = H_j(t + H_j^2)^{-1/2}$. In the notation of Lemma 27.1, the linearized reconstruction in physical units is $B\widehat{L}_j$, since

$$\widehat{L}_j\widehat{\theta}_{j-1}/a_j = (B\widehat{L}_j)\widehat{\theta}_{j-1}/H_j.$$

The difference of the two interpolations in that lemma is therefore the one in r_j. The bounds of that lemma on the rows of the heat-evolved difference and of its first differences imply

$$\|r_j(t)\|_\infty \le C\bar{b}_{j-1}H_j^{-1}d_{H_j}(t), \qquad \|D_a r_j(t)\|_\infty \le C\bar{b}_{j-1}H_j^{-1}(t+H_j^2)^{-1/2}d_{H_j}(t).$$

Since $a \le H_j$ and $\alpha < 1$, each of the two weighted terms in $\mathsf{X}^{a}_{\alpha}(r_j)$ is at most $C\bar{b}_{j-1}H_j^{\alpha-1}$: we replace $t + a^2$ by $t + H_j^2$ in the weights, and the remaining power $(t+H_j^2)^{(\alpha-1)/2}$ is largest at $t = 0$. This proves the pathwise bound R_j.

Step 4: the predictable term. The field U_j^0 is the nonlinear part of the zero-fluctuation reconstruction, in physical units, plus the conditional mean correction under the conditional fluctuation kernel. It is the field of Theorem 27.7. That theorem bounds the

moments of $\mathsf{H}^a_\sigma(I_{a_j\to a}U^0_j)$, including the first differences at times $t \le a^2$, and of $\mathsf{X}^a_\alpha(u_j)$, and it gives the pathwise polynomial bound. Its moment bound has $H^\tau_j = a^\tau_{j-1}$ in place of a^τ_j; the fixed factor B^τ is absorbed into $C_{p,\tau}$. This gives $U_{p,j}$ and the pathwise bound on $\mathcal{U}_j$.

Step 5: the product inequality. Statement (2) is Lemma 24.6, with the two orders kept separately. Its hypotheses hold, since the side $N = nB^K \ge 4$ is an integer, $a = L/N$, $1/2 < \alpha < 2/3$, and $0 < \sigma < 1 - \alpha$. For $a \le 1$ its heat-kernel proof applies. For $1 < a \le L/4$ we use its clause for coarse meshes, which we check directly. Lemma 24.2 gives the finite stencil of shifted brackets on every periodic lattice of side $N \ge 4$. Its bilinear polynomial defines $\mathcal{N}_a$ on arbitrary arrays; this does not require the smallness of link logarithms needed for the nonlinear equation. A physical first difference has supremum operator norm at most $2/a$, translations are isometries, and $\mathcal{N}_a$ is a fixed finite sum of brackets with one difference. Together with heat contraction, these facts imply that

$$\|\mathcal{Q}_a(v,w)(t)\|_\infty \le Cta^{-1}\|v_0\|_\infty\|w_0\|_\infty, \qquad \|D_a\mathcal{Q}_a(v,w)(t)\|_\infty \le Cta^{-2}\|v_0\|_\infty\|w_0\|_\infty;$$

for the second bound we commute the output difference with the outer heat semigroup and use the bound $2/a$ on the source. The reversed order has the same bound. Evaluating the input norms at $t = 0$, we find $\|v_0\|_\infty \le a^{-\alpha}\mathsf{X}^a_\alpha(v)$ and $\|w_0\|_\infty \le a^{-\sigma}\mathsf{H}^a_\sigma(w_0)$. Since $a > 1$, the sum of the two output norms is at most

$$CT\big[(T + (L/4)^2)^\beta + (T + (L/4)^2)^\alpha\big]\mathsf{X}^a_\alpha(v)\mathsf{H}^a_\sigma(w_0). \tag{28.7}$$

If $L \le 4$ this case does not occur. For $a \le 1$ the factor $(T + a^2)^{(\alpha-\sigma)/2}$ in Lemma 24.6 is bounded by a constant depending only on L, T, α, and σ, since $\alpha > \sigma$ and $a \le L/4$. Taking the larger of the two constants proves (2) for every mesh that occurs, independently of the coupling, the coarse side, and K. In particular, if $x = P^a_\cdot f$ for an array f on the mesh a, then x and u_j are heat paths on the same mesh, and (2) applies to $\mathcal{Q}_a(x, u_j)$ and $\mathcal{Q}_a(u_j, x)$ with the norm $\mathsf{H}^a_\sigma(I_{a_j\to a}U^0_j)$ bounded in Theorem 27.7.

Step 6: the sequence W_j. Let W_j be the nonnegative deterministic bound of Theorem 27.6, with $\rho_0 = r$ and with the amplitude bounds of that theorem replaced by $\bar{b}_i$. After enlarging its constant to account for this choice, both conclusions of that theorem remain valid:

$$\mathbb{E}\mathsf{Y}^a_\alpha(\mathcal{Q}_a(r_j, r_j)) \le W_j, \qquad \mathbb{E}\mathsf{Y}^a_\alpha(\mathcal{B}_a(x_{j-1}, r_j)) \le 4W_j. \tag{28.8}$$

The tail estimate of that theorem, for any $\zeta < \min(2\alpha - 1, \lambda_{\mathrm{anc}})$, implies that $W_j \le \sum_{i\ge j} W_i \le C_\zeta a_j^\zeta$, which is stronger than (4). Both quantities in (28.8) include all contributions of the coarse field and all propagated conditional means.

Step 7: all products except those of s_j and r_j. The bound on $\mathcal{Q}_a(s_j, s_j)$, including its conditional mean at a nonzero coarser field, is Theorem 26.12; the expectation it gives is at most CF_j.

For $\mathcal{Q}_a(s_j, z_j)$ and $\mathcal{Q}_a(z_j, s_j)$ we put s_j in the argument measured by X^a_α and $z_j(0)$ in the argument measured by H^a_σ, in (2). By the Cauchy–Schwarz inequality under the same measure, we have

$$\mathbb{E}\big[\mathsf{X}^a_\alpha(s_j)\mathsf{H}^a_\sigma(z_j(0))\big] \le S_{2,j}Z^-_{2,j}.$$

For $\mathcal{Q}_a(z_j, z_j)$ the same argument with z_j in the X^a_α argument gives $Z^+_{2,j}Z^-_{2,j}$.

For $\mathcal{Q}_a(z_j, r_j)$ and $\mathcal{Q}_a(r_j, z_j)$ we use the pathwise bound R_j on $\mathsf{X}^a_\alpha(r_j)$ and the bound on $\mathsf{H}^a_\sigma(z_j(0))$; each expectation is at most $CR_jZ^-_{1,j}$, in either order, without any symmetry of $\mathcal{Q}_a$.

For $\mathcal{Q}_a(s_j, u_j)$ and $\mathcal{Q}_a(u_j, s_j)$, statement (2) gives the bound $C\mathsf{X}^a_\alpha(s_j)\mathcal{U}_j$. Here $\mathcal{U}_j$ is $\mathcal{F}_{j-1}$-measurable, and the conditional first moment of $\mathsf{X}^a_\alpha(s_j)$ is at most $S_{1,j}$ for every value of the coarser field. Taking first the conditional expectation and then the expectation, we obtain $CS_{1,j}U_{1,j}$. For $\mathcal{Q}_a(z_j, u_j)$ and $\mathcal{Q}_a(u_j, z_j)$ we put u_j in the X^a_α argument and $z_j(0)$ in the H^a_σ argument; conditioning on $\mathcal{F}_{j-1}$ and using the conditional bound $Z^-_{1,j}$ gives $CU_{1,j}Z^-_{1,j}$. In these four products one factor is $\mathcal{F}_{j-1}$-measurable and the other has a conditional moment bound that is uniform in the coarser field; this is why their bounds are products of first moments.

For $\mathcal{Q}_a(r_j, u_j)$ and $\mathcal{Q}_a(u_j, r_j)$, statement (2) gives $CR_j\mathcal{U}_j$, hence $CR_jU_{1,j}$ after taking expectations. For $\mathcal{Q}_a(u_j, u_j)$, the splitting of the heat semigroup in Theorem 27.7 gives $\mathsf{X}^a_\alpha(u_j) \le C\mathcal{U}_j$, so the product is at most $C\mathcal{U}_j^2$. By the pathwise bound on $\mathcal{U}_j$, we have

$$\mathbb{E}\,\mathcal{U}_j^2 \le C(1+j)^b\,\mathbb{E}\,\mathcal{U}_j \le C(1+j)^bU_{1,j}.$$

Finally, $\mathcal{Q}_a(r_j, r_j)$ is bounded by the first estimate in (28.8).

Step 8: the products of s_j and r_j. For $\mathcal{Q}_a(s_j, r_j)$ and $\mathcal{Q}_a(r_j, s_j)$ we use the clause of Theorem 26.10 on predictable heat backgrounds on the common fine mesh. To check its hypotheses, let $h = a_j$, $H_j = a_{j-1} = Bh$, $\delta = a_K$, $\theta = \widehat{\theta}_{j-1}$, and $\mathsf{P}_j = B\widehat{L}_j$, and let E_{H_j} be the identity map of arrays on the lattice of mesh H_j. Then the background in that clause is exactly

$$r_j(t) = H_j^{-1}P^\delta_t\big(I_{h\to\delta}\mathsf{P}_j - I_{H_j\to\delta}\mathsf{E}_{H_j}\big)\theta, \qquad R^\sharp_j = C_J\|\theta\|_\infty H_j^{\alpha-1}. \tag{28.9}$$

It is a predictable heat path on the whole torus, and both input meshes are kept. That clause gives, for each integer $0 \le q \le 4$ and every choice of differences and shifts,

$$\|D^q_\delta r_j(t)\|_\infty \le C_qR^\sharp_j(t+h^2)^{-(\alpha+q)/2}, \tag{28.10}$$

which are the hypotheses on higher differences in its class of backgrounds, including times $t < h^2$. The finitely many constants C_q depend on the ratio B, the cardinal basis,

the stencil, L, T, and the exponents, but not on the coarser field or on δ/h. Since r_j is a heat path, $\partial_t D_\delta^q r_j = \Delta_\delta D_\delta^q r_j$; the time increments of its values and of its first and second differences involve differences of order at most four, which are controlled by (28.10). The same clause gives a bounded diameter for the normalized coefficient metric and the covering bound $N(v) \le C(1+L/h)^P v^{-Q}$, $0 < v \le 1$, independently of $\delta/h = B^{-(K-j)}$. Thus the conditional maximum is taken over a cover at the input mesh h.

In the expansion at $w = 0$ used in that theorem, the linear coefficient is $\mathsf{E}_{j,\theta}$ and the centered vector is $w_j - \bar{w}_j$, so the theorem concerns the same s_j and the same conditional fluctuation kernel as here. For each order separately let V_j be the norm Y_α^δ of the product. For each value of the coarser field, the theorem implies that

$$\mathbb{P}\Big\{ V_j > C g_3 h^\beta R_j^\sharp \sqrt{\log(2+L/h) + u} \;\Big|\; \mathcal{F}_{j-1} \Big\} \le C e^{-u}, \qquad u \ge 0.$$

If $R_j^\sharp = 0$, then $r_j = 0$ and the product vanishes. Otherwise we integrate this tail. With $\Lambda_h = \log(2 + L/h)$ and $A_j^\sharp = C g_3 h^\beta R_j^\sharp$, the layer-cake formula and the substitution $v = A_j^\sharp \sqrt{\Lambda_h + u}$ give

$$\mathbb{E}[V_j \mid \mathcal{F}_{j-1}] \le A_j^\sharp \sqrt{\Lambda_h} + \frac{C A_j^\sharp}{2} \int_0^\infty \frac{e^{-u}}{\sqrt{\Lambda_h + u}}\, du \le C g_3 h^\beta \big(\sqrt{\Lambda_h} + 1\big) R_j^\sharp,$$

where we used $\Lambda_h \ge \log 2$. The amplitude bound gives $R_j^\sharp \le C_J \bar{b}_{j-1} H_j^{\alpha-1} \le C R_j$. By the tower property we obtain, separately for the two orders,

$$\mathbb{E}\mathsf{Y}_\alpha^a(\mathcal{Q}_a(s_j, r_j)) \le C S_{1,j} R_j, \qquad \mathbb{E}\mathsf{Y}_\alpha^a(\mathcal{Q}_a(r_j, s_j)) \le C S_{1,j} R_j. \tag{28.11}$$

No cancellation between the two orders is used. This completes the proof of (5).

Step 9: summation. Let $\kappa_{\mathrm{cap}} = -\log a_{\mathrm{cap}} / \log B$, so that $\kappa_{\mathrm{cap}} > 1/2$. For $j \ge 2$, by the amplitude bounds, (28.5), and (28.4), we have

$$\begin{gathered} \bar{b}_{j-1} \le C(1+j)^b a_j^{1/2}, \\ R_j \le C(1+j)^b a_j^\beta, \qquad S_{p,j} \le C_p (1+j)^b a_j^\beta, \\ Z_{p,j}^- \le C_p (1+j)^b a_j^\sigma, \qquad Z_{p,j}^+ \le C_p (1+j)^b a_j^{1/2}, \qquad U_{p,j} \le C_{p,\tau} (1+j)^b a_j^\tau. \end{gathered} \tag{28.12}$$

In the first line, the contribution of the coarse field is $r a_{\mathrm{cap}}^{j-1} = r(a_{j-1}/H)^{\kappa_{\mathrm{cap}}}$, whose exponent is strictly larger than $1/2$. The first level $j = 1$ is absorbed into a constant. In

the bounds on $Z^{\pm}_{p,j}$ we used $\sigma < 1-\alpha < 1/2$ and $\alpha > 1/2$. By (28.12), the slowest powers of a_j in the sixteen entries of the table are

	s	z	r	u
s	2β	$\beta+\sigma$	2β	$\beta+\tau$
z	$\beta+\sigma$	$1/2+\sigma$	$\beta+\sigma$	$\tau+\sigma$
r	2β	$\beta+\sigma$	ζ	$\beta+\tau$
u	$\beta+\tau$	$\tau+\sigma$	$\beta+\tau$	τ

where we choose ζ strictly between ζ_2 and $\min(2\beta, \lambda_{\mathrm{anc}})$; such a choice exists since $\zeta_2 < 2\beta$ and $\zeta_2 < \tau < \lambda_{\mathrm{anc}}$. The term of F_j that contains the conditional mean has power $1/2 > 2\beta$. Every displayed power exceeds ζ_2. Absorbing the fixed powers of H into the constant, we obtain (6). □

28.4 Truncation estimates

We now bound the truncation error $(x_K - x_{k-1}, y_K - y_{k-1})$. Its first component is the sum of the increments d_j, $k \le j \le K$, which we control by the factor bounds of Lemma 28.5. By (28.6), its second component is the sum of the cross terms $\mathcal{M}_j = \mathcal{B}_a(x_{j-1}, d_j)$ and of the quadratic increments $\mathcal{Q}_a(d_j, d_j)$. The quadratic increments are controlled by Lemma 28.5(6). For the cross terms we use the splitting $\mathcal{M}_j = \mathcal{M}^s_j + \mathcal{M}^z_j + \mathcal{M}^r_j + \mathcal{M}^u_j$ with $\mathcal{M}^v_j = \mathcal{B}_a(x_{j-1}, v_j)$. The terms $\mathcal{M}^s_j$ and $\mathcal{M}^z_j$ are conditionally centered fields with the predictable background x_{j-1}; we bound them by the conditional estimates of Section 26, in terms of a radius of the background at the input scale a_j. The bound on $\mathcal{M}^r_j$ is part of (28.8), and $\mathcal{M}^u_j$ is bounded by the product inequality and the Cauchy–Schwarz inequality. In every case we bound the first moments of the norms and sum them over j; the norms themselves are not centered.

Theorem 28.6. *Assume the standing assumptions, and use the notation of Lemma 28.5. For every fixed $p \ge 1$ and every $0 < \zeta_1 < \min\{\beta, \tau\}$,*

$$\sup_{K \ge m \ge l \ge k} \Big\| \mathsf{X}^{a_K}_{\alpha}\Big(\sum_{j=l}^{m} d_j\Big)\Big\|_p \le C_{p,\zeta_1} a_k^{\zeta_1}, \qquad \sup_{K \ge 0} \sup_{0 \le j \le K} \|\mathsf{X}^{a_K}_{\alpha}(x_j)\|_p \le C_p.$$

For $1 \le j \le K$ the radius of the background x_{j-1} at the input scale,

$$\mathcal{R}_{j-1} = \sup_{0 \le t \le T} \big\{ (t + a_j^2)^{\alpha/2} \|x_{j-1}(t)\|_\infty + (t + a_j^2)^{(\alpha+1)/2} \|D_{a_K} x_{j-1}(t)\|_\infty \big\},$$

satisfies $\sup_{K,j}\|\mathcal{R}_{j-1}\|_p \le C_p$. *With* $\mathcal{M}_j^v = \mathcal{B}_{a_K}(x_{j-1}, v_j)$ *we have*

$$\sup_{K\ge m\ge l\ge k} \mathbb{E}\,\mathsf{Y}_\alpha^{a_K}\Big(\sum_{j=l}^{m}\mathcal{M}_j^s\Big) \le C g_3 \sum_{j\ge k} a_j^\beta \sqrt{1+j},$$

$$\sup_{K\ge m\ge l\ge k} \mathbb{E}\,\mathsf{Y}_\alpha^{a_K}\Big(\sum_{j=l}^{m}\mathcal{M}_j^z\Big) \le C \sum_{j\ge k}\big[g_3^2 a_j^\alpha (1+j) + g_3^3 a_j^{1/2} p_j^3\big],$$

$$\sup_{K\ge m\ge l\ge k} \mathbb{E}\,\mathsf{Y}_\alpha^{a_K}\Big(\sum_{j=l}^{m}\mathcal{M}_j^r\Big) \le 4 \sum_{j\ge k} W_j.$$

All three right sides tend to zero as $k\to\infty$. *The fields* $\mathcal{M}_j^s$ *and* $\mathcal{M}_j^z$ *have zero conditional mean given* $\mathcal{F}_{j-1}$, *while* $\mathcal{M}_j^r$ *includes all propagated conditional means. These statements include the level* $j = K$, *for which* $I_{a_K\to a_K}$ *is the identity.*

Proof. Let $a = a_K$. By the triangle inequality for the path norm and Minkowski's inequality,

$$\|\mathsf{X}_\alpha^a(d_j)\|_p \le S_{p,j} + Z_{p,j}^+ + R_j + U_{p,j}.$$

By (28.12) and (28.4), the sum of the right side over $j \ge k$ is at most $C_{p,\zeta_1} a_k^{\zeta_1}$ for every $\zeta_1 < \min(\beta, \tau)$. This proves the first bound, uniformly over all partial sums and all K.

On the coarse lattice, of mesh $H = L/n$, we have $\|A_0\|_\infty \le r/H$. By (28.3) and the bounded horizon,

$$\mathsf{X}_\alpha^a(x_0) \le (T+H^2)^{\alpha/2} r/H + C(T+H^2)^{(\alpha+1)/2} r/H^2.$$

This bound is deterministic and independent of K. Adding the sum of the increments from level 1 to level j proves the uniform moment bound on x_j, including the case $K = j = 0$.

For the radius $\mathcal{R}_{j-1}$, let $h = a_j$ and $H_j = Bh$. For $t \ge h^2$ we have $t + h^2 \le 2(t + a^2)$, so the two weighted terms in $\mathcal{R}_{j-1}$ are bounded by a fixed multiple of $\mathsf{X}_\alpha^a(x_{j-1})$. For $t < h^2$, (28.3) applied to A_{j-1}, an array on the mesh H_j with $\|A_{j-1}\|_\infty \le \bar b_{j-1}/H_j$, bounds the sum of the two terms by

$$C\big(h^\alpha + h^{\alpha+1}/H_j\big)\bar b_{j-1}/H_j \le C_B \bar b_{j-1} H_j^{\alpha-1}.$$

This bound is deterministic, bounded uniformly in j, and decays at fine levels by (28.12); at the first level it uses only $\bar b_0 = r$ and the fixed mesh H. Taking pth moments proves the bound on $\mathcal{R}_{j-1}$. At short times the weight is the one at the input scale a_j, not the one at the output mesh a.

For $\mathcal{M}_j^s$ note that the background x_{j-1} is the heat path of a cardinal interpolation from the coarser mesh, so it satisfies the hypothesis on backgrounds of Theorem 26.10. The coarser mesh is $a_{j-1} = Ba_j$, the common fine mesh is a_K, an integer refinement of both

input meshes, the input is A_{j-1}, the radius of the background is $\mathcal{R}_{j-1}$, and the horizon is T. The clause of that theorem on cardinal backgrounds from the coarser mesh gives the bounds on their higher differences with fixed geometric constants; the larger class of backgrounds is needed only for (28.9). The conditional kernel and the convention of Taylor expansion at $w = 0$ are those matched in Step 2 of the proof of Lemma 28.5. For each value of the coarser field we use $\mathcal{R}_{j-1}$ as the number R in the conditional tail bound of that theorem. Integrating the tail in u, which is justified since $\int_0^\infty e^{-u}(1+\sqrt{u})\,du < \infty$, we obtain

$$\mathbb{E}\big[\mathsf{Y}_\alpha^a(\mathcal{M}_j^s) \mid \mathcal{F}_{j-1}\big] \le C g_3 \mathcal{R}_{j-1} a_j^\beta \big[\sqrt{\log(2 + L/a_j)} + 1\big].$$

Using the bound on the first moment of $\mathcal{R}_{j-1}$ and $\log(2 + L/a_j) \le C(1+j)$, we obtain the bound on each summand. The triangle inequality and expectation then sum it over $l \le j \le m$, and the sum is at most the sum over $j \ge k$.

The same background satisfies the hypotheses of Theorem 26.11. Integrating its conditional tail, which has a quadratic part and a remainder, we obtain

$$\mathbb{E}\big[\mathsf{Y}_\alpha^a(\mathcal{M}_j^z) \mid \mathcal{F}_{j-1}\big] \le C\mathcal{R}_{j-1}\big[g_3^2 a_j^\alpha (1+j) + g_3^3 a_j^{1/2} p_j^3\big].$$

Taking expectations and using the moment bound on $\mathcal{R}_{j-1}$, we obtain the second bound. The fields $\mathcal{M}_j^s$ and $\mathcal{M}_j^z$ have zero conditional mean, since their backgrounds are predictable and $s_j(0)$ and $z_j(0)$ have zero conditional mean.

For $\mathcal{M}_j^r$ we use the second estimate in (28.8), which is a conclusion of Theorem 27.6; the triangle inequality gives $4\sum_{j\ge k} W_j$. All three right sides tend to zero as $k \to \infty$, by (28.4), (28.5), and the summability of W_j in Theorem 27.6. At the last level $I_{a_K \to a_K}$ is the identity, and none of these estimates omits that level. □

Theorem 28.7. *Assume the standing assumptions, use the notation of Lemma 28.5, and write $\mathcal{M}_j = \mathcal{B}_{a_K}(x_{j-1}, d_j)$ and $\mathcal{M}_j^v = \mathcal{B}_{a_K}(x_{j-1}, v_j)$ for $v \in \{s, z, r, u\}$. There exist $b \ge 0$ and $C < \infty$ such that, for every $k \ge 1$,*

$$\sup_{K \ge m \ge l \ge k} \mathbb{E}_{\nu_{nBK,n,\gamma}} \mathsf{Y}_\alpha^{a_K}\Big(\sum_{j=l}^m \mathcal{M}_j^u\Big) \le C(1+k)^b a_k^\tau.$$

Further, both

$$\sup_{K \ge m \ge l \ge k} \mathbb{E}\,\mathsf{Y}_\alpha^{a_K}\Big(\sum_{j=l}^m \mathcal{M}_j\Big) \qquad \text{and} \qquad \sup_{K \ge m \ge l \ge k} \mathbb{E}\,\mathsf{Y}_\alpha^{a_K}\Big(\sum_{j=l}^m \mathcal{Q}_{a_K}(d_j, d_j)\Big)$$

tend to zero as $k \to \infty$. In particular, the sums $\sum_{j=l}^m \mathcal{M}_j$ and $\sum_{j=l}^m \mathcal{Q}_{a_K}(d_j, d_j)$ tend to zero in probability in the norm $\mathsf{Y}_\alpha^{a_K}$, at every fixed threshold, uniformly over $K \ge m \ge l \ge k$. The constants have the dependence stated in Section 28.1, with the exponents added.

Proof. Let $a = a_K$, $X_j = \mathsf{X}^a_\alpha(x_{j-1})$, and $\mathcal{U}_j = \mathsf{H}^a_\sigma(I_{a_j \to a} U^0_j)$. The case $p = 2$ of Theorem 28.6 gives $\|X_j\|_2 \le C_2$, uniformly in K and j. The case $p = 2$ of Theorem 27.7, applied under the same conditioned measure to the predictable term, which includes the contributions of the coarse field, the cubic part of the zero-fluctuation reconstruction, and the conditional mean correction, gives

$$\|\mathcal{U}_j\|_2 \le C(1+j)^b a_{j-1}^\tau \le C'(1+j)^b a_j^\tau,$$

since $a_{j-1} = B a_j$ with B fixed. Neither input uses an estimate of Section 34.

We apply Lemma 24.6 separately to $\mathcal{Q}_a(x_{j-1}, u_j)$ and $\mathcal{Q}_a(u_j, x_{j-1})$; both arguments are heat paths on the mesh a, and the argument measured by H^a_σ has initial datum $I_{a_j \to a} U^0_j$. The factor $(T + a^2)^{(\alpha-\sigma)/2}$ in that lemma is bounded uniformly, since $a \le L/n$. By this inequality and the Cauchy–Schwarz inequality under the same measure, we have

$$\mathbb{E}\, \mathsf{Y}^a_\alpha(\mathcal{M}^u_j) \le C\, \mathbb{E}(X_j \mathcal{U}_j) \le C \|X_j\|_2 \|\mathcal{U}_j\|_2 \le C(1+j)^b a_j^\tau.$$

By the triangle inequality for the path norm and (28.4), for every $K \ge m \ge l \ge k$,

$$\mathbb{E}\, \mathsf{Y}^a_\alpha\Big(\sum_{j=l}^{m} \mathcal{M}^u_j\Big) \le C \sum_{j \ge k} (1+j)^b a_j^\tau \le C'(1+k)^b a_k^\tau.$$

This proves the first assertion.

By Lemma 28.5(1), $\mathcal{M}_j = \mathcal{M}^s_j + \mathcal{M}^z_j + \mathcal{M}^r_j + \mathcal{M}^u_j$. Combining the three bounds on first moments in Theorem 28.6 with the bound just proved, we obtain a bound on the first moment of the sum of the $\mathcal{M}_j$ that tends to zero as $k \to \infty$. The means of $\mathcal{M}^r_j$ are included in W_j by Theorem 27.6. At fixed coupling $p_j \le C(1+j)$ by (28.5), so all the sums of powers converge and their tails tend to zero, and the same holds for $\sum_j W_j$ by Theorem 27.6.

For the quadratic increments, let $0 < \zeta_2 < \min(2\alpha - 1, \tau)$. By Lemma 28.5(6), at every level the sum of the expected norms of the sixteen ordered products is at most $C_{\zeta_2}(1+j)^b a_j^{\zeta_2}$. We expand $\mathcal{Q}_a(d_j, d_j)$ by bilinearity, using $d_j = s_j + z_j + r_j + u_j$, apply the triangle inequality, and sum the geometric series (28.4). This proves the last limit, with the bound $C_{\zeta_2}(1+k)^b a_k^{\zeta_2}$. Markov's inequality at a fixed positive threshold implies the two statements in probability. All these operations take place under the same conditioned measure and on the same fine mesh, including the level $j = K$. □

The next theorem combines the two previous results. Here the finite-scale reconstruction A_j is extended to the fine mesh by cardinal interpolation, not by further reconstruction with zero fluctuations, and the quadratic operator is the same ordered $\mathcal{Q}_{a_K}$ for every j.

Theorem 28.8. *Assume the standing assumptions, with the norms taken on a fixed horizon* $[0, T_0]$, $0 < T_0 < \infty$, *and let* $\nu_K = \nu_{nB^K,n,\gamma}$. *Let the arrays* A_j, $0 \le j \le K$, *be those of the coarse-to-fine reconstruction of the fine configuration, at the classical depths* $J(K-j)$, *and put*

$$\mathcal{E}_{K,j} = (x_j, y_j), \qquad x_j = P^{a_K}_{\cdot} I_{a_j \to a_K} A_j, \qquad y_j = \mathcal{Q}_{a_K}(x_j, x_j),$$

$d_j = x_j - x_{j-1}$ *for* $1 \le j \le K$, *and* $\mathsf{R}_K = \|\mathcal{E}_{K,K}\|_{\mathrm{fl},a_K} = \mathsf{X}^{a_K}_\alpha(x_K) + \mathsf{Y}^{a_K}_\alpha(y_K)$, *as in Definition 24.3. Then, for* $1 \le k \le K$,

$$\mathcal{E}_{K,K} - \mathcal{E}_{K,k-1} = \Big(\sum_{j=k}^{K} d_j,\ \sum_{j=k}^{K} \big\{ \mathcal{B}_{a_K}(x_{j-1}, d_j) + \mathcal{Q}_{a_K}(d_j, d_j) \big\} \Big). \tag{28.13}$$

For every $\eta > 0$,

$$\lim_{k \to \infty} \sup_{K \ge k} \nu_K \big\{ \mathsf{X}^{a_K}_\alpha(x_K - x_{k-1}) + \mathsf{Y}^{a_K}_\alpha(y_K - y_{k-1}) > \eta \big\} = 0, \tag{28.14}$$

and $\rho(R) = \sup_{K \ge 0} \nu_K\{\mathsf{R}_K > R\}$ *tends to zero as* $R \to \infty$. *All norms include the physical first differences and shifts. For each fixed* q, *the laws of the arrays* A_q *converge, on a fixed compact set, to the law* π_q *(Theorem 15.13), and their smooth cardinal interpolants satisfy the uniform consistency and the deterministic norm bound of Lemma 28.4. For every sample, the configuration* $e^{a_K A_K}$ *on* Λ_{nB^K} *is gauge equivalent to the sampled configuration* U. *The constants and the rates of convergence in these statements are taken at fixed coarse side and fixed coupling, uniformly in* K.

Proof. Summing (28.6) over $k \le j \le K$ gives (28.13). We apply Theorems 28.6 and 28.7 with $T = T_0$. The first component in (28.14) tends to zero by Markov's inequality and Theorem 28.6. For the second component we apply Theorem 28.7 separately to the two sums in (28.13), with threshold $\eta/3$ for each of the three contributions, and use a union bound. This proves (28.14).

For the tightness, let $\varepsilon > 0$ and choose k such that the probability in (28.14) with $\eta = 1$ is less than ε for all $K \ge k$. The array A_{k-1} lies in a fixed compact set, and $x_{k-1} = P^{a_K}_{\cdot}(f|_{a_K})$, where f is its cardinal interpolant. By Lemma 28.4, $\|\mathcal{E}_{K,k-1}\|_{\mathrm{fl},a_K} \le C_k$ for a deterministic C_k that does not depend on K. By the triangle inequality, $\mathsf{R}_K \le C_k + 1$ outside the exceptional event. The finitely many $K < k$ have bounded arrays, so their linear heat evolutions and quadratic Picard terms have a common finite bound. This proves that $\rho(R) \to 0$. The remaining assertions, including the gauge equivalence of $e^{a_K A_K}$ and U, are Theorem 15.13 applied to the family $\widehat{\mathscr{R}}$ at the classical depths $J(K-j)$. □

28.5 Convergence under the conditioned measure

We first state an approximation lemma for laws, a version of the standard approximation theorem for weak convergence (compare [28, Theorem 3.2]). In that theorem the limit of

the approximating laws is assumed to exist; here its existence is part of the conclusion, so we include the proof. For a metric space E let $\mathscr{L}$ be the set of functions $f : E \to \mathbb{R}$ with $\|f\|_\infty \le 1$ and Lipschitz constant at most one, and let $d_{\mathrm{BL}}(\mu,\nu) = \sup_{f\in\mathscr{L}} \left|\int f\,d\mu - \int f\,d\nu\right|$ be the bounded-Lipschitz distance between probability measures on E.

Lemma 28.9. *Let (E,d) be a complete separable metric space. For each $K \ge 0$ let Z_K be an E-valued random variable with law μ_K, and for $0 \le k \le K$ let $Z_{K,k}$ be E-valued random variables defined on the same probability space as Z_K. Suppose that for each fixed k the laws of $Z_{K,k}$ converge weakly as $K \to \infty$, that the union over $K \ge k$ of the supports of these laws has compact closure, that each μ_K has compact support, and that for every $\eta > 0$*

$$\lim_{k\to\infty} \sup_{K\ge k} \mathbb{P}\{d(Z_K, Z_{K,k}) > \eta\} = 0.$$

Then μ_K converges weakly to a probability measure on E.

Proof. We show that (μ_K) is a Cauchy sequence for d_{BL} and is tight, and then identify the limit.

Step 1: the Cauchy property. For $f \in \mathscr{L}$ and $\eta > 0$ we have

$$|\mathbb{E} f(Z_K) - \mathbb{E} f(Z_{K,k})| \le \eta + 2\mathbb{P}\{d(Z_K, Z_{K,k}) > \eta\}.$$

Fix k, and let C_k be the compact closure of the union of the supports of the laws of $Z_{K,k}$, $K \ge k$. Given $\varepsilon > 0$, let $x_1, \dots, x_N$ be a finite ε-net of C_k. We group the functions $f \in \mathscr{L}$ according to the values $f(x_1), \dots, f(x_N)$ rounded to multiples of ε. There are finitely many groups, and two functions in the same group differ by at most 3ε on C_k. Choosing one representative in each group and applying weak convergence to the finitely many representatives, we find that $\sup_{f\in\mathscr{L}} |\mathbb{E} f(Z_{K,k}) - \mathbb{E} f(Z_{K',k})| \le 6\varepsilon + o(1)$ as $K, K' \to \infty$. Combining the two bounds and letting first $K, K' \to \infty$ with k fixed, then $k \to \infty$, and then $\eta, \varepsilon \to 0$, we obtain that (μ_K) is a Cauchy sequence for d_{BL}.

Step 2: tightness. Fix $\varepsilon > 0$, and choose increasing k_h such that $\mathbb{P}\{d(Z_K, Z_{K,k_h}) > 2^{-h}\} \le \varepsilon 2^{-h}$ for all $K \ge k_h$. Let C_h be the compact closure of the union of the supports of the laws of Z_{K,k_h}, $K \ge k_h$, together with the supports of the finitely many μ_K with $K < k_h$. For $K \ge k_h$ we have $d(Z_K, C_h) \le 2^{-h}$ unless $d(Z_K, Z_{K,k_h}) > 2^{-h}$, and for $K < k_h$ we have $Z_K \in C_h$ almost surely. By the union bound, the closed set

$$C_\varepsilon = \bigcap_{h\ge 1} \{z : d(z, C_h) \le 2^{-h}\}$$

has μ_K-mass at least $1 - \varepsilon$ for every K. It is totally bounded: for a given tolerance, choose h large and a finite net of the compact set C_h. Since C_ε is closed in a complete space, it is compact. Hence (μ_K) is tight.

Step 3: the limit. By Prokhorov's theorem [138], in the direction that holds on every metric space [28, Theorem 5.1], every subsequence of (μ_K) has a further subsequence that converges weakly to a probability measure. By Step 1, $\int f\,d\mu_K$ converges for every $f \in \mathscr{L}$, so any two subsequential limits μ and μ' satisfy $\int f\,d\mu = \int f\,d\mu'$ for all bounded Lipschitz f. For a closed set F, the functions $\max(0, 1 - m\,d(\cdot, F))$ decrease to $\mathbf{1}_F$ as $m \to \infty$, so $\mu(F) = \mu'(F)$. Closed sets form a π-system generating the Borel σ-algebra, so $\mu = \mu'$. Since every subsequence has a further subsequence converging weakly to the same limit, the whole sequence converges weakly to it. □

Proof of Theorem 28.1. We compare W_K with the Wilson loops of the flow started from the finite-scale reconstruction from the first q RG steps. By Theorem 28.8 and the stability of flowed Wilson loops, this approximation is uniform in K. We then identify the limit law of the approximation at fixed q and apply Lemma 28.9. We fix a horizon $T_0 > 0$ and use Theorem 28.8 with it.

Step 1: comparison with finite-scale reconstructions. On the same probability space, let $W_K^{[q]}$ be the vector of the same characters of holonomies for the Wilson flow on Λ_{nB^K} started from

$$U_K^{[q]} = \exp\bigl(a_K I_{a_q \to a_K} A_q\bigr),$$

and let $D_{K,q} = \|\mathcal{E}_{K,K} - \mathcal{E}_{K,q}\|_{\mathrm{fl},a_K}$. The Wilson flow commutes with lattice gauge transformations and characters of holonomies are gauge invariant. Since $e^{a_K A_K}$ is gauge equivalent to U (Theorem 28.8), W_K is also the vector of Wilson loops of the flow started from $e^{a_K A_K} = e^{a_K x_K(0)}$, while $U_K^{[q]} = e^{a_K x_q(0)}$. On the event $\{\mathrm{R}_K \le R,\ D_{K,q} \le \eta \le 1\}$ both flow data $\mathcal{E}_{K,K}$ and $\mathcal{E}_{K,q}$ have norm at most $R + 1$. By Corollary 25.11 there exist $a_* > 0$ and C_R such that $|W_K - W_K^{[q]}| \le C_R \eta$ on this event whenever $a_K \le a_*$. These constants depend on R, the loops, the characters, the times, and the deterministic data of the flow, but not on q, on the higher derivatives of the finite-scale reconstruction, or on the coupling. Thus for $f \in \mathscr{L}$ the replacement error is at most

$$C_R \eta + 2\rho(R) + 2\nu_K\{D_{K,q} > \eta\}. \tag{28.15}$$

We choose R first, then η, and then q, increasing q until $a_q \le a_*$; then $a_K \le a_*$ for all $K \ge q$. By Theorem 28.8, the last term in (28.15) tends to zero as $q \to \infty$, uniformly in $K \ge q$. In particular, for every $\eta' > 0$, $\lim_{q\to\infty} \sup_{K \ge q} \nu_K\{|W_K - W_K^{[q]}| > \eta'\} = 0$.

Step 2: the limit at fixed q. Fix q. The vector $W_K^{[q]}$ is a function of the array A_q, which lies in a fixed compact set. By Lemma 28.4, applied with $h = a_q$, these functions converge as $K \to \infty$, uniformly on that compact set, to the continuous map G_q of the statement. By Theorem 15.13 the law of A_q converges weakly to π_q. Together, these two facts imply that $\mathrm{Law}(W_K^{[q]}) \Rightarrow (G_q)_\# \pi_q$. All the vectors W_K and $W_K^{[q]}$ take values in the compact

box $\prod_i \{w \in \mathbb{C} : |w| \le \chi_i(I)\}$. We apply Lemma 28.9 in this box, with $Z_K = W_K$ and $Z_{K,q} = W_K^{[q]}$; its hypotheses were verified in Step 1 and above. Hence the law of W_K converges weakly. Applying the same argument to the union of two lists of loops proves the consistency under finite unions and projections. Taking $K \to \infty$ at fixed q in the bound (28.15), we find that the d_{BL}-distance between the limit law of W_K and $(G_q)_\# \pi_q$ is at most $C_R \eta + 2\rho(R) + 2 \sup_{K \ge q} \nu_K\{D_{K,q} > \eta\}$. This tends to zero in the order R, η, q chosen above, which identifies the limit law of W_K as the limit of $(G_q)_\# \pi_q$.

Step 3: moment generating functions. Let $D \ge \max_i \chi_i(I)$. On the box, $|\exp(z \cdot w)| \le e^{D\|z\|_1}$, and on $\{\|z\|_1 \le Z\}$ the function $z \mapsto \exp(z \cdot w)$ is Lipschitz with constant at most De^{DZ}, uniformly in w. Weak convergence of the laws of W_K implies convergence of the moment generating functions at each z. With the common Lipschitz constant and a finite net of z, we obtain uniform convergence on $\{\|z\|_1 \le Z\}$. On the ball where $e^{D\|z\|_1} - 1 < 1/2$, we have $|\mathbb{E} \exp(z \cdot W_K) - 1| \le e^{D\|z\|_1} - 1 < 1/2$, so the logarithm with value zero at the origin is defined and holomorphic. Cauchy's formula on a smaller ball gives the convergence of every finite logarithmic derivative, that is, of the cumulants. □

28.6 Removal of the small-field conditioning

Proof of Theorem 28.2. We first bound the small-field failure probability, then compare conditioned and unconditioned expectations at the same fine lattice, and finally pass to the limit, first in K and then in r.

Step 1: the failure probability. Theorem 23.9 implies that, for every $u \in 4\mathbb{N}$ with $u \ge n_0$ and every $K' \ge 0$, we have $\mu_{uB^{K'},\gamma}(S^c_{uB^{K'},u,\gamma}) \le F_\gamma(u)$, uniformly in K'. For fixed $\gamma > 0$ both terms of $F_\gamma(u)$ tend to zero as $u \to \infty$. For the second term, with $v = \log u$, the exponent is $3v - (a_{\mathrm{w}}/4)(v + \log(M/\gamma))^2$, which tends to $-\infty$. Let r_0 be such that $n_r \ge n_0$ and $F_\gamma(n_r) < 1$ for $r \ge r_0$. For $r \ge r_0$ and $K \ge r$ the conditioned hierarchy with coarse side n_r has $K - r$ RG steps and fine side $n_r B^{K-r} = n_* B^K = N_K$. Applying the bound with $u = n_r$ and $K' = K - r$, we obtain $e_{K|r} \le F_\gamma(n_r)$. This bounds the probability of the complement of the event that defines $\nu_{K|r}$, including its initial test and the tests at the end of each RG step.

Step 2: conditioning at a fixed fine lattice. For $K \ge r$, the observable $W_K = (W_{K,1}, \dots, W_{K,m})$ under $\nu_{K|r}$ is the same measurable function of the configuration on Λ_{N_K} as under $\mu_{N_K,\gamma}$. Lemma 5.5, applied to the measure $\mu_{N_K,\gamma}$, the event $S_{N_K,n_r,\gamma}$, and this map, gives, for every bounded complex measurable f on the range of W_K,

$$\left|\mathbb{E}_{\mu_{N_K,\gamma}} f(W_K) - \mathbb{E}_{\nu_{K|r}} f(W_K)\right| \le 2\|f\|_\infty e_{K|r}. \tag{28.16}$$

The events for different r need not be nested.

Step 3: the Cauchy property of the limits. We take $f(w) = \exp(\sum_i z_i w_i)$, so that $\|f\|_\infty \le e^{DR}$ on $\{\sum_i |z_i| \le R\}$. For $K \ge \max(r, r')$, we compare both conditioned expectations with the same unconditioned expectation in (28.16) and obtain

$$\sup_{\sum_i |z_i| \le R} \left| \mathbb{E}_{\nu_{K|r}} e^{\sum_i z_i W_{K,i}} - \mathbb{E}_{\nu_{K|r'}} e^{\sum_i z_i W_{K,i}} \right| \le 2e^{DR} \left(e_{K|r} + e_{K|r'} \right).$$

At fixed r and r' the conditioned moment generating functions converge uniformly by Theorem 28.1, applied with coarse side n_r, respectively $n_{r'}$. Letting $K \to \infty$ and using Step 1, we obtain the estimate of the theorem for Z_r^{sel} and $Z_{r'}^{\mathrm{sel}}$. In particular these functions form a uniform Cauchy family on every ball as $r, r' \to \infty$.

Step 4: the limit law. We show that the laws of W_K under $\mu_{N_K,\gamma}$ converge. For $r \ge r_0$ let ρ_r be the weak limit of the laws of W_K under $\nu_{K|r}$, which exists by Theorem 28.1. For $f \in \mathscr{L}$, by (28.16) and Step 1, we have $|\mathbb{E}_{\mu_{N_K,\gamma}} f(W_K) - \mathbb{E}_{\nu_{K|r}} f(W_K)| \le 2F_\gamma(n_r)$ for all $K \ge r$. Letting $K \to \infty$, we obtain

$$\limsup_{K,K' \to \infty} \left| \mathbb{E}_{\mu_{N_K,\gamma}} f(W_K) - \mathbb{E}_{\mu_{N_{K'},\gamma}} f(W_{K'}) \right| \le 4F_\gamma(n_r),$$

and the right side tends to zero as $r \to \infty$. Hence $\mathbb{E}_{\mu_{N_K,\gamma}} f(W_K)$ converges for every bounded Lipschitz f. The vectors W_K take values in the compact box $\prod_i \{w \in \mathbb{C} : |w| \le \chi_i(I)\}$, so every subsequence of their laws has a weakly convergent further subsequence. All these subsequential limits agree on bounded Lipschitz functions, and therefore coincide, as in Step 3 of the proof of Lemma 28.9. Hence the laws of W_K under $\mu_{N_K,\gamma}$ converge weakly to a unique probability measure ρ on the box. Passing to the limit $K \to \infty$ in the bound above also gives $d_{\mathrm{BL}}(\rho, \rho_r) \le 2F_\gamma(n_r)$.

Step 5: moment generating functions and consistency. Weak convergence to ρ implies pointwise convergence of the unconditioned moment generating functions. Their common Lipschitz constant on $\{\sum_i |z_i| \le R\}$ is at most De^{DR}, so the finite-net argument of the proof of Theorem 28.1 gives uniform convergence on this set. Further, letting $K \to \infty$ in (28.16) at fixed r, we find

$$\sup_{\sum_i |z_i| \le R} \left| \int e^{\sum_i z_i w_i} \, d\rho(w) - Z_r^{\mathrm{sel}}(z) \right| \le 2e^{DR} F_\gamma(n_r),$$

which shows how the error at a fixed coarse side disappears as $r \to \infty$. For a subfamily of the loops, the laws at each fixed K are coordinate projections of the laws of the larger family. Coordinate projection is continuous on the compact box, so the weak limits have the same relation. This proves the consistency under taking subfamilies. Along the limit only the coarse side n_r changes, within the single sequence $n_* B^r$. □

28.7 Compact intervals of couplings and couplings below a finite bound

The results above use the parameters of Theorem 23.2, chosen for couplings $0 < \gamma \le \gamma_0 \le 1$, and their constants depend on the fixed coupling. This section gives two extensions. The first concerns the parameters of Theorem 23.5, chosen for couplings up to an arbitrary finite bound Γ, and shows that the truncation and tightness estimates hold uniformly for γ in a compact interval $[\gamma_-, \gamma_+] \subset (0, \Gamma]$. This uniformity requires checking how each constant depends on the coupling; taking the supremum of constants obtained at fixed coupling would not suffice. The second extension states that all fixed-coupling results of this section hold for these parameters at every fixed $\gamma \le \Gamma$.

Lemma 28.10. *Fix $L > 0$, an odd integer $M \ge 3$, and $0 < \Gamma < \infty$, and let J, $B = M^J$, Θ, g_*, and n_Γ be as in Theorem 23.5. Fix a horizon $T_0 > 0$ and an interval $I = [\gamma_-, \gamma_+] \subset (0, \Gamma]$ with $\gamma_- > 0$, and fix exponents $1/2 < \alpha < 2/3$, $\beta = \alpha - 1/2$, $0 < \sigma < 1 - \alpha$, and $0 < \tau < \min\{\sigma, 1/2, \lambda_{\mathrm{anc}}\}$. There is a threshold $n_I \ge n_\Gamma$, which does not depend on $\gamma \in I$, such that the following holds for every coarse side $n \in 4\mathbb{N}$ with $n \ge n_I$. Put $H = L/n$ and $a_j = HB^{-j}$. Under $\nu_{nB^k,n,\gamma}$ use the arrays $A_j = \widehat{\theta}_j / a_j$ of the coarse-to-fine reconstruction, and on the fine mesh $a = a_k$ set*

$$x_j = P^a_\cdot I_{a_j \to a} A_j, \qquad y_j = \mathcal{Q}_a(x_j, x_j), \qquad d_{k,q} = \mathsf{X}^a_\alpha(x_k - x_q) + \mathsf{Y}^a_\alpha(y_k - y_q), \quad 0 \le q \le k,$$

with norms on $[0, T_0]$ that include all first differences and shifts. Then every estimate of Lemma 28.5 and Theorems 28.6 and 28.7 holds with constants that are uniform for $\gamma \in I$, and with the coupling factors displayed there. In particular, for each $\eta > 0$ there exist $b < \infty$, $C_{I,\eta} < \infty$, $0 < \xi < \tau$, and $\delta > 0$ such that

$$\sup_{\gamma \in I} \sup_{k \ge q} \nu_{nB^k,n,\gamma}\{d_{k,q} > \eta\} \le C_{I,\eta}\big[(1+q)^b (a_q^\delta + a_q^{\tau-\xi}) + a_q^\xi\big], \tag{28.17}$$

and the right side tends to zero as $q \to \infty$. Further, the norms of the flow data are tight uniformly in $\gamma \in I$:

$$\lim_{R \to \infty} \sup_{\gamma \in I,\, k \ge 0} \nu_{nB^k,n,\gamma}\big\{\mathsf{X}^{a_k}_\alpha(x_k) + \mathsf{Y}^{a_k}_\alpha(y_k) > R\big\} = 0. \tag{28.18}$$

All constants may depend on I, L, n, M, J, Θ, T_0, the exponents, and the moment orders. They do not depend on $\gamma \in I$, k, the classical depth, or the ratio of the fine mesh to a_j. For a one-point interval $I = \{\gamma\}$ this gives the assertions at every fixed $\gamma \in (0, \Gamma]$.

Proof. We go through the inputs of Lemma 28.5 and Theorems 28.6 and 28.7 and check that their constants, including the constants in the conditional maximal estimates, are uniform on I. Let $g_\pm = \sqrt{\gamma_\pm / L}$ and $\kappa = a_{\mathrm{cap}}\sqrt{B} < 1$.

Step 1: cutoffs and amplitude bounds. The cutoffs in the definition of the small-field event differ from $p(t_j) = A_{\mathrm{w}} \log(1/t_j)$ by at most a fixed c_J. Let $p_{\max,j}$ be the largest of them in the jth RG step. We have

$$t_j^2 = \frac{\gamma}{MnB^{j-1}}, \qquad p(t_j) = \frac{A_{\mathrm{w}}}{2}\big[\log(Mn/\gamma) + (j-1)\log B\big], \qquad p_{\max,j} \le P_I(1+j),$$
$$\frac{\bar b_j}{\sqrt{a_j}} \le C\big[rH^{-1/2}\kappa^j + g_+ P_I(1+j)\big], \qquad \frac{\bar b_j}{g_j} \le C\big[r(g_-\sqrt{H})^{-1}\kappa^j + P_I(1+j)\big]. \tag{28.19}$$

Here the amplitude bounds are those of Lemmas 17.1 and 15.10; we used $ra_{\mathrm{cap}}^j/\sqrt{a_j} = rH^{-1/2}\kappa^j$, and the first level is included by enlarging C. For every fixed integer v we have $p_{\max,j}^v \le C_{I,v} a_j^{-1}$, since $\sup_j HB^{-j}(1+j)^v < \infty$. None of these bounds requires $\gamma_+ \le 1$.

Step 2: hypotheses of the conditional estimates. The conditional fluctuation potential, the lower bound $I/2$ on its Hessian in w, the conditional means, the local bounds on derivatives of the conditional log-density, and the volume-normalized column sums are given by Theorems 13.8, 13.10, and 15.12 and Lemma 26.1. Their hypotheses are the fixed domains and finitely many smallness conditions on $t[p(t) + c_J]^v$, including (26.1). We choose a small $t_* > 0$ such that these conditions hold for $0 < t \le t_*$. We require $n \ge \gamma_+/(Mt_*^2)$, so that $t_1^2 = \gamma/(Mn) \le t_*^2$ for all $\gamma \in I$, together with the thresholds on the coarse side and on the levels in Theorem 23.5; this defines n_I. Then every conditional fluctuation kernel at a finer level satisfies these hypotheses, for every value of the coarser field. Factors that depend only on coarser fields cancel in a normalized conditional kernel; no factor that depends on w is dropped.

Step 3: uniform maximal constants. In Lemma 26.9 the deterministic moduli contain only the fixed coefficients, powers of g_3, and powers of $p_{\max,j}$. By Step 1 they give the same normalized modulus $C_I a_j^{-30}$ in space and time as in the proof of that lemma. A physical net of spacing $c_I a_j^{30}$ has at most $C_I(1 + La_j^{-30})^3(1 + T_0 a_j^{-30})$ points, and the logarithm of this number is at most $C_I \log(2 + L/a_j)$. The part with short times is estimated at the outputs on the mesh a_j before this net is used. The conditional linear and quadratic concentration of Theorem 5.4, with the Hessian lower bound $1/2$, followed by a union bound and the deterministic modulus, gives maximal constants that are uniform on I. Alternatively, Lemmas 26.7 and 26.8 already give entropy bounds for the coefficients that do not depend on the coupling.

Let $\widetilde{x}_j = x_{j-1} + r_j$. The three ordered terms $\mathcal{Q}_a(\widetilde{x}_j, r_j)$, $\mathcal{Q}_a(r_j, \widetilde{x}_j)$, and $\mathcal{Q}_a(r_j, r_j)$ of Theorem 27.6, with their propagation through the coarser levels as in that theorem, are estimated separately there. The cross term satisfies $\mathcal{B}_a(x_{j-1}, r_j) = \mathcal{Q}_a(\widetilde{x}_j, r_j) + \mathcal{Q}_a(r_j, \widetilde{x}_j) - 2\mathcal{Q}_a(r_j, r_j)$, which explains the factor 4 in (28.8). For each of these three terms the ratio of the contribution of short times to the scale of the estimate is at most $C_I(1+j)a_{j-1}^4$ (see (27.28)). A single threshold on the level makes it small, and the finitely many coarser

levels are handled by increasing C_I. The normalized modulus is $C_I a_{j-1}^{-7}$, and a net of spacing $c_I a_{j-1}^{12}$ has at most $C_I a_{j-1}^{-48}$ points (see (27.29)). Conditional concentration, the union bound, and the modulus give the same conditional moments as in that theorem, now uniformly on I. In Theorem 27.7 the normalized modulus is $C_I a_j^{-4}$, and a net of spacing $c_I a_j^8$ has at most $C_I a_j^{-32}$ points (see the estimates following (27.32)); this covers its centered terms. The estimates of Lemma 27.1 and the coefficient estimates of Theorem 27.5 are deterministic under the same amplitude bounds. Their conditional drifts keep the explicit terms

$$a_{\mathrm{cap}}^2 \bar{b}_{i-1}^3 + g_i^2 \bar{b}_{i-1} + g_i^3 p_{\max,i}^3.$$

These arguments use $g_- > 0$ only to bound the inverse coupling in $\bar{b}_j / g_j$, and $g_+ < \infty$ only in upper bounds.

Step 4: the factor bounds. By Steps 1–3, the four-term decomposition of Lemma 26.2 and Theorems 26.10 to 26.12 give, in the notation of Lemma 28.5,

$$\begin{aligned}
S_{p,j} &\le C_{I,p}(1+j)^{1/2} a_j^{\beta}, \\
Z_{p,j}^{+} &\le C_{I,p}(1+j)^{3}(a_j^{\alpha} + a_j^{1/2}), \\
Z_{p,j}^{-} &\le C_{I,p}(1+j)^{3}(a_j^{\sigma} + a_j^{1/2}), \\
R_j &\le C_I(1+j) a_j^{\beta}, \\
U_{p,j} &\le C_{I,p}(1+j)^{b} a_j^{\tau}, \\
\sum_{j \ge q} W_j &\le C_{I,\zeta}(1+q)^{b} a_q^{\zeta} \qquad (0 < \zeta < \min(2\beta, \lambda_{\mathrm{anc}})).
\end{aligned}$$

For the last line we insert (28.19) into the sum over earlier levels in Theorem 27.6, with factor q_{anc}^{j-1-i}, keeping the term from the coarse field. A convolution of two geometric sequences multiplied by polynomials decays with every exponent strictly below the smaller of the two exponents. Explicitly, if $\lambda = -\log q_{\mathrm{anc}} / \log B$, $\varrho > \zeta$, and $\lambda > \zeta$, then

$$\frac{q_{\mathrm{anc}}^{j-1-i} a_i^{\varrho}}{a_j^{\zeta}} = H^{\varrho-\zeta} B^{\zeta} B^{-(\lambda-\zeta)(j-1-i)} B^{-(\varrho-\zeta)i}.$$

We bound the sum over i by dropping the first geometric factor, which is at most one, and summing the second. The polynomial factors contribute at most $(1+j)^b$. The term from the coarse field is geometric by itself.

Step 5: the ordered products. With these factors, each of the sixteen ordered products is bounded by a constant, uniform on I, times the entry of the table in Lemma 28.5(5). Indeed, the proof of that table uses only the product inequality, whose constant does not depend on the coupling, and the following inputs. For (s, s) it uses Theorem 26.12,

including the conditional mean; for (s,r) and (r,s), the two ordered estimates with a predictable background from Theorem 26.10; for (r,r), the separate estimate for $\mathcal{Q}_a(r_j,r_j)$ in Theorem 27.6; for (u,u), the pathwise bound on $\mathcal{U}_j$ and one moment; and for the remaining products, conditional estimates, the deterministic bound on r_j, or the Cauchy–Schwarz inequality under the same measure. No product of first moments of unrelated quantities occurs. The hypotheses of all these inputs were verified above, without an upper bound $\gamma \le 1$. Every entry is summable with a positive power of a_j, after an arbitrarily small loss in the exponent.

Step 6: the truncation error. By Minkowski's inequality applied to $d_j = s_j + z_j + r_j + u_j$, we obtain the bounds on partial sums of the increments of the linear heat evolution, and all fixed moments of $\mathsf{X}^q_\alpha(x_j)$, uniformly on I. The term of the coarse field is bounded deterministically by $C_{H,T_0,\alpha} r/H$. At the fixed order $p = 2$ the factor bounds have constants uniform on I, and the constant of the product inequality does not depend on the coupling. Hence the proof of Theorem 28.7 gives its bounds on first moments uniformly on I: the cross terms with s_j and z_j are bounded by their conditional estimates times the moments of the background radius, the cross term with r_j is bounded by $4W_j$, and the cross term with u_j is bounded, by the Cauchy–Schwarz inequality, by $C_I(1+j)^b a_j^\tau$. By (28.4), for every $v > 0$,

$$\sum_{j\ge q} (1+j)^b a_j^v \le (1+q)^b a_q^v \sum_{h\ge 0} (1+h)^b B^{-hv} < \infty.$$

We apply this to every cross term and every ordered product, with $\delta > 0$ smaller than all the positive exponents that occur, and with $\delta \le \tau$.

The identity

$$y_k - y_q = \sum_{j=q+1}^{k} \{\mathcal{B}_a(x_{j-1},d_j) + \mathcal{Q}_a(d_j,d_j)\}$$

follows from (28.6). Together with $x_k - x_q = \sum_{j=q+1}^k d_j$ and Markov's inequality, it implies that $\sup_{\gamma\in I} \sup_{k\ge q} \nu_{nB^k,n,\gamma}\{d_{k,q} > \eta\} \le C_{I,\eta}(1+q)^b a_q^\delta$. Since $a_q \le H$, the right side of (28.17) is at least a constant multiple of $(1+q)^b a_q^\delta$, for every choice of $0 < \xi < \tau$. This proves (28.17).

Step 7: tightness. For fixed q the array A_q lies in a deterministic compact ball of a finite-dimensional space, uniformly for $\gamma \in I$, by (28.19). By Lemma 28.4, its flow data on every finer lattice have a deterministic norm bound C_q. We choose q so that the probability in (28.17) with $\eta = 1$ is as small as desired, uniformly for $\gamma \in I$ and $k \ge q$. Outside this event the norm $\mathsf{X}^{a_k}_\alpha(x_k) + \mathsf{Y}^{a_k}_\alpha(y_k)$ is at most $C_q + 1$. The finitely many $k < q$ have bounded arrays and bounded norms, uniformly on I. This proves (28.18). □

Lemma 28.11. *Fix $L > 0$, an odd integer $M \ge 3$, and $0 < \Gamma < \infty$, let J, $B = M^J$, Θ, g_*, and n_Γ be as in Theorem 23.5, and fix $0 < \gamma \le \Gamma$. Then Lemmas 28.4 and 28.5 and Theorems 28.1,*

28.2, and 28.6 to 28.8 hold with these parameters in place of those of Theorem 23.2, for coarse sides $n \in 4\mathbb{N}$ with $n \geq n_\Gamma$ above finitely many further thresholds, which may depend on γ. They hold for the same conditional fluctuation kernels, conditional means, laws of finite-scale reconstructions, reconstruction maps, and configurations of the Wilson measure, and their constants have the dependence on the fixed coupling stated in Section 28.1. In Theorem 28.2 the constants of F_γ are those of Theorem 23.10. No bound on the norms of the flow data, and no rate of convergence, that is uniform over $\gamma \in (0, \Gamma]$ is asserted.

Proof. From Theorems 23.5 and 23.10 we use only the common smallness conditions and the bound on the small-field failure probability. The convergence of the laws of finite-scale reconstructions is Theorem 15.13. The smallness condition at the first RG step is $t_1^2 = \gamma/(Mn) \leq g_*^2$, and $t_j = t_1 B^{-(j-1)/2}$ for $j \geq 1$. The local estimates on coefficients and means in Lemmas 27.1 and 26.2 and Theorems 26.10 to 26.12, 27.6, and 27.7 use these conditions and the same product domains. Their summations over levels use only $p_j \leq C_{\gamma,n,J}(1+j)$, which holds at fixed positive coupling. Thus the factor bounds of Lemma 28.5 and Theorems 28.6 and 28.7 keep their proofs and constants. Lemma 28.4 is deterministic and does not involve the coupling. The proofs of Theorems 28.1 and 28.8 use only these results, the convergence of the laws of finite-scale reconstructions, and deterministic flow estimates, which do not require a bound on the coupling. The proof of Theorem 28.2 uses Theorem 23.10 in place of Theorem 23.9, at coarse sides n_r tending to infinity; the bound (23.13) has the same form as the bound used there. □

PART IX

The continuum measure

SECTION 29

The continuum measure on gauge orbits

In this section we construct the continuum measure $\lambda_{L,\gamma,[n]_M}$ on the trajectory space $\mathcal{X}_L$ of Definition 2.2 and prove Theorem 2.3. The main result is Theorem 29.28. Fix $L > 0$, $\gamma > 0$, an odd integer $M \geq 3$, and $n \in 4\mathbb{N}$. Along the full sequence of lattice sides $N_k = nM^k$, every finite vector of flowed Wilson loops, evaluated on admissible approximations of loops in the sense of Definition 2.1, converges in law under the Wilson measure $\mu_{N_k,\gamma}$, jointly with the lattice energies at positive times. The limits are the joint laws of the corresponding continuum observables under one Borel probability measure on $\mathcal{X}_L$. This measure is the limit of the images of $\mu_{N_k,\gamma}$ under Borel maps π_k that are invariant under all lattice gauge transformations. The energies of the trajectories $\pi_k(U)$ are tight at every positive time, and the measure depends only on L, γ, and the class $[n]_M$ of n under the relation $nM^a = n'M^b$.

There are three inputs. The first is the convergence of the joint laws of flowed Wilson loops under the conditioned Wilson measures at a fixed coarse side, proved in Section 28, together with the description of these limits through continuum flows started from finite-scale reconstructions (Theorems 28.1 and 28.8). Combined with the bounds of Section 23 on the small-field failure probability, these results give convergence of finite families of flowed loops under the Wilson measures (Theorem 28.2). The second is the deterministic theory of the Wilson flow and its comparison with the Yang–Mills heat flow, from Sections 24 and 25. The third is the trajectory-space framework of Cao and Chatterjee [43]: compactness modulo gauge, smoothing by the heat flow at positive times, and the Borel structure of the space of trajectories.

Convergence of finite families of flowed loops does not by itself produce a measure on $\mathcal{X}_L$. The determining coordinates w_j of Theorem 29.3 define a metric d_c that determines Borel probability measures on $\mathcal{X}_L$, but the energy $X \mapsto E_L(X(t))$ is not known to be continuous for d_c or for the Wilson topology. We therefore also use the topology of Cao and Chatterjee on trajectories of Sobolev connections modulo gauge. Compactness of trajectories comes from simultaneous bounds on their energies at the times $s_j = L^2 2^{-j}$. Energy at each positive time is continuous in the Sobolev trajectory topology because the heat flow regularizes weakly convergent data (Lemma 29.7), and this identifies the limits of the lattice energies. To realize the limit as a limit of the lattice measures themselves, we construct the maps π_k explicitly, by a search over a countable family of smooth trajectories subject to one-sided energy bounds (Theorem 29.19).

Section 29.1 collects the properties of $\mathcal{X}_L$ that we use, and Section 29.2 the deterministic lattice estimates. Sections 29.3 to 29.5 prove joint convergence of energies and loops under the conditioned Wilson measures, construct the maps π_K, and remove the conditioning, for the parameters of Theorem 23.2; Section 29.6 extends this to couplings below an arbitrary fixed bound and treats general admissible approximations. Section 29.7 states the resulting limit theorems along the sequences $N_K = n_* M^{JK}$, and Section 29.8 passes to the full sequence nM^k and completes the proof of Theorem 2.3. The remaining parts of Theorem 2.5 are proved in Section 30. The results of this section are used in Section 32 and in the weak-coupling analysis of Sections 34 to 36, mainly through Theorems 29.23 to 29.25 and Corollary 29.26.

Conventions. Constants in deterministic flow estimates depend only on the data listed in each statement, with the dependence described in item (7) of Definition 4.14. Stochastic convergence moduli and the number of constraints $m(K)$ used to define π_K may in addition depend on the fixed coupling, on the parameters of the renormalization group hierarchy, on the base side, and on the observables. We claim no rate of convergence, no convergence of moments of the energy, and no uniformity as $\gamma \to 0$. We use the Lie-algebra norm $|\cdot|$ of Section 4 on connections, and for a lattice potential A at mesh a we write

$$\|A\|_{C^1_a} = \|A\|_\infty + \max_{j\in\{1,2,3\},\ \epsilon\in\{+,-\}} \|D^\epsilon_{a,j} A\|_\infty,$$

where $D^\pm_{a,j}$ are the forward and backward physical differences. The finite-difference norms of Definition 4.8 likewise include both directions and all translates.

29.1 The trajectory space and its Borel structure

Recall from Definition 2.2 the space $\mathcal{A}_L = C^\infty(\mathbb{T}^3_L; \mathfrak{su}(2)^3)$ of smooth connections, the group $\mathcal{G}_L = C^\infty(\mathbb{T}^3_L; \mathrm{SU}(2))$ of smooth periodic gauge transformations (including all its connected components), acting by $A^k = kAk^{-1} - (dk)k^{-1}$, and the Yang–Mills heat semigroup $(\Phi_t)_{t\ge 0}$ on $\mathcal{A}_L/\mathcal{G}_L$. Global existence of smooth solutions for smooth initial data is due to Råde [139]; we use it in the form of [43, Theorem 3.14, p. 25], which also relies on Charalambous and Gross [47]. The trajectory space is

$$\mathcal{X}_L = \big\{ X : (0,\infty) \to \mathcal{A}_L/\mathcal{G}_L :\ X(t) = \Phi_{t-s} X(s) \text{ for all } 0 < s < t \big\},$$

the Wilson topology $\mathcal{T}_W$ is the weakest topology for which all the functions $W_{\ell,\chi,t}(X) = \chi(\mathrm{Hol}_\ell\, X(t))$ are continuous, where ℓ ranges over piecewise C^1 loops, χ over characters of finite-dimensional representations of $\mathrm{SU}(2)$, and t over $(0,\infty)$. The energy E_L is defined in (2.3). Our parallel transport solves $H'(s) = H(s)A(\ell(s))\dot\ell(s)$ with $H(0) = I$, as in Definition 4.10, and our gauge variable is the inverse of the one in [43, (1.1), p. 2].

In [43], Cao and Chatterjee introduced trajectories of the heat flow at positive times as a state space for three-dimensional Yang–Mills theory, following the proposal of Charalambous and Gross [47] to regularize Wilson loops by the heat flow. We work in their framework. The separation of smooth gauge orbits by holonomies is due to Sengupta [144]; we use it in the formulation of [43, Lemma 1.2, p. 4]. The compactness of energy sublevel sets modulo gauge is the form of gauge compactness proved in [43, Theorem 3.8 and Theorem 3.51], which rests on Uhlenbeck's theorem [151]. We add a countable determining metric, a comparison of its Borel structure with that of the Wilson topology, and the continuity of the energy after a positive time in the form needed below.

Sobolev connections modulo gauge

Definition 29.1. Let $\mathcal{G}_L^{2,2}$ be the group of all periodic maps $\mathbb{T}_L^3 \to \mathrm{SU}(2)$ of class $W^{2,2}$, including all connected components, and set

$$\mathcal{H}_L = H^1(\mathbb{T}_L^3; \mathfrak{su}(2)^3)/\mathcal{G}_L^{2,2}.$$

For $[A_n], [A] \in \mathcal{H}_L$ we write $[A_n] \rightharpoonup_q [A]$ if there exist representatives B_n of $[A_n]$ and B of $[A]$ such that $B_n \rightharpoonup B$ weakly in H^1. The *sequential weak quotient topology* $\mathcal{T}_q$ on $\mathcal{H}_L$ is defined by declaring a set $\mathcal{U}$ open if every sequence $[A_n] \rightharpoonup_q [A]$ with $[A] \in \mathcal{U}$ eventually lies in $\mathcal{U}$. We write $\Phi_t^{1,2}$, $t \geq 0$, for the Yang–Mills heat semigroup on $\mathcal{H}_L$ of Cao and Chatterjee [43]; the properties of it that we use are recalled, with locators, in Lemma 29.2. Let $\mathcal{X}_L^{1,2}$ be the set of trajectories $Z : (0, \infty) \to \mathcal{H}_L$ with $Z(t) = \Phi_{t-s}^{1,2} Z(s)$ for all $0 < s < t$, with the weakest topology $\mathcal{T}_{1,2}$ for which every evaluation $Z \mapsto Z(t)$, $t > 0$, is continuous into $(\mathcal{H}_L, \mathcal{T}_q)$. This is the point-evaluation topology of [43, Definition 3.70, p. 41]; it is not a topology of uniform weak H^1 convergence on time intervals. Finally, let $i^{1,2} : \mathcal{X}_L \to \mathcal{X}_L^{1,2}$ send a trajectory of smooth orbits to the trajectory of the Sobolev orbits containing them.

For a finite energy bound R, we write $\Omega_R = \{[A] \in \mathcal{H}_L : E_L(A) \leq R\}$. We also need a countable family of regularized Wilson loop functions on $\mathcal{H}_L$: let $\Psi_0 : \mathcal{H}_L \to \mathbb{C}^{\mathbb{N}}$ be the map whose components are the functions $[A] \mapsto \chi(\mathrm{Hol}_\ell \Phi_v^{1,2}[A])$, where ℓ ranges over the countable family of loops of Theorem 29.3 below, v over $L^2\mathbb{Q}_{>0}$, and χ over the characters of the irreducible representations of $\mathrm{SU}(2)$. The following facts are taken from [43].

Lemma 29.2. *Let $L > 0$. Then the following statements hold.*

1. *The group $\mathcal{G}_L^{2,2}$ coincides with the set of continuous maps $k : \mathbb{T}_L^3 \to \mathrm{SU}(2)$ with a weak derivative such that $k^{-1}dk \in H^1$, and with the set of maps $s \exp \xi$, where $s : \mathbb{T}_L^3 \to \mathrm{SU}(2)$ is smooth and $\xi \in H^2(\mathbb{T}_L^3; \mathfrak{su}(2))$.*
2. *Every $A \in H^1(\mathbb{T}_L^3; \mathfrak{su}(2)^3)$ has finite energy, and $\|F_A\|_2 \leq C_L(\|A\|_{H^1} + \|A\|_{H^1}^2)$ with C_L depending only on L.*

3. *The space $(\mathcal{H}_L, \mathcal{T}_q)$ is Hausdorff, each Ω_R is closed and compact metrizable, and the map Ψ_0 is continuous and injective, with $\Psi_0(F \cap \Omega_R)$ compact for every closed set $F \subset \mathcal{H}_L$ and every R.*
4. *For every $v \geq 0$ the map $\Phi_v^{1,2}$ is continuous on $(\mathcal{H}_L, \mathcal{T}_q)$.*
5. *Restriction to times in $L^2\mathbb{Q}_{>0}$ is a homeomorphism of $\mathcal{X}_L^{1,2}$ onto the corresponding space of compatible trajectories at these times, and the Borel σ-algebra of $(\mathcal{X}_L^{1,2}, \mathcal{T}_{1,2})$ is generated by the evaluations at these times.*
6. *For every $t > 0$, every orbit $Z(t)$ of a trajectory $Z \in \mathcal{X}_L^{1,2}$ has a smooth representative, and a $W^{2,2}$ gauge transformation relating two smooth connections is smooth, so that the map $j^{1,2} : \mathcal{X}_L^{1,2} \to \mathcal{X}_L$ that replaces each orbit by its smooth representatives is well defined. This map satisfies $j^{1,2} i^{1,2} = \mathrm{Id}$ and $i^{1,2} j^{1,2} = \mathrm{Id}$, it is continuous from $\mathcal{T}_{1,2}$ to $\mathcal{T}_W$, and it preserves every function $W_{\ell,\chi,t}$.*
7. *Every function $W_{\ell,\chi,t} \circ j^{1,2}$ is continuous on $(\mathcal{X}_L^{1,2}, \mathcal{T}_{1,2})$.*

Proof. On the unit torus, (1) is [43, Definition 3.5 and (3.2), p. 22] (we verify the identification below), (2) is [43, Lemma 3.7, p. 23], (3) is [43, Lemma 3.49, p. 35; Definition 3.50 and Theorem 3.51, pp. 35–36], (4) is [43, Lemma 3.68, p. 40], (5) is [43, Definitions 3.70 and 3.74, pp. 41–42; Lemmas 3.75 and 3.78, pp. 42–44], (6) is [43, Lemma 3.10, p. 24; Lemmas 3.84–3.85, pp. 47–48; Definition 3.86 and Lemmas 3.87–3.89, pp. 48–49], and (7) is [43, Definition 3.82, p. 47]. In (3) the countable family of regularized Wilson loop functions may be chosen as above, from our dense loop family, rational positive delays, and all characters. The identity $i^{1,2} j^{1,2} = \mathrm{Id}$ in (6) holds because at each time the smooth representative lies in the given Sobolev orbit. It remains to check the hypotheses of these results and to transfer them to side L.

The bundle is trivial, the base is the periodic three-torus, and the group $\mathrm{SU}(2)$ belongs to the class allowed in [43, Section 1.1, p. 2]. All connected components of the gauge group are included. For $y = x/L$ let $\widetilde{A}(y) = LA(Ly)$. Differentiating, we find

$$F_{ij}(\widetilde{A})(y) = L^2 F_{ij}(A)(Ly), \qquad E_1(\widetilde{A}) = L E_L(A), \qquad \widetilde{t} = t/L^2,$$

where $\widetilde{t}$ is the time of the rescaled flow. Gauge transformations transform by composition with $y \mapsto Ly$, and inversion converts the gauge convention of [43] into ours. All results on the unit torus therefore apply at every fixed L to the spaces defined here. For $f_L(y) = f(Ly)$ we have

$$\|f_L\|_{L^p(\mathbb{T}_1^3)} = L^{-3/p} \|f\|_{L^p(\mathbb{T}_L^3)}, \qquad \|f_L\|_{H^1(\mathbb{T}_1^3)}^2 = L^{-3} \|f\|_{L^2(\mathbb{T}_L^3)}^2 + L^{-1} \|\nabla f\|_{L^2(\mathbb{T}_L^3)}^2.$$

Applying the Sobolev inequality $\|f\|_p \leq C_p \|f\|_{H^1}$ on the unit torus for $1 \leq p \leq 6$, with compact embedding for $p < 6$ [43, Theorem 3.3, p. 21], componentwise to Lie-algebra and matrix coefficients, we obtain at each fixed $L > 0$

$$\|f\|_{L^p(\mathbb{T}_L^3)} \leq C_p \big(L^{3/p - 3/2} \|f\|_{L^2(\mathbb{T}_L^3)} + L^{3/p - 1/2} \|\nabla f\|_{L^2(\mathbb{T}_L^3)} \big), \qquad 1 \leq p \leq 6, \tag{29.1}$$

and the compactness for $p < 6$ transfers under the same linear rescaling. Below we use $p = 6$ for the gauge group and $p = 4$ for the curvature. The constants depend only on L, p, and the fixed coefficient norms.

We now verify (1), that is, that $\mathcal{G}_L^{2,2}$ is the full group used in [43]. The description there, as the set of maps $s \exp \xi$ with s smooth (ranging over all components) and $\xi \in H^2$, and as the set of continuous maps with $k^{-1}dk \in H^1$, transfers to side L by composition with $y \mapsto Ly$. A periodic H^2 matrix function is continuous, since its Fourier series converges absolutely and uniformly by the Cauchy–Schwarz inequality and $\sum_{p \in \mathbb{Z}^3} (1 + |p|^2)^{-2} < \infty$. For a $W^{2,2}$ group-valued map k we have $dk \in H^1 \subset L^6$ by (29.1). The derivative of $k^{-1}dk$ consists of a bounded matrix times $d^2k \in L^2$ and products of two factors dk, which lie in $L^3 \subset L^2$. This shows that $k^{-1}dk \in H^1$. Conversely, let k be continuous with values in the compact group, with $\alpha = k^{-1}dk \in H^1$. Then $dk = k\alpha \in L^6$, the derivative of this identity consists of $dk\,\alpha \in L^3$ and $k\,d\alpha \in L^2$, and we conclude that $k \in W^{2,2}$. These product identities hold weakly, by smooth convolution and convergence in the displayed Lebesgue norms. This identifies the two groups, including all periodic components. Finally, for (2) we apply [43, Lemma 3.7] to $\widetilde{A}$; its only hypothesis is $A \in H^1$ on the unit torus. We then use the scaling of curvature and energy displayed above. The constant C_L depends only on L and the fixed coefficient norms. □

The determining metric

The trajectory space carries the Wilson topology, which is defined by uncountably many functions. The following theorem replaces it, for measure-theoretic purposes, by a countable family.

Theorem 29.3. *Let $L > 0$. On piecewise C^1 loops $\ell, \ell' : [0,1] \to \mathbb{T}_L^3$ consider the metric*

$$d_{\mathcal{L}}(\ell, \ell') = \sup_{u \in [0,1]} d_{\mathbb{T}_L^3}(\ell(u), \ell'(u)) + \int_0^1 |\dot{\ell}(u) - \dot{\ell}'(u)|\, du.$$

Fix a countable family of piecewise C^1 loops that is dense for $d_{\mathcal{L}}$ and contains loops of every winding class (such families exist; see the proof), and enumerate its product with $L^2\mathbb{Q}_{>0}$ as $(\ell_j, t_j)_{j \geq 1}$. Set

$$w_j(X) = \tfrac{1}{2} \operatorname{tr} \operatorname{Hol}_{\ell_j} X(t_j), \qquad d_c(X, Y) = \sum_{j \geq 1} 2^{-j} |w_j(X) - w_j(Y)|, \qquad X, Y \in \mathcal{X}_L.$$

Then the following statements hold.

1. *d_c is a metric on $\mathcal{X}_L$, and $(\mathcal{X}_L, d_c)$ is separable.*
2. *$(\mathcal{X}_L, \mathcal{T}_W)$ is separable, and the identity map from $(\mathcal{X}_L, \mathcal{T}_W)$ to $(\mathcal{X}_L, d_c)$ is continuous.*
3. *The Borel σ-algebras satisfy*

$$\mathcal{B}(d_c) = \sigma(w_j : j \geq 1) = \mathcal{B}(\mathcal{T}_W).$$

4. *Two Borel probability measures on $\mathcal{X}_L$ under which all finite vectors $(w_1, \dots, w_b)$ have the same laws are equal. In particular, a Borel probability measure on $\mathcal{X}_L$ is determined by the joint laws of all finite families of the functions $W_{\ell,\chi,t}$.*

Remark 29.4. We do not claim that the two topologies coincide, that d_c is complete, or that $\mathcal{X}_L$ or $\mathcal{H}_L$ is a Polish space. Weak convergence on $\mathcal{X}_L$ is always stated in two forms: against bounded $\mathcal{T}_W$-continuous functions, and weakly with respect to d_c.

We call the functions w_j the determining coordinates. Each is one half of a regularized Wilson loop for the fundamental character. The proof of Theorem 29.3 uses the reduction of all characters of $\mathrm{SU}(2)$ to the fundamental trace. Let P_p be the polynomials defined by

$$P_0(u) = 1, \qquad P_1(u) = u, \qquad P_{p+1}(u) = uP_p(u) - P_{p-1}(u).$$

Lemma 29.5. *The character of every finite-dimensional representation of* $\mathrm{SU}(2)$ *is a finite sum of the functions $H \mapsto P_p(\operatorname{tr} H)$, $p \geq 0$.*

Proof. This is the classical highest-weight classification; we include the short argument. If H has eigenvalues z, z^{-1}, the recursion shows that $P_p(z + z^{-1}) = z^p + z^{p-2} + \cdots + z^{-p}$ for $z \neq \pm 1$, and by continuity the identity also holds for $z = \pm 1$. Averaging a Hermitian inner product over the Haar probability measure makes a finite-dimensional representation ρ unitary. An invariant subspace then has an invariant orthogonal complement, and by induction on the dimension ρ is a direct sum of irreducible representations. If ρ is only continuous, convolving its matrix coefficients with a smooth bump supported near the identity gives a matrix arbitrarily close to I, and in particular an invertible one. Translating the bump on the left shows that ρ multiplied by this matrix is smooth, which implies that ρ is smooth. In the complexified differential of an irreducible ρ we set $H = d\rho(h)$, $E = d\rho(e)$, $F = d\rho(f)$ for

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H$, $H^* = H$, and $E^* = F$, the last two because ih, $e - f$, and $i(e + f)$ are anti-Hermitian. Since $\exp(2\pi i h) = I$, the eigenvalues of H are integers. Let n be the largest one and $Hv = nv$ with $v \neq 0$; then $Ev = 0$, and by induction, using the commutation relations, we obtain

$$EF^q v = q(n - q + 1)F^{q-1}v, \qquad \|F^{q+1}v\|^2 = (q + 1)(n - q)\|F^q v\|^2.$$

The case $q = 0$ shows that $n \geq 0$. The vectors $v, Fv, \dots, F^n v$ are nonzero with distinct weights $n, n - 2, \dots, -n$, and $F^{n+1}v = 0$. Their span is invariant under H, E, F. It is also invariant under the group, since every element of $\mathrm{SU}(2)$ is an exponential after unitary diagonalization. By irreducibility the span is the whole space, and the character on $\operatorname{diag}(z, z^{-1})$ is $P_n(z + z^{-1})$. Unitary diagonalization covers every group element. □

Proof of Theorem 29.3. The proof has six steps.

Step 1: a dense family. The continuity and separability results [43, Lemmas 3.24–3.25 and 3.28, p. 28] apply to every fixed smooth connection and character here. A dense countable family of loops can be chosen among polygonal loops whose lifts have vertices in $L\mathbb{Q}^3$ and rational subdivision parameters, with final vertex equal to the initial vertex plus Lz, for every $z \in \mathbb{Z}^3$. To see that this family is dense for $d_{\mathcal{L}}$, we approximate each C^1 piece of a loop by its polygonal interpolant and then perturb the finitely many vertices and subdivision parameters within these dense sets. The first approximation converges uniformly, with derivatives in L^1, by uniform continuity of the derivative on each piece; for a fixed interpolant the second does so because the velocities are constant on finitely many intervals. The final vertex is imposed using the same Lz. This family contains every winding class and the constant loops at rational points.

Step 2: separation. Suppose that $w_j(X) = w_j(Y)$ for all j. Fix $t \in L^2\mathbb{Q}_{>0}$. Then $\operatorname{tr}\operatorname{Hol}_\ell X(t) = \operatorname{tr}\operatorname{Hol}_\ell Y(t)$ for every loop ℓ of the dense family, and by the continuity of holonomies in $d_{\mathcal{L}}$ [43, Lemmas 3.24–3.25] for every piecewise C^1 loop. By Lemma 29.5, all characters of $\operatorname{Hol}_\ell X(t)$ and $\operatorname{Hol}_\ell Y(t)$ agree on every piecewise C^1 loop. The holonomy separation theorem [43, Lemma 1.2, p. 4], whose hypotheses are exactly these, implies that $X(t) = Y(t)$. For an arbitrary $t > 0$, let $s \in L^2\mathbb{Q}_{>0}$ with $s < t$; by compatibility we have $X(t) = \Phi_{t-s}X(s) = \Phi_{t-s}Y(s) = Y(t)$. No backward uniqueness is used. Thus d_c separates points.

Step 3: metric properties. Symmetry and the triangle inequality hold termwise. Since $|w_j| \le 1$, the tail $\sum_{j>b} 2^{-j}|w_j(X) - w_j(Y)|$ is at most 2^{1-b}. This implies that convergence in d_c is equivalent to convergence of every coordinate, and the coordinate map $X \mapsto (w_j(X))_{j\ge1}$ is a homeomorphism of $(\mathcal{X}_L, d_c)$ onto a subspace of $[-1,1]^{\mathbb{N}}$. The countable base of cylinder sets of the product restricts to a countable base, and choosing a point in each nonempty member, we obtain a countable dense set. The same observation shows that $\mathcal{B}(d_c) = \sigma(w_j : j \ge 1)$. The partial sums defining d_c are $\mathcal{T}_W$-continuous and the tails are uniformly small. Then the coordinate map is continuous from $\mathcal{T}_W$ into the product, and the identity from $(\mathcal{X}_L, \mathcal{T}_W)$ to $(\mathcal{X}_L, d_c)$ is continuous.

Step 4: the reverse Borel inclusion. Since every w_j is $\mathcal{T}_W$-continuous, it remains to show that $\mathcal{B}(\mathcal{T}_W) \subset \sigma(w_j : j \ge 1)$. We first show that the coordinates of Ψ_0 generate the Borel σ-algebra of $(\mathcal{H}_L, \mathcal{T}_q)$. By Lemma 29.2(2), every element of $\mathcal{H}_L$ has finite energy, and the sets Ω_b, $b \in \mathbb{N}$, exhaust $\mathcal{H}_L$. For a closed set $F \subset \mathcal{H}_L$ we therefore have

$$\Psi_0(F) = \bigcup_{b\ge1} \Psi_0(F \cap \Omega_b). \tag{29.2}$$

By Lemma 29.2(3), the right side is a countable union of compact, hence closed, subsets of the Hausdorff space $\mathbb{C}^{\mathbb{N}}$, and in particular it is Borel. Since Ψ_0 is injective,

$F = \Psi_0^{-1}(\Psi_0(F))$ lies in the σ-algebra generated by the coordinates of Ψ_0; the converse inclusion holds because Ψ_0 is continuous. This argument uses the compactness of the images $\Psi_0(F \cap \Omega_b)$; injectivity and measurability of Ψ_0 alone would not give measurability of its inverse.

Now let $u \in L^2\mathbb{Q}_{>0}$ and $X \in \mathcal{X}_L$. Each component of $\Psi_0(i^{1,2}X(u))$ is of the form $\chi(\mathrm{Hol}_\ell X(u+v))$ with ℓ in the dense family and $v \in L^2\mathbb{Q}_{>0}$. By Lemma 29.5 it is a polynomial in $(\operatorname{tr}\mathrm{Hol}_\ell X(u+v))/2$, which is one of the functions w_j, since $u+v \in L^2\mathbb{Q}_{>0}$. Since the coordinates of Ψ_0 generate the Borel σ-algebra of $\mathcal{H}_L$, the map $X \mapsto i^{1,2}X(u)$ is measurable from $\sigma(w_j : j \geq 1)$ into the Borel σ-algebra of $\mathcal{H}_L$. By Lemma 29.2(5), the map $i^{1,2}$ is measurable from $\sigma(w_j : j \geq 1)$ into the Borel σ-algebra of $\mathcal{X}_L^{1,2}$. Let $\mathcal{B} \in \mathcal{B}(\mathcal{T}_W)$. Since $j^{1,2}$ is continuous (Lemma 29.2(6)), the set $(j^{1,2})^{-1}(\mathcal{B})$ is Borel in $\mathcal{X}_L^{1,2}$, and since $j^{1,2}i^{1,2} = \mathrm{Id}$,

$$\mathcal{B} = (i^{1,2})^{-1}\big((j^{1,2})^{-1}(\mathcal{B})\big) \in \sigma(w_j : j \geq 1).$$

This proves the Borel identity in (3).

Step 5: separability of the Wilson topology. We consider the countable set of trajectories $t \mapsto \Phi_t[b]$, where b ranges over the trigonometric polynomial one-forms with rational coefficients. Let $X \in \mathcal{X}_L$, and fix a basic Wilson neighborhood of X determined by finitely many functions $W_{\ell,\chi,t}$. The shifted trajectories $X_\epsilon(t) = X(t+\epsilon)$ approach X in every finite list of these functions as $\epsilon \downarrow 0$, since X is the smooth solution started at any earlier positive time. Each X_ϵ is the trajectory started from a smooth representative of $X(\epsilon)$. Approximating this representative in C^3 by a rational trigonometric polynomial and using Theorem 24.10 on a finite horizon containing the listed times, we find a trajectory of the countable set in the given neighborhood.

Step 6: determination of probability measures. If two measures have the same finite joint laws of the w_j, then their image measures on $[-1,1]^{\mathbb{N}}$ agree on the cylinder sets. The class of sets on which the two image measures agree is a Dynkin class containing the class of cylinder sets, which is closed under intersections. By Dynkin's π–λ theorem, it contains the σ-algebra generated by the cylinders. Pulling back and using (3), we find that the two measures agree on $\mathcal{B}(\mathcal{T}_W)$. The joint laws of finite families of the functions $W_{\ell,\chi,t}$ include those of the w_j, which proves the last assertion. This also verifies the hypotheses of the uniqueness statement [43, Lemma 1.4, p. 7]. □

Compactness of energy sublevel sets

Theorem 29.6. *Let $L > 0$. On the Borel space of Theorem 29.3, the function $X \mapsto E_L(X(t))$ is measurable for every $t > 0$, and $t \mapsto E_L(X(t))$ is nonincreasing for every $X \in \mathcal{X}_L$. Let $s_j = L^2 2^{-j}$ for $j \geq 1$. For every sequence $R = (R_j)_{j\geq 1}$ of finite numbers $R_j \geq 0$, the set*

$$\mathcal{K}(R) = \big\{X \in \mathcal{X}_L : E_L(X(s_j)) \leq R_j \text{ for all } j \geq 1\big\}$$

is compact for $\mathcal{T}_W$ and for d_c. Let $\{\lambda_\iota\}$ be a family of Borel probability measures on $\mathcal{X}_L$ such that

$$\lim_{R\to\infty} \sup_{\iota} \lambda_\iota\{E_L(X(s_j)) > R\} = 0 \qquad \text{for every } j \geq 1.$$

Then for every $\eta > 0$ there exists R with $\lambda_\iota(\mathcal{K}(R)) \geq 1-\eta$ for all ι, and every sequence in the family has a subsequence that converges, against every bounded $\mathcal{T}_W$-continuous function and weakly with respect to d_c, to a Borel probability measure on $\mathcal{X}_L$. Finally, if A is a smooth connection with $\|A\|_{C^1} \leq R$, then

$$E_L(\Phi_t[A]) \leq E_L(A) \leq CL^3(R+R^2)^2 \qquad (t \geq 0),$$

where C depends only on the normalization of the Lie-algebra norm.

No moment condition on the measures λ_ι is needed. Subsequential compactness from energy tightness is due to Cao and Chatterjee [43, Theorem 1.6, p. 7]. The theorem adds the explicit compact sets $\mathcal{K}(R)$, which are used in Section 29.4. We say that a family of Borel probability measures on $\mathcal{X}_L$ has *compact containment* if for every $\eta > 0$ some set $\mathcal{K}(R)$ has mass at least $1-\eta$ under every member of the family. Since the sets $\mathcal{K}(R)$ are compact for both $\mathcal{T}_W$ and d_c, this is compact containment in both topologies.

Proof. For probability measures on the trajectory space with its Wilson Borel σ-algebra, the energy at each positive time is measurable and nonincreasing in time [43, Lemma 1.5, p. 7; Lemma 4.1, p. 50]. The hypotheses on the group, the bundle, the scaling, and the gauge group were checked in Lemma 29.2, and the energy in the statement has exactly this normalization.

We consider the product $\prod_{j\geq 1} \Omega_{R_j}$ of subsets of $\mathcal{H}_L$. It is compact and metrizable, by Lemma 29.2(3) and a diagonal extraction of subsequences in the factors, using a summable bounded product metric. We impose the compatibility conditions

$$x_i = \Phi^{1,2}_{s_i - s_j} x_j \qquad (j > i). \tag{29.3}$$

Each of these equations defines a closed set, since $\mathcal{H}_L$ is Hausdorff and $\Phi^{1,2}_v$ is continuous for $\mathcal{T}_q$ (Lemma 29.2(3),(4)). The compatible subset is therefore compact. Given a compatible sequence (x_j), we define $Z(t) = \Phi^{1,2}_{t-s_j} x_j$ for any j with $s_j < t$. By (29.3) and the semigroup property, $Z(t)$ does not depend on j, and Z is compatible at all times. Fixing one j with $s_j < t$ for each t, we see that $Z(t)$ is a continuous function of x_j, and therefore the map $(x_j) \mapsto Z$ is continuous into $(\mathcal{X}_L^{1,2}, \mathcal{T}_{1,2})$. By Lemma 29.2(6), composing with $j^{1,2}$, we obtain a continuous map into $(\mathcal{X}_L, \mathcal{T}_W)$.

The image of this map is exactly $\mathcal{K}(R)$. To see this, let $k > j$; then $Z(s_j) = \Phi^{1,2}_{s_j - s_k} x_k = x_j$ by (29.3). Every x_j is therefore the orbit at time s_j of the trajectory, it has a smooth representative, and its energy is at most R_j. Conversely, the values $i^{1,2}X(s_j)$ of any

$X \in \mathcal{K}(R)$ form such a compatible sequence. As a continuous image of a compact set, $\mathcal{K}(R)$ is $\mathcal{T}_W$-compact, and by Theorem 29.3(2) it is also d_c-compact. It is Borel because the energies are measurable. The argument also covers an empty compatible set and thresholds equal to zero.

Let $\eta > 0$, and for each j fix R_j such that $\sup_\iota \lambda_\iota\{E_L(X(s_j)) > R_j\} \le \eta 2^{-j}$. By the union bound, we have $\lambda_\iota(\mathcal{K}(R)) \ge 1 - \eta$ for every ι. For any $t > 0$, choosing $s_j < t$ and using monotonicity, we see that the energy laws at time t are also tight. The subsequence theorem [43, Theorem 1.6, p. 7] applies to probability measures on this Borel space with tight energy laws at each positive time, and it yields a subsequence converging against every bounded $\mathcal{T}_W$-continuous function. Every bounded d_c-continuous function is $\mathcal{T}_W$-continuous by Theorem 29.3(2), and the same subsequence converges weakly for d_c.

Finally, $|[X, Y]| \le |X||Y|$ in our normalization, which implies that $|F_{ij}(A)| \le C(R + R^2)$ if $\|A\|_{C^1} \le R$. There are three pairs $i < j$ and the volume is L^3, which gives the bound on $E_L(A)$; the bound along the flow follows from monotonicity. □

Continuity of energy after a positive time

The determining coordinates w_j do not control the energy, and we do not know whether $X \mapsto E_L(X(t))$ is continuous for d_c or for $\mathcal{T}_W$. The heat flow provides a substitute: on the Sobolev quotient, the energy after any positive time is a continuous function of the initial orbit. This property lets us identify limits of lattice energies with energies of the limiting trajectories in Theorem 29.18.

Lemma 29.7. *Let $L > 0$ and $u > 0$. The function $[A] \mapsto E_L(\Phi_u^{1,2}[A])$ is continuous on $(\mathcal{H}_L, \mathcal{T}_q)$. Moreover, for every $t > 0$ the function $Z \mapsto E_L(Z(t))$ is continuous on $(\mathcal{X}_L^{1,2}, \mathcal{T}_{1,2})$. Here the energy of an orbit at a positive time is the energy of its smooth representatives, which does not depend on the representative.*

The lemma does not assert that $[A] \mapsto E_L(A)$ itself is continuous on $(\mathcal{H}_L, \mathcal{T}_q)$.

Proof. We work on the unit torus first. Let $A_n \rightharpoonup A$ weakly in H^1, and set $R_0 = \sup_n \|A_n\|_{H^1} + \|A\|_{H^1}$, which is finite. Fix $0 < v < \min(u, \tau_0(R_0))$, where $\tau_0(R_0)$ is the existence time of [43, Lemma 3.23, pp. 27–28]. By that lemma, the local solutions B_n, B of the gauge-fixed heat equation (in the DeTurck gauge) started at A_n, A satisfy $B_n(v) \to B(v)$ strongly in H^1. These endpoints are smooth, by [43, Proposition 3.19, pp. 26–27], and they represent the orbits $\Phi_v[A_n]$ and $\Phi_v[A]$, by [43, Lemma 3.21, p. 27]. Applying [43, Lemma 3.15, p. 25] to the smooth, strongly convergent initial data $B_n(v) \to B(v)$, we obtain strong H^1 convergence of representatives of the solutions of the covariant flow at time $u - v$, that is, of representatives of $\Phi_u[A_n]$ and $\Phi_u[A]$. The smooth-data lemma is applied only after the smoothing step, and never to the weakly convergent data A_n.

Strong H^1 convergence implies convergence of the energy, by the embedding $H^1 \subset L^4$:

$$\|F(C) - F(D)\|_2 \le C\big(1 + \|C\|_{H^1} + \|D\|_{H^1}\big)\|C - D\|_{H^1}.$$

Indeed, the derivative part of F is linear, and $[C_i, C_j] - [D_i, D_j] = [C_i - D_i, C_j] + [D_i, C_j - D_j]$ is bounded in L^2 by Hölder's inequality and the embedding. Multiplying by $\|F(C)\|_2 + \|F(D)\|_2$, we obtain convergence of the squared norms. By gauge invariance, $E_1(\Phi_u[A_n]) \to E_1(\Phi_u[A])$.

If $[A_n] \rightharpoonup_q [A]$, we apply this argument to weakly convergent representatives. Since $\mathcal{T}_q$ is generated by exactly this sequential convergence, [43, Lemma 3.38 and Corollary 3.39, p. 32; Lemma 3.41, p. 33] give continuity into $\mathbb{R}$; we do not replace weakly convergent nets by bounded sequences. For trajectories, we fix $s \in (0, t)$ and factor the energy as $Z \mapsto Z(s) \mapsto E_1(\Phi^{1,2}_{t-s} Z(s))$. Finally, the rescaling $\widetilde{A}(y) = LA(Ly)$, $\widetilde{t} = t/L^2$, $E_1(\widetilde{A}) = LE_L(A)$ of Lemma 29.2 transfers every assertion to side L. □

A countable dense family of trajectories and finite-dimensional approximation

Fix an enumeration $(b_q)_{q \ge 1}$ of the real $\mathfrak{su}(2)$-valued trigonometric polynomial one-forms on $\mathbb{T}^3_L$ with rational coefficients in a fixed basis, with $b_1 = 0$, and set

$$X_q(t) = \Phi_t[b_q], \qquad t > 0.$$

In particular, X_1 is the zero trajectory, $X_1(t) = [0]$ for all t. The trajectories X_q are the candidates among which the maps π_K of Section 29.4 choose.

Lemma 29.8. *Let $X \in \mathcal{X}_L$, let $W_{\ell_1,\chi_1,t_1}, \dots, W_{\ell_b,\chi_b,t_b}$ be finitely many Wilson loop functions, let $v_1, \dots, v_p > 0$, and let $\delta > 0$. Then there exists q such that*

$$|W_{\ell_h,\chi_h,t_h}(X_q) - W_{\ell_h,\chi_h,t_h}(X)| < \delta \quad (h \le b), \qquad |E_L(X_q(v_i)) - E_L(X(v_i))| < \delta \quad (i \le p).$$

Proof. Rational trigonometric polynomials are dense in C^3 in the space of smooth real one-forms. Indeed, integration by parts bounds the Fourier coefficients of a smooth form by $C_p(1 + |k|)^{-p}$, so for $p > 6$ the Fourier truncations converge in C^3, and the finitely many real sine and cosine coefficients of a truncation can be approximated by rational numbers. Since the frequencies are $2\pi k/L$, no rationality condition on L is needed.

For $\eta > 0$, let c_η be a smooth representative of $X(\eta)$. By compatibility, $\Phi_t[c_\eta] = X(t + \eta)$ for $t > 0$. Smooth forward evolution from an earlier positive time shows that the energy and the holonomies are continuous in time at each of the finitely many listed times. Then the listed values on $X(\cdot + \eta)$ converge to those on X as $\eta \downarrow 0$. Fix η such that all these errors are less than $\delta/2$. By Theorem 24.10, the listed energies and loops of the flow are continuous functions of the initial datum in C^3 on the finite horizon. Approximating c_η in C^3 by some b_q, we make the remaining errors less than $\delta/2$. The representative c_η is used only to prove the existence of q. □

Lemma 29.9. *Let $\mathcal{C} \subset \mathcal{X}_L$ be $\mathcal{T}_W$-compact, and let $h : \mathcal{X}_L \to \mathbb{R}$ be $\mathcal{T}_W$-continuous with $|h| \le D$. For every $\delta > 0$ there exist $b < \infty$ and a continuous function $G : [-1,1]^b \to [-D, D]$ such that*

$$\sup_{X \in \mathcal{C}} |h(X) - G(w_1(X), \dots, w_b(X))| < \delta.$$

This applies in particular to the compact sets $\mathcal{K}(R)$ of Theorem 29.6 and to $h = f(W_{\ell_1,\chi_1,t_1}, \dots, W_{\ell_m,\chi_m,t_m})$ with f bounded and continuous.

Proof. If $\mathcal{C}$ is empty, we take $G = 0$. Otherwise, by Theorem 29.3(2) the identity from $(\mathcal{C}, \mathcal{T}_W)$ to $(\mathcal{C}, d_c)$ is a continuous bijection from a compact space to a Hausdorff space, and it is a homeomorphism: closed subsets of the compact domain have compact, hence closed, images. Thus h is uniformly continuous on $(\mathcal{C}, d_c)$. Let $\rho > 0$ be such that $d_c(X, Y) < \rho$ with $X, Y \in \mathcal{C}$ implies $|h(X) - h(Y)| < \delta$. Fix b with $2^{1-b} < \rho/4$ and a finite $\rho/8$-net $X^1, \dots, X^l$ of $(\mathcal{C}, d_c)$. For $u, v \in [-1,1]^b$ and $X \in \mathcal{X}_L$ write

$$d_b(u,v) = \sum_{j \le b} 2^{-j} |u_j - v_j|, \qquad p_b X = (w_j(X))_{j \le b}, \qquad h_\alpha(u) = \big(\rho/2 - d_b(u, p_b X^\alpha)\big)_+,$$

and let $h_0(u)$ be the d_b-distance from u to the compact set $p_b\mathcal{C}$. The function $h_0 + \sum_\alpha h_\alpha$ is positive on the cube: off $p_b\mathcal{C}$ we have $h_0 > 0$, and on $p_b\mathcal{C}$ the net property gives some $h_\alpha > 0$. We define

$$G(u) = \frac{\sum_\alpha h_\alpha(u) h(X^\alpha)}{h_0(u) + \sum_\alpha h_\alpha(u)}.$$

Then G is continuous and $|G| \le D$. For $X \in \mathcal{C}$ we have $h_0(p_b X) = 0$, and $h_\alpha(p_b X) > 0$ implies $d_c(X, X^\alpha) < \rho/2 + 2^{1-b} < \rho$. Thus $G(p_b X)$ is a convex combination of values $h(X^\alpha)$ within δ of $h(X)$. □

29.2 Lattice observables and deterministic estimates

Lattice approximations of loops

Let $\ell, \ell' : [0,1] \to \mathbb{T}_L^3$ be piecewise C^1 loops with base points $x = \ell(0)$ and $x' = \ell'(0)$, and let c be a piecewise C^1 path from x to x' (the constant path if $x = x'$). A *comparison homotopy* between ℓ and ℓ' with connector c is a piecewise C^1 map $H : [0,1]^2 \to \mathbb{T}_L^3$ such that $H(\cdot, 0) = \ell$, $H(\cdot, 1)$ is a parametrization of the concatenation $c * \ell' * c^{-1}$, and $H(0, v) = H(1, v) = x$ for all $v \in [0,1]$. Its swept area is

$$\mathrm{Area}(H) = \int_0^1 \int_0^1 |\partial_u H \wedge \partial_v H| \, du \, dv,$$

where for $v, w \in \mathbb{R}^3$ we write $|v \wedge w|$ for the area of the parallelogram spanned by v and w. The swept area is counted with multiplicity: parts of the homotopy with opposite

orientations do not cancel. Since transport along a concatenation is the product of the transports, $\mathrm{Hol}_{c*\ell'*c^{-1}} A = (\mathrm{Hol}_c A)(\mathrm{Hol}_{\ell'} A)(\mathrm{Hol}_c A)^{-1}$, and the characters of $\mathrm{Hol}_{\ell'} A$ and $\mathrm{Hol}_{c*\ell'*c^{-1}} A$ coincide. In this language, a sequence of closed lattice paths (ℓ_k), with ℓ_k in Λ_{N_k} and $N_k \to \infty$, is an admissible approximation of ℓ in the sense of Definition 2.1 if the physical lengths of the ℓ_k are bounded and, for all sufficiently large k, there exists a comparison homotopy H_k between ℓ and ℓ_k (with some connector) such that $\mathrm{Area}(H_k) \to 0$.

Lemma 29.10. *Let $L > 0$, let ℓ be a piecewise C^1 loop in $\mathbb{T}^3_L$, and let $a = L/N$ with an integer $N \geq 4$. There exist a closed lattice path ℓ_a in Λ_N and a comparison homotopy H_a between ℓ and ℓ_a, whose connector is a straight segment of length at most $\sqrt{3}a/2$, such that*

$$\mathrm{length}(\ell_a) \leq C\big(\mathrm{length}(\ell) + L\big), \qquad \mathrm{Area}(H_a) \leq Ca\big(\mathrm{length}(\ell) + L\big), \tag{29.4}$$

where C is an absolute constant. The path ℓ_a has the same winding class as ℓ. The homotopy H_a is piecewise C^1, and piecewise smooth if ℓ is polygonal. If ℓ is constant, then ℓ_a is constant and $\mathrm{Area}(H_a) = 0$. The path ℓ_a and the homotopy H_a depend only on the parametrized loop ℓ and on a.

In particular, for every sequence of sides $N_k \to \infty$ the paths $\ell_{a_{N_k}}$ form an admissible approximation of ℓ, with swept areas $O(a_{N_k})$.

Proof. Let $x : [0,1] \to \mathbb{R}^3$ be the lift of ℓ with $x(0) \in [0,L)^3$; then $x(1) = x(0) + Lz$ for a unique $z \in \mathbb{Z}^3$. Set $\mathfrak{l} = \mathrm{length}(\ell)$. Suppose first that $\mathfrak{l} > 0$, and set $p = \lceil \mathfrak{l}/a \rceil$, so that $p \leq 1 + \mathfrak{l}/a$. We subdivide ℓ by arclength into p pieces of length $\mathfrak{l}/p \leq a$. At an interior subdivision value we take the earliest parameter with that accumulated length, and we keep the two endpoint parameters; this fixes the choice also when x is stationary on some subintervals. We write x_j, $0 \leq j \leq p$, for the subdivision points.

We round each coordinate by the rule

$$q_j^i = a\lfloor x_j^i/a + \tfrac{1}{2} \rfloor, \qquad \text{so that} \qquad |q_j - x_j| \leq \tfrac{\sqrt{3}}{2} a.$$

Since $L/a = N \in \mathbb{N}$, this rule commutes with translations by $L\mathbb{Z}^3$, and therefore $q_p = q_0 + Lz$. Let σ_j be the lattice path from q_j to q_{j+1} that changes first the first coordinate, then the second, and then the third, each monotonically, and let ℓ_a be the projection of the concatenation of the σ_j to the torus. It is closed and has the same winding class as ℓ. Its length satisfies

$$\sum_{j=0}^{p-1} |q_{j+1} - q_j|_1 \leq \sqrt{3} \sum_{j=0}^{p-1} |x_{j+1} - x_j| + 3pa \leq (\sqrt{3} + 3)\mathfrak{l} + 3a,$$

where we used $|q_{j+1} - q_j|_1 \leq \sqrt{3}|q_{j+1} - q_j| \leq \sqrt{3}|x_{j+1} - x_j| + 3a$ and $pa \leq \mathfrak{l} + a$. Since $a \leq L/4$, this proves the first bound in (29.4), including coarse meshes.

Let η_j be the jth subarc of the lift, from x_j to x_{j+1}, let $c_j = [x_j, q_j]$ be the straight segments, and let $\delta_j = c_j * \sigma_j * c_{j+1}^{-1}$, which also runs from x_j to x_{j+1}. The curves η_j, σ_j, c_j, and c_{j+1} lie in a ball of radius Ca about x_j. We parametrize η_j and δ_j on $[0,1]$, piecewise C^1, and consider the strip $H_j(u,v) = (1-v)\eta_j(u) + v\delta_j(u)$, whose endpoints are fixed. Since $|\partial_v H_j| = |\delta_j(u) - \eta_j(u)| \le Ca$, its swept area satisfies

$$\begin{aligned}\int_0^1\int_0^1 |\partial_u H_j \wedge \partial_v H_j|\,du\,dv &\le Ca\int_0^1\int_0^1 \big((1-v)|\eta_j'(u)| + v|\delta_j'(u)|\big)\,du\,dv\\ &\le Ca\big(\operatorname{length}(\eta_j) + \operatorname{length}(\delta_j)\big) \le Ca^2.\end{aligned}$$

For polygonal loops we may instead triangulate these strips by coning their polygonal boundaries to x_j; each triangle has area at most Ca times the length of its boundary edge, which gives the same bound and piecewise smooth strips. We concatenate the strips. The interior pairs $c_j^{-1} * c_j$ are removed by retracing homotopies, whose differentials have rank at most one and which therefore have zero swept area. The final connector is the initial one translated by Lz. After projection to the torus we obtain a comparison homotopy H_a between ℓ and ℓ_a with connector c_0, and since swept areas add,

$$\operatorname{Area}(H_a) \le Cpa^2 \le Ca(\mathfrak{l} + a) \le Ca(\mathfrak{l} + L).$$

All constructions take place in the lift before projection, and they are valid across the periodic identifications and at self-intersections. For piecewise C^1 loops the strips are piecewise C^1, which suffices for the holonomy variation formula of Lemma 29.11. If $\mathfrak{l} = 0$, we round $x(0)$ by the same rule and take the constant lattice loop at the rounded point; the connector followed by its inverse and the constant loop are related by a retracing homotopy of zero area. These rules involve only ℓ and a. □

Lemma 29.11. *Let $L > 0$ and let A be a smooth connection on $\mathbb{T}^3_L$.*

1. *Let $H : [0,1]^2 \to \mathbb{T}^3_L$ be piecewise C^1, with $H(0,v)$ and $H(1,v)$ independent of v. Then*

$$\big\|\operatorname{Hol}_{H(\cdot,1)} A - \operatorname{Hol}_{H(\cdot,0)} A\big\|_{\mathrm{op}} \le C\|F_A\|_\infty \operatorname{Area}(H),$$

   where C is an absolute constant. In particular, if H is a comparison homotopy between loops ℓ and ℓ' with connector c, then $\operatorname{Hol}_\ell A$ and $(\operatorname{Hol}_c A)(\operatorname{Hol}_{\ell'} A)(\operatorname{Hol}_c A)^{-1}$ differ by at most $C\|F_A\|_\infty \operatorname{Area}(H)$ in operator norm, and for every character χ and every A with $\|A\|_{C^1} \le R$,

$$|\chi(\operatorname{Hol}_\ell A) - \chi(\operatorname{Hol}_{\ell'} A)| \le C_\chi(R + R^2)\operatorname{Area}(H).$$

2. *Let $a = L/N$, and let U^A be the lattice configuration $U^A_{(x,x+ae_i)} = \exp(aA_i(x))$. If $\|A\|_{C^1} \le R$ and ℓ_a is a closed lattice path of physical length at most $\mathfrak{l}$, then $\|\operatorname{Hol}_{\ell_a} A - \operatorname{Hol}_{\ell_a} U^A\|_{\mathrm{op}} \le C_R \mathfrak{l} a$, and $|\chi(\operatorname{Hol}_{\ell_a} A) - \chi(\operatorname{Hol}_{\ell_a} U^A)| \le C_{\chi,R}\mathfrak{l} a$.*

3. *Let $\ell, \ell_K : [0,1] \to \mathbb{T}^3_L$ be piecewise C^1 loops with $\sup_K \operatorname{length}(\ell_K) < \infty$ and $\delta_K = \sup_u d_{\mathbb{T}^3_L}(\ell(u), \ell_K(u)) \to 0$. If $\delta_K < L/2$, then ℓ_K has the same winding class as ℓ, and there exists a comparison homotopy H_K between ℓ and ℓ_K, whose connector is the shortest segment from $\ell(0)$ to $\ell_K(0)$, such that*

$$\operatorname{Area}(H_K) \le \frac{\delta_K}{2}\big(\operatorname{length}(\ell) + \operatorname{length}(\ell_K)\big). \tag{29.5}$$

In particular, a sequence of closed lattice paths of bounded physical length that converges uniformly to ℓ, as parametrized loops, is an admissible approximation of ℓ.

The constants C_χ, C_R, and $C_{\chi,R}$ depend only on the indicated quantities.

Proof. (1) We first consider a piece of H on which H is C^2, and write $B = A(H)\partial_u H$ and $\widetilde{C} = A(H)\partial_v H$ for the pulled-back components of the connection. Let P solve $\partial_u P = PB$ with $P(0,v) = I$, so that $P(1,v) = \operatorname{Hol}_{H(\cdot,v)} A$, and set $Q = \partial_v P - P\widetilde{C} + \widetilde{C}(0,v)P$. Differentiating, we find

$$\partial_u Q = QB - P\big(\partial_u \widetilde{C} - \partial_v B + [B, \widetilde{C}]\big), \qquad Q(0,v) = 0.$$

The expression in parentheses is the curvature $F_A(H)(\partial_u H, \partial_v H)$. Since $\partial_u(QP^{-1}) = -PF_A(H)(\partial_u H, \partial_v H)P^{-1}$ and all transports are unitary, we obtain

$$\|Q(1,v)\|_{\mathrm{op}} \le C\|F_A\|_\infty \int_0^1 |\partial_u H \wedge \partial_v H|\, du. \tag{29.6}$$

If $H(0,v)$ and $H(1,v)$ do not depend on v, then $\widetilde{C}(0,v) = \widetilde{C}(1,v) = 0$, so $Q(1,v) = \partial_v P(1,v)$. Integrating (29.6) over $v \in [0,1]$, we obtain the first assertion. For a piece that is only C^1, we extend a C^1 lift slightly beyond its parameter rectangle by reflection, $f(-u,v) = 2f(0,v) - f(u,v)$ at the boundary $u = 0$ and analogously at the other sides, and mollify. The mollified maps and their first derivatives converge uniformly. By their integral form, the solutions of the transport equations then converge uniformly, and the integrands in (29.6) converge in L^1. The integrated identity and the inequality therefore pass to the C^1 piece. For a piecewise C^1 map we apply this on a finite subdivision of the parameter square into rectangles and cancel the internal transport terms, which refer to the same boundary curves. This holds irrespective of crossings and orientations. For a comparison homotopy, $H(0,v) = H(1,v)$ is the base point of ℓ, and $H(\cdot,1)$ is the concatenation $c * \ell' * c^{-1}$. A fixed character is Lipschitz on the compact group, and $\|F_A\|_\infty \le C(R + R^2)$ if $\|A\|_{C^1} \le R$.

(2) Let P be the exact transport along the edge from x to $x + ae_i$, that is, $P'(v) = P(v)A_i(x + ve_i)$, $P(0) = I$. By Duhamel's formula and unitarity,

$$\|P(a) - e^{aA_i(x)}\|_{\mathrm{op}} \le C \int_0^a |A_i(x + ve_i) - A_i(x)|\, dv \le C_R a^2.$$

Reversed edges satisfy the same bound. A closed lattice path of physical length at most ℓ has at most ℓ/a edges, and by telescoping the product of unitary matrices we obtain the bound $C_R \ell a$. The character bound follows since χ is Lipschitz on $\mathrm{SU}(2)$.

(3) Since $\delta_K < L/2$, the shortest displacement from $\ell(u)$ to $\ell_K(u)$ is unique and has a unique continuous lift $d_K(u) \in \mathbb{R}^3$ with $|d_K(u)| \le \delta_K$. Let q be a lift of ℓ, and set $q_K = q + d_K$, a lift of ℓ_K. Since $\ell(1) = \ell(0)$ and $\ell_K(1) = \ell_K(0)$, we have $d_K(1) = d_K(0)$. The two lifts therefore have the same displacement $q(1) - q(0)$ and the loops have the same winding class. The ruled homotopy $G_K(u, v) = q(u) + v d_K(u)$ descends to the torus. Differentiating piecewise, we find

$$|\partial_v G_K| \le \delta_K, \qquad \partial_u G_K = (1 - v) q'(u) + v q_K'(u),$$

so $\mathrm{Area}(G_K) \le \delta_K \int_0^1 \int_0^1 ((1 - v)|q'| + v|q_K'|)\, du\, dv$, which is the right side of (29.5). The endpoints $G_K(0, v) = G_K(1, v)$ move along the segment $c_K(v) = q(0) + v d_K(0)$ from $\ell(0)$ to $\ell_K(0)$. The homotopy $v \mapsto c_K|_{[0,v]} * G_K(\cdot, v) * (c_K|_{[0,v]})^{-1}$, suitably parametrized, is a comparison homotopy between ℓ and ℓ_K with connector c_K; the added retracing parts have differentials of rank at most one and contribute no swept area. This proves (29.5). The finitely many paths with $\delta_K \ge L/2$ may have another winding class; this does not affect convergence statements. □

Remark 29.12. Uniform convergence of the paths alone does not suffice for convergence of holonomies; the bound on the lengths in Definition 2.1 is needed. Consider a smooth periodic connection that agrees in a coordinate ball with $A = b x_1 \tau_3\, dx_2$, where $b \notin 4\pi\mathbb{Z}$, and the loops that traverse a square of side δ based at the origin $m_\delta = \lfloor \delta^{-2} \rfloor$ times. These loops converge uniformly to the constant loop, but their lengths $4 m_\delta \delta$ tend to infinity, and their holonomies $\exp(b m_\delta \delta^2 \tau_3)$ converge to $\exp(b\tau_3) \ne I$. Taking the sides of the squares to be integer multiples of the mesh gives the same example with lattice loops. The length and area conditions of Definition 2.1 exclude it.

Lattice loop observables

Definition 29.13. Let $N \ge 4$, let ℓ be a piecewise C^1 loop, let χ be a character of a finite-dimensional representation of $\mathrm{SU}(2)$, let $t > 0$, and let ℓ_N be a closed lattice path in Λ_N. Set

$$W^{\mathrm{lat}}_{N,\ell,\chi,t}(U) = \chi\big(\mathrm{Hol}_{\ell_N}(\Psi^{(N)}_t U)\big), \qquad U \in \mathcal{U}_N, \tag{29.7}$$

where the dependence on ℓ_N is suppressed. For the loops and times (ℓ_j, t_j) of Theorem 29.3, let ℓ_{j,a_N} be the lattice paths of Lemma 29.10, fixed once and for all j and N, and set

$$W^{\mathrm{lat}}_{N,j}(U) = \tfrac{1}{2} \operatorname{tr} \mathrm{Hol}_{\ell_{j,a_N}}\big(\Psi^{(N)}_{t_j} U\big), \qquad U \in \mathcal{U}_N. \tag{29.8}$$

On a sequence of sides $(N_K)_K$ we write $W^{\mathrm{lat}}_{K,\ell,\chi,t}$ and $W^{\mathrm{lat}}_{K,j}$ for these functions on $\mathcal{U}_{N_K}$.

The paths ℓ_{j,a_N} depend only on the loop and the mesh, and not on the configuration, on any parameter of the renormalization group, or on the coarse comparison side used below. By Lemma 29.10, along every sequence $N_K \to \infty$ they form admissible approximations of the loops ℓ_j, with swept areas $O(a_{N_K})$.

The Wilson flow and the holonomies of closed lattice paths are continuous and covariant under lattice gauge transformations, and a character removes the conjugation at the base point. The functions (29.7) and (29.8) are therefore continuous, in particular Borel, and invariant under all lattice gauge transformations on the whole configuration space. By unitarity, $|W^{\mathrm{lat}}_{N,\ell,\chi,t}| \le \chi(I)$, and the functions $W^{\mathrm{lat}}_{N,j}$ are real with absolute value at most one. In particular,

$$|w_j(X) - W^{\mathrm{lat}}_{N,j}(U)| \le 2 \qquad \text{for all } X \in \mathcal{X}_L,\ U \in \mathcal{U}_N. \tag{29.9}$$

The results of Section 28 allow characters of arbitrary finite-dimensional representations, including the unnormalized fundamental character. Whenever we apply Theorems 28.1 and 28.2 to the functions $W^{\mathrm{lat}}_{N,j}$, we apply them to the vector of fundamental characters and multiply its entries by $1/2$. This continuous linear map preserves joint convergence in law, and for complex source functionals it replaces the sources z_i by $z_i/2$. The limits are the corresponding joint laws of the determining coordinates w_j.

The lattice energy

Recall the physical lattice energy $\mathsf{E}_a(U) = (8/a)\sum_{p\in\mathcal{P}_N} q(U_p)$ of (2.2), $a = L/N$, in which each positive plaquette is counted once. In terms of the Wilson action S_W of Definition 4.1, which equals E_0, we have $\mathsf{E}_a = 2S_W/a$. For a lattice potential A at mesh a we write e^{aA} for the configuration with $(e^{aA})_{(x,x+ae_i)} = \exp(aA_i(x))$, and for a smooth one-form b on $\mathbb{T}^3_L$ we write $S_a b$ for its restriction to the lattice, $(S_a b)_i(x) = b_i(x)$.

Lemma 29.14. *Let $L > 0$ and $a = L/N$ with $N \ge 4$.*

1. *The function E_a is continuous on $\mathcal{U}_N$, invariant under lattice gauge transformations, and nonincreasing along the Wilson flow $\Psi^{(N)}_t = \Phi^{\mathrm{W}}_{4t/a^2}$.*
2. *For every $R > 0$ there exists $C_{L,R} < \infty$, independent of N, such that for lattice potentials A, B at mesh a with $\|A\|_{C^1_a}, \|B\|_{C^1_a} \le R$,*

$$|\mathsf{E}_a(e^{aA}) - \mathsf{E}_a(e^{aB})| \le C_{L,R}\|A - B\|_{C^1_a}, \qquad 0 \le \mathsf{E}_a(e^{aA}) \le C_{L,R}.$$

3. *For every smooth periodic one-form b we have $\mathsf{E}_a(e^{aS_a b}) \to E_L(b)$ as $a \to 0$. More precisely, if $\|b\|_{C^2} \le R$ and a is sufficiently small, then $|\mathsf{E}_a(e^{aS_a b}) - E_L(b)| \le C_{L,R}a$.*
4. *Let $\mathcal{E}_a(U) = (4/a)\sum_{p\in\mathcal{P}_N} q(U_p)$ be the normalization used in Section 24. Then $\mathsf{E}_a = 2\mathcal{E}_a$ on $\mathcal{U}_N$.*

The factor $8/a$ is chosen so that the limit in (3) is the energy E_L of (2.3), without a factor $1/2$; the exponent in (2.1) equals $\mathsf{E}_{a_N}/(2g_3^2)$. The point of (2) is that the Lipschitz constant does not depend on the mesh: a plaquette logarithm is $O(a^2)$, so a plaquette energy and its variation are $O(a^4)$, which compensates the factor a^{-1} and the $3(L/a)^3$ plaquettes.

Proof. (1) Continuity follows from the finite products and the trace. A lattice gauge transformation conjugates each plaquette matrix, which leaves $q(U_p)$ unchanged. Since E_a is a positive multiple of the Wilson action, it is nonincreasing along the negative gradient flow Φ^{W}, and the positive change of time $s = 4t/a^2$ does not affect this.

(2) We consider one plaquette $p_{ij}(x)$ and set

$$v = \big(aA_i(x),\, aA_j(x+ae_i),\, -aA_i(x+ae_j),\, -aA_j(x)\big).$$

For aR small, the expansion of the logarithm of a product of four exponentials in Lemma 24.2 reads

$$f_A = \operatorname{Log} U_{p_{ij}(x)} = \sum_{k=1}^{4} v_k + \frac{1}{2}\sum_{k<l}[v_k, v_l] + \mathcal{R}(v), \qquad |\mathcal{R}(v)| \le C|v|^3, \quad \|D\mathcal{R}(v)\| \le C|v|^2.$$

The bound on $D\mathcal{R}$ follows from Taylor's integral formula for the analytic logarithm of a word of fixed length: the remainder has vanishing Taylor coefficients up to order two, and its third derivative is bounded on a smaller fixed chart. The linear sum equals $a^2(D_i^+A_j - D_j^+A_i)(x)$. If we freeze the shifted entries at x, the quadratic sum equals $a^2[A_i(x), A_j(x)]$, and changing one entry back costs at most $C_R a^3$. We conclude that

$$a^{-2}f_A = D_i^+A_j - D_j^+A_i + [A_i, A_j] + O_R(a), \qquad |f_A| \le C_R a^2. \tag{29.10}$$

Let $\delta = \|A - B\|_{C_a^1}$. By the exact cancellation in the linear term, we have $|\sum v(A) - \sum v(B)| \le Ca^2\delta$, the difference of the quadratic terms is at most $C_R a^2\delta$, and the difference of the remainders is at most $C_R a^3\delta$. Thus $|f_A - f_B| \le C_R a^2\delta$. By (4.2), $q(e^f) = |f|^2/8 + O(|f|^4)$, and the differential of $f \mapsto q(e^f)$ has norm at most $C|f|$ on this chart. Hence the plaquette energies differ by at most $C_R a^4\delta$. There are $3(L/a)^3$ positive plaquettes, and multiplication by $8/a$ gives the bound $24L^3C_R\delta$, with no remaining power of the mesh. This proves the Lipschitz bound for $a \le a_0(L,R)$. Meshes $a > a_0$ correspond to finitely many sides $N < L/a_0$; for them, the differential of the unitary exponential and telescoping give the plaquette bound $Ca\|A - B\|_\infty$, which yields a Lipschitz constant $24CN^3 \le 24C(L/a_0)^3$. We enlarge $C_{L,R}$ to cover these meshes. Since $\mathsf{E}_a(I) = 0$, taking $B = 0$, we obtain $\mathsf{E}_a(e^{aA}) \le C_{L,R}R$, and we absorb R into the constant. Nonnegativity follows from $q \ge 0$ on $\mathrm{SU}(2)$.

(3) For $A = S_a b$, forward differences differ from derivatives by $O_R(a)$ on the ball $\|b\|_{C^2} \le R$. By (29.10) and the expansion of q,

$$\mathsf{E}_a(e^{aS_a b}) = a^3 \sum_{x\in\Lambda_N} \sum_{i<j} |F_{ij}(b)(x)|^2 + O_{L,R}(a).$$

Since the summand has bounded first derivatives on this ball, replacing the Riemann sum by the integral costs a further $O_{L,R}(a)$. Finally, since $|X|^2 = -2\,\mathrm{tr}\,X^2$ and $F_{ji} = -F_{ij}$, we have $\sum_{i,j} -\mathrm{tr}\,F_{ij}^2 = \sum_{i<j}|F_{ij}|^2$, and the limit is exactly $E_L(b)$.

(4) This is immediate from the definitions. □

Stability along the Wilson flow

We recall the setting of Sections 24 and 25. Fix $1/2 < \alpha < 2/3$ and a horizon $T_0 > 0$. For a lattice potential A_0 at mesh a, its linear and quadratic flow data are $x(t) = P_t^a A_0$ and $y = \mathcal{Q}_a(x,x)$. We measure them by the flow-data norm $\mathsf{X}_\alpha^a(x) + \mathsf{Y}_\alpha^a(y)$, with the norms of Definition 4.8 on $[0,T_0]$, including all physical first differences and shifts. For two potentials A_0, A_0' in the same gauge, with flow data (x,y) and (x',y'), we write

$$d_a = \mathsf{X}_\alpha^a(x-x') + \mathsf{Y}_\alpha^a(y-y').$$

On the smaller link coordinate domain used in Lemma 24.2, the Wilson flow has damped representatives: they are obtained by a time-dependent lattice gauge transformation and solve the gauge-fixed lattice equation of that lemma, whose linear part is the lattice heat equation. A finite reference family for a smooth solution of the Yang–Mills heat flow on a time interval $[s,T]$ consists of a finite partition of $[s,T]$, smooth solutions of the DeTurck equation on the subintervals, and fixed smooth gauge transformations at the partition times, as provided by Theorem 24.10. Each of these DeTurck solutions is a reference solution.

The first lemma transfers a small error at a positive time s to all later times up to a fixed horizon. It follows directly from the comparison with a fixed smooth reference in Lemma 25.5.

Lemma 29.15. *Let $0 < s < T$, and let b be a smooth solution of the Yang–Mills heat flow on $[s,T]$ with a finite reference family, consisting of DeTurck solutions $b^{(i)}$ on intervals I_i. For lattice meshes $a \to 0$, let V_a be configurations with representatives $V_a = e^{aA_a(s)}$ such that*

$$e_a = \|A_a(s) - S_a b(s)\|_\infty + \|D_a A_a(s) - S_a \partial b(s)\|_\infty \longrightarrow 0,$$

where D_a and ∂ denote all first differences and the corresponding derivatives. Then for all sufficiently small a, the Wilson flow $t \mapsto \Psi_{t-s}^{(N)} V_a$, $t \in [s,T]$, has representatives $A_a^{(i)}$ solving that gauge-fixed lattice equation on the intervals I_i, in the gauges of the reference family, such that

$$\max_i \sup_{t\in I_i} \|A_a^{(i)}(t) - S_a b^{(i)}(t)\|_{C_a^1} \le C(e_a + a).$$

The constant C and the smallness threshold depend only on L, the horizon $T - s$, the smooth bounds of the reference family, its gauge transformations, and its chart margins. They do not depend on the mesh, the coupling, or any probability measure, and the convergence $e_a \to 0$ may be arbitrarily slow.

Proof. We translate the initial time to zero, so that the horizon is $T - s$. By Taylor's formula in our difference conventions, $\|D_a S_a b(s) - S_a \partial b(s)\|_\infty \le Ca\|b(s)\|_{C^2}$, so

$$\eta_a = \|A_a(s) - S_a b(s)\|_{C^1_a} \le e_a + Ca\|b(s)\|_{C^2}.$$

Every reference solution of the family is smooth on its closed space-time domain, and in particular its spatial derivatives up to order five are bounded. The finitely many reference solutions and fixed smooth gauge transformations have common bounds, which are part of the data of the reference family and not bounds on higher derivatives of nearby lattice fields. The fixed-reference and finite-family parts of Lemma 25.5 choose the constants and the thresholds for the mesh and the initial error from these bounds, before the incoming error is made small. At the ith partition time, using (25.14) together with the change of gauge, we obtain a recursion

$$\eta_{i+1} \le G_i C_i(\eta_i + a) + d_i a$$

for the error η_i at the start of the ith interval, with finitely many constants G_i, C_i, d_i depending only on the reference family. Iterating it finitely many times, and taking the minimum of the resulting thresholds, we obtain

$$\max_i \sup_{t \in I_i} \|A_a^{(i)}(t) - S_a b^{(i)}(t)\|_{C^1_a} \le C(\eta_a + a) \le C'(e_a + a).$$

The same lemma keeps the representatives inside the coordinate domain of Lemma 24.2, with the required positive distance from its boundary. It also identifies the solution of the gauge-fixed lattice equation on each interval with the Wilson flow, whose time normalization $4t/a^2$ is that of Lemma 24.2. Since $e_a \to 0$ and $a \to 0$, the thresholds are eventually satisfied. □

Theorem 29.16. *Fix $L, T_0, R > 0$, $1/2 < \alpha < 2/3$, $0 < \tau \le T < \infty$, and u as in Lemma 25.9, together with finitely many characters $\chi_1, \dots, \chi_b$ and lengths $\mathfrak{l}_1, \dots, \mathfrak{l}_b > 0$. There exist $a_* > 0$ and $C < \infty$, depending only on these data, with the following property. Let $a = L/N \le a_*$, and let U, U' be the configurations of two lattice potentials at mesh a in the same gauge, whose flow data have norms at most R on $[0, T_0]$ and discrepancy d_a. Then*

$$\sup_{\tau \le t \le T} \Big(\big|\mathsf{E}_a(\Psi_t^{(N)} U) - \mathsf{E}_a(\Psi_t^{(N)} U')\big| + \sum_{i=1}^{b} \big|\chi_i(\mathrm{Hol}_{\ell_{i,a}} \Psi_t^{(N)} U) - \chi_i(\mathrm{Hol}_{\ell_{i,a}} \Psi_t^{(N)} U')\big| \Big) \le C d_a \tag{29.11}$$

for all closed lattice paths $\ell_{i,a}$ in Λ_N of physical length at most $\mathfrak{l}_i$, and

$$\sup_{\tau\le t\le T} \mathsf{E}_a(\Psi_t^{(N)}U) \le C.$$

The constants do not depend on the coupling, on any probability measure, or on the number of averaging steps. The estimate requires no convergence of the configurations as $a \to 0$, no condition on swept areas, and no bound on higher derivatives, and it has no additive error term in the mesh. This allows us to apply it to finite-scale reconstructions with a growing number of scales.

Proof. Let δ, K be the constants of Lemma 25.9. Suppose first that $d_a \le \delta$. By that lemma, the two Wilson flows have damped representatives B, B', solving the gauge-fixed lattice equation on the intervals of a common reference family. They satisfy the individual bounds $\|B(t)\|_{C_a^1}, \|B'(t)\|_{C_a^1} \le K$ and the difference bound $\|B(t) - B'(t)\|_{C_a^1} \le Cd_a$. By gauge invariance and Lemma 29.14(2), the energies differ by at most $C_{L,K}Cd_a$, and each energy is bounded on every interval. Explicitly, a plaquette logarithm is $O_K(a^2)$, the difference of two of them is $O_K(a^2\|B - B'\|_{C_a^1})$, and, since the differential of $f \mapsto q(e^f)$ is $O(|f|)$, the difference of two plaquette energies is $O_K(a^4\|B - B'\|_{C_a^1})$. The normalization cancels as

$$\frac{8}{a}\cdot 3\Big(\frac{L}{a}\Big)^3 C_K a^4 \|B - B'\|_{C_a^1} = 24L^3C_K\|B - B'\|_{C_a^1}.$$

This is the energy E_a, not $\mathcal{E}_a$. For the loops we apply Corollary 25.11 at each $t \in [\tau, T]$; its constant is uniform over the times in $[\tau, T]$ and over all paths within the length bounds. The observables are continuous in t, which joins the finitely many intervals. If $d_a > \delta$, we use the individual energy bound and $|\chi_i| \le \chi_i(I)$: the left side of (29.11) is at most $2C + 2\sum_i \chi_i(I) \le (2C + 2\sum_i \chi_i(I))d_a/\delta$, and we enlarge C accordingly. All constants were fixed before the two configurations were chosen. □

The last deterministic statement concerns a fixed compact family of smooth fields: the cardinal interpolants, in the sense of Lemma 16.3, of arrays on a fixed coarse mesh. Here the constants may depend on all derivatives of the family, unlike those of Theorem 29.16.

Lemma 29.17. *Fix a coarse mesh $h > 0$, a bound $M_h < \infty$, a finite horizon, and the cardinal basis, and let f range over the cardinal interpolants of arrays on the lattice of mesh h with supremum at most M_h, as in Lemma 28.4. Let $0 < \tau < T < \infty$, and let $\ell_1, \dots, \ell_b$ be loops with characters $\chi_1, \dots, \chi_b$ and time-independent admissible approximations $(\ell_{i,a})$, with comparison homotopies of swept area $\mathcal{A}_{i,a} \to 0$. Then, as $a \to 0$ along meshes compatible with h, the paths*

$$t \mapsto \Big(\mathsf{E}_a\big(\Psi_t^{(N)}e^{af|_a}\big),\ \big(\chi_i(\mathrm{Hol}_{\ell_{i,a}}\Psi_t^{(N)}e^{af|_a})\big)_{i\le b}\Big), \qquad t \in [\tau, T],$$

converge uniformly in $t \in [\tau, T]$, and uniformly over the family, to the paths $t \mapsto (E_L(\Phi_t[f]), (\chi_i(\mathrm{Hol}_{\ell_i}\Phi_t[f]))_{i\le b})$. More precisely, the uniform distance between these paths is at most

$$C_{h,M_h,\tau,T,\mathscr{W}}\big(a + \max_i \mathcal{A}_{i,a}\big), \tag{29.12}$$

where $\mathscr{W}$ denotes the list of loops, characters, and lengths. The limit map from arrays to $C([\tau,T];\mathbb{R}\times\mathbb{C}^b)$ is continuous.

Proof. We use the finite reference families and the consistency of the lattice flow with the sampled smooth flow in the proof of Lemma 28.4. On each reference interval the C^1_a distance between the damped representative and the sampled reference solution tends to zero uniformly over the family. The Lipschitz bound and the consistency for smooth samples in Lemma 29.14 bound the error in the energy by $C\|A_a - S_a b\|_{C^1_a} + Ca$. The reference bounds are uniform over this compact family and its finitely many reference families. The error in the loops is at most $C(a + \mathcal{A}_{i,a})$, by Lemma 29.11(1),(2) (as in the proof of Lemma 28.4), plus the propagated error in the potential. All these bounds are uniform in time, which gives (29.12). The continuity of the limit map follows from the smooth dependence of the flow on its initial datum and the continuity of the energy in C^1. □

29.3 Joint convergence under the conditioned measures

Standing assumptions. In Sections 29.3 to 29.5 we fix $L > 0$ and an odd integer $M \ge 3$, together with the integer J, the remaining parameters Θ of the small-field event, and the coupling bound $\gamma_0 \in (0,1]$ of Theorem 23.2, for which Theorem 23.9 holds. We fix $\gamma \in (0,\gamma_0]$ and $n_* \in 4\mathbb{N}$, and set $B = M^J$, $N_K = n_* B^K$, $a_K = L/N_K$, and $n_r = n_* B^r$. For $0 \le r \le K$ let

$$\nu_{K|r} = \nu_{N_K,n_r,\gamma} = \mu_{N_K,\gamma}\big(\,\cdot \mid S_{N_K,n_r,\gamma}\big), \qquad e_{K|r} = \mu_{N_K,\gamma}\big(S^c_{N_K,n_r,\gamma}\big)$$

be the conditioned Wilson measure with coarse side n_r and the corresponding small-field failure probability. Let $\mathcal{R}$ be the set of indices r for which Theorem 28.8, and hence Theorems 28.1 and 28.2, apply at the coarse side n_r; this set is infinite. By Theorem 23.9, as recorded in Theorem 28.2, for $r \in \mathcal{R}$ we have

$$e_{K|r} \le \epsilon_K + F_\gamma(n_r), \qquad \epsilon_K \to 0 \ (K\to\infty), \qquad F_\gamma(n_r) \to 0 \ (r\to\infty), \tag{29.13}$$

and one may take $\epsilon_K = 0$ and $F_\gamma(u) = C' u^3 e^{-\bar{c}u/\gamma} + C_{\mathrm{w}} u^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mu/\gamma}}$.

The conditioning is removed by the following elementary identity. If $e_{K|r} < 1$, then $\mu_{N_K,\gamma} = (1 - e_{K|r})\nu_{K|r} + e_{K|r}\rho_{K|r}$ with $\rho_{K|r} = \mu_{N_K,\gamma}(\,\cdot \mid S^c_{N_K,n_r,\gamma})$ (any probability measure if $e_{K|r} = 0$). In particular,

$$\mu_{N_K,\gamma}(\mathcal{B}) \le \nu_{K|r}(\mathcal{B}) + e_{K|r} \qquad \text{for every Borel event } \mathcal{B}, \tag{29.14}$$

and $|\mathbb{E}_{\mu_{N_K,\gamma}} F - \mathbb{E}_{\nu_{K|r}} F| \le D e_{K|r}$ for every measurable F with $0 \le F \le D$, and $\le 2 e_{K|r} \|F\|_\infty$ for every bounded complex measurable F (Lemma 5.5). We always use these bounds in the following order: first $K \to \infty$ at a fixed $r \in \mathcal{R}$, and then $r \to \infty$ in $\mathcal{R}$.

Finite-scale reconstructions. Fix $r \in \mathcal{R}$ and write $n = n_r$ and $\nu_K = \nu_{nB^K,n,\gamma}$, so that $\nu_K = \nu_{K+r|r}$. We recall the following objects from Theorem 28.8, with input horizon $T_0 = L^2$; they are all defined on the probability space $(\mathcal{U}_{nB^K}, \nu_K)$. Backward reconstruction of U with the reconstruction maps of Theorem 15.12 gives arrays A_j, $0 \le j \le K$, on the lattices of mesh $a_j = (L/n)B^{-j}$, and the configuration $e^{a_K A_K}$ is a gauge representative of U. For $0 \le q \le K$, the finite-scale reconstruction from the first q renormalization group steps from the coarse side is the fine configuration $U_K^{[q]} = \exp(a_K I_{a_q \to a_K} A_q)$, where $I_{a_q \to a_K}$ is the cardinal interpolation of Lemma 16.3. Its flow data at mesh a_K are $x_q = P^{a_K} I_{a_q \to a_K} A_q$ and $y_q = \mathcal{Q}_{a_K}(x_q, x_q)$, and those of U are x_K, y_K. We set

$$\mathsf{R}_K = \mathsf{X}_\alpha^{a_K}(x_K) + \mathsf{Y}_\alpha^{a_K}(y_K), \qquad \rho(R) = \sup_{K \ge 0} \nu_K\{\mathsf{R}_K > R\},$$
$$D_{K,q} = \mathsf{X}_\alpha^{a_K}(x_K - x_q) + \mathsf{Y}_\alpha^{a_K}(y_K - y_q).$$

By Theorem 28.8, we have $\rho(R) \to 0$ as $R \to \infty$ and

$$\lim_{q \to \infty} \sup_{K \ge q} \nu_K\{D_{K,q} > \eta\} = 0 \qquad \text{for every } \eta > 0. \tag{29.15}$$

For each fixed q, there exists a compact set $\mathcal{C}_q$ of arrays, independent of $K \ge q$, that supports the law of A_q under every ν_K. These laws converge weakly as $K \to \infty$ to a law supported on $\mathcal{C}_q$ (Theorem 15.13). Let f_q be the cardinal interpolant of A_q on the torus, a one-form, and let $\lambda_q = \lambda_{r,q}$ be the law of the trajectory $t \mapsto \Phi_t[f_q]$ when A_q has this limiting law.

Theorem 29.18. *Assume the standing assumptions, and let $r \in \mathcal{R}$. There exists a Borel probability measure λ_r on $\mathcal{X}_L$, written $\lambda_{r,\gamma}$ when the coupling is displayed, with the following properties.*

1. *As $q \to \infty$, every finite vector of energies $E_L(X(v))$, $v > 0$, and functions $W_{\ell,\chi,t}$ under $X \sim \lambda_q$ converges in law to the same vector under $X \sim \lambda_r$, and $(i^{1,2})_\# \lambda_q \to (i^{1,2})_\# \lambda_r$ against bounded $\mathcal{T}_{1,2}$-continuous functions.*
2. *Let $v_1, \dots, v_p > 0$. For $h \le b$ let ℓ_h be loops with admissible approximations $(\ell_{h,K})_K$ along $(N_K)_K$, let χ_h be characters, let $t_h > 0$, and write $W_{K,h}^{\mathrm{lat}} = \chi_h(\mathrm{Hol}_{\ell_{h,K}} \Psi_{t_h}^{(N_K)} U)$. Then, under $\nu_{K|r}$, as $K \to \infty$,*

$$\Big(\big(\mathsf{E}_{a_K}(\Psi_{v_i}^{(N_K)} U)\big)_{i \le p}, \big(W_{K,h}^{\mathrm{lat}}\big)_{h \le b} \Big) \Longrightarrow \Big(\big(E_L(X(v_i))\big)_{i \le p}, \big(W_{\ell_h,\chi_h,t_h}(X)\big)_{h \le b} \Big),$$

where $X \sim \lambda_r$.

3. *Suppose that the paths $\ell_{h,K}$ do not depend on the time, and let $0 < \tau < T < \infty$. Then, under $\nu_{K|r}$, the processes*

$$t \mapsto \Big(\mathsf{E}_{a_K}(\Psi_t^{(N_K)}U), \big(\chi_h(\mathrm{Hol}_{\ell_{h,K}}\Psi_t^{(N_K)}U)\big)_{h\le b}\Big), \qquad t \in [\tau, T],$$

converge in law in $C([\tau,T];\mathbb{R}\times\mathbb{C}^b)$ to $t \mapsto (E_L(X(t)), (W_{\ell_h,\chi_h,t}(X))_{h\le b})$ with $X \sim \lambda_r$. If $(\ell'_{h,K})_K$ are other time-independent admissible approximations of the same loops, then

$$\lim_{K\to\infty} \mathbb{E}_{\nu_{K|r}} \sup_{\tau\le t\le T} \max_{h\le b} \big|\chi_h(\mathrm{Hol}_{\ell_{h,K}}\Psi_t^{(N_K)}U) - \chi_h(\mathrm{Hol}_{\ell'_{h,K}}\Psi_t^{(N_K)}U)\big| = 0.$$

4. *For admissible approximations with swept areas $O(a_K)$, in particular those of Definition 29.13, the limit laws of the loop vectors in (2) are those of Theorem 28.1.*
5. *For every $t > 0$,*

$$\lim_{A\to\infty} \sup_{K\ge r} \nu_{K|r}\big\{\mathsf{E}_{a_K}(\Psi_t^{(N_K)}U) > A\big\} = 0, \qquad \lim_{A\to\infty} \sup_{K\ge 0} \mu_{N_K,\gamma}\big\{\mathsf{E}_{a_K}(\Psi_t^{(N_K)}U) > A\big\} = 0.$$

All convergence moduli are at the fixed coupling γ and, for the conditioned measures, at the fixed index r.

Proof. We fix $r \in \mathcal{R}$ and use the reindexed measures ν_K and the notation introduced above; the statements for $\nu_{K|r}$ follow by the shift $K \mapsto K + r$. We first construct the laws λ_q and show that the lattice observables of U are close in law to those of the finite-scale reconstructions, whose limits are continuum observables under λ_q; this produces joint limits of the lattice observables. We then prove that the energies under λ_q are tight, extract a limit λ_r using the compactness theorem of Cao and Chatterjee for Sobolev trajectories, identify λ_r by its Wilson loops, and use the continuity of energies after a positive time (Lemma 29.7) to identify the joint limits of energies and loops.

Step 1: the laws λ_q. The map from arrays A_q to the initial datum f_q is linear and continuous in every smooth norm, and by Theorem 24.10 the smooth flow depends continuously on its initial datum on every finite horizon. Hence every determining coordinate w_j of the trajectory $t \mapsto \Phi_t[f_q]$ is a continuous function of the array. Since the Borel σ-algebra of $\mathcal{X}_L$ is generated by the w_j (Theorem 29.3), the map from arrays to trajectories is measurable, and λ_q is a Borel probability measure on $\mathcal{X}_L$. Its image $(i^{1,2})_\#\lambda_q$ is a Borel probability measure on $\mathcal{X}_L^{1,2}$, by Step 4 of the proof of Theorem 29.3.

Step 2: approximation by finite-scale reconstructions. Fix a finite list as in (2), respectively a time-independent list and an interval $[\tau, T]$ as in (3). Let Z_K be the corresponding finite vector, respectively path, of observables of U, with values in the complete separable metric space $E = \mathbb{R}^p \times \mathbb{C}^b$, respectively $E = C([\tau,T];\mathbb{R}\times\mathbb{C}^b)$, which we equip with the

bounded metric $d = \min(2, \|\cdot\|_\infty)$. Let $Z_K^{[q]}$ be the same observables of the finite-scale reconstruction $U_K^{[q]}$, on the same probability space $(\mathcal{U}_{nB^K}, \nu_K)$. Since $e^{a_K A_K}$ is a gauge representative of U and the observables are gauge invariant, Z_K is the observable of the conditioned configuration itself, not merely a random variable with the same law.

On the event $\{\mathsf{R}_K \le R,\ D_{K,q} \le \eta\}$ with $\eta \le 1$, both flow data have norm at most $R+1$, and the two potentials are in the same gauge. By Theorem 29.16 at radius $R+1$, applied on an interval containing all times of the list, we have $d(Z_K, Z_K^{[q]}) \le C_R \eta$ whenever $a_K \le a_*(R+1)$. We conclude that, for every test function ϕ on E bounded by one and with Lipschitz constant at most one,

$$\left|\mathbb{E}_{\nu_K}\phi(Z_K) - \mathbb{E}_{\nu_K}\phi(Z_K^{[q]})\right| \le C_R\eta + 2\rho(R) + 2\nu_K\{D_{K,q} > \eta\} \tag{29.16}$$

for all K with $a_K \le a_*(R+1)$. This threshold does not depend on q. By (29.15), the right side of (29.16) tends to zero, uniformly in $K \ge q$, if we choose first R large, then η small, and then q large, as in the proof of Theorem 28.1.

Fix q, and let G be the map that sends a trajectory to the continuum energies and Wilson loops of the list. By Lemma 29.17, applied with the coarse mesh a_q and the compact set $\mathcal{C}_q$, the observables of $U_K^{[q]}$ converge as $K \to \infty$, uniformly for $A_q \in \mathcal{C}_q$, to the continuous function $A_q \mapsto G(t \mapsto \Phi_t[f_q])$. Combining this with the weak convergence of the laws of A_q, which are all supported on $\mathcal{C}_q$, we obtain $Z_K^{[q]} \Rightarrow G_\# \lambda_q$. The observable maps converge uniformly on $\mathcal{C}_q$ to a continuous map, and each fixed-K map is continuous on $\mathcal{C}_q$. By uniform convergence, compactness of $\mathcal{C}_q$, and completeness of E, the union of their images has compact closure, and so does the union of the supports of the laws of $Z_K^{[q]}$, $K \ge q$. Each Z_K has compact support, since the compact configuration space is mapped continuously into E by the finite-dimensional Wilson flow. By Lemma 28.9, a version of the converging-together principle [28, Theorem 3.2] that also constructs the limit, the laws of Z_K converge to a probability measure ρ_G on E. Letting $K \to \infty$ at fixed q in (29.16), we find that $G_\#\lambda_q \to \rho_G$ as $q \to \infty$. The argument applies to every finite list, and in particular to the union of any two lists. The limits are therefore consistent under unions and projections, and we never combine separately obtained marginal limits.

Step 3: tightness of the energies under λ_q. Let $t > 0$ and $\varepsilon > 0$. Fix R with $\rho(R) < \varepsilon$. By (29.15), for all sufficiently large q we have $\sup_{K \ge q} \nu_K\{D_{K,q} > 1\} < \varepsilon$, and we may also assume $a_q \le a_*(R+1)$, so that $a_K \le a_*(R+1)$ for all $K \ge q$. On the event $\{\mathsf{R}_K \le R,\ D_{K,q} \le 1\}$ the flow data of $U_K^{[q]}$ have norm at most $R+1$, and by the individual bound in Theorem 29.16 we have $\mathsf{E}_{a_K}(\Psi_t^{(nB^K)} U_K^{[q]}) \le C_{R,t}$ for a deterministic $C_{R,t}$. Let ϕ be continuous, with $\phi = 0$ on $(-\infty, C_{R,t}]$, $\phi = 1$ on $[C_{R,t}+1, \infty)$, and $0 \le \phi \le 1$. Then $\mathbb{E}_{\nu_K}\phi(\mathsf{E}_{a_K}(\Psi_t^{(nB^K)} U_K^{[q]})) \le 2\varepsilon$ for $K \ge q$. Letting $K \to \infty$, by Lemma 29.17 and the

convergence of the law of A_q, we obtain

$$\lambda_q\{E_L(X(t)) > C_{R,t} + 1\} \leq \mathbb{E}_{\lambda_q}\phi\big(E_L(X(t))\big) \leq 2\varepsilon.$$

The finitely many smaller q have laws supported on compact sets of smooth initial data, and their energies at time t are bounded. We conclude that the energies at time t are tight under $\{\lambda_q\}_{q\geq 0}$, for every $t > 0$.

Step 4: the limit λ_r. The compactness theorem for Sobolev trajectories [43, Theorem 3.73, pp. 41–42] applies to Borel $\mathcal{X}_L^{1,2}$-valued random variables whose energy laws at each positive time are tight, which we have just verified for $(i^{1,2})_\#\lambda_q$. For fixed L we use the rescaling of Lemma 29.2. We obtain a subsequence along which $(i^{1,2})_\#\lambda_q$ converges, against bounded $\mathcal{T}_{1,2}$-continuous functions, to a Borel probability measure η on $\mathcal{X}_L^{1,2}$. We set $\lambda_r = (j^{1,2})_\#\eta$, a Borel probability measure on $\mathcal{X}_L$. By Lemma 29.2(6),(7), $j^{1,2}$ preserves the functions $W_{\ell,\chi,t}$, and these are continuous on $\mathcal{X}_L^{1,2}$. Hence the finite joint laws of the functions $W_{\ell,\chi,t}$ under λ_r are the limits of those under λ_q along the subsequence. By Step 2, these are the marginals of the limits ρ_G, so they do not depend on the subsequence. By Theorem 29.3(4), λ_r does not depend on the subsequence. Since $i^{1,2}j^{1,2} = \mathrm{Id}$, we have $\eta = (i^{1,2})_\#\lambda_r$, so the limit η is unique. The whole sequence $(i^{1,2})_\#\lambda_q$ therefore converges to $(i^{1,2})_\#\lambda_r$: otherwise a subsequence would stay away from it, and a further subsequence would converge to $(i^{1,2})_\#\lambda_r$, a contradiction.

Step 5: identification of the energies. By Lemma 29.7, for every $v > 0$ the function $Z \mapsto E_L(Z(v))$ is continuous on $(\mathcal{X}_L^{1,2}, \mathcal{T}_{1,2})$; one factors the evaluation at time v through an earlier time $s \in (0, v)$. The functions $W_{\ell,\chi,t} \circ j^{1,2}$ are continuous as well. Applying bounded continuous functions to the whole vector, we see from Step 4 that every finite vector of energies and Wilson loops under λ_q converges in law to the same vector under λ_r. This proves (1). By Step 2 applied to finite vectors, the same vectors under λ_q converge to the limit law ρ_G of the lattice observables. We conclude that ρ_G is the law of the continuum vector under λ_r, which proves (2). This step uses the continuity of the energy after a positive time; measurability of the energy, or convergence of the energy marginals separately, would not suffice.

Step 6: paths and comparison of approximations. In the setting of (3), the limit ρ_G is a probability measure on continuous paths, and by Step 5 its finite-dimensional distributions at rational times coincide with those of $G_\#\lambda_r$. Evaluations at rational times generate the Borel σ-algebra of $C([\tau, T]; \mathbb{R} \times \mathbb{C}^b)$ and determine probability measures on it, and the continuum path map G is Borel for the same reason. This implies that $\rho_G = G_\#\lambda_r$. For two time-independent admissible approximations, we apply this to the union of the two lists. In the limit, the corresponding loop paths coincide. The supremum over $[\tau, T]$ of their maximal difference is a continuous function of the joint path, bounded by

$2\max_h \chi_h(I)$. It converges in law to zero, and since it is bounded, its expectation tends to zero. This proves (3).

For approximations with swept areas $O(a_K)$ and lists consisting only of loops, Step 2 is the argument of Theorem 28.1, and both limits are the limits as $q \to \infty$ of the laws $G_\# \lambda_q$; this proves (4).

Step 7: tails of the lattice energies. The tails under the conditioned measures follow from the tightness of R_K and the individual energy bound of Theorem 29.16, applied with $U' = U$. The finitely many meshes above the threshold a_* are covered by the deterministic bound $\mathsf{E}_a(U) \le (8/a) \cdot 3N^3 \cdot 2 = 48N^3/a$, since $0 \le q \le 2$. For the Wilson measures we use (29.14) and (29.13):

$$\mu_{N_K,\gamma}\{\mathsf{E}_{a_K}(\Psi_t^{(N_K)}U) > A\} \le \nu_{K|r}\{\mathsf{E}_{a_K}(\Psi_t^{(N_K)}U) > A\} + \epsilon_K + F_\gamma(n_r).$$

We take the limit superior as $K \to \infty$, then let $A \to \infty$ at fixed r, and finally let $r \to \infty$ in $\mathcal{R}$. This shows that $\lim_{A\to\infty} \limsup_{K\to\infty} \mu_{N_K,\gamma}\{\mathsf{E}_{a_K}(\Psi_t^{(N_K)}U) > A\} = 0$. The finitely many remaining sides N_K have the deterministic bound $48N_K^3/a_K$, which proves (5). □

29.4 Measurable approximation by smooth trajectories

We now construct Borel maps π_K that approximate the lattice Wilson loop coordinates by smooth trajectories, subject to one-sided energy bounds. The maps choose from the countable family of trajectories X_q of Lemma 29.8, so no measurable choice of gauges is needed. The energy bounds transfer the tightness of the lattice energies to the trajectories $\pi_K(U)$. The map π_K is an auxiliary measurable approximation, not a canonical continuum interpolation of a lattice configuration.

Theorem 29.19. *Assume the standing assumptions of Section 29.3. Let $(X_q)_{q\ge1}$ be the trajectories of Lemma 29.8, let w_j and $W^{\mathrm{lat}}_{K,j}$ be as in Theorem 29.3 and Definition 29.13, and set $s_i = L^2 2^{-i}$ and $\varepsilon_m = 2^{-m}$. For $m \ge 1$, $K \ge 0$, and $U \in \mathcal{U}_{N_K}$, we say that q satisfies the level-m conditions at U if*

$$\begin{aligned} |w_j(X_q) - W^{\mathrm{lat}}_{K,j}(U)| &< \varepsilon_m && (j \le m),\\ E_L(X_q(s_i)) &< \mathsf{E}_{a_K}(\Psi_{s_i}^{(N_K)}U) + \varepsilon_m && (i \le m). \end{aligned} \tag{29.17}$$

Let $\pi_{K,m}(U) = X_q$ for the least q satisfying the level-m conditions at U, let $\pi_{K,m}(U) = X_1$, the zero trajectory, if there is no such q, and let $\pi_{K,0} = X_1$. Then the following statements hold.

1. *Each map $\pi_{K,m} : \mathcal{U}_{N_K} \to \mathcal{X}_L$ is Borel and invariant under all lattice gauge transformations.*
2. *For each fixed $m \ge 1$, the probability that no q satisfies the level-m conditions tends to zero as $K \to \infty$, both under $\nu_{K|r}$ for every $r \in \mathcal{R}$ and under $\mu_{N_K,\gamma}$.*

3. *There exists a nondecreasing sequence of integers $m(K) \to \infty$, depending only on the data fixed in the standing assumptions and not on U, such that the maps $\pi_K = \pi_{K,m(K)}$ satisfy, for every $r \in \mathcal{R}$, every $j \geq 1$, and every $t > 0$,*

$$\lim_{K\to\infty} \mathbb{E}_{\nu_{K|r}} |w_j(\pi_K(U)) - W^{\mathrm{lat}}_{K,j}(U)| = 0, \qquad \lim_{R\to\infty} \sup_{K\geq r} \nu_{K|r}\big\{E_L(\pi_K(U)(t)) > R\big\} = 0,$$

and such that the image measures $(\pi_K)_{\#}\nu_{K|r}$ converge to λ_r against every bounded $\mathcal{T}_W$-continuous function and weakly with respect to d_c.

4. *Let $\mathscr{W}$ be a finite list of loops with admissible approximations, characters, and positive times as in Theorem 29.18(2), with lattice vector $\mathscr{W}^{\mathrm{lat}}_K(U)$ and continuum vector $\mathscr{W}(X)$. Then for every bounded continuous function f of this vector and every $r \in \mathcal{R}$,*

$$\lim_{K\to\infty} \mathbb{E}_{\nu_{K|r}} \big| f(\mathscr{W}(\pi_K(U))) - f(\mathscr{W}^{\mathrm{lat}}_K(U)) \big| = 0.$$

The sequence $m(K)$ may depend on L, γ, M, J, Θ, n_*, the enumeration (b_q), the loops and times (ℓ_j, t_j), and the convergence moduli in (2) for the countably many indices $r \in \mathcal{R}$. The maps π_K do not depend on the index r that is later used to study their laws.

Proof. We prove (1) directly from the definition, (2) from the joint convergence of Theorem 29.18 and an open-set argument, and then choose $m(K)$ by a diagonal argument. The energy constraint gives the tails in (3), the compactness of energy sublevel sets and the determining property of the w_j identify the limits, and an approximation by functions of finitely many coordinates gives (4).

Step 1: measurability and invariance. Each condition in (29.17) is a strict inequality between continuous gauge-invariant functions of U. Then the set $F_{K,m,q}$ of configurations at which q satisfies the level-m conditions is open, in particular Borel, and invariant under lattice gauge transformations. The event that q is the least such index is $F_{K,m,q} \setminus \bigcup_{q'<q} F_{K,m,q'}$, and the event that no index satisfies the conditions is the complement of $\mathcal{S}_{K,m} = \bigcup_q F_{K,m,q}$. For a Borel set $\mathcal{B} \subset \mathcal{X}_L$, the preimage $\pi^{-1}_{K,m}(\mathcal{B})$ is the countable union of the events that q is the least index, over those q with $X_q \in \mathcal{B}$, together with the failure event if $X_1 \in \mathcal{B}$. This proves that $\pi_{K,m}$ is Borel. Gauge invariance holds event by event.

Step 2: success at a fixed level. Fix $m \geq 1$. Let $\mathcal{O}_m \subset \mathbb{R}^{2m}$ be the union over q of the sets

$$\big\{(u,v) \in \mathbb{R}^m \times \mathbb{R}^m : \ |w_j(X_q) - u_j| < \varepsilon_m,\ E_L(X_q(s_i)) < v_i + \varepsilon_m \ (i,j \leq m)\big\}.$$

It is open, and by Lemma 29.8 it contains the vector $((w_j(X))_{j\leq m}, (E_L(X(s_i)))_{i\leq m})$ for every $X \in \mathcal{X}_L$. By Theorem 29.18(2), applied to the joint vector of the first m lattice coordinates and the lattice energies at the times $s_1, \dots, s_m$, and by the portmanteau theorem for open sets [28, Theorem 2.1],

$$\liminf_{K\to\infty} \nu_{K|r}(\mathcal{S}_{K,m}) \geq \lambda_r\big\{\big((w_j(X))_{j\leq m}, (E_L(X(s_i)))_{i\leq m}\big) \in \mathcal{O}_m\big\} = 1.$$

This uses the joint convergence of energies and loops on the same configuration. By (29.14) and (29.13),

$$\limsup_{K\to\infty} \mu_{N_K,\gamma}(\mathcal{S}^c_{K,m}) \le F_\gamma(n_r),$$

and letting $r \to \infty$ in $\mathcal{R}$, we obtain (2) for the Wilson measures.

Step 3: the choice of $m(K)$ and the coordinates. We enumerate $\mathcal{R}$ as $r(1), r(2), \dots$. We choose strictly increasing integers $K_m \ge \max\{m, r(1), \dots, r(m)\}$ such that for every $K \ge K_m$ the probability of $\mathcal{S}^c_{K,m}$ is at most 2^{-m}, under $\mu_{N_K,\gamma}$ and under each of $\nu_{K|r(1)}, \dots, \nu_{K|r(m)}$. This is possible by Step 2, since each K_m involves only finitely many requirements. We set $m(K) = \max\{m : K_m \le K\}$, and $m(K) = 0$ if this set is empty. This is a single deterministic definition along the sequence N_K. Fix $j \ge 1$ and $r = r(l) \in \mathcal{R}$. For all sufficiently large K we have $m(K) \ge \max(j, l)$, and then, both under $\nu_{K|r}$ and under $\mu_{N_K,\gamma}$,

$$\mathbb{E}|w_j(\pi_K(U)) - W^{\mathrm{lat}}_{K,j}(U)| \le 2^{-m(K)} + 2\,\mathbb{P}(\mathcal{S}^c_{K,m(K)}) \le 3 \cdot 2^{-m(K)}. \tag{29.18}$$

Here we used the conditions (29.17) with $j \le m(K)$ on the event $\mathcal{S}_{K,m(K)}$, and (29.9) on its complement.

Step 4: energies. Let $i \le m(K)$. On the event $\mathcal{S}_{K,m(K)}$ the energy condition in (29.17) holds, and on its complement the zero trajectory has energy zero. We conclude that for every configuration $U \in \mathcal{U}_{N_K}$,

$$E_L(\pi_K(U)(s_i)) \le \mathsf{E}_{a_K}(\Psi^{(N_K)}_{s_i} U) + 2^{-m(K)} \qquad (i \le m(K)). \tag{29.19}$$

Fix i. By Theorem 29.18(5), the lattice energies at time s_i have tails that are uniform in $K \ge r$ under $\nu_{K|r}$ and uniform in $K \ge 0$ under $\mu_{N_K,\gamma}$. By (29.19) the same holds for $E_L(\pi_K(U)(s_i))$ for all K with $m(K) \ge i$. For each of the finitely many other K, every output $\pi_K(U)$ is one of the smooth trajectories X_q. Then $E_L(\pi_K(U)(s_i))$ is a finite measurable function of U, and its tail probabilities tend to zero by continuity from above of the measure. Combining the two cases, we find that for every i the energies $E_L(\pi_K(U)(s_i))$ have tails that are uniform in $K \ge r$ under $\nu_{K|r}$ and uniform in $K \ge 0$ under $\mu_{N_K,\gamma}$. For an arbitrary $t > 0$ we choose $s_i \le t$ and use the monotonicity of the continuum energy. This proves the tail bound in (3).

Step 5: convergence of the image measures. Fix $r \in \mathcal{R}$. By Step 4 and Theorem 29.6, the image measures $(\pi_K)_\# \nu_{K|r}$ have compact containment, and every subsequence has a further subsequence converging against bounded $\mathcal{T}_W$-continuous functions and weakly for d_c. By Theorem 29.18(2), the finite vectors $(W^{\mathrm{lat}}_{K,j})_{j\le b}$ converge in law under $\nu_{K|r}$ to $(w_j)_{j\le b}$ under λ_r. By (29.18), Markov's inequality, and uniform continuity of bounded continuous functions on the cube $[-1,1]^b$, the vectors $(w_j(\pi_K(U)))_{j\le b}$ have the same

limit laws. Then every subsequential limit of the image measures has the same finite joint laws of the w_j as λ_r, and by Theorem 29.3(4) it equals λ_r. Since every subsequence has a further subsequence with this limit, every bounded $\mathcal{T}_W$-continuous function converges along the whole sequence. A bounded d_c-continuous function is $\mathcal{T}_W$-continuous, which gives the weak convergence for d_c. This completes the proof of (3).

Step 6: additional lists. We now prove (4); the argument uses only compact sets and does not require the functions $W_{\ell,\chi,t}$ of the list to be d_c-continuous. It suffices to treat real f; for complex f we apply the result to the real and imaginary parts. Let $h(X) = f(\mathscr{W}(X))$ and $D = \|f\|_\infty$; only the values of f on the compact range of the character vector matter. Let $\eta > 0$. By Step 4, Theorem 29.6, and the finiteness of the energies under λ_r, there exists a compact set $\mathcal{C} = \mathcal{K}(R)$ with mass at least $1 - \eta$ under every image measure $(\pi_K)_\# \nu_{K|r}$ and under λ_r. Let $\delta > 0$. By Lemma 29.9 there exist b and a continuous $G : [-1,1]^b \to [-D, D]$ with $|h - G \circ p_b| < \delta$ on $\mathcal{C}$, where $p_b X = (w_j(X))_{j \le b}$. Then

$$\mathbb{E}_{\nu_{K|r}} |h(\pi_K(U)) - G(p_b \pi_K(U))| \le \delta + 2D\eta.$$

By (3) and the uniform continuity of G, $\mathbb{E}_{\nu_{K|r}} |G(p_b \pi_K(U)) - G((W^{\mathrm{lat}}_{K,j}(U))_{j \le b})| \to 0$. Next we apply Theorem 29.18(2) to the single joint list consisting of the first b coordinates and the whole list $\mathscr{W}$. Its limit is the corresponding vector under λ_r, and applying the bounded continuous function $|G(u) - f(v)|$ to the joint vector, we obtain

$$\lim_{K \to \infty} \mathbb{E}_{\nu_{K|r}} \big| G\big((W^{\mathrm{lat}}_{K,j}(U))_{j \le b}\big) - f(\mathscr{W}^{\mathrm{lat}}_K(U)) \big| = \mathbb{E}_{\lambda_r} |G(p_b X) - h(X)| \le \delta + 2D\eta.$$

By the triangle inequality, the limit superior in (4) is at most $2\delta + 4D\eta$, and we let $\delta, \eta \to 0$. □

29.5 Removal of the conditioning

Theorem 29.20. *Assume the standing assumptions of Section 29.3, and let π_K be the maps of Theorem 29.19. Then the following statements hold.*

1. *For every $t > 0$,*
$$\lim_{R \to \infty} \sup_{K \ge 0} \mu_{N_K,\gamma} \big\{ E_L(\pi_K(U)(t)) > R \big\} = 0.$$
2. *The image measures $(\pi_K)_\# \mu_{N_K,\gamma}$ have compact containment, and they converge, against every bounded $\mathcal{T}_W$-continuous function and weakly with respect to d_c, to a Borel probability measure $\lambda_{L,\gamma,[n_*]}$ on $\mathcal{X}_L$.*
3. *For every $j \ge 1$, $\lim_{K \to \infty} \mathbb{E}_{\mu_{N_K,\gamma}} |w_j(\pi_K(U)) - W^{\mathrm{lat}}_{K,j}(U)| = 0$.*

4. *For every finite list $\mathscr{W}$ as in Theorem 29.19(4) and every bounded continuous function f of its vector,*

$$\lim_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} f\big(\mathscr{W}_K^{\mathrm{lat}}(U)\big) = \int f(\mathscr{W}(X))\, d\lambda_{L,\gamma,[n_*]}(X),$$
$$\lim_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} \big|f(\mathscr{W}(\pi_K(U))) - f(\mathscr{W}_K^{\mathrm{lat}}(U))\big| = 0.$$

All statements concern the whole sequence $N_K = n_* B^K$ at the fixed coupling γ, and the convergence moduli and $m(K)$ may depend on γ and on this sequence.

Proof. Statement (1) and the bound (29.18) under $\mu_{N_K,\gamma}$, which gives (3), were proved in Steps 3 and 4 of the proof of Theorem 29.19. By (1) and Theorem 29.6, the image measures have compact containment, and every subsequence has a further subsequence converging against bounded $\mathcal{T}_W$-continuous functions and weakly for d_c. By Theorem 28.2, applied to the fundamental characters and multiplied by $1/2$, every finite vector $(W_{K,j}^{\mathrm{lat}})_{j\le b}$ converges in law under $\mu_{N_K,\gamma}$ along the whole sequence. By (3), every subsequential limit of the image measures has these limits as the finite joint laws of the w_j, and by Theorem 29.3(4) all subsequential limits coincide. Applying the compactness to a subsequence that stays away from this common limit gives a contradiction, which proves (2).

Fix a finite list $\mathscr{W}$ and a bounded continuous f. By Theorem 29.19(4), the nonnegative discrepancy

$$D_K = \big|f(\mathscr{W}(\pi_K(U))) - f(\mathscr{W}_K^{\mathrm{lat}}(U))\big| \le 2\|f\|_\infty$$

satisfies $\mathbb{E}_{\nu_{K|r}} D_K \to 0$ for every $r \in \mathcal{R}$. By Lemma 5.5 and (29.13), for every $r \in \mathcal{R}$,

$$\limsup_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} D_K \le 2\|f\|_\infty F_\gamma(n_r),$$

and letting $r \to \infty$ shows that the expected absolute discrepancy tends to zero. Since $f \circ \mathscr{W}$ is bounded and $\mathcal{T}_W$-continuous, by (2) we have $\mathbb{E}_{\mu_{N_K,\gamma}} f(\mathscr{W}(\pi_K(U))) \to \int f(\mathscr{W}(X))\, d\lambda_{L,\gamma,[n_*]}(X)$, and the first assertion of (4) follows. This argument covers every finite list of loops with admissible approximations, since Theorem 29.18 and Theorem 29.19(4) apply to such lists; no comparison of paths under the Wilson measures is needed. □

29.6 Couplings below a fixed bound

Sections 29.3 to 29.5 use the parameters of Theorem 23.2, which are adapted to small couplings $\gamma \le \gamma_0$. For an arbitrary fixed coupling we use instead the parameters of Theorem 23.5, which are chosen for all couplings below a given bound Γ. In this subsection we check that the construction carries over, and we collect the statements about general admissible approximations of loops.

The setting below a coupling bound. Fix $0 < \Gamma < \infty$, $L > 0$, and an odd integer $M \geq 3$. Let J, the parameters Θ, and the minimum coarse side $n_\Gamma \in 4\mathbb{N}$ be as in Theorem 23.5; they are chosen before the coupling, and Theorem 23.10 holds for all $\gamma \in (0, \Gamma]$, $u \in 4\mathbb{N}$ with $u \geq n_\Gamma$, and $K \geq 0$. Fix $\gamma \in (0, \Gamma]$ and $n_* \in 4\mathbb{N}$, and define B, N_K, a_K, n_r, $\nu_{K|r}$, and $e_{K|r}$ as in the standing assumptions of Section 29.3. Let $\mathcal{R}$ be an infinite set of indices r for which Lemma 28.11 applies at the coarse side n_r. By Theorem 23.10, the bound (29.13) holds for $r \in \mathcal{R}$, with $\epsilon_K = 0$ and $F_\gamma(u)$ equal to the right side of (23.13). We refer to these assumptions as the setting below the coupling bound Γ.

Theorem 29.21. *Assume the setting below the coupling bound Γ. Then all conclusions of Theorems 29.18 to 29.20 hold in this setting, with the same lattice energy E_a, the same trajectories X_q, the same level-m conditions, and the zero trajectory when no index satisfies them. In particular, there exist Borel maps $\pi_K : \mathcal{U}_{N_K} \to \mathcal{X}_L$, invariant under all lattice gauge transformations and independent of r, and Borel probability measures λ_r, $r \in \mathcal{R}$, and $\lambda_{L,\gamma,[n_*]}$ on $\mathcal{X}_L$, with the following properties.*

1. *For every $r \in \mathcal{R}$, the joint limits of lattice energies and loops under $\nu_{K|r}$ in Theorem 29.18(2),(3) are the laws of the corresponding continuum vectors under λ_r, and the image measures $(\pi_K)_\# \nu_{K|r}$ converge to λ_r, with tight energies at positive times and L^1 convergence of the coordinates $w_j(\pi_K(U)) - W_{K,j}^{\mathrm{lat}}(U) \to 0$.*
2. *Under $\mu_{N_K,\gamma}$, the image measures $(\pi_K)_\# \mu_{N_K,\gamma}$ converge to $\lambda_{L,\gamma,[n_*]}$, the energies of $\pi_K(U)$ at every positive time have tails that are uniform in $K \geq 0$, the coordinates converge in L^1, and every finite list of loops with admissible approximations, characters, and positive times converges in law to its continuum vector under $\lambda_{L,\gamma,[n_*]}$.*

These are statements at each fixed coupling; the success probabilities at a fixed level and the sequence $m(K)$ may depend on γ.

Proof. We fix γ after J, Θ, and n_Γ, and we go through the proofs of the three theorems. The results of Sections 29.1 and 29.2 do not involve the coupling.

The proof of Theorem 29.18 used the following stochastic inputs: the reconstruction arrays, the gauge representative property, the tightness $\rho(R) \to 0$, the tail bound (29.15), and the convergence of the laws of the arrays A_q, all from Theorem 28.8; the argument of Theorem 28.1; and Lemma 28.4. In the present setting, Lemma 28.11 provides all of them, for the same conditional fluctuation kernels, reconstruction maps, and configurations, with constants depending on the fixed coupling. Steps 1–6 of that proof therefore apply without change and construct the measures λ_r with properties (1)–(4) of Theorem 29.18. Step 7 used Theorem 23.9 only through (29.13), which holds here by Theorem 23.10; property (5) therefore holds as well.

In the proof of Theorem 29.19, Step 1 involves neither the coupling nor a measure. In Step 2, the set $\mathcal{O}_m$ is defined by the trajectories X_q, which do not depend on γ. It

contains the vector of every trajectory, and in particular of λ_r-almost every one. By the joint convergence just established, we have $\nu_{K|r}(\mathcal{S}^c_{K,m}) \to 0$ for every fixed m and $r \in \mathcal{R}$, and (29.14) with (29.13) implies that $\limsup_K \mu_{N_K,\gamma}(\mathcal{S}^c_{K,m}) \le F_\gamma(n_r)$, which tends to zero as $r \to \infty$. The choice of K_m and $m(K)$ in Step 3 is the same; it may now depend on the fixed coupling, and (29.18) holds under $\nu_{K|r}$ and under $\mu_{N_K,\gamma}$. The bound (29.19) holds on every configuration. Step 4 then gives the energy tails under both measures, using property (5) for the lattice energies and the finiteness of the energies of the finitely many other outputs. Steps 5 and 6 use only these facts and the measures λ_r constructed above.

In the proof of Theorem 29.20, compact containment follows from the energy tails. The whole-sequence limits of the finite vectors $(W^{\mathrm{lat}}_{K,j})_{j\le b}$ under $\mu_{N_K,\gamma}$ are given by Theorem 28.2, which holds in this setting by Lemma 28.11. The identification of all subsequential limits of the image measures with one measure $\lambda_{L,\gamma,[n_*]}$ is then the same. So is the treatment of additional lists: the discrepancy $D_K \le 2\|f\|_\infty$ has conditioned expectations tending to zero, which gives $\limsup_K \mathbb{E}_{\mu_{N_K,\gamma}} D_K \le 2\|f\|_\infty F_\gamma(n_r)$ for every $r \in \mathcal{R}$, and we let $r \to \infty$. □

The next lemma records the role of the deterministic estimates of Lemma 29.11 for general admissible approximations of loops, below a coupling bound.

Lemma 29.22. *The following statements hold.*

1. *The estimates of Lemma 29.11 hold with constants that do not depend on the coupling or on any probability measure.*
2. *On every compact family of cardinal interpolants as in Lemma 29.17, they give the bound* (29.12) *for every finite list of loops with time-independent admissible approximations.*
3. *Assume the setting below a coupling bound* Γ*, and let* $r \in \mathcal{R}$*. For every finite list of positive energy times and of loops with admissible approximations, characters, and positive times, the lattice vector converges in law under* $\nu_{K|r}$ *to the corresponding continuum vector under the measure* λ_r *of Theorem 29.21. For time-independent admissible approximations, the corresponding processes converge in law in* $C([\tau,T];\mathbb{R}\times\mathbb{C}^b)$ *for every* $0<\tau<T<\infty$*.*

Proof. Statement (1) is Lemma 29.11, whose proof is purely geometric. For (2), at a fixed coarse mesh the family of cardinal interpolants is compact with bounds on all derivatives. The comparison of the lattice flow with the sampled smooth flow, together with (1), then gives Lemma 29.17, including general admissible approximations. Statement (3) is Theorem 29.18(2),(3) in the setting below the coupling bound, which holds by Theorem 29.21. There is no circularity: the proof of Theorem 29.18 uses (2) through Lemma 29.17, which rests on (1) and on Lemma 28.4, and neither of these uses (3). □

Theorem 29.23. *Let* ℓ *be a piecewise* C^1 *loop in* $\mathbb{T}^3_L$*, let* $a_K \to 0$*, and let* (ℓ_K) *be an admissible approximation of* ℓ *along these meshes:* $\sup_K \mathrm{length}(\ell_K) \le \mathfrak{l} < \infty$*, and for all sufficiently*

large K there exist comparison homotopies H_K between ℓ and ℓ_K, with connectors c_K, such that $A_K = \mathrm{Area}(H_K) \to 0$.

1. *For every smooth connection A we have $\|\mathrm{Hol}_\ell A - (\mathrm{Hol}_{c_K} A)(\mathrm{Hol}_{\ell_K} A)(\mathrm{Hol}_{c_K} A)^{-1}\|_{\mathrm{op}} \le C\|F_A\|_\infty A_K$. If $\|A\|_{C^1} \le R$, then $|\chi(\mathrm{Hol}_\ell A) - \chi(\mathrm{Hol}_{\ell_K} A)| \le C_\chi(R + R^2)A_K$, and replacing the exact transports along the edges of ℓ_K by $\exp(a_K A_i(x))$ changes $\chi(\mathrm{Hol}_{\ell_K} A)$ by at most $C_{\chi,R}\,a_K$. These constants do not depend on the coupling or on K.*
2. *Assume the setting below a coupling bound Γ, with $a_K = L/N_K$. Every finite list of loops with admissible approximations, characters, and positive times converges in law under $\nu_{K|r}$, $r \in \mathcal{R}$, to its continuum vector under λ_r, and under $\mu_{N_K,\gamma}$ to its continuum vector under the measure $\lambda_{L,\gamma,[n_*]}$ of Theorem 29.21.*
3. *In the setting of (2), let $(\ell_{h,K})$ and $(\ell'_{h,K})$, $h \le b$, be two admissible approximations of the same loops, with lattice vectors $\mathscr{W}_K^{\mathrm{lat}}$ and $\mathscr{W}_K'^{\mathrm{lat}}$ for fixed characters and times. Then $\mathbb{E}_{\mu_{N_K,\gamma}}|\mathscr{W}_K^{\mathrm{lat}}(U) - \mathscr{W}_K'^{\mathrm{lat}}(U)| \to 0$, and $\mathbb{E}_{\mu_{N_K,\gamma}}|f(\mathscr{W}_K^{\mathrm{lat}}(U)) - f(\mathscr{W}_K'^{\mathrm{lat}}(U))| \to 0$ for every bounded continuous f.*
4. *In the setting of (2), for every $R < \infty$,*

$$\lim_{K\to\infty} \sup_{\sum_h |z_h| \le R} \Big| \mathbb{E}_{\mu_{N_K,\gamma}} \exp\Big(\sum_h z_h W_{K,h}^{\mathrm{lat}}\Big) - \int \exp\Big(\sum_h z_h W_{\ell_h,\chi_h,t_h}(X)\Big)\, d\lambda_{L,\gamma,[n_*]}(X) \Big| = 0.$$

5. *In the setting of (2), for time-independent admissible approximations the path convergence of Theorem 29.18(3) under $\nu_{K|r}$ holds on every interval $[\tau, T] \subset (0,\infty)$, and for two such approximations*

$$\lim_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} \sup_{\tau \le t \le T} \max_{h \le b} \big| \chi_h(\mathrm{Hol}_{\ell_{h,K}} \Psi_t^{(N_K)} U) - \chi_h(\mathrm{Hol}_{\ell'_{h,K}} \Psi_t^{(N_K)} U) \big| = 0.$$

6. *A sequence of closed lattice paths of bounded physical length that converges uniformly to ℓ, as parametrized loops, is an admissible approximation of ℓ.*

Proof. Statements (1) and (6) are Lemma 29.11, and the convergence under $\nu_{K|r}$ in (2) and (5) is Lemma 29.22(3). The convergence under $\mu_{N_K,\gamma}$ in (2) is Theorem 29.21(2). We give the comparison argument, which also proves (3)–(5). Let $D = \max_h \chi_h(I)$ for a nonempty list, and let $G_K = |\mathscr{W}_K^{\mathrm{lat}}(U) - \mathscr{W}_K'^{\mathrm{lat}}(U)|_\infty$, so that $0 \le G_K \le 2D$. By Lemma 29.22(3), applied to the union of the two lists, the two vectors converge jointly under $\nu_{K|r}$ to the same continuum vector, and $\mathbb{E}_{\nu_{K|r}} G_K \to 0$. By Lemma 5.5 and (29.13),

$$\limsup_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} G_K \le 2D F_\gamma(n_r).$$

We take the limit in K first at this fixed r, and then let $r \to \infty$ in $\mathcal{R}$; the right side tends to zero. This proves the first assertion of (3) for the maximum norm, hence for any norm. Taking for $(\ell'_{h,K})$ the lattice paths of Lemma 29.10, whose limit under $\mu_{N_K,\gamma}$ is identified

by Theorem 29.20 in the setting of Theorem 29.21, we obtain the limit in (2) for every admissible approximation. The second assertion of (3) follows from uniform continuity of f on the compact range of the character vectors and Markov's inequality. For (4), the vectors are bounded by D, and for $\sum_h |z_h| \le R$,

$$\left|e^{z\cdot v} - e^{z\cdot v'}\right| \le R e^{DR} \max_h |v_h - v'_h|.$$

Then (3) transfers the uniform convergence of source functionals for the paths of Lemma 29.10, which holds by Theorem 28.2 (in this setting by Lemma 28.11), to every admissible approximation, and (2) identifies the limit. For the supremum in (5), we apply Theorem 29.18(3), in the setting below the coupling bound, to the two time-independent approximations simultaneously. Their joint limit has identical loop paths, and the supremum of the difference is bounded by $2D$, so its conditioned expectation tends to zero. The bound of Lemma 5.5 at the same lattice side, followed by the limit $r \to \infty$, gives the conclusion under $\mu_{N_K,\gamma}$. Two arbitrary time-independent approximations are compared through the paths of Lemma 29.10, which do not depend on time. □

29.7 The continuum limit along the sequences $n_* M^{JK}$

Theorem 29.24. *Let $0 < \Gamma < \infty$, $L > 0$, and $M \ge 3$ odd. There exist $J \in \mathbb{N}$, parameters Θ, and $n_\Gamma \in 4\mathbb{N}$, chosen before the coupling, such that the following holds for every $\gamma \in (0, \Gamma]$ and every $n_* \in 4\mathbb{N}$ with $n_* \ge n_\Gamma$. Let $B = M^J$ and $N_K = n_* B^K$, and write $\mu_K = \mu_{N_K,\gamma}$. Then, as $K \to \infty$:*

1. *For loops $\ell_1, \dots, \ell_m$ with admissible approximations $(\ell_{i,K})_K$, characters $\chi_1, \dots, \chi_m$ of finite-dimensional representations of $\mathrm{SU}(2)$, and times $t_1, \dots, t_m > 0$, the vector $(\chi_i(\mathrm{Hol}_{\ell_{i,K}} \Psi_{t_i}^{(N_K)} U))_{i \le m}$ converges in law under μ_K, and for every $R < \infty$ the expectations $\mathbb{E}_{\mu_K} \exp(\sum_i z_i \chi_i(\mathrm{Hol}_{\ell_{i,K}} \Psi_{t_i}^{(N_K)} U))$ converge uniformly in $z \in \mathbb{C}^m$ with $\sum_i |z_i| \le R$.*
2. *There exist Borel maps $\pi_K : \mathcal{U}_{N_K} \to \mathcal{X}_L$, invariant under all lattice gauge transformations, such that the image measures $(\pi_K)_\# \mu_K$ have compact containment and converge, against every bounded $\mathcal{T}_W$-continuous function and weakly with respect to d_c, to a Borel probability measure $\lambda_{L,\gamma,[n_*]}$ on $\mathcal{X}_L$.*
3. *For every $t > 0$ and every $j \ge 1$,*

$$\lim_{R\to\infty} \sup_{K \ge 0} \mu_K\{E_L(\pi_K(U)(t)) > R\} = 0, \qquad \lim_{K\to\infty} \mathbb{E}_{\mu_K} |w_j(\pi_K(U)) - W_{K,j}^{\mathrm{lat}}(U)| = 0,$$

   and the limit law in (1) is the law of $(W_{\ell_i,\chi_i,t_i}(X))_{i \le m}$ under $X \sim \lambda_{L,\gamma,[n_*]}$.
4. *The limits in (1) do not depend on the choice of the admissible approximations, and for two admissible approximations of the same loops the difference of the vectors in (1) tends to zero in $L^1(\mu_K)$.*

Here the Wilson flow runs for the lattice time $4t/a_{N_K}^2$, and the characters need not be normalized; they are bounded by the dimensions of the representations. The convergence moduli, the maps π_K, and the measure may depend on γ, on the sequence N_K, and on the observables. Every fixed $\gamma > 0$ lies below some bound Γ, so the theorem applies to every $\gamma > 0$. By Corollary 29.26 below, its conclusions hold for every $n_* \in 4\mathbb{N}$, not only for $n_* \geq n_\Gamma$.

Proof. Let J, Θ, and n_Γ be given by Theorems 23.5 and 23.10; they are chosen before the coupling. Fix $\gamma \in (0, \Gamma]$ and $n_* \geq n_\Gamma$. By Lemma 28.11, the conditioned estimates of Section 28 hold at all coarse sides n_r with r in an infinite set $\mathcal{R}$. The minimum coarse sides there, the bounds on the flow data, and the convergence moduli may depend on the fixed coupling. We are therefore in the setting below the coupling bound Γ of Section 29.6. Theorem 29.21 gives (2) and (3), including the energy tails with supremum over all $K \geq 0$ and the identification of every finite list of loops with admissible approximations under the same measure $\lambda_{L,\gamma,[n_*]}$. By Theorem 29.23(2)–(4), we obtain (1) and (4), including the uniform convergence of the source functionals. The maps π_K are defined on the configuration spaces of all sides N_K, $K \geq 0$. For the finitely many K with $m(K) = 0$ their value is the zero trajectory, which adds a single compact point and does not affect compact containment, the energy tails, or any limit. If the conditioned estimates require coarse sides larger than a bound depending on γ, we use only indices $r \in \mathcal{R}$ above it. No parameter of the hierarchy is chosen again. □

The case $\Gamma = 1$ is used in the weak-coupling analysis, where the parameters of Theorem 23.2 enter. We state it separately, with the quantifiers in the form in which it is used.

Theorem 29.25. *Let $L > 0$ and $M \geq 3$ odd. There exist $J \in \mathbb{N}$ and parameters Θ, chosen before the coupling, the lattice sides, and the number of steps, such that for every $\gamma \in (0, 1]$ there exists $n_0 < \infty$, which may depend on L, γ, M, J, and Θ, with the following property. For every $n_* \in 4\mathbb{N}$ with $n_* \geq n_0$, with $B = M^J$ and $N_K = n_* B^K$, the conclusions (1)–(4) of Theorem 29.24 hold along the whole sequence $K \to \infty$, for the Wilson measures $\mu_{N_K,\gamma}$ of Definition 4.1.*

All bounds that are uniform in the lattice side range over every $K \geq 0$.

Proof. Fix L and M, and choose the bound $\Gamma = 1$ in Theorems 23.5 and 23.10; the construction there keeps the parameters of Theorem 23.2 and provides the corresponding minimum side. The proof of Theorem 29.24 establishes conclusions (1)–(4) directly from Lemma 28.11 and Theorems 29.21 and 29.23, without using the present theorem. Specializing it to $\Gamma = 1$, with $n_0 = n_1$, we obtain the theorem; in fact n_0 can be chosen independently of γ, and larger minimum sides depending on γ are also allowed. No choice of $m(K)$ uniform in γ is claimed. □

The next corollary removes the lower bound on the base side and records how the limit is used when a comparison side is fixed before the coupling, as in Sections 34 to 36.

Corollary 29.26. *Let $0 < \Gamma < \infty$, $L > 0$, and $M \geq 3$ odd, let J, Θ, and n_Γ be as in Theorem 29.24, and set $B = M^J$. Fix $n_* \in 4\mathbb{N}$ before the coupling.*

1. *Let $\gamma \in (0, \Gamma]$, and let $a_\gamma \geq 0$ be an integer with $n_* B^{a_\gamma} \geq n_\Gamma$. The measure given by Theorem 29.24 for the base side $n_* B^{a_\gamma}$ does not depend on the choice of a_γ; we denote it by $\lambda_{L,\gamma,[n_*]}$. Let π'_K be the maps of that theorem for the base side $n_* B^{a_\gamma}$, and set $\pi_K = \pi'_{K-a_\gamma}$ for $K \geq a_\gamma$ and $\pi_K = X_1$, the zero trajectory, for $K < a_\gamma$. Then the conclusions (1)–(4) of Theorem 29.24 hold along the sequence $N_K = n_* B^K$, $K \geq 0$, for these maps and this measure, including the energy tails with supremum over all $K \geq 0$.*
2. *Let $n = n_* B^a$ be a comparison side, fixed before the coupling, with $n \geq n_\Gamma$, and possibly subject to further side conditions used in later sections, such as the margin* (23.52) *for the flat connections and the conditions on the analytic radius. Then the bound of Theorem 23.10 on $\mu_{nB^k,\gamma}(S^c_{nB^k,n,\gamma})$ holds uniformly in $\gamma \in (0, \Gamma]$ and $k \geq 0$. Moreover, for each fixed $\gamma \in (0, \Gamma]$, the limits along $N = nB^k$ of the expectations of bounded continuous functions of finite loop vectors as in Theorem 29.24(1) are the corresponding expectations under $\lambda_{L,\gamma,[n_*]}$, also when $a_\gamma > a$.*

In (2), every further estimate at the side n keeps its own range of couplings, and neither the side n nor a fixed number of scales used in a later weak-coupling estimate is replaced by a side depending on γ. The coupling bound γ_0 of Theorem 23.2 is not changed. We claim no choice of $m(K)$ independent of γ, no interchange of the limits $K \to \infty$ and $\gamma \to 0$, and no comparison between different classes $[n]_M$.

Proof. We fix Γ, the parameters, and n_Γ first, and then n_*, before the coupling. Since $B > 1$, there exists an integer $a_\gamma \geq 0$ with $n_* B^{a_\gamma} \geq n_\Gamma$. For fixed $\gamma \in (0, \Gamma]$ we apply Theorem 29.24 with the base side $u = n_* B^{a_\gamma}$. That theorem allows further minimum sides depending on γ internally, and it gives its conclusions along the sequence $uB^K = N_{K+a_\gamma}$, $K \geq 0$. This sequence is a tail of the sequence $(N_K)_K$, with the same configuration spaces, Wilson measures, Wilson flow, and lattice paths. Hence every finite loop vector, and every expectation of a bounded continuous function of it, has the same limit along $(N_K)_K$. Two choices of a_γ give sequences that share a common tail. The finite joint laws of the limits of the coordinates $W^{\mathrm{lat}}_{K,j}$ therefore coincide, and by Theorem 29.3(4) the two measures coincide.

The maps π_K are Borel and gauge invariant. Changing finitely many terms of a sequence does not affect convergence or the L^1 convergence of the coordinates. Adding the zero trajectory to a compact set does not affect compactness in either topology, so compact containment is preserved. The added image measures have zero energy at

every positive time, and the energy tails with supremum over all $K \geq 0$ are therefore unchanged. This proves (1).

For (2), the uniform bound on the small-field failure probability follows from Theorem 23.10, since $n \geq n_\Gamma$. The reindexing in (1) does not change any measure, event, or observable at the side n. For fixed γ, all sufficiently large sides nB^k belong to the tail $(uB^K)_K$ used above, so the expectations of bounded continuous functions of finite loop vectors converge to the expectations under $\lambda_{L,\gamma,[n_*]}$, also when $a_\gamma > a$. Every further estimate at the side n keeps its own range of couplings, and no parameter of the comparison side or bound on the coupling was changed. □

29.8 The sequence nM^k

The results of Section 29.7 give convergence along the subsequences n_*M^{JK} for one fixed J. To obtain convergence along the full sequence nM^k, we use two consecutive values J and $J+1$, both of which are available by Corollary 23.15. Each residue class of k modulo J shares infinitely many lattice sides with the sequence of depth $J+1$, and at a common lattice side the Wilson measures and the lattice observables are identical. This identifies the limits along all residue classes.

Lemma 29.27. *Let $J \geq 1$, $n \in \mathbb{N}$, and $r \in \{0, \dots, J-1\}$. For every integer $s \geq 0$ let $K = r + (J+1)s$ and $K' = r + Js$. Then $r + JK = (J+1)K'$, so that*

$$nM^{r+JK} = nM^{(J+1)K'}, \tag{29.20}$$

and $K, K' \to \infty$ as $s \to \infty$.

Proof. We have $r + J(r + (J+1)s) = (J+1)r + J(J+1)s = (J+1)(r+Js)$. □

Theorem 29.28. *Let $0 < \Gamma < \infty$, $L > 0$, $M \geq 3$ odd, and $n \in 4\mathbb{N}$. There exist two choices of the parameters of Theorem 23.5, with consecutive numbers J and $J+1$ of averaging steps per renormalization group step, made before the coupling, such that for every $\gamma \in (0, \Gamma]$ the following statements hold, with $N_k = nM^k$ and $a_k = L/N_k$.*

1. *For loops $\ell_1, \dots, \ell_m$ with admissible approximations $(\ell_{i,k})_k$ along $(N_k)_k$, characters $\chi_1, \dots, \chi_m$, and times $t_1, \dots, t_m > 0$, the vector $(\chi_i(\mathrm{Hol}_{\ell_{i,k}} \Psi_{t_i}^{(N_k)} U))_{i \leq m}$ converges in law under $\mu_{N_k,\gamma}$ as $k \to \infty$, and for every $R < \infty$ its source functionals $\mathbb{E}_{\mu_{N_k,\gamma}} \exp(\sum_i z_i \chi_i(\mathrm{Hol}_{\ell_{i,k}} \Psi_{t_i}^{(N_k)} U))$ converge uniformly in $z \in \mathbb{C}^m$ with $\sum_i |z_i| \leq R$.*
2. *There exist Borel maps $\pi_k : \mathcal{U}_{N_k} \to \mathcal{X}_L$, invariant under all lattice gauge transformations, such that the image measures $(\pi_k)_\# \mu_{N_k,\gamma}$ have compact containment and converge, against every bounded $\mathcal{T}_W$-continuous function and weakly with respect to d_c, to a Borel probability measure $\lambda_{L,\gamma,[n]_M}$ on $\mathcal{X}_L$. Here $[n]_M$ is the class of n for the relation*

$$n \sim_M n' \quad \Longleftrightarrow \quad nM^a = n'M^b \text{ for some integers } a, b \geq 0.$$

3. *For every $t > 0$ and every $j \geq 1$,*

$$\lim_{R\to\infty} \sup_{k\geq 0} \mu_{N_k,\gamma}\{E_L(\pi_k(U)(t)) > R\} = 0, \qquad \lim_{k\to\infty} \mathbb{E}_{\mu_{N_k,\gamma}} |w_j(\pi_k(U)) - W^{\mathrm{lat}}_{N_k,j}(U)| = 0,$$

   and the limit law in (1) is the law of $(W_{\ell_i,\chi_i,t_i}(X))_{i\leq m}$ under $X \sim \lambda_{L,\gamma,[n]_M}$.
4. *The measure $\lambda_{L,\gamma,[n]_M}$ depends only on L, γ, and $[n]_M$: for every $n' \in [n]_M$, every bound $\Gamma' \geq \gamma$, and every choice of the parameters of Theorem 23.5 for Γ', the measures obtained from n' and these parameters in Corollary 29.26 and in (2) coincide with $\lambda_{L,\gamma,[n]_M}$. For two admissible approximations of the same loops, the difference of the vectors in (1) tends to zero in $L^1(\mu_{N_k,\gamma})$.*
5. *The conclusions of Theorems 30.4 to 30.6 (convergence of the processes of lattice energies and loops on compact intervals of positive times, agreement of time-independent admissible approximations in expected supremum, convergence of the images in the Sobolev trajectory space, and invariance under spatial isometries and center symmetries) hold along the full sequence $N_k = nM^k$, for the maps π_k and the measure $\lambda_{L,\gamma,[n]_M}$.*

Since M is odd and $4 \mid n$, every $n' \in [n]_M$ is also divisible by four. The maps and convergence moduli may depend on the fixed coupling, on n, and on the two parameter choices. The proof of (5), which applies the arguments of Section 30 on each residue class and then the assembly below, is given in Section 30.

Remark 29.29. The theorem identifies the limit within one class $[n]_M$. It does not compare the measures for different classes or for different odd integers M. It gives no rate of convergence, no convergence of moments of the energy, and no uniformity as $\gamma \to 0$, as $\Gamma \to \infty$, or in J. The statements about conditioned measures remain statements along the sequences $n_* B^K$ at fixed coarse sides; we do not identify the limits of the conditioned measures along different residue classes. Likewise, the bounds on the small-field failure probability refer to the sequences uB^K and the coarse sides u for which they are stated.

Proof of Theorem 29.28(1)–(4). We prove the convergence along each residue class modulo J by Corollary 29.26, identify the limits through the common lattice sides of Lemma 29.27, and then interlace the maps.

Step 1: the two parameter choices. By Corollary 23.15 there exists J_0 such that for every $J \geq J_0$ there exist parameters Θ_J and a minimum side, fixed before the coupling, with all conclusions of Theorems 23.5 and 23.10 for $0 < \gamma \leq \Gamma$. We fix such choices for two consecutive values J and $J + 1$. This use of the corollary is not circular. Its proof first applies Lemma 23.14 with the bound one and then the construction of Theorem 23.5, which changes only the constants in the failure bound and the minimum sides and keeps J and the structural parameters. Now fix $\gamma \in (0, \Gamma]$. For each of the two depths, Theorem 29.24, with Lemma 28.11 and Theorem 29.21, provides maps with all the stated

properties along the corresponding sequences. These maps choose among the trajectories X_q, with value X_1 when no index satisfies the conditions and with a sequence $m(K)$ depending on the fixed coupling, and the energy constraint uses the lattice energy of Lemma 29.14. The additional minimum side depending on γ is finite and does not change J or the parameters. No modulus of convergence and no sequence $m(K)$ is claimed to be uniform in the coupling or in J.

Step 2: the residue classes. For each $r \in \{0, \dots, J-1\}$ we apply Corollary 29.26(1) with depth J and base side nM^r. This is allowed even if nM^r is below the minimum side, since the corollary provides a finite shift and the zero trajectory for the finitely many omitted sides. Let $\lambda^{(r)}$ be the resulting measure and $\pi_K^{(r)}$ the resulting maps along the sequence $nM^rB^K = nM^{r+JK}$. We also apply the corollary with depth $J+1$ and base side n, and let λ' be the resulting measure along the sequence $nM^{(J+1)K'}$.

Step 3: identification of the limits. For the comparison we use the lattice paths of Lemma 29.10, which depend only on the loop and the mesh and not on the parameters of the hierarchy. By Lemma 29.27, for every r the sequences nM^{r+JK} and $nM^{(J+1)K'}$ have infinitely many common terms. At each common side N, the Wilson measure $\mu_{N,\gamma}$, the Wilson flow, the times, and the lattice loop vectors are identical. The two whole-sequence limits of every finite vector $(W^{\mathrm{lat}}_{N,j})_{j\le b}$ therefore agree. By Theorem 29.3(4), $\lambda^{(r)} = \lambda'$ for every r; we write $\lambda_{L,\gamma,[n]_M}$ for this common measure. No conditioned measure, density, or reconstruction has been compared.

Step 4: interlacing. For $k = r + JK$ with $r \in \{0, \dots, J-1\}$ and $K \ge 0$, set $\pi_k = \pi_K^{(r)}$. Since each side N_k belongs to exactly one residue class, these maps are Borel and gauge invariant. For every bounded $\mathcal{T}_W$-continuous function, the expectations along each of the J residue sequences converge to its expectation under $\lambda_{L,\gamma,[n]_M}$. Taking the largest of the finitely many thresholds, we obtain convergence along the whole sequence. The same argument gives the weak convergence for d_c and the L^1 convergence of the coordinates. If $\mathcal{K}(R^{(r)})$ has mass at least $1-\eta$ under all image measures of residue r, then $\mathcal{K}(R)$ with $R_j = \max_r R_j^{(r)}$ contains all these sets and has mass at least $1-\eta$ for all k. Moreover,

$$\sup_{k\ge 0} \mu_{N_k,\gamma}\{E_L(\pi_k(U)(t)) > R\} \le \max_{0\le r<J} \sup_{K\ge 0} \mu_{nM^{r+JK},\gamma}\{E_L(\pi_K^{(r)}(U)(t)) > R\},$$

which tends to zero as $R \to \infty$. The convergence of finite loop vectors, the uniform convergence of the source functionals, and the agreement of admissible approximations follow by the same finite argument from Corollary 29.26(1) on each residue class, with the limit identified by Step 3. This assembly uses different sequences $m(K)$ on the finitely many residue classes; it never identifies their values, the indices q chosen, or the conditioned measures.

Step 5: dependence only on the class. Let $n' \in [n]_M$ with $nM^a = n'M^b$, and consider a construction as in Corollary 29.26 with base side n', depth J', and parameters for a bound $\Gamma' \geq \gamma$. Its sides satisfy $n'M^{J'K} = nM^{a-b+J'K}$ for all sufficiently large K, so they form a subsequence of a tail of $(nM^k)_k$. Its loop limits therefore coincide with those along the full sequence, and Theorem 29.3(4) identifies its measure with $\lambda_{L,\gamma,[n]_M}$. The full sequence $n'M^k = nM^{a-b+k}$, $k \geq b - a$, is a shifted tail of the full sequence nM^k, and the measure obtained in (2) from n' is the same. In particular, constructions with two different bounds $\Gamma, \Gamma' \geq \gamma$ give the same measure. This proves (4) and completes the proof of (1)–(4). □

Proof of Theorem 2.3. Let $L > 0$, $\gamma > 0$, $M \geq 3$ odd, and $n \in 4\mathbb{N}$, and apply Theorem 29.28 with any bound $\Gamma \geq \gamma$. Part (1) of Theorem 2.3 is Theorem 29.28(1); the independence of the limit from the admissible approximations follows from the L^1 agreement in Theorem 29.28(4). For part (2), Theorem 29.28(3) shows that every limit law in part (1) is the law of the corresponding continuum vector under $\lambda = \lambda_{L,\gamma,[n]_M}$, and uniqueness follows from Theorem 29.3(4). Part (3) is Theorem 29.28(2),(3) for the paths of Definition 29.13; for any other fixed admissible approximation $(\ell_{j,k})_k$ of ℓ_j it follows from the L^1 agreement in Theorem 29.28(4) and the triangle inequality. Part (4) is the energy bound in Theorem 29.28(3), and part (5) is Theorem 29.28(4).

SECTION 30

Further properties of the continuum measure

In this section, we prove the properties of the continuum measure stated in Theorems 2.5 and 2.6. In Section 29, we constructed the measure $\lambda = \lambda_{L,\gamma,[n]_M}$ on the trajectory space $\mathcal{X}_L$ at each fixed $\gamma > 0$, together with Borel, gauge-invariant maps π_k from lattice configurations to $\mathcal{X}_L$, and showed that every finite family of flowed lattice Wilson loops converges in law to the corresponding regularized Wilson loops under λ (Theorem 29.28). We prove the following further properties.

1. The measures on tori of different side lengths are related by an exact dilation (Theorem 30.1).
2. The joint processes of lattice energy and flowed Wilson loops converge in distribution on every compact time interval in $(0, \infty)$, in the uniform topology (Theorem 30.4). A deterministic input is a uniform Lipschitz bound in time for these processes (Lemma 30.3).
3. The image laws also converge in the finer Sobolev topology of Cao and Chatterjee (Theorem 30.5).
4. The measure is invariant under the isometries $x \mapsto Qx + y$ of $\mathbb{T}_L^3$, with Q a signed permutation matrix, and under the center symmetries; in particular, products of Wilson loops with odd total winding have mean zero (Theorem 30.6).
5. These conclusions hold along the full sequence of lattice sides nM^k (Section 30.6).
6. The map $\gamma \mapsto \lambda_{L,\gamma,[n]_M}$ is weakly continuous. On every compact interval $I \subset (0, \infty)$, each fixed finite vector of flowed lattice Wilson loops converges uniformly in the bounded-Lipschitz distance. The approximating maps can be chosen independently of $\gamma \in I$, with uniform convergence against each bounded Wilson-continuous test and in the bounded-Lipschitz distance for d_c, and with uniform tails of the energies of the image trajectories (Theorem 30.10). Uniformity in γ is not asserted for the path laws or for the laws in the Sobolev topology.
7. At each fixed $0 < \gamma \leq 1$, along the sequence $N_K = n_* B^K$ of Theorem 29.25, replacing the Wilson flow by the gradient flow of $\sum_p [q(U_p) + cq(U_p)^2]$, with $0 \leq c \leq C_0 < \infty$, preserves the limits of finite vectors of Wilson loops and the limits of the joint processes of energies and Wilson loops at positive times (Theorems 30.13 and 30.15).

The inputs are the results of Section 29 on the trajectory space, the energy, the joint limits under conditioned measures, and the approximating maps; the deterministic

estimates for the Wilson flow and its modifications in Sections 24 and 25; the finite-mesh control of the linear and quadratic flow data in Section 28; and the bound on the small-field failure probability in Section 23. The convergence of the energy and loop processes is used in Sections 34 and 36, and the convergence in the Sobolev topology in Sections 32 and 34.

Most proofs below have the same structure. Fix a coarse side $n_r = n_* B^r$. Under the conditioned Wilson measure $\nu_{N_K, n_r, \gamma}$, the configuration is, up to a gauge transformation, the field built by the reconstruction maps of Section 15, and the norm of its linear and quadratic flow data is tight in K (Theorem 28.8). On the event that this norm is at most R, the deterministic flow estimates of Sections 24 and 25 apply. We then remove the conditioning using the elementary inequality (30.2) below, taking first the limit $K \to \infty$ at fixed r and then $r \to \infty$. Since the lattice energy is unbounded, only bounded functions of it, such as indicators of tail events or bounded continuous path tests, are transferred in this way. No convergence of moments of the energy is claimed in this section.

30.1 Setting and notation

Fix $L > 0$, an odd integer $M \geq 3$, and $0 < \Gamma < \infty$. Before the coupling is chosen, we fix the integer J, the parameter tuple Θ, and the minimum coarse side n_Γ of Theorem 29.24, that is, of Theorem 23.5. Then fix $0 < \gamma \leq \Gamma$ and $n_* \in 4\mathbb{N}$, and set

$$B = M^J, \qquad N_K = n_* B^K, \qquad a_K = L/N_K.$$

If $n_* < n_\Gamma$, we use the convention of Corollary 29.26: the results are applied to the shifted base $n_* B^{a_\gamma} \geq n_\Gamma$, and at the finitely many cutoffs $K < a_\gamma$ the maps are the stationary zero trajectory. We write $\lambda = \lambda_{L,\gamma,[n_*]}$ for the continuum measure along $(N_K)_{K \geq 0}$, and π_K for the maps of Theorem 29.21. For the full sequence $N_k = nM^k$ of Theorem 29.28, the maps are instead those of that theorem, which are obtained by interlacing finitely many sequences of the previous form, and the measure is $\lambda_{L,\gamma,[n]_M}$. By part (4) of that theorem, $\lambda = \lambda_{L,\gamma,[n_*]_M}$. The two consecutive values J and $J+1$ and their parameter tuples are given by Corollary 23.15, and no constant uniform in J is used. Every assertion in Sections 30.2 to 30.6 is made at one fixed coupling; Section 30.7 treats compact intervals of couplings.

All gauge transformations are periodic, and all homotopy classes are included. Deterministic constants have the dependence described in item (7) of Definition 4.14. They may depend on the stated bounds on the flow data, the waiting times and horizons, the lengths of the lattice paths, the representations, and the margins of the smooth reference solutions, but not on the coupling, the cutoff, the depth of the hierarchy, or the law. Stochastic constants and convergence moduli may in addition depend on γ, on the coarse side, and on the parameters. No estimate is uniform as the waiting time or the coupling tends to zero.

By Lemma 28.11, the results of Section 28 that we use hold at each fixed $0 < \gamma \le \Gamma$, with Theorem 23.10 as the bound on the small-field failure probability. These are the finite-mesh control of the flow data (Theorem 28.8), the consistency estimate for a fixed finite-scale reconstruction (Lemma 28.4), and the removal of conditioning (Theorem 28.2). By Theorem 29.21, the conclusions of Theorems 29.18 to 29.20 hold in the present setting, with their fixed-coupling dependence.

Conditioned measures. For an integer $r \ge 0$ and $K \ge r$ write

$$n_r = n_* B^r, \qquad \nu_{K|r} = \nu_{N_K,n_r,\gamma} = \mu_{N_K,\gamma}(\,\cdot \mid S_{N_K,n_r,\gamma}), \qquad e_{K|r} = \mu_{N_K,\gamma}(S^c_{N_K,n_r,\gamma}).$$

For all sufficiently large r, by Theorems 23.10 and 28.2, we have

$$e_{K|r} \le \epsilon_K^{\mathrm{sf}} + F_\gamma(n_r), \qquad \epsilon_K^{\mathrm{sf}} \longrightarrow 0, \qquad F_\gamma(u) = C' u^3 e^{-\bar{c}u/\gamma} + C_{\mathrm{w}} u^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mu/\gamma}}, \tag{30.1}$$

and $F_\gamma(n_r) \to 0$ as $r \to \infty$ at fixed $\gamma > 0$; here ϵ_K^{sf} is the remainder denoted by ϵ_K in (29.13). Let r_0 be so large that, for every $r \ge r_0$, we have $n_r \ge n_\Gamma$, all additional thresholds on the coarse side in Lemma 28.11 are satisfied for the fixed coupling and input horizon, and (30.1) holds. All conditioned trajectory laws and conditioned estimates below are used only for these indices. We write λ_r for the trajectory law of Theorem 29.18, which is the limit of the flowed observables under $\nu_{K|r}$ as $K \to \infty$, and Y^r for the identity map of $\mathcal{X}_L$ under λ_r. Since $\nu_{K|r}$ is $\mu_{N_K,\gamma}$ conditioned on an event of probability $1 - e_{K|r}$, there exists a probability measure $\rho_{K|r}$ (arbitrary if $e_{K|r} = 0$) such that $\mu_{N_K,\gamma} = (1 - e_{K|r})\nu_{K|r} + e_{K|r}\rho_{K|r}$. This implies that, for every bounded complex Borel function F, every Borel function $0 \le G \le D$, and every Borel event A,

$$\begin{gathered} \bigl|\mathbb{E}_{\mu_{N_K,\gamma}} F - \mathbb{E}_{\nu_{K|r}} F\bigr| \le 2 e_{K|r} \|F\|_\infty, \\ \mathbb{E}_{\mu_{N_K,\gamma}} G \le \mathbb{E}_{\nu_{K|r}} G + D e_{K|r}, \qquad \mu_{N_K,\gamma}(A) \le \nu_{K|r}(A) + e_{K|r}. \end{gathered} \tag{30.2}$$

This is Lemma 5.5; the last inequality is (29.14).

Observables. For a mesh $a = L/N$ we use the lattice energy

$$\mathsf{E}_a(U) = \frac{8}{a} \sum_{p \in \mathcal{P}_N} q(U_p)$$

of (2.2) and Lemma 29.14, with each positive plaquette counted once. Its smooth limit is the energy E_L of (2.3). For a closed lattice path ℓ_a and a character χ, the flowed lattice Wilson loop is $\chi(\mathrm{Hol}_{\ell_a} \Psi_t^{(N)} U)$, where $\Psi_t^{(N)}$ is the Wilson flow run for the lattice time $4t/a^2$. We use the determining coordinates $w_j(X) = (1/2) \operatorname{tr} \mathrm{Hol}_{\ell_j} X(t_j)$ and the metric d_c of Theorem 29.3, and their lattice counterparts $W_{K,j}^{\mathrm{lat}}$ of Definition 29.13. On lattice one-forms, $\|A\|_{C_a^1} = \|A\|_\infty + \max_{j,\epsilon} \|D_{a,j}^\epsilon A\|_\infty$ is the supremum norm plus the maximum of the forward and backward differences divided by a, and $S_a b$ denotes the sampling of a smooth one-form b at the lattice vertices.

Admissible approximations. Throughout this section, the lattice paths approximating a loop ℓ form an admissible approximation in the sense of Definition 2.1. That is, they are closed lattice paths ℓ_K in Λ_{N_K}, fixed in time and of uniformly bounded physical length, such that for all sufficiently large K there exists a based piecewise C^1 homotopy between ℓ_K and ℓ, together with a connecting path between the basepoints when these differ, whose swept area A_K, counted with multiplicity, tends to zero. No homotopy is required at finitely many initial indices, and all comparisons below are made on the tail. This is the class of Theorem 29.23, and it contains the paths of Lemma 29.10.

Flow data and reference solutions. For an initial potential A_0 on mesh a, its linear and quadratic flow data are the linear heat evolution $x_a(t) = P_t^a A_0$ and the quadratic Picard term $y_a = \mathcal{Q}_a(x_a, x_a)$. We measure them on a horizon $[0, T_0]$ by the norm $\mathsf{X}_\alpha^a(x_a) + \mathsf{Y}_\alpha^a(y_a)$ of Definitions 24.3 and 4.8, which includes the difference components. Under $\nu_{K|r}$, we write $\mathsf{R}_{K|r}$ for this norm of the flow data of the gauge representative of U given by the reconstruction maps (Theorem 15.13). By Theorem 28.8, we have

$$\lim_{R\to\infty} \sup_{K\geq r} \nu_{K|r}\{\mathsf{R}_{K|r} > R\} = 0 \tag{30.3}$$

for every fixed $r \geq r_0$. By Theorem 24.10, every smooth trajectory has, on a finite horizon, a finite reference family: smooth solutions of the Yang–Mills heat equation in DeTurck gauge, that is, of the gauge-fixed equation of that theorem, on consecutive time intervals, together with fixed smooth gauge transformations at the transition times. We call each of these solutions a reference solution. On the lattice, after a time-dependent gauge transformation that adds a damping term of the same kind, the Wilson flow is represented by the solution of a damped lattice equation whose linear part is the lattice heat equation (Lemma 24.2). As in Section 25, we call these solutions damped representatives and say that they are in the damped gauge. We compare them with the reference solutions using Lemma 25.5.

30.2 Exact dilation

Once the lattice sites are indexed by $(\mathbb{Z}/N\mathbb{Z})^3$, the Wilson measure (2.1) does not depend on L: the side length enters only through the physical units of time and length. The following theorem transfers this to the continuum measures. We write $\mathcal{X}_1$ for the trajectory space of the unit torus.

Theorem 30.1. *Fix an odd integer $M \geq 3$, an equivalence class $[n]_M$, and $\gamma > 0$, and let $\lambda_{L,\gamma,[n]_M}$ be the measure of Theorem 29.28, constructed with any $\Gamma \geq \gamma$. For $L > 0$ define $D_L : \mathcal{X}_1 \to \mathcal{X}_L$ on representatives by*

$$(D_L A)(t, x) = L^{-1} A(t/L^2, x/L).$$

Then D_L is a bijection, a homeomorphism for the Wilson topologies, and a Borel isomorphism. The measures satisfy

$$\lambda_{L,\gamma,[n]_M} = (D_L)_{\#}\lambda_{1,\gamma,[n]_M}. \tag{30.4}$$

For every $X \in \mathcal{X}_1$ and $t > 0$,

$$LE_L((D_L X)(t)) = E_1(X(t/L^2)). \tag{30.5}$$

In particular, for loops ℓ_i on the unit torus, characters χ_i, and times $\theta_i > 0$, the joint law under $\lambda_{L,\gamma,[n]_M}$ of the Wilson loops $W_{L\ell_i,\chi_i,L^2\theta_i}$ and the energies $LE_L(X(L^2\theta_i))$ does not depend on L.

This independence holds exactly at every finite cutoff for the flowed lattice Wilson loops on the dilated lattice paths. Let $\pi_{k,1}$ be the maps of Theorem 29.28 on the unit torus, and set $\pi_{k,L} = D_L \circ \pi_{k,1}$. These maps are Borel and invariant under lattice gauge transformations on the full configuration spaces of side L, and at every cutoff they have the same dimensionless energy tails and the same coordinate errors $\mathbb{E}|w_j(\pi_{k,L}) - W^{\mathrm{lat}}_{k,j}|$ as on the unit torus, when the determining family on side L is the dilation of that on the unit torus. Consequently, the convergence errors are uniform in $L > 0$ for each fixed bounded Wilson-continuous or bounded d_c-continuous test transported from the unit torus, and for each fixed bounded continuous test of a finite vector of flowed lattice Wilson loops. For other bounded Borel tests the transported errors equal the unit-side errors, and they tend to zero only if the unit-side errors do. No convergence of the energy laws of the maps $\pi_{k,L}$ is asserted, and no uniformity in γ, or for parameters chosen independently at different L, is asserted.

Proof. We first prove the identities at a finite cutoff, then check that D_L is a well-defined map of trajectory spaces that transforms energies and Wilson loops as stated, and then identify the limiting measures. Finally, we construct the maps $\pi_{k,L}$.

Step 1: identities at a finite cutoff. We index the sites by $(\mathbb{Z}/N\mathbb{Z})^3$ and the links by a site and a direction. Under this identification, the finite measures, including their normalizing constants, are identical for all L, since the product Haar measure and the exponent $-4N\gamma^{-1}\sum_p q(U_p)$ in (2.1) do not involve L. The Wilson flow is the gradient flow of $\sum_p q(U_p)$ for the product bi-invariant metric, which does not involve L either. By the definition $\Psi_t^{(N)} = \Phi^{\mathrm{W}}_{4t/a_N^2}$, the physical time t on side L corresponds to the lattice time $4tN^2/L^2$; at $t = L^2\theta$, this is the lattice time $4\theta N^2$ of the physical time θ on the unit torus. Dilation by L sends each edge of the lattice in the unit torus to the corresponding edge of the lattice in $\mathbb{T}_L^3$ and preserves the order of the edges in a path. We conclude that every joint identity for the flowed lattice Wilson loops stated in the theorem holds configuration by configuration, before any expectation is taken.

Step 2: the continuum dilation. The map is well defined on gauge orbits. If k is a periodic gauge transformation of the unit torus, then $k_L(x) = k(x/L)$ is periodic on side

L, and direct differentiation gives $(D_L A)^{k_L} = D_L(A^k)$ at each time. Further,

$$F_{ij}(D_L A)(t,x) = L^{-2} F_{ij}(A)(t/L^2, x/L), \qquad \partial_t (D_L A)(t,x) = L^{-3}(\partial_t A)(t/L^2, x/L),$$

and the covariant divergence $\sum_j D_j F_{ji}$ of the curvature scales in the same way as $\partial_t A$, with the factor L^{-3}. Thus D_L maps solutions of the Yang–Mills heat flow to solutions and preserves the compatibility relation $X(t) = \Phi_{t-s} X(s)$. By the uniqueness of smooth covariant solutions (Theorem 24.10), D_L is a map of trajectory spaces; its inverse is $X \mapsto ((t,y) \mapsto LX(L^2 t, Ly))$. Parallel transport along $u \mapsto L\ell(u)$ has coefficient $L^{-1} A(\ell(u)) \cdot L\dot{\ell}(u) = A(\ell(u)) \cdot \dot{\ell}(u)$, so

$$W_{L\ell,\chi,L^2\theta}(D_L X) = W_{\ell,\chi,\theta}(X). \tag{30.6}$$

Composition with the dilation therefore permutes bijectively the regularized Wilson loops, which generate the Wilson topologies, so D_L is a homeomorphism for these topologies. By Theorem 29.3, it is a Borel isomorphism. Squaring the curvature and changing variables $x = Ly$, we obtain $E_L(D_L A) = L^{-1} E_1(A)$, which is (30.5).

Step 3: identification of the measures. We use the same sequence $N_k = nM^k$ and the dilated lattice paths in the identities of Step 1. Since dilation multiplies lengths by L and swept areas by L^2, for each fixed L the dilation of an admissible approximation is an admissible approximation; the same holds for the paths of Lemma 29.10. Theorem 29.28 identifies the limiting joint laws of finitely many Wilson loops on both tori. By (30.6) and the determination of probabilities by these laws (Theorem 29.3), we obtain (30.4).

Step 4: a simultaneous choice of maps. We identify the finite link spaces as in Step 1 and set $\pi_{k,L} = D_L \circ \pi_{k,1}$. These maps are Borel and gauge invariant on the full configuration spaces. Since continuous images of compact sets are compact, the image laws have compact containment for each L, and they converge to $(D_L)_{\#}\lambda_{1,\gamma,[n]_M}$. We choose the determining family on side L by dilating the loops of the unit family by L and its rational times by L^2. Then $d_{c,L}(D_L X, D_L Y) = d_{c,1}(X,Y)$, and the coordinate errors at each cutoff agree exactly. Further, for every $\theta > 0$ and R,

$$\mu_{N_k,\gamma}\{LE_L(\pi_{k,L}(L^2\theta)) > R\} = \mu_{N_k,\gamma}\{E_1(\pi_{k,1}(\theta)) > R\}.$$

For a fixed bounded test F on $\mathcal{X}_1$, let $F_L = F \circ D_L^{-1}$ on $\mathcal{X}_L$. Its error at each cutoff, against the limiting measure, is exactly the unit-side error of F. By Theorem 29.28, this error tends to zero for bounded Wilson-continuous F and for bounded d_c-continuous F. The same identity and the same convergence hold for a fixed bounded continuous test of a finite vector of flowed lattice Wilson loops. Finally, the joint identities for energies and Wilson loops under the limiting measures follow from (30.4), (30.5), and (30.6), which do not involve the convergence of lattice energies. □

Remark 30.2. At fixed γ, the continuum coupling $g_3^2 = \gamma/L$ changes with L. At fixed $g_3^2 > 0$, the dimensionless coupling $\gamma = g_3^2 L$ eventually exceeds any fixed Γ as $L \to \infty$, and Theorem 29.24 gives no uniformity as $\Gamma \to \infty$. Thus Theorem 30.1 is not a statement about large volumes: it yields no infinite-volume limit, no mass gap, no uniform bound on the unscaled energies, and no comparison between different equivalence classes $[n]_M$. The continuity in L at fixed g_3^2 is Corollary 30.12.

30.3 Energy and loop processes at positive times

In this section, we prove that the processes $t \mapsto \mathsf{E}_{a_K}(\Psi_t^{(N_K)} U)$ and $t \mapsto \chi_i(\mathrm{Hol}_{\ell_{i,K}} \Psi_t^{(N_K)} U)$ converge jointly, on every compact interval $[\tau, T] \subset (0, \infty)$ and in the uniform topology, to the corresponding processes under λ. The main difficulty is the energy. The function $X \mapsto E_L(X(t))$ is Borel, but it is not known to be continuous for d_c or for the Wilson topology. The convergence of the image laws $(\pi_K)_\# \mu_{N_K,\gamma}$ therefore does not by itself identify the limit of the lattice energies. We use instead the joint convergence under the conditioned measures in Theorem 29.18, whose proof identifies the energies through the Sobolev topology (Lemma 29.7), together with a comparison of the conditioned and unconditioned limits that holds for all bounded Borel tests.

Lemma 30.3. *Fix $L, T_0, R > 0$, $1/2 < \alpha < 2/3$, $0 < \tau < T < \infty$, finitely many characters χ_i, and lengths $\ell_i^* > 0$. For a mesh $a = L/N$ let $\ell_{i,a}$ be closed lattice paths of physical length at most ℓ_i^*, fixed in time. Let $U \in \mathcal{U}_N$ have a gauge representative whose flow data on $[0, T_0]$ have norm at most R, and set $U(t) = \Psi_t^{(N)} U$. There are $a_0 > 0$ and $C < \infty$ such that, for all $a \le a_0$ and $t, u \in [\tau, T]$,*

$$\begin{aligned} \sup_{\tau \le t \le T} \mathsf{E}_a(U(t)) &\le C, \\ \big|\mathsf{E}_a(U(t)) - \mathsf{E}_a(U(u))\big| + \sum_i \big|\chi_i(\mathrm{Hol}_{\ell_{i,a}} U(t)) - \chi_i(\mathrm{Hol}_{\ell_{i,a}} U(u))\big| &\le C|t-u|. \end{aligned} \tag{30.7}$$

The constants may depend on $L, T_0, R, \alpha, \tau, T$, the characters, the lengths ℓ_i^*, the fixed difference stencil and interpolations, and the margins of the reference solutions used in the proof, but not on the coupling, the cutoff, the depth of the hierarchy, or the particular configuration.

Suppose in addition that, along a sequence $a \to 0$, the paths $\ell_{i,a}$ form admissible approximations of loops ℓ_i, and that at a fixed time $s < \tau$ a smooth one-form b and the reference solutions of its trajectory are given, such that a representative $A_a(s)$ of $U(s)$, in the gauge of the first reference solution, satisfies $e_a = \|A_a(s) - S_a b\|_{C_a^1} \to 0$. Then $\mathsf{E}_a(U(t))$ and $\chi_i(\mathrm{Hol}_{\ell_{i,a}} U(t))$ converge, uniformly for $t \in [\tau, T]$, to the energy and the Wilson loops at time $t - s$ of the continuum trajectory started at b.

Proof. At a small positive time s, the lattice solutions have bounded differences of every order, so they lie close to samples of a compact family of smooth fields. Finitely many

reference solutions for this compact family control the lattice flow up to time T with uniform bounds on three differences. The equation of the flow then bounds the time derivative in C^1_a, and the energy and the loops inherit a Lipschitz bound.

Step 1: smoothing at a small positive time. Let T_R be the existence time of Theorem 24.5 for flow data of norm at most R, and fix $0 < s < \min(T_R, T_0, \tau/4)$, with a positive waiting interval before s. The local solution in the damped gauge and its identification with the Wilson flow follow from Theorem 24.5 and Lemma 24.2. By the first part of Theorem 25.7, every fixed order of differences at time s is bounded, uniformly over the configurations in the statement.

These bounds produce a compact family of smooth limiting fields, uniformly over the configurations. We interpolate each value array and each difference array by piecewise multilinear interpolation. A bound on the next difference bounds the modulus of continuity, also across the cells of the lattice. Then the Arzelà–Ascoli theorem and a diagonal extraction give uniform limits of all these arrays along every sequence of meshes tending to zero. Discrete summation by parts identifies the limit of $D^\nu_a A_a$ with the corresponding distributional derivative of the limit of the value arrays, so the limits are smooth. Their bounds at every order imply that the set of all such limiting fields is compact in each fixed C^q norm and consists of smooth fields; closedness follows from a diagonal sequence of approximating lattice fields. Further, as $a \to 0$, the lattice fields at time s are uniformly close, in value and first differences, to samples of fields in this compact family. Otherwise, an offending sequence and the preceding extraction would give a contradiction.

Step 2: continuation to the time T. We apply Theorem 24.10, with a regularity index at least six, to this compact family of smooth fields. We use the open neighborhood given there, not a claim that families with finitely many bounded derivatives have smooth closure. By the part of Lemma 25.5 on compact families of reference solutions, the restarted lattice solutions in the damped gauge satisfy a uniform C^1_a bound. The compact set in that lemma is the family just constructed, which is compact in C^3 and in every higher norm, and its thresholds and its finitely many reference solutions are fixed before the configuration or the mesh is chosen. For a quantitative comparison with a given smooth limiting trajectory, we apply the fixed-reference estimate of the same lemma on each time interval; the regularity index six provides the five derivatives that it requires. The incoming error in Lemma 29.15 is measured with sampled first derivatives. By Taylor's formula, we have $\|D_a S_a b - S_a \partial b\|_\infty \le Ca\|b\|_{C^2}$, so this error differs from the C^1_a error by $O(a)$. Thus Lemma 29.15 is an instance of Lemma 25.5. Only after the C^1_a bound is available do we apply the second part of Theorem 25.7, which gives uniform bounds on three differences.

The waiting times leave no gap at the transition times. By the local existence statement in C^6 of Theorem 24.10, each reference solution extends a fixed positive amount beyond

the right endpoint of its interval. Since the finitely many families of endpoints are bounded, we can choose a common extension time. The comparison in a common gauge extends the lattice representative over this overlap. We choose the waiting time on the next interval shorter than the overlap, so that the bounds of the second part of Theorem 25.7 on the next interval overlap those on the preceding one, and the first waiting interval ends before τ. We obtain finitely many overlapping representations in the damped gauge that cover $[\tau, T]$, each with uniform bounds on three differences. The fixed gauge transformations at the transition times preserve each of these difference bounds. Indeed, for a smooth gauge transformation k, the function $a^{-1}\operatorname{Log}(k(x)e^{aZ}k(x+ae_i)^{-1})$ extends smoothly to $a = 0$, and the exact product rule gives the bound at each order, using finitely many derivatives of k. These derivatives are part of the data of the reference family.

Step 3: the time derivative. In each of these representations, the equation of the flow is exactly

$$\partial_t A_a = \Delta_a A_a + \mathcal{N}_a(A_a, A_a) + a^{-3} R_3(aA_a),$$

by Lemma 24.2. One difference of the Laplacian uses at most three differences of A_a, and one difference of the quadratic term uses at most two. A term of degree $d \geq 3$ in the last summand carries the factor a^{d-3}. Using the absolute convergence of the analytic series and the shifted product rule, we have

$$\|\partial_t A_a\|_{C^1_a} \leq C \tag{30.8}$$

on the bounds of Step 2, with no inverse power of the mesh. In particular, the time derivative of a link variable $e^{aA_{a,i}}$ has norm at most $Ca\|\partial_t A_{a,i}\|_\infty$. Differentiating a path with n_i links and using unitarity, we find that the time derivative of its character is at most $C_{\chi_i} a n_i \|\partial_t A_a\|_\infty \leq C_{\chi_i} \ell_i^* C$, since $an_i \leq \ell_i^*$. By the Lipschitz estimate of Lemma 29.14 and (30.8), the energy difference is at most $C|t-u|$, and the same lemma bounds the energy itself. Both observables are invariant under the damping gauge transformations and under the fixed gauge transformations at the transition times. Since the lattice flow is continuous in time, the finitely many estimates on the intervals concatenate with the largest constant. This proves (30.7).

Step 4: comparison with a given smooth trajectory. The last assertion uses the given incoming error e_a rather than recovering a limiting field from the flow data. Applying the fixed-reference estimate of Lemma 25.5 on the consecutive intervals of the reference solutions, we find the error $C(e_a + a)$ in C^1_a up to time T, as in Lemma 29.15. By Lemma 29.14, the energy error is bounded in the same way. For the loops, comparing exact edge transports with sampled link variables and using the curvature-area estimate of Theorem 29.23, we obtain the error $C(e_a + a + A_{i,a})$, where $A_{i,a}$ is the swept area of the comparison homotopy for $\ell_{i,a}$. These errors tend to zero uniformly in $t \in [\tau, T]$. □

Theorem 30.4. *Work in the setting of Section 30.1: fix $0 < \gamma \le \Gamma$, the sequence $N_K = n_* B^K$, the maps π_K of Theorem 29.21, and the measure λ. Let $0 < \tau < T < \infty$, let $\ell_1, \dots, \ell_m$ be loops, $\chi_1, \dots, \chi_m$ characters of finite-dimensional representations of $\mathrm{SU}(2)$, and $(\ell_{i,K})_K$ admissible approximations of ℓ_i. Then, under $\mu_{N_K,\gamma}$,*

$$\Big(t \mapsto \big(\mathsf{E}_{a_K}(\Psi_t^{(N_K)} U), (\chi_i(\mathrm{Hol}_{\ell_{i,K}} \Psi_t^{(N_K)} U))_{i=1}^m\big)\Big) \\ \Longrightarrow \Big(t \mapsto \big(E_L(X(t)), (W_{\ell_i,\chi_i,t}(X))_{i=1}^m\big)\Big), \qquad X \sim \lambda,$$

in distribution in $C([\tau,T]; \mathbb{R} \times \mathbb{C}^m)$ with the uniform norm. If $(\ell_{i,K}^{(1)})_K$ and $(\ell_{i,K}^{(2)})_K$ are two admissible approximations of the same loops, with flowed lattice Wilson loops $W_{K,i}^{(1)}(t)$ and $W_{K,i}^{(2)}(t)$, then

$$\lim_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} \sup_{\tau \le t \le T} \max_i \big| W_{K,i}^{(1)}(t) - W_{K,i}^{(2)}(t) \big| = 0.$$

In particular, every finite joint vector of lattice energies and flowed lattice Wilson loops at times in $(0,\infty)$ converges in law to the corresponding vector under λ. The same path convergence and the same comparison of two admissible approximations hold along the full sequence $N_k = nM^k$, for the maps and the measure $\lambda_{L,\gamma,[n]_M}$ of Theorem 29.28; this is proved in Section 30.6.

The convergence is tested only by bounded continuous functions on the path space. In particular, the theorem gives no moment bound for time derivatives under the Wilson measure.

Proof. We first recall that the joint path convergence holds under the conditioned measures, with limit λ_r. Then we show that λ_r and λ are close for all bounded Borel tests, which lets us pass to the Wilson measure. The comparison of two approximations is a consequence.

Step 1: conditioned measures. Let $r \ge r_0$, so that (30.1) holds. At fixed r, by Theorem 29.18, the energy and loop paths under $\nu_{K|r}$ converge jointly to the image of λ_r under the continuum path map. The proof of that theorem compares the lattice flows of the full field and of a finite-scale reconstruction, identifies the energies through the Sobolev topology, and uses Theorem 28.8 for the bounds on the flow data; no limiting initial potential is constructed. That theorem applies to every finite union of admissible approximations, so two approximations of the same loops have identical limiting loop paths.

Step 2: comparison of the limits for Borel tests. We now identify the limit of the lattice energies under the Wilson measure with the same λ; this does not follow from the agreement of the loop marginals alone. By Theorems 29.19 and 29.20, the same maps π_K

have image limit λ_r under $\nu_{K|r}$ and image limit λ under $\mu_{N_K,\gamma}$, for bounded d_c-continuous tests. We claim that

$$\left| \int f \, d\lambda_r - \int f \, d\lambda \right| \le 2\|f\|_\infty F_\gamma(n_r) \tag{30.9}$$

for every bounded Borel function f on $\mathcal{X}_L$. For bounded d_c-continuous f, this follows by applying (30.2) to $f(\pi_K)$ at the same cutoff, using (30.1), and passing to the two image limits. By Theorems 29.3 and 29.6, both limits are tight probability measures on the separable metric space $(\mathcal{X}_L, d_c)$, and in particular inner regular. A bounded Borel function on a metric space can then be approximated in $L^1(\lambda + \lambda_r)$ by bounded continuous functions with the same bound on the supremum. Indeed, we first approximate it uniformly by a simple function $\sum_k c_k \mathbf{1}_{A_k}$ with disjoint A_k and $|c_k| \le \|f\|_\infty$. We then choose disjoint compact sets $K_k \subset A_k$ that carry all but an arbitrarily small part of the mass of $\lambda + \lambda_r$, then disjoint open neighborhoods $O_k \supset K_k$ of arbitrarily small additional mass, and continuous functions $0 \le \phi_k \le 1$ equal to one on K_k and supported in O_k. The function $\sum_k c_k \phi_k$ has the required bound and approximation property. This proves (30.9) for every bounded Borel f, without any continuity of the energy for d_c.

Step 3: the Wilson measure. Let $G(X)$ be the continuum path $t \mapsto (E_L(X(t)), (W_{\ell_i,\chi_i,t}(X))_i)$ on $[\tau, T]$, and $G_K(U)$ the corresponding lattice path. The map G is Borel into $C([\tau, T]; \mathbb{R} \times \mathbb{C}^m)$. Indeed, its Wilson-loop coordinates at rational times are Borel by Theorem 29.3, its energy coordinates are Borel by Theorem 29.6, each trajectory gives continuous paths because the flow is smooth, and evaluations at rational times generate the Borel σ-algebra of the path space. Then (30.9) applies to $f \circ G$ for every bounded continuous path test f. Using (30.2) at the cutoff N_K and then Step 1, we obtain

$$\limsup_{K\to\infty} \left| \mathbb{E}_{\mu_{N_K,\gamma}} f(G_K(U)) - \int f(G(X)) \, d\lambda(X) \right| \le 4\|f\|_\infty F_\gamma(n_r).$$

Letting $r \to \infty$ proves the convergence along the whole sequence. The tightness of the lattice paths under the Wilson measure can also be seen directly. By Lemma 30.3, on the event $\{\mathsf{R}_{K|r} \le R\}$, the paths lie, for small meshes, in a compact set of the path space given by the Arzelà–Ascoli theorem. We first make $F_\gamma(n_r)$ small, then the probability in (30.3) small, and then take K large. By (30.2), the complement of the compact set has Wilson probability at most its conditioned probability plus $e_{K|r}$. The finitely many remaining cutoffs have compact path images.

The source functionals of admissible approximations also converge locally uniformly. Indeed, path convergence implies the convergence of every finite joint vector of Wilson loops, and the vector is bounded by the dimensions of the representations. On each ball $\{\sum_i |z_i| \le R'\}$, the functions $e^{\sum_i z_i w_i}$ are bounded and have a common Lipschitz constant in z, so a finite net of the ball gives uniform convergence.

Step 4: two approximations. Applying Theorem 29.18 to the union of the two approximations, we find that the limiting joint path has identical loop paths. Let Δ be the supremum over $t \in [\tau, T]$ of the maximal loop discrepancy; it is continuous in the uniform topology. If the list of loops is empty, there is nothing to prove; otherwise, let $D = \max_i \chi_i(I)$. The bounded continuous test $\min(2D, \Delta)$ equals Δ on all lattice and limiting paths of characters, and it vanishes under the limiting law. Its expectation under $\nu_{K|r}$ then tends to zero. Applying (30.2) at the same cutoff and then letting $r \to \infty$, we obtain the conclusion under the Wilson measure. □

30.4 Convergence in the Sobolev topology

In [43], Cao and Chatterjee also consider trajectories of Sobolev gauge orbits. Recall from Definition 29.1 that $\mathcal{H}_L = H^1(\mathbb{T}^3_L; \mathfrak{su}(2)^3)/\mathcal{G}^{2,2}_L$ carries the sequential weak quotient topology $\mathcal{T}_q$, that $\mathcal{X}^{1,2}_L$ is the space of trajectories $Z : (0, \infty) \to \mathcal{H}_L$ with $Z(t) = \Phi^{1,2}_{t-s} Z(s)$ for $0 < s < t$, with the topology $\mathcal{T}_{1,2}$ of pointwise evaluation [43, Definition 3.70, p. 41], and that $i^{1,2} : \mathcal{X}_L \to \mathcal{X}^{1,2}_L$ sends a smooth orbit at each time to its Sobolev orbit. The compactness behind this topology is the weak compactness of gauge orbits with bounded energy, which goes back to Uhlenbeck [151]; we use it only through the results of Cao and Chatterjee cited below. We consider $\mathcal{T}_{1,2}$ because, for each $t > 0$, the energy $Z \mapsto E_L(Z(t))$ is continuous for it (Lemma 29.7); the energy is not known to be continuous for the Wilson topology. Convergence in $\mathcal{T}_{1,2}$ therefore carries the convergence of the energies of the approximating trajectories.

Theorem 30.5. *Work in the setting of Section 30.1. The maps $i^{1,2} \circ \pi_K$ are Borel on the full configuration spaces, and*

$$(i^{1,2} \circ \pi_K)_\# \mu_{N_K,\gamma} \Longrightarrow (i^{1,2})_\# \lambda$$

against every bounded $\mathcal{T}_{1,2}$-continuous real function. The topology on $\mathcal{X}_L$ pulled back by $i^{1,2}$ is finer than $\mathcal{T}_W$ and has the same Borel σ-algebra. For each fixed $r \geq r_0$, the same conclusion holds under the conditioned measures $\nu_{K|r}$, with limit $(i^{1,2})_\# \lambda_r$. The conclusion under the Wilson measures also holds for the maps of Theorem 29.28 along $N_k = nM^k$, with limit $(i^{1,2})_\# \lambda_{L,\gamma,[n]_M}$. The conclusion under conditioned measures is asserted only along the sequences $(n_* B^K)_K$; no sequence of conditioned measures along nM^k is considered, and no conditioned limits along different residue classes are compared. In each case, every finite joint vector of energies at positive times and Wilson loops of the image trajectories converges in distribution.

More generally, let X_K, $K \geq 0$, be Borel $\mathcal{X}_L$-valued random variables such that, for every $t > 0$, the laws of $E_L(X_K(t))$ are tight, and such that the finite joint laws of the Wilson loops of X_K converge to those of a Borel probability λ' on $\mathcal{X}_L$. Then the laws of $i^{1,2}(X_K)$ converge to $(i^{1,2})_\# \lambda'$ against every bounded $\mathcal{T}_{1,2}$-continuous real function.

Each assertion has exactly the uniformity of its input convergence: the theorem adds no uniformity in the coupling and no rate.

Proof. We first show that $i^{1,2}$ is a Borel bijection with a Borel inverse that preserves Wilson loops, then prove the general assertion by compactness, and finally check its hypotheses for the maps π_K.

By [43, Lemma 3.84, pp. 47–48], $i^{1,2}$ is Borel. Every trajectory in $\mathcal{X}_L^{1,2}$ has a smooth representative at every positive time [43, Lemma 3.85, p. 48], and choosing its smooth orbit defines a map $j^{1,2} : \mathcal{X}_L^{1,2} \to \mathcal{X}_L$ [43, Definition 3.86, p. 48]; a Sobolev gauge transformation between two smooth connections is smooth [43, Lemma 3.10, p. 24], so this choice is unambiguous. We have $j^{1,2} \circ i^{1,2} = \mathrm{Id}$ [43, Lemma 3.87, p. 48], and $i^{1,2} \circ j^{1,2} = \mathrm{Id}$ directly at each time. By [43, Lemmas 3.88–3.89, pp. 48–49], $j^{1,2}$ preserves all Wilson loops and is continuous into $\mathcal{T}_W$. Thus $i^{1,2}$ and $j^{1,2}$ are inverse Borel bijections. Since $j^{1,2}$ is continuous, every $\mathcal{T}_W$-open set $V = (j^{1,2} \circ i^{1,2})^{-1}(V)$ is open for the pulled-back topology, which is therefore finer than $\mathcal{T}_W$. The measurability of $i^{1,2}$ gives the reverse inclusion of Borel σ-algebras; the continuity of $i^{1,2}$ is not used.

We now consider the general sequence. The variables $i^{1,2}(X_K)$ are measurable $\mathcal{X}_L^{1,2}$-valued, and their energies at every positive time have tight laws, since passing to the Sobolev orbit does not change the energy. These are the hypotheses of [43, Theorem 3.73, pp. 41–42]. Applying this theorem to an arbitrary subsequence, we obtain a further subsequence converging against all bounded $\mathcal{T}_{1,2}$-continuous functions to a Borel probability measure η on $\mathcal{X}_L^{1,2}$. The regularized Wilson loops on $\mathcal{X}_L^{1,2}$ are bounded and continuous [43, Definition 3.82, p. 47], and they become the usual Wilson loops at positive times after applying $j^{1,2}$ [43, Lemma 3.88, p. 48]. Hence every finite joint law of the Wilson loops under $j^{1,2}_{\#}\eta$ equals that under λ'. Since these laws determine probability measures (Theorem 29.3), we have $j^{1,2}_{\#}\eta = \lambda'$. Applying the inverse Borel map $i^{1,2}$, we find $\eta = (i^{1,2})_{\#}\lambda'$, independently of the subsequence. If the expectations of a bounded continuous test did not converge to the claimed limit, some subsequence would stay at a positive distance from it, which contradicts what we just proved. This proves the general assertion.

For the image laws under the Wilson measures, the hypotheses are the energy tails and the convergence of the finite joint laws of the Wilson loops in Theorems 29.20 and 29.21, which hold in the present setting. For the conditioned measures, they are the corresponding conditioned statements in Theorem 29.19. For the maps along $N_k = nM^k$, they are the convergence of the image laws and the energy tails at every positive time asserted in Theorem 29.28. No other property of the construction of π_K is used. The convergence of finite joint vectors of energies of the image trajectories follows from Lemma 29.7, and the Wilson loops are continuous for $\mathcal{T}_{1,2}$ by the preceding paragraph. The rescaling to side L and all measurability statements are those of Theorem 29.3. □

30.5 Spatial isometries and center symmetry

Two groups act on the trajectory space. The first consists of the isometries $f(x) = Qx + y$ of $\mathbb{T}_L^3$ with $Q \in O(3,\mathbb{Z})$, the signed permutation matrices, and $y \in \mathbb{T}_L^3$; they act at every time by pullback of one-forms, and then on gauge orbits. The second is the group $\{\pm I\}^3$ of center symmetries. On the lattice, a center symmetry multiplies the link variables crossing one coordinate plane by a central element; this is the familiar center symmetry of pure gauge theories on a torus (compare 't Hooft [149]). Both groups preserve the Wilson measure and commute with the Wilson flow, and we pass these exact identities to the limit. For a loop ℓ whose lift to $\mathbb{R}^3$ runs from x to $x + Lw(\ell)$, the vector $w(\ell) \in \mathbb{Z}^3$ is its winding vector; the winding vector of a closed lattice path is defined in the same way. This is the vector denoted by $\nu(\ell)$ in Theorem 2.5; it is distinct from the determining coordinates w_j.

Theorem 30.6. *Work in the setting of Section 30.1. The measure λ is invariant under the pullback f^* by every isometry $f(x) = Qx + y$ with $Q \in O(3,\mathbb{Z})$ and $y \in \mathbb{T}_L^3$. It is also invariant under the following action of $\{\pm I\}^3$. For $z = (z_1, z_2, z_3) \in \{\pm I\}^3$ choose a smooth map $h_z : \mathbb{R}^3 \to \mathrm{SU}(2)$ with $h_z(x + Le_i) = z_i h_z(x)$, and set*

$$C_z[A] = [A^{h_z}], \qquad A^{h_z} = h_z A h_z^{-1} - (dh_z)h_z^{-1}.$$

The connection A^{h_z} is periodic, the class $C_z[A]$ does not depend on the choice of h_z or of the representative A modulo periodic smooth gauge transformations, and C_z defines a group action on $\mathcal{A}_L/\mathcal{G}_L$ that commutes with the Yang–Mills heat flow, hence an action on $\mathcal{X}_L$. The spatial and center maps are homeomorphisms for the Wilson topology and Borel automorphisms, and they preserve the energy.

For normalized fundamental Wilson loops $W_{\ell,t}(X) = (1/2)\,\mathrm{tr}\,\mathrm{Hol}_\ell X(t)$, integers $b_1, \dots, b_m \geq 0$, and times $t_1, \dots, t_m > 0$,

$$\mathbb{E}_\lambda \prod_{j=1}^m W_{\ell_j,t_j}^{b_j} = 0 \qquad \text{if} \qquad \sum_{j=1}^m b_j w(\ell_j) \neq 0 \text{ in } (\mathbb{Z}/2\mathbb{Z})^3.$$

For characters of the irreducible representations of highest weights n_j, the same holds with b_j replaced by $b_j n_j$ in the parity condition. The corresponding selection rule holds at every finite cutoff, with the winding vectors of the lattice paths. All these conclusions also hold for the measure $\lambda_{L,\gamma,[n]_M}$ of Theorem 29.28.

The center symmetry acts on periodic gauge orbits; it is not an additional identification in the quotient by periodic gauge transformations, since h_z is not periodic when $z \neq (I, I, I)$.

Proof. We first prove exact identities on the lattice, then construct the continuum actions and compute their effect on Wilson loops, and finally pass to the limit.

Step 1: lattice symmetries. A lattice translation or a signed permutation of the coordinates relabels the directed edges, inverting a link variable when the orientation of its edge is reversed. It preserves the product Haar measure and the product bi-invariant Riemannian metric. A plaquette path is mapped to a plaquette path, up to cyclic permutation and inversion. Since $q(V) = q(V^{-1})$ and q is a class function, the Wilson action, and hence the Wilson measure (2.1), are preserved. The invariance of Haar measure under inversion also follows from Lemma 5.1 and the uniqueness of the Haar probability measure.

Let $c_e \in \{\pm I\}$, with $c_{\bar{e}} = c_e^{-1}$, be a center-valued function on links such that $\prod_{e \in \partial p} c_e = I$ for every plaquette p. The map $U_e \mapsto c_e U_e$ preserves the product Haar measure, the product metric, and the trace of every plaquette variable, so it preserves the Wilson measure. An explicit choice with monodromy z is $c_e = z_i$ on the positive direction-i links from the plane $x_i = L - a$ to the plane $x_i = 0$, and $c_e = I$ on all other links. Its product around every plaquette is I, and for a closed lattice path ℓ_a,

$$c(\ell_a) = \prod_{i=1}^{3} z_i^{w_i(\ell_a)}, \qquad \mathrm{Hol}_{\ell_a}(cU) = c(\ell_a)\,\mathrm{Hol}_{\ell_a}(U). \tag{30.10}$$

Every such flat center-valued function acts, modulo lattice gauge transformations, as one of these explicit choices. Indeed, we lift the lattice to $a\mathbb{Z}^3$. On the lift, the product of a flat function along a path with fixed endpoints does not depend on the path, since any closed lattice path can be reduced to the trivial path by cancellations and by moves across elementary squares. Thus the lifted function is the lattice differential of a center-valued function on the vertices of $a\mathbb{Z}^3$, whose three monodromies under the periods Le_i are the z_i. For two flat functions with the same monodromies, the ratio of these vertex functions is periodic, so their actions on periodic lattice gauge orbits agree.

An isometry that preserves a smooth function maps its Riemannian gradient to its gradient. All the maps above are isometries of $\mathcal{U}_N$ that preserve the Wilson action, so they commute with the Wilson flow, which exists and is unique globally on the compact configuration space; the rescaling of time by $4/a^2$ preserves this commutation. We conclude that the exact identities of joint laws, and (30.10), hold simultaneously for any finite list of lattice paths, characters, and positive times, without conditioning and without choosing representatives.

Step 2: the continuum actions. Write $z_i = (-I)^{\varepsilon_i}$ with $\varepsilon_i \in \{0, 1\}$, and let

$$h_z(x) = \exp\Big(\frac{2\pi}{L}\tau_3 \sum_i \varepsilon_i x_i\Big),$$

where $\tau_3 = -i\sigma_3/2$. Since $e^{2\pi\tau_3} = -I$, this map has the required monodromy. Since the z_i are central, A^{h_z} is periodic and smooth whenever A is. If A is replaced by A^k with k

periodic, then $(A^k)^{h_z} = (A^{h_z})^{h_z k h_z^{-1}}$, and $h_z k h_z^{-1}$ is periodic. If h_z' has the same monodromy as h_z, then $h_z' h_z^{-1}$ is periodic and $A^{h_z'} = (A^{h_z})^{h_z' h_z^{-1}}$. This proves both independence claims, for gauge transformations in all homotopy classes. The product $h_z h_{z'}$ has monodromy zz', so the induced maps on orbits form a group action.

The curvature transforms as $F(A^{h_z}) = h_z F(A) h_z^{-1}$, so the energy is preserved. The Yang–Mills heat equation is covariant under the time-independent smooth gauge transformation h_z on $\mathbb{R}^3$, and both the transformed initial datum and the transformed solution are periodic. By the uniqueness of periodic smooth solutions (Theorem 24.10), C_z commutes with the flow. The same facts for the pullback f^* follow from the chain rule, the orthogonality of Q, the invariance of volume, and the same uniqueness. Hence both kinds of maps act on trajectories.

For a loop ℓ whose lift runs from x to $x + Lw(\ell)$, the transformation rule $H^k(s) = k(\ell(0))H(s)k(\ell(s))^{-1}$ for parallel transport gives

$$\mathrm{Hol}_\ell(A^{h_z}) = h_z(x)\,\mathrm{Hol}_\ell(A) h_z(x + Lw(\ell))^{-1} = \Big(\prod_i z_i^{w_i(\ell)}\Big) h_z(x)\,\mathrm{Hol}_\ell(A) h_z(x)^{-1}. \tag{30.11}$$

For $f(x) = Qx + y$, parallel transport gives $W_{\ell,\chi,t}(f^*X) = W_{f\ell,\chi,t}(X)$. The character of the irreducible representation of highest weight n satisfies $\chi_n(-H) = (-1)^n \chi_n(H)$, since $-I$ acts as $(-1)^n$ on the nth symmetric power of the fundamental representation. For a general finite-dimensional representation ρ, the two eigenspaces of the involution $\rho(-I)$ are invariant subrepresentations, and the transformed character is the difference of their characters. It follows that every Wilson loop composed with either kind of map is a finite linear combination of Wilson loops. Since $\mathcal{T}_W$ is the initial topology of the Wilson loops, both kinds of maps and their inverses are continuous, and by Theorem 29.3 they are Borel.

Step 3: spatial invariance. Fix $y \in \mathbb{T}_L^3$ and $Q \in O(3,\mathbb{Z})$. Let y_K be lattice translations with $d(y_K, y) \le \sqrt{3}a_K/2$, and set $f_K(x) = Qx + y_K$, which maps the lattice to itself. The exact identity of Step 1 for f_K applies to a finite list of admissible approximations $(\ell_{i,K})_K$. The transformed paths have the same lengths, and the comparison homotopies transformed by f_K connect them to $f_K\ell_i$ with the same swept areas, which tend to zero. We append the translation strip between $f_K\ell_i$ and $f\ell_i$, together with a connecting path between their basepoints. On the universal cover, its swept area is at most $|y_K - y|\,\mathrm{length}(\ell_i)$, and the same bound holds after projection to the torus, for the swept area counted with multiplicity. Thus the transformed paths form an admissible approximation of $f\ell_i$, with swept areas $A_{i,K} + O(a_K) \to 0$. Using the identification of the limiting finite joint laws of the Wilson loops in Theorem 29.20 (along nM^k, in Theorem 29.28) together with Theorem 29.23, we pass the exact identities to every finite joint law of the Wilson loops of f^*X and of X under λ. By Theorem 29.3, we have $(f^*)_\# \lambda = \lambda$. This proves invariance under continuous as well as lattice translations.

Step 4: center invariance and the selection rule. We now use the exact center identity of Step 1. Since a based homotopy preserves the integer winding vector, for large K the central sign in (30.10) for $\ell_{i,K}$ is the sign in (30.11) for ℓ_i. By the convergence of joint loop laws, the identity passes to every finite joint vector under λ. Since these laws determine probability measures, we have $(C_z)_{\#}\lambda = \lambda$. For a product of normalized fundamental Wilson loops raised to the powers b_j, this invariance multiplies the expectation by $\prod_i z_i^{\sum_j b_j w_i(\ell_j)}$. If the vector $\sum_j b_j w(\ell_j)$ has a nonzero component modulo two, we choose $z_i = -I$ at such a component and $z_i = I$ elsewhere. Then the multiplier is -1, so the expectation equals its negative and vanishes; the variables are bounded, so no integrability issue arises. The rule for irreducible characters follows in the same way, with the additional parities n_j, and the rule at a finite cutoff follows directly from the exact identities of Step 1. For the measure $\lambda_{L,\gamma,[n]_M}$ along the full sequence, see Section 30.6. □

30.6 Collected conclusions and the full sequence

We collect the conclusions of Sections 30.3 to 30.5 in the setting of Theorem 29.21, and then prove their extension to the full sequence $N_k = nM^k$.

Corollary 30.7. *At every fixed $0 < \gamma \leq \Gamma$, in the setting of Theorem 29.21, all conclusions of Theorems 30.4 to 30.6 hold, with admissible approximations fixed in time. In particular, the joint processes of lattice energies E_a and flowed lattice Wilson loops converge in distribution in the uniform topology on $[\tau, T]$ to the corresponding processes under the same measure λ, and the expected supremum of the loop discrepancy between two admissible approximations tends to zero. The image laws of the maps π_K in the Sobolev topology converge, together with every finite joint vector of energies and Wilson loops of the image trajectories. The measure λ is invariant under the spatial and center symmetries, and the winding selection rules hold. These conclusions hold at each fixed coupling; no modulus of convergence for the paths or in the Sobolev topology is uniform in the coupling.*

Proof. Theorems 30.4 to 30.6 are formulated and proved at every fixed $0 < \gamma \leq \Gamma$, for the maps and the measure of Theorem 29.21, and their inputs have no further restriction on γ. The deterministic Lemma 30.3 uses Theorems 24.5 and 25.7 and Lemma 25.5, which have no restriction on the coupling, together with the energy estimates of Lemma 29.14; its Lipschitz bound in time uses only the lengths of the paths. The bounds on the flow data are tight by Lemma 28.11. The joint loop limits used in the proof of Theorem 30.6 are given by Theorems 29.21 and 29.23; the strip added when a continuous translation is approximated by lattice translations has vanishing swept area and is covered by Theorem 29.23. The Sobolev assertion requires only the energy tails and the convergence of the finite joint laws of the Wilson loops, which are part of Theorem 29.21.

We also record a second proof of the convergence of the conditioned path laws used in Step 1 of the proof of Theorem 30.4. By Lemma 30.3, on each event $\{\mathsf{R}_{K|r} \le R\}$ and for small meshes, the joint energy and loop paths are bounded and Lipschitz in time with constants depending on R, and the consistency of the loop paths for admissible approximations follows from (29.12). By (30.3) and the Arzelà–Ascoli theorem, the conditioned path laws are tight. By Theorems 29.21 and 29.23, their joint laws at rational times are those of the paths under λ_r, and these laws determine a probability measure on the continuous path space. Hence the conditioned path laws converge to the image of λ_r under the continuum path map, which is Borel because its coordinates at rational times are Borel and its paths are continuous. This argument does not require the energy to be continuous in the Wilson topology. The passage to the Wilson measure is then Steps 2 and 3 of the proof of Theorem 30.4. □

We now prove the assertions on the full sequence $N_k = nM^k$ in Theorems 30.4 to 30.6, which are also the additional assertions of Theorem 29.28 on paths, the Sobolev topology, and symmetries.

Proof of the assertions on the full sequence. We take the consecutive integers $J, J+1$ of Corollary 23.15 and the maps used in Theorem 29.28. In this paragraph, $r \in \{0, \dots, J-1\}$ denotes a residue index. For each residue r, the subsequence nM^{r+Jl}, $l \ge 0$, is a sequence of the form n_*B^l with base $nM^r \in 4\mathbb{N}$, and the maps of Theorem 29.28 on it are those of Corollary 29.26.

On every such residue-class subsequence, Theorem 30.4, applied through Theorem 29.21, yields the convergence of the energy and loop paths and the comparison of two admissible approximations. By Theorem 29.28, the limiting measure is the same $\lambda_{L,\gamma,[n]_M}$ for every residue. The continuum path map is Borel: its energy coordinates at fixed times are Borel by Theorem 29.6, its Wilson-loop coordinates are Borel by Theorem 29.3, its paths are continuous, and the Borel σ-algebra of the uniform path space is generated by the evaluations at a countable dense set of times. Hence the limiting path law is the same for every residue. For a bounded continuous path test, the expectations converge along each of the finitely many residue classes, and taking the largest of the finitely many thresholds proves the convergence along the whole sequence. The same finite maximum proves the comparison of two admissible approximations.

For the Sobolev assertion, the general part of Theorem 30.5 applies directly to the interlaced maps, since Theorem 29.28 gives the tightness of the energies of the image trajectories at every positive time and the convergence of all joint laws of determining coordinates. Alternatively, the Sobolev limit along each residue class is the same Borel image $(i^{1,2})_{\#}\lambda_{L,\gamma,[n]_M}$, and the finite maximum argument above applies to every bounded continuous test for $\mathcal{T}_{1,2}$. Lemma 29.7 identifies the finite joint vectors of energies and Wilson loops of the image trajectories.

Finally, the limiting measure equals, for example, the measure along the sequence $n(M^{J+1})^I$ used in the proof of Theorem 29.28. Theorem 30.6 for that sequence therefore gives the spatial and center invariance of this measure, and the selection rule is an identity for this measure. The proof uses only fixed couplings and finitely many values of J, so no uniformity in the coupling, in Γ, or in J is involved. □

30.7 Continuity and uniform convergence in the coupling

In this section, the coupling varies in a compact interval $I = [\gamma_-, \gamma_+] \subset (0, \infty)$, $\gamma_- > 0$. We show that the convergence statements for finite vectors of Wilson loops and for the image laws specified in Theorem 30.10 hold uniformly for $\gamma \in I$, with approximating maps that do not depend on γ, and that $\gamma \mapsto \lambda_{L,\gamma,[n]_M}$ is weakly continuous. The main point is the order of the choices: the parameters, the minimum coarse side, and the sequence $m(K)$ that determines the maps must be fixed before $\gamma \in I$ is chosen. To that end, we first prove, for each fixed number of constraints, that the probability that no element of the countable family of Lemma 29.8 satisfies the constraints tends to zero uniformly in $\gamma \in I$. Only then do we choose $m(K)$.

Constants with a subscript I may depend on both endpoints $\gamma_\pm$, on the coarse side, on L, M, J, Θ, on the analytic radii, the exponents, and the moment orders, and on the stated lists of times and observables. They do not depend on $\gamma \in I$, on the depth of the hierarchy, or on the finer mesh; that is, they have the dependence described in item (7) of Definition 4.14, with the additional dependence on I. Lattice paths are admissible approximations as in Section 30.1; they and their swept areas do not depend on γ. The first of two lemmas restates the deterministic comparison of Theorem 29.16 at finitely many observation times.

Lemma 30.8. *Fix $L, T_0, R > 0$, $1/2 < \alpha < 2/3$, $0 < \tau \le T < \infty$, and a finite list $(\ell_{h,a}, \chi_h, t_h)$ of closed lattice paths, characters, and times $t_h \in [\tau, T]$, where the paths have physical lengths bounded by fixed numbers. For $n = L/a \in \mathbb{N}$ set*

$$W_h^{\mathrm{lat}}(U) = \chi_h\big(\mathrm{Hol}_{\ell_{h,a}} \Psi_{t_h}^{(n)} U\big).$$

Consider two flow data (x_a, y_a) and (x'_a, y'_a) on the same lattice and in the same gauge, of initial configurations U and U', with norms on $[0, T_0]$ at most R, and set $d_a = \mathsf{X}_\alpha^a(x_a - x'_a) + \mathsf{Y}_\alpha^a(y_a - y'_a)$, where both norms include their difference components. There are $a_* > 0$, $d_* > 0$, and $C_R < \infty$, depending only on the displayed deterministic data, such that $d_a \le d_*$ and $a \le a_*$ imply

$$\sup_{\tau \le t \le T} \big|\mathsf{E}_a(\Psi_t^{(n)} U) - \mathsf{E}_a(\Psi_t^{(n)} U')\big| + \max_h \big|W_h^{\mathrm{lat}}(U) - W_h^{\mathrm{lat}}(U')\big| \le C_R d_a,$$

and each of these energies is at most C_R on $[\tau, T]$. The constants do not depend on the coupling, the law, the cutoff, the depth of the hierarchy, the index of a finite-scale reconstruction, or the

derivatives of its arrays. For a fixed finite-scale reconstruction, interpolated by the cardinal interpolation of Lemma 16.3, the joint vector of lattice energies and flowed lattice Wilson loops converges, uniformly over the bounded set of its arrays on the coarse mesh, to the corresponding vector of the continuum flow from the interpolated field, for admissible approximations.

Proof. We work in the common gauge in which d_a is measured and apply Theorem 29.16 with the window $[\tau, T]$ and the given length bounds. After enlarging the constant, the supremum over time in that theorem bounds the discrepancy of each loop at its own time t_h, as well as the energy discrepancy, and its bound on the individual energies gives the bound C_R. We take d_* below the threshold for the discrepancy in Lemma 25.9. These are deterministic results with no restriction on the coupling. For the bounded set of coarse arrays, the uniform consistency of the joint energy and loop paths in Lemma 29.17 implies the stated consistency of the finite vector. Only this consistency constant may depend on the derivatives of the arrays of the fixed finite-scale reconstruction. □

For the second lemma, fix a coarse side n and write $a_q = (L/n)B^{-q}$. Under $\nu_{nB^k,n,\gamma}$, let $\widehat{\theta}_q$ be the link logarithms of the field reconstructed at scale q by the reconstruction maps (Theorem 15.13), and let $f_q = I_{a_q \to 0}(\widehat{\theta}_q / a_q)$ be its cardinal interpolation to a smooth one-form on $\mathbb{T}_L^3$ (Lemma 16.3). We write $X^{[q]}$ for the continuum trajectory started at f_q, and $d_{k,q}$ for the norm of the difference between the flow data of the fields reconstructed at scales k and q, computed on the lattice of mesh a_k, as in Lemma 28.10.

Lemma 30.9. *Fix $L > 0$, $I = [\gamma_-, \gamma_+] \subset (0, \infty)$, $\Gamma \geq \gamma_+$, the parameters of Theorem 23.5, and a coarse side n above the common threshold of Lemma 28.10. Fix a finite list of energies at positive times and of admissible approximations of loops with characters and positive times. Under $\nu_{nB^k,n,\gamma}$, let $Z_{k,\gamma}$ be the joint vector of lattice energies and flowed lattice Wilson loops, and let $Z_{k,\gamma}^{[q]}$ be the joint vector of continuum energies and Wilson loops of $X^{[q]}$, computed on the same sample. Then*

$$\lim_{q \to \infty} \limsup_{k \to \infty} \sup_{\gamma \in I} \mathbb{E}_{\nu_{nB^k,n,\gamma}} \min\bigl(1, |Z_{k,\gamma} - Z_{k,\gamma}^{[q]}|\bigr) = 0. \tag{30.12}$$

For each fixed q, the laws of $Z_{k,\gamma}^{[q]}$ converge as $k \to \infty$, uniformly in $\gamma \in I$, to laws that depend continuously on γ and have a common compact support. For each fixed $t > 0$, the tails of the lattice energy at time t under $\nu_{nB^k,n,\gamma}$ are uniform over $\gamma \in I$ and over all cutoffs.

Proof. For fixed q, the array on the coarse mesh lies in a fixed compact set, and the continuum vector is a continuous function of this array by Theorem 24.10. The assertion about the laws at fixed q, including the continuity in γ and the compact support, then follows from Theorem 15.16.

Let $\rho_I(R)$ be the tail of the norm of the flow data in (28.18), and let $d_*(R+1)$ be the threshold for the discrepancy in Lemma 30.8 at the radius $R+1$. Fix $0 < \delta \leq$

$\min\{1, d_*(R+1)\}$. On the event that the flow data of the full field have norm at most R and $d_{k,q} \le \delta$, the flow data of the full field and of the field reconstructed at scale q both have norm at most $R+1$, and the hypotheses of Lemma 30.8 hold for all sufficiently small meshes. Applying that lemma, followed by its consistency statement for the fixed scale q, we obtain

$$|Z_{k,\gamma} - Z_{k,\gamma}^{[q]}| \le C_R\delta + \omega_q(a_k), \qquad \omega_q(a_k) \longrightarrow 0,$$

uniformly over the bounded set of coarse arrays and over γ; the mesh threshold for the first comparison does not depend on q. This implies that the expectation in (30.12) is at most

$$\rho_I(R) + \sup_{\gamma\in I} \nu_{nB^k,n,\gamma}\{d_{k,q} > \delta\} + C_R\delta + \omega_q(a_k).$$

With R and δ fixed, we first take the limit superior in k at fixed q, then let $q \to \infty$ using (28.17), then let $\delta \to 0$, and finally let $R \to \infty$. This proves (30.12). The energy tails follow from the uniform tail $\rho_I(R)$ and the deterministic bound on the energy for flow data of norm at most R in Lemma 30.8. The finitely many coarse cutoffs below the mesh threshold are included by the bound $\mathsf{E}_a \le 48N^3/a$, which holds because $0 \le q(U_p) \le 2$ for every plaquette and there are $3N^3$ positive plaquettes. This yields the supremum over all cutoffs. □

Theorem 30.10. *Fix $L > 0$, an odd integer $M \ge 3$, and $I = [\gamma_-, \gamma_+] \subset (0,\infty)$ with $\gamma_- > 0$. Choose $\Gamma \ge \gamma_+$ and the parameters of Theorem 23.5. There is a minimum coarse side n_I such that, on every sequence $N_K = n_* B^K$ with $n_* \in 4\mathbb{N}$, after a finite shift of the base, common to all $\gamma \in I$, if $n_* < n_I$, the following statements hold, with $\lambda_\gamma = \lambda_{L,\gamma,[n_*]_M}$.*

1. *For every fixed finite vector of flowed lattice Wilson loops on admissible approximations, the laws under $\mu_{N_K,\gamma}$ converge to the corresponding law under λ_γ in the bounded-Lipschitz distance, uniformly for $\gamma \in I$. The complex source functionals and each of their fixed derivatives converge uniformly for $\gamma \in I$ and z in each fixed finite ball.*
2. *There are Borel maps π_K on the full configuration spaces, invariant under periodic lattice gauge transformations and independent of γ and of any coarse side chosen afterwards, such that for each fixed j and $t > 0$,*

$$\sup_{\gamma\in I} \mathbb{E}_{\mu_{N_K,\gamma}} |w_j(\pi_K) - W_{K,j}^{\mathrm{lat}}| \longrightarrow 0, \qquad \lim_{R\to\infty} \sup_{K\ge 0,\, \gamma\in I} \mu_{N_K,\gamma}\{E_L(\pi_K(t)) > R\} = 0.$$

For every further fixed finite vector $\mathscr{W}$ of Wilson loops on admissible approximations, with lattice counterpart $\mathscr{W}_K^{\mathrm{lat}}$, and every bounded continuous test f of that vector,

$$\sup_{\gamma\in I} \mathbb{E}_{\mu_{N_K,\gamma}} \big| f(\mathscr{W}(\pi_K)) - f(\mathscr{W}_K^{\mathrm{lat}}) \big| \longrightarrow 0.$$

The same statements on determining coordinates and energy tails hold uniformly in $\gamma \in I$ under every fixed conditioned measure $\nu_{N_K,n_r,\gamma}$ with $n_r = n_* B^r \ge n_I$.

3. *For every bounded $\mathcal{T}_W$-continuous function F on $\mathcal{X}_L$,*

$$\sup_{\gamma\in I}\left|\mathbb{E}_{\mu_{N_K,\gamma}}F(\pi_K)-\lambda_\gamma(F)\right|\longrightarrow 0.$$

   The convergence is also uniform in the bounded-Lipschitz distance of $(\mathcal{X}_L,d_c)$. The image laws and the limiting measures have common compact containment in the Wilson topology and in d_c.
4. *The map $\gamma\mapsto\lambda_\gamma$ is weakly continuous on I for both topologies, and the law of every fixed finite vector of Wilson loops and its source functional depend continuously on γ. These statements hold on all of $(0,\infty)$, by covering it with compact intervals.*

The same conclusions under the Wilson measures hold along the full sequence $N_k=n_*M^k$. The parameters, n_I, and the sequence $m(K)$ defining the maps are fixed before $\gamma\in I$ is chosen; the constants may depend on I and on the sequence.

Remark 30.11. The constants may diverge as $\gamma_-\to 0$ or $\gamma_+\to\infty$, and no rate of convergence is asserted. The uniformity concerns each fixed finite list and each fixed bounded test; it is not asserted for the path laws of Theorem 30.4, for the Sobolev laws of Theorem 30.5, or for the energy laws of the image trajectories, whose convergence at each fixed γ follows from Theorem 30.5. The maps chosen for two overlapping intervals may differ, but their limiting measures agree on the overlap, since these are determined by the joint laws of Wilson loops. Finally, if $I\subset(0,\gamma_0]$ for the coupling bound $\gamma_0\le 1$ of Theorem 23.2, one may take $\Gamma=\gamma_0$; by the clause of Theorem 23.5 that retains a prescribed J, the parameter tuple of Theorem 23.2 can then be kept, with one common enlargement of the minimum coarse side.

Proof of Theorem 30.10. The proof has six steps. We fix the parameters and n_I, and then prove uniform convergence of finite vectors of Wilson loops. Next, we prove that the failure probability at each fixed level tends to zero uniformly, choose the maps by a diagonal argument, and treat the early cutoffs. Finally, we deduce the convergence of the image laws, the continuity in γ, and the statement along the full sequence.

Step 1: parameters. Theorem 23.5 gives the common parameter tuple. We fix the horizon $T_0=L^2$ and the exponents of the norms of the flow data before any list of observables is chosen. All later lists use this horizon; their positive waiting times and later horizons affect the deterministic flow constants and the mesh thresholds, but not the minimum coarse side. Let $t_*>0$ be so small that the finitely many smallness conditions of Lemma 28.10 on the largest effective coupling $t_1=(\gamma/(Mn))^{1/2}$ hold whenever $t_1\le t_*$, and take $n_I\ge\max(n_\Gamma,\gamma_+/(Mt_*^2))$, rounded up to an element of $4\mathbb{N}$; then $t_1\le t_*$ for all $\gamma\in I$ and $n\ge n_I$. All other conditions are already included in n_Γ. No local constant is chosen using n_I or a later failure-probability constant, and every coarse side $n\ge n_I$ in the sequence is allowed with the same tuple.

Step 2: finite vectors of Wilson loops. At a fixed coarse side, the Wilson loops lie in a fixed compact box, since each is bounded by the dimension of its representation. By Lemma 30.9, the approximation errors are uniformly small, and for fixed q the approximating laws converge uniformly and depend continuously on γ. For a test function f with $\max(\|f\|_\infty, \operatorname{Lip} f) \le 1$, that is, with bounded-Lipschitz norm at most one, the triangle inequality between two cutoffs shows that the expectations under the conditioned measures form a Cauchy sequence uniformly in $\gamma \in I$; passing to a common finer cutoff, we find that the same holds for the limiting approximating laws at two indices q, q'. The probability measures on the compact box form a compact, hence complete, space for the bounded-Lipschitz distance. To see this, we partition the box into nested finite families of half-open cubes with diameters tending to zero, assigning the outer boundary to the last cells. From any sequence of laws, we extract by a diagonal argument a subsequence along which the mass of every cell converges. The limiting weights are consistent under refinement and sum to one. We subdivide $[0,1]$ into nested intervals with these lengths, so that the intervals of the children partition that of the parent. Apart from the countably many endpoints of these intervals, each point of $[0,1]$ selects a nested chain of cells, whose closures determine one point of the box. This map is measurable as a limit of step functions, and its image of Lebesgue measure is a probability measure on the box. At each level of the partition, the integral of a test with bounded-Lipschitz norm at most one against either the laws of the subsequence or this law differs from the corresponding weighted sum of values at fixed representatives of the cells by at most the largest cell diameter. Since the cell masses converge, the subsequence converges to this law in the bounded-Lipschitz distance. Hence for each coarse index r, the laws of the vector under the conditioned measures have a unique limit P^r_γ, and the convergence is uniform on I. The same approximation shows that P^r_γ is continuous in γ, as a uniform limit of the continuous laws at fixed q. The limits for different finite lists are compatible, since the laws at each cutoff are.

We write $n_r = n_* B^r$ and $\nu_{K|r,\gamma} = \nu_{N_K,n_r,\gamma}$. For $u \ge n_\Gamma$, the right side of (23.13) is nondecreasing in $\gamma \in (0,\Gamma]$, since $Mu/\gamma \ge Mn_\Gamma/\Gamma > 1$. Let $F_\Gamma(u)$ denote its value at $\gamma = \Gamma$. By Theorem 23.10, we have

$$\sup_{K \ge r,\, \gamma \in I} \mu_{N_K,\gamma}(S^c_{N_K,n_r,\gamma}) \le F_\Gamma(n_r), \qquad F_\Gamma(n_r) \longrightarrow 0 \quad (r \to \infty).$$

By (30.2), at each cutoff, the expectation of a test with bounded-Lipschitz norm at most one under the Wilson measure differs from that under the conditioned measure by at most $2F_\Gamma(n_r)$. Taking the uniform limit in K at fixed r, and then letting $r \to \infty$, we obtain uniform convergence of the vector laws under the Wilson measures. By Theorems 29.23 and 29.24, their pointwise limit is the law of the vector under λ_γ. The continuity in γ follows from the continuity of the conditioned limits and the same uniform error. For the source functionals, since the coordinates are bounded, the exponentials and

their fixed derivatives in z are uniformly bounded and uniformly continuous on a fixed complex ball; a finite net of the ball gives the stated uniformity. In particular, fixed moments and cumulants converge uniformly and depend continuously on γ. On a common small ball of sources, the logarithm with value zero at the origin is defined, since $|Z(z) - 1| \le e^{D \sum_h |z_h|} - 1 < 1/2$ there, where D bounds the dimensions of the representations. The uniform convergence of the fixed derivatives then implies the convergence of the derivatives of the logarithms of the source functionals at the origin, with no additional nonvanishing hypothesis.

The tails of the lattice energy under the Wilson measures are also uniform. At fixed r, the conditioned tail is uniform by Lemma 30.9, and by (30.2), the Wilson probability of a tail event is at most its conditioned probability plus $F_\Gamma(n_r)$. We take first the limits in the cutoff and in the size of the tail at fixed r, and then let $r \to \infty$. The finitely many earlier cutoffs are included by the bound $\mathsf{E}_a \le 48N^3/a$.

Step 3: uniform feasibility at a fixed level. Let $(X_b)_{b \ge 1}$ be the countable family of continuum trajectories of Lemma 29.8, started at the trigonometric polynomials with rational coefficients, with $X_1 = 0$; it does not depend on γ. Set $s_i = L^2 2^{-i}$ and $\epsilon_m = 2^{-m}$, and let $\pi_{K,m}(U)$ be the first X_b that satisfies

$$\begin{aligned} \big|w_j(X_b) - W^{\mathrm{lat}}_{K,j}(U)\big| &< \epsilon_m \qquad (j \le m), \\ E_L(X_b(s_i)) &< \mathsf{E}_{a_K}(\Psi^{(N_K)}_{s_i} U) + \epsilon_m \qquad (i \le m), \end{aligned} \tag{30.13}$$

and the zero trajectory if there is no such b; this is the construction of Theorem 29.19. We say that level m fails at U if no b satisfies (30.13).

Fix m, a coarse index r with $n_r \ge n_I$, and $\eta > 0$. By Lemma 30.9, applied to the m determining coordinates and the m energies in (30.13), there exists a fixed q such that, for all sufficiently large K and uniformly in $\gamma \in I$, the lattice vector differs from the continuum vector of $X^{[q]}$ on the same sample by less than $\epsilon_m/4$ with conditioned probability at least $1 - \eta$. To deduce this from the truncated expectation in (30.12), we apply Markov's inequality with the threshold $\min(1, \epsilon_m/4)$ and make that expectation smaller than $\eta \min(1, \epsilon_m/4)$. By Lemma 29.8, for every realization of $X^{[q]}$, there exists an element of the countable family whose joint vector differs from that of $X^{[q]}$ by less than $\epsilon_m/4$. Then the strict inequalities (30.13) hold, since every loop error is less than $\epsilon_m/2$ and every energy of the family element is less than the lattice energy plus $\epsilon_m/2$. Thus the conditioned probability that level m fails is at most η, uniformly in $\gamma \in I$. No measurable choice of the approximating element is needed, since its existence places the configuration in the countable union of the sets where (30.13) holds. This proves uniform feasibility at each fixed level directly, without a limiting initial potential and without using the compactness of I to make a pointwise statement uniform.

For the failure event under the Wilson measure, (30.2) bounds the limit superior in K by $F_\Gamma(n_r)$ for every fixed r with $n_r \ge n_I$, uniformly in $\gamma \in I$. Letting $r \to \infty$ after this

limit, we find that the Wilson probability that level m fails also tends to zero uniformly. For each K and m, the conditions in (30.13) are strict inequalities between continuous gauge-invariant functions of the configuration. Therefore the events that a given b is the first to satisfy them, and the failure event, are Borel, and $\pi_{K,m}$ is Borel and invariant under periodic lattice gauge transformations. The conditions do not involve γ.

Step 4: the diagonal choice. We enumerate the coarse indices r with $n_r \geq n_I$ as $r(1), r(2), \ldots$. By Step 3, we can choose increasing integers $K_m \geq \max(m, r(1), \ldots, r(m))$ such that for every $K \geq K_m$ the probability that level m fails is at most 2^{-m}, uniformly in $\gamma \in I$, under the Wilson measure and under each of the first m conditioned measures. Set $m(K) = \max\{m : K_m \leq K\}$, or $m(K) = 0$ if this set is empty, and $\pi_K = \pi_{K,m(K)}$. When level $m(K)$ does not fail, the error of each coordinate w_j with $j \leq m(K)$ is less than $2^{-m(K)}$; when it fails, the error is at most two. Thus each fixed coordinate has error in L^1 at most $3 \cdot 2^{-m(K)}$, uniformly in $\gamma \in I$, under the Wilson measure and eventually under each fixed conditioned measure. On every configuration and for $i \leq m(K)$, we have

$$E_L(\pi_K(s_i)) \leq \mathsf{E}_{a_K}(\Psi_{s_i}^{(N_K)} U) + 2^{-m(K)}, \tag{30.14}$$

since on failure the output is the zero trajectory, whose energy is zero. Combined with the uniform tails of the lattice energies from Step 2, this yields uniform tails of the energies of the image trajectories for all sufficiently large K. By the monotonicity of the continuum energy along the flow, these tails hold at every $t > 0$.

Step 5: the early cutoffs. It remains to include the finitely many cutoffs below this threshold, uniformly in γ; the finiteness of the energy of each image trajectory, pointwise in γ, would not suffice. On a fixed finite link space, the densities of the Wilson measures with respect to Haar measure are positive and continuous on the product of the compact link space with I, hence bounded above and below by positive constants. For a fixed conditioned measure, the small-field event contains one open neighborhood of the identity configuration for all $\gamma \in I$. Indeed, its finitely many conditions on charts, constraint Jacobians, and fluctuation variables hold with strict margins at the identity, and the normalized fluctuation variables are continuous near the identity and vanish there. Further, every cutoff radius $A_{\mathrm{w}} \log(1/t_k)$ has a positive minimum over I because $t_k \leq t_* < 1$. We shrink one neighborhood O so that it satisfies all these finitely many conditions. Then $\mu_{N_K,\gamma}(S_{N_K,n_r,\gamma}) \geq c\, H_{N_K}(O) > 0$ uniformly in $\gamma \in I$, and the densities of the conditioned measures are bounded above by $C_{K,r,I}$ times Haar measure. For each configuration, $\pi_K(U)$ is an element of the countable family or the zero trajectory, and has finite energy. Its tail under Haar measure tends to zero by continuity from above, and the domination by Haar measure makes this convergence uniform in γ. Hence the finitely many early cutoffs can be included, and the energy tails hold with the supremum over

all cutoffs, without assuming a deterministic bound on the energies of all elements of the countable family.

Step 6: convergence of the image laws, continuity, and the full sequence. By Theorem 29.6, the image laws under the Wilson measures have common compact containment. At each fixed γ, the coordinate errors and the convergence of finite vectors identify every subsequential limit of the image laws with λ_γ, by the determination of probabilities in Theorems 29.3 and 29.20. The image laws therefore converge to λ_γ. By the inequality for closed sets in the portmanteau theorem, each limiting measure inherits the uniform lower bound on the mass of each closed energy sublevel set, so the family of limits has the same compact containment.

Let C be a common compact set whose complement has probability at most η under all image laws and all limits. On C, the Wilson topology and the d_c topology agree. By Lemma 29.9, a bounded Wilson-continuous F with $|F| \le D$ is approximated within δ on C by $G(w_1, \dots, w_b)$, with G continuous on $[-1,1]^b$ and $|G| \le D$. Using the coordinate errors and the uniform continuity of G, we compare the expectation of G on the image with its expectation on the lattice coordinates, uniformly in γ, and the latter converges uniformly by Step 2. The two complements of C contribute at most $4D\eta$. Therefore

$$\limsup_{K\to\infty} \sup_{\gamma\in I} \left| \mathbb{E}_{\mu_{N_K,\gamma}} F(\pi_K) - \lambda_\gamma(F) \right| \le 2\delta + 4D\eta,$$

and we let $\delta, \eta \to 0$. For the class of functions with bounded-Lipschitz norm at most one for d_c, the restrictions to C have a finite uniform net. To construct it, we take a finite fine net of C, round the values of each test at its points, and choose one test realizing each rounded list of values. The Lipschitz bound controls the error between net points. The uniform convergence for these finitely many tests and the common bound on the complements give uniform convergence in the bounded-Lipschitz distance.

For a further fixed finite vector $\mathscr{W}$ and a bounded continuous test f, the function $h = f \circ \mathscr{W}$ is bounded and Wilson-continuous; let G be its approximation on C as above, and write $p_b X = (w_j(X))_{j\le b}$ and $\mathbf{W}^{\mathrm{lat}}_{K,b} = (W^{\mathrm{lat}}_{K,j})_{j\le b}$. Then

$$\begin{aligned} \left| f(\mathscr{W}(\pi_K)) - f(\mathscr{W}^{\mathrm{lat}}_K) \right| \le{}& \left| h(\pi_K) - G(p_b \pi_K) \right| + \left| G(p_b \pi_K) - G(\mathbf{W}^{\mathrm{lat}}_{K,b}) \right| \\ &+ \left| G(\mathbf{W}^{\mathrm{lat}}_{K,b}) - f(\mathscr{W}^{\mathrm{lat}}_K) \right|. \end{aligned}$$

The expectation of the first term is at most $\delta + 2D\eta$, and that of the second tends to zero uniformly by the coordinate errors. The last term is a bounded continuous test of the single joint lattice vector consisting of the determining coordinates and $\mathscr{W}^{\mathrm{lat}}_K$. By Step 2, its expectation converges uniformly to $\mathbb{E}_{\lambda_\gamma} |G \circ p_b - h| \le \delta + 2D\eta$. This proves the statement for further vectors without combining separately given marginals. For complex tests, we apply the argument to the real and imaginary parts.

At each fixed N, the density $Z_{N,\gamma}^{-1}\exp(-4N\gamma^{-1}\sum_p q(U_p))$ of the Wilson measure is continuous in $L^1(H_N)$ as a function of $\gamma \in I$, since the numerator is bounded and jointly continuous and the normalizing constant is positive and continuous. Hence the finite measures are continuous in total variation. Since the maps π_K do not depend on γ, the expectation of every bounded Borel function of π_K is continuous in γ. Combined with the uniform convergence for bounded Wilson-continuous F, this proves the weak continuity of $\gamma \mapsto \lambda_\gamma$ for the Wilson topology. Since the identity map from the Wilson topology to d_c is continuous, we also obtain weak continuity for d_c. Similarly, the uniform convergence of the finite vectors implies the continuity of their laws and source functionals. The measures λ_γ do not depend on I, so continuity on every compact interval gives continuity on $(0,\infty)$.

In this paragraph, $r \in \{0,\dots,J-1\}$ denotes a residue index. For the full sequence n_*M^k, the statements above apply to the finitely many residue bases n_*M^r, $0 \le r < J$, with one common shift of the base. By Theorem 29.28, their pointwise limits are the same λ_γ, and the maximum of finitely many uniform errors tends to zero. Let $m_r(l)$ be the diagonal sequences of the finitely many residue classes. On the full sequence, the maps are defined by (30.13) at the level $m(k) = \min_{0\le r<J} m_r(\lfloor k/J \rfloor)$, which is a nondecreasing sequence tending to infinity. A configuration that does not fail at some level does not fail at any lower level, since there are fewer conditions with larger tolerances. Hence the failure probabilities at level $m(k)$ are bounded by those of the residue classes, and all estimates on the coordinate errors and the energies persist. The finitely many initial zero outputs affect no limit and no tail. □

Corollary 30.12. *Fix a physical coupling $c = g_3^2 > 0$, an odd integer $M \ge 3$, and an equivalence class $[n]_M$, and identify $\mathcal{X}_L$ with $\mathcal{X}_1$ by D_L^{-1}. Then the map*

$$L \longmapsto (D_L^{-1})_{\#}\lambda_{L,cL,[n]_M} = \lambda_{1,cL,[n]_M}$$

is weakly continuous on $(0,\infty)$ for both topologies on $\mathcal{X}_1$. On each compact interval of side lengths in $(0,\infty)$, the limits of the dilated finite vectors of flowed lattice Wilson loops, and the convergence of the laws of the common maps, are uniform in L. The same holds jointly in (c,L) on compact rectangles in $(0,\infty)^2$, after transporting tests and times by dilation.

Proof. The notation c is local to this corollary. The displayed identity is (30.4), and the identities of Theorem 30.1 at a finite cutoff include the measure, the lattice time, and the ordered lattice paths. A compact rectangle of pairs (c,L) is mapped by $(c,L) \mapsto cL$ onto a compact interval of couplings on the unit torus. We apply Theorem 30.10 on that interval and compose the continuous functions $\gamma \mapsto \lambda_{1,\gamma,[n]_M}(F)$ with $(c,L) \mapsto cL$. We then transport the maps on the unit torus, which do not depend on the coupling, by D_L; their dimensionless errors and energy tails are exactly those on the unit torus. No

uniformity as $L \to \infty$, or for physical observation windows that are not transported by dilation, is obtained. □

30.8 Modified smoothing flows

The flowed Wilson loops are defined with the Wilson flow, the gradient flow of $\sum_p q(U_p)$ introduced by Lüscher [128]. We now show that the limits do not change when the flow is replaced by the gradient flow of a modified plaquette function, within a bounded family. For $c \geq 0$ and $a = L/N$, let

$$\mathcal{A}_c(U) = \sum_{p \in \mathcal{P}_N} \{q(U_p) + c\, q(U_p)^2\}, \qquad \Psi_t^{(N,c)} = \Phi^{\mathcal{A}_c}_{4t/a^2},$$

where $\Phi^{\mathcal{A}_c}$ is the flow of the vector field $-\nabla \mathcal{A}_c$ for the product bi-invariant metric, as in Section 24 (see Lemma 24.11). In particular, $\Psi_t^{(N,0)} = \Psi_t^{(N)}$. Since the vector field is smooth on the compact configuration space, the flow exists for all times, and it commutes with lattice gauge transformations because $\mathcal{A}_c$ is gauge invariant and the metric is invariant. Only the smoothing changes: the measure $\mu_{N,\gamma}$, the block averaging map, and all conditional kernels are unchanged. Near the identity, $q(U_p)^2$ is quartic in the plaquette logarithm, and in the damped gauge the modification adds to the equation of the flow only terms of degree at least three in the field (Lemma 24.11).

In this section and the next we use the setting of Theorem 29.25: the parameters, the order of the quantifiers, and the sequence $N_K = n_* B^K$ are those of that theorem, so the coupling is fixed in $(0, 1]$, and $\lambda_{L,\gamma,[n_*]}$ denotes the corresponding continuum measure. We define the integer r_0 of Section 30.1 for these parameters, enlarged if necessary so that the conditioned estimates used below hold for all $r \geq r_0$.

Theorem 30.13. *Work in the setting of Theorem 29.25, and fix $0 \leq C_0 < \infty$. Under the Wilson measure $\mu_{N_K,\gamma}$ let $U_K^c(t) = \Psi_t^{(N_K,c)} U$. For loops ℓ_i with admissible approximations $(\ell_{i,K})_K$, characters χ_i, and times $t_i > 0$, $1 \leq i \leq m$, set*

$$W_{K,i}^c = \chi_i\big(\mathrm{Hol}_{\ell_{i,K}} U_K^c(t_i)\big).$$

Then

$$\lim_{K \to \infty} \mathbb{E}_{\mu_{N_K,\gamma}} \sup_{0 \leq c \leq C_0} \max_i \big|W_{K,i}^c - W_{K,i}^0\big| = 0. \tag{30.15}$$

Consequently, the laws of $(W_{K,i}^c)_i$ converge to the joint law of $(W_{\ell_i,\chi_i,t_i})_i$ under $\lambda_{L,\gamma,[n_*]}$, uniformly in $c \in [0, C_0]$ in the bounded-Lipschitz distance. For every finite Z, the complex source functionals $\mathbb{E}_{\mu_{N_K,\gamma}} \exp(\sum_i z_i W_{K,i}^c)$ converge uniformly on $[0, C_0] \times \{z : \sum_i |z_i| \leq Z\}$ to the limit of those of the Wilson flow. In particular, these conclusions hold for every bounded sequence $c = c_K \geq 0$.

The maps π_K of Theorem 29.25 can be used for every c. Their compact containment, their convergence in both topologies, their limit, and the tightness of their energies over all K are

unchanged, and for the normalized fundamental coordinates $W^c_{K,j} = (1/2)\,\mathrm{tr}\,\mathrm{Hol}_{\ell_{j,K}}\,U^c_K(t_j)$ *of the determining family,*

$$\lim_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} \sup_{0\le c\le C_0} \bigl|w_j(\pi_K) - W^c_{K,j}\bigr| = 0 \qquad (j \ge 1). \tag{30.16}$$

Every finite vector of Wilson loops for the modified flows converges jointly to the corresponding vector under $\lambda_{L,\gamma,[n_*]}$*, for every choice of admissible approximations.*

The number C_0 *is chosen after the parameters and the coupling, and neither the parameter tuple nor the minimum side is changed. Deterministic mesh thresholds may depend on* C_0 *and on the bounds on the flow data; convergence moduli may in addition depend on* γ*, on the coarse side, and on the observables, as in item (7) of Definition 4.14. No uniformity as* $\gamma \to 0$ *or* $C_0 \to \infty$ *is asserted.*

Proof. If the list of loops is empty, there is nothing to prove, so we assume it is not. We first prove (30.15) under the conditioned measures, using the deterministic comparison of Lemma 25.15, then remove the conditioning, and finally deduce the remaining assertions.

Step 1: conditioned measures. Fix $r \ge r_0$, write $u_r = n_* B^r$ and $\nu_{K|r} = \mu_{N_K,\gamma}(\,\cdot \mid S_{N_K,u_r,\gamma})$. Under $\nu_{K|r}$, the field given by the reconstruction maps of Theorem 15.13 is a gauge representative of the same configuration U, and the norm $\mathsf{R}_{K|r}$ of its flow data is tight by Theorem 28.8. Since each flow $\Psi^{(N_K,c)}$ commutes with gauge transformations, the Wilson loops of the flow of U equal those of the flow of this representative, for every c, and the supremum of the discrepancies is gauge invariant.

Let $D = \max_i \chi_i(I)$ and

$$\Delta_K = \sup_{0\le c\le C_0} \max_i \bigl|W^c_{K,i} - W^0_{K,i}\bigr| \le 2D.$$

On each finite lattice, the smooth flow depends continuously on c and on the initial configuration. This follows by subtracting the integral equations for two values of the parameters and applying the Lipschitz bound of the vector field on the compact configuration space. Hence the supremum over c equals the supremum over a countable dense subset of $[0, C_0]$, and Δ_K is Borel. For fixed R, the early time interval of Theorem 24.12 for flow data of norm at most R is deterministic, and the norm of the flow data on it is at most $\mathsf{R}_{K|r}$. The admissible approximations have a common bound on their lengths. Then Lemma 25.15, which applies to any fixed finite set of characters, yields a deterministic $\omega_R(a) \to 0$ such that

$$\mathbb{E}_{\nu_{K|r}}\Delta_K \le \omega_R(a_K) + 2D\,\nu_{K|r}\{\mathsf{R}_{K|r} > R\}.$$

We let $K \to \infty$ at fixed r, R, and C_0, and then $R \to \infty$. By (30.3), we conclude that $\mathbb{E}_{\nu_{K|r}}\Delta_K \to 0$.

Step 2: the Wilson measure. At the same cutoff, by Theorem 23.9, applied with the coupling bound one as in Theorem 29.25, we have $\mu_{N_K,\gamma}(S^c_{N_K,u_r,\gamma}) \le F_\gamma(u_r)$ for all $K \ge r$, with $F_\gamma(u_r) \to 0$ as $r \to \infty$ at fixed $\gamma > 0$. The second inequality in (30.2) then implies that

$$\mathbb{E}_{\mu_{N_K,\gamma}} \Delta_K \le \mathbb{E}_{\nu_{K|r}} \Delta_K + 2DF_\gamma(u_r).$$

Letting first $K \to \infty$ and then $r \to \infty$ proves (30.15), with the same order of limits as in Theorem 28.2.

Step 3: consequences. For a test with bounded-Lipschitz norm, the difference of its expectations for the modified flow and for the Wilson flow is at most its Lipschitz constant times $\mathbb{E}\Delta_K$, uniformly in c. By Theorem 29.25, the vector for the Wilson flow converges to its law under $\lambda_{L,\gamma,[n_*]}$, which proves the uniform convergence in the bounded-Lipschitz distance. For a bounded continuous test f with modulus of continuity ϖ_f on the compact range of the vector, Markov's inequality implies that the error is at most $\varpi_f(\eta) + 2\|f\|_\infty \mathbb{E}\Delta_K/\eta$ for every $\eta > 0$. For the source functionals, we have

$$\sup_{\sum_i |z_i| \le Z} \left| e^{\sum_i z_i W^c_{K,i}} - e^{\sum_i z_i W^0_{K,i}} \right| \le Z e^{DZ} \Delta_K,$$

since both exponents are bounded by DZ in absolute value and differ by at most $Z\Delta_K$. Combined with the convergence of the source functionals of the Wilson flow in Theorem 29.25, this implies the uniform convergence. The uniformity in c covers every bounded deterministic sequence c_K.

The maps π_K are those of Theorems 29.20 and 29.25; their definition does not involve c, so their law, gauge invariance, Borel measurability, energy tails, and compact containment are unchanged. For each determining coordinate,

$$\mathbb{E} \sup_c \left| w_j(\pi_K) - W^c_{K,j} \right| \le \mathbb{E} \left| w_j(\pi_K) - W^0_{K,j} \right| + \mathbb{E} \sup_c \left| W^c_{K,j} - W^0_{K,j} \right| \longrightarrow 0,$$

which is (30.16). For every other finite list, (30.15) and the identification of the limit of the Wilson-flow vector in Theorem 29.20 identify its limit under the same $\lambda_{L,\gamma,[n_*]}$; this identifies the law of the joint vector, not only its marginals. By Theorems 29.23 and 29.25, all admissible approximations have the same limits for the Wilson flow, and applying the comparison to each of them proves the independence of the choice of approximation. □

30.9 Energy and loop processes for the modified flows

We now extend Theorem 30.4 to the modified flows, jointly in $c \in [0, C_0]$ and $t \in [\tau, T]$, including the lattice energy. The loops are controlled by Lemma 25.15; the new deterministic input is the comparison of the energies.

Lemma 30.14. *Fix $L > 0$, $1/2 < \alpha < 2/3$, $0 \le R, C_0 < \infty$, $0 < \tau < T < \infty$, finitely many characters χ_i, and lengths $\ell_i^* > 0$. Consider any initial configuration on the lattice of mesh $a = L/N$ whose flow data, on the common early time interval of Theorem 24.12, have norm at most R, and let $U_a^c(t) = \Psi_t^{(N,c)} U$ for $0 \le c \le C_0$. Let $\ell_{i,a}$ be closed lattice paths of physical length at most ℓ_i^*. There is a deterministic function $\omega_{R,C_0,\tau,T}(a) \to 0$ as $a \to 0$ such that, uniformly over all such configurations and paths,*

$$\sup_{\substack{0\le c\le C_0\\ \tau\le t\le T}} \Big\{ \big|\mathsf{E}_a(U_a^c(t)) - \mathsf{E}_a(U_a^0(t))\big| + \max_i \big|\chi_i(\mathrm{Hol}_{\ell_{i,a}} U_a^c(t)) - \chi_i(\mathrm{Hol}_{\ell_{i,a}} U_a^0(t))\big| \Big\} \le \omega_{R,C_0,\tau,T}(a).$$

The maximum over an empty family of loops is zero. The function ω and the mesh thresholds depend only on the displayed data, the representations, and the margins of the charts and of the difference stencil, and not on the coupling or the depth of the hierarchy. No rate is asserted.

Proof. The assertion for the loops is Lemma 25.15; we prove the assertion for the energy on the same configurations, uniformly in both c and t. Fix $s > 0$ smaller than τ and than the length of the common early time interval, with a positive waiting interval before s. By Theorem 24.12, we have

$$\sup_{0\le c\le C_0} \|A_a^c(s) - A_a^0(s)\|_{C_a^1} \longrightarrow 0$$

uniformly over the configurations, where A_a^c is the solution in the damped gauge for the flow with parameter c. The bootstrap of Theorem 25.7 bounds every fixed order of differences of the Wilson-flow solution on the waiting interval before s, and the equation of Lemma 24.2 bounds its time derivatives at each fixed order of differences by the next two orders.

Suppose that the assertion for the energy fails. Then there exist meshes $a_j \to 0$, configurations, parameters $c_j \in [0, C_0]$, and times $t_j \in [\tau, T]$ with a fixed positive discrepancy of the energies. We interpolate the value and difference arrays of the Wilson-flow solution at time s by piecewise multilinear interpolation. The bounds on the next differences give equicontinuity across the cells. Then the Arzelà–Ascoli theorem with a diagonal extraction gives uniform limits of the arrays of all orders, and discrete summation by parts identifies them as the derivatives of one smooth periodic one-form b_s. In particular, $\|A_{a_j}^0(s) - S_{a_j} b_s\|_{C_{a_j}^1} \to 0$, and by the early comparison, the same convergence holds for all $c \in [0, C_0]$, uniformly.

We apply Theorem 24.10 to the single field b_s up to time $T - s$, with at least five derivatives and strict chart margins. This gives a finite reference family, whose fixed gauge transformations we sample at the transition times. Next, Lemma 25.14 applies on

the consecutive intervals, for $c = 0$ and for every $c \le C_0$. If the initial errors are $\eta_j \to 0$, this lemma gives errors at most $C(\eta_j + a_j) \to 0$ in $C^1_{a_j}$ throughout the reference solutions. At a transition time, the fixed smooth gauge transformation k changes the C^1_a norm by a bounded factor on the chart, since $a^{-1}\operatorname{Log}(k(x)e^{aA_i(x)}k(x+ae_i)^{-1})$ extends smoothly to $a = 0$. Thus the errors propagate across every transition, with constants uniform in c.

On each interval, both lattice representatives have a common bound on their C^1_a norms, and their distance tends to zero. By the Lipschitz estimate of Lemma 29.14, which is uniform in the mesh, the difference of their energies is at most this distance times a fixed constant. By the gauge invariance of E_a, these are the energies of the two flows, also at the transition times. This contradicts the assumed discrepancy at t_j. Therefore the supremum over the configurations, c, and t defines a function of a that tends to zero, and combining it with the function for the loops proves the lemma. The only reference solutions used for a hypothetical offending sequence are those of its extracted smooth limit b_s, so no derivatives of finite-scale reconstructions of growing index enter the constants. □

Theorem 30.15. *Work in the setting of Theorem 30.13; in particular $\gamma \in (0,1]$ is fixed. Fix $0 \le C_0 < \infty$, $0 < \tau < T < \infty$, loops $\ell_1, \dots, \ell_m$ with admissible approximations $(\ell_{i,K})_K$ fixed in time, and characters $\chi_1, \dots, \chi_m$. Under the Wilson measure $\mu_{N_K,\gamma}$ let*

$$U^c_K(t) = \Psi_t^{(N_K,c)}U, \qquad W^c_{K,i}(t) = \chi_i\big(\operatorname{Hol}_{\ell_{i,K}} U^c_K(t)\big).$$

Then

$$D_K = \sup_{\substack{c\in[0,C_0]\\ t\in[\tau,T]}} \Big\{ \big|\mathsf{E}_{a_K}(U^c_K(t)) - \mathsf{E}_{a_K}(U^0_K(t))\big| + \max_i \big|W^c_{K,i}(t) - W^0_{K,i}(t)\big| \Big\} \longrightarrow 0 \tag{30.17}$$

in $\mu_{N_K,\gamma}$-probability. The random continuous fields $(c,t) \mapsto (\mathsf{E}_{a_K}(U^c_K(t)), (W^c_{K,i}(t))_i)$ converge in distribution in $C([0,C_0]\times[\tau,T];\mathbb{R}\times\mathbb{C}^m)$, with the uniform norm, to the field that does not depend on c and has the time path

$$t \longmapsto \big(E_L(X(t)), (W_{\ell_i,\chi_i,t}(X))_i\big), \qquad X \sim \lambda_{L,\gamma,[n_*]}.$$

Moreover $\mathbb{E}_{\mu_{N_K,\gamma}} \sup_{c,t} \max_i |W^c_{K,i}(t) - W^0_{K,i}(t)| \to 0$. The number C_0, the time interval, and the observables are fixed before $K \to \infty$, and the convergence moduli have the dependence described in item (7) of Definition 4.14, with these additional data.

The theorem asserts convergence in probability of the energy discrepancy, not convergence of its expectation: only the indicator of the event $\{D_K > \eta\}$ is transferred from the conditioned measures.

Proof. We first recall the tightness of the norm of the flow data under the conditioned measures, then apply Lemma 30.14 on the event where this norm is bounded and transfer the probability of the complementary event, and finally deduce the convergence of the fields.

Step 1: tightness of the flow data. Fix $r \geq r_0$, write $u_r = n_* B^r$ and $\nu_{K|r} = \mu_{N_K,\gamma}(\cdot \mid S_{N_K,u_r,\gamma})$. By Theorem 15.13, the field given by the reconstruction maps at the finest scale is a gauge representative of the same configuration. Let x_j be the linear heat evolutions, on the finest lattice, of the cardinal interpolations of the fields reconstructed at the scales j, and $y_j = \mathcal{Q}_{a_K}(x_j, x_j)$, as in Theorem 28.8. The tightness (30.3) of $\mathsf{R}_{K|r} = \mathsf{X}_\alpha^{a_K}(x_{K-r}) + \mathsf{Y}_\alpha^{a_K}(y_{K-r})$ is part of Theorem 28.8; we recall the argument, since it uses only finite-scale reconstructions. Lemma 26.2 gives the exact identity

$$(x_k, y_k) - (x_{h-1}, y_{h-1}) = \Big(\sum_{j=h}^{k} d_j,\ \sum_{j=h}^{k} \{ \mathcal{B}_{a_K}(x_{j-1}, d_j) + \mathcal{Q}_{a_K}(d_j, d_j) \} \Big), \qquad d_j = x_j - x_{j-1},$$

where $\mathcal{B}_a(u, v) = \mathcal{Q}_a(u, v) + \mathcal{Q}_a(v, u)$. By the estimate for the linear heat evolution in Theorem 28.6 and the two tail estimates in Theorem 28.7, this difference is small in probability as $h \to \infty$, uniformly in the remaining depth. For each fixed h, the arrays of the field reconstructed at scale $h-1$ range over a bounded finite-dimensional set, by the construction of the reconstruction maps and the cutoffs on the normalized fluctuation variables, and their cardinal interpolations have bounded derivatives of every fixed order. Using the estimates for cardinal interpolation and the heat semigroup in Lemmas 16.1 and 16.3, we bound x_{h-1} and its fixed differences on every finer lattice. The exact shifted quadratic term of Lemma 24.2, the contraction of the heat semigroup, and the Duhamel integral then bound y_{h-1} and its first difference by $C_h t$. Hence the norm of the flow data of the field reconstructed at scale $h-1$ has a deterministic bound C_h, independent of the finest mesh. We choose h so that the probability that the difference above exceeds one is less than a given $\varepsilon > 0$, and then a bound $R > C_h + 1$. The finitely many remaining depths have bounded arrays and are included by enlarging R. This proves (30.3) at the fixed r and γ. Restricting the horizon to the common early time interval does not increase the norm.

Step 2: the discrepancy. The flows are gauge equivariant and the energies and Wilson loops are gauge invariant. On the event $\{\mathsf{R}_{K|r} \leq R\}$, Lemma 30.14 implies that $D_K \leq \omega_R(a_K)$. The random variable D_K is Borel, since the finite flow and its dependence on c and on the initial configuration are continuous, so the supremum over (c, t) equals the supremum over a countable dense set. For every fixed $\eta > 0$ and all sufficiently large K, we have $\omega_R(a_K) < \eta$ and hence

$$\nu_{K|r}\{D_K > \eta\} \leq \nu_{K|r}\{\mathsf{R}_{K|r} > R\}.$$

Applying the last inequality in (30.2) to the event $\{D_K > \eta\}$, and using Theorem 23.9 with the coupling bound one, we obtain

$$\limsup_{K \to \infty} \mu_{N_K,\gamma}\{D_K > \eta\} \leq \sup_{K \geq r} \nu_{K|r}\{\mathsf{R}_{K|r} > R\} + F_\gamma(u_r), \qquad F_\gamma(u_r) \longrightarrow 0.$$

Letting first $R \to \infty$ at the fixed r and then $r \to \infty$ proves (30.17).

Step 3: convergence of the fields. The finite flow is smooth in (c, t) on its compact time interval, so each field is continuous. By Theorem 30.4, the path of the Wilson flow, $c = 0$, converges in distribution. Its image under the continuous map that sends a time path to the field that is constant in c therefore converges to the stated field. With the maximum norm on $\mathbb{R} \times \mathbb{C}^m$, the uniform distance between the field for the modified flows and this image is at most D_K. For a test f with bounded-Lipschitz norm, the difference of the two expectations is at most $\mathrm{Lip}(f)\eta + 2\|f\|_\infty \mu_{N_K,\gamma}\{D_K > \eta\}$, which proves convergence for such tests. For bounded continuous tests, we also need the compact containment of the paths of the Wilson flow. It follows from the bounds on the values and the Lipschitz bound in time of Lemma 30.3, with the same localization to $\{\mathsf{R}_{K|r} \le R\}$ and the same removal of the conditioning as in Step 2. By a finite covering argument, a bounded continuous test is uniformly continuous between the points of a compact set and all sufficiently nearby paths, and outside the compact set we use the bound on the test. This proves the convergence in distribution of the fields. Finally, the supremum of the loop discrepancy is bounded by twice the largest dimension of the representations, so its convergence to zero in probability implies the convergence of its expectation. □

Remark 30.16. The results concern parameters c in a bounded interval and fixed positive observation times. No assertion is made for unbounded c_K, for other modifications of the flow, for observation times tending to zero with the lattice spacing, or for different equivalence classes $[n]_M$, and no rate and no uniformity as $\gamma \to 0$ is asserted. Statements formulated only in terms of $\lambda_{L,\gamma,[n_*]}$, such as those in Sections 32 and 34 to 36, concern the same measure; the theorems above do not give new proofs of the estimates at finite cutoff on which those statements rest.

Proof of Theorems 2.5 and 2.6. The measure $\lambda = \lambda_{L,\gamma,[n]_M}$ and the maps π_k are those of Theorem 29.28. Part (1) of Theorem 2.5 is the assertion along the full sequence nM^k in Theorem 30.4, proved in Section 30.6. Part (2) is the assertion under the Wilson measures along nM^k in Theorem 30.5. Part (3) is Theorem 30.6, applied to $\lambda_{L,\gamma,[n]_M}$ as shown in Section 30.6; the winding vectors $w(\ell_j)$ of that theorem are the vectors $\nu(\ell_j)$ of Theorem 2.5. Part (4) is (30.4) in Theorem 30.1. Part (5) is Theorem 30.10. Items (1)–(3) of that theorem, together with its assertion along the full sequence, give the uniform convergence of finite vectors in the bounded-Lipschitz distance and the choice of maps independent of $\gamma \in I$ for which the convergence in Theorem 2.3(3) and the tightness in Theorem 2.3(4) hold uniformly. Item (4) gives the weak continuity of $\gamma \mapsto \lambda_{L,\gamma,[n]_M}$ on $(0, \infty)$ for both topologies. Theorem 2.6 is Theorems 30.13 and 30.15, along the sequence $N_K = n_* B^K$ of Theorem 29.25. □

PART X

Gaussian short-distance limits

SECTION 31

Reconstruction estimates and divergence-free currents

This section and Section 32 prove the short-distance theorem, Theorem 2.8. We study the heat-smoothed circulations defined in Definition 31.1: circulations, around small lattice loops of size ε, of the linear heat evolution of the reconstructed fine field, divided by $\sqrt{g_3^2\varepsilon}$. They are linear in the link logarithms of one fixed gauge representative; flowed Wilson loops are compared with them in Lemma 32.11.

The main result of this section is an exact decomposition of the heat-smoothed circulation $I^{K,s}$ under the conditioned Wilson measure, proved in Lemma 31.8:

$$I^{K,s} = T_s + \sum_{j=1}^{K-s} \Gamma_j(\widehat{\theta}_{j-1})(w_j - \bar{w}_j) + \mathsf{N}_{K,s}.$$

Here T_s is the contribution of the coarse field on the lattice of side n_s; the j-th summand is linear in the centered normalized fluctuation variables of the j-th RG step, so that the sum is a martingale; and $\mathsf{N}_{K,s}$ collects conditional means and nonlinear terms. For $a \le \varepsilon \le \varepsilon_0(s)$ and $\eta = g_3^2\varepsilon$, we show that

$$|T_s| + \mathbb{E}|\mathsf{N}_{K,s}| \le \frac{Cr}{g_3 H_s}\sqrt{\varepsilon} + Cg_3 D_s\sqrt{\varepsilon} + C\sqrt{\eta}\,[1+\log(1/\eta)]^2,$$

uniformly in the number K of RG steps, where $H_s = L/n_s$ is the coarse mesh and D_s is bounded uniformly in s. Theorem 31.6 identifies the covariance of the martingale increments at the zero field:

$$\sum_{j=1}^{K-s} \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}} = C^{a,\varepsilon,L} \otimes I_3 - C^{\mathrm{ret}}_{s,a,\varepsilon}.$$

In the linearized Gaussian theory, $C^{a,\varepsilon,L} \otimes I_3$ is the covariance of the circulation of the lattice Maxwell Gaussian field, and $C^{\mathrm{ret}}_{s,a,\varepsilon}$ is the covariance of its conditional expectation given the linearized coarse field. As $a \to 0$ and then $\varepsilon \to 0$, with s fixed, the right side converges to $\Sigma \otimes I_3$, where Σ is the covariance in (2.6). Section 32 uses these facts to show that the martingale is asymptotically Gaussian.

The normalization by $\sqrt{g_3^2\varepsilon}$ comes from the Gaussian Maxwell field. If A has covariance $g_3^2(-\Delta)^{-1}$ in three dimensions, then $e^{\sigma\varepsilon^2\Delta}A$ has typical size $g_3\varepsilon^{-1/2}$, and its circulation around a loop of length of order ε has typical size $g_3\sqrt{\varepsilon} = \sqrt{\eta}$.

Two ideas enter the proof of the decomposition. The first concerns the zero-fluctuation reconstruction, in which every fluctuation variable is set to zero. Let $\widehat{L}_i$ be the linearization at the zero field of the i-th zero-fluctuation reconstruction map; it takes link logarithms on the lattice of mesh a_{i-1} to link logarithms on the lattice of mesh a_i. Each $\widehat{L}_i$ is a constrained minimization of the Maxwell quadratic form. By Theorem 31.3, a product of them agrees, up to a periodic gradient, with a single constrained minimization over all the averaging steps at once:

$$a^{-1}\widehat{L}_k \cdots \widehat{L}_{j+1} v = a_j^{-1} E_j^a v + d_a f_j(v).$$

The circulation around a closed lattice loop annihilates the gradient exactly, and E_j^a obeys bounds that do not depend on $k - j$. The nonlinear Taylor remainders $\widehat{\mathcal{R}}_i$ of the reconstruction maps are inserted at their own scale and then propagated to the fine lattice by the linearizations:

$$\text{mesh } a_j \xrightarrow{\widehat{P}_{i-1}\circ\cdots\circ\widehat{P}_{j+1}} \text{mesh } a_{i-1} \xrightarrow{\widehat{\mathcal{R}}_i} \text{mesh } a_i \xrightarrow{\widehat{L}_k\cdots\widehat{L}_{i+1}} \text{mesh } a.$$

Converting link logarithms at mesh a_i into a potential at the scale a_j introduces the factor $a_j/a_i = B^{i-j}$. Since $\widehat{\mathcal{R}}_i$ vanishes to second order, every monomial contains at least two inputs, each carrying a contraction factor α_* for every preceding step. For $i-1-j$ preceding steps, the physical rescaling is therefore controlled by

$$\frac{a_j}{a_i}\,\alpha_*^{2(i-1-j)+1} = C_*\,(B\alpha_*^2)^{i-1-j}, \qquad B\alpha_*^2 \le \tfrac{1}{4}.$$

The propagated remainders are therefore summable. The result is an analytic field $\widehat{\mathcal{E}}_j^a$ on the fine lattice with the same circulations as the reconstruction, whose coefficients are bounded uniformly in the number of steps; we need no such bound for the composed gauge representative itself.

The second idea concerns conditional means. The conditional law of the fluctuation variables is interacting. Using its density, the local covariance estimates of Section 13, and global color-rotation symmetry, we show that the conditional means of the nonlinear terms are of order t_j^2 times the size of the coarse field. Their conditional variances are bounded independently of the number of fluctuation variables.

We keep four kinds of variables apart. The link logarithms $\widehat{\theta}_j$ on the lattice of mesh a_j are dimensionless; the potentials $\widehat{\theta}_j/a_j$ are in physical units; the normalized fluctuation variables w_j are distributed according to the conditional fluctuation kernel k_j of

Theorem 31.5, which is supported on a product of balls; and independent standard Gaussian coordinates occur only in the linearized calculation of Theorem 31.6. All circulations are computed for the same gauge representative and with the componentwise heat semigroup.

The inputs come from Sections 4, 8, 10 to 13, 15, 16, and 23. Section 31.1 fixes notation. Section 31.2 defines the circulations and computes their covariance under the lattice Maxwell Gaussian field. Section 31.3 compares linearized reconstructions and controls the nonlinear remainders. Section 31.4 estimates the coefficients and the conditional means. Section 31.5 proves the covariance identity, Section 31.6 proves the decomposition and sums over scales, and Section 31.7 records that all estimates hold at every fixed positive coupling. All results are used in Section 32.

31.1 Setting and conventions

Fix $L > 0$, an odd integer $M \geq 3$, and an upper bound $\Gamma > 0$ on the dimensionless coupling. Let J, $B = M^J$, the parameter tuple Θ, the reconstruction maps, and the minimal coarse side n_Γ be as in Theorem 23.5; they are chosen before the coupling. Fix $0 < \gamma \leq \Gamma$ and set $g_3^2 = \gamma / L$. Fix $n_* \in 4\mathbb{N}$ and the sequence of lattice sides $N_K = n_* B^K$. For a coarse index $s \geq 0$ with $n_s \geq n_\Gamma$ and for $K \geq s$, set

$$n_s = n_* B^s, \qquad H_s = L / n_s, \qquad k = K - s,$$

$$a_j = H_s B^{-j} \quad (0 \leq j \leq k), \qquad a = a_k = L / N_K,$$

$$d_j = J(k - j), \qquad g_j = g_3 \sqrt{a_j}, \qquad \lambda_B = M^{(J-1)/2}, \qquad t_j = \lambda_B g_j, \qquad p_j = A_{\mathrm{w}} \log(1/t_j).$$

Here a_j is the mesh of the lattice reached after j RG steps of the reconstruction, starting from the coarse lattice of side n_s. The integer d_j is the number of averaging steps between the fine lattice and the lattice of mesh a_j; it determines the effective classical action used at that scale. In the notation of Section 26, d_j is the depth $\mathfrak{d}_j$, not the increment $x_j - x_{j-1}$ of the heat path. The number t_j is the largest coupling among the J intermediate scales of the j-th RG step, and p_j is the fluctuation cutoff radius, with the amplitude A_{w} fixed in Θ. The couplings $g_{j,\ell}$ of the intermediate scales satisfy $g_{j,\ell} \leq t_j$, and the ratios $g_{j,\ell}/t_j$ depend only on M and J.

Let $\nu_{K,s} = \nu_{N_K, n_s, \gamma} = \mu_{N_K,\gamma}(\,\cdot \mid S_{N_K, n_s, \gamma})$ be the conditioned Wilson measure. Under $\nu_{K,s}$ the fine configuration is described by its backward reconstruction sequence (Sections 4 and 15). The coarse field $\widehat{\theta}_0$ is the array of link logarithms, on the lattice of side n_s, of the chosen representative of the coarse configuration; it satisfies $\|\widehat{\theta}_0\|_\infty \leq r$. For $1 \leq j \leq k$,

$$\widehat{\theta}_j = \widehat{\Phi}_j(\widehat{\theta}_{j-1}, g_{j,\cdot} w_j),$$

where $\widehat{\Phi}_j$ is the reconstruction map of Theorem 15.12 for the j-th RG step, at the classical recursion index d_j, the $w_j = (w_{j,\ell})_\ell$ are the normalized fluctuation variables of that step,

and $g_{j,\cdot} w_j$ is the vector with components $g_{j,\ell} w_{j,\ell}$. By Definition 4.9, the heat-smoothed circulation is $I^{K,s} = \mathcal{I}_a(\widehat{\theta}_k)$, with $\mathcal{I}_a$ as in Definition 31.1 below. We write

$$\mathcal{F}_j = \sigma(\widehat{\theta}_0, w_1, \dots, w_j), \qquad \bar{w}_j = \mathbb{E}[w_j \mid \mathcal{F}_{j-1}], \qquad b_j = \|\widehat{\theta}_j\|_\infty.$$

The zero-fluctuation reconstruction maps and their linear and nonlinear parts are

$$\widehat{P}_j(v) = \widehat{\Phi}_j(v, 0), \qquad \widehat{L}_j = D\widehat{P}_j(0), \qquad \widehat{\mathcal{R}}_j(v) = \widehat{P}_j(v) - \widehat{L}_j v,$$

and for $i \geq j$ we set $\widehat{\Pi}_{i \leftarrow j} = \widehat{P}_i \circ \cdots \circ \widehat{P}_{j+1}$, which is the identity if $i = j$. For $1 \leq j \leq k$, a field θ on the lattice of mesh a_{j-1}, and fluctuation variables w of the j-th step, define

$$G_j(\theta, w) = \mathcal{I}_a\big(\widehat{\Pi}_{k \leftarrow j} \widehat{\Phi}_j(\theta, g_{j,\cdot} w)\big), \qquad \Gamma_j(\theta) = D_w G_j(\theta, 0), \qquad \omega_j = \min\{a_j/\varepsilon, \sqrt{\varepsilon/a_j}\},$$

$$\mathsf{R}_j^{\mathrm{cur}} = G_j(\widehat{\theta}_{j-1}, w_j) - G_j(\widehat{\theta}_{j-1}, 0) - \Gamma_j(\widehat{\theta}_{j-1}) w_j.$$

The matrix $\Gamma_j(\widehat{\theta}_{j-1})$ is $\mathcal{F}_{j-1}$-measurable, and the circulation Taylor remainder $\mathsf{R}_j^{\mathrm{cur}}$ is the nonlinear part of the change in the circulation caused by the fluctuations of the j-th step.

Coefficient norms. We use the norms of Definition 4.6. For a matrix $T = (T_{eu})$ from link variables on a lattice of mesh h to link variables on a lattice of mesh a', the row norm is $\sup_e \sum_u |T_{eu}|$ and the volume-normalized column norm is $\sup_u (a'^3/h^3) \sum_e |T_{eu}|$; we write $\|T\|_{\mathrm{row+col}}$ for the larger of the two. For a tensor of higher degree, a column sum fixes one input index, called the distinguished input, sums over all outputs and all other inputs, and carries the same volume factor. Taylor coefficients are divided by the corresponding factorials, and every coefficient is weighted by $e^{a_{\mathrm{sp}} \mathsf{t}}$, where t is the length of a shortest connected network, with internal vertices allowed, joining the locations of its indices. For an analytic map F vanishing at the origin and a power series ϕ with nonnegative coefficients, we write $F \prec \phi$ if, for every $m \geq 1$, the row norm of the homogeneous degree-m part of F is at most the coefficient of z^m in ϕ, and each of its column norms with a distinguished input is at most the coefficient of z^{m-1} in ϕ'. These bounds on absolute values of coefficients are stronger than bounds on the values of F, and they allow compositions of arbitrary length below.

Parameters and constants. By Lemma 15.15 and Theorem 23.5, there exists a constant C_* bounding the first derivative of the zero-fluctuation reconstruction maps, fixed before J is chosen, such that

$$\alpha_* = C_*/B < 1, \qquad q = B\alpha_*^2 = C_*^2/B \leq \tfrac{1}{4}.$$

Further, there exist a constant β_J and radii r_{eff} and r, chosen in this order, with

$$\frac{\beta_J r_{\mathrm{eff}}}{1 - \alpha_*} < \frac{1}{8}, \qquad r < \frac{r_{\mathrm{eff}}}{4},$$

such that all reconstructed fields $\widehat{\theta}_j$ and all their zero-fluctuation continuations lie in the ball $\{v : \|v\|_\infty \le r_{\mathrm{eff}}\}$, on which the bounds of Lemma 15.15 hold; we call it the reconstruction domain. The deterministic coefficient bounds below hold on this domain. The conditional fluctuation kernels, and all statements about their expectations, concern coarse fields θ with $e^\theta \in B_r$, the small-link domain of Definition 4.4, together with the product-domain and support conditions of Theorem 13.8. On the support of $\nu_{K,s}$ these conditions hold at every step: the coarse configurations of all RG steps lie in D_r, and the thresholds in Theorem 15.12 keep the images of the reconstruction maps inside B_r, so that $e^{\widehat{\theta}_j} \in B_r$ for $0 \le j \le k$. We write E_{q} for the bound, fixed in Θ, on the weighted analytic norm of the input effective interaction, which is produced by the integrations at finer scales.

Unless stated otherwise, C and c denote positive constants that may depend on L, γ, M, J, Θ, n_*, the analytic domains of Theorem 23.5, the loops, and σ; this is the dependence described in item (5) of Definition 4.14. They do not depend on K, j, d_j, the point x, the lattice offset (defined below), or the coarse field. We say explicitly when a constant does not depend on s; otherwise it may depend on s through H_s and n_s. All estimates are on the periodic torus. In the limit statements $\gamma > 0$ and s are fixed, the number of steps K tends to infinity first, $\varepsilon \downarrow 0$ second, and no rate uniform as $\gamma \downarrow 0$ is claimed.

31.2 Divergence-free currents and the Maxwell covariance

We define the circulations and the limiting covariance Σ, compute the covariance of the circulation of the lattice Maxwell Gaussian field, and show that it converges to Σ when the lattice spacing and then the loop size tend to zero.

Definition 31.1. Fix $\sigma > 0$ and finitely many smooth closed loops $\ell_i : [0,1] \to \mathbb{R}^3$, $1 \le i \le m$. For $\varepsilon > 0$ put $\eta = g_3^2 \varepsilon$, and for $p \in \mathbb{R}^3$ define

$$\widehat{J}_i(p) = \int_0^1 \dot{\ell}_i(v) e^{-ip\cdot \ell_i(v)}\, dv, \qquad \Sigma_{ih} = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} \frac{e^{-2\sigma|p|^2}}{|p|^2}\, \overline{\widehat{J}_i(p)} \cdot \widehat{J}_h(p)\, dp.$$

Let $G = (G_i^c)_{1\le i\le m,\, 1\le c\le 3}$ be a centered Gaussian vector with covariance $\Sigma \otimes I_3$, write $G_i = (G_i^1, G_i^2, G_i^3)$, and set $\mathsf{Z}_i = |G_i|^2/8$. In Theorem 2.8 and Section 32 this vector is denoted C, and $(\mathsf{Z}_i)_i$ is denoted Z_G.

For a lattice mesh $a = L/N \le \varepsilon$ and $x \in \mathbb{T}_L^3$, let $\ell_{i,\varepsilon,a,x}$ be a closed oriented lattice cycle approximating the loop $x + \varepsilon \ell_i$ in $\mathbb{T}_L^3$, such that its physical length and diameter are at most $C_\ell \varepsilon$, it has zero winding when $\varepsilon \le \varepsilon_{\mathrm{loop}}$, and there is a based homotopy between it and $x + \varepsilon \ell_i$ whose swept area, counted with multiplicity, is at most $C_\ell \varepsilon a$. The constants C_ℓ and $\varepsilon_{\mathrm{loop}}$ depend only on the loops; they are uniform in x and in the choice of rounding

of the loop to the lattice, which we call the lattice offset. For a link potential A on the lattice of mesh a, in physical units, set

$$(\mathscr{C}_{a,\varepsilon,x}A)_i^c = \frac{a}{\sqrt{\eta}} \sum_{e \in \ell_{i,\varepsilon,a,x}} \operatorname{sgn}(e)\, (P^a_{\sigma\varepsilon^2}A)^c_{\operatorname{dir}(e)}(x_{\text{positive}}(e)), \qquad \mathcal{I}_a(v) = \mathscr{C}_{a,\varepsilon,x}(v/a),$$

for $1 \le i \le m$ and $1 \le c \le 3$. Here $P^a_t = e^{t\Delta_a}$ is the lattice heat semigroup, acting componentwise; $\operatorname{dir}(e)$ is the coordinate direction of e; $x_{\text{positive}}(e)$ is the starting vertex of the positively oriented link underlying e; and $\operatorname{sgn}(e) = -1$ if e is traversed against its positive orientation, so that a reversed link contributes minus the value on its positive orientation. The coefficient array $e \mapsto a \operatorname{sgn}(e)$ is a divergence-free lattice current, since the cycle is closed. We call $\mathscr{C}_{a,\varepsilon,x}A$ the heat-smoothed circulation of A, and $\mathcal{I}_a(v)$ that of the link logarithms v.

Finally, for $k \in (2\pi/L)\mathbb{Z}^3$ (a frequency) set

$$J_{i,\varepsilon,a,x}(k) = a \sum_{e \in \ell_{i,\varepsilon,a,x}} \operatorname{sgn}(e)\, e_{\operatorname{dir}(e)}\, e^{-ik\cdot x_{\text{positive}}(e)}, \qquad \lambda_a(k) = 4a^{-2} \sum_{r=1}^{3} \sin^2(ak_r/2),$$

$$C^{a,\varepsilon,L}_{ih} = \frac{1}{\varepsilon L^3} \sum_{k \neq 0} \frac{e^{-2\sigma\varepsilon^2\lambda_a(k)}}{\lambda_a(k)}\, \overline{J_{i,\varepsilon,a,x}(k)} \cdot J_{h,\varepsilon,a,x}(k),$$

where k ranges over representatives of $(2\pi/L)\mathbb{Z}^3$ modulo $(2\pi/a)\mathbb{Z}^3$ with $|ak_r| \le \pi$, and

$$C^{\varepsilon,L}_{ih} = \Big(\frac{\varepsilon}{L}\Big)^3 \sum_{k \in (2\pi/L)\mathbb{Z}^3 \setminus \{0\}} \frac{e^{-2\sigma|\varepsilon k|^2}}{|\varepsilon k|^2}\, \overline{\widehat{J}_i(\varepsilon k)} \cdot \widehat{J}_h(\varepsilon k).$$

The matrix Σ is the covariance of heat-smoothed circulations of a massless transverse field on $\mathbb{R}^3$: before smoothing, the Fourier covariance of that field is $|p|^{-2}(I - pp^{\mathsf{T}}/|p|^2)$ in each of three independent colors, and the transverse projection drops out against $\widehat{J}_i$ because $p \cdot \widehat{J}_i(p) = 0$.

Lemma 31.2. *In the notation of Definition 31.1, the following statements hold.*

1. *We have $p \cdot \widehat{J}_i(p) = 0$ and $|\widehat{J}_i(p)| \le C_\ell \min\{|p|, 1\}$. The integral defining Σ converges, and Σ is a real symmetric positive semidefinite matrix; it may be singular.*
2. *For $v_1, \dots, v_m \ge 0$, with $V = \operatorname{diag}(v_i)$,*

$$\mathbb{E} e^{-\sum_i v_i \mathsf{Z}_i} = \det\big(I + \tfrac{1}{4}\Sigma^{1/2} V \Sigma^{1/2}\big)^{-3/2},$$

$$\mathbb{E}\mathsf{Z}_i = \tfrac{3}{8}\Sigma_{ii}, \qquad \operatorname{Cov}(\mathsf{Z}_i, \mathsf{Z}_h) = \tfrac{3}{32}\Sigma^2_{ih}.$$

3. *The functional $\mathscr{C}_{a,\varepsilon,x}$ vanishes on every periodic lattice gradient. If $\varepsilon \le \varepsilon_{\text{loop}}$, it also vanishes on every constant field. For the initial potential $A^{K,s} = a_{N_K}^{-1}\widehat{\theta}_{K-s}$ of Definition 4.9, we have $I^{K,s} = \mathcal{I}_{a_{N_K}}(\widehat{\theta}_{K-s})$.*

4. *Let $\varepsilon \le \varepsilon_{\mathrm{loop}}$, and let A be the physical Maxwell Gaussian field on the lattice of mesh a, in three independent colors: the centered Gaussian field on link potentials modulo gradients and constants whose density is proportional to $\exp(-\|d\theta\|_{\ell^2}^2/(2g_3^2 a))$, where $\theta = aA$ and d is the dimensionless plaquette coboundary. Equivalently, its covariance is $g_3^2(-\Delta_a)^{+}P_{\mathrm{tr},a}$, where $P_{\mathrm{tr},a}$ is the projection onto lattice divergence-free fields and $(-\Delta_a)^{+}$ is the inverse of $-\Delta_a$ on fields with zero mean. Then $\mathbb{E}[(\mathscr{C}_{a,\varepsilon,x}A)_i^c(\mathscr{C}_{a,\varepsilon,x}A)_h^{c'}] = \delta_{cc'}C_{ih}^{a,\varepsilon,L}$. The longitudinal part of the covariance drops out by the exact closedness of the cycles, before any limit is taken.*
5. *For each fixed $\varepsilon \in (0, \varepsilon_{\mathrm{loop}}]$, $C^{a,\varepsilon,L} \to C^{\varepsilon,L}$ as $a \downarrow 0$, uniformly in x and the lattice offset. Further, $C^{\varepsilon,L} \to \Sigma$ as $\varepsilon \downarrow 0$.*

The constants in these statements depend only on the loops, L, σ, C_ℓ, and $\varepsilon_{\mathrm{loop}}$. The covariances $C^{a,\varepsilon,L}$, $C^{\varepsilon,L}$, and Σ do not depend on g_3.

Proof. We verify the statements in order.

Step 1: the continuum current. Let $l_i = \int_0^1 |\dot\ell_i(v)|\,dv$, and let y_i be a point on the i-th loop. Integrating the derivative of $v \mapsto e^{-ip\cdot\ell_i(v)}$ over the closed loop, we find $p \cdot \widehat{J}_i(p) = 0$. Since $\int_0^1 \dot\ell_i = 0$, we also have

$$\widehat{J}_i(p) = \int_0^1 \dot\ell_i(v)\big[e^{-ip\cdot\ell_i(v)} - e^{-ip\cdot y_i}\big]\,dv, \qquad |\widehat{J}_i(p)| \le l_i \min\{2, |p|\operatorname{diam}\ell_i\}.$$

The integrand defining Σ is therefore bounded near zero and bounded by $C|p|^{-2}e^{-2\sigma|p|^2}$ for $|p| \ge 1$; both bounds are integrable in three dimensions. For real numbers u_i, the quadratic form of Σ is

$$(2\pi)^{-3}\int_{\mathbb{R}^3} \frac{e^{-2\sigma|p|^2}}{|p|^2}\Big|\sum_i u_i \widehat{J}_i(p)\Big|^2\,dp \ge 0.$$

The substitution $p \mapsto -p$, together with $\widehat{J}_i(-p) = \overline{\widehat{J}_i(p)}$, shows that the entries are real, and interchanging i and h shows that Σ is symmetric.

Step 2: the Gaussian comparison vector. Let Z^1, Z^2, Z^3 be independent standard Gaussian vectors in $\mathbb{R}^m$ and set $G^c = \Sigma^{1/2}Z^c$; this constructs G also when Σ is singular. Then $\sum_i v_i \mathsf{Z}_i = (1/8)\sum_c (Z^c)^{\mathsf{T}}\Sigma^{1/2}V\Sigma^{1/2}Z^c$. Diagonalizing the positive semidefinite matrix $\Sigma^{1/2}V\Sigma^{1/2}$ by an orthogonal matrix reduces the Laplace transform to a product, over its eigenvalues λ and the three colors, of

$$(2\pi)^{-1/2}\int_{\mathbb{R}} e^{-(1+\lambda/4)z^2/2}\,dz = (1+\lambda/4)^{-1/2},$$

which gives the determinant formula. For a centered Gaussian pair (X, Y) we have $\mathbb{E}(X^2Y^2) = \mathbb{E}X^2\,\mathbb{E}Y^2 + 2(\mathbb{E}XY)^2$; this is Isserlis' formula [112]. Applying it to the coordinates of G, we obtain $\mathbb{E}\mathsf{Z}_i = (3/8)\Sigma_{ii}$ and $\operatorname{Cov}(\mathsf{Z}_i, \mathsf{Z}_h) = (1/64)\sum_c 2\Sigma_{ih}^2 = (3/32)\Sigma_{ih}^2$.

Step 3: gradients and constants. For a vertex function f, let $d_a f$ be the field whose value on the positive link from y to $y + ae_r$ is $(f(y + ae_r) - f(y))/a$. The heat semigroup commutes with d_a, since the lattice Laplacian is a sum of translations, which commute. The oriented sum of $a(P_t^a d_a f)_{\mathrm{dir}(e)}(x_{\mathrm{positive}}(e))$ around the lattice cycle telescopes to zero. The heat semigroup fixes constant fields, and the sum of the oriented displacement vectors $a\,\mathrm{sgn}(e)e_{\mathrm{dir}(e)}$ vanishes for a cycle of zero winding; hence the current also annihilates constants. Both assertions use the convention that a reversed link is evaluated on its positive orientation, at the starting vertex of that positive link. The identity $I^{K,s} = \mathcal{I}_{a_{N_K}}(\widehat{\theta}_{K-s})$ is the substitution $A = a_{N_K}^{-1}\widehat{\theta}_{K-s}$ in Definition 4.9; the gauge representative is not changed.

Step 4: the lattice covariance. We take the representatives k with $|ak_r| \le \pi$ and define

$$d_r(k) = a^{-1}(e^{iak_r} - 1), \qquad \text{so that} \qquad \lambda_a(k) = \sum_r |d_r(k)|^2.$$

The Fourier transform of the boundary of the cycle is zero, which is the identity

$$\sum_r \overline{d_r(k)}\, J_{i,\varepsilon,a,x,r}(k) = 0 \tag{31.1}$$

at every frequency, with repeated links counted with multiplicity. Indeed, the left side equals $\sum_e \mathrm{sgn}(e)(e^{-ik\cdot y_+(e)} - e^{-ik\cdot y_-(e)})$, where $y_\pm(e)$ are the endpoints of the positive link underlying e, and this sum telescopes around the closed cycle. At $k = 0$, zero winding implies $J_i(0) = 0$. On each nonzero frequency, the algebraic identity

$$\sum_{r<q} |d_r A_q - d_q A_r|^2 = \lambda_a |A|^2 - \Big|\sum_r \overline{d_r} A_r\Big|^2$$

identifies the inverse of the Maxwell quadratic form, restricted to transverse fields, as $\lambda_a^{-1} P_{\mathrm{tr},a}$. This normalization agrees with the Wilson normalization of Theorem 8.3: writing $\theta = aA$, the quadratic exponent is

$$\frac{\|d\theta\|_{\ell^2}^2}{2g_3^2 a} = \frac{a^3}{2g_3^2} \sum_y \sum_{r<q} |(d_a A)_{rq}(y)|^2,$$

where d on the left is dimensionless and d_a on the right is in physical units. With $\widehat{A}(k) = a^3 \sum_y e^{-ik\cdot y} A(y)$, the covariance of the Fourier coefficients is $g_3^2 L^3 \mathbf{1}_{k'=-k} \lambda_a(k)^{-1} P_{\mathrm{tr},a}(k)$. Pairing twice with the cycle, applying the two heat factors $e^{-\sigma\varepsilon^2\lambda_a(k)}$, and dividing by η, we obtain the covariance

$$\frac{g_3^2}{\eta L^3} \sum_{k\neq 0} \frac{e^{-2\sigma\varepsilon^2\lambda_a(k)}}{\lambda_a(k)}\, \overline{J_i(k)} \cdot J_h(k),$$

in each color separately. Here (31.1) removed the projection $P_{\mathrm{tr},a}$. Since $g_3^2/\eta = 1/\varepsilon$, this is the formula for $C^{a,\varepsilon,L}$.

Step 5: the limit $a \downarrow 0$. We lift the small cycle and the loop $x + \varepsilon\ell_i$ to a ball of diameter $C\varepsilon$ in $\mathbb{R}^3$. For a C^1 one-form f, replacing its integral over each oriented link by the value at the starting vertex of the positive link changes the integral by at most $a^2\|Df\|_\infty/2$ per link; summing over at most $C_\ell\varepsilon/a$ links gives an error at most $C\varepsilon a\|Df\|_\infty$. For a piecewise smooth based homotopy $H(u,v)$ between the polygon and the loop, differentiating $\int f(H)\partial_v H\,dv$ in u and integrating by parts in v gives

$$\oint_{\text{polygon}} f - \oint_{x+\varepsilon\ell_i} f = \int_0^1\int_0^1 df(H)(\partial_u H, \partial_v H)\,du\,dv.$$

The endpoint terms cancel because the paths are closed and the homotopy is based; for a piecewise smooth homotopy the formula holds piece by piece, and the internal boundaries cancel. The swept area bounds this difference by $C\varepsilon a\|df\|_\infty$. Applying both estimates to $f(y) = e^{-ik\cdot y}dy_r$, and using $\oint_{x+\varepsilon\ell_i} e^{-ik\cdot y}dy = \varepsilon e^{-ik\cdot x}\widehat{J_i}(\varepsilon k)$, we obtain for each fixed ε and k

$$\left|J_{i,\varepsilon,a,x}(k) - \varepsilon e^{-ik\cdot x}\widehat{J_i}(\varepsilon k)\right| \le C\varepsilon a|k|, \tag{31.2}$$

uniformly in x and the lattice offset; a rounding of the common base point is included in the same estimate. Subtracting from each exponential in the definition of J_i its value at one vertex of the cycle, which is allowed since $J_i(0) = 0$, and using the bounds on length and diameter, we find

$$|J_i(k)| \le C\varepsilon\min\{1, \varepsilon|k|\}, \qquad \frac{4}{\pi^2}|k|^2 \le \lambda_a(k) \le |k|^2. \tag{31.3}$$

The second assertion follows in each coordinate from $2|u|/\pi \le |\sin u| \le |u|$ for $|u| \le \pi/2$. For fixed $\varepsilon > 0$, we extend the summand defining $C^{a,\varepsilon,L}$ by zero outside the chosen representatives. By (31.3) its absolute value is bounded by a summable Gaussian function of k on the fixed reciprocal lattice $(2\pi/L)\mathbb{Z}^3$, uniformly in a, x, and the offset. Given a tolerance, we first truncate this Gaussian tail. On the finitely many remaining frequencies, (31.2) and $\lambda_a(k) \to |k|^2$ give uniform convergence. Substituting $J_i(k) = \varepsilon e^{-ik\cdot x}\widehat{J_i}(\varepsilon k)$, the two translation phases cancel and the prefactor becomes $\varepsilon^2/(\varepsilon L^3|k|^2) = (\varepsilon/L)^3|\varepsilon k|^{-2}$. This proves $C^{a,\varepsilon,L} \to C^{\varepsilon,L}$.

Step 6: the limit $\varepsilon \downarrow 0$. Let $\delta = 2\pi\varepsilon/L$ and, for $p \neq 0$, define

$$f_{ih}(p) = \frac{e^{-2\sigma|p|^2}}{|p|^2}\,\overline{\widehat{J_i}(p)}\cdot\widehat{J_h}(p).$$

By Step 1, f_{ih} is bounded near zero, smooth away from zero, and has a Gaussian tail. We have $C_{ih}^{\varepsilon,L} = (2\pi)^{-3}\delta^3\sum_{v\in\mathbb{Z}^3\setminus\{0\}} f_{ih}(\delta v)$. For $r > 0$, the contribution of the points with $|\delta v| < r$ is at most $C(r+\delta)^3$, as one sees by counting the disjoint cubes of side δ centered at these points inside the ball of radius $r + \sqrt{3}\delta$; the corresponding integral is at most

Cr^3. On a fixed annulus $r \le |p| \le R$, uniform continuity bounds the difference between the integral over each cube and the sampled value by the modulus of continuity at $\sqrt{3}\delta$, and the boundary cubes have total volume tending to zero. The tails are uniform for $0 < \delta \le 1$: for $R \ge 2$, the absolute sum over the shell $2^q R \le |p| < 2^{q+1}R$ is at most $C(2^{q+1}R+1)^3 e^{-c_\sigma (2^q R)^2}$, and these bounds are summable in q with sum tending to zero as $R \to \infty$. Letting $\delta \downarrow 0$, then $r \downarrow 0$, and then $R \to \infty$, we obtain $C_{ih}^{\varepsilon,L} \to (2\pi)^{-3} \int f_{ih} = \Sigma_{ih}$. The constants used depend only on the stated data, and g_3 appeared only through $g_3^2/\eta = 1/\varepsilon$. □

31.3 Linearized reconstruction and nonlinear error terms

This section constructs the field $\widehat{\mathcal{E}}_j^a$ described at the beginning of the section and proves the deterministic bounds that convert coefficient bounds into bounds for circulations. The linear comparison follows the constrained Gaussian calculations for Bałaban's averaging operation [10, Section A, (1.1)–(1.8), pp. 18–19, and Section D, (1.47)–(1.49), p. 26, and (1.64)–(1.67), p. 29]; the averaging map, the effective actions, and the gauge representatives used here are those of Sections 8, 10, 11, and 15, and the comparison is proved below for them.

Theorem 31.3. *Assume the setting of Section 31.1 and the loop data of Definition 31.1. Fix a coarse index s with $n_s \ge n_\Gamma$ and $K \ge s$. Then the following statements hold.*

1. *On the larger analytic domain fixed in Theorem 23.5, the zero-fluctuation reconstruction maps satisfy*

$$\widehat{P}_j \prec \phi(z) = \frac{\alpha_* z}{1-\beta_J z}, \qquad \big(v \mapsto \widehat{P}_j(v) - D\widehat{P}_j(0)v\big) \prec \frac{\alpha_* \beta_J z^2}{1-\beta_J z},$$

   with output rows, every volume-normalized column with a distinguished input, and a fixed spatial weight. The constant β_J is chosen on that domain before r_{eff} and r.

2. *There exist deterministic matrices E_j^a, $0 \le j \le k$, with $E_k^a = I$, each a scalar multiple of the identity on the color factor, with rows and volume-normalized columns (with factor a^3/a_j^3) bounded by a constant, and with a fixed positive spatial decay rate measured in units of a_j, such that*

$$a^{-1}\widehat{L}_k \cdots \widehat{L}_{j+1} v - a_j^{-1} E_j^a v = d_a f_j(v)$$

   for a periodic vertex function $f_j(v)$ on the fine lattice depending linearly on v. The difference has no constant part, because the block averaging constraint fixes the componentwise means in physical units. These bounds do not depend on $k - j$.

3. *Let T be a matrix from link variables on a lattice of mesh h to link variables on the fine lattice, with row norm and volume-normalized column norm at most A_0, and with at most a fixed number of input coordinates per cell of side h. Then for $0 \le t \le \sigma\varepsilon_{\mathrm{loop}}^2$,*

$$\sup_y \|\operatorname{row}_y(P_t^a T)\|_{\ell^2} \le C A_0 \min\{1, h^{3/2}(t+a^2)^{-3/4}\}, \qquad \|P_t^a T\|_{\ell^\infty \to \ell^\infty} \le A_0.$$

Consequently, for $a \le \varepsilon \le \varepsilon_{\mathrm{loop}}$,

$$\left\| \mathscr{C}_{a,\varepsilon,x} \frac{g_3\sqrt{h}}{h} T \right\|_{\ell^2 \to \mathbb{R}^{3m}} \le C A_0 \min\{h/\varepsilon, \sqrt{\varepsilon/h}\},$$

$$\left\| \mathscr{C}_{a,\varepsilon,x} \frac{g_3\sqrt{h}}{h} T \right\|_{\ell^\infty \to \mathbb{R}^{3m}} \le C A_0 \sqrt{\varepsilon/h},$$

where ℓ^2 and ℓ^∞ refer to the counting norms on the inputs.

4. *For $i > j$ put $r' = i - 1 - j$ and $D = \beta_J/(1-\alpha_*)$. Then*

$$\big(v \mapsto \widehat{\mathcal{R}}_i(\widehat{\Pi}_{i-1\leftarrow j}(v))\big) \prec C\alpha_*^{2r'+1} \frac{z^2}{(1-Dz)^2},$$

for rows and for columns with the normalization a_i^3/a_j^3. Derivatives with a distinguished input are bounded by derivatives of the same series, with the same spatial decay rate.

5. *Define the field on the fine lattice*

$$\widehat{\mathcal{E}}_j^a(v) = E_j^a v + \sum_{i=j+1}^{k} \frac{a_j}{a_i} E_i^a \, \widehat{\mathcal{R}}_i\big(\widehat{\Pi}_{i-1\leftarrow j}(v)\big).$$

For $\|v\|_\infty \le r_{\mathrm{eff}}$, it is equivariant under global color rotations, vanishes at $v = 0$, and satisfies

$$\mathcal{I}_a\big(\widehat{\Pi}_{k\leftarrow j}(v)\big) = \mathscr{C}_{a,\varepsilon,x}\big(a_j^{-1}\widehat{\mathcal{E}}_j^a(v)\big).$$

Its rows and its columns with normalization a^3/a_j^3 are bounded by a power series independent of k, j, and d_j. In particular, on the reconstruction domain,

$$\|D\widehat{\mathcal{E}}_j^a(v)\|_{\mathrm{row+col}} \le C, \qquad \|D\widehat{\mathcal{E}}_j^a(v) - D\widehat{\mathcal{E}}_j^a(0)\|_{\mathrm{row+col}} \le C\|v\|_\infty,$$

and every fixed higher derivative obeys the corresponding bound. The conversion to physical units is governed by the identities

$$\frac{a_j}{a_i}\alpha_*^{2r'+1} = B^{r'+1}\alpha_*^{2r'+1} = C_* q^{r'}, \qquad \frac{a^3}{a_i^3}\cdot\frac{a_i^3}{a_j^3} = \frac{a^3}{a_j^3}.$$

The constants in the coefficient bounds depend only on M, J, Θ, the analytic domains and their ratios, the spatial decay rates, and the number of coordinates per cell; the constants in the heat and circulation bounds may in addition depend on L, σ, C_ℓ, and $\varepsilon_{\mathrm{loop}}$. None depends on K, s, j, d_j, or the coarse field in the reconstruction domain.

Remark 31.4. The identity in part (5) is an identity between circulations. It does not say that the composed nonlinear reconstruction $\widehat{\Pi}_{k\leftarrow j}$ coincides with a single reconstruction over all intermediate scales, and no inverse of the Maxwell operator on the coarse lattice is used.

Proof. The proof has six steps. Step 1 recalls the analytic input. Step 2 proves the linear comparison of part (2), which is the source of the exact cancellation of gradients. Step 3 proves the heat and circulation bounds of part (3). Steps 4 and 5 show that coefficient bounds survive compositions of arbitrary length and prove part (4). Step 6 assembles the field $\widehat{\mathcal{E}}_j^a$.

Step 1: the analytic input. The coefficient bound on the larger domain and the order in which the constants are chosen are given by Lemma 15.15. The identification of C_* with the constant C_{CP} of that lemma and the choice of J are part of Theorem 23.5; in particular we use $q \le 1/4$, and no constant of an all-order bound at fixed J is used to choose J. Restricting the joint coefficient series of Lemma 15.15 to zero fluctuation variables gives the two majorants of part (1), with the stated output and distinguished-input conventions. Its domain inclusions hold on the reconstruction domain.

Step 2: the linear comparison. For $j < k$ we apply Lemma 11.9 once, with $J(k-j)$ averaging steps, preceding classical recursion index zero, coarse mesh a_j, and relative fine spacing $u = a/a_j$. The logarithm of the representative given by that lemma is $u\overline{A}_u(v)$, and we define

$$E_j^a = D\overline{A}_u(0).$$

By the first-derivative clause of that lemma, we have the row bound and the column factor $u^3 = a^3/a_j^3$, with one positive spatial decay rate in units of a_j, uniformly in the number $J(k-j)$ of averaging steps. We use no bound on the dependence on fluctuation variables at this growing number of steps. By global color equivariance, each 3×3 color block of E_j^a commutes with every rotation. Commuting with the rotations by π about the coordinate axes makes the block diagonal, and commuting with the rotations that permute the axes makes its three diagonal entries equal. Hence E_j^a is a scalar multiple of the identity on the color factor. We set $E_k^a = I$.

We now show that uE_j^a represents the composed linearization for circulations. First let v be real; all minimizations and comparisons of nonnegative quadratic forms in this paragraph are on real tangent spaces. By the quadratic part of the nested minimization

identity of Theorem 10.10, each linearized reconstruction minimizes the inherited nonnegative Maxwell quadratic form over the free coordinates, on which this form is positive definite. If a quadratic form $Q(x,y,z)$ has positive blocks in the constrained variables y and z, then

$$\inf_z \inf_y Q(x,y,z) = \inf_{(y,z)} Q(x,y,z).$$

Indeed, every value of the problem on the right bounds the left side from above, and for each fixed z the inner minimizer realizes the infimum over y. Completing the squares in the positive blocks shows that both infima are attained at the classical tangents. We iterate this equality over the finitely many averaging steps. Since the indices $d_j = J(k-j)$ are compatible, the inherited forms are exactly the successive Schur complements. Thus the composed tangent $\widehat{L}_k \cdots \widehat{L}_{j+1} v$ and the tangent $uE_j^a v$ minimize the same quadratic form on the fine lattice subject to the same linearized averaging constraint v. The gauge transformations of Theorem 15.12 depend only on the coarse field and equal the identity at the tree roots; they preserve the constraint and the gauge orbit of the minimizer. Their derivatives at zero add periodic gradients, which do not affect this conclusion.

Let X and Y be these two minimizers, and let Q_0 be the Maxwell quadratic form. Since the constraint is linear, the midpoint satisfies it. By the parallelogram identity, we have

$$Q_0\Big(\frac{X+Y}{2}\Big) = \frac{1}{2}Q_0(X) + \frac{1}{2}Q_0(Y) - \frac{1}{4}Q_0(X-Y).$$

Minimality and nonnegativity imply $Q_0(X-Y) = 0$; in particular $d(X-Y) = 0$. We determine the kernel. Let $Z = X - Y$ and fix a nonzero lattice frequency, with $b_r = e^{iak_r} - 1$. Zero curl says $b_r \widehat{Z}_q = b_q \widehat{Z}_r$. Choosing an index r with $b_r \neq 0$, we have $\widehat{Z}_q = b_q(\widehat{Z}_r / b_r)$ for every q; that is, every nonzero-frequency part of Z is a periodic gradient. The zero-frequency part is a constant field, and its componentwise means in physical units are its constant components.

This constant vanishes for $X - Y$. By Lemma 8.1, the derivative of the composed block averaging map is the box and segment average plus a periodic gradient on the coarse lattice, and its version in physical units preserves each componentwise mean. Explicitly, if $m' = a_j/a$ is the total blocking factor and $K_{m'}$ the box and segment average, the box and segment formula counts each fine link m' times before its factor m'^{-3}, so that

$$\sum_Y (K_{m'} Z)_r(Y) = m'^{-2} \sum_y Z_r(y), \qquad (m'a)^3 \sum_Y \frac{(K_{m'} Z)_r(Y)}{m'a} = a^3 \sum_y \frac{Z_r(y)}{a}.$$

The coarse gradient has mean zero. Since $X - Y$ satisfies the homogeneous constraint, its mean in physical units is zero. The same argument applies to coarse fields with nonzero constant components, which are fixed by the constraint. We conclude that $a^{-1}(X-Y) = d_a f_j(v)$ for a periodic vertex function, chosen with mean zero, which depends linearly on v by the frequency formula above. This proves part (2), and by

Lemma 31.2 the circulation annihilates the gradient. The maps in this identity are real linear; their complex-linear extensions give the same identity for complex v.

Step 3: heat and circulation bounds. Let $R(T)$ and $C(T)$ be the row norm and the volume-normalized column norm of T. For a vector z of inputs, by the Cauchy–Schwarz inequality with the weights $|T_{eu}|$, we have

$$\sum_e \Big|\sum_u T_{eu} z_u\Big|^2 \le R(T) \sum_{e,u} |T_{eu}||z_u|^2 \le R(T)C(T)(h/a)^3 \sum_u |z_u|^2,$$

since $\sum_e |T_{eu}| \le (h/a)^3 C(T)$. Thus $\|T\|_{\ell^2\to\ell^2} \le A_0(h/a)^{3/2}$. We use Lemma 16.1 with the fixed horizon $T_{\mathrm{heat}} = \sigma\varepsilon_{\mathrm{loop}}^2$ and difference order zero. Its hypotheses hold on the periodic fine lattice of mesh $a = L/N_K$, with $N_K \ge 4$, and all times considered lie in $[0, T_{\mathrm{heat}}]$. Since the heat kernel of that lemma is a density with respect to $a^3\sum_y$, the row of P_t^a at y, as a vector in the counting norm, is $a^3 p_t^a(y,\cdot)$, and its ℓ^2 norm is at most $Ca^{3/2}(t+a^2)^{-3/4}$. Applying this row to T, we obtain

$$\|\operatorname{row}_y(P_t^a T)\|_{\ell^2} \le \|a^3 p_t^a(y,\cdot)\|_{\ell^2}\|T\|_{\ell^2\to\ell^2} \le CA_0 h^{3/2}(t+a^2)^{-3/4}.$$

Since the heat kernel is positive with unit mass, the absolute row sums of $P_t^a T$ are at most A_0; this bounds both the ℓ^2 norm of each row and the $\ell^\infty \to \ell^\infty$ norm, including at $t = 0$. The finitely many directions and input types are absorbed into C. Since the horizon is fixed, the constants depend on its ratio to L^2, and the constant mode of the heat semigroup is included.

For $\varepsilon \le \varepsilon_{\mathrm{loop}}$ the time $t = \sigma\varepsilon^2$ lies in $[0, T_{\mathrm{heat}}]$, and $a \le \varepsilon$ implies $(t+a^2)^{-3/4} \le C_\sigma \varepsilon^{-3/2}$. The circulation sums the rows over at most $C_\ell \varepsilon/a$ links with the weight $a/\sqrt{\eta}$, so the prefactor multiplying a single row is

$$\frac{\varepsilon}{\sqrt{\eta}} \cdot \frac{g_3\sqrt{h}}{h} = \sqrt{\varepsilon/h}.$$

Multiplying it by $\min\{1, C(h/\varepsilon)^{3/2}\}$ proves the ℓ^2 circulation bound, and multiplying it by the absolute row bound proves the ℓ^∞ bound. Summing over the fixed number of loop and color components gives a fixed factor.

Step 4: compositions of coefficient bounds. In the composition of two tensors, fixing an output and summing over all inputs uses the row bound of the outer tensor and a row bound of the inner tensor at every input slot of the outer tensor. If instead we fix one input of the inner tensor, we follow the chain of factors through which this input enters. Along this chain we use the column bound of the inner tensor and the column bound of the outer tensor with the corresponding distinguished input; all other factors use rows. For example, let B be a bilinear tensor from mesh a_{i-1} to mesh a_i and G a linear tensor from

mesh a_j to mesh a_{i-1}, and consider $(B(Gv, Gv))_e = \sum_{y,y',u,u'} B_{e;yy'} G_{yu} G_{y'u'} v_u v_{u'}$. Fixing u in the first slot, we have

$$\sum_{e,u'} \Big| \sum_{y,y'} B_{e;yy'} G_{yu} G_{y'u'} \Big| \le \sum_{y} |G_{yu}| \sum_{e,y'} |B_{e;yy'}| \sum_{u'} |G_{y'u'}| \le \frac{a_j^3}{a_{i-1}^3} C(G) \cdot \frac{a_{i-1}^3}{a_i^3} C_1(B) \cdot R(G),$$

where $C_1(B)$ is the column norm of B with its first input distinguished. After multiplication by the volume factor a_i^3 / a_j^3, the volume factors cancel:

$$\frac{a_i^3}{a_{i-1}^3} \cdot \frac{a_{i-1}^3}{a_j^3} = \frac{a_i^3}{a_j^3}.$$

In general there is exactly one chain of factors carrying the fixed input; summing over its possible positions corresponds to differentiating the scalar power series. The Taylor coefficients $D^n / n!$ distribute the input indices among the input slots of the outer tensor with exactly the multiplicities of scalar substitution: ordering the slots first and then dividing by their factorials counts each partition once. We conclude that composition of the scalar majorants majorizes these tensor contractions, also with a distinguished input, and that no constant is lost at successive levels.

The spatial weights are treated in the same contraction. We take the union of the connecting networks of the participating coefficients, joined at the contracted indices. It is a connected network joining all the external indices, so its length is at least their minimal network length. Every later input mesh is at most a_j, so the weight of each part of the network at its own scale dominates its weight in units of a_j. Multiplying the weights bounds the weight of the union with the initially fixed decay rate. This holds for compositions of arbitrary length, without a reduction of the decay rate at each step.

Step 5: iterated majorants and the remainder bound. Let $S_r = \sum_{v=0}^{r-1} \alpha_*^v$, with $S_0 = 0$. By induction, using the formula for ϕ, we have the identity

$$\phi^{\circ r}(z) = \frac{\alpha_*^r z}{1 - \beta_J S_r z}. \tag{31.4}$$

For the inductive step, we substitute the right side into $\alpha_* y / (1 - \beta_J y)$ and use $S_{r+1} = S_r + \alpha_*^r$. By Step 4, $\widehat{\Pi}_{i-1 \leftarrow j} \prec \phi^{\circ r'}$ with $r' = i - 1 - j$, and the remainder satisfies, coefficientwise,

$$\begin{aligned} \widehat{\mathcal{R}}_i \circ \widehat{\Pi}_{i-1 \leftarrow j} \prec \frac{\alpha_* \beta_J (\phi^{\circ r'}(z))^2}{1 - \beta_J \phi^{\circ r'}(z)} &= \frac{\beta_J \alpha_*^{2r'+1} z^2}{(1 - \beta_J S_{r'} z)(1 - \beta_J S_{r'+1} z)} \\ &\preceq \frac{\beta_J \alpha_*^{2r'+1} z^2}{(1 - Dz)^2}, \qquad D = \frac{\beta_J}{1 - \alpha_*}. \end{aligned} \tag{31.5}$$

Here $\preceq$ denotes coefficientwise inequality of power series. In the equality we used $1-\beta_J\phi^{\circ r'}(z) = (1-\beta_J S_{r'+1}z)/(1-\beta_J S_{r'}z)$, and the coefficientwise inequality holds because $\beta_J S_r \le D$ for every r. This proves the row, column, and distinguished-derivative assertions of part (4), with $C=\beta_J$, enlarged once for the fixed convention on components. The bound $Dr_{\mathrm{eff}} < 1/8$ makes all these substitutions convergent on the reconstruction domain. The two inputs that every monomial of the remainder contains each contribute a factor $\alpha_*^{r'}$; this is the origin of the exponent $2r'+1$.

Step 6: the field $\widehat{\mathcal{E}}_j^a$. For a finite composition we have the identity

$$\widehat{\Pi}_{k\leftarrow j}(v) = \widehat{L}_k\cdots\widehat{L}_{j+1}v + \sum_{i=j+1}^{k} \widehat{L}_k\cdots\widehat{L}_{i+1}\,\widehat{\mathcal{R}}_i\big(\widehat{\Pi}_{i-1\leftarrow j}(v)\big), \tag{31.6}$$

which follows by induction on $k-j$: to pass from $k-1$ to k, apply $\widehat{L}_k$ to the identity at $k-1$ and add $\widehat{\mathcal{R}}_k(\widehat{\Pi}_{k-1\leftarrow j}(v))$. We apply the linear functional $\mathcal{I}_a$ to (31.6) and use the linear comparison of Step 2 on each summand, with the remainder $\widehat{\mathcal{R}}_i(\widehat{\Pi}_{i-1\leftarrow j}(v))$ as input at scale i. Since the circulation annihilates the gradients, we obtain

$$\mathcal{I}_a\big(\widehat{\Pi}_{k\leftarrow j}(v)\big) = \mathscr{C}_{a,\varepsilon,x}\Big(a_j^{-1}E_j^a v + \sum_{i=j+1}^{k} a_i^{-1}E_i^a\,\widehat{\mathcal{R}}_i\big(\widehat{\Pi}_{i-1\leftarrow j}(v)\big)\Big),$$

which is the identity of part (5). Each remainder is equivariant and vanishes at zero, and each E_i^a is a scalar multiple of the identity on the color factor; hence $\widehat{\mathcal{E}}_j^a$ has both properties.

Let C_E be a common bound for the rows and volume-normalized columns of all E_i^a. Composing E_i^a with a remainder multiplies its column normalization a_i^3/a_j^3 by a^3/a_i^3, which gives a^3/a_j^3. The factor a_j/a_i in the definition of $\widehat{\mathcal{E}}_j^a$ combines with (31.5) to give

$$\frac{a_j}{a_i}\alpha_*^{2r'+1} = B^{r'+1}\alpha_*^{2r'+1} = C_* q^{r'}.$$

Summing over i, a common majorant for $\widehat{\mathcal{E}}_j^a$ is

$$\Psi(z) = C_E z + \frac{C_E\beta_J C_*}{1-q}\,\frac{z^2}{(1-Dz)^2}. \tag{31.7}$$

By the fixed-input calculation of Step 4, its derivatives bound the columns with a distinguished input. It converges on a disk of radius larger than r_{eff}. Its first derivative bounds $D\widehat{\mathcal{E}}_j^a$; its second derivative, integrated along the segment from 0 to v, gives $\|D\widehat{\mathcal{E}}_j^a(v) - D\widehat{\mathcal{E}}_j^a(0)\|_{\mathrm{row+col}} \le C\|v\|_\infty$; and each fixed further derivative is bounded by the corresponding derivative of Ψ, since $Dr_{\mathrm{eff}} < 1/8$. All constants are functions of the coefficient data listed in the theorem; only Step 3 introduced the loops, the heat horizon, and the torus. This completes the proof. □

31.4 Conditional current coefficients and means

We now bring in the conditioned Wilson measure. The next theorem collects the properties of the conditional fluctuation kernels that we need, bounds the coefficient matrices Γ_j and their dependence on the coarse field, and bounds the conditional mean and the conditional variance of the circulation Taylor remainders $\mathsf{R}_j^{\mathrm{cur}}$. The conditional kernels satisfy a uniform convexity bound on a product of balls. The resulting dimension-free Poincaré, logarithmic Sobolev, and covariance estimates are in the tradition of Brascamp and Lieb [33], Bakry and Émery [8], and Helffer and Sjöstrand [110]; their versions on products of balls are proved in Sections 5 and 13.

Theorem 31.5. *Assume the setting of Section 31.1 and the loop data of Definition 31.1, with one family of reconstruction maps on the whole sequence of lattices. There exists $s_0 < \infty$ such that for every integer $s \geq s_0$, every $K \geq s$, every $1 \leq j \leq K - s$, and every ε with $a = a_{K-s} \leq \varepsilon \leq \varepsilon_0(s)$, the following statements hold. One may take $\varepsilon_0(s) \leq \min\{cH_s, \varepsilon_{\mathrm{loop}}, c/g_3^2\}$ for a fixed small $c > 0$.*

1. *Under $\nu_{K,s}$, the conditional law of w_j given $\mathcal{F}_{j-1}$ is $k_j(\widehat{\theta}_{j-1}, dw)$, where*

$$k_j(\theta, dw) = \frac{\mathbf{1}_{\Omega_j}(w) e^{-V_j(\theta,w)}\, dw}{\int_{\Omega_j} e^{-V_j(\theta,w)}\, dw}, \qquad \Omega_j = \prod_{\ell=1}^{m_j} \overline{B}_{\mathbb{R}^3}(0, p_{j,\ell}), \qquad p_j \leq p_{j,\ell} \leq p_j + C_J.$$

   The potential V_j contains the part of the classical action beyond its quadratic approximation, the input effective interaction, the Haar disintegration density, and the determinants of the triangular change of variables. The restrictions at coarser scales do not change this kernel once θ is fixed. On all of Ω_j,

$$\|D_w^2 V_j - I\|_{\mathrm{row+col}} \leq \vartheta_j = C\big[t_j p_j + t_j^2(1 + \mathsf{E}_{\mathrm{q}})\big], \qquad D_w^2 V_j \geq \tfrac{1}{2} I,$$

$$\mathrm{Var}_{k_j} f \leq 2\mathbb{E}_{k_j}|\nabla f|^2, \qquad \mathrm{Ent}_{k_j}(f^2) \leq 4\mathbb{E}_{k_j}|\nabla f|^2.$$

   The number of three-dimensional blocks satisfies $m_j \leq C_J(L/a_j)^3 \leq C_{J,L,\gamma} t_j^{-6}$, and the conditional means satisfy $\sup_\ell |\mathbb{E}_\theta w_{j,\ell}| \leq C t_j \|\theta\|_\infty$; in particular they vanish at $\theta = 0$. Here $\mathbb{E}_\theta$ denotes expectation under $k_j(\theta, \cdot)$, for a coarse field with $e^\theta \in B_r$ as in Section 31.1. The threshold s_0 can be chosen so that, in addition, $p_j \geq 1$, $6m_j e^{-p_j^2/192} \leq 1/2$, and $|\mathbb{E}_\theta w_{j,\ell}^c| \leq p_j/(16\sqrt{3})$ at all these levels, for every color component c and every coarse field with $e^\theta \in B_r$.

2. *For every θ in the reconstruction domain, with $b = \|\theta\|_\infty$,*

$$\|\Gamma_j(\theta)\|_{\ell^2 \to \mathbb{R}^{3m}} \leq C\omega_j, \qquad \|\Gamma_j(\theta)\|_{\ell^\infty \to \mathbb{R}^{3m}} \leq C\sqrt{\varepsilon/a_j},$$

$$\|\Gamma_j(\theta) - \Gamma_j(0)\|_{\ell^2 \to \mathbb{R}^{3m}} \leq Cb\,\omega_j, \qquad \|\Gamma_j(\theta)\Gamma_j(\theta)^{\mathsf{T}} - \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}}\| \leq Cb\,\omega_j^2,$$

   where ℓ^2 and ℓ^∞ refer to the counting norms on the fluctuation variables.

3. *Let $t = t_j$, and recall that the couplings $g_{j,\ell} \le t_j$ of the intermediate scales have fixed ratios to t_j. Let $F_e(\theta, q)$ be an analytic function of the coarse field θ and the coupling-scaled fluctuation inputs q on the common larger coarse/free domain, with values in $\mathbb{R}^3$, equivariant under global color rotations, such that $F(0,0) = 0$, such that its absolute coefficient row sum at the fixed output e is at most a given constant $C_F < \infty$, and such that, after the substitution $q_\ell = g_{j,\ell} w_\ell$, every term of degree l in w carries a factor t^l, also after one derivative in θ. The constants in this part may depend on C_F and on the radius of the larger domain, but not on the output resolution. For $D_e(\theta, w) = F_e(\theta, g_{j,\cdot} w) - F_e(\theta, 0)$ and every coarse field θ with $e^\theta \in B_r$, we have*

$$|\mathbb{E}_\theta D_e| \le C t^2 \|\theta\|_\infty, \qquad \mathbb{E}_0 D_e = 0.$$

For $\theta_u = u\theta$, $0 \le u \le 1$, and the potential $V_{\theta_u} = V_j(\theta_u, \cdot)$,

$$\frac{d}{du} \mathbb{E}_{\theta_u} D_e(\theta_u, w) = \mathbb{E}_{\theta_u} \partial_u D_e - \mathrm{Cov}_{\theta_u}(D_e, \partial_u V_{\theta_u}),$$

and the local estimates

$$\sup_l \|\partial_{w_l} \partial_u V_{\theta_u}\|_{L^2(k_j(\theta_u, \cdot))} \le C t \|\theta\|_\infty, \qquad \sum_l \|\partial_{w_l} D_e\|_{L^2(k_j(\theta_u, \cdot))} \le C t$$

hold. The derivative of the normalization, the input effective interaction, the Haar disintegration density, and all determinants are included in these statements. At $\theta = 0$ the covariance of the fluctuation variables has the form $\mathbb{E}_0[w_l^a w_h^b] = S_{lh} \delta_{ab}$, with all spatial correlations S_{lh} retained, and its contraction with any equivariant bilinear coefficient with values in $\mathbb{R}^3$ is zero. This is a consequence of global color equivariance, not of parity.

4. *With $b_{j-1} = \|\widehat{\theta}_{j-1}\|_\infty$,*

$$\big|\Gamma_j(\widehat{\theta}_{j-1}) \bar{w}_j + \mathbb{E}[\mathsf{R}_j^{\mathrm{cur}} \mid \mathcal{F}_{j-1}]\big| \le C t_j b_{j-1} \sqrt{\varepsilon / a_j},$$

$$\mathbb{E}\big[|\mathsf{R}_j^{\mathrm{cur}} - \mathbb{E}(\mathsf{R}_j^{\mathrm{cur}} \mid \mathcal{F}_{j-1})|^2 \,\big|\, \mathcal{F}_{j-1}\big] \le C t_j^2 p_j^2 \omega_j^2.$$

The same mean bound holds separately for $|\Gamma_j(\widehat{\theta}_{j-1}) \bar{w}_j|$ and for $|\mathbb{E}[\mathsf{R}_j^{\mathrm{cur}} \mid \mathcal{F}_{j-1}]|$, and both vanish when $\widehat{\theta}_{j-1} = 0$.

All constants in parts (1)–(4) are independent of $s \ge s_0$, K, j, d_j, the output resolution, the point x, and the lattice offset. They may depend on L, γ, M, J, Θ, the loops, σ, the analytic domains, and, in part (3), on C_F and the larger domain.

Proof. We first identify the conditional kernels and choose s_0. We then bound the coefficients at a general coarse field (Step 2), prove the moment and coupling bounds needed for general outputs (Steps 3–5), prove part (3) (Step 6), and apply it to the circulation (Step 7).

Step 1: the kernels and the threshold. By Theorem 15.12, the reconstruction maps preserve the conditional law and its product of balls at every index d_j. We apply Theorem 13.8 to this law; its bound on the input effective interaction is E_{q}, and its largest coupling is t_j. The finitely many cutoff radii within one RG step differ by at most a fixed C_J; write $p_j^+ = p_j + C_J$. For $p_j \geq 1$ this is at most $(1 + C_J)p_j$, so replacing p by p_j^+ in the rational estimate of that theorem changes only its constant, and choosing the coarse side large bounds its denominator away from zero. This gives the bound ϑ_j, and the convexity, Poincaré, and entropy statements on the closed product. Every factor coming from coarser restrictions is a function of the coarse field and cancels in the normalized kernel, as stated in the same theorem. Neither this cancellation nor the choice of gauge representatives removes from V_j a term that depends on w.

The count of free coordinates in Theorem 12.13, summed over the J intermediate scales, bounds the number of three-dimensional blocks by $C_J(L/a_j)^3$. Since $t_j = \lambda_B g_3 \sqrt{a_j}$, we have $(L/a_j)^3 = L^3(\lambda_B g_3)^6 t_j^{-6}$. The mean bound of Theorem 13.10, applied to this product with its individual radii, implies that $\sup_\ell |\mathbb{E}_\theta w_{j,\ell}| \leq C t_j b$; in particular the means vanish at $\theta = 0$.

At coarse index s the largest coupling is

$$t_1(s) = g_3 \sqrt{H_s / M}, \qquad t_j(s) = t_1(s) B^{-(j-1)/2}, \tag{31.8}$$

and $t_1(s) \to 0$ as $s \to \infty$. All the domain, analytic, curvature, and moment conditions required here are finitely many fixed inequalities in t and in its cutoff radius $p(t) = A_{\mathrm{w}} \log(1/t)$. We also require $n_s \geq n_\Gamma$ for every $s \geq s_0$. We increase s_0 until all these inequalities hold for every $0 < t \leq t_1(s_0)$. This is possible also for the condition $6m_j e^{-p_j^2/192} \leq 1/2$, which involves the number of blocks: with $u = \log(1/t)$ its left side is at most

$$C_{J,L,\gamma} \exp\bigl(6u - A_{\mathrm{w}}^2 u^2/192\bigr) \longrightarrow 0 \qquad (u \to \infty).$$

The same reasoning bounds the quantity $(1 + 3m_j)(6m_j)^{1/2} e^{-p_j^2/384}$, which appears in the cutoff estimates of Lemma 32.6. As a function of $h = a_j$, we have

$$(1 + 3m_j)(6m_j)^{1/2} e^{-p_j^2/384} \leq C\bigl(1 + (L/h)^3\bigr)(L/h)^{3/2} e^{-c \log^2(1/(c' g_3 \sqrt{h}))}, \tag{31.9}$$

and for every fixed $v > 0$ the right side is $O(h^v)$ as $h \downarrow 0$, since after taking logarithms the negative quadratic term dominates every fixed linear term. The bound on the mean in part (1) follows from $C t_j r \leq p_j/(16\sqrt{3})$ after another increase of s_0, since $\|\theta\|_\infty < r$ when $e^\theta \in B_r$. These inequalities persist at every later s and every level j. Finally, we choose $\varepsilon_0(s)$ as in the statement, with c small enough that $\eta < 1$ and the small lattice cycles have zero winding. None of these choices changes the reconstruction maps, their thresholds, or the small-field event at any side.

Step 2: coefficients at a general coarse field. We suppress a, ε, and x from $\mathscr{C} = \mathscr{C}_{a,\varepsilon,x}$, and define the map to the fine lattice

$$Z_j(\theta, w) = \widehat{\mathcal{E}}_j^a\big(\widehat{\Phi}_j(\theta, g_{j,\cdot} w)\big). \tag{31.10}$$

By the identity of Theorem 31.3(5), we have $G_j = \mathscr{C}(a_j^{-1} Z_j)$. We compose the joint coefficient series of Theorem 15.12 with the majorant (31.7). The column bound for compositions, proved in Step 4 of the proof of Theorem 31.3, applies with one fluctuation input distinguished, and we obtain

$$D_w Z_j(\theta, 0) = g_j T_j(\theta), \qquad \|T_j(\theta)\|_{\mathrm{row+col}} \le C, \qquad \|T_j(\theta) - T_j(0)\|_{\mathrm{row+col}} \le Cb. \tag{31.11}$$

Here the fluctuation variables are referred to the mesh a_j. Their meshes differ from a_j by the fixed factors $1, M, \dots, M^{J-1}$, and their couplings differ from g_j by the corresponding fixed square-root factors, so converting their columns and densities to the mesh a_j introduces only constants depending on J. A derivative in θ introduces a factor depending only on the fixed gap between the larger and the smaller domain, and integrating it from 0 to θ proves the last bound; no depth-dependent derivative bound or new domain is used. Since $\Gamma_j(\theta) = \mathscr{C}\big((g_3\sqrt{a_j}/a_j) T_j(\theta)\big)$, Theorem 31.3(3), applied to $T_j(\theta)$ and to $T_j(\theta) - T_j(0)$ with $h = a_j$, proves the three bounds on Γ_j in part (2). For the Gram matrices we use, with $A = \Gamma_j(\theta)$ and $A_0 = \Gamma_j(0)$,

$$\|AA^{\mathsf{T}} - A_0 A_0^{\mathsf{T}}\| \le \|A - A_0\|\,(\|A\| + \|A_0\|),$$

which gives the bound $Cb\,\omega_j^2$.

Step 3: local moments. We fix one of these products of balls, set $\theta_u = u\theta$, and write $\mathbb{E}_u$ for expectation under $k_j(\theta_u, \cdot)$. By the mean bound and the inequalities of Step 1, we have $\sup_l |\mathbb{E}_u w_l| \le 1$. For a scalar coordinate let $\xi = w_l^c - \mathbb{E}_u w_l^c$. Applying the Poincaré inequality first to ξ and then to ξ^2, we obtain

$$\mathbb{E}_u \xi^2 \le 2, \qquad \mathbb{E}_u \xi^4 - (\mathbb{E}_u \xi^2)^2 \le 2\mathbb{E}_u |\nabla \xi^2|^2 = 8\mathbb{E}_u \xi^2 \le 16.$$

Hence all fourth moments of scalar coordinates are bounded by a fixed constant, and the inequality $(x_1^2 + x_2^2 + x_3^2)^2 \le 3\sum_c x_c^4$ gives the same for each three-dimensional block. These bounds are uniform in u, the dimension of the product, and the output mesh.

Step 4: coupling powers in the potential. We read off the powers of the coupling in the potential from Theorem 12.13. The first derivative of each classical action with respect to the free coordinates vanishes at its conditional constrained minimizer. Its quadratic term becomes exactly $|w_i|^2/2$, because the normalization uses $K_i^{-1/2}$, the inverse square root of the constrained Hessian. Although the minimizer and the square root depend on the fluctuation variables of the coarser intermediate scales, this quadratic expression

does not. After subtracting $|w|^2/2$, the classical part has total degree at least three in w. The coupling-scaled fluctuation inputs are $q_i = g_i w_i = t\lambda_i w_i$, with fixed λ_i, and the free logarithmic coordinates are $z_i = z_i^* + K_i^{-1/2} q_i$, where z_i^* and K_i depend on the coarser fields. We expand the analytic reconstruction in the inputs q_i, keeping the conditional constrained minimizers and the Hessian factors inside it. Each term of total degree r then contributes t^r after the substitution, and the classical divisor g_i^2 removes two powers of t. The differences of the Haar disintegration density and of the determinants have positive degree and no inverse power of the coupling. Writing H_r for the part of degree r in w of the interaction generated at the current integration step, there are tensors A_r and B_r (unrelated to the small-link domain B_r) such that

$$H_r(\theta, w; t) = t^{r-2} A_r(\theta)[w^r] + t^r B_r(\theta)[w^r], \qquad A_1 = A_2 = 0. \tag{31.12}$$

The differences of the Haar density and of the determinants belong to the terms B_r, and their dependence on the fluctuation variables of the coarser intermediate scales is kept in this decomposition.

Step 5: the mixed derivative of the potential. Fix a level j and condition on $\mathcal{F}_{j-1}$. We apply the local clause of Theorem 13.10 with

$$t = t_j, \qquad \theta = \widehat{\theta}_{j-1}, \qquad E_* = \mathsf{E}_{\mathrm{q}}, \qquad \theta_u = u\theta \quad (0 \le u \le 1),$$

and check its hypotheses. The coarse field satisfies $e^\theta \in B_r$ on the support of $\nu_{K,s}$, as noted in Section 31.1; its radial segment remains in B_r and in the larger analytic domain. The ratios $g_{j,\ell}/t_j$ are the fixed ratios of that theorem. The product is exactly Ω_j, with its radii $p_{j,\ell}$, which depend on the level and the coupling but not on u. The conditions on the product, including the smallness of $t_j(p_j + C_J)^3$, were imposed in the choice of s_0, and the input effective interaction has norm at most E_{q} at every index d_j. By Theorem 15.12, the change of gauge representative preserves the fluctuation variables, the product of balls, the part of the classical action beyond its quadratic approximation, the Haar disintegration density, the determinants, the normalization constant, and the normalized conditional kernel; hence the theorem applies to this kernel. A difference between two choices of the potential that depends only on the coarse field has zero mixed derivative. We conclude that

$$\sup_l \|\partial_{w_l} \partial_u V_j(\theta_u, \cdot)\|_{L^2(k_j(\theta_u, \cdot))} \le C t_j \|\theta\|_\infty, \tag{31.13}$$

with a constant independent of the side, j, K, d_j, and u. This is a bound at each fluctuation index, with a summable spatial profile; no global norm of the extensive derivative $\partial_u V$ is used. The Gaussian part, the dependence of the determinants on the coarser fluctuation variables, and the derivatives of the input effective interaction through the chain rule are included.

For the mean bounds below the following weaker estimate suffices. Summing the pointwise clause for the mixed derivative in Theorem 13.10 against θ, we obtain

$$\sup_l \|\partial_{w_l}\partial_u V_{\theta_u}\|_{L^2} \le \sup_l \|\partial_{w_l}\partial_u V_{\theta_u}\|_\infty \le Ct(p^+)^2 b, \qquad p^+ = p + C_J; \tag{31.14}$$

the exponential decay in the coarse index is summable since the number of coordinates per cell is bounded. We increase s_0 once more so that $t(p^+)^2 \le 1$ at every level. Both estimates concern the same potential.

Step 6: general equivariant outputs. Fix C_F and the larger domain, and let F range over the class described in part (3). Since F is bounded on the larger domain, its absolute coefficient series on the smaller domain also bounds one derivative in θ. All constants in this step may depend on C_F, the radius, and the fixed ratios of radii, but not on the number of output sites or on the resolution. The part of degree r in w of $D_e = F_e(\theta, t\lambda w) - F_e(\theta, 0)$ has row coefficients bounded by $C_1 C_2^r t^r$, also after a derivative in θ. We split $D_e = D_e^{(1)} + D_e^{(\ge 2)}$ into its part of degree one and the sum of all higher degrees. By the mean bound and the absolute output row, we have

$$|\mathbb{E}_\theta D_e^{(1)}| \le Ct \sup_l |\mathbb{E}_\theta w_l| \le Ct^2 b.$$

For the terms of degree $r \ge 2$ in $\partial_u D_e^{(\ge 2)}$, we bound two of the fluctuation factors in L^4 and the others by p^+. For a derivative in one fluctuation variable instead, one factor is bounded in L^2 and the remaining $r-2$ by p^+. Summing the absolute coefficient series, we obtain

$$|\mathbb{E}_u \partial_u D_e^{(\ge 2)}| \le Cbt^2, \qquad \sum_l \|\partial_{w_l} D_e^{(\ge 2)}\|_{L^2} \le Ct^2; \tag{31.15}$$

for example, the second sum is at most $C\sum_{r\ge 2} r C_2^r t^r (p^+)^{r-2} \le Ct^2$. The derivative of the part of degree one has output row at most Ct; hence $\sum_l \|\partial_{w_l} D_e\|_{L^2} \le Ct$ for the whole increment. All sums are at the single fixed output e, so the number of outputs does not appear.

The local covariance inequality of Theorem 13.8 has kernel $Ce^{-b'\,\mathrm{dist}(l,h)}$. Its rows sum to a fixed constant, since the number of fluctuation coordinates per cell is bounded and $\sum_{q\ge 0}(q+1)^3 e^{-b'q} < \infty$. Combining it with (31.14) and (31.15), we find

$$|\operatorname{Cov}_u(D_e^{(\ge 2)}, \partial_u V_{\theta_u})| \le C\sum_{l,h} e^{-b'\,\mathrm{dist}(l,h)} \|\partial_{w_l} D_e^{(\ge 2)}\|_{L^2} \|\partial_{w_h}\partial_u V_{\theta_u}\|_{L^2} \le Ct^3(p^+)^2 b \le Ct^2 b.$$

For each finite product, differentiation under the integral is justified: the domain is compact and fixed, and the functions and their derivatives in the coarse field are continuous on an open neighborhood. Differentiating the numerator and the denominator separately gives exactly

$$\frac{d}{du}\mathbb{E}_u D_e = \mathbb{E}_u \partial_u D_e - \operatorname{Cov}_u(D_e, \partial_u V_{\theta_u}).$$

Thus the derivative of the normalization is present, with the minus sign in the statement, and all Haar, determinant, and effective-interaction terms are those of the same V_j.

The same identity holds with $D_e^{(\geq 2)}$ in place of D_e. At $\theta = 0$ the conditional law is invariant under simultaneous color rotation of all blocks. Both $D_e(0, w)$ and $D_e^{(\geq 2)}(0, w)$ transform as vectors in the adjoint representation, since taking a homogeneous part of fixed degree commutes with color rotations. Each of their expectations is fixed by every rotation, and a vector in $\mathbb{R}^3$ fixed by the rotations by π about two perpendicular axes is zero. Hence both expectations vanish at $\theta = 0$. Integrating the identity for $D_e^{(\geq 2)}$ from $u = 0$ to $u = 1$ and using the two bounds above, we obtain $|\mathbb{E}_\theta D_e^{(\geq 2)}| \leq Ct^2 b$. Adding the bound for the part of degree one, we find $|\mathbb{E}_\theta D_e| \leq Ct^2 b$ for the whole increment. This argument does not use (31.13).

The same reasoning proves the cancellation at $\theta = 0$ without discarding spatial correlations. Let $S_{la,hb} = \mathbb{E}_0[w_l^a w_h^b]$. Invariance makes each 3×3 color matrix $(S_{la,hb})_{a,b}$ commute with every rotation, so $S_{la,hb} = S_{lh}\delta_{ab}$ by the argument used for E_j^a in Step 2 of the proof of Theorem 31.3. If B_e is an equivariant bilinear coefficient with values in $\mathbb{R}^3$, then $\sum_{l,h,a} S_{lh} B_e(e_{la}, e_{ha})$ is fixed by every color rotation and hence zero. Every homogeneous equivariant term with values in $\mathbb{R}^3$ has zero expectation for the same reason. This proves the cancellation for the whole series, which converges absolutely on the product.

Step 7: the remainder of the circulation. We write $\mathbb{E}_{j-1} = \mathbb{E}[\,\cdot \mid \mathcal{F}_{j-1}]$. The map Z_j in (31.10) belongs to one fixed class of outputs as in part (3). Indeed, we choose its larger coarse/free domain inside the domain of Theorem 15.12, with image in the reconstruction domain on which $\widehat{\mathcal{E}}_j^a$ is defined, and we compose the joint coefficient series with (31.7). The composition bound fixes one constant C_F and one larger radius for all j, k, and fine output sites; a distinguished coarse input costs one fixed restriction of the domain. Equivariance and the value zero at $(0, 0)$ are preserved. The substitution $q_{j,\ell} = g_{j,\ell} w_\ell$ gives one factor t_j for each fluctuation factor, also after a derivative in the coarse field. Thus neither C_F nor the radius depends on the remaining number of steps or on the output resolution. By part (3), the increment $Z_j(\theta, w) - Z_j(\theta, 0)$ has conditional mean at most $Ct_j^2 b$ at each output site. Since $G_j(\theta, w) - G_j(\theta, 0) = \mathscr{C}(a_j^{-1}(Z_j(\theta, w) - Z_j(\theta, 0)))$, and since heat smoothing does not increase supremum norms while the cycle has length at most $C_\ell \varepsilon$, we obtain

$$|\Gamma_j \bar{w}_j + \mathbb{E}_{j-1} \mathsf{R}_j^{\mathrm{cur}}| \leq C \frac{\varepsilon}{\sqrt{\eta}} \cdot \frac{t_j^2}{a_j} b = C\lambda_B t_j b \sqrt{\varepsilon / a_j},$$

and we absorb λ_B into C. By the mean bound of part (1) and the ℓ^∞ bound of part (2), we have $|\Gamma_j \bar{w}_j| \leq Ct_j b \sqrt{\varepsilon / a_j}$ separately, and subtracting it from the preceding bound gives

the bound for $|\mathbb{E}_{j-1}\mathsf{R}_j^{\mathrm{cur}}|$. Both vanish at zero coarse field, by the color argument of Step 6 and the vanishing of the conditional mean of w_j.

Finally, every term of $D_w Z_j(\theta, w) - D_w Z_j(\theta, 0)$ has one differentiated fluctuation factor and at least one further fluctuation factor. In the joint coefficient series their sizes are t_j and at most $t_j p_j^+$, respectively. Summing all remaining factors on the fixed larger domain, and using the column bound with a fluctuation input distinguished, we find

$$\|D_w Z_j(\theta, w) - D_w Z_j(\theta, 0)\|_{\mathrm{row+col}} \le C t_j^2 p_j.$$

We divide this matrix by g_j and apply Theorem 31.3(3); since $t_j/g_j = \lambda_B$, we obtain

$$\|D_w \mathsf{R}_j^{\mathrm{cur}}\|_{\ell^2 \to \mathbb{R}^{3m}} \le C t_j p_j \omega_j.$$

The conditional Poincaré inequality bounds the conditional variance of each of the $3m$ components of $\mathsf{R}_j^{\mathrm{cur}}$ by twice its expected squared gradient. Summing over the components and using $\|D_w \mathsf{R}_j^{\mathrm{cur}}\|_{\mathrm{HS}}^2 \le 3m \|D_w \mathsf{R}_j^{\mathrm{cur}}\|_{\mathrm{op}}^2$ proves the variance bound of part (4), with no dependence on the number of fluctuation variables. Every constant was fixed on the common larger domains or by the choice of s_0; none depends on a later s, on K, j, x, the lattice offset, or d_j. Together with (31.13), this proves all claims. □

31.5 Covariance identity for the linearized reconstruction

The coefficients $\Gamma_j(0)$ at the zero field are derivatives of the reconstruction at the origin in all its coordinates. We show that, in the Gaussian theory obtained by linearizing the reconstruction, the fluctuation variables become independent standard Gaussian vectors. Then the sum of the Gram matrices $\Gamma_j(0)\Gamma_j(0)^{\mathsf{T}}$ is the covariance of the fine circulation minus the covariance of its conditional mean given the coarse field. The calculation is Gaussian conditioning applied to the linearized reconstruction; the corresponding constrained Gaussian calculation for Bałaban's averaging operation is in [10, Section D, (1.47)–(1.49), p. 26, and (1.64)–(1.67), p. 29].

Theorem 31.6. *Assume the setting of Section 31.1, and let $s \ge s_0$, $K \ge s$, $k = K - s$, $H_s = L/n_s$, and $a = L/N_K$. Let X_a be the real finite-dimensional space of link fields with three colors on the fine lattice modulo periodic gradients and constant fields, and let $\mathbb{G}_a$ be the Gaussian measure on X_a with density proportional to $\exp(-\|d\theta\|_{\ell^2}^2/(2g_a^2))$, where $g_a^2 = g_3^2 a$, d is the dimensionless plaquette coboundary, and the norm counts each positive plaquette once. Let $Q_{a,H_s} : X_a \to X_{H_s}$ be the derivative at the identity of the composed block averaging map from the fine lattice to the lattice of side n_s, on the corresponding quotients, and let $\mathcal{C}_{s,a} = \sigma(Q_{a,H_s}\theta)$.*

The differential of the reconstruction at zero coarse field and zero fluctuation variables,

$$L(z, w) = L_0 z + \sum_{j=1}^{k} L_j^{\mathrm{w}} w_j,$$

is a bijection from $X_{H_s} \times \prod_j \mathbb{R}^{\dim w_j}$ *onto* X_a*, and it satisfies*

$$\frac{1}{g_a^2}\|dL(z,w)\|_{\ell^2}^2 = \langle z, \mathbf{H}_{s,a} z\rangle + \sum_{j=1}^{k} |w_j|^2, \qquad Q_{a,H_s} L(z,w) = z.$$

Here $\mathbf{H}_{s,a}$ *is the Schur complement of the quadratic form on the coarse lattice given by the nested minimization of Theorem 10.10, including its coupling; it is positive definite on* X_{H_s}*, and its inverse is used only at this fixed* s*. For every linear map* $F : X_a \to \mathbb{R}^d$*,*

$$\sum_{j=1}^{k} (FL_j^{\mathrm{w}})(FL_j^{\mathrm{w}})^{\mathsf{T}} = \mathrm{Cov}_{\mathbf{G}_a}(F\theta) - \mathrm{Cov}_{\mathbf{G}_a}\big(\mathbb{E}[F\theta \mid \mathcal{C}_{s,a}]\big).$$

For the circulation $F = \mathcal{I}_a$ *this reads*

$$\sum_{j=1}^{k} \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}} = C^{a,\varepsilon,L} \otimes I_3 - C^{\mathrm{ret}}_{s,a,\varepsilon}, \qquad C^{\mathrm{ret}}_{s,a,\varepsilon} = \mathrm{Cov}_{\mathbf{G}_a}\big(\mathbb{E}[\mathcal{I}_a\theta \mid \mathcal{C}_{s,a}]\big) \geq 0.$$

Uniformly in K*,* x*, and the lattice offset,*

$$\|C^{\mathrm{ret}}_{s,a,\varepsilon}\| \leq C n_s^2 \varepsilon / H_s \qquad (a \leq \varepsilon \leq \varepsilon_0(s)).$$

Together with Lemma 31.2, this gives, for every fixed $s \geq s_0$*,*

$$\lim_{\varepsilon \downarrow 0} \limsup_{K\to\infty} \sup_{x,\, \mathit{offset}} \Big\| \sum_{j=1}^{K-s} \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}} - \Sigma \otimes I_3 \Big\| = 0.$$

The constants may depend on L*,* γ*,* M*,* J*,* Θ*, the loops,* σ*, and* n_s*, but not on* K *or* d_j*.*

Remark 31.7. The Gaussian coordinates of this theorem are independent only in the linearized calculation; the conditional fluctuation kernels k_j are not Gaussian. The theorem concerns the differential of the reconstruction at the origin in all coordinates. It is not a statement about the nonlinear reconstruction conditioned on $\widehat{\theta}_0 = 0$, whose later fields would remain random. Gradients and constant fields are removed only for this Gaussian identity; they are not removed from the Wilson measure.

Proof. We first set up coordinates on the quotients and prove the energy identity, then prove that L is bijective, and only then change variables in the Gaussian integral. Finally we bound the coarse contribution. All quotient spaces and Gaussian coordinates in this proof are real.

Step 1: coordinates. We choose linear coordinates on each quotient by taking the representatives with mean zero and zero lattice divergence. By the calculation of the kernel of the curl in Step 2 of the proof of Theorem 31.3, these representatives identify exactly the quotient by gradients and constants. The Maxwell form is positive definite on this space: a field of zero energy has zero curl, and its gradient and constant parts have been removed. All Gaussian integrals below are proper finite-dimensional integrals.

We differentiate the reconstruction at its origin and write the resulting linear map, with values in X_a, as $L(z,w) = L_0 z + \sum_j L_j^{\mathrm{w}} w_j$. Here z uses the coordinates on X_{H_s} just described, and w_j ranges over the space of free fluctuation coordinates of Theorem 12.13; the factors $K_i^{-1/2}$ defining them exist by Theorem 10.10. We do not yet claim that L is injective. The gauge transformations of Theorem 15.12, which depend only on the coarse field and equal the identity at the tree roots, add gradients at the identity and therefore do not change this map with values in X_a. The linearized averaging of Lemma 8.1 maps gradients to gradients and constant fields to constant fields, so the differential of the block averaging map descends to the map Q_{a,H_s} between the quotients.

Step 2: the energy identity. We differentiate the exact classical decomposition of Theorem 12.13 at zero coarse field and zero fluctuation variables and take its homogeneous quadratic part. At every intermediate scale the normalized quadratic free action is $|w_i|^2/2$. All remaining classical terms have degree at least three in the fluctuation variables, as shown in (31.12), for every coarse field. They contribute nothing in total degree two, also after composition with the coarser fields. The value and the first derivative of the classical action at the identity vanish. It follows that the quadratic part of the fine action, evaluated on the nonlinear reconstruction, is its quadratic form evaluated on L, with no contribution from second derivatives of the reconstruction. Iterating over all RG steps, we obtain

$$\frac{\|dL(z,w)\|_{\ell^2}^2}{2g_a^2} = \frac{1}{2}\langle z, \mathbf{H}_{s,a} z\rangle + \frac{1}{2}\sum_j |w_j|^2. \tag{31.16}$$

By Theorem 10.10, the form $\langle z, \mathbf{H}_{s,a} z\rangle$ is the nonnegative Schur-complement form, including its coupling. Differentiating the identity expressing that the reconstruction has the prescribed coarse field, we find

$$Q_{a,H_s} L(z,w) = z. \tag{31.17}$$

The Haar, Haar disintegration, determinant, and effective-interaction terms enter the nonlinear conditional law of Theorem 31.5; they are not part of this identity for the classical quadratic form.

Step 3: bijectivity and positivity. If $L(z,w) = 0$ in X_a, its Maxwell energy is zero. Both terms on the right of (31.16) are nonnegative, so every $w_j = 0$, and (31.17) then implies $z = 0$ in X_{H_s}. Thus L is injective. For a lattice of side n, the space of link

fields with three colors has dimension $9n^3$, the gradients have dimension $3(n^3-1)$, and the constant fields have dimension nine, so the quotient has dimension $6n^3-6$. By Theorem 12.13, the number of free fluctuation coordinates from side Mn to side n is $6(M^3-1)n^3$. Summing these numbers over all averaging steps gives $6(N_K^3-n_s^3)$, and adding the dimension $6n_s^3-6$ of X_{H_s} gives exactly $\dim X_a$. Injectivity in equal finite dimensions implies surjectivity. Further, for nonzero z, injectivity makes $L(z,0)$ nonzero in X_a; positivity of its Maxwell energy and (31.16) show that $\mathbf{H}_{s,a}$ is positive definite on X_{H_s}. We do not claim a lower bound on $\mathbf{H}_{s,a}$ independent of the side.

Step 4: the Gaussian change of variables. We make the linear change of variables $\theta = L(z,w)$ in the normalized Gaussian integral on X_a. Its Jacobian is constant and cancels between numerator and denominator. By (31.16), in these coordinates

$$z \sim N(0, \mathbf{H}_{s,a}^{-1}), \qquad w_j \sim N(0, I),$$

and all these vectors are independent; the inverse exists on X_{H_s} at this fixed side. For a linear map F, (31.17) gives

$$\mathbb{E}[F\theta \mid \mathcal{C}_{s,a}] = FL_0 z, \qquad F\theta = FL_0 z + \sum_j FL_j^{\mathrm{w}} w_j.$$

By independence, expanding the covariance of $F\theta$ proves the general covariance identity, and the subtracted covariance is positive semidefinite.

For $F = \mathcal{I}_a$, the closed cycles and their zero winding make F a well-defined functional on X_a, by Lemma 31.2(3). By the chain rule at the origin of the reconstruction, we have

$$FL_j^{\mathrm{w}} = D_w \mathcal{I}_a\big(\widehat{\Pi}_{k\leftarrow j}\widehat{\Phi}_j(0, g_{j,\cdot}w)\big)\big|_{w=0} = \Gamma_j(0).$$

The finite-lattice Fourier calculation of Lemma 31.2(4) identifies $\mathrm{Cov}_{\mathbb{G}_a}(\mathcal{I}_a\theta)$ as $C^{a,\varepsilon,L} \otimes I_3$, since the density of $\mathbb{G}_a$ is that of the Maxwell field of that lemma written in the variable $\theta = aA$. This proves the identity for $\sum_j \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}}$.

Step 5: the coarse contribution. We write $H = H_s$ and $n = n_s$. At $j = 0$, the linear comparison of Theorem 31.3(2) implies that, on circulations, $FL_0 = \mathscr{C}_{a,\varepsilon,x}(H^{-1}E_0^a)$. Using the ℓ^2 circulation bound of Theorem 31.3(3) with input mesh H, dividing out the normalization $g_3\sqrt{H}$, and taking the branch $\sqrt{\varepsilon/H}$, we obtain

$$\|FL_0\|_{\ell^2 \to \mathbb{R}^{3m}} \le \frac{C\sqrt{\varepsilon}}{g_3 H}. \tag{31.18}$$

In unitary Fourier coordinates on the coarse lattice, Theorem 8.3 bounds the covariance of the coarse link variables by $Cg_H^2/\lambda(\vartheta)$ on each nonzero transverse frequency ϑ, uniformly in the number of eliminated scales; the gradient term in the linearized block averaging

map disappears on the quotient by Lemma 8.1. For a nonzero frequency on a lattice of side $n \geq 4$,

$$\lambda(\vartheta) \geq 4\sin^2(\pi/n) \geq 16/n^2,$$

where the second inequality uses $\sin u \geq 2u/\pi$ for $0 \leq u \leq \pi/2$. Since $g_H^2 = g_3^2 H$, this proves

$$\|\mathbf{H}_{s,a}^{-1}\|_{\ell^2 \to \ell^2} \leq C g_3^2 H n_s^2. \tag{31.19}$$

Only the factor n_s^2 of the fixed side has been used. Multiplying the square of (31.18) by (31.19), we obtain

$$\|C_{s,a,\varepsilon}^{\mathrm{ret}}\| = \|(FL_0)\mathbf{H}_{s,a}^{-1}(FL_0)^{\mathsf{T}}\| \leq C n_s^2 \varepsilon / H_s,$$

uniformly in the number of eliminated scales, x, and the lattice offset. The factor n_s^2 is allowed to grow with the fixed coarse side.

Step 6: the limits. The covariance identity implies

$$\Big\|\sum_j \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}} - \Sigma \otimes I_3\Big\| \leq \|C^{a,\varepsilon,L} - C^{\varepsilon,L}\| + \|C^{\varepsilon,L} - \Sigma\| + C n_s^2 \varepsilon / H_s.$$

For each fixed $\varepsilon > 0$ the first term tends to zero as $K \to \infty$, uniformly in x and the lattice offset, by Lemma 31.2(5). After this limit, the second term tends to zero as $\varepsilon \downarrow 0$ by the same lemma, and the last term tends to zero at fixed s. This is the iterated limit in the statement. At $k = 0$ the same change of variables has no fluctuation variables and the whole covariance is the coarse covariance, so the identities also hold in that case. □

31.6 The coarse-field contribution and summation over scales

We now prove the decomposition stated at the beginning of the section and sum the estimates of Theorem 31.5 over all scales. The sums are controlled by a deterministic bound on the size b_j of the reconstructed fields, which we prove first; its constant does not depend on the coarse mesh H_s, which matters because s is later sent to infinity.

Lemma 31.8. *Assume the setting of Theorem 31.5, with s_0 enlarged if necessary depending only on the data fixed in Section 31.1. For every $s \geq s_0$ the following deterministic bounds hold on the support of $\nu_{K,s}$, uniformly in $K \geq s$:*

$$b_0 \leq r, \qquad b_j \leq C\big[r\alpha_*^j + g_3\sqrt{a_j}\,p_j\big],$$

$$\sum_{j=1}^{K-s} b_{j-1} \leq D_s, \qquad D_s = C\Big[\frac{r}{1-\alpha_*} + g_3\sqrt{H_s}\,(p_1(s)+1)\Big], \qquad \sup_{s \geq s_0} D_s < \infty,$$

where $p_1(s) = p_1$ is the cutoff radius of the first RG step. Put

$$T_s = \mathcal{I}_a\big(\widehat{\Pi}_{k\leftarrow 0}\widehat{\theta}_0\big), \qquad \mathsf{N}_{K,s} = \sum_{j=1}^{k}\big[\Gamma_j(\widehat{\theta}_{j-1})\bar{w}_j + \mathsf{R}_j^{\mathrm{cur}}\big].$$

Then we have the exact decomposition

$$I^{K,s} = T_s + \sum_{j=1}^{k}\Gamma_j(\widehat{\theta}_{j-1})(w_j - \bar{w}_j) + \mathsf{N}_{K,s}.$$

For $a \le \varepsilon \le \varepsilon_0(s)$,

$$\sup_{\|\widehat{\theta}_0\|_\infty \le r}|T_s| \le \frac{Cr}{g_3 H_s}\sqrt{\varepsilon},$$

$$\sup_{x,\,\mathit{offset}}\mathbb{E}_{\nu_{K,s}}|\mathsf{N}_{K,s}| + \sup_{\|\widehat{\theta}_0\|_\infty \le r}|T_s| \le \frac{Cr}{g_3 H_s}\sqrt{\varepsilon} + Cg_3 D_s\sqrt{\varepsilon} + C\sqrt{\eta}\,[1+\log(1/\eta)]^2.$$

The constants C and the threshold s_0 have the dependence stated in Theorem 31.5, and C does not depend on s; the quantity D_s has the displayed additional dependence on s. The coarse-field term is allowed to grow like H_s^{-1}. For each fixed $s \ge s_0$, the right side of the last display tends to zero as $\varepsilon \downarrow 0$, uniformly in K; no limit in which s and ε vary together is asserted.

Proof. We first prove the bounds on b_j (Steps 1–3) and their sum (Step 4), then the decomposition (Step 5), and finally the bounds on T_s and $\mathsf{N}_{K,s}$ (Steps 6 and 7).

Step 1: a recurrence. The inputs $q_j = g_{j,\cdot}w_j$ of the j-th reconstruction map satisfy $\|q_j\|_\infty \le t_j(p_j + C_J)$, since $g_{j,\ell} \le t_j$ and $|w_{j,\ell}| \le p_{j,\ell} \le p_j + C_J$. On the fixed larger coarse/free domain let $C_{\mathrm{fr},J}$ bound the derivative of $\widehat{\Phi}_j$ in the coupling-scaled fluctuation inputs; this bound does not depend on the side or on d_j. By the row majorant of Theorem 31.3(1), we have $\|\widehat{P}_j(v)\|_\infty \le \phi(\|v\|_\infty)$. Integrating this derivative along the segment from 0 to q_j, we obtain, with $\kappa = C_{\mathrm{fr},J}(1 + C_J)$ and $p_j \ge 1$,

$$b_j \le \frac{\alpha_* b_{j-1}}{1 - \beta_J b_{j-1}} + \kappa t_j p_j \le a_{\mathrm{map}} b_{j-1} + \kappa t_j p_j, \tag{31.20}$$

as long as $\widehat{\theta}_{j-1}$ lies in the reconstruction domain. Here $a_{\mathrm{map}} = \alpha_*/(1 - \beta_J r_{\mathrm{eff}})^2$ is the bound of Lemma 15.15 on the derivative of the zero-fluctuation maps on the reconstruction domain, which satisfies $a_{\mathrm{map}}\sqrt{B} \le 32/49 < 1$ by Theorem 23.5.

Step 2: a first bound and the sum. Let $\rho_B = B^{-1/2}$ and $d_p = A_{\mathrm{w}} \log B/2$. Since $t_i = t_j B^{(j-i)/2}$ and $p_i \le p_j$ for $i \le j$, iterating the second inequality in (31.20), we obtain

$$b_j \le r a_{\mathrm{map}}^j + \kappa' t_j p_j, \qquad \kappa' = \frac{\kappa}{1 - a_{\mathrm{map}}\sqrt{B}}.$$

For small t the function $t \mapsto tp(t)$ is increasing, so a fixed upper bound on t_1 with $\kappa' t_1 p_1 \le r_{\mathrm{eff}}/4$ keeps this bound below $r + r_{\mathrm{eff}}/4 < r_{\mathrm{eff}}/2$, which justifies the use of (31.20) at every step by induction. Increasing s_0 enforces this bound for every later s, without changing any radius of the construction. The geometric series and its derivative give

$$\sum_{l=0}^{k-1} b_l \le \frac{r_{\mathrm{eff}}}{4(1-a_{\mathrm{map}})} + \kappa' t_1 \Big[\frac{p_1}{1-\rho_B} + \frac{d_p \rho_B}{(1-\rho_B)^2} \Big] \le S_*.$$

Here we fix S_* by replacing $t_1(p_1+1)$ by its supremum over the allowed range $0 < t_1 \le t_1(s_0)$; this supremum is finite since $t\log(1/t) \to 0$. Thus S_* depends neither on H_s nor on s.

Step 3: the rate α_*^j. For $0 \le \beta_J b \le \beta_J r_{\mathrm{eff}} < 1/8$ we have

$$\frac{\alpha_*}{1-\beta_J b} \le \alpha_* \exp(8\beta_J b/7).$$

This implies that every product over h consecutive steps of the multiplicative coefficients $\alpha_*/(1-\beta_J b_l)$ in the first recurrence of (31.20) is at most $C_{\mathrm{mult}}\alpha_*^h$, where $C_{\mathrm{mult}} = \exp(8\beta_J S_*/7)$. Unrolling that recurrence, we find

$$b_j \le C_{\mathrm{mult}} \Big[r\alpha_*^j + \kappa \sum_{i=1}^{j} \alpha_*^{j-i} t_i p_i \Big] \le C_{\mathrm{mult}} \Big[r\alpha_*^j + \frac{\kappa t_j p_j}{1-\alpha_*\sqrt{B}} \Big],$$

where the denominator is positive because $\alpha_*\sqrt{B} = \sqrt{q} \le 1/2$. Substituting $t_j = \lambda_B g_3 \sqrt{a_j}$, we obtain

$$b_j \le C_{\mathrm{cap}} \big(r\alpha_*^j + g_3 \sqrt{a_j}\, p_j \big), \tag{31.21}$$

with C_{cap} independent of H_s and s. This is the first bound of the lemma. The rate α_*^j does not follow from the first bound of Step 2, where the contraction factor is a_{map}; the product argument is needed.

Step 4: the sum of the bounds. By the definition of p_j and $a_j = H_s B^{-j}$,

$$p_j = p_1(s) + (j-1)d_p. \tag{31.22}$$

Summing (31.21), including the separate term $b_0 \le r$, and using $\sqrt{a_l} = \sqrt{H_s}\rho_B^l$, we obtain

$$\begin{aligned} \sum_{j=1}^{k} b_{j-1} &\le \frac{Cr}{1-\alpha_*} + C g_3 \sqrt{H_s} \sum_{l \ge 1} \rho_B^l \big[p_1(s) + (l-1)d_p \big] \\ &\le C \Big[\frac{r}{1-\alpha_*} + g_3 \sqrt{H_s}\, (p_1(s)+1) \Big]. \end{aligned}$$

In the last inequality we used $\sum_{l\ge1}\rho_B^l=\rho_B/(1-\rho_B)$ and $\sum_{l\ge1}(l-1)\rho_B^l=\rho_B^2/(1-\rho_B)^2$. By (31.8), $p_1(s)=A_{\mathrm{w}}\log(\sqrt{M}/(g_3\sqrt{H_s}))$ is affine in s, while $\sqrt{H_s}=\sqrt{L/n_*}\,B^{-s/2}$ decays geometrically, so their product is bounded for $s\ge s_0$. More directly, $g_3\sqrt{H_s}(p_1(s)+1)=\sqrt{M}\,t_1(s)(p_1(s)+1)$ is bounded by the supremum used in Step 2. Since the constant in (31.21) does not depend on s, this proves $\sup_s D_s<\infty$.

Step 5: the decomposition. For $0\le j\le k$, let $F_j=\mathcal{I}_a(\widehat{\Pi}_{k\leftarrow j}\widehat{\theta}_j)$. Then $F_0=T_s$ and $F_k=I^{K,s}$. Since $\widehat{\Pi}_{k\leftarrow j-1}=\widehat{\Pi}_{k\leftarrow j}\circ\widehat{\Phi}_j(\cdot,0)$, and all fields involved lie in the reconstruction domain, we have

$$F_j-F_{j-1}=G_j(\widehat{\theta}_{j-1},w_j)-G_j(\widehat{\theta}_{j-1},0)=\Gamma_j(\widehat{\theta}_{j-1})w_j+\mathsf{R}_j^{\mathrm{cur}}.$$

Summing this finite identity over j, and subtracting and adding $\Gamma_j(\widehat{\theta}_{j-1})\bar{w}_j$ in each summand, gives the decomposition. The coefficients $\Gamma_j(\widehat{\theta}_{j-1})$ are $\mathcal{F}_{j-1}$-measurable and the increments $w_j-\bar{w}_j$ are conditionally centered, so the middle sum is a martingale for the filtration $(\mathcal{F}_j)$.

Step 6: the coarse-field contribution. By the row bound on $D\widehat{\mathcal{E}}_0^a$ in Theorem 31.3(5) and $\widehat{\mathcal{E}}_0^a(0)=0$, we have

$$\|H_s^{-1}\widehat{\mathcal{E}}_0^a(\widehat{\theta}_0)\|_\infty\le Cr/H_s.$$

By the identity of Theorem 31.3(5), the fact that heat smoothing does not increase supremum norms, and the length bound on the cycles, we obtain

$$|T_s|\le\frac{C\varepsilon r}{\sqrt{\eta}\,H_s}=\frac{Cr}{g_3H_s}\sqrt{\varepsilon}. \tag{31.23}$$

This bound is uniform over all coarse fields with $\|\widehat{\theta}_0\|_\infty\le r$ and over K, including $k=0$.

Step 7: conditional means and centered remainders. We write $\mathsf{R}_j^{\mathrm{cur}}=\mathbb{E}_{j-1}\mathsf{R}_j^{\mathrm{cur}}+\zeta_j$, where $\mathbb{E}_{j-1}$ denotes conditional expectation given $\mathcal{F}_{j-1}$. By Theorem 31.5(4), using $t_j\sqrt{\varepsilon/a_j}=\lambda_B g_3\sqrt{\varepsilon}$, we have

$$\sum_{j=1}^{k}\bigl|\Gamma_j(\widehat{\theta}_{j-1})\bar{w}_j+\mathbb{E}_{j-1}\mathsf{R}_j^{\mathrm{cur}}\bigr|\le Cg_3\sqrt{\varepsilon}\sum_{j=1}^{k}b_{j-1}\le Cg_3D_s\sqrt{\varepsilon}.$$

For the centered terms we use the triangle inequality and the conditional Cauchy–Schwarz inequality, without any independence:

$$\mathbb{E}\Bigl|\sum_{j=1}^{k}\zeta_j\Bigr|\le C\sum_{j=1}^{k}t_jp_j\omega_j. \tag{31.24}$$

We bound the right side uniformly in s. Shrinking the constant c in $\varepsilon_0(s)$ if necessary, we may assume $u = \log(1/\eta) \geq 1$. For a scale with $a_j > \varepsilon$,

$$t_j p_j \omega_j = \lambda_B \sqrt{\eta}\, p_j, \qquad p_j \leq C(1+u);$$

the second inequality follows from $\log(1/t_j) = (1/2)\log(1/(g_3^2 a_j)) - \log \lambda_B$ and $a_j > \varepsilon$. The number of such scales is at most $1 + \log(L/\varepsilon)/\log B \leq C(1+u)$, with a constant depending only on L, g_3, and B, since $H_s \leq L/4$. Their contribution is at most $C\sqrt{\eta}(1+u)^2$. If there exists a first index j_ε with $a_{j_\varepsilon} \leq \varepsilon$, let $h_* = a_{j_\varepsilon}$; then $\varepsilon/B < h_* \leq \varepsilon$, since $\varepsilon < cH_s$. Using (31.22), the contribution of the finer scales is at most

$$\begin{aligned}\sum_{r\geq 0} \frac{\lambda_B g_3}{\varepsilon} (h_* B^{-r})^{3/2} \big(p_{j_\varepsilon} + r d_p\big) &\leq C g_3 \sqrt{\varepsilon}\, (p_{j_\varepsilon} + 1) \\ &\leq C\sqrt{\eta}\, (1+u).\end{aligned}$$

In the last inequality we used $h_* > \varepsilon/B$ in the formula for $\log(1/t_{j_\varepsilon})$. Extending the finite sum to all $r \geq 0$ is allowed since all summands are nonnegative, and if there is no such index this contribution is zero. Hence

$$\sum_j t_j p_j \omega_j \leq C\sqrt{\eta}\, [1 + \log(1/\eta)]^2.$$

Combining this with (31.24), the bound on the conditional means, and (31.23) proves the last estimate of the lemma, uniformly in x, the lattice offset, and K in the stated range. For fixed s the first constant is finite and $\sqrt{\eta}(1+\log(1/\eta))^2 \to 0$ as $\varepsilon \downarrow 0$ at fixed $g_3 > 0$. □

Remark 31.9. The bound (31.21) also follows from the forced-recursion estimate of Lemma 15.15, which applies to inputs of size at most $tp(t)$, $t = \lambda_B g \sqrt{a_j}$, for all t up to a fixed threshold τ, with constants uniform in the coarse mesh and in the coupling g in this range. To apply it, we increase s_0 so that $p_1 \geq 1 + C_J + 2A_{\mathrm{w}} \log 2$ and $2t_1 \leq \tau$, and set $\widetilde{t}_j = 2t_j$ and $\widetilde{p}_j = A_{\mathrm{w}} \log(1/\widetilde{t}_j) = p_j - A_{\mathrm{w}} \log 2$. Then $\widetilde{p}_j \geq 1$ and

$$t_j(p_j + C_J) \leq \widetilde{t}_j \widetilde{p}_j \qquad \text{for every } j \geq 1, \tag{31.25}$$

since the difference divided by t_j is $p_j - C_J - 2A_{\mathrm{w}} \log 2 \geq 1$. This is only a deterministic upper bound for the same inputs; the coupling, the measure, the maps, the fluctuation variables, and the cutoff radii are unchanged. Since the constant of that lemma is uniform in H and for the auxiliary coupling in the stated range, the lemma applies with $g = 2g_3$; as $\widetilde{p}_j \leq p_j$, it gives (31.21). Steps 1–3 above verify the independence of H_s directly. No stochastic estimate with a constant depending on a fixed coarse mesh is used.

31.7 Validity at each fixed positive coupling

Lemma 31.10. *Let the parameters be as in Theorem 23.5. Fix $0 < \gamma \le \Gamma$ and one sequence $N_K = n_* B^K$ with its reconstruction maps, and let n_s, H_s, a_j, and t_j be as in Section 31.1. There exists $s_0 < \infty$ such that every $s \ge s_0$ satisfies all conclusions of Lemmas 31.2 and 31.8 and Theorems 31.3, 31.5, and 31.6, with the same quantifiers and the same dependence of the constants. The constants in the bounds on coefficients, potential derivatives, conditional means, reconstructed fields, and conditional remainders do not depend on $s \ge s_0$, while the factor $1/(g_3 H_s)$ in the bound on the coarse-field contribution and the factor n_s^2/H_s in the bound on the coarse covariance remain explicit.*

Proof. We check that each argument of this section uses only the smallness of t_j and never an upper bound on γ such as $\gamma \le 1$.

Deterministic estimates. The zero-fluctuation maps, their coefficient series, the identities expressing that they have the prescribed coarse field, and their column bounds are those of Theorem 23.5 and Lemma 15.15. The proof of Theorem 31.3 uses Lemmas 16.1 and 11.9 and the majorant $\alpha_* z/(1-\beta_J z)$, with $B\alpha_*^2 < 1$; no step uses $\gamma \le 1$. In each circulation coefficient the factor g_3 cancels against $\sqrt{g_3^2\varepsilon}$, and the only remaining scale is $\omega_j = \min\{a_j/\varepsilon, \sqrt{\varepsilon/a_j}\}$. Thus the deterministic coefficient bounds and the expansion in nonlinear remainders hold on the same domains.

Probabilistic estimates. The conditional potential is that of Theorem 13.8, and Theorem 15.12 preserves its coordinates, its balls, and its density. Theorem 13.10, applied to this product and to the radial segment of the coarse field, gives the conditional means of order $t_j b$ and the local bound $\sup_l \|\partial_{w_l}\partial_u V\|_{L^2} \le C t_j b$, with all effective-interaction, determinant, and Haar disintegration terms included. The class of outputs used for the circulation is fixed by the coefficient series on one larger domain (Step 7 of the proof of Theorem 31.5), so its constant C_F depends neither on the remaining number of steps, nor on the output resolution, nor on s. The differentiation formula and the local covariance bound therefore give the mean and remainder conclusions of Theorem 31.5. Global color equivariance at zero coarse field gives the vanishing mean and the cancellation of equivariant bilinear terms. The Poincaré and logarithmic Sobolev inequalities hold for the same conditional kernel, with dimension-free constants.

The threshold. We have exactly

$$t_1(s)^2 = \frac{\gamma}{M n_s}, \qquad t_j = t_1(s) B^{-(j-1)/2},$$

so $t_1(s) \downarrow 0$ as $s \to \infty$ for fixed $\gamma > 0$. Every analytic, domain, Hessian, potential-derivative, and moment condition is a fixed inequality of the form $t[p(t) + c_J]^v \le c'$

or $t^2(1+\mathsf{E}_{\mathrm{q}}) \le c'$, with a fixed finite v, and all of them hold on an interval $0 < t \le \tau_*$. We also require $n_s \ge n_\Gamma$. There are at most $m_j \le C_J(L/a_j)^3 = C_J L^3(\lambda_B g_3)^6 t_j^{-6}$ three-dimensional blocks, and with $u = \log(1/t)$ the quantity $6m_j e^{-p_j^2/192}$ is at most $C_{J,L,\gamma}\exp(6u - A_{\mathrm{w}}^2 u^2/192)$, which tends to zero as $u \to \infty$. Thus a single s_0 enforces $p_j \ge 1$, $6m_j e^{-p_j^2/192} \le 1/2$, and $Ct_j r \le p_j/(16\sqrt{3})$ for all j and all later s, as in Step 1 of the proof of Theorem 31.5. The cutoff error $(1+3m_j)(6m_j)^{1/2}e^{-p_j^2/384}$, which retains the number of blocks, satisfies (31.9); as a function of $h = a_j$ it is $O(h^v)$ for every fixed $v > 0$, and in particular it is summable over the scales.

The bounds on the reconstructed fields. The recurrence (31.20) and Steps 1–4 of the proof of Lemma 31.8 use only the smallness of $t_1(s)$; equivalently, by Remark 31.9, the forced-recursion estimate of Lemma 15.15 applies with the auxiliary input bound (31.25), without changing the measure. They give

$$b_j \le C\big[r\alpha_*^j + g_3\sqrt{a_j}\,p_j\big], \qquad \sum_{j=1}^{K-s} b_{j-1} \le D_s = C\Big\{\frac{r}{1-\alpha_*} + g_3\sqrt{H_s}\,(p_1(s)+1)\Big\},$$

with a constant independent of H_s and of s. Since $p_1(s)$ is affine in s and $\sqrt{H_s}$ decays geometrically, $\sup_{s\ge s_0} D_s < \infty$.

The covariance identity. The identity of Theorem 31.6 is the exact quadratic part of Theorems 10.10 and 12.13 at the origin of the reconstruction, and the bound on $\mathbf{H}_{s,a}^{-1}$ comes from Theorem 8.3. The independent Gaussian coordinates and the coarse Schur complement therefore give

$$\sum_j \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}} = C^{a,\varepsilon,L}\otimes I_3 - C^{\mathrm{ret}}_{s,a,\varepsilon}, \qquad \|C^{\mathrm{ret}}_{s,a,\varepsilon}\| \le Cn_s^2\varepsilon/H_s,$$

and the coupling cancels in the normalized covariance.

All results of this section have now been checked at every $s \ge s_0$. □

SECTION 32

Gaussian short-distance limits and asymptotic independence

This section completes the proof of the short-distance theorem, Theorem 2.8. Fix smooth closed loops $\ell_1, \dots, \ell_m$ in $\mathbb{R}^3$ and $\sigma > 0$, set $g_3^2 = \gamma/L$, and for $x \in \mathbb{T}_L^3$ and $\varepsilon > 0$ let

$$Z_\varepsilon(x) = \Big(\frac{1 - W_{x+\varepsilon\ell_i, \sigma\varepsilon^2}}{g_3^2 \varepsilon} \Big)_{i=1}^m,$$

where $W_{x+\varepsilon\ell_i,\sigma\varepsilon^2}$ is the normalized fundamental Wilson loop of the trajectory at time $\sigma\varepsilon^2$. We prove that under the continuum measure λ of Theorem 2.3, the law of $Z_\varepsilon(x)$ converges to the law of $Z_G = (|C_i|^2/8)_{i=1}^m$, uniformly in x (Theorem 32.13). Here the three color components of C are independent centered Gaussian vectors with the covariance Σ of Definition 31.1, the covariance of heat-smoothed circulations of a Gaussian Maxwell field; in the notation of that definition, $C_i = G_i$ and $(Z_G)_i = \mathsf{Z}_i$. We deduce a lower bound on the variance of small Wilson loops (Corollary 32.14), and we prove that $Z_\varepsilon(x)$ is asymptotically independent of every fixed bounded Borel function of the trajectory (Theorems 32.16 and 32.17). All results are at a fixed positive coupling; no weak-coupling limit is taken.

The form of the limit has a short heuristic explanation. By the computation at the beginning of Section 31, the circulation of the heat-smoothed Gaussian Maxwell field around $x + \varepsilon\ell_i$ is a centered Gaussian vector with covariance of order $g_3^2 \varepsilon \Sigma$ in each color. A Wilson loop sees only the squared norm of this circulation, since for $X \in \mathfrak{su}(2)$

$$1 - \tfrac{1}{2} \operatorname{tr} e^X = |X|^2/8 + O(|X|^4).$$

Dividing by $g_3^2\varepsilon$ therefore produces $|C_i|^2/8$. The proof shows that the nonlinear terms, the coarse field, and the conditioning on the small-field event do not change this picture.

The proof proceeds as follows. By Lemma 31.8, the heat-smoothed circulation $I^{K,s}$ is the sum of a coarse-field term, a martingale whose increments are linear in the centered fluctuation variables, and a small remainder. For each RG step we compare the conditional law of the linear increment with a Gaussian law by an integration by parts with a cutoff, in the spirit of Stein's method [147] (Lemma 32.4), and we telescope the conditional characteristic functions (Lemma 32.5). The cutoff error grows with the number of fluctuation variables, which is polynomial in $1/t_j$, but the probability that the

cutoff is active decays like $\exp(-c\log^2(1/t_j))$ (Lemma 32.6). We then replace the coefficient matrices by their values at the zero field, whose Gram sum converges to $\Sigma \otimes I_3$ by Theorem 31.6. This proves a Gaussian limit for the circulations under the conditioned measure (Theorem 32.8), and the same proof applies conditionally on the first q RG steps, uniformly in their values (Lemma 32.10). Next we compare flowed Wilson loops with circulations (Lemma 32.11), remove the conditioning using the bound of Theorem 23.10, and pass to the continuum measure using Section 29. For asymptotic independence we approximate macroscopic flowed Wilson loops by functions of the first q RG steps (Lemma 32.15).

Four length scales appear: the coarse mesh H_s, the mesh $a_q = H_{s+q}$ after q RG steps, the loop size ε, and the fine mesh a, with $a \le \varepsilon \ll a_q \le H_s$. The estimates split at the scales with $a_j \approx \varepsilon$; the meshes H_s and a_q stay fixed while $\varepsilon \to 0$. The limits are taken in the following order. For the Gaussian limit,

$$K \to \infty, \qquad \varepsilon \downarrow 0 \ \ (s, R \text{ fixed}), \qquad R \to \infty, \qquad s \to \infty,$$

where R is the radius of a ball for the linear and quadratic flow data; for asymptotic independence,

$$K \to \infty, \qquad \varepsilon \downarrow 0 \ \ (s, q, R, \delta \text{ fixed}), \qquad q \to \infty, \qquad \delta \downarrow 0, \qquad R \to \infty, \qquad s \to \infty,$$

where δ is a tolerance. No estimate allows these parameters to vary simultaneously.

Besides Section 31, we use the bound for conditioning of Lemma 5.5, the identity for finite-scale reconstructions of Theorem 15.13, and the bound on the small-field failure probability of Theorem 23.10. We also use the local theory of the Wilson flow and its stability, Theorem 24.5 and Corollary 25.11, the estimates on the linear and quadratic flow data of Lemmas 28.4 and 28.10, and the construction of the continuum measure in Section 29. The section ends with the proof of Theorem 2.8.

32.1 Setting and inputs

Definition 32.1. Fix $L > 0$, $\Gamma > 0$, $0 < \gamma \le \Gamma$, and the parameters of Theorem 23.5, chosen before γ: an odd integer $M \ge 3$, the integer J, $B = M^J$, the reconstruction maps, and their analytic domains. Fix $n_* \in 4\mathbb{N}$, smooth closed loops $\ell_1, \dots, \ell_m$ in $\mathbb{R}^3$, and $\sigma > 0$. We use the notation of Section 31.1: $N_K = n_* B^K$, $n_s = n_* B^s$, $H_s = L/n_s$, $k = K - s$, $a_j = H_s B^{-j}$, $a = a_k$, $g_3^2 = \gamma/L$, $\eta = g_3^2\varepsilon$, $t_j = M^{(J-1)/2} g_3 \sqrt{a_j}$, and $p_j = A_{\mathrm{w}} \log(1/t_j)$. The probability measure is the conditioned Wilson measure

$$\nu_{K,s} = \nu_{N_K, n_s, \gamma} = \mu_{N_K,\gamma}(\,\cdot \mid S_{N_K, n_s, \gamma}).$$

Its backward reconstruction sequence is

$$\widehat{\theta}_j = \widehat{\Phi}_j(\widehat{\theta}_{j-1}, g_{j,\cdot} w_j), \qquad \mathcal{F}_j = \sigma(\widehat{\theta}_0, w_1, \dots, w_j),$$

where $g_{j,\ell}$ are the couplings of the intermediate scales of the j-th RG step, and, conditionally on $\mathcal{F}_{j-1}$, the vector w_j has the conditional fluctuation kernel $k_j(\widehat{\theta}_{j-1},\cdot)$ of Theorem 31.5, whose potential includes the input effective interaction produced by the integrations at finer scales, the Haar disintegration density, and the determinants of the change of variables. We write $\bar{w}_j = \mathbb{E}(w_j \mid \mathcal{F}_{j-1})$. All maps, zero-fluctuation continuations, and circulations use the same reconstruction maps and the same choice of representative of the coarse field. Put

$$\widehat{\Pi}_{k\leftarrow j} = \widehat{\Phi}_k(\cdot,0)\circ\cdots\circ\widehat{\Phi}_{j+1}(\cdot,0), \qquad \widehat{\Pi}_{k\leftarrow k} = \mathrm{id}\,.$$

The circulation map $\mathcal{I}$ is the map denoted $\mathcal{I}_a$ in Definition 31.1, with lattice cycles of physical length at most $C_\ell\varepsilon$, zero winding, and swept area at most $C_\ell\varepsilon a$, which tends to zero at fixed ε as $a\to 0$. Thus $I^{K,s} = \mathcal{I}(\widehat{\theta}_k) \in \mathbb{R}^{3m}$. As in Section 31.1 we put

$$G_j(v,w) = \mathcal{I}\big(\widehat{\Pi}_{k\leftarrow j}\widehat{\Phi}_j(v, g_{j,\cdot}w)\big), \qquad \Gamma_j(v) = D_w G_j(v,0), \qquad \omega_j = \min\{a_j/\varepsilon, \sqrt{\varepsilon/a_j}\},$$

$$Q_j^{\mathrm{Tay}} = G_j(\widehat{\theta}_{j-1}, w_j) - G_j(\widehat{\theta}_{j-1},0) - \Gamma_j(\widehat{\theta}_{j-1})w_j.$$

The Taylor remainder Q_j^{Tay} is the quantity denoted $\mathsf{R}_j^{\mathrm{cur}}$ in Section 31.1, and the vector C below is the vector G of Definition 31.1. Unless stated otherwise, constants may depend on L, γ, M, J, n_*, Θ, the analytic domains, the loops, σ, and the coarse index s (the dependence described in item (5) of Definition 4.14); they are independent of K, j, x, the lattice offset, and the classical recursion index $d_j = J(K-s-j)$. A constant indexed by s may depend on H_s. All regions are the full periodic torus.

The threshold s_0 below depends on the fixed data and is chosen before the coarse index s; the reconstruction maps do not change with s. The classical recursion index d_j is the depth $\mathfrak{d}_j$ of Section 26, not the increment $x_j - x_{j-1}$ of the heat path there.

Remark 32.2. The bounds of Theorem 31.5(4) on the circulation Taylor remainders Q_j^{Tay} come from the bound for general outputs in part (3) of that theorem, applied to

$$Z_j^a(\theta,q) = \widehat{\mathcal{E}}_j^a\big(\widehat{\Phi}_j(\theta,q)\big), \qquad G_j(\theta,w) = \mathscr{C}_{a,\varepsilon,x}\big(a_j^{-1}Z_j^a(\theta, g_{j,\cdot}w)\big),$$

with the field $\widehat{\mathcal{E}}_j^a$ of Theorem 31.3. That bound is useful only if the larger coarse/free radius, which we denote ρ^+_{joint}, and the coefficient bound, which we denote C_F^{cur}, are fixed before K, s, j, and the output resolution vary. We explain why. The only composition of growing length in Z_j^a is the field $\widehat{\mathcal{E}}_j^a$. Let C_E bound the rows and volume-normalized columns of the matrices E_i^a, let C_{src} be the constant in Theorem 31.3(4), and recall $D = \beta_J/(1-\alpha_*)$ and $q = B\alpha_*^2 < 1$. A common majorant for $\widehat{\mathcal{E}}_j^a$ is

$$\mathcal{M}_{\mathrm{ext}}(z) = C_E z + \frac{C_E C_{\mathrm{src}} C_*}{1-q}\,\frac{z^2}{(1-Dz)^2}. \tag{32.1}$$

Indeed, a remainder at scale $i > j$, with $r' = i - 1 - j$, contributes at most $C_E C_{\mathrm{src}} C_* q^{r'} z^2/(1-Dz)^2$, since $(a_j/a_i)\alpha_*^{2r'+1} = C_* q^{r'}$; for a distinguished input its column factors multiply to a^3/a_j^3, so the same series and its derivatives bound the columns. We fix a radius ρ_E inside the larger domain, with $D\rho_E < 1$, such that the reconstruction images lie strictly inside the disk of radius ρ_E. Then $\mathcal{M}_{\mathrm{ext}}(\rho_E) < \infty$, and its derivatives on the smaller domain are fixed constants. We then choose the larger coarse/free domain of radius ρ^+_{joint} so that its image under $\widehat{\Phi}_j$ lies in this disk, uniformly in the indices, as in Step 7 of the proof of Theorem 31.5. The majorant $\mathcal{M}_{\mathrm{ext}}$ and its derivatives on the smaller domain, composed with the joint coefficient bound for $\widehat{\Phi}_j$, then give the fixed constant C_F^{cur}. Both maps are equivariant and vanish at zero, so $Z_j^a(0,0) = 0$. After the substitution $q_{j,\ell} = g_{j,\ell} w_{j,\ell}$, each fluctuation factor contributes its coupling $g_{j,\ell} \le t_j$, and one derivative in the coarse field costs a fixed factor on the larger domain, which we include in C_F^{cur}. Thus neither C_F^{cur} nor ρ^+_{joint} depends on the remaining number of steps or on s; the bound for general outputs is not applied to an output chosen anew at each cutoff. This is Step 7 of the proof of Theorem 31.5.

The following proposition collects the estimates of Section 31 that are used below, together with the dependence of their constants.

Proposition 32.3. *In the setting of Definition 32.1, let s_0 be the threshold of Theorem 31.5, enlarged as in Lemma 31.8. For every $s \ge s_0$, $K \ge s$, $1 \le j \le k$, and $a \le \varepsilon \le \varepsilon_0(s)$, the following statements hold. The constants do not depend on $s \ge s_0$, except through the factors in H_s and n_s that are displayed.*

1. *Conditionally on $\mathcal{F}_{j-1}$, the vector w_j has the law $k_j(\widehat{\theta}_{j-1}, \cdot)$ on $\Omega_j = \prod_\ell \overline{B}_{\mathbb{R}^3}(0, p_{j,\ell})$, with $p_j \le p_{j,\ell} \le p_j + C_J$; this kernel does not depend on the restrictions at coarser scales once $\widehat{\theta}_{j-1}$ is fixed. Its potential satisfies $D_w^2 V_j \ge I/2$ and $\|D_w^2 V_j - I\|_{\mathrm{row+col}} \le \vartheta_j \le C[t_j p_j + t_j^2(1 + \mathsf{E}_{\mathrm{q}})]$, and $\operatorname{Var} f \le 2\mathbb{E}|\nabla f|^2$ and $\operatorname{Ent}(f^2) \le 4\mathbb{E}|\nabla f|^2$ conditionally on $\mathcal{F}_{j-1}$. Further, $m_j \le C_J(L/a_j)^3$, $|\mathbb{E}[w_{j,\ell} \mid \mathcal{F}_{j-1}]| \le C t_j b_{j-1}$, $p_j \ge 1$, $6m_j e^{-p_j^2/192} \le 1/2$, and $|\mathbb{E}[w^c_{j,\ell} \mid \mathcal{F}_{j-1}]| \le p_j/(16\sqrt{3})$.*
2. *For θ in the reconstruction domain, $\|\Gamma_j(\theta)\|_{\ell^2 \to \mathbb{R}^{3m}} \le C\omega_j$, $\|\Gamma_j(\theta)\|_{\ell^\infty \to \mathbb{R}^{3m}} \le C\sqrt{\varepsilon/a_j}$, and $\|\Gamma_j(\theta) - \Gamma_j(0)\|_{\ell^2 \to \mathbb{R}^{3m}} \le C\|\theta\|_\infty \omega_j$.*
3. *We have $|\Gamma_j(\widehat{\theta}_{j-1})\bar{w}_j + \mathbb{E}[Q_j^{\mathrm{Tay}} \mid \mathcal{F}_{j-1}]| \le C t_j b_{j-1}\sqrt{\varepsilon/a_j}$, the same bound for $|\mathbb{E}[Q_j^{\mathrm{Tay}} \mid \mathcal{F}_{j-1}]|$, and $\mathbb{E}[|Q_j^{\mathrm{Tay}} - \mathbb{E}(Q_j^{\mathrm{Tay}} \mid \mathcal{F}_{j-1})|^2 \mid \mathcal{F}_{j-1}] \le C t_j^2 p_j^2 \omega_j^2$.*
4. *Pathwise, $b_0 \le r$, $b_j \le C[r\alpha_*^j + g_3\sqrt{a_j}\, p_j]$, and $\sum_{j=1}^k b_{j-1} \le D_s$, with D_s as in Lemma 31.8 and $\sup_{s \ge s_0} D_s < \infty$. With T_s as in Lemma 31.8 and $\widetilde{N}$ the quantity denoted $\mathsf{N}_{K,s}$ there, $\sup_{\|\widehat{\theta}_0\|_\infty \le r} |T_s| \le Cr\sqrt{\varepsilon}/(g_3 H_s)$ and $\mathbb{E}|\widetilde{N}| + \sup_{\|\widehat{\theta}_0\|_\infty \le r} |T_s| \le Cr\sqrt{\varepsilon}/(g_3 H_s) + C g_3 D_s \sqrt{\varepsilon} + C\sqrt{\eta}[1 + \log(1/\eta)]^2$.*

5. *We have* $\sum_j \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}} = C^{a,\varepsilon,L} \otimes I_3 - C^{\mathrm{ret}}_{s,a,\varepsilon}$ *with* $\|C^{\mathrm{ret}}_{s,a,\varepsilon}\| \le C n_s^2 \varepsilon / H_s$. *As* $a \downarrow 0$ *at fixed* ε, $C^{a,\varepsilon,L} \to C^{\varepsilon,L}$ *uniformly in* x *and the lattice offset, and* $C^{\varepsilon,L} \to \Sigma$ *as* $\varepsilon \downarrow 0$.

Proof. Parts (1)–(3) are parts (1), (2), and (4) of Theorem 31.5; the constants in part (3) are fixed by Remark 32.2. Part (4) is Lemma 31.8, and part (5) is Theorem 31.6 together with Lemma 31.2(5). By Lemma 31.10, all of these hold at the fixed coupling γ, with constants independent of $s \ge s_0$ apart from the displayed factors. □

32.2 Integration by parts and conditional characteristic functions

We now prove two finite-dimensional lemmas. The first compares one conditional increment with a Gaussian vector by an integration by parts against the density of the conditional fluctuation kernel, with a cutoff that vanishes near the boundary of the product of balls; this is the characteristic-function form of Stein's Gaussian integration by parts [147]. The second lemma assembles the one-step comparisons by telescoping conditional characteristic functions.

Lemma 32.4. *Let* $\mathcal{K} \subset \mathbb{R}^D$ *be compact and convex with nonempty interior, let* V *be a* C^2 *function on a neighborhood of* $\mathcal{K}$ *with* $D^2V \ge \kappa I$ *for some* $\kappa > 0$, *and let* $d\zeta = Z^{-1}\mathbf{1}_{\mathcal{K}} e^{-V}\,dw$. *Assume the Poincaré inequality* $\mathrm{Var}_\zeta f \le \kappa^{-1}\mathbb{E}_\zeta|\nabla f|^2$ *for every real* C^1 *function* f. *Let* Ψ *be a real* $d \times D$ *matrix with rows* ψ_b, *and set*

$$\mu = \mathbb{E}_\zeta w, \qquad \xi = \Psi(w-\mu), \qquad S = \Psi\Psi^{\mathsf{T}}, \qquad R_b = \langle \nabla V - (w-\mu), \psi_b \rangle.$$

Then for every $\chi \in C^1_c(\operatorname{int}\mathcal{K}; [0,1])$, *with* $\vartheta = \sup_{\operatorname{supp}\chi} \|D^2V - I\|_{\mathrm{op}}$, *every* $u \in \mathbb{R}^d$, *and* $1 \le b \le d$,

$$\begin{aligned}\left|\mathbb{E}_\zeta[(\xi_b - i(Su)_b)e^{iu\cdot\xi}]\right| \le{}& \frac{\sqrt{2}}{\kappa}\|\Psi\|_{\mathrm{op}}|u|\big(\vartheta|\psi_b| + \|R_b\nabla\chi\|_{L^2(\zeta)}\big)\\ &+ 2\mathbb{E}_\zeta|D_{\psi_b}\chi| + 2\mathbb{E}_\zeta[(1-\chi)|\xi_b|] + |(Su)_b|\,\zeta\{\chi < 1\}.\end{aligned}$$

The numerical coefficients do not depend on D, *on the shape of* $\mathcal{K}$, *or on its radii.*

By Proposition 32.3(1), the conditional fluctuation kernels satisfy the Poincaré hypothesis with $\kappa = 1/2$.

Proof. We write $\mathbb{E}$ for $\mathbb{E}_\zeta$ and $e_u(w) = e^{iu\cdot\xi(w)}$. Since χ has compact support in the interior of $\mathcal{K}$, we may integrate the directional derivative of $\chi e_u e^{-V}$, extended by zero, over $\mathbb{R}^D$; there is no boundary contribution. Since

$$D_{\psi_b}(u\cdot\xi) = (Su)_b, \qquad D_{\psi_b}V = \xi_b + R_b,$$

the resulting identity is

$$\mathbb{E}[\chi(\xi_b + R_b)e_u] = i(Su)_b\mathbb{E}[\chi e_u] + \mathbb{E}[(D_{\psi_b}\chi)e_u]. \tag{32.2}$$

Rearranging, we obtain

$$\mathbb{E}[(\xi_b - i(Su)_b)e_u] = \mathbb{E}[(1-\chi)(\xi_b - i(Su)_b)e_u] - \mathbb{E}[\chi R_b e_u] + \mathbb{E}[(D_{\psi_b}\chi)e_u]. \tag{32.3}$$

At $u = 0$, the identity (32.2) and $\mathbb{E}\xi_b = 0$ imply

$$\mathbb{E}[\chi R_b] = \mathbb{E}[D_{\psi_b}\chi] + \mathbb{E}[(1-\chi)\xi_b]. \tag{32.4}$$

Here it matters that R_b is defined with the mean μ of ζ.

For a real function F and a complex function H, we have $|\operatorname{Cov}(F,H)| \le (\operatorname{Var} F)^{1/2}(\mathbb{E}|H - \mathbb{E}H|^2)^{1/2}$ by the Cauchy–Schwarz inequality. We apply the Poincaré inequality to $F = \chi R_b$ and separately to $\Re e_u$ and $\Im e_u$; the sum of the squared gradients of the latter two functions is $|\Psi^{\mathsf T}u|^2 \le \|\Psi\|_{\mathrm{op}}^2|u|^2$. Further,

$$\nabla(\chi R_b) = \chi(D^2V - I)\psi_b + R_b\nabla\chi, \qquad \|\nabla(\chi R_b)\|_{L^2} \le \vartheta|\psi_b| + \|R_b\nabla\chi\|_{L^2},$$

and only the Hessian on $\operatorname{supp}\chi$ enters the last bound. Thus

$$|\operatorname{Cov}(\chi R_b, e_u)| \le \kappa^{-1}\|\Psi\|_{\mathrm{op}}|u|\big(\vartheta|\psi_b| + \|R_b\nabla\chi\|_{L^2}\big).$$

In (32.3) we write $\mathbb{E}[\chi R_b e_u] = \operatorname{Cov}(\chi R_b, e_u) + \mathbb{E}[\chi R_b]\mathbb{E}e_u$ and use (32.4). The two occurrences of each cutoff term are bounded by the corresponding absolute expectations, and the remaining term with $(Su)_b$ is bounded by $|(Su)_b|\zeta\{\chi < 1\}$. This proves the stated inequality, even with 1 in place of $\sqrt{2}$ in the first coefficient. □

Lemma 32.5. *Let $(\mathcal{F}_j)_{j=0}^k$ be a finite filtration, and let $\xi_j \in \mathbb{R}^d$ be bounded, $\mathcal{F}_j$-measurable, and conditionally centered, $\mathbb{E}[\xi_j \mid \mathcal{F}_{j-1}] = 0$. Let $S_j^0 \ge 0$ be deterministic matrices, and let $\rho_j, \eta_j \ge 0$ be integrable $\mathcal{F}_{j-1}$-measurable random variables. Suppose that for every $u \in \mathbb{R}^d$*

$$\big|\mathbb{E}[(\xi_j - iS_j^0u)e^{iu\cdot\xi_j} \mid \mathcal{F}_{j-1}]\big| \le \rho_j|u| + \eta_j(1+|u|).$$

For $0 \le q \le k$ put $M_{q,k}^0 = \sum_{j=q+1}^k \xi_j$. Then almost surely

$$\Big|\mathbb{E}[e^{iu\cdot M_{q,k}^0} \mid \mathcal{F}_q] - \exp\Big(-\frac{1}{2}u^{\mathsf T}\sum_{j=q+1}^k S_j^0 u\Big)\Big| \le \sum_{j=q+1}^k \mathbb{E}\big[\tfrac{1}{2}\rho_j|u|^2 + \eta_j|u|(1+|u|) \mid \mathcal{F}_q\big].$$

For every integrable random vector N, replacing $M_{q,k}^0$ by $M_{q,k}^0 + N$ adds at most $|u|\mathbb{E}[|N| \mid \mathcal{F}_q]$ to the bound. No independence or smallness of the increments is assumed, and the numerical constants do not depend on k, q, or the dimensions.

Proof. We set

$$\varphi_j(v) = \mathbb{E}[e^{iv\cdot\xi_j} \mid \mathcal{F}_{j-1}], \qquad \Delta_j(v) = \mathbb{E}[(\xi_j - iS_j^0v)e^{iv\cdot\xi_j} \mid \mathcal{F}_{j-1}].$$

Since ξ_j is bounded, we may differentiate along $v = \tau u$, and we find

$$\frac{d}{d\tau}\varphi_j(\tau u) = -\tau u^{\mathsf{T}} S_j^0 u\, \varphi_j(\tau u) + iu \cdot \Delta_j(\tau u).$$

The integrating factor from τ to 1 is $\exp[-(1-\tau^2)u^{\mathsf{T}} S_j^0 u/2] \le 1$. Since $\varphi_j(0) = 1$, integrating from 0 to 1, we obtain

$$|\varphi_j(u) - \varphi_j^0(u)| \le \mathsf{e}_j(u), \qquad \varphi_j^0(u) = e^{-u^{\mathsf{T}} S_j^0 u/2}, \qquad \mathsf{e}_j(u) = \tfrac{1}{2}\rho_j |u|^2 + \eta_j |u|(1+|u|). \tag{32.5}$$

Fix q, and for $q \le j \le k$ set

$$H_j = \exp\Big(iu \cdot \sum_{l=q+1}^{j} \xi_l\Big) \prod_{l=j+1}^{k} \varphi_l^0(u).$$

Then $H_q = \prod_{l=q+1}^{k} \varphi_l^0(u)$ is deterministic and $H_k = e^{iu \cdot M_{q,k}^0}$. Conditioning the difference $H_j - H_{j-1}$ first on $\mathcal{F}_{j-1}$ and then on $\mathcal{F}_q$, and using (32.5), we bound its conditional expectation given $\mathcal{F}_q$ in modulus by $\mathbb{E}[\mathsf{e}_j(u) \mid \mathcal{F}_q]$; here we used that the exponential of the past increments has modulus one and that every remaining deterministic factor lies in $[0,1]$. Telescoping proves the assertion. Finally, $|e^{iu\cdot(z+n)} - e^{iu\cdot z}| \le |u||n|$ implies the statement about N; no measurability or independence requirement on N is needed. □

32.3 Telescoping circulations and dimension-dependent cutoffs

The next lemma applies Lemma 32.4 to the conditional fluctuation kernels. The cutoff χ_j must vanish near the boundary of the product of balls, and its gradient grows like the square root of the number D_j of fluctuation variables, so the error terms carry a polynomial factor in D_j. This factor is harmless because the cutoff radius grows logarithmically in $1/t_j$, so that the probability that the cutoff is active decays faster than any power of the mesh.

Lemma 32.6. *In the setting of Definition 32.1, for every finite $K \ge s$,*

$$I^{K,s} = T_s + \sum_{j=1}^{k} \Gamma_j(\widehat{\theta}_{j-1})(w_j - \bar{w}_j) + \widetilde{N},$$

where

$$T_s = \mathcal{I}(\widehat{\Pi}_{k \leftarrow 0}\widehat{\theta}_0), \qquad \widetilde{N} = \sum_{j=1}^{k} \big\{\Gamma_j(\widehat{\theta}_{j-1})\bar{w}_j + Q_j^{\mathrm{Tay}}\big\}.$$

All zero-fluctuation continuations lie in the reconstruction domain, and the middle sum is a sum of martingale increments under $\nu_{K,s}$.

There is $s_0 < \infty$, depending only on L, γ, M, J, n_*, Θ, the analytic domains and column bounds, the loops, and σ, and not on the later index s, such that for $s \ge s_0$ the following holds at each RG step. The kernel k_j lives on m_j balls in $\mathbb{R}^3$, with $D_j = 3m_j \le 3C_J(L/a_j)^3$, and $p_j \ge 1$, $D_w^2 V_j \ge I/2$, and

$$\|D_w^2 V_j - I\|_{\mathrm{op}} \le \vartheta_j \le C\{t_j p_j + t_j^2(1+\mathsf{E}_{\mathrm{q}})\} \le \tfrac{1}{2}.$$

There is a smooth function $\chi_j : \Omega_j \to [0,1]$, equal to one on the product of the balls of half the radii and to zero outside the product of the balls of three quarters of the radii, with $|\nabla\chi_j| \le C\sqrt{D_j}/p_j$ and $\nabla\chi_j$ supported in $\{\chi_j < 1\}$. The conditional probability that the cutoff is active,

$$q_j^{\mathrm{cut}} = \mathbb{P}\{\chi_j(w_j) < 1 \mid \mathcal{F}_{j-1}\},$$

satisfies $q_j^{\mathrm{cut}} \le 6m_j e^{-p_j^2/192} \le 1/2$. Put $\widehat{\tau}_j = (1+3m_j)(6m_j)^{1/2}e^{-p_j^2/384}$. Let Ψ_j be either $\Gamma_j(\widehat{\theta}_{j-1})$ or $\Gamma_j(0)$, and let $\xi_j = \Psi_j(w_j - \bar{w}_j)$ and $S_j = \Psi_j\Psi_j^{\mathsf{T}}$. Then for every $u \in \mathbb{R}^{3m}$,

$$\big|\mathbb{E}[(\xi_j - iS_j u)e^{iu\cdot\xi_j} \mid \mathcal{F}_{j-1}]\big| \le \rho_j|u| + \eta_j(1+|u|),$$

with

$$\rho_j \le C_{3m}\vartheta_j\|\Psi_j\|_{\mathrm{op}}^2, \qquad \eta_j \le C_{3m}\|\Psi_j\|_{\mathrm{op}}(1+\|\Psi_j\|_{\mathrm{op}})\widehat{\tau}_j,$$

where C_{3m} depends only on m, J, and the ratios of the radii, and not on D_j, p_j, or the coarse field. There is a deterministic function T of the mesh such that

$$\widehat{\tau}_j \le T(a_j) \le C\big(1+(L/a_j)^3\big)(L/a_j)^{3/2}\exp\Big\{-\frac{A_{\mathrm{w}}^2}{384}\log^2\frac{1}{M^{(J-1)/2}g_3\sqrt{a_j}}\Big\}.$$

For every $r' > 0$ we have $T(h) \le C_{r'}h^{r'}$ as $h \downarrow 0$. In particular, with $A_s^{\mathrm{cut}} = \sum_{j\ge1} a_j^{-1/2}T(a_j)$ and $\Theta_{\mathrm{cut}}(v) = \sup_{0<h\le v} T(h)$,

$$\sum_{j=1}^{k}\omega_j(1+\omega_j)\widehat{\tau}_j \le C\sqrt{\varepsilon}\,A_s^{\mathrm{cut}} + C_B\Theta_{\mathrm{cut}}(\varepsilon) \longrightarrow 0 \qquad (\varepsilon \downarrow 0),$$

uniformly in K, x, the lattice offset, and d_j, and $A_s^{\mathrm{cut}} \le A_{s_0}^{\mathrm{cut}} < \infty$. The number of blocks is $m_j = O_{J,L,\gamma}(t_j^{-6})$.

Proof. The decomposition is that of Lemma 31.8, proved in Step 5 of its proof: for $F_j = \mathcal{I}(\widehat{\Pi}_{k\leftarrow j}\widehat{\theta}_j)$ we have $F_0 = T_s$, $F_k = I^{K,s}$, and $F_j - F_{j-1} = \Gamma_j(\widehat{\theta}_{j-1})w_j + Q_j^{\mathrm{Tay}}$, since $\widehat{\Pi}_{k\leftarrow j-1} = \widehat{\Pi}_{k\leftarrow j}\circ\widehat{\Phi}_j(\cdot,0)$ on the reconstruction domain, which contains all sampled fields and all zero-fluctuation continuations by Theorem 31.3. The coefficients are $\mathcal{F}_{j-1}$-measurable and $\bar{w}_j$ is the conditional mean under the same law, so the middle sum is a martingale.

We next fix the threshold, uniformly for all later coarse indices; all data in the dependence list are fixed before this choice. The largest coupling of each RG step and the cutoff radii satisfy

$$t_1(s) = g_3\sqrt{H_s/M}, \qquad t_j(s) = t_1(s)B^{-(j-1)/2}, \qquad p_j = p_1(s) + \tfrac{1}{2}A_{\mathrm{w}}(j-1)\log B. \tag{32.6}$$

By Proposition 32.3(1), the kernel k_j lives on m_j balls with radii $p_j \le p_{j,\ell} \le p_j + C_J$ and has the stated Hessian, dimension, and mean bounds. The bound on rows and columns implies the bound on the operator norm, since for a finite matrix A

$$\|A\|^2_{\ell^2\to\ell^2} \le \Big(\sup_i \sum_l |A_{il}|\Big)\Big(\sup_l \sum_i |A_{il}|\Big),$$

as follows by applying the weighted Cauchy–Schwarz inequality in each row and then summing. The Hessian error therefore tends to zero with $t_j p_j + t_j^2(1+\mathsf{E}_{\mathrm{q}})$. Since $t_1(s) \downarrow 0$, one enlargement of s_0 enforces $p_j \ge 1$ and $\vartheta_j \le 1/2$ at every level. The number of blocks satisfies

$$m_j \le C_J(L/a_j)^3 \le C_{J,L,\gamma}t_j^{-6}, \qquad a_j = t_j^2/(\lambda_B^2 g_3^2),$$

with $\lambda_B = M^{(J-1)/2}$, so $6m_j e^{-p_j^2/192} \to 0$ uniformly over all sufficiently small t_j. The same s_0 enforces the bound $1/2$ on this quantity and the bound on the means

$$|\mathbb{E}[w^c_{j,\ell} \mid \mathcal{F}_{j-1}]| \le p_j/(16\sqrt{3}), \qquad 1 \le c \le 3. \tag{32.7}$$

These are restrictions on the coarse side only; the cutoff radii and the maps are not changed.

The cutoff. Fix a smooth function $\psi : [0,\infty) \to [0,1]$, equal to one on $[0,1/2]$ and to zero on $[3/4,\infty)$, and set

$$\chi_j(w) = \prod_{\ell=1}^{m_j} \psi(|w_\ell|/p_{j,\ell}).$$

It is smooth, since each factor is constant near $w_\ell = 0$, and it has compact support in the interior of the product. The gradient of each factor has norm at most C/p_j, so $|\nabla\chi_j| \le C\sqrt{m_j}/p_j \le C\sqrt{D_j}/p_j$, and $\nabla\chi_j$ vanishes wherever $\chi_j = 1$. These bounds depend only on ψ, not on the number of balls.

The probability that the cutoff is active. We use the logarithmic Sobolev inequality $\mathrm{Ent}(f^2) \le 4\mathbb{E}|\nabla f|^2$ of Proposition 32.3(1) on the product itself, by Herbst's argument (see [123]). For a scalar coordinate $F(w) = w^c_\ell$, let $H(\lambda) = \log \mathbb{E}e^{\lambda F}$. Applying the inequality to $f = e^{\lambda F/2}$, we find $\lambda H'(\lambda) - H(\lambda) \le \lambda^2$. Integrating $(H/\lambda)' \le 1$ from zero for $\lambda > 0$, and applying the same argument to $-F$, we obtain

$$\mathbb{E}e^{\lambda(F-\mathbb{E}F)} \le e^{\lambda^2} \quad (\lambda \in \mathbb{R}), \qquad \mathbb{P}\{|F-\mathbb{E}F| \ge v\} \le 2e^{-v^2/4},$$

where the tail bound follows by taking $\lambda = v/2$ in $e^{\lambda^2 - \lambda v}$ for each of the two tails. If $\chi_j < 1$, then some $|w_\ell| > p_{j,\ell}/2 \geq p_j/2$, so some coordinate has absolute value larger than $p_j/(2\sqrt{3})$. By (32.7), its deviation from its conditional mean is at least $p_j/(4\sqrt{3})$. A union bound over the $3m_j$ coordinates gives

$$q_j^{\mathrm{cut}} = \mathbb{P}\{\chi_j < 1 \mid \mathcal{F}_{j-1}\} \leq 6m_j e^{-p_j^2/192}. \tag{32.8}$$

The error terms with the dimension retained. We suppress the index j and write $D = 3m_j$, $p = p_j$, $q = \zeta\{\chi < 1\} \leq 1/2$, $\Psi = \Psi_j$, and ζ for the conditional kernel. Both choices of Ψ are linear maps whose rows are fixed once the coarse field is fixed; neither choice changes the law, its mean, the cutoff, or the score vector $\nabla V - (w - \mu)$. The contraction $R_b = (\nabla V - (w - \mu)) \cdot \psi_b$ retains its dependence on the chosen row ψ_b. The product of balls has diameter at most $C_J p\sqrt{D}$, and by convexity the mean $\mu = \mathbb{E}w$ lies in the product. Since $\nabla R_b = (D^2 V - I)\psi_b$, with $c_b = R_b(\mu)$ we have

$$\sup_w |R_b(w) - c_b| \leq C_J \vartheta p \sqrt{D}\, |\psi_b|. \tag{32.9}$$

By the Poincaré inequality, $\mathbb{E}|\xi_b|^2 \leq 2|\psi_b|^2$. Further,

$$\mathbb{E}|D_{\psi_b}\chi| \leq C\sqrt{D}\, |\psi_b|\, q/p, \qquad \mathbb{E}[(1-\chi)|\xi_b|] \leq \sqrt{2}\, |\psi_b| \sqrt{q}.$$

Using (32.4), $\mathbb{E}\chi \geq 1/2$, and (32.9), we obtain

$$|c_b| \leq C_J |\psi_b| \big\{ \vartheta p \sqrt{D} + (\sqrt{D}/p) q + \sqrt{q} \big\}.$$

Since $\nabla\chi$ is supported in $\{\chi < 1\}$, multiplying the resulting bound on $\sup |R_b|$ by the gradient bound and by $\sqrt{q}$, and using $p \geq 1$ and $\vartheta \leq 1/2$, we find

$$\|R_b \nabla\chi\|_{L^2} \leq C_J (1 + D) |\psi_b| \sqrt{q}. \tag{32.10}$$

In particular, R_b is not bounded independently of D.

We insert these estimates into Lemma 32.4 with $\kappa = 1/2$, and use $|(Su)_b| \leq |\psi_b| \|\Psi\|_{\mathrm{op}} |u|$ and $(\sum_b |\psi_b|^2)^{1/2} \leq \sqrt{3m}\, \|\Psi\|_{\mathrm{op}}$. The Euclidean norm of the vector of errors is at most

$$C_{3m} \vartheta \|\Psi\|_{\mathrm{op}}^2 |u| + C_{3m} \|\Psi\|_{\mathrm{op}} (1 + \|\Psi\|_{\mathrm{op}})(1 + D)\sqrt{q}\, (1 + |u|).$$

All constants may depend on J and the ratios of the radii, but not on D, p, or the coarse field. By (32.8), $(1 + D)\sqrt{q} \leq \widehat{\tau}_j$, which gives the stated ρ_j and η_j.

The deterministic envelope. On $0 < h \le H_{s_0}/B$ we take

$$T(h) = C\big(1 + (L/h)^3\big)(L/h)^{3/2} \exp\Big[-\frac{A_{\mathrm{w}}^2}{384} \log^2 \frac{1}{\lambda_B g_3 \sqrt{h}}\Big], \tag{32.11}$$

with one sufficiently large C; then $\widehat{\tau}_j \le T(a_j)$, since $m_j \le C_J (L/a_j)^3$ and $p_j = A_{\mathrm{w}} \log(1/(\lambda_B g_3 \sqrt{a_j}))$. This definition uses the mesh itself, not the index j or the coarse index. To verify the decay, let $h_* = H_{s_0}/B$ and $z = \log(h_*/h) \ge 0$. The logarithm of the polynomial prefactor is at most $C + (9/2)z$. The exponent is a quadratic polynomial in z with leading coefficient $-A_{\mathrm{w}}^2/1536$. For every $r' > 0$, subtracting $\log h^{r'}$ adds only $r'z$ and a constant, and the resulting quadratic polynomial is bounded above. This proves $T(h) \le C_{r'} h^{r'}$ and $\Theta_{\mathrm{cut}}(v) \to 0$, with constants of the stated dependence.

Taking $r' = 2$ shows that $\sum_{j \ge 1} a_j^{-1/2} T(a_j) < \infty$. For $s \ge s_0$ this series is a tail of the same series along the fixed sequence of meshes, so $A_s^{\mathrm{cut}} \le A_{s_0}^{\mathrm{cut}}$. Finally $\omega_j \le 1$, and

$$\sum_{a_j > \varepsilon} \omega_j (1 + \omega_j) T(a_j) \le 2\sqrt{\varepsilon}\, A_s^{\mathrm{cut}}, \qquad \sum_{a_j \le \varepsilon} \omega_j (1 + \omega_j) T(a_j) \le \frac{2\Theta_{\mathrm{cut}}(\varepsilon)}{1 - B^{-1}}.$$

For the second bound we used that the meshes form a geometric sequence, so that $\sum_{a_j \le \varepsilon} a_j/\varepsilon \le (1 - B^{-1})^{-1}$. Truncating either sum at k only decreases it. This proves the summability and uniformity assertions. □

32.4 Gaussian approximation of the circulations

We first record how characteristic-function estimates are converted into estimates in the bounded-Lipschitz distance d_{BL}, which uses test functions bounded by one with Lipschitz constant at most one. The argument also covers a singular limiting covariance.

Lemma 32.7. *Let P' and P be probability measures on $\mathbb{R}^d$, with characteristic functions $\widehat{P}'(u) = \int e^{iu\cdot z} P'(dz)$ and $\widehat{P}$. For all $A > 0$ and $\delta > 0$,*

$$d_{\mathrm{BL}}(P', P) \le P'\{|z| > A\} + P\{|z| > A\} + 4\delta \mathbb{E}|Z_d| + \frac{|B_{\mathbb{R}^d}(0, A+1)|}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-\delta^2 |u|^2/2} |\widehat{P}'(u) - \widehat{P}(u)|\, du, \tag{32.12}$$

where Z_d is a standard Gaussian vector in $\mathbb{R}^d$. In particular, if probability measures P_l are uniformly tight and their characteristic functions converge pointwise to that of P, then $d_{\mathrm{BL}}(P_l, P) \to 0$. If $d = 3m$, the same convergence holds after applying $\mathfrak{q}(v) = (|v_i|^2/8)_{i=1}^m$, provided the tails of the P_l and of P vanish uniformly as the radius tends to infinity; quantitatively, for every $A > 0$,

$$d_{\mathrm{BL}}(\mathfrak{q}_* P', \mathfrak{q}_* P) \le \max\{1, A/4\}\, d_{\mathrm{BL}}(P', P) + 2P'\{|v| > A\} + 2P\{|v| > A\}. \tag{32.13}$$

The constants depend only on d and on the chosen radii A and δ.

Proof. Let f satisfy $\|f\|_\infty \le 1$ and $\operatorname{Lip} f \le 1$, and let $\chi_A(z) = \min\{1, (A+1-|z|)_+\}$ and $h = f\chi_A$. Then h is supported in the ball of radius $A+1$, is bounded by one, and has Lipschitz constant at most two. If φ_δ is the centered Gaussian density with covariance $\delta^2 I_d$, then $\|h * \varphi_\delta - h\|_\infty \le 2\delta \mathbb{E}|Z_d|$. The Fourier transform of a Gaussian density gives $\varphi_\delta(z) = (2\pi)^{-d} \int e^{iu\cdot z} e^{-\delta^2|u|^2/2}\, du$. Inserting this formula into the convolution and using Fubini's theorem, which is justified since h and the Gaussian in u are integrable, we obtain, with $\widehat{h}(u) = \int h(z) e^{-iu\cdot z}\, dz$,

$$P'(h*\varphi_\delta) - P(h*\varphi_\delta) = (2\pi)^{-d} \int \widehat{h}(u) e^{-\delta^2|u|^2/2} \big(\widehat{P}'(u) - \widehat{P}(u)\big)\, du.$$

Since $|\widehat{h}(u)| \le |B_{\mathbb{R}^d}(0, A+1)|$, the modulus of the right side is at most the last term of (32.12). Restoring h costs $4\delta\mathbb{E}|Z_d|$, and restoring f costs at most $P'\{|z|>A\} + P\{|z|>A\}$, since $|f-h| \le \mathbf{1}_{\{|z|>A\}}$. The right side does not depend on f, which proves (32.12). Under the hypotheses on P_l, dominated convergence (the characteristic functions differ by at most two) shows that the last term tends to zero for fixed A and δ; we then let $A \to \infty$ and $\delta \downarrow 0$.

For the quadratic map, let π_A be the nearest-point projection onto the closed ball of radius A. It is 1-Lipschitz: for $z, z' \in \mathbb{R}^d$, the inequalities $\langle z - \pi_A z, y - \pi_A z\rangle \le 0$ for y in the ball, applied with $y = \pi_A z'$ and symmetrically, give $|\pi_A z - \pi_A z'|^2 \le \langle z - z', \pi_A z - \pi_A z'\rangle$. On the ball,

$$|\mathfrak{q}(v) - \mathfrak{q}(w)| \le \tfrac{1}{8}(|v| + |w|)|v - w| \le (A/4)|v - w|.$$

Thus, for a test function f on $\mathbb{R}^m$ bounded by one with Lipschitz constant at most one, the function $f \circ \mathfrak{q} \circ \pi_A$ is bounded by one and has Lipschitz constant at most $A/4$. Further, $f \circ \mathfrak{q}$ differs from it only outside the ball, by at most two. This proves (32.13). For the convergence, take $P' = P_l$, let $l \to \infty$, and then let $A \to \infty$. □

Theorem 32.8. *In the setting of Definition 32.1, there is an integer s_0 such that for every fixed integer $s \ge s_0$,*

$$\lim_{\varepsilon \downarrow 0} \limsup_{K \to \infty} \sup_{x \in \mathbb{T}^3_L,\, \textit{offset}} d_{\mathrm{BL}}\Big(\mathrm{Law}_{\nu_{K,s}}\, I^{K,s}(\varepsilon, x),\, N(0, \Sigma \otimes I_3)\Big) = 0,$$

where Σ is the covariance of Definition 31.1; it may be singular. The underlying estimates hold for every $K \ge s$, $1 \le j \le K - s$, and $a \le \varepsilon \le \varepsilon_0(s)$, where one may take $\varepsilon_0(s) \le \min\{cH_s, \varepsilon_{\mathrm{loop}}, c/g_3^2\}$. The constants in the bounds on the coefficients, on the variation of the covariance, and on the conditional remainders can be chosen independently of $s \ge s_0$. More explicitly, let $\kappa_* = -\log \alpha_* / \log B$, which exceeds $1/2$ since $\alpha_* < B^{-1/2}$, and let

$$\Psi_\kappa(u) = \begin{cases} u^\kappa, & \kappa < 1, \\ u(1 + |\log u|), & \kappa = 1, \\ u, & \kappa > 1, \end{cases} \qquad 0 < u \le 1.$$

Then the variation of the coefficients and the remainder satisfy

$$\sum_j \|\widehat{\theta}_{j-1}\|_\infty \omega_j^2 \le Cr\Psi_{\kappa_*}(\varepsilon/H_s) + C\sqrt{\eta}\,[1+|\log\eta|],$$

$$\mathbb{E}|\widetilde{N}| + |T_s| \le \frac{Cr}{g_3 H_s}\sqrt{\varepsilon} + Cg_3 D_s\sqrt{\varepsilon} + C\sqrt{\eta}\,[1+|\log\eta|]^2,$$

where $D_s = C\{r/(1-\alpha_*) + g_3\sqrt{H_s}(p_1(s)+1)\}$ *and* $\sup_{s\ge s_0} D_s < \infty$. *The coarse-field term may grow like* H_s^{-1}, $\varepsilon_0(s)$ *may decrease with* s, *and* s *is fixed before both limits.*

Proof. We first sum the one-step bounds over the scales (Steps 1–3), then replace the coefficients by their values at the zero field (Step 4), then bound the characteristic functions (Step 5), and finally pass to the bounded-Lipschitz distance (Step 6).

Step 0: the threshold. We take the threshold of Lemma 32.6 and enlarge it, if necessary, to that of Proposition 32.3. All constants in the coefficient, potential-derivative, and conditional variance bounds are independent of the later s. Indeed, the maps for coarse index s are those of the fixed sequence at the absolute levels $s+j$, with recursion indices $J(K-s-j)$, and the output class is that of Remark 32.2. We use only the mean and variance bounds of Proposition 32.3(3); the local bound (31.13) is not needed. The domain and zero-fluctuation statements come from Theorem 31.3. The law of the coarse field may depend on s; since every estimate holds for every coarse field, no comparison of these laws is needed. We take the range of ε in the statement, shrinking c if necessary so that $\eta < 1$ and $p(\varepsilon) = A_{\mathrm{w}}\log(1/(\lambda_B g_3\sqrt{\varepsilon})) \ge 1$; this is possible with c depending only on M, J, and A_{w}.

Step 1: sums over the scales. Since the meshes a_j decrease by the ratio B, splitting at $a_j = \varepsilon$ gives

$$\sum_{j=1}^{k} \omega_j^2 \le \frac{1}{1-B^{-1}} + \frac{1}{1-B^{-2}} = C_B^{(2)}. \tag{32.14}$$

For $a_j > \varepsilon$ the summands are ε/a_j, and reading them from the last coarse scale backward bounds their sum by the first geometric series; for $a_j \le \varepsilon$ they are a_j^2/ε^2, bounded by the second series. The estimate holds also if either part is empty. For $h \le \varepsilon$ write

$$p(h) = p(\varepsilon) + \tfrac{1}{2}A_{\mathrm{w}}\log(\varepsilon/h),$$

so that $p_j = p(a_j)$. For each fixed $v > 0$, geometric summation gives

$$\sum_{a_j \le \varepsilon} (a_j/\varepsilon)^v (1+p(a_j)) \le C_{B,v,A_{\mathrm{w}}}(1+p(\varepsilon)). \tag{32.15}$$

Indeed, the first ratio $z = a_{j_0}/\varepsilon \le 1$ need not be one, and the sum is at most $z^v \sum_{l\ge 0} B^{-vl}[1 + p(\varepsilon) + (A_{\mathrm{w}}/2)(\log(1/z) + l\log B)]$; we use $z^v \le 1$,

$\sup_{0<z\le 1} z^v \log(1/z) < \infty$, and the convergence of $\sum_l B^{-vl}$ and $\sum_l lB^{-vl}$. For $a_j > \varepsilon$ we have $p(a_j) \le p(\varepsilon)$ and $\sum_{a_j>\varepsilon} a_j^{-1/2} \le C_B \varepsilon^{-1/2}$. It follows that

$$g_3 \sum_j \sqrt{a_j}(1+p_j)\omega_j^2 \le C g_3 \sqrt{\varepsilon}\,(1+p(\varepsilon)) \le C\sqrt{\eta}\,(1+|\log\eta|), \tag{32.16}$$

where on the fine scales we used (32.15) with $v = 5/2$, and on the coarse scales the sum of inverse square roots. Here and below the constants may depend on γ and on H_{s_0}, but not on the later s. Since $p_j \ge 1$ and $t_j \le t_1(s_0)$, the Hessian error satisfies $\vartheta_j \le C g_3 \sqrt{a_j}(1+p_j)$. Thus

$$\sum_j \vartheta_j \omega_j^2 \le C\sqrt{\eta}\,(1+|\log\eta|). \tag{32.17}$$

Step 2: the variation of the coefficients. We write $b_j = \|\widehat{\theta}_j\|_\infty$. The bound of Proposition 32.3(4) is pathwise, with a coefficient independent of H_s and of all later $s \ge s_0$; we use this uniform coefficient rather than a bound with a constant depending on H_s. Its conditions hold here: all fields and zero-fluctuation continuations lie in the reconstruction domain, and $t_1(s) \le t_1(s_0)$ satisfies the required upper bound for every later s. With $b_0 \le r$, it implies

$$b_{j-1} \le C\big\{ r(a_j/H_s)^{\kappa_*} + g_3\sqrt{a_j}(1+p_j)\big\}, \qquad 1 \le j \le k. \tag{32.18}$$

For $j \ge 2$ we used $a_{j-1} = Ba_j$, $p_{j-1} \le p_j$, and $\alpha_*^{j-1} = \alpha_*^{-1}(a_j/H_s)^{\kappa_*}$; for $j = 1$ we used $b_0 \le r$ and absorbed α_*^{-1} into C. The deterministic part of the weighted sum $\sum_j b_{j-1}\omega_j^2$ over the coarse scales is

$$Cr\varepsilon H_s^{-\kappa_*} \sum_{a_j>\varepsilon} a_j^{\kappa_*-1}.$$

If $\kappa_* < 1$, summing from the last coarse mesh bounds this by $Cr(\varepsilon/H_s)^{\kappa_*}$. If $\kappa_* = 1$, every summand is one and there are at most $C_B(1+\log(H_s/\varepsilon))$ of them, which gives the logarithmic case of Ψ_{κ_*}; this case cannot be bounded by $C\varepsilon/H_s$. If $\kappa_* > 1$, summing from the largest mesh bounds it by $Cr\varepsilon/H_s$. On the fine scales the sum is

$$CrH_s^{-\kappa_*}\varepsilon^{-2} \sum_{a_j\le\varepsilon} a_j^{\kappa_*+2} \le Cr(\varepsilon/H_s)^{\kappa_*}.$$

In all three cases this is bounded by $Cr\Psi_{\kappa_*}(\varepsilon/H_s)$. For the fluctuation part of (32.18) we use (32.16). This proves, pathwise, the first estimate of the theorem:

$$\sum_j b_{j-1}\omega_j^2 \le Cr\Psi_{\kappa_*}(\varepsilon/H_s) + C\sqrt{\eta}\,(1+|\log\eta|). \tag{32.19}$$

Step 3: the remainder. Let $\mathcal{D}_j = Q_j^{\mathrm{Tay}} - \mathbb{E}[Q_j^{\mathrm{Tay}} \mid \mathcal{F}_{j-1}]$. By the conditional mean and variance bounds of Proposition 32.3(3), the tower property, $\mathbb{E}|\mathcal{D}_j| \le (\mathbb{E}|\mathcal{D}_j|^2)^{1/2}$, and the triangle inequality, we have, without any independence,

$$\mathbb{E}|\widetilde{N}| \le C \sum_j t_j \, \mathbb{E} b_{j-1} \sqrt{\varepsilon / a_j} + C \sum_j t_j p_j \omega_j. \tag{32.20}$$

In the first sum we apply the pathwise bound inside the expectation:

$$\sum_j t_j \, \mathbb{E} b_{j-1} \sqrt{\varepsilon / a_j} = \lambda_B g_3 \sqrt{\varepsilon} \, \mathbb{E}\Big[\sum_j b_{j-1}\Big] \le \lambda_B g_3 \sqrt{\varepsilon} \, D_s.$$

On the fine scales the second sum is $\lambda_B g_3 \varepsilon^{-1} \sum_j a_j^{3/2} p_j$, which is at most $C\sqrt{\eta}(1 + p(\varepsilon))$ by (32.15). On the coarse scales each term is $\lambda_B \sqrt{\eta}\, p_j$; there are at most $C_B[1 + \log(H_{s_0}/\varepsilon)]$ such scales, and each $p_j \le p(\varepsilon)$. For fixed g_3 and H_{s_0} both factors are at most $C(1 + |\log \eta|)$. Adding the bound on T_s from Proposition 32.3(4), we obtain

$$\mathbb{E}|\widetilde{N}| + \sup_{\|\widehat{\theta}_0\|_\infty \le r} |T_s| \le \frac{Cr}{g_3 H_s} \sqrt{\varepsilon} + C g_3 D_s \sqrt{\varepsilon} + C\sqrt{\eta}\,(1 + |\log \eta|)^2, \tag{32.21}$$

which is also the estimate of Lemma 31.8. The quantity D_s is the sum of the bounds on b_{j-1}, with a coefficient independent of s, and $\sup_{s \ge s_0} D_s < \infty$ follows from (32.6): $p_1(s)$ grows linearly in s, and $\sqrt{H_s}$ decreases by the factor $B^{-1/2}$ at each increment of s.

Step 4: coefficients at the zero field. We now replace the coefficient matrices by their values at the zero field, keeping the centered fluctuation variables $e_j = w_j - \bar{w}_j$ of the conditioned measure. Let

$$M_k = \sum_j \Gamma_j(\widehat{\theta}_{j-1}) e_j, \qquad M_k^0 = \sum_j \Gamma_j(0) e_j, \qquad S_j^0 = \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}}.$$

The reconstruction domain gives a fixed deterministic bound $b_j \le b_{\max} < \infty$. By the conditional Poincaré inequality and the bound on $\Gamma_j(\theta) - \Gamma_j(0)$ in Proposition 32.3(2), we have

$$\mathbb{E}\big[|(\Gamma_j(\widehat{\theta}_{j-1}) - \Gamma_j(0))e_j|^2 \mid \mathcal{F}_{j-1}\big] \le C_{3m} b_{j-1}^2 \omega_j^2.$$

Since these differences are martingale increments for the same measure, conditioning on the earlier variables makes all cross terms vanish. With (32.19) we obtain

$$\mathbb{E}|M_k - M_k^0|^2 \le C_{3m} \sum_j \mathbb{E} b_{j-1}^2 \omega_j^2 \le C_{3m} b_{\max} \big\{ Cr \Psi_{\kappa_*}(\varepsilon / H_s) + C\sqrt{\eta}\,(1 + |\log \eta|) \big\} \longrightarrow 0, \tag{32.22}$$

uniformly in K at fixed s.

Step 5: characteristic functions. We apply Lemma 32.6 with $\Psi_j = \Gamma_j(0)$, under the same conditional kernels at the coarse fields $\widehat{\theta}_{j-1}$; the kernel at the zero field is not used. The bound $\|\Gamma_j(0)\|_{\mathrm{op}} \le C\omega_j$ implies the deterministic bounds

$$\rho_j^0 \le C\vartheta_j\omega_j^2, \qquad \eta_j^0 \le C\omega_j(1+\omega_j)\widehat{\tau}_j,$$

whose sums tend to zero by (32.17) and Lemma 32.6. Lemma 32.5 with $q = 0$, followed by (32.22) and (32.21), bounds the error in the characteristic function of the circulation by

$$\begin{aligned}\Big|\mathbb{E}e^{iu\cdot I^{K,s}} - e^{-u^{\mathsf{T}}(\Sigma\otimes I_3)u/2}\Big| \le |u|\Big\{\mathbb{E}|T_s + \widetilde{N}| + (\mathbb{E}|M_k - M_k^0|^2)^{1/2}\Big\} \\ + \sum_j \Big\{\tfrac{1}{2}\rho_j^0|u|^2 + \eta_j^0|u|(1+|u|)\Big\} \\ + \tfrac{1}{2}|u|^2\Big\|\sum_j S_j^0 - \Sigma\otimes I_3\Big\|_{\mathrm{op}}.\end{aligned} \tag{32.23}$$

In the last term we used $|e^{-x} - e^{-y}| \le |x-y|$ for $x, y \ge 0$. The last term is deterministic. By Proposition 32.3(5),

$$\sum_j S_j^0 = C^{a,\varepsilon,L}\otimes I_3 - C^{\mathrm{ret}}_{s,a,\varepsilon}, \qquad \|C^{\mathrm{ret}}_{s,a,\varepsilon}\| \le Cn_s^2\varepsilon/H_s,$$

and the first covariance converges to $C^{\varepsilon,L}\otimes I_3$ as $K\to\infty$ and then to $\Sigma\otimes I_3$ as $\varepsilon\downarrow 0$, uniformly in x and the lattice offset. This identification concerns the linearization of the reconstruction at the origin and the linearized coarse field; the nonlinear reconstruction is never conditioned to have zero coarse field. At each fixed s, every term on the right of (32.23) therefore vanishes in the stated iterated limit, uniformly in x and the lattice offset. In particular,

$$\lim_{\varepsilon\downarrow 0}\limsup_{K\to\infty}\sup_{x,\,\mathrm{offset}}\Big|\mathbb{E}e^{iu\cdot I^{K,s}} - e^{-u^{\mathsf{T}}(\Sigma\otimes I_3)u/2}\Big| = 0. \tag{32.24}$$

No random covariance and no compensated exponential is needed.

Step 6: the bounded-Lipschitz distance. By the conditional Poincaré inequality, we have $\mathbb{E}[|\Gamma_j(0)e_j|^2 \mid \mathcal{F}_{j-1}] \le 2\,\mathrm{tr}\,S_j^0$, and orthogonality together with (32.14) gives $\mathbb{E}|M_k^0|^2 \le C_{3m}$. With (32.22) and (32.21), this gives a bound on $\mathbb{E}|I^{K,s}|$ that is uniform in K, x, the lattice offset, and ε in a sufficiently small range at the fixed s; by Markov's inequality the tails of $I^{K,s}$ vanish uniformly. We apply (32.12) with $P' = \mathrm{Law}_{\nu_{K,s}} I^{K,s}$ and $P = N(0, \Sigma\otimes I_3)$. Its last term is bounded by the integral against $e^{-\delta^2|u|^2/2}$ of the minimum of two and the right side of (32.23), which does not depend on x or the lattice offset. By dominated convergence, this integral vanishes in the iterated limit $K\to\infty$, then $\varepsilon\downarrow 0$, for fixed A and δ. Letting then $A\to\infty$ and $\delta\downarrow 0$ proves the theorem, also for singular Σ. The dependence on s of the coarse-field constant and of the range of ε is exactly the one in the statement; no choice of s depending on ε has been made. □

Remark 32.9. Lemmas 32.4 and 32.5 are abstract finite-dimensional statements. Their application to the circulations fixes the coupling $\gamma > 0$ before s_0 is chosen, and gives estimates with the dependence described in item (5) of Definition 4.14; it does not give constants uniform as $\gamma \to 0$ at a fixed side, as would be needed for item (6). The proof of Theorem 32.8 uses no tightness of the linear and quadratic flow data; these enter in Lemma 32.11 and in the passage to the continuum measure.

The same proof applies conditionally on the first q RG steps, with error bounds that do not depend on the values of these steps. This uniformity is what makes the asymptotic independence in Section 32.8 possible; it does not follow from Theorem 32.8 itself.

Lemma 32.10. *In the setting of Definition 32.1, fix $s \geq s_0$. Write $\|h\|_{\mathrm{BL}} = \max\{\|h\|_\infty, \operatorname{Lip} h\}$. Fix an integer $q \geq 0$, and for $K - s \geq q$ let $\mathcal{F}_q = \sigma(\widehat{\theta}_0, w_1, \dots, w_q)$ be the σ-field generated by the coarse field and the fluctuation variables of the first q RG steps. Let $C \sim N(0, \Sigma \otimes I_3)$ and $\mathfrak{q}(v) = (|v_i|^2/8)_{i=1}^m$. Regular conditional probabilities can be chosen so that*

$$\lim_{\varepsilon \downarrow 0} \limsup_{K \to \infty} \sup_{x,\, \mathit{offset}} \operatorname*{ess\,sup}_{\nu_{K,s}} d_{\mathrm{BL}}\Big(\operatorname{Law}\big(I^{K,s}(\varepsilon, x) \mid \mathcal{F}_q\big), \operatorname{Law}(C)\Big) = 0. \tag{32.25}$$

The same holds with $I^{K,s}$ and C replaced by $\mathfrak{q}(I^{K,s})$ and $\mathfrak{q}(C)$. In particular, if $|B_{K,q}| \leq 1$ is any real $\mathcal{F}_q$-measurable random variable, then

$$\lim_{\varepsilon \downarrow 0} \limsup_{K \to \infty} \sup_{x,\, \mathit{offset}} \sup_{\|f\|_{\mathrm{BL}} \leq 1} \Big| \mathbb{E}_{\nu_{K,s}}\big[f(\mathfrak{q}(I^{K,s}))B_{K,q}\big] - \mathbb{E} f(\mathfrak{q}(C))\, \mathbb{E}_{\nu_{K,s}} B_{K,q} \Big| = 0. \tag{32.26}$$

The constants may depend on s and q and otherwise have the dependence of item (5) of Definition 4.14. The variable $B_{K,q}$ may depend on K, and no regularity in the values of the first q steps is required. Nothing is asserted uniformly in q.

Proof. We follow the proof of Theorem 32.8 conditionally on $\mathcal{F}_q$, and check that every error bound is deterministic and does not depend on the values of the first q steps.

Step 1: the decomposition after q steps. Since all spaces of coarse fields and fluctuation variables are finite-dimensional compact metric spaces, the explicit normalized kernels give regular conditional probabilities. Fix a realization of the first q steps. By Proposition 32.3(1), the later conditional kernels are the same kernels as before conditioning, since the restrictions at coarser scales do not alter them once their coarse field is fixed. Every such field lies in the reconstruction domain and satisfies $e^{\widehat{\theta}_{j-1}} \in B_r$, as noted in Section 31.1. The maps at the absolute levels $s + q + 1, \dots, K$ are the same reconstruction maps, with the same recursion indices. By Theorem 31.3, every zero-fluctuation continuation is defined. Thus the decomposition starting at q is

$$I^{K,s} = T_q + \sum_{j=q+1}^{K-s} \Gamma_j(\widehat{\theta}_{j-1})(w_j - \bar{w}_j) + N_q,$$

where

$$T_q = \mathcal{I}\big(\widehat{\Pi}_{K-s\leftarrow q}\widehat{\theta}_q\big), \qquad N_q = \sum_{j=q+1}^{K-s} \{\Gamma_j(\widehat{\theta}_{j-1})\bar{w}_j + Q_j^{\mathrm{Tay}}\},$$

and $\bar{w}_j$ is still the conditional mean under the conditioned Wilson measure. This is a restriction of the same telescoping sum, with no change of measure.

Step 2: bounds uniform in the first q steps. The field $\widehat{\mathcal{E}}_q^a$ of Theorem 31.3 vanishes at zero and has bounded derivative on the reconstruction domain, so $\|\widehat{\mathcal{E}}_q^a(\widehat{\theta}_q)\|_\infty \le C b_{\max}$. By its identity for circulations, the contraction of the heat semigroup, and the length bound, we have, with $a_q = H_{s+q}$,

$$|T_q| \le \frac{C b_{\max}}{g_3 a_q}\sqrt{\varepsilon}.$$

Here $b_{\max}$ is the fixed bound for the reconstruction domain. We keep the pathwise bounds of Proposition 32.3(4) and (32.19), which hold after conditioning as well; they are not restarted at $\widehat{\theta}_q$. Using the mean and variance bounds of Proposition 32.3(3), which hold for every coarse field, and the tower property, we obtain

$$\mathbb{E}[|N_q| \mid \mathcal{F}_q] \le C g_3 D_s \sqrt{\varepsilon} + C \sum_{j>q} t_j p_j \omega_j \le C g_3 D_s \sqrt{\varepsilon} + C\sqrt{\eta}\,(1 + |\log \eta|)^2.$$

For the means we used $t_j\sqrt{\varepsilon/a_j} = \lambda_B g_3 \sqrt{\varepsilon}$ and the deterministic bound on $\sum_j b_{j-1}$; for the centered part, the conditional variance and the conditional Cauchy–Schwarz inequality. These bounds vanish at fixed s and q and do not depend on the values of the first q steps. The cutoff estimates of Lemma 32.6 hold at every later coarse field. For the replacement of the coefficients by their values at the zero field, we set

$$B_{s,\varepsilon} = C r \Psi_{\kappa_*}(\varepsilon/H_s) + C\sqrt{\eta}\,(1 + |\log \eta|).$$

By the conditional Poincaré inequality and the orthogonality of martingale increments, we have

$$\begin{aligned}\mathbb{E}\Big[\Big|\sum_{j>q}\big(\Gamma_j(\widehat{\theta}_{j-1}) - \Gamma_j(0)\big)(w_j - \bar{w}_j)\Big|^2 \;\Big|\; \mathcal{F}_q\Big] &\le C\,\mathbb{E}\Big[\sum_{j>q} b_{j-1}^2 \omega_j^2 \;\Big|\; \mathcal{F}_q\Big] \\ &\le C b_{\max} B_{s,\varepsilon}.\end{aligned}$$

The last inequality follows from the pathwise bound (32.19), which holds on every sample path; its right side is deterministic and tends to zero at fixed s. Lemma 32.5 applies to this tail of the sum.

Step 3: the covariance of the tail. The coefficients $\Gamma_j(0)$ for $j > q$ are exactly the coefficients at the zero field for the coarse index $s+q$, since their defining maps are the same restrictions of the fixed family. By Theorem 31.6 at index $s+q$,

$$\sum_{j>q} \Gamma_j(0)\Gamma_j(0)^{\mathsf{T}} = C^{a,\varepsilon,L} \otimes I_3 - C^{\mathrm{ret}}_{s+q,a,\varepsilon}, \qquad \|C^{\mathrm{ret}}_{s+q,a,\varepsilon}\| \le C n^2_{s+q} \varepsilon / H_{s+q},$$

and by Lemma 31.2 the first term has the same limit $\Sigma \otimes I_3$. For every u, the conditional error in the characteristic function is therefore bounded almost surely by the sum of these deterministic remainder, coefficient, cutoff, and covariance errors, as in (32.23). This bound does not depend on the first q steps, and it vanishes in the stated order at fixed s and q. The factors $1/a_q$ and n^2_{s+q}/H_{s+q} grow with q, which is why q is fixed before the limits.

Step 4: conditional laws. By the conditional Poincaré inequality, the conditional second moment of the tail $\sum_{j>q} \Gamma_j(0)(w_j - \bar{w}_j)$ also has a deterministic bound, and the conditional first-moment bounds of Step 2 then give uniform conditional tails for $I^{K,s}$. We apply (32.12) to these conditional laws: its Fourier integral is bounded by the same deterministic envelope for the characteristic functions, and its truncation terms by these conditional tails. Choosing one null set for all rational u and using the continuity of each conditional characteristic function, we obtain an almost sure bound that does not depend on the version of the conditional law or on the test function. This proves (32.25).

Step 5: the quadratic map. By (32.13) applied to the conditional laws, with the projection onto a ball of radius A on which $\mathfrak{q}$ has Lipschitz constant at most $A/4$, and with the conditional tails outside the ball vanishing uniformly as $A \to \infty$, the statement for $\mathfrak{q}(I^{K,s})$ follows; we let $K \to \infty$ and $\varepsilon \downarrow 0$ before $A \to \infty$. Multiplying by $B_{K,q}$ and using the tower property proves (32.26). □

32.5 Wilson flow and removal of the conditioning

We now compare the normalized Wilson-loop deficits of the flowed fine configuration with the squared circulations, and remove the conditioning on the small-field event. The comparison has two parts. The first is an expansion of the trace of a product of $\mathrm{SU}(2)$ matrices to second order, with a remainder controlled by the total variation and independent of the number of links. The second is the local theory of the Wilson flow of Theorem 24.5, which writes the flowed field as the linear heat evolution plus the quadratic Picard term plus a smaller remainder. The Wilson flow is the one introduced by Lüscher [128], with time measured in physical units as in Section 2.1.

Lemma 32.11. *The following statements hold.*

1. *For every finite sequence $v_e \in \mathbb{R}^3$, with $D = \sum_e |v_e|$ and $Y = \sum_e v_e$, and every $s' > 0$,*

$$\Big| \frac{1 - \frac{1}{2}\operatorname{tr} \prod_e e^{s' v_e^a \tau_a}}{(s')^2} - \frac{|Y|^2}{8} \Big| \le \frac{s' D^3}{48} e^{s' D/2}.$$

The same bound holds for an ordered exponential along a path of finite length, with D the total variation of the integrated coefficient. The bound does not depend on the number of links.

2. *Fix $1/2 < \alpha_{\mathrm{fl}} < 2/3$ and a horizon $T_0 > 0$, and put $\beta_{\mathrm{fl}} = \alpha_{\mathrm{fl}} - 1/2$ and $\kappa_{\mathrm{fl}} = 1 - 3\alpha_{\mathrm{fl}}/2$. For the reconstruction of Definition 32.1, let $\mathfrak{x} = P^a_\cdot A^{K,s}$ be the linear heat evolution of the initial potential $A^{K,s} = a^{-1}\widehat{\theta}_k$, let $\mathfrak{y} = \mathcal{Q}_a(\mathfrak{x},\mathfrak{x})$ be its quadratic Picard term, and let $\mathcal{B}_{K,s} = \mathsf{X}^a_{\alpha_{\mathrm{fl}}}(\mathfrak{x}) + \mathsf{Y}^a_{\alpha_{\mathrm{fl}}}(\mathfrak{y})$ be the norm of the linear and quadratic flow data of Definition 4.8. For every $R < \infty$ there are $T_R \le T_0$, $a_R > 0$, $\varepsilon_R > 0$, and C_R such that, on $\{\mathcal{B}_{K,s} \le R\}$ and for $a \le \varepsilon \le \varepsilon_R$, $a \le a_R$, and $\sigma\varepsilon^2 < T_R$, the representative A_{fl} of the Wilson flow in the damping gauge of Theorem 24.5 satisfies*

$$A_{\mathrm{fl}} = \mathfrak{x} + \mathfrak{y} + v, \qquad \|v(t)\|_\infty \le C_R (t + a^2)^{\kappa_{\mathrm{fl}}}.$$

Write $Y_i = (a/\sqrt{\eta}) \sum_{e \in \ell_{i,\varepsilon,a,x}} \operatorname{sgn}(e) A_{\mathrm{fl},e}(\sigma\varepsilon^2)$ and $D_i = (a/\sqrt{\eta}) \sum_{e \in \ell_{i,\varepsilon,a,x}} |A_{\mathrm{fl},e}(\sigma\varepsilon^2)|$. Then, uniformly in x, the lattice offset, K, and s, whenever the same radius and domains apply,

$$\max_i |Y_i - I_i^{K,s}| \le C_R \varepsilon^{3/2 - 2\alpha_{\mathrm{fl}}}, \qquad \max_i \big(D_i + |I_i^{K,s}|\big) \le C_R \varepsilon^{-\beta_{\mathrm{fl}}},$$

$$\max_i \Big| \frac{1 - \frac{1}{2}\operatorname{tr} \operatorname{Hol}_{\ell_{i,\varepsilon,a,x}} \Psi^{(N_K)}_{\sigma\varepsilon^2} U}{\eta} - \frac{|I_i^{K,s}|^2}{8} \Big| \le C_R \varepsilon^{2 - 3\alpha_{\mathrm{fl}}}.$$

The representative stays in the logarithmic chart on the whole time interval:

$$a \sup_{t \le T_R} \|A_{\mathrm{fl}}(t)\|_\infty \le C_R \big\{ a^{1-\alpha_{\mathrm{fl}}} + a^{2-2\alpha_{\mathrm{fl}}} + a (T_R + a^2)^{\kappa_{\mathrm{fl}}} \big\} \longrightarrow 0 \qquad (a \to 0).$$

3. *Enlarge s_0 so that n_{s_0} also satisfies the requirement on the coarse side in Lemma 28.10 for the interval $\{\gamma\}$, the horizon T_0, and the exponent α_{fl}. For each fixed $s \ge s_0$, $\rho_s(R) = \sup_{K \ge s} \nu_{K,s}\{\mathcal{B}_{K,s} > R\} \to 0$ as $R \to \infty$. Let*

$$Z_{\varepsilon,K} = \Big(\frac{1 - \frac{1}{2}\operatorname{tr} \operatorname{Hol}_{\ell_{i,\varepsilon,a,x}} \Psi^{(N_K)}_{\sigma\varepsilon^2} U}{\eta} \Big)_{i=1}^{m}$$

be the vector of normalized Wilson-loop deficits on the fine lattice, and $\mathfrak{q}(I) = (|I_i|^2/8)_{i=1}^m$. For every $f : \mathbb{R}^m \to \mathbb{R}$ with $\|f\|_\infty \le 1$ and $\operatorname{Lip} f \le 1$,

$$\big| \mathbb{E}_{\mu_{N_K,\gamma}} f(Z_{\varepsilon,K}) - \mathbb{E}_{\nu_{K,s}} f(\mathfrak{q}(I^{K,s})) \big| \le 2\mu_{N_K,\gamma}(S^c_{N_K,n_s,\gamma}) + 2\rho_s(R) + C_R \varepsilon^{2 - 3\alpha_{\mathrm{fl}}}.$$

The constants have the dependence described in items (5) and (7) of Definition 4.14, with additional dependence on R, T_0, and α_{fl}. No expectation of an unbounded function is transferred between the two measures.

Proof. Part (1) is an algebraic expansion (Step 1). Part (2) follows from the local theory of the Wilson flow and part (1) (Steps 2–4), and part (3) combines part (2) with the tightness of the flow data and Lemma 5.5 (Step 5).

Step 1: the trace expansion. Let $B_e = v_e^a \tau_a$. Then $\|B_e\|_{\mathrm{op}} = |v_e|/2$, and $|\operatorname{tr} T|/2 \le \|T\|_{\mathrm{op}}$ for every 2×2 matrix T. In the product of the exponential series, the sum of the norms of all terms that are homogeneous of degree p is at most

$$\sum_{\sum_e p_e = p} \prod_e \frac{(s'|v_e|/2)^{p_e}}{p_e!} = \frac{(s'D/2)^p}{p!},$$

by the multinomial identity; this bound has no constant depending on the number of links. The trace of the term of degree one vanishes. The identity $\operatorname{tr}(B_e B_f)/2 = -v_e \cdot v_f/4$ gives the half-trace of the terms of degree two,

$$-\frac{(s')^2}{8} \sum_e |v_e|^2 - \frac{(s')^2}{4} \sum_{e<f} v_e \cdot v_f = -\frac{(s')^2}{8} |Y|^2.$$

The norms of all terms of degree at least three sum to at most $(s'D/2)^3 e^{s'D/2}/6$. Dividing by $(s')^2$ gives the first assertion. For an integrable connection coefficient along a path of finite length, the term of degree p of the ordered exponential series has norm at most $(s'D/2)^p/p!$, since the ordered integral is bounded by $1/p!$ times the integral of the symmetric product of the absolute coefficients. The linear term has trace zero, and the half-trace of the quadratic term is $-(s')^2 |\int v|^2/8$, since the dot product is symmetric on the two ordered halves of the square. The same remainder bound therefore proves the assertion for ordered exponentials, with $D = \int |v|$.

Step 2: the flow. Fix R, T_0, and α_{fl}, and abbreviate $\alpha = \alpha_{\mathrm{fl}}$, $\beta = \beta_{\mathrm{fl}} = \alpha - 1/2$, and $\kappa = \kappa_{\mathrm{fl}} = 1 - 3\alpha/2$. The event $\{\mathcal{B}_{K,s} \le R\}$ controls exactly the linear heat evolution, the quadratic Picard term, and all first differences required in Theorem 24.5. Indeed, the linear part is $P_\cdot^a A^{K,s}$, and the quadratic part is the ordered quadratic term of the damped Wilson equation on the same fine lattice. Restricting the horizon only decreases these norms, and the difference arrays include every forward and backward difference in the norm required by that theorem. We use its bound on the remainder of one solution and its identification with the Wilson flow, reducing its lifespan to $T_R \le T_0$ if necessary. It gives the decomposition and the bound on v, and it identifies $e^{aA_{\mathrm{fl}}}$ with the Wilson flow up to a time-dependent periodic lattice gauge transformation. The trace of a closed loop is unchanged by this gauge transformation.

The chart condition holds on the whole interval, not only at time zero. Indeed, by the two bounds on the flow data and the remainder bound, we have

$$a \sup_{t \le T_R} \|A_{\mathrm{fl}}(t)\|_\infty \le C_R\{a^{1-\alpha} + a^{1-2\beta} + a(T_R + a^2)^\kappa\}.$$

Since $1 - 2\beta = 2 - 2\alpha > 0$ and $\kappa > 0$, all terms tend to zero as $a \to 0$. The last term takes its maximum at the final time, where it is $O_R(a)$ for fixed T_R. We choose a_R below the threshold of the chart and $\varepsilon_R \le 1$ with $\sigma\varepsilon_R^2 < T_R$.

Step 3: the circulation of the flowed field. At $t_\varepsilon = \sigma\varepsilon^2$ and $a \le \varepsilon$ we have $\sigma\varepsilon^2 \le t_\varepsilon + a^2 \le (\sigma + 1)\varepsilon^2$. By the definitions of the norms and the remainder bound, we have

$$\|\mathfrak{x}(t_\varepsilon)\|_\infty \le C_\sigma R\varepsilon^{-\alpha}, \qquad \|\mathfrak{y}(t_\varepsilon)\|_\infty \le C_\sigma R\varepsilon^{-2\beta}, \qquad \|v(t_\varepsilon)\|_\infty \le C_{R,\sigma}\varepsilon^{2\kappa}. \tag{32.27}$$

The physical length of the cycle divided by $\sqrt{\eta}$ is at most $C_\ell\sqrt{\varepsilon}/g_3$. Since $I_i^{K,s} = (a/\sqrt{\eta})\sum_e \mathrm{sgn}(e)\mathfrak{x}_e(t_\varepsilon)$, multiplying the last two estimates by this factor, we obtain

$$\max_i |Y_i - I_i^{K,s}| \le C_R\{\varepsilon^{1/2-2\beta} + \varepsilon^{1/2+2\kappa}\} \le C_R\varepsilon^{3/2-2\alpha},$$

where we used $1/2 - 2\beta = 3/2 - 2\alpha$ and $1/2 + 2\kappa = 5/2 - 3\alpha > 3/2 - 2\alpha$. Multiplying all three estimates in (32.27) by the same factor, we find $D_i + |I_i^{K,s}| \le C_R\varepsilon^{-\beta}$. These constants depend on the fixed $g_3 > 0$, as allowed.

Step 4: the Wilson-loop deficit. For each oriented link we take $v_e = \mathrm{sgn}(e)aA_{\mathrm{fl},e}/\sqrt{\eta}$ and $s' = \sqrt{\eta}$ in the trace expansion of Step 1. A link traversed against its orientation contributes the inverse of the positive link variable, whose logarithm is the negative of the positive one. The resulting D and Y are exactly D_i and Y_i. In particular,

$$\sqrt{\eta}\, D_i \le C_R\varepsilon^{1-\alpha}, \qquad \sqrt{\eta}\, D_i^3 \le C_R\varepsilon^{2-3\alpha},$$

and the exponential in the trace bound is bounded for small ε. Further,

$$\big||Y_i|^2 - |I_i^{K,s}|^2\big| \le (|Y_i| + |I_i^{K,s}|)|Y_i - I_i^{K,s}| \le C_R\varepsilon^{2-3\alpha}.$$

This proves the bound on the deficit. It is deterministic on the event $\{\mathcal{B}_{K,s} \le R\}$ and uses only the length bound on the cycles, so it is uniform over x, the lattice offset, K, and s for which the same radius and domains apply.

Step 5: radius tightness and removal of the conditioning. For each fixed $s \ge s_0$, we apply Lemma 28.10, in particular (28.18), with coarse side n_s, remaining depth $K - s$, and $I = \{\gamma\}$; the enlargement of s_0 in part (3) makes n_s an admissible coarse side for that lemma. Its fine potential is $A^{K,s} = a^{-1}\widehat{\theta}_{K-s}$, by the definition of the reconstruction, and no interpolation is involved at the fine lattice. Its tightness statement on the horizon T_0 is exactly $\rho_s(R) \to 0$, at fixed s and fixed positive coupling.

Finally, $Z_{\varepsilon,K}$ is a gauge-invariant function of the fine configuration, so on the small-field event it takes the same value whether it is evaluated on the link variables of the fine configuration or on the reconstructed representative. Lemma 5.5, applied to the bounded function $f(Z_{\varepsilon,K})$, shows that replacing $\mu_{N_K,\gamma}$ by $\nu_{K,s}$ costs at most $2\mu_{N_K,\gamma}(S^c_{N_K,n_s,\gamma})$; the event has positive probability, since $\nu_{K,s}$ is defined. Under $\nu_{K,s}$, the difference between $f(Z_{\varepsilon,K})$ and $f(\mathfrak{q}(I^{K,s}))$ is at most $C_R\varepsilon^{2-3\alpha}$ on $\{\mathcal{B}_{K,s} \le R\}$, where the dimension m is absorbed into C_R, and at most two on its complement. Adding these bounds proves the last inequality. □

The next lemma restates the Gaussian limits in the form in which they are applied below. Since every input of the preceding proofs is available at each fixed coupling $0 < \gamma \le \Gamma$ by Lemma 31.10, it requires no new argument.

Lemma 32.12. *In the setting of Lemma 31.10, every fixed $s \ge s_0$ satisfies the conclusion of Theorem 32.8. Moreover, for every fixed $q \ge 0$ the conclusions of Lemma 32.10 hold, including the essential supremum over the values of the first q RG steps, with $\mathcal{F}_q = \sigma(\widehat{\theta}_0, w_1, \dots, w_q)$. In particular, with $C \sim N(0, \Sigma \otimes I_3)$,*

$$\lim_{\varepsilon\downarrow 0} \limsup_{K\to\infty} \sup_{x,\,\mathit{offset}} \operatorname{ess\,sup} d_{\mathrm{BL}}\big(\mathrm{Law}(I^{K,s} \mid \mathcal{F}_q), \mathrm{Law}(C)\big) = 0,$$

and the same holds after applying $\mathfrak{q}(v) = (|v_i|^2/8)_{i=1}^m$. The constants have the dependence described in item (5) of Definition 4.14, with additional dependence on q; nothing is asserted for q growing with ε or K. Definition 32.1 and the conclusions of Lemmas 32.4 to 32.6 and 32.11 also hold in this setting; for the last one, only the radius tightness at a fixed coarse side is used.

Proof. We check the inputs of the local tools. Lemmas 32.4 and 32.5 are finite-dimensional statements; the conditional fluctuation kernels satisfy the Poincaré hypothesis by Proposition 32.3(1). The decomposition in Lemma 32.6 is an exact finite sum, and its cutoff estimates use the conditions on the number of blocks and on the means, and the decay of the cutoff error, which hold by Step 1 of the proof of Theorem 31.5 and (31.9). Hence every cutoff estimate holds with the same constants. Lemma 32.11 uses Theorem 24.5 for the linear and quadratic flow data on the fine lattice, the identity of Theorem 15.13 between the reconstructed field and the fine configuration, and the trace expansion; these deterministic arguments need no bound on γ, and the removal of the radius uses (28.18). None of these checks uses a limiting random variable for the flow data.

By Lemma 31.10, all five inputs of Proposition 32.3 hold at this fixed coupling, with the stated independence of s. The proofs of Theorem 32.8 and Lemma 32.10 therefore apply with these inputs. In the conditional proof, the pathwise bounds of Proposition 32.3(4) are kept, not restarted; the coarse-field term after q steps is bounded by $Cb_{\max}\sqrt{\varepsilon}/(g_3 a_q)$, and the coarse covariance by $Cn_{s+q}^2\varepsilon/H_{s+q}$. Thus q remains fixed before both limits. □

32.6 The continuum limit

We now pass to the continuum measure, with the limits in the order displayed at the beginning of the section. The final limit removes the conditioning error, whose available bound $F_\Gamma(n_s)$ tends to zero as $s \to \infty$ and is independent of K. In the theorem below, the class $[n_*]$ in the subscript of $\lambda_{L,\gamma,[n_*]}$ is the class $[n_*]_M$ of Theorem 29.28.

Theorem 32.13. *Fix $L > 0$, an odd integer $M \geq 3$, an equivalence class $[n_*]_M$ of lattice sides, and any $\gamma > 0$. Choose $\Gamma \geq \gamma$ and the parameters of Theorem 23.5, and let $\lambda_{L,\gamma,[n_*]}$ be the continuum measure of Theorem 2.3. For every finite family of smooth closed loops $\ell_1, \dots, \ell_m$ in $\mathbb{R}^3$ and every $\sigma > 0$, put $g_3^2 = \gamma / L$ and*

$$Z_\varepsilon(x) = \Big(\frac{1 - W_{x+\varepsilon\ell_i, \sigma\varepsilon^2}}{g_3^2 \varepsilon} \Big)_{i=1}^m .$$

Then

$$\lim_{\varepsilon \downarrow 0} \sup_{x \in \mathbb{T}_L^3} d_{\mathrm{BL}}\big(\mathrm{Law}_{\lambda_{L,\gamma,[n_*]}} Z_\varepsilon(x), \mathrm{Law}\, Z_G \big) = 0, \qquad Z_G = (|C_i|^2/8)_{i=1}^m,$$

where the three color components of C are independent centered Gaussian vectors with covariance

$$\Sigma_{ih} = (2\pi)^{-3} \int_{\mathbb{R}^3} \frac{e^{-2\sigma|k|^2}}{|k|^2} \overline{\widehat{J}_i(k)} \cdot \widehat{J}_h(k)\, dk, \qquad \widehat{J}_i(k) = \int_0^1 \dot{\ell}_i(v) e^{-ik\cdot \ell_i(v)}\, dv.$$

Degenerate covariances and repeated or reversed loops are allowed. Every sufficiently large fixed coarse index satisfies the conclusion of Lemma 32.12 for the circulations. The limits are taken in the order described above. The constants have the dependence described in item (5) of Definition 4.14; no rate uniform in the coupling, in the coarse index, or as $\sigma \downarrow 0$ is asserted.

Proof. By Lemma 31.2(1), $k \cdot \widehat{J}_i(k) = 0$ and $|\widehat{J}_i(k)| \leq C \min\{|k|, 1\}$, so Σ is finite, real, symmetric, and positive semidefinite, and its positive square root constructs the three color vectors. This is the covariance of Definition 31.1, and Lemma 31.10 shows that the results of Section 31 apply at this coupling.

Let $F_\Gamma(n_s)$ be the supremum over $0 < \gamma' \leq \Gamma$ of the bound in Theorem 23.10 at coarse side n_s; by that theorem, $F_\Gamma(n_s) \to 0$ as $s \to \infty$. At fixed s, let $\rho_s(R)$ be the radius tail of Lemma 32.11, which tends to zero by (28.18) with $I = \{\gamma\}$. Let $Z_{\varepsilon,K}(x)$ be the vector of normalized Wilson-loop deficits of Lemma 32.11 for the lattice cycles approximating $x + \varepsilon\ell_i$. By that lemma and the triangle inequality, for every function f with $\|f\|_\infty \leq 1$ and $\mathrm{Lip}\, f \leq 1$,

$$\big| \mathbb{E}_{\mu_{N_K,\gamma}} f(Z_{\varepsilon,K}(x)) - \mathbb{E} f(Z_G) \big| \leq 2F_\Gamma(n_s) + 2\rho_s(R) + C_R \varepsilon^{2-3\alpha_{\mathrm{fl}}} + d_{\mathrm{BL}}\big(\mathrm{Law}_{\nu_{K,s}}\, \mathfrak{q}(I^{K,s}), \mathrm{Law}\, Z_G \big).$$

For each fixed $\varepsilon > 0$ and x, the joint convergence of flowed Wilson loops, Theorems 29.23 and 29.24, implies that $\mathbb{E}_{\mu_{N_K,\gamma}} f(Z_{\varepsilon,K}(x)) \to \mathbb{E}_{\lambda_{L,\gamma,[n_*]}} f(Z_\varepsilon(x))$ as $K \to \infty$; the lattice cycles are admissible approximations of the loops $x + \varepsilon\ell_i$. The bound above is uniform in x and the lattice offset. To see that the identification at fixed ε is compatible with this uniformity, we translate the lattice sequence by a lattice vector so that the base point is near a vertex, and use the allowed lattice approximation of the translated loops; the additional swept area is $O(\varepsilon a)$, a connecting path and its inverse do not change the trace, and the laws of the fine lattice and of the continuum are invariant under translations. At fixed ε all deficits lie in the compact cube $[0, 2/(g_3^2\varepsilon)]^m$, so a finite net of test functions with $\|f\|_\infty \le 1$ and $\operatorname{Lip} f \le 1$ makes this passage to the limit uniform over all such test functions. We let $K \to \infty$, and then $\varepsilon \downarrow 0$, using the statement for $\mathfrak{q}(I^{K,s})$ in Lemma 32.12 and $2 - 3\alpha_{\mathrm{fl}} > 0$. What remains is $2F_\Gamma(n_s) + 2\rho_s(R)$. We let $R \to \infty$ at fixed s, and only then $s \to \infty$. This proves the theorem. For the full class $[n_*]_M$, the continuum measure was identified in Theorem 29.28; the argument above needs only one subsequence $N_K = n_* B^K$. □

32.7 A uniform short-distance variance lower bound

Corollary 32.14. *In Theorem 32.13 take $m = 1$, write $\ell = \ell_1$ and $v = \Sigma_{11}$, and let $D_{\varepsilon,x} = (1 - W_{x+\varepsilon\ell,\sigma\varepsilon^2})/(g_3^2\varepsilon)$. Then*

$$\liminf_{\varepsilon\downarrow 0} \inf_x \operatorname{Var}_{\lambda_{L,\gamma,[n_*]}} D_{\varepsilon,x} = \liminf_{\varepsilon\downarrow 0} \inf_x \frac{\operatorname{Var}_{\lambda_{L,\gamma,[n_*]}} W_{x+\varepsilon\ell,\sigma\varepsilon^2}}{(g_3^2\varepsilon)^2} \ge \frac{3v^2}{32}. \tag{32.28}$$

If $\widehat{J}_1 \not\equiv 0$, then $v > 0$, and for every sufficiently small fixed ε the observable $W_{x+\varepsilon\ell,\sigma\varepsilon^2}$ is not almost surely constant, for every x. The threshold for ε may depend on the data listed in item (5) of Definition 4.14.

Proof. For independent copies D and D' of a random variable with finite variance, $\operatorname{Var} D = \mathbb{E}(D - D')^2/2$. At each fixed scale the variance is finite, since $0 \le D_{\varepsilon,x} \le 2/(g_3^2\varepsilon)$. By Theorem 32.13, the laws of pairs of independent copies converge, uniformly in x, to the law of independent copies Z and Z' of $v\chi_3^2/8$, where χ_3^2 is a chi-square variable with three degrees of freedom: on compact sets, approximate a bounded continuous function of two variables by finite sums of products, and use the uniform tails of the marginals on the complement. Applying this to the bounded continuous function $\min\{A, (u-u')^2\}$ and then letting $A \to \infty$, we find that the left side of (32.28) is at least $\mathbb{E}(Z - Z')^2/2 = \operatorname{Var} Z = 3v^2/32$. A nonzero continuous function $\widehat{J}_1$ is nonzero on an open set, on which the positive weight in the integral defining v has positive integral, so $v > 0$. The strict lower bound then gives a common threshold below which the observable is not constant. No moment is transferred from the Wilson measure. □

32.8 Asymptotic independence from macroscopic Wilson loops

The small-loop deficits are asymptotically independent of fixed macroscopic flowed Wilson loops. The reason is that a macroscopic flowed Wilson loop is approximated, on the same probability space, by a function of the first q RG steps, while the Gaussian limit of Lemma 32.10 holds uniformly conditionally on these steps. The approximation must have constants independent of the higher derivatives of the finite-scale reconstruction, which grow with q; this is provided by the stability estimate Corollary 25.11.

Lemma 32.15. *In the setting of Definition 32.1, fix $s \geq s_0$, finitely many piecewise C^1 closed loops Γ_r in $\mathbb{T}^3_L$, times $t_r > 0$, and characters χ_r of finite-dimensional representations of $\mathrm{SU}(2)$, $1 \leq r \leq b$. Use lattice approximations $\Gamma_{r,K}$ of bounded physical length and swept area $O(a)$, and let*

$$V_{K,r} = \chi_r\big(\mathrm{Hol}_{\Gamma_{r,K}} \Psi_{t_r}^{(N_K)}(U)\big), \qquad \mathbf{V}_K = (V_{K,r})_{r=1}^{b}$$

be the vector of flowed Wilson loops of the fine configuration under $\nu_{K,s}$. There are $\mathcal{F}_q$-measurable random vectors $\mathbf{V}_K^{[q]}$ on the same probability space such that

$$\lim_{q\to\infty} \limsup_{K\to\infty} \mathbb{E}_{\nu_{K,s}} \min\big\{2, |\mathbf{V}_K - \mathbf{V}_K^{[q]}|\big\} = 0. \tag{32.29}$$

The vector $\mathbf{V}_K^{[q]}$ consists of the flowed Wilson loops of the finite-scale reconstruction $U_K^{[q]} = \exp(a I_{a_q \to a} A_q)$, with the same times and lattice paths, where $A_q = \widehat{\theta}_q / a_q$, $a_q = H_s B^{-q}$, and $I_{a_q\to a}$ is the cardinal interpolation of Lemma 28.10. The approximation does not depend on ε, x, or any test function of the small loops. The constants have the dependence described in item (7) of Definition 4.14, at fixed coupling and with s fixed.

Proof. We use the cardinal interpolation and the notation of Lemma 28.10. On the fine lattice the linear and quadratic flow data of the finite-scale reconstruction are

$$x_q = P_\cdot^a I_{a_q\to a} A_q, \qquad y_q = \mathcal{Q}_a(x_q, x_q).$$

This is the cardinal interpolation of the field at index q; we do not claim that it equals the nonlinear zero-fluctuation continuation. Its link variables and their Wilson flow are defined without any small-field event of their own. At fixed q the field A_q ranges over a fixed compact set and its cardinal interpolation is smooth, as in Lemma 28.4. The construction uses only $\widehat{\theta}_q$, so it is $\mathcal{F}_q$-measurable.

Fix $1/2 < \alpha < 2/3$ and a horizon $T_0 > 0$, and set

$$d_{K,q} = \mathsf{X}_\alpha^a(x_{K-s} - x_q) + \mathsf{Y}_\alpha^a(y_{K-s} - y_q), \qquad \mathcal{B}_{K,s} = \mathsf{X}_\alpha^a(x_{K-s}) + \mathsf{Y}_\alpha^a(y_{K-s}).$$

Both norms include every first difference on the common lattice. The exact bilinear decomposition of $y_{K-s} - y_q$ and the estimates of Lemma 28.10 on the linear term and on the two mixed terms and the quadratic term imply, for every $\delta > 0$,

$$\lim_{q\to\infty} \sup_{K \geq s+q} \nu_{K,s}\{d_{K,q} > \delta\} = 0. \tag{32.30}$$

The linear part is bounded by Markov's inequality and the tail estimate for the linear term; the quadratic part is bounded by the sum of the tail estimates for the mixed terms and the quadratic term, each with threshold $\delta/3$. The radius tail $\rho_s(R) = \sup_{K \geq s} \nu_{K,s}\{\mathcal{B}_{K,s} > R\}$ tends to zero by (28.18).

Fix R and $0 < \delta \leq 1$. On the event $\{\mathcal{B}_{K,s} \leq R,\ d_{K,q} \leq \delta\}$, both pairs of flow data lie in the ball of radius $R+1$. Let

$$0 < u_R < \tfrac{1}{4} \min\{T_{R+1}, T_0, \min_r t_r\}.$$

By the estimate of Theorem 24.5 for differences on the same lattice, we have, at this early time,

$$\|A_a(u_R) - A_a^{[q]}(u_R)\|_{C_a^1} \leq C_{R,u_R} d_{K,q}, \qquad \|A_a(u_R)\|_{C_a^1} + \|A_a^{[q]}(u_R)\|_{C_a^1} \leq K_R < \infty.$$

Both damping gauges start at the identity. By Theorems 15.13 and 24.5, the first damped solution represents the Wilson flow of the fine configuration, and the second represents the Wilson flow started from $U_K^{[q]}$. Since each character is gauge invariant, both flows may be restarted from these damped fields at time u_R for the purpose of evaluating their later traced loops.

We apply Corollary 25.11 with radius $R+1$, input horizon T_0, early time $u = u_R$, $\tau = \min_r t_r$, $T = \max_r t_r$, the given characters, and the fixed bounds on the lengths of the paths. Both pairs of flow data lie in that ball on the stated event, in the same initial gauge, and the discrepancy in that corollary is exactly $d_{K,q}$. It gives

$$|\mathbf{V}_K - \mathbf{V}_K^{[q]}| \leq C_{R,u_R,\mathbf{t},\mathbf{\Gamma},\chi}\, d_{K,q}, \tag{32.31}$$

for the whole vector of characters, including the normalized fundamental traces. The constant in (32.31) depends on the radius and on the fixed waiting time, but not on the higher derivatives of the finite-scale reconstruction, which grow with q; the threshold for the mesh depends on R, u_R, and the loops, but not on q. For all sufficiently fine meshes it follows that

$$\mathbb{E} \min\{2, |\mathbf{V}_K - \mathbf{V}_K^{[q]}|\} \leq 2\rho_s(R) + 2\nu_{K,s}\{d_{K,q} > \delta\} + C_R\delta. \tag{32.32}$$

We let $q \to \infty$ at fixed s, R, and δ, using (32.30), then $\delta \downarrow 0$, and finally $R \to \infty$. This proves the claim on the same finite probability spaces, without coupling different cutoffs. □

Theorem 32.16. *Assume the setting of Theorem 32.13. Fix, independently of ε and x, a finite vector $\mathbf{V} = (\chi_r(\mathrm{Hol}_{\Gamma_r} X(t_r)))_{r=1}^b$ of characters of holonomies of the trajectory X around piecewise C^1 closed loops Γ_r in $\mathbb{T}_L^3$ at times $t_r > 0$. Then*

$$\limsup_{\varepsilon \downarrow 0} \sup_x \sup_{\|f\|_{\mathrm{BL}}, \|g\|_{\mathrm{BL}} \leq 1} \left| \mathbb{E}_{\lambda_{L,\gamma,[n_*]}}[f(Z_\varepsilon(x)) g(\mathbf{V})] - \mathbb{E} f(Z_G)\, \mathbb{E}_{\lambda_{L,\gamma,[n_*]}} g(\mathbf{V}) \right| = 0, \tag{32.33}$$

where $\|h\|_{\mathrm{BL}} = \max\{\|h\|_\infty, \operatorname{Lip} h\}$. *The constants have the dependence described in items (5) and (7) of Definition 4.14. Nothing is asserted uniformly in a growing family of macroscopic loops, as* $\gamma \downarrow 0$, *as* $\sigma \downarrow 0$, *or as* $\min_r t_r \downarrow 0$.

Proof. On the probability space of $\nu_{K,s}$, let $\mathbf{V}_K$ be the vector of flowed Wilson loops of the fine configuration and $\mathbf{V}_K^{[q]}$ the $\mathcal{F}_q$-measurable approximation of Lemma 32.15. By (32.32),

$$a_{K,s,q} = \mathbb{E}_{\nu_{K,s}} \min\{2, |\mathbf{V}_K - \mathbf{V}_K^{[q]}|\} \le 2\rho_s(R) + 2\nu_{K,s}\{d_{K,q} > \delta\} + C_R\delta.$$

Let $\Delta_{K,s,q}(\varepsilon)$ be the deterministic bound on the conditional bounded-Lipschitz distance for the squared circulations from Lemma 32.12. Multiplying by $g(\mathbf{V}_K^{[q]})$, which is $\mathcal{F}_q$-measurable and bounded by one, and using the tower property, we obtain an error of at most $\Delta_{K,s,q}(\varepsilon)$ against the independent Gaussian factor. Replacing $g(\mathbf{V}_K^{[q]})$ by $g(\mathbf{V}_K)$, in the product and in its separate expectation, costs at most $2a_{K,s,q}$, since $|g(\mathbf{V}_K) - g(\mathbf{V}_K^{[q]})| \le \min\{2, |\mathbf{V}_K - \mathbf{V}_K^{[q]}|\}$. Replacing $f(\mathfrak{q}(I^{K,s}))$ by the deficits $f(Z_{\varepsilon,K}(x))$ costs at most $2\rho_s(R) + C_R\varepsilon^{2-3\alpha_{\mathrm{fl}}}$, by Lemma 32.11. Finally, two applications of Lemma 5.5, to the product and to the expectation of $g(\mathbf{V}_K)$, transfer from $\nu_{K,s}$ to $\mu_{N_K,\gamma}$ at a cost of at most $4F_\Gamma(n_s)$. Thus the error under the Wilson measure is at most

$$4F_\Gamma(n_s) + 2\rho_s(R) + C_R\varepsilon^{2-3\alpha_{\mathrm{fl}}} + 2a_{K,s,q} + \Delta_{K,s,q}(\varepsilon).$$

For fixed ε, x, f, and g, the joint law of the deficits and of $\mathbf{V}_K$ under $\mu_{N_K,\gamma}$ converges to the corresponding joint law under $\lambda_{L,\gamma,[n_*]}$, by Theorems 29.23 and 29.24 applied to the union of the two families of loops. The bound above does not depend on x, f, or g, so it also bounds their supremum after this identification. We let $K \to \infty$, then $\varepsilon \downarrow 0$ at fixed s, R, q, and δ. Next we let $q \to \infty$, using (28.17) to bound $\sup_{K \ge s+q} \nu_{K,s}\{d_{K,q} > \delta\}$, then $\delta \downarrow 0$, then $R \to \infty$ at this s, and finally $s \to \infty$. The bound vanishes. □

32.9 Asymptotic independence from bounded Borel functions

The last theorem extends the asymptotic independence to every fixed bounded Borel function on the trajectory space. In the terminology of Rényi, as discussed by Aldous and Eagleson [4], it says that $Z_\varepsilon(x)$ converges mixingly relative to the Borel σ-field of $\mathcal{X}_L$, uniformly in x and over test functions with $\|f\|_{\mathrm{BL}} \le 1$: the limit is independent of that σ-field. This only names the mode of convergence.

Theorem 32.17. *Assume the setting of Theorem 32.16, with* $Z_\varepsilon(x)$ *and* Z_G *as there. For every bounded Borel function* $H : \mathcal{X}_L \to \mathbb{R}$, *chosen independently of* ε *and* x,

$$\lim_{\varepsilon \downarrow 0} \sup_{x \in \mathbb{T}_L^3} \sup_{\|f\|_{\mathrm{BL}} \le 1} \left| \mathbb{E}_{\lambda_{L,\gamma,[n_*]}}[f(Z_\varepsilon(x))H] - \mathbb{E} f(Z_G)\, \mathbb{E}_{\lambda_{L,\gamma,[n_*]}} H \right| = 0.$$

In particular this holds for $H = h \circ i^{1,2}$, for every bounded Borel function h on the space of Sobolev trajectories, where $i^{1,2}$ is the map of Definition 29.1. The approximations in the proof may depend on H and $\lambda_{L,\gamma,[n_*]}$. There is no supremum over H, and nothing is asserted for H depending on ε or x.

Proof. Let $\lambda = \lambda_{L,\gamma,[n_*]}$. By Theorem 29.3, the Borel σ-field of $\mathcal{X}_L$ is generated by the determining coordinates w_j, $j \geq 1$, which are normalized fundamental Wilson loops. We first approximate H in $L^1(\lambda)$ by a bounded Lipschitz function of finitely many of these coordinates.

The events A whose indicators can be approximated in $L^1(\lambda)$ by $[0,1]$-valued Borel functions of finitely many coordinates form a σ-field. It contains the events determined by finitely many coordinates. It is closed under complements, and under finite unions by the Lipschitz map $(s_1, s_2) \mapsto \max\{s_1, s_2\}$ on $[0,1]^2$; for a countable union, we first truncate to a finite union whose indicator has small L^1 error, by continuity of λ from below. Hence it contains every Borel event, and approximating a bounded H by simple functions gives approximations bounded by $\|H\|_\infty$.

Next, a bounded Borel function of $(w_1, \dots, w_q)$ can be approximated by a bounded Lipschitz function. The law λ_q of $(w_1, \dots, w_q)$ is a Borel probability measure on the compact cube $[-1,1]^q$, so each Borel set A has a compact subset K_A and a relatively open superset O_A with $\lambda_q(O_A \setminus K_A)$ arbitrarily small; indeed, the sets with this property contain the compact sets, by continuity from above along decreasing open neighborhoods, and they form a σ-field, by choosing the outer errors summably and keeping finitely many inner compact sets. If the relative complement of O_A is nonempty, it has positive distance d_A from K_A, and $\max\{0, 1 - \operatorname{dist}(\cdot, K_A)/d\}$ with $0 < d < d_A$ is Lipschitz, equals one on K_A, and vanishes outside O_A; in the remaining cases we use constants. Its $L^1(\lambda_q)$ distance to $\mathbf{1}_A$ is at most $\lambda_q(O_A \setminus K_A)$. Applying this to the indicators of a simple function, extending the resulting Lipschitz function from the cube to $\mathbb{R}^q$ by the distance-cone extension, and truncating at $\|H\|_\infty$, we find for every $\eta' > 0$ a bounded Lipschitz function $g_{\eta'} : \mathbb{R}^q \to \mathbb{R}$ with

$$H_{\eta'} = g_{\eta'}(w_1, \dots, w_q), \qquad \mathbb{E}_\lambda |H - H_{\eta'}| < \eta'.$$

Both q and the Lipschitz constant are fixed before the loops shrink.

Let $\mathbf{V} = (2w_1, \dots, 2w_q)$. Its components are fundamental characters, so $\mathbf{V}$ satisfies the hypotheses of Theorem 32.16. We apply that theorem to the function $z \mapsto g_{\eta'}(z/2)/A_{\eta'}$, where $A_{\eta'} = \max\{1, \|g_{\eta'}\|_\infty, \operatorname{Lip} g_{\eta'}\}$. This function is bounded by one and has Lipschitz constant at most one, and its value at $\mathbf{V}$ is $H_{\eta'}/A_{\eta'}$. Multiplying by $A_{\eta'}$, we find that the expression in the statement with $H_{\eta'}$ in place of H tends to zero, uniformly in x and in f with $\|f\|_{\mathrm{BL}} \leq 1$. Replacing $H_{\eta'}$ by H costs at most

$$\mathbb{E}_\lambda |H - H_{\eta'}| + |\mathbb{E} f(Z_G)| \, \mathbb{E}_\lambda |H - H_{\eta'}| \leq 2\eta',$$

uniformly in these variables and in ε. We let $\varepsilon \downarrow 0$ and then $\eta' \downarrow 0$. Finally, by Theorem 30.5 the map $i^{1,2}$ is Borel, so $h \circ i^{1,2}$ is an admissible H; no continuity of $i^{1,2}$ is used. □

Proof of Theorem 2.8. Let $\lambda = \lambda_{L,\gamma,[n]_M}$ with n divisible by four. Part (1) is Theorem 32.13, applied with $n_* = n$ and any $\Gamma \geq \gamma$; its covariance Σ is the matrix in (2.6). Part (2) is Corollary 32.14. Part (3) is Theorem 32.17, applied to the bounded Borel function G. □

PART XI

Weak-coupling asymptotics near flat connections

SECTION 33

Laplace asymptotics near flat connections

This section begins the analysis of weak coupling. We fix the side n of the coarse lattice and let $\gamma \to 0$. Then every scale of the renormalization group is weakly coupled. The law of the coarsest field V^{JK} under the conditioned Wilson measure is the stopped measure $\widehat{\nu}$ of Theorem 14.8, and it concentrates near the flat configurations on Λ_n. We study $\widehat{\nu}$ by Laplace's method on the finite-dimensional space $\mathcal{U}_n$, uniformly in the number of averaging steps that precede the coarse lattice. The results are used in Sections 34 to 36 to prove Theorems 2.9 to 2.11; nothing in Sections 31 and 32 depends on this section.

We prove four main results. Throughout, $g^2 = \gamma/n$ and $E_0 = 4\sum_p q(V_p)$ is the Wilson action on Λ_n.

1. Theorem 33.10 shows that the partition function of $\widehat{\nu}$ is comparable to γ^{3n^3-1}, uniformly in the depth, for sufficiently small $\gamma \in (0, \gamma_0]$, and that the integral of $e^{-\kappa E_0/g^2}$ over all of $\mathcal{U}_n$ is $O(g^{6n^3-2})$.
2. Theorem 33.8 constructs coordinates near the trivial connection in which the $6n^3 - 6$ nonconstant directions transverse to the gauge orbits are nondegenerate for the effective classical action E_d, uniformly in d, and integrates these directions out by Laplace's method. What remains is an integral over the nine constant coordinates in the Lie algebra, in which E_d has no positive quadratic part; this integral is analyzed in Section 36.
3. Theorem 33.11 computes, in the continuum and on the lattice, the covariance Σ_θ of the linearization of the flowed cycle logarithms at a regular flat connection, under the Gaussian measure defined by the quadratic part of the Yang–Mills action.
4. Proposition 33.12 shows that, transverse to a compact patch of regular flat configurations, $\widehat{\nu}$ is Gaussian up to errors of order $\sqrt{\gamma}$. It also shows that the lattice covariance Σ_θ^N decomposes exactly into a contribution of the coarse lattice and contributions of the normalized fluctuation variables of each RG step.

The tools are an analysis of the cone of commuting triples in $\mathfrak{su}(2)^3$ (Lemma 33.2) and a Laplace estimate in normal coordinates that is uniform in an arbitrary tangential weight (Lemma 33.3 and Corollary 33.4).

The inputs from earlier sections are the stopped density and its estimates (Theorem 14.8), the description of flat configurations and of their refinements (Lemma 14.9), bounds and symmetries of the effective classical actions (Theorems 10.11 and 11.2

and Lemma 10.8), the triangular change of variables of one RG step (Theorem 12.13), and the reconstruction maps $\widehat{\mathscr{R}}$ (Theorem 15.12).

The exponent. We work modulo based gauge transformations, that is, lattice gauge transformations equal to the identity at a fixed root. This quotient still carries the simultaneous conjugation of the cycle holonomies; see Section 33.1. The exponent $6n^3 - 2$ in Theorem 33.10 comes from the following count of directions.

Directions	Dimension	Factor in $\widehat{Z}$
Based gauge quotient of $\mathcal{U}_n$	$6n^3 + 3$	ambient space
Regular flat configurations	$5 = 3 + 2$	tangential volume
Normal to a regular flat configuration	$6n^3 - 2$	g^{6n^3-2}
Nonconstant transverse directions near I	$6n^3 - 6$	g^{6n^3-6}
Normal to a nonzero commuting triple	4	g^4
Constant modes with $\lvert X\rvert \le \sqrt{g}$	9	$g^{9/2}$

The five flat directions are the three holonomy angles and the two directions in which simultaneous conjugation moves the common axis of the holonomies. Near a regular flat configuration the normal integral directly gives g^{6n^3-2}. Near the trivial connection the count is less direct. We write the constant modes as $X \in \mathfrak{su}(2)^3 \simeq \mathbb{R}^9$. On the constant configurations $e^{X/n}$ the Wilson action is comparable to $Q(X)/n$, where $Q(X) = 2^{-1}\sum_{i<j}|x_i \times x_j|^2$, and Q vanishes exactly on the cone $\mathcal{M}$ of commuting triples. Near a commuting triple of radius ϱ, the volume element of $\mathcal{M}$ is proportional to $\varrho^4\, d\varrho$, and in each of the four normal directions Q grows like $\varrho^2|b|^2/2$ (Lemma 33.2). Each normal Gaussian width is then of order g/ϱ, and the product $\varrho^4(g/\varrho)^4\, d\varrho = g^4\, d\varrho$ integrates to $O(g^4)$ over a bounded range of radii. The ball of radius $\sqrt{g}$ contributes only $O(g^{9/2})$. With the factor g^{6n^3-6} from the nonconstant directions, we again obtain $g^{6n^3-2} = (\gamma/n)^{3n^3-1}$. The same count indicates that the measure $e^{-Q(Y)}dY$ on $\mathbb{R}^9$ has infinite total mass. This divergence of the bosonic three-matrix SU(2) integral is known [7, 121, 122]; see also Section 36. In this section e^{-Q} is only integrated over bounded sets.

Two normalizations. Two normalizations of the exponent occur. In Section 33.5 we write the exponent of $\widehat{\nu}$ as E_d/g^2 and use the Laplace parameter $g^2 = \gamma/n$. In Sections 33.6 and 33.8 we write it as nE_d/γ and use the Laplace parameter γ. These are the same exponent, since

$$\frac{E_d}{g^2} = \frac{nE_d}{\gamma}.$$

Accordingly the normal Hessians H_d in Lemma 33.9 and Proposition 33.12 are Hessians of nE_d.

Constants. Constants in results about $\widehat{\nu}$ may depend on n, L, M, J, the parameter tuple Θ, the coupling bound γ_0, and on fixed patches, charts, and test functions, all chosen before

γ; this is the dependence described in item (6) of Definition 4.14. They do not depend on the depth d or on $\gamma \in (0, \gamma_0]$. All spectral gaps are proved at the fixed side n, and we prove no bound that is uniform in n. The general Laplace estimates of Section 33.4 state their own finite-dimensional dependence, as in item (1) of Definition 4.14.

Earlier work. Laplace's method in the directions normal to a manifold of critical points is rooted in Bott's theory of nondegenerate critical manifolds [31]; see Wong [155] for the asymptotic analysis of integrals. We use no specific theorem from these works. The estimates we need are uniform in the depth, the coupling, and an arbitrary tangential weight; they are proved in Section 33.4, and the singular tip of the cone of commuting triples is treated separately. The special role of the constant modes on a finite periodic torus in weak-coupling analysis is well known in the physics literature; see Lüscher [127] and the computation of the zero-momentum contribution to Wilson loops by Coste, González-Arroyo, Jurkiewicz, and Korthals Altes [62]. For the gradient flow on a four-dimensional periodic torus, Fodor, Holland, Kuti, Nógrádi, and Wong [91] separate constant and nonconstant modes and their flowed observables. These works do not treat the fixed-side three-dimensional normalization or the covariance of flowed based cycle logarithms. The decomposition of the covariance across scales in Proposition 33.12 is completion of squares for Gaussian measures; the work lies in matching it with the constrained coordinates of the renormalization group.

33.1 Preliminaries

Lie-algebra conventions. We use the conventions of Section 4. Let $\tau_a = -i\sigma_a/2$, where σ_a are the Pauli matrices, and write $X = x^a \tau_a \in \mathfrak{su}(2)$. Then $[\tau_a, \tau_b] = \epsilon_{abc}\tau_c$, $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$, and $|X| = |x|$, so that the bracket corresponds to the cross product in $\mathbb{R}^3$. We identify $\mathfrak{su}(2)^3$ with $\mathbb{R}^9$ by $X = (x_1, x_2, x_3)$ and use the Euclidean norm $|X|$. In exponential coordinates

$$dH(e^X) = \frac{1}{16\pi^2}\,\mathrm{sinc}^2(|X|/2)\,d^3X, \qquad q(e^X) = 1 - \cos(|X|/2) = \frac{|X|^2}{8} + O(|X|^4),$$

by (4.2). The logarithm Log is the inverse of exp on $\{|X| < 2\pi\}$.

Based gauge transformations. Fix a root $x_* \in \Lambda_n$. A lattice gauge transformation k is *based* if $k_{x_*} = I$. The group of based gauge transformations acts freely on $\mathcal{U}_n$, and we call the quotient the based quotient. Constant gauge transformations are not divided out: they act on the based quotient, and on the holonomies around the three coordinate cycles based at x_* they act by simultaneous conjugation. In the proof of Theorem 33.8 we show, using a spanning tree, that the based quotient is diffeomorphic to $\mathrm{SU}(2)^{2n^3+1}$, of dimension $6n^3 + 3$, and that the based gauge transformations carry Haar measure of total mass one. A configuration V is flat if $V_p = I$ for every plaquette p.

Smooth charts. We use the following routine facts about local coordinates several times. We include the short proofs because the dependence of the constants matters below.

Lemma 33.1. *1. Let F be a smooth map from a neighborhood of $p \in \mathbb{R}^k$ into $\mathbb{R}^k$ such that $T = DF(p)$ is invertible. Then there exist a closed ball $\overline{B}$ about p and an open neighborhood W of $F(p)$ such that F maps an open subset of $\overline{B}$ diffeomorphically onto W. The radii of $\overline{B}$ and W and the bounds for every fixed derivative of the inverse depend only on $\|T^{-1}\|$ and on bounds for the derivatives of F on a fixed neighborhood of p.*

2. Let $\mathcal{Q}$ be a smooth Riemannian manifold, for instance an open subset of $\mathbb{R}^k$ with the Euclidean metric or a compact Lie group with a bi-invariant metric, and let Exp *be its exponential map. Let $\Sigma \subset \mathcal{Q}$ be a smooth embedded submanifold and let $P \subset \Sigma$ be compact. Then there exists $r_0 > 0$ such that the normal exponential map $(y, v) \mapsto \mathrm{Exp}_y(v)$, defined for y in a compact neighborhood of P in Σ and $v \perp T_y\Sigma$ with $|v| < r_0$, is injective and is a local diffeomorphism with smooth positive Jacobian. Further, there exist finitely many open sets $W_l \Subset W_l^+ \subset \Sigma$ with $P \subset \bigcup_l W_l$ and smooth orthonormal frames $(\nu_{l,1}, \dots, \nu_{l,k'})$ of the normal bundle over W_l^+ such that all derivatives of fixed order of the maps $(y, v) \mapsto \mathrm{Exp}_y(\sum_\alpha v_\alpha \nu_{l,\alpha}(y))$, $v \in \mathbb{R}^{k'}$, and of their inverses are bounded on the closures of $\{(y, v) : y \in W_l,\ |v| \le r_0/2\}$. The transitions between these trivializations preserve y and are orthogonal in v, so integrals over the tube $\{\mathrm{Exp}_y(v) : y \in P,\ |v| < r_0/2\}$ can be computed with a partition of unity depending only on y. In $\mathbb{R}^k$ with the Euclidean metric, $\mathrm{Exp}_y(v) = y + v$.*

Proof. For (1), we translate p to zero and write $T^{-1}(F(h) - F(0)) = h + r(h)$, where $Dr(0) = 0$. On a sufficiently small closed ball $\overline{B}$ we have $\|Dr\| \le 1/2$. For $|z|$ less than half the radius of $\overline{B}$, the iterates $h_{k+1} = z - r(h_k)$, $h_0 = 0$, stay in $\overline{B}$, and their successive differences decrease by the factor $1/2$. Their limit is the unique solution h in $\overline{B}$ of $h + r(h) = z$. Differentiating this identity, we find that the inverse is smooth with $\|Dh\| \le 2$, and repeated differentiation bounds every fixed derivative in terms of those of r.

For (2), we take a point of P, an orthonormal tangent frame of Σ near it, and fixed smooth vector fields of $\mathcal{Q}$ whose projections onto the normal spaces are linearly independent at this point. The Gram–Schmidt formulas applied to these projections give a smooth orthonormal normal frame, since their denominators stay positive near the point. Since the differential of Exp_y at 0 is the identity, the derivative of $(y, v) \mapsto \mathrm{Exp}_y(\sum_\alpha v_\alpha \nu_\alpha(y))$ at $v = 0$ consists of the tangent and normal frames. It is invertible, with determinant one in orthonormal coordinates. By (1), applied in a chart of $\mathcal{Q}$, the map is a local diffeomorphism near $v = 0$, and its Jacobian is smooth and positive. Since P is compact, finitely many such charts cover a compact neighborhood P' of P in Σ, and we choose open sets $W_l \Subset W_l^+$ inside them, covering P, and a common normal radius. All maps extend smoothly past the closures of the smaller sets, and their derivatives of every fixed order are bounded there. It remains to prove injectivity for a common radius. If it failed

on $\{(y,v) : y \in P',\ |v| < 1/k\}$ for every k, there would be distinct points (y_k, v_k) and (y'_k, v'_k) of the normal bundle with $|v_k|, |v'_k| < 1/k$ and $\mathrm{Exp}_{y_k}(v_k) = \mathrm{Exp}_{y'_k}(v'_k)$. Passing to subsequences, $y_k \to y$ and $y'_k \to y'$ in P', and the common image converges both to y and to y'. Then $y = y'$, and both preimages converge to $(y, 0)$, near which the map is a local diffeomorphism and hence injective; this is a contradiction. Thus, for a small r_0, the base point y is a function on the tube, the transitions between the orthonormal frames act on v by orthogonal matrices depending on y, and a partition of unity in y handles overlaps. □

33.2 The stopped measure at a fixed side

Fix $L > 0$, $n \in 4\mathbb{N}$, an odd integer $M \geq 3$, an integer $J \geq 1$, and the tuple Θ of construction parameters of Theorem 23.2. Let ρ, r, and E_* be the analytic radius, the radius of the small-field domains $B_r \subset D_r$, and the bound for the effective interactions in Theorem 14.8. For a depth $d = kJ$, $k \geq 0$, and $\gamma > 0$, we set $g^2 = \gamma/n$ and consider the stopped measure

$$d\widehat{\nu}_{n,d,\gamma}(V) = \widehat{Z}^{-1}_{n,d,\gamma} \mathbf{1}_{D_r}(V) \exp\Big(-\frac{E_{d,\mathrm{sat}}(V)}{g^2} - \overline{R}_{d,\gamma}(V)\Big) dH_n(V), \qquad \overline{R}_{d,\gamma} = R_{d,\gamma} - R_{d,\gamma}(I), \tag{33.1}$$

of Theorem 14.8, with the indicator of D_r and the Haar disintegration factors of the construction. Here $E_{d,\mathrm{sat}}$ and the subscript sat in general denote the gauge-invariant extension from B_r to D_r, $R_{d,\gamma}$ is the effective interaction, $\overline{R}_{d,\gamma}$ is the centered effective interaction, and $\widehat{Z}_{n,d,\gamma}$ is the partition function. The depths $d = kJ$ are those at which the recursion of Theorem 14.8 defines an effective interaction. Subtracting $R_{d,\gamma}(I)$ fixes the normalization constant. If Z^{raw} denotes the partition function before centering, that is, the integral with $R_{d,\gamma}$ in place of $\overline{R}_{d,\gamma}$, then $\widehat{Z}_{n,d,\gamma} = e^{R_{d,\gamma}(I)} Z^{\mathrm{raw}}$ (Theorem 33.10).

For $d = JK$ and $N = nB^K$, the measure $\widehat{\nu}_{n,JK,\gamma}$ is the law of the stopping field V^{JK} under the conditioned Wilson measure $\nu_{N,n,\gamma}$ (Theorem 14.8). For a stopped measure at an intermediate side this identification fails, since the conditioned Wilson measure also imposes the remaining coarser restrictions (Definition 4.4). Our results concern the density (33.1) and make no identification of the latter kind.

We use the following properties. All constants and domains in them are independent of d and of $\gamma \in (0, \gamma_0]$.

(P1) There exists $\kappa > 0$ such that $E_{d,\mathrm{sat}} \geq \kappa E_0$ on D_r for every integer $d \geq 0$, whether or not an effective interaction is attached to d (Theorem 14.8; see also Theorem 11.2).

(P2) The effective classical action satisfies $E_d \geq 0$ (Theorem 11.2). It is invariant under lattice gauge transformations and exactly invariant under lattice translations on a fixed chart (Lemma 10.8). It vanishes at constant commuting configurations satisfying

the smallness condition of Lemma 33.7, as proved there. Its derivatives through order five are bounded uniformly in d on a fixed larger link chart (Theorem 10.11).

(P3) The centered effective interaction satisfies $|\overline{R}_{d,\gamma}| \le C_n$ on D_r (Theorem 14.8), and its first derivatives in fixed charts are bounded on the larger chart (Lemma 33.6).

(P4) The coupling bound $\gamma_0 > 0$ satisfies $\sqrt{\gamma_0/n} \le g_*$, where g_* is the coupling bound of Theorem 14.8, and, with $t_0 = \sqrt{\gamma_0/(Mn)}$ and $p(t) = A_{\mathrm{w}} \log(1/t)$,

$$p(t) \ge p_0, \qquad C_J t p(t) < \frac{1}{2}, \qquad \frac{C_J(tp(t)^3 + tp(t))}{1 - C_J t p(t)} \le \sigma_0 \qquad (0 < t \le t_0), \tag{33.2}$$

where p_0, C_J, and σ_0 are the constants of Theorem 12.13.

Since the largest coupling of an RG step with output side $m \ge n$ is $\sqrt{\gamma/(Mm)} \le t_0$, the conditions (33.2) contain the coupling conditions of Theorem 12.13 for all these steps. Once n, M, J, and Θ are fixed, (P4) holds for all sufficiently small $\gamma_0 > 0$: the function p is decreasing, $t \log^a(1/t) \to 0$ as $t \to 0$ for $a = 1, 3$, and these two functions are increasing in t when $\log(1/t) > a$. Thus γ_0 is chosen after n, M, J, and Θ, and before γ, and no separate condition is imposed for an individual coupling or depth.

33.3 The cone of commuting triples

Since two elements of $\mathfrak{su}(2)$ commute if and only if they are parallel, the commuting triples in $\mathfrak{su}(2)^3 \simeq \mathbb{R}^9$ are the triples $(\vartheta_1 u, \vartheta_2 u, \vartheta_3 u)$ with $\boldsymbol{\vartheta} \in \mathbb{R}^3$ and $u \in S^2$. If we regard $X \in \mathbb{R}^9$ as the real 3×3 matrix with columns x_1, x_2, x_3, these are exactly the matrices of rank at most one. The following lemma collects the geometry of this cone and the behavior of the quartic Q near it. The distance to the set of matrices of rank at most one is computed from the singular values, as in the rank-one case of the theorem of Eckart and Young [80].

Lemma 33.2. *Set*

$$Q(X) = \tfrac{1}{2} \sum_{i<j} |x_i \times x_j|^2, \qquad \mathcal{M} = \{(\vartheta_i u)_{i=1}^3 : \boldsymbol{\vartheta} \in \mathbb{R}^3,\ u \in S^2\}.$$

Then the following statements hold.

1. *The set $\mathcal{M} \setminus \{0\}$ is a smooth submanifold of $\mathbb{R}^9$ of dimension five. The map $(\boldsymbol{\vartheta}, u) \mapsto (\vartheta_i u)_i$ identifies exactly $(\boldsymbol{\vartheta}, u)$ with $(-\boldsymbol{\vartheta}, -u)$. At $y = (\vartheta_i u)_i$ with $\varrho = |\boldsymbol{\vartheta}| > 0$ and $\omega = \boldsymbol{\vartheta}/\varrho$, the normal space and the induced volume are*

$$\mathcal{N}_y = \Big\{ b = (b_1, b_2, b_3) : b_i \perp u,\ \sum_i \vartheta_i b_i = 0 \Big\}, \qquad \dim \mathcal{N}_y = 4,$$

$$d\mathrm{vol}_{\mathcal{M}} = \tfrac{1}{2} \varrho^4 \, d\varrho \, d\Omega_\omega \, d\Omega_u.$$

Note that $|y| = \varrho$.

2. *There exists an absolute constant $c \in (0,1)$ such that $(y,b) \mapsto y+b$, for $y \in \mathcal{M} \setminus \{0\}$, $b \in \mathcal{N}_y$, and $|b| \le c\varrho$, is injective and is a local diffeomorphism (a normal tube), with Euclidean Jacobian $1 + O(|b|/\varrho)$. The implicit constant is absolute; in particular it is uniform in the directions ω and u.*
3. *For $y \in \mathcal{M} \setminus \{0\}$ and $b \in \mathcal{N}_y$,*

$$Q(y+b) = \tfrac{1}{2}\varrho^2|b|^2 + \tfrac{1}{2}\sum_{i<j}|b_i \times b_j|^2,$$

and for every $X \in \mathbb{R}^9$,

$$Q(X) \ge \tfrac{1}{6}|X|^2 \operatorname{dist}(X,\mathcal{M})^2.$$

4. *Let f be a nonnegative C^5 function on a fixed ball about 0 that vanishes on $\mathcal{M}$ in that ball. Then $Df(0) = 0$, $D^2f(0) = 0$, and $D^3f(0) = 0$. On a smaller fixed ball,*

$$\|D^2f(X)\| \le C|X|^2, \qquad \|D^3f(X)\| \le C|X|,$$

and, for $y \in \mathcal{M} \setminus \{0\}$ and $b \in \mathcal{N}_y$ with $y+b$ in the smaller ball,

$$f(y+b) = \tfrac{1}{2}b^{\mathsf{T}}D^2_\perp f(y)b + O(\varrho|b|^3 + |b|^4),$$

where $D^2_\perp f(y)$ is the Hessian of f at y restricted to $\mathcal{N}_y$. If f_4 is the homogeneous quartic Taylor polynomial of f at 0 and $y = \varrho\widehat{y}$ with $|\widehat{y}| = 1$, then $D^2_\perp f(\varrho\widehat{y}) = \varrho^2 D^2_\perp f_4(\widehat{y}) + O(\varrho^3)$. If, in addition, $f \ge c_0 Q$ on the ball, then $D^2_\perp f_4(\widehat{y}) \ge c_0 I_4$. The constants depend only on the radii of the two balls, on $\|f\|_{C^5}$, and, in the last assertion, on c_0.

The last hypothesis $f \ge c_0 Q$ cannot be omitted: the function Q^2 is nonnegative and vanishes on $\mathcal{M}$, but its normal Hessian vanishes on all of $\mathcal{M}$.

Proof. We begin with part (1). The tangent vectors at $y_i = \vartheta_i u \neq 0$ have the form

$$t_i = \eta_i u + \vartheta_i v, \qquad \eta \in \mathbb{R}^3,\ v \perp u.$$

These five parameters are independent, and orthogonality to every η and every v gives exactly the asserted normal space. In the parameters $(\boldsymbol{\vartheta}, u)$ the induced metric is

$$\sum_i |d(\vartheta_i u)|^2 = |d\boldsymbol{\vartheta}|^2 + \varrho^2|du|^2.$$

Each nonzero matrix of rank one has exactly two preimages: a nonzero column determines the common line, and the two unit vectors on this line determine all three coefficients. We obtain the volume element $\varrho^2\, d^3\boldsymbol{\vartheta}\, d\Omega_u/2$, which is the displayed formula after polar coordinates in $\boldsymbol{\vartheta}$.

For part (2), the tube is taken over the nonzero stratum, and its radius is proportional to $|y|$. We first restrict to $1/2 < |y| < 2$. By Lemma 33.1(2), finitely many coordinate patches

of this compact piece of $\mathcal{M}$ carry smooth orthonormal normal frames with bounded first derivatives on smaller patches. In such a frame the derivative of $(y, b) \mapsto y + b$ at $b = 0$ is the orthogonal sum of the tangent and normal frames, and its volume determinant is one. At $b \neq 0$ its tangent columns differ by $O(|b|)$, and the determinant is $1 + O(|b|)$. We now check that the base point is unique. If $b \in \mathcal{N}_y$, the matrix $X = y + b$ satisfies

$$XX^{\mathsf{T}} = \varrho^2 uu^{\mathsf{T}} + bb^{\mathsf{T}}, \qquad bb^{\mathsf{T}}u = 0,$$

since $\sum_i \vartheta_i b_i = 0$ and $b_i \perp u$. If $|b| < c\varrho$ with $c < 1$, the largest eigenvalue ϱ^2 of XX^{T} is simple, and its eigenline is exactly $\mathbb{R}u$. Projecting the columns of X onto this line recovers y, and the two choices of sign of u give the same matrix y. Together with the local invertibility of the derivative, this proves that the map is a normal tube. Since the map commutes with the dilations $(y, b) \mapsto (\lambda y, \lambda b)$, the bound on the unit stratum implies the Jacobian bound $1 + O(|b|/\varrho)$ with an absolute constant.

For part (3), let $b \in \mathcal{N}_y$ and expand

$$(\vartheta_i u + b_i) \times (\vartheta_j u + b_j) = u \times (\vartheta_i b_j - \vartheta_j b_i) + b_i \times b_j.$$

The first summand is perpendicular to u and the second is parallel to u, so they are orthogonal. Since $\vartheta_i b_j - \vartheta_j b_i \perp u$, we have $|u \times (\vartheta_i b_j - \vartheta_j b_i)| = |\vartheta_i b_j - \vartheta_j b_i|$, and the Lagrange identity gives

$$\sum_{i<j} |\vartheta_i b_j - \vartheta_j b_i|^2 = |\boldsymbol{\vartheta}|^2 |b|^2 - \Big|\sum_i \vartheta_i b_i\Big|^2 = \varrho^2 |b|^2.$$

This proves the formula for $Q(y + b)$. For the global bound, let $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0$ be the singular values of X. Since $|x_i \times x_j|^2$ is the sum of the squares of the 2×2 minors in the columns i and j, the quantity $2Q(X)$ is the sum of the squares of all 2×2 minors of X. This sum is invariant under left and right multiplication by orthogonal matrices, and evaluating it on a diagonal matrix gives

$$2Q(X) = \sigma_1^2\sigma_2^2 + \sigma_1^2\sigma_3^2 + \sigma_2^2\sigma_3^2.$$

For a fixed unit vector u, the closest matrix with all columns parallel to u has columns $(u \cdot x_i)u$, and its squared distance to X is $|X|^2 - u^{\mathsf{T}}XX^{\mathsf{T}}u$. Diagonalizing XX^{T}, we see that the minimum over u is $\sigma_2^2 + \sigma_3^2 = \operatorname{dist}(X, \mathcal{M})^2$. Since $\sigma_1^2 \geq |X|^2/3$, we obtain

$$Q(X) \geq \tfrac{1}{2}\sigma_1^2(\sigma_2^2 + \sigma_3^2) \geq \tfrac{1}{6}|X|^2 \operatorname{dist}(X, \mathcal{M})^2.$$

For part (4), since the origin is a minimum of f, we have $Df(0) = 0$, and $D^2 f(0)$ is positive semidefinite. Since $\mathcal{M}$ is a cone, the restriction of f to every line through a matrix of rank one vanishes near zero, and the quadratic form $D^2 f(0)$ vanishes on every matrix

of rank one. If a positive semidefinite form H satisfies $v^{\mathsf{T}} H v = 0$, then $Hv = 0$, as one sees by diagonalizing H. Since the matrices with a single nonzero entry have rank one and span $\mathbb{R}^9$, we conclude that $D^2 f(0) = 0$. The cubic Taylor term vanishes as well, that is, $D^3 f(0) = 0$: on each line through the origin it would otherwise take opposite signs on the two sides of zero, contradicting nonnegativity, and a cubic form that vanishes on every line is zero.

By Taylor's formula with integral remainder, we have, on a smaller concentric ball,

$$\|D^2 f(X)\| \le C|X|^2, \qquad \|D^3 f(X)\| \le C|X|, \qquad D^2 f(X) = D^2 f_4(X) + O(|X|^3).$$

For example, the last error is at most $\sup\|D^5 f\||X|^3/6$; the other constants use the bounds for the fourth and fifth derivatives on the larger ball. Since $f(y) = 0$ is a minimum at every $y \in \mathcal{M}$ in the ball, we have $Df(y) = 0$. Using Taylor's formula on the segment from y to $y + b$ and the bound for $D^3 f$, we find a cubic remainder at most $C(\varrho + |b|)|b|^3$, which proves the normal expansion. The Hessian of the quartic f_4 is homogeneous of degree two, and $\mathcal{N}_{\varrho\widehat{y}} = \mathcal{N}_{\widehat{y}}$; this proves the expansion at $y = \varrho\widehat{y}$. If $f \ge c_0 Q$, the function $f - c_0 Q$ is nonnegative and vanishes at y, and its Hessian at this point is positive semidefinite. By the exact normal formula in part (3), the normal Hessian of Q at y is $\varrho^2 I_4$, and we obtain

$$D^2_\perp f(\varrho\widehat{y}) \ge c_0 \varrho^2 I_4.$$

Dividing by ϱ^2 and letting $\varrho \downarrow 0$, we obtain $D^2_\perp f_4(\widehat{y}) \ge c_0 I_4$. This completes the proof. □

33.4 Laplace asymptotics in normal coordinates

We now prove the Laplace estimates used in the normal directions of the regular flat configurations and in the nonconstant directions near the trivial connection. The main point is uniformity. The phase, the amplitude, and the tangential weight may depend on an arbitrary index, which in the applications records the depth and the coupling, and the tangential weight need not converge or have any regularity. We therefore prove the estimates for each fixed tangential point and then integrate them against the tangential marginal of the measure itself.

Lemma 33.3. *Let P be a compact patch of a smooth manifold with a smooth volume measure $d\mathrm{vol}$, and let a Euclidean vector bundle of rank $d \ge 1$ over P be given with a finite atlas of smooth orthonormal trivializations. Fix $0 < b_1 < b_0$ and a cutoff $\chi_y(b)$ with $0 \le \chi \le 1$, jointly Borel in every trivialization, equal to one for $|b| \le b_1$ and supported in a compact subset of the tube $\{|b| < b_0\}$. Let ι range over an arbitrary index set and let $0 < \delta \le \delta_0$. For each ι and δ let $V_\iota(y, b)$ and $a_{\iota,\delta}(y, b)$ be real functions on the tube, jointly Borel in every trivialization, and*

assume that the Hessian $L_\iota(y) = D_b^2 V_\iota(y,0)$ exists and is a measurable function of y. Assume that on the support of χ

$$\begin{aligned} V_\iota(y,0) = D_b V_\iota(y,0) = 0, \qquad & c_0 I \le L_\iota(y) \le C_0 I, \\ |V_\iota(y,b) - \tfrac{1}{2} b^{\mathsf T} L_\iota(y) b| \le C_0 |b|^3, \qquad & V_\iota(y,b) \ge c_0 |b|^2, \\ c_0 \le a_{\iota,\delta}(y,b) \le C_0, \qquad & |a_{\iota,\delta}(y,b) - a_{\iota,\delta}(y,0)| \le C_0 |b|. \end{aligned}$$

The amplitude $a_{\iota,\delta}$ includes the Jacobian of the tube coordinates, so that the reference measure is $d\mathrm{vol}(y)\, db$, where db is Lebesgue measure in an orthonormal trivialization. Let $h_{\iota,\delta} \ge 0$ be any measurable function on P with $0 < \int_P h_{\iota,\delta}\, d\mathrm{vol} < \infty$; no regularity in y and no convergence as ι or δ varies is assumed. Let

$$d\mathsf P_{\iota,\delta} = Z_{\iota,\delta}^{-1} h_{\iota,\delta}(y) \chi_y(b) a_{\iota,\delta}(y,b) e^{-V_\iota(y,b)/\delta}\, d\mathrm{vol}(y)\, db$$

be the normalized measure, and set

$$z = \delta^{-1/2} L_\iota(y)^{1/2} b, \qquad \alpha_{\iota,\delta}(y) = \frac{a_{\iota,\delta}(y,0)}{\sqrt{\det L_\iota(y)}}.$$

Let G_d be a standard Gaussian vector in $\mathbb R^d$. Then for every family $P_y(z)$ of polynomials of a fixed degree whose coefficients are bounded measurable functions of y, and every bounded measurable function f on P,

$$\begin{aligned} Z_{\iota,\delta} \mathbb E_{\mathsf P_{\iota,\delta}}[f(y) P_y(z)] &= (2\pi\delta)^{d/2} \Big[\int_P f\, h_{\iota,\delta}\, \alpha_{\iota,\delta}\, \mathbb E P_y(G_d)\, d\mathrm{vol} + \mathcal R_{\iota,\delta}(f,P) \Big], \\ |\mathcal R_{\iota,\delta}(f,P)| &\le C_P \sqrt{\delta} \int_P |f|\, h_{\iota,\delta}\, d\mathrm{vol}. \end{aligned} \tag{33.3}$$

The error is additive, also when $\mathbb E P_y(G_d) = 0$. For $y \in P$ let $\mathsf P_{\iota,\delta}(\cdot \mid y)$ be the probability measure on the fiber with density proportional to $\chi_y(b) a_{\iota,\delta}(y,b) e^{-V_\iota(y,b)/\delta}$. It is defined for every y, also where $h_{\iota,\delta}(y) = 0$, and it is a version of the conditional law of b given y under $\mathsf P_{\iota,\delta}$. Moreover, the following hold.

1. *For every $y \in P$, $|\mathbb E_{\mathsf P_{\iota,\delta}}[P_y(z) \mid y] - \mathbb E P_y(G_d)| \le C_P \sqrt{\delta}$, and $\mathbb E_{\mathsf P_{\iota,\delta}}[|z|^k \mid y] \le C_k$ for every fixed real $k \ge 0$.*
2. *Let $d = 4$ and $q_\iota = b^{\mathsf T} L_\iota(y) b/2$. Then for $j = 0, 1, 2$,*

$$|\mathbb E_{\mathsf P_{\iota,\delta}}[f(y)(q_\iota/\delta)^j] - m_j \mathbb E_{\mathsf P_{\iota,\delta}} f(y)| \le C_j \|f\|_\infty \sqrt{\delta}, \qquad (m_0, m_1, m_2) = (1, 2, 6).$$

3. *Let $d = 4$, let L_∞ be a measurable field of symmetric matrices with $c_0 I \le L_\infty(y) \le C_0 I$, and let $q_\infty = b^{\mathsf T} L_\infty(y) b/2$. Then the estimate in (2) holds with q_∞ in place of q_ι, up to the additional error $C_j \|f\|_\infty \|L_\iota - L_\infty\|_\infty$.*

The constants depend only on d, on b_0, b_1, and the atlas, on c_0 and C_0, and on the degree and the coefficient bounds of P_y (respectively on k and j). They are independent of ι, δ, and δ_0, and of $h_{\iota,\delta}$, its total mass, and its variation. In applications at a fixed side n, the geometric data may depend on n.

Corollary 33.4. *Assume the hypotheses of Lemma 33.3. For $y \in P$ let $r_{\iota,\delta,y}$ be the Lebesgue density of $z = \delta^{-1/2} L_\iota(y)^{1/2} b$ under $\mathsf{P}_{\iota,\delta}(\cdot \mid y)$, and let $\phi_d(z) = (2\pi)^{-d/2} e^{-|z|^2/2}$. For every real $s \geq 0$ there exists $C_s < \infty$ such that, for all $0 < \delta \leq \delta_0$,*

$$\sup_{\iota,y} \int_{\mathbb{R}^d} (1 + |z|^s) |r_{\iota,\delta,y}(z) - \phi_d(z)| \, dz \leq C_s \sqrt{\delta}. \tag{33.4}$$

The density $r_{\iota,\delta,y}$ is defined by its normal integral at every y, also where the tangential weight vanishes, and assertions integrated against the tangential marginal of $\mathsf{P}_{\iota,\delta}$ hold almost everywhere with respect to that marginal; no assumption on the behavior of this marginal as ι and δ vary is made. The constant C_s has the dependence described in Lemma 33.3, and in addition depends on s.

Suppose in addition that $V_\iota(y,b) \geq c|b|^2$ and $c \leq a_{\iota,\delta}(y,b) \leq C$ for all $|b| < b_0$, with $c, C > 0$ independent of ι, y, and δ. In the following whole-tube conclusions, the constants C_s may additionally depend on c and C; they remain independent of ι, y, δ, δ_0, and the tangential weight. Then there exists $c' > 0$, depending only on the data of Lemma 33.3 and on c and C, such that for $0 < \delta \leq 1$

$$\int_{|b|<b_0} (1 + |z|^s)(1 - \chi_y(b)) a_{\iota,\delta}(y,b) e^{-V_\iota(y,b)/\delta} \, db \leq C_s \delta^{d/2} e^{-c'/\delta}. \tag{33.5}$$

Let $r^{\mathrm{tube}}_{\iota,\delta,y}$ be defined as $r_{\iota,\delta,y}$, but with χ_y replaced by the indicator of the tube $\{|b| < b_0\}$. Then $\int (1 + |z|^s) |r^{\mathrm{tube}}_{\iota,\delta,y} - r_{\iota,\delta,y}| \, dz \leq C_s e^{-c'/\delta}$ for $0 < \delta \leq 1$, and (33.4) holds for $r^{\mathrm{tube}}_{\iota,\delta,y}$ for all $0 < \delta \leq \delta_0$.

In the application of Proposition 33.12, the rank is $6n^3 - 2$, $\delta = \gamma$, the phase is $nE_{JK}(V(y,v))$, and $L = H^{\mathrm{act}}_K(y) = D^2_v[nE_{JK}(V(y,v))]|_{v=0}$. In this application the additional hypotheses on the whole tube hold, and all constants are independent of K.

Both results follow from an estimate at a fixed tangential point, which we prove first.

Lemma 33.5. *Assume the hypotheses of Lemma 33.3. Fix a trivialization of the atlas, a point y in its domain, and the parameters ι and δ. Write $L = L_\iota(y)$ and $a_0 = a_{\iota,\delta}(y,0)$. Let*

$$u_y(z) = \chi_y(b) a_{\iota,\delta}(y,b) e^{-V_\iota(y,b)/\delta}, \qquad b = \delta^{1/2} L^{-1/2} z,$$

extended by zero outside the tube, and let $Z_y = \int u_y(z) \, dz$ and $r_y = u_y / Z_y$. Then $Z_y > 0$, and for every real $s \geq 0$ there exist $C_s < \infty$ and $c > 0$, with the dependence described in Lemma 33.3 and additionally on s, such that the following hold for all $0 < \delta \leq \delta_0$.

1. $\int (1+|z|^s)\,|u_y(z) - a_0 e^{-|z|^2/2}|\,dz \le C_s\sqrt{\delta}$.
2. *If* $\delta \le 1$, *then* $Z_y \ge c$.
3. $\int (1+|z|^s)\,|r_y(z) - \phi_d(z)|\,dz \le C_s\sqrt{\delta}$.
4. $\int |z|^s r_y(z)\,dz \le C_s$.

Proof. Let $\varepsilon = \sqrt{\delta}$. On the support of χ we set

$$A(z) = V_\iota(y, \varepsilon L^{-1/2} z)/\delta, \qquad B(z) = |z|^2/2.$$

Since $|b| \le \varepsilon|z|/\sqrt{c_0}$ and $|z|^2 = b^{\mathsf T} L b/\delta$, the hypotheses imply, with constants determined by c_0 and C_0, that

$$A(z) \ge c|z|^2, \qquad B(z) \ge c|z|^2, \qquad |A(z) - B(z)| \le C\varepsilon|z|^3, \qquad |a_{\iota,\delta}(y,b) - a_0| \le C\varepsilon|z|.$$

The identity $e^{-A} - e^{-B} = -(A-B)\int_0^1 e^{-[(1-t)B + tA]}\,dt$ yields

$$|e^{-A} - e^{-B}| \le |A - B| e^{-\min(A,B)}. \tag{33.6}$$

This bound does not require $|A-B|$ to be small. Here the lower bound $V_\iota \ge c_0|b|^2$ on the whole support of χ enters, since the cubic Taylor bound alone would not control the integrand far from $b = 0$. On the support of χ we obtain

$$|a_{\iota,\delta}e^{-A} - a_0 e^{-B}| \le a_{\iota,\delta}|e^{-A} - e^{-B}| + |a_{\iota,\delta} - a_0|e^{-B} \le C\varepsilon(|z| + |z|^3)e^{-c|z|^2}.$$

We have $u_y - a_0 e^{-B} = \chi(a_{\iota,\delta}e^{-A} - a_0 e^{-B}) - (1-\chi)a_0 e^{-B}$, and the last term is supported where $|b| > b_1$, hence where $|z| \ge \sqrt{c_0}\, b_1/\varepsilon$. For $0 < \delta \le 1$ we have

$$\begin{aligned}\int (1+|z|^s)\,|u_y - a_0 e^{-B}|\,dz &\le C\varepsilon \int (1+|z|^s)(|z| + |z|^3)e^{-c|z|^2}\,dz \\ &\quad + C_0 \int_{|z| \ge \sqrt{c_0} b_1/\varepsilon} (1+|z|^s) e^{-|z|^2/2}\,dz \le C_s\varepsilon.\end{aligned}$$

In the last inequality, we used polar coordinates and $\int_0^\infty \varrho^m e^{-c\varrho^2}\,d\varrho < \infty$ for $m > -1$ for the first integral; for the second, we absorbed half of the exponential into the polynomial to obtain the bound $C_s e^{-c'/\delta}$, and we used that $u^{-1}e^{-c'/u^2}$ is bounded for $u \in (0,1]$. This proves (1) for $\delta \le 1$. If $\delta > 1$, then u_y is supported in $|z| \le C_0^{1/2} b_0$ and $0 \le u_y \le C_0$, since $V_\iota \ge 0$ on the support of χ. Both $\int (1+|z|^s)u_y\,dz$ and $a_0\int (1+|z|^s)e^{-|z|^2/2}\,dz$ are then bounded by a constant C_s, and $C_s \le C_s\sqrt{\delta}$. This proves (1) in general. Note that no dependence on δ_0 enters.

For (2), let $0 < \delta \le 1$ and set $\varrho_0 = \min(b_1/2, \sqrt{\delta})$. On the ball $|b| \le \varrho_0$ we have $\chi = 1$, $a_{\iota,\delta} \ge c_0$, and $V_\iota \le C_0|b|^2/2 + C_0|b|^3 \le C_0(1/2 + b_0)\delta$. Changing variables, we obtain

$$Z_y = \delta^{-d/2}(\det L)^{1/2} \int \chi_y a_{\iota,\delta} e^{-V_\iota/\delta}\,db \ge \delta^{-d/2} c_0^{d/2}\, c_0\, e^{-C_0(1/2+b_0)}\, \omega_d \varrho_0^d \ge c,$$

where ω_d is the volume of the unit ball and we used $\varrho_0^d \delta^{-d/2} \geq \min(b_1/2, 1)^d$. The same integral over $|b| \leq b_1/2$ shows that $Z_y > 0$ for every δ.

For (3) with $\delta \leq 1$, let $A_0 = a_0(2\pi)^{d/2}$, so that $a_0 e^{-|z|^2/2} = A_0 \phi_d(z)$. Then

$$r_y - \phi_d = \frac{u_y - a_0 e^{-|z|^2/2}}{Z_y} + a_0 e^{-|z|^2/2}\Big(\frac{1}{Z_y} - \frac{1}{A_0}\Big).$$

By (1) with $s = 0$ we have $|Z_y - A_0| \leq C\sqrt{\delta}$, and $Z_y \geq c$ by (2) and $A_0 \geq c_0(2\pi)^{d/2}$. Inserting these bounds and (1) into the last display, we obtain (3). For $\delta > 1$, we use that r_y and ϕ_d are probability densities and that r_y is supported in $|z| \leq C_0^{1/2} b_0$. Then the left side of (3) is at most $1 + (C_0^{1/2} b_0)^s + \int (1 + |z|^s)\phi_d \leq C_s \leq C_s\sqrt{\delta}$. Finally, (4) follows from (3) for $\delta \leq 1$, and from the bound on the support of r_y for $\delta > 1$. □

Proof of Lemma 33.3. We first check that all densities and conditional laws are well defined. On the spectral interval $[c_0, C_0]$ the maps $L \mapsto L^{1/2}$ and $L \mapsto L^{-1/2}$ are continuous: their binomial series in $I - L/C_0$ converge uniformly because $\|I - L/C_0\| \leq 1 - c_0/C_0 < 1$, and all partial sums are polynomials in the matrix entries. Since the Hessian is measurable, so are z, $\det L_\iota$, and $\alpha_{\iota,\delta}$. Since V_ι, $a_{\iota,\delta}$, and χ are jointly Borel, every normal integrand below is jointly measurable. Integrals over the fiber of nonnegative integrands are measurable functions of y, by integration of nonnegative simple functions followed by monotone limits; signed polynomial integrands are treated through their positive and negative parts, once the bounds below establish absolute integrability. Thus Fubini's theorem applies to the weight $h_{\iota,\delta}$ without any continuity in y.

Next we prove the formula at a fixed tangential point. Fix a trivialization and y, and let u_y, Z_y, and r_y be as in Lemma 33.5. The change of variables $b = \delta^{1/2} L^{-1/2} z$ has Jacobian $\delta^{d/2}(\det L)^{-1/2}$, so with

$$I_y(P) = \int \chi_y(b) a_{\iota,\delta}(y,b) e^{-V_\iota(y,b)/\delta} P_y(z)\, db$$

we have $I_y(P) = \delta^{d/2}(\det L)^{-1/2} \int u_y P_y\, dz$. Since $|P_y(z)| \leq C_P(1 + |z|^{\deg P})$, part (1) of Lemma 33.5 implies that $|\int u_y P_y\, dz - a_0(2\pi)^{d/2} \mathbb{E} P_y(G_d)| \leq C_P\sqrt{\delta}$. Since $(\det L)^{-1/2} \leq c_0^{-d/2}$, we obtain, uniformly in y and ι,

$$I_y(P) = (2\pi\delta)^{d/2}\big[\alpha_{\iota,\delta}(y)\mathbb{E} P_y(G_d) + \mathcal{E}_y(P)\big], \qquad |\mathcal{E}_y(P)| \leq C_P\sqrt{\delta}. \tag{33.7}$$

We multiply (33.7) by $f(y)h_{\iota,\delta}(y)$ and integrate. By Fubini's theorem, the left side becomes $Z_{\iota,\delta}\mathbb{E}_{\mathsf{P}_{\iota,\delta}}[f P_y(z)]$ on the domain of the trivialization, and the error is bounded by $C_P\sqrt{\delta}\int |f| h_{\iota,\delta}$, also when the Gaussian mean vanishes. Orthogonal changes of the normal frame preserve db, the law of G_d, and the positive square root, and they act on z by orthogonal maps. Hence the formula does not depend on the trivialization, and a finite nonnegative partition of unity on P subordinate to the atlas proves (33.3). We

obtain such a partition by choosing coordinate balls whose smaller concentric balls cover P, placing a smooth bump equal to one on each smaller ball, and dividing by the positive sum of the bumps; only the atlas enters its constants. No derivative or convergence of $h_{\iota,\delta}$ has been used.

For the conditional statements, note that $I_y(1) > 0$ for every y by Lemma 33.5(2). The density $\chi_y a_{\iota,\delta} e^{-V_\iota/\delta}/I_y(1)$ of $\mathsf{P}_{\iota,\delta}(\cdot \mid y)$ is jointly Borel, and its product with the tangential density $Z_{\iota,\delta}^{-1} h_{\iota,\delta}(y) I_y(1)$ is the joint density of $\mathsf{P}_{\iota,\delta}$. Thus $\mathsf{P}_{\iota,\delta}(\cdot \mid y)$ is a version of the conditional law, including on the set where $h_{\iota,\delta}$ vanishes. Since z has density r_y under this law, we have

$$|\mathbb{E}[P_y(z) \mid y] - \mathbb{E}P_y(G_d)| \le C_P \int (1 + |z|^{\deg P}) |r_y - \phi_d| \, dz \le C_P \sqrt{\delta}$$

by Lemma 33.5(3), and the moment bound in (1) is Lemma 33.5(4).

For (2), let $d = 4$. Then $q_\iota/\delta = |z|^2/2$ exactly. For independent one-dimensional standard Gaussians, integration by parts gives $\mathbb{E}G_i^2 = 1$ and $\mathbb{E}G_i^4 = 3$. This implies

$$\mathbb{E}(|G_4|^2/2) = 2, \qquad \mathbb{E}(|G_4|^2/2)^2 = \tfrac{1}{4}(4 \cdot 3 + 12) = 6.$$

By (1), the conditional expectations of $(q_\iota/\delta)^j$ given y differ from m_j by at most $C_j\sqrt{\delta}$, uniformly in y. Integrating against the tangential marginal of $\mathsf{P}_{\iota,\delta}$, multiplied by f, we obtain (2).

For (3), let $\eta = \|L_\iota - L_\infty\|_\infty$. Since $|b|^2 \le \delta |z|^2/c_0$, we have pointwise

$$|q_\iota - q_\infty|/\delta \le C\eta |z|^2, \qquad |q_\iota|/\delta + |q_\infty|/\delta \le C|z|^2,$$

and both constants depend only on c_0 and C_0, because L_∞ obeys the same spectral bounds as L_ι. The case $j = 0$ is trivial. For $j = 1$ the first bound and the bounded second moment give the additional error $C\|f\|_\infty \eta$. For $j = 2$ we write $x^2 - y^2 = (x - y)(x + y)$ and use the bounded fourth moment from (1). All constants have the stated dependence. This completes the proof. □

Proof of Corollary 33.4. In a trivialization of the atlas, the density of z under $\mathsf{P}_{\iota,\delta}(\cdot \mid y)$ is $r_{\iota,\delta,y} = r_y$ of Lemma 33.5, and (33.4) is part (3) of that lemma. Since orthogonal changes of the normal frame act on z by orthogonal maps and preserve ϕ_d and $|z|$, the left side of (33.4) does not depend on the trivialization. The density is jointly measurable by the Borel hypotheses and integration over the fiber, as in the proof of Lemma 33.3. Since the estimate holds at every y, integrating it against the tangential marginal of $\mathsf{P}_{\iota,\delta}$ requires no assumption on this marginal. We note that the constant in part (3) of Lemma 33.5 does not depend on δ_0: for $\delta \le 1$ the proof uses the lower bound $Z_y \ge c$, valid on all of $(0, 1]$, and for $\delta > 1$ it uses only that r_y is supported in a fixed ball.

We turn to the additional statements on the whole tube. Let $0 < \delta \le 1$, and let Z^{tube} and Z^χ be the integrals of $a_{\iota,\delta} e^{-V_\iota/\delta}$ over $|b| < b_0$, without and with the factor χ_y. By the

proof of Lemma 33.5(2) we have $Z^{\mathrm{tube}} \geq Z^{\chi} \geq c\delta^{d/2}$. On the region where $\chi_y(b) \neq 1$ we have $|b| \geq b_1$, and the lower bound $V_\iota \geq c|b|^2$ on the whole tube bounds the integrand by $C(1+|z|^s)e^{-c|b|^2/\delta}$, with $|z| \leq C|b|/\sqrt{\delta}$. Absorbing half of the exponential, we obtain

$$\begin{aligned}\int_{b_1 \leq |b| < b_0} (1 + C\delta^{-s/2}|b|^s) C e^{-c|b|^2/\delta}\, db &\leq C(1+\delta^{-s/2}) e^{-cb_1^2/(2\delta)} \int e^{-c|b|^2/(2\delta)}\, db \\ &\leq C_s \delta^{d/2} e^{-c'/\delta},\end{aligned}$$

which is (33.5). Let p^{tube} and p^{χ} be the two normalized fiber densities. Then

$$\int w|p^{\mathrm{tube}} - p^{\chi}| \leq \frac{1}{Z^{\mathrm{tube}}} \int w(1-\chi_y) a_{\iota,\delta} e^{-V_\iota/\delta} + \frac{|Z^{\mathrm{tube}} - Z^{\chi}|}{Z^{\mathrm{tube}}} \int w p^{\chi},$$

with $w = 1 + |z|^s$. By (33.5), the lower bound for Z^{tube}, and the bounded moments of p^{χ} from Lemma 33.5(4), the right side is at most $C_s e^{-c'/\delta}$. Since $e^{-c'/\delta} \leq C\sqrt{\delta}$ for $\delta \leq 1$, (33.4) holds for r^{tube} when $\delta \leq 1$. When $\delta > 1$, the density r^{tube} is supported in $|z| \leq C_0^{1/2} b_0$, and the argument for part (3) of Lemma 33.5 with $\delta > 1$ applies.

The final assertion concerns the stopped measures near a compact patch of regular flat configurations. We prove it at the end of Section 33.8, after the normal coordinates of that application have been constructed. That proof uses Lemma 33.3, but not the present corollary. □

33.5 Based coordinates near the trivial connection

In this section we construct coordinates on the based quotient near the class of the identity, separate the nine constant directions from the $6n^3 - 6$ nonconstant directions transverse to the gauge orbits, and integrate out the latter by Lemma 33.3. We first record the derivative bounds for the effective interaction and the domain conditions for small commuting configurations that the integration requires.

Lemma 33.6. *Let $R = R_{d,\gamma}$ be the effective interaction in* (33.1)*, at a depth $d = kJ$ and with $0 < \gamma \leq \gamma_0$, and let $\overline{R} = R - R(I)$. Fix $r < r_1 < \min(\rho, \pi)$ and put $\eta = 1 - r_1/\rho$. There exist constants C_j, independent of d and γ, such that in link logarithm coordinates $\xi = (\xi_e)$ with $\max_e |\xi_e| \leq r_1$,*

$$\|D_\xi^j \overline{R}(e^\xi)\|_{\mathrm{root}} \leq C_j E_* \rho^{2-j} \eta^{-j}, \qquad j \geq 1, \tag{33.8}$$

where $\|\cdot\|_{\mathrm{root}}$ is the weighted absolute coefficient norm of Definition 4.6 with one index fixed, followed by the supremum over the fixed index. In particular, let $K_1 = C_1 E_* \rho \eta^{-1}$. Then at every point of the coordinates $V(X,s) = e^{X/n+s}$ of Theorem 33.8 whose link logarithms lie in B_{r_1},

$$|D_X \overline{R}(e^{X/n+s})| \leq 3n^2 K_1, \qquad |D_s \overline{R}(e^{X/n+s})| \leq 3n^{3/2} K_1, \tag{33.9}$$

where both norms are Euclidean and the derivative in s is taken in the subspace $\mathcal{S}_n$.

Further, let $\mathcal{K} \subset D_r$ be compact. There exist finitely many based gauge transformations and an open cover of $\mathcal{K}$ such that each member of the cover, together with a larger neighborhood, is mapped into B_{r_1} by one of these gauge transformations. Let $W \Subset W^+$ be open sets of parameters and let $\Psi : W^+ \to \mathcal{U}_n$ be smooth, extend smoothly to a neighborhood of $\overline{W}$, and satisfy $\Psi(\overline{W}) \subset \mathcal{K}$. Then for every $j \geq 1$,

$$\|D^j(\overline{R}_{\mathrm{sat}} \circ \Psi)\|_{L^\infty(W)} \leq C_{n,j,\mathcal{K},\Psi,r_1,\rho,\eta} E_*, \tag{33.10}$$

where the constant depends on the suprema over $\overline{W}$ of the derivatives of Ψ through order j, which are finite by smooth extension and compactness, but not on d or γ. The gauge transformations, the subcharts, and the constants are chosen independently of d and γ. For charts of the based quotient, Ψ denotes a fixed smooth lift to configurations. These bounds concern only the effective interactions $R_{d,\gamma}$, $d = kJ$, of the stopped densities (33.1).

Proof. By Theorem 14.8, the effective interaction of (33.1) satisfies $\mathcal{B}_{\rho,a_{\mathrm{sp}}}(R) \leq E_*$ and $DR(I) = 0$, and its local expressions in different gauges agree on the overlaps of their domains in the common outer ball. That theorem also contains the bounds (33.8) and their transfer to fixed charts. We recall the argument and verify the conditions on the charts in the form used below. The constant $R(I)$ does not enter derivatives of positive order. Set

$$A_k = \frac{\rho^{k-2}}{k!} \|D^k R(0)\|_{a_{\mathrm{sp}},\mathrm{root}}, \qquad \sum_{k \geq 2} A_k \leq E_*, \qquad q = r_1/\rho = 1 - \eta.$$

We fix one index of a differentiated coefficient, sum over the contracted indices, and use $|\xi_e| \leq r_1$. The spatial weights are at least one, and deleting contracted indices does not increase the length of a shortest connecting network of the remaining indices. By the absolute Taylor expansion, we have

$$\|D^j R(e^\xi)\|_{\mathrm{root}} \leq C_j \rho^{2-j} \sum_{k \geq \max(2,j)} A_k \frac{k!}{(k-j)!} q^{k-j},$$

where the fixed conventions for color coordinates contribute only the factor C_j. For $k \geq j$ the binomial identity implies

$$\binom{k}{j} \eta^j q^{k-j} \leq \sum_{l=0}^{k} \binom{k}{l} \eta^l q^{k-l} = 1,$$

so that $k! q^{k-j}/(k-j)! \leq j! \eta^{-j}$. This summable majorant also justifies differentiating the series term by term in the interior, and it proves (33.8).

There are $9n^3$ scalar link and color coordinates, and each first partial derivative of $\overline{R}$ is at most K_1, by (33.8) with $j = 1$. Then the Euclidean gradient is at most $3n^{3/2} K_1$, and its restriction to the subspace $\mathcal{S}_n$ has no larger norm. In the coordinates (X, s) we have

$\xi_i(x) = X_i/n + s_i(x)$. For each of the nine components of X, the chain rule expresses the derivative as a sum of n^3 partial derivatives with coefficient $1/n$, which is at most n^2K_1 in absolute value. The Euclidean norm of the gradient in X is then at most $3n^2K_1$. This proves (33.9).

For the transfer, let $U_0 \in \mathcal{K}$ and choose a lattice gauge transformation k_0 with $U_0^{k_0} \in B_r$. It can be taken based: replacing $k_0(x)$ by $k_0(x_*)^{-1}k_0(x)$ conjugates the transformed configuration by one constant matrix, which preserves every link logarithm norm. We keep this gauge transformation fixed. By continuity there exist neighborhoods $O_0 \Subset O_0^+$ of U_0, with closures inside an ambient smooth coordinate chart, whose images under k_0 lie in B_{r_1} even on $\overline{O_0^+}$. To find them, we take an open neighborhood with image in B_{r_1} and then two smaller coordinate balls with closures inside it. Finitely many of the sets O_0 cover $\mathcal{K}$. They depend on $\mathcal{K}$ and on $r < r_1$, but not on the effective interaction. The action of a fixed gauge transformation is smooth on all of $\mathcal{U}_n$ and an isometry for the product metric; in particular, it extends past each of these closures.

By the agreement of the local expressions, we have, on the larger neighborhood O_l^+ of a member of this cover,

$$\overline{R}_{\mathrm{sat}}(U) = R\big(e^{\xi^{(l)}(U)}\big) - R(I), \qquad \xi^{(l)}(U)_e = \mathrm{Log}\big((U^{k_l})_e\big).$$

Since $r_1 < \pi$, the logarithm and all its derivatives of fixed order are bounded on $\{|\mathrm{Log}| \le r_1\}$. Indeed, the transverse eigenvalue of the differential of the exponential map has modulus $2\sin(|\xi|/2)/|\xi|$, which is bounded away from zero on this closed ball. Repeated differentiation of the inverse function identity then bounds every higher derivative. We refine the parameter subcharts so that the closure of each is contained in $W^+ \cap \Psi^{-1}(O_l^+)$ for one l; this is possible because the sets O_l cover $\Psi(\overline{W})$ and $\overline{W}$ is compact. The maps $\xi^{(l)} \circ \Psi$ then extend smoothly past these closures, and for every fixed j their derivatives through order j are bounded there. This verifies, at each use, the condition on compact subcharts in Theorem 14.8. By the chain rule, the derivative of order j of $\overline{R}_{\mathrm{sat}} \circ \Psi$ is a finite sum of contractions of $D^a R$, $1 \le a \le j$, with derivatives of $\xi^{(l)} \circ \Psi$. The bound (33.8) and the conversion from the norm $\|\cdot\|_{\mathrm{root}}$ to Euclidean norms at the fixed side n bound every term. Since the local expressions agree on overlaps, these bounds concern one smooth function $\overline{R}_{\mathrm{sat}}$ on a neighborhood of $\mathcal{K}$. This proves (33.10). We note that no derivative of a gauge transformation chosen as a function of U was taken. □

The next lemma is a local version of Lemma 14.9 for small constant commuting configurations. Unlike that result, it requires no condition on the side m. It is used for the chart of Theorem 33.8 and for the lower bound in Theorem 33.10.

Lemma 33.7. *There exists $\epsilon_{\mathrm{loc}} > 0$, depending only on the fixed domains of the classical background fields, of the local constraint coordinates and Haar disintegration densities, and of the based gauge transformations used in Theorem 12.13, and in particular independent of the*

preceding depth and of γ, with the following property. Let $m \geq 4$, let $H_1, H_2, H_3 \in \mathfrak{su}(2)$ commute, and suppose that $\epsilon = \max_i |H_i|/m < \epsilon_{\mathrm{loc}}$. Let c be the configuration on Λ_m with $c_e = e^{H_i/m}$ on every link e in direction i. Then c and all its commuting refinements, together with their representatives in the block axial gauge, lie in the interior of these domains for one RG step, with a margin independent of m, of the depth, and of γ. For every depth $d \geq 0$ we have $E_d(c) = 0$. In the free logarithmic coordinates relative to the coarse-field lift $U_0(c)$, the refinement has $z = 0$; this point is the conditional constrained minimizer $z_d^*(c)$, and the corresponding normalized fluctuation variable is $w = 0$. The lemma imposes no condition on the other flat configurations on Λ_m.

Proof. We check the domain conditions directly, without using a neighborhood of the small-field event. We use the identities for commuting roots in Lemma 14.9, but not its condition on the minimum side. For one averaging step, let

$$a_i = e^{H_i/(Mm)}, \qquad c_i = a_i^M,$$

where a denotes the constant configuration on Λ_{Mm} with these link variables. In the block averaging map (4.3), every translated path with endpoint connectors $F_{c,x}$ has the same transport a_i^M as the reference path C_c, since all factors commute and the paths have the same displacement. Hence the logarithms of all based loop holonomies vanish, $T_M a = c$, and no exceptional value of T_M occurs.

We write $M = 2R_M + 1$. In a centered block with center v, the based gauge transformation to the block axial gauge at $x = v + r$ is $k(x) = \prod_j a_j^{r_j}$, where $|r_j| \leq R_M$. By commutativity, we have

$$|\mathrm{Log}\, k(x)| \leq \frac{3R_M}{M}\epsilon, \qquad \max_e |\mathrm{Log}\, a_e| \leq \epsilon/M, \qquad \max_e |\mathrm{Log}\, U_0(c)_e| \leq \epsilon. \tag{33.11}$$

A transformed link in the interior of a block is the identity. On a link between two blocks in direction i, the offsets r_i at its endpoints differ by $1 - M$, and the transformed link is $a_i^M = c_i$. Thus the configuration in the block axial gauge is exactly the coarse-field lift $U_0(c)$ of Section 4, and all free logarithmic coordinates and the logarithms on the eliminated edges e_c^* vanish. Since the derivative of the block averaging constraint with respect to the eliminated edge is $M^{-2} I_3$ there, the point lies in the domain of the local constraint coordinates, with positive Jacobian. The Haar factors are positive on the small link ball.

We now choose the margin. Let $b_* > 0$ be smaller than r, than the radii r_* and r_{sym} and the constant κ_{sym} of Theorem 10.11 and Lemma 10.8, than the radius r_{lift} of the C^1 ball of fine potentials in Lemma 10.8, and than the fixed radii of the coarse-field, constraint, Haar disintegration, and gauge domains of one RG step in Theorem 12.13. These are the radii used in the choice of r_{flat} in Lemma 14.9. There are only finitely many types of words in the construction of one RG step. Let $C_{\mathrm{geo},J} \geq 1 + 3R_M/M$ be a bound for their

lengths and for the sums of the offsets in their based gauge transformations, measured in units of the output lattice of the RG step. This bound does not depend on the side, since the words are translates of the words in one fixed block, and there are only J successive block sizes. Finally, we require

$$4C_{\mathrm{geo},J}\epsilon_{\mathrm{loc}} < b_*.$$

The logarithm of a product of commuting small factors is bounded by its word length times the largest logarithm of a factor. Thus (33.11) and this choice verify every intermediate condition on coarse fields, tree gauges, and gauge transformations, with a fixed margin. At a finer side mM^l the link bound is ϵM^{-l}, and none of these inequalities becomes worse with the number of averaging steps. The gauge identity of Lemma 10.8 applies, because both representatives lie in its ball of radius r_{sym} and the logarithms of the gauge transformations are below κ_{sym}.

We prove that the effective classical action vanishes by induction on d, simultaneously for all sides $m \geq 4$ and all constant commuting configurations as in the statement. The comparison below concerns real configurations and real free coordinates. No minimum and no positive Hessian is asserted for complex coarse fields. The continuation in a complex parameter below concerns only the analytic identity for the minimizer, after it has been proved near real parameter values. For $d = 0$, the Wilson action vanishes at the flat configuration c. Suppose that the claim holds at depth d. The refinement a has side Mm and satisfies the smaller bound ϵ/M. By the induction hypothesis, we have $E_d(a) = 0$, and by the gauge invariance just checked, $E_d(U_0(c)) = E_d(a) = 0$. By the definition of the effective classical action in Section 4, for the actions and minimizers of Theorem 12.13, we have

$$E_{d+1}(c) = MF_d(c, z_d^*(c)) = M \min_z F_d(c, z), \qquad F_d(c, z) = E_d(b(c, z)),$$

where the minimum is over the domain of the classical background field. The free point $z = 0$ lies in this domain by the margins verified above; this does not presuppose that $z = 0$ lies in a fluctuation cutoff domain centered at $z_d^*(c)$. Since $F_d(c, 0) = 0$, the minimum is at most zero, and nonnegativity (Theorem 11.2) implies that $E_{d+1}(c) = 0$. This completes the induction, and it also shows that $E_d(U_0(c)) = 0$ at every depth. In other words, the iteration uses exact commuting roots, whose potential in units of the coarse lattice is H_i/m at every side and whose first spatial differences vanish; no loss of domain accumulates.

It remains to identify $z = 0$ with the conditional constrained minimizer. In the coordinates of Theorem 12.13, z_d^* is the analytic branch of constrained minimizers through $z_d^*(I) = 0$. Since the point $z = 0$ attains the minimum of $F_d(c, \cdot)$, we have $D_z F_d(c, 0) = 0$. At $c = I$ the constrained Hessian $K_d(I)$ is positive. By Lemma 33.1(1), applied to $D_z F_d$, the critical point near 0 is then locally unique for c near I. For each fixed side and depth, we find $z_d^*(e^{tH/m}) = 0$ for real t near zero. The minimizer is holomorphic on the common

domain of coarse fields of Theorem 12.13, and the factor four in the choice of ϵ_{loc} keeps $e^{tH/m}$ in this domain for complex $|t| < 2$. Each component of $t \mapsto z_d^*(e^{tH/m})$ is therefore holomorphic on $|t| < 2$ and vanishes on an interval around zero. Its Taylor coefficients at zero vanish, and continuation through overlapping discs gives $z_d^*(e^{tH/m}) = 0$ for $|t| < 2$, in particular at $t = 1$. The radius of local uniqueness at the origin need not be uniform; the common domain and the margins are. Since the normalization $z - z_d^*(c) = g_d K_d^{-1/2} w$ is invertible, $z = z_d^*(c)$ corresponds to $w = 0$, which lies in the interior of every fluctuation cutoff domain. By Theorem 15.12, the reconstruction maps $\widehat{\mathscr{R}}$, whose gauge transformations are based at the coarse vertices, preserve these properties. This completes the proof. □

Theorem 33.8. *Consider the stopped measures of Section 33.2, with L, $n \in 4\mathbb{N}$, M, J, Θ, and γ_0 fixed before γ, and let $g = \sqrt{\gamma/n}$. Let d_0 denote the forward lattice gradient on Λ_n (not a depth) and d_0^* its adjoint, the discrete divergence, both acting componentwise on $\mathfrak{su}(2)$-valued functions and link arrays, and let*

$$\mathcal{S}_n = \Big\{ s = (s_i(x)) \in \mathfrak{su}(2)^{\mathcal{E}_n} : d_0^* s = 0,\ \sum_x s_i(x) = 0 \ (i = 1,2,3) \Big\},$$

and $d_n = \dim \mathcal{S}_n = 6n^3 - 6$. There exist $\rho_n, s_n, c_n, C_n > 0$ such that the following hold.

1. *The based quotient of $\mathcal{U}_n$ is a compact manifold of dimension $6n^3 + 3$, and the based gauge transformations carry Haar measure of total mass one. The map $(X, s) \mapsto V(X, s) = e^{X/n+s}$, which assigns $\exp(X_i/n + s_i(x))$ to the link from x to $x + a_n e_i$, defined for $X \in \mathbb{R}^9$ with $|X| < \rho_n$ and $s \in \mathcal{S}_n$ with $|s| < s_n$, induces coordinates on a neighborhood of the class of the identity in the based quotient. In these coordinates the image of H_n is $J_n(X, s)\, dX\, ds$, where J_n is smooth and strictly positive. The configurations $V(X, s)$ lie in $B_r \subset D_r$. Simultaneous conjugation at the root is not divided out.*
2. *Let $F_d(X, s) = E_d(e^{X/n+s})$, $f_d(X) = F_d(X, 0)$, and $A_d(X) = D_s^2 F_d(X, 0)$. For every integer $d \geq 0$,*

$$D_s F_d(X, 0) = 0, \qquad F_d(X, s) = f_d(X) + \tfrac{1}{2}\langle s, A_d(X) s \rangle + O_n(|s|^3),$$
$$c_n I \leq A_d(X) \leq C_n I, \qquad F_d(X, s) \geq f_d(X) + c_n |s|^2.$$

3. *For $d = kJ$ and $0 < \gamma \leq \gamma_0$, write $\overline{R}_{d,\gamma}(X, s) = \overline{R}_{d,\gamma}(V(X, s))$. Then, uniformly in X, d, and γ,*

$$\int_{|s|<s_n} e^{-F_d(X,s)/g^2 - \overline{R}_{d,\gamma}(X,s)} J_n(X, s)\, ds = (2\pi)^{d_n/2} g^{d_n} e^{-f_d(X)/g^2} a_{d,\gamma}(X) \big[1 + O_n(g)\big],$$

$$a_{d,\gamma}(X) = \frac{e^{-\overline{R}_{d,\gamma}(X,0)} J_n(X, 0)}{\sqrt{\det A_d(X)}}.$$

The amplitude $a_{d,\gamma}$ has positive upper and lower bounds and a C^1 bound, uniformly in d and γ.

The constants may depend on n, L, M, J, Θ, γ_0, and on the larger chart and the bounds in (P2) and (P3), but not on d, on the total number of averaging steps, or on $\gamma \in (0, \gamma_0]$.

No positive Hessian is asserted in the nine constant directions X: at the identity the quadratic part of E_0 vanishes on constant configurations.

Proof. The proof has five steps. We construct the based quotient and the coordinates (X, s), fix the radii, prove the stationarity in s and the gap at the fixed side n, and then apply Lemma 33.3 in the variable s at each fixed X.

Step 1: inputs. All bounds in this proof are at the fixed side n. The lower bound (P1) holds at every integer $d \geq 0$, whether or not an effective interaction is attached to d. In the statements involving $\overline{R}_{d,\gamma}$, we have $d = kJ$, and $R_{d,\gamma}$ is the effective interaction of the density (33.1), with the value bound and the analytic norm of that density. Its first derivatives in the chart are obtained from Lemma 33.6, on the relatively compact subcharts chosen in Step 3 and with the identity gauge transformation, which extends past their closures. They do not follow from the value bound. The derivative bounds for the classical actions in fixed charts come from Theorem 10.11: restricted to the finitely many coordinates at side n, its coefficient bounds give bounds for ordinary derivatives, with constants depending on n. The exact translation invariance of E_d is Lemma 10.8. The vanishing of E_d and the domain conditions for small constant commuting configurations are proved in Lemma 33.7, with no condition on the minimum side. Finally, (P4) allows us to apply Theorems 12.13 and 14.8 for every $0 < \gamma \leq \gamma_0$.

Step 2: the based quotient. We fix a periodic spanning tree of the lattice graph Λ_n. There exists a unique based gauge transformation that makes every tree link the identity. It is obtained successively from the root, by prescribing its value at a new vertex from the tree link leading to that vertex. The remaining links are $3n^3 - (n^3 - 1) = 2n^3 + 1$ independent non-tree links. This change of variables has an exact interpretation for the measure. We condition on the preceding tree variables and use a left or right translation of Haar measure at the next tree link. This replaces that link by the value of the gauge transformation at the new vertex, with Haar measure. At each non-tree link the two endpoint factors are determined by the tree variables, and its change of variables is again a Haar translation. By Fubini's theorem, the image of H_n is the product of Haar probability measures in the $n^3 - 1$ values of the based gauge transformation and in the $2n^3 + 1$ non-tree link variables. Integrating out the gauge variables gives one. In particular the based quotient is the compact product $\mathrm{SU}(2)^{2n^3+1}$, of dimension $6n^3 + 3$, and no conjugation at the root has been divided out from the non-tree link variables.

Step 3: the coordinates and the radii. The gradient is injective on $\mathfrak{su}(2)$-valued vertex functions that vanish at the root: if the gradient of such a function vanishes, the function is constant on the connected graph and hence zero. The image of these gradients has

dimension $3(n^3-1)$, and its orthogonal complement in the $9n^3$-dimensional space of link arrays is $\ker d_0^*$, of dimension $6n^3+3$. The nine arrays that are constant in position lie in this complement, and their orthogonal complement within it is $\mathcal{S}_n$, of dimension $6n^3-6$. The orthogonal decomposition can be constructed explicitly. For an array A, the equation $d_0^* d_0 \xi = d_0^* A$ has a unique solution ξ with mean zero, by diagonalization of this positive matrix on mean-zero vertex functions. Subtracting $d_0\xi$ and then the spatial means of the three components, we obtain the component in $\mathcal{S}_n$.

We consider the map that sends (X,s) to $e^{X/n+s}$ and then to the non-tree link variables of its representative in the tree gauge. At $(0,0)$ its derivative is the projection along gradients onto the non-tree link coordinates. Indeed, the linearized tree gauge fixing sends a link logarithm array A to $A - d_0\xi$, where ξ vanishes at the root and $A - d_0\xi$ vanishes on the tree, and then reads off the non-tree links. The kernel of the derivative is the image of d_0 on functions vanishing at the root. The derivative is therefore injective on the complement $\mathbb{R}^9 \oplus \mathcal{S}_n$ constructed above, and invertible there, since the dimensions agree. By Lemma 33.1(1), the map is a diffeomorphism near $(0,0)$. Pulling back the product of Haar measures on the non-tree links, whose density in exponential coordinates is smooth and positive near the identity, we obtain a smooth strictly positive density $J_n(X,s)$. Its bounds and the bounds for its derivatives of fixed order on smaller closed coordinate balls depend only on n and on these balls.

We now fix the radii, independently of d and γ. We shrink the balls so that $\max_{x,i} |X_i/n + s_i(x)| < r/2$, and choose slightly larger balls with the same property. We also require $2\rho_n/n < \epsilon_{\mathrm{loc}}$, with ϵ_{loc} from Lemma 33.7, and put the larger balls inside the charts of radii r_* and r_{sym} of Theorem 10.11 and Lemma 10.8. More precisely, we choose the product $\mathcal{C}$ of the two coordinate balls in the statement and a larger product $\mathcal{C}^+$ with $\overline{\mathcal{C}} \subset \mathcal{C}^+$, such that $\overline{\mathcal{C}^+}$ lies in the chart just constructed and satisfies all these strict bounds on link logarithms. These finitely many conditions are compatible with positive radii. The map $(X,s) \mapsto e^{X/n+s}$ and the quotient chart extend smoothly to a neighborhood of $\overline{\mathcal{C}^+}$, and the gauge transformation used here is the identity. Every derivative of fixed order of the chart is therefore bounded on this closure, with constants depending only on n. This verifies the condition on compact subcharts in Theorem 14.8 and Lemma 33.6, both for the integration in s and for the derivative of the amplitude in X. The configurations $V(X,s)$ lie in $B_r \subset D_r$, with link logarithms in the ball B_{r_1} of Lemma 33.6, for a fixed $r_1 \in (r, \min(\rho,\pi))$; in particular $\eta = 1 - r_1/\rho$ is fixed. Every commuting point X used here satisfies $\max_i |X_i|/n < \epsilon_{\mathrm{loc}}$, so E_d vanishes there at every depth.

Step 4: stationarity and the gap. Fix a constant array of logarithms $A_i(x) = X_i/n$. Translation by any lattice vector fixes this array. Differentiating the exact translation invariance of E_d, we find that the gradient $DE_d(e^A)$ in link logarithm coordinates is constant in position for each direction and color. Its pairing with every array of spatial

mean zero vanishes, and

$$D_s F_d(X,0) = 0 \tag{33.12}$$

for every d and every X in the chart. This argument concerns E_d only, and it does not set any derivative of the effective interaction or of the Haar density to zero.

We now prove the gap at the fixed side. The expansion of a plaquette variable at the identity in link logarithms is $d_0 A + O(|A|^2)$, and $4q(e^Y) = |Y|^2/2 + O(|Y|^4)$. It follows that

$$E_0(e^A) = \tfrac{1}{2} \sum_{x,\, i<j} |A_j(x+e_i) - A_j(x) - A_i(x+e_j) + A_i(x)|^2 + O_n(|A|^3),$$

where e_i denotes one lattice step. At the Fourier mode $m \in \mathbb{Z}^3/n\mathbb{Z}^3$, let $\eta_i(m) = e^{2\pi i m_i/n} - 1$. For each color,

$$\sum_{i<j} |\eta_i a_j - \eta_j a_i|^2 = |\eta|^2|a|^2 - |\eta^* a|^2.$$

Since the zero mode is absent on $\mathcal{S}_n$ and $\eta^* a = 0$ there, the Hessian of E_0 at the identity is at least $\lambda_n I$ on $\mathcal{S}_n$, where $\lambda_n = 4\sin^2(\pi/n) > 0$. All inequalities between functions and Hessians here concern real configurations and real tangent vectors. Since $E_d - \kappa E_0 \geq 0$ near the identity by (P1) and vanishes at the identity, its Hessian at this point is positive semidefinite. This implies

$$D_s^2 F_d(0,0) \geq \kappa \lambda_n I.$$

By the uniform bound on third derivatives in (P2), we have

$$\|D_s^2 F_d(X,s) - D_s^2 F_d(0,0)\| \leq C_n(|X| + |s|).$$

We choose the radii so that the right side is at most $\kappa\lambda_n/2$; the upper derivative bounds in (P2) give a uniform upper bound. Then $A_d(X) = D_s^2 F_d(X,0) \geq \kappa\lambda_n I/2$, and Taylor's formula with integral remainder and (33.12) give

$$F_d(X,s) = f_d(X) + \tfrac{1}{2}\langle s, A_d(X)s\rangle + O(C_n|s|^3), \qquad F_d(X,s) - f_d(X) \geq \tfrac{1}{4}\kappa\lambda_n|s|^2.$$

This proves part (2), after decreasing c_n.

Step 5: integration in s and the amplitude. On the larger s-ball we apply Lemma 33.3, at each fixed X, with

$$V(X,s) = F_d(X,s) - f_d(X), \qquad \delta = g^2, \qquad a(X,s) = e^{-\overline{R}_{d,\gamma}(X,s)} J_n(X,s),$$

the rank d_n, and the Hessian $L = A_d(X)$. All these functions and the Hessian field are smooth in (X,s) on the fixed real chart, hence jointly Borel, and we choose a fixed smooth cutoff in s, equal to one on a smaller fixed ball and supported in the ball $|s| < s_n$. The

phase bounds were proved in Step 4. For the amplitude, let $B_n = \sup_{D_r} |\overline{R}_{d,\gamma}|$, which is bounded by (P3). By (33.9) and the mean value theorem on the convex s-ball,

$$|a(X,s) - a(X,0)| \le e^{B_n}\big(\|J_n\|_\infty 3n^{3/2}K_1 + \|D_s J_n\|_\infty\big)|s|.$$

The positive lower and upper bounds follow from $e^{-B_n} \le e^{-\overline{R}_{d,\gamma}} \le e^{B_n}$ and the positivity of J_n. Every hypothesis of Lemma 33.3 therefore holds with constants c_0, C_0 depending only on the fixed data. The case $P = 1$ of (33.7) gives the integral with the cutoff. The additive error $O(g)$ becomes a relative error because the amplitude $a(X,0)/\sqrt{\det A_d(X)}$ has a fixed positive lower bound. The lower bound $F_d - f_d \ge c_n|s|^2$ holds on the whole ball $|s| < s_n$, and so do the amplitude bounds. By (33.5), for $g \le 1$ and after factoring out $e^{-f_d(X)/g^2}$, the difference between the integral with the cutoff and the integral over the whole ball is at most $Cg^{d_n}e^{-c'/g^2}$, which is $O_n(g^{d_n+1})$. If $\gamma_0/n > 1$, then for $1 < g^2 \le \gamma_0/n$ both the integral and the main term are bounded above and below by positive constants depending only on the fixed data, and the asserted formula holds there after enlarging the constant. We obtain

$$(2\pi)^{d_n/2} g^{d_n} e^{-f_d(X)/g^2} \frac{e^{-\overline{R}_{d,\gamma}(X,0)} J_n(X,0)}{\sqrt{\det A_d(X)}} \big[1 + O_n(g)\big].$$

For the C^1 bound of the amplitude, we have

$$D_X \log a_{d,\gamma}(X) = -D_X \overline{R}_{d,\gamma}(X,0) + D_X \log J_n(X,0) - \tfrac{1}{2}\operatorname{tr}\big(A_d(X)^{-1} D_X A_d(X)\big).$$

The first term is bounded by $3n^2K_1$, by (33.9); the value bound alone would not give this estimate. The trace is bounded by $d_n \|A_d^{-1}\| \|D_X A_d\|$, and every factor is bounded uniformly, by the fixed chart, the lower bound for the Hessian, and the bound on third derivatives. This proves the bounds for the amplitude, with the phase f_d at finite depth. In particular, no limit of the classical actions as $d \to \infty$ at fixed coupling has been inserted into an exponent divided by g^2. All constants have the dependence stated in the theorem. □

33.6 Flat configurations and the partition function

We now describe the flat configurations on Λ_n away from the central triples, that is, the triples of central cycle holonomies. We then prove the two-sided bound for the partition function. Recall from Lemma 14.9 that the flat configurations modulo all periodic gauge transformations are classified by their commuting cycle holonomies modulo simultaneous conjugation. In the based quotient, where simultaneous conjugation is not divided out, the flat set is the set of commuting triples $(H_1, H_2, H_3) \in \mathrm{SU}(2)^3$. We call a flat configuration regular if its cycle holonomies are not all central. A compact regular patch is a compact subset of the flat set in the based quotient containing none of the eight configurations with central cycle holonomies.

Lemma 33.9. *1. In the based quotient at side $n \geq 4$, the set of regular flat configurations is a smooth embedded submanifold of dimension five. The Hessian of E_0 is positive definite in the normal directions at every regular flat configuration. On every compact regular patch it has a positive lower bound, and there is a finite atlas of normal coordinates $V(y,v)$, with y in the flat manifold and $v \in \mathbb{R}^{6n^3-2}$, in which the image of H_n has a smooth positive density $J(y,v)$ with respect to $d\mathrm{vol}(y)\,dv$.*

2. Consider the stopped measures of Section 33.2 and a compact regular patch. Assume either that the patch consists of configurations gauge equivalent to constant commuting links $e^{H_i/n}$ with $\max_i |H_i|/n < \epsilon_{\mathrm{loc}}$, as in Lemma 33.7, or that $2\pi/n < r_{\mathrm{flat}}$, where r_{flat} is the radius of Lemma 14.9. Assume also that the patch lies in a common larger classical link chart, after fixed gauge transformations. Then there is a normal tube over the patch, fixed independently of d and γ, on which, for every integer $d \geq 0$,

$$nE_d(V(y,v)) = \tfrac{1}{2} v^{\mathsf{T}} H_d(y) v + O(C_n|v|^3),$$
$$c_n I \leq H_d(y) \leq C_n I, \qquad nE_d(V(y,v)) \geq c_n |v|^2.$$

The normal rank is $6n^3 - 2$. The constants may depend on n, the patch, the larger chart, the constant κ of (P1), the derivative bounds of (P2), and the margins of the tube.

Proof. We use the based quotient $\mathrm{SU}(2)^{2n^3+1}$ of non-tree link variables constructed in Step 2 of the proof of Theorem 33.8. By Lemma 14.9, a flat point of the based quotient is determined by its three commuting cycle holonomies. This identification and its inverse are locally smooth. Indeed, each holonomy is a finite product of non-tree link variables. Conversely, after fixing paths in the tree, the flat transport along every non-tree link is the word in the three commuting cycle transports given by the winding numbers of the closed path that it forms with the tree. Such a word is a smooth product of integer powers.

At a regular triple we choose a noncentral member. Its quaternion vector is nonzero and, after fixing its sign, determines a smooth unit axis on a small patch. Every member of a commuting triple has its vector parallel to that axis. We choose local lifts of the three angles on the corresponding circles and two local coordinates for the axis. Differentiation in the three angles and the two axis coordinates is injective, since the chosen member has nonzero vector. Projection onto five suitable ambient coordinates therefore has invertible derivative. By Lemma 33.1(1), the regular flat set is a smooth embedded five-dimensional manifold, and its only possible singular points are the eight triples whose members are all central.

Next we determine the kernel of the Hessian, including the count of based gauge directions. By a periodic gauge transformation and a constant conjugation, we conjugate a flat representative to the constant commuting links $U_i^0 = e^{\vartheta_i \tau_3/n}$. Such transformations change neither the rank of the Hessian nor its quadratic form on the gauge quotient.

We write a variation of the links in left coordinates as $e^{ta_i(x)}U_i^0$. The derivative of the plaquette logarithm is

$$(d_U a)_{ij}(x) = \mathrm{Ad}_{U_i^0}\, a_j(x+e_i) - a_j(x) - \mathrm{Ad}_{U_j^0}\, a_i(x+e_j) + a_i(x),$$

as one sees by differentiating its four ordered factors, since the background plaquette variable is the identity. Since the quadratic part of E_0 is $\sum |d_U a|^2/2$, the kernel of the Hessian is $\{d_U a = 0\}$.

We split the colors into $\mathbb{R}\tau_3$, the neutral color, and its orthogonal plane, the charged color, which we identify with $\mathbb{C}$ so that ad_{τ_3} acts as multiplication by i. In neutral color the Fourier difference vector is $\eta_i = e^{2\pi i m_i/n} - 1$. If $\eta \neq 0$, the equations $\eta_i a_j - \eta_j a_i = 0$ imply $a = \eta\phi$, with $\phi = a_j/\eta_j$ for an index j with $\eta_j \neq 0$. These are exactly the gradient directions, and the zero mode gives three real constant neutral directions. In charged color the difference vector is

$$\eta_i = e^{i(2\pi m_i + \vartheta_i)/n} - 1.$$

It cannot vanish in all three components, since that would force every ϑ_i into $2\pi\mathbb{Z}$ and all three holonomies to be central. The same algebra $a = \eta\phi$ shows that every charged mode in the kernel is a covariant gradient. This argument includes the cases in which some ϑ_i vanish, without division by a possibly vanishing component.

The covariant gradient on vertex functions has a one-dimensional kernel, consisting of the constant neutral functions, since its charged Fourier vectors never vanish and its neutral kernel consists of constants. Its real image has dimension $3n^3 - 1$. Its restriction to functions vanishing at the root is injective, since a covariantly constant function vanishing at one point vanishes everywhere by transport along links. Hence the image of the based gradients has dimension $3n^3 - 3$, and after dividing out the based gradients, the kernel has dimension

$$(3n^3 - 1) + 3 - (3n^3 - 3) = 5.$$

Two of these directions are infinitesimal simultaneous conjugations at the root, and three are the constant neutral directions. They are exactly the tangent directions of the five-dimensional manifold parametrized above. Since the Hessian is a nonnegative symmetric matrix with no further kernel, it is positive definite on a normal complement. Its least normal eigenvalue has a positive minimum on a compact regular patch, since continuous unit normal vectors over that compact set form a compact set on which the quadratic form does not vanish.

For integration, we give the based quotient $\mathrm{SU}(2)^{2n^3+1}$ its product Riemannian metric, which is bi-invariant, and let $\mathcal{F}$ be the manifold of regular flat configurations. We use the normal exponential map

$$V(y, v) = \mathrm{Exp}_y(v), \qquad y \in \mathcal{F},\ v \perp T_y\mathcal{F},$$

of Lemma 33.1(2). Over a compact neighborhood of a compact regular patch, it is injective for a common small normal radius. We choose finitely many smooth orthonormal trivializations of the normal bundle of $\mathcal{F}$, with working subcharts relatively compact in larger charts. Since their transitions preserve y and are orthogonal in v, a partition of unity on the base extends consistently to the tube. The maps $V(y,v)$, their lifts to configurations, and the normal frames extend smoothly past the closures of the working subcharts, so all derivatives of fixed order of the charts are bounded there. Since the Haar measure of the product group is a constant multiple of its Riemannian volume, its pullback $J(y,v)$ is smooth and positive with respect to $d\mathrm{vol}(y)\,dv$. This proves part (1).

For part (2), on patches of small commuting configurations, the vanishing of the classical actions is proved in Lemma 33.7. On a general patch at a side with $2\pi/n < r_{\mathrm{flat}}$, it is the conclusion of Lemma 14.9 for all flat configurations: every averaging step has output side $m \geq n$, and $2\pi/m \leq 2\pi/n < r_{\mathrm{flat}}$ verifies the condition on the side at each step. By gauge invariance, these zeros transfer to the smooth based representatives used here. Since $E_d \geq 0$, its first derivative vanishes on the flat manifold. Since $E_d - \kappa E_0 \geq 0$ by (P1) and vanishes there, the Hessian of $E_d - \kappa E_0$ is positive semidefinite on real normal vectors. The lower bound of part (1) then implies $H_d = D_v^2(nE_d)(y,0) \geq c_n I$. The uniform derivative bounds of (P2) on the common larger chart give the upper bound and a common cubic Taylor error. If a quotient chart is not itself a representative with small links, we fix a gauge transformation taking its center into B_r and shrink the neighborhood inside the fixed larger classical chart. By compactness, finitely many such fixed gauge transformations suffice. Their derivatives and the bounds of Theorem 10.11 give the same bound by the chain rule, and no gauge transformation depending on the point is needed. We choose the radius of the tube so that the cubic error is at most one quarter of the lower bound for the Hessian times $|v|^2$. This proves the lower bound for nE_d on the whole smaller tube. All these choices depend only on the listed data. The normal rank is the dimension $6n^3+3$ of the based quotient minus five. □

Theorem 33.10. *The following statements hold.*

1. *For every integer $n \geq 4$ and every $\kappa > 0$ there exists $C_{n,\kappa} < \infty$ such that for $0 < g \leq 1$*

$$\int_{\mathcal{U}_n} e^{-\kappa E_0(V)/g^2}\, dH_n(V) \leq C_{n,\kappa} g^{6n^3-2}, \qquad E_0(V) = 4 \sum_{p \in \mathcal{P}_n} q(V_p).$$

   This is an integral over the whole configuration space, including neighborhoods of all eight flat configurations with central cycle holonomies. Divisibility of n by four is not needed, and $C_{n,\kappa}$ depends only on n and κ.

2. *Consider the stopped measures of Section 33.2. There exist $g_0(n) > 0$ and $0 < c_n \leq C_n < \infty$, with the dependence described in Theorem 33.8, such that for all depths $d = kJ$ and all $0 < \gamma \leq \gamma_0$ with $g = \sqrt{\gamma/n} \leq g_0(n)$,*

$$c_n g^{d_n+4} \leq \widehat{Z}_{n,d,\gamma} \leq C_n g^{d_n+4}, \qquad d_n = 6n^3 - 6.$$

Equivalently, $\widehat{Z}_{n,d,\gamma}$ is comparable to γ^{p_n}, $p_n = 3n^3 - 1$, uniformly in d and γ in this range, at the fixed side n. Here $\widehat{Z}_{n,d,\gamma}$ is the partition function of (33.1)*, with the centered effective interaction. If Z^{raw} is defined with $R_{d,\gamma}$ in place of $\overline{R}_{d,\gamma}$, then $\widehat{Z}_{n,d,\gamma} = e^{R_{d,\gamma}(I)} Z^{\mathrm{raw}}$ exactly.*

3. *For every $\rho > 0$ and $c > 0$ there exists $C_{\rho,c}$ such that for $0 < g \le 1$*

$$\int_{|X|<\rho} e^{-cQ(X)/g^2}\, dX \le C_{\rho,c} g^4.$$

Here ρ is an arbitrary radius, not the analytic radius. More precisely, for $g \le \min(1, \rho^2)$ the ball $|X| \le \sqrt{g}$ contributes $O(g^{9/2})$, and each annulus $\varrho/2 < |X| < \varrho$ with $\sqrt{g} \le \varrho \le \rho$ contributes $O(g^4 \varrho)$.

The lower bound in (2) comes from a fixed annulus of regular constant commuting configurations in the chart of Theorem 33.8, and it uses only the bound on the values of the centered effective interaction.

Proof. We prove the upper bound in (1) by covering the based quotient by neighborhoods of the eight central flat configurations, normal tubes over the remaining compact regular part of the flat set, and a compact set without flat configurations. Near the central configurations, we integrate the nonconstant directions with the gap of Theorem 33.8, and the constant directions with the estimate in (3). We then deduce (2) from (1), (P1), and (P3).

Step 1: constant configurations. We need an exact lower bound for the Wilson action on constant configurations. We write $U = aI + 2v^a\tau_a$ and $W = bI + 2w^a\tau_a$ with $a^2 + |v|^2 = b^2 + |w|^2 = 1$. The multiplication rule

$$(a, v)(b, w) = (ab - v \cdot w,\ aw + bv + v \times w)$$

follows directly from the products of the matrices τ_a. Conjugation by (a, v) sends w to $(a^2 - |v|^2)w + 2(v \cdot w)v + 2a(v \times w)$. Thus the scalar part of $UWU^{-1}W^{-1}$ is

$$b^2 + \left[(a^2 - |v|^2)w + 2(v \cdot w)v + 2a(v \times w)\right] \cdot w = 1 - 2|v \times w|^2,$$

which implies $q(UWU^{-1}W^{-1}) = 2|v \times w|^2$ exactly. For $U_i = e^{x_i/n}$ we have $v_i = \sin(|x_i|/(2n))\widehat{x}_i$, and summing over the n^3 plaquettes in each of the three planes gives, continuously also when some x_i vanish,

$$\begin{aligned} E_0(e^{X/n}) &= 8n^3 \sum_{i<j} \sin^2(|x_i|/(2n)) \sin^2(|x_j|/(2n)) |\widehat{x}_i \times \widehat{x}_j|^2 \\ &= \frac{1}{2n} \sum_{i<j} |x_i \times x_j|^2 \operatorname{sinc}^2(|x_i|/(2n)) \operatorname{sinc}^2(|x_j|/(2n)). \end{aligned} \tag{33.13}$$

Since the factors sinc^2 have positive lower and upper bounds on a fixed small ball, $E_0(e^{X/n})$ is comparable to $Q(X)/n$ there. This is a relative bound, valid also in almost commuting directions.

Step 2: the neighborhood of the identity. Applying the proof of Theorem 33.8 to the Wilson action alone, we obtain, on a fixed neighborhood of the identity in the based quotient,

$$E_0(e^{X/n+s}) \geq E_0(e^{X/n}) + c_n|s|^2 \geq c_n'(Q(X) + |s|^2), \tag{33.14}$$

where the second inequality is (33.13). This uses only the translation invariance of E_0, the gap at side n, and the smoothness of E_0, all verified explicitly in Steps 3 and 4 of that proof. It does not use the stopped density or the divisibility of n by four, and it holds at every integer $n \geq 4$.

Step 3: the nine-dimensional estimate. We prove (3). The ball $|X| \leq \sqrt{g}$ has volume $C_9 g^{9/2}$, which bounds its contribution. We now consider an annulus $\varrho/2 < |X| < \varrho$ with $\varrho \geq \sqrt{g}$, and set $X = \varrho Y$ and $\epsilon = g/\varrho^2 \leq 1$. Since Q is homogeneous of degree four, the integral over the annulus is

$$\varrho^9 \int_{1/2<|Y|<1} e^{-cQ(Y)/\epsilon^2}\, dY.$$

On the fixed closed annulus $1/2 \leq |Y| \leq 1$, the zero set of Q is a compact piece of the nonzero cone $\mathcal{M}$. In a fixed small normal tube around it the radius of the base point is bounded below, and by Lemma 33.2 we have $Q(y+b) \geq c'|b|^2$ and a bounded Jacobian. The tangential volume is finite by the volume formula of Lemma 33.2. Integrating in the four normal directions, we see that the tube contributes at most $C\epsilon^4$. Off the tube, Q has a positive minimum q_* on the annulus: a sequence with $Q \to 0$ would have a limit point in the zero set, hence in the tube. Since $\sup_{u>0} u^2 e^{-cq_* u} < \infty$, the integrand off the tube is at most $C\epsilon^4$. The annulus therefore contributes at most

$$C\varrho^9\epsilon^4 = Cg^4\varrho. \tag{33.15}$$

We take the radii $\rho, \rho/2, \rho/4, \ldots$ until the last annulus reaches $\sqrt{g}$. The sum of these radii is at most 2ρ, and the remaining ball has radius at most $2\sqrt{g}$ and contributes $O(g^{9/2})$. This proves the bound $C_{\rho,c}g^4$ for $g \leq \min(1, \rho^2)$. If $\rho^2 \leq g \leq 1$, the volume of the ball, $C\rho^9$, is at most $C\rho g^4$. The statements about the ball and the annuli follow from the same estimates.

Step 4: the whole configuration space. Integrating (33.14) against the bounded density of the chart, first in s and then in X, we find that the neighborhood of the identity contributes at most

$$C_{n,\kappa} g^{d_n} \int_{|X|<\rho_n} e^{-c_{n,\kappa} Q(X)/g^2}\, dX \leq C_{n,\kappa} g^{d_n+4},$$

by Step 3. To cover the other seven central points, we multiply, for each of the three directions, the links crossing one wrapping coordinate plane in that direction by either

I or $-I$, and call the resulting array of central link variables z. Since each plaquette contains either no such factor or two canceling ones, $z_p = I$ for every plaquette. The three cycle signs of z are arbitrary, and they give the eight central triples. The map $V \mapsto zV$ preserves every plaquette variable and the product Haar measure, and it commutes with based gauge transformations because z is central. It maps the identity point of the flat set in the based quotient to the corresponding central flat point. Transporting the chart just used, we obtain the same bound near every central point, without dividing by the volume of a stabilizer.

We choose smaller open neighborhoods inside these eight charts; the flat set outside them is a compact regular patch. By Lemma 33.9(1), it is covered by finitely many normal coordinate charts with bounded Haar density. By Taylor's formula as in the proof of part (2) of that lemma, with E_0 in place of E_d, we have $E_0 \geq c_n|v|^2$ on these tubes after shrinking them. Each such tube contributes at most Cg^{6n^3-2}, since its normal rank is $6n^3 - 2$. The complement of the union of these tubes and of the central neighborhoods is compact and contains no flat configuration. Since $E_0 \geq 0$ and its zero set consists exactly of the flat configurations, E_0 has a positive minimum e_n on this complement. The complement therefore contributes at most $e^{-\kappa e_n/g^2} \leq C_{n,\kappa} g^{6n^3-2}$ for $0 < g \leq 1$. Summing over the finite cover, and recalling that the based gauge transformations have mass one, we obtain (1).

Step 5: the stopped measure. For the upper bound in (2), we use (P1), which holds at every integer $d \geq 0$, and the bound $|\overline{R}_{d,\gamma}| \leq C_n$ of (P3) at the depths $d = kJ$. Enlarging the domain of integration from D_r to $\mathcal{U}_n$ and applying (1), we obtain

$$\widehat{Z}_{n,d,\gamma} \leq e^{C_n} \int_{\mathcal{U}_n} e^{-\kappa E_0/g^2}\, dH_n \leq C_n g^{6n^3-2}.$$

Only an upper integral has been enlarged, and the partition function is that of the centered density (33.1).

For the lower bound, we choose a small fixed annulus of constant commuting configurations $X = (\vartheta_i u)_i$, with $|\boldsymbol{\vartheta}|$ between two fixed positive radii, strictly inside the chart of Theorem 33.8, with positive tangential volume and with $\max_i |X_i|/n < \epsilon_{\mathrm{loc}}/2$. Such an annulus exists at every fixed side, since its radii can be taken arbitrarily small. Its commuting refinements and their representatives in the block axial gauge satisfy the domain conditions with the margins of Lemma 33.7, which are independent of the depth; no condition on the side is needed for this annulus. We shrink its fixed normal tube so that it stays inside B_r. On this compact annulus, Lemma 33.9 implies that $E_d \leq C_n|v|^2$ and that the Haar density has positive upper and lower bounds. The lower bound for the density times e^{-C_n} bounds the amplitude $e^{-\overline{R}_{d,\gamma}J}$ from below. We integrate only over $|v| \leq g$ and over a fixed smaller tangential subpatch of positive volume. For g below the

fixed radius of the tube, we obtain

$$\widehat{Z}_{n,d,\gamma} \geq c_n \int_{|v|\leq g} e^{-C_n|v|^2/g^2}\, dv \geq c'_n g^{6n^3-2}.$$

All the sets used here are subsets of D_r. Neither bound requires derivative bounds for the centered effective interaction or any convergence of its restriction to the flat directions as $\gamma \to 0$.

Since $d_n + 4 = 6n^3 - 2 = 2(3n^3 - 1)$ and $g^2 = \gamma/n$, the two bounds have the asserted form γ^{p_n} at the fixed side n. Finally,

$$e^{-E_d/g^2-(R_{d,\gamma}-R_{d,\gamma}(I))} = e^{R_{d,\gamma}(I)} e^{-E_d/g^2-R_{d,\gamma}}$$

on the same domain, which proves the relation between the two normalizations. All constants in the lower and upper bounds for $\widehat{Z}_{n,d,\gamma}$ are uniform in d and γ; their dependence on the side has not been bounded. □

33.7 Linearized flowed cycle logarithms

We now compute the covariance Σ_θ of (4.12), which appears in Theorem 2.10. It is the covariance, divided by γ, of the linear response of the flowed holonomy logarithms at a regular flat connection, under the Gaussian measure defined by the quadratic part of the Yang–Mills action. The computation has three ingredients: the covariance of the transverse Gaussian field at the flat background; the derivative of the flowed cycle logarithms, including the differential of the logarithm; and the projection onto the normal space $\mathcal{N}_y$, which removes the longitudinal part. We carry out the computation in the continuum and on the lattice, and compare the two. The lattice flow is the Wilson flow of Lüscher [128], with the normalization of Section 4.

Theorem 33.11. *Fix $L > 0$, $\theta > 0$, and a compact regular patch of commuting triples $y = (\vartheta_i u)_{i=1}^3$ with $u \in S^2$ and $0 < r_0 \leq |\vartheta| \leq r_1 < \pi$. Work in the variables of the unit torus, $\widetilde{A}(x) = LA(Lx)$, in which the flat background is $\widetilde{A}^0_i = \vartheta_i u$ and the time $t = \theta L^2$ becomes θ. Let $D_i = \partial_i + \vartheta_i \operatorname{ad}_u$ be the covariant derivative of the background, acting on $\mathfrak{su}(2)$-valued one-forms $a = (a_1, a_2, a_3)$ on $\mathbb{T}^3_1$, and consider the quadratic form*

$$\frac{1}{2\gamma} \int_{\mathbb{T}^3_1} \sum_{i<j} |D_i a_j - D_j a_i|^2\, dx.$$

Its kernel $\mathcal{K}$ consists of the covariant gradients $D\phi$ and the constant one-forms with values in $\mathbb{R}u$. The form is positive definite on the quotient by $\mathcal{K}$. Let a be the centered Gaussian field with distributional values on this quotient, realized on the L^2-orthogonal complement of $\mathcal{K}$, whose covariance is γ times the inverse of the positive operator associated with $\int_{\mathbb{T}^3_1} \sum_{i<j} |D_i a_j - D_j a_i|^2\, dx$

on this complement; it is constructed from its Fourier coefficients in Step 2 of the proof. Let $\mathcal{L}$ be the derivative at $\widetilde{A}^0$ of the map that sends a connection to the logarithms of the holonomies of its Yang–Mills heat flow at time θ around the three coordinate cycles based at 0, followed by the orthogonal projection onto $\mathcal{N}_y$. Then $\mathcal{L}$ vanishes on $\mathcal{K}$, and the covariance of $\mathcal{L}(a)$ is $\gamma\Sigma_\theta(y)$, where $\Pi_{\boldsymbol{\vartheta}} = I - \boldsymbol{\vartheta}\boldsymbol{\vartheta}^{\mathsf{T}}/|\boldsymbol{\vartheta}|^2$ and

$$k_m = 2\pi m + \boldsymbol{\vartheta}, \qquad \lambda_m = |k_m|^2, \qquad R_i(m,\boldsymbol{\vartheta}) = \begin{cases} 1, & m_i = 0, \\ \vartheta_i/(2\pi m_i + \vartheta_i), & m_i \neq 0, \end{cases}$$

$$d_i = \sum_{m\in\mathbb{Z}^3} \frac{e^{-2\theta\lambda_m}}{\lambda_m} R_i(m,\boldsymbol{\vartheta})^2, \qquad D = \operatorname{diag}(d_1,d_2,d_3), \qquad \Sigma_\theta(y) = (\Pi_{\boldsymbol{\vartheta}} D \Pi_{\boldsymbol{\vartheta}})\big|_{\boldsymbol{\vartheta}^\perp} \otimes I_{u^\perp}.$$

Here $\mathcal{N}_y$ is identified with $\boldsymbol{\vartheta}^\perp \otimes u^\perp$, and these formulas include the cases in which some coordinates of $\boldsymbol{\vartheta}$ vanish. Moreover, the following hold.

1. *On the patch, $cI_4 \le \Sigma_\theta \le CI_4$, and Σ_θ and its inverse are smooth, covariant under color rotations, and invariant under $(\boldsymbol{\vartheta}, u) \mapsto (-\boldsymbol{\vartheta}, -u)$. The inverse is*

$$\Big[D^{-1} - \frac{D^{-1}\boldsymbol{\vartheta}\boldsymbol{\vartheta}^{\mathsf{T}}D^{-1}}{\boldsymbol{\vartheta}^{\mathsf{T}}D^{-1}\boldsymbol{\vartheta}}\Big]\Big|_{\boldsymbol{\vartheta}^\perp} \otimes I_{u^\perp}.$$

   The constant charged mode $m = 0$ alone contributes $e^{-2\theta|\boldsymbol{\vartheta}|^2}|\boldsymbol{\vartheta}|^{-2} I_4$.

2. *Before the flow is applied, the derivative of the cycle logarithms is*

$$T_i a = J_i^{-1} \int_0^1 e^{v\vartheta_i \operatorname{ad}_u} a_i(ve_i)\, dv, \qquad J_i = \int_0^1 e^{v\vartheta_i \operatorname{ad}_u}\, dv,$$

   where J_i^{-1} is the differential of the logarithm. Its normal projection annihilates covariant gradients and the neutral color $\mathbb{R}u$, and the linearized flow multiplies the charged transverse Fourier mode m by $e^{-\theta\lambda_m}$.

3. *For $N \ge 4$ let $\mathcal{B}_N$ be the set of centered representatives of $\mathbb{Z}^3/N\mathbb{Z}^3$, and set*

$$\delta_i^N = N\big(e^{i(2\pi m_i + \vartheta_i)/N} - 1\big), \qquad \lambda_m^N = \sum_i |\delta_i^N|^2, \qquad R_i^N = i\vartheta_i/\delta_i^N,$$

   with $R_i^N = 1$ if $m_i = \vartheta_i = 0$ and $R_i^N = 0$ if $\vartheta_i = 0 \neq m_i$. On the lattice Λ_N of mesh $1/N$ in $\mathbb{T}^3_1$, consider the background links $e^{\vartheta_i u/N}$, the Gaussian measure defined by the quadratic part of the Wilson exponent $(4N/\gamma)\sum_p q(U_p)$ on the quotient by its kernel, and the derivative of the cycle logarithms after the Wilson flow at lattice time $4\theta N^2$, projected onto $\mathcal{N}_y$. The covariance of this derivative, divided by γ, is exactly

$$\Sigma_\theta^N(y) = \Big(\Pi_{\boldsymbol{\vartheta}} \operatorname{diag}\Big(\sum_{m\in\mathcal{B}_N} \frac{e^{-2\theta\lambda_m^N}}{\lambda_m^N} |R_i^N|^2\Big)\Pi_{\boldsymbol{\vartheta}}\Big)\Big|_{\boldsymbol{\vartheta}^\perp} \otimes I_{u^\perp},$$

   and $\sup_y \|\Sigma_\theta^N(y) - \Sigma_\theta(y)\| \le CN^{-2}$.

4. *As* $\varrho = |\boldsymbol{\vartheta}| \downarrow 0$, *with* $\varrho < r_1 < \pi$,

$$\Sigma_\theta(y) = \varrho^{-2} I_{\mathcal{N}_y} + O_\theta(1), \qquad \varrho^2 \Sigma_\theta(y) \to I_{\mathcal{N}_y},$$

uniformly in the directions of $\boldsymbol{\vartheta}$ *and* u.

The constants, and the bounds for any fixed number of derivatives on the patch, depend on r_0, r_1, θ, *the order of the derivatives, and the margins of the charts, but not on* N, *on* γ, *or on any depth. The bounds on the patch are not uniform as* $|\boldsymbol{\vartheta}| \to 0$.

The positive time is needed: for $\theta = 0$ the series defining d_i diverges logarithmically, since $R_i = 1$ on the plane $m_i = 0$. Part (4) is a statement about the explicit matrix Σ_θ; it is used in Section 36 only for a comparison measure.

Proof. The proof follows the three ingredients described above: Steps 1 and 2 construct the Gaussian field, Steps 3–5 compute the response, and Step 6 performs the projection. Steps 7–10 prove the bounds, the lattice formula, the comparison with the continuum, and the central asymptotics.

Step 1: normalization. The map $\widetilde{A}(x) = LA(Lx)$ sends the flat connection in physical units to $\widetilde{A}_i^0 = \vartheta_i u$ and the time $t = \theta L^2$ to θ. Since the curvature is multiplied by L^2, the volume by L^{-3}, and $g_3^2 L = \gamma$, the quadratic part of the exponent $E_L/(2g_3^2)$ becomes exactly the form in the statement.

Step 2: the Gaussian field. We fix an oriented orthonormal basis of $u^\perp$ in which $v \mapsto u \times v$ is multiplication by i, and regard the two real charged colors as one complex color. On the Fourier mode $e^{2\pi i m \cdot x}$, the operator D_i acts on charged fields as multiplication by $ik_{m,i}$, and the quadratic form on that mode is

$$\sum_{i<j} |k_{m,i} a_j - k_{m,j} a_i|^2 = \lambda_m |a|^2 - |k_m^{\mathsf{T}} a|^2. \tag{33.16}$$

Since $|\boldsymbol{\vartheta}| \geq r_0$ and $r_1 < \pi$,

$$|k_0| \geq r_0, \qquad |k_m| \geq 2\pi |m| - r_1 \quad (m \neq 0),$$

so $k_m \neq 0$ for every m. By (33.16) the charged kernel consists of the multiples of k_m, which are the charged covariant gradients. In neutral color the form at mode m is $|2\pi m|^2 |a|^2 - |2\pi m \cdot a|^2$, whose kernel consists of gradients for $m \neq 0$ and of all constants for $m = 0$. This proves the description of $\mathcal{K}$. The form is positive definite on the quotient by $\mathcal{K}$. We realize the quotient on the L^2-orthogonal complement of $\mathcal{K}$, namely the covariantly transverse one-forms, $\sum_j D_j a_j = 0$, that are orthogonal to the constant neutral one-forms. We show in Step 4 that $\mathcal{L}$ vanishes on $\mathcal{K}$. Since the form vanishes on $\mathcal{K}$ as well, the projection along $\mathcal{K}$ identifies any two complements isometrically for

the form and does not change $\mathcal{L}$. In particular, the law of $\mathcal{L}(a)$ does not depend on the choice of the complement. On the chosen complement we have $k_m^{\mathsf{T}} a = 0$ at each charged mode, and the covariance divided by γ is

$$\lambda_m^{-1} P(k_m), \qquad P(k) = I - kk^{\mathsf{T}}/|k|^2. \tag{33.17}$$

This follows by diagonalizing the finite-dimensional form (33.16) and using $\int e^{-x^2/(2\sigma^2)} e^{i\tau x}\, dx = e^{-\sigma^2\tau^2/2} \int e^{-x^2/(2\sigma^2)}\, dx$. We construct all the Fourier Gaussians in this way and take their sum as a distribution: for every $s > 1/2$ the expected squared norm in the Sobolev space H^{-s} is finite, since $\sum_{m\neq 0} |m|^{-2-2s} < \infty$. Since $\theta > 0$, every functional considered below is an absolutely summable Gaussian series.

The complex notation does not double the covariance. We regard the complex Hilbert space of charged fields as a real Hilbert space with inner product $\operatorname{Re} \int a^* b$. For each orthonormal complex Fourier vector e, the pair e, ie is real orthonormal, and a precision eigenvalue λ assigns the variance γ/λ to each of the two real coefficients. The realification of a Hermitian covariance C is therefore

$$\begin{pmatrix} \operatorname{Re} C & -\operatorname{Im} C \\ \operatorname{Im} C & \operatorname{Re} C \end{pmatrix},$$

with no extra scalar factor. When C is real, this is $C \otimes I_{u^\perp}$ in spatial and color notation.

Step 3: the derivative of the cycle logarithms. Let $Y_i = \vartheta_i u$. With the convention $H'(v) = H(v) A(\ell(v)) \dot{\ell}(v)$ of Definition 4.10, the transport along the ith cycle at the background is e^{vY_i} at the parameter v. Differentiating the transport equation, we find the right-trivialized derivative

$$\delta H_i \, H_i^{-1} = \int_0^1 e^{v\vartheta_i \operatorname{ad}_u} a_i(ve_i)\, dv.$$

For the exponential map the corresponding derivative is $\delta(e^{Y_i}) e^{-Y_i} = J_i \, \delta Y_i$ with $J_i = \int_0^1 e^{v\vartheta_i \operatorname{ad}_u}\, dv$. In neutral color $J_i = 1$, and in charged color $J_i = (e^{i\vartheta_i} - 1)/(i\vartheta_i)$, with value one at $\vartheta_i = 0$. It is invertible on the whole patch because $|\vartheta_i| < \pi$. The differential of the logarithm is therefore J_i^{-1}, which proves the formula for T_i in part (2). On the charged Fourier mode m this derivative is the scalar

$$\frac{\int_0^1 e^{i(2\pi m_i + \vartheta_i)v}\, dv}{\int_0^1 e^{i\vartheta_i v}\, dv}.$$

For $\vartheta_i \neq 0$, canceling $e^{i\vartheta_i} - 1$ gives $\vartheta_i/(2\pi m_i + \vartheta_i)$. If $m_i = 0$, the two integrals are equal, also when $\vartheta_i = 0$. If $\vartheta_i = 0$ and $m_i \neq 0$, the numerator vanishes and the denominator is one. These are exactly the cases in the definition of R_i.

Step 4: the normal projection. Let ϕ be a smooth periodic gauge parameter. Integrating the derivative of $v \mapsto e^{v\vartheta_i \operatorname{ad}_u}\phi(ve_i)$ over $[0,1]$ and using periodicity, we find

$$T_i(D\phi) = \vartheta_i \operatorname{ad}_u \phi(0).$$

As a triple indexed by i, this is $\boldsymbol{\vartheta}$ tensored with a vector of $u^\perp$, an infinitesimal simultaneous conjugation at the root, and the normal projection $\Pi_{\boldsymbol{\vartheta}} \otimes (I - uu^{\mathsf{T}})$ annihilates it. Every response in neutral color is parallel to u and is annihilated as well. Hence the constant neutral one-forms and all other flat tangent directions contribute zero, $\mathcal{L}$ vanishes on $\mathcal{K}$, and the normal response is well defined on the quotient, independently of the representative used to compute it.

Step 5: the linearized flow. At a flat background $[D_i, D_j] = 0$, and the linearization of the Yang–Mills heat flow $\partial_t A_i = \sum_j D_j F_{ji}$ is

$$\partial_\theta a_i = \sum_j D_j(D_j a_i - D_i a_j).$$

Covariant gradients are stationary for this equation. On covariantly transverse fields, $\sum_j D_j a_j = 0$, it reads $\partial_\theta a = \sum_j D_j^2 a$. The Fourier multiplier on charged modes is $-\lambda_m$, and the linearized flow multiplies the mode m by $e^{-\theta\lambda_m}$. A linearized gauge damping term can only change a covariant gradient, whose normal logarithmic response vanishes by Step 4. This derives the response of the Yang–Mills heat flow itself on the quotient, rather than replacing it by the scalar heat flow at a charged background.

Step 6: the covariance. Let $R_m = \operatorname{diag}(R_i(m, \boldsymbol{\vartheta}))$. Applying the response of Steps 3–5 to (33.17), we find the spatial covariance

$$\sum_m \frac{e^{-2\theta\lambda_m}}{\lambda_m} \big(\Pi_{\boldsymbol{\vartheta}} R_m P(k_m) R_m \Pi_{\boldsymbol{\vartheta}}\big)\big|_{\boldsymbol{\vartheta}^\perp}.$$

For each i, including every case in which a coordinate vanishes, $R_i k_{m,i} = \vartheta_i$. Then $R_m k_m = \boldsymbol{\vartheta}$, and the longitudinal term $R_m k_m k_m^{\mathsf{T}} R_m / \lambda_m$ is annihilated exactly by $\Pi_{\boldsymbol{\vartheta}}$. What remains is $\Pi_{\boldsymbol{\vartheta}} D \Pi_{\boldsymbol{\vartheta}}$ restricted to $\boldsymbol{\vartheta}^\perp$. Since this matrix is real, the realification of Step 2 gives the factor $I_{u^\perp}$. This proves the formula for Σ_θ.

Step 7: bounds, inverse, and invariance. Every R_i is bounded on the patch. For any fixed number of derivatives in $\boldsymbol{\vartheta}$, the terms of the series are bounded by $C(1+|m|)^q e^{-c\theta|m|^2}$, with constants depending on the order, on r_0 and r_1, and on θ. The sum of these bounds is finite, as one sees by separating the three coordinates and comparing $\sum_{j\in\mathbb{Z}}(1+|j|)^q e^{-c\theta j^2}$ with a finite initial segment and a convergent integral. Termwise differentiation is therefore valid. The term $m = 0$ has $R_i = 1$ and contributes

$$e^{-2\theta|\boldsymbol{\vartheta}|^2}\big|\boldsymbol{\vartheta}\big|^{-2} I_4.$$

This term gives a positive lower bound on the compact patch, and the summable series gives an upper bound; in particular, all d_i are positive. To invert the form, let $b, v \perp \boldsymbol{\vartheta}$ with $\Pi_{\boldsymbol{\vartheta}} D v = b$. Then $Dv = b + \alpha\boldsymbol{\vartheta}$ for a scalar α, and the condition $\boldsymbol{\vartheta}^{\mathsf{T}} v = 0$ gives

$$\alpha = -\frac{\boldsymbol{\vartheta}^{\mathsf{T}} D^{-1} b}{\boldsymbol{\vartheta}^{\mathsf{T}} D^{-1} \boldsymbol{\vartheta}}.$$

This proves the formula for the inverse. Its denominator is positive, and the bounds just proved give smoothness and the two-sided bound. Color rotations act by the corresponding orthogonal maps of $u^{\perp}$. The reversal $(\boldsymbol{\vartheta}, u) \mapsto (-\boldsymbol{\vartheta}, -u)$ is implemented in the series by $m \mapsto -m$ and complex conjugation of the charged color, since $R_i(m, -\boldsymbol{\vartheta}) = R_i(-m, \boldsymbol{\vartheta})$ and $\lambda_m(-\boldsymbol{\vartheta}) = \lambda_{-m}(\boldsymbol{\vartheta})$. Thus the covariance and its inverse have the asserted invariance. This proves part (1).

Step 8: the lattice covariance. We compute the lattice covariance directly, without using the continuum. At the background links $U_i^0 = e^{\vartheta_i u/N}$, we write a perturbation as $e^{a_i(x)/N} U_i^0$, where x is now an integer lattice coordinate. Differentiating the four factors of the plaquette variable, we find, in charged color,

$$\delta \operatorname{Log} U_{p_{ij}(x)} = N^{-1}\big\{ e^{i\vartheta_i/N} a_j(x + e_i) - a_j(x) - e^{i\vartheta_j/N} a_i(x + e_j) + a_i(x) \big\}.$$

On the Fourier mode m this is $N^{-2}(\delta_i^N a_j - \delta_j^N a_i)$. The coefficient $4N/\gamma$ of the Wilson exponent and $q(e^v) = |v|^2/8 + O(|v|^4)$ give the quadratic exponent

$$\frac{1}{2\gamma N^3} \sum_x \sum_{i<j} |\delta_i^N a_j - \delta_j^N a_i|^2. \tag{33.18}$$

Fourier orthonormality uses the normalized volume $N^{-3}\sum_x$. With $\delta = (\delta_i^N)_i$, the corresponding transverse projection is $P_\delta = I - \delta\delta^*/\lambda_m^N$, and the covariance divided by γ is $(\lambda_m^N)^{-1} P_\delta$. We treat the quotient by the kernel and its independence of the complement as in Step 2, using the description of the kernel in the proof of Lemma 33.9.

We check the flow factor in the same coordinates. In the Lie-algebra coordinates $b = a/N$ of the links, the quadratic part of $\sum_p q(U_p)$ is $\|d_U b\|^2/8$, and its Hessian is $d_U^* d_U/4$. Since the left group coordinates are orthonormal for the group metric at the critical flat configuration, the linearized negative gradient equation is $\partial_s b = -d_U^* d_U b/4$. On the transverse mode m its eigenvalue is $-\lambda_m^N/(4N^2)$, and at the lattice time $s = 4\theta N^2$ the multiplier is exactly $e^{-\theta\lambda_m^N}$.

The derivative of the cycle product, including the differential of its final logarithm, is

$$J_i^{-1} \frac{1}{N} \sum_{l=0}^{N-1} e^{i(2\pi m_i + \vartheta_i) l/N}.$$

For $\vartheta_i \neq 0$, the formula for a geometric sum and the formula for J_i reduce this to

$$\frac{i\vartheta_i}{N\big(e^{i(2\pi m_i+\vartheta_i)/N}-1\big)} = R_i^N.$$

For $\vartheta_i = 0$ the sum is one if $m_i = 0$ and zero otherwise, which are the prescribed values. We conclude that, with $R_m^N = \operatorname{diag}(R_i^N)$,

$$R_m^N \delta = i\boldsymbol{\vartheta}, \qquad \Pi_{\boldsymbol{\vartheta}} R_m^N P_\delta (R_m^N)^* \Pi_{\boldsymbol{\vartheta}} = \Pi_{\boldsymbol{\vartheta}} \operatorname{diag}(|R_i^N|^2) \Pi_{\boldsymbol{\vartheta}}.$$

This equality uses the complex phases of R_i^N before the longitudinal part cancels; the phases can be discarded only afterwards. Summing over the modes in the real covariance, as in Step 2, we obtain the formula for Σ_θ^N.

Step 9: comparison with the continuum. For centered representatives and $N \geq 4$ we have $|k_{m,i}/(2N)| \leq \pi/2 + r_1/8 < 5\pi/8$. On this interval the function $\sin x/x$ has a positive minimum, and Taylor's formula gives $|4\sin^2(x/2) - x^2| \leq C|x|^4$. Since $\lambda_m^N = \sum_i 4N^2 \sin^2(k_{m,i}/(2N))$, we obtain

$$c|k_m|^2 \leq \lambda_m^N \leq |k_m|^2, \qquad |\lambda_m^N - |k_m|^2| \leq CN^{-2}|k_m|^4. \tag{33.19}$$

Also

$$|R_i^N|^2 = R_i^2 \Big(\frac{k_{m,i}/(2N)}{\sin(k_{m,i}/(2N))}\Big)^2, \qquad \big||R_i^N|^2 - R_i^2\big| \leq CN^{-2} R_i^2 k_{m,i}^2 \leq CN^{-2},$$

where the continuous extensions at the vanishing cases are understood. In particular $R_i^2 k_{m,i}^2 = \vartheta_i^2$, also when $m_i = 0$, and all these multipliers are bounded. For $F(x) = e^{-2\theta x}/x$ we have $|F'(x)| = e^{-2\theta x}(2\theta/x + x^{-2})$. By the mean value theorem and (33.19), the difference of corresponding summands is at most

$$CN^{-2} e^{-c\theta|k_m|^2} \big(1 + |k_m|^2 + |k_m|^{-2}\big),$$

which is summable uniformly on the patch. The continuum modes not in $\mathcal{B}_N$ satisfy $|m_i| \geq N/2$ for some i, and their Gaussian tail is at most $Ce^{-c'\theta N^2} \leq C_\theta N^{-2}$. Summing, we obtain the uniform bound CN^{-2} in part (3). Fixed derivatives in the parameters obey the same bound. Indeed, differentiating the Taylor remainders in the sine expressions leaves a factor N^{-2} times a fixed polynomial in $|k_m|$. All inverse powers of λ_m^N and λ_m are bounded by the gap on the patch, and the differentiated exponentials add only polynomial factors. The same summable majorant applies, with constants depending also on the order, and smooth local normal frames introduce only constants depending on the margins of the patch.

Step 10: the central asymptotics. Let $\varrho = |\boldsymbol{\vartheta}| \downarrow 0$ with $\varrho < r_1 < \pi$. The nonzero modes still satisfy $|k_m| \geq 2\pi|m| - r_1$, and their multipliers R_i are bounded uniformly in all directions, since $|2\pi m_i + \vartheta_i| \geq \pi$ for $m_i \neq 0$. Their total contribution is $O_\theta(1)$. The zero mode contributes

$$e^{-2\theta\varrho^2}\varrho^{-2} I_4 = \varrho^{-2} I_4 + O_\theta(1).$$

Since the projection onto the normal space has norm one, this proves both assertions of part (4), uniformly in the directions. It also shows why the bounds on the regular patch cannot be extended uniformly to the central point. □

33.8 The stopped measure near regular flat configurations

We now combine the preceding results. The first part of the next proposition is the Laplace approximation of the stopped measure transverse to a compact patch of regular flat configurations. The second part is an exact decomposition of the covariance Σ_θ^N of Theorem 33.11 at the flat reconstruction, obtained by completion of squares for Gaussian measures in the constrained coordinates of Theorem 12.13. In Section 35 this decomposition identifies the covariance of the Gaussian approximation obtained at finitely many scales with Σ_θ.

Proposition 33.12. *Consider the stopped measures of Section 33.2 at the side n, a compact regular patch, and a fixed normal tube over a larger compact regular patch whose closure lies in D_r. Assume that $2\pi/n < r_{\mathrm{flat}}$, where r_{flat} is the radius of Lemma 14.9, which lies below the fixed margins of all classical, constraint, Haar disintegration, and gauge domains. For a patch of small commuting configurations as in Lemma 33.7, the margins of that lemma may be used instead of this condition.*

1. *There is a finite atlas of normal coordinates $V(y,v)$ over the patch, of normal rank $\mathfrak{d}_n = 6n^3 - 2 = d_n + 4$, in which the image of H_n has a smooth positive density $J(y,v)$, and for every integer $d \geq 0$*

$$nE_d(V(y,v)) = \tfrac{1}{2} v^{\mathsf{T}} H_d(y) v + O_n(|v|^3),$$
$$c_n I \leq H_d(y) \leq C_n I, \qquad nE_d(V(y,v)) \geq c_n |v|^2.$$

Let χ be a fixed normal cutoff equal to one near $v = 0$, let f be a bounded measurable function of y with compact support in the patch, and let P_y be a family of polynomials of a fixed degree with bounded measurable coefficients. Put

$$z_0 = H_d(y)^{1/2} v/\sqrt{\gamma}, \qquad a_{d,\gamma}(y,v) = e^{-\overline{R}_{d,\gamma}(V(y,v))} J(y,v).$$

Then for $d = kJ$ and $0 < \gamma \le \gamma_0$, with the partition of unity of the atlas understood and the insertion extended by zero outside the charts,

$$\widehat{Z}_{n,d,\gamma}\mathbb{E}_{\widehat{\nu}_{n,d,\gamma}}[f(y)\chi(v)P_y(z_0)] = (2\pi\gamma)^{\mathfrak{d}_n/2}\Big[\int f(y)\frac{a_{d,\gamma}(y,0)}{\sqrt{\det H_d(y)}}\mathbb{E}P_y(G_{\mathfrak{d}_n})\,d\mathrm{vol}(y) + O_{n,f,P}(\sqrt{\gamma})\Big].$$

The error is uniform in d and γ. Given y, the moments of z_0 under the normal conditional law, normalized by its own normal integral, differ from those of $G_{\mathfrak{d}_n}$ by $O_n(\sqrt{\gamma})$, and all fixed moments of z_0 are bounded. The tangential density may vary with γ within its bounds, without converging. Here $\widehat{Z}_{n,d,\gamma}$ is the partition function of Theorem 33.10. For every nonnegative f as above with $\int f\,d\mathrm{vol} > 0$ there exists $c_{n,f} > 0$ such that $\mathbb{E}_{\widehat{\nu}_{n,d,\gamma}}[f(y)\chi(v)] \ge c_{n,f}$ for all depths $d = kJ$ and all sufficiently small γ.

2. *Suppose in addition that the cycle holonomies of the patch have logarithms $y = (\vartheta_i u)_i$ with $0 < r_0 \le |\boldsymbol{\vartheta}| \le r_1 < \pi$, and fix $\theta > 0$. Let $K \ge 0$, $N = nB^K$, $n_j = nB^j$, $d = JK$, and $H_K^{\mathrm{act}}(y) = D_v^2(nE_{JK})(y,0)$. Consider the reconstruction of a configuration on Λ_N from the stopping field V on Λ_n and the normalized fluctuation variables $w_j = (w_{j,l})_{0\le l<J}$ of the RG steps $j = 1,\dots,K$, given by the triangular coordinates*

$$z_{j,l} = z_{j,l}^{*} + g_{j,l}K_{j,l}^{-1/2}w_{j,l}$$

of Theorem 12.13, computed in smooth local charts in which the expressions in different gauges agree, and satisfying $T_M^{JK}U = V$. At the flat reconstruction of the flat stopping field $V(y,0)$, where all w_j vanish, differentiate the composition of this reconstruction with the map that sends a configuration on Λ_N to the logarithms of the cycle holonomies, based at the root, of its Wilson flow at physical time θL^2, followed by the orthogonal projection onto $\mathcal{N}_y$. Normalize the normal variable of the stopping field by $v = \sqrt{\gamma}(H_K^{\mathrm{act}})^{-1/2}z_0$. After division by $\sqrt{\gamma}$, denote the resulting linear maps by $A_{0,K}(y) : \mathbb{R}^{\mathfrak{d}_n} \to \mathcal{N}_y$ and $A_{j,K}(y) : \mathbb{R}^{D_j} \to \mathcal{N}_y$, $1 \le j \le K$, where $D_j = 6(n_j^3 - n_{j-1}^3)$ is the number of real normalized fluctuation variables of the jth RG step. Then for every finite K,

$$A_{0,K}A_{0,K}^{\mathsf{T}} + \sum_{j=1}^{K} A_{j,K}A_{j,K}^{\mathsf{T}} = \Sigma_\theta^N(y), \qquad \|A_{0,K}\|_{\mathrm{HS}}^2 + \sum_{j=1}^{K}\|A_{j,K}\|_{\mathrm{HS}}^2 \le C.$$

Each $A_{j,K}$ includes the factors $g_{j,l}/\sqrt{\gamma}$, the inverse square roots of the constrained Hessians at the flat background, and the derivative of the composition with all finer reconstruction steps. If T_K denotes the response of the normal variable of the stopping field before normalization and $H_K^E = D_v^2E_{JK}(y,0)$ in the same normal coordinates, then

$$H_K^{\mathrm{act}} = nH_K^E, \qquad A_{0,K}A_{0,K}^{\mathsf{T}} = n^{-1}T_K(H_K^E)^{-1}T_K^{\mathsf{T}}.$$

The same identities hold for the reconstruction maps $\widehat{\mathscr{R}}$ of Theorem 15.12.

The constants depend on n, L, θ, M, J, Θ, the patch, the larger tube, the bounds for the polynomials, and the fixed analytic domains, with the dependence described in item (6) of Definition 4.14. They are independent of K, of the depth, and of $\gamma \in (0, \gamma_0]$.

The identities in part (2) concern the Gaussian quadratic model at the flat reconstruction; they do not assert that the normalized fluctuation variables are independent under the interacting measure. The normal responses of the stopping field, including those of the constant charged modes, are part of the identity: for $K = 0$ the block $A_{0,0}$ gives the whole covariance.

Proof of part (1) of Proposition 33.12. We first check that the flat configurations and their refinements lie in the domains of the construction, separately from the normal integral. Then we apply Lemma 33.3 in the normal coordinates of Lemma 33.9.

Step 1: flat refinements. First suppose that $2\pi/n < r_{\mathrm{flat}}$. Fix the side n before the coupling, and let

$$\delta_{\mathrm{flat}} = r_{\mathrm{flat}} - 2\pi/n > 0. \tag{33.20}$$

The radius r_{flat} lies below all the classical, constraint, Haar disintegration, and gauge margins of one RG step, with the strict margin required in Lemma 14.9. These margins include the fixed conversions for the words of the block axial gauge. A flat triple of cycle holonomies has commuting logarithms H_i with $|H_i| \le 2\pi$. At every averaging step with output side $m \ge n$, its constant representative and its refinement at the next finer side have link logarithms H_i/m and $H_i/(Mm)$, and

$$\max_i |H_i|/m \le 2\pi/n = r_{\mathrm{flat}} - \delta_{\mathrm{flat}}.$$

By the computation for commuting products leading to (33.11), the representative in the block axial gauge is $U_0(c)$, and the free logarithmic coordinates and the logarithms on the eliminated edges vanish. The conclusion of Lemma 14.9 for all flat configurations, with smallest side n, verifies every intermediate classical, constraint, based gauge, and small-field condition, for every K. The classical action of the free coordinates, which is nonnegative, vanishes at this point. By the uniqueness of the conditional constrained minimizer in the same conclusion, the point is $z_i^*(c) = 0$, and each normalized fluctuation variable vanishes. This verifies the hypothesis of part (2) at all the depths involved. The margins and the radii of the fluctuation cutoff domains are fixed by the parameters. No neighborhood of the flat set depending on the cutoffs of the small-field event is used as a normal tube. Under the alternative local hypothesis, we apply Lemma 33.7 on a slightly larger compact patch satisfying its strict inequality. It gives the same conclusions at every side $m \ge n$ and every depth, and its positive domain margins replace δ_{flat} below.

Step 2: phase and amplitude. The finite atlas and the estimates for the phase are those of Lemma 33.9, applied on the larger patch and then shrunk. The normal rank is $\mathfrak{d}_n = 6n^3 + 3 - 5 = 6n^3 - 2 = d_n + 4$. The vanishing of E_d on the flat set, the lower bound (P1), and the uniform bounds on third derivatives were verified there. The centered effective interaction of Theorem 14.8 is bounded on D_r by (P3). For its derivatives, we take the normal atlas on the larger tangential patch with a larger normal radius, and finitely many relatively compact parameter subcharts $W_l \Subset W_l^+$ covering the working tube. The maps $\Psi_l(y, v) = V(y, v)$ and their based lifts are smooth on W_l^+, as constructed in Lemma 33.9, and extend past $\overline{W}_l$. Let $\mathcal{K}$ be the compact union of the images of the closures in the larger tube; it lies in D_r by hypothesis and does not depend on d or γ. On these compact subcharts we apply (33.10). Finitely many fixed based gauge transformations map neighborhoods of $\mathcal{K}$ into B_{r_1}, with $r < r_1 < \rho$ and the fixed margin η. We refine the subcharts so that each closure lies in one of the larger neighborhoods in the proof of Lemma 33.6. Each gauge transformation acts smoothly on all of $\mathcal{U}_n$. The chart composed with it extends to a neighborhood of the closure of the subchart, and its derivatives of every fixed order are bounded on these finitely many compact closures, independently of d and γ. These are the conditions on subcharts in Theorem 14.8, at each chart of the atlas. The chain rule then gives $|D_{y,v}\overline{R}_{d,\gamma}(V(y, v))| \le C_n E_*$ on each compact parameter closure, uniformly in the depth and the coupling. The amplitude

$$a_{d,\gamma}(y, v) = e^{-\overline{R}_{d,\gamma}(V(y,v))} J(y, v)$$

therefore has fixed positive upper and lower bounds and a fixed Lipschitz bound in v; it includes the Haar density of the based quotient. Since all insertions in the charts are supported in D_r, the indicator of D_r equals one where these coordinates are used. On each normal fiber, with $B_n = \sup_{D_r} |\overline{R}_{d,\gamma}|$, the mean value theorem gives

$$|a_{d,\gamma}(y, v) - a_{d,\gamma}(y, 0)| \le e^{B_n} \big(\|J\|_\infty C_n E_* + \|D_v J\|_\infty \big) |v|.$$

This is the Lipschitz bound in the normal direction required by Lemma 33.3; a bound on the values of the effective interaction alone would not imply it.

Step 3: the Laplace approximation. We apply Lemma 33.3 with $\delta = \gamma$, the phase nE_d, the Hessian $L = H_d$, and the weight h given by f times the partition of unity of the atlas; the formula for general f follows by splitting f into its positive and negative parts. The phase, the amplitude, and the Hessian are smooth in each real normal trivialization, hence jointly Borel, and we choose the normal cutoff smooth as well. The tests and the polynomial coefficients are measurable, as required in that lemma. We decrease the constant c_0 of that lemma and increase C_0 to include the bounds for the normal Hessian,

the lower bound for the phase, the bounds for the amplitude, and the Lipschitz constant just verified. No matrix L_∞ is substituted in this application. We obtain

$$(2\pi\gamma)^{\mathfrak{d}_n/2}\Big[\int f(y)\frac{a_{d,\gamma}(y,0)}{\sqrt{\det H_d(y)}}\mathbb{E}P_y(G_{\mathfrak{d}_n})\,d\mathrm{vol}(y)+O_{n,f,P}(\sqrt{\gamma})\Big].$$

Since the exponent of (33.1) is $E_d/g^2 = nE_d/\gamma$, the left side of (33.3) is exactly the integral of the insertion extended by zero against the unnormalized stopped density, namely $\widehat{Z}_{n,d,\gamma}\mathbb{E}_{\widehat{v}_{n,d,\gamma}}[f(y)\chi(v)P_y(z_0)]$. This proves the displayed identity, with the partition function of Theorem 33.10. The statements about conditional moments and bounded moments are part (1) of Lemma 33.3. If the conditional law is taken on the whole tube instead of with the cutoff, the additional part lies outside a fixed smaller normal ball. By Corollary 33.4 and the bounds on the whole tube verified below, it is exponentially small, and the conditional statements are unchanged.

Step 4: lower bounds for tangential tests. Let $f \geq 0$ with $\int f\,d\mathrm{vol} > 0$. By the case $P = 1$ of the formula, with the error $C\sqrt{\gamma}\int f$, the numerator is at least $c_{n,f}\gamma^{\mathfrak{d}_n/2}$ for all sufficiently small γ. The upper bound of Theorem 33.10 is $\widehat{Z}_{n,d,\gamma} \leq C_n g^{\mathfrak{d}_n} = C_n n^{-\mathfrak{d}_n/2}\gamma^{\mathfrak{d}_n/2}$, with n fixed. Dividing, we obtain the uniform positive lower bound for the expectation. For an event expressed by a bounded insertion, dividing this expectation by the supremum of the insertion gives a lower bound for the probability as well. The tangential amplitude may depend on γ arbitrarily within its uniform bounds. □

Proof of the last assertion of Corollary 33.4. We verify the additional hypotheses on the whole tube in the setting of part (1) of Proposition 33.12, with $d = JK$. On the fixed working tube, Step 2 and Lemma 33.9 give $nE_{JK}(V(y,v)) \geq c_n|v|^2$, the quadratic expansion, and the bounds for the spectrum of H_K^{act}. The amplitude $a_{JK,\gamma}(y,v) = e^{-\overline{R}_{JK,\gamma}(V(y,v))}J(y,v)$ has uniform positive upper and lower bounds there: the centered effective interaction is bounded on D_r, and the Haar density of the based quotient is smooth and positive on the fixed compact subcharts. These bounds are used before the partition of unity of the atlas is absorbed into the tangential weight. Then the normal integral over the whole tube is at least $c_n\gamma^{\mathfrak{d}_n/2}$, uniformly in K and y, and the statements of Corollary 33.4 on the whole tube apply with $\delta = \gamma$, $L = H_K^{\mathrm{act}}$, and rank $\mathfrak{d}_n = 6n^3 - 2$, with constants independent of K. □

Proof of part (2) of Proposition 33.12. We first count the variables. Then we expand the classical action to second order along the flat reconstruction, which gives an identity $S_K^{\mathsf{T}}\mathcal{H}_N S_K = I$ for the differential S_K of the reconstruction. Finally, we apply the derivative of the observable and Theorem 33.11.

Step 5: the count of variables. By Theorem 12.13, one averaging step from side Mm to side m has $6((Mm)^3 - m^3)$ real free coordinates. Summing over the J averaging steps of the jth RG step, from side n_j to side n_{j-1}, we find

$$D_j = 6(n_j^3 - n_{j-1}^3), \qquad \mathfrak{d}_n + \sum_{j=1}^{K} D_j = 6N^3 - 2. \tag{33.21}$$

These are counts on based quotients. The five flat tangent directions belong to the stopping field, and they are held fixed in the quadratic computation. The right side $6N^3 - 2$ is the dimension of the based tangent space at side N, $6N^3 + 3$, minus the five-dimensional kernel of the Hessian of the Wilson action at a regular flat configuration (Lemma 33.9).

Step 6: the flat reconstruction. At a flat stopping field, Step 1 gives the flat refinement with vanishing normalized fluctuation variables at every averaging step, with the same based cycle holonomies and zero classical action. This uses Lemma 14.9 or, under the alternative local hypothesis, Lemma 33.7. The conditional constrained minimizers are these flat refinements. In particular, the flat zero-fluctuation reconstruction is critical for each classical action, and it is stationary for the Wilson flow on the finest lattice. The fixed domain conditions give smooth constraint coordinates and Haar disintegration densities near this reconstruction. By Theorem 15.12, the reconstruction maps $\widehat{\mathscr{R}}$ preserve the coarse field, the orbit of the minimizer, the free coordinates, the square roots of the constrained Hessians, and the density in these coordinates. Their gauge transformations are smooth on these local charts and equal to the identity at the coarse vertices. No gauge transformation chosen measurably as a function of the configuration is differentiated. At each RG step we use the nested domains $\mathscr{D}_{\mathrm{in}} \Subset \mathscr{D}_{\mathrm{mid}} \Subset \mathscr{D}_{\mathrm{out}}$ of Theorem 15.12. The flat coarse field lies below its margin by the domain margins verified in Step 1, and every free coordinate vanishes. Hence a joint neighborhood of the flat reconstruction lies in $\mathscr{D}_{\mathrm{in}}$ before the outer domain of holomorphy is used. The fluctuation cutoff domains satisfy (33.2), and the reconstruction maps keep these domains, their kernels, and their identities at the coarse vertices. These checks use only the fixed parameters and the integer sides mM^i, $m \geq n \geq 4$, and no domain depending on the depth.

Step 7: completion of squares. We consider one averaging step with coarse field c and free coordinate z, and set

$$F(c,z) = E_{\mathrm{old}}(b(c,z)), \qquad z = z^*(c) + g_{\mathrm{old}} K(c)^{-1/2} w, \qquad K(c) = D_z^2 F(c, z^*(c)),$$

where E_{old} and g_{old} are the effective classical action and the coupling on the finer side. The minimizer satisfies $D_z F(c, z^*(c)) = 0$. By Taylor's formula and the definition of $K(c)$, we have

$$\frac{F(c,z)}{g_{\mathrm{old}}^2} = \frac{F(c,z^*(c))}{g_{\mathrm{old}}^2} + \tfrac{1}{2}|w|^2 + U^{\mathrm{cl}}(c,w), \qquad D_w^\alpha U^{\mathrm{cl}}(c,0) = 0 \quad (|\alpha| \leq 2).$$

The exact identity for the classical actions in Theorem 12.13 identifies the first term with $E_{\mathrm{new}}(c)/g_{\mathrm{new}}^2$, and these identities compose over the averaging steps. Since the displayed derivatives vanish for every coarse field c in the domain, U^{cl} has no joint Taylor terms of total degree at most two in the variation of the coarse field and the normalized fluctuation variables at the flat reconstruction. Hence, at the flat reconstruction, the quadratic part of the exponent is exactly

$$\frac{1}{2\gamma}\langle v, H_K^{\mathrm{act}} v\rangle + \frac{1}{2}\sum_{j=1}^{K}|w_j|^2, \qquad H_K^{\mathrm{act}} = D_v^2(nE_{JK})(y,0), \tag{33.22}$$

where we used that the exponent at the stopping side is $E_{JK}/g_n^2 = nE_{JK}/\gamma$. Second derivatives of the changes of coordinates do not affect this conclusion. In the second derivative of a composed scalar function, they are multiplied by the first derivative of the function, which vanishes at a flat configuration. Since the dependence of the positive square roots on the background is included in the differential of the coordinates, it creates no missing quadratic cross term. The Haar disintegration factors and the effective interaction remain in the amplitude and are not part of this identity for the classical quadratic form.

Step 8: Gaussian linear algebra. Let $\mathcal{H}_N$ be the Hessian of NE_0 at the finest flat configuration, on the based tangent space modulo its five-dimensional kernel; it is positive definite there by Lemma 33.9. By the count in the proof of that lemma, this quotient is the quotient of the full space of link variations by the kernel of the Hessian, which consists of all covariant gradients and the three constant neutral directions. Thus it is the space on which the lattice covariance of Theorem 33.11(3) is computed, and the exponent NE_0/γ is the Wilson exponent $(4N/\gamma)\sum_p q(U_p)$ used there. As in the statement, $v = \sqrt{\gamma}(H_K^{\mathrm{act}})^{-1/2} z_0$. The normalized fluctuation variable $w_{j,l}$ of each averaging step enters through the differential $g_{j,l}K_{j,l}^{-1/2}w_{j,l}$ of the free coordinate. Composing with all finer reconstruction steps, we obtain the differential of the configuration on the finest lattice. After extracting $\sqrt{\gamma}$, we denote by S_K this linear map from $(z_0, w_1, \dots, w_K)$ to the quotient on the finest lattice. It does not depend on γ: the remaining coupling factors are $g_{j,l}/\sqrt{\gamma}$, and the classical maps and Hessians do not depend on the coupling. Since the quadratic part of NE_0/γ at the finest flat configuration, composed with the reconstruction, is (33.22), we have

$$S_K^{\mathsf{T}} \mathcal{H}_N S_K = I. \tag{33.23}$$

This makes S_K injective. Since its domain and codomain have the same dimension by (33.21), S_K is invertible, and

$$S_K S_K^{\mathsf{T}} = \mathcal{H}_N^{-1}.$$

Equivalently, consider independent standard real Gaussian variables in the normal coordinates of the stopping field and in all normalized fluctuation variables. In the auxiliary quadratic model, the differential of the reconstruction sends them to the Gaussian field on the finest lattice with covariance $\gamma \mathcal{H}_N^{-1}$.

Step 9: the observable. Let ℓ_N be the derivative of the flowed based cycle logarithms on the finest lattice, followed by the projection onto $\mathcal{N}_y$. By the computations for gauge directions and flat tangent directions in the proof of Theorem 33.11, ℓ_N vanishes on the kernel of the quotient. Applied to the columns of S_K, it gives exactly the response maps of the statement:

$$\ell_N S_K = (A_{0,K}, A_{1,K}, \dots, A_{K,K}).$$

The finest flat reconstruction has the same based cycle holonomies as the constant links $e^{\vartheta_i u/N}$. To justify the use of this Fourier representative, we fix a spanning tree and compare the flat transports of the reconstruction with those of the constant links. Their transports along the tree determine a periodic vertex gauge transformation, equal to the identity at the root, because the three cycle holonomies agree. This gauge transformation identifies the two flat configurations. The Wilson action and the Wilson flow commute with it, and the based holonomies are the same before and after it. If such a gauge transformation depends on parameters, its derivative contributes a covariant gradient, whose normal logarithmic response vanishes by the same computation. The covariance of ℓ_N is therefore exactly the one computed in Theorem 33.11(3), and

$$\sum_{j=0}^{K} A_{j,K} A_{j,K}^{\mathsf{T}} = \ell_N S_K S_K^{\mathsf{T}} \ell_N^{\mathsf{T}} = \ell_N \mathcal{H}_N^{-1} \ell_N^{\mathsf{T}} = \Sigma_\theta^N(y).$$

Taking traces and using the uniform upper bound for Σ_θ^N on the patch, which follows from Theorem 33.11(1) and (3), we obtain $\sum_{j=0}^{K} \|A_{j,K}\|_{\mathrm{HS}}^2 \le 4C$.

Step 10: the block of the stopping field. If T_K is the response of the normal variable of the stopping field before normalization, then

$$H_K^{\mathrm{act}} = n H_K^E, \qquad A_{0,K} = T_K (H_K^{\mathrm{act}})^{-1/2} = n^{-1/2} T_K (H_K^E)^{-1/2}.$$

Multiplying by the transpose, we obtain the last identity of the statement. For $K = 0$ this block gives the whole covariance; in particular, it cannot be omitted. Since the gauge transformations of the reconstruction maps $\widehat{\mathscr{R}}$ are equal to the identity at the coarse vertices and contribute only gauge variations, which are annihilated, the proof applies to these maps without a new choice of representatives. No growth in the number of scales enters the trace bound or the error of the Laplace approximation. □

Remark 33.13. The convergence of the effective classical actions as the depth tends to infinity (Theorem 11.2) does not allow one to replace the phase at finite depth in an

exponent divided by g^2, uniformly as $\gamma \to 0$. None of the proofs above makes such a replacement. The Gaussian covariance computed here is the covariance of the quadratic model at a flat background; transferring its moments to the flowed observables under the interacting measures requires the additional estimates of Sections 34 and 35. The identification of the limiting observables uses Theorem 29.25, and Lemma 5.6 identifies the continuous functions of the cycle holonomies that are invariant under simultaneous conjugation as functions of the seven traces; neither identification replaces a comparison of normal moments.

33.9 An Abelian comparison measure

For later comparison with the measures λ_γ we record one explicit comparison measure in which all cycle holonomies commute: the Abelian Maxwell measure with Haar cycle holonomies of Definition 4.13. It is used, together with the other comparison measures, in Sections 35 and 36.

Lemma 33.14. *Embed the subgroup $\{e^{v\tau_3} : v \in \mathbb{R}\}$ of $\mathrm{SU}(2)$. Take a centered transverse Gaussian Maxwell field with one real color on $\mathbb{T}^3_L$, with covariance $g_3^2(-\Delta)^+P_{\mathrm{tr}}$, where $(-\Delta)^+$ is the inverse of $-\Delta$ on the nonconstant Fourier modes and P_{tr} the projection onto transverse fields, and three independent phases, uniformly distributed in $(\mathbb{R}/4\pi\mathbb{Z})^3$ and independent of the field. Smooth the field by the scalar heat flow at a fixed positive time, add the flat connection with these cycle phases, and form the three cycle holonomies in the embedded subgroup. Then the following hold.*

1. *The smoothed field is almost surely smooth, so the holonomies are defined. Conditionally on the Maxwell field, the three cycle angles are distributed according to the Haar probability measure on $(\mathbb{R}/4\pi\mathbb{Z})^3$; hence their joint law is exactly Haar. The three holonomies commute, and an independent common random conjugation does not change the value of any function invariant under simultaneous conjugation.*
2. *Let R_{ab} be the Euclidean norm of the three logarithms of the cycle holonomies, taken in the principal chart. For every nonnegative $\varphi \in C_c^\infty((1,2))$ and every δ with $0 < 4\delta < 2\pi$,*

$$\mathbb{E}_{\mathrm{ab}}\varphi(R_{\mathrm{ab}}/\delta) = (4\pi)^{-3}\,4\pi\delta^3\int_1^2 \varphi(v)v^2\,dv.$$

This lemma specifies a comparison probability measure and makes no assertion about the Wilson measure.

Proof. In the Fourier modes $k = 2\pi m/L$, $m \neq 0$, we use the transverse multiplier $g_3^2|k|^{-2}(I - kk^{\mathsf{T}}/|k|^2)$ in one real color, with real sine and cosine bases orthonormal in $L^2(\mathbb{T}^3_L, dx)$. Independent real Gaussian coefficients in a transverse orthonormal basis define this field. For every $s > 1/2$, the expected sum of the squared coefficients with

the weight $(1+|k|^2)^{-s}$ is finite, since $\sum_{m\neq 0}|m|^{-2-2s}<\infty$. The weighted sum of squared coefficients is then a nonnegative random variable with finite expectation, and the Fourier series converges almost surely in this weighted Hilbert norm. This constructs the field as a distribution. After any positive heat time, the expected sum of the absolute values of the differentiated Fourier coefficients is finite at every order of differentiation, because of the factor $e^{-t|k|^2}$. The smoothed field is therefore almost surely smooth, and the three cycle integrals are defined.

We add the independent flat connection whose three cycle integrals are the independent uniform phases $\alpha_i \in \mathbb{R}/4\pi\mathbb{Z}$. In the embedded Abelian subgroup, each smoothed cycle angle is $\alpha_i + l_i$ modulo 4π, where l_i is a function of the Maxwell field. Since a translation of a uniform variable on the circle preserves its law, the three angles are distributed, conditionally on the Maxwell field, according to the Haar probability measure. If $\beta_i \in [-2\pi, 2\pi)$ are their representatives, the distance of $e^{\beta_i \tau_3}$ from the identity is $|\beta_i|$, and $R_{\mathrm{ab}} = |\beta|$. Since the support of the insertion lies in the Euclidean ball of radius $2\delta < 2\pi$, polar coordinates in $\mathbb{R}^3$ give

$$(4\pi)^{-3}\int_{\mathbb{R}^3}\varphi(|\beta|/\delta)\,d\beta = (4\pi)^{-3}\,4\pi\delta^3\int_1^2 \varphi(v)v^2\,dv.$$

All group elements here belong to the same Abelian subgroup. An independent common conjugation therefore preserves the values of all functions invariant under simultaneous conjugation, and it preserves commutativity. This completes the proof. □

SECTION 34

Uniform fluctuation bounds and comparison of holonomies

In this section, we prove estimates for the conditional refinement of the coarse field to the fine field that are uniform as the dimensionless coupling $\gamma = g_3^2 L$ tends to zero, and we transfer them to the holonomy matrices of the Wilson flow at a positive time. The main application is Theorem 34.13, which is Theorem 2.9: every subsequential limit of the continuum measures λ_γ as $\gamma \to 0$ is supported on stationary flat trajectories. The holonomy estimates are also the inputs for the normal fluctuations near regular commuting triples in Section 35 and for the quartic asymptotics near the central triples in Section 36.

The basic comparison is the following. We fix the coarse field and the normalized fluctuation variables of the first m renormalization group steps from the coarse side, and we set the normalized fluctuation variables w_j of all later steps equal to zero. Applying the same reconstruction maps gives a configuration on the finest lattice, which we evolve by the same Wilson flow as the reconstructed field. We then compare the holonomies of the two flowed configurations around fixed loops, as matrices in one common initial gauge frame. The comparison configuration is neither the interpolation of the field on the m-th lattice nor a conditional mean: setting $w_j = 0$ is different from setting the centered variable $w_j - \mathbb{E}(w_j \mid \mathcal{F}_{j-1})$ to zero.

We use three probability settings, summarized in Table 34.1.

Two estimates explain the exponent conditions used throughout. First, under the conditioned measure the reconstructed field at mesh ℓ_j has supremum norm at most of order $r\varrho^j + g_3\sqrt{\ell_j}\, p_j$, where $\varrho < B^{-1/2}$ is the contraction factor of the zero-fluctuation reconstruction maps and p_j is the logarithmic cutoff on the fluctuation variables. The signed difference between the linearized reconstruction and the interpolation of the coarser field costs a factor $\ell_j^{\alpha-1}$ in the norm of the linear heat evolution. Its random part therefore has size $g_3 \ell_j^{\alpha-1/2}$, which is summable over the geometric sequence of meshes because $\alpha > 1/2$. A classical quadratic term has size ϱ^{2j}/ℓ_j, which is summable because $B\varrho^2 < 1$. The second point concerns maxima. The supremum over the finest lattice of a linear function of the fluctuation variables would cost a factor $\sqrt{\log(L/a)}$, which grows as the finest mesh $a \to 0$. Instead, we cover the coefficient arrays at the mesh ℓ_j of their inputs, in metrics of bounded diameter, and chain in these metrics. The loss is then

Setting	What is fixed	Random variables	Conclusion
Conditioned Wilson measure (Theorems 34.9 and 34.10)	The parameters; the coarse field is random	Fluctuation variables with their conditional fluctuation kernels	Holonomy differences have moments of order $\gamma^{p/2}$, and the leading term decays in m
Prescribed coarse field (Theorem 34.11)	Every initial field $v = e^{\theta_0}$ with $\|\theta_0\|_\infty \le r$	The same kernels, iterated from v	Moments of order $\gamma^{p/2}$ of the supremum over a time interval, uniformly in v
Prescribed flat field (Lemma 34.12)	A regular flat coarse field y	Independent standard Gaussian vectors inserted into derivatives of the reconstruction	The omitted Gaussian covariance decays geometrically in m

Table 34.1. The three probability settings of this section.

$\sqrt{\log(2 + L/\ell_j)}$, which is summable against the positive powers of ℓ_j. Combined with dimension-free concentration for the conditional fluctuation kernels, this gives moment bounds of order $\sqrt{\gamma}$ without a factor $|\log \gamma|$ in the leading term.

The proofs use the following inputs. The concentration estimates for the conditional fluctuation kernels are those of Theorem 5.4, which adapt the methods of Brascamp and Lieb [33] and Bakry and Émery [8] to products of balls with reflecting boundary. The quadratic estimate there is of Hanson–Wright type [108], in the setting of dependent coordinates exemplified by Adamczak [1]. Neither independence nor Gaussianity of the fluctuation variables is used. The maximal estimates use increment chaining in the tradition of Dudley [79] and Dirksen [74, Theorems 3.2 and 3.5], in the elementary form of Lemma 16.4. The reconstruction maps, their coefficient bounds, and the conditional kernels come from Section 15; the zero-fluctuation maps are built from constrained minimizers in the tradition of Bałaban's background fields [10, 11, 13, 16]. The parameters are those of Theorem 23.2. The coefficient bounds for the four parts of a reconstruction increment and for their ordered quadratic products are those of Sections 26 to 28. The decomposition of a rough initial datum into its linear heat evolution, its quadratic Picard term, and a smoother remainder follows Da Prato and Debussche [63] and Cao and Chatterjee [42]; the lattice flow theory is in Sections 24 and 25, and the flow is Lüscher's Wilson flow [128]. The final theorem uses the continuum measures and the energy compactness of Section 29, which rest on the state space of Cao and Chatterjee [43] and on Uhlenbeck's compactness theorem [151], and the stopped measures on the coarse lattice of Section 33.

Sections 34.1 and 34.2 fix the notation and collect the deterministic inputs: heat estimates, covering estimates for the coefficient arrays, the zero-fluctuation continuation, and two elementary probability facts. Sections 34.3 to 34.6 prove conditional bounds uniform in γ, recall the stability of based holonomy matrices, and compare the two reconstructions, first through their flow data and then through their holonomies, under the conditioned measure and for every prescribed coarse field. Sections 34.7 and 34.8 contain the applications: the Gaussian estimate at a prescribed flat field and the flat support of the weak-coupling limits.

34.1 Setting and notation

Throughout this section we fix $L > 0$, $\theta > 0$, a finite horizon $T \geq \theta L^2$, an odd integer $M \geq 3$, an integer $J \geq 1$, $B = M^J$, and a coupling ceiling $0 < \gamma_0 \leq 1$. All parameters of the small-field event, the analytic domains, and the coefficient bounds of the reconstruction are fixed as in Theorem 23.2, before $\gamma \in (0, \gamma_0]$ is chosen. We fix an integer $n \in 4\mathbb{N}$ in the lattice refinement sequence under consideration, at least the minimum coarse side of Theorem 23.2, and such that every flat coarse field has a representative well inside the small-field domain. Precisely, we fix $r_{\mathrm{flat}} < r$ with $2\pi/n < r_{\mathrm{flat}}/4$, so that Lemma 14.9 applies; here r is the radius of the small-link domain B_r. We use the exponents

$$\tfrac{1}{2} < \alpha = \alpha_{\mathrm{fl}} < \tfrac{2}{3}, \qquad \beta = \alpha - \tfrac{1}{2}, \qquad \kappa_{\mathrm{fl}} = 1 - \tfrac{3\alpha}{2}, \qquad 0 < \sigma = \sigma_{\mathrm{sm}} < 1 - \alpha,$$

and the weighted norms X^a_α, Y^a_α, H^a_σ, and V^a_σ of Definition 4.8 on the time interval $[0, T]$. These norms include every component, every physical first difference, and every shift of the Wilson stencil required there. For a random norm such as $\mathsf{X}^a_\alpha(f)$, we write $\|\mathsf{X}^a_\alpha(f)\|_p$ for the L^p norm of this scalar random variable, and similarly for conditional L^p norms. The moment order p is unrelated to the cutoff radii p_j defined below.

Definition 34.1. Fix an integer $K \geq 0$ and set $N = nB^K$. For $0 \leq j \leq K$ put

$$\ell_j = \frac{L}{n} B^{-j}, \quad a = \ell_K, \quad \mathfrak{g}_j^2 = \frac{\gamma}{nB^j} = g_3^2 \ell_j, \quad t_j = M^{(J-1)/2} \mathfrak{g}_j, \quad p_j = A_{\mathrm{w}} \log(1/t_j).$$

1. *Conditional refinement.* Let $\nu = \nu_{N,n,\gamma}$ be the conditioned Wilson measure. Let $\widehat{\theta}_0$ be the link logarithm of the coarse field on Λ_n in the gauge of the reconstruction maps, and for $1 \leq j \leq K$ let
$$\widehat{\theta}_j = \widehat{\Phi}_j\big(\widehat{\theta}_{j-1}, (g_{j,r} w_{j,r})_r\big),$$
where $\widehat{\Phi}_j$ is the reconstruction map of Theorem 15.12 for the j-th renormalization group step counted from the coarse side, from mesh ℓ_{j-1} to mesh ℓ_j. Here $w_j = (w_{j,r})_r$ are the normalized fluctuation variables of this step, r runs over its J intermediate scales, and $g_{j,r}$ are the couplings of these scales; their ratios to t_j

lie between $M^{-(J-1)/2}$ and 1. Let $\mathcal{F}_j = \sigma(\widehat{\theta}_0, w_1, \dots, w_j)$ be the backward filtration. Conditionally on $\mathcal{F}_{j-1}$, the vector w_j has the law $k_j(\widehat{\theta}_{j-1}, dw_j)$ of the conditional fluctuation kernel (4.8) of this step, on its fluctuation cutoff domain and at prior depth $J(K-j)$ (Lemma 15.10 and Theorem 15.9).

2. *Zero-fluctuation maps.* Let $P_j(v) = \widehat{\Phi}_j(v, 0)$ and $L_j = DP_j(0)$. There exists a convex working ball, given by Lemma 15.15, which contains all the fields $\widehat{\theta}_j$ and all the zero-fluctuation continuations below, and on which

$$\sup_j \|DP_j\| \le a_{\mathrm{map}}, \qquad a_{\mathrm{map}}\sqrt{B} \le \tfrac{32}{49}, \qquad \mathfrak{q} = Ba_{\mathrm{map}}^2 < 1, \qquad \kappa = -\frac{\log a_{\mathrm{map}}}{\log B} > \tfrac{1}{2}.$$

We write $\varrho = a_{\mathrm{map}}$. The absolute coefficient bounds of the reconstruction maps (on rows and on volume-normalized columns), their bounds in the weighted heat norms with first differences, the bounds on the three ordered quadratic forms of Theorem 27.5, the bounds on the conditional means, and the bounds on the derivatives of the normalized conditional log-densities with respect to the incoming field, $\partial \log k = -\partial V + \mathbb{E}_k \partial V$ (with the parameters and bounds of Theorems 13.10 and 15.9), are fixed on a larger domain before the working ball is chosen. Since $q_{\mathrm{anc}} \le \mathfrak{q}$, the ratio $\mathfrak{q}$ may be used in place of q_{anc} in all the estimates of Section 27.

3. *Zero-fluctuation continuation.* For $0 \le m \le K$ let $\widehat{\theta}_j^0 = \widehat{\theta}_j$ for $j \le m$, and $\widehat{\theta}_j^0 = P_j \cdots P_{m+1}\widehat{\theta}_m$ for $m < j \le K$.

4. *Flow data.* Let $I_{\ell_j \to a}$ be the cardinal interpolation of Lemma 16.3 from mesh ℓ_j to mesh a, with $I_{a\to a}$ the identity, and let P_t^a be the lattice heat semigroup. On the finest lattice define

$$x_j = P_\cdot^a I_{\ell_j\to a}(\widehat{\theta}_j/\ell_j), \qquad y_j = \mathcal{Q}_a(x_j, x_j),$$
$$x_j^0 = P_\cdot^a I_{\ell_j\to a}(\widehat{\theta}_j^0/\ell_j), \qquad y_j^0 = \mathcal{Q}_a(x_j^0, x_j^0),$$

where $\mathcal{Q}_a$ is the bilinear Duhamel term of the Wilson flow with gauge damping, in the order displayed in Section 4; it is not symmetric.

5. *Increments.* Let $A_j = \widehat{\theta}_j/\ell_j$, $H = \ell_{j-1}$, and $\zeta_j = w_j - \mathbb{E}(w_j \mid \mathcal{F}_{j-1})$. Then $d_j = x_j - x_{j-1}$ satisfies $d_j = s_j + z_j + r_j + u_j$, where

$$\begin{aligned} s_j &= \frac{\mathfrak{g}_j}{\ell_j} P_\cdot^a I_{\ell_j\to a} E_{\widehat{\theta}_{j-1}} \zeta_j, \qquad z_j = P_\cdot^a I_{\ell_j\to a}\big(A_j - \mathbb{E}(A_j \mid \mathcal{F}_{j-1})\big) - s_j, \\ r_j &= P_\cdot^a \Big(I_{\ell_j\to a}\frac{L_j\widehat{\theta}_{j-1}}{\ell_j} - I_{H\to a}\frac{\widehat{\theta}_{j-1}}{H}\Big), \\ u_j &= P_\cdot^a I_{\ell_j\to a} U_j^0, \qquad U_j^0 = \mathbb{E}(A_j \mid \mathcal{F}_{j-1}) - \frac{L_j\widehat{\theta}_{j-1}}{\ell_j}. \end{aligned} \tag{34.1}$$

Here E_θ is the normalized first Taylor coefficient of the reconstruction at the conditional mean, in the centered fluctuation variables, as in Lemma 26.2. Thus s_j is the

centered linear fluctuation term, z_j the centered nonlinear heat term, r_j the signed interpolation defect, and u_j the predictable term. For $j > m$ the continuation has increment $d_j^0 = x_j^0 - x_{j-1}^0 = r_j^0 + u_j^{\mathrm{cl},0}$, where r_j^0 is r_j with $\widehat{\theta}_{j-1}$ replaced by $\widehat{\theta}_{j-1}^0$, and $u_j^{\mathrm{cl},0} = P_\cdot^a I_{\ell_j \to a} U_j^{\mathrm{cl}}(\widehat{\theta}_{j-1}^0)$ with $U_j^{\mathrm{cl}}(v) = \ell_j^{-1}(P_j(v) - L_j v)$.

6. *Holonomies.* Let $\mathcal{H}^K = (\mathcal{H}_i^K)_{i=1}^3$ be the based holonomies around the three coordinate cycles of $\mathbb{T}_L^3$ of the configuration $\Psi_{\theta L^2}^{(N)}(\exp \widehat{\theta}_K)$, that is, of $\exp \widehat{\theta}_K$ after the Wilson flow for the lattice flow time $4\theta L^2/a^2$. Let $\Xi_m^K = (\Xi_{m,i}^K)_{i=1}^3$ be the same triple for $\exp \widehat{\theta}_K^0$. Both triples are computed in the common initial frame of the reconstruction maps.

The two configurations on the finest lattice compared in this section are

$$U_K = \exp \widehat{\theta}_K, \qquad U_{K,m}^0 = \exp\bigl(P_K \cdots P_{m+1} \widehat{\theta}_m\bigr).$$

Above level m, the first is refined with the random fluctuation variables and the second with zero fluctuation variables:

$$\begin{aligned}
&\widehat{\theta}_0 \ \to \ \cdots \ \to \ \widehat{\theta}_m \ \xrightarrow{\widehat{\Phi}_{m+1}(\cdot, g w_{m+1})} \ \widehat{\theta}_{m+1} \ \to \cdots \to \ \widehat{\theta}_K, \\
&\widehat{\theta}_0 \ \to \ \cdots \ \to \ \widehat{\theta}_m \ \xrightarrow{\qquad P_{m+1} \qquad} \ \widehat{\theta}_{m+1}^0 \ \to \cdots \to \ \widehat{\theta}_K^0.
\end{aligned}$$

We have $U_j^0 = U_j^{\mathrm{cl}}(\widehat{\theta}_{j-1}) + m_j^{\mathrm{ph}}(\widehat{\theta}_{j-1})$, where the conditional mean correction $m_j^{\mathrm{ph}}(v) = \mathbb{E}(A_j \mid \mathcal{F}_{j-1}) - \ell_j^{-1} P_j(v)$, at $\widehat{\theta}_{j-1} = v$, is the difference between the conditional mean and the zero-fluctuation value. It enters $d_j - d_j^0$ and is estimated separately, never as part of a centered term.

We call $\mathcal{H}^K$ the flowed holonomy triple and Ξ_m^K the zero-fluctuation continuation triple. With $a_j = \ell_j$, $g_j = \mathfrak{g}_j$, the common mesh $\delta = a$, and the depth $\mathfrak{d}_j = J(K-j)$, the four terms in (34.1) are those of Lemma 26.2. There the logarithmic reconstruction map is $\mathscr{R}_j(\theta, w_j) = \widehat{\Phi}_j(\theta, (g_{j,r} w_{j,r})_r)$; for an incoming link logarithm v, the zero-fluctuation reconstructed lattice potential is $C_j(v) = \ell_j^{-1} P_j(v)$; and the linearization on lattice potentials is $\mathsf{P}_j = B L_j$, so that $\ell_j^{-1} L_j v = H^{-1} \mathsf{P}_j v$. The expansion is taken at the conditional mean, not at zero (see Remark 26.3). In particular, the centered nonlinear heat term is $z_j = P_\cdot^a I_{\ell_j \to a} Z_j$, the heat evolution of the conditionally centered nonlinear reconstruction residual Z_j of that lemma.

Constants. Unless stated otherwise, constants $C, c, \dots$ have the dependence described in item (6) of Definition 4.14, and may in addition depend on T, α, σ, the lengths of the loops considered, the margins of the reference neighborhoods of the flow, the fixed coefficient bounds of the reconstruction, and the fixed bounds on the conditional fluctuation kernels, their conditional means, and the derivatives of their log-densities. These data are fixed before the coupling is varied. The constants are independent of γ, K, N, the prior depths,

the indices m, j, i, and the finest mesh, unless an index is displayed. A subscript p allows dependence on a fixed moment order. All estimates concern the torus itself; no block region or frozen coordinate is varied. Recall that $g_3^2 = \gamma/L$, that $\mathfrak{g}_j = g_3\sqrt{\ell_j}$ is the coupling at mesh ℓ_j, that t_j is the largest coupling of the intermediate scales of the j-th step, and that p_j is the cutoff radius of its fluctuation variables.

Quadratic forms pulled back to earlier levels. The quadratic Picard term of the signed defect and of the predictable term is controlled through quadratic forms in the fields of earlier levels, as in Section 27. For $0 \le i \le j-1$ let $L_{j-1,i} = L_{j-1} \circ \cdots \circ L_{i+1}$, with $L_{j-1,j-1}$ the identity, and for an array v at level i we define

$$\mathsf{U}_{j,i}v = P_\cdot^a I_{\ell_j \to a} \frac{L_j L_{j-1,i} v}{\ell_j}, \qquad \mathsf{V}_{j,i}v = P_\cdot^a \Big(I_{\ell_j \to a} \frac{L_j L_{j-1,i} v}{\ell_j} - I_{\ell_{j-1} \to a} \frac{L_{j-1,i} v}{\ell_{j-1}} \Big).$$

The three ordered quadratic forms are

$$T_{j,i}^{\mathsf{UV}}(v,v) = \mathcal{Q}_a(\mathsf{U}_{j,i}v, \mathsf{V}_{j,i}v), \qquad T_{j,i}^{\mathsf{VU}}(v,v) = \mathcal{Q}_a(\mathsf{V}_{j,i}v, \mathsf{U}_{j,i}v),$$
$$T_{j,i}^{\mathsf{VV}}(v,v) = \mathcal{Q}_a(\mathsf{V}_{j,i}v, \mathsf{V}_{j,i}v),$$

and we write $\omega \in \{\mathsf{UV}, \mathsf{VU}, \mathsf{VV}\}$ for the label. Since $\mathsf{V}_{j,j-1}\widehat{\theta}_{j-1} = r_j$ and $\mathsf{U}_{j,j-1}\widehat{\theta}_{j-1} - \mathsf{V}_{j,j-1}\widehat{\theta}_{j-1} = x_{j-1}$, we have, at $(\widehat{\theta}_{j-1}, \widehat{\theta}_{j-1})$,

$$\mathcal{Q}_a(r_j, r_j) = T_{j,j-1}^{\mathsf{VV}}, \qquad \mathcal{Q}_a(x_{j-1}, r_j) = T_{j,j-1}^{\mathsf{UV}} - T_{j,j-1}^{\mathsf{VV}}, \qquad \mathcal{Q}_a(r_j, x_{j-1}) = T_{j,j-1}^{\mathsf{VU}} - T_{j,j-1}^{\mathsf{VV}}. \tag{34.2}$$

The same identities hold for the continuation, with r_j^0, x_{j-1}^0, and $\widehat{\theta}_{j-1}^0$. Let $S_{j,j-1}(v,v)$ be the quadratic part of $I_{\ell_j \to a} U_j^{\mathrm{cl}}(v)$, and $S_{j,i}(v,v) = S_{j,j-1}(L_{j-1,i}v, L_{j-1,i}v)$; we call $S_{j,i}$ the quadratic initial forms, and we call H_σ^a the initial-array norm. By construction both kinds of forms are pulled back exactly by the linearized maps:

$$T_{j,i-1}^{\omega}(v,v) = T_{j,i}^{\omega}(L_i v, L_i v), \qquad S_{j,i-1}(v,v) = S_{j,i}(L_i v, L_i v). \tag{34.3}$$

By Theorem 27.5, the coefficient norm of $T_{j,i}^{\omega}$ with output in the norm Y_α^a is at most $C(1+j)^c \mathfrak{q}^{j-1-i}/\ell_i$, and by Theorem 27.7 the coefficient norm of $S_{j,i}$ with output in H_σ^a is at most $C\mathfrak{q}^{j-1-i}/\ell_i$. We write $K_{ji} = \mathfrak{q}^{j-1-i}/\ell_i$.

The identities (34.3) give a cancellation that we use repeatedly. For a bilinear form A and $P = L + E$ with L linear and $E(v)$ of order $|v|^2$, we have

$$A(Pv, Pv) - A(Lv, Lv) = A(Lv, Ev) + A(Ev, Lv) + A(Ev, Ev). \tag{34.4}$$

The right side has degree at least three in v. Its difference at two points v, v' is bounded by $(|v| + |v'|)^2 |v - v'|$ times the coefficient norm of A, and two undifferentiated factors remain. With $A = T_{j,i}^{\omega}$ or $S_{j,i}$ and $P = P_i$, $L = L_i$, this identity reduces the comparison of quadratic forms along the two reconstructions to centered fluctuation terms, classical terms of degree at least three, and conditional means.

The bilinear expansion of the quadratic Picard term. Since $\mathcal{Q}_a$ is bilinear, we have

$$y_l - y_m = \sum_{j=m+1}^{l} \big\{ \mathcal{Q}_a(x_{j-1}, d_j) + \mathcal{Q}_a(d_j, x_{j-1}) + \mathcal{Q}_a(d_j, d_j) \big\}. \tag{34.5}$$

Inserting $d_j = s_j + z_j + r_j + u_j$ produces, for each j, the sixteen ordered products $\mathcal{Q}_a(v_j, w_j)$ with $v, w \in \{s, z, r, u\}$, and the eight cross terms $\mathcal{Q}_a(x_{j-1}, v_j)$ and $\mathcal{Q}_a(v_j, x_{j-1})$ with the preceding field. We never estimate a product of two rough heat paths in the norm X^a_α directly. Instead, each term is handled by one of four tools: the estimates for the quadratic expression in the centered linear term in Theorem 26.12; the mixed linear rows of Theorem 26.10, for a product of s_j with a predictable heat path; the forms (34.2); and the ordered product inequality of Lemma 28.5, which states that for two heat paths v and $w = P^a_\cdot w_0$,

$$\mathsf{Y}^a_\alpha\big(\mathcal{Q}_a(v, w)\big) + \mathsf{Y}^a_\alpha\big(\mathcal{Q}_a(w, v)\big) \le C\, \mathsf{X}^a_\alpha(v)\, \mathsf{H}^a_\sigma(w_0), \tag{34.6}$$

separately for the two orders, with a constant independent of the coupling, the coarse side, and the depth. For differences between the two reconstructions we use the identity

$$\mathcal{Q}_a(v, u) - \mathcal{Q}_a(v^0, u^{\mathrm{cl},0}) = \mathcal{Q}_a(v, u - u^{\mathrm{cl},0}) + \mathcal{Q}_a(v - v^0, u^{\mathrm{cl},0}), \tag{34.7}$$

where $v \in \{s, z, r, u\}$ and $v^0 = 0, 0, r^0, u^{\mathrm{cl},0}$ respectively, together with the identity in which both arguments are interchanged; for the cross terms the same identity holds with $(v, v^0) = (x_{j-1}, x^0_{j-1})$. Table 34.2 lists the tool used for each term in the three proofs where the expansion appears.

34.2 Heat estimates on a common mesh and fluctuation tails

We first record the elementary calculations needed to apply the coefficient bounds of Sections 15 and 26 to 28 uniformly in the coupling, and we specify which part of each earlier estimate at fixed coupling is used.

All uses below satisfy the following domain conditions. Each renormalization group step consists of J averaging steps; the lattice at level j has side nB^j, and the prior depth at level j is $J(K - j)$. Since $t_1^2 = \gamma/(Mn)$ and $t_j = t_1 B^{-(j-1)/2}$, the minimum-side condition of Theorem 23.2 puts every t_j below the upper bound on the couplings in the conditional kernel theorem. The inputs $g_{j,r} w_{j,r}$ of the reconstruction maps have size at most $t_j p_j$, and the larger coefficient domain and the working ball are those of Lemma 15.15, fixed before the coupling. In particular the deterministic amplitude bound of that lemma applies to these inputs, and the bounds with the factor a_{map} used below require no stochastic estimate uniform in the mesh. The heat horizon is always T. Every refinement ratio $\ell_i/a = B^{K-i}$ is an integer, and every order of spatial differences used below is an integer. The side n satisfies $2\pi/n < r_{\mathrm{flat}}/4$, hence the condition on flat fields in Lemma 14.9.

These conditions let us evaluate the coefficients at zero fluctuation for a fixed incoming field, including a flat one; they do not identify a stopped measure with a marginal of the conditioned Wilson measure.

The linearized maps have the geometric contraction ratio $q_{\mathrm{anc}} = C_{\mathrm{CP}}^2/B$ of Lemma 15.15, and by the choice in Definition 34.1 we have $q_{\mathrm{anc}} \le \mathfrak{q} < 1$. All geometric losses below are taken from the positive gap $-\log \mathfrak{q}/\log B$ and the other positive exponents displayed in this section. The two ratios remain distinct in the normalization of differentiated coefficient rows; see (34.21).

In every use of the weighted coefficients of Theorem 27.5, the constants may depend on the coarse mesh $H_0 = L/n$. Indeed, for $H = \ell_{j-1} = H_0 B^{-(j-1)}$ and $j \ge 1$ we have

$$\begin{aligned} 1 + \log(1 + T/H^2) &\le 1 + \log(1 + T/H_0^2) + 2(j-1)\log B \\ &\le C_{T,H_0,B}(1+j), \end{aligned} \tag{34.8}$$

where in the first inequality we used $1 + B^{2(j-1)}T/H_0^2 \le B^{2(j-1)}(1 + T/H_0^2)$. This constant is independent of the coupling, of all level indices, of the prior depth, and of the finest mesh. The coefficient rows without time weights keep the dependence stated in Theorem 27.5.

Lemma 34.2. *Put $H_0 = L/n$ and $\lambda = -\log \mathfrak{q}/\log B = 2\kappa - 1 > 0$. The following bounds hold.*

1. *For cardinal interpolation from any input mesh $H \ge a$ compatible with the common lattice, that is, with H/a an integer, and every fixed integer $q \ge 0$,*

$$\|D_a^q P_t^a I_{H\to a} v\|_\infty \le C_q (t + H^2)^{-q/2} \|v\|_\infty. \tag{34.9}$$

2. *Let E be a coefficient array of a reconstruction map with respect to normalized fluctuation variables at input mesh H, whose output-fixed absolute coefficient row sums and volume-normalized input-fixed column sums satisfy the bounds of Lemma 15.15. Then*

$$\sup_x \|D_a^q P_t^a I_{H\to a} E(x,\cdot)\|_{\ell^2(w)} \le C_q H^{3/2} (t + H^2)^{-3/4 - q/2}, \tag{34.10}$$

   where $\ell^2(w)$ is the counting ℓ^2 norm in the fluctuation variables, and the operator norm of $D_a^q P_t^a I_{H\to a} E$ from $\ell^2(w)$ to the physical space L_a^2, with measure a^3 times counting measure, is at most $C_q H^{3/2}(t + H^2)^{-q/2}$. The constants may depend on the fixed ratios of the reference volumes of the coordinate types within one renormalization group step.

3. *Let $H = \ell_{j-1}$, and let $\mathscr{D}_j v = I_{\ell_j \to a}(L_j v/\ell_j) - I_{H\to a}(v/H)$ be the initial value of the signed defect r_j when the incoming field is v. Then*

$$\mathsf{X}_\alpha^a(P_\cdot^a \mathscr{D}_j v) \le C H^{\alpha - 1} \|v\|_\infty. \tag{34.11}$$

4. *Let $V_j(v)$ be the part of degree at least three of the initial value $I_{\ell_j\to a}U_j^{\mathrm{cl}}(v)$ (this V_j is unrelated to the coarse configuration V of Section 34.8), so that $I_{\ell_j\to a}U_j^{\mathrm{cl}}(v) = S_{j,j-1}(v,v) + V_j(v)$. On the working ball,*

$$\|V_j(v) - V_j(v')\|_\infty \le \frac{C}{H}(\|v\|_\infty + \|v'\|_\infty)^2\|v - v'\|_\infty. \tag{34.12}$$

5. *The conditional mean correction satisfies*

$$\|m_j^{\mathrm{ph}}(v)\|_\infty \le Cg_3^2\|v\|_\infty, \qquad U_j^0 = U_j^{\mathrm{cl}}(\widehat{\theta}_{j-1}) + m_j^{\mathrm{ph}}(\widehat{\theta}_{j-1}). \tag{34.13}$$

Every bound on the supremum norm of an initial array in items (4) and (5) implies the corresponding bound in the full norm H_σ^a, with a fixed factor.

6. *If $x = P_\cdot^a I_{H\to a}(v/H)$ and $h = H/B$, then*

$$\sup_{t\le T}\big\{(t+h^2)^{\alpha/2}\|x(t)\|_\infty + (t+h^2)^{(\alpha+1)/2}\|D_a x(t)\|_\infty\big\} \le C\big\{\mathsf{X}_\alpha^a(x) + \|v\|_\infty H^{\alpha-1}\big\}. \tag{34.14}$$

Each fixed higher difference $D_a^q x$ satisfies the corresponding bound with the weight $(t+h^2)^{(\alpha+q)/2}$.

7. *For $b \ge 0$, $e > 0$, and $0 < s < \min(e,\lambda)$,*

$$\sum_{i>m}(1+i)^b B^{-ei} \le C_{b,e}(1+m)^b B^{-em}, \tag{34.15}$$

$$\sum_{j>m}\sum_{i=m+1}^{j-1}(1+j)^b \mathfrak{q}^{j-1-i}B^{-ei} \le C_{b,e,s}(1+m)^b B^{-sm}. \tag{34.16}$$

In (34.16) *an initial term $(1+j)^b\mathfrak{q}^j$ satisfies the same bound. These constants depend on b, e, s and not on m.*

Proof. We begin with the cardinal interpolation. It has tensor-product basis functions $\psi((y-x)/H)$, where $\psi = \phi^{\otimes 3}$ and $\phi(u) = b_0(u)/\sum_{k\in\mathbb{Z}} b_0(u-k)$ for the fixed even smooth bump b_0 of Section 16.2, which is supported in $[-1,1]$, positive inside, and flat at its endpoints; in the notation there, $b_0 = b$, $\phi = \zeta$, and $\psi = \zeta_3$. Its translates sum to one. Grouping the fine vertices according to their offset in a coarse cell, we find

$$a^3\sum_{y\in\Lambda_N}\psi((y-x)/H) = H^3, \qquad a^3\sum_y \mathrm{dist}(y,x)|\psi((y-x)/H)| \le CH^4.$$

Taking q differences of the smooth basis costs $C_q H^{-q}$, independently of a/H. In particular these statements also hold for $H = a$, where the interpolation is the identity on arrays.

The scalar heat estimates follow from the symbol

$$\omega_a(k) = 4a^{-2}\sum_{i=1}^3 \sin^2(\pi a k_i/L), \qquad k\in\mathbb{Z}^3,\ |k_i| \le L/(2a).$$

On this frequency cube $\omega_a(k) \geq c|k|^2/L^2$, and the symbol of D_a^q is at most $C_q(|k|/L)^q$. By Parseval's identity and comparison of each unit frequency cube with the corresponding Gaussian integral, we have

$$\|D_a^q P_t^a(x,\cdot)\|_{L_a^2} \leq C_{q,L,T}(t+a^2)^{-3/4-q/2},$$

where the kernel is taken with respect to the physical measure. The corresponding L^1 bound is $C_{q,L,T}(t+a^2)^{-q/2}$. To prove it without discarding the constant mode, we periodize the heat kernel p_t of the infinite lattice. By the Cauchy–Schwarz inequality with the weight $(1+|x|^2/(t+a^2))^2$ and Parseval's identity, the L^1 norm of $D_a^q p_t$ is at most

$$C(t+a^2)^{3/4}\big\|(1+|x|^2/(t+a^2))D_a^q p_t(x)\big\|_{L_a^2} \leq C_q(t+a^2)^{-q/2}.$$

In the last inequality, we used that multiplication by x_i corresponds to differentiation of the Fourier multiplier. Differentiating $e^{-t\omega_a}$ once or twice produces $t\partial_i\omega_a$ or $t\partial_i^2\omega_a + t^2(\partial_i\omega_a)^2$, and differentiating a difference symbol lowers its frequency power by the number of derivatives. The bounds $|\partial_i\omega_a| \leq C\sqrt{\omega_a}$ and $|\partial_i^2\omega_a| \leq C$, together with the Gaussian integrals above, give the displayed powers. For $t \leq a^2$, the difference stencil and the facts that P_t^a is positive and has mass one (which follow from the Poisson series of the nearest-neighbor walk) give the bound $C_q a^{-q}$ instead. Periodization does not increase the L^1 norm, and the contribution of the constant mode to the L^2 norm is bounded on the horizon T by a constant depending on L and T. Splitting at $t = H^2$ now proves (34.9): for $t \leq H^2$ we put the differences on the cardinal basis and use that P_t^a has mass one, and for $t \geq H^2$ we use the L^1 bound and $t + a^2 \geq (t+H^2)/2$.

Next, we prove item (2). By Lemma 15.15, the array E has bounded output-fixed absolute coefficient row sums and bounded volume-normalized input-fixed column sums, and it is exponentially localized in units of the input mesh. Let $\mathcal{J} = I_{H\to a}E$. Its absolute row sums are at most C, and each of its columns has physical mass at most CH^3. For an input ξ in counting ℓ^2, by the weighted Cauchy–Schwarz inequality we have

$$a^3\sum_y\Big|\sum_\lambda \mathcal{J}_{y\lambda}\xi_\lambda\Big|^2 \leq a^3\sum_{y,\lambda}|\mathcal{J}_{y\lambda}||\xi_\lambda|^2\sum_\mu|\mathcal{J}_{y\mu}| \leq CH^3\sum_\lambda|\xi_\lambda|^2.$$

For $t \leq H^2$, the absolute row of the differenced cardinal interpolation bounds the ℓ^2 row in (34.10) by CH^{-q}, and its volume-normalized column sum gives the same Schur bound for the operator norm with the extra factor H^{-q}. For $t \geq H^2$, we compose $\mathcal{J}\colon \ell^2(w) \to L_a^2$ with the physical L_a^2 row of $D_a^q P_t^a$ computed above. The two norms multiply to $CH^{3/2}t^{-3/4-q/2}$, which proves (34.10). Composition with the operator $D_a^q P_t^a$ on L_a^2, whose norm is at most $Ct^{-q/2}$, gives $CH^{3/2}t^{-q/2}$ and proves the operator bound. Conversion to the counting output norm would introduce a factor $a^{-3/2}$, which is canceled by the conversion of the physical heat-kernel coefficient row. Coordinate types at the other $J-1$ meshes of the step change these factors only by fixed powers of M.

For item (3), we use the signed mass identity of Lemma 15.15. The column of the linearized refinement and the column of the parent cardinal interpolation have equal physical mass, separately for every input orientation and color, and their difference has first absolute moment at most CH^4 about the input location. In a heat convolution we subtract the kernel value at the input location before summing such a column. Using the first difference of the kernel and this moment, we obtain, for every fixed integer $q \geq 0$,

$$\|D_a^q P_t^a \mathscr{D}_j v\|_\infty \leq \frac{C_q \|v\|_\infty}{H} (t + H^2)^{-q/2} \frac{H}{\sqrt{t + H^2}}.$$

For $t \leq H^2$ the same estimate follows from the unsigned cardinal rows. For $t \geq H^2$ the row bound requires a sampling estimate. On a cube of side H, repeated one-dimensional telescoping and averaging bounds the supremum of a sampled scalar function f by $CH^{-3} \sum_{|\eta| \leq 3} H^{|\eta|} \|D_a^\eta f\|_{L^1_a}$ on the enlarged cube, with at most one difference in each coordinate. In one dimension, we choose a vertex where $|f|$ is at most its average and telescope from that vertex; this is the discrete form of $|f(x)| \leq H^{-1} \int |f| + \int |Df|$. We apply it successively in the three coordinates. Applied to a differenced heat-kernel coefficient row, it shows that the sum over the cubes of H^3 times the local supremum is at most the L^1 bounds of its differences through three further orders. When $t \geq H^2$, these are at most $C(t + H^2)^{-(q+1)/2}$, since $H/\sqrt{t + H^2} \leq 1$. The first moment of the column gives the remaining factor H. For columns outside a fixed enlarged cube we apply this estimate on successive shells, whose polynomial volumes are summable against the exponential decay of the coefficients. This proves the row bound without counting output sites. Multiplying by the two weights of X_α^a and using $a \leq H$ proves (34.11); the weighted maxima are at most $CH^{\alpha - 1}$ since $\alpha < 1$.

For item (4), we expand $P_j = L_j + P_{j,2} + P_{j,\geq 3}$ on the larger domain, where $P_{j,2}$ is quadratic and $P_{j,\geq 3}$ has degree at least three. The coefficientwise majorant series of Lemma 15.15 bounds the difference of the part of degree at least three by $C(\|v\| + \|v'\|)^2 \|v - v'\|$ on the working ball. Division by $\ell_j = H/B$ and interpolation prove (34.12), with B fixed.

For item (5), the bound on m_j^{ph} and the decomposition of U_j^0 are the conditional-mean assertions of Theorem 27.7 and Lemma 28.5, together with Lemma 26.2. For a fixed incoming field v, the amplitude bound in those statements may be taken equal to $\|v\|_\infty$. This proves (34.13). Since the heat semigroup contracts the supremum norm, the value part of $\mathsf{H}_\sigma^a(f)$ for an initial array f is at most $C_{T,\sigma} \|f\|_\infty$; the bound $\|D_a P_t^a f\|_\infty \leq C(t + a^2)^{-1/2} \|f\|_\infty$ proved above gives the same bound for its first-difference part.

For (34.14), when $t \geq H^2$ we have $t + h^2 \leq 2(t + a^2)$, and the weights are bounded by those of X_α^a. When $t \leq H^2$, we use (34.9): for instance $(t + h^2)^{\alpha/2} \|x(t)\|_\infty \leq C(2H^2)^{\alpha/2} H^{-1} \|v\|_\infty$. For a higher difference, above H^2 we put the additional differences on a half heat step, and below H^2 we put them on the cardinal basis. This proves the assertion about higher differences.

To prove (34.15), we write $i = m + k$ and use $1 + m + k \le (1 + m)(1 + k)$; the remaining series is $\sum_{k \ge 1} (1 + k)^b B^{-ek} < \infty$. For (34.16), we set $j = i + 1 + k$ and sum first in k. The factor $(1 + j)^b$ is at most $(2 + i)^b (1 + k)^b$, and $\sum_{k \ge 0} (1 + k)^b \mathfrak{q}^k < \infty$. This proves the bound with exponent e for the restricted double sum in (34.16). If an earlier level $i \le m$ is included, its convolution with $j > m$ is bounded by the same calculation after factoring out B^{-sm}; the two remaining geometric ratios are $B^{-(e-s)}$ and $B^{-(\lambda - s)}$, since $\mathfrak{q} = B^{-\lambda}$. This also proves the assertion for the initial term. □

The next lemma prepares the chaining arguments. Each component of a reconstruction increment is a random field indexed by time, position, and finitely many labels, and it is a linear, quadratic, or nonlinear function of the normalized fluctuation variables of one step, with deterministic coefficients. We divide each coefficient by the scale at which it is expected to be bounded, and we measure the distance between two indices by the distance between their normalized coefficients. In the lemma, we show that these coefficient metrics have bounded diameter and covering numbers polynomial in L/h_*, where h_* is the input mesh of the coefficients and not the finest mesh a. This is why the maximal estimates below contain $\log(2 + L/\ell_j)$ rather than $\log(L/a)$.

Lemma 34.3. *Fix the cutoff radii, an incoming field in the reconstruction domain, and a common mesh a no coarser than all input meshes in use. Each process below is indexed by the time $t \in [0, T]$, the position on the lattice of mesh a, and the finitely many labels (components, shifts, and the choice between value and first difference) required by its norm. Write $h = \ell_j$, $H = \ell_{j-1}$, and for $1 \le i \le j - 1$ put $G = \ell_i = B^k H$ with $k = j - 1 - i$, and*

$$b_i(v) = C_J \big(a_{\mathrm{map}} \|v\|_\infty + \mathfrak{g}_i p_i \big).$$

The quantity $b_i(v)$ bounds $\|\widehat{\theta}_i\|_\infty$ on the whole conditional fiber over the incoming field $v = \widehat{\theta}_{i-1}$. We consider the following families.

1. *The coefficient of $w_j - \mathbb{E}(w_j \mid \mathcal{F}_{j-1})$ in s_j, in the norm X^a_α, with scale A_s.*
2. *The quadratic part of the initial value of z_j, in the norm H^a_σ, with scale $A_{z,\sigma}$, and of the heat path z_j, in the norm X^a_α, with scale $A_{z,\alpha}$.*
3. *The quadratic part of $\mathcal{Q}_a(s_j, s_j)$, in the norm Y^a_α, with scale A_{ss}.*
4. *The two ordered terms $\mathcal{Q}_a(b, s_j)$ and $\mathcal{Q}_a(s_j, b)$, in the norm Y^a_α, where b is a predictable heat path with $\|D^q_a b(t)\|_\infty \le C_q \mathcal{R} (t + \ell_j^2)^{-(\alpha+q)/2}$ for $0 \le q \le 4$; the scale is $C g_3 \mathcal{R} h^{\alpha - 1/2}$.*
5. *For each ω separately, the gradient in w_i of $T^\omega_{j,i}(\widehat{\theta}_i, \widehat{\theta}_i)$ on the whole conditional fiber, in the norm Y^a_α, with scale $C(1 + j)^c K_{ji} \mathfrak{g}_i b_i(v) G^{2\alpha - 1}$.*
6. *The gradient in w_i of $S_{j,i}(\widehat{\theta}_i, \widehat{\theta}_i)$ on the whole conditional fiber, in the weighted value norm V^a_σ, with scale $C K_{ji} \mathfrak{g}_i b_i(v) G^\sigma$.*

In items (1)–(3) the coefficients are real vectors or real symmetric matrices in the centered vector $w_j - \mathbb{E}(w_j \mid \mathcal{F}_{j-1})$*, before subtraction of any quadratic mean, and*

$$\begin{aligned} A_s &= C\mathfrak{g}_j h^{\alpha-1}, \qquad A_{z,\sigma} = C g_3^2 h^{\sigma}, \\ A_{z,\alpha} &= C g_3^2 h^{\alpha}, \qquad A_{ss} = C g_3^2 h^{2\alpha-1}. \end{aligned} \tag{34.17}$$

In items (5)–(6) the gradients are taken before subtraction of the conditional mean. Put $h_* = H$ *in item (5) and* $h_* = h$ *otherwise. Let* S *be the applicable scale, and after division by* S *let* d_* *be the counting* ℓ^2 *distance of coefficient rows for a linear family, the maximum of the Hilbert–Schmidt and operator distances for a quadratic family, and* $\sup_w \|\nabla_w F_z - \nabla_w F_{z'}\|_2$ *for a gradient family. If* $S = 0$*, the corresponding evaluations vanish and are omitted. If the fixed factor in* S *is large enough, then there exist a fixed dimensionless space-time distance* d_0*, which separates the finitely many labels, and fixed* $P_0 < \infty$*,* $\zeta > 0$*, and* $P, Q_1 < \infty$ *such that*

$$\begin{aligned} d_*(z,z') &\le C h_*^{-P_0} d_0(z,z')^{\zeta}, \\ \operatorname{diam}(d_*) \le 2, \qquad N(v, d_*) &\le C(1 + L/h_*)^P v^{-Q_1} \quad (0 < v \le 1). \end{aligned} \tag{34.18}$$

At every fixed cutoff, the normalized coefficient or gradient images have continuous compact completions. The constants may depend also on the constants C_q *of the background in item (4); they are independent of the coupling, the cutoff, the separation* $j - i$*, the incoming field, and the finer common mesh.*

Proof. All assertions concern deterministic functions on a fixed conditional fiber, that is, for a fixed incoming field.

We begin with the scales. For the centered linear family, by (34.10) and the operator bound of Lemma 34.2, the rows have size

$$C(\mathfrak{g}_j/h) h^{3/2} (t + h^2)^{-3/4-\iota/2}, \qquad \iota = 0, 1,$$

for the value and the first difference. Multiplication by $(t + a^2)^{(\alpha+\iota)/2}$ gives at most $C\mathfrak{g}_j h^{\alpha-1}$: for $t \le h^2$ we use $t + a^2 \le 2h^2$, and for $t \ge h^2$ the remaining power $t^{\alpha/2-3/4}$ is decreasing. The quadratic coefficient rows of Theorem 26.11 have size $C g_3^2 h^{3/2}(t + h^2)^{-3/4-\iota/2}$; the same two-range calculation, with norm exponent σ or α, gives the two scales of the centered nonlinear heat term. The Hilbert–Schmidt and operator rows of the Duhamel term in Theorem 26.12 contain the factor $t/(t + h^2)$, which vanishes at microscopic times, and give the scale A_{ss} in the norm Y_α^a. None of these steps uses a maximal estimate from those results.

Next we consider the covers. Every input of a coefficient lies on the input lattice of mesh h, which carries at most $C_J(L/h)^3$ scalar coordinates. The derivative bounds of the cardinal interpolation in Lemma 16.3 cost $h^{-\iota}$ at each fixed spatial order ι. We apply the exact shifted product rule of Lemma 24.2 to the source of a quadratic coefficient. Time

derivatives are two physical differences of the smooth input, and the time derivative of a Duhamel term is its source plus its Laplacian, which we commute onto the source. Absolute coefficient rows bound the nuclear norm, hence the Hilbert–Schmidt and operator norms, of these differentiated matrices. In these crude estimates we may even sum over all inputs, since their number is a fixed power of h^{-1}. Thus the derivative bounds cost fixed powers of h^{-1}, keep the coupling prefactor, and contain no inverse volume of the finest mesh. Division by the scale cancels the prefactor. For a weight exponent $0 < \eta \le 1$ we have

$$|(t + a^2)^\eta - (s + a^2)^\eta| \le |t - s|^\eta.$$

Together with the differentiated rows, this gives for each normalized coefficient a modulus $Ch^{-P_0}(\operatorname{dist}(x, y) + |t - s|^\zeta)$ with fixed P_0 and $\zeta > 0$, including at $t = 0$. Different scalar labels are covered by a finite union. Each normalized coefficient has norm at most one, which proves that the diameter is at most two, independently of the modulus.

For the mixed terms of item (4), the deterministic part of Theorem 26.10 and Lemma 26.7 applies with input mesh h and the four differences assumed for the background. It gives the scale $g_3 \mathcal{R} h^{\alpha - 1/2}$, the normalized cover, and the diameter bound. Its two output row bounds are

$$C g_3 \mathcal{R} h \int_0^t (s + h^2)^{-(\alpha+1)/2} (t - s + h^2)^{-\mathfrak{e}}\, ds, \qquad \mathfrak{e} \in \{3/4, 5/4\}. \tag{34.19}$$

These are the deterministic ordered rows of that theorem, in its version for a predictable heat background with four differences. Splitting the integral at h^2 and at $t/2$, we find that the two weights of the norm give the stated scale. In later applications, the background is the signed defect or the field of the preceding level. The signed defect has the required four differences by the bound on $D_a^q P_t^a \mathscr{D}_j$ in the proof of Lemma 34.2, and the inequality

$$(t + H^2)^{\alpha/2} H (t + H^2)^{-1/2} \le H^\alpha \qquad (\alpha < 1)$$

gives the background bound $\mathcal{R} \le C \|\widehat{\theta}_{j-1}\|_\infty H^{\alpha - 1}$ on its two input lattices. The heat equation gives its time increments.

Integrating the bounds on the rows of the reconstruction maps with respect to the fluctuation inputs in Lemma 15.15 along the segment of fluctuation inputs, we obtain $\|\widehat{\theta}_i\|_\infty \le \varrho \|\widehat{\theta}_{i-1}\|_\infty + C \mathfrak{g}_i p_i$. Thus $b_i(v)$ bounds $\|\widehat{\theta}_i\|_\infty$ on the whole conditional fiber. Iterating, and using $p_l \le p_i$ for $l \le i$ and $\varrho \sqrt{B} < 1$, we obtain

$$\|\widehat{\theta}_i\|_\infty \le r \varrho^i + C \sum_{l=1}^{i} \varrho^{i-l} \mathfrak{g}_l p_l \le C \{ r \varrho^i + g_3 \sqrt{\ell_i}\, p_i \}. \tag{34.20}$$

In the last inequality we used $\mathfrak{g}_l = \mathfrak{g}_i B^{(i-l)/2}$, so that the sum is at most $\mathfrak{g}_i p_i \sum_{k \ge 0} (\varrho \sqrt{B})^k$.

For the gradients of item (5) we use the gradient rows and the differentiated gradient rows of Theorem 27.5, without time weights. Their weighted size is the scale of item (5).

The crude differentiated rows there have the prefactor $B_* = C_{\mathrm{CP}}^{2k}\mathfrak{g}_i b_i(v)G^{-2}$ and cost fixed powers of H^{-1}. Since $C_{\mathrm{CP}}^2 = Bq_{\mathrm{anc}}$ and $G = B^k H$, both the coupling and the fiber bound cancel:

$$\frac{B_*}{S} \le C\Big(\frac{q_{\mathrm{anc}}}{\mathfrak{q}}\Big)^k H^{-1}G^{1-2\alpha} \le CH^{-2\alpha}. \tag{34.21}$$

In the last inequality we used $q_{\mathrm{anc}} \le \mathfrak{q}$ and $G^{1-2\alpha} \le H^{1-2\alpha}$, which holds since $G \ge H$ and $\alpha > 1/2$. The factor $(1+j)^c$ in S only decreases this ratio. The supremum over the fiber of the gradient differences therefore has the same polynomial modulus. Each normalized gradient has norm at most one on the whole fiber, and the diameter is at most two. This verifies the increment condition in the gradient metric directly, rather than deducing it from pointwise concentration.

For the gradients of item (6), the deterministic rows without time weights in Theorem 27.7 give the stated scale. Their crude derivatives have the prefactor $CK_{ji}\mathfrak{g}_i b_i(v)$ and cost fixed powers of ℓ_j^{-1}. After division by S their cost is at most $C\ell_j^{-P_0-\sigma}$, since $G \ge \ell_j$. The weight, with exponent $\sigma/2 > 0$, is continuous at time zero. This proves the modulus and the diameter bound for the weighted value process. The norm with first differences is obtained afterwards from the deterministic heat-splitting inequality of Theorem 27.7, and no additional cover is needed.

Finally, a modulus $Ch_*^{-P_0}(\mathrm{dist}(x,y) + |t-s|^{\zeta})$ is covered at radius v by the spatial step $cvh_*^{P_0}$ and the time step $(cvh_*^{P_0})^{1/\zeta}$. If the spatial step is smaller than a, we use every lattice point, whose number is then smaller than the proposed count. Otherwise we choose lattice representatives in spatial cubes; lattice paths between them give the same modulus. The resulting number of points is at most a fixed power of $1 + L/h_*$ times a fixed power of v^{-1}. All the coefficient maps are continuous in time and have finitely many spatial outputs. In the gradient case, since the fiber is compact, the supremum of the gradient norm keeps this continuity. The images therefore have compact completions with countable dense subsets. This proves every claimed cover with the stated dependence of the constants. □

The following lemma controls the zero-fluctuation continuation from any coarse field. It is used for the comparison configuration in the rest of this section, and in Sections 35 and 36, where its holomorphic version gives Cauchy estimates.

Lemma 34.4. *Let θ_0 be a link logarithm on the lattice of mesh $H_0 = L/n$ in the reconstruction domain with $\|\theta_0\|_\infty \le r$, and let $\theta_j = P_j\theta_{j-1}$ for $j \ge 1$. Let x_j and y_j be the flow data of θ_j as in Definition 34.1. For arbitrary compatible prior depths, every $l \ge m \ge 0$, and every compatible common mesh $a \le \ell_l$, the following hold; here compatibility means that the refinement ratios ℓ_i/a, $i \le l$, and the ratios of the intermediate meshes to a are integers.*

1. *We have*

$$\|\theta_j\|_\infty \le r\varrho^j, \qquad \sup_j\big\{\mathsf{X}_\alpha^a(x_j) + \mathsf{Y}_\alpha^a(y_j)\big\} \le R_{\mathrm{cl}}.$$

2. *We have*

$$\mathsf{X}^a_\alpha(x_l - x_m) + \mathsf{Y}^a_\alpha(y_l - y_m) \le C(1+m)^c \vartheta^m, \qquad \vartheta = \max\{\varrho B^{1-\alpha}, \mathfrak{q}\} < 1.$$

3. *There exist radii $r < r_1 < r_2$, fixed together with the working ball, such that the maps $\theta_0 \mapsto \theta_j, x_j, y_j$ extend holomorphically to the polydisc $\mathbb{D}_{r_2}$ of complex link logarithms θ_0 whose $D = 9n^3$ real coordinates are complexified with modulus less than r_2, and, with the norms applied to complex arrays,*

$$\sup_{\theta_0 \in \mathbb{D}_{r_2}} \left\{ \mathsf{X}^a_\alpha(x_j) + \mathsf{Y}^a_\alpha(y_j) \right\} \le C,$$
$$\sup_{\theta_0 \in \mathbb{D}_{r_2}} \left\{ \mathsf{X}^a_\alpha(x_l - x_m) + \mathsf{Y}^a_\alpha(y_l - y_m) \right\} \le C(1+m)^c \vartheta^m.$$

Write $x_j(\theta_0) = \sum_\mu f_{j,\mu} \theta_0^\mu$ and $y_j(\theta_0) = \sum_\mu h_{j,\mu} \theta_0^\mu$ for the Taylor expansions at zero in these D coordinates. Then

$$\sum_\mu r_1^{|\mu|} \left\{ \mathsf{X}^a_\alpha(f_{j,\mu}) + \mathsf{Y}^a_\alpha(h_{j,\mu}) \right\} \le C,$$
$$\sum_\mu r_1^{|\mu|} \left\{ \mathsf{X}^a_\alpha(f_{l,\mu} - f_{m,\mu}) + \mathsf{Y}^a_\alpha(h_{l,\mu} - h_{m,\mu}) \right\} \le C(1+m)^c \vartheta^m.$$

These bounds are uniform in the prior depths, the length of the continuation, and the compatible common mesh, and the constants may depend in addition on r_1 and r_2. If $\theta_0 = F(u)$ is the image of a parameter u under a fixed finite-scale reconstruction F, the same assertions hold on a complex neighborhood of the real parameters that may depend on F, but not on the length l of the continuation.

The constants R_{cl}, C, c depend on H_0, r, and the data of item (6) of Definition 4.14, and are independent of the coupling and of the length of the sequence.

Proof. We begin with item (1) and the first component in item (2). Since $P_j(0) = 0$, integrating DP_j along the segment from 0 to v, we obtain $\|P_j v\|_\infty \le \varrho \|v\|_\infty$. Thus $b_j = r\varrho^j$ bounds the sequence. Its increments split as $d_j^0 = r_j^0 + u_j^{\mathrm{cl},0}$, with no conditional mean, and

$$\mathsf{X}^a_\alpha(r_j^0) \le C b_{j-1} \ell_{j-1}^{\alpha-1}, \qquad \mathsf{H}^a_\sigma(u_j^{\mathrm{cl},0}(0)) \le C b_{j-1}^2 / \ell_{j-1}. \tag{34.22}$$

The first inequality is (34.11). The second follows from the series of P_j in degrees at least two and from item (5) of Lemma 34.2. The two geometric ratios are $\varrho B^{1-\alpha} < 1$ and $B\varrho^2 = \mathfrak{q} < 1$. The initial field x_0 has bounded norm $\mathsf{X}^a_\alpha(x_0)$ by (34.9). Summing these bounds, and using the inequality $\mathsf{X}^a_\alpha(P^a_\cdot f) \le C \mathsf{H}^a_\sigma(f)$ (which holds since $\sigma < \alpha$ and T is finite) for the terms $u_j^{\mathrm{cl},0}$, we find the uniform bound on $\mathsf{X}^a_\alpha(x_j)$ and the tail $\mathsf{X}^a_\alpha(x_l - x_m) \le C\vartheta^m$.

We now turn to the quadratic component. We fix one of the three ordered forms and denote it by $T_{j,i}$. Its coefficient norm with output in Y^a_α is at most $C(1+j)^c K_{ji}$. By the pullback identity (34.3), we have the finite identity

$$\begin{aligned} T_{j,j-1}(\theta_{j-1},\theta_{j-1}) &= T_{j,0}(\theta_0,\theta_0) \\ &\quad + \sum_{i=1}^{j-1} \big\{ T_{j,i}(P_i\theta_{i-1}, P_i\theta_{i-1}) - T_{j,i}(L_i\theta_{i-1}, L_i\theta_{i-1}) \big\}. \end{aligned} \tag{34.23}$$

Here $P_i - L_i$ starts at degree two. The coefficientwise majorant series gives $\|P_i v - L_i v\| \le C\varrho\|v\|^2$ and $\|P_i v\| + \|L_i v\| \le C\varrho\|v\|$ on the working ball. By (34.4) and the coefficient bound, we have

$$\mathsf{Y}^a_\alpha\big(T_{j,j-1}(\theta_{j-1},\theta_{j-1})\big) \le C(1+j)^c \Big\{ \frac{r^2 \mathfrak{q}^{j-1}}{H_0} + \sum_{i=1}^{j-1} \frac{\mathfrak{q}^{j-1-i}}{\ell_i} b_{i-1}^3 \Big\}. \tag{34.24}$$

The sum is at most $Cr^3 H_0^{-1}\mathfrak{q}^{j-1}$. Indeed, since $\ell_i^{-1} = H_0^{-1}B^i$ and $\mathfrak{q} = B\varrho^2$, its i-th term divided by $\mathfrak{q}^{j-1}$ is $H_0^{-1} r^3 \varrho^{i-3}$; the factor ϱ^{-3} is fixed and $\sum_{i\ge 1}\varrho^i$ is finite.

By (34.2) for the continuation, $\mathcal{Q}_a(r_j^0, r_j^0)$, $\mathcal{Q}_a(x_{j-1}, r_j^0)$, and $\mathcal{Q}_a(r_j^0, x_{j-1})$ are T^{VV}, $T^{\mathsf{UV}} - T^{\mathsf{VV}}$, and $T^{\mathsf{VU}} - T^{\mathsf{VV}}$ at $(\theta_{j-1},\theta_{j-1})$. Applying (34.24) to each order gives summable bounds for these three terms. Every remaining term of

$$y_l - y_m = \sum_{j=m+1}^{l} \big\{ \mathcal{Q}_a(x_{j-1}, d_j^0) + \mathcal{Q}_a(d_j^0, x_{j-1}) + \mathcal{Q}_a(d_j^0, d_j^0) \big\}$$

contains $u_j^{\mathrm{cl},0}$ as one argument. We apply the two ordered inequalities (34.6), using the bounded norms $\mathsf{X}^a_\alpha(x_j)$ just proved and (34.22). Their total tail is $C(1+m)^c\mathfrak{q}^m$ by (34.15), and the three forms above contribute at most $C(1+m)^c\mathfrak{q}^m$ as well. The initial term y_0 also has a fixed bound: its smooth cardinal input has bounded values and first two differences. Heat contraction and the stencil of the quadratic term, which contains a single difference, then bound its Duhamel value and its first difference on $[0,T]$. This proves the bound R_{cl} and both tails.

For item (3), we note that the argument above uses only absolute values of coefficients. The coefficientwise majorant series of Lemma 15.15, evaluated on a slightly larger polydisc of complex inputs still inside the working domain, gives the same bounds for complex inputs. Indeed, by the strict choice of the working radius, its derivative series is bounded by ϱ there. No positivity of a real field was used in (34.23). The quadratic coefficient bounds include the volume-normalized columns and all output differences, and every absolute sum above remains valid. Since r is smaller than a fixed fraction of the working radius, we may choose $r < r_1 < r_2$ such that $\mathbb{D}_{r_2}$ lies inside that larger polydisc, independently of the subsequent number of steps. This proves the two bounds

on $\mathbb{D}_{r_2}$, with r replaced by a multiple of r_2 in the constants. For the Taylor coefficients we apply the Cauchy estimates in each of the D coordinates on $\mathbb{D}_{r_2}$: for a holomorphic map g with values in a normed space and $\sup_{\theta_0\in\mathbb{D}_{r_2}}\|g(\theta_0)\|\le A$, its Taylor coefficients satisfy $\|g_\mu\|\le Ar_2^{-|\mu|}$. Summing, we find

$$\sum_{\mu} r_1^{|\mu|}\|g_\mu\| \le A\sum_{\mu}(r_1/r_2)^{|\mu|} = A(1-r_1/r_2)^{-D}.$$

Applied to x_j, y_j, and the differences x_l-x_m, y_l-y_m in the norms X^a_α and Y^a_α, this gives the coefficient bounds, with the fixed factor $(1-r_1/r_2)^{-D}$. For a fixed finite-scale input, we first take a complex neighborhood whose image under the fixed finite reconstruction has positive distance from the boundary of the working domain, and then apply this argument to the composed map. This proves item (3). □

The chaining estimate we use is the following specialization of Lemma 16.4.

Lemma 34.5. *Let $(F_z)_{z\in\mathcal{T}}$ be a centered real process indexed by a compact pseudometric space $(\mathcal{T},d_*)$, continuous in d_* at each fixed cutoff. Let $z_0\in\mathcal{T}$, $A>0$, and $c_0>0$, and suppose that one of the following two sets of conditions holds for every $u>0$ and $z,z'\in\mathcal{T}$:*

1. *$\mathbb{P}(|F_z-F_{z'}|>Ad_*(z,z')u)\le 2e^{-u^2/c_0}$ and $\mathbb{P}(|F_{z_0}|>Au)\le 2e^{-u^2/c_0}$;*
2. *$\mathbb{P}(|F_z-F_{z'}|>Ad_*(z,z')(\sqrt{u}+u))\le 2e^{-u/c_0}$ and $\mathbb{P}(|F_{z_0}|>A(\sqrt{u}+u))\le 2e^{-u/c_0}$.*

Assume that for some $H>0$

$$\operatorname{diam}(\mathcal{T},d_*)\le C_0,\qquad N(v,d_*)\le C(1+L/H)^P v^{-Q_0}\quad(0<v\le 1).$$

Then for every $1\le p<\infty$,

$$\Big\|\sup_z|F_z|\Big\|_{L^p}\le C_pA\sqrt{1+\log(2+L/H)}$$

under the conditions (1), and

$$\Big\|\sup_z|F_z|\Big\|_{L^p}\le C_pA\big[1+\log(2+L/H)\big]$$

under the conditions (2). Here C_p depends only on p,c_0,C_0,C,P,Q_0. The same holds under a conditional probability kernel when the hypotheses hold for each fixed value of the conditioning variables.

By Lemma 34.3, the normalized coefficient metrics for linear rows, for quadratic rows (with the maximum of the Hilbert–Schmidt and operator distances), and for gradients have diameter at most two and covering numbers of this form at their input mesh.

Proof. We apply Lemma 16.4 with the metric d_*, the diameter bound C_0, the cover prefactor $C(1+L/H)^P$, and the exponent Q_0. Under the conditions (1) we take $s = A\sqrt{c_0}$ and $b = 0$ in that lemma, and under the conditions (2) we take $s = A\sqrt{c_0}$ and $b = Ac_0$. These choices correspond to the substitutions $u = \sqrt{c_0 v}$ and $u = c_0 v$ in the two tails, which turn them into the form $\mathbb{P}\{|F_z - F_{z'}| > d_*(z,z')(s\sqrt{v} + bv)\} \le 2e^{-v}$ required there. The conditions at the reference point z_0 take the same form. Since

$$\log\bigl(2C(1+L/H)^P\bigr) + Q_0 \le C'\bigl[1 + \log(2 + L/H)\bigr],$$

the moment bound of Lemma 16.4 implies the two stated bounds, with the stated dependence of the constants. The conditional version is the corresponding clause of that lemma. □

Lemma 34.6. *We have $p_j \le C(1 + j + |\log\gamma|)$. Further, for every $\varepsilon > 0$ and $d \ge 0$,*

$$\sup_{0<\gamma\le\gamma_0} \gamma^{\varepsilon} p_j^d \le C_{\varepsilon,d}(1+j)^d,$$

where $C_{\varepsilon,d}$ depends in addition on ε and d.

In every later application a logarithm of the coupling is absorbed in this way only together with a positive power of γ.

Proof. By the definition of t_j, we have

$$p_j = \frac{A_{\mathrm{w}}}{2}\bigl\{\log(n/\gamma) + j\log B - (J-1)\log M\bigr\},$$

which proves the first bound. Since $\gamma \le 1$, we may write $v = |\log\gamma|$. For any $d \ge 0$ we have $(1+j+v)^d \le (1+j)^d(1+v)^d$, and $\sup_{v\ge0} e^{-\varepsilon v}(1+v)^d < \infty$, as one sees from the logarithmic derivative $-\varepsilon + d/(1+v)$ of the last function. This proves the second bound. □

34.3 Conditional bounds uniform in the coupling

We now bound the four parts of a reconstruction increment, conditionally on the preceding levels and with constants independent of γ. The leading terms carry no factor $|\log\gamma|$: the logarithm of the coupling enters only through the cutoff radii p_j, in terms of higher order in g_3. The theorem also bounds the quadratic forms pulled back to earlier levels and the conditional means in the predictable term. For $1 \le i \le j-1$ we write

$$Z^{\omega}_{j,i} = T^{\omega}_{j,i}(\widehat{\theta}_i, \widehat{\theta}_i) - \mathbb{E}\bigl[T^{\omega}_{j,i}(\widehat{\theta}_i, \widehat{\theta}_i) \mid \mathcal{F}_{i-1}\bigr], \qquad F_{j,i} = S_{j,i}(\widehat{\theta}_i, \widehat{\theta}_i) - \mathbb{E}\bigl[S_{j,i}(\widehat{\theta}_i, \widehat{\theta}_i) \mid \mathcal{F}_{i-1}\bigr]$$

for the centered pulled-back forms, and

$$e^{\omega}_{j,i} = \mathbb{E}\bigl[T^{\omega}_{j,i}(\widehat{\theta}_i, \widehat{\theta}_i) \mid \mathcal{F}_{i-1}\bigr] - T^{\omega}_{j,i-1}(\widehat{\theta}_{i-1}, \widehat{\theta}_{i-1})$$

for the conditional drifts, with the analogous drift $e^{S}_{j,i}$ for the quadratic initial forms. By (34.3), the drift $e^{\omega}_{j,i}$ is the sum of the classical part $T^{\omega}_{j,i}(P_i v, P_i v) - T^{\omega}_{j,i}(L_i v, L_i v)$ at $v = \widehat{\theta}_{i-1}$ and of the conditional-mean part $\mathbb{E}[T^{\omega}_{j,i}(\widehat{\theta}_i, \widehat{\theta}_i) \mid \mathcal{F}_{i-1}] - T^{\omega}_{j,i}(P_i v, P_i v)$.

Theorem 34.7. *Assume the setting of Definition 34.1. For every fixed $1 \le p < \infty$, all $K \ge j \ge 1$, every incoming field $\widehat{\theta}_{j-1}$ in the reconstruction domain, and every finer common mesh $a \le \ell_j$, the following conditional estimates hold. Let $h = \ell_j$ and let $\mathcal{Z}_j$ be the initial value of z_j. Then*

$$\big\|\mathsf{X}^a_\alpha(s_j)\big\|_{L^p(\cdot|\mathcal{F}_{j-1})} \le C\mathfrak{g}_j h^{\alpha-1}\big[\sqrt{\log(2+L/h)} + \sqrt{p}\big],$$

$$\big\|\mathsf{H}^a_\sigma(\mathcal{Z}_j)\big\|_{L^p(\cdot|\mathcal{F}_{j-1})} \le C_p\big\{g_3^2 h^{\sigma}(1+j) + g_3^3 h^{1/2} p_j^3\big\},$$

$$\big\|\mathsf{X}^a_\alpha(z_j)\big\|_{L^p(\cdot|\mathcal{F}_{j-1})} \le C_p\big\{g_3^2 h^{\alpha}(1+j) + g_3^3 h^{1/2} p_j^3\big\}.$$

The centered part of $\mathcal{Q}_a(s_j, s_j)$ has conditional $L^p(\mathsf{Y}^a_\alpha)$ norm at most $C_p g_3^2 h^{2\alpha-1}(1+j)$, and its conditional mean has norm Y^a_α at most $C_p g_3^2 \|\widehat{\theta}_{j-1}\|_\infty$. Fix constants $C_0, \dots, C_4 < \infty$, independently of the coupling, the level indices, and the common mesh. Let b be an $\mathcal{F}_{j-1}$-measurable heat path on the lattice of mesh a such that, for $0 \le q \le 4$ and $0 \le t \le T$,

$$\|D_a^q b(t)\|_\infty \le C_q \mathcal{R}(t + \ell_j^2)^{-(\alpha+q)/2}.$$

Then each of the two ordered terms $\mathcal{Q}_a(b, s_j)$ and $\mathcal{Q}_a(s_j, b)$ has conditional $L^p(\mathsf{Y}^a_\alpha)$ norm at most $C_p g_3 h^{\alpha-1/2}(1+j)^c \mathcal{R}$. The constant in this mixed estimate may also depend on $C_0, \dots, C_4$.

For each ω separately, each $1 \le i \le j-1$, and the incoming field $v = \widehat{\theta}_{i-1}$, put $b_i(v) = C_J(a_{\mathrm{map}}\|v\|_\infty + \mathfrak{g}_i p_i)$. Then

$$\big\|\mathsf{Y}^a_\alpha(Z^{\omega}_{j,i})\big\|_{L^p(\cdot|\mathcal{F}_{i-1})} \le C_p(1+j)^c \mathfrak{q}^{j-1-i} \mathfrak{g}_i b_i(v) \ell_i^{2\alpha-2},$$

$$\big\|\mathsf{H}^a_\sigma(F_{j,i})\big\|_{L^p(\cdot|\mathcal{F}_{i-1})} \le C_p(1+j)^c \frac{\mathfrak{q}^{j-1-i}}{\ell_i} \mathfrak{g}_i b_i(v) \ell_i^{\sigma}.$$

The deterministic bound $b_i \le C\{r a^i_{\mathrm{map}} + g_3\sqrt{\ell_i}\, p_i\}$ holds. The conditional drifts satisfy

$$\mathsf{Y}^a_\alpha(e^{\omega}_{j,i}) + \mathsf{H}^a_\sigma(e^{S}_{j,i}) \le C(1+j)^c \frac{\mathfrak{q}^{j-1-i}}{\ell_i}\big\{a_{\mathrm{map}}^2 b_{i-1}^3 + \mathfrak{g}_i^2 b_{i-1} + \mathfrak{g}_i^3 p_i^3\big\},$$

and their conditional-mean parts are bounded by the last two terms alone. All norms include every required first difference and Wilson shift. Further, there exist $c, \epsilon > 0$ such that for every $z > 0$,

$$\sup_{0<\gamma\le\gamma_0} \sup_{K\ge m} \nu\big\{\mathsf{X}^a_\alpha(x_K - x_m) + \mathsf{Y}^a_\alpha(y_K - y_m) > z\big\} \le C_z(1+m)^c B^{-\epsilon m}.$$

In particular this tail tends to zero in the order $K \to \infty$, then $\gamma \to 0$, then $m \to \infty$.

Here C_p depends in addition on p, and C_z on z; no factor $\log(1/t_j)$ is hidden in the leading constants.

Proof. The proof has six steps. We first check that the concentration estimates of Theorem 5.4 apply to the conditional kernels. We combine them with the covers of Lemma 34.3 to bound the centered linear and quadratic terms, and we treat the remainder of higher degree and the conditional means separately. Finally, we sum over levels with explicit powers of ℓ_i and estimate the products in the bilinear expansion (34.5).

Step 1: concentration. All coefficient estimates are evaluated on the larger fixed domain, and only then restricted to a fixed incoming field. The conditional potential of the fluctuation variables is C^2 on a neighborhood of its convex product of balls. The Hessian condition of Theorem 23.2 implies that its Hessian is bounded below there by a fixed $m_* > 0$: the perturbation of the identity Hessian of the normalized variables is at most $8/27$. Then Theorem 5.4 applies with one constant m_*, independently of the radii of the balls, of the dimension, and of the prior depth.

Step 2: coefficient metrics and conditional maxima. We use Lemma 34.3 with $h = \ell_j$ and with its parameters for the pulled-back forms, followed by Theorem 5.4. Note that a difference of coefficient rows is linear in the centered vector ζ_j, and a difference of coefficient matrices is quadratic in it. By the Lipschitz concentration and the quadratic estimate of Theorem 5.4, their increments satisfy the tail conditions (1) and (2) of Lemma 34.5, respectively, with $A = CS$, where S is the scale. For the gradients of a nonlinear function of the fluctuation variables we have

$$\operatorname{Lip}_w(F_z - F_{z'}) \le S d_*(z, z'), \qquad (F_z - \mathbb{E}F_z) - (F_{z'} - \mathbb{E}F_{z'}) = (F_z - F_{z'}) - \mathbb{E}(F_z - F_{z'}).$$

The Lipschitz concentration bound therefore applies to this difference, and also at the reference point of the chaining, with scale CS. This proves the increment conditions under the conditional fluctuation kernel. The continuity required in the gradient metric also follows directly: for $g = F_z - F_{z'}$ on the convex finite fiber Ω with conditional law μ, we have

$$|g(w) - \mathbb{E}g| \le \int_\Omega |g(w) - g(v')|\, d\mu(v') \le \operatorname{diam}(\Omega)\, S d_*(z, z').$$

The diameter of Ω is used only for continuity at a fixed cutoff, never in the constant of a maximal bound. The linear and quadratic coefficient images have the same continuity by the Cauchy–Schwarz inequality in finite dimension, applied to their centered evaluations. This verifies the countable separable realization required for chaining. Lemma 16.4, applied as in the proof of Lemma 34.5 but keeping track of the moment order, gives the losses

$$C\big[\sqrt{\log(2 + L/h_*)} + \sqrt{p}\big] \qquad \text{and} \qquad C\big[\log(2 + L/h_*) + p\big]$$

for linear and quadratic rows, respectively. Since $\log(2 + L/\ell_j) \le C(1+j)$, these losses imply the first bound of the theorem, the quadratic parts of the bounds on $\mathcal{Z}_j$ and z_j, the bound on the centered part of $\mathcal{Q}_a(s_j, s_j)$, the bound on the mixed terms with the background condition of item (4) of Lemma 34.3, and the bounds on $Z^{\omega}_{j,i}$ and $F_{j,i}$. For the last two, the scales of items (5) and (6) of that lemma are $C(1+j)^c \mathfrak{q}^{j-1-i} \mathfrak{g}_i b_i(v) \ell_i^{2\alpha-2}$ and $C\mathfrak{q}^{j-1-i}\ell_i^{-1}\mathfrak{g}_i b_i(v)\ell_i^{\sigma}$, since $K_{ji} = \mathfrak{q}^{j-1-i}/\ell_i$ and $G = \ell_i$.

Step 3: higher-degree remainder, conditional means, and first differences. The centered remainder of degree at least three in the fluctuation variables has supremum norm at most $C g_3^3 h^{1/2} p_j^3$ by Theorem 26.11, and the heat-splitting inequality of Theorem 27.7 bounds its first difference. The conditional mean of $\mathcal{Q}_a(s_j, s_j)$ is the mean in Theorem 26.12, which is at most $C g_3^2 \|\widehat{\theta}_{j-1}\|_\infty$ in the norm Y^a_α.

To verify that the bound on $\mathcal{Z}_j$ holds in the norm H^a_σ, which includes first differences, let f be an initial array. If $t \ge a^2$, we have

$$\|D_a P^a_t f\|_\infty \le C t^{-1/2} \|P^a_{t/2} f\|_\infty \le C t^{-(\sigma+1)/2} \mathsf{V}^a_\sigma(f).$$

If $t \le a^2$, we use $\|f\|_\infty \le a^{-\sigma}\mathsf{V}^a_\sigma(f)$ and $\|D_a P^a_t\|_{\infty\to\infty} \le C/a$. Multiplication by $(t + a^2)^{(\sigma+1)/2}$ cancels these factors. This proves $\mathsf{H}^a_\sigma(f) \le C\mathsf{V}^a_\sigma(f)$, uniformly in the finer mesh, and we apply it to the centered initial field.

Step 4: the conditional drifts. The splitting of the conditional drift into its classical part and its conditional-mean part is given in Theorem 27.5 for the ordered forms and in Theorem 27.7 for the quadratic initial forms. Its three bounds are

$$C(1+j)^c K_{ji}\big(\varrho^2 b_{i-1}^3 + \mathfrak{g}_i^2 b_{i-1} + \mathfrak{g}_i^3 p_i^3\big).$$

The first term bounds the classical part, which is the error of the zero-fluctuation pullback. The last two terms bound the conditional-mean part, which contains the conditional mean of the fluctuation variables and the covariance correction.

Step 5: summation over levels. We record the powers of the mesh needed to check summability. Fixed powers of H_0 are included in C. Using (34.20), $\varrho^i = (\ell_i/H_0)^\kappa$, and $\mathfrak{g}_i = g_3\sqrt{\ell_i}$, we find that the centered pulled-back bounds and the conditional drifts split before the convolution in $\mathfrak{q}$ as

$$\begin{aligned}
\mathfrak{g}_i b_i \ell_i^{2\alpha-2} &\le C\big\{g_3 r \ell_i^{\kappa+2\alpha-3/2} + g_3^2 \ell_i^{2\alpha-1} p_i\big\},\\
\mathfrak{g}_i b_i \ell_i^{\sigma-1} &\le C\big\{g_3 r \ell_i^{\kappa+\sigma-1/2} + g_3^2 \ell_i^{\sigma} p_i\big\},\\
\frac{\varrho^2 b_{i-1}^3 + \mathfrak{g}_i^2 b_{i-1} + \mathfrak{g}_i^3 p_i^3}{\ell_i} &\le C\big\{r^3 \ell_i^{3\kappa-1} + g_3^2 r \ell_i^{\kappa} + g_3^3 \ell_i^{1/2}(p_i+1)^3\big\}.
\end{aligned}$$

In the last line we used $(u+v)^3 \le 4(u^3+v^3)$. Every exponent of ℓ_i is positive.

The initial value of the predictable term decomposes as $u_j(0) = S_{j,j-1}(\widehat{\theta}_{j-1}, \widehat{\theta}_{j-1}) + V_j(\widehat{\theta}_{j-1}) + I_{\ell_j \to a} m_j^{\mathrm{ph}}(\widehat{\theta}_{j-1})$, and telescoping the first term by (34.3), we obtain

$$S_{j,j-1}(\widehat{\theta}_{j-1}, \widehat{\theta}_{j-1}) = S_{j,0}(\widehat{\theta}_0, \widehat{\theta}_0) + \sum_{i=1}^{j-1} (F_{j,i} + e_{j,i}^S).$$

The initial term is at most $Cr^2(1+j)^c \mathfrak{q}^{j-1}/H_0$. The other two terms of $u_j(0)$ are bounded by (34.12) and (34.13), and their bounds have the same powers as the third line above. Combining the chaining bounds of Step 2, the absorption of the logarithms of the coupling by Lemma 34.6, and the convolution bound (34.16), we obtain $\tau > 0$, a finite b, and constants independent of the coupling such that

$$\|\mathsf{H}_\sigma^a(u_j(0))\|_p + \|\mathsf{X}_\alpha^a(u_j)\|_p \le C_p (1+j)^b B^{-\tau j}. \tag{34.25}$$

For instance, we may take τ strictly below $\min(\sigma, 1/2, \lambda, \kappa + \sigma - 1/2)$ and reduce it for the finitely many other positive exponents listed above. The bound on $\mathsf{X}_\alpha^a(u_j)$ follows from the one on $\mathsf{H}_\sigma^a(u_j(0))$, since $\mathsf{X}_\alpha^a(P_\cdot^a f) \le C\mathsf{H}_\sigma^a(f)$. All fixed moments follow from the conditional bounds by the conditional Minkowski inequality and then the ordinary Minkowski inequality under ν.

Combining the bounds on s_j and z_j with (34.11), (34.20), and (34.25), we find

$$\sup_{K,j} \|\mathsf{X}_\alpha^a(x_j)\|_p \le C_p, \qquad \Big\| \mathsf{X}_\alpha^a\Big(\sum_{j>m} d_j \Big) \Big\|_p \le C_p (1+m)^b B^{-\tau_1 m} \tag{34.26}$$

for some $\tau_1 > 0$. Here the bound on s_j is of order $g_3 \ell_j^{\alpha - 1/2} \sqrt{1+j}$, the bound on r_j is $CH^{\alpha-1}\|\widehat{\theta}_{j-1}\|_\infty$ with $H = \ell_{j-1}$, and the initial field costs only $C_{H_0} r$. The correction in (34.14) also has bounded fixed moments: by (34.20) its powers of ℓ_{j-1} are $\kappa + \alpha - 1 > 0$ and $\alpha - 1/2 > 0$, with positive powers of the coupling on every remaining factor p_{j-1}.

Step 6: products and the probability tail. We now bound $y_K - y_m$ through (34.5), following the second column of Table 34.2. We write S_j for the bound on $\mathsf{X}_\alpha^a(s_j)$, Z_j^- and Z_j^+ for the bounds on $\mathsf{H}_\sigma^a(\mathcal{Z}_j)$ and $\mathsf{X}_\alpha^a(z_j)$, R_j for the bound on $\mathsf{X}_\alpha^a(r_j)$ given by (34.11), and U_j for the bound in (34.25), always using moments of order $2p$ in a product. Up to a fixed factor, the sixteen ordered products $\mathcal{Q}_a(v_j, w_j)$, with v indexing rows and w indexing columns, are bounded in $L^p(\mathsf{Y}_\alpha^a)$ by the entries of

	s	z	r	u
s	F_j	$S_j Z_j^-$	$S_j R_j$	$S_j U_j$
z	$S_j Z_j^-$	$Z_j^+ Z_j^-$	$R_j Z_j^-$	$U_j Z_j^-$
r	$R_j S_j$	$R_j Z_j^-$	W_j	$R_j U_j$
u	$U_j S_j$	$U_j Z_j^-$	$R_j U_j$	U_j^2

Here $F_j = C_p g_3^2[\ell_j^{2\alpha-1}(1+j)+b_{j-1}]$ includes the conditional mean of $\mathcal{Q}_a(s_j,s_j)$, and W_j is the bound on $\mathsf{Y}_\alpha^a(T^{\mathsf{VV}}_{j,j-1}(\widehat\theta_{j-1},\widehat\theta_{j-1}))$ obtained by telescoping as above: the initial term, the centered terms $Z^{\mathsf{VV}}_{j,i}$, and both the classical and the conditional-mean parts of the conditional drifts $e^{\mathsf{VV}}_{j,i}$. The entries with z or u use (34.6) and Hölder's inequality under ν. The entries (s,r) and (r,s) use (34.19), conditionally on $\mathcal{F}_{j-1}$, with the predictable background r_j, which has the four differences required there and $\mathcal{R}\le CR_j$. The entry (r,r) is the form T^{VV}, not a difference of uncontrolled forms. Thus every entry is justified in L^p. The slowest powers of ℓ_j that occur include $2\alpha-1$, $\kappa+2\alpha-3/2$, $\kappa+\alpha+\sigma-1$, and $\sigma+\alpha-1/2$; the entries with u also keep the decay $B^{-\tau j}$. All these exponents are positive. The parts of degree at least three in the fluctuation variables carry an additional factor $\ell_j^{1/2}$ and create no smaller exponent.

For the eight cross terms with x_{j-1}, we use (34.19) and the bounded norm (34.14) of x_{j-1} at its input mesh for s; (34.6) and Z_j^- for z; the forms $T^{\mathsf{UV}}-T^{\mathsf{VV}}$ and $T^{\mathsf{VU}}-T^{\mathsf{VV}}$ of (34.2) for r; and (34.6) with (34.25) and (34.26) for u. This covers both orders of all four arguments. By (34.15) and (34.16), the sums over $j>m$ of these bounds and of the sixteen products are at most $C_p(1+m)^b B^{-\epsilon m}$ for some smaller $\epsilon>0$. The expansion (34.5) then bounds $\|\mathsf{Y}_\alpha^a(y_K-y_m)\|_p$ by the same expression. By Markov's inequality and (34.26), we obtain the probability tail, after enlarging C_z to cover the finitely many small indices. □

34.4 The initial gauge frame along the flow

A based holonomy transforms by conjugation at its basepoint. Two configurations evolved with independent time-dependent gauge transformations can therefore have holonomies with equal traces but very different matrices. The Wilson flow is analyzed in a damped gauge, in which its linear part is the heat equation. The damping gauge transformations solve an ordinary differential equation started at the identity, and the difference of their generators is integrable in time uniformly in the mesh. Hence the holonomy matrices of two configurations that start in the same frame can be compared directly. We use the following form of Theorem 25.10.

Remark 34.8. Fix $R<\infty$, $0<t_-\le t_+\le T$, and finitely many based piecewise C^1 loops $\mathsf{c}_1,\dots,\mathsf{c}_k$ with lattice approximations $\mathsf{c}_{i,a}$ of uniformly bounded physical length. Let $U=e^{aA_0}$ and $U'=e^{aA_0'}$ be two configurations on the same lattice of mesh $a=L/N$, in the reconstruction domain and represented in the same initial vertex and basepoint frame, whose flow data (x,y) and (x',y') on $[0,T]$ have norms $\mathsf{X}_\alpha^a+\mathsf{Y}_\alpha^a$ at most R. Let $\mathcal{D}=\mathsf{X}_\alpha^a(x-x')+\mathsf{Y}_\alpha^a(y-y')$. By Theorem 25.10, applied with radius $\max(1,R)$ and input horizon T, there exists a flow stability constant $\Lambda_{R,t_-,T}<\infty$ such that

$$\sup_{t_-\le t\le t_+}\max_i\big\|\mathrm{Hol}_{\mathsf{c}_{i,a}}\Psi_t^{(N)}(U)-\mathrm{Hol}_{\mathsf{c}_{i,a}}\Psi_t^{(N)}(U')\big\|_{\mathrm{op}}\le\Lambda_{R,t_-,T}\,\mathcal{D},$$

where $\Psi_t^{(N)}$ is the Wilson flow run for the lattice flow time $4t/a^2$. The constant depends only on R, t_-, T, the loops, the fixed analytic domains, and the margins of the reference neighborhoods of the continuum flow in Theorem 24.10, with the dependence described in item (7) of Definition 4.14; it is independent of the mesh, the prior depth, and the coupling, and it covers the finitely many meshes that are too coarse for the small-mesh condition of the local theory. Let T_R and a_R be the constants of Theorem 24.5 for radius $\max(1, R)$, and put $h_R = \min(T_R, T)$. After decreasing a_R if necessary, for $a \le a_R$ and $0 \le s \le h_R$ the damped potentials A, A' and their damping gauge transformations k, k', with $k(0) = k'(0) = I$, satisfy the following estimates, where $v_a(s) = s + a^2$.

$$\|D_a(A - A')(s)\|_\infty \le C_R \mathcal{D}\{v_a(s)^{-(\alpha+1)/2} + v_a(s)^{-\alpha} + v_a(s)^{\kappa_{\mathrm{fl}}-1/2}\},$$

$$\|k(s, x_0) - k'(s, x_0)\|_{\mathrm{op}} \le C \int_0^s \|D_a(A - A')(u)\|_\infty \, du \le C_R \mathcal{D}.$$

The three powers of $v_a(s)$ are integrable at zero uniformly in the mesh, since $(\alpha+1)/2 < 1$, $\alpha < 1$, and $1/2 - \kappa_{\mathrm{fl}} < 1$. For $a \le a_*$ and $\mathcal{D} \le \delta$, with a_* and δ supplied by Lemma 25.9 with $T_0 = T$ and $\tau = t_-$, both flows can use a common reference solution and the same fixed restarts on the later intervals. For other pairs or meshes, the matrix inequality follows from the unitary diameter bound or the finite-dimensional Wilson flow, as in Theorem 25.10. All matrices are expressed in their initial frames; no conjugation of either matrix is optimized.

34.5 Comparison of linear and quadratic flow data

We now compare the flow data of the reconstruction with those of its zero-fluctuation continuation after level m. The comparison is exact on the finest lattice: both fields are interpolated by the identity there, and no interpolation error enters at level K. The two fields agree at level m.

Theorem 34.9. *Assume the setting of Definition 34.1. For every fixed $p \ge 1$ and all $K \ge j \ge m \ge 0$, under the conditioned Wilson measure ν,*

$$\big\| \|\widehat{\theta}_j - \widehat{\theta}_j^0\|_\infty \big\|_p \le C_p \mathfrak{g}_j (1+j)^{1/2}, \qquad \big\| \|\widehat{\theta}_j\|_\infty \big\|_p \le C_p \{ r a_{\mathrm{map}}^j + \mathfrak{g}_j (1+j)^{1/2} \}.$$

On the finest lattice, writing $\Delta x_j = x_j - x_j^0$, we have

$$\Delta x_K = \sum_{j=m+1}^{K} (d_j - d_j^0), \qquad y_K - y_K^0 = \sum_{j=m+1}^{K} \big(y_j - y_{j-1} - y_j^0 + y_{j-1}^0\big).$$

For each ordered form $T_{j,i}^\omega$ and each quadratic initial form $S_{j,i}$, the pullback (34.3) *is exact. In particular, with $L_i = DP_i(0)$, the function*

$$R_{j,i}^\omega(v) = T_{j,i}^\omega(P_i v, P_i v) - T_{j,i}^\omega(L_i v, L_i v)$$

has degree at least three and, on the working ball,

$$\mathsf{Y}^a_\alpha\big(R^\omega_{j,i}(v)-R^\omega_{j,i}(v')\big)\le C(1+j)^c\,\frac{\mathfrak{q}^{\,j-1-i}}{\ell_i}\big(\|v\|_\infty+\|v'\|_\infty\big)^2\|v-v'\|_\infty.$$

The corresponding estimate for the quadratic initial forms holds in the full initial-array norm H^a_σ. *There exist* $s>0$ *and* $c,d<\infty$, *independent of* K, m, *and* γ, *such that for every fixed* $p\ge 1$,

$$\begin{aligned}\sup_{m\le l\le K}\|\mathsf{X}^a_\alpha(\Delta x_l)\|_p+\sum_{j=m+1}^{K}\big\|\mathsf{H}^a_\sigma\big(u_j(0)-u_j^{\mathrm{cl},0}(0)\big)\big\|_p\\ \le C_p\big[\sqrt{\gamma}+\gamma(1+|\log\gamma|)^d\big](1+m)^cB^{-sm},\end{aligned}$$

$$\big\|\mathsf{X}^a_\alpha(x_K-x_K^0)+\mathsf{Y}^a_\alpha(y_K-y_K^0)\big\|_p\le C_p\sqrt{\gamma}(1+m)^cB^{-sm}+C_p\gamma(1+|\log\gamma|)^d.$$

Further, for each ω *let*

$$\mathcal{T}^\omega_{K,m}=\sum_{j=m+1}^{K}\big\{T^\omega_{j,j-1}(\widehat\theta_{j-1},\widehat\theta_{j-1})-T^\omega_{j,j-1}(\widehat\theta^0_{j-1},\widehat\theta^0_{j-1})\big\}.$$

Then $\|\mathsf{Y}^a_\alpha(\mathcal{T}^\omega_{K,m})\|_p\le C_p[\sqrt{\gamma}+\gamma(1+|\log\gamma|)^d](1+m)^cB^{-sm}$. *One may choose* s *strictly below each of the positive numbers*

$$\alpha-\tfrac12,\quad \sigma,\quad 2\alpha-1,\quad \kappa+\sigma-\tfrac12,\quad \kappa+2\alpha-\tfrac32,\quad 2\kappa-1,\quad \kappa+\alpha+\sigma-1,\quad -\frac{\log\mathfrak{q}}{\log B},$$

and below the fixed smaller exponents of the product estimates in Lemma 28.5.

Proof. We first bound the moments of the fields, before any product is formed. We then use (34.4) to write the difference of each pulled-back quadratic form along the two reconstructions as a sum of centered terms, classical terms of degree at least three, and conditional means. We record the size of each term, counting the classical part of degree at least three of the predictable term once. Summation over levels then gives the bounds on the predictable term, on the linear heat evolution, and on the three ordered forms. Finally, we bound both orders of every product in (34.5), following the third column of Table 34.2, with Hölder's inequality for higher moments under ν in place of independence. The only delicate product is that of Δx_{j-1} with the classical predictable term, whose sum over levels is not uniformly small in m before it is multiplied by the decaying factor. Throughout, we write $\varrho=a_{\mathrm{map}}$ and $\delta_j=\widehat\theta_j-\widehat\theta^0_j$, and all conditional expectations are for the conditional kernels of ν.

Step 1: moments of the fields. We define the innovation about the zero-fluctuation value by

$$\epsilon_j=\widehat\Phi_j(\widehat\theta_{j-1},g_{j,\cdot}w_j)-P_j(\widehat\theta_{j-1}).$$

By the coefficient bounds of the reconstruction maps, the Euclidean gradient of each scalar output with respect to the fluctuation variables is at most $C_J \mathfrak{g}_j$, and there are at most $C_J (nB^j)^3$ such outputs. Applying Theorem 5.4 for a fixed incoming field and taking a union bound over the outputs, we obtain

$$\mathbb{P}\big\{ \|\epsilon_j - \mathbb{E}_{j-1}\epsilon_j\|_\infty > C\mathfrak{g}_j(\sqrt{1+j} + \sqrt{u}) \;\big|\; \mathcal{F}_{j-1} \big\} \le Ce^{-u},$$

where $\mathbb{E}_{j-1} = \mathbb{E}(\cdot \mid \mathcal{F}_{j-1})$. Integrating the tail, we obtain the bound $C_p \mathfrak{g}_j \sqrt{1+j}$ in every fixed L^p, without a factor of the cutoff. Since $\mathbb{E}_{j-1}\epsilon_j = \ell_j m_j^{\mathrm{ph}}(\widehat{\theta}_{j-1})$, (34.13) implies that $\|\mathbb{E}_{j-1}\epsilon_j\|_\infty \le C\mathfrak{g}_j^2 \|\widehat{\theta}_{j-1}\|_\infty \le Cr\mathfrak{g}_j^2$, where in the last inequality we used that the working ball has a fixed radius. The exact recursion, with $\delta_m = 0$, is

$$\delta_j = P_j(\widehat{\theta}_{j-1}) - P_j(\widehat{\theta}^0_{j-1}) + \epsilon_j.$$

The segment between the two inputs lies in the convex working ball. By the Minkowski inequality and the bound on DP_j, we have

$$\big\| \|\delta_j\|_\infty \big\|_p \le C_p \sum_{i=m+1}^{j} \varrho^{j-i} \big[\mathfrak{g}_i \sqrt{1+i} + r\mathfrak{g}_i^2 \big].$$

Since $\mathfrak{g}_i = \mathfrak{g}_j B^{(j-i)/2}$, the first sum is at most $C_p \mathfrak{g}_j \sqrt{1+j}/(1-\varrho\sqrt{B})$. For the second sum, after division by $\mathfrak{g}_j$, we use $\mathfrak{g}_i \le \mathfrak{g}_{m+1}$ to bound it by $r\mathfrak{g}_{m+1}/(1-\varrho\sqrt{B})$, and $\mathfrak{g}_{m+1} \le \mathfrak{g}_1$ is bounded uniformly by the coupling ceiling. This proves the first estimate. The second follows by comparing the reconstruction with the zero-fluctuation continuation from the same coarse field (the case $m = 0$), whose norm is at most $r\varrho^j$ by Lemma 34.4. By the same argument, for every $i \ge m$ we have

$$\begin{aligned} \big\| \|\widehat{\theta}_i\|_\infty \big\|_p + \big\| \|\widehat{\theta}^0_i\|_\infty \big\|_p &\le C_p \big[r\varrho^i + \mathfrak{g}_i \sqrt{1+i} \big], \\ \big\| \|\delta_i\|_\infty \big\|_p &\le C_p \mathfrak{g}_i \sqrt{1+i}. \end{aligned} \tag{34.27}$$

For the continuation in the first line we apply $\|P_i \cdots P_{m+1} v\|_\infty \le \varrho^{i-m}\|v\|_\infty$ to the moment at level m just proved, and use $\varrho^{i-m}\mathfrak{g}_m \le \mathfrak{g}_i$.

Step 2: exact telescopes and the classical cubic difference. At level m the two fields and their interpolations coincide, and $I_{a\to a}$ is the identity at the finest level. Subtracting the two finite telescopes proves the two identities for Δx_K and $y_K - y_K^0$ in the statement, without an interpolation error at level m.

Let $A_{j,i}$ denote any one of $T^{\mathsf{UV}}_{j,i}$, $T^{\mathsf{VU}}_{j,i}$, $T^{\mathsf{VV}}_{j,i}$, or $S_{j,i}$. Its coefficient norm is at most $C(1+j)^c K_{ji}$, where the polynomial factor is needed only for the ordered forms, and the pullback (34.3) is exact by the construction in Theorems 27.5 and 27.7. Set $P_i = L_i + E_i$, where E_i starts at degree two. By (34.4), we have

$$C_{j,i}(v) = A_{j,i}(P_i v, P_i v) - A_{j,i}(L_i v, L_i v) = A_{j,i}(L_i v, E_i v) + A_{j,i}(E_i v, L_i v) + A_{j,i}(E_i v, E_i v).$$

By the coefficientwise majorant series of Lemma 15.15, we have $\|E_i v\| \le C\|v\|^2$, $\|DE_i(v)\| \le C\|v\|$, and $\|L_i\| \le \varrho$ on the working ball. Applying the product rule to the last identity, we obtain

$$\|DC_{j,i}(v)[w]\|_{\mathrm{out}} \le C(1+j)^c K_{ji}\|v\|_\infty^2\|w\|_\infty. \tag{34.28}$$

Here the output norm is Y^a_α, or, for a quadratic initial form, the initial-array norm H^a_σ after cardinal interpolation. In the latter case, the coefficient row bound of the quadratic initial form controls the supremum of the initial value, and item (5) of Lemma 34.2 then controls the first differences. Equivalently, in the expansion in homogeneous coefficients one distinguishes one argument, keeps the other two, and sums the remaining degrees inside the fixed radius; the differentiated series converges there, and the constant is therefore independent of the length of the sequence. Integrating (34.28) along the segment from v' to v proves the difference estimate in the statement, including first differences. It also proves that $R^\omega_{j,i} = C_{j,i}$ has no quadratic part.

Step 3: decomposition of the pulled-back forms. We define

$$\begin{aligned} Z_{j,i} &= A_{j,i}(\widehat\theta_i,\widehat\theta_i) - \mathbb{E}_{i-1}A_{j,i}(\widehat\theta_i,\widehat\theta_i),\\ M_{j,i}(v) &= \mathbb{E}\big[A_{j,i}\big(\widehat\Phi_i(v,g_{i,\cdot}w),\widehat\Phi_i(v,g_{i,\cdot}w)\big)\;\big|\;\widehat\theta_{i-1}=v\big] - A_{j,i}(P_iv,P_iv). \end{aligned}$$

By the splitting of the conditional means in Theorems 27.5 and 27.7, we have

$$\|M_{j,i}(v)\|_{\mathrm{out}} \le C(1+j)^c K_{ji}\big[\mathfrak{g}_i^2\|v\|_\infty + \mathfrak{g}_i^3 p_i^3\big]. \tag{34.29}$$

The classical part of degree three is contained in $C_{j,i}$ and does not appear again in (34.29). The pullback identity (34.3) implies that $A_{j,i}(\widehat\theta_i,\widehat\theta_i) - A_{j,i-1}(\widehat\theta_{i-1},\widehat\theta_{i-1}) = Z_{j,i} + C_{j,i}(\widehat\theta_{i-1}) + M_{j,i}(\widehat\theta_{i-1})$. For the continuation, the same difference equals $C_{j,i}(\widehat\theta^0_{i-1})$ when $i > m$. Summing over i, we obtain the identity

$$\begin{aligned} &A_{j,j-1}(\widehat\theta_{j-1},\widehat\theta_{j-1}) - A_{j,j-1}(\widehat\theta^0_{j-1},\widehat\theta^0_{j-1})\\ &\quad= \sum_{i=m+1}^{j-1}\big\{Z_{j,i} + C_{j,i}(\widehat\theta_{i-1}) - C_{j,i}(\widehat\theta^0_{i-1}) + M_{j,i}(\widehat\theta_{i-1})\big\}. \end{aligned} \tag{34.30}$$

The terms at level m cancel identically, since both arguments are $\widehat\theta_m$.

Step 4: sizes of the terms and summation. We give the quantitative bounds before suppressing positive powers of the mesh. In this proof we set $q_i = 1 + i + |\log\gamma|$; then $p_i \le Cq_i$ by Lemma 34.6. Conditionally on the incoming field, the fiber bound in

Theorem 34.7 is $C(\varrho\|\widehat{\theta}_{i-1}\|_\infty + \mathfrak{g}_i p_i)$. Taking its outer moments with (34.27), we obtain the following bounds, up to the factor $C_p(1+j)^c \mathfrak{q}^{j-1-i}$:

term	bound before the convolution in $\mathfrak{q}$
$Z_{j,i}$, quadratic initial form	$\sqrt{\gamma}\, r\ell_i^{\kappa+\sigma-1/2} + \gamma\ell_i^{\sigma} q_i$
$Z_{j,i}$, ordered form	$\sqrt{\gamma}\, r\ell_i^{\kappa+2\alpha-3/2} + \gamma\ell_i^{2\alpha-1} q_i$
$C_{j,i}(\widehat{\theta}_{i-1}) - C_{j,i}(\widehat{\theta}^0_{i-1})$	$\sqrt{\gamma}\, r^2\ell_i^{2\kappa-1/2}(1+i)^{1/2} + \gamma r\ell_i^{\kappa} q_i^2 + \gamma^{3/2}\ell_i^{1/2} q_i^3$
$M_{j,i}(\widehat{\theta}_{i-1})$	$\gamma r\ell_i^{\kappa} + \gamma^{3/2}\ell_i^{1/2} q_i^3$

(34.31)

Fixed factors depending on L, n, B, and r are allowed in these bounds. To verify the third line, we use (34.28) and Hölder's inequality with moments of order $3p$, which bound the difference by

$$\frac{C_p}{\ell_i}\big[r\varrho^{i-1} + \mathfrak{g}_{i-1}\sqrt{1+i}\big]^2 \mathfrak{g}_{i-1}\sqrt{1+i}.$$

Expanding the square and using $\varrho^i = H_0^{-\kappa}\ell_i^{\kappa}$ and $\mathfrak{g}_i^2 = \gamma\ell_i/L$ gives the listed powers. The first two lines follow by multiplying the conditional fiber bound by $\mathfrak{g}_i\ell_i^{\sigma-1}$ or by $\mathfrak{g}_i\ell_i^{2\alpha-2}$, respectively, as in Theorem 34.7. The last line follows from (34.29) and (34.27). Every power multiplying $\sqrt{\gamma}$ is free of $|\log\gamma|$, and every term that contains this logarithm has coupling order at least γ and a strictly positive power of the mesh.

We apply (34.30) to the quadratic initial term of $u_j(0) - u_j^{\mathrm{cl},0}(0)$. The remaining classical part of this difference is $V_j(\widehat{\theta}_{j-1}) - V_j(\widehat{\theta}^0_{j-1})$, whose bound by (34.12) is the third line of (34.31) without the factor $\mathfrak{q}^{j-1-i}$. The remaining mean is $I_{\ell_j\to a} m_j^{\mathrm{ph}}(\widehat{\theta}_{j-1})$, which by (34.13) and (34.27) is at most $C_p[\gamma r\ell_{j-1}^{\kappa} + \gamma^{3/2}\ell_{j-1}^{1/2}\sqrt{1+j}]$. All these bounds hold in the initial-array norm H^a_σ; for the centered quadratic initial terms, the first-difference part was proved by heat splitting in Theorem 34.7. By (34.15) and (34.16), we obtain, for some $s_0 > 0$,

$$\sum_{j=m+1}^{K} \big\|\mathsf{H}^a_\sigma\big(u_j(0) - u_j^{\mathrm{cl},0}(0)\big)\big\|_p \le C_p\big[\sqrt{\gamma} + \gamma(1+|\log\gamma|)^d\big](1+m)^c B^{-s_0 m}. \tag{34.32}$$

This bounds the predictable term in the norm of its initial value, which is stronger than a bound in X^a_α.

For the linear heat evolution, the difference of the increments splits exactly as

$$d_j - d_j^0 = s_j + z_j + (r_j - r_j^0) + (u_j - u_j^{\mathrm{cl},0}).$$

The bounds (34.11) and (34.27) imply that

$$\|\mathsf{X}^a_\alpha(r_j - r_j^0)\|_p \le C_p\sqrt{\gamma}(1+j)^{1/2}\ell_{j-1}^{\alpha-1/2}.$$

The centered linear term has the same leading power by Theorem 34.7, and the centered nonlinear heat term contributes $C_p[\gamma \ell_j^{\alpha}(1+j) + \gamma^{3/2}\ell_j^{1/2} p_j^3]$. The last part is bounded by (34.32) and the inequality $\mathsf{X}_{\alpha}^{a}(P_{\cdot}^{a} f) \le C\mathsf{H}_{\sigma}^{a}(f)$. Thus

$$\Big\| \sup_{m \le l \le K} \mathsf{X}_{\alpha}^{a}(\Delta x_l) \Big\|_p \le C_p\big[\sqrt{\gamma} + \gamma(1+|\log\gamma|)^d\big](1+m)^c B^{-s_0 m}. \tag{34.33}$$

By the triangle inequality, this bound holds with the supremum inside the expectation, and the sum over j of the $L^p(\mathsf{X}_{\alpha}^{a})$ norms of each of the four parts of $d_j - d_j^0$ is bounded by the same right side.

Applying (34.30) to each of the three ordered forms and using (34.31), we obtain

$$\begin{aligned} \sum_{j=m+1}^{K} \Big\| \mathsf{Y}_{\alpha}^{a}\big(T_{j,j-1}^{\omega}(\widehat{\theta}_{j-1}, \widehat{\theta}_{j-1}) - T_{j,j-1}^{\omega}(\widehat{\theta}_{j-1}^{0}, \widehat{\theta}_{j-1}^{0})\big) \Big\|_p \\ \le C_p\big[\sqrt{\gamma} + \gamma(1+|\log\gamma|)^d\big](1+m)^c B^{-s_0 m}, \end{aligned} \tag{34.34}$$

separately for $\omega = \mathsf{UV}, \mathsf{VU}, \mathsf{VV}$, with all conditional means included. We choose s_0 below the positive exponents listed in the statement and below the smaller exponents of the product estimates cited there. The other exponents in (34.31), such as $2\kappa - 1/2$ and κ, are positive and are accommodated by reducing s_0 further if necessary. The constants of all geometric sums depend on these fixed positive gaps, not on m.

Step 5: the cross products with the predictable term. There are two such products in each order. For the one containing $u_j - u_j^{\mathrm{cl},0}$, we use (34.6), Hölder's inequality, (34.26), and (34.32); its summed norm is at most a fixed multiple of the right side of (34.32). For the other one, the continuation satisfies the deterministic bound

$$\sum_{j>m} \mathsf{H}_{\sigma}^{a}\big(u_j^{\mathrm{cl},0}(0)\big) \le \frac{C\|\widehat{\theta}_m\|_{\infty}^2}{\ell_m} \le C\big[r^2 \ell_m^{2\kappa-1} + \gamma p_m^2\big], \tag{34.35}$$

given its input at level m. The first inequality sums the quadratic series of the zero-fluctuation maps with ratio $B\varrho^2 = \mathfrak{q}$, and the second uses (34.20). At $m = 0$ the last expression may be replaced by the constant Cr^2/H_0. The right side is not bounded uniformly in m at fixed γ, because of the term γp_m^2. We therefore multiply (34.35) by (34.33), using (34.6), before absorbing anything. The resulting term of order larger than γ is at most

$$C_p\big[\gamma^{3/2} p_m^2 + \gamma^2 p_m^2 (1+|\log\gamma|)^d\big](1+m)^c B^{-s_0 m}.$$

Since $p_m \le C(1+m)(1+|\log\gamma|)$, its additional polynomial in m is absorbed by $B^{-(s_0-s)m}$ for any fixed $0 < s < s_0$, and after enlarging d the term has coupling order at least γ. The deterministic part $r^2\ell_m^{2\kappa-1}$ of (34.35) keeps the positive power $2\kappa - 1$. This bounds both orders of the other cross product with the predictable term. In particular γp_m^2 is never bounded uniformly in m before its decaying factor has been restored.

Step 6: the remaining products. We follow the third column of Table 34.2. Each term is a difference from the zero-fluctuation counterpart, which is zero whenever s or z occurs, and the factor $(1+j)^c$ is suppressed in this step.

- (s,s): the centered part is at most $C_p\gamma\ell_j^{2\alpha-1}(1+j)$, and the conditional mean is at most $C_p\gamma[r\varrho^{j-1}+\mathfrak{g}_{j-1}\sqrt{1+j}]$, by Theorem 34.7 with the field moments (34.27).
- (s,z), (z,s), (z,z): the inequality (34.6) and Hölder's inequality under ν give the products of the $2p$-moment bounds $S_jZ_j^-$ or $Z_j^+Z_j^-$ of Theorem 34.7. Their coupling orders are at least $\gamma^{3/2}$ or γ^2, and their powers of ℓ_j include $\beta+\sigma>0$ and $\alpha+\sigma>0$. The factors of higher degree carry an additional $\ell_j^{1/2}$.
- (s,r), (r,s): conditionally on $\mathcal{F}_{j-1}$ we use the two mixed linear rows with the predictable background r_j. Its background bound has moments $C_p[r\ell_{j-1}^{\kappa+\alpha-1}+\sqrt{\gamma}\ell_{j-1}^{\alpha-1/2}\sqrt{1+j}]$. The resulting powers are $\sqrt{\gamma}\,r\ell_j^{\kappa+2\alpha-3/2}$ and $\gamma\ell_j^{2\alpha-1}$, each with a fixed polynomial factor.
- (z,r), (r,z): we put the norm H^a_σ of the initial value of z_j in (34.6) and use the moment bound for r_j just stated. The leading term is $\gamma r\ell_j^{\kappa+\alpha+\sigma-1}$; the rest has order at least $\gamma^{3/2}$ and the power $\alpha+\sigma-1/2>0$.
- (r,r): this is the difference of T^{VV} in (34.34), including its classical and conditional-mean parts.
- The seven pairs containing u: we use (34.7) and the identity with both arguments interchanged; for (u,u) this gives $\mathcal{Q}_a(u,u-u^{\mathrm{cl},0})+\mathcal{Q}_a(u-u^{\mathrm{cl},0},u^{\mathrm{cl},0})$. The individual norms of s_j, z_j, r_j, and u_j have bounded moments of order $2p$ by Theorem 34.7, and the sums of the moments of the differences $v-v^0$ are bounded by the statement following (34.33). We use (34.32) for the first term of (34.7), and (34.35) with the decaying factor of (34.33) for the second, as in Step 5.

There are also eight cross terms with x_{j-1}. Those with s are bounded by (34.19), the bounded moments of the norm (34.14) of x_{j-1}, and $\sqrt{\gamma}\ell_j^\beta$. Those with z use the norm H^a_σ of the initial value of z_j and $\mathsf{X}^a_\alpha(x_{j-1})$; they have order at least γ and a positive power of the mesh. For those with r, the two exact identities $T^{\mathsf{UV}}-T^{\mathsf{VV}}$ and $T^{\mathsf{VU}}-T^{\mathsf{VV}}$ of (34.2) reduce the differences to (34.34). For the two orders with u we use the identity

$$\mathcal{Q}_a(x_{j-1},u_j)-\mathcal{Q}_a(x^0_{j-1},u^{\mathrm{cl},0}_j)=\mathcal{Q}_a(x_{j-1},u_j-u^{\mathrm{cl},0}_j)+\mathcal{Q}_a(\Delta x_{j-1},u^{\mathrm{cl},0}_j)$$

and its reversed order, which were bounded in Step 5. None of these terms requires a bilinear estimate of two rough heat paths in the norm X^a_α; the products of z_j with r_j use only the initial-array norm of the centered nonlinear heat term, the norm of the signed defect, and (34.6).

By bilinearity, $y_K-y^0_K$ is the sum over $m<j\le K$ of these differences, and we apply the Minkowski inequality to this finite sum. The terms of order $\sqrt{\gamma}$ listed above keep a

positive power of the mesh and contain no logarithm of the coupling; their sum is at most $C_p\sqrt{\gamma}(1+m)^c B^{-sm}$. All terms containing a logarithm of the coupling have order at least γ, and the geometric sums above bound them by $C_p\gamma(1+|\log\gamma|)^d$, where dropping a remaining decaying factor only weakens the bound. Together with (34.33), this proves the final estimate of the theorem, and (34.32), (34.33), and (34.34) prove the other estimates. Each application of Hölder's inequality used higher fixed moments under the same measure; for a final moment p it suffices to use moments up to order $6p$, since a cubic difference has three factors and a product has two. These moments were proved in Theorem 34.7 and in Step 1. The exponents and all constants are fixed before γ, and they are independent of K, m, and the prior depths. □

34.6 Holonomy comparison at the weak-coupling scale

We now transfer the comparison of flow data to the holonomy matrices of the flowed configurations. For a pair $\mathcal{E} = (x, y)$ of heat paths on the lattice of mesh a we write $\|\mathcal{E}\|_{\mathrm{fl}} = \mathsf{X}^a_\alpha(x) + \mathsf{Y}^a_\alpha(y)$ for the flow-data norm of Definition 24.3, with the fixed mesh a suppressed, and we abbreviate $\mathscr{L}_\gamma = 1 + |\log\gamma|$.

Theorem 34.10. *For the triples of Definition 34.1, let $D_{K,m} = \max_i \|\mathcal{H}_i^K - \Xi_{m,i}^K\|_{\mathrm{op}}$. There exist $C, c, \epsilon, d > 0$ and, for every fixed $p \geq 1$, a constant $C_p < \infty$, all independent of $K \geq m \geq 0$ and $0 < \gamma \leq \gamma_0$, such that*

$$\mathbb{E}_\nu \min(1, D_{K,m})^2 \leq C\gamma\delta_m + C\gamma^2\mathscr{L}_\gamma^{2d}, \qquad \delta_m = (1+m)^c B^{-\epsilon m},$$

$$\mathbb{E}_\nu \min(1, D_{K,m})^p \leq C_p\gamma^{p/2}.$$

For every fixed m, the second term in the first bound is $o(\gamma)$ as $\gamma \to 0$, uniformly in K.

In particular the moments of order at most eight used in Section 35 have constants independent of m. The estimates hold under the conditioned measure, whose coarse field ranges over neighborhoods of all flat connections, regular and central.

Proof. Let $\mathcal{E} = (x_K, y_K)$, let $\mathcal{E}_m^0 = (x_K^0, y_K^0)$ be the flow data of the zero-fluctuation continuation after level m, and let $\mathcal{E}_{\mathrm{cl}}$ be the flow data of the zero-fluctuation continuation from the same coarse field, that is, the case $m = 0$. All three live on the finest lattice. By Lemma 34.4, $\|\mathcal{E}_{\mathrm{cl}}\|_{\mathrm{fl}} \leq R_{\mathrm{cl}}$ deterministically for every coarse field in the reconstruction domain, including every flat coarse field.

By Theorem 34.9, for every fixed p we have

$$\big\|\,\|\mathcal{E} - \mathcal{E}_m^0\|_{\mathrm{fl}}\big\|_p \leq C_p\sqrt{\gamma}(1+m)^c B^{-sm} + C_p\gamma\mathscr{L}_\gamma^d. \tag{34.36}$$

The function $\gamma^{1/2}\mathscr{L}_\gamma^d$ is bounded on $(0, \gamma_0]$ by the calculation in the proof of Lemma 34.6, and $\sup_m (1+m)^c B^{-sm} < \infty$ by (34.15). Thus the right side of (34.36) is at most $C_p\sqrt{\gamma}$,

uniformly in m. Its case $m = 0$ bounds $\mathcal{E} - \mathcal{E}_{\mathrm{cl}}$ by the same quantity, and the triangle inequality then bounds $\mathcal{E}^0_m - \mathcal{E}_{\mathrm{cl}}$ by $C_p\sqrt{\gamma}$ as well.

Let $R = R_{\mathrm{cl}} + 2$. For any real $q \geq 1$, the preceding bounds at moment $2q$ and Markov's inequality imply that

$$\nu\{\|\mathcal{E}\|_{\mathrm{fl}} > R \text{ or } \|\mathcal{E}^0_m\|_{\mathrm{fl}} > R\} \leq C_q\gamma^q. \tag{34.37}$$

This bound holds for each m with the same constant; there is no union over m. On the complementary event we apply Remark 34.8 with $t_- = t_+ = \theta L^2$ to the configurations $\exp\widehat{\theta}_K$ and $\exp\widehat{\theta}^0_K$. They have the common initial frame by construction, and their difference of flow data is exactly the one in (34.36). Hence

$$D_{K,m} \leq \Lambda_{R,\theta L^2,T}\|\mathcal{E} - \mathcal{E}^0_m\|_{\mathrm{fl}}$$

on that event. On the exceptional event we use $\min(1, D_{K,m}) \leq 1$. Squaring (34.36), using $(u+v)^2 \leq 2u^2 + 2v^2$, and taking $q = 2$ in (34.37), we obtain

$$\mathbb{E}_\nu \min(1, D_{K,m})^2 \leq C\gamma(1+m)^{2c}B^{-2sm} + C\gamma^2\mathscr{L}_\gamma^{2d} + C\gamma^2.$$

After increasing d to a positive value if necessary, the last term is absorbed into the second, and relabeling $2c$ and $2s$ as c and ϵ gives the first assertion. For a fixed $p \geq 1$ we use $(u+v)^p \leq 2^{p-1}(u^p + v^p)$, choose $q \geq \max(1, p/2)$ in (34.37), and use the bound $C_p\sqrt{\gamma}$ on the right side of (34.36); this gives $\mathbb{E}_\nu \min(1, D_{K,m})^p \leq C_p\gamma^{p/2}$. The stability estimate of Remark 34.8 covers the finitely many coarse meshes; no further event depending on the mesh has been discarded. Finally $\gamma\mathscr{L}_\gamma^{2d} \to 0$, again because $e^{-v}(1+v)^{2d} \to 0$ as $v = |\log\gamma| \to \infty$. Thus the second term in the bound on the mean square is $o(\gamma)$ uniformly in K for every fixed m; in fact its constant is independent of m. □

The next theorem gives the same comparison for every prescribed coarse field, uniformly over a time interval. It is used in Section 36, where the conditional estimate is multiplied by the small probability of a neighborhood of a central triple.

Theorem 34.11. *Fix the data of Definition 34.1 and $0 < \tau \leq T < \infty$, with the same three coordinate cycles. For each $K \geq 0$ and every coarse field $v = e^{\theta_0}$ in the reconstruction domain with $\|\theta_0\|_\infty \leq r$, define the probability*

$$\mathbb{Q}^K_v(dw_1 \cdots dw_K) = \prod_{j=1}^{K} k^K_j(\theta_{j-1}, dw_j), \qquad \theta_j = \widehat{\Phi}_j(\theta_{j-1}, g_{j,\cdot}w_j),$$

as an iterated integral: w_1 has the law $k^K_1(\theta_0, \cdot)$, and given $w_1, \ldots, w_{j-1}$ the vector w_j has the law $k^K_j(\theta_{j-1}, \cdot)$. Here k^K_j are the conditional fluctuation kernels of Theorem 15.9 and Lemma 15.10 at prior depth $J(K-j)$, written in the gauge of the reconstruction maps of Theorem 15.12; for

$K = 0$ the product is the unit point mass. Let $\theta_j^0 = P_j \cdots P_1 \theta_0$, and let $\mathcal{H}_t^K$ and $\mathcal{H}_{0,t}^K(v)$ be the based holonomy triples of $\Psi_t^{(N)}(\exp \theta_K)$ and $\Psi_t^{(N)}(\exp \theta_K^0)$, in their common initial frame. Set

$$d_{K,[\tau,T]} = \min\Big(1, \sup_{\tau \le t \le T} \max_i \|\mathcal{H}_{t,i}^K - \mathcal{H}_{0,t,i}^K(v)\|_{\mathrm{op}}\Big).$$

There exists $\gamma_W \in (0, \gamma_0]$, independent of K and v, such that for every fixed $p \ge 1$ there exists $C_{p,\tau,T} < \infty$ with

$$\sup_{K \ge 0} \sup_v \mathbb{E}_{\mathbb{Q}_v^K} d_{K,[\tau,T]}^p \le C_{p,\tau,T} \gamma^{p/2} \qquad (0 < \gamma \le \gamma_W).$$

The constants depend in addition on p, τ, T, and the loops, but not on the coarse field, which ranges over the set of initial fields specified in the theorem, including every flat coarse field. The ceiling γ_W is chosen before the moment order. The theorem is a statement about the probability $\mathbb{Q}_v^K$ itself, and it does not follow by conditioning Theorem 34.10 on the coarse field, which would only give a statement for almost every coarse field under the coarse marginal of ν.

Proof. Given the coarse field, the conditioned Wilson measure has the same conditional structure as $\mathbb{Q}_v^K$. We first construct $\mathbb{Q}_v^K$ for every v and check that all conditional estimates of Theorem 34.7 hold under it. The proof of Theorem 34.9 then applies with $m = 0$ and a deterministic coarse field; we repeat its steps in this setting, since the averaged statement cannot be quoted. The only change is that the sum of the classical predictable terms is now bounded by a deterministic constant. Finally we localize to a fixed ball of flow data and apply the matrix stability over the whole interval $[\tau, T]$.

Step 1: construction and coupling ceiling. The initial fields in the statement satisfy $\|\theta_0\|_\infty \le r$. The kernel formula of Theorem 15.9 extends continuously to this closed ball: its product with the fluctuation cutoff domain lies strictly inside the common analytic domains, and its positive normalizer is the integral of a bounded positive density. The gauge change of Theorem 15.12 changes neither this density nor its product of balls. By item (2) of Theorem 15.12, with the closed-product image conditions verified in its proof, the reconstruction maps this product into B_r. Inductively, every later incoming field lies in B_r, and Theorem 15.9 gives each successive kernel. Iterated integration therefore defines $\mathbb{Q}_v^K$, including at initial fields to which the coarse marginal of ν assigns no mass. By the amplitude bound in Lemma 15.10, all intermediate fields also stay in the working ball, and

$$\|\theta_j\|_\infty \le C\{r\varrho^j + \mathfrak{g}_j p_j\}, \qquad p_j \le C(1 + j + |\log \gamma|).$$

We reduce γ_W so that the Hessian lower bound and the bounds on the conditional means of the kernels hold at all these fields. This is possible because $t_j \le t_1 = \sqrt{\gamma/(Mn)}$ and $t^{a'} \log^{b'}(1/t) \to 0$ as $t \to 0$ for $a' > 0$. The hierarchy and the cutoffs of the fluctuation variables are not changed.

Step 2: transfer of conditional estimates. Let $\mathcal{F}_j = \sigma(w_1, \dots, w_j)$ on this probability space; the coarse field v is deterministic. Given $\mathcal{F}_{j-1}$, the law of the next vector is exactly $k_j^K(\theta_{j-1}, \cdot)$. Thus every estimate of Theorem 34.7 for a fixed incoming field holds here with the same constants. Explicitly, if $\mathbb{E}(|A|^q \mid \mathcal{F}_{i-1}) \le C_q^q b(\theta_{i-1})^q$ is one of those estimates, then the tower property implies that

$$\|A\|_{L^q(\mathbb{Q}_v^K)} \le C_q \|b(\theta_{i-1})\|_{L^q(\mathbb{Q}_v^K)}. \tag{34.38}$$

No coarse marginal enters this inequality.

Step 3: moments of the fields. We write $\delta_j = \theta_j - \theta_j^0$ and $\epsilon_j = \widehat{\Phi}_j(\theta_{j-1}, g_{j,\cdot} w_j) - P_j \theta_{j-1}$. The bound on the gradient in the fluctuation variables and Theorem 5.4, applied for a fixed incoming field to the at most $C_J(nB^j)^3$ scalar outputs, imply that

$$\big\| \|\epsilon_j - \mathbb{E}(\epsilon_j \mid \mathcal{F}_{j-1})\|_\infty \big\|_{L^q(\cdot \mid \mathcal{F}_{j-1})} \le C_q \mathfrak{g}_j \sqrt{1+j}.$$

The conditional mean is bounded by (34.13): $\|\mathbb{E}(\epsilon_j \mid \mathcal{F}_{j-1})\|_\infty \le C r \mathfrak{g}_j^2$. Since the segment between the two incoming fields lies in the convex working ball, we have

$$\big\| \|\delta_j\|_\infty \big\|_q \le \varrho \big\| \|\delta_{j-1}\|_\infty \big\|_q + C_q \mathfrak{g}_j \sqrt{1+j} + C r \mathfrak{g}_j^2.$$

We start at $\delta_0 = 0$ and sum, using $\mathfrak{g}_i = \mathfrak{g}_j B^{(j-i)/2}$, $\varrho\sqrt{B} < 1$, and, for the quadratic term, $\mathfrak{g}_i^2 \le \mathfrak{g}_1 \mathfrak{g}_i$. For every fixed $q < \infty$ we obtain

$$\big\| \|\delta_j\|_\infty \big\|_q \le C_q \mathfrak{g}_j \sqrt{1+j}, \qquad \big\| \|\theta_j\|_\infty \big\|_q + \|\theta_j^0\|_\infty \le C_q \{ r\varrho^j + \mathfrak{g}_j \sqrt{1+j} \}. \tag{34.39}$$

The continuation is deterministic and $\|\theta_j^0\|_\infty \le r\varrho^j$.

Step 4: the pulled-back forms. Let $A_{j,i}$ be the quadratic initial form $S_{j,i}$ or one of the three ordered forms $T_{j,i}^\omega$, and recall $K_{ji} = \mathfrak{q}^{j-1-i}/\ell_i$; fixed polynomial factors in $1+j$ are allowed below. With $L_i = DP_i(0)$, we define

$$\begin{aligned} Z_{j,i} &= A_{j,i}(\theta_i, \theta_i) - \mathbb{E}\big[A_{j,i}(\theta_i, \theta_i) \mid \mathcal{F}_{i-1}\big], \\ C_{j,i}(u) &= A_{j,i}(P_i u, P_i u) - A_{j,i}(L_i u, L_i u), \\ M_{j,i}(u) &= \mathbb{E}_{k_i^K(u,\cdot)} A_{j,i}(\theta_i, \theta_i) - A_{j,i}(P_i u, P_i u), \end{aligned}$$

where in the last line $\theta_i = \widehat{\Phi}_i(u, g_{i,\cdot} w_i)$ with w_i distributed according to $k_i^K(u, \cdot)$. By the difference estimate of Theorem 34.9, which follows from (34.28), we have

$$\|C_{j,i}(u) - C_{j,i}(u')\|_{\mathrm{out}} \le C(1+j)^c K_{ji} (\|u\|_\infty + \|u'\|_\infty)^2 \|u - u'\|_\infty.$$

By the splitting into classical and conditional-mean parts in Theorems 27.5 and 27.7, we have

$$\|M_{j,i}(u)\|_{\mathrm{out}} \le C(1+j)^c K_{ji} \big(\mathfrak{g}_i^2 \|u\|_\infty + \mathfrak{g}_i^3 p_i^3 \big). \tag{34.40}$$

Here the output of the quadratic initial form uses the initial-array norm H^a_σ and the ordered outputs use the norm Y^a_α; there is no classical cubic term in $M_{j,i}$. As in Step 3 of the proof of Theorem 34.9, the exact pullback telescopes to

$$A_{j,j-1}(\theta_{j-1},\theta_{j-1}) - A_{j,j-1}(\theta^0_{j-1},\theta^0_{j-1}) = \sum_{i=1}^{j-1}\big\{Z_{j,i} + C_{j,i}(\theta_{i-1}) - C_{j,i}(\theta^0_{i-1}) + M_{j,i}(\theta_{i-1})\big\}. \tag{34.41}$$

The term at level zero cancels because both starting fields are θ_0.

Step 5: sizes and summation. As before, let $q_i = 1 + i + |\log\gamma|$. Applying (34.38), (34.39), the conditional fiber bound $b_i(u) = C_J(\varrho\|u\|_\infty + \mathfrak{g}_i p_i)$, and Hölder's inequality under $\mathbb{Q}^K_v$, we find that the four bounds of (34.31) hold in $L^q(\mathbb{Q}^K_v)$, up to the factor $C_q(1+j)^c\mathfrak{q}^{j-1-i}$, for $Z_{j,i}$ (for the quadratic initial form and each ordered form), for $C_{j,i}(\theta_{i-1}) - C_{j,i}(\theta^0_{i-1})$, and for $M_{j,i}(\theta_{i-1})$. The verification is the one given after (34.31). For example, at moment $3q$ the difference of the classical terms is at most $C_q\ell_i^{-1}[r\varrho^{i-1} + \mathfrak{g}_{i-1}\sqrt{1+i}]^2\mathfrak{g}_{i-1}\sqrt{1+i}$; the bounds on $Z_{j,i}$ follow from the conditional bounds $\mathfrak{g}_i b_i \ell_i^{\sigma-1}$ and $\mathfrak{g}_i b_i \ell_i^{2\alpha-2}$; and the bound on $M_{j,i}$ is (34.40).

Since all the exponents of the mesh are positive, the geometric sums of Lemma 34.2 sum (34.41) separately for the quadratic initial form and for each of the ordered forms. The remaining local terms of the predictable part are the classical cubic term and the conditional mean of (34.12) and (34.13), which are bounded by the third and fourth lines of (34.31). The first differences in the initial-array norm are controlled by Theorem 27.7, including for $t \le a^2$. We obtain

$$\sum_j \big\|\mathsf{H}^a_\sigma\big(u_j(0) - u_j^{\mathrm{cl},0}(0)\big)\big\|_q + \sum_j \|\mathsf{X}^a_\alpha(d_j - d_j^0)\|_q + \sum_{j,\omega}\|\mathsf{Y}^a_\alpha(\Delta T_j^\omega)\|_q \le C_q\big[\sqrt{\gamma} + \gamma\mathscr{L}^d_\gamma\big], \tag{34.42}$$

where ΔT_j^ω is the left side of (34.41) for the form T^ω, and every sum runs over $1 \le j \le K$. For the linear heat evolution, the bound on s_j is $C_q\sqrt{\gamma}\ell_j^{\alpha-1/2}\sqrt{1+j}$, the bound on z_j is $C_q(\gamma\ell_j^\alpha(1+j) + \gamma^{3/2}\ell_j^{1/2}p_j^3)$, the bound on $r_j - r_j^0$ follows from (34.11) and (34.39), and the inequality $\mathsf{X}^a_\alpha(P^a_\cdot f) \le C\mathsf{H}^a_\sigma(f)$ bounds $u_j - u_j^{\mathrm{cl},0}$. Since these are the four parts of $d_j - d_j^0$, (34.42) also bounds $\|\sup_j \mathsf{X}^a_\alpha(x_j - x_j^0)\|_q$.

Step 6: products. We use only the pointwise ordered inequality (34.6); the averaged conclusions at fixed coupling of Lemma 28.5 are not needed. We denote the q-moment

bounds by

$$\begin{aligned} S_j &= C_q\sqrt{\gamma}\ell_j^{\alpha-1/2}\sqrt{1+j}, \\ Z_j^- &= C_q(\gamma\ell_j^{\sigma}(1+j)+\gamma^{3/2}\ell_j^{1/2}p_j^3), \\ Z_j^+ &= C_q(\gamma\ell_j^{\alpha}(1+j)+\gamma^{3/2}\ell_j^{1/2}p_j^3), \\ R_j &= C_q(r\ell_{j-1}^{\kappa+\alpha-1}+\sqrt{\gamma}\ell_{j-1}^{\alpha-1/2}\sqrt{1+j}), \end{aligned}$$

and use moments of order $2q$ in products. The norms $\mathsf{H}^a_\sigma(u_j(0))$ and the norms (34.14) of x_{j-1} at its input mesh have uniformly bounded moments. This follows from the same estimates for the quadratic initial terms without subtraction, after adding the deterministic initial term and the classical cubic term, whose powers of the mesh are $2\kappa-1>0$ and $3\kappa-1>0$, besides the positive powers already listed. The initial cardinal field costs only $C_{H_0}r$. The sixteen ordered differences $\mathcal{Q}_a(v_j,w_j)-\mathcal{Q}_a(v_j^0,w_j^0)$ and the eight cross terms are then estimated as in Step 6 of the proof of Theorem 34.9, following the third column of Table 34.2, with the following bounds.

- (s,s): the centered part is at most $C_q\gamma\ell_j^{2\alpha-1}(1+j)$ and the conditional mean at most $C_q\gamma[r\varrho^{j-1}+\mathfrak{g}_{j-1}\sqrt{1+j}]$, by Theorem 34.7.
- (s,z), (z,s), (z,z): respectively $CS_jZ_j^-$, $CS_jZ_j^-$, and $CZ_j^+Z_j^-$, by (34.6) and Hölder's inequality under $\mathbb{Q}_v^K$.
- (s,r), (r,s): CS_jR_j, by the conditional mixed linear rows of Theorem 34.7; the powers are $\kappa+2\alpha-3/2$ and $2\alpha-1$.
- (z,r), (r,z): $CZ_j^-R_j$ in each order, using the initial-array norm of the centered nonlinear heat term. The leading powers include $\kappa+\alpha+\sigma-1$ and $\alpha+\sigma-1/2$.
- (r,r): the difference of the form T^{VV} in (34.41) and (34.42).
- The seven pairs containing u: we use (34.7) and the identity with both arguments interchanged; for (u,u) this gives $\mathcal{Q}_a(u,u-u^{\mathrm{cl},0})+\mathcal{Q}_a(u-u^{\mathrm{cl},0},u^{\mathrm{cl},0})$. The first part is summed using the bound on the predictable term in (34.42) and the bounded norms. For the second part, the quadratic series of the zero-fluctuation maps and $B\varrho^2<1$ give the deterministic bound

$$\sum_{j\ge1}\mathsf{H}^a_\sigma\big(u_j^{\mathrm{cl},0}(0)\big)\le C\|\theta_0\|_\infty^2/H_0\le Cr^2/H_0,$$

  which we multiply by the summed bounds on the first component in (34.42). Since $m=0$ and the coarse field is deterministic, there is no term γp_m^2 here.

There are eight cross terms with x_{j-1}, two for each argument. For s, the conditional mixed linear rows give $C_q\sqrt{\gamma}\ell_j^{\alpha-1/2}$ times a bounded norm of x_{j-1}. For z, the two orders

of (34.6) give $C_q Z_j^-$ times that norm. For r, the two exact differences $T^{\mathsf{UV}} - T^{\mathsf{VV}}$ and $T^{\mathsf{VU}} - T^{\mathsf{VV}}$ are bounded in (34.42). For u we use

$$\mathcal{Q}_a(x_{j-1}, u_j) - \mathcal{Q}_a(x_{j-1}^0, u_j^{\mathrm{cl},0}) = \mathcal{Q}_a(x_{j-1}, u_j - u_j^{\mathrm{cl},0}) + \mathcal{Q}_a(x_{j-1} - x_{j-1}^0, u_j^{\mathrm{cl},0})$$

and its reversed order, which are bounded by the two estimates on the predictable terms just used. This accounts for all eight cross terms and all sixteen ordered products.

Step 7: the difference of flow data. The smallest exponents of the mesh in these sums are

$$\alpha - \tfrac{1}{2}, \quad \sigma, \quad 2\alpha - 1, \quad \kappa + \sigma - \tfrac{1}{2}, \quad \kappa + 2\alpha - \tfrac{3}{2}, \quad 2\kappa - 1, \quad \kappa + \alpha + \sigma - 1,$$

and $-\log \mathfrak{q} / \log B$, all strictly positive. Every term with a logarithm of the coupling has order at least γ, and Lemma 34.2 sums each fixed polynomial times its geometric decay. By bilinearity and the Minkowski inequality, under $\mathbb{Q}_v^K$,

$$\big\| \|\mathcal{E}_K - \mathcal{E}_K^0\|_{\mathrm{fl}} \big\|_q \le C_q \big(\sqrt{\gamma} + \gamma \mathscr{L}_\gamma^d \big) \le C_q' \sqrt{\gamma}, \tag{34.43}$$

where $\mathcal{E}_K = (x_K, y_K)$ and $\mathcal{E}_K^0 = (x_K^0, y_K^0)$, and in the last inequality we used that $\sqrt{\gamma}\mathscr{L}_\gamma^d$ is bounded. The bookkeeping of moments uses at most order $6q$: three factors in a cubic difference and two in a product. Thus the final moments $q = 1, 2$ require only the finitely many moment bounds up to order 12, all of which were obtained for fixed incoming fields before the tower integration.

Step 8: localization and the interval estimate. By Lemma 34.4, $\|\mathcal{E}_K^0\|_{\mathrm{fl}} \le R_{\mathrm{cl}}$ uniformly in v. With $R = R_{\mathrm{cl}} + 2$, (34.43) at moment q and Markov's inequality imply that

$$\mathbb{Q}_v^K \big\{ \|\mathcal{E}_K\|_{\mathrm{fl}} > R \big\} \le C_q \gamma^{q/2}.$$

On the complementary event we apply Remark 34.8 directly with $t_- = \tau$ and $t_+ = T$. It bounds the supremum over $t \in [\tau, T]$ by $\Lambda_{R,\tau,T} \|\mathcal{E}_K - \mathcal{E}_K^0\|_{\mathrm{fl}}$. On the exceptional event we use $d_{K,[\tau,T]} \le 1$. Taking $q \ge p$ and using (34.43) at moment p, we obtain the asserted bound $C_{p,\tau,T} \gamma^{p/2}$. The finitely many coarse meshes are included in the stability estimate. For $p = 1, 2$, the choice $q = 2$ suffices for the localization, and moments up to order 12 suffice for the whole argument. □

34.7 Linear Gaussian response near a fixed flat connection

In this section, we prove the estimate used in Section 35 to calibrate the normal covariance near regular commuting triples. We fix a regular flat coarse field and differentiate the deterministic reconstruction and the Wilson flow at this field with respect to the fluctuation inputs of the steps after level m. We divide the coupling factors by $\sqrt{\gamma}$,

and only then insert independent standard Gaussian vectors in place of the fluctuation variables. The lemma bounds the covariance, in the normal space, of the resulting Gaussian vector. Three points need care. First, the inputs $q_j = g_{j,\cdot} w_j$ are the arguments of the reconstruction maps. The normalization by the inverse square root of the constrained Hessian, which depends on the field, is already inside these maps, and the derivative therefore includes all normalizing matrices and their dependence on the field through the later reconstruction. Second, we recall from Definition 4.11 that at a point $y = (\vartheta_i u)_{i=1}^3$ of a compact regular patch of the cone of commuting triples, the normal space is $\mathcal{N}_y = \{b = (b_i)_{i=1}^3 : b_i \perp u,\ \sum_i \vartheta_i b_i = 0\}$, of dimension four, and $P_{\mathcal{N}_y}$ denotes the Euclidean orthogonal projection onto it in the fixed coordinates of cycle logarithms. Third, the identity $\mathbb{E}_G|\sum_j \mathsf{A}_{j,K} G_j|^2 = \sum_j \operatorname{tr}(\mathsf{A}_{j,K}\mathsf{A}_{j,K}^{\mathsf{T}})$ concerns only the contribution of the omitted steps. The covariance of the normal fluctuations of the coarse field is a separate contribution, which is not set to zero, and for this reason the lemma cannot be replaced by an averaged estimate for the conditioned measure.

Lemma 34.12. *Use the hierarchy, the side n, the time θL^2, and the reconstruction maps of Definition 34.1. Fix a compact regular patch of the cone of commuting triples and margins for the logarithm and for the normal coordinates as in Theorem 33.11. For a flat coarse field V_y whose cycle holonomies lie in this patch, let $\mathcal{H}_K(y,q)$ be the based holonomy triple, around the three coordinate cycles, of the configuration on the finest lattice reconstructed from V_y with fluctuation inputs $q = (q_1, \dots, q_K)$, flowed for the time θL^2. At $q = 0$ it equals $\mathrm{Hol}(V_y)$, in the original frame. Put*

$$\mathsf{A}_{j,K}(y) = P_{\mathcal{N}_y} D_{q_j}(\mathrm{Log} \circ \mathcal{H}_K)(y,0)\ \mathrm{diag}_{r,e}(g_{j,r}/\sqrt{\gamma}), \qquad 1 \le j \le K,$$

where the diagonal matrix has the entry $g_{j,r}/\sqrt{\gamma}$ at every scalar coordinate (r,e) of the fluctuation variables of step j; these ratios do not depend on γ. Equivalently, $\mathsf{A}_{j,K} = \gamma^{-1/2} P_{\mathcal{N}_y} D_{w_j}(\mathrm{Log} \circ \mathcal{H}_K)$ at zero fluctuation, with the full zero-fluctuation continuation at the later levels and every normalizing matrix included. Let G_j be independent standard real Gaussian vectors indexed like w_j. Then there exist $C, c, \epsilon > 0$ such that

$$\sup_y \mathbb{E}_G \Big| \sum_{j=m+1}^{K} \mathsf{A}_{j,K}(y) G_j \Big|^2 \le C(1+m)^c B^{-\epsilon m} \qquad (K \ge m \ge 0).$$

The constants may depend on the fixed n, L, θ, M, J, the parameters Θ of Theorem 23.2, the patch, and the fixed bounds on the coefficients and the flow, but not on m, K, the prior depths, or $\gamma \in (0, \gamma_0]$. Before the flow is applied, the first-order variation of the flow data in these directions, measured in both full norms with their first differences, has a bound $C(1+m)^c B^{-sm}$ in $L^2(G)$ for some $s > 0$.

Only linear maps are evaluated at the unbounded Gaussian vectors; no nonlinear reconstruction is evaluated outside its cutoff domain. For $j \ge 1$, the matrices $\mathsf{A}_{j,K}$ are the

linear response matrices $A_{j,K}$ of Definition 35.7, on the same normal patch and with the same gauge and basepoint, as the second expression for $\mathsf{A}_{j,K}$ shows; the coarse response $A_{0,K}$ there is a separate contribution.

Proof. The proof uses only deterministic coefficient estimates and independent Gaussian vectors; in particular, it does not use Theorems 34.9 and 34.10 as probability estimates. Throughout, we keep the zero-fluctuation base point and the direction of differentiation separate. We first set up and bound the linear recursion for the derivative of the reconstruction. We then write the first-order variation of the flow data as a sum over levels, bound each term by deterministic rows and Gaussian chaining, and sum. Finally, we pass to the holonomies through the derivative of the flow and the logarithm.

Step 1: the base point and the linear recursion. We write $\varrho = a_{\mathrm{map}}$ and $H_0 = L/n$. In one of the finitely many fixed charts covering the compact patch, let $v_0(y)$ be the logarithmic representative of V_y in the reconstruction domain, and set

$$v_i = P_i v_{i-1}, \qquad b_i^{\mathrm{fl}} = r\varrho^i, \qquad X_i = P_\cdot^a I_{\ell_i \to a}(v_i/\ell_i), \qquad Y_i = \mathcal{Q}_a(X_i, X_i). \tag{34.44}$$

The condition on flat fields was checked at the beginning of Section 34.2. By Lemma 14.9, at every side and prior depth, each constrained minimizer is an exact flat lift, its free coordinates vanish, and the gauges of the reconstruction maps preserve the prescribed cycle holonomies. The Wilson flow is stationary at each resulting flat field. It follows that $\mathcal{H}_K(y,0) = \mathrm{Hol}(V_y)$ exactly, in the original frame; this uses the refinement of flat fields, not a property of cardinal interpolation. By the bound on DP_i and Lemma 34.4,

$$\|v_i\|_\infty \le b_i^{\mathrm{fl}}, \qquad \sup_{K,i,y} \{\mathsf{X}_\alpha^a(X_i) + \mathsf{Y}_\alpha^a(Y_i)\} \le R_{\mathrm{cl}}. \tag{34.45}$$

The same domains serve every chart of the finite cover.

Let D_i be the diagonal matrix with the entries $g_{i,r}/\sqrt{\gamma}$ at all scalar coordinates of the fluctuation variables of step i. We write

$$W_i = D_q \widehat{\Phi}_i(v_{i-1}, 0) D_i, \qquad \eta_m = 0, \qquad \eta_i = DP_i(v_{i-1})\eta_{i-1} + W_i G_i \quad (i > m), \tag{34.46}$$

and we set $\eta_i = 0$ for $i < m$ as well. By the coefficient bounds of the reconstruction maps, we have

$$\|DP_i(v_{i-1})\|_{\infty\to\infty} \le \varrho, \qquad \sup_e \|W_i(e,\cdot)\|_2 \le C_J d_i, \qquad d_i = \mathfrak{g}_i/\sqrt{\gamma} = \sqrt{\ell_i/L}.$$

The ratios of the couplings within one step are kept in D_i; replacing them by d_i costs only a constant depending on J. All these linear maps are independent of the coupling. Their derivatives contain the positive inverse square roots of the constrained Hessians used to

normalize the fluctuation variables, including the dependence of these matrices on the field through the subsequent zero-fluctuation maps.

A scalar Gaussian row $c \cdot G$ has the Laplace transform $\mathbb{E}e^{tc\cdot G} = e^{t^2|c|_2^2/2}$, by completing the square in each one-dimensional Gaussian integral. Hence its two-sided tail at the threshold $u|c|_2$ is at most $2e^{-u^2/2}$. There are at most $C_J(nB^i)^3$ scalar outputs at level i. By the finite-family clause of Lemma 16.4, with $N \le C_J(nB^i)^3$, scale Cd_i, $b = 0$, and $e = 0$, we have $\|\|W_iG_i\|_\infty\|_{L^2(G)} \le Cd_i\sqrt{1+i}$. Applying the Minkowski inequality to (34.46), we obtain

$$\big\|\|\eta_i\|_\infty\big\|_{L^2(G)} \le C \sum_{l=m+1}^{i} \varrho^{i-l} d_l \sqrt{1+l} \le C\sqrt{\ell_i}(1+i)^{1/2}, \tag{34.47}$$

where we used $d_l = d_i B^{(i-l)/2}$ and $\varrho\sqrt{B} < 1$. This bound is uniform in m, the cutoff, and the flat field.

Step 2: first-order identities on the common lattice. In this proof V_i and Z_i denote first variations of the flow data; they are unrelated to the flat coarse field V_y. Define

$$V_i = P_\cdot^a I_{\ell_i\to a}(\eta_i/\ell_i), \qquad Z_i = \mathcal{Q}_a(X_i, V_i) + \mathcal{Q}_a(V_i, X_i). \tag{34.48}$$

Both are linear in $G_{m+1}, \dots, G_i$. They are the first derivatives of the flow data (x_i, y_i) when only the fluctuation inputs of the steps after m vary, in the directions D_lG_l. This is an identity between finite-dimensional linear maps: we first differentiate at zero in deterministic directions, and only afterwards apply the derivatives to the Gaussian vectors.

We use the uncentered deterministic Taylor expansion in the fluctuation inputs,

$$\widehat{\Phi}_i(v, q) = P_i v + D_q\widehat{\Phi}_i(v, 0)q + O(q^2).$$

Along the deterministic reconstruction $\theta_i(q) = \widehat{\Phi}_i(\theta_{i-1}(q), q_i)$ from $\theta_0 = v_0$, with flow data $x_i(q)$ and $y_i(q)$, the increment splits exactly into four uncentered summands,

$$x_i(q) - x_{i-1}(q) = \tilde{s}_i + \tilde{z}_i + \tilde{r}_i + \tilde{u}_i,$$

where, with $\theta = \theta_{i-1}(q)$,

$$\tilde{s}_i = \ell_i^{-1} P_\cdot^a I_{\ell_i\to a} D_q\widehat{\Phi}_i(\theta, 0)q_i, \qquad \tilde{z}_i = \ell_i^{-1} P_\cdot^a I_{\ell_i\to a}\big[\widehat{\Phi}_i(\theta, q_i) - P_i\theta - D_q\widehat{\Phi}_i(\theta, 0)q_i\big],$$
$$\tilde{r}_i = P_\cdot^a \mathscr{D}_i\theta, \qquad \tilde{u}_i = P_\cdot^a I_{\ell_i\to a} U_i^{\mathrm{cl}}(\theta).$$

These are not the centered summands s_i and z_i of (34.1), which are defined under the conditioned measure by an expansion at the conditional mean; the present expansion is at zero, and no conditional mean enters. At $q = 0$ the four summands are $(0, 0, R_i, U_i)$, defined below, and their first variations in the directions D_lG_l are $(\xi_i, 0, \rho_i, v_i)$. Thus the first-order part of the increment splits exactly as

$$V_i - V_{i-1} = \xi_i + \rho_i + v_i, \tag{34.49}$$

where

$$\xi_i = \ell_i^{-1} P_\cdot^a I_{\ell_i \to a} W_i G_i, \qquad \rho_i = P_\cdot^a \mathscr{D}_i \eta_{i-1}, \qquad v_i = P_\cdot^a I_{\ell_i \to a} D U_i^{\mathrm{cl}}(v_{i-1})[\eta_{i-1}].$$

Here $\mathscr{D}_i$ and U_i^{cl} are the signed defect and the classical predictable term of Lemma 34.2, including the quadratic terms and the terms of higher degree of the zero-fluctuation map. At order zero the same split is $X_i - X_{i-1} = R_i + U_i$, with $R_i = P_\cdot^a \mathscr{D}_i v_{i-1}$ and $U_i = P_\cdot^a I_{\ell_i \to a} U_i^{\mathrm{cl}}(v_{i-1})$. Terms with two fluctuation factors have zero first derivative at zero. The zero-fluctuation continuation agrees with the base point through level m and has zero derivative afterwards, because $\eta_m = 0$. Thus

$$V_l = \sum_{i=m+1}^{l} (\xi_i + \rho_i + v_i), \qquad Z_l = \sum_{i=m+1}^{l} (Z_i - Z_{i-1}),$$

and all initial terms at level m vanish exactly.

Step 3: linear rows and the background on two lattices. We apply Lemma 34.5 to the deterministic linear pullbacks, with the coefficient covers of Lemma 34.3. The values at the reference point have the Gaussian tail bounds just described. A difference of two linear Gaussian rows is a centered Gaussian variable whose variance is the squared coefficient distance, and it satisfies the conditions (1) of Lemma 34.5 exactly. We use only the deterministic coefficient part of Theorem 26.10, at zero fluctuation: we evaluate the coefficient at $(v_{i-1}, 0)$, divide its derivative in the fluctuation variables by $\sqrt{\gamma}$, and use the matrix D_i. Its weighted linear row scale is $C\ell_i^{\alpha - 1/2}$. The normalized coefficient metric has bounded diameter and a cover at the input mesh, and the Gaussian Laplace transform gives the tails of the increments and at the reference point. Applying Lemma 34.5 under the Gaussian measure, we obtain

$$\|\mathsf{X}_\alpha^a(\xi_i)\|_{L^2(G)} \le C(1+i)^c \ell_i^{\alpha - 1/2}, \qquad \|\mathsf{X}_\alpha^a(\rho_i)\|_{L^2(G)} \le C(1+i)^{1/2} \ell_{i-1}^{\alpha - 1/2}. \tag{34.50}$$

The second bound follows from (34.11) and (34.47).

We verify the stronger background conditions for the mixed rows used below. Let $H = \ell_{i-1} = B\ell_i$. Lemma 27.1 gives the signed heat estimate for difference orders 0, 1, and 2. For orders 3 and 4 and $t \ge H^2$, we put the additional differences on a half heat step in the estimate of order two, using Lemma 34.2. For $t \le H^2$, we put all differences on the two smooth cardinal inputs, whose input meshes are H and H/B; since their ratio is the fixed B, this costs only a constant $C_{J,q}$. For every $0 \le q \le 4$ this proves

$$\|D_a^q R_i(t)\|_\infty \le C_q \frac{b_{i-1}^{\mathrm{fl}}}{H} (t + H^2)^{-q/2} \frac{H}{\sqrt{t + H^2}} \le C_{J,q} b_{i-1}^{\mathrm{fl}} H^{\alpha - 1} (t + \ell_i^2)^{-(\alpha+q)/2}. \tag{34.51}$$

The last inequality follows by maximizing $(t + H^2)^{\alpha/2} H / \sqrt{t + H^2}$, whose maximum is H^α since $\alpha < 1$. Since every refinement ratio of the common lattice is an integer, the

statements about signed columns and cardinal interpolation apply. This background is a heat path built from two lattices, not a single cardinal interpolation of the parent field. The other background X_{i-1} has bounded norm at its input mesh by (34.14) and (34.45), and its differences through order four are bounded by the last assertion of item (6) of Lemma 34.2. Its correction term in (34.14) is $Cb_{i-1}^{\mathrm{fl}}H^{\alpha-1}$, which is bounded since $\kappa+\alpha-1>0$.

Thus both orders of the version of Theorem 26.10 with four differences apply, with the deterministic background bounds C for X_{i-1} and $Cb_{i-1}^{\mathrm{fl}}H^{\alpha-1}$ for R_i. Its envelopes of order four give the normalized space-time cover, independently of a/ℓ_i, and the heat equation gives the time increments. Using its separate linear rows and Lemma 34.5, we obtain, for each order separately,

$$\|\mathsf{Y}_\alpha^a(\mathcal{Q}_a(X_{i-1},\xi_i))\|_2+\|\mathsf{Y}_\alpha^a(\mathcal{Q}_a(\xi_i,X_{i-1}))\|_2\le C(1+i)^c\ell_i^{\alpha-1/2}, \tag{34.52}$$

$$\|\mathsf{Y}_\alpha^a(\mathcal{Q}_a(R_i,\xi_i))\|_2+\|\mathsf{Y}_\alpha^a(\mathcal{Q}_a(\xi_i,R_i))\|_2\le Cr(1+i)^c\ell_i^{\kappa+2\alpha-3/2}, \tag{34.53}$$

where $\|\cdot\|_2$ is the norm of $L^2(G)$. These are not the conditional maxima at fixed coupling of Theorem 26.10: we use only its deterministic rows and normalized covers, and the maximum is taken here for a linear Gaussian process.

Step 4: the pulled-back forms at first order. Let $T_{j,i}^{\omega}$ be the three ordered forms and $S_{j,i}$ the quadratic initial form, including its physical prefactor and its cardinal interpolation. For either type we write $F_{j,i}(v)=A_{j,i}(v,v)$, and we set

$$C_{j,i}(v)=F_{j,i}(P_iv)-F_{j,i}(L_iv),\qquad B_{j,i}=DF_{j,i}(v_i)W_i.$$

The calculation that proved (34.28) implies that

$$\|DC_{j,i}(v)[w]\|_{\mathrm{out}}\le C(1+j)^c\frac{\mathfrak{q}^{j-1-i}}{\ell_i}\|v\|_\infty^2\|w\|_\infty. \tag{34.54}$$

The output norm is Y_α^a for an ordered form and the initial-array norm H_σ^a for a quadratic initial form. Differentiating the finite telescope given by (34.3), or applying the chain rule term by term, we obtain exactly

$$DF_{j,j-1}(v_{j-1})[\eta_{j-1}]=\sum_{i=m+1}^{j-1}\big\{B_{j,i}G_i+DC_{j,i}(v_{i-1})[\eta_{i-1}]\big\}. \tag{34.55}$$

Indeed, the part of the chain rule that comes from P_i is $DF_{j,i}(v_i)DP_i(v_{i-1})\eta_{i-1}$; subtracting the derivative of the pullback, $DF_{j,i}(L_iv_{i-1})L_i\eta_{i-1}$, leaves $DC_{j,i}(v_{i-1})[\eta_{i-1}]$, and the remaining part is $DF_{j,i}(v_i)W_iG_i$. The derivative of $F_{j,m}(v_m)$ is zero because $\eta_m=0$.

We now bound the Gaussian norm of $B_{j,i}G_i$ by deterministic rows at the base point. Let $H=\ell_{j-1}$, $G=\ell_i=B^kH$ with $k=j-1-i$, and $K_{ji}=\mathfrak{q}^k/G$. For the form T^{UV} the two terms in the derivative of the source are

$$\mathcal{N}_a(\mathsf{U}W_iG_i,\mathsf{V}v_i)+\mathcal{N}_a(\mathsf{U}v_i,\mathsf{V}W_iG_i),$$

where $\mathsf{U} = \mathsf{U}_{j,i}$, $\mathsf{V} = \mathsf{V}_{j,i}$, and $\mathcal{N}_a$ is the quadratic source of the Duhamel term. For T^{VU} we interchange U and V, and for T^{VV} we put V in both arguments of these two terms. Thus each of the three derivatives has one differentiated argument and one deterministic argument bounded by b_i^{fl}. By the product inequalities for rows and volume-normalized columns in Lemma 27.1, followed by the bounds on the time-indexed rows of heat-kernel coefficients in Theorem 27.5, we have

$$\begin{aligned}
\sup_x \|B_{j,i}(t,x,\cdot)\|_2 &\le C K_{ji} d_i b_i^{\mathrm{fl}} \log(1 + t/H^2), \qquad t \le G^2,\\
\sup_x \|D_a B_{j,i}(t,x,\cdot)\|_2 &\le C K_{ji} d_i b_i^{\mathrm{fl}} J_H(t), \qquad t \le G^2,\\
\sup_x \|D_a^{\iota} B_{j,i}(t,x,\cdot)\|_2 &\le C K_{ji} d_i b_i^{\mathrm{fl}} G^{3/2} t^{-3/4-\iota/2} \ell_H(T), \qquad t \ge G^2,\ \iota = 0,1.
\end{aligned}$$

Here J_H and ℓ_H are the functions defined in Theorem 27.5. The conversion of the Schur row-column estimate with volume-normalized columns uses the input volume G^3 and the output volume a^3; its factor $(G/a)^{3/2}$ cancels the factor $a^{3/2}$ of the row of heat-kernel coefficients and leaves $G^{3/2}$. The undifferentiated argument is evaluated at v_i, and only its deterministic size enters this calculation. We do not use b_i^{fl} as a bound on a conditional fiber with nonzero fluctuation variables, which is why no term $d_i p_i$ appears. The Jacobian at zero fluctuation has been substituted in the differentiated argument, with $\mathfrak{g}_i$ replaced by d_i through the diagonal matrix D_i; all other heat and signed-column operations are unchanged.

Taking the two weighted maxima of these early and late row bounds gives the scale $C(1+j)^c K_{ji} d_i b_i^{\mathrm{fl}} G^{2\alpha-1}$. The crude space and time derivatives of the coefficients have the amplitude $C_{\mathrm{CP}}^{2k} d_i b_i^{\mathrm{fl}}/G^2$ and cost fixed powers of H^{-1}. Their ratio to the weighted amplitude is bounded as in (34.21), with the same factor $(q_{\mathrm{anc}}/\mathfrak{q})^k \le 1$. Therefore their normalized coefficient metrics have bounded diameter and polynomial covering numbers at the input mesh, including at time zero, and Lemma 34.5 proves, for each ordered form separately,

$$\|\mathsf{Y}_\alpha^a(B_{j,i}G_i)\|_2 \le C r (1+j)^c \mathfrak{q}^{j-1-i} \ell_i^{\kappa+2\alpha-3/2}. \tag{34.56}$$

For the quadratic initial form we use the rows of Theorem 27.7 with respect to the fluctuation variables, with the same differentiated and deterministic arguments. Its value row is at most

$$C K_{ji} d_i b_i^{\mathrm{fl}} \min\bigl\{1, G^{3/2}(t+G^2)^{-3/4}\bigr\}.$$

After the weight with exponent $\sigma/2$ its scale is $C K_{ji} d_i b_i^{\mathrm{fl}} G^{\sigma}$. Its crude parameter derivatives keep the same factor $K_{ji} d_i b_i^{\mathrm{fl}}$, and division gives a fixed power of ℓ_j^{-1} and the cover at the input mesh. Lemma 34.5 for the value process, followed by the deterministic half-splitting of the heat semigroup for its linear initial field, proves

$$\|\mathsf{H}_\sigma^a(B_{j,i}G_i)\|_2 \le C r (1+j)^c \mathfrak{q}^{j-1-i} \ell_i^{\kappa+\sigma-1/2}. \tag{34.57}$$

The half-splitting above a^2 and the bound by the initial value below a^2 are the two deterministic inequalities of Step 3 of the proof of Theorem 34.7, and this bound includes the first difference on every finer lattice.

Finally, (34.54) and (34.47) imply that, for both types of output,

$$\big\| \|DC_{j,i}(v_{i-1})[\eta_{i-1}]\|_{\mathrm{out}} \big\|_2 \le Cr^2(1+j)^c \mathfrak{q}^{j-1-i} \ell_i^{2\kappa-1/2}. \tag{34.58}$$

The part of degree at least three of the predictable term at level j satisfies the same bound with its own coefficient, by (34.12). There is no conditional-mean remainder in (34.55).

Step 5: summation of both components. Let $s_0 > 0$ be strictly smaller than each of

$$\alpha - \tfrac{1}{2}, \quad \kappa + \sigma - \tfrac{1}{2}, \quad \kappa + 2\alpha - \tfrac{3}{2}, \quad 2\kappa - \tfrac{1}{2}, \quad -\frac{\log \mathfrak{q}}{\log B}.$$

All these numbers are positive, and we reduce s_0 further if a fixed exponent of the product estimates requires it. For the predictable term, the initial value of its first-order variation is exactly

$$v_j(0) = DS_{j,j-1}(v_{j-1}, v_{j-1})[\eta_{j-1}] + I_{\ell_j \to a} DC_j^{\mathrm{loc}}(v_{j-1})[\eta_{j-1}],$$

where C_j^{loc} is the part of degree at least three of U_j^{cl} at its input mesh, so that $I_{\ell_j \to a} C_j^{\mathrm{loc}} = V_j$ in the notation of Lemma 34.2. The first term is given by the identity (34.55) for the quadratic initial form and the second by the bound on the part of degree at least three, with no conditional mean added to either. Summing both in the norm of the initial value, using (34.57), (34.58), and geometric summation, we obtain

$$\sum_{j=m+1}^{K} \|\mathsf{H}_\sigma^a(v_j(0))\|_2 \le C(1+m)^c B^{-s_0 m}. \tag{34.59}$$

Combining this with (34.50), we obtain

$$\Big\| \sup_{m \le l \le K} \mathsf{X}_\alpha^a(V_l) \Big\|_2 + \sum_{j=m+1}^{K} \big[\|\mathsf{X}_\alpha^a(\xi_j)\|_2 + \|\mathsf{X}_\alpha^a(\rho_j)\|_2 + \|\mathsf{X}_\alpha^a(v_j)\|_2 \big] \le C(1+m)^c B^{-s_0 m}. \tag{34.60}$$

For the ordered forms, (34.55), (34.56), and (34.58) imply the same summed tail, separately for each of the three forms. These estimates use the Minkowski inequality, not independence between an earlier field and a later coefficient.

We now treat the products, following the fourth column of Table 34.2, for the uncentered summands of Step 2. Their values at zero fluctuation are $(0, 0, R_j, U_j)$ and their first variations are $(\xi_j, 0, \rho_j, v_j)$. Thus the product $(\tilde{s}, \tilde{s})$ and all seven products containing $\tilde{z}$ vanish at first order. The two products $(\tilde{s}, \tilde{r})$ and $(\tilde{r}, \tilde{s})$ are exactly (34.53). The product $(\tilde{r}, \tilde{r})$ and the two cross terms with $\tilde{r}$ are the derivatives of T^{VV}, $T^{\mathsf{UV}} - T^{\mathsf{VV}}$, and

$T^{\mathsf{VU}} - T^{\mathsf{VV}}$, which have the separate tails just proved. For the seven products containing $\tilde{u}$, each nonzero derivative has one linear factor and one deterministic classical factor. For example, the derivative of $\mathcal{Q}_a(\tilde{r}, \tilde{u})$ is $\mathcal{Q}_a(\rho, U) + \mathcal{Q}_a(R, v)$, and the reversed order reverses both brackets; the derivative of $\mathcal{Q}_a(\tilde{u}, \tilde{u})$ is $\mathcal{Q}_a(v, U) + \mathcal{Q}_a(U, v)$; the derivatives for $(\tilde{s}, \tilde{u})$ and $(\tilde{u}, \tilde{s})$ pair ξ with U; and those for $(\tilde{z}, \tilde{u})$ and $(\tilde{u}, \tilde{z})$ vanish, as already counted among the products with $\tilde{z}$. This accounts for all sixteen products.

We use the two ordered inequalities (34.6). The deterministic norms of X_j and R_j are bounded by (34.45) and (34.11), and the linear predictable terms are summed by (34.59). For the factors paired with a deterministic predictable term, the series of the zero-fluctuation maps gives the sharper deterministic sum

$$\sum_{j=m+1}^{K} \mathsf{H}^a_\sigma(U_j(0)) \le C \sum_{j>m} \frac{r^2 \varrho^{2(j-1)}}{\ell_{j-1}} \le C \frac{r^2 \varrho^{2m}}{\ell_m} = C r^2 H_0^{-2\kappa} \ell_m^{2\kappa-1}. \tag{34.61}$$

The ratio in this sum is $B\varrho^2 = \mathfrak{q} < 1$, and there is no term $d_m p_m$ at this base point. Multiplication by (34.60) bounds all the remaining products. It also bounds both orders of the cross terms with the predictable term,

$$\mathcal{Q}_a(V_{j-1}, U_j) + \mathcal{Q}_a(X_{j-1}, v_j) \qquad \text{and} \qquad \mathcal{Q}_a(U_j, V_{j-1}) + \mathcal{Q}_a(v_j, X_{j-1}).$$

The two cross terms with s are (34.52), and the two cross terms with z vanish. This covers both orders of every cross term. The constant in (34.6) is valid at every mesh $a = L/N$, including the finitely many coarse meshes at fixed L; no condition $a \le 1$ is needed.

Differentiating the finite telescope (34.5) and summing these bounds geometrically, we obtain, with $s < s_0$ if a polynomial factor is absorbed,

$$\big\| \mathsf{X}^a_\alpha(V_K) + \mathsf{Y}^a_\alpha(Z_K) \big\|_{L^2(G)} \le C(1+m)^c B^{-sm}. \tag{34.62}$$

This proves the asserted bound on the first-order variation of the flow data, before the flow is applied.

Step 6: the derivative of the flow. We check the hypotheses of Theorem 25.13 without assuming a uniform bound on second derivatives for an unbounded Gaussian family. We fix a deterministic finite vector of inputs of the steps after m, let η_K be the solution of (34.46) with G replaced by this vector, and let $v = I_{a\to a}(\eta_K/a)$ be the corresponding direction of the initial potential. Let $V = P^a_\cdot v$ and $Z = \mathcal{Q}_a(X_K, V) + \mathcal{Q}_a(V, X_K)$; these are V_K and Z_K when the deterministic vector is subsequently replaced by G. We write $\mathscr{H}_a(A_0)$ for the based cycle triple of the Wilson flow started from $\exp(aA_0)$. By the chain rule, the derivative of $\mathcal{H}_K(y, q)$ in these directions is $D\mathscr{H}_a(v_K/a)[\eta_K/a]$.

At a fixed cutoff, we consider the family of initial potentials $v_K/a + \varsigma\tau v$, where $\varsigma \in (-2, 2)$ and $\tau > 0$ is deterministic. Its flow data are

$$\big(X_K + \varsigma\tau V,\ Y_K + \varsigma\tau Z + \varsigma^2\tau^2 \mathcal{Q}_a(V, V)\big).$$

We choose τ so that $\tau[\mathsf{X}_\alpha^a(V)+\mathsf{Y}_\alpha^a(Z)]\le 1/16$, $\tau^2\mathsf{Y}_\alpha^a(\mathcal{Q}_a(V,V))\le 1/16$, and the perturbation of the initial link variables stays within one quarter of the fixed margin of the logarithmic chart. Such a $\tau>0$ exists for each deterministic direction, and the zero direction needs no restriction. The flow data and their first two derivatives in ς are then bounded by $R_2=R_{\mathrm{cl}}+C$ on $U_1=(-2,2)$, uniformly in the mesh, and we take $U_0=(-1,1)$. The parameter space has dimension one, the parameter domains and chart margins are fixed, the loops and the positive observation time are the given ones, and all damping gauges start at the identity in the same frame. These are the hypotheses of Theorem 25.13. Its restart at a positive time uses the local chart and the C_a^1 bound already proved, and its fixed restarts include the conjugations at the basepoint. Thus its real differential is the differential of the based holonomy matrices of the Wilson flow used here.

To obtain a bound independent of the rescaling τ, we apply Remark 34.8 on the fixed ball of radius $R_{\mathrm{cl}}+1$ to the configurations of the family at ς and at 0, divide by $|\varsigma|\tau$, and let $\varsigma\to 0$. The difference of the flow data divided by $|\varsigma|\tau$ tends in both norms to (V,Z); the remaining term is bounded by $|\varsigma|\tau\mathsf{Y}_\alpha^a(\mathcal{Q}_a(V,V))$ and tends to zero at this fixed cutoff. We obtain

$$\max_i\big|D\mathscr{H}_{a,i}(v_K/a)[\eta_K/a]\big|\le C\big\{\mathsf{X}_\alpha^a(V_K)+\mathsf{Y}_\alpha^a(Z_K)\big\}. \tag{34.63}$$

The constant depends on R_{cl} and the fixed time and reference data, not on the rescaling, the direction, the cutoff, or m. The finitely many coarse meshes are included in both results on the flow. Differentiating the reconstruction instead of the straight line of initial potentials gives the same first derivative by the chain rule, and higher derivatives at the initial time do not enter (34.63).

Step 7: the logarithm and the conclusion. At the base point the cycle holonomies are exactly $\mathrm{Hol}(V_y)$. The compact regular patch has the margin for the logarithm of Theorem 33.11, and the differential of Log is uniformly bounded there. In the notation of that theorem, its differential on the right-trivialized variation $\delta H_i H_i^{-1}$ is J_i^{-1}, with $J_i=\int_0^1 e^{s\vartheta_i\,\mathrm{ad}_u}\,ds$, where $u\in S^2$ is the color axis of the background. The projection $P_{\mathcal{N}_y}$ has norm at most one. Composing (34.63) with this differential and this projection, we obtain the pointwise bound

$$\Big|\sum_{j=m+1}^{K}\mathsf{A}_{j,K}(y)G_j\Big|\le C\big\{\mathsf{X}_\alpha^a(V_K)+\mathsf{Y}_\alpha^a(Z_K)\big\}.$$

By the definition of the diagonal matrices D_j in (34.46), these are exactly the blocks of the statement. Equivalently, the chain rule gives $D_{w_j}(\mathrm{Log}\circ\mathcal{H}_K)=D_{q_j}(\mathrm{Log}\circ\mathcal{H}_K)\,\mathrm{diag}_{r,e}(g_{j,r})$, and multiplication by $\gamma^{-1/2}P_{\mathcal{N}_y}$ gives exactly the first expression for $\mathsf{A}_{j,K}$ in the statement. Squaring, taking the Gaussian expectation, and using (34.62), we obtain the claim with $\epsilon=2s$, after relabeling the polynomial exponent.

For comparison with the covariance notation of Section 35, independence and $\mathbb{E} G_j G_l^{\mathsf{T}} = \mathbf{1}_{j=l} I$ give exactly

$$\mathbb{E}_G \Big| \sum_{j=m+1}^{K} \mathsf{A}_{j,K} G_j \Big|^2 = \sum_{j=m+1}^{K} \operatorname{tr}\big(\mathsf{A}_{j,K} \mathsf{A}_{j,K}^{\mathsf{T}}\big).$$

The transpose here is the Euclidean transpose in the counting coordinates of the fluctuation variables, not an adjoint with respect to physical volumes; the physical volumes are already contained in the deterministic rows. Every bound above is uniform over the finite cover of the compact patch, the prior depths, K, m, and the coupling. This completes the proof. □

34.8 Flat support of weak-coupling subsequential limits

In this section, we prove that the continuum measures concentrate on flat connections as $\gamma \to 0$. We fix the data of Definition 34.1 before varying $0 < \gamma \le \gamma_0 \le 1$; the hierarchy is the one fixed by Theorem 23.2. Here we choose the common input horizon $T = T_0 \ge \max\{1, L^2, \theta L^2\}$, and all flow-data norms and the preceding comparison estimates are evaluated on $[0, T_0]$. Later observation times need not lie in this interval: they are reached by the monotonicity of the energy after a short initial comparison. Let $n_* \in 4\mathbb{N}$ define the lattice refinement sequence $n_* B^K$, $K \ge 0$. We use a comparison side $n = n_* B^{k_0}$ of this sequence which satisfies the minimum-side condition, the condition on flat fields, and the conditions on the working radius above; in particular $2\pi/n < r_{\mathrm{flat}}/4$. The sequence, the hierarchy, the side n, and all the bounds are fixed before the coupling and do not change as $\gamma \to 0$. We write $N_K = nB^K$, $a_K = L/N_K$, and $\nu_{K,\gamma} = \nu_{N_K,n,\gamma}$. The law of the coarse field $V = V^{JK}$ under $\nu_{K,\gamma}$ is the stopped measure $\widehat{\nu}_{n,JK,\gamma}$, by item (9) of Theorem 14.8. Let $\lambda_\gamma = \lambda_{L,\gamma,[n_*]}$ be the continuum measure of Corollary 29.26. Constants have the dependence described in items (6) and (7) of Definition 4.14.

Theorem 34.13. *Fix $L > 0$, the hierarchy, $n_* \in 4\mathbb{N}$, and a ceiling $0 < \gamma_0 \le 1$, all before varying the coupling, and consider the lattice refinement sequence $n_* B^K$. Comparison sides $n = n_* B^{k_0}$ are required to satisfy the conditions of Definition 34.1. Let $\lambda_\gamma = \lambda_{L,\gamma,[n_*]}$ be the continuum measures of Corollary 29.26. Then the following statements hold.*

1. *For every $t > 0$ and $\delta > 0$,*

$$\lambda_\gamma\{X : E_L(X(t)) > \delta\} \longrightarrow 0 \qquad (\gamma \to 0). \tag{34.64}$$

   More precisely, for every comparison side n as above, fixed before the coupling, there exist $\gamma_1 > 0$ and $C_{n,t,\delta} < \infty$ such that this probability is at most $C_{n,t,\delta}(\sqrt{\gamma} + \gamma)$ for $0 < \gamma \le \gamma_1$.
2. *For every $\eta > 0$ there exist finite $R_j \ge 0$ such that*

$$\inf_{0<\gamma\le\gamma_0} \lambda_\gamma\big\{X : E_L(X(L^2 2^{-j})) \le R_j \text{ for all } j \ge 1\big\} \ge 1 - \eta.$$

The set in this display is compact for $\mathcal{T}_W$ and for d_c. Every sequence $\gamma_k \to 0$ has a subsequence along which λ_{γ_k} converges, against all bounded $\mathcal{T}_W$-continuous functions and weakly for d_c, to the same Borel probability measure. Along this subsequence the images of λ_{γ_k} in the Sobolev trajectory space of Definition 29.1 also converge against every bounded $\mathcal{T}_{1,2}$-continuous function.

3. *Every subsequential limit, in either of the topologies of item (2), is supported on the set*

$$\mathcal{F}_L^{\mathrm{stat}} = \big\{X : X(t) = [A] \text{ for all } t > 0,\ F_A = 0\big\}$$

of stationary flat trajectories. This set is compact, and the cycle holonomies identify it with

$$\mathcal{F}_L \simeq \mathrm{Hom}(\mathbb{Z}^3, \mathrm{SU}(2))/\mathrm{SU}(2) \simeq (\mathbb{R}/4\pi\mathbb{Z})^3/\{\boldsymbol{\vartheta} \sim -\boldsymbol{\vartheta}\},$$

including the eight central triples. Every subsequential limit is the image of a Borel probability measure on this compact space.

The constants in item (1) have the dependence described in item (6) of Definition 4.14, with the displayed additional parameters, and the ultraviolet limit has been taken at each fixed coupling before $\gamma \to 0$. The theorem does not identify the probability measure on the moduli space: it asserts neither uniqueness of the subsequential limit nor a density on the moduli space, and it makes no assertion about unbounded moments of the energy or about ultraviolet convergence uniform down to zero coupling.

The proof combines four facts. The effective classical action of the coarse field has moments of order γ. The zero-fluctuation reconstruction from the coarse field has lattice energy exactly proportional to this action. The flow data of the reconstruction differ from those of the zero-fluctuation reconstruction by $O(\sqrt{\gamma})$ in every moment. Finally, a short-time comparison of the two Wilson flows, followed by the monotonicity of the energy, transfers this to the energy at positive times. We first prove two lemmas.

The coarse effective action and the zero-fluctuation reconstruction

Lemma 34.14. *Use the data above and the families of coarse densities of Theorem 33.8. On the support of the stopped measure write $E_d(V) = E_{d,\mathrm{sat}}(V)$, and put $g^2 = \gamma/n$ and $D_n = 6n^3 - 2$. There exists $\gamma_1 \in (0, \gamma_0]$, independent of the depth, such that for every fixed real $p > 0$ and every depth d allowed in Theorem 33.8,*

$$\mathbb{E}_{\widehat{\nu}_{n,d,\gamma}} E_d^p \le C_{n,p} g^{2p}, \qquad \mathbb{E}_{\widehat{\nu}_{n,d,\gamma}} E_0^p \le C_{n,p} g^{2p} \qquad (0 < \gamma \le \gamma_1),$$

where $E_0 = S_W$ is the Wilson action of Definition 4.1 on the lattice of side n. At $d = JK$, start from the same representative $\widehat{\theta}_0(V)$ of the coarse field as the reconstruction, and set $\widehat{\theta}_j^0 = P_j \cdots P_1 \widehat{\theta}_0(V)$. Then the configuration $U_K^0(V) = \exp \widehat{\theta}_K^0$ on the finest lattice satisfies

$$E_0(U_K^0(V)) = B^{-K} E_{JK}(V), \qquad \mathsf{E}_{a_K}(U_K^0(V)) = \frac{2n}{L} E_{JK}(V), \tag{34.65}$$

where on the left E_0 is the Wilson action on the finest lattice and E_{a_K} is the lattice energy (2.2)*. In particular, the expected initial lattice energy of $U_K^0(V)$ under the stopped measure is at most $C_n\gamma$, and the expectation of its p-th power is at most $C_{n,p}\gamma^p$.*

The constants depend in addition on p and on the fixed bounds on the coarse density, and they are independent of the coupling and of the depth. The identities (34.65) are exact for every finite K, including $K = 0$, with the prior depths $J(K-j)$ ending at depth zero.

Proof. By Theorem 33.8, the stopped measure has the density

$$\widehat{Z}^{-1}\mathbf{1}_{D_r}e^{-E_d/g^2-\overline{R}_{d,\gamma}}\,dH_n, \qquad E_d \geq \kappa E_0 \geq 0, \quad |\overline{R}_{d,\gamma}| \leq C_n,$$

where $\kappa > 0$ is the constant of Theorem 33.8; in this proof κ denotes this constant, not the contraction exponent of Definition 34.1. These bounds apply to the law of the coarse field under the conditioned measure itself, not to an interior stopped measure in its place. The lower bound of Theorem 33.10 is $\widehat{Z} \geq c_n g^{D_n}$ for $g \leq g_0(n)$. We choose $\gamma_1 \leq \min\{\gamma_0, ng_0(n)^2\}$, so that also $g \leq 1$. Since $u^p e^{-u} \leq C_p e^{-u/2}$ for $u \geq 0$, we have

$$\begin{aligned}\mathbb{E}_{\widehat{\nu}}E_d^p &\leq C_{n,p}g^{2p-D_n}\int_{D_r} e^{-E_d/(2g^2)}\,dH_n \\ &\leq C_{n,p}g^{2p-D_n}\int_{\mathcal{U}_n} e^{-\kappa E_0/(2g^2)}\,dH_n \leq C_{n,p}g^{2p}.\end{aligned}$$

In the first inequality we used the bound on $\overline{R}_{d,\gamma}$ and the lower bound on $\widehat{Z}$. In the last inequality we used the upper bound for the integral over the compact configuration space in Theorem 33.10, with the coefficient $\kappa/2$. The lower bound $E_d \geq \kappa E_0$ then implies the bound for E_0^p. In particular, the stopped measure concentrates near the set of all flat coarse fields: outside any fixed open neighborhood of $\{E_0 = 0\}$ the continuous function E_0 has a positive minimum, and Markov's inequality applies.

We now check that the configuration in the identity is the zero-fluctuation continuation of Definition 34.1. At level j the prior depth is $d_j = J(K-j)$. By Theorems 15.12 and 15.13, P_j is the constrained minimization of the renormalization group step followed by its based gauge transformation, which depends only on the coarse field. It preserves the gauge orbit of the minimizer and the value of the effective classical action. Iterating the nested minimization identity $E_{d+1}(c) = M\min_z E_d(b(c,z))$ of Theorem 10.10 through the J averaging steps of the renormalization group step, we obtain, at every zero-fluctuation input in the reconstruction domain,

$$E_{d_j}\big(e^{\widehat{\theta}_j^0}\big) = B^{-1}E_{d_j+J}\big(e^{\widehat{\theta}_{j-1}^0}\big).$$

For the chosen representative of the coarse field, the initial value on the right is $E_{JK,\mathrm{sat}}(V)$; the extension $E_{JK,\mathrm{sat}}$ agrees with E_{JK} on the support and does not change it. All changes

of representative are compact gauge transformations, which preserve this value. Iterating over $j = 1, \dots, K$ ends at $d_K = 0$ and proves the first identity. The continuation sets the fluctuation variables w to zero, not the centered variables $w - \mathbb{E}(w \mid \mathcal{F})$, and no conditional mean is substituted for the minimizer. Finally, by Lemma 29.14 we have $\mathsf{E}_{a_K} = 2E_0/a_K$. Since $a_K = (L/n)B^{-K}$, this proves the second identity, and the moment bounds follow from the first part of the lemma and $g^2 = \gamma/n$. The identity also covers $K = 0$, without iteration. □

Early-time comparison and energy monotonicity

Lemma 34.15. *Fix $L, R > 0$, $1/2 < \alpha < 2/3$, $T_0 \geq 1$, and $\tau > 0$. There exist $a_* > 0$ and $C < \infty$ with the following property. Let $a = L/N \leq a_*$, and let $U = e^{aA_0}$ and $U' = e^{aA'_0}$ be configurations on the lattice of mesh a, represented in the same initial frame, whose flow data $\mathcal{E}_a = (x_a, y_a)$ and $\mathcal{E}'_a = (x'_a, y'_a)$ as in Definition 24.3 have norms at most R on $[0, T_0]$. Put*

$$d_a = \mathsf{X}^a_\alpha(x_a - x'_a) + \mathsf{Y}^a_\alpha(y_a - y'_a),$$

with both full norms on this lattice and this horizon. Then, for the Wilson flow run for the lattice flow time $4t/a^2$,

$$\sup_{t \geq \tau} \mathsf{E}_a\big(\Psi^{(N)}_t U\big) \leq \mathsf{E}_a(U') + C d_a, \qquad \sup_{t \geq \tau} \mathsf{E}_a\big(\Psi^{(N)}_t U\big) \leq C.$$

The second bound requires only the bound on the norm of $\mathcal{E}_a$.

The constants have the dependence described in item (7) of Definition 4.14 on the displayed parameters and on the fixed chart and stencil; they are independent of the coupling, of any probability measure, of the depth, and of the configurations.

Proof. Let $T_R \leq 1$ be the lifetime in Theorem 24.5, and choose $s = \min\{\tau, T_R\}/2 > 0$. Applied to the two flow data, that theorem yields local damped representatives A_a and A'_a with

$$\|A_a(s) - A'_a(s)\|_{C^1_a} \leq C_{R,s} d_a, \qquad \|A_a(s)\|_{C^1_a} + \|A'_a(s)\|_{C^1_a} \leq K_{R,s}.$$

The second bound follows from $A = x + y + r$, the two weighted norms, and the bound on the remainder r in that theorem at the fixed positive time s. Its constants and its restriction on the mesh are independent of the coupling. The exponentials of $A_a(s)$ and $A'_a(s)$ are gauge representatives of the two Wilson flows at time s started from U and U'. By gauge invariance and the C^1_a estimate for the lattice energy in Lemma 29.14,

$$\mathsf{E}_a\big(\Psi^{(N)}_s U\big) \leq \mathsf{E}_a\big(\Psi^{(N)}_s U'\big) + C d_a \leq \mathsf{E}_a(U') + C d_a,$$

where in the last inequality we used that the energy does not increase along the flow. The same monotonicity from s to every $t \geq \tau$ proves the first assertion. The uniform bound on the energy on a bounded C^1_a ball in Lemma 29.14 gives $\mathsf{E}_a(\Psi^{(N)}_s U) \leq C$, and monotonicity proves the second assertion. □

Proof of the support theorem

Proof of Theorem 34.13. Step 1: the coarse action and the zero-fluctuation reconstruction. By Lemma 34.14, for $0 < \gamma \le \gamma_1$ the expectation of $(2n/L)E_{JK}(V)$ under $\nu_{K,\gamma}$ is at most $C_n\gamma$, uniformly in K, and this quantity is exactly the initial lattice energy of the zero-fluctuation reconstruction $U_K^0(V)$.

Step 2: the distance between the flow data. We take $m = 0$ in Theorem 34.9 and denote by

$$D_K = \mathsf{X}_\alpha^{a_K}(x_K - x_K^0) + \mathsf{Y}_\alpha^{a_K}(y_K - y_K^0)$$

the distance between the flow data of the reconstruction and of the zero-fluctuation continuation from the coarse field. By that theorem, for every fixed $p \ge 1$ we have

$$\|D_K\|_{L^p(\nu_{K,\gamma})} \le C_{n,p}\big\{\sqrt{\gamma} + \gamma(1 + |\log\gamma|)^d\big\} \le C'_{n,p}\sqrt{\gamma},$$

uniformly in K and in the coupling; the last inequality uses the bound on $\sqrt{\gamma}(1+|\log\gamma|)^d$ from the proof of Lemma 34.6. By Lemma 34.4, the norm of the flow data of the continuation is at most R_{cl}, deterministically for every coarse field in the reconstruction domain and independently of K and γ.

Step 3: comparison of energies. On the event $G_K = \{D_K \le 1\}$ both flow data have norm at most $R_{\mathrm{cl}} + 1$. The two configurations to which we apply Lemma 34.15 are $\exp\widehat{\theta}_K$ and $U_K^0(V)$, in the same initial frame on the finest lattice. Indeed $I_{a_K \to a_K}$ is the identity in Definition 34.1, and both pairs consist of the linear heat evolution and the quadratic Picard term of these fine arrays. The zero-fluctuation configuration is the one in (34.65), with the same representative $\widehat{\theta}_0(V)$ of the coarse field; it is neither a flow started on the lattice of side n nor a separately chosen flat approximation. For every K large enough that $a_K \le a_*$, a threshold independent of the coupling, Lemma 34.15 and (34.65) imply that

$$\mathsf{E}_{a_K}\big(\Psi_t^{(N_K)}U\big) \le \frac{2n}{L}E_{JK}(V) + C_{n,t}D_K \qquad \text{on } G_K.$$

The configuration $\exp\widehat{\theta}_K$ is a gauge transform of U by Theorem 15.13, and the inequality concerns the gauge-invariant lattice energy of U. By Step 1, Step 2, and Markov's inequality,

$$\nu_{K,\gamma}\big\{\mathsf{E}_{a_K}\big(\Psi_t^{(N_K)}U\big) > \delta\big\} \le \nu_{K,\gamma}(G_K^c) + \delta^{-1}\mathbb{E}_{\nu_{K,\gamma}}\Big[\frac{2n}{L}E_{JK}(V) + C_{n,t}D_K\Big] \le C_{n,t,\delta}(\sqrt{\gamma} + \gamma).$$

Only the coarse action and the distance between flow data were integrated in this estimate; no unbounded quantity was transferred from the Wilson measure.

Step 4: transfer to the Wilson measure and to the continuum. For a Borel event A, the definition of the conditioned measure gives $\mu_{N_K,\gamma}(A) \le \nu_{K,\gamma}(A) + \mu_{N_K,\gamma}(S^c_{N_K,n,\gamma})$. By Corollary 23.11 with $q = 1$, at this fixed n, the last probability is at most $C_n\gamma$. We apply

this to the event in Step 3. For the ultraviolet limit we use the bounded continuous function

$$f_\delta(v) = \min\{1, \max\{0, 2v/\delta - 1\}\}, \qquad v \geq 0,$$

which equals one for $v \geq \delta$ and vanishes for $v \leq \delta/2$. By Theorem 30.4, at each fixed γ the lattice energy at time t converges in distribution to $E_L(X(t))$ under λ_γ, and the sequences based at n and at n_* have the same limit by Corollary 29.26. Hence

$$\lambda_\gamma\{E_L(X(t)) > \delta\} \leq \mathbb{E}_{\lambda_\gamma} f_\delta(E_L(X(t))) = \lim_{K\to\infty} \mathbb{E}_{\mu_{N_K,\gamma}} f_\delta\big(\mathsf{E}_{a_K}(\Psi_t^{(N_K)} U)\big) \leq C_{n,t,\delta}(\sqrt{\gamma} + \gamma),$$

where in the last inequality we used Step 3 with $\delta/2$ in place of δ and the transfer just described. The limit in the middle is taken at fixed γ, after the uniform estimates at finite K. This proves item (1).

Step 5: common compact containment. We argue directly on the whole interval of small couplings. For each fixed comparison side n, Step 2 implies that

$$\lim_{R\to\infty} \sup_{K\geq 0,\, 0<\gamma\leq\gamma_0} \nu_{K,\gamma}\big\{\|(x_K, y_K)\|_{\mathrm{fl}} > R\big\} = 0,$$

with constants that may depend on n. The second assertion of Lemma 34.15, applied on the event where this norm is at most R, bounds the lattice energy at each fixed $t > 0$ by a finite constant on all sufficiently fine lattices. The finitely many remaining lattices have a deterministic finite maximum of the energy on their compact configuration spaces. Therefore, for each fixed n,

$$\lim_{H\to\infty} \sup_{K\geq 0,\, 0<\gamma\leq\gamma_0} \nu_{K,\gamma}\big\{\mathsf{E}_{a_K}\big(\Psi_t^{(N_K)} U\big) > H\big\} = 0.$$

The comparison side may now be enlarged to reach a prescribed tolerance, always before the coupling is chosen. By Theorem 23.9 and Corollary 23.11,

$$\epsilon_n = \sup_{K\geq 0,\, 0<\gamma\leq\gamma_0} \mu_{nB^K,\gamma}\big(S^c_{nB^K,n,\gamma}\big) \longrightarrow 0 \qquad (n = n_* B^{k_0},\ k_0 \to \infty).$$

Indeed, the bound of Theorem 23.9 is the sum of $C' n^3 e^{-\bar{c} n/\gamma}$ and $C_{\mathrm{w}} n^3 e^{-a_{\mathrm{w}} \log^2 \sqrt{Mn/\gamma}}$; both terms are largest at $\gamma = \gamma_0$ and tend to zero as $n \to \infty$. We choose n so that ϵ_n is small, and then the radius and the energy threshold for the conditioned measure. The transfer of bounded events gives uniform tails of the lattice energy under the Wilson measure along the sequence nB^K, and the finitely many initial terms of the sequence $n_* B^K$ below n have deterministic finite energy bounds as well. Thus

$$\lim_{H\to\infty} \sup_{K\geq 0,\, 0<\gamma\leq\gamma_0} \mu_{n_* B^K,\gamma}\big\{\mathsf{E}_{L/(n_* B^K)}\big(\Psi_t^{(n_* B^K)} U\big) > H\big\} = 0.$$

For each γ separately, the convergence of the energy at fixed coupling used in Step 4 and the portmanteau inequality for the open set $\{v > H\}$ give the same uniform tails for $\lambda_\gamma\{E_L(X(t)) > H\}$. No ultraviolet limit uniform in the coupling was used.

We now apply Theorem 29.6. We choose R_j such that the energy tail at time $s_j = L^2 2^{-j}$ is smaller than $\eta 2^{-j-1}$, uniformly in the coupling, and use the union bound. The compact set $\mathcal{K}(R)$ of that theorem, with $R = (R_j)_j$, gives the asserted common containment and the extraction of subsequences for $\mathcal{T}_W$ and for d_c. This proves item (2), except for the Sobolev images, which we treat in the next step.

Step 6: identification of the support. Let λ_* be a subsequential limit obtained from this extraction. All finite joint laws of regularized Wilson loops converge along the subsequence, since these coordinates are bounded and $\mathcal{T}_W$-continuous. The general sequence clause of Theorem 30.5 applies: the energies at every positive time are tight, as just proved, and the joint limit of the Wilson loops is given by the Borel probability measure λ_*. That clause gives convergence of the Sobolev images to $(i^{1,2})_\# \lambda_*$. By Lemma 29.7, for any $s, \delta > 0$ the function

$$Z \longmapsto \min\{1, E_L(Z(s))/\delta\}$$

is bounded and continuous in the Sobolev trajectory topology. Its expectation under λ_γ tends to zero by item (1): for every $\varepsilon > 0$, it is at most $\varepsilon/\delta + \lambda_\gamma\{E_L(X(s)) > \varepsilon\}$. Its expectation under the limit is therefore zero. At each time s_j we obtain $E_L(X(s_j)) = 0$, λ_*-almost surely. We take the countable intersection of these events of full measure. Since the energy does not increase along the flow (Theorem 29.6) and $s_j \downarrow 0$, we have $E_L(X(t)) = 0$ for every $t > 0$ on this event.

Since the energy is the sum of the squared norms of the curvature components, every smooth orbit at these times is flat. For a flat smooth connection the right side of the Yang–Mills heat equation vanishes, and the uniqueness of smooth solutions in Theorem 24.10 makes its future orbit constant. The trajectory relation then makes X a single stationary flat trajectory on $(0, \infty)$.

This proves the support property for every subsequential limit for $\mathcal{T}_W$. If a subsequence converges only weakly for d_c, we apply the common compact containment again to extract a further subsequence that converges for $\mathcal{T}_W$. Since the identity map from $\mathcal{T}_W$ to d_c is continuous by Theorem 29.3, the limit for d_c coincides with the limit for $\mathcal{T}_W$; equivalently, their finite-dimensional laws of the determining coordinates agree, and the same theorem identifies the Borel probability measures. The limit for d_c is therefore supported on stationary flat trajectories as well.

Step 7: the moduli space. We include the classification of flat connections on the torus; it is standard, and we check its normalization. A smooth flat connection lifted to the universal cover $\mathbb{R}^3$ has a path-independent transport $P_A(x)$ from a fixed basepoint, with $dP_A = P_A A$ and $P_A(x + Le_i) = h_i P_A(x)$. Flatness makes the three matrices h_i commute.

Conversely, a commuting triple in $\mathrm{SU}(2)$ is simultaneously diagonalizable, and it is realized by the constant flat connection $A_i = \vartheta_i \tau_3 / L$ with $h_i = e^{\vartheta_i \tau_3}$. If two flat connections A and B have the same holonomies at the basepoint, then $k(x) = P_B(x)^{-1} P_A(x)$ is smooth and periodic and gives $A^k = B$, by the conventions for gauge transformations and transport in Definition 4.10. Conjugate holonomies are handled by a constant gauge transformation at the basepoint. Two simultaneously diagonal triples are conjugate exactly when they agree or are related by the common Weyl inversion $\boldsymbol{\vartheta} \mapsto -\boldsymbol{\vartheta}$, and each angle has period 4π. This proves the two displayed identifications, including the central points.

The map from the compact quotient of angles to stationary trajectories is continuous for $\mathcal{T}_W$, since every holonomy of a constant commuting connection around a loop depends continuously on the three cycle holonomies, through the integer winding vector of the loop. It is a bijection onto the stationary flat trajectories. The target is Hausdorff, since d_c separates points by Theorem 29.3. A continuous bijection from a compact space onto a Hausdorff space is a homeomorphism, so the identification holds for $\mathcal{T}_W$ and also for d_c. Alternatively, the set of stationary flat trajectories is exactly $\mathcal{K}(0)$ in Theorem 29.6. Pulling back a probability measure supported on this set gives the asserted probability measure on the moduli space, which proves item (3). □

Proof of Theorem 2.9. Fix $L > 0$, an odd integer $M \geq 3$, and n divisible by four, as in Theorem 2.9, and set $n_* = n$. Let the hierarchy, in particular the integer J, and a ceiling $\gamma_0 \in (0, 1]$ be as in Theorem 23.2. We choose an integer $k_0 \geq 0$ such that the comparison side $\bar{n} = n_* B^{k_0}$ satisfies the minimum-side condition and the condition on flat fields of Definition 34.1, and we apply Theorem 34.13 with the original base n_* and this comparison side. All these choices precede the coupling. By item (4) of Theorem 29.28, for every $0 < \gamma \leq \gamma_0$ the measure $\lambda_{L,\gamma,[n]_M}$ of Theorem 2.3 coincides with the measure $\lambda_{L,\gamma,[n_*]}$ of Corollary 29.26 built from this hierarchy. Parts (1), (2), and (3) of Theorem 2.9 are therefore items (1), (2), and (3) of Theorem 34.13. □

Terms	The terms themselves (Theorem 34.7)	Difference from the zero-fluctuation counterpart (Theorems 34.9 and 34.11)	First derivative at a flat base point, for the uncentered summands (Lemma 34.12)
(s,s)	Quadratic expression in s_j: centered part and conditional mean	The same; the counterpart is zero	Zero (two fluctuation factors)
(s,z), (z,s), (z,z)	(34.6) with z_j in H^a_σ, and Hölder's inequality	The same; the counterpart is zero	Zero
(s,r), (r,s)	Mixed linear rows with background r_j, conditionally on $\mathcal{F}_{j-1}$	The same; the counterpart is zero	Mixed linear rows with the deterministic background
(z,r), (r,z)	(34.6) with z_j in H^a_σ	The same; the counterpart is zero	Zero
(r,r)	The form T^{VV}	The difference of T^{VV}	The derivative of T^{VV}
The seven pairs containing u	(34.6) with u_j in H^a_σ	(34.7) and (34.6)	Zero for (z,u), (u,z); otherwise one linear and one deterministic classical factor, and (34.6)
x_{j-1} with s	Mixed linear rows with background x_{j-1}	The same; the counterpart is zero	Mixed linear rows
x_{j-1} with z	(34.6)	The same; the counterpart is zero	Zero
x_{j-1} with r	$T^{\mathsf{UV}} - T^{\mathsf{VV}}$ and $T^{\mathsf{VU}} - T^{\mathsf{VV}}$	Differences of these forms	Derivatives of these forms
x_{j-1} with u	(34.6)	(34.7) with $x_{j-1} - x^0_{j-1}$, and (34.6)	One linear and one deterministic classical factor, and (34.6)

Table 34.2. The tools used for the sixteen ordered products and the eight cross terms in (34.5). Both orders of every pair are estimated separately. In the fourth column, s, z, r, and u stand for the uncentered summands $\tilde{s}_j$, $\tilde{z}_j$, $\tilde{r}_j$, and $\tilde{u}_j$ of the deterministic reconstruction in the proof of Lemma 34.12, whose values at zero fluctuation are $(0,0,R_j,U_j)$ and whose first variations are $(\xi_j,0,\rho_j,v_j)$; they are not the centered summands of (34.1).

SECTION 35

Normal fluctuations near regular commuting holonomies

This section studies the continuum measure λ_γ of Theorem 2.3 at weak coupling near a regular point of the space of flat connections, and proves Theorem 2.10. We observe the holonomies $H = (H_1, H_2, H_3)$ of a trajectory at the fixed time $t = \theta L^2$ around the three coordinate cycles of $\mathbb{T}^3_L$, and their logarithms $X = \operatorname{Log} H \in \mathfrak{su}(2)^3 \simeq \mathbb{R}^9$ in a fixed identity chart. Commuting triples correspond to the cone $\mathcal{M}$. Away from the origin it is a five-dimensional manifold whose normal space $\mathcal{N}_y$ at y has dimension four, and near a regular point we write $X = y + b$ with $y \in \mathcal{M}$ and $b \in \mathcal{N}_y$. We prove three results. After localization to a fixed regular patch, the normalized normal component $\gamma^{-1/2}\Sigma_\theta(y)^{-1/2}b$ is compared, in the weighted normal-bundle sense, with a standard Gaussian vector in $\mathcal{N}_y$ and the same γ-dependent tangential marginal, where Σ_θ is the explicit covariance of (4.12) (Theorem 35.4). The localized quadratic form $q_*/\gamma = b^{\mathsf{T}}\Sigma_\theta(y)^{-1}b/(2\gamma)$ has the limit law with density se^{-s} on $[0,\infty)$ (Theorem 35.3). The normalized regression residual variance $\mathscr{C}_\gamma$ of Theorem 2.10 equals $2\gamma^2 + o(\gamma^2)$ (Theorem 35.2).

The following heuristic explains the coefficient. If $z = \gamma^{-1/2}\Sigma_\theta(y)^{-1/2}b$ were exactly a standard Gaussian vector in four dimensions, independent of y, then $q_*/\gamma = |z|^2/2$ would have mean 2, second moment 6, and variance 2. The observables of Theorem 2.10 are $B_{\mathrm{obs}} = \psi$ and $A_{\mathrm{obs}} = \psi q_*$, where $\psi = \psi_0(y)\chi(b)$ is a fixed tangential cutoff times a normal cutoff equal to one near $b = 0$. Omitting the normal cutoff, we set $B = \psi_0(y)$ and $A = \gamma B|z|^2/2$. If z is standard Gaussian and independent of y, the best affine predictor of A from B is $2\gamma B$, with mean square error $2\gamma^2\mathbb{E}B^2$, and dividing by $\mathbb{E}B^2$ gives $2\gamma^2$. The proofs control the effect of the normal cutoff. The coefficient 2 is a property of four-dimensional Gaussian vectors, and a conditional normal Gaussian measure with the same covariance reproduces it (Lemma 35.17). The theorems show that the continuum measure has this Gaussian behavior in the normal directions, with the precise covariance $\gamma\Sigma_\theta(y)$, while its tangential marginal is not identified and may depend on γ. For comparison, the same statistic vanishes for the Abelian Maxwell measure with Haar cycle holonomies, and it is a positive constant independent of γ for Gaussian holonomies with independent Haar factors (Lemma 35.17).

Near a regular point the flat connections form a nondegenerate critical manifold of the action in the sense of Bott [31], and the $\sqrt{\gamma}$ scaling in the normal directions is

the usual Laplace asymptotics transverse to such a manifold (see [155]). The uniform versions on the coarse lattice that we use are proved in Section 33. The treatment of zero-momentum contributions to Wilson loops in finite periodic volume by Coste, González-Arroyo, Jurkiewicz, and Korthals Altes [62] is a related antecedent; the observables, normalization, and regression functional used here are defined below.

The proof works at a finite lattice cutoff, under the conditioned Wilson measure $\nu = \nu_{N,n,\gamma}$ with $N = nB^K$ and the coarse side n fixed before γ varies. It rests on two ideas. First, we compare with a Gaussian only finitely many scale variables: the coarse field and the normalized fluctuation variables of the first m renormalization group steps, counted from the coarse side. Their joint law is close to a Gaussian law in a space of fixed dimension, with error of order $\sqrt{\gamma}|\log\gamma|$ (Lemma 35.8). Second, the remaining steps are controlled directly at the scale $\sqrt{\gamma}$ by Section 34: Theorem 34.10 compares the flowed holonomies of the fine configuration with those of the finite-scale reconstruction, and Lemma 34.12 bounds the part of the Gaussian covariance coming from the omitted steps. The Gaussian covariance of all K steps is exactly the normal covariance of the finest lattice (Proposition 33.12), which converges to Σ_θ as $N \to \infty$ (Theorem 33.11). The resulting normalized moment errors have the form

$$C\big(\delta_m^{1/2} + \delta_m + N^{-2}\big) + e_m(\gamma), \qquad e_m(\gamma) \to 0 \text{ as } \gamma \downarrow 0 \text{ for each fixed } m,$$

where $\delta_m \to 0$ is a fixed geometric envelope and C does not depend on m. We then let $K \to \infty$ at fixed γ and m, after removing the conditioning with Corollary 23.11 and passing to the continuum with Theorem 29.25; then $\gamma \downarrow 0$ at fixed m; and finally $m \to \infty$. The constant C_m in the finite-dimensional comparison may be very large; this is harmless because m is fixed before $\gamma \to 0$. The other inputs are the conditional fluctuation kernels and their means (Theorems 13.8 and 13.10), the reconstruction maps (Theorem 15.12), the flat refinements (Lemma 14.9), the parameter derivatives of the Wilson flow (Theorem 25.13), the Laplace estimates and the normalization on the coarse lattice (Proposition 33.12, Corollary 33.4, , and Theorem 33.10), and the trace coordinates and conditioning identities of Lemmas 5.5 and 5.6.

Besides Theorem 2.10, two results of this section are used in Section 36: the uniform regularity of the finite-scale flowed holonomy map near every flat configuration, including the central triples (Theorem 35.9), and the comparison measures (Lemmas 35.16 and 35.17).

The three main theorems are stated in Section 35.1 and proved in Sections 35.6 to 35.8, after the auxiliary results of Sections 35.2 to 35.5; the section ends with the comparison measures (Section 35.9) and the proof of Theorem 2.10.

35.1 Setting and main results

Fixed data. Throughout this section we fix $L > 0$, $\theta > 0$, a horizon $T \geq \theta L^2$, an odd integer $M \geq 3$, an integer $J \geq 1$, $B = M^J$, a coupling ceiling $0 < \gamma_0 \leq 1$, and the construction parameters of Definition 34.1: the cutoff amplitude A_{w}, the radii $r < \rho$, the reconstruction maps of Theorem 15.12, the flow exponents $1/2 < \alpha_{\mathrm{fl}} < 2/3$ and σ_{sm}, and the contraction constant a_{map} with $a_{\mathrm{map}}\sqrt{B} < 1$ and $\mathfrak{q} = Ba_{\mathrm{map}}^2 < 1$. All of these are chosen before γ varies. We fix $n \in 4\mathbb{N}$ satisfying the minimum-side conditions of Definition 34.1; in particular

$$2\pi/n < r_{\mathrm{flat}}/4,$$

where $r_{\mathrm{flat}} < r$ is the flat margin of Lemma 14.9. For $K \geq 0$ we set $N = nB^K$, let $\nu = \nu_{N,n,\gamma}$ be the conditioned Wilson measure of Definition 4.4, and let $\mu = \mu_{N,\gamma}$ be the Wilson measure. The renormalization group steps of the hierarchy are indexed by $j = 1, \dots, K$, counted from the coarse side: the j-th step relates the lattices of sides nB^{j-1} and nB^j. We write $w_j = (w_{j,r,e})_{r,e}$ for the normalized fluctuation variables of the j-th step, indexed by its J intermediate scales r and the links e, and $g_{j,r}$ for the coupling at the intermediate scale r, so that $g_{j,r}^2/\gamma$ is the reciprocal of a fixed lattice side.

Let $\mathcal{H}^K = (\mathcal{H}_i^K)_{i=1}^3$ be the triple of coordinate-cycle holonomies, based at the origin, of the Wilson-flowed fine configuration at time θL^2 (lattice time $4\theta L^2/a_N^2 = 4\theta N^2$), computed for the reconstructed configuration $\exp \widehat{\theta}_K$ of Definition 34.1 in the gauge frame at the basepoint fixed by the reconstruction maps. We call $\mathcal{H}^K$ the flowed holonomy triple. For $0 \leq m \leq K$ let Ξ_m^K be the same triple for the finite-scale reconstruction $\exp \widehat{\theta}_K^0$, in which the coarse field and the normalized fluctuation variables of the first m steps are those of the same sample and all normalized fluctuation variables of the steps after the m-th are set to zero. Both triples are random variables under ν, defined in the same gauge frame at the basepoint, and both are flowed on the finest lattice.

Finally, $\lambda_\gamma = \lambda_{L,\gamma,[n_*]}$ denotes the continuum measure on the trajectory space associated with the sequence nB^K (Corollary 29.26); by Theorem 29.28 it is the measure $\lambda_{L,\gamma,[n]_M}$ of Theorem 2.3. For a trajectory $X \in \mathcal{X}_L$, the coordinate-cycle holonomies of $X(\theta L^2)$ are defined up to simultaneous conjugation, and so are the coordinates introduced below; all observables we evaluate are invariant under simultaneous conjugation.

Constants. Unless stated otherwise, constants c, C and thresholds in this section have the dependence described in item (6) of Definition 4.14. They may depend on n, L, θ, M, J, the construction parameters, the horizon, the flow exponents, the loop lengths, and the patches, tubes, test functions, and extensions fixed below, but not on K, N, the preceding classical depths, the index m, or γ below the relevant threshold. A subscript m, as in C_m or γ_m, allows additional dependence on m (through the first m steps and their finite dimensions); constants without this subscript do not depend on m. Further subscripts indicate dependence on moment orders, powers of γ, or test functions.

Regular coordinates. In the fixed identity chart we write $X = (x_1, x_2, x_3) = \operatorname{Log} H \in \mathbb{R}^9$, identifying $\mathfrak{su}(2)$ with $\mathbb{R}^3$ so that the bracket becomes the cross product. The cone of commuting triples is

$$\mathcal{M} = \{y = (\vartheta_1 u, \vartheta_2 u, \vartheta_3 u) : \boldsymbol{\vartheta} \in \mathbb{R}^3,\ u \in S^2\}, \qquad (\boldsymbol{\vartheta}, u) \sim (-\boldsymbol{\vartheta}, -u).$$

On the regular region $0 < r_0 \le |\boldsymbol{\vartheta}| \le r_1 < \pi$ it is a smooth five-dimensional manifold, parametrized by the three angles $\boldsymbol{\vartheta}$ and the direction u. Its normal space at $y = (\vartheta_i u)_i$ is

$$\mathcal{N}_y = \Big\{ b = (b_1, b_2, b_3) :\ b_i \perp u,\ \sum_i \vartheta_i b_i = 0 \Big\}.$$

The conditions $b_i \perp u$ leave six dimensions, and the relation $\sum_i \vartheta_i b_i = 0$ removes two of them, which gives $\dim \mathcal{N}_y = 4$. Simultaneous conjugation $H_i \mapsto g H_i g^{-1}$ acts on $\mathbb{R}^9$ by the same rotation $\mathrm{Ad}(g)$ in each component; it preserves $\mathcal{M}$ and maps $\mathcal{N}_y$ isometrically onto $\mathcal{N}_{\mathrm{Ad}(g)y}$. Two of the five tangential directions of $\mathcal{M}$ are orbits of this action, which is not removed by fixing a gauge at the basepoint.

We write $\Pi_{\boldsymbol{\vartheta}} = I - \boldsymbol{\vartheta}\boldsymbol{\vartheta}^{\mathsf{T}}/|\boldsymbol{\vartheta}|^2$ and let $\Sigma_\theta(y)$ be the positive definite matrix on $\mathcal{N}_y$ defined in (4.12). By Theorem 33.11, $\gamma\Sigma_\theta(y)$ is the covariance of the linearized flowed cycle logarithms, projected to $\mathcal{N}_y$, under the Gaussian measure of the Maxwell quadratic form around the flat connection with holonomy logarithms y. It does not depend on γ, depends smoothly on y, is equivariant under simultaneous conjugation, and satisfies $cI \le \Sigma_\theta \le CI$ on the regular region. For comparisons of vectors in $\mathbb{R}^9$ we extend the inverse covariance to the ambient matrix $P_{\mathcal{N}_y}\Sigma_\theta(y)^{-1}P_{\mathcal{N}_y}$, which is smooth on the regular region and has the same quadratic form on $\mathcal{N}_y$. Normal matrices are expressed in smooth orthonormal frames of a finite bundle atlas.

For a positive semidefinite matrix Σ we write G_Σ for a centered Gaussian vector with covariance Σ, realized as a linear image of a standard Gaussian vector, so that Σ may be singular. We write G_d for a standard Gaussian vector in a d-dimensional Euclidean space and Γ_y for the standard Gaussian measure on $\mathcal{N}_y$. For a bounded Lipschitz function f we write $\|f\|_{\mathrm{BL}} = \|f\|_\infty + \operatorname{Lip} f$. Its dual distance is not the bounded-Lipschitz distance of Section 4.10, which uses the maximum of the two terms, but differs from it by at most a factor two.

Definition 35.1. Fix compact regular patches $\mathsf{P} \subset \mathsf{P}^+ \subset \mathsf{P}^{++}$ in $\mathcal{M} \cap \{r_0 \le |\boldsymbol{\vartheta}| \le r_1\}$, invariant under simultaneous conjugation, each with nonempty relative interior and contained in the relative interior of the next, and radii $0 < b_{\mathrm{o}} < b^+ < b^{++}$. Put

$$\mathcal{U} = \{y + b :\ y \in \mathsf{P},\ b \in \mathcal{N}_y,\ |b| < b_{\mathrm{o}}\},$$

and define $\mathcal{U}^+$ and $\mathcal{U}^{++}$ in the same way with (P^+, b^+) and $(\mathsf{P}^{++}, b^{++})$. The radius b^{++} is so small that $X = y + b$ are unique coordinates on $\mathcal{U}^{++}$ and that the closure of $\mathcal{U}^{++}$ lies in the identity chart at positive distance from its boundary. The following objects are fixed once, before γ varies.

1. On the whole tube $\mathcal{U}^{++}$ put
$$q_*(y+b) = \frac{1}{2}\, b^{\mathsf{T}} \Sigma_\theta(y)^{-1} b.$$
Outside $\mathcal{U}^{++}$, q_* is a fixed bounded continuous nonnegative function, invariant under simultaneous conjugation; on $\mathrm{SU}(2)^3$ it is evaluated through $X = \operatorname{Log} H$ and extended outside the identity chart by the same bounded function.
2. Let $\psi_0 : \mathsf{P} \to [0,1]$ be smooth and invariant, with compact support in the relative interior of P, and suppose that there exist nonempty open subsets O_0, O_+ of P, invariant under simultaneous conjugation, and $b_* > 0$ with $\psi_0 = 0$ on O_0 and $\psi_0 \geq 2b_*$ on O_+. Let χ be a fixed smooth invariant normal cutoff with $0 \leq \chi \leq 1$, equal to one near the zero section and supported in $\{|b| < b_{\mathrm{o}}\}$; it is a smooth function on the normal bundle. Put $\psi(y+b) = \psi_0(y)\chi(b)$ on $\mathcal{U}$ and $\psi = 0$ elsewhere.
3. Put $B_{\mathrm{obs}} = \psi$ and $A_{\mathrm{obs}} = \psi q_*$, regarded as functions on $\mathrm{SU}(2)^3$ through $X = \operatorname{Log} H$ and extended by zero outside the identity chart.
4. A localizing weight is a function $\widetilde{w}(y+b) = w(y)\chi_w(b)$, extended by zero outside $\mathcal{U}$, where $w \geq 0$ is smooth and invariant with compact support in the relative interior of P, and $0 \leq \chi_w \leq 1$ is smooth, invariant, equal to one near $b = 0$, and supported in $\{|b| < b_{\mathrm{o}}\}$.

For a probability measure P and bounded real random variables A, B with $\mathbb{E}_P B^2 > 0$ and $\operatorname{Var}_P B > 0$, define
$$\mathscr{C}(P; A, B) = \frac{\operatorname{Var}_P A - \operatorname{Cov}_P(A,B)^2 / \operatorname{Var}_P B}{\mathbb{E}_P B^2}, \tag{35.1}$$
and set $\mathscr{C}_\gamma = \mathscr{C}(\lambda_\gamma; A_{\mathrm{obs}}, B_{\mathrm{obs}})$, with $A_{\mathrm{obs}}, B_{\mathrm{obs}}$ evaluated at the coordinate-cycle holonomies of $X(\theta L^2)$.

Every localizing weight, ψ, and ψ^2 are supported in $\mathcal{U} \subset \mathcal{U}^{++}$, so the products $\widetilde{w} q_*^l$, ψq_*^l, and $\psi^2 q_*^l$ involve only the exact quadratic form $b^{\mathsf{T}} \Sigma_\theta(y)^{-1} b/2$, which is nonnegative. The extension of q_* outside $\mathcal{U}^{++}$ plays no role in any of the results. All further patches and tubes used in the proofs (the coarse tubes, the output tubes, and the auxiliary tubes) are chosen before γ varies, nested and with positive margins, between $\mathcal{U}$ and $\mathcal{U}^{++}$ (respectively over patches contained in P^{++}). By Lemma 35.12 below, $\mathscr{C}(P; A, B)$ is the minimal mean square error of an affine predictor of A from B, divided by $\mathbb{E}_P B^2$.

The envelope δ_m. We fix $c, \epsilon > 0$ such that
$$\delta_m = (1+m)^c B^{-\epsilon m}, \qquad m \geq 0,$$
dominates the envelopes in Theorem 34.10 and Lemma 34.12, which have the same form. Then $\sup_m \delta_m < \infty$ and $\delta_m \to 0$ as $m \to \infty$. This choice does not change the dependence of any constant on m.

Main results.

Theorem 35.2. *Evaluate $A_{\mathrm{obs}} = \psi q_*$ and $B_{\mathrm{obs}} = \psi$ at the coordinate-cycle holonomies of $X(\theta L^2)$. These are bounded continuous functions of the seven normalized traces of Lemma 5.6, and they do not depend on γ. There exists $c > 0$ such that, for all sufficiently small $\gamma > 0$,*

$$\mathbb{E}_{\lambda_\gamma} B_{\mathrm{obs}}^2 \geq c, \qquad \mathrm{Var}_{\lambda_\gamma} B_{\mathrm{obs}} \geq c.$$

For $f = \psi$ and $f = \psi^2$,

$$\mathbb{E}_{\lambda_\gamma}[f q_*] = 2\gamma\, \mathbb{E}_{\lambda_\gamma} f + o(\gamma), \qquad \mathbb{E}_{\lambda_\gamma}[f q_*^2] = 6\gamma^2\, \mathbb{E}_{\lambda_\gamma} f + o(\gamma^2),$$

and the normalized regression residual variance satisfies

$$\mathscr{C}_\gamma = 2\gamma^2 + o(\gamma^2) \qquad (\gamma \downarrow 0).$$

Further, there exists C such that for every fixed $m \geq 0$,

$$\limsup_{\gamma \downarrow 0} \Big| \frac{\mathscr{C}_\gamma}{\gamma^2} - 2 \Big| \leq C\big(\delta_m^{1/2} + \delta_m\big),$$

and the right side tends to zero as $m \to \infty$.

Theorem 35.3. *Let $\widetilde{w} = w\chi_w$ be a localizing weight with w not identically zero, evaluated at the coordinate-cycle holonomies of $X(\theta L^2)$. There exist $c_w, \gamma_w > 0$ such that $D_\gamma = \mathbb{E}_{\lambda_\gamma} \widetilde{w} \geq c_w$ for $0 < \gamma \leq \gamma_w$. For such γ define the probability measure*

$$\pi_\gamma(A) = \frac{\mathbb{E}_{\lambda_\gamma}\big[\widetilde{w}\, \mathbf{1}_{\{q_*/\gamma \in A\}}\big]}{\mathbb{E}_{\lambda_\gamma} \widetilde{w}}, \qquad A \subset [0,\infty) \text{ Borel}.$$

As $\gamma \downarrow 0$, π_γ converges weakly to the law of $|G_4|^2/2$, which has density $s e^{-s}$ on $[0,\infty)$. Equivalently, for every bounded continuous $f : [0,\infty) \to \mathbb{R}$,

$$\frac{\mathbb{E}_{\lambda_\gamma}[\widetilde{w}\, f(q_*/\gamma)]}{\mathbb{E}_{\lambda_\gamma} \widetilde{w}} \longrightarrow \mathbb{E} f(|G_4|^2/2) = \int_0^\infty f(s)\, s e^{-s}\, ds.$$

Further, there exist C_w, independent of m, and for each fixed $m \geq 0$ a function $e_{m,w}$ with $e_{m,w}(\gamma) \to 0$ as $\gamma \downarrow 0$, such that for every bounded Lipschitz f and all sufficiently small γ, depending on m,

$$\Big| \int f\, d\pi_\gamma - \mathbb{E} f(|G_4|^2/2) \Big| \leq \|f\|_{\mathrm{BL}} \big[C_w(\delta_m^{1/2} + \delta_m) + e_{m,w}(\gamma) \big].$$

Since $q_* \geq 0$ on the support of $\widetilde{w}$, the measure π_γ is a probability measure on $[0,\infty)$.

Theorem 35.4. *Let $\widetilde{w}$, D_γ, c_w, and γ_w be as in Theorem 35.3. On $\{\widetilde{w} \neq 0\}$ write the logarithm of the coordinate-cycle holonomy triple of $X(\theta L^2)$ as $y_\gamma + b_\gamma$ and set*

$$z_\gamma = \gamma^{-1/2} \Sigma_\theta(y_\gamma)^{-1/2} b_\gamma \in \mathcal{N}_{y_\gamma},$$

where $\Sigma_\theta(y)^{-1/2}$ is the positive inverse square root on $\mathcal{N}_y$. These coordinates are defined up to simultaneous conjugation, and any representative may be used in the invariant expressions below. Let $\mathsf{P}' \subset \mathsf{P}^{++}$ be a compact invariant regular patch containing the support of w in its relative interior. For a bounded continuous function F on the normal bundle $\{(y,z) : y \in \mathsf{P}',\ z \in \mathcal{N}_y\} \subset \mathbb{R}^9 \times \mathbb{R}^9$, invariant under $(y,z) \mapsto (\mathrm{Ad}(g)y, \mathrm{Ad}(g)z)$ for $g \in \mathrm{SU}(2)$, put

$$\overline{F}(y) = \int_{\mathcal{N}_y} F(y,z)\, \Gamma_y(dz).$$

Products with $\widetilde{w}$ are defined to be zero where $\widetilde{w} = 0$. Then

$$\frac{\mathbb{E}_{\lambda_\gamma}[\widetilde{w}\, F(y_\gamma, z_\gamma)] - \mathbb{E}_{\lambda_\gamma}[\widetilde{w}\, \overline{F}(y_\gamma)]}{D_\gamma} \longrightarrow 0 \qquad (\gamma \downarrow 0). \tag{35.2}$$

Let $\mathsf{P}'_{\mathrm{orb}} = \mathsf{P}' / \mathrm{Ad}(\mathrm{SU}(2))$, with its quotient Borel σ-algebra, and write $[y]$ for the class of y. The tangential marginal of the localized measure is the probability measure

$$\rho_\gamma(A) = D_\gamma^{-1}\, \mathbb{E}_{\lambda_\gamma}\big[\widetilde{w}\, \mathbf{1}_{\{[y_\gamma] \in A\}}\big], \qquad A \subset \mathsf{P}'_{\mathrm{orb}} \text{ Borel},$$

the function $\overline{F}(y)$ depends only on $[y]$, and

$$D_\gamma^{-1}\, \mathbb{E}_{\lambda_\gamma}[\widetilde{w}\, \overline{F}(y_\gamma)] = \int_{\mathsf{P}'_{\mathrm{orb}}} \overline{F}(y)\, \rho_\gamma(d[y]).$$

If F is bounded Lipschitz for the Euclidean distance of $\mathbb{R}^9 \times \mathbb{R}^9$, then the absolute value of the left side of (35.2) *is at most*

$$\|F\|_{\mathrm{BL}} \big[C_w(\delta_m^{1/2} + \delta_m) + e_{m,w}(\gamma)\big] \tag{35.3}$$

for every sufficiently large fixed m and all sufficiently small γ, depending on m, where C_w does not depend on m or F, and $e_{m,w}(\gamma) \to 0$ as $\gamma \downarrow 0$ for each fixed m.

Remark 35.5. In the three theorems the continuum limit is taken first, at fixed γ. The proofs take the limits in the order $K \to \infty$ at fixed γ and m, then $\gamma \downarrow 0$ at fixed m, then $m \to \infty$. Only bounded observables, not divided by powers of γ, are compared between ν and μ, and no rate of convergence uniform in γ is used or asserted. The tangential marginal ρ_γ is not identified: Theorem 35.4 states that the normal fluctuations are Gaussian over this marginal, not that the pair (y_γ, z_γ) converges to a fixed limit law. Finally, Theorem 35.2 identifies no corrections to the coefficient 2, which is the coefficient of a four-dimensional Gaussian vector and is reproduced by the conditional normal Gaussian measure of Lemma 35.17(4).

Remark 35.6. Theorem 35.4 contains more information than the law of $|z_\gamma|^2/2 = q_*/\gamma$. For example,

$$F(y,z) = \frac{|z_1|^2 - |z_2|^2}{1+|z|^2}$$

is bounded, Lipschitz, and invariant, and it requires no choice of a color frame. At $y = (0,0,ru)$ with $r_0 < r < r_1$, its values at $(z_1,z_2,z_3) = (sv,0,0)$ and at $(0,sv,0)$, with $v \perp u$, are opposite, although the two points have the same radius. In general, writing $R = |G_4|$ and $\Pi = \Pi_{\vartheta}$,

$$\overline{F}(y) = \frac{\Pi_{11} - \Pi_{22}}{2}\,\mathbb{E}\frac{R^2}{1+R^2}.$$

Indeed, $\mathcal{N}_y = \vartheta^\perp \otimes u^\perp$, so the quadratic form $b \mapsto |b_i|^2$ has trace $2\Pi_{ii}$ on the four-dimensional space $\mathcal{N}_y$, and the conditional second moment of $|z_i|^2$ given $|z| = R$ under Γ_y is $R^2\Pi_{ii}/2$. In the theorem, both $\overline{F}$ and the normalization of b_γ are evaluated at y_γ, and the normalization is by D_γ.

35.2 The coarse marginal and the finite-dimensional Gaussian comparison

Definition 35.7. Let $\widehat{\nu} = (T_M^{JK})_\# \nu$ be the law of the coarse field $V = T_M^{JK} U$ under ν. It has the density

$$d\widehat{\nu} = \widehat{Z}^{-1} \mathbf{1}_{D_r}\, e^{-nE_{JK,\mathrm{sat}}/\gamma - \overline{R}_{K,\gamma}}\, dH_n, \qquad \overline{R}_{K,\gamma} = R_{K,\gamma} - R_{K,\gamma}(I),$$

where $E_{JK,\mathrm{sat}}$ is the gauge-invariant extension of the effective classical action E_{JK} from B_r to D_r of Theorem 11.6 (that is, $E_{JK,\mathrm{sat}}(U) = E_{JK}(U^k)$ whenever $U^k \in B_r$), $R_{K,\gamma}$ is the effective interaction after K renormalization group steps, of classical depth JK, $\overline{R}_{K,\gamma}$ is the centered effective interaction, and $\widehat{Z}$ is the normalizing constant of this density; it is the normalizing constant of Theorem 33.10.

Near the flat configurations of $\mathcal{U}_n$ whose holonomy logarithms lie in P^{++} we use the finite normal atlas $V = V(y,v)$ of Proposition 33.12. Here $y \in \mathcal{M}$ is the logarithm of the holonomy triple of the flat configuration $V(y,0)$, and v is a normal variable of dimension

$$d_{\mathrm{reg}} = 6n^3 - 2,$$

whereas $\dim \mathcal{N}_y = 4$. A coarse regular tube is a set $\{V(y,v) : y \in \mathsf{P}_{\mathrm{c}},\ |v| \le \varepsilon_{\mathrm{c}}\}$, where $\mathsf{P}_{\mathrm{c}} \subset \mathsf{P}^{++}$ is a compact invariant regular patch and $\varepsilon_{\mathrm{c}} > 0$ is fixed. Put

$$H_K(y) = D_v^2(nE_{JK})(V(y,0)), \qquad z_0 = \gamma^{-1/2} H_K(y)^{1/2} v.$$

Let $\mathscr{H}_K$ denote the triple of coordinate-cycle holonomies of the Wilson-flowed fine configuration at time θL^2, regarded as a function of the coarse coordinates (y,v) and of all normalized fluctuation variables $(w_1,\dots,w_K)$. Evaluate derivatives at the zero-fluctuation

reconstruction of the flat coarse field $V(y,0)$, that is, at the sequence of flat refinements of $V(y,0)$ obtained with $w=0$. Let $T_{0,K}$ be the derivative in v of $P_{\mathcal{N}_y}(\mathrm{Log}\circ\mathscr{H}_K)$ there, and define

$$A_{0,K}=T_{0,K}H_K^{-1/2},\qquad A_{j,K}=\gamma^{-1/2}P_{\mathcal{N}_y}D_{w_j}(\mathrm{Log}\circ\mathscr{H}_K)\big|_{\mathrm{flat}},\qquad 1\le j\le K.$$

The Hessian normalization of each step at the flat background and the factors $g_{j,r}/\sqrt{\gamma}$ are included in $A_{j,K}$, exactly once. These deterministic matrices do not depend on γ. Put

$$\Sigma_m^K(y)=A_{0,K}A_{0,K}^{\mathsf{T}}+\sum_{j=1}^{m}A_{j,K}A_{j,K}^{\mathsf{T}},\qquad 0\le m\le K.$$

The matrix $\Sigma_m^K(y)$ is the covariance of the image, under the linearized normal response, of independent standard Gaussian vectors placed in the coarse normal variable and in the first m steps. We call the matrices $A_{j,K}$ the linear response matrices. On a sample, $\mathscr{H}_K$ takes the value $\mathcal{H}^K$, the flowed holonomy triple of Definition 34.1, in the same gauge frame at the basepoint. In the notation of Proposition 33.12, $H_K=H_K^{\mathrm{act}}$, the Hessian of the action phase nE_{JK}. For $j\ge 1$ the matrices $A_{j,K}$ coincide with the matrices $\mathsf{A}_{j,K}$ of Lemma 34.12, as shown in the proof of Theorem 35.10.

Hypotheses of the earlier results. We record how the hypotheses of the results used below are met. None of these choices changes as γ varies.

Flat refinements. Since $2\pi/n<r_{\mathrm{flat}}/4$ and $r_{\mathrm{flat}}<r$ lies below every fixed margin of the constrained minimizers, of the inverse of the block averaging constraint, of the Haar disintegration densities, and of the gauge charts, every flat coarse field has a commuting representative with link logarithms of norm at most $2\pi/n$. By Lemma 14.9, its zero-fluctuation reconstruction is the sequence of flat refinements, and it lies inside all these coordinate domains with a positive margin at every depth. On the fixed coarse lattice the compact set of flat configurations has a neighborhood in D_r independent of the depth and of γ. This verifies the geometric hypothesis in every application of Proposition 33.12 below.

The coarse normal atlas. The dimension in the Laplace integral on the coarse lattice is $d_{\mathrm{reg}}=\mathfrak{d}_n=6n^3-2$, the regular normal rank of Proposition 33.12, not four, and $d_{\mathrm{reg}}/2=3n^3-1$. We take the finite normal atlas on relatively compact smooth subcharts, in rooted spanning-tree gauge coordinates, whose charts and gauge transformations extend smoothly to neighborhoods of their closures. Their representatives and the closed normal tubes lie strictly inside the outer ball B_ρ, hence inside $B_{(1-\eta)\rho}$ for a fixed $0<\eta<1$, and all chart and gauge jets are bounded on these closures. We use the derivative bounds for the amplitude of the coarse density only on these subcharts; their constants may depend on n, η, the chart and gauge jets, and the outer domains, but not on the depth or on γ. In this atlas the density of $\widehat{\nu}$ is

$$\widehat{Z}^{-1}a_{K,\gamma}(y,v)\,e^{-nE_{JK}(V(y,v))/\gamma}\,dy\,dv,\qquad a_{K,\gamma}=e^{-\overline{R}_{K,\gamma}}\,J(y,v),$$

where J is the density of the product Haar measure in these coordinates, including the partition of unity of the atlas. On each compact subpatch, before the partition is applied, $a_{K,\gamma}$ is bounded above and below by positive constants and is uniformly Lipschitz in v. These are hypotheses of the coarse family of Proposition 33.12, which also has $nE_{JK,\mathrm{sat}} \geq \kappa_n E_0$ on its support and $|\overline{R}_{K,\gamma}| \leq C_n$. The phase and amplitude are smooth in the fixed charts, and the derivative bounds on the effective interaction in that family imply the amplitude bounds uniformly in the depth and in γ. In particular all phase, amplitude, and cutoff data are jointly Borel, H_K is measurable, and the phase is coercive on each chosen cutoff support. These facts verify the measurability, spectral, Lipschitz, and coercivity hypotheses of Lemma 33.3, on which the coarse Laplace formula rests. The partition of the atlas is absorbed into the measurable tangential weights, after the positive bounds on the amplitude have been used.

Coordinate domains of the reconstruction. Every coarse field in the support of $\widehat{v}$ has a representative in B_r; the reconstruction maps take values in the inner coordinate domain, and the larger complex images lie inside $B_{\rho/2}$. The products of coarse and free coordinates lie strictly inside the nested domains $\mathscr{D}_{\mathrm{in}} \Subset \mathscr{D}_{\mathrm{mid}} \Subset \mathscr{D}_{\mathrm{out}}$ of Theorem 15.12, and every intervening constrained minimizer, inverse, logarithm, and Haar disintegration density is defined there. In terms of majorizing series, the radius of the coarse field plus the fixed radius of the free coordinates lies strictly below the radius of the outer domain and below the radius of convergence of the series. These are the hypotheses of Theorem 15.12 on the images of the reconstruction maps; no auxiliary integral is evaluated outside B_r.

Couplings and cutoffs. Let $t_* > 0$ be the common coupling ceiling in Theorems 13.8 and 13.10. At the j-th step we set

$$t_j = M^{(J-1)/2}\sqrt{\gamma/(nB^j)}, \qquad p_j = A_{\mathrm{w}} \log(1/t_j), \qquad t_1^2 = \gamma/(Mn). \tag{35.4}$$

We keep the ceiling γ_0 for the bound on the small-field failure probability and for the continuum limit theorem. At the fixed side n we fix a threshold

$$0 < \gamma_W(n) \leq \gamma_0, \tag{35.5}$$

and all estimates below are restricted to $0 < \gamma \leq \gamma_W(n)$. The threshold may depend on n and the fixed data, but not on K, m, or γ. It is so small that $t_1 \leq t_*$, $p_1 \geq 1$, $Ct_j p_j \leq 1/2$, and the Hessian bound of Theorem 13.8 and the smallness condition on $t_j p_j^3$ of Theorem 13.10 hold for every $j \geq 1$. One choice serves all j: we have $t_j \leq t_1$, and each function $t^a \log^b(1/t)$ needed here is increasing on a sufficiently small fixed interval, since its derivative has the sign of $a\log(1/t) - b$. We also require that $t_1 p_1$ fit within the fixed margin of the coordinate domains for the free coordinates. No radius, cutoff, reconstruction threshold, or side changes as γ varies. Later reductions of the threshold concern only $\gamma_W(n)$, and every threshold γ_m below is taken at most $\gamma_W(n)$.

The reconstruction maps preserve the conditional potential, the product of cutoff balls, the normalized conditional kernels, and the coarse field. In particular, the two theorems of Section 13 quoted above apply to every conditional kernel in this section, uniformly in the coarse field, the side, and the preceding depth. We write E for their common bound E_* on the effective interaction.

Disintegration. At finite K we have the backward disintegration

$$d\nu = d\widehat{\nu}(V) \prod_{j=1}^{K} k_j(V, w_1, \dots, w_{j-1}; dw_j), \tag{35.6}$$

where k_j is the conditional fluctuation kernel of the j-th step, given the coarse field and the earlier steps. Restricting ν to an event depending only on the coarse field multiplies the first factor by an indicator and leaves the kernels k_j unchanged. Each k_j still integrates to one, including its Haar disintegration density, its determinant, and the input effective interaction in its potential. Conditioning further on y changes the conditional law of v in the first factor and leaves the same finer kernels. All conditional statements are understood for a version defined for almost every y; their uniform bounds hold on the whole compact patch wherever that version is defined by the displayed smooth densities.

Lemma 35.8. *Use Definition 35.7, the disintegration* (35.6)*, and a fixed coarse regular tube, and condition on the coarse tangential coordinate y. For every fixed $m \geq 0$ there exists $\gamma_m > 0$ such that the following holds uniformly in $K \geq m$ and $0 < \gamma \leq \gamma_m$. Let $\mathcal{F}_{j-1}$ be the σ-algebra generated by the coarse field and the normalized fluctuation variables of the first $j-1$ steps, let $\bar{w}_j = \mathbb{E}(w_j \mid \mathcal{F}_{j-1})$ be the conditional mean under the conditional fluctuation kernel k_j, and put*

$$Z_{K,m} = A_{0,K}(y) z_0 + \sum_{j=1}^{m} A_{j,K}(y)(w_j - \bar{w}_j).$$

For every fixed $s \geq 0$ and every jointly Borel family of real functions F_y on $\mathcal{N}_y$ with $|F_y(z)| \leq C_F(1 + |z|^s)$ in the finite smooth bundle atlas,

$$\left|\mathbb{E}[F_y(Z_{K,m}) \mid y] - \mathbb{E}F_y(G_{\Sigma_m^K(y)})\right| \leq C_{m,s} C_F \sqrt{\gamma}\,(1 + |\log \gamma|).$$

All fixed moments of $Z_{K,m}$ and of its partial sums are bounded by constants $C_{m,s}$. More strongly, the same polynomially weighted comparison holds for the joint vector $(z_0, w_1 - \bar{w}_1, \dots, w_m - \bar{w}_m)$ against a standard Gaussian vector of the same fixed finite dimension. The restriction to the coarse tube changes only the first factor of (35.6)*; no conditional kernel and no nonlinear reconstruction is evaluated at a Gaussian argument.*

Proof. We first bound each conditional kernel and the probability that its variables come close to the boundary of the cutoff product (Steps 1 and 2). We then control the linear term of the conditional potential (Step 3), compare each conditional density with a Gaussian

density in a weighted norm (Step 4), and treat the coarse variable (Step 5). Finally, a telescoping argument replaces the variables one at a time by Gaussian variables (Step 6), and we pass to the linear image (Step 7).

Step 1: the conditional kernels. Fix the coarse field and the first $j-1$ steps in the coordinate domains. By Theorem 13.8, whose conclusions are preserved by Theorem 15.12, the kernel k_j on the product $\Omega_j \subset \mathbb{R}^{D_j}$ of cutoff balls satisfies

$$k_j(dw) = Z_j^{-1}\mathbf{1}_{\Omega_j}e^{-V_j(w)}\,dw, \qquad \|D^2V_j - I\| \le \epsilon_j,$$
$$\operatorname{Var} F \le 2\mathbb{E}|\nabla F|^2, \qquad \operatorname{Ent}(F^2) \le 4\mathbb{E}|\nabla F|^2,$$

where $\epsilon_j \le C\{t_jp_j + t_j^2(1+E)\}$ under the condition $Ct_jp_j < 1/2$, with t_j, p_j as in (35.4), and $D_j \le 9J(nB^j)^3$. By Theorem 13.10, the conditional mean satisfies

$$\|\bar w_j\|_\infty \le Ct_j\|\widehat\theta_{j-1}\|_\infty \le Ct_jr, \tag{35.7}$$

where $\widehat\theta_{j-1}$ is the logarithm of the reconstructed field after $j-1$ steps. For $j \le m$ the dimension D_j is fixed, and we have $|\bar w_j| \le C_m\sqrt{\gamma}$ and $\epsilon_j \le C_m\sqrt{\gamma}(1+|\log\gamma|)$. These bounds are uniform over all earlier variables in the coordinate domains and over K.

Step 2: concentration and an interior cutoff. Let f be a 1-Lipschitz function of w, and let $\Lambda(\lambda) = \log \mathbb{E}e^{\lambda(f-\mathbb{E}f)}$. Inserting $e^{\lambda f/2}$ into the logarithmic Sobolev inequality, we find $\lambda\Lambda' - \Lambda \le \lambda^2$, which implies $\Lambda(\lambda) \le \lambda^2$. Applying this also to $-f$ and choosing $\lambda = u/2$ in Markov's inequality, we obtain $\mathbb{P}\{|f-\mathbb{E}f| \ge u\} \le 2e^{-u^2/4}$; this is the Herbst argument (see [123, Section 2.3]). Lipschitz functions are covered by smooth approximation on the compact product. Since $t_j = c_{n,M,J}\sqrt{\gamma}B^{-j/2}$ and $p_j = A_{\mathrm w}\log(1/t_j)$, we may decrease γ_m so that $Ct_jr \le p_j/(4\sqrt{3})$ and $p_j \ge 1$ for $j \le m$. If a three-dimensional block w_a has $|w_a| \ge p_j/2$, then one of its scalar coordinates deviates from its conditional mean by at least $p_j/(4\sqrt{3})$. A union bound over the scalar coordinates yields

$$\mathbb{P}\Big\{\max_a |w_a| \ge p_j/2\Big\} \le 2D_je^{-p_j^2/192}.$$

Let $0 \le \chi \le 1$ be the product radial cutoff equal to one on the balls of half the cutoff radius and zero outside the balls of three quarters of it. Then $\chi \in C_c^\infty(\Omega_j)$, $|\nabla\chi| \le C\sqrt{D_j}/p_j$, and $q_j^{\mathrm{cut}} = \mathbb{P}(\chi < 1) \le CD_je^{-cp_j^2}$. At fixed m we have $p_j \ge c|\log\gamma| - C_m$. This implies that q_j^{cut} and the standard Gaussian mass outside Ω_j are $O_{m,q}(\gamma^q)$ for every fixed $q > 0$; polynomially weighted Gaussian tails have the same property, with the fixed moment order in the constant. Decreasing γ_m further, we may assume $q_j^{\mathrm{cut}} \le 1/2$, $|\bar w_j| \le 1$, and $\epsilon_j \le 1/4$ for $j \le m$.

Step 3: the linear term. We write $D = D_j$, $\mu = \bar{w}_j$, $\epsilon = \epsilon_j$, and

$$V_j(w) - V_j(0) = |w|^2/2 + \ell \cdot w + R(w), \qquad \ell = \nabla V_j(0),$$
$$|\nabla R(w)| \le \epsilon |w|, \qquad |R(w)| \le \epsilon |w|^2/2.$$

The bounds on R follow from Taylor's formula along radial segments, since the product is convex and contains zero. Applying the Poincaré inequality to each coordinate, we have $\mathbb{E}|w|^2 \le 2D + |\mu|^2$. Integrating $\nabla(\chi e^{-V_j})$ over Ω_j, which gives zero because χ has compact support in the interior, even at the corners of the product, we obtain

$$\ell\, \mathbb{E}\chi = \mathbb{E}\nabla\chi - \mu + \mathbb{E}[(1-\chi)w] - \mathbb{E}[\chi \nabla R(w)]. \tag{35.8}$$

Since $\nabla\chi$ is supported on $\{\chi < 1\}$, we have $|\mathbb{E}\nabla\chi| \le C\sqrt{D}\, q_j^{\mathrm{cut}}/p_j$. By the Cauchy–Schwarz inequality, $|\mathbb{E}[(1-\chi)w]| \le \sqrt{q_j^{\mathrm{cut}}}(\mathbb{E}|w|^2)^{1/2}$, and $|\mathbb{E}[\chi\nabla R(w)]| \le \epsilon(\mathbb{E}|w|^2)^{1/2}$. Since $\mathbb{E}\chi \ge 1/2$, we obtain

$$|\ell| \le C_m\{\sqrt{\gamma} + \epsilon + \sqrt{q_j^{\mathrm{cut}}}\} \le C_m\sqrt{\gamma}(1 + |\log\gamma|).$$

Here we used the bound on the conditional mean μ from Step 1, not that μ vanishes. A bound on $D^2V_j - I$ alone would not control the linear coefficient ℓ.

Step 4: weighted density comparison. Let $U = V_j - V_j(0)$. When $\epsilon \le 1/4$, the preceding bounds imply $U(w) \ge |w|^2/4 - C|\ell|^2$. After a further reduction of γ_m we have $|\ell| \le 1$, and the exponential-difference identity (33.6), used in the proof of Corollary 33.4, gives on Ω_j

$$\bigl|e^{-U(w)} - e^{-|w|^2/2}\bigr| \le C\bigl(|\ell||w| + \epsilon|w|^2\bigr)e^{-c|w|^2}.$$

We extend e^{-U} by zero outside Ω_j, integrate against $1 + |w|^s$, bound the Gaussian mass of the complement of Ω_j by Step 2, and compare the two normalizing constants. Since the Gaussian integral is $(2\pi)^{D/2}$, both normalizing constants have a common positive lower bound at this fixed D. Then the normalized conditional density r_j of w_j satisfies

$$\int (1 + |w|^s)|r_j(w) - \phi_D(w)|\, dw \le C_{m,s}\sqrt{\gamma}(1 + |\log\gamma|), \tag{35.9}$$

where ϕ_D is the standard Gaussian density in $\mathbb{R}^D$. In particular all fixed uncentered conditional moments are bounded. The same bound holds for $x = w - \mu$ against $\phi_D(x)$. Indeed, translation changes the polynomial weights by a bounded factor when $|\mu| \le 1$, and

$$\phi_D(x+\mu) - \phi_D(x) = \int_0^1 \mu \cdot \nabla\phi_D(x + u\mu)\, du$$

has weighted L^1 norm at most $C_{D,s}|\mu|$. No derivative of the mean in the earlier variables is needed.

Step 5: the coarse variable. We condition on y in the coarse tube. The weighted comparison of the conditional density of z_0 with $\phi_{d_{\mathrm{reg}}}$ is given by Corollary 33.4, together with the extension to the whole tube established in its proof. That proof verifies the coercivity of the phase on the whole tube, the bounds on the amplitude, and the lower bound on the normalizing constant of the tube for the coarse family of Proposition 33.12, uniformly in K. The chart and amplitude checks recorded above supply the data on the subcharts. The comparison error for the coarse variable is $C_{n,s}\sqrt{\gamma}$. As noted after (35.6), the restriction to the coarse tube leaves the finer kernels unchanged; this also follows from the fixed-target statement of Theorem 13.8. All bounds of Steps 1–4 hold under this conditional probability.

Step 6: substitution of Gaussian variables. Let $X_0 = z_0$ and $X_j = w_j - \bar{w}_j$ for $1 \le j \le m$, and let F be a Borel function of $(X_0, \dots, X_m)$ bounded by $1 + |(X_0, \dots, X_m)|^s$. For $0 \le j \le m$ let I_j be the expectation of F when the arguments $0, \dots, j$ are $X_0, \dots, X_j$ and the arguments $j+1, \dots, m$ are replaced by independent standard Gaussian vectors, independent of everything else. For $j \ge 1$, integrating out these Gaussian vectors gives a function of the current variable bounded by

$$C_{m,s}\big(1 + |(X_0, \dots, X_{j-1})|^s + |X_j|^s\big).$$

For fixed values of the earlier variables, (35.9) in its translated form bounds the change from a Gaussian j-th argument to X_j by $C_{m,s}\sqrt{\gamma}(1 + |\log\gamma|)$ times $1 + |(X_0, \dots, X_{j-1})|^s$. Taking expectations, we find that $|I_j - I_{j-1}|$ is at most $C_{m,s}\sqrt{\gamma}(1 + |\log\gamma|)$ times $\mathbb{E}(1 + |(X_0, \dots, X_{j-1})|^s)$. This expectation is uniformly bounded: the moments of the coarse variable follow from the corollary of Step 5, each conditional variable has all fixed moments by Step 4, and finite-dimensional norm inequalities bound the whole vector. The centered variables are measurable functions of the earlier variables, so no inverse transformation of a Gaussian argument is required. Finally the coarse argument in I_0 is replaced by a Gaussian one using Step 5. Summing the finitely many differences proves the joint weighted comparison. The Gaussian arguments enter only F, never a reconstruction, a conditional mean, or a conditional kernel.

Step 7: the linear image. The exact Gaussian decomposition of Proposition 33.12 includes the bound

$$\|A_{0,K}\|_{\mathrm{HS}}^2 + \sum_{j=1}^{K} \|A_{j,K}\|_{\mathrm{HS}}^2 \le C. \tag{35.10}$$

This implies that the linear map $(X_0, \dots, X_m) \mapsto A_{0,K}X_0 + \sum_{j=1}^{m} A_{j,K}X_j$ has operator norm bounded uniformly in K and m. Applying the joint comparison to F_y composed with this map, which has polynomial growth with a fixed bound, proves the stated estimate. The Gaussian image has covariance $\Sigma_m^K(y)$ by definition, and the argument covers singular covariances. The same norm bound and the moments of each argument give the bounds

on all moments of the partial sums. The kernels and the conditional means are Borel, as normal integrals of smooth conditional densities, and the Borel hypothesis on F_y together with Fubini's theorem justifies every substitution of a test function and every integration in y. □

35.3 Finite-scale reconstructions and the normal covariance

The proofs in this subsection use coordinates $q_{j,r} = g_{j,r} w_{j,r}$ on fixed finite lattices; as coordinates they do not depend on γ, and all domains below are chosen independently of γ and of the number of later steps.

Theorem 35.9. *Fix $m \geq 0$, and express the triple Ξ_m^K as a function of the coarse coordinates and of the coordinates $q_{j,r} = g_{j,r} w_{j,r}$, $1 \leq j \leq m$, of the first m steps. There exist a neighborhood of its flat inputs (the inputs whose coarse field is flat and whose coordinates of the first m steps vanish) and a constant C_m such that, on this neighborhood,*

$$\sup_{K \geq m} \|\Xi_m^K\|_{C^{1,1}} \leq C_m.$$

The derivatives are taken in smooth rooted spanning-tree gauge coordinates; no derivative of a measurable choice of gauge representatives is taken. The neighborhood and C_m may depend on m and on the fixed lattices of the first m steps, but not on K, on γ, or on the preceding and later classical depths.

Moreover, Ξ_m^K extends to a fixed neighborhood of the set of all inputs $(V_0, 0)$ with $V_0 \in \mathcal{U}_n$ flat, and there exists a function ω_m with $\omega_m(s) \to 0$ as $s \downarrow 0$ such that, for every flat V_0,

$$\max_i \|\Xi_{m,i}^K(V, z_{\leq m}) - \mathrm{Hol}_i(V_0)\|_{\mathrm{op}} \leq \omega_m\big(d_{\mathrm{in}}((V, z_{\leq m}), (V_0, 0))\big).$$

Here $z_{\leq m}$ denotes the physical coordinates $q_{j,r}$ of the first m steps. For this comparison with the zero section, $d_{\mathrm{in}}((V, z_{\leq m}), (V_0, 0))$ is the sum of the distance between the based gauge classes of V and V_0 in one fixed product Riemannian metric on the compact coarse quotient in rooted spanning-tree coordinates and the Euclidean norm of $z_{\leq m}$. The triangular orthogonal transitions preserve this norm. This distance to a flat input is uniformly equivalent, on the finite cover used below, to the corresponding local product distances, with constants independent of K. Every chart uses the same gauge frame at the basepoint. Finally, $\Xi_m^K(V_0, 0) = \mathrm{Hol}(V_0)$ for every finite $K \geq m$ and every flat V_0, including those whose holonomy triple is one of the central triples $(\pm I, \pm I, \pm I)$.

Proof. Fix m. All parameter norms in this proof are norms on the coarse lattice and on the finite lattices of the first m steps; their dimensions and the constants comparing different norms may depend on m, n, M, J. We write u for these parameters. As arguments of the successive Hessian normalization we use the coordinates $q_{j,r} = g_{j,r} w_{j,r}$, which do not depend on γ. The changes of variables between these coordinates and the free logarithmic

coordinates have uniformly bounded charts and inverse charts on each fixed set of first m steps. On the coarse lattice we use analytic link coordinates or rooted spanning-tree gauge coordinates, keeping the initial gauge frame at the basepoint. The smooth normal atlas is composed with these coordinates only after the real derivative bounds have been obtained.

Step 1: a domain for the inputs. By Lemma 14.9 and the condition $2\pi/n < r_{\mathrm{flat}}/4$, each of the first m zero-fluctuation steps at a flat input is flat and lies strictly inside every margin of the constrained minimizers, of the inverse of the block averaging constraint, of the Haar disintegration densities, and of the gauge charts. Each fixed finite composition of reconstruction maps has uniformly bounded weighted analytic coefficients on the nested domains of Theorem 15.12; in particular its first derivative is bounded there independently of the later depth. Let $F_m^K(u)$ be the field at the end of the m-th step, which is the input of the zero-fluctuation reconstruction of the later steps. At each flat chart center u_0 we start from a small commuting representative with link logarithm of norm at most $2\pi/n$, keeping the gauge frame at the basepoint. The reconstruction maps preserve flatness and the based cycle holonomies. Their zero-fluctuation maps P_j satisfy $P_j(0) = 0$ and $\|DP_j\| \le a_{\mathrm{map}} < 1$ on the convex working ball, by Lemma 15.15 and Definition 34.1. Integrating along segments and iterating therefore gives

$$\|F_m^K(u_0)\|_\infty \le a_{\mathrm{map}}^m \frac{2\pi}{n} \le \frac{2\pi}{n} < \frac{r_{\mathrm{flat}}}{4} < \frac{r}{4}.$$

This bound applies to the actual reconstructed representative, which may differ from the constant commuting refinement by a based gauge transformation. The derivative bound just obtained gives a constant $C_{\mathrm{in},m}$ for $D_u F_m^K$, uniform in K and in the depths, before any of the later steps is applied. Let U_{larger} be a real neighborhood of the flat inputs, of radius ϵ_m, such that $C_{\mathrm{in},m}\epsilon_m < r/4$ and the derivative bound times ϵ_m is less than half of every image margin of the first m steps. By the mean value formula, we have, in these fixed charts,

$$\sup_{K \ge m,\, u \in U_{\mathrm{larger}}} \|F_m^K(u)\|_\infty \le r/4 + C_{\mathrm{in},m}\epsilon_m < r/2 < r, \tag{35.11}$$

and every intermediate image stays in its coordinate domain. The same holomorphic coefficient bounds allow a larger complex polydisc, on which the complex image has norm below $3r/4$, and two smaller polydiscs. Indeed, the complex derivative bound times the added radius may be chosen below $r/4$ and below half of the remaining image margins. A finite cover of the compact set of flat inputs gives common positive radii and margins. These choices depend on m and on the coefficient bounds, but not on K, on γ, or on the number of later steps. In particular, (35.11) verifies the bound $\|\theta_0\|_\infty \le r$ on the initial field required in Lemma 34.4, independently of the larger radius and of the cutoffs.

Step 2: the flow data of the zero-fluctuation reconstruction. Let $a = \ell_K = (L/n)B^{-K}$. Let $\mathcal{E}_K(u)$ be the linear and quadratic flow data, in the sense of Definition 4.8, of the zero-fluctuation reconstruction from $F_m^K(u)$ down to the mesh a, including both weighted components and their first differences. We apply parts (1)–(3) of Lemma 34.4, including the analytic coefficient estimates and the finite-scale composition clause in part (3), starting at the mesh ℓ_m with this finite input. Its hypotheses are the bound (35.11), the contraction condition $a_{\mathrm{map}}\sqrt{B} < 1$, the complex domain of Step 1, and zero fluctuation variables at all later steps; they hold by Step 1 and Definition 34.1. We obtain a fixed smaller complex domain, a bound R_m, and a number $0 < \tau < 1$ such that, for every fixed $l \geq m$,

$$\sup_{K\geq l,\,u} \|\mathcal{E}_K(u)\|_{\mathrm{fl},a} \leq R_m, \qquad \sup_{K\geq l,\,u} \|\mathcal{E}_K(u) - \mathcal{E}_{K,l}(u)\|_{\mathrm{fl},a} \leq C_m(1+l)^c\tau^{l-m}. \tag{35.12}$$

Here $\mathcal{E}_{K,l}$ is the flow data of the field at the end of the l-th step, embedded on the same finest lattice, and $\|\cdot\|_{\mathrm{fl},a}$ is the sum of the four weighted value and first-difference norms of Definition 4.8. One can take $\tau = \max\{a_{\mathrm{map}}B^{1-\alpha_{\mathrm{fl}}}, \mathfrak{q}\}$; both terms are less than one, since $\alpha_{\mathrm{fl}} > 1/2$, $a_{\mathrm{map}}\sqrt{B} < 1$, and $\mathfrak{q} = Ba_{\mathrm{map}}^2 < 1$. Starting at ℓ_m only changes C_m and R_m, and the smaller complex domain does not shrink with the number of later steps.

Step 3: parameter derivatives of the flow data. Let $U_0 \Subset U_1$ be real bounded parameter domains inside that complex domain, and let $s_m > 0$ be a common complex distance from U_1 to its boundary. Applying Cauchy's integral formula for Banach-valued holomorphic functions in each input coordinate, we obtain

$$\sup_{K,\,u\in U_1} \sum_{r=0}^{2} \|D_u^r\mathcal{E}_K(u)\|_{\mathrm{fl},a} \leq C_{\dim u}R_m\big(1 + s_m^{-1} + s_m^{-2}\big) = R_{2,m}. \tag{35.13}$$

For a mixed second derivative we use two coordinate circles of radius at most $s_m/2$, which only changes the finite-dimensional constant. The estimate covers the first-difference components as well as the values. Complex continuation is used only for the reconstruction and the flow data, not for the nonlinear Wilson flow.

Step 4: real variations of the Wilson flow. We apply Theorem 25.13 with the parameter domains U_0, U_1, dimension $\dim u$, bound $R_{2,m}$ on the flow data, horizon T, and smallest observation time $t_* = \theta L^2/2$. The flow data are twice continuously differentiable by (35.13) and are bounded in the norms required there. Every damping gauge transformation starts at the identity in the same initial gauge frame at the basepoint. The early parameter variations given by that theorem satisfy the first-difference bound

$$C_m\big\{(s+a^2)^{-(\alpha_{\mathrm{fl}}+1)/2} + (s+a^2)^{-\alpha_{\mathrm{fl}}} + (s+a^2)^{\kappa_{\mathrm{fl}}-1/2}\big\},$$

which is integrable in s near zero, uniformly in a, since $(\alpha_{\mathrm{fl}}+1)/2<1$, $\alpha_{\mathrm{fl}}<1$, and $\kappa_{\mathrm{fl}}>0$. The condition keeping the links in the compact chart is

$$a\sup_{s\le T_{R_{2,m}}}\|A(s)\|_\infty\le C_m\{a^{1-\alpha_{\mathrm{fl}}}+a^{2-2\alpha_{\mathrm{fl}}}+a(T_{R_{2,m}}+a^2)^{\kappa_{\mathrm{fl}}}\}\longrightarrow 0\qquad(a\to 0).$$

At a positive early restart time the same theorem gives smooth bounds at the restart and a finite reference family of smooth DeTurck solutions, joined by fixed gauge transformations, on an open neighborhood of the compact family of smooth restart configurations. We check its hypothesis that this family has a compact smooth closure, using the gauge transformations at the restart and the bounds on all fixed spatial differences given by the smoothing at positive time. For each restart array, the coefficients of its periodic trigonometric interpolant are bounded by $C_q(1+|k|)^{-q}$ for every fixed q. To see this, we apply q differences in a coordinate with maximal $|k_i|$ and use $|a^{-1}(e^{2\pi i a k_i/L}-1)|\ge c_L|k_i|$ on the centered Brillouin representatives, and at $k=0$ we use the bound on the values. The tails of the coefficients therefore tend to zero uniformly in each C^s norm, taking $q>s+3$. Finite Fourier truncations have compact sets of coefficients, and successive finite nets in these truncations, followed by a diagonal subsequence, give a compact smooth closure, uniformly over the compact parameter domains. Taking real parts at the unpaired Nyquist modes preserves real arrays and the estimates. This verifies the compactness hypothesis. We use the fixed gauge transformations of that theorem at the restart for the varying input, keeping their conjugations at the basepoint. The theorem then bounds the first and second real parameter derivatives of the three cycle matrices at time θL^2 by C_m. Since the initial configurations are the finest configurations of the zero-fluctuation reconstruction that define Ξ_m^K, and the lattice flow time is $4\theta L^2/a^2$, this bounds the derivatives of the map Ξ_m^K itself. The finitely many meshes for which the smallness conditions of that theorem fail are covered by its clause on compact smooth ordinary differential equations. Integrating the second derivative on smaller convex coordinate balls, we obtain $\sup_{K\ge m}\|\Xi_m^K\|_{C^{1,1}}\le C_m$.

Step 5: all flat configurations. We repeat Steps 1–4 in ambient charts given by rooted spanning-tree gauges, covering the set of flat configurations, which is compact on the coarse lattice. Its quotient by gauge transformations fixed at the basepoint is parametrized by finitely many matrices on the non-tree links, because rooted spanning-tree gauges are unique and have no residual stabilizer. Hence smooth ambient charts exist also at flat configurations with central holonomies, although the set of flat configurations is singular there.

For each flat chart center V_*, fix a periodic coarse gauge transformation h_* with $h_*(0)=I$ taking V_* to a small constant commuting representative, as in Lemma 14.9, and use this same fixed transformation on its whole ambient neighborhood. The strict margins at the center from Step 1 and the uniform coefficient bounds for the first m

steps give a neighborhood independent of K. On an overlap, transport both the coarse field and the first m fluctuation coordinates by the triangular transitions $q'_{j,r} = O_{j,r} q_{j,r}$ of Theorems 13.4 and 14.3, where $O_{j,r}$ depends on the coarser inputs. By (15.28), the corresponding reconstructions are related by a fine gauge transformation whose restriction to the coarse vertices is the original coarse transformation. It therefore equals I at the common basepoint. Iterating this identity transports the whole prefix, and every subsequent zero fluctuation coordinate remains zero. The finest zero-fluctuation configurations and their Wilson flows thus have identical based cycle matrices on the overlap. These transitions also preserve the cutoff balls, the normalized conditional kernels, and the selected event. The uniform coefficient bounds on the first m fixed lattices control the finite input charts; exact equality of the based matrices requires no derivative of a gauge transformation composed through the later steps.

We take finitely many larger charts covering the compact flat set and smaller charts with compact closures inside them. Steps 1–4 then give a common constant C_m and a common positive neighborhood on this finite cover. This construction chooses each center gauge once and uses neither a logarithm of a varying central matrix nor a regular normal Hessian.

Step 6: exact values and the modulus. At a flat input with zero fluctuation variables, Lemma 14.9 identifies the successive constrained minimizers up to based gauge transformations with commuting flat refinements, which preserve the cycle holonomies. By Theorem 15.12, the gauge transformations of the reconstruction maps are the identity at the roots and therefore preserve these values. The gradient of the Wilson action vanishes on the refinements, so the Wilson flow does not move them. We conclude that $\Xi_m^K(V_0, 0) = \mathrm{Hol}(V_0)$ for every finite K, for central as well as regular flat configurations.

It remains to obtain the modulus from the uniform first-derivative bounds. We take coordinate balls whose doubled closures lie in the larger charts, cover the set of flat configurations by finitely many of the corresponding half-balls, and let $\varrho_m > 0$ be a common smaller radius. An input and a flat zero-section input at distance at most ϱ_m lie in one of these charts, where the local product distance is bounded by a fixed multiple of d_{in}. The norm of the transported fluctuation input is unchanged, so the first-derivative estimate and the fixed gauge frame at the basepoint bound their matrix difference by $C'_m d_{\mathrm{in}}$. For larger distances the difference is at most two. After increasing C'_m so that $C'_m \geq 2/\varrho_m$, the function $\omega_m(s) = \min(2, C'_m s)$ is a common modulus. On overlaps the gauge transformations keep the same initial gauge frame at the basepoint. This proves the statement near all flat configurations. □

Theorem 35.10. *For y in the regular patch P^{++} and $N = nB^K$ we have the exact identity*

$$\Sigma_K^K(y) = \Sigma_\theta^N(y),$$

where $\Sigma_\theta^N(y)$ is the normal covariance, divided by γ, of the linearized flowed cycle logarithms under the Gaussian measure of the lattice Maxwell form of the finest lattice around the flat configuration y (Theorem 33.11 and Proposition 33.12). There exists C, independent of m, K, and y, such that

$$0 \le \Sigma_\theta^N(y) - \Sigma_m^K(y) \le C\delta_m\, I_{\mathcal{N}_y} \qquad (K \ge m).$$

Put $\widetilde{\Sigma}_m^K(y) = \Sigma_\theta(y)^{-1/2}\Sigma_m^K(y)\Sigma_\theta(y)^{-1/2}$. With constants independent of m, K, and y,

$$\sup_y \|\widetilde{\Sigma}_m^K(y) - I\|_{\mathrm{op}} \le C\delta_m + C(nB^K)^{-2}, \qquad K \ge m. \tag{35.14}$$

The continuum covariance satisfies $cI \le \Sigma_\theta \le CI$ and $\Sigma_\theta(y) \ge |\boldsymbol{\vartheta}|^{-2}e^{-2\theta|\boldsymbol{\vartheta}|^2} I_{\mathcal{N}_y}$. For every fixed m, in the finite smooth normal atlas,

$$\sup_{K\ge m} \sum_{j=0}^{m} \big(\|A_{j,K}\|_\infty + \mathrm{Lip}_y\, A_{j,K}\big) \le C_m. \tag{35.15}$$

Finally, if $H_K^E = D_v^2 E_{JK}(V(y,0))$, then $H_K = nH_K^E$ and

$$A_{0,K}A_{0,K}^{\mathsf{T}} = n^{-1}T_{0,K}(H_K^E)^{-1}T_{0,K}^{\mathsf{T}}.$$

Proof. The exact identity is the Gaussian quadratic identity of Proposition 33.12, applied to the linear response matrices of Definition 35.7. Its hypotheses concern the same reconstruction maps, the Hessians at the flat background, the coarse normal variable, and the derivative of the flowed cycle logarithms. In particular,

$$\Sigma_\theta^N - \Sigma_m^K = \sum_{j=m+1}^{K} A_{j,K}A_{j,K}^{\mathsf{T}} \ge 0. \tag{35.16}$$

If G_j are independent standard Gaussian vectors, this positive matrix has trace

$$\mathrm{tr}\big(\Sigma_\theta^N - \Sigma_m^K\big) = \mathbb{E}\Big|\sum_{j=m+1}^{K} A_{j,K}G_j\Big|^2. \tag{35.17}$$

To bound this trace we identify our matrices with those of Lemma 34.12. Its flat coarse field V_y is $V(y,0)$ here. Its coordinates $q_{j,r}$ are our $g_{j,r}w_{j,r}$, taken before the Hessian normalization of each step at the flat background; that normalization and the zero-fluctuation reconstruction of all finer steps are included in the differentiated reconstruction on both sides. The two response matrices therefore agree:

$$\mathsf{A}_{j,K} = P_{\mathcal{N}_y}D_{q_j}(\mathrm{Log}\circ\mathscr{H}_K)\ \mathrm{diag}_{r,e}(g_{j,r}/\sqrt{\gamma}) = \gamma^{-1/2}P_{\mathcal{N}_y}D_{w_j}(\mathrm{Log}\circ\mathscr{H}_K)\big|_{\mathrm{flat}} = A_{j,K}.$$

The diagonal contains the coupling of every intermediate scale and every scalar color component. Its entries do not depend on γ, since each $g_{j,r}^2/\gamma$ is the reciprocal of a fixed

lattice side. Both response maps use the Wilson flow at time θL^2 (lattice time $4\theta N^2$), the same based coordinate cycles, and the same initial gauge frame at the basepoint. Our patch is regular and has the margins of the identity chart and of the normal charts required by that lemma. Its flat refinements satisfy the coordinate-domain conditions by the condition $2\pi/n < r_{\mathrm{flat}}/4$ and Lemma 14.9, and $D\,\mathrm{Log}$ is the bounded differential of the identity chart. Thus every hypothesis of Lemma 34.12 holds, and we obtain

$$\sup_{K\geq m,\,y} \mathbb{E}\Big| \sum_{j=m+1}^{K} A_{j,K}(y) G_j \Big|^2 \leq C\delta_m. \tag{35.18}$$

Only a linear map acts on the independent Gaussian vectors, and the bound holds at each fixed flat y; it is not deduced from an averaged estimate for the interacting measure. A positive semidefinite matrix is bounded above by its trace times the identity, as one sees by diagonalizing it. Combining (35.16), (35.17), and (35.18), we obtain the bound $0 \leq \Sigma_\theta^N - \Sigma_m^K \leq C\delta_m I$.

We next prove the derivative bounds at finite cutoff. The spectral bounds $c_n I \leq H_K \leq C_n I$ follow from Proposition 33.12. By the uniform bounds on the derivatives of the effective classical action in Theorem 10.11, composed with the fixed compact normal atlas, we have $\|H_K\|_{C^1(y)} \leq C_n$; only three derivatives of the action are needed. For the steps $j \geq 1$ we use the coordinates $q_{j,r} = g_{j,r} w_{j,r}$ of Theorem 12.13. The Hessian normalization of each step is part of the map in these coordinates and is preserved by Theorem 15.12; these are the coordinates used in the proof of Theorem 35.9, and no limit of a normalization matrix or of its derivative is needed.

For the inverse square root, let S be a symmetric matrix with $cI \leq S \leq CI$, choose $C' > C$ and write $S = C'(I - Q)$, so that $\|Q\| \leq 1 - c/C' < 1$. The binomial series for $(I-Q)^{-1/2}$ and its term-by-term derivative converge uniformly: the derivative of Q^k is a sum of k terms bounded by $\|Q\|^{k-1}\|DQ\|$, and the coefficients of the series are bounded. This bounds $S^{-1/2}$ and its first tangential derivative without constructing any limit. Applied to $S = H_K$, it bounds $H_K^{-1/2}$ and its first tangential derivative.

At the flat input, the derivative of $\mathrm{Log} \circ \Xi_m^K$ in the variables of a step $j \leq m$ equals the corresponding derivative of $\mathrm{Log} \circ \mathscr{H}_K$ with all other fluctuation variables set to zero. Indeed, setting the later variables to zero does not change the derivative in an earlier variable at the flat refinements. For $j \geq 1$ this gives the identity

$$A_{j,K} = P_{\mathcal{N}_y} D_{q_j} (\mathrm{Log} \circ \Xi_m^K)\big|_{\mathrm{flat}}\, \mathrm{diag}_{r,e}(g_{j,r}/\sqrt{\gamma}),$$

with a fixed diagonal independent of y, K, and γ. Using the $C^{1,1}$ bounds of Theorem 35.9 in these coordinates, the bounds on the first two derivatives of the logarithm on the identity chart, and the smoothness of the embedding of the flat configurations, we bound each $A_{j,K}$ and its first tangential derivative. For $A_{0,K}$ we compose the corresponding coarse derivative with $H_K^{-1/2}$ and use the preceding bounds. Each step carries exactly one

Hessian normalization, already included in the differentiated map, and the derivatives of the orthonormal frames and of the projections in the finite atlas are bounded. This proves (35.15), and hence a Lipschitz bound for Σ_m^K at each fixed m.

By Theorem 33.11 we have $\|\Sigma_\theta^{nB^K} - \Sigma_\theta\| \le C(nB^K)^{-2}$, together with the spectral bounds $cI \le \Sigma_\theta \le CI$. Adding this lattice error to the bound on the positive remainder proved above and conjugating by $\Sigma_\theta^{-1/2}$, we obtain (35.14). By the same theorem, the constant off-diagonal color modes alone contribute $e^{-2\theta|\vartheta|^2}|\vartheta|^{-2} I_{\mathcal{N}_y}$ to $\Sigma_\theta(y)$; since all other contributions are positive semidefinite, this gives the stated lower bound.

Finally, $H_K = D_v^2(nE_{JK}) = nH_K^E$, so

$$A_{0,K}A_{0,K}^{\mathsf{T}} = T_{0,K}H_K^{-1}T_{0,K}^{\mathsf{T}} = n^{-1}T_{0,K}(H_K^E)^{-1}T_{0,K}^{\mathsf{T}}.$$

In particular the contribution of the coarse normal variable is included in Σ_m^K for every m. □

35.4 Finite-scale moments and localization

This subsection combines the Gaussian comparison of Lemma 35.8 with the deterministic results of Section 35.3. Two facts about the coarse lattice drive the argument. First, the normalizing constant $\widehat{Z}$ is of order γ^{3n^3-1} (Theorem 33.10). This is also the order of the Laplace mass of any fixed regular patch, so a fixed regular neighborhood has probability bounded below, uniformly in γ. Second, the action has a positive gap at a fixed distance from the flat configurations. Together with the uniform modulus of Theorem 35.9, this shows that, with overwhelming probability, an output in a fixed regular tube comes from a coarse field near the corresponding regular flat configurations, and not from a neighborhood of other flat configurations, such as those with central holonomies.

Theorem 35.11. *Fix nested tubes as in Section 35.1. For every fixed $m \ge 0$ there exist $K_0(m)$ and $\gamma_m > 0$ such that the following statements hold uniformly for $K \ge K_0(m)$ and $0 < \gamma \le \gamma_m$.*

1. *Fix bounded Borel functions on $\mathrm{SU}(2)^3$ that agree with the local maps $H \mapsto b(\operatorname{Log} H)$ and $H \mapsto y(\operatorname{Log} H)$ on a smaller fixed regular output tube; we denote them again by $b(\operatorname{Log} H)$ and $y(\operatorname{Log} H)$. Under the restriction of ν to a larger coarse regular tube, normalized by its own mass, we have for every fixed finite r'*

$$\frac{b(\operatorname{Log}\Xi_m^K)}{\sqrt{\gamma}} = A_{0,K}(y)z_0 + \sum_{j=1}^{m} A_{j,K}(y)w_j + \mathcal{R}_{K,m},$$

$$\|\mathcal{R}_{K,m}\|_{L^{r'}} \le C_{m,r'}\sqrt{\gamma}(1+|\log\gamma|)^2, \qquad \|y(\operatorname{Log}\Xi_m^K) - y\|_{L^{r'}} \le C_{m,r'}\sqrt{\gamma}(1+|\log\gamma|).$$

The same integrated estimates hold for the expectation under ν with any bounded measurable multiplicative factor supported in the coarse tube. Replacing w_j by $w_j - \bar{w}_j$ changes $\mathcal{R}_{K,m}$ by a term of norm $O_m(\sqrt{\gamma})$ in $L^{r'}$.

2. *Let w be any fixed real smooth tangential weight compactly supported in the relative interior of P, and let χ_w be a fixed smooth normal cutoff supported in $\{|b| < b_{\mathrm{o}}\}$ and equal to one near the zero section. Set $\widetilde{w}(y+b) = w(y)\chi_w(b)$, with the fixed zero extension; for this part, nonnegativity and invariance of w and χ_w are not required. Choose a smooth normal cutoff ζ equal to one near the support of χ_w and supported in $\{|b| < b^+\}$, and set $w^+(H) = w(y(\operatorname{Log} H))\zeta(b(\operatorname{Log} H))$, extended by zero. For $l = 0, 1, 2$,*

$$\begin{aligned}&\mathbb{E}_\nu\Big[\widetilde{w}(\operatorname{Log}\Xi_m^K)\Big(\frac{q_*(\operatorname{Log}\Xi_m^K)}{\gamma}\Big)^l\Big]\\&\qquad = \mathbb{E}_\nu\Big[w^+(\Xi_m^K)\,\mathfrak{m}_l\big(\widetilde{\Sigma}_m^K(y(\operatorname{Log}\Xi_m^K))\big)\Big] + O_{m,w}\big(\sqrt{\gamma}(1+|\log\gamma|)^{k_m}\big),\end{aligned}$$

where k_m is a fixed exponent, $\widetilde{\Sigma}_m^K = \Sigma_\theta^{-1/2}\Sigma_m^K\Sigma_\theta^{-1/2}$, and

$$\mathfrak{m}_0(C) = 1, \qquad \mathfrak{m}_1(C) = \frac{1}{2}\operatorname{tr} C, \qquad \mathfrak{m}_2(C) = \frac{1}{4}\big\{(\operatorname{tr} C)^2 + 2\operatorname{tr}(C^2)\big\}. \tag{35.19}$$

Only values in the support of w^+ are used on the right.

3. *For every fixed r', the normal moments of the coarse variable satisfy $\mathbb{E}_\nu[|v|^{r'}\mathbf{1}_{\{V \text{ in the coarse tube}\}}] \le C_{r'}\gamma^{r'/2}$, and for every fixed larger regular output tube $\mathcal{V}$,*

$$\mathbb{E}_\nu\big[|b(\operatorname{Log}\Xi_0^K)|^{r'}\mathbf{1}_{\{\Xi_0^K\in\mathcal{V}\}}\big] \le C_{r'}\gamma^{r'/2},$$

where $C_{r'}$ depends on $\mathcal{V}$ but not on m.

4. *There exists $c_* > 0$, independent of K, m, and γ, such that, after possibly decreasing γ_m,*

$$\nu\{\psi(\operatorname{Log}\Xi_m^K) = 0\} \ge c_*, \qquad \nu\{\psi(\operatorname{Log}\Xi_m^K) \ge b_*\} \ge c_*.$$

5. *For every compact regular output tube $\mathcal{O}$ over a patch in the relative interior of P^{++} there exists a larger coarse regular tube $\mathcal{T}^+$ such that, for every $q > 0$,*

$$\nu\{\Xi_m^K \in \mathcal{O},\ V \notin \mathcal{T}^+\} \le C_{m,q}\gamma^q.$$

This excludes contributions from neighborhoods of all other flat configurations, including those with central holonomies, as well as from nonflat coarse fields.

Throughout, the normalizing constant satisfies $c_n\gamma^{3n^3-1} \le \widehat{Z} \le C_n\gamma^{3n^3-1}$.

Proof. The cases of finitely many small K are covered by the bounds of Section 35.3, so we may take $K_0(m) = m$; any larger choice is also allowed. We first record the Laplace estimates on the coarse lattice (Step 1) and prove the localization (Step 2), then construct the bounded output coordinates and a good event (Steps 3 and 4), prove the expansion of part (1) (Steps 5–7), and finally prove parts (2), (3), and (4) (Steps 8–10).

Step 1: Laplace estimates on the coarse lattice. By the first part of Proposition 33.12, with its exact conditional normalizing constant, we have

$$\mathbb{E}\big[|v|^r \mid y,\ V \text{ in the coarse tube}\big] \le C_r \gamma^{r/2} \qquad (r < \infty). \tag{35.20}$$

To obtain the mass of a patch, we apply the additive Laplace formula of the same lemma with a nonnegative smooth tangential bump h, supported strictly inside the patch, and a normal cutoff equal to one on a smaller ball. Positivity and the bounds on the amplitude and on the Hessian give

$$c_h \gamma^{3n^3-1} \le \widehat{Z}\, \mathbb{E}_{\widehat{\nu}}[h(y)\chi(v)] \le C_h \gamma^{3n^3-1}$$

for all sufficiently small γ, uniformly in K; here the additive error $O_h(\sqrt{\gamma})$ has been absorbed into the positive leading integral, and we used $(2\pi\gamma)^{d_{\mathrm{reg}}/2} = (2\pi\gamma)^{3n^3-1}$. By Theorem 33.10, $\widehat{Z}$ is bounded above and below by constant multiples of the same power γ^{3n^3-1}. Dividing the two bounds, we find that every fixed nonempty open regular subpatch has coarse probability bounded below by a positive constant, uniformly in K and γ. No convergence of the tangential amplitude is used. Integrating (35.20) over the coarse tube proves the first assertion of part (3), both under the normalized restriction to the tube and with any bounded multiplicative factor under ν.

Step 2: localization (part (5)). We work on the compact quotient of the coarse configuration space by gauge transformations fixed at the basepoint. The zero set $\mathcal{F}_n$ of E_0 is the set of flat configurations, as in Section 36. For any fixed $\eta > 0$, compactness and continuity give

$$c_\eta = \inf\{\kappa_n E_0(V) : \operatorname{dist}(V, \mathcal{F}_n) \ge \eta\} > 0,$$

since otherwise a convergent sequence in this closed set would have a flat limit in the set. Since the centered effective interaction is bounded, $nE_{JK,\mathrm{sat}} \ge \kappa_n E_0$, and the product Haar measure has mass one, the lower bound on $\widehat{Z}$ implies

$$\widehat{\nu}\{\operatorname{dist}(V, \mathcal{F}_n) \ge \eta\} \le C_n \gamma^{-(3n^3-1)} e^{-c_\eta/\gamma} \le C_{n,\eta,q}\gamma^q \qquad (q > 0). \tag{35.21}$$

The last bound follows by maximizing $x^{3n^3-1+q}e^{-c_\eta x}$ over $x \ge 1/\gamma_0$.

Let $\mathcal{O}$ be the given compact output tube. The flat holonomies meeting a sufficiently small neighborhood of $\mathcal{O}$ stay away from the central triples and from the boundary of the identity chart. We may therefore choose a strictly larger compact set of regular commuting holonomies in the relative interior of P^{++}, and a coarse regular tube $\mathcal{T}^+$ over it. On the compact complement, within the set of flat configurations, of the flat configurations with holonomies in this larger patch, the distance of the flat holonomies from $\mathcal{O}$ is at least some $a_0 > 0$. By the modulus of Theorem 35.9, there exists $\eta_m > 0$ such that a change of the input by less than $2\eta_m$ changes every flat output by less than

$a_0/2$. For the finitely many steps $j \le m$, the coordinates of the first m steps satisfy deterministically

$$|z_{\le m}| \le C_m \sqrt{\gamma}(1 + |\log \gamma|). \tag{35.22}$$

Indeed each coordinate is a fixed ratio times $\sqrt{\gamma}\, w_{j,r}$, followed by the bounded Hessian normalization on these finite lattices, and $|w_{j,r}| \le p_j$. For small enough γ_m the right side of (35.22) is less than η_m. Decreasing η_m if necessary, we may also assume that a coarse field within distance η_m of a flat configuration with holonomies in the larger patch belongs to $\mathcal{T}^+$ whenever the holonomies of that flat configuration are within $a_0/2$ of $\mathcal{O}$. Now suppose that $\Xi_m^K \in \mathcal{O}$, $V \notin \mathcal{T}^+$, and $\operatorname{dist}(V, \mathcal{F}_n) < \eta_m$, and let V_0 be flat with $\operatorname{dist}(V, V_0) < \eta_m$. The input distance between $(V, z_{\le m})$ and $(V_0, 0)$ is less than $2\eta_m$, so $\operatorname{Hol}(V_0)$ is within $a_0/2$ of $\Xi_m^K \in \mathcal{O}$. By the choice of a_0, the holonomies of V_0 lie in the larger patch, and then $V \in \mathcal{T}^+$, a contradiction. We conclude that an output in $\mathcal{O}$ coming from outside $\mathcal{T}^+$ forces $\operatorname{dist}(V, \mathcal{F}_n) \ge \eta_m$, and (35.21) proves part (5). The flat configurations with central holonomies lie in the excluded part; the argument does not use that small curvature forces trivial holonomy.

Step 3: bounded output coordinates. We fix regular output tubes $\mathcal{O}_1 \Subset \mathcal{O}_2 \Subset \mathcal{O}_3$ with positive margins, such that $\mathcal{O}_1$ contains the flat outputs of the larger coarse patch being used. Let ϕ_{out} be a smooth function equal to one on $\mathcal{O}_2$ and supported in $\mathcal{O}_3$. On $\mathcal{O}_3$ we multiply the local coordinate maps by ϕ_{out}, and we set both products equal to zero elsewhere on $\mathrm{SU}(2)^3$. This gives the bounded Borel maps

$$b_{\mathrm{ext}}(H) = \phi_{\mathrm{out}}(H)\, b(\operatorname{Log} H), \qquad y_{\mathrm{ext}}(H) = \phi_{\mathrm{out}}(H)\, y(\operatorname{Log} H), \tag{35.23}$$

where the right sides are used only on $\mathcal{O}_3$. They agree with the normal and tangential coordinates on $\mathcal{O}_2$, and they are the extensions used in part (1). They do not alter the invariant observables, which were extended by zero.

Step 4: a good event. We consider the restriction of ν to the coarse tube, normalized by its own mass. By the uniform first-derivative bound and the exact flat values of Theorem 35.9, there exists $\varepsilon_m > 0$, inside the input neighborhood of that theorem, such that $|v| + |z_{\le m}| \le 2\varepsilon_m$ implies $\Xi_m^K \in \mathcal{O}_2$, uniformly in K. By (35.22), after decreasing γ_m we have $|z_{\le m}| \le \varepsilon_m$ deterministically. Let $G_m = \{|v| \le \varepsilon_m\}$ inside the coarse tube. Then

$$\mathbb{P}(G_m^c \mid V \text{ in the coarse tube}) \le C_m \gamma^{-(3n^3-1)} e^{-c_m/\gamma}. \tag{35.24}$$

Indeed, the normal phase is at least $c_n|v|^2$ throughout the coarse tube, the amplitude is bounded, and the conditional normalizing constant at each y is bounded below by $c\gamma^{3n^3-1}$ by the coarse Laplace formula; we then integrate over the tangential distribution. The same estimate holds conditionally at each y in the compact subpatch. Since no departure from $\mathcal{O}_2$ occurs on G_m, (35.24) also bounds the probability of such a departure.

Step 5: Taylor expansion on the good event. On G_m we use the local coordinates and write $u = (v, z_{\le m})$. By the uniform $C^{1,1}$ bound and the smoothness of the normal projection, Taylor's formula along the segment su, $0 \le s \le 1$, gives

$$\Big| b(\operatorname{Log} \Xi_m^K) - T_{0,K} v - \sqrt{\gamma} \sum_{j=1}^{m} A_{j,K} w_j \Big| \le C_m \big(|v| + |z_{\le m}| \big)^2. \tag{35.25}$$

The derivatives here are those of the flowed triple $\mathscr{H}_K$ of the fine configuration, since setting the later fluctuation variables to zero does not change the derivative in an earlier variable at the flat refinements. Substituting $v = \sqrt{\gamma} H_K^{-1/2} z_0$ gives the coarse term $\sqrt{\gamma} A_{0,K} z_0$. The normal moments (35.20) and the deterministic bound (35.22) then bound the remainder divided by $\sqrt{\gamma}$, on G_m, by $C_{m,r'} \sqrt{\gamma} (1 + |\log \gamma|)^2$ in $L^{r'}$. Using the first-order bound for the tangential coordinate, we similarly obtain the bound $C_{m,r'} \sqrt{\gamma} (1 + |\log \gamma|)$ for its displacement.

Step 6: the exceptional event. On G_m^c we define $\mathcal{R}_{K,m}$ by subtracting the displayed linear terms from $b_{\mathrm{ext}}(\Xi_m^K) / \sqrt{\gamma}$. Both extended coordinate maps are bounded, the coarse tube has bounded v, and the fluctuation terms are bounded by $C_m(1 + |\log \gamma|)$ there. This remainder is at most $C_m \gamma^{-1/2}$ for small γ, and (35.24) implies, for every fixed $r' < \infty$ and every $q > 0$,

$$\| \mathbf{1}_{G_m^c} \mathcal{R}_{K,m} \|_{L^{r'}} \le C_{m,r',q} \gamma^q, \qquad \| \mathbf{1}_{G_m^c} \big(y_{\mathrm{ext}}(\Xi_m^K) - y \big) \|_{L^{r'}} \le C_{m,r',q} \gamma^q;$$

here we took the r'-th root of the exponential bound and absorbed the fixed negative power of γ. Together with Step 5 this proves both estimates of part (1). Multiplying the r'-th powers by the mass of the coarse tube, and then by the norm of a bounded multiplicative factor, proves the integrated versions. Any other fixed bounded Borel extensions agreeing with the coordinates on $\mathcal{O}_2$ satisfy the same estimates, with their own bound in the constant.

Step 7: recentering. For $j \le m$ the field at the start of the j-th step is a coarse field in B_r satisfying the coordinate-domain conditions. Since the conditional fluctuation kernel and its cutoff are preserved by Theorem 15.12, Theorem 13.10 applies with $t = t_j$. By (35.7),

$$|\bar{w}_j| \le C \sqrt{D_j}\, t_j r \le C_{m,n,M,J,r} \sqrt{\gamma} \qquad (1 \le j \le m).$$

The inequality holds for every value of the earlier variables, hence also in $L^{r'}$ under the restriction to the coarse tube. Using (35.10) and the Cauchy–Schwarz inequality, we obtain

$$\Big\| \sum_{j=1}^{m} A_{j,K} \bar{w}_j \Big\|_{L^{r'}} \le C \Big(\sum_{j=1}^{m} D_j t_j^2 r^2 \Big)^{1/2} \le C_{m,r'} \sqrt{\gamma}. \tag{35.26}$$

The finite sum may grow with m, as allowed. This is the mean under the conditional fluctuation kernel, not under an enclosing product measure. Replacing w_j by $w_j - \bar{w}_j$ changes the scaled remainder by exactly this quantity, which completes the proof of part (1).

Step 8: the moment identities (part (2)). At fixed m all uncentered moments of the normalized fluctuation variables are bounded, since their centered moments were bounded in Lemma 35.8 and their means are $O_m(\sqrt{\gamma})$. Let $Z = Z_{K,m}$, $b' = b_{\mathrm{ext}}(\Xi_m^K)$, and $y' = y_{\mathrm{ext}}(\Xi_m^K)$. On G_m these are the local coordinates, so $X' = y' + b' = \operatorname{Log} \Xi_m^K$ there. By Taylor's formula, the bound on the exceptional event, and (35.26), we have, under the whole restriction to the coarse tube,

$$b'/\sqrt{\gamma} = Z + R, \qquad \|R\|_{L^r} \le C_{m,r}\sqrt{\gamma}(1 + |\log \gamma|)^2.$$

Every fixed moment of Z is bounded, by Lemma 35.8, and the error R has bounded fixed moments for small γ. The inverse covariance Σ_θ^{-1} has a fixed Lipschitz norm on the larger regular tube. Expanding the quadratic form in $Z + R$, and using $\|y' - y\|_{L^{2r}} \le C_{m,r}\sqrt{\gamma}(1 + |\log \gamma|)$ and Hölder's inequality with higher fixed moments of Z and R for each product, we obtain on G_m

$$\frac{q_*(X')}{\gamma} = \frac{1}{2} Z^{\mathsf{T}} \Sigma_\theta(y)^{-1} Z + O_{L^r,m}\big(\sqrt{\gamma}(1 + |\log \gamma|)^2\big).$$

For the square we use $a^2 - b^2 = (a - b)(a + b)$ and Hölder's inequality, which gives the same order with a possibly larger power of the logarithm. On G_m^c, each bounded weighted observable divided by γ^l has expectation at most $C\gamma^{-l}\mathbb{P}(G_m^c)$, which is smaller than any power of γ by (35.24). The corresponding polynomial in Z has a bounded second moment, so its contribution from G_m^c is at most $C_m \mathbb{P}(G_m^c)^{1/2}$; for $l \le 2$ this uses moments of Z only up to order eight. Thus the comparison extends to the weighted expectations, without evaluating a logarithm or the quadratic form at a point where they are not defined.

The normal cutoff $\chi_w(b')$ differs from one only at a fixed positive normal radius. Using Taylor's formula, the coarse Laplace estimates, and the moments of the first m steps, we find $\mathbb{E}[|b'|^r \mathbf{1}_{\{\Xi_m^K \text{ in the larger tube}\}}] \le C_{m,r}\gamma^{r/2}$ for every fixed r. Markov's inequality with $r > 2(q + l)$ bounds the effect of replacing $\chi_w(b')$ by one, even after division by γ^l, by $C_{m,q}\gamma^q$, and part (5) controls the complement of the larger tube. Next we replace the smooth tangential weight $w(y')$ by $w(y)$; the error is bounded by its Lipschitz norm times $|y' - y|$, multiplied by a power of the scaled quadratic form that is bounded in L^2. We then apply Lemma 35.8 to the resulting polynomial of degree $2l$, conditionally on y, and integrate over the tangential distribution of the coarse field. The coefficient fields of this polynomial are smooth on the fixed compact bundle subcharts, hence bounded and Borel,

as required by that lemma. If $Z \sim G_{\Sigma_m^K(y)}$, then $Z^{\mathsf{T}}\Sigma_\theta(y)^{-1}Z/2$ has the law of $G^{\mathsf{T}}CG/2$ with $C = \widetilde{\Sigma}_m^K(y)$ and G standard Gaussian in four dimensions, and diagonalizing the nonnegative symmetric matrix C gives

$$\mathbb{E}\Big(\frac{1}{2}G^{\mathsf{T}}CG\Big) = \frac{1}{2}\operatorname{tr} C, \qquad \mathbb{E}\Big(\frac{1}{2}G^{\mathsf{T}}CG\Big)^2 = \frac{1}{4}\big\{(\operatorname{tr} C)^2 + 2\operatorname{tr} C^2\big\}.$$

Here we used $\mathbb{E}G_i^2 = 1$, $\mathbb{E}G_i^4 = 3$, and, by independence, $\mathbb{E}G_i^2G_j^2 = 1$ for $i \neq j$ (special cases of Isserlis' theorem [112]); a zero eigenvalue simply deletes the corresponding summand. This identifies the polynomials $\mathfrak{m}_l$ of (35.19).

Step 9: return to the output coordinates. By (35.15), Σ_m^K has a uniformly bounded Lipschitz norm at each fixed m. The field Σ_θ and its inverse square root are smooth, with fixed positive spectral bounds, on the larger regular patch by Theorem 33.11, and the binomial series used in the proof of Theorem 35.10 also bounds the derivative of the inverse square root. Then $\widetilde{\Sigma}_m^K$ has a uniform Lipschitz bound at each fixed m, and the tangential test $w(y)\mathfrak{m}_l(\widetilde{\Sigma}_m^K(y))$ can be moved back to y' with error $C_{m,w}\sqrt{\gamma}(1+|\log\gamma|)^{k_m}$. Finally we multiply by the normal cutoff $\zeta(b')$; as before, $1-\zeta(b')$ is supported at a fixed positive normal radius, so this costs $C_{m,q}\gamma^q$, and part (5) controls the complement of the larger tube. The test on the right side is then exactly $w^+(\Xi_m^K)\mathfrak{m}_l(\widetilde{\Sigma}_m^K(y'))$, and the tube coordinates are never evaluated outside the support of w^+. This proves part (2).

Step 10: normal moments for $m = 0$ and the two value ranges of ψ (parts (3) and (4)). For $m = 0$ there are no fluctuation variables. By the exact flat values and the uniform first derivative, we have $|b(\operatorname{Log}\Xi_0^K)| \le C|v|$ in the larger coarse tube on G_0, and (35.20) gives the claimed moment bound on G_0. On its complement inside the larger coarse tube, the normal radius of an output in the given regular tube is bounded and (35.24) is smaller than any power of γ. Outside the coarse tube, part (5) gives the same bound. These three cases prove the second assertion of part (3); the constant depends on $\mathcal{V}$ but, since only $m = 0$ is used, not on m.

For part (4), let $h_0, h_+ \le 1$ be nonnegative smooth tangential bumps, with positive integrals, supported strictly inside the open sets O_0 (where $\psi_0 = 0$) and O_+ (where $\psi_0 \ge 2b_*$), and fix a smaller normal ball on which $\chi = 1$. By Step 1, the coarse probability of each of the two corresponding sets of coarse fields is at least some $a > 0$. These compact supports have a fixed positive margin to the boundaries of O_0 and O_+ and to the boundary of $\{\chi = 1\}$. Markov's inequality, applied to the tangential and normal displacement bounds of part (1), makes the probability of crossing these margins less than $a/2$, after decreasing γ_m. On the first remaining event $\psi(\operatorname{Log}\Xi_m^K) = 0$, and on the second $\psi(\operatorname{Log}\Xi_m^K) \ge 2b_* \ge b_*$. Thus $c_* = a/2$ works, independently of m, K, and γ; only the required threshold γ_m depends on m. This completes the proof. □

35.5 Regression identities, moment comparison, and removal of the conditioning

This subsection collects the regression identities (Lemma 35.12), a comparison estimate for localized moments at nearby triples (Lemma 35.13), the removal of the conditioning together with the passage to the continuum (Lemma 35.14), and lower bounds for fixed nonnegative observables (Lemma 35.15).

Lemma 35.12. *Let P be a probability measure and let A, B be bounded real random variables with $s = \mathbb{E}_P B^2 > 0$ and $v = \operatorname{Var}_P B > 0$. Then*

$$\mathscr{C}(P; A, B) = \frac{\operatorname{Var}_P A - \operatorname{Cov}_P(A,B)^2/v}{s} = \frac{\inf_{\alpha,c\in\mathbb{R}} \mathbb{E}_P(A - \alpha B - c)^2}{s}.$$

If $0 \le B \le 1$, $P\{B = 0\} \ge p_0 > 0$, and $P\{B \ge b_*\} \ge p_1 > 0$, then $s \ge p_1 b_*^2$ and $v \ge p_0 p_1 b_*^2$. Further, put $a = \mathbb{E}_P B$ and define e_1, e_2, e_3 by

$$\mathbb{E}_P A = \gamma(2a + e_1), \qquad \mathbb{E}_P[AB] = \gamma(2s + e_2), \qquad \mathbb{E}_P A^2 = \gamma^2(6s + e_3)$$

for some $\gamma > 0$. Then

$$\frac{\mathscr{C}(P; A, B)}{\gamma^2} = 2 + \frac{e_3 - 4e_2 - e_1^2 - (e_2 - a e_1)^2/v}{s}. \tag{35.27}$$

In particular, if $P = P_\gamma$ depends on γ, the bounds p_0, p_1, and b_* hold uniformly, and $e_1, e_2, e_3 \to 0$ as $\gamma \downarrow 0$, then $\mathscr{C}(P_\gamma; A, B) = 2\gamma^2 + o(\gamma^2)$, even if $\mathbb{E}_{P_\gamma} B$ and $\mathbb{E}_{P_\gamma} B^2$ have no limits.

Proof. We minimize first over c. At a given α the unique minimizer is $c = \mathbb{E}A - \alpha \mathbb{E}B$, and the remaining quadratic is $\operatorname{Var} A - 2\alpha \operatorname{Cov}(A,B) + \alpha^2 v$. Since $v > 0$, its minimizer is $\alpha = \operatorname{Cov}(A,B)/v$, and its minimum is the numerator in the first expression. This proves the regression identity.

For an independent copy B' of B we have $\operatorname{Var} B = \mathbb{E}(B - B')^2/2$. The two events $\{B = 0,\ B' \ge b_*\}$ and $\{B' = 0,\ B \ge b_*\}$ each have probability at least $p_0 p_1$, and $(B - B')^2 \ge b_*^2$ on each of them, which implies $v \ge p_0 p_1 b_*^2$. Integrating B^2 over $\{B \ge b_*\}$ gives $s \ge p_1 b_*^2$.

For (35.27), note that $v = s - a^2$, that $\operatorname{Cov}(A,B)/\gamma = 2v + e_2 - a e_1$, and that $\operatorname{Var} A/\gamma^2 = 6s + e_3 - (2a + e_1)^2$. Subtracting the square of the first expression divided by v from the second, the terms linear in $a e_1$ cancel, and we obtain $2s + e_3 - 4e_2 - e_1^2 - (e_2 - a e_1)^2/v$; dividing by s gives (35.27). Uniform positive lower bounds on s and v, together with $|a| \le 1$, make the correction in (35.27) tend to zero when $e_1, e_2, e_3 \to 0$, without any limit for a or s. □

Lemma 35.13. *Let $\mathcal{U} \Subset \mathcal{U}^+ \Subset \mathcal{U}^{++}$ be the tubes of Definition 35.1, and let $\widetilde{f}$ be a smooth invariant weight supported strictly inside $\mathcal{U}$ (for example ψ, ψ^2, or a localizing weight $\widetilde{w}$).*

Assume that the inverse covariance Σ_θ^{-1} is bounded and locally Lipschitz on $\mathcal{U}^{++}$ (it is in fact smooth there). Put $f_l = \widetilde{f} q_*^l$, with the fixed bounded zero extensions. For two random triples H, H' in a common gauge frame at the basepoint let

$$D = \max_i \|H_i - H_i'\|_{\mathrm{op}}, \qquad d = \min(1, D), \qquad \beta = |b(\operatorname{Log} H')| \mathbf{1}_{\{H' \in \mathcal{U}^+\}}.$$

There exist $C < \infty$ and $0 < \eta_0 < 1$, depending only on the tubes and their margins, on $\|\Sigma_\theta^{-1}\|_\infty + \operatorname{Lip}(\Sigma_\theta^{-1})$ on $\mathcal{U}^{++}$, and on the weight, such that for every $p \geq 1$ and $l = 1, 2$,

$$|\mathbb{E} f_l(H) - \mathbb{E} f_l(H')| \leq C \big[\mathbb{E}(\beta + d)^{4l-2}\big]^{1/2} \big[\mathbb{E} d^2\big]^{1/2} + C \eta_0^{-p} \, \mathbb{E} d^p,$$

where the constant in front of η_0^{-p} does not depend on p. For $l = 0$ the bound is $C \, \mathbb{E} d$. Applied under ν to $H = \mathcal{H}^K$ and $H' = \Xi_m^K$, with $p = 2l + 2 \leq 6$, these bounds give

$$\limsup_{K \to \infty} \gamma^{-l} |\mathbb{E}_\nu f_l(\mathcal{H}^K) - \mathbb{E}_\nu f_l(\Xi_m^K)| \leq C \delta_m^{1/2} + r_m(\gamma), \qquad l = 1, 2,$$

where $r_m(\gamma) \to 0$ as $\gamma \downarrow 0$ for each fixed m, and C does not depend on m.

Proof. We first prove the comparison inequality without any probabilistic assumption on H, H'. In the coordinates $X = y + b$ write $\mathcal{L}(y) = \Sigma_\theta(y)^{-1}$. The boundedness and local Lipschitz continuity of $\mathcal{L}$, and the smoothness of y and b on the tubes, give

$$|q_*(X) - q_*(X')| \leq C\big(|b(X)| + |b(X')|\big)|X - X'|.$$

Indeed, we expand the difference of the two quadratic forms, first changing b with $\mathcal{L}$ fixed and then changing $\mathcal{L}$; the second term is at most $C|b(X')|^2 |y(X) - y(X')|$, which is bounded by the displayed expression since the tube radius is bounded. For $l = 1, 2$, the factorization of a difference of l-th powers and the Lipschitz bound on $\widetilde{f}$ give

$$|f_l(X) - f_l(X')| \leq C\big(|b(X')| + |X - X'|\big)^{2l-1} |X - X'| \tag{35.28}$$

whenever the segment between the two points lies in $\mathcal{U}^{++}$. The derivative of the weight contributes a term of order $|b|^{2l}$, which we absorbed using the fixed upper bound on $|b|$.

We choose $\eta_0 < 1$ so small that any two triples at matrix distance at most η_0, one of which lies in the support of f_l, both lie in $\mathcal{U}^+$ and have their logarithmic segment in $\mathcal{U}^{++}$. Such a choice exists: we cover the compact support by finitely many coordinate balls whose doubled closures lie in $\mathcal{U}^{++}$ and use the positive margins to the complement of $\mathcal{U}^+$. The logarithm and the tube coordinates have bounded derivatives on these balls, so $|X - X'| \leq CD$ there. On $\{D \leq \eta_0\}$, if both values of f_l vanish there is no difference; otherwise (35.28) gives

$$|f_l(H) - f_l(H')| \leq C(\beta + d)^{2l-1} d.$$

On $\{D > \eta_0\}$ the difference is at most $2\|f_l\|_\infty \le 2\|f_l\|_\infty \eta_0^{-p} d^p$. Taking expectations and applying the Cauchy–Schwarz inequality on the first event proves the inequality; the constant in front of η_0^{-p} does not depend on p. For $l = 0$, the fixed smooth zero extension of $\widetilde{f}$ is globally Lipschitz for the bounded distance d, and the same argument gives $C\,\mathbb{E}d$.

We now prove the application, and first bound the moments of β uniformly in m. Let $d_m = \min(1, D_{K,m})$ and $d_0 = \min(1, D_{K,0})$, where $D_{K,m} = \max_i \|\mathcal{H}_i^K - \Xi_{m,i}^K\|_{\mathrm{op}}$ as in Theorem 34.10, and let $e_m = \min(1, \max_i \|\Xi_{m,i}^K - \Xi_{0,i}^K\|_{\mathrm{op}})$. By the triangle inequality, $e_m \le d_m + d_0$. On $\{e_m \le \eta_1\}$, with η_1 fixed sufficiently small, an endpoint $\Xi_m^K \in \mathcal{U}^+$ has Ξ_0^K in a fixed larger regular tube $\mathcal{V} \subset \mathcal{U}^{++}$, and $|b(\operatorname{Log}\Xi_m^K)| \le |b(\operatorname{Log}\Xi_0^K)| + Ce_m$. On $\{e_m > \eta_1\}$, the bounded normal radius is at most $C\eta_1^{-1}e_m$. Combining the two cases, we find, for every fixed $r \le 8$,

$$\mathbb{E}\beta^r \le C_r\,\mathbb{E}\big[|b(\operatorname{Log}\Xi_0^K)|^r \mathbf{1}_{\{\Xi_0^K \in \mathcal{V}\}}\big] + C_r\,\mathbb{E}(d_m + d_0)^r \le C_r\gamma^{r/2}. \tag{35.29}$$

In the last step we used only part (3) of Theorem 35.11 with $m = 0$ and the bounds on high moments in Theorem 34.10. Both are estimates under ν, at the fixed side and in the same gauge frame at the basepoint. The tube $\mathcal{V}$ is chosen before m, and both constants for the distance moments are independent of m, so the constant in (35.29) does not depend on m.

For $l = 1, 2$ we take $r = 4l - 2$, which is two or six. By (35.29) and the moment bound for d_m, we have $\mathbb{E}(\beta + d_m)^{4l-2} \le C\gamma^{2l-1}$. The second-moment bound of Theorem 34.10 gives

$$(\mathbb{E}d_m^2)^{1/2} \le C\sqrt{\gamma}\,\delta_m^{1/2} + C\gamma(1 + |\log\gamma|)^k,$$

where k denotes the exponent written d in that theorem. With $p = 2l + 2$, the exceptional term is at most $C_p\gamma^{l+1}$ by the same theorem. The comparison inequality therefore gives, at every finite $K \ge m$,

$$\gamma^{-l}|\mathbb{E}f_l(\mathcal{H}^K) - \mathbb{E}f_l(\Xi_m^K)| \le C\delta_m^{1/2} + C\sqrt{\gamma}(1 + |\log\gamma|)^k + C\gamma. \tag{35.30}$$

This proves the claim, with $r_m(\gamma) = C\sqrt{\gamma}(1 + |\log\gamma|)^k + C\gamma$. For $l = 2$ the sixth normal moment is used in an essential way. □

Lemma 35.14. *Fix $0 < \gamma \le \gamma_W(n)$, and let $F : \mathrm{SU}(2)^3 \to \mathbb{R}$ be bounded, continuous, and invariant under simultaneous conjugation; F may depend on γ. Write $e = \mu_{N,\gamma}(S_{N,n,\gamma}^c)$, with $N = nB^K$. Then, for every $q > 0$,*

$$|\mathbb{E}_{\mu_{N,\gamma}}F(\mathcal{H}^K) - \mathbb{E}_\nu F(\mathcal{H}^K)| \le 2\|F\|_\infty e \le 2\|F\|_\infty C_{n,q}\gamma^q,$$

uniformly in K, and

$$\lim_{K\to\infty} \mathbb{E}_{\mu_{N,\gamma}}F(\mathcal{H}^K) = \mathbb{E}_{\lambda_\gamma}F(H),$$

where H denotes the coordinate-cycle holonomy triple of $X(\theta L^2)$.

Proof. The value $F(\mathcal{H}^K)$ depends only on the gauge orbit of the fine configuration, since F is invariant, the reconstruction maps produce a gauge-equivalent configuration, and the Wilson flow commutes with lattice gauge transformations. Hence $F(\mathcal{H}^K)$ equals F evaluated at the coordinate-cycle holonomy triple of $\Psi^{(N)}_{\theta L^2}U$, which is a bounded measurable function on the whole configuration space; this is how $F(\mathcal{H}^K)$ is understood under $\mu_{N,\gamma}$. For any bounded F and any event S with $e = \mu(S^c)$ we have

$$\mathbb{E}_\mu F = (1-e)\mathbb{E}_\nu F + e\,\mathbb{E}_\mu(F \mid S^c), \qquad |\mathbb{E}_\mu F - \mathbb{E}_\nu F| \le 2\|F\|_\infty e, \tag{35.31}$$

where the last term of the identity is set to zero when $e = 0$; see also Lemma 5.5. By Corollary 23.11, at the fixed side n we have $e \le C_{n,q}\gamma^q$ for every fixed $q > 0$, uniformly in K. This proves the first assertion.

For the second, Lemma 5.6 writes $F = f \circ \mathfrak{p}$, where $\mathfrak{p}$ is the vector of the seven normalized traces of H_1, H_2, H_3, H_1H_2, H_1H_3, H_2H_3, and $H_1H_2H_3$, and f is continuous on $[-1,1]^7$ with $\|f\|_\infty \le \|F\|_\infty$. These traces are the normalized fundamental Wilson loops of the three coordinate cycles and of their concatenations, which are fixed piecewise C^1 based loops of bounded length. On each lattice of the sequence they are exact lattice loops, which form admissible approximations. By Theorem 29.25, in the form of Corollary 29.26, the joint law of these seven traces under $\mu_{N,\gamma}$, at the flow time θL^2, converges as $K \to \infty$ to their joint law under λ_γ. At fixed γ the corollary may discard finitely many initial terms of the sequence nB^K; this does not affect the limit, and the side n is not changed. Since f is bounded and continuous, the expectations converge. □

Lemma 35.15. *Let $\mathsf{K}_1 \subset \mathsf{P}^{++}$ be a compact invariant set with nonempty relative interior, contained in the relative interior of P^{++}, and let $r_c > 0$ be such that $\mathcal{T}_c = \{y + b : y \in \mathsf{K}_1,\ b \in \mathcal{N}_y,\ |b| \le r_c\} \subset \mathcal{U}^{++}$. Let $\varphi : \mathrm{SU}(2)^3 \to [0,\infty)$ be bounded, invariant under simultaneous conjugation, and Lipschitz for the matrix distance, and assume $\varphi(H) \ge c_1 > 0$ whenever $\operatorname{Log} H \in \mathcal{T}_c$. Then there exist $a, \gamma_1 > 0$ such that, for $0 < \gamma \le \gamma_1$ and $K \ge K_0(0)$,*

$$\mathbb{E}_\nu \varphi(\Xi_0^K) \ge c_1 a, \qquad \mathbb{E}_\nu \varphi(\mathcal{H}^K) \ge c_1 a/2, \qquad \mathbb{E}_{\lambda_\gamma} \varphi(H) \ge c_1 a/4.$$

Proof. Let K_1' be a compact invariant subset of the relative interior of K_1 with nonempty relative interior. At the flat coarse field $V(y,0)$ we have $\operatorname{Log}\Xi_0^K = y$, by the exact flat values of Theorem 35.9. By the uniform first-derivative bound of that theorem with $m = 0$, there exists $\varepsilon' > 0$ such that, for every coarse field $V = V(y,v)$ with $y \in \mathsf{K}_1'$ and $|v| \le \varepsilon'$, the tangential component of $\operatorname{Log}\Xi_0^K$ lies in K_1 and its normal component has norm at most r_c; hence $\operatorname{Log}\Xi_0^K \in \mathcal{T}_c$, uniformly in K. Taking in Step 1 of the proof of Theorem 35.11 a tangential bump supported in K_1' and a normal cutoff supported in $\{|v| \le \varepsilon'\}$, we find that the set $\{V(y,v) : y \in \mathsf{K}_1',\ |v| \le \varepsilon'\}$ has $\widehat{\nu}$-probability at least some $a > 0$, uniformly in K, for all sufficiently small γ. This gives $\mathbb{E}_\nu \varphi(\Xi_0^K) \ge c_1 a$.

Since φ is Lipschitz for the bounded matrix distance, Theorem 34.10 with $m = 0$ gives

$$|\mathbb{E}_\nu \varphi(\mathcal{H}^K) - \mathbb{E}_\nu \varphi(\Xi_0^K)| \le \operatorname{Lip}(\varphi)\,\big(\mathbb{E}_\nu \min(1, D_{K,0})^2\big)^{1/2} \le C\sqrt{\gamma},$$

so $\mathbb{E}_\nu \varphi(\mathcal{H}^K) \geq c_1 a/2$ for small γ, uniformly in K. Finally, by Lemma 35.14 with a fixed $q > 0$, we have $\mathbb{E}_{\mu_{N,\gamma}} \varphi(\mathcal{H}^K) \geq c_1 a/2 - 2C_{n,q}\gamma^q$ for every K, and letting $K \to \infty$ gives $\mathbb{E}_{\lambda_\gamma} \varphi \geq c_1 a/2 - 2C_{n,q}\gamma^q \geq c_1 a/4$ for small γ. □

35.6 Proof of the regression coefficient

Proof of Theorem 35.2. We first check that the observables are admissible continuum observables (Step 1) and prove the lower bounds on the denominators (Step 2). We then prove the normalized moment estimate at finite cutoff (Steps 3 and 4), take the limits in the stated order (Step 5), and conclude with Lemma 35.12 (Step 6).

Step 1: the observables. The inverse covariance Σ_θ^{-1}, the tube coordinates, and the cutoffs are invariant under simultaneous conjugation. The supports of ψ and ψ^2 have positive distance from the boundary of the identity chart, so their zero extensions, and their products with q_* and q_*^2, are bounded and continuous on $\mathrm{SU}(2)^3$; they contain no coupling parameter. The continuous invariant factorization of Lemma 5.6 applies on the whole compact space of triples, including central triples and triples of deficient rank. In particular, these observables are continuous functions of the seven normalized traces, as stated.

Step 2: lower bounds on the denominators. We choose compact invariant sets $\mathsf{K}_0 \subset O_0$ and $\mathsf{K}_+ \subset O_+$ with nonempty relative interiors, and a radius $r_c > 0$ such that $\chi(b) = 1$ whenever $|b| \leq 2r_c$. Let $\rho_0, \rho_+ : \mathrm{SU}(2)^3 \to [0,1]$ be smooth invariant functions, equal to one on the tubes of radius r_c over K_0 and K_+ respectively, with ρ_0 supported in the interior of $\{\psi = 0\}$ and ρ_+ supported inside $\{\psi > b_*\}$. The latter set has room for such a choice because $\psi_0 \geq 2b_*$ on O_+ and $\chi = 1$ on the normal ball of radius $2r_c$. In finite coordinates such functions are obtained from a smooth function of the squared distance in finitely many small balls, by adding these bumps and truncating smoothly to one on their cores. Averaging over simultaneous conjugation with respect to the Haar probability measure makes them invariant; since the relevant sets are invariant, the support and core properties are preserved. Both functions are Lipschitz for the matrix distance. By Lemma 35.15, there exists $a' > 0$ such that $\mathbb{E}_{\lambda_\gamma} \rho_0 \geq a'$ and $\mathbb{E}_{\lambda_\gamma} \rho_+ \geq a'$ for all sufficiently small γ. By the support properties, $\lambda_\gamma\{B_{\mathrm{obs}} = 0\} \geq a'$ and $\lambda_\gamma\{B_{\mathrm{obs}} \geq b_*\} \geq a'$. Since $0 \leq B_{\mathrm{obs}} \leq 1$, Lemma 35.12 implies $\mathbb{E}_{\lambda_\gamma} B_{\mathrm{obs}}^2 \geq a' b_*^2$ and $\mathrm{Var}_{\lambda_\gamma} B_{\mathrm{obs}} \geq (a')^2 b_*^2$, which proves the first assertion.

Step 3: the Gaussian moment polynomials. On any bounded set of nonnegative 4×4 matrices we have

$$|\mathfrak{m}_1(C) - 2| + |\mathfrak{m}_2(C) - 6| \leq C' \|C - I\|.$$

Indeed, $|\mathrm{tr}(C - I)| \leq 4\|C - I\|$, $|\mathrm{tr}\, C^2 - 4| = |\mathrm{tr}((C - I)(C + I))| \leq 4\|C - I\|(\|C\| + 1)$, and $(\mathrm{tr}\, C)^2 - 16 = (\mathrm{tr}\, C - 4)(\mathrm{tr}\, C + 4)$. By (35.16), $0 \leq \Sigma_m^K \leq \Sigma_\theta^N$, and together with

the spectral bounds of Theorem 33.11 this bounds the norms of all $\widetilde{\Sigma}_m^K$ by a constant independent of m. Then (35.14) compares the moment polynomials at finite K directly with $\mathfrak{m}_1(I) = 2$ and $\mathfrak{m}_2(I) = 6$.

Step 4: the normalized moments at finite cutoff. We apply part (2) of Theorem 35.11 with the tangential weights ψ_0 and ψ_0^2 and the normal cutoffs χ and χ^2, so that $\widetilde{w} = \psi$ or ψ^2. On its right side, w^+ and $\widetilde{w}$ differ only at a fixed positive normal radius. The normal moment bounds of Step 8 of its proof, Markov's inequality, and the boundedness of $\mathfrak{m}_l(\widetilde{\Sigma}_m^K)$ show that replacing w^+ by $\widetilde{w}$ changes the expectation by $O_{m,q}(\gamma^q)$ for every fixed q; the polynomial comparison in that theorem is supplied by Lemma 35.8. We take $K_0(m) \geq \max\{m, K_0(0)\}$, as allowed. For $f = \psi$ and $f = \psi^2$ and $l = 1, 2$, we compare $\gamma^{-l}\mathbb{E}_\nu[f q_*^l(\mathcal{H}^K)]$ with $\gamma^{-l}\mathbb{E}_\nu[f q_*^l(\Xi_m^K)]$ by (35.30), then with $\mathbb{E}_\nu[f(\Xi_m^K)\mathfrak{m}_l(\widetilde{\Sigma}_m^K(y'))]$ by part (2), then with $\mathfrak{m}_l(I)\mathbb{E}_\nu f(\Xi_m^K)$ by Step 3 and (35.14), and finally with $\mathfrak{m}_l(I)\mathbb{E}_\nu f(\mathcal{H}^K)$ by the case $l = 0$ of Lemma 35.13, which costs $C\,\mathbb{E}d_m \leq C\sqrt{\gamma}$. We obtain

$$\begin{aligned} &\bigl|\gamma^{-l}\mathbb{E}_\nu[f q_*^l(\mathcal{H}^K)] - \mathfrak{m}_l(I)\mathbb{E}_\nu[f(\mathcal{H}^K)]\bigr| \\ &\qquad \leq C\bigl\{\delta_m^{1/2} + \delta_m + (nB^K)^{-2}\bigr\} + e_m(\gamma), \qquad K \geq K_0(m), \end{aligned} \tag{35.32}$$

where $e_m(\gamma) \to 0$ as $\gamma \downarrow 0$ for each fixed m; one may take for e_m a finite sum of terms $C_m\sqrt{\gamma}(1 + |\log\gamma|)^{k_m}$, $C\sqrt{\gamma}(1 + |\log\gamma|)^k$, and $C_m\gamma$. The constants in front of $\delta_m^{1/2}$, δ_m, and the lattice error do not depend on m.

Step 5: the limits. We apply Lemma 35.14 to each bounded product $f q_*^l$ and to f, before dividing by γ^l. For a fixed $q > 2$, the cost of replacing ν by $\mu_{N,\gamma}$ is $O(\gamma^{q-l}) = o(1)$ after division by γ^l, for every $l \leq 2$. Then (35.32) holds for $\mu_{N,\gamma}$, with this additional error. At each fixed γ, Lemma 35.14 gives the limits as $K \to \infty$ of these expectations under $\mu_{N,\gamma}$, which are the corresponding expectations under λ_γ. When $K \to \infty$, the lattice error disappears, without any division by γ. Next we let $\gamma \downarrow 0$ at the fixed m, which gives

$$\limsup_{\gamma\downarrow 0}\bigl|\gamma^{-l}\mathbb{E}_{\lambda_\gamma}[f q_*^l] - \mathfrak{m}_l(I)\mathbb{E}_{\lambda_\gamma} f\bigr| \leq C\bigl(\delta_m^{1/2} + \delta_m\bigr).$$

Letting $m \to \infty$, we obtain the two moment identities. The side n and the sequence nB^K are fixed throughout; only finitely many initial terms of the sequence may be discarded in the inner limit $K \to \infty$, as allowed by Corollary 29.26.

Step 6: the statistic. With $A_{\mathrm{obs}} = \psi q_*$ and $B_{\mathrm{obs}} = \psi$, the three errors of Lemma 35.12 are

$$e_1 = \gamma^{-1}\mathbb{E}[\psi q_*] - 2\mathbb{E}\psi, \qquad e_2 = \gamma^{-1}\mathbb{E}[\psi^2 q_*] - 2\mathbb{E}\psi^2, \qquad e_3 = \gamma^{-2}\mathbb{E}[\psi^2 q_*^2] - 6\mathbb{E}\psi^2,$$

with all expectations under λ_γ. By Step 5 they tend to zero. Combining the lower bounds of Step 2 with (35.27), we obtain $\mathscr{C}_\gamma = 2\gamma^2 + o(\gamma^2)$. Further, Step 5 shows that at each fixed m the upper limits of $|e_1|, |e_2|, |e_3|$ are at most $C(\delta_m^{1/2} + \delta_m)$, and the exact identity

(35.27) then implies the stated bound on $\mathscr{C}_\gamma/\gamma^2 - 2$, with a constant independent of m. The quadratic errors e_1^2 and $(e_2 - ae_1)^2$ are absorbed using that δ_m is bounded in m and that $|a| \le 1$. □

35.7 A weighted chi-square limit with four degrees of freedom

Proof of Theorem 35.3. Only the values of q_* on the support of $\widetilde{w}$ enter, where q_* is the exact nonnegative quadratic form; the extension of q_* has no effect on π_γ. We first prove a finite-scale version of the statement for bounded Lipschitz f (Steps 1–3) and replace the finite-scale triple by the triple of the fine configuration (Step 4). We then prove the lower bound on D_γ (Step 5), pass to the continuum and take the limits (Step 6), and finally extend the result to bounded continuous f and identify the limit law (Step 7).

Step 1: the finite-scale expansion. Fix m, let $K \ge K_0(m) \ge \max\{m, K_0(0)\}$, let γ be below the threshold of Theorem 35.11, and write $H_m = \Xi_m^K$ and $a_f = \mathbb{E} f(|G_4|^2/2)$. We choose a larger coarse regular tube before m: its flat holonomies cover a fixed larger regular patch and are separated from the compact support of $\widetilde{w}$ on the complementary part of the set of flat configurations. The modulus of Theorem 35.9 has an input margin depending on m, which changes the coupling threshold but not this tube. By part (5) of Theorem 35.11, there exists a larger coarse regular tube $\mathcal{T}^+$ such that an output in the support of $\widetilde{w}$ coming from outside $\mathcal{T}^+$ has probability $O_{m,q}(\gamma^q)$ for each fixed $q > 0$. Inside the coarse tube we use the restriction of ν normalized by its own mass, and write y for the tangential coordinate of the coarse field. Part (1) of the same theorem, with its bounded output coordinates, gives

$$b(H_m)/\sqrt{\gamma} = Z_{K,m} + \mathcal{R}_{K,m}, \qquad \|\mathcal{R}_{K,m}\|_{L^r} \le C_{m,r}\sqrt{\gamma}(1 + |\log\gamma|)^2,$$

and $\|y(H_m) - y\|_{L^r} \le C_{m,r}\sqrt{\gamma}(1 + |\log\gamma|)$ for every fixed finite r, where the recentering by the conditional means is included in $\mathcal{R}_{K,m}$. These formulas are local coordinate identities only on the good event G_m of Step 4 of that proof, whose complement has probability $O_{m,q}(\gamma^q)$ for every fixed $q > 0$ by (35.24). By Lemma 35.8, all required moments of $Z_{K,m}$ are bounded at this fixed m.

Step 2: comparison with a Gaussian quadratic form. The ambient inverse covariance $P_{\mathcal{N}_y}\Sigma_\theta(y)^{-1}P_{\mathcal{N}_y}$ is smooth on the larger patch by Theorem 33.11. Expanding the quadratic form and using Hölder's inequality, we obtain on the good event

$$\Big\| \frac{q_*(H_m)}{\gamma} - \frac{1}{2} Z_{K,m}^{\mathsf{T}} \Sigma_\theta(y)^{-1} Z_{K,m} \Big\|_{L^1} \le C_m \sqrt{\gamma}(1 + |\log\gamma|)^{k_m}.$$

Explicitly, the difference is bounded by a constant times $|\mathcal{R}|(|Z| + |\mathcal{R}|) + |y(H_m) - y|(|Z| + |\mathcal{R}|)^2$, and moments of order at most six suffice. Since f is Lipschitz, the same

bound times $\operatorname{Lip} f$ controls the change of $f(q_*(H_m)/\gamma)$. The smooth weight $w(y(H_m))$ can be moved to $w(y)$, and back, with an error of the same order. The normal cutoff χ_w differs from one only at a fixed positive normal radius, so arbitrarily high fixed moments of the normal coordinate of H_m (Step 8 of the proof of Theorem 35.11) make the effect of replacing χ_w by one $O_{m,q}(\gamma^q)$. On the exceptional events we use only that f and the weight are bounded, without taking a logarithm there. These comparisons hold after multiplication by the mass of the coarse tube, which is at most one.

Step 3: the Gaussian value. Conditionally on y, we apply Lemma 35.8 to the bounded Borel function $z \mapsto f(z^{\mathsf T}\Sigma_\theta(y)^{-1}z/2)$. This gives the Gaussian value

$$\kappa_f\big(\widetilde\Sigma_m^K(y)\big) = \mathbb{E} f\Big(\frac{1}{2} G_4^{\mathsf T}\widetilde\Sigma_m^K(y) G_4\Big),$$

with error $C_m\|f\|_\infty\sqrt{\gamma}(1+|\log\gamma|)$. The equality of the laws of the two Gaussian quadratic forms follows by diagonalizing the nonnegative covariance matrix; no inverse of Σ_m^K is used. Coupling the two quadratic forms through the same G_4, we obtain

$$|\kappa_f(C) - a_f| \le \frac{1}{2}\operatorname{Lip} f\, \|C-I\|\, \mathbb{E}|G_4|^2 = 2\operatorname{Lip} f\, \|C-I\|.$$

Integrating against the tangential law of the coarse field and using (35.14), we obtain

$$\big|\mathbb{E}_\nu[\widetilde w(H_m) f(q_*(H_m)/\gamma)] - a_f \mathbb{E}_\nu \widetilde w(H_m)\big| \le C_w\|f\|_{\mathrm{BL}}\big\{\delta_m + (nB^K)^{-2}\big\} + \|f\|_{\mathrm{BL}}\, e_{m,w}(\gamma), \tag{35.33}$$

with $e_{m,w}(\gamma) \to 0$ as $\gamma \downarrow 0$. One may take for $e_{m,w}$ a finite sum of terms $C_{m,w}\sqrt{\gamma}(1+|\log\gamma|)^{k_m}$ and of arbitrarily high fixed powers of γ. The constant C_w does not depend on m: only the bound on the weight and the uniform covariance error enter it.

Step 4: comparison with the triple of the fine configuration. We now replace H_m by $H = \mathcal{H}^K$ in the bounded test $F_\gamma(H) = \widetilde w(H) f(q_*(H)/\gamma)$. This needs a separate estimate, because the Lipschitz constant of $f(q_*/\gamma)$ grows as $\gamma \downarrow 0$. We use the nested tubes of Lemma 35.13 and set

$$d_m = \min\Big(1, \max_i \|H_i - H_{m,i}\|_{\mathrm{op}}\Big), \qquad \beta_m = |b(\operatorname{Log} H_m)|\mathbf{1}_{\{H_m \in \mathcal{U}^+\}}.$$

The normal coordinate is used only on the indicated tube. For a fixed $0 < \eta < 1$, if $d_m \le \eta$ and at least one of the two weights is nonzero, then both triples lie in $\mathcal{U}^+$ and their logarithmic segment lies in $\mathcal{U}^{++}$. On this event the smooth coordinates and the Lipschitz bound on Σ_θ^{-1} give

$$|q_*(H) - q_*(H_m)| \le C(\beta_m + d_m) d_m;$$

indeed we expand the two quadratic forms and use $|b(\operatorname{Log} H)| \le |b(\operatorname{Log} H_m)| + C d_m$. Therefore, on this event,

$$|F_\gamma(H) - F_\gamma(H_m)| \le C_w\|f\|_\infty d_m + C_w \operatorname{Lip} f\, \gamma^{-1}(\beta_m + d_m) d_m.$$

If both weights vanish, the difference is zero. On $\{d_m > \eta\}$ the boundedness of F_γ gives the bound $C_w\|f\|_\infty \eta^{-4} d_m^4$.

We bound the second normal moment with a constant independent of m. We compare H_m with $H_0 = \Xi_0^K$ and let $d_0 = \min(1, \max_i \|H_i - H_{0,i}\|_{\mathrm{op}})$; the truncated distance between H_m and H_0 is at most $d_m + d_0$. When this distance is small and $H_m \in \mathcal{U}^+$, H_0 lies in a fixed larger regular tube $\mathcal{V}$ and $\beta_m \le |b(\operatorname{Log} H_0)| + C(d_m + d_0)$. When the distance exceeds a fixed margin, the boundedness of the normal radius gives $\beta_m \le C(d_m + d_0)$. Combining the two cases, we obtain

$$\mathbb{E}_\nu \beta_m^2 \le C\,\mathbb{E}_\nu\big[|b(\operatorname{Log} H_0)|^2 \mathbf{1}_{\{H_0 \in \mathcal{V}\}}\big] + C\,\mathbb{E}_\nu (d_m + d_0)^2 \le C\gamma,$$

where we used part (3) of Theorem 35.11 with $m = 0$ and the moment bounds of Theorem 34.10, which hold for all m with the same constants. The tube $\mathcal{V}$ is fixed before m. By the Cauchy–Schwarz inequality and Theorem 34.10, we have, uniformly for $K \ge K_0(m)$,

$$|\mathbb{E}_\nu F_\gamma(H) - \mathbb{E}_\nu F_\gamma(H_m)| \le C_w \|f\|_{\mathrm{BL}} \big\{\delta_m^{1/2} + \sqrt{\gamma}(1 + |\log\gamma|)^k + \gamma^2\big\}. \tag{35.34}$$

Indeed $(\mathbb{E}(\beta_m + d_m)^2)^{1/2} \le C\sqrt{\gamma}$ and $(\mathbb{E} d_m^2)^{1/2} \le C\sqrt{\gamma}\,\delta_m^{1/2} + C\gamma(1 + |\log\gamma|)^k$, so the factor γ^{-1} is compensated by the size of the quadratic form; the fourth moment on the far event is at most $C\gamma^2$. For the weight alone, its global Lipschitz continuity gives the simpler bound $|\mathbb{E}\widetilde{w}(H) - \mathbb{E}\widetilde{w}(H_m)| \le C_w\sqrt{\gamma}$.

Step 5: the lower bound on D_γ. Since w is continuous, nonnegative, and not identically zero, there exist a compact invariant set K_1 with nonempty relative interior in the relative interior of P and $c_1 > 0$ with $w \ge c_1$ on K_1; let $r_c > 0$ be such that $\chi_w = 1$ on $\{|b| \le r_c\}$. Then $\widetilde{w} \ge c_1$ on the corresponding tube $\mathcal{T}_c$, and $\widetilde{w}$ is smooth, invariant, and Lipschitz for the matrix distance. By Lemma 35.15, $\mathbb{E}_\nu \widetilde{w}(H) \ge c_w'$ uniformly in K and $D_\gamma = \mathbb{E}_{\lambda_\gamma}\widetilde{w} \ge c_w > 0$ for all sufficiently small γ.

Step 6: the continuum and the limits. For each fixed $\gamma > 0$, F_γ is a bounded continuous invariant function on the compact space of triples, with its fixed zero extension, and its supremum $\|\widetilde{w}\|_\infty \|f\|_\infty$ does not depend on γ. By Lemma 35.14, the expectations of $F_\gamma(\mathcal{H}^K)$ and of $\widetilde{w}(\mathcal{H}^K)$ under ν and under $\mu_{N,\gamma}$ differ by at most $C_{w,q}\|f\|_\infty \gamma^q$ for any fixed $q > 0$, and the expectations under $\mu_{N,\gamma}$ converge to the expectations under λ_γ as $K \to \infty$. This uses only the supremum of the test, never its Lipschitz constant, so no negative power of γ appears. Combining (35.33) and (35.34) with the bound for the weight alone, letting $K \to \infty$ at fixed γ and m, and dividing by $D_\gamma \ge c_w$, we obtain the quantitative bound in the statement, after absorbing the terms from Step 4 and from the conditioning into $e_{m,w}$. The lattice error vanishes in this first limit. The estimates at finite cutoff were needed only for $K \ge K_0(m)$, which suffices for this limit. Letting $\gamma \downarrow 0$ at fixed m, and then $m \to \infty$, we obtain $\int f\, d\pi_\gamma \to \mathbb{E} f(|G_4|^2/2)$ for every bounded Lipschitz f.

Step 7: bounded continuous tests and the limit law. We extend the convergence to bounded continuous f by tightness, without using convergence of moments. Let η_R be zero on $[0, R]$, linear on $[R, R+1]$, and one on $[R+1, \infty)$. It is bounded and Lipschitz, so the convergence already proved gives

$$\limsup_{\gamma \downarrow 0} \pi_\gamma([R+1, \infty)) \le \mathbb{E}\eta_R(|G_4|^2/2) \le \mathbb{P}\{|G_4|^2/2 > R\} \longrightarrow 0 \qquad (R \to \infty).$$

On each compact interval a bounded continuous f has uniformly accurate piecewise linear approximations, which we extend by constants outside the interval; they are bounded Lipschitz functions. The two tail probabilities bound the error outside the interval by a fixed multiple of $\|f\|_\infty$. Increasing first the interval and then decreasing the approximation error, we obtain the convergence for f. Finally, each of $(G_1^2 + G_2^2)/2$ and $(G_3^2 + G_4^2)/2$ has density $e^{-u}\mathbf{1}_{\{u \ge 0\}}$, by polar coordinates in the plane. They are independent, and their convolution has density $ue^{-u}\mathbf{1}_{\{u \ge 0\}}$, which identifies the law of $|G_4|^2/2$ and completes the proof. □

35.8 Gaussian approximation on the regular normal bundle

Proof of Theorem 35.4. We first verify the geometric and measurability statements (Step 1) and extend a Lipschitz test to a larger bundle (Step 2). We then prove the quantitative bound for bounded Lipschitz F at finite cutoff, first for the finite-scale triple (Steps 3–5) and then for the triple of the fine configuration (Step 6), and pass to the continuum (Step 7). Finally we extend the result to bounded continuous F (Step 8).

Step 1: geometry of the quotient. All vector comparisons below use the Euclidean embedding of the normal spaces in $\mathbb{R}^9$; no global frame is introduced. On a finite smooth orthonormal bundle atlas, Σ_θ and its inverse square root are smooth with bounded derivatives, by Theorems 33.11 and 35.10, and they are equivariant under simultaneous conjugation, as are the normal coordinates. For $g \in \mathrm{SU}(2)$ the isometry $\mathrm{Ad}(g) : \mathcal{N}_y \to \mathcal{N}_{\mathrm{Ad}(g)y}$ therefore carries Γ_y to $\Gamma_{\mathrm{Ad}(g)y}$, and the invariance of F gives $\overline{F}(\mathrm{Ad}(g)y) = \overline{F}(y)$. The quotient of the compact invariant patch P' is compact and metrizable, for example by

$$d_{\mathrm{orb}}([y], [y']) = \min_{g \in \mathrm{SU}(2)} |y - \mathrm{Ad}(g)y'|.$$

Since the group is compact, the minimum is attained and zero distance identifies exactly one orbit; the isometric action gives the triangle inequality. Its topology is the quotient topology, since the continuous bijection from the compact quotient to this metric space is a homeomorphism. On the compact support of $\widetilde{w}$, the tangential projection followed by the quotient map is continuous and invariant. By Lemma 5.6, the seven traces separate the classes of simultaneous conjugacy; their map on this compact support is a continuous

surjection onto its Hausdorff image, hence a quotient map, and the tangential class factors continuously through this image. In particular $[y_\gamma]$ is a Borel function on the support of $\widetilde{w}$; elsewhere we may assign any fixed point of the quotient, since the weight vanishes there. Thus ρ_γ is a probability measure on the Borel quotient, and the identity $D_\gamma^{-1}\mathbb{E}[\widetilde{w}\,\overline{F}(y_\gamma)] = \int \overline{F}\,d\rho_\gamma$ holds and does not depend on the choice of representatives. For the regularity of the fiber integral, let $E_y : \mathbb{R}^4 \to \mathcal{N}_y$ be a smooth orthonormal frame in one chart and write $\overline{F}(y) = \mathbb{E}F(y, E_y G_4)$. The bounded derivative of the frame gives

$$|\overline{F}(y) - \overline{F}(y')| \le C \operatorname{Lip} F\,(1 + \mathbb{E}|G_4|)|y - y'|,$$

and a finite cover with positive interior margins gives the same bound on the compact patch. Hence $\overline{F}$ is bounded Lipschitz with $\|\overline{F}\|_{\mathrm{BL}} \le C\|F\|_{\mathrm{BL}}$. The same formula and dominated convergence show that $\overline{F}$ is continuous when F is only continuous.

Step 2: extension of a Lipschitz test. Let F be bounded Lipschitz. In the proof the test is evaluated at the tangential coordinate y of the coarse field and at triples where the weight may vanish, and these tangential coordinates may lie outside P'. We therefore first extend F. Let $\widehat{F}(x) = \inf_s\{F(s) + \operatorname{Lip} F\,|x - s|\}$, the infimum over s in the normal bundle over P', be the McShane extension to $\mathbb{R}^9 \times \mathbb{R}^9$; it has the same Lipschitz constant. Truncating it at $\pm\|F\|_\infty$ preserves the Lipschitz constant and gives the bound $\|F\|_\infty$. Averaging the result with respect to the Haar probability measure over the diagonal action $(y, z) \mapsto (\mathrm{Ad}(g)y, \mathrm{Ad}(g)z)$, which acts by isometries of $\mathbb{R}^9 \times \mathbb{R}^9$, preserves the bound and the Lipschitz constant and gives an invariant function. Since F is invariant and the bundle over P' is invariant, the averaged function agrees with F there. We denote this extension, restricted to the normal bundle over P^{++}, again by F, and define $\overline{F}$ by the same formula on P^{++}; Step 1 applies on P^{++} as well. Since $\widetilde{w}$ vanishes unless the tangential coordinate lies in P', the quantities in the theorem are unchanged.

Step 3: the finite-scale expansion. We work under ν, with $H = \mathcal{H}^K$ and $H_m = \Xi_m^K$, the triple of the finite-scale reconstruction from the first m steps on the same sample. All estimates are for $K \ge K_0(m)$, as allowed by Theorem 35.11. Under the restriction of ν to its larger coarse regular tube, normalized by its own mass, let y be the tangential coordinate of the coarse field. That theorem and Lemma 35.8 give

$$\gamma^{-1/2} b(H_m) = Z_{K,m} + \mathcal{R}_{K,m}, \qquad \|\mathcal{R}_{K,m}\|_{L^r} \le C_{m,r}\sqrt{\gamma}(1 + |\log\gamma|)^2,$$

$$\|y(H_m) - y\|_{L^r} \le C_{m,r}\sqrt{\gamma}(1 + |\log\gamma|), \qquad Z_{K,m} = A_{0,K} z_0 + \sum_{j=1}^{m} A_{j,K}(w_j - \bar{w}_j),$$

where $\bar{w}_j$ is the conditional mean of w_j under the conditional fluctuation kernel and the error of the recentering is included in $\mathcal{R}_{K,m}$. All fixed moments of $Z_{K,m}$ are bounded uniformly in K at this m. These are local coordinate identities only on the output tube,

with the bounded extensions of part (1) of Theorem 35.11. By part (5) of that theorem, outputs in the support of the weight coming from outside the larger coarse tube have probability at most $C_{m,q}\gamma^q$ for every fixed $q > 0$; inside, the exceptional event (35.24) has the same bound. Bounded tests on these exceptional events therefore contribute only $C_{m,q}\|F\|_\infty\gamma^q$.

Step 4: comparison of the normalized normal vectors. On the good event, using the bounded derivatives of the inverse square root, Hölder's inequality, and the preceding moments, we obtain

$$\mathbb{E}\big|\gamma^{-1/2}\Sigma_\theta(y(H_m))^{-1/2}b(H_m) - \Sigma_\theta(y)^{-1/2}Z_{K,m}\big| \le C_m\sqrt{\gamma}(1+|\log\gamma|)^{k_m};$$

indeed the difference is bounded by $C|\mathcal{R}_{K,m}| + C|y(H_m)-y||Z_{K,m}|$. Since F is Lipschitz in both arguments, $F(y(H_m),z(H_m))$ may be replaced by $F(y,\Sigma_\theta(y)^{-1/2}Z_{K,m})$ with an error $\operatorname{Lip}F$ times this bound plus $\operatorname{Lip}F\,\mathbb{E}|y(H_m)-y|$. Replacing $w(y(H_m))$ by $w(y)$ gives an error of the same type. The normal cutoff χ_w differs from one only at a fixed positive distance from zero, and arbitrarily high fixed normal moments from the same theorem bound the effect of replacing it by one by $C_{m,q}\gamma^q$.

Step 5: the Gaussian value. Conditionally on y, we apply Lemma 35.8 to the function $v \mapsto F(y,\Sigma_\theta(y)^{-1/2}v)$. Since it is a jointly Borel bounded function of the whole vector, the lemma applies without any radial restriction, and the Gaussian value is $\mathbb{E}F(y,(\widetilde{\Sigma}_m^K(y))^{1/2}G_y)$ with $G_y \sim \Gamma_y$. For every nonnegative self-adjoint C we have

$$\|C^{1/2}-I\|_{\mathrm{op}} \le \|C-I\|_{\mathrm{op}}, \qquad |\mathbb{E}F(y,C^{1/2}G_y)-\overline{F}(y)| \le 2\operatorname{Lip}F\,\|C-I\|_{\mathrm{op}}.$$

The first inequality follows on each eigenvalue from $|\sqrt{r}-1| = |r-1|/(\sqrt{r}+1)$, including $r=0$; the second uses $\mathbb{E}|G_y| \le (\mathbb{E}|G_y|^2)^{1/2} = 2$. Moving $\overline{F}(y)$ back to $\overline{F}(y(H_m))$ with the Lipschitz bound of Step 1 and the weight back to $\widetilde{w}(H_m)$, and using the covariance error (35.14), we obtain

$$\begin{aligned}\Big|\mathbb{E}_\nu\big\{\widetilde{w}(H_m)\big[F(y(H_m),z(H_m))-\overline{F}(y(H_m))\big]\big\}\Big| \\ \le C_w\|F\|_{\mathrm{BL}}\big\{\delta_m + (nB^K)^{-2}\big\} + \|F\|_{\mathrm{BL}}\,e^1_{m,w}(\gamma),\end{aligned} \tag{35.35}$$

where $z(H') = \gamma^{-1/2}\Sigma_\theta(y(H'))^{-1/2}b(H')$ and $e^1_{m,w}(\gamma) \to 0$ as $\gamma \downarrow 0$ at fixed m. Multiplying by the mass of the coarse tube, which is at most one, justifies the use of the normalized conditional estimates. The tangential integration in this step is always against the law of the coarse field, and the constant in front of the covariance error does not depend on m.

Step 6: comparison with the triple of the fine configuration. We replace H_m by H on the same sample. Let $d_m = \min(1,\max_i\|H_i - H_{m,i}\|_{\mathrm{op}})$, and fix nested regular output

tubes with positive margins. There exists $\eta > 0$, independent of m, such that if $d_m \le \eta$ and at least one of the two weights is nonzero, both triples lie in the larger identity chart. On that event the normal coordinate, the tangential coordinate, and the map $H' \mapsto \Sigma_\theta(y(H'))^{-1/2} b(H')$ have bounded derivatives, as one sees by differentiating the product on the fixed bounded tube. Hence, for $J_\gamma(H') = \widetilde{w}(H') F(y(H'), z(H'))$,

$$|J_\gamma(H) - J_\gamma(H_m)| \le C_w \|F\|_{\mathrm{BL}} (1 + \gamma^{-1/2}) d_m.$$

If both weights vanish, the difference is zero; on $\{d_m > \eta\}$ we use the bound $2\|\widetilde{w}\|_\infty \|F\|_\infty$. By Theorem 34.10,

$$\gamma^{-1/2} \mathbb{E}_\nu d_m \le C\delta_m^{1/2} + C\sqrt{\gamma}(1 + |\log \gamma|)^k, \qquad \nu\{d_m > \eta\} \le C_\eta \gamma^2.$$

The same argument for $\widetilde{w}(H')\overline{F}(y(H'))$ involves no factor $\gamma^{-1/2}$. Combining these bounds with (35.35), we obtain the same estimate for H in place of H_m, with an additional term $C_w \|F\|_{\mathrm{BL}} \delta_m^{1/2}$ and an additional error at fixed m that tends to zero with γ.

Step 7: the continuum and the limits. For each fixed $\gamma > 0$, both tests J_γ and $\widetilde{w}\,\overline{F}(y(\cdot))$ are bounded continuous invariant functions on $\mathrm{SU}(2)^3$, with the stated zero extensions. By Lemma 35.14, the expectations under ν and $\mu_{N,\gamma}$ differ by at most $C_{w,q}\|F\|_{\mathrm{BL}}\gamma^q$ for any fixed $q > 0$, and the expectations under $\mu_{N,\gamma}$ converge to those under λ_γ as $K \to \infty$. This uses only the supremum of the tests, not their Lipschitz constants, which depend on γ. Letting $K \to \infty$ at fixed γ and m and dividing by $D_\gamma \ge c_w$ (Theorem 35.3), we obtain (35.3). Letting $\gamma \downarrow 0$ at fixed m, and then $m \to \infty$, proves (35.2) for bounded Lipschitz F.

Step 8: bounded continuous tests. Let F be bounded and continuous. On the support of the weight we have $|z_\gamma|^2/2 = q_*/\gamma$, so Theorem 35.3 shows that the localized probabilities of $\{|z_\gamma| > R\}$ satisfy $\limsup_{\gamma\downarrow 0} \pi_\gamma((R^2/2, \infty)) \to 0$ as $R \to \infty$. For the comparison integral, the inner probability $\Gamma_y\{|z| > R\} = \mathbb{P}\{|G_4| > R\}$ does not depend on y or on the representative, and integrating it against ρ_γ gives the same tail bound. The set $\{(y, z) : y \in \mathsf{P}',\ |z| \le R\}$ is compact, and a continuous function on it admits uniformly accurate bounded Lipschitz approximations. To construct them, we take a sufficiently fine finite net, form the upper and lower cones of a sufficiently large finite slope through the values on the net, and truncate at the supremum bound. The uniform continuity of F gives the approximation as the net is refined and the slope is increased, and these cone functions are defined on the whole ambient space. Averaging them over simultaneous conjugations with respect to the Haar probability measure makes them invariant, preserves their bound and their Lipschitz constant, and preserves the approximation on this invariant compact set. We apply the bounded Lipschitz case to each approximant; the two tail probabilities control the errors outside the compact set. Letting first the approximation error tend to zero and then $R \to \infty$, we obtain (35.2). □

35.9 Comparison probability measures

This subsection evaluates the observables of Definition 35.1 under the four comparison probability measures of Definition 4.13: the Abelian Maxwell measure with Haar cycle holonomies (item (1) there, constructed in Lemma 33.14), Gaussian holonomies with independent Haar factors (item (2)), Gaussian holonomies with Gaussian constant modes (item (3)), and the conditional normal Gaussian measure (item (4)). These measures are different constructions, and their hypotheses are part of their definitions. The comparisons show how to read the conclusion of Theorem 35.2. The conditional normal Gaussian measure has the same coefficient $2\gamma^2$. The two holonomy models without a non-Abelian flat background give a vanishing statistic or a statistic of order one, and the Gaussian holonomies with Gaussian constant modes give the regular patch an exponentially small mass.

For the third measure we use Gaussian concentration. Its classical antecedents are the Gaussian isoperimetric inequalities of Borell [30] and of Sudakov and Tsirelson [148], and the sample-function formulation of Cirel'son, Ibragimov, and Sudakov [60]; we use the formulation of Adler and Taylor [3].

Lemma 35.16. *Let $d \geq 1$, let G be a standard Gaussian vector in $\mathbb{R}^d$, and let $F : \mathbb{R}^d \to \mathbb{R}$ be Lipschitz with constant ℓ for the Euclidean distance. Then for every $\lambda \in \mathbb{R}$,*

$$\log \mathbb{E} e^{\lambda (F(G) - \mathbb{E}F(G))} \leq \lambda^2 \ell^2 / 2.$$

Let $\mathcal{T}$ be a nonempty compact metric space and $(Z_t)_{t \in \mathcal{T}}$ a centered real Gaussian process, whose finite-dimensional covariances may be degenerate, with a separable realization having almost surely continuous paths, and assume $\sigma_Z^2 = \sup_{t \in \mathcal{T}} \mathbb{E} Z_t^2 < \infty$. For any fixed countable dense set $\mathcal{T}_0 \subset \mathcal{T}$, the random variable $S_Z = \sup_{t \in \mathcal{T}} Z_t = \sup_{t \in \mathcal{T}_0} Z_t$ is measurable, almost surely finite, and integrable. If $\sigma_Z^2 > 0$, then, writing $m_Z = \mathbb{E} S_Z$,

$$\log \mathbb{E} e^{\lambda (S_Z - m_Z)} \leq \lambda^2 \sigma_Z^2 / 2 \quad (\lambda \in \mathbb{R}), \qquad \mathbb{P}\{S_Z - m_Z \geq u\} \leq e^{-u^2/(2\sigma_Z^2)} \quad (u > 0).$$

If $\sigma_Z^2 = 0$, the process and its supremum vanish almost surely. In particular this applies to $\mathcal{T} = \mathbb{T}_L^3 \times \{1,2,3\} \times S^2$ and $Z_{x,i,v} = \langle v, G_i(x) \rangle$ for a continuous centered Gaussian field $G = (G_i)$ with this realization; its supremum is $\max_i \sup_x |G_i(x)|$. The only constants are the displayed Lipschitz constant and supremum variance; no dimension, cardinality of the index set, or parameter of the Yang–Mills measure enters.

Proof. We use two statements of Adler and Taylor. By [3, Theorem 2.1.1, p. 50, (2.1.5)], the supremum of an almost surely bounded centered Gaussian process has finite mean and satisfies an upper tail bound. By [3, Lemma 2.1.5, p. 54, (2.1.13)], with the boundedness assumption added in the authors' correction list (entry 54-6), we have $\mathbb{E} e^{t h(G)} \leq e^{t^2/2}$ for $t > 0$, every bounded C^2 function h with $\mathrm{Lip}(h) \leq 1$ and $\mathbb{E} h(G) = 0$, and a vector G of

independent standard real Gaussian variables. The reference to Lemma 2.1.4 specifies this law of G.

The case $\lambda = 0$ is immediate, and if $\ell = 0$ then F is constant. Suppose $\ell > 0$ and $\lambda \neq 0$. Let $F_R = \max(-R, \min(F, R))$ and let $F_{R,\varepsilon}$ be its convolution with a nonnegative smooth probability kernel supported in a ball of radius ε. Then $F_{R,\varepsilon}$ is bounded and smooth with bounded derivatives, $\operatorname{Lip}(F_{R,\varepsilon}) \le \ell$, and $|F_{R,\varepsilon} - F_R| \le \ell\varepsilon$. Applying the corrected lemma to $h = \pm(F_{R,\varepsilon} - \mathbb{E}F_{R,\varepsilon}(G))/\ell$, with $t = |\lambda|\ell$ and the sign of λ, gives the asserted exponential moment bound for $F_{R,\varepsilon}$. At fixed R, uniform convergence as $\varepsilon \downarrow 0$ gives the bound for F_R; the centered functions differ by at most $2\ell\varepsilon$. Next $F_R \to F$ pointwise as $R \to \infty$ and $|F_R| \le |F|$. The bound $|F(g)| \le |F(0)| + \ell|g|$ implies integrability of $F(G)$ and every $e^{c|F(G)|}$, since $c\ell|g| \le |g|^2/4 + c^2\ell^2$. In particular,

$$e^{\lambda(F_R(G) - \mathbb{E}F_R(G))} \le e^{|\lambda|(|F(G)| + \mathbb{E}|F(G)|)},$$

an integrable bound independent of R. Dominated convergence for the expectations and centered exponentials proves the first assertion.

For the process, we set all values equal to zero on the exceptional null set where the paths are not continuous; this does not change the finite-dimensional laws. Continuity on the compact metric space identifies S_Z with the supremum over $\mathcal{T}_0$ and makes it measurable and almost surely finite. These properties, the centered Gaussian finite-dimensional laws, and separability are the hypotheses of the cited theorem, which gives a finite mean. Since $S_Z \ge Z_{t_1}$ for any fixed t_1, the negative part of S_Z is bounded by $|Z_{t_1}|$, so S_Z is integrable. We enumerate $\mathcal{T}_0 = \{t_1, t_2, \dots\}$, repeating points if it is finite, and let $M_N = \max_{i \le N} Z_{t_i}$. Let C_N be the positive semidefinite covariance of $(Z_{t_i})_{i \le N}$, take a square root $C_N^{1/2}$, and denote its rows by v_i. Then $(Z_{t_i})_{i \le N}$ has the law of $(v_i \cdot G)_{i \le N}$, since both characteristic functions equal $\exp(-\xi^{\mathsf{T}} C_N \xi/2)$. Since $|v_i|^2 = (C_N)_{ii} \le \sigma_Z^2$, the map $g \mapsto \max_i v_i \cdot g$ is σ_Z-Lipschitz, and the first assertion gives its centered exponential moment bound uniformly in N; no inverse covariance is used, so singular covariances are included. Since $M_N - Z_{t_1} \uparrow S_Z - Z_{t_1} \ge 0$, we have $\mathbb{E}M_N \to m_Z$ by monotone convergence, and Fatou's lemma yields, for each $\lambda \in \mathbb{R}$,

$$\mathbb{E}e^{\lambda(S_Z - m_Z)} \le \liminf_{N \to \infty} \mathbb{E}e^{\lambda(M_N - \mathbb{E}M_N)} \le e^{\lambda^2\sigma_Z^2/2}.$$

If $\sigma_Z > 0$, Markov's inequality with $\lambda = u/\sigma_Z^2$ gives the stated upper tail. If $\sigma_Z = 0$, all coordinates on $\mathcal{T}_0$ vanish on one event of probability one, and continuity extends this to every index. Finally, the directional index space $\mathbb{T}_L^3 \times \{1,2,3\} \times S^2$ is compact, and $\sup_{|v|=1} \langle v, G_i(x) \rangle = |G_i(x)|$. □

Lemma 35.17. *Let ψ, q_*, A_{obs}, and B_{obs} be as in Definition 35.1, evaluated at the coordinate-cycle holonomy triple H at the fixed observation time, under each of the following probability measures of Definition 4.13.*

1. *Under the Abelian Maxwell measure with Haar cycle holonomies, all three holonomies commute,* $A_{\mathrm{obs}} = 0$*, the statistic vanishes exactly,* $\mathscr{C}_{\mathrm{Ab}} = 0$*, and* $\mathbb{E}B_{\mathrm{obs}}^2$*,* $\operatorname{Var} B_{\mathrm{obs}} \geq c_{\mathrm{Ab}} > 0$*, independently of the coupling.*
2. *Under Gaussian holonomies with independent Haar factors, the triple* H *has the law* $H^{\otimes 3}$ *for every coupling, and its statistic is the fixed number* $\mathscr{C}(H^{\otimes 3}; A_{\mathrm{obs}}, B_{\mathrm{obs}}) > 0$*. More precisely, let* K_1 *be a compact tangential subset of the regular patch* P *with positive relative volume and* $\psi_0 \geq b_* > 0$ *on* K_1*, let* $r_c > 0$ *with* $\chi(b) = 1$ *whenever* $|b| \leq r_c$*, and put* $\mathcal{T}_c = \{y + b : y \in \mathsf{K}_1,\ b \in \mathcal{N}_y,\ |b| \leq r_c\}$*, a set contained in the tube* $\mathcal{U}$ *on which* q_* *is the exact quadratic form. If* P *is a probability measure on* $\mathrm{SU}(2)^3$ *with* $\operatorname{Var}_P B_{\mathrm{obs}} > 0$ *whose law of* $X = \operatorname{Log} H$ *dominates* $c_0\, d^9X$ *on* $\mathcal{T}_c$*, then* $\mathscr{C}(P; A_{\mathrm{obs}}, B_{\mathrm{obs}}) \geq c_0\kappa_{\mathrm{c}} > 0$*, where* κ_{c} *depends only on* $\mathcal{T}_c$*, the Jacobian of the tube coordinates,* Σ_θ^{-1}*, and the cutoffs.*
3. *Under Gaussian holonomies with Gaussian constant modes, assume the realization of Definition 4.13(3): with* $\mathsf{S}_t a$ *the smoothed field at the observation time, the centered scalar Gaussian process* $\langle v, (\mathsf{S}_t a)_i(x)/\sqrt{\gamma}\rangle$*, indexed by* $(x, i, v) \in \mathbb{T}_L^3 \times \{1,2,3\} \times S^2$*, has a separable realization with continuous paths and*

$$\sigma_*^2 = \sup_{x,i,|v|=1} \operatorname{Var}\langle v, (\mathsf{S}_t a)_i(x)/\sqrt{\gamma}\rangle < \infty.$$

Put $Z_* = \sup_{x,i,|v|=1}\langle v, (\mathsf{S}_t a)_i(x)/\sqrt{\gamma}\rangle$ *and* $m_* = \mathbb{E}Z_*$*. Then* $m_* < \infty$ *and, by Lemma 35.16,* $\mathbb{P}(Z_* - m_* > u) \leq e^{-u^2/(2\sigma_*^2)}$ *for* $u > 0$ *if* $\sigma_* > 0$*. Let* $r_* = \inf_{\operatorname{supp}\psi}|\operatorname{Log} H| > 0$ *and* $a_* = r_*/(\sqrt{3}L)$*. If* $\sqrt{\gamma}m_* \leq a_*/2$ *and* $\sigma_* > 0$*, then*

$$\mathbb{P}\{\operatorname{Log} H \in \operatorname{supp}\psi\} \leq \exp\Big\{-\frac{(a_* - \sqrt{\gamma}m_*)^2}{2\gamma\sigma_*^2}\Big\} \leq e^{-a_*^2/(8\gamma\sigma_*^2)},$$

and if $\sigma_* = 0$ *the regular patch has mass zero. In particular* $\mathbb{E}\psi = O(e^{-c/\gamma})$*. The constants may depend on the fixed covariance and smoothing of this measure, as well as on the fixed data.*
4. *Under the conditional normal Gaussian measure, a fixed smooth positive probability density on the regular patch, charging* O_0 *and* O_+*, is followed, given* y*, by the normalized Gaussian law* $b \sim N(0, \gamma\Sigma_\theta(y))$ *on* $\mathcal{N}_y$ *and by the map* $X = y + b$*; a fixed tube cutoff equal to one near* $b = 0$*, followed by normalization, changes all moments below only by* $O(e^{-c/\gamma})$*. This measure has the moment identities of Theorem 35.2, with errors* $O(e^{-c/\gamma})$*, and* $\mathscr{C} = 2\gamma^2 + O(e^{-c'/\gamma})$*.*

Proof. Item (1). In the embedded Abelian theory every value of the connection lies in the same one-dimensional subalgebra, so path ordering reduces to exponentiation of the integral and the three cycle matrices commute. Their logarithms in the identity chart lie in $\mathcal{M}$, so their normal coordinate is zero and $q_* = 0$ on the tube; off the chart the observables vanish by definition. Thus $A_{\mathrm{obs}} = 0$ everywhere for this measure. Conditionally on the

field of the Abelian Maxwell measure, the three cycle angles are translates of the three independent uniformly distributed phases on $\mathbb{R}/4\pi\mathbb{Z}$, and their joint law is therefore the Haar probability measure on this three-dimensional torus, independently of the coupling (Lemma 33.14). The invariant open sets O_0 and O_+ contain smaller open sets of angles of positive Haar measure, and the invariance of ψ_0 makes these probabilities independent of the color axis of the embedded subgroup. At $b = 0$ the normal cutoff equals one. If $a_0, a_+ > 0$ are the probabilities of these two sets of angles, then $\mathbb{P}\{B_{\mathrm{obs}} = 0\} \geq a_0$ and $\mathbb{P}\{B_{\mathrm{obs}} \geq b_*\} \geq a_+$, and Lemma 35.12 implies $\mathbb{E}B_{\mathrm{obs}}^2 \geq a_+ b_*^2$ and $\operatorname{Var} B_{\mathrm{obs}} \geq a_0 a_+ b_*^2$. This proves item (1).

Item (2): the law of the triple. We condition on the Gaussian field. Each factor $\mathrm{Hol}_i(\mathsf{S}_t a)$ is then deterministic, right multiplication of the Haar distributed U_i^0 by it preserves the Haar measure, and the three U_i^0 remain conditionally independent. For a bounded product test $\prod_i f_i(H_i)$ the conditional expectation is therefore $\prod_i \int f_i \, dH$, independently of the field. Taking indicators of rectangles, we find that the triple has the law $H^{\otimes 3}$ at every coupling.

Item (2): the lower bound. On $\mathcal{T}_c$ the density of d^9X in the tube coordinates, $d^9X = J(y,b)\, d\mathrm{vol}(y)\, db$, has a positive lower bound J_c. For $y \in \mathsf{K}_1$ we define

$$\overline{q}_y = \frac{1}{|B_{r_c}^4|} \int_{|b| \leq r_c} q_*(y+b)\, db, \qquad V_c(y) = \int_{|b| \leq r_c} \big(q_*(y+b) - \overline{q}_y\big)^2 \, db,$$

where $B_{r_c}^4$ is the ball of radius r_c in $\mathcal{N}_y$. The positive definite quadratic form $q_*(y + \cdot)$ is not constant on this ball, so $V_c(y) > 0$. Since its coefficients vary continuously and the ball is fixed in any orthonormal bundle trivialization, V_c is continuous on K_1 and has a positive minimum by compactness. On $\mathcal{T}_c$ we have $B_{\mathrm{obs}} = \psi_0(y) \geq b_*$ and $A_{\mathrm{obs}} = \psi_0(y) q_*$. For all real α and c, subtracting the normal average shows

$$\int_{|b| \leq r_c} \big(\psi_0(y)(q_*(y+b) - \alpha) - c\big)^2 \, db \geq \psi_0(y)^2 V_c(y).$$

If the law of X under P dominates $c_0\, d^9X$ on $\mathcal{T}_c$, the infimum in Lemma 35.12 is therefore at least $c_0 \kappa_{\mathrm{c}}$, where one choice is

$$\kappa_{\mathrm{c}} = J_c b_*^2 \int_{\mathsf{K}_1} V_c(y)\, d\mathrm{vol}(y) > 0.$$

Since $\mathbb{E}_P B_{\mathrm{obs}}^2 \leq 1$, the same lower bound holds for the statistic, whenever its denominators are positive, as assumed. In the identity chart the density of $H^{\otimes 3}$ is $(16\pi^2)^{-3} \prod_{i=1}^3 \operatorname{sinc}^2(|x_i|/2)$ by (4.2), which is smooth and strictly positive on $\mathcal{T}_c$ for any choice of K_1 as above, for instance a compact subset of O_+ with positive relative volume. The open sets of triples over O_0 and O_+ near $b = 0$ have positive Haar measure, so $\operatorname{Var} B_{\mathrm{obs}} > 0$ and $\mathbb{E}B_{\mathrm{obs}}^2 > 0$ under $H^{\otimes 3}$, and the lower bound just proved shows that the statistic of $H^{\otimes 3}$ is a strictly positive number. This proves item (2).

Item (3). We consider the scalar process $f_{x,i,v} = \langle v, (\mathsf{S}_t a)_i(x)/\sqrt{\gamma}\rangle$ on the nonempty compact metric space $\mathbb{T}^3_L \times \{1,2,3\} \times S^2$. Since the smoothing is linear and fixed, every finite vector of this process is centered Gaussian, possibly with singular covariance. Its continuous separable realization and the finiteness of σ_*^2 are the hypotheses imposed in the statement. By Lemma 35.16, we obtain measurability, almost sure finiteness, $m_* < \infty$, and the displayed tail bound above the mean; the supremum variance is $\sup \operatorname{Var} f_{x,i,v} = \sigma_*^2$ and the supremum is $Z_* = \max_i \sup_x |(\mathsf{S}_t a)_i(x)|/\sqrt{\gamma}$. The covariance before smoothing is γC with C fixed, so the law of the normalized field, m_*, and σ_*^2 do not depend on γ.

We check the deterministic bound on the holonomies. Along a coordinate cycle ℓ_i, the transport equation $H' = H(\mathsf{S}_t a)_i$ has speed $|(\mathsf{S}_t a)_i|$ for the invariant metric with the norm $|\cdot|$ on $\mathfrak{su}(2)$. The distance of its endpoint from the identity is at most the length of this path, which is at most $L \sup_{x,j} |(\mathsf{S}_t a)_j(x)|$, and on the identity chart this distance is $|\operatorname{Log} H_i|$. Taking the Euclidean norm of the three components, we obtain

$$|\operatorname{Log} H| \le \sqrt{3} L \sup_{x,i} |(\mathsf{S}_t a)_i(x)| = \sqrt{3} L \sqrt{\gamma}\, Z_*,$$

where the equality uses the supremum over all unit vectors v, which includes both signs. Thus a triple with $\operatorname{Log} H \in \operatorname{supp} \psi$ forces $Z_* \ge a_*/\sqrt{\gamma}$. If $\sqrt{\gamma} m_* \le a_*/2$, this threshold lies strictly above the mean, and the tail bound of Lemma 35.16 with $u = a_*/\sqrt{\gamma} - m_* > 0$ gives both displayed exponential bounds; then $\mathbb{E}\psi \le \mathbb{P}\{\operatorname{Log} H \in \operatorname{supp} \psi\}$. If $\sigma_*^2 = 0$, the zero-variance clause of Lemma 35.16 shows that the directional process and Z_* vanish almost surely; the field vanishes, the holonomies are the identity, which lies outside the regular patch, and the patch has mass zero.

Item (4). In an orthonormal frame of the normal space let G be a standard Gaussian vector in four dimensions, independent of y, and let $b = \sqrt{\gamma}\Sigma_\theta(y)^{1/2} G$. The frame may change between charts, but the conditional law and the following quadratic form on the normal bundle are intrinsic. We define, on the whole normal bundle over the patch,

$$q_{\mathrm{coord}}(y,b) = \frac{1}{2} b^{\mathsf{T}} \Sigma_\theta(y)^{-1} b = \gamma Q_0, \qquad Q_0 = |G|^2/2. \tag{35.36}$$

We choose $b_{\mathrm{t}} > 0$ so that $\{|b| < b_{\mathrm{t}}\}$ lies in a fixed smaller normal tube, where the tube coordinates are unique and the normal cutoff of ψ equals one. Only on this event do we identify $q_*(y+b) = q_{\mathrm{coord}}(y,b)$ and $\psi(y+b) = \psi_0(y)$; outside it, q_* and ψ denote their fixed bounded extensions, not the unbounded quadratic form q_{coord}. Gaussian integration by parts in one dimension gives $\mathbb{E}G_i^2 = 1$ and $\mathbb{E}G_i^4 = 3$, and expanding the square and using independence yields

$$\mathbb{E}Q_0 = 2, \qquad \mathbb{E}Q_0^2 = 6, \qquad \operatorname{Var} Q_0 = 2.$$

The compact patch has spectral bounds $cI \le \Sigma_\theta(y) \le CI$, by Theorem 33.11. We fix $b_1 > 0$ smaller than b_{t} and than every radius on which a tube cutoff equals one. All

departures from the preceding local identities, or from these cutoffs being one, occur on $\{|b| \geq b_1\}$, which requires $|G| \geq b_1/\sqrt{C\gamma}$. For every fixed r, the Gaussian polynomial tail on this event is at most $C_r e^{-c/\gamma}$: the function $|G|^r e^{-|G|^2/8}$ is bounded, and the remaining Gaussian density integrates to a finite constant after extracting $e^{-b_1^2/(8C\gamma)}$. These bounds are uniform in y. For $k = 1, 2$ and $j = 0, 1, 2$, the difference between $\psi(y+b)^k q_*(y+b)^j$ and $\psi_0(y)^k q_{\mathrm{coord}}(y,b)^j$ vanishes on $\{|b| < b_1\}$; on its complement the first is bounded by a fixed constant and the second by $C\gamma^j |G|^{2j}$. The polynomial tail bound therefore makes the expectation of the difference exponentially small, uniformly in y. Inserting the fixed tube cutoff also changes these moments by exponentially small errors, and its total mass is $1 + O(e^{-c/\gamma})$, so dividing by this mass preserves the errors; any polynomial factor in γ^{-1} is absorbed by decreasing c.

Up to these errors, $B_{\mathrm{obs}} = \psi_0(y)$ is independent of Q_0. The positive tangential density charges O_0 and O_+, and together with the normal ball near $b = 0$ this gives fixed positive lower bounds on $\mathbb{E}B_{\mathrm{obs}}^2$ and $\operatorname{Var} B_{\mathrm{obs}}$ for small γ, by Lemma 35.12. For $f = \psi$ and $f = \psi^2$ we obtain

$$\mathbb{E}[f q_*] = 2\gamma \mathbb{E} f + O(e^{-c/\gamma}), \qquad \mathbb{E}[f q_*^2] = 6\gamma^2 \mathbb{E} f + O(e^{-c/\gamma}).$$

The identity (35.27) of Lemma 35.12 now implies $\mathscr{C} = 2\gamma^2 + O(e^{-c'/\gamma})$ for a smaller $c' > 0$. □

35.10 Proof of the theorem on regular commuting triples

Proof of Theorem 2.10. Let L, M, θ, and $n \in 4\mathbb{N}$ be as in Theorem 2.10, fix J and the construction parameters as in Section 35.1, and choose $a \geq 0$ so large that $n' = nB^a$ satisfies the minimum-side conditions of Section 35.1, including $2\pi/n' < r_{\mathrm{flat}}/4$. Since $[n']_M = [n]_M$, the measure $\lambda_\gamma = \lambda_{L,\gamma,[n]_M}$ of Theorem 2.10 is the measure of Section 35.1 for the side n' (Corollary 29.26 and Theorem 29.28), so the results of this section apply with n' in place of n. The function ψ of Theorem 2.10, with a sufficiently small tube, is a function as in Definition 35.1(2). If necessary, we replace its normal cutoff by the average over simultaneous conjugation, which does not change ψ because ψ and its tangential factor are invariant. The tangential factor ψ_0 vanishes on a nonempty open set and is positive on another, hence at least $2b_*$ on a smaller nonempty open set for some $b_* > 0$; these sets may be taken invariant since ψ_0 is invariant. On the tube, q_* is the quadratic form of Theorem 2.10. Part (1) of Theorem 2.10 is Theorem 35.2. Part (2) is Theorem 35.3, since w and χ_w satisfy the invariance and support conditions of Definition 35.1(4); these invariant normal cutoffs, like the normal cutoff χ of ψ, need not be radial. Part (3) is Theorem 35.4, applied to the restriction of F to the normal bundle over a compact invariant regular patch containing the support of w in its relative interior. The constants depend on L, M, n, θ, and the test functions, but not on γ, and in each case the continuum limit is taken first, at fixed γ. □

SECTION 36

Quartic asymptotics near central holonomies

This section describes the continuum measures λ_γ near the eight central triples $(\pm I, \pm I, \pm I)$ as $\gamma \to 0$, and proves Theorem 2.11. We observe the holonomies $H = (H_1, H_2, H_3)$ around the three coordinate cycles, based at the origin, at a fixed flow time $t = \theta L^2$. Near the identity we write their logarithms as $X \in \mathfrak{su}(2)^3 \simeq \mathbb{R}^9$. Near the trivial flat connection, the Hessian of the Wilson action vanishes in the nine constant directions, and the leading term in these directions is the constant-mode quartic form

$$Q(Y) = \frac{1}{2} \sum_{i<j} |y_i \times y_j|^2, \qquad Y = (y_1, y_2, y_3) \in \mathbb{R}^9,$$

where the cross product corresponds to the Lie bracket of $\mathfrak{su}(2)$. We study central windows of radius $r_\gamma = \gamma^{1/4}$ and show that compactly supported tests of X/r_γ have leading expectations proportional to their integrals against the quartic reference measure $e^{-Q(Y)}\, dY$. This measure has infinite total mass, and all statements concern compactly supported tests. A continuous, nonnegative, nonzero, compactly supported invariant test has expectation comparable to r_γ. Ratios of expectations converge to ratios of integrals against $e^{-Q(Y)}\, dY$, with an error $O(\gamma^{1/4})$ for Lipschitz tests, and the dilation statistic tends to zero. We also treat the eight central triples, finitely many flow times, and holonomy paths on a compact time interval, and we compute the dilation statistic for four comparison measures.

The block averaging constraint and the effective classical action come from Sections 6, 9, and 10, and the flat refinements, the reconstruction maps, and the disintegration of the conditioned Wilson measure from Sections 14 and 15. The coordinates near the trivial flat connection, the transverse Hessian, and the order of the normalization constant come from Section 33, and the regularity of the flowed zero-fluctuation reconstruction and the comparison of flowed holonomies from Sections 25, 34, and 35. Finally, we use the bound on the small-field failure probability from Section 23 and the continuum limit at fixed coupling from Section 29. The covariance of Theorem 33.11 enters only in the last comparison measure. The results of this section are not used in later sections; they prove Theorem 2.11 and the further statements about central triples described after it in Section 2.

Section 36.1 fixes the setting and explains the scaling. The main steps are the exact quartic term of the effective classical action (Section 36.2), the comparison of flowed

holonomies (Section 36.3), and the integration over the transverse modes (Section 36.5). The passage to the continuum measure and the further results occupy Sections 36.6 to 36.9, and Section 36.10 deduces Theorem 2.11.

The separation of constant modes, with a quartic commutator interaction, from the nonconstant modes on a torus is an established idea in finite-volume Yang–Mills theory; see Lüscher [127] and Coste, González-Arroyo, Jurkiewicz, and Korthals Altes [62]. For a direct comparison in four-dimensional perturbation theory in finite volume, see Fodor, Holland, Kuti, Nógrádi, and Wong [91, Section 2, (2.1)–(2.7)], who separate the constant modes, treat their quartic interaction separately, and integrate the nonconstant modes. What we add is the exact coefficient Q/n for the constrained minimizers of our block averaging constraint at every finite depth, with bounds uniform in the depth, and a uniform comparison on windows of size $\gamma^{1/4}$. Up to normalization, Q is the action of the bosonic reduced SU(2) Yang–Mills matrix integral with three matrices, whose divergence is known: see Krauth, Nicolai, and Staudacher [122, (24)], Krauth and Staudacher [121, (2)], and, for a rigorous treatment, Austing and Wheater [7, Section 3 and Appendix A]. The integration over the transverse modes is a Laplace argument with degenerate directions (see Wong [155] for background); the main work is to verify its hypotheses uniformly. The center symmetry of Lemma 36.11 is the standard center symmetry of periodic lattice gauge theory, which goes back to 't Hooft [149].

36.1 Setting and notation

We fix the comparison side n and all parameters of the construction before the coupling varies. Below, K counts RG steps, and the underlying classical depth is JK. The quantity a_{map} below is the bound on the derivatives of the zero-fluctuation reconstruction maps fixed in Definition 34.1.

Definition 36.1. Fix $L > 0$, $\theta > 0$, an odd integer $M \geq 3$, an integer $J \geq 1$ with $B = M^J$, and the remaining parameters of the construction, such that $a_{\mathrm{map}}\sqrt{B} \leq 32/49$. Fix $n_* \in 4\mathbb{N}$ and a comparison side $n = n_* B^a$ on the sequence $(n_* B^K)_{K\geq 0}$, large enough for the lower bounds on the coarse side in Corollaries 23.11 and 29.26 and Definition 34.1, and in particular for (36.1) below. These choices are made before γ varies in $(0, \gamma_0]$, where $0 < \gamma_0 \leq 1$ is a fixed ceiling. For $K \geq 0$ set

$$N = nB^K, \qquad g^2 = \frac{\gamma}{n}, \qquad \nu = \nu_{N,n,\gamma}, \qquad \widehat{\nu} = (T_M^{JK})_{\#}\nu, \qquad r_\gamma = \gamma^{1/4},$$

and let Q be the quartic form on $\mathbb{R}^9$ displayed at the beginning of this section.

1. The stopped measure $\widehat{\nu}$ is the law of the coarse configuration $V = T_M^{JK} U$ on Λ_n when U has law ν. It has density

$$d\widehat{\nu} = \widehat{Z}^{-1} \mathbf{1}_{D_r} e^{-E_{JK,\mathrm{sat}}/g^2 - \overline{R}_{K,\gamma}}\, dH_n, \qquad \overline{R}_{K,\gamma} = R_{K,\gamma} - R_{K,\gamma}(I),$$

where $\widehat{Z}$ is the normalization constant, $R_{K,\gamma}$ is the effective interaction, and $E_{JK,\mathrm{sat}}$ is the gauge-invariant extension of the effective classical action E_{JK} from B_r to D_r of Theorem 11.6, that is, $E_{JK,\mathrm{sat}}(U) = E_{JK}(U^k)$ whenever $U^k \in B_r$. In particular $E_{JK,\mathrm{sat}} = E_{JK}$ on B_r.

2. Near the trivial orbit modulo based gauge transformations (gauge transformations equal to the identity at the origin) we use the coordinates of Theorem 33.8,

$$V(X,s) = e^{X/n+s}, \qquad X \in \mathbb{R}^9, \qquad d_0^* s = 0, \qquad \sum_x s_i(x) = 0,$$

with $d_{\mathrm{st}} = \dim s = 6n^3 - 6$, where X/n denotes the constant configuration and d_0^* is the lattice divergence on Λ_n. We write $J_n(X,s)\,dX\,ds$ for the image of the product Haar measure in the based quotient, and set

$$F_K(X,s) = E_{JK}(V(X,s)), \qquad f_K(X) = F_K(X,0), \qquad A_K(X) = D_s^2 F_K(X,0).$$

We write $\overline{R}_{K,\gamma}(X,s)$ for $\overline{R}_{K,\gamma}(V(X,s))$.

3. Let $\mathcal{H}^K = (\mathcal{H}_i^K)_{i=1}^3$ be the holonomies, around the three coordinate cycles based at the origin, of the fine configuration flowed by the Wilson flow $\Psi_{\theta L^2}^{(N)}$, where the fine configuration is represented through the reconstruction maps of Theorem 15.12. Let $\mathcal{H}_0^K(V)$ be the same triple for the flowed zero-fluctuation reconstruction of V (the map Ξ_0^K of Definition 34.1), in the same gauge frame at the basepoint. Let $\mathcal{H}_n(V) = \mathrm{Hol}(V)$ be the triple of coordinate-cycle holonomies of V itself, without flow.

4. Fix a neighborhood $\mathcal{V}$ of (I,I,I) with compact closure in $(\mathrm{SU}(2) \setminus \{-I\})^3$, on which Log denotes the componentwise principal logarithm. For a bounded Borel function $\varphi : \mathbb{R}^9 \to \mathbb{R}$ with compact support, define

$$\mathcal{I}_{\gamma,\varphi}(H) = \begin{cases} \varphi(\mathrm{Log}\, H / r_\gamma), & H \in \mathcal{V}, \\ 0, & H \notin \mathcal{V}. \end{cases}$$

5. A radial cutoff is a function $\chi \in C_c^\infty([0,\infty))$ that is nonnegative, not identically zero, and constant near zero. For a radial cutoff χ we write $\mathcal{I}_{\gamma,\chi(|\cdot|)}$ for the observable of the test $Y \mapsto \chi(|Y|)$, and we set

$$W_\chi(Y) = (4Q(Y) - 9)\chi(|Y|) - |Y|\chi'(|Y|), \qquad \mathcal{S}_{\gamma,P}(\chi) = \frac{\mathbb{E}_P \mathcal{I}_{\gamma,W_\chi}}{\mathbb{E}_P \mathcal{I}_{\gamma,\chi(|\cdot|)}}$$

for a probability P under which the denominator is positive.

6. For a probability P of a random vector $X \in \mathbb{R}^9$ we use instead $\mathcal{I}_{\gamma,\varphi}^{\mathrm{coord}}(X) = \mathbf{1}_{\{X \in \mathrm{Log}\,\mathcal{V}\}} \varphi(X/r_\gamma)$, and define $\mathcal{S}_{\gamma,P}(\chi)$ with $\mathcal{I}^{\mathrm{coord}}$ in place of $\mathcal{I}$.

We call $\mathcal{S}_{\gamma,P}(\chi)$ the dilation statistic; the name refers to the dilation integration-by-parts identity $\int W_\chi\, e^{-Q}\, dY = 0$ of Lemma 36.4. For each fixed bound on the support of φ and all sufficiently small γ, the map $Y \mapsto r_\gamma Y$ sends the support of $Y \mapsto \varphi(Y)$ into a compact subset of $\operatorname{Log}\mathcal{V}$. Then $\mathcal{I}_{\gamma,\varphi}$ is a continuous function on $\mathrm{SU}(2)^3$ for continuous φ, and the indicator in $\mathcal{I}^{\mathrm{coord}}_{\gamma,\varphi}$ equals one on the support of $\varphi(\cdot/r_\gamma)$. The observables are defined by extension by zero because we never use a logarithm outside $\mathcal{V}$.

We apply Theorem 33.8 at depth JK; in the depth notation of that theorem, $F_K = F_{JK}$, $A_K = A_{JK}$, and $d_{\mathrm{st}} = d_n$. The coordinates $V(X,s)$ are defined on a fixed neighborhood $\{|X| < \rho_n,\ |s| < s_n\}$ whose image lies in $B_r \subset D_r$, and the density J_n is smooth and strictly positive. The gauge directions along a based spanning tree integrate to one, while the conjugation at the root remains in the measure and is not divided out. The theorem also gives

$$D_s F_K(X,0) = 0, \qquad a_n I \le A_K(X) \le C_n I,$$

where the Hessian lower bound $a_n > 0$ is a fixed constant (in this section a_n does not denote the mesh L/n). The first derivative of $\overline{R}_{K,\gamma}$ remains in the density. The measure $\widehat{\nu}$ is the stopped measure of Definition 4.4 at side n and depth JK. Since no coarser restriction remains at the side n, it is the law of V under ν by Theorem 14.8(9).

Constants. Unless stated otherwise, constants $C, c, C_n, c_n, \dots$ in this section may depend on n, L, θ, M, J, the remaining parameters, γ_0, the charts, the neighborhoods, and the gauge frames fixed in this section, and the test functions, but not on K, N, the number of preceding averaging steps, or $\gamma \in (0, \gamma_0]$. This is the dependence described in item (6) of Definition 4.14. Additional dependence is indicated by subscripts. All estimates are on the periodic torus of side L.

Margins. Fix a radius $r_{\mathrm{flat}} < r$ below the margins of the classical, block-averaging, and gauge domains of the construction. The lower bound on n in Definition 36.1 includes

$$\frac{2\pi}{n} < \frac{r_{\mathrm{flat}}}{4}. \tag{36.1}$$

By Lemma 14.9, every flat configuration on Λ_n is gauge equivalent to constant commuting links $e^{H_i/n}$ with commuting $H_i \in \mathfrak{su}(2)$, $|H_i| \le 2\pi$. Its flat refinements to side nM^ℓ have links $e^{H_i/(nM^\ell)}$, and $|H_i|/(nM^\ell) \le 2\pi/n < r_{\mathrm{flat}}/4$ for every $\ell \ge 0$. The flat refinements, their vanishing conditional constrained minimizers, and their vanishing normalized fluctuation variables therefore lie in all coordinate domains of the construction with the same margins at every depth. This holds near each of the eight central triples.

We also fix nested coordinate balls $\mathcal{C} \Subset \mathcal{C}^+$ in the chart of Theorem 33.8, with

$$\frac{\rho_{\mathrm{c}}^+}{n} + s_{\mathrm{c}}^+ < \frac{1}{2}\min(r, r_{\mathrm{flat}}), \qquad \mathcal{C}^+ = \{|X| < \rho_{\mathrm{c}}^+,\ |s| < s_{\mathrm{c}}^+\}, \tag{36.2}$$

where the radii may be reduced, and such that the closure of $\mathcal{C}^+$ lies in the chart. Since $\max_{x,i}|X_i/n + s_i(x)| \le |X|/n + |s|$, the bound (36.2) gives a fixed chart of link logarithms on which the derivative bounds for $\overline{R}_{K,\gamma}$ in Lemma 33.6 hold. On $\{s = 0\}$, it also places the commuting constant configurations in the domains above. The coordinate map extends analytically past the closure of $\mathcal{C}^+$, and by compactness its derivatives of each fixed order are bounded there. These choices concern only the coordinates used for integration; the conditioned Wilson measure and the restriction to D_r in $\widehat{v}$ are unchanged.

Heuristics. In the coordinates $V(X,s)$, the stopped measure has density proportional to $e^{-F_K(X,s)/g^2}$ times the smooth positive factor $e^{-\overline{R}_{K,\gamma}} J_n$. By Theorem 36.2 below, $f_K(X) = Q(X)/n + O(|X|^5)$, and by Theorem 33.8, $F_K(X,s) = f_K(X) + (1/2)s^{\mathsf{T}} A_K(X) s + O(|s|^3)$. Since $ng^2 = \gamma = r_\gamma^4$, the substitution

$$X = r_\gamma Y, \qquad s = gz, \qquad g = \frac{r_\gamma^2}{\sqrt{n}},$$

gives

$$\frac{F_K(X,s)}{g^2} = Q(Y) + \frac{1}{2} z^{\mathsf{T}} A_K(0) z + (\text{an error of order } r_\gamma \text{ after integration}).$$

The Jacobian of the substitution is $r_\gamma^9 g^{d_{\mathrm{st}}}$. By Theorem 33.10, the normalization constant is of order $g^{d_{\mathrm{st}}+4}$, where the extra power g^4 comes from a neighborhood of the cone of commuting triples, along which Q vanishes. Since $r_\gamma^9/g^4 = n^2 r_\gamma$, the expectation of a compactly supported test of X/r_γ is of order r_γ. More precisely, for a bounded test φ supported in $\{|Y| \le R\}$ and either output map of Theorem 36.6, its expectation is $r_\gamma \kappa_{K,\gamma} \{ \int \varphi(Y) e^{-Q(Y)}\, dY + O_{n,R}(\|\varphi\|_\infty r_\gamma)\}$, where $\kappa_{K,\gamma}$ is positive and bounded above and below. The factor $\kappa_{K,\gamma}$ does not depend on the test and cancels in ratios, so we never need its limit. The measure $e^{-Q(Y)}\, dY$ has infinite mass because Q vanishes on the five-dimensional cone of commuting triples and grows only quadratically in the four normal directions, with a coefficient proportional to the square of the distance from the origin (Lemma 36.4). After rescaling, most of the mass near the cone lies at commuting triples far from the central triple. In the unscaled variables these triples give the dominant contribution to the normalization constant, but tests supported on the scale r_γ do not see them. This is why only compactly supported tests appear.

To pass from the coarse configuration to the flowed holonomies of the fine configuration, we compare $\mathcal{H}^K$ with the deterministic triple $\mathcal{H}_0^K(V)$. The matrix difference is of order $\sqrt{\gamma} = r_\gamma^2$, while the window $\{|\operatorname{Log} \mathcal{H}_0^K(V)| \le R r_\gamma\}$ has probability of order r_γ. An average estimate of the difference does not suffice, since its contribution could be large relative to the small probability of the window. Instead, we use an estimate conditional on the coarse configuration V (Theorem 34.11). Suppose that the window is measurable

with respect to V and has probability $O(r_\gamma)$, and that the conditional expected difference is $O(r_\gamma^2)$ uniformly in V. Then the expected difference on the window is $O(r_\gamma^3)$, which is $O(r_\gamma^2)$ relative to the window. This is the mechanism behind the rate $\gamma^{1/4}$.

Finally, we distinguish three quotients. The integration over V takes place in the quotient by based gauge transformations, so that the constant coordinates $X \in \mathbb{R}^9$ include the simultaneous conjugation at the basepoint, which is kept in the nine-dimensional integral. Observables of the continuum measure are invariant under simultaneous conjugation of the triple, and at several flow times they are invariant under an independent conjugation at each time. On a time interval, the observable is a path in the quotient $\mathfrak{su}(2)^3 / \operatorname{Ad}\mathrm{SU}(2)$; no gauge frame common to different times is needed.

36.2 The quartic term of the effective classical action

The following theorem identifies the fourth Taylor polynomial of the effective classical action on constant configurations. Here E_d is the effective classical action at depth d on the lattice of side n, that is, the value of the constrained minimizer of Theorem 10.6 (the background field, following Bałaban [16]), with $E_0 = S_W$.

Theorem 36.2. *For every side n as in Definition 36.1 there exist $\rho_0(n) > 0$ and $C_n < \infty$ such that, for every depth $d \geq 0$, the function $f_d(X) = E_d(e^{X/n})$ satisfies*

$$D^j f_d(0) = 0 \quad (0 \leq j \leq 3), \qquad \frac{1}{4!} D^4 f_d(0)[X, X, X, X] = \frac{Q(X)}{n},$$

and

$$\left| E_d(e^{X/n}) - \frac{Q(X)}{n} \right| \leq C_n |X|^5 \qquad (|X| < \rho_0(n)). \tag{36.3}$$

The constants depend only on n, M, and the domains and derivative bounds of Theorem 10.11; they do not depend on d or on the coupling. In particular this applies to f_K of Definition 36.1, with $d = JK$.

Moreover, for every fixed d and X there exist constant fine links $e^{a_i(u)}$, analytic in u for $|u| < u_{d,X}$, such that

$$a_i(u) = uM^{-d} x_i / n + O_{d,X}(u^2), \qquad T_M^d(e^{a(u)}) = e^{uX/n},$$

and these links lie in the domain of the variational comparison of Theorem 9.7 at depth d. The radius $u_{d,X}$ need not be uniform in d.

We identify the fourth Taylor polynomial exactly, at each finite depth, by comparing the minimizer with an exactly constrained trial configuration. The proof uses two different tools. To identify the Taylor coefficients at a fixed depth d, we use an inverse of the block averaging constraint on constant configurations whose radius of definition may shrink

as d grows. This is harmless, because Taylor coefficients at the origin are determined by arbitrarily small neighborhoods. For the uniformity of the remainder in d, we use instead the bounds on the fifth derivative of E_d from Theorem 10.11, which are uniform in the depth. We also distinguish two constraints: the block averaging constraint $T_M^d U = e^\theta$ itself, and the gauge-modified constraint satisfied by the fields in the projected Landau gauge slice of Theorem 9.9. The trial configurations and the representative of the minimizer constructed in Step 2 satisfy the former, and their transforms into the slice in Step 1 satisfy the latter. The gauge transformation used in Step 5 need not preserve either constraint; we use it only to compute the action, which it leaves unchanged.

Proof. All variational comparisons use real targets, fields, and parameters; holomorphic extensions supply analytic representatives and Taylor bounds.

Step 1: the variational comparison. We set $\varepsilon_d = M^{-d}$ and measure lengths in units of the lattice of side n, so that the fine lattice has side n/ε_d and the action is

$$S_{\varepsilon_d}(U) = \frac{4}{\varepsilon_d} \sum_p q(U_p). \tag{36.4}$$

We take the minimizer and its value from Theorem 10.6, in the notation of Definition 10.1. We write Ξ_d for the constraint coordinates $\mathsf{P}_d^{\mathrm{cl}}$ and H_d for the minimizing linear extension H_d, as in Theorem 9.7. The critical point is $v_d^*(\theta) \in \mathcal{V}_d$, the minimizing field is $A_d(\theta) = \Xi_d(v_d^*(\theta) + H_d\theta)$, and

$$E_d(e^\theta) = S_{\varepsilon_d}(e^{\varepsilon_d A_d(\theta)}). \tag{36.5}$$

Fix $0 < r_{\mathrm{ph}} < r_0$ within the common margins, where r_0 and C_* are the constants of Theorem 9.7. By the bound on v_d^* in Theorem 10.6, there exists a constant $C_{\min}$, which includes the conversion from first differences to the curl, such that

$$\|v_d^*(\theta)\|_\infty + \|\mathrm{d}_{\varepsilon_d} v_d^*(\theta)\|_\infty \le C_{\min}\|\theta\|_\infty^2.$$

Let $\tau > 0$ be a target radius with

$$\tau < \min\{r_\theta/2,\ r_{\mathrm{ph}}/(2C_*)\}, \qquad C_{\min}\tau^2 < r_{\mathrm{ph}}/4, \tag{36.6}$$

where r_θ is the radius of Theorem 10.6. For $\|\theta\|_\infty < \tau$ the critical point then lies strictly inside the convex domain of potentials and curls of Theorem 9.7, which verifies the critical point hypothesis of that theorem.

Next we describe the trial configurations, which are needed only in a small neighborhood that may depend on d. Let $U = e^{\varepsilon_d A}$ satisfy $T_M^d U = e^\theta$ with $\|\theta\|_\infty < \tau$, $\|A\|_{\mathsf{X}} = \|A\|_\infty + \|\mathrm{div}_{\varepsilon_d} A\|_\infty < r_{\mathsf{X}}$, and small curl. We apply Theorem 9.9 with prescribed coarse gauge datum zero. The transformed field has vanishing projected divergence and gauge-modified target exactly e^θ. It has the same action, and its potential A' and its curl

tend to zero with those of A. The transformed field need not satisfy the block averaging constraint $T_M^d U = e^\theta$ itself; for this reason, Step 2 constructs a separate representative of the minimizer that does.

To check that A' lies in the chart of Theorem 9.7, we write the constraint coordinates as $\Xi_d(w) = w - H_d \mathscr{D}_d(w)$, with $\mathscr{D}_d(0) = D\mathscr{D}_d(0) = 0$; here the argument w is a fine field, not a vector of normalized fluctuation variables. Since H_d is a fixed linear map on this finite lattice, the first-derivative estimate of Theorem 10.11 makes $w \mapsto A' + H_d \mathscr{D}_d(w)$ a contraction on a sufficiently small ball, which implies that Ξ_d is locally invertible at A'. By the defining identities of the constraint coordinates, the gauge-modified constraint of $\Xi_d(w)$ is the linear target, and the correction $H_d \mathscr{D}_d(w)$ has vanishing projected divergence. Hence the inverse parameter has the form $w = a + H_d\theta$ with $a \in \mathcal{V}_d$, and on this finite lattice the norm of a in potential and curl tends to zero with (A', θ). In particular, a sufficiently small trial configuration satisfies

$$\|a\|_\infty + \|\mathrm{d}_{\varepsilon_d} a\|_\infty < r_{\mathrm{ph}}/2. \tag{36.7}$$

By (36.6) and (36.7), the trial parameter and the critical point lie in the same convex domain with target e^θ, and so does the segment joining them. Since the transformation into the slice preserves the action, the convexity inequality of Theorem 9.7, together with the minimum of Theorem 10.6, implies that

$$E_d(e^\theta) \le S_{\varepsilon_d}(U). \tag{36.8}$$

The smallness required in this step may depend on d and on the direction of the trial configuration.

Step 2: an analytic representative of the minimizer. The field $A_d(\theta)$ is holomorphic, and its value is (36.5). It satisfies the gauge-modified constraint of the projected Landau slice, and its accumulated gauge transformation is a finite composition of the analytic gauge maps of that slice. We take the logarithm of this gauge transformation at the vertices of the coarse lattice, extend it by any fixed linear extension to all fine vertices, and exponentiate. The result is an analytic periodic fine gauge transformation with the prescribed values at the coarse vertices. Since T_M is gauge covariant, applying this transformation changes the target of the block averaging constraint to e^θ, and it leaves (36.4) unchanged. We obtain an analytic configuration $U_*(\theta)$ with

$$U_*(0) = I, \qquad T_M^d U_*(\theta) = e^\theta, \qquad S_{\varepsilon_d}(U_*(\theta)) = E_d(e^\theta).$$

Only analyticity at the fixed depth d is used: even if the derivative bounds of this extension grow with d, it tends to the identity as $\theta \to 0$ at fixed d.

Step 3: an exactly constrained trial and the upper bound. Fix d and X. On constant configurations, the map $G_d(a) = \operatorname{Log} T_M^d(e^a)$ is analytic near zero and $DG_d(0) = \varepsilon_d^{-1} I_9$. Indeed, the derivative at the identity of the formula of Definition 6.1 is the average of the integrals of a over the block paths. On a constant configuration each of these paths has integral Ma_i, and d iterations give $M^d a_i$. To invert G_d without a radius uniform in d, we solve

$$a = \varepsilon_d u \frac{X}{n} - \varepsilon_d \{ G_d(a) - \varepsilon_d^{-1} a \}.$$

Since the derivative of the term in braces vanishes at zero, the right side is a contraction with factor at most $1/2$ on a small fixed complex ball. For small $|u|$, it also maps the ball into itself. The iterates converge uniformly on a smaller complex disc in u, and the limit is analytic in u and real for real u. A fixed point satisfies $G_d(a) = uX/n$, and we obtain

$$a_i(u) = u\varepsilon_d x_i/n + O_{d,X}(u^2), \qquad T_M^d(e^{a(u)}) = e^{uX/n}. \tag{36.9}$$

The potential $a(u)/\varepsilon_d$ is constant, has vanishing divergence and curl, and tends to zero as $u \to 0$. We reduce $u_{d,X}$ until $\|uX/n\|_\infty < \tau$, the input bound of the projected Landau slice holds, and the trial parameter satisfies (36.7). By (36.6), the critical point for the same target lies in the same convex chart, and (36.8) applies to $e^{a(u)}$. The logarithm of a plaquette of the constant configuration $e^{a(u)}$ in the (i,j) plane is $u^2[\varepsilon_d x_i/n, \varepsilon_d x_j/n] + O_{d,X}(u^3)$. Using $q(e^v) = |v|^2/8 + O(|v|^4)$ and the fact that there are $(n/\varepsilon_d)^3$ plaquettes in each plane, we obtain

$$\begin{aligned} E_d(e^{uX/n}) &\le \frac{4}{\varepsilon_d}\Big(\frac{n}{\varepsilon_d}\Big)^3 \frac{u^4}{8} \sum_{i<j} |[\varepsilon_d x_i/n, \varepsilon_d x_j/n]|^2 + O_{d,X}(u^5) \\ &= \frac{u^4}{n} Q(X) + O_{d,X}(u^5). \end{aligned} \tag{36.10}$$

Step 4: the first-order term of the minimizer. We write the link logarithms of $U_*(uX/n)$ from Step 2 as $ua + O(u^2)$, where a is a cochain on the fine torus. By Step 2, its action equals $E_d(e^{uX/n})$, which is nonnegative and, by (36.10), of order u^4. The coefficient of u^2 in its action therefore vanishes. This coefficient is

$$\frac{1}{2\varepsilon_d} \sum_x \sum_{i<j} |(\mathrm{d}a)_{ij}(x)|^2,$$

where d denotes the unscaled cochain differential on the fine torus. We conclude that $\mathrm{d}a = 0$.

We record the periodic decomposition of a closed cochain. A closed cochain has the same integral over every translate of a coordinate cycle, because the difference of two such cycles is a sum of plaquette boundaries. We subtract the constant c_i equal to this integral divided by n/ε_d. The resulting cochain is closed and has vanishing integrals over

the coordinate cycles. Its integrals over all closed paths vanish: neighboring coordinate steps can be exchanged using plaquette boundaries, adjacent inverse steps cancel, and a remaining closed path reduces to a combination of the three coordinate cycles. Hence its integral from a fixed vertex to a vertex x does not depend on the path. Applying this to a and writing $\phi(x)$ for the integral of the modified cochain from the fixed vertex to x, we have

$$a_i(x) = c_i + \phi(x + \varepsilon_d e_i) - \phi(x). \tag{36.11}$$

Each statement holds separately in each real component of $\mathfrak{su}(2)$.

At the identity, the derivative of the block averaging map is the average of the integrals over the block paths. For a closed cochain, a translated path with endpoint connectors and the reference path differ by a contractible closed path and have equal integrals. It follows that d averaging steps send the constant part of (36.11) to $\varepsilon_d^{-1} c_i$, and the gradient to the gradient of the restriction of ϕ to the coarse lattice. The linearization of the constraint $T_M^d U_*(uX/n) = e^{uX/n}$ states that this image equals x_i/n on every coarse link in direction i. When we average over all coarse links in direction i, the gradient terms cancel by periodicity, and we obtain

$$c_i = \varepsilon_d x_i / n. \tag{36.12}$$

Step 5: the lower bound. The periodic fine gauge transformation $e^{u\phi(x)}$, with ϕ normalized to vanish at the root, removes the gradient in (36.11) to first order, and it preserves the action. In this gauge the link logarithms are $uc + u^2 b + O(u^3)$ for some cochain b, and the plaquette logarithms are

$$u^2 \big\{ (\mathrm{d}b)_{ij}(x) + [c_i, c_j] \big\} + O(u^3).$$

For each $i < j$, we have $\sum_x (\mathrm{d}b)_{ij}(x) = 0$ by periodicity, and therefore

$$\sum_x |(\mathrm{d}b)_{ij}(x) + [c_i, c_j]|^2 = \sum_x |(\mathrm{d}b)_{ij}(x)|^2 + \Big(\frac{n}{\varepsilon_d}\Big)^3 |[c_i, c_j]|^2.$$

Using (36.12) and $q(e^v) = |v|^2/8 + O(|v|^4)$, we find that the coefficient of u^4 in $E_d(e^{uX/n})$ is at least

$$\frac{1}{2\varepsilon_d} \Big(\frac{n}{\varepsilon_d}\Big)^3 \sum_{i<j} |[\varepsilon_d x_i/n, \varepsilon_d x_j/n]|^2 = \frac{Q(X)}{n}.$$

Together with (36.10), this proves that the fourth Taylor coefficient of $u \mapsto E_d(e^{uX/n})$ equals $Q(X)/n$ for every direction X. The vanishing of the lower coefficients also follows from $0 \le f_d(uX) \le C_{d,X} u^4$ for both signs of small real u. By analyticity and polarization, we obtain the stated identities for the multilinear derivatives, so the fourth Taylor polynomial of f_d at zero is Q/n at every finite depth. We make no claim about the fifth Taylor polynomial.

Step 6: the uniform remainder. Fix the side n. On a common smaller ball, we restrict the bound of Theorem 10.11 on the derivatives of order five of $\theta \mapsto E_d(e^\theta)$ to the linear substitution $\theta_{x,i} = X_i/n$. At the fixed side n there are finitely many coordinates $\theta_{x,i}$. Summing the tensor entries and including the factors $1/n$, we obtain

$$\sup_{d\geq 0} \sup_{|X|<\rho_0(n)} \|D^5 f_d(X)\| \leq C_n.$$

The ball and the bound do not depend on the radii $u_{d,X}$ of the trial configurations. Since the Taylor polynomial of order four of f_d at zero is Q/n by Step 5, Taylor's formula on the segment from 0 to X gives

$$f_d(X) - \frac{Q(X)}{n} = \frac{1}{4!}\int_0^1 (1-t)^4 D^5 f_d(tX)[X,X,X,X,X]\,dt.$$

The absolute value of the right side is at most $C_n|X|^5/5!$, which proves (36.3). □

36.3 Holonomies near the trivial flat configuration

This section proves two facts about the flowed holonomies. The first is deterministic. In the central coordinates, the logarithm of the flowed zero-fluctuation holonomy triple equals X up to errors of order $|X|^2 + |s|$. Further, a configuration that is close to the set of flat configurations and whose zero-fluctuation holonomies are close to the identity lies close to the trivial flat configuration. The second fact is probabilistic: under the conditioned Wilson measure, the flowed holonomies of the configuration differ from those of its zero-fluctuation reconstruction by $O(\sqrt{\gamma})$ in mean square. Let $\mathcal{F}_n$ denote the set of flat configurations on Λ_n, regarded in the quotient by based gauge transformations.

Theorem 36.3. *After a fixed reduction of the central coordinate neighborhood and of $\mathcal{V}$, the map $Z_K(X,s) = \operatorname{Log} \mathcal{H}_0^K(V(X,s))$ is defined on that neighborhood for every K, and*

$$\sup_K \|Z_K\|_{C^{1,1}} \leq C_n, \qquad |Z_K(X,s) - X| \leq C_n(|X|^2 + |s|), \qquad |Z_K(X,s)| \geq \tfrac{3}{4}|X| - C_n|s|,$$

$$\|D_X Z_K(X,s) - I\|_{\mathrm{op}} \leq C_n(|X| + |s|).$$

The maps $\mathcal{H}_0^K$ are equicontinuous, in a fixed finite family of smooth charts of the based quotient, on a neighborhood of $\mathcal{F}_n$, and $\mathcal{H}_0^K = \mathcal{H}_n$ on $\mathcal{F}_n$ for every finite K. Fix a continuous distance d on the based quotient. For every open neighborhood $\mathcal{U}$ of the trivial based orbit there exists $\delta_{\mathcal{U}} > 0$, independent of K and γ, such that

$$d(V,\mathcal{F}_n) \leq \delta_{\mathcal{U}} \quad \text{and} \quad \Big(\sum_{i=1}^{3} \rho_G(\mathcal{H}_{0,i}^K(V))^2\Big)^{1/2} \leq \delta_{\mathcal{U}} \qquad \text{imply} \qquad V \in \mathcal{U}.$$

The constant $\delta_{\mathcal{U}}$ may depend on d and $\mathcal{U}$.

Let $D_K = \max_i \|\mathcal{H}_i^K - \mathcal{H}_{0,i}^K(V)\|_{\mathrm{op}}$. Then, uniformly in K and $0 < \gamma \le \gamma_0$,

$$\mathbb{E}_\nu \min(1, D_K)^2 \le C_n \gamma.$$

For every neighborhood $\mathcal{V}'$ with compact closure in $\mathcal{V}$,

$$\mathbb{E}_\nu \Big[\mathbf{1}_{\{\mathcal{H}^K, \mathcal{H}_0^K(V) \in \mathcal{V}'\}} \min\big(1, |\operatorname{Log} \mathcal{H}^K - \operatorname{Log} \mathcal{H}_0^K(V)|\big)^2 \Big] \le C_{n,\mathcal{V}'} \gamma.$$

The estimates on D_K are averages over the whole conditioned Wilson measure; they are not conditioned on the configuration being near a central triple.

Proof. Steps 1–3 prove the deterministic statements, and Step 4 the probabilistic ones.

Step 1: deterministic regularity. We apply Theorem 35.9 with $m = 0$ to the parameters and the side n of Definition 36.1. With $m = 0$ there are no fluctuation variables among the inputs. The input is the coarse configuration alone, and the map of that theorem is $\Xi_0^K = \mathcal{H}_0^K$, with the same gauge frame at the basepoint and the same flow time. The part of Theorem 35.9 concerning a neighborhood of all flat configurations applies, because (36.1) places every commuting constant representative strictly inside the classical, block-averaging, and gauge margins, as checked in Section 36.1. We obtain a neighborhood of the trivial flat configuration on which $\sup_K \|\mathcal{H}_0^K\|_{C^{1,1}} \le C_n$, a modulus of continuity of $\mathcal{H}_0^K$ near $\mathcal{F}_n$ that is common to all K, and the identity

$$\mathcal{H}_0^K(V_0) = \mathrm{Hol}(V_0) \qquad (V_0 \in \mathcal{F}_n) \tag{36.13}$$

at every finite K. Independently, Lemma 14.9 explains (36.13): the zero-fluctuation refinements of a flat configuration are flat refinements, and the Wilson flow is stationary on flat configurations. These are identities of matrices; the conjugation at the basepoint has not been divided out. The neighborhoods and the modulus are those of the deterministic theorem, on its finite cover of charts, and they do not depend on K or γ. In particular, they are not chosen using the small-field event.

Step 2: the logarithm. At $V = I$ the output is (I, I, I). By the common modulus, for any prescribed smaller neighborhood of the identity in $\mathcal{V}$, there exists a neighborhood of the input whose outputs lie in it. The logarithm and its first two derivatives are bounded on the closure of that neighborhood. Composing with the maps of Step 1, we obtain $\sup_K \|Z_K\|_{C^{1,1}} \le C_n$. Let $b_K(X, s) = Z_K(X, s) - X$. If $s = 0$ and the three components of X are parallel, the links of $V(X, 0)$ commute, the configuration is flat, and (36.13) implies that $b_K(X, 0) = 0$. Each of the nine coordinate directions of $\mathbb{R}^9$ is of this form, and these directions span $\mathbb{R}^9$. Differentiating along them, we find

$$b_K(0, 0) = 0, \qquad D_X b_K(0, 0) = 0.$$

Integrating the Lipschitz bound on $D_X b_K$ along $v \mapsto (vX, 0)$, we obtain $|b_K(X,0)| \le C_n|X|^2$, and the bound on the derivative in s gives $|b_K(X,s) - b_K(X,0)| \le C_n|s|$. The Lipschitz bound on the derivative also implies $\|D_X Z_K(X,s) - I\| \le C_n(|X| + |s|)$. We reduce the radius in X so that $C_n|X| \le 1/4$. Then, by the triangle inequality, $|Z_K(X,s)| \ge |X| - C_n|X|^2 - C_n|s| \ge (3/4)|X| - C_n|s|$. The same reduction works for every K and every coupling.

Step 3: localization near the trivial flat configuration. The based quotient is compact, since it is the quotient of a finite product of copies of $\mathrm{SU}(2)$ by a compact group, and $\mathcal{F}_n$ is a closed subset. On a flat configuration, the transport along a path is unchanged by elementary plaquette moves and depends only on the homotopy class of the path. Then, on the universal covering lattice, the transports from the root define a periodic gauge transformation to the trivial configuration if and only if all three cycle holonomies are the identity. Thus the trivial orbit is the only flat based orbit whose cycle triple is (I, I, I); this is the based version of Lemma 14.9.

Let $\mathcal{U}$ be given, and let $\mathcal{U}_1$ be an open set containing the trivial orbit with $\overline{\mathcal{U}_1} \subset \mathcal{U}$. Suppose first that $\mathcal{F}_n \setminus \mathcal{U}_1$ is nonempty. Since it is compact and contains no flat configuration with identity cycle triple,

$$m = \min_{V_0 \in \mathcal{F}_n \setminus \mathcal{U}_1} \Big(\sum_i \rho_G(\mathrm{Hol}_i(V_0))^2 \Big)^{1/2} > 0.$$

By the common modulus of Step 1, (36.13), and the continuity of ρ_G, there exists $\delta_1 > 0$ such that, for every K, if V_0 is flat and $d(V, V_0) < \delta_1$, then the quantities $(\sum_i \rho_G(\mathcal{H}^K_{0,i}(V))^2)^{1/2}$ and $(\sum_i \rho_G(\mathrm{Hol}_i(V_0))^2)^{1/2}$ differ by less than $m/2$. Let $\delta_2 > 0$ be smaller than the distance from $\overline{\mathcal{U}_1}$ to the complement of $\mathcal{U}$, and let $\delta_{\mathcal{U}}$ be smaller than $\delta_1/2$, $\delta_2/2$, $m/4$, and the radius of the charts of Step 1. Suppose that V satisfies the two hypotheses, and let $V_0 \in \mathcal{F}_n$ be a closest flat point, which exists by compactness. Then the cycle radius of V_0, that is, the second of the two quantities above, is less than $\delta_{\mathcal{U}} + m/2 < m$. Hence $V_0 \notin \mathcal{F}_n \setminus \mathcal{U}_1$, that is, $V_0 \in \mathcal{U}_1$, and since $d(V, V_0) < \delta_2$, we have $V \in \mathcal{U}$. If $\mathcal{F}_n \setminus \mathcal{U}_1$ is empty, then $\mathcal{F}_n$ has positive distance from the complement of $\mathcal{U}$, and the first hypothesis alone implies the conclusion. This argument includes all flat configurations, in particular those near the other seven central triples.

Step 4: the probabilistic estimates. We apply Theorem 34.10 with $m = 0$ and $p = 2$. Its comparison triple is $\mathcal{H}^K_0(V)$, and its probability measure is the whole conditioned Wilson measure ν. By the moment bound of that theorem, we have

$$\mathbb{E}_\nu \min(1, D_K)^2 \le C_n \gamma.$$

This uses the condition $a_{\mathrm{map}}\sqrt{B} \le 32/49$ imposed in Definition 36.1. That theorem includes the conditional means, the matrices at the basepoint, and all flat configurations.

Finally, Log is Lipschitz, as a map of the three matrices, on the compact closure of $\mathcal{V}'$. To obtain a single constant for all pairs, we cover the closure by finitely many neighborhoods on which Log has bounded derivatives. Pairs closer than a Lebesgue number of this cover satisfy the derivative estimate in one of these neighborhoods. For the remaining pairs, we use the boundedness of Log and the positive lower bound on their distance. We obtain

$$\min\bigl(1, |\operatorname{Log} H - \operatorname{Log} H'|\bigr) \le C_{\mathcal{V}'} \min\bigl(1, \max_i \|H_i - H_i'\|_{\mathrm{op}}\bigr) \qquad (H, H' \in \mathcal{V}').$$

Squaring, restricting to the event in the statement, and using the bound on $\mathbb{E}_\nu \min(1, D_K)^2$, we obtain the last estimate. All reductions used fixed neighborhoods, the fixed side, and the bounds of the cited theorems, so the constants have the stated dependence. □

36.4 The quartic reference measure

We write m_Q for the quartic reference measure $dm_Q(Y) = e^{-Q(Y)}\, d^9Y$ on $\mathbb{R}^9$. The following lemma records its basic properties. The divergence of its total mass is known in the matrix-integral literature cited at the beginning of this section; we give a direct proof in the coordinates of Lemma 33.2.

Lemma 36.4. *The measure m_Q is locally finite and has infinite total mass. For every $\chi \in C_c^\infty([0,\infty))$ that is constant near zero and every $\lambda > 0$,*

$$\int \chi(\lambda|Y|)\, dm_Q(Y) = \lambda^{-9} \int \chi(|Y|) e^{-\lambda^{-4} Q(Y)}\, d^9Y,$$

and

$$\int \bigl\{(4Q(Y) - 9)\chi(|Y|) - |Y|\chi'(|Y|)\bigr\}\, dm_Q(Y) = 0.$$

If $\chi \ge 0$ and χ is not identically zero, then $\int \chi(|Y|)\, dm_Q(Y)$ is finite and strictly positive.

We call the second identity the dilation integration-by-parts identity. It states that $\int W_\chi\, dm_Q = 0$ for every radial cutoff χ, and it uses only the compact support of the test; no normalization of m_Q is involved.

Proof. Since $Q \ge 0$ is a polynomial, $0 < e^{-Q} \le 1$ everywhere, which proves local finiteness.

Infinite mass. We use Lemma 33.2, with its homogeneous cone, its injective normal tube, and its bound on the Jacobian, at every radius. We write the commuting triples as $y_i = \rho a_i u$ with $\rho \ge 1$ and $a, u \in S^2$, where (a, u) and $(-a, -u)$ give the same triple, and we set $\zeta_i = a_i u$. The normal space at ζ is

$$\mathcal{B}_{a,u} = \Bigl\{ b = (b_1, b_2, b_3) : b_i \perp u,\ \sum_i a_i b_i = 0 \Bigr\}, \qquad \dim \mathcal{B}_{a,u} = 4.$$

We claim that

$$Q(\zeta + b) = \tfrac{1}{2}|b|^2 + Q(b) \qquad (b \in \mathcal{B}_{a,u}). \tag{36.14}$$

Indeed, $(\zeta_i + b_i) \times (\zeta_j + b_j) = u \times (a_i b_j - a_j b_i) + b_i \times b_j$. The first term lies in $u^\perp$, while $b_i \times b_j$ is parallel to u because $b_i, b_j \perp u$. The two terms are therefore orthogonal. Since $|u \times w| = |w|$ for $w \perp u$, and

$$\sum_{i<j} |a_i b_j - a_j b_i|^2 = |a|^2 \sum_i |b_i|^2 - \Big|\sum_i a_i b_i\Big|^2 = |b|^2,$$

we obtain (36.14). The same computation with $y = \rho\zeta$ gives $Q(y + b) = \rho^2|b|^2/2 + Q(b)$.

Let $\eta > 0$ be smaller than the relative radius of the tube in Lemma 33.2 and so small that the Jacobian of the tube is at least $1/2$ when $|b|/\rho \le \eta$. We restrict to $|b| \le \eta/\rho$. For $\rho \ge 1$, this set lies in the tube, and $|b|/\rho \le \eta$. Since

$$Q(b) \le \frac{1}{2} \sum_{i<j} |b_i|^2 |b_j|^2 \le \frac{1}{4} |b|^4,$$

we have $Q(y + b) \le \eta^2/2 + \eta^4/4$, which we denote by C_η, on this whole unbounded tube. By Lemma 33.2, the volume of the cone is $(1/2)\rho^4 \, d\rho \, d\Omega_a \, d\Omega_u$, where the factor $1/2$ accounts for the double parametrization, and the tube is injective in the point y of the cone. The normal ball has volume $v_4 \eta^4 \rho^{-4}$, where v_4 is the volume of the unit ball in $\mathbb{R}^4$. For every $T > 1$, we obtain

$$\int_{\mathbb{R}^9} e^{-Q(X)} \, dX \ge \frac{1}{4} e^{-C_\eta} (4\pi)^2 v_4 \eta^4 \int_1^T \rho^4 \rho^{-4} \, d\rho = c_\eta (T - 1).$$

Letting $T \to \infty$ proves that the total mass is infinite. All radii and constants in this argument are absolute.

Scaling, the dilation integration-by-parts identity, and positivity. The change of variables $Z = \lambda Y$, together with $Q(Z/\lambda) = \lambda^{-4} Q(Z)$, gives the scaling formula. For λ in a compact neighborhood of 1, the integrands and their derivatives in λ are supported in a common compact set and have a common integrable bound. We may therefore differentiate at $\lambda = 1$, which gives the dilation integration-by-parts identity. Equivalently, by the homogeneity relation $Y \cdot \nabla Q = 4Q$, we have

$$\operatorname{div}\big\{ Y \chi(|Y|) e^{-Q(Y)} \big\} = \big\{ 9\chi(|Y|) + |Y| \chi'(|Y|) - 4Q(Y) \chi(|Y|) \big\} e^{-Q(Y)}.$$

The vector field on the left is smooth at zero because χ is constant there, and it vanishes outside a compact set. The integral of each of its coordinate derivatives over a box containing its support is zero, which proves the dilation integration-by-parts identity with the stated sign. Finally, a nonnegative smooth radial test that is not identically zero is positive on an interval of radii, which corresponds to a subset of $\mathbb{R}^9$ of positive volume. Since the density e^{-Q} is positive, the integral is positive, and it is finite by compact support. □

Remark 36.5. For $F_Q(\rho) = m_Q\{|Y| \le \rho\}$ and $f_Q = F_Q'$ we have, as $\rho \downarrow 0$,

$$f_Q(\rho) = \frac{32\pi^4}{105}\rho^8\big[1 + O(\rho^4)\big], \qquad F_Q(\rho) = \frac{32\pi^4}{945}\rho^9\big[1 + O(\rho^4)\big],$$

with absolute constants for the Euclidean norm and the normalization of Q used here. Indeed, polar coordinates give

$$f_Q(\rho) = \rho^8 \int_{S^8} e^{-\rho^4 Q(\omega)}\, d\omega. \tag{36.15}$$

Since Q is bounded on S^8, the exponential is $1 + O(\rho^4)$ uniformly. The area of S^8 is $2\pi^{9/2}/\Gamma(9/2) = 32\pi^4/105$, by $\Gamma(1/2) = \sqrt{\pi}$ and $\Gamma(v+1) = v\Gamma(v)$. Integrating the density from zero gives the asymptotics of F_Q. These formulas are not used below.

36.5 Integration over the transverse modes

The next theorem, our main integration result, computes the expectation, under the stopped measure, of a compactly supported test of the rescaled logarithm of either the unflowed coarse holonomy triple $\mathcal{H}_n(V)$ or the flowed zero-fluctuation holonomy triple $\mathcal{H}_0^K(V)$.

Theorem 36.6. *For every $R < \infty$ there exist $C_{n,R} < \infty$ and $\gamma_R > 0$ such that, for every bounded Borel function φ supported in $\{|Y| \le R\}$, every $K \ge 0$, and every $0 < \gamma \le \gamma_R$, both $T = \mathcal{H}_n(V)$ and $T = \mathcal{H}_0^K(V)$ satisfy*

$$\mathbb{E}_{\widehat{\nu}}\mathcal{I}_{\gamma,\varphi}(T) = r_\gamma \kappa_{K,\gamma}\Big\{\int \varphi\, dm_Q + \mathcal{E}_{K,\gamma,T,\varphi}\Big\}, \qquad |\mathcal{E}_{K,\gamma,T,\varphi}| \le C_{n,R}\|\varphi\|_\infty r_\gamma,$$

where the factor $\kappa_{K,\gamma}$ is the same for both maps and all tests, namely

$$\kappa_{K,\gamma} = (2\pi)^{d_{\mathrm{st}}/2} n^2 \frac{e^{-\overline{R}_{K,\gamma}(0,0)} J_n(0,0)}{\sqrt{\det A_K(0)}}\, \frac{g^{d_{\mathrm{st}}+4}}{\widehat{Z}}, \qquad 0 < c_n \le \kappa_{K,\gamma} \le C_n.$$

Here $\widehat{Z}$ is the normalization constant of $\widehat{\nu}$ in Definition 36.1, and $c_n g^{d_{\mathrm{st}}+4} \le \widehat{Z} \le C_n g^{d_{\mathrm{st}}+4}$.

Further, for $T_K(X,s) = \mathrm{Log}\, T(V(X,s))$ and both maps, the change of variables $Z = T_K(X,s)$ is invertible on a fixed smaller ball of outputs, and its inverse $X = H_K(Z,s)$ satisfies

$$|H_K(Z,s) - Z| \le C_n(|Z|^2 + |s|), \qquad |\det D_Z H_K(Z,s) - 1| \le C_n(|Z| + |s|).$$

The error bound depends on φ only through $\|\varphi\|_\infty$, and no modulus of continuity of φ enters. It accounts for the density factor $e^{-\overline{R}_{K,\gamma}} J_n$, including the first derivatives of $\overline{R}_{K,\gamma}$, for the Gaussian determinant of the transverse modes, and for the contributions of configurations away from the trivial flat configuration. We do not assert that $\kappa_{K,\gamma}$ has a limit. Since $\kappa_{K,\gamma}$ depends only on the stopped measure, it does not depend on the flow time at which $\mathcal{H}_0^K$ is observed.

Proof. Throughout the proof we write $D = d_{\mathrm{st}}$ and $\epsilon = r_\gamma$, and we set

$$a_{K,\gamma}(X,s) = e^{-\overline{R}_{K,\gamma}(X,s)} J_n(X,s), \qquad a^0_{K,\gamma} = a_{K,\gamma}(0,0).$$

Steps 1–3 localize to the central coordinates and invert the output map, Step 4 rescales so that the test becomes exactly $\varphi(Y)$, and Steps 5–6 compare the density with its Gaussian–quartic model and integrate.

Step 1: bounds on the density. We apply Theorem 33.8 to the stopped measure at depth JK, with the side, the parameters, and the upper bound γ_0 on the coupling from Definition 36.1. The chart $\mathcal{C}^+$ of (36.2) is a fixed chart of link logarithms, independent of the depth and the coupling. On it, Lemma 33.6, with $d = JK$ and link logarithms $X/n + s$, together with the residual value bound (P3) in Section 33.2, gives

$$\sup_{K,\,0<\gamma\le\gamma_0} \sup_{\mathcal{C}^+} \big(|\overline{R}_{K,\gamma}| + \|D_{X,s}\overline{R}_{K,\gamma}\|\big) \le C_n.$$

The smooth positive density J_n has positive upper and lower bounds and bounded first derivatives on the closure of a slightly smaller chart that still contains $\overline{\mathcal{C}}$, and we replace $\mathcal{C}^+$ by this chart. The identity

$$D_{X,s} a_{K,\gamma} = e^{-\overline{R}_{K,\gamma}} \big(D_{X,s} J_n - J_n D_{X,s}\overline{R}_{K,\gamma}\big)$$

then gives, on this fixed chart,

$$0 < c_n \le a_{K,\gamma}(X,s) \le C_n, \qquad \|D_{X,s} a_{K,\gamma}(X,s)\| \le C_n. \tag{36.16}$$

This uses the first derivatives of the centered effective interaction in both X and s. Together with the uniform classical derivative bounds of Theorem 10.11, Theorem 33.8 also gives

$$\begin{aligned} c_n I \le A_K(X) \le C_n I, \qquad \|A_K(X) - A_K(0)\| \le C_n|X|, \\ F_K(X,s) = f_K(X) + \tfrac{1}{2} s^{\mathsf{T}} A_K(X) s + O_n(|s|^3), \qquad F_K(X,s) \ge f_K(X) + c_n|s|^2. \end{aligned} \tag{36.17}$$

The density in Theorem 33.8 at depth JK is the stopped measure of Definition 36.1, so its normalization constant is $\widehat{Z}$. By Theorem 33.10, after reducing the upper bound on the coupling if necessary, we have

$$c_n g^{D+4} \le \widehat{Z} \le C_n g^{D+4}. \tag{36.18}$$

Step 2: reduction to a neighborhood of the trivial flat configuration. Fix a neighborhood $\mathcal{U}$ of the trivial based orbit whose closure lies in the coordinate chart of (36.17). By Theorem 36.3, applied to the map $\mathcal{H}^K_0$, there exists $\delta > 0$ such that a configuration within distance δ of $\mathcal{F}_n$ whose cycle radius $(\sum_i \rho_G(\mathcal{H}^K_{0,i}(V))^2)^{1/2}$ is at most δ lies in $\mathcal{U}$. For the map $\mathcal{H}_n = \mathrm{Hol}$, the same statement follows from the argument of Step 3 of that proof,

with the continuous map Hol in place of $\mathcal{H}_0^K$. On the complement of $\{d(V,\mathcal{F}_n)\le\delta\}$, we have $E_0\ge c_{n,\delta}>0$ by the compactness of the configuration space and $E_0^{-1}(0)=\mathcal{F}_n$. The lower bound $E_{JK,\mathrm{sat}}\ge\kappa E_0$ on D_r and the bound $|\overline{R}_{K,\gamma}|\le C_n$ on D_r, properties (P1) and (P3) in Section 33.2, together with (36.18), give

$$\widehat{\nu}\{d(V,\mathcal{F}_n)>\delta\}\le C_n g^{-(D+4)}e^{-c_n/g^2}. \tag{36.19}$$

Since the product Haar measure has total mass one, no volume factor depending on K appears in this bound. Suppose that $R\epsilon<\delta$ and that the image of $\{|Y|\le R\}$ under $Y\mapsto\epsilon Y$ lies in $\operatorname{Log}\mathcal{V}$. If $\mathcal{I}_{\gamma,\varphi}(T)\neq 0$, then $|\operatorname{Log}T|\le R\epsilon$, and the cycle radius is at most $R\epsilon<\delta$, since $\rho_G(H_i)\le|\operatorname{Log}H_i|$. Every contribution to either observable from $V\notin\mathcal{U}$ is therefore bounded by (36.19). The flat configurations near the other seven central triples, and all other flat configurations, still contribute to $\widehat{Z}$.

Step 3: the output coordinates. For $T=\mathcal{H}_0^K$, Theorem 36.3 gives

$$|T_K(X,s)-X|\le C_n(|X|^2+|s|),\qquad \|D_XT_K(X,s)-I\|\le C_n(|X|+|s|). \tag{36.20}$$

For $T=\mathrm{Hol}$, each cycle holonomy is a product of n link matrices of $V(X,s)$, which are analytic functions of (X,s) at the fixed side n, and at $s=0$ the i-th holonomy is exactly e^{x_i}. Then $T_K(X,0)=X$ and $D_XT_K(X,0)=I$. Since these finite products and Log have bounded derivatives, (36.20) holds for this map as well, with constants independent of K.

Fix a convex ball $\{|X|\le\rho\}$ and a ball in s on which $\|D_XT_K-I\|\le 1/4$ for both maps. We set $b_K=T_K-X$ and restrict $|s|$ further so that $|b_K(0,s)|\le\rho/8$. For $|Z|<\rho/4$, the map $X\mapsto Z-b_K(X,s)$ sends the closed ball of radius $\rho/2$ into itself, since $|Z-b_K(X,s)|\le|Z|+|b_K(0,s)|+|X|/4\le\rho/2$, and it is a contraction with factor at most $1/4$. Its iterates converge to a unique fixed point $X=H_K(Z,s)$, which solves $T_K(X,s)=Z$. The inequality

$$|T_K(X,s)-T_K(X',s)|\ge\tfrac{3}{4}|X-X'|$$

on the larger convex ball excludes any other preimage there. Differentiating the fixed point equation, or taking difference quotients and using the uniformly convergent Neumann series for $(I+D_Xb_K)^{-1}$, we obtain

$$D_ZH_K=\big(I+D_Xb_K(H_K,s)\big)^{-1}.$$

The fixed point equation first gives $|H_K|\le|Z|+|b_K(0,s)|+|H_K|/4$, which implies $|H_K|\le C_n(|Z|+|s|)$, and then

$$|H_K(Z,s)-Z|\le C_n(|Z|^2+|s|),\qquad |\det D_ZH_K(Z,s)-1|\le C_n(|Z|+|s|). \tag{36.21}$$

For the determinant, we expand about I_9 and use the bound on D_Xb_K; the determinant is positive since D_ZH_K stays close to I_9. This proves the second part of the theorem. We may

take the neighborhood $\mathcal{U}$ of Step 2 inside this domain, so that points outside it whose output lies in the compact window are controlled by (36.19). Inside $\mathcal{U}$, after a further fixed reduction of ρ, (36.20) implies the lower bound $|T_K(X,s)| \geq (3/4)|X| - C_n|s|$ for both maps. Because of its term in s, we use it together with the tail bound of Step 4.

Step 4: the transverse tail and the change of variables. Since $f_K \geq 0$, (36.17) and the bounds on the density factor give

$$\widehat{\nu}\big(\{|s| > \epsilon\} \cap \{V \text{ in the central coordinates}\}\big) \leq C_n g^{-4} e^{-c_n \epsilon^2/g^2} = C_n g^{-4} e^{-c_n'/\sqrt{\gamma}}. \tag{36.22}$$

To see this, we split $e^{-c_n|s|^2/g^2}$ into two equal factors and bound one of them by $e^{-c_n\epsilon^2/(2g^2)}$ on $\{|s| > \epsilon\}$. Integrating the other over $\mathbb{R}^D$ gives $C_n g^D$. The X-integral is over a bounded set, and we divide by (36.18). In the last equality, we used $\epsilon^2/g^2 = n/\sqrt{\gamma}$. Both (36.19) and (36.22) are smaller than every fixed power of ϵ: with $v = 1/\sqrt{\gamma}$, each is a fixed power of v times e^{-cv} or e^{-cv^2}.

On $\{|s| \leq \epsilon\}$, an output with $|T_K| \leq R\epsilon$ satisfies $|X| \leq C_{n,R}\epsilon$ by the lower bound of Step 3. For γ_R small, this lies in the domain of the inverse. Conversely, the inverse is defined for all $|Z| \leq R\epsilon$ and $|s| \leq \epsilon$, and the bound $|H_K| \leq C_{n,R}\epsilon$ places its values in $\mathcal{U}$ for γ_R small. The part of the unnormalized integral of the observable over $\mathcal{U} \cap \{|s| \leq \epsilon\}$ is therefore transformed exactly by

$$Z = \epsilon Y, \qquad s = gz, \qquad X = H_K(\epsilon Y, gz), \qquad dX\, ds = \epsilon^9 g^D \det D_Z H_K(\epsilon Y, gz)\, dY\, dz. \tag{36.23}$$

In these variables the test is exactly $\varphi(Y)$, and no regularity of φ is needed. The variable z ranges over the ball $\{|z| \leq \epsilon/g\}$, and the test restricts Y to $\{|Y| \leq R\}$. Since $g = \epsilon^2/\sqrt{n}$, (36.21) gives on this domain

$$\Big|\frac{X}{\epsilon} - Y\Big| \leq C_{n,R}\epsilon(1+|z|), \qquad |\det D_Z H_K - 1| \leq C_{n,R}\epsilon(1+|z|), \qquad \frac{|X|}{\epsilon} \leq C_{n,R}. \tag{36.24}$$

The last bound uses $|s| \leq \epsilon$, so it holds on the whole truncated ball in z, even when $|z|$ is large.

Step 5: comparison of the density. By Theorem 36.2 and $ng^2 = \epsilon^4$, we have $f_K(X)/g^2 = Q(X/\epsilon) + O(|X|^5/g^2)$, and therefore

$$\Big|\frac{f_K(X)}{g^2} - Q(X/\epsilon)\Big| \leq C_n \frac{|X|^5}{g^2} \leq C_{n,R}\epsilon.$$

Since X/ϵ and Y lie in a fixed ball, on which Q has bounded first derivatives, the first bound in (36.24) gives

$$\Big|\frac{f_K(X)}{g^2} - Q(Y)\Big| \leq C_{n,R}\epsilon(1+|z|). \tag{36.25}$$

This uses only $|X| \le C_{n,R}\epsilon$, which holds on the whole truncated domain. In particular, it bounds the absolute error on the compact rescaled window, including the directions of commuting triples, where Q vanishes. The expansion in (36.17) gives

$$\Big|\frac{F_K(X, gz) - f_K(X)}{g^2} - \frac{1}{2} z^{\mathsf{T}} A_K(0) z\Big| \le C_{n,R}\big(\epsilon|z|^2 + g|z|^3\big), \tag{36.26}$$

where we used $\|A_K(X) - A_K(0)\| \le C_n|X| \le C_{n,R}\epsilon$ and the cubic remainder in $s = gz$. We also use the elementary inequality

$$|e^{-A} - e^{-B}| \le |A - B|\big(e^{-A} + e^{-B}\big) \qquad (A, B \in \mathbb{R}), \tag{36.27}$$

which holds because the integral of the derivative of $t \mapsto e^{-t}$ between A and B is at most $|A - B| \max(e^{-A}, e^{-B})$.

Let $P_K(Y, z) = F_K(H_K(\epsilon Y, gz), gz)/g^2$ and $P_K^0(Y, z) = Q(Y) + z^{\mathsf{T}} A_K(0) z/2$. Both are at least $c_n|z|^2$, by $f_K \ge 0$, $Q \ge 0$, and the lower bounds in (36.17). Combining (36.25) and (36.26), and using $g \le C_n\epsilon^2 \le C_n\epsilon$, we find

$$|P_K - P_K^0| \le C_{n,R}\epsilon\big(1 + |z| + |z|^2 + |z|^3\big).$$

By (36.27), $|e^{-P_K} - e^{-P_K^0}|$ is at most $C_{n,R}\epsilon$ times a fixed polynomial in $|z|$ times $e^{-c_n|z|^2}$, which is integrable over $\mathbb{R}^D$ uniformly in K and γ; the phase error itself need not be uniformly small on the whole truncated ball in z. For the density factor and the Jacobian, we write $b_{K,\gamma}(Y, z) = a_{K,\gamma}(X, gz) \det D_Z H_K(\epsilon Y, gz)$ with $X = H_K(\epsilon Y, gz)$. By (36.16) and (36.24), these quantities are bounded on our domain and

$$|b_{K,\gamma}(Y, z) - a_{K,\gamma}^0| \le C_{n,R}\epsilon(1 + |z|).$$

Splitting the difference as $(b_{K,\gamma} - a_{K,\gamma}^0)e^{-P_K} + a_{K,\gamma}^0(e^{-P_K} - e^{-P_K^0})$ and using that $t \mapsto (1 + t + t^2 + t^3)e^{-c_n t^2/2}/(1 + t)$ is bounded on $[0, \infty)$, we obtain

$$\Big|b_{K,\gamma}(Y, z)e^{-P_K(Y,z)} - a_{K,\gamma}^0 e^{-P_K^0(Y,z)}\Big| \le C_{n,R}\epsilon(1 + |z|)e^{-c_n'|z|^2} \qquad (|Y| \le R,\ |z| \le \epsilon/g), \tag{36.28}$$

with $c_n' = c_n/2$.

Step 6: integration and the common factor. The Gaussian integral of $a_{K,\gamma}^0 e^{-P_K^0}$ over $\{|z| > \epsilon/g\}$ is exponentially small, since $\epsilon/g = \sqrt{n}/\epsilon \to \infty$. Integrating (36.28) first in z and then over $\{|Y| \le R\}$, we obtain

$$\widehat{Z}\,\mathbb{E}_{\widehat{\nu}}\mathcal{I}_{\gamma,\varphi}(T) = g^D\epsilon^9\Big\{a_{K,\gamma}^0\frac{(2\pi)^{D/2}}{\sqrt{\det A_K(0)}}\int \varphi(Y)e^{-Q(Y)}\,dY + O_{n,R}\big(\|\varphi\|_\infty\epsilon\big)\Big\}. \tag{36.29}$$

The contributions discarded in Steps 2 and 4 satisfy the same bound after reducing γ_R: after normalization by $\widehat{Z}$ they are $O_{n,R}(\|\varphi\|_\infty\epsilon^2)$, while the prefactor of the claimed expansion is bounded above and below by fixed multiples of ϵ.

Finally, since $\epsilon^4 = \gamma = ng^2$, we have the identity

$$\frac{g^D \epsilon^9}{\widehat{Z}} = \epsilon n^2 \frac{g^{D+4}}{\widehat{Z}}.$$

Thus the coefficient of $\int \varphi \, dm_Q$ in (36.29), divided by $\widehat{Z}$, is exactly $\epsilon \kappa_{K,\gamma}$. The bounds $c_n \le \kappa_{K,\gamma} \le C_n$ follow from (36.18), $c_n \le a^0_{K,\gamma} \le C_n$, and $c_n I \le A_K(0) \le C_n I$ in the fixed dimension D. Dividing by $\epsilon \kappa_{K,\gamma}$, we obtain the stated bound on $\mathcal{E}_{K,\gamma,T,\varphi}$. The leading term involves only $a^0_{K,\gamma}$ and $A_K(0)$, which are the same for both maps; the factor is therefore the same for both maps and every test. Every reduction used a radius or an upper bound on the coupling depending only on the fixed data and R. □

36.6 Passage to the continuum measure

We now pass from the stopped measure to the conditioned Wilson measure, then to the Wilson measure, and finally to the continuum measure. The order of limits is as follows: at each fixed γ we let $K \to \infty$ with the comparison side n fixed, and only afterwards do we let $\gamma \to 0$. Throughout, $\lambda_\gamma = \lambda_{L,\gamma,[n_*]_M}$ denotes the continuum measure for the sequence of sides $(n_* B^K)_{K \ge 0}$ (Theorem 29.25 and Corollary 29.26). Since $n = n_* B^a$, the classes $[n]_M$ and $[n_*]_M$ coincide. We observe the holonomies of a trajectory around the three coordinate cycles based at the origin, at the time $t = \theta L^2$. Under λ_γ this triple is defined up to simultaneous conjugation, and we use invariant tests.

Theorem 36.7. *Fix the data of Definition 36.1 before varying γ. For every continuous compactly supported function $\varphi : \mathbb{R}^9 \to \mathbb{R}$ that is invariant under simultaneous conjugation and every radial cutoff χ,*

$$\lim_{\gamma \downarrow 0} \frac{\mathbb{E}_{\lambda_\gamma} \mathcal{I}_{\gamma,\varphi}}{\mathbb{E}_{\lambda_\gamma} \mathcal{I}_{\gamma,\chi(|\cdot|)}} = \frac{\int \varphi \, dm_Q}{\int \chi(|Y|) \, dm_Q(Y)}.$$

For every such φ that is nonnegative and not identically zero, there exist $0 < c_\varphi \le C_\varphi < \infty$ and $\gamma_\varphi > 0$ such that

$$c_\varphi r_\gamma \le \mathbb{E}_{\lambda_\gamma} \mathcal{I}_{\gamma,\varphi} \le C_\varphi r_\gamma \qquad (0 < \gamma \le \gamma_\varphi).$$

In particular the denominator above is positive for small γ, and

$$\lim_{\gamma \downarrow 0} \mathcal{S}_{\gamma,\lambda_\gamma}(\chi) = 0.$$

Further, for any two smooth compactly supported functions χ_1, χ_2 on $[0,\infty)$ that are not identically zero, with $A_i = \mathcal{I}_{\gamma,\chi_i(|\cdot|)}$,

$$\frac{\mathrm{Var}_{\lambda_\gamma}(A_1)}{\mathrm{Var}_{\lambda_\gamma}(A_2)} \longrightarrow \frac{\int \chi_1(|Y|)^2 \, dm_Q(Y)}{\int \chi_2(|Y|)^2 \, dm_Q(Y)} \qquad (\gamma \downarrow 0).$$

The tests χ_1, χ_2 in the variance statement need not be nonnegative. The loops and the flow time are fixed, while the window shrinks with γ. Each expectation under λ_γ is the limit, as $K \to \infty$ at fixed γ, of the expectation of the corresponding observable of the flowed lattice holonomies under the Wilson measure $\mu_{nB^K,\gamma}$. By Corollary 23.11, the small-field failure probability satisfies $\sup_K \mu_{nB^K,\gamma}(S^c_{nB^K,n,\gamma}) = O_{n,q}(\gamma^q)$ for every fixed $q > 0$, which is $o(r_\gamma)$.

Proof. For a compactly supported bounded test, we write $m(\varphi) = \int \varphi \, dm_Q$, and we set $\epsilon = r_\gamma$. Until Step 4, where we pass to the continuum measure, we work at finite K, uniformly in K.

Step 1: comparison under the conditioned Wilson measure. Let the support of φ lie in the ball of radius R, and let $0 < \eta \le 1$. For $\eta\epsilon < 1$, Markov's inequality and the bound on $\mathbb{E}_\nu \min(1, D_K)^2$ in Theorem 36.3 give

$$\nu\{D_K > \eta\epsilon\} \le C_n \eta^{-2} \frac{\gamma}{\epsilon^2} = C_n \eta^{-2} \gamma^{1/2}. \tag{36.30}$$

On the complementary event, if either of $\mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K)$ and $\mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K_0(V))$ is nonzero, the corresponding triple is within distance $C_R\epsilon$ of (I, I, I) as a triple of matrices, and the other triple is within an additional distance $\eta\epsilon$. For γ small, both triples therefore lie in a fixed smaller neighborhood of the identity, independently of K. Since Log is Lipschitz there, their rescaled logarithms differ by at most $C_n\eta$, and the rescaled logarithm of $\mathcal{H}^K_0(V)$ lies in the ball of radius $R + C_n$.

Let

$$\omega_\varphi(v) = \sup\bigl\{|\varphi(Y) - \varphi(Y')| : |Y - Y'| \le v\bigr\},$$

which tends to zero as $v \downarrow 0$ since φ is uniformly continuous. Theorem 36.6, applied to the indicator function of the ball of radius $R + C_n$, shows that the event that the rescaled logarithm of $\mathcal{H}^K_0(V)$ lies in this ball has $\widehat{\nu}$-probability at most $C_{n,R}\epsilon$. Since $\mathcal{H}^K_0(V)$ depends only on V, and the law of V under ν is $\widehat{\nu}$, we obtain

$$\Bigl|\mathbb{E}_\nu \mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K) - \mathbb{E}_{\widehat{\nu}} \mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K_0)\Bigr| \le C_{n,R}\epsilon\, \omega_\varphi(C_n\eta) + C_n \|\varphi\|_\infty \eta^{-2}\gamma^{1/2}. \tag{36.31}$$

Here the first term bounds the difference on $\{D_K \le \eta\epsilon\}$, and the second term bounds it on the event in (36.30). After division by ϵ, the right side tends to zero if we first let $\gamma \downarrow 0$ at fixed η and then let $\eta \downarrow 0$, uniformly in K. The factor ϵ in front of the modulus of continuity is essential; it comes from the small probability of the window, and uniform continuity alone would not give it.

Step 2: removing the conditioning. Let $e_{K,\gamma} = \mu_{nB^K,\gamma}(S^c_{nB^K,n,\gamma})$. At our fixed side n, we have $e_{K,\gamma} \le C_{n,q}\gamma^q$ for every fixed $q > 0$, uniformly in K, by Corollary 23.11. For any bounded observable F, conditioning on the small-field event gives

$$\mathbb{E}_\mu F = (1 - e_{K,\gamma})\mathbb{E}_\nu F + e_{K,\gamma}\mathbb{E}_{\mu(\cdot|S^c)} F, \qquad |\mathbb{E}_\mu F - \mathbb{E}_\nu F| \le 2\|F\|_\infty e_{K,\gamma},$$

where $\mu = \mu_{nB^K,\gamma}$; if $e_{K,\gamma} = 0$ the inequality holds without the second conditional expectation. For $q > 1/4$, this error is $o(\epsilon)$ uniformly in K. Combining this with (36.31) and Theorem 36.6, we obtain, for each fixed continuous compactly supported test,

$$\mathbb{E}_{\mu_{nB^K,\gamma}} \mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K) = \epsilon \kappa_{K,\gamma} m(\varphi) + \epsilon \delta^{\varphi}_{K,\gamma}, \qquad \sup_K |\delta^{\varphi}_{K,\gamma}| \longrightarrow 0 \quad (\gamma \downarrow 0). \tag{36.32}$$

The signed error $\delta^{\varphi}_{K,\gamma}$ converges by the explicit bound with two parameters above. Given a tolerance, we first choose η to control the term with the modulus of continuity, and then γ to control the remaining terms. No choice of the comparison side or of the parameters depends on γ.

Step 3: bounds and ratios at finite K. If $\varphi \geq 0$ is continuous and not identically zero, it is positive on a nonempty open ball, and since e^{-Q} is positive everywhere, $m(\varphi) > 0$. By the bounds on $\kappa_{K,\gamma}$ in Theorem 36.6 and (36.32), there exist constants $0 < c_\varphi \leq C_\varphi$ with

$$c_\varphi \epsilon \leq \mathbb{E}_{\mu_{nB^K,\gamma}} \mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K) \leq C_\varphi \epsilon \tag{36.33}$$

for all K and all sufficiently small γ. For a general φ and the fixed radial cutoff χ, let $b = m(\varphi)/m(\chi(|\cdot|))$. We apply (36.32) to $\varphi - b\chi(|\cdot|)$, using $\mathcal{I}_{\gamma,\varphi - b\chi(|\cdot|)} = \mathcal{I}_{\gamma,\varphi} - b\mathcal{I}_{\gamma,\chi(|\cdot|)}$. The leading term vanishes, and the expectation of this observable is $o(\epsilon)$ uniformly in K. Since (36.33) gives a lower bound $c\epsilon$ for the denominator, uniformly in K, the ratio of the expectations under the Wilson measure tends to b as $\gamma \downarrow 0$, uniformly in K.

Step 4: passage to the continuum measure. We fix γ and identify the limit as $K \to \infty$. For small γ, the support of the observable lies strictly inside the logarithmic chart, and $\mathcal{I}_{\gamma,\varphi}$, extended by zero, is a continuous function of the triple. It is invariant under simultaneous conjugation: the logarithm is conjugation covariant in its chart, φ is invariant, and a fixed conjugation-invariant neighborhood of the identity contained in $\mathcal{V}$ contains the shrinking support for small γ. This holds even if $\mathcal{V}$ itself is not conjugation invariant. By Lemma 5.6, $\mathcal{I}_{\gamma,\varphi}$ is therefore a continuous function of the normalized traces of H_1, H_2, H_3, H_1H_2, H_1H_3, H_2H_3, and $H_1H_2H_3$, and convergence in law of these seven traces implies convergence of the expectation of $\mathcal{I}_{\gamma,\varphi}$. The seven traces are characters, divided by two, of the holonomies of loops obtained by concatenating the three based coordinate cycles, observed at the same time. These loops and their lattice versions are covered by Theorem 29.25.

To apply that theorem at a given γ, we may have to drop finitely many terms of the sequence $n_* B^K$. By Corollary 29.26, the shortened sequence has the same tail and hence the same limit of the joint law of the flowed loops at fixed coupling, and neither the comparison side in (36.32) nor its constants change.

We now let $K \to \infty$ at fixed γ. The bounds (36.33) pass to the limit, and the expectation of the observable of $\varphi - b\chi(|\cdot|)$ remains $o(\epsilon)$, since that bound was uniform in K. Dividing

by the limiting denominator and then letting $\gamma \downarrow 0$, we obtain the ratio limit and the two-sided bounds for λ_γ.

Step 5: the dilation statistic and the variances. The function W_χ is continuous, compactly supported, and invariant, and $m(W_\chi) = 0$ by Lemma 36.4. Applying the ratio limit to W_χ, we obtain $\mathcal{S}_{\gamma,\lambda_\gamma}(\chi) \to 0$.

For the variances, let $m_i = m(\chi_i(|\cdot|)^2) > 0$ and $b = m_1/m_2$. Since $A_i^2 = \mathcal{I}_{\gamma,\chi_i(|\cdot|)^2}$, (36.32) applied to $\chi_1(|\cdot|)^2 - b\chi_2(|\cdot|)^2$ shows that $\mathbb{E}A_1^2 - b\mathbb{E}A_2^2 = o(\epsilon)$ under the Wilson measure, uniformly in K. By the same formula, each mean $\mathbb{E}A_i$ is $O(\epsilon)$, and subtracting the squared means changes this difference by $O(\epsilon^2)$. Further, since $\mathbb{E}A_2^2 \geq c\epsilon m_2$ and $(\mathbb{E}A_2)^2 = O(\epsilon^2)$, we have $\operatorname{Var} A_2 \geq c'\epsilon$ for small γ, uniformly in K. We conclude that $\operatorname{Var} A_1 / \operatorname{Var} A_2 - b \to 0$ uniformly in K. At fixed γ, the first and second moments pass to λ_γ by the identification of Step 4, and the ratio of the variances tends to b. □

36.7 Rates, central triples, and several times

This section strengthens Theorem 36.7 in three directions: a rate $\gamma^{1/4}$ for Lipschitz tests, the extension to the eight central triples, and joint statements at finitely many flow times.

Definition 36.8. For a bounded Borel function φ on $\mathbb{R}^9$ with compact support we write $m(\varphi) = \int_{\mathbb{R}^9} \varphi(Y) e^{-Q(Y)}\, d^9Y$. For a bounded function f on $(\mathbb{R}^9)^\ell$, $\ell \geq 1$, we write

$$\|f\|_{\mathrm{BL}} = \|f\|_\infty + \operatorname{Lip} f, \qquad \|(Y^{(1)}, \dots, Y^{(\ell)})\|_{\max} = \max_{1 \leq j \leq \ell} |Y^{(j)}|,$$

where $\operatorname{Lip} f$ is the global Lipschitz constant with respect to the metric of $\|\cdot\|_{\max}$; for $\ell = 1$ this is the Euclidean metric on $\mathbb{R}^9$. We write B_R for the closed Euclidean ball of radius R about the origin in $\mathbb{R}^9$. The coupling-dependent scale is always written r_γ; the radius r in the domains B_r and D_r of configurations keeps its fixed meaning from Definition 36.1.

The norm $\|f\|_{\mathrm{BL}}$ is a sum. Its unit ball is contained in the set of functions with $\|f\|_\infty \leq 1$ and $\operatorname{Lip} f \leq 1$ that defines the bounded-Lipschitz distance d_{BL} used in Lemma 28.9, and it contains that set scaled by $1/2$. The distance dual to $\|f\|_{\mathrm{BL}}$ is therefore equivalent to d_{BL}, within a factor two, but not identical to it.

Rates for compactly supported Lipschitz tests

The next lemma is the quantitative version of Step 1 of the proof of Theorem 36.7. It uses the comparison of holonomies conditional on the coarse configuration, Theorem 34.11. For a coarse configuration v represented in B_r, let $\mathbb{Q}_v^K$ be the conditional law of the normalized fluctuation variables of all K renormalization group steps given $V = v$, that is, the product of the conditional fluctuation kernels of Lemma 15.10. By that lemma, ν disintegrates as $\nu(d\,\cdot) = \int \mathbb{Q}_v^K(d\,\cdot)\, \widehat{\nu}(dv)$.

Lemma 36.9. *For every $R < \infty$ there exist $C_R, \gamma_R > 0$, independent of K and of the test, such that every globally Lipschitz function $\varphi : \mathbb{R}^9 \to \mathbb{R}$ supported in B_R satisfies*

$$\left|\mathbb{E}_\nu \mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K) - \mathbb{E}_{\widehat{\nu}} \mathcal{I}_{\gamma,\varphi}(\mathcal{H}_0^K)\right| \le C_R \|\varphi\|_{\mathrm{BL}} r_\gamma^2 \tag{36.34}$$

for all K and $0 < \gamma \le \gamma_R$.

The proof is the mechanism described in Section 36.1: an event determined by V of probability $O(r_\gamma)$, a conditional matrix difference $O(r_\gamma^2)$, and a loss r_γ^{-1} from the rescaling of the logarithm.

Proof. Let $d = \min(1, \max_i \|\mathcal{H}_i^K - \mathcal{H}_{0,i}^K(v)\|_{\mathrm{op}})$. By Theorem 34.11, applied with $\tau = T = \theta L^2$ and $p = 1, 2$, we have, for every coarse configuration v in the support of $\widehat{\nu}$ and every K,

$$\mathbb{E}_{\mathbb{Q}_v^K} d \le C r_\gamma^2, \qquad \mathbb{E}_{\mathbb{Q}_v^K} d^2 \le C r_\gamma^4,$$

for γ below the upper bound on the coupling in that theorem. On the event $\{d \le r_\gamma\}$, with $r_\gamma < 1$, if either observable is nonzero, then both triples lie in a fixed compact subset of the logarithmic chart, and their rescaled logarithms differ by at most Cd/r_γ. In this case, the zero-fluctuation triple also lies in the event

$$E_{R'} = \{\mathcal{H}_0^K(v) \in \mathcal{V},\ |\mathrm{Log}\, \mathcal{H}_0^K(v)| \le R' r_\gamma\},$$

where R' and the upper bound on γ depend only on R and the chart. The event $E_{R'}$ is determined by v. Theorem 36.6, applied to the indicator function of $B_{R'}$, implies that $\widehat{\nu}(E_{R'}) \le C_R r_\gamma$. By the disintegration of Lemma 15.10,

$$\mathbb{E}_\nu[d \mathbf{1}_{E_{R'}}] = \int_{E_{R'}} \mathbb{E}_{\mathbb{Q}_v^K} d \, \widehat{\nu}(dv) \le C r_\gamma^2 \widehat{\nu}(E_{R'}) \le C_R r_\gamma^3.$$

The contribution of $\{d \le r_\gamma\}$ to the left side of (36.34) is therefore at most $C \,\mathrm{Lip}\, \varphi \, r_\gamma^{-1} \mathbb{E}_\nu[d \mathbf{1}_{E_{R'}}] \le C_R (\mathrm{Lip}\, \varphi) r_\gamma^2$. On $\{d > r_\gamma\}$ we use the bound $2\|\varphi\|_\infty$ and

$$\nu(d > r_\gamma) \le r_\gamma^{-2} \mathbb{E}_\nu d^2 = r_\gamma^{-2} \int \mathbb{E}_{\mathbb{Q}_v^K} d^2 \, \widehat{\nu}(dv) \le C r_\gamma^2.$$

This proves (36.34). □

Theorem 36.10. *Fix the data and the measures λ_γ of Theorem 36.7, and a radial cutoff χ, and set $b_\chi = \int \chi(|Y|)\, dm_{\mathrm{Q}}(Y) > 0$. For every $R < \infty$ there exist $C_{R,\chi}, \gamma_{R,\chi} > 0$ such that every globally Lipschitz function φ supported in B_R and invariant under simultaneous conjugation satisfies*

$$\left|\frac{\mathbb{E}_{\lambda_\gamma} \mathcal{I}_{\gamma,\varphi}}{\mathbb{E}_{\lambda_\gamma} \mathcal{I}_{\gamma,\chi(|\cdot|)}} - \frac{\int \varphi \, dm_{\mathrm{Q}}}{b_\chi}\right| \le C_{R,\chi} \big(\|\varphi\|_\infty + \mathrm{Lip}\, \varphi\big) \gamma^{1/4} \qquad (0 < \gamma \le \gamma_{R,\chi}),$$

and the denominator is at least $c_\chi r_\gamma$. The same bounds hold for the expectations of $\mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K)$ and $\mathcal{I}_{\gamma,\chi(|\cdot|)}(\mathcal{H}^K)$ under the Wilson measures $\mu_{nB^K,\gamma}$, uniformly in K. In particular $|\mathcal{S}_{\gamma,\lambda_\gamma}(\chi)| \le C_\chi \gamma^{1/4}$. For any two fixed smooth compactly supported functions χ_1, χ_2 on $[0,\infty)$ that are not identically zero, the ratio of variances in Theorem 36.7 differs from $m(\chi_1(|\cdot|)^2)/m(\chi_2(|\cdot|)^2)$ by at most $C_{\chi_1,\chi_2}\gamma^{1/4}$, and the denominator $\operatorname{Var} A_2$ *is at least $c_{\chi_2} r_\gamma$; this holds for λ_γ and, uniformly in K, for $\mu_{nB^K,\gamma}$.*

Proof. We write $m(\varphi) = \int \varphi \, dm_Q$ and, in this proof only,

$$A_K(\varphi) = \mathbb{E}_{\mu_{nB^K,\gamma}} \mathcal{I}_{\gamma,\varphi}(\mathcal{H}^K).$$

We combine Lemma 36.9, Theorem 36.6, and the bound of Corollary 23.11 with the exponent $q = 1$. Since $\gamma = r_\gamma^4$, the error from removing the conditioning is at most $2\|\varphi\|_\infty C_{n,1}\gamma = 2C_{n,1}\|\varphi\|_\infty r_\gamma^4$, and we obtain

$$A_K(\varphi) = r_\gamma \kappa_{K,\gamma} m(\varphi) + O_R\big(\|\varphi\|_{\mathrm{BL}} r_\gamma^2\big). \tag{36.35}$$

Applying (36.35) to $\chi(|\cdot|)$, which is globally Lipschitz, and using $\kappa_{K,\gamma} \ge c_n$, we find $A_K(\chi(|\cdot|)) \ge c_\chi r_\gamma$ for small γ. When we subtract $m(\varphi)A_K(\chi(|\cdot|))/b_\chi$ from $A_K(\varphi)$, the leading terms cancel at finite K. Since $|m(\varphi)| \le m(\mathbf{1}_{B_R})\|\varphi\|_\infty$, the difference is $O_{R,\chi}(\|\varphi\|_{\mathrm{BL}} r_\gamma^2)$. Dividing by $A_K(\chi(|\cdot|)) \ge c_\chi r_\gamma$ proves the assertion at finite cutoff, with one upper bound on γ for the whole class of tests.

At fixed γ, these observables are continuous functions of the triple, supported strictly inside the logarithmic chart, and invariant under simultaneous conjugation. By Lemma 5.6, they factor continuously through the seven traces, which are traces of holonomies of concatenations of the three based cycles. We apply Theorem 29.25 and Corollary 29.26 to the combination $A_K(\varphi) - m(\varphi)A_K(\chi(|\cdot|))/b_\chi$ and to the denominator; the comparison side remains valid on the sequence $n_* B^K$, as in Step 4 of the proof of Theorem 36.7. Letting $K \to \infty$ at fixed γ proves the assertion for λ_γ.

The function W_χ is globally Lipschitz with a fixed support, and $m(W_\chi) = 0$ by Lemma 36.4. The ratio bound then gives the bound on the dilation statistic. For the variances, the argument is the same before and after the limit $K \to \infty$. The two means are $O(r_\gamma)$, and their squares are $O(r_\gamma^2)$. Applying (36.35) to the Lipschitz function $\chi_1(|\cdot|)^2 - b\chi_2(|\cdot|)^2$, we obtain

$$\mathbb{E}A_1^2 - b\mathbb{E}A_2^2 = O(r_\gamma^2), \qquad b = \frac{m(\chi_1(|\cdot|)^2)}{m(\chi_2(|\cdot|)^2)} > 0.$$

The second moment $\mathbb{E}A_2^2$ is at least $c r_\gamma$ by (36.35) and the lower bound on $\kappa_{K,\gamma}$. Subtracting the squared mean preserves a lower bound $c' r_\gamma$ for small γ. Subtracting the squared means in the combination above and dividing by this lower bound proves the rate for the variances. □

The eight central triples

The next lemma is the center symmetry of the periodic lattice Wilson measure [149], with a verification that it commutes with the Wilson flow. Here $\mu_{N,\gamma}$ is the Wilson measure of Definition 4.1, and the winding number $w_i(\ell) \in \mathbb{Z}$ of a closed lattice loop ℓ is the signed number of its crossings of the hyperplane $\{x_i = N - 1/2\}$ in lattice units.

Lemma 36.11. *Let $z = (z_1, z_2, z_3) \in \{I, -I\}^3$. There exists a central link configuration c^z, with all plaquette products equal to I and coordinate-cycle products equal to z_i, such that the map $\Gamma_z U = (c^z_e U_e)_e$ preserves $\mu_{N,\gamma}$ and commutes with the Wilson flow at every time. For every based closed lattice loop ℓ,*

$$\mathrm{Hol}_\ell\big(\Psi_t^{(N)}(\Gamma_z U)\big) = \Big(\prod_{i=1}^{3} z_i^{w_i(\ell)}\Big)\,\mathrm{Hol}_\ell\big(\Psi_t^{(N)} U\big).$$

In particular, for any finite list of times, the vector of coordinate-cycle holonomies of the flowed configuration has the same law under $\mu_{N,\gamma}$ after the i-th holonomy is multiplied by z_i at every listed time. These are exact identities for every $N \geq 4$ and $\gamma > 0$; they do not concern the small-field event.

Proof. We represent the vertices by $x \in \{0, \dots, N-1\}^3$ and define, on the positive link from x in direction i,

$$c^z_{x,i} = \begin{cases} z_i, & x_i = N-1, \\ I, & x_i \neq N-1, \end{cases} \qquad c^z_{\bar e} = (c^z_e)^{-1}.$$

In an (i,j) plaquette, the two links in direction i carry the same factor, because a displacement in direction j does not change x_i, and similarly for the two links in direction j. Since the factors are central, they cancel against the oppositely oriented factors. Hence every plaquette product of c^z is I, including the plaquettes that cross a periodic seam. A positive coordinate cycle in direction i crosses the seam in its direction once, and its product is z_i. For a general closed loop, the signed number of crossings of the i-th seam is $w_i(\ell)$. Since the factors are central, they can be moved past every link matrix, and the product of all factors is $\prod_i z_i^{w_i(\ell)}$.

The map Γ_z is a smooth involution of the configuration space, and the product Haar measure is invariant under it, since Haar measure is invariant under multiplication by a fixed element (Lemma 5.1). Every plaquette matrix is unchanged, and so are every $q(U_p)$, the Wilson action, and the normalization constant. This proves that $\mu_{N,\gamma}$ is invariant.

We use the product of the bi-invariant Riemannian metrics, as in the definition of the Wilson flow. Since multiplication by a fixed central element is an isometry, so is Γ_z. Let $S(U) = \sum_p q(U_p)$, so that $S \circ \Gamma_z = S$. For every tangent vector v at U,

$$\langle \nabla S(\Gamma_z U), D\Gamma_z v\rangle_{\Gamma_z U} = dS_{\Gamma_z U}(D\Gamma_z v) = dS_U(v) = \langle D\Gamma_z \nabla S(U), D\Gamma_z v\rangle_{\Gamma_z U},$$

which implies $\nabla S(\Gamma_z U) = D\Gamma_z \nabla S(U)$. The smooth vector field $-\nabla S$ on the compact configuration space has a unique global flow, and Γ_z maps its trajectories to trajectories. This proves commutation at every lattice time, and hence at every physical time $4t/a_N^2$. The identity for the holonomies follows from the identity for the products of the factors. Integrating any bounded Borel function of the finite vector of holonomies, we obtain the identity in law. □

Theorem 36.12. *Choose $\mathcal{V}$ in Definition 36.1 invariant under simultaneous conjugation. For $z \in \{I, -I\}^3$ let $\mathcal{V}_z = z\mathcal{V}$, and set*

$$\mathcal{I}^z_{\gamma,\varphi}(H) = \begin{cases} \varphi\big((\mathrm{Log}(z_i^{-1} H_i))_{i=1}^3 / r_\gamma\big), & H \in \mathcal{V}_z, \\ 0, & H \notin \mathcal{V}_z, \end{cases}$$

so that $\mathcal{I}^{(I,I,I)}_{\gamma,\varphi} = \mathcal{I}_{\gamma,\varphi}$. For every $\gamma > 0$, every positive observation time, and every bounded Borel compactly supported test φ invariant under simultaneous conjugation,

$$\mathbb{E}_{\lambda_\gamma} \mathcal{I}^z_{\gamma,\varphi} = \mathbb{E}_{\lambda_\gamma} \mathcal{I}_{\gamma,\varphi}.$$

The corresponding identity holds under $\mu_{nB^K,\gamma}$ for the flowed lattice holonomies at every finite K, for every bounded Borel compactly supported test. Consequently the ratio limits, the two-sided bounds, the limits of the ratios of variances, and the limit of the dilation statistic in Theorem 36.7, and the rates of Theorem 36.10, hold at each of the eight central triples, with expectations equal to those at (I, I, I) and with constants and ceilings on γ independent of z.

Proof. Let H denote the triple of flowed coordinate-cycle holonomies. At finite cutoff, Lemma 36.11 shows that H and $z^{-1}H$ have the same law. By definition, $\mathcal{I}^z_{\gamma,\varphi}(H) = \mathcal{I}_{\gamma,\varphi}(z^{-1}H)$ pointwise, including on the complements of the charts, since $H \in z\mathcal{V}$ if and only if $z^{-1}H \in \mathcal{V}$. This proves the identity at finite cutoff, without any assumption of regularity or invariance on the test.

For the continuum measure, let F be any continuous function on $\mathrm{SU}(2)^3$ invariant under simultaneous conjugation. Since z is central, $F(z^{-1}\cdot)$ is also continuous and invariant, and both factor continuously through the seven traces by Lemma 5.6. Passing the identity $\mathbb{E}F(H) = \mathbb{E}F(z^{-1}H)$ at finite cutoff to the limit at every fixed $\gamma > 0$ by Theorems 29.28 and 30.6, we obtain $\mathbb{E}_{\lambda_\gamma} F(H) = \mathbb{E}_{\lambda_\gamma} F(z^{-1}H)$ for every such F. By the same lemma, the compact quotient of $\mathrm{SU}(2)^3$ by simultaneous conjugation is homeomorphic to the image of the seven traces. The law of the conjugacy class of H under λ_γ, a Borel probability on this compact metric space, is therefore invariant under the continuous map induced by multiplication by z^{-1}. Indeed, continuous functions determine Borel probabilities on a metric space. For a closed set C, the continuous functions $\max(0, 1 - j\,\mathrm{dist}(\cdot, C))$ decrease to $\mathbf{1}_C$, so the two probabilities agree on closed sets, and hence on all Borel sets by the monotone class theorem. The identity then extends to bounded Borel functions on

the quotient. Since $\mathcal{V}$ is invariant, the logarithm is conjugation covariant on it, and φ is invariant, the observable $\mathcal{I}^z_{\gamma,\varphi}$ is a bounded Borel function on the quotient. This proves the identity for the continuum measure, even if φ is not continuous.

Applying the identity to each numerator and denominator in Theorems 36.7 and 36.10, and to the squares of the observables for the variances, we obtain the remaining assertions. Since all eight expectations are equal, so are the constants. □

Joint asymptotics at finitely many times

We now observe the holonomies at finitely many flow times. The observables are invariant under an independent conjugation at each time; no relation between the gauge frames at different times is used or asserted.

Definition 36.13. Fix $\ell \geq 1$, positive times $t_j = \theta_j L^2$ for $1 \leq j \leq \ell$, and the same three based coordinate cycles at each time, before varying γ. Choose the parameters, the side n, and the margins as in Definition 36.1, so that the finitely many lower bounds required at each of the times t_j hold. Write $H^{(j)}$ for the triple of cycle holonomies at time t_j. Let $\mathcal{V}$ be a neighborhood of (I, I, I) as in Definition 36.1, invariant under simultaneous conjugation and small enough for all listed times. For $\Phi \in C_c((\mathbb{R}^9)^\ell; \mathbb{R})$ define

$$\mathcal{J}_{\gamma,\Phi}(H^{(1)}, \dots, H^{(\ell)}) = \begin{cases} \Phi\big(\operatorname{Log} H^{(1)}/r_\gamma, \dots, \operatorname{Log} H^{(\ell)}/r_\gamma\big), & H^{(j)} \in \mathcal{V} \text{ for every } j, \\ 0, & \text{otherwise.} \end{cases}$$

The class of tests consists of the functions Φ that are invariant under independent conjugations at the different times:

$$\Phi\big(\operatorname{Ad}(k_1)Y^{(1)}, \dots, \operatorname{Ad}(k_\ell)Y^{(\ell)}\big) = \Phi\big(Y^{(1)}, \dots, Y^{(\ell)}\big) \qquad \text{for all } k_1, \dots, k_\ell \in \mathrm{SU}(2),$$

where each $\operatorname{Ad}(k_j)$ acts simultaneously on the three components of $Y^{(j)}$. Put $\Phi_\Delta(Y) = \Phi(Y, \dots, Y)$ and $m_\Delta(\Phi) = \int \Phi_\Delta \, dm_Q$. We write $\mathcal{H}^K_t$ and $\mathcal{H}^K_{0,t}(V)$ for the triples of Definition 36.1 observed at time t.

The following lemma localizes the transverse coordinates on the event that the flowed zero-fluctuation holonomies at the first time are in a window of size r_γ.

Lemma 36.14. *Use the stopped measure $\widehat{\nu}$, the coordinates $V(X, s)$, and the zero-fluctuation holonomies at the first time t_1 of Definition 36.13. Set*

$$E_R = \big\{\mathcal{H}^K_{0,t_1} \in \mathcal{V},\ |\operatorname{Log} \mathcal{H}^K_{0,t_1}| \leq R r_\gamma\big\}, \qquad D = d_{\mathrm{st}}, \qquad g^2 = \gamma/n.$$

There is a fixed neighborhood $\mathcal{U}$ of the trivial based orbit, contained in the central coordinates, such that, for every fixed $R < \infty$, all sufficiently small γ, and every K,

$$\widehat{\nu}(E_R \setminus \mathcal{U}) \leq C g^{-(D+4)} e^{-c/g^2}, \qquad \mathbb{E}_{\widehat{\nu}}\big[\mathbf{1}_{E_R \cap \mathcal{U}} |s|^a\big] \leq C_{R,a} r_\gamma g^a \quad (a = 0, 1, 2).$$

Further, $\widehat{\nu}(\mathcal{U} \cap \{|s| > r_\gamma\}) \leq C g^{-4} e^{-c r_\gamma^2/g^2}$.

Proof. We choose $\mathcal{U}$ with closure strictly inside the chart of Theorem 33.8 and inside the domains of the deterministic results at all listed times. By the localization statement of Theorem 36.3, applied with θ_1 in place of θ, there exists a fixed $\delta > 0$ such that a coarse configuration within distance δ of $\mathcal{F}_n$ whose holonomies at time t_1 have cycle radius at most δ lies in $\mathcal{U}$. For $Rr_\gamma < \delta$, every point of $E_R \setminus \mathcal{U}$ therefore lies outside the fixed neighborhood $\{d(V, \mathcal{F}_n) \le \delta\}$. There $E_0 \ge c_{n,\delta} > 0$, by compactness of the configuration space at side n and $E_0^{-1}(0) = \mathcal{F}_n$, which follows from the classification of flat configurations in Lemma 14.9. The first estimate follows as in (36.19). We use properties (P1) and (P3) in Section 33.2, namely the lower bound $E_{JK,\mathrm{sat}} \ge \kappa E_0$ on D_r and the bound on the centered effective interaction, together with the normalization of H_n and the lower bound on the normalization constant in Theorem 33.10.

On $\mathcal{U}$, Theorem 36.3 at time t_1 gives $|\operatorname{Log} \mathcal{H}^K_{0,t_1}| \ge (3/4)|X| - C|s|$, which implies

$$|X| \le C(Rr_\gamma + |s|) \qquad \text{on } E_R \cap \mathcal{U}.$$

The density factor $e^{-\overline{R}_{K,\gamma}} J_n$ is bounded above, and $F_K(X,s) \ge f_K(X) + c|s|^2 \ge c|s|^2$ by Theorem 33.8. For fixed s, the possible values of X form a set of volume at most $C_R(r_\gamma + |s|)^9$. Substituting $s = gz$, we find

$$\begin{aligned}\mathbb{E}_{\widehat{v}}\big[\mathbf{1}_{E_R \cap \mathcal{U}} |s|^a\big] &\le \frac{C_R}{\widehat{Z}} \int_{\mathbb{R}^D} |s|^a (r_\gamma + |s|)^9 e^{-c|s|^2/g^2}\, ds \\ &= \frac{C_R g^{D+a} r_\gamma^9}{\widehat{Z}} \int_{\mathbb{R}^D} |z|^a \Big(1 + \frac{g}{r_\gamma}|z|\Big)^9 e^{-c|z|^2}\, dz.\end{aligned}$$

Since $g/r_\gamma = r_\gamma/\sqrt{n} \le 1$, the Gaussian integral is bounded in the fixed dimension D. Since $\widehat{Z} \ge c g^{D+4}$ and $r_\gamma^9 g^{-4} = n^2 r_\gamma$, the right side is at most $C_{R,a} r_\gamma g^a$. The conjugation at the basepoint has not been divided out: the nine-dimensional volume in X and the Haar density are kept.

For the last estimate there is no restriction on the output, and the volume in X is bounded by a fixed constant. We split the Gaussian factor into two factors, bound one of them on $\{|s| > r_\gamma\}$ by $e^{-cr_\gamma^2/(2g^2)}$, and integrate the other over $\mathbb{R}^D$. After division by $\widehat{Z} \ge c g^{D+4}$, we obtain the bound $C g^{-4} e^{-c' r_\gamma^2/g^2}$, and renaming c' proves the claim. Since $g = r_\gamma^2/\sqrt{n}$, the first and last bounds are smaller than every fixed power of r_γ. □

Theorem 36.15. *Use Definition 36.13, and fix a radial cutoff χ. For every continuous compactly supported test Φ in the class of Definition 36.13,*

$$\lim_{\gamma \downarrow 0} \frac{\mathbb{E}_{\lambda_\gamma} \mathcal{J}_{\gamma,\Phi}}{\mathbb{E}_{\lambda_\gamma} \mathcal{I}_{\gamma,\chi(|\cdot|)}(H^{(1)})} = \frac{\int_{\mathbb{R}^9} \Phi(Y, \dots, Y) e^{-Q(Y)}\, dY}{\int_{\mathbb{R}^9} \chi(|Y|) e^{-Q(Y)}\, dY},$$

and the denominator is at least $c_\chi r_\gamma$ for small γ. If $\Phi \ge 0$ and $m_\Delta(\Phi) > 0$, the numerator is bounded above and below by positive multiples of r_γ; if $m_\Delta(\Phi) = 0$, the numerator is $o(r_\gamma)$.

More precisely, with the factor $\kappa_{K,\gamma}$ of Theorem 36.6,

$$\mathbb{E}_{\mu_{nB^K,\gamma}}\mathcal{J}_{\gamma,\Phi}(\mathcal{H}^K_{t_1},\dots,\mathcal{H}^K_{t_\ell}) = r_\gamma\kappa_{K,\gamma}\{m_\Delta(\Phi)+\delta^\Phi_{K,\gamma}\},\qquad \sup_K|\delta^\Phi_{K,\gamma}|\longrightarrow 0\quad(\gamma\downarrow 0),$$

and the expectation of $\mathcal{J}_{\gamma,\Phi}(\mathcal{H}^K_{0,t_1},\dots,\mathcal{H}^K_{0,t_\ell})$ under $\widehat{\nu}$ has the same expansion with the same factor. For every $R<\infty$ there exist $C_{R,\chi,\boldsymbol{t}}$ and $\gamma_{R,\chi,\boldsymbol{t}}>0$ such that, for every globally Lipschitz test Φ in the class with $\operatorname{supp}\Phi\subset(B_R)^\ell$ *and $0<\gamma\le\gamma_{R,\chi,\boldsymbol{t}}$, the difference between the ratio above (for λ_γ, or for $\mu_{nB^K,\gamma}$ at any finite K) and $m_\Delta(\Phi)/m(\chi(|\cdot|))$ is at most $C_{R,\chi,\boldsymbol{t}}(\|\Phi\|_\infty+\operatorname{Lip}\Phi)\gamma^{1/4}$. Here* Lip *refers to the metric of $\|\cdot\|_{\max}$, and $\boldsymbol{t}=(t_1,\dots,t_\ell)$.*

A nonnegative test that is not identically zero but vanishes near the diagonal has $m_\Delta(\Phi)=0$, and its numerator is $o(r_\gamma)$: to leading order, the rescaled logarithms at the different times coincide. The diagonal is described through tests that are invariant under independent conjugations at the different times. Uniform statements on a time interval are proved in Theorem 36.18. The constants depend additionally on the list of times, on ℓ, R, and χ.

Proof. Step 1 reduces the joint observable of the zero-fluctuation holonomies to the single-time observable of Φ_Δ, Step 2 compares with the holonomies of the configuration, and Steps 3–5 treat Lipschitz tests, continuous tests, and the continuum measure. All estimates use the same coarse configuration V and the same reconstruction.

Step 1: the zero-fluctuation holonomies near the diagonal. For each listed time we apply Theorem 35.9 with $m=0$. By its bound in $C^{1,1}$ at fixed time and its identity on flat configurations, we have, on a common smaller central neighborhood,

$$Z_j(X,s)=\operatorname{Log}\mathcal{H}^K_{0,t_j}(V(X,s))=X+O_{\boldsymbol{t}}(|X|^2+|s|),\qquad \|D_XZ_j-I\|\le C_{\boldsymbol{t}}(|X|+|s|).\tag{36.36}$$

Indeed, each of the nine coordinate directions in X consists of commuting triples. The identity on flat configurations therefore gives $Z_j(ue_a,0)=ue_a$, and hence $D_XZ_j(0,0)=I$. Integrating the Lipschitz bound on the derivative along $v\mapsto(vX,0)$, and then using the bounded derivative in s, we obtain (36.36). This is the statement of Theorem 36.3 at each listed time. We intersect finitely many domains and take the maximum of finitely many constants.

Let Φ be globally Lipschitz with support in $(B_R)^\ell$. We compare $\mathcal{J}_{\gamma,\Phi}(\mathcal{H}^K_{0,t_1},\dots,\mathcal{H}^K_{0,t_\ell})$ with $\mathcal{I}_{\gamma,\Phi_\Delta}(\mathcal{H}^K_{0,t_1})$. If either is nonzero, then the event E_R of Lemma 36.14 holds. Outside the fixed neighborhood $\mathcal{U}$ of that lemma its probability is smaller than every power of r_γ, and inside $\mathcal{U}$ we discard $\{|s|>r_\gamma\}$, again at a cost smaller than every power of r_γ. On the remaining set, the lower bound for Z_1 implies $|X|\le C_Rr_\gamma$. By (36.36), all outputs lie in $\mathcal{V}$ for small γ, and

$$\max_j\frac{|Z_j-Z_1|}{r_\gamma}\le C_R\Big(r_\gamma+\frac{|s|}{r_\gamma}\Big).$$

On this set, both observables are given by their formulas in coordinates, $\Phi(Z_1/r_\gamma,\dots,Z_\ell/r_\gamma)$ and $\Phi(Z_1/r_\gamma,\dots,Z_1/r_\gamma)$, including when one of them vanishes. Lemma 36.14 gives, for any fixed $A \geq 2$,

$$\begin{aligned}
&\Big|\mathbb{E}_{\widehat{\nu}}\mathcal{J}_{\gamma,\Phi}(\mathcal{H}^K_{0,t_1},\dots,\mathcal{H}^K_{0,t_\ell}) - \mathbb{E}_{\widehat{\nu}}\mathcal{I}_{\gamma,\Phi_\Delta}(\mathcal{H}^K_{0,t_1})\Big| \\
&\qquad \leq C_R(\operatorname{Lip}\Phi)\Big\{r_\gamma\widehat{\nu}(E_R\cap\mathcal{U}) + r_\gamma^{-1}\mathbb{E}_{\widehat{\nu}}\big[|s|\mathbf{1}_{E_R\cap\mathcal{U}}\big]\Big\} + O_A\big(\|\Phi\|_\infty r_\gamma^A\big) \\
&\qquad \leq C_R\|\Phi\|_{\mathrm{BL}} r_\gamma^2,
\end{aligned} \tag{36.37}$$

where in the last step we used $g = r_\gamma^2/\sqrt{n}$. The diagonal test Φ_Δ is globally Lipschitz, supported in B_R, and invariant under simultaneous conjugation. Theorem 36.6 at time t_1 therefore implies that

$$\mathbb{E}_{\widehat{\nu}}\mathcal{J}_{\gamma,\Phi}(\mathcal{H}^K_{0,t_1},\dots,\mathcal{H}^K_{0,t_\ell}) = r_\gamma\kappa_{K,\gamma}m_\Delta(\Phi) + O_R\big(\|\Phi\|_{\mathrm{BL}}r_\gamma^2\big). \tag{36.38}$$

The factor is the same $\kappa_{K,\gamma}$ as at a single time, and no factor depending on the times appears.

Step 2: comparison at all listed times. Let $[\tau,T]$ be an interval containing all listed times, and let

$$d = \min\Big(1,\ \sup_{\tau\leq t\leq T}\max_i \|\mathcal{H}^K_{t,i} - \mathcal{H}^K_{0,t,i}(v)\|_{\mathrm{op}}\Big).$$

By Theorem 34.11 with $p = 1,2$, we have $\mathbb{E}_{\mathbb{Q}^K_v} d \leq Cr_\gamma^2$ and $\mathbb{E}_{\mathbb{Q}^K_v} d^2 \leq Cr_\gamma^4$ for every coarse configuration v and every K, where $\mathbb{Q}^K_v$ is the conditional law of Lemma 36.9. On $\{d \leq r_\gamma\}$, if either joint observable is nonzero, then all flowed holonomies of the configuration and of the zero-fluctuation reconstruction at the listed times lie in a fixed compact subset of the logarithmic chart, and their rescaled logarithms differ by at most Cd/r_γ in the metric of $\|\cdot\|_{\max}$. In this case, the zero-fluctuation holonomies at time t_1 also lie in the enlarged window $E_{R'}$ of Lemma 36.14, which is determined by v and has probability $O_R(r_\gamma)$ by Theorem 36.6. The argument of Lemma 36.9, with the metric of $\|\cdot\|_{\max}$, gives

$$\Big|\mathbb{E}_\nu\mathcal{J}_{\gamma,\Phi}(\mathcal{H}^K_{t_1},\dots,\mathcal{H}^K_{t_\ell}) - \mathbb{E}_{\widehat{\nu}}\mathcal{J}_{\gamma,\Phi}(\mathcal{H}^K_{0,t_1},\dots,\mathcal{H}^K_{0,t_\ell})\Big| \leq C_{R,\boldsymbol{t}}\|\Phi\|_{\mathrm{BL}}r_\gamma^2. \tag{36.39}$$

The same argument at the time t_1 alone, applied to a Lipschitz function equal to one on B_{R+1}, shows that the event that $\mathcal{H}^K_{t_1}$ lies in $\mathcal{V}$ and its rescaled logarithm lies in B_{R+1} has ν-probability $O_R(r_\gamma)$.

Step 3: Lipschitz tests at finite cutoff. We remove the conditioning with Corollary 23.11 at $q = 1$, which costs at most $2\|\Phi\|_\infty C_{n,1}\gamma = 2C_{n,1}\|\Phi\|_\infty r_\gamma^4$. Combining this with (36.38) and (36.39), we obtain

$$\Big|\mathbb{E}_{\mu_{nB^K,\gamma}}\mathcal{J}_{\gamma,\Phi}(\mathcal{H}^K_{t_1},\dots,\mathcal{H}^K_{t_\ell}) - r_\gamma\kappa_{K,\gamma}m_\Delta(\Phi)\Big| \leq C_{R,\boldsymbol{t}}\|\Phi\|_{\mathrm{BL}}r_\gamma^2. \tag{36.40}$$

For the denominator, (36.35) at time t_1 holds with the same factor $\kappa_{K,\gamma}$, and the denominator is at least $c_\chi r_\gamma$. When we subtract $m_\Delta(\Phi)/m(\chi(|\cdot|))$ times the denominator, the factor $\kappa_{K,\gamma}$ cancels at finite K. Dividing by $c_\chi r_\gamma$, we obtain the rate $\gamma^{1/4} = r_\gamma$ for the ratio at finite cutoff, with one upper bound on γ for the whole class of tests supported in $(B_R)^\ell$.

Step 4: continuous tests. Let Φ be continuous, compactly supported, and in the class, with support in $(B_R)^\ell$. There exist smooth compactly supported functions Φ_η, $\eta > 0$, with support in $(B_{R+1})^\ell$ and $\|\Phi_\eta - \Phi\|_\infty \to 0$. For example, we may convolve Φ with the product of ℓ copies of a nonnegative radial smooth approximate identity on $\mathbb{R}^9$ of total mass one, and use the uniform continuity of Φ. Radial convolution preserves the invariance under independent conjugations, as we see by changing each integration variable by the orthogonal map $\mathrm{Ad}(k_j)$. Each Φ_η is globally Lipschitz. The difference of the joint observables of Φ_η and Φ is nonzero only if the first output lies in a fixed enlarged compact window, which has probability $O_R(r_\gamma)$, both for the holonomies of the configuration under ν (by Step 2) and for the zero-fluctuation holonomies under $\widehat{\nu}$ (by Theorem 36.6). The corresponding expectations therefore differ by at most $C_R r_\gamma \|\Phi_\eta - \Phi\|_\infty$. For the Wilson measure, we add at most $2C_{n,a}\gamma^a\|\Phi_\eta - \Phi\|_\infty$, with any fixed $a > 1/4$, from Corollary 23.11. Further,

$$|m_\Delta(\Phi_\eta) - m_\Delta(\Phi)| \le m(\mathbf{1}_{B_{R+1}})\|\Phi_\eta - \Phi\|_\infty.$$

At fixed η we apply (36.40) and (36.38) to Φ_η and let $\gamma \downarrow 0$ uniformly in K; then we let $\eta \downarrow 0$. Since $c \le \kappa_{K,\gamma} \le C$, this proves $\sup_K |\delta^\Phi_{K,\gamma}| \to 0$, both for the Wilson measure and for the zero-fluctuation expectation. For continuous tests we do not obtain a rate.

Step 5: the continuum measure. At fixed small γ, the joint observable is a bounded continuous function on $(\mathrm{SU}(2)^3)^\ell$, since its support lies strictly inside the logarithmic charts. By the invariance under independent conjugations, it factors through the product of the ℓ maps $\mathfrak{p}$ of seven traces of Lemma 5.6: two points with the same vector of traces at every time differ by one conjugation at each time. The factored function f is continuous on the compact image C of this product map, since every convergent sequence of trace vectors has convergent subsequences of representatives, and all possible limits give the same value. On C, the functions $\inf_{u\in C}\{f(u) + k|v - u|\}$, truncated to the range of f, are bounded Lipschitz functions on the ambient Euclidean space that converge uniformly to f on C as $k \to \infty$. This follows from the uniform continuity of f, by considering separately the points u with $|v - u|$ small and large. Joint convergence in law of the traces therefore suffices for these observables. The seven words at each time are concatenations of the three based cycles at that time, and the whole finite vector of their traces is covered by Theorem 29.25, with the comparison side justified by Corollary 29.26. Letting $K \to \infty$ at fixed γ in the lower bound on the denominator and in the combination in which $\kappa_{K,\gamma}$ cancels proves the ratio limit for continuous tests and the rate for Lipschitz tests under

λ_γ. The expansions and $c \le \kappa_{K,\gamma} \le C$ imply that the numerator is comparable with r_γ when $m_\Delta(\Phi) > 0$ and $\Phi \ge 0$, and that it is $o(r_\gamma)$ when $m_\Delta(\Phi) = 0$. These bounds are uniform in K and pass to the limit. □

36.8 Holonomy paths on a time interval

We now observe the holonomies on a whole compact time interval $[\tau, T] \subset (0, \infty)$. The observable is the path of rescaled holonomy triples modulo simultaneous conjugation. After weighting with a compactly supported cutoff at the initial time, we show that this path converges to a constant path whose value has density proportional to the cutoff times e^{-Q}. The first ingredient is a version of Theorem 36.3 that is uniform in time.

Lemma 36.16. *Fix the data of Definition 36.1 and an interval $0 < \tau < T < \infty$ before varying γ, choosing the side n and the margins for the whole interval. There exist a central coordinate neighborhood $\mathcal{U}$ and a neighborhood of the identity in $\mathcal{V}$ such that*

$$Z_{K,t}(X, s) = \operatorname{Log} \mathcal{H}^K_{0,t}(V(X, s))$$

is defined on $\mathcal{U}$ for every K and $t \in [\tau, T]$, is continuous in t, and satisfies

$$\sup_{K,t} \|Z_{K,t}\|_{C^{1,1}(\mathcal{U})} \le C_{\tau,T}, \qquad \sup_t |Z_{K,t}(X, s) - X| \le C_{\tau,T}(|X|^2 + |s|),$$

$$\sup_t \|D_X Z_{K,t}(X, s) - I\| \le C_{\tau,T}(|X| + |s|).$$

For every flat coarse configuration, the flowed zero-fluctuation holonomy triple equals the holonomy triple of the coarse configuration, for every K and t. The neighborhoods and constants do not depend on K, γ, or the number of preceding averaging steps.

Proof. We work in the smooth chart $V(X, s)$ of the based quotient, not with a Borel selection of representatives; the parameters (X, s) have a fixed dimension at the side n. On a neighborhood of the trivial flat configuration, the zero-fluctuation reconstructions stay within the input and image bounds of the reconstruction maps of Theorem 15.12. Indeed, they vanish at zero input, and the uniform bounds on the coefficients together with the contraction condition $a_{\mathrm{map}}\sqrt{B} \le 32/49 < 1$ of Definition 36.1 give one smaller neighborhood, independent of the number of steps. More precisely, items (1) and (3) of Lemma 34.4 bound the linear and quadratic flow data $\mathcal{E}_K(u)$ of the zero-fluctuation reconstruction, in the weighted norms X^a_α and Y^a_α of Definition 4.8, by a constant R on a fixed complex domain of inputs. Here $a = L/(nB^K)$ is the finest mesh, and α is the fixed regularity exponent of Definition 34.1; the bound is uniform in K with this mesh. We compose with the analytic map $u = (X, s) \mapsto V(X, s)$ of finitely many variables and shrink the domain once. Let $U_0 \Subset U_1$ be real domains whose complex neighborhoods of

radius $\delta > 0$ lie in this domain. The Cauchy integral formula on one coordinate circle, or on two circles for mixed derivatives, gives

$$\sup_{K,\,u\in U_1}\sum_{j=0}^{2}\|D_u^j\mathcal{E}_K(u)\|_{\mathrm{fl}} \le C_{n,\delta}R\bigl(1+\delta^{-1}+\delta^{-2}\bigr),$$

where $\|\cdot\|_{\mathrm{fl}}$ is the flow-data norm at the finest mesh a, which we suppress from the notation. It is the sum of the two weighted norms, which include the first differences of both components. This bound is deterministic and does not depend on the coupling.

We apply Theorem 25.13 on $U_0 \subset U_1$, with waiting time $t_* = \tau$, horizon T, and damping gauge transformations started at the identity in the common gauge frame at the basepoint. Its hypotheses are the displayed bound on derivatives and the margins of the based charts. It yields bounds, uniform in K, on the first and second derivatives in the parameters of the flowed cycle holonomies, simultaneously for all $t \in [\tau, T]$, including the finitely many remaining meshes. Continuity in t follows from the smoothness of the flow on a finite lattice. At $u = 0$ every output is the identity. By the uniform bound on the first derivatives, the outputs on a smaller central neighborhood lie in one compact subset of the logarithmic chart, simultaneously for all K and t. Since Log has bounded first and second derivatives there, we obtain the bound in $C^{1,1}$.

By Lemma 14.9, the zero-fluctuation refinements of a flat configuration are flat and have the same based cycle holonomies, and the Wilson flow is stationary on flat configurations. The flowed holonomies therefore equal the coarse holonomies at every time, in the given gauge frame. For $s = 0$ and X in one of the nine coordinate directions, the configuration consists of commuting links, and the identity on flat configurations gives $Z_{K,t}(ue_a, 0) = ue_a$. Since these directions span $\mathbb{R}^9$, we have $Z_{K,t}(0,0) = 0$ and $D_X Z_{K,t}(0,0) = I$ for all K and t. Integrating the uniform Lipschitz bound on the derivative along $v \mapsto (vX, 0)$ gives the remainder $C|X|^2$ and the error $C|X|$ for the derivative. The bounded derivative in s and the Lipschitz bound on the first derivative add $C|s|$ to these two errors. All choices were made before the coupling and are uniform on the interval. □

Definition 36.17. Let

$$\mathfrak{C} = \mathfrak{su}(2)^3/\operatorname{Ad}\mathrm{SU}(2), \qquad \rho([Y],[Y']) = \min_{k\in\mathrm{SU}(2)}|Y - \operatorname{Ad}(k)Y'|,$$
$$\mathscr{P} = C([\tau,T];\mathfrak{C}), \qquad \rho_\infty(z,z') = \sup_{\tau\le t\le T}\rho(z(t),z'(t)).$$

Fix a smooth map $\eta : \mathrm{SU}(2)^3 \to \mathfrak{su}(2)^3$, equivariant under simultaneous conjugation, that agrees with the componentwise logarithm on a fixed smaller neighborhood of the

identity. For the triple $H(t)$ of cycle holonomies of a trajectory, defined up to simultaneous conjugation, set $Z_\gamma(t) = [\eta(H(t))/r_\gamma]$. For a radial cutoff χ let

$$D_\gamma = \mathbb{E}_{\lambda_\gamma} \mathcal{I}_{\gamma,\chi(|\cdot|)}(H(\tau)), \qquad \Pi_\gamma(F) = \frac{\mathbb{E}_{\lambda_\gamma}\big[\mathcal{I}_{\gamma,\chi(|\cdot|)}(H(\tau))F(Z_\gamma)\big]}{D_\gamma}.$$

For small γ, $D_\gamma > 0$ by Theorem 36.10, and Π_γ is a probability on $\mathscr{P}$. Define $\Pi_{K,\gamma}$ by the same formula under the Wilson measure $\mu_{nB^K,\gamma}$, with the flowed lattice holonomies and with its own denominator.

Only the conjugacy class of $H(t)$ is needed to define Z_γ; we never choose representatives depending on t.

Theorem 36.18. *Fix L, an interval $0 < \tau < T < \infty$, the parameters, a sufficiently large side n on the sequence, and the central margins for that interval, all before varying γ, and use Definition 36.17. Let $b_\chi = m(\chi(|\cdot|))$, and define the probability*

$$\Pi_0 = \big(Y \mapsto (t \mapsto [Y])\big)_{\#}\Big(\frac{\chi(|Y|)e^{-Q(Y)}\,d^9Y}{b_\chi}\Big)$$

on $\mathscr{P}$. There exist $C_{\tau,T,\chi,\eta}$ and $\gamma_{\tau,T,\chi,\eta} > 0$ such that, for every bounded function $F : \mathscr{P} \to \mathbb{R}$ that is globally Lipschitz for ρ_∞, and every $0 < \gamma \le \gamma_{\tau,T,\chi,\eta}$,

$$|\Pi_\gamma(F) - \Pi_0(F)| \le C_{\tau,T,\chi,\eta}\big(\|F\|_\infty + \operatorname{Lip} F\big)\gamma^{1/4}.$$

The same bound holds for $\Pi_{K,\gamma}$, uniformly in K, and every denominator is at least $c_\chi r_\gamma$. At each fixed small γ, $\Pi_{K,\gamma} \to \Pi_\gamma$ weakly as $K \to \infty$. In particular $\Pi_\gamma \to \Pi_0$ weakly on $(\mathscr{P}, \rho_\infty)$ as $\gamma \downarrow 0$, after the continuum limit, and the limit does not depend on the choice of η.

Proof. The core of the proof is Steps 2–3, which reduce the weighted path to the single-time test $\varphi_F(Y) = \chi(|Y|)F(c_{[Y]})$ defined below.

Step 1: quotients and paths. The minimum defining ρ is attained because $\mathrm{SU}(2)$ is compact, and it vanishes exactly when the two orbits coincide. The symmetry of ρ follows from the invariance of the Euclidean norm under the orthogonal maps $\mathrm{Ad}(k)$, and composing minimizers gives the triangle inequality. The quotient map is 1-Lipschitz. Since $\rho([Y],[0]) = |Y|$, closed bounded balls of $\mathfrak{C}$ are continuous images of compact Euclidean balls. They are therefore compact, and $\mathfrak{C}$ is complete. The classes of vectors with rational coordinates are dense, and $\mathfrak{C}$ is separable. Since uniform limits of continuous paths are continuous, $\mathscr{P}$ is complete. For separability, let $\varepsilon > 0$. Given a path, by uniform continuity there exist a rational subdivision of $[\tau, T]$ on whose intervals the path oscillates by at most ε, and balls of radius ε, centered in a fixed countable dense set, containing its values at the subdivision points. We consider the classes of paths determined by

a rational subdivision and a finite list of such balls, with oscillation at most ε on each interval. There are countably many classes, each of diameter at most 4ε, and we choose one path in each nonempty class. For $\varepsilon = 1/j$, $j \geq 1$, the chosen paths form a countable dense set, and $\mathscr{P}$ is Polish.

To see that a map η exists, we multiply the componentwise logarithm by a smooth cutoff, invariant under conjugation, supported strictly inside the logarithmic chart, and equal to one on a smaller neighborhood, and we extend the product by zero. The result is smooth and equivariant, and its global Lipschitz constant is finite by compactness of $\mathrm{SU}(2)^3$, for the maximum of the operator norms of the three components. By Lemma 5.6, the map $\mathfrak{p}$ of seven traces identifies the quotient of $\mathrm{SU}(2)^3$ by simultaneous conjugation homeomorphically with $C = \mathfrak{p}(\mathrm{SU}(2)^3)$. The map

$$f_\eta : C \to \mathfrak{C}, \qquad f_\eta(\mathfrak{p}(H)) = [\eta(H)],$$

is therefore well defined, by the equivariance of η, and continuous. Composing with this uniformly continuous map at each time carries a continuous path of traces to a continuous path in $\mathfrak{C}$, continuously for the uniform metrics, and scaling by $1/r_\gamma$ is continuous at fixed γ. This defines Z_γ under λ_γ.

Step 2: comparison with the zero-fluctuation path. We abbreviate $I(H) = \mathcal{I}_{\gamma,\chi(|\cdot|)}(H(\tau))$, and we write Z and Z^0 for the paths of the flowed holonomies of the configuration and of the zero-fluctuation reconstruction. Let $d = d_{K,[\tau,T]}$ be the quantity of Theorem 34.11, that is, the minimum of 1 and the supremum over $t \in [\tau, T]$ of $\max_i \|\mathcal{H}^K_{t,i} - \mathcal{H}^K_{0,t,i}(v)\|_{\mathrm{op}}$. Since η is globally Lipschitz and the quotient map is 1-Lipschitz,

$$\rho_\infty(Z, Z^0) \leq C_\eta \frac{d}{r_\gamma} \qquad \text{on } \{d \leq r_\gamma\},\ r_\gamma < 1.$$

On this event, if either $I(\mathcal{H}^K)$ or $I(\mathcal{H}^K_0)$ is nonzero, the zero-fluctuation holonomies at time τ lie in an enlarged window E_R as in Lemma 36.14, and

$$|I(\mathcal{H}^K) - I(\mathcal{H}^K_0)| \leq C_\chi \frac{d}{r_\gamma}.$$

These two assertions use the logarithm only at time τ, in its fixed compact subchart; at the other times we use η. Writing the difference of products as $(I(\mathcal{H}^K) - I(\mathcal{H}^K_0))F(Z) + I(\mathcal{H}^K_0)(F(Z) - F(Z^0))$, we find, on the same event,

$$|I(\mathcal{H}^K)F(Z) - I(\mathcal{H}^K_0)F(Z^0)| \leq C_{\chi,\eta}\big(\|F\|_\infty + \operatorname{Lip} F\big)\frac{d}{r_\gamma}\mathbf{1}_{E_R}.$$

Integrating with the disintegration of Lemma 15.10, we have $\mathbb{E}_\nu[d\mathbf{1}_{E_R}] \leq C r_\gamma^2 \widehat{\nu}(E_R) \leq C_R r_\gamma^3$, exactly as in the proof of Lemma 36.9. On $\{d > r_\gamma\}$ we use the bound $2\|\chi\|_\infty\|F\|_\infty$ and $\nu(d > r_\gamma) \leq C r_\gamma^2$. We conclude that

$$\Big|\mathbb{E}_\nu I(\mathcal{H}^K)F(Z) - \mathbb{E}_{\widehat{\nu}} I(\mathcal{H}^K_0)F(Z^0)\Big| \leq C_{\chi,\eta}\|F\|_{\mathrm{BL}} r_\gamma^2. \tag{36.41}$$

Step 3: the weighted zero-fluctuation path is nearly constant. We use Lemma 36.14 with $t_1 = \tau$, choosing its neighborhood $\mathcal{U}$ with closure in the domain of Lemma 36.16; the localization argument in the proof of that lemma allows any such fixed reduction. Outside $\mathcal{U}$ the contribution of the window at time τ is smaller than every power of r_γ, and so is the contribution of $\{|s| > r_\gamma\}$. On the remaining part of the weighted set, the lower bound on the output at time τ implies $|X| \le C_\chi r_\gamma$. By Lemma 36.16, for small γ every output lies in the region where $\eta = \mathrm{Log}$, and

$$\rho_\infty\big(Z^0, c_{[Z_{K,\tau}(X,s)/r_\gamma]}\big) \le \sup_{\tau \le t \le T} \frac{|Z_{K,t} - Z_{K,\tau}|}{r_\gamma} \le C_\chi\Big(r_\gamma + \frac{|s|}{r_\gamma}\Big),$$

where $c_{[Y]}$ denotes the constant path with value $[Y]$. The localized moments give $\widehat{\nu}(E_R) \le C_R r_\gamma$ and $\mathbb{E}_{\widehat{\nu}}[|s|\mathbf{1}_{E_R \cap \mathcal{U}}] \le C_R r_\gamma g$, with $g = r_\gamma^2/\sqrt{n}$. Since $I(\mathcal{H}_0^K)F(c_{[Z_{K,\tau}/r_\gamma]}) = \mathcal{I}_{\gamma,\varphi_F}(\mathcal{H}_{0,\tau}^K)$ on this set, we obtain

$$\Big|\mathbb{E}_{\widehat{\nu}} I(\mathcal{H}_0^K)F(Z^0) - \mathbb{E}_{\widehat{\nu}}\mathcal{I}_{\gamma,\varphi_F}(\mathcal{H}_{0,\tau}^K)\Big| \le C_\chi \|F\|_{\mathrm{BL}} r_\gamma^2, \tag{36.42}$$

where $\varphi_F(Y) = \chi(|Y|)F(c_{[Y]})$. This test is invariant under simultaneous conjugation, supported in the fixed ball containing the support of $\chi(|\cdot|)$, and globally Lipschitz, with

$$\|\varphi_F\|_\infty \le \|\chi\|_\infty \|F\|_\infty, \qquad \mathrm{Lip}\,\varphi_F \le \mathrm{Lip}(\chi(|\cdot|))\|F\|_\infty + \|\chi\|_\infty \mathrm{Lip}\,F,$$

since the quotient map and the map $[Y] \mapsto c_{[Y]}$ are 1-Lipschitz. We apply Theorem 36.6 to φ_F at time τ. Combining (36.41) and (36.42), and removing the conditioning with Corollary 23.11 at $q = 1$, we obtain

$$A_K(F) = \mathbb{E}_{\mu_{nB^K,\gamma}} I(\mathcal{H}^K)F(Z) = r_\gamma \kappa_{K,\gamma} m(\varphi_F) + O\big(\|F\|_{\mathrm{BL}} r_\gamma^2\big). \tag{36.43}$$

Here $A_K(F)$, notation local to this proof, is the numerator of $\Pi_{K,\gamma}(F)$. The same factor appears in $A_K(1)$, with leading coefficient $b_\chi > 0$, which implies $A_K(1) \ge c_\chi r_\gamma$. Since $m(\varphi_F) = b_\chi \Pi_0(F)$, subtracting $\Pi_0(F)A_K(1)$ from $A_K(F)$ cancels the leading terms. Dividing by $A_K(1)$ gives the bound for $\Pi_{K,\gamma} = A_K(\cdot)/A_K(1)$, uniformly in K.

Step 4: the continuum limit in the uniform topology. At fixed γ we apply Theorem 30.4 to the seven loops, concatenations of the three based coordinate cycles, that define $\mathfrak{p}$. Each loop has a single lattice approximation independent of time, of uniformly bounded length. The comparison side is covered by Corollary 29.26; at fixed coupling only finitely many terms of the sequence of sides are dropped. The paths of traces converge jointly in law in $C([\tau, T]; \mathbb{R}^7)$. All their values lie in the compact set C, and $C([\tau, T]; C)$ is a closed subset. Applying the continuous map of paths induced by f_η, followed by scaling, we obtain convergence in law of the lattice paths Z to Z_γ, jointly with the vector of traces at time τ. For small fixed γ, $I(H(\tau))$ is a continuous invariant function of the triple at time

τ, and by Lemma 5.6 it is a continuous function of its seven traces. For every bounded continuous F on $\mathscr{P}$, the product $I(H(\tau))F(Z_\gamma)$ is then a bounded continuous function on the space of trace paths with values in C. By the continuous mapping theorem on this closed subspace, which carries all the measures involved, $A_K(F)$ converges to the numerator of $\Pi_\gamma(F)$, and $A_K(1)$ converges to D_γ. Dividing by the positive limit of the denominators proves $\Pi_{K,\gamma} \to \Pi_\gamma$ weakly, and the bound of Step 3 passes to the limit.

Step 5: weak convergence. The bound for bounded Lipschitz functions implies weak convergence on $\mathscr{P}$. Indeed, for a closed set $A \subset \mathscr{P}$, the functions $f_j(z) = (1 - j\,\mathrm{dist}(z, A))_+$, $j \geq 1$, are bounded and Lipschitz, and $\Pi_\gamma(f_j) \to \Pi_0(f_j)$ for each fixed j. This implies $\limsup_{\gamma\downarrow 0} \Pi_\gamma(A) \leq \Pi_0(f_j)$, and letting $j \to \infty$, by monotone convergence, $\limsup_{\gamma\downarrow 0} \Pi_\gamma(A) \leq \Pi_0(A)$. By the portmanteau theorem, $\Pi_\gamma \to \Pi_0$ weakly. The same argument applies to any sequence of finite cutoffs with $\gamma \downarrow 0$, since the bound is uniform in K. The argument works for each such η and gives the same Π_0; the limit therefore does not depend on η. □

36.9 Dilation statistics of comparison measures

To show that the value zero in Theorem 36.7 reflects the structure of λ_γ near a central triple, we compute the dilation statistic for the comparison probabilities of Definition 4.13, observed at the same cycles and the same flow time. For $k \in \{1, 2, 3, 5\}$ we write P_k for the probability of item (k) of Definition 4.13. Here P_1 is the Abelian Maxwell measure with Haar cycle holonomies, P_2 the Gaussian holonomies with independent Haar factors, P_3 the Gaussian holonomies with Gaussian constant modes, and P_5 the mixture of normalized conditional normal Gaussian measures over the cone of commuting triples. For P_5 we use the observables $\mathcal{I}^{\mathrm{coord}}$ of Definition 36.1. The local mass scales differ. A compact window has probability of order r_γ^3 under P_1, r_γ^9 under P_2, and r_γ^5 under P_5, while under P_3 the rescaled logarithm tends to zero. For λ_γ the scale is r_γ.

Recall that $\mathcal{M} = \{(\vartheta_1 u, \vartheta_2 u, \vartheta_3 u) : \boldsymbol{\vartheta} \in \mathbb{R}^3,\ u \in S^2\}$ is the cone of commuting triples, that $\mathcal{N}_y = \{b : b_i \perp u,\ \sum_i \vartheta_i b_i = 0\}$ is the normal space at $y = (\vartheta_i u)_i \neq 0$, and that Σ_θ is the normal covariance of Theorem 33.11.

Theorem 36.19. *Let χ be a radial cutoff with $\chi(0) > 0$. Under each of P_1, P_2, P_3, with the hypotheses of Definition 4.13 and Lemma 35.17, the denominator of the dilation statistic is positive for all sufficiently small γ, and*

$$\lim_{\gamma\downarrow 0} \mathcal{S}_{\gamma,P_1}(\chi) = -6, \qquad \lim_{\gamma\downarrow 0} \mathcal{S}_{\gamma,P_2}(\chi) = 4\,\frac{\int_{\mathbb{R}^9} Q(Y)\chi(|Y|)\,dY}{\int_{\mathbb{R}^9} \chi(|Y|)\,dY} > 0, \qquad \lim_{\gamma\downarrow 0} \mathcal{S}_{\gamma,P_3}(\chi) = -9.$$

For P_3 the hypotheses are that the field is a centered Gaussian field with covariance γC for a fixed covariance C, smoothed by a fixed linear operator, and that the smoothed field divided by $\sqrt{\gamma}$ has

a centered separable realization with continuous paths, a measurable and almost surely bounded supremum, and finite supremum variance, as specified there.

For P_5, let the prior be a fixed probability $\alpha(y)\,d\mathrm{vol}_{\mathcal{M}}(y)$ on a small neighborhood of the tip of $\mathcal{M}$, with α continuous at the tip and $\alpha(0) > 0$. Conditionally on $y \neq 0$, let b be the normalized Gaussian $N(0, \gamma\Sigma(y))$ on $\mathcal{N}_y$, and put $X = y + b$. Assume that Σ is measurable and positive definite, with

$$c|y|^{-2}I \le \Sigma(y) \le C|y|^{-2}I, \qquad |y|^2\Sigma(y) \longrightarrow I_{\mathcal{N}_y} \quad (|y| \downarrow 0),$$

uniformly in the angular directions. The covariance $\Sigma = \Sigma_\theta$ has these properties; in fact $\Sigma_\theta(y) = |y|^{-2}I + O(1)$ uniformly for small $|y|$. Then, for every continuous compactly supported φ,

$$r_\gamma^{-5}\mathbb{E}_{P_5}\varphi(X/r_\gamma) \longrightarrow \alpha(0)\int \varphi(y+b)\,d\mu_{13}(y,b),$$

where

$$d\mu_{13}(y,b) = (2\pi)^{-2}|y|^4 e^{-|y|^2|b|^2/2}\,d\mathrm{vol}_{\mathcal{M}}(y)\,db.$$

The weighted normal-bundle measure μ_{13} is locally finite, and y and b are bounded on compact sets of outputs $y + b$ because $y \perp b$. The denominator of $\mathcal{S}_{\gamma,P_5}(\chi)$ is positive for all sufficiently small γ. Writing $\langle\cdot\rangle_{\chi,\mu_{13}}$ for the integral against μ_{13} divided by $\int \chi(|y+b|)\,d\mu_{13}$,

$$\lim_{\gamma\downarrow 0}\mathcal{S}_{\gamma,P_5}(\chi) = 4 + 2\Big\langle \sum_{i<j}|b_i \times b_j|^2\Big\rangle_{\chi,\mu_{13}} > 4.$$

The constants in these comparisons may depend on the fixed priors, the bounds on the covariances, the smoothing, and the test, but not on γ.

The factor $|y|^4$ in μ_{13} comes from the normalization of the four-dimensional conditional Gaussian. The weighted volume element $(2\pi)^{-2}|y|^4\,d\mathrm{vol}_{\mathcal{M}}(y)\,db$ is homogeneous of degree $13 = 5 + 4 + 4$ under dilations: five from the volume of the cone, four from the normal fibers, and four from the factor $|y|^4$. The remaining factor of μ_{13} is e^{-Q_2}, where $Q_2(y,b) = |y|^2|b|^2/2$ is homogeneous of degree four. The degree 13, compared with the degree 9 of Lebesgue measure on $\mathbb{R}^9$, produces the constant 4. Without the normalization of the conditional Gaussians, both the mixture and the value would change. The conclusions concern only these specified priors and covariances.

Proof. We write $\epsilon = r_\gamma$. Each numerator is the observable of the bounded compactly supported function W_χ.

The Abelian Maxwell measure with Haar cycle holonomies. We condition on the smoothed Abelian field and on the color direction $u \in S^2$ of the embedded subgroup. The circulations of the field translate the three independent uniformly distributed phases in $(\mathbb{R}/4\pi\mathbb{Z})^3$, and conditionally the cycle phases are still independent and uniformly

distributed. In a small neighborhood of the identity their logarithms have the form (x_1u, x_2u, x_3u), with density $(4\pi)^{-3}d^3x$. For radial tests, a further common conjugation does not change the integrals. The triple commutes, and $Q = 0$ on it. Once the compact support lies in the logarithmic chart, the denominator is exactly

$$(4\pi)^{-3}\epsilon^3 \int_{\mathbb{R}^3} \chi(|x|)\, dx > 0.$$

The divergence of $x\chi(|x|)$ on $\mathbb{R}^3$ has integral zero, which gives $\int |x|\chi'(|x|)\, dx = -3\int \chi(|x|)\, dx$. The dilation statistic therefore equals $-9 + 3 = -6$ for every sufficiently small γ.

Gaussian holonomies with independent Haar factors. By Lemma 35.17, the triple has the law of three independent Haar distributed matrices for every γ. We write $H = \cos(|x|/2)I + 2\sin(|x|/2)(x/|x|)^a\tau_a$. The Haar probability is the normalized volume of S^3, and its polar factor $\sin^2(|x|/2)\, d(|x|/2)/(2\pi^2)$ gives $dH(e^x) = (16\pi^2)^{-1}\operatorname{sinc}^2(|x|/2)\, d^3x$, as in (4.2). For every bounded test ψ supported in $\{|Y| \le R\}$, we obtain

$$\mathbb{E}_{P_2}\mathcal{I}_{\gamma,\psi} = \epsilon^9(16\pi^2)^{-3}\Big\{\int \psi(Y)\, dY + O_R\big(\|\psi\|_\infty \epsilon^2\big)\Big\}.$$

The denominator has a positive limiting coefficient. In $\mathbb{R}^9$ the divergence of $Y\chi(|Y|)$ has integral zero, that is, $\int (9\chi(|Y|) + |Y|\chi'(|Y|))\, dY = 0$. The limit of the statistic is therefore $4\int Q\chi / \int \chi$. To see that it is strictly positive, take a radius at which $\chi > 0$ and, at that radius, a triple with two nonzero orthogonal components. Then $Q > 0$ there, and a neighborhood of this point has positive measure, $\chi > 0$, and $Q > 0$.

Gaussian holonomies with Gaussian constant modes. At the observation time $t = \theta L^2$, let $G_i(x) = (\mathsf{S}_t a)_i(x)/\sqrt{\gamma}$ be the smoothed field divided by $\sqrt{\gamma}$, and consider the real process

$$Z_{x,i,v} = \langle v, G_i(x)\rangle, \qquad (x, i, v) \in \mathcal{T} = \mathbb{T}^3_L \times \{1, 2, 3\} \times S^2.$$

The index set $\mathcal{T}$ is nonempty, compact, and metrizable, and the process is centered Gaussian since the smoothing is fixed and linear. By the hypotheses of Definition 4.13 and Lemma 35.17, there exists a separable realization with almost surely continuous paths on $\mathcal{T}$, and $\sigma_*^2 = \sup_{x,i,v} \mathbb{E}Z^2_{x,i,v} < \infty$. These are the hypotheses of Lemma 35.16, which allows degenerate covariances; that lemma is the Borell–TIS inequality in the formulation of Adler and Taylor [3, Theorem 2.1.1]. It provides a measurable, almost surely finite, and integrable supremum

$$Z_* = \sup_{\mathcal{T}} Z_{x,i,v} = \max_i \sup_x |G_i(x)|, \qquad m_* = \mathbb{E}Z_* < \infty,$$

where the equality follows by taking the supremum over all unit vectors v. The law of G, and hence m_* and σ_*^2, do not depend on γ, since the field has covariance γC and the covariance and the smoothing are fixed. When $\sigma_*^2 > 0$, the same lemma implies that

$$\mathbb{P}\{Z_* - m_* \geq u\} \leq e^{-u^2/(2\sigma_*^2)} \qquad (u > 0). \tag{36.44}$$

The group distance of the transport along a path from the identity is at most the integral of the norm of its generator. This follows by partitioning the path, using the triangle inequality for the bi-invariant metric, and passing to the limit in the transport equation. Since each coordinate cycle has length L, we obtain

$$\Big(\sum_i \rho_G(H_i)^2\Big)^{1/2} \leq \sqrt{3}L\sqrt{\gamma}Z_*.$$

Since Z_* is almost surely finite and its law does not depend on γ, all three holonomies lie in the logarithmic chart with probability tending to one. There the rescaled logarithm has norm at most $\sqrt{3}L\epsilon Z_*$, which tends to zero in probability. Quantitatively, fix $\delta > 0$ and suppose that $\sigma_*^2 > 0$. For small γ we have $\sqrt{3}L\epsilon m_* \leq \delta/2$, and the ball of radius $\delta\epsilon$ about the identity, for the product of the group distances, lies in $\mathcal{V}$. By the transport bound and (36.44), we have

$$\begin{aligned}\mathbb{P}\big\{H \notin \mathcal{V} \text{ or } (H \in \mathcal{V},\ |\operatorname{Log} H|/\epsilon > \delta)\big\} &\leq \exp\Big\{-\frac{(\delta/(\sqrt{3}L\epsilon) - m_*)^2}{2\sigma_*^2}\Big\} \\ &\leq \exp\Big\{-\frac{\delta^2}{24L^2\sigma_*^2\sqrt{\gamma}}\Big\},\end{aligned} \tag{36.45}$$

with constants depending only on δ, L, the covariance, and the smoothing. If $\sigma_*^2 = 0$, Lemma 35.16 shows that the continuous process vanishes almost surely, and then $G = 0$ and $H = (I, I, I)$ almost surely. Since the two tests are bounded and continuous at zero, we obtain

$$\mathbb{E}_{P_3}\mathcal{I}_{\gamma,\chi(|\cdot|)} \longrightarrow \chi(0) > 0, \qquad \mathbb{E}_{P_3}\mathcal{I}_{\gamma,W_\chi} \longrightarrow W_\chi(0) = -9\chi(0);$$

to see this, we split each expectation according to whether the rescaled logarithm has norm at most δ, use continuity at zero on this event, and bound the expectation on its complement by twice the supremum of the test times its probability. This gives the value -9 without any statement about densities of the nonlinear cycle logarithms.

The mixture over the cone of commuting triples. Here P_5 is the law of the vector $X = y + b \in \mathbb{R}^9$, and the statistic uses $\mathcal{I}^{\mathrm{coord}}$. Fix $\zeta > 0$ with $\{|X| < \zeta\} \subset \operatorname{Log}\mathcal{V}$. For a bounded test ψ supported in $\{|Y| \leq R\}$ and $R\epsilon < \zeta$, we have the identity on all of $\mathbb{R}^9$

$$\mathcal{I}^{\mathrm{coord}}_{\gamma,\psi}(X) = \psi(X/\epsilon). \tag{36.46}$$

Indeed, if the right side is nonzero, then $|X| \le R\epsilon < \zeta$ and the indicator equals one; if it is zero, both sides vanish. This applies to $\psi = \chi(|\cdot|)$ and $\psi = W_\chi$ simultaneously, with one bound on the supports. The observables therefore agree with the expectations computed below, even though the conditional Gaussians have unbounded support.

We record the geometry and the normalization. Away from the tip, we write $y_i = \vartheta_i u$ and $s = |\boldsymbol{\vartheta}| = |y| > 0$; in this proof and in Remark 36.20, s denotes this radius, not a transverse coordinate. For variations of $\boldsymbol{\vartheta}$ and u, the induced metric is $|d\boldsymbol{\vartheta}|^2 + s^2|du|^2$, since $u \cdot du = 0$, and the parametrization is two-to-one under the simultaneous change of sign. The volume of the cone is therefore

$$d\mathrm{vol}_{\mathcal{M}}(y) = \tfrac{1}{2} s^4 \, ds \, d\omega \, du, \qquad \omega = \boldsymbol{\vartheta}/s \in S^2, \quad u \in S^2, \tag{36.47}$$

which is homogeneous of degree five and gives zero volume to the tip. The normal space $\mathcal{N}_y$ has dimension four, and $\langle y, b\rangle = \sum_i \vartheta_i \, u \cdot b_i = 0$, which gives

$$|y + b|^2 = |y|^2 + |b|^2. \tag{36.48}$$

The computation proving (36.14), applied with $a = \boldsymbol{\vartheta}/s$, gives

$$Q(y + b) = Q_2(y, b) + \frac{1}{2} \sum_{i<j} |b_i \times b_j|^2, \qquad Q_2(y, b) = \frac{1}{2} |y|^2 |b|^2. \tag{36.49}$$

These are identities on the normal bundle; they do not require uniqueness of the decomposition $X = y + b$.

The conditional density in the fiber $\mathcal{N}_{y_0}$ is

$$(2\pi\gamma)^{-2} \det \Sigma(y_0)^{-1/2} \exp\{ -b_0^{\mathsf{T}} \Sigma(y_0)^{-1} b_0 / (2\gamma) \} \, db_0.$$

We set $y_0 = \epsilon y$, $b_0 = \epsilon b$, and $C_\epsilon(y) = \epsilon^2 \Sigma(\epsilon y)$, using that the normal spaces are constant along rays. Then $d\mathrm{vol}(y_0) = \epsilon^5 \, d\mathrm{vol}(y)$ and $db_0 = \epsilon^4 db$, and since $\gamma = \epsilon^4$ we obtain the exact formula

$$\epsilon^{-5} \mathbb{E}_{P_5} \varphi(X/\epsilon) = \int \varphi(y + b) \, \alpha(\epsilon y) (2\pi)^{-2} \det C_\epsilon(y)^{-1/2} e^{-b^{\mathsf{T}} C_\epsilon(y)^{-1} b/2} \, d\mathrm{vol}_{\mathcal{M}}(y) \, db, \tag{36.50}$$

where the prior is set to zero outside its neighborhood; on a compact set of outputs, this neighborhood eventually contains every ϵy that occurs. By the assumptions on the covariance, we have, uniformly in the angular directions and for $y \neq 0$,

$$C_\epsilon(y) \longrightarrow |y|^{-2} I, \qquad c|y|^{-2} I \le C_\epsilon(y) \le C|y|^{-2} I.$$

Hence the Gaussian density in (36.50) tends to $|y|^4 e^{-Q_2(y,b)}$ and is bounded by $C|y|^4$. If φ is supported in the ball of radius R, then (36.48) bounds both $|y|$ and $|b|$ by R. The continuity of α at the tip bounds $\alpha(\epsilon y)$ there and gives the limit $\alpha(0)$. The dominating

integral is finite, since by (36.47) its radial factor near zero is $s^8\,ds$, and each fiber is restricted to a bounded ball in four dimensions. In finitely many angular trivializations of the normal bundle, or with the intrinsic Gaussian density on each fiber, the dominated convergence theorem applies. Then (36.50) gives the claimed limit with the normalization $(2\pi)^{-2}$ of μ_{13}.

For $\Sigma = \Sigma_\theta$, the explicit formula (4.12) gives $\Sigma_\theta(y) = |y|^{-2}I + O_\theta(1)$ uniformly in the angular directions: the term $m = 0$ of each d_i equals $e^{-2\theta|\boldsymbol{\vartheta}|^2}|\boldsymbol{\vartheta}|^{-2}$, since $R_i(0,\boldsymbol{\vartheta}) = 1$, and the sum of the terms with $m \neq 0$ is bounded for small $|\boldsymbol{\vartheta}|$, since then $\lambda_m \geq c|m|^2$ and $|R_i| \leq 1$. After a fixed reduction of the neighborhood carrying the prior, this implies both bounds on the covariance and the relative limit, and Theorem 33.11 shows that Σ_θ is positive definite and measurable there.

By the same domination and (36.48), μ_{13} is locally finite as a measure of the output $y + b$. Its integral against $\chi(|y+b|)$ is finite and strictly positive, for instance by considering an open set of the bundle with $|y| > 0$ and output radius in an interval on which $\chi > 0$. By (36.46), the denominator of the statistic satisfies

$$\mathbb{E}_{P_5}\mathcal{I}^{\mathrm{coord}}_{\gamma,\chi(|\cdot|)}(X) \sim \epsilon^5\alpha(0)\int \chi(|y+b|)\,d\mu_{13} > 0.$$

Under the dilation $(y,b) \mapsto (\lambda y, \lambda b)$, the measure $d\mathrm{vol}_{\mathcal{M}}\,db$ is homogeneous of degree $5+4=9$, the factor $|y|^4$ of degree four, and Q_2 of degree four. Changing variables in the compactly supported integral of $\chi(\lambda|y+b|)$ and differentiating at $\lambda = 1$, we obtain

$$\int \big\{(4Q_2 - 13)\chi(\rho) - \rho\chi'(\rho)\big\}\,d\mu_{13} = 0, \qquad \rho = |y+b|. \tag{36.51}$$

The differentiation is justified on a common compact set of outputs by (36.48), with the integrable radial bound $Cs^8\,ds$. This change of variables includes the tip of the cone and needs no boundary term there. We apply the limit of the rescaled expectations to W_χ and $\chi(|\cdot|)$, using (36.46), and divide by the positive limit of the denominators. Combining (36.49) and (36.51), we find

$$\lim_{\gamma\downarrow 0}\mathcal{S}_{\gamma,P_5}(\chi) = 4 + 2\,\frac{\int \sum_{i<j}|b_i \times b_j|^2\chi(\rho)\,d\mu_{13}}{\int \chi(\rho)\,d\mu_{13}}.$$

The second term is strictly positive. To see this, take $\boldsymbol{\vartheta}$ in the third coordinate direction, a fixed color direction u, linearly independent $b_1, b_2 \in u^\perp$, and $b_3 = 0$. Then $b \in \mathcal{N}_y$ and $b_1 \times b_2 \neq 0$. Scaling y and b together, we place the output radius in an interval on which $\chi > 0$. A neighborhood of this point in the bundle has positive measure, and on it $\chi > 0$ and the polynomial $\sum_{i<j}|b_i \times b_j|^2$ is positive. This proves the strict inequality. □

Remark 36.20. With the normalization of μ_{13} in Theorem 36.19,

$$\mu_{13}\{|y+b| \leq \rho\} = \frac{8\pi^2}{1287}\rho^{13}\big[1 + O(\rho^4)\big] \qquad (\rho \downarrow 0).$$

Indeed, on the ball $\{|y+b| \le \rho\}$ we have $0 \le Q_2 \le \rho^4/8$ by (36.48), and $e^{-Q_2} = 1 + O(\rho^4)$ uniformly there. The angular integral in (36.47) is $(4\pi)^2/2 = 8\pi^2$, and the ball of radius $(\rho^2 - s^2)^{1/2}$ in the four-dimensional fiber has volume $(\pi^2/2)(\rho^2 - s^2)^2$. Hence

$$\begin{aligned}\mu_{13}\{|y+b| \le \rho\} &= \pi^2 \int_0^\rho s^8 (\rho^2 - s^2)^2 \, ds \, \big[1 + O(\rho^4)\big] \\ &= \pi^2 \rho^{13} \Big(\frac{1}{9} - \frac{2}{11} + \frac{1}{13}\Big) \big[1 + O(\rho^4)\big] = \frac{8\pi^2}{1287} \rho^{13} \big[1 + O(\rho^4)\big].\end{aligned}$$

This formula is not used elsewhere.

36.10 The main result on central holonomies

We conclude by deducing Theorem 2.11 from the results of this section.

Proof of Theorem 2.11. Fix $L > 0$, an odd integer $M \ge 3$, an integer n divisible by four, and $\theta > 0$. By Theorem 2.3 (part 5), proved in Theorem 29.28, the measure $\lambda_\gamma = \lambda_{L,\gamma,[n]_M}$ depends only on L, γ, and the class $[n]_M$. We choose J and the remaining parameters as in Corollary 29.26 and Definition 34.1, so that $a_{\mathrm{map}}\sqrt{B} \le 32/49$, and take $n_* = n$ and a comparison side nB^a satisfying the lower bounds of Definition 36.1. Then λ_γ is the continuum measure of the sequence $(nB^K)_{K \ge 0}$, and the results of this section apply with this comparison side in place of n; all choices depend only on L, M, n, and θ. The observable $\mathcal{I}_{\gamma,\varphi}$ of Theorem 2.11 is the observable of Definition 36.1, with the extension by zero outside the fixed neighborhood $\mathcal{V}$ of the identity. With $\rho = |X|/r_\gamma$, the statistic of part (3) of Theorem 2.11 is the dilation statistic $\mathcal{S}_{\gamma,\lambda_\gamma}(\chi)$.

Part (1) of Theorem 2.11 is the ratio limit and the two-sided bound of Theorem 36.7. Parts (2) and (3) are the rates of Theorem 36.10, for invariant Lipschitz tests supported in the ball of radius R and for the dilation statistic. Part (4) follows from Theorem 36.12, with $\mathcal{V}$ chosen invariant under simultaneous conjugation: at each fixed γ the expectations near the eight central triples are equal, and hence parts (1)–(3) hold near each of them with the same constants. In all these results, the continuum limit is taken first, at fixed γ, and then $\gamma \to 0$. The constants depend on L, M, n, θ, and the test functions, but not on γ. The further statements after Theorem 2.11 are Theorems 36.15, 36.18, and 36.19. □

References

[1] Radosław Adamczak. A Note on the Hanson–Wright Inequality for Random Vectors with Dependencies. *Electronic Communications in Probability*, 20(72):1–13, 2015. doi: 10.1214/ECP.v20-3829. URL https://arxiv.org/abs/1409.8457.

[2] Arka Adhikari and Sky Cao. Correlation decay for finite lattice gauge theories at weak coupling. *The Annals of Probability*, 53(1):140–174, 2025. doi: 10.1214/24-AOP1702. URL https://doi.org/10.1214/24-AOP1702.

[3] Robert J. Adler and Jonathan E. Taylor. *Random Fields and Geometry*. Springer Monographs in Mathematics. Springer, New York, 2007. ISBN 978-0-387-48112-8. doi: 10.1007/978-0-387-48116-6. URL https://doi.org/10.1007/978-0-387-48116-6. Theorem 2.1.1, printed p. 50, (2.1.5); Lemma 2.1.5, printed p. 54, (2.1.13), proof pp. 54–55. Used with the authors' correction list of 7 October 2015, entries 54-6, 55-11, and 55-4: https://robert.net.technion.ac.il/files/2016/08/corrections.pdf. Gaussian reference only.

[4] David J. Aldous and Geoffrey K. Eagleson. On mixing and stability of limit theorems. *Annals of Probability*, 6(2):325–331, 1978. doi: 10.1214/aop/1176995577.

[5] Thomas Appelquist and Robert D. Pisarski. High-temperature Yang–Mills theories and three-dimensional quantum chromodynamics. *Physical Review D*, 23(10):2305–2317, 1981. doi: 10.1103/PhysRevD.23.2305.

[6] Michael F. Atiyah and Raoul Bott. The Yang–Mills equations over Riemann surfaces. *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 308(1505):523–615, 1983. doi: 10.1098/rsta.1983.0017. URL https://doi.org/10.1098/rsta.1983.0017.

[7] Peter Austing and John F. Wheater. The convergence of Yang-Mills integrals. *Journal of High Energy Physics*, 2001(02):028, 2001. doi: 10.1088/1126-6708/2001/02/028. URL https://arxiv.org/abs/hep-th/0101071.

[8] Dominique Bakry and Michel Émery. Diffusions hypercontractives. In *Séminaire de Probabilités XIX 1983/84*, volume 1123 of *Lecture Notes in Mathematics*, pages 177–206. Springer, 1985. doi: 10.1007/BFb0075847. URL https://www.numdam.org/item/SPS_1985__19__177_0/.

[9] Tadeusz Bałaban. Regularity and decay of lattice Green's functions. *Communications in Mathematical Physics*, 89(4):571–597, 1983. doi: 10.1007/BF01214744. URL https://doi.org/10.1007/BF01214744.

[10] Tadeusz Bałaban. Propagators and renormalization transformations for lattice gauge theories. I. *Communications in Mathematical Physics*, 95(1):17–40, 1984. doi: 10.1007/BF01215753. URL https://doi.org/10.1007/BF01215753. Proposition 1.1, p. 33; formulas (1.11), (1.18), (1.39)–(1.44), (1.60)–(1.69), (1.103).

[11] Tadeusz Bałaban. Propagators and renormalization transformations for lattice gauge theories. II. *Communications in Mathematical Physics*, 96(2):223–250, 1984. doi: 10.1007/BF01240221. URL https://doi.org/10.1007/BF01240221. Proposition 2.7 and Corollary 2.8, p. 249.

[12] Tadeusz Bałaban. Averaging operations for lattice gauge theories. *Communications in Mathematical Physics*, 98(1):17–51, 1985. doi: 10.1007/BF01211042. URL https://doi.org/10.1007/BF01211042. The reference-transport logarithmic averaging construction is equation (15), p. 19.

[13] Tadeusz Bałaban. Propagators for lattice gauge theories in a background field. *Communications in Mathematical Physics*, 99(3):389–434, 1985. doi: 10.1007/BF01240355. URL https://doi.org/10.1007/BF01240355. Theorems 3.1–3.3, pp. 397–399; Hölder norms (3.40), p. 397, and nonendpoint first-gradient Hölder bounds (3.43), p. 398; extension (3.133), p. 422, and Theorem 3.12, p. 423; constrained covariance (3.148), (3.153), pp. 425–426; Theorem 3.13, p. 426. Only the stated identity-background portions are used.

[14] Tadeusz Bałaban. Spaces of regular gauge field configurations on a lattice and gauge fixing conditions. *Communications in Mathematical Physics*, 99(1):75–102, 1985. doi: 10.1007/BF01466594. URL https://doi.org/10.1007/BF01466594.

[15] Tadeusz Bałaban. Ultraviolet stability of three-dimensional lattice pure gauge field theories. *Communications in Mathematical Physics*, 102(2):255–275, 1985. doi: 10.1007/BF01229380. URL https://doi.org/10.1007/BF01229380.

[16] Tadeusz Bałaban. The variational problem and background fields in renormalization group method for lattice gauge theories. *Communications in Mathematical Physics*, 102(2):277–309, 1985. doi: 10.1007/BF01229381. URL https://doi.org/10.1007/BF01229381.

[17] Tadeusz Bałaban. Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions. *Communications in Mathematical Physics*, 109(2):249–301, 1987. doi: 10.1007/BF01215223. URL https://doi.org/10.1007/BF01215223.

[18] Tadeusz Bałaban. Convergent renormalization expansions for lattice gauge theories. *Communications in Mathematical Physics*, 119(2):243–285, 1988. doi: 10.1007/BF01217741. URL https://doi.org/10.1007/BF01217741.

[19] Tadeusz Bałaban. Renormalization group approach to lattice gauge field theories. II. Cluster expansions. *Communications in Mathematical Physics*, 116(1):1–22, 1988. doi: 10.1007/BF01239022. URL https://doi.org/10.1007/BF01239022.

[20] Tadeusz Bałaban. Large field renormalization. I. The basic step of the R operation. *Communications in Mathematical Physics*, 122(2):175–202, 1989. doi: 10.1007/BF01257412. URL https://doi.org/10.1007/BF01257412.

[21] Tadeusz Bałaban. Large field renormalization. II. Localization, exponentiation, and bounds for the R operation. *Communications in Mathematical Physics*, 122(3):355–392, 1989. doi: 10.1007/BF01238433. URL https://doi.org/10.1007/BF01238433.

[22] Tadeusz Bałaban and Arthur Jaffe. Constructive gauge theory. In *Fundamental Problems of Gauge Field Theory*, volume 141 of *NATO ASI Series B*, pages 207–263. Plenum Press, 1986. doi: 10.1007/978-1-4757-0363-4_5. URL https://link.springer.com/chapter/10.1007/978-1-4757-0363-4_5.

[23] Tadeusz Bałaban, John Imbrie, and Arthur Jaffe. Exact renormalization group for gauge theories. In *Progress in Gauge Field Theory*, volume 115 of *NATO ASI Series B*, pages 79–103. Plenum Press, 1984. doi: 10.1007/978-1-4757-0280-4_4. URL https://link.springer.com/chapter/10.1007/978-1-4757-0280-4_4.

[24] Tadeusz Bałaban, John Imbrie, and Arthur Jaffe. Renormalization of the Higgs model: Minimizers, propagators and the stability of mean field theory. *Communications in Mathematical Physics*, 97(1–2):299–329, 1985. doi: 10.1007/BF01206191. URL https://link.springer.com/article/10.1007/BF01206191.

[25] Tadeusz Bałaban, John Z. Imbrie, and Arthur Jaffe. Effective action and cluster properties of the abelian Higgs model. *Communications in Mathematical Physics*, 114 (2):257–315, 1988. doi: 10.1007/BF01225038. URL https://link.springer.com/article/10.1007/BF01225038.

[26] Riddhipratim Basu and Shirshendu Ganguly. $SO(N)$ lattice gauge theory, planar and beyond. *Communications on Pure and Applied Mathematics*, 71(10):2016–2064, 2018. doi: 10.1002/cpa.21788. URL https://doi.org/10.1002/cpa.21788.

[27] Rajendra Bhatia. *Matrix Analysis*, volume 169 of *Graduate Texts in Mathematics*. Springer, New York, 1997. doi: 10.1007/978-1-4612-0653-8. URL https://doi.org/10.1007/978-1-4612-0653-8.

[28] Patrick Billingsley. *Convergence of Probability Measures*. Wiley Series in Probability and Statistics. John Wiley & Sons, New York, second edition, 1999. doi: 10.1002/9780470316962.

[29] Marek Biskup. Reflection positivity and phase transitions in lattice spin models. In Roman Kotecký, editor, *Methods of Contemporary Mathematical Statistical Physics*, volume 1970 of *Lecture Notes in Mathematics*, pages 1–86. Springer, Berlin, Heidelberg, 2009. doi: 10.1007/978-3-540-92796-9_1. URL https://doi.org/10.1007/

978-3-540-92796-9_1. arXiv version 2, 27 April 2009. Theorem 5.8, (5.34)–(5.35), printed p. 44; proof pp. 45–46.

[30] Christer Borell. The Brunn-Minkowski inequality in Gauss space. *Inventiones Mathematicae*, 30(2):207–216, 1975. doi: 10.1007/BF01425510. URL https://doi.org/10.1007/BF01425510.

[31] Raoul Bott. Nondegenerate critical manifolds. *Annals of Mathematics*, 60(2):248–261, 1954. doi: 10.2307/1969631. URL https://doi.org/10.2307/1969631.

[32] Jean Bourgain. Invariant measures for the 2D-defocusing nonlinear Schrödinger equation. *Communications in Mathematical Physics*, 176(2):421–445, 1996. doi: 10.1007/BF02099556. URL https://doi.org/10.1007/BF02099556.

[33] Herm Jan Brascamp and Elliott H. Lieb. On Extensions of the Brunn–Minkowski and Prékopa–Leindler Theorems, Including Inequalities for Log Concave Functions, and with an Application to the Diffusion Equation. *Journal of Functional Analysis*, 22(4):366–389, 1976. doi: 10.1016/0022-1236(76)90004-5. URL https://doi.org/10.1016/0022-1236(76)90004-5.

[34] Christian Brennecke. On the Leading Order Term of the Lattice Yang–Mills Free Energy. *Journal of Statistical Physics*, 193(7):83, 2026. doi: 10.1007/s10955-026-03649-4. URL https://doi.org/10.1007/s10955-026-03649-4.

[35] Bjoern Bringmann and Sky Cao. A para-controlled approach to the stochastic Yang–Mills equation in two dimensions. *Memoirs of the American Mathematical Society*, 319(1624):v+125, 2026. doi: 10.1090/memo/1624. URL https://doi.org/10.1090/memo/1624. Publication metadata corroborated by the zbMATH-linked record DE 8212947 and author publication lists; direct AMS page inaccessible during verification.

[36] David Brydges, Jürg Fröhlich, and Erhard Seiler. Diamagnetic and critical properties of Higgs lattice gauge theories. *Nuclear Physics B*, 152(3–4):521–532, 1979. doi: 10.1016/0550-3213(79)90095-6. URL https://www.sciencedirect.com/science/article/pii/0550321379900956.

[37] David Brydges, Jürg Fröhlich, and Erhard Seiler. On the construction of quantized gauge fields. I. general results. *Annals of Physics*, 121(1–2):227–284, 1979. doi: 10.1016/0003-4916(79)90098-8. URL https://www.sciencedirect.com/science/article/abs/pii/0003491679900988.

[38] David C. Brydges. A short course on cluster expansions. In Konrad Osterwalder and Raymond Stora, editors, *Critical Phenomena, Random Systems, Gauge Theories, Part I*, Les Houches, Session XLIII, 1984, pages 129–183. North-Holland, Amsterdam, 1986. URL https://inspirehep.net/literature/211093. School held in 1984; proceedings published in 1986. No DOI or arXiv identifier found.

[39] David C. Brydges. Lectures on the renormalisation group. In Scott Sheffield and Thomas Spencer, editors, *Statistical Mechanics*, volume 16 of *IAS/Park City Mathematics Series*, pages 7–93. American Mathematical Society, 2009. doi: 10.1090/pcms/016/02. URL https://doi.org/10.1090/pcms/016/02.

[40] David C. Brydges, Jürg Fröhlich, and Erhard Seiler. Construction of quantised gauge fields. II. convergence of the lattice approximation. *Communications in Mathematical Physics*, 71(2):159–205, 1980. doi: 10.1007/BF01197918. URL https://link.springer.com/article/10.1007/BF01197918.

[41] David C. Brydges, Jürg Fröhlich, and Erhard Seiler. On the construction of quantized gauge fields. III. the two-dimensional abelian Higgs model without cutoffs. *Communications in Mathematical Physics*, 79(3):353–399, 1981. doi: 10.1007/BF01208500. URL https://link.springer.com/article/10.1007/BF01208500.

[42] Sky Cao and Sourav Chatterjee. The Yang–Mills heat flow with random distributional initial data. *Communications in Partial Differential Equations*, 48(2):209–251, 2023. doi: 10.1080/03605302.2023.2169937. URL https://arxiv.org/abs/2111.10652v4. All numbered and printed-page locators refer to arXiv:2111.10652v4, 25 August 2022, not journal pagination. Only deterministic smooth-data results are used: Theorem 1.2 and Lemma 1.3, p. 3; Lemmas C.2–C.3, pp. 55–56; Corollary C.6, p. 58; Lemmas C.12–C.13, p. 61.

[43] Sky Cao and Sourav Chatterjee. A state space for 3D euclidean Yang–Mills theories. *Communications in Mathematical Physics*, 405(1):3, 2024. doi: 10.1007/s00220-023-04870-y. URL https://arxiv.org/abs/2111.12813v4. Article 3, 69 pp. All numbered and printed-page locators refer to arXiv:2111.12813v4, 20 November 2023, not journal pagination, including Lemmas 1.4–1.5 and Theorem 1.6, p. 7; Theorem 3.14, p. 25; Theorem 3.51, pp. 35–36; Lemma 3.68, p. 40; the complete exercised leaf contracts, including the supporting Sobolev and trajectory leaves, are registered in the manuscript BLUEPRINT, section 6.2.

[44] Sky Cao, Ron Nissim, and Scott Sheffield. Expanded regimes of area law for lattice Yang–Mills theories, 2025. URL https://arxiv.org/abs/2505.16585. Preprint; no published version verified.

[45] Ajay Chandra, Ilya Chevyrev, Martin Hairer, and Hao Shen. Langevin Dynamic for the 2D Yang–Mills Measure. *Publications mathématiques de l'IHÉS*, 136:1–147, 2022. doi: 10.1007/s10240-022-00132-0. URL https://arxiv.org/abs/2006.04987.

[46] Ajay Chandra, Ilya Chevyrev, Martin Hairer, and Hao Shen. Stochastic Quantisation of Yang–Mills–Higgs in 3D. *Inventiones mathematicae*, 237(2):541–696, 2024. doi: 10.1007/s00222-024-01264-2. URL https://arxiv.org/abs/2201.03487.

[47] Nelia Charalambous and Leonard Gross. The Yang–Mills Heat Semigroup on Three-Manifolds with Boundary. *Communications in Mathematical Physics*, 317(3):727–785,

2013. doi: 10.1007/s00220-012-1558-0. URL https://arxiv.org/abs/1004.1639.

[48] Nelia Charalambous and Leonard Gross. Neumann domination for the Yang–Mills heat equation. *Journal of Mathematical Physics*, 56(7):073505, 2015. doi: 10.1063/1.4927250. URL https://doi.org/10.1063/1.4927250.

[49] Nelia Charalambous and Leonard Gross. Initial behavior of solutions to the Yang–Mills heat equation. *Journal of Mathematical Analysis and Applications*, 451(2):873–905, 2017. doi: 10.1016/j.jmaa.2017.02.027. URL https://doi.org/10.1016/j.jmaa.2017.02.027.

[50] Sourav Chatterjee. The Leading Term of the Yang–Mills Free Energy. *Journal of Functional Analysis*, 271(10):2944–3005, 2016. doi: 10.1016/j.jfa.2016.04.032. URL https://arxiv.org/abs/1602.01222v6.

[51] Sourav Chatterjee. Yang–Mills for Probabilists. In Peter Friz, Wolfgang König, Chiranjib Mukherjee, and Stefano Olla, editors, *Probability and Analysis in Interacting Physical Systems*, volume 283 of *Springer Proceedings in Mathematics & Statistics*, pages 1–16. Springer, 2019. doi: 10.1007/978-3-030-15338-0_1. URL https://arxiv.org/abs/1803.01950.

[52] Sourav Chatterjee. Rigorous solution of strongly coupled $SO(N)$ lattice gauge theory in the large N limit. *Communications in Mathematical Physics*, 366(1):203–268, 2019. doi: 10.1007/s00220-019-03353-3. URL https://doi.org/10.1007/s00220-019-03353-3.

[53] Sourav Chatterjee. A scaling limit of SU(2) lattice Yang–Mills–Higgs theory. *Probability and Mathematical Physics*, 7(2):339–381, 2026. doi: 10.2140/pmp.2026.7.339. URL https://arxiv.org/abs/2401.10507v4.

[54] Sourav Chatterjee and Jafar Jafarov. The $1/N$ expansion for $SO(N)$ lattice gauge theory at strong coupling, 2016. URL https://inspirehep.net/literature/1448303. Preprint; no published version verified.

[55] Ilya Chevyrev. Yang–Mills Measure on the Two-Dimensional Torus as a Random Distribution. *Communications in Mathematical Physics*, 372(3):1027–1058, 2019. doi: 10.1007/s00220-019-03567-5. URL https://arxiv.org/abs/1808.09196.

[56] Ilya Chevyrev. Stochastic Quantization of Yang–Mills. *Journal of Mathematical Physics*, 63(9):091101, 2022. doi: 10.1063/5.0089431. URL https://arxiv.org/abs/2202.13359.

[57] Ilya Chevyrev and Hao Shen. Invariant Measure and Universality of the 2D Yang–Mills Langevin Dynamic. *Communications on Pure and Applied Mathematics*, 79 (8):1973–2102, 2026. doi: 10.1002/cpa.70043. URL https://arxiv.org/abs/2302.12160v3.

[58] Ilya Chevyrev and Hao Shen. Uniqueness of Gauge Covariant Renormalisation of Stochastic 3D Yang–Mills–Higgs. *Archive for Rational Mechanics and Analysis*, 250

(1):11, 2026. doi: 10.1007/s00205-025-02163-3. URL https://arxiv.org/abs/2503.03060v2.

[59] Ilya Chevyrev, Yahui Qu, and Hao Shen. Scaling limit of the 3D abelian Yang–Mills langevin dynamics, 2026. URL https://arxiv.org/abs/2608.27828v2. Preprint, version 2, 17 September 2026.

[60] B. S. Cirel'son, I. A. Ibragimov, and V. N. Sudakov. Norms of Gaussian sample functions. In Gisiro Maruyama and Jurii V. Prokhorov, editors, *Proceedings of the Third Japan–USSR Symposium on Probability Theory*, volume 550 of *Lecture Notes in Mathematics*, pages 20–41. Springer, 1976. doi: 10.1007/BFb0077482. URL https://doi.org/10.1007/BFb0077482.

[61] Jean-Michel Combes and Lawrence Thomas. Asymptotic Behaviour of Eigenfunctions for Multiparticle Schrödinger Operators. *Communications in Mathematical Physics*, 34(4):251–270, 1973. doi: 10.1007/BF01646473. URL https://doi.org/10.1007/BF01646473.

[62] A. Coste, Antonio González-Arroyo, J. Jurkiewicz, and C. P. Korthals Altes. Zero-momentum contribution to Wilson loops in periodic boxes. *Nuclear Physics B*, 262(1):67–94, 1985. doi: 10.1016/0550-3213(85)90064-1. URL https://doi.org/10.1016/0550-3213(85)90064-1.

[63] Giuseppe Da Prato and Arnaud Debussche. Strong Solutions to the Stochastic Quantization Equations. *The Annals of Probability*, 31(4):1900–1916, 2003. doi: 10.1214/aop/1068646370. URL https://doi.org/10.1214/aop/1068646370.

[64] Paulo A. Faria da Veiga and Michael O'Carroll. On Thermodynamic and Ultraviolet Stability of Yang–Mills, 2019. URL https://arxiv.org/abs/1903.09829v2. Preprint, version 2, 7 October 2019. No journal publication verified as of 16 October 2026.

[65] Paulo A. Faria da Veiga and Michael O'Carroll. On Yang–Mills Stability Bounds and Plaquette Field Generating Function. *Reports on Mathematical Physics*, 95(3):303–380, 2025. doi: 10.1016/S0034-4877(25)00035-7. URL https://arxiv.org/abs/2205.07376v2.

[66] Dennis M. DeTurck. Deforming Metrics in the Direction of Their Ricci Tensors. *Journal of Differential Geometry*, 18(1):157–162, 1983. doi: 10.4310/jdg/1214509286. URL https://doi.org/10.4310/jdg/1214509286.

[67] Jonathan Dimock. The renormalization group according to Balaban. II. Large fields. *Journal of Mathematical Physics*, 54(9):092301, 2013. doi: 10.1063/1.4821275. URL https://doi.org/10.1063/1.4821275.

[68] Jonathan Dimock. The renormalization group according to Balaban—I. small fields. *Reviews in Mathematical Physics*, 25(7):1330010, 2013. doi: 10.1142/S0129055X13300100. URL https://arxiv.org/abs/1108.1335v2. All numbered

and printed-page locators refer to arXiv:1108.1335v2, 21 December 2012, not journal pagination: Appendix B, Theorem 27, p. 39, and connected hard-core graph estimate (323), p. 42. Only the finite graph bound is imported.

[69] Jonathan Dimock. The Renormalization Group According to Balaban III. Convergence. *Annales Henri Poincaré*, 15(11):2133–2175, 2014. doi: 10.1007/s00023-013-0303-3. URL https://doi.org/10.1007/s00023-013-0303-3.

[70] Jonathan Dimock. Covariant Axial Gauge. *Letters in Mathematical Physics*, 105(7):959–987, 2015. doi: 10.1007/s11005-015-0763-0. URL https://doi.org/10.1007/s11005-015-0763-0.

[71] Jonathan Dimock. Nonperturbative renormalization of scalar quantum electrodynamics in $d = 3$. *Journal of Mathematical Physics*, 56(10):102304, 2015. doi: 10.1063/1.4933224. URL https://pubs.aip.org/aip/jmp/issue/56/10.

[72] Jonathan Dimock. Ultraviolet regularity for QED in $d = 3$. *Journal of Mathematical Physics*, 59(1):012301, 2018. doi: 10.1063/1.5009458. URL https://arxiv.org/abs/1512.04373.

[73] Jonathan Dimock. Multiscale block averaging for QED in $d = 3$. *Journal of Mathematical Physics*, 61(3):032302, 2020. doi: 10.1063/1.5134439. URL https://arxiv.org/abs/1712.10029.

[74] Sjoerd Dirksen. Tail Bounds via Generic Chaining. *Electronic Journal of Probability*, 20(53):1–29, 2015. doi: 10.1214/EJP.v20-3760. URL https://arxiv.org/abs/1309.3522.

[75] Simon K. Donaldson. A new proof of a theorem of Narasimhan and Seshadri. *Journal of Differential Geometry*, 18(2):269–277, 1983. doi: 10.4310/jdg/1214437664. URL https://doi.org/10.4310/jdg/1214437664.

[76] Simon K. Donaldson. Anti Self-Dual Yang–Mills Connections over Complex Algebraic Surfaces and Stable Vector Bundles. *Proceedings of the London Mathematical Society*, 50(1):1–26, 1985. doi: 10.1112/plms/s3-50.1.1. URL https://doi.org/10.1112/plms/s3-50.1.1.

[77] Michael R. Douglas. The Yang–Mills millennium problem. *Nature Reviews Physics*, 8(2):86–97, 2026. doi: 10.1038/s42254-025-00909-2. URL https://www.nature.com/articles/s42254-025-00909-2.

[78] Bruce K. Driver. YM_2: Continuum expectations, lattice convergence, and lassos. *Communications in Mathematical Physics*, 123(4):575–616, 1989. doi: 10.1007/BF01218586. URL https://doi.org/10.1007/BF01218586.

[79] Richard M. Dudley. The Sizes of Compact Subsets of Hilbert Space and Continuity of Gaussian Processes. *Journal of Functional Analysis*, 1(3):290–330, 1967. doi: 10.1016/0022-1236(67)90017-1. URL https://doi.org/10.1016/0022-1236(67)90017-1.

[80] Carl Eckart and Gale Young. The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3):211–218, 1936. doi: 10.1007/BF02288367.

[81] Paul Federbush. A phase cell approach to Yang–Mills theory. I. modes, lattice-continuum duality. *Communications in Mathematical Physics*, 107(2):319–329, 1986. doi: 10.1007/BF01209397. URL https://link.springer.com/article/10.1007/BF01209397.

[82] Paul Federbush. A phase cell approach to Yang–Mills theory. III. local stability, modified renormalization group transformation. *Communications in Mathematical Physics*, 110(2):293–309, 1987. doi: 10.1007/BF01207369. URL https://link.springer.com/article/10.1007/BF01207369.

[83] Paul Federbush. A phase cell approach to Yang–Mills theory. VI. non-abelian lattice-continuum duality. *Annales de l'Institut Henri Poincaré, Physique théorique*, 47(1):17–23, 1987. URL https://www.numdam.org/item/AIHPA_1987__47_1_17_0/. Bibliographic record verified in Numdam; no DOI or arXiv identifier located.

[84] Paul Federbush. A phase cell approach to Yang–Mills theory. IV. the choice of variables. *Communications in Mathematical Physics*, 114(2):317–343, 1988. doi: 10.1007/BF01225039. URL https://link.springer.com/article/10.1007/BF01225039.

[85] Paul Federbush. A phase cell approach to Yang–Mills theory. V. analysis of a chunk. *Communications in Mathematical Physics*, 127(3):433–457, 1990. doi: 10.1007/BF02104497. URL https://link.springer.com/article/10.1007/BF02104497.

[86] Paul Federbush and Calvin Williamson. A phase cell approach to Yang–Mills theory. II. analysis of a mode. *Journal of Mathematical Physics*, 28(6):1416–1419, 1987. doi: 10.1063/1.527495. URL https://pubs.aip.org/aip/jmp/issue/28/6.

[87] Roberto Fernández and Aldo Procacci. Cluster Expansion for Abstract Polymer Models. New Bounds from an Old Approach. *Communications in Mathematical Physics*, 274(1):123–140, 2007. doi: 10.1007/s00220-007-0279-2. URL https://doi.org/10.1007/s00220-007-0279-2.

[88] Richard P. Feynman. The qualitative behavior of Yang–Mills theory in 2+1 dimensions. *Nuclear Physics B*, 188(3):479–512, 1981. doi: 10.1016/0550-3213(81)90005-5.

[89] Dana S. Fine. Quantum Yang–Mills on the two-sphere. *Communications in Mathematical Physics*, 134(2):273–292, 1990. doi: 10.1007/BF02097703. URL https://doi.org/10.1007/BF02097703.

[90] Dana S. Fine. Quantum Yang–Mills on a Riemann surface. *Communications in Mathematical Physics*, 140(2):321–338, 1991. doi: 10.1007/BF02099502. URL https://doi.org/10.1007/BF02099502.

[91] Zoltan Fodor, Kieran Holland, Julius Kuti, Daniel Nogradi, and Chik Him Wong. The Yang-Mills gradient flow in finite volume. *Journal of High Energy Physics*, 2012

(11):007, 2012. doi: 10.1007/JHEP11(2012)007. URL https://arxiv.org/abs/1208.1051.

[92] Jürg Fröhlich, Robert Israel, Elliott H. Lieb, and Barry Simon. Phase Transitions and Reflection Positivity. I. General Theory and Long Range Lattice Models. *Communications in Mathematical Physics*, 62(1):1–34, 1978. doi: 10.1007/BF01940327. URL https://doi.org/10.1007/BF01940327.

[93] Jürg Fröhlich, Robert B. Israel, Elliott H. Lieb, and Barry Simon. Phase Transitions and Reflection Positivity. II. Lattice Systems with Short-Range and Coulomb Interactions. *Journal of Statistical Physics*, 22(3):297–347, 1980. doi: 10.1007/BF01014646. URL https://doi.org/10.1007/BF01014646.

[94] Krzysztof Gawędzki and Antti Kupiainen. A rigorous block spin approach to massless lattice theories. *Communications in Mathematical Physics*, 77(1):31–64, 1980. doi: 10.1007/BF01205038. URL https://link.springer.com/article/10.1007/BF01205038.

[95] Krzysztof Gawędzki and Antti Kupiainen. Block spin renormalization group for dipole gas and $(\nabla\varphi)^4$. *Annals of Physics*, 147(1):198–243, 1983. doi: 10.1016/0003-4916(83)90071-4. URL https://www.sciencedirect.com/science/article/pii/0003491683900714.

[96] Krzysztof Gawędzki and Antti Kupiainen. Rigorous renormalization group and asymptotic freedom. In *Scaling and Self-Similarity in Physics*, volume 7 of *Progress in Physics*, pages 227–262. Birkhäuser, 1983. doi: 10.1007/978-1-4899-6762-6_7. URL https://link.springer.com/chapter/10.1007/978-1-4899-6762-6_7.

[97] Krzysztof Gawędzki and Antti Kupiainen. Continuum limit of the hierarchical $O(N)$ non-linear σ-model. *Communications in Mathematical Physics*, 106(4):533–550, 1986. doi: 10.1007/BF01463394. URL https://link.springer.com/article/10.1007/BF01463394.

[98] Paul H. Ginsparg. First order and second order phase transitions in gauge theories at finite temperature. *Nuclear Physics B*, 170:388–408, 1980. doi: 10.1016/0550-3213(80)90418-6.

[99] James Glimm and Arthur Jaffe. *Quantum Physics: A Functional Integral Point of View*. Springer, 2 edition, 1987. doi: 10.1007/978-1-4612-4728-9. URL https://link.springer.com/book/10.1007/978-1-4612-4728-9.

[100] Antonio González-Arroyo. Yang–Mills fields on the 4-dimensional torus. part I: Classical theory, 1998. URL https://arxiv.org/abs/hep-th/9807108. First part of lectures given at the 1997 Advanced School on Non-perturbative Quantum Physics, Peñiscola; cited version: arXiv:hep-th/9807108v1.

[101] Antonio González-Arroyo, J. Jurkiewicz, and C. P. Korthals-Altes. Ground state metamorphosis for Yang–Mills fields on a finite periodic lattice. In J. Honerkamp,

K. Pohlmeyer, and H. Römer, editors, *Structural Elements in Particle Physics and Statistical Mechanics*, pages 339–358. Plenum Press, New York, 1983. doi: 10.1007/978-1-4613-3509-2_25. Freiburg 1981 lectures. Year follows the publisher's original-volume record (1983); the chapter landing page inconsistently displays 1987.

[102] David J. Gross and Frank Wilczek. Ultraviolet behavior of non-abelian gauge theories. *Physical Review Letters*, 30(26):1343–1346, 1973. doi: 10.1103/PhysRevLett.30.1343.

[103] Leonard Gross. Logarithmic Sobolev Inequalities. *American Journal of Mathematics*, 97(4):1061–1083, 1975. doi: 10.2307/2373688. URL https://doi.org/10.2307/2373688.

[104] Leonard Gross. Convergence of $U(1)_3$ Lattice Gauge Theory to Its Continuum Limit. *Communications in Mathematical Physics*, 92(2):137–162, 1983. doi: 10.1007/BF01210842. URL https://doi.org/10.1007/BF01210842.

[105] Leonard Gross. The Yang–Mills heat equation with finite action in three dimensions. *Memoirs of the American Mathematical Society*, 275(1349):v+111, 2022. doi: 10.1090/memo/1349. URL https://doi.org/10.1090/memo/1349.

[106] Leonard Gross, Christopher King, and Ambar Sengupta. Two dimensional Yang–Mills theory via stochastic differential equations. *Annals of Physics*, 194(1):65–112, 1989. doi: 10.1016/0003-4916(89)90032-8. URL https://doi.org/10.1016/0003-4916(89)90032-8.

[107] Martin Hairer. A theory of regularity structures. *Inventiones mathematicae*, 198(2):269–504, 2014. doi: 10.1007/s00222-014-0505-4. URL https://doi.org/10.1007/s00222-014-0505-4.

[108] D. L. Hanson and F. T. Wright. A Bound on Tail Probabilities for Quadratic Forms in Independent Random Variables. *The Annals of Mathematical Statistics*, 42(3):1079–1083, 1971. doi: 10.1214/aoms/1177693335. URL https://doi.org/10.1214/aoms/1177693335.

[109] Bernard Helffer. Remarks on Decay of Correlations and Witten Laplacians: Brascamp–Lieb Inequalities and Semiclassical Limit. *Journal of Functional Analysis*, 155(2):571–586, 1998. doi: 10.1006/jfan.1997.3239. URL https://doi.org/10.1006/jfan.1997.3239.

[110] Bernard Helffer and Johannes Sjöstrand. On the Correlation for Kac-like Models in the Convex Case. *Journal of Statistical Physics*, 74(1–2):349–409, 1994. doi: 10.1007/BF02186817. URL https://doi.org/10.1007/BF02186817.

[111] Robert D. Horowitz. Characters of free groups represented in the two-dimensional special linear group. *Communications on Pure and Applied Mathematics*, 25(6):635–649, 1972. doi: 10.1002/cpa.3160250602. URL https://doi.org/10.1002/cpa.3160250602.

[112] Leon Isserlis. On a formula for the product-moment coefficient of any order of a normal frequency distribution in any number of variables. *Biometrika*, 12(1-2): 134–139, 1918. doi: 10.1093/biomet/12.1-2.134. URL https://doi.org/10.1093/biomet/12.1-2.134.

[113] R. Jackiw and S. Templeton. How super-renormalizable interactions cure their infrared divergences. *Physical Review D*, 23(10):2291–2304, 1981. doi: 10.1103/PhysRevD.23.2291.

[114] Arthur Jaffe and Edward Witten. Quantum Yang–Mills theory. In James Carlson, Arthur Jaffe, and Andrew Wiles, editors, *The Millennium Prize Problems*, pages 129–152. Clay Mathematics Institute and American Mathematical Society, 2006. URL https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf. Published version, pp. 129–152; official Clay PDF also available. No chapter DOI or arXiv identifier verified.

[115] Dimitra Karabali, Chanju Kim, and V. P. Nair. Planar Yang–Mills theory: Hamiltonian, regulators and mass gap. *Nuclear Physics B*, 524:661–694, 1998. doi: 10.1016/S0550-3213(98)00309-5.

[116] Dimitra Karabali, Chanju Kim, and V. P. Nair. On the vacuum wavefunction and string tension of Yang–Mills theories in (2+1) dimensions. *Physics Letters B*, 434 (1–2):103–109, 1998. doi: 10.1016/S0370-2693(98)00751-5.

[117] Christopher King. The U(1) Higgs model. I. the continuum limit. *Communications in Mathematical Physics*, 102(4):649–677, 1986. doi: 10.1007/BF01221651. URL https://link.springer.com/article/10.1007/BF01221651.

[118] Christopher King. The U(1) Higgs model. II. the infinite volume limit. *Communications in Mathematical Physics*, 103(2):323–349, 1986. doi: 10.1007/BF01206942. URL https://link.springer.com/article/10.1007/BF01206942.

[119] Jeffrey Koller and Pierre van Baal. A non-perturbative analysis in finite volume gauge theory. *Nuclear Physics B*, 302(1):1–64, 1988. doi: 10.1016/0550-3213(88)90665-7.

[120] Roman Kotecký and David Preiss. Cluster expansion for abstract polymer models. *Communications in Mathematical Physics*, 103(3):491–498, 1986. doi: 10.1007/BF01211762. URL https://doi.org/10.1007/BF01211762.

[121] Werner Krauth and Matthias Staudacher. Finite Yang-Mills integrals. *Physics Letters B*, 435(3-4):350–355, 1998. doi: 10.1016/S0370-2693(98)00814-4. URL https://arxiv.org/abs/hep-th/9804199.

[122] Werner Krauth, Hermann Nicolai, and Matthias Staudacher. Monte Carlo approach to M-theory. *Physics Letters B*, 431(1-2):31–41, 1998. doi: 10.1016/S0370-2693(98)00557-7. URL https://arxiv.org/abs/hep-th/9803117.

[123] Michel Ledoux. Concentration of Measure and Logarithmic Sobolev Inequalities. In *Séminaire de Probabilités XXXIII*, volume 1709 of *Lecture Notes in Mathematics*, pages 120–216. Springer, 1999. doi: 10.1007/BFb0096511. URL https://www.numdam.org/item/SPS_1999__33__120_0/.

[124] Thierry Lévy. Yang–Mills measure on compact surfaces. *Memoirs of the American Mathematical Society*, 166(790):xiv+122, 2003. doi: 10.1090/memo/0790. URL https://doi.org/10.1090/memo/0790.

[125] Thierry Lévy. Two-dimensional Markovian holonomy fields. *Astérisque*, 329:vi+172, 2010. URL https://www.numdam.org/item/AST_2010__329__R1_0/.

[126] A. D. Linde. Infrared problem in the thermodynamics of the Yang–Mills gas. *Physics Letters B*, 96(3–4):289–292, 1980. doi: 10.1016/0370-2693(80)90769-8.

[127] Martin Lüscher. Some analytic results concerning the mass spectrum of Yang-Mills gauge theories on a torus. *Nuclear Physics B*, 219(1):233–261, 1983. doi: 10.1016/0550-3213(83)90436-4. URL https://doi.org/10.1016/0550-3213(83)90436-4.

[128] Martin Lüscher. Properties and Uses of the Wilson Flow in Lattice QCD. *Journal of High Energy Physics*, 2010(08):071, 2010. doi: 10.1007/JHEP08(2010)071. URL https://arxiv.org/abs/1006.4518. Erratum: Journal of High Energy Physics 2014, no. 03, article 092, doi:10.1007/JHEP03(2014)092.

[129] Martin Lüscher. Trivializing maps, the Wilson flow and the HMC algorithm. *Communications in Mathematical Physics*, 293(3):899–919, 2010. doi: 10.1007/s00220-009-0953-7. URL https://doi.org/10.1007/s00220-009-0953-7.

[130] Martin Lüscher and Gernot Münster. Weak-coupling expansion of the low-lying energy values in the SU(2) gauge theory on a torus. *Nuclear Physics B*, 232(3):445–472, 1984. doi: 10.1016/0550-3213(84)90038-5.

[131] Martin Lüscher and Peter Weisz. Perturbative analysis of the gradient flow in non-abelian gauge theories. *Journal of High Energy Physics*, 2011(02):051, 2011. doi: 10.1007/JHEP02(2011)051. URL https://doi.org/10.1007/JHEP02(2011)051.

[132] Jacques Magnen, Vincent Rivasseau, and Roland Sénéor. Construction of YM_4 with an Infrared Cutoff. *Communications in Mathematical Physics*, 155(2):325–383, 1993. doi: 10.1007/BF02097397. URL https://doi.org/10.1007/BF02097397.

[133] Rajamani Narayanan and Herbert Neuberger. Infinite N phase transitions in continuum Wilson loop operators. *Journal of High Energy Physics*, 2006(03):064, 2006. doi: 10.1088/1126-6708/2006/03/064. URL https://doi.org/10.1088/1126-6708/2006/03/064.

[134] Konrad Osterwalder and Erhard Seiler. Gauge Field Theories on a Lattice. *Annals of Physics*, 110(2):440–471, 1978. doi: 10.1016/0003-4916(78)90039-8. URL https://doi.org/10.1016/0003-4916(78)90039-8.

[135] Giorgio Parisi and Yong-shi Wu. Perturbation theory without gauge fixing. *Scientia Sinica*, 24(4):483–496, 1981. URL https://chimera.roma1.infn.it/GIORGIO/papers.html.

[136] Oliver Penrose. Convergence of fugacity expansions for classical systems. In Thor A. Bak, editor, *Statistical Mechanics: Foundations and Applications*, pages 101–109. W. A. Benjamin, New York, 1967.

[137] H. David Politzer. Reliable perturbative results for strong interactions? *Physical Review Letters*, 30(26):1346–1349, 1973. doi: 10.1103/PhysRevLett.30.1346.

[138] Yu. V. Prokhorov. Convergence of random processes and limit theorems in probability theory. *Theory of Probability and Its Applications*, 1(2):157–214, 1956. doi: 10.1137/1101016. URL https://epubs.siam.org/doi/10.1137/1101016. English translation; Russian original: Teoriya Veroyatnostei i ee Primeneniya 1(2), 177–238 (1956).

[139] Johan Råde. On the Yang–Mills Heat Equation in Two and Three Dimensions. *Journal für die reine und angewandte Mathematik*, 431:123–163, 1992. doi: 10.1515/crll.1992.431.123. URL https://doi.org/10.1515/crll.1992.431.123. Publisher metadata gives pages 123–164, including a blank final verso; substantive text ends at page 163 (as documented in the retained citation audit).

[140] Vincent Rivasseau. *From Perturbative to Constructive Renormalization*. Princeton Series in Physics. Princeton University Press, 1991. doi: 10.1515/9781400862085. URL https://www.degruyterbrill.com/document/doi/10.1515/9781400862085/html. Originally published in 1991; DOI refers to the electronic reissue (2014).

[141] Erhard Seiler. *Gauge Theories as a Problem of Constructive Quantum Field Theory and Statistical Mechanics*, volume 159 of *Lecture Notes in Physics*. Springer, 1982. doi: 10.1007/3-540-11559-5. URL https://link.springer.com/book/10.1007/3-540-11559-5.

[142] Ambar Sengupta. The Yang–Mills measure for S^2. *Journal of Functional Analysis*, 108(2):231–273, 1992. doi: 10.1016/0022-1236(92)90025-E. URL https://doi.org/10.1016/0022-1236(92)90025-E.

[143] Ambar Sengupta. Quantum gauge theory on compact surfaces. *Annals of Physics*, 221(1):17–52, 1993. doi: 10.1006/aphy.1993.1002. URL https://doi.org/10.1006/aphy.1993.1002.

[144] Ambar Sengupta. Gauge Invariant Functions of Connections. *Proceedings of the American Mathematical Society*, 121(3):897–905, 1994. doi: 10.1090/S0002-9939-1994-1215205-7. URL https://doi.org/10.1090/S0002-9939-1994-1215205-7.

[145] Ambar Sengupta. Gauge theory on compact surfaces. *Memoirs of the American Mathematical Society*, 126(600):viii+85, 1997. doi: 10.1090/memo/0600. URL https://doi.org/10.1090/memo/0600.

[146] Hao Shen, Rongchan Zhu, and Xiangchan Zhu. A stochastic analysis approach to lattice Yang–Mills at strong coupling. *Communications in Mathematical Physics*, 400(2):805–851, 2023. doi: 10.1007/s00220-022-04609-1. URL https://doi.org/10.1007/s00220-022-04609-1.

[147] Charles Stein. A bound for the error in the normal approximation to the distribution of a sum of dependent random variables. In Lucien M. Le Cam, Jerzy Neyman, and Elizabeth L. Scott, editors, *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Volume II: Probability Theory*, pages 583–602. University of California Press, 1972. URL https://digicoll.lib.berkeley.edu/record/113245.

[148] V. N. Sudakov and B. S. Tsirel'son. Extremal properties of half-spaces for spherically invariant measures. *Journal of Soviet Mathematics*, 9(1):9–18, 1978. doi: 10.1007/BF01086099. URL https://doi.org/10.1007/BF01086099. English translation. Russian original: Zapiski Nauchnykh Seminarov LOMI 41 (1974), 14–24, 165.

[149] Gerard 't Hooft. A property of electric and magnetic flux in non-abelian gauge theories. *Nuclear Physics B*, 153:141–160, 1979. doi: 10.1016/0550-3213(79)90595-9.

[150] Michael J. Teper. SU(N) gauge theories in 2+1 dimensions. *Physical Review D*, 59(1): 014512, 1999. doi: 10.1103/PhysRevD.59.014512. Issue dated 1 January 1999; first published online 9 December 1998.

[151] Karen K. Uhlenbeck. Connections with L^p Bounds on Curvature. *Communications in Mathematical Physics*, 83(1):31–42, 1982. doi: 10.1007/BF01947069. URL https://doi.org/10.1007/BF01947069.

[152] Pierre van Baal. QCD in a finite volume. In M. Shifman, editor, *At the Frontier of Particle Physics: Handbook of QCD*, volume 2, pages 683–760. World Scientific, 2001. doi: 10.1142/9789812810458_0021.

[153] Kenneth G. Wilson. Confinement of Quarks. *Physical Review D*, 10(8):2445–2459, 1974. doi: 10.1103/PhysRevD.10.2445. URL https://doi.org/10.1103/PhysRevD.10.2445.

[154] Edward Witten. On quantum gauge theories in two dimensions. *Communications in Mathematical Physics*, 141(1):153–209, 1991. doi: 10.1007/BF02100009. URL https://doi.org/10.1007/BF02100009.

[155] Roderick Wong. *Asymptotic Approximations of Integrals*, volume 34 of *Classics in Applied Mathematics*. Society for Industrial and Applied Mathematics, 2001. doi: 10.1137/1.9780898719260. URL https://doi.org/10.1137/1.9780898719260. SIAM reprint of the 1989 Academic Press edition.

[156] C. N. Yang and R. L. Mills. Conservation of Isotopic Spin and Isotopic Gauge Invariance. *Physical Review*, 96(1):191–195, 1954. doi: 10.1103/PhysRev.96.191. URL

https://doi.org/10.1103/PhysRev.96.191.

[157] Daniel Zwanziger. Covariant Quantization of Gauge Fields without Gribov Ambiguity. *Nuclear Physics B*, 192(1):259–269, 1981. doi: 10.1016/0550-3213(81)90202-9. URL https://doi.org/10.1016/0550-3213(81)90202-9.